

EUCLID'S ELEMENTS OF GEOMETRY

The Greek text of J.L. Heiberg (1883–1885)

from *Euclidis Elementa, edidit et Latine interpretatus est I.L. Heiberg, in aedibus
B.G. Teubneri, 1883–1885*

edited, and provided with a modern English translation, by
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Introduction

Euclid's *Elements* is by far the most famous mathematical work of classical antiquity, and also has the distinction of being the world's oldest continuously used mathematical textbook. Little is known about the author, beyond the fact that he lived in Alexandria around 300 BCE. The main subjects of the work are geometry, proportion, and number theory.

Most of the theorems appearing in the *Elements* were not discovered by Euclid himself, but were the work of earlier Greek mathematicians such as Pythagoras (and his school), Hippocrates of Chios, Theaetetus of Athens, and Eudoxus of Cnidos. However, Euclid is generally credited with arranging these theorems in a logical manner, so as to demonstrate (admittedly, not always with the rigour demanded by modern mathematics) that they necessarily follow from five simple axioms. Euclid is also credited with devising a number of particularly ingenious proofs of previously discovered theorems: e.g., Theorem 48 in Book 1.

The geometrical constructions employed in the *Elements* are restricted to those which can be achieved using a straight-rule and a compass. Furthermore, empirical proofs by means of measurement are strictly forbidden: i.e., any comparison of two magnitudes is restricted to saying that the magnitudes are either equal, or that one is greater than the other.

The *Elements* consists of thirteen books. Book 1 outlines the fundamental propositions of plane geometry, including the three cases in which triangles are congruent, various theorems involving parallel lines, the theorem regarding the sum of the angles in a triangle, and the Pythagorean theorem. Book 2 is commonly said to deal with “geometric algebra”, since most of the theorems contained within it have simple algebraic interpretations. Book 3 investigates circles and their properties, and includes theorems on tangents and inscribed angles. Book 4 is concerned with regular polygons inscribed in, and circumscribed around, circles. Book 5 develops the arithmetic theory of proportion. Book 6 applies the theory of proportion to plane geometry, and contains theorems on similar figures. Book 7 deals with elementary number theory: e.g., prime numbers, greatest common denominators, etc. Book 8 is concerned with geometric series. Book 9 contains various applications of results in the previous two books, and includes theorems on the infinitude of prime numbers, as well as the sum of a geometric series. Book 10 attempts to classify incommensurable (i.e., irrational) magnitudes using the so-called “method of exhaustion”, an ancient precursor to integration. Book 11 deals with the fundamental propositions of three-dimensional geometry. Book 12 calculates the relative volumes of cones, pyramids, cylinders, and spheres using the method of exhaustion. Finally, Book 13 investigates the five so-called Platonic solids.

This edition of Euclid's *Elements* presents the definitive Greek text—i.e., that edited by J.L. Heiberg (1883–1885)—accompanied by a modern English translation, as well as a Greek-English lexicon. Neither the spurious books 14 and 15, nor the extensive scholia which have been added to the *Elements* over the centuries, are included. The aim of the translation is to make the mathematical argument as clear and unambiguous as possible, whilst still adhering closely to the meaning of the original Greek. Text within square parenthesis (in both Greek and English) indicates material identified by Heiberg as being later interpolations to the original text (some particularly obvious or unhelpful interpolations have been omitted altogether). Text within round parenthesis (in English) indicates material which is implied, but not actually present, in the Greek text.

My thanks to Mariusz Wodzicki (Berkeley) for typesetting advice, and to Sam Watson & Jonathan Fenno (U. Mississippi), and Gregory Wong (UCSD) for pointing out a number of errors in Book 1.

In the second edition of this work I have opted for more compact (and nicer looking) Greek and English fonts. I have also made some minor improvements and corrections to my translation.

ELEMENTS BOOK 1
*Fundamentals of Plane Geometry Involving
Straight-Lines*

<Ὅροι.

1. Σημεῖόν ἐστιν, οὐ μέρος οὐθέν.
2. Γραμμὴ δὲ μήκος ἀπλατέξ.
3. Γραμμῆς δὲ πέρατα σημεῖα.
4. Εὐθεῖα γραμμὴ ἐστίν, <ἥτις ἐχ>ίσου τοῖς ἐφ> ἐαυτῆς σημεῖοις κεῖται.
5. Ἐπιφάνεια δὲ ἐστίν, <ὃ μήκος καὶ πλάτος μόνον>έθει.
6. Ἐπιφανείας δὲ πέρατα γραμμαί.
7. Ἐπίπεδοξ ἐπιφάνεια ἐστίν, <ἥτις ἐχ>ίσου ταῖς ἐφ> ἐαυτῆς εὐθείαις κεῖται.
8. Ἐπίπεδοξ δὲ γωνία ἐστίν ἢ ἐν ἐπιπέδῳ δύο γραμμῶν ἀπτομένων ἀλλήλων καὶ μὴ ἐπ> εὐθείας κειμένων πρὸς ἀλλήλας τῶν γραμμῶν κλίσις.
9. <Ὅταν δὲ αἱ περιέθουσαι τὴν γωνίαν γραμμαί εὐθεῖαι>ῶσιν, εὐθύγραμμοξ καλεῖται ἡ γωνία.
10. <Ὅταν δὲ εὐθεῖα ἐπ> εὐθεῖαν σταθεῖσα τὰς ἐφεχῆς γωνίας>ῖσας ἀλλήλαις ποιῇ, ὀρθὴ ἑκάτερα τῶν>ῖσων γωνιῶν ἐστί, καὶ ἡ ἐφεστηκυῖα εὐθεῖα κάθετοξ καλεῖται, ἐφ> <ἣν ἐφέστηκεν.
11. Ἀμβλεῖα γωνία ἐστίν ἢ μείζων ὀρθῆς.
12. Ὀχεῖα δὲ ἢ ἐλάσσων ὀρθῆς.
13. <Ὅροξ ἐστίν, <ὃ τινός ἐστι πέραξ.
14. Σθῆμά ἐστι τὸ ὑπὸ τινος>ἢ τινων <όρων περιεχόμενον.
15. Κύκλοξ ἐστὶ σθῆμα ἐπίπεδον ὑπὸ μιᾶς γραμμῆς περιεχόμενον <ἣ καλεῖται περιφέρεια], πρὸς <ἣν ἄφ> ἐνὸς σημείου τῶν ἐντὸς τοῦ σθήματος κειμένων πᾶσαι αἱ προσπίπτουσαι εὐθεῖαι [πρὸς τὴν τοῦ κύκλου περιφέρειαν]>ῖσαι ἀλλήλαις εἰσίν.
16. Κέντρον δὲ τοῦ κύκλου τὸ σημεῖον καλεῖται.
17. Διάμετροξ δὲ τοῦ κύκλου ἐστίν εὐθεῖα τιξ διὰ τοῦ κέντρου ἡγμένη καὶ περατουμένη ἐφ> ἑκάτερα τὰ μέρη ὑπὸ τῆς τοῦ κύκλου περιφέρειας, <ἥτις καὶ δίθα τέμνει τὸν κύκλον.
18. Ἡμικύκλιον δὲ ἐστὶ τὸ περιεχόμενον σθῆμα ὑπὸ τε τῆς διαμέτρου καὶ τῆς ἀπολαμβανομένης ὑπ> αὐτῆς περιφέρειας. κέντρον δὲ τοῦ ἡμικυκλίου τὸ αὐτό, <ὃ καὶ τοῦ κύκλου ἐστίν.
19. Σθήματα εὐθύγραμμά ἐστί τὰ ὑπὸ εὐθειῶν περιεθόμενα, τρίπλευρα μὲν τὰ ὑπὸ τριῶν, τετράπλευρα δὲ τὰ ὑπὸ τεσσάρων, πολύπλευρα δὲ τὰ ὑπὸ πλείονων>ἢ τεσσάρων εὐθειῶν περιεθόμενα.
20. Τῶν δὲ τριπλεύρων σθημάτων ἰσόπλευρον μὲν τρίγωνόν ἐστι τὸ τὰς τρεῖς>ῖσας>έθον πλευράς, ἰσοσκελεξ δὲ τὸ τὰς δύο μόνας>ῖσας>έθον πλευράς, σκαληνὸν δὲ τὸ τὰς τρεῖς ἀνίσουξ>έθον πλευράς.
21. >Ἐτι δὲ τῶν τριπλεύρων σθημάτων ὀρθογώνιον μὲν τρίγωνόν ἐστι τὸ >έθον ὀρθὴν γωνίαν, ἀμβλυγώνιον δὲ τὸ >έθον ἀμβλεῖαν γωνίαν, ὀχυγώνιον δὲ τὸ τὰς τρεῖς ὀχεῖας>έθον γωνίας.
22. Τῶν δὲ τετραπλεύρων σθημάτων τετράγωνον μὲν ἐστίν, <ὃ ἰσόπλευρόν τε ἐστὶ καὶ ὀρθογώνιον, ἐτερόμηκεξ δὲ, <ὃ ὀρθογώνιον μὲν, οὐκ ἰσόπλευρον δὲ, <ρόμβοξ δὲ, <ὃ ἰσόπλευρον μὲν, οὐκ ὀρθογώνιον δὲ, <ρομβοειδέξ δὲ τὸ τὰς ἀπεναντίον πλευράς τε καὶ γωνίας>ῖσας ἀλλήλαις>έθον, <ὃ ο>ὔτε ἰσόπλευρόν ἐστιν ο>ὔτε ὀρθογώνιον; τὰ δὲ παρὰ ταῦτα τετράπλευρα τραπέζια καλεῖσθω.
23. Παράλληλοι εἰσὶν εὐθεῖαι, α<ίτινεξ ἐν τῷ| αὐτῷ| ἐπιπέδῳ ο>ὔσαι καὶ ἐκβαλλόμεναι εἰς>ἄπειρον ἐφ> ἑκάτερα τὰ μέρη ἐπὶ μηδέτερα συμπίπτουσιν ἀλλήλαις.

Definitions

1. A point is that of which there is no part.
2. And a line is a length without breadth.
3. And the extremities of a line are points.
4. A straight-line is (any) one which lies evenly with points on itself.
5. And a surface is that which has length and breadth only.
6. And the extremities of a surface are lines.
7. A plane surface is (any) one which lies evenly with the straight-lines on itself.
8. And a plane angle is the inclination of the lines to one another, when two lines in a plane meet one another, and are not lying in a straight-line.
9. And when the lines containing the angle are straight then the angle is called rectilinear.
10. And when a straight-line stood upon (another) straight-line makes adjacent angles (which are) equal to one another, each of the equal angles is a right-angle, and the former straight-line is called a perpendicular to that upon which it stands.
11. An obtuse angle is one greater than a right-angle.
12. And an acute angle (is) one less than a right-angle.
13. A boundary is that which is the extremity of something.
14. A figure is that which is contained by some boundary or boundaries.
15. A circle is a plane figure contained by a single line [which is called a circumference], (such that) all of the straight-lines radiating towards [the circumference] from one point amongst those lying inside the figure are equal to one another.
16. And the point is called the center of the circle.
17. And a diameter of the circle is any straight-line, being drawn through the center, and terminated in each direction by the circumference of the circle. (And) any such (straight-line) also cuts the circle in half.[†]
18. And a semi-circle is the figure contained by the diameter and the circumference cut off by it. And the center of the semi-circle is the same (point) as (the center of) the circle.
19. Rectilinear figures are those (figures) contained by straight-lines: trilateral figures being those contained by three straight-lines, quadrilateral by four, and multilateral by more than four.
20. And of the trilateral figures: an equilateral triangle is that having three equal sides, an isosceles (triangle) that having only two equal sides, and a scalene (triangle) that having three unequal sides.
21. And further of the trilateral figures: a right-angled triangle is that having a right-angle, an obtuse-angled (triangle) that having an obtuse angle, and an acute-angled (triangle) that having three acute angles.
22. And of the quadrilateral figures: a square is that which is right-angled and equilateral, a rectangle that which is right-angled but not equilateral, a rhombus that which is equilateral but not right-angled, and a rhomboid that having opposite sides and angles equal to one another which is neither right-angled nor equilateral. And let quadrilateral figures besides these be called trapezia.
23. Parallel lines are straight-lines which, being in the same plane, and being produced to infinity in each direction, meet

with one another in neither (of these directions).

† This should really be counted as a postulate, rather than as part of a definition.

Αιτήματα.

1. Ἦιτήσθω ἀπὸ παντὸς σημείου ἐπὶ πᾶν σημεῖον εὐθεῖαν γραμμὴν ἀγαγεῖν.
2. Καὶ πεπερασμένην εὐθεῖαν κατὰ τὸ συνεθὲς ἐπ' εὐθείας ἐκβαλεῖν.
3. Καὶ παντὶ κέντρῳ καὶ διαστήματι κύκλον γράφεσθαι.
4. Καὶ πάσαξ τὰξ ὀρθὰξ γωνίαξ >ίσαξ ἀλλήλαιξ ε>ῖναι.
5. Καὶ ἔαν εἰξ δύο εὐθείας εὐθεῖα ἐμπίπτουσα τὰξ ἐντὸξ καὶ ἐπὶ τὰ αὐτὰ μέρη γωνίαξ δύο ὀρθῶν ἐλάσσοναξ ποιῇ], ἐκβαλλομέναξ τὰξ δύο εὐθείας ἐπ' >άπειρον συμπίπτειν, ἐφ' <α μέρη εἰσὶν αἱ τῶν δύο ὀρθῶν ἐλάσσονεξ.

Postulates

1. Let it be postulated[†] (that it is possible) to draw a straight-line from any point to any point;
2. And to produce a finite straight-line continuously in a straight-line;
3. And to draw a circle with any center and radius.
4. And that all right-angles are equal to one another.
5. And that if a straight-line falling across two (other) straight-lines makes internal angles on the same side (of itself whose sum is) less than two right-angles, (then) the two (other) straight-lines, being produced to infinity, meet on that side (of the original straight-line) that the (sum of the internal angles) is less than two right-angles (and do not meet on the other side).[‡]

† Literally, "let it have been postulated". The Greek present perfect tense indicates a past action with present significance. Hence, the 3rd-person present perfect imperative ἡιτήσθω means "let it be postulated" in the sense "let it stand as postulated", but not "let the postulate be now brought forward". This peculiar Greek idiom is used throughout the Elements.

‡ This postulate effectively specifies that we are dealing with the geometry of flat, rather than curved, space.

Κοινὰ >έννοιαι.

1. Τὰ τῶ| αὐτῶ| >ίσα καὶ ἀλλήλοισι ἐστὶν >ίσα.
2. Καὶ ἔαν >ίσοισι >ίσα προστεθῇ], τὰ <όλα ἐστὶν >ίσα.
3. Καὶ ἔαν ἀπὸ >ίσων >ίσα ἀφαιρεθῇ], τὰ καταλειπόμενά ἐστὶν >ίσα.
4. Καὶ τὰ ἐφαρμόζοντα ἐπ' ἀλλήλα >ίσα ἀλλήλοισι ἐστὶν.
5. Καὶ τὸ <όλον τοῦ μέρους μείζον [ἐστὶν].

Common Notions

1. Things equal to the same thing are also equal to one another.
2. And if equal things are added to equal things (then) the wholes are equal.
3. And if equal things are subtracted from equal things (then) the remainders are equal.[†]
4. And things coinciding with one another are equal to one another.
5. And the whole [is] greater than the part.

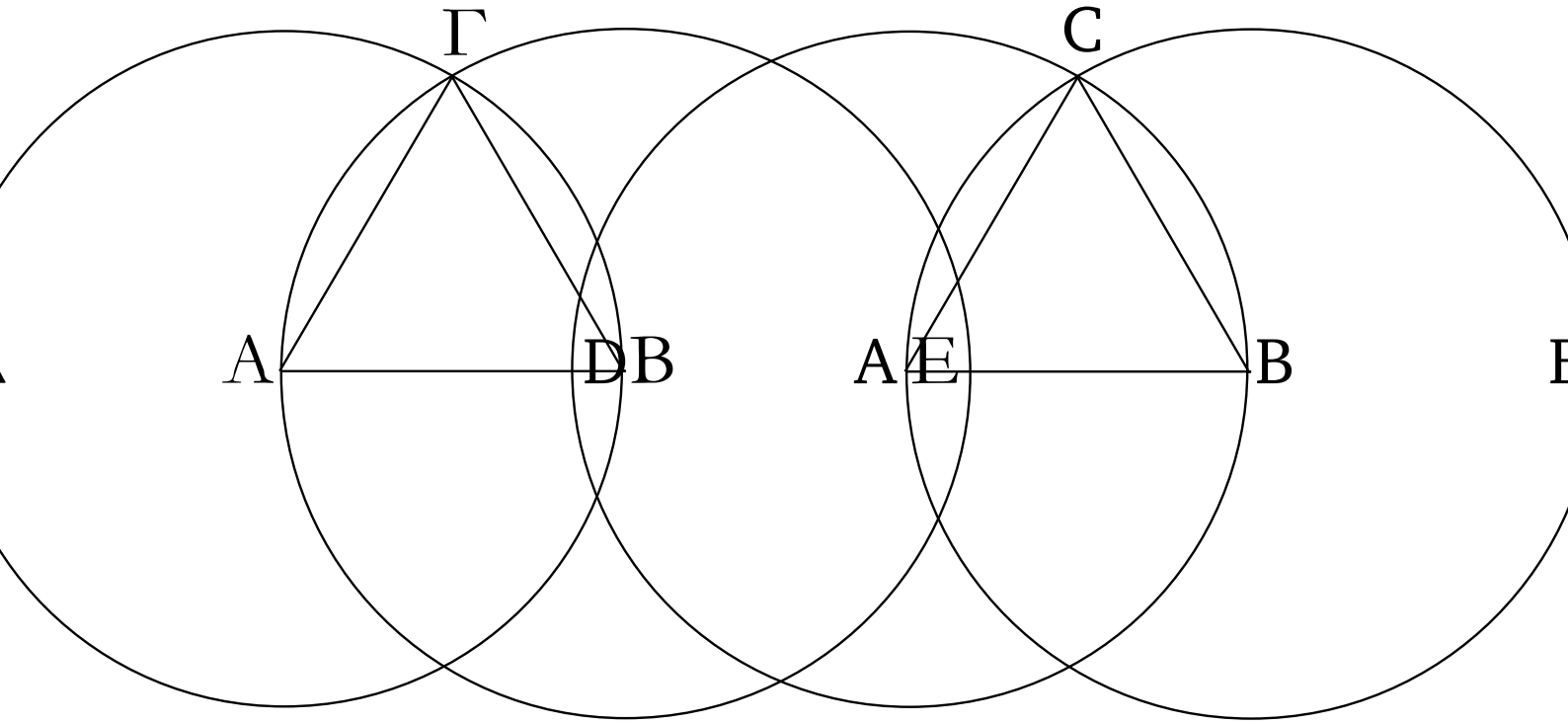
† As an obvious extension of C.N.s 2 & 3—if equal things are added or subtracted from the two sides of an inequality then the inequality remains an inequality of the same type.

1.

Ἐπὶ τῇξ δοθείσῃξ εὐθείας πεπερασμένηξ τρίγωνον ἰσόπλευρον συστήσασθαι.
>Ἐστω ἡ δοθεῖσα εὐθεῖα πεπερασμένη ἡ AB.
Δεῖ δὴ ἐπὶ τῇξ AB εὐθείας τρίγωνον ἰσόπλευρον συστήσασθαι.

Proposition 1

To construct an equilateral triangle on a given finite straight-line.
Let *AB* be the given finite straight-line.
So it is required to construct an equilateral triangle on the straight-line *AB*.



Κέντρον μὲν τῷ A διαστήματι δὲ τῷ AB κύκλος γεγράφθω ὁ BΓΔ, καὶ πάλιν κέντρον μὲν τῷ B διαστήματι δὲ τῷ BA κύκλος γεγράφθω ὁ ΑΓΕ, καὶ ἀπὸ τοῦ Γ σημείου, καθ' ὃ τέμνουσιν ἀλλήλους οἱ κύκλοι, ἐπὶ τὰ Α, Β σημεία ἐπεζεύθωσαν εὐθεῖαι αἱ ΓΑ, ΓΒ.

Καὶ ἐπεὶ τὸ Α σημεῖον κέντρον ἐστὶ τοῦ ΓΔΒ κύκλου, >ίση ἐστὶν ἡ ΑΓ τῇ ΑΒ; πάλιν, ἐπεὶ τὸ Β σημεῖον κέντρον ἐστὶ τοῦ ΓΑΕ κύκλου, >ίση ἐστὶν ἡ ΒΓ τῇ ΒΑ. ἐδείκθη δὲ καὶ ἡ ΓΑ τῇ ΑΒ >ίση; ἑκατέρα >άρα τῶν ΓΑ, ΓΒ τῇ ΑΒ ἐστὶν >ίση. τὰ δὲ τῷ αὐτῷ >ίσα καὶ ἀλλήλοις ἐστὶν >ίσα; καὶ ἡ ΓΑ >άρα τῇ ΓΒ ἐστὶν >ίση; αἱ τρεῖς >άρα αἱ ΓΑ, ΑΒ, ΒΓ >ίσαι ἀλλήλαις εἰσίν.

Ἰσόπλευρον >άρα ἐστὶ τὸ ΑΒΓ τρίγωνον. καὶ συνέσταται ἐπὶ τῇ δοθείσῃ εὐθείᾳ πεπερασμένης τῇ ΑΒ. <ὅπερ >έδει ποιῆσαι.

Let the circle BCD with center A and radius AB be drawn[†] [Post. 3], and again let the circle ACE with center B and radius BA be drawn [Post. 3]. And let the straight-lines CA and CB be joined from the point C , where the circles cut one another,[‡] to the points A and B (respectively) [Post. 1]. And since the point A is the center of the circle CDB , AC is equal to AB [Def. 1.15]. Again, since the point B is the center of the circle CAE , BC is equal to BA [Def. 1.15]. But CA was also shown (to be) equal to AB . Thus, CA and CB are each equal to AB . But things equal to the same thing are also equal to one another [C.N. 1]. Thus, CA is also equal to CB . Thus, the three (straight-lines) CA , AB , and BC are equal to one another.

Thus, the triangle ABC is equilateral, and has been constructed on the given finite straight-line AB . (Which is) the very thing it was required to do.

[†] Literally, "have been drawn". The use of this peculiar Greek idiom will pass without further comment.

[‡] The assumption that the circles do indeed cut one another should be counted as an additional postulate. There is also an implicit assumption that two straight-lines cannot share a common segment.

2.

Πρὸς τῷ δοθέντι σημείῳ τῇ δοθείσῃ εὐθείᾳ >ίσην εὐθεῖαν θέσθαι.

>Ἐστω τὸ μὲν δοθὲν σημεῖον τὸ Α, ἡ δὲ δοθεῖσα εὐθεῖα ἡ ΒΓ; δεῖ δὴ πρὸς τῷ Α σημείῳ τῇ δοθείσῃ εὐθείᾳ τῇ ΒΓ >ίσην εὐθεῖαν θέσθαι.

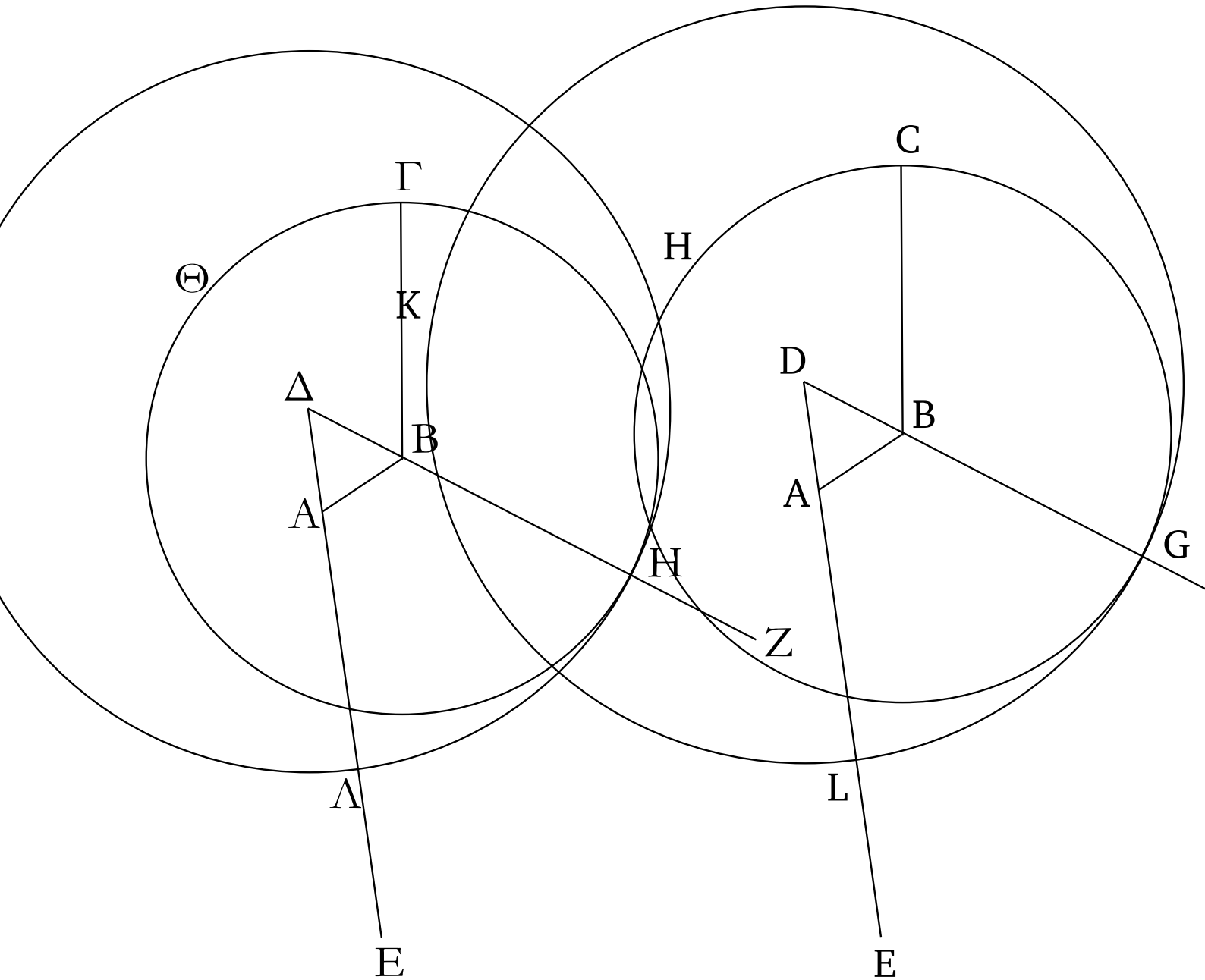
Ἐπεζεύθω γὰρ ἀπὸ τοῦ Α σημείου ἐπὶ τὸ Β σημεῖον εὐθεῖα ἡ ΑΒ, καὶ συνεστάτω ἐπ' αὐτῇ τρίγωνον ἰσόπλευρον τὸ ΔΑΒ, καὶ ἐκβεβλήσθωσαν ἐπ' εὐθείᾳ ταῖς ΔΑ, ΔΒ εὐθεῖαι αἱ ΑΕ, ΒΖ, καὶ κέντρον μὲν τῷ Β διαστήματι δὲ τῷ ΒΓ κύκλος γεγράφθω ὁ ΓΗΘ, καὶ πάλιν κέντρον τῷ Δ καὶ διαστήματι τῷ ΔΗ κύκλος γεγράφθω ὁ ΗΚΛ.

Proposition 2[†]

To place a straight-line equal to a given straight-line at a given point (as an extremity).

Let A be the given point, and BC the given straight-line. So it is required to place a straight-line at point A equal to the given straight-line BC .

For let the straight-line AB be joined from point A to point B [Post. 1], and let the equilateral triangle DAB be constructed upon it [Prop. 1.1]. And let the straight-lines AE and BF be produced in a straight-line with DA and DB (respectively) [Post. 2]. And let the circle CGH with center B and radius BC be drawn [Post. 3], and again let the circle GKL with center D and radius DG be drawn [Post. 3].



Ἐπεὶ ὁὖν τὸ Β σημεῖον κέντρον ἐστὶ τοῦ ΓΗΘ, ὅτι ἐστὶν ἡ ΒΓ τῇ ΒΗ. πάλιν, ἐπεὶ τὸ Δ σημεῖον κέντρον ἐστὶ τοῦ ΗΚΛ κύκλου, ὅτι ἐστὶν ἡ ΔΛ τῇ ΔΗ, ὥν ἡ ΔΑ τῇ ΔΒ ὅτι ἐστὶν. λοιπὴ ἄρα ἡ ΑΛ λοιπῇ τῇ ΒΗ ἐστὶν ὅτι. ἐδείχθη δὲ καὶ ἡ ΒΓ τῇ ΒΗ ὅτι; ἑκάτερα ἄρα τῶν ΑΛ, ΒΓ τῇ ΒΗ ἐστὶν ὅτι. τὰ δὲ τῶν αὐτῶν ὅτι καὶ ἀλλήλοις ἐστὶν ὅτι; καὶ ἡ ΑΛ ἄρα τῇ ΒΓ ἐστὶν ὅτι. Πρὸς ἄρα τῶν δοθέντων σημείων τῶν Α τῇ δοθείσῃ εὐθείᾳ τῇ ΒΓ ὅτι εὐθεῖα κεῖται ἡ ΑΛ; ὅπερ ἔδει ποιῆσαι.

Therefore, since the point B is the center of (the circle) ΓGH , BC is equal to BG [Def. 1.15]. Again, since the point D is the center of the circle ΓKL , DL is equal to DG [Def. 1.15]. And within these, DA is equal to DB . Thus, the remainder AL is equal to the remainder BG [C.N. 3]. But BC was also shown (to be) equal to BG . Thus, AL and BC are each equal to BG . But things equal to the same thing are also equal to one another [C.N. 1]. Thus, AL is also equal to BC .

Thus, the straight-line AL , equal to the given straight-line BC , has been placed at the given point A . (Which is) the very thing it was required to do.

† This proposition admits of a number of different cases, depending on the relative positions of the point A and the line BC . In such situations, Euclid invariably only considers one particular case—usually, the most difficult—and leaves the remaining cases as exercises for the reader.

3.

Proposition 3

Δύο δοθεισῶν εὐθειῶν ἀνίσων ἀπὸ τῆς μείζονος τῇ ἐλάσσονι ὅτι εὐθεῖαν ἀφελεῖν.

>Ἐστωσαν αἱ δοθεῖσαι δύο εὐθεῖαι >άνισοι αἱ ΑΒ, Γ, ὥν

For two given unequal straight-lines, to cut off from the greater a straight-line equal to the lesser.

Let AB and C be the two given unequal straight-lines, of

μείζων >έστω ἡ AB ; δεῖ δὴ ἀπὸ τῆς μείζονος τῆς AB τῇ ἐλάσσονι τῇ Γ >ίσην εὐθεΐαν ἀφελεῖν.

Κείσθω πρὸς τῷ A σημείῳ τῇ Γ εὐθεία >ίση ἡ AD ; καὶ κέντρῳ μὲν τῷ A διαστήματι δὲ τῷ AD κύκλος γεγράφθω ὁ ΔEZ .

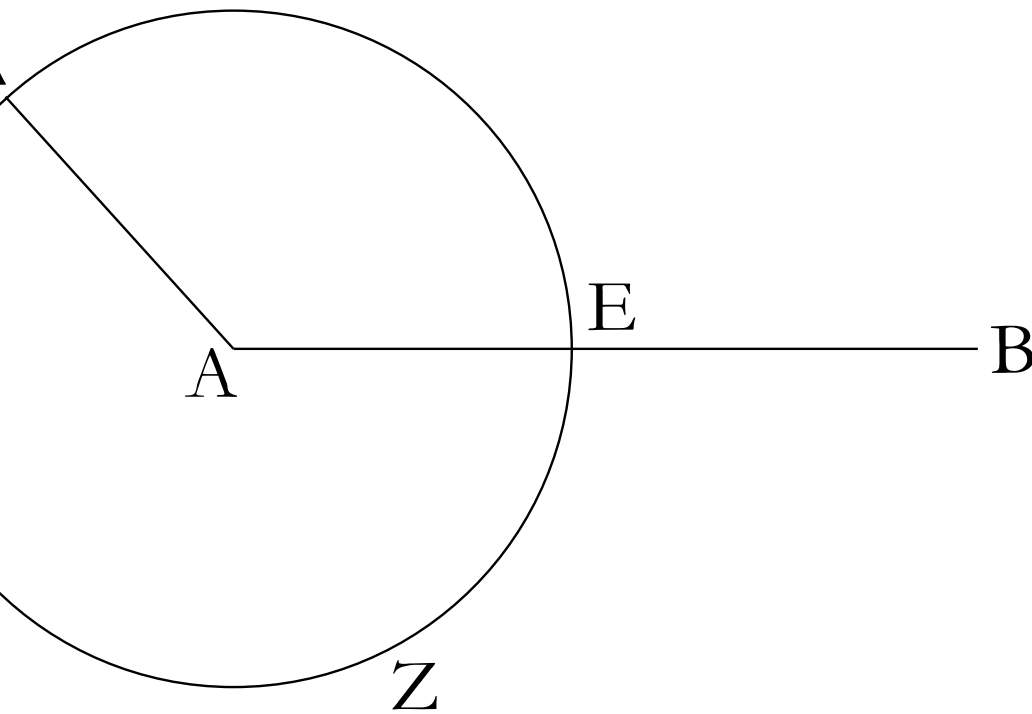
Καὶ ἐπεὶ τὸ A σημεῖον κέντρον ἐστὶ τοῦ ΔEZ κύκλου, >ίση ἐστὶν ἡ AE τῇ AD ; ἀλλὰ καὶ ἡ Γ τῇ AD ἐστὶν >ίση. ἑκατέρα >άρα τῶν AE , Γ τῇ AD ἐστὶν >ίση; <ώστε καὶ ἡ AE τῇ Γ ἐστὶν >ίση.

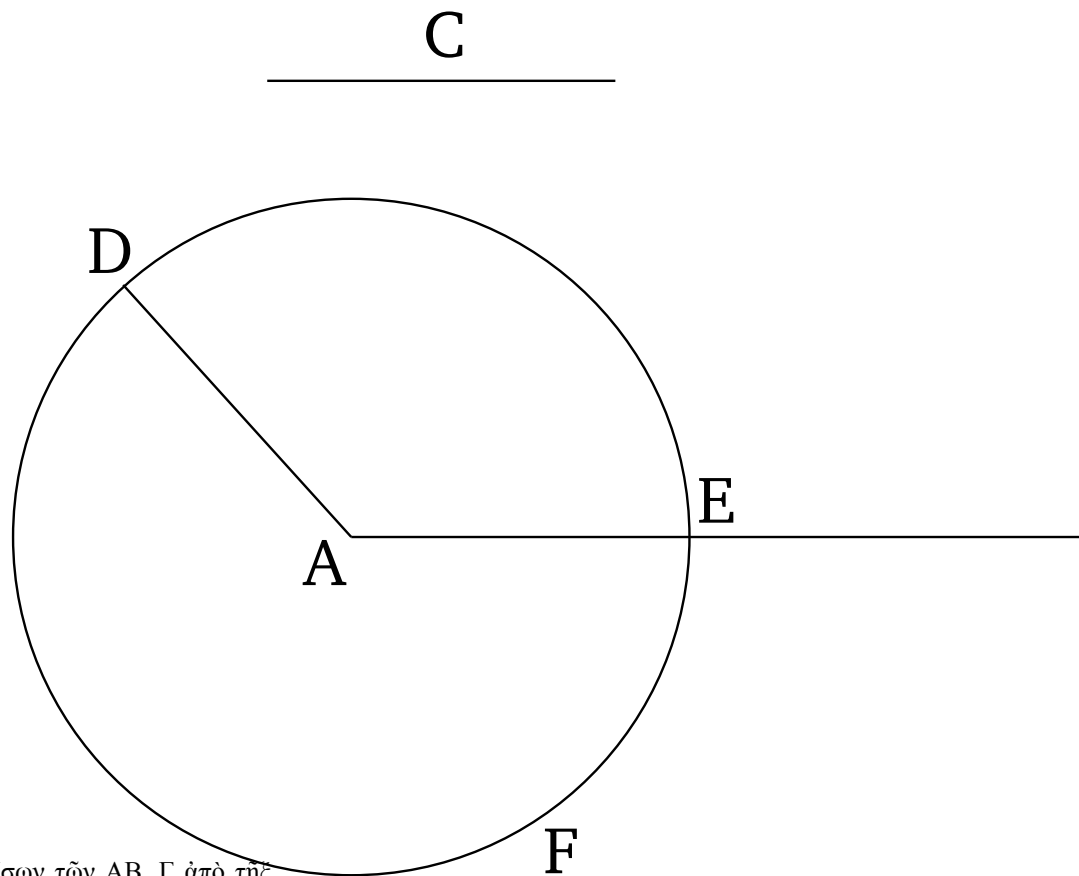
which let the greater be AB . So it is required to cut off a straight-line equal to the lesser C from the greater AB .

Let the line AD , equal to the straight-line C , be placed at point A [Prop. 1.2]. And let the circle DEF be drawn with center A and radius AD [Post. 3].

And since point A is the center of circle DEF , AE is equal to AD [Def. 1.15]. But, C is also equal to AD . Thus, AE and C are each equal to AD . So AE is also equal to C [C.N. 1].

Γ





Δύο >άρα δοθεισῶν εὐθειῶν ἀνίσων τῶν AB , Γ ἀπὸ τῆς μείζονος τῆς AB τῇ ἐλάσσονι τῇ Γ >ίση ἀφή|ρηται ἡ AE ; <όπερ >έδει ποιῆσαι.

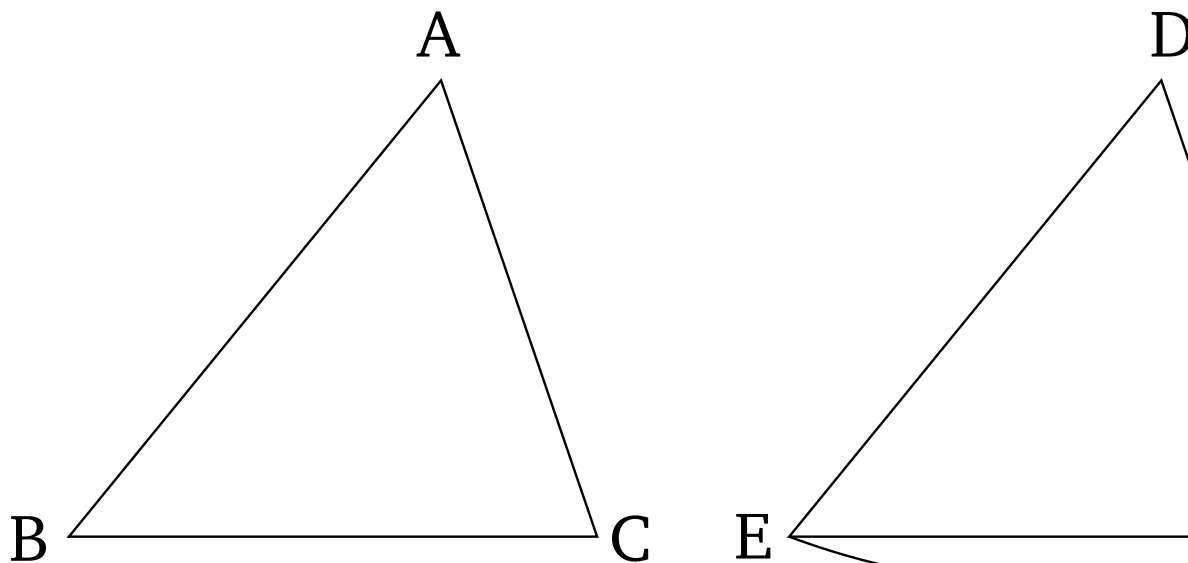
Thus, for two given unequal straight-lines, AB and C , the (straight-line) AE , equal to the lesser C , has been cut off from the greater AB . (Which is) the very thing it was required to do.

4.

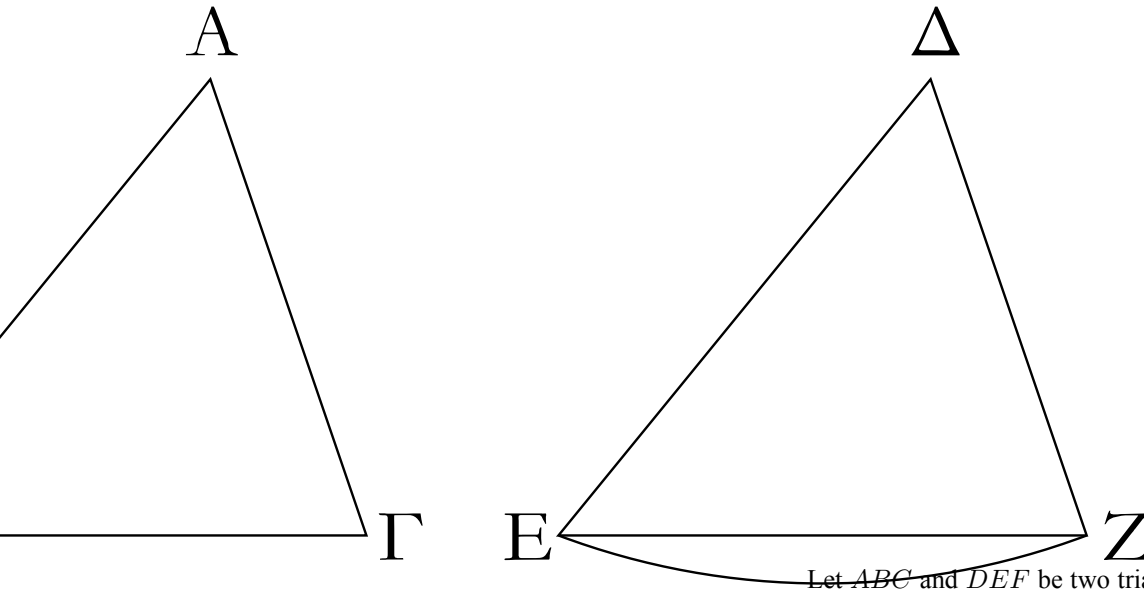
Proposition 4

Ἐὰν δύο τρίγωνα τὰς δύο πλευρὰς [ταῖς] δυσὶ πλευραῖς >ίσαξ >έθῃ ἑκατέραν ἑκατέρα καὶ τὴν γωνίαν τῇ γωνία >ίσην >έθῃ τὴν ὑπὸ τῶν >ίσων εὐθειῶν περιεθομένην, καὶ τὴν βάσιν τῇ βάσει >ίσην <έχει, καὶ τὸ τρίγωνον τῷ τριγώνῳ >ίσον >έσται, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις >ίσαι >έσονται ἑκατέρα ἑκατέρα, ὅφ <ὰς αἱ >ίσαι

If two triangles have two sides equal to two sides, respectively, and have the angle(s) enclosed by the equal straight-lines equal, (then) they will also have the base equal to the base, and the triangle will be equal to the triangle, and the remaining angles subtended by the equal sides will be equal to the corresponding remaining angles.



πλευραὶ ὑποτείνουσιν.



>Ἐστω δύο τρίγωνα τὰ ΑΒΓ, ΔΕΖ τὰς δύο πλευρὰς τὰς ΑΒ, ΑΓ ταῖς δυοῖς πλευραῖς ταῖς ΔΕ, ΔΖ >ῖσαξ >έθοντα ἑκατέραν ἑκατέρῃ τὴν μὲν ΑΒ τῇ ΔΕ τὴν δὲ ΑΓ τῇ ΔΖ καὶ γωνίαν τὴν ὑπὸ ΒΑΓ γωνία τῇ ὑπὸ ΕΔΖ >ῖσην. λέγω, <ὅτι καὶ βάσιξ ἡ ΒΓ βάσει τῇ ΕΖ >ῖση ἔστί, καὶ τὸ ΑΒΓ τρίγωνον τῷ ΔΕΖ τριγώνῳ >ῖσον >έσται, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαιξ >ῖσαι >έσονται ἑκατέρα ἑκατέρῃ, ὅφ <ὰξ αἱ >ῖσαι πλευραὶ ὑποτείνουσιν, ἢ μὲν ὑπὸ ΑΒΓ τῇ ὑπὸ ΔΕΖ, ἢ δὲ ὑπὸ ΑΓΒ τῇ ὑπὸ ΔΖΕ.

Ἐφαρμοζομένου γὰρ τοῦ ΑΒΓ τριγώνου ἐπὶ τὸ ΔΕΖ τρίγωνον καὶ τιθεμένου τοῦ μὲν Α σημείου ἐπὶ τὸ Δ σημείον τῆς δὲ ΑΒ εὐθείας ἐπὶ τὴν ΔΕ, ἐφαρμόσει καὶ τὸ Β σημείον ἐπὶ τὸ Ε διὰ τὸ >ῖσην εἶναι τὴν ΑΒ τῇ ΔΕ; ἐφαρμοσάσης δὲ τῆς ΑΒ ἐπὶ τὴν ΔΕ ἐφαρμόσει καὶ ἡ ΑΓ εὐθεῖα ἐπὶ τὴν ΔΖ διὰ τὸ >ῖσην εἶναι τὴν ὑπὸ ΒΑΓ γωνίαν τῇ ὑπὸ ΕΔΖ; <ώστε καὶ τὸ Γ σημείον ἐπὶ τὸ Ζ σημείον ἐφαρμόσει διὰ τὸ >ῖσην πάλιν εἶναι τὴν ΑΓ τῇ ΔΖ. ἀλλὰ μὴν καὶ τὸ Β ἐπὶ τὸ Ε ἐφηρμόκει; <ώστε βάσιξ ἡ ΒΓ ἐπὶ βάσιν τὴν ΕΖ ἐφαρμόσει. εἰ γὰρ τοῦ μὲν Β ἐπὶ τὸ Ε ἐφαρμόσαντοξ τοῦ δὲ Γ ἐπὶ τὸ Ζ ἢ ΒΓ βάσιξ ἐπὶ τὴν ΕΖ οὐκ ἐφαρμόσει, δύο εὐθεῖαι θωρίον περιέχουσιν; <όπερ ἔστιν ἀδύνατον. ἐφαρμόσει >άρα ἡ ΒΓ βάσιξ ἐπὶ τὴν ΕΖ καὶ >ῖση αὐτῇ >έσται; <ώστε καὶ <όλον τὸ ΑΒΓ τρίγωνον ἐπὶ <όλον τὸ ΔΕΖ τρίγωνον ἐφαρμόσει καὶ >ῖσον αὐτῷ >έσται, καὶ αἱ λοιπαὶ γωνίαι ἐπὶ τὰς λοιπὰς γωνίαξ ἐφαρμόσουσι καὶ >ῖσαι αὐταῖς >έσονται, ἢ μὲν ὑπὸ ΑΒΓ τῇ ὑπὸ ΔΕΖ ἢ δὲ ὑπὸ ΑΓΒ τῇ ὑπὸ ΔΖΕ.

Ἐὰν >άρα δύο τρίγωνα τὰς δύο πλευρὰς [ταῖς] δύο πλευραῖς >ῖσαξ >έθῃ ἑκατέραν ἑκατέρῃ καὶ τὴν γωνίαν τῇ γωνία >ῖσην >έθῃ τὴν ὑπὸ τῶν >ῖσων εὐθειῶν περιεθρομένην, καὶ τὴν βάσιν τῇ βάσει >ῖσην <έχει, καὶ τὸ τρίγωνον τῷ τριγώνῳ >ῖσον >έσται, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαιξ >ῖσαι >έσονται ἑκατέρα ἑκατέρῃ, ὅφ <ὰξ αἱ >ῖσαι πλευραὶ ὑποτείνουσιν; <όπερ >έδει δεῖξαι.

† The application of one figure to another should be counted as an additional postulate.

‡ Since Post. 1 implicitly assumes that the straight-line joining two given points is unique.

Let ABC and DEF be two triangles having the two sides AB and AC equal to the two sides DE and DF , respectively. (That is) AB to DE , and AC to DF . And (let) the angle BAC (be) equal to the angle EDF . I say that the base BC is also equal to the base EF , and triangle ABC will be equal to triangle DEF , and the remaining angles subtended by the equal sides will be equal to the corresponding remaining angles. (That is) ABC to DEF , and ACB to DFE .

For if triangle ABC is applied to triangle DEF ,[†] the point A being placed on the point D , and the straight-line AB on DE , (then) the point B will also coincide with E , on account of AB being equal to DE . So (because of) AB coinciding with DE , the straight-line AC will also coincide with DF , on account of the angle BAC being equal to EDF . So the point C will also coincide with the point F , again on account of AC being equal to DF . But, point B certainly also coincided with point E , so that the base BC will coincide with the base EF . For if B coincides with E , and C with F , and the base BC does not coincide with EF , (then) two straight-lines will encompass an area. The very thing is impossible [Post. 1].[‡] Thus, the base BC will coincide with EF , and will be equal to it [C.N. 4]. So the whole triangle ABC will coincide with the whole triangle DEF , and will be equal to it [C.N. 4]. And the remaining angles will coincide with the remaining angles, and will be equal to them [C.N. 4]. (That is) ABC to DEF , and ACB to DFE [C.N. 4].

Thus, if two triangles have two sides equal to two sides, respectively, and have the angle(s) enclosed by the equal straight-lines equal, (then) they will also have the base equal to the base, and the triangle will be equal to the triangle, and the remaining angles subtended by the equal sides will be equal to the corresponding remaining angles. (Which is) the very thing it was required to show.

5.

Τῶν ἰσοσκελῶν τριγώνων αἱ πρὸξ τῇ βάσει γωνίαι >ῖσαι ἀλλήλαιξ εἰσίν, καὶ προσεκβληθειςῶν τῶν >ῖσων εὐθειῶν αἱ ὑπὸ τὴν βάσιν γωνίαι >ῖσαι ἀλλήλαιξ >έσονται.

>Ἐστω τρίγωνον ἰσοσκελεξ τὸ ΑΒΓ >ῖσην >έθον τὴν ΑΒ

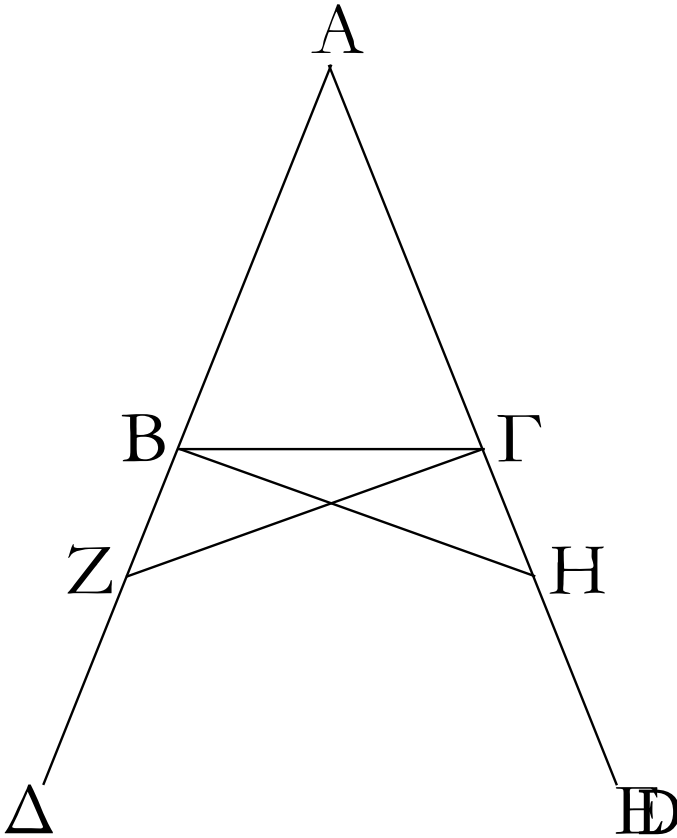
Proposition 5

For isosceles triangles, the angles at the base are equal to one another, and if the equal sides are produced (then) the angles under the base will be equal to one another.

Let ABC be an isosceles triangle having the side AB equal

πλευρὰν τῇ $ΑΓ$ πλευρᾷ, καὶ προσεκβεβλήσθωσαν ἐπ' εὐθείᾳ ταῖς $ΑΒ$, $ΑΓ$ εὐθεῖαι αἱ $ΒΔ$, $ΓΕ$; λέγω, <ὅτι ἡ μὲν ὑπὸ $ΑΒΓ$ γωνία τῇ ὑπὸ $ΑΓΒ$ >ίση ἐστίν, ἡ δὲ ὑπὸ $ΓΒΔ$ τῇ ὑπὸ $ΒΓΕ$.

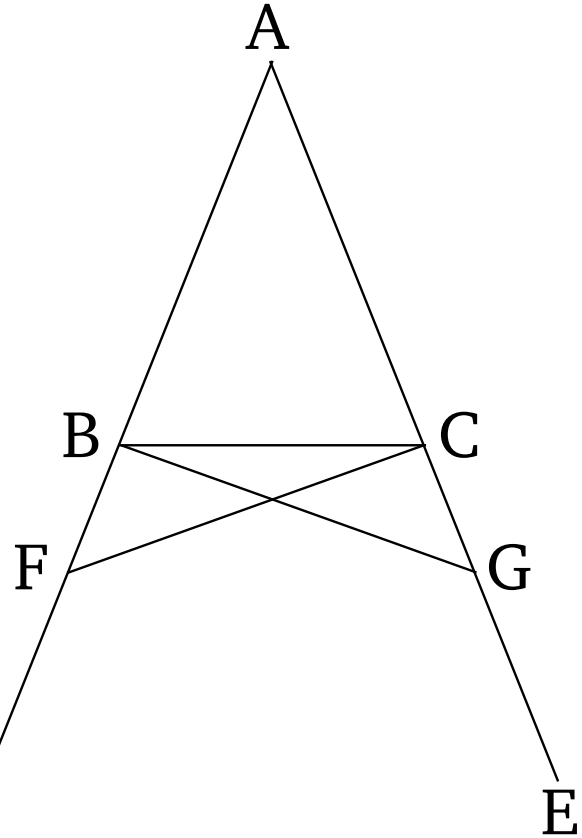
Εἰλήφθω γὰρ ἐπὶ τῇ $ΒΔ$ τυθὸν σημεῖον τὸ $Ζ$, καὶ ἀφηρήσθω ἀπὸ τῇς μείζονος τῇς $ΑΕ$ τῇ ἐλάσσονι τῇ $ΑΖ$ >ίση ἡ $ΑΗ$, καὶ ἐπεξεύθθωσαν αἱ $ΖΓ$, $ΗΒ$ εὐθεῖαι.



Ἐπεὶ οὖν >ίση ἐστίν ἡ μὲν $ΑΖ$ τῇ $ΑΗ$ ἡ δὲ $ΑΒ$ τῇ $ΑΓ$, δύο δὲ αἱ $ΖΑ$, $ΑΓ$ δυοὶ ταῖς $ΗΑ$, $ΑΒ$ >ίσαι εἰσὶν ἑκατέρω ἑκατέρω; καὶ γωνίαν κοινὴν περιέθουσι τὴν ὑπὸ $ΖΑΗ$; βάσεις >άρα ἡ $ΖΓ$ βάσει τῇ $ΗΒ$ >ίση ἐστίν, καὶ τὸ $ΑΖΓ$ τρίγωνον τῷ $ΑΗΒ$ τριγώνῳ >ίσον >έσται, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις >ίσαι >έσονται ἑκατέρα ἑκατέρω, ὅφ' <ὰς αἱ >ίσαι πλευραὶ ὑποτείνουσιν, ἡ μὲν ὑπὸ $ΑΓΖ$ τῇ ὑπὸ $ΑΒΗ$, ἡ δὲ ὑπὸ $ΑΖΓ$ τῇ ὑπὸ $ΑΗΒ$. καὶ ἐπεὶ <ὅλη ἡ $ΑΖ$ <ὅλη τῇ $ΑΗ$ ἐστίν >ίση, <ὢν ἡ $ΑΒ$ τῇ $ΑΓ$ ἐστίν >ίση, λοιπὴ >άρα ἡ $ΒΖ$ λοιπῇ τῇ $ΓΗ$ ἐστίν >ίση. ἐδείκθη δὲ καὶ ἡ $ΖΓ$ τῇ $ΗΒ$ >ίση; δύο δὲ αἱ $ΒΖ$, $ΖΓ$ δυοὶ ταῖς $ΓΗ$, $ΗΒ$ >ίσαι εἰσὶν ἑκατέρα ἑκατέρω; καὶ γωνία ἡ ὑπὸ $ΒΖΓ$ γωνία τῇ ὑπὸ $ΓΗΒ$ >ίση, καὶ βάσις αὐτῶν κοινὴ ἡ $ΒΓ$; καὶ τὸ $ΒΖΓ$ >άρα τρίγωνον τῷ $ΓΗΒ$ τριγώνῳ >ίσον >έσται, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις >ίσαι >έσονται ἑκατέρα ἑκατέρω, ὅφ' <ὰς αἱ >ίσαι πλευραὶ ὑποτείνουσιν; >ίση >άρα ἐστίν ἡ μὲν ὑπὸ $ΒΖΓ$ τῇ ὑπὸ $ΗΓΒ$ ἡ δὲ ὑπὸ $ΒΓΖ$ τῇ ὑπὸ $ΓΒΗ$. ἐπεὶ οὖν <ὅλη ἡ ὑπὸ $ΑΒΗ$ γωνία <ὅλη τῇ ὑπὸ $ΑΓΖ$ γωνία ἐδείκθη >ίση, <ὢν ἡ ὑπὸ $ΓΒΗ$ τῇ ὑπὸ $ΒΓΖ$ >ίση, λοιπὴ >άρα ἡ ὑπὸ $ΑΒΓ$ λοιπῇ τῇ ὑπὸ $ΑΓΒ$ ἐστίν >ίση; καὶ εἰσι πρὸς τῇ βάσει τοῦ $ΑΒΓ$ τριγώνου. ἐδείκθη δὲ καὶ ἡ ὑπὸ $ΒΖΓ$ τῇ ὑπὸ $ΗΓΒ$ >ίση; καὶ εἰσιν ὑπὸ τὴν βάσιν. Τῶν >άρα ἰσοσκελῶν τριγώνων αἱ πρὸς τῇ βάσει γωνίαι >ίσαι ἀλλήλαις εἰσὶν, καὶ προσεκβληθεῖσιν τῶν >ίσων εὐθειῶν αἱ ὑπὸ τὴν βάσιν γωνίαι >ίσαι ἀλλήλαις >έσονται; <ὅπερ >έδει δεῖχαι.

to the side AC , and let the straight-lines BD and CE be produced in a straight-line with AB and AC (respectively) [Post. 2]. I say that the angle ABC is equal to ACB , and (angle) CBD to BCE .

For let the point F be taken at random on BD , and let AG be cut off from the greater AE , equal to the lesser AF [Prop. 1.3]. Also, let the straight-lines FC and GB be joined [Post. 1].



In fact, since AF is equal to AG , and AB to AC , the two (straight-lines) FA , AC are equal to the two (straight-lines) GA , AB , respectively. They also encompass a common angle, FAG . Thus, the base FC is equal to the base GB , and the triangle AFC will be equal to the triangle AGB , and the remaining angles subtended by the equal sides will be equal to the corresponding remaining angles [Prop. 1.4]. (That is) ACF to ABG , and AFC to AGB . And since the whole of AF is equal to the whole of AG , within which AB is equal to AC , the remainder BF is thus equal to the remainder CG [C.N. 3]. But FC was also shown (to be) equal to GB . So the two (straight-lines) BF , FC are equal to the two (straight-lines) CG , GB , respectively, and the angle BFC (is) equal to the angle CGB , and the base BC is common to them. Thus, the triangle BFC will be equal to the triangle CGB , and the remaining angles subtended by the equal sides will be equal to the corresponding remaining angles [Prop. 1.4]. Thus, FBC is equal to GCB , and BCF to CBG . Therefore, since the whole angle ABG was shown (to be) equal to the whole angle ACF , within which CBG is equal to BCF , the remainder ABC is thus equal to the remainder ACB [C.N. 3]. And they are at the base of triangle ABC . And FBC was also shown (to be) equal to GCB . And they are under the base.

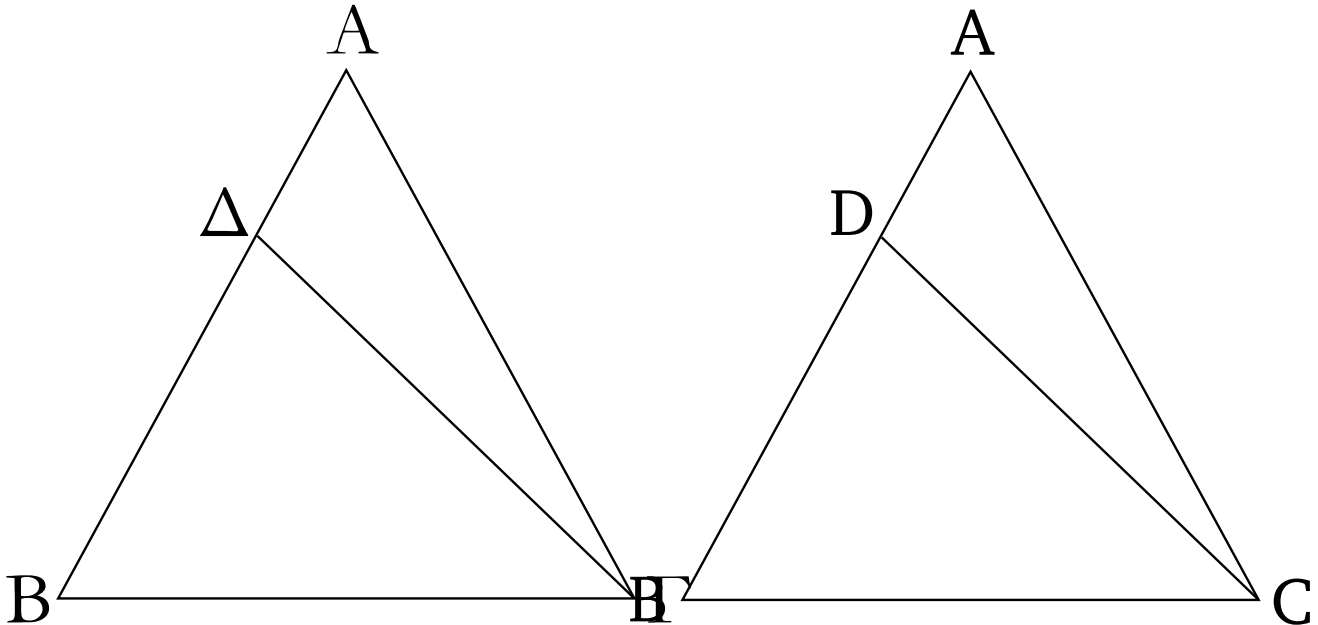
Thus, for isosceles triangles, the angles at the base are equal to one another, and if the equal sides are produced (then) the

angles under the base will be equal to one another. (Which is) the very thing it was required to show.

6.

Ἐάν τριγώνου αἱ δύο γωνίαι >ίσαι ἀλλήλαις >ῶσιν, καὶ αἱ ὑπὸ τὰς >ίσαξ γωνίαξ ὑποτείνουσαι πλευραὶ >ίσαι ἀλλήλαις >έσονται.

>Ἐστω τρίγωνον τὸ ΑΒΓ >ίσην >έθον τὴν ὑπὸ ΑΒΓ γωνίαν τῇ ὑπὸ ΑΓΒ γωνίᾳ; λέγω, <ὅτι καὶ πλευρὰ ἡ ΑΒ πλευρᾷ τῇ ΑΓ ἐστὶν >ίση.



Εἰ γὰρ >άνισός ἐστιν ἡ ΑΒ τῇ ΑΓ, ἡ ἑτέρα αὐτῶν μείζων ἐστίν. >έστω μείζων ἡ ΑΒ, καὶ ἀφηρήσθω ἀπὸ τῆς μείζονος τῆς ΑΒ τῇ ἐλάττω τῇ ΑΓ >ίση ἡ ΔΒ, καὶ ἐπεζεύθω ἡ ΔΓ. Ἐπεὶ οὖν >ίση ἐστὶν ἡ ΔΒ τῇ ΑΓ κοινὴ δὲ ἡ ΒΓ, δύο δὲ αἱ ΔΒ, ΒΓ δύο ταῖς ΑΓ, ΓΒ >ίσαι εἰσὶν ἑκατέρα ἑκατέρα, καὶ γωνία ἡ ὑπὸ ΔΒΓ γωνία τῇ ὑπὸ ΑΓΒ ἐστὶν >ίση; βάσει >άρα ἡ ΔΓ βάσει τῇ ΑΒ >ίση ἐστίν, καὶ τὸ ΔΒΓ τρίγωνον τῷ ΑΓΒ τριγώνῳ >ίσον >έσται, τὸ >έλασσον τῷ μείζονι; <όπερ >άτοπον; οὐκ >άρα >άνισός ἐστιν ἡ ΑΒ τῇ ΑΓ; >ίση >άρα.

Ἐάν >άρα τριγώνου αἱ δύο γωνίαι >ίσαι ἀλλήλαις >ῶσιν, καὶ αἱ ὑπὸ τὰς >ίσαξ γωνίαξ ὑποτείνουσαι πλευραὶ >ίσαι ἀλλήλαις >έσονται; <όπερ >έδει δεῖχαι.

Proposition 6

If a triangle has two angles equal to one another (then) the sides subtending the equal angles will also be equal to one another.

Let ABC be a triangle having the angle ABC equal to the angle ACB . I say that side AB is also equal to side AC .

For if AB is unequal to AC (then) one of them is greater. Let AB be greater. And let DB , equal to the lesser AC , be cut off from the greater AB [Prop. 1.3]. And let DC be joined [Post. 1].

Therefore, since DB is equal to AC , and BC (is) common, the two sides DB , BC are equal to the two sides AC , CB , respectively, and the angle DBC is equal to the angle ACB . Thus, the base DC is equal to the base AB , and the triangle DBC will be equal to the triangle ACB [Prop. 1.4], the lesser to the greater. The very notion (is) absurd [C.N. 5]. Thus, AB is not unequal to AC . Thus, (it is) equal.[†]

Thus, if a triangle has two angles equal to one another (then) the sides subtending the equal angles will also be equal to one another. (Which is) the very thing it was required to show.

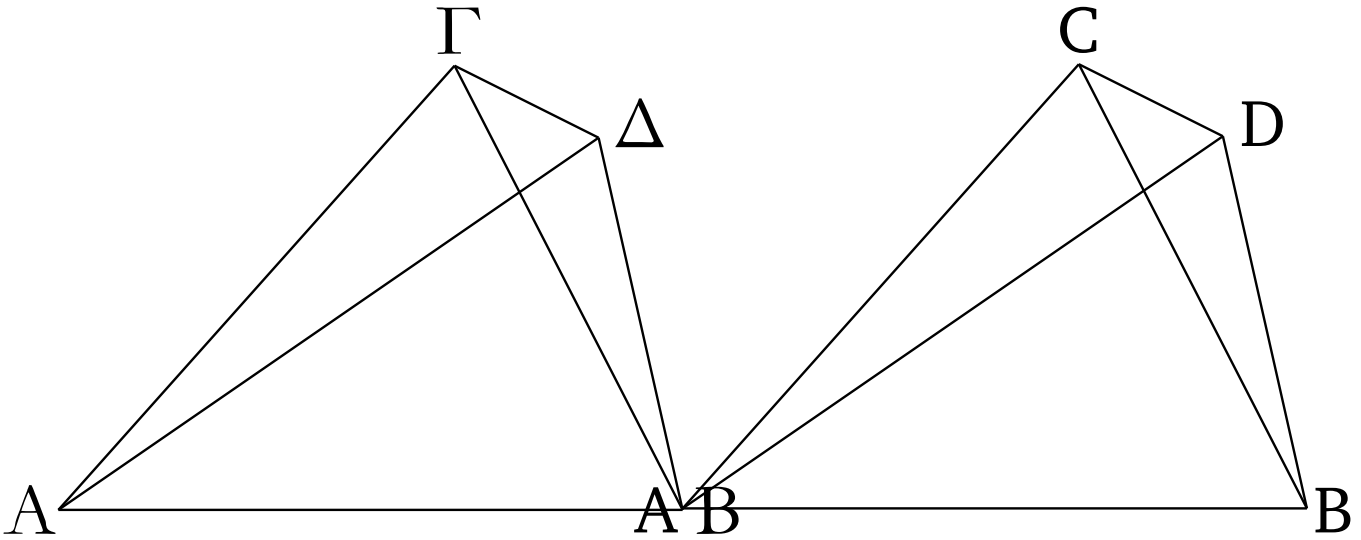
[†] Here, use is made of the previously unmentioned common notion that if two quantities are not unequal then they must be equal. Later on, use is made of the closely related common notion that if two quantities are not greater than or less than one another, respectively, then they must be equal to one another.

7.

Ἐπὶ τῆς αὐτῆς εὐθείας δύο ταῖς αὐταῖς εὐθείαις >άλλαι δύο εὐθεῖαι >ίσαι ἑκατέρα ἑκατέρα οὐ συσταθήσονται πρὸς >άλλω καὶ >άλλω σημείῳ ἐπὶ τὰ αὐτὰ μέρη τὰ αὐτὰ πέρατα >έθουσαι ταῖς ἐκ ἀρθῆς εὐθείαις.

Proposition 7

On the same straight-line, two other straight-lines equal, respectively, to two (given) straight-lines (which meet) cannot be constructed (meeting) at a different point on the same side (of the straight-line), but having the same ends as the given straight-lines.



Εἰ γὰρ δυνατόν, ἐπὶ τῇ αὐτῇ εὐθείᾳ τῇ AB δύο ταῖς αὐταῖς εὐθείαις ταῖς AG , GB ἄλλαι δύο εὐθεῖαι αἱ AD , DB ἴσαι ἑκατέρα ἑκατέρᾳ συνεστάνωσαν πρὸς ἄλλῃ καὶ ἄλλῃ σημείῳ τῷ τε Γ καὶ Δ ἐπὶ τὰ αὐτὰ μέρη τὰ αὐτὰ πέρατα ἔθουσιν, ὥστε ἴσην εἶναι τὴν μὲν GA τῇ DA τὸ αὐτὸ πέραξ ἔθουσιν αὐτῇ τὸ A , τὴν δὲ GB τῇ DB τὸ αὐτὸ πέραξ ἔθουσιν αὐτῇ τὸ B , καὶ ἐπεξεύθω ἡ GD .

Ἐπεὶ οὖν ἴση ἐστὶν ἡ AG τῇ AD , ἴση ἐστὶ καὶ γωνία ἡ ὑπὸ AGD τῇ ὑπὸ ADG ; μείζων ἄρα ἡ ὑπὸ ADG τῇ ὑπὸ AGB ; πολλῶν ἄρα ἡ ὑπὸ GAB μείζων ἐστὶ τῇ ὑπὸ AGB . πάλιν ἐπεὶ ἴση ἐστὶν ἡ GB τῇ DB , ἴση ἐστὶ καὶ γωνία ἡ ὑπὸ GAB γωνία τῇ ὑπὸ ADB . ἐδείκθη δὲ αὐτῇ καὶ πολλῶν μείζων; ὅπερ ἐστὶν ἀδύνατον.

Οὐκ ἄρα ἐπὶ τῇ αὐτῇ εὐθείᾳ δύο ταῖς αὐταῖς εὐθείαις ἄλλαι δύο εὐθεῖαι ἴσαι ἑκατέρα ἑκατέρᾳ συσταθήσονται πρὸς ἄλλῃ καὶ ἄλλῃ σημείῳ ἐπὶ τὰ αὐτὰ μέρη τὰ αὐτὰ πέρατα ἔθουσιν ταῖς ἐκ ἀρθῆς εὐθείαις; ὅπερ ἔδει δεῖξαι.

For, if possible, let the two straight-lines AC , CB , equal to two other straight-lines AD , DB , respectively, be constructed on the same straight-line AB , meeting at different points, C and D , on the same side (of AB), and having the same ends (on AB). So CA is equal to DA , having the same end A as it, and CB is equal to DB , having the same end B as it. And let CD be joined [Post. 1].

Therefore, since AC is equal to AD , the angle ACD is also equal to angle ADC [Prop. 1.5]. Thus, ADC (is) greater than DCB [C.N. 5]. Thus, CDB is much greater than DCB [C.N. 5]. Again, since CB is equal to DB , the angle CDB is also equal to angle DCB [Prop. 1.5]. But it was shown that the former (angle) is also much greater (than the latter). The very thing is impossible.

Thus, on the same straight-line, two other straight-lines equal, respectively, to two (given) straight-lines (which meet) cannot be constructed (meeting) at a different point on the same side (of the straight-line), but having the same ends as the given straight-lines. (Which is) the very thing it was required to show.

8.

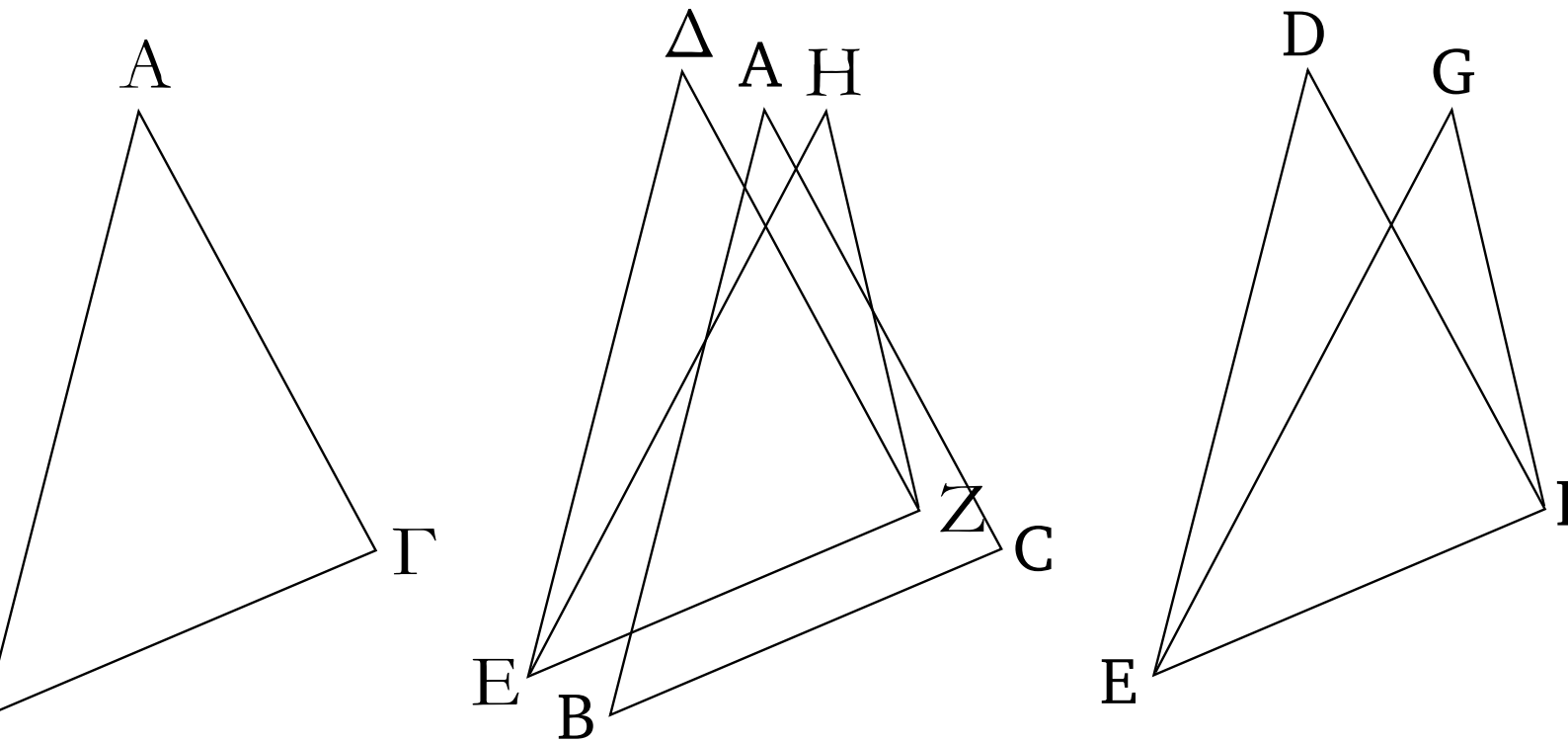
Ἐὰν δύο τρίγωνα τὰς δύο πλευρὰς [ταῖς] δύο πλευραῖς ἴσαξ ἔθῃ ἑκατέραν ἑκατέρᾳ, ἔθῃ δὲ καὶ τὴν βάσιν τῇ βάσει ἴσην, καὶ τὴν γωνίαν τῇ γωνίᾳ ἴσην ἔχει τὴν ὑπὸ τῶν ἴσων εὐθειῶν περιεθρομένην.

>Ἐστω δύο τρίγωνα τὰ $AB\Gamma$, ΔEZ τὰς δύο πλευρὰς τὰς AB , $A\Gamma$ ταῖς δύο πλευραῖς ταῖς ΔE , ΔZ ἴσαξ ἔθοντα ἑκατέραν ἑκατέρᾳ, τὴν μὲν AB τῇ ΔE τὴν δὲ $A\Gamma$ τῇ ΔZ ; ἐθέτω δὲ καὶ βάσιν τὴν $B\Gamma$ βάσει τῇ EZ ἴσην; λέγω, ὅτι καὶ γωνία ἡ ὑπὸ BAG γωνία τῇ ὑπὸ EDZ ἐστὶν ἴση.

Proposition 8

If two triangles have two sides equal to two sides, respectively, and also have the base equal to the base, (then) they will also have equal the angles encompassed by the equal straight-lines.

Let ABC and DEF be two triangles having the two sides AB and AC equal to the two sides DE and DF , respectively. (That is) AB to DE , and AC to DF . Let them also have the base BC equal to the base EF . I say that the angle BAC is also equal to the angle EDF .



Ἐφαρμοζομένου γὰρ τοῦ ABΓ τριγώνου ἐπὶ τὸ ΔEZ τρίγωνον καὶ τιθεμένου τοῦ μὲν B σημείου ἐπὶ τὸ E σημείον τῆς δὲ BΓ εὐθείας ἐπὶ τὴν EZ ἐφαρμόσει καὶ τὸ Γ σημείον ἐπὶ τὸ Z διὰ τὸ >ίσην εἶναι τὴν BΓ τῇ EZ; ἐφαρμοσάσης δὲ τῆς BΓ ἐπὶ τὴν EZ ἐφαρμόσουσι καὶ αἱ BA, ΓA ἐπὶ τὰς EA, ΔZ. εἰ γὰρ βάσις μὲν ἡ BΓ ἐπὶ βάσιν τὴν EZ ἐφαρμόσει, αἱ δὲ BA, AG πλευραὶ ἐπὶ τὰς EA, ΔZ οὐκ ἐφαρμόσουσιν ἀλλὰ παραλλάχουσιν ὥς αἱ EH, HZ, συσταθήσονται ἐπὶ τῆς αὐτῆς εὐθείας δύο ταῖς αὐταῖς εὐθείαις >άλλαι δύο εὐθεῖαι >ίσαι ἑκατέρα ἑκατέρα πρὸς >άλλω καὶ >άλλω σημείῳ ἐπὶ τὰ αὐτὰ μέρη τὰ αὐτὰ πέρατα >έθουσai. οὐ συνίστανται δέ; οὐκ >άρα ἐφαρμοζομένης τῆς BΓ βάσεως ἐπὶ τὴν EZ βάσιν οὐκ ἐφαρμόσουσι καὶ αἱ BA, AG πλευραὶ ἐπὶ τὰς EA, ΔZ. ἐφαρμόσουσιν >άρα; <ώστε καὶ γωνία ἡ ὑπὸ BΑΓ ἐπὶ γωνίαν τὴν ὑπὸ EΔZ ἐφαρμόσει καὶ >ίση αὐτῇ >έσται. Ἐὰν >άρα δύο τρίγωνα τὰς δύο πλευρὰς [ταῖς] δύο πλευραῖς >ίσας >έθῃ ἑκατέραν ἑκατέρα καὶ τὴν βάσιν τῇ βάσει >ίσην >έθῃ, καὶ τὴν γωνίαν τῇ γωνία >ίσην <έχει τὴν ὑπὸ τῶν >ίσων εὐθειῶν περιεθομένην; <όπερ >έδει δεῖξαι.

For if triangle ABC is applied to triangle DEF , the point B being placed on point E , and the straight-line BC on EF , (then) point C will also coincide with F , on account of BC being equal to EF . So (because of) BC coinciding with EF , (the sides) BA and CA will also coincide with ED and DF (respectively). For if base BC coincides with base EF , but the sides AB and AC do not coincide with ED and DF (respectively), but miss like EG and GF (in the above figure), then we will have constructed upon the same straight-line two other straight-lines equal, respectively, to two (given) straight-lines, and (meeting) at a different point on the same side (of the straight-line), but having the same ends. But (such straight-lines) cannot be constructed [Prop. 1.7]. Thus, the base BC being applied to the base EF , the sides BA and AC cannot not coincide with ED and DF (respectively). Thus, they will coincide. So the angle BAC will also coincide with angle EDF , and will be equal to it [C.N. 4].

Thus, if two triangles have two sides equal to two side, respectively, and have the base equal to the base, (then) they will also have equal the angles encompassed by the equal straight-lines. (Which is) the very thing it was required to show.

9.

Τὴν δοθεῖσαν γωνίαν εὐθύγραμμον δίθα τεμεῖν.
>Ἐστω ἡ δοθεῖσα γωνία εὐθύγραμμοξ ἡ ὑπὸ BΑΓ. δεῖ δὲ αὐτὴν δίθα τεμεῖν.
Εἰλήφθω ἐπὶ τῆς AB τυθὸν σημείον τὸ Δ, καὶ ἀφηρήσθω ἀπὸ τῆς AG τῇ AD >ίση ἡ AE, καὶ ἐπεζεύθθω ἡ DE, καὶ συνεστάτω ἐπὶ τῆς DE τρίγωνον ἰσόπλευρον τὸ ΔEZ, καὶ ἐπεζεύθθω ἡ AZ; λέγω, <ὅτι ἡ ὑπὸ BΑΓ γωνία δίθα τέτμηται ὑπὸ τῆς AZ εὐθείας.
Ἐπεὶ γὰρ >ίση ἐστὶν ἡ AD τῇ AE, κοινὴ δὲ ἡ AZ, δύο δὲ αἱ ΔA, AZ δυοὶ ταῖς EA, AZ >ίσαι εἰσὶν ἑκατέρα ἑκατέρα. καὶ βάσις ἡ ΔZ βάσει τῇ EZ >ίση ἐστίν; γωνία >άρα ἡ ὑπὸ ΔAZ γωνία τῇ ὑπὸ EAZ >ίση ἐστίν.

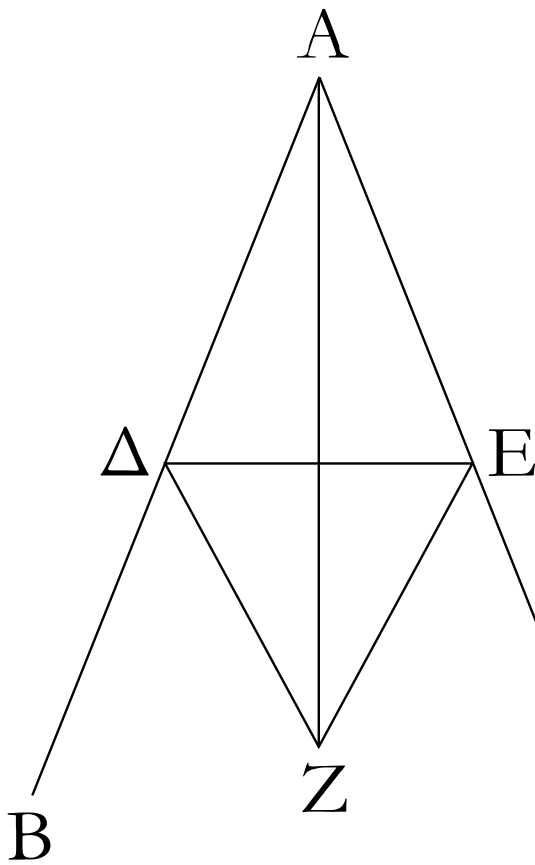
Proposition 9

To cut a given rectilinear angle in half.

Let BAC be the given rectilinear angle. So it is required to cut it in half.

Let the point D be taken at random on AB , and let AE , equal to AD , be cut off from AC [Prop. 1.3], and let DE be joined. And let the equilateral triangle DEF be constructed upon DE [Prop. 1.1], and let AF be joined. I say that the angle BAC has been cut in half by the straight-line AF .

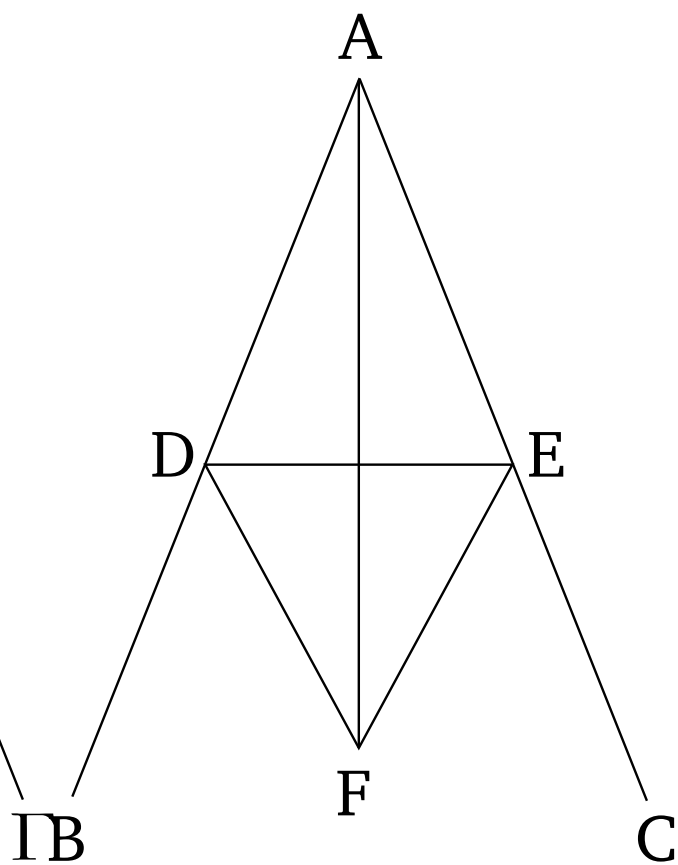
For since AD is equal to AE , and AF is common, the two (straight-lines) DA , AF are equal to the two (straight-lines) EA , AF , respectively. And the base DF is equal to the base EF . Thus, angle DAF is equal to angle EAF [Prop. 1.8].



Ἡ >ἀρα δοθεῖσα γωνία εὐθύγραμμοξ ἡ ὑπὸ BAG δίθρα τέτμηται ὑπὸ τῆξ AZ εὐθείαξ; <όπερ >έδει ποιῆσαι.

10.

Τὴν δοθεῖσαν εὐθεῖαν πεπερασμένην δίθρα τεμεῖν.
>Ἐστω ἡ δοθεῖσα εὐθεῖα πεπερασμένη ἡ AB ; δεῖ δὴ τὴν AB εὐθεῖαν πεπερασμένην δίθρα τεμεῖν.
Συνεστάτω ἐπ' αὐτῆξ τρίγωνον ἰσόπλευρον τὸ $ABΓ$, καὶ τετμήσθω ἡ ὑπὸ $ΑΓΒ$ γωνία δίθρα τῇ $ΓΔ$ εὐθεία; λέγω, <ὅτι ἡ AB εὐθεῖα δίθρα τέτμηται κατὰ τὸ $Δ$ σημεῖον.



Thus, the given rectilinear angle BAC has been cut in half by the straight-line AF . (Which is) the very thing it was required to do.

Proposition 10

To cut a given finite straight-line in half.
Let AB be the given finite straight-line. So it is required to cut the finite straight-line AB in half.
Let the equilateral triangle ABC be constructed upon (AB) [Prop. 1.1], and let the angle ACB be cut in half by the straight-line CD [Prop. 1.9]. I say that the straight-line AB has been cut in half at point D .

τὸ $Z\Delta E$, καὶ ἐπεζεύθω ἡ $Z\Gamma$; λέγω, <ὅτι τῇ δοθείσῃ εὐθείᾳ τῇ AB ἀπὸ τοῦ πρὸς αὐτῇ δοθέντος σημείου τοῦ Γ πρὸς ὀρθὰς γωνίαξ εὐθεῖα γραμμὴ >ῆκται ἡ $Z\Gamma$.

Ἐπεὶ γὰρ >ίση ἐστὶν ἡ $\Delta\Gamma$ τῇ ΓE , κοινὴ δὲ ἡ ΓZ , δύο δὲ αἱ $\Delta\Gamma$, ΓZ δυσὶ ταῖς $E\Gamma$, ΓZ >ίσαι εἰσὶν ἑκατέρα ἑκατέρᾳ; καὶ βάσις ἡ ΔZ βάσει τῇ ZE >ίση ἐστίν; γωνία >άρα ἡ ὑπὸ $\Delta\Gamma Z$ γωνία τῇ ὑπὸ $E\Gamma Z$ >ίση ἐστίν; καὶ εἰσὶν ἐφεχῆς. <ὅταν δὲ εὐθεῖα ἐπ' εὐθεῖαν σταθεῖσα τὰς ἐφεχῆς γωνίαξ >ίσαξ ἀλλήλαις ποιῇ, ὀρθὴ ἑκατέρα τῶν >ίσων γωνιῶν ἐστίν; ὀρθὴ >άρα ἐστὶν ἑκατέρα τῶν ὑπὸ $\Delta\Gamma Z$, $Z\Gamma E$.

Τῇ >άρα δοθείσῃ εὐθείᾳ τῇ AB ἀπὸ τοῦ πρὸς αὐτῇ δοθέντος σημείου τοῦ Γ πρὸς ὀρθὰς γωνίαξ εὐθεῖα γραμμὴ >ῆκται ἡ ΓZ ; <ὅπερ >έδει ποιῆσαι.

angle FDE be constructed on DE [Prop. 1.1], and let FC be joined. I say that the straight-line FC has been drawn at right-angles to the given straight-line AB from the given point C on it.

For since DC is equal to CE , and CF is common, the two (straight-lines) DC , CF are equal to the two (straight-lines), EC , CF , respectively. And the base DF is equal to the base FE . Thus, the angle DCF is equal to the angle ECF [Prop. 1.8], and they are adjacent. But when a straight-line stood on a(nother) straight-line makes the adjacent angles equal to one another, each of the equal angles is a right-angle [Def. 1.10]. Thus, each of the (angles) DCF and FCE is a right-angle.

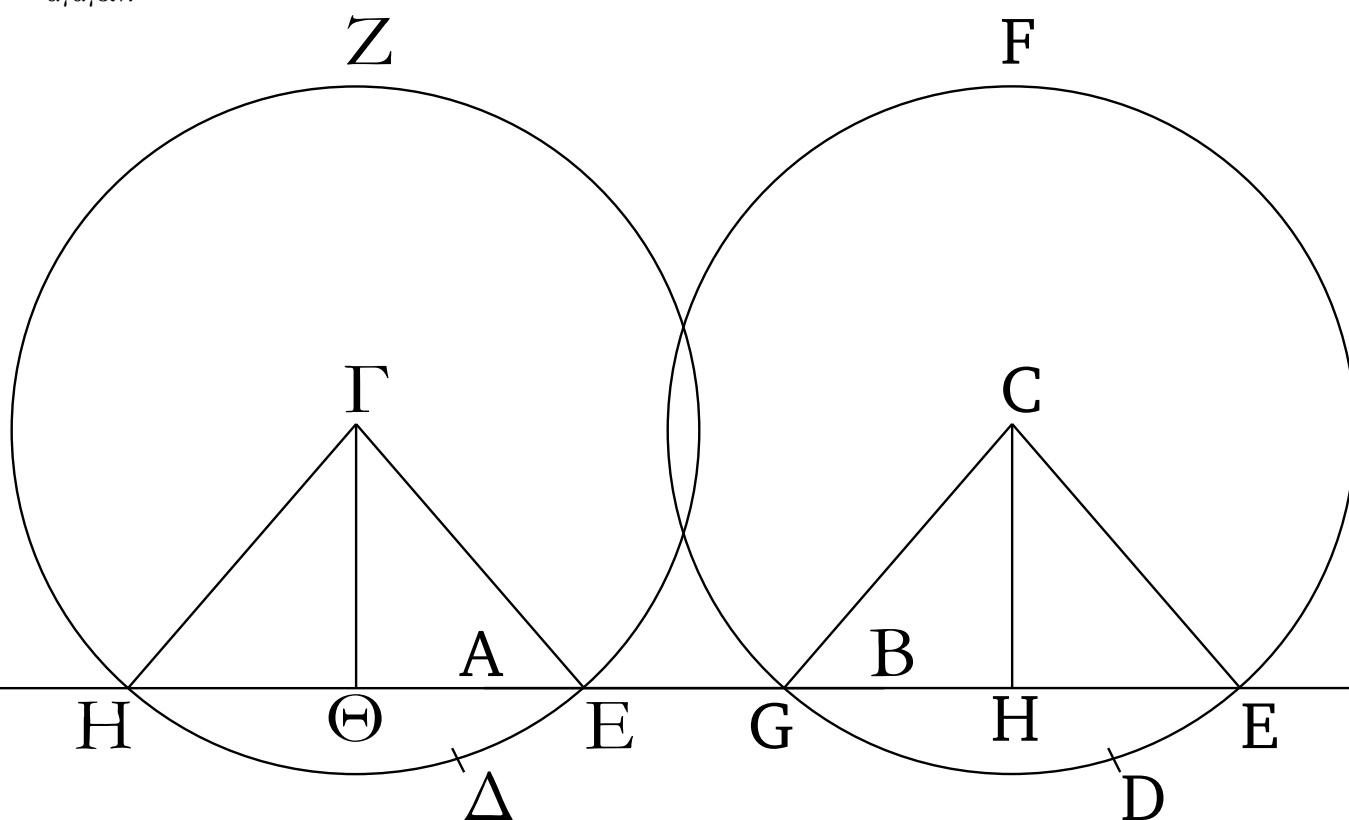
Thus, the straight-line CF has been drawn at right-angles to the given straight-line AB from the given point C on it. (Which is) the very thing it was required to do.

12.

Proposition 12

Ἐπὶ τὴν δοθεῖσαν εὐθεῖαν >άπειρον ἀπὸ τοῦ δοθέντος σημείου, <ὃ μὴ ἐστὶν ἐπ' αὐτῇ, κάθετον εὐθεῖαν γραμμὴν ἀγαγεῖν.

To draw a straight-line perpendicular to a given infinite straight-line from a given point which is not on it.



>Ἐστω ἡ μὲν δοθεῖσα εὐθεῖα >άπειρος ἡ AB τὸ δὲ δοθὲν σημεῖον, <ὃ μὴ ἐστὶν ἐπ' αὐτῇ, τὸ Γ ; δεῖ δὲ ἐπὶ τὴν δοθεῖσαν εὐθεῖαν >άπειρον τὴν AB ἀπὸ τοῦ δοθέντος σημείου τοῦ Γ , <ὃ μὴ ἐστὶν ἐπ' αὐτῇ, κάθετον εὐθεῖαν γραμμὴν ἀγαγεῖν.

Εἰλήφθω γὰρ ἐπὶ τὰ <έτερα μέρη τῆς AB εὐθείας τυθὸν σημεῖον τὸ Δ , καὶ κέντρῳ μὲν τῷ Γ διαστήματι δὲ τῷ $\Gamma\Delta$ κύκλος γεγράφθω ὁ EZH , καὶ τετμήσθω ἡ EH εὐθεῖα δίθᾳ κατὰ τὸ Θ , καὶ ἐπεζεύθωσαν αἱ ΓH , $\Gamma\Theta$, ΓE εὐθεῖαι; λέγω, <ὅτι ἐπὶ τὴν δοθεῖσαν εὐθεῖαν >άπειρον τὴν AB ἀπὸ τοῦ δοθέντος σημείου τοῦ Γ , <ὃ μὴ ἐστὶν ἐπ' αὐτῇ, κάθετος >ῆκται ἡ $\Gamma\Theta$.

Ἐπεὶ γὰρ >ίση ἐστὶν ἡ $H\Theta$ τῇ ΘE , κοινὴ δὲ ἡ $\Theta\Gamma$, δύο δὲ αἱ

Let AB be the given infinite straight-line and C the given point, which is not on it. So it is required to draw a straight-line perpendicular to the given infinite straight-line AB from the given point C , which is not on it.

For let point D be taken at random on the other side (to C) of the straight-line AB , and let the circle EFG be drawn with center C and radius CD [Post. 3], and let the straight-line EG be cut in half at (point) H [Prop. 1.10], and let the straight-lines CG , CH , and CE be joined. I say that the (straight-line) CH has been drawn perpendicular to the given infinite straight-line AB from the given point C , which is not on it.

For since GH is equal to HE , and HC (is) common, the

ΗΘ, ΘΓ δύο ταῖς ΕΘ, ΘΓ >ίσαι εἰσὶν ἑκατέρα ἑκατέρα; καὶ βάσις ἡ ΓΗ βάσει τῇ ΓΕ ἐστὶν >ίση; γωνία >άρα ἡ ὑπὸ ΓΘΗ γωνία τῇ ὑπὸ ΕΘΓ ἐστὶν >ίση. καὶ εἰσὶν ἐφεχῆς. <ὅταν δὲ εὐθεῖα ἐπ> εὐθεῖαν σταθεῖσα τὰς ἐφεχῆς γωνίας >ίσαξ ἀλλήλαις ποιῇ, ὀρθὴ ἑκατέρα τῶν >ίσων γωνιῶν ἐστὶν, καὶ ἡ ἐφεστηκυῖα εὐθεῖα κάθετος καλεῖται ἐφ> <ὴν ἐφέστηκεν. Ἐπὶ τὴν δοθεῖσαν >άρα εὐθεῖαν >άπειρον τὴν ΑΒ ἀπὸ τοῦ δοθέντος σημείου τοῦ Γ, <ὃ μὴ ἐστὶν ἐπ> αὐτῇ, κάθετος ἡκται ἡ ΓΘ; <ὅπερ >έδει ποιῆσαι.

two (straight-lines) GH, HC are equal to the two (straight-lines) EH, HC , respectively, and the base CG is equal to the base CE . Thus, the angle CHG is equal to the angle EHC [Prop. 1.8], and they are adjacent. But when a straight-line stood on a(nother) straight-line makes the adjacent angles equal to one another, each of the equal angles is a right-angle, and the former straight-line is called a perpendicular to that upon which it stands [Def. 1.10].

Thus, the (straight-line) CH has been drawn perpendicular to the given infinite straight-line AB from the given point C , which is not on it. (Which is) the very thing it was required to do.

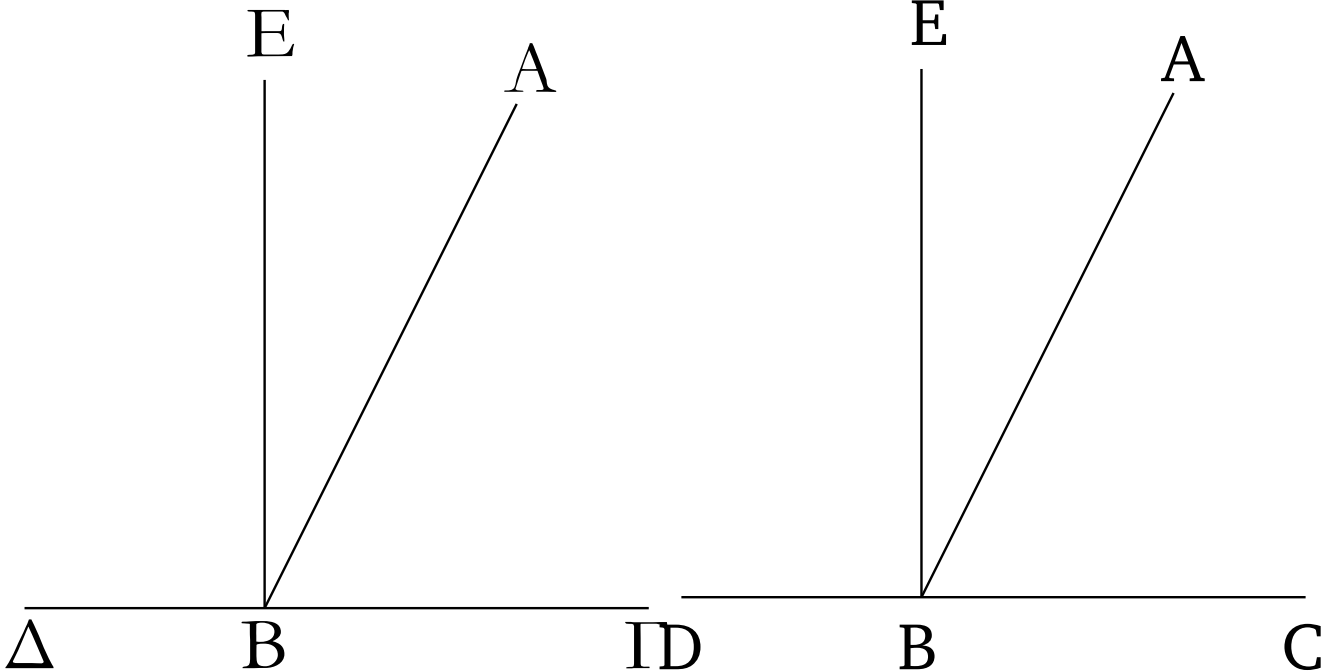
13.

Ἐὰν εὐθεῖα ἐπ> εὐθεῖαν σταθεῖσα γωνίαξ ποιῇ, >ήτοι δύο ὀρθὰξ >ἢ δυσὶν ὀρθαῖξ >ίσαξ ποιήσῃ. Εὐθεῖα γάρ τις ἡ ΑΒ ἐπ> εὐθεῖαν τὴν ΓΔ σταθεῖσα γωνίαξ ποιείτω τὰς ὑπὸ ΓΒΑ, ΑΒΔ; λέγω, <ὅτι αἱ ὑπὸ ΓΒΑ, ΑΒΔ γωνίαι >ήτοι δύο ὀρθαὶ εἰσὶν >ἢ δυσὶν ὀρθαῖξ >ίσαι.

Proposition 13

If a straight-line stood on a(nother) straight-line makes angles, it will certainly either make two right-angles, or (angles whose sum is) equal to two right-angles.

For let some straight-line AB stood on the straight-line CD make the angles CBA and ABD . I say that the angles CBA and ABD are certainly either two right-angles, or (have a sum) equal to two right-angles.



Εἰ μὲν ο>ὺν >ίση ἐστὶν ἡ ὑπὸ ΓΒΑ τῇ ὑπὸ ΑΒΔ, δύο ὀρθαὶ εἰσὶν. εἰ δὲ ο>ύ, >ήθω ἀπὸ τοῦ Β σημείου τῇ ΓΔ [εὐθείᾳ] πρὸξ ὀρθὰξ ἡ ΒΕ; αἱ >άρα ὑπὸ ΓΒΕ, ΕΒΔ δύο ὀρθαὶ εἰσὶν; καὶ ἐπεὶ ἡ ὑπὸ ΓΒΕ δυσὶ ταῖς ὑπὸ ΓΒΑ, ΑΒΕ >ίση ἐστίν, κοινὴ προσκείσθω ἡ ὑπὸ ΕΒΔ; αἱ >άρα ὑπὸ ΓΒΕ, ΕΒΔ τρισὶ ταῖς ὑπὸ ΓΒΑ, ΑΒΕ, ΕΒΔ >ίσαι εἰσὶν. πάλιν, ἐπεὶ ἡ ὑπὸ ΔΒΑ δυσὶ ταῖς ὑπὸ ΔΒΕ, ΕΒΑ >ίση ἐστίν, κοινὴ προσκείσθω ἡ ὑπὸ ΑΒΓ; αἱ >άρα ὑπὸ ΔΒΑ, ΑΒΓ τρισὶ ταῖς ὑπὸ ΔΒΕ, ΕΒΑ, ΑΒΓ >ίσαι εἰσὶν. ἐδείκθησαν δὲ καὶ αἱ ὑπὸ ΓΒΕ, ΕΒΔ τρισὶ ταῖς αὐταῖς >ίσαι; τὰ δὲ τῶ| αὐτῶ| >ίσα καὶ ἀλλήλοισι ἐστὶν >ίσα; καὶ αἱ ὑπὸ ΓΒΕ, ΕΒΔ >άρα ταῖς ὑπὸ ΔΒΑ, ΑΒΓ >ίσαι εἰσὶν; ἀλλὰ αἱ ὑπὸ ΓΒΕ, ΕΒΔ δύο ὀρθαὶ εἰσὶν; καὶ αἱ ὑπὸ ΔΒΑ, ΑΒΓ >άρα δυσὶν ὀρθαῖξ >ίσαι εἰσὶν. Ἐὰν >άρα εὐθεῖα ἐπ> εὐθεῖαν σταθεῖσα γωνίαξ ποιῇ, >ήτοι δύο ὀρθὰξ >ἢ δυσὶν ὀρθαῖξ >ίσαξ ποιήσῃ; <ὅπερ >έδει δείχαι.

In fact, if CBA is equal to ABD (then) they are two right-angles [Def. 1.10]. But, if not, let BE be drawn from the point B at right-angles to [the straight-line] CD [Prop. 1.11]. Thus, CBE and EBD are two right-angles. And since CBE is equal to the two (angles) CBA and ABE , let EBD be added to both. Thus, the (sum of the angles) CBE and EBD is equal to the (sum of the) three (angles) CBA , ABE , and EBD [C.N. 2]. Again, since DBA is equal to the two (angles) DBE and EBA , let ABC be added to both. Thus, the (sum of the angles) DBA and ABC is equal to the (sum of the) three (angles) DBE , EBA , and ABC [C.N. 2]. But (the sum of) CBE and EBD was also shown (to be) equal to the (sum of the) same three (angles). And things equal to the same thing are also equal to one another [C.N. 1]. Therefore, (the sum of) CBE and EBD is also equal to (the sum of) DBA and ABC . But, (the sum of) CBE and EBD is two right-angles. Thus, (the sum of) ABD and ABC is also equal to two right-

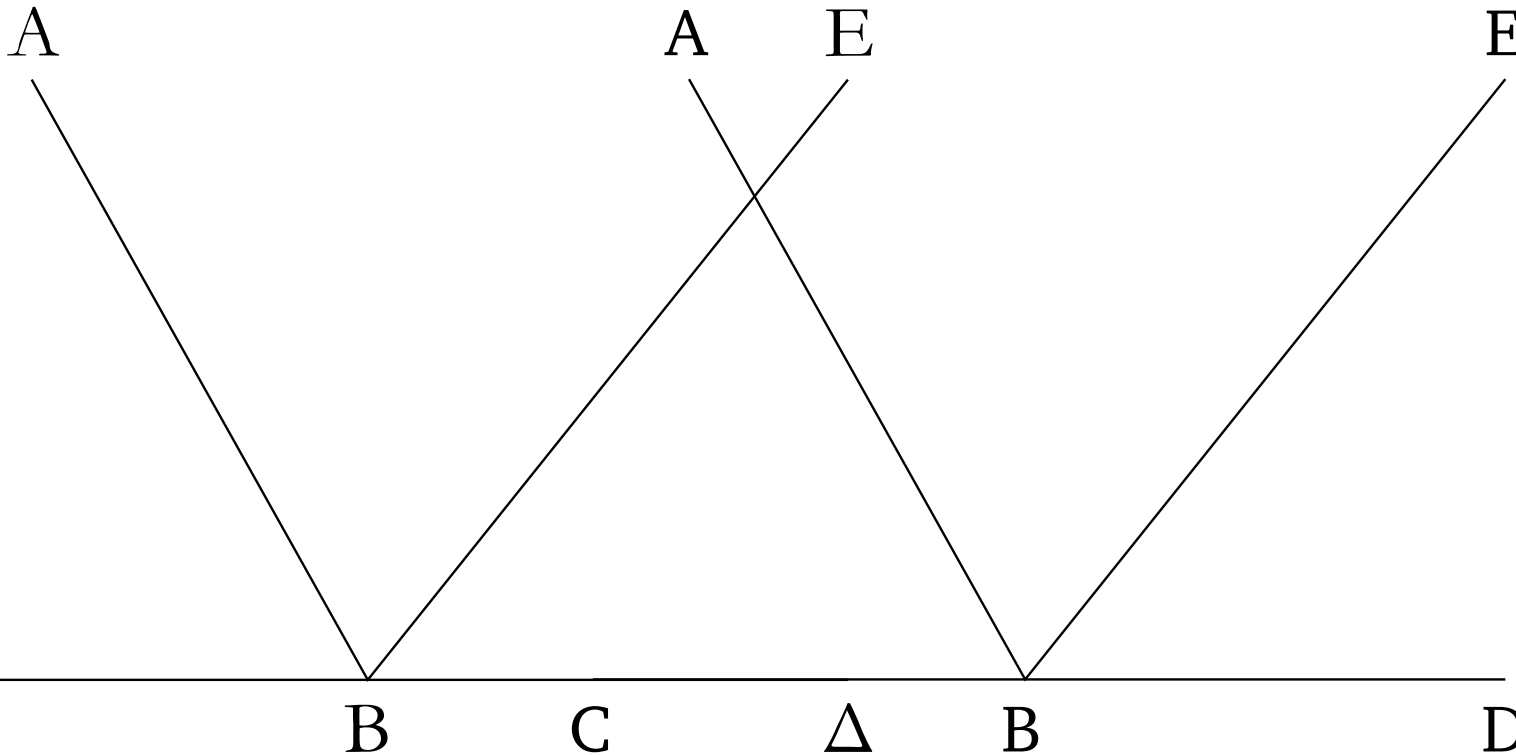
angles.

Thus, if a straight-line stood on a(nother) straight-line makes angles, it will certainly either make two right-angles, or (angles whose sum is) equal to two right-angles. (Which is) the very thing it was required to show.

14.

Ἐὰν πρὸς τινι εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείω δύο εὐθεῖαι μὴ ἐπὶ τὰ αὐτὰ μέρη κείμεναι τὰς ἐφεχῆς γωνίας δυσὶν ὀρθαῖς >ίσαζ ποιῶσιν, ἐπ' εὐθείας >έσσονται ἀλλήλαις αἱ εὐθεῖαι.

Πρὸς γάρ τινι εὐθείᾳ τῇ AB καὶ τῷ πρὸς αὐτῇ σημείω τῷ B δύο εὐθεῖαι αἱ BF , BD μὴ ἐπὶ τὰ αὐτὰ μέρη κείμεναι τὰς ἐφεχῆς γωνίας τὰς ὑπὸ ABF , ABD δύο ὀρθαῖς >ίσαζ ποιείτωσαν; λέγω, <ὅτι ἐπ' εὐθείας ἐστὶ τῇ GB ἢ BD .



Εἰ γὰρ μὴ ἐστὶ τῇ BF ἐπ' εὐθείας ἢ BD , >έστω τῇ GB ἐπ' εὐθείας ἢ BE .

Ἐπεὶ οὖν εὐθεῖα ἡ AB ἐπ' εὐθεῖαν τὴν GBE ἐφέστηκεν, αἱ >άρα ὑπὸ ABF , ABE γωνίαι δύο ὀρθαῖς >ίσαι εἰσὶν; εἰσὶ δὲ καὶ αἱ ὑπὸ ABF , ABD δύο ὀρθαῖς >ίσαι; αἱ >άρα ὑπὸ GBA , ABE ταῖς ὑπὸ GBA , ABD >ίσαι εἰσὶν. κοινὴ ἀφηρήσθω ἡ ὑπὸ GBA ; λοιπὴ >άρα ἢ ὑπὸ ABE λοιπῇ τῇ ὑπὸ ABD ἐστὶν >ίση, ἡ ἐλάσσων τῇ μείζονι; <ὅπερ ἐστὶν ἀδύνατον. οὐκ >άρα ἐπ' εὐθείας ἐστὶν ἡ BE τῇ GB . ὁμοίως δὲ δείχομεν, <ὅτι οὐδὲ >άλλη τις πλὴν τῆς BD ; ἐπ' εὐθείας >άρα ἐστὶν ἡ GB τῇ BD .

Ἐὰν >άρα πρὸς τινι εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείω δύο εὐθεῖαι μὴ ἐπὶ αὐτὰ μέρη κείμεναι τὰς ἐφεχῆς γωνίας δυσὶν ὀρθαῖς >ίσαζ ποιῶσιν, ἐπ' εὐθείας >έσσονται ἀλλήλαις αἱ εὐθεῖαι; <ὅπερ >έδει δεῖχαι.

Proposition 14

If two straight-lines, not lying on the same side, make adjacent angles (whose sum is) equal to two right-angles with some straight-line, at a point on it, (then) the two straight-lines will be straight-on (with respect) to one another.

For let two straight-lines BC and BD , not lying on the same side, make adjacent angles ABC and ABD (whose sum is) equal to two right-angles with some straight-line AB , at the point B on it. I say that BD is straight-on with respect to CB .

For if BD is not straight-on to BC (then) let BE be straight-on to CB .

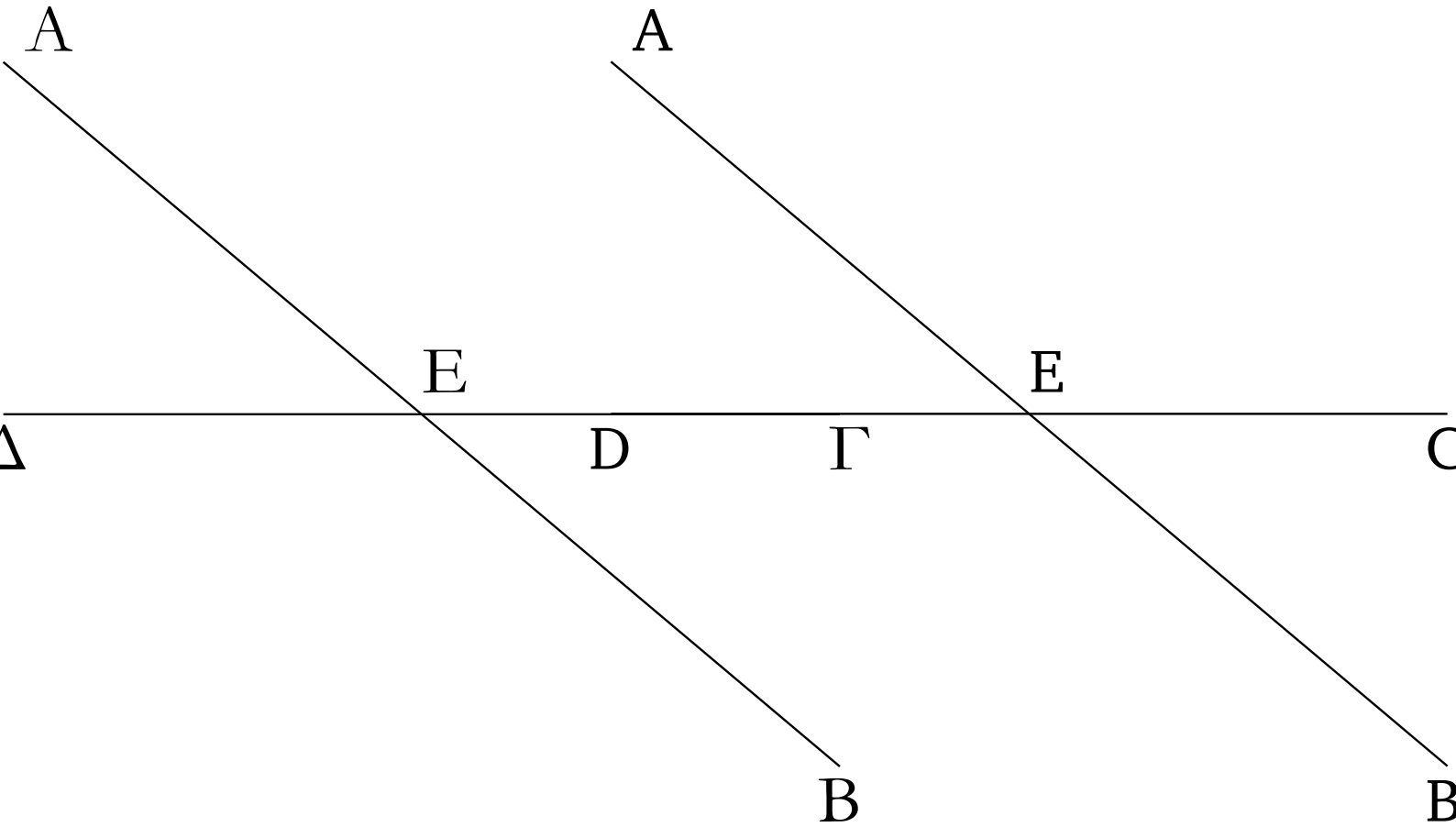
Therefore, since the straight-line AB stands on the straight-line CBE , the (sum of the) angles ABC and ABE is thus equal to two right-angles [Prop. 1.13]. But (the sum of) ABC and ABD is also equal to two right-angles. Thus, (the sum of angles) CBA and ABE is equal to (the sum of angles) CBA and ABD [C.N. 1]. Let (angle) CBA be subtracted from both. Thus, the remainder ABE is equal to the remainder ABD [C.N. 3], the lesser to the greater. The very thing is impossible. Thus, BE is not straight-on with respect to CB . Similarly, we can show that neither (is) any other (straight-line) than BD . Thus, CB is straight-on with respect to BD .

Thus, if two straight-lines, not lying on the same side, make adjacent angles (whose sum is) equal to two right-angles with some straight-line, at a point on it, (then) the two straight-lines will be straight-on (with respect) to one another. (Which is) the very thing it was required to show.

15.

Ἐὰν δύο εὐθεῖαι τέμνωσιν ἀλλήλας, τὰς κατὰ κορυφὴν γωνίας >ίσας ἀλλήλαις ποιοῦσιν.

Δύο γὰρ εὐθεῖαι αἱ AB , $ΓΔ$ τεμνέτωσαν ἀλλήλας κατὰ τὸ E σημεῖον; λέγω, <ὅτι >ίση ἐστὶν ἡ μὲν ὑπὸ $ΑΕΓ$ γωνία τῇ ὑπὸ $ΔΕΒ$, ἡ δὲ ὑπὸ $ΓΕΒ$ τῇ ὑπὸ $ΑΕΔ$.



Ἐπεὶ γὰρ εὐθεῖα ἡ AE ἐπ' εὐθεῖαν τὴν $ΓΔ$ ἐφέστηκε γωνίας ποιοῦσα τὰς ὑπὸ $ΓΕΑ$, $ΑΕΔ$, αἱ >άρα ὑπὸ $ΓΕΑ$, $ΑΕΔ$ γωνίαι δυσὶν ὀρθαῖς >ίσαι εἰσὶν. πάλιν, ἐπεὶ εὐθεῖα ἡ $ΔΕ$ ἐπ' εὐθεῖαν τὴν AB ἐφέστηκε γωνίας ποιοῦσα τὰς ὑπὸ $ΑΕΔ$, $ΔΕΒ$, αἱ >άρα ὑπὸ $ΑΕΔ$, $ΔΕΒ$ γωνίαι δυσὶν ὀρθαῖς >ίσαι εἰσὶν. ἐδείκθησαν δὲ καὶ αἱ ὑπὸ $ΓΕΑ$, $ΑΕΔ$ δυσὶν ὀρθαῖς >ίσαι; αἱ >άρα ὑπὸ $ΓΕΑ$, $ΑΕΔ$ ταῖς ὑπὸ $ΑΕΔ$, $ΔΕΒ$ >ίσαι εἰσὶν. κοινὴ ἀφηρήσθω ἡ ὑπὸ $ΑΕΔ$; λοιπὴ >άρα ἡ ὑπὸ $ΓΕΑ$ λοιπῇ τῇ ὑπὸ $ΒΕΔ$ >ίση ἐστίν; ὁμοίως δὲ δεῖθῇσεται, <ὅτι καὶ αἱ ὑπὸ $ΓΕΒ$, $ΔΕΑ$ >ίσαι εἰσὶν.

Ἐὰν >άρα δύο εὐθεῖαι τέμνωσιν ἀλλήλας, τὰς κατὰ κορυφὴν γωνίας >ίσας ἀλλήλαις ποιοῦσιν; <ὅπερ >έδει δεῖξαι.

16.

Παντὸς τριγώνου μιᾶς τῶν πλευρῶν προσεκβληθείσης ἡ ἐκτὸς γωνία ἑκατέρωξ τῶν ἐντὸς καὶ ἀπεναντίον γωνιῶν μείζων ἐστίν.

>Ἐστω τρίγωνον τὸ $ABΓ$, καὶ προσεκβεβλήσθω αὐτοῦ μία πλευρὰ ἡ $ΒΓ$ ἐπὶ τὸ $Δ$; λέγω, <ὅτι ἡ ἐκτὸς γωνία ἡ ὑπὸ $ΑΓΔ$ μείζων ἐστὶν ἑκατέρωξ τῶν ἐντὸς καὶ ἀπεναντίον τῶν ὑπὸ $ΓΒΑ$, $ΒΑΓ$ γωνιῶν.

Τετμήσθω ἡ $ΑΓ$ δίῃα κατὰ τὸ E , καὶ ἐπιζευθθεῖσα ἡ BE

Proposition 15

If two straight-lines cut one another (then) they make the vertically opposite angles equal to one another.

For let the two straight-lines AB and CD cut one another at the point E . I say that angle AEC is equal to (angle) DEB , and (angle) CEB to (angle) AED .

For since the straight-line AE stands on the straight-line CD , making the angles CEA and AED , the (sum of the) angles CEA and AED is thus equal to two right-angles [Prop. 1.13]. Again, since the straight-line DE stands on the straight-line AB , making the angles AED and DEB , the (sum of the) angles AED and DEB is thus equal to two right-angles [Prop. 1.13]. But (the sum of) CEA and AED was also shown (to be) equal to two right-angles. Thus, (the sum of) CEA and AED is equal to (the sum of) AED and DEB [C.N. 1]. Let AED be subtracted from both. Thus, the remainder CEA is equal to the remainder DEB [C.N. 3]. Similarly, it can be shown that CEB and DEA are also equal.

Thus, if two straight-lines cut one another (then) they make the vertically opposite angles equal to one another. (Which is) the very thing it was required to show.

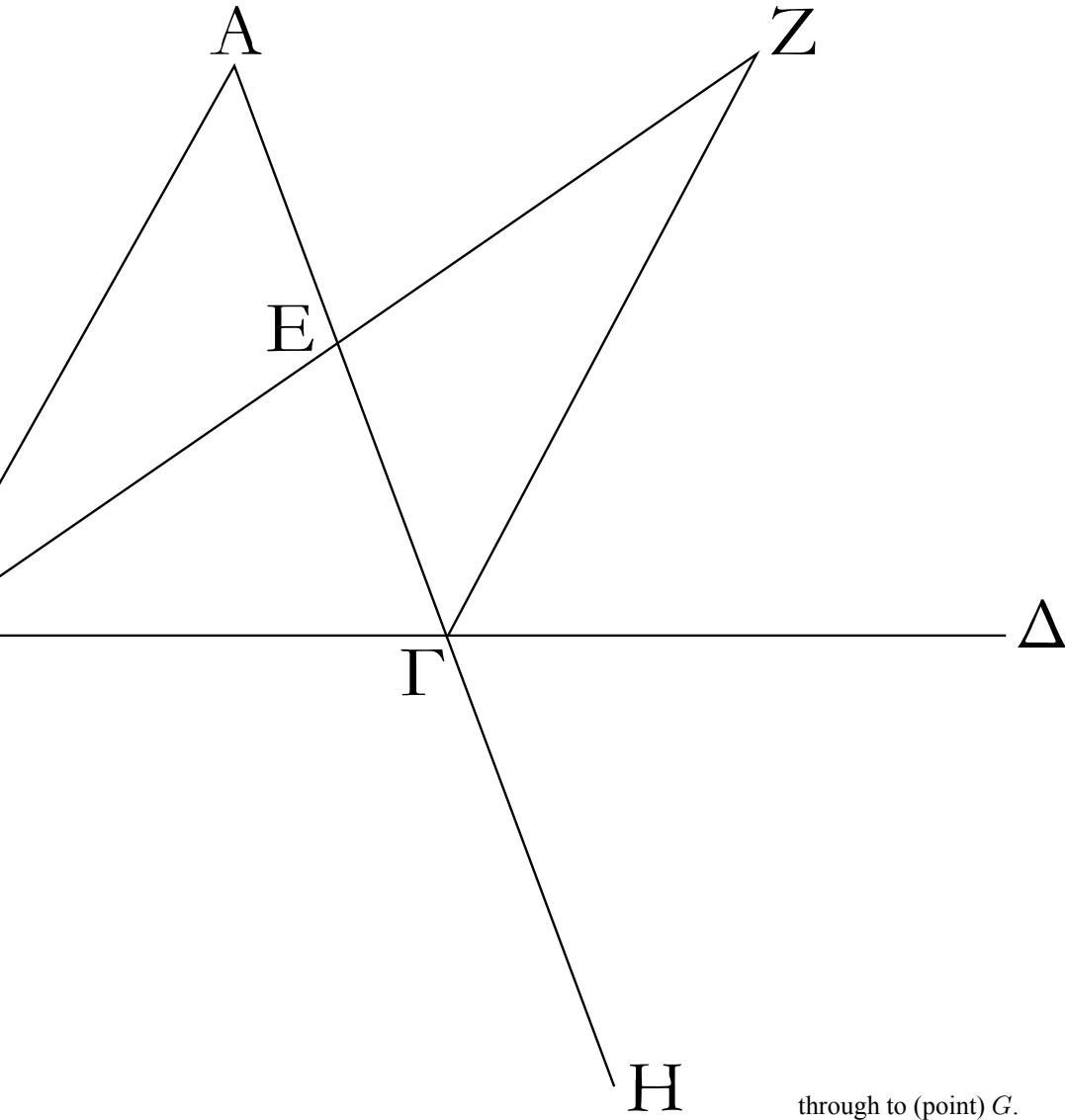
Proposition 16

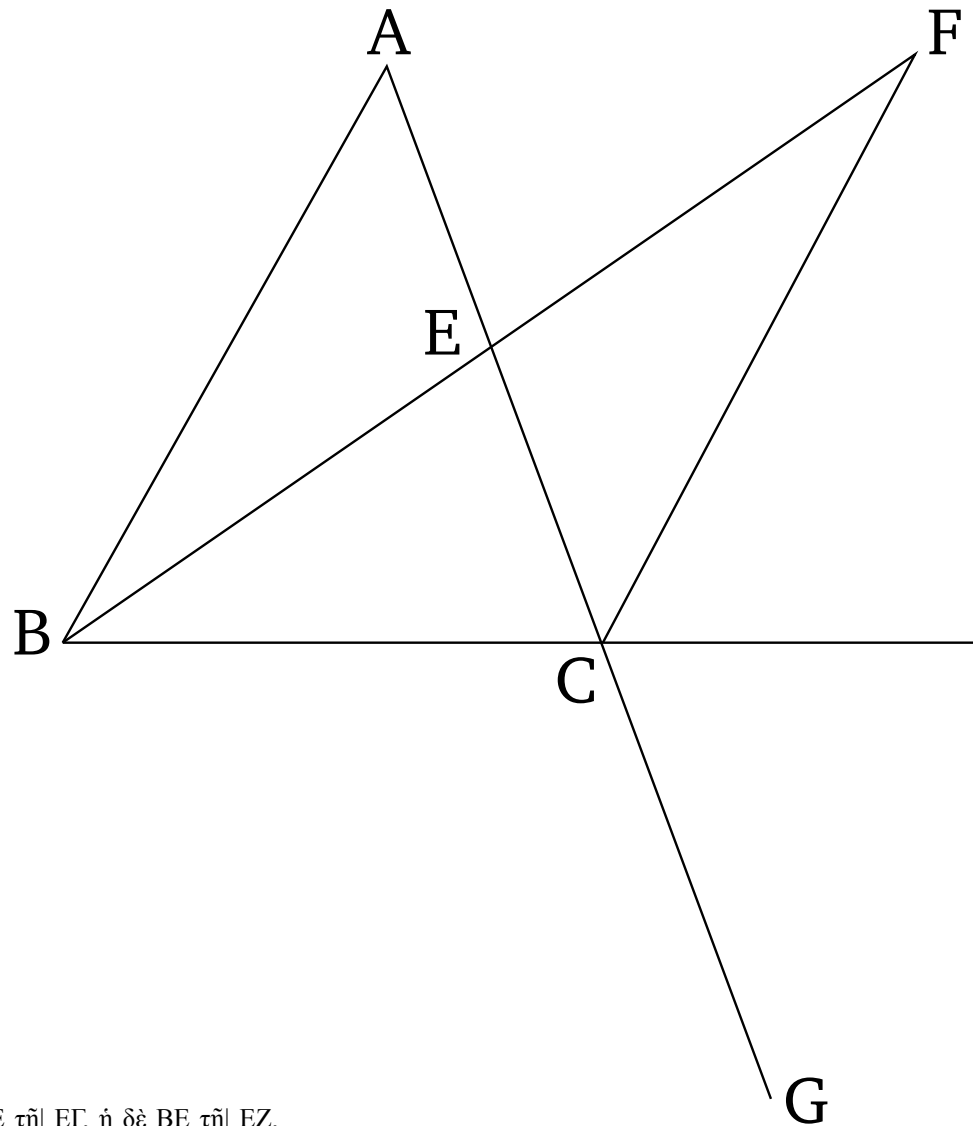
For any triangle, when one of the sides is produced, the external angle is greater than each of the internal and opposite angles.

Let ABC be a triangle, and let one of its sides BC be produced to D . I say that the external angle ACD is greater than each of the internal and opposite angles, CBA and BAC .

Let the (straight-line) AC be cut in half at (point) E

ἐκβεβλήσθω ἐπ' εὐθείᾳ ἐπὶ τὸ Z , καὶ κείσθω τῇ BE ὡς ἰσὴ [Prop. 1.10]. And BE being joined, let it be produced in a straight-line to (point) F .[†] And let EF be made equal to BE [Prop. 1.3], and let FC be joined, and let AC be drawn





Ἐπεὶ οὖν ὅτι $AE = EC$, καὶ $BE = EF$, οἱ δύο (ῥα) AE, EB ὅσοι ταῖς CE, EF , καὶ ἡ γωνία $\angle AEB$ ὅση τῇ $\angle FEC$ ὡς ἐναντίαι. ὥστε τὰ τρίγωνα ABE, FEC ὅμοια, καὶ ἡ AB ὅση τῇ FC , καὶ ἡ $\angle BAE$ ὅση τῇ $\angle ECF$. ὅτι $\angle ECD$ ἰσχυρότερος τοῦ $\angle ECF$. ὥστε ἡ $\angle ACD$ ἰσχυρότερος τοῦ $\angle BAE$. ὁμοίως δὲ καὶ ἡ $\angle BCG$ ἰσχυρότερος τοῦ $\angle ABC$. ὥστε ἡ $\angle ACD$ ἰσχυρότερος τοῦ $\angle ABC$. ὥστε ἡ $\angle ACD$ ἰσχυρότερος τοῦ $\angle ABC$.

Therefore, since AE is equal to EC , and BE to EF , the two (straight-lines) AE, EB are equal to the two (straight-lines) CE, EF , respectively. Also, angle AEB is equal to angle FEC , for (they are) vertically opposite [Prop. 1.15]. Thus, the base AB is equal to the base FC , and the triangle ABE is equal to the triangle FEC , and the remaining angles subtended by the equal sides are equal to the corresponding remaining angles [Prop. 1.4]. Thus, BAE is equal to ECF . But ECD is greater than ECF . Thus, ACD is greater than BAE . Similarly, by having cut BC in half, it can be shown (that) BCG —that is to say, ACD —(is) also greater than ABC .

Thus, for any triangle, when one of the sides is produced, the external angle is greater than each of the internal and opposite angles. (Which is) the very thing it was required to show.

† The implicit assumption that the point F lies in the interior of the angle ABC should be counted as an additional postulate.

17.

Proposition 17

Παντὸν τριγώνου αἱ δύο γωνίαι δύο ὀρθῶν ἐλάσσονές εἰσι πάντῃ μεταλαμβανόμεναι.

>Ἐστω τρίγωνον τὸ ABG ; λέγω, ὅτι τοῦ ABG τριγώνου αἱ δύο γωνίαι δύο ὀρθῶν ἐλάττωσεν εἰσι πάντῃ μεταλαμβανόμενα. Ἐκβεβλήσθω γὰρ ἡ BG ἐπὶ τὸ Δ .

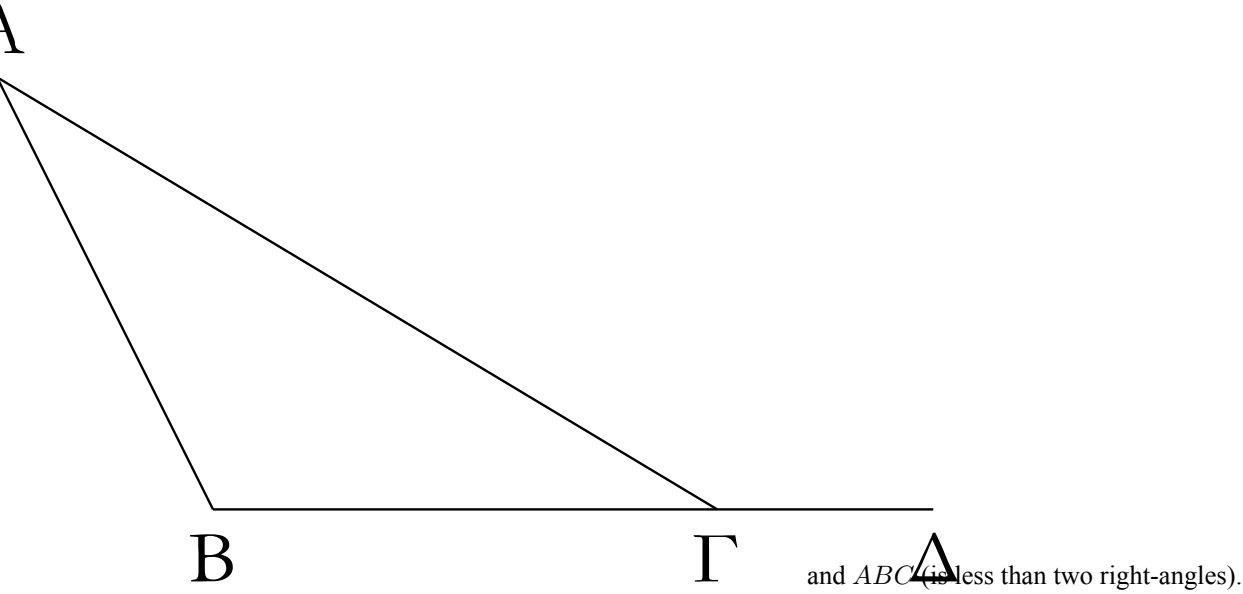
For any triangle, (the sum of) two angles taken together in any (possible way) is less than two right-angles.

Let ABC be a triangle. I say that (the sum of) two angles of triangle ABC taken together in any (possible way) is less than two right-angles.

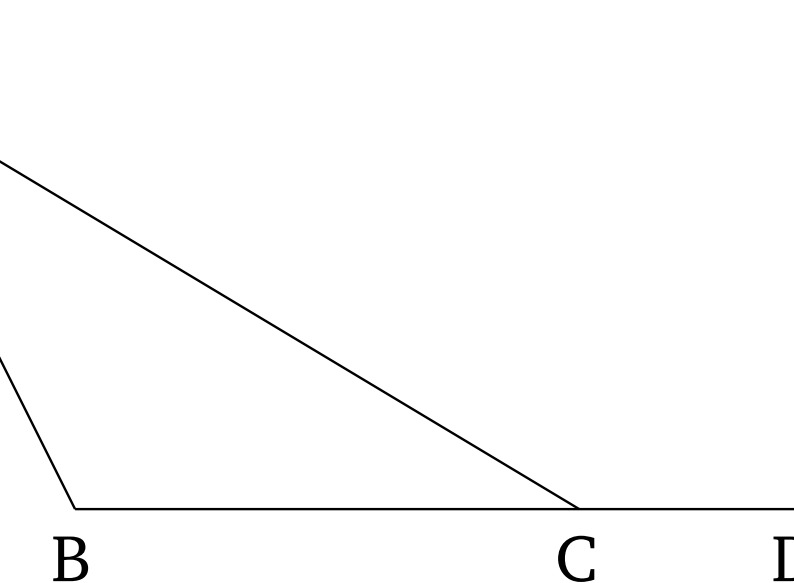
Καὶ ἐπεὶ τριγώνου τοῦ $AB\Gamma$ ἐκτόξ ἐστὶ γωνία ἢ ὑπὸ $ΑΓΔ$, μείζων ἐστὶ τῆς ἐντὸς καὶ ἀπεναντίον τῆς ὑπὸ $AB\Gamma$. κοινὴ προσκείσθω ἡ ὑπὸ $ΑΓΒ$; αἱ >άρα ὑπὸ $ΑΓΔ$, $ΑΓΒ$ τῶν ὑπὸ $AB\Gamma$, $ΒΓΑ$ μείζονές εἰσιν. ἄλλ> αἱ ὑπὸ $ΑΓΔ$, $ΑΓΒ$ δύο ὀρθαῖς >ίσαι εἰσιν; αἱ >άρα ὑπὸ $AB\Gamma$, $ΒΓΑ$ δύο ὀρθῶν ἐλάσσονές εἰσιν. ὁμοίωξ δὴ δείχομεν, <ὅτι καὶ αἱ ὑπὸ $ΒΑΓ$, $ΑΓΒ$ δύο ὀρθῶν ἐλάσσονές εἰσι καὶ >έτι αἱ ὑπὸ $ΓΑΒ$, $AB\Gamma$.

For let BC be produced to D .

And since the angle ACD is external to triangle ABC , it is greater than the internal and opposite angle ABC [Prop. 1.16]. Let ACB be added to both. Thus, the (sum of the angles) ACD and ACB is greater than the (sum of the angles) ABC and BCA . But, (the sum of) ACD and ACB is equal to two right-angles [Prop. 1.13]. Thus, (the sum of) ABC and BCA is less than two right-angles. Similarly, we can show that (the sum of) BAC and ACB is also less than two right-angles, and further (that the sum of) CAB



Παντὸν >άρα τριγώνου αἱ δύο γωνίαι δύο ὀρθῶν ἐλάσσονές εἰσι πάντῃ| μεταλαμβανόμεναι; <ὅπερ >έδει δεῖχαι.



Thus, for any triangle, (the sum of) two angles taken together in any (possible way) is less than two right-angles. (Which is) the very thing it was required to show.

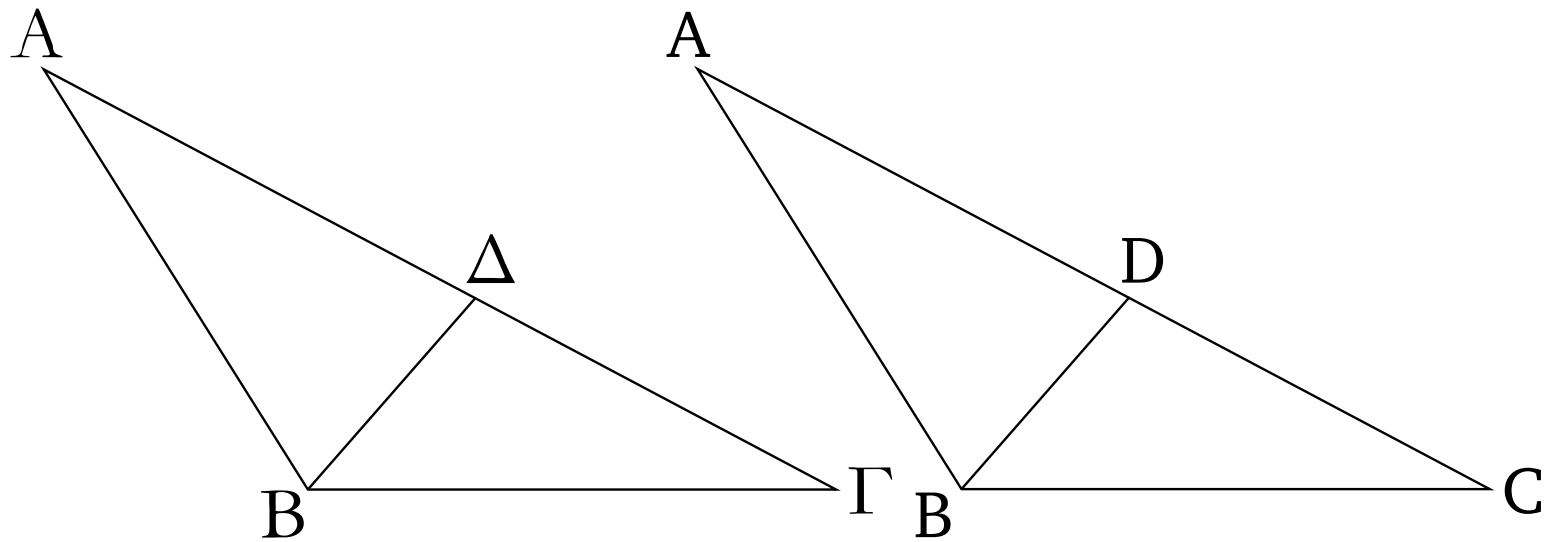
18.

Παντὸξ τριγώνου ἡ μείζων πλευρὰ τὴν μείζονα γωνίαν ὑποτείνει.

>Ἐστω γὰρ τρίγωνον τὸ $AB\Gamma$ μείζονα >έθον τὴν $ΑΓ$ πλευρὰν τῆς AB ; λέγω, <ὅτι καὶ γωνία ἢ ὑπὸ $AB\Gamma$ μείζων ἐστὶ τῆς ὑπὸ $ΒΓΑ$.

Proposition 18

In any triangle, the greater side subtends the greater angle. For let ABC be a triangle having side AC greater than AB . I say that angle ABC is also greater than BCA .



Ἐπεὶ γὰρ μείζων ἐστὶν ἡ ΑΓ τῆς ΑΒ, κείσθω τῇ ΑΒ ὀρθὴ ἡ ΑΔ, καὶ ἐπεζεύθω ἡ ΒΔ.

Καὶ ἐπεὶ τριγώνου τοῦ ΒΓΔ ἐκτόξ ἐστὶ γωνία ἡ ὑπὸ ΑΔΒ, μείζων ἐστὶ τῆς ἐντόξ καὶ ἀπεναντίον τῆς ὑπὸ ΔΓΒ; ὀρθὴ δὲ ἡ ὑπὸ ΑΔΒ τῇ ὑπὸ ΑΒΔ, ἐπεὶ καὶ πλευρὰ ἡ ΑΒ τῇ ΑΔ ἐστὶν ὀρθή; μείζων ἄρα καὶ ἡ ὑπὸ ΑΒΔ τῆς ὑπὸ ΑΓΒ; πολλῶν ἄρα ἡ ὑπὸ ΑΒΓ μείζων ἐστὶ τῆς ὑπὸ ΑΓΒ.

Παντὸς ἄρα τριγώνου ἡ μείζων πλευρὰ τὴν μείζονα γωνίαν ὑποτείνει; ὅπερ ὀφείδει δεῖξαι.

19.

Παντὸς τριγώνου ὑπὸ τὴν μείζονα γωνίαν ἡ μείζων πλευρὰ ὑποτείνει.

Ἐστω τρίγωνον τὸ ΑΒΓ μείζονα ἔχον τὴν ὑπὸ ΑΒΓ γωνίαν τῆς ὑπὸ ΒΓΑ; λέγω, ὅτι καὶ πλευρὰ ἡ ΑΓ πλευρᾷ τῆς ΑΒ μείζων ἐστίν.

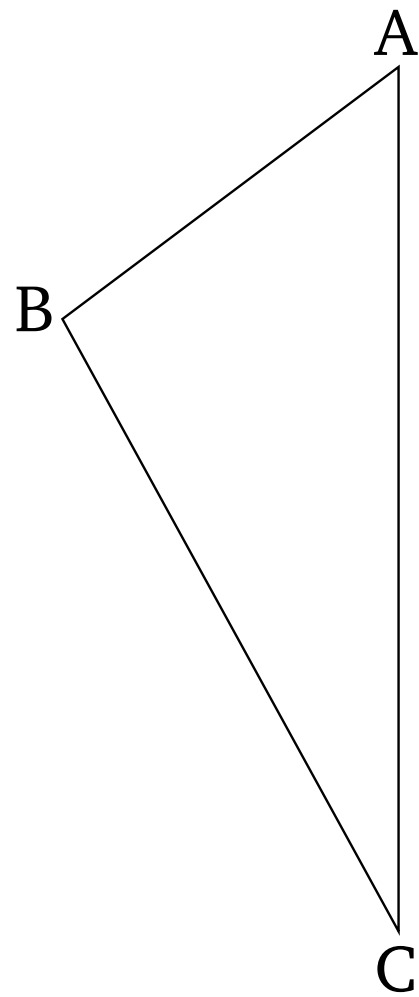
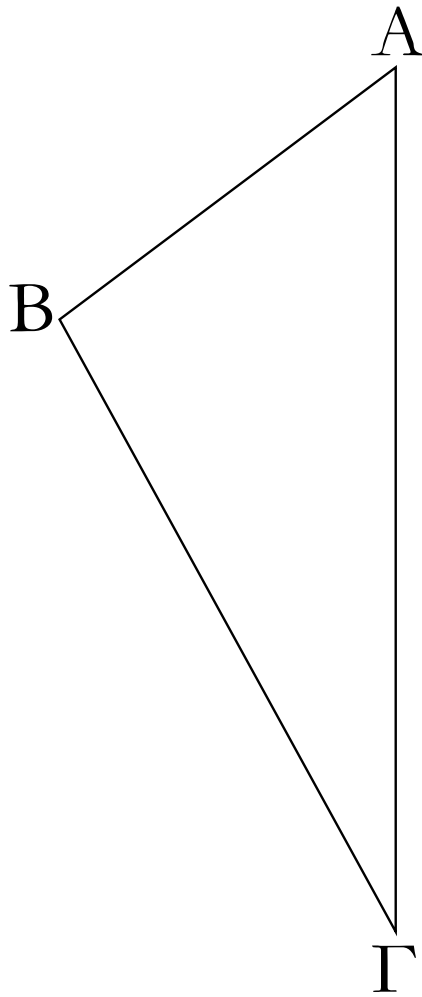
For since AC is greater than AB , let AD be made equal to AB [Prop. 1.3], and let BD be joined.

And since angle ADB is external to triangle BCD , it is greater than the internal and opposite (angle) DCB [Prop. 1.16]. But ADB (is) equal to ABD , since side AB is also equal to side AD [Prop. 1.5]. Thus, ABD is also greater than ACB . Thus, ABC is much greater than ACB . Thus, in any triangle, the greater side subtends the greater angle. (Which is) the very thing it was required to show.

Proposition 19

In any triangle, the greater angle is subtended by the greater side.

Let ABC be a triangle having the angle ABC greater than BCA . I say that side AC is also greater than side AB .



Εἰ γὰρ μή, >ήτοι >ίση ἐστὶν ἡ ΑΓ τῇ ΑΒ >ἢ ἐλάσσων; >ίση μὲν ο>ὺν οὐκ >έστιν ἡ ΑΓ τῇ ΑΒ; >ίση γὰρ >ὰν >ῆν καὶ γωνία ἡ ὑπὸ ΑΒΓ τῇ ὑπὸ ΑΓΒ; οὐκ >έστι δέ; οὐκ >άρα >ίση ἐστὶν ἡ ΑΓ τῇ ΑΒ. οὐδὲ μὴν ἐλάσσων ἐστὶν ἡ ΑΓ τῇ ΑΒ; ἐλάσσων γὰρ >ὰν >ῆν καὶ γωνία ἡ ὑπὸ ΑΒΓ τῇ ὑπὸ ΑΓΒ; οὐκ >έστι δέ; οὐκ >άρα ἐλάσσων ἐστὶν ἡ ΑΓ τῇ ΑΒ. ἐδείκθη δέ, <ὅτι οὐδὲ >ίση ἐστὶν. μείζων >άρα ἐστὶν ἡ ΑΓ τῇ ΑΒ.

Παντὸς >άρα τριγώνου ὑπὸ τὴν μείζονα γωνίαν ἡ μείζων πλευρὰ ὑποτείνεται; <όπερ >έδει δεῖξαι.

20.

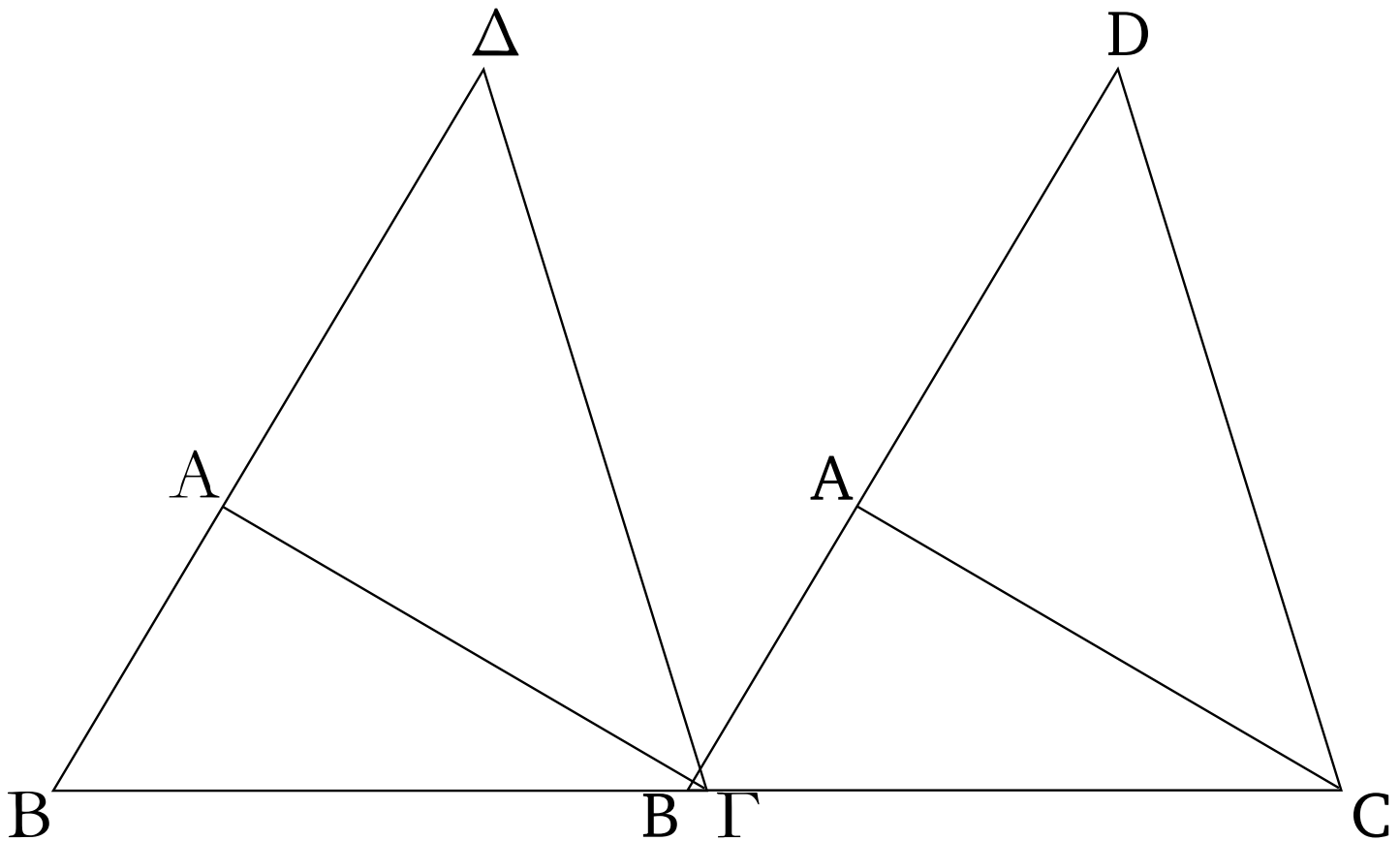
Παντὸς τριγώνου αἱ δύο πλευραὶ τῇς λοιπῇς μείζονές εἰσι πάντη μεταλαμβανόμεναι.

For if not, AC is certainly either equal to, or less than, AB . In fact, AC is not equal to AB . For then angle ABC would also be equal to ACB [Prop. 1.5]. But it is not. Thus, AC is not equal to AB . Neither, indeed, is AC less than AB . For then angle ABC would also be less than ACB [Prop. 1.18]. But it is not. Thus, AC is not less than AB . But it was shown that (AC) is not equal (to AB) either. Thus, AC is greater than AB .

Thus, in any triangle, the greater angle is subtended by the greater side. (Which is) the very thing it was required to show.

Proposition 20

In any triangle, (the sum of) two sides taken together in any (possible way) is greater than the remaining (side).



>Ἐστω γάρ τρίγωνον τὸ $AB\Gamma$; λέγω, <ὅτι τοῦ $AB\Gamma$ τριγώνου αἱ δύο πλευραὶ τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι αἰ μὲν BA , AG τῆς $B\Gamma$, αἱ δὲ AB , $B\Gamma$ τῆς AG , αἱ δὲ $B\Gamma$, GA τῆς AB .

Διήθθω γὰρ ἡ BA ἐπὶ τὸ Δ σημεῖον, καὶ κείσθω τῇ GA ἴση ἡ $A\Delta$, καὶ ἐπεζεύθθω ἡ $\Delta\Gamma$.

Ἐπεὶ οὖν ἴση ἐστὶν ἡ ΔA τῇ AG , ἴση ἐστὶ καὶ γωνία ἡ ὑπὸ $A\Delta\Gamma$ τῇ ὑπὸ $AG\Delta$; μείζων >άρα ἡ ὑπὸ $B\Gamma\Delta$ τῆς ὑπὸ $A\Delta\Gamma$; καὶ ἐπεὶ τρίγωνόν ἐστι τὸ $\Delta\Gamma B$ μείζονα >έθον τὴν ὑπὸ $B\Gamma\Delta$ γωνίαν τῆς ὑπὸ $B\Delta\Gamma$, ὑπὸ δὲ τὴν μείζονα γωνίαν ἡ μείζων πλευρὰ ὑποτείνει, ἡ ΔB >άρα τῆς $B\Gamma$ ἐστὶ μείζων. ἴση δὲ ἡ ΔA τῇ AG ; μείζονες >άρα αἱ BA , AG τῆς $B\Gamma$; ὁμοίως δὲ δείχομεν, <ὅτι καὶ αἱ μὲν AB , $B\Gamma$ τῆς GA μείζονές εἰσιν, αἱ δὲ $B\Gamma$, GA τῆς AB .

Παντὸς >άρα τριγώνου αἱ δύο πλευραὶ τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι; <ὅπερ >έδει δεῖχαι.

21.

Ἐὰν τριγώνου ἐπὶ μιᾷ τῶν πλευρῶν ἀπὸ τῶν περάτων δύο εὐθεῖαι ἐντὸς συσταθῶσιν, αἱ συσταθεῖσαι τῶν λοιπῶν τοῦ τριγώνου δύο πλευρῶν ἐλάττονες μὲν >έσονται, μείζονα δὲ γωνίαν περιέχουσιν.

Τριγώνου γάρ τοῦ $AB\Gamma$ ἐπὶ μιᾷ τῶν πλευρῶν τῆς $B\Gamma$ ἀπὸ τῶν περάτων τῶν B , Γ δύο εὐθεῖαι ἐντὸς συνεστάτωσαν αἱ BD , $\Delta\Gamma$; λέγω, <ὅτι αἱ BD , $\Delta\Gamma$ τῶν λοιπῶν τοῦ τριγώνου δύο πλευρῶν τῶν BA , AG ἐλάττονες μὲν εἰσιν, μείζονα δὲ γωνίαν περιέθουσι τὴν ὑπὸ $B\Delta\Gamma$ τῆς ὑπὸ BAG .

For let ABC be a triangle. I say that in triangle ABC (the sum of) two sides taken together in any (possible way) is greater than the remaining (side). (So), (the sum of) BA and AC (is greater) than BC , (the sum of) AB and BC than AC , and (the sum of) BC and CA than AB .

For let BA be drawn through to point D , and let AD be made equal to CA [Prop. 1.3], and let DC be joined.

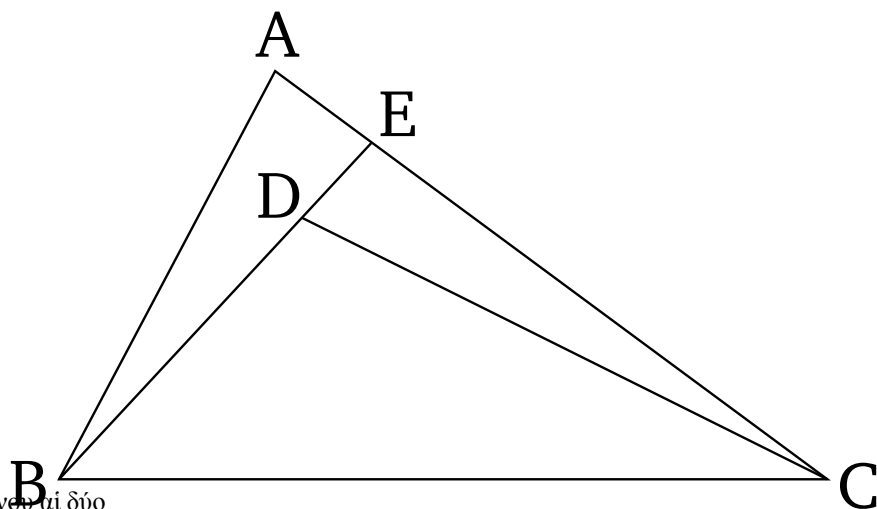
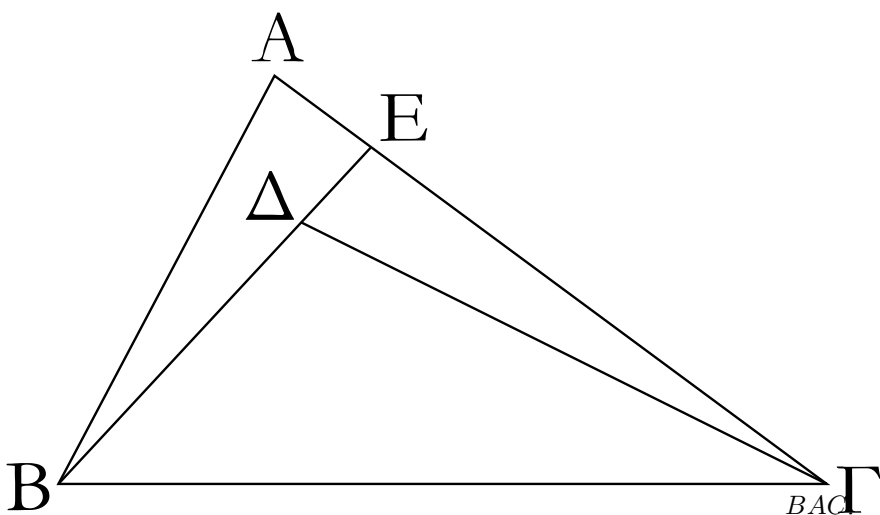
Therefore, since DA is equal to AC , the angle ADC is also equal to ACD [Prop. 1.5]. Thus, BCD is greater than ADC . And since DCB is a triangle having the angle BCD greater than BDC , and the greater angle subtends the greater side [Prop. 1.19], DB is thus greater than BC . But DA is equal to AC . Thus, (the sum of) BA and AC is greater than BC . Similarly, we can show that (the sum of) AB and BC is also greater than CA , and (the sum of) BC and CA than AB .

Thus, in any triangle, (the sum of) two sides taken together in any (possible way) is greater than the remaining (side). (Which is) the very thing it was required to show.

Proposition 21

If two internal straight-lines are constructed on one of the sides of a triangle, from its ends, the constructed (straight-lines) will be less than the two remaining sides of the triangle, but will encompass a greater angle.

For let the two internal straight-lines BD and DC be constructed on one of the sides BC of the triangle ABC , from its ends B and C (respectively). I say that BD and DC are less than the (sum of the) two remaining sides of the triangle BA and AC , but encompass an angle BDC greater than



Διήθθω γὰρ ἡ ΒΔ ἐπὶ τὸ Ε. καὶ ἐπεὶ παντὸς τριγώνου αἱ δύο πλευραὶ τῆς λοιπῆς μείζονές εἰσιν, τοῦ ΑΒΕ > ἄρα τριγώνου αἱ δύο πλευραὶ αἱ ΑΒ, ΑΕ τῆς ΒΕ μείζονές εἰσιν; κοινὴ προσκείσθω ἡ ΕΓ; αἱ > ἄρα ΒΑ, ΑΓ τῶν ΒΕ, ΕΓ μείζονές εἰσιν. πάλιν, ἐπεὶ τοῦ ΓΕΔ τριγώνου αἱ δύο πλευραὶ αἱ ΓΕ, ΕΔ τῆς ΓΔ μείζονές εἰσιν, κοινὴ προσκείσθω ἡ ΔΒ; αἱ ΓΕ, ΕΒ > ἄρα τῶν ΓΔ, ΔΒ μείζονές εἰσιν. ἀλλὰ τῶν ΒΕ, ΕΓ μείζονες ἐδείκθησαν αἱ ΒΑ, ΑΓ; πολλῶν > ἄρα αἱ ΒΑ, ΑΓ τῶν ΒΔ, ΔΓ μείζονές εἰσιν.

Πάλιν, ἐπεὶ παντὸς τριγώνου ἡ ἐκτὸς γωνία τῆς ἐντὸς καὶ ἀπεναντίον μείζων ἐστίν, τοῦ ΓΔΕ > ἄρα τριγώνου ἡ ἐκτὸς γωνία ἡ ὑπὸ ΒΔΓ μείζων ἐστὶ τῆς ὑπὸ ΓΕΔ. διὰ ταῦτὰ τοίνυν καὶ τοῦ ΑΒΕ τριγώνου ἡ ἐκτὸς γωνία ἡ ὑπὸ ΓΕΒ μείζων ἐστὶ τῆς ὑπὸ ΒΑΓ. ἀλλὰ τῆς ὑπὸ ΓΕΒ μείζων ἐδείκθη ἡ ὑπὸ ΒΔΓ; πολλῶν > ἄρα ἡ ὑπὸ ΒΔΓ μείζων ἐστὶ τῆς ὑπὸ ΒΑΓ.

Ἐὰν > ἄρα τριγώνου ἐπὶ μιᾷ τῶν πλευρῶν ἀπὸ τῶν περάτων δύο εὐθεῖαι ἐντὸς συσταθῶσιν, αἱ συσταθεῖσαι τῶν λοιπῶν τοῦ τριγώνου δύο πλευρῶν ἐλάττονες μὲν εἰσιν, μείζονα δὲ γωνίαν περιέθουσιν; < ὅπερ > εἶδει δεῖχαι.

For let BD be drawn through to E . And since in any triangle (the sum of any) two sides is greater than the remaining (side) [Prop. 1.20], in triangle ABE the (sum of the) two sides AB and AE is thus greater than BE . Let EC be added to both. Thus, (the sum of) BA and AC is greater than (the sum of) BE and EC . Again, since in triangle CED the (sum of the) two sides CE and ED is greater than CD , let DB be added to both. Thus, (the sum of) CE and EB is greater than (the sum of) CD and DB . But, (the sum of) BA and AC was shown (to be) greater than (the sum of) BE and EC . Thus, (the sum of) BA and AC is much greater than (the sum of) BD and DC .

Again, since in any triangle the external angle is greater than the internal and opposite (angles) [Prop. 1.16], in triangle CDE the external angle BDC is thus greater than CED . Accordingly, for the same (reason), the external angle CEB of the triangle ABE is also greater than BAC . But, BDC was shown (to be) greater than CEB . Thus, BDC is much greater than BAC .

Thus, if two internal straight-lines are constructed on one of the sides of a triangle, from its ends, the constructed (straight-lines) are less than the two remaining sides of the triangle, but encompass a greater angle. (Which is) the very thing it was required to show.

22.

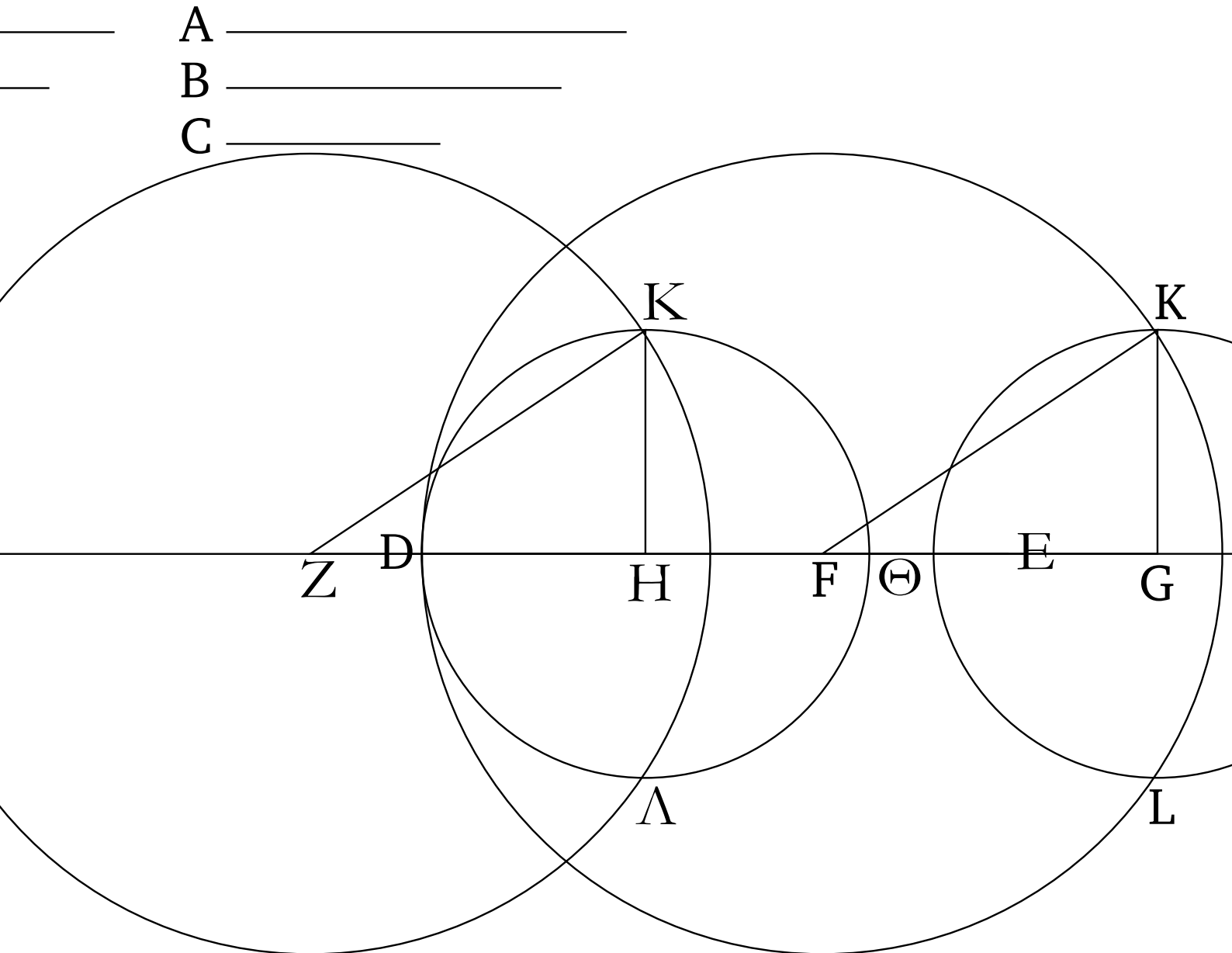
Proposition 22

Ἐκ τριῶν εὐθειῶν, αἱ εἰσιν > ἴσαι τρισὶ ταῖς δοθείσαις [εὐθείαις], τρίγωνον συστήσασθαι; δεῖ δὲ τὰς δύο τῆς

To construct a triangle from three straight-lines which are equal to three given [straight-lines]. It is necessary for (the

λοιπῆς μείζονα εἶναι πάντη μεταλαμβανομένης [διὰ τὸ καὶ παντὸς τριγώνου τὰς δύο πλευρὰς τῆς λοιπῆς μείζονα εἶναι πάντη μεταλαμβανομένης].

sum of) two (of the straight-lines) taken together in any (possible way) to be greater than the remaining (one), [on account of the (fact that) in any triangle (the sum of) two sides taken together in any (possible way) is greater than the remaining (one) [Prop. 1.20]].



>Ἐστωσαν αἱ δοθεῖσαι τρεῖς εὐθεῖαι αἱ A, B, Γ , <ὥν αἱ δύο τῆς λοιπῆς μείζονες >έστωσαν πάντη μεταλαμβανόμεναι, αἱ μὲν A, B τῆς Γ , αἱ δὲ A, Γ τῆς B , καὶ >έτι αἱ B, Γ τῆς A ; δεῖ δὴ ἐκ τῶν >ίσων ταῖς A, B, Γ τρίγωνον συστήσασθαι. Ἐκκεῖσθω τις εὐθεῖα ἡ ΔE πεπερασμένη μὲν κατὰ τὸ Δ >άπειρος δὲ κατὰ τὸ E , καὶ κείσθω τῇ μὲν A >ῖση ἡ ΔZ , τῇ δὲ B >ῖση ἡ ZH , τῇ δὲ Γ >ῖση ἡ $H\Theta$; καὶ κέντρῳ μὲν τῷ Z , διαστήματι δὲ τῷ ΔZ κύκλος γεγράφθω ὁ $\Delta K\Lambda$; πάλιν κέντρῳ μὲν τῷ H , διαστήματι δὲ τῷ $H\Theta$ κύκλος γεγράφθω ὁ $K\Lambda\Theta$, καὶ ἐπεεὺθώσαν αἱ KZ, KH ; λέγω, <ὅτι ἐκ τριῶν εὐθειῶν τῶν >ίσων ταῖς A, B, Γ τρίγωνον συνέσταται τὸ KZH .

Ἐπεὶ γὰρ τὸ Z σημεῖον κέντρον ἐστὶ τοῦ $\Delta K\Lambda$ κύκλου, >ῖση ἐστὶν ἡ ΔZ τῇ ZK ; ἀλλὰ ἡ ΔZ τῇ A ἐστὶν >ῖση. καὶ ἡ KZ >άρα τῇ A ἐστὶν >ῖση. πάλιν, ἐπεὶ τὸ H σημεῖον κέντρον ἐστὶ τοῦ $K\Lambda\Theta$ κύκλου, >ῖση ἐστὶν ἡ $H\Theta$ τῇ HK ; ἀλλὰ ἡ $H\Theta$ τῇ Γ ἐστὶν >ῖση; καὶ ἡ KH >άρα τῇ Γ ἐστὶν >ῖση. ἐστὶ δὲ

Let A, B , and C be the three given straight-lines, of which let (the sum of) two taken together in any (possible way) be greater than the remaining (one). (Thus), (the sum of) A and B (is greater) than C , (the sum of) A and C than B , and also (the sum of) B and C than A . So it is required to construct a triangle from (straight-lines) equal to A, B , and C .

Let some straight-line DE be set out, terminated at D , and infinite in the direction of E . And let DF made equal to A , and FG equal to B , and GH equal to C [Prop. 1.3]. And let the circle DKL be drawn with center F and radius FD . Again, let the circle KLH be drawn with center G and radius GH . And let KF and KG be joined. I say that the triangle KFG has been constructed from three straight-lines equal to A, B , and C .

For since point F is the center of the circle DKL , FD is equal to FK . But, FD is equal to A . Thus, KF is also equal to A . Again, since point G is the center of the circle

καὶ ἡ ZH τῇ B ὀϊση; αἱ τρεῖς ὅρα εὐθεῖαι αἱ KZ , ZH , HK τρισὶ ταῖς A , B , Γ ὀϊσαι εἰσίν.
Ἐκ τριῶν ὅρα εὐθειῶν τῶν KZ , ZH , HK , αἱ εἰσιν ὀϊσαι τρισὶ ταῖς δοθείσαις εὐθείαις ταῖς A , B , Γ , τριγώνον συνέσταται τὸ KZH ; ὅπερ ὀδεῖ ποιῆσαι.

LKH , GH is equal to GK . But, GH is equal to C . Thus, KG is also equal to C . And FG is also equal to B . Thus, the three straight-lines KF , FG , and GK are equal to A , B , and C (respectively).

Thus, the triangle KFG has been constructed from the three straight-lines KF , FG , and GK , which are equal to the three given straight-lines A , B , and C (respectively). (Which is) the very thing it was required to do.

23.

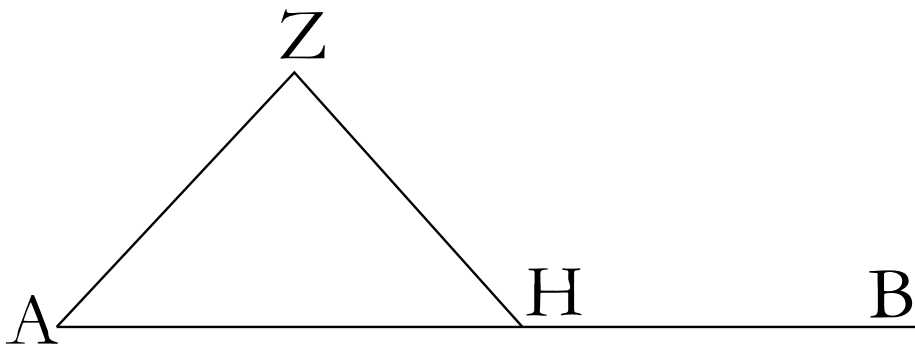
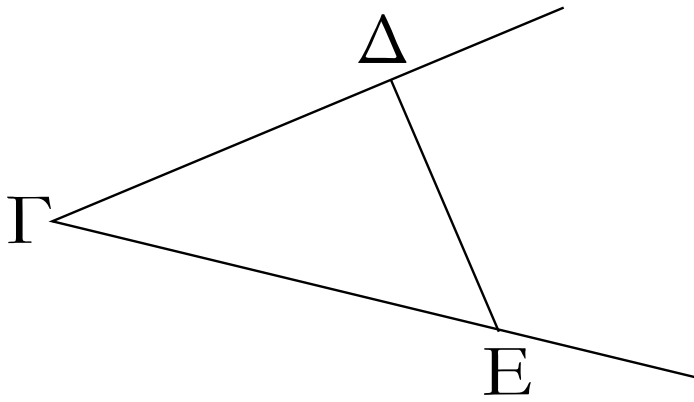
Πρὸς τῇ δοθείσῃ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῇ δοθείσῃ γωνίᾳ εὐθύγραμμῳ ὀϊσην γωνίαν εὐθύγραμμον συστήσασθαι.

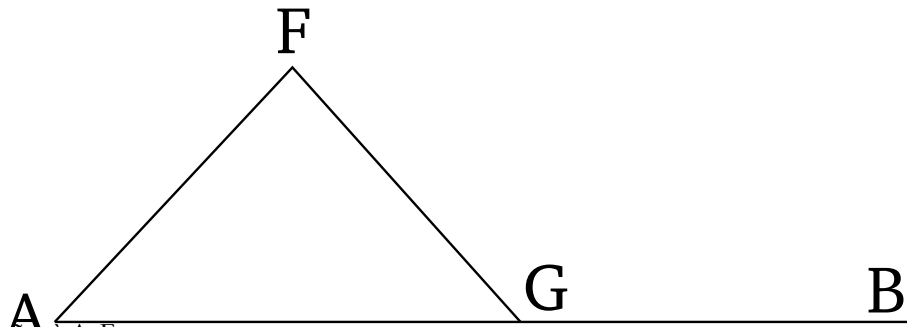
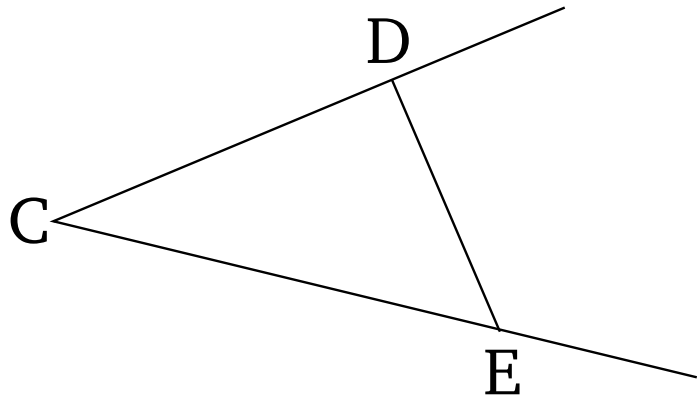
Ἐστω ἡ μὲν δοθεῖσα εὐθεῖα ἡ AB , τὸ δὲ πρὸς αὐτῇ σημείον τὸ A , ἡ δὲ δοθεῖσα γωνία εὐθύγραμμοξ ἡ ὑπὸ $\Delta\Gamma E$; δεῖ δὲ πρὸς τῇ δοθείσῃ εὐθείᾳ τῇ AB καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ A τῇ δοθείσῃ γωνίᾳ εὐθύγραμμῳ τῇ ὑπὸ $\Delta\Gamma E$ ὀϊσην γωνίαν εὐθύγραμμον συστήσασθαι.

Proposition 23

To construct a rectilinear angle equal to a given rectilinear angle at a (given) point on a given straight-line.

Let AB be the given straight-line, A the (given) point on it, and DCE the given rectilinear angle. So it is required to construct a rectilinear angle equal to the given rectilinear angle DCE at the (given) point A on the given straight-line AB .





Εἰλήφθω ἐφ' ἑκατέρᾳ τῶν ΓΔ, ΓΕ τυθόντα σημεῖα Δ, Ε, καὶ ἐπεζεύθθω ἡ ΔΕ; καὶ ἐκ τριῶν εὐθειῶν, αἱ εἰσὶν >ῖσαι τρισὶ ταῖς ΓΔ, ΔΕ, ΓΕ, τρίγωνον συνεστάτω τὸ ΑΖΗ, <ώστε >ῖσιν ἐῖναι τὴν μὲν ΓΔ τῇ ΑΖ, τὴν δὲ ΓΕ τῇ ΑΗ, καὶ >έτι τὴν ΔΕ τῇ ΖΗ.

Ἐπεὶ οὖν δύο αἱ ΔΓ, ΓΕ δύο ταῖς ΖΑ, ΑΗ >ῖσαι εἰσὶν ἑκατέρα ἑκατέρα, καὶ βάσις ἡ ΔΕ βάσει τῇ ΖΗ >ῖση, γωνία >άρα ἡ ὑπὸ ΔΓΕ γωνία τῇ ὑπὸ ΖΑΗ ἐστὶν >ῖση.

Πρὸς >άρα τῇ δοθείσῃ εὐθείᾳ τῇ ΑΒ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ Α τῇ δοθείσῃ γωνίᾳ εὐθυγράμμω τῇ ὑπὸ ΔΓΕ >ῖση γωνίᾳ εὐθύγραμμος συνέσταται ἡ ὑπὸ ΖΑΗ; <όπερ >έδει ποιῆσαι.

Let the points D and E be taken at random on each of the (straight-lines) CD and CE (respectively), and let DE be joined. And let the triangle AFG be constructed from three straight-lines which are equal to CD , DE , and CE , such that CD is equal to AF , CE to AG , and further DE to FG [Prop. 1.22].

Therefore, since the two (straight-lines) DC , CE are equal to the two (straight-lines) FA , AG , respectively, and the base DE is equal to the base FG , the angle DCE is thus equal to the angle FAG [Prop. 1.8].

Thus, the rectilinear angle FAG , equal to the given rectilinear angle DCE , has been constructed at the (given) point A on the given straight-line AB . (Which is) the very thing it was required to do.

24.

Ἐὰν δύο τρίγωνα τὰς δύο πλευρὰς [ταῖς] δύο πλευραῖς >ῖσαξ >έθῃ ἑκατέραν ἑκατέρα, τὴν δὲ γωνίαν τῆς γωνίαξ μείζονα >έθῃ τὴν ὑπὸ τῶν >ῖσων εὐθειῶν περιεθρομένην, καὶ τὴν βάσιν τῆς βάσεως μείζονα <έχει.

>Ἐστω δύο τρίγωνα τὰ ΑΒΓ, ΔΕΖ τὰς δύο πλευρὰς τὰς ΑΒ, ΑΓ ταῖς δύο πλευραῖς ταῖς ΔΕ, ΔΖ >ῖσαξ >έθοντα ἑκατέραν ἑκατέρα, τὴν μὲν ΑΒ τῇ ΔΕ τὴν δὲ ΑΓ τῇ ΔΖ, ἡ δὲ πρὸς τῷ Α γωνία τῆς πρὸς τῷ Δ γωνίαξ μείζων >έστω; λέγω, <ὅτι καὶ βάσις ἡ ΒΓ βάσεως τῆς ΕΖ μείζων ἐστίν.

Ἐπεὶ γὰρ μείζων ἡ ὑπὸ ΒΑΓ γωνία τῇ ὑπὸ ΕΔΖ γωνίαξ, συνεστάτω πρὸς τῇ ΔΕ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ Δ τῇ ὑπὸ ΒΑΓ γωνίᾳ >ῖση ἡ ὑπὸ ΕΔΗ, καὶ κείσθω ὁποτέρᾳ τῶν ΑΓ, ΔΖ >ῖση ἡ ΔΗ, καὶ ἐπεζεύθθωσαν αἱ ΕΗ, ΖΗ.

Proposition 24

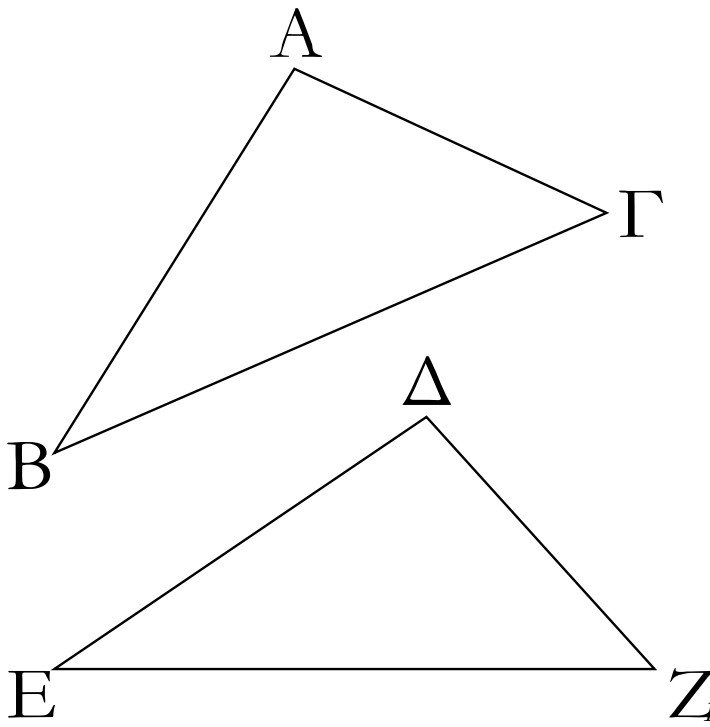
If two triangles have two sides equal to two sides, respectively, but (one) has the angle encompassed by the equal straight-lines greater than the (corresponding) angle (in the other), (then the former triangle) will also have a base greater than the base (of the latter).

Let ABC and DEF be two triangles having the two sides AB and AC equal to the two sides DE and DF , respectively. (That is), AB (equal) to DE , and AC to DF . Let them also have the angle at A greater than the angle at D . I say that the base BC is also greater than the base EF .

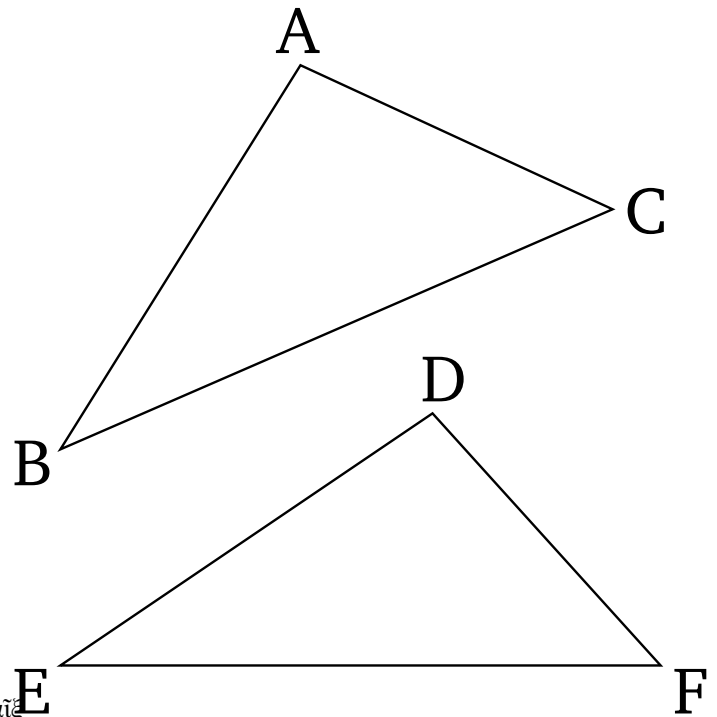
For since angle BAC is greater than angle EDF , let (angle) EDG , equal to angle BAC , be constructed at the point D on the straight-line DE [Prop. 1.23]. And let DG be made equal to either of AC or DF [Prop. 1.3], and let EG and FG be joined.

ἢ ὑπὸ $BA\Gamma$ γωνία τῆς ὑπὸ $E\Delta Z$. ἐδείκθη δέ, <ὅτι οὐδὲ >ίση;
μείζων >ἀρα ἐστὶν ἢ ὑπὸ $BA\Gamma$ τῆς ὑπὸ $E\Delta Z$.

BC would also be less than the base EF [Prop. 1.24]. But it is not. Thus, angle BAC is not less than EDF . But it was shown that (BAC) is not equal (to EDF) either. Thus,



$\angle BAC$ is greater than EDF .



Ἐὰν >ἀρα δύο τρίγωνα τὰς δύο πλευρὰς δυσὶ πλευραῖς
>ίσαξ >έθῃ ἐκατέραν ἐκάτερα, τὴν δὲ βασὶν τῆς βάσεως
μείζονα >έθῃ, καὶ τὴν γωνίαν τῆς γωνίας μείζονα <έχει τὴν
ὑπὸ τῶν >ίσων εὐθειῶν περιεθομένην; <ὅπερ >έδει δεῖχαι.

Thus, if two triangles have two sides equal to two sides, respectively, but (one) has a base greater than the base (of the other), (then the former triangle) will also have the angle encompassed by the equal straight-lines greater than the (corresponding) angle (in the latter). (Which is) the very thing it was required to show.

26.

Proposition 26

Ἐὰν δύο τρίγωνα τὰς δύο γωνίαξ δυσὶ γωνίαξ >ίσαξ >έθῃ
ἐκατέραν ἐκάτερα καὶ μίαν πλευρὰν μιᾷ | πλευρᾷ | >ίσην
>ήτοι τὴν πρὸς ταῖς >ίσαιξ γωνίαξ >ἢ τὴν ὑποτείνουσιν
ὑπὸ μίαν τῶν >ίσων γωνιῶν, καὶ τὰς λοιπὰς πλευρὰς ταῖς

If two triangles have two angles equal to two angles, respectively, and one side equal to one side—in fact, either that by the equal angles, or that subtending one of the equal angles—(then the triangles) will also have the remaining

λοιπαῖς πλευραῖς >ῖσαι <έχει [ἐκατέραν ἐκατέρα] καὶ τὴν λοιπὴν γωνίαν τῇ λοιπῇ γωνίᾳ.

>Ἐστω δύο τρίγωνα τὰ $ABΓ$, $ΔΕΖ$ τὰς δύο γωνίας τὰς ὑπὸ $ABΓ$, $ΒΓΑ$ δυσὶ ταῖς ὑπὸ $ΔΕΖ$, $ΕΖΔ$ >ῖσαι >έθοντα ἐκατέραν ἐκατέρα, τὴν μὲν ὑπὸ $ABΓ$ τῇ ὑπὸ $ΔΕΖ$, τὴν δὲ ὑπὸ $ΒΓΑ$ τῇ ὑπὸ $ΕΖΔ$; ἐθέτω δὲ καὶ μίαν πλευρὰν μιᾷ πλευρᾷ >ῖσιν, πρότερον τὴν πρὸς ταῖς >ῖσαις γωνίαις τὴν $ΒΓ$ τῇ $ΕΖ$; λέγω, <ὅτι καὶ τὰς λοιπὰς πλευρὰς ταῖς λοιπαῖς πλευραῖς >ῖσαι <έχει ἐκατέραν ἐκατέρα, τὴν μὲν AB τῇ $ΔΕ$ τὴν δὲ $ΑΓ$ τῇ $ΔΖ$, καὶ τὴν λοιπὴν γωνίαν τῇ λοιπῇ γωνίᾳ, τὴν ὑπὸ $BAΓ$ τῇ ὑπὸ $ΕΔΖ$.

Εἰ γὰρ >άνισός ἐστιν ἡ AB τῇ $ΔΕ$, μία αὐτῶν μείζων ἐστίν. >έστω μείζων ἡ AB , καὶ κείσθω τῇ $ΔΕ$ >ῖση ἡ BH , καὶ ἐπεζεύθω ἡ $ΗΓ$.

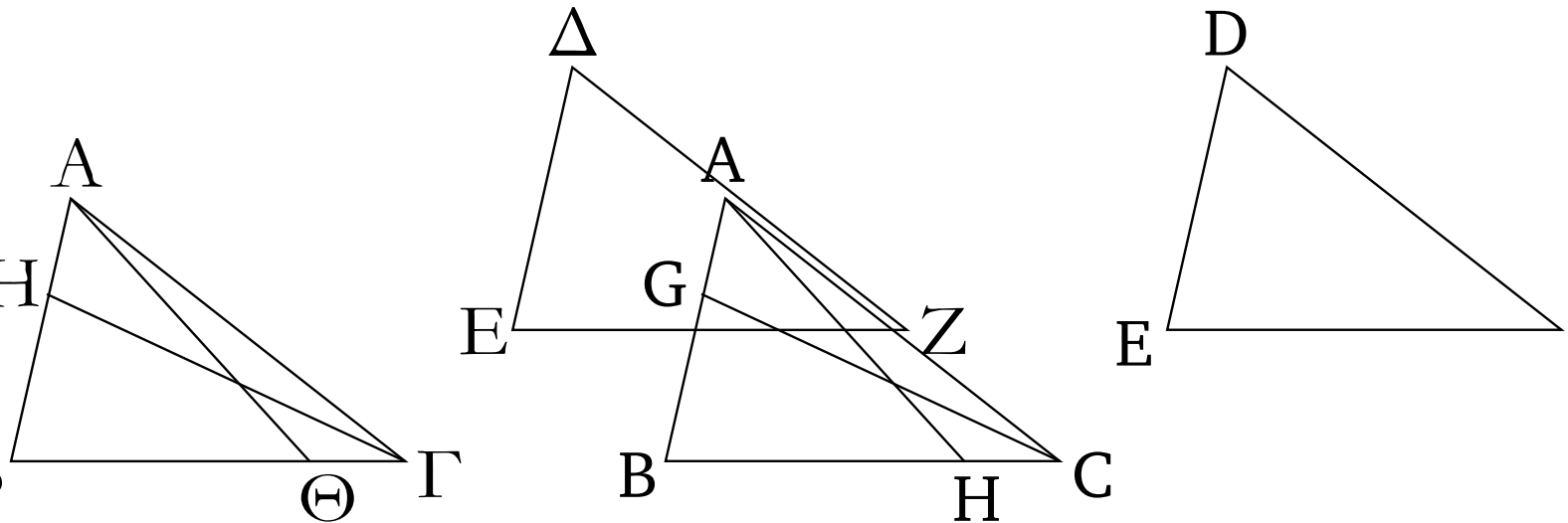
Ἐπεὶ ο>ὺν >ῖση ἐστίν ἡ μὲν BH τῇ $ΔΕ$, ἡ δὲ $ΒΓ$ τῇ $ΕΖ$, δύο δὴ αἱ BH , $ΒΓ$ δυσὶ ταῖς $ΔΕ$, $ΕΖ$ >ῖσαι εἰσὶν ἐκατέρα ἐκατέρα; καὶ γωνία ἡ ὑπὸ $HBΓ$ γωνία τῇ ὑπὸ $ΔΕΖ$ >ῖση ἐστίν; βάσις >άρα ἡ $ΗΓ$ βάσει τῇ $ΔΖ$ >ῖση ἐστίν, καὶ τὸ $HBΓ$ τρίγωνον τῷ $ΔΕΖ$ τριγώνῳ >ῖσον ἐστίν, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις >ῖσαι >έσσονται, ὅφ> <ὰς αἱ >ῖσαι πλευραὶ ὑποτείνουσιν; >ῖση >άρα ἡ ὑπὸ $ΗΓΒ$ γωνία τῇ ὑπὸ $ΔΖΕ$. ἀλλὰ ἡ ὑπὸ $ΔΖΕ$ τῇ ὑπὸ $ΒΓΑ$ ὑπόκειται >ῖση; καὶ ἡ ὑπὸ $ΒΓΗ$ >άρα τῇ ὑπὸ $ΒΓΑ$ >ῖση ἐστίν, ἡ ἐλάσσων τῇ μείζονι; <όπερ ἀδύνατον. οὐκ >άρα >άνισός ἐστιν ἡ AB τῇ $ΔΕ$. >ῖση >άρα. >έστι δὲ καὶ ἡ $ΒΓ$ τῇ $ΕΖ$ >ῖση; δύο δὴ αἱ AB , $ΒΓ$ δυσὶ ταῖς $ΔΕ$, $ΕΖ$ >ῖσαι εἰσὶν ἐκατέρα ἐκατέρα; καὶ γωνία ἡ ὑπὸ $ABΓ$ γωνία τῇ ὑπὸ $ΔΕΖ$ ἐστίν >ῖση; βάσις >άρα ἡ $ΑΓ$ βάσει τῇ $ΔΖ$ >ῖση ἐστίν, καὶ λοιπὴ γωνία ἡ ὑπὸ $BAΓ$ τῇ λοιπῇ γωνίᾳ τῇ ὑπὸ $ΕΔΖ$ >ῖση ἐστίν.

sides equal to the [corresponding] remaining sides, and the remaining angle (equal) to the remaining angle.

Let ABC and DEF be two triangles having the two angles ABC and BCA equal to the two (angles) DEF and EFD , respectively. (That is) ABC (equal) to DEF , and BCA to EFD . And let them also have one side equal to one side. First of all, the (side) by the equal angles. (That is) BC (equal) to EF . I say that they will have the remaining sides equal to the corresponding remaining sides. (That is) AB (equal) to DE , and AC to DF . And (they will have) the remaining angle (equal) to the remaining angle. (That is) BAC (equal) to EDF .

For if AB is unequal to DE (then) one of them is greater. Let AB be greater, and let BG be made equal to DE [Prop. 1.3], and let GC be joined.

Therefore, since BG is equal to DE , and BC to EF , the two (straight-lines) GB , BC † are equal to the two (straight-lines) DE , EF , respectively. And angle GBC is equal to angle DEF . Thus, the base GC is equal to the base DF , and triangle GBC is equal to triangle DEF , and the remaining angles subtended by the equal sides will be equal to the (corresponding) remaining angles [Prop. 1.4]. Thus, GCB (is equal) to DFE . But, DFE was assumed (to be) equal to BCA . Thus, BCG is also equal to BCA , the lesser to the greater. The very thing (is) impossible. Thus, AB is not unequal to DE . Thus, (it is) equal. And BC is also equal to EF . So the two (straight-lines) AB , BC are equal to the two (straight-lines) DE , EF , respectively. And angle ABC is equal to angle DEF . Thus, the base AC is equal to the base DF , and the remaining angle BAC is equal to the remaining angle EDF [Prop. 1.4].



Ἀλλὰ δὴ πάλιν >έστωσαν αἱ ὑπὸ τὰς >ῖσαις γωνίαις πλευραὶ ὑποτείνουσαι >ῖσαι, ὥς ἡ AB τῇ $ΔΕ$; λέγω πάλιν, <ὅτι καὶ αἱ λοιπαὶ πλευραὶ ταῖς λοιπαῖς πλευραῖς >ῖσαι >έσσονται, ἡ μὲν $ΑΓ$ τῇ $ΔΖ$, ἡ δὲ $ΒΓ$ τῇ $ΕΖ$ καὶ >έτι ἡ λοιπὴ γωνία ἡ ὑπὸ $BAΓ$ τῇ λοιπῇ γωνίᾳ τῇ ὑπὸ $ΕΔΖ$ >ῖση ἐστίν.

Εἰ γὰρ >άνισός ἐστιν ἡ $ΒΓ$ τῇ $ΕΖ$, μία αὐτῶν μείζων ἐστίν. >έστω μείζων, εἰ δυνατόν, ἡ $ΒΓ$, καὶ κείσθω τῇ $ΕΖ$ >ῖση ἡ $BΘ$, καὶ ἐπεζεύθω ἡ $ΑΘ$. καὶ ἐπεὶ >ῖση ἐστίν ἡ μὲν $BΘ$ τῇ $ΕΖ$ ἡ δὲ AB τῇ $ΔΕ$, δύο δὴ αἱ AB , $BΘ$ δυσὶ ταῖς $ΔΕ$, $ΕΖ$ >ῖσαι εἰσὶν ἐκατέρα ἐκατέρα; καὶ γωνία >ῖσαι περιέουσιν; βάσις >άρα ἡ $ΑΘ$ βάσει τῇ $ΔΖ$ >ῖση ἐστίν, καὶ τὸ $ABΘ$ τρίγωνον τῷ $ΔΕΖ$ τριγώνῳ >ῖσον ἐστίν, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις >ῖσαι >έσσονται, ὅφ>

But, again, let the sides subtending the equal angles be equal: for instance, (let) AB (be equal) to DE . Again, I say that the remaining sides will be equal to the remaining sides. (That is) AC (equal) to DF , and BC to EF . Furthermore, the remaining angle BAC is equal to the remaining angle EDF .

For if BC is unequal to EF (then) one of them is greater. If possible, let BC be greater. And let BH be made equal to EF [Prop. 1.3], and let AH be joined. And since BH is equal to EF , and AB to DE , the two (straight-lines) AB , BH are equal to the two (straight-lines) DE , EF , respectively. And the angles they encompass (are also equal). Thus, the base AH is equal to the base DF , and the triangle

<ἂν αἱ >ίσαὶ πλευραὶ ὑποτείνουσιν; >ίση >ἄρα ἐστὶν ἡ ὑπὸ BΘΑ γωνία τῇ ὑπὸ EZΔ. ἀλλὰ ἡ ὑπὸ EZΔ τῇ ὑπὸ BΓΑ ἐστὶν >ίση; τριγώνου δὲ τοῦ AΘΓ ἡ ἐκτὸς γωνία ἡ ὑπὸ BΘΑ >ίση ἐστὶ τῇ ἐντὸς καὶ ἀπεναντίον τῇ ὑπὸ BΓΑ; <ὅπερ ἀδύνατον. οὐκ >ἄρα >ἀνισόξ ἐστὶν ἡ BΓ τῇ EZ; >ίση >ἄρα. ἐστὶ δὲ καὶ ἡ AB τῇ ΔΕ >ίση. δύο δὲ αἱ AB, BΓ δύο ταῖς ΔΕ, EZ >ίσαι εἰσὶν ἑκατέρα ἑκατέρα; καὶ γωνίαξ >ίσαξ περιέθουσι; βάσιξ >ἄρα ἡ AΓ βάσει τῇ ΔΖ >ίση ἐστὶν, καὶ τὸ ABΓ τρίγωνον τῷ ΔΕΖ τριγώνῳ >ίσον καὶ λοιπὴ γωνία ἡ ὑπὸ BΑΓ τῇ λοιπῇ γωνίᾳ τῇ ὑπὸ EΔΖ >ίση.
Ἐὰν >ἄρα δύο τρίγωνα τὰξ δύο γωνίαξ δυσι γωνίαξ >ίσαξ >έθῃ ἑκατέραν ἑκατέρα καὶ μίαν πλευρὰν μιᾷ πλευρᾷ >ίσην >ήτοι τὴν πρὸς ταῖς >ίσαιξ γωνίαξ, >ή τὴν ὑποτείνουσα ὑπὸ μίαν τῶν >ίσων γωνιῶν, καὶ τὰξ λοιπάξ πλευράξ ταῖς λοιπαῖξ πλευραῖξ >ίσαξ <έχει καὶ τὴν λοιπὴν γωνίαν τῇ λοιπῇ γωνίᾳ; <ὅπερ >έδει δεῖχαι.

ABH is equal to the triangle DEF , and the remaining angles subtended by the equal sides will be equal to the (corresponding) remaining angles [Prop. 1.4]. Thus, angle BHA is equal to EFD . But, EFD is equal to BCA . So, in triangle AHC , the external angle BHA is equal to the internal and opposite angle BCA . The very thing (is) impossible [Prop. 1.16]. Thus, BC is not unequal to EF . Thus, (it is) equal. And AB is also equal to DE . So the two (straight-lines) AB , BC are equal to the two (straight-lines) DE , EF , respectively. And they encompass equal angles. Thus, the base AC is equal to the base DF , and triangle ABC (is) equal to triangle DEF , and the remaining angle BAC (is) equal to the remaining angle EDF [Prop. 1.4].

Thus, if two triangles have two angles equal to two angles, respectively, and one side equal to one side—in fact, either that by the equal angles, or that subtending one of the equal angles—(then the triangles) will also have the remaining sides equal to the (corresponding) remaining sides, and the remaining angle (equal) to the remaining angle. (Which is) the very thing it was required to show.

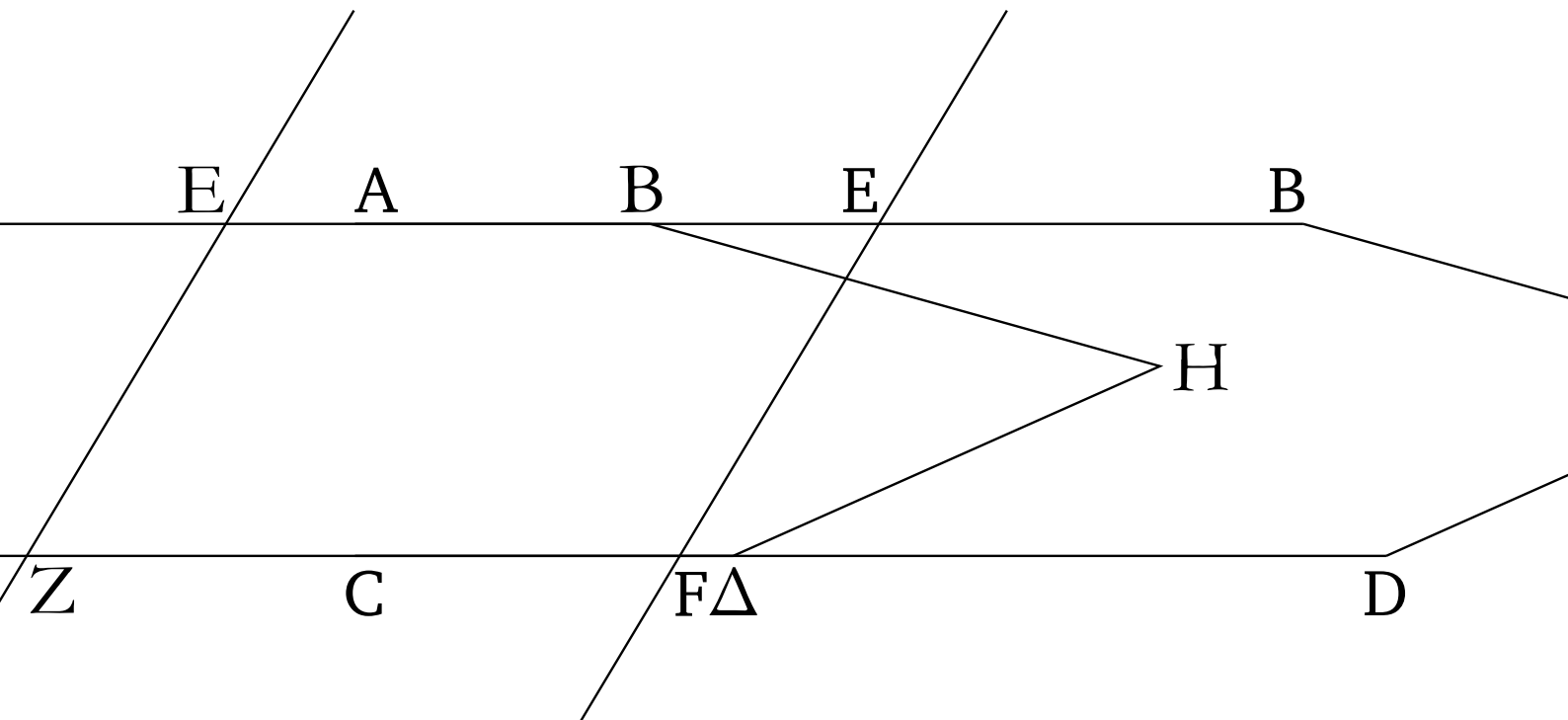
† The Greek text has “ BG , BC ”, which is obviously a mistake.

27.

Ἐὰν εἰξ δύο εὐθείαξ εὐθεῖα ἐμπίπτουσα τὰξ ἐναλλάξ γωνίαξ >ίσαξ ἀλλήλαιξ ποιῇ, παράλληλοι >έσσονται ἀλλήλαιξ αἱ εὐθεῖαι.

Proposition 27

If a straight-line falling across two straight-lines makes the alternate angles equal to one another (then) the (two) straight-lines will be parallel to one another.



Εἰξ γὰρ δύο εὐθείαξ τὰξ AB, ΓΔ εὐθεῖα ἐμπίπτουσα ἡ EZ τὰξ ἐναλλάξ γωνίαξ τὰξ ὑπὸ AEZ, EZΔ >ίσαξ ἀλλήλαιξ ποιέτω; λέγω, <ὅτι παράλληλόξ ἐστὶν ἡ AB τῇ ΓΔ.
Εἰ γὰρ μή, ἐκβαλλόμεναι αἱ AB, ΓΔ συμπεσοῦνται >ήτοι ἐπὶ τὰ B, Δ μέρη >ή ἐπὶ τὰ A, Γ. ἐκβεβλήσθωσαν καὶ συμπιτέτωσαν ἐπὶ τὰ B, Δ μέρη κατὰ τὸ H. τριγώνου δὲ τοῦ HEZ ἡ ἐκτὸς γωνία ἡ ὑπὸ AEZ >ίση ἐστὶ τῇ ἐντὸς καὶ ἀπεναντίον τῇ ὑπὸ EZH; <ὅπερ ἐστὶν ἀδύνατον; οὐκ >ἄρα αἱ AB, ΔΓ ἐκβαλλόμεναι συμπεσοῦνται ἐπὶ τὰ B, Δ μέρη. ὁμοίωξ δὲ δεῖθῃσεται, <ὅτι οὐδὲ ἐπὶ τὰ A, Γ; αἱ

For let the straight-line EF , falling across the two straight-lines AB and CD , make the alternate angles AEF and EFD equal to one another. I say that AB and CD are parallel.

For if not, being produced, AB and CD will certainly meet together: either in the direction of B and D , or (in the direction) of A and C [Def. 1.23]. Let them be produced, and let them meet together in the direction of B and D at (point) G . So, for the triangle GEF , the external angle AEF is equal to the interior and opposite (angle) EFG . The very

δὲ ἐπὶ μηδέτερα τὰ μέρη συμπίπτουσαι παράλληλοί εἰσιν; παράλληλος >άρα ἐστὶν ἡ AB τῇ $\Gamma\Delta$.
 Ἐὰν >άρα εἰς δύο εὐθείας εὐθεῖα ἐμπίπτουσα τὰς ἐναλλάχ
 γωνίας >ίσαξ ἀλλήλαις ποιῇ, παράλληλοι >έσονται αἱ
 εὐθεῖαι; <όπερ >έδει δεῖχαι.

28.

Ἐὰν εἰς δύο εὐθείας εὐθεῖα ἐμπίπτουσα τὴν ἐκτὸς γωνίαν
 τῇ ἐντὸς καὶ ἀπεναντίον καὶ ἐπὶ τὰ αὐτὰ μέρη >ίσην ποιῇ
 >ἢ τὰς ἐντὸς καὶ ἐπὶ τὰ αὐτὰ μέρη δυσὶν ὀρθαῖς >ίσαξ,
 παράλληλοι >έσονται ἀλλήλαις αἱ εὐθεῖαι.
 Εἰς γὰρ δύο εὐθείας τὰς AB , $\Gamma\Delta$ εὐθεῖα ἐμπίπτουσα ἡ EZ
 τὴν ἐκτὸς γωνίαν τὴν ὑπὸ EHB τῇ ἐντὸς καὶ ἀπεναντίον
 γωνία τῇ ὑπὸ $H\Theta\Delta$ >ίσην ποιεῖτω >ἢ τὰς ἐντὸς καὶ ἐπὶ τὰ
 αὐτὰ μέρη τὰς ὑπὸ $BH\Theta$, $H\Theta\Delta$ δυσὶν ὀρθαῖς >ίσαξ; λέγω,
 <ὅτι παράλληλός ἐστιν ἡ AB τῇ $\Gamma\Delta$.
 Ἐπεὶ γὰρ >ίση ἐστὶν ἡ ὑπὸ EHB τῇ ὑπὸ $H\Theta\Delta$, ἀλλὰ ἡ ὑπὸ
 EHB τῇ ὑπὸ $AH\Theta$ ἐστὶν >ίση, καὶ ἡ ὑπὸ $AH\Theta$ >άρα τῇ
 ὑπὸ $H\Theta\Delta$ ἐστὶν >ίση; καὶ εἰσιν ἐναλλάχ; παράλληλος >άρα
 ἐστὶν ἡ AB τῇ $\Gamma\Delta$.
 Πάλιν, ἐπεὶ αἱ ὑπὸ $BH\Theta$, $H\Theta\Delta$ δύο ὀρθαῖς >ίσαι εἰσίν, εἰς
 δὲ καὶ αἱ ὑπὸ $AH\Theta$, $BH\Theta$ δυσὶν ὀρθαῖς >ίσαι, αἱ >άρα
 ὑπὸ $AH\Theta$, $BH\Theta$ ταῖς ὑπὸ $BH\Theta$, $H\Theta\Delta$ >ίσαι εἰσίν; κοινὴ
 ἀφηρήσθω ἡ ὑπὸ $BH\Theta$; λοιπὴ >άρα ἡ ὑπὸ $AH\Theta$ λοιπῇ τῇ
 ὑπὸ $H\Theta\Delta$ ἐστὶν >ίση; καὶ εἰσιν ἐναλλάχ; παράλληλος >άρα
 ἐστὶν ἡ AB τῇ $\Gamma\Delta$.

thing is impossible [Prop. 1.16]. Thus, being produced, AB
 and CD will not meet together in the direction of B and D .
 Similarly, it can be shown that neither (will they meet to-
 gether) in (the direction of) A and C . But (straight-lines)
 meeting in neither direction are parallel [Def. 1.23]. Thus,
 AB and CD are parallel.

Thus, if a straight-line falling across two straight-lines
 makes the alternate angles equal to one another (then) the
 (two) straight-lines will be parallel (to one another). (Which
 is) the very thing it was required to show.

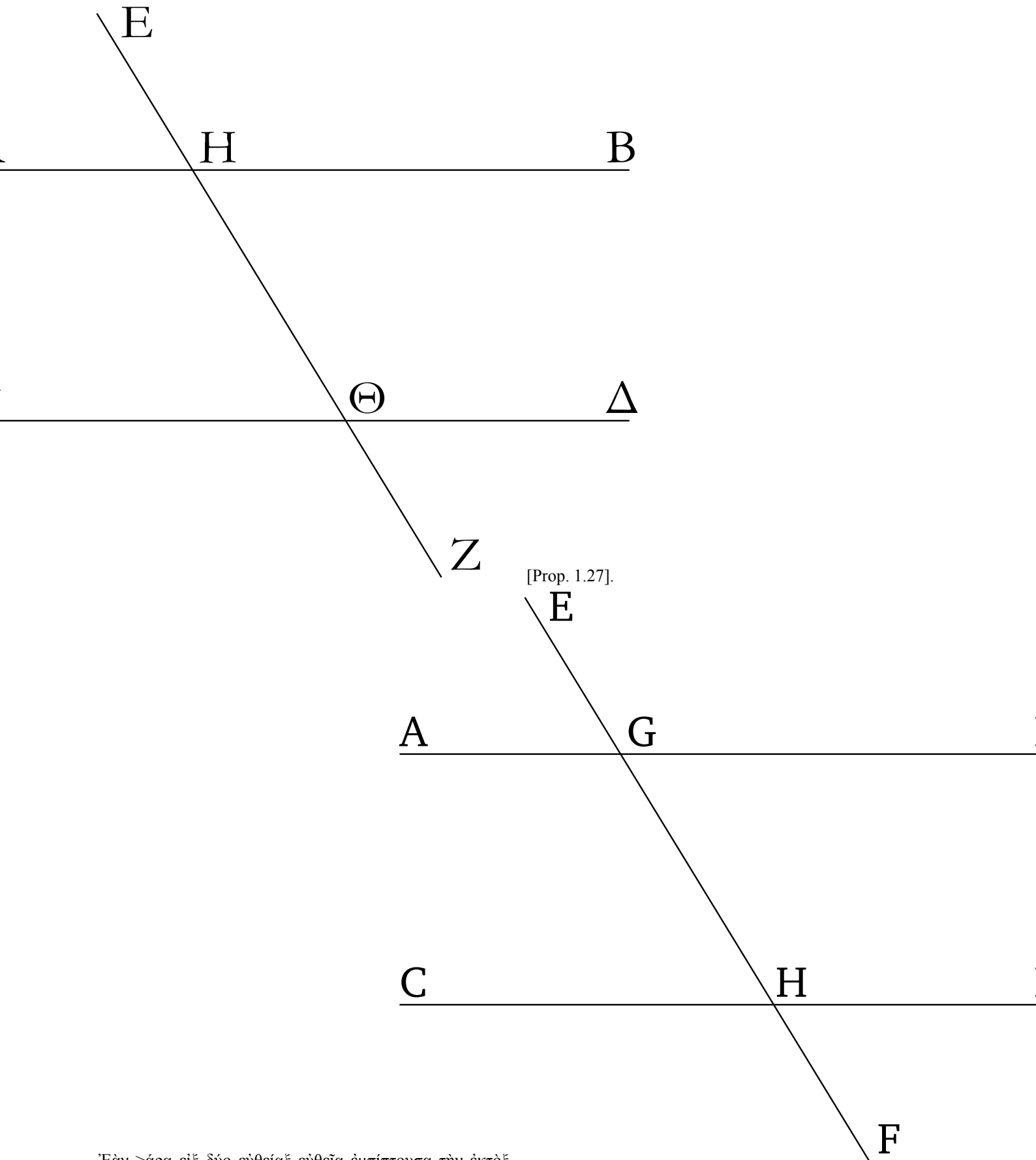
Proposition 28

If a straight-line falling across two straight-lines makes the
 external angle equal to the internal and opposite angle on
 the same side, or (makes) the (sum of the) internal (angles)
 on the same side equal to two right-angles, (then) the (two)
 straight-lines will be parallel to one another.

For let EF , falling across the two straight-lines AB and
 CD , make the external angle EGB equal to the internal
 and opposite angle GHD , or the (sum of the) internal (an-
 gles) on the same side, BGH and GHD , equal to two right-
 angles. I say that AB is parallel to CD .

For since (in the first case) EGB is equal to GHD , but
 EGB is equal to AGH [Prop. 1.15], AGH is thus also equal
 to GHD . And they are alternate (angles). Thus, AB is
 parallel to CD [Prop. 1.27].

Again, since (in the second case, the sum of) BGH and
 GHD is equal to two right-angles, and (the sum of) AGH
 and BGH is also equal to two right-angles [Prop. 1.13],
 (the sum of) AGH and BGH is thus equal to (the sum
 of) BGH and GHD . Let BGH be subtracted from both.
 Thus, the remainder AGH is equal to the remainder GHD .
 And they are alternate (angles). Thus, AB is parallel to CD



Ἐάν >άρα εἰς δύο εὐθείας εὐθεῖα ἐμπίπτουσα τὴν ἐκτὸς
γωνίαν τῇ ἐντὸς καὶ ἀπεναντίον καὶ ἐπὶ τὰ αὐτὰ μέρη >ίσην
ποιῇ >ῇ τὰς ἐντὸς καὶ ἐπὶ τὰ αὐτὰ μέρη δυσὶν ὀρθαῖς >ίσας,
παράλληλοι >έσσονται αἱ εὐθεῖαι; <όπερ >έδει δεῖχαι.

Thus, if a straight-line falling across two straight-lines
makes the external angle equal to the internal and oppo-
site angle on the same side, or (makes) the (sum of the)
internal (angles) on the same side equal to two right-angles,
(then) the (two) straight-lines will be parallel (to one an-

29.

Ἡ εἰς τὰς παραλλήλους εὐθείας εὐθεῖα ἐμπίπτουσα τὰς τε ἐναλλάχ γωνίας >ίσαξ ἀλλήλαις ποιεῖ καὶ τὴν ἐκτὸς τῇ ἐντὸς καὶ ἀπεναντίον >ίσην καὶ τὰς ἐντὸς καὶ ἐπὶ τὰ αὐτὰ μέρη δυσὶν ὀρθαῖς >ίσαξ.

Εἰς γὰρ παραλλήλους εὐθείας τὰς AB , $\Gamma\Delta$ εὐθεῖα ἐμπίπτέτω ἡ EZ ; λέγω, <ὅτι τὰς ἐναλλάχ γωνίας τὰς ὑπὸ $AH\Theta$, $H\Theta\Delta$ >ίσαξ ποιεῖ καὶ τὴν ἐκτὸς γωνίαν τὴν ὑπὸ EHB τῇ ἐντὸς καὶ ἀπεναντίον τῇ ὑπὸ $H\Theta\Delta$ >ίσην καὶ τὰς ἐντὸς καὶ ἐπὶ τὰ αὐτὰ μέρη τὰς ὑπὸ $BH\Theta$, $H\Theta\Delta$ δυσὶν ὀρθαῖς >ίσαξ.

Εἰ γὰρ >άνισός ἐστιν ἡ ὑπὸ $AH\Theta$ τῇ ὑπὸ $H\Theta\Delta$, μία αὐτῶν μείζων ἐστίν. >έστω μείζων ἡ ὑπὸ $AH\Theta$; κοινὴ προσκείσθω ἡ ὑπὸ $BH\Theta$; αἱ >άρα ὑπὸ $AH\Theta$, $BH\Theta$ τῶν ὑπὸ $BH\Theta$, $H\Theta\Delta$ μείζονές εἰσιν. ἀλλὰ αἱ ὑπὸ $AH\Theta$, $BH\Theta$ δυσὶν ὀρθαῖς >ίσαι εἰσιν. [καὶ] αἱ >άρα ὑπὸ $BH\Theta$, $H\Theta\Delta$ δύο ὀρθῶν ἐλάσσονές εἰσιν. αἱ δὲ ἀπ> ἐλασσόνων >ἡ δύο ὀρθῶν ἐκβαλλόμεναι εἰς >άπειρον συμπίπτουσιν; αἱ >άρα AB , $\Gamma\Delta$ ἐκβαλλόμεναι εἰς >άπειρον συμπεσοῦνται; οὐ συμπίπτουσι δὲ διὰ τὸ παραλλήλους αὐτὰς ὑποκεῖσθαι; οὐκ >άρα >άνισός ἐστιν ἡ ὑπὸ $AH\Theta$ τῇ ὑπὸ $H\Theta\Delta$; >ίση >άρα. ἀλλὰ ἡ ὑπὸ $AH\Theta$ τῇ ὑπὸ EHB ἐστίν >ίση; καὶ ἡ ὑπὸ EHB >άρα τῇ ὑπὸ $H\Theta\Delta$ ἐστίν >ίση; κοινὴ προσκείσθω ἡ ὑπὸ $BH\Theta$; αἱ >άρα ὑπὸ EHB , $BH\Theta$ ταῖς ὑπὸ $BH\Theta$, $H\Theta\Delta$ >ίσαι εἰσιν. ἀλλὰ αἱ ὑπὸ EHB , $BH\Theta$ δύο ὀρθαῖς >ίσαι εἰσιν; καὶ αἱ ὑπὸ $BH\Theta$, $H\Theta\Delta$ >άρα δύο ὀρθαῖς >ίσαι εἰσιν.

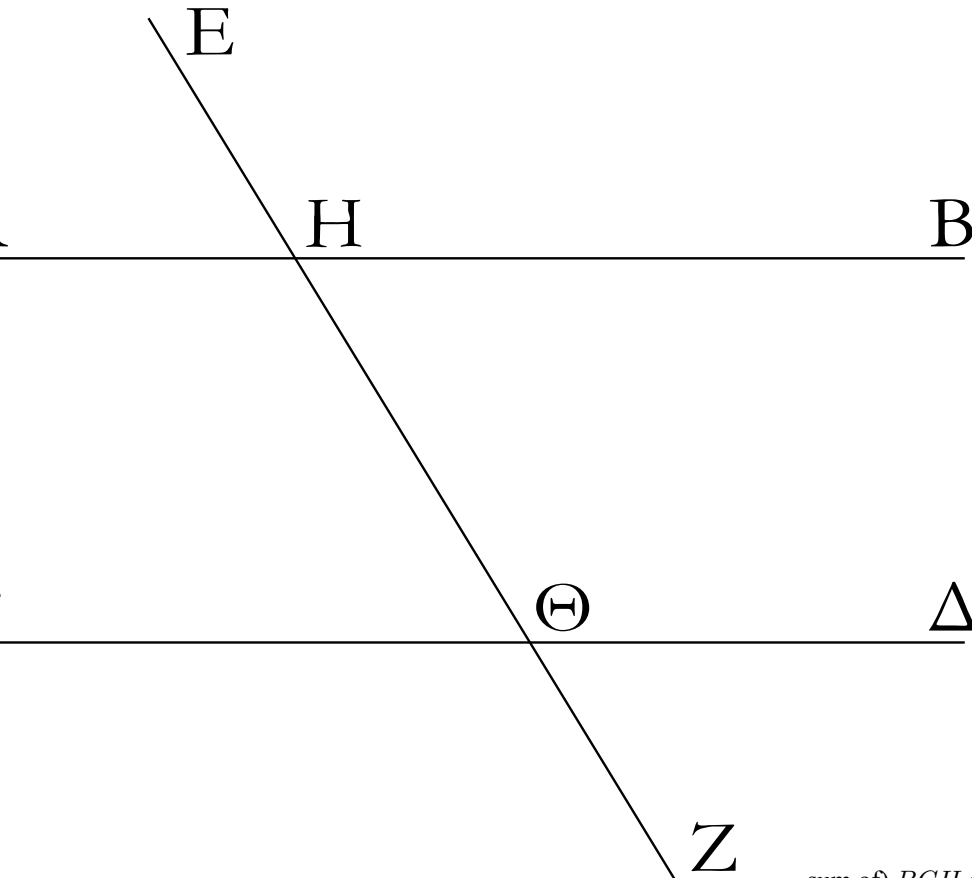
other). (Which is) the very thing it was required to show.

Proposition 29

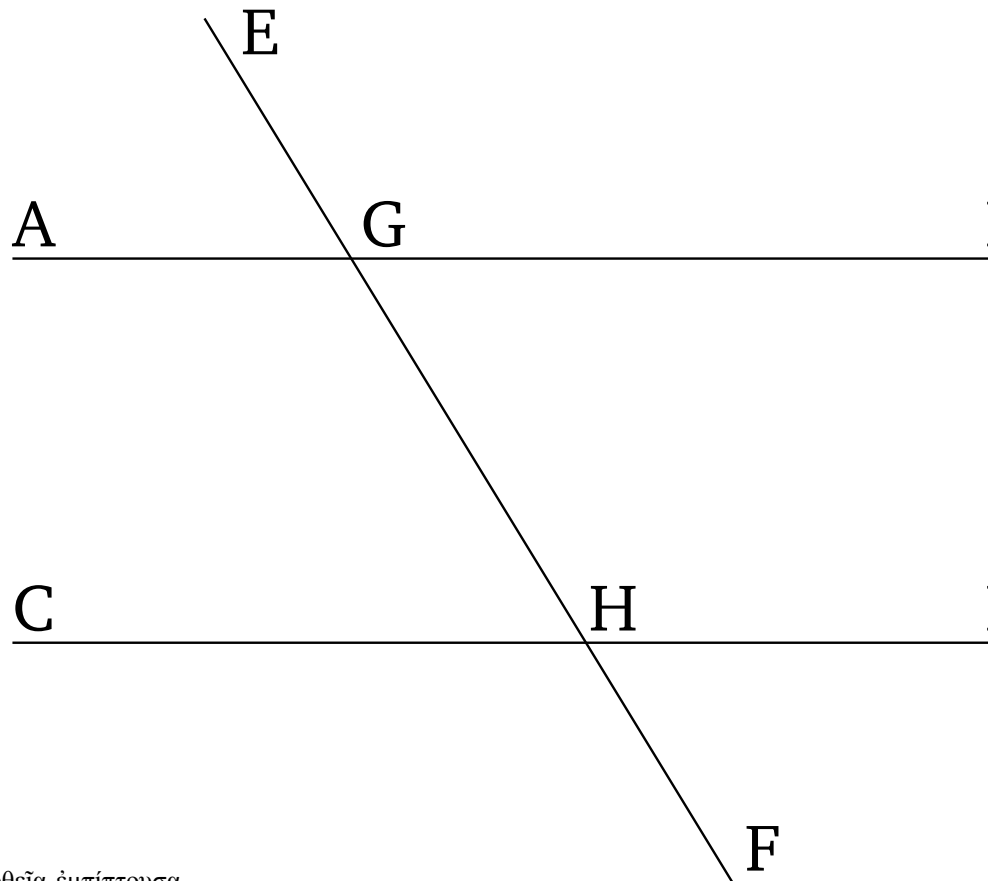
A straight-line falling across parallel straight-lines makes the alternate angles equal to one another, the external (angle) equal to the internal and opposite (angle), and the (sum of the) internal (angles) on the same side equal to two right-angles.

For let the straight-line EF fall across the parallel straight-lines AB and CD . I say that it makes the alternate angles, AGH and GHD , equal, the external angle EGB equal to the internal and opposite (angle) GHD , and the (sum of the) internal (angles) on the same side, BGH and GHD , equal to two right-angles.

For if AGH is unequal to GHD (then) one of them is greater. Let AGH be greater. Let BGH be added to both. Thus, (the sum of) AGH and BGH is greater than (the sum of) BGH and GHD . But, (the sum of) AGH and BGH is equal to two right-angles [Prop 1.13]. Thus, (the sum of) BGH and GHD is [also] less than two right-angles. But (straight-lines) being produced to infinity from (internal angles whose sum is) less than two right-angles meet together [Post. 5]. Thus, AB and CD , being produced to infinity, will meet together. But they do not meet, on account of them (initially) being assumed parallel (to one another) [Def. 1.23]. Thus, AGH is not unequal to GHD . Thus, (it is) equal. But, AGH is equal to EGB [Prop. 1.15]. And EGB is thus also equal to GHD . Let BGH be added to both. Thus, (the sum of) EGB and BGH is equal to (the sum of) BGH and GHD . But, (the sum of) EGB and BGH is equal to two right-angles [Prop. 1.13]. Thus, (the



sum of) BGH and GHD is also equal to two right-angles.



Ἡ ὥρα εἰς τὰς παραλλήλους εὐθείας εὐθεῖα ἐμπίπτουσα
τὰς τε ἐναλλάξ γωνίας ὀρθὰς ἀλλήλαις ποιεῖ καὶ τὴν ἐκτὸς
τῇ ἐντὸς καὶ ἀπεναντίον ὀρθὴν καὶ τὰς ἐντὸς καὶ ἐπὶ τὰ αὐτὰ
μέρη δυσὶν ὀρθαῖς ὀρθὰς; ὅπερ ὅδε δεῖξει.

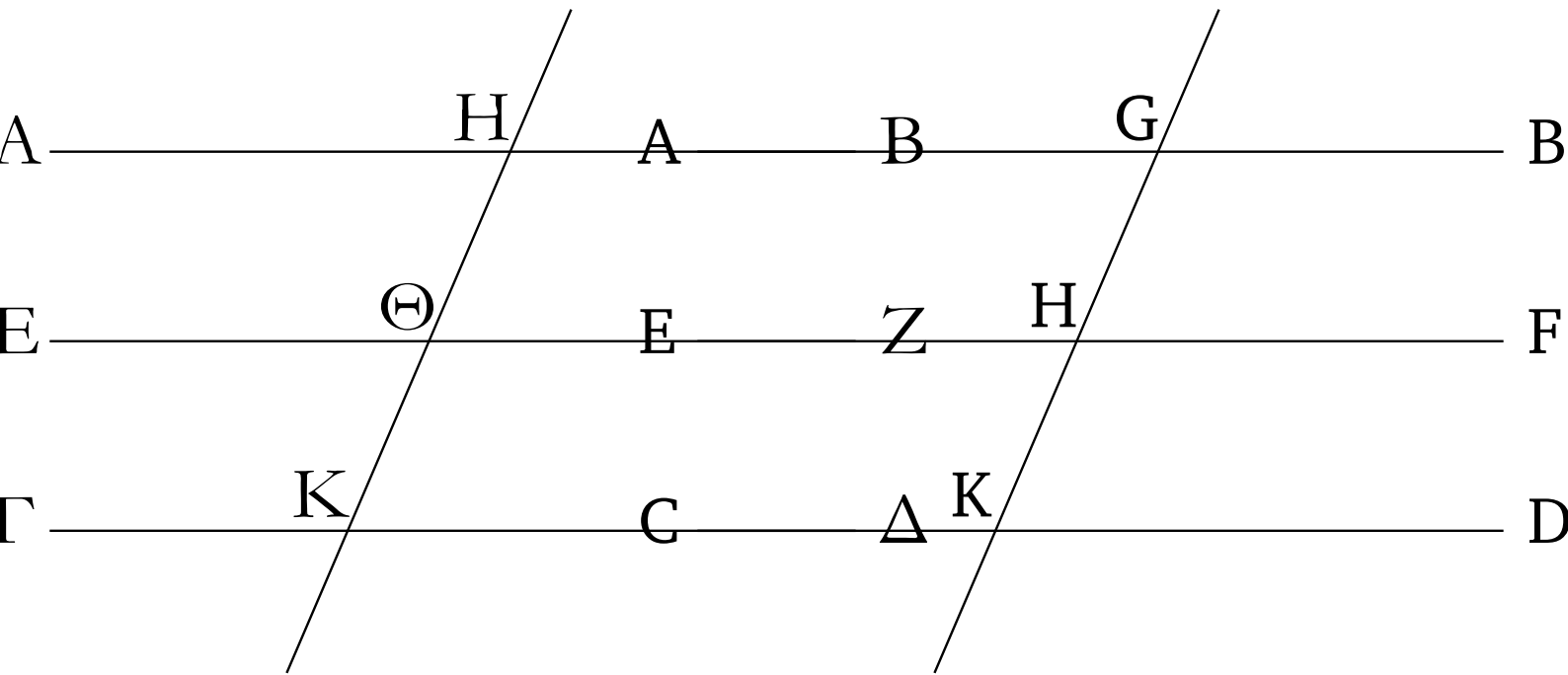
Thus, a straight-line falling across parallel straight-lines
makes the alternate angles equal to one another, the exter-
nal (angle) equal to the internal and opposite (angle), and
the (sum of the) internal (angles) on the same side equal to
two right-angles. (Which is) the very thing it was required
to show.

30.

Αἱ τῇ αὐτῇ εὐθείᾳ παράλληλοι καὶ ἀλλήλαις εἰσὶ παράλλη-
λοι.
>Ἐστω ἑκατέρα τῶν AB , $ΓΔ$ τῇ EZ παράλληλος; λέγω, <ὅτι
καὶ ἡ AB τῇ $ΓΔ$ ἐστὶ παράλληλος.
Ἐμπίπτει γὰρ εἰς αὐτὰς εὐθεῖα ἡ HK .

Proposition 30

(Straight-lines) parallel to the same straight-line are also
parallel to one another.
Let each of the (straight-lines) AB and CD be parallel to
 EF . I say that AB is also parallel to CD .
For let the straight-line GK fall across (AB , CD , and EF).



Καὶ ἐπεὶ εἰς παραλλήλουξ εὐθείαξ τὰξ AB, EZ εὐθεῖα ἐμπέπτωκεν ἡ HK, >ίση >άρα ἡ ὑπὸ AHK τῇ ὑπὸ HΘZ. πάλιν, ἐπεὶ εἰς παραλλήλουξ εὐθείαξ τὰξ EZ, ΓΔ εὐθεῖα ἐμπέπτωκεν ἡ HK, >ίση ἐστὶν ἡ ὑπὸ HΘZ τῇ ὑπὸ HKΔ. ἐδείθθη δὲ καὶ ἡ ὑπὸ AHK τῇ ὑπὸ HΘZ >ίση. καὶ ἡ ὑπὸ AHK >άρα τῇ ὑπὸ HKΔ ἐστὶν >ίση; καὶ εἰσιν ἐναλλάχ. παράλληλοξ >άρα ἐστὶν ἡ AB τῇ ΓΔ.

[Αἱ >άρα τῇ αὐτῇ εὐθεία παράλληλοι καὶ ἀλλήλαιξ εἰσι παράλληλοι;] <ὅπερ >έδει δεῖξαι.

And since the straight-line GK has fallen across the parallel straight-lines AB and EF , (angle) AGK (is) thus equal to GHF [Prop. 1.29]. Again, since the straight-line GK has fallen across the parallel straight-lines EF and CD , (angle) GHF is equal to GKD [Prop. 1.29]. But AGK was also shown (to be) equal to GHF . Thus, AGK is also equal to GKD . And they are alternate (angles). Thus, AB is parallel to CD [Prop. 1.27].

[Thus, (straight-lines) parallel to the same straight-line are also parallel to one another.] (Which is) the very thing it was required to show.

31.

Διὰ τοῦ δοθέντοξ σημείου τῇ δοθείσῃ εὐθείᾳ παράλληλον εὐθεῖαν γραμμὴν ἀγαγεῖν.

>Ἐστω τὸ μὲν δοθὲν σημεῖον τὸ A, ἡ δὲ δοθεῖσα εὐθεῖα ἡ BΓ; δεῖ δὴ διὰ τοῦ A σημείου τῇ BΓ εὐθείᾳ παράλληλον εὐθεῖαν γραμμὴν ἀγαγεῖν.

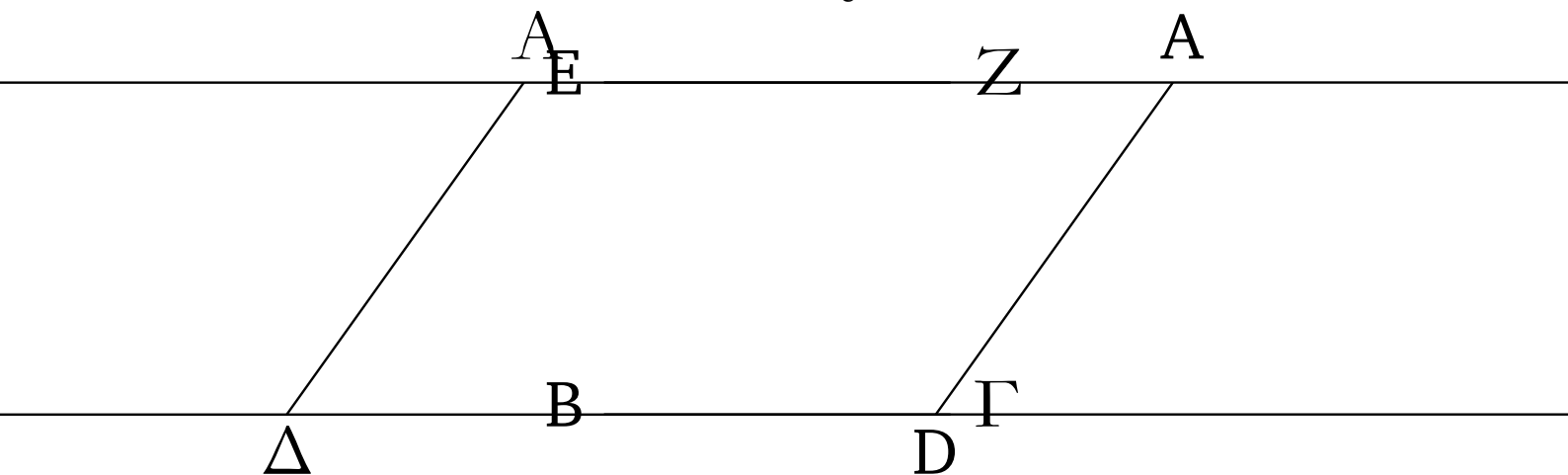
Εἰλήφθω ἐπὶ τῇ BΓ τυθὸν σημεῖον τὸ Δ, καὶ ἐπεζεύθθω ἡ AΔ; καὶ συνεστάτω πρὸξ τῇ ΔA εὐθείᾳ καὶ τῷ πρὸξ αὐτῇ σημείῳ τῷ A τῇ ὑπὸ AΔΓ γωνία >ίση ἡ ὑπὸ ΔAE; καὶ ἐκβεβλήσθω ἐπ' εὐθείαξ τῇ EA εὐθεῖα ἡ AZ.

Proposition 31

To draw a straight-line parallel to a given straight-line, through a given point.

Let A be the given point, and BC the given straight-line. So it is required to draw a straight-line parallel to the straight-line BC , through the point A .

Let the point D be taken a random on BC , and let AD be joined. And let (angle) DAE , equal to angle ADC , be constructed on the straight-line DA at the point A on it [Prop. 1.23]. And let the straight-line AF be produced in a straight-line with EA .



Καὶ ἐπεὶ εἰς δύο εὐθείας τὰς ΒΓ, ΕΖ εὐθεῖα ἐμπίπτουσα ἡ ΑΔ τὰς ἐναλλάχ γωνίας τὰς ὑπὸ ΕΑΔ, ΑΔΓ >ίσαξ ἀλλήλαις πεποίηκεν, παράλληλος >άρα ἐστὶν ἡ ΕΑΖ τῇ ΒΓ.
Διὰ τοῦ δοθέντος >άρα σημείου τοῦ Α τῇ δοθείσῃ εὐθείᾳ τῇ ΒΓ παράλληλος εὐθεῖα γραμμὴ >ῆκεται ἡ ΕΑΖ; <όπερ >έδει ποιῆσαι.

And since the straight-line AD , (in) falling across the two straight-lines BC and EF , has made the alternate angles EAD and ADC equal to one another, EAF is thus parallel to BC [Prop. 1.27].

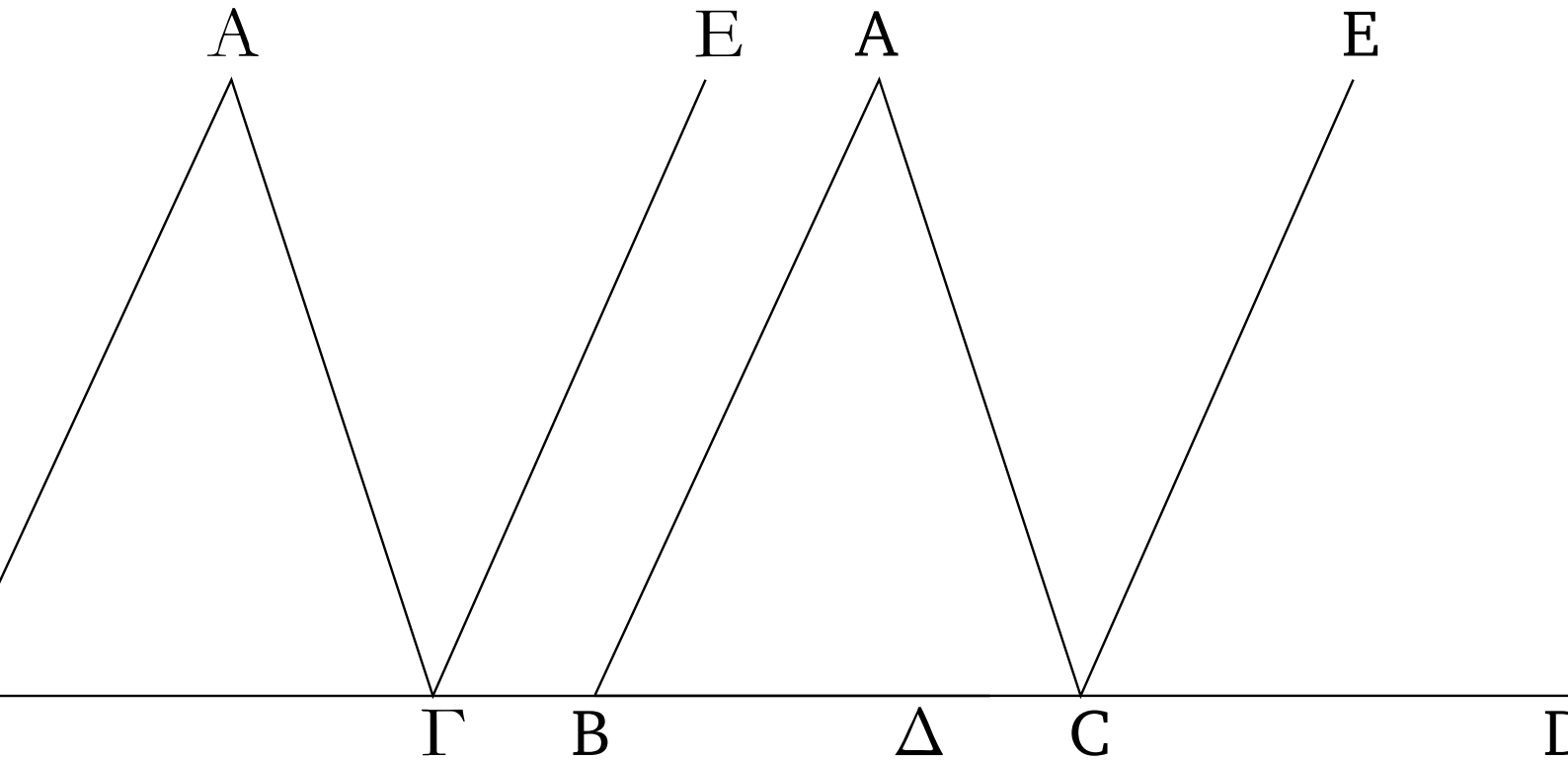
Thus, the straight-line EAF has been drawn parallel to the given straight-line BC , through the given point A . (Which is) the very thing it was required to do.

32.

Παντὸς τριγώνου μιᾷ τῶν πλευρῶν προσεκβληθείσης ἡ ἐκτὸς γωνία δυοὶ ταῖς ἐντὸς καὶ ἀπεναντίον >ίση ἐστίν, καὶ αἱ ἐντὸς τοῦ τριγώνου τρεῖς γωνίαι δυοὶν ὀρθαῖς >ίσαι εἰσίν.

Proposition 32

In any triangle, (if) one of the sides (is) produced (then) the external angle is equal to the (sum of the) two internal and opposite (angles), and the (sum of the) three internal angles of the triangle is equal to two right-angles.



>Ἐστω τρίγωνον τὸ ΑΒΓ, καὶ προσεκβεβλήσθω αὐτοῦ μία πλευρὰ ἡ ΒΓ ἐπὶ τὸ Δ; λέγω, <ὅτι ἡ ἐκτὸς γωνία ἡ ὑπὸ ΑΓΔ >ίση ἐστὶ δυοὶ ταῖς ἐντὸς καὶ ἀπεναντίον ταῖς ὑπὸ ΓΑΒ, ΑΒΓ, καὶ αἱ ἐντὸς τοῦ τριγώνου τρεῖς γωνίαι αἱ ὑπὸ ΑΒΓ, ΒΓΑ, ΓΑΒ δυοὶν ὀρθαῖς >ίσαι εἰσίν.

>Ἦθθω γὰρ διὰ τοῦ Γ σημείου τῇ ΑΒ εὐθείᾳ παράλληλος ἡ ΓΕ.

Καὶ ἐπεὶ παράλληλός ἐστιν ἡ ΑΒ τῇ ΓΕ, καὶ εἰς αὐτὰς ἐμπίπτωκεν ἡ ΑΓ, αἱ ἐναλλάχ γωνίαι αἱ ὑπὸ ΒΑΓ, ΑΓΕ >ίσαι ἀλλήλαις εἰσίν. πάλιν, ἐπεὶ παράλληλός ἐστιν ἡ ΑΒ τῇ ΓΕ, καὶ εἰς αὐτὰς ἐμπίπτωκεν εὐθεῖα ἡ ΒΔ, ἡ ἐκτὸς γωνία ἡ ὑπὸ ΕΓΔ >ίση ἐστὶ τῇ ἐντὸς καὶ ἀπεναντίον τῇ ὑπὸ ΑΒΓ. ἐδείχθη δὲ καὶ ἡ ὑπὸ ΑΓΕ τῇ ὑπὸ ΒΑΓ >ίση; <ὅλη> >άρα ἡ ὑπὸ ΑΓΔ γωνία >ίση ἐστὶ δυοὶ ταῖς ἐντὸς καὶ ἀπεναντίον ταῖς ὑπὸ ΒΑΓ, ΑΒΓ.

Κοινὴ προσκείσθω ἡ ὑπὸ ΑΓΒ; αἱ >άρα ὑπὸ ΑΓΔ, ΑΓΒ τρισὶ ταῖς ὑπὸ ΑΒΓ, ΒΓΑ, ΓΑΒ >ίσαι εἰσίν. ὅλλ> αἱ ὑπὸ ΑΓΔ, ΑΓΒ δυοὶν ὀρθαῖς >ίσαι εἰσίν; καὶ αἱ ὑπὸ ΑΓΒ, ΓΒΑ, ΓΑΒ >άρα δυοὶν ὀρθαῖς >ίσαι εἰσίν.

Παντὸς >άρα τριγώνου μιᾷ τῶν πλευρῶν προσεκβληθείσης ἡ ἐκτὸς γωνία δυοὶ ταῖς ἐντὸς καὶ ἀπεναντίον >ίση ἐστίν, καὶ αἱ ἐντὸς τοῦ τριγώνου τρεῖς γωνίαι δυοὶν ὀρθαῖς >ίσαι εἰσίν; <όπερ >έδει δεῖξαι.

Let ABC be a triangle, and let one of its sides BC be produced to D . I say that the external angle ACD is equal to the (sum of the) two internal and opposite angles CAB and ABC , and the (sum of the) three internal angles of the triangle— ABC , BCA , and CAB —is equal to two right-angles.

For let CE be drawn through point C parallel to the straight-line AB [Prop. 1.31].

And since AB is parallel to CE , and AC has fallen across them, the alternate angles BAC and ACE are equal to one another [Prop. 1.29]. Again, since AB is parallel to CE , and the straight-line BD has fallen across them, the external angle ECD is equal to the internal and opposite (angle) ABC [Prop. 1.29]. But ACE was also shown (to be) equal to BAC . Thus, the whole angle ACD is equal to the (sum of the) two internal and opposite (angles) BAC and ABC . Let ACB be added to both. Thus, (the sum of) ACD and ACB is equal to the (sum of the) three (angles) ABC , BCA , and CAB . But, (the sum of) ACD and ACB is equal to two right-angles [Prop. 1.13]. Thus, (the sum of) ACB , CBA , and CAB is also equal to two right-angles. Thus, in any triangle, (if) one of the sides (is) produced (then) the external angle is equal to the (sum of the) two

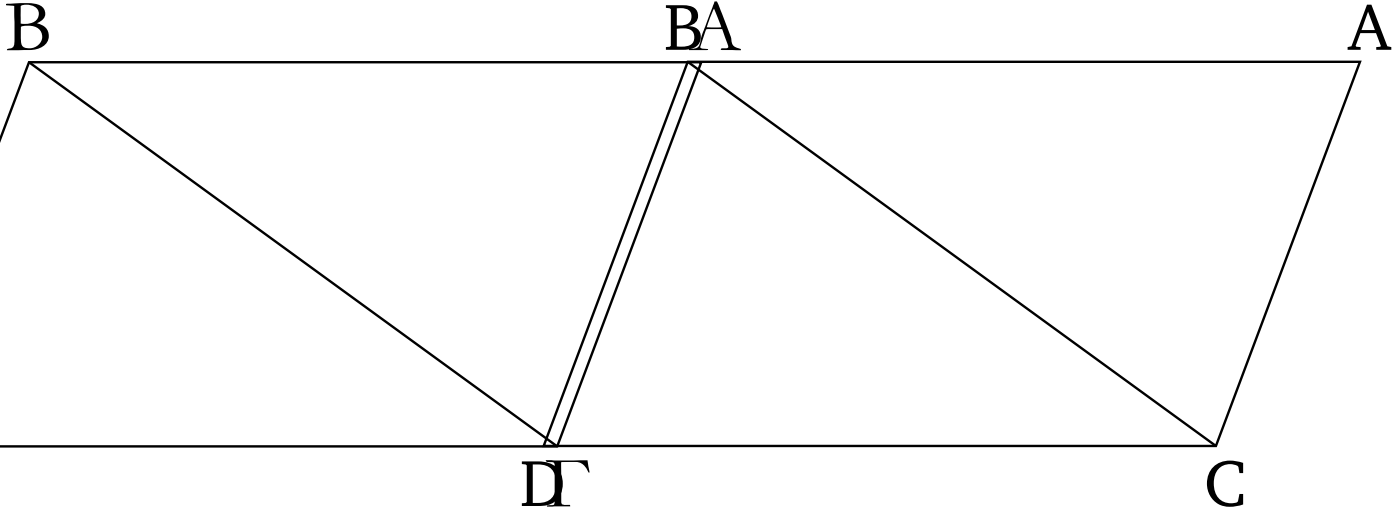
internal and opposite (angles), and the (sum of the) three internal angles of the triangle is equal to two right-angles. (Which is) the very thing it was required to show.

33.

Αἱ τὰς >ίσας τε καὶ παραλλήλους ἐπὶ τὰ αὐτὰ μέρη ἐπιζευγνύουσαι εὐθεῖαι καὶ αὐταὶ >ίσαι τε καὶ παράλληλοί εἰσιν.
>Ἐστῶσαν >ίσαι τε καὶ παράλληλοι αἱ AB , $\Gamma\Delta$, καὶ ἐπιζευγνύονται αὐτὰς ἐπὶ τὰ αὐτὰ μέρη εὐθεῖαι αἱ $ΑΓ$, $ΒΔ$; λέγω, <ὅτι καὶ αἱ $ΑΓ$, $ΒΔ$ >ίσαι τε καὶ παράλληλοί εἰσιν.

Proposition 33

Straight-lines joining equal and parallel (straight-lines) on the same sides are themselves also equal and parallel.
Let AB and CD be equal and parallel (straight-lines), and let the straight-lines AC and BD join them on the same sides. I say that AC and BD are also equal and parallel.



Ἐπεξεύθθω ἡ $ΒΓ$. καὶ ἐπεὶ παράλληλός ἐστιν ἡ AB τῇ $\Gamma\Delta$, καὶ εἰς αὐτὰς ἐμπίπτωκεν ἡ $ΒΓ$, αἱ ἐναλλάξ γωνίαι αἱ ὑπὸ $AB\Gamma$, $ΒΓ\Delta$ >ίσαι ἀλλήλαις εἰσίν. καὶ ἐπεὶ >ίση ἐστὶν ἡ AB τῇ $\Gamma\Delta$ κοινὴ δὲ ἡ $ΒΓ$, δύο δὴ αἱ AB , $ΒΓ$ δύο ταῖς $ΒΓ$, $\Gamma\Delta$ >ίσαι εἰσίν; καὶ γωνία ἡ ὑπὸ $AB\Gamma$ γωνία τῇ ὑπὸ $ΒΓ\Delta$ >ίση; βάσεις >άρα ἡ $ΑΓ$ βάσει τῇ $ΒΔ$ ἐστὶν >ίση, καὶ τὸ $AB\Gamma$ τρίγωνον τῷ $ΒΓ\Delta$ τριγώνῳ >ίσον ἐστίν, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις >ίσαι >έσονται ἑκατέρω ἑκατέρω, ὅφρ' <ἂν αἱ >ίσαι πλευραὶ ὑποτείνουσιν; >ίση >άρα ἡ ὑπὸ $ΑΓΒ$ γωνία τῇ ὑπὸ $ΓΒΔ$. καὶ ἐπεὶ εἰς δύο εὐθείας τὰς $ΑΓ$, $ΒΔ$ εὐθεῖα ἐμπίπτουσα ἡ $ΒΓ$ τὰς ἐναλλάξ γωνίας >ίσας ἀλλήλαις πεποίηκεν, παράλληλός >άρα ἐστὶν ἡ $ΑΓ$ τῇ $ΒΔ$. ἐδείκθη δὲ αὐτῇ καὶ >ίση.

Αἱ >άρα τὰς >ίσας τε καὶ παραλλήλους ἐπὶ τὰ αὐτὰ μέρη ἐπιζευγνύουσιν εὐθεῖαι καὶ αὐταὶ >ίσαι τε καὶ παράλληλοί εἰσιν; <ὅπερ >έδει δεῖχαι.

Let BC be joined. And since AB is parallel to CD , and BC has fallen across them, the alternate angles ABC and BCD are equal to one another [Prop. 1.29]. And since AB is equal to CD , and BC is common, the two (straight-lines) AB , BC are equal to the two (straight-lines) DC , CB .[†] And the angle ABC is equal to the angle BCD . Thus, the base AC is equal to the base BD , and triangle ABC is equal to triangle DCB ,[‡] and the remaining angles will be equal to the corresponding remaining angles subtended by the equal sides [Prop. 1.4]. Thus, angle ACB is equal to CBD . Also, since the straight-line BC , (in) falling across the two straight-lines AC and BD , has made the alternate angles (ACB and CBD) equal to one another, AC is thus parallel to BD [Prop. 1.27]. And (AC) was also shown (to be) equal to (BD).

Thus, straight-lines joining equal and parallel (straight-lines) on the same sides are themselves also equal and parallel. (Which is) the very thing it was required to show.

[†] The Greek text has " BC , CD ", which is obviously a mistake.

[‡] The Greek text has " DCB ", which is obviously a mistake.

34.

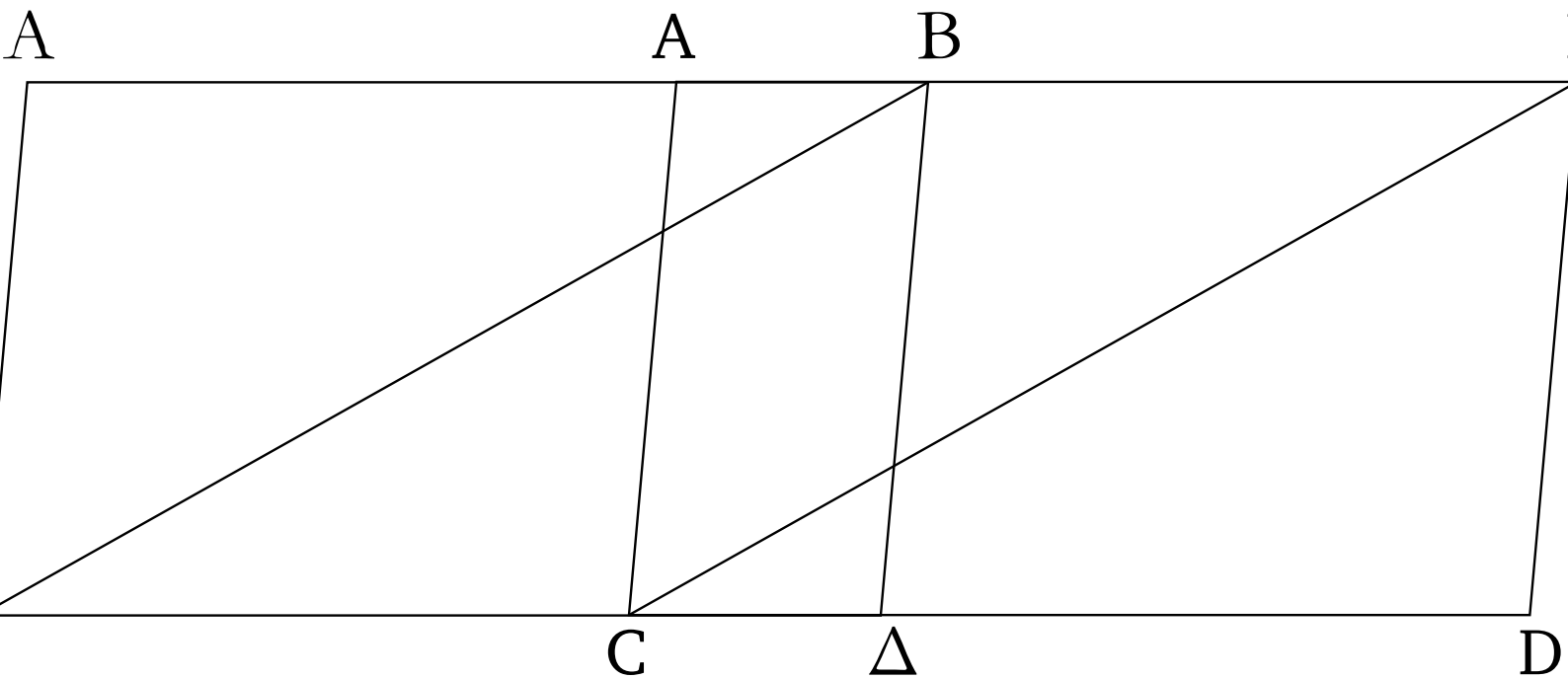
Τῶν παραλληλογράμμων θωρίων αἱ ἀπεναντίον πλευραὶ τε καὶ γωνίαι >ίσαι ἀλλήλαις εἰσίν, καὶ ἡ διάμετρος αὐτὰ διῆα τέμνει.

>Ἐστω παραλληλόγραμμον θωρίον τὸ $ΑΓΔΒ$, διάμετρος δὲ αὐτοῦ ἡ $ΒΓ$; λέγω, <ὅτι τοῦ $ΑΓΔΒ$ παραλληλογράμμου αἱ ἀπεναντίον πλευραὶ τε καὶ γωνίαι >ίσαι ἀλλήλαις εἰσίν, καὶ ἡ $ΒΓ$ διάμετρος αὐτὸ διῆα τέμνει.

Proposition 34

In parallelogramic figures, the opposite sides and angles are equal to one another, and a diagonal cuts them in half.

Let $ACDB$ be a parallelogramic figure, and BC its diagonal. I say that for parallelogram $ACDB$, the opposite sides and angles are equal to one another, and the diagonal BC cuts it in half.



Ἐπει γὰρ παράλληλός ἐστιν ἡ AB τῇ $\Gamma\Delta$, καὶ εἰς αὐτὰς ἐμπεπτωκεν εὐθεῖα ἡ $B\Gamma$, αἱ ἐναλλάχ γωνίαι αἱ ὑπὸ $AB\Gamma$, $B\Gamma\Delta$ ὀρθαὶ ἀλλήλαις εἰσίν. πάλιν ἐπει παράλληλός ἐστιν ἡ $A\Gamma$ τῇ $B\Delta$, καὶ εἰς αὐτὰς ἐμπεπτωκεν ἡ $B\Gamma$, αἱ ἐναλλάχ γωνίαι αἱ ὑπὸ $A\Gamma B$, $\Gamma B\Delta$ ὀρθαὶ ἀλλήλαις εἰσίν. δύο δὲ τρίγωνά ἐστι τὰ $AB\Gamma$, $B\Gamma\Delta$ τὰς δύο γωνίας τὰς ὑπὸ $AB\Gamma$, $B\Gamma\Delta$ δυοὶ ταῖς ὑπὸ $B\Gamma\Delta$, $\Gamma B\Delta$ ὀρθαὶ ἔχοντα ἑκατέραν ἑκατέρα καὶ μίαν πλευρὰν μιᾷ πλευρᾷ ὀρθὴν τὴν πρὸς ταῖς ὀρθαῖς γωνίαις κοινὴν αὐτῶν τὴν $B\Gamma$; καὶ τὰς λοιπὰς ὀρθαὶ πλευρὰς ταῖς λοιπαῖς ὀρθαὶ ἔχει ἑκατέραν ἑκατέρα καὶ τὴν λοιπὴν γωνίαν τῇ λοιπῇ γωνίᾳ; ὀρθὴ ὀρθαὶ ἢ μὲν AB πλευρὰ τῇ $\Gamma\Delta$, ἢ δὲ $A\Gamma$ τῇ $B\Delta$, καὶ ὅτι ὀρθὴ ἐστὶν ἡ ὑπὸ $BA\Gamma$ γωνία τῇ ὑπὸ $\Gamma\Delta B$. καὶ ἐπει ὀρθὴ ἐστὶν ἡ μὲν ὑπὸ $AB\Gamma$ γωνία τῇ ὑπὸ $B\Gamma\Delta$, ἢ δὲ ὑπὸ $\Gamma B\Delta$ τῇ ὑπὸ $A\Gamma B$, ὅλη ὀρθαὶ ἢ ὑπὸ $AB\Delta$ ὅλη τῇ ὑπὸ $A\Gamma\Delta$ ἐστὶν ὀρθὴ. ἐδείχθη δὲ καὶ ἡ ὑπὸ $BA\Gamma$ τῇ ὑπὸ $\Gamma\Delta B$ ὀρθὴ.

Τῶν ὀρθαὶ παραλληλογράμμων θωρίων αἱ ἀπεναντίον πλευραὶ τε καὶ γωνίαι ὀρθαὶ ἀλλήλαις εἰσίν.

Λέγω δὲ, ὅτι καὶ ἡ διάμετρος αὐτὰ διίθα τέμνει. ἐπει γὰρ ὀρθὴ ἐστὶν ἡ AB τῇ $\Gamma\Delta$, κοινὴ δὲ ἡ $B\Gamma$, δύο δὲ αἱ AB , $B\Gamma$ δυοὶ ταῖς $\Gamma\Delta$, $B\Gamma$ ὀρθαὶ εἰσὶν ἑκατέρα ἑκατέρα; καὶ γωνία ἢ ὑπὸ $AB\Gamma$ γωνία τῇ ὑπὸ $B\Gamma\Delta$ ὀρθὴ. καὶ βάσις ὀρθαὶ ἢ $A\Gamma$ τῇ ΔB ὀρθὴ. καὶ τὸ $AB\Gamma$ [ὀρθαὶ] τρίγωνον τῷ $B\Gamma\Delta$ τριγώνῳ ὀρθὸν ἐστίν.

Ἦ ὀρθαὶ $B\Gamma$ διάμετρος διίθα τέμνει τὸ $AB\Gamma\Delta$ παραλληλόγραμμόν, ὅπερ ὀρθὸν δεῖχαι.

For since AB is parallel to CD , and the straight-line BC has fallen across them, the alternate angles ABC and BCD are equal to one another [Prop. 1.29]. Again, since AC is parallel to BD , and BC has fallen across them, the alternate angles ACB and CBD are equal to one another [Prop. 1.29]. So ABC and BCD are two triangles having the two angles ABC and BCA equal to the two (angles) BCD and CBD , respectively, and one side equal to one side—the (one) by the equal angles and common to them, (namely) BC . Thus, they will also have the remaining sides equal to the corresponding remaining (sides), and the remaining angle (equal) to the remaining angle [Prop. 1.26]. Thus, side AB is equal to CD , and AC to BD . Furthermore, angle BAC is equal to CDB . And since angle ABC is equal to BCD , and CBD to ACB , the whole (angle) ABD is thus equal to the whole (angle) ACD . And BAC was also shown (to be) equal to CDB .

Thus, in parallelogramic figures, the opposite sides and angles are equal to one another.

And, I also say that a diagonal cuts them in half. For since AB is equal to CD , and BC (is) common, the two (straight-lines) AB , BC are equal to the two (straight-lines) DC , CB [†], respectively. And angle ABC is equal to angle BCD . Thus, the base AC (is) also equal to DB , and triangle ABC equal to triangle BCD [Prop. 1.4].

Thus, the diagonal BC cuts the parallelogram $ACDB$ [‡] in half. (Which is) the very thing it was required to show.

[†] The Greek text has “ CD , BC ”, which is obviously a mistake.

[‡] The Greek text has “ $ABCD$ ”, which is obviously a mistake.

35.

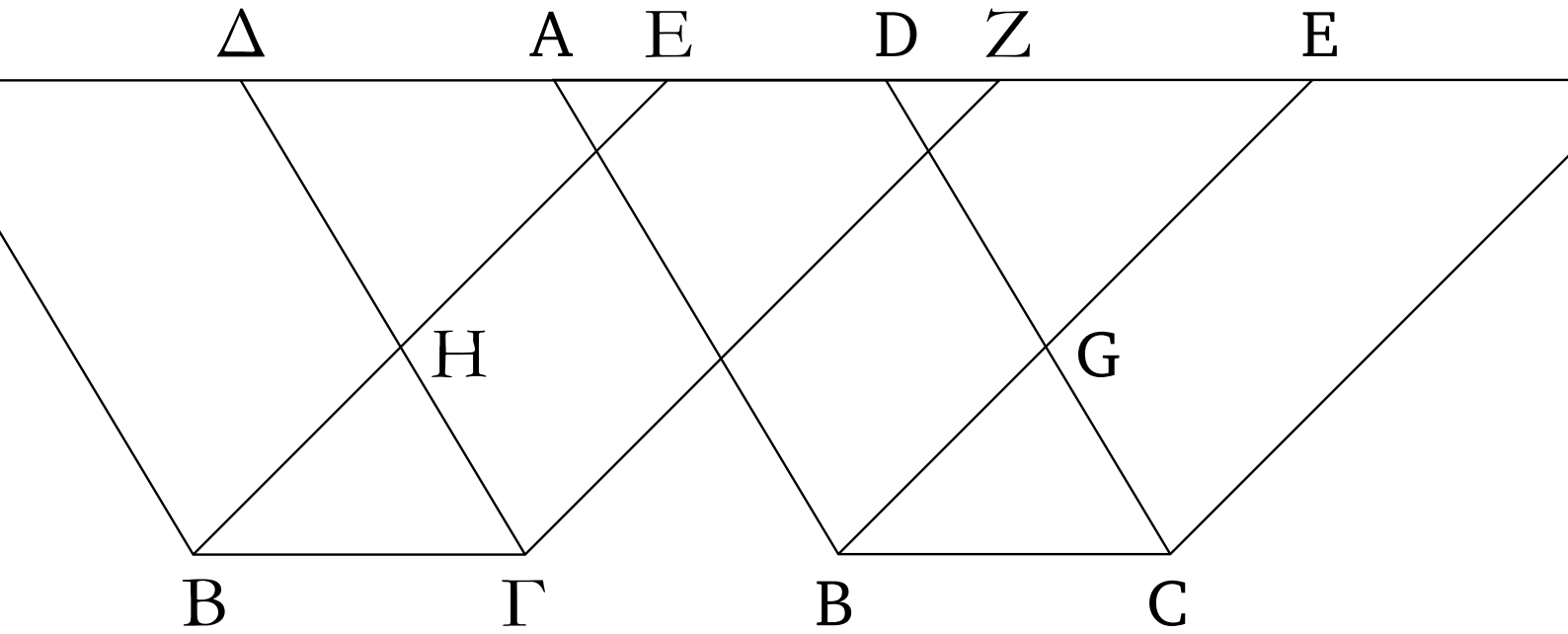
Τὰ παραλληλόγραμμα τὰ ἐπὶ τῇ αὐτῇ βάσει ὄντα καὶ ἐν ταῖς αὐταῖς παραλλήλοις ὀρθαὶ ἀλλήλοις ἐστίν.

Ἐστω παραλληλόγραμμα τὰ $AB\Gamma\Delta$, $EB\Gamma Z$ ἐπὶ τῇ αὐτῇ βάσει τῇ $B\Gamma$ καὶ ἐν ταῖς αὐταῖς παραλλήλοις ταῖς AZ , $B\Gamma$; λέγω, ὅτι ὀρθὸν ἐστὶ τὸ $AB\Gamma\Delta$ τῷ $EB\Gamma Z$ παραλληλόγραμμῳ.

Proposition 35

Parallelograms which are on the same base and between the same parallels are equal[†] to one another.

Let $ABCD$ and $EBCF$ be parallelograms on the same base BC , and between the same parallels AF and BC . I say that $ABCD$ is equal to parallelogram $EBCF$.



Ἐπεὶ γὰρ παραλληλόγραμμον ἐστὶ τὸ $ABGD$, ὅση ἐστὶν ἡ AD τῇ BG . διὰ τὰ αὐτὰ δὲ καὶ ἡ EZ τῇ BG ἐστὶν ὅση; ὥστε καὶ ἡ AD τῇ EZ ἐστὶν ὅση; καὶ κοινὴ ἡ DE ; ὅλη ἄρα ἡ AE ὅλη τῇ DZ ἐστὶν ὅση. ὅστις δὲ καὶ ἡ AB τῇ DG ὅση; δύο δὲ αἱ EA , AB δύο ταῖς DZ , DG ὅσαι εἰσὶν ἑκατέρω ἑκατέρω; καὶ γωνία ἡ ὑπὸ ZDG γωνία τῇ ὑπὸ EAB ἐστὶν ὅση ἢ ἐκτὸς τῇ ἐντὸς; βάσει ἄρα ἡ EB βάσει τῇ ZG ὅση ἐστίν, καὶ τὸ EAB τρίγωνον τῷ DZG τριγώνῳ ὅσον ὅσται; κοινὸν ἀφηρήσθω τὸ DHE ; λοιπὸν ἄρα τὸ $ABHD$ τραπέζιον λοιπῷ τῷ $EHGZ$ τραπέζιῳ ἐστὶν ὅσον; κοινὸν προσκείσθω τὸ HBG τρίγωνον; ὅλον ἄρα τὸ $ABGD$ παραλληλόγραμμον ὅλῳ τῷ $EBGZ$ παραλληλογράμῳ ὅσον ἐστίν.

Τὰ ἄρα παραλληλόγραμμα τὰ ἐπὶ τῆς αὐτῆς βάσεως ὄντα καὶ ἐν ταῖς αὐταῖς παραλλήλοις ὅσα ἀλλήλοις ἐστίν; ὅπερ ὀφείδει δεῖξαι.

For since $ABGD$ is a parallelogram, AD is equal to BC [Prop. 1.34]. So, for the same (reasons), EF is also equal to BC . So AD is also equal to EF . And DE is common. Thus, the whole (straight-line) AE is equal to the whole (straight-line) DF . And AB is also equal to DC . So the two (straight-lines) EA , AB are equal to the two (straight-lines) FD , DC , respectively. And angle FDC is equal to angle EAB , the external to the internal [Prop. 1.29]. Thus, the base EB is equal to the base FC , and triangle EAB will be equal to triangle DFC [Prop. 1.4]. Let DGE be taken away from both. Thus, the remaining trapezium $ABGD$ is equal to the remaining trapezium $EGCF$. Let triangle GBC be added to both. Thus, the whole parallelogram $ABCD$ is equal to the whole parallelogram $EBCF$.

Thus, parallelograms which are on the same base and between the same parallels are equal to one another. (Which is) the very thing it was required to show.

† Here, for the first time, “equal” means “equal in area”, rather than “congruent”.

36.

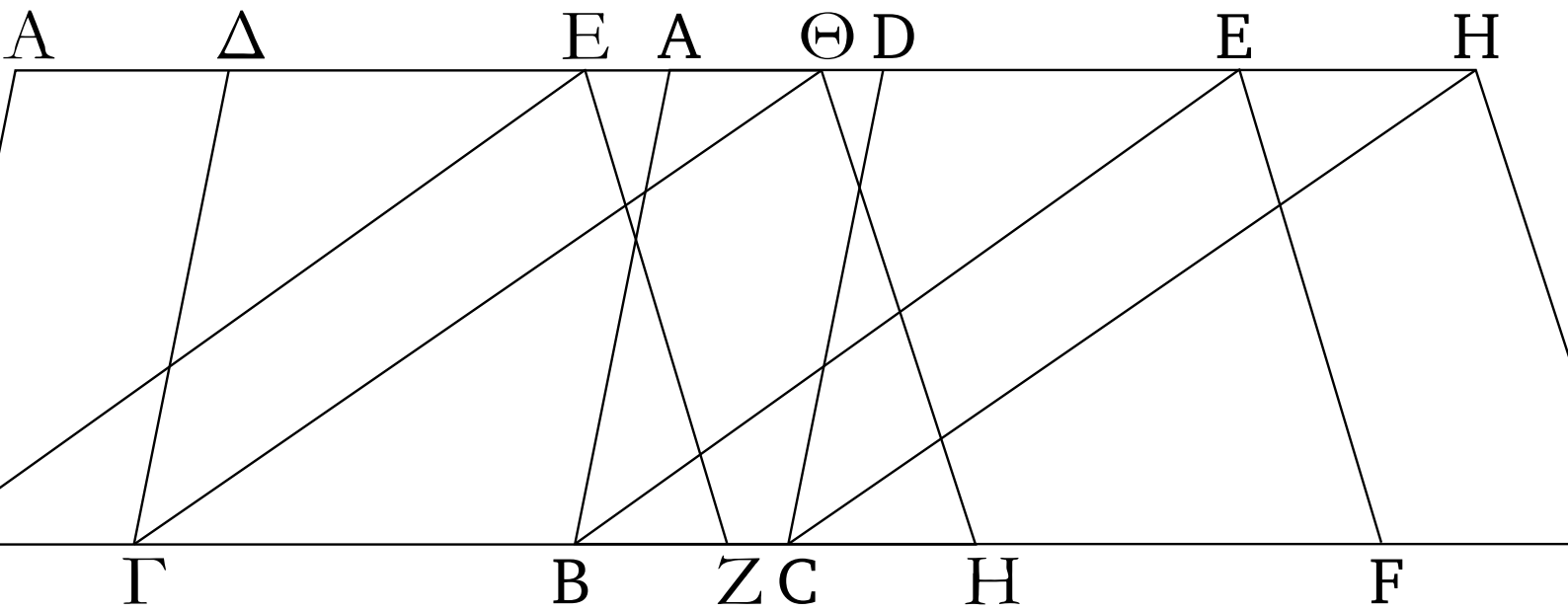
Τὰ παραλληλόγραμμα τὰ ἐπὶ ὅσων βάσεων ὄντα καὶ ἐν ταῖς αὐταῖς παραλλήλοις ὅσα ἀλλήλοις ἐστίν.

Ἐστω παραλληλόγραμμα τὰ $ABGD$, $EZHΘ$ ἐπὶ ὅσων βάσεων ὄντα τῶν BG , ZH καὶ ἐν ταῖς αὐταῖς παραλλήλοις ταῖς $ΑΘ$, BH ; λέγω, ὅτι ὅσον ἐστὶ τὸ $ABGD$ παραλληλόγραμμο τῷ $EZHΘ$.

Proposition 36

Parallelograms which are on equal bases and between the same parallels are equal to one another.

Let $ABCD$ and $EFGH$ be parallelograms which are on the equal bases BC and FG , and (are) between the same parallels AH and BG . I say that the parallelogram $ABCD$ is equal to $EFGH$.



Ἐπεξεύθθωσαν γὰρ αἱ BE , $\Gamma\Theta$. καὶ ἐπεὶ ὀρθὴ ἐστὶν ἡ $B\Gamma$ τῇ ZH , ἀλλὰ ἡ ZH τῇ $E\Theta$ ἐστὶν ὀρθή, καὶ ἡ $B\Gamma$ ὀρθὰ τῇ $E\Theta$ ἐστὶν ὀρθή. εἰσὶ δὲ καὶ παράλληλοι. καὶ ἐπιζευγνύουσιν αὐτάς αἱ EB , $\Theta\Gamma$; αἱ δὲ τὰς ὀρθὰς τε καὶ παραλλήλους ἐπὶ τὰ αὐτὰ μέρη ἐπιζευγνύουσαι ὀρθαί τε καὶ παράλληλοι εἰσι [καὶ αἱ EB , $\Theta\Gamma$ ὀρθὰς ὀρθαί τε εἰσι καὶ παράλληλοι]. παραλληλόγραμμον ὀρθὰ ἐστὶ τὸ $EB\Gamma\Theta$. καὶ ἐστὶν ὀρθὸν τῷ $AB\Gamma\Delta$; βάσιν τε γὰρ αὐτῶ τὴν αὐτὴν ὀρθὴν τὴν $B\Gamma$, καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἐστὶν αὐτῶ ταῖς $B\Gamma$, $A\Theta$. διὰ τὰ αὐτὰ δὴ καὶ τὸ $EZH\Theta$ τῷ αὐτῶ τῷ $EB\Gamma\Theta$ ἐστὶν ὀρθόν; ὥστε καὶ τὸ $AB\Gamma\Delta$ παραλληλόγραμμον τῷ $EZH\Theta$ ἐστὶν ὀρθόν.

Τὰ ὀρθὰ παραλληλόγραμμα τὰ ἐπὶ ὀρθῶν βάσεων ὀρθὰ καὶ ἐν ταῖς αὐταῖς παραλλήλοις ὀρθὰ ἀλλήλοις ἐστίν; ὅπερ ὀφείδει δεῖξαι.

For let BE and CH be joined. And since BC is equal to FG , but FG is equal to EH [Prop. 1.34], BC is thus equal to EH . And they are also parallel, and EB and HC join them. But (straight-lines) joining equal and parallel (straight-lines) on the same sides are (themselves) equal and parallel [Prop. 1.33] [thus, EB and HC are also equal and parallel]. Thus, $EBCH$ is a parallelogram [Prop. 1.34], and is equal to $ABCD$. For it has the same base, BC , (as $ABCD$), and is between the same parallels, BC and AH , (as $ABCD$) [Prop. 1.35]. So, for the same (reasons), $EFGH$ is also equal to the same (parallelogram) $EBCH$ [Prop. 1.34]. So that the parallelogram $ABCD$ is also equal to $EFGH$.

Thus, parallelograms which are on equal bases and between the same parallels are equal to one another. (Which is) the very thing it was required to show.

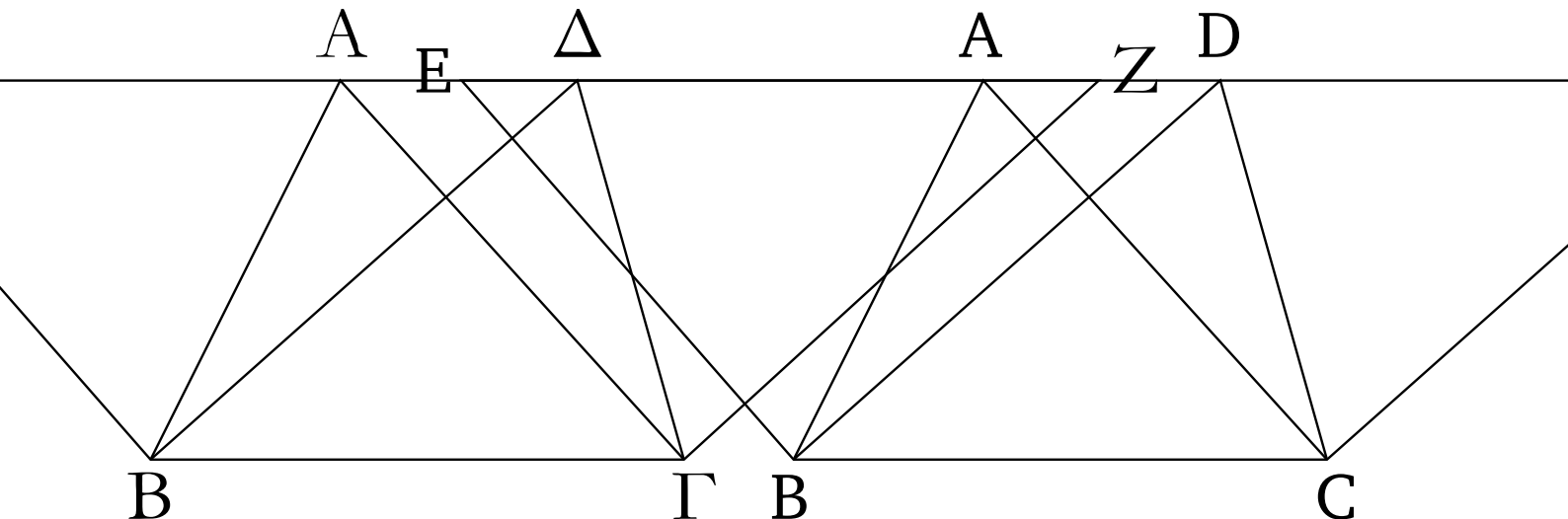
37.

Τὰ τρίγωνα τὰ ἐπὶ τῆς αὐτῆς βάσεως ὀρθὰ καὶ ἐν ταῖς αὐταῖς παραλλήλοις ὀρθὰ ἀλλήλοις ἐστίν.
>Ἐστω τρίγωνα τὰ $AB\Gamma$, $\Delta B\Gamma$ ἐπὶ τῆς αὐτῆς βάσεως τῆς $B\Gamma$ καὶ ἐν ταῖς αὐταῖς παραλλήλοις ταῖς AD , $B\Gamma$; λέγω, ὅτι ὀρθὸν ἐστὶ τὸ $AB\Gamma$ τρίγωνον τῷ $\Delta B\Gamma$ τριγώνῳ.

Proposition 37

Triangles which are on the same base and between the same parallels are equal to one another.

Let ABC and DBC be triangles on the same base BC , and between the same parallels AD and BC . I say that triangle ABC is equal to triangle DBC .



Ἐκβεβλήσθω ἡ AD ὀρθὴ ἑκάτερα τὰ μέρη ἐπὶ τὰ E , Z , καὶ

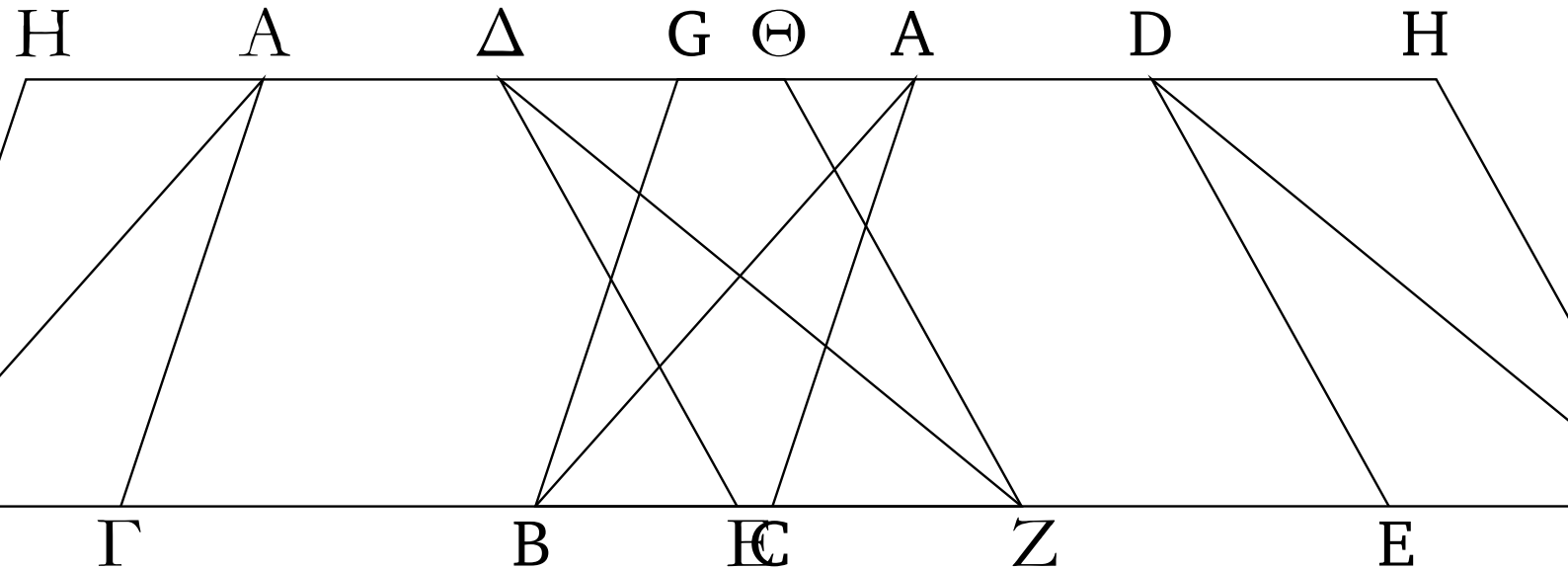
Let AD be produced in both directions to E and F , and

διὰ μὲν τοῦ B τῇ ΓA παράλληλος $>\eta\theta\omega$ ἢ BE , διὰ δὲ τοῦ Γ τῇ $B\Delta$ παράλληλος $>\eta\theta\omega$ ἢ ΓZ . παραλληλόγραμμον $>\acute{\alpha}\rho\alpha$ ἐστὶν ἑκάτερον τῶν $EB\Gamma A$, $\Delta B\Gamma Z$; καὶ εἰσιν $>\acute{\iota}\sigma\alpha$; ἐπὶ τε γὰρ τῆς αὐτῆς βάσεως εἰσι τῆς $B\Gamma$ καὶ ἐν ταῖς αὐταῖς παραλλήλοις ταῖς $B\Gamma$, EZ ; καὶ ἐστὶ τοῦ μὲν $EB\Gamma A$ παραλληλογράμμου $<\acute{\eta}\mu\iota\sigma\upsilon$ τὸ $AB\Gamma$ τρίγωνον; ἢ γὰρ AB διάμετρος αὐτὸ δίθα τέμνει; τοῦ δὲ $\Delta B\Gamma Z$ παραλληλογράμμου $<\acute{\eta}\mu\iota\sigma\upsilon$ τὸ $\Delta B\Gamma$ τρίγωνον; ἢ γὰρ $\Delta\Gamma$ διάμετρος αὐτὸ δίθα τέμνει. [τὰ δὲ τῶν $>\acute{\iota}\sigma\omega\upsilon$ ἡμίση $>\acute{\iota}\sigma\alpha$ ἀλλήλοις ἐστίν]. $>\acute{\iota}\sigma\omega\upsilon$ $>\acute{\alpha}\rho\alpha$ ἐστὶ τὸ $AB\Gamma$ τρίγωνον τῷ $\Delta B\Gamma$ τριγώνῳ.
Τὰ $>\acute{\alpha}\rho\alpha$ τρίγωνα τὰ ἐπὶ τῆς αὐτῆς βάσεως $>\acute{\omicron}\nu\tau\alpha$ καὶ ἐν ταῖς αὐταῖς παραλλήλοις $>\acute{\iota}\sigma\alpha$ ἀλλήλοις ἐστίν; $<\acute{\omicron}\pi\epsilon\rho$ $>\acute{\epsilon}\delta\epsilon\iota$ δεῖχαι.

† This is an additional common notion.

38.

Τὰ τρίγωνα τὰ ἐπὶ $>\acute{\iota}\sigma\omega\upsilon$ βάσεων $>\acute{\omicron}\nu\tau\alpha$ καὶ ἐν ταῖς αὐταῖς παραλλήλοις $>\acute{\iota}\sigma\alpha$ ἀλλήλοις ἐστίν.
 $>\acute{\epsilon}\sigma\tau\omega$ τρίγωνα τὰ $AB\Gamma$, ΔEZ ἐπὶ $>\acute{\iota}\sigma\omega\upsilon$ βάσεων τῶν $B\Gamma$, EZ καὶ ἐν ταῖς αὐταῖς παραλλήλοις ταῖς BZ , AD ; λέγω, $<\acute{\omicron}\tau\iota$ $>\acute{\iota}\sigma\omega\upsilon$ ἐστὶ τὸ $AB\Gamma$ τρίγωνον τῷ ΔEZ τριγώνῳ.



Ἐκβεβλήσθω γὰρ ἡ AD ἐφ' ἑκάτερα τὰ μέρη ἐπὶ τὰ H , Θ , καὶ διὰ μὲν τοῦ B τῇ ΓA παράλληλος $>\eta\theta\omega$ ἢ BH , διὰ δὲ τοῦ Z τῇ ΔE παράλληλος $>\eta\theta\omega$ ἢ $Z\Theta$. παραλληλόγραμμον $>\acute{\alpha}\rho\alpha$ ἐστὶν ἑκάτερον τῶν $HB\Gamma A$, $\Delta EZ\Theta$; καὶ $>\acute{\iota}\sigma\omega\upsilon$ τὸ $HB\Gamma A$ τῷ $\Delta EZ\Theta$; ἐπὶ τε γὰρ $>\acute{\iota}\sigma\omega\upsilon$ βάσεων εἰσι τῶν $B\Gamma$, EZ καὶ ἐν ταῖς αὐταῖς παραλλήλοις ταῖς BZ , $H\Theta$; καὶ ἐστὶ τοῦ μὲν $HB\Gamma A$ παραλληλογράμμου $<\acute{\eta}\mu\iota\sigma\upsilon$ τὸ $AB\Gamma$ τρίγωνον. ἢ γὰρ AB διάμετρος αὐτὸ δίθα τέμνει; τοῦ δὲ $\Delta EZ\Theta$ παραλληλογράμμου $<\acute{\eta}\mu\iota\sigma\upsilon$ τὸ $ZE\Delta$ τρίγωνον; ἢ γὰρ ΔZ διάμετρος αὐτὸ δίθα τέμνει [τὰ δὲ τῶν $>\acute{\iota}\sigma\omega\upsilon$ ἡμίση $>\acute{\iota}\sigma\alpha$ ἀλλήλοις ἐστίν]. $>\acute{\iota}\sigma\omega\upsilon$ $>\acute{\alpha}\rho\alpha$ ἐστὶ τὸ $AB\Gamma$ τρίγωνον τῷ ΔEZ τριγώνῳ.

Τὰ $>\acute{\alpha}\rho\alpha$ τρίγωνα τὰ ἐπὶ $>\acute{\iota}\sigma\omega\upsilon$ βάσεων $>\acute{\omicron}\nu\tau\alpha$ καὶ ἐν ταῖς αὐταῖς παραλλήλοις $>\acute{\iota}\sigma\alpha$ ἀλλήλοις ἐστίν; $<\acute{\omicron}\pi\epsilon\rho$ $>\acute{\epsilon}\delta\epsilon\iota$ δεῖχαι.

39.

Proposition 38

Triangles which are on equal bases and between the same parallels are equal to one another.

Let ABC and DEF be triangles on the equal bases BC and EF , and between the same parallels BF and AD . I say that triangle ABC is equal to triangle DEF .

For let AD be produced in both directions to G and H , and let the (straight-line) BG be drawn through B parallel to CA [Prop. 1.31], and let the (straight-line) FH be drawn through F parallel to DE [Prop. 1.31]. Thus, $GBCA$ and $DEFH$ are each parallelograms. And $GBCA$ is equal to $DEFH$. For they are on the equal bases BC and EF , and between the same parallels BF and GH [Prop. 1.36]. And triangle ABC is half of the parallelogram $GBCA$. For the diagonal AB cuts the latter in half [Prop. 1.34]. And triangle FED (is) half of parallelogram $DEFH$. For the diagonal DF cuts the latter in half. [And the halves of equal things are equal to one another.] Thus, triangle ABC is equal to triangle DEF .

Thus, triangles which are on equal bases and between the same parallels are equal to one another. (Which is) the very thing it was required to show.

Proposition 39

Τὰ >ῖσα τρίγωνα τὰ ἐπὶ τῆς αὐτῆς βάσεως >όντα καὶ ἐπὶ τὰ αὐτὰ μέρη καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἐστίν.

>Εστω >ῖσα τρίγωνα τὰ $AB\Gamma$, $\Delta B\Gamma$ ἐπὶ τῆς αὐτῆς βάσεως >όντα καὶ ἐπὶ τὰ αὐτὰ μέρη τῆς $B\Gamma$; λέγω, <ὅτι καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἐστίν.

Ἐπεξεύθθω γὰρ ἡ AD ; λέγω, <ὅτι παράλληλός ἐστιν ἡ AD τῇ $B\Gamma$.

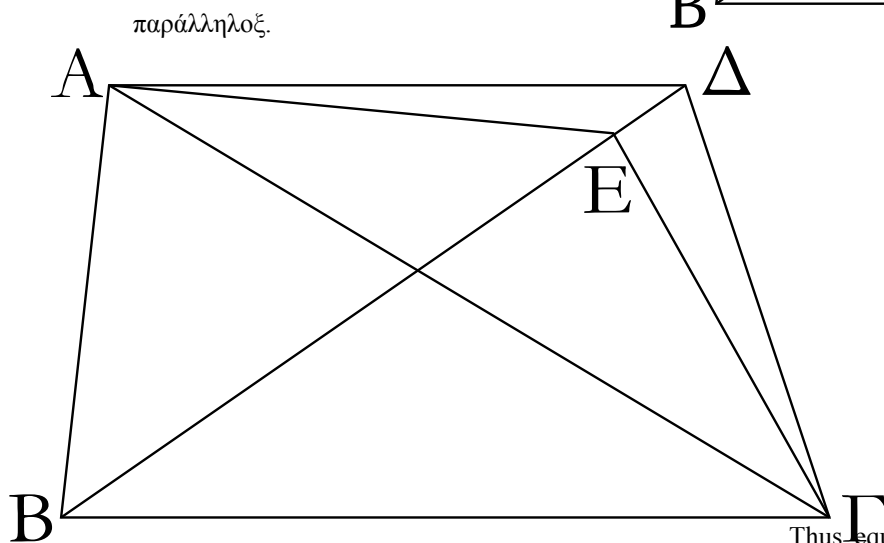
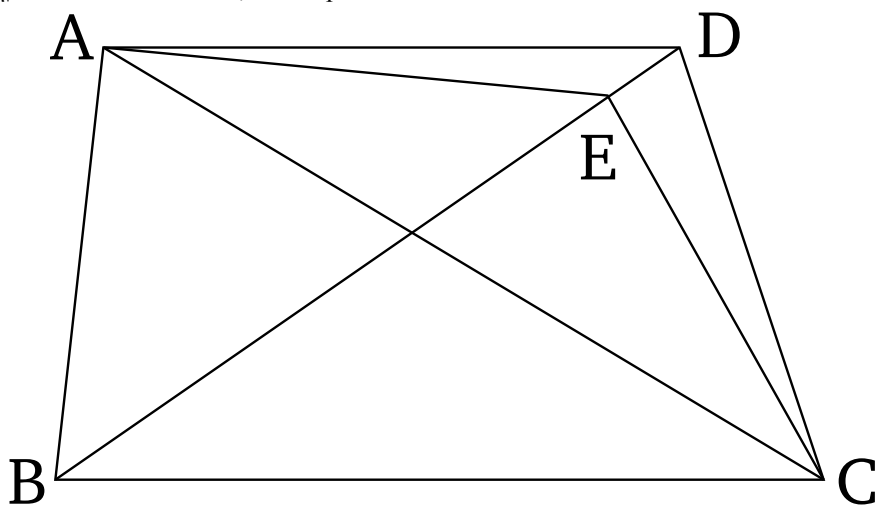
Εἰ γὰρ μή, >ῆθθω διὰ τοῦ A σημείου τῇ $B\Gamma$ εὐθεία παράλληλος ἡ AE , καὶ ἐπεξεύθθω ἡ EF . >ῖσον >άρα ἐστὶ τὸ $AB\Gamma$ τριγώνον τῷ $EB\Gamma$ τριγώνῳ; ἐπὶ τε γὰρ τῆς αὐτῆς βάσεως ἐστὶν αὐτῷ τῆς $B\Gamma$ καὶ ἐν ταῖς αὐταῖς παραλλήλοις. ἀλλὰ τὸ $AB\Gamma$ τῷ $\Delta B\Gamma$ ἐστὶν >ῖσον; καὶ τὸ $\Delta B\Gamma$ >άρα τῷ $EB\Gamma$ >ῖσον ἐστὶ τὸ μείζον τῷ ἐλάσσονι; <ὅπερ ἐστὶν ἀδύνατον; οὐκ >άρα παράλληλός ἐστιν ἡ AE τῇ $B\Gamma$. ὁμοίως δὲ δείχμεν, <ὅτι οὐδ' >άλλη τις πλὴν τῆς AD ; ἡ AD >άρα τῇ $B\Gamma$ ἐστι

Equal triangles which are on the same base, and on the same side, are also between the same parallels.

Let ABC and DBC be equal triangles which are on the same base BC , and on the same side (of it). I say that they are also between the same parallels.

For let AD be joined. I say that AD and BC are parallel.

For, if not, let AE be drawn through point A parallel to the straight-line BC [Prop. 1.31], and let EC be joined. Thus, triangle ABC is equal to triangle EBC . For it is on the same base as it, BC , and between the same parallels [Prop. 1.37]. But ABC is equal to DBC . Thus, DBC is also equal to EBC , the greater to the lesser. The very thing is impossible. Thus, AE is not parallel to BC . Similarly, we can show that neither (is) any other (straight-line) than AD . Thus, AD is parallel to BC .



Τὰ >άρα >ῖσα τρίγωνα τὰ ἐπὶ τῆς αὐτῆς βάσεως >όντα καὶ ἐπὶ τὰ αὐτὰ μέρη καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἐστίν; <ὅπερ >έδει δεῖχαι.

Thus, equal triangles which are on the same base, and on the same side, are also between the same parallels. (Which is) the very thing it was required to show.

40.

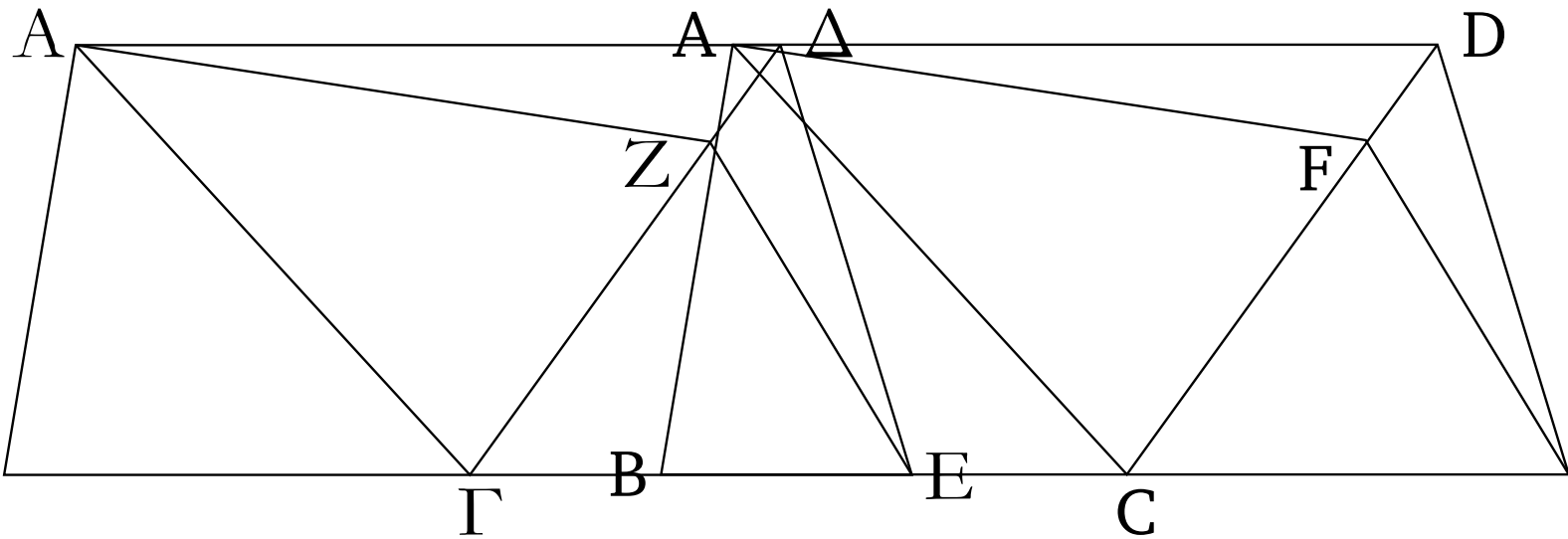
Τὰ >ῖσα τρίγωνα τὰ ἐπὶ >ῖσων βάσεων >όντα καὶ ἐπὶ τὰ αὐτὰ μέρη καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἐστίν.

>Εστω >ῖσα τρίγωνα τὰ $AB\Gamma$, $\Gamma\Delta E$ ἐπὶ >ῖσων βάσεων τῶν $B\Gamma$, ΓE καὶ ἐπὶ τὰ αὐτὰ μέρη. λέγω, <ὅτι καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἐστίν.

Proposition 40[†]

Equal triangles which are on equal bases, and on the same side, are also between the same parallels.

Let ABC and CDE be equal triangles on the equal bases BC and CE (respectively), and on the same side (of BE). I say that they are also between the same parallels.



Ἐπεζεύθθω γὰρ ἡ AD ; λέγω, <ὅτι παράλληλός ἐστιν ἡ AD τῇ BE .

Εἰ γὰρ μή, >ήθθω διὰ τοῦ A τῇ BE παράλληλος ἡ AZ , καὶ ἐπεζεύθθω ἡ ZE . >ίσον >άρα ἐστὶ τὸ $ABΓ$ τρίγωνον τῷ $ZΓΕ$ τριγώνῳ; ἐπὶ τε γὰρ >ίσων βάσεων εἰσι τῶν $BΓ$, $ΓΕ$ καὶ ἐν ταῖς αὐταῖς παραλλήλοις ταῖς BE , AZ . ἀλλὰ τὸ $ABΓ$ τρίγωνον >ίσον ἐστὶ τῷ $ΔΓΕ$ [τριγώνῳ]; καὶ τὸ $ΔΓΕ$ >άρα [τρίγωνον] >ίσον ἐστὶ τῷ $ZΓΕ$ τριγώνῳ τὸ μείζον τῷ ἐλάσσονι; <ὅπερ ἐστὶν ἀδύνατον; οὐκ >άρα παράλληλος ἡ AZ τῇ BE . ὁμοίως δὲ δείχομεν, <ὅτι οὐδ> >άλλη τις πλὴν τῆς AD ; ἡ AD >άρα τῇ BE ἐστὶ παράλληλος.

Τὰ >άρα >ίσα τρίγωνα τὰ ἐπὶ >ίσων βάσεων >όντα καὶ ἐπὶ τὰ αὐτὰ μέρη καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἐστίν; <ὅπερ >έδει δεῖχαι.

For let AD be joined. I say that AD is parallel to BE .

For if not, let AF be drawn through A parallel to BE [Prop. 1.31], and let FE be joined. Thus, triangle ABC is equal to triangle FCE . For they are on equal bases, BC and CE , and between the same parallels, BE and AF [Prop. 1.38]. But, triangle ABC is equal to [triangle] DCE . Thus, [triangle] DCE is also equal to triangle FCE , the greater to the lesser. The very thing is impossible. Thus, AF is not parallel to BE . Similarly, we can show that neither (is) any other (straight-line) than AD . Thus, AD is parallel to BE .

Thus, equal triangles which are on equal bases, and on the same side, are also between the same parallels. (Which is) the very thing it was required to show.

† This whole proposition is regarded by Heiberg as a relatively early interpolation to the original text.

41.

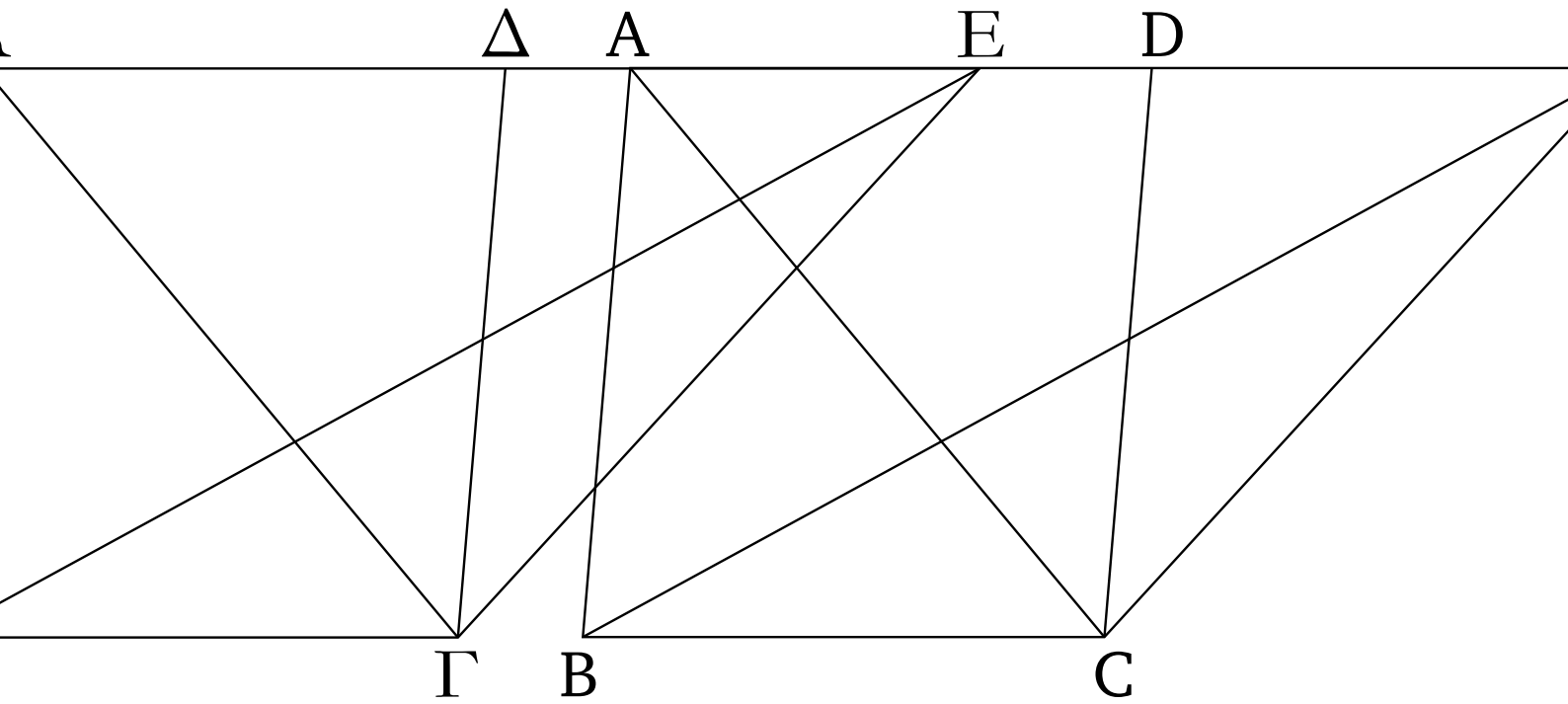
Ἐὰν παραλληλόγραμμον τριγώνῳ βάσιν τε >έθῃ τὴν αὐτὴν καὶ ἐν ταῖς αὐταῖς παραλλήλοις >ῇ, διπλάσιόν ἐστὶ τὸ παραλληλόγραμμον τοῦ τριγώνου.

Παραλληλόγραμμον γὰρ τὸ $ABΓΔ$ τριγώνῳ τῷ $EBΓ$ βάσιν τε ἐθέτω τὴν αὐτὴν τὴν $BΓ$ καὶ ἐν ταῖς αὐταῖς παραλλήλοις >έστω ταῖς $BΓ$, AE ; λέγω, <ὅτι διπλάσιόν ἐστὶ τὸ $ABΓΔ$ παραλληλόγραμμον τοῦ BEG τριγώνου.

Proposition 41

If a parallelogram has the same base as a triangle, and is between the same parallels, (then) the parallelogram is double (the area) of the triangle.

For let parallelogram $ABCD$ have the same base BC as triangle EBC , and let it be between the same parallels, BC and AE . I say that parallelogram $ABCD$ is double (the area) of triangle BEC .



Ἐπεξεύθθω γὰρ ἡ ΑΓ. >ῖσον δὴ ἐστὶ τὸ ΑΒΓ τρίγωνον τῷ ΕΒΓ τριγώνῳ; ἐπὶ τε γὰρ τῇ αὐτῇ βάσει ὡς ἐστὶν αὐτῷ τῇ ΒΓ καὶ ἐν ταῖς αὐταῖς παραλλήλοις ταῖς ΒΓ, ΑΕ. ἀλλὰ τὸ ΑΒΓΔ παραλληλόγραμμον διπλάσιόν ἐστι τοῦ ΑΒΓ τριγώνου; ἡ γὰρ ΑΓ διάμετρος αὐτὸ διῖθα τέμνει; <ώστε τὸ ΑΒΓΔ παραλληλόγραμμον καὶ τοῦ ΕΒΓ τριγώνου ἐστὶ διπλάσιον.

Ἐὰν >άρα παραλληλόγραμμον τριγώνῳ βάσιν τε >έθῃ τὴν αὐτὴν καὶ ἐν ταῖς αὐταῖς παραλλήλοις >ῇ, διπλάσιόν ἐστὶ τὸ παραλληλόγραμμον τοῦ τριγώνου; <όπερ >έδει δεῖξαι.

For let AC be joined. So triangle ABC is equal to triangle EBC . For it is on the same base, BC , (as EBC), and between the same parallels, BC and AE [Prop. 1.37]. But, parallelogram $ABCD$ is double (the area) of triangle ABC . For the diagonal AC cuts the former in half [Prop. 1.34]. So parallelogram $ABCD$ is also double (the area) of triangle EBC .

Thus, if a parallelogram has the same base as a triangle, and is between the same parallels, (then) the parallelogram is double (the area) of the triangle. (Which is) the very thing it was required to show.

42.

Τῷ δοθέντι τριγώνῳ >ῖσον παραλληλόγραμμον συστήσασθαι ἐν τῇ δοθείσῃ γωνίᾳ εὐθυγράμμῳ.

>Ἐστω τὸ μὲν δοθέν τρίγωνον τὸ ΑΒΓ, ἡ δὲ δοθεῖσα γωνία εὐθύγραμμος ἡ Δ; δεῖ δὴ τῷ ΑΒΓ τριγώνῳ >ῖσον παραλληλόγραμμον συστήσασθαι ἐν τῇ Δ γωνίᾳ εὐθυγράμμῳ.

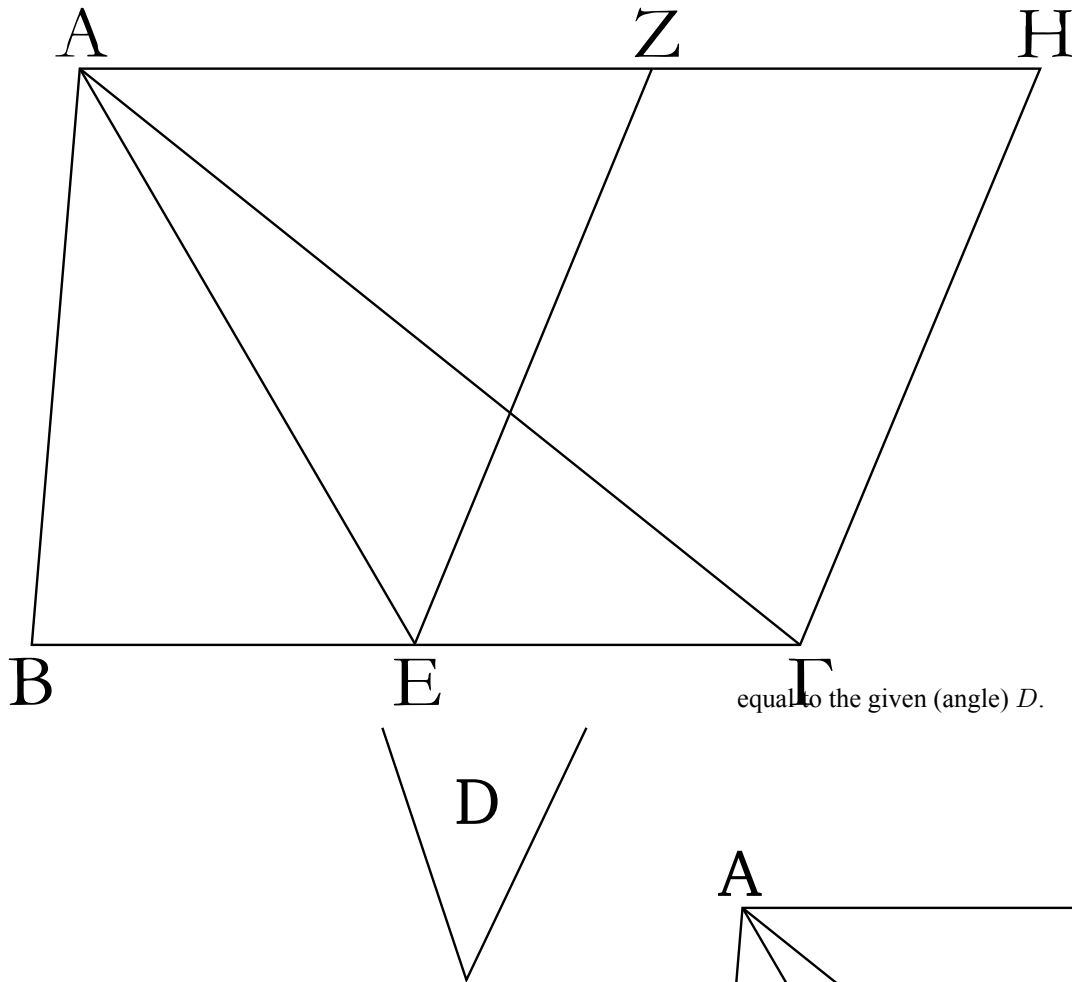
Τετμήσθω ἡ ΒΓ δίθα κατὰ τὸ Ε, καὶ ἐπεξεύθθω ἡ ΑΕ, καὶ συνεστάτω πρὸς τῇ ΕΓ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ Ε τῇ Δ γωνία >ῖση ἡ ὑπὸ ΓΕΖ, καὶ διὰ μὲν τοῦ Α τῇ ΕΓ παράλληλος >ῇθθω ἡ ΑΗ, διὰ δὲ τοῦ Γ τῇ ΕΖ παράλληλος >ῇθθω ἡ ΓΗ; παραλληλόγραμμον >άρα ἐστὶ τὸ ΖΕΓΗ. καὶ ἐπεὶ >ῖση ἐστὶν ἡ ΒΕ τῇ ΕΓ, >ῖσον ἐστὶ καὶ τὸ ΑΒΕ τρίγωνον τῷ ΑΕΓ τριγώνῳ; ἐπὶ τε γὰρ >ῖσων βάσεων εἰσι τῶν ΒΕ, ΕΓ καὶ ἐν ταῖς αὐταῖς παραλλήλοις ταῖς ΒΓ, ΑΗ; διπλάσιον >άρα ἐστὶ τὸ ΑΒΓ τρίγωνον τοῦ ΑΕΓ τριγώνου. >έστι δὲ καὶ τὸ ΖΕΓΗ παραλληλόγραμμον διπλάσιον τοῦ ΑΕΓ τριγώνου; βάσιν τε γὰρ αὐτῷ τὴν αὐτὴν >έθει καὶ ἐν ταῖς αὐταῖς ἐστὶν αὐτῷ παραλλήλοις; >ῖσον >άρα ἐστὶ τὸ ΖΕΓΗ παραλληλόγραμμον τῷ ΑΒΓ τριγώνῳ. καὶ >έθει τὴν ὑπὸ ΓΕΖ γωνίαν >ῖσην τῇ δοθείσῃ τῇ Δ.

Proposition 42

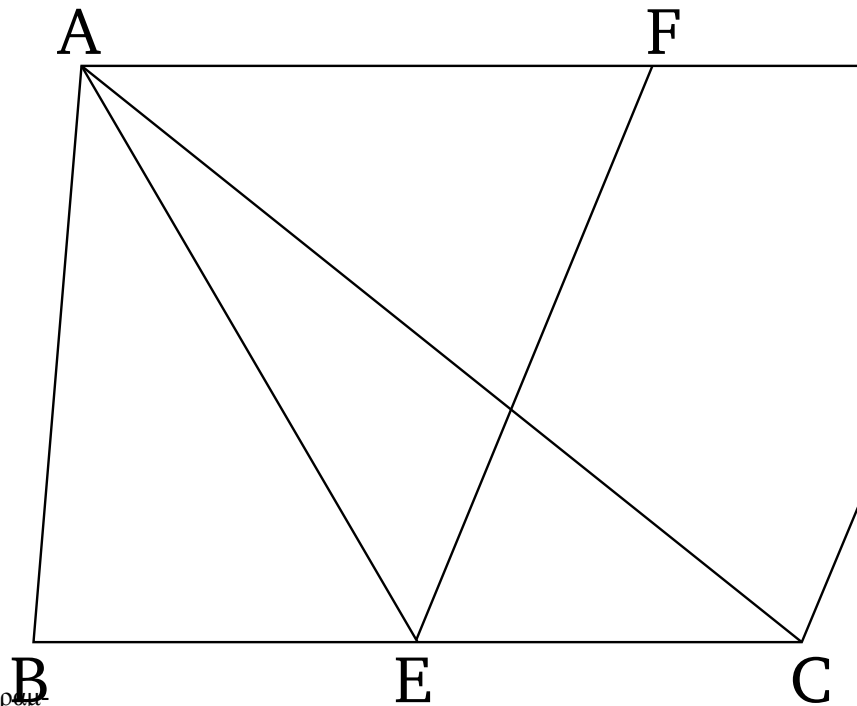
To construct a parallelogram equal to a given triangle in a given rectilinear angle.

Let ABC be the given triangle, and D the given rectilinear angle. So it is required to construct a parallelogram equal to triangle ABC in the rectilinear angle D .

Let BC be cut in half at E [Prop. 1.10], and let AE be joined. And let (angle) CEF , equal to angle D , be constructed at the point E on the straight-line EC [Prop. 1.23]. And let AG be drawn through A parallel to EC [Prop. 1.31], and let CG be drawn through C parallel to EF [Prop. 1.31]. Thus, $FECG$ is a parallelogram. And since BE is equal to EC , triangle ABE is also equal to triangle AEC . For they are on the equal bases, BE and EC , and between the same parallels, BC and AG [Prop. 1.38]. Thus, triangle ABC is double (the area) of triangle AEC . And parallelogram $FECG$ is also double (the area) of triangle AEC . For it has the same base (as AEC), and is between the same parallels (as AEC) [Prop. 1.41]. Thus, parallelogram $FECG$ is equal to triangle ABC . ($FECG$) also has the angle CEF



equal to the given (angle) D .



Τῷ >ἀρα δοθέντι τριγώνῳ τῷ $ABΓ$ >ἴσον παραλληλόγραμ-
μον συνέσταται τὸ $ΖΕΓΗ$ ἐν γωνίᾳ τῇ ὑπὸ $ΓΕΖ$, <ἥτις ἐστὶν
>ἴση τῇ $Δ$; <ὅπερ >έδει ποιῆσαι.

Thus, parallelogram $FECΓ$, equal to the given triangle
 ABC , has been constructed in the angle CEF , which is
equal to D . (Which is) the very thing it was required to do.

43.

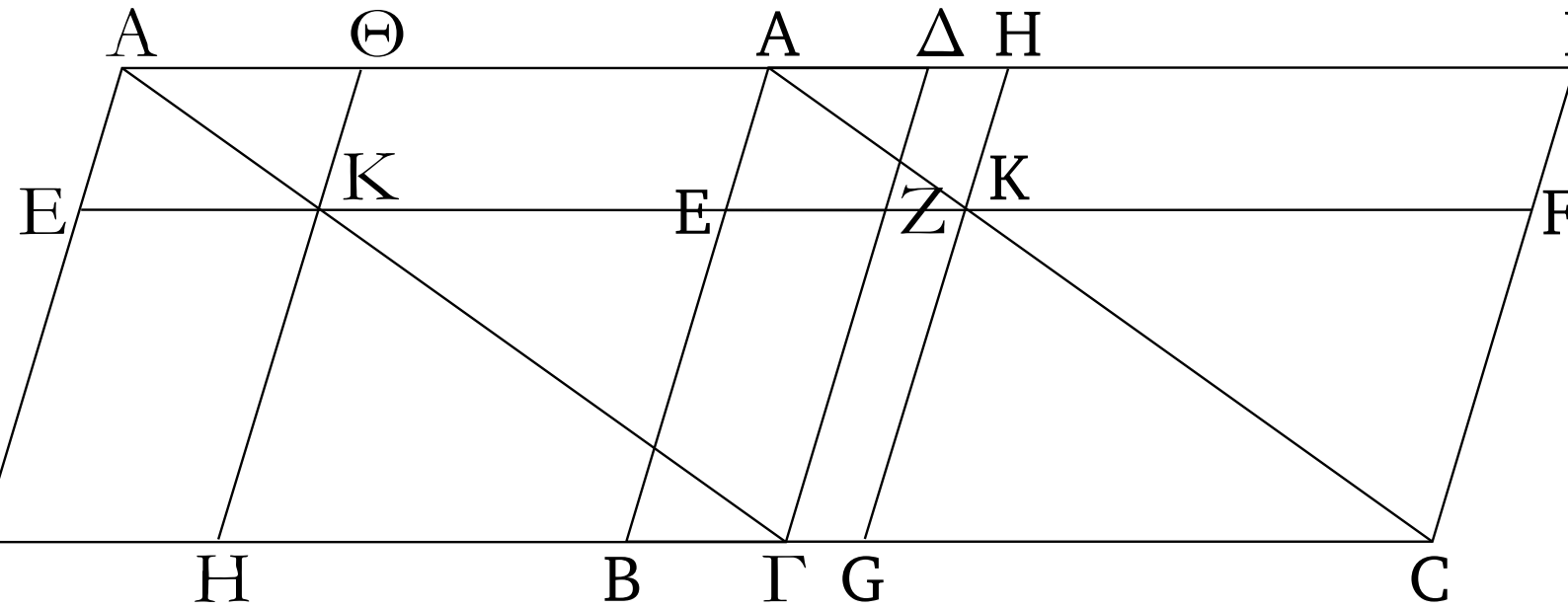
Proposition 43

Παντὸς παραλληλογράμμου τῶν περὶ τὴν διάμετρον παραλληλογράμμου
τὰ παραπληρώματα >ἴσα ἀλλήλοις ἐστίν.
>Ἐστω παραλληλόγραμμον τὸ $ABΓΔ$, διάμετρος δὲ αὐτοῦ

For any parallelogram, the complements of the parallelo-
grams about the diagonal are equal to one another.
Let $ABCD$ be a parallelogram, and AC its diagonal. And

ἡ ΑΓ, περὶ δὲ τὴν ΑΓ παραλληλόγραμμα μὲν >έστω τὰ ΕΘ, ΖΗ, τὰ δὲ λεγόμενα παραπληρώματα τὰ ΒΚ, ΚΔ; λέγω, <ὅτι >ίσον ἐστὶ τὸ ΒΚ παραπλήρωμα τῷ ΚΔ παραπληρώματι.

let EH and FG be the parallelograms about AC , and BK and KD the so-called complements (about AC). I say that the complement BK is equal to the complement KD .



Ἐπεὶ γὰρ παραλληλόγραμμον ἐστὶ τὸ ΑΒΓΔ, διάμετροξ δὲ αὐτοῦ ἡ ΑΓ, >ίσον ἐστὶ τὸ ΑΒΓ τρίγωνον τῷ ΑΓΔ τριγώνῳ. πάλιν, ἐπεὶ παραλληλόγραμμον ἐστὶ τὸ ΕΘ, διάμετροξ δὲ αὐτοῦ ἐστὶν ἡ ΑΚ, >ίσον ἐστὶ τὸ ΑΕΚ τρίγωνον τῷ ΑΘΚ τριγώνῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ ΚΖΓ τρίγωνον τῷ ΚΗΓ ἐστὶν >ίσον. ἐπεὶ οὖν τὸ μὲν ΑΕΚ τρίγωνον τῷ ΑΘΚ τριγώνῳ ἐστὶν >ίσον, τὸ δὲ ΚΖΓ τῷ ΚΗΓ, τὸ ΑΕΚ τρίγωνον μετὰ τοῦ ΚΗΓ >ίσον ἐστὶ τῷ ΑΘΚ τριγώνῳ μετὰ τοῦ ΚΖΓ; >έστι δὲ καὶ <όλον τὸ ΑΒΓ τρίγωνον <όλω τῷ ΑΔΓ >ίσον; λοιπὸν >άρα τὸ ΒΚ παραπλήρωμα λοιπῷ τῷ ΚΔ παραπληρώματι ἐστὶν >ίσον.

Παντὸξ >άρα παραλληλογράμμου θωρίου τῶν περὶ τὴν διάμετρον παραλληλογράμμων τὰ παραπληρώματα >ίσα ἀλλήλοιξ ἐστίν; <όπερ >έδει δεῖχαι.

For since $ABCD$ is a parallelogram, and AC its diagonal, triangle ABC is equal to triangle ACD [Prop. 1.34]. Again, since EH is a parallelogram, and AK is its diagonal, triangle AEK is equal to triangle AHK [Prop. 1.34]. So, for the same (reasons), triangle KFC is also equal to (triangle) KGC . Therefore, since triangle AEK is equal to triangle AHK , and KFC to KGC , triangle AEK plus KGC is equal to triangle AHK plus KFC . And the whole triangle ABC is also equal to the whole (triangle) ADC . Thus, the remaining complement BK is equal to the remaining complement KD .

Thus, for any parallelogramic figure, the complements of the parallelograms about the diagonal are equal to one another. (Which is) the very thing it was required to show.

44.

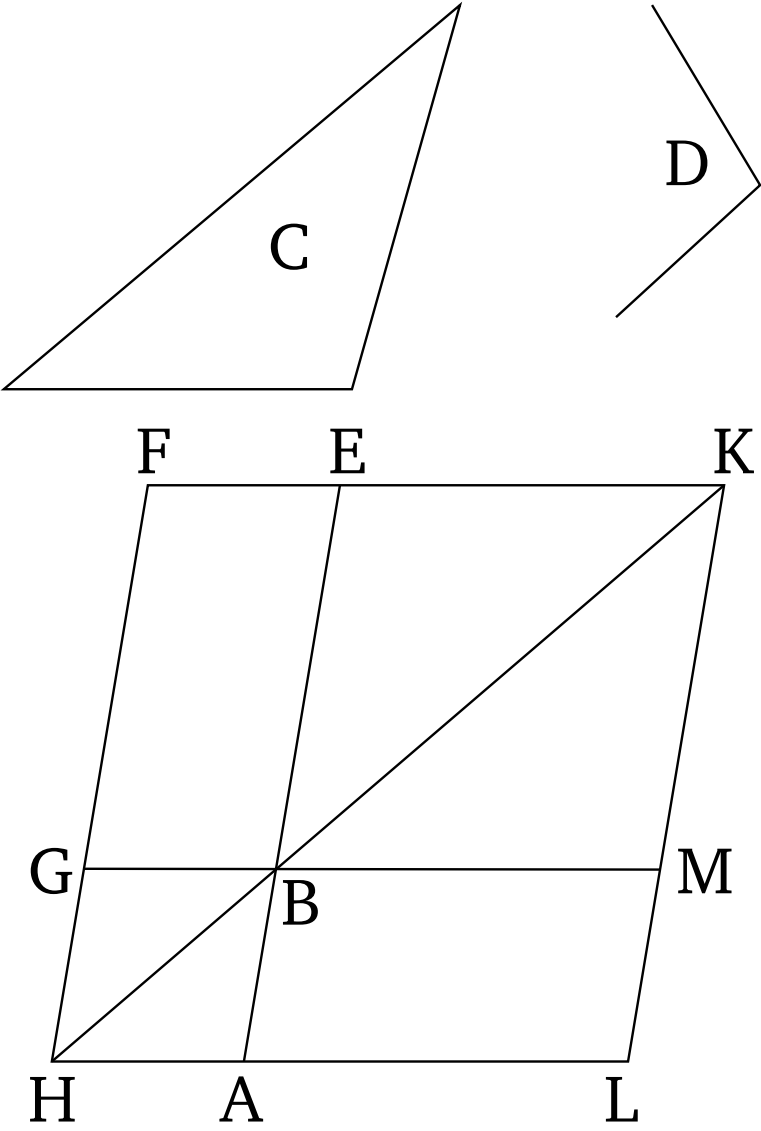
Proposition 44

Παρὰ τὴν δοθεῖσαν εὐθεῖαν τῷ δοθέντι τριγώνῳ >ίσον παραλληλόγραμμον παραβαλεῖν ἐν τῇ δοθείσῃ γωνίᾳ εὐθυγράμμω.

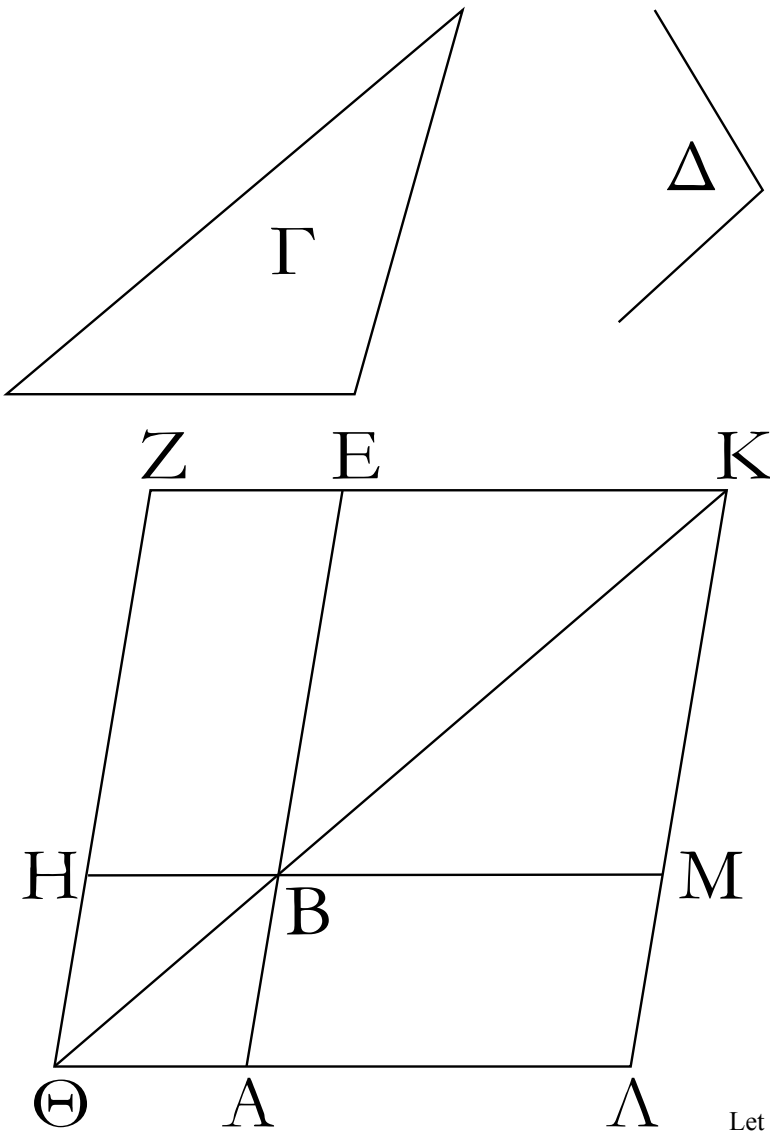
>Ἐστω ἡ μὲν δοθεῖσα εὐθεῖα ἡ ΑΒ, τὸ δὲ δοθὲν τρίγωνον τὸ Γ, ἡ δὲ δοθεῖσα γωνία εὐθύγραμμοξ ἡ Δ; δεῖ δὴ παρὰ τὴν δοθεῖσαν εὐθεῖαν τὴν ΑΒ τῷ δοθέντι τριγώνῳ τῷ Γ >ίσον

To apply a parallelogram equal to a given triangle to a given straight-line in a given rectilinear angle.

Let AB be the given straight-line, C the given triangle, and D the given rectilinear angle. So it is required to apply a parallelogram equal to the given triangle C to the given straight-line AB in an angle equal to (angle) D .



παράλληλόγραμμον παραβαλεῖν ἐν >ίση τῇ Δ γωνίᾳ.



Συνεστάτω τῷ Γ τριγώνῳ ὀκτὼ παραλληλόγραμμον τὸ ΒΕΖΗ ἐν γωνίᾳ τῇ ὑπὸ ΕΒΗ, <ἢ ἐστὶν ὀκτὼ τῇ Δ; καὶ κείσθω <ὥστε ἐπ> εὐθείᾳ εἶναι τὴν ΒΕ τῇ ΑΒ, καὶ διήθθω ἡ ΖΗ ἐπὶ τὸ Θ, καὶ διὰ τοῦ Α ὁποτέρᾳ τῶν ΒΗ, ΕΖ παράλληλοξ >ἡθθω ἡ ΑΘ, καὶ ἐπεξεύθθω ἡ ΘΒ. καὶ ἐπεὶ εἰς παραλλήλουξ τὰξ ΑΘ, ΕΖ εὐθεῖα ἐνέπεσεν ἡ ΘΖ, αἱ >άρα ὑπὸ ΑΘΖ, ΘΖΕ γωνίαι δυσὶν ὀρθαῖξ εἰσὶν ὀκτὼ. αἱ >άρα ὑπὸ ΒΘΗ, ΗΖΕ δύο ὀρθῶν ἐλάσσονέξ εἰσιν; αἱ δὲ ἀπὸ ἐλασσόνων >ἡ δύο ὀρθῶν εἰξ >άπειρον ἐκβαλλόμεναι συμπύπτουσιν; αἱ ΘΒ, ΖΕ >άρα ἐκβαλλόμεναι συμπεσοῦνται. ἐκβεβλήσθωσαν καὶ συμπίπτέωσαν κατὰ τὸ Κ, καὶ διὰ τοῦ Κ σημείου ὁποτέρᾳ τῶν ΕΑ, ΖΘ παράλληλοξ >ἡθθω ἡ ΚΛ, καὶ ἐκβεβλήσθωσαν αἱ ΘΑ, ΗΒ ἐπὶ τὰ Λ, Μ σημεία. παραλληλόγραμμον >άρα ἐστὶ τὸ ΘΑΚΖ, διάμετροξ δὲ αὐτοῦ ἡ ΘΚ, περὶ δὲ τὴν ΘΚ παραλληλόγραμμα μὲν τὰ ΑΗ, ΜΕ, τὰ δὲ λεγόμενα παραπληρώματα τὰ ΛΒ, ΒΖ; ὀκτὼ ἐστὶ τὸ ΛΒ τῷ ΒΖ. ἀλλὰ τὸ ΒΖ τῷ Γ τριγώνῳ ἐστὶν ὀκτὼ; καὶ τὸ ΛΒ >άρα τῷ Γ ἐστὶν ὀκτὼ. καὶ ἐπεὶ ὀκτὼ ἐστὶν ἡ ὑπὸ ΗΒΕ γωνία τῇ ὑπὸ ΑΒΜ, ἀλλὰ ἡ ὑπὸ ΗΒΕ τῇ Δ ἐστὶν ὀκτὼ, καὶ ἡ ὑπὸ ΑΒΜ >άρα τῇ Δ γωνία ἐστὶν ὀκτὼ. Παρὰ τὴν δοθεῖσαν >άρα εὐθεῖαν τὴν ΑΒ τῷ δοθέντι τριγώνῳ τῷ Γ ὀκτὼ παραλληλόγραμμον παραβέβληται τὸ ΛΒ ἐν γωνίᾳ τῇ ὑπὸ ΑΒΜ, <ἢ ἐστὶν ὀκτὼ τῇ Δ; <ὁπερ >έδει ποιῆσαι.

Let the parallelogram $BEFG$, equal to the triangle C , be constructed in the angle EBG , which is equal to D [Prop. 1.42]. And let it be placed so that BE is straight-on to AB .[†] And let FG be drawn through to H , and let AH be drawn through A parallel to either of BG or EF [Prop. 1.31], and let HB be joined. And since the straight-line HF falls across the parallels AH and EF , the (sum of the) angles AHF and HFE is thus equal to two right-angles [Prop. 1.29]. Thus, (the sum of) BHG and GFE is less than two right-angles. And (straight-lines) produced to infinity from (internal angles whose sum is) less than two right-angles meet together [Post. 5]. Thus, being produced, HB and FE will meet together. Let them be produced, and let them meet together at K . And let KL be drawn through point K parallel to either of EA or FH [Prop. 1.31]. And let HA and GB be produced to points L and M (respectively). Thus, $HLKF$ is a parallelogram, and HK its diagonal. And AG and ME (are) parallelograms, and LB and BF the so-called complements, about HK . Thus, LB is equal to BF [Prop. 1.43]. But, BF is equal to triangle C . Thus, LB is also equal to C . Also, since angle GBE is equal to ABM [Prop. 1.15], but GBE is equal to D , ABM is thus also equal to angle D .

Thus, the parallelogram LB , equal to the given triangle C , has been applied to the given straight-line AB in the angle ABM , which is equal to D . (Which is) the very thing it was

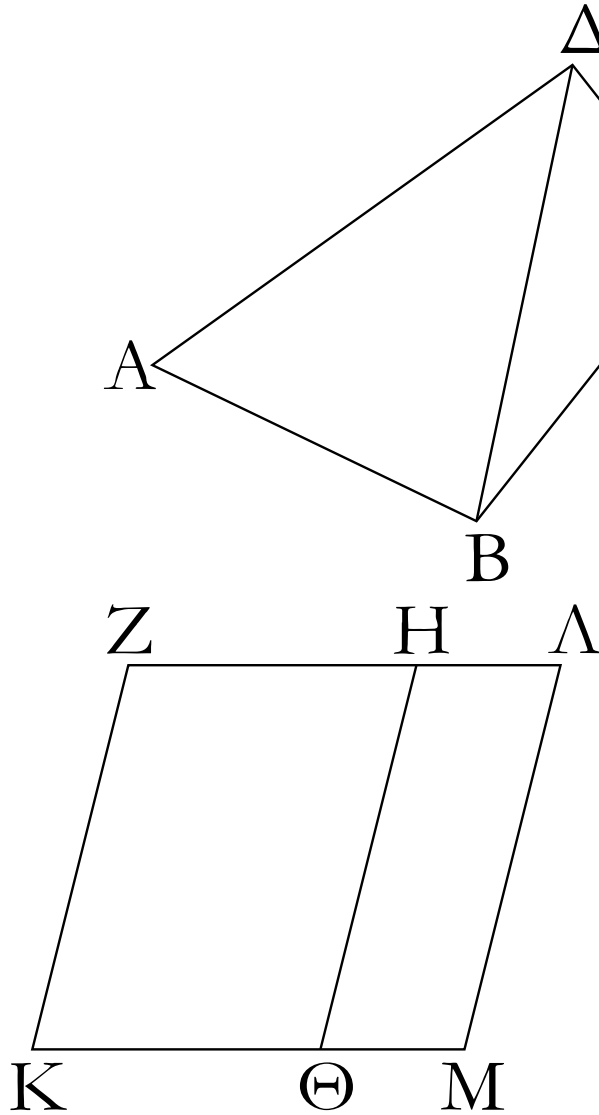
required to do.

† This can be achieved using Props. 1.3, 1.23, and 1.31.

45.

Τῷ δοθέντι εὐθύγραμμῳ ὀλίγον παραλληλόγραμμον συστήσασθαι ἐν τῇ δοθείσῃ γωνίᾳ εὐθύγραμμῳ.

>Ἐστω τὸ μὲν δοθὲν εὐθύγραμμον τὸ $AB\Gamma\Delta$, ἡ δὲ δοθεῖσα γωνία εὐθύγραμμοξ ἡ E ; δεῖ δὴ τῷ $AB\Gamma\Delta$ εὐθύγραμμῳ ὀλίγον παραλληλόγραμμον συστήσασθαι ἐν τῇ δοθείσῃ γωνίᾳ τῇ E .

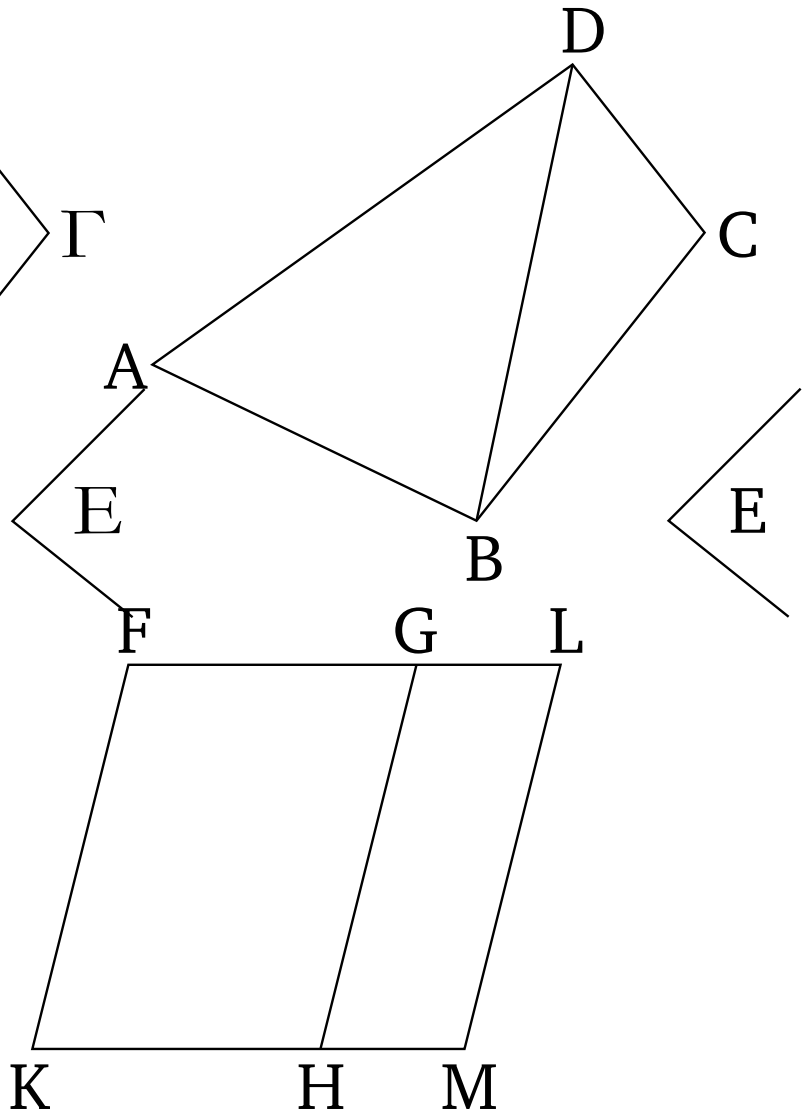


Ἐπεξεύθθω ἡ ΔB , καὶ συνεστάτω τῷ $\Delta B\Delta$ τριγώνῳ ὀλίγον παραλληλόγραμμον τὸ $Z\Theta$ ἐν τῇ ὑπὸ ΘKZ γωνίᾳ, <ἡ ἔστιν ὀλίγη τῇ E ; καὶ παραβελήσθω παρὰ τὴν $H\Theta$ εὐθεῖαν τῷ $\Delta B\Gamma$ τριγώνῳ ὀλίγον παραλληλόγραμμον τὸ $H\Theta$ ἐν τῇ ὑπὸ $H\Theta M$ γωνίᾳ, <ἡ ἔστιν ὀλίγη τῇ E . καὶ ἐπεὶ ἡ E γωνία ἑκατέρω τῶν ὑπὸ ΘKZ , $H\Theta M$ ἔστιν ὀλίγη, καὶ ἡ ὑπὸ ΘKZ ὀλίγη τῇ ὑπὸ $H\Theta M$ ἔστιν ὀλίγη. κοινὴ προσκεῖσθω ἡ ὑπὸ $K\Theta H$; αἱ ὀλίγαι ὑπὸ $ZK\Theta$, $K\Theta H$ ταῖς ὑπὸ $K\Theta H$, $H\Theta M$ ὀλίγαι εἰσὶν. ἀλλ' αἱ ὑπὸ $ZK\Theta$, $K\Theta H$ δυσὶν ὀρθαῖς ὀλίγαι εἰσὶν; καὶ αἱ ὑπὸ $K\Theta H$, $H\Theta M$ ὀλίγαι δύο ὀρθαῖς ὀλίγαι εἰσὶν. πρὸς δὲ τινὶ εὐθείᾳ τῇ $H\Theta$ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ Θ δύο εὐθεῖαι αἱ $K\Theta$, ΘM μὴ ἐπὶ τὰ αὐτὰ μέρη κείμεναι τὰς ἐφεχθῆς γωνίας δύο ὀρθαῖς ὀλίγαις ποιοῦσιν; ἐπ' εὐθείας ὀλίγαι εἰσὶν ἡ $K\Theta$ τῇ ΘM ; καὶ ἐπεὶ εἰς παραλλήλους τὰς KM , ZH εὐθεῖας

Proposition 45

To construct a parallelogram equal to a given rectilinear figure in a given rectilinear angle.

Let $ABCD$ be the given rectilinear figure,[†] and E the given rectilinear angle. So it is required to construct a parallelogram equal to the rectilinear figure $ABCD$ in the given angle E .



Let DB be joined, and let the parallelogram FH , equal to the triangle ABD , be constructed in the angle HKF , which is equal to E [Prop. 1.42]. And let the parallelogram GM , equal to the triangle DBC , be applied to the straight-line GH in the angle GHM , which is equal to E [Prop. 1.44]. And since angle E is equal to each of (angles) HKF and GHM , (angle) HKF is thus also equal to GHM . Let KHG be added to both. Thus, (the sum of) FKH and KHG is equal to (the sum of) KHG and GHM . But, (the sum of) FKH and KHG is equal to two right-angles [Prop. 1.29]. Thus, (the sum of) KHG and GHM is also equal to two right-angles. So two straight-lines, KH and HM , not lying on the same side, make adjacent angles with some straight-line GH , at the point H on it, (whose sum is)

>ίση; αἱ τέσσαρες >άρα αἱ BA, AD, DE, EB >ίσαι ἀλλήλαις εἰσίν; ἰσόπλευρον >άρα ἐστὶ τὸ $ADEB$ παραλληλόγραμμον. λέγω δὴ, <ὅτι καὶ ὀρθογώνιον. ἐπεὶ γὰρ εἰς παραλλήλους τὰς AB, DE εὐθεῖα ἐνέπεσεν ἡ AD , αἱ >άρα ὑπὸ BAD, ADE γωνίαι δύο ὀρθαῖς >ίσαι εἰσίν. ὀρθὴ δὲ ἡ ὑπὸ BAD ; ὀρθὴ >άρα καὶ ἡ ὑπὸ ADE . τῶν δὲ παραλληλογράμμων θωρίων αἱ ἀπεναντίον πλευραὶ τε καὶ γωνίαι >ίσαι ἀλλήλαις εἰσίν; ὀρθὴ >άρα καὶ ἑκατέρα τῶν ἀπεναντίον τῶν ὑπὸ ABE, BED γωνιῶν; ὀρθογώνιον >άρα ἐστὶ τὸ $ADEB$. ἐδείκθη δὲ καὶ ἰσόπλευρον.

Τετράγωνον >άρα ἐστίν; καὶ ἐστὶν ἀπὸ τῆς AB εὐθείας ἀναγεγραμμένον; <ὅπερ >έδει ποιῆσαι.

47.

Ἐν τοῖς ὀρθογωνίοις τριγώνοις τὸ ἀπὸ τῆς τὴν ὀρθὴν γωνίαν ὑποτείνουσῃς πλευρᾷς τετράγωνον >ίσον ἐστὶ τοῖς ἀπὸ τῶν τὴν ὀρθὴν γωνίαν περιθεουσῶν πλευρῶν τετραγώνοις.

>Ἐστω τρίγωνον ὀρθογώνιον τὸ $ABΓ$ ὀρθὴν >έθον τὴν ὑπὸ BAG γωνίαν; λέγω, <ὅτι τὸ ἀπὸ τῆς $BΓ$ τετράγωνον >ίσον

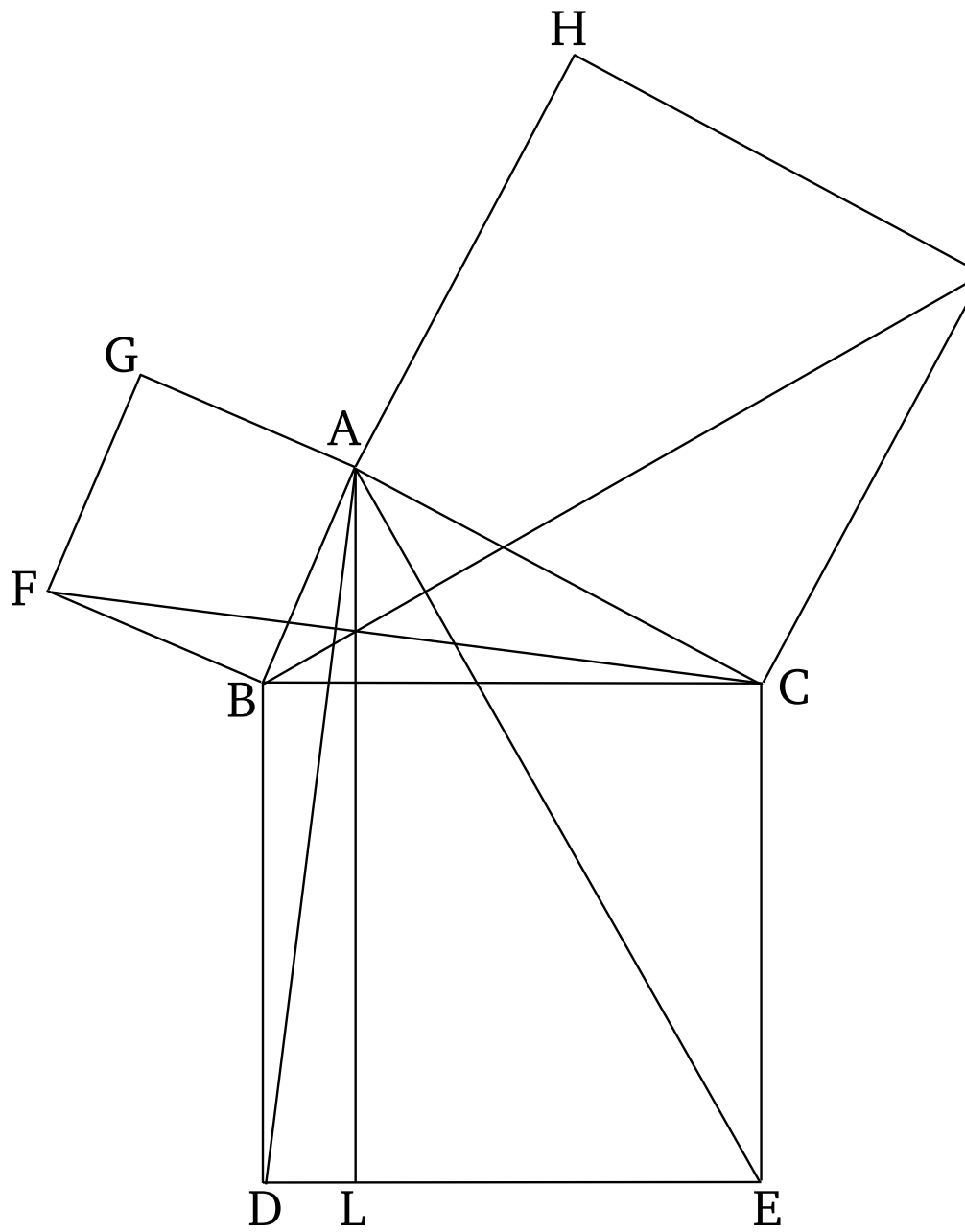
to BE [Prop. 1.34]. But, AB is equal to AD . Thus, the four (sides) BA, AD, DE, EB are equal to one another. Thus, the parallelogram $ADEB$ is equilateral. So I say that (it is) also right-angled. For since the straight-line AD falls across the parallels AB and DE , the (sum of the) angles BAD and ADE is equal to two right-angles [Prop. 1.29]. But BAD (is a) right-angle. Thus, ADE (is) also a right-angle. And for parallelogramic figures, the opposite sides and angles are equal to one another [Prop. 1.34]. Thus, each of the opposite angles ABE and BED (are) also right-angles. Thus, $ADEB$ is right-angled. And it was also shown (to be) equilateral.

Thus, ($ADEB$) is a square [Def. 1.22]. And it is described on the straight-line AB . (Which is) the very thing it was required to do.

Proposition 47

In right-angled triangles, the square on the side subtending the right-angle is equal to the (sum of the) squares on the sides containing the right-angle.

Let ABC be a right-angled triangle having the angle BAC a right-angle. I say that the square on BC is equal to the (sum of the) squares on BA and AC .



ἐστὶ τοῖξ ἀπὸ τῶν ΒΑ, ΑΓ τετραγώνοιξ.

εἰσι παραλλήλοις ταῖς $ΒΔ$, $ΑΛ$; τοῦ δὲ $ΖΒΓ$ τριγώνου διπλάσιον τὸ $ΗΒ$ τετράγωνον; βάσιν τε γὰρ πάλιν τὴν αὐτὴν >έθουσι τὴν $ΖΒ$ καὶ ἐν ταῖς αὐταῖς εἰσι παραλλήλοις ταῖς $ΖΒ$, $ΗΓ$. [τὰ δὲ τῶν >ίσων διπλάσια >ῖσα ἀλλήλοις ἐστίν;] >ῖσον >άρα ἐστὶ καὶ τὸ $ΒΛ$ παραλληλόγραμμον τῷ $ΗΒ$ τετραγώνῳ. ὁμοίως δὲ ἐπιζευγνυμένων τῶν $ΑΕ$, $ΒΚ$ δειθθήσεται καὶ τὸ $ΓΛ$ παραλληλόγραμμον >ῖσον τῷ $ΘΓ$ τετραγώνῳ; <όλον >άρα τὸ $ΒΔΕΓ$ τετράγωνον δυοῖς τοῖς $ΗΒ$, $ΘΓ$ τετραγώνοις >ῖσον ἐστίν. καὶ ἐστὶ τὸ μὲν $ΒΔΕΓ$ τετράγωνον ἀπὸ τῆς $ΒΓ$ ἀναγραφέν, τὰ δὲ $ΗΒ$, $ΘΓ$ ἀπὸ τῶν $ΒΑ$, $ΑΓ$. τὸ >άρα ἀπὸ τῆς $ΒΓ$ πλευρᾶς τετράγωνον >ῖσον ἐστὶ τοῖς ἀπὸ τῶν $ΒΑ$, $ΑΓ$ πλευρῶν τετραγώνοις. Ἐν >άρα τοῖς ὀρθογωνίοις τριγώνοις τὸ ἀπὸ τῆς τὴν ὀρθὴν γωνίαν ὑποτείνουσιν πλευρᾶς τετράγωνον >ῖσον ἐστὶ τοῖς ἀπὸ τῶν τὴν ὀρθὴν [γωνίαν] περιεθουσῶν πλευρῶν τετραγώνοις; <όπερ >έδει δεῖξαι.

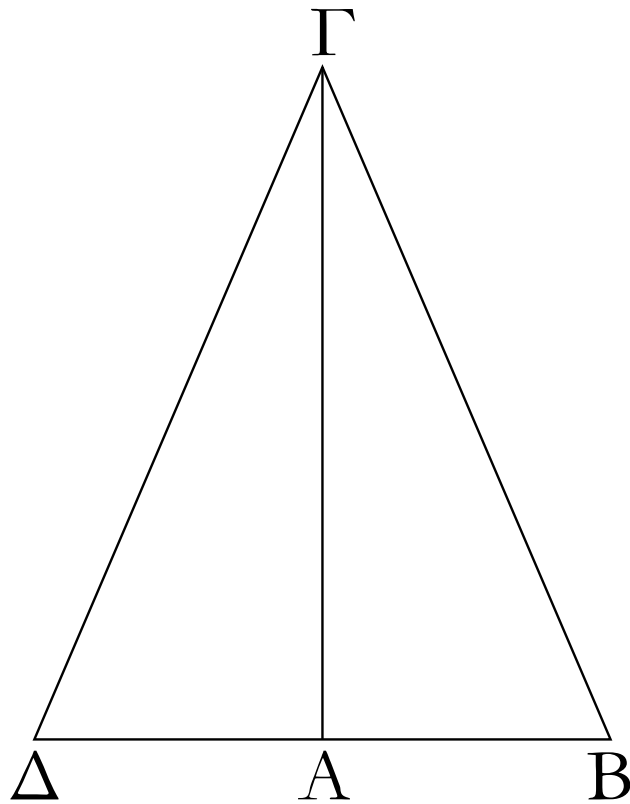
† The Greek text has “ FB , BC ”, which is obviously a mistake.

‡ This is an additional common notion.

the same base, BD , and are between the same parallels, BD and AL [Prop. 1.41]. And square GB is double (the area) of triangle FBC . For again they have the same base, FB , and are between the same parallels, FB and GC [Prop. 1.41]. [And the doubles of equal things are equal to one another.][†] Thus, the parallelogram BL is also equal to the square GB . So, similarly, AE and BK being joined, the parallelogram CL can be shown (to be) equal to the square HC . Thus, the whole square $BDEC$ is equal to the (sum of the) two squares GB and HC . And the square $BDEC$ is described on BC , and the (squares) GB and HC on BA and AC (respectively). Thus, the square on the side BC is equal to the (sum of the) squares on the sides BA and AC . Thus, in right-angled triangles, the square on the side subtending the right-angle is equal to the (sum of the) squares on the sides surrounding the right-[angle]. (Which is) the very thing it was required to show.

48.

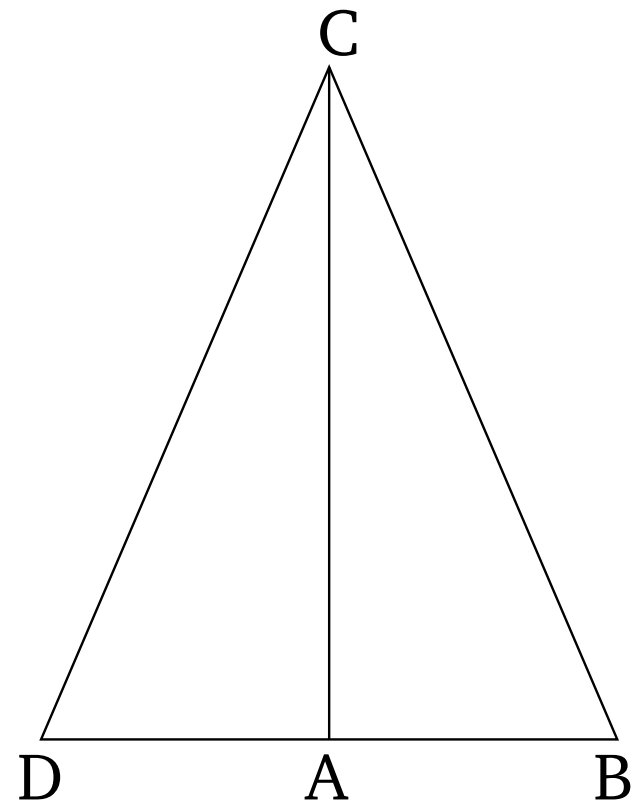
Ἐὰν τριγώνου τὸ ἀπὸ μιᾶς τῶν πλευρῶν τετράγωνον >ῖσον >ῇ τοῖς ἀπὸ τῶν λοιπῶν τοῦ τριγώνου δύο πλευρῶν τετραγώνοις ἢ περιεθομένη γωνία ὑπὸ τῶν λοιπῶν τοῦ τριγώνου δύο πλευρῶν ὀρθὴ ἐστίν. Τριγώνου γὰρ τοῦ $ΑΒΓ$ τὸ ἀπὸ μιᾶς τῆς $ΒΓ$ πλευρᾶς τετράγωνον >ῖσον >έστω τοῖς ἀπὸ τῶν $ΒΑ$, $ΑΓ$ πλευρῶν τετραγώνοις; λέγω, <ὅτι ὀρθὴ ἐστὶν ἡ ὑπὸ $ΒΑΓ$ γωνία.



>Ἡθθω γὰρ ἀπὸ τοῦ $Α$ σημείου τῇ $ΑΓ$ εὐθείᾳ πρὸς ὀρθὰς ἢ $ΑΔ$ καὶ κείσθω τῇ $ΒΑ$ >ῖση ἢ $ΑΔ$, καὶ ἐπεζεύθθω ἡ $ΔΓ$. ἐπεὶ >ῖση ἐστὶν ἡ $ΔΑ$ τῇ $ΑΒ$, >ῖσον ἐστὶ καὶ τὸ ἀπὸ τῆς $ΔΑ$ τετράγωνον τῷ ἀπὸ τῆς $ΑΒ$ τετραγώνῳ. κοινὸν προσκείσθω τὸ ἀπὸ τῆς $ΑΓ$ τετράγωνον; τὰ >άρα ἀπὸ τῶν $ΔΑ$, $ΑΓ$ τετράγωνα >ῖσα ἐστὶ τοῖς ἀπὸ τῶν $ΒΑ$, $ΑΓ$

Proposition 48

If the square on one of the sides of a triangle is equal to the (sum of the) squares on the two remaining sides of the triangle (then) the angle contained by the two remaining sides of the triangle is a right-angle. For let the square on one of the sides, BC , of triangle ABC be equal to the (sum of the) squares on the sides BA and AC . I say that angle BAC is a right-angle.



For let AD be drawn from point A at right-angles to the straight-line AC [Prop. 1.11], and let AD be made equal to BA [Prop. 1.3], and let DC be joined. Since DA is equal to AB , the square on DA is thus also equal to the square on AB .[†] Let the square on AC be added to both. Thus, the (sum of the) squares on DA and AC is equal to the (sum

τετραγώνοιζ. ἀλλὰ τοῖς μὲν ἀπὸ τῶν ΔA , $A\Gamma$ >ίσον ἐστὶ τὸ ἀπὸ τῆς $\Delta\Gamma$; ὀρθὴ γάρ ἐστιν ἡ ὑπὸ $\Delta A\Gamma$ γωνία; τοῖς δὲ ἀπὸ τῶν BA , $A\Gamma$ >ίσον ἐστὶ τὸ ἀπὸ τῆς $B\Gamma$; ὑπόκειται γάρ; τὸ >άρα ἀπὸ τῆς $\Delta\Gamma$ τετράγωνον >ίσον ἐστὶ τῷ ἀπὸ τῆς $B\Gamma$ τετραγώνῳ; <ώστε καὶ πλευρὰ ἡ $\Delta\Gamma$ τῇ $B\Gamma$ ἐστὶν >ίση; καὶ ἐπεὶ >ίση ἐστὶν ἡ ΔA τῇ AB , κοινὴ δὲ ἡ $A\Gamma$, δύο δὴ αἱ ΔA , $A\Gamma$ δύο ταῖς BA , $A\Gamma$ >ίσαι εἰσίν; καὶ βάσις ἡ $\Delta\Gamma$ βάσει τῇ $B\Gamma$ >ίση; γωνία >άρα ἡ ὑπὸ $\Delta A\Gamma$ γωνία τῇ ὑπὸ $B A\Gamma$ [ἐστὶν] >ίση. ὀρθὴ δὲ ἡ ὑπὸ $\Delta A\Gamma$; ὀρθὴ >άρα καὶ ἡ ὑπὸ $B A\Gamma$. Ἐὰν ἀρὰ τριγώνου τὸ ἀπὸ μιᾶς τῶν πλευρῶν τετράγωνον >ίσον >ῃ τοῖς ἀπὸ τῶν λοιπῶν τοῦ τριγώνου δύο πλευρῶν τετραγώνοιζ, ἡ περιεθομένη γωνία ὑπὸ τῶν λοιπῶν τοῦ τριγώνου δύο πλευρῶν ὀρθὴ ἐστίν; <όπερ >έδει δεῖχαι.

of the) squares on BA and AC . But, the (square) on DC is equal to the (sum of the squares) on DA and AC . For angle DAC is a right-angle [Prop. 1.47]. But, the (square) on BC is equal to (sum of the squares) on BA and AC . For (that) was assumed. Thus, the square on DC is equal to the square on BC . So side DC is also equal to (side) BC . And since DA is equal to AB , and AC (is) common, the two (straight-lines) DA , AC are equal to the two (straight-lines) BA , AC . And the base DC is equal to the base BC . Thus, angle DAC [is] equal to angle BAC [Prop. 1.8]. But DAC is a right-angle. Thus, BAC is also a right-angle. Thus, if the square on one of the sides of a triangle is equal to the (sum of the) squares on the remaining two sides of the triangle (then) the angle contained by the remaining two sides of the triangle is a right-angle. (Which is) the very thing it was required to show.

† Here, use is made of the additional common notion that the squares of equal things are themselves equal. Later on, the inverse notion is used.

ELEMENTS BOOK 2
Fundamentals of Geometric Algebra

<Όροι.

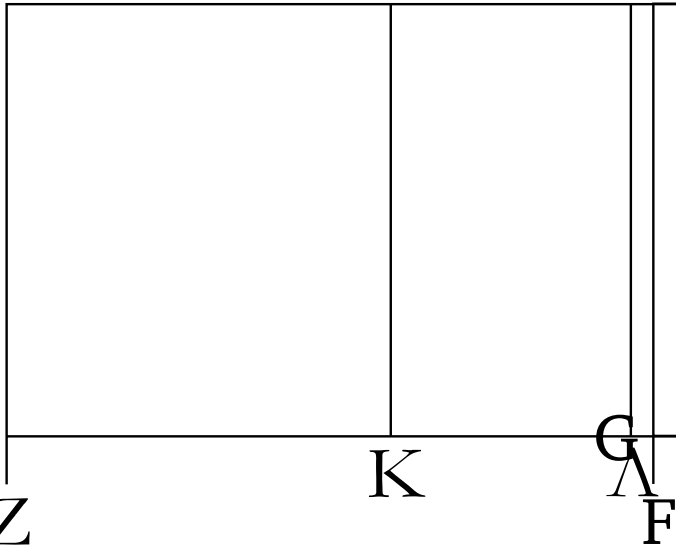
1. Πᾶν παραλληλόγραμμον ὀρθογώνιον περιέθεσθαι λέγεται ὑπὸ δύο τῶν τὴν ὀρθὴν γωνίαν περιεουσῶν εὐθειῶν.
2. Παντὸς δὲ παραλληλογράμμου θωρίου τῶν περὶ τὴν διάμετρον αὐτοῦ παραλληλογράμμων <ἐν ὁποιοῦν σὺν τοῖς δυσὶ παραπληρώμασι γνῶμων καλεῖσθω.

1.

Ἐὰν >ῶσι δύο εὐθεῖαι, τμηθῇ δὲ ἡ ἑτέρα αὐτῶν εἰς ὅσαδηποτοῦν τμήματα, τὸ περιεχόμενον ὀρθογώνιον ὑπὸ τῶν δύο εὐθειῶν >ῖσον ἐστὶ τοῖς ὑπὸ τε τῆς ἀτμήτου καὶ ἐκάστου τῶν τμημάτων περιεχομένοις ὀρθογωνίοις.

A —————

B Δ EB



>Ἐστωσαν δύο εὐθεῖαι αἱ A, BΓ, καὶ τετμήσθω ἡ BΓ, ὥς >έτυθεν, κατὰ τὰ Δ, E σημεία; λέγω, <ὅτι τὸ ὑπὸ τῶν A, BΓ περιεχομένον ὀρθογώνιον >ῖσον ἐστὶ τῶν ὑπὸ τῶν A, BΔ περιεχομένῳ ὀρθογωνίῳ καὶ τῶν ὑπὸ τῶν A, ΔE καὶ >έτι τῶν ὑπὸ τῶν A, EΓ.

>Ἡθῶ γὰρ ἀπὸ τοῦ B τῇ BΓ πρὸς ὀρθὰς ἡ BZ, καὶ κείσθω τῇ A >ῖση ἡ BH, καὶ διὰ μὲν τοῦ H τῇ BΓ παράλληλος >ῇθῶ ἡ HΘ, διὰ δὲ τῶν Δ, E, Γ τῇ BH παράλληλοι >ῇθῶσαν αἱ ΔK, EΛ, ΓΘ.

>Ἦσον δὲ ἐστὶ τὸ BΘ τοῖς BK, ΔΛ, EΘ. καὶ ἐστὶ τὸ μὲν BΘ τὸ ὑπὸ τῶν A, BΓ; περιέθεται μὲν γὰρ ὑπὸ τῶν HB, BΓ, >ῖση δὲ ἡ BH τῇ A; τὸ δὲ BK τὸ ὑπὸ τῶν A, BΔ; περιέθεται μὲν γὰρ ὑπὸ τῶν HB, BΔ, >ῖση δὲ ἡ BH τῇ A. τὸ δὲ ΔΛ τὸ ὑπὸ τῶν A, ΔE; >ῖση γὰρ ἡ ΔK, τουτέστιν ἡ BH, τῇ A. καὶ >έτι ὁμοίως τὸ EΘ τὸ ὑπὸ τῶν A, EΓ; τὸ >άρα ὑπὸ τῶν A, BΓ >ῖσον ἐστὶ τῶν ὑπὸ A, BΔ καὶ τῶν ὑπὸ A, ΔE καὶ >έτι τῶν ὑπὸ A, EΓ.

Ἐὰν >άρα >ῶσι δύο εὐθεῖαι, τμηθῇ δὲ ἡ ἑτέρα αὐτῶν εἰς ὅσαδηποτοῦν τμήματα, τὸ περιεχόμενον ὀρθογώνιον ὑπὸ τῶν δύο εὐθειῶν >ῖσον ἐστὶ τοῖς ὑπὸ τε τῆς ἀτμήτου καὶ ἐκάστου τῶν τμημάτων περιεχομένοις ὀρθογωνίοις; <ὅπερ >έδει δεῖξαι.

Definitions

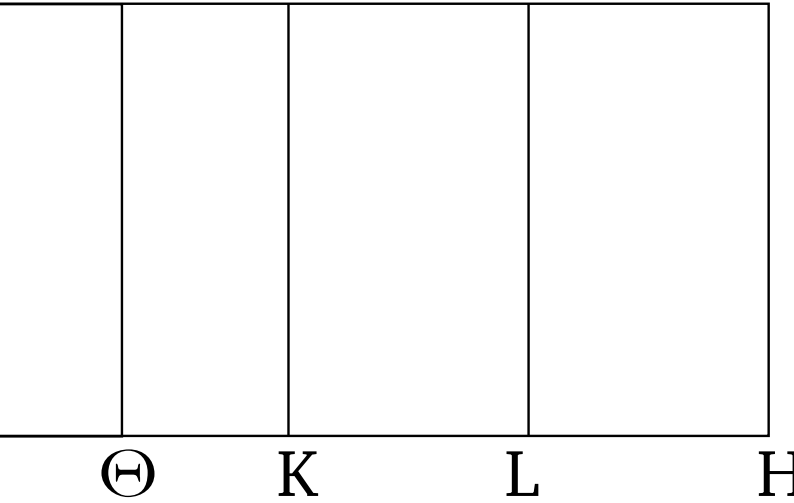
1. Any rectangular parallelogram is said to be contained by the two straight-lines containing the right-angle.
2. And in any parallelogrammic figure, let any one whatsoever of the parallelograms about its diagonal, (taken) with its two complements, be called a gnomon.

Proposition 1[†]

If there are two straight-lines, and one of them is cut into any number of pieces whatsoever, (then) the rectangle contained by the two straight-lines is equal to the (sum of the) rectangles contained by the uncut (straight-line), and every one of the pieces (of the cut straight-line).

A —————

A Γ D E C



Let A and BC be the two straight-lines, and let BC be cut, at random, at points D and E. I say that the rectangle contained by A and BC is equal to the rectangle(s) contained by A and BD, by A and DE, and, finally, by A and EC. For let BF be drawn from point B, at right-angles to BC [Prop. 1.11], and let BG be made equal to A [Prop. 1.3], and let GH be drawn through (point) G, parallel to BC [Prop. 1.31], and let DK, EL, and CH be drawn through (points) D, E, and C (respectively), parallel to BG [Prop. 1.31].

So the (rectangle) BH is equal to the (rectangles) BK, DL, and EH. And BH is the (rectangle contained) by A and BC. For it is contained by GB and BC, and BG (is) equal to A. And BK (is) the (rectangle contained) by A and BD. For it is contained by GB and BD, and BG (is) equal to A. And DL (is) the (rectangle contained) by A and DE. For DK, that is to say BG [Prop. 1.34], (is) equal to A. Similarly, EH (is) also the (rectangle contained) by A and EC. Thus, the (rectangle contained) by A and BC is equal to the (rectangles contained) by A and BD, by A and DE, and, finally, by A and EC.

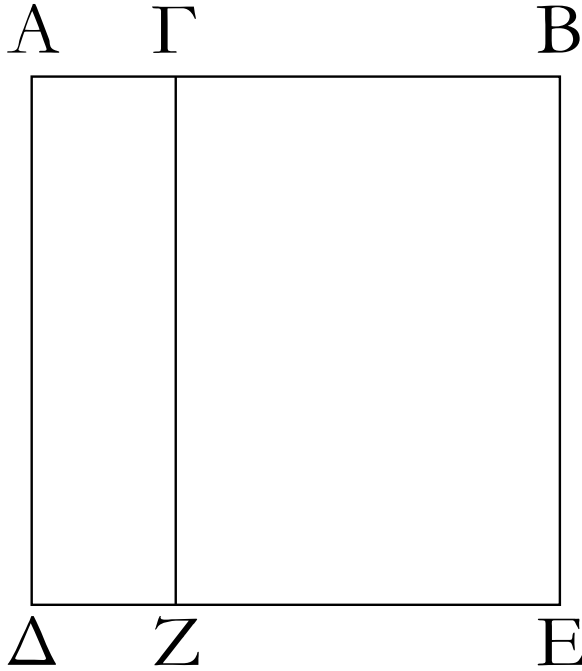
Thus, if there are two straight-lines, and one of them is cut into any number of pieces whatsoever, (then) the rectangle contained by the two straight-lines is equal to the (sum of

the) rectangles contained by the uncut (straight-line), and every one of the pieces (of the cut straight-line). (Which is) the very thing it was required to show.

† This proposition is a geometric version of the algebraic identity: $a(b + c + d + \dots) = ab + ac + ad + \dots$.

2.

Ἐάν εὐθεῖα γραμμὴ τμηθῇ, ὥς >έτυθεν, τὸ ὑπὸ τῆς <όλης καὶ ἑκατέρου τῶν τμημάτων περιεχόμενον ὀρθογώνιον >ῖσον ἐστὶ τῷ ἀπὸ τῆς <όλης τετραγώνῳ.



Εὐθεῖα γὰρ ἡ AB τετμήσθω, ὥς >έτυθεν, κατὰ τὸ Γ σημεῖον; λέγω, <ὅτι τὸ ὑπὸ τῶν AB, ΒΓ περιεχόμενον ὀρθογώνιον μετὰ τοῦ ὑπὸ BA, ΑΓ περιεχομένου ὀρθογωνίου >ῖσον ἐστὶ τῷ ἀπὸ τῆς AB τετραγώνῳ.

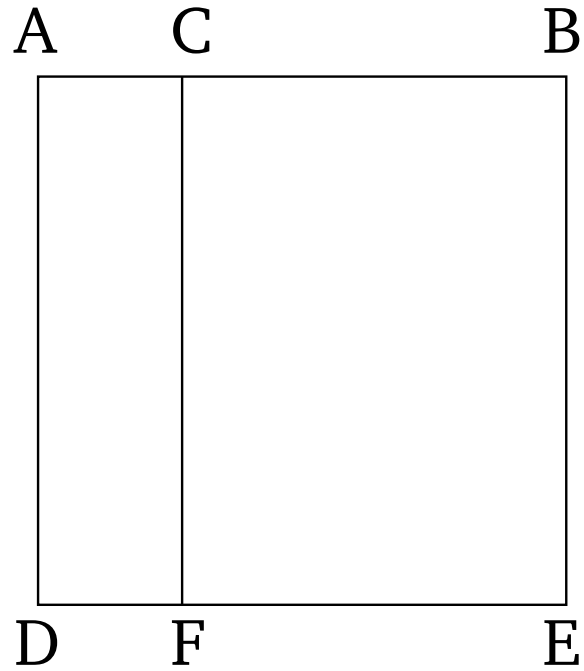
Ἀναγεγράφθω γὰρ ἀπὸ τῆς AB τετράγωνον τὸ ΑΔΕΒ, καὶ >ήθθω διὰ τοῦ Γ ὁποτέρῳ τῶν ΑΔ, ΒΕ παράλληλος ἡ ΓΖ.

>Ἴσον δὴ ἐστὶ τὸ ΑΕ τοῖς ΑΖ, ΓΕ. καὶ ἐστὶ τὸ μὲν ΑΕ τὸ ἀπὸ τῆς AB τετράγωνον, τὸ δὲ ΑΖ τὸ ὑπὸ τῶν BA, ΑΓ περιεχόμενον ὀρθογώνιον; περιέθεται μὲν γὰρ ὑπὸ τῶν ΔΑ, ΑΓ, >ῖση δὲ ἡ ΑΔ τῇ AB; τὸ δὲ ΓΕ τὸ ὑπὸ τῶν AB, ΒΓ; >ῖση γὰρ ἡ ΒΕ τῇ AB. τὸ >άρα ὑπὸ τῶν BA, ΑΓ μετὰ τοῦ ὑπὸ τῶν AB, ΒΓ >ῖσον ἐστὶ τῷ ἀπὸ τῆς AB τετραγώνῳ.

Ἐάν >άρα εὐθεῖα γραμμὴ τμηθῇ, ὥς >έτυθεν, τὸ ὑπὸ τῆς <όλης καὶ ἑκατέρου τῶν τμημάτων περιεχόμενον ὀρθογώνιον >ῖσον ἐστὶ τῷ ἀπὸ τῆς <όλης τετραγώνῳ; <όπερ >έδει δεῖχαι.

Proposition 2[†]

If a straight-line is cut at random, (then) the (sum of the) rectangle(s) contained by the whole (straight-line), and each of the pieces (of the straight-line), is equal to the square on the whole.



For let the straight-line AB be cut, at random, at point C . I say that the rectangle contained by AB and BC , plus the rectangle contained by BA and AC , is equal to the square on AB .

For let the square $ADEB$ be described on AB [Prop. 1.46], and let CF be drawn through C , parallel to either of AD or BE [Prop. 1.31].

So the (square) AE is equal to the (rectangles) AF and CE . And AE is the square on AB . And AF (is) the rectangle contained by the (straight-lines) BA and AC . For it is contained by DA and AC , and AD (is) equal to AB . And CE (is) the (rectangle contained) by AB and BC . For BE (is) equal to AB . Thus, the (rectangle contained) by BA and AC , plus the (rectangle contained) by AB and BC , is equal to the square on AB .

Thus, if a straight-line is cut at random, (then) the (sum of the) rectangle(s) contained by the whole (straight-line), and each of the pieces (of the straight-line), is equal to the square on the whole. (Which is) the very thing it was required to show.

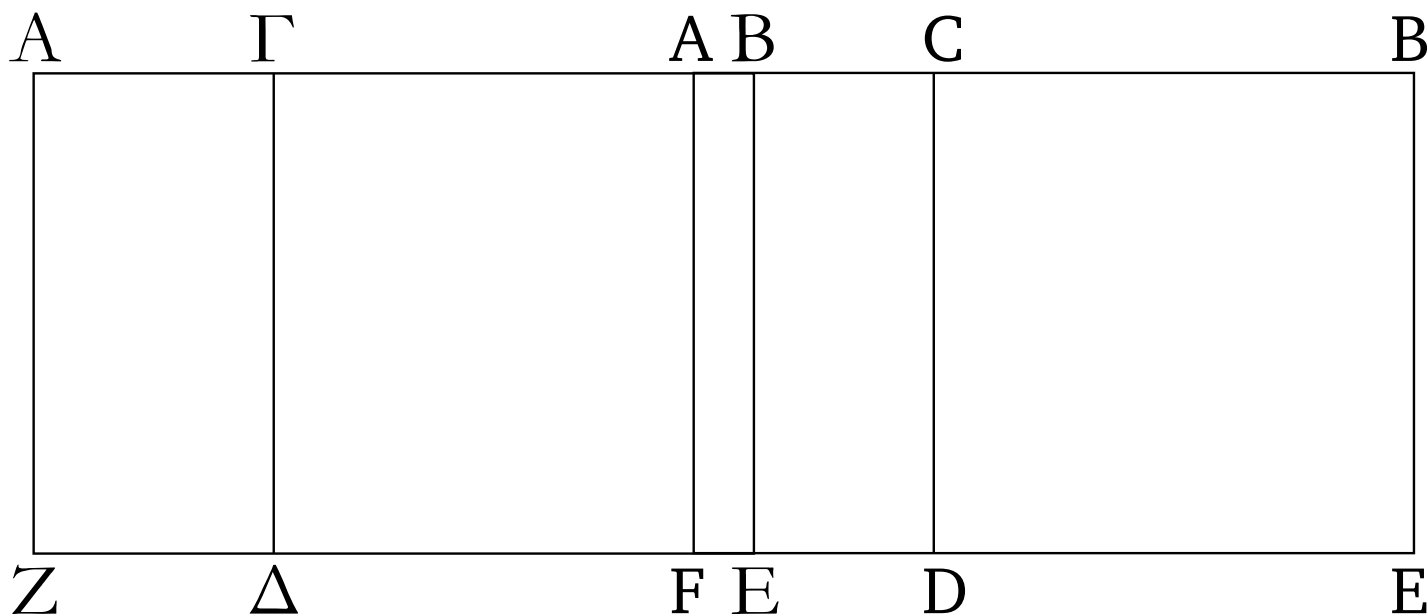
† This proposition is a geometric version of the algebraic identity: $ab + ac = a^2$ if $a = b + c$.

3.

Ἐάν εὐθεῖα γραμμὴ τμηθῇ, ὥς >έτυθεν, τὸ ὑπὸ τῆς <όλης καὶ ἐνὸς τῶν τμημάτων περιεχόμενον ὀρθογώνιον >ῖσον ἐστὶ τῷ τε ὑπὸ τῶν τμημάτων περιεχομένῳ ὀρθογωνίῳ καὶ τῷ ἀπὸ τοῦ προειρημένου τμήματος τετραγώνῳ.

Proposition 3[†]

If a straight-line is cut at random, (then) the rectangle contained by the whole (straight-line), and one of the pieces (of the straight-line), is equal to the rectangle contained by (both of) the pieces, and the square on the aforementioned piece.



Εὐθεῖα γὰρ ἡ ΑΒ τετμήσθω, ὥς >έτυθεν, κατὰ τὸ Γ; λέγω, <ὅτι τὸ ὑπὸ τῶν ΑΒ, ΒΓ περιεχόμενον ὀρθογώνιον >ῖσον ἐστὶ τῷ τε ὑπὸ τῶν ΑΓ, ΓΒ περιεχομένῳ ὀρθογώνιῳ μετὰ τοῦ ἀπὸ τῆς ΒΓ τετραγώνου.

Ἀναγεγράφθω γὰρ ἀπὸ τῆς ΓΒ τετράγωνον τὸ ΓΔΕΒ, καὶ διήθθω ἡ ΕΔ ἐπὶ τὸ Ζ, καὶ διὰ τοῦ Α ὁποτέρῃ τῶν ΓΔ, ΒΕ παράλληλος >ήθθω ἡ ΑΖ. >ῖσον δὴ ἐστὶ τὸ ΑΕ τοῖς ΑΔ, ΓΕ; καὶ ἐστὶ τὸ μὲν ΑΕ τὸ ὑπὸ τῶν ΑΒ, ΒΓ περιεχόμενον ὀρθογώνιον; περιέθεται μὲν γὰρ ὑπὸ τῶν ΑΒ, ΒΕ, >ῖση δὲ ἡ ΒΕ τῇ ΒΓ; τὸ δὲ ΑΔ τὸ ὑπὸ τῶν ΑΓ, ΓΒ; >ῖση γὰρ ἡ ΔΓ τῇ ΓΒ; τὸ δὲ ΔΒ τὸ ἀπὸ τῆς ΓΒ τετράγωνον; τὸ >άρα ὑπὸ τῶν ΑΒ, ΒΓ περιεχόμενον ὀρθογώνιον >ῖσον ἐστὶ τῷ ὑπὸ τῶν ΑΓ, ΓΒ περιεχομένῳ ὀρθογώνιῳ μετὰ τοῦ ἀπὸ τῆς ΒΓ τετραγώνου.

Ἐὰν >άρα εὐθεῖα γραμμὴ τμηθῇ, ὥς >έτυθεν, τὸ ὑπὸ τῆς <όλης καὶ ἐνδὲς τῶν τμημάτων περιεχόμενον ὀρθογώνιον >ῖσον ἐστὶ τῷ τε ὑπὸ τῶν τμημάτων περιεχομένῳ ὀρθογώνιῳ καὶ τῷ ἀπὸ τοῦ προειρημένου τμήματος τετραγώνῳ; <όπερ >έδει δεῖχαι.

For let the straight-line AB be cut, at random, at (point) C . I say that the rectangle contained by AB and BC is equal to the rectangle contained by AC and CB , plus the square on BC .

For let the square $CDEB$ be described on CB [Prop. 1.46], and let ED be drawn through to F , and let AF be drawn through A , parallel to either of CD or BE [Prop. 1.31]. So the (rectangle) AE is equal to the (rectangle) AD and the (square) CE . And AE is the rectangle contained by AB and BC . For it is contained by AB and BE , and BE (is) equal to BC . And AD (is) the (rectangle contained) by AC and CB . For DC (is) equal to CB . And DB (is) the square on CB . Thus, the rectangle contained by AB and BC is equal to the rectangle contained by AC and CB , plus the square on BC .

Thus, if a straight-line is cut at random, (then) the rectangle contained by the whole (straight-line), and one of the pieces (of the straight-line), is equal to the rectangle contained by (both of) the pieces, and the square on the aforementioned piece. (Which is) the very thing it was required to show.

† This proposition is a geometric version of the algebraic identity: $(a + b)a = ab + a^2$.

4.

Ἐὰν εὐθεῖα γραμμὴ τμηθῇ, ὥς >έτυθεν, τὸ ἀπὸ τῆς <όλης τετράγωνον >ῖσον ἐστὶ τοῖς τε ἀπὸ τῶν τμημάτων τετραγώνοις καὶ τῷ δι᾽ ὑπὸ τῶν τμημάτων περιεχομένῳ ὀρθογώνιῳ.

Εὐθεῖα γὰρ γραμμὴ ἡ ΑΒ τετμήσθω, ὥς >έτυθεν, κατὰ τὸ Γ. λέγω, <ὅτι τὸ ἀπὸ τῆς ΑΒ τετράγωνον >ῖσον ἐστὶ τοῖς τε ἀπὸ τῶν ΑΓ, ΓΒ τετραγώνοις καὶ τῷ δι᾽ ὑπὸ τῶν ΑΓ, ΓΒ περιεχομένῳ ὀρθογώνιῳ.

Ἀναγεγράφθω γὰρ ἀπὸ τῆς ΑΒ τετράγωνον τὸ ΑΔΕΒ, καὶ ἐπεζεύθθω ἡ ΒΔ, καὶ διὰ μὲν τοῦ Γ ὁποτέρῃ τῶν ΑΔ, ΕΒ παράλληλος >ήθθω ἡ ΓΖ, διὰ δὲ τοῦ Η ὁποτέρῃ τῶν ΑΒ, ΔΕ παράλληλος >ήθθω ἡ ΘΚ. καὶ ἐπεὶ παράλληλός ἐστιν ἡ ΓΖ τῇ ΑΔ, καὶ εἰς αὐτὰς ἐμπίπτωκεν ἡ ΒΔ, ἡ ἐκτὸς γωνία ἡ ὑπὸ ΓΗΒ >ῖση ἐστὶ τῇ ἐντὸς καὶ ἀπεναντίον τῇ ὑπὸ ΑΔΒ. ἄλλ> ἡ ὑπὸ ΑΔΒ τῇ ὑπὸ ΑΒΔ ἐστὶν >ῖση, ἐπεὶ καὶ πλευρὰ ἡ ΒΑ τῇ ΑΔ ἐστὶν >ῖση; καὶ ἡ ὑπὸ ΓΗΒ >άρα γωνία τῇ ὑπὸ ΗΒΓ ἐστὶν >ῖση; <ώστε καὶ πλευρὰ ἡ ΒΓ πλευρᾷ τῇ ΓΗ ἐστὶν >ῖση; ἄλλ> ἡ μὲν ΓΒ τῇ ΗΚ ἐστὶν >ῖση, ἡ δὲ ΓΗ τῇ

Proposition 4†

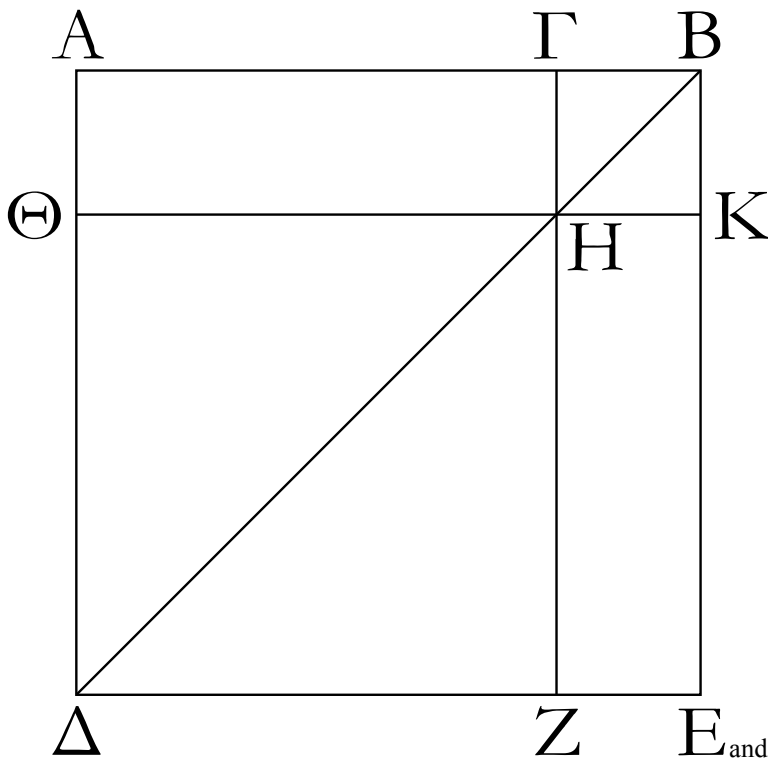
If a straight-line is cut at random, (then) the square on the whole (straight-line) is equal to the (sum of the) squares on the pieces (of the straight-line), and twice the rectangle contained by the pieces.

For let the straight-line AB be cut, at random, at (point) C . I say that the square on AB is equal to the (sum of the) squares on AC and CB , and twice the rectangle contained by AC and CB .

For let the square $ADEB$ be described on AB [Prop. 1.46], and let BD be joined, and let CF be drawn through C , parallel to either of AD or EB [Prop. 1.31], and let HK be drawn through G , parallel to either of AB or DE [Prop. 1.31]. And since CF is parallel to AD , and BD has fallen across them, the external angle CGB is equal to the internal and opposite (angle) ADB [Prop. 1.29]. But, ADB is equal to ABD , since the side BA is also equal to AD [Prop. 1.5]. Thus, angle CGB is also equal to GBC .

KB; καὶ ἡ HK > ἄρα τῇ KB ἐστὶν > ἴση; ἰσόπλευρον > ἄρα ἐστὶ τὸ ΓΗΚΒ. λέγω δὴ, < ὅτι καὶ ὀρθογώνιον. ἐπεὶ γὰρ παράλληλός ἐστιν ἡ ΓΗ τῇ BK [καὶ εἰς αὐτὰς ἐμπίπτωκεν εὐθεῖα ἡ ΓΒ], αἱ > ἄρα ὑπὸ ΚΒΓ, ΗΓΒ γωνίαι δύο ὀρθαῖς εἰσιν > ἴσαι. ὀρθὴ δὲ ἡ ὑπὸ ΚΒΓ; ὀρθὴ > ἄρα καὶ ἡ ὑπὸ ΒΓΗ; < ὥστε καὶ αἱ ἀπεναντίον αἱ ὑπὸ ΓΗΚ, ΗΚΒ ὀρθαὶ εἰσιν. ὀρθογώνιον > ἄρα ἐστὶ τὸ ΓΗΚΒ; ἐδείκθη δὲ καὶ ἰσόπλευρον; τετράγωνον > ἄρα ἐστίν; καὶ ἐστὶν ἀπὸ τῆς ΓΒ. διὰ τὰ αὐτὰ δὴ καὶ τὸ ΘΖ τετράγωνόν ἐστιν; καὶ ἐστὶν ἀπὸ τῆς ΘΗ, τουτέστιν [ἀπὸ] τῆς ΑΓ; τὰ > ἄρα ΘΖ, ΚΓ τετράγωνα ἀπὸ τῶν ΑΓ, ΓΒ εἰσιν. καὶ ἐπεὶ > ἴσον ἐστὶ τὸ ΑΗ τῷ ΗΕ, καὶ ἐστὶ τὸ ΑΗ τὸ ὑπὸ τῶν ΑΓ, ΓΒ; > ἴση γὰρ ἡ ΗΓ τῇ ΓΒ; καὶ τὸ ΗΕ > ἄρα > ἴσον ἐστὶ τῷ ὑπὸ ΑΓ, ΓΒ; τὰ > ἄρα ΑΗ, ΗΕ > ἴσα ἐστὶ τῷ διζ ὑπὸ τῶν ΑΓ, ΓΒ. > ἐστὶ δὲ καὶ τὰ ΘΖ, ΓΚ τετράγωνα ἀπὸ τῶν ΑΓ, ΓΒ; τὰ > ἄρα τέσσαρα τὰ ΘΖ, ΓΚ, ΑΗ, ΗΕ > ἴσα ἐστὶ τοῖς τε ἀπὸ τῶν ΑΓ, ΓΒ τετραγώνοις καὶ τῷ διζ ὑπὸ τῶν ΑΓ, ΓΒ περιεθρομένῳ ὀρθογώνιῳ. ἀλλὰ τὰ ΘΖ, ΓΚ, ΑΗ, ΗΕ < ὅλον ἐστὶ τὸ ΑΔΕΒ, < ὅ ἐστιν ἀπὸ τῆς ΑΒ τετράγωνον; τὸ > ἄρα ἀπὸ τῆς ΑΒ τετράγωνον > ἴσον ἐστὶ τοῖς τε ἀπὸ τῶν ΑΓ, ΓΒ τετραγώνοις καὶ τῷ διζ ὑπὸ τῶν ΑΓ, ΓΒ περιεθρομένῳ ὀρθογώνιῳ.

So the side BC is equal to the side CG [Prop. 1.6]. But, CB is equal to GK , and CG to KB [Prop. 1.34]. Thus, GK is also equal to KB . Thus, $CGKB$ is equilateral. So I say that (it is) also right-angled. For since CG is parallel to BK [and the straight-line CB has fallen across them], the angles KBC and GCB are thus equal to two right-angles [Prop. 1.29]. But KBC (is) a right-angle. Thus, BCG (is) also a right-angle. So the opposite (angles) CGK and GKB are also right-angles [Prop. 1.34]. Thus, $CGKB$ is right-angled. And it was also shown (to be) equilateral. Thus, it is a square. And it is on CB . So, for the same (reasons), HF is also a square. And it is on HG , that is to say [on] AC [Prop. 1.34]. Thus, the squares HF and KC are on AC and CB (respectively). And the (rectangle) AG is equal to the (rectangle) GE [Prop. 1.43]. And AG is the (rectangle contained) by AC and CB . For GC (is) equal to CB . Thus, GE is also equal to the (rectangle contained) by AC and CB . Thus, the (rectangles) AG and GE are equal to twice the (rectangle contained) by AC and CB . And HF and CK are the squares on AC and CB (respectively). Thus, the four (figures) HF , CK , AG , and GE are equal to the (sum of the) squares on AC and BC , and twice the rectangle contained by AC and CB . But, the (figures) HF , CK , AG , and GE are (equivalent to) the whole of $ADEB$, which is the square on AB . Thus, the square on AB is equal to the (sum of the) squares on AC and CB ,



and twice the rectangle contained by AC and CB .

Ἀναγεγράφθω γὰρ ἀπὸ τῆς ΓΒ τετράγωνον τὸ ΓΕΖΒ, καὶ ἐπεζεύθθω ἡ ΒΕ, καὶ διὰ μὲν τοῦ Δ ὁποτέρᾳ τῶν ΓΕ, ΒΖ παράλληλος >ήθθω ἡ ΔΗ, διὰ δὲ τοῦ Θ ὁποτέρᾳ τῶν ΑΒ, ΕΖ παράλληλος πάλιν >ήθθω ἡ ΚΜ, καὶ πάλιν διὰ τοῦ Α ὁποτέρᾳ τῶν ΓΛ, ΒΜ παράλληλος >ήθθω ἡ ΑΚ. καὶ ἐπεὶ >ίσον ἐστὶ τὸ ΓΘ παραπλήρωμα τῶ| ΘΖ παραπληρώματι, κοινὸν προσκείσθω τὸ ΔΜ; <όλον >άρα τὸ ΓΜ <όλω τῶ| ΔΖ >ίσον ἐστίν. ἀλλὰ τὸ ΓΜ τῶ| ΑΛ >ίσον ἐστίν, ἐπεὶ καὶ ἡ ΑΓ τῇ| ΓΒ ἐστίν >ίση; καὶ τὸ ΑΛ >άρα τῶ| ΔΖ >ίσον ἐστίν. κοινὸν προσκείσθω τὸ ΓΘ; <όλον >άρα τὸ ΑΘ τῶ| ΜΝΧ[†] γνῶμονι >ίσον ἐστίν. ἀλλὰ τὸ ΑΘ τὸ ὑπὸ τῶν ΑΔ, ΔΒ ἐστίν; >ίση γὰρ ἡ ΔΘ τῇ| ΔΒ; καὶ ὁ ΜΝΧ >άρα γνῶμων >ίσοξ ἐστὶ τῶ| ὑπὸ ΑΔ, ΔΒ. κοινὸν προσκείσθω τὸ ΛΗ, <ὅ ἐστίν >ίσον τῶ| ἀπὸ τῆς ΓΔ; ὁ >άρα ΜΝΧ γνῶμων καὶ τὸ ΛΗ >ίσα ἐστὶ τῶ| ὑπὸ τῶν ΑΔ, ΔΒ περιεχομένῳ ὀρθογώνιῳ καὶ τῶ| ἀπὸ τῆς ΓΔ τετραγώνῳ. ἀλλὰ ὁ ΜΝΧ γνῶμων καὶ τὸ ΛΗ <όλον ἐστὶ τὸ ΓΕΖΒ τετράγωνον, <ὅ ἐστίν ἀπὸ τῆς ΓΒ; τὸ >άρα ὑπὸ τῶν ΑΔ, ΔΒ περιεχόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τῆς ΓΔ τετραγώνου >ίσον ἐστὶ τῶ| ἀπὸ τῆς ΓΒ τετραγώνῳ. Ἐὰν >άρα εὐθεῖα γραμμὴ τμηθῇ| εἰς >ίσα καὶ >άνισα, τὸ ὑπὸ τῶν ἀνίσων τῆς <όλης τμημάτων περιεχόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τῆς μεταχὺ τῶν τομῶν τετραγώνου >ίσον ἐστὶ τῶ| ἀπὸ τῆς ἡμισείας τετραγώνῳ. <όπερ >έδει δεῖξαι.

and let BE be joined, and let DG be drawn through D , parallel to either of CE or BF [Prop. 1.31], and again let KM be drawn through H , parallel to either of AB or EF [Prop. 1.31], and again let AK be drawn through A , parallel to either of CL or BM [Prop. 1.31]. And since the complement CH is equal to the complement HF [Prop. 1.43], let the (square) DM be added to both. Thus, the whole (rectangle) CM is equal to the whole (rectangle) DF . But, (rectangle) CM is equal to (rectangle) AL , since AC is also equal to CB [Prop. 1.36]. Thus, (rectangle) AL is also equal to (rectangle) DF . Let (rectangle) CH be added to both. Thus, the whole (rectangle) AH is equal to the gnomon NOP . But, AH is the (rectangle contained) by AD and DB . For DH (is) equal to DB . Thus, the gnomon NOP is also equal to the (rectangle contained) by AD and DB . Let LG , which is equal to the (square) on CD , be added to both. Thus, the gnomon NOP and the (square) LG are equal to the rectangle contained by AD and DB , and the square on CD . But, the gnomon NOP and the (square) LG is (equivalent to) the whole square $CEFB$, which is on CB . Thus, the rectangle contained by AD and DB , plus the square on CD , is equal to the square on CB .

Thus, if a straight-line is cut into equal and unequal (pieces, then) the rectangle contained by the unequal pieces of the whole (straight-line), plus the square on the (difference) between the (equal and unequal) pieces, is equal to the square on half (of the straight-line). (Which is) the very thing it was required to show.

[†] Note the (presumably mistaken) double use of the label M in the Greek text.

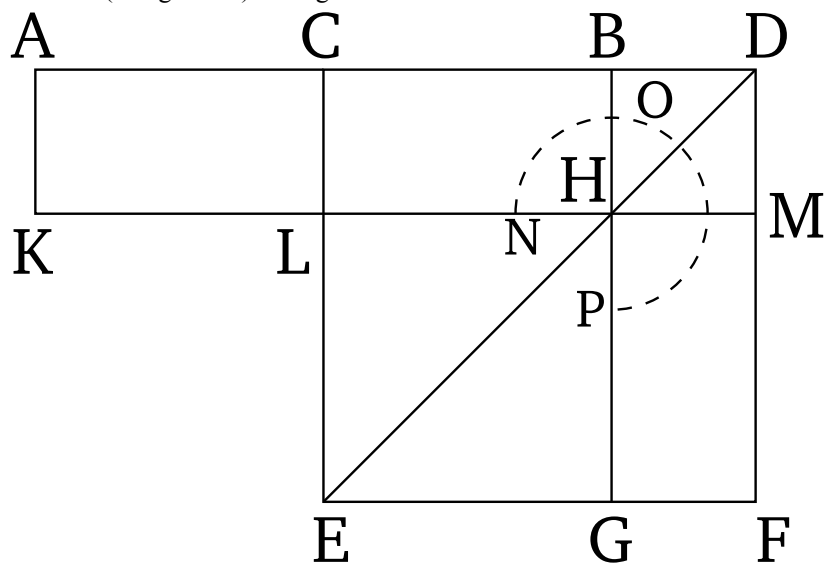
[‡] This proposition is a geometric version of the algebraic identity: $a b + [(a + b)/2 - b]^2 = [(a + b)/2]^2$.

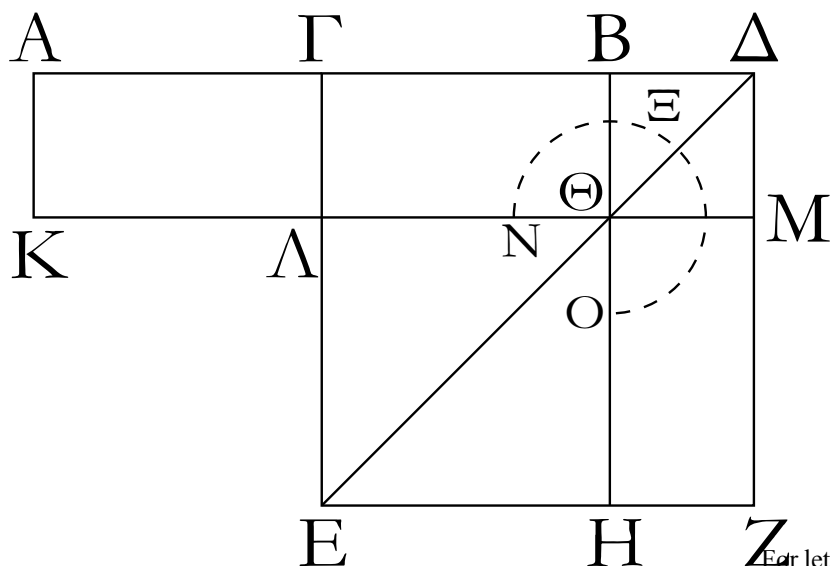
6.

Proposition 6[†]

Ἐὰν εὐθεῖα γραμμὴ τμηθῇ| δίθῃ, προστεθῇ| δὲ τιξ αὐτῇ| εὐθεῖα ἐπ> εὐθείᾳ, τὸ ὑπὸ τῆς <όλης σὺν τῇ| προσκειμένῃ καὶ τῆς προσκειμένης περιεχόμενον ὀρθόγώνιον μετὰ τοῦ ἀπὸ τῆς ἡμισείας τετραγώνου >ίσον ἐστὶ τῶ| ἀπὸ τῆς συγκεκριμένης >έκ τε τῆς ἡμισείας καὶ τῆς προσκειμένης τετραγώνῳ.

If a straight-line is cut in half, and any straight-line added to it straight-on, (then) the rectangle contained by the whole (straight-line) with the (straight-line) having being added, and the (straight-line) having being added, plus the square on half (of the original straight-line), is equal to the square on the sum of half (of the original straight-line) and the (straight-line) having been added.





Εὐθεῖα γάρ τις ἡ AB τετμήσθω δίθῃ κατὰ τὸ Γ σημείον, προσκείσθω δὲ τις αὐτῇ εὐθεῖα ἐπ' εὐθείᾳς ἡ BD ; λέγω, <ὅτι τὸ ὑπὸ τῶν AD , DB περιεχόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τῆς GB τετραγώνου >ῖσον ἐστὶ τῷ ἀπὸ τῆς GA τετραγώνῳ.

Ἀναγεγράφθω γὰρ ἀπὸ τῆς GA τετράγωνον τὸ $GEZD$, καὶ ἐπέξευθθῶ ἡ DE , καὶ διὰ μὲν τοῦ B σημείου ὁποτέρᾳ τῶν EG , DZ παράλληλοξ >ήθθῳ ἡ BH , διὰ δὲ τοῦ Θ σημείου ὁποτέρᾳ τῶν AB , EZ παράλληλοξ >ήθθῳ ἡ KM , καὶ >έτι διὰ τοῦ A ὁποτέρᾳ τῶν GA , DM παράλληλοξ >ήθθῳ ἡ AK . Ἐπεὶ οὖν >ῖση ἐστὶν ἡ AG τῇ GB , >ῖσον ἐστὶ καὶ τὸ AA τῷ $\Gamma\Theta$. ἀλλὰ τὸ $\Gamma\Theta$ τῷ ΘZ >ῖσον ἐστίν. καὶ τὸ AA >άρα τῷ ΘZ ἐστὶν >ῖσον. κοινὸν προσκείσθω τὸ GM ; <όλον >άρα τὸ AM τῷ NXO γνώμονι ἐστὶν >ῖσον. ἀλλὰ τὸ AM ἐστὶ τὸ ὑπὸ τῶν AD , DB ; >ῖση γάρ ἐστὶν ἡ DM τῇ DB ; καὶ ὁ NXO >άρα γνώμων >ῖσοξ ἐστὶ τῷ ὑπὸ τῶν AD , DB [περιεχομένῳ ὀρθογώνιῳ]. κοινὸν προσκείσθω τὸ LH , <ό ἐστὶν >ῖσον τῷ ἀπὸ τῆς BF τετραγώνῳ; τὸ >άρα ὑπὸ τῶν AD , DB περιεχόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τῆς GB τετραγώνου >ῖσον ἐστὶ τῷ NXO γνώμονι καὶ τῷ LH . ἀλλὰ ὁ NXO γνώμων καὶ τὸ LH <όλον ἐστὶ τὸ $GEZD$ τετράγωνον, <ό ἐστὶν ἀπὸ τῆς GA ; τὸ >άρα ὑπὸ τῶν AD , DB περιεχόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τῆς GB τετραγώνου >ῖσον ἐστὶ τῷ ἀπὸ τῆς GA τετραγώνῳ.

Ἐὰν >άρα εὐθεῖα γραμμὴ τμηθῇ, δίθῃ, προστεθῇ δὲ τις αὐτῇ εὐθεῖα ἐπ' εὐθείᾳς, τὸ ὑπὸ τῆς <όληξ συν τῇ προσκειμένῃ καὶ τῆς προσκειμένης περιεχόμενον ὀρθόγώνιον μετὰ τοῦ ἀπὸ τῆς ἡμισείας τετραγώνου >ῖσον ἐστὶ τῷ ἀπὸ τῆς συγκεκριμένης >έκ τε τῆς ἡμισείας καὶ τῆς προσκειμένης τετραγώνῳ; <όπερ >έδει δεῖχαι.

For let any straight-line AB be cut in half at point C , and let any straight-line BD be added to it straight-on. I say that the rectangle contained by AD and DB , plus the square on CB , is equal to the square on CD .

For let the square $CEFD$ be described on CD [Prop. 1.46], and let DE be joined, and let BG be drawn through point B , parallel to either of EC or DF [Prop. 1.31], and let KM be drawn through point H , parallel to either of AB or EF [Prop. 1.31], and finally let AK be drawn through A , parallel to either of CL or DM [Prop. 1.31].

Therefore, since AC is equal to CB , (rectangle) AL is also equal to (rectangle) CH [Prop. 1.36]. But, (rectangle) CH is equal to (rectangle) HF [Prop. 1.43]. Thus, (rectangle) AL is also equal to (rectangle) HF . Let (rectangle) CM be added to both. Thus, the whole (rectangle) AM is equal to the gnomon NOP . But, AM is the (rectangle contained) by AD and DB . For DM is equal to DB . Thus, gnomon NOP is also equal to the [rectangle contained] by AD and DB . Let LG , which is equal to the square on BC , be added to both. Thus, the rectangle contained by AD and DB , plus the square on CB , is equal to the gnomon NOP and the (square) LG . But the gnomon NOP and the (square) LG is (equivalent to) the whole square $CEFD$, which is on CD . Thus, the rectangle contained by AD and DB , plus the square on CB , is equal to the square on CD .

Thus, if a straight-line is cut in half, and any straight-line added to it straight-on, (then) the rectangle contained by the whole (straight-line) with the (straight-line) having being added, and the (straight-line) having being added, plus the square on half (of the original straight-line), is equal to the square on the sum of half (of the original straight-line) and the (straight-line) having been added. (Which is) the very thing it was required to show.

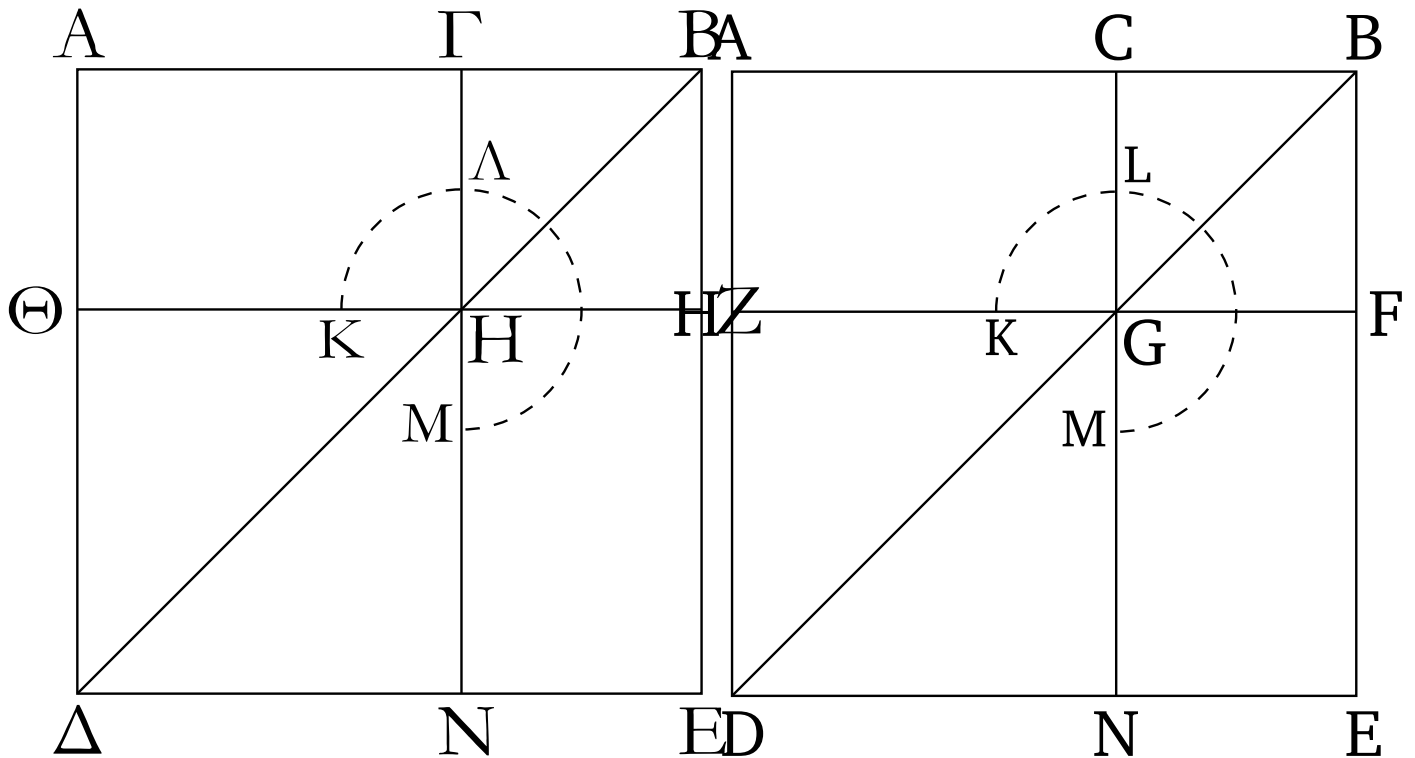
† This proposition is a geometric version of the algebraic identity: $(2a + b)b + a^2 = (a + b)^2$.

7.

Ἐὰν εὐθεῖα γραμμὴ τμηθῇ, ὥς >έτυθεν, τὸ ἀπὸ τῆς <όληξ καὶ τὸ ἀφ' ἐνὸς τῶν τμημάτων τὰ συναμφοτέρα τετράγωνα >ῖσα ἐστὶ τῷ τε διξ ὑπὸ τῆς <όληξ καὶ τοῦ εἰρημένου τμήματος περιεχομένῳ ὀρθογώνιῳ καὶ τῷ ἀπὸ τοῦ λοιποῦ τμήματος τετραγώνῳ.

Proposition 7†

If a straight-line is cut at random, (then) the sum of the squares on the whole (straight-line), and one of the pieces (of the straight-line), is equal to twice the rectangle contained by the whole, and the said piece, and the square on the remaining piece.



Εὐθεῖα γάρ τις ἡ AB τεμήσθω, ὥς >έτυθεν, κατὰ τὸ Γ σημείον; λέγω, <ὅτι τὰ ἀπὸ τῶν AB , $B\Gamma$ τετράγωνα >ῖσα ἐστὶ τῷ τε διζ ὑπὸ τῶν AB , $B\Gamma$ περιεθρομένῳ ὀρθογωνίῳ καὶ τῷ ἀπὸ τῆς ΓA τετραγώνῳ.

Ἀναγεγράφθω γὰρ ἀπὸ τῆς AB τετράγωνον τὸ $ADEB$; καὶ καταγεγράφθω τὸ σῆμα.

Ἐπεὶ ο>ὖν >ῖσον ἐστὶ τὸ AH τῷ HE , κοινὸν προσκείσθω τὸ ΓZ ; <ὅλον >ᾶρα τὸ AZ <ὀλῳ τῷ ΓE >ῖσον ἐστίν; τὰ >ᾶρα AZ , ΓE διπλάσιά ἐστι τοῦ AZ . ἀλλὰ τὰ AZ , ΓE ὁ KAM ἐστὶ γνῶμων καὶ τὸ ΓZ τετράγωνον; ὁ KAM >ᾶρα γνῶμων καὶ τὸ ΓZ διπλάσιά ἐστι τοῦ AZ . >ἐστὶ δὲ τοῦ AZ διπλάσιον καὶ τὸ διζ ὑπὸ τῶν AB , $B\Gamma$; >ῖση γὰρ ἡ BZ τῇ $B\Gamma$; ὁ >ᾶρα KAM γνῶμων καὶ τὸ ΓZ τετράγωνον >ῖσον ἐστὶ τῷ διζ ὑπὸ τῶν AB , $B\Gamma$. κοινὸν προσκείσθω τὸ ΔH , <ὁ ἐστὶν ἀπὸ τῆς $A\Gamma$ τετράγωνον; ὁ >ᾶρα KAM γνῶμων καὶ τὰ BH , $H\Delta$ τετράγωνα >ῖσα ἐστὶ τῷ τε διζ ὑπὸ τῶν AB , $B\Gamma$ περιεθρομένῳ ὀρθογωνίῳ καὶ τῷ ἀπὸ τῆς $A\Gamma$ τετραγώνῳ. ἀλλὰ ὁ KAM γνῶμων καὶ τὰ BH , $H\Delta$ τετράγωνα <ὅλον ἐστὶ τὸ $ADEB$ καὶ τὸ ΓZ , <ᾶ ἐστὶν ἀπὸ τῶν AB , $B\Gamma$ τετράγωνα; τὰ >ᾶρα ἀπὸ τῶν AB , $B\Gamma$ τετράγωνα >ῖσα ἐστὶ τῷ [τε] διζ ὑπὸ τῶν AB , $B\Gamma$ περιεθρομένῳ ὀρθογωνίῳ μετὰ τοῦ ἀπὸ τῆς $A\Gamma$ τετραγώνου.

Ἐὰν >ᾶρα εὐθεῖα γραμμὴ τμηθῇ, ὥς >έτυθεν, τὸ ἀπὸ τῆς <ὀληξ καὶ τὸ ᾀφ> ἐνδὲς τῶν τμημάτων τὰ συναμφότερα τετράγωνα >ῖσα ἐστὶ τῷ τε διζ ὑπὸ τῆς <ὀληξ καὶ τοῦ εἰρημένου τμήματος περιεθρομένῳ ὀρθογωνίῳ καὶ τῷ ἀπὸ τοῦ λοιποῦ τμήματος τετραγώνῳ; <ὅπερ >έδει δεῖχαι.

For let any straight-line AB be cut, at random, at point C . I say that the (sum of the) squares on AB and BC is equal to twice the rectangle contained by AB and BC , and the square on CA .

For let the square $ADEB$ be described on AB [Prop. 1.46], and let the (rest of) the figure be drawn.

Therefore, since (rectangle) AG is equal to (rectangle) GE [Prop. 1.43], let the (square) CF be added to both. Thus, the whole (rectangle) AF is equal to the whole (rectangle) CE . Thus, (rectangle) AF plus (rectangle) CE is double (rectangle) AF . But, (rectangle) AF plus (rectangle) CE is the gnomon KLM , and the square CF . Thus, the gnomon KLM , and the square CF , is double the (rectangle) AF . But double the (rectangle) AF is also twice the (rectangle contained) by AB and BC . For BF (is) equal to BC . Thus, the gnomon KLM , and the square CF , are equal to twice the (rectangle contained) by AB and BC . Let DG , which is the square on AC , be added to both. Thus, the gnomon KLM , and the squares BG and GD , are equal to twice the rectangle contained by AB and BC , and the square on AC . But, the gnomon KLM and the squares BG and GD is (equivalent to) the whole of $ADEB$ and CF , which are the squares on AB and BC (respectively). Thus, the (sum of the) squares on AB and BC is equal to twice the rectangle contained by AB and BC , and the square on AC .

Thus, if a straight-line is cut at random, (then) the sum of the squares on the whole (straight-line), and one of the pieces (of the straight-line), is equal to twice the rectangle contained by the whole, and the said piece, and the square on the remaining piece. (Which is) the very thing it was required to show.

† This proposition is a geometric version of the algebraic identity: $(a + b)^2 + a^2 = 2(a + b)a + b^2$.

8.

Ἐὰν εὐθεῖα γραμμὴ τμηθῇ, ὥς >έτυθεν, τὸ τετράκις ὑπὸ τῆς <ὀληξ καὶ ἐνδὲς τῶν τμημάτων περιεχόμενον ὀρθογώνιον

Proposition 8†

If a straight-line is cut at random, (then) four times the rectangle contained by the whole (straight-line), and one of the

μετὰ τοῦ ἀπὸ τοῦ λοιποῦ τμήματος τετραγώνου ὅσον ἐστὶ τῷ ἀπὸ τε τῆς ὅλης καὶ τοῦ εἰρημένου τμήματος ὥς ἀπὸ μιᾶς ἀναγραφέντι τετραγώνῳ.

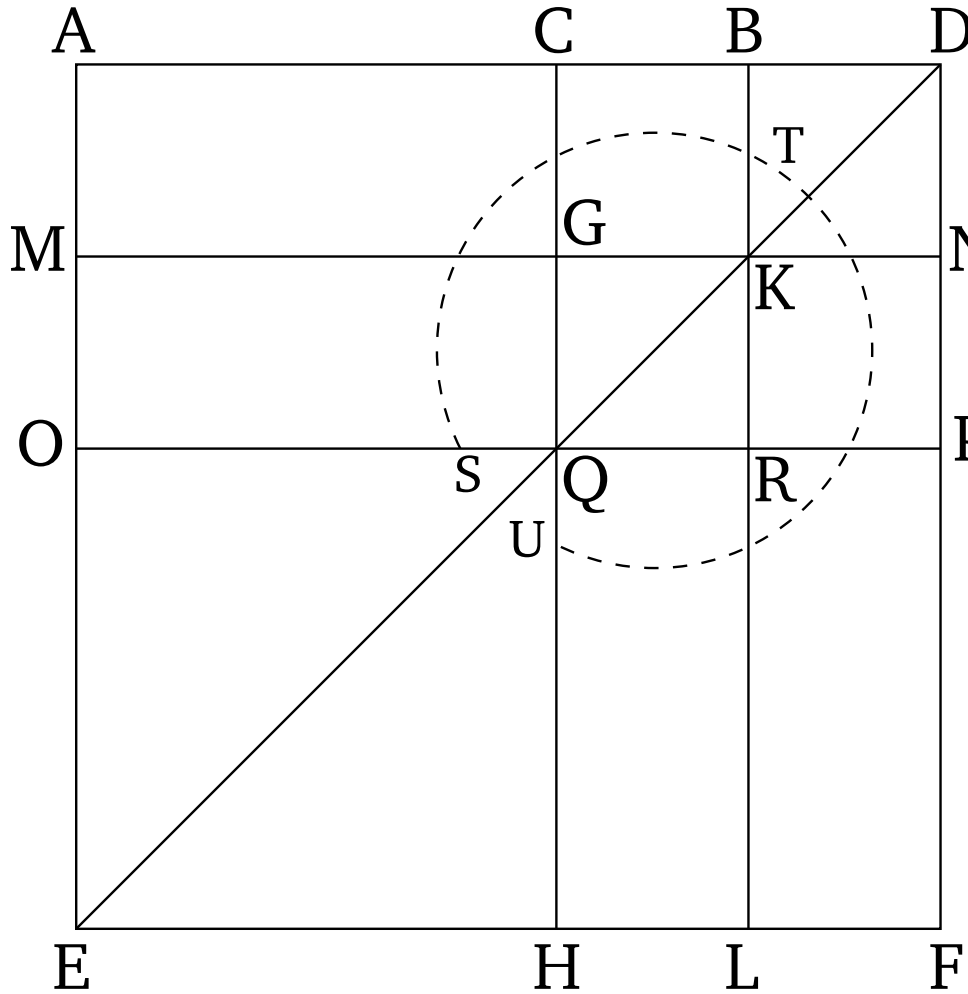
Εὐθεῖα γάρ τις ἢ AB τετμήσθω, ὥς >ετυθεν, κατὰ τὸ Γ σημεῖον; λέγω, <ὅτι τὸ τετράκις ὑπὸ τῶν AB , $B\Gamma$ περιεχόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τῆς AG τετραγώνου ὅσον ἐστὶ τῷ ἀπὸ τῆς AB , $B\Gamma$ ὥς ἀπὸ μιᾶς ἀναγραφέντι τετραγώνῳ.

Ἐκβεβλήσθω γὰρ ἐπ' εὐθείᾳ [τῇ AB εὐθείᾳ] ἡ BD , καὶ κείσθω τῇ ΓB ὅση ἡ BD , καὶ ἀναγεγράφθω ἀπὸ τῆς AD τετράγωνον τὸ $AEZ\Delta$, καὶ καταγεγράφθω διπλοῦν τὸ

pieces (of the straight-line), plus the square on the remaining piece, is equal to the square described on the whole and the former piece, as on one (complete straight-line).

For let any straight-line AB be cut, at random, at point C . I say that four times the rectangle contained by AB and BC , plus the square on AC , is equal to the square described on AB and BC , as on one (complete straight-line).

For let BD be produced in a straight-line [with the straight-line AB], and let BD be made equal to CB [Prop. 1.3], and let the square $AEFD$ be described on AD [Prop. 1.46], and let the (rest of the) figure be drawn double.



σθῆμα.

Therefore, since CB is equal to BD , but CB is equal to

ἐστὶν ἀπὸ τῆς AD ; τὸ $>$ άρα τετράκις ὑπὸ τῶν AB , BD μετὰ τοῦ ἀπὸ AG $>$ ίσον ἐστὶ τῶ| ἀπὸ AD τετραγώνω; $>$ ίση δὲ ἡ BD τῇ BG . τὸ $>$ άρα τετράκις ὑπὸ τῶν AB , BG περιεχόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ AG τετραγώνου $>$ ίσον ἐστὶ τῶ| ἀπὸ τῆς AD , τουτέστι τῶ| ἀπὸ τῆς AB καὶ BG ὡς ἀπὸ μιᾶς ἀναγραφέντι τετραγώνω.

Ἐὰν $>$ άρα εὐθεῖα γραμμὴ τμηθῇ, ὡς $>$ έτυθεν, τὸ τετράκις ὑπὸ τῆς $<$ όλης καὶ ἐνδὲς τῶν τμημάτων περιεχόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τοῦ λοιποῦ τμήματος τετραγώνου $>$ ίσον ἐστὶ τῶ| ἀπὸ τε τῆς $<$ όλης καὶ τοῦ εἰρημένου τμήματος ὡς ἀπὸ μιᾶς ἀναγραφέντι τετραγώνω; $<$ όπερ $>$ έδει δεῖχαι.

OH , which is equal to the square on AC , be added to both. Thus, four times the rectangle contained by AB and BD , plus the square on AC , is equal to the gnomon STU , and the (square) OH . But, the gnomon STU and the (square) OH is (equivalent to) the whole square $AEFD$, which is on AD . Thus, four times the (rectangle contained) by AB and BD , plus the (square) on AC , is equal to the square on AD . And BD (is) equal to BC . Thus, four times the rectangle contained by AB and BC , plus the square on AC , is equal to the (square) on AD , that is to say the square described on AB and BC , as on one (complete straight-line).

Thus, if a straight-line is cut at random, (then) four times the rectangle contained by the whole (straight-line), and one of the pieces (of the straight-line), plus the square on the remaining piece, is equal to the square described on the whole and the former piece, as on one (complete straight-line). (Which is) the very thing it was required to show.

† This proposition is a geometric version of the algebraic identity: $4(a+b)a + b^2 = [(a+b) + a]^2$.

9.

Ἐὰν εὐθεῖα γραμμὴ τμηθῇ εἰς $>$ ίσα καὶ $>$ άνισα, τὰ ἀπὸ τῶν ἀνίσων τῆς $<$ όλης τμημάτων τετράγωνα διπλάσια ἐστὶ τοῦ τε ἀπὸ τῆς ἡμίσειας καὶ τοῦ ἀπὸ τῆς μεταχὺ τῶν τομῶν τετραγώνου.

Εὐθεῖα γάρ τις ἡ AB τεμήσθω εἰς μὲν $>$ ίσα κατὰ τὸ Γ , εἰς δὲ $>$ άνισα κατὰ τὸ Δ ; λέγω, $<$ ότι τὰ ἀπὸ τῶν AD , ΔB τετράγωνα διπλάσια ἐστὶ τῶν ἀπὸ τῶν AG , $\Gamma\Delta$ τετραγώνων. $>$ Ἡθθω γὰρ ἀπὸ τοῦ Γ τῇ AB πρὸς ὀρθὰς ἡ ΓE , καὶ κείσθω $>$ ίση ἑκατέρα τῶν AG , ΓB , καὶ ἐπεζεύθωσαν αἱ EA , EB , καὶ διὰ μὲν τοῦ Δ τῇ EG παράλληλος ἡ ΔZ , διὰ δὲ τοῦ Z τῇ AB ἡ ZH , καὶ ἐπεζεύθω ἡ AZ . καὶ ἐπεὶ $>$ ίση ἐστὶν ἡ AG τῇ ΓE , $>$ ίση ἐστὶ καὶ ἡ ὑπὸ EAG γωνία τῇ ὑπὸ AEG . καὶ ἐπεὶ ὀρθὴ ἐστὶν ἡ πρὸς τῶ| Γ , λοιπαὶ $>$ άρα αἱ ὑπὸ EAG , AEG μιᾷ ὀρθῇ $>$ ίσαι εἰσίν; καὶ εἰσιν $>$ ίσαι; ἡμίσεια $>$ άρα ὀρθῆς ἐστὶν ἑκατέρα τῶν ὑπὸ GEA , GAE . διὰ τὰ αὐτὰ δὴ καὶ ἑκατέρα τῶν ὑπὸ $ΓEB$, $EBΓ$ ἡμίσεια ἐστὶν ὀρθῆς; $<$ όλη $>$ άρα ἡ ὑπὸ AEB ὀρθὴ ἐστὶν. καὶ ἐπεὶ ἡ ὑπὸ HEZ ἡμίσεια ἐστὶν ὀρθῆς, ὀρθὴ δὲ ἡ ὑπὸ EHZ ; $>$ ίση γὰρ ἐστὶ τῇ ἐντὸς καὶ ἀπεναντίον τῇ ὑπὸ $ΕΓB$; λοιπὴ $>$ άρα ἡ ὑπὸ EZH ἡμίσεια ἐστὶν ὀρθῆς; $>$ ίση $>$ άρα [ἐστὶν] ἡ ὑπὸ HEZ γωνία τῇ ὑπὸ EZH ; $<$ ώστε καὶ πλευρὰ ἡ EH τῇ HZ ἐστὶν $>$ ίση. πάλιν ἐπεὶ ἡ πρὸς τῶ| B γωνία ἡμίσεια ἐστὶν ὀρθῆς, ὀρθὴ δὲ ἡ ὑπὸ ZDB ; $>$ ίση γὰρ πάλιν ἐστὶ τῇ ἐντὸς καὶ ἀπεναντίον τῇ ὑπὸ $ΕΓB$; λοιπὴ $>$ άρα ἡ ὑπὸ $BZ\Delta$ ἡμίσεια ἐστὶν ὀρθῆς; $>$ ίση $>$ άρα ἡ πρὸς τῶ| B γωνία τῇ ὑπὸ ΔZB ; $<$ ώστε καὶ πλευρὰ ἡ $Z\Delta$ πλευρᾷ τῇ ΔB ἐστὶν $>$ ίση. καὶ ἐπεὶ $>$ ίση ἐστὶν ἡ AG τῇ ΓE , $>$ ίσον ἐστὶ καὶ τὸ ἀπὸ AG τῶ| ἀπὸ ΓE ; τὰ $>$ άρα ἀπὸ τῶν AG , ΓE τετράγωνα διπλάσια ἐστὶ τοῦ ἀπὸ AG . τοῖς δὲ ἀπὸ τῶν AG , ΓE $>$ ίσον ἐστὶ τὸ ἀπὸ τῆς EA τετράγωνον; ὀρθὴ γὰρ ἡ ὑπὸ AGE γωνία; τὸ $>$ άρα ἀπὸ τῆς EA διπλάσιον ἐστὶ τοῦ ἀπὸ τῆς AG . πάλιν, ἐπεὶ $>$ ίση ἐστὶν ἡ EH τῇ HZ , $>$ ίσον καὶ τὸ ἀπὸ τῆς EH τῶ| ἀπὸ τῆς HZ ; τὰ $>$ άρα ἀπὸ τῶν EH , HZ τετράγωνα διπλάσια ἐστὶ τοῦ ἀπὸ τῆς HZ τετραγώνου. τοῖς δὲ ἀπὸ τῶν EH , HZ τετραγώνοις $>$ ίσον ἐστὶ τὸ ἀπὸ τῆς EZ τετράγωνον; τὸ $>$ άρα ἀπὸ τῆς EZ τετράγωνον διπλάσιον ἐστὶ τοῦ ἀπὸ τῆς HZ . $>$ ίση δὲ ἡ HZ τῇ $\Gamma\Delta$; τὸ $>$ άρα ἀπὸ τῆς EZ διπλάσιον ἐστὶ τοῦ ἀπὸ τῆς $\Gamma\Delta$. $>$ έστι δὲ καὶ τὸ ἀπὸ τῆς EA διπλάσιον τοῦ ἀπὸ τῆς AG ; τὰ $>$ άρα ἀπὸ τῶν AE , EZ τετράγωνα διπλάσια ἐστὶ τῶν ἀπὸ

Proposition 9†

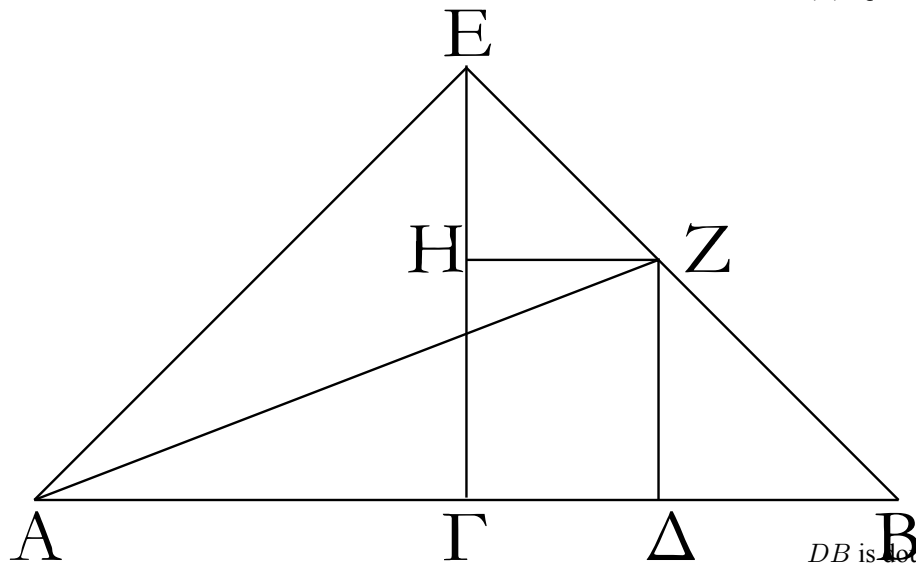
If a straight-line is cut into equal and unequal (pieces, then) the (sum of the) squares on the unequal pieces of the whole (straight-line) is double the (sum of the) square on half (the straight-line) and (the square) on the (difference) between the (equal and unequal) pieces.

For let any straight-line AB be cut—equally at C , and unequally at D . I say that the (sum of the) squares on AD and DB is double the (sum of the squares) on AC and CD .

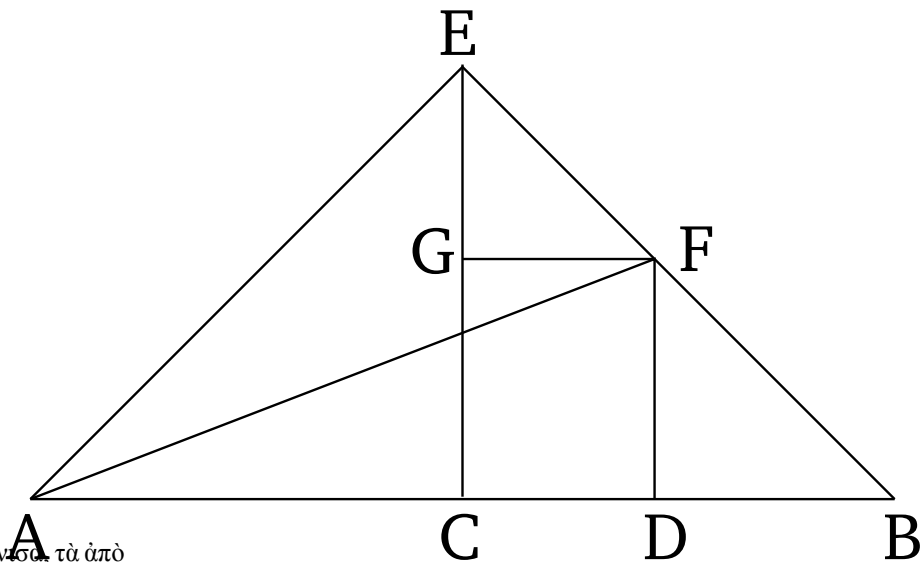
For let CE be drawn from (point) C , at right-angles to AB [Prop. 1.11], and let it be made equal to each of AC and CB [Prop. 1.3], and let EA and EB be joined. And let DF be drawn through (point) D , parallel to EC [Prop. 1.31], and (let) FG (be drawn) through (point) F , (parallel) to AB [Prop. 1.31]. And let AF be joined. And since AC is equal to CE , the angle EAC is also equal to the (angle) AEC [Prop. 1.5]. And since the (angle) at C is a right-angle, the (sum of the) remaining angles (of triangle AEC), EAC and AEC , is thus equal to one right-angle [Prop. 1.32]. And they are equal. Thus, (angles) CEA and CAE are each half a right-angle. So, for the same (reasons), (angles) CEB and EBC are also each half a right-angle. Thus, the whole (angle) AEB is a right-angle. And since GEF is half a right-angle, and EGF (is) a right-angle—for it is equal to the internal and opposite (angle) ECB [Prop. 1.29]—the remaining (angle) EFG is thus half a right-angle [Prop. 1.32]. Thus, angle GEF [is] equal to EFG . So the side EG is also equal to the (side) GF [Prop. 1.6]. Again, since the angle at B is half a right-angle, and (angle) FDB (is) a right-angle—for again it is equal to the internal and opposite (angle) ECB [Prop. 1.29]—the remaining (angle) BFD is half a right-angle [Prop. 1.32]. Thus, the angle at B (is) equal to DFB . So the side FD is also equal to the side DB [Prop. 1.6]. And since AC is equal to CE , the (square) on AC (is) also equal to the (square) on CE . Thus, the (sum of the) squares on AC and CE is double the (square) on AC . And the square on EA is equal to the (sum of the) squares on AC and CE . For angle ACE (is) a right-angle [Prop. 1.47]. Thus, the (square) on EA is double the (square) on AC . Again, since EG is equal to GF , the (square) on EG (is)

τῶν $ΑΓ, ΓΔ$ τετραγώνων. τοῖς δὲ ἀπὸ τῶν $ΑΕ, ΕΖ$ ὀρθῶν ἔστι τὸ ἀπὸ τῆς $ΑΖ$ τετραγώνου; ὁρθὴ γάρ ἐστιν ἡ ὑπὸ $ΑΕΖ$ γωνία; τὸ ὅρα ἀπὸ τῆς $ΑΖ$ τετραγώνου διπλάσιόν ἐστι τῶν ἀπὸ τῶν $ΑΓ, ΓΔ$. τῷ δὲ ἀπὸ τῆς $ΑΖ$ ὀρθῶν ἔστι τὰ ἀπὸ τῶν $ΑΔ, ΔΖ$; ὁρθὴ γάρ ἡ πρὸς τῷ $Δ$ γωνία; τὰ ὅρα ἀπὸ τῶν $ΑΔ, ΔΖ$ διπλάσιά ἐστι τῶν ἀπὸ τῶν $ΑΓ, ΓΔ$ τετραγώνων. ὀρθῶν δὲ ἡ $ΔΖ$ τῇ $ΔΒ$; τὰ ὅρα ἀπὸ τῶν $ΑΔ, ΔΒ$ τετραγώνων διπλάσιά ἐστι τῶν ἀπὸ τῶν $ΑΓ, ΓΔ$ τετραγώνων.

also equal to the (square) on GF . Thus, the (sum of the squares) on EG and GF is double the square on GF . And the square on EF is equal to the (sum of the) squares on EG and GF [Prop. 1.47]. Thus, the square on EF is double the (square) on GF . And GF (is) equal to CD [Prop. 1.34]. Thus, the (square) on EF is double the (square) on CD . And the (square) on EA is also double the (square) on AC . Thus, the (sum of the) squares on AE and EF is double the (sum of the) squares on AC and CD . And the square on AF is equal to the (sum of the squares) on AE and EF . For the angle AEF is a right-angle [Prop. 1.47]. Thus, the square on AF is double the (sum of the squares) on AC and CD . And the (sum of the squares) on AD and DF (is) equal to the (square) on AF . For the angle at D is a right-angle [Prop. 1.47]. Thus, the (sum of the squares) on AD and DF is double the (sum of the) squares on AC and CD . And DF (is) equal to DB . Thus, the (sum of the) squares on AD and



DB is double the (sum of the) squares on AC and CD .



Ἐάν ὅρα εὐθεῖα γραμμὴ τμηθῇ εἰς ὀρθῶν καὶ ἄνισων, τὰ ἀπὸ τῶν ἄνισων τῆς ὀρθῆς τμημάτων τετραγώνων διπλάσιά ἐστι τοῦ τε ἀπὸ τῆς ἡμισείας καὶ τοῦ ἀπὸ τῆς μεταχὺ τῶν τομῶν τετραγώνου; ὅπερ ὀφείδει δεῖξαι.

Thus, if a straight-line is cut into equal and unequal (pieces, then) the (sum of the) squares on the unequal pieces of the whole (straight-line) is double the (sum of the) square on half (the straight-line) and (the square) on the (difference) between the (equal and unequal) pieces. (Which is) the very thing it was required to show.

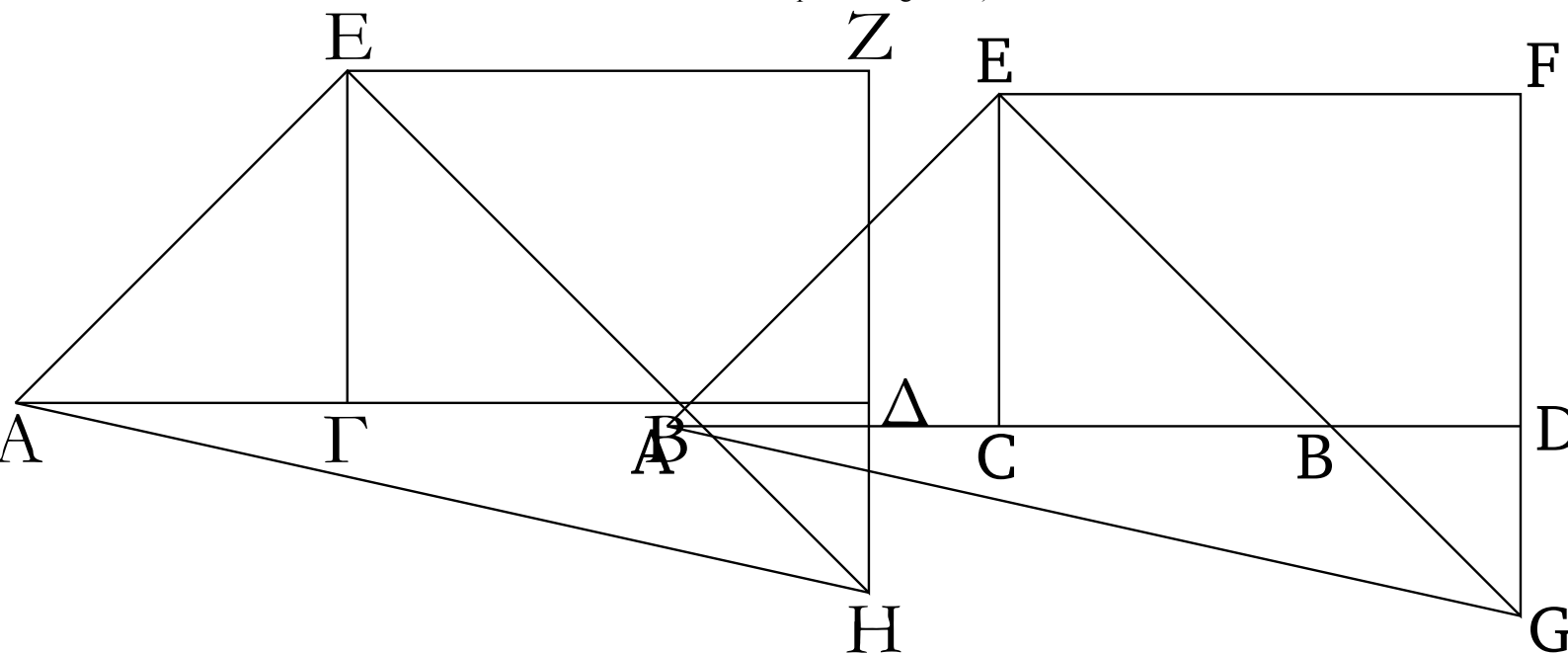
[†] This proposition is a geometric version of the algebraic identity: $a^2 + b^2 = 2 \left[([a+b]/2)^2 + ([a+b]/2 - b)^2 \right]$.

10.

Ἐὰν εὐθεῖα γραμμὴ τηθηῇ| διῖθα, προστεθῇ| δέ τις ᾠτῇ|
εὐθεῖα ἐπ> εὐθείαις, τὸ ἀπὸ τῆς <όλης σὺν τῇ| προσκειμένη
καὶ τὸ ἀπὸ τῆς προσκειμένης τὰ συναμφότερα τετραγώνω
διπλάσιά ἐστι τοῦ τε ἀπὸ τῆς ἡμισείας καὶ τοῦ ἀπὸ τῆς
συγκεκλιμένης >ἐκ τε τῆς ἡμισείας καὶ τῆς προσκειμένης ὥς
ἀπὸ μιᾶς ἀναγραφέντος τετραγώνου.

Proposition 10[†]

If a straight-line is cut in half, and any straight-line added to it straight-on, (then) the sum of the square on the whole (straight-line) with the (straight-line) having been added, and the (square) on the (straight-line) having been added, is double the (sum of the square) on half (the straight-line), and the square described on the sum of half (the straight-line) and (straight-line) having been added, as on one (complete straight-line).



Εὐθεΐα γάρ τις ἢ ΑΒ τετμήσθω διῖθα κατὰ τὸ Γ, προσκείσθω δέ τις αὐτῇ| εὐθεΐα ἐπ> εὐθεΐας ἡ ΒΔ; λέγω, <ὅτι τὰ ἀπὸ τῶν ΑΔ, ΔΒ τετράγωνα διπλάσιά ἐστι τῶν ἀπὸ τῶν ΑΓ, ΓΔ τετραγώνων.

>H0θω γάρ ἀπὸ τοῦ Γ σημείου τῇ| AB πρὸς ὀρθὰς ἡ ΓΕ, καὶ κείσθω >ίση ἑκάτερα τῶν ΑΓ, ΓΒ, καὶ ἐπεζεύθωσαν αἱ ΕΑ, ΕΒ; καὶ διὰ μὲν τοῦ Ε τῇ| ΑΔ παράλληλῳς >ήθθω ἡ ΕΖ, διὰ δὲ τοῦ Δ τῇ| ΓΕ παράλληλῳς >ήθθω ἡ ΖΔ. καὶ ἐπεὶ εἰς παραλλήλωνς εὐθείαις τὰς ΕΓ, ΖΔ εὐθεῖα τις ἐνέπεσεν ἡ ΕΖ, αἱ ὑπὸ ΓΕΖ, ΕΖΔ >άρα δυσὶν ὀρθαῖς >ίσαι εἰσὶν; αἱ >άρα ὑπὸ ΖΕΒ, ΕΖΔ δύο ὀρθῶν ἐλάσσονές εἰσιν; αἱ δὲ ἀπ> ἐλασσόνων >ἡ δύο ὀρθῶν ἐκβαλλόμεναι συμπίπτουσιν; αἱ >άρα ΕΒ, ΖΔ ἐκβαλλόμεναι ἐπὶ τὰ Β, Δ μέρη συμπεσόνται. ἐκβεβλήσθωσαν καὶ συμπιπτέωσαν κατὰ τὸ Η, καὶ ἐπεζεύθω ἡ ΑΗ. καὶ ἐπεὶ >ίση ἐστὶν ἡ ΑΓ τῇ| ΓΕ, >ίση ἐστὶ καὶ γωνία ἡ ὑπὸ ΕΑΓ τῇ| ὑπὸ ΑΕΓ; καὶ ὀρθή ἡ πρὸς τῷ Γ; ἡμίσεια >άρα ὀρθῆς [ἐστὶν] ἑκάτερα τῶν ὑπὸ ΕΑΓ, ΑΕΓ. διὰ τὰ αὐτὰ δὴ καὶ ἑκάτερα τῶν ὑπὸ ΓΕΒ, ΕΒΓ ἡμίσεια ἐστὶν ὀρθῆς; ὀρθή >άρα ἐστὶν ἡ ὑπὸ ΑΕΒ. καὶ ἐπεὶ ἡμίσεια ὀρθῆς ἐστὶν ἡ ὑπὸ ΕΒΓ, ἡμίσεια >άρα ὀρθῆς καὶ ἡ ὑπὸ ΔΒΗ. >έστι δὲ καὶ ἡ ὑπὸ ΒΔΗ ὀρθή; >ίση γάρ ἐστὶ τῇ| ὑπὸ ΔΓΕ; ἐναλλάχ γάρ; λοιπὴ <άρα ἡ ὑπὸ ΔΗΒ ἡμίσειά ἐστὶν ὀρθῆς; ἡ >άρα ὑπὸ ΔΗΒ τῇ| ὑπὸ ΔΒΗ ἐστὶν >ίση; <ώστε καὶ πλευρὰ ἡ ΒΔ πλευρᾷ| τῇ| ΗΔ ἐστὶν >ίση. πάλιν, ἐπεὶ ἡ ὑπὸ ΕΗΖ ἡμίσειά ἐστὶν ὀρθῆς, ὀρθή δὲ ἡ πρὸς τῷ Ζ; >ίση γάρ ἐστὶ τῇ| ἀπεναντίον τῇ| πρὸς τῷ Γ; λοιπὴ >άρα ἡ ὑπὸ ΖΕΗ ἡμίσειά ἐστὶν ὀρθῆς; >ίση >άρα ἡ ὑπὸ ΕΗΖ γωνία τῇ| ὑπὸ ΖΕΗ; <ώστε καὶ πλευρὰ ἡ ΗΖ πλευρᾷ| τῇ| ΕΖ ἐστὶν

For let any straight-line AB be cut in half at (point) C , and let any straight-line BD be added to it straight-on. I say that the (sum of the) squares on AD and DB is double the (sum of the) squares on AC and CD .

For let CE be drawn from point C , at right-angles to AB [Prop. 1.11], and let it be made equal to each of AC and CB [Prop. 1.3], and let EA and EB be joined. And let EF be drawn through E , parallel to AD [Prop. 1.31], and let FD be drawn through D , parallel to CE [Prop. 1.31]. And since some straight-line EF falls across the parallel straight-lines EC and FD , the (internal angles) CEF and EFD are thus equal to two right-angles [Prop. 1.29]. Thus, FEB and EFD are less than two right-angles. And (straight-lines) produced from (internal angles whose sum is) less than two right-angles meet together [Post. 5]. Thus, being produced in the direction of B and D , the (straight-lines) EB and FD will meet. Let them be produced, and let them meet together at G , and let AG be joined. And since AC is equal to CE , angle EAC is also equal to (angle) AEC [Prop. 1.5]. And the (angle) at C (is) a right-angle. Thus, EAC and AEC [are] each half a right-angle [Prop. 1.32]. So, for the same (reasons), CEB and EBC are also each half a right-angle. Thus, (angle) AEB is a right-angle. And since EBC is half a right-angle, DBG (is) thus also half a right-angle [Prop. 1.15]. And BDG is also a right-angle. For it is equal to DCE . For (they are) alternate (angles) [Prop. 1.29]. Thus, the remaining (angle) DGB is half a right-angle. Thus, DGB is equal to DBG . So side BD

>ίση. καὶ ἐπεὶ [$>ίση$ ἐστὶν ἡ $ΕΓ$ τῇ $ΓΑ$], $>ίσον$ ἐστὶ [καὶ] τὸ ἀπὸ τῆς $ΕΓ$ τετραγώνου τῷ ἀπὸ τῆς $ΓΑ$ τετραγώνου; τὰ $>άρα$ ἀπὸ τῶν $ΕΓ$, $ΓΑ$ τετράγωνα διπλάσιά ἐστι τοῦ ἀπὸ τῆς $ΓΑ$ τετραγώνου. τοῖς δὲ ἀπὸ τῶν $ΕΓ$, $ΓΑ$ $>ίσον$ ἐστὶ τὸ ἀπὸ τῆς $ΕΑ$; τὸ $>άρα$ ἀπὸ τῆς $ΕΑ$ τετράγωνον διπλάσιόν ἐστι τοῦ ἀπὸ τῆς $ΑΓ$ τετραγώνου. πάλιν, ἐπεὶ $>ίση$ ἐστὶν ἡ $ΖΗ$ τῇ $ΕΖ$, $>ίσον$ ἐστὶ καὶ τὸ ἀπὸ τῆς $ΖΗ$ τῷ ἀπὸ τῆς $ΖΕ$; τὰ $>άρα$ ἀπὸ τῶν $ΗΖ$, $ΖΕ$ διπλάσιά ἐστι τοῦ ἀπὸ τῆς $ΕΖ$. τοῖς δὲ ἀπὸ τῶν $ΗΖ$, $ΖΕ$ $>ίσον$ ἐστὶ τὸ ἀπὸ τῆς $ΕΗ$; τὸ $>άρα$ ἀπὸ τῆς $ΕΗ$ διπλάσιόν ἐστι τοῦ ἀπὸ τῆς $ΕΖ$. $>ίση$ δὲ ἡ $ΕΖ$ τῇ $ΓΔ$; τὸ $>άρα$ ἀπὸ τῆς $ΕΗ$ τετράγωνον διπλάσιόν ἐστι τοῦ ἀπὸ τῆς $ΓΔ$. ἐδείκθη δὲ καὶ τὸ ἀπὸ τῆς $ΕΑ$ διπλάσιον τοῦ ἀπὸ τῆς $ΑΓ$; τὰ $>άρα$ ἀπὸ τῶν $ΑΕ$, $ΕΗ$ τετράγωνα διπλάσιά ἐστι τῶν ἀπὸ τῶν $ΑΓ$, $ΓΔ$ τετραγώνων. τοῖς δὲ ἀπὸ τῶν $ΑΕ$, $ΕΗ$ τετραγώνοις $>ίσον$ ἐστὶ τὸ ἀπὸ τῆς $ΑΗ$ τετράγωνον; τὸ $>άρα$ ἀπὸ τῆς $ΑΗ$ διπλάσιόν ἐστι τῶν ἀπὸ τῶν $ΑΓ$, $ΓΔ$. τῷ δὲ ἀπὸ τῆς $ΑΗ$ $>ίσα$ ἐστὶ τὰ ἀπὸ τῶν $ΑΔ$, $ΔΗ$; τὰ $>άρα$ ἀπὸ τῶν $ΑΔ$, $ΔΗ$ [τετράγωνα] διπλάσιά ἐστι τῶν ἀπὸ τῶν $ΑΓ$, $ΓΔ$ [τετραγώνων]. $>ίση$ δὲ ἡ $ΔΗ$ τῇ $ΔΒ$; τὰ $>άρα$ ἀπὸ τῶν $ΑΔ$, $ΔΒ$ [τετράγωνα] διπλάσιά ἐστι τῶν ἀπὸ τῶν $ΑΓ$, $ΓΔ$ τετραγώνων.

Ἐὰν $>άρα$ εὐθεῖα γραμμὴ τμηθῇ δίθρα, προστεθῇ δὲ τις αὐτῇ εὐθεῖα ἐπ' εὐθείᾳ, τὸ ἀπὸ τῆς $<όλης$ συν τῇ προσκειμένης καὶ τὸ ἀπὸ τῆς προσκειμένης τὰ συναμφότερα τετράγωνα διπλάσιά ἐστι τοῦ τε ἀπὸ τῆς ἡμισείας καὶ τοῦ ἀπὸ τῆς συγκειμένης $>έκ$ τε τῆς ἡμισείας καὶ τῆς προσκειμένης ὥς ἀπὸ μιᾶς ἀναγραφέντος τετραγώνου; $<όπερ >έδει$ δεῖχαι.

is also equal to side GD [Prop. 1.6]. Again, since EGF is half a right-angle, and the (angle) at F (is) a right-angle, for it is equal to the opposite (angle) at C [Prop. 1.34], the remaining (angle) FEF is thus half a right-angle. Thus, angle EGF (is) equal to FEF . So the side GF is also equal to the side EF [Prop. 1.6]. And since [EC is equal to CA] the square on EC is [also] equal to the square on CA . Thus, the (sum of the) squares on EC and CA is double the square on CA . And the (square) on EA is equal to the (sum of the squares) on EC and CA [Prop. 1.47]. Thus, the square on EA is double the square on AC . Again, since FG is equal to EF , the (square) on FG is also equal to the (square) on FE . Thus, the (sum of the squares) on GF and FE is double the (square) on EF . And the (square) on EG is equal to the (sum of the squares) on GF and FE [Prop. 1.47]. Thus, the (square) on EG is double the (square) on EF . And EF (is) equal to CD [Prop. 1.34]. Thus, the square on EG is double the (square) on CD . But it was also shown that the (square) on EA (is) double the (square) on AC . Thus, the (sum of the) squares on AE and EG is double the (sum of the) squares on AC and CD . And the square on AG is equal to the (sum of the) squares on AE and EG [Prop. 1.47]. Thus, the (square) on AG is double the (sum of the squares) on AC and CD . And the (sum of the squares) on AD and DG is equal to the (square) on AG [Prop. 1.47]. Thus, the (sum of the) [squares] on AD and DG is double the (sum of the) [squares] on AC and CD . And DG (is) equal to DB . Thus, the (sum of the) [squares] on AD and DB is double the (sum of the) squares on AC and CD .

Thus, if a straight-line is cut in half, and any straight-line added to it straight-on, (then) the sum of the square on the whole (straight-line) with the (straight-line) having been added, and the (square) on the (straight-line) having been added, is double the (sum of the square) on half (the straight-line), and the square described on the sum of half (the straight-line) and (straight-line) having been added, as on one (complete straight-line). (Which is) the very thing it was required to show.

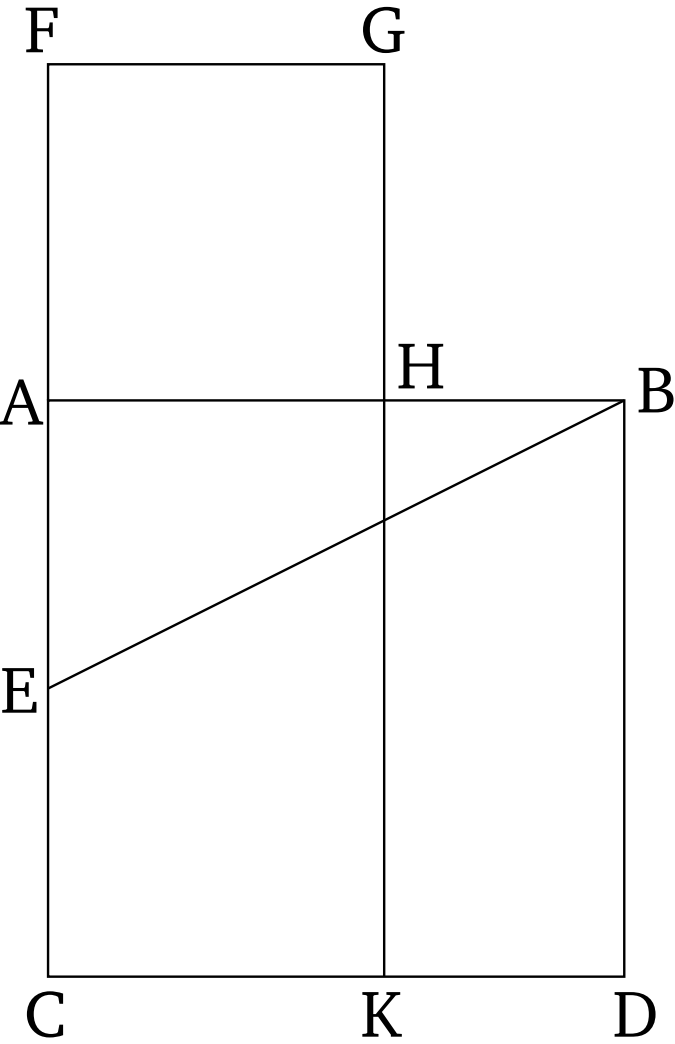
† This proposition is a geometric version of the algebraic identity: $(2a + b)^2 + b^2 = 2[a^2 + (a + b)^2]$.

11.

Τὴν δοθεῖσαν εὐθεῖαν τεμεῖν $<ώστε$ τὸ ὑπὸ τῆς $<όλης$ καὶ τοῦ ἐτέρου τῶν τμημάτων περιεχόμενον ὀρθογώνιον $>ίσον$ εἶναι τῷ ἀπὸ τοῦ λοιποῦ τμήματος τετραγώνου.
>Ἐστω ἡ δοθεῖσα εὐθεῖα ἡ $ΑΒ$; δεῖ δὴ τὴν $ΑΒ$ τεμεῖν $<ώστε$ τὸ ὑπὸ τῆς $<όλης$ καὶ τοῦ ἐτέρου τῶν τμημάτων περιεχόμενον ὀρθογώνιον $>ίσον$ εἶναι τῷ ἀπὸ τοῦ λοιποῦ τμήματος τετραγώνου.

Proposition 11†

To cut a given straight-line such that the rectangle contained by the whole (straight-line), and one of the pieces (of the straight-line), is equal to the square on the remaining piece. Let AB be the given straight-line. So it is required to cut AB such that the rectangle contained by the whole (straight-line), and one of the pieces (of the straight-line), is equal to the square on the remaining piece.



Thus, the given straight-line AB has been cut at (point) H such as to make the rectangle contained by AB and BH equal to the square on HA . (Which is) the very thing it was required to do.

† This manner of cutting a straight-line—so that the ratio of the whole to the larger piece is equal to the ratio of the larger to the smaller piece—is sometimes called the “Golden Section”.

12.

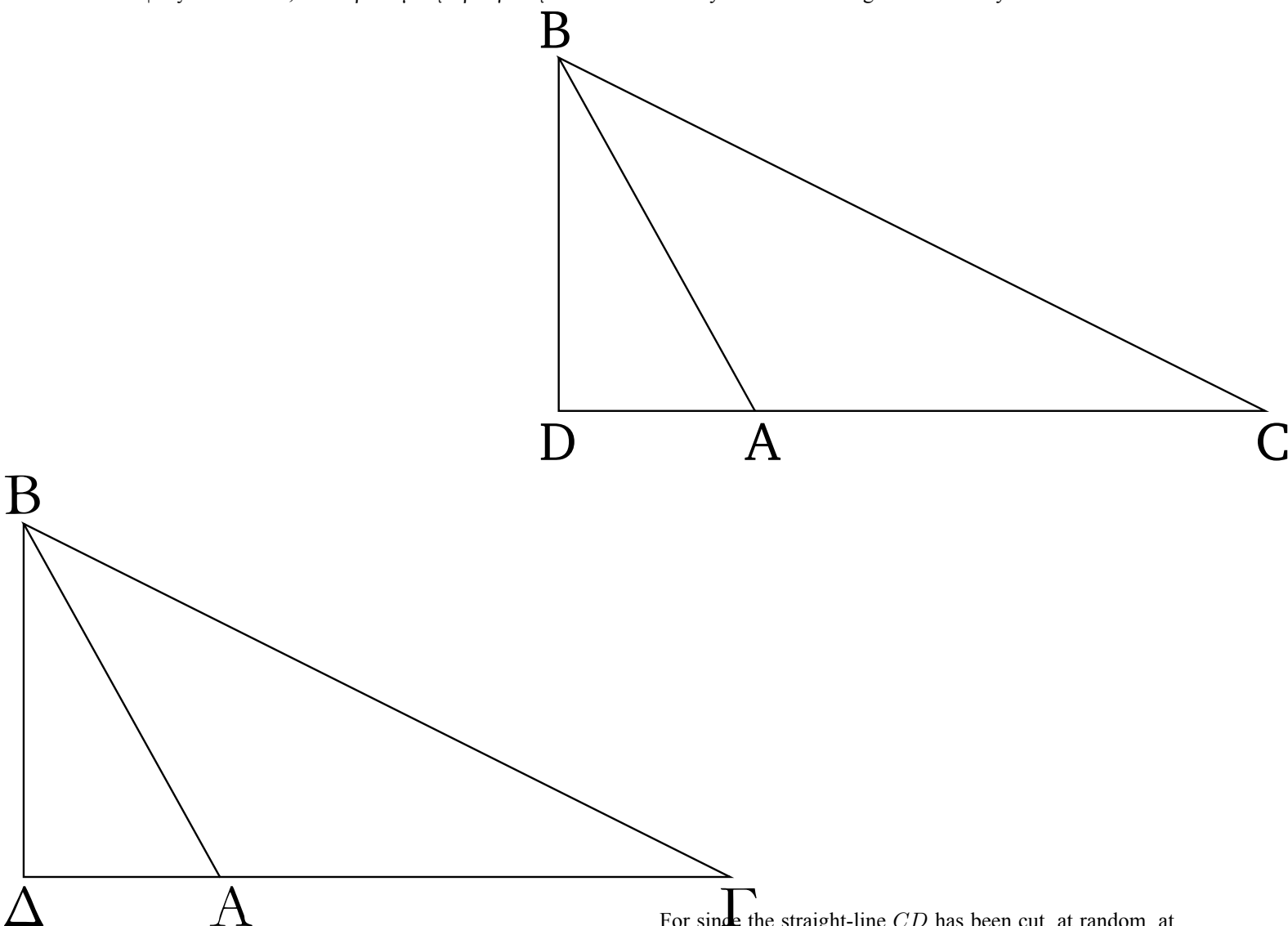
Proposition 12[†]

Ἐν τοῖς ἀμβλυγωνίοις τριγώνοις τὸ ἀπὸ τῆς τὴν ἀμβλεῖαν γωνίαν ὑποτείνουσῃς πλευρᾶς τετράγωνον μείζον ἔστι τῶν ἀπὸ τῶν τὴν ἀμβλεῖαν γωνίαν περιεουσῶν πλευρῶν τετραγώνων τῷ περιεομένῳ δις ὑπὸ τε μιᾶς τῶν περὶ τὴν ἀμβλεῖαν γωνίαν, ἐφ' ἣν ἡ κάθετος πίπτει, καὶ τῆς ἀπολαμβανομένης ἐκτὸς ὑπὸ τῆς καθέτου πρὸς τῇ ἀμβλεῖᾳ γωνίᾳ.

>Ἐστω ἀμβλυγώνιον τρίγωνον τὸ $AB\Gamma$ ἀμβλεῖαν >έθον τὴν ὑπὸ $BA\Gamma$, καὶ >ῆθθω ἀπὸ τοῦ B σημείου ἐπὶ τὴν ΓA ἐκβληθεῖσαν κάθετος ἡ BD . λέγω, <ὅτι τὸ ἀπὸ τῆς $B\Gamma$ τετράγωνον μείζον ἔστι τῶν ἀπὸ τῶν BA , $A\Gamma$ τετραγώνων τῷ δις ὑπὸ τῶν ΓA , $A\Delta$ περιεομένῳ ὀρθογωνίῳ.

In obtuse-angled triangles, the square on the side subtending the obtuse angle is greater than the (sum of the) squares on the sides containing the obtuse angle by twice the (rectangle) contained by one of the sides around the obtuse angle, to which a perpendicular (straight-line) falls, and the (straight-line) cut off outside (the triangle) by the perpendicular (straight-line) towards the obtuse angle.

Let ABC be an obtuse-angled triangle, having the angle BAC obtuse. And let BD be drawn from point B , perpendicular to CA produced [Prop. 1.12]. I say that the square on BC is greater than the (sum of the) squares on BA and AC by twice the rectangle contained by CA and AD .



Ἐπεὶ γὰρ εὐθεῖα ἡ ΓA τέμνεται, ὥς >έτυθεν, κατὰ τὸ A σημεῖον, τὸ >άρα ἀπὸ τῆς $\Delta\Gamma$ >ίσον ἔστι τοῖς ἀπὸ τῶν ΓA , $A\Delta$ τετραγώνοις καὶ τῷ δις ὑπὸ τῶν ΓA , $A\Delta$ περιεομένῳ ὀρθογωνίῳ. κοινὸν προσκείσθω τὸ ἀπὸ τῆς ΔB ; τὰ >άρα

For since the straight-line CD has been cut, at random, at point A , the (square) on DC is thus equal to the (sum of the) squares on CA and AD , and twice the rectangle contained by CA and AD [Prop. 2.4]. Let the (square) on DB be added to both. Thus, the (sum of the squares) on CD

ἀπὸ τῶν $\Gamma\Delta$, ΔB >ίσα ἐστὶ τοῖς τε ἀπὸ τῶν ΓA , $A\Delta$, ΔB τετραγώνοις καὶ τῷ διζ ὑπὸ τῶν ΓA , $A\Delta$ [περιεθομένῳ ὀρθογωνίῳ]. ἀλλὰ τοῖς μὲν ἀπὸ τῶν $\Gamma\Delta$, ΔB >ίσον ἐστὶ τὸ ἀπὸ τῆς ΓB ; ὀρθή γάρ ἡ πρὸς τῷ Δ γωνία; τοῖς δὲ ἀπὸ τῶν $A\Delta$, ΔB >ίσον τὸ ἀπὸ τῆς AB ; τὸ >άρα ἀπὸ τῆς ΓB τετράγωνον >ίσον ἐστὶ τοῖς τε ἀπὸ τῶν ΓA , AB τετραγώνοις καὶ τῷ διζ ὑπὸ τῶν ΓA , $A\Delta$ περιεθομένῳ ὀρθογωνίῳ; <ώστε τὸ ἀπὸ τῆς ΓB τετράγωνον τῶν ἀπὸ τῶν ΓA , AB τετραγώνων μείζον ἐστὶ τῷ διζ ὑπὸ τῶν ΓA , $A\Delta$ περιεθομένῳ ὀρθογωνίῳ.

Ἐν >άρα τοῖς ἀμβλυγωνίοις τριγώνοις τὸ ἀπὸ τῆς τὴν ἀμβλείαν γωνίαν ὑποτείνουσας πλευρᾶς τετράγωνον μείζον ἐστὶ τῶν ἀπὸ τῶν τὴν ἀμβλείαν γωνίαν περιεουσῶν πλευρῶν τετραγώνων τῷ περιεθομένῳ διζ ὑπὸ τε μιᾶς τῶν περὶ τὴν ἀμβλείαν γωνίαν, ἐφ' <ὴν ἡ κάθετος πίπτει, καὶ τῆς ἀπολαμβανομένης ἐκτὸς ὑπὸ τῆς καθέτου πρὸς τῇ ἀμβλείᾳ γωνίᾳ; <όπερ >έδει δεῖχαι.

and DB is equal to the (sum of the) squares on CA , AD , and DB , and twice the [rectangle contained] by CA and AD . But, the (square) on CB is equal to the (sum of the squares) on CD and DB . For the angle at D (is) a right-angle [Prop. 1.47]. And the (square) on AB (is) equal to the (sum of the squares) on AD and DB [Prop. 1.47]. Thus, the square on CB is equal to the (sum of the) squares on CA and AB , and twice the rectangle contained by CA and AD . So the square on CB is greater than the (sum of the) squares on CA and AB by twice the rectangle contained by CA and AD .

Thus, in obtuse-angled triangles, the square on the side subtending the obtuse angle is greater than the (sum of the) squares on the sides containing the obtuse angle by twice the (rectangle) contained by one of the sides around the obtuse angle, to which a perpendicular (straight-line) falls, and the (straight-line) cut off outside (the triangle) by the perpendicular (straight-line) towards the obtuse angle. (Which is) the very thing it was required to show.

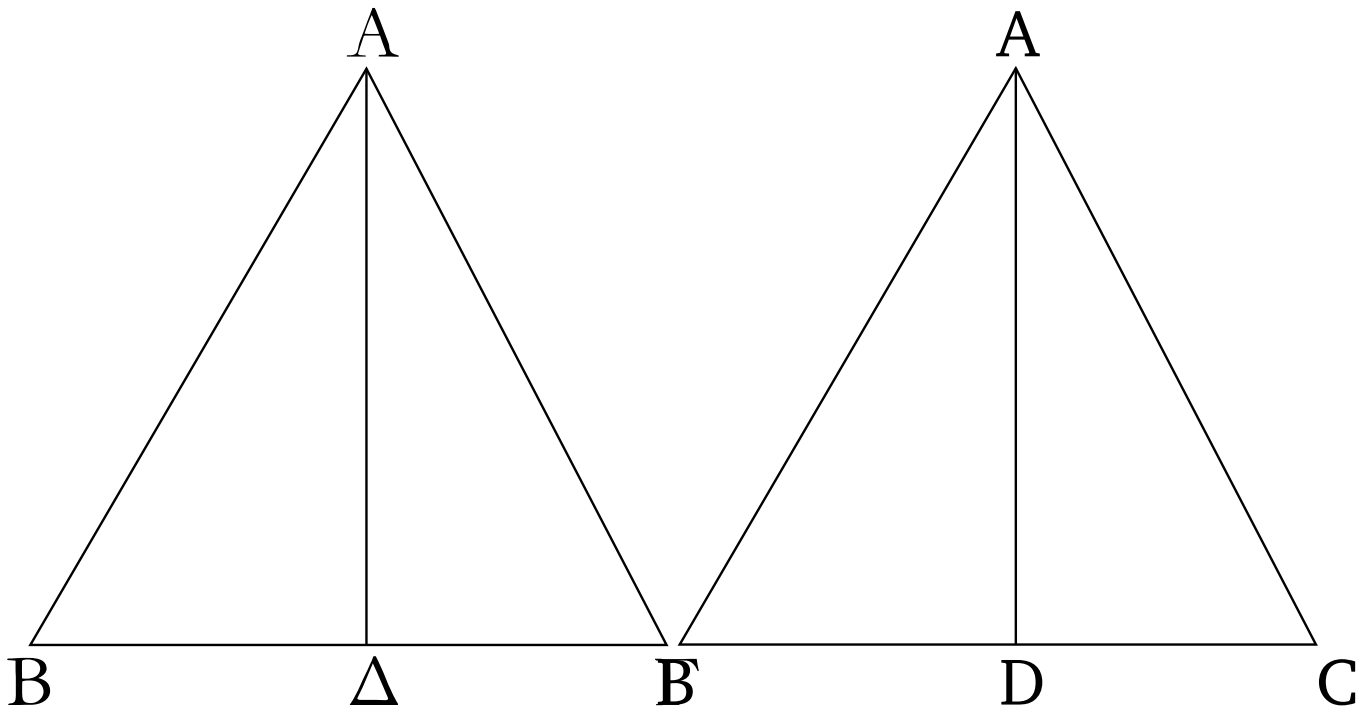
† This proposition is equivalent to the well-known cosine formula: $BC^2 = AB^2 + AC^2 - 2 AB AC \cos BAC$, since $\cos BAC = -AD/AB$.

13.

Proposition 13[†]

Ἐν τοῖς ὀξυγωνίοις τριγώνοις τὸ ἀπὸ τῆς τὴν ὀχείαν γωνίαν ὑποτείνουσας πλευρᾶς τετράγωνον >έλαττον ἐστὶ τῶν ἀπὸ τῶν τὴν ὀχείαν γωνίαν περιεουσῶν πλευρῶν τετραγώνων τῷ περιεθομένῳ διζ ὑπὸ τε μιᾶς τῶν περὶ τὴν ὀχείαν γωνίαν, ἐφ' <ὴν ἡ κάθετος πίπτει, καὶ τῆς ἀπολαμβανομένης ἐντὸς ὑπὸ τῆς καθέτου πρὸς τῇ ὀχείᾳ γωνίᾳ.

In acute-angled triangles, the square on the side subtending the acute angle is less than the (sum of the) squares on the sides containing the acute angle by twice the (rectangle) contained by one of the sides around the acute angle, to which a perpendicular (straight-line) falls, and the (straight-line) cut off inside (the triangle) by the perpendicular (straight-line) towards the acute angle.



>Ἐστω ὀξυγώνιον τρίγωνον τὸ $AB\Gamma$ ὀχείαν >έθον τὴν πρὸς τῷ B γωνίαν, καὶ >ήθθω ἀπὸ τοῦ A σημείου ἐπὶ τὴν $B\Gamma$ κάθετος ἡ AD ; λέγω, <ὅτι τὸ ἀπὸ τῆς AG τετράγωνον >έλαττον ἐστὶ τῶν ἀπὸ τῶν ΓB , BA τετραγώνων τῷ διζ ὑπὸ τῶν ΓB , BD περιεθομένῳ ὀρθογωνίῳ.

Ἐπεὶ γὰρ εὐθεῖα ἡ ΓB τέμνεται, ὡς >έτυθεν, κατὰ τὸ Δ , τὰ >άρα ἀπὸ τῶν ΓB , BD τετράγωνα >ίσα ἐστὶ τῷ τε διζ ὑπὸ τῶν ΓB , BD περιεθομένῳ ὀρθογωνίῳ καὶ τῷ ἀπὸ

Let ABC be an acute-angled triangle, having the angle at (point) B acute. And let AD be drawn from point A , perpendicular to BC [Prop. 1.12]. I say that the square on AC is less than the (sum of the) squares on CB and BA by twice the rectangle contained by CB and BD .

For since the straight-line CB has been cut, at random, at (point) D , the (sum of the) squares on CB and BD is thus equal to twice the rectangle contained by CB and BD , and

δὲ >άνισα κατὰ τὸ E , τὸ >άρα ὑπὸ τῶν BE, EZ περιεχόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τῆς EH τετραγώνου >ίσον ἐστὶ τῷ ἀπὸ τῆς HZ τετραγώνῳ. >ίση δὲ ἡ HZ τῇ $H\Theta$; τὸ >άρα ὑπὸ τῶν BE, EZ μετὰ τοῦ ἀπὸ τῆς HE >ίσον ἐστὶ τῷ ἀπὸ τῆς $H\Theta$. τῷ δὲ ἀπὸ τῆς $H\Theta$ >ίσα ἐστὶ τὰ ἀπὸ τῶν $\Theta E, EH$ τετράγωνα; τὸ >άρα ὑπὸ τῶν BE, EZ μετὰ τοῦ ἀπὸ HE >ίσα ἐστὶ τοῖς ἀπὸ τῶν $\Theta E, EH$. κοινὸν ἀφηρήσθω τὸ ἀπὸ τῆς HE τετράγωνον; λοιπὸν >άρα τὸ ὑπὸ τῶν BE, EZ περιεχόμενον >όρθογώνιον >ίσον ἐστὶ τῷ ἀπὸ τῆς $E\Theta$ τετραγώνῳ. ἀλλὰ τὸ ὑπὸ τῶν BE, EZ τὸ $B\Delta$ ἐστίν; >ίση γὰρ ἡ EZ τῇ $E\Delta$; τὸ >άρα $B\Delta$ παραλληλόγραμμον >ίσον ἐστὶ τῷ ἀπὸ τῆς ΘE τετραγώνῳ. >ίσον δὲ τὸ $B\Delta$ τῷ A εὐθυγράμμῳ. καὶ τὸ A >άρα εὐθύγραμμον >ίσον ἐστὶ τῷ ἀπὸ τῆς $E\Theta$ ἀναγραφισμένῳ τετραγώνῳ. Τῷ >άρα δοθέντι εὐθυγράμμῳ τῷ A >ίσον τετράγωνον συνέσταται τὸ ἀπὸ τῆς $E\Theta$ ἀναγραφισόμενον; <όπερ >έδει ποιῆσαι.

be drawn. And let DE be produced to H , and let GH be joined.

Therefore, since the straight-line BF has been cut—equally at G , and unequally at E —the rectangle contained by BE and EF , plus the square on EG , is thus equal to the square on GF [Prop. 2.5]. And GF (is) equal to GH . Thus, the (rectangle contained) by BE and EF , plus the (square) on GE , is equal to the (square) on GH . And the (sum of the) squares on HE and EG is equal to the (square) on GH [Prop. 1.47]. Thus, the (rectangle contained) by BE and EF , plus the (square) on GE , is equal to the (sum of the) squares on HE and EG . Let the square on GE be taken from both. Thus, the remaining rectangle contained by BE and EF is equal to the square on EH . But, BD is the (rectangle contained) by BE and EF . For EF (is) equal to ED . Thus, the parallelogram BD is equal to the square on HE . And BD (is) equal to the rectilinear figure A . Thus, the rectilinear figure A is also equal to the square (which) can be described on EH .

Thus, a square—(namely), that (which) can be described on EH —has been constructed, equal to the given rectilinear figure A . (Which is) the very thing it was required to do.

ELEMENTS BOOK 3
*Fundamentals of Plane Geometry Involving
Circles*

<Ὅροι.

1. >ἴσοι κύκλοι εἰσίν, <ὧν αἱ διάμετροι >ίσαι εἰσίν, >ἢ <ὧν αἱ ἐκ τῶν κέντρων >ίσαι εἰσίν.
2. Εὐθεῖα κύκλου ἐφάπτεσθαι λέγεται, <ἥτις ἀπτομένη τοῦ κύκλου καὶ ἐκβαλλομένη οὐ τέμνει τὸν κύκλον.
3. Κύκλοι ἐφάπτεσθαι ἀλλήλων λέγονται ο<ίτιν ἐξ ἀπτομένων ἀλλήλων οὐ τέμνουσιν ἀλλήλους.
4. Ἐν κύκλῳ >ίσον ἀπέθιν ἀπὸ τοῦ κέντρου εὐθεῖαι λέγονται, <ὅταν αἱ ἀπὸ τοῦ κέντρου ἐπ> αὐτὰς κάθετοι ἀγόμεναι >ίσαι >ῶσιν.
5. Μείζων δὲ ἀπέθιν λέγεται, ἐφ> <ἣν ἢ μείζων κάθετος πίπτει.
6. Τμήμα κύκλου ἐστὶ τὸ περιεχόμενον σθῆμα ὑπὸ τε εὐθείας καὶ κύκλου περιφερείας.
7. Τμήματοξ δὲ γωνία ἐστὶν ἡ περιεθρομένη ὑπὸ τε εὐθείας καὶ κύκλου περιφερείας.
8. Ἐν τμήματι δὲ γωνία ἐστίν, <ὅταν ἐπὶ τῇ περιφερείᾳ τοῦ τμήματος ληφθῇ τι σημεῖον καὶ ἀπ> αὐτοῦ ἐπὶ τὰ πέρατα τῇ εὐθείᾳ, <ἢ ἐστὶ βάσις τοῦ τμήματος, ἐπιζευθῶσιν εὐθεῖαι, ἡ περιεθρομένη γωνία ὑπὸ τῶν ἐπιζευθεισῶν εὐθειῶν.
9. <Ὅταν δὲ αἱ περιέθουσιν τὴν γωνίαν εὐθεῖαι ἀπολαμβάνωσιν τινὰ περιφέρειαν, ἐπ> ἐκείνῃ λέγεται βεβηκέναι ἡ γωνία.
10. Τομεὺς δὲ κύκλου ἐστίν, <ὅταν πρὸς τῷ κέντρῳ τοῦ κύκλου συσταθῇ γωνία, τὸ περιεχόμενον σθῆμα ὑπὸ τε τῶν τὴν γωνίαν περιέθουσῶν εὐθειῶν καὶ τῇ ἀπολαμβανομένηξ ὑπ> αὐτῶν περιφερείας.
11. <Ὅμοια τμήματα κύκλων ἐστὶ τὰ δεθόμενα γωνίαξ >ίσαξ, >ἢ ἐν ο<ίξ αἱ γωνίαι >ίσαι ἀλλήλαιξ εἰσίν.

1.

Τοῦ δοθέντος κύκλου τὸ κέντρον εὐρεῖν.

>Ἐστω ὁ δοθεὶς κύκλος ὁ $AB\Gamma$; δεῖ δὴ τοῦ $AB\Gamma$ κύκλου τὸ κέντρον εὐρεῖν.

Διήθθω τιξ εἰς αὐτόν, ὡς >έτυθεν, εὐθεῖα ἡ AB , καὶ τετμήσθω δίθα κατὰ τὸ Δ σημεῖον, καὶ ἀπὸ τοῦ Δ τῇ AB πρὸς ὀρθὰξ >ήθθω ἡ $\Delta\Gamma$ καὶ διήθθω ἐπὶ τὸ E , καὶ τετμήσθω ἡ ΓE δίθα κατὰ τὸ Z ; λέγω, <ὅτι τὸ Z κέντρον ἐστὶ τοῦ $AB\Gamma$ [κύκλου].

μὴ γάρ, ἀλλ> εἰ δυνατόν, >έστω τὸ H , καὶ ἐπεζεύθωσαν αἱ HA , $H\Delta$, HB . καὶ ἐπεὶ >ίση ἐστὶν ἡ $\Delta\Gamma$ τῇ ΔB , κοινὴ δὲ ἡ ΔH , δύο δὴ αἱ $\Delta\Delta$, ΔH δύο ταῖξ $H\Delta$, ΔB >ίσαι εἰσὶν ἑκατέρα ἑκατέρᾳ; καὶ βάσις ἡ HA βάσει τῇ HB ἐστὶν >ίση; ἐκ κέντρου γάρ; γωνία >άρα ἡ ὑπὸ $\Delta\Delta H$ γωνία τῇ ὑπὸ $H\Delta B$ >ίση ἐστίν. <ὅταν δὲ εὐθεῖα ἐπ> εὐθεῖαν σταθεῖσα τὰξ ἐφεχῇξ γωνίαξ >ίσαξ ἀλλήλαιξ ποιῇ, ὀρθὴ ἑκατέρα τῶν >ίσων γωνιῶν ἐστίν; ὀρθὴ >άρα ἐστὶν ἡ ὑπὸ $H\Delta B$. ἐστὶ δὲ καὶ ἡ ὑπὸ $Z\Delta B$ ὀρθή; >ίση >άρα ἡ ὑπὸ $Z\Delta B$ τῇ ὑπὸ $H\Delta B$, ἢ μείζων τῇ ἐλάττω; <ὅπερ ἐστὶν ἀδύνατον. οὐκ >άρα τὸ H κέντρον ἐστὶ τοῦ $AB\Gamma$ κύκλου. ὁμοίωξ δὴ δείχομεν, <ὅτι οὐδ> >άλλο τι πλὴν τοῦ Z .

Definitions

1. Equal circles are (circles) whose diameters are equal, or whose (distances) from the centers (to the circumferences) are equal (i.e., whose radii are equal).
2. A straight-line said to touch a circle is any (straight-line) which, meeting the circle and being produced, does not cut the circle.
3. Circles said to touch one another are any (circles) which, meeting one another, do not cut one another.
4. In a circle, straight-lines are said to be equally far from the center when the perpendiculars drawn to them from the center are equal.
5. And (that straight-line) is said to be further (from the center) on which the greater perpendicular falls (from the center).
6. A segment of a circle is the figure contained by a straight-line and a circumference of a circle.
7. And the angle of a segment is that contained by a straight-line and a circumference of a circle.
8. And the angle in a segment is the angle contained by the joined straight-lines, when any point is taken on the circumference of a segment, and straight-lines are joined from it to the ends of the straight-line which is the base of the segment.
9. And when the straight-lines containing an angle cut off some circumference, the angle is said to stand upon that (circumference).
10. And a sector of a circle is the figure contained by the straight-lines surrounding an angle, and the circumference cut off by them, when the angle is constructed at the center of a circle.
11. Similar segments of circles are those accepting equal angles, or in which the angles are equal to one another.

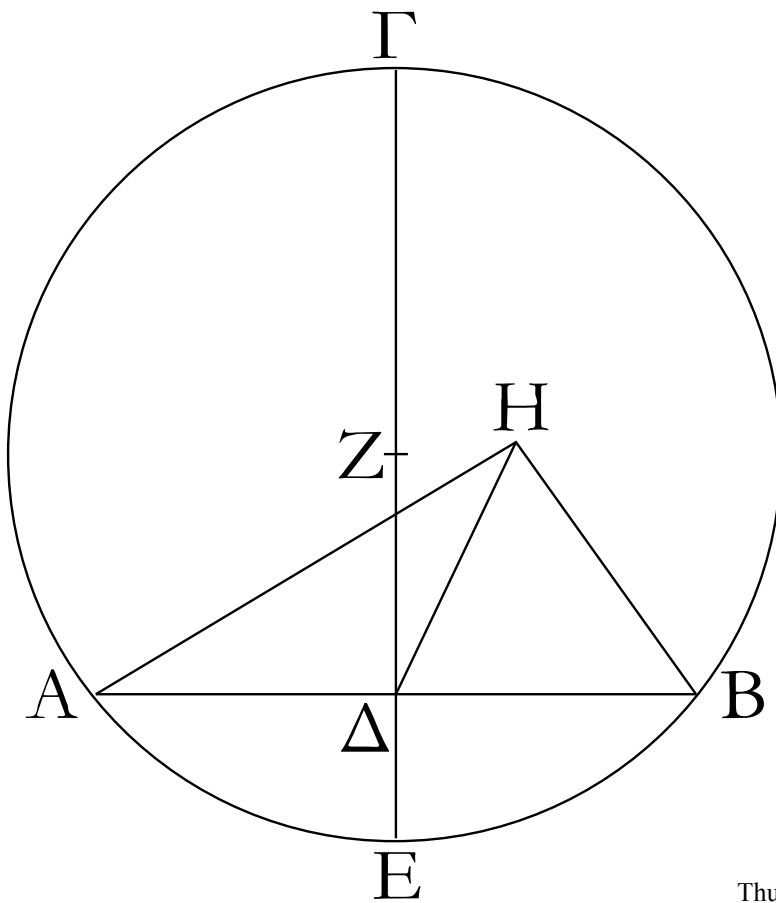
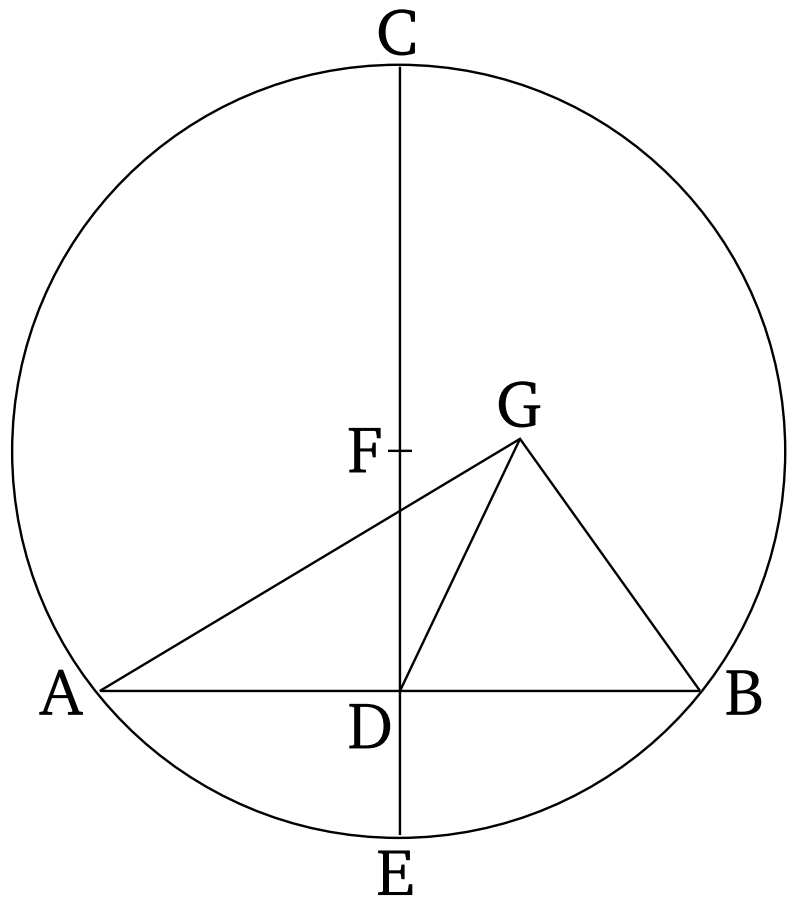
Proposition 1

To find the center of a given circle.

Let ABC be the given circle. So it is required to find the center of circle ABC .

Let some straight-line AB be drawn through (ABC), at random, and let (AB) be cut in half at point D [Prop. 1.9]. And let DC be drawn from D , at right-angles to AB [Prop. 1.11]. And let (CD) be drawn through to E . And let CE be cut in half at F [Prop. 1.9]. I say that (point) F is the center of the [circle] ABC .

For (if) not (then), if possible, let G (be the center of the circle), and let GA , GD , and GB be joined. And since AD is equal to DB , and DG (is) common, the two (straight-lines) AD , DG are equal to the two (straight-lines) BD , DG ,[†] respectively. And the base GA is equal to the base GB . For (they are both) radii. Thus, angle ADG is equal to angle GDB [Prop. 1.8]. And when a straight-line stood upon (another) straight-line make adjacent angles (which are) equal to one another, each of the equal angles is a right-angle [Def. 1.10]. Thus, GDB is a right-angle. And FDB is also a right-angle. Thus, FDB (is) equal to GDB , the greater to the lesser. The very thing is impossible. Thus, (point) G is not the center of the circle ABC . So, similarly, we can show that neither is any other (point) except F .



Τὸ Z ἄρα σημεῖον κέντρον ἐστὶ τοῦ $AB\Gamma$ [κύκλου].

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, <ὅτι ἔαν ἐν κύκλῳ εὐθεῖα τιξ

Thus, point F is the center of the [circle] ABC .

Corollary

So, from this, (it is) manifest that if any straight-line in a circle cuts any (other) straight-line in half, and at right-angles,

εὐθεϊάν τινα δίθα καὶ πρὸς ὀρθὰς τέμνη, ἐπὶ τῇς τεμνουσῆς
ἐστὶ τὸ κέντρον τοῦ κύκλου. — <ὅπερ >έδει ποιῆσαι.

† The Greek text has “ GD, DB ”, which is obviously a mistake.

2.

Ἐὰν κύκλου ἐπὶ τῇς περιφερείας ληφθῇ δύο τυθόντα
σημεῖα, ἢ ἐπὶ τὰ σημεῖα ἐπιζευγνυμένη εὐθεῖα ἐντὸς πεσεῖται
τοῦ κύκλου.

>Ἐστω κύκλος ὁ $AB\Gamma$, καὶ ἐπὶ τῇς περιφερείας αὐτοῦ
εἰλήφθω δύο τυθόντα σημεῖα τὰ A, B ; λέγω, <ὅτι ἢ ἀπὸ
τοῦ A ἐπὶ τὸ B ἐπιζευγνυμένη εὐθεῖα ἐντὸς πεσεῖται τοῦ
κύκλου.

μὴ γάρ, ἀλλ> εἰ δυνατόν, πιπτέτω ἐκτὸς ὡς ἡ AEB , καὶ
εἰλήφθω τὸ κέντρον τοῦ $AB\Gamma$ κύκλου, καὶ >έστω τὸ Δ , καὶ
ἐπεζεύθωσαν αἱ $\Delta A, \Delta B$, καὶ διήθω ἡ ΔZE .

Ἐπεὶ ο>ὦν >ίση ἐστὶν ἡ ΔA τῇ ΔB , >ίση >άρα καὶ γωνία
ἡ ὑπὸ ΔAE τῇ ὑπὸ ΔBE ; καὶ ἐπεὶ τριγώνου τοῦ ΔAE μία
πλευρὰ προσεκβέβληται ἡ AEB , μείζων >άρα ἡ ὑπὸ ΔEB
γωνία τῇς ὑπὸ ΔAE . >ίση δὲ ἡ ὑπὸ ΔAE τῇ ὑπὸ ΔBE ;
μείζων >άρα ἡ ὑπὸ ΔEB τῇς ὑπὸ ΔBE . ὑπὸ δὲ τὴν μείζονα
γωνίαν ἡ μείζων πλευρὰ ὑποτείνει; μείζων >άρα ἡ ΔB τῇς
 ΔE . >ίση δὲ ἡ ΔB τῇ ΔZ . μείζων >άρα ἡ ΔZ τῇς ΔE ἢ
ἐλάττων τῇς μείζονος; <ὅπερ ἐστὶν ἀδύνατον. οὐκ >άρα ἢ
ἀπὸ τοῦ A ἐπὶ τὸ B ἐπιζευγνυμένη εὐθεῖα ἐκτὸς πεσεῖται
τοῦ κύκλου. ὁμοίως δὲ δείχομεν, <ὅτι οὐδὲ ἐπ> αὐτῇς τῇς
περιφερείας; ἐντὸς >άρα.

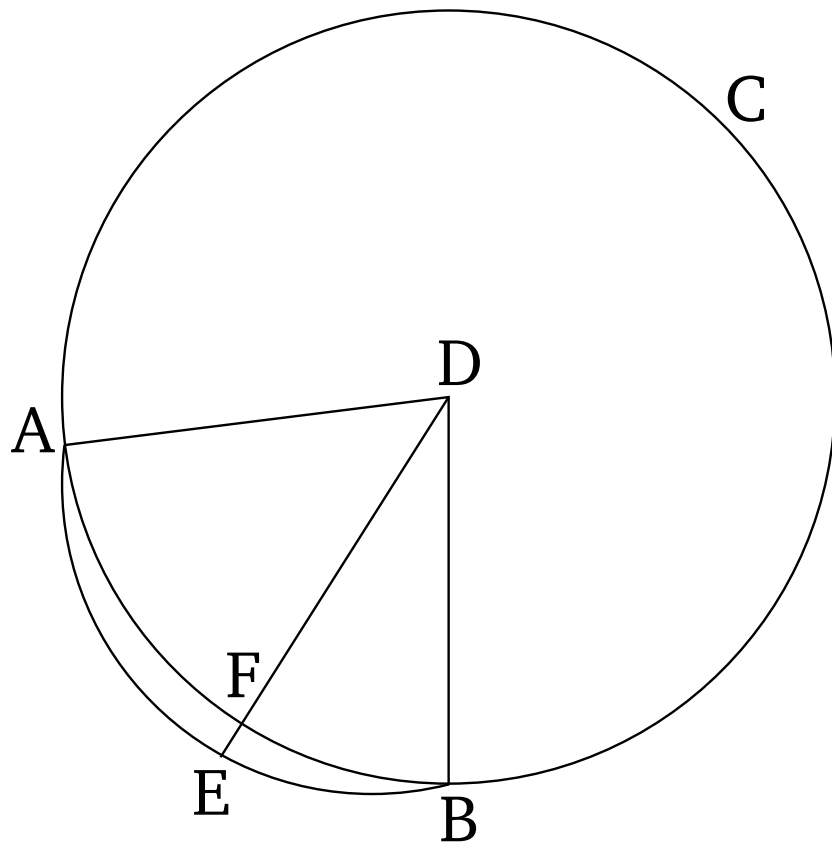
(then) the center of the circle is on the former (straight-line).
— (Which is) the very thing it was required to do.

Proposition 2

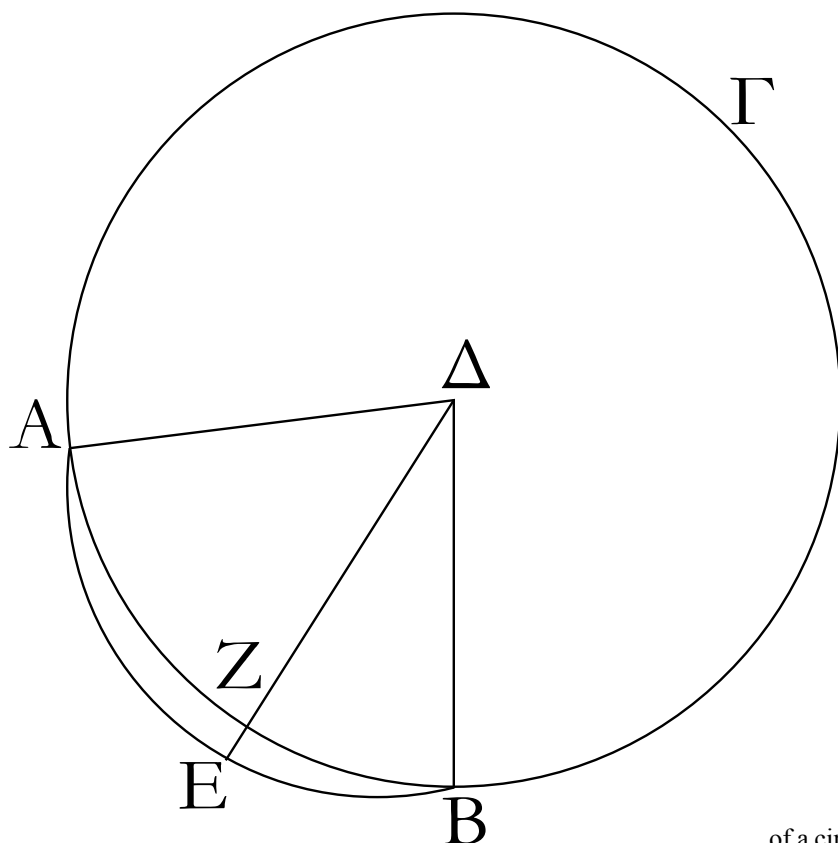
If two points are taken at random on the circumference of
a circle, (then) the straight-line joining the points will fall
inside the circle.

Let ABC be a circle, and let two points A and B be taken
at random on its circumference. I say that the straight-line
joining A to B will fall inside the circle.

For (if) not (then), if possible, let it fall outside (the circle),
like AEB (in the figure). And let the center of the circle
 ABC be found [Prop. 3.1], and let it be (at point) D . And
let DA and DB be joined, and let DFE be drawn through.
Therefore, since DA is equal to DB , the angle DAE (is)
thus also equal to DBE [Prop. 1.5]. And since in triangle
 DAE the one side, AEB , has been produced, angle DEB
(is) thus greater than DAE [Prop. 1.16]. And DAE (is)
equal to DBE [Prop. 1.5]. Thus, DEB (is) greater than
 DBE . And the greater angle is subtended by the greater
side [Prop. 1.19]. Thus, DB (is) greater than DE . And DB
(is) equal to DF . Thus, DF (is) greater than DE , the lesser
than the greater. The very thing is impossible. Thus, the
straight-line joining A to B will not fall outside the circle.
So, similarly, we can show that neither (will it fall) on the
circumference itself. Thus, (it will fall) inside (the circle).



Thus, if two points are taken at random on the circumference



Ἐάν >άρα κύκλου ἐπὶ τῇ περιφερείᾳ ληφθῇ δύο τυθόντα σημεῖα, ἢ ἐπὶ τὰ σημεῖα ἐπιζευγνυμένη εὐθεῖα ἐντὸς πεσεῖται τοῦ κύκλου; <όπερ >έδει δεῖξαι.

3.

Ἐάν ἐν κύκλῳ εὐθεῖα τις διὰ τοῦ κέντρου εὐθεῖαν τινα μὴ διὰ τοῦ κέντρου διῆα τέμνη, καὶ πρὸς ὀρθὰς αὐτὴν τέμνει; καὶ ἔαν πρὸς ὀρθὰς αὐτὴν τέμνη, καὶ διῆα αὐτὴν τέμνει.

>Ἐστω κύκλος ὁ ΑΒΓ, καὶ ἐν αὐτῷ εὐθεῖα τις διὰ τοῦ κέντρου ἡ ΓΔ εὐθεῖαν τινα μὴ διὰ τοῦ κέντρου τὴν ΑΒ διῆα τεμνέτω κατὰ τὸ Ζ σημεῖον; λέγω, <ὅτι καὶ πρὸς ὀρθὰς αὐτὴν τέμνει.

Εἰλήφθω γάρ τὸ κέντρον τοῦ ΑΒΓ κύκλου, καὶ >έστω τὸ Ε, καὶ ἐπεζεύθωσαν αἱ ΕΑ, ΕΒ.

Καὶ ἐπεὶ >ῖση ἐστὶν ἡ ΑΖ τῇ ΖΒ, κοινὴ δὲ ἡ ΖΕ, δύο δυσὶν >ῖσαι [εἰσὶν]; καὶ βάσεις ἡ ΕΑ βάσει τῇ ΕΒ >ῖση; γωνία >άρα ἡ ὑπὸ ΑΖΕ γωνία τῇ ὑπὸ ΒΖΕ >ῖση ἐστίν. <ὅταν δὲ εὐθεῖα ἐπ' εὐθεῖαν σταθεῖσα τὰς ἐφεχθῆς γωνίας >ῖσαξ ἀλλήλαις ποιῇ, ὀρθὴ ἑκατέρα τῶν >ῖσων γωνιῶν ἐστίν; ἑκατέρα >άρα τῶν ὑπὸ ΑΖΕ, ΒΖΕ ὀρθή ἐστίν. ἡ ΓΔ >άρα διὰ τοῦ κέντρου ο>ύσα τὴν ΑΒ μὴ διὰ τοῦ κέντρου ο>ύσαν διῆα τέμνουσα καὶ πρὸς ὀρθὰς τέμνει.

Ἀλλὰ δὴ ἡ ΓΔ τὴν ΑΒ πρὸς ὀρθὰς τεμνέτω; λέγω, <ὅτι καὶ διῆα αὐτὴν τέμνει, τουτέστιν, <ὅτι >ῖση ἐστὶν ἡ ΑΖ τῇ ΖΒ. Τῶν γὰρ αὐτῶν κατασκευασθέντων, ἐπεὶ >ῖση ἐστὶν ἡ ΕΑ τῇ ΕΒ, >ῖση ἐστὶ καὶ γωνία ἡ ὑπὸ ΕΑΖ τῇ ὑπὸ ΕΒΖ. ἐστὶ δὲ καὶ ὀρθὴ ἡ ὑπὸ ΑΖΕ ὀρθὴ τῇ ὑπὸ ΒΖΕ >ῖση; δύο >άρα τρίγωνά ἐστι ΕΑΖ, ΕΒΖ τὰς δύο γωνίας δυσὶ γωνίαῖς >ῖσαξ >έθοντα καὶ μίαν πλευρὰν μιᾷ πλευρᾷ >ῖσην κοινὴν αὐτῶν τὴν ΕΖ ὑποτείνουσιν ὑπὸ μίαν τῶν >ῖσων γωνιῶν; καὶ τὰς λοιπὰς >άρα πλευρὰς ταῖς λοιπαῖς πλευραῖς >ῖσαξ <έχει; >ῖση >άρα ἡ ΑΖ τῇ ΖΒ.

of a circle, (then) the straight-line joining the points will fall inside the circle. (Which is) the very thing it was required to show.

Proposition 3

In a circle, if any straight-line through the center cuts in half any straight-line not through the center, (then) it also cuts it at right-angles. And (conversely) if it cuts it at right-angles, (then) it also cuts it in half.

Let ABC be a circle, and, within it, let some straight-line through the center, CD , cut in half some straight-line not through the center, AB , at the point F . I say that (CD) also cuts (AB) at right-angles.

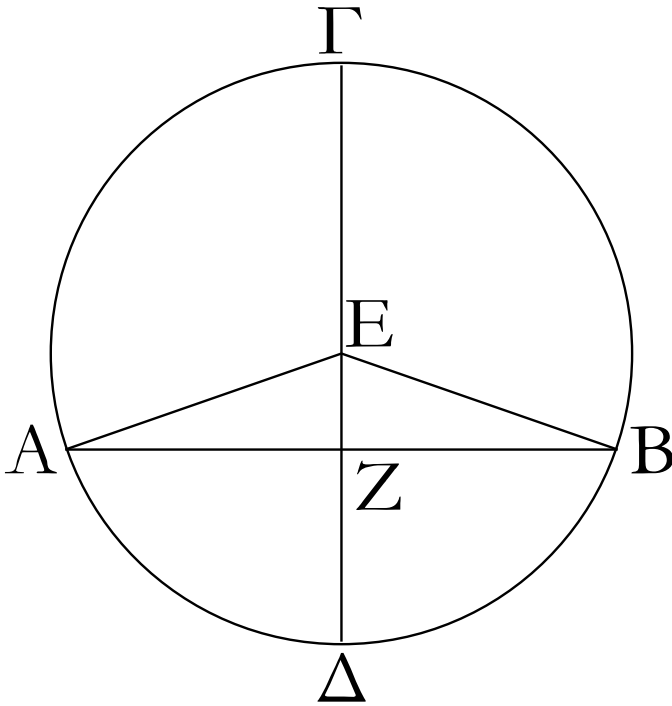
For let the center of the circle ABC be found [Prop. 3.1], and let it be (at point) E , and let EA and EB be joined.

And since AF is equal to FB , and FE (is) common, two (sides of triangle AFE) [are] equal to two (sides of triangle BFE). And the base EA (is) equal to the base EB . Thus, angle AFE is equal to angle BFE [Prop. 1.8]. And when a straight-line stood upon (another) straight-line makes adjacent angles (which are) equal to one another, each of the equal angles is a right-angle [Def. 1.10]. Thus, AFE and BFE are each right-angles. Thus, the (straight-line) CD , which is through the center and cuts in half the (straight-line) AB , which is not through the center, also cuts (AB) at right-angles.

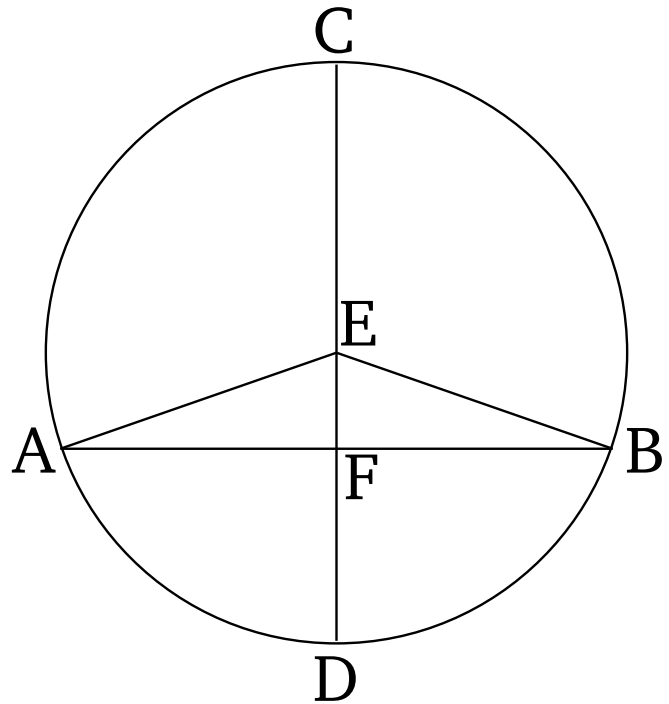
And so let CD cut AB at right-angles. I say that it also cuts (AB) in half. That is to say, that AF is equal to FB .

For, with the same construction, since EA is equal to EB , angle EAF is also equal to EBF [Prop. 1.5]. And the right-angle AFE is also equal to the right-angle BFE . Thus, EAF and EBF are two triangles having two angles equal to two angles, and one side equal to one side—(namely), their common (side) EF , subtending one of the equal an-

gles. Thus, they will also have the remaining sides equal to the (corresponding) remaining sides [Prop. 1.26]. Thus,



AF (is) equal to FB .



Ἐάν >άρα ἐν κύκλῳ εὐθεῖά τις διὰ τοῦ κέντρου εὐθεῖάν τινα μὴ διὰ τοῦ κέντρου δίθῃα τέμνῃ, καὶ πρὸς ὀρθὰς αὐτὴν τέμνῃ; καὶ ἔαν πρὸς ὀρθὰς αὐτὴν τέμνῃ, καὶ δίθῃα αὐτὴν τέμνῃ; <όπερ >έδει δεῖξαι.

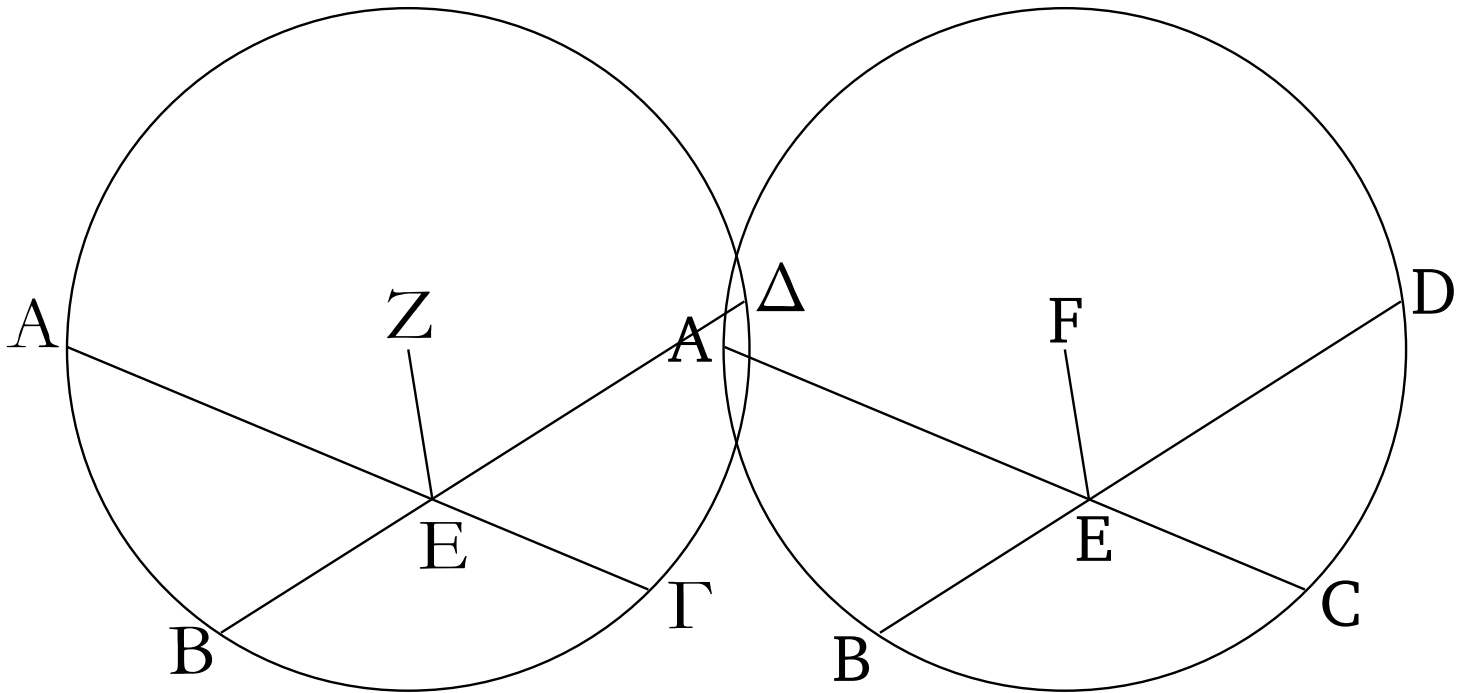
Thus, in a circle, if any straight-line through the center cuts in half any straight-line not through the center, (then) it also cuts it at right-angles. And (conversely) if it cuts it at right-angles, (then) it also cuts it in half. (Which is) the very thing it was required to show.

4.

Ἐάν ἐν κύκλῳ δύο εὐθεῖαι τέμνωσιν ἀλλήλας μὴ διὰ τοῦ κέντρου ο>ῦσαι, οὐ τέμνουσιν ἀλλήλας δίθῃα.

Proposition 4

In a circle, if two straight-lines, which are not through the center, cut one another, (then) they do not cut one another in half.



>Ἐστω κύκλος ὁ ΑΒΓΔ, καὶ ἐν αὐτῷ δύο εὐθεῖαι αἱ ΑΓ, ΒΔ τεμνέτωσαν ἀλλήλας κατὰ τὸ Ε μὴ διὰ τοῦ κέντρου οὕσαι; λέγω, <ὅτι οὐ τέμνουσιν ἀλλήλας δίθα.

Εἰ γὰρ δυνατόν, τεμνέτωσαν ἀλλήλας δίθα <ώστε >ίσην εἶναι τὴν μὲν ΑΕ τῇ ΕΓ, τὴν δὲ ΒΕ τῇ ΕΔ; καὶ εἰλήφθω τὸ κέντρον τοῦ ΑΒΓΔ κύκλου, καὶ >έστω τὸ Ζ, καὶ ἐπεζεύθω ἡ ΖΕ.

Ἐπεὶ οὕν εὐθεῖα τις διὰ τοῦ κέντρου ἡ ΖΕ εὐθειάν τινα μὴ διὰ τοῦ κέντρου τὴν ΑΓ δίθα τέμνει, καὶ πρὸς ὀρθὰς αὐτὴν τέμνει; ὀρθὴ >άρα ἐστὶν ἡ ὑπὸ ΖΕΑ; πάλιν, ἐπεὶ εὐθεῖα τις ἡ ΖΕ εὐθειάν τινα τὴν ΒΔ δίθα τέμνει, καὶ πρὸς ὀρθὰς αὐτὴν τέμνει; ὀρθὴ >άρα ἡ ὑπὸ ΖΕΒ. ἐδείκθη δὲ καὶ ἡ ὑπὸ ΖΕΑ ὀρθή; >ίση >άρα ἡ ὑπὸ ΖΕΑ τῇ ὑπὸ ΖΕΒ ἢ ἐλάττων τῇ μείζονι; <ὅπερ ἐστὶν ἀδύνατον. οὐκ >άρα αἱ ΑΓ, ΒΔ τέμνουσιν ἀλλήλας δίθα.

Ἐὰν >άρα ἐν κύκλῳ δύο εὐθεῖαι τέμνωσιν ἀλλήλας μὴ διὰ τοῦ κέντρου οὕσαι, οὐ τέμνουσιν ἀλλήλας δίθα; <ὅπερ >έδει δεῖχαι.

Let $ABCD$ be a circle, and within it, let two straight-lines, AC and BD , which are not through the center, cut one another at (point) E . I say that they do not cut one another in half.

For, if possible, let them cut one another in half, such that AE is equal to EC , and BE to ED . And let the center of the circle $ABCD$ be found [Prop. 3.1], and let it be (at point) F , and let FE be joined.

Therefore, since some straight-line through the center, FE , cuts in half some straight-line not through the center, AC , it also cuts it at right-angles [Prop. 3.3]. Thus, FEA is a right-angle. Again, since some straight-line FE cuts in half some straight-line BD , it also cuts it at right-angles [Prop. 3.3]. Thus, FEB (is) a right-angle. But FEA was also shown (to be) a right-angle. Thus, FEA (is) equal to FEB , the lesser to the greater. The very thing is impossible. Thus, AC and BD do not cut one another in half.

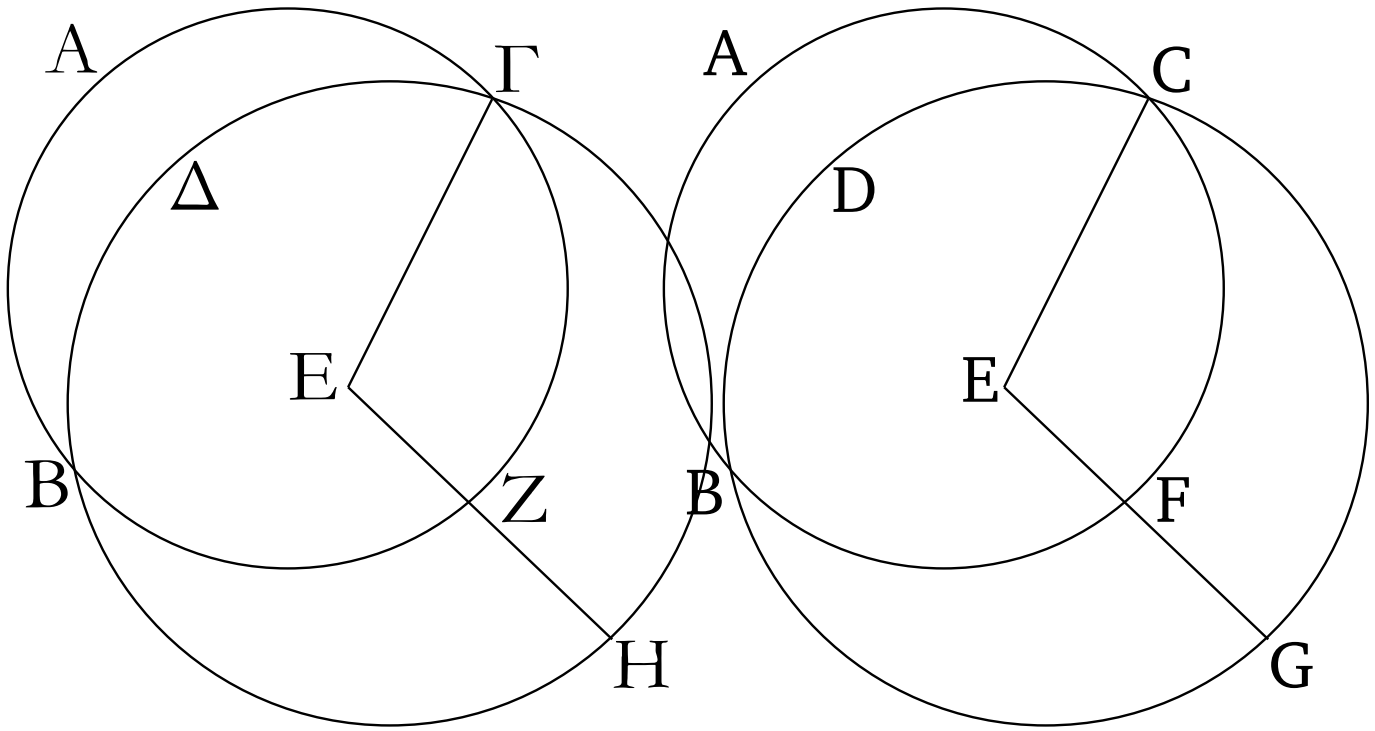
Thus, in a circle, if two straight-lines, which are not through the center, cut one another, (then) they do not cut one another in half. (Which is) the very thing it was required to show.

5.

Ἐὰν δύο κύκλοι τέμνωσιν ἀλλήλους, οὐκ >έσται αὐτῶν τὸ αὐτὸ κέντρον.

Proposition 5

If two circles cut one another, (then) they will not have the same center.



Δύο γὰρ κύκλοι οἱ $AB\Gamma$, $\Gamma\Delta\Theta$ τεμνέτωσαν ἀλλήλους κατὰ τὰ B , Γ σημεία. λέγω, <ὅτι οὐκ >έσται αὐτῶν τὸ αὐτὸ κέντρον. Εἰ γὰρ δυνατόν, >έστω τὸ E , καὶ ἐπεζεύθω ἡ $E\Gamma$, καὶ διήθω ἡ EZH , ὥς >έτυθεν. καὶ ἐπεὶ τὸ E σημεῖον κέντρον ἐστὶ τοῦ $AB\Gamma$ κύκλου, <ίση ἐστὶν ἡ $E\Gamma$ τῇ EZ . πάλιν, ἐπεὶ τὸ E σημεῖον κέντρον ἐστὶ τοῦ $\Gamma\Delta\Theta$ κύκλου, >ίση ἐστὶν ἡ $E\Gamma$ τῇ EH ; ἐδείθθη δὲ ἡ $E\Gamma$ καὶ τῇ EZ >ίση; καὶ ἡ EZ >άρα τῇ EH ἐστὶν >ίση ἢ ἐλάσσων τῇ μείζονι; <όπερ ἐστὶν ἀδύνατον. οὐκ >άρα τὸ E σημεῖον κέντρον ἐστὶ τῶν $AB\Gamma$, $\Gamma\Delta\Theta$ κύκλων. Ἐὰν >άρα δύο κύκλοι τέμνωσιν ἀλλήλους, οὐκ >έστιν αὐτῶν τὸ αὐτὸ κέντρον; <όπερ >έδει δεῖξαι.

For let the two circles ABC and CDG cut one another at points B and C . I say that they will not have the same center. For, if possible, let E be (the common center), and let EC be joined, and let EFG be drawn through (the two circles), at random. And since point E is the center of the circle ABC , EC is equal to EF . Again, since point E is the center of the circle CDG , EC is equal to EG . But EC was also shown (to be) equal to EF . Thus, EF is also equal to EG , the lesser to the greater. The very thing is impossible. Thus, point E is not the (common) center of the circles ABC and CDG .

Thus, if two circles cut one another, (then) they will not have the same center. (Which is) the very thing it was required to show.

6.

Ἐὰν δύο κύκλοι ἐφάπτονται ἀλλήλων, οὐκ >έσται αὐτῶν τὸ αὐτὸ κέντρον.

Δύο γὰρ κύκλοι οἱ $AB\Gamma$, $\Gamma\Delta E$ ἐφαπτέσθωσαν ἀλλήλων κατὰ τὸ Γ σημεῖον; λέγω, <ὅτι οὐκ >έσται αὐτῶν τὸ αὐτὸ κέντρον. Εἰ γὰρ δυνατόν, >έστω τὸ Z , καὶ ἐπεζεύθω ἡ $Z\Gamma$, καὶ διήθω, ὥς >έτυθεν, ἡ ZEB .

Ἐπεὶ οὖν τὸ Z σημεῖον κέντρον ἐστὶ τοῦ $AB\Gamma$ κύκλου, >ίση ἐστὶν ἡ $Z\Gamma$ τῇ ZB . πάλιν, ἐπεὶ τὸ Z σημεῖον κέντρον ἐστὶ τοῦ $\Gamma\Delta E$ κύκλου, >ίση ἐστὶν ἡ $Z\Gamma$ τῇ ZE . ἐδείθθη δὲ ἡ $Z\Gamma$ τῇ ZB >ίση; καὶ ἡ ZE >άρα τῇ ZB ἐστὶν >ίση, ἢ ἐλάττων τῇ μείζονι; <όπερ ἐστὶν ἀδύνατον. οὐκ >άρα τὸ Z σημεῖον κέντρον ἐστὶ τῶν $AB\Gamma$, $\Gamma\Delta E$ κύκλων.

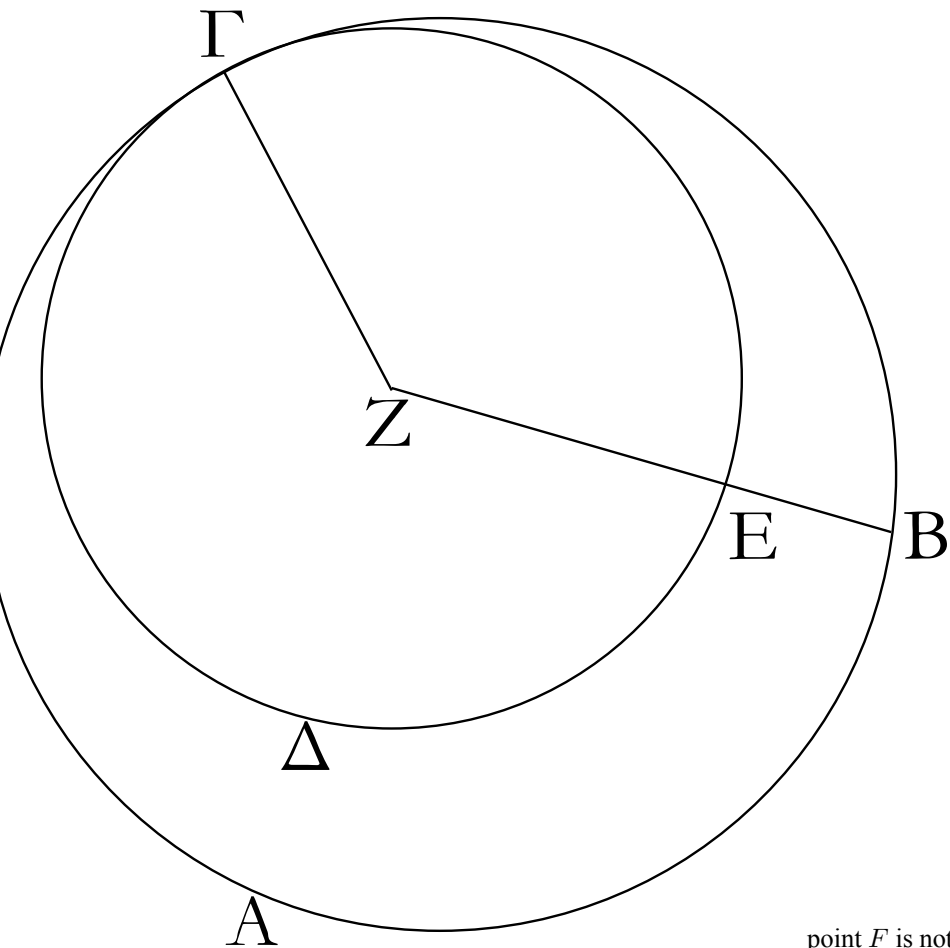
Proposition 6

If two circles touch one another, (then) they will not have the same center.

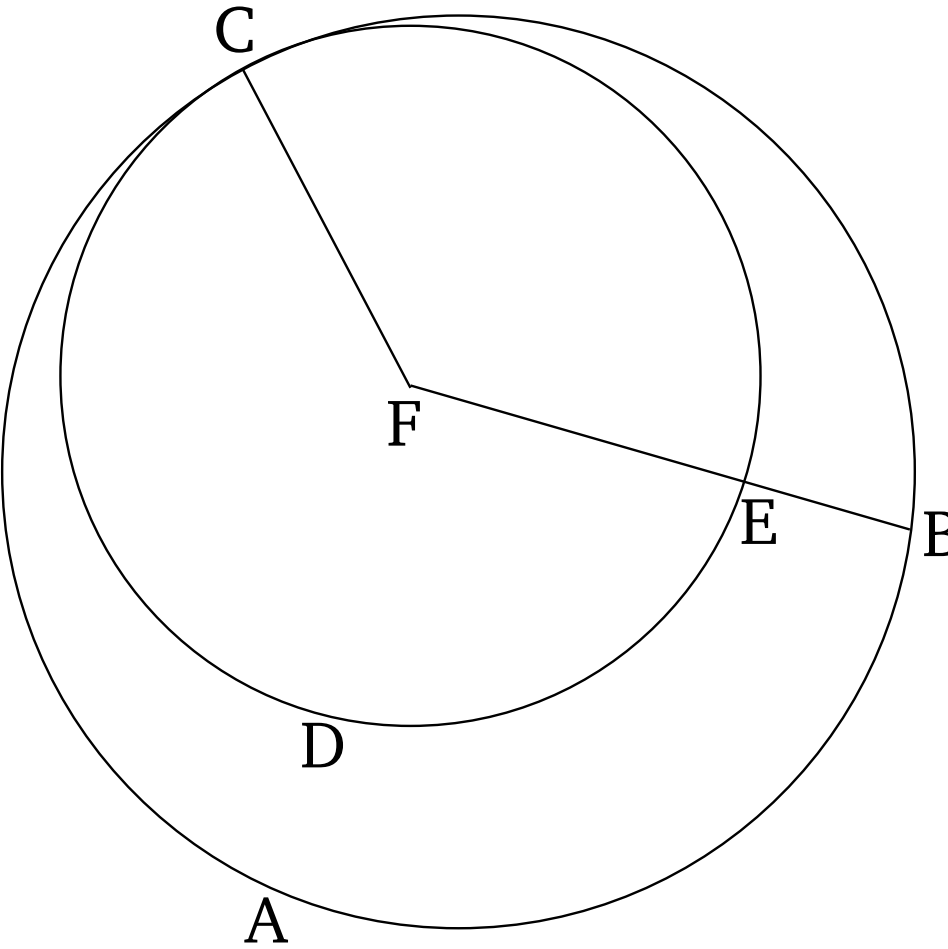
For let the two circles ABC and CDE touch one another at point C . I say that they will not have the same center.

For, if possible, let F be (the common center), and let FC be joined, and let FEB be drawn through (the two circles), at random.

Therefore, since point F is the center of the circle ABC , FC is equal to FB . Again, since point F is the center of the circle CDE , FC is equal to FE . But FC was shown (to be) equal to FB . Thus, FE is also equal to FB , the lesser to the greater. The very thing is impossible. Thus,



Ἐὰν ᾤκηρα δύο κύκλοι ἐφάπτωνται ἀλλήλων, οὐκ ᾤκηται ἡ κοινὴ κέντρον. point *F* is not the (common) center of the circles *ABC* and *CDE*.



αὐτῶν τὸ αὐτὸ κέντρον; <ὅπερ> ᾤδει δεῖχαι.

7.

Ἐὰν κύκλου ἐπὶ τῆς διαμέτρου ληφθῇ τι σημεῖον, <ὃ μὴ ἔστι κέντρον τοῦ κύκλου, ἀπὸ δὲ τοῦ σημείου πρὸς τὸν κύκλον προσπίπτωσιν εὐθεῖαι τινεῖς, μεγίστη μὲν >έσται, ἐφ' <ἣς τὸ κέντρον, ἐλαθίστη δὲ ἡ λοιπή, τῶν δὲ >άλλων αἰ ἡ >έγγιον τῆς διὰ τοῦ κέντρου τῆς ἀπώτερον μείζων ἔστί, δύο δὲ μόνον >ῖσαι ἀπὸ τοῦ σημείου προσπεσοῦνται πρὸς τὸν κύκλον ἐφ' ἑκάτερα τῆς ἐλαθίστης.

>Ἐστω κύκλος ὁ ΑΒΓΔ, διάμετρος δὲ αὐτοῦ >έστω ἡ ΑΔ, καὶ ἐπὶ τῆς ΑΔ εἰλήφθω τι σημεῖον τὸ Ζ, <ὃ μὴ ἔστι κέντρον τοῦ κύκλου, κέντρον δὲ τοῦ κύκλου >έστω τὸ Ε, καὶ ἀπὸ τοῦ Ζ πρὸς τὸν ΑΒΓΔ κύκλον προσπιπτέτωσαν εὐθεῖαι τινεῖς αἰ ΖΒ, ΖΓ, ΖΗ; λέγω, <ὅτι μεγίστη μὲν ἔστιν ἡ ΖΑ, ἐλαθίστη δὲ ἡ ΖΔ, τῶν δὲ >άλλων ἡ μὲν ΖΒ τῆς ΖΓ μείζων, ἡ δὲ ΖΓ τῆς ΖΗ.

Ἐπεξεύθθωσαν γὰρ αἱ ΒΕ, ΓΕ, ΗΕ. καὶ ἐπεὶ παντὸς τριγώνου αἱ δύο πλευραὶ τῆς λοιπῆς μείζονές εἰσιν, αἱ >άρα ΕΒ, ΕΖ τῆς ΒΖ μείζονές εἰσιν. >ῖση δὲ ἡ ΑΕ τῇ ΒΕ [αἱ >άρα ΒΕ, ΕΖ >ῖσαι εἰσὶ τῇ ΑΖ]; μείζων >άρα ἡ ΑΖ τῆς ΒΖ. πάλιν, ἐπεὶ >ῖση ἔστιν ἡ ΒΕ τῇ ΓΕ, κοινὴ δὲ ἡ ΖΕ, δύο δὲ αἱ ΒΕ, ΕΖ δυσὶ ταῖς ΓΕ, ΕΖ >ῖσαι εἰσίν. ἀλλὰ καὶ γωνία ἡ ὑπὸ ΒΕΖ γωνία τῆς ὑπὸ ΓΕΖ μείζων; βάσις >άρα ἡ ΒΖ βάσεως τῆς ΓΖ μείζων ἔστί. διὰ τὰ αὐτὰ δὲ καὶ ἡ ΓΖ τῆς ΖΗ μείζων ἔστί.

Πάλιν, ἐπεὶ αἱ ΗΖ, ΖΕ τῆς ΕΗ μείζονές εἰσιν, >ῖση δὲ ἡ ΕΗ τῇ ΕΔ, αἱ >άρα ΗΖ, ΖΕ τῆς ΕΔ μείζονές εἰσιν. κοινὴ ἀφηρήσθω ἡ ΕΖ; λοιπὴ >άρα ἡ ΗΖ λοιπῆς τῆς ΖΔ μείζων ἔστί. μεγίστη μὲν >άρα ἡ ΖΑ, ἐλαθίστη δὲ ἡ ΖΔ, μείζων δὲ ἡ μὲν ΖΒ τῆς ΖΓ, ἡ δὲ ΖΓ τῆς ΖΗ.

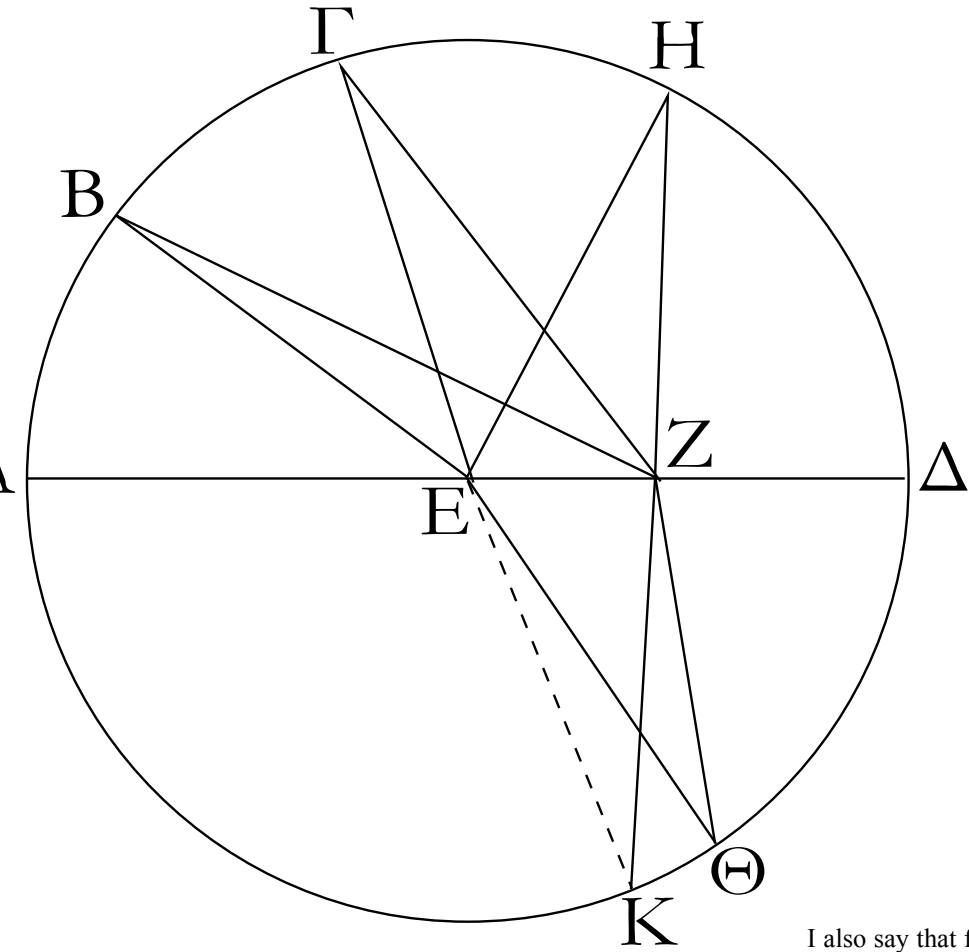
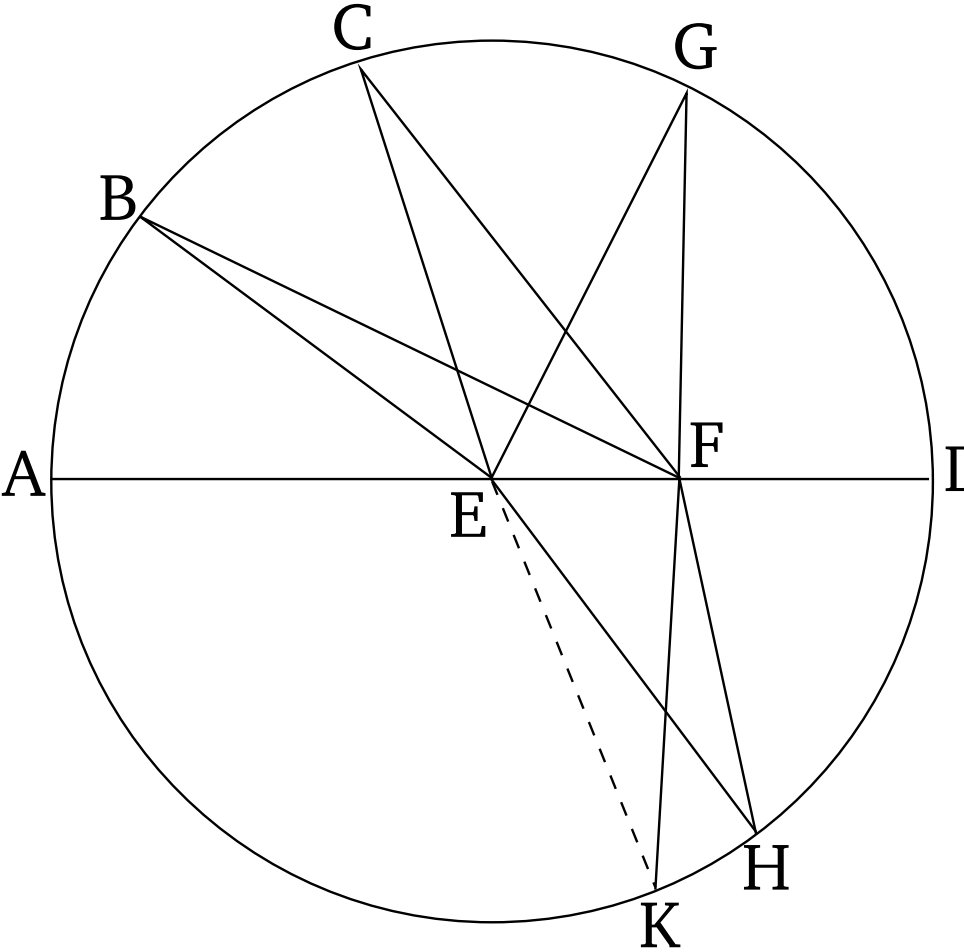
Thus, if two circles touch one another, (then) they will not have the same center. (Which is) the very thing it was required to show.

Proposition 7

If some point, which is not the center of the circle, is taken on the diameter of a circle, and some straight-lines radiate from the point towards the (circumference of the) circle, (then) the greatest (straight-line) will be that on which the center (lies), and the least the remainder (of the same diameter). And for the others, a (straight-line) nearer[†] to the (straight-line) through the center is always greater than a (straight-line) further away. And only two equal (straight-lines) will radiate from the point towards the (circumference of the) circle, (one) on each (side) of the least (straight-line). Let $ABCD$ be a circle, and let AD be its diameter, and let some point F , which is not the center of the circle, be taken on AD . Let E be the center of the circle. And let some straight-lines, FB , FC , and FG , radiate from F towards (the circumference of) circle $ABCD$. I say that FA is the greatest (straight-line), FD the least, and of the others, FB (is) greater than FC , and FC than FG .

For let BE , CE , and GE be joined. And since for every triangle (any) two sides are greater than the remaining (side) [Prop. 1.20], EB and EF is thus greater than BF . And AE (is) equal to BE [thus, BE and EF is equal to AF]. Thus, AF (is) greater than BF . Again, since BE is equal to CE , and FE (is) common, the two (straight-lines) BE , EF are equal to the two (straight-lines) CE , EF (respectively). But, angle BEF (is) also greater than angle CEF .[‡] Thus, the base BF is greater than the base CF . Thus, the base BF is greater than the base CF [Prop. 1.24]. So, for the same (reasons), CF is also greater than FG .

Again, since GF and FE are greater than EG [Prop. 1.20], and EG (is) equal to ED , GF and FE are thus greater than ED . Let EF be taken from both. Thus, the remainder GF is greater than the remainder FD . Thus, FA (is) the greatest (straight-line), FD the least, and FB (is) greater than FC , and FC than FG .



Λέγω, <ότι και από τοῦ Z σημείου δύο μόνον>ῖσαι προσπεσοῦν will radiate towards (the circumference of) circle $ABCD$,

πρὸς τὸν ΑΒΓΔ κύκλον ἐφ' ἑκάτερα τῆς ΖΔ ἐλαθίστης. συνεστάτω γὰρ πρὸς τῇ ΕΖ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ Ε τῇ ὑπὸ ΗΕΖ γωνίᾳ >ίση ἢ ὑπὸ ΖΕΘ, καὶ ἐπεζεύθθω ἡ ΖΘ. ἐπεὶ οὖν >ίση ἐστὶν ἡ ΗΕ τῇ ΕΘ, κοινὴ δὲ ἡ ΕΖ, δύο δὲ αἱ ΗΕ, ΕΖ δυοὶ ταῖς ΘΕ, ΕΖ >ίσαι εἰσὶν; καὶ γωνία ἡ ὑπὸ ΗΕΖ γωνία τῇ ὑπὸ ΘΕΖ >ίση; βάσις >άρα ἡ ΖΗ βάσει τῇ ΖΘ >ίση ἐστίν. λέγω δὲ, <ὅτι τῇ ΖΗ >άλλη >ίση οὐ προσπесеῖται πρὸς τὸν κύκλον ἀπὸ τοῦ Ζ σημείου. εἰ γὰρ δυνατόν, προσπιπτέτω ἡ ΖΚ. καὶ ἐπεὶ ἡ ΖΚ τῇ ΖΗ >ίση ἐστίν, ἀλλὰ ἡ ΖΘ τῇ ΖΗ [>ίση ἐστίν], καὶ ἡ ΖΚ >άρα τῇ ΖΘ ἐστὶν >ίση, ἡ >έγγιον τῆς διὰ τοῦ κέντρου τῇ ἀπώτερον >ίση; <ὅπερ ἀδύνατον. οὐκ >άρα ἀπὸ τοῦ Ζ σημείου ἑτέρα τις προσπесеῖται πρὸς τὸν κύκλον >ίση τῇ ΗΖ; μία >άρα μόνη.

Ἐὰν >άρα κύκλου ἐπὶ τῆς διαμέτρου ληφθῇ τι σημεῖον, <ὃ μὴ ἐστὶ κέντρον τοῦ κύκλου, ἀπὸ δὲ τοῦ σημείου πρὸς τὸν κύκλον προσπίπτουσιν εὐθεῖαι τινεῖς, μεγίστη μὲν >έσται, ἐφ' <ῆς τὸ κέντρον, ἐλαθίστη δὲ ἡ λοιπή, τῶν δὲ >άλλων αἰ ἡ >έγγιον τῆς διὰ τοῦ κέντρου τῆς ἀπώτερον μείζων ἐστίν, δύο δὲ μόνον >ίσαι ἀπὸ τοῦ αὐτοῦ σημείου προσπесοῦνται πρὸς τὸν κύκλον ἐφ' ἑκάτερα τῆς ἐλαθίστης; <ὅπερ >έδει δεῖχαι.

(one) on each (side) of the least (straight-line) FD . For let the (angle) FEH , equal to angle GEF , be constructed on the straight-line EF , at the point E on it [Prop. 1.23], and let FH be joined. Therefore, since GE is equal to EH , and EF (is) common, the two (straight-lines) GE , EF are equal to the two (straight-lines) HE , EF (respectively). And angle GEF (is) equal to angle HEF . Thus, the base FG is equal to the base FH [Prop. 1.4]. So I say that another (straight-line) equal to FG will not radiate towards (the circumference of) the circle from point F . For, if possible, let FK (so) radiate. And since FK is equal to FG , but FH [is equal] to FG , FK is thus also equal to FH , the nearer to the (straight-line) through the center equal to the further away. The very thing (is) impossible. Thus, another (straight-line) equal to GF will not radiate from the point F towards (the circumference of) the circle. Thus, (there is) only one (such straight-line).

Thus, if some point, which is not the center of the circle, is taken on the diameter of a circle, and some straight-lines radiate from the point towards the (circumference of the) circle, (then) the greatest (straight-line) will be that on which the center (lies), and the least the remainder (of the same diameter). And for the others, a (straight-line) nearer to the (straight-line) through the center is always greater than a (straight-line) further away. And only two equal (straight-lines) will radiate from the same point towards the (circumference of the) circle, (one) on each (side) of the least (straight-line). (Which is) the very thing it was required to show.

† Presumably, in an angular sense.

‡ This is not proved, except by reference to the figure.

8.

Ἐὰν κύκλου ληφθῇ τι σημεῖον ἐκτός, ἀπὸ δὲ τοῦ σημείου πρὸς τὸν κύκλον διαθῶσιν εὐθεῖαι τινεῖς, <ὧν μία μὲν διὰ τοῦ κέντρου, αἱ δὲ λοιπαί, ὧς >έτυθεν, τῶν μὲν πρὸς τὴν κοίλην περιφέρειαν προσπιπτουσῶν εὐθειῶν μεγίστη μὲν ἐστὶν ἡ διὰ τοῦ κέντρου, τῶν δὲ >άλλων αἰ ἡ >έγγιον τῆς διὰ τοῦ κέντρου τῆς ἀπώτερον μείζων ἐστίν, τῶν δὲ πρὸς τὴν κυρτὴν περιφέρειαν προσπιπτουσῶν εὐθειῶν ἐλαθίστη μὲν ἐστὶν ἡ μεταχὺ τοῦ τε σημείου καὶ τῆς διαμέτρου, τῶν δὲ >άλλων αἰ ἡ >έγγιον τῆς ἐλαθίστης τῆς ἀπώτερον ἐστὶν ἐλάττων, δύο δὲ μόνον >ίσαι ἀπὸ τοῦ σημείου προσπесοῦνται πρὸς τὸν κύκλον ἐφ' ἑκάτερα τῆς ἐλαθίστης.

>Ἐστω κύκλος ὁ ΑΒΓ, καὶ τοῦ ΑΒΓ εἰλήφθω τι σημεῖον ἐκτός τὸ Δ, καὶ ἀπ' αὐτοῦ διήθθωσαν εὐθεῖαι τινεῖς αἱ ΔΑ, ΔΕ, ΔΖ, ΔΓ, >έστω δὲ ἡ ΔΑ διὰ τοῦ κέντρου. λέγω, <ὅτι τῶν μὲν πρὸς τὴν ΑΕΖΓ κοίλην περιφέρειαν προσπιπτουσῶν εὐθειῶν μεγίστη μὲν ἐστὶν ἡ διὰ τοῦ κέντρου ἡ ΔΑ, μείζων δὲ ἡ μὲν ΔΕ τῆς ΔΖ ἢ δὲ ΔΖ τῆς ΔΓ, τῶν δὲ πρὸς τὴν ΘΑΚΗ κυρτὴν περιφέρειαν προσπιπτουσῶν εὐθειῶν ἐλαθίστη μὲν ἐστὶν ἡ ΔΗ ἢ μεταχὺ τοῦ σημείου καὶ τῆς διαμέτρου τῆς ΑΗ, αἰ δὲ ἡ >έγγιον τῆς ΔΗ ἐλαθίστης ἐλάττων ἐστὶ τῆς ἀπώτερον, ἡ μὲν ΔΚ τῆς ΔΛ, ἡ δὲ ΔΛ τῆς ΔΘ.

Εἰλήφθω γὰρ τὸ κέντρον τοῦ ΑΒΓ κύκλου καὶ >έστω τὸ Μ; καὶ ἐπεζεύθθωσαν αἱ ΜΕ, ΜΖ, ΜΓ, ΜΚ, ΜΛ, ΜΘ.

Καὶ ἐπεὶ >ίση ἐστὶν ἡ ΑΜ τῇ ΕΜ, κοινὴ προσκείσθω ἡ ΜΔ; ἡ >άρα ΑΔ >ίση ἐστὶ ταῖς ΕΜ, ΜΔ. ἀλλ' αἱ ΕΜ, ΜΔ τῆς ΕΔ μείζονές εἰσιν; καὶ ἡ ΑΔ >άρα τῆς ΕΔ μείζων ἐστίν.

Proposition 8

If some point is taken outside a circle, and some straight-lines are drawn from the point to the (circumference of the) circle, one of which (passes) through the center, the remainder (being) random, (then) for the straight-lines radiating towards the concave (part of the) circumference, the greatest is that (passing) through the center. For the others, a (straight-line) nearer[†] to the (straight-line) through the center is always greater than one further away. For the straight-lines radiating towards the convex (part of the) circumference, the least is that between the point and the diameter. For the others, a (straight-line) nearer to the least (straight-line) is always less than one further away. And only two equal (straight-lines) will radiate from the point towards the (circumference of the) circle, (one) on each (side) of the least (straight-line).

Let ABC be a circle, and let some point D be taken outside ABC , and from it let some straight-lines, DA , DE , DF , and DC , be drawn through (the circle), and let DA be through the center. I say that for the straight-lines radiating towards the concave (part of the) circumference, AFC , the greatest is the one (passing) through the center, (namely) AD , and (that) DE (is) greater than DF , and DF than DC . For the straight-lines radiating towards the convex (part of the) circumference, $HLKG$, the least is the one between the point and the diameter AG , (namely) DG , and a (straight-line) nearer to the least (straight-line) DG is always less than one farther away, (so that) DK (is less) than DL , and DL

πάλιν, ἐπεὶ $\angle$ ME τῇ MZ , κοινὴ δὲ ἡ MA , αἱ EM , MA $\angle$ ZM , MA $\angle$ EM εἰσὶν; καὶ γωνία ἡ ὑπὸ EMA γωνία τῇ ZMA μείζων ἐστίν. βάσις $\angle$ EA βάσεως τῇ ZA μείζων ἐστίν. ὁμοίως δὲ δείχμεν, $\angle$ ZA τῇ GA μείζων ἐστίν; μεγίστη μὲν $\angle$ DA , μείζων δὲ ἡ μὲν DE τῇ DZ , ἡ δὲ DZ τῇ DA .

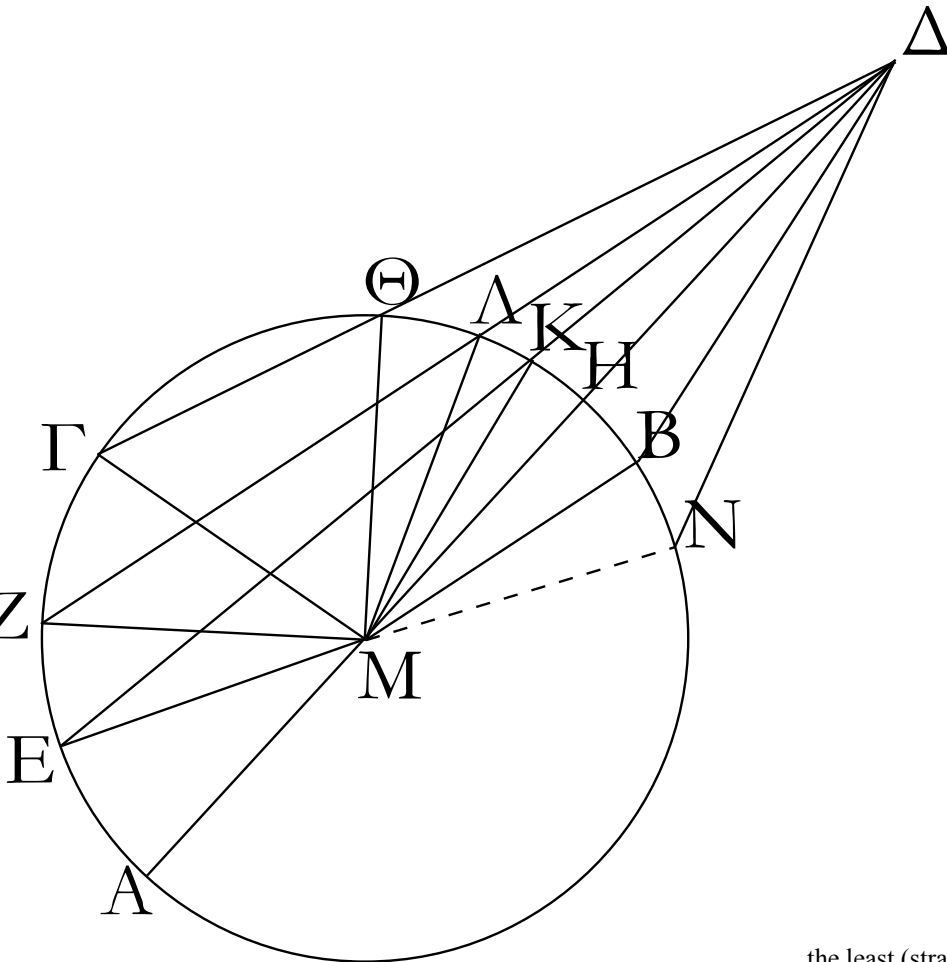
Καὶ ἐπεὶ αἱ MK , KA τῇ MA μείζονές εἰσιν, $\angle$ MA δὲ ἡ MA τῇ MK , λοιπὴ $\angle$ KA λοιπῇ τῇ HA μείζων ἐστίν; $\angle$ HA τῇ KA ἐλάττων ἐστίν; καὶ ἐπεὶ τριγώνου τοῦ MAA ἐπὶ μιᾷ τῶν πλευρῶν τῇ MA δύο εὐθεῖαι ἐντὸς συνεστάθησαν αἱ MK , KA , αἱ $\angle$ MK , KA τῶν MA , AA ἐλάττονές εἰσιν; $\angle$ MA δὲ ἡ MK τῇ MA ; λοιπὴ $\angle$ KA λοιπῇ τῇ HA ἐλάττων ἐστίν. ὁμοίως δὲ δείχμεν, $\angle$ HA τῇ DA ἐλάττων ἐστίν; ἐλαθίστη μὲν $\angle$ DA , ἐλάττων δὲ ἡ μὲν DA τῇ DA ἡ δὲ DA τῇ DA .

than than DH .

For let the center of the circle be found [Prop. 3.1], and let it be (at point) M [Prop. 3.1]. And let ME , MF , MC , MK , ML , and mh be joined.

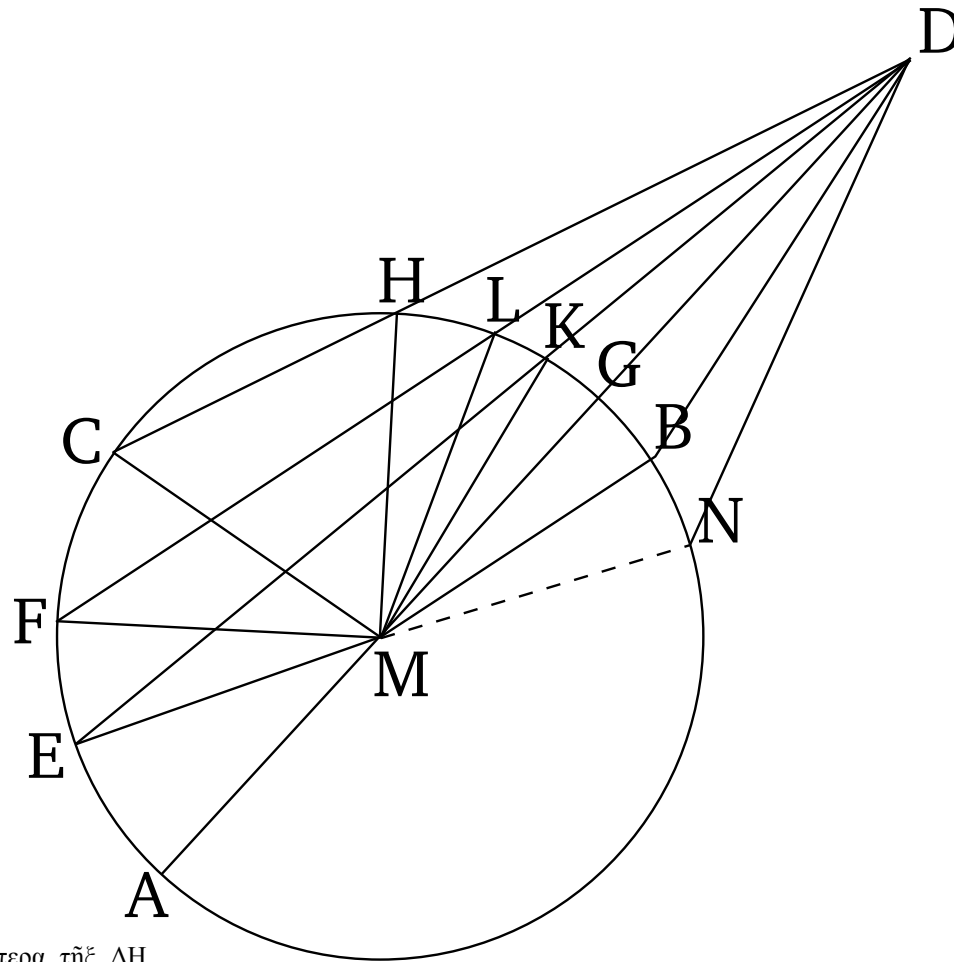
And since AM is equal to EM , let MD be added to both. Thus, AD is equal to EM and MD . But, EM and MD is greater than ED [Prop. 1.20]. Thus, AD is also greater than ED . Again, since ME is equal to MF , and MD (is) common, the (straight-lines) EM , MD are thus equal to FM , MD . And angle EMD is greater than angle FMD .[‡] Thus, the base ED is greater than the base FD [Prop. 1.24]. So, similarly, we can show that FD is also greater than CD . Thus, AD (is) the greatest (straight-line), and DE (is) greater than DF , and DF than DC .

And since MK and KD is greater than MD [Prop. 1.20], and MG (is) equal to MK , the remainder KD is thus greater than the remainder GD . So GD is less than KD . And since in triangle MLD , the two internal straight-lines MK and KD were constructed on one of the sides, MD , (then) MK and KD are thus less than ML and LD [Prop. 1.21]. And MK (is) equal to ML . Thus, the remainder KD is less than the remainder DL . So, similarly, we can show that DL is also less than DH . Thus, DG (is)



Λέγω, $\angle$ DA καὶ δύο μόνον $\angle$ DA ἀπὸ τοῦ Δ σημείου

the least (straight-line), and DK (is) less than DL , and DL than DH .



προσπεσούνται πρὸς τὸν κύκλον ἐφ' ἑκάτερα τῆς ΔΗ ἐλαθίστης; συνεστάτω πρὸς τῇ ΜΔ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ Μ τῇ ὑπὸ ΚΜΔ γωνία > ἴση γωνία ἢ ὑπὸ ΔΜΒ, καὶ ἐπεζεύθω ἡ ΔΒ. καὶ ἐπεὶ > ἴση ἐστὶν ἡ ΜΚ τῇ ΜΒ, κοινὴ δὲ ἡ ΜΔ, δύο δὲ αἱ ΚΜ, ΜΔ δύο ταῖς ΒΜ, ΜΔ > ἴσαι εἰσὶν ἑκάτερα ἑκατέρᾳ; καὶ γωνία ἢ ὑπὸ ΚΜΔ γωνία τῇ ὑπὸ ΒΜΔ > ἴση; βάσις > ἄρα ἡ ΔΚ βάσει τῇ ΔΒ > ἴση ἐστίν. λέγω [δὴ], <ὅτι τῇ ΔΚ εὐθείᾳ > ἄλλη > ἴση οὐ προσπεσεῖται πρὸς τὸν κύκλον ἀπὸ τοῦ Δ σημείου. εἰ γὰρ δυνατόν, προσπιπτέτω καὶ > ἔστω ἡ ΔΝ. ἐπεὶ οὖν ἡ ΔΚ τῇ ΔΝ ἐστὶν > ἴση, ἀλλ' ἡ ΔΚ τῇ ΔΒ ἐστὶν > ἴση, καὶ ἡ ΔΒ > ἄρα τῇ ΔΝ ἐστὶν > ἴση, ἡ > ἔγγιον τῆς ΔΗ ἐλαθίστης τῇ ἀπώτερον [ἐστίν] > ἴση; <ὅπερ ἀδύνατον ἐδείχθη. οὐκ > ἄρα πλείους > ἢ δύο > ἴσαι πρὸς τὸν ΑΒΓ κύκλον ἀπὸ τοῦ Δ σημείου ἐφ' ἑκάτερα τῆς ΔΗ ἐλαθίστης προσπεσούνται. Ἐὰν > ἄρα κύκλου ληφθῇ τι σημεῖον ἐκτόξ, ἀπὸ δὲ τοῦ σημείου πρὸς τὸν κύκλον διαθῶσιν εὐθεῖαι τινεῖς, <ὧν μία μὲν διὰ τοῦ κέντρου αἱ δὲ λοιπαί, ὥς > ἔτυθεν, τῶν μὲν πρὸς τὴν κοίλην περιφέρειαν προσπιπτουσῶν εὐθειῶν μεγίστη μὲν ἐστὶν ἡ διὰ τοῦ κέντρου, τῶν δὲ > ἄλλων αἰεὶ ἡ > ἔγγιον τῆς διὰ τοῦ κέντρου τῆς ἀπώτερον μείζων ἐστίν, τῶν δὲ πρὸς τὴν κυρτὴν περιφέρειαν προσπιπτουσῶν εὐθειῶν ἐλαθίστη μὲν ἐστὶν ἡ μεταχὺ τοῦ τε σημείου καὶ τῆς διαμέτρου, τῶν δὲ > ἄλλων αἰεὶ ἡ > ἔγγιον τῆς ἐλαθίστης τῆς ἀπώτερον ἐστὶν ἐλάττων, δύο δὲ μόνον > ἴσαι ἀπὸ τοῦ σημείου προσπεσούνται πρὸς τὸν κύκλον ἐφ' ἑκάτερα τῆς ἐλαθίστης; <ὅπερ > ἔδει δεῖχαι.

I also say that only two equal (straight-lines) will radiate from point D towards (the circumference of) the circle, (one) on each (side) on the least (straight-line), DG . Let the angle DMB , equal to angle KMD , be constructed on the straight-line MD , at the point M on it [Prop. 1.23], and let DB be joined. And since MK is equal to MB , and MD (is) common, the two (straight-lines) KM , MD are equal to the two (straight-lines) BM , MD , respectively. And angle KMD (is) equal to angle BMD . Thus, the base DK is equal to the base DB [Prop. 1.4]. [So] I say that another (straight-line) equal to DK will not radiate towards the (circumference of the) circle from point D . For, if possible, let (such a straight-line) radiate, and let it be DN . Therefore, since DK is equal to DN , but DK is equal to DB , (then) DB is thus also equal to DN , (so that) a (straight-line) nearer to the least (straight-line) DG [is] equal to one further away. The very thing was shown (to be) impossible. Thus, not more than two equal (straight-lines) will radiate towards (the circumference of) circle ABC from point D , (one) on each side of the least (straight-line) DG .

Thus, if some point is taken outside a circle, and some straight-lines are drawn from the point to the (circumference of the) circle, one of which (passes) through the center, the remainder (being) random, (then) for the straight-lines radiating towards the concave (part of the) circumference, the greatest is that (passing) through the center. For the others, a (straight-line) nearer to the (straight-line) through the center is always greater than one further away. For the straight-lines radiating towards the convex (part of the) circumference, the least is that between the point and the di-

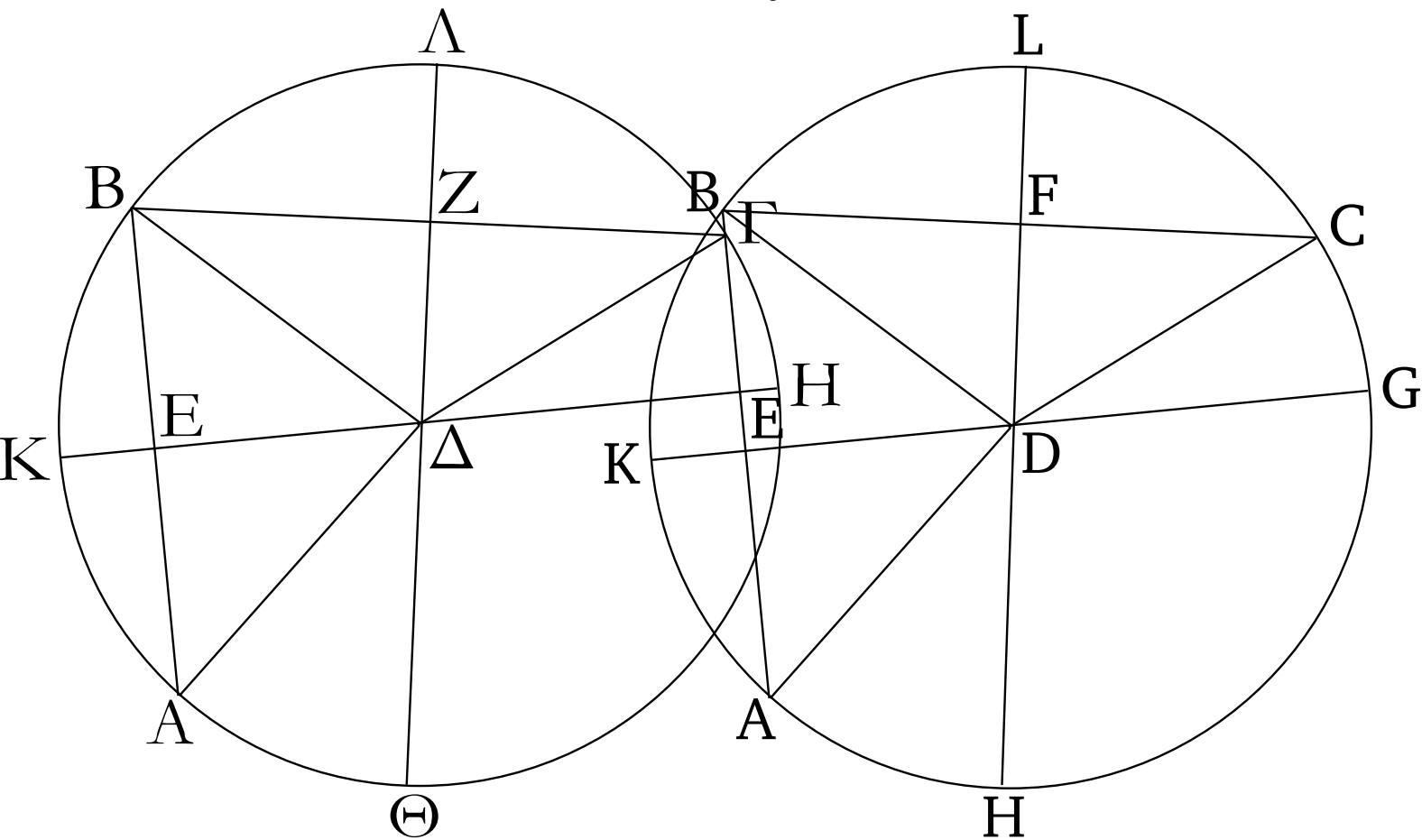
† Presumably, in an angular sense.

‡ This is not proved, except by reference to the figure.

9.

Ἐάν κύκλου ληφθῇ τι σημεῖον ἐντόξ, ἀπο δὲ τοῦ σημείου πρὸς τὸν κύκλον προσπίπτωσι πλείους > ἢ δύο > ίσαι εὐθεῖαι, τὸ ληφθὲν σημεῖον κέντρον ἐστὶ τοῦ κύκλου.

> Ἐστω κύκλος ὁ ΑΒΓ, ἐντόξ δὲ αὐτοῦ σημεῖον τὸ Δ, καὶ ἀπὸ τοῦ Δ πρὸς τὸν ΑΒΓ κύκλον προσπιπτέτωσαν πλείους > ἢ δύο > ίσαι εὐθεῖαι αἱ ΔΑ, ΔΒ, ΔΓ; λέγω, < ὅτι τὸ Δ σημεῖον κέντρον ἐστὶ τοῦ ΑΒΓ κύκλου.



Ἐπεξεύθθωσαν γὰρ αἱ ΑΒ, ΒΓ καὶ τεμήσθωσαν δίθαι κατὰ τὰ Ε, Ζ σημεία, καὶ ἐπιζευθεῖσαι αἱ ΕΔ, ΖΔ διήθθωσαν ἐπὶ τὰ Η, Κ, Θ, Λ σημεία.

Ἐπεὶ οὖν > ίση ἐστὶν ἡ ΑΕ τῇ ΕΒ, κοινὴ δὲ ἡ ΕΔ, δύο δὲ αἱ ΑΕ, ΕΔ δύο ταῖς ΒΕ, ΕΔ > ίσαι εἰσίν; καὶ βάσει ἡ ΔΑ βάσει τῇ ΔΒ > ίση; γωνία > άρα ἡ ὑπὸ ΑΕΔ γωνία τῇ ὑπὸ ΒΕΔ > ίση ἐστίν; ὀρθὴ > άρα ἑκάτερα τῶν ὑπὸ ΑΕΔ, ΒΕΔ γωνιῶν; ἡ ΗΚ > άρα τὴν ΑΒ τέμνει δίθαι καὶ πρὸς ὀρθάξ. καὶ ἐπεὶ, ἔαν ἐν κύκλῳ εὐθεῖα τις εὐθεῖαν τινα δίθαι τε καὶ πρὸς ὀρθάξ τέμνῃ, ἐπὶ τῇς τεμνομένης ἐστὶ τὸ κέντρον τοῦ κύκλου, ἐπὶ τῇς ΗΚ > άρα ἐστὶ τὸ κέντρον τοῦ κύκλου. διὰ τὰ αὐτὰ δὲ καὶ ἐπὶ τῇς ΘΛ ἐστὶ τὸ κέντρον τοῦ ΑΒΓ κύκλου.

ameter. For the others, a (straight-line) nearer to the least (straight-line) is always less than one further away. And only two equal (straight-lines) will radiate from the point towards the (circumference of the) circle, (one) on each (side) of the least (straight-line). (Which is) the very thing it was required to show.

Proposition 9

If some point is taken inside a circle, and more than two equal straight-lines radiate from the point towards the (circumference of the) circle, (then) the point taken is the center of the circle.

Let ABC be a circle, and D a point inside it, and let more than two equal straight-lines, DA , DB , and DC , radiate from D towards (the circumference of) circle ABC . I say that point D is the center of circle ABC .

For let AB and BC be joined, and (then) be cut in half at points E and F (respectively) [Prop. 1.10]. And ED and FD being joined, let them be drawn through to points G , K , H , and L .

Therefore, since AE is equal to EB , and ED (is) common, the two (straight-lines) AE , ED are equal to the two (straight-lines) BE , ED (respectively). And the base DA (is) equal to the base DB . Thus, angle AED is equal to angle BED [Prop. 1.8]. Thus, angles AED and BED (are) each right-angles [Def. 1.10]. Thus, GK cuts AB in half, and at right-angles. And since, if some straight-line in a circle cuts some (other) straight-line in half, and

καὶ οὐδὲν <έτερον κοινὸν> ἔθουσιν αἱ HK , $ΘΛ$ εὐθεῖαι >ῆ τὸ $Δ$ σημεῖον; τὸ $Δ$ >άρα σημεῖον κέντρον ἐστὶ τοῦ $ABΓ$ κύκλου.

Ἐὰν >άρα κύκλου ληφθῇ| τι σημεῖον ἐντόξ, ἀπὸ δὲ τοῦ σημείου πρὸς τὸν κύκλον προσπίπτωσι πλείουξ >ῆ δύο >ίσαι εὐθεῖαι, τὸ ληφθὲν σημεῖον κέντρον ἐστὶ τοῦ κύκλου; <όπερ> ἐδει δεῖχαι.

at right-angles, (then) the center of the circle is on the former (straight-line) [Prop. 3.1 corr.], the center of the circle is thus on GK . So, for the same (reasons), the center of circle ABC is also on HL . And the straight-lines GK and HL have no common (point) other than point D . Thus, point D is the center of circle ABC .

Thus, if some point is taken inside a circle, and more than two equal straight-lines radiate from the point towards the (circumference of the) circle, (then) the point taken is the center of the circle. (Which is) the very thing it was required to show.

10.

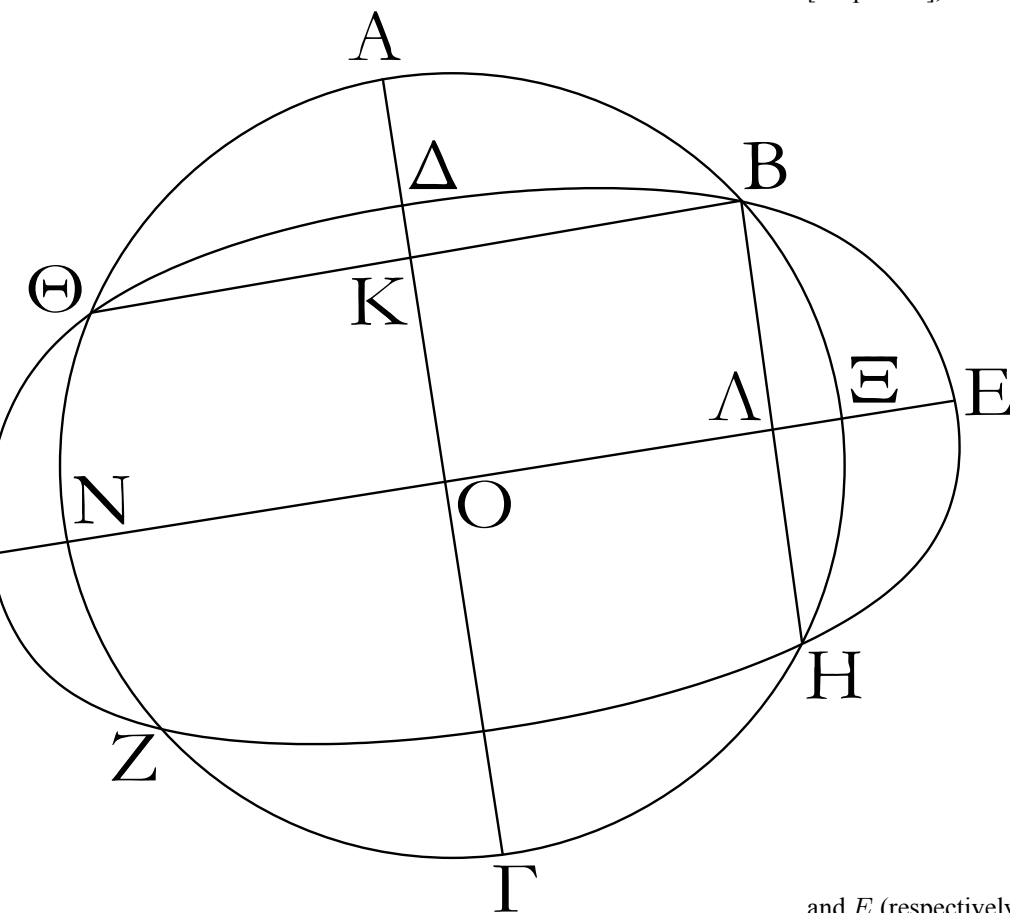
Κύκλος κύκλον οὐ τέμνει κατὰ πλείονα σημεία >ῆ δύο.

Εἰ γὰρ δυνατόν, κύκλος ὁ $ABΓ$ κύκλον τὸν $ΔΕΖ$ τεμνέτω κατὰ πλείονα σημεία >ῆ δύο τὰ $B, H, Z, Θ$, καὶ ἐπιευθεῖσαι αἱ $BΘ, BH$ δίθρα τεμνέσθωσαν κατὰ τὰ $K, Λ$ σημεία; καὶ ἀπὸ τῶν $K, Λ$ ταῖξ $BΘ, BH$ πρὸς ὀρθὰξ ἀθθεῖσαι αἱ $KΓ, ΛΜ$ διήθθωσαν ἐπὶ τὰ $A, Ε$ σημεία.

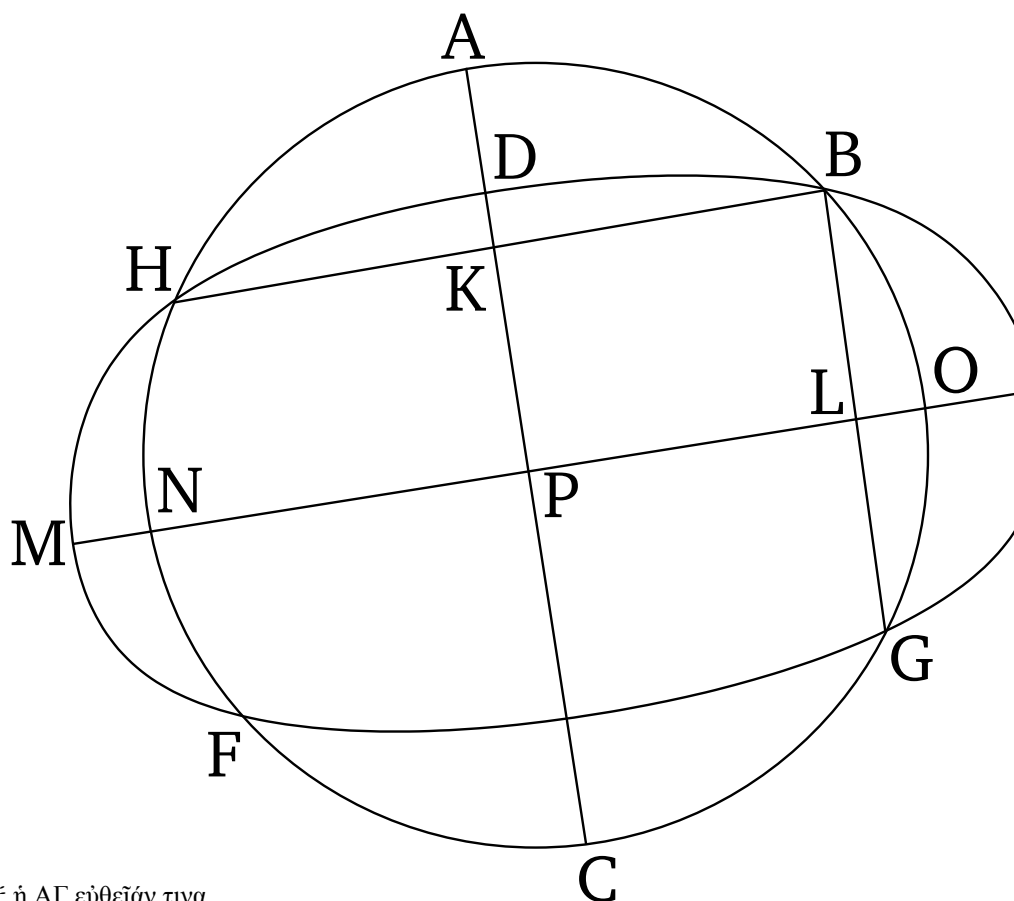
Proposition 10

A circle does not cut a(nother) circle at more than two points.

For, if possible, let the circle ABC cut the circle DEF at more than two points, B, G, F , and H . And BH and BG being joined, let them (then) be cut in half at points K and L (respectively). And KC and LM being drawn at right-angles to BH and BG from K and L (respectively) [Prop. 1.11], let them (then) be drawn through to points A



and E (respectively).



Ἐπει οὖν ἐν κύκλῳ τῷ ΑΒΓ εὐθεῖα τις ἡ ΑΓ εὐθεῖαν τινα τὴν ΒΘ δίθαι καὶ πρὸς ὀρθᾶς τέμνει, ἐπὶ τῆς ΑΓ >άρα ἐστὶ τὸ κέντρον τοῦ ΑΒΓ κύκλου. πάλιν, ἐπεὶ ἐν κύκλῳ τῷ αὐτῷ τῷ ΑΒΓ εὐθεῖα τις ἡ ΝΧ εὐθεῖαν τινα τὴν ΒΗ δίθαι καὶ πρὸς ὀρθᾶς τέμνει, ἐπὶ τῆς ΝΧ >άρα ἐστὶ τὸ κέντρον τοῦ ΑΒΓ κύκλου. ἐδείθθη δὲ καὶ ἐπὶ τῆς ΑΓ, καὶ κατ' οὐδὲν συμβάλλουσιν αἱ ΑΓ, ΝΧ εὐθεῖαι >ἡ κατὰ τὸ Ο; τὸ Ο >άρα σημεῖον κέντρον ἐστὶ τοῦ ΑΒΓ κύκλου. ὁμοίως δὴ δείχομεν, <ὅτι καὶ τοῦ ΔΕΖ κύκλου κέντρον ἐστὶ τὸ Ο; δύο >άρα κύκλων τεμνόντων ἀλλήλους τῶν ΑΒΓ, ΔΕΖ τὸ αὐτὸ ἐστὶ κέντρον τὸ Ο; <ὅπερ ἐστὶν ἀδύνατον.

Οὐκ >άρα κύκλος κύκλον τέμνει κατὰ πλείονα σημεία >ἡ δύο; <ὅπερ >έδει δεῖξαι.

Therefore, since in circle ABC some straight-line AC cuts some (other) straight-line BH in half, and at right-angles, the center of circle ABC is thus on AC [Prop. 3.1 corr.]. Again, since in the same circle ABC some straight-line NO cuts some (other straight-line) BG in half, and at right-angles, the center of circle ABC is thus on NO [Prop. 3.1 corr.]. And it was also shown (to be) on AC . And the straight-lines AC and NO meet at no other (point) than P . Thus, point P is the center of circle ABC . So, similarly, we can show that P is also the center of circle DEF . Thus, two circles cutting one another, ABC and DEF , have the same center P . The very thing is impossible [Prop. 3.5]. Thus, a circle does not cut a(nother) circle at more than two points. (Which is) the very thing it was required to show.

11.

Ἐὰν δύο κύκλοι ἐφάπτονται ἀλλήλων ἐντός, καὶ ληφθῇ
αὐτῶν τὰ κέντρα, ἢ ἐπὶ τὰ κέντρα αὐτῶν ἐπιζυγνυμένη
εὐθεῖα καὶ ἐκβαλλομένη ἐπὶ τὴν συναφὴν πεσεῖται τῶν
κύκλων.

Δύο γὰρ κύκλοι οἱ ΑΒΓ, ΑΔΕ ἐφαπτέσθωσαν ἀλλήλων ἐντὸς κατὰ τὸ Α σημεῖον, καὶ εἰλήφθω τοῦ μὲν ΑΒΓ κύκλου κέντρον τὸ Ζ, τοῦ δὲ ΑΔΕ τὸ Η· λέγω, <ὅτι ἡ ἀπὸ τοῦ Η ἐπὶ τὸ Ζ ἐπιξυγγυμένη εὐθεῖα ἐκβαλλομένη ἐπὶ τὸ Α πεσεῖται. μὴ γάρ, ἄλλ> εἰ δυνατόν, πιπτέτω ὥς ἡ ΖΗΘ, καὶ ἐπεζεύθωσιν ΑΖ, ΑΗ.

Ἐπεὶ οὖν αἱ ΑΗ, ΗΖ τῆς ΖΑ, τουτέστι τῆς ΖΘ, μείζονές
 εἰσιν, κοινὴ ἀφηρήσθω ἡ ΖΗ; λοιπὴ >άρα ἡ ΑΗ λοιπῆς τῆς
 ΗΘ μείζων ἐστίν. >ίση δὲ ἡ ΑΗ τῇ ΗΔ; καὶ ἡ ΗΔ >άρα
 τῆς ΗΘ μείζων ἐστίν ἡ ἐλάττων τῆς μείζονος; <όπερ ἐστίν
 ἀδύνατον; οὐκ >άρα ἡ ἀπὸ τοῦ Ζ ἐπὶ τὸ Η ἐπιτεταμένη
 εὐθεῖα ἐκτὸς πεσεῖται; κατὰ τὸ Α >άρα ἐπὶ τῆς συναφῆς

Proposition 11

If two circles touch one another internally, and their centers are found, (then) the straight-line joining their centers, being produced, will fall upon the point of union of the circles.

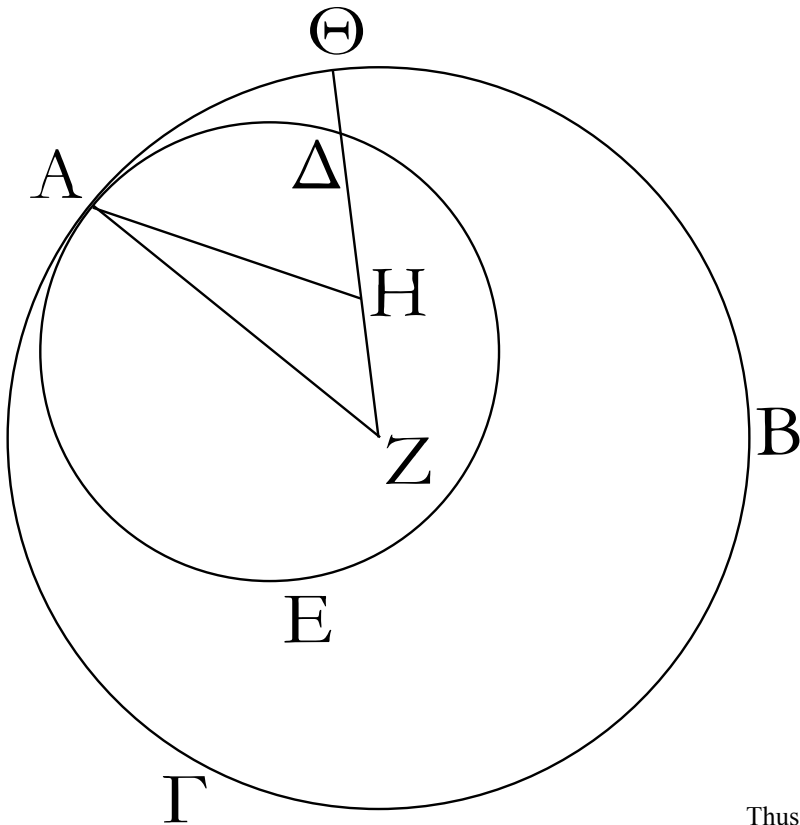
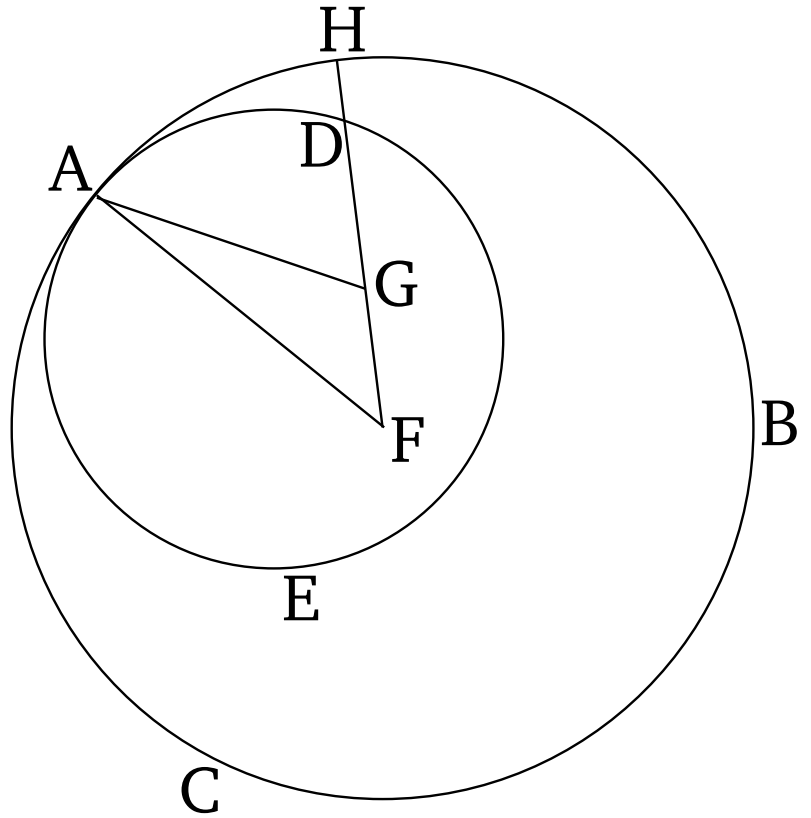
For let two circles, ABC and ADE , touch one another internally at point A , and let the center F of circle ABC be found [Prop. 3.1], and (the center) G of (circle) ADE [Prop. 3.1]. I say that the straight-line joining G to F , being produced, will fall on A .

For (if) not (then), if possible, let it fall like FGH (in the figure), and let AF and AG be joined.

Therefore, since AG and GF is greater than FA , that is to say FH [Prop. 1.20], let FG be taken from both. Thus, the remainder AG is greater than the remainder GH . And AG (is) equal to GD . Thus, GD is also greater than GH , the lesser than the greater. The very thing is impossible. Thus, the straight-line joining F to G will not fall outside (one

πεσεῖται.

circle but inside the other). Thus, it will fall upon the point of union (of the circles) at point *A*.

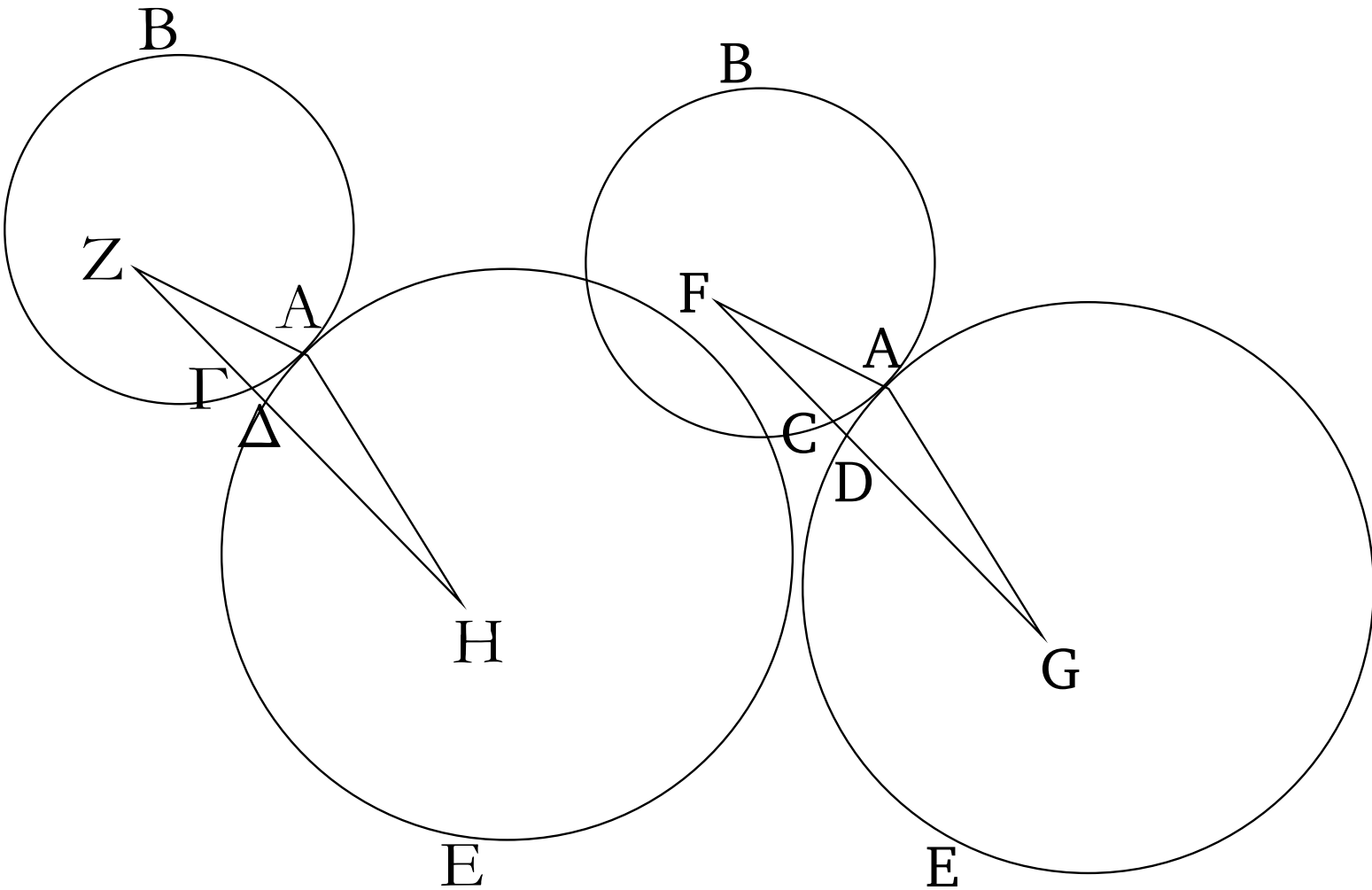


Ἐάν >άρα δύο κύκλοι ἐφάπτονται ἀλλήλων ἐντός, [καὶ ληφθῇ] αὐτῶν τὰ κέντρα], ἢ ἐπὶ τὰ κέντρα αὐτῶν ἐπιζευγνυμένων εὐθεῖα [καὶ ἐκβαλλομένη] ἐπὶ τὴν συναφὴν πεσεῖται τῶν κύκλων; <ὅπερ >έδει δεῖχαι.

Thus, if two circles touch one another internally, [and their centers are found], (then) the straight-line joining their centers, [being produced], will fall upon the point of union of the circles. (Which is) the very thing it was required to show.

Ἐάν δύο κύκλοι ἐφάπτονται ἀλλήλων ἐκτός, ἢ ἐπὶ τὰ κέντρα αὐτῶν ἐπιζευγνυμένη διὰ τῆς ἐπαφῆς ἐλεύσεται.

If two circles touch one another externally, (then) the (straight-line) joining their centers will go through the point of union.



Δύο γὰρ κύκλοι οἱ ABΓ, AΔΕ ἐφαπτέσθωσαν ἀλλήλων ἐκτός κατὰ τὸ A σημεῖον, καὶ εἰλήφθω τοῦ μὲν ABΓ κέντρον τὸ Z, τοῦ δὲ AΔΕ τὸ H; λέγω, <ὅτι ἢ ἀπὸ τοῦ Z ἐπὶ τὸ H ἐπιζευγνυμένη εὐθεῖα διὰ τῆς κατὰ τὸ A ἐπαφῆς ἐλεύσεται.

μὴ γάρ, ἀλλ' εἰ δυνατόν, ἐρθέσθω ὥς ἡ ZΓΔΗ, καὶ ἐπεζεύθωσαν αἱ AZ, AH.

Ἐπεὶ οὖν τὸ Z σημεῖον κέντρον ἐστὶ τοῦ ABΓ κύκλου, >ίση ἐστὶν ἡ ZA τῇ ZΓ. πάλιν, ἐπεὶ τὸ H σημεῖον κέντρον ἐστὶ τοῦ AΔΕ κύκλου, >ίση ἐστὶν ἡ HA τῇ ΗΔ. ἐδείχθη δὲ καὶ ἡ ZA τῇ ZΓ >ίση; αἱ >άρα ZA, AH ταῖς ZΓ, ΗΔ >ίσαι εἰσίν; <ὥστε <ὅλη ἡ ZH τῶν ZA, AH μείζων ἐστίν; ἀλλὰ καὶ ἐλάττω; <ὅπερ ἐστὶν ἀδύνατον. οὐκ >άρα ἢ ἀπὸ τοῦ Z ἐπὶ τὸ H ἐπιζευγνυμένη εὐθεῖα διὰ τῆς κατὰ τὸ A ἐπαφῆς οὐκ ἐλεύσεται; δι' αὐτῆς >άρα.

Ἐάν >άρα δύο κύκλοι ἐφάπτονται ἀλλήλων ἐκτός, ἢ ἐπὶ τὰ κέντρα αὐτῶν ἐπιζευγνυμένη [εὐθεῖα] διὰ τῆς ἐπαφῆς ἐλεύσεται; <ὅπερ >έδει δεῖχαι.

For let two circles, ABC and ADE , touch one another externally at point A , and let the center F of ABC be found [Prop. 3.1], and (the center) G of ADE [Prop. 3.1]. I say that the straight-line joining F to G will go through the point of union at A .

For (if) not (then), if possible, let it go like $FCDG$ (in the figure), and let AF and AG be joined.

Therefore, since point F is the center of circle ABC , FA is equal to FC . Again, since point G is the center of circle ADE , GA is equal to GD . And FA was also shown (to be) equal to FC . Thus, the (straight-lines) FA and AG are equal to the (straight-lines) FC and GD . So the whole of FG is greater than FA and AG . But, (it is) also less [Prop. 1.20]. The very thing is impossible. Thus, the straight-line joining F to G cannot not go through the point of union at A . Thus, (it will go) through it.

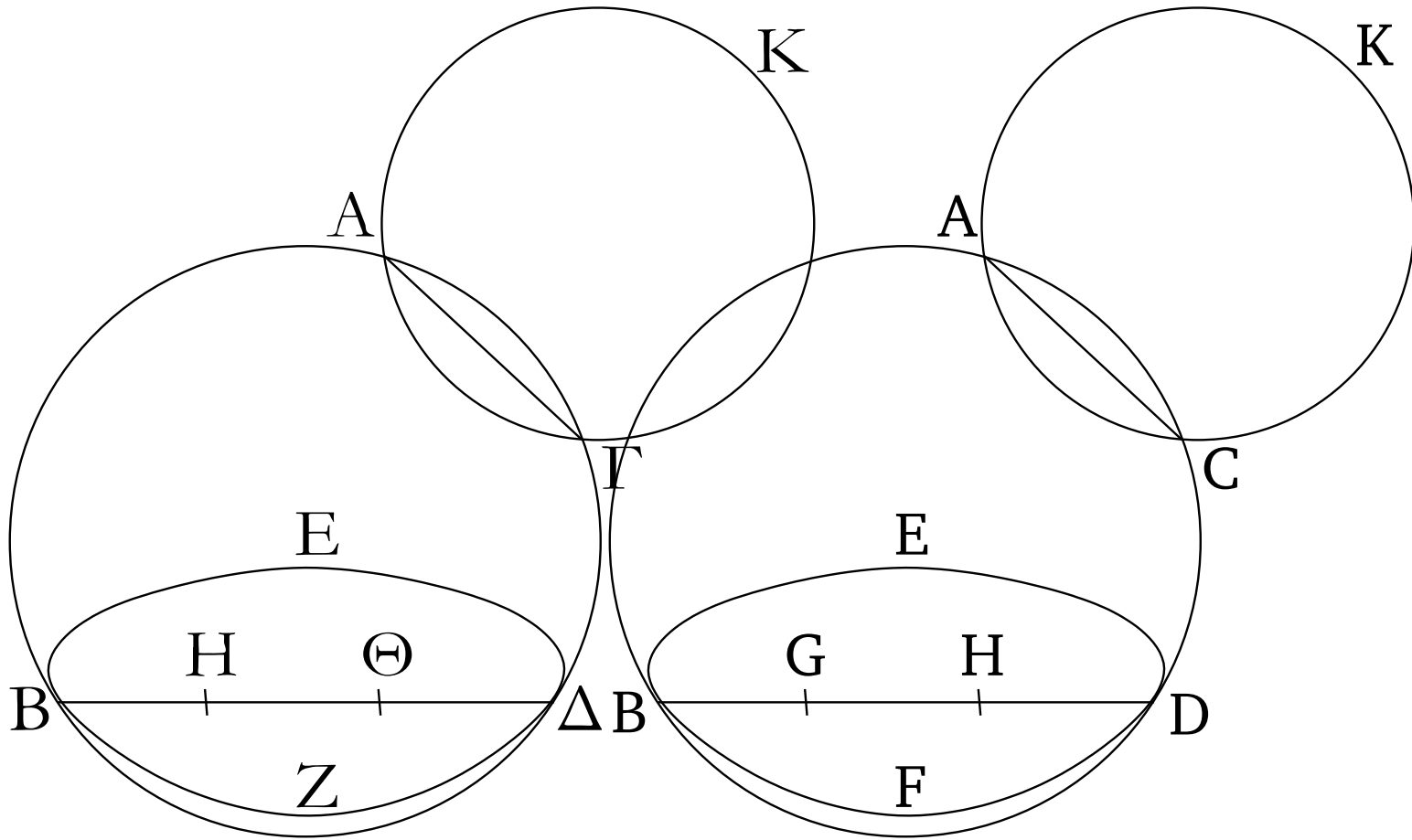
Thus, if two circles touch one another externally, (then) the [straight-line] joining their centers will go through the point of union. (Which is) the very thing it was required to show.

13.

Proposition 13

Κύκλος κύκλου οὐκ ἐφάπτεται κατὰ πλείονα σημεία >ἢ καθ' <έν, ἔαν τε ἐντὸς ἔαν τε ἐκτός ἐφάπτηται.

A circle does not touch a(nother) circle at more than one point, whether they touch internally or externally.



Εἰ γὰρ δυνατόν, κύκλος ὁ ΑΒΓΔ κύκλου τοῦ ΕΒΖΔ ἐφαπτέσθω
πρότερον ἐντὸς κατὰ πλείονα σημεῖα > ἢ < ἐν τὰ Δ, Β.

Καὶ εἰλήφθω τοῦ μὲν ΑΒΓΔ κύκλου κέντρον τὸ Η, τοῦ δὲ
ΕΒΖΔ τὸ Θ.

Ἡ > ἄρα ἀπὸ τοῦ Η ἐπὶ τὸ Θ ἐπιζευγνυμένη ἐπὶ τὰ Β, Δ
πεσεῖται. πιπτέτω ὡς ἡ ΒΗΘΔ. καὶ ἐπεὶ τὸ Η σημεῖον
κέντρον ἐστὶ τοῦ ΑΒΓΔ κύκλου, > ἴση ἐστὶν ἡ ΒΗ τῇ ΗΔ;
μείζων > ἄρα ἡ ΒΗ τῇς ΘΔ; πολλῶν > ἄρα μείζων ἡ ΒΘ
τῇς ΘΔ. πάλιν, ἐπεὶ τὸ Θ σημεῖον κέντρον ἐστὶ τοῦ ΕΒΖΔ
κύκλου, > ἴση ἐστὶν ἡ ΒΘ τῇ ΘΔ; ἐδείχθη δὲ αὐτῇς καὶ
πολλῶν μείζων; < ὅπερ ἀδύνατον; οὐκ > ἄρα κύκλος κύκλου
ἐφάπτεται ἐντὸς κατὰ πλείονα σημεῖα > ἢ < ἐν.

Λέγω δὴ, < ὅτι οὐδὲ ἐκτόξ.

Εἰ γὰρ δυνατόν, κύκλος ὁ ΑΓΚ κύκλου τοῦ ΑΒΓΔ ἐφαπτέσθω
ἐκτόξ κατὰ πλείονα σημεῖα > ἢ < ἐν τὰ Α, Γ, καὶ ἐπεξεύθθω
ἡ ΑΓ.

> Ἐπεὶ οὖν κύκλων τῶν ΑΒΓΔ, ΑΓΚ εὐληπται ἐπὶ τῇς
περιφερείας ἑκατέρου δύο τυθόντα σημεῖα τὰ Α, Γ, ἡ ἐπὶ
τὰ σημεῖα ἐπιζευγνυμένη εὐθεῖα ἐντὸς ἑκατέρου πεσεῖται;
ἀλλὰ τοῦ μὲν ΑΒΓΔ ἐντὸς > ἔπεσεν, τοῦ δὲ ΑΓΚ ἐκτόξ;
< ὅπερ > ἀτοπον; οὐκ > ἄρα κύκλος κύκλου ἐφάπτεται ἐκτόξ
κατὰ πλείονα σημεῖα > ἢ < ἐν. ἐδείχθη δὲ, < ὅτι οὐδὲ ἐντόξ.
Κύκλος > ἄρα κύκλου οὐκ ἐφάπτεται κατὰ πλείονα σημεῖα
> ἢ [καθ'] < ἐν, ἔαν τε ἐντὸς ἔαν τε ἐκτόξ ἐφάπτηται; < ὅπερ
> ἔδει δεῖξαι.

For, if possible, let circle $ABDC$ [†] touch circle $EBFD$ —
first of all, internally—at more than one point, D and B .

And let the center G of circle $ABDC$ be found [Prop. 3.1],
and (the center) H of $EBFD$ [Prop. 3.1].

Thus, the (straight-line) joining G and H will fall on B and
 D [Prop. 3.11]. Let it fall like $BGHD$ (in the figure). And
since point G is the center of circle $ABDC$, BG is equal to
 GD . Thus, BG (is) greater than HD . Thus, BH (is) much
greater than HD . Again, since point H is the center of circle
 $EBFD$, BH is equal to HD . But it was also shown (to be)
much greater than it. The very thing (is) impossible. Thus, a
circle does not touch a(nother) circle internally at more than
one point.

So, I say that neither (does it touch) externally (at more than
one point).

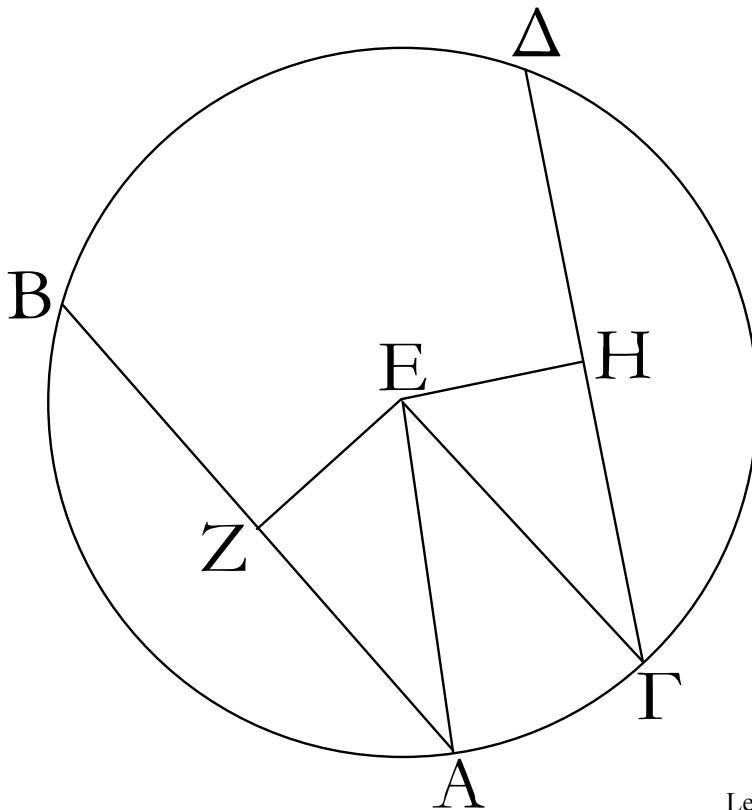
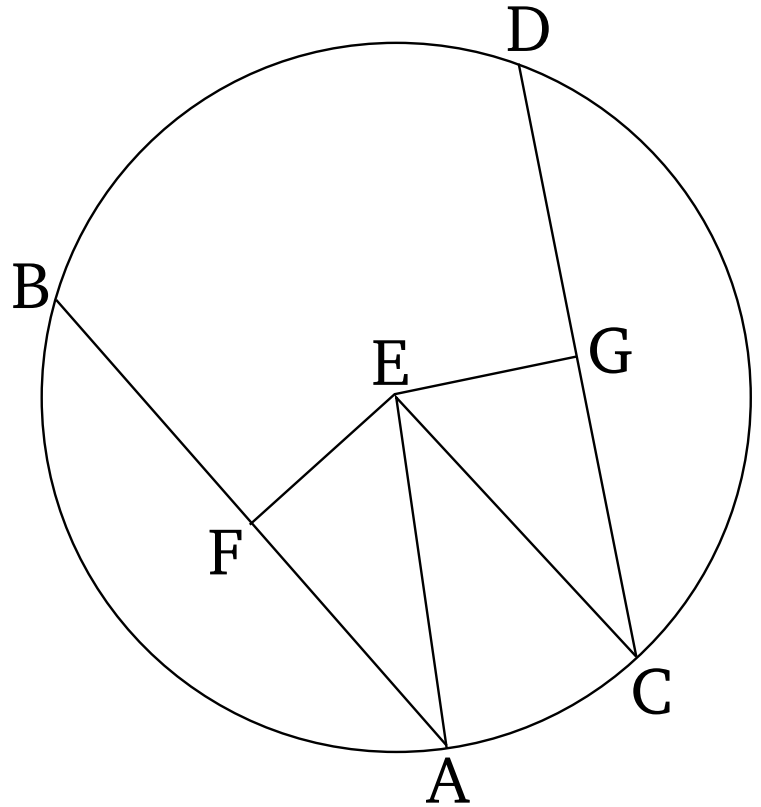
For, if possible, let circle ACK touch circle $ABDC$ ex-
ternally at more than one point, A and C . And let AC be
joined.

Therefore, since two points, A and C , be taken at random on
the circumference of each of the circles $ABDC$ and ACK ,
the straight-line joining the points will fall inside each (circle)
[Prop. 3.2]. But, it fell inside $ABDC$, and outside
 ACK [Def. 3.3]. The very thing (is) absurd. Thus, a circle
does not touch a(nother) circle externally at more than
one point. And it was shown that neither (does it) internally.
Thus, a circle does not touch a(nother) circle at more than
one point, whether they touch internally or externally.
(Which is) the very thing it was required to show.

[†] The Greek text has “ $ABCD$ ”, which is obviously a mistake.

Ἐν κύκλῳ αἱ >ίσαι εὐθεῖαι >ίσον ἀπέθουσιν ἀπὸ τοῦ κέντρου, καὶ αἱ >ίσον ἀπέθουσαι ἀπὸ τοῦ κέντρου >ίσαι ἀλλήλαις εἰσίν.

In a circle, equal straight-lines are equally far from the center, and (straight-lines) which are equally far from the center are equal to one another.



>Ἐστω κύκλος ὁ ΑΒΓΔ, καὶ ἐν αὐτῷ >ίσαι εὐθεῖαι >έστωσαν αἱ ΑΒ, ΓΔ; λέγω, <ὅτι αἱ ΑΒ, ΓΔ >ίσον ἀπέθουσιν ἀπὸ τοῦ κέντρου.
Εἰλήφθω γὰρ τὸ κέντρον τοῦ ΑΒΓΔ κύκλου καὶ >έστω τὸ Ε, καὶ ἀπὸ τοῦ Ε ἐπὶ τὰς ΑΒ, ΓΔ κάθετοι >ήθθωσαν αἱ ΕΖ, ΕΗ, καὶ ἐπεζεύθθωσαν αἱ ΑΕ, ΕΓ.

Let $ABDC^{\dagger}$ be a circle, and let AB and CD be equal straight-lines within it. I say that AB and CD are equally far from the center.

For let the center of circle $ABDC$ be found [Prop. 3.1], and let it be (at) E . And let EF and EG be drawn from (point) E , perpendicular to AB and CD (respectively) [Prop. 1.12]. And let AE and EC be joined.

Ἐπεὶ οὖν εὐθεῖα τις δια τοῦ κέντρου ἢ EZ εὐθεῖαν τινα μὴ διὰ τοῦ κέντρου τὴν AB πρὸς ὀρθᾷ τέμνει, καὶ δίθα αὐτὴν τέμνει. >ῖση >άρα ἢ AZ τῇ ZB ; διπλῇ >άρα ἢ AB τῇ AZ . διὰ τὰ αὐτὰ δὲ καὶ ἢ $ΓΔ$ τῇ $ΓΗ$ ἐστὶ διπλῇ; καὶ ἐστὶν >ῖση ἢ AB τῇ $ΓΔ$; >ῖση >άρα καὶ ἢ AZ τῇ $ΓΗ$. καὶ ἐπεὶ >ῖση ἐστὶν ἢ AE τῇ $ΕΓ$, >ῖσον καὶ τὸ ἀπὸ τῇ AE τῷ ἀπὸ τῇ $ΕΓ$. ἀλλὰ τῷ μὲν ἀπὸ τῇ AE >ῖσα τὰ ἀπὸ τῶν AZ , EZ ; ὀρθὴ γὰρ ἢ πρὸς τῷ Z γωνία; τῷ δὲ ἀπὸ τῇ $ΕΓ$ >ῖσα τὰ ἀπὸ τῶν EH , $ΗΓ$; ὀρθὴ γὰρ ἢ πρὸς τῷ H γωνία; τὰ >άρα ἀπὸ τῶν AZ , ZE >ῖσα ἐστὶ τοῖς ἀπὸ τῶν $ΓΗ$, HE , <ὦν τὸ ἀπὸ τῇ AZ >ῖσον ἐστὶ τῷ ἀπὸ τῇ $ΓΗ$; >ῖση γὰρ ἐστὶν ἢ AZ τῇ $ΓΗ$; λοιπὸν >άρα τὸ ἀπὸ τῇ ZE τῷ ἀπὸ τῇ EH >ῖσον ἐστίν; >ῖση >άρα ἢ EZ τῇ EH . ἐν δὲ κύκλῳ >ῖσον ἀπέθουν ἀπὸ τοῦ κέντρου εὐθεῖαι λέγονται, <όταν αἱ ἀπὸ τοῦ κέντρου ἐπ> αὐτὰς κάθετοι ἀγόμεναι >ῖσαι >ῶσιν; αἱ >άρα AB , $ΓΔ$ >ῖσον ἀπέθουν ἀπὸ τοῦ κέντρου.

Ἀλλὰ δὲ αἱ AB , $ΓΔ$ εὐθεῖαι >ῖσον ἀπεθέωσαν ἀπὸ τοῦ κέντρου, τουτέστιν >ῖση >έστω ἢ EZ τῇ EH . λέγω, <ότι >ῖση ἐστὶ καὶ ἢ AB τῇ $ΓΔ$.

Τῶν γὰρ αὐτῶν κατασκευασθέντων ὁμοίως δείχομεν, <ότι διπλῇ ἐστὶν ἢ μὲν AB τῇ AZ , ἢ δὲ $ΓΔ$ τῇ $ΓΗ$; καὶ ἐπεὶ >ῖση ἐστὶν ἢ AE τῇ $ΓΕ$, >ῖσον ἐστὶ τὸ ἀπὸ τῇ AE τῷ ἀπὸ τῇ $ΓΕ$; ἀλλὰ τῷ μὲν ἀπὸ τῇ AE >ῖσα ἐστὶ τὰ ἀπὸ τῶν EZ , ZA , τῷ δὲ ἀπὸ τῇ $ΓΕ$ >ῖσα τὰ ἀπὸ τῶν EH , $ΗΓ$. τὰ >άρα ἀπὸ τῶν EZ , ZA >ῖσα ἐστὶ τοῖς ἀπὸ τῶν EH , $ΗΓ$; <ὦν τὸ ἀπὸ τῇ EZ τῷ ἀπὸ τῇ EH ἐστὶν >ῖσον; >ῖση γὰρ ἢ EZ τῇ EH ; λοιπὸν >άρα τὸ ἀπὸ τῇ AZ >ῖσον ἐστὶ τῷ ἀπὸ τῇ $ΓΗ$; >ῖση >άρα ἢ AZ τῇ $ΓΗ$; καὶ ἐστὶ τῇ μὲν AZ διπλῇ ἢ AB , τῇ δὲ $ΓΗ$ διπλῇ ἢ $ΓΔ$; >ῖση >άρα ἢ AB τῇ $ΓΔ$.

Ἐν κύκλῳ >άρα αἱ >ῖσαι εὐθεῖαι >ῖσον ἀπέθουν ἀπὸ τοῦ κέντρου, καὶ αἱ >ῖσον ἀπέθουν ἀπὸ τοῦ κέντρου >ῖσαι ἀλλήλαις εἰσίν; <όπερ >έδει δεῖχαι.

Therefore, since some straight-line, EF , through the center (of the circle), cuts some (other) straight-line, AB , not through the center, at right-angles, it also cuts it in half [Prop. 3.3]. Thus, AF (is) equal to FB . Thus, AB (is) double AF . So, for the same (reasons), CD is also double CG . And AB is equal to CD . Thus, AF (is) also equal to CG . And since AE is equal to EC , the (square) on AE (is) also equal to the (square) on EC . But, the (sum of the squares) on AF and EF (is) equal to the (square) on AE . For the angle at F (is) a right-angle [Prop. 1.47]. And the (sum of the squares) on EG and GC (is) equal to the (square) on EC . For the angle at G (is) a right-angle [Prop. 1.47]. Thus, the (sum of the squares) on AF and FE is equal to the (sum of the squares) on CG and GE , of which the (square) on AF is equal to the (square) on CG . For AF is equal to CG . Thus, the remaining (square) on FE is equal to the (remaining square) on EG . Thus, EF (is) equal to EG . And straight-lines in a circle are said to be equally far from the center when perpendicular (straight-lines) which are drawn to them from the center are equal [Def. 3.4]. Thus, AB and CD are equally far from the center.

So, let the straight-lines AB and CD be equally far from the center. That is to say, let EF be equal to EG . I say that AB is also equal to CD .

For, with the same construction, we can, similarly, show that AB is double AF , and CD (double) CG . And since AE is equal to CE , the (square) on AE is equal to the (square) on CE . But, the (sum of the squares) on EF and FA is equal to the (square) on AE [Prop. 1.47]. And the (sum of the squares) on EG and GC (is) equal to the (square) on CE [Prop. 1.47]. Thus, the (sum of the squares) on EF and FA is equal to the (sum of the squares) on EG and GC , of which the (square) on EF is equal to the (square) on EG . For EF (is) equal to EG . Thus, the remaining (square) on AF is equal to the (remaining square) on CG . Thus, AF (is) equal to CG . And AB is double AF , and CD double CG . Thus, AB (is) equal to CD .

Thus, in a circle, equal straight-lines are equally far from the center, and (straight-lines) which are equally far from the center are equal to one another. (Which is) the very thing it was required to show.

† The Greek text has “ $ABCD$ ”, which is obviously a mistake.

15.

Ἐν κύκλῳ μεγίστη μὲν ἢ διάμετρος, τῶν δὲ >άλλων αἰεὶ ἢ >έγγιον τοῦ κέντρου τῇ ἀπώτερον μείζων ἐστίν.

>Ἐστω κύκλος ὁ $ABΓΔ$, διάμετρος δὲ αὐτοῦ >έστω ἢ AD , κέντρον δὲ τὸ E , καὶ >έγγιον μὲν τῇ AD διαμέτρου >έστω ἢ $BΓ$, ἀπώτερον δὲ ἢ ZH ; λέγω, <ότι μεγίστη μὲν ἐστὶν ἢ AD , μείζων δὲ ἢ $BΓ$ τῇ ZH .

>Ἡθώσαν γὰρ ἀπὸ τοῦ E κέντρου ἐπὶ τὰς $BΓ$, ZH κάθετοι αἱ $EΘ$, $EΚ$. καὶ ἐπεὶ >έγγιον μὲν τοῦ κέντρου ἐστὶν ἢ $BΓ$, ἀπώτερον δὲ ἢ ZH , μείζων >άρα ἢ $EΚ$ τῇ $EΘ$. κείσθω τῇ $EΘ$ >ῖση ἢ $ΕΛ$, καὶ διὰ τοῦ $Λ$ τῇ $EΚ$ πρὸς ὀρθᾷ ἀθροῖσα ἢ $ΛΜ$ διήθω ἐπὶ τὸ N , καὶ ἐπεξεύθωσαν αἱ ME , EN , ZE , EH .

Καὶ ἐπεὶ >ῖση ἐστὶν ἢ $EΘ$ τῇ $ΕΛ$, >ῖση ἐστὶ καὶ ἢ $BΓ$ τῇ MN . πάλιν, ἐπεὶ >ῖση ἐστὶν ἢ μὲν AE τῇ EM , ἢ δὲ $ΕΔ$ τῇ EN , ἢ >άρα AD ταῖς ME , EN >ῖση ἐστίν. ἀλλ> αἱ μὲν ME , EN τῇ MN μείζονές εἰσιν [καὶ ἢ AD τῇ MN μείζων ἐστίν], >ῖση

Proposition 15

In a circle, a diameter (is) the greatest (straight-line), and for the others, a (straight-line) nearer to the center is always greater than one further away.

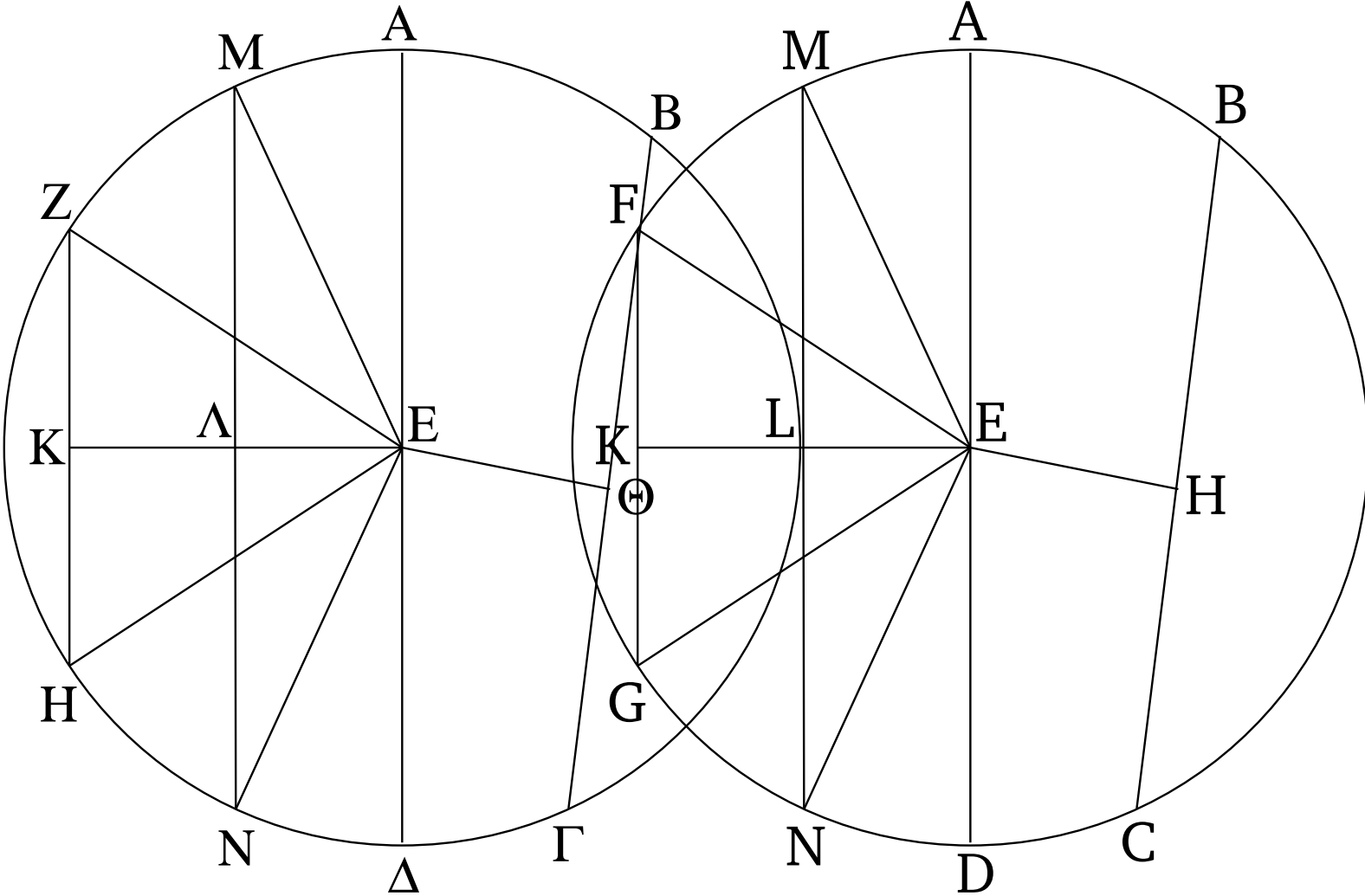
Let $ABCD$ be a circle, and let AD be its diameter, and E (its) center. And let BC be nearer to the diameter AD ,[†] and FG further away. I say that AD is the greatest (straight-line), and BC (is) greater than FG .

For let EH and $EΚ$ be drawn from the center E , at right-angles to BC and FG (respectively) [Prop. 1.12]. And since BC is nearer to the center, and FG further away, $EΚ$ (is) thus greater than EH [Def. 3.5]. Let EL be made equal to EH [Prop. 1.3]. And LM being drawn through L , at right-angles to $EΚ$ [Prop. 1.11], let it be drawn through to N . And let ME , EN , FE , and EG be joined.

And since EH is equal to EL , BC is also equal to MN [Prop. 3.14]. Again, since AE is equal to EM , and ED

δὲ ἡ MN τῇ $BΓ$; ἡ $AD >$ ἄρα τῇ $BΓ$ μείζων ἐστίν. καὶ ἐπεὶ δύο αἱ ME , EN δύο ταῖς ZE , $EH >$ ἴσαι εἰσίν, καὶ γωνία ἡ ὑπὸ MEN γωνία τῇ $\text{ὑπὸ } ZEH$ μείζων [ἐστίν], βάσις $>$ ἄρα ἡ MN βάσεως τῇ ZH μείζων ἐστίν. ἀλλὰ ἡ MN τῇ $BΓ$ ἐδείκθη $>$ ἴση [καὶ ἡ $BΓ$ τῇ ZH μείζων ἐστίν]. μεγίστη μὲν $>$ ἄρα ἡ AD διάμετρος, μείζων δὲ ἡ $BΓ$ τῇ ZH .

to EN , AD is thus equal to ME and EN . But, ME and EN is greater than MN [Prop. 1.20] [also AD is greater than MN], and MN (is) equal to BC . Thus, AD is greater than BC . And since the two (straight-lines) ME , EN are equal to the two (straight-lines) FE , EG (respectively), and angle MEN [is] greater than angle FEG ,[‡] the base MN is thus greater than the base FG [Prop. 1.24]. But, MN was shown (to be) equal to BC [(so) BC is also greater than FG]. Thus, the diameter AD (is) the greatest (straight-line), and BC (is) greater than FG .



Ἐν κύκλῳ $>$ ἄρα μεγίστη μὲν ἐστίν ἡ διάμετρος, τῶν δὲ $>$ ἄλλων αἰεὶ ἡ $>$ ἐγγιον τοῦ κέντρου τῇ $\text{ἄπωτερον μείζων ἐστίν}$; $<$ ὅπερ $>$ ἐδει δεῖχαι.

Thus, in a circle, a diameter (is) the greatest (straight-line), and for the others, a (straight-line) nearer to the center is always greater than one further away. (Which is) the very thing it was required to show.

[†] Euclid should have said “to the center”, rather than “to the diameter AD ”, since BC , AD and FG are not necessarily parallel.

[‡] This is not proved, except by reference to the figure.

16.

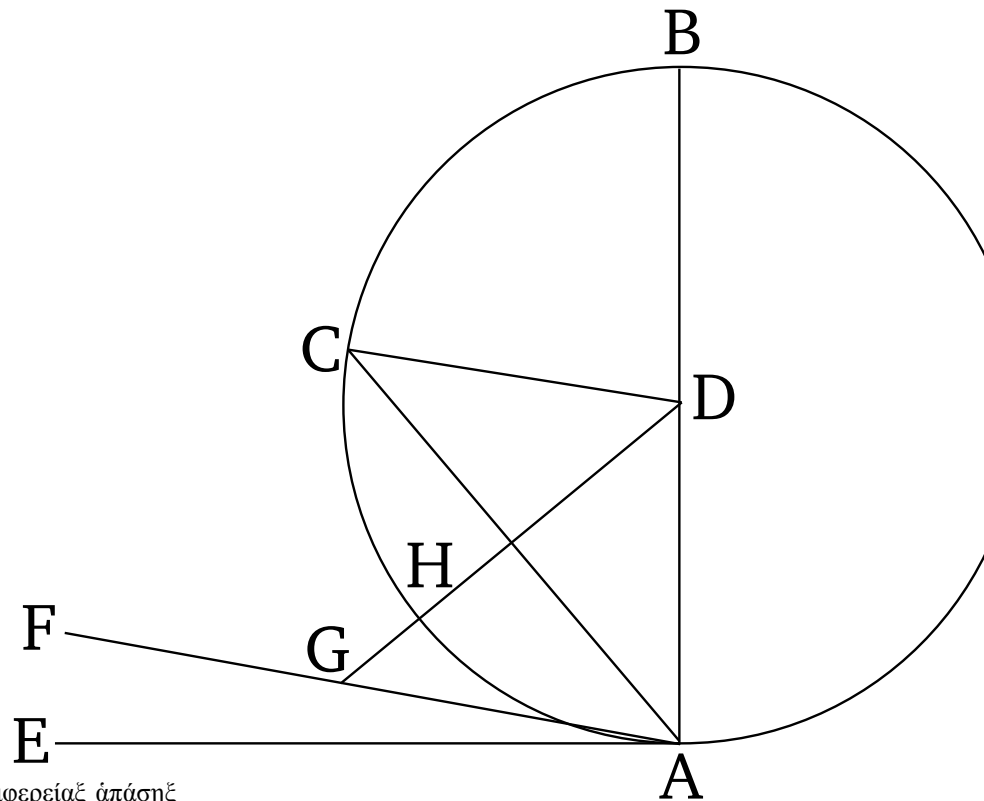
Ἡ τῇ διαμέτρῳ τοῦ κύκλου πρὸς ὀρθὰς ἀπ' $>$ ἀκρὰς ἀγομένη ἐκτὸς πεσεῖται τοῦ κύκλου, καὶ εἰς τὸν μεταχὼ τόπον τῇ $\text{τε εὐθείᾳ καὶ τῇ περιφερείᾳ ἑτέρα εὐθεῖα οὐ παρεμπεσείται}$, καὶ ἡ μὲν τοῦ ἡμικυκλίου γωνία ἀπάσης γωνίας ὀχείας εὐθυγράμμου μείζων ἐστίν, ἡ δὲ λοιπὴ ἐλάττω.

$>$ Ἐστω κύκλος ὁ $ABΓ$ περὶ κέντρον τὸ Δ καὶ διάμετρον τὴν AB ; λέγω, $<$ ὅτι ἡ ἀπὸ τοῦ A τῇ AB πρὸς ὀρθὰς ἀπ' $>$ ἀκρὰς ἀγομένη ἐκτὸς πεσεῖται τοῦ κύκλου.

Proposition 16

A (straight-line) drawn at right-angles to the diameter of a circle, from its end, will fall outside the circle. And another straight-line cannot be inserted into the space between the (aforementioned) straight-line and the circumference. And the angle of the semi-circle is greater than any acute rectilinear angle whatsoever, and the remaining (angle is) less (than any acute rectilinear angle).

Let ABC be a circle around the center D and the diameter AB . I say that the (straight-line) drawn from A , at right-



ὑπὸ τε τῆς BA εὐθείας καὶ τῆς ΓΘΑ περιφερείας ἀπάσης
γωνίας ὀχείας εὐθυγράμμου μείζων ἐστίν, ἢ δὲ λοιπὴ ἢ
περιεθομένη ὑπὸ τε τῆς ΓΘΑ περιφερείας καὶ τῆς AE
εὐθείας ἀπάσης γωνίας ὀχείας εὐθυγράμμου ἐλάττων ἐστίν.
Εἰ γὰρ ἐστί τις γωνία εὐθύγραμμος μείζων μὲν τῆς περιεθομένης
ὑπὸ τε τῆς BA εὐθείας καὶ τῆς ΓΘΑ περιφερείας, ἐλάττων
δὲ τῆς περιεθομένης ὑπὸ τε τῆς ΓΘΑ περιφερείας καὶ τῆς
AE εὐθείας, εἰς τὸν μεταχὺ τόπον τῆς τε ΓΘΑ περιφερείας
καὶ τῆς AE εὐθείας εὐθεῖα παρεμπεσεῖται, <ἥτις ποιήσει
μείζονα μὲν τῆς περιεθομένης ὑπὸ τε τῆς BA εὐθείας καὶ
τῆς ΓΘΑ περιφερείας ὑπὸ εὐθειῶν περιεθομένην, ἐλάττονα
δὲ τῆς περιεθομένης ὑπὸ τε τῆς ΓΘΑ περιφερείας καὶ τῆς
AE εὐθείας. οὐ παρεμπίπτει δέ; οὐκ >άρα τῆς περιεθομένης
γωνίας ὑπὸ τε τῆς BA εὐθείας καὶ τῆς ΓΘΑ περιφερείας
>έσται μείζων ὀχείᾳ ὑπὸ εὐθειῶν περιεθομένη, οὐδὲ μὴν
ἐλάττων τῆς περιεθομένης ὑπὸ τε τῆς ΓΘΑ περιφερείας καὶ
τῆς AE εὐθείας.

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, <ὅτι ἡ τῇ| διαμέτρῳ τοῦ κύκλου
πρὸς ὀρθὰς ἀπ> >άκρας ἀγομένη ἐφάπτεται τοῦ κύκλου
[καὶ <ὅτι εὐθεῖα κύκλου καθ> <ἐν μόνον ἐφάπτεται σημείον,
ἐπειδήπερ καὶ ἡ κατὰ δύο αὐτῶ| συμβάλλουσα ἐντὸς αὐτοῦ
πίπτουσα ἐδείχθη]; <ὅπερ >έδει δεῖχαι.

And I also say that the semi-circular angle contained by
the straight-line BA and the circumference CHA is greater
than any acute rectilinear angle whatsoever, and the re-
maining (angle) contained by the circumference CHA and
the straight-line AE is less than any acute rectilinear angle
whatsoever.

For if any rectilinear angle is greater than the (angle) con-
tained by the straight-line BA and the circumference CHA ,
or less than the (angle) contained by the circumference
 CHA and the straight-line AE , (then) a straight-line can be
inserted into the space between the circumference CHA and
the straight-line AE —anything which will make (an angle)
contained by straight-lines greater than the angle contained
by the straight-line BA and the circumference CHA , or less
than the (angle) contained by the circumference CHA and
the straight-line AE . But (such a straight-line) cannot be
inserted. Thus, an acute (angle) contained by straight-lines
cannot be greater than the angle contained by the straight-
line BA and the circumference CHA , neither (can it be)
less than the (angle) contained by the circumference CHA
and the straight-line AE .

Corollary

So, from this, (it is) manifest that a (straight-line) drawn at
right-angles to the diameter of a circle, from its extremity,
touches the circle [and that the straight-line touches the cir-
cle at a single point, inasmuch as it was also shown that a
(straight-line) meeting (the circle) at two (points) falls inside
it [Prop. 3.2]]. (Which is) the very thing it was required to
show.

17.

Ἀπὸ τοῦ δοθέντος σημείου τοῦ δοθέντος κύκλου ἐφαπτομένην
εὐθεῖαν γραμμὴν ἀγαγεῖν.

>Ἐστω τὸ μὲν δοθὲν σημεῖον τὸ A , ὁ δὲ δοθεὶς κύκλος ὁ

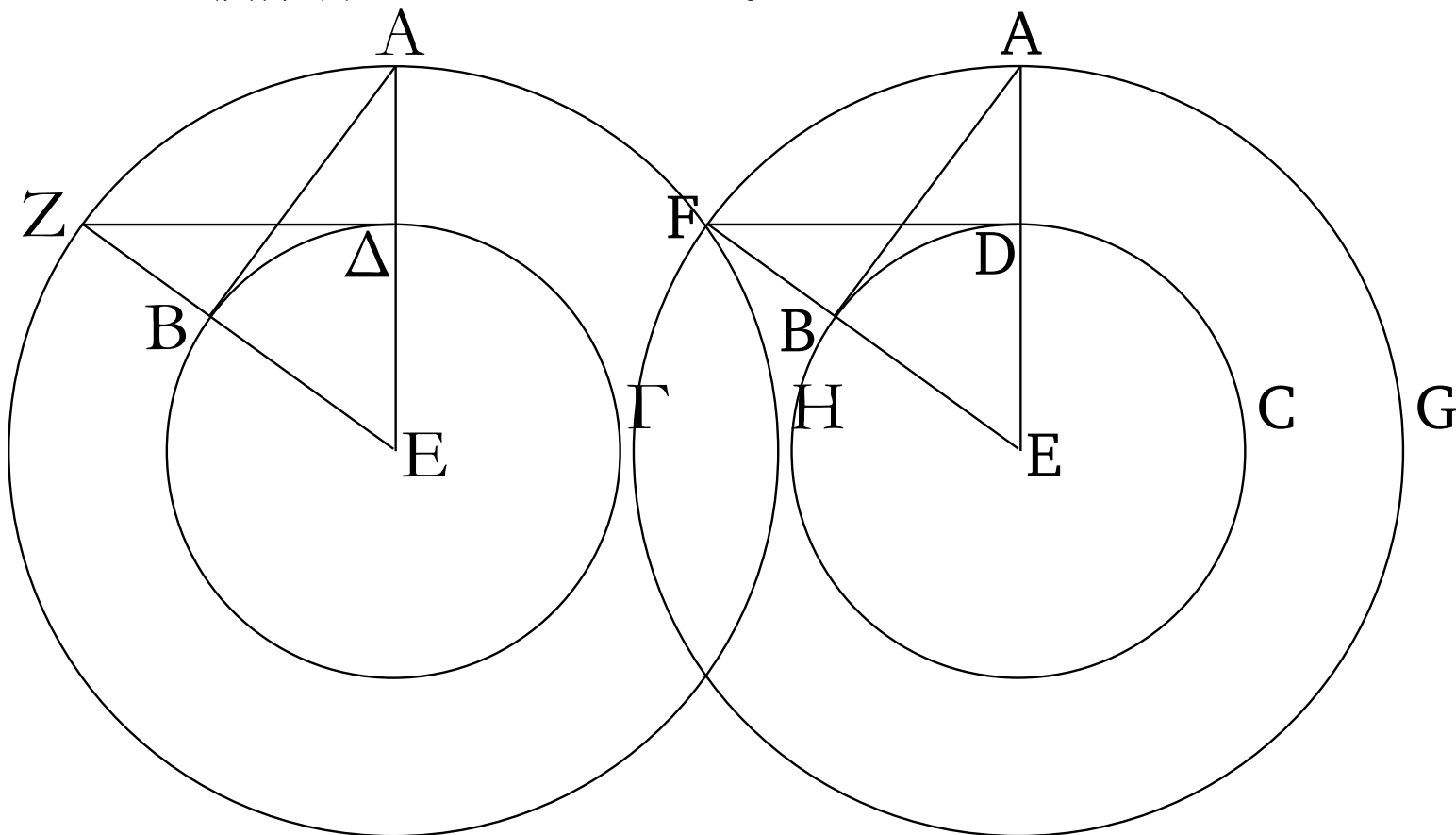
Proposition 17

To draw a straight-line touching a given circle from a given
point.

Let A be the given point, and BCD the given circle. So it is

ΒΓΔ; δεῖ δὴ ἀπὸ τοῦ Α σημείου τοῦ ΒΓΔ κύκλου ἐφαπτομένην εὐθεΐαν γραμμὴν ἀγαγεῖν.

required to draw a straight-line touching circle BCD from point A .



Εἰλήφθω γὰρ τὸ κέντρον τοῦ κύκλου τὸ Ε, καὶ ἐπεζεύθθω ἡ ΑΕ, καὶ κέντρῳ μὲν τῷ Ε διαστήματι δὲ τῷ ΕΑ κύκλοξ γεγράφθω ὁ ΑΖΗ, καὶ ἀπὸ τοῦ Δ τῇ ΕΑ πρὸξ ὀρθὰξ >ήθθω ἡ ΔΖ, καὶ ἐπεζεύθθωσαν αἱ ΕΖ, ΑΒ; λέγω, <ὅτι ἀπὸ τοῦ Α σημείου τοῦ ΒΓΔ κύκλου ἐφαπτομένη >ῆκται ἡ ΑΒ.

Ἐπεὶ γὰρ τὸ Ε κέντρον ἐστὶ τῶν ΒΓΔ, ΑΖΗ κύκλων, >ίση >άρα ἐστὶν ἡ μὲν ΕΑ τῇ ΕΖ, ἡ δὲ ΕΔ τῇ ΕΒ; δύο δὲ αἱ ΑΕ, ΕΒ δύο ταῖς ΖΕ, ΕΔ >ίσαι εἰσίν; καὶ γωνίαν κοινὴν περιέθουσι τὴν πρὸξ τῷ Ε; βάσειξ >άρα ἡ ΔΖ βάσει τῇ ΑΒ >ίση ἐστίν, καὶ τὸ ΔΕΖ τρίγωνον τῷ ΕΒΑ τριγώνῳ >ίσον ἐστίν, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαιξ; >ίση >άρα ἡ ὑπὸ ΕΔΖ τῇ ὑπὸ ΕΒΑ. ὀρθὴ δὲ ἡ ὑπὸ ΕΔΖ; ὀρθὴ >άρα καὶ ἡ ὑπὸ ΕΒΑ. καὶ ἐστὶν ἡ ΕΒ ἐκ τοῦ κέντρου; ἡ δὲ τῇ διαμέτρῳ τοῦ κύκλου πρὸξ ὀρθὰξ ἀπ> >άκραξ ἀγομένη ἐφάπτεται τοῦ κύκλου; ἡ ΑΒ >άρα ἐφάπτεται τοῦ ΒΓΔ κύκλου.

Ἀπὸ τοῦ >άρα δοθέντοξ σημείου τοῦ Α τοῦ δοθέντοξ κύκλου τοῦ ΒΓΔ ἐφαπτομένη εὐθεΐα γραμμὴ >ῆκται ἡ ΑΒ; <ὅπερ >έδει ποιῆσαι.

For let the center E of the circle be found [Prop. 3.1], and let AE be joined. And let (the circle) AFG be drawn with center E and radius EA . And let DF be drawn from (point) D , at right-angles to EA [Prop. 1.11]. And let EF and AB be joined. I say that the (straight-line) AB has been drawn from point A touching circle BCD .

For since E is the center of circles BCD and AFG , EA is thus equal to EF , and ED to EB . So the two (straight-lines) AE , EB are equal to the two (straight-lines) FE , ED (respectively). And they contain a common angle at E . Thus, the base DF is equal to the base AB , and triangle DEF is equal to triangle EBA , and the remaining angles (are equal) to the (corresponding) remaining angles [Prop. 1.4]. Thus, (angle) EDF (is) equal to EBA . And EDF (is) a right-angle. Thus, EBA (is) also a right-angle. And EB is a radius. And a (straight-line) drawn at right-angles to the diameter of a circle, from its extremity, touches the circle [Prop. 3.16 corr.]. Thus, AB touches circle BCD . Thus, the straight-line AB has been drawn touching the given circle BCD from the given point A . (Which is) the very thing it was required to do.

18.

Ἐὰν κύκλου ἐφάπτηται τιξ εὐθεΐα, ἀπὸ δὲ τοῦ κέντρου ἐπὶ τὴν ἀφὴν ἐπιζευθθῇ τιξ εὐθεΐα, ἡ ἐπιζευθθεῖσα κάθετοξ >έσται ἐπὶ τὴν ἐφαπτομένην.

Κύκλου γὰρ τοῦ ΑΒΓ ἐφαπτέσθω τιξ εὐθεΐα ἡ ΔΕ κατὰ τὸ Γ σημεῖον, καὶ εἰλήφθω τὸ κέντρον τοῦ ΑΒΓ κύκλου τὸ Ζ, καὶ ἀπὸ τοῦ Ζ ἐπὶ τὸ Γ ἐπεζεύθθω ἡ ΖΓ; λέγω, <ὅτι ἡ ΖΓ κάθετόξ ἐστὶν ἐπὶ τὴν ΔΕ.

Proposition 18

If some straight-line touches a circle, and some (other) straight-line is joined from the center (of the circle) to the point of contact, (then) the (straight-line) so joined will be perpendicular to the tangent.

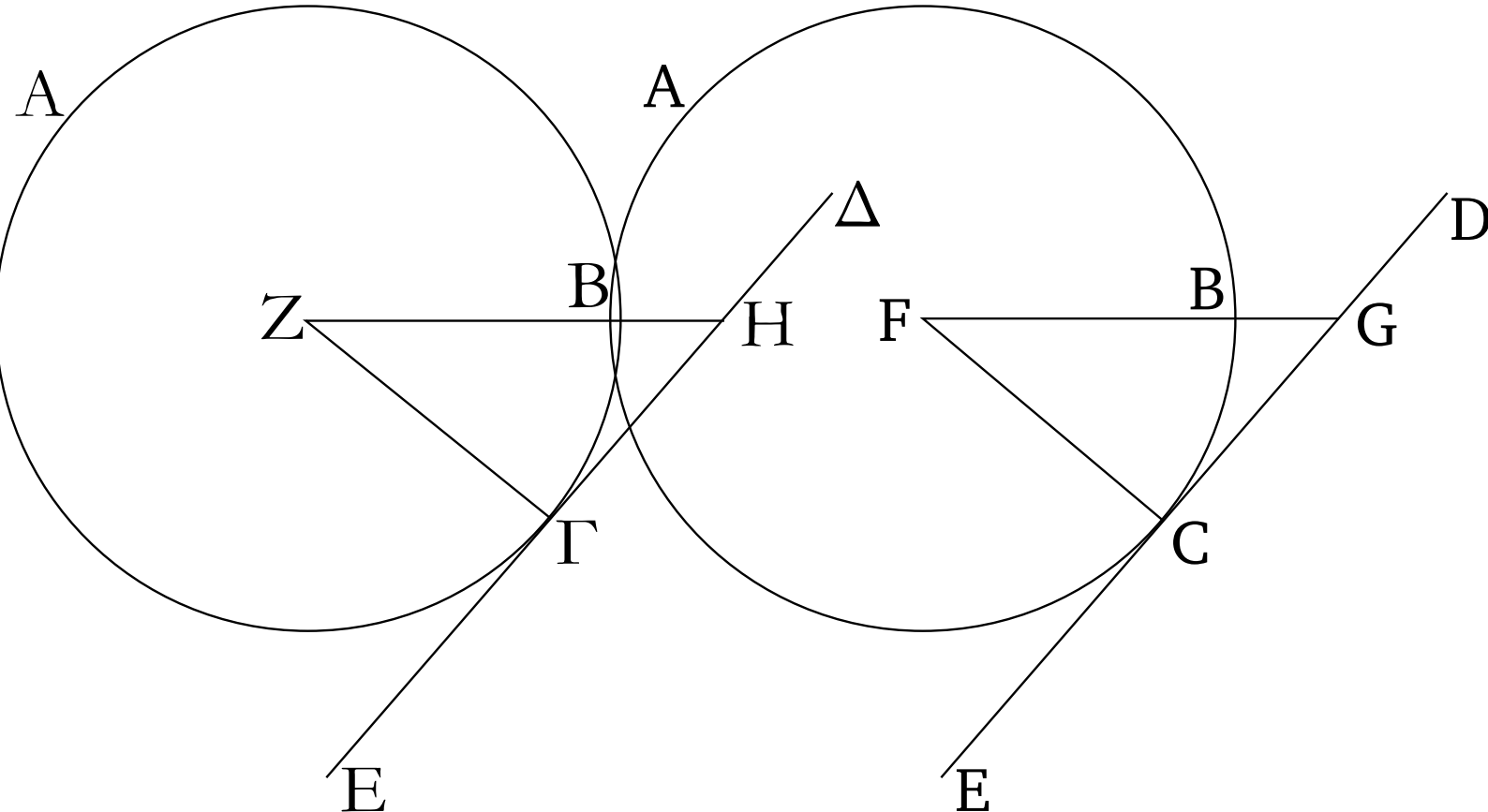
For let some straight-line DE touch the circle ABC at point C , and let the center F of circle ABC be found [Prop. 3.1], and let FC be joined from F to C . I say that FC is perpen-

Εἰ γὰρ μή, >ήθω ἀπὸ τοῦ Z ἐπὶ τὴν ΔE κάθετος ἡ ZH .
 Ἐπεὶ οὖν ἡ ὑπὸ ZHG γωνία ὀρθή ἐστιν, ὀχρεῖα >άρα ἐστὶν ἡ ὑπὸ ZGH ; ὑπὸ δὲ τὴν μείζονα γωνίαν ἡ μείζων πλευρὰ ὑποτείνει; μείζων >άρα ἡ ZG τῆς ZH ; >ίση δὲ ἡ ZG τῇ ZB ; μείζων >άρα καὶ ἡ ZB τῆς ZH ἢ ἐλάττων τῆς μείζονος; <ὅπερ ἐστὶν ἀδύνατον. οὐκ >άρα ἡ ZH κάθετός ἐστιν ἐπὶ τὴν ΔE . ὁμοίως δὲ δείχομεν, <ὅτι οὐδ> >άλλη τις πλὴν τῆς ZG ; ἡ ZG >άρα κάθετός ἐστιν ἐπὶ τὴν ΔE .

dicular to DE .

For if not, let FG be drawn from F , perpendicular to DE [Prop. 1.12].

Therefore, since angle FGC is a right-angle, (angle) FCG is thus acute [Prop. 1.17]. And the greater angle is subtended by the greater side [Prop. 1.19]. Thus, FC (is) greater than FG . And FC (is) equal to FB . Thus, FB (is) also greater than FG , the lesser than the greater. The very thing is impossible. Thus, FG is not perpendicular to DE . So, similarly, we can show that neither (is) any other (straight-line) except FC . Thus, FC is perpendicular to DE .



Ἐὰν >άρα κύκλου ἐφάπτηται τις εὐθεῖα, ἀπὸ δὲ τοῦ κέντρου ἐπὶ τὴν ἀφὴν ἐπιζευθθῇ τις εὐθεῖα, ἡ ἐπιζευθθεῖσα κάθετος >έσται ἐπὶ τὴν ἐφαπτομένην; <ὅπερ> >έδει δεῖχαι.

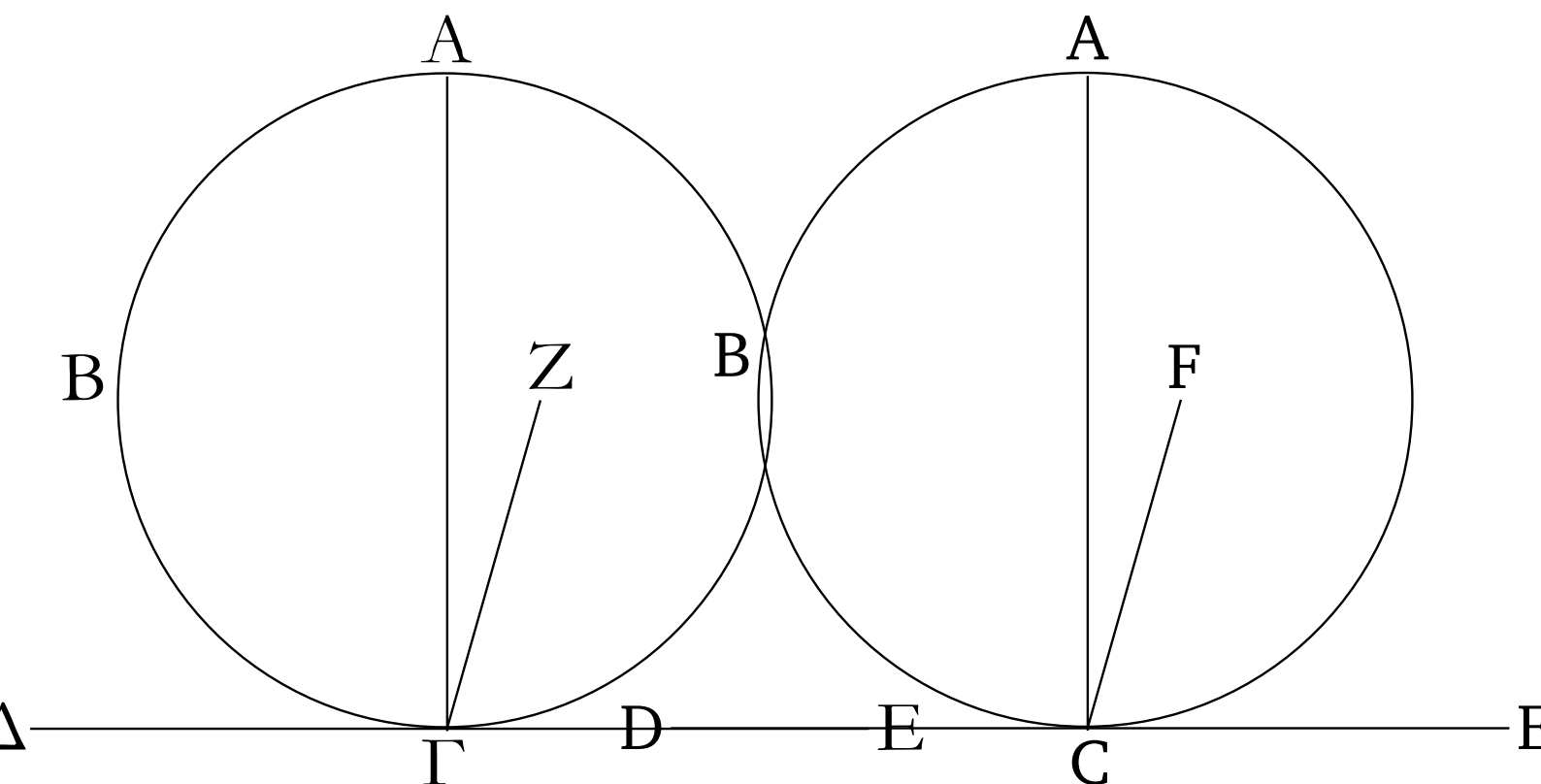
Thus, if some straight-line touches a circle, and some (other) straight-line is joined from the center (of the circle) to the point of contact, (then) the (straight-line) so joined will be perpendicular to the tangent. (Which is) the very thing it was required to show.

19.

Ἐὰν κύκλου ἐφάπτηται τις εὐθεῖα, ἀπὸ δὲ τῆς ἀφῆς τῇ ἐφαπτομένῃ πρὸς ὀρθὰς [γωνίαξ] εὐθεῖα γραμμὴ ἀθθῇ, ἐπὶ τῆς ἀθθεῖσης >έσται τὸ κέντρον τοῦ κύκλου.

Proposition 19

If some straight-line touches a circle, and a straight-line is drawn from the point of contact, at right-[angles] to the tangent, (then) the center (of the circle) will be on the (straight-line) so drawn.



Κύκλου γάρ τοῦ $AB\Gamma$ ἐφαπτέσθω τις εὐθεΐα ἡ ΔE κατὰ τὸ Γ σημεῖον, καὶ ἀπὸ τοῦ Γ τῇ ΔE πρὸς ὀρθὰς >ῆθθω ἡ ΓA ; λέγω, <ὅτι ἐπὶ τῇς $A\Gamma$ ἐστὶ τὸ κέντρον τοῦ κύκλου.

μὴ γάρ, ἀλλ> εἰ δυνατόν, >έστω τὸ Z , καὶ ἐπεζεύθθω ἡ ΓZ . Ἐπεὶ [ο>ὖν] κύκλου τοῦ $AB\Gamma$ ἐφάπτεται τις εὐθεΐα ἡ ΔE , ἀπὸ δὲ τοῦ κέντρου ἐπὶ τὴν ἀφὴν ἐπέζευκται ἡ $Z\Gamma$, ἡ $Z\Gamma$ >άρα κάθετόξ ἐστὶν ἐπὶ τὴν ΔE ; ὀρθή>άρα ἐστὶν ἡ ὑπὸ $Z\Gamma E$. ἐστὶ δὲ καὶ ἡ ὑπὸ $A\Gamma E$ ὀρθή; >ῖση >άρα ἐστὶν ἡ ὑπὸ $Z\Gamma E$ τῇ ὑπὸ $A\Gamma E$ ἡ ἐλάττων τῇ μείζονι; <ὅπερ ἐστὶν ἀδύνατον. οὐκ >άρα τὸ Z κέντρον ἐστὶ τοῦ $AB\Gamma$ κύκλου. ὁμοίωξ δὴ δείχομεν, <ὅτι οὐδ> >άλλο τι πλὴν ἐπὶ τῇς $A\Gamma$.

Ἐὰν >άρα κύκλου ἐφάπτηται τις εὐθεΐα, ἀπὸ δὲ τῆς ἀφῆς τῇ ἐφαπτομένη πρὸς ὀρθὰς εὐθεΐα γραμμὴ ἀθθῇ, ἐπὶ τῇς ἀθθείσης >έσται τὸ κέντρον τοῦ κύκλου; <ὅπερ >έδει δείχαι.

For let some straight-line DE touch the circle ABC at point C . And let CA be drawn from C , at right-angles to DE [Prop. 1.11]. I say that the center of the circle is on AC .

For (if) not, if possible, let F be (the center of the circle), and let CF be joined.

[Therefore], since some straight-line DE touches the circle ABC , and FC has been joined from the center to the point of contact, FC is thus perpendicular to DE [Prop. 3.18]. Thus, FCE is a right-angle. And ACE is also a right-angle. Thus, FCE is equal to ACE , the lesser to the greater. The very thing is impossible. Thus, F is not the center of circle ABC . So, similarly, we can show that neither is any (point) other (than one) on AC .

Thus, if some straight-line touches a circle, and a straight-line is drawn from the point of contact, at right-angles to the tangent, (then) the center (of the circle) will be on the (straight-line) so drawn. (Which is) the very thing it was required to show.

20.

Ἐν κύκλῳ ἡ πρὸς τῷ κέντρῳ γωνία διπλασίῳ ἐστὶ τῇς πρὸς τῇ περιφερείᾳ, <ὅταν τὴν αὐτὴν περιφέρειαν βάσιν>έθωσιν αἱ γωνίαι.

>Ἐστω κύκλος ὁ $AB\Gamma$, καὶ πρὸς μὲν τῷ κέντρῳ αὐτοῦ γωνία >έστω ἡ ὑπὸ $BE\Gamma$, πρὸς δὲ τῇ περιφερείᾳ ἡ ὑπὸ $BA\Gamma$, ἐθέτωσαν δὲ τὴν αὐτὴν περιφέρειαν βάσιν τὴν $B\Gamma$; λέγω, <ὅτι διπλασίῳ ἐστὶν ἡ ὑπὸ $BE\Gamma$ γωνία τῇς ὑπὸ $BA\Gamma$.

Ἐπιζευθεῖσα γάρ ἡ AE διήθθω ἐπὶ τὸ Z .

Ἐπεὶ ο>ὖν >ῖση ἐστὶν ἡ EA τῇ EB , >ῖση καὶ γωνία ἡ ὑπὸ EAB τῇ ὑπὸ EBA ; αἱ >άρα ὑπὸ EAB , EBA γωνίαι τῇς ὑπὸ EAB διπλασίους εἰσίν. >ῖση δὲ ἡ ὑπὸ BEZ ταῖς ὑπὸ EAB , EBA ; καὶ ἡ ὑπὸ BEZ >άρα τῇς ὑπὸ EAB ἐστὶ διπλῇ. διὰ τὰ αὐτὰ δὲ καὶ ἡ ὑπὸ ZEG τῇς ὑπὸ EAG ἐστὶ διπλῇ. <ὅλη>άρα ἡ ὑπὸ $BE\Gamma$ <όλης τῇς ὑπὸ $BA\Gamma$ ἐστὶ διπλῇ.

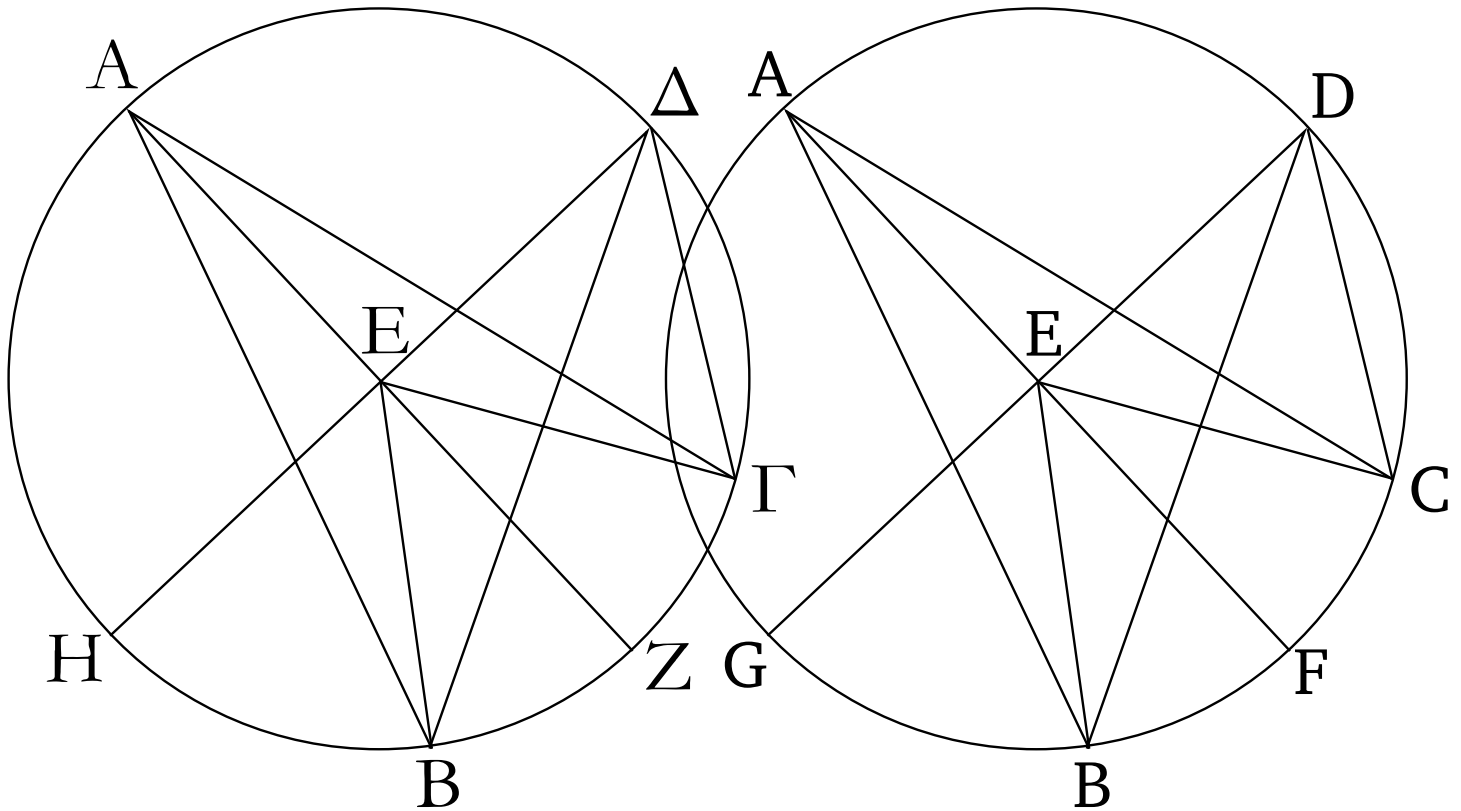
Proposition 20

In a circle, the angle at the center is double that at the circumference, when the angles have the same circumference base.

Let ABC be a circle, and let BEC be an angle at its center, and BAC (one) at (its) circumference. And let them have the same circumference base BC . I say that angle BEC is double (angle) BAC .

For being joined, let AE be drawn through to F .

Therefore, since EA is equal to EB , angle EAB (is) also equal to EBA [Prop. 1.5]. Thus, angle EAB and EBA is double (angle) EAB . And BEF (is) equal to EAB and EBA [Prop. 1.32]. Thus, BEF is also double EAB . So, for the same (reasons), FEC is also double EAC . Thus, the whole (angle) BEC is double the whole (angle) BAC .



Κεκλάσθω δὴ πάλιν, καὶ ᾗ ἐστὼ ἑτέρα γωνία ἢ ὑπὸ $B\Delta\Gamma$, καὶ ἐπιζευθεῖσα ἡ ΔE ἐκβεβλήσθω ἐπὶ τὸ H . ὁμοίως δὲ δείχομεν, <ὅτι διπλὴ ἐστὶν ἡ ὑπὸ $HE\Gamma$ γωνία τῇς ὑπὸ $E\Delta\Gamma$, <ὡν ἡ ὑπὸ HEB διπλὴ ἐστὶ τῇς ὑπὸ $E\Delta B$; λοιπὴ ᾗ ὑπὸ $BE\Gamma$ διπλὴ ἐστὶ τῇς ὑπὸ $B\Delta\Gamma$.

Ἐν κύκλῳ ᾗ ᾖ ἡ πρὸς τῷ κέντρῳ γωνία διπλασίον ἐστὶ τῇς πρὸς τῇ περιφερείᾳ, <ὅταν τὴν αὐτὴν περιφέρειαν βάσιν ᾗ ἔθωσιν [αἱ γωνίαι]; <ὅπερ ᾗ δεῖ δεῖξαι.

So let another (straight-line) be inflected, and let there be another angle, BDC . And DE being joined, let it be produced to G . So, similarly, we can show that angle GEC is double EDC , of which GEB is double EDB . Thus, the remaining (angle) BEC is double the (remaining angle) BDC .

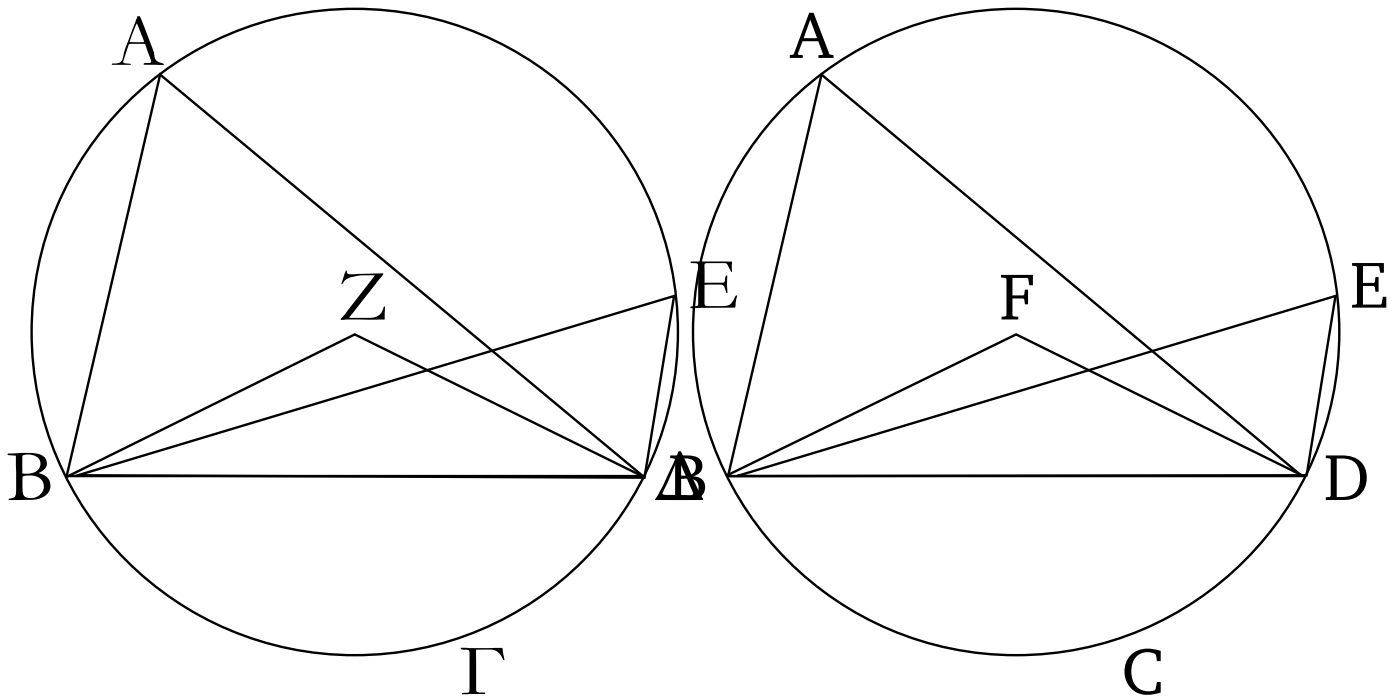
Thus, in a circle, the angle at the center is double that at the circumference, when [the angles] have the same circumference base. (Which is) the very thing it was required to show.

21.

Ἐν κύκλῳ αἱ ἐν τῷ αὐτῷ τμήματι γωνίαι ᾗσαι ἀλλήλαις εἰσίν.

Proposition 21

In a circle, angles in the same segment are equal to one another.



>Ἐστω κύκλος ὁ ΑΒΓΔ, καὶ ἐν τῷ αὐτῷ τμήματι τῷ ΒΑΕΔ γωνίαι >έστωσαν αἱ ὑπὸ ΒΑΔ, ΒΕΔ; λέγω, <ὅτι αἱ ὑπὸ ΒΑΔ, ΒΕΔ γωνίαι >ίσαι ἀλλήλαις εἰσίν.

Εἰλήφθω γὰρ τοῦ ΑΒΓΔ κύκλου τὸ κέντρον, καὶ >έστω τὸ Ζ, καὶ ἐπέξευθώσαν αἱ ΒΖ, ΖΔ.

Καὶ ἐπεὶ ἡ μὲν ὑπὸ ΒΖΔ γωνία πρὸς τῷ κέντρῳ ἐστίν, ἡ δὲ ὑπὸ ΒΑΔ πρὸς τῇ περιφερείᾳ, καὶ >έθουσι τὴν αὐτὴν περιφέρειαν βάσιν τὴν ΒΓΔ, ἡ >άρα ὑπὸ ΒΖΔ γωνία διπλασίῳ ἐστὶ τῇ ὑπὸ ΒΑΔ. διὰ τὰ αὐτὰ δὲ ἡ ὑπὸ ΒΖΔ καὶ τῇ ὑπὸ ΒΕΔ ἐστὶ διπλασίῳ; >ίση >άρα ἡ ὑπὸ ΒΑΔ τῇ ὑπὸ ΒΕΔ.

Ἐν κύκλῳ >άρα αἱ ἐν τῷ αὐτῷ τμήματι γωνίαι >ίσαι ἀλλήλαις εἰσίν; <ὅπερ >έδει δεῖχαι.

22.

Τῶν ἐν τοῖς κύκλοις τετραπλεύρων αἱ ἀπεναντίον γωνίαι δυσὶν ὀρθαῖς >ίσαι εἰσίν.

Let $ABCD$ be a circle, and let BAD and BED be angles in the same segment $BAED$. I say that angles BAD and BED are equal to one another.

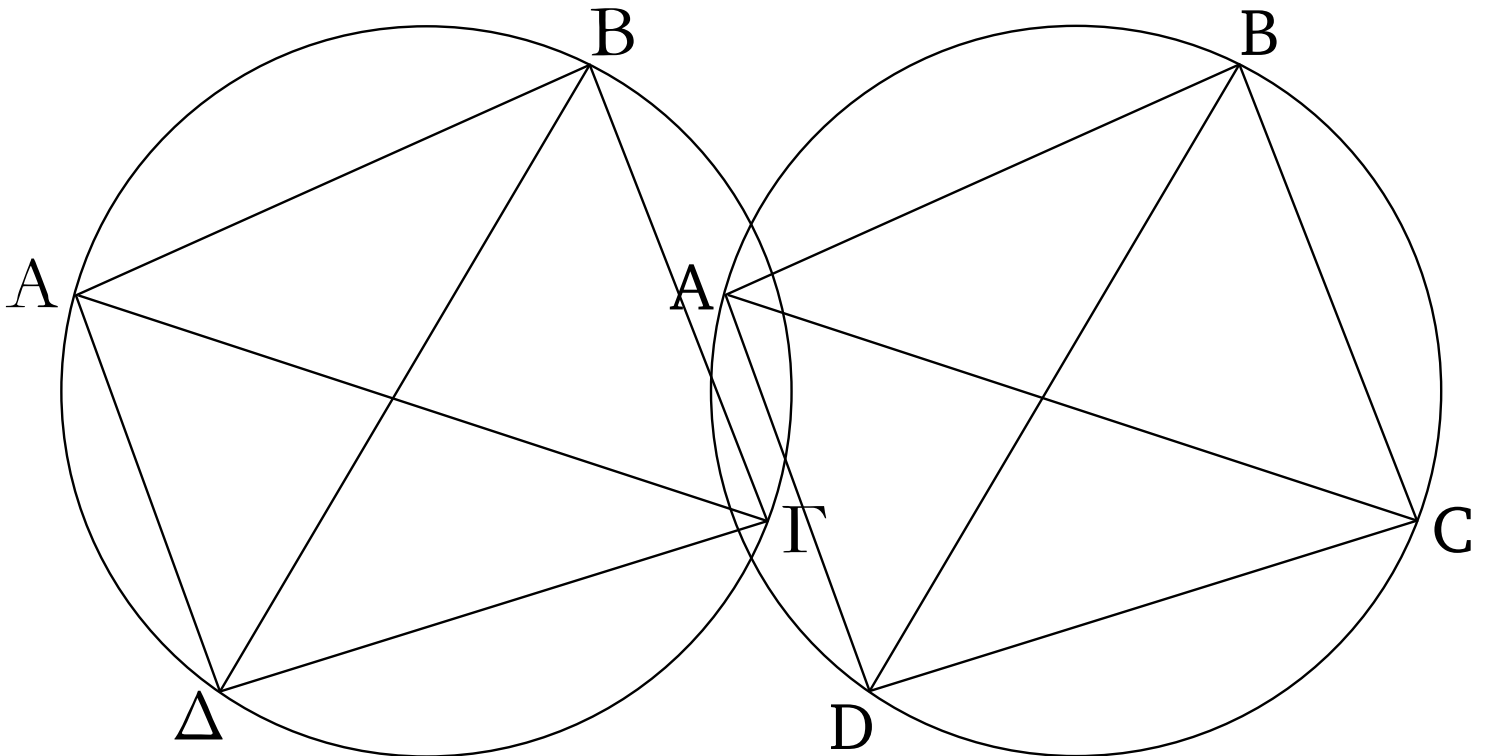
For let the center of circle $ABCD$ be found [Prop. 3.1], and let it be (at point) F . And let BF and FD be joined.

And since angle BFD is at the center, and BAD at the circumference, and they have the same circumference base BCD , angle BFD is thus double BAD [Prop. 3.20]. So, for the same (reasons), BFD is also double BED . Thus, BAD (is) equal to BED .

Thus, in a circle, angles in the same segment are equal to one another. (Which is) the very thing it was required to show.

Proposition 22

For quadrilaterals within circles, the (sum of the) opposite angles is equal to two right-angles.



>Ἐστω κύκλος ὁ ΑΒΓΔ, καὶ ἐν αὐτῷ τετράπλευρον >ἔστω τὸ ΑΒΓΔ; λέγω, <ὅτι αἱ ἀπεναντίον γωνίαι δυσὶν ὀρθαῖς >ίσαι εἰσίν.

Ἐπεξεύθωσαν αἱ ΑΓ, ΒΔ.

Ἐπεὶ ο>ὖν παντὸς τριγώνου αἱ τρεῖς γωνίαι δυσὶν ὀρθαῖς >ίσαι εἰσίν, τοῦ ΑΒΓ >άρα τριγώνου αἱ τρεῖς γωνίαι αἱ ὑπὸ ΓΑΒ, ΑΒΓ, ΒΓΑ δυσὶν ὀρθαῖς >ίσαι εἰσίν. >ίση δὲ ἡ μὲν ὑπὸ ΓΑΒ τῇ ὑπὸ ΒΔΓ; ἐν γὰρ τῷ αὐτῷ τμήματι εἰσι τῷ ΒΑΔΓ; ἡ δὲ ὑπὸ ΑΓΒ τῇ ὑπὸ ΑΔΒ; ἐν γὰρ τῷ αὐτῷ τμήματι εἰσι τῷ ΑΔΓΒ; <ὅλη >άρα ἡ ὑπὸ ΑΔΓ ταῖς ὑπὸ ΒΑΓ, ΑΓΒ >ίση ἔστιν. κοινὴ προσκείσθω ἡ ὑπὸ ΑΒΓ; αἱ >άρα ὑπὸ ΑΒΓ, ΒΑΓ, ΑΓΒ ταῖς ὑπὸ ΑΒΓ, ΑΔΓ >ίσαι εἰσίν. ἀλλ> αἱ ὑπὸ ΑΒΓ, ΒΑΓ, ΑΓΒ δυσὶν ὀρθαῖς >ίσαι εἰσίν. καὶ αἱ ὑπὸ ΑΒΓ, ΑΔΓ >άρα δυσὶν ὀρθαῖς >ίσαι εἰσίν. ὁμοίως δὲ δείχομεν, <ὅτι καὶ αἱ ὑπὸ ΒΑΔ, ΔΓΒ γωνίαι δυσὶν ὀρθαῖς >ίσαι εἰσίν. Τῶν >άρα ἐν τοῖς κύκλοις τετραπλεύρων αἱ ἀπεναντίον γωνίαι δυσὶν ὀρθαῖς >ίσαι εἰσίν; <ὅπερ >έδει δεῖχαι.

Let $ABCD$ be a circle, and let $ABCD$ be a quadrilateral within it. I say that the (sum of the) opposite angles is equal to two right-angles.

Let AC and BD be joined.

Therefore, since the three angles of any triangle are equal to two right-angles [Prop. 1.32], the three angles CAB , ABC , and BCA of triangle ABC are thus equal to two right-angles. And CAB (is) equal to BDC . For they are in the same segment $BADC$ [Prop. 3.21]. And ACB (is equal) to ADB . For they are in the same segment $ADCB$ [Prop. 3.21]. Thus, the whole of ADC is equal to BAC and ACB . Let ABC be added to both. Thus, ABC , BAC , and ACB are equal to ABC and ADC . But, ABC , BAC , and ACB are equal to two right-angles. Thus, ABC and ADC are also equal to two right-angles. Similarly, we can show that angles BAD and DCB are also equal to two right-angles.

Thus, for quadrilaterals within circles, the (sum of the) opposite angles is equal to two right-angles. (Which is) the very thing it was required to show.

23.

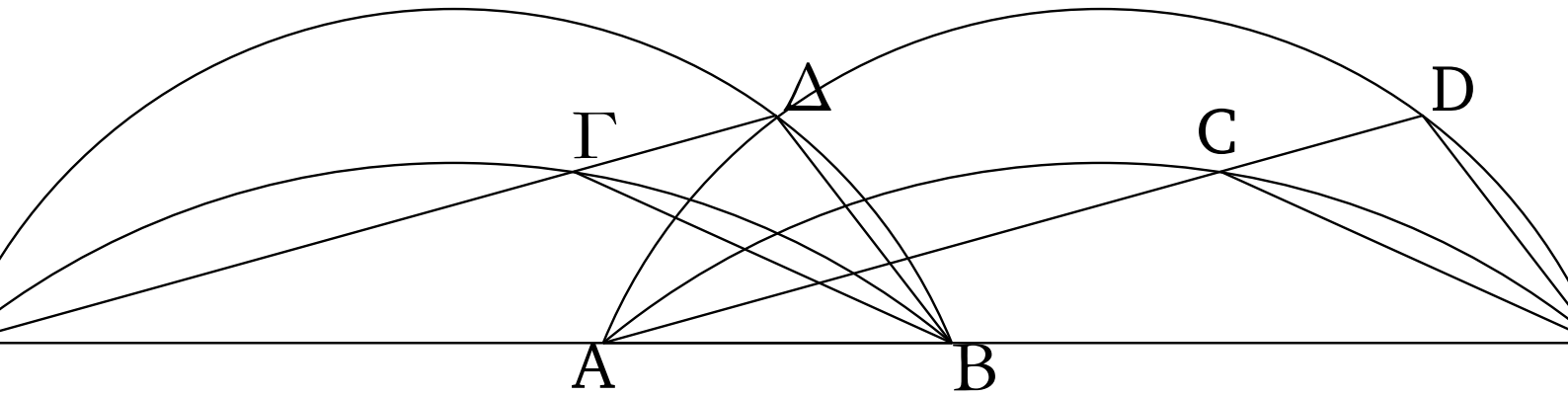
Ἐπὶ τῇ αὐτῇ εὐθείᾳ δύο τμήματα κύκλων <ὅμοια καὶ >άνισα οὐ συσταθήσεται ἐπὶ τὰ αὐτὰ μέρη.

Εἰ γὰρ δυνατόν, ἐπὶ τῇ αὐτῇ εὐθείᾳ τῇ ΑΒ δύο τμήματα κύκλων <ὅμοια καὶ >άνισα συνεστήτω ἐπὶ τὰ αὐτὰ μέρη τὰ ΑΓΒ, ΑΔΒ, καὶ διήθθω ἡ ΑΓΔ, καὶ ἐπεξεύθωσαν αἱ ΓΒ, ΔΒ.

Proposition 23

Two similar and unequal segments of circles cannot be constructed on the same side of the same straight-line.

For, if possible, let the two similar and unequal segments of circles, ACB and ADB , be constructed on the same side of the same straight-line AB . And let ACD be drawn through (the segments), and let CB and DB be joined.



Ἐπεὶ οὖν <όμοιόν ἐστι τὸ ΑΓΒ τμήμα τῷ ΑΔΒ τμήματι,
<όμοια δὲ τμήματα κύκλων ἐστὶ τὰ δεθόμενα γωνίαξ>ῖσαξ,
>ῖση >άρα ἐστὶν ἡ ὑπὸ ΑΓΒ γωνία τῇ ὑπὸ ΑΔΒ ἢ ἐκτὸς τῇ
ἐντὸς; <όπερ ἐστὶν ἀδύνατον.

Οὐκ >άρα ἐπὶ τῇξ αὐτῇξ εὐθείαξ δύο τμήματα κύκλων
<όμοια καὶ >άνισα συσταθήσεται ἐπὶ τὰ αὐτὰ μέρη; <όπερ
>έδει δεῖξαι.

Therefore, since segment ACB is similar to segment ADB ,
and similar segments of circles are those accepting equal
angles [Def. 3.11], angle ACB is thus equal to ADB ,
the external to the internal. The very thing is impossible
[Prop. 1.16].

Thus, two similar and unequal segments of circles cannot be
constructed on the same side of the same straight-line.

24.

Τὰ ἐπὶ >ῖσων εὐθειῶν <όμοια τμήματα κύκλων>ῖσα ἀλλήλοιξ
ἐστίν.

>Ἐστωσαν γὰρ ἐπὶ >ῖσων εὐθειῶν τῶν ΑΒ, ΓΔ <όμοια
τμήματα κύκλων τὰ ΑΕΒ, ΓΖΔ; λέγω, <ότι> ῖσον ἐστὶ τὸ
ΑΕΒ τμήμα τῷ ΓΖΔ τμήματι.

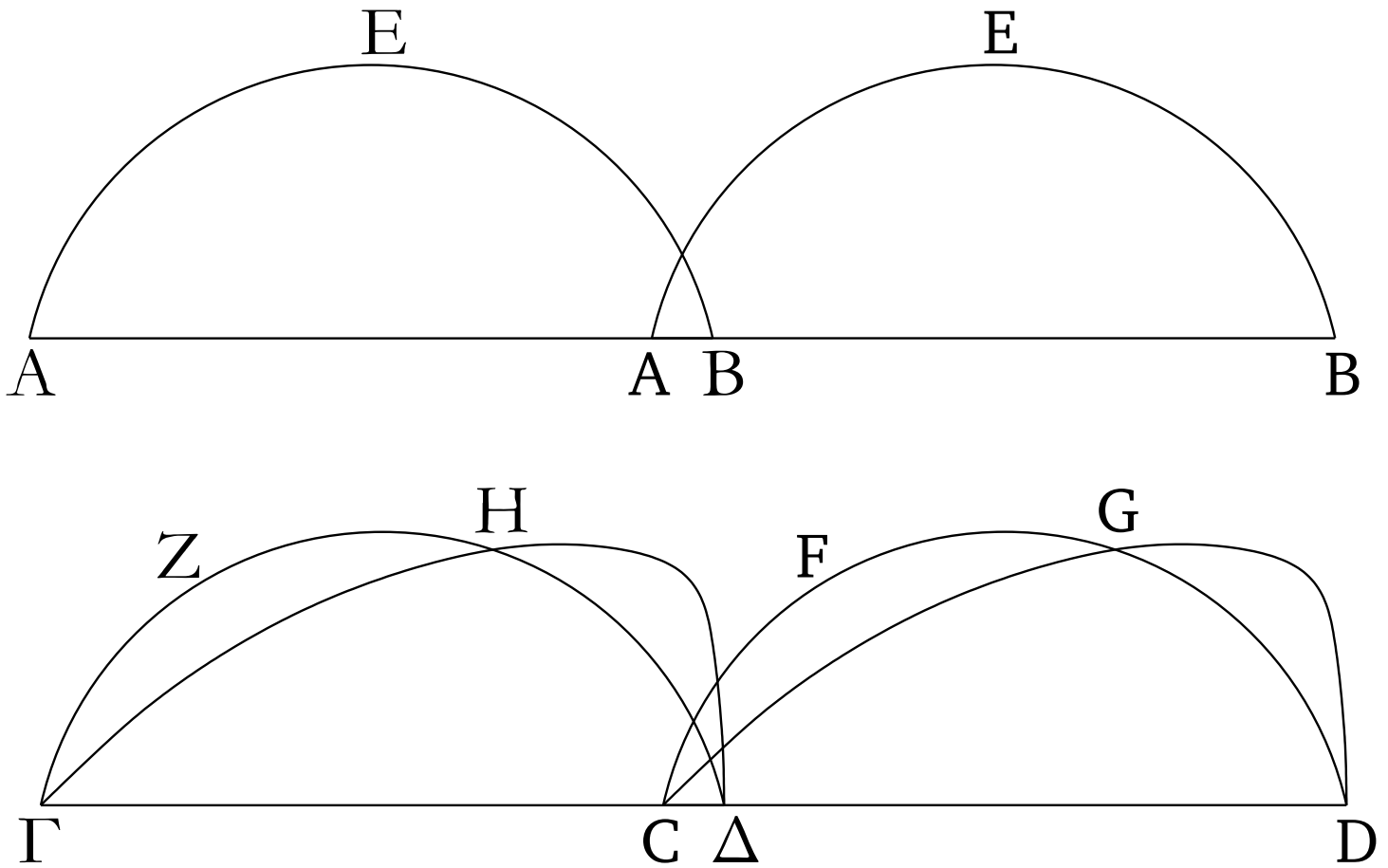
Ἐφαρμοζομένου γὰρ τοῦ ΑΕΒ τμήματος ἐπὶ τὸ ΓΖΔ καὶ
τιθεμένου τοῦ μὲν Α σημείου ἐπὶ τὸ Γ τῇξ δὲ ΑΒ εὐθείαξ
ἐπὶ τὴν ΓΔ, ἐφαρμόσει καὶ τὸ Β σημεῖον ἐπὶ τὸ Δ σημεῖον
διὰ τὸ >ῖσην εἶναι τὴν ΑΒ τῇ ΓΔ; τῇξ δὲ ΑΒ ἐπὶ τὴν ΓΔ
ἐφαρμοσάσης ἐφαρμόσει καὶ τὸ ΑΕΒ τμήμα ἐπὶ τὸ ΓΖΔ. εἰ
γὰρ ἡ ΑΒ εὐθεῖα ἐπὶ τὴν ΓΔ ἐφαρμόσει, τὸ δὲ ΑΕΒ τμήμα ἐπὶ
τὸ ΓΖΔ μὴ ἐφαρμόσει, >ήτοι ἐντὸς αὐτοῦ πεσεῖται >ἢ ἐκτὸς
>ἢ παραλλάξει, ὥς τὸ ΓΗΔ, καὶ κύκλος κύκλον τέμνει κατὰ
πλείονα σημεία >ἢ δύο; <όπερ ἐστὶν ἀδύνατον. οὐκ >άρα
ἐφαρμοζομένηξ τῇξ ΑΒ εὐθείαξ ἐπὶ τὴν ΓΔ οὐκ ἐφαρμόσει
καὶ τὸ ΑΕΒ τμήμα ἐπὶ τὸ ΓΖΔ; ἐφαρμόσει >άρα, καὶ >ῖσον
αὐτῷ >έσται.

Proposition 24

Similar segments of circles on equal straight-lines are equal
to one another.

For let AEB and CFD be similar segments of circles on
the equal straight-lines AB and CD (respectively). I say
that segment AEB is equal to segment CFD .

For if the segment AEB is applied to the segment CFD ,
and point A is placed on (point) C , and the straight-line AB
on CD , (then) point B will also coincide with point D , on
account of AB being equal to CD . And if AB coincides
with CD , (then) the segment AEB will also coincide with
 CFD . For if the straight-line AB coincides with CD , and
the segment AEB does not coincide with CFD , (then) it
will surely either fall inside it, outside (it),[†] or it will miss
like CGD (in the figure), and a circle (will) cut (another)
circle at more than two points. The very thing is impossi-
ble [Prop. 3.10]. Thus, if the straight-line AB is applied
to CD , the segment AEB cannot not also coincide with
 CFD . Thus, it will coincide, and will be equal to it [C.N. 4].



Τὰ >άρα ἐπὶ >ίσεων εὐθειῶν <όμοια τμήματα κύκλων >ίσα ἀλλήλοις ἐστίν; <όπερ >έδει δεῖχαι.

Thus, similar segments of circles on equal straight-lines are equal to one another. (Which is) the very thing it was required to show.

† Both this possibility, and the previous one, are precluded by Prop. 3.23.

25.

Κύκλου τμήματοξ δοθέντοξ προσαναγράψαι τὸν κύκλον, ο<υ̐πέρ ἐστι τμήμα.

>Ἐστω τὸ δοθέν τμήμα κύκλου τὸ ABΓ; δεῖ δὴ τοῦ ABΓ τμήματοξ προσαναγράψαι τὸν κύκλον, ο<υ̐πέρ ἐστι τμήμα. Τετμήσθω γὰρ ἡ AΓ δίθα κατὰ τὸ Δ, καὶ >ήθθω ἀπὸ τοῦ Δ σημείου τῇ AΓ πρὸξ ὀρθῶξ ἡ ΔB, καὶ ἐπεζεύθθω ἡ AB; ἡ ὑπὸ ABA γωνία >άρα τῇξ ὑπὸ BAA >ήτοι μείζων ἐστίν >ἡ >ίση >ἡ ἐλάττων.

>Ἐστω πρότερον μείζων, καὶ συνεστάτω πρὸξ τῇ BA εὐθείᾳ καὶ τῷ πρὸξ αὐτῇ σημείῳ τῷ A τῇ ὑπὸ ABA γωνία >ίση ἡ ὑπὸ BAE, καὶ διήθθω ἡ ΔB ἐπὶ τὸ E, καὶ ἐπεζεύθθω ἡ EΓ. ἐπεὶ ο<υ̐ν >ίση ἐστίν ἡ ὑπὸ ABE γωνία τῇ ὑπὸ BAE, >ίση >άρα ἐστὶ καὶ ἡ EB εὐθεῖα τῇ EA. καὶ ἐπεὶ >ίση ἐστίν ἡ AΔ τῇ ΔΓ, κοινὴ δὲ ἡ ΔE, δύο δὴ αἱ AΔ, ΔE δύο ταῖξ ΓΔ, ΔE >ίσαι εἰσὶν ἑκατέρα ἑκατέρω; καὶ γωνία ἡ ὑπὸ AΔE γωνία τῇ ὑπὸ ΓΔE ἐστίν >ίση; ὀρθὴ γὰρ ἑκατέρα; βάσιξ >άρα ἡ AE βάσει τῇ ΓE ἐστίν >ίση. ἀλλὰ ἡ AE τῇ BE ἔδειθθη >ίση; καὶ ἡ BE >άρα τῇ ΓE ἐστίν >ίση; αἱ τρεῖξ >άρα αἱ AE, EB, EΓ >ίσαι ἀλλήλαιξ εἰσίν; ὁ >άρα κέντρῳ τῷ E διαστήματι δὲ ἐνὶ τῶν AE, EB, EΓ κύκλοξ γραφόμενοξ <ήχει καὶ διὰ τῶν λοιπῶν σημείων καὶ >έσται προσαναγεγραμμένοξ. κύκλου >άρα τμήματοξ δοθέντοξ προσαναγράφεται ὁ κύκλοξ. καὶ δῆλον, ὥξ τὸ ABΓ τμήμα >έλαττόν ἐστιν ἡμικυκλίου διὰ τὸ τὸ E κέντρον ἐκτὸξ αὐτοῦ τυγθάνειν.

Proposition 25

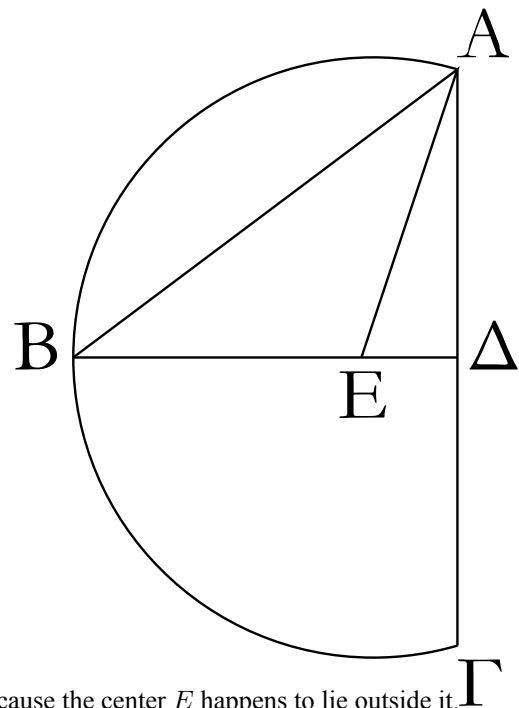
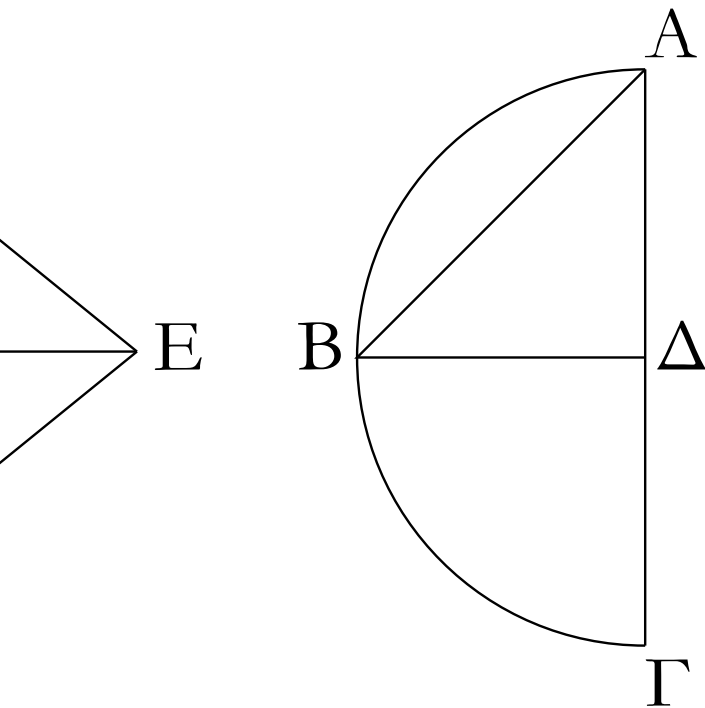
For a given segment of a circle, to complete the circle, the very one of which it is a segment.

Let ABC be the given segment of a circle. So it is required to complete the circle for segment ABC , the very one of which it is a segment.

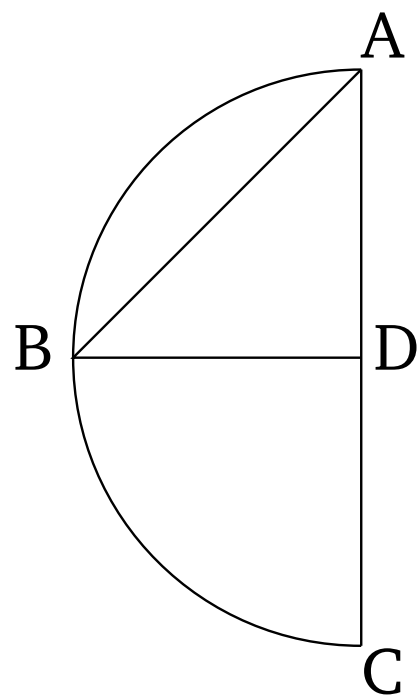
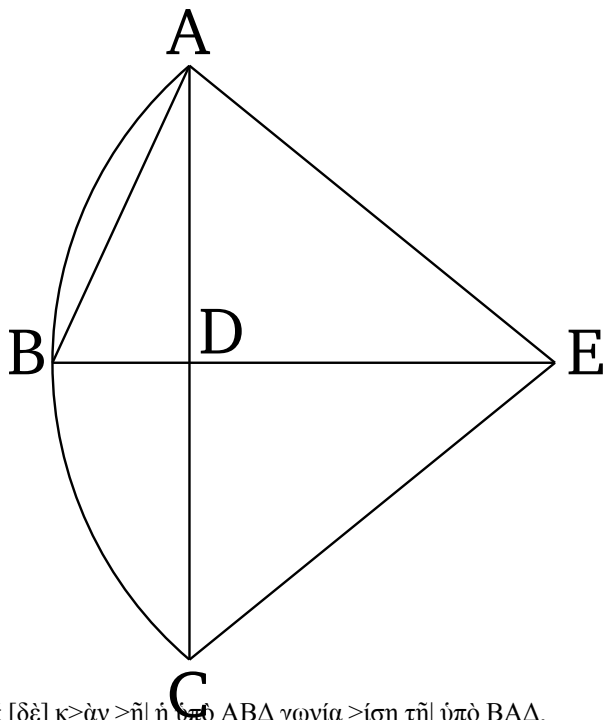
For let AC be cut in half at (point) D [Prop. 1.10], and let DB be drawn from point D , at right-angles to AC [Prop. 1.11]. And let AB be joined. Thus, angle ABD is surely either greater than, equal to, or less than (angle) BAD .

First of all, let it be greater. And let (angle) BAE , equal to angle ABD , be constructed on the straight-line BA , at the point A on it [Prop. 1.23]. And let DB be drawn through to E , and let EC be joined. Therefore, since angle ABE is equal to BAE , the straight-line EB is thus also equal to EA [Prop. 1.6]. And since AD is equal to DC , and DE (is) common, the two (straight-lines) AD , DE are equal to the two (straight-lines) CD , DE , respectively. And angle ADE is equal to angle CDE . For each (is) a right-angle. Thus, the base AE is equal to the base CE [Prop. 1.4]. But, AE was shown (to be) equal to BE . Thus, BE is also equal to CE . Thus, the three (straight-lines) AE , EB , and EC are equal to one another. Thus, if a circle is drawn with center E , and radius one of AE , EB , or EC , it will also go through the remaining points (of the segment), and the (as-

sociated circle) will be completed [Prop. 3.9]. Thus, a circle has been completed from the given segment of a circle. And (it is) clear that the segment ABC is less than a semi-circle,



because the center E happens to lie outside it.



Ὅμοιως [δὲ] κ' ἂν >ῃ ἡ ὑπὸ ABD γωνία > ἰση τῇ ὑπὸ BAD , τῇ AD > ἰση γενομένης ἑκατέρω τῶν BD , $ΔΓ$ αἱ τρεῖς αἱ $ΔA$, $ΔB$, $ΔΓ$ > ἰσαι ἀλλήλαις > ἔσονται, καὶ > ἔσται τὸ $Δ$ κέντρον τοῦ προσαναπληρωμένου κύκλου, καὶ δηλαδὴ > ἔσται τὸ $ABΓ$ ἡμικύκλιον.

Ἐὰν δὲ ἡ ὑπὸ ABD ἐλάττων >ῃ τῇ ὑπὸ BAD , καὶ συστησώμεθα πρὸς τῇ BA εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ A τῇ ὑπὸ ABD γωνία > ἰσην, ἐντὸς τοῦ $ABΓ$ τμήματος πεσεῖται τὸ κέντρον ἐπὶ τῇ $ΔB$, καὶ > ἔσται δηλαδὴ τὸ $ABΓ$ τμήμα μείζον ἡμικυκλίου.

Κύκλου > ἄρα τμήματος δοθέντος προσαναγράφεται ὁ κύκλος; < ὅπερ > ἔδει ποιῆσαι.

[And], similarly, even if angle ABD is equal to BAD , (since) AD becomes equal to each of BD [Prop. 1.6] and DC , the three (straight-lines) DA , DB , and DC will be equal to one another. And point D will be the center of the completed circle. And ABC will manifestly be a semi-circle.

And if ABD is less than BAD , and we construct (angle BAE), equal to angle ABD , on the straight-line BA , at the point A on it [Prop. 1.23], then the center will fall on DB , inside the segment ABC . And segment ABC will manifestly be greater than a semi-circle.

Thus, a circle has been completed from the given segment

of a circle. (Which is) the very thing it was required to do.

26.

Ἐν τοῖς ὕλοις κύκλοις αἱ ὕλοις γωνίαι ἐπὶ ὕλων περιφερειῶν βεβήκασιν, ἔαν τε πρὸς τοῖς κέντροις ἔαν τε πρὸς ταῖς περιφερείαις ὡς βεβηκυῖαι.

>Ἐστωσαν ὕλοις κύκλοι οἱ $ABΓ$, $ΔEZ$ καὶ ἐν αὐτοῖς ὕλοις γωνίαι >έστωσαν πρὸς μὲν τοῖς κέντροις αἱ ὑπὸ $BHΓ$, $ΕΘΖ$, πρὸς δὲ ταῖς περιφερείαις αἱ ὑπὸ $BAΓ$, $ΕΔΖ$; λέγω, <ὅτι >ἴση ἐστὶν ἡ $BKΓ$ περιφέρεια τῇ $ΕΛΖ$ περιφερείᾳ.

Ἐπεξεύθθωσαν γὰρ αἱ $BΓ$, EZ .

Καὶ ἐπεὶ ὕλοις εἰσὶν οἱ $ABΓ$, $ΔEZ$ κύκλοι, ὕλοις εἰσὶν αἱ ἐκ τῶν κέντρων; δύο δὲ αἱ BH , $HΓ$ δύο ταῖς $ΕΘ$, $ΘΖ$ ὕλοις; καὶ γωνία ἡ πρὸς τῷ H γωνία τῇ πρὸς τῷ $Θ$ ὕλοις; βάσεις ὅρα ἡ $BΓ$ βάσει τῇ EZ ἐστὶν ὕλοις. καὶ ἐπεὶ ὕλοις ἐστὶν ἡ πρὸς τῷ A γωνία τῇ πρὸς τῷ $Δ$, <ὅμοιον ὅρα ἐστὶ τὸ $BAΓ$ τμήμα τῷ $ΕΔΖ$ τμήματι; καὶ εἰσὶν ἐπὶ ὕλων εὐθειῶν [τῶν $BΓ$, EZ]; τὰ δὲ ἐπὶ ὕλων εὐθειῶν <ὅμοια τμήματα κύκλων ὕλοις ἀλλήλοις ἐστίν; ὕλον ὅρα τὸ $BAΓ$ τμήμα τῷ $ΕΔΖ$. >έστι δὲ καὶ <ὅλος ὁ $ABΓ$ κύκλος <ὅλος τῷ $ΔEZ$ κύκλῳ ὕλοις; λοιπὴ ὅρα ἡ $BKΓ$ περιφέρεια τῇ $ΕΛΖ$ περιφερείᾳ ἐστὶν ὕλοις.

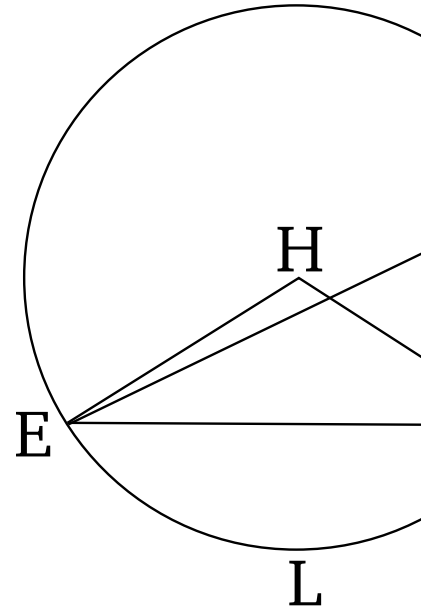
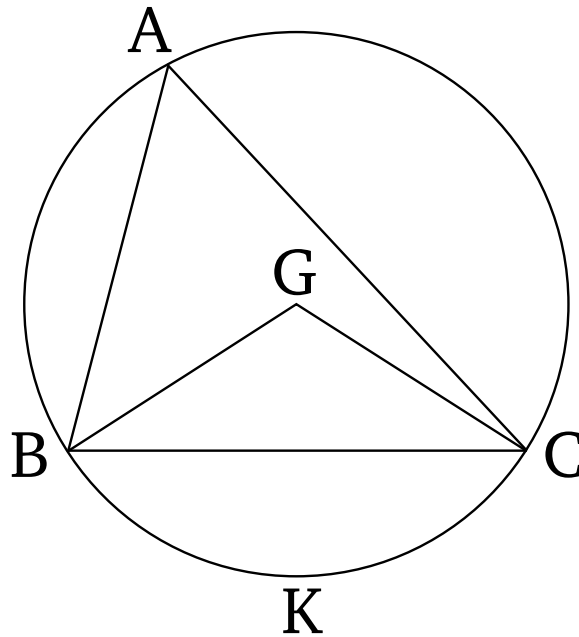
Proposition 26

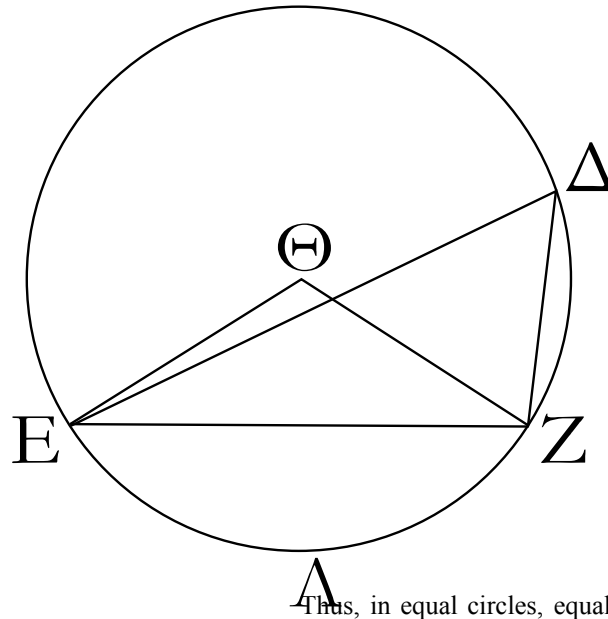
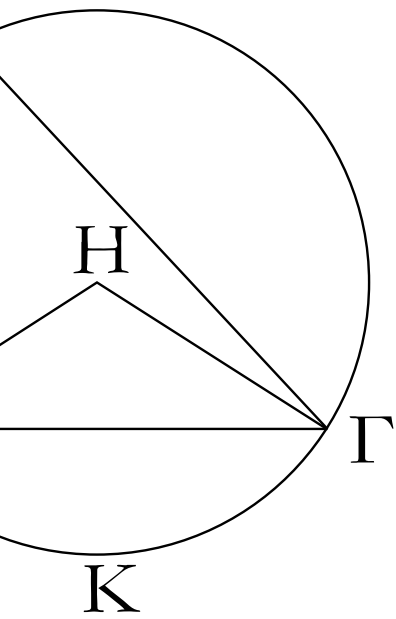
In equal circles, equal angles stand upon equal circumferences whether they are standing at the center or at the circumference.

Let ABC and DEF be equal circles, and within them let BGC and EHF be equal angles at the center, and BAC and EDF (equal angles) at the circumference. I say that circumference BKC is equal to circumference ELF .

For let BC and EF be joined.

And since circles ABC and DEF are equal, their radii are equal. So the two (straight-lines) BG , GC (are) equal to the two (straight-lines) EH , HF (respectively). And the angle at G (is) equal to the angle at H . Thus, the base BC is equal to the base EF [Prop. 1.4]. And since the angle at A is equal to the (angle) at D , the segment BAC is thus similar to the segment EDF [Def. 3.11]. And they are on equal straight-lines [BC and EF]. And similar segments of circles on equal straight-lines are equal to one another [Prop. 3.24]. Thus, segment BAC is equal to (segment) EDF . And the whole circle ABC is also equal to the whole circle DEF . Thus, the remaining circumference BKC is equal to the (remaining) circumference ELF .





Ἐν >άρα τοῖς >ίσοις κύκλοις αἱ >ίσαι γωνίαι ἐπὶ >ίσων περιφερειῶν βεβήκασιν, ἔαν τε πρὸς τοῖς κέντροις ἔαν τε πρὸς ταῖς περιφερείαις >ῶσι βεβηκυῖαι; <όπερ >έδει δεῖξαι.

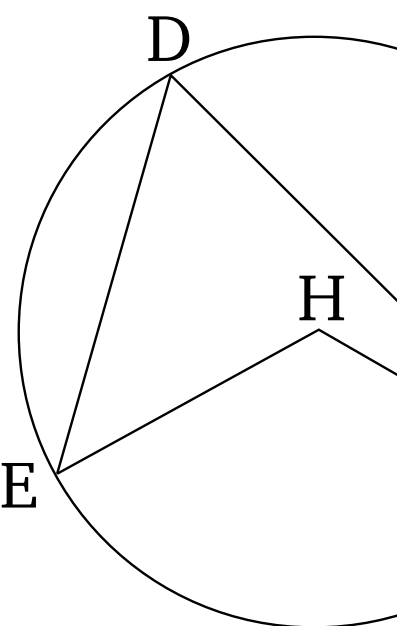
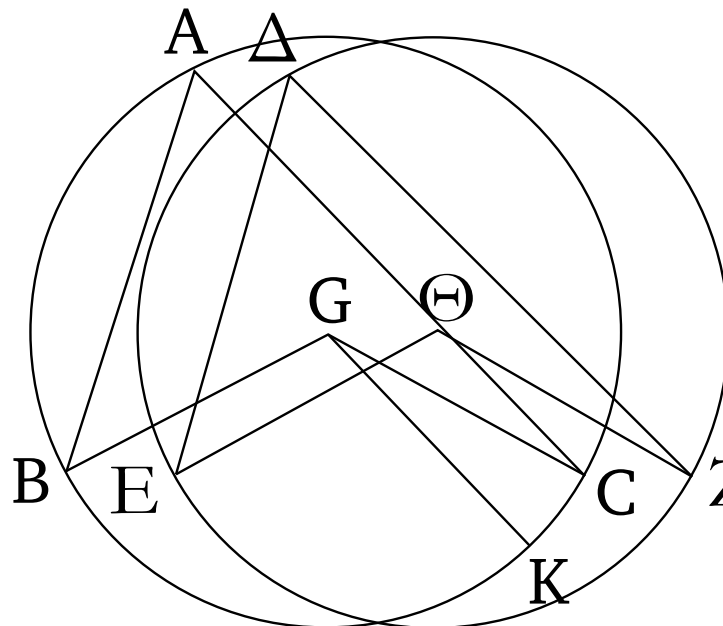
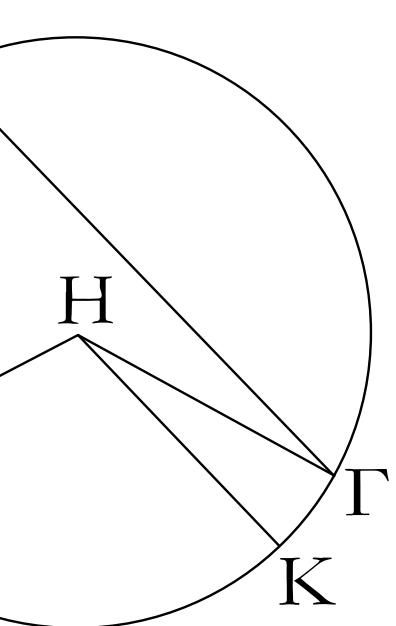
Thus, in equal circles, equal angles stand upon equal circumferences, whether they are standing at the center or at the circumference. (Which is) the very thing which it was required to show.

27.

Proposition 27

Ἐν τοῖς >ίσοις κύκλοις αἱ ἐπὶ >ίσων περιφερειῶν βεβηκυῖαι γωνίαι >ίσαι ἀλλήλαις εἰσίν, ἔαν τε πρὸς τοῖς κέντροις ἔαν τε πρὸς ταῖς περιφερείαις >ῶσι βεβηκυῖαι.

In equal circles, angles standing upon equal circumferences are equal to one another, whether they are standing at the center or at the circumference.



Ἐν γὰρ >ίσοις κύκλοις τοῖς ABΓ, ΔΕΖ ἐπὶ >ίσων περιφερειῶν τῶν ΒΓ, ΕΖ πρὸς μὲν τοῖς Η, Θ κέντροις γωνία βεβηκέτωσαν αἱ ὑπὸ ΒΗΓ, ΕΘΖ, πρὸς δὲ ταῖς περιφερείαις αἱ ὑπὸ ΒΑΓ, ΕΔΖ; λέγω, <ότι ἡ μὲν ὑπὸ ΒΗΓ γωνία τῇ ὑπὸ ΕΘΖ ἐστὶν >ίση, ἡ δὲ ὑπὸ ΒΑΓ τῇ ὑπὸ ΕΔΖ ἐστὶν >ίση. Εἰ γὰρ >άνισός ἐστιν ἡ ὑπὸ ΒΗΓ τῇ ὑπὸ ΕΘΖ, μία αὐτῶν μείζων ἐστίν. >έστω μείζων ἡ ὑπὸ ΒΗΓ, καὶ συνεστάτω πρὸς τῇ ΒΗ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ Η τῇ ὑπὸ ΕΘΖ γωνία >ίση ἡ ὑπὸ ΒΗΚ; αἱ δὲ >ίσαι γωνίαι ἐπὶ >ίσων περιφερειῶν βεβήκασιν, <όταν πρὸς τοῖς κέντροις >ῶσιν; >ίση >άρα ἡ ΒΚ περιφέρεια τῇ ΕΖ περιφερείᾳ. ἀλλὰ ἡ ΕΖ τῇ ΒΓ ἐστὶν >ίση; καὶ ἡ ΒΚ >άρα τῇ ΒΓ ἐστὶν >ίση

For let the angles BGC and EHF at the centers G and H , and the (angles) BAC and EDF at the circumferences, stand upon the equal circumferences BC and EF , in the equal circles ABC and DEF (respectively). I say that angle BGC is equal to (angle) EHF , and BAC is equal to EDF .

For if BGC is unequal to EHF , one of them is greater. Let BGC be greater, and let the (angle) BGK , equal to angle EHF , be constructed on the straight-line BG , at the point G on it [Prop. 1.23]. But equal angles (in equal circles) stand upon equal circumferences, when they are at the centers [Prop. 3.26]. Thus, circumference BK (is) equal to

ἢ ἐλάττων τῇ μείζονι; <όπερ ἐστὶν ἀδύνατον. οὐκ >άρα >άνισός ἐστιν ἡ ὑπὸ BHG γωνία τῇ ὑπὸ $EΘZ$; >ίση >άρα. καὶ ἐστὶ τῆς μὲν ὑπὸ BHG ἡμίσεια ἢ πρὸς τῷ A , τῆς δὲ ὑπὸ $EΘZ$ ἡμίσεια ἢ πρὸς τῷ Δ ; >ίση >άρα καὶ ἡ πρὸς τῷ A γωνία τῇ πρὸς τῷ Δ .

Ἐν >άρα τοῖς >ίσοις κύκλοις αἱ ἐπὶ >ίσων περιφερειῶν βεβηκυῖαι γωνίαι >ίσαι ἀλλήλαις εἰσὶν, ἔαν τε πρὸς τοῖς κέντροις ἔαν τε πρὸς ταῖς περιφερείαις >ῶσι βεβηκυῖαι; <όπερ >έδει δεῖχαι.

circumference EF . But, EF is equal to BC . Thus, BK is also equal to BC , the lesser to the greater. The very thing is impossible. Thus, angle BGC is not unequal to EHF . Thus, (it is) equal. And the (angle) at A is half BGC , and the (angle) at D half EHF [Prop. 3.20]. Thus, the angle at A (is) also equal to the (angle) at D .

Thus, in equal circles, angles standing upon equal circumferences are equal to one another, whether they are standing at the center or at the circumference. (Which is) the very thing it was required to show.

28.

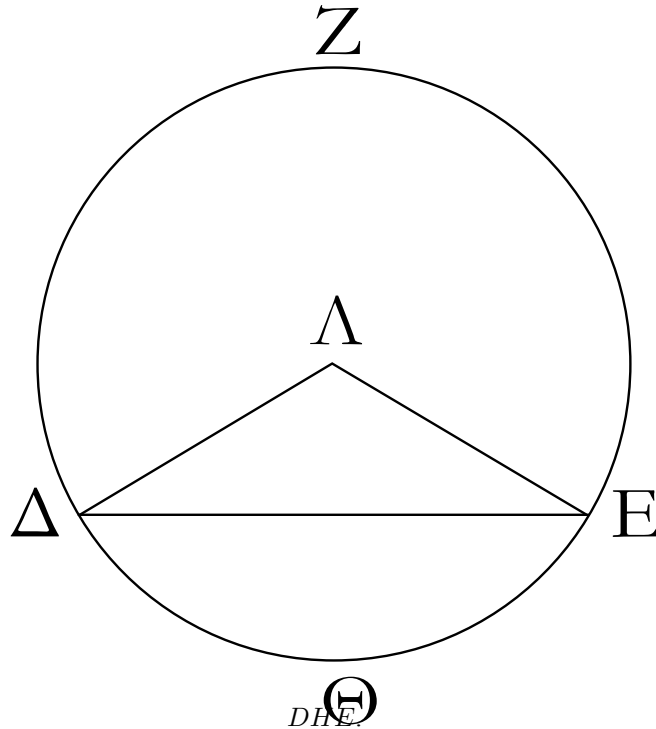
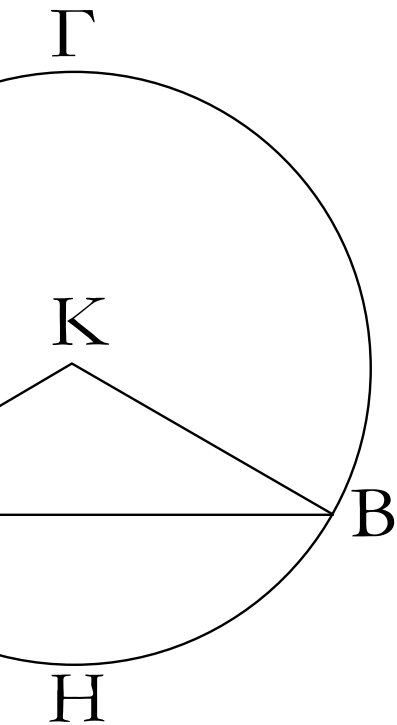
Ἐν τοῖς >ίσοις κύκλοις αἱ >ίσαι εὐθεῖαι >ίσαξ περιφερείας ἀφαιροῦσι τὴν μὲν μείζονα τῇ μείζονι τὴν δὲ ἐλάττονα τῇ ἐλάττονι.

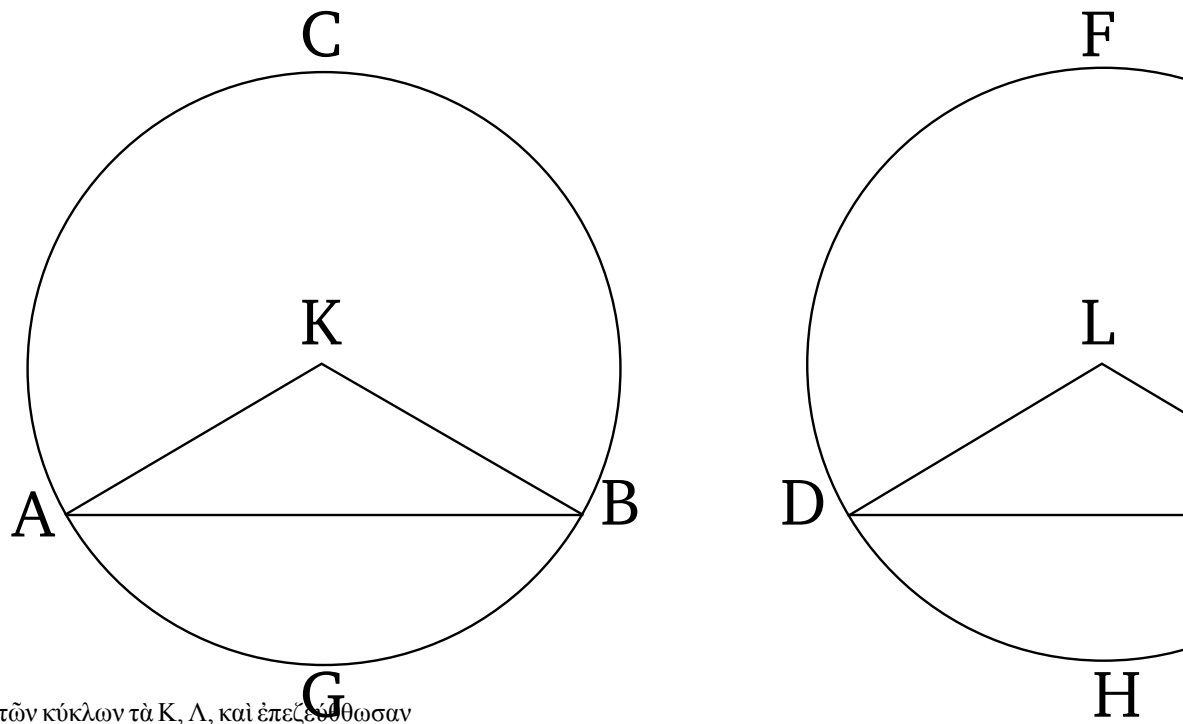
>Ἐστῶσαν >ίσοι κύκλοι οἱ $AB\Gamma$, ΔEZ , καὶ ἐν τοῖς κύκλοις >ίσαι εὐθεῖαι >έστωσαν αἱ AB , ΔE τὰς μὲν AGB , AZE περιφερείας μείζονας ἀφαιροῦσαι τὰς δὲ AHB , $\Delta\Theta E$ ἐλάττονας, λέγω, <ότι ἡ μὲν AGB μείζων περιφέρεια >ίση ἐστὶ τῇ ΔZE μείζονι περιφέρειᾳ ἡ δὲ AHB ἐλάττων περιφέρεια τῇ $\Delta\Theta E$.

Proposition 28

In equal circles, equal straight-lines cut off equal circumferences, the greater (circumference being equal) to the greater, and the lesser to the lesser.

Let ABC and DEF be equal circles, and let AB and DE be equal straight-lines in these circles, cutting off the greater circumferences ACB and DFE , and the lesser (circumferences) AGB and DHE (respectively). I say that the greater circumference ACB is equal to the greater circumference DFE , and the lesser circumference AGB to (the lesser)





Εἰλήφθω γὰρ τὰ κέντρα τῶν κύκλων τὰ K, Λ , καὶ ἐπεζεύθωσαν αἱ $AK, KB, \Delta\Lambda, \Lambda E$.

Καὶ ἐπεὶ ὅμοιοι κύκλοι εἰσὶν, ὅμοιοι εἰσὶ καὶ αἱ ἐκ τῶν κέντρων; δύο δὲ αἱ AK, KB δυοὶ ταῖς $\Delta\Lambda, \Lambda E$ ὅμοιοι εἰσὶν; καὶ βάσις ἡ AB βάσει τῇ ΔE ὅμοιος; γωνία ὅρα ἡ ὑπὸ AKB γωνία τῇ ὑπὸ $\Delta\Lambda E$ ὅμοιος ἐστίν. αἱ δὲ ὅμοιοι γωνίαι ἐπὶ ὅμοιων περιφερειῶν βεβήκασιν, ὅταν πρὸς τοῖς κέντροις ᾤσιν; ὅμοιος ὅρα ἡ AHB περιφέρεια τῇ $\Delta\Theta E$. ἐστὶ δὲ καὶ ὁλόξ ὁ $AB\Gamma$ κύκλος ὁλόξ τῷ $\Delta E Z$ κύκλῳ ὅμοιος; καὶ λοιπὴ ὅρα ἡ $A\Gamma B$ περιφέρεια λοιπῇ τῇ $\Delta Z E$ περιφερίᾳ ὅμοιος ἐστίν. Ἐν ὅρα τοῖς ὅμοιοις κύκλοις αἱ ὅμοιοι εὐθεῖαι ὅμοιας περιφερείας $B\Gamma C$ καὶ $E\Gamma H$ ἀφαιροῦσι τὴν μὲν μείζονα τῇ μείζονι τὴν δὲ ἐλάττονα τῇ ἐλάττονι; ὅπερ ἔδει δεῖξαι.

For let the centers of the circles, K and L , be found [Prop. 3.1], and let AK, KB, DL , and LE be joined.

And since $(ABC$ and $DEF)$ are equal circles, their radii are also equal [Def. 3.1]. So the two (straight-lines) AK, KB are equal to the two (straight-lines) DL, LE (respectively). And the base AB (is) equal to the base DE . Thus, angle AKB is equal to angle DLE [Prop. 1.8]. And equal angles stand upon equal circumferences, when they are at the centers [Prop. 3.26]. Thus, circumference AGB (is) equal to circumference DHE . And the whole circle ABC is also equal to the whole circle DEF . Thus, the remaining circumference ACB is also equal to the remaining circumference DFE .

Thus, in equal circles, equal straight-lines cut off equal circumferences, the greater (circumference being equal) to the greater, and the lesser to the lesser. (Which is) the very thing it was required to show.

29.

Ἐν τοῖς ὅμοιοις κύκλοις τὰς ὅμοιας περιφερείας ὅμοιοι εὐθεῖαι ὑποτείνουσιν.

>Ἐστῶσαν ὅμοιοι κύκλοι οἱ $AB\Gamma, \Delta E Z$, καὶ ἐν αὐτοῖς ὅμοιοι περιφερίαι ἀπειλήφθωσαν αἱ $B\Gamma C, E\Gamma H$, καὶ ἐπεζεύθωσαν αἱ $B\Gamma, E Z$ εὐθεῖαι; λέγω, ὅτι ὅμοιος ἐστὶν ἡ $B\Gamma$ τῇ $E Z$.

Εἰλήφθω γὰρ τὰ κέντρα τῶν κύκλων, καὶ ἔστω τὰ K, Λ , καὶ ἐπεζεύθωσαν αἱ $BK, K\Gamma, E\Lambda, \Lambda Z$.

Καὶ ἐπεὶ ὅμοιος ἐστὶν ἡ $B\Gamma C$ περιφέρεια τῇ $E\Gamma H$ περιφερίᾳ, ὅμοιος ἐστὶ καὶ γωνία ἡ ὑπὸ $BK\Gamma$ τῇ ὑπὸ $E\Lambda Z$. καὶ ἐπεὶ ὅμοιοι εἰσὶν οἱ $AB\Gamma, \Delta E Z$ κύκλοι, ὅμοιοι εἰσὶ καὶ αἱ ἐκ τῶν κέντρων; δύο δὲ αἱ $BK, K\Gamma$ δυοὶ ταῖς $E\Lambda, \Lambda Z$ ὅμοιοι εἰσὶν; καὶ γωνίαξ ὅμοιας περιέθουσιν; βάσις ὅρα ἡ $B\Gamma$ βάσει τῇ $E Z$ ὅμοιος ἐστίν;

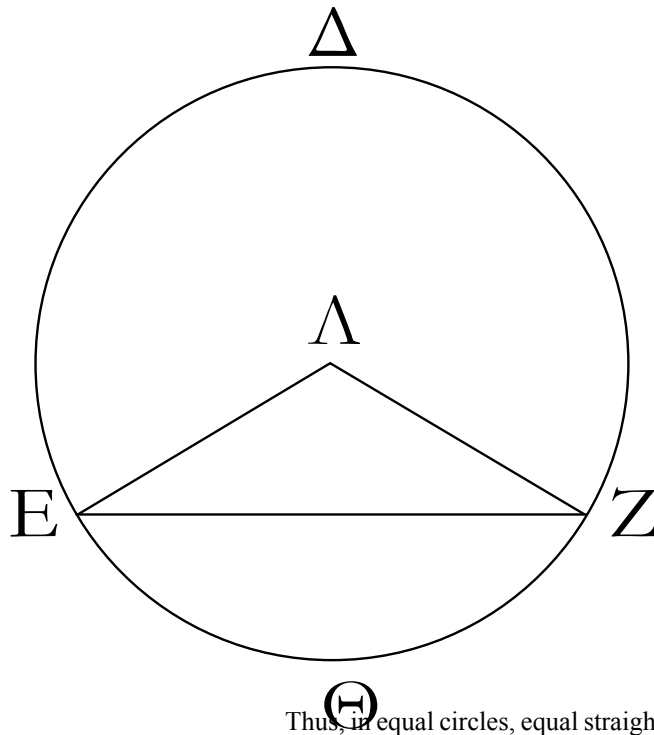
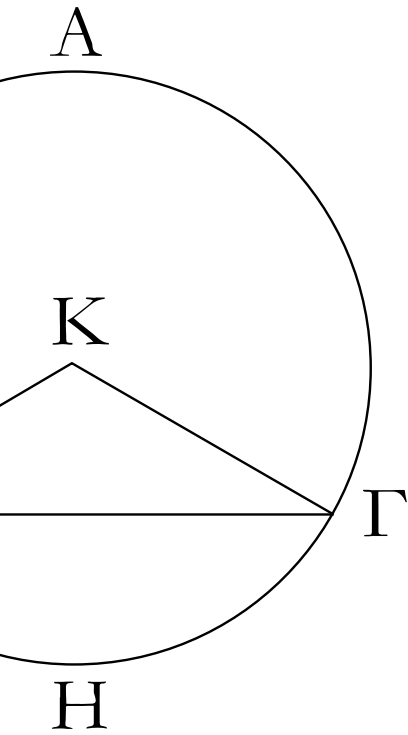
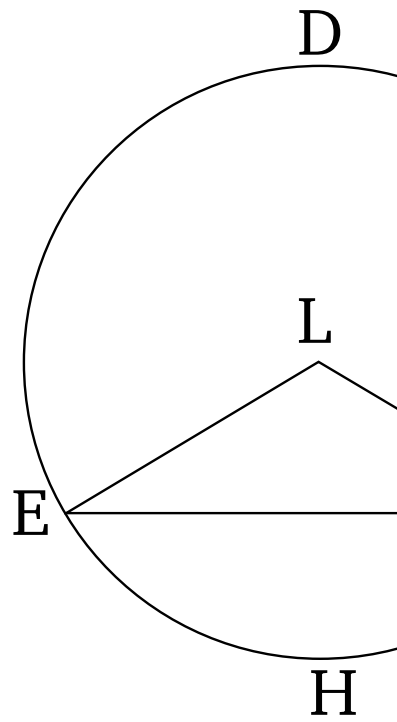
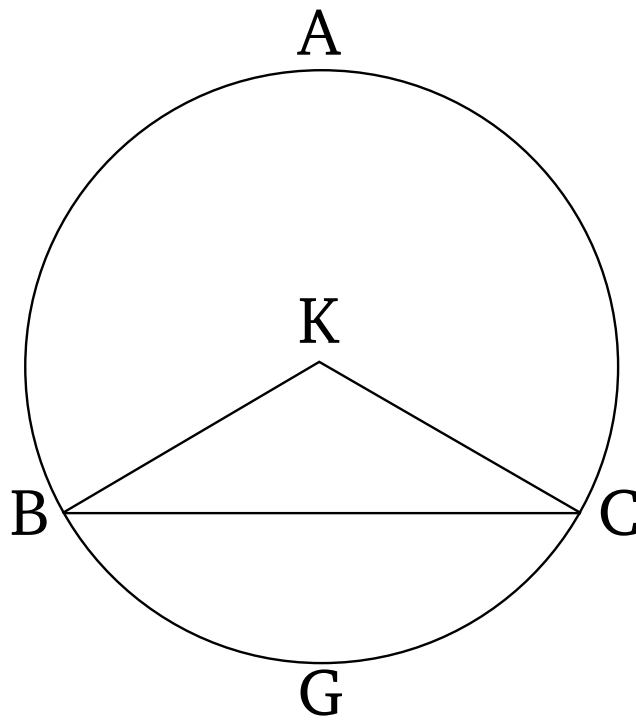
Proposition 29

In equal circles, equal straight-lines subtend equal circumferences.

Let ABC and DEF be equal circles, and within them let the equal circumferences BGC and EHF be cut off. And let the straight-lines BC and EF be joined. I say that BC is equal to EF .

For let the centers of the circles be found [Prop. 3.1], and let them be (at) K and L . And let BK, KC, EL , and LF be joined.

And since the circumference BGC is equal to the circumference EHF , the angle BKC is also equal to (angle) ELF [Prop. 3.27]. And since the circles ABC and DEF are equal, their radii are also equal [Def. 3.1]. So the two (straight-lines) BK, KC are equal to the two (straight-lines) EL, LF (respectively). And they contain equal angles. Thus, the base BC is equal to the base EF [Prop. 1.4].



Ἐν >άρα τοῖς >ίσοις κύκλοις τὰς >ίσας περιφερείας >ίσαι
εὐθεῖαι ὑποτείνουνσιν; <όπερ >έδει δεῖχαι.

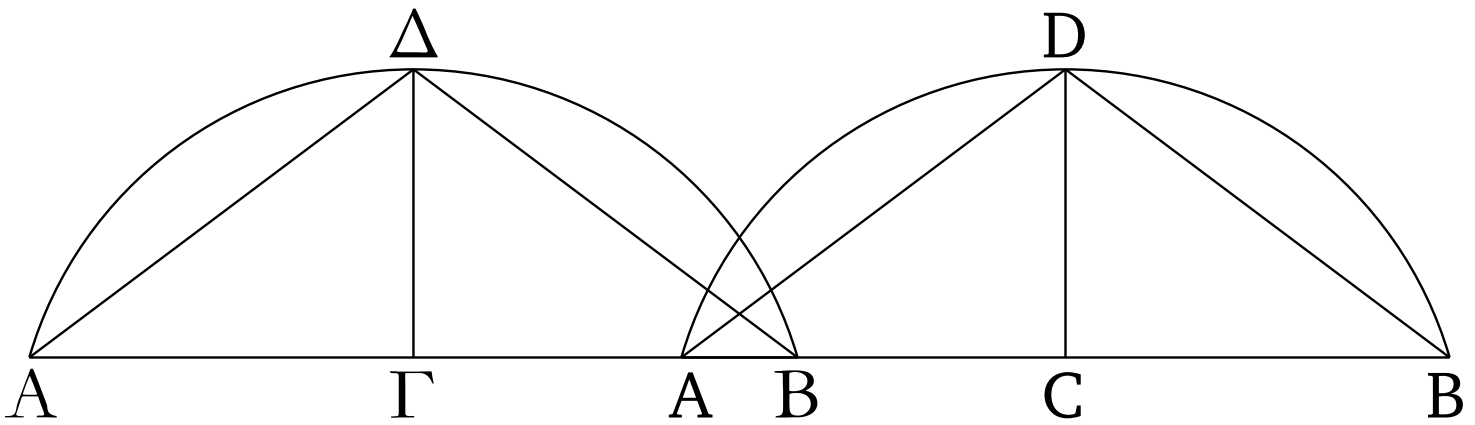
Thus, in equal circles, equal straight-lines subtend equal cir-
cumferences. (Which is) the very thing it was required to
show.

30.

Τὴν δοθεῖσαν περιφέρειαν δίθᾳ τεμεῖν.
>Ἐστω ἡ δοθεῖσα περιφέρεια ἡ ΑΔΒ; δεῖ δὴ τὴν ΑΔΒ
περιφέρειαν δίθᾳ τεμεῖν.
Ἐπεζεύθῃω ἡ ΑΒ, καὶ τετμήσθῃ δίθᾳ κατὰ τὸ Γ, καὶ ἀπὸ
τοῦ Γ σημείου τῇ ΑΒ εὐθείᾳ πρὸς ὀρθὰς >ήθῃω ἡ ΓΔ, καὶ
ἐπεζεύθῃωσαν αἱ ΑΔ, ΔΒ.

Proposition 30

To cut a given circumference in half.
Let ADB be the given circumference. So it is required to
cut circumference ADB in half.
Let AB be joined, and let it be cut in half at (point) C
[Prop. 1.10]. And let CD be drawn from point C , at right-
angles to AB [Prop. 1.11]. And let AD , and DB be joined.



Καὶ ἐπεὶ ὅση ἐστὶν ἡ ΑΓ τῇ ΓΒ, κοινὴ δὲ ἡ ΓΔ, δύο δὲ αἱ ΑΓ, ΓΔ δυσὶ ταῖς ΒΓ, ΓΔ ὅσαι εἰσὶν; καὶ γωνία ἡ ὑπὸ ΑΓΔ γωνία τῇ ὑπὸ ΒΓΔ ὅση; ὀρθὴ γὰρ ἑκατέρα; βάσις ὅρα ἡ ΑΔ βάσει τῇ ΔΒ ὅση ἐστίν. αἱ δὲ ὅσαι εὐθεῖαι ὅσαζ περιφερείας ἀφαιροῦσι τὴν μὲν μείζονα τῇ μείζονι τὴν δὲ ἐλάττωνα τῇ ἐλάττω; καὶ ἐστὶν ἑκατέρα τῶν ΑΔ, ΔΒ περιφερειῶν ἐλάττων ἡμικυκλίου; ὅση ὅρα ἡ ΑΔ περιφέρεια τῇ ΔΒ περιφέρειᾳ.
Ἡ ὅρα δοθεῖσα περιφέρεια δίθᾳ τέμνεται κατὰ τὸ Δ σημεῖον; ὅπερ ὅδει ποιῆσαι.

And since AC is equal to CB , and CD (is) common, the two (straight-lines) AC , CD are equal to the two (straight-lines) BC , CD (respectively). And angle ACD (is) equal to angle BCD . For (they are) each right-angles. Thus, the base AD is equal to the base DB [Prop. 1.4]. And equal straight-lines cut off equal circumferences, the greater (circumference being equal) to the greater, and the lesser to the lesser [Prop. 1.28]. And the circumferences AD and DB are each less than a semi-circle. Thus, circumference AD (is) equal to circumference DB .

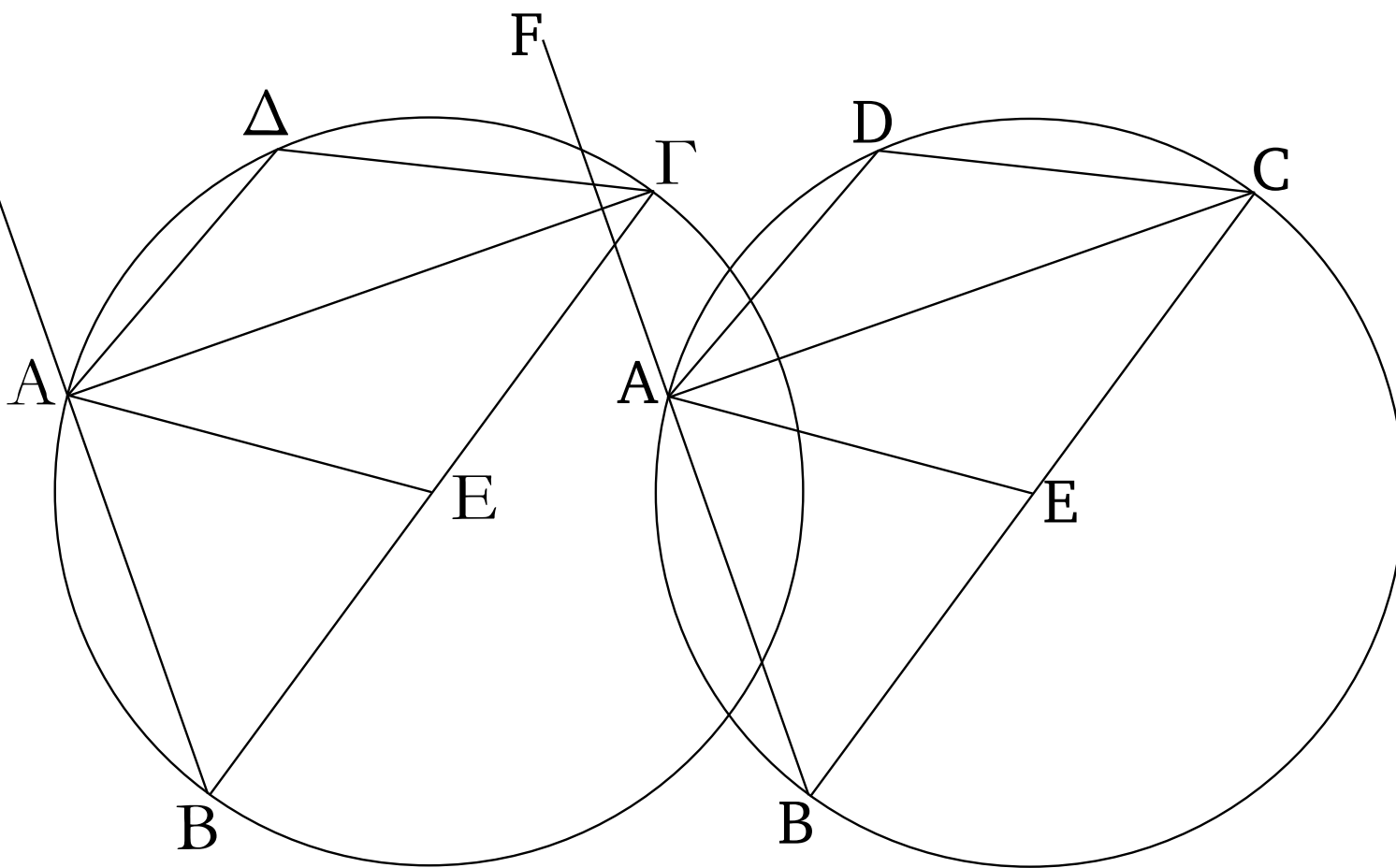
Thus, the given circumference has been cut in half at point D . (Which is) the very thing it was required to do.

31.

Ἐν κύκλῳ ἡ μὲν ἐν τῷ ἡμικυκλίῳ γωνία ὀρθή ἐστίν, ἡ δὲ ἐν τῷ μείζονι τμήματι ἐλάττων ὀρθῆς, ἡ δὲ ἐν τῷ ἐλάττονι τμήματι μείζων ὀρθῆς; καὶ ὅτι ἡ μὲν τοῦ μείζονος τμήματος γωνία μείζων ἐστὶν ὀρθῆς, ἡ δὲ τοῦ ἐλάττονος τμήματος γωνία ἐλάττων ὀρθῆς.

Proposition 31

In a circle, the angle in a semi-circle is a right-angle, and that in a greater segment (is) less than a right-angle, and that in a lesser segment (is) greater than a right-angle. And, further, the angle of a segment greater (than a semi-circle) is greater than a right-angle, and the angle of a segment less (than a semi-circle) is less than a right-angle.



>Ἐστω κύκλος ὁ ΑΒΓΔ, διάμετρος δὲ αὐτοῦ >ἔστω ἡ ΒΓ, κέντρον δὲ τὸ Ε, καὶ ἐπεξεύθωσαν αἱ ΒΑ, ΑΓ, ΑΔ, ΔΓ; λέγω, <ὅτι ἡ μὲν ἐν τῷ ΑΒΓ ἡμικυκλίῳ γωνία ἡ ὑπὸ ΒΑΓ ὀρθή ἐστίν, ἡ δὲ ἐν τῷ ΑΒΓ μείζονι τοῦ ἡμικυκλίου τμήματι γωνία ἡ ὑπὸ ΑΒΓ ἐλάττων ἐστίν ὀρθῆς, ἡ δὲ ἐν τῷ ΑΔΓ ἐλάττονι τοῦ ἡμικυκλίου τμήματι γωνία ἡ ὑπὸ ΑΔΓ μείζων ἐστίν ὀρθῆς.

Ἐπεξεύθω ἡ ΑΕ, καὶ διήθω ἡ ΒΑ ἐπὶ τὸ Ζ.

Καὶ ἐπεὶ >ἴση ἐστὶν ἡ ΒΕ τῇ ΑΕ, >ἴση ἐστὶ καὶ γωνία ἡ ὑπὸ ΑΒΕ τῇ ὑπὸ ΒΑΕ. πάλιν, ἐπεὶ >ἴση ἐστὶν ἡ ΓΕ τῇ ΑΕ, >ἴση ἐστὶ καὶ ἡ ὑπὸ ΑΓΕ τῇ ὑπὸ ΓΑΕ; <ὅλη >άρα ἡ ὑπὸ ΒΑΓ δυοὶ ταῖς ὑπὸ ΑΒΓ, ΑΓΒ >ἴση ἐστίν. ἐστὶ δὲ καὶ ἡ ὑπὸ ΖΑΓ ἐκτὸς τοῦ ΑΒΓ τριγώνου δυοὶ ταῖς ὑπὸ ΑΒΓ, ΑΓΒ γωνίαις >ἴση; >ἴση >άρα καὶ ἡ ὑπὸ ΒΑΓ γωνία τῇ ὑπὸ ΖΑΓ; ὀρθή >άρα ἑκατέρω; ἡ >άρα ἐν τῷ ΑΒΓ ἡμικυκλίῳ γωνία ἡ ὑπὸ ΒΑΓ ὀρθή ἐστίν.

Καὶ ἐπεὶ τοῦ ΑΒΓ τριγώνου δύο γωνίαι αἱ ὑπὸ ΑΒΓ, ΒΑΓ δύο ὀρθῶν ἐλάττονές εἰσιν, ὀρθή δὲ ἡ ὑπὸ ΒΑΓ, ἐλάττων >άρα ὀρθῆς ἐστὶν ἡ ὑπὸ ΑΒΓ γωνία; καὶ ἐστὶν ἐν τῷ ΑΒΓ μείζονι τοῦ ἡμικυκλίου τμήματι.

Καὶ ἐπεὶ ἐν κύκλῳ τετράπλευρόν ἐστι τὸ ΑΒΓΔ, τῶν δὲ ἐν τοῖς κύκλοις τετραπλεύρων αἱ ἀπεναντίον γωνίαι δυοὶν ὀρθαῖς >ἴσαι εἰσὶν [αἱ >άρα ὑπὸ ΑΒΓ, ΑΔΓ γωνίαι δυοὶν ὀρθαῖς >ἴσαι εἰσὶν], καὶ ἐστὶν ἡ ὑπὸ ΑΒΓ ἐλάττων ὀρθῆς; λοιπὴ >άρα ἡ ὑπὸ ΑΔΓ γωνία μείζων ὀρθῆς ἐστίν; καὶ ἐστὶν ἐν τῷ ΑΔΓ ἐλάττονι τοῦ ἡμικυκλίου τμήματι.

Λέγω, <ὅτι καὶ ἡ μὲν τοῦ μείζονος τμήματος γωνία ἡ περιεθρομένη ὑπὸ [τε] τῆς ΑΒΓ περιφερείας καὶ τῆς ΑΓ εὐθείας μείζων ἐστὶν ὀρθῆς, ἡ δὲ τοῦ ἐλάττονος τμήματος γωνία ἡ περιεθρομένη ὑπὸ [τε] τῆς ΑΔ[Γ] περιφερείας καὶ τῆς ΑΓ εὐθείας ἐλάττων ἐστὶν ὀρθῆς. καὶ ἐστὶν αὐτόθεν

Let $ABCD$ be a circle, and let BC be its diameter, and E its center. And let BA , AC , AD , and DC be joined. I say that the angle BAC in the semi-circle BAC is a right-angle, and the angle ABC in the segment ABC , (which is) greater than a semi-circle, is less than a right-angle, and the angle ADC in the segment ADC , (which is) less than a semi-circle, is greater than a right-angle.

Let AE be joined, and let BA be drawn through to F .

And since BE is equal to EA , angle ABE is also equal to BAE [Prop. 1.5]. Again, since CE is equal to EA , ACE is also equal to CAE [Prop. 1.5]. Thus, the whole (angle) BAC is equal to the two (angles) ABC and ACB . And FAC , (which is) external to triangle ABC , is also equal to the two angles ABC and ACB [Prop. 1.32]. Thus, angle BAC (is) also equal to FAC . Thus, (they are) each right-angles. [Def. 1.10]. Thus, the angle BAC in the semi-circle BAC is a right-angle.

And since the two angles ABC and BAC of triangle ABC are less than two right-angles [Prop. 1.17], and BAC is a right-angle, angle ABC is thus less than a right-angle. And it is in segment ABC , (which is) greater than a semi-circle. And since $ABCD$ is a quadrilateral within a circle, and for quadrilaterals within circles the (sum of the) opposite angles is equal to two right-angles [Prop. 3.22] [angles ABC and ADC are thus equal to two right-angles], and (angle) ABC is less than a right-angle. The remaining angle ADC is thus greater than a right-angle. And it is in segment ADC , (which is) less than a semi-circle.

I also say that the angle of the greater segment, (namely) that contained by the circumference ABC and the straight-line AC , is greater than a right-angle. And the angle of the

φανερὸν. ἐπεὶ γὰρ ἡ ὑπὸ τῶν BA , AG εὐθειῶν ὀρθή ἐστίν, ἡ $\angle$ ὑπὸ τῆς ABG περιφερείας καὶ τῆς AG εὐθείας περιεθρομένη μείζων ἐστὶν ὀρθῆς. πάλιν, ἐπεὶ ἡ ὑπὸ τῶν AG , AZ εὐθειῶν ὀρθή ἐστίν, ἡ $\angle$ ὑπὸ τῆς GAZ εὐθείας καὶ τῆς $AD[G]$ περιφερείας περιεθρομένη ἐλάττω ἐστὶν ὀρθῆς. Ἐν κύκλῳ $\angle$ ἡ μὲν ἐν τῷ ἡμικυκλίῳ γωνία ὀρθή ἐστίν, ἡ δὲ ἐν τῷ μείζονι τμήματι ἐλάττων ὀρθῆς, ἡ δὲ ἐν τῷ ἐλάττονι [τμήματι] μείζων ὀρθῆς; καὶ $\angle$ ἐπὶ ἡ μὲν τοῦ μείζονος τμήματος [γωνία] μείζων [ἐστὶν] ὀρθῆς, ἡ δὲ τοῦ ἐλάττονος τμήματος [γωνία] ἐλάττων ὀρθῆς; $\angle$ ὅπερ $\angle$ ἐδει δεῖχαι.

lesser segment, (namely) that contained by the circumference $AD[C]$ and the straight-line AC , is less than a right-angle. And this is immediately apparent. For since the (angle contained by) the two straight-lines BA and AC is a right-angle, the (angle) contained by the circumference ABC and the straight-line AC is thus greater than a right-angle. Again, since the (angle contained by) the straight-lines AC and AF is a right-angle, the (angle) contained by the circumference $AD[C]$ and the straight-line CA is thus less than a right-angle.

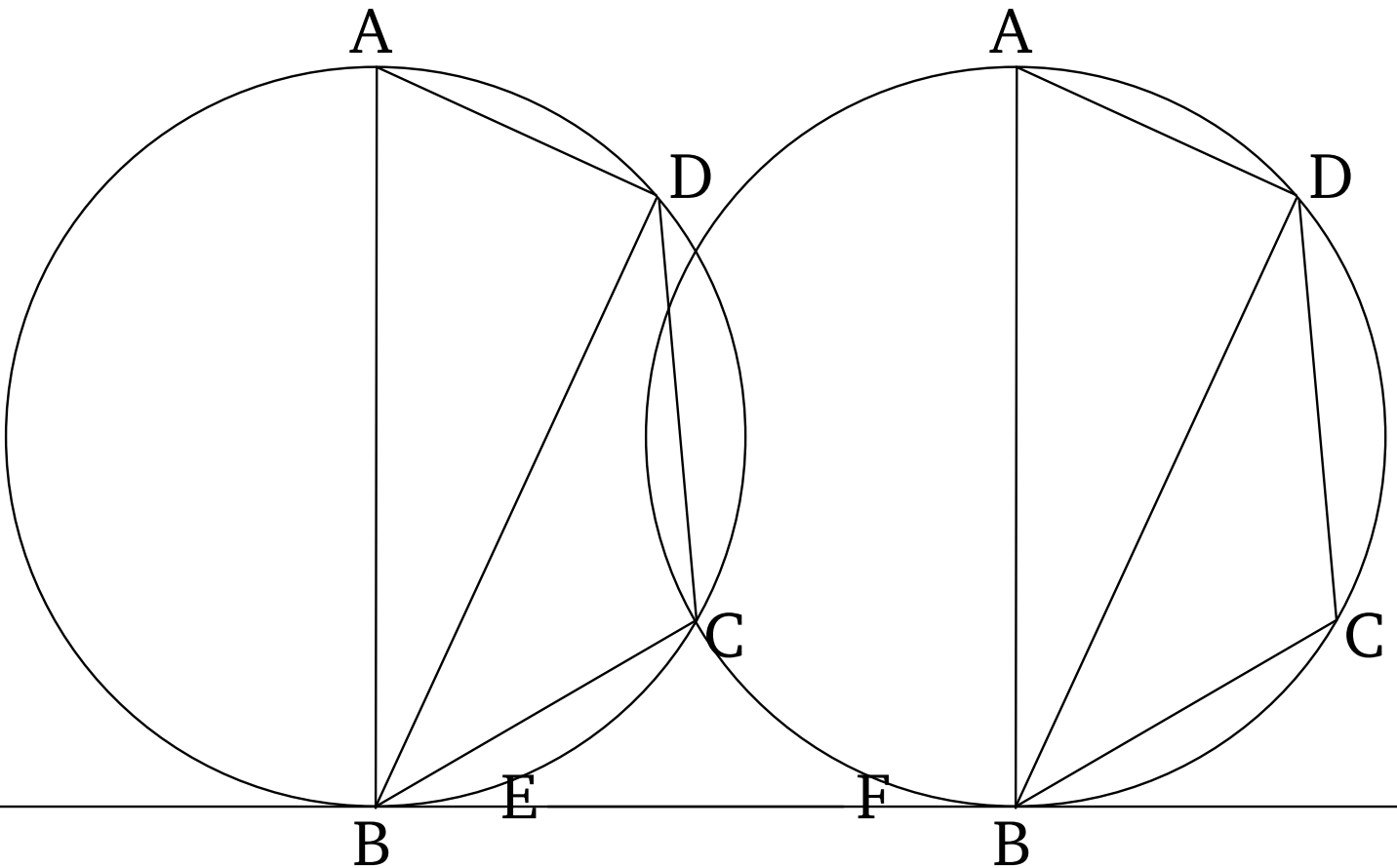
Thus, in a circle, the angle in a semi-circle is a right-angle, and that in a greater segment (is) less than a right-angle, and that in a lesser [segment] (is) greater than a right-angle. And, further, the [angle] of a segment greater (than a semi-circle) [is] greater than a right-angle, and the [angle] of a segment less (than a semi-circle) is less than a right-angle. (Which is) the very thing it was required to show.

32.

Ἐὰν κύκλου ἐφάπτηται τις εὐθεῖα, ἀπὸ δὲ τῆς ἀφῆς εἰς τὸν κύκλον διαθῇ τις εὐθεῖα τέμνουσα τὸν κύκλον, $\angle$ αὖ ποιεῖ γωνίαξ πρὸς τῇ ἐφαπτομένῃ, $\angle$ ἴσαι $\angle$ ἔσονται ταῖς ἐν τοῖς ἐναλλάχ τοῦ κύκλου τμήμασι γωνίαξ.

Proposition 32

If some straight-line touches a circle, and some (other) straight-line is drawn across, from the point of contact into the circle, cutting the circle (in two), (then) those angles the (straight-line) makes with the tangent will be equal to the angles in the alternate segments of the circle.



Κύκλου γὰρ τοῦ $ABGD$ ἐφαπτέσθω τις εὐθεῖα ἡ EZ κατὰ τὸ B σημεῖον, καὶ ἀπὸ τοῦ B σημείου διήθθω τις εὐθεῖα εἰς τὸν $ABGD$ κύκλον τέμνουσα αὐτὸν ἡ BD . λέγω, $\angle$ ὅτι $\angle$ αὖ ποιεῖ γωνίαξ ἡ BD μετὰ τῆς EZ ἐφαπτομένης, $\angle$ ἴσαι $\angle$ ἔσονται ταῖς ἐν τοῖς ἐναλλάχ τμήμασι τοῦ κύκλου γωνίαξ, τουτέστιν, $\angle$ ὅτι ἡ μὲν ὑπὸ ZBD γωνία $\angle$ ἴση ἐστὶ τῇ ἐν τῷ BAD τμήματι συνισταμένη γωνία, ἡ δὲ ὑπὸ EBD γωνία $\angle$ ἴση ἐστὶ τῇ ἐν τῷ DGB τμήματι συνισταμένη γωνία.

For let some straight-line EF touch the circle $ABCD$ at the point B , and let some (other) straight-line BD be drawn from point B into the circle $ABCD$, cutting it (in two). I say that the angles BD makes with the tangent EF will be equal to the angles in the alternate segments of the circle. That is to say, that angle FBD is equal to the angle constructed in segment BAD , and angle EBD is equal to the angle constructed in segment DCB .

>Ἐστω ἡ δοθεῖσα εὐθεῖα ἡ AB , ἡ δὲ δοθεῖσα γωνία εὐθύγραμμος ἢ πρὸς τῷ Γ ; δεῖ δὲ ἐπὶ τῇ δοθείσῃ εὐθείᾳ τῆς AB γράψαι τμήμα κύκλου δεθόμενον γωνίαν >ίσην τῇ πρὸς τῷ Γ .
Ἡ δὲ πρὸς τῷ Γ [γωνία] >ῆται ὀξεῖα ἐστὶν >ῆ ὀρθῇ >ῆ ὀβελείᾳ; >έστω πρότερον ὀξεῖα, καὶ ὥς ἐπὶ τῇ πρώτῃ καταγραφῇ συνεστάτω πρὸς τῇ AB εὐθείᾳ καὶ τῷ A σημείῳ τῇ πρὸς τῷ Γ γωνία >ίση ἢ ὑπὸ BAD ; ὀξεῖα >άρα ἐστὶ καὶ ἡ ὑπὸ BAD . >ῆθω τῇ DA πρὸς ὀρθᾶς ἡ AE , καὶ τεμήσθω ἡ AB δίθα κατὰ τὸ Z , καὶ >ῆθω ἀπὸ τοῦ Z σημείου τῇ AB πρὸς ὀρθᾶς ἡ ZH , καὶ ἐπεζεύθω ἡ HB . Καὶ ἐπεὶ >ίση ἐστὶν ἡ AZ τῇ ZB , κοινὴ δὲ ἡ ZH , δύο δὲ αἱ AZ , ZH δύο ταῖς BZ , ZH >ίσαι εἰσὶν; καὶ γωνία ἡ ὑπὸ AZH [γωνία] τῇ ὑπὸ BZH >ίση; βάσις >άρα ἡ AH βάσει τῇ BH >ίση ἐστίν. ὁ >άρα κέντρον μὲν τῷ H διαστήματι δὲ τῷ HA κύκλος γραφόμενος <ῆχει καὶ διὰ τοῦ B . γεγράφθω καὶ >έστω ὁ ABE , καὶ ἐπεζεύθω ἡ EB . ἐπεὶ ο>ὐν ἀπ> >άκρας τῆς AE διαμέτρου ἀπὸ τοῦ A τῇ AE πρὸς ὀρθᾶς ἐστὶν ἡ AD , ἡ AD >άρα ἐφάπτεται τοῦ ABE κύκλου; ἐπεὶ ο>ὐν κύκλου τοῦ ABE ἐφάπτεται τὶς εὐθεῖα ἡ AD , καὶ ἀπὸ τῆς κατὰ τὸ A ἀφῆς εἰς τὸν ABE κύκλον διῆκται τὶς εὐθεῖα ἡ AB , ἡ >άρα ὑπὸ ΔAB γωνία >ίση ἐστὶ τῇ ἐν τῷ ἐναλλάχ τοῦ κύκλου τμήματι γωνία τῇ ὑπὸ AEB . ἀλλ> ἡ ὑπὸ ΔAB τῇ πρὸς τῷ Γ ἐστὶν >ίση; καὶ ἡ πρὸς τῷ Γ >άρα γωνία >ίση ἐστὶ τῇ ὑπὸ AEB .

Ἐπὶ τῇ δοθείσῃ >άρα εὐθείᾳ τῆς AB τμήμα κύκλου γέγραπται τὸ ABE δεθόμενον γωνίαν τὴν ὑπὸ AEB >ίσην τῇ δοθείσῃ τῇ πρὸς τῷ Γ .

Ἀλλὰ δὲ ὀρθῇ >έστω ἡ πρὸς τῷ Γ ; καὶ δέον πάλιν >έστω ἐπὶ τῇ AB γράψαι τμήμα κύκλου δεθόμενον γωνίαν >ίσην τῇ πρὸς τῷ Γ ὀρθῇ [γωνία]. συνεστάτω [πάλιν] τῇ πρὸς τῷ Γ ὀρθῇ γωνία >ίση ἢ ὑπὸ BAD , ὥς >έθει ἐπὶ τῇ δευτέρᾳ καταγραφῇ, καὶ τεμήσθω ἡ AB δίθα κατὰ τὸ Z , καὶ κέντρον τῷ Z , διαστήματι δὲ ὅποτέρω τῶν ZA , ZB , κύκλος γεγράφθω ὁ ABE .

Ἐφάπτεται >άρα ἡ AD εὐθεῖα τοῦ ABE κύκλου διὰ τὸ ὀρθὴν ε>ῖναι τὴν πρὸς τῷ A γωνίαν. καὶ >ίση ἐστὶν ἡ ὑπὸ BAD γωνία τῇ ἐν τῷ ABE τμήματι; ὀρθῇ γὰρ καὶ αὐτὴ ἐν ἡμικυκλίῳ ο>ῦσα. ἀλλὰ καὶ ἡ ὑπὸ BAD τῇ πρὸς τῷ Γ >ίση ἐστίν. καὶ ἡ ἐν τῷ ABE >άρα >ίση ἐστὶ τῇ πρὸς τῷ Γ .

Γέγραπται >άρα πάλιν ἐπὶ τῇ AB τμήμα κύκλου τὸ ABE δεθόμενον γωνίαν >ίσην τῇ πρὸς τῷ Γ .

Ἀλλὰ δὲ ἡ πρὸς τῷ Γ ὀβελεία >έστω; καὶ συνεστάτω αὐτῇ >ίση πρὸς τῇ AB εὐθείᾳ καὶ τῷ A σημείῳ ἡ ὑπὸ BAD , ὥς >έθει ἐπὶ τῇ τρίτῃ καταγραφῇ, καὶ τῇ AD πρὸς ὀρθᾶς >ῆθω ἡ AE , καὶ τεμήσθω πάλιν ἡ AB δίθα κατὰ τὸ Z , καὶ τῇ AB πρὸς ὀρθᾶς >ῆθω ἡ ZH , καὶ ἐπεζεύθω ἡ HB .

Καὶ ἐπεὶ πάλιν >ίση ἐστὶν ἡ AZ τῇ ZB , καὶ κοινὴ ἡ ZH , δύο δὲ αἱ AZ , ZH δύο ταῖς BZ , ZH >ίσαι εἰσὶν; καὶ γωνία ἡ ὑπὸ AZH γωνία τῇ ὑπὸ BZH >ίση; βάσις >άρα ἡ AH βάσει τῇ BH >ίση ἐστίν; ὁ >άρα κέντρον μὲν τῷ H διαστήματι δὲ τῷ HA κύκλος γραφόμενος <ῆχει καὶ διὰ τοῦ B . ἐρθέσθω ὥς ὁ ABE . καὶ ἐπεὶ τῇ AE διαμέτρῳ ἀπ> >άκρας πρὸς ὀρθᾶς ἐστὶν ἡ AD , ἡ AD >άρα ἐφάπτεται τοῦ ABE κύκλου. καὶ ἀπὸ τῆς κατὰ τὸ A ἐπαφῆς διῆκται ἡ AB ; ἡ >άρα ὑπὸ BAD γωνία >ίση ἐστὶ τῇ ἐν τῷ ἐναλλάχ τοῦ κύκλου τμήματι τῷ $A\Theta B$ συνισταμένῃ γωνίᾳ. ἀλλ> ἡ ὑπὸ BAD γωνία τῇ πρὸς τῷ Γ >ίση ἐστίν. καὶ ἡ ἐν τῷ $A\Theta B$ >άρα τμήματι γωνία >ίση ἐστὶ τῇ πρὸς τῷ Γ .

Ἐπὶ τῇ >άρα δοθείσῃ εὐθείᾳ τῆς AB γέγραπται τμήμα κύκλου τὸ $A\Theta B$ δεθόμενον γωνίαν >ίσην τῇ πρὸς τῷ Γ ; <όπερ >έδει ποιῆσαι.

Let AB be the given straight-line, and C the given rectilinear angle. So it is required to draw a segment of a circle, accepting an angle equal to C , on the given straight-line AB . So the [angle] C is surely either acute, a right-angle, or obtuse. First of all, let it be acute. And, as in the first diagram (from the left), let (angle) BAD , equal to angle C , be constructed on the straight-line AB , at the point A (on it) [Prop. 1.23]. Thus, BAD is also acute. Let AE be drawn, at right-angles to DA [Prop. 1.11]. And let AB be cut in half at F [Prop. 1.10]. And let FG be drawn from point F , at right-angles to AB [Prop. 1.11]. And let GB be joined. And since AF is equal to FB , and FG (is) common, the two (straight-lines) AF , FG are equal to the two (straight-lines) BF , FG (respectively). And angle AFG (is) equal to [angle] BFG . Thus, the base AG is equal to the base BG [Prop. 1.4]. Thus, the circle drawn with center G , and radius GA , will also go through B (as well as A). Let it be drawn, and let it be (denoted) ABE . And let EB be joined. Therefore, since AD is at the extremity of diameter AE , (namely, point) A , at right-angles to AE , the (straight-line) AD thus touches the circle ABE [Prop. 3.16 corr.]. Therefore, since some straight-line AD touches the circle ABE , and some (other) straight-line AB has been drawn across from the point of contact A into circle ABE , angle DAB is thus equal to the angle AEB in the alternate segment of the circle [Prop. 3.32]. But, DAB is equal to C . Thus, angle C is also equal to AEB .

Thus, a segment ABE of a circle, accepting the angle AEB (which is) equal to the given (angle) C , has been drawn on the given straight-line AB .

And so let C be a right-angle. And let it again be necessary to draw a segment of a circle on AB , accepting an angle equal to the right-[angle] C . Let the (angle) BAD [again] be constructed, equal to the right-angle C [Prop. 1.23], as in the second diagram (from the left). And let AB be cut in half at F [Prop. 1.10]. And let the circle ABE be drawn with center F , and radius either FA or FB .

Thus, the straight-line AD touches the circle ABE , on account of the angle at A being a right-angle [Prop. 3.16 corr.]. And angle BAD is equal to the angle in segment AEB . For (the latter angle), being in a semi-circle, is also a right-angle [Prop. 3.31]. But, BAD is also equal to C . Thus, the (angle) in (segment) AEB is also equal to C .

Thus, a segment ABE of a circle, accepting an angle equal to C , has again been drawn on AB .

And so let (angle) C be obtuse. And let (angle) BAD , equal to (C), be constructed on the straight-line AB , at the point A (on it) [Prop. 1.23], as in the third diagram (from the left). And let AE be drawn, at right-angles to AD [Prop. 1.11]. And let AB again be cut in half at F [Prop. 1.10]. And let FG be drawn, at right-angles to AB [Prop. 1.10]. And let GB be joined.

And again, since AF is equal to FB , and FG (is) common, the two (straight-lines) AF , FG are equal to the two (straight-lines) BF , FG (respectively). And angle AFG (is) equal to angle BFG . Thus, the base AG is equal to the base BG [Prop. 1.4]. Thus, a circle of center G , and radius GA , being drawn, will also go through B (as well as A). Let it go like ABE (in the third diagram from the left). And since AD is at right-angles to the diameter AE , at its extremity, AD thus touches circle ABE

[Prop. 3.16 corr.]. And AB has been drawn across (the circle) from the point of contact A . Thus, angle BAD is equal to the angle constructed in the alternate segment AHB of the circle [Prop. 3.32]. But, angle BAD is equal to C . Thus, the angle in segment AHB is also equal to C .

Thus, a segment AHB of a circle, accepting an angle equal to C , has been drawn on the given straight-line AB . (Which is) the very thing it was required to do.

34.

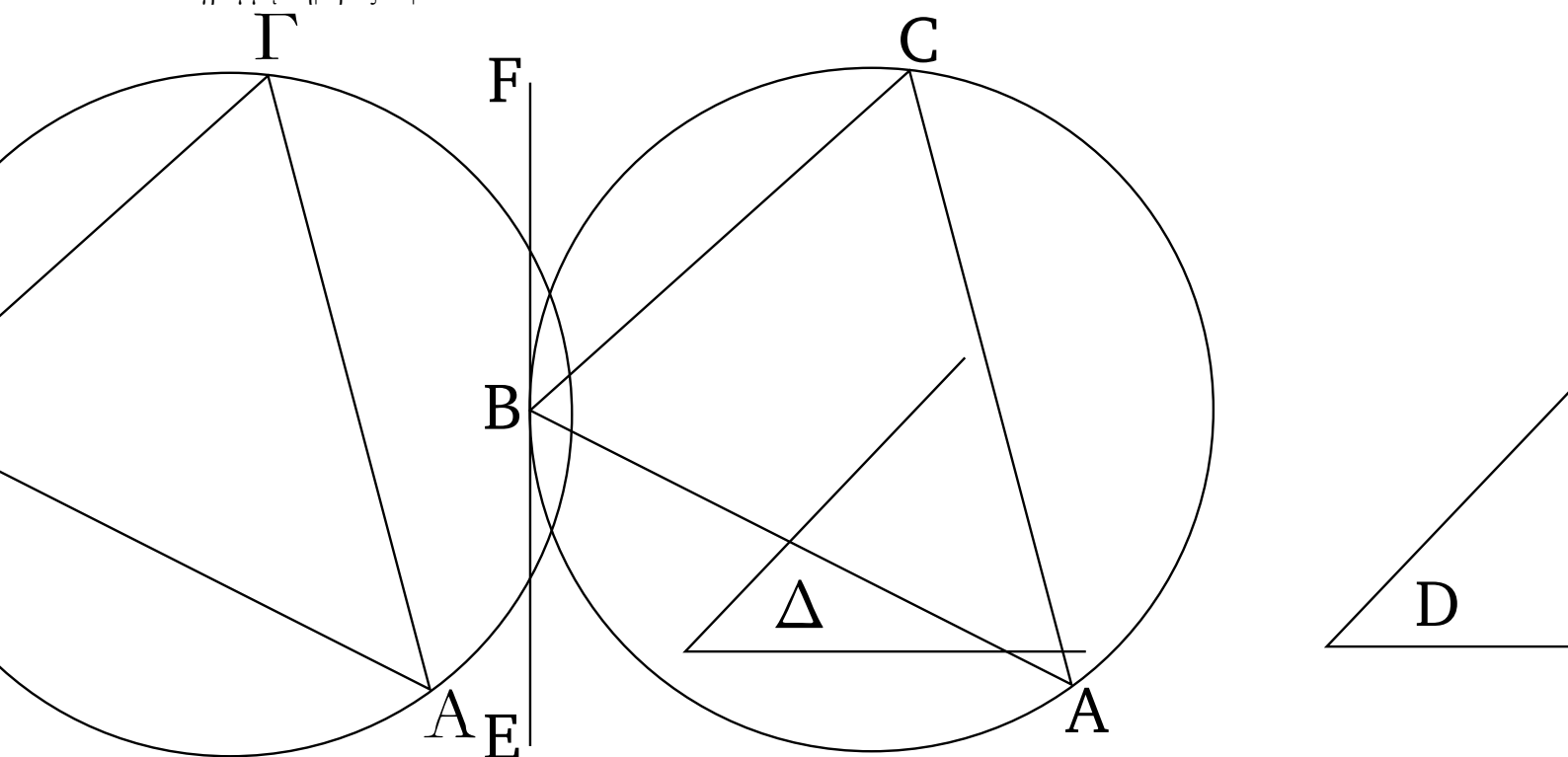
Ἀπὸ τοῦ δοθέντος κύκλου τμήμα ἀφελεῖν δεθόμενον γωνίαν
 ὅσην τῇ δοθείσῃ γωνίᾳ εὐθυγράμμω.

>Εστω ὁ δοθεὶς κύκλος ὁ ΑΒΓ, ἡ δὲ δοθεῖσα γωνία
εὐθύγραμμος ἡ πρὸς τῷ Δ; δεῖ δὴ ἀπὸ τοῦ ΑΒΓ κύκλου
τμημα ἀφελεῖν δεθόμενον γωνίαν >ίσην τῇ| δοθείσῃ γωνίᾳ
εὐθυγράμμω τῇ| πρὸς τῷ Δ.

Proposition 34

To cut off a segment, accepting an angle equal to a given rectilinear angle, from a given circle.

Let ABC be the given circle, and D the given rectilinear angle. So it is required to cut off a segment, accepting an angle equal to the given rectilinear angle D , from the given circle ABC .



>H0θω τοῦ ΑΒΓ ἔφαπτομένη ἡ ΕΖ κατὰ τὸ Β σημεῖον, καὶ
 συνεστάτω πρὸς τῇ ΖΒ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημεῖω
 τῷ Β τῇ πρὸς τῷ Δ γωνία >ιση ἡ ὑπὸ ΖΒΓ.

Ἐπεὶ οὖν κύκλου τοῦ ΑΒΓ ἐφάπτεται τῆς εὐθείας ἡ ΕΖ, καὶ ἀπὸ τῆς κατὰ τὸ Β ἐπαφῆς διῆκται ἡ ΒΓ, ἡ ὑπὸ ΖΒΓ > ἄρα γωνία > ἴση ἐστὶ τῇ ἐν τῷ ΒΑΓ ἐναλλάξ τμήματι συνισταμένη γωνίᾳ. ἄλλ> ἡ ὑπὸ ΖΒΓ τῇ πρὸς τῷ Δ ἐστὶν > ἴση; καὶ ἡ ἐν τῷ ΒΑΓ > ἄρα τμήματι > ἴση ἐστὶ τῇ πρὸς τῷ Δ [γωνίᾳ].

Ἀπὸ τοῦ δοθέντος >άρα κύκλου τοῦ ΑΒΓ τμήμα ἀφή|ρηται
τὸ ΒΑΓ δεθόμενον γωνίαν >ίσην τῇ|δοθείσῃ γωνίᾳ εὐθυγράμ-
μω τῇ|πρὸς τῷ Δ: <όπερ >έδει ποιῆσαι.

Let EF be drawn touching ABC at point B .[†] And let (angle) FBC , equal to angle D , be constructed on the straight-line FB , at the point B on it [Prop. 1.23].

Therefore, since some straight-line EF touches the circle ABC , and BC has been drawn across (the circle) from the point of contact B , angle FBC is thus equal to the angle constructed in the alternate segment BAC [Prop. 1.32]. But, FBC is equal to D . Thus, the (angle) in the segment BAC is also equal to [angle] D .

Thus, the segment BAC , accepting an angle equal to the given rectilinear angle D , has been cut off from the given circle ABC . (Which is) the very thing it was required to do.

† Presumably, by finding the center of ABC [Prop. 3.1], drawing a straight-line between the center and point B , and then drawing EF through point B , at right-angles to the aforementioned straight-line [Prop. 1.11].

35.

Ἐὰν ἐν κύκλῳ δύο εὐθεῖαι τέμνωσιν ἀλλήλας, τὸ ὑπὸ τῶν
τῆς μιᾶς τμημάτων περιεχόμενον ὀρθογώνιον > ἴσον ἐστὶ τῷ

Proposition 35

If two straight-lines in a circle cut one another, (then) the rectangle contained by the pieces of one is equal to the rect-

ὑπὸ τῶν τῆς ἐτέρας τμημάτων περιεθρομένῳ ὀρθογωνίῳ.

Ἐν γὰρ κύκλῳ τῷ $AB\Gamma\Delta$ δύο εὐθεῖαι αἱ $ΑΓ$, $ΒΔ$ τεμνέτωσαν ἀλλήλας κατὰ τὸ E σημεῖον; λέγω, <ὅτι τὸ ὑπὸ τῶν $ΑΕ$, $ΕΓ$ περιεχόμενον ὀρθογώνιον >ῖσον ἐστὶ τῷ ὑπὸ τῶν $ΔΕ$, $ΕΒ$ περιεθρομένῳ ὀρθογωνίῳ.

Εἰ μὲν ο>ὖν αἱ $ΑΓ$, $ΒΔ$ διὰ τοῦ κέντρου εἰσὶν <ώστε τὸ E κέντρον ε>ῖναι τοῦ $AB\Gamma\Delta$ κύκλου, φανερόν, <ὅτι >ῖσων οὐσῶν τῶν $ΑΕ$, $ΕΓ$, $ΔΕ$, $ΕΒ$ καὶ τὸ ὑπὸ τῶν $ΑΕ$, $ΕΓ$ περιεχόμενον ὀρθογώνιον >ῖσον ἐστὶ τῷ ὑπὸ τῶν $ΔΕ$, $ΕΒ$ περιεθρομένῳ ὀρθογωνίῳ.

μὴ >έστωσαν δὴ αἱ $ΑΓ$, $ΔΒ$ διὰ τοῦ κέντρου, καὶ εἰλήφθω τὸ κέντρον τοῦ $AB\Gamma\Delta$, καὶ >έστω τὸ Z , καὶ ἀπὸ τοῦ Z ἐπὶ τὰς $ΑΓ$, $ΔΒ$ εὐθείας κάθετοι >ήθθωσαν αἱ ZH , $ZΘ$, καὶ ἐπεξεύθθωσαν αἱ ZB , $ZΓ$, ZE .

Καὶ ἐπεὶ εὐθεῖα τις διὰ τοῦ κέντρου ἢ HZ εὐθεῖαν τινα μὴ διὰ τοῦ κέντρου τὴν $ΑΓ$ πρὸς ὀρθὰς τέμνει, καὶ δίθα αὐτὴν τέμνει, >ῖση >άρα ἢ AH τῇ $HΓ$. ἐπεὶ ο>ὖν εὐθεῖα ἢ $ΑΓ$ τέτμηται εἰς μὲν >ῖσα κατὰ τὸ H , εἰς δὲ >άνισα κατὰ τὸ E , τὸ >άρα ὑπὸ τῶν $ΑΕ$, $ΕΓ$ περιεχόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τῆς $ΕΗ$ τετραγώνου >ῖσον ἐστὶ τῷ ἀπὸ τῆς $HΓ$; [κοινὸν] προσκείσθω τὸ ἀπὸ τῆς HZ ; τὸ >άρα ὑπὸ τῶν $ΑΕ$, $ΕΓ$ μετὰ τῶν ἀπὸ τῶν HE , HZ >ῖσον ἐστὶ τοῖς ἀπὸ τῶν $ΓΗ$, HZ . ἀλλὰ τοῖς μὲν ἀπὸ τῶν $ΕΗ$, HZ >ῖσον ἐστὶ τὸ ἀπὸ τῆς ZE , τοῖς δὲ ἀπὸ τῶν $ΓΗ$, HZ >ῖσον ἐστὶ τὸ ἀπὸ τῆς $ZΓ$; τὸ >άρα ὑπὸ τῶν $ΑΕ$, $ΕΓ$ μετὰ τοῦ ἀπὸ τῆς ZE >ῖσον ἐστὶ τῷ ἀπὸ τῆς $ZΓ$. >ῖση δὲ ἢ $ZΓ$ τῇ ZB ; τὸ >άρα ὑπὸ τῶν $ΑΕ$, $ΕΓ$ μετὰ τοῦ ἀπὸ τῆς EZ >ῖσον ἐστὶ τῷ ἀπὸ τῆς ZB . διὰ τὰ αὐτὰ δὴ καὶ τὸ ὑπὸ τῶν $ΔΕ$, $ΕΒ$ μετὰ τοῦ ἀπὸ τῆς ZE ἰσον ἐστὶ τῷ ἀπὸ τῆς ZB . ἐδείθθη δὲ καὶ τὸ ὑπὸ τῶν $ΑΕ$, $ΕΓ$ μετὰ τοῦ ἀπὸ τῆς ZE >ῖσον τῷ ἀπὸ τῆς ZB ; τὸ >άρα ὑπὸ τῶν $ΑΕ$, $ΕΓ$ μετὰ τοῦ ἀπὸ τῆς ZE >ῖσον ἐστὶ τῷ ὑπὸ τῶν $ΔΕ$, $ΕΒ$ μετὰ τοῦ ἀπὸ τῆς ZE . κοινὸν ἀφῆ|ρήσθω τὸ ἀπὸ τῆς ZE ; λοιπὸν >άρα τὸ ὑπὸ τῶν $ΑΕ$, $ΕΓ$ περιεχόμενον ὀρθογώνιον >ῖσον ἐστὶ τῷ ὑπὸ τῶν $ΔΕ$, $ΕΒ$ περιεθρομένῳ ὀρθογωνίῳ.

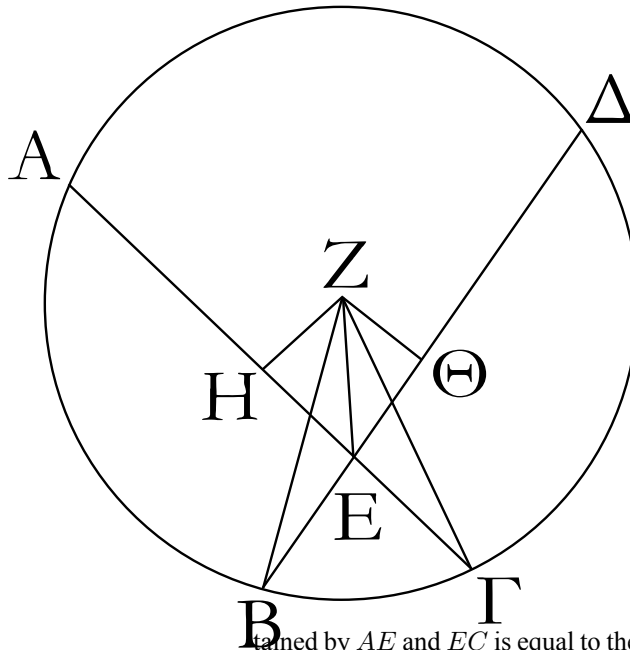
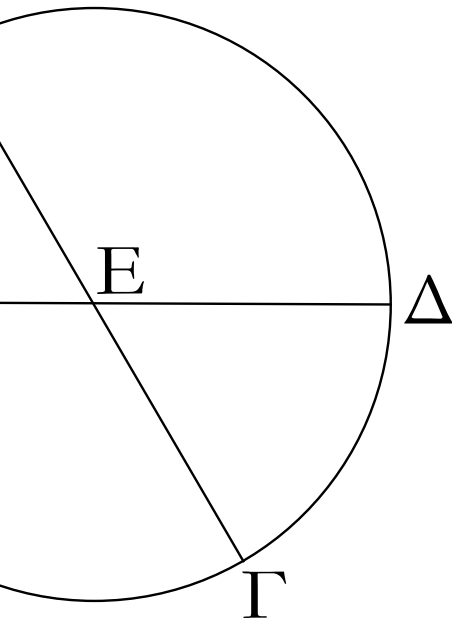
angle contained by the pieces of the other.

For let the two straight-lines AC and BD , in the circle $ABCD$, cut one another at point E . I say that the rectangle contained by AE and EC is equal to the rectangle contained by DE and EB .

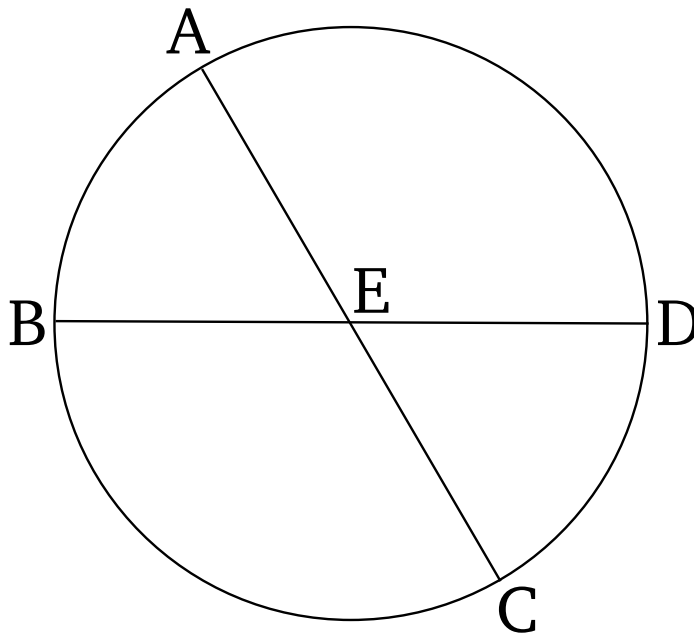
In fact, if AC and BD are through the center (as in the first diagram from the left), so that E is the center of circle $ABCD$, (then it is) clear that, AE , EC , DE , and EB being equal, the rectangle contained by AE and EC is also equal to the rectangle contained by DE and EB .

So let AC and DB not be through the center (as in the second diagram from the left), and let the center of $ABCD$ be found [Prop. 3.1], and let it be (at) F . And let FG and FH be drawn from F , perpendicular to the straight-lines AC and DB (respectively) [Prop. 1.12]. And let FB , FC , and FE be joined.

And since some straight-line, GF , through the center, cuts at right-angles some (other) straight-line, AC , not through the center, (then) it also cuts it in half [Prop. 3.3]. Thus, AG (is) equal to GC . Therefore, since the straight-line AC is cut equally at G , and unequally at E , the rectangle contained by AE and EC plus the square on EG is thus equal to the (square) on GC [Prop. 2.5]. Let the (square) on GF be added [to both]. Thus, the (rectangle contained) by AE and EC plus the (sum of the squares) on GE and GF is equal to the (sum of the squares) on CG and GF . But, the (square) on FE is equal to the (sum of the squares) on EG and GF [Prop. 1.47], and the (square) on FC is equal to the (sum of the squares) on CG and GF [Prop. 1.47]. Thus, the (rectangle contained) by AE and EC plus the (square) on FE is equal to the (square) on FC . And FC (is) equal to FB . Thus, the (rectangle contained) by AE and EC plus the (square) on FE is equal to the (square) on FB . So, for the same (reasons), the (rectangle contained) by DE and EB plus the (square) on FE is equal to the (square) on FB . And the (rectangle contained) by AE and EC plus the (square) on FE was also shown (to be) equal to the (square) on FB . Thus, the (rectangle contained) by AE and EC plus the (square) on FE is equal to the (rectangle contained) by DE and EB plus the (square) on FE . Let the (square) on FE be taken from both. Thus, the remaining rectangle con-



Ἐάν >άρα ἐν κύκλῳ εὐθεῖαι δύο τέμνωσιν ἀλλήλας, τὸ >ῖσον ἐστὶ τῷ ὑπὸ τῶν τῆς ἐτέρας τμημάτων περιεχομένῳ ὀρθογωνίῳ; <όπερ >έδει δεῖξαι.



ὑπὸ τῶν τῆς μιᾶς τμημάτων περιεχόμενον ὀρθογώνιον >ῖσον ἐστὶ τῷ ὑπὸ τῶν τῆς ἐτέρας τμημάτων περιεχομένῳ ὀρθογωνίῳ; <όπερ >έδει δεῖξαι.

Thus, if two straight-lines in a circle cut one another, (then) the rectangle contained by the pieces of one is equal to the rectangle contained by the pieces of the other. (Which is) the very thing it was required to show.

36.

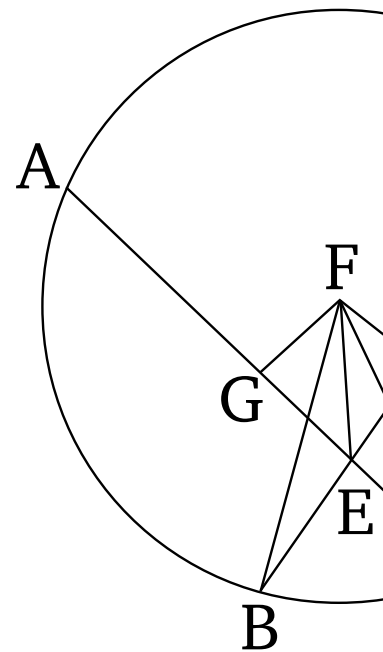
Proposition 36

Ἐάν κύκλου ληφθῇ τι σημεῖον ἐκτόξ, καὶ ἀπ' αὐτοῦ πρὸς τὸν κύκλον προσπίπτωσι δύο εὐθεῖαι, καὶ ἡ μὲν αὐτῶν τέμνη τὸν κύκλον, ἡ δὲ ἐφάπτεται, >έσται τὸ ὑπὸ <όλης τῆς τεμνούσης καὶ τῆς ἐκτόξ ἀπολαμβανομένης μεταχὺ τοῦ τε σημείου καὶ τῆς κυρτῆς περιφερείας >ῖσον τῷ ἀπὸ τῆς ἐφαπτομένης τετραγώνῳ.

Κύκλου γὰρ τοῦ ΑΒΓ εἰλήθω τι σημεῖον ἐκτόξ τὸ Δ, καὶ ἀπὸ τοῦ Δ πρὸς τὸν ΑΒΓ κύκλον προσπιπέτωσαν δύο εὐθεῖαι αἱ ΔΓ[Α], ΔΒ; καὶ ἡ μὲν ΔΓΑ τεμνέτω τὸν ΑΒΓ κύκλον, ἡ δὲ ΒΔ ἐφαπτέσθω; λέγω, <ότι τὸ ὑπὸ τῶν ΑΔ, ΔΓ περιεχόμενον ὀρθογώνιον >ῖσον ἐστὶ τῷ ἀπὸ τῆς ΔΒ

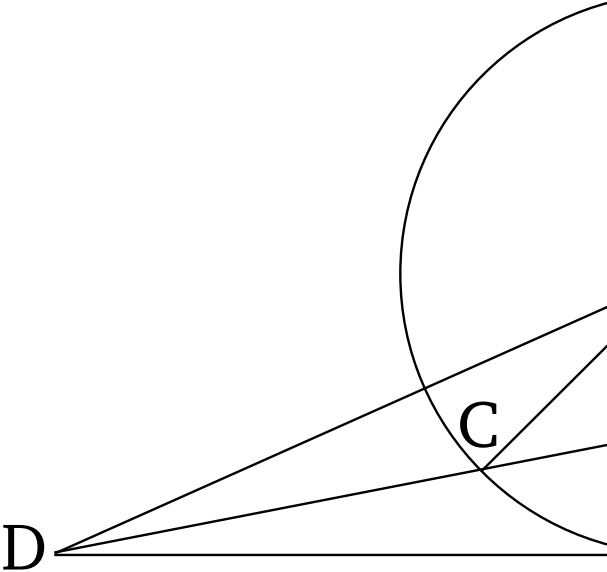
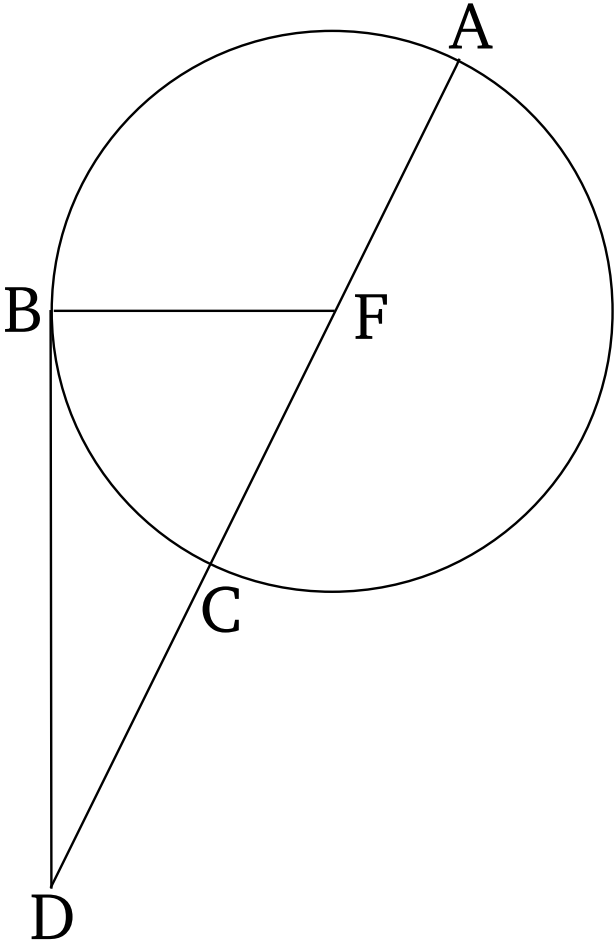
If some point is taken outside a circle, and two straight-lines radiate from it towards the circle, and (one) of them cuts the circle, and the (other) touches (it), (then) the (rectangle contained) by the whole (straight-line) cutting (the circle), and the (part of it) cut off outside (the circle), between the point and the convex circumference, will be equal to the square on the tangent (line).

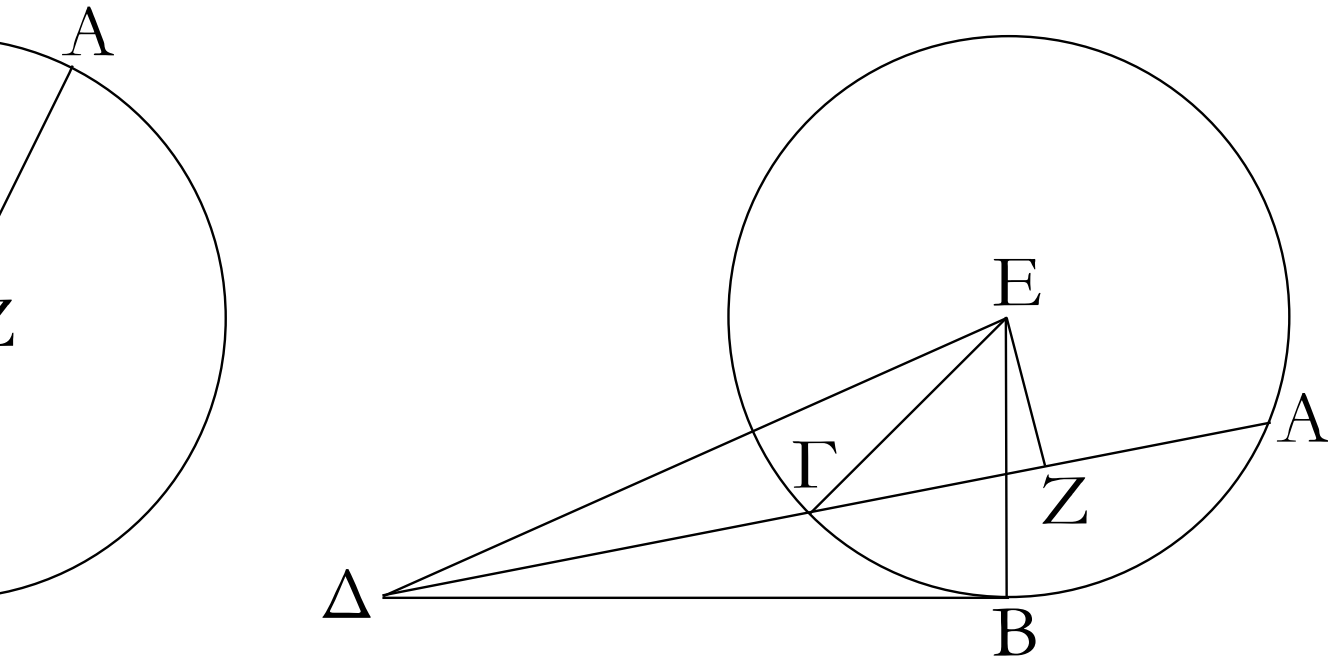
For let some point D be taken outside circle ABC , and let two straight-lines, $DC[A]$ and DB , radiate from D towards circle ABC . And let DCA cut circle ABC , and let BD touch (it). I say that the rectangle contained by AD and



τετραγώνω.

DC is equal to the square on DB .





Ἡ >άρα [Δ]ΓΑ >ήτοι διὰ τοῦ κέντρου ἐστὶν >ή ο>ύ. >έστω πρότερον διὰ τοῦ κέντρου, καὶ >έστω τὸ Ζ κέντρον τοῦ ΑΒΓ κύκλου, καὶ ἐπεζεύθω ἡ ΖΒ; ὀρθὴ >άρα ἐστὶν ἡ ὑπὸ ΖΒΔ. καὶ ἐπεὶ εὐθεῖα ἡ ΑΓ δίθα τέτμηται κατὰ τὸ Ζ, πρόσκειται δὲ αὐτῇ ἡ ΓΔ, τὸ >άρα ὑπὸ τῶν ΑΔ, ΔΓ μετὰ τοῦ ἀπὸ τῆς ΖΓ >ίσον ἐστὶ τῶ ἀπὸ τῆς ΖΔ. >ίση δὲ ἡ ΖΓ τῇ ΖΒ; τὸ >άρα ὑπὸ τῶν ΑΔ, ΔΓ μετὰ τοῦ ἀπὸ τῆς ΖΒ >ίσον ἐστὶ τῶ ἀπὸ τῆς ΖΔ. τῶ δὲ ἀπὸ τῆς ΖΔ >ίσα ἐστὶ τὰ ἀπὸ τῶν ΖΒ, ΒΔ; τὸ >άρα ὑπὸ τῶν ΑΔ, ΔΓ μετὰ τοῦ ἀπὸ τῆς ΖΒ >ίσον ἐστὶ τοῖς ἀπὸ τῶν ΖΒ, ΒΔ. κοινὸν ἀφηρήσθω τὸ ἀπὸ τῆς ΖΒ; λοιπὸν >άρα τὸ ὑπὸ τῶν ΑΔ, ΔΓ >ίσον ἐστὶ τῶ ἀπὸ τῆς ΔΒ ἐφαπτομένης.

Ἀλλὰ δὴ ἡ ΔΓΑ μὴ >έστω διὰ τοῦ κέντρου τοῦ ΑΒΓ κύκλου, καὶ εἰλήφθω τὸ κέντρον τὸ Ε, καὶ ἀπὸ τοῦ Ε ἐπὶ τὴν ΑΓ κάθετος >ήθω ἡ ΕΖ, καὶ ἐπεζεύθωσαν αἱ ΕΒ, ΕΓ, ΕΔ; ὀρθὴ >άρα ἐστὶν ἡ ὑπὸ ΕΒΔ. καὶ ἐπεὶ εὐθεῖα τίς διὰ τοῦ κέντρου ἡ ΕΖ εὐθειάν τινα μὴ διὰ τοῦ κέντρου τὴν ΑΓ πρὸς ὀρθὰς τέμνει, καὶ δίθα αὐτὴν τέμνει; ἡ ΑΖ >άρα τῇ ΖΓ ἐστὶν >ίση. καὶ ἐπεὶ εὐθεῖα ἡ ΑΓ τέτμηται δίθα κατὰ τὸ Ζ σημεῖον, πρόσκειται δὲ αὐτῇ ἡ ΓΔ, τὸ >άρα ὑπὸ τῶν ΑΔ, ΔΓ μετὰ τοῦ ἀπὸ τῆς ΖΓ >ίσον ἐστὶ τῶ ἀπὸ τῆς ΖΔ. κοινὸν προσκείσθω τὸ ἀπὸ τῆς ΖΕ; τὸ >άρα ὑπὸ τῶν ΑΔ, ΔΓ μετὰ τῶν ἀπὸ τῶν ΓΖ, ΖΕ >ίσον ἐστὶ τοῖς ἀπὸ τῶν ΖΔ, ΖΕ. τοῖς δὲ ἀπὸ τῶν ΓΖ, ΖΕ >ίσον ἐστὶ τὸ ἀπὸ τῆς ΕΓ; ὀρθὴ γὰρ [ἐστὶν] ἡ ὑπὸ ΕΖΓ [γωνία]; τοῖς δὲ ἀπὸ τῶν ΔΖ, ΖΕ >ίσον ἐστὶ τὸ ἀπὸ τῆς ΕΔ; τὸ >άρα ὑπὸ τῶν ΑΔ, ΔΓ μετὰ τοῦ ἀπὸ τῆς ΕΓ >ίσον ἐστὶ τῶ ἀπὸ τῆς ΕΔ. >ίση δὲ ἡ ΕΓ τῇ ΕΒ; τὸ >άρα ὑπὸ τῶν ΑΔ, ΔΓ μετὰ τοῦ ἀπὸ τῆς ΕΒ >ίσον ἐστὶ τῶ ἀπὸ τῆς ΕΔ. τῶ δὲ ἀπὸ τῆς ΕΔ >ίσα ἐστὶ τὰ ἀπὸ τῶν ΕΒ, ΒΔ; ὀρθὴ γὰρ ἡ ὑπὸ ΕΒΔ γωνία; τὸ >άρα ὑπὸ τῶν ΑΔ, ΔΓ μετὰ τοῦ ἀπὸ τῆς ΕΒ >ίσον ἐστὶ τοῖς ἀπὸ τῶν ΕΒ, ΒΔ. κοινὸν ἀφηρήσθω

[D]CA is surely either through the center, or not. Let it first of all be through the center, and let F be the center of circle ABC , and let FB be joined. Thus, (angle) FBD is a right-angle [Prop. 3.18]. And since straight-line AC is cut in half at F , let CD be added to it. Thus, the (rectangle contained) by AD and DC plus the (square) on FC is equal to the (square) on FD [Prop. 2.6]. And FC (is) equal to FB . Thus, the (rectangle contained) by AD and DC plus the (square) on FB is equal to the (square) on FD . And the (square) on FD is equal to the (sum of the squares) on FB and BD [Prop. 1.47]. Thus, the (rectangle contained) by AD and DC plus the (square) on FB is equal to the (sum of the squares) on FB and BD . Let the (square) on FB be subtracted from both. Thus, the remaining (rectangle contained) by AD and DC is equal to the (square) on the tangent DB .

And so let DCA not be through the center of circle ABC , and let the center E be found, and let EF be drawn from E , perpendicular to AC [Prop. 1.12]. And let EB , EC , and ED be joined. (Angle) EBD (is) thus a right-angle [Prop. 3.18]. And since some straight-line, EF , through the center, cuts some (other) straight-line, AC , not through the center, at right-angles, it also cuts it in half [Prop. 3.3]. Thus, AF is equal to FC . And since the straight-line AC is cut in half at point F , let CD be added to it. Thus, the (rectangle contained) by AD and DC plus the (square) on FC is equal to the (square) on FD [Prop. 2.6]. Let the (square) on FE be added to both. Thus, the (rectangle contained) by AD and DC plus the (sum of the squares) on CF and FE is equal to the (sum of the squares) on FD and FE . But the (square) on EC is equal to the (sum of the squares) on CF and FE . For [angle] EFC [is] a right-

τὸ ἀπὸ τῆς EB ; λοιπὸν $>$ ἄρα τὸ ὑπὸ τῶν $AD, \Delta\Gamma$ $>$ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΔB .

Ἐὰν $>$ ἄρα κύκλου ληφθῇ τι σημεῖον ἐκτόξ, καὶ ἀπ' αὐτοῦ πρὸς τὸν κύκλον προσπίπτωσι δύο εὐθεῖαι, καὶ ἡ μὲν αὐτῶν τέμνῃ τὸν κύκλον, ἡ δὲ ἐφάπτεται, $>$ ἐστὶ τὸ ὑπὸ $<$ ὅλης τῆς τεμνουσῆς καὶ τῆς ἐκτόξ ἀπολαμβανομένης μεταχὺ τοῦ τε σημείου καὶ τῆς κυρτῆς περιφερείας $>$ ἴσον τῷ ἀπὸ τῆς ἐφαπτομένης τετραγώνῳ; $<$ ὅπερ $>$ ἔδει δεῖξαι.

angle [Prop. 1.47]. And the (square) on ED is equal to the (sum of the squares) on DF and FE [Prop. 1.47]. Thus, the (rectangle contained) by AD and DC plus the (square) on EC is equal to the (square) on ED . And EC (is) equal to EB . Thus, the (rectangle contained) by AD and DC plus the (square) on EB is equal to the (square) on ED . And the (sum of the squares) on EB and BD is equal to the (square) on ED . For EBD (is) a right-angle [Prop. 1.47]. Thus, the (rectangle contained) by AD and DC plus the (square) on EB is equal to the (sum of the squares) on EB and BD . Let the (square) on EB be subtracted from both. Thus, the remaining (rectangle contained) by AD and DC is equal to the (square) on BD .

Thus, if some point is taken outside a circle, and two straight-lines radiate from it towards the circle, and (one) of them cuts the circle, and (the other) touches (it), (then) the (rectangle contained) by the whole (straight-line) cutting (the circle), and the (part of it) cut off outside (the circle), between the point and the convex circumference, will be equal to the square on the tangent (line). (Which is) the very thing it was required to show.

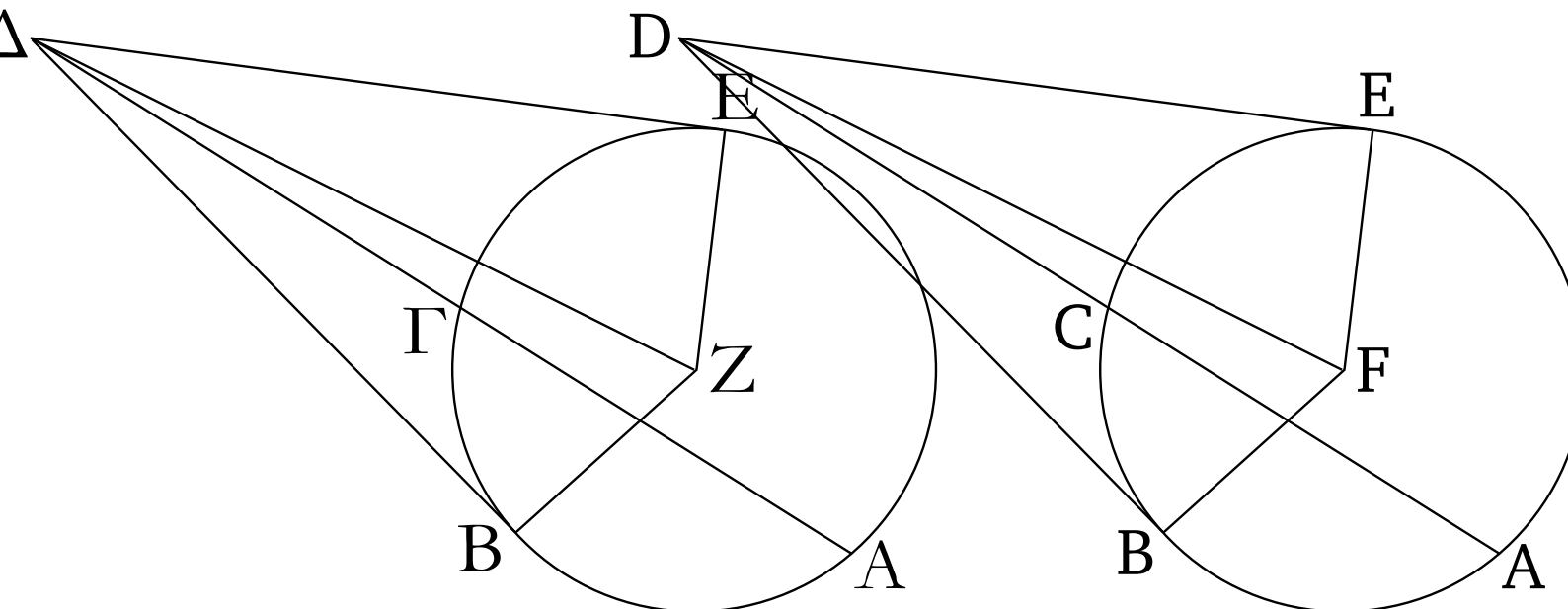
37.

Ἐὰν κύκλου ληφθῇ τι σημεῖον ἐκτόξ, ἀπὸ δὲ τοῦ σημείου πρὸς τὸν κύκλον προσπίπτωσι δύο εὐθεῖαι, καὶ ἡ μὲν αὐτῶν τέμνῃ τὸν κύκλον, ἡ δὲ προσπίπτῃ, $>$ ἡ δὲ τὸ ὑπὸ [τῆς] $<$ ὅλης τῆς τεμνουσῆς καὶ τῆς ἐκτόξ ἀπολαμβανομένης μεταχὺ τοῦ τε σημείου καὶ τῆς κυρτῆς περιφερείας $>$ ἴσον τῷ ἀπὸ τῆς προσπιπτούσης, ἡ προσπίπτουσα ἐφάπεται τοῦ κύκλου.

Κύκλου γὰρ τοῦ $AB\Gamma$ εἰλήφθω τι σημεῖον ἐκτόξ τὸ Δ , καὶ ἀπὸ τοῦ Δ πρὸς τὸν $AB\Gamma$ κύκλον προσπιπτέωσαν δύο εὐθεῖαι αἱ $\Delta\Gamma A, \Delta B$, καὶ ἡ μὲν $\Delta\Gamma A$ τεμνέτω τὸν κύκλον, ἡ δὲ ΔB προσπιπτέτω, $>$ ἐστω δὲ τὸ ὑπὸ τῶν $AD, \Delta\Gamma$ $>$ ἴσον τῷ ἀπὸ τῆς ΔB . λέγω, $<$ ὅτι ἡ ΔB ἐφάπτεται τοῦ $AB\Gamma$ κύκλου.

Proposition 37

If some point is taken outside a circle, and two straight-lines radiate from the point towards the circle, and one of them cuts the circle, and the (other) meets (it), and the (rectangle contained) by the whole (straight-line) cutting (the circle), and the (part of it) cut off outside (the circle), between the point and the convex circumference, is equal to the (square) on the (straight-line) meeting (the circle), (then) the (straight-line) meeting (the circle) will touch the circle. For let some point D be taken outside circle ABC , and let two straight-lines, DCA and DB , radiate from D towards circle ABC , and let DCA cut the circle, and let DB meet (the circle). And let the (rectangle contained) by AD and DC be equal to the (square) on DB . I say that DB touches circle ABC .



$>$ Ἡθθα γὰρ τοῦ $AB\Gamma$ ἐφαπτομένη ἡ ΔE , καὶ εἰλήφθω τὸ κέντρον τοῦ $AB\Gamma$ κύκλου, καὶ $>$ ἐστω τὸ Z , καὶ ἐπεζεύθωσαν αἱ ZE, ZB, ZA . ἡ $>$ ἄρα ὑπὸ $ZE\Delta$ ὀρθή ἐστίν. καὶ ἐπεὶ ἡ ΔE ἐφάπτεται τοῦ $AB\Gamma$ κύκλου, τέμνει δὲ ἡ $\Delta\Gamma A$, τὸ $>$ ἄρα ὑπὸ

For let DE be drawn touching ABC [Prop. 3.17], and let the center of the circle ABC be found, and let it be (at) F . And let FE, FB , and FD be joined. (Angle) FED is thus a right-angle [Prop. 3.18]. And since DE touches circle

τῶν $ΑΔ, ΔΓ$ ὀρθόν ἐστι τῶν ἀπὸ τῆς $ΔΕ$. ὁρθὸν δὲ καὶ τὸ ὑπὸ τῶν $ΑΔ, ΔΓ$ ὀρθόν ἀπὸ τῆς $ΔΒ$; τὸ ὀρθὸν ἀπὸ τῆς $ΔΕ$ ὀρθόν ἐστι τῶν ἀπὸ τῆς $ΔΒ$; ὀρθὸν ὅρα ἢ $ΔΕ$ τῇ $ΔΒ$. ἐστὶ δὲ καὶ ἡ $ΖΕ$ τῇ $ΖΒ$ ὀρθή; δύο δὲ αἱ $ΔΕ, ΕΖ$ δύο ταῖς $ΔΒ, ΒΖ$ ὀρθαὶ εἰσὶν; καὶ βάσις αὐτῶν κοινὴ ἡ $ΖΔ$; γωνία ὀρθὴ ἢ ὑπὸ $ΔΕΖ$ γωνία τῇ ὑπὸ $ΔΒΖ$ ἐστὶν ὀρθή. ὁρθὴ δὲ ἢ ὑπὸ $ΔΕΖ$; ὁρθὴ ὀρθὴ καὶ ἢ ὑπὸ $ΔΒΖ$. καὶ ἐστὶν ἡ $ΖΒ$ ἐκβαλλομένη διάμετρος; ἢ δὲ τῇ διαμέτρῳ τοῦ κύκλου πρὸς ὁρθὰς ἀπὸ ὀρθὰς ἀγομένη ἐφάπτεται τοῦ κύκλου; ἢ $ΔΒ$ ὀρθὴ ἐφάπτεται τοῦ $ΑΒΓ$ κύκλου. ὁμοίως δὲ δεῖθῃσεται, κἂν τὸ κέντρον ἐπὶ τῆς $ΑΓ$ τυθῇ.

Ἐὰν ὀρθὴ κύκλου ληφθῇ τι σημεῖον ἐκτός, ἀπὸ δὲ τοῦ σημείου πρὸς τὸν κύκλον προσπίπτωσι δύο εὐθεῖαι, καὶ ἡ μὲν αὐτῶν τέμνῃ τὸν κύκλον, ἡ δὲ προσπίπτῃ, ὅτι δὲ τὸ ὑπὸ ὀρθῆς τῆς τεμνομένης καὶ τῆς ἐκτός ἀπολαμβανομένης μεταχὺ τοῦ τε σημείου καὶ τῆς κυρτῆς περιφερείας ὀρθόν τῶν ἀπὸ τῆς προσπίπτουσας ἐφάπεται τοῦ κύκλου; ὅπερ ὀφείδει δεῖχαι.

ABC , and DCA cuts (it), the (rectangle contained) by AD and DC is thus equal to the (square) on DE [Prop. 3.36]. And the (rectangle contained) by AD and DC was also equal to the (square) on DB . Thus, the (square) on DE is equal to the (square) on DB . Thus, DE (is) equal to DB . And FE is also equal to FB . So the two (straight-lines) DE, EF are equal to the two (straight-lines) DB, BF (respectively). And their base, FD , is common. Thus, angle DEF is equal to angle DBF [Prop. 1.8]. And DEF (is) a right-angle. Thus, DBF (is) also a right-angle. And FB produced is a diameter, And a (straight-line) drawn at right-angles to a diameter of a circle, at its extremity, touches the circle [Prop. 3.16 corr.]. Thus, DB touches circle ABC . Similarly, (the same thing) can be shown, even if the center happens to be on AC .

Thus, if some point is taken outside a circle, and two straight-lines radiate from the point towards the circle, and one of them cuts the circle, and the (other) meets (it), and the (rectangle contained) by the whole (straight-line) cutting (the circle), and the (part of it) cut off outside (the circle), between the point and the convex circumference, is equal to the (square) on the (straight-line) meeting (the circle), (then) the (straight-line) meeting (the circle) will touch the circle. (Which is) the very thing it was required to show.

ELEMENTS BOOK 4
*Construction of Rectilinear Figures In and
Around Circles*

<Όροι.

1. Σθῆμα εὐθύγραμμον εἰς σθῆμα εὐθύγραμμον ἐγγράφεσθαι λέγεται, <όταν ἐκάστη τῶν τοῦ ἐγγραφομένου σθῆματος γωνιῶν ἐκάστης πλευρᾷ τοῦ, εἰς <ὃ ἐγγράφεται, <ἀπτηται.
2. Σθῆμα δὲ ὁμοίως περὶ σθῆμα περιγράφεσθαι λέγεται, <όταν ἐκάστη πλευρὰ τοῦ περιγραφομένου ἐκάστης γωνίας τοῦ, περὶ <ὃ περιγράφεται, <ἀπτηται.
3. Σθῆμα εὐθύγραμμον εἰς κύκλον ἐγγράφεσθαι λέγεται, <όταν ἐκάστη γωνία τοῦ ἐγγραφομένου <ἀπτηται τῇς τοῦ κύκλου περιφερείᾳ.
4. Σθῆμα δὲ εὐθύγραμμον περὶ κύκλον περιγράφεσθαι λέγεται, <όταν ἐκάστη πλευρὰ τοῦ περιγραφομένου ἐφάπτηται τῇς τοῦ κύκλου περιφερείᾳ.
5. Κύκλος δὲ εἰς σθῆμα ὁμοίως ἐγγράφεσθαι λέγεται, <όταν ἢ τοῦ κύκλου περιφέρεια ἐκάστης πλευρᾷ τοῦ, εἰς <ὃ ἐγγράφεται, <ἀπτηται.
6. Κύκλος δὲ περὶ σθῆμα περιγράφεσθαι λέγεται, <όταν ἢ τοῦ κύκλου περιφέρεια ἐκάστης γωνίας τοῦ, περὶ <ὃ περιγράφεται, <ἀπτηται.
7. Εὐθεῖα εἰς κύκλον ἐναρμόζεσθαι λέγεται, <όταν τὰ πέρατα αὐτῆς ἐπὶ τῇς περιφερείᾳ >ῇ τοῦ κύκλου.

Definitions

1. A rectilinear figure is said to be inscribed in a(nother) rectilinear figure when the respective angles of the inscribed figure touch the respective sides of the (figure) in which it is inscribed.
2. And, similarly, a (rectilinear) figure is said to be circumscribed about a(nother rectilinear) figure when the respective sides of the circumscribed (figure) touch the respective angles of the (figure) about which it is circumscribed.
3. A rectilinear figure is said to be inscribed in a circle when each angle of the inscribed (figure) touches the circumference of the circle.
4. And a rectilinear figure is said to be circumscribed about a circle when each side of the circumscribed (figure) touches the circumference of the circle.
5. And, similarly, a circle is said to be inscribed in a (rectilinear) figure when the circumference of the circle touches each side of the (figure) in which it is inscribed.
6. And a circle is said to be circumscribed about a rectilinear (figure) when the circumference of the circle touches each angle of the (figure) about which it is circumscribed.
7. A straight-line is said to be inserted into a circle when its extremities are on the circumference of the circle.

1.

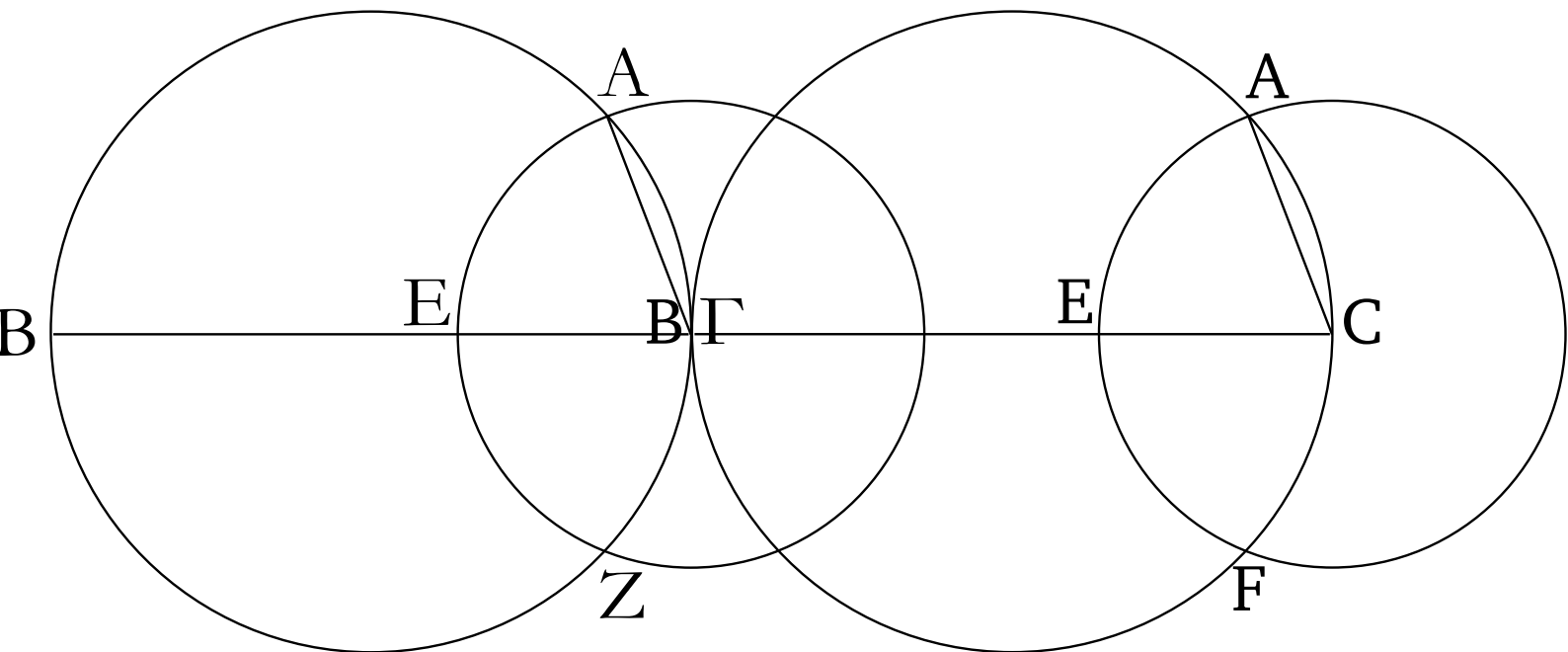
Εἰς τὸν δοθέντα κύκλον τῇ δοθείσῃ εὐθείᾳ μὴ μείζονι οὐσίῃ τῇς τοῦ κύκλου διαμέτρου >ίσην εὐθεῖαν ἐναρμόσαι.

Proposition 1

To insert a straight-line equal to a given straight-line into a circle, (the latter straight-line) not being greater than the diameter of the circle.

Δ

D



>Ἐστω ὁ δοθείς κύκλος ὁ ABΓ, ἡ δὲ δοθεῖσα εὐθεῖα μὴ μείζων τῇς τοῦ κύκλου διαμέτρου ἡ Δ. δεῖ δὴ εἰς τὸν ABΓ κύκλον τῇ Δ εὐθείᾳ >ίσην εὐθεῖαν ἐναρμόσαι.

>Ἡθθω τοῦ ABΓ κύκλου διάμετρος ἡ BΓ. εἰ μὲν οὖν

Let ABC be the given circle, and D the given straight-line (which is) not greater than the diameter of the circle. So it is required to insert a straight-line, equal to the straight-line D , into the circle ABC .

ΔEZ ἔστιν $>$ ίση. διὰ τὰ αὐτὰ δὴ καὶ ἡ ὑπὸ $\Lambda \Gamma B$ τῇ ὑπὸ ΔZE ἔστιν $>$ ίση; καὶ λοιπὴ $>$ άρα ἡ ὑπὸ $\Lambda \Gamma B$ λοιπῇ τῇ ὑπὸ ΔEZ ἔστιν $>$ ίση [ἰσογώνιον $>$ άρα ἔστι τὸ $\Lambda \Gamma B$ τρίγωνον τῷ ΔEZ τριγώνῳ, καὶ ἐγγέγραπται εἰς τὸν $\Lambda \Gamma B$ κύκλον].
Εἰς τὸν δοθέντα $>$ άρα κύκλον τῷ δοθέντι τριγώνῳ ἰσογώνιον τρίγωνον ἐγγέγραπται; $<$ όπερ $>$ έδει ποιῆσαι.

ABC is also equal to DEF . So, for the same (reasons), ACB is also equal to DFE . Thus, the remaining (angle) BAC is equal to the remaining (angle) EDF [Prop. 1.32]. [Thus, triangle ABC is equiangular with triangle DEF , and has been inscribed in circle ABC].

Thus, a triangle, equiangular with the given triangle, has been inscribed in the given circle. (Which is) the very thing it was required to do.

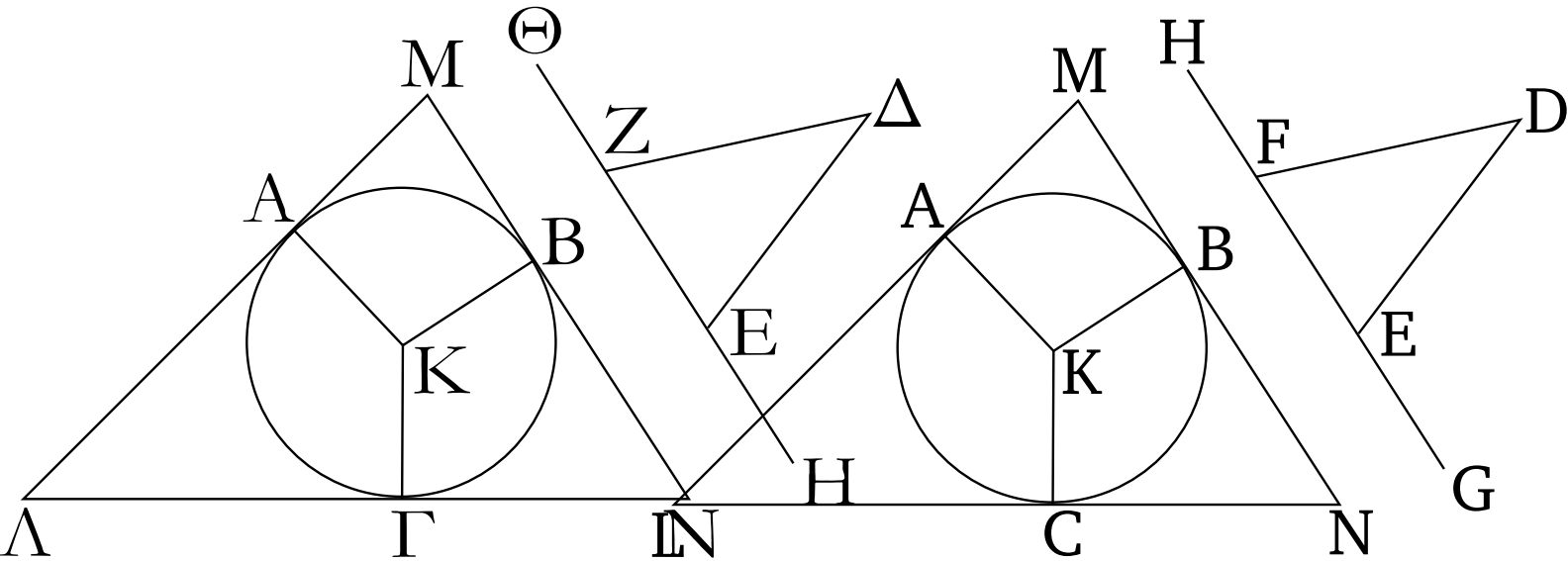
† See the footnote to Prop. 3.34.

3.

Περὶ τὸν δοθέντα κύκλον τῷ δοθέντι τριγώνῳ ἰσογώνιον τρίγωνον περιγράψαι.

Proposition 3

To circumscribe a triangle, equiangular with a given triangle, about a given circle.



$>$ Ἐστω ὁ δοθεὶς κύκλος ὁ $\Lambda \Gamma B$, τὸ δὲ δοθὲν τρίγωνον τὸ ΔEZ ; δεῖ δὴ περὶ τὸν $\Lambda \Gamma B$ κύκλον τῷ ΔEZ τριγώνῳ ἰσογώνιον τρίγωνον περιγράψαι.

Ἐκβεβλήσθω ἡ EZ ἔφ' ἑκάτερα τὰ μέρη κατὰ τὰ H , Θ σημεία, καὶ εἰλήφθω τοῦ $\Lambda \Gamma B$ κύκλου κέντρον τὸ K , καὶ διήθθω, ὥς $>$ έτυθεν, εὐθεῖα ἡ KB , καὶ συνεστάτω πρὸς τῇ KB εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ K τῇ μὲν ὑπὸ ΔEH γωνία $>$ ίση ἡ ὑπὸ BKA , τῇ δὲ ὑπὸ $\Delta Z\Theta$ $>$ ίση ἡ ὑπὸ BKG , καὶ διὰ τῶν A , B , Γ σημείων $>$ ήθθωσαν ἐφαπτόμεναι τοῦ $\Lambda \Gamma B$ κύκλου αἱ ΛM , MBN , $N\Gamma A$.

Καὶ ἐπεὶ ἐφάπτονται τοῦ $\Lambda \Gamma B$ κύκλου αἱ ΛM , MN , $N\Lambda$ κατὰ τὰ A , B , Γ σημεία, ἀπὸ δὲ τοῦ K κέντρου ἐπὶ τὰ A , B , Γ σημεία ἐπεξευγμέναι εἰσὶν αἱ KA , KB , $K\Gamma$, ὁρθαὶ $>$ άρα εἰσὶν αἱ πρὸς τοῖς A , B , Γ σημείοις γωνίαι. καὶ ἐπεὶ τοῦ ΛMBK τετραπλεύρου αἱ τέσσαρες γωνίαι τέτρασιν ὁρθαῖς $>$ ίσαι εἰσὶν, ἐπειδήπερ καὶ εἰς δύο τρίγωνα διαιρεῖται τὸ ΛMBK , καὶ εἰσὶν ὁρθαὶ αἱ ὑπὸ KAM , KBM γωνίαι, λοιπαὶ $>$ άρα αἱ ὑπὸ AKB , AMB δυσὶν ὁρθαῖς $>$ ίσαι εἰσὶν. εἰσὶ δὲ καὶ αἱ ὑπὸ ΔEH , ΔEZ δυσὶν ὁρθαῖς $>$ ίσαι; αἱ $>$ άρα ὑπὸ AKB , AMB ταῖς ὑπὸ ΔEH , ΔEZ $>$ ίσαι εἰσὶν, $<$ ὦν ἡ ὑπὸ AKB τῇ ὑπὸ ΔEH ἔστιν $>$ ίση; λοιπὴ $>$ άρα ἡ ὑπὸ AMB λοιπῇ τῇ ὑπὸ ΔEZ ἔστιν $>$ ίση. ὁμοίως δὴ δειθθήσεται, $<$ ότι καὶ ἡ ὑπὸ ΛNB τῇ ὑπὸ ΔZE ἔστιν $>$ ίση; καὶ λοιπὴ $>$ άρα ἡ ὑπὸ ΛMN [λοιπῇ] τῇ ὑπὸ ΔZ ἔστιν $>$ ίση. ἰσογώνιον $>$ άρα ἔστι τὸ ΛMN τρίγωνον τῷ ΔEZ τριγώνῳ; καὶ περιέγραπται περὶ τὸν $\Lambda \Gamma B$ κύκλον.

Περὶ τὸν δοθέντα $>$ άρα κύκλον τῷ δοθέντι τριγώνῳ ἰσογώνιον τρίγωνον περιέγραπται; $<$ όπερ $>$ έδει ποιῆσαι.

Let ABC be the given circle, and DEF the given triangle. So it is required to circumscribe a triangle, equiangular with triangle DEF , about circle ABC .

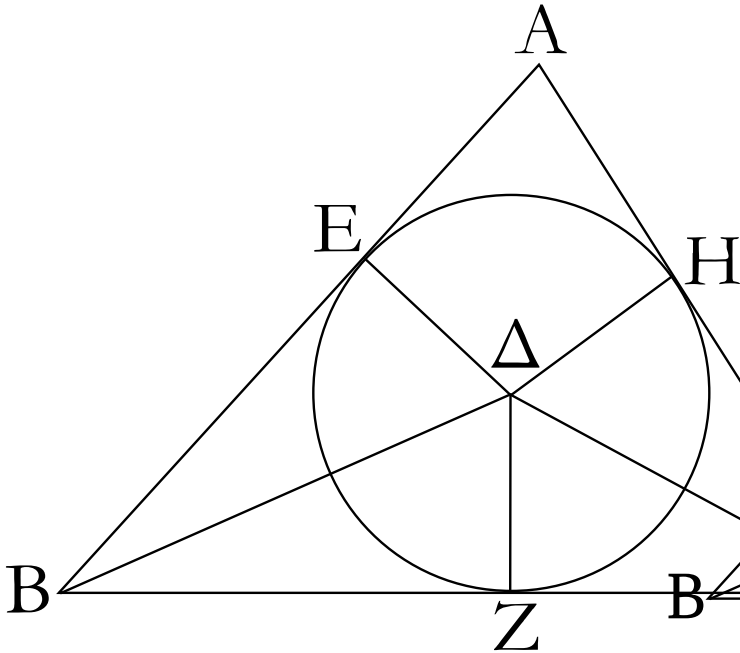
Let EF be produced in each direction to points G and H . And let the center K of circle ABC be found [Prop. 3.1]. And let the straight-line KB be drawn, at random, across (ABC). And let (angle) BKA , equal to angle DEG , be constructed on the straight-line KB at the point K on it, and (angle) BKC , equal to DFH [Prop. 1.23]. And let the (straight-lines) LAM , MBN , and NCL be drawn through the points A , B , and C (respectively), touching the circle ABC .†

And since LM , MN , and NL touch circle ABC at points A , B , and C (respectively), and KA , KB , and KC are joined from the center K to points A , B , and C (respectively), the angles at points A , B , and C are thus right-angles [Prop. 3.18]. And since the (sum of the) four angles of quadrilateral $AMBK$ is equal to four right-angles, inasmuch as $AMBK$ (can) also (be) divided into two triangles [Prop. 1.32], and angles KAM and KBM are (both) right-angles, the (sum of the) remaining (angles), AKB and AMB , is thus equal to two right-angles. And DEG and DEF is also equal to two right-angles [Prop. 1.13]. Thus, AKB and AMB is equal to DEG and DEF , of which AKB is equal to DEG . Thus, the remainder AMB is equal to the remainder DEF . So, similarly, it can be shown that LNK is also equal to DFE . Thus, the remaining (angle) MLN is also equal to the [remaining] (angle) EDF [Prop. 1.32]. Thus, triangle LMN is equiangular with tri-

† See the footnote to Prop. 3.34.

4.

Εἰς τὸ δοθὲν τρίγωνον κύκλον ἐγγράψαι.



>Ἐστω τὸ δοθὲν τρίγωνον τὸ ABΓ; δεῖ δὴ εἰς τὸ ABΓ τρίγωνον κύκλον ἐγγράψαι.

Τετμήσθωσαν αἱ ὑπὸ ABΓ, AΓB γωνίαι διῖθα ταῖς BΔ, ΓΔ εὐθείαις, καὶ συμβαλλέτωσαν ἀλλήλαις κατὰ τὸ Δ σημεῖον, καὶ >ήθθωσαν ἀπὸ τοῦ Δ ἐπὶ ταῖς AB, BΓ, ΓA εὐθείαις κάθετοι αἱ ΔE, ΔZ, ΔH.

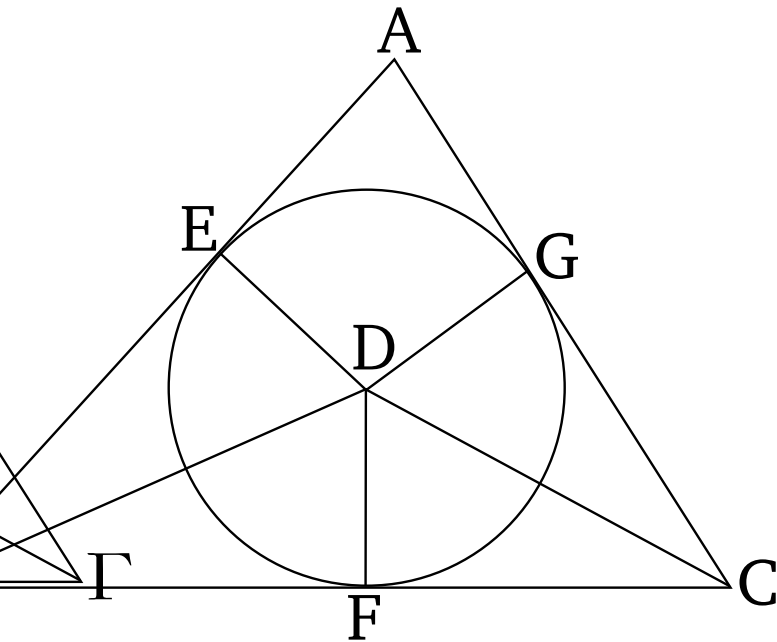
Καὶ ἐπεὶ >ῖση ἐστὶν ἡ ὑπὸ ABD γωνία τῇ ὑπὸ ΓBD, ἐστὶ δὲ καὶ ὀρθὴ ἡ ὑπὸ BEΔ ὀρθὴ τῇ ὑπὸ BZΔ >ῖση, δύο δὴ τρίγωνά ἐστι τὰ EBD, ZBD τὰς δύο γωνίας ταῖς δυσὶ γωνίαις >ῖσαζ >έθοντα καὶ μίαν πλευρὰν μιᾷ πλευρᾷ >ῖσην τὴν ὑποτείνουσιν ὑπὸ μίαν τῶν >ῖσων γωνιῶν κοινὴν αὐτῶν τὴν BΔ; καὶ τὰς λοιπὰς >άρα πλευρὰς ταῖς λοιπαῖς πλευραῖς >ῖσαζ <έχουσιν; >ῖση >άρα ἡ ΔE τῇ ΔZ. διὰ τὰ αὐτὰ δὴ καὶ ἡ ΔH τῇ ΔZ ἐστὶν >ῖση. αἱ τρεῖς >άρα εὐθεῖαι αἱ ΔE, ΔZ, ΔH >ῖσαι ἀλλήλαις εἰσίν; ὁ >άρα κέντρω τῷ Δ καὶ διαστήματι ἐνὶ τῶν E, Z, H κύκλος γραφόμενος <ήχει καὶ διὰ τῶν λοιπῶν σημείων καὶ ἐφάπεται τῶν AB, BΓ, ΓA εὐθειῶν διὰ τὸ ὀρθὰς ε>ῖναι τὰς πρὸς τοῖς E, Z, H σημείοις γωνίας. εἰ γὰρ τεμεῖ αὐτάς, >έσται ἡ τῇ διαμέτρῳ τοῦ κύκλου πρὸς ὀρθὰς ἀπ> >άκραζ ἀγομένη ἐντὸς πίπτουσα τοῦ κύκλου; <όπερ >άτοπον ἐδείχθη; οὐκ >άρα ὁ κέντρω τῷ Δ διαστήματι δὲ ἐνὶ τῶν E, Z, H γραφόμενος κύκλος τεμεῖ τὰς AB, BΓ, ΓA εὐθείας; ἐφάπνεται >άρα αὐτῶν, καὶ >έσται ὁ κύκλος ἐγγεγραμμένος εἰς τὸ ABΓ τρίγωνον. ἐγγεγράφθω ὥς ὁ ZHE.

Εἰς >άρα τὸ δοθὲν τρίγωνον τὸ ABΓ κύκλος ἐγγέγραπται ὁ EZH; <όπερ >έδει ποιῆσαι.

angle DEF . And it has been drawn around circle ABC . Thus, a triangle, equiangular with the given triangle, has been circumscribed about the given circle. (Which is) the very thing it was required to do.

Proposition 4

To inscribe a circle in a given triangle.



Let ABC be the given triangle. So it is required to inscribe a circle in triangle ABC .

Let the angles ABC and ACB be cut in half by the straight-lines BD and CD (respectively) [Prop. 1.9], and let them meet one another at point D , and let DE , DF , and DG be drawn from point D , perpendicular to the straight-lines AB , BC , and CA (respectively) [Prop. 1.12].

And since angle ABD is equal to CBD , and the right-angle BED is also equal to the right-angle BFD , EBD and FBD are thus two triangles having two angles equal to two angles, and one side equal to one side—the (one) subtending one of the equal angles (which is) common to the (triangles)—(namely), BD . Thus, they will also have the remaining sides equal to the (corresponding) remaining sides [Prop. 1.26]. Thus, DE (is) equal to DF . So, for the same (reasons), DG is also equal to DF . Thus, the three straight-lines DE , DF , and DG are equal to one another. Thus, the circle drawn with center D , and radius one of E , F , or G ,[†] will also go through the remaining points, and will touch the straight-lines AB , BC , and CA , on account of the angles at E , F , and G being right-angles. For if it cuts (one of) them then it will be a (straight-line) drawn at right-angles to a diameter of the circle, from its extremity, falling inside the circle. The very thing was shown (to be) absurd [Prop. 3.16]. Thus, the circle drawn with center D , and radius one of E , F , or G , does not cut the straight-lines AB , BC , and CA . Thus, it will touch them, and will be the circle inscribed in triangle ABC . Let it be (so) inscribed, like FGE (in the figure).

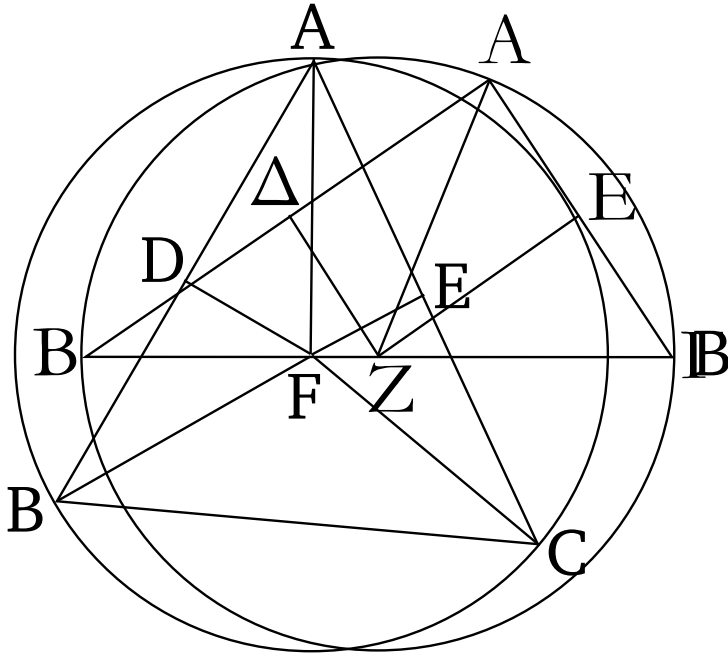
Thus, the circle EFG has been inscribed in the given triangle ABC . (Which is) the very thing it was required to

do.

† Here, and in the following propositions, it is understood that the radius is actually one of DE , DF , or DG .

5.

Περὶ τὸ δοθὲν τρίγωνον κύκλον περιγράφαι.



>Ἐστω τὸ δοθὲν τρίγωνον τὸ $AB\Gamma$; δεῖ δὲ περὶ τὸ δοθὲν τρίγωνον τὸ $AB\Gamma$ κύκλον περιγράφαι.

Τετμήσθωσαν αἱ AB , $A\Gamma$ εὐθεῖαι δίθῃ κατὰ τὰ Δ , E σημεία, καὶ ἀπὸ τῶν Δ , E σημείων ταῖς AB , $A\Gamma$ πρὸς ὀρθὰς >ήθωσαν αἱ ΔZ , EZ ; συμπεσοῦνται δὴ >ήτοι ἐντὸς τοῦ $AB\Gamma$ τριγώνου >ή ἐπὶ τῇς $B\Gamma$ εὐθείας >ή ἐκτὸς τῇς $B\Gamma$.

Συμπιπτέωσαν πρότερον ἐντὸς κατὰ τὸ Z , καὶ ἐπεζεύθωσαν αἱ ZB , $Z\Gamma$, ZA . καὶ ἐπεὶ >ῖση ἐστὶν ἡ $A\Delta$ τῇ ΔB , κοινὴ δὲ καὶ πρὸς ὀρθὰς ἡ ΔZ , βάσις >άρα ἡ AZ βάσει τῇ ZB ἐστὶν >ῖση. ὁμοίως δὴ δείχομεν, <ὅτι καὶ ἡ ΓZ τῇ AZ ἐστὶν >ῖση; <ώστε καὶ ἡ ZB τῇ $Z\Gamma$ ἐστὶν >ῖση; αἱ τρεῖς >άρα αἱ ZA , ZB , $Z\Gamma$ >ῖσαι ἀλλήλαις εἰσὶν. ὁ >άρα κέντρον τῶν Z διαστήματι δὲ ἐνὶ τῶν A , B , Γ κύκλος γραφόμενος <ήχει καὶ διὰ τῶν λοιπῶν σημείων, καὶ >έσται περιγεγραμμένος ὁ κύκλος περὶ τὸ $AB\Gamma$ τρίγωνον. περιγεγράφθω ὡς ὁ $AB\Gamma$.

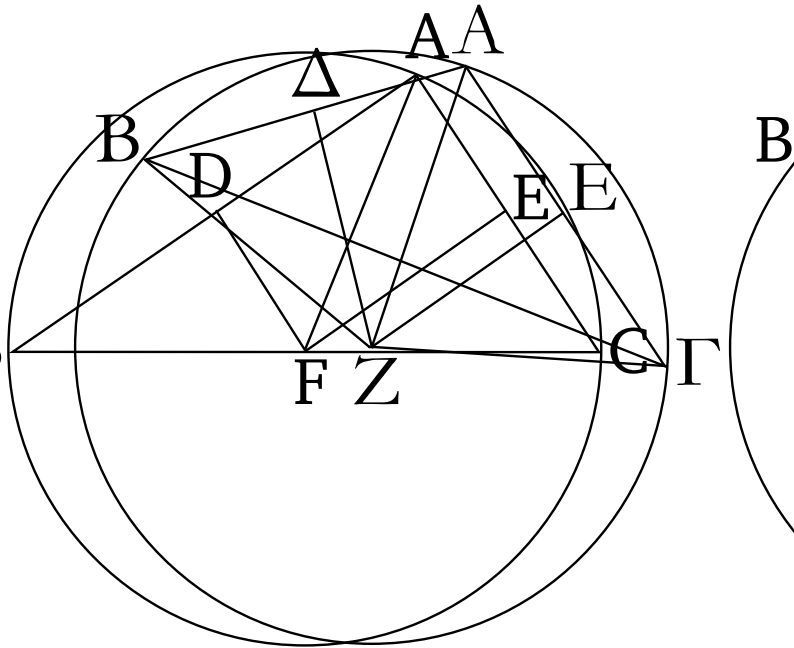
Ἀλλὰ δὴ αἱ ΔZ , EZ συμπιπτέωσαν ἐπὶ τῇς $B\Gamma$ εὐθείας κατὰ τὸ Z , ὡς >έθει ἐπὶ τῇς δευτέρας καταγραφῆς, καὶ ἐπεζεύθω ἡ AZ . ὁμοίως δὴ δείχομεν, <ὅτι τὸ Z σημεῖον κέντρον ἐστὶ τοῦ περὶ τὸ $AB\Gamma$ τρίγωνον περιγεγραμμένου κύκλου.

Ἀλλὰ δὴ αἱ ΔZ , EZ συμπιπτέωσαν ἐκτὸς τοῦ $AB\Gamma$ τριγώνου κατὰ τὸ Z πάλιν, ὡς >έθει ἐπὶ τῇς τρίτης καταγραφῆς, καὶ ἐπεζεύθωσαν αἱ AZ , BZ , ΓZ . καὶ ἐπεὶ πάλιν >ῖση ἐστὶν ἡ $A\Delta$ τῇ ΔB , κοινὴ δὲ καὶ πρὸς ὀρθὰς ἡ ΔZ , βάσις >άρα ἡ AZ βάσει τῇ BZ ἐστὶν >ῖση. ὁμοίως δὴ δείχομεν, <ὅτι καὶ ἡ ΓZ τῇ AZ ἐστὶν >ῖση; <ώστε καὶ ἡ BZ τῇ $Z\Gamma$ ἐστὶν >ῖση; ὁ >άρα [πάλιν] κέντρον τῶν Z διαστήματι δὲ ἐνὶ τῶν ZA , ZB , $Z\Gamma$ κύκλος γραφόμενος <ήχει καὶ διὰ τῶν λοιπῶν σημείων, καὶ >έσται περιγεγραμμένος περὶ τὸ $AB\Gamma$ τρίγωνον.

Περὶ τὸ δοθὲν >άρα τρίγωνον κύκλος περιγράφεται; <όπερ >έδει ποιῆσαι.

Proposition 5

To circumscribe a circle about a given triangle.



Let ABC be the given triangle. So it is required to circumscribe a circle about the given triangle ABC .

Let the straight-lines AB and AC be cut in half at points D and E (respectively) [Prop. 1.10]. And let DF and EF be drawn from points D and E , at right-angles to AB and AC (respectively) [Prop. 1.11]. So (DF and EF) will surely either meet inside triangle ABC , on the straight-line BC , or beyond BC .

Let them, first of all, meet inside (triangle ABC) at (point) F , and let FB , FC , and FA be joined. And since AD is equal to DB , and DF is common and at right-angles, the base AF is thus equal to the base FB [Prop. 1.4]. So, similarly, we can show that CF is also equal to AF . So that FB is also equal to FC . Thus, the three (straight-lines) FA , FB , and FC are equal to one another. Thus, the circle drawn with center F , and radius one of A , B , or C , will also go through the remaining points. And the circle will be circumscribed about triangle ABC . Let it be (so) circumscribed, like ABC (in the first diagram from the left).

And so, let DF and EF meet on the straight-line BC at (point) F , like in the second diagram (from the left). And let AF be joined. So, similarly, we can show that point F is the center of the circle circumscribed about triangle ABC . And so, let DF and EF meet outside triangle ABC , again at (point) F , like in the third diagram (from the left). And let AF , BF , and CF be joined. And, again, since AD is equal to DB , and DF is common and at right-angles, the base AF is thus equal to the base BF [Prop. 1.4]. So, similarly, we can show that CF is also equal to AF . So that BF is also equal to FC . Thus, [again] the circle drawn with center F , and radius one of FA , FB , and FC , will also go through the remaining points. And it will be circumscribed about triangle ABC .

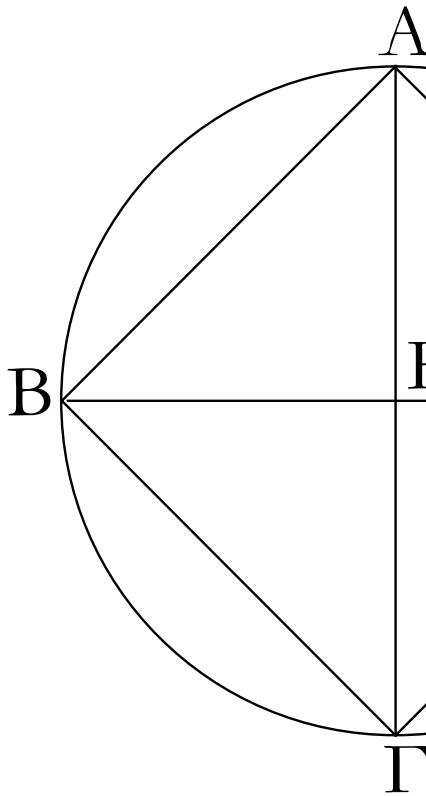
Thus, a circle has been circumscribed about the given trian-

6.

Εἰς τὸν δοθέντα κύκλον τετράγωνον ἐγγράψαι.

>Ἐστω ἡ δοθεὶς κύκλος ὁ ΑΒΓΔ; δεῖ δὴ εἰς τὸν ΑΒΓΔ κύκλον τετράγωνον ἐγγράψαι.

>Ἡθθωσαν τοῦ ΑΒΓΔ κύκλου δύο διάμετροι πρὸς ὀρθὰς ἀλλήλαις αἱ ΑΓ, ΒΔ, καὶ ἐπεζεύθθωσαν αἱ ΑΒ, ΒΓ, ΓΔ, ΔΑ.



Καὶ ἐπεὶ >ῖση ἐστὶν ἡ ΒΕ τῇ ΕΔ; κέντρον γὰρ τὸ Ε; κοινὴ δὲ καὶ πρὸς ὀρθὰς ἡ ΕΑ, βάσει >άρα ἡ ΑΒ βάσει τῇ ΑΔ >ῖση ἐστὶν. διὰ τὰ αὐτὰ δὴ καὶ ἑκατέρω τῶν ΒΓ, ΓΔ ἑκατέρω τῶν ΑΒ, ΑΔ >ῖση ἐστὶν; ἰσόπλευρον >άρα ἐστὶ τὸ ΑΒΓΔ τετράπλευρον. λέγω δὴ, <ὅτι καὶ ὀρθογώνιον. ἐπεὶ γὰρ ἡ ΒΔ εὐθεῖα διάμετρος ἐστὶ τοῦ ΑΒΓΔ κύκλου, ἡμικύκλιον >άρα ἐστὶ τὸ ΒΑΔ; ὀρθὴ >άρα ἡ ὑπὸ ΒΑΔ γωνία. διὰ τὰ αὐτὰ δὴ καὶ ἑκάστη τῶν ὑπὸ ΑΒΓ, ΒΓΔ, ΓΔΑ ὀρθὴ ἐστὶν; ὀρθογώνιον >άρα ἐστὶ τὸ ΑΒΓΔ τετράπλευρον. ἐδείθθη δὲ καὶ ἰσόπλευρον; τετράγωνον >άρα ἐστὶν. καὶ ἐγγέγραπται εἰς τὸν ΑΒΓΔ κύκλον.

Εἰς >άρα τὸν δοθέντα κύκλον τετράγωνον ἐγγέγραπται τὸ ΑΒΓΔ; <ὅπερ >έδει ποιῆσαι.

7.

Περὶ τὸν δοθέντα κύκλον τετράγωνον περιγράψαι.

>Ἐστω ὁ δοθεὶς κύκλος ὁ ΑΒΓΔ; δεῖ δὴ περὶ τὸν ΑΒΓΔ κύκλον τετράγωνον περιγράψαι.

>Ἡθθωσαν τοῦ ΑΒΓΔ κύκλου δύο διάμετροι πρὸς ὀρθὰς ἀλλήλαις αἱ ΑΓ, ΒΔ, καὶ διὰ τῶν Α, Β, Γ, Δ σημείων >ῆθθωσαν ἐφαπτόμεναι τοῦ ΑΒΓΔ κύκλου αἱ ΖΗ, ΗΘ, ΘΚ,

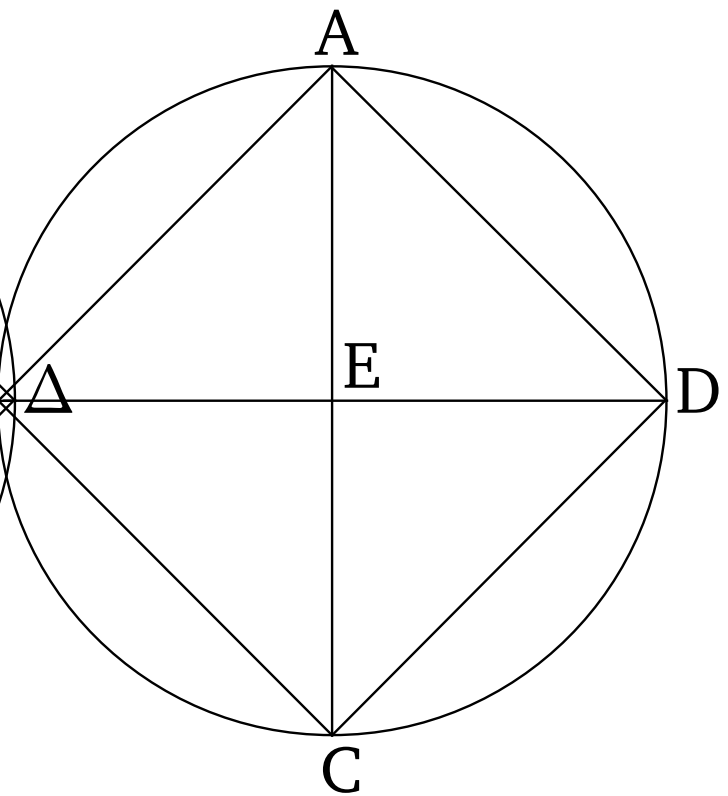
gle. (Which is) the very thing it was required to do.

Proposition 6

To inscribe a square in a given circle.

Let $ABCD$ be the given circle. So it is required to inscribe a square in circle $ABCD$.

Let two diameters of circle $ABCD$, AC and BD , be drawn at right-angles to one another.[†] And let AB , BC , CD , and DA be joined.



And since BE is equal to ED , for E (is) the center (of the circle), and EA is common and at right-angles, the base AB is thus equal to the base AD [Prop. 1.4]. So, for the same (reasons), each of BC and CD is equal to each of AB and AD . Thus, the quadrilateral $ABCD$ is equilateral. So I say that (it is) also right-angled. For since the straight-line BD is a diameter of circle $ABCD$, BAD is thus a semi-circle. Thus, angle BAD (is) a right-angle [Prop. 3.31]. So, for the same (reasons), (angles) ABC , BCD , and CDA are also each right-angles. Thus, the quadrilateral $ABCD$ is right-angled. And it was also shown (to be) equilateral. Thus, it is a square [Def. 1.22]. And it has been inscribed in circle $ABCD$.

Thus, the square $ABCD$ has been inscribed in the given circle. (Which is) the very thing it was required to do.

[†] Presumably, by finding the center of the circle [Prop. 3.1], drawing a line through it, and then drawing a second line through it, at right-angles to the first [Prop. 1.11].

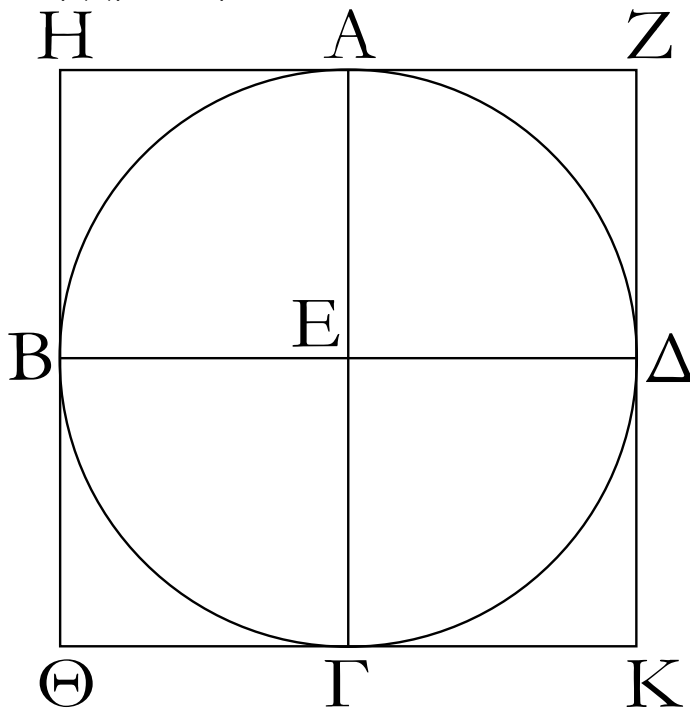
Proposition 7

To circumscribe a square about a given circle.

Let $ABCD$ be the given circle. So it is required to circumscribe a square about circle $ABCD$. Let two diameters of circle $ABCD$, AC and BD , be drawn at right-angles to one another.[†] And let FG , GH , HK , and KF be drawn through points A , B , C , and D (respectively), touching cir-

KZ.

Ἐπεὶ οὖν ἐφάπτεται ἡ ΖΗ τοῦ ΑΒΓΔ κύκλου, ἀπὸ δὲ τοῦ Ε κέντρου ἐπὶ τὴν κατὰ τὸ Α ἐπαφὴν ἐπέξενεται ἡ ΕΑ, αἱ >άρα πρὸς τῷ Α γωνίαι ὀρθαί εἰσιν. διὰ τὰ αὐτὰ δὴ καὶ αἱ πρὸς τοῖς Β, Γ, Δ σημείοις γωνίαι ὀρθαί εἰσιν. καὶ ἐπεὶ ὀρθή ἐστὶν ἡ ὑπὸ ΑΕΒ γωνία, ἐστὶ δὲ ὀρθή καὶ ἡ ὑπὸ ΕΒΗ, παράλληλος >άρα ἐστὶν ἡ ΗΘ τῇ ΑΓ. διὰ τὰ αὐτὰ δὴ καὶ ἡ ΑΓ τῇ ΖΚ ἐστὶ παράλληλος. <ώστε καὶ ἡ ΗΘ τῇ ΖΚ ἐστὶ παράλληλος. ὁμοίως δὴ δείχομεν, <ὅτι καὶ ἑκάτερα τῶν ΗΖ, ΘΚ τῇ ΒΕΔ ἐστὶ παράλληλος. παραλληλόγραμμα >άρα ἐστὶ τὰ ΗΚ, ΗΓ, ΑΚ, ΖΒ, ΒΚ; >ίση >άρα ἐστὶν ἡ μὲν ΗΖ τῇ ΘΚ, ἡ δὲ ΗΘ τῇ ΖΚ. καὶ ἐπεὶ >ίση ἐστὶν ἡ ΑΓ τῇ ΒΔ, ἀλλὰ καὶ ἡ μὲν ΑΓ ἑκάτερα τῶν ΗΘ, ΖΚ, ἡ δὲ ΒΔ ἑκάτερα τῶν ΗΖ, ΘΚ ἐστὶν >ίση [καὶ ἑκάτερα >άρα τῶν ΗΘ, ΖΚ ἑκάτερα τῶν ΗΖ, ΘΚ ἐστὶν >ίση], ἰσόπλευρον >άρα ἐστὶ τὸ ΖΗΘΚ τετράπλευρον. λέγω δὴ, <ὅτι καὶ ὀρθογώνιον. ἐπεὶ γὰρ παραλληλόγραμμόν ἐστὶ τὸ ΗΒΕΑ, καὶ ἐστὶν ὀρθή ἡ ὑπὸ ΑΕΒ, ὀρθή >άρα καὶ ἡ ὑπὸ ΑΗΒ. ὁμοίως δὴ δείχομεν, <ὅτι καὶ αἱ πρὸς τοῖς Θ, Κ, Ζ γωνίαι ὀρθαί εἰσιν. ὀρθογώνιον >άρα ἐστὶ τὸ ΖΗΘΚ. ἐδείκθη δὲ καὶ ἰσόπλευρον; τετράγωνον >άρα ἐστίν. καὶ περιέγραπται περὶ τὸν ΑΒΓΔ κύκλον.

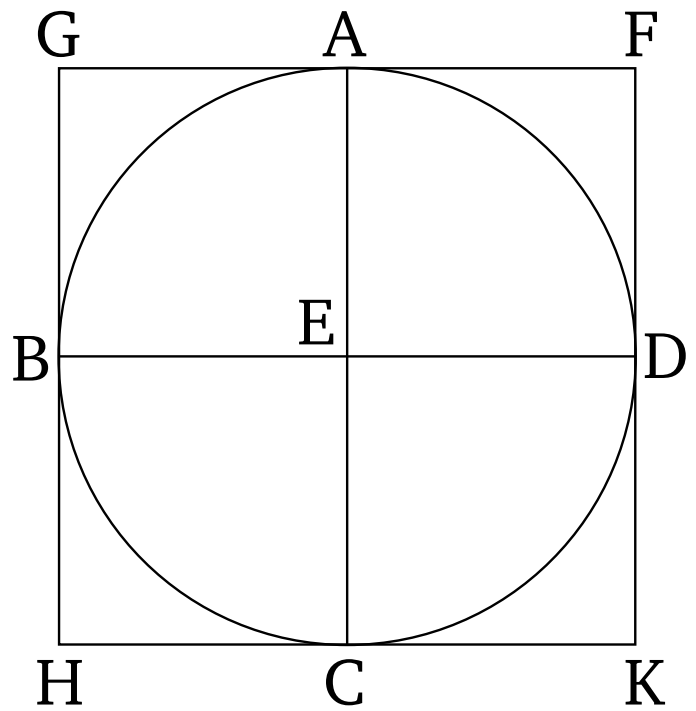


Περὶ τὸν δοθέντα >άρα κύκλον τετράγωνον περιέγραπται;

cle $ABCD$.[‡]

Therefore, since FG touches circle $ABCD$, and EA has been joined from the center E to the point of contact A , the angles at A are thus right-angles [Prop. 3.18]. So, for the same (reasons), the angles at points B , C , and D are also right-angles. And since angle AEB is a right-angle, and EBG is also a right-angle, GH is thus parallel to AC [Prop. 1.29]. So, for the same (reasons), AC is also parallel to FK . So that GH is also parallel to FK [Prop. 1.30]. So, similarly, we can show that GF and HK are each parallel to BED . Thus, GK , GC , AK , FB , and BK are (all) parallelograms. Thus, GF is equal to HK , and GH to FK [Prop. 1.34]. And since AC is equal to BD , but AC (is) also (equal) to each of GH and FK , and BD is equal to each of GF and HK [Prop. 1.34] [and each of GH and FK is thus equal to each of GF and HK], the quadrilateral $FGHK$ is thus equilateral. So I say that (it is) also right-angled. For since $GBEA$ is a parallelogram, and AEB is a right-angle, AGB is thus also a right-angle [Prop. 1.34]. So, similarly, we can show that the angles at H , K , and F are also right-angles. Thus, $FGHK$ is right-angled. And it was also

shown (to be) equilateral. Thus, it is a square [Def. 1.22]. And it has been circumscribed about circle $ABCD$.



<όπερ >έδει ποιῆσαι.

Thus, a square has been circumscribed about the given circle. (Which is) the very thing it was required to do.

† See the footnote to the previous proposition.

‡ See the footnote to Prop. 3.34.

8.

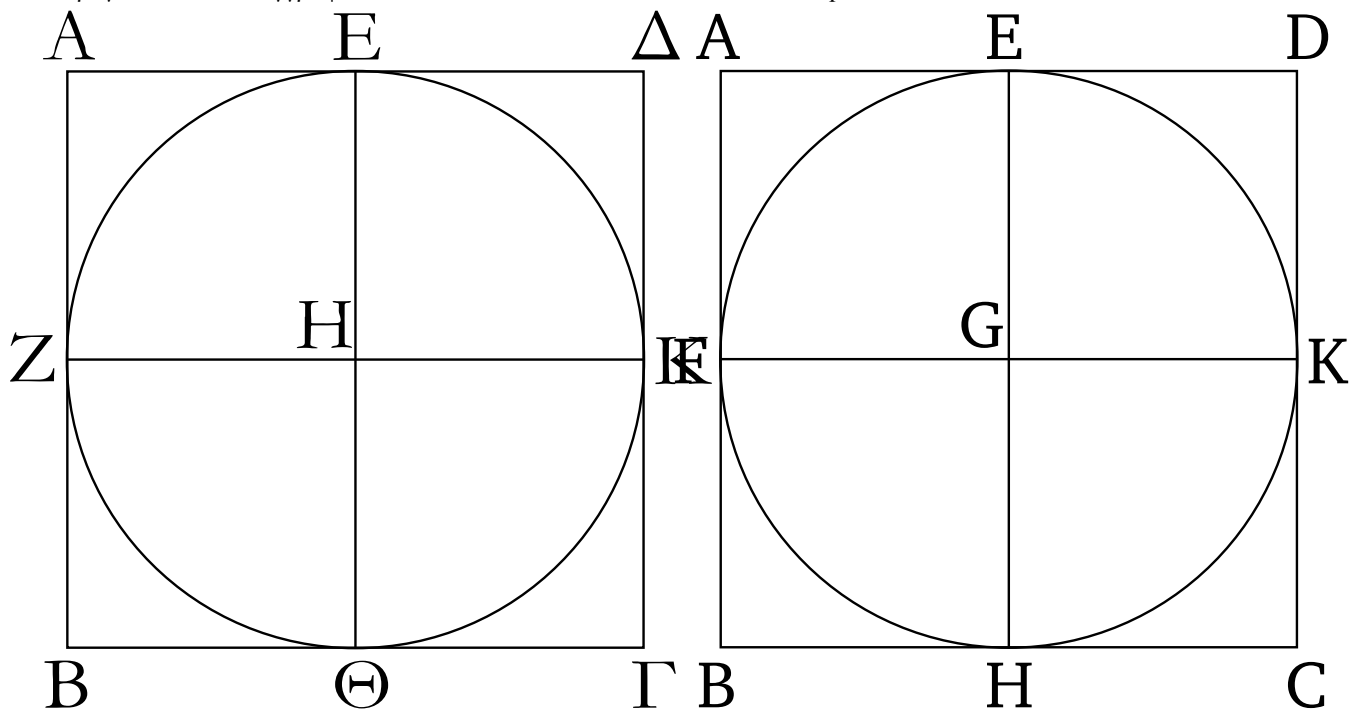
Proposition 8

Εἰς τὸ δοθὲν τετράγωνον κύκλον ἐγγράψαι.

>Ἐστω τὸ δοθὲν τετράγωνον τὸ ΑΒΓΔ; δεῖ δὴ εἰς τὸ ΑΒΓΔ τετράγωνον κύκλον ἐγγράψαι.

To inscribe a circle in a given square.

Let the given square be $ABCD$. So it is required to inscribe a circle in square $ABCD$.



Τεμήσθω ἑκατέρα τῶν ΑΔ, ΑΒ δίθα κατὰ τὰ Ε, Ζ σημεῖα, καὶ διὰ μὲν τοῦ Ε ὁποτέρᾳ τῶν ΑΒ, ΓΔ παράλληλοξ >ήθθω ὁ ΕΘ, διὰ δὲ τοῦ Ζ ὁποτέρᾳ τῶν ΑΔ, ΒΓ παράλληλοξ >ήθθω ἡ ΖΚ; παραλληλόγραμμον >ἀρα ἐστὶν <έκαστον τῶν ΑΚ, ΚΒ, ΑΘ, ΘΔ, ΑΗ, ΗΓ, ΒΗ, ΗΔ, καὶ αἱ ἀπεναντίον αὐτῶν πλευραὶ δηλονότι >ίσαι [εἰσὶν]. καὶ ἐπεὶ >ίση ἐστὶν ἡ ΑΔ τῇ

Let AD and AB each be cut in half at points E and F (respectively) [Prop. 1.10]. And let EH be drawn through E , parallel to either of AB or CD , and let FK be drawn through F , parallel to either of AD or BC [Prop. 1.31]. Thus, AK , KB , AH , HD , AG , GC , BG , and GD are each parallelograms, and their opposite sides [are] mani-

AB, καὶ ἐστὶ τῆς μὲν AD ἡμίσεια ἢ AE, τῆς δὲ AB ἡμίσεια ἢ AZ, >ίση >άρα καὶ ἡ AE τῇ AZ; <ώστε καὶ αἱ ἀπεναντίον; >ίση >άρα καὶ ἡ ZH τῇ HE. ὁμοίως δὲ δείχομεν, <ὅτι καὶ ἑκάτερα τῶν HΘ, HK ἑκάτερα τῶν ZH, HE ἐστὶν >ίση; αἱ τέσσαρες >άρα αἱ HE, HZ, HΘ, HK >ίσαι ἀλλήλαις [εἰσὶν]. ὁ >άρα κέντρω μὲν τῷ H διαστήματι δὲ ἐνὶ τῶν E, Z, Θ, K κύκλος γραφόμενος <ήχει καὶ διὰ τῶν λοιπῶν σημείων; καὶ ἐφάπεται τῶν AB, BG, ΓΔ, ΔΑ εὐθειῶν διὰ τὸ ὀρθὰς εἶναι τὰς πρὸς τοῖς E, Z, Θ, K γωνίας; εἰ γὰρ τεμεῖ ὁ κύκλος τὰς AB, BG, ΓΔ, ΔΑ, ἢ τῇ διαμέτρῳ τοῦ κύκλου πρὸς ὀρθὰς ἀπ> >άκρας ἀγομένη ἐντὸς πεσεῖται τοῦ κύκλου; <όπερ >άτοπον ἐδείκθη. οὐκ >άρα ὁ κέντρω τῷ H διαστήματι δὲ ἐνὶ τῶν E, Z, Θ, K κύκλος γραφόμενος τεμεῖ τὰς AB, BG, ΓΔ, ΔΑ εὐθείας. ἐφάπεται >άρα αὐτῶν καὶ >έσται ἐγγεγραμμένος εἰς τὸ ABΓΔ τετράγωνον. Εἰς >άρα τὸ δοθὲν τετράγωνον κύκλος ἐγγέγραπται; <όπερ >έδει ποιῆσαι.

festly equal [Prop. 1.34]. And since AD is equal to AB , and AE is half of AD , and AF half of AB , AE (is) thus also equal to AF . So that the opposite (sides are) also (equal). Thus, FG (is) also equal to GE . So, similarly, we can also show that each of GH and GK is equal to each of FG and GE . Thus, the four (straight-lines) GE , GF , GH , and GK [are] equal to one another. Thus, the circle drawn with center G , and radius one of E , F , H , or K , will also go through the remaining points. And it will touch the straight-lines AB , BC , CD , and DA , on account of the angles at E , F , H , and K being right-angles. For if the circle cuts AB , BC , CD , or DA , (then) a (straight-line) drawn at right-angles to a diameter of the circle, from its extremity, will fall inside the circle. The very thing was shown (to be) absurd [Prop. 3.16]. Thus, the circle drawn with center G , and radius one of E , F , H , or K , does not cut the straight-lines AB , BC , CD , or DA . Thus, it will touch them, and will be inscribed in the square $ABCD$.

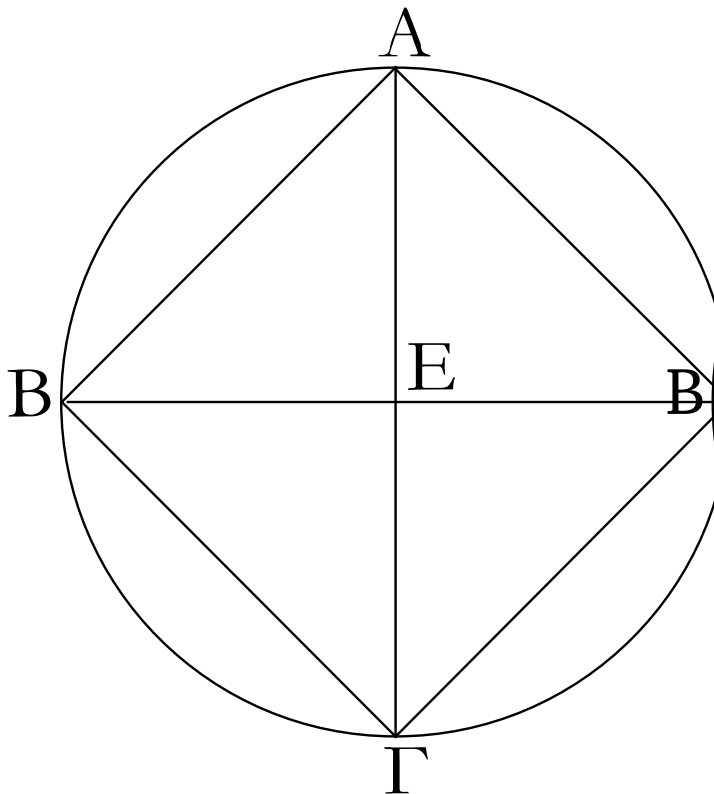
Thus, a circle has been inscribed in the given square. (Which is) the very thing it was required to do.

9.

Περὶ τὸ δοθὲν τετράγωνον κύκλον περιγράψαι.

>Ἐστω τὸ δοθὲν τετράγωνον τὸ ABΓΔ; δεῖ δὲ περὶ τὸ ABΓΔ τετράγωνον κύκλον περιγράψαι.

Ἐπιζευθεῖσαι γὰρ αἱ ΑΓ, ΒΔ τεμνέτωσαν ἀλλήλας κατὰ τὸ E.



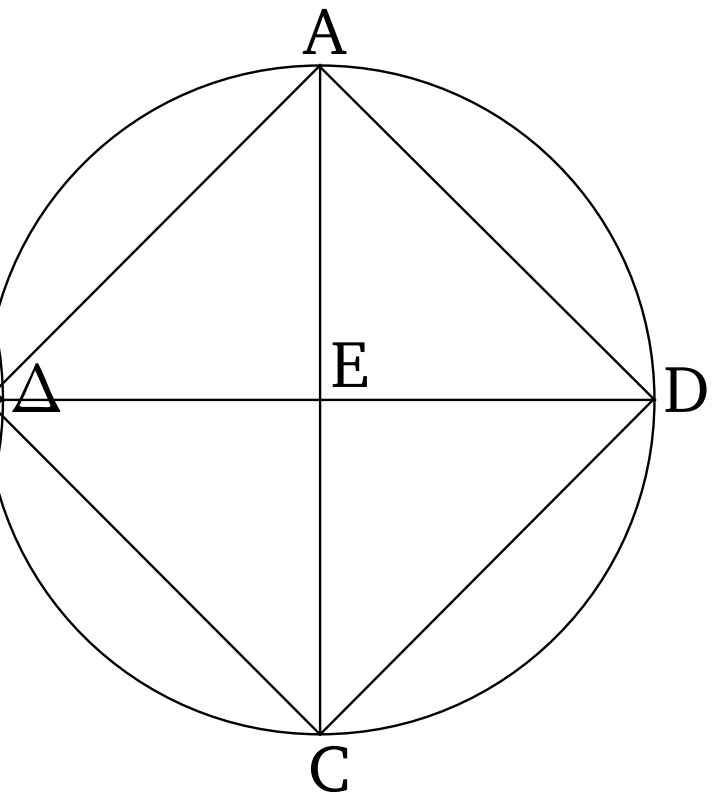
Καὶ ἐπεὶ >ίση ἐστὶν ἡ ΔΑ τῇ AB, κοινὴ δὲ ἡ ΑΓ, δύο δὲ αἱ ΔΑ, ΑΓ δυσὶ ταῖς ΒΑ, ΑΓ >ίσαι εἰσὶν; καὶ βάσει τῇ ΒΓ βάσει τῇ ΒΓ >ίση; γωνία >άρα ἡ ὑπὸ ΔΑΓ γωνία τῇ ὑπὸ ΒΑΓ >ίση ἐστίν; ἡ >άρα ὑπὸ ΔΑΒ γωνία δίθρα τέτμηται ὑπὸ τῆς ΑΓ. ὁμοίως δὲ δείχομεν, <ὅτι καὶ ἑκάστη τῶν ὑπὸ ΑΒΓ, ΒΓΔ, ΓΔΑ δίθρα τέτμηται ὑπὸ τῶν ΑΓ, ΒΔ εὐθειῶν. καὶ ἐπεὶ >ίση ἐστὶν ἡ ὑπὸ ΔΑΒ γωνία τῇ ὑπὸ ΑΒΓ, καὶ ἐστὶ τῆς μὲν ὑπὸ ΔΑΒ ἡμίσεια ἢ ὑπὸ ΕΑΒ, τῆς δὲ ὑπὸ ΑΒΓ ἡμίσεια ἢ

Proposition 9

To circumscribe a circle about a given square.

Let $ABCD$ be the given square. So it is required to circumscribe a circle about square $ABCD$.

AC and BD being joined, let them cut one another at E .



And since DA is equal to AB , and AC (is) common, the two (straight-lines) DA , AC are thus equal to the two (straight-lines) BA , AC . And the base DC (is) equal to the base BC . Thus, angle DAC is equal to angle BAC [Prop. 1.8]. Thus, the angle DAB has been cut in half by AC . So, similarly, we can show that ABC , BCD , and CDA have each been cut in half by the straight-lines AC and DB . And since angle DAB is equal to ABC , and EAB is half of DAB ,

ὑπὸ EBA , καὶ ἡ ὑπὸ $EAB >$ ἄρα τῇ ὑπὸ EBA ἐστὶν $>$ ἴση; $<$ ὥστε καὶ πλευρὰ ἡ EA τῇ EB ἐστὶν $>$ ἴση. ὁμοίως δὲ δείχομεν, $<$ ὅτι καὶ ἑκατέρα τῶν EA, EB [εὐθειῶν] ἑκατέρα τῶν $EC, ED >$ ἴση ἐστὶν. αἱ τέσσαρες $>$ ἄρα αἱ $EA, EB, EC, ED >$ ἴσαι ἀλλήλαις εἰσὶν. ὁ $>$ ἄρα κέντρω τῷ E καὶ διαστήματι ἐνὶ τῶν A, B, C, D κύκλος γραφόμενος $<$ ἴχει καὶ διὰ τῶν λοιπῶν σημείων καὶ $>$ ἐστὶ περιγεγραμμένος περὶ τὸ $ABCD$ τετράγωνον. περιγεγράφθω ὡς ὁ $ABCD$.
Περὶ τὸ δοθὲν $>$ ἄρα τετράγωνον κύκλος περιγράφεται, $<$ ὅπερ $>$ ἐδει ποιῆσαι.

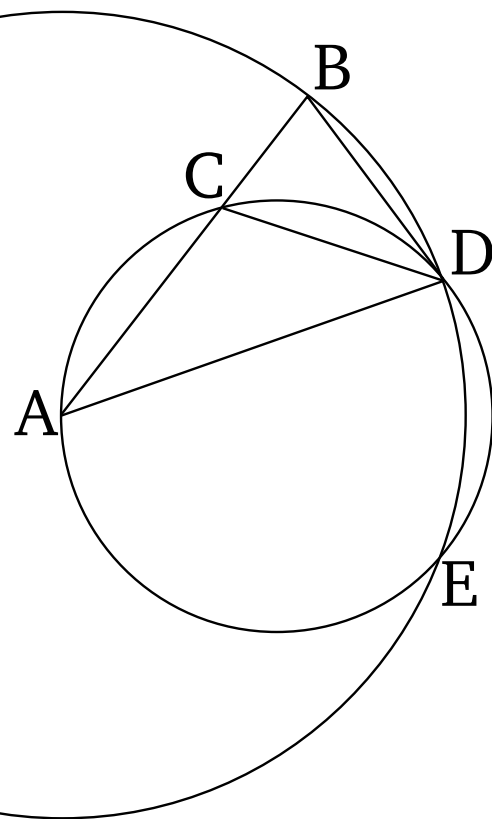
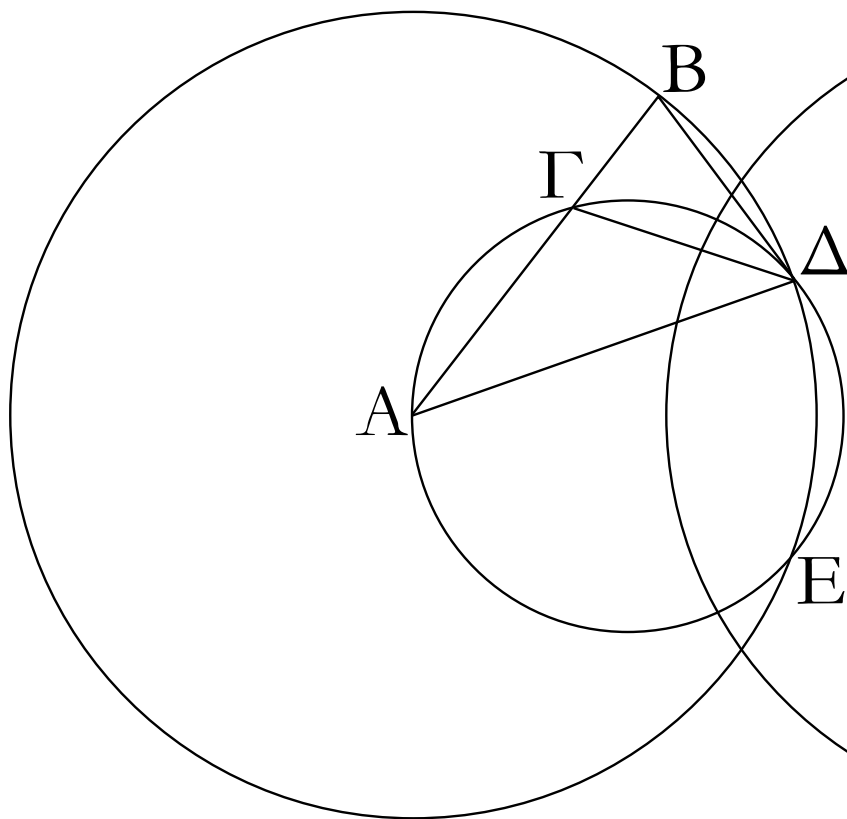
and EBA half of ABC , EAB is thus also equal to EBA . So that side EA is also equal to EB [Prop. 1.6]. So, similarly, we can show that each of the [straight-lines] EA and EB are also equal to each of EC and ED . Thus, the four (straight-lines) EA, EB, EC , and ED are equal to one another. Thus, the circle drawn with center E , and radius one of A, B, C , or D , will also go through the remaining points, and will be circumscribed about the square $ABCD$. Let it be (so) circumscribed, like $ABCD$ (in the figure).
Thus, a circle has been circumscribed about the given square. (Which is) the very thing it was required to do.

10.

Ἰσοσκελὲς τρίγωνον συστήσασθαι $>$ ἐθον ἑκατέραν τῶν πρὸς τῇ βάσει γωνιῶν διπλασίονα τῆς λοιπῆς.

Proposition 10

To construct an isosceles triangle having each of the angles at the base double the remaining (angle).



Ἐκκεῖσθω τις εὐθεῖα ἡ AB , καὶ τετμήσθω κατὰ τὸ Γ σημεῖον, $<$ ὥστε τὸ ὑπὸ τῶν AB, BG περιεχόμενον ὀρθογώνιον $>$ ἴσον εἶναι τῷ ἀπὸ τῆς GA τετραγώνῳ; καὶ κέντρω τῷ A καὶ διαστήματι τῷ AB κύκλος γεγράφθω ὁ BDE , καὶ ἐνηρμόσθω εἰς τὸν BDE κύκλον τῇ AG εὐθείᾳ μὴ μείζονι $>$ ἴση τῆς τοῦ BDE κύκλου διαμέτρου $>$ ἴση εὐθεῖα ἡ BD ; καὶ ἐπεζεύθθωσαν αἱ AD, DG , καὶ περιγεγράφθω περὶ τὸ AGD τρίγωνον κύκλος ὁ AGD .

Καὶ ἐπεὶ τὸ ὑπὸ τῶν $AB, BG >$ ἴσον ἐστὶ τῷ ἀπὸ τῆς AG , $>$ ἴση δὲ ἡ AG τῇ BD , τὸ $>$ ἄρα ὑπὸ τῶν $AB, BG >$ ἴσον ἐστὶ τῷ ἀπὸ τῆς BD . καὶ ἐπεὶ κύκλου τοῦ AGD $>$ ἴληπται τι σημεῖον ἐκτὸς τὸ B , καὶ ἀπὸ τοῦ B πρὸς τὸν AGD κύκλον προσπεπτώκασιν δύο εὐθεῖαι αἱ BA, BD , καὶ ἡ μὲν αὐτῶν τέμνει, ἡ δὲ προσπίπτει, καὶ ἐστὶ τὸ ὑπὸ τῶν $AB, BG >$ ἴσον τῷ ἀπὸ τῆς BD , ἡ $BD >$ ἄρα ἐφάπτεται τοῦ AGD κύκλου. ἐπεὶ $>$ ὅν ἐφάπτεται μὲν ἡ BD , ἀπὸ δὲ τῆς κατὰ τὸ D ἐπαφῆς διῆκται ἡ DG , ἡ $>$ ἄρα ὑπὸ BAG γωνία $>$ ἴση ἐστὶ τῇ ἐν τῷ ἐναλλάχ τοῦ κύκλου τμήματι γωνίᾳ τῇ ὑπὸ DAG . ἐπεὶ $>$ ὅν $>$ ἴση ἐστὶν ἡ ὑπὸ BAG τῇ ὑπὸ DAG , κοινὴ προσκεῖσθω ἡ

Let some straight-line AB be taken, and let it be cut at point C so that the rectangle contained by AB and BC is equal to the square on CA [Prop. 2.11]. And let the circle BDE be drawn with center A , and radius AB . And let the straight-line BD , equal to the straight-line AC , being not greater than the diameter of circle BDE , be inserted into circle BDE [Prop. 4.1]. And let AD and DC be joined. And let the circle ACD be circumscribed about triangle ACD [Prop. 4.5].

And since the (rectangle contained) by AB and BC is equal to the (square) on AC , and AC (is) equal to BD , the (rectangle contained) by AB and BC is thus equal to the (square) on BD . And since some point B has been taken outside of circle ACD , and two straight-lines BA and BD have radiated from B towards the circle ACD , and (one) of them cuts (the circle), and (the other) meets (the circle), and the (rectangle contained) by AB and BC is equal to the (square) on BD , BD thus touches circle ACD [Prop. 3.37]. Therefore, since BD touches (the circle), and DC has been drawn

ὑπὸ ΓΔΑ; <ὅλη >άρα ἢ ὑπὸ ΒΔΑ >ίση ἐστὶ δυσὶ ταῖς ὑπὸ ΓΔΑ, ΔΑΓ. ἀλλὰ ταῖς ὑπὸ ΓΔΑ, ΔΑΓ >ίση ἐστὶν ἢ ἐκτὸς ἢ ὑπὸ ΒΓΔ; καὶ ἢ ὑπὸ ΒΔΑ >άρα >ίση ἐστὶ τῇ ὑπὸ ΒΓΔ. ἀλλὰ ἢ ὑπὸ ΒΔΑ τῇ ὑπὸ ΓΒΔ ἐστὶν >ίση, ἐπεὶ καὶ πλευρὰ ἢ ΑΔ τῇ ΑΒ ἐστὶν >ίση; <ὥστε καὶ ἢ ὑπὸ ΔΒΑ τῇ ὑπὸ ΒΓΔ ἐστὶν >ίση. αἱ τρεῖς >άρα αἱ ὑπὸ ΒΔΑ, ΔΒΑ, ΒΓΑ >ίσαι ἀλλήλαις εἰσὶν. καὶ ἐπεὶ >ίση ἐστὶν ἢ ὑπὸ ΔΒΓ γωνία τῇ ὑπὸ ΒΓΔ, >ίση ἐστὶ καὶ πλευρὰ ἢ ΒΔ πλευρᾷ τῇ ΔΓ. ἀλλὰ ἢ ΒΔ τῇ ΓΑ ὑπόκειται >ίση; καὶ ἢ ΓΑ >άρα τῇ ΓΔ ἐστὶν >ίση; <ὥστε καὶ γωνία ἢ ὑπὸ ΓΔΑ γωνία τῇ ὑπὸ ΔΑΓ ἐστὶν >ίση; αἱ >άρα ὑπὸ ΓΔΑ, ΔΑΓ τῆς ὑπὸ ΔΑΓ εἰσι διπλασίονες. >ίση δὲ ἢ ὑπὸ ΒΓΔ ταῖς ὑπὸ ΓΔΑ, ΔΑΓ; καὶ ἢ ὑπὸ ΒΓΔ >άρα τῆς ὑπὸ ΓΑΔ ἐστὶ διπλῇ. >ίση δὲ ἢ ὑπὸ ΒΓΔ ἑκατέρᾳ τῶν ὑπὸ ΒΔΑ, ΔΒΑ; καὶ ἑκατέρα >άρα τῶν ὑπὸ ΒΔΑ, ΔΒΑ τῆς ὑπὸ ΔΑΒ ἐστὶ διπλῇ.

Ἰσοσκελεὲς >άρα τρίγωνον συνέσταται τὸ ΑΒΔ >έθον ἑκατέραν τῶν πρὸς τῇ ΔΒ βάσει γωνιῶν διπλασίονα τῆς λοιπῆς; <ὅπερ >έδει ποιῆσαι.

across (the circle) from the point of contact D , the angle BDC is thus equal to the angle DAC in the alternate segment of the circle [Prop. 3.32]. Therefore, since BDC is equal to DAC , let CDA be added to both. Thus, the whole of BDA is equal to the two (angles) CDA and DAC . But, the external (angle) BCD is equal to CDA and DAC [Prop. 1.32]. Thus, BDA is also equal to BCD . But, BDA is equal to CBD , since the side AD is also equal to AB [Prop. 1.5]. So that DBA is also equal to BCD . Thus, the three (angles) BDA , DBA , and BCD are equal to one another. And since angle DBC is equal to BCD , side BD is also equal to side DC [Prop. 1.6]. But, BD was assumed (to be) equal to CA . Thus, CA is also equal to CD . So that angle CDA is also equal to angle DAC [Prop. 1.5]. Thus, CDA and DAC is double DAC . But BCD (is) equal to CDA and DAC . Thus, BCD is also double CAD . And BCD (is) equal to each of BDA and DBA . Thus, BDA and DBA are each double DAB .

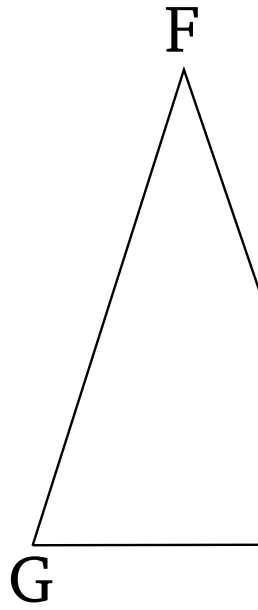
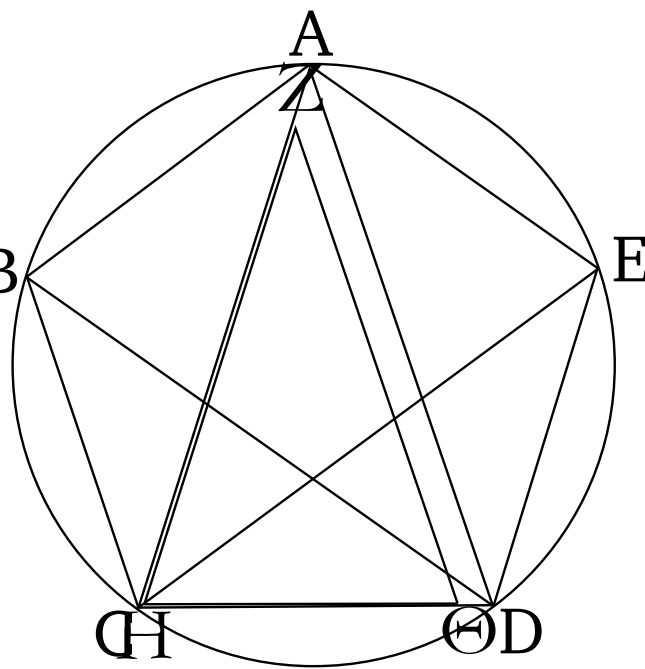
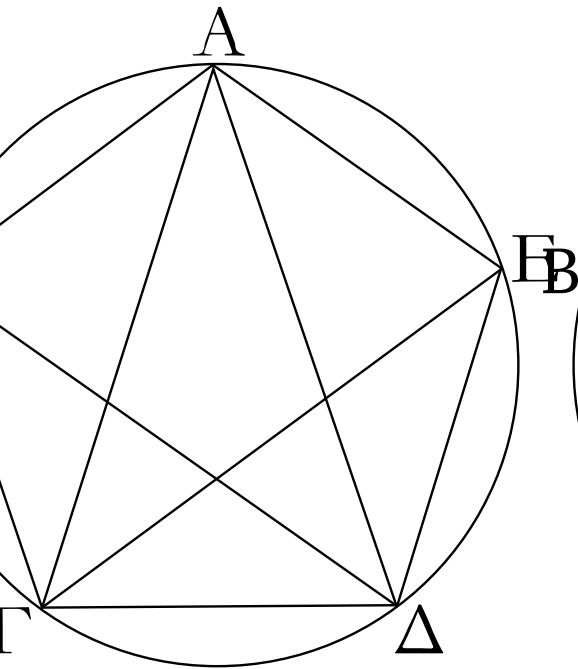
Thus, the isosceles triangle ABD has been constructed having each of the angles at the base BD double the remaining (angle). (Which is) the very thing it was required to do.

11.

Proposition 11

Εἰς τὸν δοθέντα κύκλον πεντάγωνον ἰσόπλευρόν τε καὶ ἰσογώνιον ἐγγράψαι.

To inscribe an equilateral and equiangular pentagon in a given circle.



>Ἐστω ὁ δοθεὶς κύκλος ὁ ΑΒΓΔΕ; δεῖ δὲ εἰς τὸν ΑΒΓΔΕ κύκλον πεντάγωνον ἰσόπλευρόν τε καὶ ἰσογώνιον ἐγγράψαι. Ἐκκείσθω τρίγωνον ἰσοσκελεὲς τὸ ΖΗΘ διπλασίονα >έθον ἑκατέραν τῶν πρὸς τοῖς Η, Θ γωνιῶν τῆς πρὸς τῷ Ζ, καὶ ἐγγεγράφθω εἰς τὸν ΑΒΓΔΕ κύκλον τῷ ΖΗΘ τρίγωνῳ ἰσογώνιον τρίγωνον τὸ ΑΓΔ, <ὥστε τῇ μὲν πρὸς τῷ Ζ γωνίᾳ >ίσην εἶναι τὴν ὑπὸ ΓΑΔ, ἑκατέραν δὲ τῶν πρὸς τοῖς Η, Θ >ίσην ἑκατέρα τῶν ὑπὸ ΑΓΔ, ΓΔΑ; καὶ ἑκατέρα >άρα τῶν ὑπὸ ΑΓΔ, ΓΔΑ τῆς ὑπὸ ΓΑΔ ἐστὶ διπλῇ. τεμήσθω δὲ ἑκατέρα τῶν ὑπὸ ΑΓΔ, ΓΔΑ δίθᾳ ὑπὸ ἑκατέρας τῶν ΓΕ, ΔΒ εὐθειῶν, καὶ ἐπεζεύθωσαν αἱ ΑΒ, ΒΓ, ΔΕ, ΕΑ.

Ἐπεὶ οὖν ἑκατέρα τῶν ὑπὸ ΑΓΔ, ΓΔΑ γωνιῶν διπλασίον ἐστὶ τῆς ὑπὸ ΓΑΔ, καὶ τετμημένα εἰσι δίθᾳ ὑπὸ τῶν ΓΕ, ΔΒ εὐθειῶν, αἱ πέντε >άρα γωνίαι αἱ ὑπὸ ΔΑΓ, ΑΓΕ, ΕΓΔ,

Let $ABCDE$ be the given circle. So it is required to inscribed an equilateral and equiangular pentagon in circle $ABCDE$.

Let the the isosceles triangle FGH be set up having each of the angles at G and H double the (angle) at F [Prop. 4.10]. And let triangle ACD , equiangular to FGH , be inscribed in circle $ABCDE$, such that CAD is equal to the angle at F , and the (angles) at G and H (are) equal to ACD and CDA , respectively [Prop. 4.2]. Thus, ACD and CDA are each double CAD . So let ACD and CDA be cut in half by the straight-lines CE and DB , respectively [Prop. 1.9]. And let AB , BC , DE and EA be joined.

Therefore, since angles ACD and CDA are each double CAD , and are cut in half by the straight-lines CE and DB ,

ΓΔΒ, ΒΔΑ >ίσαι ἀλλήλαις εἰσίν. αἱ δὲ >ίσαι γωνίαι ἐπὶ >ίσων περιφερειῶν βεβήκασιν; αἱ πέντε >άρα περιφέρειαι αἱ ΑΒ, ΒΓ, ΓΔ, ΔΕ, ΕΑ >ίσαι ἀλλήλαις εἰσίν. ὑπὸ δὲ τὰς >ίσαζ περιφερείας >ίσαι εὐθεῖαι ὑποτείνουσιν; αἱ πέντε >άρα εὐθεῖαι αἱ ΑΒ, ΒΓ, ΓΔ, ΔΕ, ΕΑ >ίσαι ἀλλήλαις εἰσίν; ἰσόπλευρον >άρα ἐστὶ τὸ ΑΒΓΔΕ πεντάγωνον. λέγω δὴ, <ὅτι καὶ ἰσογώνιον. ἐπεὶ γὰρ ἡ ΑΒ περιφέρεια τῇ ΔΕ περιφερείᾳ ἐστὶν >ίση, κοινὴ προσκείσθω ἡ ΒΓΔ; <ὅλη >άρα ἡ ΑΒΓΔ περιφέρεια <ὅλη τῇ ΕΔΓΒ περιφερείᾳ ἐστὶν >ίση. καὶ βέβηκεν ἐπὶ μὲν τῇ ΑΒΓΔ περιφερείᾳ γωνία ἡ ὑπὸ ΑΕΔ, ἐπὶ δὲ τῇ ΕΔΓΒ περιφερείᾳ γωνία ἡ ὑπὸ ΒΑΕ; καὶ ἡ ὑπὸ ΒΑΕ >άρα γωνία τῇ ὑπὸ ΑΕΔ ἐστὶν >ίση. διὰ τὰ αὐτὰ δὴ καὶ ἐκάστη τῶν ὑπὸ ΑΒΓ, ΒΓΔ, ΓΔΕ γωνιῶν ἐκατέρᾳ τῶν ὑπὸ ΒΑΕ, ΑΕΔ ἐστὶν >ίση; ἰσογώνιον >άρα ἐστὶ τὸ ΑΒΓΔΕ πεντάγωνον. ἐδείκθη δὲ καὶ ἰσόπλευρόν τε καὶ ἰσογώνιον ἐγγέγραπται; <ὅπερ >έδει ποιῆσαι.

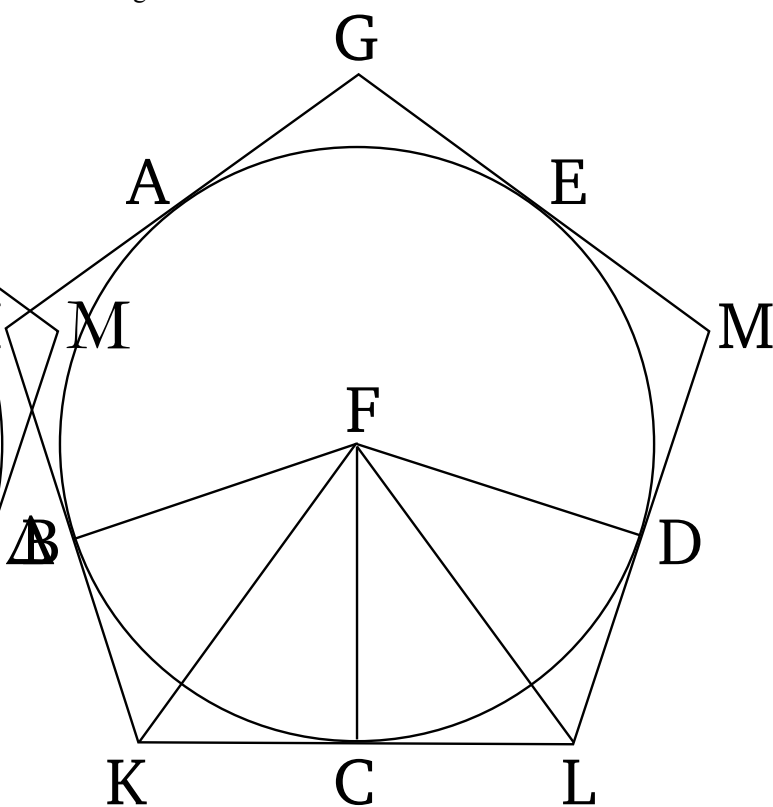
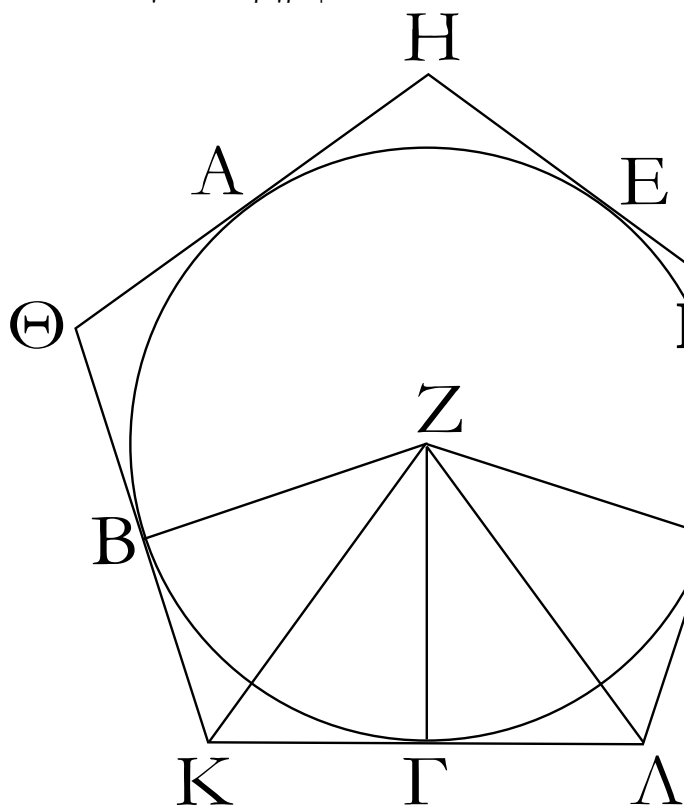
the five angles DAC , ACE , ECD , CDB , and BDA are thus equal to one another. And equal angles stand upon equal circumferences [Prop. 3.26]. Thus, the five circumferences AB , BC , CD , DE , and EA are equal to one another [Prop. 3.29]. Thus, the pentagon $ABCDE$ is equilateral. So I say that (it is) also equiangular. For since the circumference AB is equal to the circumference DE , let BCD be added to both. Thus, the whole circumference $ABCD$ is equal to the whole circumference $EDCB$. And the angle AED stands upon circumference $ABCD$, and angle BAE upon circumference $EDCB$. Thus, angle BAE is also equal to AED [Prop. 3.27]. So, for the same (reasons), each of the angles ABC , BCD , and CDE is also equal to each of BAE and AED . Thus, pentagon $ABCDE$ is equiangular. And it was also shown (to be) equilateral. Thus, an equilateral and equiangular pentagon has been inscribed in the given circle. (Which is) the very thing it was required to do.

12.

Proposition 12

Περὶ τὸν δοθέντα κύκλον πεντάγωνον ἰσόπλευρόν τε καὶ ἰσογώνιον περιγράψαι.

To circumscribe an equilateral and equiangular pentagon about a given circle.



>Ἐστω ὁ δοθεὶς κύκλος ὁ ΑΒΓΔΕ; δεῖ δὲ περὶ τὸν ΑΒΓΔΕ κύκλον πεντάγωνον ἰσόπλευρόν τε καὶ ἰσογώνιον περιγράψαι. Νενοήσθω τοῦ ἐγγεγραμμένου πενταγώνου τῶν γωνιῶν σημεῖα τὰ Α, Β, Γ, Δ, Ε, <ὥστε >ίσαζ ε>ῖναι τὰς ΑΒ, ΒΓ, ΓΔ, ΔΕ, ΕΑ περιφερείας; καὶ διὰ τῶν Α, Β, Γ, Δ, Ε >ήθθωσαν τοῦ κύκλου ἐφαπτόμεναι αἱ ΗΘ, ΘΚ, ΚΛ, ΛΜ, μῆ, καὶ εἰλήφθω τοῦ ΑΒΓΔΕ κύκλου κέντρον τὸ Ζ, καὶ ἐπεζεύθωσαν αἱ ΖΒ, ΖΚ, ΖΓ, ΖΛ, ΖΔ.

Καὶ ἐπεὶ ἡ μὲν ΚΛ εὐθεῖα ἐφάπτεται τοῦ ΑΒΓΔΕ κατὰ τὸ Γ, ἀπὸ δὲ τοῦ Ζ κέντρου ἐπὶ τὴν κατὰ τὸ Γ ἐπαφὴν ἐπέzeugκται ἡ ΖΓ, ἡ ΖΓ >άρα κάθετόξ ἐστὶν ἐπὶ τὴν ΚΛ; ὁρθή >άρα ἐστὶν ἐκατέρᾳ τῶν πρὸς τῷ Γ γωνιῶν. διὰ τὰ αὐτὰ δὴ καὶ αἱ πρὸς τοῖς Β, Δ σημείοιξ γωνίαι ὁρθαὶ εἰσίν. καὶ ἐπεὶ ὁρθή ἐστὶν

Let $ABCDE$ be the given circle. So it is required to circumscribe an equilateral and equiangular pentagon about circle $ABCDE$.

Let A , B , C , D , and E be conceived as the angular points of a pentagon having been inscribed (in circle $ABCDE$) [Prop. 3.11], such that the circumferences AB , BC , CD , DE , and EA are equal. And let GH , HK , KL , LM , and MG be drawn through (points) A , B , C , D , and E (respectively), touching the circle.[†] And let the center F of the circle $ABCDE$ be found [Prop. 3.1]. And let FB , FK , FC , FL , and FD be joined.

And since the straight-line KL touches (circle) $ABCDE$ at C , and FC has been joined from the center F to the point

ἡ ὑπὸ ΖΓΚ γωνία, τὸ >άρα ἀπὸ τῆς ΖΚ >ίσον ἐστὶ τοῖς ἀπὸ τῶν ΖΓ, ΓΚ. διὰ τὰ αὐτὰ δὴ καὶ τοῖς ἀπὸ τῶν ΖΒ, ΒΚ >ίσον ἐστὶ τὸ ἀπὸ τῆς ΖΚ; <ώστε τὰ ἀπὸ τῶν ΖΓ, ΓΚ τοῖς ἀπὸ τῶν ΖΒ, ΒΚ ἐστὶν >ίσα, <ὥν τὸ ἀπὸ τῆς ΖΓ τῷ ἀπὸ τῆς ΖΒ ἐστὶν >ίσον; λοιπὸν >άρα τὸ ἀπὸ τῆς ΓΚ τῷ ἀπὸ τῆς ΒΚ ἐστὶν >ίσον. >ίση >άρα ἡ ΒΚ τῇ ΓΚ. καὶ ἐπεὶ >ίση ἐστὶν ἡ ΖΒ τῇ ΖΓ, καὶ κοινὴ ἡ ΖΚ, δύο δὴ αἱ ΒΖ, ΖΚ δυοὶ ταῖς ΓΖ, ΖΚ >ίσαι εἰσίν; καὶ βάσις ἡ ΒΚ βάσει τῇ ΓΚ [ἐστὶν] >ίση; γωνία >άρα ἡ μὲν ὑπὸ ΒΖΚ [γωνία] τῇ ὑπὸ ΚΖΓ ἐστὶν >ίση; ἡ δὲ ὑπὸ ΒΚΖ τῇ ὑπὸ ΖΚΓ; διπλῇ >άρα ἡ μὲν ὑπὸ ΒΖΓ τῆς ὑπὸ ΚΖΓ, ἡ δὲ ὑπὸ ΒΚΓ τῆς ὑπὸ ΖΚΓ. διὰ τὰ αὐτὰ δὴ καὶ ἡ μὲν ὑπὸ ΓΖΔ τῆς ὑπὸ ΓΖΛ ἐστὶ διπλῇ, ἡ δὲ ὑπὸ ΔΛΓ τῆς ὑπὸ ΖΛΓ. καὶ ἐπεὶ >ίση ἐστὶν ἡ ΒΓ περιφέρεια τῇ ΓΔ, >ίση ἐστὶ καὶ γωνία ἡ ὑπὸ ΒΖΓ τῇ ὑπὸ ΓΖΔ. καὶ ἐστὶν ἡ μὲν ὑπὸ ΒΖΓ τῆς ὑπὸ ΚΖΓ διπλῇ, ἡ δὲ ὑπὸ ΔΖΓ τῆς ὑπὸ ΛΖΓ; >ίση >άρα καὶ ἡ ὑπὸ ΚΖΓ τῇ ὑπὸ ΛΖΓ; ἐστὶ δὲ καὶ ἡ ὑπὸ ΖΓΚ γωνία τῇ ὑπὸ ΖΓΛ >ίση. δύο δὴ τρίγωνά ἐστὶ τὰ ΖΚΓ, ΖΛΓ τὰς δύο γωνίας ταῖς δυοὶ γωνίαις >ίσαζ >έθοντα καὶ μίαν πλευρὰν μὲν [πλευρὰ] >ίσην κοινὴν αὐτῶν τὴν ΖΓ; καὶ τὰς λοιπὰς >άρα πλευρὰς ταῖς λοιπαῖς πλευραῖς >ίσαζ <έχει καὶ τὴν λοιπὴν γωνίαν τῇ λοιπῇ γωνία; >ίση >άρα ἡ μὲν ΚΓ εὐθεῖα τῇ ΓΛ, ἡ δὲ ὑπὸ ΖΚΓ γωνία τῇ ὑπὸ ΖΛΓ. καὶ ἐπεὶ >ίση ἐστὶν ἡ ΚΓ τῇ ΓΛ, διπλῇ >άρα ἡ ΚΛ τῆς ΚΓ. διὰ τὰ αὐτὰ δὴ δευθρήσεται καὶ ἡ ΘΚ τῆς ΒΚ διπλῇ. καὶ ἐστὶν ἡ ΒΚ τῇ ΚΓ >ίση; καὶ ἡ ΘΚ >άρα τῇ ΚΛ ἐστὶν >ίση. ὁμοίωζ δὴ δευθρήσεται καὶ ἐκάστη τῶν ΘΗ, ΗΜ, ΜΛ ἐκατέρᾳ τῶν ΘΚ, ΚΛ >ίση; ἰσόπλευρον >άρα ἐστὶ τὸ ΗΘΚΛΜ πεντάγωνον. λέγω δὴ, <ὅτι καὶ ἰσογώνιον. ἐπεὶ γὰρ >ίση ἐστὶν ἡ ὑπὸ ΖΚΓ γωνία τῇ ὑπὸ ΖΛΓ, καὶ ἐδειθθῇ τῆς μὲν ὑπὸ ΖΚΓ διπλῇ ἡ ὑπὸ ΘΚΛ, τῆς δὲ ὑπὸ ΖΛΓ διπλῇ ἡ ὑπὸ ΚΛΜ, καὶ ἡ ὑπὸ ΘΚΛ >άρα τῇ ὑπὸ ΚΛΜ ἐστὶν >ίση. ὁμοίωζ δὴ δευθρήσεται καὶ ἐκάστη τῶν ὑπὸ ΚΘΗ, ΘΗΜ, ΗΜΛ ἐκατέρᾳ τῶν ὑπὸ ΘΚΛ, ΚΛΜ >ίση; αἱ πέντε >άρα γωνίαι αἱ ὑπὸ ΗΘΚ, ΘΚΛ, ΚΛΜ, Λμη, μηΘ >ίσαι ἀλλήλαις εἰσίν. ἰσογώνιον >άρα ἐστὶ τὸ ΗΘΚΛΜ πεντάγωνον. ἐδειθθῇ δὲ καὶ ἰσόπλευρον, καὶ περιγέγραπται περὶ τὸν ΑΒΓΔΕ κύκλον.

[Per'i t'on doj'enta >'ara k'uklon pent'agwnon >is'opleur'on te ka'i >isog'wnion perig'egraftai]; <'oper >'edei poi hsai.

of contact C , FC is thus perpendicular to KL [Prop. 3.18]. Thus, each of the angles at C is a right-angle. So, for the same (reasons), the angles at B and D are also right-angles. And since angle FCK is a right-angle, the (square) on FK is thus equal to the (sum of the squares) on FC and CK [Prop. 1.47]. So, for the same (reasons), the (square) on FK is also equal to the (sum of the squares) on FB and BK . So that the (sum of the squares) on FC and CK is equal to the (sum of the squares) on FB and BK , of which the (square) on FC is equal to the (square) on FB . Thus, the remaining (square) on CK is equal to the remaining (square) on BK . Thus, BK (is) equal to CK . And since FB is equal to FC , and FK (is) common, the two (straight-lines) BF , FK are equal to the two (straight-lines) CF , FK . And the base BK [is] equal to the base CK . Thus, angle BFK is equal to [angle] KFC [Prop. 1.8]. And BKF (is) equal to FKC [Prop. 1.8]. Thus, BFC (is) double KFC , and BKC (is) double FKC . So, for the same (reasons), CFD is also double CFL , and DLC (is) also double FLC . And since circumference BC is equal to CD , angle BFC is also equal to CFD [Prop. 3.27]. And BFC is double KFC , and DFC (is) double LFC . Thus, KFC is also equal to LFC . And angle FCK is also equal to FCL . So, FKC and FLC are two triangles having two angles equal to two angles, and one side equal to one side, (namely) their common (side) FC . Thus, they will also have the remaining sides equal to the (corresponding) remaining sides, and the remaining angle to the remaining angle [Prop. 1.26]. Thus, the straight-line KC (is) equal to CL , and the angle FKC to FLC . And since KC is equal to CL , KL (is) thus double KC . So, for the same (reasons), it can be shown that HK (is) also double BK . And BK is equal to KC . Thus, HK is also equal to KL . So, similarly, each of HG , GM , and ML can also be shown (to be) equal to each of HK and KL . Thus, pentagon $GHKLM$ is equilateral. So I say that (it is) also equiangular. For since angle FKC is equal to FLC , and HKL was shown (to be) double FKC , and KLM double FLC , HKL is thus also equal to KLM . So, similarly, each of KHG , HGM , and GML can also be shown (to be) equal to each of HKL and KLM . Thus, the five angles GHK , HKL , KLM , LMG , and MGH are equal to one another. Thus, the pentagon $GHKLM$ is equiangular. And it was also shown (to be) equilateral, and has been circumscribed about circle $ABCDE$.

[Thus, an equilateral and equiangular pentagon has been circumscribed about the given circle]. (Which is) the very thing it was required to do.

† See the footnote to Prop. 3.34.

13.

Εἰς τὸ δοθὲν πεντάγωνον, <ὅ ἐστὶν ἰσόπλευρόν τε καὶ ἰσογώνιον, κύκλον ἐγγράψαι.

>'Εστω τὸ δοθὲν πεντάγωνον ἰσόπλευρόν τε καὶ ἰσογώνιον τὸ ΑΒΓΔΕ; δεῖ δὴ εἰς τὸ ΑΒΓΔΕ πεντάγωνον κύκλον ἐγγράψαι.

Τεμήσθω γὰρ ἐκατέρᾳ τῶν ὑπὸ ΒΓΔ, ΓΔΕ γωνιῶν δίθῃ ὑπὸ ἐκατέρᾳ τῶν ΓΖ, ΔΖ εὐθειῶν; καὶ ἀπὸ τοῦ Ζ σημείου, καθ' <ὃ συμβάλλουσιν ἀλλήλαις αἱ ΓΖ, ΔΖ εὐθεῖαι, ἐπεξεύθωσαν αἱ ΖΒ, ΖΑ, ΖΕ εὐθεῖαι. καὶ ἐπεὶ >ίση ἐστὶν ἡ ΒΓ τῇ ΓΔ, κοινὴ δὲ ἡ ΓΖ, δύο δὴ αἱ ΒΓ, ΓΖ δυοὶ ταῖς ΔΓ, ΓΖ >ίσαι

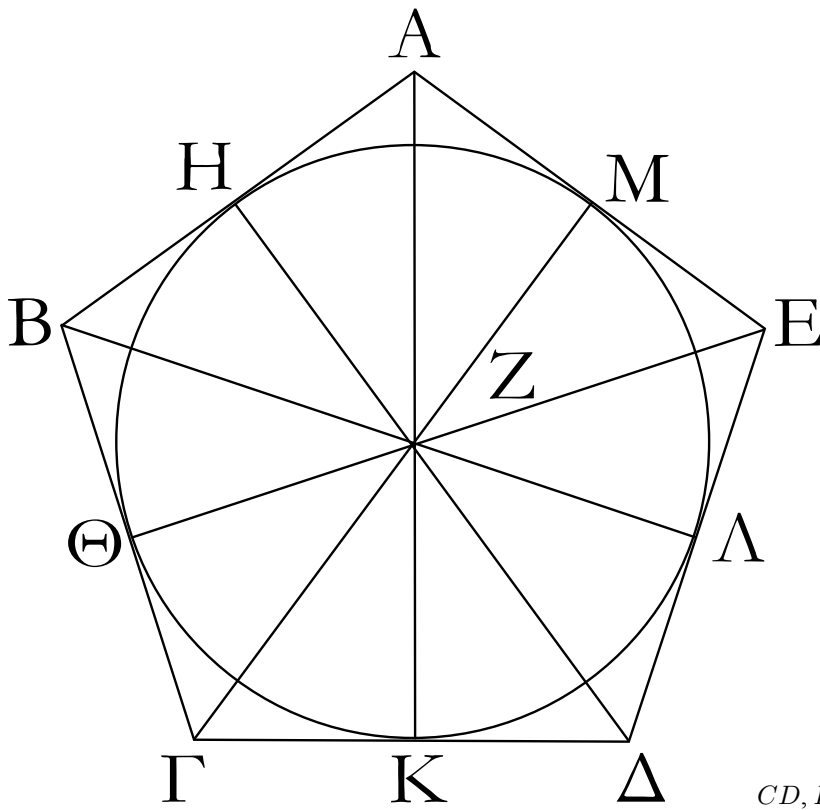
To inscribe a circle in a given pentagon, which is equilateral and equiangular.

Let $ABCDE$ be the given equilateral and equiangular pentagon. So it is required to inscribe a circle in pentagon $ABCDE$.

For let angles BCD and CDE each be cut in half by each of the straight-lines CF and DF (respectively) [Prop. 1.9]. And from the point F , at which the straight-lines CF and DF meet one another, let the straight-lines FB , FA , and FE be joined. And since BC is equal to CD , and CF (is)

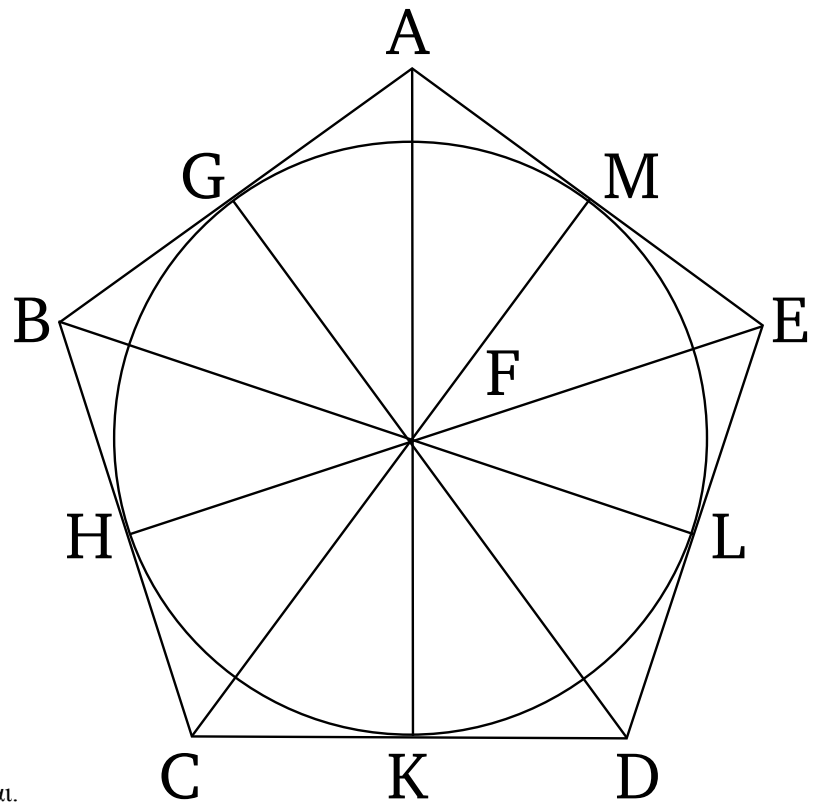
εἰσίν; καὶ γωνία ἡ ὑπὸ BFZ γωνία τῇ ὑπὸ ΔGZ [ἐστίν] $>$ ίση; βάσις $>$ άρα ἡ BZ βάσει τῇ ΔZ ἐστίν $>$ ίση, καὶ τὸ BFZ τρίγωνον τῷ ΔGZ τριγώνῳ ἐστίν $>$ ίσον, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις $>$ ίσαι $>$ έσονται, ὅφρ $<$ ὰς αἱ $>$ ίσαι πλευραὶ ὑποτείνουσιν; $>$ ίση $>$ άρα ἡ ὑπὸ ΓBZ γωνία τῇ ὑπὸ $\Gamma \Delta Z$. καὶ ἐπεὶ διπλῇ ἐστίν ἡ ὑπὸ $\Gamma \Delta E$ τῇ ὑπὸ $\Gamma \Delta Z$, $>$ ίση δὲ ἡ μὲν ὑπὸ $\Gamma \Delta E$ τῇ ὑπὸ $AB\Gamma$, ἡ δὲ ὑπὸ $\Gamma \Delta Z$ τῇ ὑπὸ ΓBZ , καὶ ἡ ὑπὸ $\Gamma B A$ $>$ άρα τῇ ὑπὸ $\Gamma B Z$ ἐστὶ διπλῇ; $>$ ίση $>$ άρα ἡ ὑπὸ ABZ γωνία τῇ ὑπὸ $ZB\Gamma$; ἡ $>$ άρα ὑπὸ $AB\Gamma$ γωνία δίθα τέτμηται ὑπὸ τῇ BZ εὐθείᾳ. ὁμοίως δὲ δειθθήσεται, $<$ ὅτι καὶ ἑκατέρα τῶν ὑπὸ BAE , $AE\Delta$ δίθα τέτμηται ὑπὸ ἑκατέρας τῶν ZA , ZE εὐθειῶν. $>$ ήθωσαν δὲ ἀπὸ τοῦ Z σημείου ἐπὶ τὰς AB , $B\Gamma$, $\Gamma \Delta$, ΔE , EA εὐθείας κάθετοι αἱ ZH , $Z\Theta$, ZK , $Z\Lambda$, ZM . καὶ ἐπεὶ $>$ ίση ἐστίν ἡ ὑπὸ $\Theta \Gamma Z$ γωνία τῇ ὑπὸ $K \Gamma Z$, ἐστὶ δὲ καὶ ὀρθὴ ἡ ὑπὸ $Z\Theta \Gamma$ [ὀρθῇ] τῇ ὑπὸ $ZK \Gamma$ $>$ ίση, δύο δὲ τρίγωνα ἐστὶ τὰ $Z\Theta \Gamma$, $ZK \Gamma$ τὰς δύο γωνίας δυσὶ γωνίαις $>$ ίσας $>$ έθοντα καὶ μίαν πλευρὰν μιᾷ πλευρᾷ $>$ ίσην κοινὴν αὐτῶν τὴν $Z\Gamma$ ὑποτείνουσιν ὑπὸ μίαν τῶν $>$ ίσων γωνιών; καὶ τὰς λοιπὰς $>$ άρα πλευρὰς ταῖς λοιπαῖς πλευραῖς $>$ ίσας $<$ έχει; $>$ ίση $>$ άρα ἡ $Z\Theta$ κάθετος τῇ ZK καθετῷ. ὁμοίως δὲ δειθθήσεται, $<$ ὅτι καὶ ἑκάστη τῶν $Z\Lambda$, ZM , ZH ἑκατέρα τῶν $Z\Theta$, ZK $>$ ίση ἐστίν; αἱ πέντε $>$ άρα εὐθεῖαι αἱ ZH , $Z\Theta$, ZK , $Z\Lambda$, ZM $>$ ίσαι ἀλλήλαις εἰσίν. ὁ $>$ άρα κέντρον τῷ Z διαστήματι δὲ ἐνὶ τῶν H , Θ , K , Λ , M κύκλος γραφόμενος $<$ ήχει καὶ διὰ τῶν λοιπῶν σημείων καὶ ἐφάπεται τῶν AB , $B\Gamma$, $\Gamma \Delta$, ΔE , EA εὐθειῶν διὰ τὸ ὀρθὰς εἶναι τὰς πρὸς τοῖς H , Θ , K , Λ , M σημείοις γωνίας. εἰ γὰρ οὐκ ἐφάπεται αὐτῶν, ἀλλὰ τεμεῖ αὐτάς, συμβήσεται τὴν τῇ διαμέτρῳ τοῦ κύκλου πρὸς ὀρθὰς ἀπὸ $>$ άκρας ἀγομένην ἐντὸς πίπτειν τοῦ κύκλου; $<$ όπερ $>$ άτοπον ἐδείχθη. οὐκ $>$ άρα ὁ κέντρον τῷ Z διαστήματι δὲ ἐνὶ τῶν H , Θ , K , Λ , M σημείων γραφόμενος κύκλος τεμεῖ τὰς AB , $B\Gamma$, $\Gamma \Delta$, ΔE , EA εὐθείας; ἐφάπεται $>$ άρα αὐτῶν. γεγράφθω ὡς ὁ $H\Theta K\Lambda M$.

common, the two (straight-lines) BC , CF are equal to the two (straight-lines) DC , CF . And angle BCF [is] equal to angle DCF . Thus, the base BF is equal to the base DF , and triangle BCF is equal to triangle DCF , and the remaining angles will be equal to the (corresponding) remaining angles which the equal sides subtend [Prop. 1.4]. Thus, angle CBF (is) equal to CDF . And since CDE is double CDF , and CDE (is) equal to ABC , and CDF to CBF , CBA is thus also double CBF . Thus, angle ABF is equal to FBC . Thus, angle ABC has been cut in half by the straight-line BF . So, similarly, it can be shown that BAE and AED have been cut in half by the straight-lines FA and FE , respectively. So let FG , FH , FK , FL , and FM be drawn from point F , perpendicular to the straight-lines AB , BC , CD , DE , and EA (respectively) [Prop. 1.12]. And since angle HCF is equal to KCF , and the right-angle FHC is also equal to the [right-angle] FKC , FHC and FKC are two triangles having two angles equal to two angles, and one side equal to one side, (namely) their common (side) FC , subtending one of the equal angles. Thus, they will also have the remaining sides equal to the (corresponding) remaining sides [Prop. 1.26]. Thus, the perpendicular FH (is) equal to the perpendicular FK . So, similarly, it can be shown that FL , FM , and FG are each equal to each of FH and FK . Thus, the five straight-lines FG , FH , FK , FL , and FM are equal to one another. Thus, the circle drawn with center F , and radius one of G , H , K , L , or M , will also go through the remaining points, and will touch the straight-lines AB , BC , CD , DE , and EA , on account of the angles at points G , H , K , L , and M being right-angles. For if it does not touch them, but cuts them, it follows that a (straight-line) drawn at right-angles to the diameter of the circle, from its extremity, falls inside the circle. The very thing was shown (to be) absurd [Prop. 3.16]. Thus, the circle drawn with center F , and radius one of G , H , K , L , or M , does not cut the straight-lines AB , BC ,



Εἰς ὅρα τὸ δοθὲν πεντάγωνον, ὅ ἐστιν ἰσόπλευρόν τε καὶ

CD , DE , or EA . Thus, it will touch them. Let it be drawn, like $GHKLM$ (in the figure).



ἰσογώνιον, κύκλος ἐγγέγραπται; ὅπερ ἔδει ποιῆσαι.

Thus, a circle has been inscribed in the given pentagon which is equilateral and equiangular. (Which is) the very thing it was required to do.

14.

Proposition 14

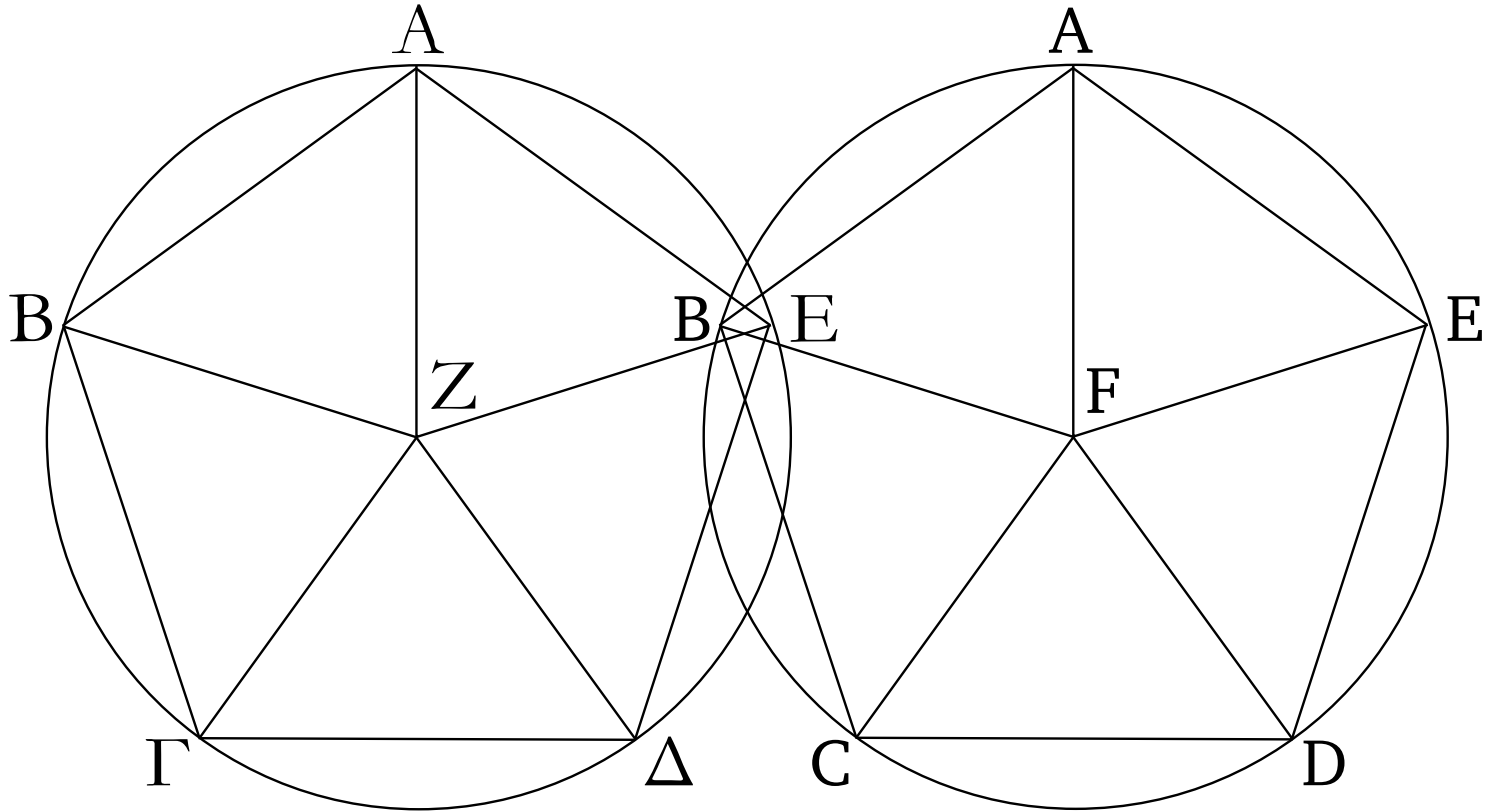
Περὶ τὸ δοθὲν πεντάγωνον, ὅ ἐστιν ἰσόπλευρόν τε καὶ ἰσογώνιον, κύκλον περιγράψαι.

> Ἐστω τὸ δοθὲν πεντάγωνον, ὅ ἐστιν ἰσόπλευρόν τε καὶ ἰσογώνιον, τὸ $ABCDE$; δεῖ δὲ περὶ τὸ $ABCDE$ πεντάγωνον

To circumscribe a circle about a given pentagon which is equilateral and equiangular.

Let $ABCDE$ be the given pentagon which is equilateral and equiangular. So it is required to circumscribe a circle about

κύκλον περιγράψαι.

the pentagon $ABCDE$.

Τετμήσθω δὴ ἑκάτερα τῶν ὑπὸ $B\Gamma\Delta$, $\Gamma\Delta\epsilon$ γωνιῶν δίθρα ὑπὸ ἑκατέρᾳ τῶν ΓZ , ΔZ , καὶ ἀπὸ τοῦ Z σημείου, καθ' $\angle$ ο συμβάλλουσιν αἱ εὐθεῖαι, ἐπὶ τὰ B , A , ϵ σημεία ἐπεξεύθθωσαν εὐθεῖαι αἱ ZB , ZA , ZE . ὁμοίως δὴ τῷ $\angle$ πρὸ τούτου δεῖσθῆσεται, $\angle$ ότι καὶ ἑκάστη τῶν ὑπὸ $\Gamma B A$, $B A \epsilon$, $A \epsilon \Delta$ γωνιῶν δίθρα τέτμηται ὑπὸ ἑκάστης τῶν ZB , ZA , ZE εὐθειῶν. καὶ ἐπεὶ $\angle$ ίση ἐστὶν ἡ ὑπὸ $B\Gamma\Delta$ γωνία τῇ $\angle$ ὑπὸ $\Gamma\Delta\epsilon$, καὶ ἐστὶ τῇς μὲν ὑπὸ $B\Gamma\Delta$ ἡμίσεια ἡ ὑπὸ $\Gamma Z\Delta$, τῇς δὲ ὑπὸ $\Gamma\Delta\epsilon$ ἡμίσεια ἡ ὑπὸ $\Gamma\Delta Z$, καὶ ἡ ὑπὸ $\Gamma Z\Delta$ $\angle$ άρα τῇ $\angle$ ὑπὸ $Z\Delta\Gamma$ ἐστὶν $\angle$ ίση; $\angle$ ώστε καὶ πλευρὰ ἡ $Z\Gamma$ πλευρᾷ τῇ $Z\Delta$ ἐστὶν $\angle$ ίση. ὁμοίως δὴ δεῖσθῆσεται, $\angle$ ότι καὶ ἑκάστη τῶν ZB , ZA , ZE ἑκατέρᾳ τῶν $Z\Gamma$, $Z\Delta$ ἐστὶν $\angle$ ίση; αἱ πέντε $\angle$ άρα εὐθεῖαι αἱ ZA , ZB , $Z\Gamma$, $Z\Delta$, ZE $\angle$ ίσαι ἀλλήλαις εἰσίν. ὁ $\angle$ άρα κέντρῳ τῷ Z καὶ διαστήματι ἐνὶ τῶν ZA , ZB , $Z\Gamma$, $Z\Delta$, ZE κύκλος γραφόμενος $\angle$ ήχει καὶ διὰ τῶν λοιπῶν σημείων καὶ $\angle$ έσται περιγεγραμμένος. περιγεγράφθω καὶ $\angle$ έστω ὁ $AB\Gamma\Delta\epsilon$. Περὶ $\angle$ άρα τὸ δοθὲν πεντάγωνον, $\angle$ ό ἐστὶν ἰσόπλευρόν τε καὶ ἰσογώνιον, κύκλος περιγράφεται; $\angle$ όπερ $\angle$ έδει ποιῆσαι.

So let angles BCD and CDE have been cut in half by the (straight-lines) CF and DF , respectively [Prop. 1.9]. And let the straight-lines FB , FA , and FE be joined from point F , at which the straight-lines meet, to the points B , A , and E (respectively). So, similarly, to the (proposition) before this (one), it can be shown that angles CBA , BAE , and AED have also been cut in half by the straight-lines FB , FA , and FE , respectively. And since angle BCD is equal to CDE , and FCD is half of BCD , and CDF half of CDE , FCD is thus also equal to FDC . So that side FC is also equal to side FD [Prop. 1.6]. So, similarly, it can be shown that FB , FA , and FE are also each equal to each of FC and FD . Thus, the five straight-lines FA , FB , FC , FD , and FE are equal to one another. Thus, the circle drawn with center F , and radius one of FA , FB , FC , FD , or FE , will also go through the remaining points, and will be circumscribed. Let it be (so) circumscribed, and let it be $ABCDE$. Thus, a circle has been circumscribed about the given pentagon, which is equilateral and equiangular. (Which is) the very thing it was required to do.

15.

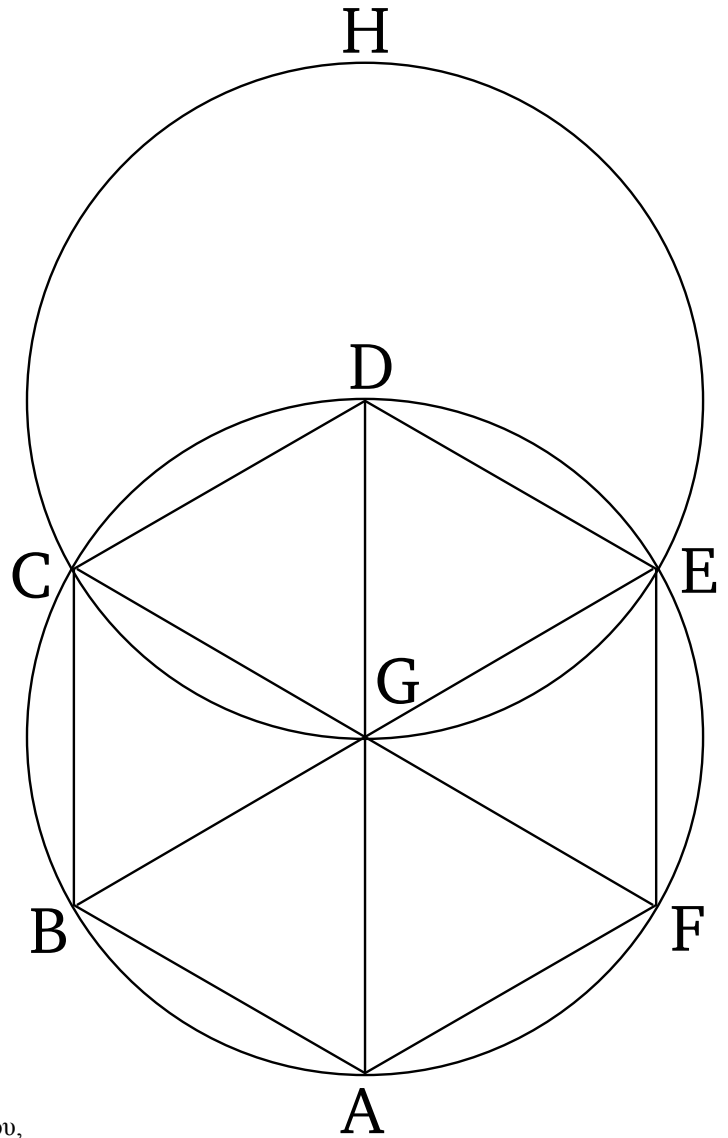
Proposition 15

Εἰς τὸν δοθέντα κύκλον ἑχάγωνον ἰσόπλευρόν τε καὶ ἰσογώνιον ἐγγράψαι.
 $\angle$ Εστω ὁ δοθείς κύκλος ὁ $AB\Gamma\Delta\epsilon Z$; δεῖ δὴ εἰς τὸν $AB\Gamma\Delta\epsilon Z$ κύκλον ἑχάγωνον ἰσόπλευρόν τε καὶ ἰσογώνιον ἐγγράψαι.
 $\angle$ Ἡθθω τοῦ $AB\Gamma\Delta\epsilon Z$ κύκλου διάμετρος ἡ $A\Delta$, καὶ εἰλήφθω τὸ κέντρον τοῦ κύκλου τὸ H , καὶ κέντρῳ μὲν τῷ Δ διαστήματι δὲ τῷ ΔH κύκλος γεγράφθω ὁ $E\eta\Gamma\Theta$, καὶ ἐπιεϋθθεῖσαι αἱ $E\eta$, $\Gamma\eta$ διήθθωσαν ἐπὶ τὰ B , Z σημεία, καὶ ἐπεξεύθθωσαν αἱ AB , $B\Gamma$, $\Gamma\Delta$, $\Delta\epsilon$, $E Z$, ZA ; λέγω, $\angle$ ότι τὸ $AB\Gamma\Delta\epsilon Z$ ἑχάγωνον ἰσόπλευρόν τε ἐστὶ καὶ ἰσογώνιον.

To inscribe an equilateral and equiangular hexagon in a given circle.

Let $ABCDEF$ be the given circle. So it is required to inscribe an equilateral and equiangular hexagon in circle $ABCDEF$.

Let the diameter AD of circle $ABCDEF$ be drawn,[†] and let the center G of the circle be found [Prop. 3.1]. And let the circle $EGCH$ be drawn, with center D , and radius DG . And EG and CG being joined, let them be drawn across (the circle) to points B and F (respectively). And let AB , BC , CD , DE , EF , and FA be joined. I say that the hexagon



Ἐπεὶ γὰρ τὸ H σημεῖον κέντρον ἐστὶ τοῦ $ABΓΔΕΖ$ κύκλου, $>ίση$ ἐστὶν ἡ HE τῇ HA . πάλιν, ἐπεὶ τὸ Δ σημεῖον κέντρον ἐστὶ τοῦ $HΓΘ$ κύκλου, $>ίση$ ἐστὶν ἡ ΔE τῇ ΔH . ἀλλ' ἡ HE τῇ HA ἐδείκθη $>ίση$; καὶ ἡ $HE >άρα$ τῇ $ED >ίση$ ἐστίν; ἰσόπλευρον $>άρα$ ἐστὶ τὸ $EH\Delta$ τρίγωνον; καὶ αἱ τρεῖς $>άρα$ αὐτοῦ γωνίαι αἱ ὑπὸ $EH\Delta$, $H\Delta E$, ΔEH $>ίσαι$ ἀλλήλαις εἰσίν, ἐπειδὴ περ τῶν ἰσοσκελῶν τριγώνων αἱ πρὸς τῇ βάσει γωνίαι $>ίσαι$ ἀλλήλαις εἰσίν; καὶ εἰσιν αἱ τρεῖς τοῦ τριγώνου γωνίαι δυσὶν ὀρθαῖς $>ίσαι$; ἡ $>άρα$ ὑπὸ $EH\Delta$ γωνία τρίτον ἐστὶ δύο ὀρθῶν. ὁμοίως δὴ δειθθήσεται καὶ ἡ ὑπὸ $\Delta HΓ$ τρίτον δύο ὀρθῶν. καὶ ἐπεὶ ἡ $ΓH$ εὐθεῖα ἐπὶ τὴν EB σταθεῖσα τὰς ἐφερχῆς γωνίας τὰς ὑπὸ $EHΓ$, $ΓHB$ δυσὶν ὀρθαῖς $>ίσα$ ς ποιεῖ, καὶ λοιπὴ $>άρα$ ἡ ὑπὸ $ΓHB$ τρίτον ἐστὶ δύο ὀρθῶν; αἱ $>άρα$ ὑπὸ $EH\Delta$, $\Delta HΓ$, $ΓHB$ γωνίαι $>ίσαι$ ἀλλήλαις εἰσίν; $<ώστε$ καὶ αἱ κατὰ κορυφὴν αὐταῖς αἱ ὑπὸ BHA , AHZ , ZHE $>ίσαι$ εἰσίν [ταῖς ὑπὸ $EH\Delta$, $\Delta HΓ$, $ΓHB$]. αἱ $<ἐχ$ $>άρα$ γωνίαι αἱ ὑπὸ $EH\Delta$, $\Delta HΓ$, $ΓHB$, BHA , AHZ , ZHE $>ίσαι$ ἀλλήλαις εἰσίν. αἱ δὲ $>ίσαι$ γωνίαι ἐπὶ $>ίσων$ περιφερειῶν βεβήκασιν; αἱ $<ἐχ$ $>άρα$ περιφέρειαι αἱ AB , $ΒΓ$, $ΓΔ$, ΔE , EZ , $ΖΑ$ $>ίσαι$ ἀλλήλαις εἰσίν. ὑπὸ δὲ τὰς $>ίσα$ ς περιφερείας αἱ $>ίσαι$ εὐθεῖαι ὑποτείνουσιν; αἱ $<ἐχ$ $>άρα$ εὐθεῖαι $>ίσαι$ ἀλλήλαις εἰσίν; ἰσόπλευρον $>άρα$ ἐστὶ το $ABΓΔΕΖ$ ἑξάγωνον. λέγω δὴ, $<ότι$ καὶ ἰσογώνιον. ἐπεὶ γὰρ $>ίση$ ἐστὶν ἡ $ΖΑ$ περιφέρεια τῇ $ΕΔ$ περιφέρειᾳ, κοινὴ προσκείσθω ἡ $ABΓΔ$ περιφέρεια; $<όλη$ $>άρα$ ἡ $ZABΓΔ$ $<όλη$ τῇ $ΕΔΓΒΑ$ ἐστὶν $>ίση$; καὶ βέβηκεν ἐπὶ μὲν τῆς

For since point G is the center of circle $ABCDEF$, GE is equal to GD . Again, since point D is the center of circle GCH , DE is equal to DG . But, GE was shown (to be) equal to GD . Thus, GE is also equal to ED . Thus, triangle EGD is equilateral. Thus, its three angles EGD , GDE , and DEG are also equal to one another, inasmuch as the angles at the base of isosceles triangles are equal to one another [Prop. 1.5]. And the three angles of the triangle are equal to two right-angles [Prop. 1.32]. Thus, angle EGD is one third of two right-angles. So, similarly, DGC can also be shown (to be) one third of two right-angles. And since the straight-line CG , standing on EB , makes adjacent angles EGC and CGB equal to two right-angles [Prop. 1.13], the remaining angle CGB is thus also one third of two right-angles. Thus, angles EGD , DGC , and CGB are equal to one another. And hence the (angles) opposite to them BGA , AGF , and FGE are also equal [to EGD , DGC , and CGB (respectively)] [Prop. 1.15]. Thus, the six angles EGD , DGC , CGB , BGA , AGF , and FGE are equal to one another. And equal angles stand on equal circumferences [Prop. 3.26]. Thus, the six circumferences AB , BC , CD , DE , EF , and FA are equal to one another. And equal circumferences are subtended by equal straight-lines [Prop. 3.29]. Thus, the six straight-lines (AB , BC , CD , DE , EF , and FA) are equal to one an-

$ZAB\Gamma\Delta$ περιφερείᾳ ἢ ὑπὸ $ZE\Delta$ γωνία, ἐπὶ δὲ τῇ $ΕΔΓΒΑ$ περιφερείᾳ ἢ ὑπὸ AZE γωνία; $>$ ίση $>$ άρα ἢ ὑπὸ AZE γωνία τῇ ὑπὸ ΔEZ . ὁμοίωξ δὴ δεῖθῇσεται, $<$ ὅτι καὶ αἱ λοιπαὶ γωνίαι τοῦ $AB\Gamma\Delta EZ$ ἐχάγωνου κατὰ μίαν $>$ ίσαι εἰσὶν ἑκατέρᾳ τῶν ὑπὸ AZE , $ZE\Delta$ γωνιῶν; ἰσογώνιον $>$ άρα ἐστὶ τὸ $AB\Gamma\Delta EZ$ ἐχάγωνον. ἐδείκθη δὲ καὶ ἰσόπλευρον; καὶ ἐγγέγραπται εἰς τὸν $AB\Gamma\Delta EZ$ κύκλον. Εἰς $>$ άρα τὸν δοθέντα κύκλον ἐχάγωνον ἰσόπλευρόν τε καὶ ἰσογώνιον ἐγγέγραπται; $<$ όπερ $>$ έδει ποιῆσαι.

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, $<$ ὅτι ἢ τοῦ ἐχάγωνου πλευρὰ $>$ ίση ἐστὶ τῇ ἑκ τοῦ κέντρου τοῦ κύκλου.

Ὅμοίωξ δὲ τοῖς ἐπὶ τοῦ πενταγώνου ἔαν διὰ τῶν κατὰ τὸν κύκλον διαιρέσεων ἐφαπτομένας τοῦ κύκλου ἀγάγωμεν, περιγραφίσεται περὶ τὸν κύκλον ἐχάγωνον ἰσόπλευρόν τε καὶ ἰσογώνιον ἀκολουθῶς τοῖς ἐπὶ τοῦ πενταγώνου εἰρημένοις. καὶ $>$ έτι διὰ τῶν ὁμοίων τοῖς ἐπὶ τοῦ πενταγώνου εἰρημένοις εἰς τὸ δοθὲν ἐχάγωνον κύκλον ἐγγράψομεν τε καὶ περιγράψομεν; $<$ όπερ $>$ έδει ποιῆσαι.

other. Thus, hexagon $ABCDEF$ is equilateral. So, I say that (it is) also equiangular. For since circumference FA is equal to circumference ED , let circumference $ABCD$ be added to both. Thus, the whole of $FABCD$ is equal to the whole of $EDCBA$. And angle FED stands on circumference $FABCD$, and angle AFE on circumference $EDCBA$. Thus, angle AFE is equal to DEF [Prop. 3.27]. Similarly, it can also be shown that the remaining angles of hexagon $ABCDEF$ are individually equal to each of the angles AFE and FED . Thus, hexagon $ABCDEF$ is equiangular. And it was also shown (to be) equilateral. And it has been inscribed in circle $ABCDE$.

Thus, an equilateral and equiangular hexagon has been inscribed in the given circle. (Which is) the very thing it was required to do.

Corollary

So, from this, (it is) manifest that a side of the hexagon is equal to the radius of the circle.

And similarly to a pentagon, if we draw tangents to the circle through the (sixfold) divisions of the (circumference of the) circle, an equilateral and equiangular hexagon can be circumscribed about the circle, analogously to the aforementioned pentagon. And, further, by (means) similar to the aforementioned pentagon, we can inscribe and circumscribe a circle in (and about) a given hexagon. (Which is) the very thing it was required to do.

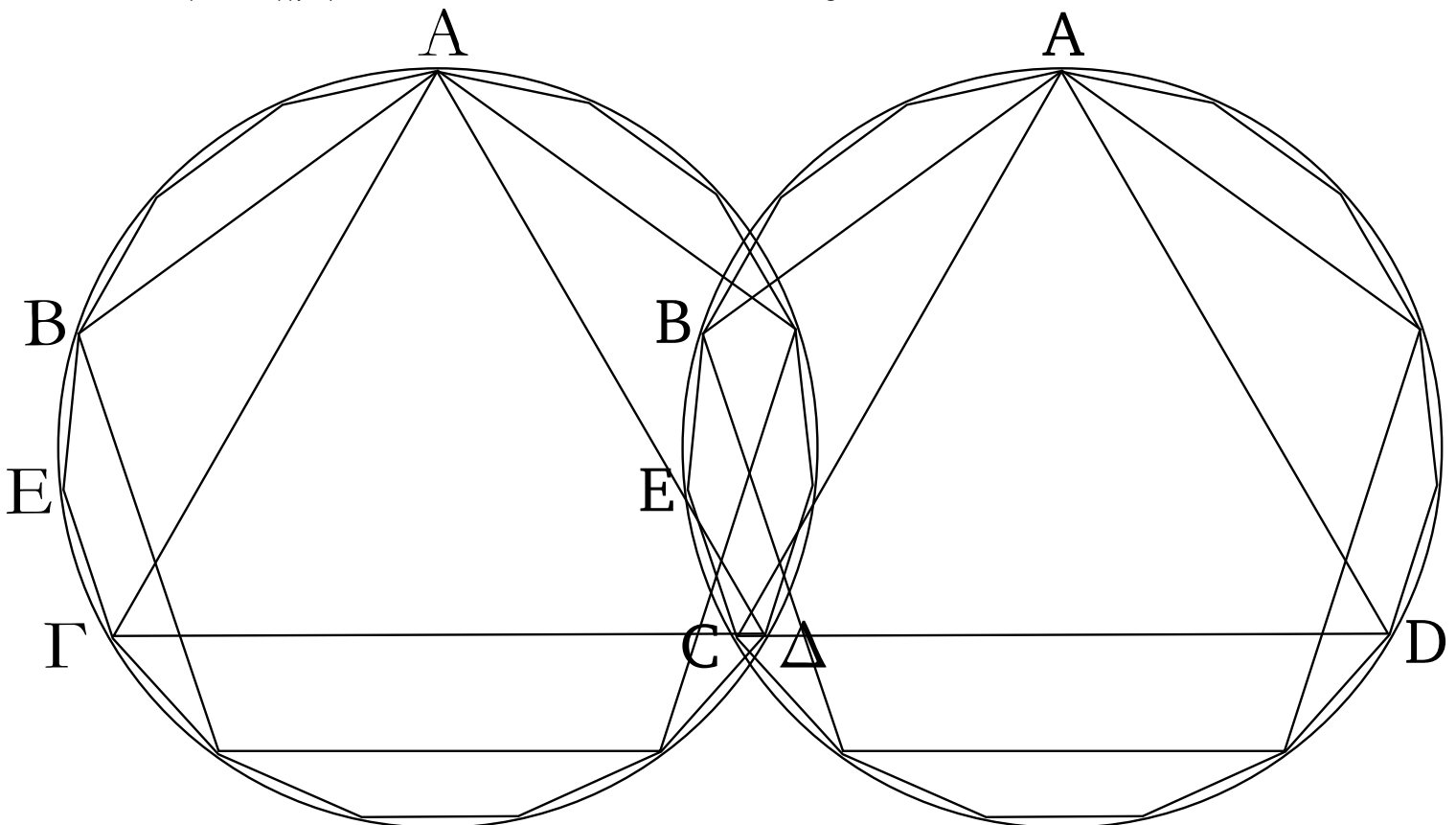
[†] See the footnote to Prop. 4.6.

16.

Εἰς τὸν δοθέντα κύκλον πεντεκαίδεκάγωνον ἰσόπλευρόν τε καὶ ἰσογώνιον ἐγγράψαι.

Proposition 16

To inscribe an equilateral and equiangular fifteen-sided figure in a given circle.



>Ἐστω ὁ δοθεὶς κύκλος ὁ $AB\Gamma\Delta$; δεῖ δὴ εἰς τὸν $AB\Gamma\Delta$ κύκλον πεντεκαιδεκάγωνον ἰσόπλευρόν τε καὶ ἰσογώνιον ἐγγράψαι.

Ἐγγεγράφθω εἰς τὸν $AB\Gamma\Delta$ κύκλον τριγώνου μὲν ἰσοπλεύρου τοῦ εἰς αὐτὸν ἐγγραφομένου πλευρὰ ἡ $ΑΓ$, πενταγώνου δὲ ἰσοπλεύρου ἡ AB ; οἷων >άρα ἐστὶν ὁ $AB\Gamma\Delta$ κύκλος >ίσων τμήματων δεκαπέντε, τοιούτων ἡ μὲν $AB\Gamma$ περιφέρεια τρίτον ο>ῦσα τοῦ κύκλου >έσται πέντε, ἡ δὲ AB περιφέρεια πέμpton ο>ῦσα τοῦ κύκλου >έσται τριῶν; λοιπὴ >άρα ἡ $B\Gamma$ τῶν >ίσων δύο. τεμήσθω ἡ $B\Gamma$ δίθα κατὰ τὸ E ; ἑκατέρα >άρα τῶν BE , $E\Gamma$ περιφερειῶν πεντεκαιδεκάτον ἐστὶ τοῦ $AB\Gamma\Delta$ κύκλου.

Ἐὰν >άρα ἐπιζεύχαντες τὰς BE , $E\Gamma$ >ίσαξ αὐταῖς κατὰ τὸ συνεθὲς εὐθείας ἐναρμόσωμεν εἰς τὸν $AB\Gamma\Delta[E]$ κύκλον, >έσται εἰς αὐτὸν ἐγγεγραμμένον πεντεκαιδεκάγωνον ἰσόπλευρόν τε καὶ ἰσογώνιον; <όπερ >έδει ποιῆσαι.

Ὅμοίως δὲ τοῖς ἐπὶ τοῦ πενταγώνου ἔαν διὰ τῶν κατὰ τὸν κύκλον διαιρέσεων ἐφαπτομένας τοῦ κύκλου ἀγάγωμεν, περιγραφῆσεται περὶ τὸν κύκλον πεντεκαιδεκάγωνον ἰσόπλευρός τε καὶ ἰσογώνιος. >έτι δὲ διὰ τῶν ὁμοίων τοῖς ἐπὶ τοῦ πενταγώνου δείξεων καὶ εἰς τὸ δοθὲν πεντεκαιδεκάγωνον κύκλον ἐγγράψομεν τε καὶ περιγράψομεν; <όπερ >έδει ποιῆσαι.

Let $ABCD$ be the given circle. So it is required to inscribe an equilateral and equiangular fifteen-sided figure in circle $ABCD$.

Let the side AC of an equilateral triangle inscribed in (the circle) [Prop. 4.2], and (the side) AB of an (inscribed) equilateral pentagon [Prop. 4.11], be inscribed in circle $ABCD$. Thus, just as the circle $ABCD$ is (made up) of fifteen equal pieces, the circumference ABC , being a third of the circle, will be (made up) of five such (pieces), and the circumference AB , being a fifth of the circle, will be (made up) of three. Thus, the remainder BC (will be made up) of two equal (pieces). Let (circumference) BC be cut in half at E [Prop. 3.30]. Thus, each of the circumferences BE and EC is one fifteenth of the circle $ABCDE$.

Thus, if, joining BE and EC , we continuously insert straight-lines equal to them into circle $ABCD[E]$ [Prop. 4.1], (then) an equilateral and equiangular fifteen-sided figure will be inserted into (the circle). (Which is) the very thing was required to do.

And similarly to the pentagon, if we draw tangents to the circle through the (fifteenfold) divisions of the (circumference of the) circle, we can circumscribe an equilateral and equiangular fifteen-sided figure about the circle. And, further, through similar proofs to the pentagon, we can also inscribe and circumscribe a circle in (and about) a given fifteen-sided figure. (Which is) the very thing it was required to do.

ELEMENTS BOOK 5

Proportion[†]

[†]The theory of proportion set out in this book is generally attributed to Eudoxus of Cnidus. The novel feature of this theory is its ability to deal with irrational magnitudes, which had hitherto been a major stumbling block for Greek mathematicians. Throughout the footnotes in this book, α , β , γ , etc., denote general (possibly irrational) magnitudes, whereas m , n , l , etc., denote positive integers.

<Ὅροι.

1. Μέρος ἐστὶ μέγεθος μεγέθους τὸ >έλασσον τοῦ μείζονος, <ὅταν καταμετρηῇ τὸ μείζον.
2. Πολλαπλάσιον δὲ τὸ μείζον τοῦ ἐλάττονος, <ὅταν καταμετρηθῇ ὑπὸ τοῦ ἐλάττονος.
3. Λόγος ἐστὶ δύο μεγεθῶν ὁμογενῶν ἢ κατὰ πηλικότητά ποια σθέσις.
4. Λόγον >έθειν πρὸς >άλληλα μεγέθη λέγεται, <ἂ δύναται πολλαπλασιαζόμενα ἀλλήλων ὑπερέθειν.
5. Ἐν τῷ αὐτῷ λόγῳ μεγέθη λέγεται ε>ῖναι πρῶτον πρὸς δεῦτερον καὶ τρίτον πρὸς τέταρτον, <ὅταν τὰ τοῦ πρώτου καὶ τρίτου ἰσάκις πολλαπλάσια τῶν τοῦ δευτέρου καὶ τετάρτου ἰσάκις πολλαπλασίων καθ> ὅποιον οὖν πολλαπλασιασθῇ τὰ πρῶτον καὶ τὸ τρίτον ἑκάτερον >ἢ <άμα ὑπερέθῃ >ἢ <άμα >ῖσα >ἢ >ἢ <άμα ἐλλείπῃ ληφθέντα κατάλληλα.
6. Τὰ δὲ τὸν αὐτὸν >έθοντα λόγον μεγέθη ἀνάλογον καλεῖσθαι.
7. <Ὅταν δὲ τῶν ἰσάκις πολλαπλασίων τὸ μὲν τοῦ πρώτου πολλαπλάσιον ὑπερέθῃ τοῦ τοῦ δευτέρου πολλαπλασίου, τὸ δὲ τοῦ τρίτου πολλαπλάσιον μὴ ὑπερέθῃ τοῦ τοῦ τετάρτου πολλαπλασίου, τότε τὸ πρῶτον πρὸς τὸ δεῦτερον μείζονα λόγον >έθειν λέγεται, >ἢ περ τὸ τρίτον πρὸς τὸ τέταρτον.
8. Ἀναλογία δὲ ἐν τρισὶν <όροις ἐλαθίστη ἐστίν.
9. <Ὅταν δὲ τρία μεγέθη ἀνάλογον >ἢ, τὸ πρῶτον πρὸς τὸ τρίτον διπλασίονα λόγον >έθειν λέγεται >ἢ περ πρὸς τὸ δεῦτερον.
10. <Ὅταν δὲ τέσσαρα μεγέθη ἀνάλογον >ἢ, τὸ πρῶτον πρὸς τὸ τέταρτον τριπλασίονα λόγον >έθειν λέγεται >ἢ περ πρὸς τὸ δεῦτερον, καὶ αἰεὶ ἐχῆς ὁμοίωξ, ὥς >ἂν ἡ ἀναλογία ὑπάρθῃ.
11. Ὁμόλογα μεγέθη λέγεται τὰ μὲν ἡγούμενα τοῖς ἡγουμένοις τὰ δὲ ἐπόμενα τοῖς ἐπομένοις.
12. Ἐναλλάχ λόγος ἐστὶ λήψις τοῦ ἡγουμένου πρὸς τὸ ἡγούμενον καὶ τοῦ ἐπομένου πρὸς τὸ ἐπόμενον.
13. Ἀνάπαλιν λόγος ἐστὶ λήψις τοῦ ἐπομένου ὡς ἡγουμένου πρὸς τὸ ἡγούμενον ὡς ἐπόμενον.
14. Σύνθεσις λόγου ἐστὶ λήψις τοῦ ἡγουμένου μετὰ τοῦ ἐπομένου ὡς ἐνδὸς πρὸς αὐτὸ τὸ ἐπόμενον.
15. Διαίρεσις λόγου ἐστὶ λήψις τῆς ὑπεροθῆς, <ἢ ὑπερέθει τὸ ἡγούμενον τοῦ ἐπομένου, πρὸς αὐτὸ τὸ ἐπόμενον.
16. Ἀναστροφή λόγου ἐστὶ λήψις τοῦ ἡγουμένου πρὸς τὴν ὑπεροθὴν, ἢ ὑπερέθει τὸ ἡγούμενον τοῦ ἐπομένου.
17. Δι> ῖσου λόγος ἐστὶ πλειόνων >όντων μεγεθῶν καὶ >άλλων αὐτοῖς >ῖσων τὸ πλῆθος σύνδυο λαμβανομένων καὶ ἐν τῷ αὐτῷ λόγῳ, <ὅταν >ἢ ὥς ἐν τοῖς πρώτοις μεγέθεσι τὸ πρῶτον πρὸς τὸ >έσθαι, ο<ύτως ἐν τοῖς δευτέροις μεγέθεσι τὸ πρῶτον πρὸς τὸ >έσθαι, >ἢ >άλλωξ; λήψις τῶν >άκρων καθ> ὑπεχαίρεσιν τῶν μέσων.
18. Τεταραγμένη δὲ ἀναλογία ἐστίν, <ὅταν τριῶν >όντων μεγεθῶν καὶ >άλλων αὐτοῖς >ῖσων τὸ πλῆθος γίνηται ὥς μὲν ἐν τοῖς πρώτοις μεγέθεσιν ἡγούμενον πρὸς ἐπόμενον, ο<ύτως ἐν τοῖς δευτέροις μεγέθεσιν ἡγούμενον πρὸς ἐπόμενον, ὥς δὲ ἐν τοῖς πρώτοις μεγέθεσιν ἐπόμενον πρὸς >άλλο τι, ο<ύτως ἐν τοῖς δευτέροις >άλλο τι πρὸς ἡγούμενον.

Definitions

1. A magnitude is a part of a(nother) magnitude, the lesser of the greater, when it measures the greater.[†]
2. And the greater (magnitude is) a multiple of the lesser when it is measured by the lesser.
3. A ratio is a certain type of relation with respect to size of two magnitudes of the same kind.[‡]
4. (Those) magnitudes are said to have a ratio with respect to one another which, being multiplied, are capable of exceeding one another.[§]
5. Magnitudes are said to be in the same ratio, the first to the second, and the third to the fourth, when equal multiples of the first and the third either both exceed, are both equal to, or are both less than, equal multiples of the second and the fourth, respectively, being taken in corresponding order, according to any kind of multiplication whatever.[¶]
6. And let magnitudes having the same ratio be called proportional.*
7. And when for equal multiples (as in Def. 5), the multiple of the first (magnitude) exceeds the multiple of the second, and the multiple of the third (magnitude) does not exceed the multiple of the fourth, (then) the first (magnitude) is said to have a greater ratio to the second than the third (magnitude has) to the fourth.
8. And a proportion in three terms is the smallest (possible).[§]
9. And when three magnitudes are proportional, the first is said to have to the third the squared^b ratio of that (it has) to the second.^{††}
10. And when four magnitudes are (continuously) proportional, the first is said to have to the fourth the cubed^{‡‡} ratio of that (it has) to the second.^{§§} And so on, similarly, in successive order, whatever the (continuous) proportion might be.
11. These magnitudes are said to be corresponding (magnitudes): the leading to the leading (of two ratios), and the following to the following.
12. An alternate ratio is a taking of the (ratio of the) leading (magnitude) to the leading (of two equal ratios), and (setting it equal to) the (ratio of the) following (magnitude) to the following.^{¶¶}
13. An inverse ratio is a taking of the (ratio of the) following (magnitude) as the leading and the leading (magnitude) as the following.^{**}
14. A composition of a ratio is a taking of the (ratio of the) leading plus the following (magnitudes), as one, to the following (magnitude) by itself.^{§§}
15. A separation of a ratio is a taking of the (ratio of the) excess by which the leading (magnitude) exceeds the following to the following (magnitude) by itself.^{bb}
16. A conversion of a ratio is a taking of the (ratio of the) leading (magnitude) to the excess by which the leading (magnitude) exceeds the following.^{†††}
17. There being several magnitudes, and other (magnitudes) of equal number to them, (which are) also in the same ratio taken two by two, a ratio via equality (i.e., ex aequali) occurs when as the first is to the last in the first (set of) magnitudes, so the first (is) to the last in the second (set of) magnitudes. Or alternately, (it is) a taking of the (ratio of the) outer (mag-

nitudes) by the removal of the inner (magnitudes).^{†††}

18. There being three magnitudes, and other (magnitudes) of equal number to them, a perturbed proportion occurs when as the leading is to the following in the first (set of) magnitudes, so the leading (is) to the following in the second (set of) magnitudes, and as the following (is) to some other (i.e., the remaining magnitude) in the first (set of) magnitudes, so some other (is) to the leading in the second (set of) magnitudes.^{§§§}

[†] In other words, α is said to be a part of β if $\beta = m\alpha$.

[‡] In modern notation, the ratio of two magnitudes, α and β , is denoted $\alpha : \beta$.

[§] In other words, α has a ratio with respect to β if $m\alpha > \beta$ and $n\beta > \alpha$, for some m and n .

[¶] In other words, $\alpha : \beta :: \gamma : \delta$ if and only if $m\alpha > n\beta$ whenever $m\gamma > n\delta$, and $m\alpha = n\beta$ whenever $m\gamma = n\delta$, and $m\alpha < n\beta$ whenever $m\gamma < n\delta$, for all m and n . This definition is the kernel of Eudoxus' theory of proportion, and is valid even if α, β , etc., are irrational.

* Thus if α and β have the same ratio as γ and δ then they are proportional. In modern notation, $\alpha : \beta :: \gamma : \delta$.

[§] In modern notation, a proportion in three terms— α, β , and γ —is written: $\alpha : \beta :: \beta : \gamma$.

^ᵇ Literally, “double”.

^{††} In other words, if $\alpha : \beta :: \beta : \gamma$ then $\alpha : \gamma :: \alpha^2 : \beta^2$.

^{‡‡} Literally, “triple”.

^{§§} In other words, if $\alpha : \beta :: \beta : \gamma :: \gamma : \delta$ then $\alpha : \delta :: \alpha^3 : \beta^3$.

^{¶¶} In other words, if $\alpha : \beta :: \gamma : \delta$ then the alternate ratio corresponds to $\alpha : \gamma :: \beta : \delta$.

** In other words, if $\alpha : \beta$ then the inverse ratio corresponds to $\beta : \alpha$.

^{§§} In other words, if $\alpha : \beta$ then the composed ratio corresponds to $\alpha + \beta : \beta$.

^{ᵇᵇ} In other words, if $\alpha : \beta$ then the separated ratio corresponds to $\alpha - \beta : \beta$.

^{†††} In other words, if $\alpha : \beta$ then the converted ratio corresponds to $\alpha : \alpha - \beta$.

^{‡‡‡} In other words, if α, β, γ are the first set of magnitudes, and δ, ϵ, ζ the second set, and $\alpha : \beta : \gamma :: \delta : \epsilon : \zeta$, then the ratio via equality (i.e. ex aequali) corresponds to $\alpha : \gamma :: \delta : \zeta$.

^{§§§} In other words, if α, β, γ are the first set of magnitudes, and δ, ϵ, ζ the second set, and $\alpha : \beta :: \delta : \epsilon$ as well as $\beta : \gamma :: \zeta : \delta$, then the proportion is said to be perturbed.

1.

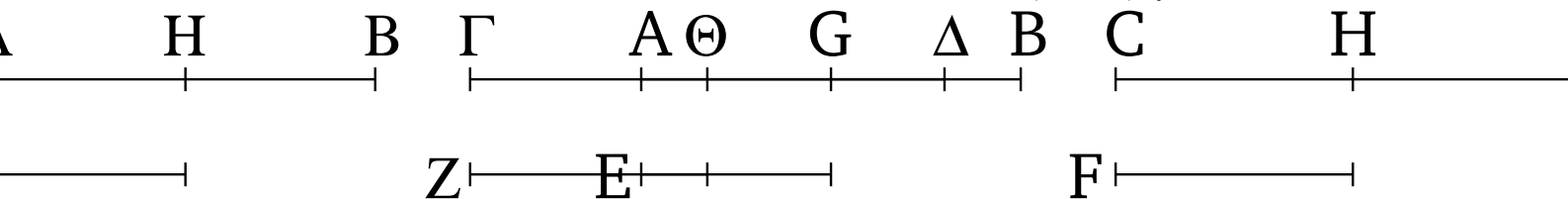
Proposition 1[†]

Ἐὰν >ῆ| ὅποσαοῦν μεγέθη ὅποσωνοῦν μεγεθῶν >ίσων τὸ πληθὸς <έκαστον ἐκάστου ισάκις πολλαπλάσιον, ὅσαπλάσιόν ἐστιν <ἐν τῶν μεγεθῶν ἐνόξ, τοσαυταπλάσια >έσται καὶ τὰ πάντα τῶν πάντων.

>Ἐστω ὅποσαοῦν μεγέθη τὰ AB, ΓΔ ὅποσωνοῦν μεγεθῶν τῶν E, Z >ίσων τὸ πληθὸς <έκαστον ἐκάστου ισάκις πολλαπλάσιον; λέγω, <ὅτι ὅσαπλάσιόν ἐστι τὸ AB τοῦ E, τοσαυταπλάσια >έσται καὶ τὰ AB, ΓΔ τῶν E, Z.

If there are any number of magnitudes whatsoever (which are) equal multiples, respectively, of some (other) magnitudes, of equal number (to them), (then) as many times as one of the (first) magnitudes is (divisible) by one (of the second), so many times will all (of the first magnitudes) also (be divisible) by all (of the second).

Let there be any number of magnitudes whatsoever, AB, CD , (which are) equal multiples, respectively, of some (other) magnitudes, E, F , of equal number (to them). I say that as many times as AB is (divisible) by E , so many times will AB, CD also be (divisible) by E, F .



Ἐπεὶ γὰρ ισάκις ἐστὶ πολλαπλάσιον τὸ AB τοῦ E καὶ τὸ ΓΔ τοῦ Z, <όσα >άρα ἐστὶν ἐν τῶ| AB μεγέθη >ίσα τῶ| E, τοσαῦτα καὶ ἐν τῶ| ΓΔ >ίσα τῶ| Z. διηρήσθω τὸ μὲν AB εἰς τὰ τῶ| E μεγέθη >ίσα τὰ AH, HB, τὸ δὲ ΓΔ εἰς τὰ τῶ| Z >ίσα τὰ ΓΘ, ΘΔ; >έσται δὴ >ίσον τὸ πληθὸς τῶν AH, HB τῶ| πληθὸς τῶν ΓΘ, ΘΔ. καὶ ἐπεὶ >ίσον ἐστὶ τὸ μὲν AH τῶ| E, τὸ δὲ ΓΘ τῶ| Z, >ίσον >άρα τὸ AH τῶ| E, καὶ τὰ AH, ΓΘ τοῖς E, Z. διὰ τὰ αὐτὰ δὴ >ίσον ἐστὶ τὸ HB τῶ| E, καὶ τὰ HB, ΘΔ τοῖς E, Z; <όσα >άρα ἐστὶν ἐν τῶ| AB >ίσα τῶ| E, τοσαῦτα

For since AB, CD are equal multiples of E, F , thus as many magnitudes as (there are) in AB equal to E , so many (are there) also in CD equal to F . Let AB be divided into magnitudes AG, GB , equal to E , and CD into (magnitudes) CH, HD , equal to F . So, the number of (divisions) AG, GB will be equal to the number of (divisions) CH, HD . And since AG is equal to E , and CH to F , AG (is) thus equal to E , and AG, CH to E, F . So, for the same (reasons), GB is equal to E , and GB, HD to E, F . Thus,

καὶ ἐν τοῖς AB , $\Gamma\Delta$ ἴσα τοῖς E , Z ; ὅσαπλάσιον ἄρα ἐστὶ τὸ AB τοῦ E , τοσαυταπλάσια ἔσται καὶ τὰ AB , $\Gamma\Delta$ τῶν E , Z .

Ἐὰν ἄρα $>\eta|$ ὅποσαοῦν μεγέθη ὅποσωνοῦν μεγεθῶν ἴσων τὸ πλῆθος ἑκάστων ἰσάκις πολλαπλάσιον, ὅσαπλάσιόν ἐστιν ἐν τῶν μεγεθῶν ἐνός, τοσαυταπλάσια ἔσται καὶ τὰ πάντα τῶν πάντων; ὅπερ >έδει δεῖχαι.

as many (magnitudes) as (there) are in AB equal to E , so many (are there) also in AB , CD equal to E , F . Thus, as many times as AB is (divisible) by E , so many times will AB , CD also be (divisible) by E , F .

Thus, if there are any number of magnitudes whatsoever (which are) equal multiples, respectively, of some (other) magnitudes, of equal number (to them), (then) as many times as one of the (first) magnitudes is (divisible) by one (of the second), so many times will all (of the first magnitudes) also (be divisible) by all (of the second). (Which is) the very thing it was required to show.

† In modern notation, this proposition reads $m\alpha + m\beta + \dots = m(\alpha + \beta + \dots)$.

2.

Ἐὰν πρῶτον δευτέρου ἰσάκις $>\eta|$ πολλαπλάσιον καὶ τρίτον τετάρτου, $>\eta|$ δὲ καὶ πέμπτου δευτέρου ἰσάκις πολλαπλάσιον καὶ ἑκτον τετάρτου, καὶ συντεθὲν πρῶτον καὶ πέμπτου δευτέρου ἰσάκις ἔσται πολλαπλάσιον καὶ τρίτον καὶ ἑκτον τετάρτου.

Πρῶτον γὰρ τὸ AB δευτέρου τοῦ Γ ἰσάκις ἔστω πολλαπλάσιον καὶ τρίτον τὸ ΔE τετάρτου τοῦ Z , ἔστω δὲ καὶ πέμπτου τὸ BH δευτέρου τοῦ Γ ἰσάκις πολλαπλάσιον καὶ ἑκτον τὸ $E\Theta$ τετάρτου τοῦ Z ; λέγω, ὅτι καὶ συντεθὲν πρῶτον καὶ πέμπτου τὸ AH δευτέρου τοῦ Γ ἰσάκις ἔσται πολλαπλάσιον καὶ τρίτον καὶ ἑκτον τὸ $\Delta\Theta$ τετάρτου τοῦ Z .

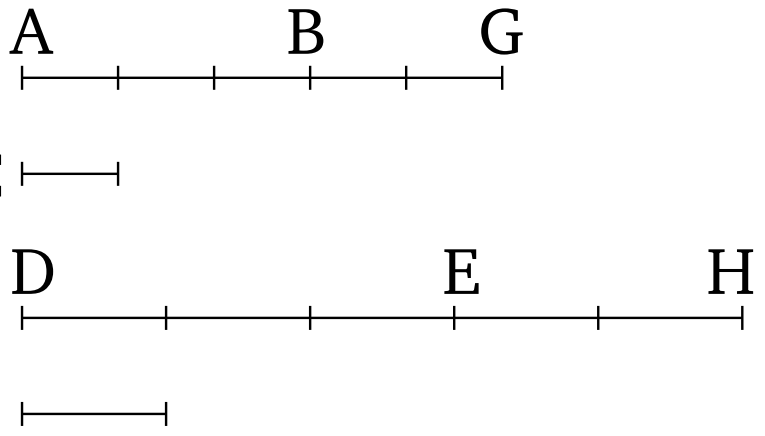
Ἐπεὶ γὰρ ἰσάκις ἐστὶ πολλαπλάσιον τὸ AB τοῦ Γ καὶ τὸ ΔE τοῦ Z , ὅσα ἄρα ἐστὶν ἐν τῷ AB ἴσα τῷ Γ , τοσαῦτα καὶ ἐν τῷ ΔE ἴσα τῷ Z . διὰ τὰ αὐτὰ δὴ καὶ ὅσα ἐστὶν ἐν τῷ BH ἴσα τῷ Γ , τοσαῦτα καὶ ἐν τῷ $E\Theta$ ἴσα τῷ Z ; ὅσα ἄρα ἐστὶν ἐν ὅλῳ τῷ AH ἴσα τῷ Γ , τοσαῦτα καὶ ἐν ὅλῳ τῷ $\Delta\Theta$ ἴσα τῷ Z ; ὅσαπλάσιον ἄρα ἐστὶ τὸ AH τοῦ Γ , τοσαυταπλάσιον ἔσται καὶ τὸ $\Delta\Theta$ τοῦ Z . καὶ συντεθὲν ἄρα πρῶτον καὶ πέμπτου τὸ AH δευτέρου τοῦ Γ ἰσάκις ἔσται πολλαπλάσιον καὶ τρίτον καὶ ἑκτον τὸ $\Delta\Theta$ τετάρτου τοῦ Z .

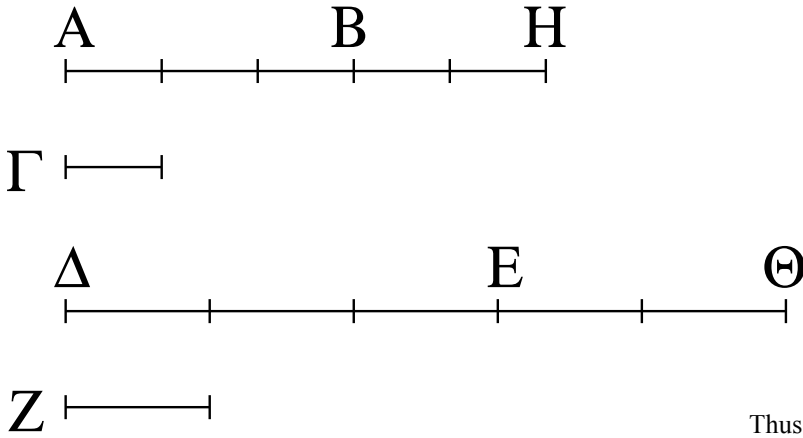
Proposition 2†

If a first (magnitude) and a third are equal multiples of a second and a fourth (respectively), and a fifth (magnitude) and a sixth (are) also equal multiples of the second and fourth (respectively), (then) the first (magnitude) and the fifth, being added together, and the third and the sixth, (being added together), will also be equal multiples of the second (magnitude) and the fourth (respectively).

For let a first (magnitude) AB and a third DE be equal multiples of a second C and a fourth F (respectively). And let a fifth (magnitude) BG and a sixth EH also be (other) equal multiples of the second C and the fourth F (respectively). I say that the first (magnitude) and the fifth, being added together, (to give) AG , and the third (magnitude) and the sixth, (being added together, to give) DH , will also be equal multiples of the second (magnitude) C and the fourth F (respectively).

For since AB and DE are equal multiples of C and F (respectively), thus as many (magnitudes) as (there) are in AB equal to C , so many (are there) also in DE equal to F . And so, for the same (reasons), as many (magnitudes) as (there) are in BG equal to C , so many (are there) also in EH equal to F . Thus, as many (magnitudes) as (there) are in the whole of AG equal to C , so many (are there) also in the whole of DH equal to F . Thus, as many times as AG is (divisible) by C , so many times will DH also be divisible by F . Thus, the first (magnitude) and the fifth, being added together, (to give) AG , and the third (magnitude) and the sixth, (being added together, to give) DH , will also be equal multiples of the second (magnitude) C and the fourth F (respectively).





Ἐὰν >άρα πρῶτον δευτέρου ισάκις >ῆ|| πολλαπλάσιον καὶ τρίτον τετάρτου, >ῆ|| δὲ καὶ πέμπτον δευτέρου ισάκις πολλαπλάσιον καὶ <έκτον τετάρτου, καὶ συντεθὲν πρῶτον καὶ πέμπτον δευτέρου ισάκις >έσται πολλαπλάσιον καὶ τρίτον καὶ <έκτον τετάρτου; <όπερ >έδει δεῖξαι.

† In modern notation, this proposition reads $m\alpha + n\alpha = (m+n)\alpha$.

3.

Ἐὰν πρῶτον δευτέρου ισάκις >ῆ|| πολλαπλάσιον καὶ τρίτον τετάρτου, ληφθῇ|| δὲ ισάκις πολλαπλάσια τοῦ τε πρώτου καὶ τρίτου, καὶ δι> >ίσου τῶν ληφθέντων ἐκάτερον ἐκατέρου ισάκις >έσται πολλαπλάσιον τὸ μὲν τοῦ δευτέρου τὸ δὲ τοῦ τετάρτου.

Πρῶτον γὰρ τὸ Α δευτέρου τοῦ Β ισάκις >έστω πολλαπλάσιον καὶ τρίτον τὸ Γ τετάρτου τοῦ Δ, καὶ εἰλήφθω τῶν Α, Γ ισάκις πολλαπλάσια τὰ ΕΖ, ΗΘ; λέγω, <ότι ισάκις ἐστὶ πολλαπλάσιον τὸ ΕΖ τοῦ Β καὶ τὸ ΗΘ τοῦ Δ.

Ἐπεὶ γὰρ ισάκις ἐστὶ πολλαπλάσιον τὸ ΕΖ τοῦ Α καὶ τὸ ΗΘ τοῦ Γ, <όσα >άρα ἐστὶν ἐν τῷ ΕΖ >ίσα τῷ Α, τοσαῦτα καὶ ἐν τῷ ΗΘ >ίσα τῷ Γ. διηρήσθω τὸ μὲν ΕΖ εἰς τὰ τῷ Α μεγέθη >ίσα τὰ ΕΚ, ΚΖ, τὸ δὲ ΗΘ εἰς τὰ τῷ Γ >ίσα τὰ ΗΛ, ΛΘ; >έσται δὴ >ίσον τὸ πλῆθος τῶν ΕΚ, ΚΖ τῷ πλῆθει τῶν ΗΛ, ΛΘ. καὶ ἐπεὶ ισάκις ἐστὶ πολλαπλάσιον τὸ Α τοῦ Β καὶ τὸ Γ τοῦ Δ, >ίσον δὲ τὸ μὲν ΕΚ τῷ Α, τὸ δὲ ΗΛ τῷ Γ, ισάκις >άρα ἐστὶ πολλαπλάσιον τὸ ΕΚ τοῦ Β καὶ τὸ ΗΛ τοῦ Δ, διὰ τὰ αὐτὰ δὴ ισάκις ἐστὶ πολλαπλάσιον τὸ ΚΖ τοῦ Β καὶ τὸ ΛΘ τοῦ Δ. ἐπεὶ οὖν πρῶτον τὸ ΕΚ δευτέρου τοῦ Β >ισάκις ἐστὶ πολλαπλάσιον καὶ τρίτον τὸ ΗΛ τετάρτου τοῦ Δ, >έστι δὲ καὶ πέμπτον τὸ ΚΖ δευτέρου τοῦ Β ισάκις πολλαπλάσιον καὶ <έκτον τὸ ΛΘ τετάρτου τοῦ Δ, καὶ συντεθὲν >άρα πρῶτον καὶ πέμπτον τὸ ΕΖ δευτέρου τοῦ Β ισάκις ἐστὶ πολλαπλάσιον καὶ τρίτον καὶ <έκτον τὸ ΗΘ τετάρτου τοῦ Δ.

Thus, if a first (magnitude) and a third are equal multiples of a second and a fourth (respectively), and a fifth (magnitude) and a sixth (are) also equal multiples of the second and fourth (respectively), (then) the first (magnitude) and the fifth, being added together, and the third and sixth, (being added together), will also be equal multiples of the second (magnitude) and the fourth (respectively). (Which is) the very thing it was required to show.

Proposition 3†

If a first (magnitude) and a third are equal multiples of a second and a fourth (respectively), and equal multiples are taken of the first and the third, (then), via equality, the (magnitudes) taken will also be equal multiples of the second (magnitude) and the fourth, respectively.

For let a first (magnitude) A and a third C be equal multiples of a second B and a fourth D (respectively), and let the equal multiples EF and GH be taken of A and C (respectively). I say that EF and GH are equal multiples of B and D (respectively).

For since EF and GH are equal multiples of A and C (respectively), thus as many (magnitudes) as (there) are in EF equal to A , so many (are there) also in GH equal to C . Let EF be divided into magnitudes EK, KF equal to A , and GH into (magnitudes) GL, LH equal to C . So, the number of (magnitudes) EK, KF will be equal to the number of (magnitudes) GL, LH . And since A and C are equal multiples of B and D (respectively), and EK (is) equal to A , and GL to C , EK and GL are thus equal multiples of B and D (respectively). So, for the same (reasons), KF and LH are equal multiples of B and D (respectively). Therefore, since the first (magnitude) EK and the third GL are equal multiples of the second B and the fourth D (respectively), and the fifth (magnitude) KF and the sixth LH are also equal multiples of the second B and the fourth D (respectively), then the first (magnitude) and fifth, being added together, (to give) EF , and the third (magnitude) and sixth, (being added together, to give) GH , are thus also equal multiples of the second (magnitude) B and the fourth D (respectively) [Prop. 5.2].

A |-----|

B |-----|

E |-----| K |-----| F

C |-----|

D |-----|

G |-----| L |-----| H

A |-----|

B |-----|

E |-----| K |-----| Z

Γ |-----|

Δ |-----|

H |-----| Λ |-----| Θ

Ἐάν >άρα πρῶτον δευτέρου ισάκιζ >ῆ| πολλαπλάσιον καὶ τρίτον τετάρτου, ληφθῆ| δὲ τοῦ πρώτου καὶ τρίτου ισάκιζ πολλαπλάσια, καὶ δι> >ίσου τῶν ληφθέντων ἑκάτερον ἑκατέρου ισάκιζ >έσται πολλαπλάσιον τὸ μὲν τοῦ δευτέρου τὸ δὲ τοῦ τετάρτου; <όπερ >έδει δεῖχαι.

† In modern notation, this proposition reads $m(n\alpha) = (m n)\alpha$.

4.

Ἐάν πρῶτον πρὸς δευτέρον τὸν αὐτὸν >έθῃ λόγον καὶ τρίτον πρὸς τέταρτον, καὶ τὰ ισάκιζ πολλαπλάσια τοῦ τε πρώτου καὶ τρίτου πρὸς τὰ ισάκιζ πολλαπλάσια τοῦ δευτέρου καὶ τετάρτου καθ> ὁποιοῦν πολλαπλασιασμὸν τὸν αὐτὸν <έχει λόγον ληφθέντα κατάλληλα.

Πρῶτον γὰρ τὸ A πρὸς δευτέρον τὸ B τὸν αὐτὸν ἔθετο λόγον καὶ τρίτον τὸ Γ πρὸς τέταρτον τὸ Δ, καὶ εἰλήφθω τῶν μὲν A, Γ ισάκιζ πολλαπλάσια τὰ E, Z, τῶν δὲ B, Δ >άλλα, <ὰ >έτυθεν, ισάκιζ πολλαπλάσια τὰ H, Θ; λέγω, <ότι ἐστὶν ὡς τὸ E πρὸς τὸ H, ο<ύτωξ τὸ Z πρὸς τὸ Θ.

Εἰλήφθω γὰρ τῶν μὲν E, Z ισάκιζ πολλαπλάσια τὰ K, Λ, τῶν δὲ H, Θ >άλλα, <ὰ >έτυθεν, ισάκιζ πολλαπλάσια τὰ M,

Thus, if a first (magnitude) and a third are equal multiples of a second and a fourth (respectively), and equal multiples are taken of the first and the third, (then), via equality, the (magnitudes) taken will also be equal multiples of the second (magnitude) and the fourth, respectively. (Which is) the very thing it was required to show.

Proposition 4[†]

If a first (magnitude) has the same ratio to a second that a third (has) to a fourth, (then) equal multiples of the first (magnitude) and the third will also have the same ratio to equal multiples of the second and the fourth, being taken in corresponding order, according to any kind of multiplication whatsoever.

For let a first (magnitude) *A* have the same ratio to a second *B* that a third *C* (has) to a fourth *D*. And let equal multiples *E* and *F* be taken of *A* and *C* (respectively), and other random equal multiples *G* and *H* of *B* and *D* (respectively). I say that as *E* (is) to *G*, so *F* (is) to *H*.

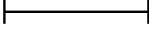
For let equal multiples *K* and *L* be taken of *E* and *F* (re-

N.

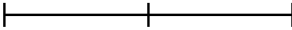
[Ka`i] >epe`i >is`akic >est`i pollapl'asion t`o m`en E to u A, t`o d`e Z to u G, ka`i e>ilhptai t wn E, Z >is`akic pollapl'asia t`a K, L, >is`akic >'ara >est`i pollapl'asion t`o K to u A ka`i t`o L to u G. di`a t`a a>ut`a d`h >is`akic >est`i pollapl'asion t`o M to u B ka`i t`o N to u D. ka`i >epe`i >est`in <wc t`o A pr`oc t`o B, o<'utwc t`o G pr`oc t`o D, ka`i e>ilhptai t wn m`en A, G >is`akic pollapl'asia t`a K, L, t wn d`e B, D >'alla, <'a >'etuqen, >is`akic pollapl'asia t`a M, N, e>i >'ara <uper'epei t`o K to u M, <uper'epei ka`i t`o L to u N, ka`i e>i >'ison, >'ison, ka`i e>i >'elatton, >'elatton. ka`i >esti t`a m`en K, L t wn E, Z >is`akic pollapl'asia, t`a d`e M, N t wn H, J >'alla, <'a >'etuqen, >is`akic pollapl'asia; >est`in >'ara <wc t`o E pr`oc t`o H, o<'utwc t`o Z pr`oc t`o J.

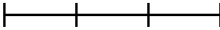
spectively), and other random equal multiples M and N of G and H (respectively).

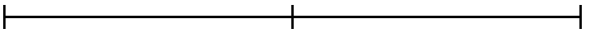
[And] since E and F are equal multiples of A and C (respectively), and the equal multiples K and L have been taken of E and F (respectively), K and L are thus equal multiples of A and C (respectively) [Prop. 5.3]. So, for the same (reasons), M and N are equal multiples of B and D (respectively). And since as A is to B , so C (is) to D , and the equal multiples K and L have been taken of A and C (respectively), and the other random equal multiples M and N of B and D (respectively), (then) if K exceeds M (then) L also exceeds N , and if (K is) equal (to M then L is also) equal (to N), and if (K is) less (than M then L is also) less (than N) [Def. 5.5]. And K and L are equal multiples of E and F (respectively), and M and N other random equal multiples of G and H (respectively). Thus, as E (is) to G , so F (is) to H [Def. 5.5].

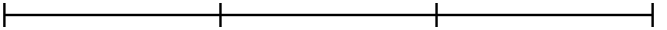
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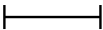
B 

E 

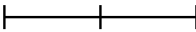
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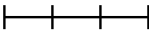
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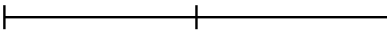
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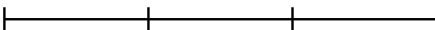
C 

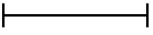
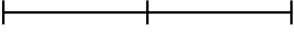
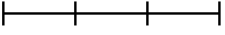
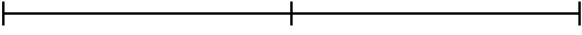
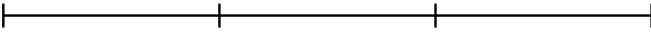
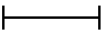
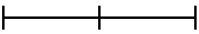
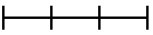
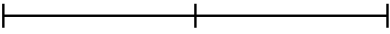
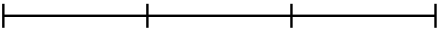
D 

F 

H 

L 

N 

A B E H K M Γ Δ Z Θ Λ N 

Ἐάν >άρα πρώτον πρὸς δεύτερον τὸν αὐτὸν >έθῃ λόγον καὶ τρίτον πρὸς τέταρτον, καὶ τὰ ἰσάκις πολλαπλάσια τοῦ τε πρώτου καὶ τρίτου πρὸς τὰ ἰσάκις πολλαπλάσια τοῦ δευτέρου καὶ τετάρτου τὸν αὐτὸν <έχει λόγον καθ> ὁποιοῦν πολλαπλασιασμὸν ληφθέντα κατάλληλα; <όπερ >έδει δεῖχαι.

† In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$ then $m\alpha : n\beta :: m\gamma : n\delta$, for all m and n .

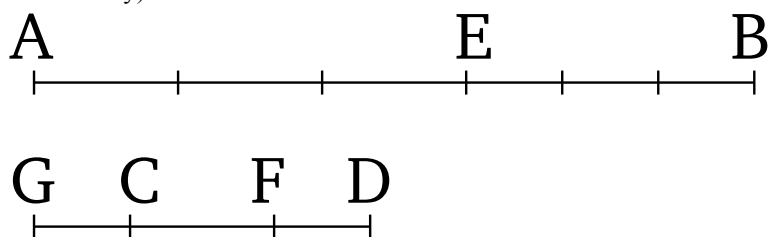
Thus, if a first (magnitude) has the same ratio to a second that a third (has) to a fourth, (then) equal multiples of the first (magnitude) and the third will also have the same ratio to equal multiples of the second and the fourth, being taken in corresponding order, according to any kind of multiplication whatsoever. (Which is) the very thing it was required to show.

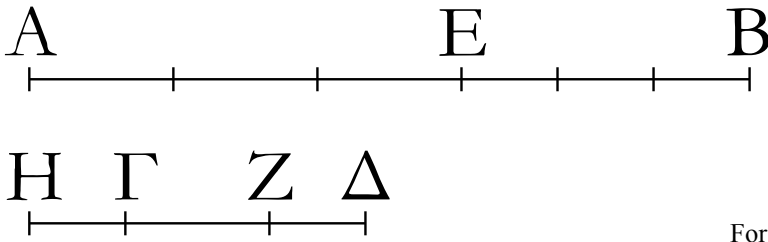
5.

Proposition 5[†]

Ἐάν μέγεθος μεγέθους ἰσάκις >ῆ πολλαπλάσιον, <όπερ ἀφαιρεθὲν ἀφαιρεθέντος, καὶ τὸ λοιπὸν τοῦ λοιποῦ ἰσάκις >έσται πολλαπλάσιον, ὅσαπλάσιόν ἐστι τὸ <όλον τοῦ <όλου.

If a magnitude is the same multiple of a magnitude that a (part) taken away (is) of a (part) taken away (respectively), (then) the remainder will also be the same multiple of the remainder as that which the whole (is) of the whole (respectively).





Μέγεθος γάρ τὸ AB μεγέθους τοῦ ΓΔ ἰσάκις >έστω πολλαπλάσιον τοῦ ΓΔ. <όπερ ἀφαιρεθὲν τὸ AE ἀφαιρεθέντος τοῦ ΓΖ; λέγω, <ὅτι καὶ λοιπὸν τὸ EB λοιποῦ τοῦ ΖΔ ἰσάκις >έσται πολλαπλάσιον, ὅσαπλάσιόν ἐστιν <ὅλον τὸ AB <όλου τοῦ ΓΔ.

Ὅσαπλάσιον γάρ ἐστι τὸ AE τοῦ ΓΖ, τοσαυταπλάσιον γεγενέτω καὶ τὸ EB τοῦ ΓΗ.

Καὶ ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ AE τοῦ ΓΖ καὶ τὸ EB τοῦ ΗΓ, ἰσάκις >άρα ἐστὶ πολλαπλάσιον τὸ AE τοῦ ΓΖ καὶ τὸ AB τοῦ ΗΖ. κεῖται δὲ ἰσάκις πολλαπλάσιον τὸ AE τοῦ ΓΖ καὶ τὸ AB τοῦ ΓΔ. ἰσάκις >άρα ἐστὶ πολλαπλάσιον τὸ AB ἑκατέρου τῶν ΗΖ, ΓΔ; >ἴσον >άρα τὸ ΗΖ τῷ ΓΔ. κοινὸν ἀφηρήσθω τὸ ΓΖ; λοιπὸν >άρα τὸ ΗΓ λοιπῷ τῷ ΖΔ >ἴσον ἐστίν. καὶ ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ AE τοῦ ΓΖ καὶ τὸ EB τοῦ ΗΓ, >ἴσον δὲ τὸ ΗΓ τῷ ΔΖ, ἰσάκις >άρα ἐστὶ πολλαπλάσιον τὸ AE τοῦ ΓΖ καὶ τὸ EB τοῦ ΖΔ. ἰσάκις δὲ ὑπόκειται πολλαπλάσιον τὸ AE τοῦ ΓΖ καὶ τὸ AB τοῦ ΓΔ; ἰσάκις >άρα ἐστὶ πολλαπλάσιον τὸ EB τοῦ ΖΔ καὶ τὸ AB τοῦ ΓΔ. καὶ λοιπὸν >άρα τὸ EB λοιποῦ τοῦ ΖΔ ἰσάκις >έσται πολλαπλάσιον, ὅσαπλάσιόν ἐστιν <ὅλον τὸ AB <όλου τοῦ ΓΔ.

Ἐὰν >άρα μέγεθος μεγέθους ἰσάκις >ἢ| πολλαπλάσιον, <όπερ ἀφαιρεθὲν ἀφαιρεθέντος, καὶ τὸ λοιπὸν τοῦ λοιποῦ ἰσάκις >έσται πολλαπλάσιον, ὅσαπλάσιόν ἐστι καὶ τὸ <ὅλον τοῦ <όλου; <όπερ >έδει δεῖχαι.

For let the magnitude AB be the same multiple of the magnitude CD that the (part) taken away AE (is) of the (part) taken away CF (respectively). I say that the remainder EB will also be the same multiple of the remainder FD as that which the whole AB (is) of the whole CD (respectively).

For as many times as AE is (divisible) by CF , so many times let EB also be made (divisible) by CG .

And since AE and EB are equal multiples of CF and GC (respectively), AE and AB are thus equal multiples of CF and GF (respectively) [Prop. 5.1]. And AE and AB are assumed (to be) equal multiples of CF and CD (respectively). Thus, AB is an equal multiple of each of GF and CD . Thus, GF (is) equal to CD . Let CF be subtracted from both. Thus, the remainder GC is equal to the remainder FD . And since AE and EB are equal multiples of CF and GC (respectively), and GC (is) equal to DF , AE and EB are thus equal multiples of CF and FD (respectively). And AE and AB are assumed (to be) equal multiples of CF and CD (respectively). Thus, EB and AB are equal multiples of FD and CD (respectively). Thus, the remainder EB will also be the same multiple of the remainder FD as that which the whole AB (is) of the whole CD (respectively).

Thus, if a magnitude is the same multiple of a magnitude that a (part) taken away (is) of a (part) taken away (respectively), (then) the remainder will also be the same multiple of the remainder as that which the whole (is) of the whole (respectively). (Which is) the very thing it was required to show.

† In modern notation, this proposition reads $m\alpha - m\beta = m(\alpha - \beta)$.

6.

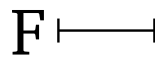
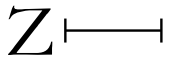
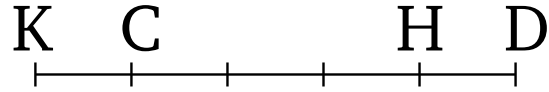
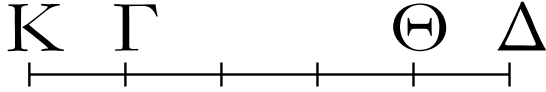
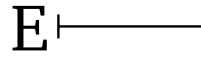
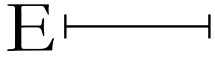
Ἐὰν δύο μεγέθη δύο μεγεθῶν ἰσάκις >ἢ| πολλαπλάσια, καὶ ἀφαιρεθέντα τινὰ τῶν αὐτῶν ἰσάκις >ἢ| πολλαπλάσια, καὶ τὰ λοιπὰ τοῖς αὐτοῖς >ήτοι >ἴσα ἐστὶν >ἢ ἰσάκις αὐτῶν πολλαπλάσια.

Δύο γὰρ μεγέθη τὰ AB, ΓΔ δύο μεγεθῶν τῶν E, Z ἰσάκις >έστω πολλαπλάσια, καὶ ἀφαιρεθέντα τὰ AH, ΓΘ τῶν αὐτῶν τῶν E, Z ἰσάκις >έστω πολλαπλάσια; λέγω, <ὅτι καὶ λοιπὰ τὰ HB, ΘΔ τοῖς E, Z >ήτοι >ἴσα ἐστὶν >ἢ ἰσάκις αὐτῶν πολλαπλάσια.

Proposition 6†

If two magnitudes are equal multiples of two (other) magnitudes, and some (parts) taken away (from the former magnitudes) are equal multiples of the latter (magnitudes, respectively), (then) the remainders are also either equal to the latter (magnitudes), or (are) equal multiples of them (respectively).

For let two magnitudes AB and CD be equal multiples of two magnitudes E and F (respectively). And let the (parts) taken away (from the former) AG and CH be equal multiples of E and F (respectively). I say that the remainders GB and HD are also either equal to E and F (respectively), or (are) equal multiples of them.



>Ἐστω γὰρ πρότερον τὸ HB τῶ| E >ῖσον; λέγω, <ὅτι καὶ τὸ ΘΔ τῶ| Z >ῖσον ἐστίν.

Κείσθω γὰρ τῶ| Z >ῖσον τὸ ΓΚ. ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ ΑΗ τοῦ Ε καὶ τὸ ΓΘ τοῦ Ζ, >ῖσον δὲ τὸ μὲν HB τῶ| Ε, τὸ δὲ ΚΓ τῶ| Ζ, ἰσάκις >άρα ἐστὶ πολλαπλάσιον τὸ ΑΒ τοῦ Ε καὶ τὸ ΚΘ τοῦ Ζ. ἰσάκις δὲ ὑπόκειται πολλαπλάσιον τὸ ΑΒ τοῦ Ε καὶ τὸ ΓΔ τοῦ Ζ; >ῖσάκις >άρα ἐστὶ πολλαπλάσιον τὸ ΚΘ τοῦ Ζ καὶ τὸ ΓΔ τοῦ Ζ. ἐπεὶ ο>ὖν ἑκάτερον τῶν ΚΘ, ΓΔ τοῦ Ζ ἰσάκις ἐστὶ πολλαπλάσιον, >ῖσον >άρα ἐστὶ τὸ ΚΘ τῶ| ΓΔ. κοινὸν ἀφηρήσθω τὸ ΓΘ; λοιπὸν >άρα τὸ ΚΓ λοιπῶ| τῶ| ΘΔ >ῖσον ἐστίν. ἀλλὰ τὸ Ζ τῶ| ΚΓ ἐστὶν >ῖσον; καὶ τὸ ΘΔ >άρα τῶ| Ζ >ῖσον ἐστίν. <ὥστε εἰ τὸ HB τῶ| Ε >ῖσον ἐστίν, καὶ τὸ ΘΔ >ῖσον >έσται τῶ| Ζ.

Ὅμοίως δὴ δείχομεν, <ὅτι, κ>ᾶν πολλαπλάσιον >ῆ| τὸ HB τοῦ Ε, τοσανταπλάσιον >έσται καὶ τὸ ΘΔ τοῦ Ζ.

Ἐὰν >άρα δύο μεγέθη δύο μεγεθῶν ἰσάκις >ῆ| πολλαπλάσια, καὶ ἀφαιρεθέντα τινὰ τῶν αὐτῶν ἰσάκις >ῆ| πολλαπλάσια, καὶ τὰ λοιπὰ τοῖς αὐτοῖς >ήτοι >ῖσα ἐστὶν >ῆ ἰσάκις αὐτῶν πολλαπλάσια; <όπερ >έδει δεῖχαι.

For let GB be, first of all, equal to E . I say that HD is also equal to F .

For let CK be made equal to F . Since AG and CH are equal multiples of E and F (respectively), and GB (is) equal to E , and KC to F , AB and KH are thus equal multiples of E and F (respectively) [Prop. 5.2]. And AB and CD are assumed (to be) equal multiples of E and F (respectively). Thus, KH and CD are equal multiples of F and F (respectively). Therefore, KH and CD are each equal multiples of F . Thus, KH is equal to CD . Let CH be taken away from both. Thus, the remainder KC is equal to the remainder HD . But, F is equal to KC . Thus, HD is also equal to F . Hence, if GB is equal to E , (then) HD will also be equal to F .

So, similarly, we can show that even if GB is a multiple of E , (then) HD will also be the same multiple of F .

Thus, if two magnitudes are equal multiples of two (other) magnitudes, and some (parts) taken away (from the former magnitudes) are equal multiples of the latter (magnitudes, respectively), (then) the remainders are also either equal to the latter (magnitudes), or (are) equal multiples of them (respectively). (Which is) the very thing it was required to show.

† In modern notation, this proposition reads $m\alpha - n\alpha = (m - n)\alpha$.

7.

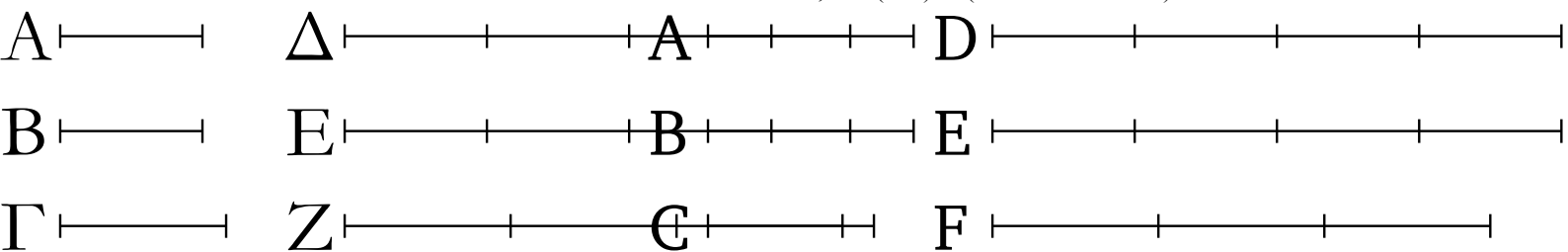
Τὰ >ῖσα πρὸς τὸ αὐτὸ τὸν αὐτὸν >έθει λόγον καὶ τὸ αὐτὸ πρὸς τὰ >ῖσα.

>Ἐστω >ῖσα μεγέθη τὰ Α, Β, >άλλο δέ τι, <ὃ >έτυθεν, μέγεθος τὸ Γ; λέγω, <ὅτι ἑκάτερον τῶν Α, Β πρὸς τὸ Γ τὸν αὐτὸν >έθει λόγον, καὶ τὸ Γ πρὸς ἑκάτερον τῶν Α, Β.

Proposition 7

Equal (magnitudes) have the same ratio to the same (magnitude), and the latter (magnitude has the same ratio) to the equal (magnitudes).

Let A and B be equal magnitudes, and C some other random magnitude. I say that A and B each have the same ratio to C , and (that) C (has the same ratio) to each of A and B .



Εἰλήφθω γὰρ τῶν μὲν Α, Β ἰσάκις πολλαπλάσια τὰ Δ, Ε, τοῦ δὲ Γ >άλλο, <ὃ >έτυθεν, πολλαπλάσιον τὸ Ζ.

Ἐπεὶ ο>ὖν ἰσάκις ἐστὶ πολλαπλάσιον τὸ Δ τοῦ Α καὶ τὸ

For let the equal multiples D and E be taken of A and B (respectively), and the other random multiple F of C .

Therefore, since D and E are equal multiples of A and B

Ε τοῦ Β, >ῖσον δὲ τὸ Α τῷ Β, >ῖσον >άρα καὶ τὸ Δ τῷ Ε. >άλλο δέ, <ὸ >έτυθεν, τὸ Ζ. εἰ >άρα ὑπερέθει τὸ Δ τοῦ Ζ, ὑπερέθει καὶ τὸ Ε τοῦ Ζ, καὶ εἰ >ῖσον, >ῖσον, καὶ εἰ >έλαττον, >έλαττον. καὶ ἐστὶ τὰ μὲν Δ, Ε τῶν Α, Β ἰσάκις πολλαπλάσια, τὸ δὲ Ζ τοῦ Γ >άλλο, <ὸ >έτυθεν, πολλαπλάσιον; >έστιν >άρα ὡς τὸ Α πρὸς τὸ Γ, ο<ύτωξ τὸ Β πρὸς τὸ Γ.

Λέγω [δὴ], <ὅτι καὶ τὸ Ε πρὸς ἐκάτερον τῶν Α, Β τὸν αὐτὸν >έθει λόγον.

Τῶν γὰρ αὐτῶν κατασκευασθέντων ὁμοίωξ δείχομεν, <ὅτι >ῖσον ἐστὶ τὸ Δ τῷ Ε; >άλλο δέ τι τὸ Ζ; εἰ >άρα ὑπερέθει τὸ Ζ τοῦ Δ, ὑπερέθει καὶ τοῦ Ε, καὶ εἰ >ῖσον, >ῖσον, καὶ εἰ >έλαττον, >έλαττον. καὶ ἐστὶ τὸ μὲν Ζ τοῦ Γ πολλαπλάσιον, τὰ δὲ Δ, Ε τῶν Α, Β >άλλα, <ὰ >έτυθεν, ἰσάκις πολλαπλάσια; >έστιν >άρα ὡς τὸ Γ πρὸς τὸ Α, ο<ύτωξ τὸ Γ πρὸς τὸ Β.

Τὰ >ῖσα >άρα πρὸς τὸ αὐτὸ τὸν αὐτὸν >έθει λόγον καὶ τὸ αὐτὸ πρὸς τὰ >ῖσα.

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, <ὅτι ἔαν μεγέθη τινὰ ἀνάλογον >ῆ|, καὶ ἀνάπαλιν ἀνάλογον >έσται. <όπερ >έδει δεῖχαι.

† The Greek text has “Ε”, which is obviously a mistake.

‡ In modern notation, this corollary reads that if $\alpha : \beta :: \gamma : \delta$ then $\beta : \alpha :: \delta : \gamma$.

8.

Τῶν ἀνίσων μεγεθῶν τὸ μείζον πρὸς τὸ αὐτὸ μείζονα λόγον >έθει >ήπερ τὸ >έλαττον. καὶ τὸ αὐτὸ πρὸς τὸ >έλαττον μείζονα λόγον >έθει >ήπερ πρὸς τὸ μείζον.

>Ἐστω >άνισα μεγέθη τὰ ΑΒ, Γ, καὶ >έστω μείζον τὸ ΑΒ, >άλλο δέ, <ὸ >έτυθεν, τὸ Δ; λέγω, <ὅτι τὸ ΑΒ πρὸς τὸ Δ μείζονα λόγον >έθει >ήπερ τὸ Γ πρὸς τὸ Δ, καὶ τὸ Δ πρὸς τὸ Γ μείζονα λόγον >έθει >ήπερ πρὸς τὸ ΑΒ.

(respectively), and A (is) equal to B , D (is) thus also equal to E . And F (is) different, at random. Thus, if D exceeds F (then) E also exceeds F , and if (D is) equal (to F then E is also) equal (to F), and if (D is) less (than F then E is also) less (than F). And D and E are equal multiples of A and B (respectively), and F another random multiple of C . Thus, as A (is) to C , so B (is) to C [Def. 5.5].

[So] I say that $C^\dagger$ also has the same ratio to each of A and B .

For, similarly, we can show, by the same construction, that D is equal to E . And F (has) some other (value). Thus, if F exceeds D then it also exceeds E , and if (F is) equal (to D then it is also) equal (to E), and if (F is) less (than D then it is also) less (than E). And F is a multiple of C , and D and E other random equal multiples of A and B . Thus, as C (is) to A , so C (is) to B [Def. 5.5].

Thus, equal (magnitudes) have the same ratio to the same (magnitude), and the latter (magnitude has the same ratio) to the equal (magnitudes).

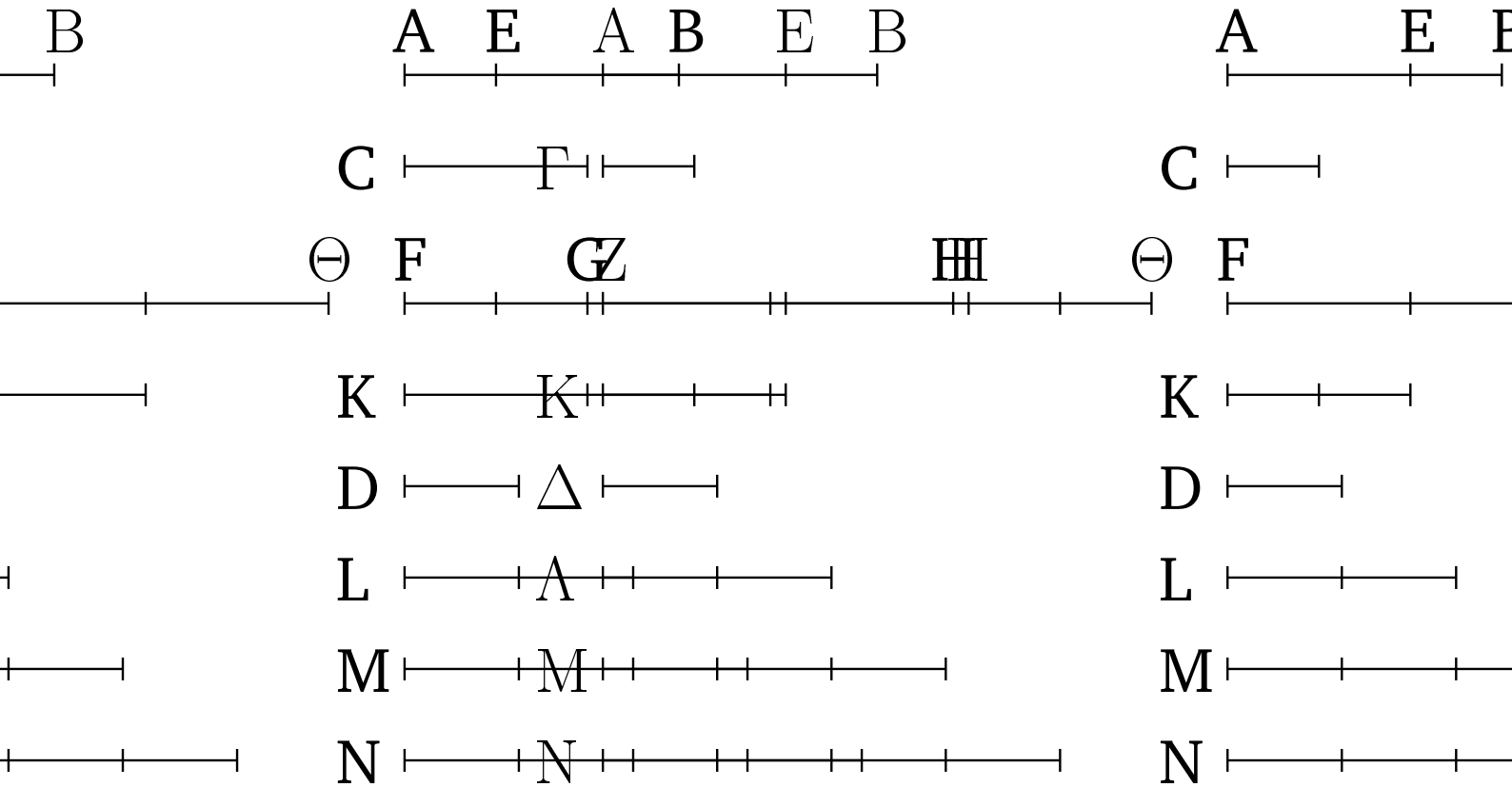
Corollary[‡]

So (it is) clear, from this, that if some magnitudes are proportional, (then) they will also be proportional inversely. (Which is) the very thing it was required to show.

Proposition 8

For unequal magnitudes, the greater (magnitude) has a greater ratio than the lesser to the same (magnitude). And the latter (magnitude) has a greater ratio to the lesser (magnitude) than to the greater.

Let AB and C be unequal magnitudes, and let AB be the greater (of the two), and D another random magnitude. I say that AB has a greater ratio to D than C (has) to D , and (that) D has a greater ratio to C than (it has) to AB .



Ἐπεὶ γὰρ μείζον ἐστὶ τὸ AB τοῦ Γ, κείσθω τῷ Γ ἴσον τὸ BE; τὸ δὲ >έλασσον τῶν AE, EB πολλαπλασιαζόμενον >έσται ποτὲ τοῦ Δ μείζον. >έστω πρότερον τὸ AE >έλαττον τοῦ EB, καὶ πεπολλαπλασιάσθω τὸ AE, καὶ >έστω αὐτοῦ πολλαπλάσιον τὸ ZH μείζον >όν τοῦ Δ, καὶ ὅσαπλάσιόν ἐστι τὸ ZH τοῦ AE, τοσαυταπλάσιον γεγόνετω καὶ τὸ μὲν HΘ τοῦ EB τὸ δὲ K τοῦ Γ; καὶ εἰλήφθω τοῦ Δ διπλάσιον μὲν τὸ Λ, τριπλάσιον δὲ τὸ Μ, καὶ ἔχῃξ ἐνὶ πλεῖον, <έωξ >άν τὸ λαμβανόμενον πολλαπλάσιον μὲν γένηται τοῦ Δ, πρώτῳξ δὲ μείζον τοῦ Κ. εἰλήφθω, καὶ >έστω τὸ Ν τετραπλάσιον μὲν τοῦ Δ, πρώτῳξ δὲ μείζον τοῦ Κ.

Ἐπεὶ ο>ύν τὸ Κ τοῦ Ν πρώτῳξ ἐστὶν >έλαττον, τὸ Κ >άρα τοῦ Μ ο>ύκ ἐστὶν >έλαττον. καὶ ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ ZH τοῦ AE καὶ τὸ HΘ τοῦ EB, ἰσάκις >άρα ἐστὶ πολλαπλάσιον τὸ ZH τοῦ AE καὶ τὸ ZΘ τοῦ AB. ἰσάκις δὲ ἐστὶ πολλαπλάσιον τὸ ZH τοῦ AE καὶ τὸ Κ τοῦ Γ; ἰσάκις >άρα ἐστὶ πολλαπλάσιον τὸ ZΘ τοῦ AB καὶ τὸ Κ τοῦ Γ. τὰ ZΘ, Κ >άρα τῶν AB, Γ ἰσάκις ἐστὶ πολλαπλάσια. πάλιν, ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ HΘ τοῦ EB καὶ τὸ Κ τοῦ Γ, >ἴσον δὲ τὸ EB τῷ Γ, >ἴσον >άρα καὶ τὸ HΘ τῷ Κ. τὸ δὲ Κ τοῦ Μ ο>ύκ ἐστὶν >έλαττον; οὐδ' >άρα τὸ HΘ τοῦ Μ >έλαττόν ἐστιν. μείζον δὲ τὸ ZH τοῦ Δ; <όλον >άρα τὸ ZΘ συναμφοτέρων τῶν Δ, Μ μείζον ἐστὶν. ἀλλὰ συναμφοτέρα τὰ Δ, Μ τῷ Ν ἐστὶν >ἴσα, ἐπειδὴ ἵππερ τὸ Μ τοῦ Δ τριπλάσιόν ἐστιν, συναμφοτέρα δὲ τὰ Μ, Δ τοῦ Δ ἐστὶ τετραπλάσια, >έστι δὲ καὶ τὸ Ν τοῦ Δ τετραπλάσιον; συναμφοτέρα >άρα τὰ Μ, Δ τῷ Ν >ἴσα ἐστὶν. ἀλλὰ τὸ ZΘ τῶν Μ, Δ μείζον ἐστὶν; τὸ ZΘ >άρα τοῦ Ν ὑπερέθει; τὸ δὲ Κ τοῦ Ν οὐθ' ὑπερέθει. καὶ ἐστὶ τὰ μὲν ZΘ, Κ τῶν AB, Γ ἰσάκις πολλαπλάσια, τὸ δὲ Ν τοῦ Δ >άλλο, <ὸ >έτυθεν, πολλαπλάσιον; τὸ AB >άρα πρὸς τὸ Δ μείζονα λόγον >έθει >ήπερ τὸ Γ πρὸς τὸ Δ.

Λέγω δὴ, <ὅτι καὶ τὸ Δ πρὸς τὸ Γ μείζονα λόγον >έθει >ήπερ τὸ Δ πρὸς τὸ AB.

Τῶν γὰρ αὐτῶν κατασκευασθέντων ὁμοίως δείχομεν, <ὅτι

For since AB is greater than C , let BE be made equal to C . So, the lesser of AE and EB , being multiplied, will sometimes be greater than D [Def. 5.4]. First of all, let AE be less than EB , and let AE be multiplied, and let FG be a multiple of it which (is) greater than D . And as many times as FG is (divisible) by AE , so many times let GH also become (divisible) by EB , and K by C . And let the double multiple L of D be taken, and the triple multiple M , and several more, (each increasing) in order by one, until the (multiple) taken becomes the first multiple of D (which is) greater than K . Let it be taken, and let it also be the quadruple multiple N of D —the first (multiple) greater than K .

Therefore, since K is less than N first, K is thus not less than M . And since FG and GH are equal multiples of AE and EB (respectively), FG and FH are thus equal multiples of AE and AB (respectively) [Prop. 5.1]. And FG and K are equal multiples of AE and C (respectively). Thus, FH and K are equal multiples of AB and C (respectively). Thus, FH , K are equal multiples of AB , C . Again, since GH and K are equal multiples of EB and C , and EB (is) equal to C , GH (is) thus also equal to K . And K is not less than M . Thus, GH not less than M either. And FG (is) greater than D . Thus, the whole of FH is greater than D and M (added) together. But, D and M (added) together is equal to N , inasmuch as M is three times D , and M and D (added) together is four times D , and N is also four times D . Thus, M and D (added) together is equal to N . But, FH is greater than M and D . Thus, FH exceeds N . And K does not exceed N . And FH , K are equal multiples of AB , C , and N another random multiple of D . Thus, AB has a greater ratio to D than C (has) to D [Def. 5.7].

So, I say that D also has a greater ratio to C than D (has) to AB .

For, similarly, by the same construction, we can show that

τὸ μὲν N τοῦ K ὑπερέθει, τὸ δὲ N τοῦ $Z\Theta$ οὐθ' ὑπερέθει. καὶ ἐστὶ τὸ μὲν N τοῦ Δ πολλαπλάσιον, τὰ δὲ $Z\Theta$, K τῶν AB , Γ >άλλα, <ὰ >έτυθεν, ἰσάκις πολλαπλάσια; τὸ Δ >άρα πρὸς τὸ Γ μείζονα λόγον >έθει >ήπερ τὸ Δ πρὸς τὸ AB .

Ἀλλὰ δὴ τὸ AE τοῦ EB μείζον >έστω. τὸ δὴ >έλαττον τὸ EB πολλαπλασιαζόμενον >έσται ποτὲ τοῦ Δ μείζον. πεπολλαπλασιάσθω, καὶ >έστω τὸ $H\Theta$ πολλαπλάσιον μὲν τοῦ EB , μείζον δὲ τοῦ Δ ; καὶ ὅσαπλάσιόν ἐστι τὸ $H\Theta$ τοῦ EB , τοσαυταπλάσιον γεγονέτω καὶ τὸ μὲν ZH τοῦ AE , τὸ δὲ K τοῦ Γ . ὁμοίως δὴ δείχομεν, <ὅτι τὰ $Z\Theta$, K τῶν AB , Γ ἰσάκις ἐστὶ πολλαπλάσια; καὶ εἰλήφθω ὁμοίως τὸ N πολλαπλάσιον μὲν τοῦ Δ , πρῶτως δὲ μείζον τοῦ ZH ; <ώστε πάλιν τὸ ZH τοῦ M ο>ύκ ἐστιν >έλασσον. μείζον δὲ τὸ $H\Theta$ τοῦ Δ ; <όλον >άρα τὸ $Z\Theta$ τῶν Δ , M , τουτέστι τοῦ N , ὑπερέθει. τὸ δὲ K τοῦ N οὐθ' ὑπερέθει, ἐπειδήπερ καὶ τὸ ZH μείζον >όν τοῦ $H\Theta$, τουτέστι τοῦ K , τοῦ N οὐθ' ὑπερέθει. καὶ ὡσαύτως κατακολουθοῦντες τοῖς ἐπάνω περαίνομεν τὴν ἀπόδειξιν.

Τῶν >άρα ἀνίσων μεγεθῶν τὸ μείζον πρὸς τὸ αὐτὸ μείζονα λόγον >έθει >ήπερ τὸ >έλαττον; καὶ τὸ αὐτὸ πρὸς τὸ >έλαττον μείζονα λόγον >έθει >ήπερ πρὸς τὸ μείζον; <όπερ >έδει δεῖξαι.

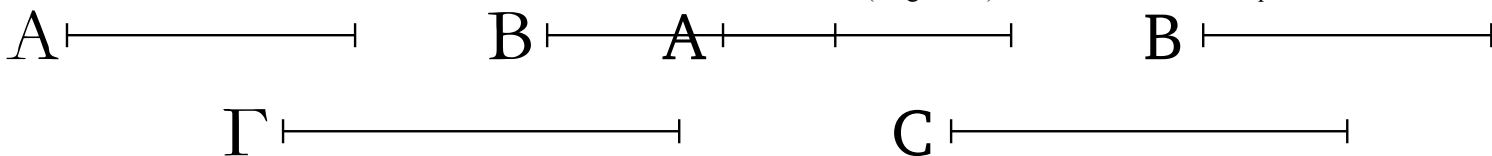
N exceeds K , and N does not exceed FH . And N is a multiple of D , and FH , K other random equal multiples of AB , C (respectively). Thus, D has a greater ratio to C than D (has) to AB [Def. 5.5].

And so let AE be greater than EB . So, the lesser, EB , being multiplied, will sometimes be greater than D . Let it be multiplied, and let GH be a multiple of EB (which is) greater than D . And as many times as GH is (divisible) by EB , so many times let FG also become (divisible) by AE , and K by C . So, similarly (to the above), we can show that FH and K are equal multiples of AB and C (respectively). And, similarly (to the above), let the multiple N of D , (which is) the first (multiple) greater than FG , be taken. So, FG is again not less than M . And GH (is) greater than D . Thus, the whole of FH exceeds D and M , that is to say N . And K does not exceed N , inasmuch as FG , which (is) greater than GH —that is to say, K —also does not exceed N . And, following the above (arguments), we (can) complete the proof in the same manner.

Thus, for unequal magnitudes, the greater (magnitude) has a greater ratio than the lesser to the same (magnitude). And the latter (magnitude) has a greater ratio to the lesser (magnitude) than to the greater. (Which is) the very thing it was required to show.

9.

Τὰ πρὸς τὸ αὐτὸ τὸν αὐτὸν >έθοντα λόγον >ῖσα ἀλλήλοις ἐστίν; καὶ πρὸς <ὰ τὸ αὐτὸ τὸν αὐτὸν <έθει λόγον, ἐκεῖνα >ῖσα ἐστίν.



Ἐθέτω γὰρ ἑκάτερον τῶν A , B πρὸς τὸ Γ τὸν αὐτὸν λόγον; λέγω, <ὅτι >ῖσον ἐστὶ τὸ A τῷ B .

Εἰ γὰρ μή, οὐκ >άν ἑκάτερον τῶν A , B πρὸς τὸ Γ τὸν αὐτὸν ε>ῖθε λόγον; >έθει δέ; >ῖσον >άρα ἐστὶ τὸ A τῷ B .

Ἐθέτω δὴ πάλιν τὸ Γ πρὸς ἑκάτερον τῶν A , B τὸν αὐτὸν λόγον; λέγω, <ὅτι >ῖσον ἐστὶ τὸ A τῷ B .

Εἰ γὰρ μή, οὐκ >άν τὸ Γ πρὸς ἑκάτερον τῶν A , B τὸν αὐτὸν ε>ῖθε λόγον; >έθει δέ; >ῖσον >άρα ἐστὶ τὸ A τῷ B .

Τὰ >άρα πρὸς τὸ αὐτὸ τὸν αὐτὸν >έθοντα λόγον >ῖσα ἀλλήλοις ἐστίν; καὶ πρὸς <ὰ τὸ αὐτὸ τὸν αὐτὸν >έθει λόγον, ἐκεῖνα >ῖσα ἐστίν; <όπερ >έδει δεῖξαι.

Proposition 9

(Magnitudes) having the same ratio to the same (magnitude) are equal to one another. And those (magnitudes) to which the same (magnitude) has the same ratio are equal.

For let A and B each have the same ratio to C . I say that A is equal to B .

For if not, A and B would not each have the same ratio to C [Prop. 5.8]. But they do. Thus, A is equal to B .

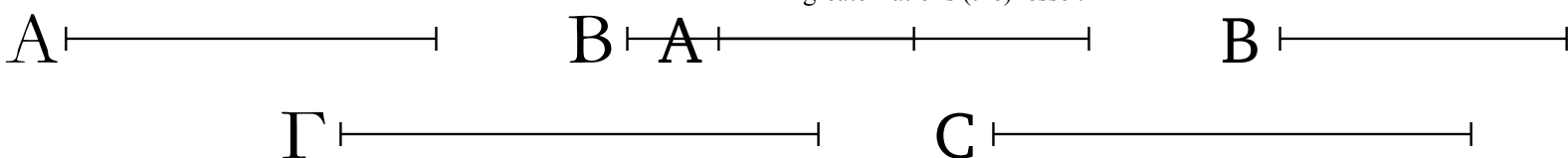
So, again, let C have the same ratio to each of A and B . I say that A is equal to B .

For if not, C would not have the same ratio to each of A and B [Prop. 5.8]. But it does. Thus, A is equal to B .

Thus, (magnitudes) having the same ratio to the same (magnitude) are equal to one another. And those (magnitudes) to which the same (magnitude) has the same ratio are equal. (Which is) the very thing it was required to show.

10.

Τῶν πρὸς τὸ αὐτὸ λόγον ἐθόντων τὸ μείζονα λόγον >έθον ἐκεῖνο μείζον ἐστίν; πρὸς <ὸ δὲ τὸ αὐτὸ μείζονα λόγον >έθει, ἐκεῖνο >έλαττόν ἐστιν.



Ἐθέτω γὰρ τὸ A πρὸς τὸ Γ μείζονα λόγον >ήπερ τὸ B πρὸς τὸ Γ ; λέγω, <ὅτι μείζον ἐστὶ τὸ A τοῦ B .

Εἰ γὰρ μή, >ήτοι >ῖσον ἐστὶ τὸ A τῷ B >ή >έλασσον. >ῖσον

Proposition 10

For (magnitudes) having a ratio to the same (magnitude), that (magnitude which) has the greater ratio is (the) greater. And that (magnitude) to which the latter (magnitude) has a greater ratio is (the) lesser.

For let A have a greater ratio to C than B (has) to C . I say that A is greater than B .

For if not, A is surely either equal to or less than B . In fact,

μὲν ο>ὕν ο>ύκ ἐστὶ τὸ Α τῷ Β; ἐκάτερον γὰρ >ὰν τῶν Α, Β πρὸς τὸ Γ τὸν αὐτὸν ε>ῖθε λόγον. οὐκ >έθει δέ; οὐκ >άρα >ῖσον ἐστὶ τὸ Α τῷ Β. οὐδὲ μὴν >έλασσόν ἐστι τὸ Α τοῦ Β; τὸ Α γὰρ >ὰν πρὸς τὸ Γ ἐλάσσονα λόγον ε>ῖθεν >ήπερ τὸ Β πρὸς τὸ Γ. οὐκ >έθει δέ; οὐκ >άρα >έλασσόν ἐστι τὸ Α τοῦ Β. ἐδειθθῇ δὲ οὐδὲ >ῖσον; μείζον >άρα ἐστὶ τὸ Α τοῦ Β. Ἐθέτω δὴ πάλιν τὸ Γ πρὸς τὸ Β μείζονα λόγον >ήπερ τὸ Γ πρὸς τὸ Α; λέγω, <ὅτι >έλασσόν ἐστι τὸ Β τοῦ Α. Εἰ γὰρ μή, >ήτοι >ῖσον ἐστὶν >ἢ μείζον. >ῖσον μὲν ο>ὕν ο>ύκ ἐστὶ τὸ Β τῷ Α; τὸ Γ γὰρ >ὰν πρὸς ἐκάτερον τῶν Α, Β τὸν αὐτὸν ε>ῖθε λόγον. οὐκ >έθει δέ; οὐκ >άρα >ῖσον ἐστὶ τὸ Α τῷ Β. οὐδὲ μὴν μείζον ἐστὶ τὸ Β τοῦ Α; τὸ Γ γὰρ >ὰν πρὸς τὸ Β ἐλάσσονα λόγον ε>ῖθεν >ήπερ πρὸς τὸ Α. οὐκ >έθει δέ; οὐκ >άρα μείζον ἐστὶ τὸ Β τοῦ Α. ἐδειθθῇ δέ, <ὅτι οὐδὲ >ῖσον; >έλαττον >άρα ἐστὶ τὸ Β τοῦ Α. Τῶν >άρα πρὸς τὸ αὐτὸ λόγον ἐθόντων τὸ μείζονα λόγον >έθον μείζον ἐστὶν; καὶ πρὸς <ὃ τὸ αὐτὸ μείζονα λόγον >έθει, ἐκείνο >έλαττον ἐστὶν; <όπερ >έδει δεῖχαι.

11.

Οἱ τῷ αὐτῷ λόγῳ οἱ αὐτοὶ καὶ ἀλλήλοις εἰσὶν οἱ αὐτοί. >Ἐστωσαν γὰρ ὡς μὲν τὸ Α πρὸς τὸ Β, ο<ύτωξ τὸ Γ πρὸς τὸ Δ, ὡς δὲ τὸ Γ πρὸς τὸ Δ, ο<ύτωξ τὸ Ε πρὸς τὸ Ζ; λέγω, <ὅτι ἐστὶν ὡς τὸ Α πρὸς τὸ Β, ο<ύτωξ τὸ Ε πρὸς τὸ Ζ. Εἰλήφθω γὰρ τῶν Α, Γ, Ε ἰσάκις πολλαπλάσια τὰ Η, Θ, Κ, τῶν δὲ Β, Δ, Ζ >άλλα, <ὰ >έτυθεν, ἰσάκις πολλαπλάσια τὰ Λ, Μ, Ν.

Γ ——— Α ——— Ε ——— C ———

Δ ——— Β ——— Ζ ——— D ———

Θ ——— G ——— K ——— H ———

M ——— L ——— N ——— M ———

E ———

F ———

K ———

N ———

Καὶ ἐπεὶ ἐστὶν ὡς τὸ Α πρὸς τὸ Β, ο<ύτωξ τὸ Γ πρὸς τὸ Δ, καὶ ε>ῖληπται τῶν μὲν Α, Γ ἰσάκις πολλαπλάσια τὰ Η, Θ, τῶν δὲ Β, Δ >άλλα, <ὰ >έτυθεν, ἰσάκις πολλαπλάσια τὰ Λ, Μ, εἰ >άρα ὑπερέθει τὸ Η τοῦ Λ, ὑπερέθει καὶ τὸ Θ τοῦ Μ, καὶ εἰ >ῖσον ἐστίν, >ῖσον, καὶ εἰ ἐλλείπει, ἐλλείπει. πάλιν, ἐπεὶ ἐστὶν ὡς τὸ Γ πρὸς τὸ Δ, ο<ύτωξ τὸ Ε πρὸς τὸ Ζ, καὶ ε>ῖληπται τῶν Γ, Ε ἰσάκις πολλαπλάσια τὰ Θ, Κ, τῶν δὲ Δ, Ζ >άλλα, <ὰ >έτυθεν, ἰσάκις πολλαπλάσια τὰ Μ, Ν, εἰ >άρα ὑπερέθει τὸ Θ τοῦ Μ, ὑπερέθει καὶ τὸ Κ τοῦ Ν, καὶ εἰ >ῖσον, >ῖσον, καὶ εἰ >έλαττον, >έλαττον. ἀλλὰ εἰ ὑπερέθει τὸ Θ τοῦ Μ, ὑπερέθει καὶ τὸ Η τοῦ Λ, καὶ εἰ >ῖσον, >ῖσον, καὶ εἰ >έλαττον, >έλαττον; <ώστε καὶ εἰ ὑπερέθει τὸ Η τοῦ Λ, ὑπερέθει καὶ τὸ Κ τοῦ Ν, καὶ εἰ >ῖσον, >ῖσον, καὶ εἰ >έλαττον, >έλαττον. καὶ ἐστὶ τὰ μὲν Η, Κ τῶν Α, Ε ἰσάκις πολλαπλάσια, τὰ δὲ Λ, Ν τῶν Β, Ζ >άλλα, <ὰ >έτυθεν, ἰσάκις πολλαπλάσια; >έστιν >άρα ὡς τὸ Α πρὸς τὸ Β, ο<ύτωξ τὸ Ε πρὸς τὸ Ζ.

Οἱ >άρα τῷ αὐτῷ λόγῳ οἱ αὐτοὶ καὶ ἀλλήλοις εἰσὶν οἱ αὐτοί;

A is not equal to *B*. For (then) *A* and *B* would each have the same ratio to *C* [Prop. 5.7]. But they do not. Thus, *A* is not equal to *B*. Neither, indeed, is *A* less than *B*. For (then) *A* would have a lesser ratio to *C* than *B* (has) to *C* [Prop. 5.8]. But it does not. Thus, *A* is not less than *B*. And it was shown not (to be) equal either. Thus, *A* is greater than *B*.

So, again, let *C* have a greater ratio to *B* than *C* (has) to *A*. I say that *B* is less than *A*.

For if not, (it is) surely either equal or greater. In fact, *B* is not equal to *A*. For (then) *C* would have the same ratio to each of *A* and *B* [Prop. 5.7]. But it does not. Thus, *A* is not equal to *B*. Neither, indeed, is *B* greater than *A*. For (then) *C* would have a lesser ratio to *B* than (it has) to *A* [Prop. 5.8]. But it does not. Thus, *B* is not greater than *A*. And it was shown that (it is) not equal (to *A*) either. Thus, *B* is less than *A*.

Thus, for (magnitudes) having a ratio to the same (magnitude), that (magnitude which) has the greater ratio is (the) greater. And that (magnitude) to which the latter (magnitude) has a greater ratio is (the) lesser. (Which is) the very thing it was required to show.

Proposition 11[†]

(Ratios which are) the same with the same ratio are also the same with one another.

For let it be that as *A* (is) to *B*, so *C* (is) to *D*, and as *C* (is) to *D*, so *E* (is) to *F*. I say that as *A* is to *B*, so *E* (is) to *F*. For let the equal multiples *G*, *H*, *K* be taken of *A*, *C*, *E* (respectively), and the other random equal multiples *L*, *M*, *N* of *B*, *D*, *F* (respectively).

And since as *A* is to *B*, so *C* (is) to *D*, and the equal multiples *G* and *H* be taken of *A* and *C* (respectively), and the other random equal multiples *L* and *M* of *B* and *D* (respectively), thus if *G* exceeds *L* (then) *H* also exceeds *M*, and if (*G* is) equal (to *L* then *H* is also) equal (to *M*), and if (*G* is) less (than *L* then *H* is also) less (than *M*) [Def. 5.5]. Again, since as *C* is to *D*, so *E* (is) to *F*, and the equal multiples *H* and *K* have been taken of *C* and *E* (respectively), and the other random equal multiples *M* and *N* of *D* and *F* (respectively), thus if *H* exceeds *M* (then) *K* also exceeds *N*, and if (*H* is) equal (to *M* then *K* is also) equal (to *N*), and if (*H* is) less (than *M* then *K* is also) less (than *N*) [Def. 5.5]. But (we saw that) if *H* was exceeding *M* (then) *G* was also exceeding *L*, and if (*H* was) equal (to *M* then *G* was also) equal (to *L*), and if (*H* was) less (than *M* then *G* was also) less (than *L*). And, hence, if *G* exceeds *L* (then) *K* also exceeds *N*, and if (*G* is) equal (to *L* then *K* is also) equal (to *N*), and if (*G* is) less (than *L* then *K* is also) less

<όπερ >έδει δεῖχαι.

(than N). And G and K are equal multiples of A and E (respectively), and L and N other random equal multiples of B and F (respectively). Thus, as A is to B , so E (is) to F [Def. 5.5].

Thus, (ratios which are) the same with the same ratio are also the same with one another. (Which is) the very thing it was required to show.

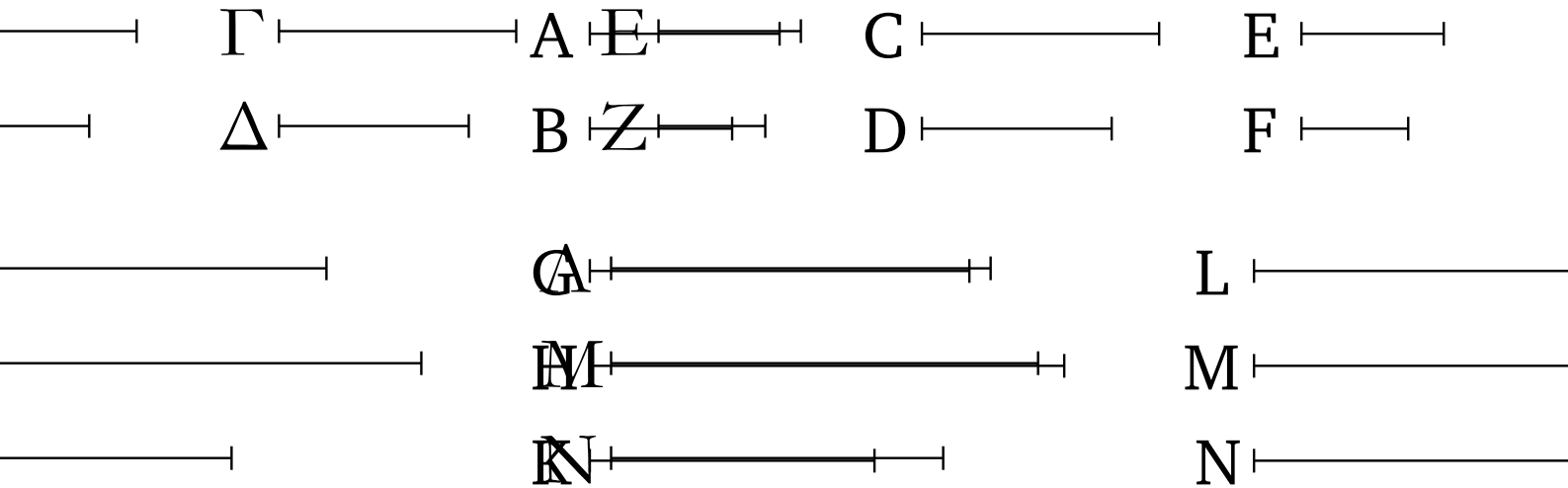
† In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$ and $\gamma : \delta :: \epsilon : \zeta$ then $\alpha : \beta :: \epsilon : \zeta$.

12.

Proposition 12[†]

Ἐάν >ἢ| ὅποσαοῦν μεγέθη ἀνάλογον, >έσται ὡς <έν τῶν ἡγουμένων πρὸς <έν τῶν ἐπομένων, ο<ύτωξ <άπαντα τὰ ἡγούμενα πρὸς <άπαντα τὰ ἐπόμενα.

If there are any number of magnitudes whatsoever (which are) proportional, (then) as one of the leading (magnitudes is) to one of the following, so will all of the leading (magnitudes) be to all of the following.



>Ἐστωσαν ὅποσαοῦν μεγέθη ἀνάλογον τὰ Α, Β, Γ, Δ, Ε, Ζ, ὡς τὸ Α πρὸς τὸ Β, ο<ύτωξ τὸ Γ πρὸς τὸ Δ, καὶ τὸ Ε πρὸς τὸ Ζ; λέγω, <ότι ἐστὶν ὡς τὸ Α πρὸς τὸ Β, ο<ύτωξ τὰ Α, Γ, Ε πρὸς τὰ Β, Δ, Ζ.

Εἰλήφθω γὰρ τῶν μὲν Α, Γ, Ε ἰσάκις πολλαπλάσια τὰ Η, Θ, Κ, τῶν δὲ Β, Δ, Ζ >άλλα, <ὰ >έτυθεν, ἰσάκις πολλαπλάσια τὰ Λ, Μ, Ν.

Καὶ ἐπεὶ ἐστὶν ὡς τὸ Α πρὸς τὸ Β, ο<ύτωξ τὸ Γ πρὸς τὸ Δ, καὶ τὸ Ε πρὸς τὸ Ζ, καὶ ε>ῖληπται τῶν μὲν Α, Γ, Ε ἰσάκις πολλαπλάσια τὰ Η, Θ, Κ τῶν δὲ Β, Δ, Ζ >άλλα, <ὰ >έτυθεν, ἰσάκις πολλαπλάσια τὰ Λ, Μ, Ν, εἰ >άρα ὑπερέθει τὸ Η τοῦ Α, ὑπερέθει καὶ τὸ Θ τοῦ Μ, καὶ τὸ Κ τοῦ Ν, καὶ εἰ >ἴσον, >ἴσον, καὶ εἰ >έλαττον, >έλαττον. <ώστε καὶ εἰ ὑπερέθει τὸ Η τοῦ Α, ὑπερέθει καὶ τὰ Η, Θ, Κ τῶν Α, Μ, Ν, καὶ εἰ >ἴσον, >ἴσα, καὶ εἰ >έλαττον, >έλαττονα. καὶ ἐστὶ τὸ μὲν Η καὶ τὰ Η, Θ, Κ τοῦ Α καὶ τῶν Α, Γ, Ε ἰσάκις πολλαπλάσια, ἐπειδὴ περ ἔαν >ἢ| ὅποσαοῦν μεγέθη ὅποσωνοῦν μεγεθῶν >ἴσων τὸ πλῆθος <έκαστον ἐκάστου ἰσάκις πολλαπλάσιον, ὅσαπλάσιόν ἐστιν <έν τῶν μεγεθῶν ἐνόξ, τοσαυταπλάσια >έσται καὶ τὰ πάντα τῶν πάντων. διὰ τὰ αὐτὰ δὴ καὶ τὸ Α καὶ τὰ Α, Μ, Ν τοῦ Β καὶ τῶν Β, Δ, Ζ ἰσάκις ἐστὶ πολλαπλάσια; >έστιν >άρα ὡς τὸ Α πρὸς τὸ Β, ο<ύτωξ τὰ Α, Γ, Ε πρὸς τὰ Β, Δ, Ζ.

Ἐάν >άρα >ἢ| ὅποσαοῦν μεγέθη ἀνάλογον, >έσται ὡς <έν τῶν ἡγουμένων πρὸς <έν τῶν ἐπομένων, ο<ύτωξ <άπαντα τὰ ἡγούμενα πρὸς <άπαντα τὰ ἐπόμενα; <όπερ >έδει δεῖχαι.

Let there be any number of magnitudes whatsoever, A, B, C, D, E, F , (which are) proportional, (so that) as A (is) to B , so C (is) to D , and E to F . I say that as A is to B , so A, C, E (are) to B, D, F .

For let the equal multiples G, H, K be taken of A, C, E (respectively), and the other random equal multiples L, M, N of B, D, F (respectively).

And since as A is to B , so C (is) to D , and E to F , and the equal multiples G, H, K have been taken of A, C, E (respectively), and the other random equal multiples L, M, N of B, D, F (respectively), thus if G exceeds L (then) H also exceeds M , and K (exceeds) N , and if (G is) equal (to L then H is also) equal (to M , and K to N), and if (G is) less (than L then H is also) less (than M , and K than N) [Def. 5.5]. And, hence, if G exceeds L (then) G, H, K also exceed L, M, N , and if (G is) equal (to L then G, H, K are also) equal (to L, M, N) and if (G is) less (than L then G, H, K are also) less (than L, M, N). And G and G, H, K are equal multiples of A and A, C, E (respectively), inasmuch as if there are any number of magnitudes whatsoever (which are) equal multiples, respectively, of some (other) magnitudes, of equal number (to them), (then) as many times as one of the (first) magnitudes is (divisible) by one (of the second), so many times will all (of the first magnitudes) also (be divisible) by all (of the second) [Prop. 5.1]. So, for the same (reasons), L and L, M, N are also equal multiples of B and B, D, F (respectively). Thus, as A is to B , so A, C, E (are) to B, D, F (respectively).

Thus, if there are any number of magnitudes whatsoever (which are) proportional, (then) as one of the leading (mag-

nitudes is) to one of the following, so will all of the leading (magnitudes) be to all of the following. (Which is) the very thing it was required to show.

† In modern notation, this proposition reads that if $\alpha : \alpha' :: \beta : \beta' :: \gamma : \gamma'$ etc. then $\alpha : \alpha' :: (\alpha + \beta + \gamma + \dots) : (\alpha' + \beta' + \gamma' + \dots)$.

13.

Proposition 13[†]

Ἐὰν πρῶτον πρὸς δεύτερον τὸν αὐτὸν >ἔθῃ λόγον καὶ τρίτον πρὸς τέταρτον, τρίτον δὲ πρὸς τέταρτον μείζονα λόγον >ἔθῃ >ῆ πέμπτων πρὸς <έκτον, καὶ πρῶτον πρὸς δεύτερον μείζονα λόγον <έχει >ῆ πέμπτων πρὸς <έκτον.

If a first (magnitude) has the same ratio to a second that a third (has) to a fourth, and the third (magnitude) has a greater ratio to the fourth than a fifth (has) to a sixth, (then) the first (magnitude) will also have a greater ratio to the second than the fifth (has) to the sixth.

Γ — A —

E — C —

E —

Δ — B —

Z — D —

F —

H — M — Θ — G — H —

K — N — Λ — K — L —

Πρῶτον γὰρ τὸ A πρὸς δεύτερον τὸ B τὸν αὐτὸν ἔθετο λόγον καὶ τρίτον τὸ Γ πρὸς τέταρτον τὸ Δ, τρίτον δὲ τὸ Γ πρὸς τέταρτον τὸ Δ μείζονα λόγον ἔθετο >ῆ πέμπτων τὸ E πρὸς <έκτον τὸ Z. λέγω, <ὅτι καὶ πρῶτον τὸ A πρὸς δεύτερον τὸ B μείζονα λόγον <έχει >ῆ περ πέμπτων τὸ E πρὸς <έκτον τὸ Z.

Ἐπεὶ γὰρ >έστι τινὰ τῶν μὲν Γ, E ἰσάκις πολλαπλάσια, τῶν δὲ Δ, Z >άλλα, <ᾶ >έτυθεν, ἰσάκις πολλαπλάσια, καὶ τὸ μὲν τοῦ Γ πολλαπλάσιον τοῦ τοῦ Δ πολλαπλασίου ὑπερέθει, τὸ δὲ τοῦ E πολλαπλάσιον τοῦ τοῦ Z πολλαπλασίου οὐθ ὑπερέθει, εἰλήφθω, καὶ >έστω τῶν μὲν Γ, E ἰσάκις πολλαπλάσια τὰ H, Θ, τῶν δὲ Δ, Z >άλλα, <ᾶ >έτυθεν, ἰσάκις πολλαπλάσια τὰ K, Λ, <ώστε τὸ μὲν H τοῦ K ὑπερέθειν, τὸ δὲ Θ τοῦ Λ μὴ ὑπερέθειν; καὶ ὅσα πλάσιον μὲν ἔστι τὸ H τοῦ Γ, τοσαυταπλάσιον >έστω καὶ τὸ M τοῦ A, ὅσα πλάσιον δὲ τὸ K τοῦ Δ, τοσαυταπλάσιον >έστω καὶ τὸ N τοῦ B.

Καὶ ἐπεὶ ἔστιν ὥς τὸ A πρὸς τὸ B, ο<ύτωξ τὸ Γ πρὸς τὸ Δ, καὶ >ίληπται τῶν μὲν A, Γ ἰσάκις πολλαπλάσια τὰ M, H, τῶν δὲ B, Δ >άλλα, <ᾶ >έτυθεν, ἰσάκις πολλαπλάσια τὰ N, K, εἰ >άρα ὑπερέθει τὸ M τοῦ N, ὑπερέθει καὶ τὸ H τοῦ K, καὶ εἰ >ίσον, >ίσον, καὶ εἰ >έλαττον, >έλαττον. ὑπερέθει δὲ τὸ H τοῦ K; ὑπερέθει >άρα καὶ τὸ M τοῦ N. τὸ δὲ Θ τοῦ Λ οὐθ ὑπερέθει; καὶ ἔστι τὰ μὲν M, Θ τῶν A, E ἰσάκις πολλαπλάσια, τὰ δὲ N, Λ τῶν B, Z >άλλα, <ᾶ >έτυθεν, ἰσάκις πολλαπλάσια; τὸ >άρα A πρὸς τὸ B μείζονα λόγον >έθει >ῆ περ τὸ E πρὸς τὸ Z.

Ἐὰν >άρα πρῶτον πρὸς δεύτερον τὸν αὐτὸν >ἔθῃ λόγον καὶ τρίτον πρὸς τέταρτον, τρίτον δὲ πρὸς τέταρτον μείζονα λόγον >ἔθῃ >ῆ πέμπτων πρὸς <έκτον, καὶ πρῶτον πρὸς δεύτερον μείζονα λόγον <έχει >ῆ πέμπτων πρὸς <έκτον; <όπερ >έδει δεῖξαι.

For let a first (magnitude) A have the same ratio to a second B that a third C (has) to a fourth D, and let the third (magnitude) C have a greater ratio to the fourth D than a fifth E (has) to a sixth F. I say that the first (magnitude) A will also have a greater ratio to the second B than the fifth E (has) to the sixth F.

For since there are some equal multiples of C and E, and other random equal multiples of D and F, (for which) the multiple of C exceeds the (multiple) of D, and the multiple of E does not exceed the multiple of F [Def. 5.7], let them be taken. And let G and H be equal multiples of C and E (respectively), and K and L other random equal multiples of D and F (respectively), such that G exceeds K, but H does not exceed L. And as many times as G is (divisible) by C, so many times let M be (divisible) by A. And as many times as K (is divisible) by D, so many times let N be (divisible) by B.

And since as A is to B, so C (is) to D, and the equal multiples M and G have been taken of A and C (respectively), and the other random equal multiples N and K of B and D (respectively), thus if M exceeds N (then) G exceeds K, and if (M is) equal (to N then G is also) equal (to K), and if (M is) less (than N then G is also) less (than K) [Def. 5.5]. And G exceeds K. Thus, M also exceeds N. And H does not exceed L. And M and H are equal multiples of A and E (respectively), and N and L other random equal multiples of B and F (respectively). Thus, A has a greater ratio to B than E (has) to F [Def. 5.7].

Thus, if a first (magnitude) has the same ratio to a second that a third (has) to a fourth, and a third (magnitude) has a greater ratio to a fourth than a fifth (has) to a sixth, (then) the first (magnitude) will also have a greater ratio to the second than the fifth (has) to the sixth. (Which is) the very thing it was required to show.

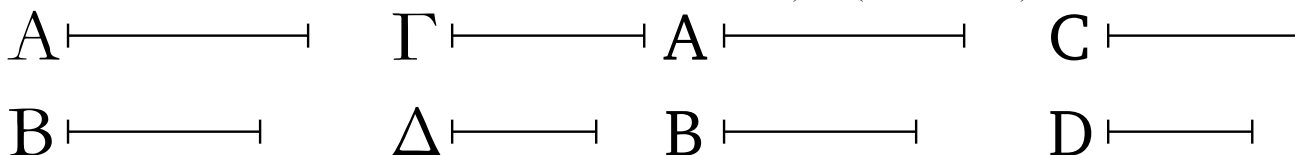
† In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$ and $\gamma : \delta > \epsilon : \zeta$ then $\alpha : \beta > \epsilon : \zeta$.

14.

Proposition 14[†]

Ἐὰν πρῶτον προξ δεύτερον τὸν αὐτὸν >έθῃ λόγον καὶ τρίτον
 προξ τέταρτον, τὸ δὲ πρῶτον τοῦ τρίτου μείζον >ῆ|, καὶ τὸ
 δεύτερον τοῦ τετάρτου μείζον >έσται, κ>ὰν >ίσον, >ίσον,
 κ>ὰν >έλαττον, >έλαττον.

If a first (magnitude) has the same ratio to a second that a third (has) to a fourth, and the first (magnitude) is greater than the third, (then) the second will also be greater than the fourth. And if (the first magnitude is) equal (to the third then the second will also be) equal (to the fourth). And if (the first magnitude is) less (than the third then the second will also be) less (than the fourth).



Πρῶτον γὰρ τὸ Α πρὸς δεῦτερον τὸ Β τὸν αὐτὸν ἐθέτω
λόγον καὶ τρίτον τὸ Γ πρὸς τέταρτον τὸ Δ, μείζον δὲ ἔστω
τὸ Α τοῦ Γ· λέγω, ὅτι καὶ τὸ Β τοῦ Δ μείζον ἐστίν.

Ἐπεὶ γὰρ τὸ Α τοῦ Γ μείζων ἐστίν, >ἄλλο δέ, <ὁ >ἐνυθεν,
[μέγεθος] τὸ Β, τὸ Α >άρα πρὸς τὸ Β μείζονα λόγον >έθει
>ἥπερ τὸ Γ πρὸς τὸ Β. ὥς δὲ τὸ Α πρὸς τὸ Β, ο<ύτως τὸ
Γ πρὸς τὸ Δ; καὶ τὸ Γ >άρα πρὸς τὸ Δ μείζονα λόγον >έθει
>ἥπερ τὸ Γ πρὸς τὸ Β. πρὸς <ὁ δὲ τὸ αὐτὸ μείζονα λόγον
>έθει, ἐκεῖνο >έλασσόν ἐστιν; >έλασσον >άρα τὸ Δ τοῦ Β;
<ώστε μείζων ἐστὶ τὸ Β τοῦ Δ.

Ὁμοίως δὴ δεῖχμεν, <ὅτι κ>άν >ίσον >ῆ| τὸ Α τῷ Γ, >ίσον >έσται καὶ τὸ Β τῷ Δ, κ>άν >έλασσον >ῆ| τὸ Α τοῦ Γ, >έλασσον >έσται καὶ τὸ Β τοῦ Δ.

Ἐάν >ἀρα πρώτον πρὸς δεύτερον τὸν αὐτὸν >έθη λόγον καὶ τρίτον πρὸς τέταρτον, τὸ δὲ πρῶτον τοῦ τρίτου μείζων >ῆ], καὶ τὸ δεύτερον τοῦ τετάρτου μείζων >έσται, κ>ὰν >ίσον, >ίσον, κ>ὰν >έλαττον, >έλαττον; <όπερ >έδει δεῖχθαι.

For let a first (magnitude) A have the same ratio to a second B that a third C (has) to a fourth D . And let A be greater than C . I say that B is also greater than D .

For since A is greater than C , and B (is) another random [magnitude], A thus has a greater ratio to B than C (has) to B [Prop. 5.8]. And as A (is) to B , so C (is) to D . Thus, C also has a greater ratio to D than C (has) to B . And that (magnitude) to which the same (magnitude) has a greater ratio is the lesser [Prop. 5.10]. Thus, D (is) less than B . Hence, B is greater than D .

So, similarly, we can show that even if A is equal to C , (then) B will also be equal to D , and even if A is less than C , (then) B will also be less than D .

Thus, if a first (magnitude) has the same ratio to a second that a third (has) to a fourth, and the first (magnitude) is greater than the third, (then) the second will also be greater than the fourth. And if (the first magnitude is) equal (to the third then the second will also be) equal (to the fourth). And if (the first magnitude is) less (than the third then the second will also be) less (than the fourth). (Which is) the very thing it was required to show.

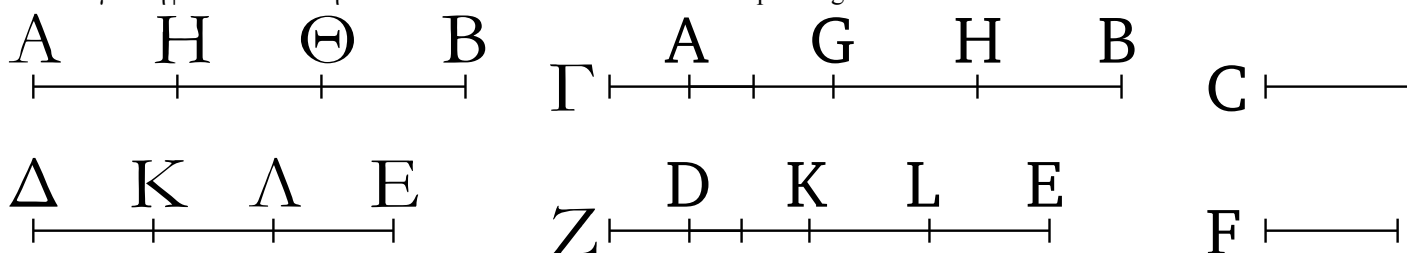
[†] In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$ then $\alpha \supseteq \gamma$ as $\beta \supseteq \delta$.

15.

Proposition 15[†]

Τὰ μέρη τοῖς ὡσαύτως πολλαπλασίοις τὸν αὐτὸν >έθει-
λόγον ληφθέντα κατάλληλα.

Parts have the same ratio as similar multiples, taken in corresponding order.



>Ἐστω γὰρ ἰσάκις πολλαπλάσιον τὸ AB τοῦ Γ καὶ τὸ ΔΕ τοῦ Ζ; λέγω, <ὅτι ἐστὶν ὡς τὸ Γ πρὸς τὸ Ζ, ο<ύτωξ τὸ AB πρὸς τὸ ΔΕ.

Ἐπει γὰρ ισάκις ἐστὶ πολλαπλάσιον τὸ ΑΒ τοῦ Γ καὶ τὸ ΔΕ τοῦ Ζ, <όσα> ἄρα ἐστὶν ἐν τῷ ΑΒ μεγέθῃ >ἴσα τῷ Γ τοσαῦτα καὶ ἐν τῷ ΔΕ >ἴσα τῷ Ζ διηρήσθω τὸ μὲν ΑΒ εἰς τὰ τῷ Γ >ἴσα τὰ ΑΗ, ΗΘ, ΘΒ, τὸ δὲ ΔΕ εἰς τὰ τῷ Ζ >ἴσα τὰ ΔΚ, ΚΛ, ΛΕ; >έσται δὴ >ἴσον τὸ πλῆθος τῶν ΑΗ, ΗΘ, ΘΒ τῷ πλήθει τῶν ΔΚ, ΚΛ, ΛΕ. καὶ ἐπεὶ >ἴσα ἐστὶ τὰ ΑΗ, ΗΘ, ΘΒ ἀλλήλοισι, >έστι δὲ καὶ τὰ ΔΚ, ΚΛ, ΛΕ >ἴσα ἀλλήλοισι, >έστιν >ἄρα ὥς τὸ ΑΗ πρὸς τὸ ΔΚ, ο<ύτωξ τὸ ΗΘ πρὸς τὸ ΚΛ, καὶ τὸ ΘΒ πρὸς τὸ ΛΕ. >έσται >ἄρα καὶ ὥς <ἐν τῶν ἡγουμένων πρὸς <ἐν τῶν ἐπομένων, ο<ύτωξ <ἀπαντα

For let AB and DE be equal multiples of C and F (respectively). I say that as C is to F , so AB (is) to DE .

For since AB and DE are equal multiples of C and F (respectively), thus as many magnitudes as there are in AB equal to C , so many (are there) also in DE equal to F . Let AB be divided into (magnitudes) AG, GH, HB , equal to C , and DE into (magnitudes) DK, KL, LE , equal to F . So, the number of (magnitudes) AG, GH, HB will equal the number of (magnitudes) DK, KL, LE . And since AG, GH, HB are equal to one another, and DK, KL, LE are also equal to one another, thus as AG is to DK , so GH (is) to KL , and HB to LE [Prop. 5.7]. And, thus (for proportional magnitudes), as one of the leading (magnitudes) will

τὰ ἡγούμενα πρὸς <άπαντα τὰ ἐπόμενα; >έστιν >άρα ὡς τὸ ΑΗ πρὸς τὸ ΔΚ, ο<ύτωξ τὸ ΑΒ πρὸς τὸ ΔΕ. >ίσον δὲ τὸ μὲν ΑΗ τῷ Γ, τὸ δὲ ΔΚ τῷ Ζ; >έστιν >άρα ὡς τὸ Γ πρὸς τὸ Ζ ο<ύτωξ τὸ ΑΒ πρὸς τὸ ΔΕ.
Τὰ >άρα μέρη τοῖς ὡσαύτωξ πολλαπλασίοις τὸν αὐτὸν >έθει λόγον ληφθέντα κατάλληλα; <όπερ >έδει δεῖχαι.

† In modern notation, this proposition reads that $\alpha : \beta :: m\alpha : m\beta$.

16.

Ἐὰν τέσσαρα μεγέθη ἀνάλογον >ῃ, καὶ ἐναλλάχ ἀνάλογον >έσται.

>Ἐστω τέσσαρα μεγέθη ἀνάλογον τὰ Α, Β, Γ, Δ, ὡς τὸ Α πρὸς τὸ Β, ο<ύτωξ τὸ Γ πρὸς τὸ Δ; λέγω, <ὅτι καὶ ἐναλλάχ [ἀνάλογον] >έσται, ὡς τὸ Α πρὸς τὸ Γ, ο<ύτωξ τὸ Β πρὸς τὸ Δ.

Εἰλήφθω γὰρ τῶν μὲν Α, Β ἰσάκις πολλαπλάσια τὰ Ε, Ζ, τῶν δὲ Γ, Δ >άλλα, <ὰ >έθυθεν, ἰσάκις πολλαπλάσια τὰ Η, Θ.

Proposition 16†

If four magnitudes are proportional, (then) they will also be proportional alternately.

Let A, B, C and D be four proportional magnitudes, (such that) as A (is) to B , so C (is) to D . I say that they will also be [proportional] alternately, (so that) as A (is) to C , so B (is) to D .

For let the equal multiples E and F be taken of A and B (respectively), and the other random equal multiples G and H of C and D (respectively).

A ———

C ———

B ———

D ———

E ———+———+———+———

G ———+———+———+———

F ———+———+———+———

H ———+———+———+———

A ———

Γ ———

B ———

Δ ———

E ———+———+———+———

Η ———+———+———+———

Z ———+———+———+———

Θ ———+———+———+———

Καὶ ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ Ε τοῦ Α καὶ τὸ Ζ τοῦ Β, τὰ δὲ μέρη τοῖς ὡσαύτωξ πολλαπλασίοις τὸν αὐτὸν >έθει λόγον, >έστιν >άρα ὡς τὸ Α πρὸς τὸ Β, ο<ύτωξ τὸ Ε πρὸς τὸ Ζ. ὡς δὲ τὸ Α πρὸς τὸ Β, ο<ύτωξ τὸ Γ πρὸς τὸ Δ; καὶ ὡς >άρα τὸ Γ πρὸς τὸ Δ, ο<ύτωξ τὸ Ε πρὸς τὸ Ζ. πάλιν, ἐπεὶ τὰ Η, Θ τῶν Γ, Δ ἰσάκις ἐστὶ πολλαπλάσια, >έστιν >άρα ὡς τὸ Γ πρὸς τὸ Δ, ο<ύτωξ τὸ Η πρὸς τὸ Θ. ὡς δὲ τὸ Γ πρὸς τὸ Δ, [ο<ύτωξ] τὸ Ε πρὸς τὸ Ζ; καὶ ὡς >άρα τὸ Ε πρὸς τὸ Ζ, ο<ύτωξ τὸ Η πρὸς τὸ Θ. ἔαν δὲ τέσσαρα μεγέθη ἀνάλογον >ῃ, τὸ δὲ πρῶτον τοῦ τρίτου μείζον >ῃ, καὶ τὸ δεύτερον τοῦ τετάρτου μείζον >έσται, κ<άν >ίσον, >ίσον, κ<άν >έλαττον, >έλαττον. εἰ >άρα ὑπερέθει τὸ Ε τοῦ Η, ὑπερέθει καὶ τὸ Ζ τοῦ Θ, καὶ εἰ >ίσον, >ίσον, καὶ εἰ >έλαττον, >έλαττον. καὶ ἐστὶ τὰ μὲν Ε, Ζ τῶν Α, Β ἰσάκις πολλαπλάσια, τὰ δὲ Η, Θ τῶν Γ, Δ >άλλα, <ὰ >έθυθεν, ἰσάκις πολλαπλάσια; >έστιν >άρα ὡς τὸ Α πρὸς τὸ Γ, ο<ύτωξ τὸ Β πρὸς τὸ Δ.

Ἐὰν >άρα τέσσαρα μεγέθη ἀνάλογον >ῃ, καὶ ἐναλλάχ ἀνάλογον >έσται; <όπερ >έδει δεῖχαι.

And since E and F are equal multiples of A and B (respectively), and parts have the same ratio as similar multiples [Prop. 5.15], thus as A is to B , so E (is) to F . But as A (is) to B , so C (is) to D . And, thus, as C (is) to D , so E (is) to F [Prop. 5.11]. Again, since G and H are equal multiples of C and D (respectively), thus as C is to D , so G (is) to H [Prop. 5.15]. But as C (is) to D , [so] E (is) to F . And, thus, as E (is) to F , so G (is) to H [Prop. 5.11]. And if four magnitudes are proportional, and the first is greater than the third, (then) the second will also be greater than the fourth, and if (the first is) equal (to the third then the second will also be) equal (to the fourth), and if (the first is) less (than the third then the second will also be) less (than the fourth) [Prop. 5.14]. Thus, if E exceeds G (then) F also exceeds H , and if (E is) equal (to G then F is also) equal (to H), and if (E is) less (than G then F is also) less (than H). And E and F are equal multiples of A and B (respectively), and G and H other random equal multiples of C and D (respectively). Thus, as A is to C , so B (is) to D [Def. 5.5].

Thus, if four magnitudes are proportional, (then) they will

also be proportional alternately. (Which is) the very thing it was required to show.

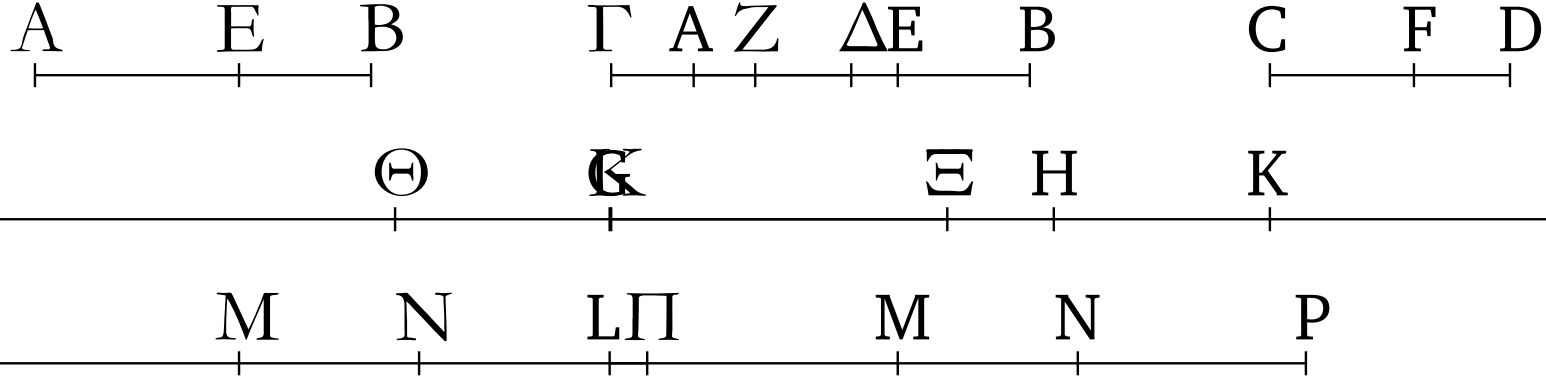
† In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$ then $\alpha : \gamma :: \beta : \delta$.

17.

Proposition 17†

Ἐὰν συγκείμενα μεγέθη ἀνάλογον >ῆ|, καὶ διαιρεθέντα ἀνάλογον >έσται.

If composed magnitudes are proportional, (then) they will also be proportional (when) separated.



>Ἐστω συγκείμενα μεγέθη ἀνάλογον τὰ AB, BE, ΓΑ, ΔΖ, ὥς τὸ AB πρὸς τὸ BE, ο<ύτωξ τὸ ΓΑ πρὸς τὸ ΔΖ; λέγω, <ὅτι καὶ διαιρεθέντα ἀνάλογον >έσται, ὥς τὸ ΑΕ πρὸς τὸ EB, ο<ύτωξ τὸ ΓΖ πρὸς τὸ ΔΖ.

Εἰλήφθω γὰρ τῶν μὲν ΑΕ, ΕΒ, ΓΖ, ΖΔ ἰσάκις πολλαπλάσια τὰ ΗΘ, ΘΚ, ΛΜ, ΜΝ, τῶν δὲ ΕΒ, ΖΔ >άλλα, <ὰ >έτυθεν, ἰσάκις πολλαπλάσια τὰ ΚΧ, ΝΠ.

Καὶ ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ ΗΘ τοῦ ΑΕ καὶ τὸ ΘΚ τοῦ ΕΒ, ἰσάκις >άρα ἐστὶ πολλαπλάσιον τὸ ΗΘ τοῦ ΑΕ καὶ τὸ ΗΚ τοῦ ΑΒ. ἰσάκις δὲ ἐστὶ πολλαπλάσιον τὸ ΗΘ τοῦ ΑΕ καὶ τὸ ΛΜ τοῦ ΓΖ; ἰσάκις >άρα ἐστὶ πολλαπλάσιον τὸ ΗΚ τοῦ ΑΒ καὶ τὸ ΛΜ τοῦ ΓΖ. πάλιν, ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ ΛΜ τοῦ ΓΖ καὶ τὸ ΜΝ τοῦ ΖΔ, ἰσάκις >άρα ἐστὶ πολλαπλάσιον τὸ ΛΜ τοῦ ΓΖ καὶ τὸ ΑΝ τοῦ ΓΑ. ἰσάκις δὲ >ῆν πολλαπλάσιον τὸ ΛΜ τοῦ ΓΖ καὶ τὸ ΗΚ τοῦ ΑΒ; ἰσάκις >άρα ἐστὶ πολλαπλάσιον τὸ ΗΚ τοῦ ΑΒ καὶ τὸ ΑΝ τοῦ ΓΑ. τὰ ΗΚ, ΑΝ >άρα τῶν ΑΒ, ΓΑ ἰσάκις ἐστὶ πολλαπλάσια. πάλιν, ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ ΘΚ τοῦ ΕΒ καὶ τὸ ΜΝ τοῦ ΖΔ, >έστι δὲ καὶ τὸ ΚΧ τοῦ ΕΒ ἰσάκις πολλαπλάσιον καὶ τὸ ΝΠ τοῦ ΖΔ, καὶ συντεθέν τὸ ΘΧ τοῦ ΕΒ ἰσάκις ἐστὶ πολλαπλάσιον καὶ τὸ ΜΠ τοῦ ΖΔ. καὶ ἐπεὶ ἐστὶν ὥς τὸ ΑΒ πρὸς τὸ BE, ο<ύτωξ τὸ ΓΑ πρὸς τὸ ΔΖ, καὶ >ίληπται τῶν μὲν ΑΒ, ΓΑ ἰσάκις πολλαπλάσια τὰ ΗΚ, ΑΝ, τῶν δὲ ΕΒ, ΖΔ ἰσάκις πολλαπλάσια τὰ ΘΧ, ΜΠ, εἰ >άρα ὑπερέθει τὸ ΗΚ τοῦ ΘΧ, ὑπερέθει καὶ τὸ ΑΝ τοῦ ΜΠ, καὶ εἰ >ῖσον, >ῖσον, καὶ εἰ >έλαττον, >έλαττον. ὑπερεθέτω δὴ τὸ ΗΚ τοῦ ΘΧ, καὶ κοινοῦ ἀφαιρεθέντος τοῦ ΘΚ ὑπερέθει >άρα καὶ τὸ ΗΘ τοῦ ΚΧ. ἄλλα εἰ ὑπερεῖθε τὸ ΗΚ τοῦ ΘΧ ὑπερεῖθε καὶ τὸ ΑΝ τοῦ ΜΠ; ὑπερέθει >άρα καὶ τὸ ΑΝ τοῦ ΜΠ, καὶ κοινοῦ ἀφαιρεθέντος τοῦ ΜΝ ὑπερέθει καὶ τὸ ΛΜ τοῦ ΝΠ; <ώστε εἰ ὑπερέθει τὸ ΗΘ τοῦ ΚΧ, ὑπερέθει καὶ τὸ ΛΜ τοῦ ΝΠ. ὁμοίως δὴ δεῖχμεν, <ὅτι κ<ὰν >ῖσον >ῆ| τὸ ΗΘ τῶ| ΚΧ, >ῖσον >έσται καὶ τὸ ΛΜ τῶ| ΝΠ, κ<ὰν >έλαττον, >έλαττον. καὶ ἐστὶ τὰ μὲν ΗΘ, ΑΜ τῶν ΑΕ, ΓΖ ἰσάκις πολλαπλάσια, τὰ δὲ ΚΧ, ΝΠ τῶν ΕΒ, ΖΔ >άλλα, <ὰ >έτυθεν, ἰσάκις πολλαπλάσια; >έστιν >άρα ὥς τὸ ΑΕ πρὸς τὸ EB, ο<ύτωξ τὸ ΓΖ πρὸς τὸ ΔΖ.

Ἐὰν >άρα συγκείμενα μεγέθη ἀνάλογον >ῆ|, καὶ διαιρεθέντα ἀνάλογον >έσται; <όπερ >έδει δεῖχαι.

Let AB , BE , CD , and DF be composed magnitudes (which are) proportional, (so that) as AB (is) to BE , so CD (is) to DF . I say that they will also be proportional (when) separated, (so that) as AE (is) to EB , so CF (is) to DF .

For let the equal multiples GH , HK , LM , and MN be taken of AE , EB , CF , and FD (respectively), and the other random equal multiples KO and NP of EB and FD (respectively).

And since GH and HK are equal multiples of AE and EB (respectively), GH and GK are thus equal multiples of AE and AB (respectively) [Prop. 5.1]. But GH and LM are equal multiples of AE and CF (respectively). Thus, GK and LM are equal multiples of AB and CF (respectively). Again, since LM and MN are equal multiples of CF and FD (respectively), LM and LN are thus equal multiples of CF and CD (respectively) [Prop. 5.1]. And LM and GK were equal multiples of CF and AB (respectively). Thus, GK and LN are equal multiples of AB and CD (respectively). Thus, GK , LN are equal multiples of AB , CD . Again, since HK and MN are equal multiples of EB and FD (respectively), and KO and NP are also equal multiples of EB and FD (respectively), (then), added together, HO and MP are also equal multiples of EB and FD (respectively) [Prop. 5.2]. And since as AB (is) to BE , so CD (is) to DF , and the equal multiples GK , LN be taken of AB , CD , and the equal multiples HO , MP of EB , FD , thus if GK exceeds HO (then) LN also exceeds MP , and if (GK is) equal (to HO then LN is also) equal (to MP), and if (GK is) less (than HO then LN is also) less (than MP) [Def. 5.5]. So let GK exceed HO , and thus, HK being taken away from both, GH exceeds KO . But (we saw that) if GK was exceeding HO (then) LN was also exceeding MP . Thus, LN also exceeds MP , and, MN being taken away from both, LM also exceeds NP . Hence, if GH exceeds KO (then) LM also exceeds NP . So, similarly, we can show that even if GH is equal to KO (then) LM will also be equal to NP , and even if (GH is) less (than KO then LM will also be) less (than NP). And GH , LM are equal multiples of AE , CF , and KO , NP other random

equal multiples of EB , FD . Thus, as AE is to EB , so CF (is) to FD [Def. 5.5].

Thus, if composed magnitudes are proportional, (then) they will also be proportional (when) separated. (Which is) the very thing it was required to show.

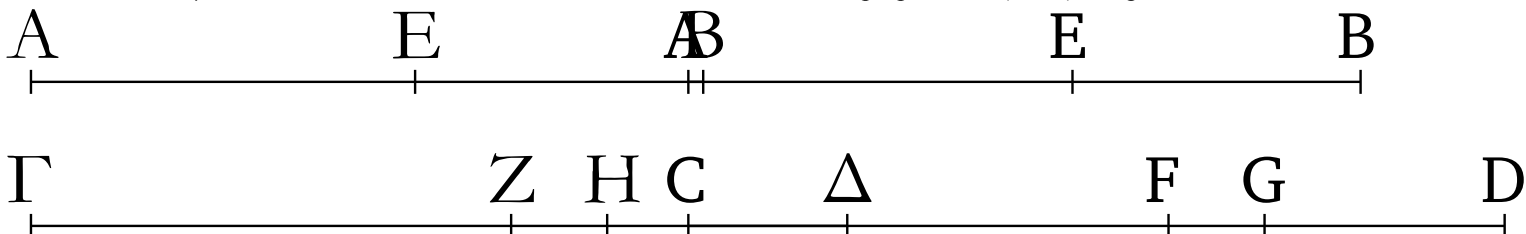
† In modern notation, this proposition reads that if $\alpha + \beta : \beta :: \gamma + \delta : \delta$ then $\alpha : \beta :: \gamma : \delta$.

18.

Proposition 18†

Ἐὰν διηρημένα μεγέθη ἀνάλογον >ῃ|, καὶ συντεθέντα ἀνάλογον >έσται.

If separated magnitudes are proportional, (then) they will also be proportional (when) composed.



>Ἐστω διηρημένα μεγέθη ἀνάλογον τὰ AE , EB , ΓZ , $Z\Delta$, ὥς τὸ AE πρὸς τὸ EB , ο<ύτως τὸ ΓZ πρὸς τὸ $Z\Delta$; λέγω, <ότι καὶ συντεθέντα ἀνάλογον >έσται, ὥς τὸ AB πρὸς τὸ BE , ο<ύτως τὸ $\Gamma\Delta$ πρὸς τὸ $Z\Delta$.

Εἰ γὰρ μὴ ἐστὶν ὥς τὸ AB πρὸς τὸ BE , ο<ύτως τὸ $\Gamma\Delta$ πρὸς τὸ $Z\Delta$, >έσται ὥς τὸ AB πρὸς τὸ BE , ο<ύτως τὸ $\Gamma\Delta$ >ήτοι πρὸς >έλασσόν τι τοῦ $Z\Delta$ >ῃ πρὸς μείζον.

>Ἐστω πρότερον πρὸς >έλασσον τὸ ΔH . καὶ ἐπεὶ ἐστὶν ὥς τὸ AB πρὸς τὸ BE , ο<ύτως τὸ $\Gamma\Delta$ πρὸς τὸ ΔH , συγκείμενα μεγέθη ἀνάλογόν ἐστιν; <ώστε καὶ διαιρεθέντα ἀνάλογον >έσται. >έστιν >άρα ὥς τὸ AE πρὸς τὸ EB , ο<ύτως τὸ ΓH πρὸς τὸ $H\Delta$. ὑπόκειται δὲ καὶ ὥς τὸ AE πρὸς τὸ EB , ο<ύτως τὸ ΓZ πρὸς τὸ $Z\Delta$. καὶ ὥς >άρα τὸ ΓH πρὸς τὸ $H\Delta$, ο<ύτως τὸ ΓZ πρὸς τὸ $Z\Delta$. μείζον δὲ τὸ πρῶτον τὸ ΓH τοῦ τρίτου τοῦ ΓZ ; μείζον >άρα καὶ τὸ δεῦτερον τὸ $H\Delta$ τοῦ τετάρτου τοῦ $Z\Delta$. ἀλλὰ καὶ >έλαττον; <όπερ ἐστὶν ἀδύνατον; οὐκ >άρα ἐστὶν ὥς τὸ AB πρὸς τὸ BE , ο<ύτως τὸ $\Gamma\Delta$ πρὸς >έλασσον τοῦ $Z\Delta$. ὁμοίως δὲ δείχομεν, <ότι οὐδὲ πρὸς μείζον; πρὸς αὐτὸ >άρα.

Ἐὰν >άρα διηρημένα μεγέθη ἀνάλογον >ῃ|, καὶ συντεθέντα ἀνάλογον >έσται; <όπερ >έδει δεῖχαι.

Let AE , EB , CF , and FD be separated magnitudes (which are) proportional, (so that) as AE (is) to EB , so CF (is) to FD . I say that they will also be proportional (when) composed, (so that) as AB (is) to BE , so CD (is) to FD .

For if (it is) not (the case that) as AB is to BE , so CD (is) to FD , (then) it will surely be (the case that) as AB (is) to BE , so CD is either to some (magnitude) less than DF , or (some magnitude) greater (than DF).[‡]

Let it, first of all, be to (some magnitude) less (than DF), (namely) DG . And since composed magnitudes are proportional, (so that) as AB is to BE , so CD (is) to DG , they will thus also be proportional (when) separated [Prop. 5.17]. Thus, as AE is to EB , so CG (is) to GD . But it was also assumed that as AE (is) to EB , so CF (is) to FD . Thus, (it is) also (the case that) as CG (is) to GD , so CF (is) to FD [Prop. 5.11]. And the first (magnitude) CG (is) greater than the third CF . Thus, the second (magnitude) GD (is) also greater than the fourth FD [Prop. 5.14]. But (it is) also less. The very thing is impossible. Thus, (it is) not (the case that) as AB is to BE , so CD (is) to less than FD . Similarly, we can show that neither (is it the case) to greater (than FD). Thus, (it is the case) to the same (as FD).

Thus, if separated magnitudes are proportional, (then) they will also be proportional (when) composed. (Which is) the very thing it was required to show.

† In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$ then $\alpha + \beta : \beta :: \gamma + \delta : \delta$.

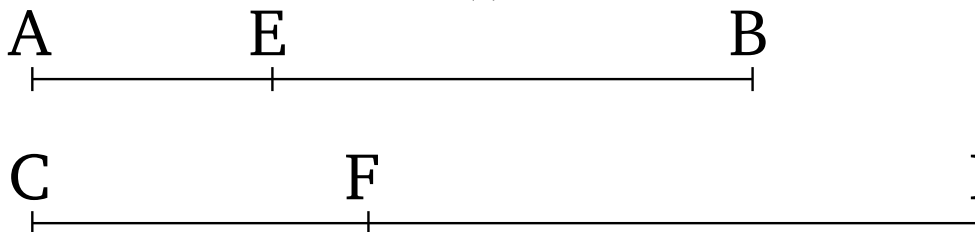
‡ Here, Euclid assumes, without proof, that a fourth magnitude proportional to three given magnitudes can always be found.

19.

Proposition 19†

Ἐὰν >ῃ| ὥς <όλον πρὸς <όλον, ο<ύτως ἀφαιρεθὲν πρὸς ἀφαιρεθὲν, καὶ τὸ λοιπὸν πρὸς τὸ λοιπὸν >έσται ὥς <όλον πρὸς <όλον.

If as the whole is to the whole so the (part) taken away is to the (part) taken away, (then) the remainder to the remainder will also be as the whole (is) to the whole.



E

B

Z

Δ

>Ἐστω γάρ ὡς <ὅλον τὸ AB πρὸς <ὅλον τὸ ΓΔ, ο<ύτως ἀφαιρεθὲν τὸ AE πρὸς ἀφαιρεθὲν τὸ ΓΖ; λέγω, <ὅτι καὶ λοιπὸν τὸ EB πρὸς λοιπὸν τὸ ΖΔ >έσται ὡς <ὅλον τὸ AB πρὸς <ὅλον τὸ ΓΔ.

Ἐπεὶ γάρ ἐστιν ὡς τὸ AB πρὸς τὸ ΓΔ, ο<ύτως τὸ AE πρὸς τὸ ΓΖ, καὶ ἐναλλάχ ὡς τὸ BA πρὸς τὸ AE, ο<ύτως τὸ ΔΓ πρὸς τὸ ΓΖ. καὶ ἐπεὶ συγκείμενα μεγέθη ἀνάλογόν ἐστιν, καὶ διαιρεθέντα ἀνάλογον >έσται, ὡς τὸ BE πρὸς τὸ EA, ο<ύτως τὸ ΔΖ πρὸς τὸ ΓΖ; καὶ ἐναλλάχ, ὡς τὸ BE πρὸς τὸ ΔΖ, ο<ύτως τὸ EA πρὸς τὸ ΖΓ. ὡς δὲ τὸ AE πρὸς τὸ ΓΖ, ο<ύτως ὑπόκειται <ὅλον τὸ AB πρὸς <ὅλον τὸ ΓΔ. καὶ λοιπὸν >άρα τὸ EB πρὸς λοιπὸν τὸ ΖΔ >έσται ὡς <ὅλον τὸ AB πρὸς <ὅλον τὸ ΓΔ.

Ἐὰν >άρα >ῆ| ὡς <ὅλον πρὸς <ὅλον, ο<ύτως ἀφαιρεθὲν πρὸς ἀφαιρεθὲν, καὶ τὸ λοιπὸν πρὸς τὸ λοιπὸν >έσται ὡς <ὅλον πρὸς <ὅλον [ὅπερ >έδει δεῖχαι].

[Ka`i>epe`i>ede`iqjh<wc t`o AB pr`oc t`o GD, ο<`utwc t`o EB pr`oc t`o ZD, ka`i>enall`ax<wc t`o AB pr`oc t`o BE ο<`utwc t`o GD pr`oc t`o ZD, sugke`imena >`ara meg`ejh>an`alog'on>estin; >ede`iqjh d`e<wc t`o BA pr`oc t`o AE, ο<`utwc t`o DG pr`oc t`o GZ; ka`i>estin>anastr`eyanti].

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, <ὅτι ἔαν συγκείμενα μεγέθη ἀνάλογον >ῆ|, καὶ ἀναστρέψαντι ἀνάλογον >έσται; ὅπερ >έδει δεῖχαι.

Corollary[‡]

So (it is) clear, from this, that if composed magnitudes are proportional, (then) they will also be proportional (when) converted. (Which is) the very thing it was required to show.

[†] In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$ then $\alpha : \beta :: \alpha - \gamma : \beta - \delta$.

[‡] In modern notation, this corollary reads that if $\alpha : \beta :: \gamma : \delta$ then $\alpha : \alpha - \beta :: \gamma : \gamma - \delta$.

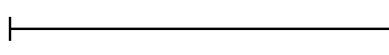
20.

Ἐὰν >ῆ| τρία μεγέθη καὶ >άλλα αὐτοῖς >ίσα τὸ πλῆθος, σύνδυο λαμβανόμενα καὶ ἐν τῷ αὐτῷ λόγῳ, δι> >ίσου δὲ τὸ πρῶτον τοῦ τρίτου μεῖζον >ῆ|, καὶ τὸ τέταρτον τοῦ <έκτου μεῖζον >έσται, κ>ὰν >ίσον, >ίσον, κ>ὰν >έλαττον, >έλαττον.

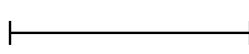
Proposition 20[†]

If there are three magnitudes, and others of equal number to them, (being) also in the same ratio taken two by two, and (if), via equality, the first is greater than the third (then) the fourth will also be greater than the sixth. And if (the first is) equal (to the third then the fourth will also be) equal (to the sixth). And if (the first is) less (than the third then the fourth will also be) less (than the sixth).

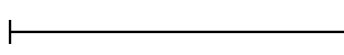
A



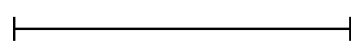
B



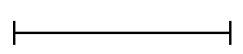
C



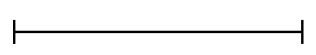
D

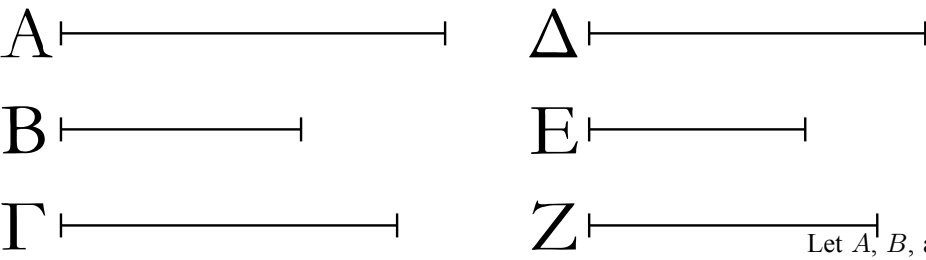


E



F





>Ἐστω τρία μεγέθη τὰ Α, Β, Γ, καὶ >άλλα αὐτοῖς >ίσα τὸ πληθοῦς τὰ Δ, Ε, Ζ, σύνδυο λαμβανόμενα ἐν τῷ αὐτῷ λόγῳ, ὥς μὲν τὸ Α πρὸς τὸ Β, ο<ύτωξ τὸ Δ πρὸς τὸ Ε, ὥς δὲ τὸ Β πρὸς τὸ Γ, ο<ύτωξ τὸ Ε πρὸς τὸ Ζ, δι> >ίσου δὲ μείζον >έστω τὸ Α τοῦ Γ; λέγω, <ὅτι καὶ τὸ Δ τοῦ Ζ μείζον >έσται, κ>ὰν >ίσον, >ίσον, κ>ὰν >έλαττον, >έλαττον.

Ἐπεὶ γὰρ μείζον ἐστὶ τὸ Α τοῦ Γ, >άλλο δέ τι τὸ Β, τὸ δὲ μείζον πρὸς τὸ αὐτὸ μείζονα λόγον >έθει >ήπερ τὸ >έλαττον, τὸ Α >άρα πρὸς τὸ Β μείζονα λόγον >έθει >ήπερ τὸ Γ πρὸς τὸ Β. ἀλλ> ὥς μὲν τὸ Α πρὸς τὸ Β [ο<ύτωξ] τὸ Δ πρὸς τὸ Ε, ὥς δὲ τὸ Γ πρὸς τὸ Β, ἀνάπαλιν ο<ύτωξ τὸ Ζ πρὸς τὸ Ε; καὶ τὸ Δ >άρα πρὸς τὸ Ε μείζονα λόγον >έθει >ήπερ τὸ Ζ πρὸς τὸ Ε. τῶν δὲ πρὸς τὸ αὐτὸ λόγον ἐθόντων τὸ μείζονα λόγον >έθον μείζον ἐστίν. μείζον >άρα τὸ Δ τοῦ Ζ. ὁμοίωξ δὴ δείχομεν, <ὅτι κ>ὰν >ίσον >ῆ| τὸ Α τῷ Γ, >ίσον >έσται καὶ τὸ Δ τῷ Ζ, κ>ὰν >έλαττον, >έλαττον.

Ἐὰν >άρα >ῆ| τρία μεγέθη καὶ >άλλα αὐτοῖς >ίσα τὸ πληθοῦς, σύνδυο λαμβανόμενα καὶ ἐν τῷ αὐτῷ λόγῳ, δι> >ίσου δὲ τὸ πρῶτον τοῦ τρίτου μείζον >ῆ|, καὶ τὸ τέταρτον τοῦ <έκτου μείζον >έσται, κ>ὰν >ίσον, >ίσον, κ>ὰν >έλαττον, >έλαττον; <όπερ >έδει δεῖχαι.

Let A , B , and C be three magnitudes, and D , E , F other (magnitudes) of equal number to them, (being) in the same ratio taken two by two, (so that) as A (is) to B , so D (is) to E , and as B (is) to C , so E (is) to F . And let A be greater than C , via equality. I say that D will also be greater than F . And if (A is) equal (to C then D will also be) equal (to F). And if (A is) less (than C then D will also be) less (than F).

For since A is greater than C , and B some other (magnitude), and the greater (magnitude) has a greater ratio than the lesser to the same (magnitude) [Prop. 5.8], A thus has a greater ratio to B than C (has) to B . But as A (is) to B , [so] D (is) to E . And, inversely, as C (is) to B , so F (is) to E [Prop. 5.7 corr.]. Thus, D also has a greater ratio to E than F (has) to E [Prop. 5.13]. And for (magnitudes) having a ratio to the same (magnitude), that having the greater ratio is greater [Prop. 5.10]. Thus, D (is) greater than F . Similarly, we can show that even if A is equal to C (then) D will also be equal to F , and even if (A is) less (than C then D will also be) less (than F).

Thus, if there are three magnitudes, and others of equal number to them, (being) also in the same ratio taken two by two, and (if), via equality, the first is greater than the third, (then) the fourth will also be greater than the sixth. And if (the first is) equal (to the third then the fourth will also be) equal (to the sixth). And (if the first is) less (than the third then the fourth will also be) less (than the sixth). (Which is) the very thing it was required to show.

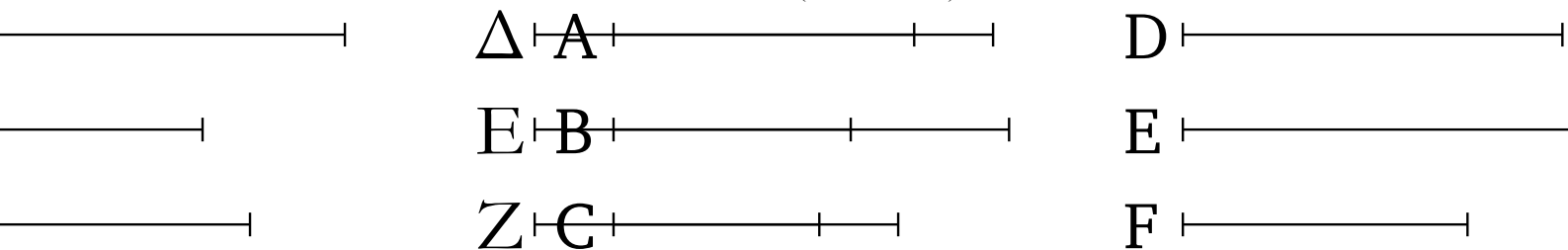
† In modern notation, this proposition reads that if $\alpha : \beta :: \delta : \epsilon$ and $\beta : \gamma :: \epsilon : \zeta$ then $\alpha : \gamma \gtrless \delta : \zeta$.

21.

Proposition 21†

Ἐὰν >ῆ| τρία μεγέθη καὶ >άλλα αὐτοῖς >ίσα τὸ πληθοῦς σύνδυο λαμβανόμενα καὶ ἐν τῷ αὐτῷ λόγῳ, >ῆ| δὲ τεταραγμένη αὐτῶν ἡ ἀναλογία, δι> >ίσου δὲ τὸ πρῶτον τοῦ τρίτου μείζον >ῆ|, καὶ τὸ τέταρτον τοῦ <έκτου μείζον >έσται, κ>ὰν >ίσον, >ίσον, κ>ὰν >έλαττον, >έλαττον.

If there are three magnitudes, and others of equal number to them, (being) also in the same ratio taken two by two, and (if) their proportion (is) perturbed, and (if), via equality, the first is greater than the third (then) the fourth will also be greater than the sixth. And if (the first is) equal (to the third then the fourth will also be) equal (to the sixth). And if (the first is) less (than the third then the fourth will also be) less (than the sixth).



>Ἐστω τρία μεγέθη τὰ Α, Β, Γ καὶ >άλλα αὐτοῖς >ίσα τὸ πληθοῦς τὰ Δ, Ε, Ζ, σύνδυο λαμβανόμενα καὶ ἐν τῷ αὐτῷ λόγῳ, >έστω δὲ τεταραγμένη αὐτῶν ἡ ἀναλογία, ὥς μὲν τὸ Α πρὸς τὸ Β, ο<ύτωξ τὸ Ε πρὸς τὸ Ζ, ὥς δὲ τὸ Β πρὸς τὸ Γ, ο<ύτωξ τὸ Δ πρὸς τὸ Ε, δι> >ίσου δὲ τὸ Α τοῦ Γ μείζον >έστω; λέγω, <ὅτι καὶ τὸ Δ τοῦ Ζ μείζον >έσται,

Let A , B , and C be three magnitudes, and D , E , F other (magnitudes) of equal number to them, (being) in the same ratio taken two by two. And let their proportion be perturbed, (so that) as A (is) to B , so E (is) to F , and as B (is) to C , so D (is) to E . And let A be greater than C , via equality. I say that D will also be greater than F . And if (A

$\kappa > \alpha\nu > \text{ισον}, > \text{ισον}, \kappa > \alpha\nu > \text{ἐλαττον}, > \text{ἐλαττον}.$

Ἐπεὶ γὰρ μείζον ἐστὶ τὸ Α τοῦ Γ, > ἄλλο δέ τι τὸ Β, τὸ Α > ἄρα πρὸς τὸ Β μείζονα λόγον > ἐθεὶ > ἤπερ τὸ Γ πρὸς τὸ Β. ἄλλ' ὥς μὲν τὸ Α πρὸς τὸ Β, ο< ὑτωξ τὸ Ε πρὸς τὸ Ζ, ὥς δὲ τὸ Γ πρὸς τὸ Β, ἀνάπαλιν ο< ὑτωξ τὸ Ε πρὸς τὸ Δ. καὶ τὸ Ε > ἄρα πρὸς τὸ Ζ μείζονα λόγον > ἐθεὶ > ἤπερ τὸ Ε πρὸς τὸ Δ. πρὸς < ὃ δὲ τὸ αὐτὸ μείζονα λόγον > ἐθεὶ, ἐκεῖνο > ἐλασσόν ἐστιν; > ἐλασσον > ἄρα ἐστὶ τὸ Ζ τοῦ Δ; μείζον > ἄρα ἐστὶ τὸ Δ τοῦ Ζ. ὁμοίως δὴ δείχομεν, < ὅτι $\kappa > \alpha\nu > \text{ισον} > \eta \mid$ τὸ Α τῷ Γ, > ἴσον > ἐστὶ καὶ τὸ Δ τῷ Ζ, $\kappa > \alpha\nu > \text{ἐλαττον}, > \text{ἐλαττον}.$ Ἐὰν > ἄρα > $\eta \mid$ τρία μεγέθη καὶ > ἄλλα αὐτοῖς > ἴσα τὸ πληθοξ, σύνδυο λαμβανόμενα καὶ ἐν τῷ αὐτῷ λόγῳ, > $\eta \mid$ δὲ τεταραγμένη αὐτῶν ἡ ἀναλογία, δι> > ἴσου δὲ τὸ πρῶτον τοῦ τρίτου μείζον > $\eta \mid$, καὶ τὸ τέταρτον τοῦ < ἔκτου μείζον > ἐστὶ, $\kappa > \alpha\nu > \text{ισον}, > \text{ισον}, \kappa > \alpha\nu > \text{ἐλαττον}, > \text{ἐλαττον};$ < ὅπερ > ἐδει δεῖχαι.

is) equal (to C then D will also be) equal (to F). And if (A is) less (than C then D will also be) less (than F).

For since A is greater than C , and B some other (magnitude), A thus has a greater ratio to B than C (has) to B [Prop. 5.8]. But as A (is) to B , so E (is) to F . And, inversely, as C (is) to B , so E (is) to D [Prop. 5.7 corr.]. Thus, E also has a greater ratio to F than E (has) to D [Prop. 5.13]. And that (magnitude) to which the same (magnitude) has a greater ratio is (the) lesser (magnitude) [Prop. 5.10]. Thus, F is less than D . Thus, D is greater than F . Similarly, we can show that even if A is equal to C (then) D will also be equal to F , and even if (A is) less (than C then D will also be) less (than F).

Thus, if there are three magnitudes, and others of equal number to them, (being) also in the same ratio taken two by two, and (if) their proportion (is) perturbed, and (if), via equality, the first is greater than the third (then) the fourth will also be greater than the sixth. And if (the first is) equal (to the third then the fourth will also be) equal (to the sixth). And if (the first is) less (than the third then the fourth will also be) less (than the sixth). (Which is) the very thing it was required to show.

† In modern notation, this proposition reads that if $\alpha : \beta :: \epsilon : \zeta$ and $\beta : \gamma :: \delta : \epsilon$ then $\alpha \gtrless \gamma$ as $\delta \gtrless \zeta$.

22.

Proposition 22†

Ἐὰν > $\eta \mid$ ὅποσαοῦν μεγέθη καὶ > ἄλλα αὐτοῖς > ἴσα τὸ πληθοξ, σύνδυο λαμβανόμενα καὶ ἐν τῷ αὐτῷ λόγῳ, καὶ δι> > ἴσου ἐν τῷ αὐτῷ λόγῳ > ἐστὶ.

If there are any number of magnitudes whatsoever, and (some) other (magnitudes) of equal number to them, (which are) also in the same ratio taken two by two, (then) they will also be in the same ratio via equality.

BA

B F

C

ED

E Z

F

KG

K M

M

ΛΗ

L N

N

> Ἐστω ὅποσαοῦν μεγέθη τὰ Α, Β, Γ καὶ > ἄλλα αὐτοῖς > ἴσα τὸ πληθοξ τὰ Δ, Ε, Ζ, σύνδυο λαμβανόμενα ἐν τῷ αὐτῷ λόγῳ, ὥς μὲν τὸ Α πρὸς τὸ Β, ο< ὑτωξ τὸ Δ πρὸς τὸ Ε, ὥς δὲ τὸ Β πρὸς τὸ Γ, ο< ὑτωξ τὸ Ε πρὸς τὸ Ζ; λέγω, < ὅτι καὶ δι> > ἴσου ἐν τῷ αὐτῷ λόγῳ > ἐστὶ.

Εἰλήφθω γὰρ τῶν μὲν Α, Δ ἰσάκις πολλαπλάσια τὰ Η, Θ, τῶν δὲ Β, Ε > ἄλλα, < ἃ > ἐτυθεν, ἰσάκις πολλαπλάσια τὰ Κ, Λ, καὶ > ἐτι τῶν Γ, Ζ > ἄλλα, < ἃ > ἐτυθεν, ἰσάκις πολλαπλάσια τὰ Μ, Ν.

Καὶ ἐπεὶ ἐστὶν ὥς τὸ Α πρὸς τὸ Β, ο< ὑτωξ τὸ Δ πρὸς τὸ Ε, καὶ > ἰληπται τῶν μὲν Α, Δ ἰσάκις πολλαπλάσια τὰ Η, Θ, τῶν δὲ Β, Ε > ἄλλα, < ἃ > ἐτυθεν, ἰσάκις πολλαπλάσια τὰ Κ, Λ, > ἐστὶν > ἄρα ὥς τὸ Η πρὸς τὸ Κ, ο< ὑτωξ τὸ Θ πρὸς τὸ Λ. διὰ τὰ αὐτὰ δὴ καὶ ὥς τὸ Κ πρὸς τὸ Μ, ο< ὑτωξ τὸ Λ πρὸς τὸ Ν. ἐπεὶ ο> ὦν τρία μεγέθη ἐστὶ τὰ Η, Κ, Μ, καὶ > ἄλλα αὐτοῖς > ἴσα τὸ πληθοξ τὰ Θ, Λ, Ν, σύνδυο λαμβανόμενα καὶ ἐν τῷ αὐτῷ λόγῳ, δι> > ἴσου > ἄρα, εἰ ὑπερέθει τὸ Η τοῦ Μ, ὑπερέθει καὶ τὸ Θ τοῦ Ν, καὶ εἰ > ἴσον, > ἴσον, καὶ εἰ > ἐλαττον, > ἐλαττον. καὶ ἐστὶ τὰ μὲν Η, Θ τῶν Α, Δ ἰσάκις

Let there be any number of magnitudes whatsoever, A , B , C , and (some) other (magnitudes), D , E , F , of equal number to them, (which are) in the same ratio taken two by two, (so that) as A (is) to B , so D (is) to E , and as B (is) to C , so E (is) to F . I say that they will also be in the same ratio via equality. (That is, as A is to C , so D is to F .)

For let the equal multiples G and H be taken of A and D (respectively), and the other random equal multiples K and L of B and E (respectively), and the yet other random equal multiples M and N of C and F (respectively).

And since as A is to B , so D (is) to E , and the equal multiples G and H have been taken of A and D (respectively), and the other random equal multiples K and L of B and E (respectively), thus as G is to K , so H (is) to L [Prop. 5.4]. And, so, for the same (reasons), as K (is) to M , so L (is) to N . Therefore, since G , K , and M are three magnitudes, and H , L , and N other (magnitudes) of equal number to them, (which are) also in the same ratio taken two by two, thus, via equality, if G exceeds M (then) H also exceeds N , and

πολλαπλάσια, τὰ δὲ M, N τῶν Γ, Z >άλλα, <ὰ >έτυθεν, ισάκις πολλαπλάσια. >έστιν >άρα ὡς τὸ A πρὸς τὸ Γ , ο<ύτως τὸ Δ πρὸς τὸ Z .

Ἐὰν >άρα >ῆ| ὅποσαοῦν μεγέθη καὶ >άλλα αὐτοῖς >ίσα τὸ πληῖθοξ, σύνδυο λαμβανόμενα ἐν τῷ αὐτῷ λόγῳ, καὶ δι> >ίσου ἐν τῷ αὐτῷ λόγῳ >έσται; <όπερ >έδει δεῖχαι.

if (G is) equal (to M then H is also) equal (to N), and if (G is) less (than M then H is also) less (than N) [Prop. 5.20]. And G and H are equal multiples of A and D (respectively), and M and N other random equal multiples of C and F (respectively). Thus, as A is to C , so D (is) to F [Def. 5.5].

Thus, if there are any number of magnitudes whatsoever, and (some) other (magnitudes) of equal number to them, (which are) also in the same ratio taken two by two, (then) they will also be in the same ratio via equality. (Which is) the very thing it was required to show.

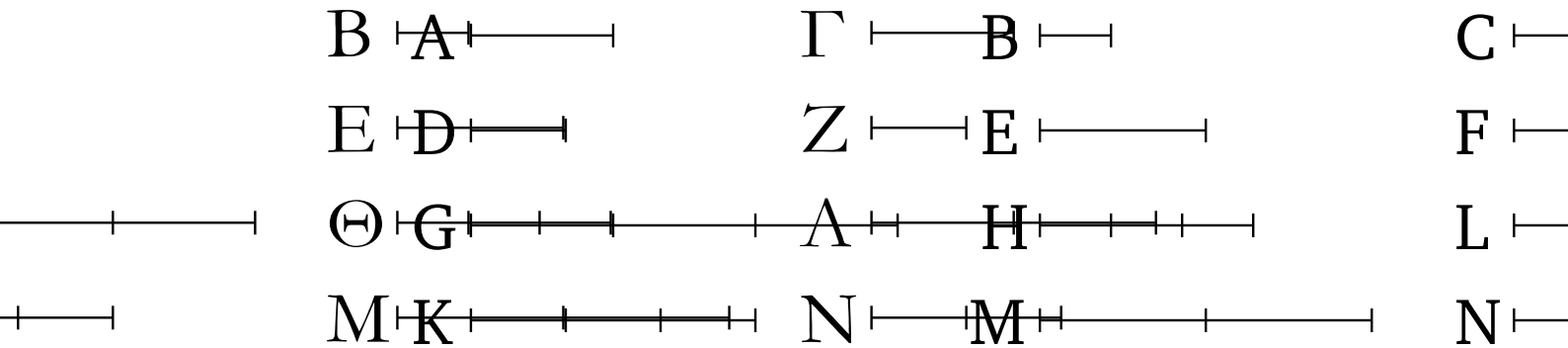
† In modern notation, this proposition reads that if $\alpha : \beta :: \epsilon : \zeta$ and $\beta : \gamma :: \zeta : \eta$ and $\gamma : \delta :: \eta : \theta$ then $\alpha : \delta :: \epsilon : \theta$.

23.

Ἐὰν >ῆ| τρία μεγέθη καὶ >άλλα αὐτοῖς >ίσα τὸ πληῖθοξ σύνδυο λαμβανόμενα ἐν τῷ αὐτῷ λόγῳ, >ῆ| δὲ τεταραγμένη αὐτῶν ἡ ἀναλογία, καὶ δι> >ίσου ἐν τῷ αὐτῷ λόγῳ >έσται.

Proposition 23†

If there are three magnitudes, and others of equal number to them, (being) in the same ratio taken two by two, and (if) their proportion is perturbed, (then) they will also be in the same ratio via equality.



>Ἐστω τρία μεγέθη τὰ A, B, Γ καὶ >άλλα αὐτοῖς >ίσα τὸ πληῖθοξ σύνδυο λαμβανόμενα ἐν τῷ αὐτῷ λόγῳ τὰ Δ, E, Z , >έστω δὲ τεταραγμένη αὐτῶν ἡ ἀναλογία, ὡς μὲν τὸ A πρὸς τὸ B , ο<ύτως τὸ E πρὸς τὸ Z , ὡς δὲ τὸ B πρὸς τὸ Γ , ο<ύτως τὸ Δ πρὸς τὸ E ; λέγω, <ότι ἐστὶν ὡς τὸ A πρὸς τὸ Γ , ο<ύτως τὸ Δ πρὸς τὸ Z .

Εἰλήφθω τῶν μὲν A, B, Δ ισάκις πολλαπλάσια τὰ H, Θ, K , τῶν δὲ Γ, E, Z >άλλα, <ὰ >έτυθεν, ισάκις πολλαπλάσια τὰ Λ, M, N .

Καὶ ἐπεὶ ισάκις ἐστὶ πολλαπλάσια τὰ H, Θ τῶν A, B , τὰ δὲ μέρη τοῖς ὡσαύτως πολλαπλασίοις τὸν αὐτὸν >έθει λόγον, >έστιν >άρα ὡς τὸ A πρὸς τὸ B , ο<ύτως τὸ H πρὸς τὸ Θ . διὰ τὰ αὐτὰ δὴ καὶ ὡς τὸ E πρὸς τὸ Z , ο<ύτως τὸ M πρὸς τὸ N ; καὶ ἐστὶν ὡς τὸ A πρὸς τὸ B , ο<ύτως τὸ E πρὸς τὸ Z ; καὶ ὡς >άρα τὸ H πρὸς τὸ Θ , ο<ύτως τὸ M πρὸς τὸ N . καὶ ἐπεὶ ἐστὶν ὡς τὸ B πρὸς τὸ Γ , ο<ύτως τὸ Δ πρὸς τὸ E , καὶ ἐναλλάχ ὡς τὸ B πρὸς τὸ Δ , ο<ύτως τὸ Γ πρὸς τὸ E . καὶ ἐπεὶ τὰ Θ, K τῶν B, Δ ισάκις ἐστὶ πολλαπλάσια, τὰ δὲ μέρη τοῖς ισάκις πολλαπλασίοις τὸν αὐτὸν >έθει λόγον, >έστιν >άρα ὡς τὸ B πρὸς τὸ Δ , ο<ύτως τὸ Θ πρὸς τὸ K . ἀλλ> ὡς τὸ B πρὸς τὸ Δ , ο<ύτως τὸ Γ πρὸς τὸ E ; καὶ ὡς >άρα τὸ Θ πρὸς τὸ K , ο<ύτως τὸ Γ πρὸς τὸ E . πάλιν, ἐπεὶ τὰ Λ, M τῶν Γ, E ισάκις ἐστὶ πολλαπλάσια, >έστιν >άρα ὡς τὸ Γ πρὸς τὸ E , ο<ύτως τὸ Λ πρὸς τὸ M . ἀλλ> ὡς τὸ Γ πρὸς τὸ E , ο<ύτως τὸ Θ πρὸς τὸ K ; καὶ ὡς >άρα τὸ Θ πρὸς τὸ K , ο<ύτως τὸ Λ πρὸς τὸ M , καὶ ἐναλλάχ ὡς τὸ Θ πρὸς τὸ Λ , τὸ K πρὸς τὸ M . ἐδείκθῃ δὲ καὶ ὡς τὸ H πρὸς τὸ Θ , ο<ύτως τὸ M πρὸς τὸ N . ἐπεὶ ο>ὖν τρία μεγέθη ἐστὶ τὰ H, Θ, Λ , καὶ >άλλα αὐτοῖς >ίσα τὸ πληῖθοξ τὰ K, M, N σύνδυο λαμβανόμενα ἐν τῷ αὐτῷ λόγῳ, καὶ ἐστὶν αὐτῶν τεταραγμένη ἡ ἀναλογία, δι> >ίσου >άρα, εἰ ὑπερέθει τὸ H τοῦ Λ , ὑπερέθει καὶ τὸ K

Let A, B , and C be three magnitudes, and D, E and F other (magnitudes) of equal number to them, (being) in the same ratio taken two by two. And let their proportion be perturbed, (so that) as A (is) to B , so E (is) to F , and as B (is) to C , so D (is) to E . I say that as A is to C , so D (is) to F .

Let the equal multiples G, H , and K be taken of A, B , and D (respectively), and the other random equal multiples L, M , and N of C, E , and F (respectively).

And since G and H are equal multiples of A and B (respectively), and parts have the same ratio as similar multiples [Prop. 5.15], thus as A (is) to B , so G (is) to H . And, so, for the same (reasons), as E (is) to F , so M (is) to N . And as A is to B , so E (is) to F . And, thus, as G (is) to H , so M (is) to N [Prop. 5.11]. And since as B is to C , so D (is) to E , also, alternately, as B (is) to D , so C (is) to E [Prop. 5.16]. And since H and K are equal multiples of B and D (respectively), and parts have the same ratio as similar multiples [Prop. 5.15], thus as B is to D , so H (is) to K . But, as B (is) to D , so C (is) to E . And, thus, as H (is) to K , so C (is) to E [Prop. 5.11]. Again, since L and M are equal multiples of C and E (respectively), thus as C is to E , so L (is) to M [Prop. 5.15]. But, as C (is) to E , so H (is) to K . And, thus, as H (is) to K , so L (is) to M [Prop. 5.11]. Also, alternately, as H (is) to L , so K (is) to M [Prop. 5.16]. And it was also shown (that) as G (is) to H , so M (is) to N . Therefore, since G, H , and L are three magnitudes, and K, M , and N other (magnitudes) of equal number to them, (being) in the same ratio taken two by two, and their proportion is perturbed, thus, via equality, if G exceeds L (then) K also exceeds N , and if (G is) equal

τοῦ N , καὶ εἰ $>$ ίσον, $>$ ίσον, καὶ εἰ $>$ έλαττον, $>$ έλαττον. καὶ ἔστι τὰ μὲν H, K τῶν A, Δ ἰσάκις πολλαπλάσια, τὰ δὲ Λ, N τῶν Γ, Z . $>$ έστιν $>$ άρα ὡς τὸ A πρὸς τὸ Γ , ο $<$ ύτως τὸ Δ πρὸς τὸ Z .

Ἐὰν $>$ άρα $>$ ῆ| τρία μεγέθη καὶ $>$ άλλα αὐτοῖς $>$ ίσα τὸ πληθοῦς σύνδυο λαμβανόμενα ἐν τῷ αὐτῷ| λόγῳ, $>$ ῆ| δὲ τεταραγμένη αὐτῶν ἡ ἀναλογία, καὶ δι $>$ $>$ ίσου ἐν τῷ| αὐτῷ| λόγῳ $>$ έσται; <όπερ $>$ έδει δεῖχαι.

(to L then K is also) equal (to N), and if (G is) less (than L then K is also) less (than N) [Prop. 5.21]. And G and K are equal multiples of A and D (respectively), and L and N of C and F (respectively). Thus, as A (is) to C , so D (is) to F [Def. 5.5].

Thus, if there are three magnitudes, and others of equal number to them, (being) in the same ratio taken two by two, and (if) their proportion is perturbed, (then) they will also be in the same ratio via equality. (Which is) the very thing it was required to show.

† In modern notation, this proposition reads that if $\alpha : \beta :: \epsilon : \zeta$ and $\beta : \gamma :: \delta : \epsilon$ then $\alpha : \gamma :: \delta : \zeta$.

24.

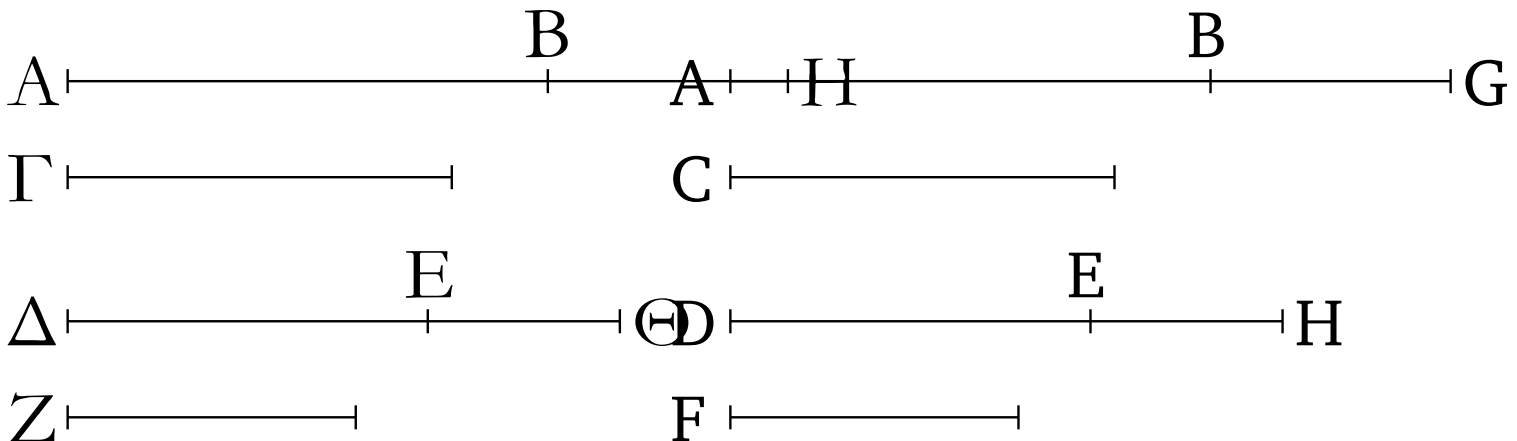
Ἐὰν πρῶτον πρὸς δεύτερον τὸν αὐτὸν $>$ έθῃ λόγον καὶ τρίτον πρὸς τέταρτον, $>$ έθῃ δὲ καὶ πέμπτον πρὸς δεύτερον τὸν αὐτὸν λόγον καὶ $<$ έκτον πρὸς τέταρτον, καὶ συντεθὲν πρῶτον καὶ πέμπτον πρὸς δεύτερον τὸν αὐτὸν $<$ έχει λόγον καὶ τρίτον καὶ $<$ έκτον πρὸς τέταρτον.

Πρῶτον γὰρ τὸ AB πρὸς δεύτερον τὸ Γ τὸν αὐτὸν ἔθετο λόγον καὶ τρίτον τὸ ΔE πρὸς τέταρτον τὸ Z , ἔθετο δὲ καὶ πέμπτον τὸ BH πρὸς δεύτερον τὸ Γ τὸν αὐτὸν λόγον καὶ $<$ έκτον τὸ $E\Theta$ πρὸς τέταρτον τὸ Z ; λέγω, $<$ ότι καὶ συντεθὲν πρῶτον καὶ πέμπτον τὸ AH πρὸς δεύτερον τὸ Γ τὸν αὐτὸν $<$ έχει λόγον, καὶ τρίτον καὶ $<$ έκτον τὸ $\Delta\Theta$ πρὸς τέταρτον τὸ Z .

Proposition 24†

If a first (magnitude) has to a second the same ratio that third (has) to a fourth, and a fifth (magnitude) also has to the second the same ratio that a sixth (has) to the fourth, (then) the first (magnitude) and the fifth, added together, will also have the same ratio to the second that the third (magnitude) and sixth (added together, have) to the fourth.

For let a first (magnitude) AB have the same ratio to a second C that a third DE (has) to a fourth F . And let a fifth (magnitude) BG also have the same ratio to the second C that a sixth EH (has) to the fourth F . I say that the first (magnitude) and the fifth, added together, AG , will also have the same ratio to the second C that the third (magnitude) and the sixth, (added together), DH , (has) to the fourth F .



Ἐπεὶ γὰρ ἔστιν ὡς τὸ BH πρὸς τὸ Γ , ο $<$ ύτως τὸ $E\Theta$ πρὸς τὸ Z , ἀνάπαλιν $>$ άρα ὡς τὸ Γ πρὸς τὸ BH , ο $<$ ύτως τὸ Z πρὸς τὸ $E\Theta$. ἐπεὶ ο $>$ ὺν ἔστιν ὡς τὸ AB πρὸς τὸ Γ , ο $<$ ύτως τὸ ΔE πρὸς τὸ Z , ὡς δὲ τὸ Γ πρὸς τὸ BH , ο $<$ ύτως τὸ Z πρὸς τὸ $E\Theta$, δι $>$ $>$ ίσου $>$ άρα ἔστιν ὡς τὸ AB πρὸς τὸ BH , ο $<$ ύτως τὸ ΔE πρὸς τὸ $E\Theta$. καὶ ἐπεὶ διηρημένα μεγέθη ἀνάλογόν ἐστιν, καὶ συντεθέντα ἀνάλογον $>$ έσται; $>$ έστιν $>$ άρα ὡς τὸ AH πρὸς τὸ HB , ο $<$ ύτως τὸ $\Delta\Theta$ πρὸς τὸ ΘE . $>$ έστι δὲ καὶ ὡς τὸ BH πρὸς τὸ Γ , ο $<$ ύτως τὸ $E\Theta$ πρὸς τὸ Z ; δι $>$ $>$ ίσου $>$ άρα ἔστιν ὡς τὸ AH πρὸς τὸ Γ , ο $<$ ύτως τὸ $\Delta\Theta$ πρὸς τὸ Z .

Ἐὰν $>$ άρα πρῶτον πρὸς δεύτερον τὸν αὐτὸν $>$ έθῃ λόγον καὶ τρίτον πρὸς τέταρτον, $>$ έθῃ δὲ καὶ πέμπτον πρὸς δεύτερον τὸν αὐτὸν λόγον καὶ $<$ έκτον πρὸς τέταρτον, καὶ συντεθὲν πρῶτον καὶ πέμπτον πρὸς δεύτερον τὸν αὐτὸν $<$ έχει λόγον καὶ τρίτον καὶ $<$ έκτον πρὸς τέταρτον; <όπερ $>$ έδει δεῖχαι.

For since as BG is to C , so EH (is) to F , thus, inversely, as C (is) to BG , so F (is) to EH [Prop. 5.7 corr.]. Therefore, since as AB is to C , so DE (is) to F , and as C (is) to BG , so F (is) to EH , thus, via equality, as AB is to BG , so DE (is) to EH [Prop. 5.22]. And since separated magnitudes are proportional, (then) they will also be proportional (when) composed [Prop. 5.18]. Thus, as AG is to GB , so DH (is) to HE . And, also, as BG is to C , so EH (is) to F . Thus, via equality, as AG is to C , so DH (is) to F [Prop. 5.22]. Thus, if a first (magnitude) has to a second the same ratio that a third (has) to a fourth, and a fifth (magnitude) also has to the second the same ratio that a sixth (has) to the fourth, (then) the first (magnitude) and the fifth, added together, will also have the same ratio to the second that the third (magnitude) and the sixth (added together, have) to the fourth. (Which is) the very thing it was required to show.

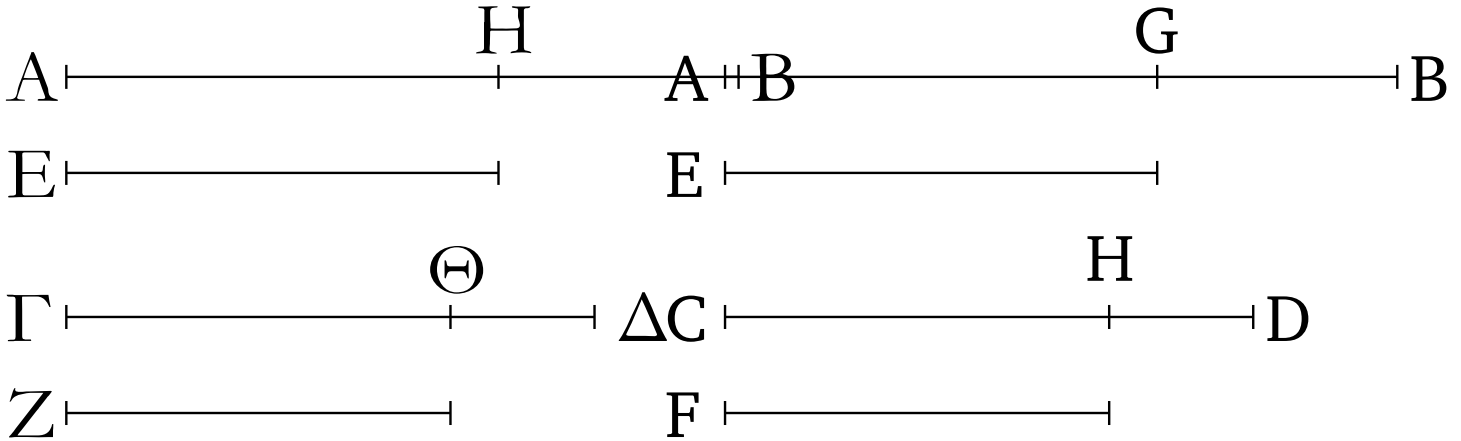
† In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$ and $\epsilon : \beta :: \zeta : \delta$ then $\alpha + \epsilon : \beta :: \gamma + \zeta : \delta$.

25.

Proposition 25[†]

Ἐὰν τέσσαρα μεγέθη ἀνάλογον >ῃ, τὸ μέγιστον [αὐτῶν] καὶ τὸ ἐλάχιστον δύο τῶν λοιπῶν μείζονά ἐστιν.

If four magnitudes are proportional, (then) the (sum of the) largest and the smallest [of them] is greater than the (sum of the) remaining two (magnitudes).



>Ἐστω τέσσαρα μεγέθη ἀνάλογον τὰ AB , $\Gamma\Delta$, E , Z , ὥς τὸ AB πρὸς τὸ $\Gamma\Delta$, οὐτως τὸ E πρὸς τὸ Z , >έστω δὲ μέγιστον μὲν αὐτῶν τὸ AB , ἐλάχιστον δὲ τὸ Z ; λέγω, <ὅτι τὰ AB , Z τῶν $\Gamma\Delta$, E μείζονά ἐστιν.

Κείσθω γὰρ τῷ μὲν E ἴσον τὸ AH , τῷ δὲ Z ἴσον τὸ $\Gamma\Theta$. Ἐπεὶ [οὖν] ἐστὶν ὥς τὸ AB πρὸς τὸ $\Gamma\Delta$, οὐτως τὸ E πρὸς τὸ Z , ἴσον δὲ τὸ μὲν E τῷ AH , τὸ δὲ Z τῷ $\Gamma\Theta$, >ἐστὶν >άρα ὥς τὸ AB πρὸς τὸ $\Gamma\Delta$, οὐτως τὸ AH πρὸς τὸ $\Gamma\Theta$. καὶ ἐπεὶ ἐστὶν ὥς <όλον τὸ AB πρὸς <όλον τὸ $\Gamma\Delta$, οὐτως ἀφαιρεθὲν τὸ AH πρὸς ἀφαιρεθὲν τὸ $\Gamma\Theta$, καὶ λοιπὸν >άρα τὸ HB πρὸς λοιπὸν τὸ $\Theta\Delta$ >έσται ὥς <όλον τὸ AB πρὸς <όλον τὸ $\Gamma\Delta$. μείζον δὲ τὸ AB τοῦ $\Gamma\Delta$; μείζον >άρα καὶ τὸ HB τοῦ $\Theta\Delta$. καὶ ἐπεὶ ἴσον ἐστὶ τὸ μὲν AH τῷ E , τὸ δὲ $\Gamma\Theta$ τῷ Z , τὰ >άρα AH , Z ἴσα ἐστὶ τοῖς $\Gamma\Theta$, E . καὶ [ἐπεὶ] ἔαν [ἀνίστοιξ] ἴσα προστεθῇ, τὰ <όλα >άνισά ἐστιν, ἔαν >άρα] τῶν HB , $\Theta\Delta$ ἀνίσων >όντων καὶ μείζονος τοῦ HB τῷ μὲν HB προστεθῇ] τὰ AH , Z , τῷ δὲ $\Theta\Delta$ προστεθῇ] τὰ $\Gamma\Theta$, E , συνάγεται τὰ AB , Z μείζονα τῶν $\Gamma\Delta$, E .

Ἐὰν >άρα τέσσαρα μεγέθη ἀνάλογον >ῃ, τὸ μέγιστον αὐτῶν καὶ τὸ ἐλάχιστον δύο τῶν λοιπῶν μείζονά ἐστιν. <όπερ >έδει δεῖχαι.

Let AB , CD , E , and F be four proportional magnitudes, (such that) as AB (is) to CD , so E (is) to F . And let AB be the greatest of them, and F the least. I say that AB and F is greater than CD and E .

For let AG be made equal to E , and CH equal to F .

[In fact,] since as AB is to CD , so E (is) to F , and E (is) equal to AG , and F to CH , thus as AB is to CD , so AG (is) to CH . And since the whole AB is to the whole CD as the (part) taken away AG (is) to the (part) taken away CH , thus the remainder GB will also be to the remainder HD as the whole AB (is) to the whole CD [Prop. 5.19]. And AB (is) greater than CD . Thus, GB (is) also greater than HD . And since AG is equal to E , and CH to F , thus AG and F is equal to CH and E . And [since] if [equal (magnitudes)] are added to unequal (magnitudes then) the wholes are unequal, thus if] AG and F are added to GB , and CH and E to HD — GB and HD being unequal, and GB greater—it is inferred that AB and F (is) greater than CD and E .

Thus, if four magnitudes are proportional, (then) the (sum of the) largest and the smallest of them is greater than the (sum of the) remaining two (magnitudes). (Which is) the very thing it was required to show.

[†] In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$, and α is the greatest and δ the least, then $\alpha + \delta > \beta + \gamma$.

ELEMENTS BOOK 6

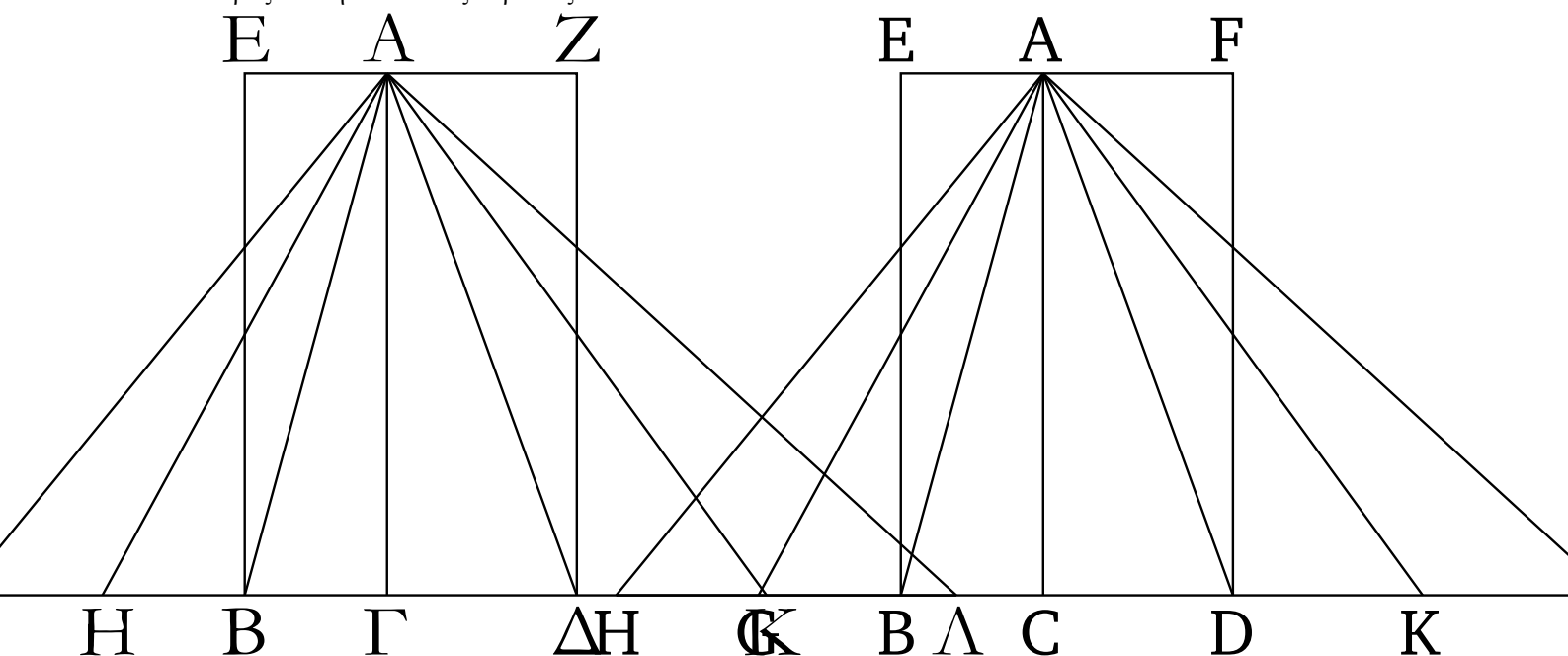
Similar Figures

<'Οροι.

1. <'Ομοια σθήματα εὐθύγραμμα ἐστίν, <όσα τάς τε γωνίας >ίσαζ >έθει κατὰ μίαν καὶ τὰς περὶ τὰς >ίσαζ γωνίας πλευράς ἀνάλογον.
2. >'Ακρον καὶ μέσον λόγον εὐθεῖα τετμηῖσθαι λέγεται, <όταν >ῆ| ὡς ἡ <όλη πρὸς τὸ μείζον τμήμα, ο<ύτως τὸ μείζον πρὸς τὸ >έλαττον.
3. <'Υψος ἐστὶ πάντοξ σθήματοξ ἡ ἀπὸ τῆς κορυφῆς ἐπὶ τὴν βάσιν κάθετοξ ἀγομένη.

1.

Τὰ τρίγωνα καὶ τὰ παραλληλόγραμμα τὰ ὑπὸ τὸ αὐτὸ <ύψοξ >όντα πρὸς >άλληλά ἐστίν ὡς αἱ βάσεις.



>Ἐστω τρίγωνα μὲν τὰ ΑΒΓ, ΑΓΔ, παραλληλόγραμμα δὲ τὰ ΕΓ, ΓΖ ὑπὸ τὸ αὐτὸ <ύψοξ τὸ ΑΓ; λέγω, <ὅτι ἐστὶν ὡς ἡ ΒΓ βάσις πρὸς τὴν ΓΔ βάσιν, ο<ύτως τὸ ΑΒΓ τρίγωνον πρὸς τὸ ΑΓΔ τρίγωνον, καὶ τὸ ΕΓ παραλληλόγραμμον πρὸς τὸ ΓΖ παραλληλόγραμμον.

Ἐκβεβλήσθω γὰρ ἡ ΒΔ ἐφ' ἐκάτερα τὰ μέρη ἐπὶ τὰ Θ, Λ σημεῖα, καὶ κείσθωσαν τῇ μὲν ΒΓ βάσει >ίσαι [ὅσα ἰσθιποτοῦν] αἱ ΒΗ, ΗΘ, τῇ δὲ ΓΔ βάσει >ίσαι ὅσα ἰσθιποτοῦν αἱ ΔΚ, ΚΛ, καὶ ἐπεζεύθωσαν αἱ ΑΗ, ΑΘ, ΑΚ, ΑΛ.

Καὶ ἐπεὶ >ίσαι εἰσὶν αἱ ΓΒ, ΒΗ, ΗΘ ἀλλήλαις, >ίσα ἐστὶ καὶ τὰ ΑΘΗ, ΑΗΒ, ΑΒΓ τρίγωνα ἀλλήλοις. ὅσα πλάσιον >άρα ἐστὶν ἡ ΘΓ βάσις τῆς ΒΓ βάσεως, τοσαυταπλάσιόν ἐστὶ καὶ τὸ ΑΘΓ τρίγωνον τοῦ ΑΒΓ τριγώνου. διὰ τὰ αὐτὰ δὴ ὅσα πλάσιον ἐστὶν ἡ ΛΓ βάσις τῆς ΓΔ βάσεως, τοσαυταπλάσιόν ἐστὶ καὶ τὸ ΑΛΓ τρίγωνον τοῦ ΑΓΔ τριγώνου; καὶ εἰ >ίση ἐστὶν ἡ ΘΓ βάσις τῇ ΓΔ βάσει, >ίσον ἐστὶ καὶ τὸ ΑΘΓ τρίγωνον τῷ ΑΓΔ τριγώνῳ, καὶ εἰ ὑπερέθει ἡ ΘΓ βάσις τῆς ΓΔ βάσεως, ὑπερέθει καὶ τὸ ΑΘΓ τρίγωνον τοῦ ΑΓΔ τριγώνου, καὶ εἰ ἐλάσσων, >έλασσον. τεσσάρων δὲ >όντων μεγεθῶν δύο μὲν βάσεων τῶν ΒΓ, ΓΔ, δύο δὲ τριγώνων τῶν ΑΒΓ, ΑΓΔ ε>ίληπται ἰσάκις πολλαπλάσια τῆς μὲν ΒΓ βάσεως καὶ τοῦ ΑΒΓ τριγώνου <ἡ τε ΘΓ βάσις καὶ τὸ ΑΘΓ τρίγωνον, τῆς δὲ ΓΔ βάσεως καὶ τοῦ ΑΓΔ τριγώνου >άλλα, <ὰ >ετυθεν, ἰσάκις πολλαπλάσια <ἡ τε ΛΓ βάσις καὶ τὸ ΑΛΓ τρίγωνον; καὶ δέδεικται, <ὅτι, εἰ ὑπερέθει ἡ ΘΓ βάσις

Definitions

1. Similar rectilinear figures are those (which) have (their) angles separately equal, and the (corresponding) sides about the equal angles proportional.
2. A straight-line is said to be cut in extreme and mean ratio when as the whole is to the greater segment so the greater (segment is) to the lesser.
3. The height of any figure is the (straight-line) drawn from the vertex perpendicular to the base.

Proposition 1[†]

Triangles and parallelograms which are of the same height are to one another as their bases.

Let ABC and ACD be triangles, and EC and CF parallelograms, of the same height AC . I say that as base BC is to base CD , so triangle ABC (is) to triangle ACD , and parallelogram EC to parallelogram CF .

For let the (straight-line) BD be produced in each direction to points H and L , and let [any number] (of straight-lines) BG and GH be made equal to base BC , and any number (of straight-lines) DK and KL equal to base CD . And let AG , AH , AK , and AL be joined.

And since CB , BG , and GH are equal to one another, triangles AHG , AGB , and ABC are also equal to one another [Prop. 1.38]. Thus, as many times as base HC is (divisible by) base BC , so many times is triangle AHC also (divisible by) triangle ABC . So, for the same (reasons), as many times as base LC is (divisible by) base CD , so many times is triangle ALC also (divisible by) triangle ACD . And if base HC is equal to base CL , (then) triangle AHC is also equal to triangle ACL [Prop. 1.38]. And if base HC exceeds base CL , (then) triangle AHC also exceeds triangle ACL .[‡] And if (HC is) less (than CL then AHC is also) less (than ACL). So, their being four magnitudes, two bases, BC and CD , and two triangles, ABC and ACD , equal multiples have been taken of base BC and triangle ABC —(namely), base HC and triangle AHC —and other random equal multiples of base CD and triangle ADC —(namely), base LC

τῆς ΓΑ βάσεως, ὑπερέθει καὶ τὸ ΑΘΓ τρίγωνον τοῦ ΑΛΓ τριγώνου, καὶ εἰ >ίση, >ίσον, καὶ εἰ >έλασσων, >έλασσον; >έστιν >άρα ὡς ἡ ΒΓ βάσις πρὸς τὴν ΓΔ βάσιν, ο<ύτωξ τὸ ΑΒΓ τρίγωνον πρὸς τὸ ΑΓΔ τρίγωνον.

Καὶ ἐπεὶ τοῦ μὲν ΑΒΓ τριγώνου διπλάσιόν ἐστι τὸ ΕΓ παραλληλόγραμμον, τοῦ δὲ ΑΓΔ τριγώνου διπλάσιόν ἐστι τὸ ΖΓ παραλληλόγραμμον, τὰ δὲ μέρη τοῖς ὡσαύτωξ πολλαπλασίοις τὸν αὐτὸν >έθει λόγον, >έστιν >άρα ὡς τὸ ΑΒΓ τρίγωνον πρὸς τὸ ΑΓΔ τρίγωνον, ο<ύτωξ τὸ ΕΓ παραλληλόγραμμον πρὸς τὸ ΖΓ παραλληλόγραμμον. ἐπεὶ ο>υν ἐδείκθη, ὡς μὲν ἡ ΒΓ βάσις πρὸς τὴν ΓΔ, ο<ύτωξ τὸ ΑΒΓ τρίγωνον πρὸς τὸ ΑΓΔ τρίγωνον, ὡς δὲ τὸ ΑΒΓ τρίγωνον πρὸς τὸ ΑΓΔ τρίγωνον, ο<ύτωξ τὸ ΕΓ παραλληλόγραμμον πρὸς τὸ ΖΓ παραλληλόγραμμον, καὶ ὡς >άρα ἡ ΒΓ βάσις πρὸς τὴν ΓΔ βάσιν, ο<ύτωξ τὸ ΕΓ παραλληλόγραμμον πρὸς τὸ ΖΓ παραλληλόγραμμον.

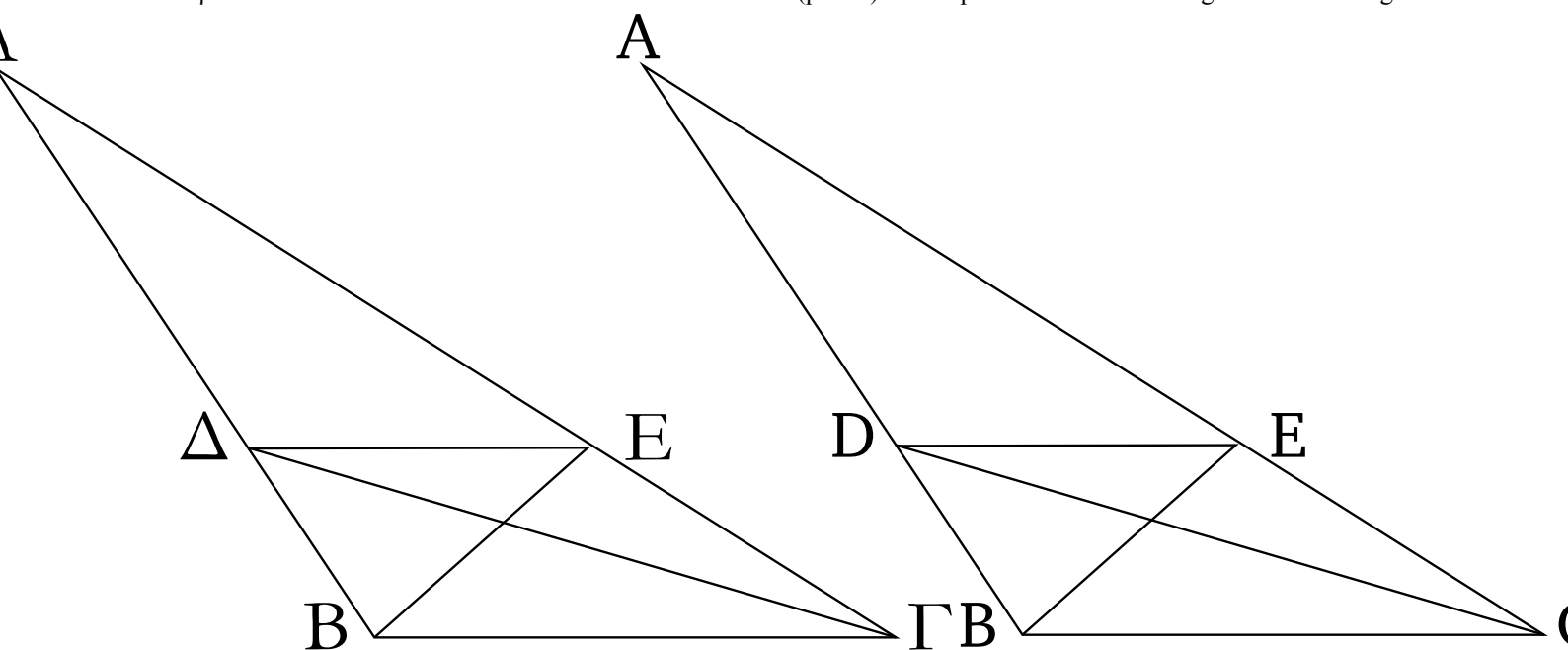
Τὰ >άρα τρίγωνα καὶ τὰ παραλληλόγραμμα τὰ ὑπὸ τὸ αὐτὸ <ύψοξ >όντα πρὸς >άλληλά ἐστιν ὡς αἱ βάσεις; <όπερ >έδει δεῖξαι.

† As is easily demonstrated, this proposition holds even when the triangles, or parallelograms, do not share a common side, and/or are not right-angled.

‡ This is a straight-forward generalization of Prop. 1.38.

2.

Ἐὰν τριγώνου παρὰ μίαν τῶν πλευρῶν ἀθθῇ| τὶς εὐθεῖα, ἀνάλογον τεμεῖ τὰξ τοῦ τριγώνου πλευράξ; καὶ ἔαν αἱ τοῦ τριγώνου πλευραὶ ἀνάλογον τμηθῶσιν, ἡ ἐπὶ τὰξ τομάξ ἐπιζευγνυμένη εὐθεῖα παρὰ τὴν λοιπὴν >έσται τοῦ τριγώνου πλευράν.



Τριγώνου γὰρ τοῦ ΑΒΓ παράλληλοξ μιᾷ| τῶν πλευρῶν τῇ ΒΓ >ήθθω ἡ ΔΕ; λέγω, <ότι ἐστὶν ὡς ἡ ΒΔ πρὸς τὴν ΔΑ, ο<ύτωξ ἡ ΓΕ πρὸς τὴν ΕΑ.

Ἐπεξεύθθωσαν γὰρ αἱ ΒΕ, ΓΔ.

>Ίσον >άρα ἐστὶ τὸ ΒΔΕ τρίγωνον τῷ ΓΔΕ τριγώνω; ἐπὶ γὰρ τῆς αὐτῆς βάσεως ἐστὶ τῆς ΔΕ καὶ ἐν ταῖς αὐταῖς παραλλήλοις ταῖς ΔΕ, ΒΓ; >άλλο δέ τι τὸ ΑΔΕ τρίγωνον. τὰ δὲ >ίσα πρὸς τὸ αὐτὸ τὸν αὐτὸν >έθει λόγον; >έστιν >άρα ὡς τὸ ΒΔΕ τρίγωνον πρὸς τὸ ΑΔΕ [τριγώνον], ο<ύτωξ τὸ ΓΔΕ τρίγωνον πρὸς τὸ ΑΔΕ τρίγωνον. ἀλλ> ὡς μὲν τὸ ΒΔΕ τρίγωνον πρὸς τὸ ΑΔΕ, ο<ύτωξ ἡ ΒΔ πρὸς τὴν ΔΑ;

and triangle ALC . And it has been shown that if base HC exceeds base CL , (then) triangle AHC also exceeds triangle ALC , and if (HC is) equal (to CL then AHC is also) equal (to ALC), and if (HC is) less (than CL then AHC is also) less (than ALC). Thus, as base BC is to base CD , so triangle ABC (is) to triangle ACD [Def. 5.5]. And since parallelogram EC is double triangle ABC , and parallelogram FC is double triangle ACD [Prop. 1.34], and parts have the same ratio as similar multiples [Prop. 5.15], thus as triangle ABC is to triangle ACD , so parallelogram EC (is) to parallelogram FC . In fact, since it was shown that as base BC (is) to CD , so triangle ABC (is) to triangle ACD , and as triangle ABC (is) to triangle ACD , so parallelogram EC (is) to parallelogram CF , thus, also, as base BC (is) to base CD , so parallelogram EC (is) to parallelogram FC [Prop. 5.11].

Thus, triangles and parallelograms which are of the same height are to one another as their bases. (Which is) the very thing it was required to show.

Proposition 2

If some straight-line is drawn parallel to one of the sides of a triangle, (then) it will cut the (other) sides of the triangle proportionally. And if (two of) the sides of a triangle are cut proportionally, (then) the straight-line joining the cutting (points) will be parallel to the remaining side of the triangle.

For let DE be drawn parallel to one of the sides BC of triangle ABC . I say that as BD is to DA , so CE (is) to EA .

For let BE and CD be joined.

Thus, triangle BDE is equal to triangle CDE . For they are on the same base DE and between the same parallels DE and BC [Prop. 1.38]. And ADE is some other triangle. And equal (magnitudes) have the same ratio to the same (magnitude) [Prop. 5.7]. Thus, as triangle BDE is to [triangle] ADE , so triangle CDE (is) to triangle ADE . But, as triangle BDE (is) to triangle ADE , so (is) BD to DA .

ὑπὸ γὰρ τὸ αὐτὸ <ὑψος>όντα τὴν ἀπὸ τοῦ E ἐπὶ τὴν AB κάθετον ἀγομένην πρὸς >ἀλλήλα εἰσιν ὡς αἱ βάσεις. διὰ τὰ αὐτὰ δὴ ὡς τὸ $\Gamma\Delta E$ τρίγωνον πρὸς τὸ $\Lambda\Delta E$, ο<ύτωξ ἡ ΓE πρὸς τὴν EA ; καὶ ὡς >άρα ἡ $B\Delta$ πρὸς τὴν ΔA , ο<ύτωξ ἡ ΓE πρὸς τὴν EA .

Ἀλλὰ δὴ αἱ τοῦ $AB\Gamma$ τριγώνου πλευραὶ αἱ AB , AG ἀνάλογον τετμήσθωσαν, ὡς ἡ $B\Delta$ πρὸς τὴν ΔA , ο<ύτωξ ἡ ΓE πρὸς τὴν EA , καὶ ἐπεξεύθθω ἡ ΔE ; λέγω, <ὅτι παράλληλός ἐστιν ἡ ΔE τῇ $B\Gamma$.

Τῶν γὰρ αὐτῶν κατασκευασθέντων, ἐπεὶ ἐστὶν ὡς ἡ $B\Delta$ πρὸς τὴν ΔA , ο<ύτωξ ἡ ΓE πρὸς τὴν EA , ἀλλ> ὡς μὲν ἡ $B\Delta$ πρὸς τὴν ΔA , ο<ύτωξ τὸ $B\Delta E$ τρίγωνον πρὸς τὸ $\Lambda\Delta E$ τρίγωνον, ὡς δὲ ἡ ΓE πρὸς τὴν EA , ο<ύτωξ τὸ $\Gamma\Delta E$ τρίγωνον πρὸς τὸ $\Lambda\Delta E$ τρίγωνον, καὶ ὡς >άρα τὸ $B\Delta E$ τρίγωνον πρὸς τὸ $\Lambda\Delta E$ τρίγωνον, ο<ύτωξ τὸ $\Gamma\Delta E$ τρίγωνον πρὸς τὸ $\Lambda\Delta E$ τρίγωνον. ἐκάτερον >άρα τῶν $B\Delta E$, $\Gamma\Delta E$ τριγώνων πρὸς τὸ $\Lambda\Delta E$ τὸν αὐτὸν >έθει λόγον. >ίσον>άρα ἐστὶ τὸ $B\Delta E$ τρίγωνον τῷ $\Gamma\Delta E$ τριγώνῳ; καὶ εἰσιν ἐπὶ τῇ αὐτῇ βάσει τῇ ΔE . τὰ δὲ >ίσα τρίγωνα καὶ ἐπὶ τῇ αὐτῇ βάσει >όντα καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἐστίν. παράλληλος >άρα ἐστὶν ἡ ΔE τῇ $B\Gamma$.

Ἐὰν >άρα τριγώνου παρὰ μίαν τῶν πλευρῶν ἀθθῇ| τὴν εὐθεῖαν, ἀνάλογον τεμεῖ τὰς τοῦ τριγώνου πλευράς; καὶ ἔαν αἱ τοῦ τριγώνου πλευραὶ ἀνάλογον τμηθῶσιν, ἡ ἐπὶ τὰς τομὰς ἐπιζευγνυμένη εὐθεῖα παρὰ τὴν λοιπὴν >έσται τοῦ τριγώνου πλευράν; <ὅπερ>έδει δεῖχαι.

3.

Ἐὰν τριγώνου ἡ γωνία δίθω τμηθῇ|, ἡ δὲ τέμνουσα τὴν γωνίαν εὐθεῖα τέμνη καὶ τὴν βάσιν, τὰ τῇς βάσεως τμήματα τὸν αὐτὸν <έχει λόγον ταῖς λοιπαῖς τοῦ τριγώνου πλευραῖς; καὶ ἔαν τὰ τῇς βάσεως τμήματα τὸν αὐτὸν >έθῃ λόγον ταῖς λοιπαῖς τοῦ τριγώνου πλευραῖς, ἡ ἀπὸ τῇς κορυφῆς ἐπὶ τὴν τομὴν ἐπιζευγνυμένη εὐθεῖα δίθω τεμεῖ τὴν τοῦ τριγώνου γωνίαν.

>Ἐστω τρίγωνον τὸ $AB\Gamma$, καὶ τετμήσθω ἡ ὑπὸ $BA\Gamma$ γωνία δίθω ὑπὸ τῇς AD εὐθείας; λέγω, <ὅτι ἐστὶν ὡς ἡ $B\Delta$ πρὸς τὴν $\Gamma\Delta$, ο<ύτωξ ἡ BA πρὸς τὴν AG .

>Ἦθθω γὰρ διὰ τοῦ Γ τῇ| ΔA παράλληλος ἡ ΓE , καὶ διαθεῖσα ἡ BA συμπίπτέτω αὐτῇ| κατὰ τὸ E .

Καὶ ἐπεὶ εἰς παραλλήλους τὰς AD , EG εὐθεῖαν ἐνέπεσεν ἡ AG , ἡ >άρα ὑπὸ AGE γωνία >ίση ἐστὶ τῇ| ὑπὸ $ΓΑΔ$. ἀλλ> ἡ ὑπὸ $ΓΑΔ$ τῇ| ὑπὸ $BA\Delta$ ὑπόκειται >ίση; καὶ ἡ ὑπὸ $BA\Delta$ >άρα τῇ| ὑπὸ AGE ἐστὶν >ίση. πάλιν, ἐπεὶ εἰς παραλλήλους τὰς AD , EG εὐθεῖαν ἐνέπεσεν ἡ BAE , ἡ ἐκτὸς γωνία ἡ ὑπὸ $BA\Delta$ >ίση ἐστὶ τῇ| ἐντὸς τῇ| ὑπὸ $AE\Gamma$. ἐδείθθη δὲ καὶ ἡ ὑπὸ AGE τῇ| ὑπὸ $BA\Delta$ >ίση; καὶ ἡ ὑπὸ AGE >άρα γωνία τῇ| ὑπὸ $AE\Gamma$ ἐστὶν >ίση; <ὥστε καὶ πλευρὰ ἡ AE πλευρᾷ| τῇ| AG ἐστὶν >ίση. καὶ ἐπεὶ τριγώνου τοῦ $B\Gamma E$ παρὰ μίαν τῶν πλευρῶν τὴν EG >ῆκται ἡ AD , ἀνάλογον >άρα ἐστὶν ὡς ἡ $B\Delta$ πρὸς τὴν $\Delta\Gamma$, ο<ύτωξ ἡ BA πρὸς τὴν AE . >ίση δὲ ἡ AE τῇ| AG ; ὡς >άρα ἡ $B\Delta$ πρὸς τὴν $\Delta\Gamma$, ο<ύτωξ ἡ BA πρὸς τὴν AG .

For, having the same height—(namely), the (straight-line) drawn from E perpendicular to AB —they are to one another as their bases [Prop. 6.1]. So, for the same (reasons), as triangle CDE (is) to ADE , so CE (is) to EA . And, thus, as BD (is) to DA , so CE (is) to EA [Prop. 5.11].

And so, let the sides AB and AC of triangle ABC be cut proportionally (such that) as BD (is) to DA , so CE (is) to EA . And let DE be joined. I say that DE is parallel to BC .

For, by the same construction, since as BD is to DA , so CE (is) to EA , but as BD (is) to DA , so triangle BDE (is) to triangle ADE , and as CE (is) to EA , so triangle CDE (is) to triangle ADE [Prop. 6.1], thus, also, as triangle BDE (is) to triangle ADE , so triangle CDE (is) to triangle ADE [Prop. 5.11]. Thus, triangles BDE and CDE each have the same ratio to ADE . Thus, triangle BDE is equal to triangle CDE [Prop. 5.9]. And they are on the same base DE . And equal triangles, which are also on the same base, are also between the same parallels [Prop. 1.39]. Thus, DE is parallel to BC .

Thus, if some straight-line is drawn parallel to one of the sides of a triangle, (then) it will cut the (other) sides of the triangle proportionally. And if (two of) the sides of a triangle are cut proportionally, (then) the straight-line joining the cutting (points) will be parallel to the remaining side of the triangle. (Which is) the very thing it was required to show.

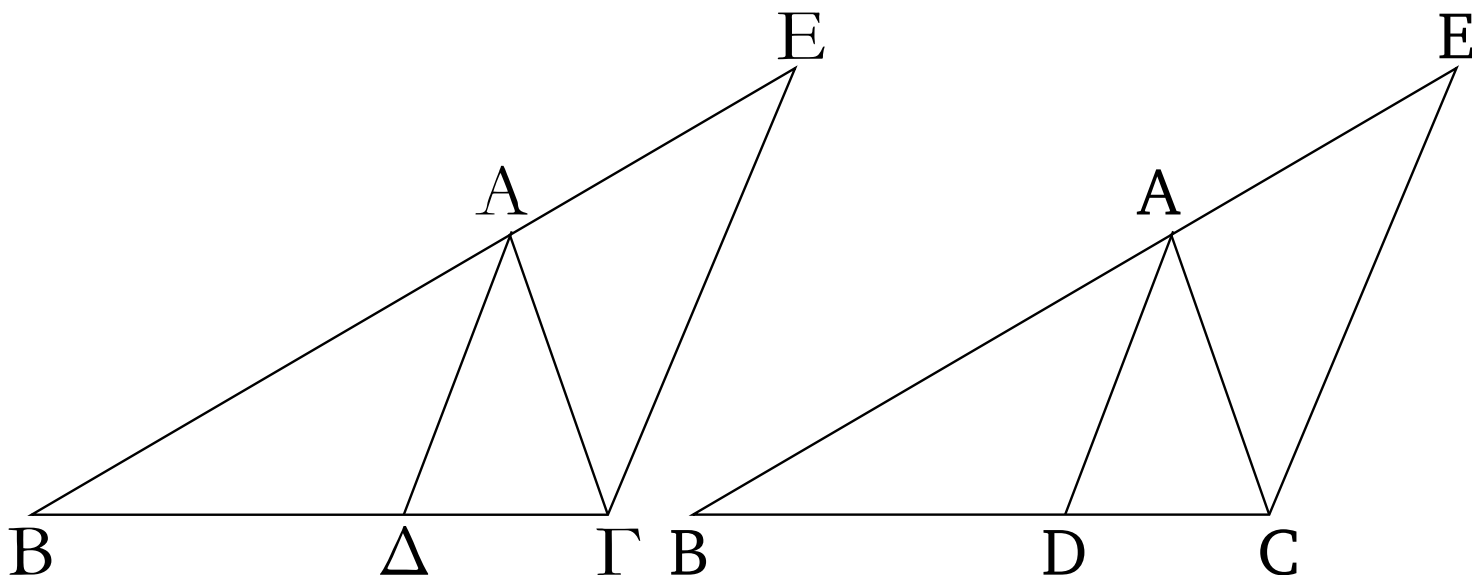
Proposition 3

If an angle of a triangle is cut in half, and the straight-line cutting the angle also cuts the base, (then) the segments of the base will have the same ratio as the remaining sides of the triangle. And if the segments of the base have the same ratio as the remaining sides of the triangle, (then) the straight-line joining the vertex to the cutting (point) will cut the angle of the triangle in half.

Let ABC be a triangle. And let the angle BAC be cut in half by the straight-line AD . I say that as BD is to CD , so BA (is) to AC .

For let CE be drawn through (point) C parallel to DA . And, BA being drawn through, let it meet (CE) at (point) E .[†]

And since the straight-line AC falls across the parallel (straight-lines) AD and EC , angle ACE is thus equal to CAD [Prop. 1.29]. But, (angle) CAD is assumed (to be) equal to BAD . Thus, (angle) BAD is also equal to ACE . Again, since the straight-line BAE falls across the parallel (straight-lines) AD and EC , the external angle BAD is equal to the internal (angle) AEC [Prop. 1.29]. And (angle) ACE was also shown (to be) equal to BAD . Thus, angle ACE is also equal to AEC . And, hence, side AE is equal to side AC [Prop. 1.6]. And since AD has been drawn parallel to one of the sides EC of triangle BCE , thus, proportionally, as BD is to DC , so BA (is) to AE [Prop. 6.2]. And AE (is) equal to AC . Thus, as BD (is) to DC , so BA (is) to AC .



Ἀλλὰ δὴ >έστω ὥς ἡ ΒΔ πρὸς τὴν ΔΓ, ο<ύτωξ ἡ ΒΑ πρὸς τὴν ΑΓ, καὶ ἐπεζεύθω ἡ ΑΔ; λέγω, <ὅτι δίθῃ τέμνεται ἡ ὑπὸ ΒΑΓ γωνία ὑπὸ τῆς ΑΔ εὐθείας.

Τῶν γὰρ αὐτῶν κατασκευασθέντων, ἐπεὶ ἐστὶν ὥς ἡ ΒΔ πρὸς τὴν ΔΓ, ο<ύτωξ ἡ ΒΑ πρὸς τὴν ΑΓ, ἀλλὰ καὶ ὥς ἡ ΒΔ πρὸς τὴν ΔΓ, ο<ύτωξ ἐστὶν ἡ ΒΑ πρὸς τὴν ΑΕ; τριγώνου γὰρ τοῦ ΒΓΕ παρὰ μίαν τὴν ΕΓ >ῆκται ἡ ΑΔ; καὶ ὥς >άρα ἡ ΒΑ πρὸς τὴν ΑΓ, ο<ύτωξ ἡ ΒΑ πρὸς τὴν ΑΕ. >ίση >άρα ἡ ΑΓ τῇ ΑΕ; <ώστε καὶ γωνία ἡ ὑπὸ ΑΕΓ τῇ ὑπὸ ΑΓΕ ἐστὶν >ίση. ἀλλ' ἡ μὲν ὑπὸ ΑΕΓ τῇ ἐκτὸς τῇ ὑπὸ ΒΑΔ [ἐστὶν] >ίση, ἡ δὲ ὑπὸ ΑΓΕ τῇ ἐναλλάχ τῇ ὑπὸ ΓΑΔ ἐστὶν >ίση; καὶ ἡ ὑπὸ ΒΑΔ >άρα τῇ ὑπὸ ΓΑΔ ἐστὶν >ίση. ἡ >άρα ὑπὸ ΒΑΓ γωνία δίθῃ τέμνεται ὑπὸ τῆς ΑΔ εὐθείας.

Ἐὰν >άρα τριγώνου ἡ γωνία δίθῃ τμηθῇ, ἡ δὲ τέμνουσα τὴν γωνίαν εὐθεῖα τέμνῃ καὶ τὴν βάσιν, τὰ τῆς βάσεως τμήματα τὸν αὐτὸν <έχει λόγον ταῖς λοιπαῖς τοῦ τριγώνου πλευραῖς; καὶ ἔαν τὰ τῆς βάσεως τμήματα τὸν αὐτὸν >έθῃ λόγον ταῖς λοιπαῖς τοῦ τριγώνου πλευραῖς, ἡ ἀπὸ τῆς κορυφῆς ἐπὶ τὴν τομὴν ἐπιζευγνυμένη εὐθεῖα δίθῃ τέμνῃ τὴν τοῦ τριγώνου γωνίαν; <όπερ >έδει δεῖξαι.

And so, let BD be to DC , as BA (is) to AC . And let AD be joined. I say that angle BAC has been cut in half by the straight-line AD .

For, by the same construction, since as BD is to DC , so BA (is) to AC , (then) also as BD (is) to DC , so BA is to AE . For AD has been drawn parallel to one (of the sides) EC of triangle BCE [Prop. 6.2]. Thus, also, as BA (is) to AC , so BA (is) to AE [Prop. 5.11]. Thus, AC (is) equal to AE [Prop. 5.9]. And, hence, angle AEC is equal to ACE [Prop. 1.5]. But, AEC [is] equal to the external (angle) BAD , and ACE is equal to the alternate (angle) CAD [Prop. 1.29]. Thus, (angle) BAD is also equal to CAD . Thus, angle BAC has been cut in half by the straight-line AD .

Thus, if an angle of a triangle is cut in half, and the straight-line cutting the angle also cuts the base, (then) the segments of the base will have the same ratio as the remaining sides of the triangle. And if the segments of the base have the same ratio as the remaining sides of the triangle, (then) the straight-line joining the vertex to the cutting (point) will cut the angle of the triangle in half. (Which is) the very thing it was required to show.

† The fact that the two straight-lines meet follows because the sum of ACE and CAE is less than two right-angles, as can easily be demonstrated. See Post. 5.

4.

Τῶν ἰσογωνίων τριγώνων ἀνάλογόν εἰσιν αἱ πλευραὶ αἱ περὶ τὰς >ίσαξ γωνίαξ καὶ ὁμόλογοι αἱ ὑπὸ τὰς >ίσαξ γωνίαξ ὑποτείνουσαι.

>Ἐστω ἰσογώνια τρίγωνα τὰ ΑΒΓ, ΔΓΕ >ίσην >έθοντα τὴν μὲν ὑπὸ ΑΒΓ γωνίαν τῇ ὑπὸ ΔΓΕ, τὴν δὲ ὑπὸ ΒΑΓ τῇ ὑπὸ ΓΔΕ καὶ >έτι τὴν ὑπὸ ΑΓΒ τῇ ὑπὸ ΓΕΔ; λέγω, <ὅτι τῶν ΑΒΓ, ΔΓΕ τριγώνων ἀνάλογόν εἰσιν αἱ πλευραὶ αἱ περὶ τὰς >ίσαξ γωνίαξ καὶ ὁμόλογοι αἱ ὑπὸ τὰς >ίσαξ γωνίαξ ὑποτείνουσαι.

Κείσθω γὰρ ἐπ' εὐθείας ἡ ΒΓ τῇ ΓΕ. καὶ ἐπεὶ αἱ ὑπὸ ΑΒΓ, ΑΓΒ γωνίαι δύο ὀρθῶν ἐλάττονές εἰσιν, >ίση δὲ ἡ ὑπὸ ΑΓΒ τῇ ὑπὸ ΔΕΓ, αἱ >άρα ὑπὸ ΑΒΓ, ΔΕΓ δύο ὀρθῶν ἐλάττονές εἰσιν; αἱ ΒΑ, ΕΔ >άρα ἐκβαλλόμεναι συμπεσοῦνται. ἐκβεβλήσθωσαν καὶ συμπιπτέτωσαν κατὰ

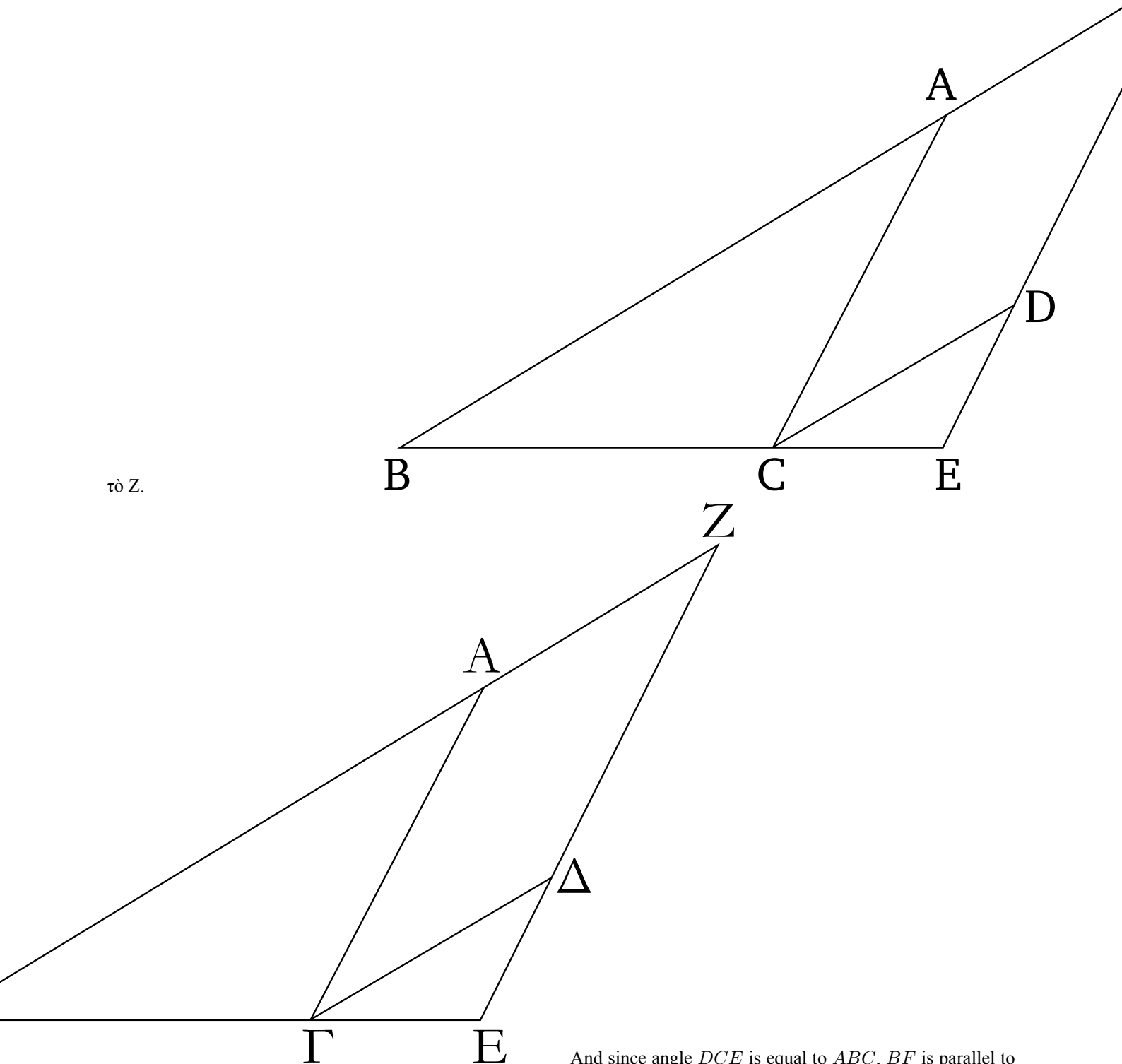
Proposition 4

In equiangular triangles the sides about the equal angles are proportional, and those (sides) subtending equal angles correspond.

Let ABC and DCE be equiangular triangles, having angle ABC equal to DCE , and (angle) BAC to CDE , and, further, (angle) ACB to CED . I say that in triangles ABC and DCE the sides about the equal angles are proportional, and those (sides) subtending equal angles correspond.

Let BC be placed straight-on to CE . And since angles ABC and ACB are less than two right-angles [Prop 1.17], and ACB (is) equal to DEC , thus ABC and DEC are less than two right-angles. Thus, BA and ED , being produced, will meet [C.N. 5]. Let them be produced, and let them meet at (point) F .

τὸ Z.



Καὶ ἐπεὶ ὀρθὴ ἐστὶν ἡ ὑπὸ ΔΓΕ γωνία τῇ ὑπὸ ΑΒΓ, παράλληλός ἐστιν ἡ ΒΖ τῇ ΓΔ. πάλιν, ἐπεὶ ὀρθὴ ἐστὶν ἡ ὑπὸ ΑΓΒ τῇ ὑπὸ ΔΕΓ, παράλληλός ἐστιν ἡ ΑΓ τῇ ΖΕ. παραλληλόγραμμον ὅρα ἐστὶ τὸ ΖΑΓΔ; ὀρθὴ ὅρα ἡ μὲν ΖΑ τῇ ΔΓ, ἡ δὲ ΑΓ τῇ ΖΔ. καὶ ἐπεὶ τριγώνου τοῦ ΖΒΕ παρὰ μίαν τὴν ΖΕ ὤκται ἡ ΑΓ, ἐστὶν ὅρα ὥς ἡ ΒΑ πρὸς τὴν ΑΖ, ὡς ἡ ΒΓ πρὸς τὴν ΓΕ. ὀρθὴ δὲ ἡ ΑΖ τῇ ΓΔ; ὥς ὅρα ἡ ΒΑ πρὸς τὴν ΓΔ, ὡς ἡ ΒΓ πρὸς τὴν ΓΕ, καὶ ἐναλλάξ ὥς ἡ ΑΒ πρὸς τὴν ΒΓ, ὡς ἡ ΔΓ πρὸς τὴν ΓΕ. πάλιν, ἐπεὶ παράλληλός ἐστιν ἡ ΓΔ τῇ ΒΖ, ὅρα ὥς

And since angle DCE is equal to ABC , BZ is parallel to CD [Prop. 1.28]. Again, since (angle) ACB is equal to DEC , AC is parallel to FE [Prop. 1.28]. Thus, $FACD$ is a parallelogram. Thus, FA is equal to DC , and AC to FD [Prop. 1.34]. And since AC has been drawn parallel to one (of the sides) FE of triangle FBE , thus as BA is to AF , so BC (is) to CE [Prop. 6.2]. And AF (is) equal to CD . Thus, as BA (is) to CD , so BC (is) to CE , and, alternately, as AB (is) to BC , so DC (is) to CE [Prop. 5.16]. Again, since CD is parallel to BZ , thus as BC (is) to CE , so FD (is) to DE [Prop. 6.2]. And FD (is) equal to AC . Thus, as

ἢ ΒΓ πρὸς τὴν ΓΕ, οὐτως ἢ ΖΔ πρὸς τὴν ΔΕ. >ίση δὲ ἢ ΖΔ τῇ ΑΓ; ὥς >άρα ἢ ΒΓ πρὸς τὴν ΓΕ, οὐτως ἢ ΑΓ πρὸς τὴν ΔΕ, καὶ ἐναλλάξ ὥς ἢ ΒΓ πρὸς τὴν ΓΑ, οὐτως ἢ ΓΕ πρὸς τὴν ΕΔ. ἐπεὶ οὖν ἐδείκθη ὥς μὲν ἢ ΑΒ πρὸς τὴν ΒΓ, οὐτως ἢ ΔΓ πρὸς τὴν ΓΕ, ὥς δὲ ἢ ΒΓ πρὸς τὴν ΓΑ, οὐτως ἢ ΓΕ πρὸς τὴν ΕΔ, δι> >ίσου >άρα ὥς ἢ ΒΑ πρὸς τὴν ΑΓ, οὐτως ἢ ΓΔ πρὸς τὴν ΔΕ.

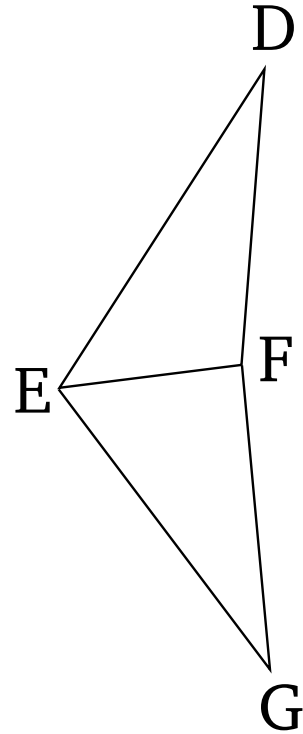
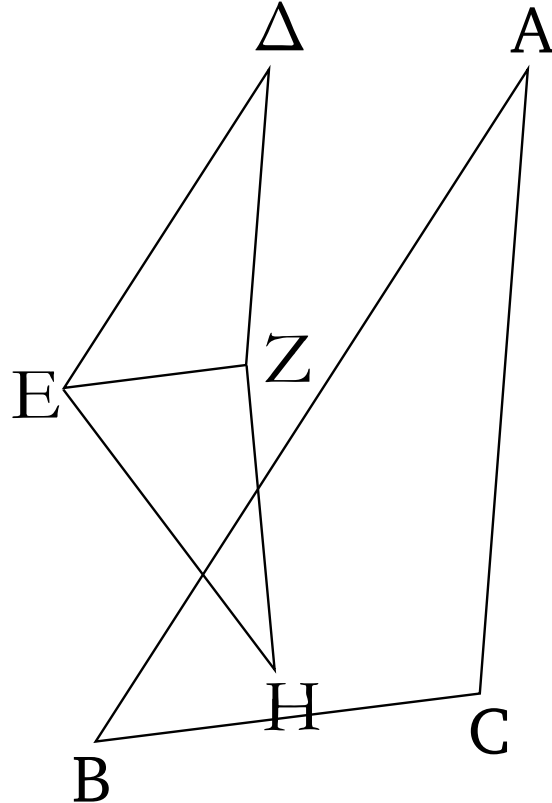
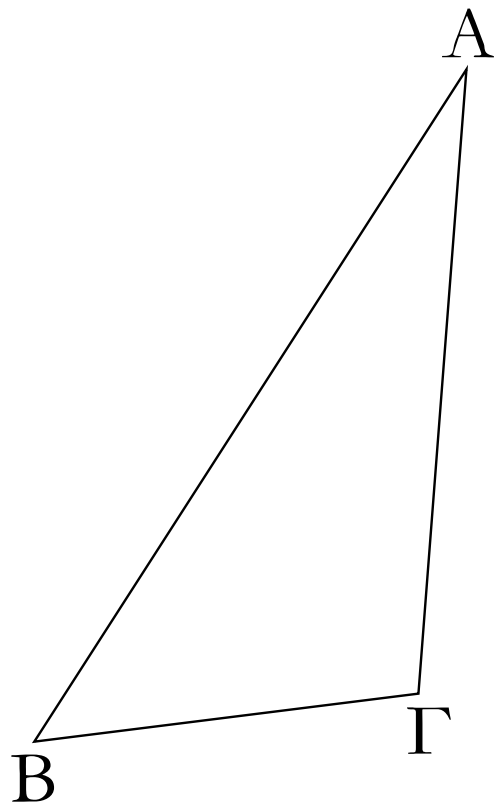
Τῶν >άρα ἰσογώνων τριγώνων ἀνάλογόν εἰσιν αἱ πλευραὶ αἱ περὶ τὰς >ίσας γωνίας καὶ ὁμόλογοι αἱ ὑπὸ τὰς >ίσας γωνίας ὑποτείνουσαι; <όπερ >έδει δεῖξαι.

5.

Ἐὰν δύο τρίγωνα τὰς πλευρὰς ἀνάλογον >έθῃ, ἰσογώνια >έσται τὰ τρίγωνα καὶ >ίσας <έχει τὰς γωνίας, ὅφ> <ὰς αἱ ὁμόλογοι πλευραὶ ὑποτείνουσιν.

>Ἐστω δύο τρίγωνα τὰ ΑΒΓ, ΔΕΖ τὰς πλευρὰς ἀνάλογον >έθοντα, ὥς μὲν τὴν ΑΒ πρὸς τὴν ΒΓ, οὐτως τὴν ΔΕ πρὸς τὴν ΕΖ, ὥς δὲ τὴν ΒΓ πρὸς τὴν ΓΑ, οὐτως τὴν ΕΖ πρὸς τὴν ΖΔ, καὶ >έτι ὥς τὴν ΒΑ πρὸς τὴν ΑΓ, οὐτως τὴν ΕΔ πρὸς τὴν ΔΖ. λέγω, <ὅτι ἰσογώνιον ἔστι τὸ ΑΒΓ τρίγωνον τῷ ΔΕΖ τριγώνῳ καὶ >ίσας <έχουσι τὰς γωνίας, ὅφ> <ὰς αἱ ὁμόλογοι πλευραὶ ὑποτείνουσιν, τὴν μὲν ὑπὸ ΑΒΓ τῇ ὑπὸ ΔΕΖ, τὴν δὲ ὑπὸ ΒΓΑ τῇ ὑπὸ ΕΖΔ καὶ >έτι τὴν ὑπὸ ΒΑΓ τῇ ὑπὸ ΕΔΖ.

Συνεστάτω γὰρ πρὸς τῇ ΕΖ εὐθείᾳ καὶ τοῖς πρὸς αὐτῇ σημείοις τοῖς Ε, Ζ τῇ μὲν ὑπὸ ΑΒΓ γωνία >ίση ἢ ὑπὸ ΖΕΗ, τῇ δὲ ὑπὸ ΑΓΒ >ίση ἢ ὑπὸ ΕΖΗ; λοιπὴ >άρα ἢ πρὸς τῷ Α λοιπῇ τῇ πρὸς τῷ Η ἔστιν >ίση.



>Ἴσογώνιον >άρα ἔστι τὸ ΑΒΓ τρίγωνον τῷ ΕΗΖ [τριγώνῳ]. τῶν >άρα ΑΒΓ, ΕΗΖ τριγώνων ἀνάλογόν εἰσιν αἱ πλευραὶ αἱ περὶ τὰς >ίσας γωνίας καὶ ὁμόλογοι αἱ ὑπὸ τὰς >ίσας γωνίας ὑποτείνουσαι; >έστιν >άρα ὥς ἢ ΑΒ πρὸς τὴν ΒΓ, [οὐτως] ἢ ΗΕ πρὸς τὴν ΕΖ. ἀλλ> ὥς ἢ ΑΒ πρὸς τὴν ΒΓ, οὐτως ὑπόκειται ἢ ΔΕ πρὸς τὴν ΕΖ; ὥς >άρα ἢ ΔΕ πρὸς τὴν ΕΖ, οὐτως ἢ ΗΕ πρὸς τὴν ΕΖ. ἑκατέρα >άρα

BC is to CE , so AC (is) to DE , and, alternately, as BC (is) to CA , so CE (is) to ED [Prop. 6.2]. Therefore, since it was shown that as AB (is) to BC , so DC (is) to CE , and as BC (is) to CA , so CE (is) to ED , thus, via equality, as BA (is) to AC , so CD (is) to DE [Prop. 5.22].

Thus, in equiangular triangles the sides about the equal angles are proportional, and those (sides) subtending equal angles correspond. (Which is) the very thing it was required to show.

Proposition 5

If two triangles have proportional sides, (then) the triangles will be equiangular, and will have the angles which corresponding sides subtend equal.

Let ABC and DEF be two triangles having proportional sides, (so that) as AB (is) to BC , so DE (is) to EF , and as BC (is) to CA , so EF (is) to FD , and, further, as BA (is) to AC , so ED (is) to DF . I say that triangle ABC is equiangular to triangle DEF , and (that the triangles) will have the angles which corresponding sides subtend equal. (That is), (angle) ABC (equal) to DEF , BCA to EFD , and, further, BAC to EDF .

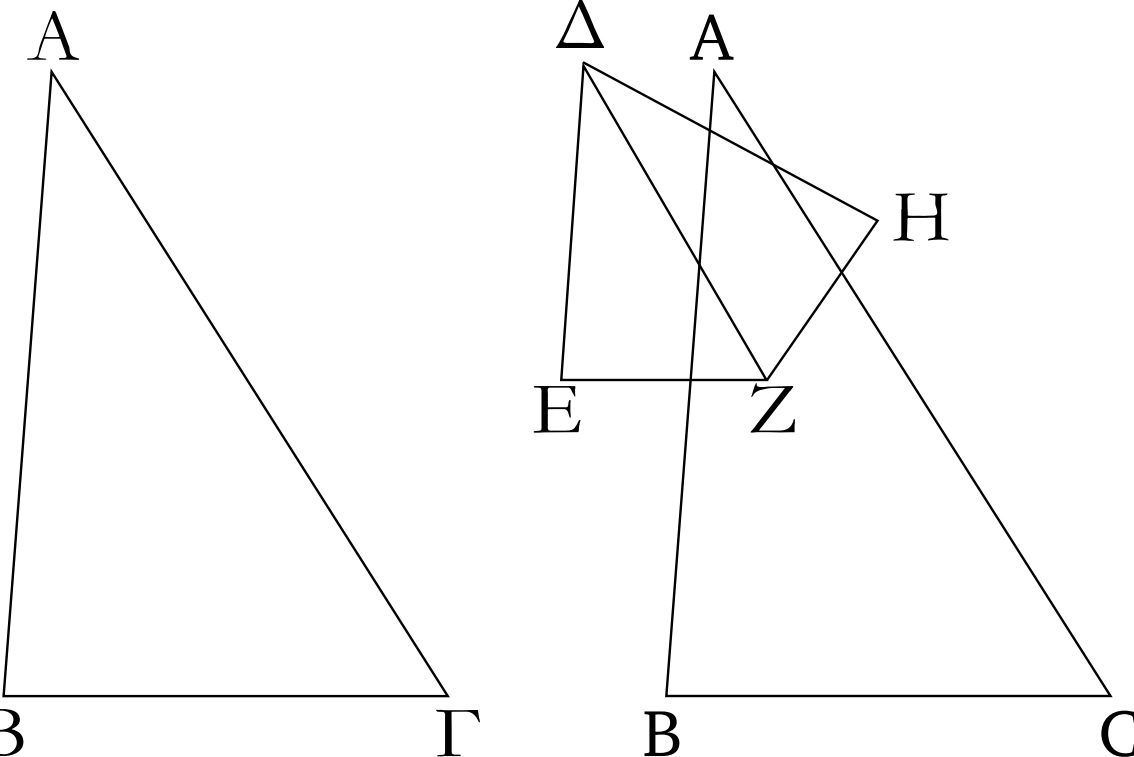
For let (angle) FEG , equal to angle ABC , and (angle) EFG , equal to ACB , be constructed on the straight-line EF at the points E and F on it (respectively) [Prop. 1.23]. Thus, the remaining (angle) at A is equal to the remaining (angle) at G [Prop. 1.32].

Thus, triangle ABC is equiangular to [triangle] EGF . Thus, for triangles ABC and EGF , the sides about the equal angles are proportional, and (those) sides subtending equal angles correspond [Prop. 6.4]. Thus, as AB is to BC , [so] GE (is) to EF . But, as AB (is) to BC , so, it was assumed, (is) DE to EF . Thus, as DE (is) to EF , so GE (is) to EF [Prop. 5.11]. Thus, DE and GE each have the

τῶν ΔΕ, ΗΕ πρὸς τὴν ΕΖ τὸν αὐτὸν >έθει λόγον; >ίση >άρα ἐστὶν ἡ ΔΕ τῇ ΗΕ. διὰ τὰ αὐτὰ δὴ καὶ ἡ ΔΖ τῇ ΗΖ ἐστὶν >ίση. ἐπεὶ ο>ὖν >ίση ἐστὶν ἡ ΔΕ τῇ ΗΕ, κοινὴ δὲ ἡ ΕΖ, δύο δὴ αἱ ΔΕ, ΕΖ δυσὶ ταῖς ΗΕ, ΕΖ >ίσαι εἰσὶν; καὶ βάσις ἡ ΔΖ βάσει τῇ ΖΗ [ἐστὶν] >ίση; γωνία >άρα ἡ ὑπὸ ΔΕΖ γωνία τῇ ὑπὸ ΗΕΖ ἐστὶν >ίση, καὶ τὸ ΔΕΖ τρίγωνον τῷ ΗΕΖ τριγώνῳ >ίσον, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις >ίσαι, ὅφ> <ὰς αἱ >ίσαι πλευραὶ ὑποτείνουσιν. >ίση >άρα ἐστὶ καὶ ἡ μὲν ὑπὸ ΔΖΕ γωνία τῇ ὑπὸ ΗΖΕ, ἡ δὲ ὑπὸ ΕΔΖ τῇ ὑπὸ ΕΗΖ. καὶ ἐπεὶ ἡ μὲν ὑπὸ ΖΕΔ τῇ ὑπὸ ΗΕΖ ἐστὶν >ίση, ὅλλ> ἡ ὑπὸ ΗΕΖ τῇ ὑπὸ ΑΒΓ, καὶ ἡ ὑπὸ ΑΒΓ >άρα γωνία τῇ ὑπὸ ΔΕΖ ἐστὶν >ίση. διὰ τὰ αὐτὰ δὴ καὶ ἡ ὑπὸ ΑΓΒ τῇ ὑπὸ ΔΖΕ ἐστὶν >ίση, καὶ >έτι ἡ πρὸς τῷ Α τῇ πρὸς τῷ Δ; ἰσογώνιον >άρα ἐστὶ τὸ ΑΒΓ τρίγωνον τῷ ΔΕΖ τριγώνῳ.
Ἐὰν >άρα δύο τρίγωνα τὰς πλευρὰς ἀνάλογον >έθῃ, ἰσογώνια >έσται τὰ τρίγωνα καὶ >ίσας <έχει τὰς γωνίας, ὅφ> <ὰς αἱ ὁμόλογοι πλευραὶ ὑποτείνουσιν; <όπερ >έδει δεῖχαι.

6.

Ἐὰν δύο τρίγωνα μίαν γωνίαν μιᾷ γωνία >ίσην >έθῃ, περὶ δὲ τὰς >ίσας γωνίας τὰς πλευρὰς ἀνάλογον, ἰσογώνια >έσται τὰ τρίγωνα καὶ >ίσας <έχει τὰς γωνίας, ὅφ> <ὰς αἱ ὁμόλογοι πλευραὶ ὑποτείνουσιν.
>Ἐστω δύο τρίγωνα τὰ ΑΒΓ, ΔΕΖ μίαν γωνίαν τὴν ὑπὸ ΒΑΓ μιᾷ γωνία τῇ ὑπὸ ΕΔΖ >ίσην >έθοντα, περὶ δὲ τὰς >ίσας γωνίας τὰς πλευρὰς ἀνάλογον, ὥς τὴν ΒΑ πρὸς τὴν ΑΓ, ο<ύτω τὴν ΕΔ πρὸς τὴν ΔΖ; λέγω, <ότι ἰσογώνιον ἐστὶ τὸ ΑΒΓ τρίγωνον τῷ ΔΕΖ τριγώνῳ καὶ >ίσην <έχει τὴν ὑπὸ ΑΒΓ γωνίαν τῇ ὑπὸ ΔΕΖ, τὴν δὲ ὑπὸ ΑΓΒ τῇ ὑπὸ ΔΖΕ.



Συνεστάτω γὰρ πρὸς τῇ ΔΖ εὐθείᾳ καὶ τοῖς πρὸς αὐτῇ σημείοις τοῖς Δ, Ζ ὁποτέρᾳ μὲν τῶν ὑπὸ ΒΑΓ, ΕΔΖ >ίση ἡ ὑπὸ ΖΔΗ, τῇ δὲ ὑπὸ ΑΓΒ >ίση ἡ ὑπὸ ΔΖΗ; λοιπὴ >άρα ἡ πρὸς τῷ Β γωνία λοιπῇ τῇ πρὸς τῷ Η >ίση ἐστὶν.
Ἰσογώνιον >άρα ἐστὶ τὸ ΑΒΓ τρίγωνον τῷ ΔΗΖ τριγώνῳ.

same ratio to EF . Thus, DE is equal to GE [Prop. 5.9]. So, for the same (reasons), DF is also equal to GF . Therefore, since DE is equal to EG , and EF (is) common, the two (sides) DE , EF are equal to the two (sides) GE , EF (respectively). And base DF [is] equal to base FG . Thus, angle DEF is equal to angle GEF [Prop. 1.8], and triangle DEF (is) equal to triangle GEF , and the remaining angles (are) equal to the remaining angles which the equal sides subtend [Prop. 1.4]. Thus, angle DFE is also equal to GFE , and (angle) EDF to EGF . And since (angle) FED is equal to GEF , and (angle) GEF to ABC , angle ABC is thus also equal to DEF . So, for the same (reasons), (angle) ACB is also equal to DFE , and, further, the (angle) at A to the (angle) at D . Thus, triangle ABC is equiangular to triangle DEF .

Thus, if two triangles have proportional sides, (then) the triangles will be equiangular, and will have the angles which corresponding sides subtend equal. (Which is) the very thing it was required to show.

Proposition 6

If two triangles have one angle equal to one angle, and the sides about the equal angles proportional, (then) the triangles will be equiangular, and will have the angles which corresponding sides subtend equal.

Let ABC and DEF be two triangles having one angle, BAC , equal to one angle, EDF (respectively), and the sides about the equal angles proportional, (so that) as BA (is) to AC , so ED (is) to DF . I say that triangle ABC is equiangular to triangle DEF , and will have angle ABC equal to DEF , and (angle) ACB to DFE .

For let (angle) FDG , equal to each of BAC and EDF , and (angle) DFG , equal to ACB , be constructed on the straight-line AF at the points D and F on it (respectively) [Prop. 1.23]. Thus, the remaining angle at B is equal to the remaining angle at G [Prop. 1.32].

ἀνάλογον >άρα ἐστὶν ὡς ἡ BA πρὸς τὴν AF , οὐτόωξ ἡ HA πρὸς τὴν AZ . ὑπόκειται δὲ καὶ ὡς ἡ BA πρὸς τὴν AF , οὐτόωξ ἡ EA πρὸς τὴν AZ ; καὶ ὡς >άρα ἡ EA πρὸς τὴν AZ , οὐτόωξ ἡ HA πρὸς τὴν AZ . >ίση >άρα ἡ EA τῇ HA ; καὶ κοινὴ ἡ AZ ; δύο δὲ αἱ EA , AZ δυοὶ ταῖς HA , AZ >ίσαξ εἰσὶν; καὶ γωνία ἡ ὑπὸ EAZ γωνία τῇ ὑπὸ HAZ [ἐστὶν] >ίση; βάσειξ >άρα ἡ EZ βάσει τῇ HZ ἐστὶν >ίση, καὶ τὸ AEZ τρίγωνον τῷ HAZ τριγώνῳ >ίσον ἐστὶν, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαιξ >ίσαξ >έσσονται, ὅφρ <ὰξ >ίσαξ πλευραὶ ὑποτείνουσιν. >ίση >άρα ἐστὶν ἡ μὲν ὑπὸ AZH τῇ ὑπὸ AZE , ἡ δὲ ὑπὸ AHZ τῇ ὑπὸ AEX . ἄλλ > ἡ ὑπὸ AZH τῇ ὑπὸ AGB ἐστὶν >ίση; καὶ ἡ ὑπὸ AGB >άρα τῇ ὑπὸ AZE ἐστὶν >ίση. ὑπόκειται δὲ καὶ ἡ ὑπὸ BAG τῇ ὑπὸ EAZ >ίση; καὶ λοιπὴ >άρα ἡ πρὸς τῷ B λοιπῇ τῇ πρὸς τῷ E >ίση ἐστὶν; ἰσογώνιον >άρα ἐστὶ τὸ ABG τρίγωνον τῷ AEZ τριγώνῳ. Ἐὰν >άρα δύο τρίγωνα μίαν γωνίαν μᾶλ γωνία >ίσην >έθῃ, περὶ δὲ τὰξ >ίσαξ γωνίαξ τὰξ πλευράξ ἀνάλογον, ἰσογώνια >έσται τὰ τρίγωνα καὶ >ίσαξ <έχει τὰξ γωνίαξ, ὅφρ <ὰξ αἱ ὁμόλογοι πλευραὶ ὑποτείνουσιν, <όπερ >έδει δεῖχαι.

7.

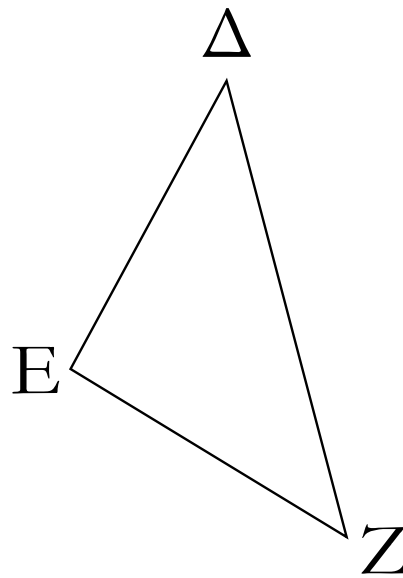
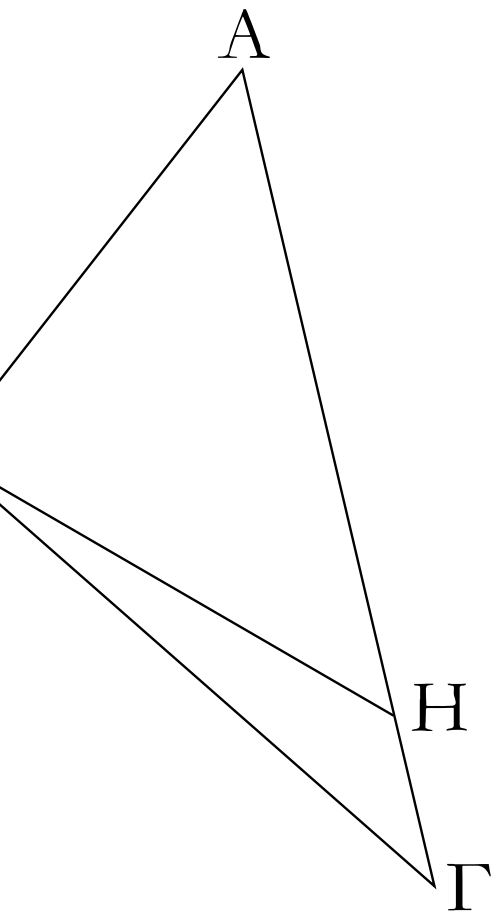
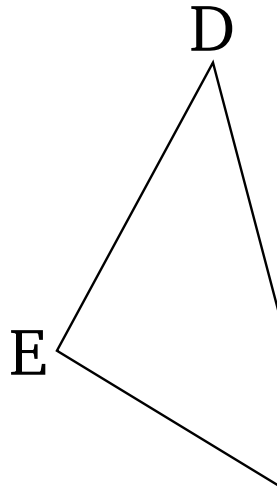
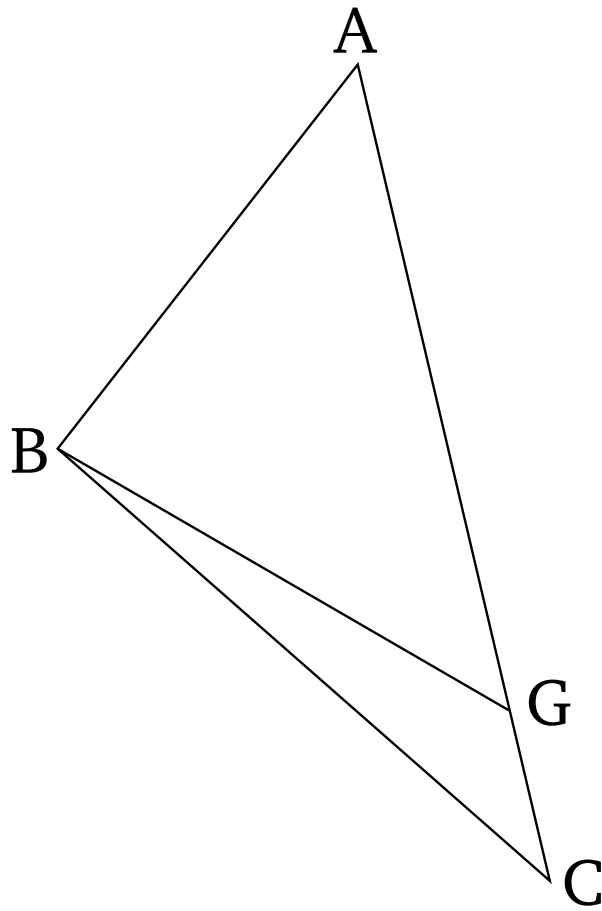
Ἐὰν δύο τρίγωνα μίαν γωνίαν μᾶλ γωνία >ίσην >έθῃ, περὶ δὲ >άλλαξ γωνίαξ τὰξ πλευράξ ἀνάλογον, τῶν δὲ λοιπῶν ἐκατέραν <άμα >ήτοι ἐλάσσονα > ἢ μὴ ἐλάσσονα ὀρθῇξ, ἰσογώνια >έσται τὰ τρίγωνα καὶ >ίσαξ <έχει τὰξ γωνίαξ, περὶ <ὰξ ἀνάλογόν εἰσιν αἱ πλευραὶ.
>Ἐστω δύο τρίγωνα τὰ ABG , AEZ μίαν γωνίαν μᾶλ γωνία >ίσην >έθοντα τὴν ὑπὸ BAG τῇ ὑπὸ EAZ , περὶ δὲ >άλλαξ γωνίαξ τὰξ ὑπὸ ABG , AEZ τὰξ πλευράξ ἀνάλογον, ὡς τὴν AB πρὸς τὴν BG , οὐτόωξ τὴν AE πρὸς τὴν EZ , τῶν δὲ λοιπῶν τῶν πρὸς τοῖς G , Z πρότερον ἐκατέραν <άμα ἐλάσσονα ὀρθῇξ; λέγω, <ότι ἰσογώνιον ἐστὶ τὸ ABG τρίγωνον τῷ AEZ τριγώνῳ, καὶ >ίση >έσται ἡ ὑπὸ ABG γωνία τῇ ὑπὸ AEZ , καὶ λοιπὴ δηλονότι ἡ πρὸς τῷ G λοιπῇ τῇ πρὸς τῷ Z >ίση.

Thus, triangle ABC is equiangular to triangle DGF . Thus, proportionally, as BA (is) to AC , so GD (is) to DF [Prop. 6.4]. And it was also assumed that as BA (is) to AC , so ED (is) to DF . And, thus, as ED (is) to DF , so GD (is) to DF [Prop. 5.11]. Thus, ED (is) equal to DG [Prop. 5.9]. And DF (is) common. So, the two (sides) ED , DF are equal to the two (sides) GD , DF (respectively). And angle EDF [is] equal to angle GDF . Thus, base EF is equal to base GF , and triangle DEF is equal to triangle GDF , and the remaining angles will be equal to the remaining angles which the equal sides subtend [Prop. 1.4]. Thus, (angle) DFG is equal to DFE , and (angle) DGF to DEF . But, (angle) DFG is equal to ACB . Thus, (angle) ACB is also equal to DFE . And (angle) BAC was also assumed (to be) equal to EDF . Thus, the remaining (angle) at B is equal to the remaining (angle) at E [Prop. 1.32]. Thus, triangle ABC is equiangular to triangle DEF .

Thus, if two triangles have one angle equal to one angle, and the sides about the equal angles proportional, (then) the triangles will be equiangular, and will have the angles which corresponding sides subtend equal. (Which is) the very thing it was required to show.

Proposition 7

If two triangles have one angle equal to one angle, and the sides about other angles proportional, and the remaining angles either both less than, or both not less than, right-angles, (then) the triangles will be equiangular, and will have the angles about which the sides are proportional equal.
Let ABC and DEF be two triangles having one angle, BAC , equal to one angle, EDF (respectively), and the sides about (some) other angles, ABC and DEF (respectively), proportional, (so that) as AB (is) to BC , so DE (is) to EF , and the remaining (angles) at C and F , first of all, both less than right-angles. I say that triangle ABC is equiangular to triangle DEF , and (that) angle ABC will be equal to DEF , and (that) the remaining (angle) at C (will be) manifestly equal to the remaining (angle) at F .



Εἰ γὰρ >άνισός ἐστιν ἡ ὑπὸ $AB\Gamma$ γωνία τῇ ὑπὸ ΔEZ ,
μία αὐτῶν μείζων ἐστίν. >έστω μείζων ἡ ὑπὸ $AB\Gamma$. καὶ
συνεστάτω πρὸς τῇ AB εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ

For if angle ABC is not equal to (angle) DEF , (then) one of
them is greater. Let ABC be greater. And let (angle) ABG ,
equal to (angle) DEF , be constructed on the straight-line
 AB at the point B on it [Prop. 1.23].

τῷ B τῇ $\angle$ ΔEZ γωνία $>$ ἴση ἢ ὑπὸ ABH .

Καὶ ἐπεὶ $>$ ἴση ἐστὶν ἢ μὲν A γωνία τῇ Δ , ἢ δὲ ὑπὸ ABH τῇ $\angle$ ΔEZ , λοιπὴ $>$ ἄρα ἢ ὑπὸ AHB λοιπῇ τῇ $\angle$ ΔZE ἐστὶν $>$ ἴση. ἰσογώνιον $>$ ἄρα ἐστὶ τὸ ABH τρίγωνον τῷ ΔEZ τριγώνῳ. $>$ ἐστὶν $>$ ἄρα ὥς ἢ AB πρὸς τὴν BH , οὐκ ὅτως ἢ ΔE πρὸς τὴν EZ . ὥς δὲ ἢ ΔE πρὸς τὴν EZ , [οὐκ ὅτως] ὑπόκειται ἢ AB πρὸς τὴν BG ; ἢ $AB >$ ἄρα πρὸς ἐκατέραν τῶν BG , BH τὸν αὐτὸν $>$ ἐθεὶ λόγον; $>$ ἴση $>$ ἄρα ἢ BG τῇ BH . $<$ ὥστε καὶ γωνία ἢ πρὸς τῷ Γ γωνία τῇ $\angle$ ΔBH ἐστὶν $>$ ἴση. ἐλάττων δὲ ὀρθῆς ὑπόκειται ἢ πρὸς τῷ Γ ; ἐλάττων $>$ ἄρα ἐστὶν ὀρθῆς καὶ ὑπὸ BHG ; $<$ ὥστε ἢ ἐφεχθῆς αὐτῇ γωνία ἢ ὑπὸ AHB μείζων ἐστὶν ὀρθῆς. καὶ ἐδείκθη $>$ ἴση οὐδὲν τῇ πρὸς τῷ Z ; καὶ ἢ πρὸς τῷ $Z >$ ἄρα μείζων ἐστὶν ὀρθῆς. ὑπόκειται δὲ ἐλάσσων ὀρθῆς; $<$ ὅπερ ἐστὶν $>$ ἀτοπον. οὐκ $>$ ἄρα $>$ ἀνισόξ ἐστὶν ἢ ὑπὸ ABG γωνία τῇ $\angle$ ΔEZ ; $>$ ἴση $>$ ἄρα. $>$ ἐστὶ δὲ καὶ ἢ πρὸς τῷ $A >$ ἴση τῇ πρὸς τῷ Δ ; καὶ λοιπὴ $>$ ἄρα ἢ πρὸς τῷ Γ λοιπῇ τῇ πρὸς τῷ $Z >$ ἴση ἐστίν. ἰσογώνιον $>$ ἄρα ἐστὶ τὸ ABG τρίγωνον τῷ ΔEZ τριγώνῳ.

Ἀλλὰ δὴ πάλιν ὑποκείσθω ἐκατέρα τῶν πρὸς τοῖς Γ , Z μὴ ἐλάσσων ὀρθῆς; λέγω πάλιν, $<$ ὅτι καὶ οὐκ ὅτως ἐστὶν ἰσογώνιον τὸ ABG τρίγωνον τῷ ΔEZ τριγώνῳ.

Τῶν γὰρ αὐτῶν κατασκευασθέντων ὁμοίως δείχομεν, $<$ ὅτι $>$ ἴση ἐστὶν ἢ BG τῇ BH ; $<$ ὥστε καὶ γωνία ἢ πρὸς τῷ Γ τῇ $\angle$ ΔBH $>$ ἴση ἐστίν. οὐκ ἐλάττων δὲ ὀρθῆς ἢ πρὸς τῷ Γ ; οὐκ ἐλάττων $>$ ἄρα ὀρθῆς οὐδὲ ἢ ὑπὸ BHG . τριγώνου δὴ τοῦ BHG αἱ δύο γωνίαι δύο ὀρθῶν οὐκ εἰσὶν ἐλάττονες; $<$ ὅπερ ἐστὶν ἀδύνατον. οὐκ $>$ ἄρα πάλιν $>$ ἀνισόξ ἐστὶν ἢ ὑπὸ ABG γωνία τῇ $\angle$ ΔEZ ; $>$ ἴση $>$ ἄρα. $>$ ἐστὶ δὲ καὶ ἢ πρὸς τῷ A τῇ πρὸς τῷ $\Delta >$ ἴση; λοιπὴ $>$ ἄρα ἢ πρὸς τῷ Γ λοιπῇ τῇ πρὸς τῷ $Z >$ ἴση ἐστίν. ἰσογώνιον $>$ ἄρα ἐστὶ τὸ ABG τρίγωνον τῷ ΔEZ τριγώνῳ.

Ἐὰν $>$ ἄρα δύο τρίγωνα μίαν γωνίαν μὴ γωνία $>$ ἴσην $>$ ἐθῇ, περὶ δὲ $>$ ἄλλαξ γωνίαξ τὰξ πλευράξ ἀνάλογον, τῶν δὲ λοιπῶν ἐκατέραν $<$ ἄμα ἐλάττονα $>$ ἢ μὴ ἐλάττονα ὀρθῆς, ἰσογώνια $>$ ἐστὶ τὰ τρίγωνα καὶ $>$ ἴσαξ $<$ ἐχει τὰξ γωνίαξ, περὶ $<$ ἄξ ἀνάλογόν εἰσιν αἱ πλευραί; $<$ ὅπερ $>$ ἐδει δεῖχαι.

8.

Ἐὰν ἐν ὀρθογωνίῳ τριγώνῳ ἀπὸ τῆς ὀρθῆς γωνίαξ ἐπὶ τὴν βάσιν κάθετοξ ἀθῇ, τὰ πρὸς τῇ καθέτῳ τρίγωνα $<$ ὁμοιά ἐστὶ τῷ τε $<$ ὅλῳ καὶ ἀλλήλοιξ.

$>$ Ἐστω τρίγωνον ὀρθογώνιον τὸ ABG ὀρθὴν $>$ ἐθὼν τὴν ὑπὸ BAG γωνίαν, καὶ $>$ ἠθῇ ἀπὸ τοῦ A ἐπὶ τὴν BG κάθετοξ ἢ AD ; λέγω, $<$ ὅτι $<$ ὁμοιόν ἐστὶν ἐκάτερον τῶν ABD , ADC τριγώνων $<$ ὅλῳ τῷ ABG καὶ $>$ ἐτι ἀλλήλοιξ.

And since angle A is equal to (angle) D , and (angle) ABG to DEF , the remaining (angle) AGB is thus equal to the remaining (angle) DFE [Prop. 1.32]. Thus, triangle ABG is equiangular to triangle DEF . Thus, as AB is to BG , so DE (is) to EF [Prop. 6.4]. And as DE (is) to EF , [so] it was assumed (is) AB to BC . Thus, AB has the same ratio to each of BC and BG [Prop. 5.11]. Thus, BC (is) equal to BG [Prop. 5.9]. And, hence, the angle at C is equal to angle BGC [Prop. 1.5]. And the angle at C was assumed (to be) less than a right-angle. Thus, (angle) BGC is also less than a right-angle. Hence, the adjacent angle to it, AGB , is greater than a right-angle [Prop. 1.13]. And (AGB) was shown to be equal to the (angle) at F . Thus, the (angle) at F is also greater than a right-angle. But it was assumed (to be) less than a right-angle. The very thing is absurd. Thus, angle ABC is not unequal to (angle) DEF . Thus, (it is) equal. And the (angle) at A is also equal to the (angle) at D . And thus the remaining (angle) at C is equal to the remaining (angle) at F [Prop. 1.32]. Thus, triangle ABC is equiangular to triangle DEF .

But, again, let each of the (angles) at C and F be assumed (to be) not less than a right-angle. I say, again, that triangle ABC is equiangular to triangle DEF in this case also.

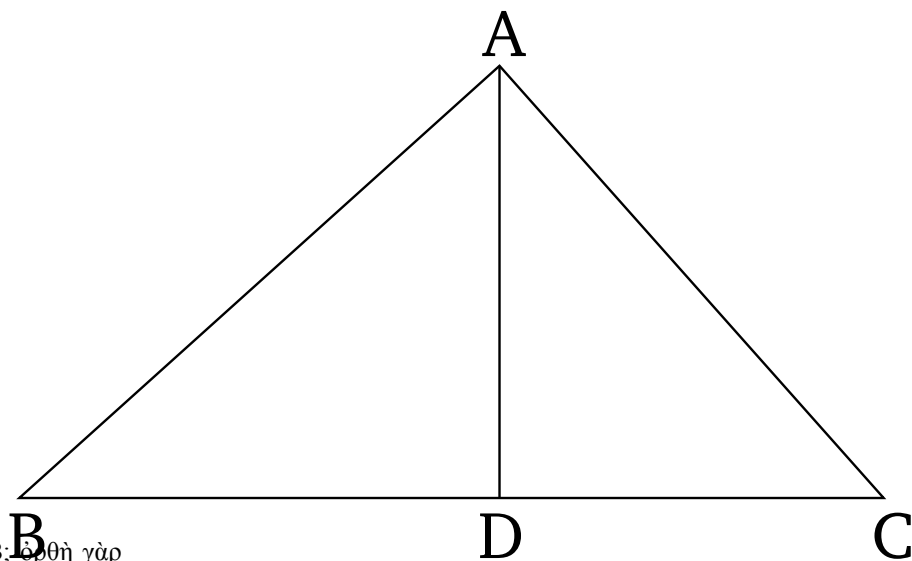
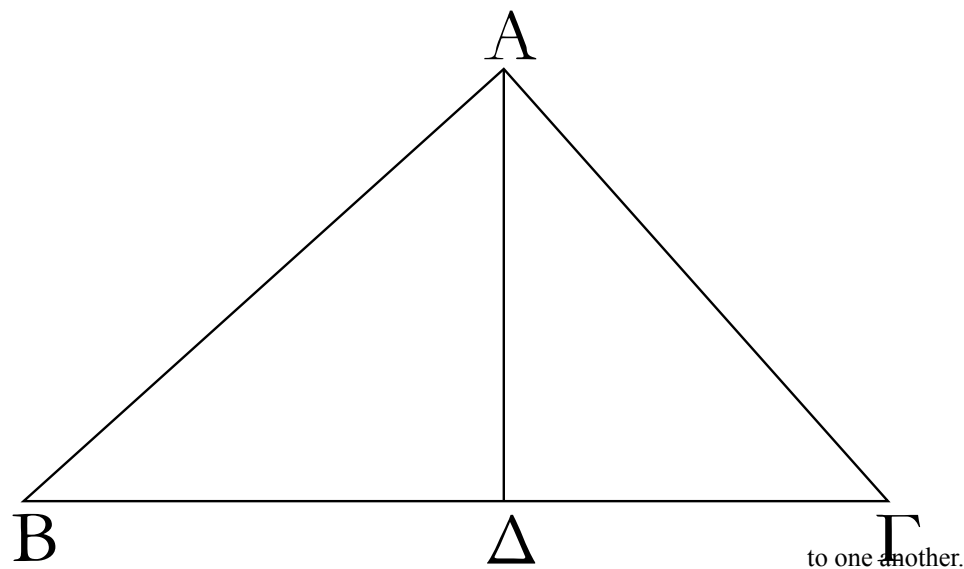
For, with the same construction, we can similarly show that BC is equal to BG . Hence, also, the angle at C is equal to (angle) BGC . And the (angle) at C (is) not less than a right-angle. Thus, BGC (is) not less than a right-angle either. So, in triangle BGC the (sum of) two angles is not less than two right-angles. The very thing is impossible [Prop. 1.17]. Thus, again, angle ABC is not unequal to DEF . Thus, (it is) equal. And the (angle) at A is also equal to the (angle) at D . Thus, the remaining (angle) at C is equal to the remaining (angle) at F [Prop. 1.32]. Thus, triangle ABC is equiangular to triangle DEF .

Thus, if two triangles have one angle equal to one angle, and the sides about other angles proportional, and the remaining angles both less than, or both not less than, right-angles, (then) the triangles will be equiangular, and will have the angles about which the sides (are) proportional equal. (Which is) the very thing it was required to show.

Proposition 8

If, in a right-angled triangle, a (straight-line) is drawn from the right-angle perpendicular to the base, (then) the triangles around the perpendicular are similar to the whole (triangle), and to one another.

Let ABC be a right-angled triangle having the angle BAC a right-angle, and let AD be drawn from A , perpendicular to BC [Prop. 1.12]. I say that triangles ABD and ADC are each similar to the whole (triangle) ABC and, further,



Ἐπεὶ γὰρ ὀρθή ἐστιν ἡ ὑπὸ ΒΑΓ τῇ ὑπὸ ΑΔΒ, ὁρθή γὰρ ἑκατέρα; καὶ κοινὴ τῶν δύο τριγώνων τοῦ τε ΑΒΓ καὶ τοῦ ΑΒΔ ἡ πρὸς τῷ Β, λοιπὴ ᾠρα ἡ ὑπὸ ΑΓΒ λοιπῇ τῇ ὑπὸ ΒΑΔ ἐστὶν ὀρθή; ἰσογώνιον ᾠρα ἐστὶ τὸ ΑΒΓ τρίγωνον τῷ ΑΒΔ τριγώνῳ. ὁ ἐστὶν ᾠρα ὡς ἡ ΒΓ ὑποτείνουσα τὴν ὀρθὴν τοῦ ΑΒΓ τριγώνου πρὸς τὴν ΒΑ ὑποτείνουσαν τὴν ὀρθὴν τοῦ ΑΒΔ τριγώνου, οὕτως αὐτὴ ἡ ΑΒ ὑποτείνουσα τὴν πρὸς τῷ Γ γωνίαν τοῦ ΑΒΓ τριγώνου πρὸς τὴν ΒΔ ὑποτείνουσαν τὴν ὀρθὴν τὴν ὑπὸ ΒΑΔ τοῦ ΑΒΔ τριγώνου, καὶ ἔτι ἡ ΑΓ πρὸς τὴν ΑΔ ὑποτείνουσαν τὴν πρὸς τῷ Β γωνίαν κοινὴν τῶν δύο τριγώνων. τὸ ΑΒΓ ᾠρα τρίγωνον τῷ ΑΒΔ τριγώνῳ ἰσογώνιον τέ ἐστι καὶ τὰς περὶ τὰς ὀρθὰς γωνίας πλευρὰς ἀνάλογον ἔχει. ὁμοίον ᾠμα [ἐστὶ] τὸ ΑΒΓ τρίγωνον τῷ ΑΒΔ τριγώνῳ. ὁμοίως δὲ δείχομεν, ὅτι καὶ τῷ ΑΔΓ τριγώνῳ ὁμοίον ἐστὶ τὸ ΑΒΓ τρίγωνον; ἑκάτερον ᾠρα τῶν ΑΒΔ, ΑΔΓ [τριγώνων] ὁμοίον ἐστὶν ὅλῳ τῷ ΑΒΓ.

Λέγω δὲ, ὅτι καὶ ἀλλήλοις ἐστὶν ὁμοία τὰ ΑΒΔ, ΑΔΓ τρίγωνα.

Ἐπεὶ γὰρ ὁρθή ἡ ὑπὸ ΒΔΑ ὁρθῇ τῇ ὑπὸ ΑΔΓ ἐστὶν ὀρθή, ἀλλὰ μὴν καὶ ἡ ὑπὸ ΒΑΔ τῇ πρὸς τῷ Γ ἐδείκθη ὀρθή, καὶ λοιπὴ ᾠρα ἡ πρὸς τῷ Β λοιπῇ τῇ ὑπὸ ΔΑΓ ἐστὶν ὀρθή; ἰσογώνιον ᾠρα ἐστὶ τὸ ΑΒΔ τρίγωνον τῷ ΑΔΓ τριγώνῳ. ὁ ἐστὶν ᾠρα ὡς ἡ ΒΔ τοῦ ΑΒΔ τριγώνου ὑποτείνουσα τὴν ὑπὸ ΒΑΔ πρὸς τὴν ΔΑ τοῦ ΑΔΓ τριγώνου ὑποτείνουσαν τὴν πρὸς τῷ Γ ὀρθὴν τῇ ὑπὸ ΒΑΔ, οὕτως

For since (angle) BAC is equal to ADB —for each (are) right-angles—and the (angle) at B (is) common to the two triangles ABC and ABD , the remaining (angle) ACB is thus equal to the remaining (angle) BAD [Prop. 1.32]. Thus, triangle ABC is equiangular to triangle ABD . Thus, as BC , subtending the right-angle in triangle ABC , is to BA , subtending the right-angle in triangle ABD , so the same AB , subtending the angle at C in triangle ABC , (is) to BD , subtending the equal (angle) BAD in triangle ABD , and, further, (so is) AC to AD , (both) subtending the angle at B common to the two triangles [Prop. 6.4]. Thus, triangle ABC is equiangular to triangle ABD , and has the sides about the equal angles proportional. Thus, triangle ABC [is] similar to triangle ABD [Def. 6.1]. So, similarly, we can show that triangle ABC is also similar to triangle ADC . Thus, [triangles] ABD and ADC are each similar to the whole (triangle) ABC .

So I say that triangles ABD and ADC are also similar to one another.

For since the right-angle BDA is equal to the right-angle ADC , and, indeed, (angle) BAD was also shown (to be) equal to the (angle) at C , thus the remaining (angle) at B is also equal to the remaining (angle) DAC [Prop. 1.32]. Thus, triangle ABD is equiangular to triangle ADC . Thus, as BD , subtending (angle) BAD in triangle ABD , is to

αὐτὴ ἡ $\Delta\Delta$ τοῦ ABD τριγώνου ὑποτείνουσα τὴν πρὸς τῷ B γωνίαν πρὸς τὴν $\Delta\Gamma$ ὑποτείνουσαν τὴν ὑπὸ $\Delta\text{A}\Gamma$ τοῦ $\Delta\text{A}\Gamma$ τριγώνου $>$ ίσην τῇ πρὸς τῷ B , καὶ $>$ έτι ἡ BA πρὸς τὴν $\text{A}\Gamma$ ὑποτείνουσαι τὰς ὀρθὰς; $<$ όμοιον $>$ άρα ἐστὶ τὸ ABD τρίγωνον τῷ $\Delta\text{A}\Gamma$ τριγώνῳ.

Ἐὰν $>$ άρα ἐν ὀρθογωνίῳ τριγώνῳ ἀπὸ τῆς ὀρθῆς γωνίας ἐπὶ τὴν βάσιν κάθετοξ ἀθθῇ, τὰ πρὸς τῇ καθέτῳ τρίγωνα $<$ όμοιά ἐστι τῷ τε $<$ όλῳ καὶ ἀλλήλοις [$<$ όπερ $>$ έδει δεῖχαι].

Πόρισμα.

Ἐκ δὲ τούτου φανερόν, $<$ ότι ἐὰν ἐν ὀρθογωνίῳ τριγώνῳ ἀπὸ τῆς ὀρθῆς γωνίας ἐπὶ τὴν βάσιν κάθετοξ ἀθθῇ, ἡ ἀθθεῖσα τῶν τῆς βάσεως τμημάτων μέση ἀνάλογόν ἐστιν; $<$ όπερ $>$ έδει δεῖχαι.

[†] In other words, the perpendicular is the geometric mean of the pieces.

9.

Τῆς δοθείσης εὐθείας τὸ προσταθθὲν μέρος ἀφελεῖν.

$>$ Ἐστω ἡ δοθεῖσα εὐθεῖα ἡ AB ; δεῖ δὲ τῆς AB τὸ προσταθθὲν μέρος ἀφελεῖν.

DA , subtending the (angle) at C in triangle ADC , (which is) equal to (angle) BAD , so (is) the same AD , subtending the angle at B in triangle ABD , to DC , subtending (angle) DAC in triangle ADC , (which is) equal to the (angle) at B , and, further, (so is) BA to AC , (each) subtending right-angles [Prop. 6.4]. Thus, triangle ABD is similar to triangle ADC [Def. 6.1].

Thus, if, in a right-angled triangle, a (straight-line) is drawn from the right-angle perpendicular to the base, (then) the triangles around the perpendicular are similar to the whole (triangle), and to one another. [(Which is) the very thing it was required to show.]

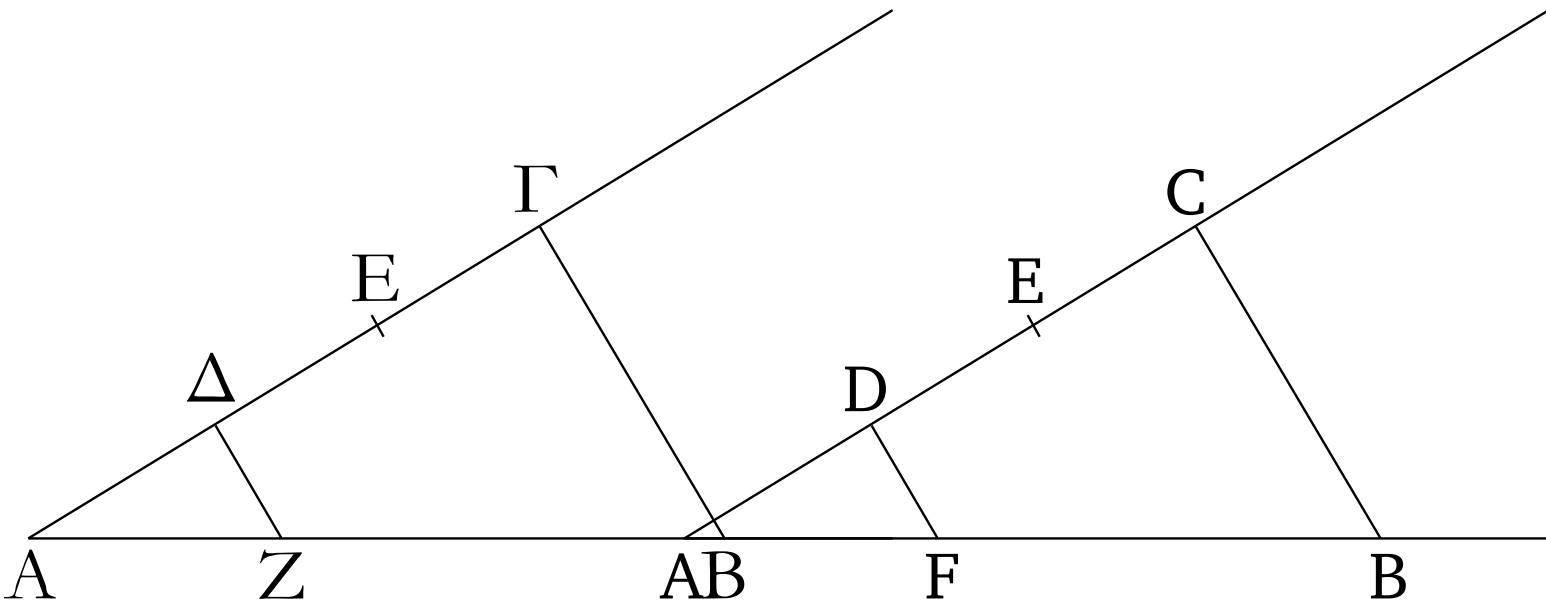
Corollary

So (it is) clear, from this, that if, in a right-angled triangle, a (straight-line) is drawn from the right-angle perpendicular to the base, (then) the (straight-line so) drawn is in mean proportion to the pieces of the base.[†] (Which is) the very thing it was required to show.

Proposition 9

To cut off a prescribed part from a given straight-line.

Let AB be the given straight-line. So it is required to cut off a prescribed part from AB .



Ἐπιτετάθθω δὲ τὸ τρίτον. [καὶ] διήθθω τιξ ἀπὸ τοῦ A εὐθεῖα ἡ $\text{A}\Gamma$ γωνίαν περιέθουσα μετὰ τῆς AB τυθοῦσαν; καὶ εἰλήφθω τυθὸν σημείον ἐπὶ τῆς $\text{A}\Gamma$ τὸ Δ , καὶ κείσθωσαν τῇ $\text{A}\Delta$ $>$ ίσαι αἱ ΔE , $\text{E}\Gamma$. καὶ ἐπεξεύθθω ἡ $\text{B}\Gamma$, καὶ διὰ τοῦ Δ παράλληλοξ αὐτῇ $>$ ήθθω ἡ ΔZ .

Ἐπεὶ ο $>$ ὺν τριγώνου τοῦ $\text{AB}\Gamma$ παρὰ μίαν τῶν πλευρῶν τὴν $\text{B}\Gamma$ $>$ ήκται ἡ $\text{Z}\Delta$, ἀνάλογον $>$ άρα ἐστὶν ὡξ ἡ $\text{I}\Delta$ πρὸς τὴν ΔA , ο $<$ ύτωξ ἡ BZ πρὸς τὴν ZA . διπλῇ δὲ ἡ $\text{I}\Delta$ τῆς ΔA ; διπλῇ $>$ άρα καὶ ἡ BZ τῆς ZA ; τριπλῇ $>$ άρα ἡ BA τῆς AZ .

Τῆς $>$ άρα δοθείσης εὐθείας τῆς AB τὸ ἐπιταθθὲν τρίτον μέρος ἀφῆλρηται τὸ AZ ; $<$ όπερ $>$ έδει ποιῆσαι.

So let a third (part) be prescribed. [And] let some straight-line AC be drawn from (point) A , encompassing a random angle with AB . And let a random point D be taken on AC . And let DE and EC be made equal to AD [Prop. 1.3]. And let BC be joined. And let DF be drawn through D parallel to it [Prop. 1.31].

Therefore, since FD has been drawn parallel to one of the sides, BC , of triangle ABC , then, proportionally, as CD is to DA , so BF (is) to FA [Prop. 6.2]. And CD (is) double DA . Thus, BF (is) also double FA . Thus, BA (is) triple AF .

Thus, the prescribed third part, AF , has been cut off from the given straight-line, AB . (Which is) the very thing it was required to do.

10.

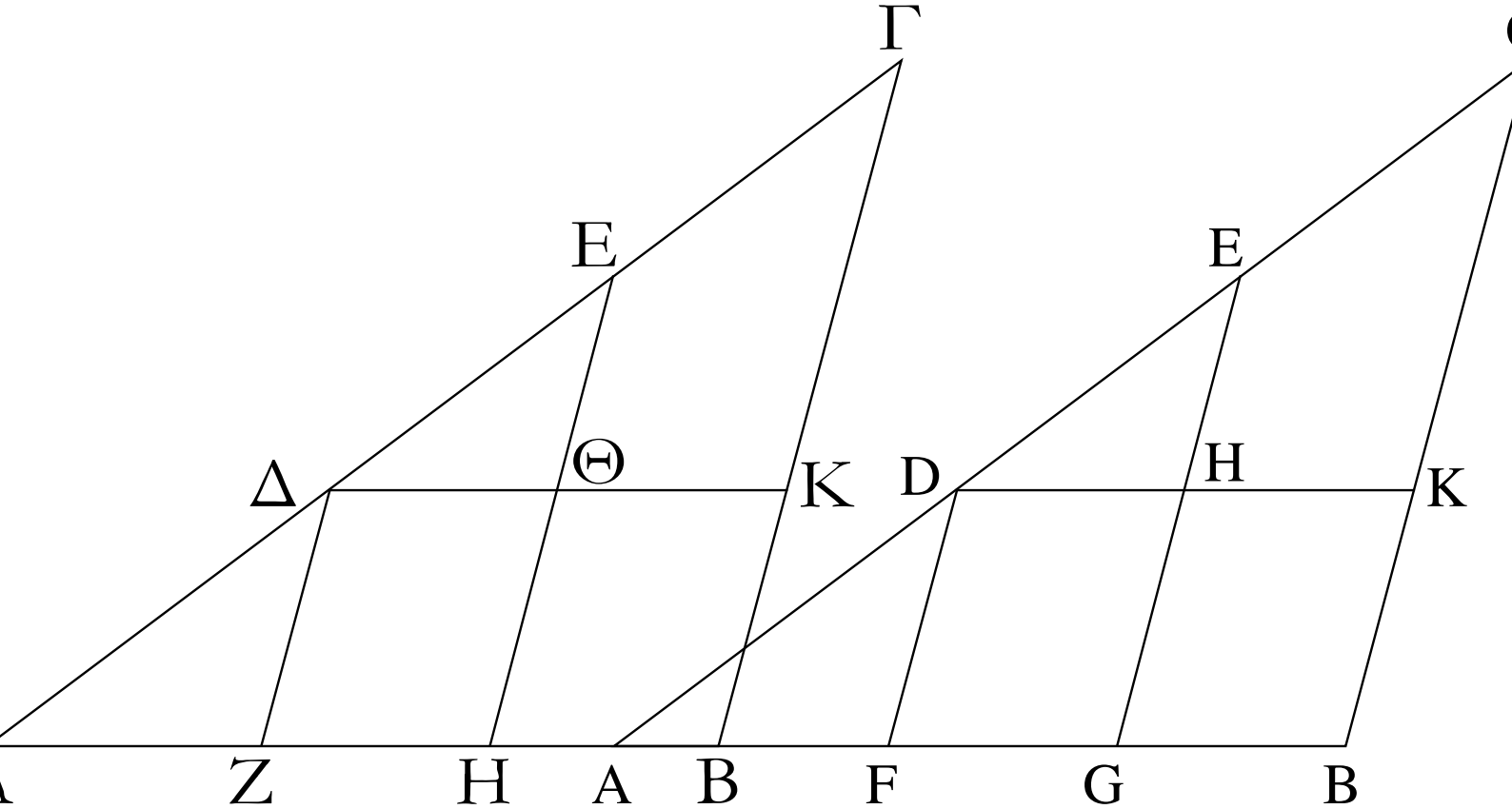
Τὴν δοθεῖσαν εὐθεῖαν >ἀτμητον τῇ δοθείσῃ τετμημένῃ ὁμοίως τεμεῖν.

>Ἐστω ἡ μὲν δοθεῖσα εὐθεῖα <ἀτμητοξ ἡ AB , ἡ δὲ τετμημένη ἡ AG κατὰ τὰ Δ , E σημεῖα, καὶ κείσθωσαν <ώστε γωνίαν τυθοῦσαν περιέθειν, καὶ ἐπεζεύθω ἡ GB , καὶ διὰ τῶν Δ , E τῇ BG παράλληλοι >ἡθθωσαν αἱ ΔZ , EH , διὰ δὲ τοῦ Δ τῇ AB παράλληλοξ >ἡθθω ἡ $\Delta\Theta K$.

Proposition 10

To cut a given uncut straight-line similarly to a given cut (straight-line).

Let AB be the given uncut straight-line, and AC a (straight-line) cut at points D and E , and let (AC) be laid down so as to encompass a random angle (with AB). And let CB be joined. And let DF and EG be drawn through (points) D and E (respectively), parallel to BC , and let DHK be drawn through (point) D , parallel to AB [Prop. 1.31].



Παραλληλόγραμμον >άρα ἐστὶν ἑκάτερον τῶν $Z\Theta$, ΘB ; >ίση >άρα ἡ μὲν $\Delta\Theta$ τῇ ZH , ἡ δὲ ΘK τῇ HB . καὶ ἐπεὶ τριγώνου τοῦ $\Delta K\Gamma$ παρὰ μίαν τῶν πλευρῶν τὴν $K\Gamma$ εὐθεῖα >ἡκται ἡ ΘE , ἀνάλογον >άρα ἐστὶν ὡς ἡ ΓE πρὸς τὴν $E\Delta$, ο<ύτωξ ἡ $K\Theta$ πρὸς τὴν $\Theta\Delta$. >ίση δὲ ἡ μὲν $K\Theta$ τῇ BH , ἡ δὲ $\Theta\Delta$ τῇ HZ . >έστιν >άρα ὡς ἡ ΓE πρὸς τὴν $E\Delta$, ο<ύτωξ ἡ BH πρὸς τὴν HZ . πάλιν, ἐπεὶ τριγώνου τοῦ AHE παρὰ μίαν τῶν πλευρῶν τὴν HE >ἡκται ἡ ΔZ , ἀνάλογον >άρα ἐστὶν ὡς ἡ $E\Delta$ πρὸς τὴν ΔA , ο<ύτωξ ἡ HZ πρὸς τὴν ZA . ἐδείκθη δὲ καὶ ὡς ἡ ΓE πρὸς τὴν $E\Delta$, ο<ύτωξ ἡ BH πρὸς τὴν HZ ; >έστιν >άρα ὡς μὲν ἡ ΓE πρὸς τὴν $E\Delta$, ο<ύτωξ ἡ BH πρὸς τὴν HZ , ὡς δὲ ἡ $E\Delta$ πρὸς τὴν ΔA , ο<ύτωξ ἡ HZ πρὸς τὴν ZA .

Ἡ >άρα δοθεῖσα εὐθεῖα >ἀτμητοξ ἡ AB τῇ δοθείσῃ εὐθείᾳ τετμημένῃ τῇ AG ὁμοίως τέτμηται; <όπερ >έδει ποιῆσαι;

Thus, FH and HB are each parallelograms. Thus, DH (is) equal to FG , and HK to GB [Prop. 1.34]. And since the straight-line HE has been drawn parallel to one of the sides, KC , of triangle DKC , thus, proportionally, as CE is to ED , so KH (is) to HD [Prop. 6.2]. And KH (is) equal to BG , and HD to GF . Thus, as CE is to ED , so BG (is) to GF . Again, since FD has been drawn parallel to one of the sides, GE , of triangle AGE , thus, proportionally, as ED is to DA , so GF (is) to FA [Prop. 6.2]. And it was also shown that as CE (is) to ED , so BG (is) to GF . Thus, as CE is to ED , so BG (is) to GF , and as ED (is) to DA , so GF (is) to FA .

Thus, the given uncut straight-line, AB , has been cut similarly to the given cut straight-line, AC . (Which is) the very thing it was required to do.

11.

Δύο δοθεισῶν εὐθειῶν τρίτην ἀνάλογον προσευρεῖν.

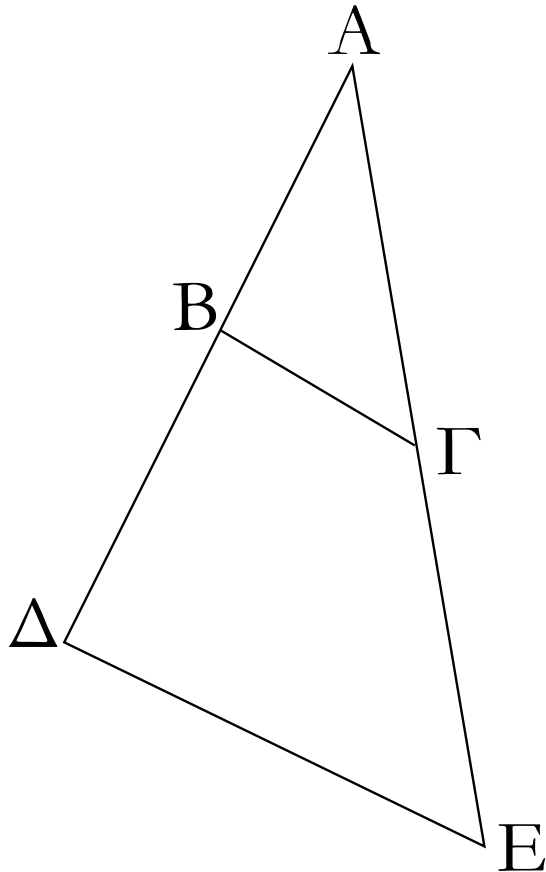
>Ἐστωσαν αἱ δοθεῖσαι [δύο εὐθεῖαι] αἱ BA , AC καὶ κείσθωσαν γωνίαν περιέθειν τυθοῦσαν. δεῖ δὴ τῶν BA , AC τρίτην ἀνάλογον προσευρεῖν. ἐκβεβλήσθωσαν γὰρ ἐπὶ τὰ Δ , E σημεῖα, καὶ κείσθω τῇ AG >ίση ἡ BD , καὶ ἐπεζεύθω ἡ $B\Gamma$, καὶ διὰ τοῦ Δ παράλληλοξ αὐτῇ >ἡθθω ἡ ΔE .

Proposition 11

To find a third (straight-line) proportional to two given

straight-lines.

Let BA and AC be the [two] given [straight-lines], and let them be laid down encompassing a random angle. So it is required to find a third (straight-line) proportional to BA and AC . For let (BA and AC) be produced to points D and E (respectively), and let BD be made equal to AC [Prop. 1.3].

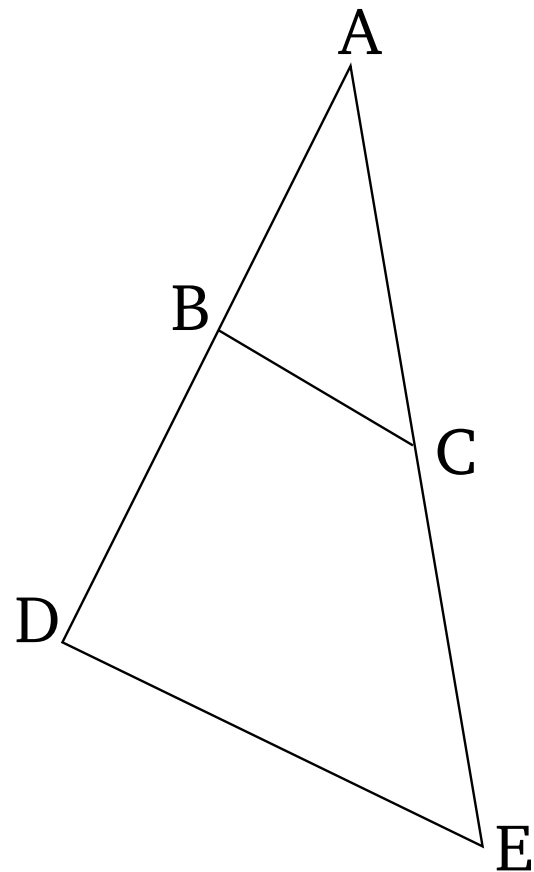


Ἐπεὶ οὖν τριγώνου τοῦ ADE παρὰ μίαν τῶν πλευρῶν τὴν AE >ῆκται ἡ $BΓ$, ἀνάλογόν ἐστιν ὥς ἡ AB πρὸς τὴν BD , οὕτως ἡ AG πρὸς τὴν GE . >ίση δὲ ἡ BD τῇ AG . >έστιν ἄρα ὥς ἡ AB πρὸς τὴν AG , οὕτως ἡ AG πρὸς τὴν GE . Δύο ἄρα δοθεισῶν εὐθειῶν τῶν AB , AG τρίτη ἀνάλογον αὐταῖς προσεύρηται ἡ GE ; <όπερ >έδει ποιῆσαι.

12.

Τριῶν δοθεισῶν εὐθειῶν τετάρτην ἀνάλογον προσευρεῖν.

And let BC be joined. And let DE be drawn through (point) D parallel to it [Prop. 1.31].

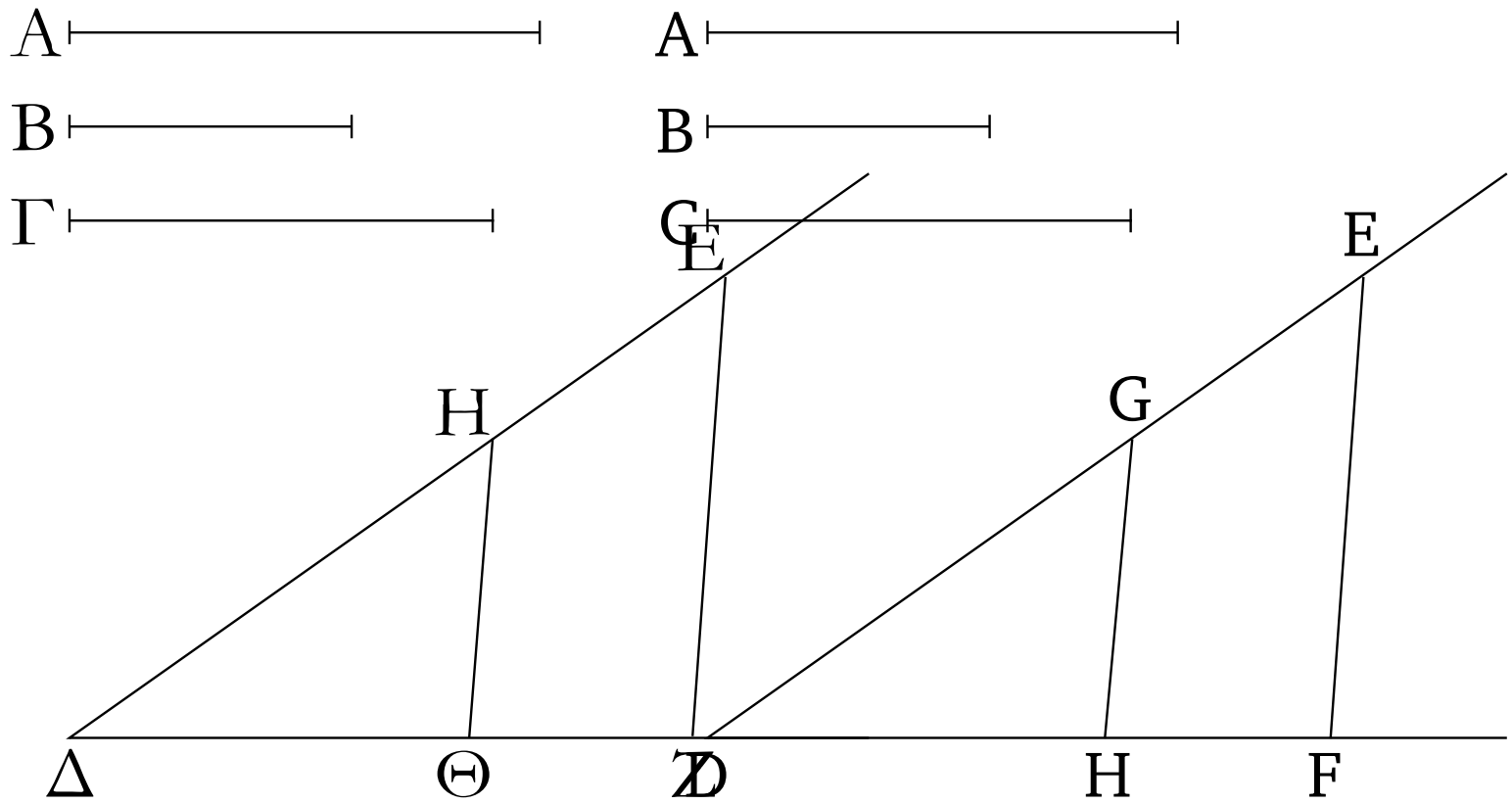


Therefore, since BC has been drawn parallel to one of the sides DE of triangle ADE , proportionally, as AB is to BD , so AC (is) to CE [Prop. 6.2]. And BD (is) equal to AC . Thus, as AB is to AC , so AC (is) to CE .

Thus, a third (straight-line), CE , has been found (which is) proportional to the two given straight-lines, AB and AC . (Which is) the very thing it was required to do.

Proposition 12

To find a fourth (straight-line) proportional to three given straight-lines.



>Ἐστωσαν αἱ δοθεῖσαι τρεῖς εὐθεῖαι αἱ A, B, Γ ; δεῖ δὴ τῶν A, B, Γ τετάρτην ἀνάλογον προσευρεῖν.

Ἐκκείσθωσαν δύο εὐθεῖαι αἱ $\Delta E, \Delta Z$ γωνίαν περιέθουσai [τυθοῦσαν] τὴν ὑπὸ $E\Delta Z$; καὶ κείσθω τῇ μὲν A >ῖση ἡ ΔH , τῇ δὲ B >ῖση ἡ HE , καὶ >έτι τῇ Γ >ῖση ἡ $\Delta\Theta$; καὶ ἐπιζευθθείσῃ τῇ ΘZ παράλληλος αὐτῇ >ήθῃ διὰ τοῦ E ἡ EZ .

Ἐπεὶ οὖν τριγώνου τοῦ ΔEZ παρὰ μίαν τὴν EZ >ῖσται ἡ ΘZ , >έστιν >άρα ὡς ἡ ΔH πρὸς τὴν HE , οὕτως ἡ $\Delta\Theta$ πρὸς τὴν ΘZ . >ῖση δὲ ἡ μὲν ΔH τῇ A , ἡ δὲ HE τῇ B , ἡ δὲ $\Delta\Theta$ τῇ Γ ; >έστιν >άρα ὡς ἡ A πρὸς τὴν B , οὕτως ἡ Γ πρὸς τὴν ΘZ .

Τριῶν >άρα δοθεισῶν εὐθειῶν τῶν A, B, Γ τετάρτη ἀνάλογον προσεύρηται ἡ ΘZ ; <όπερ >έδει ποιῆσαι.

Let A, B , and C be the three given straight-lines. So it is required to find a fourth (straight-line) proportional to A, B , and C .

Let the two straight-lines DE and DF be set out encompassing the [random] angle EDF . And let DG be made equal to A , and GE to B , and, further, DH to C [Prop. 1.3]. And GH being joined, let EF be drawn through (point) E parallel to it [Prop. 1.31].

Therefore, since GH has been drawn parallel to one of the sides EF of triangle DEF , thus as DG is to GE , so DH (is) to HF [Prop. 6.2]. And DG (is) equal to A , and GE to B , and DH to C . Thus, as A is to B , so C (is) to HF .

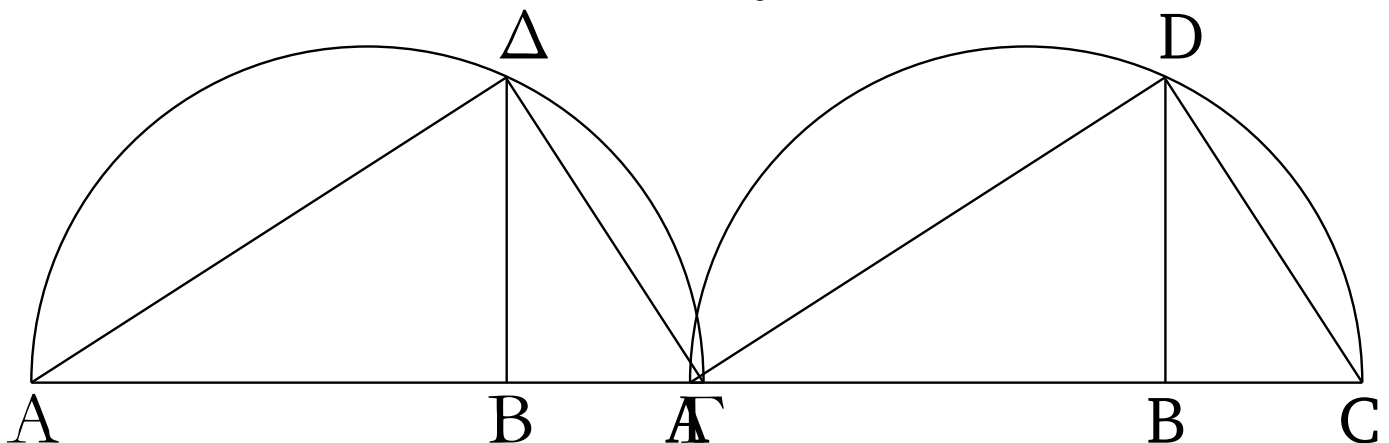
Thus, a fourth (straight-line), HF , has been found (which is) proportional to the three given straight-lines, A, B , and C . (Which is) the very thing it was required to do.

13.

Δύο δοθεισῶν εὐθειῶν μέσην ἀνάλογον προσευρεῖν.

Proposition 13

To find the (straight-line) in mean proportion to two given straight-lines.[†]



>Ἐστωσαν αἱ δοθεῖσαι δύο εὐθεῖαι αἱ AB, BC ; δεῖ δὴ τῶν

Let AB and BC be the two given straight-lines. So it is

AB, BΓ μέσην ἀνάλογον προσευρεῖν.

Κείσθωσαν ἐπ' εὐθείᾳ, καὶ γεγράφθω ἐπὶ τῇ ΑΓ ἡμικύκλιον τὸ ΑΔΓ, καὶ >ήθθω ἀπὸ τοῦ Β σημείου τῇ ΑΓ εὐθεῖα πρὸς ὀρθὰς ἡ ΒΔ, καὶ ἐπεζεύθθωσαν αἱ ΑΔ, ΔΓ.

Ἐπεὶ ἐν ἡμικυκλίῳ γωνία ἐστὶν ἡ ὑπὸ ΑΔΓ, ὀρθή ἐστιν. καὶ ἐπεὶ ἐν ὀρθογώνιῳ τριγώνῳ τῷ ΑΔΓ ἀπὸ τῇς ὀρθῆς γωνίας ἐπὶ τὴν βάσιν κάθετοξ >ῆκται ἡ ΔΒ, ἡ ΔΒ >άρα τῶν τῇς βάσεως τμημάτων τῶν ΑΒ, ΒΓ μέση ἀνάλογόν ἐστιν.

Δύο >άρα δοθεισῶν εὐθειῶν τῶν ΑΒ, ΒΓ μέση ἀνάλογον προσεύρηται ἡ ΔΒ; <όπερ >έδει ποιῆσαι.

required to find the (straight-line) in mean proportion to AB and BC.

Let (AB and BC) be laid down straight-on (with respect to one another), and let the semi-circle ADC be drawn on AC [Prop. 1.10]. And let BD be drawn from (point) B, at right-angles to AC [Prop. 1.11]. And let AD and DC be joined.

And since ADC is an angle in a semi-circle, it is a right-angle [Prop. 3.31]. And since, in the right-angled triangle ADC, the (straight-line) DB has been drawn from the right-angle perpendicular to the base, DB is thus the mean proportional to the pieces of the base, AB and BC [Prop. 6.8 corr.].

Thus, DB has been found (which is) in mean proportion to the two given straight-lines, AB and BC. (Which is) the very thing it was required to do.

† In other words, to find the geometric mean of two given straight-lines.

14.

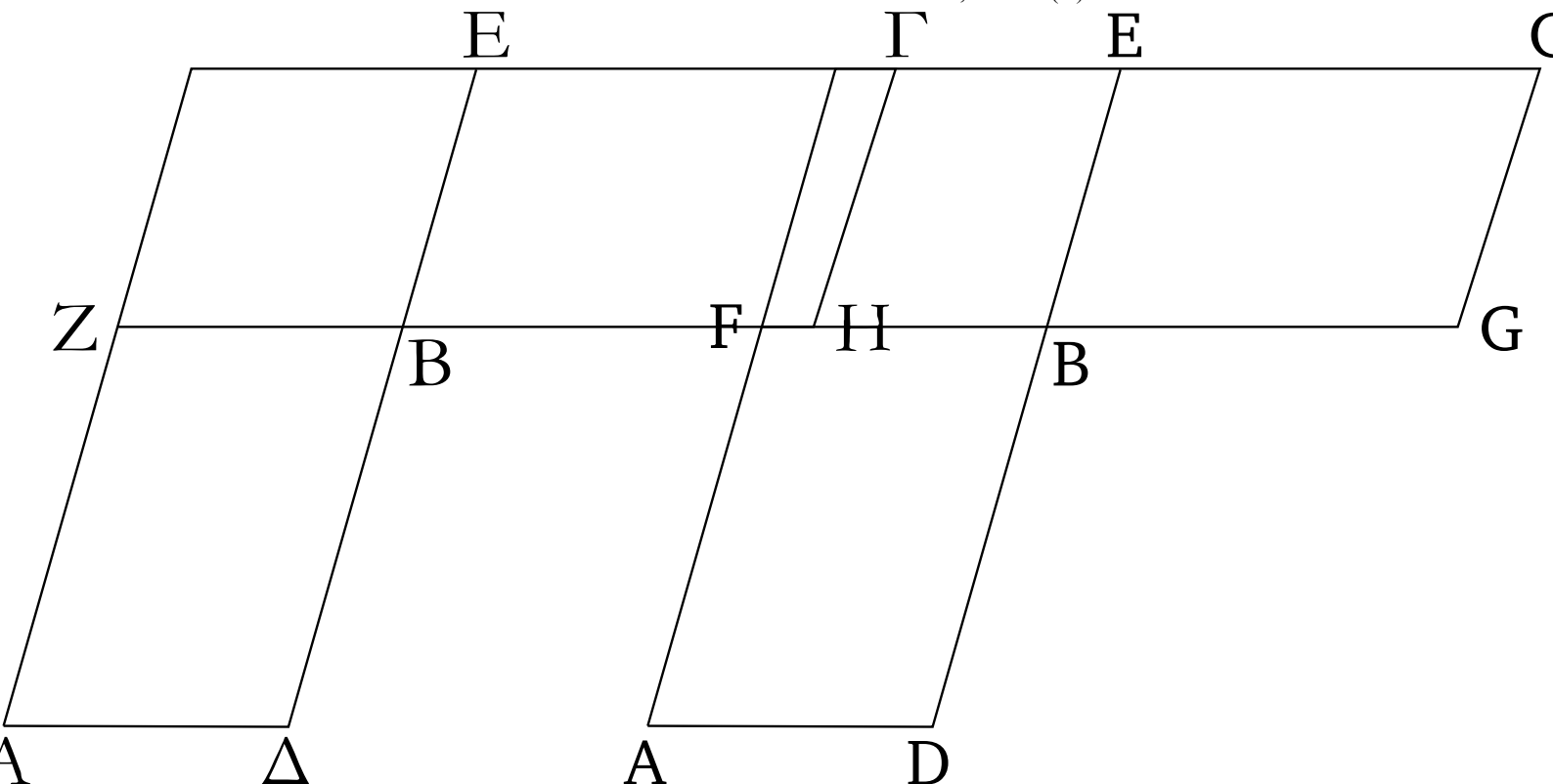
Τῶν >ίσων τε καὶ >ισογώνιων παραλληλογράμμων ἀντιτεπόνθασιν αἱ πλευραὶ αἱ περὶ τὰξ >ίσαξ γωνίαξ; καὶ <ῶν ἰσογώνιων παραλληλογράμμων ἀντιτεπόνθασιν αἱ πλευραὶ αἱ περὶ τὰξ >ίσαξ γωνίαξ, >ίσα ἐστὶν ἐκεῖνα.

>Ἐστω >ίσα τε καὶ ἰσογώνια παραλληλόγραμμα τὰ ΑΒ, ΒΓ >ίσαξ >έθοντα τὰξ πρὸξ τῷ Β γωνίαξ, καὶ κείσθωσαν ἐπ' εὐθείᾳ αἱ ΔΒ, ΒΕ; ἐπ' εὐθείᾳ >άρα εἰσὶ καὶ αἱ ΖΒ, ΒΗ. λέγω, <ὅτι τῶν ΑΒ, ΒΓ ἀντιτεπόνθασιν αἱ πλευραὶ αἱ περὶ τὰξ >ίσαξ γωνίαξ, τουτέστιν, <ὅτι ἐστὶν ὡξ ἡ ΔΒ πρὸξ τὴν ΒΕ, ο<ύτωξ ἡ ΗΒ πρὸξ τὴν ΒΖ.

Proposition 14

In equal and equiangular parallelograms, the sides about the equal angles are reciprocally proportional. And those equiangular parallelograms in which the sides about the equal angles are reciprocally proportional are equal.

Let AB and BC be equal and equiangular parallelograms having the angles at B equal. And let DB and BE be laid down straight-on (with respect to one another). Thus, FB and BG are also straight-on (with respect to one another) [Prop. 1.14]. I say that the sides of AB and BC about the equal angles are reciprocally proportional, that is to say, that as DB is to BE, so GB (is) to BF.



Συμπεπληρώσθω γὰρ τὸ ΖΕ παραλληλόγραμμον. ἐπεὶ ο<ὺν >ίσον ἐστὶ τὸ ΑΒ παραλληλόγραμμον τῷ ΒΓ παραλληλόγραμμῳ, >άλλο δέ τι τὸ ΖΕ, >έστιν >άρα ὡξ τὸ ΑΒ πρὸξ τὸ ΖΕ, ο<ύτωξ τὸ ΒΓ πρὸξ τὸ ΖΕ. ἄλλ> ὡξ μὲν τὸ ΑΒ πρὸξ τὸ ΖΕ,

For let the parallelogram FE be completed. Therefore, since parallelogram AB is equal to parallelogram BC, and FE (is) some other (parallelogram), thus as (parallelogram) AB is to FE, so (parallelogram) BC (is) to FE [Prop. 5.7].

οὐτως ἡ ΔB πρὸς τὴν BE , ὥς δὲ τὸ $B\Gamma$ πρὸς τὸ ZE , οὐτως ἡ HB πρὸς τὴν BZ ; καὶ ὥς ἡ ΔB πρὸς τὴν BE , οὐτως ἡ HB πρὸς τὴν BZ . τῶν ἄρα AB , $B\Gamma$ παραλληλογράμμων ἀντιπεπόνθασιν αἱ πλευραὶ αἱ περὶ τὰς ἴσας γωνίας. Ἀλλὰ δὴ ὥς ἡ ΔB πρὸς τὴν BE , οὐτως ἡ HB πρὸς τὴν BZ ; λέγω, ὅτι ἴσον ἐστὶ τὸ AB παραλληλόγραμμον τῷ $B\Gamma$ παραλληλόγραμμῳ.

Ἐπεὶ γάρ ἐστιν ὥς ἡ ΔB πρὸς τὴν BE , οὐτως ἡ HB πρὸς τὴν BZ , ἀλλ' ὥς μὲν ἡ ΔB πρὸς τὴν BE , οὐτως τὸ AB παραλληλόγραμμον πρὸς τὸ ZE παραλληλόγραμμον, ὥς δὲ ἡ HB πρὸς τὴν BZ , οὐτως τὸ $B\Gamma$ παραλληλόγραμμον πρὸς τὸ ZE παραλληλόγραμμον, καὶ ὥς ἡ ΔB πρὸς τὸ ZE , οὐτως τὸ $B\Gamma$ πρὸς τὸ ZE ; ἴσον ἄρα ἐστὶ τὸ AB παραλληλόγραμμον τῷ $B\Gamma$ παραλληλόγραμμῳ.

Τῶν ἄρα ἴσων τε καὶ ἰσογωνίων παραλληλογράμμων ἀντιπεπόνθασιν αἱ πλευραὶ αἱ περὶ τὰς ἴσας γωνίας; καὶ ὧν ἰσογωνίων παραλληλογράμμων ἀντιπεπόνθασιν αἱ πλευραὶ αἱ περὶ τὰς ἴσας γωνίας, ἴσα ἐστὶν ἐκεῖνα; ὅπερ ἔδει δεῖξαι.

15.

Τῶν ἴσων καὶ μίαν μὴ ἴσην ἐθόντων γωνίαν τριγώνων ἀντιπεπόνθασιν αἱ πλευραὶ αἱ περὶ τὰς ἴσας γωνίας; καὶ ὧν μίαν μὴ ἴσην ἐθόντων γωνίαν τριγώνων ἀντιπεπόνθασιν αἱ πλευραὶ αἱ περὶ τὰς ἴσας γωνίας, ἴσα ἐστὶν ἐκεῖνα.

Ἐστω ἴσα τρίγωνα τὰ $AB\Gamma$, ΔAE μίαν μὴ ἴσην ἐθόντα γωνίαν τὴν ὑπὸ BAG τῇ ὑπὸ DAE ; λέγω, ὅτι τῶν $AB\Gamma$, ΔAE τριγώνων ἀντιπεπόνθασιν αἱ πλευραὶ αἱ περὶ τὰς ἴσας γωνίας, τουτέστιν, ὅτι ἐστὶν ὥς ἡ ΓA πρὸς τὴν AD , οὐτως ἡ EA πρὸς τὴν AB .

But, as (parallelogram) AB (is) to FE , so DB (is) to BE , and as (parallelogram) BC (is) to FE , so GB (is) to BF [Prop. 6.1]. Thus, also, as DB (is) to BE , so GB (is) to BF . Thus, in parallelograms AB and BC the sides about the equal angles are reciprocally proportional.

And so, let DB be to BE , as GB (is) to BF . I say that parallelogram AB is equal to parallelogram BC .

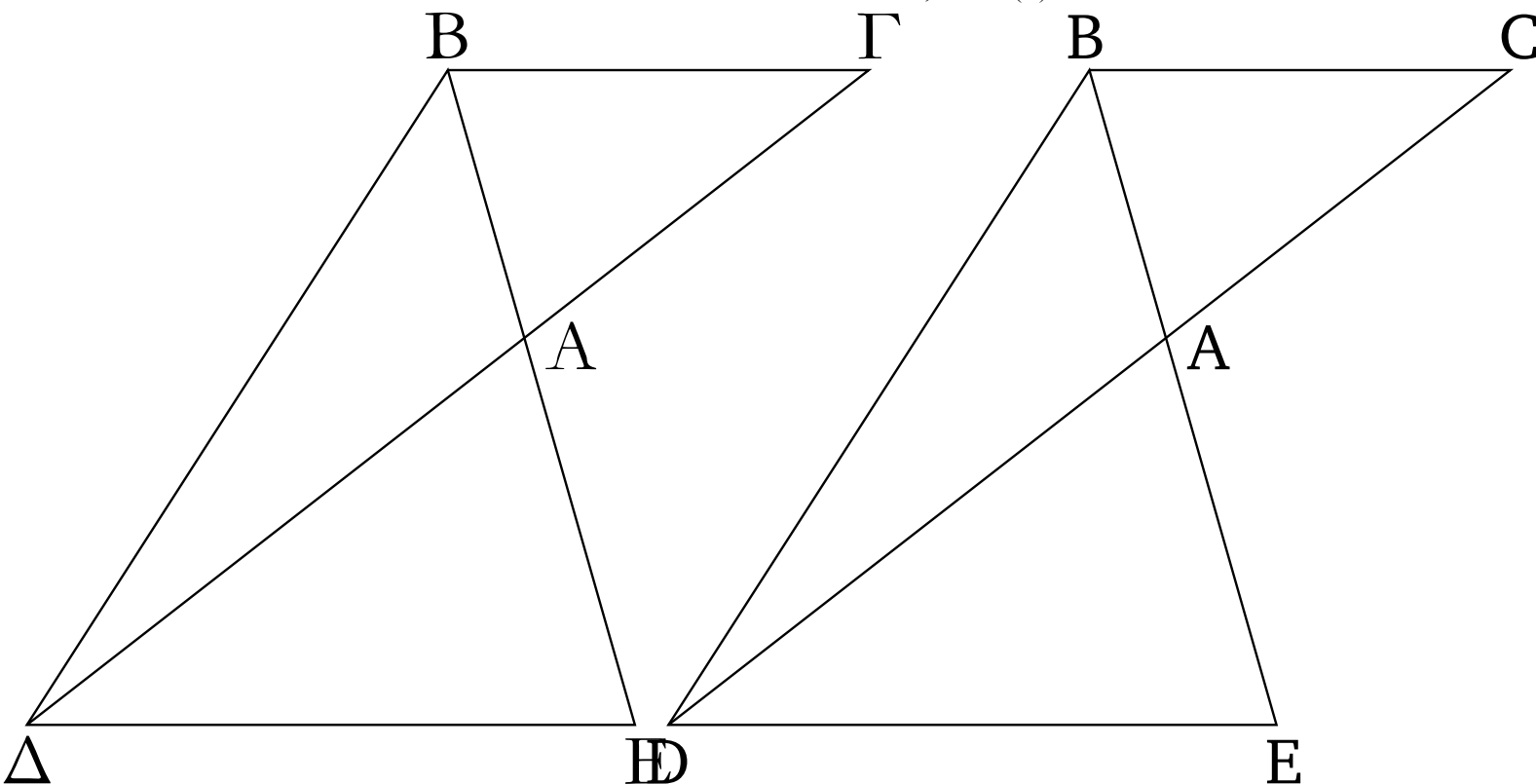
For since as DB is to BE , so GB (is) to BF , but as DB (is) to BE , so parallelogram AB (is) to parallelogram FE , and as GB (is) to BF , so parallelogram BC (is) to parallelogram FE [Prop. 6.1], thus, also, as (parallelogram) AB (is) to FE , so (parallelogram) BC (is) to FE [Prop. 5.11]. Thus, parallelogram AB is equal to parallelogram BC [Prop. 5.9].

Thus, in equal and equiangular parallelograms, the sides about the equal angles are reciprocally proportional. And those equiangular parallelograms in which the sides about the equal angles are reciprocally proportional are equal. (Which is) the very thing it was required to show.

Proposition 15

In equal triangles also having one angle equal to one (angle), the sides about the equal angles are reciprocally proportional. And those triangles having one angle equal to one angle for which the sides about the equal angles (are) reciprocally proportional are equal.

Let ABC and ADE be equal triangles having one angle equal to one (angle), (namely) BAC (equal) to DAE . I say that, in triangles ABC and ADE , the sides about the equal angles are reciprocally proportional, that is to say, that as CA is to AD , so EA (is) to AB .



Κείσθω γὰρ ὥστε ἐπ' εὐθείᾳ εἶναι τὴν ΓA τῇ AD ; ἐπ' εὐθείᾳ ἄρα ἐστὶ καὶ ἡ EA τῇ AB . καὶ ἐπεζεύθω ἡ BD . Ἐπεὶ οὖν ἴσον ἐστὶ τὸ $AB\Gamma$ τρίγωνον τῷ ΔAE τριγώνῳ, ἄλλο δὲ τι τὸ $BA\Delta$, ἔστιν ἄρα ὥς τὸ ΓAB τρίγωνον

For let CA be laid down so as to be straight-on (with respect) to AD . Thus, EA is also straight-on (with respect) to AB [Prop. 1.14]. And let BD be joined.

Therefore, since triangle ABC is equal to triangle ADE ,

πρὸς τὸ $BA\Delta$ τρίγωνον, οὐτως ὡς τὸ $EA\Delta$ τρίγωνον πρὸς τὸ $BA\Delta$ τρίγωνον. ἄλλ' ὥς μὲν τὸ ΓAB πρὸς τὸ $BA\Delta$, οὕτως ἢ ΓA πρὸς τὴν $A\Delta$, ὥς δὲ τὸ $EA\Delta$ πρὸς τὸ $BA\Delta$, οὕτως ἢ EA πρὸς τὴν AB . καὶ ὥς ἄρα ἢ ΓA πρὸς τὴν $A\Delta$, οὕτως ἢ EA πρὸς τὴν AB . τῶν $AB\Gamma$, $A\Delta E$ ἄρα τριγώνων ἀντιτεπόνθασιν αἱ πλευραὶ αἱ περὶ τὰς ἴσας γωνίας.

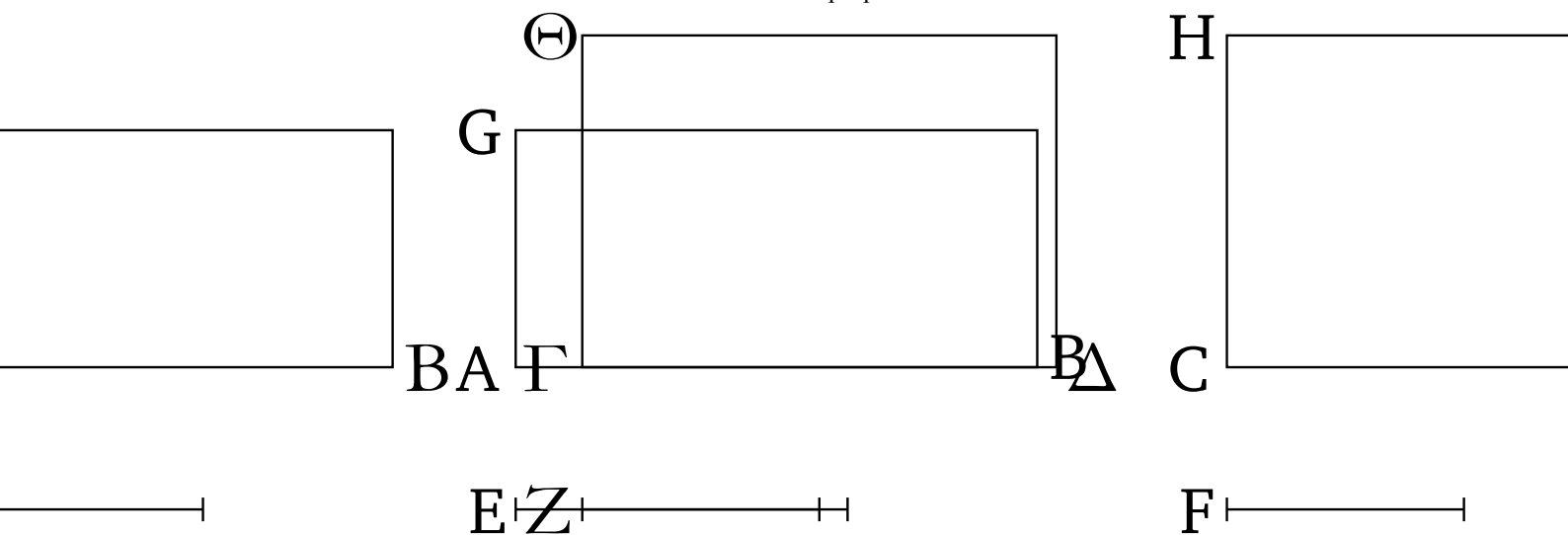
Ἀλλὰ δὴ ἀντιτεπονήτευσαν αἱ πλευραὶ τῶν $AB\Gamma$, $A\Delta E$ τριγώνων, καὶ ἔστω ὥς ἢ ΓA πρὸς τὴν $A\Delta$, οὕτως ἢ EA πρὸς τὴν AB ; λέγω, <ὅτι> ἴσον ἐστὶ τὸ $AB\Gamma$ τρίγωνον τῷ $A\Delta E$ τριγώνῳ.

Ἐπιζευθείσης γὰρ πάλιν τῆς $B\Delta$, ἐπεὶ ἐστὶν ὥς ἢ ΓA πρὸς τὴν $A\Delta$, οὕτως ἢ EA πρὸς τὴν AB , ἄλλ' ὥς μὲν ἢ ΓA πρὸς τὴν $A\Delta$, οὕτως τὸ $AB\Gamma$ τρίγωνον πρὸς τὸ $BA\Delta$ τρίγωνον, ὥς δὲ ἢ EA πρὸς τὴν AB , οὕτως τὸ $EA\Delta$ τρίγωνον πρὸς τὸ $BA\Delta$ τρίγωνον, ὥς ἄρα τὸ $AB\Gamma$ τρίγωνον πρὸς τὸ $BA\Delta$ τρίγωνον, οὕτως τὸ $EA\Delta$ τρίγωνον πρὸς τὸ $BA\Delta$ τρίγωνον. ἐκάτερον ἄρα τῶν $AB\Gamma$, $EA\Delta$ πρὸς τὸ $BA\Delta$ τὸν αὐτὸν ἔχει λόγον. ἴσον ἄρα ἐστὶ τὸ $AB\Gamma$ [τρίγωνον] τῷ $EA\Delta$ τριγώνῳ.

Τῶν ἄρα ἴσων καὶ μίαν μὲν ἴσην ἐθόντων γωνίαν τριγώνων ἀντιτεπόνθασιν αἱ πλευραὶ αἱ περὶ τὰς ἴσας γωνίας; καὶ ὥς μίαν μὲν ἴσην ἐθόντων γωνίαν τριγώνων ἀντιτεπόνθασιν αἱ πλευραὶ αἱ περὶ τὰς ἴσας γωνίας, ἐκεῖνα ἴσα ἐστίν; ὅπερ ἔδει δεῖχαι.

16.

Ἐὰν τέσσαρες εὐθεῖαι ἀνάλογον ᾖσιν, τὸ ὑπὸ τῶν ἄκρων περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ ὑπὸ τῶν μέσων περιεχομένῳ ὀρθογώνῳ; καὶ ἂν τὸ ὑπὸ τῶν ἄκρων περιεχόμενον ὀρθογώνιον ἴσον ᾖ τῷ ὑπὸ τῶν μέσων περιεχομένῳ ὀρθογώνῳ, αἱ τέσσαρες εὐθεῖαι ἀνάλογον ἔσονται.



>Ἐστωσαν τέσσαρες εὐθεῖαι ἀνάλογον αἱ AB , $\Gamma\Delta$, E , Z , ὥς ἢ AB πρὸς τὴν $\Gamma\Delta$, οὕτως ἢ E πρὸς τὴν Z ; λέγω, <ὅτι> τὸ ὑπὸ τῶν AB , Z περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ ὑπὸ τῶν $\Gamma\Delta$, E περιεχομένῳ ὀρθογώνῳ.

>Ἡθώσαν [γάρ] ἀπὸ τῶν A , Γ σημείων ταῖς AB , $\Gamma\Delta$ εὐθείαις πρὸς ὀρθὰς αἱ AH , $\Gamma\Theta$, καὶ κείσθω τῇ μὲν Z ἴση ἢ AH , τῇ δὲ E ἴση ἢ $\Gamma\Theta$. καὶ συμπληρώσθω τὰ BH , $\Delta\Theta$ παραλληλόγραμμα.

Καὶ ἐπεὶ ἐστὶν ὥς ἢ AB πρὸς τὴν $\Gamma\Delta$, οὕτως ἢ E πρὸς τὴν Z , ἴση δὲ ἢ μὲν E τῇ $\Gamma\Theta$, ἢ δὲ Z τῇ AH , ἔστιν ἄρα ὥς ἢ AB πρὸς τὴν $\Gamma\Delta$, οὕτως ἢ $\Gamma\Theta$ πρὸς τὴν AH .

and BAD (is) some other (triangle), thus as triangle CAB is to triangle BAD , so triangle EAD (is) to triangle BAD [Prop. 5.7]. But, as (triangle) CAB (is) to BAD , so CA (is) to AD , and as (triangle) EAD (is) to BAD , so EA (is) to AB [Prop. 6.1]. And thus, as CA (is) to AD , so EA (is) to AB . Thus, in triangles ABC and ADE the sides about the equal angles (are) reciprocally proportional.

And so, let the sides of triangles ABC and ADE be reciprocally proportional, and (thus) let CA be to AD , as EA (is) to AB . I say that triangle ABC is equal to triangle ADE .

For, BD again being joined, since as CA is to AD , so EA (is) to AB , but as CA (is) to AD , so triangle ABC (is) to triangle BAD , and as EA (is) to AB , so triangle EAD (is) to triangle BAD [Prop. 6.1], thus as triangle ABC (is) to triangle BAD , so triangle EAD (is) to triangle BAD . Thus, (triangles) ABC and EAD each have the same ratio to BAD . Thus, [triangle] ABC is equal to triangle EAD [Prop. 5.9].

Thus, in equal triangles also having one angle equal to one (angle), the sides about the equal angles (are) reciprocally proportional. And those triangles having one angle equal to one angle for which the sides about the equal angles (are) reciprocally proportional are equal. (Which is) the very thing it was required to show.

Proposition 16

If four straight-lines are proportional, (then) the rectangle contained by the (two) outermost is equal to the rectangle contained by the middle (two). And if the rectangle contained by the (two) outermost is equal to the rectangle contained by the middle (two), (then) the four straight-lines will be proportional.

Let AB , CD , E , and F be four proportional straight-lines, (such that) as AB (is) to CD , so E (is) to F . I say that the rectangle contained by AB and F is equal to the rectangle contained by CD and E .

[For] let AG and CH be drawn from points A and C at right-angles to the straight-lines AB and CD (respectively) [Prop. 1.11]. And let AG be made equal to F , and CH to E [Prop. 1.3]. And let the parallelograms BG and DH be completed.

And since as AB is to CD , so E (is) to F , and E (is) equal CH , and F to AG , thus as AB is to CD , so CH

τῶν BH , $\Delta\Theta$ ὅρα παραλληλογράμμων ἀντιτεπόνθασιν αἱ πλευραὶ αἱ περὶ τὰς ἴσας γωνίας. <ὦν δὲ ἰσογωνίων παραλληλογράμμων ἀντιτεπόνθασιν αἱ πλευραὶ αἱ περὶ τὰς ἴσας γωνίας, ἴσα ἐστὶν ἐκεῖνα; ἴσον ὅρα ἐστὶ τὸ BH παραλληλόγραμμον τῷ $\Delta\Theta$ παραλληλογράμμῳ. καὶ ἐστὶ τὸ μὲν BH τὸ ὑπὸ τῶν AB , Z ; ἴση γὰρ ἡ AH τῇ Z ; τὸ δὲ $\Delta\Theta$ τὸ ὑπὸ τῶν $\Gamma\Delta$, E ; ἴση γὰρ ἡ E τῇ $\Gamma\Theta$; τὸ ὅρα ὑπὸ τῶν AB , Z περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ ὑπὸ τῶν $\Gamma\Delta$, E περιεχομένῳ ὀρθογώνιῳ.

Ἀλλὰ δὴ τὸ ὑπὸ τῶν AB , Z περιεχόμενον ὀρθογώνιον ἴσον ἔστω τῷ ὑπὸ τῶν $\Gamma\Delta$, E περιεχομένῳ ὀρθογώνιῳ. λέγω, <ὅτι αἱ τέσσαρες εὐθεῖαι ἀνάλογον ἔσονται, ὥς ἡ AB πρὸς τὴν $\Gamma\Delta$, οὕτως ἡ E πρὸς τὴν Z .

Τῶν γὰρ αὐτῶν κατασκευασθέντων, ἐπεὶ τὸ ὑπὸ τῶν AB , Z ἴσον ἐστὶ τῷ ὑπὸ τῶν $\Gamma\Delta$, E , καὶ ἐστὶ τὸ μὲν ὑπὸ τῶν AB , Z τὸ BH ; ἴση γὰρ ἐστὶν ἡ AH τῇ Z ; τὸ δὲ ὑπὸ τῶν $\Gamma\Delta$, E τὸ $\Delta\Theta$; ἴση γὰρ ἡ $\Gamma\Theta$ τῇ E ; τὸ ὅρα BH ἴσον ἐστὶ τῷ $\Delta\Theta$. καὶ ἐστὶν ἰσογώνια. τῶν δὲ ἴσων καὶ ἰσογωνίων παραλληλογράμμων ἀντιτεπόνθασιν αἱ πλευραὶ αἱ περὶ τὰς ἴσας γωνίας. ἔστιν ὅρα ὥς ἡ AB πρὸς τὴν $\Gamma\Delta$, οὕτως ἡ $\Gamma\Theta$ πρὸς τὴν AH . ἴση δὲ ἡ μὲν $\Gamma\Theta$ τῇ E , ἡ δὲ AH τῇ Z ; ἔστιν ὅρα ὥς ἡ AB πρὸς τὴν $\Gamma\Delta$, οὕτως ἡ E πρὸς τὴν Z . Ἐὰν ὅρα τέσσαρες εὐθεῖαι ἀνάλογον ᾤσιν, τὸ ὑπὸ τῶν ἄκρων περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ ὑπὸ τῶν μέσων περιεχομένῳ ὀρθογώνιῳ; κ>ὰν τὸ ὑπὸ τῶν ἄκρων περιεχόμενον ὀρθογώνιον ἴσον >ῇ τῷ ὑπὸ τῶν μέσων περιεχομένῳ ὀρθογώνιῳ, αἱ τέσσαρες εὐθεῖαι ἀνάλογον ἔσονται; <ὅπερ >έδει δεῖξαι.

(is) to AG . Thus, in the parallelograms BG and DH , the sides about the equal angles are reciprocally proportional. And those equiangular parallelograms in which the sides about the equal angles are reciprocally proportional are equal [Prop. 6.14]. Thus, parallelogram BG is equal to parallelogram DH . And BG is the (rectangle contained) by AB and F . For AG (is) equal to F . And DH (is) the (rectangle contained) by CD and E . For E (is) equal to CH . Thus, the rectangle contained by AB and F is equal to the rectangle contained by CD and E .

And so, let the rectangle contained by AB and F be equal to the rectangle contained by CD and E . I say that the four straight-lines will be proportional, (so that) as AB (is) to CD , so E (is) to F .

For, with the same construction, since the (rectangle contained) by AB and F is equal to the (rectangle contained) by CD and E . And BG is the (rectangle contained) by AB and F . For AG is equal to F . And DH (is) the (rectangle contained) by CD and E . For CH (is) equal to E . BG is thus equal to DH . And they are equiangular. And in equal and equiangular parallelograms, the sides about the equal angles are reciprocally proportional [Prop. 6.14]. Thus, as AB is to CD , so CH (is) to AG . And CH (is) equal to E , and AG to F . Thus, as AB is to CD , so E (is) to F .

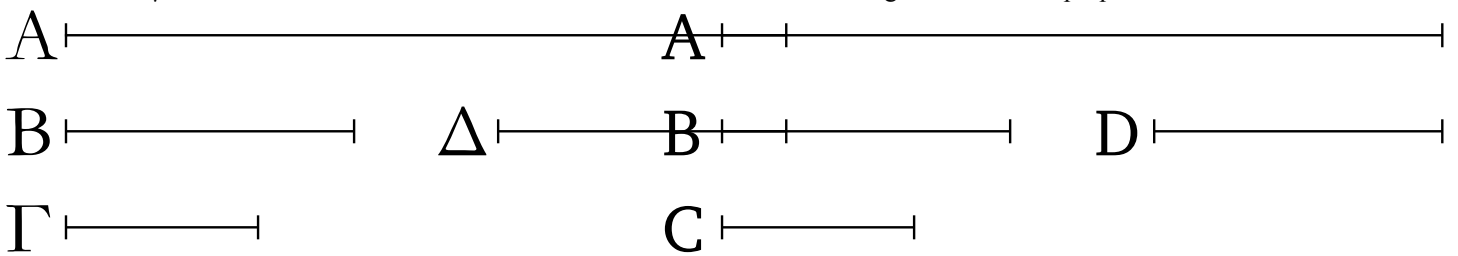
Thus, if four straight-lines are proportional, (then) the rectangle contained by the (two) outermost is equal to the rectangle contained by the middle (two). And if the rectangle contained by the (two) outermost is equal to the rectangle contained by the middle (two) then the four straight-lines will be proportional. (Which is) the very thing it was required to show.

17.

Proposition 17

Ἐὰν τρεῖς εὐθεῖαι ἀνάλογον ᾤσιν, τὸ ὑπὸ τῶν ἄκρων περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ ἀπὸ τῆς μέσης τετραγώνῳ; κ>ὰν τὸ ὑπὸ τῶν ἄκρων περιεχόμενον ὀρθογώνιον ἴσον >ῇ τῷ ἀπὸ τῆς μέσης τετραγώνῳ, αἱ τρεῖς εὐθεῖαι ἀνάλογον ἔσονται.

If three straight-lines are proportional, (then) the rectangle contained by the (two) outermost is equal to the square on the middle (one). And if the rectangle contained by the (two) outermost is equal to the square on the middle (one), (then) the three straight-lines will be proportional.



>Ἐστωσαν τρεῖς εὐθεῖαι ἀνάλογον αἱ A , B , Γ , ὥς ἡ A πρὸς τὴν B , οὕτως ἡ B πρὸς τὴν Γ ; λέγω, <ὅτι τὸ ὑπὸ τῶν A , Γ περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ ἀπὸ τῆς B τετραγώνῳ.

Κεῖσθω τῇ B ἴση ἡ Δ .

Καὶ ἐπεὶ ἐστὶν ὥς ἡ A πρὸς τὴν B , οὕτως ἡ B πρὸς τὴν Γ , ἴση δὲ ἡ B τῇ Δ , ἔστιν ὅρα ὥς ἡ A πρὸς τὴν B , ἡ Δ πρὸς τὴν Γ . Ἐὰν δὲ τέσσαρες εὐθεῖαι ἀνάλογον ᾤσιν, τὸ ὑπὸ τῶν ἄκρων περιεχόμενον [ὀρθογώνιον] ἴσον ἐστὶ τῷ ὑπὸ τῶν μέσων περιεχομένῳ ὀρθογώνιῳ. τὸ ὅρα ὑπὸ τῶν A , Γ ἴσον ἐστὶ τῷ ὑπὸ τῶν B , Δ . ἀλλὰ τὸ ὑπὸ τῶν B , Δ τὸ ἀπὸ τῆς B ἐστὶν; ἴση γὰρ ἡ B τῇ Δ ; τὸ ὅρα ὑπὸ τῶν A , Γ περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ ἀπὸ τῆς B τετραγώνῳ.

Let A , B and C be three proportional straight-lines, (such that) as A (is) to B , so B (is) to C . I say that the rectangle contained by A and C is equal to the square on B .

Let D be made equal to B [Prop. 1.3].

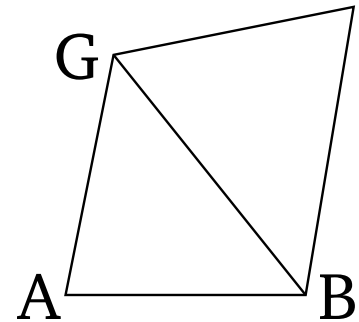
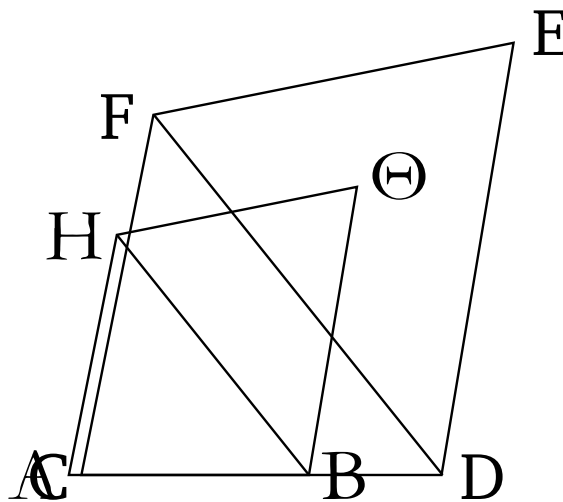
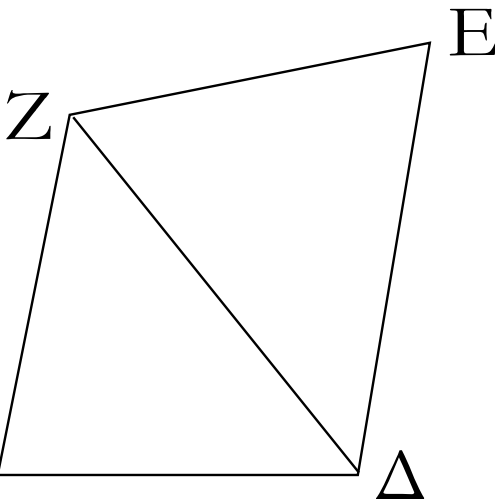
And since as A is to B , so B (is) to C , and B (is) equal to D , thus as A is to B , (so) D (is) to C . And if four straight-lines are proportional, (then) the [rectangle] contained by the (two) outermost is equal to the rectangle contained by the middle (two) [Prop. 6.16]. Thus, the (rectangle contained) by A and C is equal to the (rectangle contained) by B and D . But, the (rectangle contained) by B and D is the (square) on B . For B (is) equal to D . Thus, the rectangle contained by A and C is equal to the square on B .

And so, let the (rectangle contained) by A and C be equal

Ἀλλὰ δὴ τὸ ὑπὸ τῶν A, Γ ἴσον ἔστω τῷ ὑπὸ τῆς B ; λέγω, <ὅτι ἔστιν ὥς ἡ A πρὸς τὴν B , οὕτως ἡ B πρὸς τὴν Γ .
 Τῶν γὰρ αὐτῶν κατασκευασθέντων, ἐπεὶ τὸ ὑπὸ τῶν A, Γ ἴσον ἐστὶ τῷ ὑπὸ τῆς B , ἀλλὰ τὸ ὑπὸ τῆς B τὸ ὑπὸ τῶν B, Δ ἐστιν; ἴση γὰρ ἡ B τῇ Δ ; τὸ ἄρα ὑπὸ τῶν A, Γ ἴσον ἐστὶ τῷ ὑπὸ τῶν B, Δ . ἔαν δὲ τὸ ὑπὸ τῶν ἄκρων ἴσον ᾖ τῷ ὑπὸ τῶν μέσων, αἱ τέσσαρες εὐθεῖαι ἀνάλογον εἰσιν. ἔστιν ἄρα ὥς ἡ A πρὸς τὴν B , οὕτως ἡ Δ πρὸς τὴν Γ . ἴση δὲ ἡ B τῇ Δ ; ὥς ἄρα ἡ A πρὸς τὴν B , οὕτως ἡ B πρὸς τὴν Γ .
 Ἐὰν ἄρα τρεῖς εὐθεῖαι ἀνάλογον ᾖσιν, τὸ ὑπὸ τῶν ἄκρων περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ ὑπὸ τῆς μέσης τετραγώνῳ; κὰν τὸ ὑπὸ τῶν ἄκρων περιεχόμενον ὀρθογώνιον ἴσον ᾖ τῷ ὑπὸ τῆς μέσης τετραγώνῳ, αἱ τρεῖς εὐθεῖαι ἀνάλογον ἔσονται; ὅπερ εἶδει δεῖχαι.

18.

Ἀπὸ τῆς δοθείσης εὐθείας τῷ δοθέντι εὐθυγράμμῳ ὁμοίον τε καὶ ὁμοίως κείμενον εὐθύγραμμον ἀναγράψαι.
 Ἐστω ἡ μὲν δοθεῖσα εὐθεῖα ἡ AB , τὸ δὲ δοθὲν εὐθύγραμμον τὸ ΓE ; δεῖ δὴ ἀπὸ τῆς AB εὐθείας τῷ ΓE εὐθυγράμμῳ ὁμοίον τε καὶ ὁμοίως κείμενον εὐθύγραμμον ἀναγράψαι.



Ἐπεξεύθθω ἡ DZ , καὶ συνεστάτω πρὸς τῇ AB εὐθείᾳ καὶ τοῖς πρὸς αὐτῇ σημείοις τοῖς A, B τῇ μὲν πρὸς τῷ Γ γωνία ἴση ἡ ὑπὸ HAB , τῇ δὲ ὑπὸ $\Gamma\Delta Z$ ἴση ἡ ὑπὸ ABH . λοιπὴ ἄρα ἡ ὑπὸ $\Gamma\Delta Z$ τῇ ὑπὸ AHB ἐστὶν ἴση; ἰσογώνιον ἄρα ἐστὶ τὸ $\Gamma\Delta Z$ τρίγωνον τῷ HAB τριγώνῳ. ἀνάλογον ἄρα ἐστὶν ὥς ἡ $Z\Delta$ πρὸς τὴν HB , οὕτως ἡ $Z\Gamma$ πρὸς τὴν HA , καὶ ἡ $\Gamma\Delta$ πρὸς τὴν AB . πάλιν συνεστάτω πρὸς τῇ BH εὐθείᾳ καὶ τοῖς πρὸς αὐτῇ σημείοις τοῖς B, H τῇ μὲν ὑπὸ ΔZE γωνία ἴση ἡ ὑπὸ $BH\Theta$, τῇ δὲ ὑπὸ $Z\Delta E$ ἴση ἡ ὑπὸ $H\Theta B$. λοιπὴ ἄρα ἡ πρὸς τῷ E λοιπῇ τῇ πρὸς τῷ Θ ἐστὶν ἴση; ἰσογώνιον ἄρα ἐστὶ τὸ $Z\Delta E$ τρίγωνον τῷ $H\Theta B$ τριγώνῳ; ἀνάλογον ἄρα ἐστὶν ὥς ἡ $Z\Delta$ πρὸς τὴν HB , οὕτως ἡ ZE πρὸς τὴν $H\Theta$ καὶ ἡ $E\Delta$ πρὸς τὴν ΘB . εἰδείθην δὲ καὶ ὥς ἡ $Z\Delta$ πρὸς τὴν HB , οὕτως ἡ $Z\Gamma$ πρὸς τὴν HA καὶ ἡ $\Gamma\Delta$ πρὸς τὴν AB ; καὶ ὥς ἄρα ἡ $Z\Gamma$ πρὸς τὴν HA , οὕτως ἡ $\Gamma\Delta$ πρὸς τὴν AB καὶ ἡ ZE πρὸς τὴν $H\Theta$ καὶ ἡ $E\Delta$ πρὸς τὴν ΘB . καὶ ἐπεὶ ἴση ἐστὶν ἡ μὲν ὑπὸ $\Gamma\Delta Z$ γωνία τῇ ὑπὸ AHB , ἡ δὲ ὑπὸ ΔZE τῇ ὑπὸ $BH\Theta$, ὅλη ἄρα ἡ ὑπὸ ΓZE ὅλη τῇ ὑπὸ $AH\Theta$ ἐστὶν ἴση. διὰ τὰ αὐτὰ δὲ καὶ ἡ ὑπὸ $\Gamma\Delta E$

to the (square) on B . I say that as A is to B , so B (is) to C . For, with the same construction, since the (rectangle contained) by A and C is equal to the (square) on B . But, the (square) on B is the (rectangle contained) by B and D . For B (is) equal to D . The (rectangle contained) by A and C is thus equal to the (rectangle contained) by B and D . And if the (rectangle contained) by the (two) outermost is equal to the (rectangle contained) by the middle (two), (then) the four straight-lines are proportional [Prop. 6.16]. Thus, as A is to B , so D (is) to C . And B (is) equal to D . Thus, as A (is) to B , so B (is) to C .

Thus, if three straight-lines are proportional, (then) the rectangle contained by the (two) outermost is equal to the square on the middle (one). And if the rectangle contained by the (two) outermost is equal to the square on the middle (one), (then) the three straight-lines will be proportional. (Which is) the very thing it was required to show.

Proposition 18

To describe a rectilinear figure similar, and similarly laid down, to a given rectilinear figure on a given straight-line. Let AB be the given straight-line, and CE the given rectilinear figure. So it is required to describe a rectilinear figure similar, and similarly laid down, to the rectilinear figure CE on the straight-line AB .

Let DF be joined, and let GAB , equal to the angle at C , and ABG , equal to (angle) CDF , be constructed on the straight-line AB at the points A and B on it (respectively) [Prop. 1.23]. Thus, the remaining (angle) CFD is equal to AGB [Prop. 1.32]. Thus, triangle FCD is equiangular to triangle GAB . Thus, proportionally, as FD is to GB , so FC (is) to GA , and CD to AB [Prop. 6.4]. Again, let BGH , equal to angle DFE , and GBH equal to (angle) FDE , be constructed on the straight-line BG at the points G and B on it (respectively) [Prop. 1.23]. Thus, the remaining (angle) at E is equal to the remaining (angle) at H [Prop. 1.32]. Thus, triangle FDE is equiangular to triangle GHB . Thus, proportionally, as FD is to GB , so FE (is) to GH , and ED to HB [Prop. 6.4]. And it was also shown (that) as FD (is) to GB , so FC (is) to GA , and CD to AB . Thus, also, as FC (is) to AG , so CD (is) to AB , and FE to GH , and, further, ED to HB . And since angle CFD is equal to AGB , and DFE to BGH , thus the whole (angle) CFE is equal to the whole (angle) AGH . So, for the same

τῇ ὑπὸ $AB\Theta$ ἐστὶν ὡς. >έστι δὲ καὶ ἡ μὲν πρὸς τῷ Γ τῇ πρὸς τῷ A ὡς, ἡ δὲ πρὸς τῷ E τῇ πρὸς τῷ Θ . ἰσογώνιον >άρα ἐστὶ τὸ $A\Theta$ τῷ ΓE ; καὶ τὰς περὶ τὰς ὡσας γωνίας αὐτῶν πλευρὰς ἀνάλογον >έθει; <όμοιον >άρα ἐστὶ τὸ $A\Theta$ εὐθύγραμμον τῷ ΓE εὐθυγράμμω.

Ἀπὸ τῆς δοθείσης >άρα εὐθείας τῆς AB τῷ δοθέντι εὐθυγράμμῳ τῷ ΓE <όμοιόν τε καὶ ὁμοίως κείμενον εὐθύγραμμον ἀναγέγραπται τὸ $A\Theta$; <όπερ >έδει ποιῆσαι.

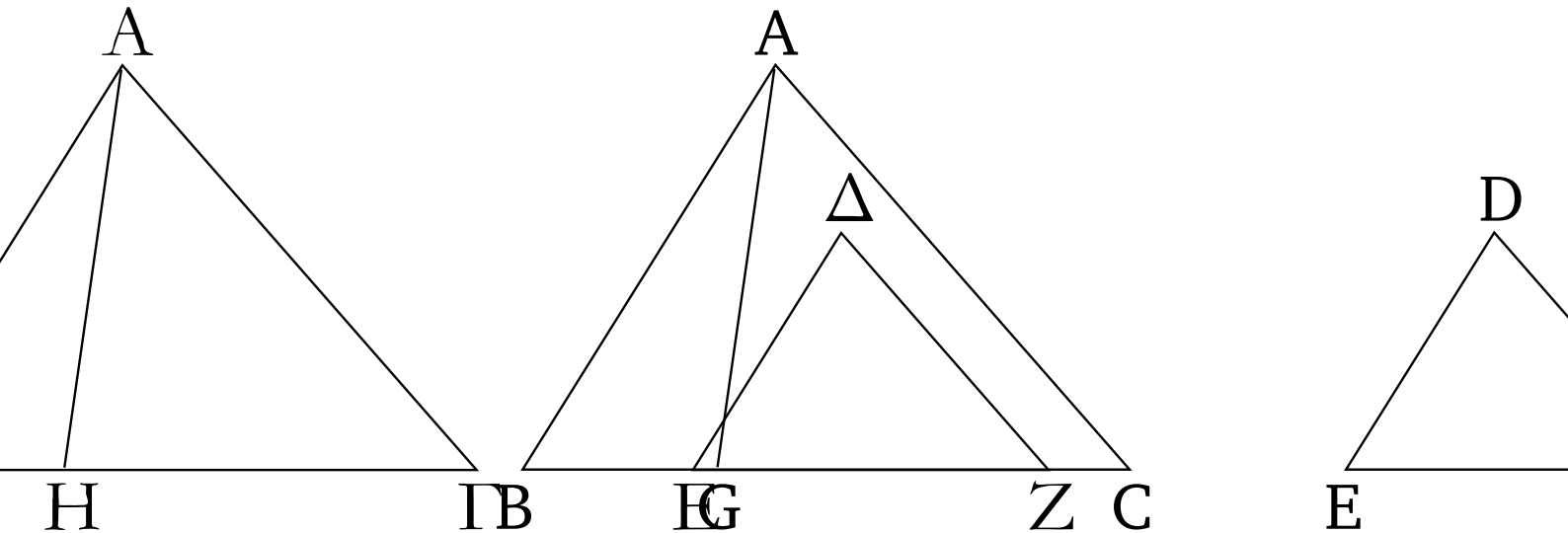
(reasons), (angle) CDE is also equal to ABH . And the (angle) at C is also equal to the (angle) at A , and the (angle) at E to the (angle) at H . Thus, (figure) AH is equiangular to CE . And (the two figures) have the sides about their equal angles proportional. Thus, the rectilinear figure AH is similar to the rectilinear figure CE [Def. 6.1].

Thus, the rectilinear figure AH , similar, and similarly laid down, to the given rectilinear figure CE has been constructed on the given straight-line AB . (Which is) the very thing it was required to do.

19.

Τὰ <όμοια τρίγωνα πρὸς >άλληλα ἐν διπλασίονι λόγῳ ἐστὶ τῶν ὁμολόγων πλευρῶν.

>Ἐστω <όμοια τρίγωνα τὰ $AB\Gamma$, ΔEZ ὡς ἔθοντα τὴν πρὸς τῷ B γωνίαν τῇ πρὸς τῷ E , ὥς δὲ τὴν AB πρὸς τὴν $B\Gamma$, οὕτως τὴν ΔE πρὸς τὴν EZ , <ώστε ὁμολόγον εἶναι τὴν $B\Gamma$ τῇ EZ ; λέγω, <ὅτι τὸ $AB\Gamma$ τρίγωνον πρὸς τὸ ΔEZ τρίγωνον διπλασίονα λόγον >έθει >ήπερ ἡ $B\Gamma$ πρὸς τὴν EZ . Εἰλήφθω γὰρ τῶν $B\Gamma$, EZ τρίτη ἀνάλογον ἡ BH , <ώστε εἶναι ὥς τὴν $B\Gamma$ πρὸς τὴν EZ , οὕτως τὴν EZ πρὸς τὴν BH ; καὶ ἐπεεῦθθω ἡ AH .



Ἐπεὶ οὖν ἐστὶν ὥς ἡ AB πρὸς τὴν $B\Gamma$, οὕτως ἡ ΔE πρὸς τὴν EZ , ἐναλλάξ >άρα ἐστὶν ὥς ἡ AB πρὸς τὴν ΔE , οὕτως ἡ $B\Gamma$ πρὸς τὴν EZ . ἀλλ' ὥς ἡ $B\Gamma$ πρὸς EZ , οὕτως ἐστὶν ἡ EZ πρὸς BH . καὶ ὥς >άρα ἡ AB πρὸς ΔE , οὕτως ἡ EZ πρὸς BH ; τῶν ABH , ΔEZ >άρα τριγώνων ἀντιπεπόνθασιν αἱ πλευραὶ αἱ περὶ τὰς ὡσας γωνίας. <ὦν δὲ μίαν μὲν ὡς ἔθοντων γωνίαν τριγώνων ἀντιπεπόνθασιν αἱ πλευραὶ αἱ περὶ τὰς ὡσας γωνίας, ὡς ἐστὶν ἐκεῖνα. ὡς >άρα ἐστὶ τὸ ABH τρίγωνον τῷ ΔEZ τριγώνῳ. καὶ ἐπεὶ ἐστὶν ὥς ἡ $B\Gamma$ πρὸς τὴν EZ , οὕτως ἡ EZ πρὸς τὴν BH , ἔαν δὲ τρεῖς εὐθεῖαι ἀνάλογον >ῶσιν, ἡ πρώτη πρὸς τὴν τρίτην διπλασίονα λόγον >έθει >ήπερ πρὸς τὴν δευτέραν, ἡ $B\Gamma$ >άρα πρὸς τὴν BH διπλασίονα λόγον >έθει >ήπερ ἡ ΓB πρὸς τὴν EZ . ὥς δὲ ἡ ΓB πρὸς τὴν BH , οὕτως τὸ $AB\Gamma$ τρίγωνον πρὸς τὸ ABH τρίγωνον; καὶ τὸ $AB\Gamma$ >άρα τρίγωνον πρὸς τὸ ABH διπλασίονα λόγον >έθει >ήπερ ἡ $B\Gamma$ πρὸς τὴν EZ . ὡς δὲ τὸ ABH τρίγωνον τῷ ΔEZ τριγώνῳ; καὶ τὸ $AB\Gamma$ >άρα τρίγωνον πρὸς τὸ ΔEZ τρίγωνον διπλασίονα λόγον >έθει >ήπερ ἡ $B\Gamma$ πρὸς τὴν EZ .

Τὰ >άρα <όμοια τρίγωνα πρὸς >άλληλα ἐν διπλασίονι λόγῳ

Proposition 19

Similar triangles are to one another in the squared[†] ratio of (their) corresponding sides.

Let ABC and DEF be similar triangles having the angle at B equal to the (angle) at E , and AB to BC , as DE (is) to EF , such that BC corresponds to EF . I say that triangle ABC has a squared ratio to triangle DEF with respect to (that side) BC (has) to EF .

For let a third (straight-line), BG , be taken (which is) proportional to BC and EF , so that as BC (is) to EF , so EF (is) to BG [Prop. 6.11]. And let AG be joined.

Therefore, since as AB is to BC , so DE (is) to EF , thus, alternately, as AB is to DE , so BC (is) to EF [Prop. 5.16]. But, as BC (is) to EF , so EF is to BG . And, thus, as AB (is) to DE , so EF (is) to BG . Thus, for triangles ABG and DEF , the sides about the equal angles are reciprocally proportional. And those triangles having one (angle) equal to one (angle) for which the sides about the equal angles are reciprocally proportional are equal [Prop. 6.15]. Thus, triangle ABG is equal to triangle DEF . And since as BC (is) to EF , so EF (is) to BG , and if three straight-lines are proportional, (then) the first has a squared ratio to the third with respect to the second [Def. 5.9], BC thus has a squared ratio to BG with respect to (that) CB (has) to EF . And as CB (is) to BG , so triangle ABC (is) to triangle ABG [Prop. 6.1]. Thus, triangle ABC also has a squared ratio to (triangle) ABG with respect to (that side) BC (has) to EF . And triangle ABG (is) equal to triangle DEF . Thus, triangle ABC also has a squared ratio to triangle DEF with respect to (that side) BC (has) to EF .

Thus, similar triangles are to one another in the squared ratio of (their) corresponding sides. [(Which is) the very thing

ἐστὶ τῶν ὁμολόγων πλευρῶν. [ὁπερ >έδει δεῖχαι.]

it was required to show].

Corollary

So it is clear, from this, that if three straight-lines are proportional, (then) as the first is to the third, so the figure (described) on the first (is) to the similar, and similarly described, (figure) on the second. (Which is) the very thing it was required to show.

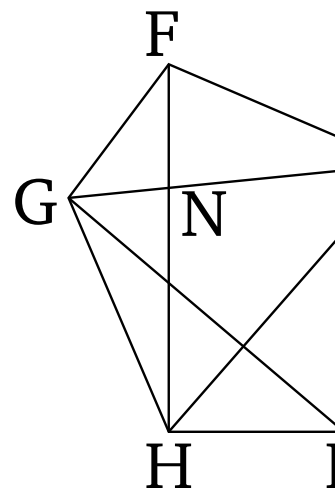
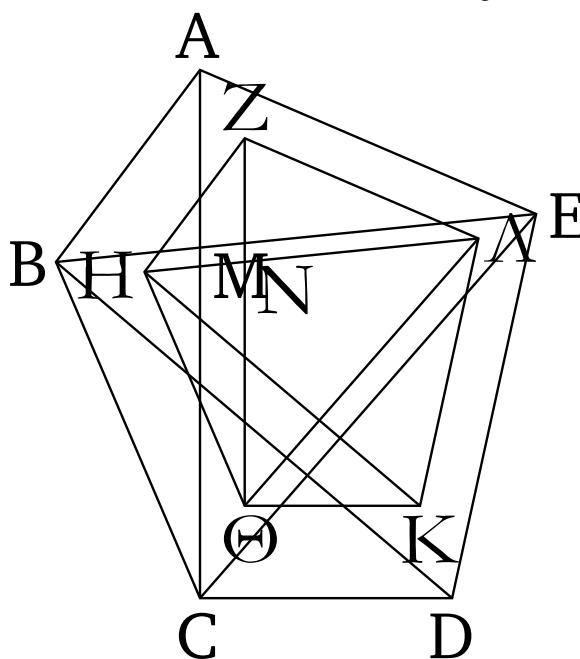
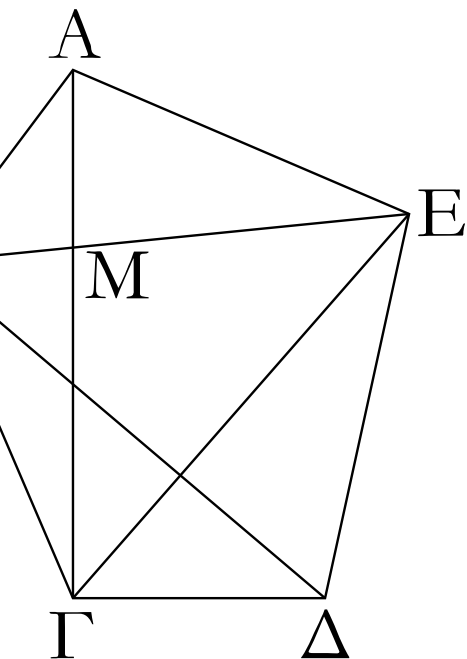
Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι, ἔαν τρεῖς εὐθεῖαι ἀνάλογον >ῶσιν, >έστιν ὥς ἡ πρώτη πρὸς τὴν τρίτην, οὕτως τὸ ἀπὸ τῆς πρώτης >ῖδοξ πρὸς τὸ ἀπὸ τῆς δευτέρας τὸ <όμοιον καὶ ὁμοίως ἀναγραφόμενον. ὁπερ >έδει δεῖχαι.

† Literally, “double”.

20.

Τὰ <όμοια πολύγωνα >ῖξ τε <όμοια τρίγωνα διαιρεῖται καὶ εἰς >ῖσα τὸ πλῆθος καὶ ὁμόλογα τοῖς <όλοις, καὶ τὸ πολύγωνον πρὸς τὸ πολύγωνον διπλασίονα λόγον >έθει >ήπερ ἡ ὁμόλογος πλευρὰ πρὸς τὴν ὁμόλογον πλευράν.
>Ἐστω <όμοια πολύγωνα τὰ ΑΒΓΔΕ, ΖΗΘΚΛ, ὁμόλογος δὲ >έστω ἡ ΑΒ τῇ ΖΗ; λέγω, ὅτι τὰ ΑΒΓΔΕ, ΖΗΘΚΛ πολύγωνα >ῖξ τε <όμοια τρίγωνα διαιρεῖται καὶ εἰς >ῖσα τὸ πλῆθος καὶ ὁμόλογα τοῖς <όλοις, καὶ τὸ ΑΒΓΔΕ πολύγωνον πρὸς τὸ ΖΗΘΚΛ πολύγωνον διπλασίονα λόγον >έθει >ήπερ ἡ ΑΒ πρὸς τὴν ΖΗ.



Ἐπεξεύθθωσαν αἱ ΒΕ, ΕΓ, ΗΛ, ΛΘ.

Καὶ ἐπεὶ <όμοιον ἐστὶ τὸ ΑΒΓΔΕ πολύγωνον τῷ ΖΗΘΚΛ πολυγώνῳ, >ῖση ἐστὶν ἡ ὑπὸ ΒΑΕ γωνία τῇ ὑπὸ ΗΖΛ. καὶ ἐστὶν ὥς ἡ ΒΑ πρὸς ΑΕ, οὕτως ἡ ΗΖ πρὸς ΖΛ. ἐπεὶ οὖν δύο τρίγωνα ἐστὶ τὰ ΑΒΕ, ΖΗΛ μίαν γωνίαν μὴ γωνία >ῖσην >έθοντα, περὶ δὲ τὰς >ῖσας γωνίας τὰς πλευρὰς ἀνάλογον, ἰσογώνιον >άρα ἐστὶ τὸ ΑΒΕ τρίγωνον τῷ ΖΗΛ τριγώνῳ; <ώστε καὶ <όμοιον; >ῖση >άρα ἐστὶν ἡ ὑπὸ ΑΒΕ γωνία τῇ ὑπὸ ΖΗΛ. >έστι δὲ καὶ <όλη ἡ ὑπὸ ΑΒΓ <όλη τῇ ὑπὸ ΖΗΘ >ῖση διὰ τὴν ὁμοιότητα τῶν πολυγώνων; λοιπὴ >άρα ἡ ὑπὸ ΕΒΓ γωνία τῇ ὑπὸ ΛΗΘ ἐστὶν >ῖση. καὶ ἐπεὶ διὰ τὴν ὁμοιότητα τῶν ΑΒΕ, ΖΗΛ τριγώνων ἐστὶν ὥς ἡ ΕΒ πρὸς ΒΑ, οὕτως ἡ ΛΗ πρὸς ΗΖ, ἀλλὰ μὴν καὶ διὰ τὴν ὁμοιότητα τῶν πολυγώνων ἐστὶν ὥς ἡ ΑΒ πρὸς ΒΓ, οὕτως ἡ ΖΗ πρὸς ΗΘ, διὸ >ῖσου >άρα ἐστὶν ὥς ἡ ΕΒ πρὸς ΒΓ, οὕτως ἡ ΛΗ πρὸς ΗΘ, καὶ περὶ τὰς >ῖσας γωνίας τὰς ὑπὸ ΕΒΓ, ΛΗΘ αἱ πλευραὶ ἀνάλογόν εἰσιν; ἰσογώνιον >άρα ἐστὶ

Let BE , EC , GL , and LH be joined.

And since polygon $ABCDE$ is similar to polygon $FGHKL$, angle BAE is equal to angle GFL , and as BA is to AE , so GF (is) to FL [Def. 6.1]. Therefore, since ABE and FGL are two triangles having one angle equal to one angle and the sides about the equal angles proportional, triangle ABE is thus equiangular to triangle FGL [Prop. 6.6]. Hence, (they are) also similar [Prop. 6.4, Def. 6.1]. Thus, angle ABE is equal to (angle) FGL . And the whole (angle) ABC is equal to the whole (angle) FGH , on account of the similarity of the polygons. Thus, the remaining angle EBC is equal to LGH . And since, on account of the similarity of triangles ABE and FGL , as EB is to BA , so LG (is) to GF , but also, on account of the similarity of the polygons, as AB is to BC , so FG (is) to GH , thus, via equality, as EB is to BC , so LG (is) to GH [Prop. 5.22], and the sides about the equal angles, EBC and LGH , are

τὸ ΕΒΓ τρίγωνον τῷ ΛΗΘ τριγώνῳ; <ὥστε καὶ <ὁμοίων ἐστὶ τὸ ΕΒΓ τρίγωνον τῷ ΛΗΘ τριγώνῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ ΕΓΔ τρίγωνον <ὁμοίων ἐστὶ τῷ ΛΘΚ τριγώνῳ. τὰ >άρα <ὁμοία πολύγωνα τὰ ΑΒΓΔΕ, ΖΗΘΚΛ ε>ἶξ τε <ὁμοία τρίγωνα διήλρηται καὶ εἶξ >ἴσα τὸ πλῆθος.

Λέγω, <ὅτι καὶ ὁμόλογα τοῖς <όλοις, τουτέστιν <ὥστε ἀνάλογον ε>ἶναι τὰ τρίγωνα, καὶ ἡγούμενα μὲν ε>ἶναι τὰ ΑΒΕ, ΕΒΓ, ΕΓΔ, ἐπόμενα δὲ αὐτῶν τὰ ΖΗΛ, ΛΗΘ, ΛΘΚ, καὶ <ὅτι τὸ ΑΒΓΔΕ πολύγωνον πρὸς τὸ ΖΗΘΚΛ πολύγωνον διπλασίονα λόγον >έθει >ἤπερ ἡ ὁμόλογος πλευρὰ πρὸς τὴν ὁμόλογον πλευράν, τουτέστιν ἡ ΑΒ πρὸς τὴν ΖΗ.

Ἐπεξεύθθησαν γὰρ αἱ ΑΓ, ΖΘ. καὶ ἐπεὶ διὰ τὴν ὁμοιότητα τῶν πολυγώνων >ἴση ἐστὶν ἡ ὑπὸ ΑΒΓ γωνία τῇ ὑπὸ ΖΗΘ, καὶ ἐστὶν ὥς ἡ ΑΒ πρὸς ΒΓ, ο<ύτωξ ἡ ΖΗ πρὸς ΗΘ, ἰσογώνιον ἐστὶ τὸ ΑΒΓ τρίγωνον τῷ ΖΗΘ τριγώνῳ; >ἴση >άρα ἐστὶν ἡ μὲν ὑπὸ ΒΑΓ γωνία τῇ ὑπὸ ΗΖΘ, ἡ δὲ ὑπὸ ΒΓΑ τῇ ὑπὸ ΗΘΖ. καὶ ἐπεὶ >ἴση ἐστὶν ἡ ὑπὸ ΒΑΜ γωνία τῇ ὑπὸ ΗΖΝ, >έστι δὲ καὶ ἡ ὑπὸ ΑΒΜ τῇ ὑπὸ ΖΗΝ >ἴση, καὶ λοιπὴ >άρα ἡ ὑπὸ ΑΜΒ λοιπὴ τῇ ὑπὸ ΖΝΗ >ἴση ἐστὶν; ἰσογώνιον >άρα ἐστὶ τὸ ΑΒΜ τρίγωνον τῷ ΖΗΝ τριγώνῳ. ὁμοίωξ δὴ δεῖχμεν, <ὅτι καὶ τὸ ΒΜΓ τρίγωνον ἰσογώνιον ἐστὶ τῷ ΗΝΘ τριγώνῳ. ἀνάλογον >άρα ἐστὶν, ὥς μὲν ἡ ΑΜ πρὸς ΜΒ, ο<ύτωξ ἡ ΖΝ πρὸς ΝΗ, ὥς δὲ ἡ ΒΜ πρὸς ΜΓ, ο<ύτωξ ἡ ΗΝ πρὸς ΝΘ; <ὥστε καὶ δι> >ἴσου, ὥς ἡ ΑΜ πρὸς ΜΓ, ο<ύτωξ ἡ ΖΝ πρὸς ΝΘ. ἀλλ> ὥς ἡ ΑΜ πρὸς ΜΓ, ο<ύτωξ τὸ ΑΒΜ [τρίγωνον] πρὸς τὸ ΜΒΓ, καὶ τὸ ΑΜΕ πρὸς τὸ ΕΜΓ; πρὸς >άλληλα γὰρ εἰσιν ὥς αἱ βάσεις. καὶ ὥς >άρα <έν τῶν ἡγουμένων πρὸς <έν τῶν ἐπόμενων, ο<ύτωξ <άπαντα τὰ ἡγούμενα πρὸς <άπαντα τὰ ἐπόμενα; ὥς >άρα τὸ ΑΜΒ τρίγωνον πρὸς τὸ ΒΜΓ, ο<ύτωξ τὸ ΑΒΕ πρὸς τὸ ΓΒΕ. ἀλλ> ὥς τὸ ΑΜΒ πρὸς τὸ ΒΜΓ, ο<ύτωξ ἡ ΑΜ πρὸς ΜΓ; καὶ ὥς >άρα ἡ ΑΜ πρὸς ΜΓ, ο<ύτωξ τὸ ΑΒΕ τρίγωνον πρὸς τὸ ΕΒΓ τρίγωνον. διὰ τὰ αὐτὰ δὴ καὶ ὥς ἡ ΖΝ πρὸς ΝΘ, ο<ύτωξ τὸ ΖΗΛ τρίγωνον πρὸς τὸ ΗΛΘ τρίγωνον. καὶ ἐστὶν ὥς ἡ ΑΜ πρὸς ΜΓ, ο<ύτωξ ἡ ΖΝ πρὸς ΝΘ; καὶ ὥς >άρα τὸ ΑΒΕ τρίγωνον πρὸς τὸ ΒΕΓ τρίγωνον, ο<ύτωξ τὸ ΖΗΛ τρίγωνον πρὸς τὸ ΗΛΘ τρίγωνον, καὶ ἐναλλάξ ὥς τὸ ΑΒΕ τρίγωνον πρὸς τὸ ΖΗΛ τρίγωνον, ο<ύτωξ τὸ ΒΕΓ τρίγωνον πρὸς τὸ ΗΛΘ τρίγωνον. ὁμοίωξ δὴ δεῖχμεν ἐπιευθθεῖσάν τῶν ΒΔ, ΗΚ, <ὅτι καὶ ὥς τὸ ΒΕΓ τρίγωνον πρὸς τὸ ΛΗΘ τρίγωνον, ο<ύτωξ τὸ ΕΓΔ τρίγωνον πρὸς τὸ ΛΘΚ τρίγωνον. καὶ ἐπεὶ ἐστὶν ὥς τὸ ΑΒΕ τρίγωνον πρὸς τὸ ΖΗΛ τρίγωνον, ο<ύτωξ τὸ ΕΒΓ πρὸς τὸ ΛΗΘ, καὶ >έτι τὸ ΕΓΔ πρὸς τὸ ΛΘΚ, καὶ ὥς >άρα <έν τῶν ἡγουμένων πρὸς <έν τῶν ἐπομένων, ο<ύτωξ <άπαντα τὰ ἡγούμενα πρὸς <άπαντα τὰ ἐπόμενα; >έστιν >άρα ὥς τὸ ΑΒΕ τρίγωνον πρὸς τὸ ΖΗΛ τρίγωνον, ο<ύτωξ τὸ ΑΒΓΔΕ πολύγωνον πρὸς τὸ ΖΗΘΚΛ πολύγωνον. ἀλλὰ τὸ ΑΒΕ τρίγωνον πρὸς τὸ ΖΗΛ τρίγωνον διπλασίονα λόγον >έθει >ἤπερ ἡ ΑΒ ὁμόλογος πλευρὰ πρὸς τὴν ΖΗ ὁμόλογον πλευράν; τὰ γὰρ <ὁμοία τρίγωνα ἐν διπλασίονι λόγῳ ἐστὶ τῶν ὁμολόγων πλευρῶν. καὶ τὸ ΑΒΓΔΕ >άρα πολύγωνον πρὸς τὸ ΖΗΘΚΛ πολύγωνον διπλασίονα λόγον >έθει >ἤπερ ἡ ΑΒ ὁμόλογος πλευρὰ πρὸς τὴν ΖΗ ὁμόλογον πλευράν. Τὰ >άρα <ὁμοία πολύγωνα ε>ἶξ τε <ὁμοία τρίγωνα διαιρεῖται καὶ εἶξ >ἴσα τὸ πλῆθος καὶ ὁμόλογα τοῖς <όλοις, καὶ τὸ πολύγωνον πρὸς τὸ πολύγωνον διπλασίονα λόγον >έθει >ἤπερ ἡ ὁμόλογος πλευρὰ πρὸς τὴν ὁμόλογον πλευράν [<ὅπερ >έδει δεῖχαι].

proportional. Thus, triangle EBC is equiangular to triangle LGH [Prop. 6.6]. Hence, triangle EBC is also similar to triangle LGH [Prop. 6.4, Def. 6.1]. So, for the same (reasons), triangle ECD is also similar to triangle LHK . Thus, the similar polygons $ABCDE$ and $FGHKL$ have been divided into equal numbers of similar triangles.

I also say that (the triangles) correspond (in proportion) to the wholes. That is to say, the triangles are proportional: ABE , EBC , and ECD are the leading (magnitudes), and their (associated) following (magnitudes are) FGL , LGH , and LHK (respectively). (I) also (say) that polygon $ABCDE$ has a squared ratio to polygon $FGHKL$ with respect to (that) a corresponding side (has) to a corresponding side—that is to say, (side) AB to FG .

For let AC and FH be joined. And since angle ABC is equal to FGH , and as AB is to BC , so FG (is) to GH , on account of the similarity of the polygons, triangle ABC is equiangular to triangle FGH [Prop. 6.6]. Thus, angle BAC is equal to GFH , and (angle) BCA to GHF . And since angle BAM is equal to GFN , and (angle) ABM is also equal to FGN (see earlier), the remaining (angle) AMB is thus also equal to the remaining (angle) FNG [Prop. 1.32]. Thus, triangle ABM is equiangular to triangle FGN . So, similarly, we can show that triangle BMC is also equiangular to triangle GNH . Thus, proportionally, as AM is to MB , so FN (is) to NG , and as BM (is) to MC , so GN (is) to NH [Prop. 6.4]. Hence, also, via equality, as AM (is) to MC , so FN (is) to NH [Prop. 5.22]. But, as AM (is) to MC , so [triangle] ABM is to MBC , and AME to EMC . For they are to one another as their bases [Prop. 6.1]. And as one of the leading (magnitudes) is to one of the following (magnitudes), so (the sum of) all the leading (magnitudes) is to (the sum of) all the following (magnitudes) [Prop. 5.12]. Thus, as triangle AMB (is) to BMC , so (triangle) ABE (is) to CBE . But, as (triangle) AMB (is) to BMC , so AM (is) to MC . Thus, also, as AM (is) to MC , so triangle ABE (is) to triangle EBC . And so, for the same (reasons), as FN (is) to NH , so triangle FGL (is) to triangle GLH . And as AM is to MC , so FN (is) to NH . Thus, also, as triangle ABE (is) to triangle BEC , so triangle FGL (is) to triangle GLH , and, alternately, as triangle ABE (is) to triangle FGL , so triangle BEC (is) to triangle GLH [Prop. 5.16]. So, similarly, we can also show, by joining BD and GK , (that) as triangle BEC (is) to triangle LGH , so triangle ECD (is) to triangle LHK . And since as triangle ABE is to triangle FGL , so (triangle) EBC (is) to LGH , and, further, (triangle) ECD to LHK , and also as one of the leading (magnitudes) is to one of the following, so (the sum of) all the leading (magnitudes) is to (the sum of) all the following [Prop. 5.12], thus as triangle ABE is to triangle FGL , so polygon $ABCDE$ (is) to polygon $FGHKL$. But, triangle ABE has a squared ratio to triangle FGL with respect to (that) the corresponding side AB (has) to the corresponding side FG . For, similar triangles are in the squared ratio of corresponding sides [Prop. 6.14]. Thus, polygon $ABCDE$ also has a squared ratio to polygon $FGHKL$ with respect to (that) the corresponding side AB (has) to the corresponding side FG .

Thus, similar polygons can be divided into equal numbers of similar triangles corresponding (in proportion) to the wholes, and one polygon has to the (other) polygon a

squared ratio with respect to (that) a corresponding side (has) to a corresponding side. [(Which is) the very thing it was required to show].

Πόρισμα.

Ὡσαύτως δὲ καὶ ἐπὶ τῶν [ὁμοίων] τετραπλεύρων δειθθή-
σεται, <ὅτι ἐν διπλασίονι λόγῳ εἰσὶ τῶν ὁμολόγων πλευρῶν.
ἐδείχθη δὲ καὶ ἐπὶ τῶν τριγώνων; <ὥστε καὶ καθόλου τὰ
<ὅμοια εὐθύγραμμα σήματα πρὸς >ἀλλήλα ἐν διπλασίονι
λόγῳ εἰσὶ τῶν ὁμολόγων πλευρῶν. <ὅπερ >ἐδει δεῖχαι.

Corollary

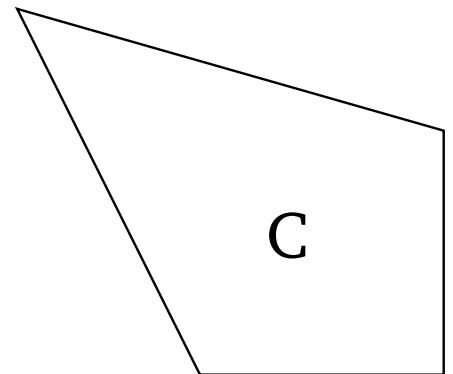
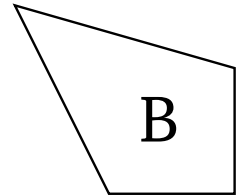
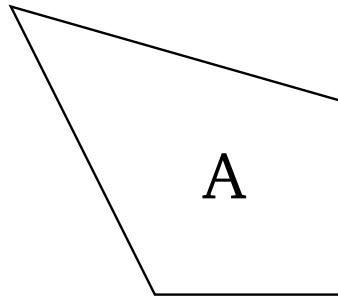
And, in the same manner, it can also be shown for [similar] quadrilaterals, (that) they are in the squared ratio of (their) corresponding sides. And it was also shown for triangles. Hence, in general, similar rectilinear figures are also to one another in the squared ratio of (their) corresponding sides. (Which is) the very thing it was required to show.

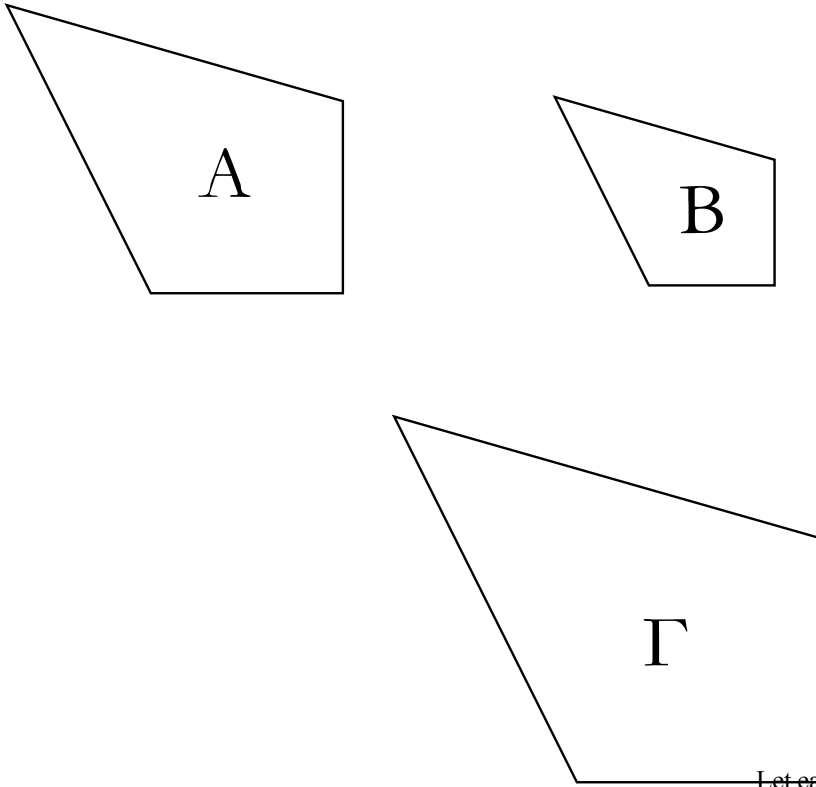
21.

Τὰ τῶ| αὐτῶ| εὐθυγράμμω <ὅμοια καὶ ἀλλήλοις ἐστὶν
<ὅμοια.

Proposition 21

(Rectilinear figures) similar to the same rectilinear figure are also similar to one another.





>Ἐστω γὰρ ἑκάτερον τῶν A, B εὐθύγραμμων τῶν Γ <όμοιον;
λέγω, <ὅτι καὶ τὸ A τῶν B ἐστὶν <όμοιον.
>Ἐπεὶ γὰρ <όμοιόν ἐστι τὸ A τῶν Γ , ἰσογώνιον τέ ἐστὶν αὐτῶν
καὶ τὰς περὶ τὰς >ἰσαξ γωνίας πλευρὰς ἀνάλογον >έθει.
πάλιν, ἐπεὶ <όμοιόν ἐστι τὸ B τῶν Γ , ἰσογώνιον τέ ἐστὶν
αὐτῶν καὶ τὰς περὶ τὰς >ἰσαξ γωνίας πλευρὰς ἀνάλογον
>έθει. ἑκάτερον >άρα τῶν A, B τῶν Γ ἰσογώνιον τέ ἐστὶ καὶ
τὰς περὶ τὰς >ἰσαξ γωνίας πλευρὰς ἀνάλογον >έθει [<ὥστε
καὶ τὸ A τῶν B ἰσογώνιον τέ ἐστὶ καὶ τὰς περὶ τὰς >ἰσαξ
γωνίας πλευρὰς ἀνάλογον >έθει]. <όμοιον >άρα ἐστὶ τὸ A
τῶν B ; <ὅπερ >έδει δεῖξαι.

22.

>Ἐὰν τέσσαρες εὐθεῖαι ἀνάλογον >ῶσιν, καὶ τὰ ἀπ' αὐτῶν
εὐθύγραμμα <ομοία τε καὶ ὁμοίως ἀναγεγραμμένα ἀνάλογον
>έσται; <ὰν τὰ ἀπ' αὐτῶν εὐθύγραμμα <ομοία τε καὶ
ὁμοίως ἀναγεγραμμένα ἀνάλογον >ῇ, καὶ αὐταὶ αἱ εὐθεῖαι
ἀνάλογον >έσονται.

>Ἐστῶσαν τέσσαρες εὐθεῖαι ἀνάλογον αἱ $AB, \Gamma\Delta, EZ, H\Theta$,
ὥς ἡ AB πρὸς τὴν $\Gamma\Delta$, ο<ύτως ἡ EZ πρὸς τὴν $H\Theta$, καὶ
ἀναγεγράφθωσαν ἀπὸ μὲν τῶν $AB, \Gamma\Delta$ <ομοία τε καὶ ὁμοίως
κείμενα εὐθύγραμμα τὰ $KAB, \Lambda\Gamma\Delta$, ἀπὸ δὲ τῶν $EZ, H\Theta$
<ομοία τε καὶ ὁμοίως κείμενα εὐθύγραμμα τὰ $MZ, N\Theta$;
λέγω, <ὅτι ἐστὶν ὡς τὸ KAB πρὸς τὸ $\Lambda\Gamma\Delta$, ο<ύτως τὸ MZ
πρὸς τὸ $N\Theta$.

Εἰλήφθω γὰρ τῶν μὲν $AB, \Gamma\Delta$ τρίτη ἀνάλογον ἡ X , τῶν δὲ
 $EZ, H\Theta$ τρίτη ἀνάλογον ἡ O . καὶ ἐπεὶ ἐστὶν ὡς μὲν ἡ AB
πρὸς τὴν $\Gamma\Delta$, ο<ύτως ἡ EZ πρὸς τὴν $H\Theta$, ὥς δὲ ἡ $\Gamma\Delta$ πρὸς
τὴν X , ο<ύτως ἡ $H\Theta$ πρὸς τὴν O , δι> >ἰσου >άρα ἐστὶν ὡς
ἡ AB πρὸς τὴν X , ο<ύτως ἡ EZ πρὸς τὴν O . ἀλλ' ὡς μὲν ἡ
 AB πρὸς τὴν X , ο<ύτως [καὶ] τὸ KAB πρὸς τὸ $\Lambda\Gamma\Delta$, ὥς δὲ
ἡ EZ πρὸς τὴν O , ο<ύτως τὸ MZ πρὸς τὸ $N\Theta$; καὶ ὡς >άρα
τὸ KAB πρὸς τὸ $\Lambda\Gamma\Delta$, ο<ύτως τὸ MZ πρὸς τὸ $N\Theta$.

Ἀλλὰ δὴ >έστω ὡς τὸ KAB πρὸς τὸ $\Lambda\Gamma\Delta$, ο<ύτως τὸ MZ
πρὸς τὸ $N\Theta$; λέγω, <ὅτι ἐστὶ καὶ ὡς ἡ AB πρὸς τὴν $\Gamma\Delta$,

Let each of the rectilinear figures A and B be similar to (the
rectilinear figure) C . I say that A is also similar to B .

For since A is similar to C , (A) is equiangular to (C), and
has the sides about the equal angles proportional [Def. 6.1].
Again, since B is similar to C , (B) is equiangular to (C), and
has the sides about the equal angles proportional [Def. 6.1].
Thus, A and B are each equiangular to C , and have the sides
about the equal angles proportional [hence, A is also equian-
gular to B , and has the sides about the equal angles propor-
tional]. Thus, A is similar to B [Def. 6.1]. (Which is) the
very thing it was required to show.

Proposition 22

If four straight-lines are proportional, (then) similar, and
similarly described, rectilinear figures (drawn) on them will
also be proportional. And if similar, and similarly described,
rectilinear figures (drawn) on them are proportional, (then)
the straight-lines themselves will also be proportional.

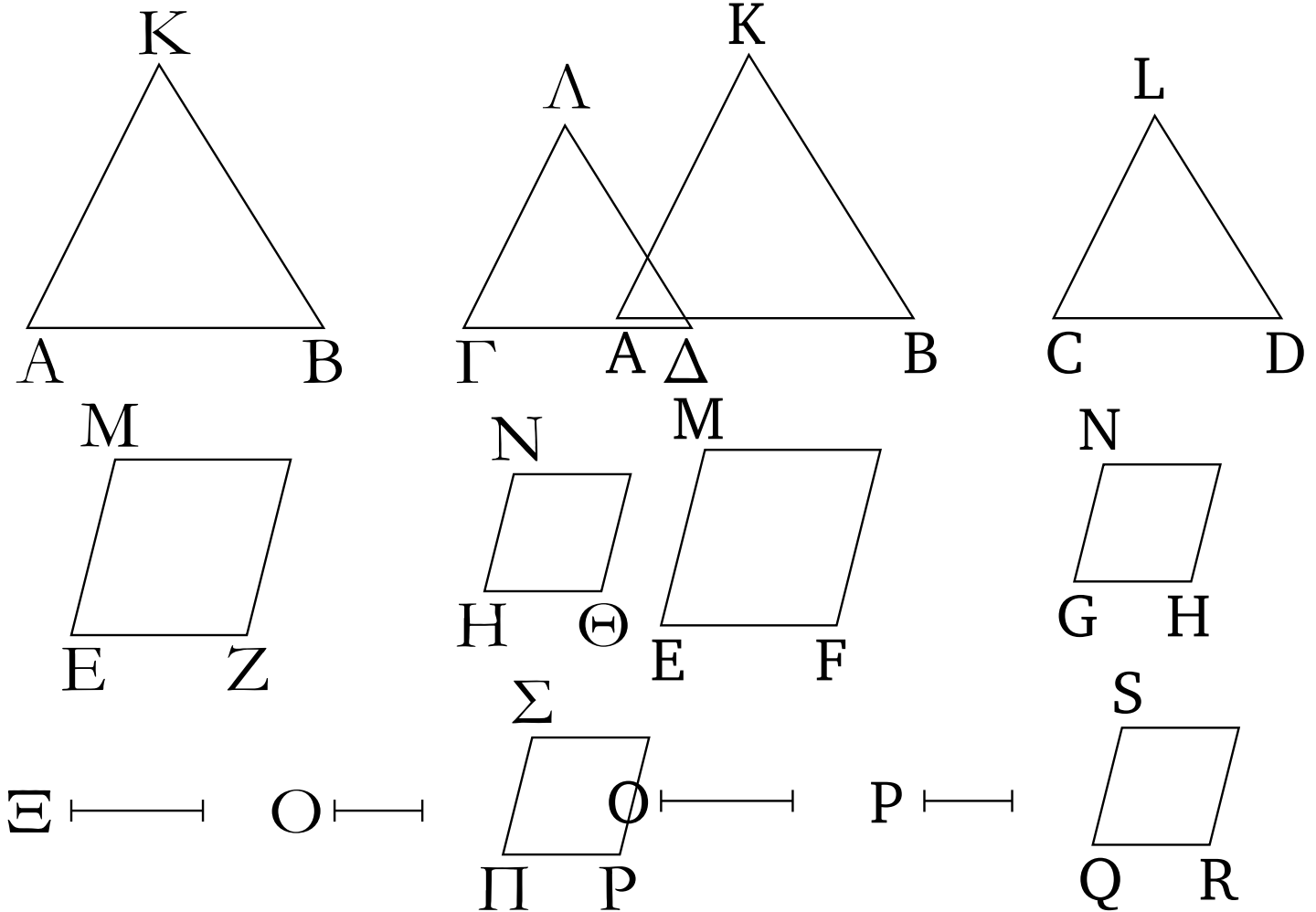
Let AB, CD, EF , and GH be four proportional straight-
lines, (such that) as AB (is) to CD , so EF (is) to GH .
And let the similar, and similarly laid out, rectilinear fig-
ures KAB and LCD be described on AB and CD (respec-
tively), and the similar, and similarly laid out, rectilinear
figures MF and NH on EF and GH (respectively). I say
that as KAB is to LCD , so MF (is) to NH .

For let a third (straight-line) O be taken (which is) propor-
tional to AB and CD , and a third (straight-line) P propor-
tional to EF and GH [Prop. 6.11]. And since as AB is
to CD , so EF (is) to GH , and as CD (is) to O , so GH
(is) to P , thus, via equality, as AB is to O , so EF (is)
to P [Prop. 5.22]. But, as AB (is) to O , so [also] KAB
(is) to LCD , and as EF (is) to P , so MF (is) to NH
[Prop. 5.19 corr.]. And, thus, as KAB (is) to LCD , so MF
(is) to NH .

And so let KAB be to LCD , as MF (is) to NH . I say also

οὐτως ἢ EZ πρὸς τὴν $H\Theta$. εἰ γὰρ μὴ ἔστιν, ὥς ἢ AB πρὸς τὴν $\Gamma\Delta$, οὐτως ἢ EZ πρὸς τὴν $H\Theta$, >έστω ὥς ἢ AB πρὸς τὴν $\Gamma\Delta$, οὐτως ἢ EZ πρὸς τὴν $\Pi\Gamma$, καὶ ἀναγεγράφθω ἀπὸ τῆς $\Pi\Gamma$ ὁποτέρῳ τῶν MZ , $N\Theta$ <όμοιόν τε καὶ ὁμοίως κείμενον εὐθύγραμμον τὸ $\Sigma\Gamma$.

that as AB is to CD , so EF (is) to GH . For if as AB is to CD , so EF (is) not to GH , let AB be to CD , as EF (is) to QR [Prop. 6.12]. And let the rectilinear figure SR , similar, and similarly laid down, to either of MF or NH , be described on QR [Props. 6.18, 6.21].



Ἐπεὶ οὖν ἔστιν ὥς ἢ AB πρὸς τὴν $\Gamma\Delta$, οὐτως ἢ EZ πρὸς τὴν $\Pi\Gamma$, καὶ ἀναγράφεται ἀπὸ μὲν τῶν AB , $\Gamma\Delta$ <όμοιά τε καὶ ὁμοίως κείμενα τὰ KAB , $\Lambda\Gamma\Delta$, ἀπὸ δὲ τῶν EZ , $\Pi\Gamma$ <όμοιά τε καὶ ὁμοίως κείμενα τὰ MZ , $\Sigma\Gamma$, >έστιν >άρα ὥς τὸ KAB πρὸς τὸ $\Lambda\Gamma\Delta$, οὐτως τὸ MZ πρὸς τὸ $\Sigma\Gamma$. ὑπόκειται δὲ καὶ ὥς τὸ KAB πρὸς τὸ $\Lambda\Gamma\Delta$, οὐτως τὸ MZ πρὸς τὸ $N\Theta$; καὶ ὥς >άρα τὸ MZ πρὸς τὸ $\Sigma\Gamma$, οὐτως τὸ MZ πρὸς τὸ $N\Theta$. τὸ MZ >άρα πρὸς ἑκάτερον τῶν $N\Theta$, $\Sigma\Gamma$ τὸν αὐτὸν >έθει λόγον; >ἴσον >άρα ἔστι τὸ $N\Theta$ τῷ $\Sigma\Gamma$. >έστι δὲ αὐτῶν καὶ <όμοιον καὶ ὁμοίως κείμενον; >ἴση >άρα ἢ $H\Theta$ τῇ $\Pi\Gamma$. καὶ ἐπεὶ ἔστιν ὥς ἢ AB πρὸς τὴν $\Gamma\Delta$, οὐτως ἢ EZ πρὸς τὴν $\Pi\Gamma$, >ἴση δὲ ἢ $\Pi\Gamma$ τῇ $H\Theta$, >έστιν >άρα ὥς ἢ AB πρὸς τὴν $\Gamma\Delta$, οὐτως ἢ EZ πρὸς τὴν $H\Theta$.

Ἐὰν >άρα τέσσαρες εὐθεῖαι ἀνάλογον >ᾶσιν, καὶ τὰ ἀπ' αὐτῶν εὐθύγραμμα <όμοιά τε καὶ ὁμοίως ἀναγεγραμμένα ἀνάλογον >έσται; <ὰν τὰ ἀπ' αὐτῶν εὐθύγραμμα <όμοιά τε καὶ ὁμοίως ἀναγεγραμμένα ἀνάλογον >ῇ, καὶ αὐτὰ αἰ εὐθεῖαι ἀνάλογον >έσονται; <όπερ >έδει δεῖξαι.

Therefore, since as AB is to CD , so EF (is) to QR , and the similar, and similarly laid out, (rectilinear figures) KAB and LCD have been described on AB and CD (respectively), and the similar, and similarly laid out, (rectilinear figures) MF and SR on EF and QR (respectively), thus as KAB is to LCD , so MF (is) to SR (see above). And it was also assumed that as KAB (is) to LCD , so MF (is) to NH . Thus, also, as MF (is) to SR , so MF (is) to NH [Prop. 5.11]. Thus, MF has the same ratio to each of NH and SR . Thus, NH is equal to SR [Prop. 5.9]. And it is also similar, and similarly laid out, to it. Thus, GH (is) equal to QR .[†] And since AB is to CD , as EF (is) to QR , and QR (is) equal to GH , thus as AB is to CD , so EF (is) to GH . Thus, if four straight-lines are proportional, (then) similar, and similarly described, rectilinear figures (drawn) on them will also be proportional. And if similar, and similarly described, rectilinear figures (drawn) on them are proportional, (then) the straight-lines themselves will also be proportional. (Which is) the very thing it was required to show.

[†] Here, Euclid assumes, without proof, that if two similar figures are equal then any pair of corresponding sides is also equal.

23.

Proposition 23

Τὰ ἰσογώνια παραλληλόγραμμα πρὸς >άλληλα λόγον >έθει

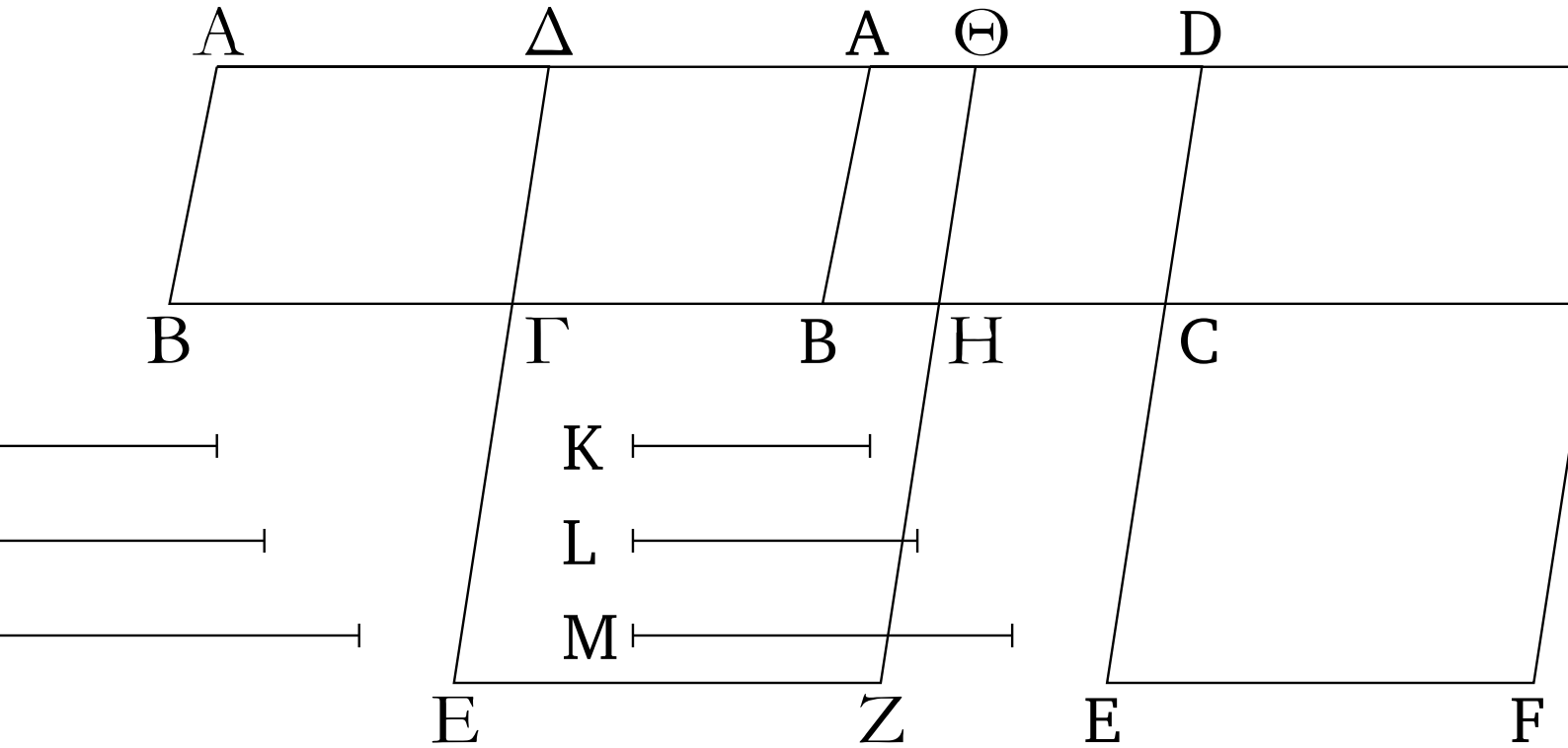
Equiangular parallelograms have to one another the ratio

τὸν συγκείμενον ἐκ τῶν πλευρῶν.

>Ἐστω ἰσογώνια παραλληλόγραμμα τὰ $ΑΓ$, $ΓΖ$ >ίσην >έθοντα τὴν ὑπὸ $ΒΓΔ$ γωνίαν τῇ ὑπὸ $ΕΓΗ$; λέγω, <ὅτι τὸ $ΑΓ$ παραλληλόγραμμον πρὸς τὸ $ΓΖ$ παραλληλόγραμμον λόγον >έθει τὸν συγκείμενον ἐκ τῶν πλευρῶν.

compounded[†] out of (the ratios of) their sides.

Let AC and CF be equiangular parallelograms having angle BCD equal to ECG . I say that parallelogram AC has to parallelogram CF the ratio compounded out of (the ratios of) their sides.



Κείσθω γὰρ <ὥστε ἐπ> εὐθείᾳ ε>ῖναι τὴν $ΒΓ$ τῇ $ΓΗ$; ἐπ> εὐθείᾳ >άρα ἐστὶ καὶ ἡ $ΔΓ$ τῇ $ΓΕ$. καὶ συμπεπληρώσθω τὸ $ΔΗ$ παραλληλόγραμμον, καὶ ἐκκείσθω τῆς εὐθείας ἡ $Κ$, καὶ γεγονέτω ὥς μὲν ἡ $ΒΓ$ πρὸς τὴν $ΓΗ$, ο<ύτωξ ἡ $Κ$ πρὸς τὴν $Λ$, ὥς δὲ ἡ $ΔΓ$ πρὸς τὴν $ΓΕ$, ο<ύτωξ ἡ $Λ$ πρὸς τὴν $Μ$.

Οἱ >άρα λόγοι τῆς τε $Κ$ πρὸς τὴν $Λ$ καὶ τῆς $Λ$ πρὸς τὴν $Μ$ οἱ αὐτοὶ εἰσι τοῖς λόγοις τῶν πλευρῶν, τῆς τε $ΒΓ$ πρὸς τὴν $ΓΗ$ καὶ τῆς $ΔΓ$ πρὸς τὴν $ΓΕ$. ἀλλ> ὁ τῆς $Κ$ πρὸς $Μ$ λόγος σύγκειται >ἐκ τε τοῦ τῆς $Κ$ πρὸς $Λ$ λόγου καὶ τοῦ τῆς $Λ$ πρὸς $Μ$; <ὥστε καὶ ἡ $Κ$ πρὸς τὴν $Μ$ λόγον >έθει τὸν συγκείμενον ἐκ τῶν πλευρῶν. καὶ ἐπεὶ ἐστὶν ὥς ἡ $ΒΓ$ πρὸς τὴν $ΓΗ$, ο<ύτωξ τὸ $ΑΓ$ παραλληλόγραμμον πρὸς τὸ $ΓΘ$, ἀλλ> ὥς ἡ $ΒΓ$ πρὸς τὴν $ΓΗ$, ο<ύτωξ ἡ $Κ$ πρὸς τὴν $Λ$, καὶ ὥς >άρα ἡ $Κ$ πρὸς τὴν $Λ$, ο<ύτωξ τὸ $ΑΓ$ πρὸς τὸ $ΓΘ$. πάλιν, ἐπεὶ ἐστὶν ὥς ἡ $ΔΓ$ πρὸς τὴν $ΓΕ$, ο<ύτωξ τὸ $ΓΘ$ παραλληλόγραμμον πρὸς τὸ $ΓΖ$, ἀλλ> ὥς ἡ $ΔΓ$ πρὸς τὴν $ΓΕ$, ο<ύτωξ ἡ $Λ$ πρὸς τὴν $Μ$, καὶ ὥς >άρα ἡ $Λ$ πρὸς τὴν $Μ$, ο<ύτωξ τὸ $ΓΘ$ παραλληλόγραμμον πρὸς τὸ $ΓΖ$ παραλληλόγραμμον. ἐπεὶ ο>ὖν ἐδείχθη, ὥς μὲν ἡ $Κ$ πρὸς τὴν $Λ$, ο<ύτωξ τὸ $ΑΓ$ παραλληλόγραμμον πρὸς τὸ $ΓΘ$ παραλληλόγραμμον, ὥς δὲ ἡ $Λ$ πρὸς τὴν $Μ$, ο<ύτωξ τὸ $ΓΘ$ παραλληλόγραμμον πρὸς τὸ $ΓΖ$ παραλληλόγραμμον, δι> >ίσου >άρα ἐστὶν ὥς ἡ $Κ$ πρὸς τὴν $Μ$, ο<ύτωξ τὸ $ΑΓ$ πρὸς τὸ $ΓΖ$ παραλληλόγραμμον. ἡ δὲ $Κ$ πρὸς τὴν $Μ$ λόγον >έθει τὸν συγκείμενον ἐκ τῶν πλευρῶν; καὶ τὸ $ΑΓ$ >άρα πρὸς τὸ $ΓΖ$ λόγον >έθει τὸν συγκείμενον ἐκ τῶν πλευρῶν.

Τὰ >άρα ἰσογώνια παραλληλόγραμμα πρὸς >άλληλα λόγον >έθει τὸν συγκείμενον ἐκ τῶν πλευρῶν; <ὅπερ >έδει δεῖξαι.

For let BC be laid down so as to be straight-on to CG . Thus, DC is also straight-on to CE [Prop. 1.14]. And let the parallelogram DG be completed. And let some straight-line K be laid down. And let it be contrived that as BC (is) to CG , so K (is) to L , and as DC (is) to CE , so L (is) to M [Prop. 6.12].

Thus, the ratios of K to L and of L to M are the same as the ratios of the sides, (namely), BC to CG and DC to CE (respectively). But, the ratio of K to M is compounded out of the ratio of K to L and (the ratio) of L to M . Hence, K also has to M the ratio compounded out of (the ratios of) the sides (of the parallelograms). And since as BC is to CG , so parallelogram AC (is) to CH [Prop. 6.1], but as BC (is) to CG , so K (is) to L , thus, also, as K (is) to L , so (parallelogram) AC (is) to CH . Again, since as DC (is) to CE , so parallelogram CH (is) to CF [Prop. 6.1], but as DC (is) to CE , so L (is) to M , thus, also, as L (is) to M , so parallelogram CH (is) to parallelogram CF . Therefore, since it was shown that as K (is) to L , so parallelogram AC (is) to parallelogram CH , and as L (is) to M , so parallelogram CH (is) to parallelogram CF , thus, via equality, as K is to M , so (parallelogram) AC (is) to parallelogram CF [Prop. 5.22]. And K has to M the ratio compounded out of (the ratios of) the sides (of the parallelograms). Thus, (parallelogram) AC also has to (parallelogram) CF the ratio compounded out of (the ratio of) their sides.

Thus, equiangular parallelograms have to one another the ratio compounded out of (the ratio of) their sides. (Which is) the very thing it was required to show.

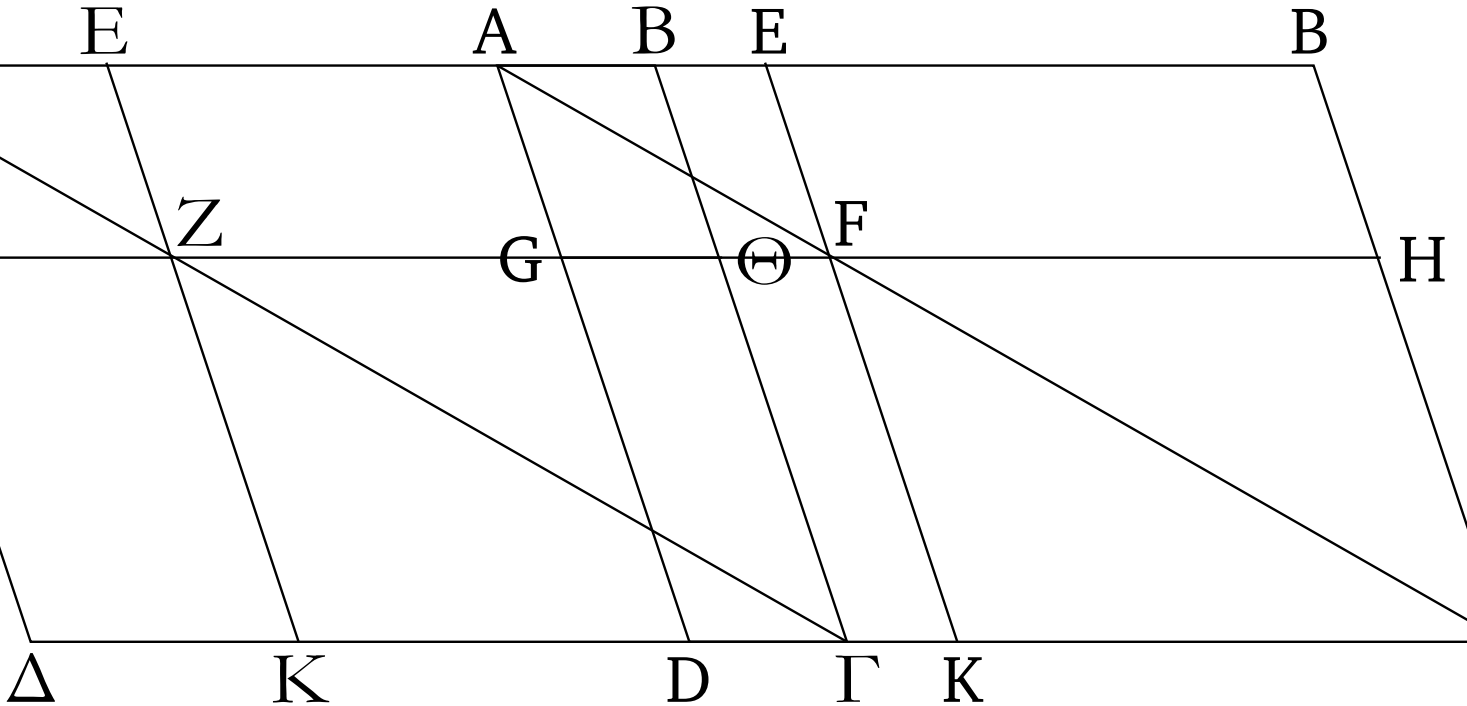
[†] In modern terminology, if two ratios are “compounded” then they are multiplied together.

24.

Proposition 24

Παντὸς παραλληλογράμμου τὰ περι τὴν διάμετρον παραλληλόγραμμοι
 <όμοιά ἐστι τῷ τε <ὅλῳ καὶ ἀλλήλοις.
 >Ἐστω παραλληλόγραμμος τὸ ΑΒΓΔ, διάμετρος δὲ αὐτοῦ
 ἡ ΑΓ, περι δὲ τὴν ΑΓ παραλληλόγραμμα >ἔστω τὰ ΕΗ,
 ΘΚ; λέγω, <ὅτι ἐκάτερον τῶν ΕΗ, ΘΚ παραλληλογράμμων
 <όμοιόν ἐστι <ὅλῳ τῷ ΑΒΓΔ καὶ ἀλλήλοις.

Any parallelogram, the parallelograms about the diagonal
 are similar to the whole, and to one another.
 Let $ABCD$ be a parallelogram, and AC its diagonal. And
 let EG and HK be parallelograms about AC . I say that the
 parallelograms EG and HK are each similar to the whole
 (parallelogram) $ABCD$, and to one another.



Ἐπεὶ γὰρ τριγώνου τοῦ ΑΒΓ παρὰ μίαν τῶν πλευρῶν τὴν
 ΒΓ >ῆται ἡ ΕΖ, ἀνάλογόν ἐστιν ὡς ἡ ΒΕ πρὸς τὴν ΕΑ,
 ο<ύτωξ ἡ ΓΖ πρὸς τὴν ΖΑ. πάλιν, ἐπεὶ τριγώνου τοῦ ΑΓΔ
 παρὰ μίαν τὴν ΓΔ >ῆται ἡ ΖΗ, ἀνάλογόν ἐστιν ὡς ἡ ΓΖ
 πρὸς τὴν ΖΑ, ο<ύτωξ ἡ ΔΗ πρὸς τὴν ΗΑ. ἀλλ> ὡς ἡ ΓΖ
 πρὸς τὴν ΖΑ, ο<ύτωξ ἡ ΔΗ πρὸς τὴν ΗΑ, καὶ
 ὡς >ἀρα ἡ ΒΕ πρὸς τὴν ΕΑ, ο<ύτωξ ἡ ΔΗ πρὸς τὴν ΗΑ, καὶ
 συνθέντι >ἀρα ὡς ἡ ΒΑ πρὸς ΑΕ, ο<ύτωξ ἡ ΔΑ πρὸς ΑΗ,
 καὶ ἐναλλάξ ὡς ἡ ΒΑ πρὸς τὴν ΑΔ, ο<ύτωξ ἡ ΕΑ πρὸς τὴν
 ΑΗ. τῶν >ἀρα ΑΒΓΔ, ΕΗ παραλληλογράμμων ἀνάλογόν
 εἰσιν αἱ πλευραὶ αἱ περι τὴν κοινὴν γωνίαν τὴν ὑπὸ ΒΑΔ.
 καὶ ἐπεὶ παράλληλός ἐστιν ἡ ΗΖ τῇ ΔΓ, >ῖση ἐστὶν ἡ μὲν
 ὑπὸ ΑΖΗ γωνία τῇ ὑπὸ ΔΓΑ; καὶ κοινὴ τῶν δύο τριγώνων
 τῶν ΑΔΓ, ΑΗΖ ἡ ὑπὸ ΔΑΓ γωνία; ἰσογώνιον >ἀρα ἐστὶ τὸ
 ΑΔΓ τρίγωνον τῷ ΑΗΖ τριγώνῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ
 ΑΓΒ τρίγωνον ἰσογώνιον ἐστὶ τῷ ΑΖΕ τριγώνῳ, καὶ <όλον
 τὸ ΑΒΓΔ παραλληλόγραμμον τῷ ΕΗ παραλληλογράμμῳ
 ἰσογώνιον ἐστὶν. ἀνάλογον >ἀρα ἐστὶν ὡς ἡ ΑΔ πρὸς τὴν
 ΔΓ, ο<ύτωξ ἡ ΑΗ πρὸς τὴν ΗΖ, ὡς δὲ ἡ ΔΓ πρὸς τὴν ΓΑ,
 ο<ύτωξ ἡ ΗΖ πρὸς τὴν ΖΑ, ὡς δὲ ἡ ΑΓ πρὸς τὴν ΓΒ,
 ο<ύτωξ ἡ ΑΖ πρὸς τὴν ΖΕ, καὶ >έτι ὡς ἡ ΓΒ πρὸς τὴν
 ΒΑ, ο<ύτωξ ἡ ΖΕ πρὸς τὴν ΕΑ. καὶ ἐπεὶ ἐδείθθη ὡς μὲν ἡ
 ΔΓ πρὸς τὴν ΓΑ, ο<ύτωξ ἡ ΗΖ πρὸς τὴν ΖΑ, ὡς δὲ ἡ ΑΓ
 πρὸς τὴν ΓΒ, ο<ύτωξ ἡ ΑΖ πρὸς τὴν ΖΕ, δι> >ῖσον >ἀρα
 ἐστὶν ὡς ἡ ΔΓ πρὸς τὴν ΓΒ, ο<ύτωξ ἡ ΗΖ πρὸς τὴν ΖΕ.
 τῶν >ἀρα ΑΒΓΔ, ΕΗ παραλληλογράμμων ἀνάλογόν εἰσιν
 αἱ πλευραὶ αἱ περι τὰς >ῖσας γωνίας; <όμοιον >ἀρα ἐστὶ
 τὸ ΑΒΓΔ παραλληλόγραμμον τῷ ΕΗ παραλληλογράμμῳ.
 διὰ τὰ αὐτὰ δὴ τὸ ΑΒΓΔ παραλληλόγραμμον καὶ τῷ ΚΘ
 παραλληλογράμμῳ <όμοιον ἐστὶν; ἐκάτερον >ἀρα τῶν ΕΗ,
 ΘΚ παραλληλογράμμων τῷ ΑΒΓΔ [παραλληλογράμμῳ]

For since EF has been drawn parallel to one of the sides
 BC of triangle ABC , proportionally, as BE is to EA , so
 CF (is) to FA [Prop. 6.2]. Again, since FG has been drawn
 parallel to one (of the sides) CD of triangle ACD , propor-
 tionally, as CF is to FA , so DG (is) to GA [Prop. 6.2].
 But, as CF (is) to FA , so it was also shown (is) BE to
 EA . And thus as BE (is) to EA , so DG (is) to GA . And,
 thus, compounding, as BA (is) to AE , so DA (is) to AG
 [Prop. 5.18]. And, alternately, as BA (is) to AD , so EA
 (is) to AG [Prop. 5.16]. Thus, in parallelograms $ABCD$
 and EG , the sides about the common angle BAD are pro-
 portional. And since GF is parallel to DC , angle AFG
 is equal to DCA [Prop. 1.29]. And angle DAC (is) com-
 mon to the two triangles ADC and AGF . Thus, triangle
 ADC is equiangular to triangle AGF [Prop. 1.32]. So, for
 the same (reasons), triangle ACB is equiangular to triangle
 AFE , and the whole parallelogram $ABCD$ is equiangu-
 lar to parallelogram EG . Thus, proportionally, as AD (is)
 to DC , so AG (is) to GF , and as DC (is) to CA , so GF
 (is) to FA , and as AC (is) to CB , so AF (is) to FE , and,
 further, as CB (is) to BA , so FE (is) to EA [Prop. 6.4].
 And since it was shown that as DC is to CA , so GF (is)
 to FA , and as AC (is) to CB , so AF (is) to FE , thus, via
 equality, as DC is to CB , so GF (is) to FE [Prop. 5.22].
 Thus, in parallelograms $ABCD$ and EG , the sides about the
 equal angles are proportional. Thus, parallelogram $ABCD$
 is similar to parallelogram EG [Def. 6.1]. So, for the same
 (reasons), parallelogram $ABCD$ is also similar to parallel-
 ogram KH . Thus, parallelograms EG and HK are each
 similar to [parallelogram] $ABCD$. And (rectilinear figures)
 similar to the same rectilinear figure are also similar to one

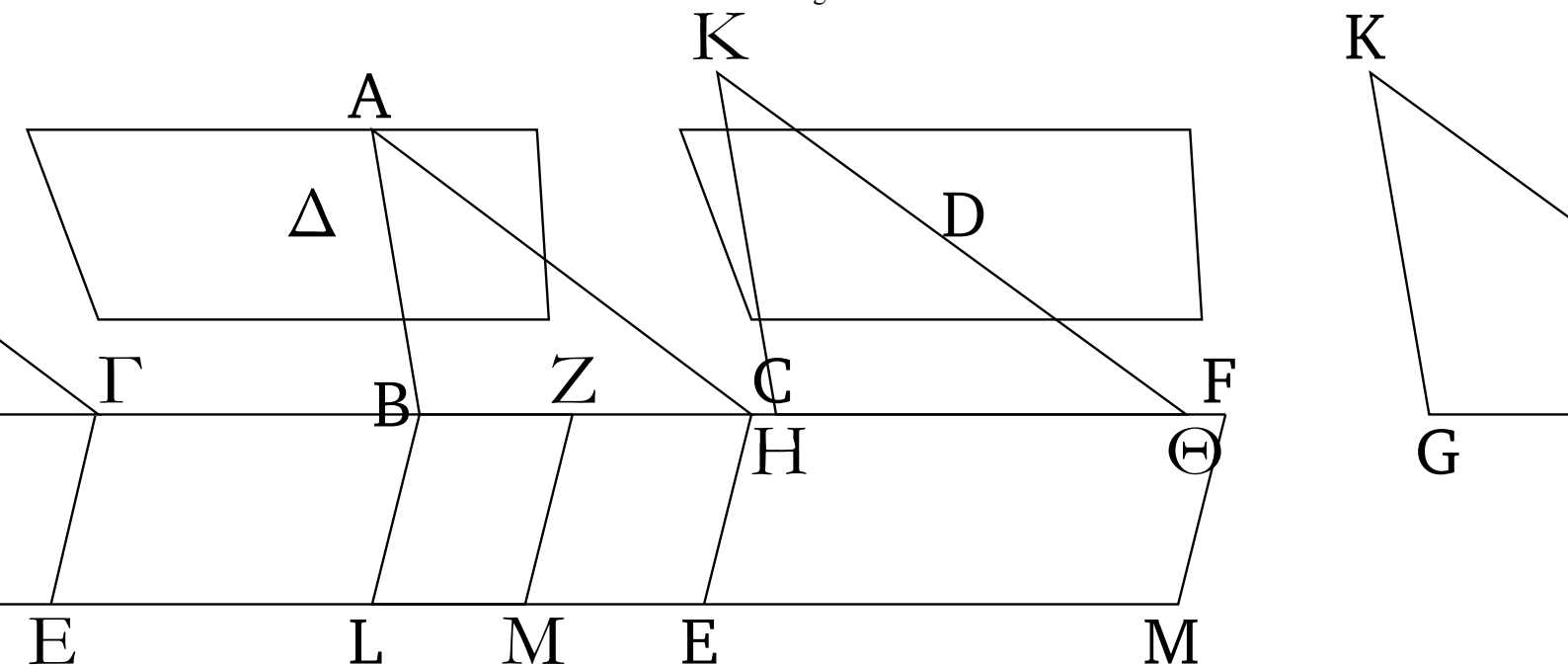
<ομοιόν ἐστιν. τὰ δὲ τῷ αὐτῷ εὐθυγράμμῳ <ομοία καὶ ἀλλήλοις ἐστὶν <ομοία; καὶ τὸ EH >ἄρα παραλληλόγραμμον τῷ ΘΚ παραλληλογράμμῳ <ομοιόν ἐστιν.
Παντὸς >ἄρα παραλληλογράμμου τὰ περὶ τὴν διάμετρον παραλληλόγραμμα <ομοιά ἐστι τῷ τε <δλω καὶ ἀλλήλοις; <ὅπερ >έδει δεῖχαι.

25.

Τῷ δοθέντι εὐθυγράμμῳ <ομοιον καὶ >ἄλλῳ τῷ δοθέντι >ῖσον τὸ αὐτὸ συστήσασθαι.

Proposition 25

To construct a single (rectilinear figure) similar to a given rectilinear figure, and equal to a different given rectilinear figure.



>Ἐστω τὸ μὲν δοθὲν εὐθύγραμμον, <ὃ δὲ <ομοιον συστήσασθαι, τὸ ABΓ, <ὃ δὲ δει >ῖσον, τὸ Δ; δεῖ δὴ τῷ μὲν ABΓ <ομοιον, τῷ δὲ Δ >ῖσον τὸ αὐτὸ συστήσασθαι.

Παραβεβλήσθω γὰρ παρὰ μὲν τὴν BΓ τῷ ABΓ τριγώνῳ >ῖσον παραλληλόγραμμον τὸ BE, παρὰ δὲ τὴν ΓΕ τῷ Δ >ῖσον παραλληλόγραμμον τὸ ΓΜ ἐν γωνίᾳ τῇ ὑπὸ ZΓΕ, <ἡ ἐστὶν >ῖση τῇ ὑπὸ ΓΒΛ. ἐπ' εὐθείᾳ >ἄρα ἐστὶν ἡ μὲν BΓ τῇ ΓΖ, ἡ δὲ ΛΕ τῇ ΕΜ. καὶ εἰλήφθω τῶν BΓ, ΓΖ μέση ἀνάλογον ἡ ΗΘ, καὶ ἀναγεγράφθω ἀπὸ τῆς ΗΘ τῷ ABΓ <ομοιόν τε καὶ ὁμοίως κείμενον τὸ ΚΗΘ.

Καὶ ἐπεὶ ἐστὶν ὡς ἡ BΓ πρὸς τὴν ΗΘ, οὕτως ἡ ΗΘ πρὸς τὴν ΓΖ, ἔαν δὲ τρεῖς εὐθεῖαι ἀνάλογον >ῶσιν, >έστιν ὡς ἡ πρώτη πρὸς τὴν τρίτην, οὕτως τὸ ἀπὸ τῆς πρώτης εἰδοῦς πρὸς τὸ ἀπὸ τῆς δευτέρας τὸ <ομοιον καὶ ὁμοίως ἀναγραφόμενον, >έστιν >ἄρα ὡς ἡ BΓ πρὸς τὴν ΓΖ, οὕτως τὸ ABΓ τρίγωνον πρὸς τὸ ΚΗΘ τρίγωνον. ἀλλὰ καὶ ὡς ἡ BΓ πρὸς τὴν ΓΖ, οὕτως τὸ BE παραλληλόγραμμον πρὸς τὸ EZ παραλληλόγραμμον. καὶ ὡς >ἄρα τὸ ABΓ τρίγωνον πρὸς τὸ ΚΗΘ τρίγωνον, οὕτως τὸ BE παραλληλόγραμμον πρὸς τὸ EZ παραλληλόγραμμον; ἐναλλάξ >ἄρα ὡς τὸ ABΓ τρίγωνον πρὸς τὸ BE παραλληλόγραμμον, οὕτως τὸ ΚΗΘ τρίγωνον πρὸς τὸ EZ παραλληλόγραμμον. >ῖσον δὲ τὸ ABΓ τρίγωνον τῷ BE παραλληλογράμῳ; >ῖσον >ἄρα καὶ τὸ ΚΗΘ τρίγωνον τῷ EZ παραλληλογράμῳ. ἀλλὰ τὸ EZ παραλληλόγραμμον τῷ Δ ἐστὶν >ῖσον; καὶ τὸ ΚΗΘ >ἄρα τῷ Δ ἐστὶν >ῖσον. >έστι δὲ τὸ ΚΗΘ καὶ τῷ ABΓ <ομοιον. Τῷ >ἄρα δοθέντι εὐθυγράμμῳ τῷ ABΓ <ομοιον καὶ >ἄλλῳ τῷ δοθέντι τῷ Δ >ῖσον τὸ αὐτὸ συνέσταται τὸ ΚΗΘ; <ὅπερ

et ABC be the given rectilinear figure to which it is required to construct a similar (rectilinear figure), and D the (rectilinear figure) to which (the constructed figure) is required (to be) equal. So it is required to construct a single (rectilinear figure) similar to ABC , and equal to D .

For let the parallelogram BE , equal to triangle ABC , be applied to (the straight-line) BC [Prop. 1.44], and the parallelogram CM , equal to D , (be applied) to (the straight-line) CE , in the angle FCE , which is equal to CBL [Prop. 1.45]. Thus, BC is straight-on to CF , and LE to EM [Prop. 1.14]. And let the mean proportion GH be taken of BC and CF [Prop. 6.13]. And let KGH , similar, and similarly laid out, to ABC , be described on GH [Prop. 6.18].

And since as BC is to GH , so GH (is) to CF , and if three straight-lines are proportional, (then) as the first is to the third, so the figure (described) on the first (is) to the similar, and similarly described, (figure) on the second [Prop. 6.19 corr.], thus as BC is to CF , so triangle ABC (is) to triangle KGH . But, also, as BC (is) to CF , so parallelogram BE (is) to parallelogram EF [Prop. 6.1]. And, thus, as triangle ABC (is) to triangle KGH , so parallelogram BE (is) to parallelogram EF . Thus, alternately, as triangle ABC (is) to parallelogram BE , so triangle KGH (is) to parallelogram EF [Prop. 5.16]. And triangle ABC (is) equal to parallelogram BE . Thus, triangle KGH (is) also equal to parallelogram EF . But, parallelogram EF is equal to D . Thus, KGH is also equal to D . And KGH is

>έδει ποιῆσαι.

also similar to ABC .

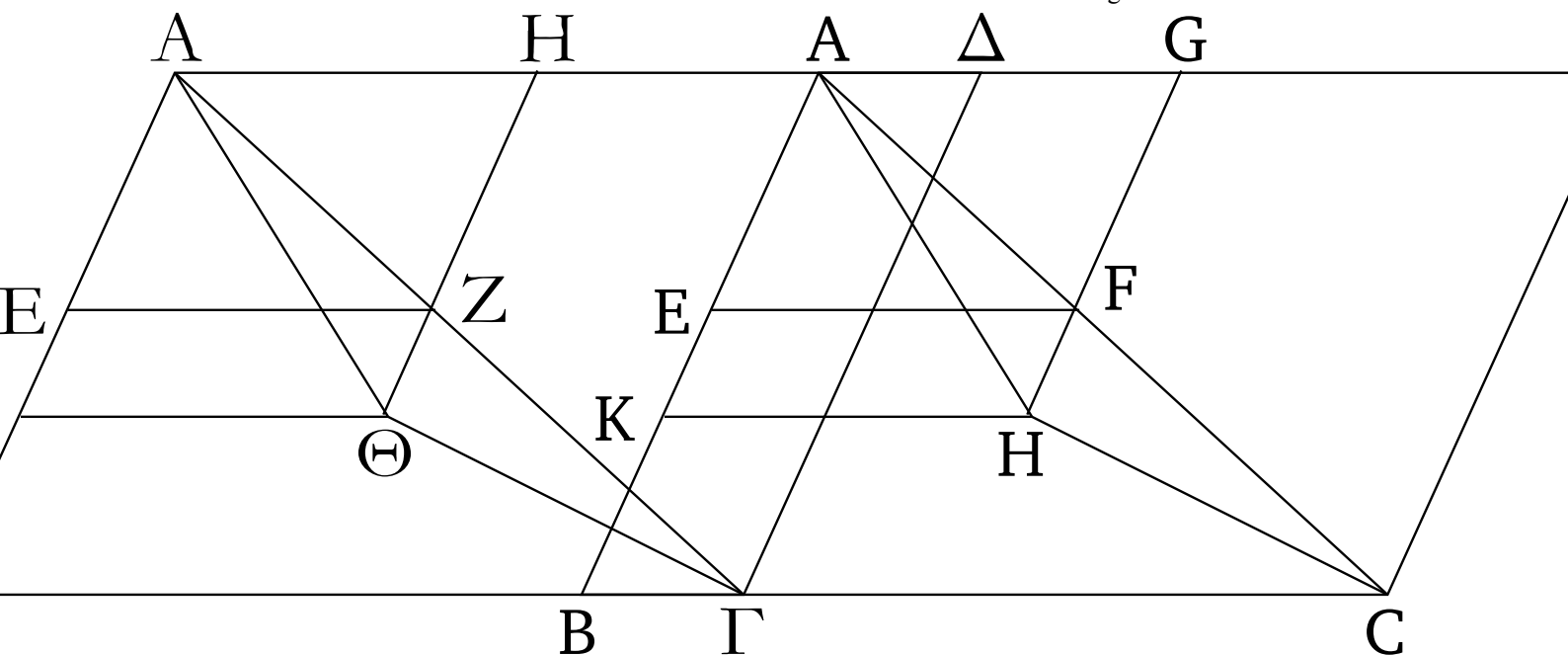
Thus, a single (rectilinear figure) KGH has been constructed (which is) similar to the given rectilinear figure ABC , and equal to a different given (rectilinear figure) D . (Which is) the very thing it was required to do.

26.

Proposition 26

Ἐὰν ἀπὸ παραλληλογράμμου παραλληλόγραμμον ἀφαιρεθῇ| <όμοιόν τε τῷ| <ὅλῳ καὶ ὁμοίῳς κείμενον κοινὴν γωνίαν| >έθον αὐτῷ|, περὶ τὴν αὐτὴν διάμετρον ἔστι τῷ| <ὅλῳ.
Ἀπὸ γὰρ παραλληλογράμμου τοῦ $ABΓΔ$ παραλληλόγραμμον ἀφαιρήσθω τὸ AZ <όμοιον τῷ| $ABΓΔ$ καὶ ὁμοίῳς κείμενον κοινὴν γωνίαν >έθον αὐτῷ| τὴν ὑπὸ $ΔAB$; λέγω, <ὅτι περὶ τὴν αὐτὴν διάμετρον ἔστι τὸ $ABΓΔ$ τῷ| AZ .

If from a parallelogram a(nother) parallelogram is subtracted (which is) similar, and similarly laid out, to the whole, having a common angle with it, (then the subtracted parallelogram) is about the same diagonal as the whole. For, from parallelogram $ABCD$, let (parallelogram) AF be subtracted (which is) similar, and similarly laid out, to $ABCD$, having the common angle DAB with it. I say that $ABCD$ is about the same diagonal as AF .



μὴ γάρ, ἀλλ> εἰ δυνατόν, >έστω [αὐτῶν] διάμετρος ἡ $AΘΓ$, καὶ ἐκβληθεῖσα ἡ HZ διήθῃ ἐπὶ τὸ $Θ$, καὶ >ήθῃ διὰ τοῦ $Θ$ ὁπορέρα τῶν $ΑΔ$, $BΓ$ παράλληλος ἡ $ΘΚ$.

Ἐπεὶ ο>ν περὶ τὴν αὐτὴν διάμετρον ἔστι τὸ $ABΓΔ$ τῷ| KH , >έστιν >άρα ὡς ἡ $ΔΑ$ πρὸς τὴν AB , ο<ύτωξ ἡ HA πρὸς τὴν AK . >έστι δὲ καὶ διὰ τὴν ὁμοιότητα τῶν $ABΓΔ$, EH καὶ ὡς ἡ $ΔΑ$ πρὸς τὴν AB , ο<ύτωξ ἡ HA πρὸς τὴν AE ; καὶ ὡς >άρα ἡ HA πρὸς τὴν AK , ο<ύτωξ ἡ HA πρὸς τὴν AE . ἡ HA >άρα πρὸς ἑκατέραν τῶν AK , AE τὸν αὐτὸν >έθει λόγον. >ίση >άρα ἔστιν ἡ AE τῇ| AK ἡ ἐλάττων τῇ| μείζων; <ὅπερ ἔστιν ἀδύνατον. οὐκ >άρα ο>ύκ ἔστι περὶ τὴν αὐτὴν διάμετρον τὸ $ABΓΔ$ τῷ| AZ ; περὶ τὴν αὐτὴν >άρα ἔστι διάμετρον τὸ $ABΓΔ$ παραλληλόγραμμον τῷ| AZ παραλληλογράμμῳ.

Ἐὰν >άρα ἀπὸ παραλληλογράμμου παραλληλόγραμμον ἀφαιρεθῇ| <όμοιόν τε τῷ| <ὅλῳ καὶ ὁμοίῳς κείμενον κοινὴν γωνίαν >έθον αὐτῷ|, περὶ τὴν αὐτὴν διάμετρον ἔστι τῷ| <ὅλῳ; <ὅπερ >έδει δεῖχαι.

For (if) not, (then), if possible, let AHC be [$ABCD$'s] diagonal. And producing GF , let it be drawn through to (point) H . And let HK be drawn through (point) H , parallel to either of AD or BC [Prop. 1.31].

Therefore, since $ABCD$ is about the same diagonal as KG , thus as DA is to AB , so GA (is) to AK [Prop. 6.24]. And, on account of the similarity of $ABCD$ and EG , also, as DA (is) to AB , so GA (is) to AE . Thus, also, as GA (is) to AK , so GA (is) to AE . Thus, GA has the same ratio to each of AK and AE . Thus, AE is equal to AK [Prop. 5.9], the lesser to the greater. The very thing is impossible. Thus, $ABCD$ is not not about the same diagonal as AF . Thus, parallelogram $ABCD$ is about the same diagonal as parallelogram AF .

Thus, if from a parallelogram a(nother) parallelogram is subtracted (which is) similar, and similarly laid out, to the whole, having a common angle with it, (then the subtracted parallelogram) is about the same diagonal as the whole. (Which is) the very thing it was required to show.

27.

Proposition 27

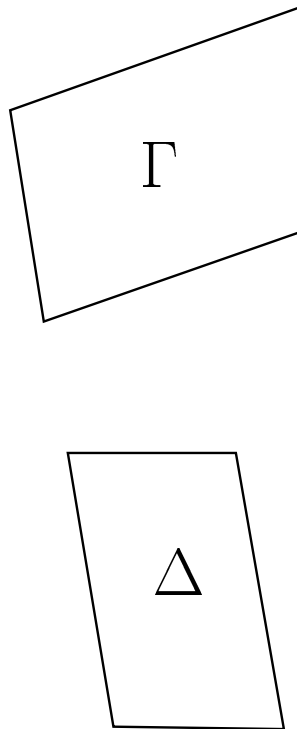
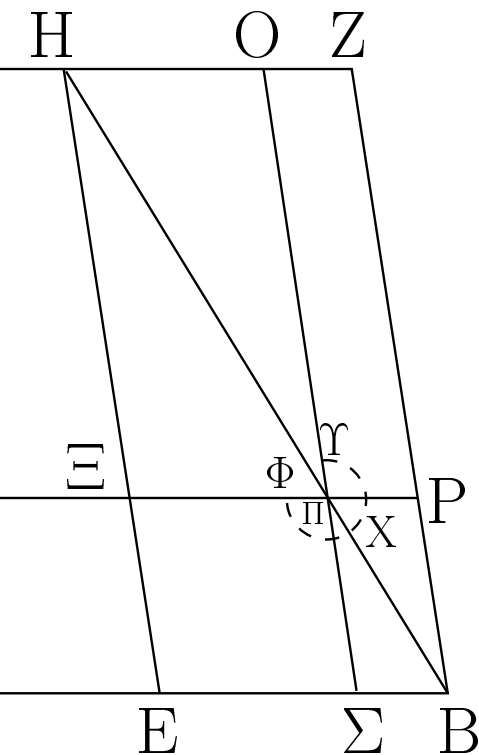
Πάντων τῶν παρὰ τὴν αὐτὴν εὐθεῖαν παραβαλλομένων παραλληλογράμμων καὶ ἐλλειπόντων ε>ίδεσι παραλληλογράμμοις ὁμοίοις τε καὶ ὁμοίῳς κειμένοις τῷ| ἀπὸ τῆς ἡμισείας

Of all the parallelograms applied to the same straight-line, and falling short by parallelogrammic figures similar, and similarly laid out, to the (parallelogram) described on half

ἡμισεία ἀναγραφομένου ὁμοίου τῷ ἑλλείμματι [τοῦ τε ἀπὸ τῆς ἡμισείας καὶ <ῶ| δεῖ <ὁμοιον ἑλλείπειν].

>Ἐστω ἡ μὲν δοθεῖσα εὐθεῖα ἡ AB , τὸ δὲ δοθὲν εὐθύγραμμον, <ῶ| δεῖ >ῖσον παρὰ τὴν AB παραβαλεῖν, τὸ Γ μὴ μείζον [>ὸν] τοῦ ἀπὸ τῆς ἡμισείας τῆς AB ἀναγραφομένου ὁμοίου τῷ ἑλλείμματι, <ῶ| δεῖ <ὁμοιον ἑλλείπειν, τὸ Δ ; δεῖ δὴ παρὰ τὴν δοθεῖσαν εὐθεῖαν τὴν AB τῷ δοθέντι εὐθύγραμμῳ τῷ Γ >ῖσον παραλληλόγραμμον παραβαλεῖν ἑλλείπον >ῖδει παραλληλογράμῳ ὁμοίῳ >όντι τῷ Δ .

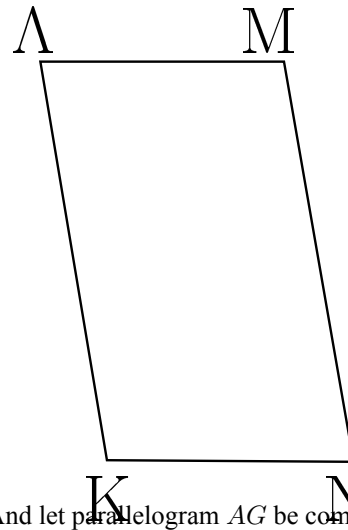
Τετμήσθω ἡ AB δίθῃ κατὰ τὸ E σημεῖον, καὶ ἀναγεγράφθω ἀπὸ τῆς EB τῷ Δ <ὁμοιον καὶ ὁμοίῳ κείμενον τὸ $EBZH$, καὶ συμπληρώσθω τὸ AH παραλληλόγραμμον.



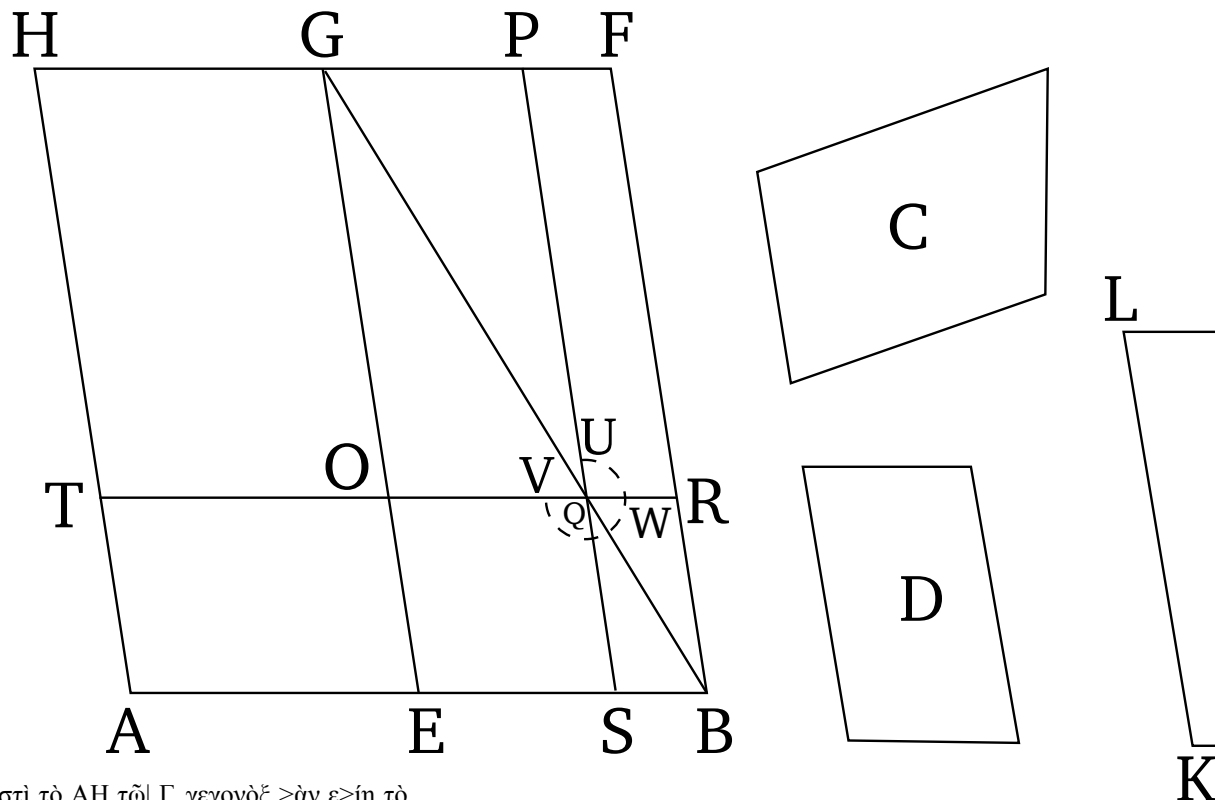
which it is required to apply an equal (parallelogram)] not to be greater than the (parallelogram) described on half (of the straight-line) and similar to the deficit.

Let AB be the given straight-line, and C the given rectilinear figure to which the (parallelogram) applied to AB is required (to be) equal, [being] not greater than the (parallelogram) described on half of AB and similar to the deficit, and D the (parallelogram) to which the deficit is required (to be) similar. So it is required to apply a parallelogram, equal to the given rectilinear figure C , to the straight-line AB , falling short by a parallelogramic figure which is similar to D .

Let AB be cut in half at point E [Prop. 1.10], and let (parallelogram) $EBFG$, (which is) similar, and similarly laid out, to (parallelogram) D , be described on EB [Prop. 6.18].



And let parallelogram AG be completed.



Εἰ μὲν οὖν ἴσον ἐστὶ τὸ ΑΗ τῷ Γ, γεγονὸς ἂν ἐπὶ τὸ ἐπιταθέν; παραβέβληται γὰρ παρὰ τὴν δοθεῖσαν εὐθεῖαν τὴν ΑΒ τῷ δοθέντι εὐθυγράμμῳ τῷ Γ ἴσον παραλληλόγραμμον τὸ ΑΗ ἐλλείπον ἐπὶ παραλληλογράμῳ τῷ ΗΒ ὁμοίῳ ὄντι τῷ Δ. εἰ δὲ οὐ, μείζον ἔστω τὸ ΘΕ τοῦ Γ. ἴσον δὲ τὸ ΘΕ τῷ ΗΒ; μείζον ἄρα καὶ τὸ ΗΒ τοῦ Γ. <ὦ δὲ μείζον ἐστὶ τὸ ΗΒ τοῦ Γ, τῷ τῇ ὑπεροθῇ ἴσον, τῷ δὲ Δ ὁμοίον καὶ ὁμοίως κείμενον τὸ αὐτὸ συνεστάτω τὸ ΚΛΜΝ. ἀλλὰ τὸ Δ τῷ ΗΒ [ἐστίν] ὁμοίον; καὶ τὸ ΚΜ ἄρα τῷ ΗΒ ἐστίν ὁμοίον. ἔστω οὖν ὁμόλογος ἡ μὲν ΚΛ τῇ ΗΕ, ἡ δὲ ΛΜ τῇ ΗΖ. καὶ ἐπεὶ ἴσον ἐστὶ τὸ ΗΒ τοῖς Γ, ΚΜ, μείζον ἄρα ἐστὶ τὸ ΗΒ τοῦ ΚΜ; μείζον ἄρα ἐστὶ καὶ ἡ μὲν ΗΕ τῇ ΚΛ, ἡ δὲ ΗΖ τῇ ΛΜ. κείσθω τῇ μὲν ΚΛ ἴση ἡ ΗΧ, τῇ δὲ ΛΜ ἴση ἡ ΗΟ, καὶ συμπεπληρώσθω τὸ ΧΗΟΠ παραλληλόγραμμον; ἴσον ἄρα καὶ ὁμοίον ἐστὶ [τὸ ΗΠ] τῷ ΚΜ [ἀλλὰ τὸ ΚΜ τῷ ΗΒ ὁμοίον ἐστίν]. καὶ τὸ ΗΠ ἄρα τῷ ΗΒ ὁμοίον ἐστίν; περὶ τὴν αὐτὴν ἄρα διάμετρον ἐστὶ τὸ ΗΠ τῷ ΗΒ. ἔστω αὐτῶν διάμετρος ἡ ΗΠΒ, καὶ καταγεγράφθω τὸ σῆμα.

Ἐπεὶ οὖν ἴσον ἐστὶ τὸ ΒΗ τοῖς Γ, ΚΜ, <ὦν τὸ ΗΠ τῷ ΚΜ ἐστίν ἴσον, λοιπὸς ἄρα ὁ ΥΘΦ γνόμεον λοιπῷ τῷ Γ ἴσοξ ἐστίν. καὶ ἐπεὶ ἴσον ἐστὶ τὸ ΟΡ τῷ ΧΣ, κοινὸν προσκείσθω τὸ ΠΒ; ὅλον ἄρα τὸ ΟΒ ὅλῳ τῷ ΧΒ ἴσον ἐστίν. ἀλλὰ τὸ ΧΒ τῷ ΤΕ ἐστίν ἴσον, ἐπεὶ καὶ πλευρὰ ἡ ΑΕ πλευρὰ τῇ ΕΒ ἐστίν ἴση; καὶ τὸ ΤΕ ἄρα τῷ ΟΒ ἐστίν ἴσον. κοινὸν προσκείσθω τὸ ΧΣ; ὅλον ἄρα τὸ ΤΣ ὅλῳ τῷ ΦΘΥ γνόμενόν ἐστιν ἴσον. ἀλλ' ὁ ΦΘΥ γνόμεον τῷ Γ ἐδείκθη ἴσοξ; καὶ τὸ ΤΣ ἄρα τῷ Γ ἐστίν ἴσον.

Παρὰ τὴν δοθεῖσαν ἄρα εὐθεῖαν τὴν ΑΒ τῷ δοθέντι εὐθυγράμμῳ τῷ Γ ἴσον παραλληλόγραμμον παραβέβληται τὸ ΣΤ ἐλλείπον ἐπὶ παραλληλογράμῳ τῷ ΠΒ ὁμοίῳ ὄντι τῷ Δ [ἐπειδὴ περὶ τὸ ΠΒ τῷ ΗΠ ὁμοίον ἐστίν]; ὅπερ ἔδει ποιῆσαι.

Therefore, if AG is equal to C , (then) the thing prescribed has happened. For a parallelogram AG , equal to the given rectilinear figure C , has been applied to the given straight-line AB , falling short by a parallelogramic figure GB which is similar to D . And if not, let HE be greater than C . And HE (is) equal to GB [Prop. 6.1]. Thus, GB (is) also greater than C . So, let (parallelogram) $KLMN$ be constructed (so as to be) both similar, and similarly laid out, to D , and equal to the excess by which GB is greater than C [Prop. 6.25]. But, GB [is] similar to D . Thus, KM is also similar to GB [Prop. 6.21]. Therefore, let KL correspond to GE , and LM to GF . And since (parallelogram) GB is equal to (figure) C and (parallelogram) KM , GB is thus greater than KM . Thus, GE is also greater than KL , and GF than LM . Let GO be made equal to KL , and GP to LM [Prop. 1.3]. And let the parallelogram $OGPQ$ be completed. Thus, $[GQ]$ is equal and similar to KM [but, KM is similar to GB]. Thus, GQ is also similar to GB [Prop. 6.21]. Thus, GQ and GB are about the same diagonal [Prop. 6.26]. Let GQB be their (common) diagonal, and let the (remainder of the) figure be described.

Therefore, since BG is equal to C and KM , of which GQ is equal to KM , the remaining gnomon UWV is thus equal to the remainder C . And since (the complement) PR is equal to (the complement) OS [Prop. 1.43], let (parallelogram) QB be added to both. Thus, the whole (parallelogram) PB is equal to the whole (parallelogram) OB . But, OB is equal to TE , since side AE is equal to side EB [Prop. 6.1]. Thus, TE is also equal to PB . Let (parallelogram) OS be added to both. Thus, the whole (parallelogram) TS is equal to the gnomon VWU . But, gnomon VWU was shown (to be) equal to C . Therefore, (parallelogram) TS is also equal to (figure) C .

Thus, the parallelogram ST , equal to the given rectilinear figure C , has been applied to the given straight-line AB , falling short by the parallelogramic figure QB ,

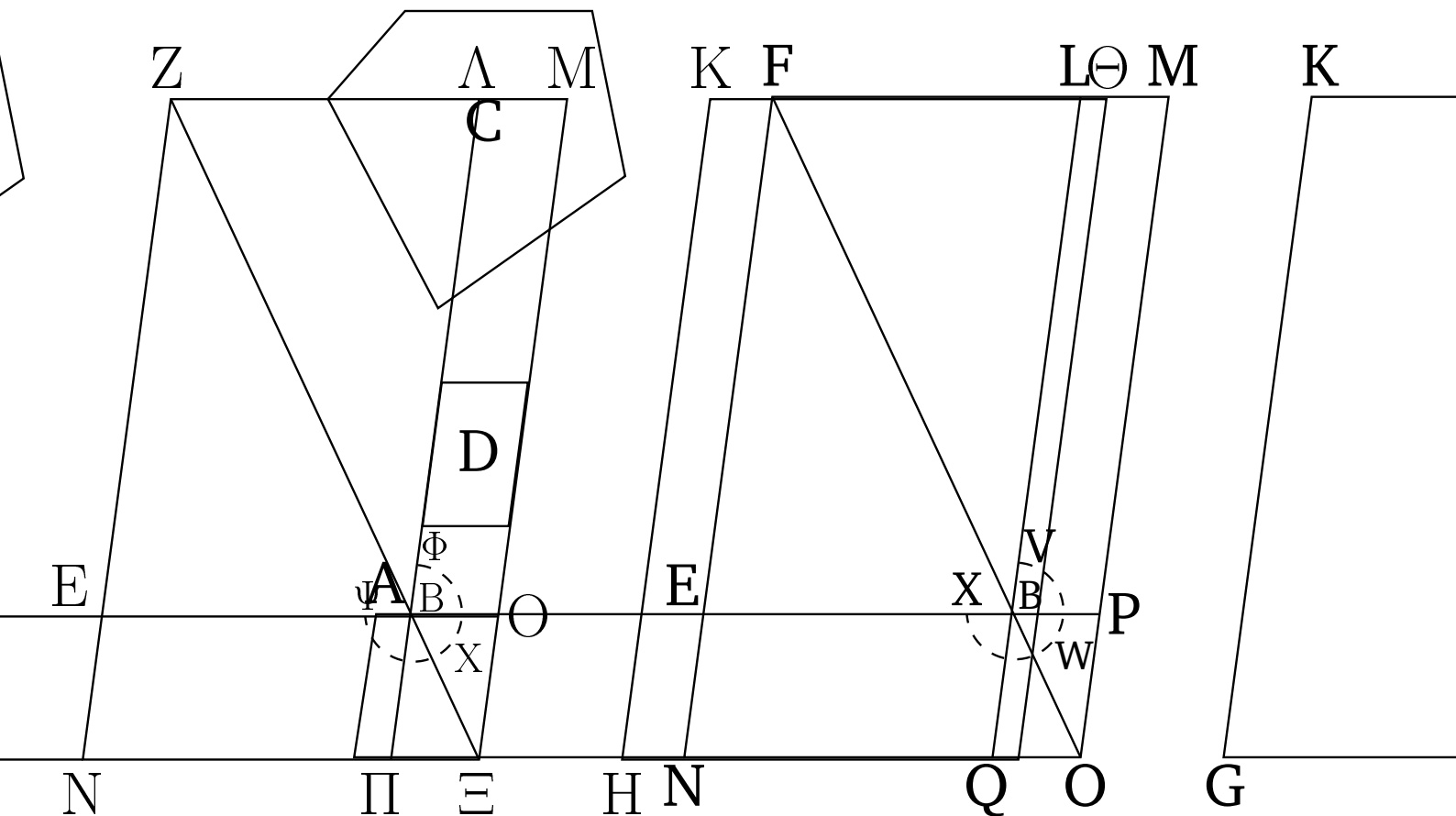
which is similar to D [inasmuch as QB is similar to GQ [Prop. 6.24]]. (Which is) the very thing it was required to do.

† This proposition is a geometric solution of the quadratic equation $x^2 - \alpha x + \beta = 0$. Here, x is the ratio of a side of the deficit to the corresponding side of figure D , α is the ratio of the length of AB to the length of that side of figure D which corresponds to the side of the deficit running along AB , and β is the ratio of the areas of figures C and D . The constraint corresponds to the condition $\beta < \alpha^2/4$ for the equation to have real roots. Only the smaller root of the equation is found. The larger root can be found by a similar method.

29.

Proposition 29†

Παρά τὴν δοθεῖσαν εὐθεῖαν τῷ δοθέντι εὐθυγράμμῳ ὅμοιον παραλληλόγραμμον παραβαλεῖν ὑπερβάλλον ἐξ ἰδὲι παραλληλογράμμου ὁμοίου τῷ δοθέντι. To apply a parallelogram, equal to a given rectilinear figure, (the applied parallelogram) overshooting by a parallelogramic figure similar to a given (parallelogram).



>Ἐστω ἡ μὲν δοθεῖσα εὐθεῖα ἡ AB , τὸ δὲ δοθὲν εὐθύγραμμον, <ὧ| δεῖ >ῖσον παρὰ τὴν AB παραβαλεῖν, τὸ Γ , <ὧ| δὲ δεῖ <όμοιον ὑπερβάλλειν, τὸ Δ ; δεῖ δὲ παρὰ τὴν AB εὐθεῖαν τῷ Γ εὐθυγράμμῳ ὅμοιον παραλληλόγραμμον παραβαλεῖν ὑπερβάλλον ἐξ ἰδὲι παραλληλογράμμου ὁμοίου τῷ Δ . Τετμήσθω ἡ AB δίθῃ κατὰ τὸ E , καὶ ἀναγεγράθω ἀπὸ τῆς EB τῷ Δ <όμοιον καὶ ὁμοίωξ κείμενον παραλληλόγραμμον τὸ BZ , καὶ συναμφοτέροισι μὲν τοῖς BZ , Γ >ῖσον, τῷ δὲ Δ <όμοιον καὶ ὁμοίωξ κείμενον τὸ αὐτὸ συνεστάτω τὸ $H\Theta$. ὁμόλογος δὲ >έστω ἡ μὲν $K\Theta$ τῇ $Z\Lambda$, ἡ δὲ KH τῇ ZE . καὶ ἐπεὶ μείζων ἐστὶ τὸ $H\Theta$ τοῦ ZB , μείζων >άρα ἐστὶ καὶ ἡ μὲν $K\Theta$ τῇ $Z\Lambda$, ἡ δὲ KH τῇ ZE . ἐκβεβλήσθωσαν αἱ $Z\Lambda$, ZE , καὶ τῇ μὲν $K\Theta$ >ῖση >έστω ἡ $Z\Lambda M$, τῇ δὲ KH >ῖση ἡ ZEN , καὶ συμπληρώσθω τὸ MN ; τὸ MN >άρα τῷ $H\Theta$ >ῖσον τέ ἐστὶ καὶ <όμοιον. ἀλλὰ τὸ $H\Theta$ τῷ $E\Lambda$ ἐστὶν <όμοιον; καὶ τὸ MN >άρα τῷ $E\Lambda$ <ομοιόν ἐστιν; περὶ τὴν αὐτὴν >άρα διάμετρον ἐστὶ τὸ $E\Lambda$ τῷ MN . >ήθθω αὐτῶν διάμετρος ἡ ZX , καὶ καταγεγράθω τὸ σθῆμα.

Let AB be the given straight-line, and C the given rectilinear figure to which the (parallelogram) applied to AB is required (to be) equal, and D the (parallelogram) to which the excess is required (to be) similar. So it is required to apply a parallelogram, equal to the given rectilinear figure C , to the given straight-line AB , overshooting by a parallelogrammic figure similar to D .

Let AB be cut in half at (point) E [Prop. 1.10], and let the parallelogram BF , (which is) similar, and similarly laid out, to D , be described on EB [Prop. 6.18]. And let (parallelogram) GH be constructed (so as to be) both similar, and similarly laid out, to D , and equal to the sum of BF and C [Prop. 6.25]. And let KH correspond to FL , and KG to FE . And since (parallelogram) GH is greater than (parallelogram) FB , KH is thus also greater than FL , and KG than FE . Let FL and FE be produced, and let FLM be (made) equal to KH , and FEN to KG [Prop. 1.3]. And let (parallelogram) MN be completed. Thus, MN is equal

Ἐπεὶ ὁμοῖον ἐστὶ τὸ $H\Theta$ τοῖς $ΕΛ$, Γ , ἀλλὰ τὸ $H\Theta$ τῷ MN ὁμοῖον ἐστίν, καὶ τὸ MN ὁμοῖον τοῖς $ΕΛ$, Γ ὁμοῖον ἐστίν. κοινὸν ἀφηρήσθω τὸ $ΕΛ$; λοιπὸς ὁ $\Psi\Theta\Phi$ γνῶμων τῷ Γ ἐστὶν ὁμοῖος. καὶ ἐπεὶ ὁμοῖον ἐστὶν ἡ $ΑΕ$ τῇ $ΕΒ$, ὁμοῖον ἐστὶ καὶ τὸ $ΑΝ$ τῷ $ΝΒ$, τουτέστι τῷ $ΛΟ$. κοινὸν προσκείσθω τὸ $ΕΧ$; ὅλον ὁμοῖον τὸ $ΑΧ$ ὁμοῖον ἐστὶ τῷ $\Phi\Theta\Psi$ γνῶμονι. ἀλλὰ ὁ $\Phi\Theta\Psi$ γνῶμων τῷ Γ ὁμοῖος ἐστίν; καὶ τὸ $ΑΧ$ ὁμοῖον τῷ Γ ὁμοῖον ἐστίν.

Παρά τὴν δοθεῖσαν ὁμοῖον εὐθεῖαν τὴν $ΑΒ$ τῷ δοθέντι εὐθυγράμμῳ τῷ Γ ὁμοῖον παραλληλόγραμμον παραβέβληται τὸ $ΑΧ$ ὑπερβάλλον ἐκείνῳ παραλληλογράμμῳ τῷ $\Pi\Theta$ ὁμοίῳ ὄντι τῷ Δ , ἐπεὶ καὶ τῷ $ΕΛ$ ἐστὶν ὁμοῖον τὸ $ΟΠ$; ὅπερ ὀφείδει ποιῆσαι.

and similar to GH . But, GH is similar to EL . Thus, MN is also similar to EL [Prop. 6.21]. EL is thus about the same diagonal as MN [Prop. 6.26]. Let their (common) diagonal FO be drawn, and let the (remainder of the) figure be described.

And since (parallelogram) GH is equal to (parallelogram) EL and (figure) C , but GH is equal to (parallelogram) MN , MN is thus also equal to EL and C . Let EL be subtracted from both. Thus, the remaining gnomon XWV is equal to (figure) C . And since AE is equal to EB , (parallelogram) AN is also equal to (parallelogram) NB [Prop. 6.1], that is to say, (parallelogram) LP [Prop. 1.43]. Let (parallelogram) EO be added to both. Thus, the whole (parallelogram) AO is equal to the gnomon VWX . But, the gnomon VWX is equal to (figure) C . Thus, (parallelogram) AO is also equal to (figure) C .

Thus, the parallelogram AO , equal to the given rectilinear figure C , has been applied to the given straight-line AB , overshooting by the parallelogramic figure QP which is similar to D , since PQ is also similar to EL [Prop. 6.24]. (Which is) the very thing it was required to do.

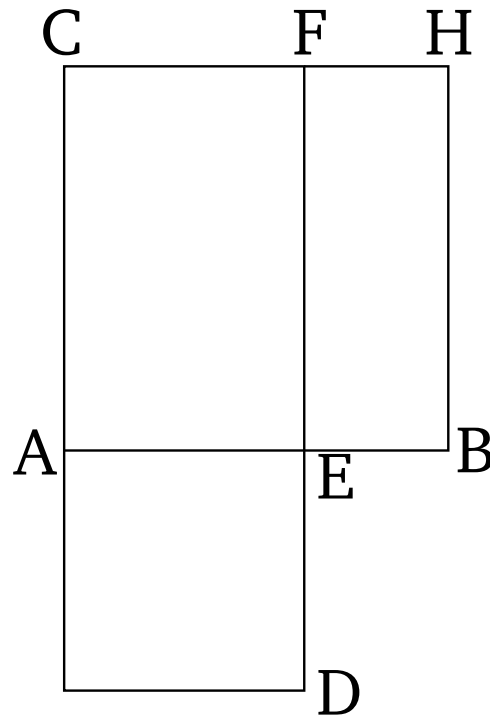
† This proposition is a geometric solution of the quadratic equation $x^2 + \alpha x - \beta = 0$. Here, x is the ratio of a side of the excess to the corresponding side of figure D , α is the ratio of the length of AB to the length of that side of figure D which corresponds to the side of the excess running along AB , and β is the ratio of the areas of figures C and D . Only the positive root of the equation is found.

30.

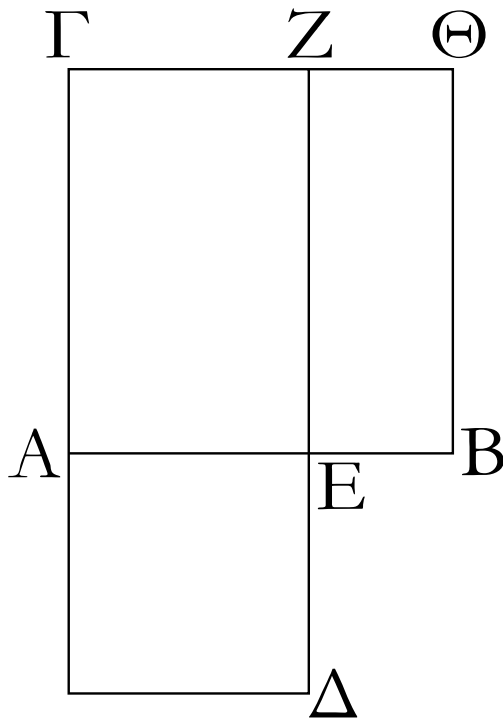
Proposition 30†

Τὴν δοθεῖσαν εὐθεῖαν πεπερασμένην ὁμοῖον καὶ μέσον

To cut a given finite straight-line in extreme and mean ratio.



λόγον τεμεῖν.



>Ἐστω ἡ δοθεῖσα εὐθεῖα πεπερασμένη ἡ AB ; δεῖ δὴ τὴν AB εὐθεῖαν >ἄκρον καὶ μέσον λόγον τεμεῖν.

Ἀναγεγράφθω ἀπὸ τῆς AB τετράγωνον τὸ $BΓ$, καὶ παραβέβλησθω τὸ παραλληλόγραμμον τὸ $ΓΔ$ ὑπερβάλλον >ἰδεῖ τῷ $ΑΔ$ ὁμοίῳ τῷ $BΓ$.

Τετράγωνον δέ ἐστι τὸ $BΓ$; τετράγωνον >ἄρα ἐστι καὶ τὸ $ΑΔ$. καὶ ἐπεὶ >ἰσον ἐστὶ τὸ $BΓ$ τῷ $ΓΔ$, κοινὸν ἀφηρήσθω τὸ $ΓΕ$; λοιπὸν >ἄρα τὸ BZ λοιπῷ τῷ $ΑΔ$ ἐστὶν >ἰσον. >ἐστι δὲ αὐτῷ καὶ ἰσογώνιον; τῶν BZ , $ΑΔ$ >ἄρα ἀντιτεπόνθασιν αἱ πλευραὶ αἱ περὶ τὰς >ἰσας γωνίας; >ἐστὶν >ἄρα ὥς ἡ ZE πρὸς τὴν $ΕΔ$, οὐ<ύτωξ ἡ $ΑΕ$ πρὸς τὴν $ΕΒ$. >ἰση δὲ ἡ μὲν ZE τῇ AB , ἡ δὲ $ΕΔ$ τῇ $ΑΕ$. >ἐστὶν >ἄρα ὥς ἡ BA πρὸς τὴν $ΑΕ$, οὐ<ύτωξ ἡ $ΑΕ$ πρὸς τὴν $ΕΒ$. μείζων δὲ ἡ AB τῆς $ΑΕ$; μείζων >ἄρα καὶ ἡ $ΑΕ$ τῆς $ΕΒ$.

Ἡ >ἄρα AB εὐθεῖα >ἄκρον καὶ μέσον λόγον τέτμηται κατὰ τὸ E , καὶ τὸ μείζων αὐτῆς τμήμα ἐστὶ τὸ $ΑΕ$; <όπερ >έδει ποιῆσαι.

Let AB be the given finite straight-line. So it is required to cut the straight-line AB in extreme and mean ratio.

Let the square BC be described on AB [Prop. 1.46], and let the parallelogram CD , equal to BC , be applied to AC , overshooting by the figure AD (which is) similar to BC [Prop. 6.29].

And BC is a square. Thus, AD is also a square. And since BC is equal to CD , let (rectangle) CE be subtracted from both. Thus, the remaining (rectangle) BF is equal to the remaining (square) AD . And it is also equiangular to it. Thus, the sides of BF and AD about the equal angles are reciprocally proportional [Prop. 6.14]. Thus, as FE is to ED , so AE (is) to EB . And FE (is) equal to AB , and ED to AE . Thus, as BA is to AE , so AE (is) to EB . And AB (is) greater than AE . Thus, AE (is) also greater than EB [Prop. 5.14].

Thus, the straight-line AB has been cut in extreme and mean ratio at E , and AE is its greater piece. (Which is) the very thing it was required to do.

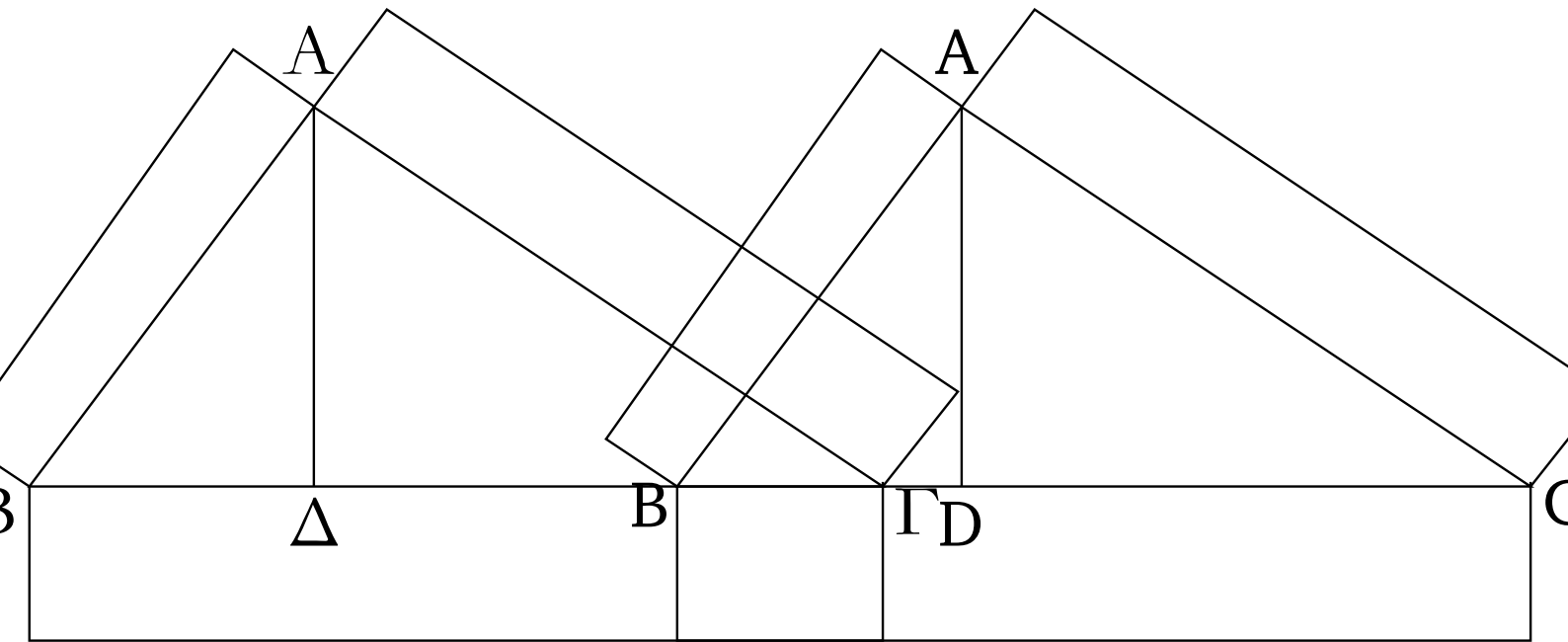
† This method of cutting a straight-line is sometimes called the “Golden Section”—see Prop. 2.11.

31.

Ἐν τοῖς ὀρθογωνίοις τριγώνοις τὸ ἀπὸ τῆς τὴν ὀρθὴν γωνίαν ὑποτείνουσας πλευρᾶς >ἴδωξ >ἰσον ἐστὶ τοῖς ἀπὸ τῶν τὴν ὀρθὴν γωνίαν περιεθουσῶν πλευρῶν >ἴδεσι τοῖς ὁμοίοις τε καὶ ὁμοίωξ ἀναγραφομένοις.

Proposition 31

In right-angled triangles, the figure (drawn) on the side subtending the right-angle is equal to the (sum of the) similar, and similarly described, figures on the sides surrounding the right-angle.



>Ἐστω τρίγωνον ὀρθογώνιον τὸ ABΓ ὀρθὴν >έθον τὴν ὑπὸ BAΓ γωνίαν; λέγω, <ὅτι τὸ ἀπὸ τῆς BΓ ε>ἰδοξ >ἴσον ἐστὶ τοῖς ἀπὸ τῶν BA, AΓ ε>ἰδεσι τοῖς ὁμοίοις τε καὶ ὁμοίως ἀναγραφόμενοις.

>Ἡθῶω κάθετοξ ἡ AΔ.

Ἐπεὶ ο>ὖν ἐν ὀρθογωνίῳ τριγώνῳ τῷ ABΓ ἀπὸ τῆς πρὸς τῷ A ὀρθῆς γωνίας ἐπὶ τὴν BΓ βάσιν κάθετοξ >ῆκται ἡ AΔ, τὰ ABΔ, AΔΓ πρὸς τῇ καθετῷ τρίγωνα <ομοιά ἐστι τῷ τε <όλῳ τῷ ABΓ καὶ ἀλλήλοις. καὶ ἐπεὶ <ομοιόν ἐστὶ τὸ ABΓ τῷ ABΔ, >έστιν >άρα ὡς ἡ ΓB πρὸς τὴν BA, ο<ύτωξ ἡ AB πρὸς τὴν BΔ. καὶ ἐπεὶ τρεῖς εὐθεῖαι ἀνάλογόν εἰσιν, >έστιν ὡς ἡ πρώτη πρὸς τὴν τρίτην, ο<ύτωξ τὸ ἀπὸ τῆς πρώτης ε>ἰδοξ πρὸς τὸ ἀπὸ τῆς δευτέρας τὸ <ομοιον καὶ ὁμοίως ἀναγραφόμενον. ὡς >άρα ἡ ΓB πρὸς τὴν BΔ, ο<ύτωξ τὸ ἀπὸ τῆς ΓB ε>ἰδοξ πρὸς τὸ ἀπὸ τῆς BA τὸ <ομοιον καὶ ὁμοίως ἀναγραφόμενον. διὰ τὰ αὐτὰ δὴ καὶ ὡς ἡ BΓ πρὸς τὴν ΓΔ, ο<ύτωξ τὸ ἀπὸ τῆς BΓ ε>ἰδοξ πρὸς τὸ ἀπὸ τῆς ΓΑ. <ώστε καὶ ὡς ἡ BΓ πρὸς τὰς BΔ, ΔΓ, ο<ύτωξ τὸ ἀπὸ τῆς BΓ ε>ἰδοξ πρὸς τὰ ἀπὸ τῶν BA, AΓ τὰ <ομοια καὶ ὁμοίως ἀναγραφόμενα. >ἴση δὲ ἡ BΓ ταῖς BΔ, ΔΓ; >ἴσον >άρα καὶ τὸ >ἀπὸ τῆς BΓ ε>ἰδοξ τοῖς ἀπὸ τῶν BA, AΓ ε>ἰδεσι τοῖς ὁμοίοις τε καὶ ὁμοίως ἀναγραφόμενοις.

Ἐν >άρα τοῖς ὀρθογωνίοις τριγώνοις τὸ ἀπὸ τῆς τὴν ὀρθὴν γωνίαν ὑποτείνουσας πλευρᾶς ε>ἰδοξ >ἴσον ἐστὶ τοῖς ἀπὸ τῶν τὴν ὀρθὴν γωνίαν περιθεουσῶν πλευρῶν ε>ἰδεσι τοῖς ὁμοίοις τε καὶ ὁμοίως ἀναγραφόμενοις; <όπερ >έδει δεῖχαι.

Let ABC be a right-angled triangle having the angle BAC a right-angle. I say that the figure (drawn) on BC is equal to the (sum of the) similar, and similarly described, figures on BA and AC .

Let the perpendicular AD be drawn [Prop. 1.12].

Therefore, since, in the right-angled triangle ABC , the (straight-line) AD has been drawn from the right-angle at A perpendicular to the base BC , the triangles ABD and ADC about the perpendicular are similar to the whole (triangle) ABC , and to one another [Prop. 6.8]. And since ABC is similar to ABD , thus as CB is to BA , so AB (is) to BD [Def. 6.1]. And since three straight-lines are proportional, as the first is to the third, so the figure (drawn) on the first is to the similar, and similarly described, (figure) on the second [Prop. 6.19 corr.]. Thus, as CB (is) to BD , so the figure (drawn) on CB (is) to the similar, and similarly described, (figure) on BA . And so, for the same (reasons), as BC (is) to CD , so the figure (drawn) on BC (is) to the (figure) on CA . Hence, also, as BC (is) to BD and DC , so the figure (drawn) on BC (is) to the (sum of the) similar, and similarly described, (figures) on BA and AC [Prop. 5.24]. And BC is equal to BD and DC . Thus, the figure (drawn) on BC (is) also equal to the (sum of the) similar, and similarly described, figures on BA and AC [Prop. 5.9].

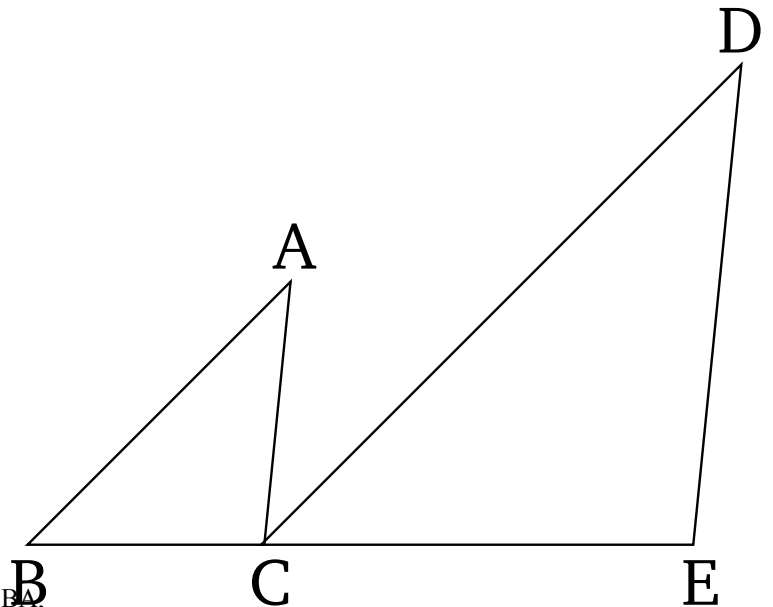
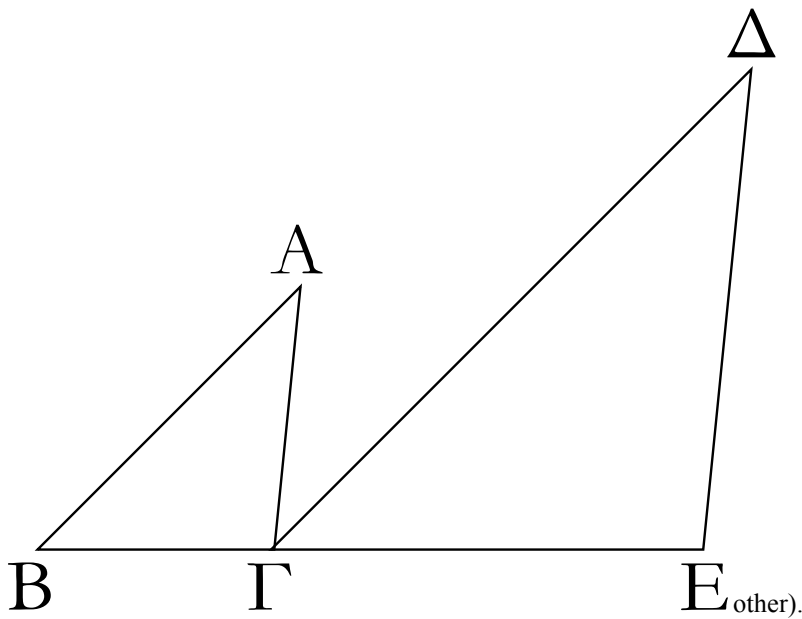
Thus, in right-angled triangles, the figure (drawn) on the side subtending the right-angle is equal to the (sum of the) similar, and similarly described, figures on the sides surrounding the right-angle. (Which is) the very thing it was required to show.

32.

Ἐὰν δύο τρίγωνα συντεθῇ κατὰ μίαν γωνίαν τὰς δύο πλευρὰς ταῖς δυσὶ πλευραῖς ἀνάλογον >έθοντα <ώστε τὰς ὁμολόγουξ αὐτῶν πλευρὰς καὶ παραλλήλουξ ε>ῖναι, αἱ λοιπαὶ τῶν τριγώνων πλευραὶ ἐπ> εὐθείας >έσσονται.

Proposition 32

If two triangles, having two sides proportional to two sides, are placed together at a single angle such that the corresponding sides are also parallel, (then) the remaining sides of the triangles will be straight-on (with respect to one an-



>Ἐστω δύο τρίγωνα τὰ $AB\Gamma$, $\Delta\Gamma\epsilon$ τὰς δύο πλευρὰς τὰς BA , $A\Gamma$ ταῖς δυσὶ πλευραῖς ταῖς $\Delta\Gamma$, $\Delta\epsilon$ ἀνάλογον >έθοντα, ὥς μὲν τὴν AB πρὸς τὴν $A\Gamma$, ο<ύτωξ τὴν $\Delta\Gamma$ πρὸς τὴν $\Delta\epsilon$, παράλληλον δὲ τὴν μὲν AB τῇ $\Delta\Gamma$, τὴν δὲ $A\Gamma$ τῇ $\Delta\epsilon$; λέγω, <ὅτι ἐπ> εὐθείας ἐστὶν ἡ $B\Gamma$ τῇ $\Gamma\epsilon$.

Ἐπεὶ γὰρ παράλληλός ἐστιν ἡ AB τῇ $\Delta\Gamma$, καὶ εἰς αὐτὰς ἐμπέπτωκεν εὐθεῖα ἡ $A\Gamma$, αἱ ἐναλλάχ γωνίαι αἱ ὑπὸ $BA\Gamma$, $A\Gamma\Delta$ >ίσαι ἀλλήλαις εἰσίν. διὰ τὰ αὐτὰ δὴ καὶ ἡ ὑπὸ $\Gamma\Delta\epsilon$ τῇ ὑπὸ $A\Gamma\Delta$ >ίση ἐστίν. <ώστε καὶ ἡ ὑπὸ $BA\Gamma$ τῇ ὑπὸ $\Gamma\Delta\epsilon$ ἐστὶν >ίση. καὶ ἐπεὶ δύο τρίγωνα ἐστὶ τὰ $AB\Gamma$, $\Delta\Gamma\epsilon$ μίαν γωνίαν τὴν πρὸς τῷ A μιᾷ γωνίᾳ τῇ πρὸς τῷ Δ >ίσην >έθοντα, περὶ δὲ τὰς >ίσαξ γωνίαξ τὰς πλευρὰξ ἀνάλογον, ὥς τὴν BA πρὸς τὴν $A\Gamma$, ο<ύτωξ τὴν $\Gamma\Delta$ πρὸς τὴν $\Delta\epsilon$, ἰσογώνιον >άρα ἐστὶ τὸ $AB\Gamma$ τρίγωνον τῷ $\Delta\Gamma\epsilon$ τριγώνῳ; >ίση >άρα ἡ ὑπὸ $AB\Gamma$ γωνία τῇ ὑπὸ $\Delta\Gamma\epsilon$. ἐδείκθη δὲ καὶ ἡ ὑπὸ $A\Gamma\Delta$ τῇ ὑπὸ $BA\Gamma$ >ίση; <όλη >άρα ἡ ὑπὸ $A\Gamma\epsilon$ δυσὶ ταῖς ὑπὸ $AB\Gamma$, $BA\Gamma$ >ίση ἐστίν. κοινὴ προσκείσθω ἡ ὑπὸ $A\Gamma\beta$; αἱ >άρα ὑπὸ $A\Gamma\epsilon$, $A\Gamma\beta$ ταῖς ὑπὸ $BA\Gamma$, $A\Gamma\beta$, $\Gamma\beta A$ >ίσαι εἰσίν. ἀλλ> αἱ ὑπὸ $BA\Gamma$, $AB\Gamma$, $A\Gamma\beta$ δυσὶν ὀρθαῖς >ίσαι εἰσίν; καὶ αἱ ὑπὸ $A\Gamma\epsilon$, $A\Gamma\beta$ >άρα δυσὶν ὀρθαῖς >ίσαι εἰσίν. πρὸξ δὴ τινὶ εὐθείᾳ τῇ $A\Gamma$ καὶ τῷ πρὸξ αὐτῇ σημείω τῷ Γ δύο εὐθεῖαι αἱ $B\Gamma$, $\Gamma\epsilon$ μὴ ἐπὶ τὰ αὐτὰ μέρη κείμεναι τὰς ἐφεχῆξ γωνίαξ τὰς ὑπὸ $A\Gamma\epsilon$, $A\Gamma\beta$ δυσὶν ὀρθαῖς >ίσαξ

Let ABC and DCE be two triangles having the two sides BA and AC proportional to the two sides DC and DE —so that as AB (is) to AC , so DC (is) to DE —and (having side) AB parallel to DC , and AC to DE . I say that (side) BC is straight-on to CE .

For since AB is parallel to DC , and the straight-line AC has fallen across them, the alternate angles BAC and ACD are equal to one another [Prop. 1.29]. So, for the same (reasons), CDE is also equal to ACD . And, hence, BAC is equal to CDE . And since ABC and DCE are two triangles having the one angle at A equal to the one angle at D , and the sides about the equal angles proportional, (so that) as BA (is) to AC , so CD (is) to DE , triangle ABC is thus equiangular to triangle DCE [Prop. 6.6]. Thus, angle ABC is equal to DCE . And (angle) ACD was also shown (to be) equal to BAC . Thus, the whole (angle) ACE is equal to the two (angles) ABC and BAC . Let ACB be added to both. Thus, ACE and ACB are equal to BAC , ACB , and CBA . But, BAC , ABC , and ACB are equal to two right-angles [Prop. 1.32]. Thus, ACE and ACB are also equal to two right-angles. Thus, the two straight-lines BC and CE , not lying on the same side, make adjacent angles

ποιοῦσιν; ἐπ' εὐθείᾳ >ἀρα ἐστὶν ἡ ΒΓ τῇ ΓΕ.

Ἐὰν >ἀρα δύο τρίγωνα συντεθῇ κατὰ μίαν γωνίαν τὰς δύο πλευρὰς ταῖς δυὶ πλευραῖς ἀνάλογον >έθοντα <ώστε τὰς ὁμολόγουσιν αὐτῶν πλευρὰς καὶ παραλλήλους ε>ῖναι, αἱ λοιπαὶ τῶν τριγώνων πλευραὶ ἐπ' εὐθείᾳ >έσσονται; <όπερ >έδει δεῖξαι.

ACE and ACB (whose sum is) equal to two right-angles with some straight-line AC , at the point C on it. Thus, BC is straight-on to CE [Prop. 1.14].

Thus, if two triangles, having two sides proportional to two sides, are placed together at a single angle such that the corresponding sides are also parallel, (then) the remaining sides of the triangles will be straight-on (with respect to one another). (Which is) the very thing it was required to show.

33.

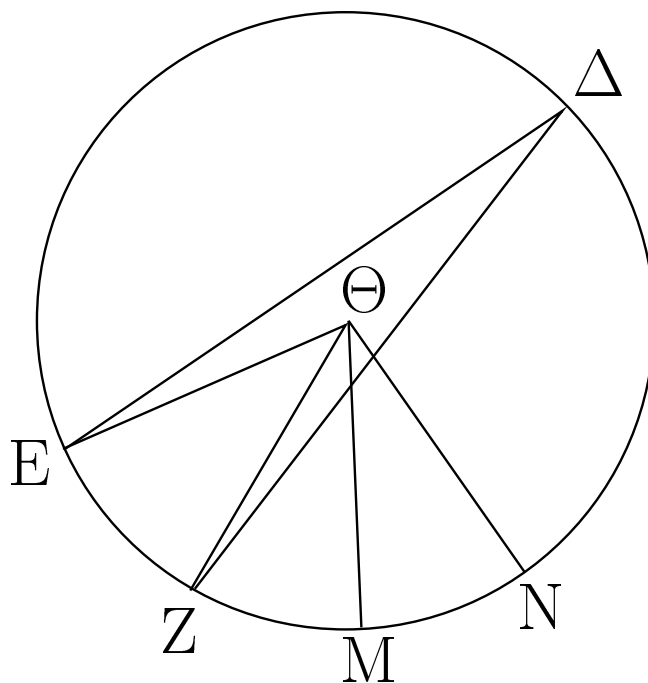
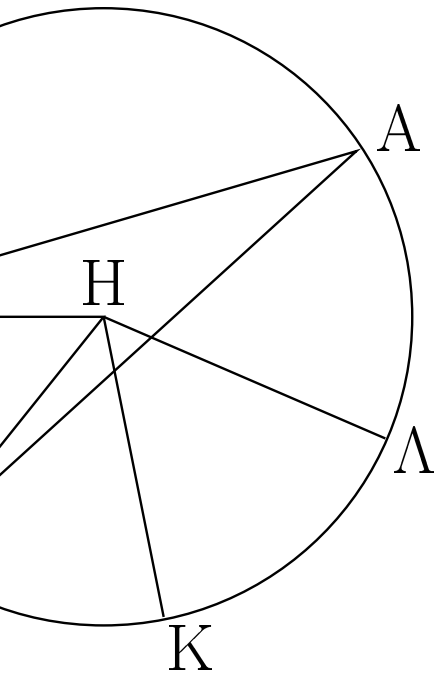
Ἐν τοῖς >ῖσοις κύκλοις αἱ γωνίαι τὸν αὐτὸν >έθουσι λόγον ταῖς περιφερείαις, ἐφ' <ῶν βεβήκασιν, ἔαν τε πρὸς τοῖς κέντροις ἔαν τε πρὸς ταῖς περιφερείαις >ῶσι βεβηκυῖαι.

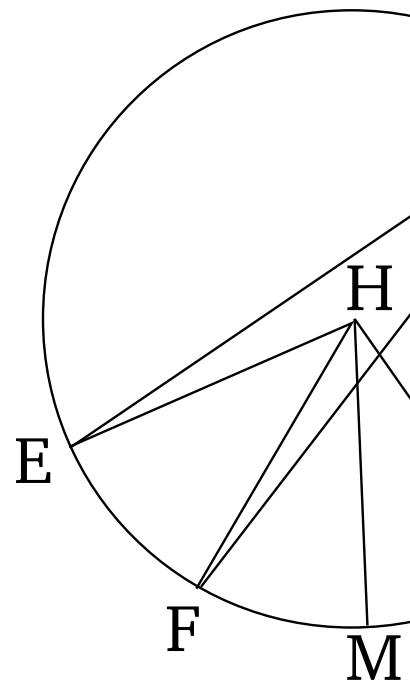
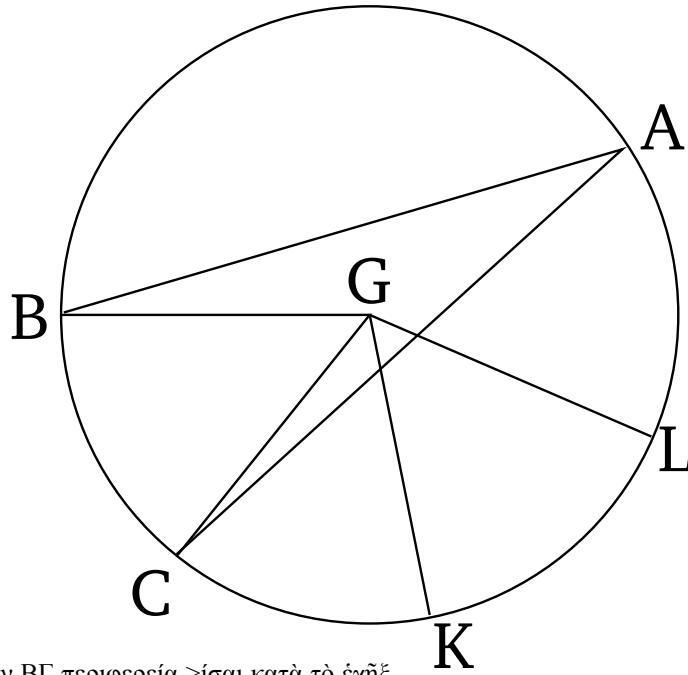
>Ἐστῶσαν >ῖσοι κύκλοι οἱ ΑΒΓ, ΔΕΖ, καὶ πρὸς μὲν τοῖς κέντροις αὐτῶν τοῖς Η, Θ γωνίαι >έστωσαν αἱ ὑπὸ ΒΗΓ, ΕΘΖ, πρὸς δὲ ταῖς περιφερείαις αἱ ὑπὸ ΒΑΓ, ΕΔΖ; λέγω, <ὅτι ἐστὶν ὡς ἡ ΒΓ περιφέρεια πρὸς τὴν ΕΖ περιφέρειαν, οὕτως <ἡ τε ὑπὸ ΒΗΓ γωνία πρὸς τὴν ὑπὸ ΕΘΖ καὶ ἡ ὑπὸ ΒΑΓ πρὸς τὴν ὑπὸ ΕΔΖ.

Proposition 33

In equal circles, angles have the same ratio as the (ratio of the) circumferences on which they stand, whether they are standing at the centers (of the circles) or at the circumferences.

Let ABC and DEF be equal circles, and let BGC and EHF be angles at their centers, G and H (respectively), and BAC and EDF (angles) at their circumferences. I say that as circumference BC is to circumference EF , so angle BGC (is) to EHF , and (angle) BAC to EDF .





Κείσθωσαν γὰρ τῇ μὲν ΒΓ περιφερεία >ῖσαι κατὰ τὸ ἐχῆξ ὁσαιδηποτοῦν αἱ ΓΚ, ΚΛ, τῇ δὲ ΕΖ περιφερεία >ῖσαι ὁσαιδηποτοῦν αἱ ΖΜ, ΜΝ, καὶ ἐπεζεύθωσαν αἱ ΗΚ, ΗΛ, ΘΜ, ΘΝ.

Ἐπεὶ οὖν >ῖσαι εἰσὶν αἱ ΒΓ, ΓΚ, ΚΛ περιφέρειαι ἀλλήλαις, >ῖσαι εἰσὶ καὶ αἱ ὑπὸ ΒΗΓ, ΓΗΚ, ΚΗΛ γωνίαι ἀλλήλαις, ὁσαπλασίων >ῖσαι ἐστὶν ἡ ΒΛ περιφέρεια τῆς ΒΓ, τοσαυταπλασίων ἐστὶ καὶ ἡ ὑπὸ ΒΗΛ γωνία τῆς ὑπὸ ΒΗΓ. διὰ τὰ αὐτὰ δὴ καὶ ὁσαπλασίων ἐστὶν ἡ ΝΕ περιφέρεια τῆς ΕΖ, τοσαυταπλασίων ἐστὶ καὶ ἡ ὑπὸ ΝΘΕ γωνία τῆς ὑπὸ ΕΘΖ. εἰ >ῖσαι ἐστὶν ἡ ΒΛ περιφέρεια τῇ ΕΝ περιφερείᾳ, >ῖση ἐστὶ καὶ γωνία ἡ ὑπὸ ΒΗΛ τῇ ὑπὸ ΕΘΝ, καὶ εἰ μείζων ἐστὶν ἡ ΒΛ περιφέρεια τῆς ΕΝ περιφερείας, μείζων ἐστὶ καὶ ἡ ὑπὸ ΒΗΛ γωνία τῆς ὑπὸ ΕΘΝ, καὶ εἰ ἐλάσσων, ἐλάσσων. τεσσάρων δὲ ὄντων μεγεθῶν, δύο μὲν περιφερειῶν τῶν ΒΓ, ΕΖ, δύο δὲ γωνιῶν τῶν ὑπὸ ΒΗΓ, ΕΘΖ, ἐκίληται τῇ μὲν ΒΓ περιφερείᾳ καὶ τῇ ὑπὸ ΒΗΓ γωνίᾳ ἰσάκις πολλαπλασίων <ῖ τε ΒΛ περιφέρεια καὶ ἡ ὑπὸ ΒΗΛ γωνία, τῆς δὲ ΕΖ περιφερείας καὶ τῆς ὑπὸ ΕΘΖ γωνίας <ῖ τε ΕΝ περιφέρεια καὶ ἡ ὑπὸ ΕΘΝ γωνία. καὶ δέδεικται, <ὅτι εἰ ὑπερέθει ἡ ΒΛ περιφέρεια τῆς ΕΝ περιφερείας, ὑπερέθει καὶ ἡ ὑπὸ ΒΗΛ γωνία τῆς ὑπὸ ΕΘΝ γωνίας, καὶ εἰ >ῖση, >ῖση, καὶ εἰ ἐλάσσων, ἐλάσσων. >ῖ ἐστιν >ῖσαι, ὥς ἡ ΒΓ περιφέρεια πρὸς τὴν ΕΖ, οὕτως ἡ ὑπὸ ΒΗΓ γωνία πρὸς τὴν ὑπὸ ΕΘΖ. ἀλλ' ὥς ἡ ὑπὸ ΒΗΓ γωνία πρὸς τὴν ὑπὸ ΕΘΖ, οὕτως ἡ ὑπὸ ΒΑΓ πρὸς τὴν ὑπὸ ΕΔΖ. διπλασία γὰρ ἑκάτερα ἑκάτερας. καὶ ὥς >ῖσαι ἡ ΒΓ περιφέρεια πρὸς τὴν ΕΖ περιφέρειαν, οὕτως <ῖ τε ὑπὸ ΒΗΓ γωνία πρὸς τὴν ὑπὸ ΕΘΖ καὶ ἡ ὑπὸ ΒΑΓ πρὸς τὴν ὑπὸ ΕΔΖ.

Ἐν >ῖσαι τοῖς >ῖσαι κύκλοις αἱ γωνίαι τὸν αὐτὸν >ῖθουσι λόγον ταῖς περιφερείαις, ἐφ' ὧν βεβήκασιν, ἔαν τε πρὸς τοῖς κέντροις ἔαν τε πρὸς ταῖς περιφερείαις >ῖσι βεβηκυῖαι, <ὅπερ >ῖδει δεῖχαι.

For let any number whatsoever of consecutive (circumferences), CK and KL , be made equal to circumference BC , and any number whatsoever, FM and MN , to circumference EF . And let GK , GL , HM , and HN be joined.

Therefore, since circumferences BC , CK , and KL are also equal to one another, angles BGC , CGK , and KGL are also equal to one another [Prop. 3.27]. Thus, as many times as circumference BL is (divisible) by BC , so many times is angle BGL also (divisible) by BGC . And so, for the same (reasons), as many times as circumference NE is (divisible) by EF , so many times is angle NHE also (divisible) by EHF . Thus, if circumference BL is equal to circumference EN (then) angle BGL is also equal to EHN [Prop. 3.27], and if circumference BL is greater than circumference EN (then) angle BGL is also greater than EHN ,[†] and if (BL is) less (than EN then BGL is also) less (than EHN). So there are four magnitudes, two circumferences BC and EF , and two angles BGC and EHF . And equal multiples have been taken of circumference BC and angle BGC , (namely) circumference BL and angle BGL , and of circumference EF and angle EHF , (namely) circumference EN and angle EHN . And it has been shown that if circumference BL exceeds circumference EN (then) angle BGL also exceeds angle EHN , and if (BL is) equal (to EN then BGL is also) equal (to EHN), and if (BL is) less (than EN then BGL is also) less (than EHN). Thus, as circumference BC (is) to EF , so angle BGC (is) to EHF [Def. 5.5]. But as angle BGC (is) to EHF , so (angle) BAC (is) to EDF [Prop. 5.15]. For the former (are) double the latter (respectively) [Prop. 3.20]. Thus, also, as circumference BC (is) to circumference EF , so angle BGC (is) to EHF , and BAC to EDF .

Thus, in equal circles, angles have the same ratio as the (ratio of the) circumferences on which they stand, whether they are standing at the centers (of the circles) or at the circumferences. (Which is) the very thing it was required to show.

[†] This is a straight-forward generalization of Prop. 3.27

ELEMENTS BOOK 7

Elementary Number Theory[†]

[†]The propositions contained in Books 7–9 are generally attributed to the school of Pythagoras.

<Ὅροι.

1. Μονὰξ ἔστιν, καθ' <ὣν <έκαστον τῶν >όντων <ἐν λέγεται.
2. Ἀριθμὸς δὲ τὸ ἐκ μονάδων συγκείμενον πλῆθος.
3. Μέρος ἔστιν ἀριθμὸς ἀριθμοῦ ὁ ἐλάσσων τοῦ μείζονος, <ὅταν καταμετρήῃ τὸν μείζονα.
4. Μέρη δέ, <ὅταν μὴ καταμετρήῃ.
5. Πολλαπλάσιοξ δὲ ὁ μείζων τοῦ ἐλάσσονος, <ὅταν καταμετρήται ὑπὸ τοῦ ἐλάσσονος.
6. >Ἀρτιοξ ἀριθμὸς ἔστιν ὁ δίθῃ διαιρούμενος.
7. Περισσὸς δὲ ὁ μὴ διαιρούμενος δίθῃ >ἢ [ὁ] μονάδι διαφέρων ἀρτίου ἀριθμοῦ.
8. Ἀρτιάκιξ >ἀρτιοξ ἀριθμὸς ἔστιν ὁ ὑπὸ ἀρτίου ἀριθμοῦ μετρούμενος κατὰ >ἀρτίον ἀριθμόν.
9. >Ἀρτιάκιξ δὲ περισσὸς ἔστιν ὁ ὑπὸ ἀρτίου ἀριθμοῦ μετρούμενος κατὰ περισσὸν ἀριθμόν.
10. Περισσάκιξ δὲ περισσὸς ἀριθμὸς ἔστιν ὁ ὑπὸ περισσοῦ ἀριθμοῦ μετρούμενος κατὰ περισσὸν ἀριθμόν.
11. Πρῶτοξ ἀριθμὸς ἔστιν ὁ μονάδι μόνῃ μετρούμενος.
12. Πρῶτοι πρὸς ἀλλήλους ἀριθμοὶ εἰσιν οἱ μονάδι μόνῃ μετρούμενοι κοινῶ| μέτρῳ.
13. Σύνθετοξ ἀριθμὸς ἔστιν ὁ ἀριθμῶ| τινι μετρούμενος.
14. Σύνθετοι δὲ πρὸς ἀλλήλους ἀριθμοὶ εἰσιν οἱ ἀριθμῶ| τινι μετρούμενοι κοινῶ| μέτρῳ.
15. Αἱ ἀριθμὸς ἀριθμὸν πολλαπλασιάζειν λέγεται, <ὅταν, <όσαι εἰσιν ἐν αὐτῶ| μονάδεξ, τοσαυτάκιξ συντεθῇ| ὁ πολλαπλασιαζόμενος, καὶ γένηται τιξ.
16. <Ὅταν δὲ δύο ἀριθμοὶ πολλαπλασιάσαντεξ ἀλλήλους ποιῶσιν τινα, ὁ γενόμενος ἐπίπεδοξ καλεῖται, πλευραὶ δὲ αὐτοῦ οἱ πολλαπλασιάσαντεξ ἀλλήλους ἀριθμοί.
17. <Ὅταν δὲ τρεῖς ἀριθμοὶ πολλαπλασιάσαντεξ ἀλλήλους ποιῶσιν τινα, ὁ γενόμενος στερεὸς ἔστιν, πλευραὶ δὲ αὐτοῦ οἱ πολλαπλασιάσαντεξ ἀλλήλους ἀριθμοί.
18. Τετράγωνος ἀριθμὸς ἔστιν ὁ ἰσάκιξ >ἴσοξ >ἢ [ὁ] ὑπὸ δύο >ἴσων ἀριθμῶν περιεθόμενος.
19. Κύβοξ δὲ ὁ ἰσάκιξ >ἴσοξ ἰσάκιξ >ἢ [ὁ] ὑπὸ τριῶν >ἴσων ἀριθμῶν περιεθόμενος.
20. Αἱ ἀριθμοὶ ἀνάλογόν εἰσιν, <ὅταν ὁ πρῶτοξ τοῦ δευτέρου καὶ ὁ τρίτοξ τοῦ τετάρτου ἰσάκιξ >ἢ| πολλαπλάσιοξ >ἢ τὸ αὐτὸ μέρος >ἢ τὰ αὐτὰ μέρη >ῶσιν.
21. <Ὅμοιοι ἐπίπεδοι καὶ στερεοὶ ἀριθμοὶ εἰσιν οἱ ἀνάλογον >έθοντεξ τὰξ πλευράξ.
22. Τέλειος ἀριθμὸς ἔστιν ὁ τοῖξ ἑαυτοῦ μέρεσιν >ἴσοξ >ῶν.

Definitions

1. A unit is (that) according to which each existing (thing) is said (to be) one.
2. And a number (is) a multitude composed of units.[†]
3. A number is part of a(nother) number, the lesser of the greater, when it measures the greater.[‡]
4. But (the lesser is) parts (of the greater) when it does not measure it.[§]
5. And the greater (number is) a multiple of the lesser when it is measured by the lesser.
6. An even number is one (which can be) divided in half.
7. And an odd number is one (which can)not (be) divided in half, or which differs from an even number by a unit.
8. An even-times-even number is one (which is) measured by an even number according to an even number.[¶]
9. And an even-times-odd number is one (which is) measured by an even number according to an odd number.*
10. And an odd-times-odd number is one (which is) measured by an odd number according to an odd number.[§]
11. A prime[‡] number is one (which is) measured by a unit alone.
12. Numbers prime to one another are those (which are) measured by a unit alone as a common measure.
13. A composite number is one (which is) measured by some number.
14. And numbers composite to one another are those (which are) measured by some number as a common measure.
15. A number is said to multiply a(nother) number when the (number being) multiplied is added (to itself) as many times as there are units in the former (number), and (thereby) some (other number) is produced.
16. And when two numbers multiplying one another make some (other number then) the (number so) created is called plane, and its sides (are) the numbers which multiply one another.
17. And when three numbers multiplying one another make some (other number then) the (number so) created is (called) solid, and its sides (are) the numbers which multiply one another.
18. A square number is an equal times an equal, or (a plane number) contained by two equal numbers.
19. And a cube (number) is an equal times an equal times an equal, or (a solid number) contained by three equal numbers.
20. Numbers are proportional when the first is the same multiple, or the same part, or the same parts, of the second that the third (is) of the fourth.
21. Similar plane and solid numbers are those having proportional sides.
22. A perfect number is that which is equal to its own parts.^{††}

[†] In other words, a “number” is a positive integer greater than unity.

[‡] In other words, a number a is part of another number b if there exists some number n such that $na = b$.

[§] In other words, a number a is parts of another number b (where $a < b$) if there exist distinct numbers, m and n , such that $na = mb$.

[¶] In other words, an even-times-even number is the product of two even numbers.

* In other words, an even-times-odd number is the product of an even and an odd number.

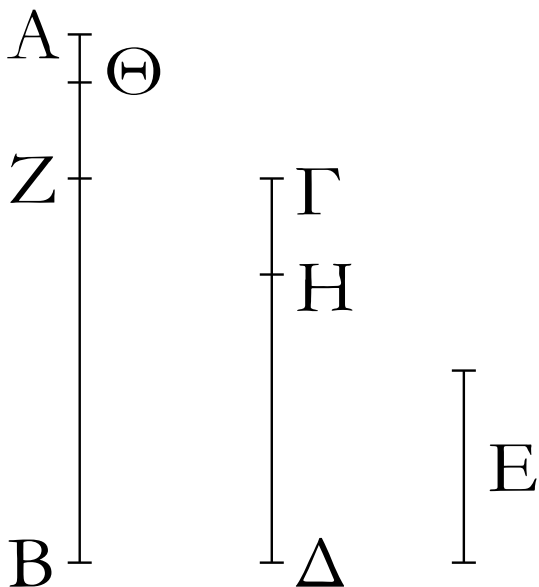
[§] In other words, an odd-times-odd number is the product of two odd numbers.

[‡] Literally, “first”.

†† In other words, a perfect number is equal to the sum of its own factors.

1.

Δύο ἀριθμῶν ἀνίσων ἐκκειμένων, ἀνθυφαιρουμένου δὲ αἰ τοῦ ἐλάσσονος ἀπὸ τοῦ μείζονος, ἔαν ὁ λειπόμενος μηδέποτε καταμετρήῃ τὸν πρὸ ἑαυτοῦ, <ἐὼς οὐ λειφθῇ| μονάξ, οἱ ἐχ' ἀρθῆξ' ἀριθμοὶ πρῶτοι πρὸς ἀλλήλους> ἐσονται.



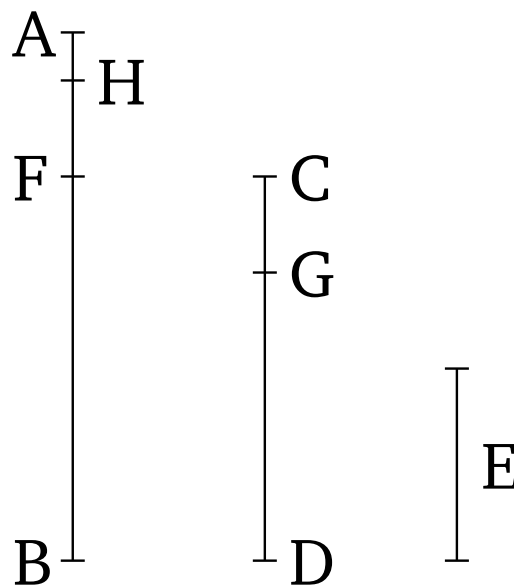
Δύο γὰρ [ἀνίσων] ἀριθμῶν τῶν AB , $\Gamma\Delta$ ἀνθυφαιρουμένου αἰ τοῦ ἐλάσσονος ἀπὸ τοῦ μείζονος ὁ λειπόμενος μηδέποτε καταμετρεῖται τὸν πρὸ ἑαυτοῦ, <ἐὼς οὐ λειφθῇ| μονάξ; λέγω, >ὅτι οἱ AB , $\Gamma\Delta$ πρῶτοι πρὸς ἀλλήλους εἰσίν, τουτέστιν <ὅτι τοὺς AB , $\Gamma\Delta$ μονάξ μόνη μετρεῖ.

Εἰ γὰρ μὴ εἰσιν οἱ AB , $\Gamma\Delta$ πρῶτοι πρὸς ἀλλήλους, μετρήσει τις αὐτοὺς ἀριθμός. μετρεῖται, καὶ >έστω ὁ E ; καὶ ὁ μὲν $\Gamma\Delta$ τὸν BZ μετρῶν λειπέτω ἑαυτοῦ ἐλάσσονα τὸν ZA , ὁ δὲ AZ τὸν ΔH μετρῶν λειπέτω ἑαυτοῦ ἐλάσσονα τὸν $H\Gamma$, ὁ δὲ $H\Gamma$ τὸν $Z\Theta$ μετρῶν λειπέτω μονάδα τὴν ΘA .

Ἐπεὶ οὐν ὁ E τὸν $\Gamma\Delta$ μετρεῖ, ὁ δὲ $\Gamma\Delta$ τὸν BZ μετρεῖ, καὶ ὁ E >άρα τὸν BZ μετρεῖ; μετρεῖ δὲ καὶ <όλον τὸν BA ; καὶ λοιπὸν >άρα τὸν AZ μετρήσει. ὁ δὲ AZ τὸν ΔH μετρεῖ; καὶ ὁ E >άρα τὸν ΔH μετρεῖ; μετρεῖ δὲ καὶ <όλον τὸν $\Delta\Gamma$; καὶ λοιπὸν >άρα τὸν ΓH μετρήσει. ὁ δὲ ΓH τὸν $Z\Theta$ μετρεῖ; καὶ ὁ E >άρα τὸν $Z\Theta$ μετρεῖ; μετρεῖ δὲ καὶ <όλον τὸν ZA ; καὶ λοιπὴν >άρα τὴν $A\Theta$ μονάδα μετρήσει ἀριθμὸς >ών; <όπερ ἐστὶν ἀδύνατον. οὐκ >άρα τοὺς AB , $\Gamma\Delta$ ἀριθμοὺς μετρήσει τις ἀριθμός; οἱ AB , $\Gamma\Delta$ >άρα πρῶτοι πρὸς ἀλλήλους εἰσίν; <όπερ >έδει δεῖχαι.

Proposition 1

Two unequal numbers (being) laid down, and the lesser being continually subtracted, in turn, from the greater, if the remainder never measures the (number) preceding it, until a unit remains, (then) the original numbers will be prime to one another.



For two [unequal] numbers, AB and CD , the lesser being continually subtracted, in turn, from the greater, let the remainder never measure the (number) preceding it, until a unit remains. I say that AB and CD are prime to one another—that is to say, that a unit alone measures (both) AB and CD .

For if AB and CD are not prime to one another, (then) some number will measure them. Let (some number) measure them, and let it be E . And let CD measuring BF leave FA less than itself, and let AF measuring DG leave GC less than itself, and let GC measuring FH leave a unit, HA .

In fact, since E measures CD , and CD measures BF , E thus also measures BF .[†] And (E) also measures the whole of BA . Thus, (E) will also measure the remainder AF .[‡] And AF measures DG . Thus, E also measures DG . And (E) also measures the whole of DC . Thus, (E) will also measure the remainder CG . And CG measures FH . Thus, E also measures FH . And (E) also measures the whole of FA . Thus, (E) will also measure the remaining unit HA , (despite) being a number. The very thing is impossible. Thus, some number does not measure (both) the numbers AB and CD . Thus, AB and CD are prime to one another. (Which is) the very thing it was required to show.

[†] Here, use is made of the unstated common notion that if a measures b , and b measures c , then a also measures c , where all symbols denote numbers.

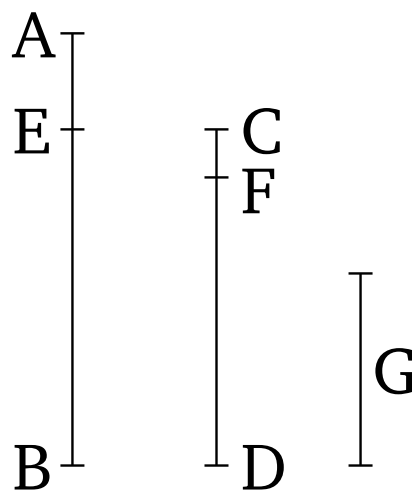
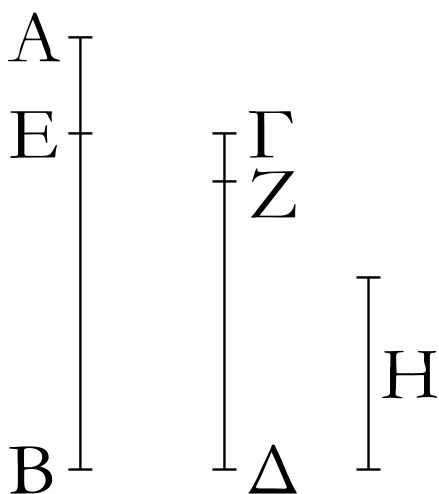
[‡] Here, use is made of the unstated common notion that if a measures b , and a measures part of b , then a also measures the remainder of b , where all symbols denote numbers.

2.

Δύο ἀριθμῶν δοθέντων μὴ πρώτων πρὸς ἀλλήλους τὸ μέγιστον αὐτῶν κοινὸν μέτρον εὑρεῖν.

Proposition 2

To find the greatest common measure of two given numbers (which are) not prime to one another.



>Ἐστωσαν οἱ δοθέντες δύο ἀριθμοὶ μὴ πρῶτοι πρὸς ἀλλήλους οἱ AB , $\Gamma\Delta$. δεῖ δὴ τῶν AB , $\Gamma\Delta$ τὸ μέγιστον κοινὸν μέτρον εὑρεῖν.

Εἰ μὲν οὖν ὁ $\Gamma\Delta$ τὸν AB μετρεῖ, μετρεῖ δὲ καὶ ἑαυτόν, ὁ $\Gamma\Delta$ >ἄρα τῶν $\Gamma\Delta$, AB κοινὸν μέτρον ἐστίν. καὶ φανερόν, <ὅτι καὶ μέγιστον; οὐδεὶς γὰρ μείζων τοῦ $\Gamma\Delta$ τὸν $\Gamma\Delta$ μετρήσει.

Εἰ δὲ οὐ μετρεῖ ὁ $\Gamma\Delta$ τὸν AB , τῶν AB , $\Gamma\Delta$ ἀνθυφαίρουμένου αἰ τοῦ ἐλάσσονος ἀπὸ τοῦ μείζονος λειφθήσεται τις ἀριθμὸς, <ὃς μετρήσει τὸν πρὸ ἑαυτοῦ. μονὰς μὲν γὰρ οὐ λειφθήσεται; εἰ δὲ μή, >έσονται οἱ AB , $\Gamma\Delta$ πρῶτοι πρὸς ἀλλήλους; <ὅπερ οὐθ' ὑπόκειται. λειφθήσεται τις >ἄρα ἀριθμὸς, <ὃς μετρήσει τὸν πρὸ ἑαυτοῦ. καὶ ὁ μὲν $\Gamma\Delta$ τὸν BE μετρῶν λειπέτω ἑαυτοῦ ἐλάσσονα τὸν EA , ὁ δὲ EA τὸν ΔZ μετρῶν λειπέτω ἑαυτοῦ ἐλάσσονα τὸν $Z\Gamma$, ὁ δὲ ΓZ τὸν AE μετρεῖτω. ἐπεὶ οὖν ὁ ΓZ τὸν AE μετρεῖ, ὁ δὲ AE τὸν ΔZ μετρεῖ, καὶ ὁ ΓZ >ἄρα τὸν ΔZ μετρήσει. μετρεῖ δὲ καὶ ἑαυτόν; καὶ <ὅλον >ἄρα τὸν $\Gamma\Delta$ μετρήσει. ὁ δὲ $\Gamma\Delta$ τὸν BE μετρεῖ; καὶ ὁ ΓZ >ἄρα τὸν BE μετρεῖ; μετρεῖ δὲ καὶ τὸν EA ; καὶ <ὅλον >ἄρα τὸν BA μετρήσει; μετρεῖ δὲ καὶ τὸν $\Gamma\Delta$; ὁ ΓZ >ἄρα τοὺς AB , $\Gamma\Delta$ μετρεῖ. ὁ ΓZ >ἄρα τῶν AB , $\Gamma\Delta$ κοινὸν μέτρον ἐστίν. λέγω δὴ, <ὅτι καὶ μέγιστον. εἰ γὰρ μή ἐστιν ὁ ΓZ τῶν AB , $\Gamma\Delta$ μέγιστον κοινὸν μέτρον, μετρήσει τις τοὺς AB , $\Gamma\Delta$ ἀριθμοὺς ἀριθμὸς μείζων >ὢν τοῦ ΓZ . μετρεῖτω, καὶ >έστω ὁ H . καὶ ἐπεὶ ὁ H τὸν $\Gamma\Delta$ μετρεῖ, ὁ δὲ $\Gamma\Delta$ τὸν BE μετρεῖ, καὶ ὁ H >ἄρα τὸν BE μετρεῖ; μετρεῖ δὲ καὶ <ὅλον τὸν BA ; καὶ λοιπὸν >ἄρα τὸν AE μετρήσει. ὁ δὲ AE τὸν ΔZ μετρεῖ; καὶ ὁ H >ἄρα τὸν ΔZ μετρήσει; μετρεῖ δὲ καὶ <ὅλον τὸν $\Delta\Gamma$; καὶ λοιπὸν >ἄρα τὸν ΓZ μετρήσει ὁ μείζων τὸν ἐλάσσονα; <ὅπερ ἐστὶν ἀδύνατον; οὐκ >ἄρα τοὺς AB , $\Gamma\Delta$ ἀριθμοὺς ἀριθμὸς τις μετρήσει μείζων >ὢν τοῦ ΓZ ; ὁ ΓZ >ἄρα τῶν AB , $\Gamma\Delta$ μέγιστόν ἐστι κοινὸν μέτρον [<ὅπερ >έδει δεῖχαι].

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, <ὅτι ἕαν ἀριθμὸς δύο ἀριθμοὺς μετρήῃ, καὶ τὸ μέγιστον αὐτῶν κοινὸν μέτρον μετρήσῃ; <ὅπερ >έδει δεῖχαι.

Let AB and CD be the two given numbers (which are) not prime to one another. So it is required to find the greatest common measure of AB and CD .

In fact, if CD measures AB , CD is thus a common measure of CD and AB , (since CD) also measures itself. And (it is) manifest that (it is) also the greatest (common measure). For nothing greater than CD can measure CD .

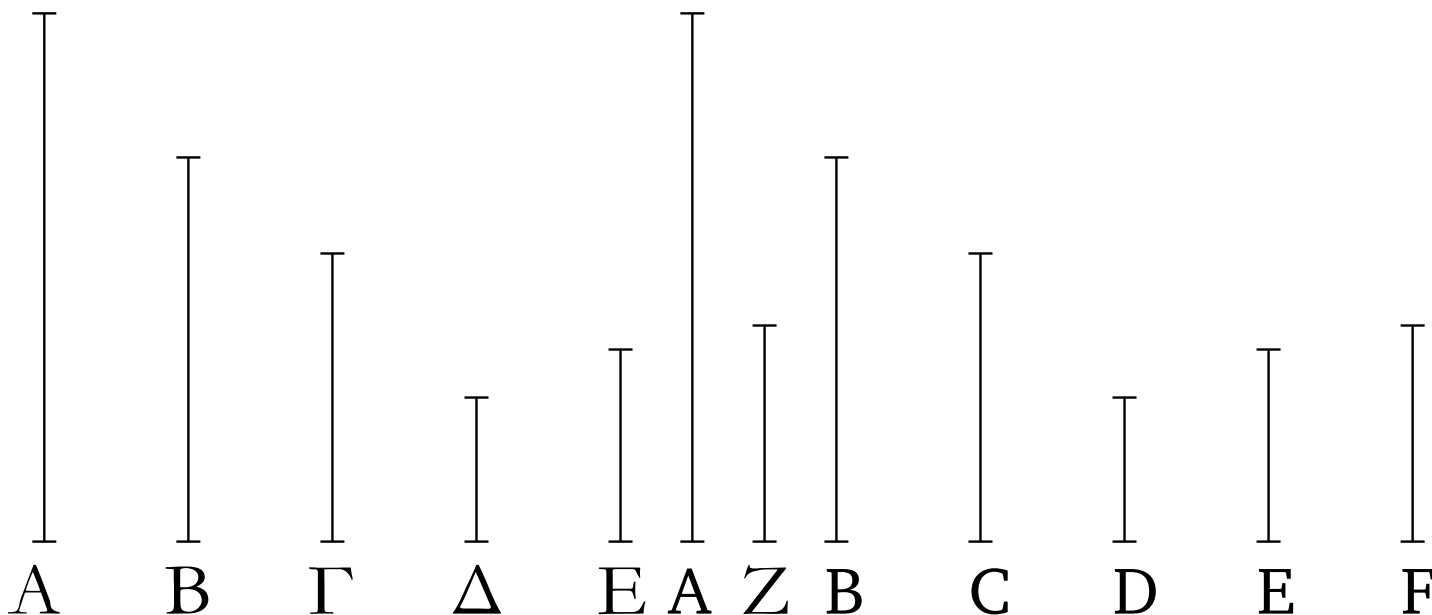
But if CD does not measure AB , (then) some number will remain from AB and CD , the lesser being continually subtracted, in turn, from the greater, which will measure the (number) preceding it. For a unit will not be left. But if not, AB and CD will be prime to one another [Prop. 7.1]. The very opposite thing was assumed. Thus, some number will remain which will measure the (number) preceding it. And let CD measuring BE leave EA less than itself, and let EA measuring DF leave FC less than itself, and let CF measure AE . Therefore, since CF measures AE , and AE measures DF , CF will thus also measure DF . And it also measures itself. Thus, it will also measure the whole of CD . And CD measures BE . Thus, CF also measures BE . And it also measures EA . Thus, it will also measure the whole of BA . And it also measures CD . Thus, CF measures (both) AB and CD . Thus, CF is a common measure of AB and CD . So I say that (it is) also the greatest (common measure). For if CF is not the greatest common measure of AB and CD , (then) some number which is greater than CF will measure the numbers AB and CD . Let it (so) measure (AB and CD), and let it be G . And since G measures CD , and CD measures BE , G thus also measures BE . And it also measures the whole of BA . Thus, it will also measure the remainder AE . And AE measures DF . Thus, G will also measure DF . And it also measures the whole of DC . Thus, it will also measure the remainder CF , the greater (measuring) the lesser. The very thing is impossible. Thus, some number which is greater than CF cannot measure the numbers AB and CD . Thus, CF is the greatest common measure of AB and CD . [(Which is) the very thing it was required to show].

Corollary

So it is manifest, from this, that if a number measures two numbers, (then) it will also measure their greatest common measure. (Which is) the very thing it was required to show.

Τριῶν ἀριθμῶν δοθέντων μὴ πρώτων πρὸς ἀλλήλους τὸ μέγιστον αὐτῶν κοινὸν μέτρον εὑρεῖν.

To find the greatest common measure of three given numbers (which are) not prime to one another.



>Ἐστωσαν οἱ δοθέντες τρεῖς ἀριθμοὶ μὴ πρῶτοι πρὸς ἀλλήλους οἱ A, B, Γ ; δεῖ δὴ τῶν A, B, Γ τὸ μέγιστον κοινὸν μέτρον εὑρεῖν.

Εἰλήφθω γὰρ δύο τῶν A, B τὸ μέγιστον κοινὸν μέτρον ὁ Δ ; ὁ δὲ Δ τὸν Γ >ήτοι μετρεῖ >ἢ οὐ μετρεῖ. μετρεῖτω πρότερον; μετρεῖ δὲ καὶ τοὺς A, B ; ὁ Δ >άρα τοὺς A, B, Γ μετρεῖ; ὁ Δ >άρα τῶν A, B, Γ κοινὸν μέτρον ἐστίν. λέγω δὴ, <ὅτι καὶ μέγιστον. εἰ γὰρ μὴ ἐστὶν ὁ Δ τῶν A, B, Γ μέγιστον κοινὸν μέτρον, μετρήσει τις τοὺς A, B, Γ ἀριθμοὺς ἀριθμὸς μείζων >ὦν τοῦ Δ . μετρεῖτω, καὶ >έστω ὁ E . ἐπεὶ ο>ὐν ὁ E τοὺς A, B, Γ μετρεῖ, καὶ τοὺς A, B >άρα μετρήσει; καὶ τὸ τῶν A, B >άρα μέγιστον κοινὸν μέτρον μετρήσει. τὸ δὲ τῶν A, B μέγιστον κοινὸν μέτρον ἐστὶν ὁ Δ ; ὁ E >άρα τὸν Δ μετρεῖ ὁ μείζων τὸν ἐλάσσονα; <ὅπερ ἐστὶν ἀδύνατον. οὐκ >άρα τοὺς A, B, Γ ἀριθμοὺς ἀριθμὸς τις μετρήσει μείζων >ὦν τοῦ Δ ; ὁ Δ >άρα τῶν A, B, Γ μέγιστόν ἐστι κοινὸν μέτρον.

μὴ μετρεῖτω δὴ ὁ Δ τὸν Γ ; λέγω πρῶτον, <ὅτι οἱ Γ, Δ ο>ὐκ εἰσι πρῶτοι πρὸς ἀλλήλους. ἐπεὶ γὰρ οἱ A, B, Γ ο>ὐκ εἰσι πρῶτοι πρὸς ἀλλήλους, μετρήσει τις αὐτοὺς ἀριθμὸς. ὁ δὲ τοὺς A, B, Γ μετρῶν καὶ τοὺς A, B μετρήσει, καὶ τὸ τῶν A, B μέγιστον κοινὸν μέτρον τὸν Δ μετρήσει; μετρεῖ δὲ καὶ τὸν Γ ; τοὺς Δ, Γ >άρα ἀριθμοὺς ἀριθμὸς τις μετρήσει; οἱ Δ, Γ >άρα ο>ὐκ εἰσι πρῶτοι πρὸς ἀλλήλους. εἰλήφθω ο>ὐν αὐτῶν τὸ μέγιστον κοινὸν μέτρον ὁ E . καὶ ἐπεὶ ὁ E τὸν Δ μετρεῖ, ὁ δὲ Δ τοὺς A, B μετρεῖ, καὶ ὁ E >άρα τοὺς A, B μετρεῖ; μετρεῖ δὲ καὶ τὸν Γ ; ὁ E >άρα τοὺς A, B, Γ μετρεῖ. ὁ E >άρα τῶν A, B, Γ κοινὸν ἐστὶ μέτρον. λέγω δὴ, <ὅτι καὶ μέγιστον. εἰ γὰρ μὴ ἐστὶν ὁ E τῶν A, B, Γ τὸ μέγιστον κοινὸν μέτρον, μετρήσει τις τοὺς A, B, Γ ἀριθμοὺς ἀριθμὸς μείζων >ὦν τοῦ E . μετρεῖτω, καὶ >έστω ὁ Z . καὶ ἐπεὶ ὁ Z τοὺς A, B, Γ μετρεῖ, καὶ τοὺς A, B μετρεῖ; καὶ τὸ τῶν A, B >άρα μέγιστον κοινὸν μέτρον μετρήσει. τὸ δὲ τῶν A, B μέγιστον κοινὸν μέτρον ἐστὶν ὁ Δ ; ὁ Z >άρα τὸν Δ μετρεῖ; μετρεῖ δὲ καὶ τὸν Γ ; ὁ Z >άρα τοὺς Δ, Γ μετρεῖ; καὶ τὸ τῶν Δ, Γ >άρα μέγιστον κοινὸν μέτρον μετρήσει. τὸ δὲ τῶν Δ, Γ μέγιστον κοινὸν μέτρον ἐστὶν ὁ E ; ὁ Z >άρα τὸν E μετρεῖ ὁ μείζων τὸν ἐλάσσονα; <ὅπερ ἐστὶν ἀδύνατον. οὐκ >άρα τοὺς A, B, Γ ἀριθμοὺς ἀριθμὸς τις μετρήσει μείζων >ὦν τοῦ E ; ὁ E >άρα τῶν A, B, Γ μέγιστόν ἐστι κοινὸν μέτρον;

Let A, B , and C be the three given numbers (which are) not prime to one another. So it is required to find the greatest common measure of A, B , and C .

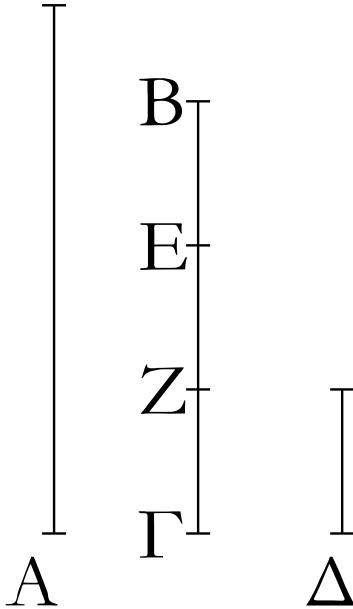
For let the greatest common measure, D , of the two (numbers) A and B be taken [Prop. 7.2]. So D either measures, or does not measure, C . First of all, let it measure (C). And it also measures A and B . Thus, D measures A, B , and C . Thus, D is a common measure of A, B , and C . So I say that (it is) also the greatest (common measure). For if D is not the greatest common measure of A, B , and C , (then) some number greater than D will measure the numbers A, B , and C . Let it (so) measure (A, B , and C), and let it be E . Therefore, since E measures A, B , and C , it will thus also measure A and B . Thus, it will also measure the greatest common measure of A and B [Prop. 7.2 corr.]. And D is the greatest common measure of A and B . Thus, E measures D , the greater (measuring) the lesser. The very thing is impossible. Thus, some number which is greater than D cannot measure the numbers A, B , and C . Thus, D is the greatest common measure of A, B , and C .

So let D not measure C . I say, first of all, that C and D are not prime to one another. For since A, B, C are not prime to one another, some number will measure them. So the (number) measuring A, B , and C will also measure A and B , and it will also measure the greatest common measure, D , of A and B [Prop. 7.2 corr.]. And it also measures C . Thus, some number will measure the numbers D and C . Thus, D and C are not prime to one another. Therefore, let their greatest common measure, E , be taken [Prop. 7.2]. And since E measures D , and D measures A and B , E thus also measures A and B . And it also measures C . Thus, E measures A, B , and C . Thus, E is a common measure of A, B , and C . So I say that (it is) also the greatest (common measure). For if E is not the greatest common measure of A, B , and C , (then) some number greater than E will measure the numbers A, B , and C . Let it (so) measure (A, B , and C), and let it be F . And since F measures A, B , and C , it also measures A and B . Thus, it will also measure the greatest common measure of A and B [Prop. 7.2 corr.].

<όπερ >έδει δεῖχαι.

4.

<'Απαξ ἀριθμὸς παντὸς ἀριθμοῦ ὁ ἐλάσσων τοῦ μείζονος >ήτοι μέρος ἐστὶν >ἢ μέρη.
>Ἐστῶσαν δύο ἀριθμοὶ οἱ Α, ΒΓ, καὶ >έστω ἐλάσσων ὁ ΒΓ; λέγω, <ὅτι ὁ ΒΓ τοῦ Α >ήτοι μέρος ἐστὶν >ἢ μέρη.
Οἱ Α, ΒΓ γὰρ >ήτοι πρῶτοι πρὸς ἀλλήλους εἰσὶν >ἢ ο>ύ.
>έστωσαν πρότερον οἱ Α, ΒΓ πρῶτοι πρὸς ἀλλήλους.
διαρθέντος δὴ τοῦ ΒΓ εἰς τὰς ἐν αὐτῷ μονάδας >έσται ἐκάστη μονὰς τῶν ἐν τῷ ΒΓ μέρος τι τοῦ Α; <ὥστε μέρος ἐστὶν ὁ ΒΓ τοῦ Α.



μὴ >έστωσαν δὴ οἱ Α, ΒΓ πρῶτοι πρὸς ἀλλήλους; ὁ δὲ ΒΓ τὸν Α >ήτοι μετρεῖ >ἢ οὐ μετρεῖ. εἰ μὲν ο>ὺν ὁ ΒΓ τὸν Α μετρεῖ, μέρος ἐστὶν ὁ ΒΓ τοῦ Α. εἰ δὲ ο>ύ, εἰλήφθω τῶν Α, ΒΓ μέγιστον κοινὸν μέτρον ὁ Δ, καὶ διηρήσθω ὁ ΒΓ εἰς τοὺς τῷ Δ >ίσοις τοὺς ΒΕ, ΕΖ, ΖΓ. καὶ ἐπεὶ ὁ Δ τὸν Α μετρεῖ, μέρος ἐστὶν ὁ Δ τοῦ Α; >ίσοις δὲ ὁ Δ ἐκάστῳ τῶν ΒΕ, ΕΖ, ΖΓ; καὶ <έκαστος >άρα τῶν ΒΕ, ΕΖ, ΖΓ τοῦ Α μέρος ἐστίν; <ὥστε μέρος ἐστὶν ὁ ΒΓ τοῦ Α.

<'Απαξ >άρα ἀριθμὸς παντὸς ἀριθμοῦ ὁ ἐλάσσων τοῦ μείζονος >ήτοι μέρος ἐστὶν >ἢ μέρη; <όπερ >έδει δεῖχαι.

5.

Ἐὰν ἀριθμὸς ἀριθμοῦ μέρος >ἢ, καὶ <έτερος ἑτέρου τὸ αὐτὸ μέρος >ἢ, καὶ συναμφοτέρως συναμφοτέρου τὸ αὐτὸ μέρος >έσται, <όπερ ὁ ε<ῖς τοῦ ἐνόξ.

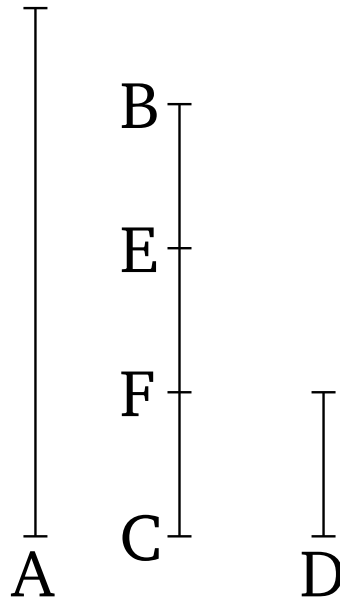
And D is the greatest common measure of A and B . Thus, F measures D . And it also measures C . Thus, F measures D and C . Thus, it will also measure the greatest common measure of D and C [Prop. 7.2 corr.]. And E is the greatest common measure of D and C . Thus, F measures E , the greater (measuring) the lesser. The very thing is impossible. Thus, some number which is greater than E does not measure the numbers A , B , and C . Thus, E is the greatest common measure of A , B , and C . (Which is) the very thing it was required to show.

Proposition 4

Any number is either part or parts of any (other) number, the lesser of the greater.

Let A and BC be two numbers, and let BC be the lesser. I say that BC is either part or parts of A .

For A and BC are either prime to one another, or not. Let A and BC , first of all, be prime to one another. So separating BC into its constituent units, each of the units in BC will be some part of A . Hence, BC is parts of A .

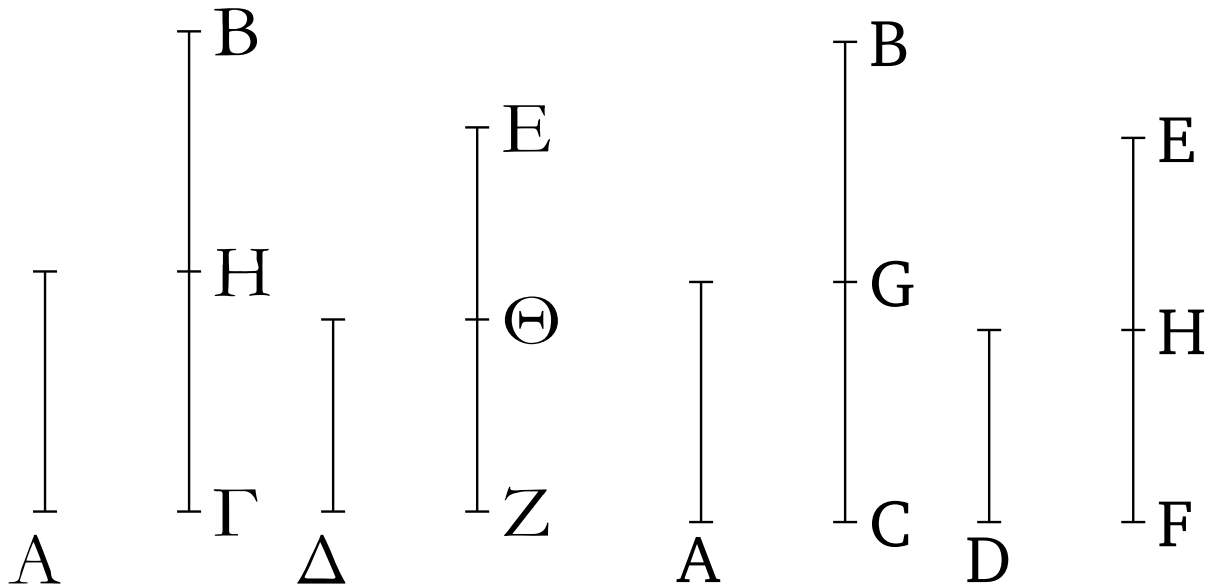


So let A and BC be not prime to one another. So BC either measures, or does not measure, A . Therefore, if BC measures A , (then) BC is part of A . And if not, let the greatest common measure, D , of A and BC be taken [Prop. 7.2], and let BC be divided into BE , EF , and FC , equal to D . And since D measures A , D is a part of A . And D is equal to each of BE , EF , and FC . Thus, BE , EF , and FC are also each part of A . Hence, BC is parts of A .

Thus, any number is either part or parts of any (other) number, the lesser of the greater. (Which is) the very thing it was required to show.

Proposition 5[†]

If a number is part of a number, and another (number) is the same part of another, (then) the sum (of the leading numbers) will also be the same part of the sum (of the following numbers) that one (number) is of another.



Ἀριθμὸς γὰρ ὁ A [ἀριθμοῦ] τοῦ $B\Gamma$ μέρος >έστω, καὶ <έτερος ὁ Δ ἑτέρου τοῦ EZ τὸ αὐτὸ μέρος, <όπερ ὁ A τοῦ $B\Gamma$; λέγω, <ὅτι καὶ συναμφοτέρος ὁ A , Δ συναμφοτέρου τοῦ $B\Gamma$, EZ τὸ αὐτὸ μέρος ἐστίν, <όπερ ὁ A τοῦ $B\Gamma$.

Ἐπεὶ γάρ, <ὁ μέρος ἐστὶν ὁ A τοῦ $B\Gamma$, τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ Δ τοῦ EZ , <όσοι >άρα εἰσὶν ἐν τῷ $B\Gamma$ ἀριθμοὶ >ῖσοι τῷ A , τοσοῦτοὶ εἰσὶ καὶ ἐν τῷ EZ ἀριθμοὶ >ῖσοι τῷ Δ . διῆ|ρήσθω ὁ μὲν $B\Gamma$ εἰς τοὺς τῷ A >ῖσους τοὺς BH , $H\Gamma$, ὁ δὲ EZ εἰς τοὺς τῷ Δ >ῖσους τοὺς $E\Theta$, ΘZ ; >έσται δὴ >ῖσον τὸ πλῆθος τῶν BH , $H\Gamma$ τῷ πλῆθει τῶν $E\Theta$, ΘZ . καὶ ἐπεὶ >ῖσοι ἐστὶν ὁ μὲν BH τῷ A , ὁ δὲ $E\Theta$ τῷ Δ , καὶ οἱ BH , $E\Theta$ >άρα τοῖς A , Δ >ῖσοι. διὰ τὰ αὐτὰ δὴ καὶ οἱ $H\Gamma$, ΘZ τοῖς A , Δ . <όσοι >άρα [εἰσὶν] ἐν τῷ $B\Gamma$ ἀριθμοὶ >ῖσοι τῷ A , τοσοῦτοὶ εἰσὶ καὶ ἐν τοῖς $B\Gamma$, EZ >ῖσοι τοῖς A , Δ . ὁσαπλασίων >άρα ἐστὶν ὁ $B\Gamma$ τοῦ A , τοσαυταπλασίων ἐστὶ καὶ συναμφοτέρος ὁ $B\Gamma$, EZ συναμφοτέρου τοῦ A , Δ . <ὁ >άρα μέρος ἐστὶν ὁ A τοῦ $B\Gamma$, τὸ αὐτὸ μέρος ἐστὶ καὶ συναμφοτέρος ὁ A , Δ συναμφοτέρου τοῦ $B\Gamma$, EZ ; <όπερ >έδει δεῖχαι.

For let a number A be part of a [number] BC , and another (number) D (be) the same part of another (number) EF that A (is) of BC . I say that the sum A , D is also the same part of the sum BC , EF that A (is) of BC .

For since which(ever) part A is of BC , D is the same part of EF , thus as many numbers as are in BC equal to A , so many numbers are also in EF equal to D . Let BC be divided into BG and GC , equal to A , and EF into EH and HF , equal to D . So the multitude of (divisions) BG , GC will be equal to the multitude of (divisions) EH , HF . And since BG is equal to A , and EH to D , thus BG , EH (is) also equal to A , D . So, for the same (reasons), GC , HF (is) also (equal) to A , D . Thus, as many numbers as [are] in BC equal to A , so many are also in BC , EF equal to A , D . Thus, as many times as BC is (divisible) by A , so many times is the sum BC , EF also (divisible) by the sum A , D . Thus, which(ever) part A is of BC , the sum A , D is also the same part of the sum BC , EF . (Which is) the very thing it was required to show.

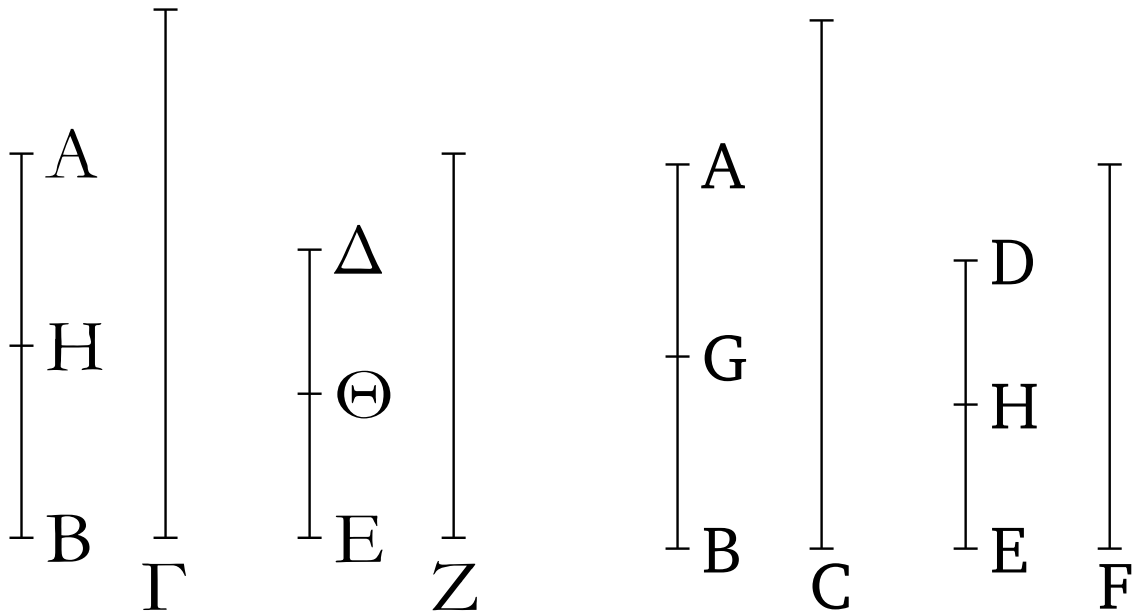
[†] In modern notation, this proposition states that if $a = (1/n)b$ and $c = (1/n)d$ then $(a + c) = (1/n)(b + d)$, where all symbols denote numbers.

6.

Ἐὰν ἀριθμὸς ἀριθμοῦ μέρη >ῆ|, καὶ <έτερος ἑτέρου τὰ αὐτὰ μέρη >ῆ|, καὶ συναμφοτέρος συναμφοτέρου τὰ αὐτὰ μέρη >έσται, <όπερ ὁ <ῖξ τοῦ ἐνόξ.

Proposition 6[†]

If a number is parts of a number, and another (number) is the same parts of another, (then) the sum (of the leading numbers) will also be the same parts of the sum (of the following numbers) that one (number) is of another.



Ἀριθμὸς γὰρ ὁ ΑΒ ἀριθμοῦ τοῦ Γ μέρη >έστω, καὶ <έτεροξ
 ὁ ΔΕ ἑτέρου τοῦ Ζ τὰ αὐτὰ μέρη, <άπερ ὁ ΑΒ τοῦ Γ; λέγω,
 <ὅτι καὶ συναμφοτέροξ ὁ ΑΒ, ΔΕ συναμφοτέρου τοῦ Γ, Ζ
 τὰ αὐτὰ μέρη ἐστίν, <άπερ ὁ ΑΒ τοῦ Γ.

Τεπεί γάρ, <ἃ μέρη ἔστιν ὁ ΑΒ τοῦ Γ, τὰ αὐτὰ μέρη καὶ ὁ ΔΕ τοῦ Ζ, <όσα> ἄρα ἔστιν ἐν τῷ ΑΒ μέρη τοῦ Γ, τοσαῦτά ἐστι καὶ ἐν τῷ ΔΕ μέρη τοῦ Ζ. διηρηθήσθω ὁ μὲν ΑΒ εἰς τὰ τοῦ Γ μέρη τὰ ΑΗ, ΗΒ, ὁ δὲ ΔΕ εἰς τὰ τοῦ Ζ μέρη τὰ ΔΘ, ΘΕ; >έσται δὴ >ῖσον τὸ πλῆθος τῶν ΑΗ, ΗΒ τῷ πλήθει τῶν ΔΘ, ΘΕ. καὶ ἐπεὶ, <ὃ μέρος ἔστιν ὁ ΑΗ τοῦ Γ, τὸ αὐτὸ μέρος ἔστι καὶ ὁ ΔΘ τοῦ Ζ, <ὃ> ἄρα μέρος ἔστιν ὁ ΑΗ τοῦ Γ, τὸ αὐτὸ μέρος ἔστι καὶ συναμφοτέροξ ὁ ΑΗ, ΔΘ συναμφοτέρου τοῦ Γ, Ζ. διὰ τὰ αὐτὰ δὴ καὶ <ὃ μέρος ἔστιν ὁ ΗΒ τοῦ Γ, τὸ αὐτὸ μέρος ἔστι καὶ συναμφοτέροξ ὁ ΗΒ, ΘΕ συναμφοτέρου τοῦ Γ, Ζ. <ἃ> ἄρα μέρη ἔστιν ὁ ΑΒ τοῦ Γ, τὰ αὐτὰ μέρη ἐστὶ καὶ συναμφοτέροξ ὁ ΑΒ, ΔΕ συναμφοτέρου τοῦ Γ, Ζ; <όπερ> ἔδει δεῖλαι.

For let a number AB be parts of a number C , and another (number) DE (be) the same parts of another (number) F that AB (is) of C . I say that the sum AB, DE is also the same parts of the sum C, F that AB (is) of C .

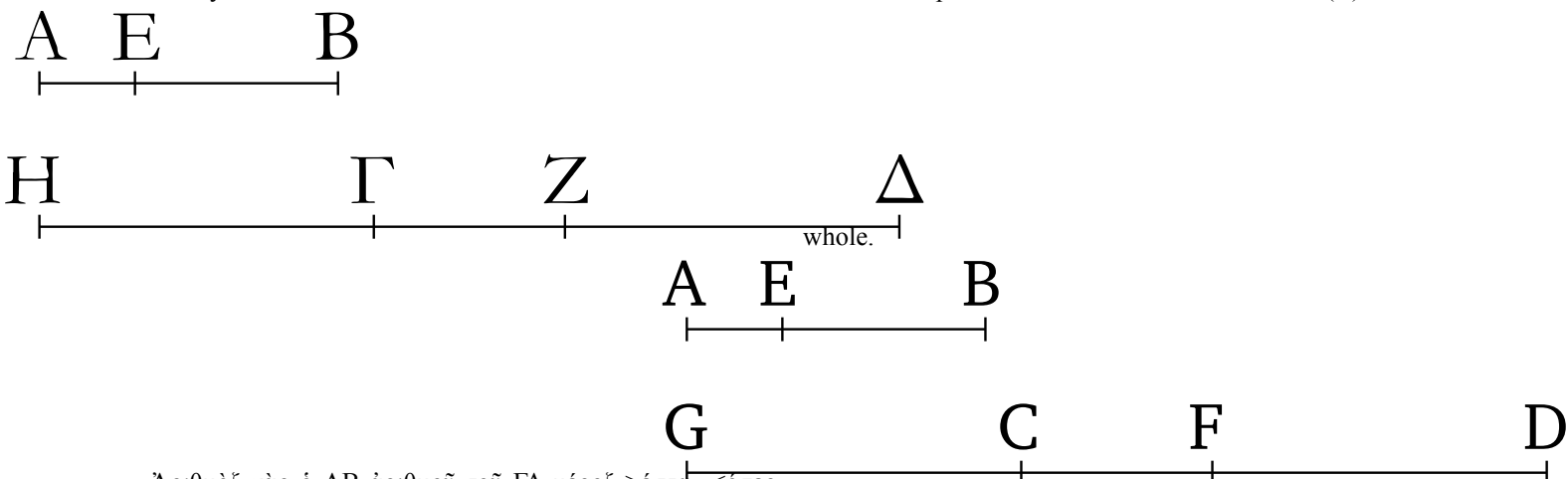
For since which(ever) parts AB is of C , DE (is) also the same parts of F , thus as many parts of C as are in AB , so many parts of F are also in DE . Let AB be divided into the parts of C , AG and GB , and DE into the parts of F , DH and HE . So the multitude of (divisions) AG , GB will be equal to the multitude of (divisions) DH , HE . And since which(ever) part AG is of C , DH is also the same part of F , thus which(ever) part AG is of C , the sum AG , DH is also the same part of the sum C , F [Prop. 7.5]. And so, for the same (reasons), which(ever) part GB is of C , the sum GB , HE is also the same part of the sum C , F . Thus, which(ever) parts AB is of C , the sum AB , DE is also the same parts of the sum C , F . (Which is) the very thing it was required to show.

[†] In modern notation, this proposition states that if $a = (m/n)b$ and $c = (m/n)d$ then $(a+c) = (m/n)(b+d)$, where all symbols denote numbers.

7.

Proposition 7[†]

Ἐὰν ἀριθμὸς ἀριθμοῦ μέρος >ῆ|, <ὅπερ ἀφαιρεθεῖς ἀφαιρεθέντι >ῆ, number is that part of a number that a (part) taken away
καὶ ὁ λοιπὸς τοῦ λοιποῦ τὸ αὐτὸ μέρος >έσται, <ὅπερ ὁ (is) of a (part) taken away, (then) the remainder will also
<όλοξ τοῦ <όλου. be the same part of the remainder that the whole (is) of the



Ἀριθμὸς γὰρ ὁ ΑΒ ἀριθμοῦ τοῦ ΓΔ μέρος >έστω, <όπερ
ἀφαιρεθεῖς ὁ ΑΕ ἀφαιρεθέντος τοῦ ΓΖ; λέγω, <ὅτι καὶ

For let a number AB be that part of a number CD that a

λοιπὸς ὁ EB λοιποῦ τοῦ $Z\Delta$ τὸ αὐτὸ μέρος ἐστίν, <ὅπερ <όλοξ ὁ AB <όλου τοῦ $\Gamma\Delta$.

<Ὁ γὰρ μέρος ἐστὶν ὁ AE τοῦ ΓZ , τὸ αὐτὸ μέρος >έστω καὶ ὁ EB τοῦ ΓH . καὶ ἐπεὶ, <ὁ μέρος ἐστὶν ὁ AE τοῦ ΓZ , τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ EB τοῦ ΓH , <ὁ >άρα μέρος ἐστὶν ὁ AE τοῦ ΓZ , τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ AB τοῦ HZ . <ὁ δὲ μέρος ἐστὶν ὁ AE τοῦ ΓZ , τὸ αὐτὸ μέρος ὑπόκειται καὶ ὁ AB τοῦ $\Gamma\Delta$; <ὁ >άρα μέρος ἐστὶ καὶ ὁ AB τοῦ HZ , τὸ αὐτὸ μέρος ἐστὶ καὶ τοῦ $\Gamma\Delta$; >ίσοξ >άρα ἐστὶν ὁ HZ τῷ $\Gamma\Delta$. κοινὸς ἀφηρησθῶ ὁ ΓZ ; λοιπὸς >άρα ὁ $H\Gamma$ λοιπῷ τῷ $Z\Delta$ ἐστὶν >ίσοξ. καὶ ἐπεὶ, <ὁ μέρος ἐστὶν ὁ AE τοῦ ΓZ , τὸ αὐτὸ μέρος [ἐστὶ] καὶ ὁ EB τοῦ $H\Gamma$, >ίσοξ δὲ ὁ $H\Gamma$ τῷ $Z\Delta$, <ὁ >άρα μέρος ἐστὶν ὁ AE τοῦ ΓZ , τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ EB τοῦ $Z\Delta$. ἀλλὰ <ὁ μέρος ἐστὶν ὁ AE τοῦ ΓZ , τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ AB τοῦ $\Gamma\Delta$; καὶ λοιπὸς >άρα ὁ EB λοιποῦ τοῦ $Z\Delta$ τὸ αὐτὸ μέρος ἐστίν, <ὅπερ <όλοξ ὁ AB <όλου τοῦ $\Gamma\Delta$; <ὅπερ >έδει δεῖχαι.

(part) taken away AE (is) of a part taken away CF . I say that the remainder EB is also the same part of the remainder FD that the whole AB (is) of the whole CD .

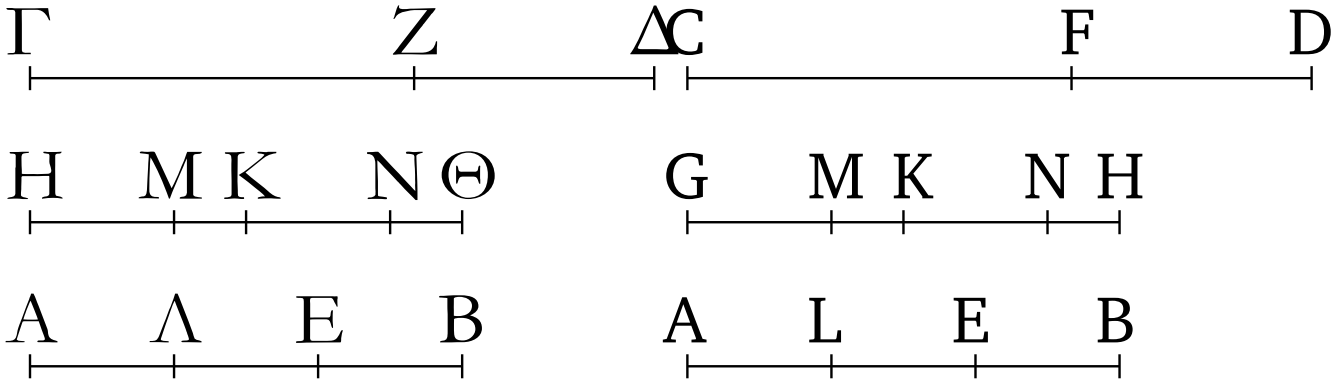
For which(ever) part AE is of CF , let EB also be the same part of CG . And since which(ever) part AE is of CF , EB is also the same part of CG , thus which(ever) part AE is of CF , AB is also the same part of GF [Prop. 7.5]. And which(ever) part AE is of CF , AB is also assumed (to be) the same part of CD . Thus, also, which(ever) part AB is of GF , (AB) is also the same part of CD . Thus, GF is equal to CD . Let CF be subtracted from both. Thus, the remainder GC is equal to the remainder FD . And since which(ever) part AE is of CF , EB [is] also the same part of GC , and GC (is) equal to FD , thus which(ever) part AE is of CF , EB is also the same part of FD . But, which(ever) part AE is of CF , AB is also the same part of CD . Thus, the remainder EB is also the same part of the remainder FD that the whole AB (is) of the whole CD . (Which is) the very thing it was required to show.

† In modern notation, this proposition states that if $a = (1/n)b$ and $c = (1/n)d$ then $(a - c) = (1/n)(b - d)$, where all symbols denote numbers.

8.

Proposition 8†

Ἐὰν ἀριθμὸς ἀριθμοῦ μέρη >ῃ, <ἀπερ ἀφαιρεθεῖς ἀφαιρεθέντος, a number is those parts of a number that a (part) taken away (is) of a (part) taken away, (then) the remainder will also be the same parts of the remainder that the whole (is) of the whole.



Ἀριθμὸς γὰρ ὁ AB ἀριθμοῦ τοῦ $\Gamma\Delta$ μέρη >έστω, <ἀπερ ἀφαιρεθεῖς ὁ AE ἀφαιρεθέντος τοῦ ΓZ ; λέγω, <ὅτι καὶ λοιπὸς ὁ EB λοιποῦ τοῦ $Z\Delta$ τὰ αὐτὰ μέρη ἐστίν, <ἀπερ <όλοξ ὁ AB <όλου τοῦ $\Gamma\Delta$.

Κείσθω γὰρ τῷ AB >ίσοξ ὁ $H\Theta$, <ὰ >άρα μέρη ἐστὶν ὁ $H\Theta$ τοῦ $\Gamma\Delta$, τὰ αὐτὰ μέρη ἐστὶ καὶ ὁ AE τοῦ ΓZ . διηρήσθω ὁ μὲν $H\Theta$ εἰς τὰ τοῦ $\Gamma\Delta$ μέρη τὰ HK , $K\Theta$, ὁ δὲ AE εἰς τὰ τοῦ ΓZ μέρη τὰ AL , LE ; >έσται δὴ >ίσον τὸ πλῆθος τῶν HK , $K\Theta$ τῷ πλῆθει τῶν AL , LE . καὶ ἐπεὶ, <ὁ μέρος ἐστὶν ὁ HK τοῦ $\Gamma\Delta$, τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ AL τοῦ ΓZ , μείζων δὲ ὁ $\Gamma\Delta$ τοῦ ΓZ , μείζων >άρα καὶ ὁ HK τοῦ AL . κείσθω τῷ AL >ίσοξ ὁ HM . <ὁ >άρα μέρος ἐστὶν ὁ HK τοῦ $\Gamma\Delta$, τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ HM τοῦ ΓZ ; καὶ λοιπὸς >άρα ὁ MK λοιποῦ τοῦ $Z\Delta$ τὸ αὐτὸ μέρος ἐστίν, <ὅπερ <όλοξ ὁ HK <όλου τοῦ $\Gamma\Delta$. πάλιν ἐπεὶ, <ὁ μέρος ἐστὶν ὁ $K\Theta$ τοῦ $\Gamma\Delta$, τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ EL τοῦ ΓZ , μείζων δὲ ὁ $\Gamma\Delta$ τοῦ ΓZ , μείζων >άρα καὶ ὁ $K\Theta$ τοῦ EL . κείσθω τῷ EL >ίσοξ ὁ KN . <ὁ >άρα μέρος ἐστὶν ὁ $K\Theta$ τοῦ $\Gamma\Delta$, τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ KN τοῦ ΓZ ; καὶ λοιπὸς >άρα ὁ $N\Theta$ λοιποῦ τοῦ $Z\Delta$ τὸ αὐτὸ μέρος ἐστίν, <ὅπερ <όλοξ ὁ $K\Theta$ <όλου τοῦ $\Gamma\Delta$. ἐδείχθη δὲ καὶ λοιπὸς ὁ MK λοιποῦ τοῦ $Z\Delta$ τὸ αὐτὸ μέρος >ών, <ὅπερ <όλοξ ὁ HK

For let a number AB be those parts of a number CD that a (part) taken away AE (is) of a (part) taken away CF . I say that the remainder EB is also the same parts of the remainder FD that the whole AB (is) of the whole CD .

For let GH be laid down equal to AB . Thus, which(ever) parts GH is of CD , AE is also the same parts of CF . Let GH be divided into the parts of CD , GK and KH , and AE into the parts of CF , AL and LE . So the multitude of (divisions) GK , KH will be equal to the multitude of (divisions) AL , LE . And since which(ever) part GK is of CD , AL is also the same part of CF , and CD (is) greater than CF , GK (is) thus also greater than AL . Let GM be made equal to AL . Thus, which(ever) part GK is of CD , GM is also the same part of CF . Thus, the remainder MK is also the same part of the remainder FD that the whole GK (is) of the whole CD [Prop. 7.5]. Again, since which(ever) part KH is of CD , EL is also the same part of CF , and CD (is) greater than CF , HK (is) thus also greater than EL . Let KN be made equal to EL . Thus, which(ever) part KH (is) of CD , KN is also the same part of CF . Thus, the remainder NH is also the same part of the remainder FD

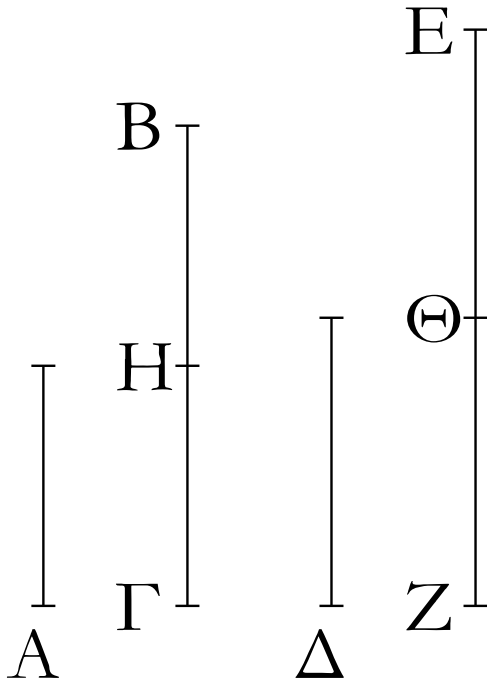
<όλου τοῦ ΓΔ; καὶ συναμφοτέροξ >άρα ὁ ΜΚ, ΝΘ τοῦ ΔΖ τὰ αὐτὰ μέρη ἐστίν, <άπερ <όλοξ ὁ ΘΗ <όλου τοῦ ΓΔ. >ίσοξ δὲ συναμφοτέροξ μὲν ὁ ΜΚ, ΝΘ τῷ ΕΒ, ὁ δὲ ΘΗ τῷ ΒΑ; καὶ λοιπὸξ >άρα ὁ ΕΒ λοιποῦ τοῦ ΖΔ τὰ αὐτὰ μέρη ἐστίν, <άπερ <όλοξ ὁ ΑΒ <όλου τοῦ ΓΔ; <όπερ >έδει δεῖξαι.

that the whole KH (is) of the whole CD [Prop. 7.5]. And the remainder MK was also shown to be the same part of the remainder FD that the whole GK (is) of the whole CD . Thus, the sum MK , NH is the same parts of DF that the whole HG (is) of the whole CD . And the sum MK , NH (is) equal to EB , and HG to BA . Thus, the remainder EB is also the same parts of the remainder FD that the whole AB (is) of the whole CD . (Which is) the very thing it was required to show.

† In modern notation, this proposition states that if $a = (m/n)b$ and $c = (m/n)d$ then $(a - c) = (m/n)(b - d)$, where all symbols denote numbers.

9.

Ἐὰν ἀριθμὸξ ἀριθμοῦ μέρος >ῆ|, καὶ <έτεροξ ἑτέρου τὸ αὐτὸ μέρος >ῆ|, καὶ ἐναλλάξ, <ὃ μέρος ἐστίν >ῆ μέρη ὁ πρῶτοξ τοῦ τρίτου, τὸ αὐτὸ μέρος >έσται >ῆ τὰ αὐτὰ μέρη καὶ ὁ δεῦτεροξ τοῦ τετάρτου.



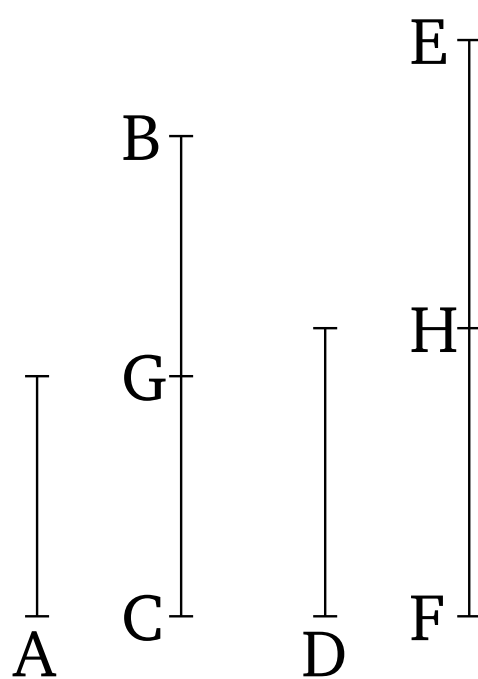
Ἀριθμὸξ γὰρ ὁ Α ἀριθμοῦ τοῦ ΒΓ μέρος >έστω, καὶ <έτεροξ ὁ Δ ἑτέρου τοῦ ΕΖ τὸ αὐτὸ μέρος, <όπερ ὁ Α τοῦ ΒΓ; λέγω, <ὅτι καὶ ἐναλλάξ, <ὃ μέρος ἐστίν ὁ Α τοῦ Δ >ῆ μέρη, τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ ΒΓ τοῦ ΕΖ >ῆ μέρη.

Ἐπεὶ γὰρ <ὃ μέρος ἐστίν ὁ Α τοῦ ΒΓ, τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ Δ τοῦ ΕΖ, <όσοι >άρα εἰσὶν ἐν τῷ ΒΓ ἀριθμοὶ >ῖσοι τῷ Α, τοσοῦτοὶ εἰσὶ καὶ ἐν τῷ ΕΖ >ῖσοι τῷ Δ. διηρήσθω ὁ μὲν ΒΓ εἰξ τοῦξ τῷ Α >ῖσουξ τοῦξ ΒΗ, ΗΓ, ὁ δὲ ΕΖ εἰξ τοῦξ τῷ Δ >ῖσουξ τοῦξ ΕΘ, ΘΖ; >έσται δὲ >ῖσον τὸ πλῆθοξ τῶν ΒΗ, ΗΓ τῷ πλῆθει τῶν ΕΘ, ΘΖ.

Καὶ ἐπεὶ >ῖσοι εἰσὶν οἱ ΒΗ, ΗΓ ἀριθμοὶ ἀλλήλοισι, εἰσὶ δὲ καὶ οἱ ΕΘ, ΘΖ ἀριθμοὶ >ῖσοι ἀλλήλοισι, καὶ ἐστὶν >ῖσον τὸ πλῆθοξ τῶν ΒΗ, ΗΓ τῷ πλῆθει τῶν ΕΘ, ΘΖ, <ὃ >άρα μέρος ἐστίν ὁ ΒΗ τοῦ ΕΘ >ῆ μέρη, τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ ΗΓ τοῦ ΘΖ >ῆ τὰ αὐτὰ μέρη; <ώστε καὶ <ὃ μέρος ἐστίν ὁ ΒΗ τοῦ ΕΘ >ῆ μέρη, τὸ αὐτὸ μέρος ἐστὶ καὶ συναμφοτέροξ ὁ ΒΓ συναμφοτέρου τοῦ ΕΖ >ῆ τὰ αὐτὰ μέρη. >ίσοξ δὲ ὁ μὲν ΒΗ τῷ Α, ὁ δὲ ΕΘ τῷ Δ; <ὃ >άρα μέρος ἐστίν ὁ Α τοῦ Δ >ῆ μέρη, τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ ΒΓ τοῦ ΕΖ >ῆ τὰ αὐτὰ μέρη; <όπερ >έδει δεῖξαι.

Proposition 9†

If a number is part of a number, and another (number) is the same part of another, also, alternately, which(ever) part, or parts, the first (number) is of the third, the second (number) will also be the same part, or the same parts, of the fourth.



For let a number A be part of a number BC , and another (number) D (be) the same part of another EF that A (is) of BC . I say that, also, alternately, which(ever) part, or parts, A is of D , BC is also the same part, or parts, of EF .

For since which(ever) part A is of BC , D is also the same part of EF , thus as many numbers as are in BC equal to A , so many are also in EF equal to D . Let BC be divided into BG and GC , equal to A , and EF into EH and HF , equal to D . So the multitude of (divisions) BG , GC will be equal to the multitude of (divisions) EH , HF .

And since the numbers BG and GC are equal to one another, and the numbers EH and HF are also equal to one another, and the multitude of (divisions) BG , GC is equal to the multitude of (divisions) EH , HF , thus which(ever) part, or parts, BG is of EH , GC is also the same part, or the same parts, of HF . And hence, which(ever) part, or parts, BG is of EH , the sum BC is also the same part, or the same parts, of the sum EF [Props. 7.5, 7.6]. And BG (is) equal to A , and EH to D . Thus, which(ever) part, or parts, A is of D , BC is also the same part, or the same parts, of EF . (Which is) the very thing it was required to show.

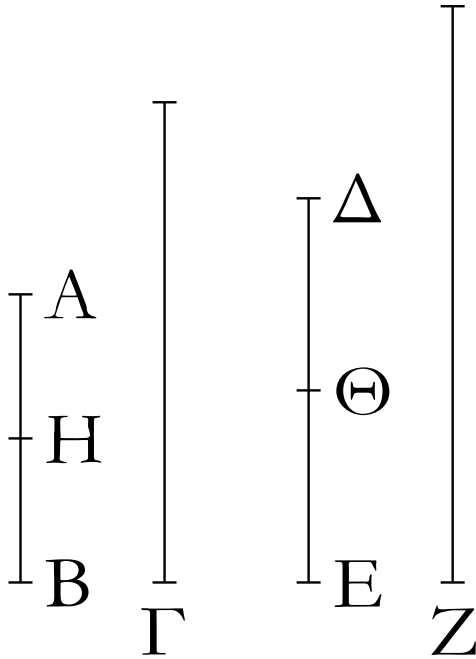
† In modern notation, this proposition states that if $a = (1/n)b$ and $c = (1/n)d$ then if $a = (k/l)c$ then $b = (k/l)d$, where all symbols denote

numbers.

10.

Ἐὰν ἀριθμὸς ἀριθμοῦ μέρη >ῆ|, καὶ <έτεροξ ἑτέρου τὰ αὐτὰ μέρη >ῆ|, καὶ ἐναλλάχ, <ὰ μέρη ἐστὶν ὁ πρῶτοξ τοῦ τρίτου >ῆ μέροςξ, τὰ αὐτὰ μέρη >έσται καὶ ὁ δεῦτεροξ τοῦ τετάρτου >ῆ τὸ αὐτὸ μέροςξ.

Ἀριθμὸξ γὰρ ὁ AB ἀριθμοῦ τοῦ Γ μέρη >έστω, καὶ <έτεροξ ὁ ΔΕ ἑτέρου τοῦ Ζ τὰ αὐτὰ μέρη; λέγω, <ὅτι καὶ ἐναλλάχ, <ὰ μέρη ἐστὶν ὁ AB τοῦ ΔΕ >ῆ μέροςξ, τὰ αὐτὰ μέρη ἐστὶ καὶ ὁ Γ τοῦ Ζ >ῆ τὸ αὐτὸ μέροςξ.

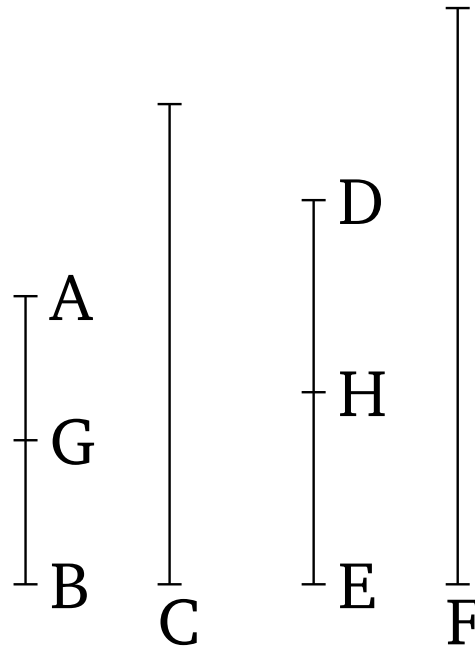


Ἐπεὶ γάρ, <ὰ μέρη ἐστὶν ὁ AB τοῦ Γ, τὰ αὐτὰ μέρη ἐστὶ καὶ ὁ ΔΕ τοῦ Ζ, <όσα >άρα ἐστὶν ἐν τῷ AB μέρη τοῦ Γ, τοσαῦτα καὶ ἐν τῷ ΔΕ μέρη τοῦ Ζ. δηρῆσθω ὁ μὲν AB εἰς τὰ τοῦ Γ μέρη τὰ ΑΗ, ΗΒ, ὁ δὲ ΔΕ εἰς τὰ τοῦ Ζ μέρη τὰ ΔΘ, ΘΕ; >έσται δὴ >ίσον τὸ πλῆθος τῶν ΑΗ, ΗΒ τῷ πλῆθει τῶν ΔΘ, ΘΕ. καὶ ἐπεὶ, <ὸ μέροςξ ἐστὶν ὁ ΑΗ τοῦ Γ, τὸ αὐτὸ μέροςξ ἐστὶ καὶ ὁ ΔΘ τοῦ Ζ, καὶ ἐναλλάχ, <ὸ μέροςξ ἐστὶν ὁ ΑΗ τοῦ ΔΘ >ῆ μέρη, τὸ αὐτὸ μέροςξ ἐστὶ καὶ ὁ Γ τοῦ Ζ >ῆ τὰ αὐτὰ μέρη. διὰ τὰ αὐτὰ δὴ καὶ, <ὸ μέροςξ ἐστὶν ὁ ΗΒ τοῦ ΘΕ >ῆ μέρη, τὸ αὐτὸ μέροςξ ἐστὶ καὶ ὁ Γ τοῦ Ζ >ῆ τὰ αὐτὰ μέρη; <ώστε καὶ [<ὸ μέροςξ ἐστὶν ὁ ΑΗ τοῦ ΔΘ >ῆ μέρη, τὸ αὐτὸ μέροςξ ἐστὶ καὶ ὁ ΗΒ τοῦ ΘΕ >ῆ τὰ αὐτὰ μέρη; καὶ <ὸ >άρα μέροςξ ἐστὶν ὁ ΑΗ τοῦ ΔΘ >ῆ μέρη, τὸ αὐτὸ μέροςξ ἐστὶ καὶ ὁ AB τοῦ ΔΕ >ῆ τὰ αὐτὰ μέρη; ἀλλ' <ὸ μέροςξ ἐστὶν ὁ ΑΗ τοῦ ΔΘ >ῆ μέρη, τὸ αὐτὸ μέροςξ ἐδείχθη καὶ ὁ Γ τοῦ Ζ >ῆ τὰ αὐτὰ μέρη, καὶ] <ὰ [>άρα] μέρη ἐστὶν ὁ AB τοῦ ΔΕ >ῆ μέροςξ, τὰ αὐτὰ μέρη ἐστὶ καὶ ὁ Γ τοῦ Ζ >ῆ τὸ αὐτὸ μέροςξ; <όπερ >έδει δεῖχαι.

Proposition 10[†]

If a number is parts of a number, and another (number) is the same parts of another, also, alternately, which(ever) parts, or part, the first (number) is of the third, the second will also be the same parts, or the same part, of the fourth.

For let a number AB be parts of a number C , and another (number) DE (be) the same parts of another F . I say that, also, alternately, which(ever) parts, or part, AB is of DE , C is also the same parts, or the same part, of F .



For since which(ever) parts AB is of C , DE is also the same parts of F , thus as many parts of C as are in AB , so many parts of F (are) also in DE . Let AB be divided into the parts of C , AG and GB , and DE into the parts of F , DH and HE . So the multitude of (divisions) AG , GB will be equal to the multitude of (divisions) DH , HE . And since which(ever) part AG is of C , DH is also the same part of F , also, alternately, which(ever) part, or parts, AG is of DH , C is also the same part, or the same parts, of F [Prop. 7.9]. And so, for the same (reasons), which(ever) part, or parts, GB is of HE , C is also the same part, or the same parts, of F [Prop. 7.9]. And so [which(ever) part, or parts, AG is of DH , GB is also the same part, or the same parts, of HE . And thus, which(ever) part, or parts, AG is of DH , AB is also the same part, or the same parts, of DE [Props. 7.5, 7.6]. But, which(ever) part, or parts, AG is of DH , C was also shown (to be) the same part, or the same parts, of F . And, thus] which(ever) parts, or part, AB is of DE , C is also the same parts, or the same part, of F . (Which is) the very thing it was required to show.

[†] In modern notation, this proposition states that if $a = (m/n)b$ and $c = (m/n)d$ then if $a = (k/l)c$ then $b = (k/l)d$, where all symbols denote numbers.

11.

Ἐὰν >ῆ| ὥξ <όλοξ πρὸξ <όλον, ο<ύτωξ ἀφαιρεθεῖξ πρὸξ ἀφαιρεθέντα, καὶ ὁ λοιποῖξ πρὸξ τὸν λοιπὸν >έσται, ὥξ <όλοξ πρὸξ <όλον.

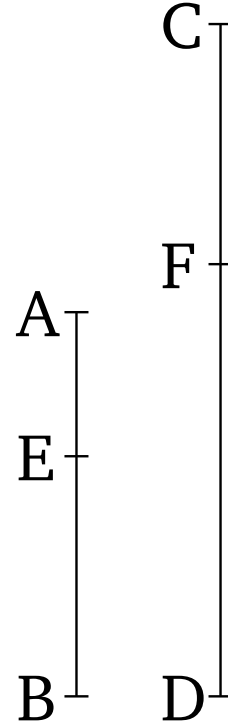
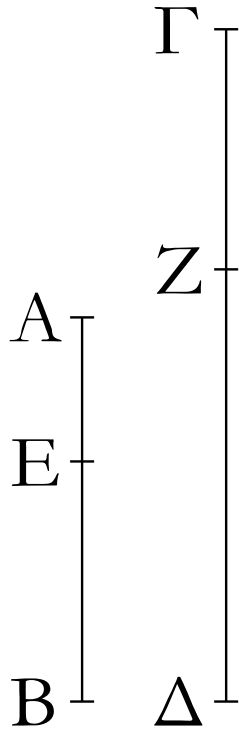
Proposition 11

If as the whole (of a number) is to the whole (of another), so a (part) taken away (is) to a (part) taken away, (then) the remainder will also be to the remainder as the whole (is) to

>Ἐστω ὡς <όλοξ ὁ AB πρὸξ <όλον τὸν $\Gamma\Delta$, ο<ύτωξ ἀφαιρεθεῖξ ὁ AE πρὸξ ἀφαιρεθέντα τὸν ΓZ ; λέγω, <ότι καὶ λοιπὸξ ὁ EB πρὸξ λοιπὸν τὸν $Z\Delta$ ἐστιν, ὡς <όλοξ ὁ AB πρὸξ <όλον τὸν $\Gamma\Delta$.

the whole.

Let the whole AB be to the whole CD as the (part) taken away AE (is) to the (part) taken away CF . I say that the remainder EB is to the remainder FD as the whole AB (is) to the whole CD .



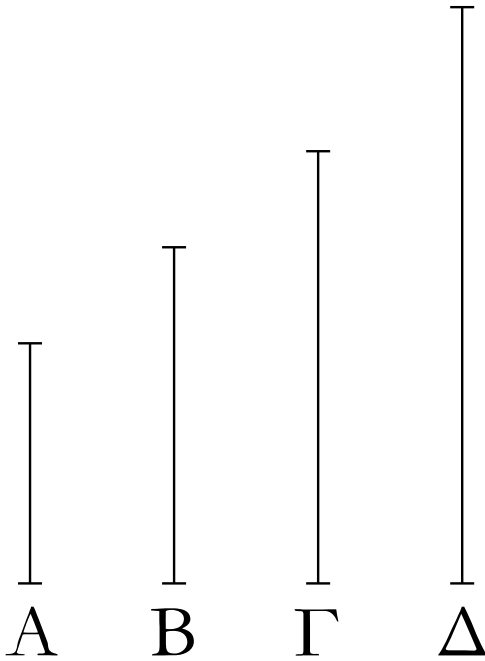
Ἐπεὶ ἐστιν ὡς ὁ AB πρὸξ τὸν $\Gamma\Delta$, ο<ύτωξ ὁ AE πρὸξ τὸν ΓZ , <ὁ >άρα μέρος ἐστιν ὁ AB τοῦ $\Gamma\Delta$ >ἢ μέρος, τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ AE τοῦ ΓZ >ἢ τὰ αὐτὰ μέρος. καὶ λοιπὸξ >άρα ὁ EB λοιποῦ τοῦ $Z\Delta$ τὸ αὐτὸ μέρος ἐστιν >ἢ μέρος, <άπερ ὁ AB τοῦ $\Gamma\Delta$. >ἐστιν >άρα ὡς ὁ EB πρὸξ τὸν $Z\Delta$, ο<ύτωξ ὁ AB πρὸξ τὸν $\Gamma\Delta$; <όπερ >έδει δεῖχαι.

(For) since as AB is to CD , so AE (is) to CF , thus which(ever) part, or parts, AB is of CD , AE is also the same part, or the same parts, of CF [Def. 7.20]. Thus, the remainder EB is also the same part, or parts, of the remainder FD that AB (is) of CD [Props. 7.7, 7.8]. Thus, as EB is to FD , so AB (is) to CD [Def. 7.20]. (Which is) the very thing it was required to show.

[†] In modern notation, this proposition states that if $a : b :: c : d$ then $a : b :: a - c : b - d$, where all symbols denote numbers.

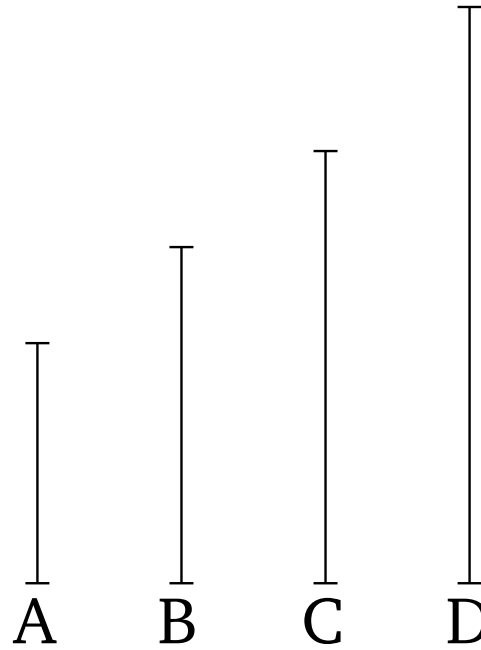
Ἐάν >ῶσιν ὅποσοιοῦν ἀριθμοὶ ἀνάλογον, >έσται ὡς ε<ῖξ
τῶν ἡγουμένων πρὸς <ένα τῶν ἐπομένων, ο<ύτωξ <άπαντεξ
οἱ ἡγούμενοι πρὸς <άπανταξ τοῦξ ἐπομένουξ.

If any multitude whatsoever of numbers are proportional,
(then) as one of the leading (numbers is) to one of the fol-
lowing so (the sum of) all of the leading (numbers) will be
to (the sum of) all of the following.



>Ἐστῶσαν ὅποσοιοῦν ἀριθμοὶ ἀνάλογον οἱ A, B, Γ, Δ, ὡξ
ὁ A πρὸς τὸν B, ο<ύτωξ ὁ Γ πρὸς τὸν Δ; λέγω, <ὅτι ἐστὶν
ὡξ ὁ A πρὸς τὸν B, ο<ύτωξ οἱ A, Γ πρὸς τοῦξ B, Δ.

Ἐπεὶ γάρ ἐστὶν ὡξ ὁ A πρὸς τὸν B, ο<ύτωξ ὁ Γ πρὸς τὸν
Δ, <ὁ >άρα μέρος ἐστὶν ὁ A τοῦ B >ῆ μέρη, τὸ αὐτὸ μέρος
ἐστὶ καὶ ὁ Γ τοῦ Δ >ῆ μέρη. καὶ συναμφοτέρως >άρα ὁ A,
Γ συναμφοτέρου τοῦ B, Δ τὸ αὐτὸ μέρος ἐστὶν >ῆ τὰ αὐτὰ
μέρη, <άπερ ὁ A τοῦ B. >έστιν >άρα ὡξ ὁ A πρὸς τὸν B,
ο<ύτωξ οἱ A, Γ πρὸς τοῦξ B, Δ; <όπερ >έδει δεῖχαι.



Let any multitude whatsoever of numbers, A, B, C, D , be
proportional, (such that) as A (is) to B , so C (is) to D . I say
that as A is to B , so A, C (is) to B, D .

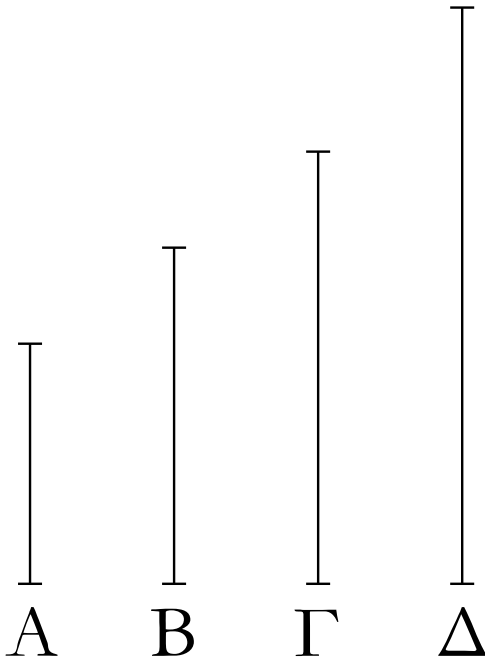
For since as A is to B , so C (is) to D , thus which(ever) part,
or parts, A is of B , C is also the same part, or parts, of D
[Def. 7.20]. Thus, the sum A, C is also the same part, or the
same parts, of the sum B, D that A (is) of B [Props. 7.5,
7.6]. Thus, as A is to B , so A, C (is) to B, D [Def. 7.20].
(Which is) the very thing it was required to show.

[†] In modern notation, this proposition states that if $a : b :: c : d$ then $a : b :: a + c : b + d$, where all symbols denote numbers.

13.

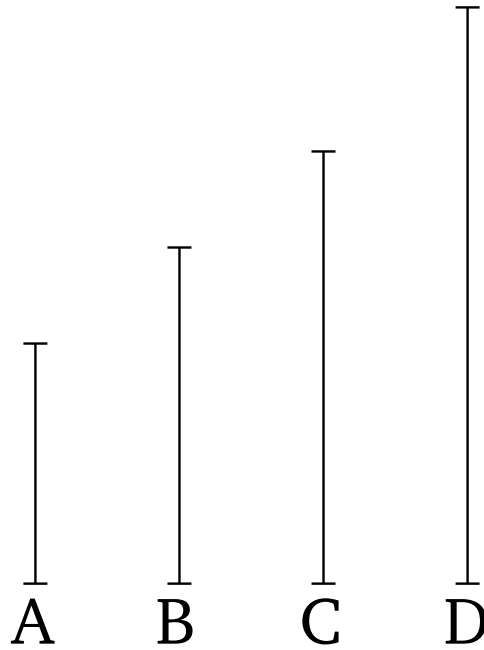
Proposition 13[†]

Ἐάν τέσσαρεξ ἀριθμοὶ ἀνάλογον >ῶσιν, καὶ ἐναλλάξ ἀνάλογον If four numbers are proportional, (then) they will also be
>έσονται. proportional alternately.



>Ἐστωσαν τέσσαρες ἀριθμοὶ ἀνάλογον οἱ A, B, Γ, Δ , ὥς ὁ A πρὸς τὸν B , οὕτως ὁ Γ πρὸς τὸν Δ ; λέγω, <ὅτι καὶ ἐναλλάχ ἀνάλογον >έσονται, ὥς ὁ A πρὸς τὸν Γ , οὕτως ὁ B πρὸς τὸν Δ .

Ἐπεὶ γάρ ἐστιν ὥς ὁ A πρὸς τὸν B , οὕτως ὁ Γ πρὸς τὸν Δ , <ὁ> ἄρα μέρος ἐστὶν ὁ A τοῦ B >ἢ μέρος, τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ Γ τοῦ Δ >ἢ τὰ αὐτὰ μέρη. ἐναλλάχ >ἄρα, <ὁ> μέρος ἐστὶν ὁ A τοῦ Γ >ἢ μέρος, τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ B τοῦ Δ >ἢ τὰ αὐτὰ μέρη. >ἐστὶν ἄρα ὥς ὁ A πρὸς τὸν Γ , οὕτως ὁ B πρὸς τὸν Δ ; <ὅπερ >εἰδει δεῖχαι.



Let the four numbers A, B, C , and D be proportional, (such that) as A (is) to B , so C (is) to D . I say that they will also be proportional alternately, (such that) as A (is) to C , so B (is) to D .

For since as A is to B , so C (is) to D , thus which(ever) part, or parts, A is of B , C is also the same part, or the same parts, of D [Def. 7.20]. Thus, alternately, which(ever) part, or parts, A is of C , B is also the same part, or the same parts, of D [Props. 7.9, 7.10]. Thus, as A is to C , so B (is) to D [Def. 7.20]. (Which is) the very thing it was required to show.

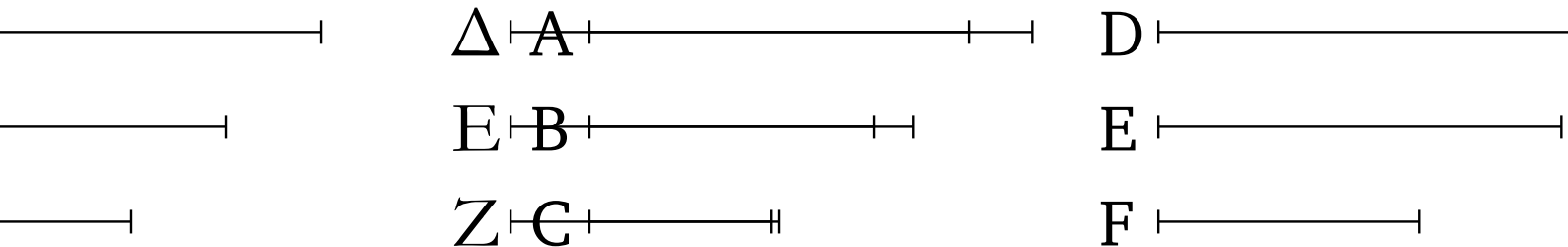
† In modern notation, this proposition states that if $a : b :: c : d$ then $a : c :: b : d$, where all symbols denote numbers.

14.

Proposition 14†

Ἐὰν >ᾧσιν ὅποσοιοῦν ἀριθμοὶ καὶ >ἄλλοι αὐτοῖς >ῖσοι τὸ πλῆθος σύνδου λαμβανόμενοι καὶ ἐν τῷ αὐτῷ λόγῳ, καὶ δι> >ῖσου ἐν τῷ αὐτῷ λόγῳ >έσονται.

If there are any multitude of numbers whatsoever, and (some) other (numbers) of equal multitude to them, (which are) also in the same ratio taken two by two, (then) they will also be in the same ratio via equality.



>Ἐστωσαν ὅποσοιοῦν ἀριθμοὶ οἱ A, B, Γ καὶ >ἄλλοι αὐτοῖς >ῖσοι τὸ πλῆθος σύνδου λαμβανόμενοι ἐν τῷ αὐτῷ λόγῳ οἱ Δ, E, Z , ὥς μὲν ὁ A πρὸς τὸν B , οὕτως ὁ Δ πρὸς τὸν E , ὥς δὲ ὁ B πρὸς τὸν Γ , οὕτως ὁ E πρὸς τὸν Z ; λέγω, <ὅτι καὶ δι> >ῖσου ἐστὶν ὥς ὁ A πρὸς τὸν Γ , οὕτως ὁ Δ πρὸς τὸν Z . Ἐπεὶ γάρ ἐστιν ὥς ὁ A πρὸς τὸν B , οὕτως ὁ Δ πρὸς τὸν E , ἐναλλάχ >ἄρα ἐστὶν ὥς ὁ A πρὸς τὸν Δ , οὕτως ὁ B πρὸς τὸν E . ἅλιν, ἐπεὶ ἐστὶν ὥς ὁ B πρὸς τὸν Γ , οὕτως ὁ E πρὸς τὸν Z , ἐναλλάχ >ἄρα ἐστὶν ὥς ὁ B πρὸς τὸν E , οὕτως ὁ Γ πρὸς τὸν Z . ὥς δὲ ὁ B πρὸς τὸν E , οὕτως ὁ A πρὸς τὸν Δ ; καὶ ὥς >ἄρα ὁ A πρὸς τὸν Δ , οὕτως ὁ Γ πρὸς τὸν Z ; ἐναλλάχ >ἄρα ἐστὶν ὥς ὁ A πρὸς τὸν Γ , οὕτως ὁ Δ πρὸς τὸν Z ; <ὅπερ >εἰδει δεῖχαι.

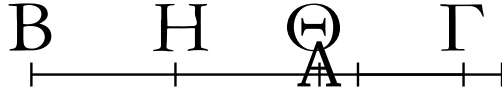
Let there be any multitude of numbers whatsoever, A, B, C , and (some) other (numbers), D, E, F , of equal multitude to them, (which are) in the same ratio taken two by two, (such that) as A (is) to B , so D (is) to E , and as B (is) to C , so E (is) to F . I say that also, via equality, as A is to C , so D (is) to F .

For since as A is to B , so D (is) to E , thus, alternately, as A is to D , so B (is) to E [Prop. 7.13]. Again, since as B is to C , so E (is) to F , thus, alternately, as B is to E , so C (is) to F [Prop. 7.13]. And as B (is) to E , so A (is) to D . Thus, also, as A (is) to D , so C (is) to F . Thus, alternately, as A is to C , so D (is) to F [Prop. 7.13]. (Which is) the very thing it was required to show.

† In modern notation, this proposition states that if $a : b :: d : e$ and $b : c :: e : f$ then $a : c :: d : f$, where all symbols denote numbers.

15.

Ἐάν μονὰξ ἀριθμὸν τινα μετρήῃ, ἰσάκις δὲ <έτεροξ ἀριθμὸξ >άλλον τινα ἀριθμὸν μετρήῃ, καὶ ἐναλλάχ ἰσάκις ἡ μονὰξ τὸν τρίτον ἀριθμὸν μετρήσει καὶ ὁ δεύτεροξ τὸν τέταρτον.

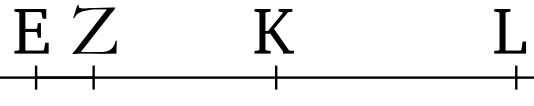
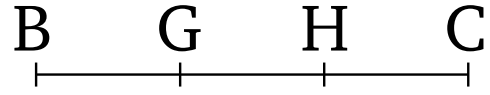


Μονὰξ γὰρ ἡ Α ἀριθμὸν τινα τὸν ΒΓ μετρεῖτω, ἰσάκις δὲ <έτεροξ ἀριθμὸξ ὁ Δ>άλλον τινα ἀριθμὸν τὸν ΕΖ μετρεῖτω; λέγω, <ὅτι καὶ ἐναλλάχ ἰσάκις ἡ Α μονὰξ τὸν Δ ἀριθμὸν μετρεῖ καὶ ὁ ΒΓ τὸν ΕΖ.

Ἐπεὶ γὰρ ἰσάκις ἡ Α μονὰξ τὸν ΒΓ ἀριθμὸν μετρεῖ καὶ ὁ Δ τὸν ΕΖ, <όσαι>άρα εἰσὶν ἐν τῷ ΒΓ μονάδεξ, τοσοῦτοί εἰσι καὶ ἐν τῷ ΕΖ ἀριθμοὶ >ἴσοι τῷ Δ. δηρῆσθω ὁ μὲν ΒΓ εἰς τὰξ ἐν ἑαυτῷ μονάδαξ τὰξ ΒΗ, ΗΘ, ΘΓ, ὁ δὲ ΕΖ εἰς τοὺξ τῷ Δ >ἴσουξ τοὺξ ΕΚ, ΚΛ, ΛΖ. >έσται δὴ >ἴσον τὸ πλῆθοξ τῶν ΒΗ, ΗΘ, ΘΓ τῷ πλῆθει τῶν ΕΚ, ΚΛ, ΛΖ. καὶ ἐπεὶ >ἴσοι εἰσὶν αἱ ΒΗ, ΗΘ, ΘΓ μονάδεξ ἀλλήλαιξ, εἰσι δὲ καὶ οἱ ΕΚ, ΚΛ, ΛΖ ἀριθμοὶ >ἴσοι ἀλλήλοισξ, καὶ ἔστιν >ἴσον τὸ πλῆθοξ τῶν ΒΗ, ΗΘ, ΘΓ μονάδων τῷ πλῆθει τῶν ΕΚ, ΚΛ, ΛΖ ἀριθμῶν, >έσται >άρα ὡξ ἡ ΒΗ μονὰξ πρὸξ τὸν ΕΚ ἀριθμὸν, ο<ύτωξ ἡ ΗΘ μονὰξ πρὸξ τὸν ΚΛ ἀριθμὸν καὶ ἡ ΘΓ μονὰξ πρὸξ τὸν ΛΖ ἀριθμὸν. >έσται >άρα καὶ ὡξ ε<ῖξ τῶν ἡγουμένων πρὸξ <ένα τῶν ἐπομένων, ο<ύτωξ <άπαντεξ οἱ ἡγούμενοι πρὸξ <άπανταξ τοὺξ ἐπομένουξ; >έστιν >άρα ὡξ ἡ ΒΗ μονὰξ πρὸξ τὸν ΕΚ ἀριθμὸν, ο<ύτωξ ὁ ΒΓ πρὸξ τὸν ΕΖ. >ἴση δὲ ἡ ΒΗ μονὰξ τῇ Α μονάδι, ὁ δὲ ΕΚ ἀριθμὸξ τῷ Δ ἀριθμῷ. >έστιν >άρα ὡξ ἡ Α μονὰξ πρὸξ τὸν Δ ἀριθμὸν, ο<ύτωξ ὁ ΒΓ πρὸξ τὸν ΕΖ. ἰσάκις >άρα ἡ Α μονὰξ τὸν Δ ἀριθμὸν μετρεῖ καὶ ὁ ΒΓ τὸν ΕΖ; <όπερ >έδει δεῖχαι.

Proposition 15

If a unit measures some number, and another number measures some other number as many times, (then), also, alternately, the unit will measure the third number as many times as the second (number measures) the fourth.



For let a unit A measure some number BC , and let another number D measure some other number EF as many times. I say that, also, alternately, the unit A also measures the number D as many times as BC (measures) EF .

For since the unit A measures the number BC as many times as D (measures) EF , thus as many units as are in BC , so many numbers are also in EF equal to D . Let BC be divided into its constituent units, BG , GH , and HC , and EF into the (divisions) EK , KL , and LF , equal to D . So the multitude of (units) BG , GH , HC will be equal to the multitude of (divisions) EK , KL , LF . And since the units BG , GH , and HC are equal to one another, and the numbers EK , KL , and LF are also equal to one another, and the multitude of the (units) BG , GH , HC is equal to the multitude of the numbers EK , KL , LF , thus as the unit BG (is) to the number EK , so the unit GH will be to the number KL , and the unit HC to the number LF . And thus, as one of the leading (numbers is) to one of the following, so (the sum of) all of the leading will be to (the sum of) all of the following [Prop. 7.12]. Thus, as the unit BG (is) to the number EK , so BC (is) to EF . And the unit BG (is) equal to the unit A , and the number EK to the number D . Thus, as the unit A is to the number D , so BC (is) to EF . Thus, the unit A measures the number D as many times as BC (measures) EF [Def. 7.20]. (Which is) the very thing it was required to show.

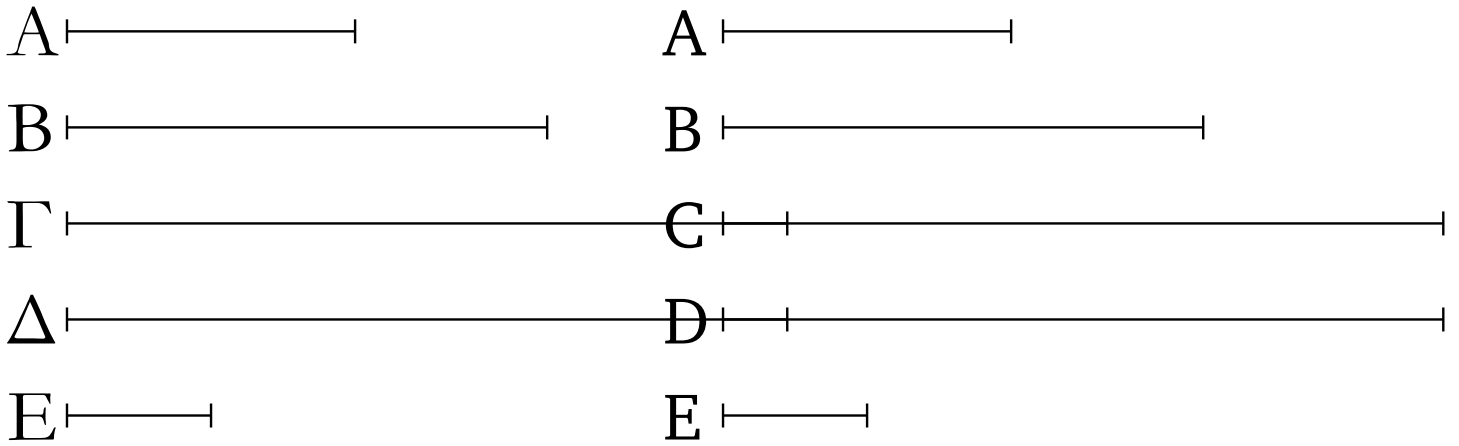
† This proposition is a special case of Prop. 7.9.

16.

Ἐάν δύο ἀριθμοὶ πολλαπλασιάσαντεξ ἀλλήλουξ ποιῶσι τιναξ, οἱ γενόμενοι ἔχ αὐτῶν >ἴσοι ἀλλήλοισξ >έσονται.

Proposition 16†

If two numbers multiplying one another make some (numbers, then) the (numbers) generated from them will be equal to one another.



>Ἐστωσαν δύο ἀριθμοὶ οἱ A , B , καὶ ὁ μὲν A τὸν B πολλαπλασιάσας τὸν Γ ποιεῖτω, ὁ δὲ B τὸν A πολλαπλασιάσας τὸν Δ ποιεῖτω; λέγω, <ὅτι >ἴσος ἐστὶν ὁ Γ τῷ Δ .
 Ἐπεὶ γὰρ ὁ A τὸν B πολλαπλασιάσας τὸν Γ πεποίηκεν, ὁ B >ἄρα τὸν Γ μετρεῖ κατὰ τὰς ἐν τῷ A μονάδας. μετρεῖ δὲ καὶ ἡ E μονὰς τὸν A ἀριθμὸν κατὰ τὰς ἐν αὐτῷ μονάδας; ἰσάκις >ἄρα ἡ E μονὰς τὸν A ἀριθμὸν μετρεῖ καὶ ὁ B τὸν Γ . ἐναλλάξ >ἄρα ἰσάκις ἡ E μονὰς τὸν B ἀριθμὸν μετρεῖ καὶ ὁ A τὸν Γ . πάλιν, ἐπεὶ ὁ B τὸν A πολλαπλασιάσας τὸν Δ πεποίηκεν, ὁ A >ἄρα τὸν Δ μετρεῖ κατὰ τὰς ἐν τῷ B μονάδας. μετρεῖ δὲ καὶ ἡ E μονὰς τὸν B κατὰ τὰς ἐν αὐτῷ μονάδας; ἰσάκις >ἄρα ἡ E μονὰς τὸν B ἀριθμὸν μετρεῖ καὶ ὁ A τὸν Δ . ἰσάκις δὲ ἡ E μονὰς τὸν B ἀριθμὸν ἐμέτρει καὶ ὁ A τὸν Γ ; ἰσάκις >ἄρα ὁ A ἐκάτερον τῶν Γ , Δ μετρεῖ. >ἴσος >ἄρα ἐστὶν ὁ Γ τῷ Δ ; <ὅπερ >έδει δεῖχαι.

Let A and B be two numbers. And let A make C (by) multiplying B , and let B make D (by) multiplying A . I say that C is equal to D .

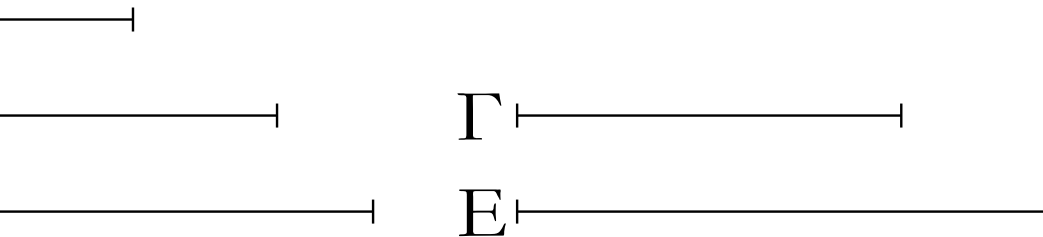
For since A has made C (by) multiplying B , B thus measures C according to the units in A [Def. 7.15]. And the unit E also measures the number A according to the units in it. Thus, the unit E measures the number A as many times as B (measures) C . Thus, alternately, the unit E measures the number B as many times as A (measures) C [Prop. 7.15]. Again, since B has made D (by) multiplying A , A thus measures D according to the units in B [Def. 7.15]. And the unit E also measures B according to the units in it. Thus, the unit E measures the number B as many times as A (measures) D . And the unit E was measuring the number B as many times as A (measures) C . Thus, A measures each of C and D an equal number of times. Thus, C is equal to D . (Which is) the very thing it was required to show.

† In modern notation, this proposition states that $ab = ba$, where all symbols denote numbers.

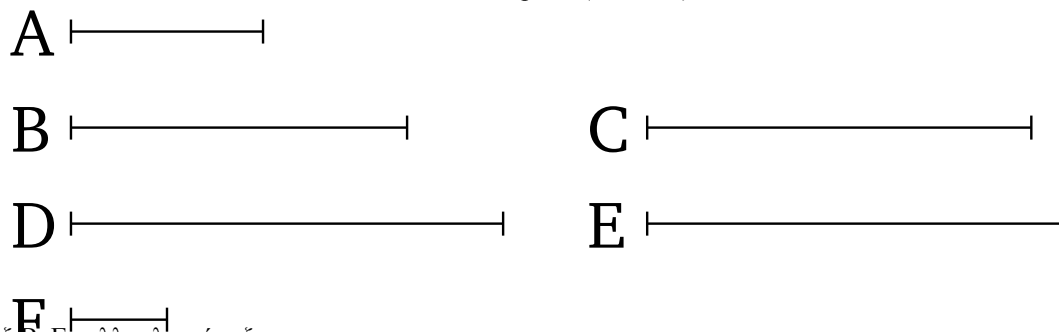
17.

Proposition 17†

Ἐὰν ἀριθμὸς δύο ἀριθμοὺς πολλαπλασιάσας ποιῇ τιναξ, οἱ γενόμενοι ἐξ αὐτῶν τὸν αὐτὸν <έχουσι λόγον τοῖς πολλαπλασιασμένοις>. If a number multiplying two numbers makes some (number), then the (numbers) generated from them will have the



same ratio as the multiplied (numbers).



Ἀριθμὸς γὰρ ὁ A δύο ἀριθμοὺς τοὺς B , Γ πολλαπλασιάσας τοὺς Δ , E ποιεῖτω; λέγω, <ὅτι ἐστὶν ὡς ὁ B πρὸς τὸν Γ , οὕτως ὁ Δ πρὸς τὸν E .

For let the number A make (the numbers) D and E (by) multiplying the two numbers B and C (respectively). I say

Ἐπεὶ γὰρ ὁ A τὸν B πολλαπλασιάσας τὸν Δ πεποίηκεν, ὁ B $>$ άρα τὸν Δ μετρεῖ κατὰ τὰς ἐν τῷ A μονάδας. μετρεῖ δὲ καὶ ἡ Z μονὰς τὸν A ἀριθμὸν κατὰ τὰς ἐν αὐτῷ μονάδας; ἰσάκις $>$ άρα ἡ Z μονὰς τὸν A ἀριθμὸν μετρεῖ καὶ ὁ B τὸν Δ . $>$ έστιν $>$ άρα ὡς ἡ Z μονὰς πρὸς τὸν A ἀριθμὸν, ο $<$ ύτως ὁ B πρὸς τὸν Δ , διὰ τὰ αὐτὰ δὴ καὶ ὡς ἡ Z μονὰς πρὸς τὸν A ἀριθμὸν, ο $<$ ύτως ὁ Γ πρὸς τὸν E ; καὶ ὡς $>$ άρα ὁ B πρὸς τὸν Δ , ο $<$ ύτως ὁ Γ πρὸς τὸν E . ἐναλλάχ $>$ άρα ἐστὶν ὡς ὁ B πρὸς τὸν Γ , ο $<$ ύτως ὁ Δ πρὸς τὸν E ; $<$ όπερ $>$ έδει δεῖχαι.

that as B is to C , so D (is) to E .

For since A has made D (by) multiplying B , B thus measures D according to the units in A [Def. 7.15]. And the unit F also measures the number A according to the units in it. Thus, the unit F measures the number A as many times as B (measures) D . Thus, as the unit F is to the number A , so B (is) to D [Def. 7.20]. And so, for the same (reasons), as the unit F (is) to the number A , so C (is) to E . And thus, as B (is) to D , so C (is) to E . Thus, alternately, as B is to C , so D (is) to E [Prop. 7.13]. (Which is) the very thing it was required to show.

† In modern notation, this proposition states that if $d = ab$ and $e = ac$ then $d : e :: b : c$, where all symbols denote numbers.

18.

Proposition 18†

Ἐὰν δύο ἀριθμοὶ ἀριθμὸν τινα πολλαπλασιάσαντες ποιῶσι τιναξ, οἱ γενόμενοι ἔχ αὐτῶν τὸν αὐτὸν $<$ έχουσι λόγον τοῖς πολλαπλασιάσασιν.

If two numbers multiplying some number make some (other numbers, then) the (numbers) generated from them will have the same ratio as the multiplying (numbers).

A ————

A ————

B ————

B ————

Γ ————

C ————

Δ ————

D ————

E ———— E ————

Δύο γὰρ ἀριθμοὶ οἱ A, B ἀριθμὸν τινα τὸν Γ πολλαπλασιάσαντες τοὺς Δ, E ποιείτωσαν; λέγω, $<$ ότι ἐστὶν ὡς ὁ A πρὸς τὸν B , ο $<$ ύτως ὁ Δ πρὸς τὸν E .

For let the two numbers A and B make (the numbers) D and E (respectively, by) multiplying some number C . I say that as A is to B , so D (is) to E .

Ἐπεὶ γὰρ ὁ A τὸν Γ πολλαπλασιάσας τὸν Δ πεποίηκεν, καὶ ὁ Γ $>$ άρα τὸν A πολλαπλασιάσας τὸν Δ πεποίηκεν. διὰ τὰ αὐτὰ δὴ καὶ ὁ Γ τὸν B πολλαπλασιάσας τὸν E πεποίηκεν. ἀριθμὸς δὴ ὁ Γ δύο ἀριθμοὺς τοὺς A, B πολλαπλασιάσας τοὺς Δ, E πεποίηκεν. $>$ έστιν $>$ άρα ὡς ὁ A πρὸς τὸν B , ο $<$ ύτως ὁ Δ πρὸς τὸν E ; $<$ όπερ $>$ έδει δεῖχαι.

For since A has made D (by) multiplying C , C has thus also made D (by) multiplying A [Prop. 7.16]. So, for the same (reasons), C has also made E (by) multiplying B . So the number C has made D and E (by) multiplying the two numbers A and B (respectively). Thus, as A is to B , so D (is) to E [Prop. 7.17]. (Which is) the very thing it was required to show.

† In modern notation, this propositions states that if $ac = d$ and $bc = e$ then $a : b :: d : e$, where all symbols denote numbers.

19.

Proposition 19†

Ἐὰν τέσσαρες ἀριθμοὶ ἀνάλογον $>$ ῶσιν, ὁ ἐκ πρώτου καὶ τετάρτου γενόμενος ἀριθμὸς $>$ ίσοξ $>$ έσται τῷ ἐκ δευτέρου καὶ τρίτου γενομένῳ ἀριθμῷ; καὶ ἔαν ὁ ἐκ πρώτου καὶ τετάρτου γενόμενος ἀριθμὸς $>$ ίσοξ $>$ ῇ τῷ ἐκ δευτέρου καὶ τρίτου, οἱ τέσσαρες ἀριθμοὶ ἀνάλογον $>$ έσονται.

If four number are proportional, (then) the number created from (multiplying) the first and fourth will be equal to the number created from (multiplying) the second and third. And if the number created from (multiplying) the first and fourth is equal to the (number created) from (multiplying) the second and third, (then) the four numbers will be proportional.

$>$ Ἐστωσαν τέσσαρες ἀριθμοὶ ἀνάλογον οἱ A, B, Γ, Δ , ὡς ὁ A πρὸς τὸν B , ο $<$ ύτως ὁ Γ πρὸς τὸν Δ , καὶ ὁ μὲν A τὸν Δ πολλαπλασιάσας τὸν E ποιείτω, ὁ δὲ B τὸν Γ πολλαπλασιάσας τὸν Z ποιείτω; λέγω, $<$ ότι $>$ ίσοξ ἐστὶν ὁ E τῷ Z .

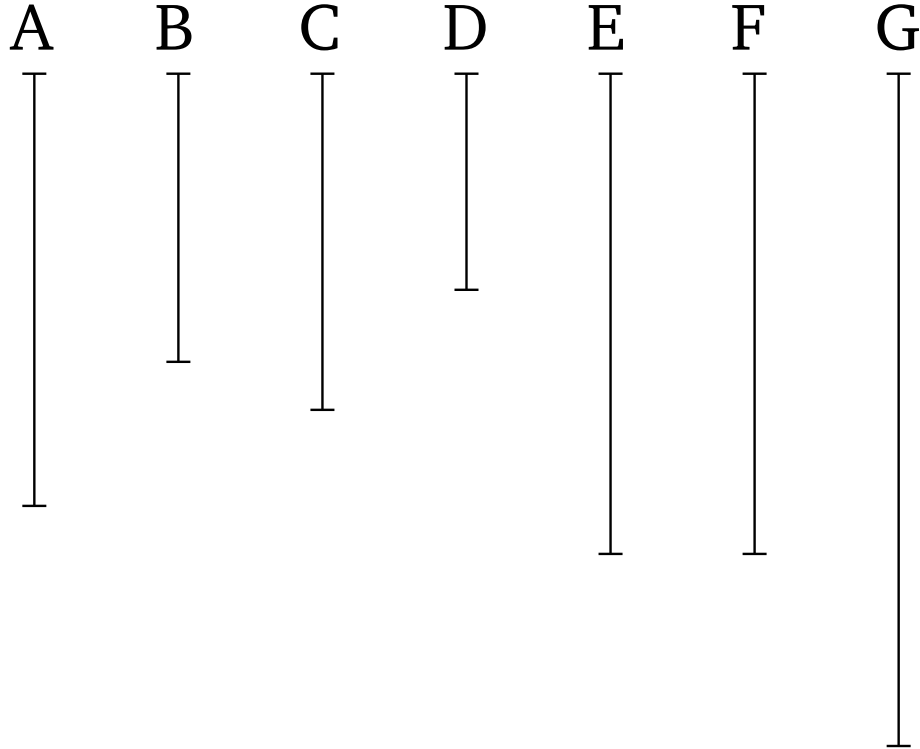
Let A, B, C , and D be four proportional numbers, (such that) as A (is) to B , so C (is) to D . And let A make E (by) multiplying D , and let B make F (by) multiplying C . I say that E is equal to F .

Ὅ γὰρ A τὸν Γ πολλαπλασιάσας τὸν H ποιείτω. ἐπεὶ ο $>$ ὖν ὁ A τὸν Γ πολλαπλασιάσας τὸν H πεποίηκεν, τὸν δὲ Δ πολλαπλασιάσας τὸν E πεποίηκεν, ἀριθμὸς δὴ ὁ A δύο ἀριθμοὺς τοὺς Γ, Δ πολλαπλασιάσας τοὺς H, E πεποίηκεν. $>$ έστιν $>$ άρα ὡς ὁ Γ πρὸς τὸν Δ , ο $<$ ύτως ὁ H πρὸς τὸν E .

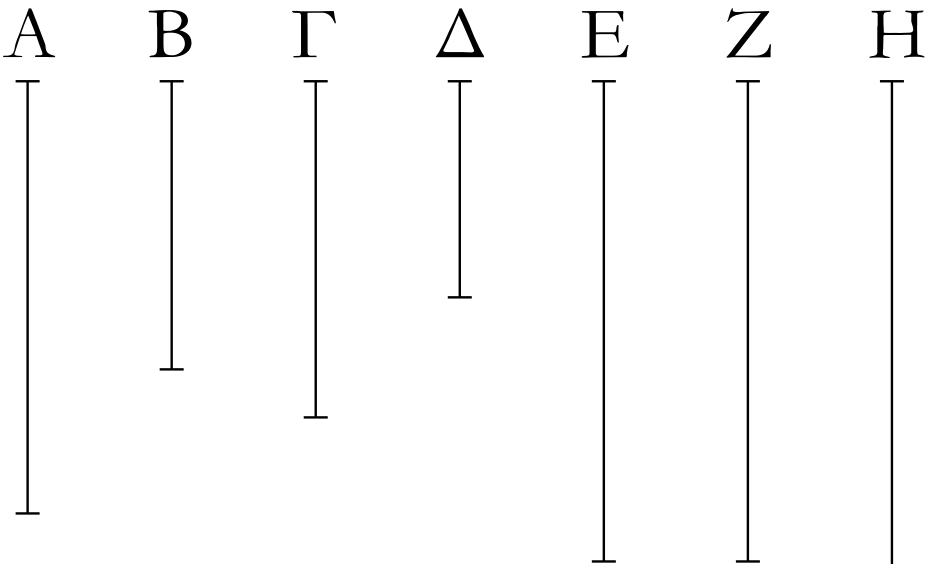
For let A make G (by) multiplying C . Therefore, since A has made G (by) multiplying C , and has made E (by) multiplying D , the number A has made G and E by multiplying the two numbers C and D (respectively). Thus, as C is to

ἀλλ' ὥς ὁ Γ πρὸς τὸν Δ, οὕτως ὁ Α πρὸς τὸν Β; καὶ ὥς
 ὡς ὁ Α πρὸς τὸν Β, οὕτως ὁ Η πρὸς τὸν Ε. πάλιν, ἐπεὶ ὁ
 Α τὸν Γ πολλαπλασιάσας τὸν Η πεποίηκεν, ἀλλὰ μὴν καὶ ὁ
 Β τὸν Γ πολλαπλασιάσας τὸν Ζ πεποίηκεν, δύο δὲ ἀριθμοὶ
 οἱ Α, Β ἀριθμὸν τινα τὸν Γ πολλαπλασιάσαντες τοὺς Η, Ζ
 πεποίηκασιν. ἔστιν ὡς ὁ Α πρὸς τὸν Β, οὕτως ὁ Η
 πρὸς τὸν Ζ. ἀλλὰ μὴν καὶ ὥς ὁ Α πρὸς τὸν Β, οὕτως ὁ Η
 πρὸς τὸν Ε; καὶ ὥς ὡς ὁ Η πρὸς τὸν Ε, οὕτως ὁ Η πρὸς
 τὸν Ζ. ὁ Η ὡς ὁ Α πρὸς τὸν Β, οὕτως ὁ Η πρὸς τὸν Ε, Ζ τὸν αὐτὸν ἔχει

D, so *G* (is) to *E* [Prop. 7.17]. But, as *C* (is) to *D*, so *A* (is)
 to *B*. Thus, also, as *A* (is) to *B*, so *G* (is) to *E*. Again, since
A has made *G* (by) multiplying *C*, but, in fact, *B* has also
 made *F* (by) multiplying *C*, the two numbers *A* and *B* have
 made *G* and *F* (respectively, by) multiplying some number
C. Thus, as *A* is to *B*, so *G* (is) to *F* [Prop. 7.18]. But, also,
 as *A* (is) to *B*, so *G* (is) to *E*. And thus, as *G* (is) to *E*, so
G (is) to *F*. Thus, *G* has the same ratio to each of *E* and *F*.
 Thus, *E* is equal to *F* [Prop. 5.9].



λόγον; ἴσος ἔστιν ὁ Ε τῷ Ζ.



Ἐστω δὲ πάλιν ἴσος ὁ Ε τῷ Ζ; λέγω, ὅτι ἔστιν ὥς ὁ Α
 πρὸς τὸν Β, οὕτως ὁ Γ πρὸς τὸν Δ.

Τῶν γὰρ αὐτῶν κατασκευασθέντων, ἐπεὶ ἴσος ἔστιν ὁ Ε
 τῷ Ζ, ἔστιν ὡς ὁ Η πρὸς τὸν Ε, οὕτως ὁ Η πρὸς

So, again, let *E* be equal to *F*. I say that as *A* is to *B*, so *C*
 (is) to *D*.

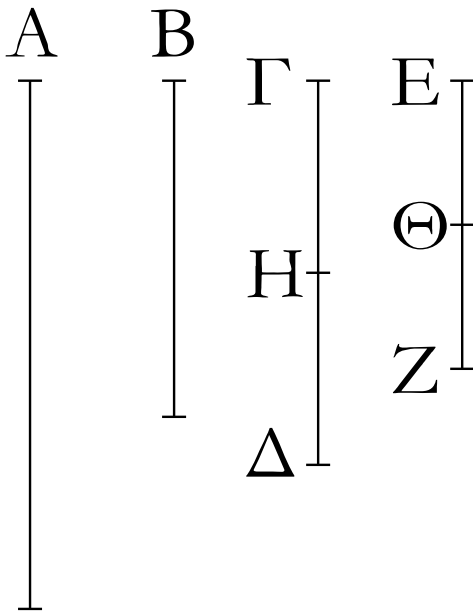
For, with the same construction, since *E* is equal to *F*, thus
 as *G* is to *E*, so *G* (is) to *F* [Prop. 5.7]. But, as *G* (is) to *E*,
 so *C* (is) to *D* [Prop. 7.17]. And as *G* (is) to *F*, so *A* (is)

τὸν Ζ. ἀλλ' ὥς μὲν ὁ Η πρὸς τὸν Ε, οὕτως ὁ Γ πρὸς τὸν Δ, ὥς δὲ ὁ Η πρὸς τὸν Ζ, οὕτως ὁ Α πρὸς τὸν Β. καὶ ὥς ἄρα ὁ Α πρὸς τὸν Β, οὕτως ὁ Γ πρὸς τὸν Δ; ὅπερ >εδει δεῖχαι.

† In modern notation, this proposition reads that if $a : b :: c : d$ then $ad = bc$, and vice versa, where all symbols denote numbers.

20.

Οἱ ἐλάχιστοι ἀριθμοὶ τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς μετροῦσι τοὺς τὸν αὐτὸν λόγον >έθονταξ ἰσάκις <ὅ τε μείζων τὸν μείζονα καὶ ὁ ἐλάσσων τὸν ἐλάσσονα.
>Ἐστωσαν γὰρ ἐλάχιστοι ἀριθμοὶ τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς Α, Β οἱ ΓΔ, ΕΖ; λέγω, <ὅτι ἰσάκις ὁ ΓΔ τὸν Α μετρεῖ καὶ ὁ ΕΖ τὸν Β.



Ὁ ΓΔ γὰρ τοῦ Α οὐκ ἐστὶ μέρη. εἰ γὰρ δυνατόν, >έστω; καὶ ὁ ΕΖ ἄρα τοῦ Β τὰ αὐτὰ μέρη ἐστίν, <ἅπερ ὁ ΓΔ τοῦ Α. <ὅσα ἄρα ἐστὶν ἐν τῷ ΓΔ μέρη τοῦ Α, τοσαῦτά ἐστι καὶ ἐν τῷ ΕΖ μέρη τοῦ Β. διηρήσθω ὁ μὲν ΓΔ εἰς τὰ τοῦ Α μέρη τὰ ΓΗ, ΗΔ, ὁ δὲ ΕΖ εἰς τὰ τοῦ Β μέρη τὰ ΕΘ, ΘΖ; >έσται δὴ ἴσον τὸ πλῆθος τῶν ΓΗ, ΗΔ τῷ πλῆθει τῶν ΕΘ, ΘΖ. καὶ ἐπεὶ ἴσοι εἰσὶν οἱ ΓΗ, ΗΔ ἀριθμοὶ ἀλλήλοισι, εἰσὶ δὲ καὶ οἱ ΕΘ, ΘΖ ἀριθμοὶ ἴσοι ἀλλήλοισι, καὶ ἐστὶν ἴσον τὸ πλῆθος τῶν ΓΗ, ΗΔ τῷ πλῆθει τῶν ΕΘ, ΘΖ, >έστιν ἄρα ὥς ὁ ΓΗ πρὸς τὸν ΕΘ, οὕτως ὁ ΗΔ πρὸς τὸν ΘΖ. >έσται ἄρα καὶ ὥς εἰς τῶν ἡγουμένων πρὸς <ένα τῶν ἐπομένων, οὕτως <ἅπαντες οἱ ἡγούμενοι πρὸς <ἅπαντας τοὺς ἐπομένους. >έστιν ἄρα ὥς ὁ ΓΗ πρὸς τὸν ΕΘ, οὕτως ὁ ΓΔ πρὸς τὸν ΕΖ; οἱ ΓΗ, ΕΘ ἄρα τοῖς ΓΔ, ΕΖ ἐν τῷ αὐτῷ λόγῳ εἰσὶν ἐλάσσονες >όντες αὐτῶν; ὅπερ ἐστὶν ἀδύνατον; ὑπόκειται γὰρ οἱ ΓΔ, ΕΖ ἐλάχιστοι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς. οὐκ ἄρα μέρη ἐστὶν ὁ ΓΔ τοῦ Α; μέρος ἄρα. καὶ ὁ ΕΖ τοῦ Β τὸ αὐτὸ μέρος ἐστίν, ὅπερ ὁ ΓΔ τοῦ Α; ἰσάκις ἄρα ὁ ΓΔ τὸν Α μετρεῖ καὶ ὁ ΕΖ τὸν Β; ὅπερ >εδει δεῖχαι.

21.

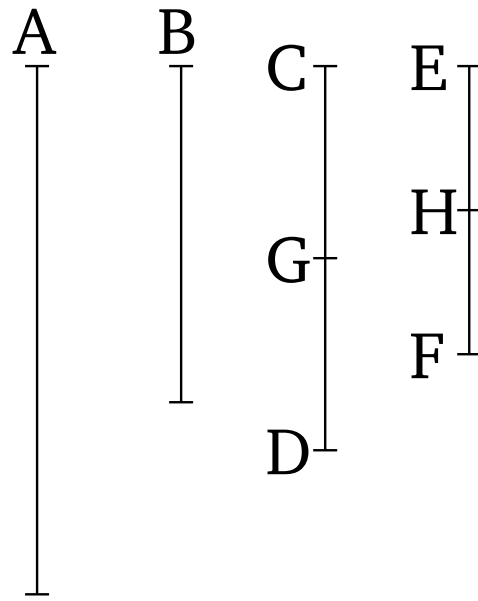
Οἱ πρῶτοι πρὸς ἀλλήλους ἀριθμοὶ ἐλάχιστοι εἰσι τῶν τὸν

to B [Prop. 7.18]. And, thus, as A (is) to B, so C (is) to D. (Which is) the very thing it was required to show.

Proposition 20

The least numbers of those (numbers) having the same ratio measure those (numbers) having the same ratio as them an equal number of times, the greater (measuring) the greater, and the lesser the lesser.

For let CD and EF be the least numbers having the same ratio as A and B (respectively). I say that CD measures A the same number of times as EF (measures) B .



For CD is not parts of A . For, if possible, let it be (parts of A). Thus, EF is also the same parts of B that CD (is) of A [Def. 7.20, Prop. 7.13]. Thus, as many parts of A as are in CD , so many parts of B are also in EF . Let CD be divided into the parts of A , CG and GD , and EF into the parts of B , EH and HF . So the multitude of (divisions) CG , GD will be equal to the multitude of (divisions) EH , HF . And since the numbers CG and GD are equal to one another, and the numbers EH and HF are also equal to one another, and the multitude of (divisions) CG , GD is equal to the multitude of (divisions) EH , HF , thus as CG is to EH , so GD (is) to HF . Thus, as one of the leading (numbers) is to one of the following, so will (the sum of) all of the leading (numbers) be to (the sum of) all of the following [Prop. 7.12]. Thus, as CG is to EH , so CD (is) to EF . Thus, CG and EH are in the same ratio as CD and EF , being less than them. The very thing is impossible. For CD and EF were assumed (to be) the least of those (numbers) having the same ratio as them. Thus, CD is not parts of A . Thus, (it is) a part (of A) [Prop. 7.4]. And EF is the same part of B that CD (is) of A [Def. 7.20, Prop. 7.13]. Thus, CD measures A the same number of times that EF (measures) B . (Which is) the very thing it was required to show.

Proposition 21

Numbers prime to one another are the least of those (num-

αὐτὸν λόγον ἐθόντων αὐτοῖς.

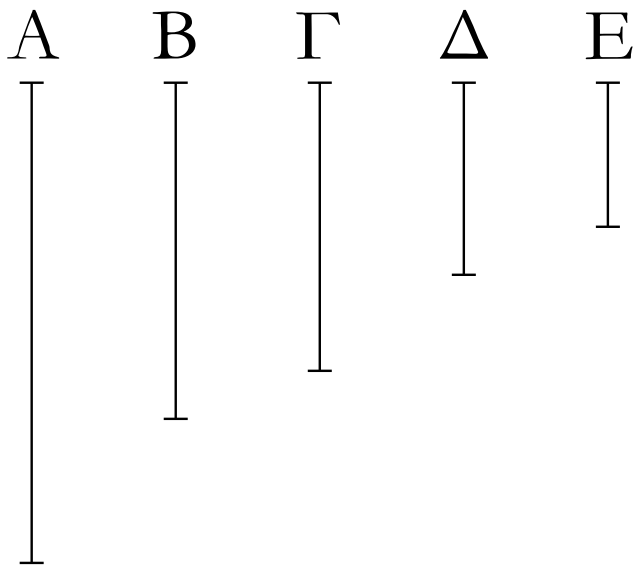
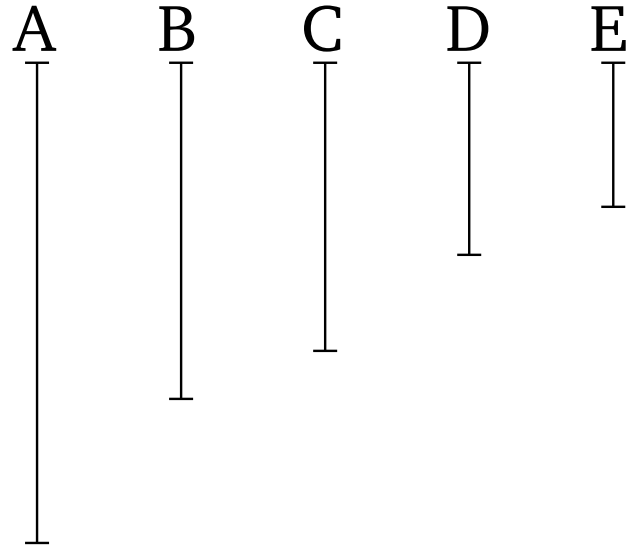
>Ἐστῶσαν πρῶτοι πρὸς ἀλλήλουξ ἀριθμοὶ οἱ A, B ; λέγω, <ὅτι οἱ A, B ἐλάχιστοί εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς.

Εἰ γὰρ μή, >έσονται τινεξ τῶν A, B ἐλάσσονεξ ἀριθμοὶ ἐν τῷ αὐτῷ λόγῳ >όντεξ τοῖς A, B . >έστῶσαν οἱ Γ, Δ .

bers) having the same ratio as them.

Let A and B be numbers prime to one another. I say that A and B are the least of those (numbers) having the same ratio as them.

For if not then there will be some numbers less than A and B which are in the same ratio as A and B . Let them be C and D .

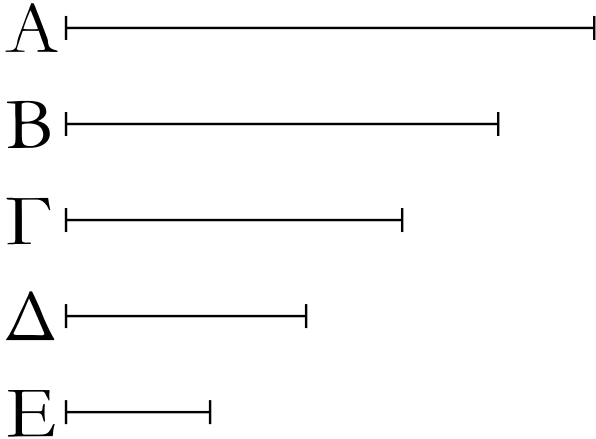


Ἐπεὶ ο>ὖν οἱ ἐλάχιστοι ἀριθμοὶ τῶν τὸν αὐτὸν λόγον ἐθόντων μετροῦσι τοὺξ τὸν αὐτὸν λόγον >έθονταξ ἰσάκιξ <ὅ τε μείζων τὸν μείζονα καὶ ὁ ἐλάττων τὸν ἐλάττονα, τουτέστιν <ὅ τε ἡγούμενοξ τὸν ἡγούμενον καὶ ὁ ἐπόμενοξ τὸν ἐπόμενον, ἰσάκιξ >άρα ὁ Γ τὸν A μετρεῖ καὶ ὁ Δ τὸν B . ὁσάκιξ δὴ ὁ Γ τὸν A μετρεῖ, τοσαῦται μονάδεξ >έστῶσαν ἐν τῷ E . καὶ ὁ Δ >άρα τὸν B μετρεῖ κατὰ τὰξ ἐν τῷ E μονάδαξ. καὶ ἐπεὶ ὁ Γ τὸν A μετρεῖ κατὰ τὰξ ἐν τῷ E μονάδαξ, καὶ ὁ E >άρα τὸν A μετρεῖ κατὰ τὰξ ἐν τῷ Γ μονάδαξ. διὰ τὰ αὐτὰ δὴ ὁ E καὶ τὸν B μετρεῖ κατὰ τὰξ ἐν τῷ Δ μονάδαξ. ὁ E >άρα τοὺξ A, B μετρεῖ πρώτουξ >όνταξ πρὸς ἀλλήλουξ; <ὅπερ ἐστὶν ἀδύνατον. οὐκ >άρα >έσονται τινεξ τῶν A, B ἐλάσσονεξ ἀριθμοὶ ἐν τῷ αὐτῷ λόγῳ >όντεξ τοῖς A, B . οἱ A, B >άρα ἐλάχιστοί εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς; <ὅπερ >έδει δεῖλαι.

Therefore, since the least numbers of those (numbers) having the same ratio measure those (numbers) having the same ratio (as them) an equal number of times, the greater (measuring) the greater, and the lesser the lesser—that is to say, the leading (measuring) the leading, and the following the following— C thus measures A the same number of times that D (measures) B [Prop. 7.20]. So as many times as C measures A , so many units let there be in E . Thus, D also measures B according to the units in E . And since C measures A according to the units in E , E thus also measures A according to the units in C [Prop. 7.16]. So, for the same (reasons), E also measures B according to the units in D [Prop. 7.16]. Thus, E measures A and B , which are prime to one another. The very thing is impossible. Thus, there cannot be any numbers less than A and B which are in the same ratio as A and B . Thus, A and B are the least of those (numbers) having the same ratio as them. (Which is) the very thing it was required to show.

22.

Οἱ ἐλάχιστοι ἀριθμοὶ τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς
πρῶτοι πρὸς ἀλλήλους εἰσίν.



>Ἐστωσαν ἐλάχιστοι ἀριθμοὶ τῶν τὸν αὐτὸν λόγον ἐθόντων
αὐτοῖς οἱ A, B ; λέγω, <ὅτι οἱ A, B πρῶτοι πρὸς ἀλλήλους
εἰσίν.

Εἰ γὰρ μὴ εἰσι πρῶτοι πρὸς ἀλλήλους, μετρήσει τις αὐτοῖς
ἀριθμὸς. μετρεῖτω, καὶ >έστω ὁ Γ . καὶ ὅσακις μὲν ὁ Γ τὸν
 A μετρεῖ, τοσαῦται μονάδες >έστωσαν ἐν τῷ Δ , ὅσακις δὲ
ὁ Γ τὸν B μετρεῖ, τοσαῦται μονάδες >έστωσαν ἐν τῷ E .

Ἐπεὶ ὁ Γ τὸν A μετρεῖ κατὰ τὰς ἐν τῷ Δ μονάδας, ὁ Γ >άρα
τὸν Δ πολλαπλασιάσας τὸν A πεποίηκεν. διὰ τὰ αὐτὰ δὴ
καὶ ὁ Γ τὸν E πολλαπλασιάσας τὸν B πεποίηκεν. ἀριθμὸς
δὴ ὁ Γ δύο ἀριθμοὺς τοῖς Δ, E πολλαπλασιάσας τοῖς A, B
πεποίηκεν; >έστιν >άρα ὥς ὁ Δ πρὸς τὸν E , ο<ύτως ὁ
 A πρὸς τὸν B ; οἱ Δ, E >άρα τοῖς A, B ἐν τῷ αὐτῷ λόγῳ
εἰσὶν ἐλάσσονες >όντες αὐτῶν; <όπερ ἐστὶν ἀδύνατον. οὐκ
>άρα τοῖς A, B ἀριθμοῖς ἀριθμὸς τις μετρήσει. οἱ A, B
>άρα πρῶτοι πρὸς ἀλλήλους εἰσίν; <όπερ >έδει δεῖχαι.

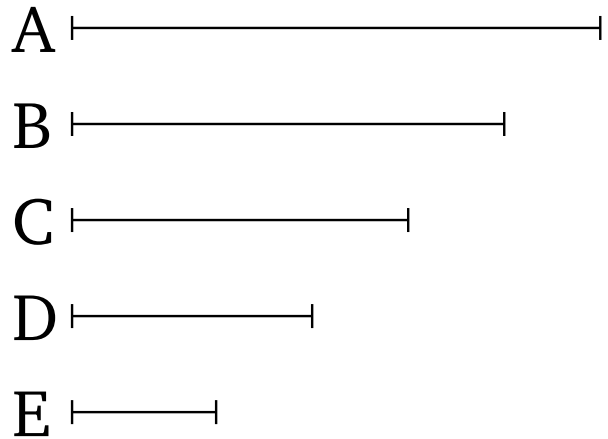
23.

Ἐὰν δύο ἀριθμοὶ πρῶτοι πρὸς ἀλλήλους >ῶσιν, ὁ τὸν <ένα
αὐτῶν μετρῶν ἀριθμὸς πρὸς τὸν λοιπὸν πρῶτος >έσται.

>Ἐστωσαν δύο ἀριθμοὶ πρῶτοι πρὸς ἀλλήλους οἱ A, B , τὸν
δὲ A μετρεῖτω τις ἀριθμὸς ὁ Γ ; λέγω, <ὅτι καὶ οἱ Γ, B πρῶτοι
πρὸς ἀλλήλους εἰσίν.

Proposition 22

The least numbers of those (numbers) having the same ratio
as them are prime to one another.



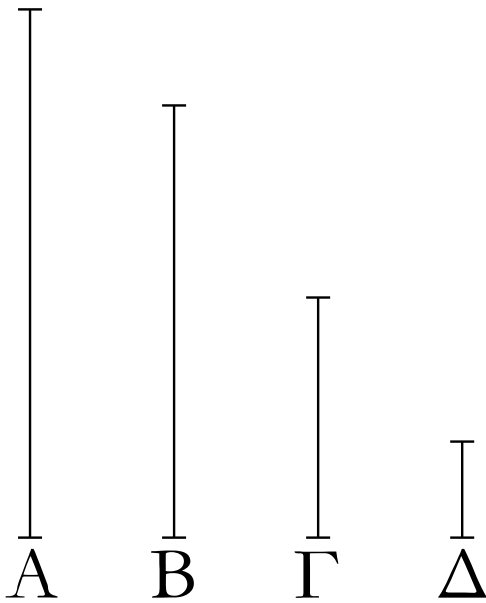
Let A and B be the least numbers of those (numbers) having
the same ratio as them. I say that A and B are prime to one
another.

For if they are not prime to one another, (then) some number
will measure them. Let it (so measure them), and let it be
 C . And as many times as C measures A , so many units let
there be in D . And as many times as C measures B , so
many units let there be in E .

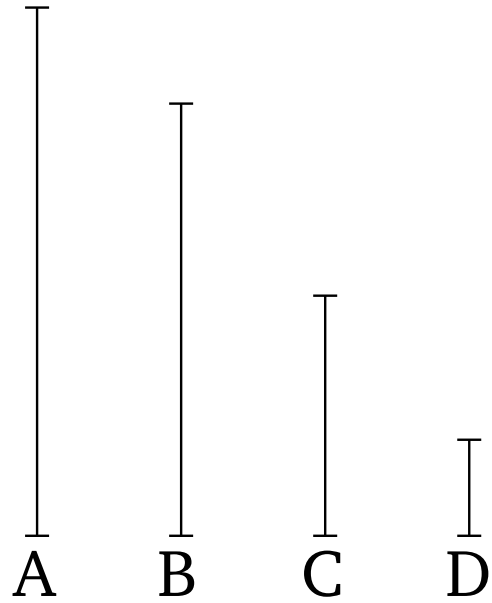
Since C measures A according to the units in D , C has thus
made A (by) multiplying D [Def. 7.15]. So, for the same
(reasons), C has also made B (by) multiplying E . So the
number C has made A and B (by) multiplying the two num-
bers D and E (respectively). Thus, as D is to E , so A (is)
to B [Prop. 7.17]. Thus, D and E are in the same ratio as
 A and B , being less than them. The very thing is impossible.
Thus, some number does not measure the numbers A and
 B . Thus, A and B are prime to one another. (Which is) the
very thing it was required to show.

Proposition 23

If two numbers are prime to one another, (then) a number
measuring one of them will be prime to the remaining (one).
Let A and B be two numbers (which are) prime to one an-
other, and let some number C measure A . I say that C and
 B are also prime to one another.



Εἰ γὰρ μὴ εἰσιν οἱ Γ , B πρῶτοι πρὸς ἀλλήλους, μετρήσει [τις] τοὺς Γ , B ἀριθμὸς. μετρεῖτω, καὶ >έστω ὁ Δ . ἐπεὶ ὁ Δ τὸν Γ μετρεῖ, ὁ δὲ Γ τὸν A μετρεῖ, καὶ ὁ Δ >άρα τὸν A μετρεῖ. μετρεῖ δὲ καὶ τὸν B ; ὁ Δ >άρα τοὺς A , B μετρεῖ πρώτους >όντας πρὸς ἀλλήλους; <όπερ ἐστὶν ἀδύνατον. οὐκ >άρα τοὺς Γ , B ἀριθμὸς ἀριθμὸς τις μετρήσει. οἱ Γ , B >άρα πρῶτοι πρὸς ἀλλήλους εἰσίν; <όπερ >έδει δεῖχαι.



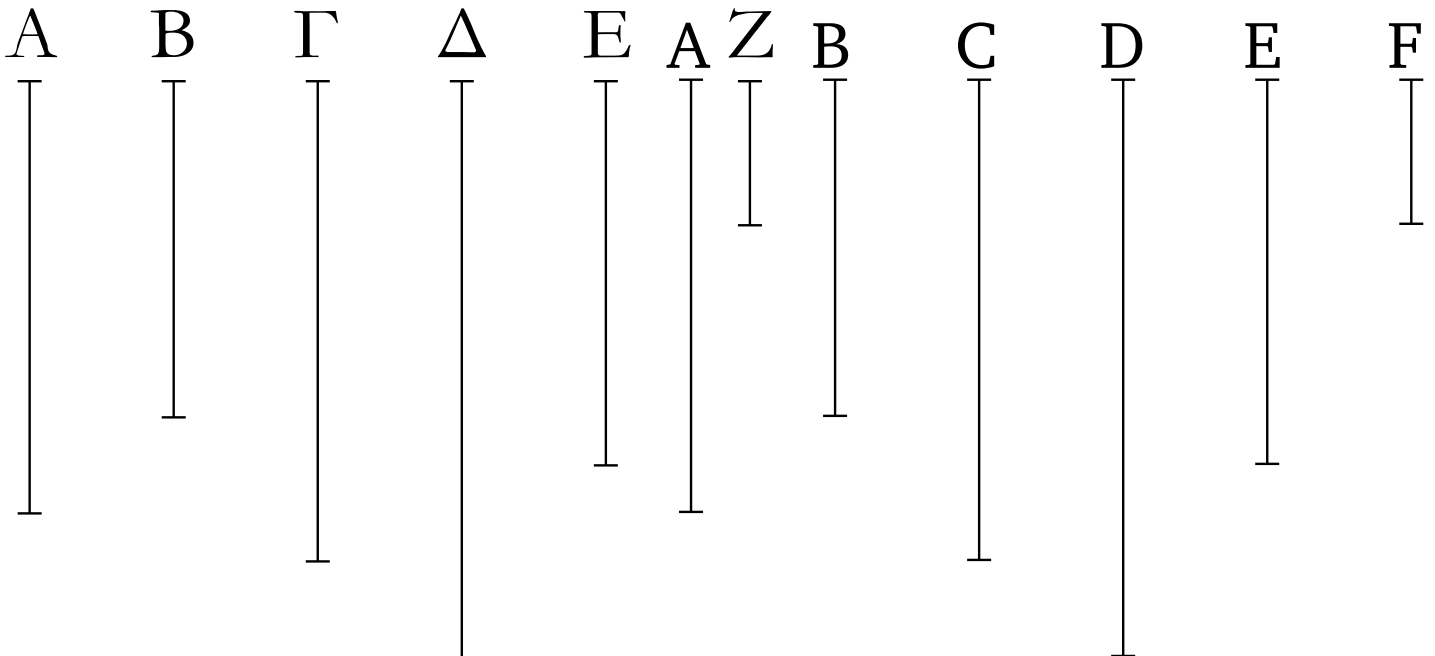
For if C and B are not prime to one another, (then) [some] number will measure C and B . Let it (so) measure (them), and let it be D . Since D measures C , and C measures A , D thus also measures A . And (D) also measures B . Thus, D measures A and B , which are prime to one another. The very thing is impossible. Thus, some number does not measure the numbers C and B . Thus, C and B are prime to one another. (Which is) the very thing it was required to show.

24.

Proposition 24

Ἐὰν δύο ἀριθμοὶ πρὸς τινὰ ἀριθμὸν πρῶτοι >ῶσιν, καὶ ὁ ἐκ αὐτῶν γενόμενος πρὸς τὸν αὐτὸν πρώτος >έσται.

If two numbers are prime to some number, (then) the number created from (multiplying) the former (two numbers) will also be prime to the latter (number).



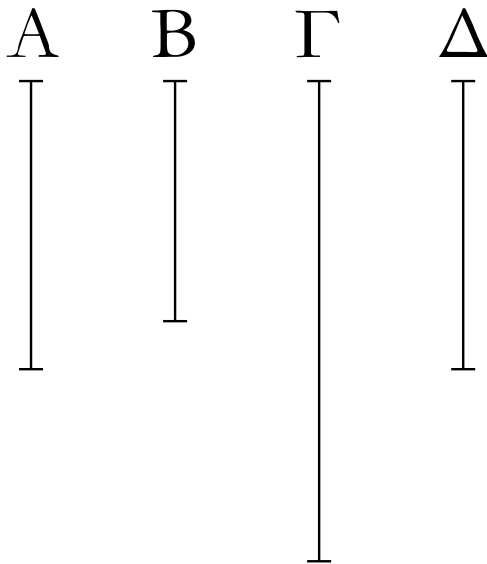
Δύο γὰρ ἀριθμοὶ οἱ A , B πρὸς τινὰ ἀριθμὸν τὸν Γ πρῶτοι >έστωσαν, καὶ ὁ A τὸν B πολλαπλασιάσας τὸν Δ ποιεῖτω; λέγω, <ὅτι οἱ Γ , Δ πρῶτοι πρὸς ἀλλήλους εἰσίν.
Εἰ γὰρ μὴ εἰσιν οἱ Γ , Δ πρῶτοι πρὸς ἀλλήλους, μετρήσει [τις] τοὺς Γ , Δ ἀριθμὸς. μετρεῖτω, καὶ >έστω ὁ E . καὶ ἐπεὶ οἱ Γ , Δ πρῶτοι πρὸς ἀλλήλους εἰσίν, τὸν δὲ Γ μετρεῖ τις ἀριθμὸς ὁ E , οἱ A , E >άρα πρῶτοι πρὸς ἀλλήλους εἰσίν.

For let A and B be two numbers (which are both) prime to some number C . And let A make D (by) multiplying B . I say that C and D are prime to one another.
For if C and D are not prime to one another, (then) [some] number will measure C and D . Let it (so) measure them, and let it be E . And since C and A are prime to one another, and some number E measures C , A and E are thus

ὁσάκις δὴ ὁ E τὸν Δ μετρεῖ, τοσαῦται μονάδες >έστωσαν ἐν τῷ Z ; καὶ ὁ Z >άρα τὸν Δ μετρεῖ κατὰ τὰς ἐν τῷ E μονάδας. ὁ E >άρα τὸν Z πολλαπλασιάσας τὸν Δ πεποίηκεν. ἀλλὰ μὴν καὶ ὁ A τὸν B πολλαπλασιάσας τὸν Δ πεποίηκεν; >ἴσος >άρα ἐστὶν ὁ ἐκ τῶν E, Z τῷ ἐκ τῶν A, B . ἔαν δὲ ὁ ὑπὸ τῶν >άκρων >ἴσος >ῆ| τῷ ὑπὸ τῶν μέσων, οἱ τέσσαρες ἀριθμοὶ ἀνάλογόν εἰσιν; >έστιν >άρα ὡς ὁ E πρὸς τὸν A , ο<ύτως ὁ B πρὸς τὸν Z . οἱ δὲ A, E πρῶτοι, οἱ δὲ πρῶτοι καὶ ἐλάχιστοι, οἱ δὲ ἐλάχιστοι ἀριθμοὶ τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς μετροῦσι τοὺς τὸν αὐτὸν λόγον >έθοντα; ἰσάκις <ὁ τε μείζων τὸν μείζονα καὶ ὁ ἐλάσσων τὸν ἐλάσσονα, τουτέστιν <ὁ τε ἡγούμενος τὸν ἡγούμενον καὶ ὁ ἐπόμενος τὸν ἐπόμενον; ὁ E >άρα τὸν B μετρεῖ. μετρεῖ δὲ καὶ τὸν Γ ; ὁ E >άρα τοὺς B, Γ μετρεῖ πρῶτους >όντα; πρὸς ἀλλήλους; <ὅπερ ἐστὶν ἀδύνατον. οὐκ >άρα τοὺς Γ, Δ ἀριθμοὺς ἀριθμὸς τις μετρήσει. οἱ Γ, Δ >άρα πρῶτοι πρὸς ἀλλήλους εἰσιν; <ὅπερ >έδει δεῖχαι.

25.

Ἐὰν δύο ἀριθμοὶ πρῶτοι πρὸς ἀλλήλους >ῶσιν, ὁ ἐκ τοῦ ἐνόξ αὐτῶν γενόμενος πρὸς τὸν λοιπὸν πρῶτος >έσται.
>Ἐστωσαν δύο ἀριθμοὶ πρῶτοι πρὸς ἀλλήλους οἱ A, B , καὶ ὁ A ἐαυτὸν πολλαπλασιάσας τὸν Γ ποιεῖτω; λέγω, <ὅτι οἱ B, Γ πρῶτοι πρὸς ἀλλήλους εἰσιν.



Κεῖσθω γὰρ τῷ A >ἴσος ὁ Δ . ἐπεὶ οἱ A, B πρῶτοι πρὸς ἀλλήλους εἰσιν, >ἴσος δὲ ὁ A τῷ Δ , καὶ οἱ Δ, B >άρα πρῶτοι πρὸς ἀλλήλους εἰσιν. ἑκάτερος >άρα τῶν Δ, A πρὸς τὸν B πρῶτός ἐστιν; καὶ ὁ ἐκ τῶν Δ, A >άρα γενόμενος πρὸς τὸν B πρῶτος >έσται. ὁ δὲ ἐκ τῶν Δ, A γενόμενος ἀριθμὸς ἐστὶν ὁ Γ . οἱ Γ, B >άρα πρῶτοι πρὸς ἀλλήλους εἰσιν; <ὅπερ >έδει δεῖχαι.

26.

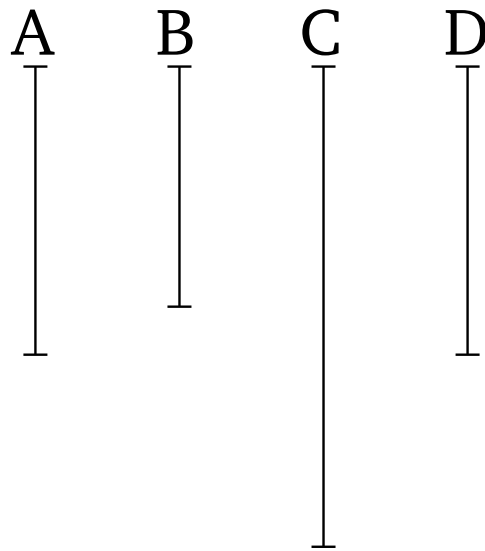
Ἐὰν δύο ἀριθμοὶ πρὸς δύο ἀριθμοὺς ἀμφοτέρωι πρὸς

prime to one another [Prop. 7.23]. So as many times as E measures D , so many units let there be in F . Thus, F also measures D according to the units in E [Prop. 7.16]. Thus, E has made D (by) multiplying F [Def. 7.15]. But, in fact, A has also made D (by) multiplying B . Thus, the (number created) from (multiplying) E and F is equal to the (number created) from (multiplying) A and B . And if the (rectangle contained) by the (two) outermost is equal to the (rectangle contained) by the middle (two then) the four numbers are proportional [Prop. 6.15]. Thus, as E is to A , so B (is) to F . And A and E (are) prime (to one another). And (numbers) prime (to one another) are also the least (of those numbers having the same ratio) [Prop. 7.21]. And the least numbers of those (numbers) having the same ratio measure those (numbers) having the same ratio as them an equal number of times, the greater (measuring) the greater, and the lesser the lesser—that is to say, the leading (measuring) the leading, and the following the following [Prop. 7.20]. Thus, E measures B . And it also measures C . Thus, E measures B and C , which are prime to one another. The very thing is impossible. Thus, some number cannot measure the numbers C and D . Thus, C and D are prime to one another. (Which is) the very thing it was required to show.

Proposition 25

If two numbers are prime to one another, (then) the number created from (squaring) one of them will be prime to the remaining (number).

Let A and B be two numbers (which are) prime to one another. And let A make C (by) multiplying itself. I say that B and C are prime to one another.



For let D be made equal to A . Since A and B are prime to one another, and A (is) equal to D , D and B are thus also prime to one another. Thus, D and A are each prime to B . Thus, the (number) created from (multiplying) D and A will also be prime to B [Prop. 7.24]. And C is the number created from (multiplying) D and A . Thus, C and B are prime to one another. (Which is) the very thing it was required to show.

Proposition 26

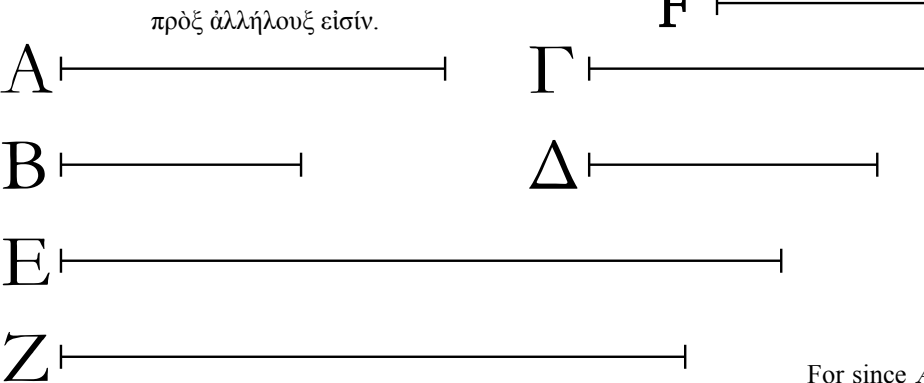
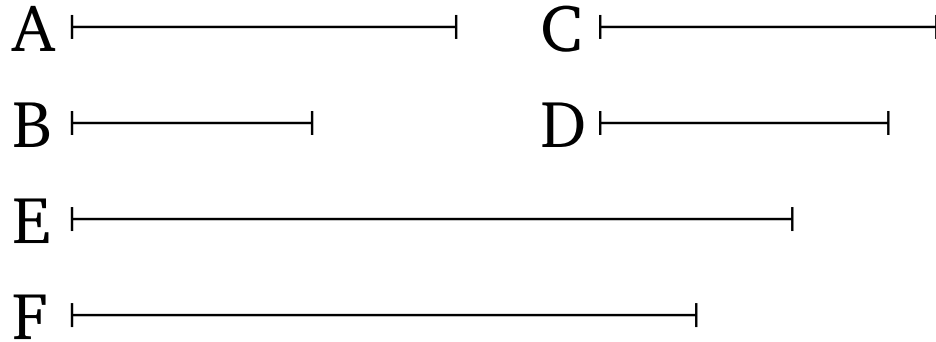
If two numbers are both prime to each of two numbers,

ἐκάτερον πρῶτοι >ῶσιν, καὶ οἱ ἐκ αὐτῶν γενόμενοι πρῶτοι πρὸς ἀλλήλουξ >έσονται.

Δύο γὰρ ἀριθμοὶ οἱ A , B πρὸς δύο ἀριθμοὺξ τοὺξ Γ , Δ ἀμφοτέρω πρὸξ ἐκάτερον πρῶτοι >έστωσαν, καὶ ὁ μὲν A τὸν B πολλαπλασιάσας τὸν E ποιείτω, ὁ δὲ Γ τὸν Δ πολλαπλασιάσας τὸν Z ποιείτω; λέγω, <ὅτι οἱ E , Z πρῶτοι

(then) the (numbers) created from (multiplying) them will also be prime to one another.

For let two numbers, A and B , both be prime to each of two numbers, C and D . And let A make E (by) multiplying B , and let C make F (by) multiplying D . I say that E and F are prime to one another.



Ἐπεὶ γὰρ ἐκάτεροξ τῶν A , B πρὸξ τὸν Γ πρῶτόξ ἐστιν, καὶ ὁ ἐκ τῶν A , B >άρα γενόμενοξ πρὸξ τὸν Γ πρῶτοξ >έσται. ὁ δὲ ἐκ τῶν A , B γενόμενόξ ἐστιν ὁ E ; οἱ E , Γ >άρα πρῶτοι πρὸξ ἀλλήλουξ εἰσίν. διὰ τὰ αὐτὰ δὴ καὶ οἱ E , Δ πρῶτοι πρὸξ ἀλλήλουξ εἰσίν. ἐκάτεροξ >άρα τῶν Γ , Δ πρὸξ τὸν E πρῶτόξ ἐστιν. καὶ ὁ ἐκ τῶν Γ , Δ >άρα γενόμενοξ πρὸξ τὸν E πρῶτοξ >έσται. ὁ δὲ ἐκ τῶν Γ , Δ γενόμενόξ ἐστιν ὁ Z . οἱ E , Z >άρα πρῶτοι πρὸξ ἀλλήλουξ εἰσίν; <ὅπερ >έδει δεῖχαι.

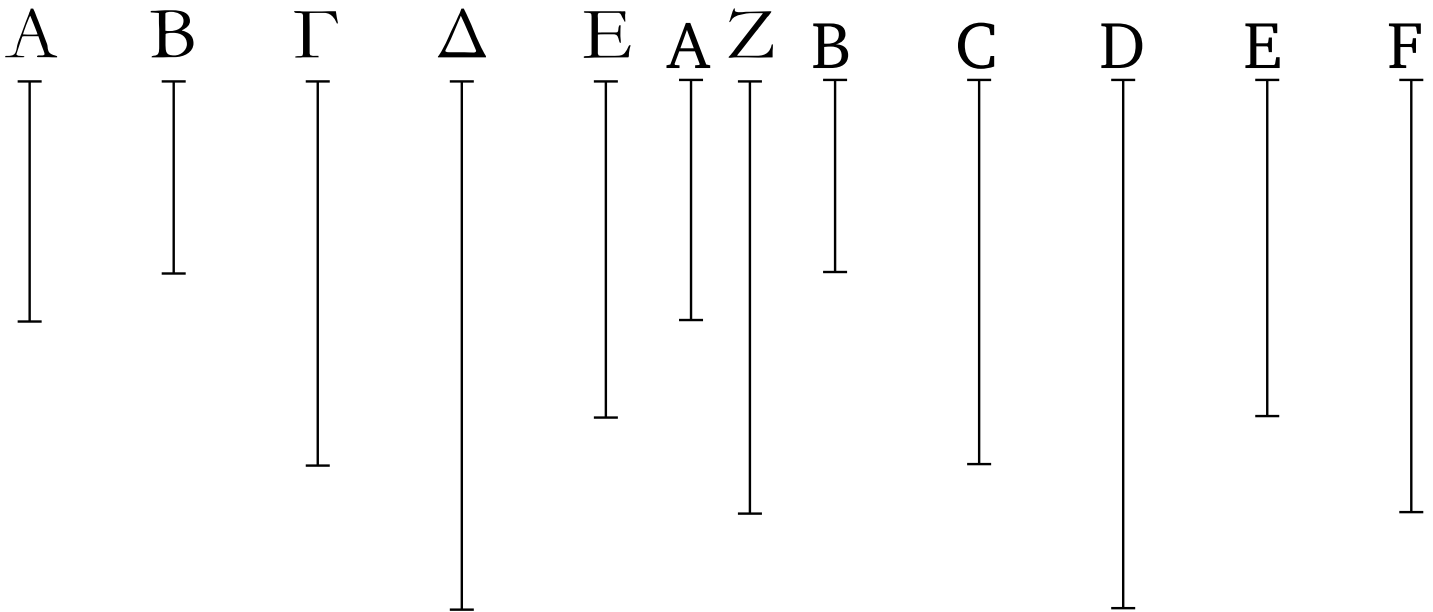
For since A and B are each prime to C , the (number) created from (multiplying) A and B will thus also be prime to C [Prop. 7.24]. And E is the (number) created from (multiplying) A and B . Thus, E and C are prime to one another. So, for the same (reasons), E and D are also prime to one another. Thus, C and D are each prime to E . Thus, the (number) created from (multiplying) C and D will also be prime to E [Prop. 7.24]. And F is the (number) created from (multiplying) C and D . Thus, E and F are prime to one another. (Which is) the very thing it was required to show.

27.

Proposition 27[†]

Ἐὰν δύο ἀριθμοὶ πρῶτοι πρὸξ ἀλλήλουξ >ῶσιν, καὶ πολλαπλασιάσας ἑαυτὸν ποιῇ| τινα, οἱ γενόμενοι ἐκ αὐτῶν πρῶτοι πρὸξ ἀλλήλουξ >έσονται, <ὰν οἱ ἐκ ἀρθῇξ τοὺξ γενομένουξ πολλαπλασιάσαντεξ ποιῶσιν τιναξ, κἀκεῖνοι πρῶτοι πρὸξ ἀλλήλουξ >έσονται [καὶ αἱ ἐπεὶ τοὺξ >άκρουξ τοῦτο συμβαίνει].

If two numbers are prime to one another and each makes some (number by) multiplying itself, (then) the numbers created from them will be prime to one another, and if the original (numbers) make some (more numbers by) multiplying the created (numbers, then) these will also be prime to one another [and this always happens with the extremes].



>Ἐστωσαν δύο ἀριθμοὶ πρῶτοι πρὸς ἀλλήλους οἱ A , B , καὶ ὁ A ἑαυτὸν μὲν πολλαπλασιάσας τὸν Γ ποιεῖτω, τὸν δὲ Γ πολλαπλασιάσας τὸν Δ ποιεῖτω, ὁ δὲ B ἑαυτὸν μὲν πολλαπλασιάσας τὸν E ποιεῖτω, τὸν δὲ E πολλαπλασιάσας τὸν Z ποιεῖτω; λέγω, <ὅτι οἱ τε Γ , E καὶ οἱ Δ , Z πρῶτοι πρὸς ἀλλήλους εἰσίν.

Ἐπεὶ γὰρ οἱ A , B πρῶτοι πρὸς ἀλλήλους εἰσίν, καὶ ὁ A ἑαυτὸν πολλαπλασιάσας τὸν Γ πεποίηκεν, οἱ Γ , B >ἀρα πρῶτοι πρὸς ἀλλήλους εἰσίν. ἐπεὶ οὖν οἱ Γ , B πρῶτοι πρὸς ἀλλήλους εἰσίν, καὶ ὁ B ἑαυτὸν πολλαπλασιάσας τὸν E πεποίηκεν, οἱ Γ , E >ἀρα πρῶτοι πρὸς ἀλλήλους εἰσίν. πάλιν, ἐπεὶ οἱ A , B πρῶτοι πρὸς ἀλλήλους εἰσίν, καὶ ὁ B ἑαυτὸν πολλαπλασιάσας τὸν E πεποίηκεν, οἱ A , E >ἀρα πρῶτοι πρὸς ἀλλήλους εἰσίν. ἐπεὶ οὖν δύο ἀριθμοὶ οἱ A , Γ πρὸς δύο ἀριθμοὺς τοὺς B , E ἀμφοτέροι πρὸς ἑκάτερον πρῶτοί εἰσιν, καὶ ὁ ἐκ τῶν A , Γ >ἀρα γενόμενος πρὸς τὸν ἐκ τῶν B , E πρῶτός ἐστιν. καὶ ἐστὶν ὁ μὲν ἐκ τῶν A , Γ ὁ Δ , ὁ δὲ ἐκ τῶν B , E ὁ Z . οἱ Δ , Z >ἀρα πρῶτοι πρὸς ἀλλήλους εἰσίν; <ὅπερ >έδει δεῖξαι.

Let A and B be two numbers prime to one another, and let A make C (by) multiplying itself, and let it make D (by) multiplying C . And let B make E (by) multiplying itself, and let it make F by multiplying E . I say that C and E , and D and F , are prime to one another.

For since A and B are prime to one another, and A has made C (by) multiplying itself, C and B are thus prime to one another [Prop. 7.25]. Therefore, since C and B are prime to one another, and B has made E (by) multiplying itself, C and E are thus prime to one another [Prop. 7.25]. Again, since A and B are prime to one another, and B has made E (by) multiplying itself, A and E are thus prime to one another [Prop. 7.25]. Therefore, since the two numbers A and C are both prime to each of the two numbers B and E , the (number) created from (multiplying) A and C is thus prime to the (number created) from (multiplying) B and E [Prop. 7.26]. And D is the (number created) from (multiplying) A and C , and F the (number created) from (multiplying) B and E . Thus, D and F are prime to one another. (Which is) the very thing it was required to show.

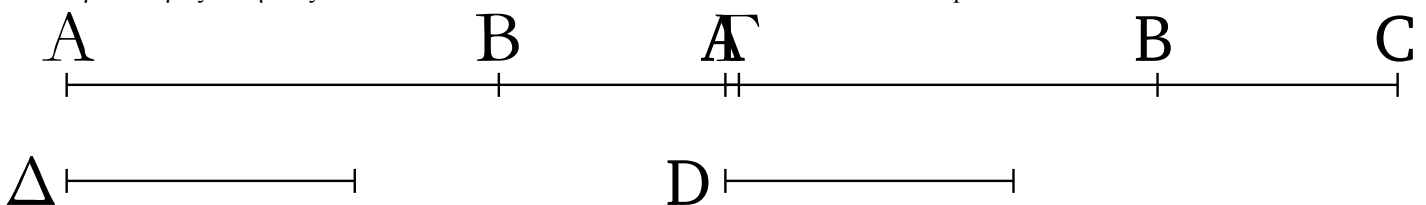
† In modern notation, this proposition states that if a is prime to b , then a^2 is also prime to b^2 , as well as a^3 to b^3 , etc., where all symbols denote numbers.

28.

Proposition 28

Ἐὰν δύο ἀριθμοὶ πρῶτοι πρὸς ἀλλήλους >ῶσιν, καὶ συναμφοτέρωσιν πρὸς ἑκάτερον αὐτῶν πρῶτος >έσται; καὶ ἔαν συναμφοτέροσιν πρὸς <ένα τινὰ αὐτῶν πρῶτος >ῇ|, καὶ οἱ ἐκ ἀρθῆς ἀριθμοὶ πρῶτοι πρὸς ἀλλήλους >έσονται.

If two numbers are prime to one another, (then) their sum will also be prime to each of them. And if the sum (of two numbers) is prime to any one of them, (then) the original numbers will also be prime to one another.



Συγκείσθωσαν γὰρ δύο ἀριθμοὶ πρῶτοι πρὸς ἀλλήλους οἱ AB , BC ; λέγω, <ὅτι καὶ συναμφοτέροσιν ὁ AC πρὸς ἑκάτερον τῶν AB , BC πρῶτός ἐστιν.

Εἰ γὰρ μὴ εἰσιν οἱ ΓA , AB πρῶτοι πρὸς ἀλλήλους, μετρήσει τις τοὺς ΓA , AB ἀριθμός. μετρεῖτω, καὶ >έστω ὁ Δ . ἐπεὶ οὖν ὁ Δ τοὺς ΓA , AB μετρεῖ, καὶ λοιπὸν >ἀρα τὸν BC μετρήσει. μετρεῖ δὲ καὶ τὸν BA ; ὁ Δ >ἀρα τοὺς AB ,

For let the two numbers, AB and BC , (which are) prime to one another, be laid down together. I say that their sum AC is also prime to each of AB and BC .

For if CA and AB are not prime to one another, (then) some number will measure CA and AB . Let it (so) measure (them), and let it be D . Therefore, since D measures CA and AB , it will thus also measure the remainder BC .

ΒΓ μετρεῖ πρώτους >όνταξ πρὸξ ἀλλήλουξ; <όπερ ἐστὶν ἀδύνατον. οὐκ >άρα τοὺξ ΓΑ, ΑΒ ἀριθμοὺξ ἀριθμόξ τιξ μετρήσει; οἱ ΓΑ, ΑΒ >άρα πρῶτοι πρὸξ ἀλλήλουξ εἰσίν. διὰ τὰ αὐτὰ δὴ καὶ οἱ ΑΓ, ΓΒ πρῶτοι πρὸξ ἀλλήλουξ εἰσίν. ὁ ΓΑ >άρα πρὸξ ἑκάτερον τῶν ΑΒ, ΒΓ πρῶτόξ ἐστίν. >Ἐστωσαν δὴ πάλιν οἱ ΓΑ, ΑΒ πρῶτοι πρὸξ ἀλλήλουξ; λέγω, <ὅτι καὶ οἱ ΑΒ, ΒΓ πρῶτοι πρὸξ ἀλλήλουξ εἰσίν. Εἰ γὰρ μὴ εἰσίν οἱ ΑΒ, ΒΓ πρῶτοι πρὸξ ἀλλήλουξ, μετρήσει τιξ τοὺξ ΑΒ, ΒΓ ἀριθμόξ. μετρεῖτω, καὶ >έστω ὁ Δ. καὶ ἐπεὶ ὁ Δ ἑκάτερον τῶν ΑΒ, ΒΓ μετρεῖ, καὶ <όλον >άρα τὸν ΓΑ μετρήσει. μετρεῖ δὲ καὶ τὸν ΑΒ; ὁ Δ >άρα τοὺξ ΓΑ, ΑΒ μετρεῖ πρώτους >όνταξ πρὸξ ἀλλήλουξ; <όπερ ἐστὶν ἀδύνατον. οὐκ >άρα τοὺξ ΑΒ, ΒΓ ἀριθμοὺξ ἀριθμόξ τιξ μετρήσει. οἱ ΑΒ, ΒΓ >άρα πρῶτοι πρὸξ ἀλλήλουξ εἰσίν; <όπερ >έδει δεῖχαι.

29.

<Ἀπαξ πρῶτοξ ἀριθμόξ πρὸξ <άπαντα ἀριθμόν, <ὸν μὴ μετρεῖ, πρῶτόξ ἐστίν. >Ἐστω πρῶτοξ ἀριθμόξ ὁ Α καὶ τὸν Β μὴ μετρεῖτω; λέγω, <ὅτι οἱ Β, Α πρῶτοι πρὸξ ἀλλήλουξ εἰσίν.

A —————

B —————

Γ —————

Εἰ γὰρ μὴ εἰσίν οἱ Β, Α πρῶτοι πρὸξ ἀλλήλουξ, μετρήσει τιξ αὐτοὺξ ἀριθμόξ. μετρεῖτω ὁ Γ. ἐπεὶ ὁ Γ τὸν Β μετρεῖ, ὁ δὲ Α τὸν Β οὐ μετρεῖ, ὁ Γ >άρα τῶ| Α ο>ύκ ἐστὶν ὁ αὐτόξ. καὶ ἐπεὶ ὁ Γ τοὺξ Β, Α μετρεῖ, καὶ τὸν Α >άρα μετρεῖ πρῶτον >όντα μὴ >ὸν αὐτῶ| ὁ αὐτόξ; <όπερ ἐστὶν ἀδύνατον. οὐκ >άρα τοὺξ Β, Α μετρήσει τιξ ἀριθμόξ. οἱ Α, Β >άρα πρῶτοι πρὸξ ἀλλήλουξ εἰσίν; <όπερ >έδει δεῖχαι.

30.

Ἐὰν δύο ἀριθμοὶ πολλαπλασιάσαντεξ ἀλλήλουξ ποιῶσι τινα, τὸν δὲ γενόμενον ἔχ αὐτῶν μετρῇ| τιξ πρῶτοξ ἀριθμόξ, καὶ <ένα τῶν ἔχ ἀρθῇξ μετρήσει.

A —————

B —————

Γ —————

Δ —————

E —————

Δύο γὰρ ἀριθμοὶ οἱ Α, Β πολλαπλασιάσαντεξ ἀλλήλουξ τὸν

And it also measures BA . Thus, D measures AB and BC , which are prime to one another. The very thing is impossible. Thus, some number cannot measure (both) the numbers CA and AB . Thus, CA and AB are prime to one another. So, for the same (reasons), AC and CB are also prime to one another. Thus, CA is prime to each of AB and BC . So, again, let CA and AB be prime to one another. I say that AB and BC are also prime to one another. For if AB and BC are not prime to one another, (then) some number will measure AB and BC . Let it (so) measure (them), and let it be D . And since D measures each of AB and BC , it will thus also measure the whole of CA . And it also measures AB . Thus, D measures CA and AB , which are prime to one another. The very thing is impossible. Thus, some number cannot measure (both) the numbers AB and BC . Thus, AB and BC are prime to one another. (Which is) the very thing it was required to show.

Proposition 29

Every prime number is prime to every number which it does not measure.

Let A be a prime number, and let it not measure B . I say that B and A are prime to one another.

A —————

B —————

C —————

For if B and A are not prime to one another, (then) some number will measure them. Let C measure (them). Since C measures B , and A does not measure B , C is thus not the same as A . And since C measures B and A , it thus also measures A , which is prime, (despite) not being the same as it. The very thing is impossible. Thus, some number cannot measure (both) B and A . Thus, A and B are prime to one another. (Which is) the very thing it was required to show.

Proposition 30

If two numbers make some (number by) multiplying one another, and some prime number measures the number (so) created from them, (then) it will also measure one of the original (numbers).

A —————

B —————

C —————

D —————

E —————

For let two numbers A and B make C (by) multiplying one

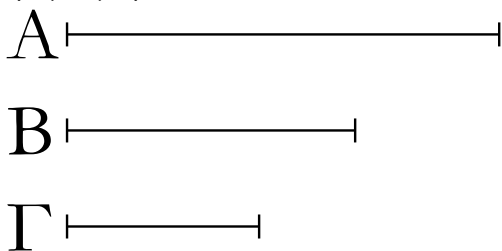
Γ ποιείτωσαν, τὸν δὲ Γ μετρεῖτω τις πρῶτοξ ἀριθμὸς ὁ Δ; λέγω, <ὅτι ὁ Δ <ένα τῶν Α, Β μετρεῖ.

Τὸν γὰρ Α μὴ μετρεῖτω; καὶ ἐστὶ πρῶτοξ ὁ Δ; οἱ Α, Δ >άρα πρῶτοι πρὸς ἀλλήλουξ εἰσίν. καὶ ὁσάκιξ ὁ Δ τὸν Γ μετρεῖ, τοσαῦται μονάδεξ >έστωσαν ἐν τῷ Ε. ἐπεὶ ο>ὺν ὁ Δ τὸν Γ μετρεῖ κατὰ τὰξ ἐν τῷ Ε μονάδαξ, ὁ Δ >άρα τὸν Ε πολλαπλασιάσαξ τὸν Γ πεποίηκεν. ἀλλὰ μὴν καὶ ὁ Α τὸν Β πολλαπλασιάσαξ τὸν Γ πεποίηκεν; >ἴσοξ >άρα ἐστὶν ὁ ἐκ τῶν Δ, Ε τῷ ἐκ τῶν Α, Β. >ἐστὶν >άρα ὡς ὁ Δ πρὸξ τὸν Α, ο<ύτωξ ὁ Β πρὸξ τὸν Ε. οἱ δὲ Δ, Α πρῶτοι, οἱ δὲ πρῶτοι καὶ ἐλάχιστοι, οἱ δὲ ἐλάχιστοι μετροῦσι τοὺξ τὸν αὐτὸν λόγον >έθονταξ ἰσάκιξ <ὁ τε μείζων τὸν μείζονα καὶ ὁ ἐλάσσων τὸν ἐλάσσονα, τουτέστιν <ὁ τε ἡγούμενοξ τὸν ἡγούμενον καὶ ὁ ἐπόμενοξ τὸν ἐπόμενον; ὁ Δ >άρα τὸν Β μετρεῖ. ὁμοίωξ δὴ δείχομεν, <ὅτι καὶ ἔαν τὸν Β μὴ μετρήῃ, τὸν Α μετρήσει. ὁ Δ >άρα <ένα τῶν Α, Β μετρεῖ; <όπερ >έδει δεῖχαι.

31.

<Ἀπαξ σύνθεντοξ ἀριθμὸξ ὑπὸ πρῶτου τινὸξ ἀριθμοῦ μετρεῖται.

>Ἐστω σύνθεντοξ ἀριθμὸξ ὁ Α; λέγω, <ὅτι ὁ Α ὑπὸ πρῶτου τινὸξ ἀριθμοῦ μετρεῖται.

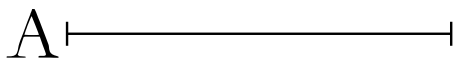


Ἐπεὶ γὰρ σύνθετόξ ἐστὶν ὁ Α, μετρήσει τιξ αὐτὸν ἀριθμόξ. μετρεῖτω, καὶ >έστω ὁ Β. καὶ εἰ μὲν πρῶτόξ ἐστὶν ὁ Β, γεγονόξ >ὰν ε>ἦ τὸ ἐπιταθέν. εἰ δὲ σύνθετοξ, μετρήσει τιξ αὐτὸν ἀριθμόξ. μετρεῖτω, καὶ >έστω ὁ Γ. καὶ ἐπεὶ ὁ Γ τὸν Β μετρεῖ, ὁ δὲ Β τὸν Α μετρεῖ, καὶ ὁ Γ >άρα τὸν Α μετρεῖ. καὶ εἰ μὲν πρῶτόξ ἐστὶν ὁ Γ, γεγονόξ >ὰν ε>ἦ τὸ ἐπιταθέν. εἰ δὲ σύνθετοξ, μετρήσει τιξ αὐτὸν ἀριθμόξ. τοιαύτηξ δὴ γινομένηξ ἐπισκέψεωξ ληφθήσεται τιξ πρῶτοξ ἀριθμόξ, <ὃξ μετρήσει. εἰ γὰρ οὐ ληφθήσεται, μετρήσουσι τὸν Α ἀριθμὸν >άπειροι ἀριθμοί, <ὢν <έτεροξ ἐτέρου ἐλάσσων ἐστίν; <όπερ ἐστὶν ἀδύνατον ἐν ἀριθμοῖξ. ληφθήσεται τιξ >άρα πρῶτοξ ἀριθμόξ, <ὃξ μετρήσει τὸν πρὸ ἑαυτοῦ, <ὃξ καὶ τὸν Α μετρήσει.

<Ἀπαξ >άρα σύνθεντοξ ἀριθμὸξ ὑπὸ πρῶτου τινὸξ ἀριθμοῦ μετρεῖται; <όπερ >έδει δεῖχαι.

32.

<Ἀπαξ ἀριθμὸξ >ἥτοι πρῶτόξ ἐστὶν >ἢ ὑπὸ πρῶτου τινὸξ ἀριθμοῦ μετρεῖται.



>Ἐστω ἀριθμὸξ ὁ Α; λέγω, <ὅτι ὁ Α >ἥτοι πρῶτόξ ἐστὶν >ἢ

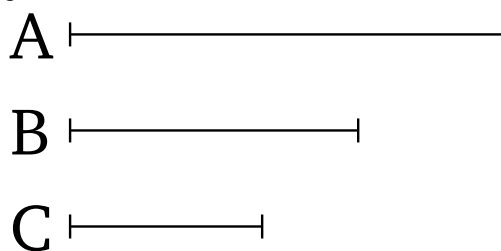
another, and let some prime number D measure C . I say that D measures one of A and B .

For let it not measure A . And since D is prime, A and D are thus prime to one another [Prop. 7.29]. And as many times as D measures C , so many units let there be in E . Therefore, since D measures C according to the units E , D has thus made C (by) multiplying E [Def. 7.15]. But, in fact, A has also made C (by) multiplying B . Thus, the (number created) from (multiplying) D and E is equal to the (number created) from (multiplying) A and B . Thus, as D is to A , so B (is) to E [Prop. 7.19]. And D and A (are) prime (to one another), and (numbers) prime (to one another are) also the least (of those numbers having the same ratio) [Prop. 7.21], and the least (numbers) measure those (numbers) having the same ratio (as them) an equal number of times, the greater (measuring) the greater, and the lesser the lesser—that is to say, the leading (measuring) the leading, and the following the following [Prop. 7.20]. Thus, D measures B . So, similarly, we can also show that if (D) does not measure B , (then) it will measure A . Thus, D measures one of A and B . (Which is) the very thing it was required to show.

Proposition 31

Every composite number is measured by some prime number.

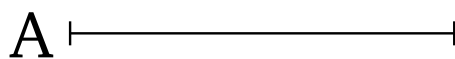
Let A be a composite number. I say that A is measured by some prime number.



For since A is composite, some number will measure it. Let it (so) measure (A), and let it be B . And if B is prime then that which was prescribed has happened. And if (B is) composite, (then) some number will measure it. Let it (so) measure (B), and let it be C . And since C measures B , and B measures A , C thus also measures A . And if C is prime then that which was prescribed has happened. And if (C is) composite, (then) some number will measure it. So, in this manner of continued investigation, some prime number will be found which will measure (the number preceding it, which will also measure A). And if (such a number) cannot be found, (then) an infinite (series of) numbers, each of which is less than the preceding, will measure the number A . The very thing is impossible for numbers. Thus, some prime number will (eventually) be found which will measure the (number) preceding it, which will also measure A . Thus, every composite number is measured by some prime number. (Which is) the very thing it was required to show.

Proposition 32

Every number is either prime, or is measured by some prime number.



Let A be a number. I say that A is either prime, or is mea-

ὑπὸ πρώτου τινὸς ἀριθμοῦ μετρεῖται.

Εἰ μὲν οὖν πρῶτός ἐστιν ὁ A , γεγονὸς >ὰν ε>ίη τὸ ἐπιταθέν. εἰ δὲ σύνθετος, μετρήσει τις αὐτὸν πρῶτος ἀριθμός.

<'Απαξ >άρα ἀριθμὸς >ήτοι πρῶτός ἐστιν >ή ὑπὸ πρώτου τινὸς ἀριθμοῦ μετρεῖται; <όπερ >έδει δεῖχαι.

33.

Ἀριθμῶν δοθέντων ὅποσωνοῦν εὑρεῖν τοὺς ἐλαθίστους τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς.

>'Εστῶσαν οἱ δοθέντες ὅποσοιοῦν ἀριθμοὶ οἱ A, B, Γ ; δεῖ δὴ εὑρεῖν τοὺς ἐλαθίστους τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς A, B, Γ .

Οἱ A, B, Γ γὰρ >ήτοι πρῶτοι πρὸς ἀλλήλους εἰσὶν >ή ο>ύ. εἰ μὲν οὖν οἱ A, B, Γ πρῶτοι πρὸς ἀλλήλους εἰσὶν, ἐλαθίστοί εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς.

sured by some prime number.

In fact, if A is prime, (then) that which was prescribed has happened. And if (it is) composite, (then) some prime number will measure it [Prop. 7.31].

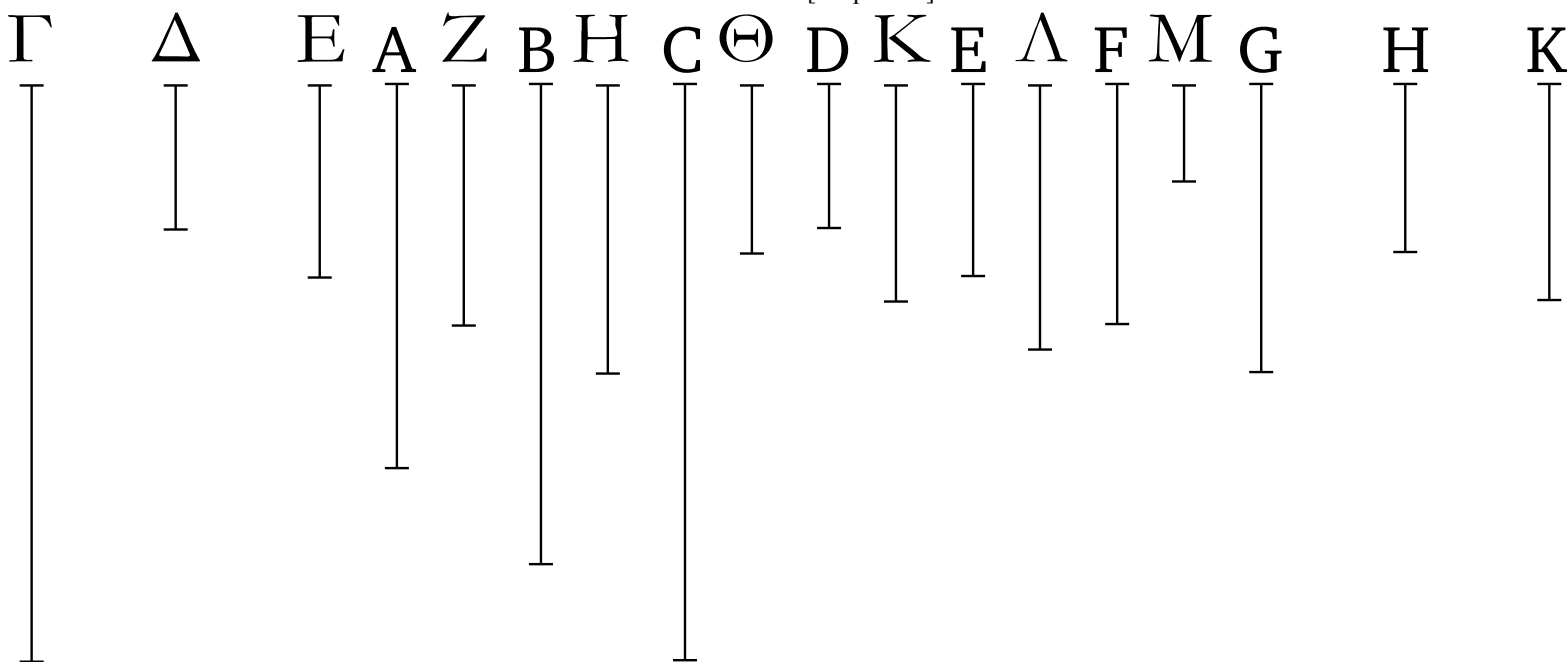
Thus, every number is either prime, or is measured by some prime number. (Which is) the very thing it was required to show.

Proposition 33

To find the least of those (numbers) having the same ratio as any given multitude of numbers.

Let A, B , and C be any given multitude of numbers. So it is required to find the least of those (numbers) having the same ratio as A, B , and C .

For A, B , and C are either prime to one another, or not. In fact, if A, B , and C are prime to one another, (then) they are the least of those (numbers) having the same ratio as them [Prop. 7.22].



Εἰ δὲ ο>ύ, εἰλήφθω τῶν A, B, Γ τὸ μέγιστον κοινὸν μέτρον ὁ Δ , καὶ ὅσάκις ὁ Δ <έκαστον τῶν A, B, Γ μετρεῖ, τοσῶται μονάδες >έστωσαν ἐν ἑκάστῳ τῶν E, Z, H . καὶ <έκαστος >άρα τῶν E, Z, H <έκαστον τῶν A, B, Γ μετρεῖ κατὰ τὰς ἐν τῷ Δ μονάδας. οἱ E, Z, H >άρα τοὺς A, B, Γ ἰσάκις μετροῦσιν; οἱ E, Z, H >άρα τοῖς A, B, Γ ἐν τῷ αὐτῷ λόγῳ εἰσὶν. λέγω δὴ, <ὅτι καὶ ἐλαθίστοι. εἰ γὰρ μὴ εἰσιν οἱ E, Z, H ἐλαθίστοι τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς A, B, Γ , >έσονται [τινὲς] τῶν E, Z, H ἐλάσσονες ἀριθμοὶ ἐν τῷ αὐτῷ λόγῳ >όντες τοῖς A, B, Γ . >έστωσαν οἱ Θ, K, Λ ; ἰσάκις >άρα ὁ Θ τὸν A μετρεῖ καὶ ἑκάτερος τῶν K, Λ ἑκάτερον τῶν B, Γ . ὅσάκις δὲ ὁ Θ τὸν A μετρεῖ, τοσῶται μονάδες >έστωσαν ἐν τῷ M ; καὶ ἑκάτερος >άρα τῶν K, Λ ἑκάτερον τῶν B, Γ μετρεῖ κατὰ τὰς ἐν τῷ M μονάδας. καὶ ἐπεὶ ὁ Θ τὸν A μετρεῖ κατὰ τὰς ἐν τῷ M μονάδας, καὶ ὁ M >άρα τὸν A μετρεῖ κατὰ τὰς ἐν τῷ Θ μονάδας. διὰ τὰ αὐτὰ δὴ ὁ M καὶ ἑκάτερον τῶν B, Γ μετρεῖ κατὰ τὰς ἐν ἑκατέρῳ τῶν K, Λ μονάδας; ὁ M >άρα τοὺς A, B, Γ μετρεῖ. καὶ ἐπεὶ ὁ Θ τὸν A μετρεῖ κατὰ τὰς ἐν τῷ M μονάδας, ὁ Θ >άρα τὸν M πολλαπλασιάσας τὸν A πεποίηκεν. διὰ τὰ αὐτὰ δὴ καὶ ὁ E τὸν Δ πολλαπλασιάσας τὸν A πεποίηκεν. >ίσοξ >άρα

And if not, let the greatest common measure, D , of A, B , and C have been taken [Prop. 7.3]. And as many times as D measures A, B, C , so many units let there be in E, F, G , respectively. And thus E, F, G measure A, B, C , respectively, according to the units in D [Prop. 7.15]. Thus, E, F, G measure A, B, C (respectively) an equal number of times. Thus, E, F, G are in the same ratio as A, B, C (respectively) [Def. 7.20]. So I say that (they are) also the least (of those numbers having the same ratio as A, B, C). For if E, F, G are not the least of those (numbers) having the same ratio as A, B, C (respectively, then) there will be [some] numbers less than E, F, G which are in the same ratio as A, B, C (respectively). Let them be H, K, L . Thus, H measures A the same number of times that K, L also measure B, C , respectively. And as many times as H measures A , so many units let there be in M . Thus, K, L measure B, C , respectively, according to the units in M . And since H measures A according to the units in M , M thus also measures A according to the units in H [Prop. 7.15]. So, for the same (reasons), M also measures B, C according to the units in K, L , respectively. Thus, M

ἐστὶν ὁ ἐκ τῶν E, Δ τῶν ἐκ τῶν Θ, M . >ἐστὶν >άρα ὡς ὁ E πρὸς τὸν Θ , ο<ύτωξ ὁ M πρὸς τὸν Δ . μείζων δὲ ὁ E τοῦ Θ ; μείζων >άρα καὶ ὁ M τοῦ Δ . καὶ μετρεῖ τοὺς A, B, Γ ; <όπερ ἐστὶν ἀδύνατον; ὑπόκειται γάρ ὁ Δ τῶν A, B, Γ τὸ μέγιστον κοινὸν μέτρον. οὐκ >άρα >έσσονται τινεξ τῶν E, Z, H ἐλάσσονεξ ἀριθμοὶ ἐν τῶν αὐτῶν λόγῳ >όντεξ τοῖς A, B, Γ . οἱ E, Z, H >άρα ἐλάχιστοι εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς A, B, Γ ; <όπερ >έδει δεῖχαι.

measures A, B , and C . And since H measures A according to the units in M , H has thus made A (by) multiplying M . So, for the same (reasons), E has also made A (by) multiplying D . Thus, the (number created) from (multiplying) E and D is equal to the (number created) from (multiplying) H and M . Thus, as E (is) to H , so M (is) to D [Prop. 7.19]. And E (is) greater than H . Thus, M (is) also greater than D [Prop. 5.13]. And (M) measures A, B , and C . The very thing is impossible. For D was assumed (to be) the greatest common measure of A, B , and C . Thus, there cannot be any numbers less than E, F, G which are in the same ratio as A, B, C (respectively). Thus, E, F, G are the least of (those numbers) having the same ratio as A, B, C (respectively). (Which is) the very thing it was required to show.

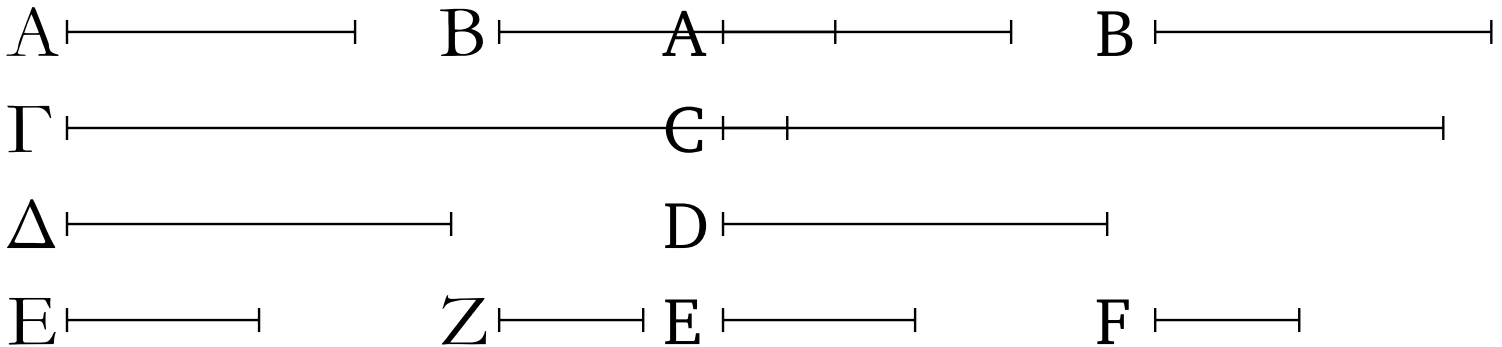
34.

Proposition 34

Δύο ἀριθμῶν δοθέντων εὑρεῖν, <ὸν ἐλάχιστον μετροῦσιν ἀριθμόν.
>Ἐστῶσαν οἱ δοθέντεξ δύο ἀριθμοὶ οἱ A, B ; δεῖ δὴ εὑρεῖν, <ὸν ἐλάχιστον μετροῦσιν ἀριθμόν.

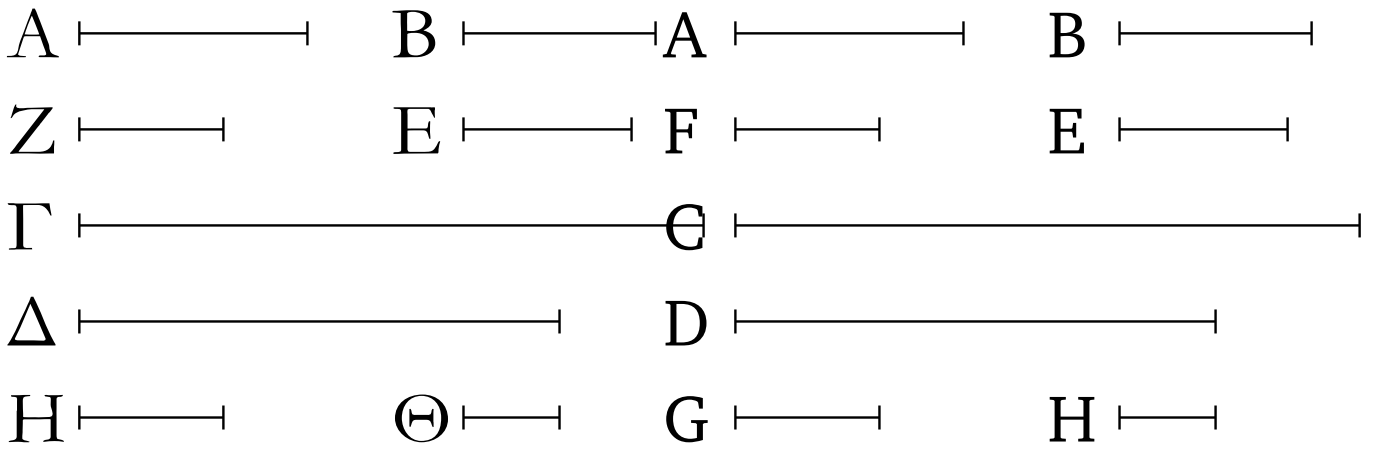
To find the least number which two given numbers (both) measure.

Let A and B be the two given numbers. So it is required to find the least number which they (both) measure.



Οἱ A, B γὰρ >ήτοι πρῶτοι πρὸς ἀλλήλουξ εἰσὶν >ή ο>ύ. >έστωσαν πρότερον οἱ A, B πρῶτοι πρὸς ἀλλήλουξ, καὶ ὁ A τὸν B πολλαπλασιάσας τὸν Γ ποιεῖτω; καὶ ὁ B >άρα τὸν A πολλαπλασιάσας τὸν Γ πεποίηκεν. οἱ A, B >άρα τὸν Γ μετροῦσιν. λέγω δὴ, <ότι καὶ ἐλάχιστον. εἰ γὰρ μή, μετρήσουσι τινα ἀριθμὸν οἱ A, B ἐλάσσονα >όντα τοῦ Γ . μετρεῖτωσαν τὸν Δ . καὶ ὁσάκιξ ὁ A τὸν Δ μετρεῖ, τοσαῦται μονάδεξ >έστωσαν ἐν τῶν E , ὁσάκιξ δὲ ὁ B τὸν Δ μετρεῖ, τοσαῦται μονάδεξ >έστωσαν ἐν τῶν Z . ὁ μὲν A >άρα τὸν E πολλαπλασιάσας τὸν Δ πεποίηκεν, ὁ δὲ B τὸν Z πολλαπλασιάσας τὸν Δ πεποίηκεν; >ίσοξ >άρα ἐστὶν ὁ ἐκ τῶν A, E τῶν ἐκ τῶν B, Z . >ἐστὶν >άρα ὡς ὁ A πρὸς τὸν B , ο<ύτωξ ὁ Z πρὸς τὸν E . οἱ δὲ A, B πρῶτοι, οἱ δὲ πρῶτοι καὶ ἐλάχιστοι, οἱ δὲ ἐλάχιστοι μετροῦσι τοὺς τὸν αὐτὸν λόγον >έθονταξ ἰσάκιξ <ό τε μείζων τὸν μείζονα καὶ ὁ ἐλάσσων τὸν ἐλάσσονα; ὁ B >άρα τὸν E μετρεῖ, ὡς ἐπόμενον. καὶ ἐπεὶ ὁ A τοὺς B, E πολλαπλασιάσας τοὺς Γ, Δ πεποίηκεν, >ἐστὶν >άρα ὡς ὁ B πρὸς τὸν E , ο<ύτωξ ὁ Γ πρὸς τὸν Δ . μετρεῖ δὲ ὁ B τὸν E ; μετρεῖ >άρα καὶ ὁ Γ τὸν Δ ὁ μείζων τὸν ἐλάσσονα; <όπερ ἐστὶν ἀδύνατον. οὐκ >άρα οἱ A, B μετροῦσι τινα ἀριθμὸν ἐλάσσονα >όντα τοῦ Γ . ὁ Γ >άρα ἐλάχιστοξ >ὼν ὑπὸ τῶν A, B μετρεῖται.

For A and B are either prime to one another, or not. Let them, first of all, be prime to one another. And let A make C (by) multiplying B . Thus, B has also made C (by) multiplying A [Prop. 7.16]. Thus, A and B (both) measure C . So I say that (C) is also the least (number which they both measure). For if not, A and B will (both) measure some (other) number which is less than C . Let them (both) measure D (which is less than C). And as many times as A measures D , so many units let there be in E . And as many times as B measures D , so many units let there be in F . Thus, A has made D (by) multiplying E , and B has made D (by) multiplying F . Thus, the (number created) from (multiplying) A and E is equal to the (number created) from (multiplying) B and F . Thus, as A (is) to B , so F (is) to E [Prop. 7.19]. And A and B are prime (to one another), and prime (numbers) are the least (of those numbers having the same ratio) [Prop. 7.21], and the least (numbers) measure those (numbers) having the same ratio (as them) an equal number of times, the greater (measuring) the greater, and the lesser the lesser [Prop. 7.20]. Thus, B measures E , as the following (number measuring) the following. And since A has made C and D (by) multiplying B and E (respectively), thus as B is to E , so C (is) to D [Prop. 7.17]. And B measures E . Thus, C also measures D , the greater (measuring) the lesser. The very thing is impossible. Thus, A and B do not (both) measure some number which is less than C . Thus, C is the least (number) which is measured by (both) A and B .



μή >έστωσαν δὴ οἱ A, B πρῶτοι πρὸς ἀλλήλους, καὶ εἰλήφθωσαν ἐλάχιστοι ἀριθμοὶ τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς A, B οἱ Z, E ; >ἴσοξ >άρα ἐστὶν ὁ ἐκ τῶν A, E τῶ| ἐκ τῶν B, Z . καὶ ὁ A τὸν E πολλαπλασιάσαξ τὸν Γ ποιείτω; καὶ ὁ B >άρα τὸν Z πολλαπλασιάσαξ τὸν Γ πεποίηκεν; οἱ A, B >άρα τὸν Γ μετροῦσιν. λέγω δὴ, <ὅτι καὶ ἐλάχιστον. εἰ γὰρ μή, μετρήσουσι τινα ἀριθμὸν οἱ A, B ἐλάσσονα >όντα τοῦ Γ . μετρεῖτωσαν τὸν Δ . καὶ ὁσάκιξ μὲν ὁ A τὸν Δ μετρεῖ, τοσαῦται μονάδεξ >έστωσαν ἐν τῶ| H , ὁσάκιξ δὲ ὁ B τὸν Δ μετρεῖ, τοσαῦται μονάδεξ >έστωσαν ἐν τῶ| Θ . ὁ μὲν A >άρα τὸν H πολλαπλασιάσαξ τὸν Δ πεποίηκεν, ὁ δὲ B τὸν Θ πολλαπλασιάσαξ τὸν Δ πεποίηκεν. >ἴσοξ >άρα ἐστὶν ὁ ἐκ τῶν A, H τῶ| ἐκ τῶν B, Θ ; >έστιν >άρα ὡς ὁ A πρὸξ τὸν B , ο<ύτωξ ὁ Θ πρὸξ τὸν H . ὡς δὲ ὁ A πρὸξ τὸν B , ο<ύτωξ ὁ Z πρὸξ τὸν E ; καὶ ὡς >άρα ὁ Z πρὸξ τὸν E , ο<ύτωξ ὁ Θ πρὸξ τὸν H . οἱ δὲ Z, E ἐλάχιστοι, οἱ δὲ ἐλάχιστοι μετροῦσι τοὺξ τὸν αὐτὸν λόγον >έθονταξ ἰσάκιξ <ὁ τε μείζων τὸν μείζονα καὶ ὁ ἐλάσσων τὸν ἐλάσσονα; ὁ E >άρα τὸν H μετρεῖ. καὶ ἐπεὶ ὁ A τοὺξ E, H πολλαπλασιάσαξ τοὺξ Γ, Δ πεποίηκεν, >έστιν >άρα ὡς ὁ E πρὸξ τὸν H , ο<ύτωξ ὁ Γ πρὸξ τὸν Δ . ὁ δὲ E τὸν H μετρεῖ; καὶ ὁ Γ >άρα τὸν Δ μετρεῖ ὁ μείζων τὸν ἐλάσσονα; <ὅπερ ἐστὶν ἀδύνατον. οὐκ >άρα οἱ A, B μετρήσουσι τινα ἀριθμὸν ἐλάσσονα >όντα τοῦ Γ . ὁ Γ >άρα ἐλάχιστοξ >ὢν ὑπὸ τῶν A, B μετρεῖται; <ὅπερ >έπει δεῖξαι.

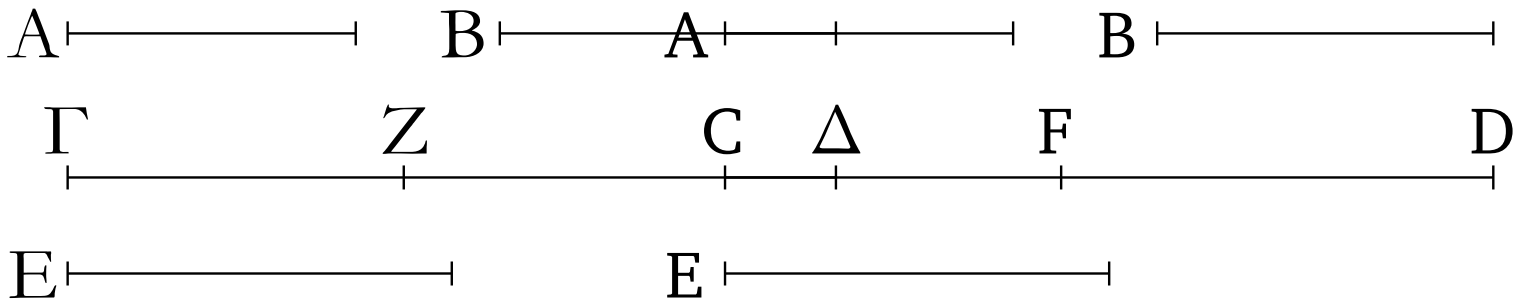
So let A and B be not prime to one another. And let the least numbers, F and E , be taken having the same ratio as A and B (respectively) [Prop. 7.33]. Thus, the (number created) from (multiplying) A and E is equal to the (number created) from (multiplying) B and F [Prop. 7.19]. And let A make C (by) multiplying E . Thus, B has also made C (by) multiplying F . Thus, A and B (both) measure C . So I say that (C) is also the least (number which they both measure). For if not, A and B will (both) measure some number which is less than C . Let them (both) measure D (which is less than C). And as many times as A measures D , so many units let there be in G . And as many times as B measures D , so many units let there be in H . Thus, A has made D (by) multiplying G , and B has made D (by) multiplying H . Thus, the (number created) from (multiplying) A and G is equal to the (number created) from (multiplying) B and H . Thus, as A is to B , so H (is) to G [Prop. 7.19]. And as A (is) to B , so F (is) to E . Thus, also, as F (is) to E , so H (is) to G . And F and E are the least (numbers having the same ratio as A and B), and the least (numbers) measure those (numbers) having the same ratio an equal number of times, the greater (measuring) the greater, and the lesser the lesser [Prop. 7.20]. Thus, E measures G . And since A has made C and D (by) multiplying E and G (respectively), thus as E is to G , so C (is) to D [Prop. 7.17]. And E measures G . Thus, C also measures D , the greater (measuring) the lesser. The very thing is impossible. Thus, A and B do not (both) measure some (number) which is less than C . Thus, C (is) the least (number) which is measured by (both) A and B . (Which is) the very thing it was required to show.

35.

Proposition 35

Ἐὰν δύο ἀριθμοὶ ἀριθμὸν τινα μετρώσιν, καὶ ὁ ἐλάχιστοξ ὑπ> αὐτῶν μετρούμενοξ τὸν αὐτὸν μετρήσει.

If two numbers (both) measure some number, (then) the least (number) measured by them will also measure the same (number).



Δύο γὰρ ἀριθμοὶ οἱ A, B ἀριθμὸν τινα τὸν $\Gamma\Delta$ μετρεῖτωσαν, ἐλάχιστον δὲ τὸν E ; λέγω, <ὅτι καὶ ὁ E τὸν $\Gamma\Delta$ μετρεῖ.

For let two numbers, A and B , (both) measure some number CD , and (let) E (be the) least (number measured by both A

Εἰ γὰρ οὐ μετρεῖ ὁ Ε τὸν ΓΔ, ὁ Ε τὸν ΔΖ μετρῶν λειπέτω ἑαυτοῦ ἐλάσσονα τὸν ΓΖ. καὶ ἐπεὶ οἱ Α, Β τὸν Ε μετροῦσιν, ὁ δὲ Ε τὸν ΔΖ μετρεῖ, καὶ οἱ Α, Β >άρα τὸν ΔΖ μετρήσουσιν. μετροῦσι δὲ καὶ <όλον τὸν ΓΔ; καὶ λοιπὸν >άρα τὸν ΓΖ μετρήσουσιν ἐλάσσονα >όντα τοῦ Ε; <όπερ ἐστὶν ἀδύνατον. οὐκ >άρα οὐ μετρεῖ ὁ Ε τὸν ΓΔ; μετρεῖ >άρα; <όπερ >έδει δεῖχαι.

36.

Τριῶν ἀριθμῶν δοθέντων εὑρεῖν, <ὸν ἐλάχιστον μετροῦσιν ἀριθμόν.

>Ἐστωσαν οἱ δοθέντες τρεῖς ἀριθμοὶ οἱ Α, Β, Γ; δεῖ δὴ εὑρεῖν, <ὸν ἐλάχιστον μετροῦσιν ἀριθμόν.

A |—————|

B |—————|

Γ |—————|

Δ |—————|

E |—————|

Z |—————|

A |—————|

B |—————|

C |—————|

D ||—————|

E |—————|

F |—————|

Εἰλήφθω γὰρ ὑπὸ δύο τῶν Α, Β ἐλάχιστοξ μετρούμενοξ ὁ Δ. ὁ δὴ Γ τὸν Δ >ήτοι μετρεῖ >ἢ οὐ μετρεῖ. μετρεῖτω πρότερον. μετροῦσι δὲ καὶ οἱ Α, Β τὸν Δ; οἱ Α, Β, Γ >άρα τὸν Δ μετροῦσιν. λέγω δὴ, <ὅτι καὶ ἐλάχιστον. εἰ γὰρ μή, μετρήσουσιν [τινα] ἀριθμόν οἱ Α, Β, Γ ἐλάσσονα >όντα τοῦ Δ. μετρεῖτωσαν τὸν Ε. ἐπεὶ οἱ Α, Β, Γ τὸν Ε μετροῦσιν, καὶ οἱ Α, Β >άρα τὸν Ε μετροῦσιν. καὶ ὁ ἐλάχιστοξ >άρα ὑπὸ τῶν Α, Β μετρούμενοξ [τὸν Ε] μετρήσει. ἐλάχιστοξ δὲ ὑπὸ τῶν Α, Β μετρούμενοξ ἐστὶν ὁ Δ; ὁ Δ >άρα τὸν Ε μετρήσει ὁ μείζων τὸν ἐλάσσονα; <όπερ ἐστὶν ἀδύνατον. οὐκ >άρα οἱ Α, Β, Γ μετρήσουσι τινα ἀριθμόν ἐλάσσονα >όντα τοῦ Δ; οἱ Α, Β, Γ >άρα ἐλάχιστον τὸν Δ μετροῦσιν.

μὴ μετρεῖτω δὴ πάλιν ὁ Γ τὸν Δ, καὶ εἰλήφθω ὑπὸ τῶν Γ, Δ ἐλάχιστοξ μετρούμενοξ ἀριθμὸξ ὁ Ε. ἐπεὶ οἱ Α, Β τὸν Δ μετροῦσιν, ὁ δὲ Δ τὸν Ε μετρεῖ, καὶ οἱ Α, Β >άρα τὸν Ε μετροῦσιν. μετρεῖ δὲ καὶ ὁ Γ [τὸν Ε; καὶ] οἱ Α, Β, Γ >άρα τὸν Ε μετροῦσιν. λέγω δὴ, <ὅτι καὶ ἐλάχιστον. εἰ γὰρ μή, μετρήσουσι τινα οἱ Α, Β, Γ ἐλάσσονα >όντα τοῦ Ε. μετρεῖτωσαν τὸν Ζ. ἐπεὶ οἱ Α, Β, Γ τὸν Ζ μετροῦσιν, καὶ οἱ Α, Β >άρα τὸν Ζ μετροῦσιν; καὶ ὁ ἐλάχιστοξ >άρα ὑπὸ τῶν Α, Β μετρούμενοξ τὸν Ζ μετρήσει. ἐλάχιστοξ δὲ ὑπὸ τῶν Α, Β μετρούμενοξ ἐστὶν ὁ Δ; ὁ Δ >άρα τὸν Ζ μετρεῖ. μετρεῖ δὲ καὶ ὁ Γ τὸν Ζ; οἱ Α, Β, Γ >άρα τὸν Ζ μετροῦσιν; <ώστε καὶ ὁ ἐλάχιστοξ ὑπὸ τῶν Α, Β μετρούμενοξ τὸν Ζ μετρήσει. ὁ δὲ ἐλάχιστοξ ὑπὸ τῶν Γ, Δ μετρούμενοξ ἐστὶν ὁ Ε; ὁ Ε >άρα τὸν Ζ μετρεῖ ὁ μείζων τὸν ἐλάσσονα; <όπερ ἐστὶν ἀδύνατον. οὐκ >άρα οἱ Α, Β, Γ μετρήσουσι τινα ἀριθμόν ἐλάσσονα >όντα τοῦ Ε. ὁ Ε >άρα ἐλάχιστοξ >ὼν ὑπὸ τῶν Α, Β, Γ

and Β). I say that Ε also measures CD.

For if Ε does not measure CD, (then) let Ε leave CF less than itself (in) measuring DF. And since Α and Β (both) measure Ε, and Ε measures DF, Α and Β will thus also measure DF. And (Α and Β) also measure the whole of CD. Thus, they will also measure the remainder CF, which is less than Ε. The very thing is impossible. Thus, Ε cannot not measure CD. Thus, (Ε) measures (CD). (Which is) the very thing it was required to show.

Proposition 36

To find the least number which three given numbers (all) measure.

Let Α, Β, and C be the three given numbers. So it is required to find the least number which they (all) measure.

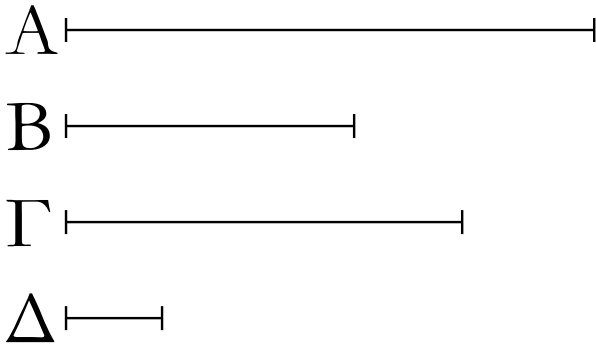
For let the least (number), D, measured by the two (numbers) Α and Β be taken [Prop. 7.34]. So C either measures, or does not measure, D. Let it, first of all, measure (D). And Α and Β also measure D. Thus, Α, Β, and C (all) measure D. So I say that (D is) also the least (number measured by Α, Β, and C). For if not, Α, Β, and C will (all) measure [some] number which is less than D. Let them measure Ε (which is less than D). Since Α, Β, and C (all) measure Ε, (then) Α and Β thus also measure Ε. Thus, the least (number) measured by Α and Β will also measure [Ε] [Prop. 7.35]. And D is the least (number) measured by Α and Β. Thus, D will measure Ε, the greater (measuring) the lesser. The very thing is impossible. Thus, Α, Β, and C cannot (all) measure some number which is less than D. Thus, Α, Β, and C (all) measure the least (number) D.

So, again, let C not measure D. And let the least number, Ε, measured by C and D be taken [Prop. 7.34]. Since Α and Β measure D, and D measures Ε, Α and Β thus also measure Ε. And C also measures [Ε]. Thus, Α, Β, and C [also] measure Ε. So I say that (Ε is) also the least (number measured by Α, Β, and C). For if not, Α, Β, and C will (all) measure some (number) which is less than Ε. Let them measure F (which is less than Ε). Since Α, Β, and C (all) measure F, Α and Β thus also measure F. Thus, the least (number) measured by Α and Β will also measure F [Prop. 7.35]. And D is the least (number) measured by Α and Β. Thus, D measures F. And C also measures F. Thus, D and C (both) measure F. Hence, the least (num-

μετρεῖται; <ὅπερ >έδει δεῖχαι.

37.

Ἐὰν ἀριθμὸς ὑπὸ τινος ἀριθμοῦ μετρήται, ὁ μετρούμενος ὁμώνυμον μέρος <έχει τῷ μετροῦντι.

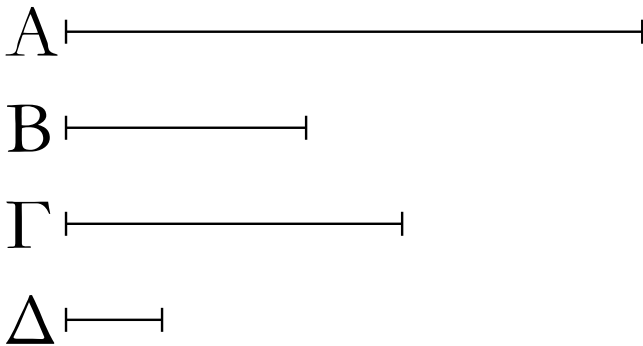


Ἀριθμὸς γὰρ ὁ Α ὑπὸ τινος ἀριθμοῦ τοῦ Β μετρεῖσθω; λέγω, <ὅτι ὁ Α ὁμώνυμον μέρος >έχει τῷ Β.

Ὅσακις γὰρ ὁ Β τὸν Α μετρεῖ, τοσαῦται μονάδες >έστωσαν ἐν τῷ Γ. ἐπεὶ ὁ Β τὸν Α μετρεῖ κατὰ τὰς ἐν τῷ Γ μονάδας, μετρεῖ δὲ καὶ ἡ Δ μονὰς τὸν Γ ἀριθμὸν κατὰ τὰς ἐν αὐτῷ μονάδας, ἰσάκις >άρα ἡ Δ μονὰς τὸν Γ ἀριθμὸν μετρεῖ καὶ ὁ Β τὸν Α. ἐναλλάξ >άρα ἰσάκις ἡ Δ μονὰς τὸν Β ἀριθμὸν μετρεῖ καὶ ὁ Γ τὸν Α; <ὃ >άρα μέρος ἐστὶν ἡ Δ μονὰς τοῦ Β ἀριθμοῦ, τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ Γ τοῦ Α. ἡ δὲ Δ μονὰς τοῦ Β ἀριθμοῦ μέρος ἐστὶν ὁμώνυμον αὐτῷ; καὶ ὁ Γ >άρα τοῦ Α μέρος ἐστὶν ὁμώνυμον τῷ Β. <ὥστε ὁ Α μέρος >έχει τὸν Γ ὁμώνυμον >όντα τῷ Β; <ὅπερ >έδει δεῖχαι.

38.

Ἐὰν ἀριθμὸς μέρος >έθῃ ὅτιοῦν, ὑπὸ ὁμώνυμου ἀριθμοῦ μετρηθήσεται τῷ μέρει.



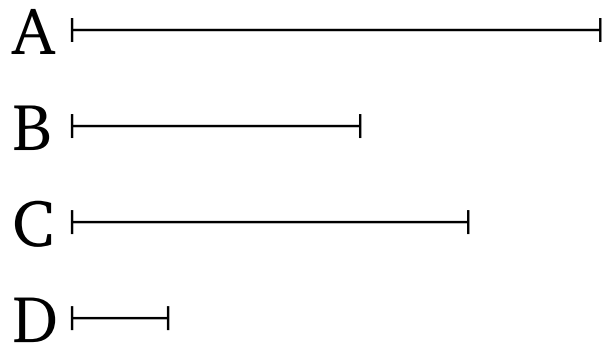
Ἀριθμὸς γὰρ ὁ Α μέρος ἐθέτω ὅτιοῦν τὸν Β, καὶ τῷ Β μέρει ὁμώνυμος >έστω [ἀριθμὸς] ὁ Γ; λέγω, <ὅτι ὁ Γ τὸν Α μετρεῖ.

Ἐπεὶ γὰρ ὁ Β τοῦ Α μέρος ἐστὶν ὁμώνυμον τῷ Γ, >έστι δὲ καὶ ἡ Δ μονὰς τοῦ Γ μέρος ὁμώνυμον αὐτῷ, <ὃ >άρα μέρος ἐστὶν ἡ Δ μονὰς τοῦ Γ ἀριθμοῦ, τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ Β τοῦ Α; ἰσάκις >άρα ἡ Δ μονὰς τὸν Γ ἀριθμὸν μετρεῖ καὶ ὁ

ber) measured by D and C will also measure F [Prop. 7.35]. And E is the least (number) measured by C and D . Thus, E measures F , the greater (measuring) the lesser. The very thing is impossible. Thus, A , B , and C cannot measure some number which is less than E . Thus, E (is) the least (number) which is measured by A , B , and C . (Which is) the very thing it was required to show.

Proposition 37

If a number is measured by some number, (then) the (number) measured will have a part called the same as the measuring (number).

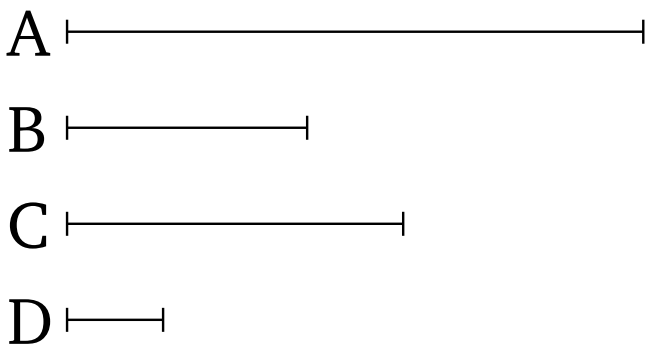


For let the number A be measured by some number B . I say that A has a part called the same as B .

For as many times as B measures A , so many units let there be in C . Since B measures A according to the units in C , and the unit D also measures C according to the units in it, the unit D thus measures the number C as many times as B (measures) A . Thus, alternately, the unit D measures the number B as many times as C (measures) A [Prop. 7.15]. Thus, which(ever) part the unit D is of the number B , C is also the same part of A . And the unit D is a part of the number B called the same as it (i.e., a B th part). Thus, C is also a part of A called the same as B (i.e., C is the B th part of A). Hence, A has a part C which is called the same as B (i.e., A has a B th part). (Which is) the very thing it was required to show.

Proposition 38

If a number has any part whatever, (then) it will be measured by a number called the same as the part.



For let the number A have any part whatever, B . And let the [number] C be called the same as the part B (i.e., B is the C th part of A). I say that C measures A .

For since B is a part of A called the same as C , and the unit D is also a part of C called the same as it (i.e., D is the C th part of C), thus which(ever) part the unit D is of the number C , B is also the same part of A . Thus, the unit

Β τὸν Α. ἐναλλάχ >άρα ἰσάκις ἢ Δ μονὰς τὸν Β ἀριθμὸν μετρεῖ καὶ ὁ Γ τὸν Α. ὁ Γ >άρα τὸν Α μετρεῖ; <όπερ >έδει δεῖχαι.

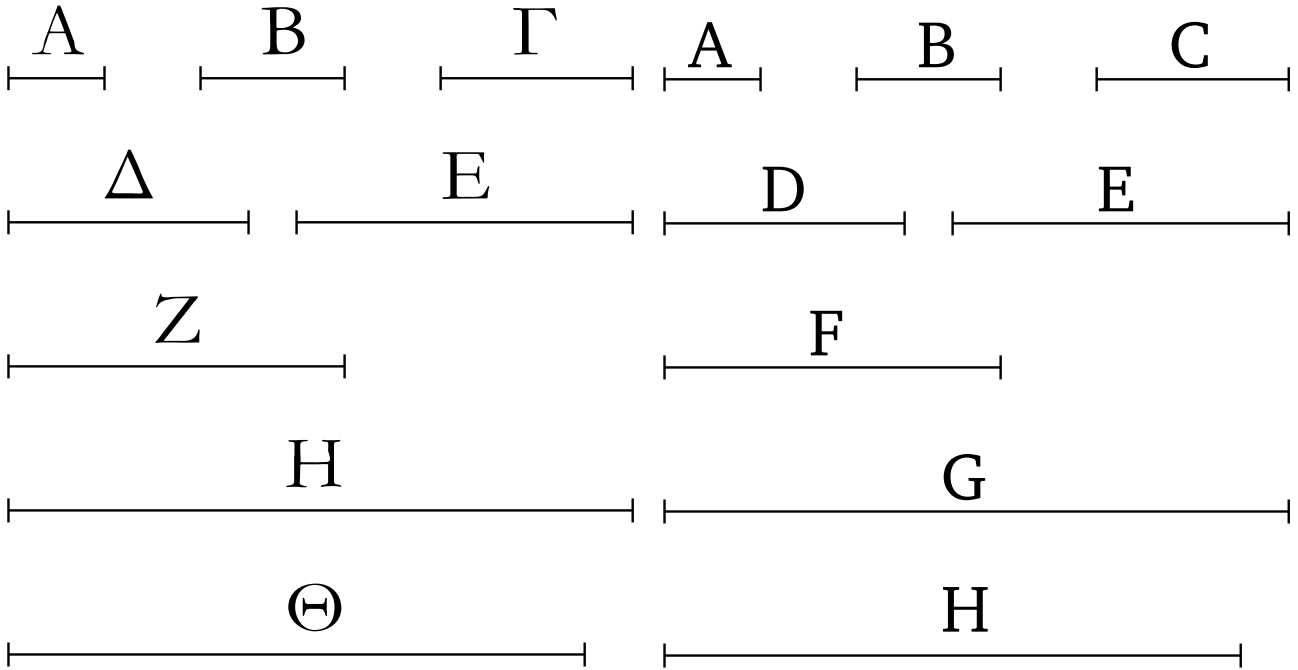
D measures the number *C* as many times as *B* (measures) *A*. Thus, alternately, the unit *D* measures the number *B* as many times as *C* (measures) *A* [Prop. 7.15]. Thus, *C* measures *A*. (Which is) the very thing it was required to show.

39.

Proposition 39

Ἀριθμὸν εὐρεῖν, <ὃς ἐλάχιστοξ >ὦν <έχει τὰ δοθέντα μέρη.

To find the least number that will have given parts.



>Ἐστω τὰ δοθέντα μέρη τὰ Α, Β, Γ; δεῖ δὴ ἀριθμὸν εὐρεῖν, <ὃς ἐλάχιστοξ >ὦν <έχει τὰ Α, Β, Γ μέρη.

>Ἐστῶσαν γὰρ τοῖς Α, Β, Γ μέρεσιν ὁμώνυμοι ἀριθμοὶ οἱ Δ, Ε, Ζ, καὶ εἰλήφθω ὑπὸ τῶν Δ, Ε, Ζ ἐλάχιστοξ μετρούμενοξ ἀριθμὸξ ὁ Η.

Ὁ Η >άρα ὁμώνυμα μέρη >έθει τοῖς Δ, Ε, Ζ. τοῖς δὲ Δ, Ε, Ζ ὁμώνυμα μέρη ἐστὶ τὰ Α, Β, Γ; ὁ Η >άρα >έθει τὰ Α, Β, Γ μέρη. λέγω δὴ, <ὅτι καὶ ἐλάχιστοξ >ὦν, εἰ γὰρ μή, >έσται τιξ τοῦ Η ἐλάσσων ἀριθμὸξ, <ὃς <έχει τὰ Α, Β, Γ μέρη. >έστω ὁ Θ. ἐπεὶ ὁ Θ >έθει τὰ Α, Β, Γ μέρη, ὁ Θ >άρα ὑπὸ ὁμωνύμων ἀριθμῶν μετρηθήσεται τοῖς Α, Β, Γ μέρεσιν. τοῖς δὲ Α, Β, Γ μέρεσιν ὁμώνυμοι ἀριθμοὶ εἰσιν οἱ Δ, Ε, Ζ; ὁ Θ >άρα ὑπὸ τῶν Δ, Ε, Ζ μετρεῖται. καὶ ἐστὶν ἐλάσσων τοῦ Η; <όπερ ἐστὶν ἀδύνατον. οὐκ >άρα >έσται τιξ τοῦ Η ἐλάσσων ἀριθμὸξ, <ὃς <έχει τὰ Α, Β, Γ μέρη; <όπερ >έδει δεῖχαι.

Let *A*, *B*, and *C* be the given parts. So it is required to find the least number which will have the parts *A*, *B*, and *C* (i.e., an *A*th part, a *B*th part, and a *C*th part).

For let *D*, *E*, and *F* be numbers having the same names as the parts *A*, *B*, and *C* (respectively). And let the least number, *G*, measured by *D*, *E*, and *F*, be taken [Prop. 7.36]. Thus, *G* has parts called the same as *D*, *E*, and *F* [Prop. 7.37]. And *A*, *B*, and *C* are parts called the same as *D*, *E*, and *F* (respectively). Thus, *G* has the parts *A*, *B*, and *C*. So I say that (*G*) is also the least (number having the parts *A*, *B*, and *C*). For if not, there will be some number less than *G* which will have the parts *A*, *B*, and *C*. Let it be *H*. Since *H* has the parts *A*, *B*, and *C*, *H* will thus be measured by numbers called the same as the parts *A*, *B*, and *C* [Prop. 7.38]. And *D*, *E*, and *F* are numbers called the same as the parts *A*, *B*, and *C* (respectively). Thus, *H* is measured by *D*, *E*, and *F*. And (*H*) is less than *G*. The very thing is impossible. Thus, there cannot be some number less than *G* which will have the parts *A*, *B*, and *C*. (Which is) the very thing it was required to show.

ELEMENTS BOOK 8

Continued Proportion[†]

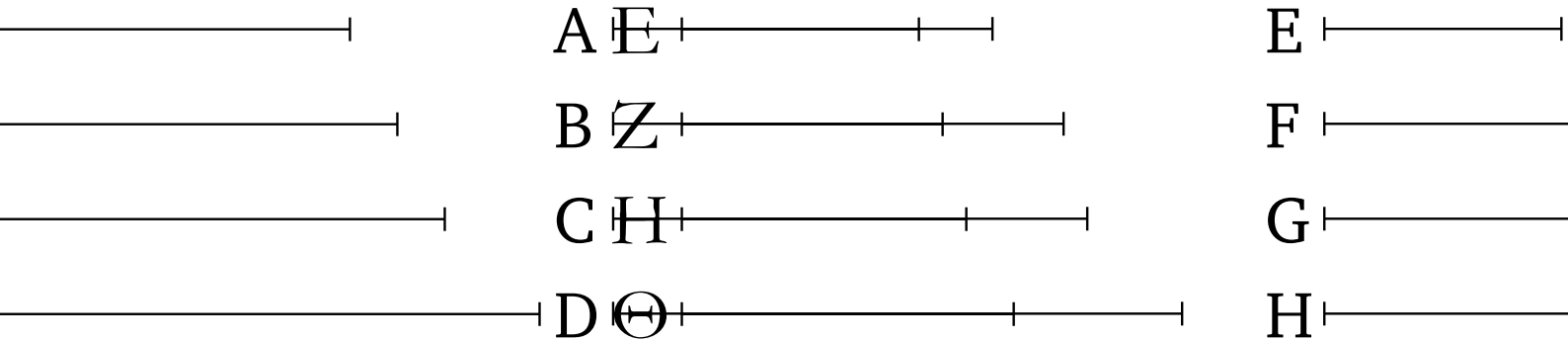
[†]The propositions contained in Books 7–9 are generally attributed to the school of Pythagoras.

1.

Proposition 1

Ἐάν >ῶσιν ὅσοιδηποτοῦν ἀριθμοὶ ἐχῆζ ἀνάλογον, οἱ δὲ >ᾱκροὶ αὐτῶν πρῶτοι πρὸς ἀλλήλουζ >ῶσιν, ἐλάχιστοι εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖζ.

If there are any multitude whatsoever of continuously proportional numbers, and the outermost of them are prime to one another, then the (numbers) are the least of those (numbers) having the same ratio as them.



>Ἐστωσαν ὅποσοιούν ἀριθμοὶ ἐχῆζ ἀνάλογον οἱ A, B, Γ, Δ, οἱ δὲ >ᾱκροὶ αὐτῶν οἱ A, Δ, πρῶτοι πρὸς ἀλλήλουζ >ἔστωσαν; λέγω, <ὅτι οἱ A, B, Γ, Δ ἐλάχιστοι εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖζ.

Εἰ γὰρ μή, >ἔστωσαν ἐλάττονεζ τῶν A, B, Γ, Δ οἱ E, Z, H, Θ ἐν τῷ αὐτῷ λόγῳ >όντεζ αὐτοῖζ. καὶ ἐπεὶ οἱ A, B, Γ, Δ ἐν τῷ αὐτῷ λόγῳ εἰσι τοῖζ E, Z, H, Θ, καὶ ἐστὶν >ἴσον τὸ πλῆθος [τῶν A, B, Γ, Δ] τῷ πλῆθει [τῶν E, Z, H, Θ], δι> ἴσου >ᾱρα ἐστὶν ὡς ὁ A πρὸς τὸν Δ, ὁ E πρὸς τὸν Θ. οἱ δὲ A, Δ πρῶτοι, οἱ δὲ πρῶτοι καὶ ἐλάχιστοι, οἱ δὲ ἐλάχιστοι ἀριθμοὶ μετροῦσι τοῖζ τὸν αὐτὸν λόγον >έθονταζ ἰσάκιζ <ὅ τε μείζων τὸν μείζονα καὶ ὁ ἐλάσσων τὸν ἐλάσσονα, τουτέστιν <ὅ τε ἡγούμενοζ τὸν ἡγούμενον καὶ ὁ ἐπόμενοζ τὸν ἐπόμενον. μετρεῖ >ᾱρα ὁ A τὸν E ὁ μείζων τὸν ἐλάσσονα; <ὅπερ ἐστὶν ἀδύνατον. οὐκ >ᾱρα οἱ E, Z, H, Θ ἐλάσσονεζ >όντεζ τῶν A, B, Γ, Δ ἐν τῷ αὐτῷ λόγῳ εἰσιν αὐτοῖζ. οἱ A, B, Γ, Δ >ᾱρα ἐλάχιστοι εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖζ; <ὅπερ >έδει δεῖχαι.

Let A, B, C, D be any multitude whatsoever of continuously proportional numbers. And let the outermost of them, A and D , be prime to one another. I say that A, B, C, D are the least of those (numbers) having the same ratio as them. For if not, let E, F, G, H be less than A, B, C, D (respectively), being in the same ratio as them. And since A, B, C, D are in the same ratio as E, F, G, H , and the multitude [of A, B, C, D] is equal to the multitude [of E, F, G, H], thus, via equality, as A is to D , (so) E (is) to H [Prop. 7.14]. And A and D (are) prime (to one another). And prime (numbers are) also the least of those (numbers having the same ratio as them) [Prop. 7.21]. And the least numbers measure those (numbers) having the same ratio (as them) an equal number of times, the greater (measuring) the greater, and the lesser the lesser—that is to say, the leading (measuring) the leading, and the following the following [Prop. 7.20]. Thus, A measures E , the greater (measuring) the lesser. The very thing is impossible. Thus, E, F, G, H , being less than A, B, C, D , are not in the same ratio as them. Thus, A, B, C, D are the least of those (numbers) having the same ratio as them. (Which is) the very thing it was required to show.

2.

Proposition 2

Αριθμοὺζ εὐρεῖν ἐχῆζ ἀνάλογον ἐλαθίστουζ, <όσουζ >ᾱν ἐπιτάχη τιζ, ἐν τῷ δοθέντι λόγῳ.

>Ἐστω ὁ δοθεῖζ λόγοζ ἐν ἐλάχιστοιζ ἀριθμοῖζ ὁ τοῦ A πρὸς τὸν B; δεῖ δὴ ἀριθμοὺζ εὐρεῖν ἐχῆζ ἀνάλογον ἐλαθίστουζ, <όσουζ >ᾱν τιζ ἐπιτάχη, ἐν τῷ τοῦ A πρὸς τὸν B λόγῳ.

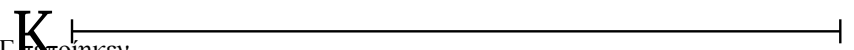
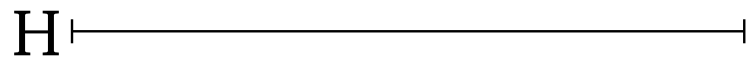
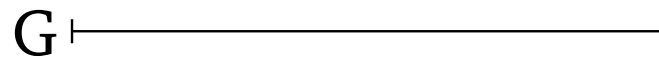
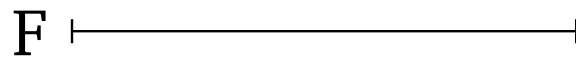
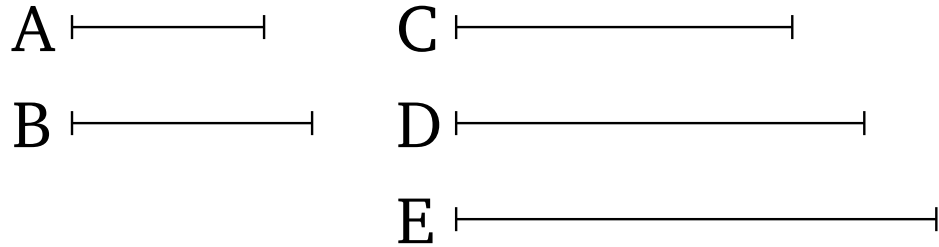
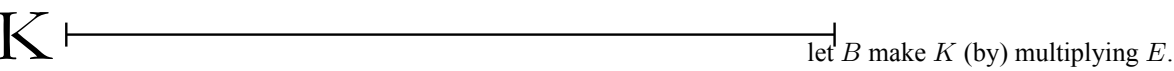
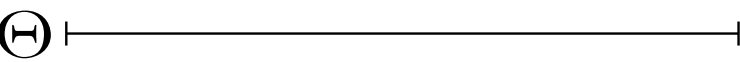
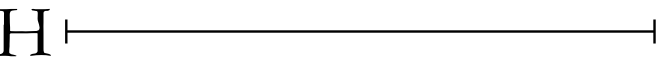
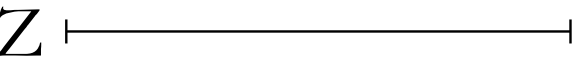
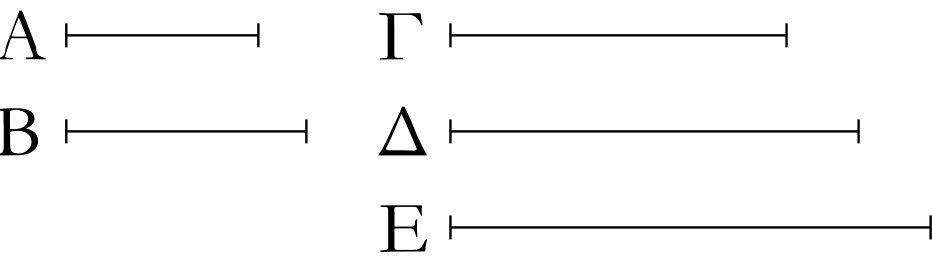
Ἐπιτετάθωσαν δὴ τέσσαρεζ, καὶ ὁ A ἐαυτὸν πολλαπλασιάσαζ B.

τὸν Γ ποιεῖτω, τὸν δὲ B πολλαπλασιάσαζ τὸν Δ ποιεῖτω, καὶ >έτι ὁ B ἐαυτὸν πολλαπλασιάσαζ τὸν E ποιεῖτω, καὶ >έτι ὁ A τοῖζ Γ, Δ, E πολλαπλασιάσαζ τοῖζ Z, H, Θ ποιεῖτω, ὁ δὲ B τὸν E πολλαπλασιάσαζ τὸν K ποιεῖτω.

To find the least numbers, as many as may be prescribed, (which are) continuously proportional in a given ratio.

Let the given ratio, (expressed) in the least numbers, be that of A to B . So it is required to find the least numbers, as many as may be prescribed, (which are) in the ratio of A to

Let four (numbers) be prescribed. And let A make C (by) multiplying itself, and let it make D (by) multiplying B . And, further, let B make E (by) multiplying itself. And, further, let A make F, G, H (by) multiplying C, D, E . And



Καὶ ἐπεὶ ὁ Α εἰς αὐτὸν μὲν πολλαπλασιάσας τὸν Γ πεποίηκεν, τὸν δὲ Β πολλαπλασιάσας τὸν Δ πεποίηκεν, >έστιν> ἄρα ὡς ὁ Α πρὸς τὸν Β, [ο<ύτωξ] ὁ Γ πρὸς τὸν Δ. πάλιν, ἐπεὶ ὁ μὲν Α τὸν Β πολλαπλασιάσας τὸν Δ πεποίηκεν, ὁ δὲ Β εἰς αὐτὸν πολλαπλασιάσας τὸν Ε πεποίηκεν, ἐκάτερος >ἄρα τῶν Α, Β τὸν Β πολλαπλασιάσας ἐκάτερον τῶν Δ, Ε πεποίηκεν. >έστιν> ἄρα ὡς ὁ Α πρὸς τὸν Β, ο<ύτωξ ὁ Δ πρὸς τὸν Ε. ἀλλ> ὡς ὁ Α πρὸς τὸν Β, ὁ Γ πρὸς τὸν Δ; καὶ ὡς >ἄρα ὁ Γ πρὸς τὸν Δ, ὁ Δ πρὸς τὸν Ε. καὶ ἐπεὶ ὁ Α τοὺς Γ, Δ πολλαπλασιάσας τοὺς Ζ, Η πεποίηκεν, >έστιν> ἄρα ὡς ὁ Γ πρὸς τὸν Δ, [ο<ύτωξ] ὁ Ζ πρὸς τὸν Η. ὡς δὲ ὁ Γ πρὸς τὸν Δ, ο<ύτωξ >ἦν ὁ Α πρὸς τὸν Β; καὶ ὡς >ἄρα ὁ Α πρὸς τὸν Β, ὁ Ζ πρὸς τὸν Η. πάλιν, ἐπεὶ ὁ Α τοὺς Δ, Ε πολλαπλασιάσας τοὺς Η, Θ πεποίηκεν, >έστιν> ἄρα ὡς ὁ Δ πρὸς τὸν Ε, ὁ Η πρὸς τὸν Θ. ἀλλ> ὡς ὁ Δ πρὸς τὸν Ε, ὁ Α πρὸς τὸν Β. καὶ ὡς >ἄρα ὁ Α πρὸς τὸν Β, ο<ύτωξ ὁ Η πρὸς τὸν Θ. καὶ ἐπεὶ οἱ Α, Β τὸν Ε πολλαπλασιάσαντες τοὺς Θ, Κ πεποιήκασιν, >έστιν> ἄρα ὡς ὁ Α πρὸς τὸν Β, ο<ύτωξ ὁ Θ πρὸς τὸν Κ. ἀλλ> ὡς ὁ Α πρὸς τὸν Β, ο<ύτωξ <ὁ τε Ζ πρὸς τὸν Η καὶ ὁ Η πρὸς τὸν Θ. καὶ ὡς >ἄρα ὁ Ζ πρὸς τὸν Η, ο<ύτωξ <ὁ τε Η πρὸς τὸν Θ καὶ ὁ Θ πρὸς τὸν Κ; οἱ Γ, Δ, Ε >ἄρα καὶ οἱ Ζ, Η, Θ, Κ ἀνάλογόν εἰσιν ἐν τῷ τοῦ Α πρὸς τὸν Β λόγῳ. λέγω

And since A has made C (by) multiplying itself, and has made D (by) multiplying B , thus as A is to B , [so] C (is) to D [Prop. 7.17]. Again, since A has made D (by) multiplying B , and B has made E (by) multiplying itself, A, B have thus made D, E , respectively, (by) multiplying B . Thus, as A is to B , so D (is) to E [Prop. 7.18]. But, as A (is) to B , (so) C (is) to D . And thus as C (is) to D , (so) D (is) to E . And since A has made F, G (by) multiplying C, D , thus as C is to D , [so] F (is) to G [Prop. 7.17]. And as C (is) to D , so A was to B . And thus as A (is) to B , (so) F (is) to G . Again, since A has made G, H (by) multiplying D, E , thus as D is to E , (so) G (is) to H [Prop. 7.17]. But, as D (is) to E , (so) A (is) to B . And thus as A (is) to B , so G (is) to H . And since A, B have made H, K (by) multiplying E , thus as A is to B , so H (is) to K . But, as A (is) to B , so F (is) to G , and G to H . And thus as F (is) to G , so G (is) to H , and H to K . Thus, C, D, E and F, G, H, K are (both continuously) proportional in the ratio of A to B . So I say that (they are) also the least (sets of numbers continuously proportional in that ratio). For since A and B are the least of those (numbers) having the same ratio as them,

δή, <ὅτι καὶ ἐλάχιστοι. ἐπεὶ γὰρ οἱ A, B ἐλάχιστοί εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς, οἱ δὲ ἐλάχιστοι τῶν τὸν αὐτὸν λόγον ἐθόντων πρῶτοι πρὸς ἀλλήλους εἰσίν, οἱ A, B >άρα πρῶτοι πρὸς ἀλλήλους εἰσίν. καὶ ἐκάτερος μὲν τῶν A, B ἑαυτὸν πολλαπλασιάσας ἐκάτερον τῶν Γ, E πεποίηκεν, ἐκάτερον δὲ τῶν Γ, E πολλαπλασιάσας ἐκάτερον τῶν Z, K πεποίηκεν; οἱ Γ, E >άρα καὶ οἱ Z, K πρῶτοι πρὸς ἀλλήλους εἰσίν. ἐὰν δὲ >ῶσιν ὅποσοι οὖν ἀριθμοὶ ἐχῇς ἀνάλογον, οἱ δὲ >άκροι αὐτῶν πρῶτοι πρὸς ἀλλήλους >ῶσιν, ἐλάχιστοί εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς. οἱ Γ, Δ, E >άρα καὶ οἱ Z, H, Θ, K ἐλάχιστοί εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς A, B ; <ὅπερ >έδει δεῖχαι.

Πόρισμα.

Ἐκ δὲ τούτου φανερόν, <ὅτι ἐὰν τρεῖς ἀριθμοὶ ἐχῇς ἀνάλογον ἐλάχιστοι >ῶσι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς, οἱ >άκρον αὐτῶν τετράγωνοί εἰσιν, ἐὰν δὲ τέσσαρες, κύβοι.

and the least of those (numbers) having the same ratio are prime to one another [Prop. 7.22], A and B are thus prime to one another. And A, B have made C, E , respectively, (by) multiplying themselves, and have made F, K by multiplying C, E , respectively. Thus, C, E and F, K are prime to one another [Prop. 7.27]. And if there are any multitude whatsoever of continuously proportional numbers, and the outermost of them are prime to one another, then the (numbers) are the least of those (numbers) having the same ratio as them [Prop. 8.1]. Thus, C, D, E and F, G, H, K are the least of those (continuously proportional sets of numbers) having the same ratio as A and B . (Which is) the very thing it was required to show.

Corollary

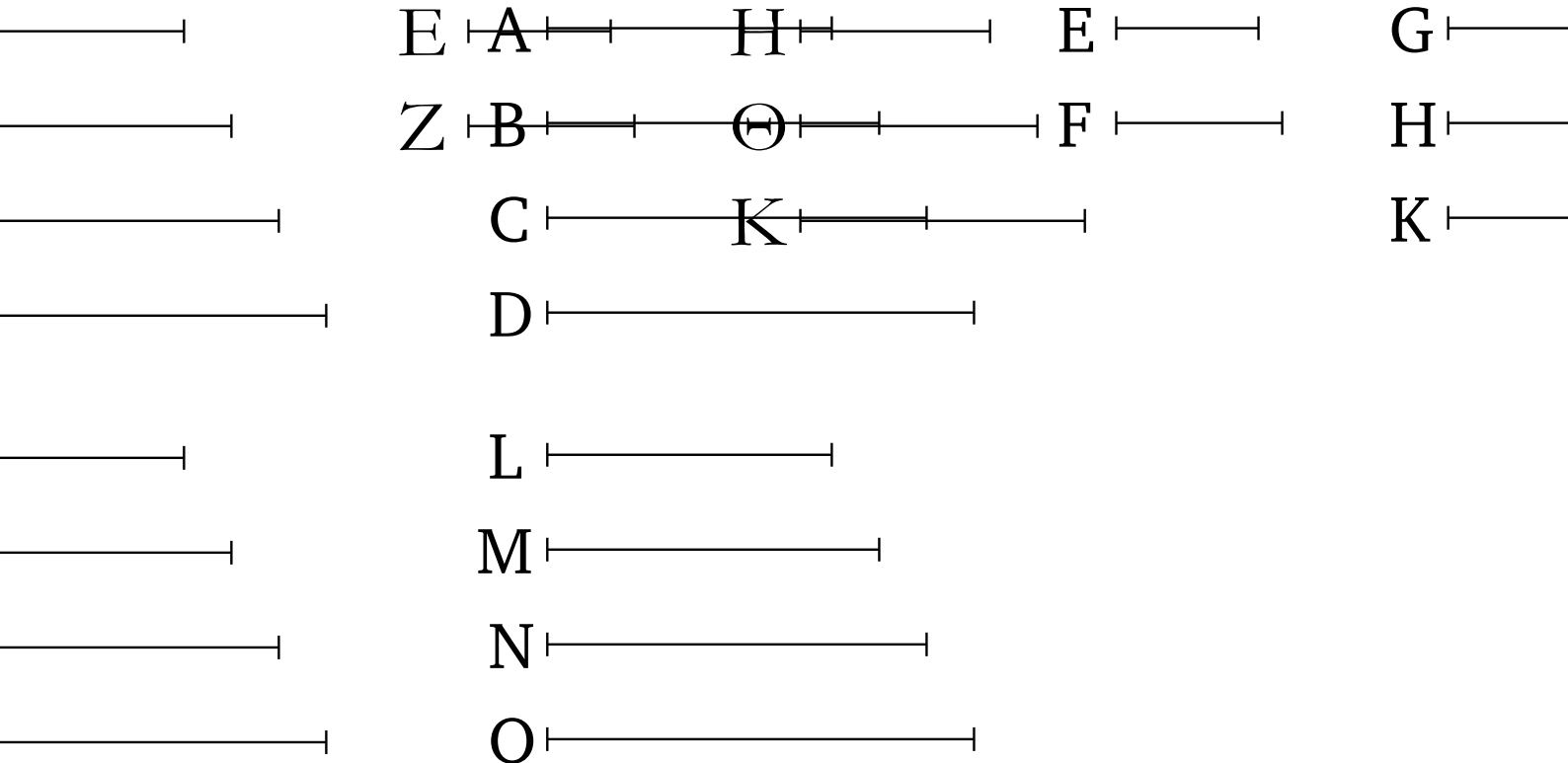
So it is clear, from this, that if three continuously proportional numbers are the least of those (numbers) having the same ratio as them then the outermost of them are square, and, if four (numbers), cube.

3.

Ἐὰν >ῶσιν ὅποσοι οὖν ἀριθμοὶ ἐχῇς ἀνάλογον ἐλάχιστοι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς, οἱ >άκροι αὐτῶν πρῶτοι πρὸς ἀλλήλους εἰσίν.

Proposition 3

If there are any multitude whatsoever of continuously proportional numbers (which are) the least of those (numbers) having the same ratio as them then the outermost of them are prime to one another.



>Ἐστωσαν ὅποσοι οὖν ἀριθμοὶ ἐχῇς ἀνάλογον ἐλάχιστοι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς οἱ A, B, Γ, Δ ; λέγω, <ὅτι οἱ >άκροι αὐτῶν οἱ A, Δ πρῶτοι πρὸς ἀλλήλους εἰσίν. Εἰλήφθωσαν γὰρ δύο μὲν ἀριθμοὶ ἐλάχιστοι ἐν τῷ τῶν A, B, Γ, Δ λόγῳ οἱ E, Z , τρεῖς δὲ οἱ H, Θ, K , καὶ ἐχῇς ἐνὶ πλείους, <έως τὸ λαμβανόμενον πληθοῦς >ίσον γένηται τῷ πλήθει τῶν A, B, Γ, Δ . εἰλήφθωσαν καὶ >έστωσαν οἱ Λ, M, N, X . Καὶ ἐπεὶ οἱ E, Z ἐλάχιστοί εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς, πρῶτοι πρὸς ἀλλήλους εἰσίν. καὶ ἐπεὶ ἐκάτερος

Let A, B, C, D be any multitude whatsoever of continuously proportional numbers (which are) the least of those (numbers) having the same ratio as them. I say that the outermost of them, A and D , are prime to one another. For let the two least (numbers) E, F (which are) in the same ratio as A, B, C, D be taken [Prop. 7.33]. And the three (least numbers) G, H, K [Prop. 8.2]. And (so on), successively increasing by one, until the multitude of (numbers) taken is made equal to the multitude of A, B, C, D . Let

τῶν E, Z ἑαυτὸν μὲν πολλαπλασιάσας ἑκάτερον τῶν H, K πεποίηκεν, ἑκάτερον δὲ τῶν H, K πολλαπλασιάσας ἑκάτερον τῶν Λ, X πεποίηκεν, καὶ οἱ $H, K >$ ἄρα καὶ οἱ Λ, X πρῶτοι πρὸς ἀλλήλους εἰσίν. καὶ ἐπεὶ οἱ A, B, Γ, Δ ἐλάχιστοι εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς, εἰσι δὲ καὶ οἱ Λ, M, N, X ἐλάχιστοι ἐν τῷ αὐτῷ λόγῳ $>$ όντες τοῖς A, B, Γ, Δ , καὶ ἐστὶν $>$ ίσον τὸ πλῆθος τῶν A, B, Γ, Δ τῷ πλῆθει τῶν Λ, M, N, X , $<$ έκαστος $>$ ἄρα τῶν A, B, Γ, Δ ἐκάστῳ τῶν $\Lambda, M, N, X >$ ίσοις ἐστίν; $>$ ίσοις $>$ ἄρα ἐστὶν ὁ μὲν A τῷ Λ , ὁ δὲ Δ τῷ X . καὶ εἰσιν οἱ Λ, X πρῶτοι πρὸς ἀλλήλους. καὶ οἱ $A, \Delta >$ ἄρα πρῶτοι πρὸς ἀλλήλους εἰσίν; $<$ όπερ $>$ έδει δεῖχαι.

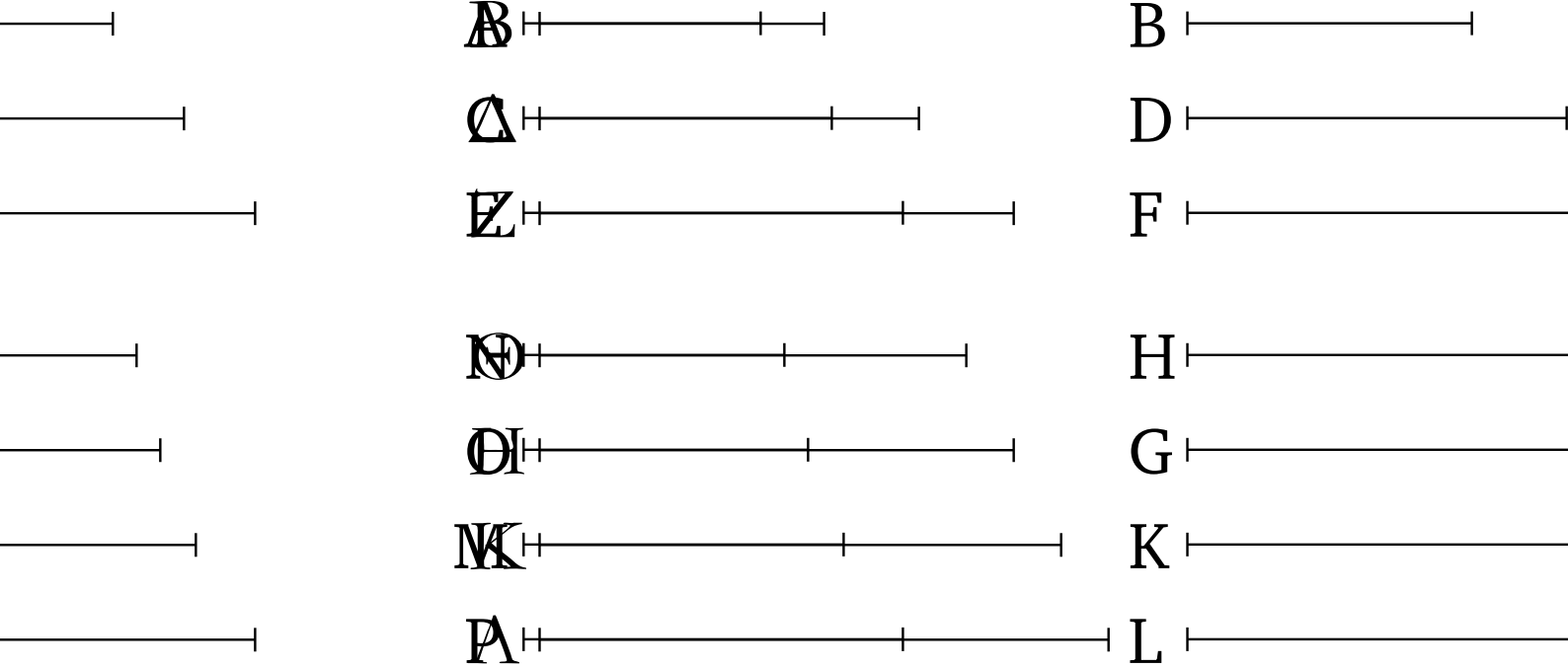
them be taken, and let them be L, M, N, O .

And since E and F are the least of those (numbers) having the same ratio as them they are prime to one another [Prop. 7.22]. And since E, F have made G, K , respectively, (by) multiplying themselves [Prop. 8.2 corr.], and have made L, O (by) multiplying G, K , respectively, G, K and L, O are thus also prime to one another [Prop. 7.27]. And since A, B, C, D are the least of those (numbers) having the same ratio as them, and L, M, N, O are also the least (of those numbers having the same ratio as them), being in the same ratio as A, B, C, D , and the multitude of A, B, C, D is equal to the multitude of L, M, N, O , thus A, B, C, D are equal to L, M, N, O , respectively. Thus, A is equal to L , and D to O . And L and O are prime to one another. Thus, A and D are also prime to one another. (Which is) the very thing it was required to show.

4.

Proposition 4

Λόγων δοθέντων ὁποσωνοῦν ἐν ἐλαθίστοις ἀριθμοῖς ἀριθμοῦν For any multitude whatsoever of given ratios, (expressed) εὑρεῖν ἐχῆς ἀνάλογον ἐλαθίστους ἐν τοῖς δοθεῖσι λόγοις. in the least numbers, to find the least numbers continuously proportional in these given ratios.



$>$ Εστωσαν οἱ δοθέντες λόγοι ἐν ἐλαθίστοις ἀριθμοῖς $<$ ό τε τοῦ A πρὸς τὸν B καὶ ὁ τοῦ Γ πρὸς τὸν Δ καὶ $>$ έτι ὁ τοῦ E πρὸς τὸν Z ; δεῖ δὴ ἀριθμοῦς εὑρεῖν ἐχῆς ἀνάλογον ἐλαθίστους $>$ έν τε τῷ τοῦ A πρὸς τὸν B λόγῳ καὶ ἐν τῷ τοῦ Γ πρὸς τὸν Δ καὶ $>$ έτι τῷ τοῦ E πρὸς τὸν Z .

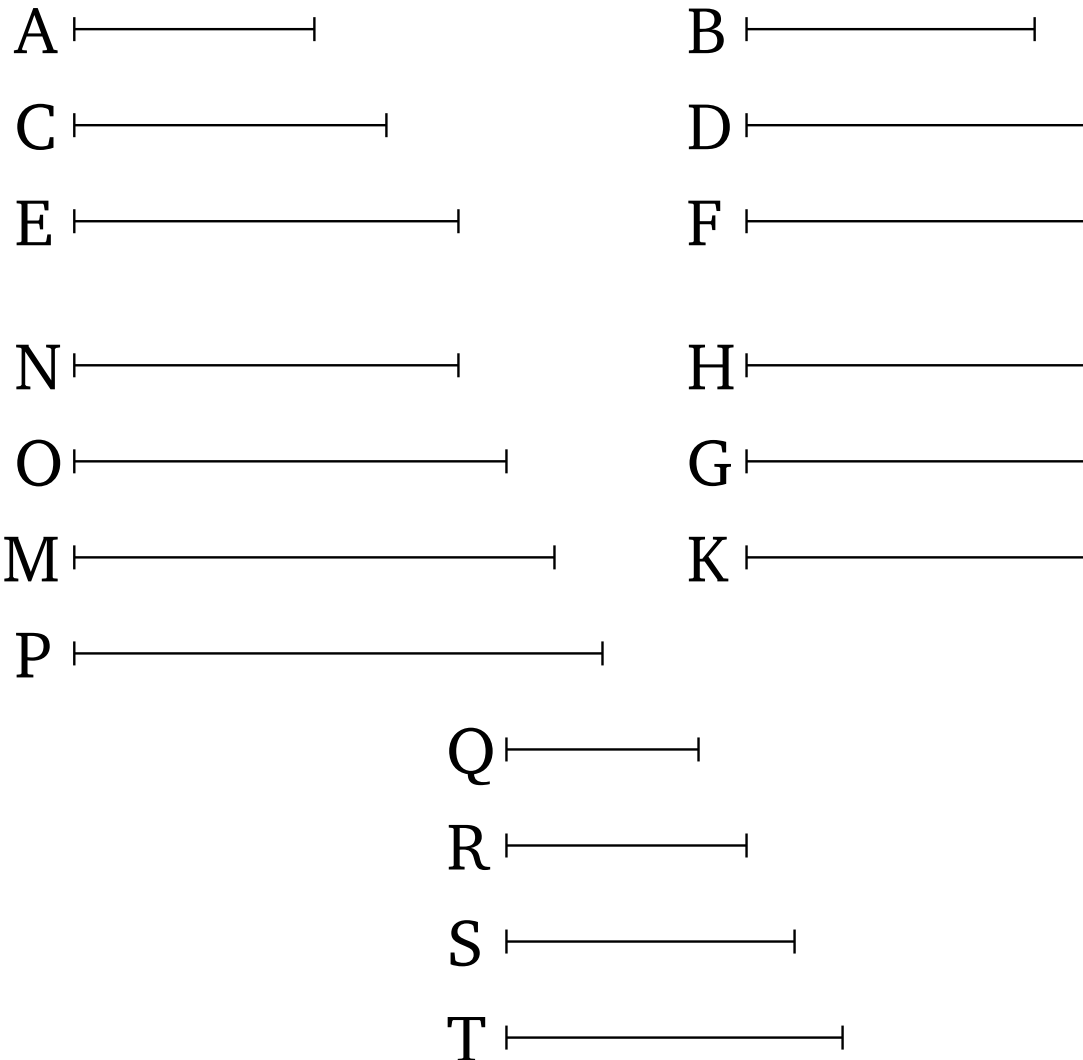
Εἰλήφθω γὰρ ὁ ὑπὸ τῶν B, Γ ἐλάχιστος μετρούμενος ἀριθμὸς ὁ H . καὶ ὡςάκις μὲν ὁ B τὸν H μετρεῖ, τοσαυτάκις καὶ ὁ A τὸν Θ μετρεῖτω, ὡςάκις δὲ ὁ Γ τὸν H μετρεῖ, τοσαυτάκις καὶ ὁ Δ τὸν K μετρεῖτω. ὁ δὲ E τὸν K $>$ ήτοι μετρεῖ $>$ ή οὐ μετρεῖ. μετρεῖτω πρότερον. καὶ ὡςάκις ὁ E τὸν K μετρεῖ, τοσαυτάκις καὶ ὁ Z τὸν Λ μετρεῖτω. καὶ ἐπεὶ ὡςάκις ὁ A τὸν Θ μετρεῖ καὶ ὁ B τὸν H , $>$ έστιν $>$ ἄρα ὡς ὁ A πρὸς τὸν B , οὕτως ὁ Θ πρὸς τὸν H . διὰ τὰ αὐτὰ δὴ καὶ ὡς ὁ Γ πρὸς τὸν Δ , οὕτως ὁ H πρὸς τὸν K , καὶ $>$ έτι ὡς ὁ E πρὸς τὸν Z , οὕτως ὁ K πρὸς τὸν Λ ; οἱ $\Theta, H, K, \Lambda >$ ἄρα ἐχῆς ἀνάλογον εἰσιν $>$ έν τε τῷ τοῦ A πρὸς τὸν B καὶ ἐν τῷ τοῦ Γ πρὸς τὸν Δ καὶ $>$ έτι ἐν τῷ τοῦ E πρὸς τὸν Z λόγῳ. λέγω δὴ, $<$ ότι καὶ ἐλάχιστοι. εἰ γὰρ μὴ εἰσιν οἱ Θ, H, K ,

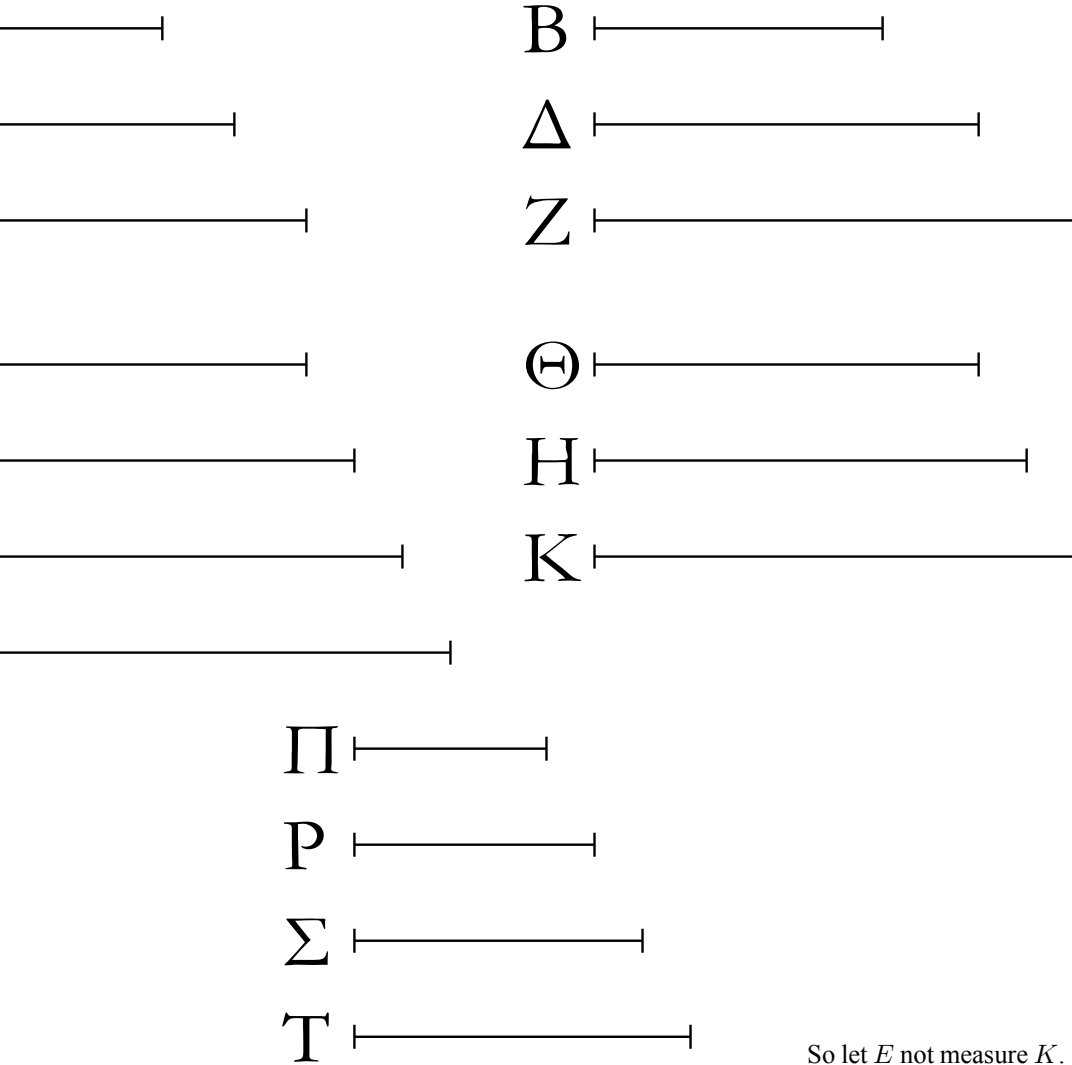
Let the given ratios, (expressed) in the least numbers, be the (ratios) of A to B , and of C to D , and, further, of E to F . So it is required to find the least numbers continuously proportional in the ratio of A to B , and of C to B , and, further, of E to F .

For let the least number, G , measured by (both) B and C have be taken [Prop. 7.34]. And as many times as B measures G , so many times let A also measure H . And as many times as C measures G , so many times let D also measure K . And E either measures, or does not measure, K . Let it, first of all, measure (K). And as many times as E measures K , so many times let F also measure L . And since A measures H the same number of times that B also (measures) G , thus as A is to B , so H (is) to G [Def. 7.20, Prop. 7.13]. And so, for the same (reasons), as C (is) to D , so G (is) to K , and, further, as E (is) to F , so K (is) to L . Thus, H, G, K, L are continuously proportional in the ratio of A to B , and of C to D , and, further, of E to F . So I say that

Λ ἐχῆς ἀνάλογον ἐλάχιστοι >έν τε τοῖς τοῦ Α πρὸς τὸν Β καὶ τοῦ Γ πρὸς τὸν Δ καὶ ἐν τῷ τοῦ Ε πρὸς τὸν Ζ λόγοις, >έστωσαν οἱ Ν, Χ, Μ, Ο. καὶ ἐπεὶ ἐστὶν ὡς ὁ Α πρὸς τὸν Β, ο<ύτωξ ὁ Ν πρὸς τὸν Χ, οἱ δὲ Α, Β ἐλάχιστοι, οἱ δὲ ἐλάχιστοι μετροῦσι τοὺς τὸν αὐτὸν λόγον >έθονταξ ἰσάκιξ <ὁ τε μείζων τὸν μείζονα καὶ ὁ ἐλάσσων τὸν ἐλάσσονα, τουτέστιν <ὁ τε ἡγούμενοξ τὸν ἡγούμενον καὶ ὁ ἐπόμενοξ τὸν ἐπόμενον, ὁ Β >άρα τὸν Χ μετρεῖ. διὰ τὰ αὐτὰ δὴ καὶ ὁ Γ τὸν Χ μετρεῖ, οἱ Β, Γ >άρα τὸν Χ μετροῦσιν; καὶ ὁ ἐλάχιστοξ >άρα ὑπὸ τῶν Β, Γ μετρούμενοξ τὸν Χ μετρήσει. ἐλάχιστοξ δὲ ὑπὸ τῶν Β, Γ μετρεῖται ὁ Η; ὁ Η >άρα τὸν Χ μετρεῖ ὁ μείζων τὸν ἐλάσσονα; <ὅπερ ἐστὶν ἀδύνατον. οὐκ >άρα >έσσονται τινεξ τῶν Θ, Η, Κ, Λ ἐλάσσονεξ ἀριθμοὶ ἐχῆς >έν τε τῷ τοῦ Α πρὸς τὸν Β καὶ τῷ τοῦ Γ πρὸς τὸν Δ καὶ >έτι τῷ τοῦ Ε πρὸς τὸν Ζ λόγῳ|.

(they are) also the least (numbers continuously proportional in these ratios). For if H, G, K, L are not the least numbers continuously proportional in the ratios of A to B , and of C to D , and of E to F , let N, O, M, P be (the least such numbers). And since as A is to B , so N (is) to O , and A and B are the least (numbers which have the same ratio as them), and the least (numbers) measure those (numbers) having the same ratio (as them) an equal number of times, the greater (measuring) the greater, and the lesser the lesser—that is to say, the leading (measuring) the leading, and the following the following [Prop. 7.20], B thus measures O . So, for the same (reasons), C also measures O . Thus, B and C (both) measure O . Thus, the least number measured by (both) B and C will also measure O [Prop. 7.35]. And G (is) the least number measured by (both) B and C . Thus, G measures O , the greater (measuring) the lesser. The very thing is impossible. Thus, there cannot be any numbers less than H, G, K, L (which are) continuously (proportional) in the ratio of A to B , and of C to D , and, further, of E to F .





μὴ μετρεῖται δὴ ὁ E τὸν K , καὶ εἰλήφθω ὑπὸ τῶν E , K ἐλάχιστος μετρούμενος ἀριθμὸς ὁ M . καὶ ὅσκις μὲν ὁ K τὸν M μετρεῖ, τοσαυτάκις καὶ ἑκάτερος τῶν Θ , H ἑκάτερον τῶν N , X μετρεῖται, ὅσκις δὲ ὁ E τὸν M μετρεῖ, τοσαυτάκις καὶ ὁ Z τὸν O μετρεῖται. ἐπεὶ ἰσάκις ὁ Θ τὸν N μετρεῖ καὶ ὁ H τὸν X , >έστιν >άρα ὡς ὁ Θ πρὸς τὸν H , ο<ύτως ὁ N πρὸς τὸν X . ὡς δὲ ὁ Θ πρὸς τὸν H , ο<ύτως ὁ A πρὸς τὸν B ; καὶ ὡς >άρα ὁ A πρὸς τὸν B , ο<ύτως ὁ N πρὸς τὸν X . διὰ τὰ αὐτὰ δὴ καὶ ὡς ὁ Γ πρὸς τὸν Δ , ο<ύτως ὁ X πρὸς τὸν M . πάλιν, ἐπεὶ ἰσάκις ὁ E τὸν M μετρεῖ καὶ ὁ Z τὸν O , >έστιν >άρα ὡς ὁ E πρὸς τὸν Z , ο<ύτως ὁ M πρὸς τὸν O ; οἱ N , X , M , O >άρα ἐχῆς ἀνάλογόν εἰσιν ἐν τοῖς τοῦ τε A πρὸς τὸν B καὶ τοῦ Γ πρὸς τὸν Δ καὶ >έτι τοῦ E πρὸς τὸν Z λόγοις. λέγω δὴ, <ὅτι καὶ ἐλάχιστοι ἐν τοῖς $A B$, $\Gamma \Delta$, $E Z$ λόγοις. εἰ γὰρ μὴ, >έσονται τινες τῶν N , X , M , O ἐλάσσονες ἀριθμοὶ ἐχῆς ἀνάλογον ἐν τοῖς $A B$, $\Gamma \Delta$, $E Z$ λόγοις. >έστωσαν οἱ Π , P , Σ , T . καὶ ἐπεὶ ἐστὶν ὡς ὁ Π πρὸς τὸν P , ο<ύτως ὁ A πρὸς τὸν B , οἱ δὲ A , B ἐλάχιστοι, οἱ δὲ ἐλάχιστοι μετροῦσι τοὺς τὸν αὐτὸν λόγον >έθοντα αὐτοῖς ἰσάκις <ὁ τε ἡγούμενος τὸν ἡγούμενον καὶ ὁ ἐπόμενος τὸν ἐπόμενον, ὁ B >άρα τὸν P μετρεῖ. διὰ τὰ αὐτὰ δὴ καὶ ὁ Γ τὸν P μετρεῖ; οἱ B , Γ >άρα τὸν P μετροῦσιν. καὶ ὁ ἐλάχιστος >άρα ὑπὸ τῶν B , Γ μετούμενος τὸν P μετρήσει. ἐλάχιστος δὲ ὑπὸ τῶν B , Γ μετρούμενος ἐστὶν ὁ H ; ὁ H >άρα τὸν P μετρεῖ. καὶ ἐστὶν ὡς ὁ H πρὸς τὸν P , ο<ύτως ὁ K πρὸς τὸν Σ ; καὶ ὁ K >άρα τὸν Σ μετρεῖ. μετρεῖ δὲ καὶ ὁ E τὸν Σ ; οἱ E , K >άρα τὸν Σ μετροῦσιν. καὶ ὁ ἐλάχιστος >άρα ὑπὸ τῶν E ,

So let E not measure K . And let the least number, M , measured by (both) E and K be taken [Prop. 7.34]. And as many times as K measures M , so many times let H , G also measure N , O , respectively. And as many times as E measures M , so many times let F also measure P . Since H measures N the same number of times as G (measures) O , thus as H is to G , so N (is) to O [Def. 7.20, Prop. 7.13]. And as H (is) to G , so A (is) to B . And thus as A (is) to B , so N (is) to O . And so, for the same (reasons), as C (is) to D , so O (is) to M . Again, since E measures M the same number of times as F (measures) P , thus as E is to F , so M (is) to P [Def. 7.20, Prop. 7.13]. Thus, N , O , M , P are continuously proportional in the ratios of A to B , and of C to D , and, further, of E to F . So I say that (they are) also the least (numbers) in the ratios of $A B$, $C D$, $E F$. For if not, then there will be some numbers less than N , O , M , P (which are) continuously proportional in the ratios of $A B$, $C D$, $E F$. Let them be Q , R , S , T . And since as Q is to R , so A (is) to B , and A and B (are) the least (numbers having the same ratio as them), and the least (numbers) measure those (numbers) having the same ratio as them an equal number of times, the leading (measuring) the leading, and the following the following [Prop. 7.20], B thus measures R . So, for the same (reasons), C also measures R . Thus, B and C (both) measure R . Thus, the least (number) measured by (both) B and C will also measure R [Prop. 7.35]. And G is the least number measured by (both) B and C . Thus, G measures R . And as G is to R , so K (is) to S . Thus,

Κ μετρούμενος τὸν Σ μετρήσει. ἐλάχιστος δὲ ὑπὸ τῶν Ε, Κ μετρούμενός ἐστιν ὁ Μ; ὁ Μ > ἄρα τὸν Σ μετρεῖ ὁ μείζων τὸν ἐλάχιστον; <ὅπερ ἐστὶν ἀδύνατον. οὐκ > ἄρα > ἐσονταί τινες τῶν Ν, Χ, Μ, Ο ἐλάσσονες ἀριθμοὶ ἐχῆξ ἀνάλογον > ἐν τε τοῖς τοῦ Α πρὸς τὸν Β καὶ τοῦ Γ πρὸς τὸν Δ καὶ > ἐτι τοῦ Ε πρὸς τὸν Ζ λόγοις; οἱ Ν, Χ, Μ, Ο > ἄρα ἐχῆξ ἀνάλογον ἐλάχιστοι εἰσιν ἐν τοῖς Α Β, Γ Δ, Ε Ζ λόγοις; <ὅπερ > ἐδει δεῖχαι.

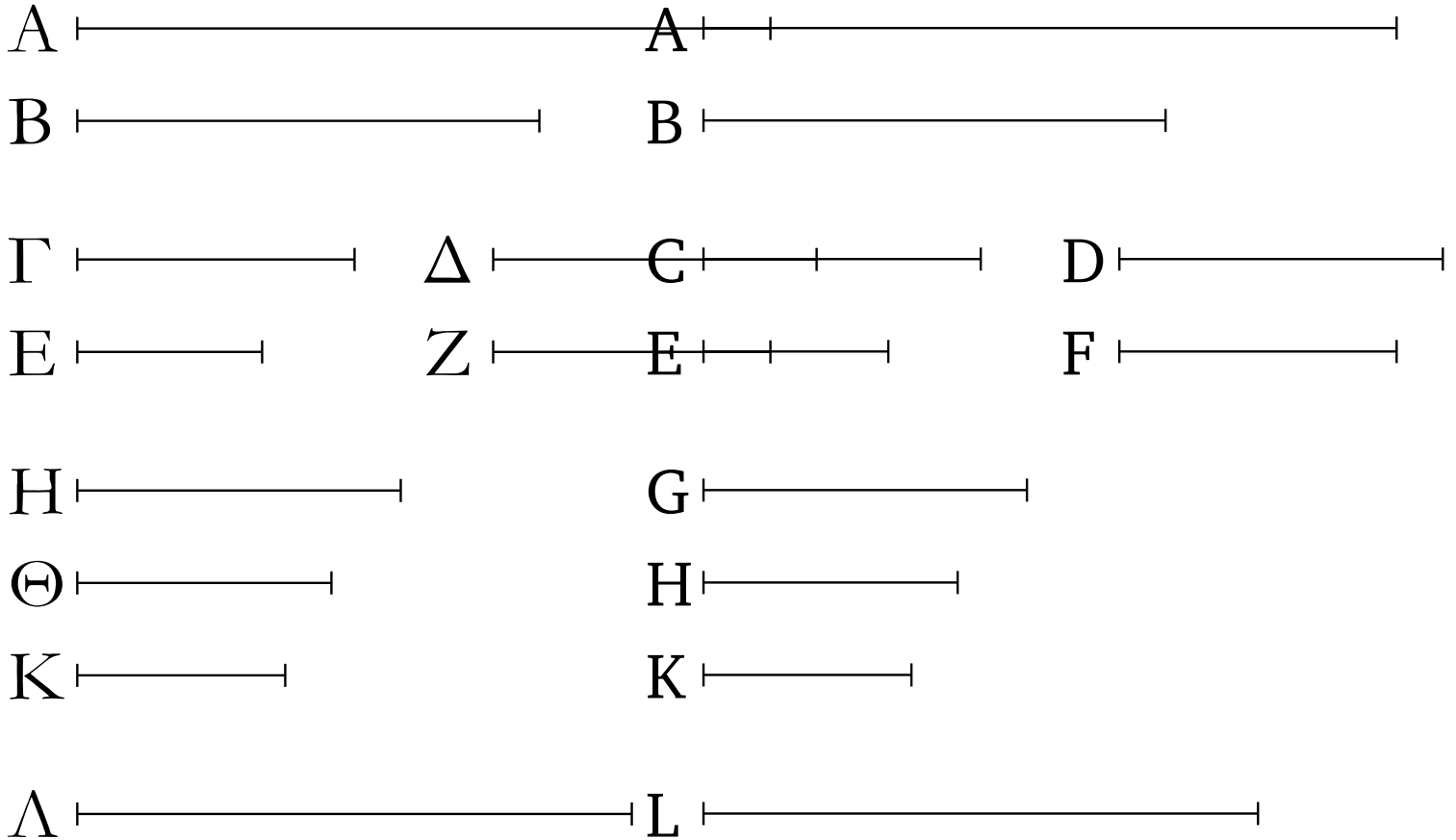
K also measures S [Def. 7.20]. And E also measures S [Prop. 7.20]. Thus, E and K (both) measure S . Thus, the least (number) measured by (both) E and K will also measure S [Prop. 7.35]. And M is the least (number) measured by (both) E and K . Thus, M measures S , the greater (measuring) the lesser. The very thing is impossible. Thus there cannot be any numbers less than N, O, M, P (which are) continuously proportional in the ratios of A to B , and of C to D , and, further, of E to F . Thus, N, O, M, P are the least (numbers) continuously proportional in the ratios of $A B, C D, E F$. (Which is) the very thing it was required to show.

5.

Οἱ ἐπίπεδοι ἀριθμοὶ πρὸς ἀλλήλουξ λόγον > ἐθουσι τὸν συγκείμενον ἐκ τῶν πλευρῶν.

Proposition 5

Plane numbers have to one another the ratio compounded[†] out of (the ratios of) their sides.



> Ἐστωσαν ἐπίπεδοι ἀριθμοὶ οἱ Α, Β, καὶ τοῦ μὲν Α πλευραὶ > ἐστώσαν οἱ Γ, Δ ἀριθμοί, τοῦ δὲ Β οἱ Ε, Ζ; λέγω, <ὅτι ὁ Α πρὸς τὸν Β λόγον > ἐθει τὸν συγκείμενον ἐκ τῶν πλευρῶν. Λόγων γὰρ δοθέντων τοῦ τε <ὸν > ἐθει ὁ Γ πρὸς τὸν Ε καὶ ὁ Δ πρὸς τὸν Ζ εἰληφθῶσαν ἀριθμοὶ ἐχῆξ ἐλάχιστοι ἐν τοῖς Γ Ε, Δ Ζ λόγοις, οἱ Η, Θ, Κ, <ὥστε ε> ἵναι ὥς μὲν τὸν Γ πρὸς τὸν Ε, ο< ὑτῶξ τὸν Η πρὸς τὸν Θ, ὥς δὲ τὸν Δ πρὸς τὸν Ζ, ο< ὑτῶξ τὸν Θ πρὸς τὸν Κ. καὶ ὁ Δ τὸν Ε πολλαπλασιάσας τὸν Λ ποιεῖτω.

Καὶ ἐπεὶ ὁ Δ τὸν μὲν Γ πολλαπλασιάσας τὸν Α πεποίηκεν, τὸν δὲ Ε πολλαπλασιάσας τὸν Λ πεποίηκεν, > ἐστὶν > ἄρα ὥς ὁ Γ πρὸς τὸν Ε, ο< ὑτῶξ ὁ Α πρὸς τὸν Λ. ὥς δὲ ὁ Γ πρὸς τὸν Ε, ο< ὑτῶξ ὁ Η πρὸς τὸν Θ; καὶ ὥς > ἄρα ὁ Η πρὸς τὸν Θ, ο< ὑτῶξ ὁ Α πρὸς τὸν Λ. πάλιν, ἐπεὶ ὁ Ε τὸν Δ πολλαπλασιάσας τὸν Λ πεποίηκεν, ἀλλὰ μὴν καὶ τὸν Ζ πολλαπλασιάσας τὸν Β πεποίηκεν, > ἐστὶν > ἄρα ὥς ὁ Δ πρὸς τὸν Ζ, ο< ὑτῶξ ὁ Α πρὸς τὸν Β. ἀλλ> ὥς ὁ Δ πρὸς

Let A and B be plane numbers, and let the numbers C, D be the sides of A , and (the numbers) E, F (the sides) of B . I say that A has to B the ratio compounded out of (the ratios of) their sides.

For given the ratios which C has to E , and D (has) to F , let the least numbers, G, H, K , continuously proportional in the ratios $C E, D F$ be taken [Prop. 8.4], so that as C is to E , so G (is) to H , and as D (is) to F , so H (is) to K . And let D make L (by) multiplying E .

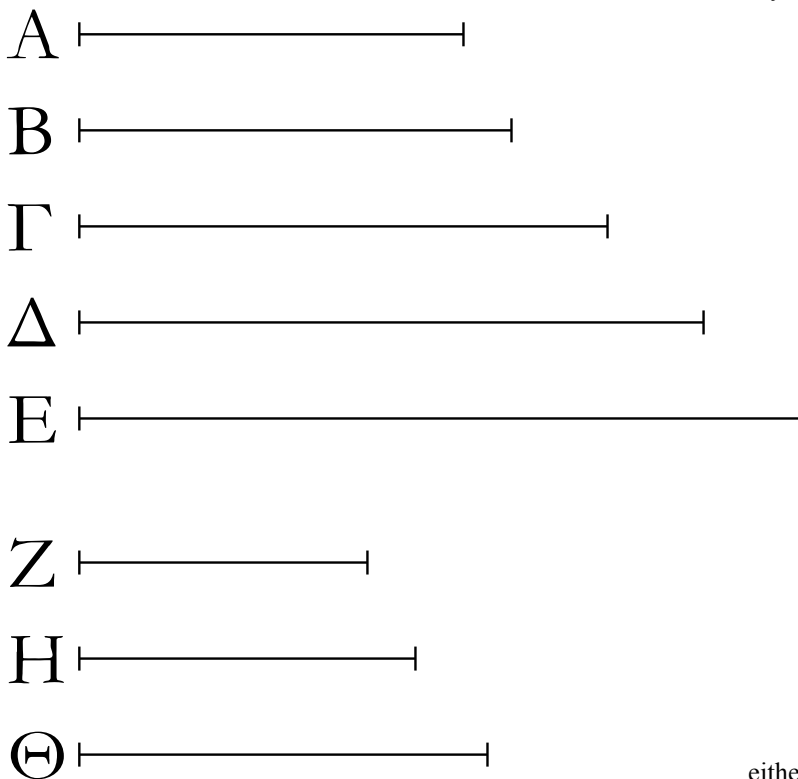
And since D has made A (by) multiplying C , and has made L (by) multiplying E , thus as C is to E , so A (is) to L [Prop. 7.17]. And as C (is) to E , so G (is) to H . And thus as G (is) to H , so A (is) to L . Again, since E has made L (by) multiplying D [Prop. 7.16], but, in fact, has also made B (by) multiplying F , thus as D is to F , so L (is) to B [Prop. 7.17]. But, as D (is) to F , so H (is) to K . And thus as H (is) to K , so L (is) to B . And it was also shown that

τὸν Ζ, οὐτως ὁ Θ πρὸς τὸν Κ; καὶ ὥς >άρα ὁ Θ πρὸς τὸν Κ, οὕτως ὁ Λ πρὸς τὸν Β. ἐδείκθη δὲ καὶ ὥς ὁ Η πρὸς τὸν Θ, οὕτως ὁ Α πρὸς τὸν Λ; δι> >ίσου >άρα ἐστὶν ὥς ὁ Η πρὸς τὸν Κ, [οὕτως] ὁ Α πρὸς τὸν Β. ὁ δὲ Η πρὸς τὸν Κ λόγον >έθει τὸν συγκείμενον ἐκ τῶν πλευρῶν; καὶ ὁ Α >άρα πρὸς τὸν Β λόγον >έθει τὸν συγκείμενον ἐκ τῶν πλευρῶν; <όπερ >έδει δεῖχαι.

† i.e., multiplied.

6.

Ἐάν >ῶσιν ὅποσοιὺν ἀριθμοὶ ἐχῇζ ἀνάλογον, ὁ δὲ πρῶτος τὸν δεύτερον μὴ μετρήῃ, οὐδὲ >άλλος οὐδεὶς οὐδένα μετρήσει. >Ἐστῶσαν ὅποσοιὺν ἀριθμοὶ ἐχῇζ ἀνάλογον οἱ Α, Β, Γ, Δ, Ε, ὁ δὲ Α τὸν Β μὴ μετρεῖτω; λέγω, <ὅτι οὐδὲ >άλλος οὐδεὶς οὐδένα μετρήσει.

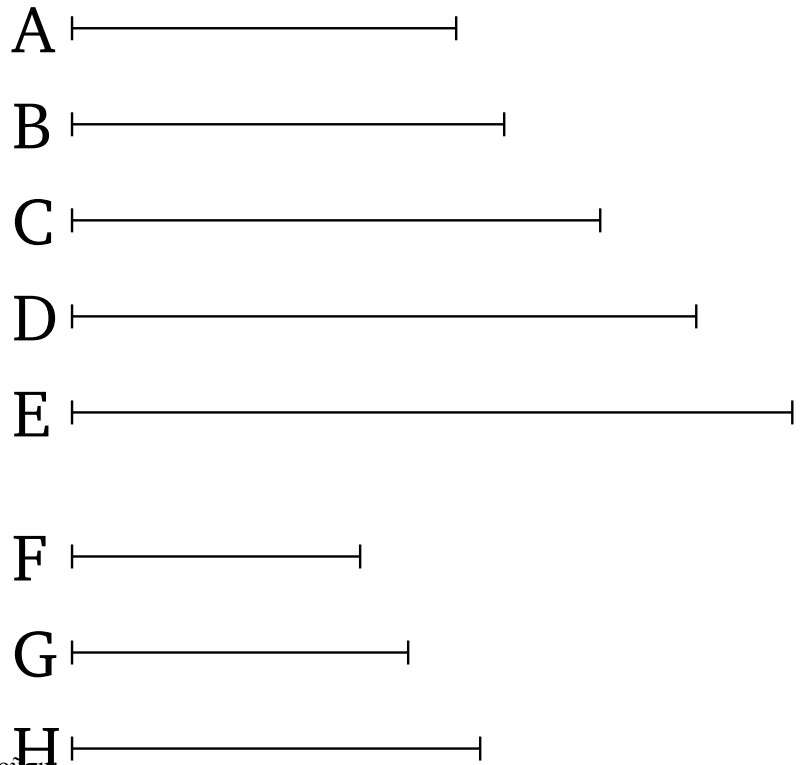


Proposition 6

If there are any multitude whatsoever of continuously proportional numbers, and the first does not measure the second, then no other (number) will measure any other (number) either.

Let A, B, C, D, E be any multitude whatsoever of continuously proportional numbers, and let A not measure B . I say that no other (number) will measure any other (number)

either.

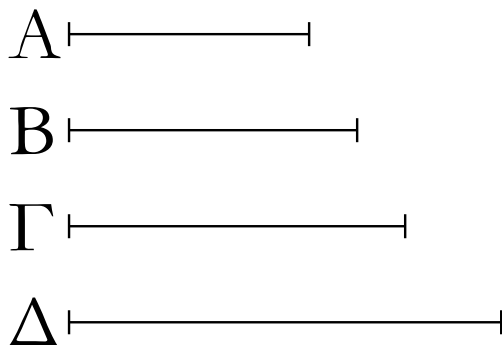


<Ὅτι μὲν οὖν οἱ A, B, Γ, Δ, E ἐχῆζ ἀλλήλους οὐ μετροῦσιν, φανερόν; οὐδὲ γὰρ ὁ A τὸν B μετρεῖ. λέγω δὴ, <ὅτι οὐδὲ >ἄλλος οὐδεὶς οὐδένα μετρήσει. εἰ γὰρ δυνατόν, μετρεῖτω ὁ A τὸν Γ . καὶ <όσοι εἰσὶν οἱ A, B, Γ , τοσοῦτοι εἰλήφθωσαν ἐλάχιστοι ἀριθμοὶ τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς A, B, Γ οἱ Z, H, Θ . καὶ ἐπεὶ οἱ Z, H, Θ ἐν τῷ αὐτῷ λόγῳ εἰσι τοῖς A, B, Γ , καὶ ἐστὶν >ῖσον τὸ πλῆθος τῶν A, B, Γ τῷ πλήθει τῶν Z, H, Θ , δι> >ῖσου >άρα ἐστὶν ὡς ὁ A πρὸς τὸν Γ , ο<ύτωξ ὁ Z πρὸς τὸν Θ . καὶ ἐπεὶ ἐστὶν ὡς ὁ A πρὸς τὸν B , ο<ύτωξ ὁ Z πρὸς τὸν H , οὐ μετρεῖ δὲ ὁ A τὸν B , οὐ μετρεῖ >άρα οὐδὲ ὁ Z τὸν H ; οὐκ >άρα μονὰξ ἐστὶν ὁ Z ; ἢ γὰρ μονὰξ πάντα ἀριθμὸν μετρεῖ. καὶ εἰσὶν οἱ Z, Θ πρῶτοι πρὸς ἀλλήλους [οὐδὲ ὁ Z >άρα τὸν Θ μετρεῖ]. καὶ ἐστὶν ὡς ὁ Z πρὸς τὸν Θ , ο<ύτωξ ὁ A πρὸς τὸν Γ ; οὐδὲ ὁ A >άρα τὸν Γ μετρεῖ. ὁμοίως δὴ δείχομεν, <ὅτι οὐδὲ >ἄλλος οὐδεὶς οὐδένα μετρήσει; <όπερ >έδει δεῖχαι.

Now, (it is) clear that A, B, C, D, E do not successively measure one another. For A does not even measure B . So I say that no other (number) will measure any other (number) either. For, if possible, let A measure C . And as many (numbers) as are A, B, C , let so many of the least numbers, F, G, H , be taken of those (numbers) having the same ratio as A, B, C [Prop. 7.33]. And since F, G, H are in the same ratio as A, B, C , and the multitude of A, B, C is equal to the multitude of F, G, H , thus, via equality, as A is to C , so F (is) to H [Prop. 7.14]. And since as A is to B , so F (is) to G , and A does not measure B , F does not measure G either [Def. 7.20]. Thus, F is not a unit. For a unit measures all numbers. And F and H are prime to one another [Prop. 8.3] [and thus F does not measure H]. And as F is to H , so A (is) to C . And thus A does not measure C either [Def. 7.20]. So, similarly, we can show that no other (number) can measure any other (number) either. (Which is) the very thing it was required to show.

7.

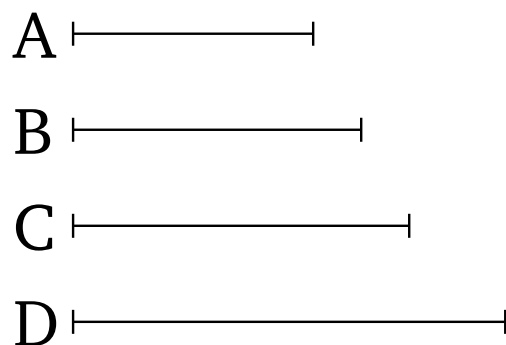
Ἐὰν >ῶσιν ὅποσοι οὖν ἀριθμοὶ [ἐχῆζ] ἀνάλογον, ὁ δὲ πρῶτος τὸν >έσθαι μετρήῃ, καὶ τὸν δεύτερον μετρήσει.



>Ἐστωσαν ὅποσοι οὖν ἀριθμοὶ ἐχῆζ ἀνάλογον οἱ A, B, Γ, Δ , ὁ δὲ A τὸν Δ μετρεῖτω; λέγω, <ὅτι καὶ ὁ A τὸν B μετρεῖ.

Proposition 7

If there are any multitude whatsoever of [continuously] proportional numbers, and the first measures the last, then (the first) will also measure the second.



Let A, B, C, D be any number whatsoever of continuously proportional numbers. And let A measure D . I say that A

Εἰ γὰρ οὐ μετρεῖ ὁ A τὸν B , οὐδὲ $>$ ἄλλοξ οὐδεὶς οὐδένα μετρήσει; μετρεῖ δὲ ὁ A τὸν Δ . μετρεῖ $>$ ἄρα καὶ ὁ A τὸν B ; $<$ ὅπερ $>$ ἔδει δεῖχαι.

also measures B .

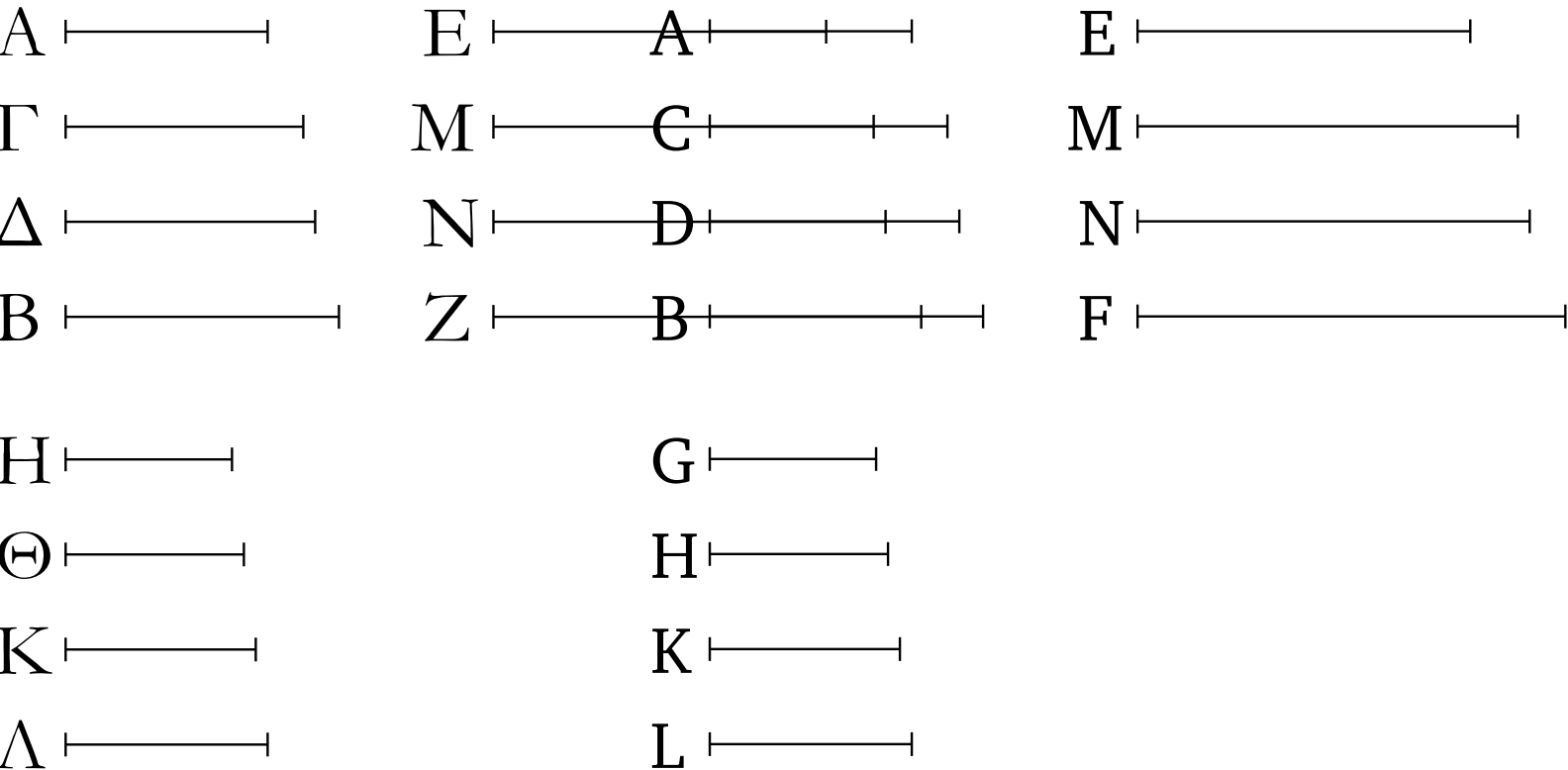
For if A does not measure B then no other (number) will measure any other (number) either [Prop. 8.6]. But A measures D . Thus, A also measures B . (Which is) the very thing it was required to show.

8.

Proposition 8

Ἐὰν δύο ἀριθμῶν μεταχὺ κατὰ τὸ συνεθεῖς ἀνάλογον ἐμπίπτωσι ἀριθμοί, $<$ ὅσοι εἰς αὐτοὺς μεταχὺ κατὰ τὸ συνεθεῖς ἀνόλογον ἐμπίπτουσιν ἀριθμοί, τοσοῦτοι καὶ εἰς τοὺς τὸν αὐτὸν λόγον $>$ ἐθονταξ [αὐτοῖς] μεταχὺ κατὰ τὸ συνεθεῖς ἀνάλογον ἐμπεσοῦνται

If between two numbers there fall (some) numbers in continued proportion then, as many numbers as fall in between them in continued proportion, so many (numbers) will also fall in between (any two numbers) having the same ratio [as them] in continued proportion.



Δύο γὰρ ἀριθμῶν τῶν A , B μεταχὺ κατὰ τὸ συνεθεῖς ἀνάλογον ἐμπίπτέτωσαν ἀριθμοὶ οἱ Γ , Δ , καὶ πεποιήσθω ὥς ὁ A πρὸς τὸν B , ο $<$ ύτωξ ὁ E πρὸς τὸν Z ; λέγω, ὅτι $<$ ὅσοι εἰς τοὺς A , B μεταχὺ κατὰ τὸ συνεθεῖς ἀνάλογον ἐμπεπτώκασιν ἀριθμοί, τοσοῦτοι καὶ εἰς τοὺς E , Z μεταχὺ κατὰ τὸ συνεθεῖς ἀνάλογον ἐμπεσοῦνται.

$<$ Ὅσοι γάρ εἰσι τῷ πλήθει οἱ A , B , Γ , Δ , τοσοῦτοι εἰλήφθωσαν ἐλάχιστοι ἀριθμοὶ τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς A , Γ , Δ , B οἱ H , Θ , K , Λ ; οἱ $>$ ἄρα $>$ ἄκροι αὐτῶν οἱ H , Λ πρῶτοι πρὸς ἀλλήλους εἰσίν. καὶ ἐπεὶ οἱ A , Γ , Δ , B τοῖς H , Θ , K , Λ ἐν τῷ αὐτῷ λόγῳ εἰσίν, καὶ ἐστὶν $>$ ἴσον τὸ πλήθος τῶν A , Γ , Δ , B τῷ πλήθει τῶν H , Θ , K , Λ , δι $>$ $>$ ἴσου $>$ ἄρα ἐστὶν ὥς ὁ A πρὸς τὸν B , ο $<$ ύτωξ ὁ H πρὸς τὸν Λ . ὥς δὲ ὁ A πρὸς τὸν B , ο $<$ ύτωξ ὁ E πρὸς τὸν Z ; καὶ ὥς $>$ ἄρα ὁ H πρὸς τὸν Λ , ο $<$ ύτωξ ὁ E πρὸς τὸν Z . οἱ δὲ H , Λ πρῶτοι, οἱ δὲ πρῶτοι καὶ ἐλάχιστοι, οἱ δὲ ἐλάχιστοι ἀριθμοὶ μετροῦσι τοὺς τὸν αὐτὸν λόγον $>$ ἐθονταξ ἰσάκιξ $<$ ὅ τε μείζων τὸν μείζονα καὶ ὁ ἐλάσσων τὸν ἐλάσσονα, τουτέστιν $<$ ὅ τε ἡγούμενος τὸν ἡγούμενον καὶ ὁ ἐπόμενος τὸν ἐπόμενον. ἰσάκιξ $>$ ἄρα ὁ H τὸν E μετρεῖ καὶ ὁ Λ τὸν Z . ὁσάκιξ δὴ ὁ H τὸν E μετρεῖ, τοσαυτάκιξ καὶ ἐκάτεροξ τῶν Θ , K ἐκάτερον τῶν M , N μετρεῖται; οἱ H , Θ , K , Λ $>$ ἄρα τοὺς E , M , N , Z ἰσάκιξ μετροῦσιν. οἱ H , Θ , K , Λ $>$ ἄρα τοῖς E , M , N , Z ἐν τῷ αὐτῷ λόγῳ εἰσίν. ἀλλὰ οἱ H , Θ , K , Λ τοῖς A , Γ , Δ , B ἐν τῷ

For let the numbers, C and D , fall between two numbers, A and B , in continued proportion, and let it be contrived (that) as A (is) to B , so E (is) to F . I say that, as many numbers as have fallen in between A and B in continued proportion, so many (numbers) will also fall in between E and F in continued proportion.

For as many as A , B , C , D are in multitude, let so many of the least numbers, G , H , K , L , having the same ratio as A , B , C , D , be taken [Prop. 7.33]. Thus, the outermost of them, G and L , are prime to one another [Prop. 8.3]. And since A , B , C , D are in the same ratio as G , H , K , L , and the multitude of A , B , C , D is equal to the multitude of G , H , K , L , thus, via equality, as A is to B , so G (is) to L [Prop. 7.14]. And as A (is) to B , so E (is) to F . And thus as G (is) to L , so E (is) to F . And G and L (are) prime (to one another). And (numbers) prime (to one another) are also the least (numbers having the same ratio as them) [Prop. 7.21]. And the least numbers measure those (numbers) having the same ratio (as them) an equal number of times, the greater (measuring) the greater, and the lesser the lesser—that is to say, the leading (measuring) the leading, and the following the following [Prop. 7.20]. Thus, G measures E the same number of times as L (measures) F . So as many times as G measures E , so many times let H , K also measure M ,

αὐτῶ| λόγῳ εἰσίν; καὶ οἱ $A, \Gamma, \Delta, B > \alpha\rho\alpha$ τοῖς E, M, N, Z ἐν τῶ| αὐτῶ| λόγῳ εἰσίν. οἱ δὲ A, Γ, Δ, B ἐχῆ|ξ ἀνάλογόν εἰσιν; καὶ οἱ $E, M, N, Z > \alpha\rho\alpha$ ἐχῆ|ξ ἀνάλογόν εἰσιν. <όσοι> ἄρα εἰς τοὺς A, B μεταχὺ κατὰ τὸ συνεθὲξ ἀνάλογον ἐμπεπτώκασιν ἀριθμοί, τοσοῦτοι καὶ εἰς τοὺς E, Z μεταχὺ κατὰ τὸ συνεθὲξ ἀνάλογον ἐμπεπτώκασιν ἀριθμοί; <όπερ> εἰδει δεῖχαι.

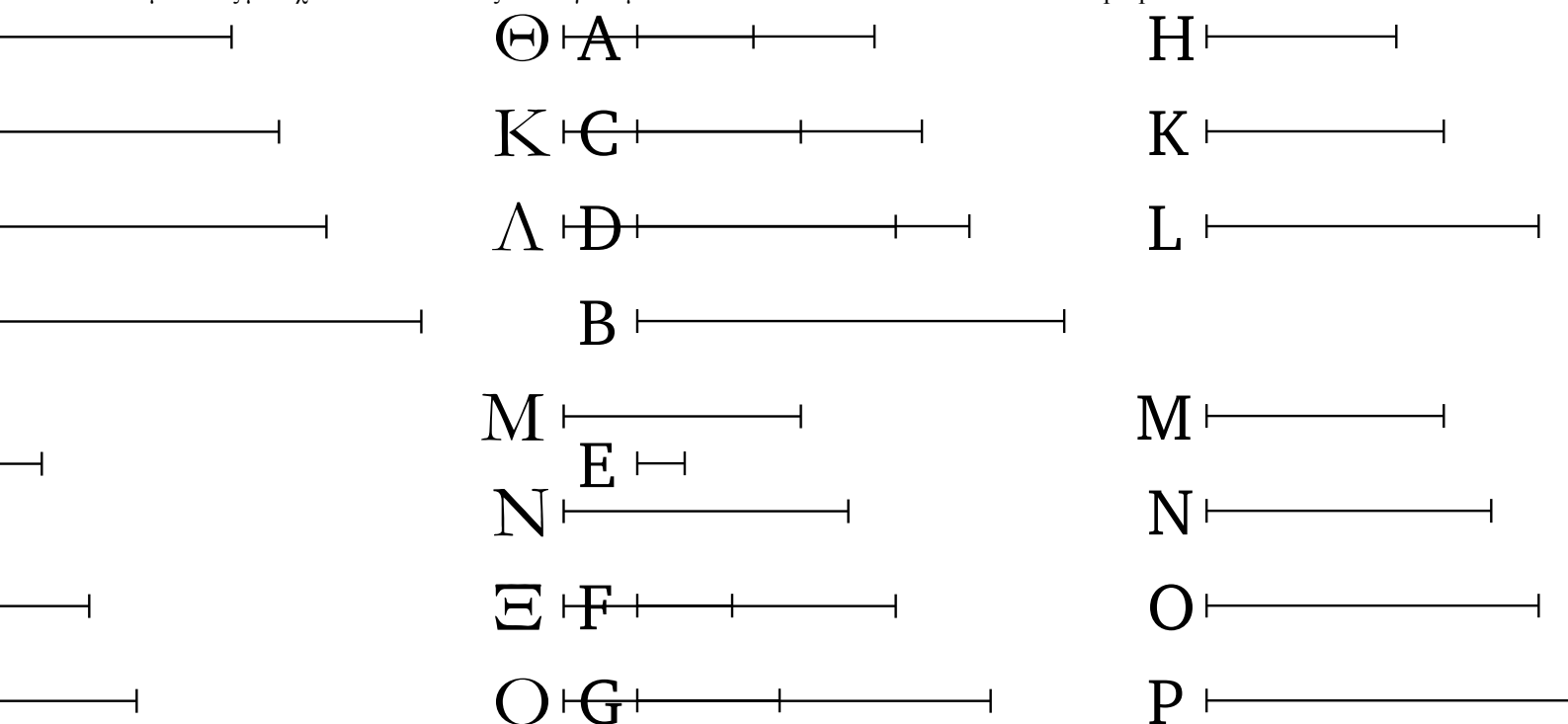
N , respectively. Thus, G, H, K, L measure E, M, N, F (respectively) an equal number of times. Thus, G, H, K, L are in the same ratio as E, M, N, F [Def. 7.20]. But, G, H, K, L are in the same ratio as A, C, D, B . Thus, A, C, D, B are also in the same ratio as E, M, N, F . And A, C, D, B are continuously proportional. Thus, E, M, N, F are also continuously proportional. Thus, as many numbers as have fallen in between A and B in continued proportion, so many numbers have also fallen in between E and F in continued proportion. (Which is) the very thing it was required to show.

9.

Proposition 9

Ἐὰν δύο ἀριθμοὶ πρῶτοι πρὸς ἀλλήλουξ >ῶσιν, καὶ εἰς αὐτοὺξ μεταχὺ κατὰ τὸ συνεθὲξ ἀνάλογον ἐμπίπτωσιν ἀριθμοί, <όσοι> εἰς αὐτοὺξ μεταχὺ κατὰ τὸ συνεθὲξ ἀνάλογον ἐμπίπτουσιν ἀριθμοί, τοσοῦτοι καὶ ἑκατέρου αὐτῶν καὶ μονάδοξ μεταχὺ κατὰ τὸ συνεθὲξ ἀνάλογον ἐμπεσοῦνται.

If two numbers are prime to one another and there fall in between them (some) numbers in continued proportion then, as many numbers as fall in between them in continued proportion, so many (numbers) will also fall between each of them and a unit in continued proportion.



>Ἐστωσαν δύο ἀριθμοὶ πρῶτοι πρὸς ἀλλήλουξ οἱ A, B , καὶ εἰς αὐτοὺξ μεταχὺ κατὰ τὸ συνεθὲξ ἀνάλογον ἐμπίπτέτωσαν οἱ Γ, Δ , καὶ ἐκκείσθω ἡ E μονάξ; λέγω, <ότι> <όσοι> εἰς τοὺς A, B μεταχὺ κατὰ τὸ συνεθὲξ ἀνάλογον ἐμπεπτώκασιν ἀριθμοί, τοσοῦτοι καὶ ἑκατέρου τῶν A, B καὶ τῆξ μονάδοξ μεταχὺ κατὰ τὸ συνεθὲξ ἀνάλογον ἐμπεσοῦνται.

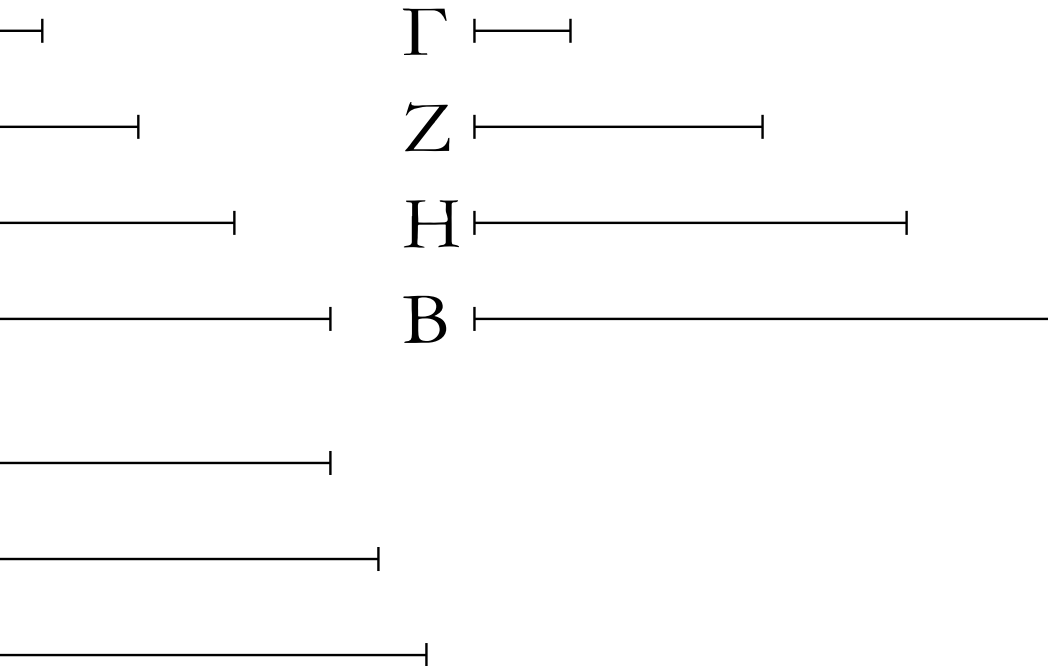
Εἰλήφθωσαν γὰρ δύο μὲν ἀριθμοὶ ἐλάχιστοι ἐν τῶ| τῶν A, Γ, Δ, B λόγῳ >όντεξ οἱ Z, H , τρεῖξ δὲ οἱ Θ, K, Λ , καὶ αἰεὶ ἐχῆ|ξ ἐνὶ πλείουξ, <έωξ> ἂν >ίσον γέννηται τὸ πληθὸξ αὐτῶν τῶ| πλήθει τῶν A, Γ, Δ, B . εἰλήφθωσαν, καὶ >έστωσαν οἱ M, N, X, O . φανερόν δὴ, <ότι> ὁ μὲν Z ἑαυτὸν πολλαπλασιάσαξ τὸν Θ πεποίηκεν, τὸν δὲ Θ πολλαπλασιάσαξ τὸν M πεποίηκεν, καὶ ὁ H ἑαυτὸν μὲν πολλαπλασιάσαξ τὸν Λ πεποίηκεν, τὸν δὲ Λ πολλαπλασιάσαξ τὸν O πεποίηκεν. καὶ ἐπεὶ οἱ M, N, X, O ἐλάχιστοι εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖξ Z, H , εἰσὶ δὲ καὶ οἱ A, Γ, Δ, B ἐλάχιστοι τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖξ Z, H , καὶ ἔστιν >ίσον τὸ πληθὸξ τῶν M, N, X, O τῶ| πλήθει τῶν A, Γ, Δ, B , <έκαστοξ> ἄρα τῶν M, N, X, O ἐκάστω τῶν $A, \Gamma, \Delta, B >ίσοξ$ ἔστιν; >ίσοξ> ἄρα

Let A and B be two numbers (which are) prime to one another, and let the (numbers) C and D fall in between them in continued proportion. And let the unit E be set out. I say that, as many numbers as have fallen in between A and B in continued proportion, so many (numbers) will also fall between each of A and B and the unit in continued proportion. For let the least two numbers, F and G , which are in the ratio of A, C, D, B , be taken [Prop. 7.33]. And the (least) three (numbers), H, K, L . And so on, successively increasing by one, until the multitude of the (least numbers taken) is made equal to the multitude of A, C, D, B [Prop. 8.2]. Let them be taken, and let them be M, N, O, P . So (it is) clear that F has made H (by) multiplying itself, and has made M (by) multiplying H . And G has made L (by) multiplying itself, and has made P (by) multiplying L [Prop. 8.2 corr.]. And since M, N, O, P are the least of those (numbers) having the same ratio as F, G , and A, C, D, B are also the least of those (numbers) having the same ratio as F, G [Prop. 8.2], and the multitude of M, N, O, P is equal to the multitude

ἐστὶν ὁ μὲν M τῷ A , ὁ δὲ O τῷ B . καὶ ἐπεὶ ὁ Z ἑαυτὸν πολλαπλασιάσας τὸν Θ πεποίηκεν, ὁ Z >άρα τὸν Θ μετρεῖ κατὰ τὰς ἐν τῷ Z μονάδας. μετρεῖ δὲ καὶ ἡ E μονὰς τὸν Z κατὰ τὰς ἐν αὐτῷ μονάδας; ἰσάκις >άρα ἡ E μονὰς τὸν Z ἀριθμὸν μετρεῖ καὶ ὁ Z τὸν Θ . >έστιν >άρα ὡς ἡ E μονὰς πρὸς τὸν Z ἀριθμὸν, ο<ύτως ὁ Z πρὸς τὸν Θ . πάλιν, ἐπεὶ ὁ Z τὸν Θ πολλαπλασιάσας τὸν M πεποίηκεν, ὁ Θ >άρα τὸν M μετρεῖ κατὰ τὰς ἐν τῷ Z μονάδας. μετρεῖ δὲ καὶ ἡ E μονὰς τὸν Z ἀριθμὸν κατὰ τὰς ἐν αὐτῷ μονάδας; ἰσάκις >άρα ἡ E μονὰς τὸν Z ἀριθμὸν μετρεῖ καὶ ὁ Θ τὸν M . >έστιν >άρα ὡς ἡ E μονὰς πρὸς τὸν Z ἀριθμὸν, ο<ύτως ὁ Θ πρὸς τὸν M . ἐδείχθη δὲ καὶ ὡς ἡ E μονὰς πρὸς τὸν Z ἀριθμὸν, ο<ύτως ὁ Z πρὸς τὸν Θ ; καὶ ὡς >άρα ἡ E μονὰς πρὸς τὸν Z ἀριθμὸν, ο<ύτως ὁ Z πρὸς τὸν Θ καὶ ὁ Θ πρὸς τὸν M . >ίσοξ δὲ ὁ M τῷ A ; >έστιν >άρα ὡς ἡ E μονὰς πρὸς τὸν Z ἀριθμὸν, ο<ύτως ὁ Z πρὸς τὸν Θ καὶ ὁ Θ πρὸς τὸν A . διὰ τὰ αὐτὰ δὴ καὶ ὡς ἡ E μονὰς πρὸς τὸν H ἀριθμὸν, ο<ύτως ὁ H πρὸς τὸν Λ καὶ ὁ Λ πρὸς τὸν B . <όσοι >άρα εἰς τοὺς A, B μεταχὺ κατὰ τὸ συνεθὲξ ἀνάλογον ἐμπεπτώκασιν ἀριθμοί, τοσοῦτοι καὶ ἑκατέρου τῶν A, B καὶ μονάδος τῆς E μεταχὺ κατὰ τὸ συνεθὲξ ἀνάλογον ἐμπεπτώκασιν ἀριθμοί; <όπερ >έδει δεῖχαι.

10.

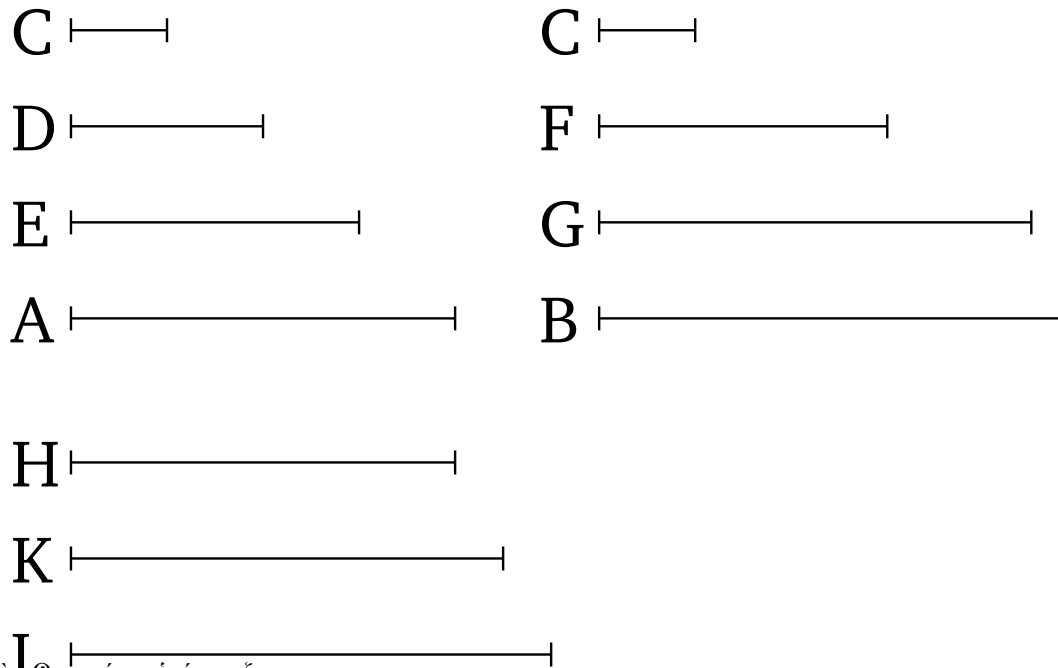
Ἐάν δύο ἀριθμῶν ἑκατέρου καὶ μονάδοξ μεταχὺ κατὰ τὸ συνεθὲξ ἀνάλογον ἐμπίπτωσιν ἀριθμοί, <όσοι ἑκατέρου αὐτῶν καὶ μονάδοξ μεταχὺ κατὰ τὸ συνεθὲξ ἀνάλογον ἐμπίπτουσιν ἀριθμοί, τοσοῦτοι καὶ εἰς αὐτοὺς μεταχὺ κατὰ τὸ συνεθὲξ ἀνάλογον ἐμπεσοῦνται. Δύο γὰρ ἀριθμῶν τῶν A, B καὶ μονάδοξ τῆς Γ μεταχὺ κατὰ τὸ συνεθὲξ ἀνάλογον ἐμπίπτέτωσαν ἀριθμοί ο<ί τε Δ, E καὶ οἱ Z, H ; λέγω, <ότι <όσοι ἑκατέρου τῶν A, B καὶ μονάδοξ τῆς Γ μεταχὺ κατὰ τὸ συνεθὲξ ἀνάλογον ἐμπεπτώκασιν ἀριθμοί, τοσοῦτοι καὶ εἰς τοὺς A, B μεταχὺ κατὰ τὸ συνεθὲξ ἀνάλογον ἐμπεσοῦνται.



of A, C, D, B , thus M, N, O, P are equal to A, C, D, B , respectively. Thus, M is equal to A , and P to B . And since F has made H (by) multiplying itself, F thus measures H according to the units in F [Def. 7.15]. And the unit E also measures F according to the units in it. Thus, the unit E measures the number F as many times as F (measures) H . Thus, as the unit E is to the number F , so F (is) to H [Def. 7.20]. Again, since F has made M (by) multiplying H , H thus measures M according to the units in F [Def. 7.15]. And the unit E also measures the number F according to the units in it. Thus, the unit E measures the number F as many times as H (measures) M . Thus, as the unit E is to the number F , so H (is) to M [Prop. 7.20]. And it was shown that as the unit E (is) to the number F , so F (is) to H . And thus as the unit E (is) to the number F , so F (is) to H , and H (is) to M . And M (is) equal to A . Thus, as the unit E is to the number F , so F (is) to H , and H to A . And so, for the same (reasons), as the unit E (is) to the number G , so G (is) to L , and L to B . Thus, as many (numbers) as have fallen in between A and B in continued proportion, so many numbers have also fallen between each of A and B and the unit E in continued proportion. (Which is) the very thing it was required to show.

Proposition 10

If (some) numbers fall between each of two numbers and a unit in continued proportion then, as many (numbers) as fall between each of the (two numbers) and the unit in continued proportion, so many (numbers) will also fall in between the (two numbers) themselves in continued proportion. For let the numbers D, E and F, G fall between the numbers A and B (respectively) and the unit C in continued proportion. I say that, as many numbers as have fallen between each of A and B and the unit C in continued proportion, so many will also fall in between A and B in continued proportion.



Ὁ Δ γὰρ τὸν Ζ πολλαπλασιάσας τὸν Θ ποιεῖτω, ἑκάτερον δὲ τῶν Δ, Ζ τὸν Θ πολλαπλασιάσας ἑκάτερον τῶν Κ, Λ ποιεῖτω.

Καὶ ἐπεὶ ἐστὶν ὡς ἡ Γ μονὰς πρὸς τὸν Δ ἀριθμὸν, οὕτως ὁ Δ πρὸς τὸν Ε, ἰσάκις ἤ Γ μονὰς τὸν Δ ἀριθμὸν μετρεῖ καὶ ὁ Δ τὸν Ε. ἡ δὲ Γ μονὰς τὸν Δ ἀριθμὸν μετρεῖ κατὰ τὰς ἐν τῷ Δ μονάδας; καὶ ὁ Δ ἄρα ἀριθμὸς τὸν Ε μετρεῖ κατὰ τὰς ἐν τῷ Δ μονάδας; ὁ Δ ἄρα ἑαυτὸν πολλαπλασιάσας τὸν Ε πεποίηκεν. πάλιν, ἐπεὶ ἐστὶν ὡς ἡ Γ [μονὰς] πρὸς τὸν Δ ἀριθμὸν, οὕτως ὁ Ε πρὸς τὸν Α, ἰσάκις ἤ Γ μονὰς τὸν Δ ἀριθμὸν μετρεῖ καὶ ὁ Ε τὸν Α. ἡ δὲ Γ μονὰς τὸν Δ ἀριθμὸν μετρεῖ κατὰ τὰς ἐν τῷ Δ μονάδας; καὶ ὁ Ε ἄρα τὸν Α μετρεῖ κατὰ τὰς ἐν τῷ Δ μονάδας; ὁ Δ ἄρα τὸν Ε πολλαπλασιάσας τὸν Α πεποίηκεν. διὰ τὰ αὐτὰ δὴ καὶ ὁ μὲν Ζ ἑαυτὸν πολλαπλασιάσας τὸν Η πεποίηκεν, τὸν δὲ Η πολλαπλασιάσας τὸν Β πεποίηκεν. καὶ ἐπεὶ ὁ Δ ἑαυτὸν μὲν πολλαπλασιάσας τὸν Ε πεποίηκεν, τὸν δὲ Ζ πολλαπλασιάσας τὸν Θ πεποίηκεν, ἔστιν ἄρα ὡς ὁ Δ πρὸς τὸν Ζ, οὕτως ὁ Ε πρὸς τὸν Θ. διὰ τὰ αὐτὰ δὴ καὶ ὡς ὁ Δ πρὸς τὸν Ζ, οὕτως ὁ Θ πρὸς τὸν Η. καὶ ὡς ἄρα ὁ Ε πρὸς τὸν Θ, οὕτως ὁ Θ πρὸς τὸν Η. πάλιν, ἐπεὶ ὁ Δ ἑκάτερον τῶν Ε, Θ πολλαπλασιάσας ἑκάτερον τῶν Α, Κ πεποίηκεν, ἔστιν ἄρα ὡς ὁ Ε πρὸς τὸν Θ, οὕτως ὁ Α πρὸς τὸν Κ. ἄλλ' ὡς ὁ Ε πρὸς τὸν Θ, οὕτως ὁ Δ πρὸς τὸν Ζ; καὶ ὡς ἄρα ὁ Δ πρὸς τὸν Ζ, οὕτως ὁ Α πρὸς τὸν Κ. πάλιν, ἐπεὶ ἑκάτερος τῶν Δ, Ζ τὸν Θ πολλαπλασιάσας ἑκάτερον τῶν Κ, Λ πεποίηκεν, ἔστιν ἄρα ὡς ὁ Δ πρὸς τὸν Ζ, οὕτως ὁ Κ πρὸς τὸν Λ. ἄλλ' ὡς ὁ Δ πρὸς τὸν Ζ, οὕτως ὁ Α πρὸς τὸν Κ; καὶ ὡς ἄρα ὁ Α πρὸς τὸν Κ, οὕτως ὁ Κ πρὸς τὸν Λ. ἔτι ἐπεὶ ὁ Ζ ἑκάτερον τῶν Θ, Η πολλαπλασιάσας ἑκάτερον τῶν Λ, Β πεποίηκεν, ἔστιν ἄρα ὡς ὁ Θ πρὸς τὸν Η, οὕτως ὁ Λ πρὸς τὸν Β. ὡς δὲ ὁ Θ πρὸς τὸν Η, οὕτως ὁ Δ πρὸς τὸν Ζ; καὶ ὡς ἄρα ὁ Δ πρὸς τὸν Ζ, οὕτως ὁ Λ πρὸς τὸν Β. ἐδείκθη δὲ καὶ ὡς ὁ Δ πρὸς τὸν Ζ, οὕτως ὁ Α πρὸς τὸν Κ καὶ ὁ Κ πρὸς τὸν Λ; καὶ ὡς ἄρα ὁ Α πρὸς τὸν Κ, οὕτως ὁ Κ πρὸς τὸν Λ καὶ ὁ Λ πρὸς τὸν Β. οἱ Α, Κ, Λ, Β ἄρα κατὰ τὸ συνθεῖς ἐχθρὸς εἰσὶν ἀνάλογον. ὅσοι ἄρα ἑκατέρου τῶν Α, Β καὶ τῆς Γ μονάδος μεταχὺ κατὰ τὸ συνθεῖς ἀνάλογον ἐμπίπτουσιν ἀριθμοί, τοσοῦτοι καὶ εἰς τοῦς Α, Β μεταχὺ κατὰ τὸ συνθεῖς

For let D make H (by) multiplying F . And let D, F make K, L , respectively, by multiplying H .

As since as the unit C is to the number D , so D (is) to E , the unit C thus measures the number D as many times as D (measures) E [Def. 7.20]. And the unit C measures the number D according to the units in D . Thus, the number D also measures E according to the units in D . Thus, D has made E (by) multiplying itself. Again, since as the [unit] C is to the number D , so E (is) to A , the unit C thus measures the number D as many times as E (measures) A [Def. 7.20]. And the unit C measures the number D according to the units in D . Thus, E also measures A according to the units in D . Thus, D has made A (by) multiplying E . And so, for the same (reasons), F has made G (by) multiplying itself, and has made B (by) multiplying G . And since D has made E (by) multiplying itself, and has made H (by) multiplying F , thus as D is to F , so E (is) to H [Prop 7.17]. And so, for the same reasons, as D (is) to F , so H (is) to G [Prop. 7.18]. And thus as E (is) to H , so H (is) to G . Again, since D has made A, K (by) multiplying E, H , respectively, thus as E is to H , so A (is) to K [Prop 7.17]. But, as E (is) to H , so D (is) to F . And thus as D (is) to F , so A (is) to K . Again, since D, F have made K, L , respectively, (by) multiplying H , thus as D is to F , so K (is) to L [Prop. 7.18]. But, as D (is) to F , so A (is) to K . And thus as A (is) to K , so K (is) to L . Further, since F has made L, B (by) multiplying H, G , respectively, thus as H is to G , so L (is) to B [Prop 7.17]. And as H (is) to G , so D (is) to F . And thus as D (is) to F , so L (is) to B . And it was also shown that as D (is) to F , so A (is) to K , and K to L . And thus as A (is) to K , so K (is) to L , and L to B . Thus, A, K, L, B are successively in continued proportion. Thus, as many numbers as fall between each of A and B and the unit C in continued proportion, so many will also fall in between A and B in continued proportion. (Which is) the very thing it was required to show.

ἐμπεσοῦνται; <όπερ >έδει δεῖχαι.

11.

Δύο τετραγώνων ἀριθμῶν ε<ίξ μέσοξ ἀνάλογόν ἐστιν ἀριθμός, καὶ ὁ τετράγωνος πρὸς τὸν τετράγωνον διπλασίονα λόγον >έθει >ήπερ ἡ πλευρὰ πρὸς τὴν πλευράν.

A —————

B —————

Γ ————— Δ —————

E —————

>Ἐστῶσαν τετράγωνοι ἀριθμοὶ οἱ A, B, καὶ τοῦ μὲν A πλευρὰ >έστω ὁ Γ, τοῦ δὲ B ὁ Δ; λέγω, <ὅτι τῶν A, B ε<ίξ μέσοξ ἀνάλογόν ἐστιν ἀριθμός, καὶ ὁ A πρὸς τὸν B διπλασίονα λόγον >έθει >ήπερ ὁ Γ πρὸς τὸν Δ.

Ὁ Γ γὰρ τὸν Δ πολλαπλασιάσας τὸν E ποιεῖτω. καὶ ἐπεὶ τετράγωνός ἐστιν ὁ A, πλευρὰ δὲ αὐτοῦ ἐστιν ὁ Γ, ὁ Γ >άρα ἑαυτὸν πολλαπλασιάσας τὸν A πεποίηκεν. διὰ τὰ αὐτὰ δὴ καὶ ὁ Δ ἑαυτὸν πολλαπλασιάσας τὸν B πεποίηκεν. ἐπεὶ ο>ὖν ὁ Γ ἐκάτερον τῶν Γ, Δ πολλαπλασιάσας ἐκάτερον τῶν A, E πεποίηκεν, >έστιν >άρα ὡς ὁ Γ πρὸς τὸν Δ, ο<ύτως ὁ A πρὸς τὸν E. διὰ τὰ αὐτὰ δὴ καὶ ὡς ὁ Γ πρὸς τὸν Δ, ο<ύτως ὁ E πρὸς τὸν B. καὶ ὡς >άρα ὁ A πρὸς τὸν E, ο<ύτως ὁ E πρὸς τὸν B. τῶν A, B >άρα ε<ίξ μέσοξ ἀνάλογόν ἐστιν ἀριθμός. Λέγω δὴ, <ὅτι καὶ ὁ A πρὸς τὸν B διπλασίονα λόγον >έθει >ήπερ ὁ Γ πρὸς τὸν Δ. ἐπεὶ γὰρ τρεῖς ἀριθμοὶ ἀνάλογόν εἰσιν οἱ A, E, B, ὁ A >άρα πρὸς τὸν B διπλασίονα λόγον >έθει >ήπερ ὁ A πρὸς τὸν E. ὡς δὲ ὁ A πρὸς τὸν E, ο<ύτως ὁ Γ πρὸς τὸν Δ. ὁ A >άρα πρὸς τὸν B διπλασίονα λόγον >έθει >ήπερ ἡ Γ πλευρὰ πρὸς τὴν Δ; <όπερ >έδει δεῖχαι.

Proposition 11

There exists one number in mean proportion to two (given) square numbers.[†] And (one) square (number) has to the (other) square (number) a squared[‡] ratio with respect to (that) the side (of the former has) to the side (of the latter).

A —————

B —————

C ————— D —————

E —————

Let A and B be square numbers, and let C be the side of A, and D (the side) of B. I say that there exists one number in mean proportion to A and B, and that A has to B a squared ratio with respect to (that) C (has) to D.

For let C make E (by) multiplying D. And since A is square, and C is its side, C has thus made A (by) multiplying itself. And so, for the same (reasons), D has made B (by) multiplying itself. Therefore, since C has made A, E (by) multiplying C, D, respectively, thus as C is to D, so A (is) to E [Prop. 7.17]. And so, for the same (reasons), as C (is) to D, so E (is) to B [Prop. 7.18]. And thus as A (is) to E, so E (is) to B. Thus, one number (namely, E) is in mean proportion to A and B.

So I say that A also has to B a squared ratio with respect to (that) C (has) to D. For since A, E, B are three (continuously) proportional numbers, A thus has to B a squared ratio with respect to (that) A (has) to E [Def. 5.9]. And as A (is) to E, so C (is) to D. Thus, A has to B a squared ratio with respect to (that) side C (has) to (side) D. (Which is) the very thing it was required to show.

[†] In other words, between two given square numbers there exists a number in continued proportion.

[‡] Literally, “double”.

12.

Δύο κύβων ἀριθμῶν δύο μέσοι ἀνάλογόν εἰσιν ἀριθμοί, καὶ ὁ κύβος πρὸς τὸν κύβον τριπλασίονα λόγον >έθει >ήπερ ἡ πλευρὰ πρὸς τὴν πλευράν.

>Ἐστῶσαν κύβοι ἀριθμοὶ οἱ A, B καὶ τοῦ μὲν A πλευρὰ >έστω ὁ Γ, τοῦ δὲ B ὁ Δ; λέγω, <ὅτι τῶν A, B δύο μέσοι ἀνάλογόν εἰσιν ἀριθμοί, καὶ ὁ A πρὸς τὸν B τριπλασίονα λόγον >έθει >ήπερ ὁ Γ πρὸς τὸν Δ.

Proposition 12

There exist two numbers in mean proportion to two (given) cube numbers.[†] And (one) cube (number) has to the (other) cube (number) a cubed[‡] ratio with respect to (that) the side (of the former has) to the side (of the latter).

Let A and B be cube numbers, and let C be the side of A, and D (the side) of B. I say that there exist two numbers in mean proportion to A and B, and that A has to B a cubed ratio with respect to (that) C (has) to D.

Ὁ γὰρ Γ ἑαυτὸν μὲν πολλαπλασιάσας τὸν Ε ποιεῖτω, τὸν δὲ Δ πολλαπλασιάσας τὸν Ζ ποιεῖτω, ὁ δὲ Δ ἑαυτὸν πολλαπλασιάσας τὸν Η ποιεῖτω, ἑκάτερος δὲ τῶν Γ, Δ τὸν Ζ πολλαπλασιάσας ἑκάτερον τῶν Θ, Κ ποιεῖτω.

Καὶ ἐπεὶ κύβος ἐστὶν ὁ Α, πλευρὰ δὲ αὐτοῦ ὁ Γ, καὶ ὁ Γ ἑαυτὸν μὲν πολλαπλασιάσας τὸν Ε πεποίηκεν, ὁ Γ >άρα ἑαυτὸν μὲν πολλαπλασιάσας τὸν Ε πεποίηκεν, τὸν δὲ Ε πολλαπλασιάσας τὸν Α πεποίηκεν. διὰ τὰ αὐτὰ δὴ καὶ ὁ Δ ἑαυτὸν μὲν πολλαπλασιάσας τὸν Η πεποίηκεν, τὸν δὲ Η πολλαπλασιάσας τὸν Β πεποίηκεν. καὶ ἐπεὶ ὁ Γ ἑκάτερον τῶν Γ, Δ πολλαπλασιάσας ἑκάτερον τῶν Ε, Ζ πεποίηκεν, >έστιν >άρα ὡς ὁ Γ πρὸς τὸν Δ, ο<ύτως ὁ Ε πρὸς τὸν Ζ. διὰ τὰ αὐτὰ δὴ καὶ ὡς ὁ Γ πρὸς τὸν Δ, ο<ύτως ὁ Ζ πρὸς τὸν Η. πάλιν, ἐπεὶ ὁ Γ ἑκάτερον τῶν Ε, Ζ πολλαπλασιάσας ἑκάτερον τῶν Α, Θ πεποίηκεν, >έστιν >άρα ὡς ὁ Ε πρὸς τὸν Ζ, ο<ύτως ὁ Α πρὸς τὸν Θ. ὡς δὲ ὁ Ε πρὸς τὸν Ζ, ο<ύτως ὁ Γ πρὸς τὸν Δ; καὶ ὡς >άρα ὁ Γ πρὸς τὸν Δ, ο<ύτως ὁ Α πρὸς τὸν Θ. πάλιν, ἐπεὶ ἑκάτερος τῶν Γ, Δ τὸν Ζ πολλαπλασιάσας ἑκάτερον τῶν Θ, Κ πεποίηκεν, >έστιν >άρα ὡς ὁ Γ πρὸς τὸν Δ, ο<ύτως ὁ Θ πρὸς τὸν Κ. πάλιν, ἐπεὶ ὁ Δ ἑκάτερον τῶν Ζ, Η πολλαπλασιάσας ἑκάτερον τῶν Κ, Β πεποίηκεν, >έστιν >άρα ὡς ὁ Ζ πρὸς τὸν Η, ο<ύτως ὁ Κ πρὸς τὸν Β. ὡς δὲ ὁ Ζ πρὸς τὸν Η, ο<ύτως ὁ Γ πρὸς τὸν Δ; καὶ ὡς >άρα ὁ Γ πρὸς τὸν Δ, ο<ύτως <ὁ τε Α πρὸς τὸν Θ καὶ ὁ Θ πρὸς τὸν Κ καὶ ὁ Κ πρὸς τὸν Β. τῶν Α, Β >άρα δύο μέσοι ἀνάλογόν εἰσιν οἱ Θ, Κ.

Λέγω δὴ, <ὅτι καὶ ὁ Α πρὸς τὸν Β τριπλασίονα λόγον >έθει >ἤπερ ὁ Γ πρὸς τὸν Δ. ἐπεὶ γὰρ τέσσαρες ἀριθμοὶ ἀνάλογόν εἰσιν οἱ Α, Θ, Κ, Β, ὁ Α >άρα πρὸς τὸν Β τριπλασίονα λόγον >έθει >ἤπερ ὁ Α πρὸς τὸν Θ. ὡς δὲ ὁ Α πρὸς τὸν Θ, ο<ύτως ὁ Γ πρὸς τὸν Δ; καὶ ὁ Α [>άρα] πρὸς τὸν Β τριπλασίονα λόγον >έθει >ἤπερ ὁ Γ πρὸς τὸν Δ; <ὅπερ >έδει δεῖχαι.

† In other words, between two given cube numbers there exist two numbers in continued proportion.

‡ Literally, “triple”.

13.

Ἐὰν >ῶσιν ὁσοῖδηποτοῦν ἀριθμοὶ ἐχῇ ἀνάλογον, καὶ πολλαπλασιάσας <έκαστος ἑαυτὸν ποιῇ| τινα, οἱ γενόμενοι ἐχ αὐτῶν ἀνάλογον >έσονται; καὶ ἔαν οἱ ἐχ ἀρθῇ τούξ γενομένουξ πολλαπλασιάσαντεξ ποιῶσί τιναξ, καὶ αὐτοὶ ἀνάλογον >έσονται [καὶ αἰεὶ περὶ τούξ >άκρουξ τοῦτο συμβαίνει].

>Ἔστωσαν ὅποσοιοῦν ἀριθμοὶ ἐχῇ ἀνάλογον, οἱ Α, Β, Γ, ὡς ὁ Α πρὸς τὸν Β, ο<ύτως ὁ Β πρὸς τὸν Γ, καὶ οἱ Α, Β, Γ ἑαυτούξ μὲν πολλαπλασιάσαντεξ τούξ Δ, Ε, Ζ ποιεῖτωσαν, τούξ δὲ Δ, Ε, Ζ πολλαπλασιάσαντεξ τούξ Η, Η,

For let *C* make *E* (by) multiplying itself, and let it make *F* (by) multiplying *D*. And let *D* make *G* (by) multiplying itself, and let *C*, *D* make *H*, *K*, respectively, (by) multiplying *F*.

And since *A* is cube, and *C* (is) its side, and *C* has made *E* (by) multiplying itself, *C* has thus made *E* (by) multiplying itself, and has made *A* (by) multiplying *E*. And so, for the same (reasons), *D* has made *G* (by) multiplying itself, and has made *B* (by) multiplying *G*. And since *C* has made *E*, *F* (by) multiplying *C*, *D*, respectively, thus as *C* is to *D*, so *E* (is) to *F* [Prop. 7.17]. And so, for the same (reasons), as *C* (is) to *D*, so *F* (is) to *G* [Prop. 7.18]. Again, since *C* has made *A*, *H* (by) multiplying *E*, *F*, respectively, thus as *E* is to *F*, so *A* (is) to *H* [Prop. 7.17]. And as *E* (is) to *F*, so *C* (is) to *D*. And thus as *C* (is) to *D*, so *A* (is) to *H*. Again, since *C*, *D* have made *H*, *K*, respectively, (by) multiplying *F*, thus as *C* is to *D*, so *H* (is) to *K* [Prop. 7.18]. Again, since *D* has made *K*, *B* (by) multiplying *F*, *G*, respectively, thus as *F* is to *G*, so *K* (is) to *B* [Prop. 7.17]. And as *F* (is) to *G*, so *C* (is) to *D*. And thus as *C* (is) to *D*, so *A* (is) to *H*, and *H* to *K*, and *K* to *B*. Thus, *H* and *K* are two (numbers) in mean proportion to *A* and *B*.

So I say that *A* also has to *B* a cubed ratio with respect to (that) *C* (has) to *D*. For since *A*, *H*, *K*, *B* are four (continuously) proportional numbers, *A* thus has to *B* a cubed ratio with respect to (that) *A* (has) to *H* [Def. 5.10]. And as *A* (is) to *H*, so *C* (is) to *D*. And [thus] *A* has to *B* a cubed ratio with respect to (that) *C* (has) to *D*. (Which is) the very thing it was required to show.

Proposition 13

If there are any multitude whatsoever of continuously proportional numbers, and each makes some (number by) multiplying itself, then the (numbers) created from them will (also) be (continuously) proportional. And if the original (numbers) make some (more numbers by) multiplying the created (numbers) then these will also be (continuously) proportional [and this always happens with the extremes]. Let *A*, *B*, *C* be any multitude whatsoever of continuously proportional numbers, (such that) as *A* (is) to *B*, so *B* (is) to *C*. And let *A*, *B*, *C* make *D*, *E*, *F* (by) multiplying

Θ, Κ ποιείωσαν; λέγω, <ότι ο<ί τε Δ, Ε, Ζ καὶ οἱ Η, Θ, Κ
ἐχῆξ ἀνάλογον εἰσιν.

themselves, and let them make G, H, K (by) multiplying
 D, E, F . I say that D, E, F and G, H, K are continuously
proportional.

A ———|

L ———|

B ———|

O ———|

C ———|

M ———|

D ———|

N ———|

E ———|

P ———|

F ———|

Q ———|

G ———|

H ———|

K ———|

Λ ———|

Ξ ———|

M ———|

N ———|

O ———|

Π ———|

For let A make L (by) multiplying B . And let A, B make

Ὁ μὲν γὰρ A τὸν B πολλαπλασιάσας τὸν Λ ποιεῖτω, ἐκάτερος δὲ τῶν A, B τὸν Λ πολλαπλασιάσας ἐκάτερον τῶν M, N ποιεῖτω. καὶ πάλιν ὁ μὲν B τὸν Γ πολλαπλασιάσας τὸν X ποιεῖτω, ἐκάτερος δὲ τῶν B, Γ τὸν X πολλαπλασιάσας ἐκάτερον τῶν O, Π ποιεῖτω.

Ὁμοίως δὴ τοῖς ἐπάνω δεῖχομεν, <ὅτι οἱ Δ, Λ, E καὶ οἱ H, M, N, Θ ἐχῆζ εἰσιν ἀνάλογον ἐν τῷ τοῦ A πρὸς τὸν B λόγῳ, καὶ >εἰ οἱ E, X, Z καὶ οἱ Θ, O, Π, K ἐχῆζ εἰσιν ἀνάλογον ἐν τῷ τοῦ B πρὸς τὸν Γ λόγῳ. καὶ ἐστὶν ὥς ὁ A πρὸς τὸν B , ο<ύτως ὁ B πρὸς τὸν Γ ; καὶ οἱ Δ, Λ, E >άρα τοῖς E, X, Z ἐν τῷ αὐτῷ λόγῳ εἰσὶ καὶ >εἰ οἱ H, M, N, Θ τοῖς Θ, O, Π, K . καὶ ἐστὶν >ίσον τὸ μὲν τῶν Δ, Λ, E πλήθος τῷ τῶν E, X, Z πλήθει, τὸ δὲ τῶν H, M, N, Θ τῷ τῶν Θ, O, Π, K ; δι> >ίσου >άρα ἐστὶν ὥς μὲν ὁ Δ πρὸς τὸν E , ο<ύτως ὁ E πρὸς τὸν Z , ὥς δὲ ὁ H πρὸς τὸν Θ , ο<ύτως ὁ Θ πρὸς τὸν K ; <όπερ >έδει δεῖχαι.

14.

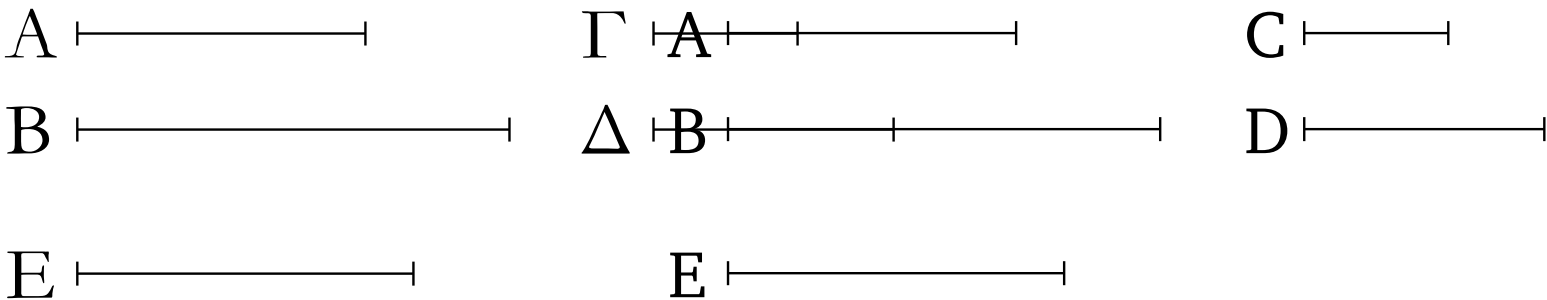
Ἐὰν τετράγωνος τετράγωνον μετρήῃ, καὶ ἡ πλευρὰ τὴν πλευρὰν μετρήσῃ; καὶ ἔαν ἡ πλευρὰ τὴν πλευρὰν μετρήῃ, καὶ ὁ τετράγωνος τὸν τετράγωνον μετρήσῃ.

>Εστώσαν τετράγωνοι ἀριθμοὶ οἱ A, B , πλευραὶ δὲ αὐτῶν >έστωσαν οἱ Γ, Δ , ὁ δὲ A τὸν B μετρεῖτω; λέγω, <ὅτι καὶ ὁ Γ τὸν Δ μετρεῖ.

Proposition 14

If a square (number) measures a(nother) square (number) then the side (of the former) will also measure the side (of the latter). And if the side (of a square number) measures the side (of another square number) then the (former) square (number) will also measure the (latter) square (number).

Let A and B be square numbers, and let C and D be their sides (respectively). And let A measure B . I say that C also measures D .



Ὁ Γ γὰρ τὸν Δ πολλαπλασιάσας τὸν E ποιεῖτω; οἱ A, E, B >άρα ἐχῆζ ἀνάλογον εἰσιν ἐν τῷ τοῦ Γ πρὸς τὸν Δ λόγῳ. καὶ ἐπεὶ οἱ A, E, B ἐχῆζ ἀνάλογον εἰσιν, καὶ μετρεῖ ὁ A τὸν B , μετρεῖ >άρα καὶ ὁ A τὸν E . καὶ ἐστὶν ὥς ὁ A πρὸς τὸν E , ο<ύτως ὁ Γ πρὸς τὸν Δ ; μετρεῖ >άρα καὶ ὁ Γ τὸν Δ .

Πάλιν δὴ ὁ Γ τὸν Δ μετρεῖτω; λέγω, <ὅτι καὶ ὁ A τὸν B μετρεῖ.

Τῶν γὰρ αὐτῶν κατασκευασθέντων ὁμοίως δεῖχομεν, <ὅτι οἱ A, E, B ἐχῆζ ἀνάλογον εἰσιν ἐν τῷ τοῦ Γ πρὸς τὸν Δ λόγῳ. καὶ ἐπεὶ ἐστὶν ὥς ὁ Γ πρὸς τὸν Δ , ο<ύτως ὁ A πρὸς τὸν E , μετρεῖ δὲ ὁ Γ τὸν Δ , μετρεῖ >άρα καὶ ὁ A τὸν E . καὶ εἰσιν οἱ A, E, B ἐχῆζ ἀνάλογον; μετρεῖ >άρα καὶ ὁ A τὸν B . Ἐὰν >άρα τετράγωνος τετράγωνον μετρήῃ, καὶ ἡ πλευρὰ τὴν πλευρὰν μετρήσῃ; καὶ ἔαν ἡ πλευρὰ τὴν πλευρὰν μετρήῃ, καὶ ὁ τετράγωνος τὸν τετράγωνον μετρήσῃ; <όπερ >έδει δεῖχαι.

15.

Ἐὰν κύβοις ἀριθμοὶς κύβον ἀριθμὸν μετρήῃ, καὶ ἡ πλευρὰ τὴν πλευρὰν μετρήσῃ; καὶ ἔαν ἡ πλευρὰ τὴν πλευρὰν μετρήῃ, καὶ ὁ κύβοις τὸν κύβον μετρήσῃ.

Κύβοις γὰρ ἀριθμοὶς ὁ A κύβον τὸν B μετρεῖτω, καὶ τοῦ μὲν A πλευρὰ >έστω ὁ Γ , τοῦ δὲ B ὁ Δ ; λέγω, <ὅτι ὁ Γ τὸν Δ

For let C make E (by) multiplying D . Thus, A, E, B are continuously proportional in the ratio of C to D [Prop. 8.11]. And since A, E, B are continuously proportional, and A measures B , A thus also measures E [Prop. 8.7]. And as A is to E , so C (is) to D . Thus, C also measures D [Def. 7.20]. So, again, let C measure D . I say that A also measures B . For similarly, with the same construction, we can show that A, E, B are continuously proportional in the ratio of C to D . And since as C is to D , so A (is) to E , and C measures D , A thus also measures E [Def. 7.20]. And A, E, B are continuously proportional. Thus, A also measures B .

Thus, if a square (number) measures a(nother) square (number) then the side (of the former) will also measure the side (of the latter). And if the side (of a square number) measures the side (of another square number) then the (former) square (number) will also measure the (latter) square (number). (Which is) the very thing it was required to show.

Proposition 15

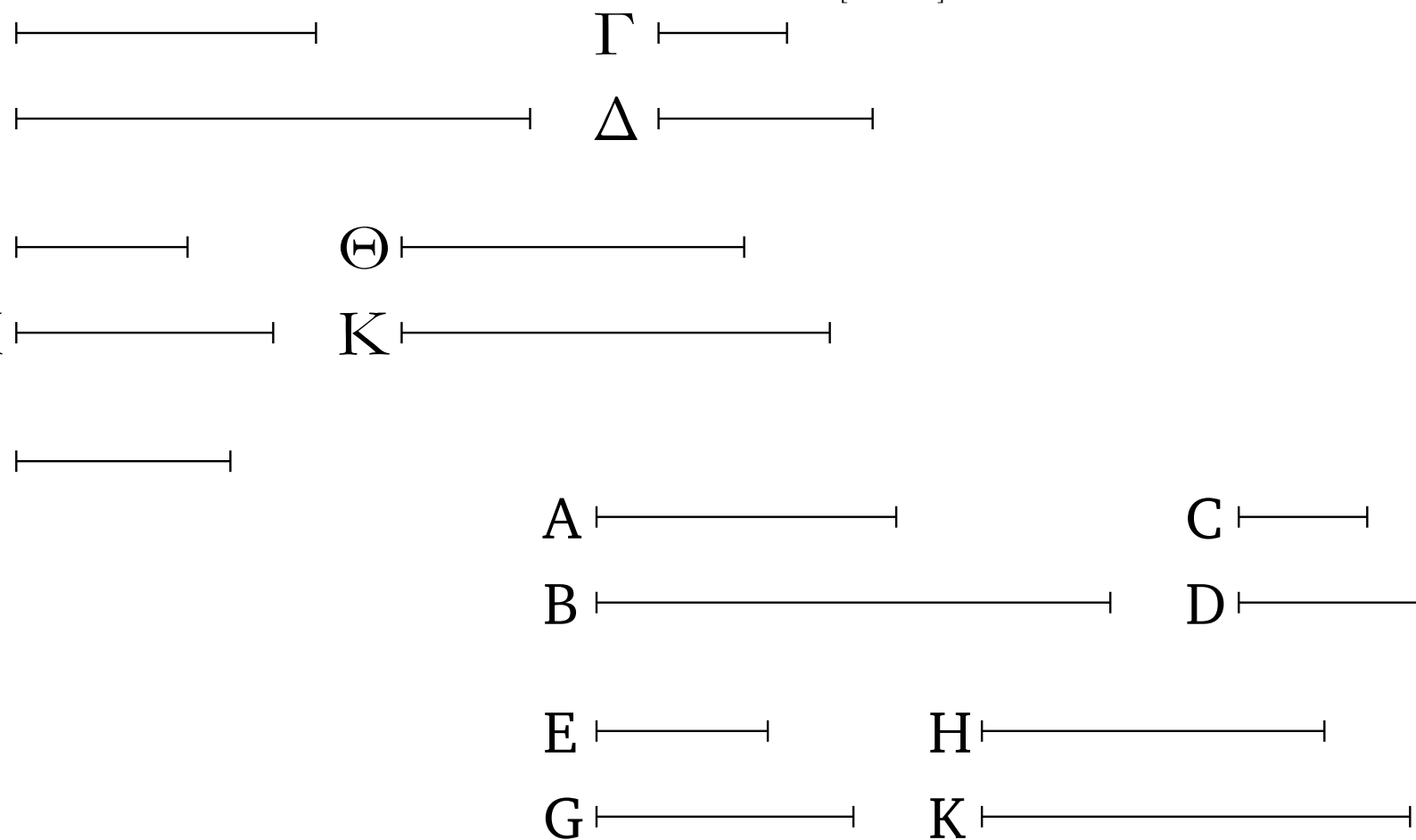
If a cube number measures a(nother) cube number then the side (of the former) will also measure the side (of the latter). And if the side (of a cube number) measures the side (of another cube number) then the (former) cube (number) will also measure the (latter) cube (number).

μετρεῖ.

Ὁ Γ γὰρ ἐαυτὸν πολλαπλασιάσας τὸν Ε ποιεῖω, ὁ δὲ Δ ἐαυτὸν πολλαπλασιάσας τὸν Η ποιεῖω, καὶ >εἶ ὁ Γ τὸν Δ πολλαπλασιάσας τὸν Ζ [ποιεῖω], ἐκάτερος δὲ τῶν Γ, Δ τὸν Ζ πολλαπλασιάσας ἐκάτερον τῶν Θ, Κ ποιεῖω. φανερόν δὴ, <ὅτι οἱ Ε, Ζ, Η καὶ οἱ Α, Θ, Κ, Β ἐχῆξ ἀνάλογόν εἰσιν ἐν τῷ| τοῦ Γ πρὸς τὸν Δ λόγῳ. καὶ ἐπεὶ οἱ Α, Θ, Κ, Β ἐχῆξ ἀνάλογόν εἰσιν, καὶ μετρεῖ ὁ Α τὸν Β, μετρεῖ >άρα καὶ τὸν Θ. καὶ ἐστὶν ὥς ὁ Α πρὸς τὸν Θ, ο<ύτως ὁ Γ πρὸς τὸν Δ; μετρεῖ >άρα καὶ ὁ Γ τὸν Δ.

For let the cube number A measure the cube (number) B , and let C be the side of A , and D (the side) of B . I say that C measures D .

For let C make E (by) multiplying itself. And let D make G (by) multiplying itself. And, further, [let] C [make] F (by) multiplying D , and let C, D make H, K , respectively, (by) multiplying F . So it is clear that E, F, G and A, H, K, B are continuously proportional in the ratio of C to D [Prop. 8.12]. And since A, H, K, B are continuously proportional, and A measures B , (A) thus also measures H [Prop. 8.7]. And as A is to H , so C (is) to D . Thus, C also measures D [Def. 7.20].



Ἀλλὰ δὴ μετρεῖτω ὁ Γ τὸν Δ; λέγω, <ὅτι καὶ ὁ Α τὸν Β μετρήσει.

Τὼν γὰρ αὐτῶν κατασκευασθέντων ὁμοίως δὴ δείχομεν,
 <ὅτι οἱ A, Θ, K, B ἐχθρὴ ἀνάλογόν εἰσιν ἐν τῷ τοῦ Γ πρὸς
 τὸν Δ λόγῳ. καὶ ἐπεὶ ὁ Γ τὸν Δ μετρεῖ, καὶ ἐστὶν ὥς ὁ Γ πρὸς
 τὸν Δ, οὕτως ὁ A πρὸς τὸν Θ, καὶ ὁ A > ἄρα τὸν Θ μετρεῖ;
 <ὥστε καὶ τὸν B μετρεῖ ὁ A; <ὅπερ >έδει δεῖλαι.

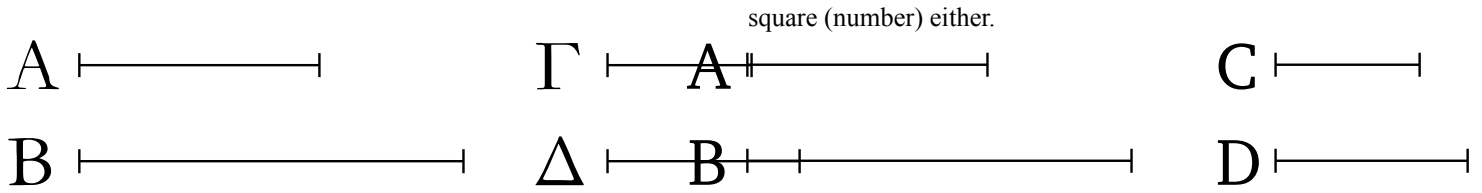
And so let C measure D . I say that A will also measure B . For similarly, with the same construction, we can show that A, H, K, B are continuously proportional in the ratio of C to D . And since C measures D , and as C is to D , so A (is) to H , A thus also measures H [Def. 7.20]. Hence, A also measures B . (Which is) the very thing it was required to show.

16.

Ἐὰν τετράγωνοξ ἀριθμὸξ τετράγωνον ἀριθμὸν μὴ μετρήῃ, οὐδὲ ἡ πλευρὰ τὴν πλευρὰν μετρήσει; κ>ὰν ἡ πλευρὰ τὴν πλευρὰν μὴ μετρήῃ, οὐδὲ ὁ τετράγωνοξ τὸν τετράγωνον μετρήσει.

Proposition 16

If a square number does not measure a(nother) square number then the side (of the former) will not measure the side (of the latter) either. And if the side (of a square number) does not measure the side (of another square number) then the (former) square (number) will not measure the (latter)



>Ἐστωσαν τετράγωνοι ἀριθμοὶ οἱ A, B , πλευραὶ δὲ αὐτῶν >ἔστωσαν οἱ Γ, Δ , καὶ μὴ μετρεῖτω ὁ A τὸν B ; λέγω, <ὅτι οὐδὲ ὁ Γ τὸν Δ μετρεῖ.
 Εἰ γὰρ μετρεῖ ὁ Γ τὸν Δ , μετρήσει καὶ ὁ A τὸν B . οὐ μετρεῖ δὲ ὁ A τὸν B ; οὐδὲ >ἀρα ὁ Γ τὸν Δ μετρήσει.
 μὴ μετρεῖτω [δὴ] πάλιν ὁ Γ τὸν Δ ; λέγω, <ὅτι οὐδὲ ὁ A τὸν B μετρήσει.
 Εἰ γὰρ μετρεῖ ὁ A τὸν B , μετρήσει καὶ ὁ Γ τὸν Δ . οὐ μετρεῖ δὲ ὁ Γ τὸν Δ ; οὐδὲ >ἀρα ὁ A τὸν B μετρήσει; <ὅπερ >ἔδει δεῖξαι.

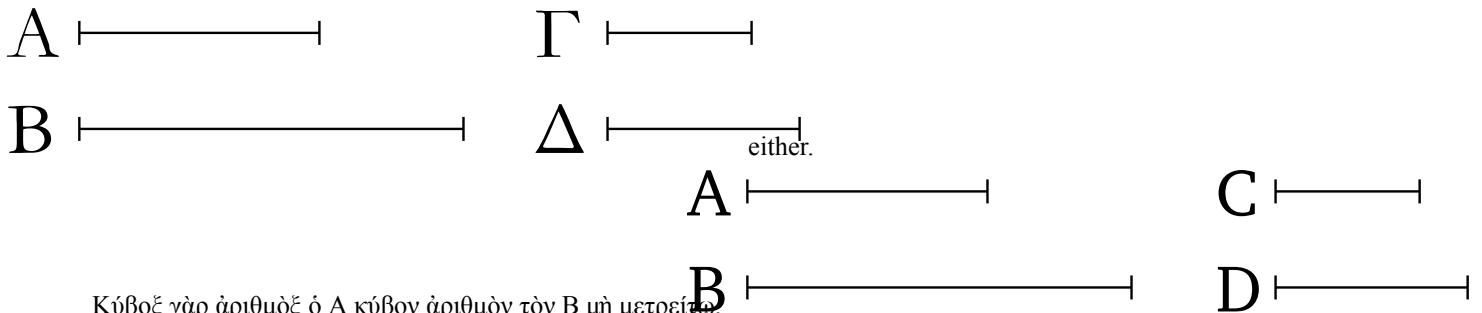
Let A and B be square numbers, and let C and D be their sides (respectively). And let A not measure B . I say that C does not measure D either.
 For if C measures D then A will also measure B [Prop. 8.14]. And A does not measure B . Thus, C will not measure D either.
 [So], again, let C not measure D . I say that A will not measure B either.
 For if A measures B then C will also measure D [Prop. 8.14]. And C does not measure D . Thus, A will not measure B either. (Which is) the very thing it was required to show.

17.

Ἐὰν κύβοξ ἀριθμὸξ κύβον ἀριθμὸν μὴ μετρήῃ, οὐδὲ ἡ πλευρὰ τὴν πλευρὰν μετρήσει; <ὅταν ἡ πλευρὰ τὴν πλευρὰν μὴ μετρήῃ, οὐδὲ ὁ κύβοξ τὸν κύβον μετρήσει.

Proposition 17

If a cube number does not measure a(nother) cube number then the side (of the former) will not measure the side (of the latter) either. And if the side (of a cube number) does not measure the side (of another cube number) then the (former) cube (number) will not measure the (latter) cube (number)



Κύβοξ γὰρ ἀριθμὸξ ὁ A κύβον ἀριθμὸν τὸν B μὴ μετρεῖτω, καὶ τοῦ μὲν A πλευρὰ >ἔστω ὁ Γ , τοῦ δὲ B ὁ Δ ; λέγω, <ὅτι ὁ Γ τὸν Δ οὐ μετρήσει.
 Εἰ γὰρ μετρεῖ ὁ Γ τὸν Δ , καὶ ὁ A τὸν B μετρήσει. οὐ μετρεῖ δὲ ὁ A τὸν B ; οὐδὲ >ἀρα ὁ Γ τὸν Δ μετρεῖ.
 Ἀλλὰ δὴ μὴ μετρεῖτω ὁ Γ τὸν Δ ; λέγω, <ὅτι οὐδὲ ὁ A τὸν B μετρήσει.
 Εἰ γὰρ ὁ A τὸν B μετρεῖ, καὶ ὁ Γ τὸν Δ μετρήσει. οὐ μετρεῖ δὲ ὁ Γ τὸν Δ ; οὐδὲ >ἀρα ὁ A τὸν B μετρήσει; <ὅπερ >ἔδει δεῖξαι.

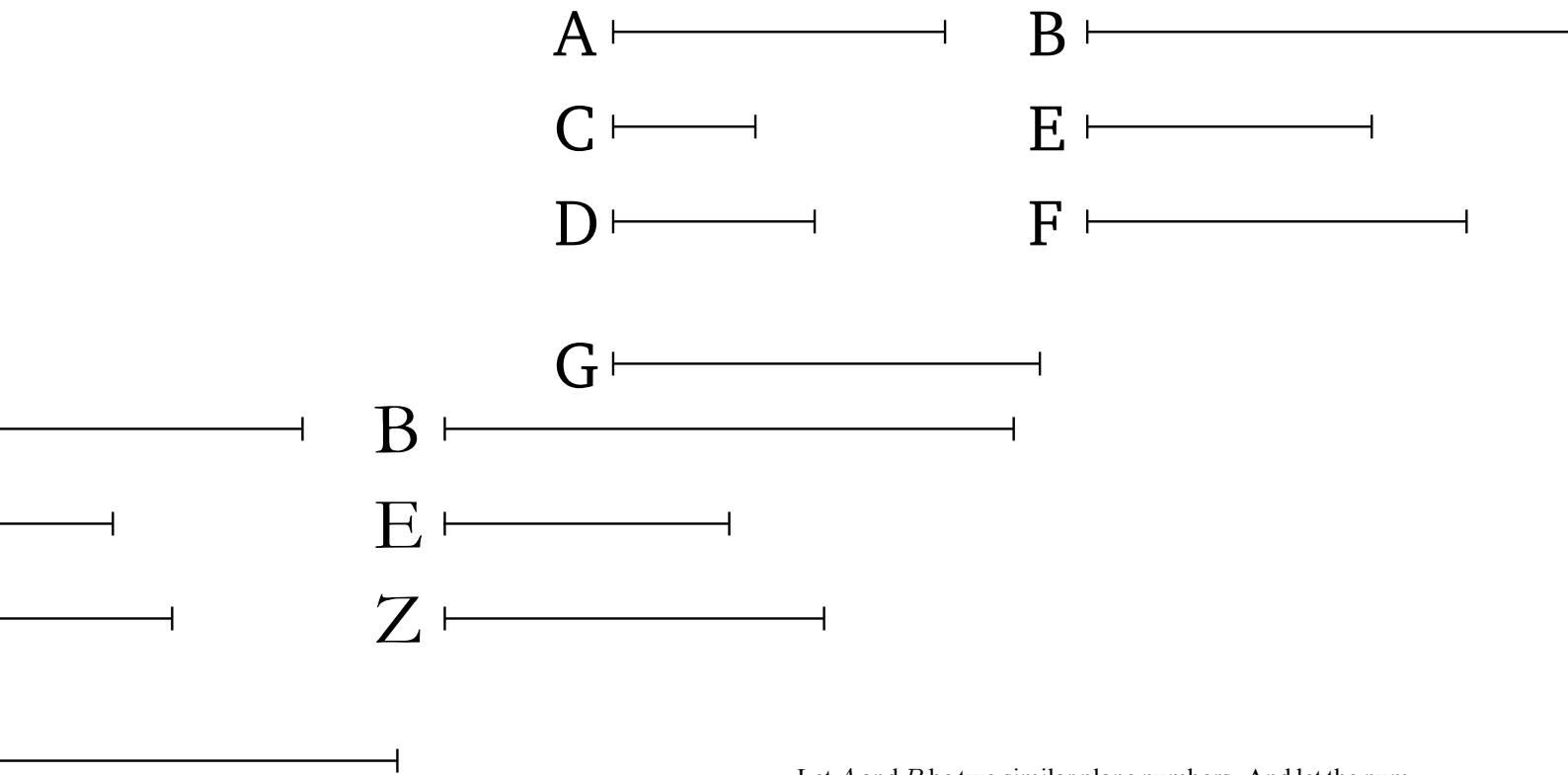
For let the cube number A not measure the cube number B . And let C be the side of A , and D (the side) of B . I say that C will not measure D .
 For if C measures D then A will also measure B [Prop. 8.15]. And A does not measure B . Thus, C does not measure D either.
 And so let C not measure D . I say that A will not measure B either.
 For if A measures B then C will also measure D [Prop. 8.15]. And C does not measure D . Thus, A will not measure B either. (Which is) the very thing it was required to show.

18.

Δύο ὁμοίων ἐπιπέδων ἀριθμῶν ε<ῖξ μέσοξ ἀνάλογόν ἐστιν ἀριθμὸξ; καὶ ὁ ἐπίπεδοξ πρὸξ τὸν ἐπίπεδον διπλασίονα λόγον >έθει >ἤπερ ἡ ὁμόλογος πλευρὰ πρὸξ τὴν ὁμόλογον πλευρὰν.

Proposition 18

There exists one number in mean proportion to two similar plane numbers. And (one) plane (number) has to the (other) plane (number) a squared[†] ratio with respect to (that) a corresponding side (of the former has) to a corresponding side (of the latter).



>Ἐστῶσαν δύο <όμοιοι ἐπίπεδοι ἀριθμοὶ οἱ A, B, καὶ τοῦ μὲν A πλευραὶ >ἕστωσαν οἱ Γ, Δ ἀριθμοί, τοῦ δὲ B οἱ E, Z. καὶ ἐπεὶ <όμοιοι ἐπίπεδοι εἰσιν οἱ ἀνάλογον >έθοντεξ τὰς πλευράς, >έστιν >άρα ὡς ὁ Γ πρὸς τὸν Δ, ο<ύτωξ ὁ E πρὸς τὸν Z. λέγω ο>ὖν, <ότι τῶν A, B ε<ῖξ μέσοξ ἀνάλογόν ἐστιν ἀριθμός, καὶ ὁ A πρὸς τὸν B διπλασίονα λόγον >έθει >ήπερ ὁ Γ πρὸς τὸν E >ή ὁ Δ πρὸς τὸν Z, τουτέστιν >ήπερ ἡ ὁμόλογοξ πλευρὰ πρὸς τὴν ὁμόλογον [πλευράν].

Καὶ ἐπεὶ ἐστιν ὡς ὁ Γ πρὸς τὸν Δ, ο<ύτωξ ὁ E πρὸς τὸν Z, ἐναλλάξ >άρα ἐστιν ὡς ὁ Γ πρὸς τὸν E, ὁ Δ πρὸς τὸν Z. καὶ ἐπεὶ ἐπίπεδόξ ἐστιν ὁ A, πλευραὶ δὲ αὐτοῦ οἱ Γ, Δ, ὁ Δ >άρα τὸν Γ πολλαπλασιάσαξ τὸν A πεποίηκεν. διὰ τὰ αὐτὰ δὴ καὶ ὁ E τὸν Z πολλαπλασιάσαξ τὸν B πεποίηκεν. ὁ Δ δὴ τὸν E πολλαπλασιάσαξ τὸν H ποιεῖτω. καὶ ἐπεὶ ὁ Δ τὸν μὲν Γ πολλαπλασιάσαξ τὸν A πεποίηκεν, τὸν δὲ E πολλαπλασιάσαξ τὸν H πεποίηκεν, >έστιν >άρα ὡς ὁ Γ πρὸς τὸν E, ο<ύτωξ ὁ A πρὸς τὸν H. ἀλλ> ὡς ὁ Γ πρὸς τὸν E, [ο<ύτωξ] ὁ Δ πρὸς τὸν Z; καὶ ὡς >άρα ὁ Δ πρὸς τὸν Z, ο<ύτωξ ὁ A πρὸς τὸν H. πάλιν, ἐπεὶ ὁ E τὸν μὲν Δ πολλαπλασιάσαξ τὸν H πεποίηκεν, τὸν δὲ Z πολλαπλασιάσαξ τὸν B πεποίηκεν, >έστιν >άρα ὡς ὁ Δ πρὸς τὸν Z, ο<ύτωξ ὁ H πρὸς τὸν B. ἐδείθθη δὲ καὶ ὡς ὁ Δ πρὸς τὸν Z, ο<ύτωξ ὁ A πρὸς τὸν H; καὶ ὡς >άρα ὁ A πρὸς τὸν H, ο<ύτωξ ὁ H πρὸς τὸν B. οἱ A, H, B >άρα ἐχῖξ ἀνάλογόν εἰσιν. τῶν A, B >άρα ε<ῖξ μέσοξ ἀνάλογόν ἐστιν ἀριθμός.

Λέγω δὴ, <ότι καὶ ὁ A πρὸς τὸν B διπλασίονα λόγον >έθει >ήπερ ἡ ὁμόλογοξ πλευρὰ πρὸς τὴν ὁμόλογον πλευράν, τουτέστιν >ήπερ ὁ Γ πρὸς τὸν E >ή ὁ Δ πρὸς τὸν Z. ἐπεὶ γάρ οἱ A, H, B ἐχῖξ ἀνάλογόν εἰσιν, ὁ A πρὸς τὸν B διπλασίονα λόγον >έθει >ήπερ πρὸς τὸν H. καὶ ἐστιν ὡς ὁ A πρὸς τὸν H, ο<ύτωξ <ό τε Γ πρὸς τὸν E καὶ ὁ Δ πρὸς τὸν Z. καὶ ὁ A >άρα πρὸς τὸν B διπλασίονα λόγον >έθει >ήπερ ὁ Γ πρὸς τὸν E >ή ὁ Δ πρὸς τὸν Z; <όπερ >έδει δεῖχαι.

Let A and B be two similar plane numbers. And let the numbers C, D be the sides of A , and E, F (the sides) of B . And since similar numbers are those having proportional sides [Def. 7.21], thus as C is to D , so E (is) to F . Therefore, I say that there exists one number in mean proportion to A and B , and that A has to B a squared ratio with respect to that C (has) to E , or D to F —that is to say, with respect to (that) a corresponding side (has) to a corresponding [side]. For since as C is to D , so E (is) to F , thus, alternately, as C is to E , so D (is) to F [Prop. 7.13]. And since A is plane, and C, D its sides, D has thus made A (by) multiplying C . And so, for the same (reasons), E has made B (by) multiplying F . So let D make G (by) multiplying E . And since D has made A (by) multiplying C , and has made G (by) multiplying E , thus as C is to E , so A (is) to G [Prop. 7.17]. But as C (is) to E , [so] D (is) to F . And thus as D (is) to F , so A (is) to G . Again, since E has made G (by) multiplying D , and has made B (by) multiplying F , thus as D is to F , so G (is) to B [Prop. 7.17]. And it was also shown that as D (is) to F , so A (is) to G . And thus as A (is) to G , so G (is) to B . Thus, A, G, B are continuously proportional. Thus, there exists one number (namely, G) in mean proportion to A and B .

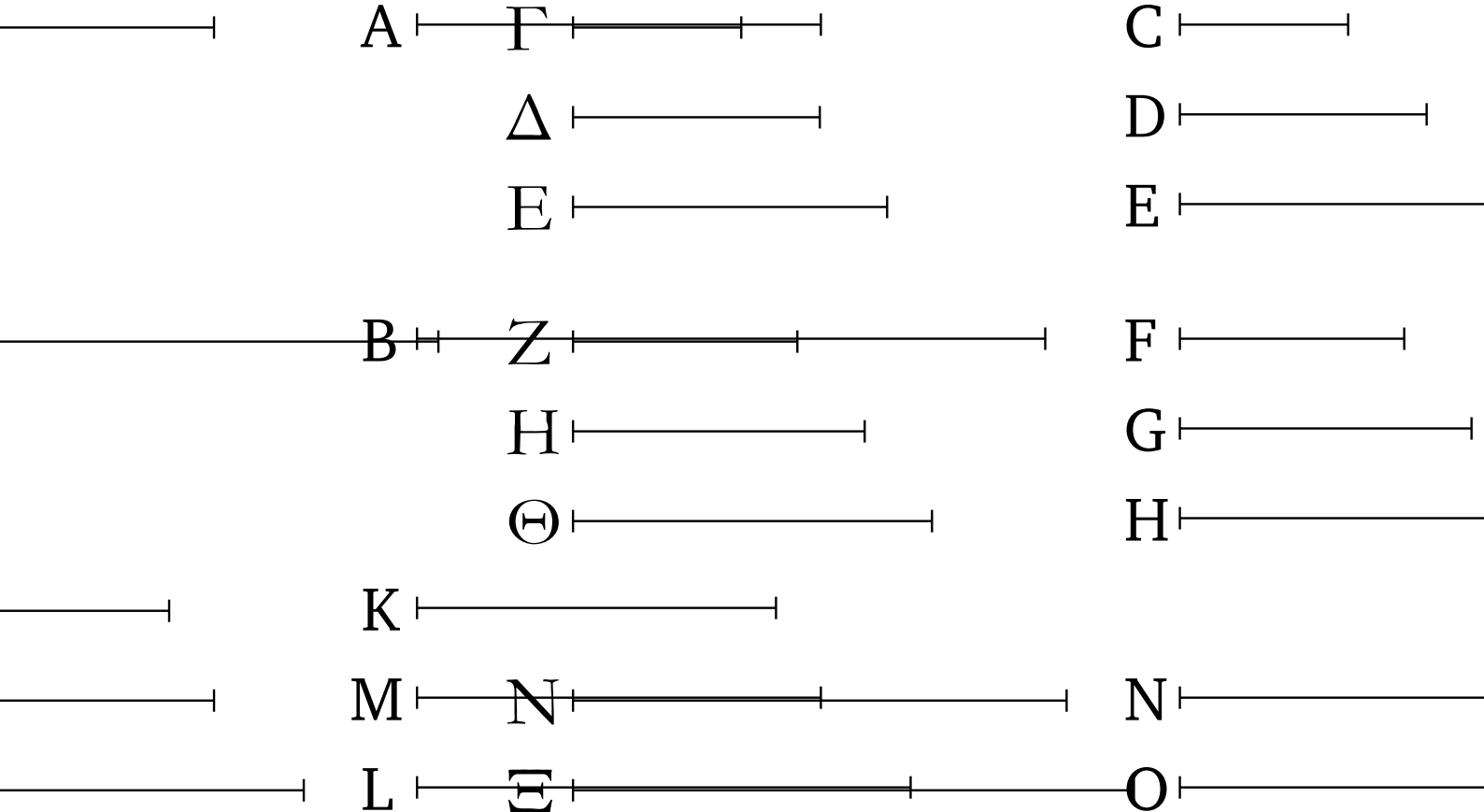
So I say that A also has to B a squared ratio with respect to (that) a corresponding side (has) to a corresponding side—that is to say, with respect to (that) C (has) to E , or D to F . For since A, G, B are continuously proportional, A has to B a squared ratio with respect to (that A has) to G [Prop. 5.9]. And as A is to G , so C (is) to E , and D to F . And thus A has to B a squared ratio with respect to (that) C (has) to E , or D to F . (Which is) the very thing it was required to show.

† Literally, “double”.

19.

Proposition 19

Δύο ὁμοίων στερεῶν ἀριθμῶν δύο μέσοι ἀνάλογον ἐμπίπτουσι Two numbers fall (between) two similar solid numbers in ἀριθμοί· καὶ ὁ στερεὸς πρὸς τὸν <ὁμοιον στερεὸν τριπλασίονα mean proportion. And a solid (number) has to a similar solid λόγον >έθει >ήπερ ἡ ὁμόλογος πλευρὰ πρὸς τὴν ὁμόλογον (number) a cubed† ratio with respect to (that) a corresponding πλευράν. side (has) to a corresponding side.



>Ἐστωσαν δύο <ὁμοιοι στερεοὶ οἱ A, B, καὶ τοῦ μὲν A πλευραὶ >έστωσαν οἱ Γ, Δ, Ε, τοῦ δὲ B οἱ Ζ, Η, Θ. καὶ ἐπεὶ <ὁμοιοι στερεοὶ εἰσιν οἱ ἀνάλογον >έθοντες τὰς πλευράς, >έστιν >άρα ὡς μὲν ὁ Γ πρὸς τὸν Δ, ο<ύτως ὁ Ζ πρὸς τὸν Η, ὡς δὲ ὁ Δ πρὸς τὸν Ε, ο<ύτως ὁ Η πρὸς τὸν Θ. λέγω, <ὅτι τῶν A, B δύο μέσοι ἀνάλογον ἐμπίπτουσιν ἀριθμοί, καὶ ὁ A πρὸς τὸν B τριπλασίονα λόγον >έθει >ήπερ ὁ Γ πρὸς τὸν Ζ καὶ ὁ Δ πρὸς τὸν Η καὶ >έτι ὁ Ε πρὸς τὸν Θ.

Ὁ Γ γὰρ τὸν Δ πολλαπλασιάσας τὸν Κ ποιεῖτω, ὁ δὲ Ζ τὸν Η πολλαπλασιάσας τὸν Λ ποιεῖτω. καὶ ἐπεὶ οἱ Γ, Δ τοῖς Ζ, Η ἐν τῷ αὐτῷ λόγῳ εἰσίν, καὶ ἐκ μὲν τῶν Γ, Δ ἔστιν ὁ Κ, ἐκ δὲ τῶν Ζ, Η ὁ Λ, οἱ Κ, Λ [>άρα] <ὁμοιοι ἐπίτεδοι εἰσιν ἀριθμοί· τῶν Κ, Λ >άρα ε<ῖς μέσοξ ἀνάλογον ἔστιν ἀριθμός. >έστω ὁ Μ. ὁ Μ >άρα ἔστιν ὁ ἐκ τῶν Δ, Ζ, ὡς ἐν τῷ πρὸ τούτου θεωρήματι ἐδείκθη. καὶ ἐπεὶ ὁ Δ τὸν μὲν Γ πολλαπλασιάσας τὸν Κ πεποίηκεν, τὸν δὲ Ζ πολλαπλασιάσας τὸν Μ πεποίηκεν, >έστιν >άρα ὡς ὁ Γ πρὸς τὸν Ζ, ο<ύτως ὁ Κ πρὸς τὸν Μ. ἀλλ> ὡς ὁ Κ πρὸς τὸν Μ, ὁ Μ πρὸς τὸν Λ. οἱ Κ, Μ, Λ >άρα ἐχῆξ εἰσιν ἀνάλογον ἐν τῷ τοῦ Γ πρὸς τὸν Ζ λόγῳ. καὶ ἐπεὶ ἔστιν ὡς ὁ Γ πρὸς τὸν Δ, ο<ύτως ὁ Ζ πρὸς τὸν Η, ἐναλλάχ>άρα ἔστιν ὡς ὁ Γ πρὸς τὸν Ζ, ο<ύτως ὁ Δ πρὸς τὸν Η. διὰ τὰ αὐτὰ δὴ καὶ ὡς ὁ Δ πρὸς τὸν Η, ο<ύτως ὁ Ε πρὸς τὸν Θ. οἱ Κ, Μ, Λ >άρα ἐχῆξ εἰσιν ἀνάλογον >ἐν τε τῷ τοῦ Γ πρὸς τὸν Ζ λόγῳ καὶ τῷ τοῦ Δ πρὸς τὸν Η καὶ >έτι τῷ τοῦ Ε πρὸς τὸν Θ. ἑκάτεροξ δὴ τῶν Ε, Θ τὸν Μ πολλαπλασιάσας ἐκάτερον τῶν Ν, Χ

Let A and B be two similar solid numbers, and let C, D, E be the sides of A , and F, G, H (the sides) of B . And since similar solid (numbers) are those having proportional sides [Def. 7.21], thus as C is to D , so F (is) to G , and as D (is) to E , so G (is) to H . I say that two numbers fall (between) A and B in mean proportion, and (that) A has to B a cubed ratio with respect to (that) C (has) to F , and D to G , and, further, E to H .

For let C make K (by) multiplying D , and let F make L (by) multiplying G . And since C, D are in the same ratio as F, G , and K is the (number created) from (multiplying) C, D , and L the (number created) from (multiplying) F, G , [thus] K and L are similar plane numbers [Def. 7.21]. Thus, there exists one number in mean proportion to K and L [Prop. 8.18]. Let it be M . Thus, M is the (number created) from (multiplying) D, F , as shown in the theorem before this (one). And since D has made K (by) multiplying C , and has made M (by) multiplying F , thus as C is to F , so K (is) to M [Prop. 7.17]. But, as K (is) to M , (so) M (is) to L . Thus, K, M, L are continuously proportional in the ratio of C to F . And since as C is to D , so F (is) to G , thus, alternately, as C is to F , so D (is) to G [Prop. 7.13]. And so, for the same (reasons), as D (is) to G , so E (is) to H . Thus, K, M, L are continuously proportional in the ratio of C to F , and of D to G , and, further, of E to H . So let E, H make N, O , respectively, (by) multiplying M . And

ποιεῖται. καὶ ἐπεὶ στερεοὺς ἐστὶν ὁ Α, πλευραὶ δὲ αὐτοῦ εἰσιν οἱ Γ, Δ, Ε, ὁ Ε >άρα τὸν ἐκ τῶν Γ, Δ πολλαπλασιάσας τὸν Α πεποίηκεν. ὁ δὲ ἐκ τῶν Γ, Δ ἐστὶν ὁ Κ; ὁ Ε >άρα τὸν Κ πολλαπλασιάσας τὸν Α πεποίηκεν. διὰ τὰ αὐτὰ δὴ καὶ ὁ Θ τὸν Λ πολλαπλασιάσας τὸν Β πεποίηκεν. καὶ ἐπεὶ ὁ Ε τὸν Κ πολλαπλασιάσας τὸν Α πεποίηκεν, ἀλλὰ μὴν καὶ τὸν Μ πολλαπλασιάσας τὸν Ν πεποίηκεν, >έστιν >άρα ὡς ὁ Κ πρὸς τὸν Μ, ο<ύτως ὁ Α πρὸς τὸν Ν. ὡς δὲ ὁ Κ πρὸς τὸν Μ, ο<ύτως ὁ Γ πρὸς τὸν Ζ καὶ ὁ Δ πρὸς τὸν Η καὶ >έτι ὁ Ε πρὸς τὸν Θ; καὶ ὡς >άρα ὁ Γ πρὸς τὸν Ζ καὶ ὁ Δ πρὸς τὸν Η καὶ ὁ Ε πρὸς τὸν Θ, ο<ύτως ὁ Α πρὸς τὸν Ν. πάλιν, ἐπεὶ ἐκάτερος τῶν Ε, Θ τὸν Μ πολλαπλασιάσας ἐκάτερον τῶν Ν, Χ πεποίηκεν, >έστιν >άρα ὡς ὁ Ε πρὸς τὸν Θ, ο<ύτως ὁ Ν πρὸς τὸν Χ. ἀλλ' ὡς ὁ Ε πρὸς τὸν Θ, ο<ύτως ὁ Γ πρὸς τὸν Ζ καὶ ὁ Δ πρὸς τὸν Η; καὶ ὡς >άρα ὁ Γ πρὸς τὸν Ζ καὶ ὁ Δ πρὸς τὸν Η καὶ ὁ Ε πρὸς τὸν Θ, ο<ύτως ὁ Α πρὸς τὸν Ν καὶ ὁ Ν πρὸς τὸν Χ. πάλιν, ἐπεὶ ὁ Θ τὸν Μ πολλαπλασιάσας τὸν Χ πεποίηκεν, ἀλλὰ μὴν καὶ τὸν Λ πολλαπλασιάσας τὸν Β πεποίηκεν, >έστιν >άρα ὡς ὁ Μ πρὸς τὸν Λ, ο<ύτως ὁ Χ πρὸς τὸν Β. ἀλλ' ὡς ὁ Μ πρὸς τὸν Λ, ο<ύτως ὁ Γ πρὸς τὸν Ζ καὶ ὁ Δ πρὸς τὸν Η καὶ ὁ Ε πρὸς τὸν Θ. καὶ ὡς >άρα ὁ Γ πρὸς τὸν Ζ καὶ ὁ Δ πρὸς τὸν Η καὶ ὁ Ε πρὸς τὸν Θ, ο<ύτως οὐ μόνον ὁ Χ πρὸς τὸν Β, ἀλλὰ καὶ ὁ Α πρὸς τὸν Ν καὶ ὁ Ν πρὸς τὸν Χ. οἱ Α, Ν, Χ, Β >άρα ἐχθῆς εἰσιν ἀνάλογον ἐν τοῖς εἰρημένοις τῶν πλευρῶν λόγοις.

Λέγω, <ὅτι καὶ ὁ Α πρὸς τὸν Β τριπλασίονα λόγον >έθει >ήπερ ἢ ὁμόλογος πλευρὰ πρὸς τὴν ὁμόλογον πλευράν, τουτέστιν >ήπερ ὁ Γ ἀριθμὸς πρὸς τὸν Ζ >ἢ ὁ Δ πρὸς τὸν Η καὶ >έτι ὁ Ε πρὸς τὸν Θ. ἐπεὶ γὰρ τέσσαρες ἀριθμοὶ ἐχθῆς ἀνάλογον εἰσιν οἱ Α, Ν, Χ, Β, ὁ Α >άρα πρὸς τὸν Β τριπλασίονα λόγον >έθει >ήπερ ὁ Α πρὸς τὸν Ν. ἀλλ' ὡς ὁ Α πρὸς τὸν Ν, ο<ύτως ἐδείκθη <ὅτε Γ πρὸς τὸν Ζ καὶ ὁ Δ πρὸς τὸν Η καὶ >έτι ὁ Ε πρὸς τὸν Θ. καὶ ὁ Α >άρα πρὸς τὸν Β τριπλασίονα λόγον >έθει >ήπερ ἢ ὁμόλογος πλευρὰ πρὸς τὴν ὁμόλογον πλευράν, τουτέστιν >ήπερ ὁ Γ ἀριθμὸς πρὸς τὸν Ζ καὶ ὁ Δ πρὸς τὸν Η καὶ >έτι ὁ Ε πρὸς τὸν Θ; <όπερ >έδει δεῖχαι.

† Literally, “triple”.

20.

Ἐὰν δύο ἀριθμῶν ε<ῖξ μέσος ἀνάλογον ἐμπίπτῃ ἀριθμὸς, <όμοιοι ἐπίπεδοι >έσονται οἱ ἀριθμοί.

Δύο γὰρ ἀριθμῶν τῶν Α, Β ε<ῖξ μέσος ἀνάλογον ἐμπιπτεύω ἀριθμὸς ὁ Γ; λέγω, <ὅτι οἱ Α, Β <όμοιοι ἐπίπεδοι εἰσιν ἀριθμοί.

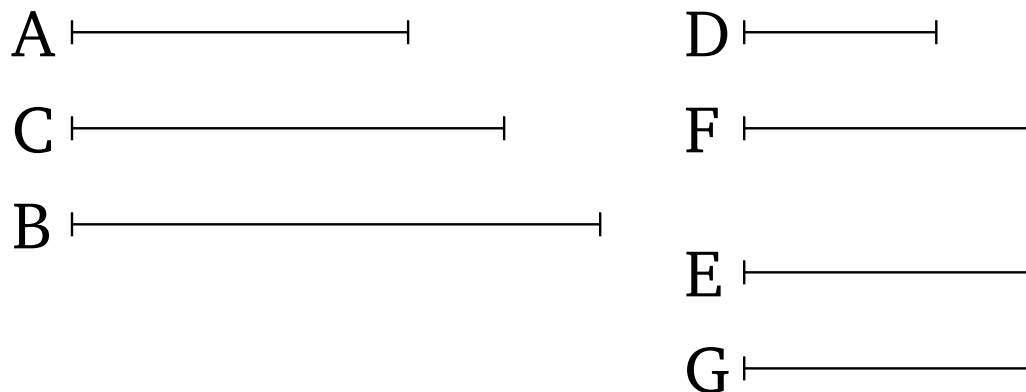
since A is solid, and C, D, E are its sides, E has thus made A (by) multiplying the (number created) from (multiplying) C, D . And K is the (number created) from (multiplying) C, D . Thus, E has made A (by) multiplying K . And so, for the same (reasons), H has made B (by) multiplying L . And since E has made A (by) multiplying K , but has, in fact, also made N (by) multiplying M , thus as K is to M , so A (is) to N [Prop. 7.17]. And as K (is) to M , so C (is) to F , and D to G , and, further, E to H . And thus as C (is) to F , and D to G , and E to H , so A (is) to N . Again, since E, H have made N, O , respectively, (by) multiplying M , thus as E is to H , so N (is) to O [Prop. 7.18]. But, as E (is) to H , so C (is) to F , and D to G . And thus as C (is) to F , and D to G , and E to H , so (is) A to N , and N to O . Again, since H has made O (by) multiplying M , but has, in fact, also made B (by) multiplying L , thus as M (is) to L , so O (is) to B [Prop. 7.17]. But, as M (is) to L , so C (is) to F , and D to G , and E to H . And thus as C (is) to F , and D to G , and E to H , so not only (is) O to B , but also A to N , and N to O . Thus, A, N, O, B are continuously proportional in the aforementioned ratios of the sides.

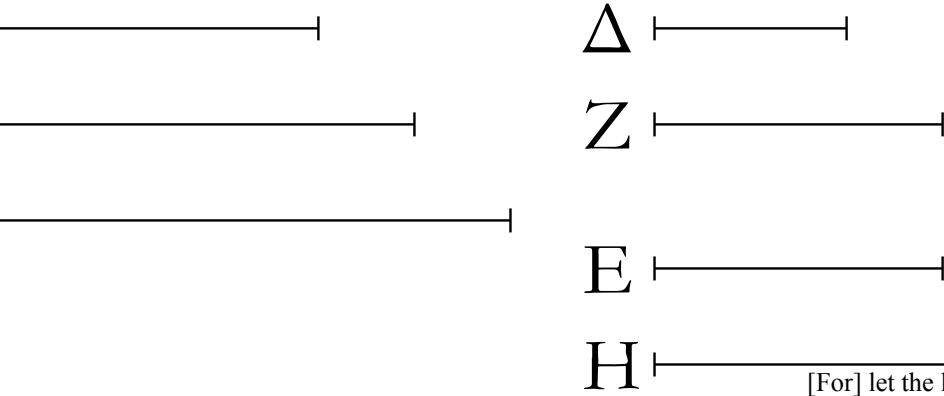
So I say that A also has to B a cubed ratio with respect to (that) a corresponding side (has) to a corresponding side—that is to say, with respect to (that) the number C (has) to F , or D to G , and, further, E to H . For since A, N, O, B are four continuously proportional numbers, A thus has to B a cubed ratio with respect to (that) A (has) to N [Def. 5.10]. But, as A (is) to N , so it was shown (is) C to F , and D to G , and, further, E to H . And thus A has to B a cubed ratio with respect to (that) a corresponding side (has) to a corresponding side—that is to say, with respect to (that) the number C (has) to F , and D to G , and, further, E to H . (Which is) the very thing it was required to show.

Proposition 20

If one number falls between two numbers in mean proportion then the numbers will be similar plane (numbers).

For let one number C fall between the two numbers A and B in mean proportion. I say that A and B are similar plane numbers.





Εἰλήφθωσαν [γάρ] ἐλάχιστοι ἀριθμοὶ τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς Α, Γ οἱ Δ, Ε; ἰσάκις >άρα ὁ Δ τὸν Α μετρεῖ καὶ ὁ Ε τὸν Γ. ὁσάκις δὴ ὁ Δ τὸν Α μετρεῖ, τοσαῦται μονάδες >έστωσαν ἐν τῷ Ζ; ὁ Ζ >άρα τὸν Δ πολλαπλασιάσας τὸν Α πεποίηκεν. <ώστε ὁ Α ἐπίπεδοξ ἐστίν, πλευραὶ δὲ αὐτοῦ οἱ Δ, Ζ. πάλιν, ἐπεὶ οἱ Δ, Ε ἐλάχιστοί εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς Γ, Β, ἰσάκις >άρα ὁ Δ τὸν Γ μετρεῖ καὶ ὁ Ε τὸν Β. ὁσάκις δὴ ὁ Ε τὸν Β μετρεῖ, τοσαῦται μονάδες >έστωσαν ἐν τῷ Η. ὁ Ε >άρα τὸν Β μετρεῖ κατὰ τὰς ἐν τῷ Η μονάδας; ὁ Η >άρα τὸν Ε πολλαπλασιάσας τὸν Β πεποίηκεν. ὁ Β >άρα ἐπίπεδοξ ἐστίν, πλευραὶ δὲ αὐτοῦ εἰσιν οἱ Ε, Η. οἱ Α, Β >άρα ἐπίπεδοί εἰσιν ἀριθμοί. λέγω δὴ, <ὅτι καὶ <όμοιοι. ἐπεὶ γὰρ ὁ Ζ τὸν μὲν Δ πολλαπλασιάσας τὸν Α πεποίηκεν, τὸν δὲ Ε πολλαπλασιάσας τὸν Γ πεποίηκεν, >έστιν >άρα ὡς ὁ Δ πρὸς τὸν Ε, ο<ύτως ὁ Α πρὸς τὸν Γ, τουτέστιν ὁ Γ πρὸς τὸν Β. πάλιν, ἐπεὶ ὁ Ε ἐκότερον τῶν Ζ, Η πολλαπλασιάσας τοὺς Γ, Β πεποίηκεν, >έστιν >άρα ὡς ὁ Ζ πρὸς τὸν Η, ο<ύτως ὁ Γ πρὸς τὸν Β. ὡς δὲ ὁ Γ πρὸς τὸν Β, ο<ύτως ὁ Δ πρὸς τὸν Ε; καὶ ὡς >άρα ὁ Δ πρὸς τὸν Ε, ο<ύτως ὁ Ζ πρὸς τὸν Η; καὶ ἐναλλάξ ὡς ὁ Δ πρὸς τὸν Ζ, ο<ύτως ὁ Ε πρὸς τὸν Η. οἱ Α, Β >άρα <όμοιοι ἐπίπεδοι ἀριθμοὶ εἰσιν; αἱ γὰρ πλευραὶ αὐτῶν ἀνάλογόν εἰσιν; <όπερ >έδει δεῖχαι.

[For] let the least numbers, D and E , having the same ratio as A and C be taken [Prop. 7.33]. Thus, D measures A as many times as E (measures) C [Prop. 7.20]. So as many times as D measures A , so many units let there be in F . Thus, F has made A (by) multiplying D [Def. 7.15]. Hence, A is plane, and D, F (are) its sides. Again, since D and E are the least of those (numbers) having the same ratio as C and B , D thus measures C as many times as E (measures) B [Prop. 7.20]. So as many times as E measures B , so many units let there be in G . Thus, E measures B according to the units in G . Thus, G has made B (by) multiplying E [Def. 7.15]. Thus, B is plane, and E, G are its sides. Thus, A and B are (both) plane numbers. So I say that (they are) also similar. For since F has made A (by) multiplying D , and has made C (by) multiplying E , thus as D is to E , so A (is) to C —that is to say, C to B [Prop. 7.17].[†] Again, since E has made C, B (by) multiplying F, G , respectively, thus as F is to G , so C (is) to B [Prop. 7.17]. And as C (is) to B , so D (is) to E . And thus as D (is) to E , so F (is) to G . And, alternately, as D (is) to F , so E (is) to G [Prop. 7.13]. Thus, A and B are similar plane numbers. For their sides are proportional [Def. 7.21]. (Which is) the very thing it was required to show.

[†] This part of the proof is defective, since it is not demonstrated that $F \times E = C$. Furthermore, it is not necessary to show that $D : E :: A : C$, because this is true by hypothesis.

21.

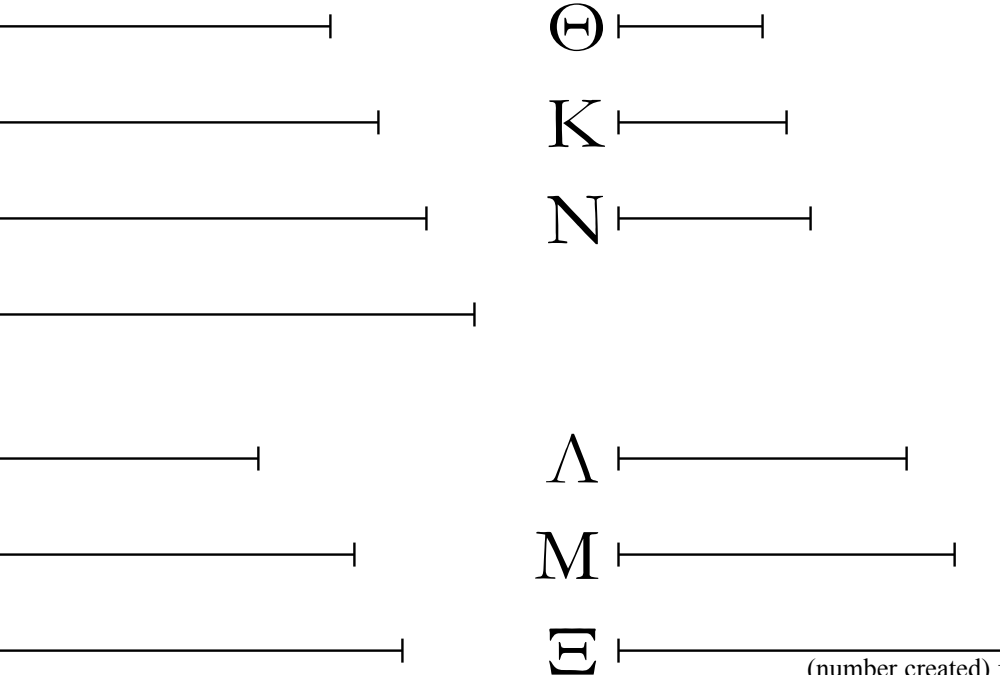
Proposition 21

Ἐὰν δύο ἀριθμῶν δύο μέσοι ἀνάλογον ἐμπίπτωσιν ἀριθμοί, <όμοιοι στερεοὶ εἰσιν οἱ ἀριθμοί. Δύο γὰρ ἀριθμῶν τῶν Α, Β δύο μέσοι ἀνάλογον ἐμπίπτέτωσαν ἀριθμοὶ οἱ Γ, Δ; λέγω, <ὅτι οἱ Α, Β <όμοιοι στερεοὶ εἰσιν. Εἰλήφθωσαν γὰρ ἐλάχιστοι ἀριθμοὶ τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς Α, Γ, Δ τρεῖς οἱ Ε, Ζ, Η; οἱ >άρα >άκροι αὐτῶν οἱ Ε, Η πρῶτοι πρὸς ἀλλήλους εἰσίν. καὶ ἐπεὶ τῶν Ε, Η ε<ῖς μέσοξ ἀνάλογον ἐμπίπτωκεν ἀριθμὸξ ὁ Ζ, οἱ Ε, Η >άρα ἀριθμοὶ <όμοιοι ἐπίπεδοί εἰσιν. >έστωσαν ο>ὗν τοῦ μὲν Ε πλευραὶ οἱ Θ, Κ, τοῦ δὲ Η οἱ Α, Μ. φανερόν >άρα ἐστίν ἐκ τοῦ πρὸ τούτου, <ὅτι οἱ Ε, Ζ, Η ἐχῆξ εἰσιν ἀνάλογον >έν τε τῷ τοῦ Θ πρὸς τὸν Α λόγῳ καὶ τῷ τοῦ Κ πρὸς τὸν Μ. καὶ ἐπεὶ οἱ Ε, Ζ, Η ἐλάχιστοί εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς Α, Γ, Δ, καὶ ἐστίν >ἰσον τὸ πλῆθος τῶν Ε, Ζ, Η τῷ πλήθει τῶν Α, Γ, Δ, δι> >ἰσου >άρα ἐστίν ὡς ὁ Ε πρὸς τὸν Η, ο<ύτως ὁ Α πρὸς τὸν Δ. οἱ δὲ Ε, Η πρῶτοι, οἱ δὲ πρῶτοι καὶ ἐλάχιστοι, οἱ δὲ ἐλάχιστοι μετροῦσι τοὺς τὸν αὐτὸν λόγον >έθονταξ αὐτοῖξ ἰσάκιξ <ὅ τε μείζων τὸν μείζονα καὶ ὁ ἐλάσσων τὸν ἐλάσσονα, τουτέστιν <ὅ τε ἡγούμενοξ τὸν ἡγούμενον καὶ ὁ ἐπόμενοξ τὸν ἐπόμενον; ἰσάκιξ >άρα ὁ Ε

If two numbers fall between two numbers in mean proportion then the (latter) are similar solid (numbers). For let the two numbers C and D fall between the two numbers A and B in mean proportion. I say that A and B are similar solid (numbers). For let the three least numbers E, F, G having the same ratio as A, C, D be taken [Prop. 8.2]. Thus, the outermost of them, E and G , are prime to one another [Prop. 8.3]. And since one number, F , has fallen (between) E and G in mean proportion, E and G are thus similar plane numbers [Prop. 8.20]. Therefore, let H, K be the sides of E , and L, M (the sides) of G . Thus, it is clear from the (proposition) before this (one) that E, F, G are continuously proportional in the ratio of H to L , and of K to M . And since E, F, G are the least (numbers) having the same ratio as A, C, D , and the multitude of E, F, G is equal to the multitude of A, C, D , thus, via equality, as E is to G , so A (is) to D [Prop. 7.14]. And E and G (are) prime (to one another), and prime (numbers) are also the least (of those numbers having the same ratio as them) [Prop. 7.21], and the least

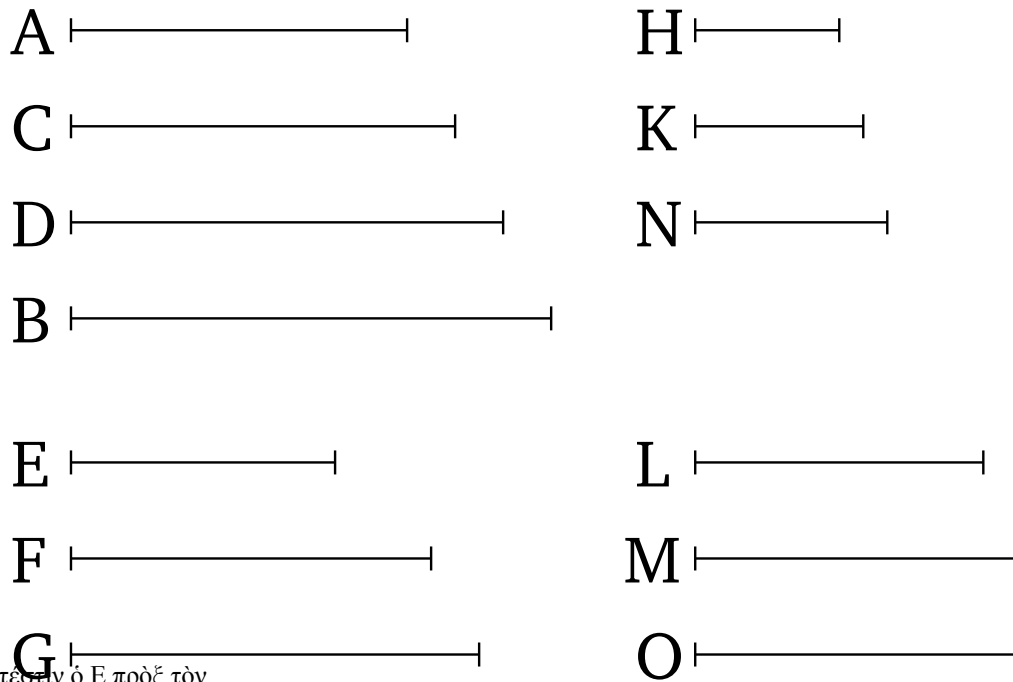
τὸν Α μετρεῖ καὶ ὁ Η τὸν Δ. ὁσάκις δὴ ὁ Ε τὸν Α μετρεῖ, τοσαῦται μονάδες >έστωσαν ἐν τῷ Ν. ὁ Ν >άρα τὸν Ε πολλαπλασιάσας τὸν Α πεποίηκεν. ὁ δὲ Ε ἐστὶν ὁ ἐκ τῶν Θ, Κ; ὁ Ν >άρα τὸν ἐκ τῶν Θ, Κ πολλαπλασιάσας τὸν Α πεποίηκεν. στερεὸς >άρα ἐστὶν ὁ Α, πλευραὶ δὲ αὐτοῦ εἰσιν οἱ Θ, Κ, Ν. πάλιν, ἐπεὶ οἱ Ε, Ζ, Η ἐλάχιστοι εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς Γ, Δ, Β, ἰσάκις >άρα ὁ Ε τὸν Γ μετρεῖ καὶ ὁ Η τὸν Β. ὁσάκις δὴ ὁ Ε τὸν Γ μετρεῖ, τοσαῦται μονάδες >έστωσαν ἐν τῷ Χ. ὁ Χ >άρα τὸν Β μετρεῖ κατὰ τὰς ἐν τῷ Χ μονάδας; ὁ Χ >άρα τὸν Η πολλαπλασιάσας τὸν Β πεποίηκεν. ὁ δὲ Η ἐστὶν ὁ ἐκ τῶν Λ, Μ; ὁ Χ >άρα τὸν ἐκ τῶν Λ, Μ πολλαπλασιάσας τὸν Β πεποίηκεν. στερεὸς >άρα ἐστὶν ὁ Β, πλευραὶ δὲ αὐτοῦ εἰσιν οἱ Λ, Μ, Χ; οἱ Α, Β >άρα στερεοὶ εἰσιν.

(numbers) measure those (numbers) having the same ratio as them an equal number of times, the greater (measuring) the greater, and the lesser the lesser—that is to say, the leading (measuring) the leading, and the following the following [Prop. 7.20]. Thus, E measures A the same number of times as G (measures) D . So as many times as E measures A , so many units let there be in N . Thus, N has made A (by) multiplying E [Def. 7.15]. And E is the (number created) from (multiplying) H and K . Thus, N has made A (by) multiplying the (number created) from (multiplying) H and K . Thus, A is solid, and its sides are H , K , N . Again, since E , F , G are the least (numbers) having the same ratio as C , D , B , thus E measures C the same number of times as G (measures) B [Prop. 7.20]. So as many times as E measures C , so many units let there be in O . Thus, G measures B according to the units in O . Thus, O has made B (by) multiplying G . And G is the (number created) from (multiplying) L and M . Thus, O has made B (by) multiplying the



Λέγω [δὴ], <ὅτι καὶ <όμοιοι. ἐπεὶ γὰρ οἱ Ν, Χ τὸν Ε πολλαπλασιάσαντες τοὺς Α, Β πεποίηκασιν, >έστιν >άρα

(number created) from (multiplying) L and M . Thus, B is solid, and its sides are L , M , O . Thus, A and B are (both) solid.



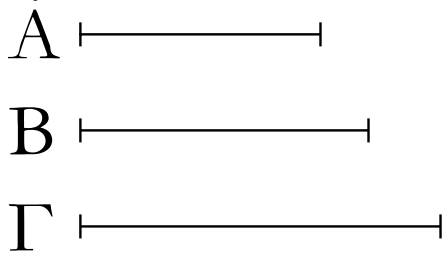
ὥς ὁ N πρὸς τὸν X , ὁ A πρὸς τὸν Γ , τουτέστιν ὁ E πρὸς τὸν Z . ἀλλ' ὥς ὁ E πρὸς τὸν Z , ὁ Θ πρὸς τὸν Λ καὶ ὁ K πρὸς τὸν M ; καὶ ὥς ἄρα ὁ Θ πρὸς τὸν Λ , οὕτως ὁ K πρὸς τὸν M καὶ ὁ N πρὸς τὸν X . καὶ εἰσιν οἱ μὲν Θ , K , N πλευραὶ τοῦ A , οἱ δὲ X , Λ , M πλευραὶ τοῦ B . οἱ A , B ἄρα ἀριθμοὶ ὁμοιοὶ στερεοὶ εἰσιν; ὅπερ ἔδει δεῖξαι.

[So] I say that (they are) also similar. For since N , O have made A , C (by) multiplying E , thus as N is to O , so A (is) to C —that is to say, E to F [Prop. 7.18]. But, as E (is) to F , so H (is) to L , and K to M . And thus as H (is) to L , so K (is) to M , and N to O . And H , K , N are the sides of A , and L , M , O the sides of B . Thus, A and B are similar solid numbers [Def. 7.21]. (Which is) the very thing it was required to show.

† The Greek text has “ O , L , M ”, which is obviously a mistake.

22.

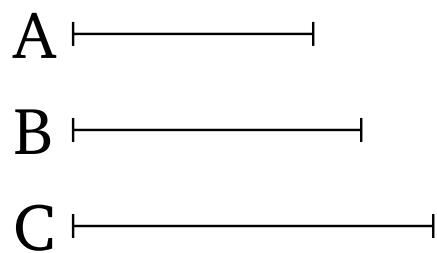
Ἐὰν τρεῖς ἀριθμοὶ ἐχῇ ἀνάλογον ὧσιν, ὁ δὲ πρῶτος τετράγωνος ᾗ, καὶ ὁ τρίτος τετράγωνος ᾗ, καὶ ὁ τρίτος τετράγωνος ᾗ. Ἐστῶσαν τρεῖς ἀριθμοὶ ἐχῇ ἀνάλογον οἱ A , B , Γ , ὁ δὲ πρῶτος ὁ A τετράγωνος ᾗ; ἔστω; λέγω, ὅτι καὶ ὁ τρίτος ὁ Γ τετράγωνος ᾗ.



Ἐπεὶ γὰρ τῶν A , Γ ἐστὶ μέσος ἀνάλογόν ἐστιν ἀριθμὸς ὁ B , οἱ A , Γ ἄρα ὁμοιοὶ ἐπίπεδοι εἰσιν. τετράγωνος δὲ ὁ A ; τετράγωνος ἄρα καὶ ὁ Γ ; ὅπερ ἔδει δεῖξαι.

Proposition 22

If three numbers are continuously proportional, and the first is square, then the third will also be square. Let A , B , C be three continuously proportional numbers, and let the first A be square. I say that the third C is also square.



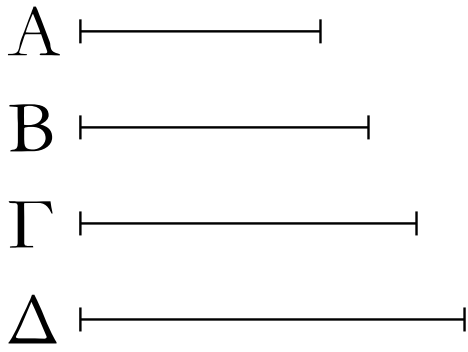
For since one number, B , is in mean proportion to A and C , A and C are thus similar plane (numbers) [Prop. 8.20]. And A is square. Thus, C is also square [Def. 7.21]. (Which is) the very thing it was required to show.

23.

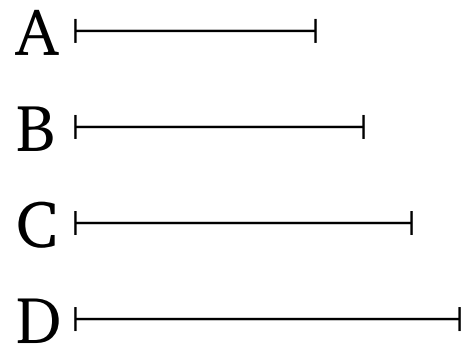
Ἐὰν τέσσαρες ἀριθμοὶ ἐχῇ ἀνάλογον ὧσιν, ὁ δὲ πρῶτος κύβος ᾗ, καὶ ὁ τέταρτος κύβος ᾗ. Ἐστῶσαν τέσσαρες ἀριθμοὶ ἐχῇ ἀνάλογον οἱ A , B , Γ , Δ , ὁ δὲ πρῶτος ὁ A κύβος ᾗ; ἔστω; λέγω, ὅτι καὶ ὁ τέταρτος ὁ Δ κύβος ᾗ.

Proposition 23

If four numbers are continuously proportional, and the first is cube, then the fourth will also be cube.



>Ἐστωσαν τέσσαρες ἀριθμοὶ ἐχῆζ ἀνάλογον οἱ A, B, Γ, Δ , ὁ δὲ A κύβοξ >έστω; λέγω, <ὅτι καὶ ὁ Δ κύβοξ ἐστίν.
Ἐπεὶ γὰρ τῶν A, Δ δύο μέσοι ἀνάλογόν εἰσιν ἀριθμοὶ οἱ B, Γ , οἱ A, Δ >ἀρα <ὅμοιοι εἰσι στερεοὶ ἀριθμοί. κύβοξ δὲ ὁ A ; κύβοξ >ἀρα καὶ ὁ Δ ; <ὅπερ >έδει δεῖχαι.



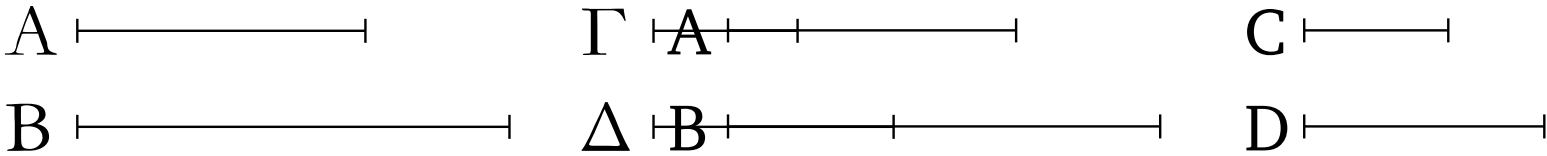
Let A, B, C, D be four continuously proportional numbers, and let A be cube. I say that D is also cube.
For since two numbers, B and C , are in mean proportion to A and D , A and D are thus similar solid numbers [Prop. 8.21]. And A (is) cube. Thus, D (is) also cube [Def. 7.21]. (Which is) the very thing it was required to show.

24.

Proposition 24

Ἐὰν δύο ἀριθμοὶ πρὸς ἀλλήλουξ λόγον >έθωσιν, <ὃν τετράγωνοξ ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν, ὁ δὲ πρῶτοξ τετράγωνοξ >ῆ|, καὶ ὁ δεῦτεροξ τετράγωνοξ >έσται.

If two numbers have to one another the ratio which a square number (has) to a(nother) square number, and the first is square, then the second will also be square.



Δύο γὰρ ἀριθμοὶ οἱ A, B πρὸς ἀλλήλουξ λόγον ἐθέτωσαν, <ὃν τετράγωνοξ ἀριθμὸξ ὁ Γ πρὸς τετράγωνον ἀριθμόν τὸν Δ , ὁ δὲ A τετράγωνοξ >έστω; λέγω, <ὅτι καὶ ὁ B τετράγωνός ἐστιν.

Ἐπεὶ γὰρ οἱ Γ, Δ τετράγωνοι εἰσιν, οἱ Γ, Δ >ἀρα <ὅμοιοι ἐπίπεδοι εἰσιν. τῶν Γ, Δ >ἀρα εἰς μέσοξ ἀνάλογον ἐμπίπτει ἀριθμὸξ. καὶ ἐστὶν ὡς ὁ Γ πρὸς τὸν Δ , ὁ A πρὸς τὸν B ; καὶ τῶν A, B >ἀρα εἰς μέσοξ ἀνάλογον ἐμπίπτει ἀριθμὸξ. καὶ ἐστὶν ὁ A τετράγωνοξ; καὶ ὁ B >ἀρα τετράγωνός ἐστιν; <ὅπερ >έδει δεῖχαι.

For let two numbers, A and B , have to one another the ratio which the square number C (has) to the square number D . And let A be square. I say that B is also square.

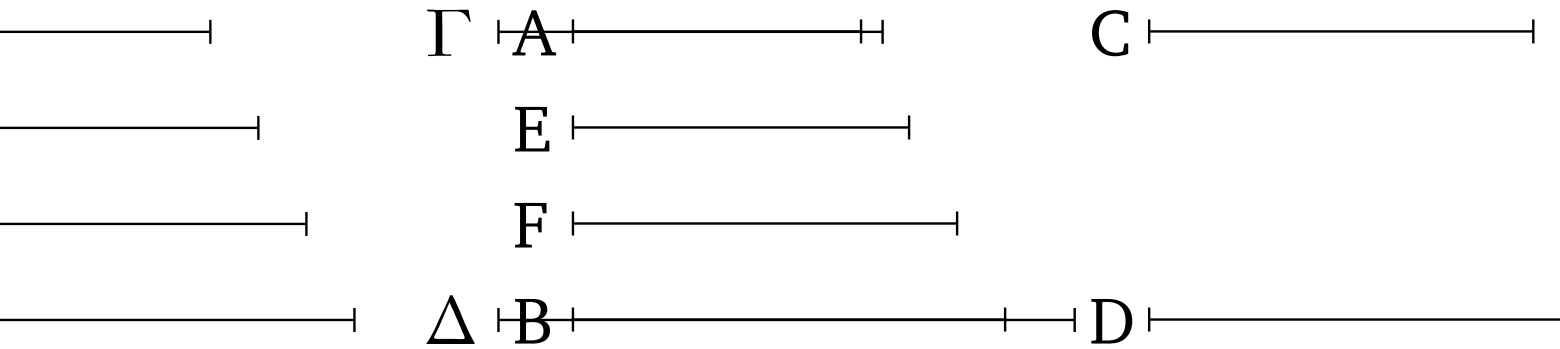
For since C and D are square, C and D are thus similar plane (numbers). Thus, one number falls (between) C and D in mean proportion [Prop. 8.18]. And as C is to D , (so) A (is) to B . Thus, one number also falls (between) A and B in mean proportion [Prop. 8.8]. And A is square. Thus, B is also square [Prop. 8.22]. (Which is) the very thing it was required to show.

25.

Proposition 25

Ἐὰν δύο ἀριθμοὶ πρὸς ἀλλήλουξ λόγον >έθωσιν, <ὃν κύβοξ ἀριθμὸξ πρὸς κύβον ἀριθμόν, ὁ δὲ πρῶτοξ κύβοξ >ῆ|, καὶ ὁ δεῦτεροξ κύβοξ >έσται.

If two numbers have to one another the ratio which a cube number (has) to a(nother) cube number, and the first is cube, then the second will also be cube.



Δύο γὰρ ἀριθμοὶ οἱ A, B πρὸς ἀλλήλουξ λόγον ἐθέτωσαν, <ὃν κύβοξ ἀριθμὸξ ὁ Γ πρὸς κύβον ἀριθμόν τὸν Δ , κύβοξ δὲ >έστω ὁ A ; λέγω [δή], <ὅτι καὶ ὁ B κύβοξ ἐστίν.

Ἐπεὶ γὰρ οἱ Γ, Δ κύβοι εἰσιν, οἱ Γ, Δ <ὅμοιοι στερεοὶ εἰσιν; τῶν Γ, Δ >ἀρα δύο μέσοι ἀνάλογον ἐμπίπτουσιν ἀριθμοί. <ὅσοι δὲ εἰς τοὺς Γ, Δ μεταχὺ κατὰ τὸ συνεθεῖς

For let two numbers, A and B , have to one another the ratio which the cube number C (has) to the cube number D . And let A be cube. [So] I say that B is also cube.

For since C and D are cube (numbers), C and D are (thus) similar solid (numbers). Thus, two numbers fall (between) C and D in mean proportion [Prop. 8.19]. And as many

ἀνάλογον ἐμπίπτουσιν, τοσοῦτοι καὶ εἰς τοὺς τὸν αὐτὸν λόγον >έθονταξ αὐτοῖς; <ώστε καὶ τῶν A, B δύο μέσοι ἀνάλογον ἐμπίπτουσιν ἀριθμοί. ἐμπιπτεύωσαν οἱ E, Z . ἐπεὶ ο>ὺν τέσσαρες ἀριθμοὶ οἱ A, E, Z, B ἐχρήξ ἀνάλογόν εἰσιν, καὶ ἐστι κύβοξ ὁ A , κύβοξ >άρα καὶ ὁ B ; <όπερ >έδει δεῖχαι.

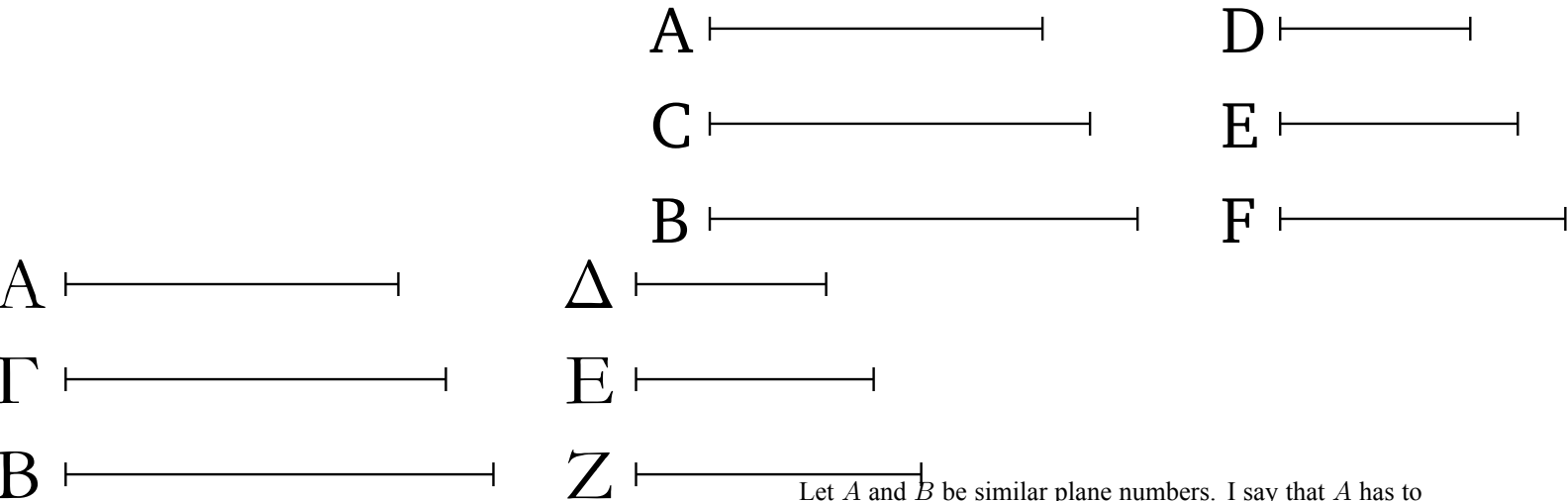
(numbers) as fall in between C and D in continued proportion, so many also (fall) in (between) those (numbers) having the same ratio as them (in continued proportion) [Prop. 8.8]. And hence two numbers fall (between) A and B in mean proportion. Let E and F (so) fall. Therefore, since the four numbers A, E, F, B are continuously proportional, and A is cube, B (is) thus also cube [Prop. 8.23]. (Which is) the very thing it was required to show.

26.

Proposition 26

Οἱ <όμοιοι ἐπίπεδοι ἀριθμοὶ πρὸς ἀλλήλουξ λόγον >έθουσιν, <ὸν τετράγωνοξ ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν.

Similar plane numbers have to one another the ratio which (some) square number (has) to a(nother) square number.



>Ἐστωσαν <όμοιοι ἐπίπεδοι ἀριθμοὶ οἱ A, B ; λέγω, <ὅτι ὁ A πρὸς τὸν B λόγον >έθει, <ὸν τετράγωνοξ ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν.

Ἐπεὶ γὰρ οἱ A, B <όμοιοι ἐπίπεδοι εἰσιν, τῶν A, B >άρα εἰς μέσοξ ἀνάλογον ἐμπίπτει ἀριθμὸξ. ἐμπιπτέτω καὶ >έστω ὁ Γ , καὶ εἰλήφθωσαν ἐλάχιστοι ἀριθμοὶ τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖξ A, Γ, B οἱ Δ, E, Z ; οἱ >άρα >άκροι αὐτῶν οἱ Δ, Z τετράγωνοι εἰσιν. καὶ ἐπεὶ ἐστὶν ὡξ ὁ Δ πρὸς τὸν Z , ο<ύτωξ ὁ A πρὸς τὸν B , καὶ εἰσιν οἱ Δ, Z τετράγωνοι, ὁ A >άρα πρὸς τὸν B λόγον >έθει, <ὸν τετράγωνοξ ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν; <όπερ >έδει δεῖχαι.

Let A and B be similar plane numbers. I say that A has to B the ratio which (some) square number (has) to a(nother) square number.

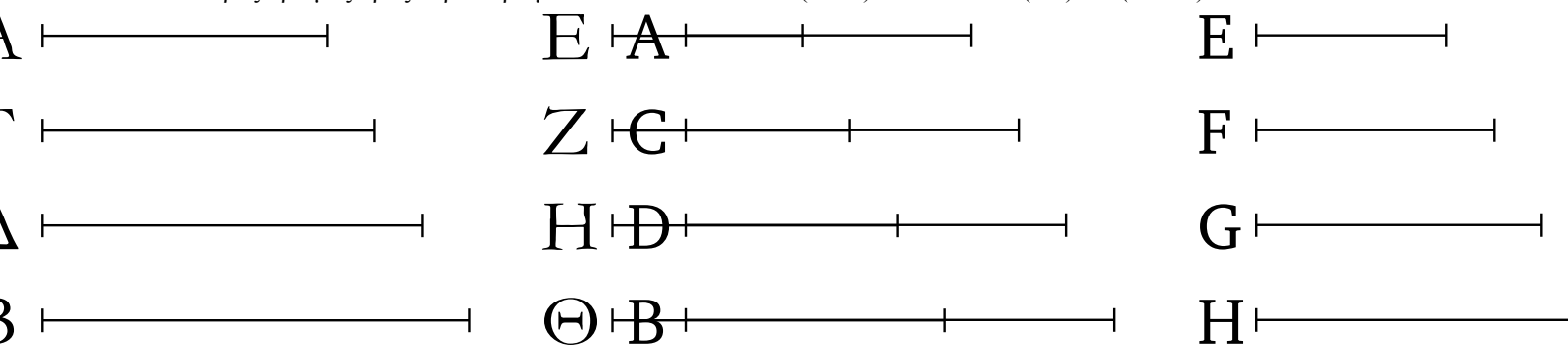
For since A and B are similar plane numbers, one number thus falls (between) A and B in mean proportion [Prop. 8.18]. Let it (so) fall, and let it be C . And let the least numbers, D, E, F , having the same ratio as A, C, B be taken [Prop. 8.2]. The outermost of them, D and F , are thus square [Prop. 8.2 corr.]. And since as D is to F , so A (is) to B , and D and F are square, A thus has to B the ratio which (some) square number (has) to a(nother) square number. (Which is) the very thing it was required to show.

27.

Proposition 27

Οἱ <όμοιοι στερεοὶ ἀριθμοὶ πρὸς ἀλλήλουξ λόγον >έθουσιν, <ὸν κύβοξ ἀριθμὸξ πρὸς κύβον ἀριθμόν.

Similar solid numbers have to one another the ratio which (some) cube number (has) to a(nother) cube number.



>Ἐστωσαν <όμοιοι στερεοὶ ἀριθμοὶ οἱ A, B ; λέγω, <ὅτι ὁ A πρὸς τὸν B λόγον >έθει, <ὸν κύβοξ ἀριθμὸξ πρὸς κύβον ἀριθμόν.

Ἐπεὶ γὰρ οἱ A, B <όμοιοι στερεοὶ εἰσιν, τῶν A, B >άρα δύο μέσοι ἀνάλογον ἐμπίπτουσιν ἀριθμοί. ἐμπιπτέτωσαν οἱ Γ, Δ , καὶ εἰλήφθωσαν ἐλάχιστοι ἀριθμοὶ τῶν τὸν αὐτὸν λόγον

Let A and B be similar solid numbers. I say that A has to B the ratio which (some) cube number (has) to a(nother) cube number.

For since A and B are similar solid (numbers), two numbers thus fall (between) A and B in mean proportion [Prop. 8.19]. Let C and D have (so) fallen. And let the least numbers,

ἐθόντων τοῖς A, Γ, Δ, B ἴσοι αὐτοῖς τὸ πλῆθος οἱ E, Z, H, Θ ; οἱ ἄρα ἄκροι αὐτῶν οἱ E, Θ κύβοι εἰσίν. καὶ ἐστὶν ὥς ὁ E πρὸς τὸν Θ , οὕτως ὁ A πρὸς τὸν B ; καὶ ὁ A ἄρα πρὸς τὸν B λόγον ἔθει, ὃν κύβοξ ἀριθμὸξ πρὸξ κύβον ἀριθμόν; ὅπερ ἔδει δεῖχαι.

E, F, G, H , having the same ratio as A, C, D, B , (and) equal in multitude to them, be taken [Prop. 8.2]. Thus, the outermost of them, E and H , are cube [Prop. 8.2 corr.]. And as E is to H , so A (is) to B . And thus A has to B the ratio which (some) cube number (has) to a(nother) cube number. (Which is) the very thing it was required to show.

ELEMENTS BOOK 9

Applications of Number Theory[†]

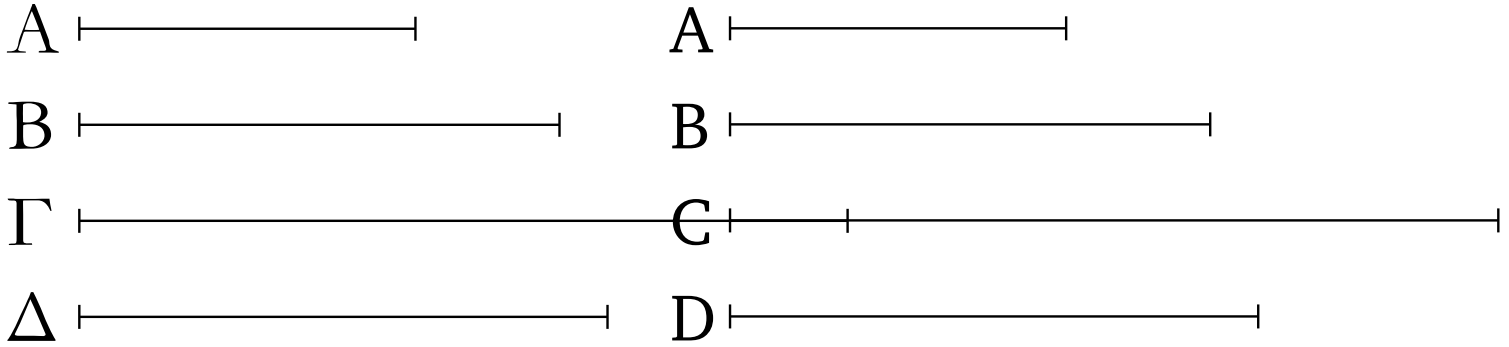
[†]The propositions contained in Books 7–9 are generally attributed to the school of Pythagoras.

1.

Proposition 1

Ἐὰν δύο <όμοιοι ἐπίπεδοι ἀριθμοὶ πολλαπλασιάσαντες ἀλλήλους ποιῶσι τινα, ὁ γεγόμενος τετράγωνος >έσται.

If two similar plane numbers make some (number by) multiplying one another then the created (number) will be square.



>Ἐστωσαν δύο <όμοιοι ἐπίπεδοι ἀριθμοὶ οἱ A, B, καὶ ὁ A τὸν B πολλαπλασιάσας τὸν Γ ποιεῖτω; λέγω, <ὅτι ὁ Γ τετράγωνός ἐστιν.

Ὅ γὰρ A ἑαυτὸν πολλαπλασιάσας τὸν Δ ποιεῖτω. ὁ Δ >ἀρα τετράγωνός ἐστιν. ἐπεὶ οὖν ὁ A ἑαυτὸν μὲν πολλαπλασιάσας τὸν Δ πεποίηκεν, τὸν δὲ B πολλαπλασιάσας τὸν Γ πεποίηκεν, >έστιν >ἀρα ὡς ὁ A πρὸς τὸν B, οὕτως ὁ Δ πρὸς τὸν Γ. καὶ ἐπεὶ οἱ A, B <όμοιοι ἐπίπεδοι εἰσιν ἀριθμοί, τῶν A, B >ἀρα εἰς μέσοξ ἀνάλογον ἐμπίπτει ἀριθμός. ἔαν δὲ δύο ἀριθμῶν μεταχὺ κατὰ τὸ συνθεῆξ ἀνάλογον ἐμπίπτωσιν ἀριθμοί, <όσοι εἰς αὐτοὺς ἐμπίπτουσι, τοσοῦτοι καὶ εἰς τοὺς τὸν αὐτὸν λόγον >έθονταξ; <ώστε καὶ τῶν Δ, Γ εἰς μέσοξ ἀνάλογον ἐμπίπτει ἀριθμός. καὶ ἐστὶ τετράγωνος ὁ Δ; τετράγωνος >ἀρα καὶ ὁ Γ; <όπερ >έδει δεῖξαι.

Let A and B be two similar plane numbers, and let A make C (by) multiplying B . I say that C is square.

For let A make D (by) multiplying itself. D is thus square.

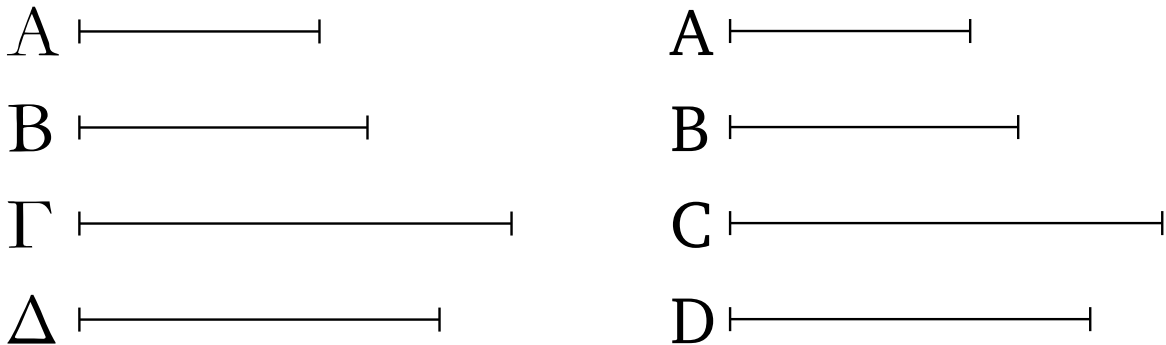
Therefore, since A has made D (by) multiplying itself, and A has made C (by) multiplying B , thus as A is to B , so D (is) to C [Prop. 7.17]. And since A and B are similar plane numbers, one number thus falls (between) A and B in mean proportion [Prop. 8.18]. And if (some) numbers fall between two numbers in continued proportion then, as many (numbers) as fall in (between) them (in continued proportion), so many also (fall) in (between numbers) having the same ratio (as them in continued proportion) [Prop. 8.8]. And hence one number falls (between) D and C in mean proportion. And D is square. Thus, C (is) also square [Prop. 8.22]. (Which is) the very thing it was required to show.

2.

Proposition 2

Ἐὰν δύο ἀριθμοὶ πολλαπλασιάσαντες ἀλλήλους ποιῶσι τετράγωνον, <όμοιοι ἐπίπεδοι εἰσιν ἀριθμοί.

If two numbers make a square (number by) multiplying one another then they are similar plane numbers.



>Ἐστωσαν δύο ἀριθμοὶ οἱ A, B, καὶ ὁ A τὸν B πολλαπλασιάσας τετράγωνον τὸν Γ ποιεῖτω; λέγω, <ὅτι οἱ A, B <όμοιοι ἐπίπεδοι εἰσιν ἀριθμοί.

Ὅ γὰρ A ἑαυτὸν πολλαπλασιάσας τὸν Δ ποιεῖτω; ὁ Δ >ἀρα τετράγωνός ἐστιν. καὶ ἐπεὶ ὁ A ἑαυτὸν μὲν πολλαπλασιάσας τὸν Δ πεποίηκεν, τὸν δὲ B πολλαπλασιάσας τὸν Γ πεποίηκεν, >έστιν >ἀρα ὡς ὁ A πρὸς τὸν B, ὁ Δ πρὸς τὸν Γ. καὶ ἐπεὶ ὁ Δ τετράγωνός ἐστιν, ἀλλὰ καὶ ὁ Γ, οἱ Δ, Γ >ἀρα <όμοιοι ἐπίπεδοι εἰσιν. τῶν Δ, Γ >ἀρα εἰς μέσοξ ἀνάλογον ἐμπίπτει. καὶ ἐστὶν ὡς ὁ Δ πρὸς τὸν Γ, οὕτως ὁ A πρὸς τὸν B; καὶ τῶν A, B >ἀρα εἰς μέσοξ ἀνάλογον ἐμπίπτει. ἔαν δὲ δύο ἀριθμῶν εἰς μέσοξ ἀνάλογον ἐμπίπτῃ, <όμοιοι ἐπίπεδοι εἰσιν [οἱ] ἀριθμοί; οἱ >ἀρα A, B <όμοιοι εἰσιν ἐπίπεδοι; <όπερ >έδει δεῖξαι.

Let A and B be two numbers, and let A make the square (number) C (by) multiplying B . I say that A and B are similar plane numbers.

For let A make D (by) multiplying itself. Thus, D is square. And since A has made D (by) multiplying itself, and has made C (by) multiplying B , thus as A is to B , so D (is) to C [Prop. 7.17]. And since D is square, and C (is) also, D and C are thus similar plane numbers. Thus, one (number) falls (between) D and C in mean proportion [Prop. 8.18]. And as D is to C , so A (is) to B . Thus, one (number) also falls (between) A and B in mean proportion [Prop. 8.8]. And if one (number) falls (between) two numbers in mean proportion then [the] numbers are similar plane (numbers) [Prop. 8.20]. Thus, A and B are similar plane (numbers). (Which is) the

very thing it was required to show.

3.

Ἐὰν κύβοξ ἀριθμὸξ ἐαυτὸν πολλαπλασιάσας ποιῇ| τινα, ὁ γενόμενος κύβοξ >έσται.

A |————|

B |————|

Γ |————|

Δ |————|

Κύβοξ γὰρ ἀριθμὸξ ὁ A ἐαυτὸν πολλαπλασιάσας τὸν B ποιείτω; λέγω, <ὅτι ὁ B κύβοξ ἐστίν.

Εἰλήφθω γὰρ τοῦ A πλευρὰ ὁ Γ, καὶ ὁ Γ ἐαυτὸν πολλαπλασιάσας τὸν Δ ποιείτω. φανερόν δὲ ἐστίν, <ὅτι ὁ Γ τὸν Δ πολλαπλασιάσας τὸν A πεποίηκεν. καὶ ἐπεὶ ὁ Γ ἐαυτὸν πολλαπλασιάσας τὸν Δ πεποίηκεν, ὁ Γ >άρα τὸν Δ μετρεῖ κατὰ τὰξ ἐν αὐτῷ| μονάδαξ. ἀλλὰ μὴν καὶ ἡ μονὰξ τὸν Γ μετρεῖ κατὰ τὰξ ἐν αὐτῷ| μονάδαξ; >έστιν >άρα ὡξ ἡ μονὰξ πρὸξ τὸν Γ, ὁ Γ πρὸξ τὸν Δ. πάλιν, ἐπεὶ ὁ Γ τὸν Δ πολλαπλασιάσας τὸν A πεποίηκεν, ὁ Δ >άρα τὸν A μετρεῖ κατὰ τὰξ ἐν τῷ| Γ μονάδαξ. μετρεῖ δὲ καὶ ἡ μονὰξ τὸν Γ κατὰ τὰξ ἐν αὐτῷ| μονάδαξ; >έστιν >άρα ὡξ ἡ μονὰξ πρὸξ τὸν Γ, ὁ Δ πρὸξ τὸν A. ἀλλ> ὡξ ἡ μονὰξ πρὸξ τὸν Γ, ὁ Γ πρὸξ τὸν Δ; καὶ ὡξ >άρα ἡ μονὰξ πρὸξ τὸν Γ, ο<ὅτωξ ὁ Γ πρὸξ τὸν Δ καὶ ὁ Δ πρὸξ τὸν A. τῆξ >άρα μονάδοξ καὶ τοῦ A ἀριθμοῦ δύο μέσοι ἀνάλογον κατὰ τὸ συνεθὲξ ἐμπεπτώκασιν ἀριθμοὶ οἱ Γ, Δ. πάλιν, ἐπεὶ ὁ A ἐαυτὸν πολλαπλασιάσας τὸν B πεποίηκεν, ὁ A >άρα τὸν B μετρεῖ κατὰ τὰξ ἐν αὐτῷ| μονάδαξ; μετρεῖ δὲ καὶ ἡ μονὰξ τὸν A κατὰ τὰξ ἐν αὐτῷ| μονάδαξ; >έστιν >άρα ὡξ ἡ μονὰξ πρὸξ τὸν A, ὁ A πρὸξ τὸν B. τῆξ δὲ μονάδοξ καὶ τοῦ A δύο μέσοι ἀνάλογον ἐμπεπτώκασιν ἀριθμοὶ; καὶ τῶν A, B >άρα δύο μέσοι ἀνάλογον ἐμπεσοῦνται ἀριθμοί. ἔαν δὲ δύο ἀριθμῶν δύο μέσοι ἀνάλογον ἐμπίπτωσιν, ὁ δὲ πρῶτοξ κύβοξ >ῇ|, καὶ ὁ δεῦτεροξ κύβοξ >έσται. καὶ ἐστίν ὁ A κύβοξ; καὶ ὁ B >άρα κύβοξ ἐστίν; <ὅπερ >έδει δεῖχαι.

4.

Ἐὰν κύβοξ ἀριθμὸξ κύβον ἀριθμὸν πολλαπλασιάσας ποιῇ| τινα, ὁ γενόμενος κύβοξ >έσται.

Κύβοξ γὰρ ἀριθμὸξ ὁ A κύβον ἀριθμὸν τὸν B πολλαπλασιάσας τὸν Γ ποιείτω; λέγω, <ὅτι ὁ Γ κύβοξ ἐστίν.

A |————|

B |————|

Γ |————|

Δ |————|

Proposition 3

If a cube number makes some (number by) multiplying itself then the created (number) will be cube.

A |————|

B |————|

C |————|

D |————|

For let the cube number A make B (by) multiplying itself. I say that B is cube.

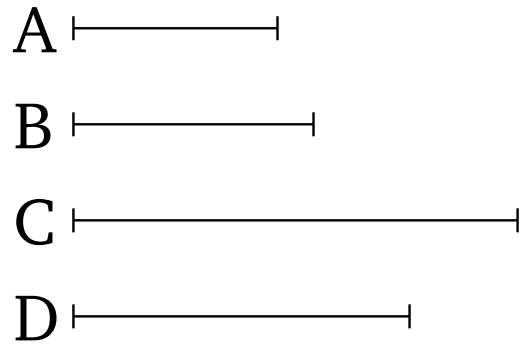
For let the side C of A be taken. And let C make D by multiplying itself. So it is clear that C has made A (by) multiplying D . And since C has made D (by) multiplying itself, C thus measures D according to the units in it [Def. 7.15]. But, in fact, a unit also measures C according to the units in it [Def. 7.20]. Thus, as a unit is to C , so C (is) to D . Again, since C has made A (by) multiplying D , D thus measures A according to the units in C . And a unit also measures C according to the units in it. Thus, as a unit is to C , so D (is) to A . But, as a unit (is) to C , so C (is) to D . And thus as a unit (is) to C , so C (is) to D , and D to A . Thus, two numbers, C and D , have fallen (between) a unit and the number A in continued mean proportion. Again, since A has made B (by) multiplying itself, A thus measures B according to the units in it. And a unit also measures A according to the units in it. Thus, as a unit is to A , so A (is) to B . And two numbers have fallen (between) a unit and A in mean proportion. Thus two numbers will also fall (between) A and B in mean proportion [Prop. 8.8]. And if two (numbers) fall (between) two numbers in mean proportion, and the first (number) is cube, then the second will also be cube [Prop. 8.23]. And A is cube. Thus, B is also cube. (Which is) the very thing it was required to show.

Proposition 4

If a cube number makes some (number by) multiplying a(nother) cube number then the created (number) will be cube.

For let the cube number A make C (by) multiplying the cube

number B . I say that C is cube.

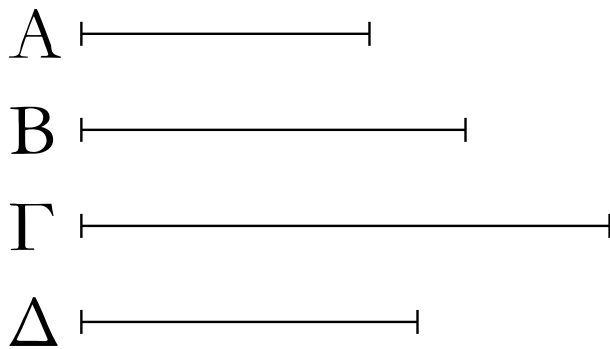


Ὁ γὰρ A ἑαυτὸν πολλαπλασιάσας τὸν Δ ποιεῖτω; ὁ Δ >άρα κύβοξ ἐστίν. καὶ ἐπεὶ ὁ A ἑαυτὸν μὲν πολλαπλασιάσας τὸν Δ πεποίηκεν, τὸν δὲ B πολλαπλασιάσας τὸν Γ πεποίηκεν, >ἐστίν >άρα ὡς ὁ A πρὸς τὸν B , ο<ύτωξ ὁ Δ πρὸς τὸν Γ . καὶ ἐπεὶ οἱ A, B κύβοι εἰσίν, <όμοιοι στερεοὶ εἰσιν οἱ A, B . τῶν A, B >άρα δύο μέσοι ἀνάλογον ἐμπίπτουσιν ἀριθμοί; <ώστε καὶ τῶν Δ, Γ δύο μέσοι ἀνάλογον ἐμπεσοῦνται ἀριθμοί. καὶ ἐστὶ κύβοξ ὁ Δ ; κύβοξ >άρα καὶ ὁ Γ ; <όπερ >έδει δεῖχαι.

For let A make D (by) multiplying itself. Thus, D is cube [Prop. 9.3]. And since A has made D (by) multiplying itself, and has made C (by) multiplying B , thus as A is to B , so D (is) to C [Prop. 7.17]. And since A and B are cube, A and B are similar solid (numbers). Thus, two numbers fall (between) A and B in mean proportion [Prop. 8.19]. Hence, two numbers will also fall (between) D and C in mean proportion [Prop. 8.8]. And D is cube. Thus, C (is) also cube [Prop. 8.23]. (Which is) the very thing it was required to show.

5.

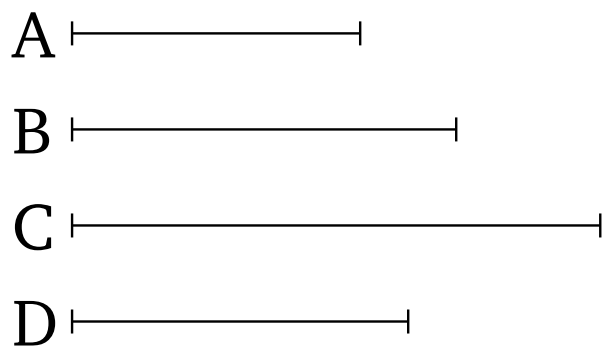
Ἐὰν κύβοξ ἀριθμὸξ ἀριθμόν τινα πολλαπλασιάσας κύβον ποιῇ, καὶ ὁ πολλαπλασιασθεὶξ κύβοξ >έσται.



Κύβοξ γὰρ ἀριθμὸξ ὁ A ἀριθμόν τινα τὸν B πολλαπλασιάσας κύβον τὸν Γ ποιεῖτω; λέγω, <ότι ὁ B κύβοξ ἐστίν. Ὁ γὰρ A ἑαυτὸν πολλαπλασιάσας τὸν Δ ποιεῖτω; κύβοξ >άρα ἐστίν ὁ Δ . καὶ ἐπεὶ ὁ A ἑαυτὸν μὲν πολλαπλασιάσας τὸν Δ πεποίηκεν, τὸν δὲ B πολλαπλασιάσας τὸν Γ πεποίηκεν, >ἐστίν ἄρα ὡς ὁ A πρὸς τὸν B , ὁ Δ πρὸς τὸν Γ . καὶ ἐπεὶ οἱ Δ, Γ κύβοι εἰσίν, <όμοιοι στερεοὶ εἰσιν. τῶν Δ, Γ >άρα δύο μέσοι ἀνάλογον ἐμπίπτουσιν ἀριθμοί. καὶ ἐστὶν ὡς ὁ Δ πρὸς τὸν Γ , ο<ύτωξ ὁ A πρὸς τὸν B ; καὶ τῶν A, B >άρα δύο μέσοι ἀνάλογον ἐμπίπτουσιν ἀριθμοί. καὶ ἐστὶ κύβοξ ὁ A ; κύβοξ >άρα ἐστὶ καὶ ὁ B ; <όπερ >έδει δεῖχαι.

Proposition 5

If a cube number makes a(nother) cube number (by) multiplying some (number) then the (number) multiplied will also be cube.



For let the cube number A make the cube (number) C (by) multiplying some number B . I say that B is cube. For let A make D (by) multiplying itself. D is thus cube [Prop. 9.3]. And since A has made D (by) multiplying itself, and has made C (by) multiplying B , thus as A is to B , so D (is) to C [Prop. 7.17]. And since D and C are (both) cube, they are similar solid (numbers). Thus, two numbers fall (between) D and C in mean proportion [Prop. 8.19]. And as D is to C , so A (is) to B . Thus, two numbers also fall (between) A and B in mean proportion [Prop. 8.8]. And A is cube. Thus, B is also cube [Prop. 8.23]. (Which is) the very thing it was required to show.

6.

Ἐὰν ἀριθμὸξ ἑαυτὸν πολλαπλασιάσας κύβον ποιῇ, καὶ αὐτὸξ κύβοξ >έσται. Ἀριθμὸξ γὰρ ὁ A ἑαυτὸν πολλαπλασιάσας κύβον τὸν B ποιεῖτω; λέγω, <ότι καὶ ὁ A κύβοξ ἐστίν.

Proposition 6

If a number makes a cube (number by) multiplying itself then it itself will also be cube. For let the number A make the cube (number) B (by) multiplying itself. I say that A is also cube.

A —————

B —————

Γ —————

Ὅ γὰρ Α τὸν Β πολλαπλασιάσας τὸν Γ ποιεῖτω. ἐπεὶ οὖν ὁ Α ἑαυτὸν μὲν πολλαπλασιάσας τὸν Β πεποίηκεν, τὸν δὲ Β πολλαπλασιάσας τὸν Γ πεποίηκεν, ὁ Γ >άρα κύβοξ ἐστίν. καὶ ἐπεὶ ὁ Α ἑαυτὸν πολλαπλασιάσας τὸν Β πεποίηκεν, ὁ Α >άρα τὸν Β μετρεῖ κατὰ τὰξ ἐν αὐτῷ μονάδαξ. μετρεῖ δὲ καὶ ἡ μονάξ τὸν Α κατὰ τὰξ ἐν αὐτῷ μονάδαξ. >έστιν >άρα ὡξ ἡ μονάξ πρὸξ τὸν Α, ο<ύτωξ ὁ Α πρὸξ τὸν Β. καὶ ἐπεὶ ὁ Α τὸν Β πολλαπλασιάσας τὸν Γ πεποίηκεν, ὁ Β >άρα τὸν Γ μετρεῖ κατὰ τὰξ ἐν τῷ Α μονάδαξ. μετρεῖ δὲ καὶ ἡ μονάξ τὸν Α κατὰ τὰξ ἐν αὐτῷ μονάδαξ. >έστιν >άρα ὡξ ἡ μονάξ πρὸξ τὸν Α, ο<ύτωξ ὁ Β πρὸξ τὸν Γ. ἀλλ> ὡξ ἡ μονάξ πρὸξ τὸν Α, ο<ύτωξ ὁ Α πρὸξ τὸν Β; καὶ ὡξ >άρα ὁ Α πρὸξ τὸν Β, ὁ Β πρὸξ τὸν Γ. καὶ ἐπεὶ οἱ Β, Γ κύβοι εἰσίν, <όμοιοι στερεοὶ εἰσιν. τῶν Β, Γ >άρα δύο μέσοι ἀνάλογόν εἰσιν ἀριθμοί. καὶ ἐστὶν ὡξ ὁ Β πρὸξ τὸν Γ, ὁ Α πρὸξ τὸν Β. καὶ τῶν Α, Β >άρα δύο μέσοι ἀνάλογόν εἰσιν ἀριθμοί. καὶ ἐστὶν κύβοξ ὁ Β; κύβοξ >άρα ἐστὶ καὶ ὁ Α; <όπερ >έδει δεῖχαι.

7.

Ἐὰν σύνθετοξ ἀριθμὸξ ἀριθμὸν τινα πολλαπλασιάσας ποιῇ τινα, ὁ γενόμενοξ στερεόξ >έσται.

A —————

B —————

Γ —————

Δ —————

E —————

Σύνθετοξ γὰρ ἀριθμὸξ ὁ Α ἀριθμὸν τινα τὸν Β πολλαπλασιάσας τὸν Γ ποιεῖτω; λέγω, <ὅτι ὁ Γ στερεόξ ἐστίν.

A —————

B —————

C —————

For let A make C (by) multiplying B . Therefore, since A has made B (by) multiplying itself, and has made C (by) multiplying B , C is thus cube. And since A has made B (by) multiplying itself, A thus measures B according to the units in (A). And a unit also measures A according to the units in it. Thus, as a unit is to A , so A (is) to B . And since A has made C (by) multiplying B , B thus measures C according to the units in A . And a unit also measures A according to the units in it. Thus, as a unit is to A , so B (is) to C . But, as a unit (is) to A , so A (is) to B . And thus as A (is) to B , (so) B (is) to C . And since B and C are cube, they are similar solid (numbers). Thus, there exist two numbers in mean proportion (between) B and C [Prop. 8.19]. And as B is to C , (so) A (is) to B . Thus, there also exist two numbers in mean proportion (between) A and B [Prop. 8.8]. And B is cube. Thus, A is also cube [Prop. 8.23]. (Which is) the very thing it was required to show.

Proposition 7

If a composite number makes some (number by) multiplying some (other) number then the created (number) will be solid.

A —————

B —————

C —————

D —————

E —————

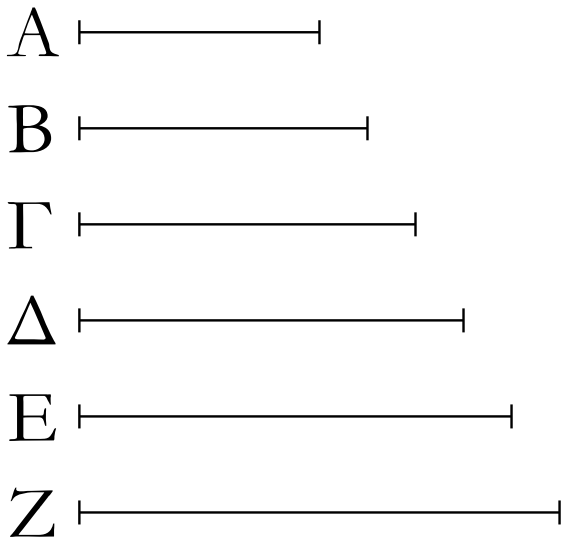
For let the composite number A make C (by) multiplying some number B . I say that C is solid.

For since A is a composite (number), it will be measured

Ἐπεὶ γὰρ ὁ A σύνθετός ἐστιν, ὑπὸ ἀριθμοῦ τινος μετρηθήσεται by some number. Let it be measured by D . And, as many times as D measures A , so many units let there be in E . Therefore, since D measures A according to the units in E , E has thus made A (by) multiplying D [Def. 7.15]. And since A has made C (by) multiplying B , and A is the (number created) from (multiplying) D , E , the (number created) from (multiplying) D , E has thus made C (by) multiplying B . Thus, C is solid, and its sides are D , E , B . (Which is) the very thing it was required to show.

8.

Ἐὰν ἀπὸ μονάδος ὅποσοι οὖν ἀριθμοὶ ἐχῇ ἀνάλογον ὦσιν, ὁ μὲν τρίτος ἀπὸ τῆς μονάδος τετράγωνος ἔσται καὶ οἱ <ένα διαλείποντες, ὁ δὲ τέταρτος κύβος καὶ οἱ δύο διαλείποντες πάντες, ὁ δὲ <έβδομος κύβος <άμα καὶ τετράγωνος καὶ οἱ πέντε διαλείποντες.

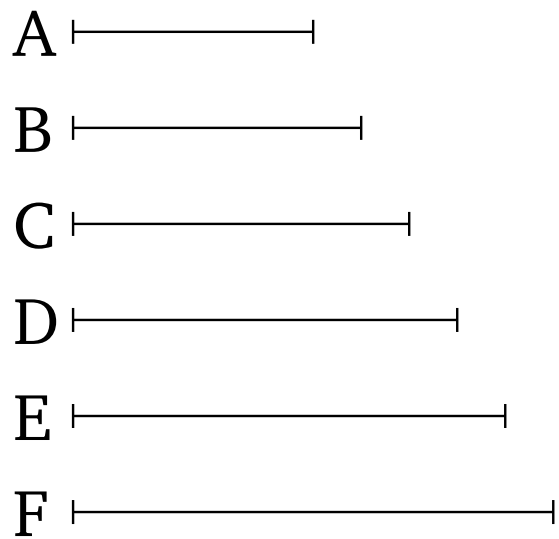


>Ἐστωσαν ἀπὸ μονάδος ὅποσοι οὖν ἀριθμοὶ ἐχῇ ἀνάλογον οἱ A , B , Γ , Δ , E , Z ; λέγω, <ὅτι ὁ μὲν τρίτος ἀπὸ τῆς μονάδος ὁ B τετράγωνός ἐστι καὶ οἱ <ένα διαλείποντες πάντες, ὁ δὲ τέταρτος ὁ Γ κύβος καὶ οἱ δύο διαλείποντες πάντες, ὁ δὲ <έβδομος ὁ Z κύβος <άμα καὶ τετράγωνος καὶ οἱ πέντε διαλείποντες πάντες.

Ἐπεὶ γὰρ ἐστὶν ὥς ἡ μονὰς πρὸς τὸν A , οὕτως ὁ A πρὸς τὸν B , ἰσάκις >άρα ἡ μονὰς τὸν A ἀριθμὸν μετρεῖ καὶ ὁ A τὸν B . ἡ δὲ μονὰς τὸν A ἀριθμὸν μετρεῖ κατὰ τὰς ἐν αὐτῷ μονάδας; καὶ ὁ A >άρα τὸν B μετρεῖ κατὰ τὰς ἐν τῷ A μονάδας. ὁ A >άρα ἐαυτὸν πολλαπλασιάσας τὸν B πεποίηκεν; τετράγωνος >άρα ἐστὶν ὁ B . καὶ ἐπεὶ οἱ B , Γ , Δ ἐχῇ ἀνάλογόν εἰσιν, ὁ δὲ B τετράγωνός ἐστιν, καὶ ὁ Δ >άρα τετράγωνός ἐστιν μὴ τὰ αὐτὰ δὴ καὶ ὁ Z τετράγωνός ἐστιν. ὁμοίως δὲ δείχομεν, <ὅτι καὶ οἱ <ένα διαλείποντες πάντες τετράγωνοι εἰσιν. λέγω δὴ, <ὅτι καὶ ὁ τέταρτος ἀπὸ τῆς μονάδος ὁ Γ κύβος ἐστὶ καὶ οἱ δύο διαλείποντες πάντες. ἐπεὶ γὰρ ἐστὶν ὥς ἡ μονὰς πρὸς τὸν A , οὕτως ὁ B πρὸς τὸν Γ , ἰσάκις >άρα ἡ μονὰς τὸν A ἀριθμὸν μετρεῖ καὶ ὁ B τὸν Γ . ἡ δὲ μονὰς τὸν A ἀριθμὸν μετρεῖ κατὰ τὰς ἐν τῷ A μονάδας; καὶ ὁ B >άρα τὸν Γ μετρεῖ κατὰ τὰς ἐν τῷ A μονάδας; ὁ A >άρα τὸν B πολλαπλασιάσας τὸν Γ πεποίηκεν. ἐπεὶ οὖν ὁ A ἐαυτὸν μὲν πολλαπλασιάσας τὸν B πεποίηκεν, τὸν δὲ B πολλαπλασιάσας τὸν Γ πεποίηκεν, κύβος >άρα

Proposition 8

If any multitude whatsoever of numbers is continuously proportional, (starting) from a unit, then the third from the unit will be square, and (all) those (numbers after that) which leave an interval of one (number), and the fourth (will be) cube, and all those (numbers after that) which leave an interval of two (numbers), and the seventh (will be) both cube and square, and (all) those (numbers after that) which leave an interval of five (numbers).



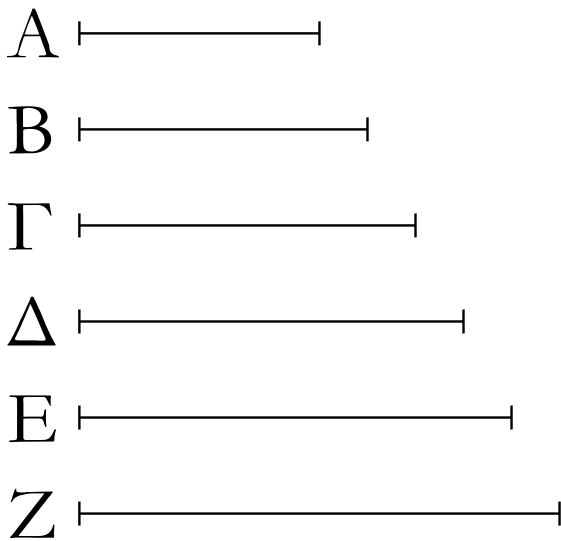
Let any multitude whatsoever of numbers, A , B , C , D , E , F , be continuously proportional, (starting) from a unit. I say that the third from the unit, B , is square, and all those (numbers after that) which leave an interval of one (number). And the fourth (from the unit), C , (is) cube, and all those (numbers after that) which leave an interval of two (numbers). And the seventh (from the unit), F , (is) both cube and square, and all those (numbers after that) which leave an interval of five (numbers).

For since as the unit is to A , so A (is) to B , the unit thus measures the number A the same number of times as A (measures) B [Def. 7.20]. And the unit measures the number A according to the units in it. Thus, A also measures B according to the units in A . A has thus made B (by) multiplying itself [Def. 7.15]. Thus, B is square. And since B , C , D are continuously proportional, and B is square, D is thus also square [Prop. 8.22]. So, for the same (reasons), F is also square. So, similarly, we can also show that all those (numbers after that) which leave an interval of one (number) are square. So I also say that the fourth (number) from the unit, C , is cube, and all those (numbers after that) which leave an interval of two (numbers). For since as the unit is to A , so B (is) to C , the unit thus measures the number A the same number of times that B (measures) C . And the unit mea-

ἐστὶν ὁ Γ. καὶ ἐπεὶ οἱ Γ, Δ, Ε, Ζ ἐχῆζ ἀνάλογόν εἰσιν, ὁ δὲ Γ κύβοξ ἐστίν, καὶ ὁ Ζ >άρα κύβοξ ἐστίν. ἐδείθθη δὲ καὶ τετράγωνοξ; ὁ >άρα <έβδομοξ ἀπὸ τῆξ μονάδοξ κύβοξ τέ ἐστι καὶ τετράγωνοξ. ὁμοίωξ δὴ δείχομεν, <ὅτι καὶ οἱ πέντε διαλείποντεξ πάντεξ κύβοι τέ εἰσι καὶ τετράγωνοι; <ὅπερ >έδει δεῖχαι.

9.

Ἐὰν ἀπὸ μονάδοξ ὅποσοιὺν ἐχῆζ κατὰ τὸ συνεθὲξ ἀριθμοὶ ἀνάλογον >ῶσιν, ὁ δὲ μετὰ τὴν μονάδα τετράγωνοξ >ῆ], καὶ οἱ λοιποὶ πάντεξ τετράγωνοι >έσσονται. καὶ ἔαν ὁ μετὰ τὴν μονάδα κύβοξ >ῆ], καὶ οἱ λοιποὶ πάντεξ κύβοι >έσσονται.



>Ἐστωσαν ἀπὸ μονάδοξ ἐχῆζ ἀνάλογον ὅσοιδηποτοῦν ἀριθμοὶ οἱ Α, Β, Γ, Δ, Ε, Ζ, ὁ δὲ μετὰ τὴν μονάδα ὁ Α τετράγωνοξ >έστω; λέγω, <ὅτι καὶ οἱ λοιποὶ πάντεξ τετράγωνοι >έσσονται.

<Ὅτι μὲν ο>ὺν ὁ τρίτοξ ἀπὸ τῆξ μονάδοξ ὁ Β τετράγωνόξ ἐστὶ καὶ οἱ <ένα διαλείποντεξ πάντεξ, δέδεικται; λέγω [δὴ], <ὅτι καὶ οἱ λοιποὶ πάντεξ τετράγωνοι εἰσιν. ἐπεὶ γάρ οἱ Α, Β, Γ ἐχῆζ ἀνάλογόν εἰσιν, καὶ ἐστὶν ὁ Α τετράγωνοξ, καὶ ὁ Γ [>άρα] τετράγωνοξ ἐστίν. πάλιν, ἐπεὶ [καὶ] οἱ Β, Γ, Δ ἐχῆζ ἀνάλογόν εἰσιν, καὶ ἐστὶν ὁ Β τετράγωνοξ, καὶ ὁ Δ [>άρα] τετράγωνόξ ἐστίν. ὁμοίωξ δὴ δείχομεν, <ὅτι καὶ οἱ λοιποὶ πάντεξ τετράγωνοι εἰσιν.

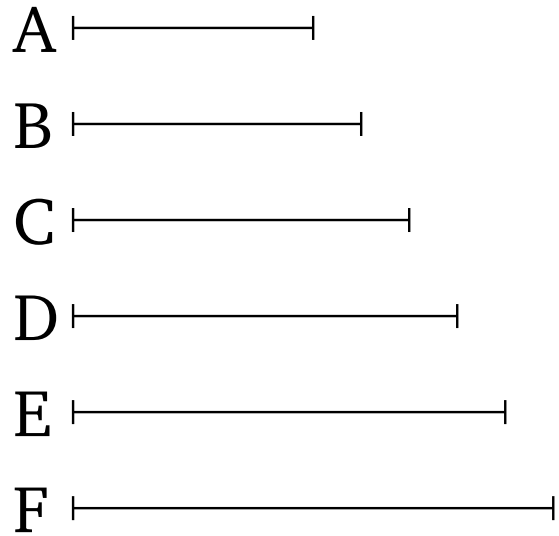
Ἀλλὰ δὴ >έστω ὁ Α κύβοξ; λέγω, <ὅτι καὶ οἱ λοιποὶ πάντεξ κύβοι εἰσιν.

<Ὅτι μὲν ο>ὺν ὁ τέταρτοξ ἀπὸ τῆξ μονάδοξ ὁ Γ κύβοξ ἐστὶ καὶ οἱ δύο διαλείποντεξ πάντεξ, δέδεικται; λέγω [δὴ], <ὅτι καὶ οἱ λοιποὶ πάντεξ κύβοι εἰσιν. ἐπεὶ γάρ ἐστιν ὡς ἡ μονὰξ πρὸξ τὸν Α, ο<ύτωξ ὁ Α πρὸξ τὸν Β, ἰσάκιξ ἄρα ἡ μονὰξ τὸν Α μετρεῖ καὶ ὁ Α τὸν Β. ἡ δὲ μονὰξ τὸν Α μετρεῖ κατὰ τὰξ ἐν αὐτῷ μονάδαξ; καὶ ὁ Α >άρα τὸν Β μετρεῖ κατὰ τὰξ ἐν αὐτῷ μονάδαξ; ὁ Α >άρα ἑαυτὸν πολλαπλασιάσαξ τὸν Β πεποίηκεν. καὶ ἐστὶν ὁ Α κύβοξ. ἔαν δὲ κύβοξ ἀριθμόξ ἑαυτὸν πολλαπλασιάσαξ ποιῇ τινα, ὁ γενόμενοξ κύβοξ ἐστίν; καὶ ὁ Β >άρα κύβοξ ἐστίν. καὶ ἐπεὶ τέσσαρεξ ἀριθμοὶ οἱ Α, Β, Γ, Δ ἐχῆζ ἀνάλογόν εἰσιν, καὶ ἐστὶν ὁ Α

measures the number A according to the units in A . And thus B measures C according to the units in A . A has thus made C (by) multiplying B . Therefore, since A has made B (by) multiplying itself, and has made C (by) multiplying B , C is thus cube. And since C, D, E, F are continuously proportional, and C is cube, F is thus also cube [Prop. 8.23]. And it was also shown (to be) square. Thus, the seventh (number) from the unit is (both) cube and square. So, similarly, we can show that all those (numbers after that) which leave an interval of five (numbers) are (both) cube and square. (Which is) the very thing it was required to show.

Proposition 9

If any multitude whatsoever of numbers is continuously proportional, (starting) from a unit, and the (number) after the unit is square, then all the remaining (numbers) will also be square. And if the (number) after the unit is cube, then all the remaining (numbers) will also be cube.



Let any multitude whatsoever of numbers, A, B, C, D, E, F , be continuously proportional, (starting) from a unit. And let the (number) after the unit, A , be square. I say that all the remaining (numbers) will also be square.

In fact, it has (already) been shown that the third (number) from the unit, B , is square, and all those (numbers after that) which leave an interval of one (number) [Prop. 9.8]. [So] I say that all the remaining (numbers) are also square. For since A, B, C are continuously proportional, and A (is) square, C is [thus] also square [Prop. 8.22]. Again, since B, C, D are [also] continuously proportional, and B is square, D is [thus] also square [Prop. 8.22]. So, similarly, we can show that all the remaining (numbers) are also square.

And so let A be cube. I say that all the remaining (numbers) are also cube.

In fact, it has (already) been shown that the fourth (number) from the unit, C , is cube, and all those (numbers after that) which leave an interval of two (numbers) [Prop. 9.8]. [So] I say that all the remaining (numbers) are also cube. For since as the unit is to A , so A (is) to B , the unit thus measures A the same number of times as A (measures) B . And the unit measures A according to the units in it. Thus, A also measures B according to the units in (A). A has thus made B (by) multiplying itself. And A is cube. And if a cube number makes some (number by) multiplying itself then the

κύβοξ, καὶ ὁ Δ >άρα κύβοξ ἐστίν. διὰ τὰ αὐτὰ δὴ καὶ ὁ E κύβοξ ἐστίν, καὶ ὁμοίωξ οἱ λοιποὶ πάντεξ κύβοι εἰσίν; <ὅπερ >έδει δεῖξαι.

10.

Ἐὰν ἀπὸ μονάδοξ ὅποσοιοῦν ἀριθμοὶ [ἐχῇξ] ἀνάλογον >ῶσιν, ὁ δὲ μετὰ τὴν μονάδα μὴ >ῇ| τετράγωνοξ, οὐδ> >άλλοξ οὐδεῖξ τετράγωνοξ >έσται θωριξ τοῦ τρίτου ἀπὸ τῇξ μονάδοξ καὶ τῶν <ένα διαλειπόντων πάντων. καὶ ἔαν ὁ μετὰ τὴν μονάδα κύβοξ μὴ >ῇ|, οὐδὲ >άλλοξ οὐδεῖξ κύβοξ >έσται θωριξ τοῦ τετάρτου ἀπὸ τῇξ μονάδοξ καὶ τῶν δύο διαλειπόντων πάντων.

A —————

B —————

Γ —————

Δ —————

E —————

Z —————

>Ἐστωσαν ἀπὸ μονάδοξ ἐχῇξ ἀνάλογον ὅσοιοιποτοῦν ἀριθμοὶ οἱ $A, B, \Gamma, \Delta, E, Z$, ὁ μετὰ τὴν μονάδα ὁ A μὴ >έστω τετράγωνοξ; λέγω, <ὅτι οὐδὲ >άλλοξ οὐδεῖξ τετράγωνοξ >έσται θωριξ τοῦ τρίτου ἀπὸ τῇξ μονάδοξ [καὶ τῶν <ένα διαλειπόντων].

Εἰ γὰρ δυνατόν, >έστω ὁ Γ τετράγωνοξ. >έστι δὲ καὶ ὁ B τετράγωνοξ; οἱ B, Γ >άρα πρὸξ ἀλλήλουξ λόγον >έθουσιν, <ὄν τετράγωνοξ ἀριθμὸξ πρὸξ τετράγωνον ἀριθμόν. καὶ ἔστιν ὡς ὁ B πρὸξ τὸν Γ , ὁ A πρὸξ τὸν B ; οἱ A, B >άρα

created (number) is cube [Prop. 9.3]. Thus, B is also cube. And since the four numbers A, B, C, D are continuously proportional, and A is cube, D is thus also cube [Prop. 8.23]. So, for the same (reasons), E is also cube, and, similarly, all the remaining (numbers) are cube. (Which is) the very thing it was required to show.

Proposition 10

If any multitude whatsoever of numbers is [continuously] proportional, (starting) from a unit, and the (number) after the unit is not square, then no other (number) will be square either, apart from the third from the unit, and all those (numbers after that) which leave an interval of one (number). And if the (number) after the unit is not cube, then no other (number) will be cube either, apart from the fourth from the unit, and all those (numbers after that) which leave an interval of two (numbers).

A —————

B —————

C —————

D —————

E —————

F —————

Let any multitude whatsoever of numbers, A, B, C, D, E, F , be continuously proportional, (starting) from a unit. And let the (number) after the unit, A , not be square. I say that no other (number) will be square either, apart from the third from the unit [and (all) those (numbers after that) which leave an interval of one (number)].

For, if possible, let C be square. And B is also square [Prop. 9.8]. Thus, B and C have to one another (the) ratio which (some) square number (has) to (some other) square number. And as B is to C , (so) A (is) to B . Thus, A and B

πρὸς ἀλλήλους λόγον >έθουσιν, <ὃν τετράγωνοξ ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν; <ὥστε οἱ A, B <ὅμοιοι ἐπίπεδοι εἰσιν. καὶ ἐστι τετράγωνοξ ὁ B; τετράγωνοξ >άρα ἐστὶ καὶ ὁ A; <ὅπερ οὐθ ὑπέκειτο. οὐκ >άρα ὁ Γ τετράγωνόξ ἐστίν. ὁμοίωξ δὴ δείχομεν, <ὅτι οὐδ> >άλλοξ οὐδεὶξ τετράγωνόξ ἐστὶ θωρίξ τοῦ τρίτου ἀπὸ τῆξ μονάδοξ καὶ τῶν <ένα διαλειπόντων.

Ἀλλὰ δὴ μὴ >έστω ὁ A κύβοξ. λέγω, <ὅτι οὐδ> >άλλοξ οὐδεὶξ κύβοξ >έσται θωρίξ τοῦ τετάρτου ἀπὸ τῆξ μονάδοξ καὶ τῶν δύο διαλειπόντων.

Εἰ γὰρ δυνατόν, >έστω ὁ Δ κύβοξ. >έστι δὲ καὶ ὁ Γ κύβοξ; τέταρτοξ γὰρ ἐστίν ἀπὸ τῆξ μονάδοξ. καὶ ἐστίν ὡξ ὁ Γ πρὸξ τὸν Δ, ὁ B πρὸξ τὸν Γ; καὶ ὁ B >άρα πρὸξ τὸν Γ λόγον >έθει, <ὃν κύβοξ πρὸξ κύβον. καὶ ἐστίν ὁ Γ κύβοξ; καὶ ὁ B >άρα κύβοξ ἐστίν. καὶ ἐπεὶ ἐστίν ὡξ ἡ μονάξ πρὸξ τὸν A, ὁ A πρὸξ τὸν B, ἡ δὲ μονάξ τὸν A μετρεῖ κατὰ τὰξ ἐν αὐτῶ| μονάδαξ, καὶ ὁ A >άρα τὸν B μετρεῖ κατὰ τὰξ ἐν αὐτῶ| μονάδαξ; ὁ A >άρα ἐαυτὸν πολλαπλασιάσαξ κύβον τὸν B πεποίηκεν. ἐὰν δὲ ἀριθμὸξ ἐαυτὸν πολλαπλασιάσαξ κύβον ποιῇ, καὶ αὐτὸξ κύβοξ >έσται. κύβοξ >άρα καὶ ὁ A; <ὅπερ οὐθ ὑπόκειται. οὐκ >άρα ὁ Δ κύβοξ ἐστίν. ὁμοίωξ δὴ δείχομεν, <ὅτι οὐδ> >άλλοξ οὐδεὶξ κύβοξ ἐστὶ θωρίξ τοῦ τετάρτου ἀπὸ τῆξ μονάδοξ καὶ τῶν δύο διαλειπόντων; <ὅπερ >έδει δείχαι.

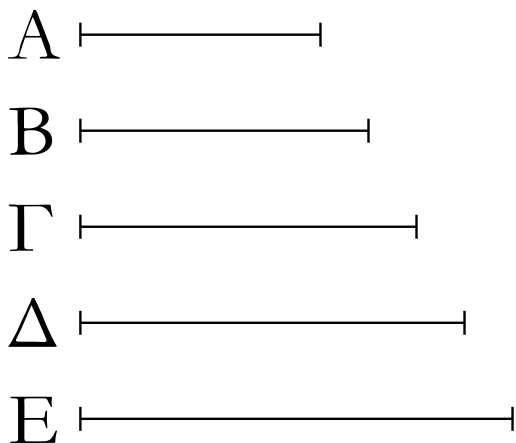
have to one another (the) ratio which (some) square number has to (some other) square number. Hence, *A* and *B* are similar plane (numbers) [Prop. 8.26]. And *B* is square. Thus, *A* is also square. The very opposite thing was assumed. *C* is thus not square. So, similarly, we can show that no other (number is) square either, apart from the third from the unit, and (all) those (numbers after that) which leave an interval of one (number).

And so let *A* not be cube. I say that no other (number) will be cube either, apart from the fourth from the unit, and (all) those (numbers after that) which leave an interval of two (numbers).

For, if possible, let *D* be cube. And *C* is also cube [Prop. 9.8]. For it is the fourth (number) from the unit. And as *C* is to *D*, (so) *B* (is) to *C*. And *B* thus has to *C* the ratio which (some) cube (number has) to (some other) cube (number). And *C* is cube. Thus, *B* is also cube [Props. 7.13, 8.25]. And since as the unit is to *A*, (so) *A* (is) to *B*, and the unit measures *A* according to the units in it, *A* thus also measures *B* according to the units in (*A*). Thus, *A* has made the cube (number) *B* (by) multiplying itself. And if a number makes a cube (number by) multiplying itself then it itself will be cube [Prop. 9.6]. Thus, *A* (is) also cube. The very opposite thing was assumed. Thus, *D* is not cube. So, similarly, we can show that no other (number) is cube either, apart from the fourth from the unit, and (all) those (numbers after that) which leave an interval of two (numbers). (Which is) the very thing it was required to show.

11.

Ἐὰν ἀπὸ μονάδοξ ὅποσοιοῦν ἀριθμοὶ ἐχῇξ ἀνάλογον >ῶσιν, ὁ ἐλάττων τὸν μείζονα μετρεῖ κατὰ τινὰ τῶν ὑπαρθόντων ἐν τοῖξ ἀνάλογον ἀριθμοῖξ.

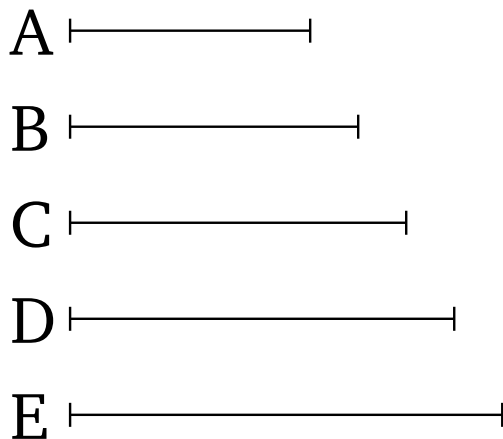


>Ἔστωσαν ἀπὸ μονάδοξ τῆξ A ὅποσοιοῦν ἀριθμοὶ ἐχῇξ ἀνάλογον οἱ B, Γ, Δ, E; λέγω, <ὅτι τῶν B, Γ, Δ, E ὁ ἐλάχιστοξ ὁ B τὸν E μετρεῖ κατὰ τινὰ τῶν Γ, Δ.

Ἐπεὶ γὰρ ἐστίν ὡξ ἡ A μονάξ πρὸξ τὸν B, ο<ύτωξ ὁ Δ πρὸξ τὸν E, ισάκιξ >άρα ἡ A μονάξ τὸν B ἀριθμόν μετρεῖ καὶ ὁ Δ τὸν E; ἐναλλάξ >άρα ισάκιξ ἡ A μονάξ τὸν Δ μετρεῖ καὶ ὁ B τὸν E. ἡ δὲ A μονάξ τὸν Δ μετρεῖ κατὰ τὰξ ἐν αὐτῶ| μονάδαξ; καὶ ὁ B >άρα τὸν E μετρεῖ κατὰ τὰξ ἐν τῶ| Δ μονάδαξ; <ὥστε ὁ ἐλάσσων ὁ B τὸν μείζονα τὸν E μετρεῖ κατὰ τινὰ ἀριθμόν τῶν ὑπαρθόντων ἐν τοῖξ ἀνάλογον ἀριθμοῖξ.

Proposition 11

If any multitude whatsoever of numbers is continuously proportional, (starting) from a unit, then a lesser (number) measures a greater according to some existing (number) among the proportional numbers.



Let any multitude whatsoever of numbers, *B*, *C*, *D*, *E*, be continuously proportional, (starting) from the unit *A*. I say that, for *B*, *C*, *D*, *E*, the least (number), *B*, measures *E* according to some (one) of *C*, *D*.

For since as the unit *A* is to *B*, so *D* (is) to *E*, the unit *A* thus measures the number *B* the same number of times as *D* (measures) *E*. Thus, alternately, the unit *A* measures *D* the same number of times as *B* (measures) *E* [Prop. 7.15]. And the unit *A* measures *D* according to the units in it. Thus, *B* also measures *E* according to the units in *D*. Hence, the lesser (number) *B* measures the greater *E* according to some existing number among the proportional numbers

Πόρισμα.

Καὶ φανερόν, <ὅτι <ἦν >έθει τάχιν ὁ μετρῶν ἀπὸ μονάδοξ, τὴν αὐτὴν >έθει καὶ ὁ καθ' <ὃν μετρεῖ ἀπὸ τοῦ μετρούμενου ἐπὶ τὸ πρὸ αὐτοῦ. <ὅπερ >έδει δεῖχαι.

(namely, D).

Corollary

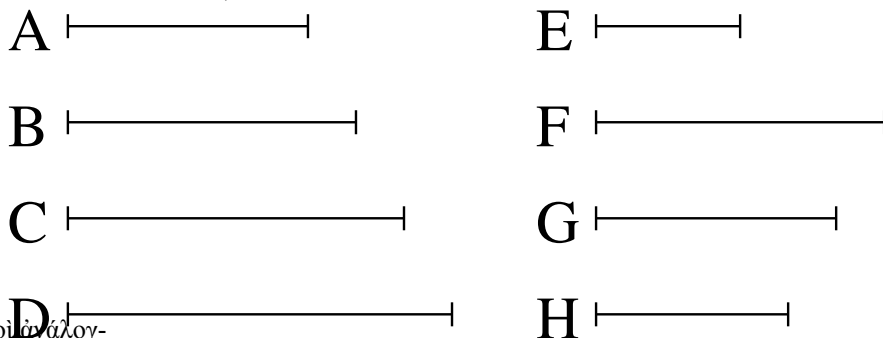
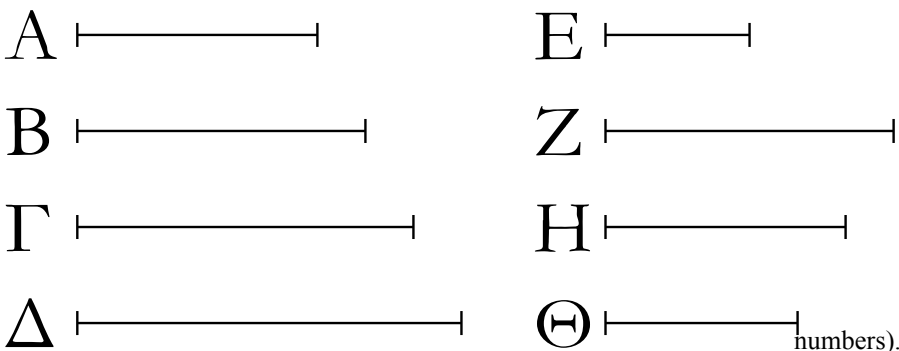
And (it is) clear that what(ever relative) place the measuring (number) has from the unit, the (number) according to which it measures has the same (relative) place from the measured (number), in (the direction of the number) before it. (Which is) the very thing it was required to show.

12.

Ἐὰν ἀπὸ μονάδοξ ὅποσοιοῦν ἀριθμοὶ ἐχῇξ ἀνάλογον >ῶσιν, ὑφ' <όσων >ὰν ὁ >έσθατοξ πρώτων ἀριθμῶν μετρήται, ὑπὸ τῶν αὐτῶν καὶ ὁ παρὰ τὴν μονάδα μετρηθήσεται.

Proposition 12

If any multitude whatsoever of numbers is continuously proportional, (starting) from a unit, then however many prime numbers the last (number) is measured by, the (number) next to the unit will also be measured by the same (prime



>Ἐστώσαν ἀπὸ μονάδοξ ὅποσοιδηποτοῦν ἀριθμοὶ ἀνάλογον οἱ A, B, Γ, Δ ; λέγω, <ὅτι ὑφ' <όσων >ὰν ὁ Δ πρώτων ἀριθμῶν μετρήται, ὑπὸ τῶν αὐτῶν καὶ ὁ A μετρηθήσεται. Μετρεῖσθω γὰρ ὁ Δ ὑπὸ τινος πρώτου ἀριθμοῦ τοῦ E ; λέγω, <ὅτι ὁ E τὸν A μετρεῖ. μὴ γάρ; καὶ ἐστὶν ὁ E πρώτοξ, <ἀπαξ δὲ πρώτοξ ἀριθμὸξ πρὸξ <άπαντα, <ὃν μὴ μετρεῖ, πρώτοξ ἐστὶν; οἱ E, A >άρα πρώτοι πρὸξ ἀλλήλουξ εἰσίν. καὶ ἐπεὶ ὁ E τὸν Δ μετρεῖ, μετρεῖτω αὐτὸν κατὰ τὸν Z ; ὁ E >άρα τὸν Z πολλαπλασιάσαξ τὸν Δ πεποίηκεν. πάλιν, ἐπεὶ ὁ A τὸν Δ μετρεῖ κατὰ τὰξ ἐν τῷ Γ μονάδαξ, ὁ A >άρα τὸν Γ πολλαπλασιάσαξ τὸν Δ πεποίηκεν. ἀλλὰ μὴν καὶ ὁ E τὸν Z πολλαπλασιάσαξ τὸν Δ πεποίηκεν; ὁ >άρα ἐκ τῶν A, Γ >ίσοξ ἐστὶ τῷ ἐκ τῶν E, Z . >έστιν >άρα ὡξ ὁ A πρὸξ τὸν E , ὁ Z πρὸξ τὸν Γ . οἱ δὲ A, E πρώτοι, οἱ δὲ πρώτοι καὶ ἐλάχιστοι, οἱ δὲ ἐλάχιστοι μετροῦσι τοῦξ τὸν αὐτὸν λόγον >έθονταξ ἰσάκιξ <ὃ τε ἡγούμενοξ τὸν ἡγούμενον καὶ ὁ ἐπόμενοξ τὸν ἐπόμενον; μετρεῖ >άρα ὁ E τὸν Γ . μετρεῖτω αὐτὸν κατὰ τὸν H ; ὁ E >άρα τὸν H πολλαπλασιάσαξ τὸν Γ πεποίηκεν. ἀλλὰ μὴν διὰ τὸ πρὸ τούτου καὶ ὁ A τὸν B πολλαπλασιάσαξ τὸν Γ πεποίηκεν. ὁ >άρα ἐκ τῶν A, B >ίσοξ ἐστὶ τῷ ἐκ τῶν E, H . >έστιν >άρα ὡξ ὁ A πρὸξ τὸν E , ὁ H πρὸξ τὸν B . οἱ δὲ A, E πρώτοι, οἱ δὲ πρώτοι καὶ ἐλάχιστοι, οἱ δὲ ἐλάχιστοι ἀριθμοὶ μετροῦσι τοῦξ τὸν αὐτὸν λόγον >έθονταξ αὐτοῖξ ἰσάκιξ <ὃ τε ἡγούμενοξ τὸν ἡγούμενον καὶ ὁ ἐπόμενοξ τὸν ἐπόμενον; μετρεῖ >άρα ὁ E τὸν B . μετρεῖτω αὐτὸν κατὰ τὸν Θ ; ὁ E

Let any multitude whatsoever of numbers, A, B, C, D , be (continuously) proportional, (starting) from a unit. I say that however many prime numbers D is measured by, A will also be measured by the same (prime numbers).

For let D be measured by some prime number E . I say that E measures A . For (suppose it does) not. E is prime, and every prime number is prime to every number which it does not measure [Prop. 7.29]. Thus, E and A are prime to one another. And since E measures D , let it measure it according to F . Thus, E has made D (by) multiplying F . Again, since A measures D according to the units in C [Prop. 9.11 corr.], A has thus made D (by) multiplying C . But, in fact, E has also made D (by) multiplying F . Thus, the (number created) from (multiplying) A, C is equal to the (number created) from (multiplying) E, F . Thus, as A is to E , (so) F (is) to C [Prop. 7.19]. And A and E (are) prime (to one another), and (numbers) prime (to one another are) also the least (of those numbers having the same ratio as them) [Prop. 7.21], and the least (numbers) measure those (numbers) having the same ratio as them an equal number of times, the leading (measuring) the leading, and the following the following [Prop. 7.20]. Thus, E measures C . Let it measure it according to G . Thus, E has made C (by) multiplying G . But, in fact, via the (proposition) before this,

>άρα τὸν Θ πολλαπλασιάσας τὸν Β πεποίηκεν. ἀλλὰ μὴν καὶ ὁ Α ἐαυτὸν πολλαπλασιάσας τὸν Β πεποίηκεν; ὁ >άρα ἐκ τῶν Ε, Θ >ίσοξ ἐστὶ τῷ ἀπὸ τοῦ Α. >έστιν >άρα ὡς ὁ Ε πρὸς τὸν Α, ὁ Α πρὸς τὸν Θ. οἱ δὲ Α, Ε πρῶτοι, οἱ δὲ πρῶτοι καὶ ἐλάχιστοι, οἱ δὲ ἐλάχιστοι μετροῦσι τοὺς τὸν αὐτὸν λόγον >έθονταξ ἰσάκιξ <ὁ ἡγούμενοξ τὸν ἡγούμενον καὶ ὁ ἐπόμενοξ τὸν ἐπόμενον; μετρεῖ >άρα ὁ Ε τὸν Α ὡς ἡγούμενοξ ἡγούμενον. ἀλλὰ μὴν καὶ οὐ μετρεῖ; <ὅπερ ἀδύνατον. οὐκ >άρα οἱ Ε, Α πρῶτοι πρὸς ἀλλήλουξ εἰσίν. σύνθετοι >άρα. οἱ δὲ σύνθετοι ὑπὸ [πρώτου] ἀριθμοῦ τινοξ μετροῦνται. καὶ ἐπεὶ ὁ Ε πρῶτοξ ὑπόκειται, ὁ δὲ πρῶτοξ ὑπὸ ἐτέρου ἀριθμοῦ οὐ μετρεῖται >ἢ ὑφ> ἐαυτοῦ, ὁ Ε >άρα τοὺς Α, Ε μετρεῖ; <ὥστε ὁ Ε τὸν Α μετρεῖ. μετρεῖ δὲ καὶ τὸν Δ; ὁ Ε >άρα τοὺς Α, Δ μετρεῖ. ὁμοίωξ δὴ δείχομεν, <ὅτι ὑφ> <όσων> ἂν ὁ Δ πρώτων ἀριθμῶν μετρήται, ὑπὸ τῶν αὐτῶν καὶ ὁ Α μετρηθῇσεται; <ὅπερ >έδει δεῖχαι.

A has also made *C* (by) multiplying *B* [Prop. 9.11 corr.]. Thus, the (number created) from (multiplying) *A*, *B* is equal to the (number created) from (multiplying) *E*, *G*. Thus, as *A* is to *E*, (so) *G* (is) to *B* [Prop. 7.19]. And *A* and *E* (are) prime (to one another), and (numbers) prime (to one another are) also the least (of those numbers having the same ratio as them) [Prop. 7.21], and the least (numbers) measure those (numbers) having the same ratio as them an equal number of times, the leading (measuring) the leading, and the following the following [Prop. 7.20]. Thus, *E* measures *B*. Let it measure it according to *H*. Thus, *E* has made *B* (by) multiplying *H*. But, in fact, *A* has also made *B* (by) multiplying itself [Prop. 9.8]. Thus, the (number created) from (multiplying) *E*, *H* is equal to the (square) on *A*. Thus, as *E* is to *A*, (so) *A* (is) to *H* [Prop. 7.19]. And *A* and *E* are prime (to one another), and (numbers) prime (to one another are) also the least (of those numbers having the same ratio as them) [Prop. 7.21], and the least (numbers) measure those (numbers) having the same ratio as them an equal number of times, the leading (measuring) the leading, and the following the following [Prop. 7.20]. Thus, *E* measures *A*, as the leading (measuring the) leading. But, in fact, (*E*) also does not measure (*A*). The very thing (is) impossible. Thus, *E* and *A* are not prime to one another. Thus, (they are) composite (to one another). And (numbers) composite (to one another) are (both) measured by some [prime] number [Def. 7.14]. And since *E* is assumed (to be) prime, and a prime (number) is not measured by another number (other) than itself [Def. 7.11], *E* thus measures (both) *A* and *E*. Hence, *E* measures *A*. And it also measures *D*. Thus, *E* measures (both) *A* and *D*. So, similarly, we can show that however many prime numbers *D* is measured by, *A* will also be measured by the same (prime numbers). (Which is) the very thing it was required to show.

13.

Proposition 13

Ἐὰν ἀπὸ μονάδοξ ὅποσοιοῦν ἀριθμοὶ ἐχῇξ ἀνάλογον >ῶσιν, ὁ δὲ μετὰ τὴν μονάδα πρῶτοξ >ῆ], ὁ μέγιστοξ ὑπ> οὐδενὸξ [>ἄλλου] μετρηθῇσεται παρὲξ τῶν ὑπαρθόντων ἐν τοῖξ ἀνάλογον ἀριθμοῖξ.
>Ἐστῶσαν ἀπὸ μονάδοξ ὅποσοιοῦν ἀριθμοὶ ἐχῇξ ἀνάλογον οἱ Α, Β, Γ, Δ, ὁ δὲ μετὰ τὴν μονάδα ὁ Α πρῶτοξ >έστω; λέγω, <ὅτι ὁ μέγιστοξ αὐτῶν ὁ Δ ὑπ> οὐδενὸξ >ἄλλου μετρηθῇσεται παρὲξ τῶν Α, Β, Γ.

If any multitude whatsoever of numbers is continuously proportional, (starting) from a unit, and the (number) after the unit is prime, then the greatest (number) will be measured by no [other] (numbers) except (numbers) existing among the proportional numbers.

Let any multitude whatsoever of numbers, *A*, *B*, *C*, *D*, be continuously proportional, (starting) from a unit. And let the (number) after the unit, *A*, be prime. I say that the greatest of them, *D*, will be measured by no other (numbers) except *A*, *B*, *C*.

Α ———|

Ε — Α ———|

Ε ———|

Β ———|

Ζ — Β ———|

Φ ———|

Γ ———|

Η — C ———|

Γ ———|

Δ ———|

Θ — D ———|

Η ———|

Εἰ γὰρ δυνατόν, μετρεῖσθω ὑπὸ τοῦ Ε, καὶ ὁ Ε μηδενὶ τῶν Α, Β, Γ >έστω ὁ αὐτόξ. φανερόν δὴ, <ὅτι ὁ Ε πρῶτοξ ο>ὐκ ἐστιν. εἰ γὰρ ὁ Ε πρῶτοξ ἐστὶ καὶ μετρεῖ τὸν Δ, καὶ τὸν Α

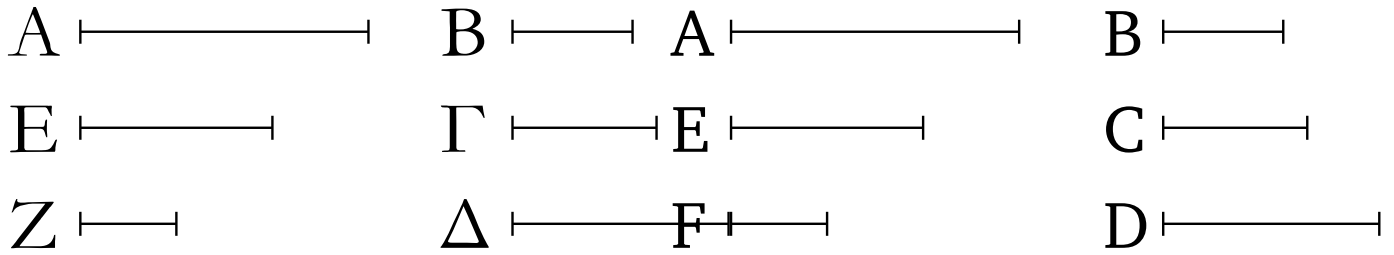
For, if possible, let it be measured by *E*, and let *E* not be the same as one of *A*, *B*, *C*. So it is clear that *E* is not prime. For if *E* is prime, and measures *D*, then it will also mea-

μετρήσει πρῶτον >όντα μὴ >ὦν αὐτῶ| ὁ αὐτόξ; <όπερ ἐστὶν ἀδύνατον. οὐκ >άρα ὁ Ε πρῶτόξ ἐστὶν. σύνθετοξ >άρα. πᾶξ δὲ σύνθετοξ ἀριθμοξ ὑπὸ πρῶτου τινὸξ ἀριθμοῦ μετρεῖται; ὁ Ε >άρα ὑπὸ πρῶτου τινὸξ ἀριθμοῦ μετρεῖται. λέγω δὴ, <ότι ὑπ> οὐδενὸξ >άλλου πρῶτου μετρηθήσεται πλὴν τοῦ Α. εἰ γὰρ ὑφ> ἐτέρου μετρεῖται ὁ Ε, ὁ δὲ Ε τὸν Δ μετρεῖ, κάκεινοξ >άρα τὸν Δ μετρήσει; <ώστε καὶ τὸν Α μετρήσει πρῶτον >όντα μὴ >ὦν αὐτῶ| ὁ αὐτόξ; <όπερ ἐστὶν ἀδύνατον. ὁ Α >άρα τὸν Ε μετρεῖ. καὶ ἐπεὶ ὁ Ε τὸν Δ μετρεῖ, μετρεῖτω αὐτὸν κατὰ τὸν Ζ. λέγω, <ότι ὁ Ζ οὐδενὶ τῶν Α, Β, Γ ἐστὶν ὁ αὐτόξ. εἰ γὰρ ὁ Ζ ἐνὶ τῶν Α, Β, Γ ἐστὶν ὁ αὐτόξ καὶ μετρεῖ τὸν Δ κατὰ τὸν Ε, καὶ ε<ίξ >άρα τῶν Α, Β, Γ τὸν Δ μετρεῖ κατὰ τὸν Ε. ἀλλὰ εἰξ τῶν Α, Β, Γ τὸν Δ μετρεῖ κατὰ τινὰ τῶν Α, Β, Γ; καὶ ὁ Ε >άρα ἐνὶ τῶν Α, Β, Γ ἐστὶν ὁ αὐτόξ; <όπερ οὐθ ὑπόκειται. οὐκ >άρα ὁ Ζ ἐνὶ τῶν Α, Β, Γ ἐστὶν ὁ αὐτόξ. ὁμοίωξ δὴ δείχομεν, <ότι μετρεῖται ὁ Ζ ὑπὸ τοῦ Α, δεικνύντεξ πάλιν, <ότι ὁ Ζ ο>ύκ ἐστι πρῶτοξ. εἰ γὰρ, καὶ μετρεῖ τὸν Δ, καὶ τὸν Α μετρήσει πρῶτον >όντα μὴ >ὦν αὐτῶ| ὁ αὐτόξ; <όπερ ἐστὶν ἀδύνατον; οὐκ >άρα πρῶτόξ ἐστὶν ὁ Ζ; σύνθετοξ >άρα. <άπαξ δὲ σύνθετοξ ἀριθμοξ ὑπὸ πρῶτου τινὸξ ἀριθμοῦ μετρεῖται; ὁ Ζ >άρα ὑπὸ πρῶτου τινὸξ ἀριθμοῦ μετρεῖται. λέγω δὴ, <ότι ὑφ> ἐτέρου πρῶτου οὐ μετρηθήσεται πλὴν τοῦ Α. εἰ γὰρ <έτερόξ τιξ πρῶτοξ τὸν Ζ μετρεῖ, ὁ δὲ Ζ τὸν Δ μετρεῖ, κάκεινοξ >άρα τὸν Δ μετρήσει; <ώστε καὶ τὸν Α μετρήσει πρῶτον >όντα μὴ >ὦν αὐτῶ| ὁ αὐτόξ; <όπερ ἐστὶν ἀδύνατον. ὁ Α >άρα τὸν Ζ μετρεῖ. καὶ ἐπεὶ ὁ Ε τὸν Δ μετρεῖ κατὰ τὸν Ζ, ὁ Ε >άρα τὸν Ζ πολλαπλασιάσαξ τὸν Δ πεποιήκεν. ἀλλὰ μὴν καὶ ὁ Α τὸν Γ πολλαπλασιάσαξ τὸν Δ πεποιήκεν; ὁ >άρα ἐκ τῶν Α, Γ >ίσοξ ἐστὶ τῶ| ἐκ τῶν Ε, Ζ. ἀνάλογον >άρα ἐστὶν ὡξ ὁ Α πρὸξ τὸν Ε, ο<ύτωξ ὁ Ζ πρὸξ τὸν Γ. ὁ δὲ Α τὸν Ε μετρεῖ; καὶ ὁ Ζ >άρα τὸν Γ μετρεῖ. μετρεῖτω αὐτὸν κατὰ τὸν Η. ὁμοίωξ δὴ δείχομεν, <ότι ὁ Η οὐδενὶ τῶν Α, Β ἐστὶν ὁ αὐτόξ, καὶ <ότι μετρεῖται ὑπὸ τοῦ Α. καὶ ἐπεὶ ὁ Ζ τὸν Γ μετρεῖ κατὰ τὸν Η, ὁ Ζ >άρα τὸν Η πολλαπλασιάσαξ τὸν Γ πεποιήκεν. ἀλλὰ μὴν καὶ ὁ Α τὸν Β πολλαπλασιάσαξ τὸν Γ πεποιήκεν; ὁ >άρα ἐκ τῶν Α, Β >ίσοξ ἐστὶ τῶ| ἐκ τῶν Ζ, Η. ἀνάλογον >άρα ὡξ ὁ Α πρὸξ τὸν Ζ, ὁ Η πρὸξ τὸν Β. μετρεῖ δὲ ὁ Α τὸν Ζ; μετρεῖ >άρα καὶ ὁ Η τὸν Β. μετρεῖτω αὐτὸν κατὰ τὸν Θ. ὁμοίωξ δὴ δείχομεν, <ότι ὁ Θ τῶ| Α οὐκ >ἐστὶν ὁ αὐτόξ. καὶ ἐπεὶ ὁ Η τὸν Β μετρεῖ κατὰ τὸν Θ, ὁ Η >άρα τὸν Θ πολλαπλασιάσαξ τὸν Β πεποιήκεν. ἀλλὰ μὴν καὶ ὁ Α ἑαυτὸν πολλαπλασιάσαξ τὸν Β πεποιήκεν; ὁ >άρα ὑπὸ Θ, Η >ίσοξ ἐστὶ τῶ| ἀπὸ τοῦ Α τετραγώνω; >ἐστὶν >άρα ὡξ ὁ Θ πρὸξ τὸν Α, ὁ Α πρὸξ τὸν Η. μετρεῖ δὲ ὁ Α τὸν Η; μετρεῖ >άρα καὶ ὁ Θ τὸν Α πρῶτον >όντα μὴ >ὦν αὐτῶ| ὁ αὐτόξ; <όπερ >άτοπον. οὐκ >άρα ὁ μέγιστοξ ὁ Δ ὑπὸ ἐτέρου ἀριθμοῦ μετρηθήσεται παρὲξ τῶν Α, Β, Γ; <όπερ >έδει δεῖχαι.

sure A , (despite A) being prime (and) not being the same as it [Prop. 9.12]. The very thing is impossible. Thus, E is not prime. Thus, (it is) composite. And every composite number is measured by some prime number [Prop. 7.31]. Thus, E is measured by some prime number. So I say that it will be measured by no other prime number than A . For if E is measured by another (prime number), and E measures D , then this (prime number) will thus also measure D . Hence, it will also measure A , (despite A) being prime (and) not being the same as it [Prop. 9.12]. The very thing is impossible. Thus, A measures E . And since E measures D , let it measure it according to F . I say that F is not the same as one of A, B, C . For if F is the same as one of A, B, C , and measures D according to E , then one of A, B, C thus also measures D according to E . But one of A, B, C (only) measures D according to some (one) of A, B, C [Prop. 9.11]. And thus E is the same as one of A, B, C . The very opposite thing was assumed. Thus, F is not the same as one of A, B, C . Similarly, we can show that F is measured by A , (by) again showing that F is not prime. For if (F is prime), and measures D , then it will also measure A , (despite A) being prime (and) not being the same as it [Prop. 9.12]. The very thing is impossible. Thus, F is not prime. Thus, (it is) composite. And every composite number is measured by some prime number [Prop. 7.31]. Thus, F is measured by some prime number. So I say that it will be measured by no other prime number than A . For if some other prime (number) measures F , and F measures D , then this (prime number) will thus also measure D . Hence, it will also measure A , (despite A) being prime (and) not being the same as it [Prop. 9.12]. The very thing is impossible. Thus, A measures F . And since E measures D according to F , E has thus made D (by) multiplying F . But, in fact, A has also made D (by) multiplying C [Prop. 9.11 corr.]. Thus, the (number created) from (multiplying) A, C is equal to the (number created) from (multiplying) E, F . Thus, proportionally, as A is to E , so F (is) to C [Prop. 7.19]. And A measures E . Thus, F also measures C . Let it measure it according to G . So, similarly, we can show that G is not the same as one of A, B , and that it is measured by A . And since F measures C according to G , F has thus made C (by) multiplying G . But, in fact, A has also made C (by) multiplying B [Prop. 9.11 corr.]. Thus, the (number created) from (multiplying) A, B is equal to the (number created) from (multiplying) F, G . Thus, proportionally, as A (is) to F , so G (is) to B [Prop. 7.19]. And A measures F . Thus, G also measures B . Let it measure it according to H . So, similarly, we can show that H is not the same as A . And since G measures B according to H , G has thus made B (by) multiplying H . But, in fact, A has also made B (by) multiplying itself [Prop. 9.8]. Thus, the (number created) from (multiplying) H, G is equal to the square on A . Thus, as H is to A , (so) A (is) to G [Prop. 7.19]. And A measures G . Thus, H also measures A , (despite A) being prime (and) not being the same as it. The very thing (is) absurd. Thus, the greatest (number) D cannot be measured by another (number) except (one of) A, B, C . (Which is) the very thing it was required to show.

Ἐὰν ἐλάχιστος ἀριθμὸς ὑπὸ πρώτων ἀριθμῶν μετρηται, ὅπ' οὐδενὸς ἄλλου πρώτου ἀριθμοῦ μετρηθήσεται παρὲξ τῶν ἐξ ἀρθῆς μετρούντων.

Ἐλάχιστος γὰρ ἀριθμὸς ὁ A ὑπὸ πρώτων ἀριθμῶν τῶν B, Γ, Δ μετρεῖσθαι; λέγω, <ὅτι ὁ A ὅπ' οὐδενὸς ἄλλου πρώτου ἀριθμοῦ μετρηθήσεται παρὲξ τῶν B, Γ, Δ .



Εἰ γὰρ δυνατόν, μετρεῖσθαι ὑπὸ πρώτου τοῦ E , καὶ ὁ E μηδενὶ τῶν B, Γ, Δ >έστω ὁ αὐτός. καὶ ἐπεὶ ὁ E τὸν A μετρεῖ, μετρεῖται αὐτὸν κατὰ τὸν Z ; ὁ E >άρα τὸν Z πολλαπλασιάσας τὸν A πεποίηκεν. καὶ μετρεῖται ὁ A ὑπὸ πρώτων ἀριθμῶν τῶν B, Γ, Δ . ἔαν δὲ δύο ἀριθμοὶ πολλαπλασιάσαντες ἀλλήλους ποιῶσιν τινα, τὸν δὲ γενόμενον ἐξ αὐτῶν μετρήῃ τις πρῶτος ἀριθμὸς, καὶ <ένα τῶν ἐξ ἀρθῆς μετρήσει; οἱ B, Γ, Δ >άρα <ένα τῶν E, Z μετρήσουσιν. τὸν μὲν $ο>$ ὦν E οὐ μετρήσουσιν; ὁ γὰρ E πρῶτός ἐστι καὶ οὐδενὶ τῶν B, Γ, Δ ὁ αὐτός. τὸν Z >άρα μετροῦσιν ἐλάσσονα >όντα τοῦ A ; <ὅπερ ἀδύνατον. ὁ γὰρ A ὑπόκειται ἐλάχιστος ὑπὸ τῶν B, Γ, Δ μετρούμενος. οὐκ >άρα τὸν A μετρήσει πρῶτος ἀριθμὸς παρὲξ τῶν B, Γ, Δ ; <ὅπερ >έδει δεῖχαι.

If a least number is measured by (some) prime numbers then it will not be measured by any other prime number except (one of) the original measuring (numbers).

For let A be the least number measured by the prime numbers B, C, D . I say that A will not be measured by any other prime number except (one of) B, C, D .

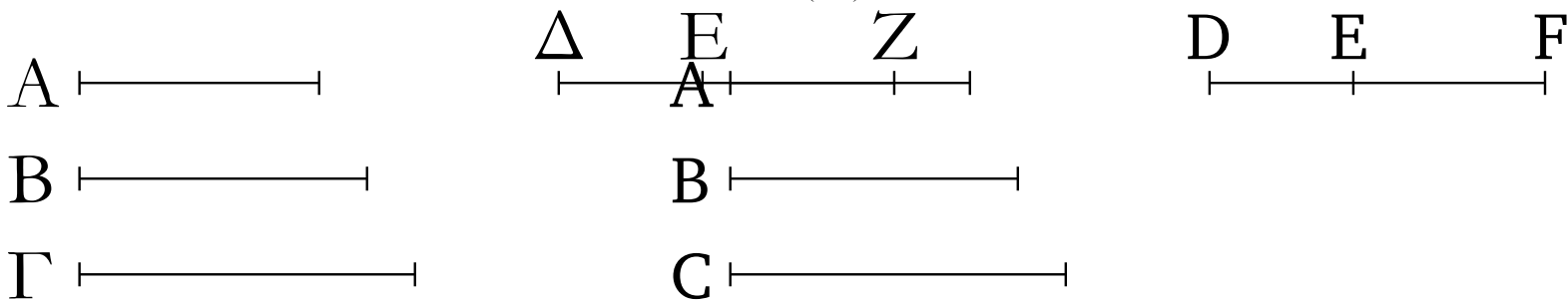
For, if possible, let it be measured by the prime (number) E . And let E not be the same as one of B, C, D . And since E measures A , let it measure it according to F . Thus, E has made A (by) multiplying F . And A is measured by the prime numbers B, C, D . And if two numbers make some (number by) multiplying one another, and some prime number measures the number created from them, then (the prime number) will also measure one of the original (numbers) [Prop. 7.30]. Thus, B, C, D will measure one of E, F . In fact, they do not measure E . For E is prime, and not the same as one of B, C, D . Thus, they (all) measure F , which is less than A . The very thing (is) impossible. For A was assumed (to be) the least (number) measured by B, C, D . Thus, no prime number can measure A except (one of) B, C, D . (Which is) the very thing it was required to show.

15.

Ἐὰν τρεῖς ἀριθμοὶ ἐχῇζ ἀνάλογον >ῶσιν ἐλάχιστοι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς, δύο ὅποιοι οὖν συντεθέντες πρὸς τὸν λοιπὸν πρῶτοί εἰσιν.

Proposition 15

If three continuously proportional numbers are the least of those (numbers) having the same ratio as them then two (of them) added together in any way are prime to the remaining (one).



>Ἐστωσαν τρεῖς ἀριθμοὶ ἐχῇζ ἀνάλογον ἐλάχιστοι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς οἱ A, B, Γ ; λέγω, <ὅτι τῶν A, B, Γ δύο ὅποιοι οὖν συντεθέντες πρὸς τὸν λοιπὸν πρῶτοί εἰσιν, οἱ μὲν A, B πρὸς τὸν Γ , οἱ δὲ B, Γ πρὸς τὸν A καὶ >έτι οἱ A, Γ πρὸς τὸν B .

Εἰλήφθωσαν γὰρ ἐλάχιστοι ἀριθμοὶ τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς A, B, Γ δύο οἱ $\Delta E, EZ$. φανερόν δὴ, <ὅτι ὁ μὲν ΔE ἑαυτὸν πολλαπλασιάσας τὸν A πεποίηκεν, τὸν δὲ EZ πολλαπλασιάσας τὸν B πεποίηκεν, καὶ >έτι ὁ EZ ἑαυτὸν πολλαπλασιάσας τὸν Γ πεποίηκεν. καὶ ἐπεὶ οἱ $\Delta E, EZ$ ἐλάχιστοι εἰσιν, πρῶτοι πρὸς ἀλλήλους εἰσίν. ἔαν δὲ δύο ἀριθμοὶ πρῶτοι πρὸς ἀλλήλους >ῶσιν, καὶ συναμρότεροξ πρὸς ἑκάτερον πρῶτός ἐστιν; καὶ ὁ ΔZ >άρα πρὸς ἑκάτερον τῶν $\Delta E, EZ$ πρῶτός ἐστιν. ἀλλὰ μὴν καὶ ὁ ΔE πρὸς τὸν EZ πρῶτός ἐστιν; οἱ $\Delta Z, \Delta E$ >άρα πρὸς τὸν EZ πρῶτοί εἰσιν.

Let A, B, C be three continuously proportional numbers (which are) the least of those (numbers) having the same ratio as them. I say that two of A, B, C added together in any way are prime to the remaining (one), (that is) A and B (prime) to C , B and C to A , and, further, A and C to B .

Let the two least numbers, DE and EF , having the same ratio as A, B, C , be taken [Prop. 8.2]. So it is clear that DE has made A (by) multiplying itself, and has made B (by) multiplying EF , and, further, EF has made C (by) multiplying itself [Prop. 8.2]. And since DE, EF are the least (of those numbers having the same ratio as them), they are prime to one another [Prop. 7.22]. And if two numbers are prime to one another then the sum (of them) is also prime to each [Prop. 7.28]. Thus, DF is also prime to each of DE, EF . But, in fact, DE is also prime to EF . Thus,

ἔαν δὲ δύο ἀριθμοὶ πρὸς τινα ἀριθμὸν πρῶτοι >ῶσιν, καὶ ὁ ἐκ αὐτῶν γενόμενος πρὸς τὸν λοιπὸν πρῶτός ἐστιν; <ὥστε ὁ ἐκ τῶν ΖΔ, ΔΕ πρὸς τὸν ΕΖ πρῶτός ἐστιν; <ὥστε καὶ ὁ ἐκ τῶν ΖΔ, ΔΕ πρὸς τὸν ἀπὸ τοῦ ΕΖ πρῶτός ἐστιν. [ἔαν γὰρ δύο ἀριθμοὶ πρῶτοι πρὸς ἀλλήλους >ῶσιν, ὁ ἐκ τοῦ ἐνὸς αὐτῶν γενόμενος πρὸς τὸν λοιπὸν πρῶτός ἐστιν]. ἀλλ> ὁ ἐκ τῶν ΖΔ, ΔΕ ὁ ἀπὸ τοῦ ΔΕ ἐστὶ μετὰ τοῦ ἐκ τῶν ΔΕ, ΕΖ; ὁ >άρα ἀπὸ τοῦ ΔΕ μετὰ τοῦ ἐκ τῶν ΔΕ, ΕΖ πρὸς τὸν ἀπὸ τοῦ ΕΖ πρῶτός ἐστιν. καὶ ἐστὶν ὁ μὲν ἀπὸ τοῦ ΔΕ ὁ Α, ὁ δὲ ἐκ τῶν ΔΕ, ΕΖ ὁ Β, ὁ δὲ ἀπὸ τοῦ ΕΖ ὁ Γ; οἱ Α, Β >άρα συντεθέντες πρὸς τὸν Γ πρῶτοι εἰσιν. ὁμοίως δὲ δείχομεν, <ὅτι καὶ οἱ Β, Γ πρὸς τὸν Α πρῶτοι εἰσιν. λέγω δὴ, <ὅτι καὶ οἱ Α, Γ πρὸς τὸν Β πρῶτοι εἰσιν. ἐπεὶ γὰρ ὁ ΔΖ πρὸς ἑκάτερον τῶν ΔΕ, ΕΖ πρῶτός ἐστιν, καὶ ὁ ἀπὸ τοῦ ΔΖ πρὸς τὸν ἐκ τῶν ΔΕ, ΕΖ πρῶτός ἐστιν. ἀλλὰ τῷ ἀπὸ τοῦ ΔΖ >ῖσοι εἰσιν οἱ ἀπὸ τῶν ΔΕ, ΕΖ μετὰ τοῦ διζ ἐκ τῶν ΔΕ, ΕΖ; καὶ οἱ ἀπὸ τῶν ΔΕ, ΕΖ >άρα μετὰ τοῦ διζ ὑπὸ τῶν ΔΕ, ΕΖ πρὸς τὸν ὑπὸ τῶν ΔΕ, ΕΖ πρῶτοι [εἰσι]. διελόντι οἱ ἀπὸ τῶν ΔΕ, ΕΖ μετὰ τοῦ <άπαχ ὑπὸ ΔΕ, ΕΖ πρὸς τὸν ὑπὸ ΔΕ, ΕΖ πρῶτοι εἰσιν. >έτι διελόντι οἱ ἀπὸ τῶν ΔΕ, ΕΖ >άρα πρὸς τὸν ὑπὸ ΔΕ, ΕΖ πρῶτοι εἰσιν. καὶ ἐστὶν ὁ μὲν ἀπὸ τοῦ ΔΕ ὁ Α, ὁ δὲ ὑπὸ τῶν ΔΕ, ΕΖ ὁ Β, ὁ δὲ ἀπὸ τοῦ ΕΖ ὁ Γ. οἱ Α, Γ >άρα συντεθέντες πρὸς τὸν Β πρῶτοι εἰσιν; <όπερ >έδει δεῖχαι.

DF , DE are (both) prime to EF . And if two numbers are (both) prime to some number then the (number) created from (multiplying) them is also prime to the remaining (number) [Prop. 7.24]. Hence, the (number created) from (multiplying) FD , DE is prime to EF . Hence, the (number created) from (multiplying) FD , DE is also prime to the (square) on EF [Prop. 7.25]. [For if two numbers are prime to one another then the (number) created from (squaring) one of them is prime to the remaining (number).] But the (number created) from (multiplying) FD , DE is the (square) on DE plus the (number created) from (multiplying) DE , EF [Prop. 2.3]. Thus, the (square) on DE plus the (number created) from (multiplying) DE , EF is prime to the (square) on EF . And the (square) on DE is A , and the (number created) from (multiplying) DE , EF (is) B , and the (square) on EF (is) C . Thus, A , B summed is prime to C . So, similarly, we can show that B , C (summed) is also prime to A . So I say that A , C (summed) is also prime to B . For since DF is prime to each of DE , EF then the (square) on DF is also prime to the (number created) from (multiplying) DE , EF [Prop. 7.25]. But, the (sum of the squares) on DE , EF plus twice the (number created) from (multiplying) DE , EF is equal to the (square) on DF [Prop. 2.4]. And thus the (sum of the squares) on DE , EF plus twice the (rectangle contained) by DE , EF [is] prime to the (rectangle contained) by DE , EF . By separation, the (sum of the squares) on DE , EF plus once the (rectangle contained) by DE , EF is prime to the (rectangle contained) by DE , EF .[†] Again, by separation, the (sum of the squares) on DE , EF is prime to the (rectangle contained) by DE , EF . And the (square) on DE is A , and the (rectangle contained) by DE , EF (is) B , and the (square) on EF (is) C . Thus, A , C summed is prime to B . (Which is) the very thing it was required to show.

[†] Since if $\alpha \beta$ measures $\alpha^2 + \beta^2 + 2\alpha\beta$ then it also measures $\alpha^2 + \beta^2 + \alpha\beta$, and vice versa.

16.

Ἐὰν δύο ἀριθμοὶ πρῶτοι πρὸς ἀλλήλους >ῶσιν, οὐκ >έσται ὥς ὁ πρῶτος πρὸς τὸν δεύτερον, ο<ύτωξ ὁ δεύτερος πρὸς >άλλον τινά.



Δύο γὰρ ἀριθμοὶ οἱ Α, Β πρῶτοι πρὸς ἀλλήλους >έστωσαν; λέγω, <ὅτι οὐκ >έστιν ὥς ὁ Α πρὸς τὸν Β, ο<ύτωξ ὁ Β πρὸς >άλλον τινά.

Εἰ γὰρ δυνατόν, >έστω ὥς ὁ Α πρὸς τὸν Β, ὁ Β πρὸς τὸν Γ. οἱ δὲ Α, Β πρῶτοι, οἱ δὲ πρῶτοι καὶ ἐλάχιστοι, οἱ δὲ ἐλάχιστοι ἀριθμοὶ μετροῦσι τοὺς τὸν αὐτὸν λόγον >έθονταξ ἰσάκιξ <ὅ τε ἡγούμενος τὸν ἡγούμενον καὶ ὁ ἐπόμενος τὸν ἐπόμενον; μετρεῖ >άρα ὁ Α τὸν Β ὥς ἡγούμενος ἡγούμενον. μετρεῖ δὲ καὶ ἐαυτόν; ὁ Α >άρα τοὺς Α, Β μετρεῖ πρώτους >όνταξ πρὸς ἀλλήλους; <όπερ >άτοπον. οὐκ >άρα >έσται ὥς ὁ Α πρὸς τὸν Β, ο<ύτωξ ὁ Β πρὸς τὸν Γ; <όπερ >έδει δεῖχαι.

Proposition 16

If two numbers are prime to one another then as the first is to the second, so the second (will) not (be) to some other (number).

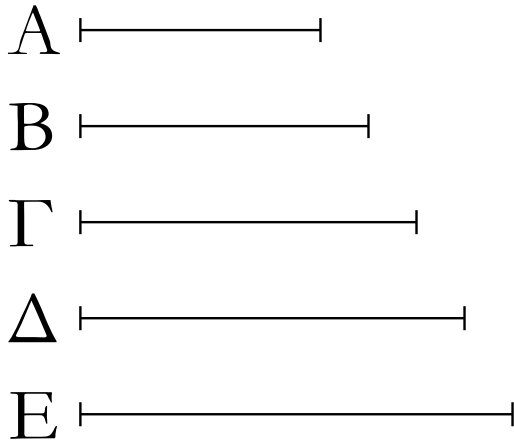


For let the two numbers A and B be prime to one another. I say that as A is to B , so B is not to some other (number). For, if possible, let it be that as A (is) to B , (so) B (is) to C . And A and B (are) prime (to one another). And (numbers) prime (to one another) are also the least (of those numbers having the same ratio as them) [Prop. 7.21]. And the least numbers measure those (numbers) having the same ratio (as them) an equal number of times, the leading (measuring) the leading, and the following the following [Prop. 7.20]. Thus, A measures B , as the leading (measuring) the leading. And (A) also measures itself. Thus, A measures A and B , which are prime to one another. The very thing (is) absurd. Thus,

17.

Ἐὰν >ῶσιν ὅσοιδηποτοῦν ἀριθμοὶ ἐχῇζ ἀνάλογον, οἱ δὲ >ᾱκροὶ αὐτῶν πρῶτοι πρὸς ἀλλήλουζ >ῶσιν, οὐκ >έσται ὡς ὁ πρῶτοζ πρὸς τὸν δεῦτερον, ο<ύτωζ ὁ >έσθατοζ πρὸς >ἄλλον τινά.

>Ἐστῶσαν ὅσοιδηποτοῦν ἀριθμοὶ ἐχῇζ ἀνάλογον οἱ A, B, Γ, Δ, οἱ δὲ >ᾱκροὶ αὐτῶν οἱ A, Δ πρῶτοι πρὸς ἀλλήλουζ >έστωσαν; λέγω, <ὅτι οὐκ >έστιν ὡς ὁ A πρὸς τὸν B, ο<ύτωζ ὁ Δ πρὸς >ἄλλον τινά.



Εἰ γὰρ δυνατόν, >έστω ὡς ὁ A πρὸς τὸν B, ο<ύτωζ ὁ Δ πρὸς τὸν E; ἐναλλάζ >άρα ἐστὶν ὡς ὁ A πρὸς τὸν Δ, ὁ B πρὸς τὸν E. οἱ δὲ A, Δ πρῶτοι, οἱ δὲ πρῶτοι καὶ ἐλάθιστοι, οἱ δὲ ἐλάθιστοι ἀριθμοὶ μετροῦσι τοὺς τὸν αὐτὸν λόγον >έθονταζ ἰσάκιζ <ὁ τε ἡγούμενοζ τὸν ἡγούμενον καὶ ὁ ἐπόμενοζ τὸν ἐπόμενον. μετρεῖ >άρα ὁ A τὸν B. καὶ ἐστὶν ὡς ὁ A πρὸς τὸν B, ὁ B πρὸς τὸν Γ. καὶ ὁ B >άρα τὸν Γ μετρεῖ; <ώστε καὶ ὁ A τὸν Γ μετρεῖ. καὶ ἐπεὶ ἐστὶν ὡς ὁ B πρὸς τὸν Γ, ὁ Γ πρὸς τὸν Δ, μετρεῖ δὲ ὁ B τὸν Γ, μετρεῖ >άρα καὶ ὁ Γ τὸν Δ. ἀλλ' ὁ A τὸν Γ ἐμέτρει; <ώστε ὁ A καὶ τὸν Δ μετρεῖ. μετρεῖ δὲ καὶ ἑαυτόν. ὁ A >άρα τοὺς A, Δ μετρεῖ πρῶτουζ >όνταζ πρὸς ἀλλήλουζ; <ὅπερ ἐστὶν ἀδύνατον. οὐκ >άρα >έσται ὡς ὁ A πρὸς τὸν B, ο<ύτωζ ὁ Δ πρὸς >ἄλλον τινά; <ὅπερ >έδει δεῖχαι.

18.

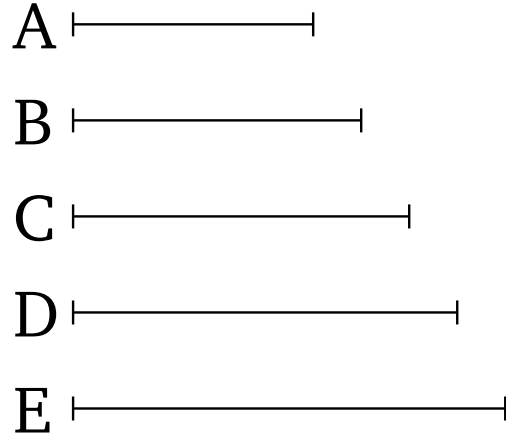
Δύο ἀριθμῶν δοθέντων ἐπισκέψασθαι, εἰ δυνατόν ἐστιν αὐτοῖζ τρίτον ἀνάλογον προσευρεῖν.

as A (is) to B , so B cannot be to C . (Which is) the very thing it was required to show.

Proposition 17

If any multitude whatsoever of numbers is continuously proportional, and the outermost of them are prime to one another, then as the first (is) to the second, so the last will not be to some other (number).

Let A, B, C, D be any multitude whatsoever of continuously proportional numbers. And let the outermost of them, A and D , be prime to one another. I say that as A is to B , so D (is) not to some other (number).



For, if possible, let it be that as A (is) to B , so D (is) to E . Thus, alternately, as A is to D , (so) B (is) to E [Prop. 7.13]. And A and D are prime (to one another). And (numbers) prime (to one another are) also the least (of those numbers having the same ratio as them) [Prop. 7.21]. And the least numbers measure those (numbers) having the same ratio (as them) an equal number of times, the leading (measuring) the leading, and the following the following [Prop. 7.20]. Thus, A measures B . And as A is to B , (so) B (is) to C . Thus, B also measures C . And hence A measures C [Def. 7.20]. And since as B is to C , (so) C (is) to D , and B measures C , C thus also measures D [Def. 7.20]. But, A was (found to be) measuring C . And hence A also measures D . And (A) also measures itself. Thus, A measures A and D , which are prime to one another. The very thing is impossible. Thus, as A (is) to B , so D cannot be to some other (number). (Which is) the very thing it was required to show.

Proposition 18

For two given numbers, to investigate whether it is possible to find a third (number) proportional to them.

$$\Gamma \quad A$$

$$C$$

$$\Delta \quad B$$

$$D$$

>Ἐστωσαν οἱ δοθέντες δύο ἀριθμοὶ οἱ A, B , καὶ δέον >έστω ἐπισκέψασθαι, εἰ δυνατόν ἐστιν αὐτοῖς τρίτον ἀνάλογον προσσευρεῖν.

Οἱ δὴ A, B >ήτοι πρῶτοι πρὸς ἀλλήλους εἰσὶν >ἢ ο>ύ. καὶ εἰ πρῶτοι πρὸς ἀλλήλους εἰσὶν, δέδεικται, <ὅτι ἀδύνατόν ἐστιν αὐτοῖς τρίτον ἀνάλογον προσσευρεῖν.

Ἀλλὰ δὴ μὴ >έστωσαν οἱ A, B πρῶτοι πρὸς ἀλλήλους, καὶ ὁ B ἐαυτὸν πολλαπλασιάσας τὸν Γ ποιεῖτω. ὁ A δὴ τὸν Γ >ήτοι μετρεῖ >ἢ οὐ μετρεῖ. μετρεῖτω πρότερον κατὰ τὸν Δ ; ὁ A >άρα τὸν Δ πολλαπλασιάσας τὸν Γ πεποίηκεν. ἀλλὰ μὴν καὶ ὁ B ἐαυτὸν πολλαπλασιάσας τὸν Γ πεποίηκεν; ὁ >άρα ἐκ τῶν A, Δ >ίσοξ ἐστὶ τῷ ἀπὸ τοῦ B . >έστιν >άρα ὡς ὁ A πρὸς τὸν B , ὁ B πρὸς τὸν Δ ; τοῖς A, B >άρα τρίτος ἀριθμὸς ἀνάλογον προσσηύρηται ὁ Δ .

Ἀλλὰ δὴ μὴ μετρεῖτω ὁ A τὸν Γ ; λέγω, <ὅτι τοῖς A, B ἀδύνατόν ἐστι τρίτον ἀνάλογον προσσευρεῖν ἀριθμόν. εἰ γὰρ δυνατόν, προσσηυρήσθω ὁ Δ . ὁ >άρα ἐκ τῶν A, Δ >ίσοξ ἐστὶ τῷ ἀπὸ τοῦ B . ὁ δὲ ἀπὸ τοῦ B ἐστὶν ὁ Γ ; ὁ >άρα ἐκ τῶν A, Δ >ίσοξ ἐστὶ τῷ Γ . <ὥστε ὁ A τὸν Δ πολλαπλασιάσας τὸν Γ πεποίηκεν; ὁ A >άρα τὸν Γ μετρεῖ κατὰ τὸν Δ . ἀλλὰ μὴν ὑπόκειται καὶ μὴ μετρῶν; <ὅπερ >άτοπον. οὐκ >άρα δυνατόν ἐστι τοῖς A, B τρίτον ἀνάλογον προσσευρεῖν ἀριθμόν, <ὅταν ὁ A τὸν Γ μὴ μετρή; <ὅπερ >έδει δεῖχαι.

Let A and B be the two given numbers. And let it be required to investigate whether it is possible to find a third (number) proportional to them.

So A and B are either prime to one another, or not. And if they are prime to one another then it has (already) been show that it is impossible to find a third (number) proportional to them [Prop. 9.16].

And so let A and B not be prime to one another. And let B make C (by) multiplying itself. So A either measures, or does not measure, C . Let it first of all measure (C) according to D . Thus, A has made C (by) multiplying D . But, in fact, B has also made C (by) multiplying itself. Thus, the (number created) from (multiplying) A, D is equal to the (square) on B . Thus, as A is to B , (so) B (is) to D [Prop. 7.19]. Thus, a third number has been found proportional to A, B , (namely) D .

And so let A not measure C . I say that it is impossible to find a third number proportional to A, B . For, if possible, let it be found, (and let it be) D . Thus, the (number created) from (multiplying) A, D is equal to the (square) on B [Prop. 7.19]. And the (square) on B is C . Thus, the (number created) from (multiplying) A, D is equal to C . Hence, A has made C (by) multiplying D . Thus, A measures C according to D . But (A) was, in fact, also assumed (to be) not measuring (C). The very thing (is) absurd. Thus, it is not possible to find a third number proportional to A, B when A does not measure C . (Which is) the very thing it was required to show.

19.

Τριῶν ἀριθμῶν δοθέντων ἐπισκέψασθαι, πότε δυνατόν ἐστιν αὐτοῖς τέταρτον ἀνάλογον προσσευρεῖν.

$$A$$

$$B$$

$$\Gamma$$

$$\Delta$$

$$E$$

>Ἐστωσαν οἱ δοθέντες τρεῖς ἀριθμοὶ οἱ A, B, Γ , καὶ δέον >έστω ἐπισκέψασθαι, πότε δυνατόν ἐστιν αὐτοῖς τέταρτον ἀνάλογον προσσευρεῖν.

>Ἦτοι ο>ὐν ο>ύκ εἰσὶν ἐχῆξ ἀνάλογον, καὶ οἱ >άκροι αὐτῶν πρῶτοι πρὸς ἀλλήλους εἰσὶν, >ἢ ἐχῆξ εἰσὶν ἀνάλογον, καὶ οἱ >άκροι αὐτῶν ο>ύκ εἰσὶ πρῶτοι πρὸς ἀλλήλους, >ἢ ο>ύτε ἐχῆξ εἰσὶν ἀνάλογον, ο>ύτε οἱ >άκροι αὐτῶν πρῶτοι πρὸς ἀλλήλους εἰσὶν, >ἢ καὶ ἐχῆξ εἰσὶν ἀνάλογον, καὶ οἱ >άκροι αὐτῶν πρῶτοι πρὸς ἀλλήλους εἰσὶν.

Proposition 19†

For three given numbers, to investigate when it is possible to find a fourth (number) proportional to them.

$$A$$

$$B$$

$$C$$

$$D$$

$$E$$

Let A, B, C be the three given numbers. And let it be required to investigate when it is possible to find a fourth (number) proportional to them.

In fact, (A, B, C) are either not continuously proportional and the outermost of them are prime to one another, or are continuously proportional and the outermost of them are not prime to one another, or are neither continuously proportional nor are the outermost of them prime to one another, or are continuously proportional and the outermost of them

Εἰ μὲν οὖν οἱ A, B, Γ ἐχῆζ· εἰσιν ἀνάλογον, καὶ οἱ >ἄκροι αὐτῶν οἱ A, Γ πρῶτοι πρὸς ἀλλήλους· εἰσίν, δέδεικται, <ὅτι ἀδύνατόν ἐστιν αὐτοῖς τέταρτον ἀνάλογον προσεῦρεῖν ἀριθμόν. μὴ >έστωσαν δὴ οἱ A, B, Γ ἐχῆζ ἀνάλογον τῶν ἀκρῶν πάλιν >όντων πρῶτων πρὸς ἀλλήλους. λέγω, <ὅτι καὶ οὕτως ἀδύνατόν ἐστιν αὐτοῖς τέταρτον ἀνάλογον προσεῦρεῖν. εἰ γὰρ δυνατόν, προσευρήσθω ὁ Δ , <ώστε εἶναι ὡς τὸν A πρὸς τὸν B , τὸν Γ πρὸς τὸν Δ , καὶ γεγενέτω ὡς ὁ B πρὸς τὸν Γ , ὁ Δ πρὸς τὸν E . καὶ ἐπεὶ ἐστὶν ὡς μὲν ὁ A πρὸς τὸν B , ὁ Γ πρὸς τὸν Δ , ὡς δὲ ὁ B πρὸς τὸν Γ , ὁ Δ πρὸς τὸν E , δι> >ῖσου >άρα ὡς ὁ A πρὸς τὸν Γ , ὁ Γ πρὸς τὸν E . οἱ δὲ A, Γ πρῶτοι, οἱ δὲ πρῶτοι καὶ ἐλάχιστοι, οἱ δὲ ἐλάχιστοι μετροῦσι τοὺς τὸν αὐτὸν λόγον >έθοντα· <ὅ τε ἡγούμενος τὸν ἡγούμενον καὶ ὁ ἐπόμενος τὸν ἐπόμενον. μετρεῖ >άρα ὁ A τὸν Γ ὡς ἡγούμενος ἡγούμενον. μετρεῖ δὲ καὶ ἐαυτόν; ὁ A >άρα τοὺς A, Γ μετρεῖ πρῶτους >όντα· πρὸς ἀλλήλους; <ὅπερ ἐστὶν ἀδύνατον. οὐκ >άρα τοῖς A, B, Γ δυνατόν ἐστι τέταρτον ἀνάλογον προσεῦρεῖν.

Ἀλλὰ δὴ πάλιν >έστωσαν οἱ A, B, Γ ἐχῆζ ἀνάλογον, οἱ δὲ A, Γ μὴ >έστωσαν πρῶτοι πρὸς ἀλλήλους. λέγω, <ὅτι δυνατόν ἐστιν αὐτοῖς τέταρτον ἀνάλογον προσεῦρεῖν. ὁ γὰρ B τὸν Γ πολλαπλασιάσας τὸν Δ ποιεῖτω; ὁ A >άρα τὸν Δ >ήτοι μετρεῖ >ῆ οὐ μετρεῖ. μετρεῖτω αὐτὸν πρότερον κατὰ τὸν E ; ὁ A >άρα τὸν E πολλαπλασιάσας τὸν Δ πεποίηκεν. ἀλλὰ μὴν καὶ ὁ B τὸν Γ πολλαπλασιάσας τὸν Δ πεποίηκεν; ὁ >άρα ἐκ τῶν A, E >ῖσο· ἐστὶ τῷ ἐκ τῶν B, Γ . ἀνάλογον >άρα [ἐστὶν] ὡς ὁ A πρὸς τὸν B , ὁ Γ πρὸς τὸν E ; τοῖς A, B, Γ >άρα τέταρτος ἀνάλογον προσήρηται ὁ E .

Ἀλλὰ δὴ μὴ μετρεῖτω ὁ A τὸν Δ ; λέγω, <ὅτι ἀδύνατόν ἐστι τοῖς A, B, Γ τέταρτον ἀνάλογον προσεῦρεῖν ἀριθμόν. εἰ γὰρ δυνατόν, προσευρήσθω ὁ E ; ὁ >άρα ἐκ τῶν A, E >ῖσο· ἐστὶ τῷ ἐκ τῶν B, Γ . ἀλλὰ ὁ ἐκ τῶν B, Γ ἐστὶν ὁ Δ ; καὶ ὁ ἐκ τῶν A, E >άρα >ῖσο· ἐστὶ τῷ Δ . ὁ A >άρα τὸν E πολλαπλασιάσας τὸν Δ πεποίηκεν; ὁ A >άρα τὸν Δ μετρεῖ κατὰ τὸν E ; <ώστε μετρεῖ ὁ A τὸν Δ . ἀλλὰ καὶ οὐ μετρεῖ; <ὅπερ >άτοπον. οὐκ >άρα δυνατόν ἐστι τοῖς A, B, Γ τέταρτον ἀνάλογον προσεῦρεῖν ἀριθμόν, <ὅταν ὁ A τὸν Δ μὴ μετρήῃ. ἀλλὰ δὴ οἱ A, B, Γ μήτε ἐχῆζ >έστωσαν ἀνάλογον μήτε οἱ >ἄκροι πρῶτοι πρὸς ἀλλήλους. καὶ ὁ B τὸν Γ πολλαπλασιάσας τὸν Δ ποιεῖτω. ὁμοίως δὴ δειθθήσεται, <ὅτι εἰ μὲν μετρεῖ ὁ A τὸν Δ , δυνατόν ἐστιν αὐτοῖς ἀνάλογον προσεῦρεῖν, εἰ δὲ οὐ μετρεῖ, ἀδύνατον; <ὅπερ >έδει δεῖχαι.

are prime to one another.

In fact, if A, B, C are continuously proportional, and the outermost of them, A and C , are prime to one another, (then) it has (already) been shown that it is impossible to find a fourth number proportional to them [Prop. 9.17]. So let A, B, C not be continuously proportional, (with) the outermost of them again being prime to one another. I say that, in this case, it is also impossible to find a fourth (number) proportional to them. For, if possible, let it be found, (and let it be) D . Hence, it will be that as A (is) to B , (so) C (is) to D . And let it be contrived that as B (is) to C , (so) D (is) to E . And since as A is to B , (so) C (is) to D , and as B (is) to C , (so) D (is) to E , thus, via equality, as A (is) to C , (so) C (is) to E [Prop. 7.14]. And A and C (are) prime (to one another). And (numbers) prime (to one another are) also the least (numbers having the same ratio as them) [Prop. 7.21]. And the least (numbers) measure those numbers having the same ratio as them (the same number of times), the leading (measuring) the leading, and the following the following [Prop. 7.20]. Thus, A measures C , (as) the leading (measuring) the leading. And it also measures itself. Thus, A measures A and C , which are prime to one another. The very thing is impossible. Thus, it is not possible to find a fourth (number) proportional to A, B, C .

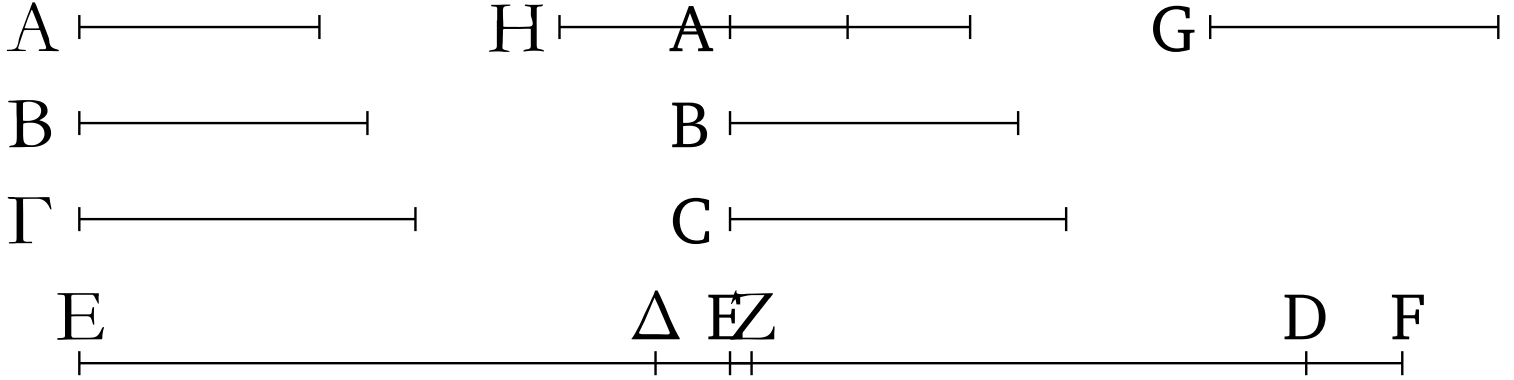
And so let A, B, C again be continuously proportional, and let A and C not be prime to one another. I say that it is possible to find a fourth (number) proportional to them. For let B make D (by) multiplying C . Thus, A either measures or does not measure D . Let it, first of all, measure (D) according to E . Thus, A has made D (by) multiplying E . But, in fact, B has also made D (by) multiplying C . Thus, the (number created) from (multiplying) A, E is equal to the (number created) from (multiplying) B, C . Thus, proportionally, as A [is] to B , (so) C (is) to E [Prop. 7.19]. Thus, a fourth (number) proportional to A, B, C has been found, (namely) E .

And so let A not measure D . I say that it is impossible to find a fourth number proportional to A, B, C . For, if possible, let it be found, (and let it be) E . Thus, the (number created) from (multiplying) A, E is equal to the (number created) from (multiplying) B, C . But, the (number created) from (multiplying) B, C is D . And thus the (number created) from (multiplying) A, E is equal to D . Thus, A has made D (by) multiplying E . Thus, A measures D according to E . Hence, A measures D . But, it also does not measure (D). The very thing (is) absurd. Thus, it is not possible to find a fourth number proportional to A, B, C when A does not measure D . And so (let) A, B, C (be) neither continuously proportional, nor (let) the outermost of them (be) prime to one another. And let B make D (by) multiplying C . So, similarly, it can be show that if A measures D then it is possible to find a fourth (number) proportional to (A, B, C), and impossible if (A) does not measure (D). (Which is) the very thing it was required to show.

† The proof of this proposition is incorrect. There are, in fact, only two cases. Either A, B, C are continuously proportional, with A and C prime to one another, or not. In the first case, it is impossible to find a fourth proportional number. In the second case, it is possible to find a fourth proportional number provided that A measures B times C . Of the four cases considered by Euclid, the proof given in the second case is incorrect, since it only demonstrates that if $A : B :: C : D$ then a number E cannot be found such that $B : C :: D : E$. The proofs given in the other three cases are correct.

Οἱ πρῶτοι ἀριθμοὶ πλείους εἰσὶ παντὸς τοῦ προτεθέντος
πλήθους πρῶτων ἀριθμῶν.

The (set of all) prime numbers is more numerous than any
assigned multitude of prime numbers.



>Ἐστῶσαν οἱ προτεθέντες πρῶτοι ἀριθμοὶ οἱ A, B, Γ ; λέγω,
<ὅτι τῶν A, B, Γ πλείους εἰσὶ πρῶτοι ἀριθμοί.

Εἰλήφθω γὰρ ὁ ὑπὸ τῶν A, B, Γ ἐλάχιστος μετρούμενος
καὶ >έστω ΔE , καὶ προσκείσθω τῷ ΔE μονὰς ἢ ΔZ . ὁ δὲ
 EZ >ήτοι πρῶτός ἐστιν >ἢ οὐ. >έστω πρότερον πρῶτος;
εὐρημένοι >άρα εἰσὶ πρῶτοι ἀριθμοὶ οἱ A, B, Γ, EZ πλείους
τῶν A, B, Γ .

Ἀλλὰ δὴ μὴ >έστω ὁ EZ πρῶτος; ὑπὸ πρῶτον >άρα τινὸς
ἀριθμοῦ μετρεῖται. μετρεῖσθω ὑπὸ πρῶτου τοῦ H ; λέγω,
<ὅτι ὁ H οὐδενὶ τῶν A, B, Γ ἐστὶν ὁ αὐτός. εἰ γὰρ δυνατόν,
>έστω. οἱ δὲ A, B, Γ τὸν ΔE μετροῦσιν; καὶ ὁ H >άρα τὸν
 ΔE μετρήσει. μετρεῖ δὲ καὶ τὸν EZ ; καὶ λοιπὴν τὴν ΔZ
μονάδα μετρήσει ὁ H ἀριθμὸς >ών; >όπερ >άτοπον. οὐκ
>άρα ὁ H ἐνὶ τῶν A, B, Γ ἐστὶν ὁ αὐτός. καὶ ὑπόκειται
πρῶτος. εὐρημένοι >άρα εἰσὶ πρῶτοι ἀριθμοὶ πλείους τοῦ
προτεθέντος πλήθους τῶν A, B, Γ οἱ A, B, Γ, H ; <όπερ >έδει
δείχαι.

Let A, B, C be the assigned prime numbers. I say that the
(set of all) primes numbers is more numerous than A, B, C .
For let the least number measured by A, B, C be taken, and
let it be DE [Prop. 7.36]. And let the unit DF be added
to DE . So EF is either prime, or not. Let it, first of all,
be prime. Thus, the (set of) prime numbers A, B, C, EF ,
(which is) more numerous than A, B, C , has been found.

And so let EF not be prime. Thus, it is measured by some
prime number [Prop. 7.31]. Let it be measured by the prime
(number) G . I say that G is not the same as any of A, B, C .
For, if possible, let it be (the same). And A, B, C (all)
measure DE . Thus, G will also measure DE . And it also
measures EF . (So) G will also measure the remainder, unit
 DF , (despite) being a number [Prop. 7.28]. The very thing
(is) absurd. Thus, G is not the same as one of A, B, C .
And it was assumed (to be) prime. Thus, the (set of) prime
numbers A, B, C, G , (which is) more numerous than the
assigned multitude (of prime numbers), A, B, C , has been
found. (Which is) the very thing it was required to show.

21.

Proposition 21

Ἐὰν >άρτιοι ἀριθμοὶ ὅποσοιὺν συντεθῶσιν, ὁ <όλος
>άρτιός ἐστιν.

If any multitude whatsoever of even numbers is added to-
gether then the whole is even.



Συγκείσθωσαν γὰρ >άρτιοι ἀριθμοὶ ὅποσοιὺν οἱ $AB, B\Gamma,$
 $\Gamma\Delta, \Delta E$; λέγω, <ὅτι <όλος ὁ AE >άρτιός ἐστιν.

Ἐπεὶ γὰρ <έκαστος τῶν $AB, B\Gamma, \Gamma\Delta, \Delta E$ >άρτιός ἐστιν,
>έθει μέρος <ήμισυ; <ώστε καὶ <όλος ὁ AE >έθει μέρος
<ήμισυ. >άρτιος δὲ ἀριθμός ἐστιν ὁ δίθα διαιρούμενος;
>άρτιος >άρα ἐστὶν ὁ AE ; <όπερ >έδει δείχαι.

For let any multitude whatsoever of even numbers, $AB,$
 BC, CD, DE , lie together. I say that the whole, AE , is
even.

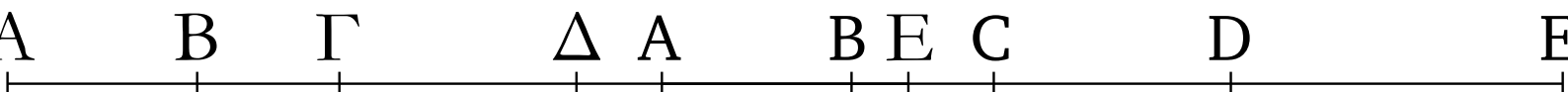
For since everyone of AB, BC, CD, DE is even, it has
a half part [Def. 7.6]. And hence the whole AE has a half
part. And an even number is one (which can be) divided in
half [Def. 7.6]. Thus, AE is even. (Which is) the very thing
it was required to show.

22.

Proposition 22

Ἐὰν περισσοὶ ἀριθμοὶ ὅποσοιὺν συντεθῶσιν, τὸ δὲ πλῆθος
αὐτῶν >άρτιον >ἢ, ὁ <όλος >άρτιος >έσται.

If any multitude whatsoever of odd numbers is added to-
gether, and the multitude of them is even, then the whole
will be even.



Συγκείσθωσαν γὰρ περισσοὶ ἀριθμοὶ ὅσοιδηποτοῦν >άρτιοι
τὸ πλῆθος οἱ $AB, B\Gamma, \Gamma\Delta, \Delta E$; λέγω, <ὅτι <όλος ὁ AE
>άρτιός ἐστιν.

For let any even multitude whatsoever of odd numbers, $AB,$
 BC, CD, DE , lie together. I say that the whole, AE , is
even.

Ἐπεὶ γὰρ <έκαστοξ τῶν $AB, BG, ΓΔ, ΔΕ$ περιττός ἐστίν, ἀφαιρεθείσης μονάδοξ ἀφ' ἐκάστου <έκαστοξ τῶν λοιπῶν> ἄρτιοξ >έσται; <ώστε καὶ ὁ συγκείμενοξ ἐξ αὐτῶν> ἄρτιοξ >έσται. >έστι δὲ καὶ τὸ πλῆθοξ τῶν μονάδων >ἄρτιον. καὶ <όλοξ> ἄρα ὁ AE >ἄρτιοξ ἐστίν; <όπερ> ἐδει δεῖχαι.

For since everyone of AB, BC, CD, DE is odd then, a unit being subtracted from each, everyone of the remainders will be (made) even [Def. 7.7]. And hence the sum of them will be even [Prop. 9.21]. And the multitude of the units is even. Thus, the whole AE is also even [Prop. 9.21]. (Which is) the very thing it was required to show.

23.

Ἐὰν περισσοὶ ἀριθμοὶ ὅποσοιοῦν συντεθῶσιν, τὸ δὲ πλῆθοξ αὐτῶν περισσὸν >ῆ|, καὶ ὁ <όλοξ περισσὸξ> >έσται.

Proposition 23

If any multitude whatsoever of odd numbers is added together, and the multitude of them is odd, then the whole will also be odd.



Συγκείσθωσαν γὰρ ὅποσοιοῦν περισσοὶ ἀριθμοί, <ὦν τὸ πλῆθοξ περισσὸν> >έστω, οἱ $AB, BG, ΓΔ$; λέγω, <ὅτι καὶ <όλοξ ὁ AD περισσὸξ ἐστίν.

Ἀφηρήσθω ἀπὸ τοῦ $ΓΔ$ μονάξ ἢ $ΔΕ$; λοιπὸξ >άρα ὁ $ΓΕ$ >ἄρτιοξ ἐστίν. >έστι δὲ καὶ ὁ $ΓΑ$ >ἄρτιοξ; καὶ <όλοξ> >άρα ὁ AE >ἄρτιοξ ἐστίν. καὶ ἐστὶ μονάξ ἢ $ΔΕ$. περισσὸξ >άρα ἐστίν ὁ AD ; <όπερ> ἐδει δεῖχαι.

For let any multitude whatsoever of odd numbers, AB, BC, CD , lie together, and let the multitude of them be odd. I say that the whole, AD , is also odd.

For let the unit DE be subtracted from CD . The remainder, CE , is thus even [Def. 7.7]. And CA is also even [Prop. 9.22]. Thus, the whole, AE , is also even [Prop. 9.21]. And DE is a unit. Thus, AD is odd [Def. 7.7]. (Which is) the very thing it was required to show.

24.

Ἐὰν ἀπὸ ἄρτίου ἀριθμοῦ >ἄρτιοξ ἀφαιρεθῇ|, ὁ λοιπὸξ >ἄρτιοξ >έσται.

Proposition 24

If an even (number) is subtracted from an (other) even number then the remainder will be even.



Ἀπὸ γὰρ ἄρτίου τοῦ AB >ἄρτιοξ ἀφηρήσθω ὁ $BΓ$; λέγω, <ὅτι ὁ λοιπὸξ ὁ $ΓΑ$ >ἄρτιοξ ἐστίν.

Ἐπεὶ γὰρ ὁ AB >ἄρτιοξ ἐστίν, >έθει μέρος <ήμισυ. διὰ τὰ αὐτὰ δὴ καὶ ὁ $BΓ$ >έθει μέρος <ήμισυ; <ώστε καὶ λοιπὸξ [ὁ $ΓΑ$ >έθει μέρος <ήμισυ] >ἄρτιοξ [>άρα] ἐστίν ὁ $ΑΓ$; <όπερ> ἐδει δεῖχαι.

For let the even (number) BC be subtracted from the even number AB . I say that the remainder, CA , is even.

For since AB is even, it has a half part [Def. 7.6]. So, for the same (reasons), BC also has a half part. And hence the remainder [CA has a half part]. [Thus,] AC is even. (Which is) the very thing it was required to show.

25.

Ἐὰν ἀπὸ ἄρτίου ἀριθμοῦ περισσὸξ ἀφαιρεθῇ|, ὁ λοιπὸξ περισσὸξ >έσται.

Proposition 25

If an odd (number) is subtracted from an even number then the remainder will be odd.



Ἀπὸ γὰρ ἄρτίου τοῦ AB περισσὸξ ἀφηρήσθω ὁ $BΓ$; λέγω, <ὅτι ὁ λοιπὸξ ὁ $ΓΑ$ περισσὸξ ἐστίν.

Ἀφηρήσθω γὰρ ἀπὸ τοῦ $BΓ$ μονάξ ἢ $ΓΔ$; ὁ $ΔΒ$ >άρα >ἄρτιοξ ἐστίν. >έστι δὲ καὶ ὁ AB >ἄρτιοξ; καὶ λοιπὸξ >άρα ὁ AD >ἄρτιοξ ἐστίν. καὶ ἐστὶ μονάξ ἢ $ΓΔ$; ὁ $ΓΑ$ >άρα περισσὸξ ἐστίν; <όπερ> ἐδει δεῖχαι.

For let the odd (number) BC be subtracted from the even number AB . I say that the remainder CA is odd.

For let the unit CD be subtracted from BC . DB is thus even [Def. 7.7]. And AB is also even. And thus the remainder AD is even [Prop. 9.24]. And CD is a unit. Thus, CA is odd [Def. 7.7]. (Which is) the very thing it was required to show.

26.

Ἐὰν ἀπὸ περισσοῦ ἀριθμοῦ περισσὸξ ἀφαιρεθῇ|, ὁ λοιπὸξ >ἄρτιοξ >έσται.

Proposition 26

If an odd (number) is subtracted from an odd number then the remainder will be even.



Ἀπὸ γὰρ περισσοῦ τοῦ AB περισσὸξ ἀφηρήσθω ὁ $BΓ$; λέγω, <ὅτι ὁ λοιπὸξ ὁ $ΓΑ$ >ἄρτιοξ ἐστίν.

For let the odd (number) BC be subtracted from the odd (number) AB . I say that the remainder, CA , is even.

Ἐπεὶ γὰρ ὁ AB περισσόξ ἐστίν, ἀφηρήσθω μονάξ ἡ BD ; λοιπὸξ >άρα ὁ AD >άρτιόξ ἐστίν. διὰ τὰ αὐτὰ δὴ καὶ ὁ $ΓΔ$ >άρτιόξ ἐστίν; <ώστε καὶ λοιπὸξ ὁ $ΓΑ$ >άρτιόξ ἐστίν; <όπερ >έδει δεῖχαι.

For since AB is odd, let the unit BD be subtracted (from it). Thus, the remainder AD is even [Def. 7.7]. So, for the same (reasons), CD is also even. And hence the remainder CA is even [Prop. 9.24]. (Which is) the very thing it was required to show.

27.

Proposition 27

Ἐὰν ἀπὸ περισσοῦ ἀριθμοῦ >άρτιοξ ἀφαιρεθῇ, ὁ λοιπὸξ περισσόξ >έσται.

If an even (number) is subtracted from an odd number then the remainder will be odd.



Ἀπὸ γὰρ περισσοῦ τοῦ AB >άρτιοξ ἀφηρήσθω ὁ BF ; λέγω, <ὅτι ὁ λοιπὸξ ὁ $ΓΑ$ περισσόξ ἐστίν. Ἀφηρήσθω [γὰρ] μονάξ ἡ AD ; ὁ $ΔB$ >άρα >άρτιόξ ἐστίν. >έστι δὲ καὶ ὁ BF >άρτιοξ; καὶ λοιπὸξ >άρα ὁ $ΓΔ$ >άρτιόξ ἐστίν. περισσόξ >άρα ὁ $ΓΑ$; <όπερ >έδει δεῖχαι.

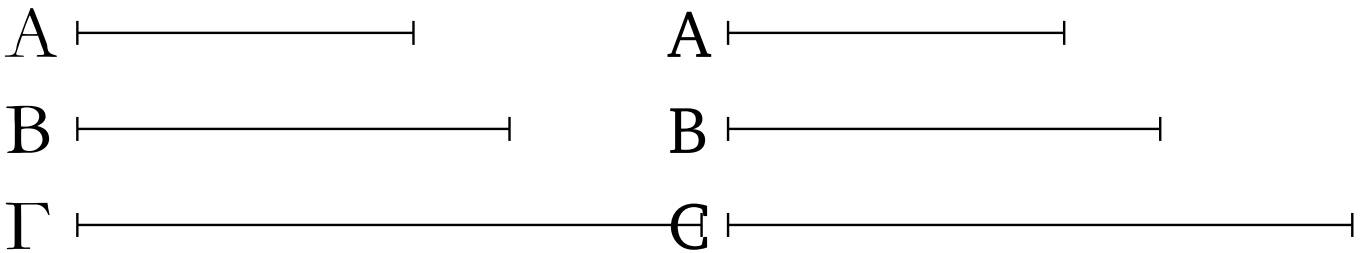
For let the even (number) BC be subtracted from the odd (number) AB . I say that the remainder, CA , is odd. [For] let the unit AD be subtracted (from AB). DB is thus even [Def. 7.7]. And BC is also even. Thus, the remainder CD is also even [Prop. 9.24]. CA (is) thus odd [Def. 7.7]. (Which is) the very thing it was required to show.

28.

Proposition 28

Ἐὰν περισσόξ ἀριθμὸξ >άρτιον πολλαπλασιάσας ποιῇ τινα, ὁ γενόμενοξ >άρτιοξ >έσται.

If an odd number makes some (number by) multiplying an even (number) then the created (number) will be even.



Περισσόξ γὰρ ἀριθμὸξ ὁ A >άρτιον τὸν B πολλαπλασιάσας τὸν $Γ$ ποιεῖται; λέγω, <ὅτι ὁ $Γ$ >άρτιόξ ἐστίν. Ἐπεὶ γὰρ ὁ A τὸν B πολλαπλασιάσας τὸν $Γ$ πεποίηκεν, ὁ $Γ$ >άρα σύγκειται ἐκ τοσοῦτων >ίσων τῷ B , <όσαι εἰσὶν ἐν τῷ A μονάδεξ. καὶ ἐστίν ὁ B >άρτιοξ; ὁ $Γ$ >άρα σύγκειται ἐξ ἄρτιων. Ἐὰν δὲ >άρτιοι ἀριθμοὶ ὅποσοιοῦν συντεθῶσιν, ὁ <όλοξ >άρτιόξ ἐστίν. >άρτιοξ >άρα ἐστίν ὁ $Γ$; <όπερ >έδει δεῖχαι.

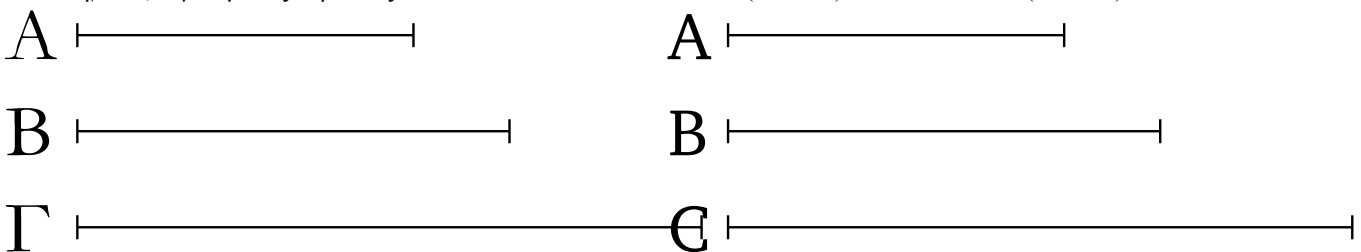
For let the odd number A make C (by) multiplying the even (number) B . I say that C is even. For since A has made C (by) multiplying B , C is thus composed out of so many (magnitudes) equal to B , as many as (there) are units in A [Def. 7.15]. And B is even. Thus, C is composed out of even (numbers). And if any multitude whatsoever of even numbers is added together then the whole is even [Prop. 9.21]. Thus, C is even. (Which is) the very thing it was required to show.

29.

Proposition 29

Ἐὰν περισσόξ ἀριθμὸξ περισσὸν ἀριθμὸν πολλαπλασιάσας ποιῇ τινα, ὁ γενόμενοξ περισσόξ >έσται.

If an odd number makes some (number by) multiplying an odd (number) then the created (number) will be odd.



Περισσόξ γὰρ ἀριθμὸξ ὁ A περισσὸν τὸν B πολλαπλασιάσας τὸν $Γ$ ποιεῖται; λέγω, <ὅτι ὁ $Γ$ περισσόξ ἐστίν. Ἐπεὶ γὰρ ὁ A τὸν B πολλαπλασιάσας τὸν $Γ$ πεποίηκεν, ὁ $Γ$ >άρα σύγκειται ἐκ τοσοῦτων >ίσων τῷ B , <όσαι εἰσὶν ἐν τῷ A μονάδεξ. καὶ ἐστίν ἐκάτεροξ τῶν A , B περισσόξ; ὁ $Γ$ >άρα σύγκειται ἐκ περισσῶν ἀριθμῶν, <ὧν τὸ πλῆθος περισσὸν ἐστίν. <ώστε ὁ $Γ$ περισσόξ ἐστίν; <όπερ >έδει δεῖχαι.

For let the odd number A make C (by) multiplying the odd (number) B . I say that C is odd. For since A has made C (by) multiplying B , C is thus composed out of so many (magnitudes) equal to B , as many as (there) are units in A [Def. 7.15]. And each of A , B is odd. Thus, C is composed out of odd (numbers), (and) the multitude of them is odd. Hence C is odd [Prop. 9.23]. (Which is) the very thing it was required to show.

δειχαι.

is) the very thing it was required to show.

30.

Proposition 30

Ἐὰν περισσὸς ἀριθμὸς >ἀρτιον ἀριθμὸν μετρήῃ, καὶ τὸν <ἡμισυν αὐτοῦ μετρήσει.

If an odd number measures an even number then it will also measure (one) half of it.

A |—————|

A |—————|

B |—————|

B |—————|

Γ |—————|

C |—————|

Περὶ σὸς γὰρ ἀριθμὸς ὁ A >ἀρτιον τὸν B μετρεῖτω; λέγω, <ὅτι καὶ τὸν <ἡμισυν αὐτοῦ μετρήσει.

For let the odd number *A* measure the even (number) *B*. I say that (*A*) will also measure (one) half of (*B*).

Ἐπεὶ γὰρ ὁ A τὸν B μετρεῖ, μετρεῖτω αὐτὸν κατὰ τὸν Γ; λέγω, <ὅτι ὁ Γ οὐκ >έστι περισσός. εἰ γὰρ δυνατόν, >έστω. καὶ ἐπεὶ ὁ A τὸν B μετρεῖ κατὰ τὸν Γ, ὁ A >άρα τὸν Γ πολλαπλασιάσας τὸν B πεποίηκεν. ὁ B >άρα σύγκειται ἐκ περισσῶν ἀριθμῶν, <ὧν τὸ πλῆθος περισσόν ἐστιν. ὁ B >άρα περισσός ἐστιν; <ὅπερ >άτοπον; ὑπόκειται γὰρ >ἀρτιος. οὐκ >άρα ὁ Γ περισσός ἐστιν; >ἀρτιος >άρα ἐστὶν ὁ Γ. <ώστε ὁ A τὸν B μετρεῖ ἀρτιάκις. διὰ δὲ τοῦτο καὶ τὸν <ἡμισυν αὐτοῦ μετρήσει; <ὅπερ >έδει δεῖχαι.

For since *A* measures *B*, let it measure it according to *C*. I say that *C* is not odd. For, if possible, let it be (odd). And since *A* measures *B* according to *C*, *A* has thus made *B* (by) multiplying *C*. Thus, *B* is composed out of odd numbers, (and) the multitude of them is odd. *B* is thus odd [Prop. 9.23]. The very thing (is) absurd. For (*B*) was assumed (to be) even. Thus, *C* is not odd. Thus, *C* is even. Hence, *A* measures *B* an even number of times. So, on account of this, (*A*) will also measure (one) half of (*B*). (Which is) the very thing it was required to show.

31.

Proposition 31

Ἐὰν περισσὸς ἀριθμὸς πρὸς τινὰ ἀριθμὸν πρῶτος >ῃ, καὶ πρὸς τὸν διπλασίονα αὐτοῦ πρῶτος >έσται.

If an odd number is prime to some number then it will also be prime to its double.

A |—————|

A |—————|

B |—————|

B |—————|

Γ |—————|

C |—————|

Δ |—————|

D |—————|

Περὶ σὸς γὰρ ἀριθμὸς ὁ A πρὸς τινὰ ἀριθμὸν τὸν B πρῶτος >έστω, τοῦ δὲ B διπλασίου >έστω ὁ Γ; λέγω, <ὅτι ὁ A [καὶ] πρὸς τὸν Γ πρῶτός ἐστιν.

For let the odd number *A* be prime to some number *B*. And let *C* be double *B*. I say that *A* is [also] prime to *C*.

Εἰ γὰρ μὴ εἰσιν [οἱ A, Γ] πρῶτοι, μετρήσει τις αὐτοῦς ἀριθμός. μετρεῖτω, καὶ >έστω ὁ Δ. καὶ ἐστὶν ὁ A περισσός; περισσὸς >άρα καὶ ὁ Δ. καὶ ἐπεὶ ὁ Δ περισσὸς >ὦν τὸν Γ μετρεῖ, καὶ ἐστὶν ὁ Γ >ἀρτιος, καὶ τὸν <ἡμισυν >άρα τοῦ Γ μετρήσει [ὁ Δ]. τοῦ δὲ Γ <ἡμισύ ἐστὶν ὁ B; ὁ Δ >άρα τὸν B μετρεῖ. μετρεῖ δὲ καὶ τὸν A. ὁ Δ >άρα τοὺς A, B μετρεῖ πρῶτους >όντας πρὸς ἀλλήλους; <ὅπερ ἐστὶν ἀδύνατον. οὐκ >άρα ὁ A πρὸς τὸν Γ πρῶτος ο>ύκ ἐστιν. οἱ A, Γ >άρα πρῶτοι πρὸς ἀλλήλους εἰσίν; <ὅπερ >έδει δεῖχαι.

For if [*A* and *C*] are not prime (to one another) then some number will measure them. Let it measure (them), and let it be *D*. And *A* is odd. Thus, *D* (is) also odd. And since *D*, which is odd, measures *C*, and *C* is even, [*D*] will thus also measure half of *C* [Prop. 9.30]. And *B* is half of *C*. Thus, *D* measures *B*. And it also measures *A*. Thus, *D* measures (both) *A* and *B*, (despite) them being prime to one another. The very thing is impossible. Thus, *A* is not unprime to *C*. Thus, *A* and *C* are prime to one another. (Which is) the very thing it was required to show.

32.

Proposition 32

Τῶν ἀπὸ δυάδος διπλασιαζομένων ἀριθμῶν <έκαστος >ἀρτιάκις >ἀρτιός ἐστι μόνον.

Each of the numbers (which is continually) doubled, (starting) from a dyad, is an even-times-even (number) only.

A ———|

B ———|

Γ ———|

Δ ———|

Ἀπὸ γὰρ δυάδοξ τῆς A δεδιπλασιάσθωσαν ὅσοιδηποτοῦν ἀριθμοὶ οἱ B, Γ, Δ; λέγω, <ὅτι οἱ B, Γ, Δ ἀρτιάκις >ἀρτιοὶ εἰσι μόνον.

<Ὅτι μὲν ο>ὺν <έκαστοξ [τῶν B, Γ, Δ] ἀρτιάκις >ἀρτιόξ ἐστίν, φανερόν; ἀπὸ γὰρ δυάδοξ ἐστὶ διπλασιασθείξ. λέγω, <ὅτι καὶ μόνον. ἐκκείσθω γὰρ μονάξ. ἐπεὶ ο>ὺν ἀπὸ μονάδοξ ὅποσοιοῦν ἀριθμοὶ ἐχῆξ ἀνάλογόν εἰσιν, ὁ δὲ μετὰ τὴν μονάδα ὁ A πρῶτόξ ἐστίν, ὁ μέγιστοξ τῶν A, B, Γ, Δ ὁ Δ ὑπ> οὐδενόξ >ἄλλου μετρηθήσεται παρὲχ τῶν A, B, Γ. καὶ ἐστίν <έκαστοξ τῶν A, B, Γ >ἀρτιοξ; ὁ Δ >ἄρα ἀρτιάκις >ἀρτιόξ ἐστὶ μόνον. ὁμοίωξ δὴ δείχομεν, <ὅτι [καὶ] ἑκάτεροξ τῶν B, Γ ἀρτιάκις >ἀρτιόξ ἐστὶ μόνον; <ὅπερ >έδει δεῖχαι.

A ———|

B ———|

C ———|

D ———|

For let any multitude of numbers whatsoever, B, C, D , be (continually) doubled, (starting) from the dyad A . I say that B, C, D are even-times-even (numbers) only.

In fact, (it is) clear that each [of B, C, D] is an even-times-even (number). For it is doubled from a dyad [Def. 7.8]. I also say that (they are even-times-even numbers) only. For let a unit be laid down. Therefore, since any multitude of numbers whatsoever are continuously proportional, starting from a unit, and the (number) A after the unit is prime, the greatest of A, B, C, D , (namely) D , will not be measured by any other (numbers) except A, B, C [Prop. 9.13]. And each of A, B, C is even. Thus, D is an even-time-even (number) only [Def. 7.8]. So, similarly, we can show that each of B, C is [also] an even-time-even (number) only. (Which is) the very thing it was required to show.

33.

Ἐὰν ἀριθμὸξ τὸν <ήμισυν >έθῃ περισσόν, ἀρτιάκιξ περισσόξ ἐστὶ μόνον.

A ———|

Ἀριθμὸξ γὰρ ὁ A τὸν <ήμισυν ἐθέτω περισσόν; λέγω, <ὅτι ὁ A ἀρτιάκιξ περισσόξ ἐστὶ μόνον.

<Ὅτι μὲν ο>ὺν ἀρτιάκιξ περισσόξ ἐστίν, φανερόν; ὁ γὰρ <ήμισυξ αὐτοῦ περισσόξ >ὢν μετρεῖ αὐτὸν ἀρτιάκιξ, λέγω δὴ, <ὅτι καὶ μόνον. εἰ γὰρ >έσται ὁ A καὶ ἀρτιάκιξ >ἀρτιοξ, μετρηθήσεται ὑπὸ ἀρτίου κατὰ >ἀρτιον ἀριθμόν; <ὥστε καὶ ὁ <ήμισυξ αὐτοῦ μετρηθήσεται ὑπὸ ἀρτίου ἀριθμοῦ περισσόξ >ὢν; <ὅπερ ἐστίν >ἀτοπον. ὁ A >ἄρα ἀρτιάκιξ περισσόξ ἐστὶ μόνον; <ὅπερ >έδει δεῖχαι.

Proposition 33

If a number has an odd half then it is an even-time-odd (number) only.

A ———|

For let the number A have an odd half. I say that A is an even-times-odd (number) only.

In fact, (it is) clear that (A) is an even-times-odd (number). For its half, being odd, measures it an even number of times [Def. 7.9]. So I also say that (it is an even-times-odd number) only. For if A is also an even-times-even (number) then it will be measured by an even (number) according to an even number [Def. 7.8]. Hence, its half will also be measured by an even number, (despite) being odd. The very thing is absurd. Thus, A is an even-times-odd (number) only. (Which is) the very thing it was required to show.

34.

Ἐὰν ἀριθμὸξ μήτε τῶν ἀπὸ δυάδοξ διπλασιαζομένων >ή], μήτε τὸν <ήμισυν >έθῃ περισσόν, ἀρτιάκιξ τε >ἀρτιόξ ἐστὶ καὶ ἀρτιάκιξ περισσόξ.

A ———|

Ἀριθμὸξ γὰρ ὁ A μήτε τῶν ἀπὸ δυάδοξ διπλασιαζομένων >έστω μήτε τὸν <ήμισυν ἐθέτω περισσόν; λέγω, <ὅτι ὁ A ἀρτιάκιξ τέ ἐστίν >ἀρτιοξ καὶ ἀρτιάκιξ περισσόξ.

<Ὅτι μὲν ο>ὺν ὁ A ἀρτιάκιξ ἐστίν >ἀρτιοξ, φανερόν; τὸν γὰρ <ήμισυν οὐκ >έθει περισσόν. λέγω δὴ, <ὅτι καὶ ἀρτιάκιξ περισσόξ ἐστίν. ἔαν γὰρ τὸν A τέμνωμεν δίθα καὶ τὸν <ήμισυν αὐτοῦ δίθα καὶ τοῦτο αἰε ποιῶμεν, καταστήσομεν ε>ίξ τινα ἀριθμόν περισσόν, <ὃξ μετρήσει τὸν A κατὰ >ἀρτιον ἀριθμόν. εἰ γὰρ ο>ύ, καταστήσομεν εἰξ δυάδα, καὶ >έσται ὁ A τῶν ἀπὸ δυάδοξ διπλασιαζομένων;

Proposition 34

If a number is neither (one) of the (numbers) doubled from a dyad, nor has an odd half, then it is (both) an even-times-even and an even-times-odd (number).

A ———|

For let the number A neither be (one) of the (numbers) doubled from a dyad, nor let it have an odd half. I say that A is (both) an even-times-even and an even-times-odd (number). In fact, (it is) clear that A is an even-times-even (number) [Def. 7.8]. For it does not have an odd half. So I say that it is also an even-times-odd (number). For if we cut A in half, and (then cut) its half in half, and we do this continually, then we will arrive at some odd number which will measure A according to an even number. For if not, we will arrive at a dyad, and A will be (one) of the (numbers) doubled from

<όπερ οὐθ ὑπόκειται. <ώστε ὁ A ἀρτιάκις περισσόν ἐστιν. ἐδείθθη δὲ καὶ ἀρτιάκις >ἀρτιόξ. ὁ A >άρα ἀρτιάκις τε >ἀρτιόξ ἐστι καὶ ἀρτιάκις περισσόξ; <όπερ >έδει δεῖχαι.

a dyad. The very opposite thing (was) assumed. Hence, A is an even-times-odd (number) [Def. 7.9]. And it was also shown (to be) an even-times-even (number). Thus, A is (both) an even-times-even and an even-times-odd (number). (Which is) the very thing it was required to show.

35.

Proposition 35[†]

Ἐάν >ῶσιν ὁσοιοδηποτοῦν ἀριθμοὶ ἐχῇξ ἀνάλογον, ἀφαιρεθῶσιIf there is any multitude whatsoever of continually proportional numbers, and (numbers) equal to the first are subtracted from (both) the second and the last, then as the excess of the second (number is) to the first, so the excess of the last will be to (the sum of) all those (numbers) before it.

If there is any multitude whatsoever of continually proportional numbers, and (numbers) equal to the first are subtracted from (both) the second and the last, then as the excess of the second (number is) to the first, so the excess of the last will be to (the sum of) all those (numbers) before it.

A

B G C

D

E L K H F

A

B H Γ

Δ

E Λ K Θ Z

>Ἐστωσαν ὅποσοιοδηποτοῦν ἀριθμοὶ ἐχῇξ ἀνάλογον οἱ A , $B\Gamma$, Δ , EZ ἀφθόμενοι ἀπὸ ἐλαθίστου τοῦ A , καὶ ἀφηρήσθω ἀπὸ τοῦ $B\Gamma$ καὶ τοῦ EZ τῶ| A >ίσοξ ἐκάτεροξ τῶν BH , $Z\Theta$; λέγω, <ὅτι ἐστὶν ὡξ ὁ $H\Gamma$ πρὸξ τὸν A , ο<ύτωξ ὁ $E\Theta$ πρὸξ τοῦξ A , $B\Gamma$, Δ .

Κείσθω γὰρ τῶ| μὲν $B\Gamma$ >ίσοξ ὁ ZK , τῶ| δὲ Δ >ίσοξ ὁ $Z\Lambda$. καὶ ἐπεὶ ὁ ZK τῶ| $B\Gamma$ >ίσοξ ἐστίν, <ῶν ὁ $Z\Theta$ τῶ| BH >ίσοξ ἐστίν, λοιπόξ >άρα ὁ ΘK λοιπῶ| τῶ| $H\Gamma$ ἐστὶν >ίσοξ. καὶ ἐπεὶ ἐστὶν ὡξ ὁ EZ πρὸξ τὸν Δ , ο<ύτωξ ὁ Δ πρὸξ τὸν $B\Gamma$ καὶ ὁ $B\Gamma$ πρὸξ τὸν A , >ίσοξ δὲ ὁ μὲν Δ τῶ| $Z\Lambda$, ὁ δὲ $B\Gamma$ τῶ| ZK , ὁ δὲ A τῶ| $Z\Theta$, >έστιν >άρα ὡξ ὁ EZ πρὸξ τὸν $Z\Lambda$, ο<ύτωξ ὁ ΛZ πρὸξ τὸν ZK καὶ ὁ ZK πρὸξ τὸν $Z\Theta$. διελόντι, ὡξ ὁ $E\Lambda$ πρὸξ τὸν ΛZ , ο<ύτωξ ὁ ΛK πρὸξ τὸν ZK καὶ ὁ $K\Theta$ πρὸξ τὸν $Z\Theta$. >έστιν >άρα καὶ ὡξ ε<ῖξ τῶν ἡγουμένων πρὸξ <ένα τῶν ἐπομένων, ο<ύτωξ <άπαντεξ οἱ ἡγούμενοι πρὸξ <άπανταξ τοῦξ ἐπομένουξ; >έστιν >άρα ὡξ ὁ $K\Theta$ πρὸξ τὸν $Z\Theta$, ο<ύτωξ οἱ $E\Lambda$, ΛK , $K\Theta$ πρὸξ τοῦξ ΛZ , ZK , ΘZ . >ίσοξ δὲ ὁ μὲν $K\Theta$ τῶ| ΓH , ὁ δὲ $Z\Theta$ τῶ| A , οἱ δὲ ΛZ , ZK , ΘZ

Let A , BC , D , EF be any multitude whatsoever of continually proportional numbers, beginning from the least A . And let BG and FH , each equal to A , be subtracted from BC and EF (respectively). I say that as GC is to A , so EH is to A , BC , D .

For let FK be made equal to BC , and FL to D . And since FK is equal to BC , of which FH is equal to BG , the remainder, HK , is thus equal to the remainder, GC . And since as EF is to D , so D (is) to BC , and BC to A [Prop. 7.13], and D (is) equal to FL , and BC to FK , and A to FH , thus as EF is to FL , so LF (is) to FK , and FK to FH . By separation, as EL (is) to LF , so LK (is) to FK , and KH to FH [Props. 7.11, 7.13]. And thus as one of the leading (numbers) is to one of the following, so (the sum of) all of the leading (numbers) is to (the sum of) all of the following [Prop. 7.12]. Thus, as KH is to FH , so EL , LK , KH (are) to LF , FK , HF . And KH (is) equal to CG , and FH to A , and LF , FK , HF to D , BC , A . Thus, as CG is to A , so EH (is) to D , BC , A . Thus, as the excess of the

τοις Δ, ΒΓ, Α; >έστιν >άρα ὡς ὁ ΓΗ πρὸς τὸν Α, ο<ύτωξ ὁ ΕΘ πρὸς τοὺς Δ, ΒΓ, Α. >έστιν >άρα ὡς ἡ τοῦ δευτέρου ὑπεροθὴ πρὸς τὸν πρῶτον, ο<ύτωξ ἡ τοῦ ἐσθάτου ὑπεροθὴ πρὸς τοὺς πρὸ αὐτοῦ πάνταξ; <όπερ >έδει δεῖχαι.

† This proposition allows us to sum a geometric series of the form $a, ar, ar^2, ar^3, \dots, ar^{n-1}$. According to Euclid, the sum, S_n , satisfies $(ar - a)/a = (ar^n - a)/S_n$. Hence, $S_n = a(r^n - 1)/(r - 1)$.

36.

Ἐὰν ἀπὸ μονάδοξ ὅποσοιὸν ἀριθμοὶ ἐχῇξ ἐκτεθῶσιν ἐν τῇ διπλασίονι ἀναλογίᾳ, <έωξ ο<ὺ ὁ σύμπαξ συντεθειξ πρῶτοξ γένηται, καὶ ὁ σύμπαξ ἐπὶ τὸν >έσθατον πολλαπλασιασθεὶξ ποιῇξ τινα, ὁ γενόμενοξ τέλειοξ >έσται.

Ἀπὸ γὰρ μονάδοξ ἐκκείσθωσαν ὁσοιδηποτοῦν ἀριθμοὶ ἐν τῇ διπλασίονι ἀναλογίᾳ, <έωξ ο<ὺ ὁ σύμπαξ συντεθειξ πρῶτοξ γένηται, οἱ Α, Β, Γ, Δ, καὶ τῶξ σύμπαντι >ίσοξ >έστω ὁ Ε, καὶ ὁ Ε τὸν Δ πολλαπλασιάσαξ τὸν ΖΗ ποιείτω. λέγω, <ότι ὁ ΖΗ τέλειόξ ἐστιν.

Α

Β

Γ

Δ

Ε

Η Ν Κ

Λ

Μ

Φ Ο

Η

Ρ

Q

<Όσοι γὰρ εἰσιν οἱ Α, Β, Γ, Δ τῶξ πλήθει, τοσοῦτοι ἀπὸ τοῦ Ε εἰλήφθωσαν ἐν τῇ διπλασίονι ἀναλογίᾳ οἱ Ε, ΘΚ, Λ, Μ; δι> >ίσου >άρα ἐστὶν ὡς ὁ Α πρὸς τὸν Δ, ο<ύτωξ ὁ Ε πρὸς τὸν Μ. ὁ >άρα ἐκ τῶν Ε, Δ >ίσοξ ἐστὶ τῶξ ἐκ τῶν Α, Μ. καὶ ἐστὶν ὁ ἐκ τῶν Ε, Δ ὁ ΖΗ; καὶ ὁ ἐκ τῶν Α, Μ

second (number) is to the first, so the excess of the last (is) to (the sum of) all those (numbers) before it. (Which is) the very thing it was required to show.

Proposition 36†

If any multitude whatsoever of numbers is set out continuously in a double proportion, (starting) from a unit, until the whole sum added together becomes prime, and the sum multiplied into the last (number) makes some (number), then the (number so) created will be perfect.

For let any multitude of numbers, A, B, C, D , be set out (continuously) in a double proportion, until the whole sum added together is made prime. And let E be equal to the sum. And let E make FG (by) multiplying D . I say that FG is a perfect (number).

Α

Β

C

D

For as many as is the multitude of A, B, C, D , let so many (numbers), E, HK, L, M , be taken in a double proportion, (starting) from E . Thus, via equality, as A is to D , so E (is) to M [Prop. 7.14]. Thus, the (number created) from (multiplying) E, D is equal to the (number created) from

>άρα ἐστὶν ὁ ΖΗ. ὁ Α >άρα τὸν Μ πολλαπλασιάσας τὸν ΖΗ πεποίηκεν; ὁ Μ >άρα τὸν ΖΗ μετρεῖ κατὰ τὰς ἐν τῷ Α μονάδας. καὶ ἐστὶ δὴ ὁ Α; διπλάσιος >άρα ἐστὶν ὁ ΖΗ τοῦ Μ. εἰσὶ δὲ καὶ οἱ Μ, Λ, ΘΚ, Ε ἐχθὺς διπλάσιοι ἀλλήλων; οἱ Ε, ΘΚ, Λ, Μ, ΖΗ >άρα ἐχθὺς ἀνάλογόν εἰσιν ἐν τῇ διπλασίονι ἀναλογία. ἀφηρήσθω δὴ ἀπὸ τοῦ δευτέρου τοῦ ΘΚ καὶ τοῦ ἐσθάτου τοῦ ΖΗ τῷ πρώτῳ τῷ Ε >ἴσος ἐκάτερος τῶν ΘΝ, ΖΧ; >ἐστὶν >άρα ὡς ἡ τοῦ δευτέρου ἀριθμοῦ ὑπεροθὴ πρὸς τὸν πρῶτον, ο<ύτως ἡ τοῦ ἐσθάτου ὑπεροθὴ πρὸς τοὺς πρὸ ἑαυτοῦ πάντα. >ἐστὶν >άρα ὡς ὁ ΝΚ πρὸς τὸν Ε, ο<ύτως ὁ ΧΗ πρὸς τοὺς Μ, Λ, ΚΘ, Ε. καὶ ἐστὶν ὁ ΝΚ >ἴσος τῷ Ε; καὶ ὁ ΧΗ >άρα >ἴσος ἐστὶ τοῖς Μ, Λ, ΘΚ, Ε. >ἐστὶ δὲ καὶ ὁ ΖΧ τῷ Ε >ἴσος, ὁ δὲ Ε τοῖς Α, Β, Γ, Δ καὶ τῇ μονάδι. <όλος >άρα ὁ ΖΗ >ἴσος ἐστὶ τοῖς τε Ε, ΘΚ, Λ, Μ καὶ τοῖς Α, Β, Γ, Δ καὶ τῇ μονάδι; καὶ μετρεῖται ὑπ> αὐτῶν. λέγω, <ὅτι καὶ ὁ ΖΗ ὑπ> οὐδενὸς >άλλου μετρηθήσεται παρέχ τῶν Α, Β, Γ, Δ, Ε, ΘΚ, Λ, Μ καὶ τῆς μονάδος. εἰ γὰρ δυνατόν, μετρεῖται τις τὸν ΖΗ ὁ Ο, καὶ ὁ Ο μηδενὶ τῶν Α, Β, Γ, Δ, Ε, ΘΚ, Λ, Μ >έστω ὁ αὐτός. καὶ ὁσάκις ὁ Ο τὸν ΖΗ μετρεῖ, τοσαῦται μονάδες >έστωσαν ἐν τῷ Π; ὁ Π >άρα τὸν Ο πολλαπλασιάσας τὸν ΖΗ πεποίηκεν. ἀλλὰ μὴν καὶ ὁ Ε τὸν Δ πολλαπλασιάσας τὸν ΖΗ πεποίηκεν; >ἐστὶν >άρα ὡς ὁ Ε πρὸς τὸν Π, ὁ Ο πρὸς τὸν Δ. καὶ ἐπεὶ ἀπὸ μονάδος ἐχθὺς ἀνάλογόν εἰσιν οἱ Α, Β, Γ, Δ, ὁ Δ >άρα ὑπ> οὐδενὸς >άλλου ἀριθμοῦ μετρηθήσεται παρέχ τῶν Α, Β, Γ. καὶ ὑπόκειται ὁ Ο οὐδενὶ τῶν Α, Β, Γ ὁ αὐτός; οὐκ >άρα μετρήσει ὁ Ο τὸν Δ. ἀλλ> ὡς ὁ Ο πρὸς τὸν Δ, ὁ Ε πρὸς τὸν Π; οὐδὲ ὁ Ε >άρα τὸν Π μετρεῖ. καὶ ἐστὶν ὁ Ε πρῶτος; πᾶς δὲ πρῶτος ἀριθμὸς πρὸς <άπαντα, <ὀν μὴ μετρεῖ, πρῶτός [ἐστὶν]. οἱ Ε, Π >άρα πρῶτοι πρὸς ἀλλήλους εἰσίν. οἱ δὲ πρῶτοι καὶ ἐλάχιστοι, οἱ δὲ ἐλάχιστοι μετροῦσι τοὺς τὸν αὐτὸν λόγον >έθονταξ ἰσάκις <ὀ τε ἡγούμενος τὸν ἡγούμενον καὶ ὁ ἐπόμενος τὸν ἐπόμενον; καὶ ἐστὶν ὡς ὁ Ε πρὸς τὸν Π, ὁ Ο πρὸς τὸν Δ; ἰσάκις >άρα ὁ Ε τὸν Ο μετρεῖ καὶ ὁ Π τὸν Δ. ὁ δὲ Δ ὑπ> οὐδενὸς >άλλου μετρεῖται παρέχ τῶν Α, Β, Γ; ὁ Π >άρα ἐνὶ τῶν Α, Β, Γ ἐστὶν ὁ αὐτός. >έστω τῷ Β ὁ αὐτός. καὶ <όσοι εἰσιν οἱ Β, Γ, Δ τῷ πληθῇ τοσοῦτοι εἰλήφθωσαν ἀπὸ τοῦ Ε οἱ Ε, ΘΚ, Λ. καὶ εἰσιν οἱ Ε, ΘΚ, Λ τοῖς Β, Γ, Δ ἐν τῷ αὐτῷ λόγῳ; δι> >ἴσου >άρα ἐστὶν ὡς ὁ Β πρὸς τὸν Δ, ὁ Ε πρὸς τὸν Λ. ὁ >άρα ἐκ τῶν Β, Λ >ἴσος ἐστὶ τῷ ἐκ τῶν Δ, Ε; ἀλλ> ὁ ἐκ τῶν Δ, Ε >ἴσος ἐστὶ τῷ ἐκ τῶν Π, Ο; καὶ ὁ ἐκ τῶν Π, Ο >άρα >ἴσος ἐστὶ τῷ ἐκ τῶν Β, Λ. >ἐστὶν >άρα ὡς ὁ Π πρὸς τὸν Β, ὁ Λ πρὸς τὸν Ο. καὶ ἐστὶν ὁ Π τῷ Β ὁ αὐτός; καὶ ὁ Λ >άρα τῷ Ο ἐστὶν ὁ αὐτός; <όπερ ἀδύνατον; ὁ γὰρ Ο ὑπόκειται μηδενὶ τῶν ἐκκειμένων ὁ αὐτός; οὐκ >άρα τὸν ΖΗ μετρήσει τις ἀριθμὸς παρέχ τῶν Α, Β, Γ, Δ, Ε, ΘΚ, Λ, Μ καὶ τῆς μονάδος. καὶ ἐδείθη ὁ ΖΗ τοῖς Α, Β, Γ, Δ, Ε, ΘΚ, Λ, Μ καὶ τῇ μονάδι >ἴσος. τέλειος δὲ ἀριθμὸς ἐστὶν ὁ τοῖς ἑαυτοῦ μέρεσιν >ἴσος >ών; τέλειος >άρα ἐστὶν ὁ ΖΗ; <όπερ >έδει δεῖχαι.

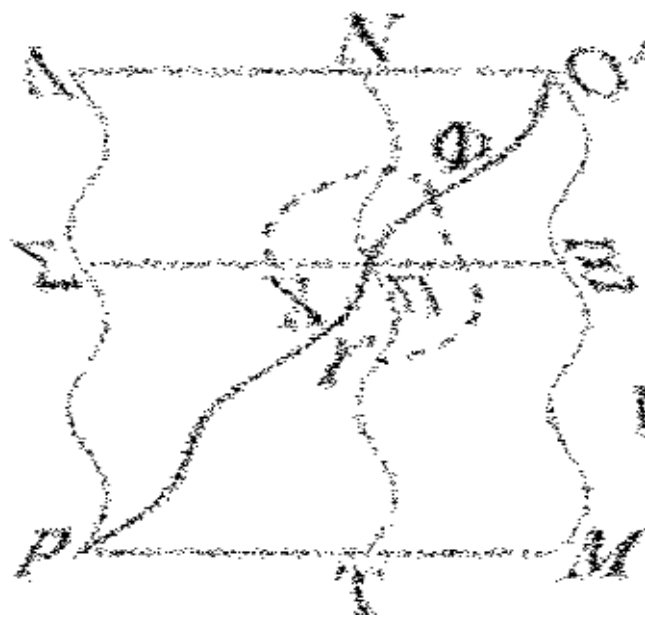
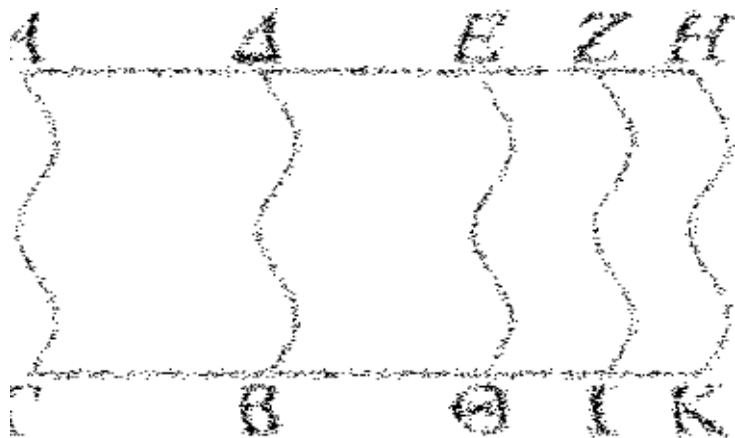
(multiplying) A, M . And FG is the (number created) from (multiplying) E, D . Thus, FG is also the (number created) from (multiplying) A, M [Prop. 7.19]. Thus, A has made FG (by) multiplying M . Thus, M measures FG according to the units in A . And A is a dyad. Thus, FG is double M . And M, L, HK, E are also continuously double one another. Thus, E, HK, L, M, FG are continuously proportional in a double proportion. So let HN and FO , each equal to the first (number) E , be subtracted from the second (number) HK and the last FG (respectively). Thus, as the excess of the second number is to the first, so the excess of the last (is) to (the sum of) all those (numbers) before it [Prop. 9.35]. Thus, as NK is to E , so OG (is) to M, L, KH, E . And NK is equal to E . And thus OG is equal to M, L, HK, E . And FO is also equal to E , and E to A, B, C, D , and a unit. Thus, the whole of FG is equal to E, HK, L, M , and A, B, C, D , and a unit. And it is measured by them. I also say that FG will be measured by no other (numbers) except A, B, C, D, E, HK, L, M , and a unit. For, if possible, let some (number) P measure FG , and let P not be the same as any of A, B, C, D, E, HK, L, M . And as many times as P measures FG , so many units let there be in Q . Thus, Q has made FG (by) multiplying P . But, in fact, E has also made FG (by) multiplying D . Thus, as E is to Q , so P (is) to D [Prop. 7.19]. And since A, B, C, D are continually proportional, (starting) from a unit, D will thus not be measured by any other numbers except A, B, C [Prop. 9.13]. And P was assumed not (to be) the same as any of A, B, C . Thus, P does not measure D . But, as P (is) to D , so E (is) to Q . Thus, E does not measure Q either [Def. 7.20]. And E is a prime (number). And every prime number [is] prime to every (number) which it does not measure [Prop. 7.29]. Thus, E and Q are prime to one another. And (numbers) prime (to one another are) also the least (of those numbers having the same ratio as them) [Prop. 7.21], and the least (numbers) measure those (numbers) having the same ratio as them an equal number of times, the leading (measuring) the leading, and the following the following [Prop. 7.20]. And as E is to Q , (so) P (is) to D . Thus, E measures P the same number of times as Q (measures) D . And D is not measured by any other (numbers) except A, B, C . Thus, Q is the same as one of A, B, C . Let it be the same as B . And as many as is the multitude of B, C, D , let so many (of the set out numbers) be taken, (starting) from E , (namely) E, HK, L . And E, HK, L are in the same ratio as B, C, D . Thus, via equality, as B (is) to D , (so) E (is) to L [Prop. 7.14]. Thus, the (number created) from (multiplying) B, L is equal to the (number created) from multiplying D, E [Prop. 7.19]. But, the (number created) from (multiplying) D, E is equal to the (number created) from (multiplying) Q, P . Thus, the (number created) from (multiplying) Q, P is equal to the (number created) from (multiplying) B, L . Thus, as Q is to B , (so) L (is) to P [Prop. 7.19]. And Q is the same as B . Thus, L is also the same as P . The very thing (is) impossible. For P was assumed not (to be) the same as any of the (numbers) set out. Thus, FG cannot be measured by any number except A, B, C, D, E, HK, L, M , and a unit. And FG was shown (to be) equal to (the sum of) A, B, C, D, E, HK, L, M , and a unit. And a perfect number is one which is equal to (the sum of) its own parts [Def. 7.22]. Thus, FG is a per-

fect (number). (Which is) the very thing it was required to show.

† This proposition demonstrates that perfect numbers take the form $2^{n-1}(2^n - 1)$ provided that $2^n - 1$ is a prime number. The ancient Greeks knew of four perfect numbers: 6, 28, 496, and 8128, which correspond to $n = 2, 3, 5$, and 7 , respectively.

ELEMENTS BOOK 10

Incommensurable Magnitudes[†]



[†]The theory of incommensurable magnitudes set out in this book is generally attributed to Theaetetus of Athens. In the footnotes throughout this book, k , k' , etc. stand for distinct ratios of positive integers.

<Ὅροι.

1. Σύμμετρα μεγέθη λέγεται τὰ τῷ αὐτῷ μετρῶ μετρούμενα, ἀσύμμετρα δέ, <ὧν μηδὲν ἐνδέχεται κοινὸν μέτρον γενέσθαι.
2. Εὐθεῖαι δυνάμει σύμμετροί εἰσιν, <ὅταν τὰ ἀπ> αὐτῶν τετράγωνα τῷ αὐτῷ θωρίῳ μετρηῇται, ἀσύμμετροι δέ, <ὅταν τοῖς ἀπ> αὐτῶν τετραγῶνοις μηδὲν ἐνδέχεται θωρίῳ κοινὸν μέτρον γενέσθαι.
3. Τούτων ὑποκειμένων δείκνυται, <ὅτι τῇ| προτεθείσῃ εὐθείᾳ ὑπάρθουσιν εὐθεῖαι πλήθει >άπειροι σύμμετροί τε καὶ ἀσύμμετροι αἱ μὲν μήκει μόνον, αἱ δὲ καὶ δυνάμει. καλείσθω ο>ὗν ἡ μὲν προτεθείσα εὐθεῖα <ρητή, καὶ αἱ αὐτῇ σύμμετροι ε>ἴτε μήκει καὶ δυνάμει ε>ἴτε δυνάμει μόνον <ρηταί, αἱ δὲ αὐτῇ ἀσύμμετροι >άλογοι καλείσθωσαν.
4. Καὶ τὸ μὲν ἀπὸ τῆς προτεθείσης εὐθείας τετράγωνον <ρητόν, καὶ τὰ τούτῳ σύμμετρα <ρητά, τὰ δὲ τούτῳ ἀσύμμετρα >άλογα καλείσθω, καὶ αἱ δυνάμεναι αὐτὰ >άλογοι, εἰ μὲν τετράγωνα ε>ῖη, αὐταὶ αἱ πλευραί, εἰ δὲ <έτερα τινὰ εὐθύγραμμα, αἱ >ῖσα αὐτοῖς τετράγωνα ἀναγράφουσαι.

Definitions

1. Those magnitudes measured by the same measure are said (to be) commensurable, but (those) of which no (magnitude) admits to be a common measure (are said to be) incommensurable.[†]
2. (Two) straight-lines are commensurable in square[‡] when the squares on them are measured by the same area, but (are) incommensurable (in square) when no area admits to be a common measure of the squares on them.[§]
3. These things being assumed, it is proved that there exist an infinite multitude of straight-lines commensurable and incommensurable with an assigned straight-line—those (incommensurable) in length only, and those also (commensurable or incommensurable) in square.[¶] Therefore, let the assigned straight-line be called rational. And (let) the (straight-lines) commensurable with it, either in length and square, or in square only, (also be called) rational. But let the (straight-lines) incommensurable with it be called irrational.^{*}
4. And let the square on the assigned straight-line be called rational. And (let areas) commensurable with it (also be called) rational. But (let areas) incommensurable with it (be called) irrational, and (let) their square-roots[§] (also be called) irrational—the sides themselves, if the (areas) are squares, and the (straight-lines) describing squares equal to them, if the (areas) are some other rectilinear (figure).^ᵇ

[†] In other words, two magnitudes α and β are commensurable if $\alpha : \beta :: 1 : k$, and incommensurable otherwise.

[‡] Literally, “in power”.

[§] In other words, two straight-lines of length α and β are commensurable in square if $\alpha : \beta :: 1 : k^{1/2}$, and incommensurable in square otherwise. Likewise, the straight-lines are commensurable in length if $\alpha : \beta :: 1 : k$, and incommensurable in length otherwise.

[¶] To be more exact, straight-lines can either be commensurable in square only, incommensurable in length only, or commensurable/incommensurable in both length and square, with an assigned straight-line.

^{*} Let the length of the assigned straight-line be unity. Then rational straight-lines have lengths expressible as k or $k^{1/2}$, depending on whether the lengths are commensurable in length, or in square only, respectively, with unity. All other straight-lines are irrational.

[§] The square-root of an area is the length of the side of an equal area square.

^ᵇ The area of the square on the assigned straight-line is unity. Rational areas are expressible as k . All other areas are irrational. Thus, squares whose sides are of rational length have rational areas, and vice versa.

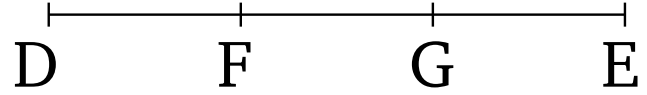
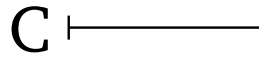
1.

Δύο μεγεθῶν ἀνίσων ἐκκειμένων, ἔαν ἀπὸ τοῦ μείζονος ἀφαιρεθῇ| μείζον >ῇ τὸ <ήμισυ καὶ τοῦ καταλειπομένου μείζον >ῇ τὸ <ήμισυ, καὶ τοῦτο ἀεὶ γίγνηται, λειφθήσεται τι μέγεθος, <ὃ >έσται >έλασσον τοῦ ἐκκειμένου ἐλάσσονος μεγέθους.
>Ἐστω δύο μεγέθη >άνισα τὰ AB, Γ, <ὧν μείζον τὸ AB; λέγω, <ὅτι, ἔαν ἀπὸ τοῦ AB ἀφαιρεθῇ| μείζον >ῇ τὸ <ήμισυ καὶ τοῦ καταλειπομένου μείζον >ῇ τὸ <ήμισυ, καὶ τοῦτο ἀεὶ γίγνηται, λειφθήσεται τι μέγεθος, <ὃ >έσται >έλασσον τοῦ Γ μεγέθους.

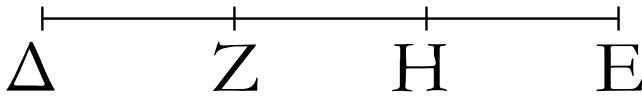
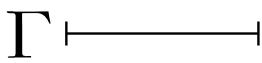
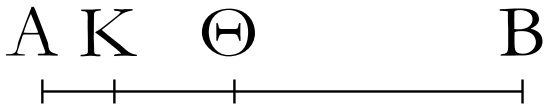
Proposition 1[†]

If, from the greater of two unequal magnitudes (which are) laid out, (a part) greater than half is subtracted, and (if from) the remainder (a part) greater than half (is subtracted), and (if) this happens continually, then some magnitude will (eventually) be left which will be less than the lesser laid out magnitude.

Let AB and C be two unequal magnitudes, of which (let) AB (be) the greater. I say that if (a part) greater than half is subtracted from AB , and (if a part) greater than half (is subtracted) from the remainder, and (if) this happens continually, then some magnitude will (eventually) be left which will be less than the magnitude C .



For C , when multiplied (by some number), will sometimes



Τὸ Γ γὰρ πολλαπλασιαζόμενον >έσται ποτὲ τοῦ AB μείζον. πεπολλαπλασιάσθω, καὶ >έστω τὸ ΔE τοῦ μὲν Γ πολλαπλάσιον†. Let DE be (both) a multiple of C , and greater than AB . And let DE be divided into the (divisions) DF , FG , GE , equal to C . And let BH , (which is) greater than half, be subtracted from AB . And (let) HK , (which is) greater than half, (be subtracted) from AH . And let this happen continually, until the divisions in AB become equal in number to the divisions in DE .

>Ἐστωσαν ο>ὗν αἱ AK , KH , HB διαιρέσειζ ἰσοπληθεῖς ο>ῦσαι ταῖς ΔZ , ZH , HE ; καὶ ἐπεὶ μείζον ἐστὶ τὸ ΔE τοῦ AB , καὶ ἀφή|ρηται ἀπὸ μὲν τοῦ ΔE >έλασσον τοῦ ἡμίσεως τὸ EH , ἀπὸ δὲ τοῦ AB μείζον >ἢ τὸ <ἡμισυ τὸ BH , λοιπὸν >άρα τὸ HA λοιποῦ τοῦ ΘA μείζον ἐστίν. καὶ ἐπεὶ μείζον ἐστὶ τὸ HA τοῦ ΘA , καὶ ἀφή|ρηται τοῦ μὲν HA <ἡμισυ τὸ HZ , τοῦ δὲ ΘA μείζον >ἢ τὸ <ἡμισυ τὸ ΘK , λοιπὸν >άρα τὸ ΔZ λοιποῦ τοῦ AK μείζον ἐστίν. >ἴσον δὲ τὸ ΔZ τῷ Γ ; καὶ τὸ Γ >άρα τοῦ AK μείζον ἐστίν. >έλασσον >άρα τὸ AK τοῦ Γ .

Καταλείπεται >άρα ἀπὸ τοῦ AB μεγέθους τὸ AK μέγεθος >έλασσον >ὄν τοῦ ἐκκειμένου ἐλάσσονος μεγέθους τοῦ Γ ; <όπερ >έδει δεῖχαι. — ὁμοίως δὲ δειθθήσεται, κ>ὰν ἡμίση >ἢ τὰ ἀφαιρούμενα.

Therefore, let the divisions (in AB) be AK , KH , HB , being equal in number to DF , FG , GE . And since DE is greater than AB , and EG , (which is) less than half, has been subtracted from DE , and BH , (which is) greater than half, from AB , the remainder GD is thus greater than the remainder HA . And since GD is greater than HA , and the half GF has been subtracted from GD , and HK , (which is) greater than half, from HA , the remainder DF is thus greater than the remainder AK . And DF (is) equal to C . C is thus also greater than AK . Thus, AK (is) less than C .

Thus, the magnitude AK , which is less than the lesser laid out magnitude C , is left over from the magnitude AB . (Which is) the very thing it was required to show. — (The theorem) can similarly be proved even if the (parts) subtracted are halves.

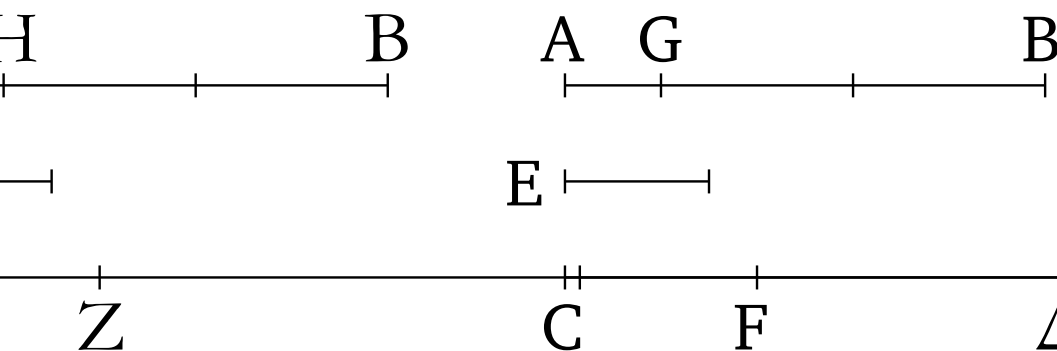
† This theorem is the basis of the so-called method of exhaustion, and is generally attributed to Eudoxus of Cnidus.

2.

Ἐὰν δύο μεγεθῶν [ἐκκειμένων] ἀνίσων ἀνθυφαιρουμένου ἀεὶ τοῦ ἐλάσσονος ἀπὸ τοῦ μείζονος τὸ καταλειπόμενον μηδέποτε καταμετρή| τὸ πρὸ ἑαυτοῦ, ἀσύμμετρα >έσται τὰ μεγέθη.

Proposition 2

If the remainder of two unequal magnitudes (which are) [laid out] never measures the (magnitude) before it, (when) the lesser (magnitude is) continually subtracted in turn from the greater, then the (original) magnitudes will be incommensurable.



Δύο γὰρ μεγεθῶν >όντων ἀνίσων τῶν AB , $\Gamma\Delta$ καὶ ἐλάσσονοξ τοῦ AB ἀνθυφαιρουμένου ἀεὶ τοῦ ἐλάσσονοξ ἀπὸ τοῦ μείζονοξ τὸ περιλειπόμενον μηδέποτε καταμετρεῖτω τὸ πρὸ ἑαυτοῦ; λέγω, <ὅτι ἀσύμμετρά ἐστι τὰ AB , $\Gamma\Delta$ μεγέθη.

Εἰ γὰρ ἐστὶ σύμμετρα, μετρήσει τι αὐτὰ μέγεθος. μετρεῖτω, εἰ δυνατόν, καὶ >έστω τὸ E ; καὶ τὸ μὲν AB τὸ $Z\Delta$ καταμετροῦν λειπέτω ἑαυτοῦ >έλασσον τὸ ΓZ , τὸ δὲ ΓZ τὸ BH καταμετροῦν λειπέτω ἑαυτοῦ >έλασσον τὸ AH , καὶ τοῦτο ἀεὶ γινέσθω, <έωξ ο<ὐ λειφθῇ τι μέγεθος, <ὅ ἐστιν >έλασσον τοῦ E . γεγονέτω, καὶ λελείφθω τὸ AH >έλασσον τοῦ E . ἐπεὶ ο>ὐν τὸ E τὸ AB μετρεῖ, ἀλλὰ τὸ AB τὸ ΔZ μετρεῖ, καὶ τὸ E >άρα τὸ $Z\Delta$ μετρήσει. μετρεῖ δὲ καὶ <όλον τὸ $\Gamma\Delta$; καὶ λοιπὸν >άρα τὸ ΓZ μετρήσει. ἀλλὰ τὸ ΓZ τὸ BH μετρεῖ; καὶ τὸ E >άρα τὸ BH μετρεῖ. μετρεῖ δὲ καὶ <όλον τὸ AB ; καὶ λοιπὸν >άρα τὸ AH μετρήσει, τὸ μείζον τὸ >έλασσον; <όπερ ἐστὶν ἀδύνατον. οὐκ >άρα τὰ AB , $\Gamma\Delta$ μεγέθη μετρήσει τι μέγεθος; ἀσύμμετρα >άρα ἐστὶ τὰ AB , $\Gamma\Delta$ μεγέθη.

Ἐὰν >άρα δύο μεγεθῶν ἀνίσων, καὶ τὰ ἐχῇξ.

For, AB and CD being two unequal magnitudes, and AB (being) the lesser, let the remainder never measure the (magnitude) before it, (when) the lesser (magnitude is) continually subtracted in turn from the greater. I say that the magnitudes AB and CD are incommensurable.

For if they are commensurable then some magnitude will measure them (both). If possible, let it (so) measure (them), and let it be E . And let AB leave CF less than itself (in) measuring FD , and let CF leave AG less than itself (in) measuring BG , and let this happen continually, until some magnitude which is less than E is left. Let (this) have occurred,[†] and let AG , (which is) less than E , be left. Therefore, since E measures AB , but AB measures DF , E will thus also measure FD . And it also measures the whole (of) CD . Thus, it will also measure the remainder CF . But, CF measures BG . Thus, E also measures BG . And it also measures the whole (of) AB . Thus, it will also measure the remainder AG , the greater (measuring) the lesser. The very thing is impossible. Thus, some magnitude cannot measure (both) the magnitudes AB and CD . Thus, the magnitudes AB and CD are incommensurable [Def. 10.1].

Thus, if ... of two unequal magnitudes, and so on

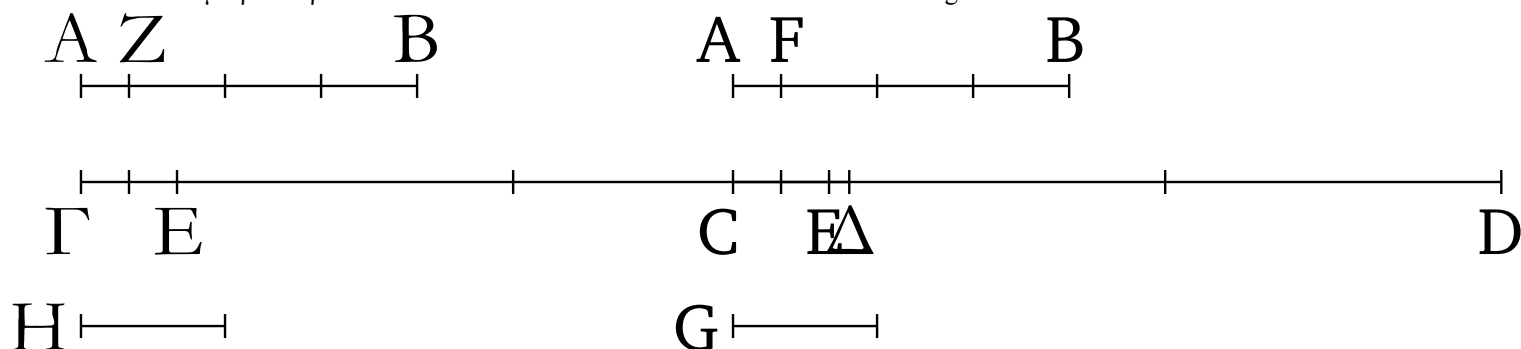
[†] The fact that this will eventually occur is guaranteed by Prop. 10.1.

3.

Proposition 3

Δύο μεγεθῶν συμμέτρων δοθέντων τὸ μέγιστον αὐτῶν κοινὸν μέτρον εὑρεῖν.

To find the greatest common measure of two given commensurable magnitudes.



>Ἐστω τὰ δοθέντα δύο μεγέθη σύμμετρα τὰ AB , $\Gamma\Delta$, <ὠν >έλασσον τὸ AB ; δεῖ δὴ τῶν AB , $\Gamma\Delta$ τὸ μέγιστον κοινὸν μέτρον εὑρεῖν.

Τὸ AB γὰρ μέγεθος >ήτοι μετρεῖ τὸ $\Gamma\Delta$ >ή ο>ύ. εἰ μὲν ο>ὐν μετρεῖ, μετρεῖ δὲ καὶ ἑαυτό, τὸ AB >άρα τῶν AB , $\Gamma\Delta$ κοινὸν μέτρον ἐστίν; καὶ φανερόν, <ὅτι καὶ μέγιστον. μείζον γὰρ τοῦ AB μεγέθους τὸ AB οὐ μετρήσει.

μὴ μετρεῖτω δὴ τὸ AB τὸ $\Gamma\Delta$. καὶ ἀνθυφαιρουμένου ἀεὶ τοῦ ἐλάσσονοξ ἀπὸ τοῦ μείζονοξ, τὸ περιλειπόμενον μετρήσει ποτὲ τὸ πρὸ ἑαυτοῦ διὰ τὸ μὴ ε>ῖναι ἀσύμμετρα τὰ AB , $\Gamma\Delta$; καὶ τὸ μὲν AB τὸ $E\Delta$ καταμετροῦν λειπέτω ἑαυτοῦ

Let AB and CD be the two given magnitudes, of which (let) AB (be) the lesser. So, it is required to find the greatest common measure of AB and CD .

For the magnitude AB either measures, or (does) not (measure), CD . Therefore, if it measures (CD), and (since) it also measures itself, AB is thus a common measure of AB and CD . And (it is) clear that (it is) also (the) greatest. For a (magnitude) greater than magnitude AB cannot measure AB .

So let AB not measure CD . And continually subtracting in turn the lesser (magnitude) from the greater, the remaining

>έλασσον τὸ ΕΓ, τὸ δὲ ΕΓ τὸ ΖΒ καταμετροῦν λειπέτω
ἑαυτοῦ >έλασσον τὸ ΑΖ, τὸ δὲ ΑΖ τὸ ΓΕ μετρεῖτω.

Ἐπεὶ οὖν τὸ ΑΖ τὸ ΓΕ μετρεῖ, ἀλλὰ τὸ ΓΕ τὸ ΖΒ μετρεῖ, καὶ
τὸ ΑΖ >άρα τὸ ΖΒ μετρήσει. μετρεῖ δὲ καὶ ἑαυτό; καὶ <όλον
>άρα τὸ ΑΒ μετρήσει τὸ ΑΖ. ἀλλὰ τὸ ΑΒ τὸ ΔΕ μετρεῖ;
καὶ τὸ ΑΖ >άρα τὸ ΕΔ μετρήσει. μετρεῖ δὲ καὶ τὸ ΓΕ; καὶ
<όλον >άρα τὸ ΓΔ μετρεῖ; τὸ ΑΖ >άρα τῶν ΑΒ, ΓΔ κοινὸν
μέτρον ἐστίν. λέγω δὴ, <ὅτι καὶ μέγιστον. εἰ γὰρ μή, >έσται
τι μέγεθος μείζον τοῦ ΑΖ, <ὃ μετρήσει τὰ ΑΒ, ΓΔ. >έστω τὸ
Η. ἐπεὶ οὖν τὸ Η τὸ ΑΒ μετρεῖ, ἀλλὰ τὸ ΑΒ τὸ ΕΔ μετρεῖ,
καὶ τὸ Η >άρα τὸ ΕΔ μετρήσει. μετρεῖ δὲ καὶ <όλον τὸ ΓΔ;
καὶ λοιπὸν >άρα τὸ ΓΕ μετρήσει τὸ Η. ἀλλὰ τὸ ΓΕ τὸ ΖΒ
μετρεῖ; καὶ τὸ Η >άρα τὸ ΖΒ μετρήσει. μετρεῖ δὲ καὶ <όλον
τὸ ΑΒ, καὶ λοιπὸν τὸ ΑΖ μετρήσει, τὸ μείζον τὸ >έλασσον;
<ὅπερ ἐστὶν ἀδύνατον. οὐκ >άρα μείζον τι μέγεθος τοῦ ΑΖ
τὰ ΑΒ, ΓΔ μετρήσει; τὸ ΑΖ >άρα τῶν ΑΒ, ΓΔ τὸ μέγιστον
κοινὸν μέτρον ἐστίν.

Δύο >άρα μεγεθῶν συμμέτρων δοθέντων τῶν ΑΒ, ΓΔ τὸ
μέγιστον κοινὸν μέτρον η<ύρηται; <ὅπερ >έδει δεῖξαι.

(magnitude) will (at) some time measure the (magnitude)
before it, on account of AB and CD not being incommen-
surable [Prop. 10.2]. And let AB leave EC less than itself
(in) measuring ED , and let EC leave AF less than itself
(in) measuring FB , and let AF measure CE .

Therefore, since AF measures CE , but CE measures FB ,
 AF will thus also measure FB . And it also measures it-
self. Thus, AF will also measure the whole (of) AB . But,
 AB measures DE . Thus, AF will also measure ED . And it
also measures CE . Thus, it also measures the whole of CD .
Thus, AF is a common measure of AB and CD . So I say
that (it is) also (the) greatest (common measure). For, if not,
there will be some magnitude, greater than AF , which will
measure (both) AB and CD . Let it be G . Therefore, since
 G measures AB , but AB measures ED , G will thus also
measure ED . And it also measures the whole of CD . Thus,
 G will also measure the remainder CE . But CE measures
 FB . Thus, G will also measure FB . And it also measures
the whole (of) AB . And (so) it will measure the remain-
der AF , the greater (measuring) the lesser. The very thing
is impossible. Thus, some magnitude greater than AF can-
not measure (both) AB and CD . Thus, AF is the greatest
common measure of AB and CD .

Thus, the greatest common measure of two given commensurable magnitudes, AB and CD , has been found. (Which is) the very thing it was required to show.

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, <ὅτι, ἔαν μέγεθος δύο μεγέθη
μετρήῃ, καὶ τὸ μέγιστον αὐτῶν κοινὸν μέτρον μετρήσει.

Corollary

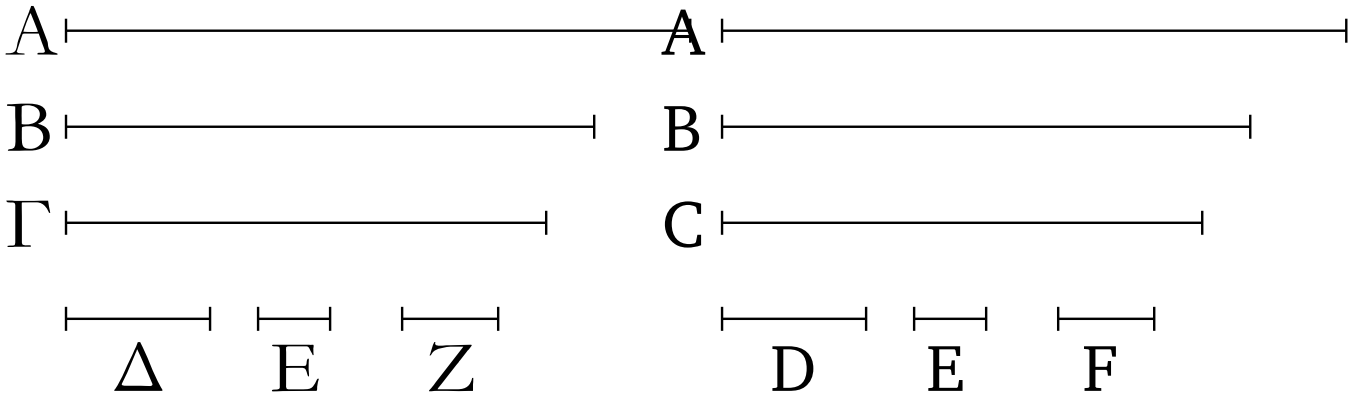
So (it is) clear, from this, that if a magnitude measures two
magnitudes then it will also measure their greatest common
measure.

4.

Τριῶν μεγεθῶν συμμέτρων δοθέντων τὸ μέγιστον αὐτῶν
κοινὸν μέτρον εὑρεῖν.

Proposition 4

To find the greatest common measure of three given com-
mensurable magnitudes.



>Ἐστω τὰ δοθέντα τρία μεγέθη σύμμετρα τὰ Α, Β, Γ; δεῖ δὴ
τῶν Α, Β, Γ τὸ μέγιστον κοινὸν μέτρον εὑρεῖν.

Εἰλήφθω γὰρ δύο τῶν Α, Β τὸ μέγιστον κοινὸν μέτρον, καὶ
>έστω τὸ Δ; τὸ δὴ Δ τὸ Γ >ήτοι μετρεῖ >ή οὐ [μετρεῖ].
μετρεῖτω πρότερον. ἐπεὶ οὖν τὸ Δ τὸ Γ μετρεῖ, μετρεῖ δὲ
καὶ τὰ Α, Β, τὸ Δ >άρα τὰ Α, Β, Γ μετρεῖ; τὸ Δ >άρα τῶν Α,
Β, Γ κοινὸν μέτρον ἐστίν. καὶ φανερόν, <ὅτι καὶ μέγιστον;
μείζον γὰρ τοῦ Δ μεγέθους τὰ Α, Β οὐ μετρεῖ.

μὴ μετρεῖτω δὴ τὸ Δ τὸ Γ. λέγω πρῶτον, <ὅτι σύμμετρά ἐστι
τὰ Γ, Δ. ἐπεὶ γὰρ σύμμετρά ἐστι τὰ Α, Β, Γ, μετρήσει τι αὐτὰ
μέγεθος, <ὃ δηλαδὴ καὶ τὰ Α, Β μετρήσει; <ώστε καὶ τὸ τῶν
Α, Β μέγιστον κοινὸν μέτρον τὸ Δ μετρήσει. μετρεῖ δὲ καὶ τὸ

Let A, B, C be the three given commensurable magnitudes.
So it is required to find the greatest common measure of A ,
 B, C .

For let the greatest common measure of the two (magni-
tudes) A and B be taken [Prop. 10.3], and let it be D . So
 D either measures, or [does] not [measure], C . Let it, first
of all, measure (C). Therefore, since D measures C , and it
also measures A and B , D thus measures A, B, C . Thus, D
is a common measure of A, B, C . And (it is) clear that (it
is) also (the) greatest (common measure). For no magnitude
larger than D measures (both) A and B .

So let D not measure C . I say, first, that C and D are com-

Γ; <ώστε τὸ εἰρημένον μέγεθος μετρήσει τὰ Γ, Δ; σύμμετρα >άρα ἐστὶ τὰ Γ, Δ. εἰλήφθω οὖν αὐτῶν τὸ μέγιστον κοινὸν μέτρον, καὶ >έστω τὸ Ε. ἐπεὶ οὖν τὸ Ε τὸ Δ μετρεῖ, ἀλλὰ τὸ Δ τὰ Α, Β μετρεῖ, καὶ τὸ Ε >άρα τὰ Α, Β μετρήσει. μετρεῖ δὲ καὶ τὸ Γ. τὸ Ε >άρα τὰ Α, Β, Γ μετρεῖ; τὸ Ε >άρα τῶν Α, Β, Γ κοινὸν ἐστὶ μέτρον. λέγω δὴ, <ὅτι καὶ μέγιστον. εἰ γὰρ δυνατόν, >έστω τι τοῦ Ε μείζον μέγεθος τὸ Ζ, καὶ μετρεῖτω τὰ Α, Β, Γ. καὶ ἐπεὶ τὸ Ζ τὰ Α, Β, Γ μετρεῖ, καὶ τὰ Α, Β >άρα μετρήσει καὶ τὸ τῶν Α, Β μέγιστον κοινὸν μέτρον μετρήσει. τὸ δὲ τῶν Α, Β μέγιστον κοινὸν μέτρον ἐστὶ τὸ Δ; τὸ Ζ >άρα τὸ Δ μετρεῖ. μετρεῖ δὲ καὶ τὸ Γ; τὸ Ζ >άρα τὰ Γ, Δ μετρεῖ; καὶ τὸ τῶν Γ, Δ >άρα μέγιστον κοινὸν μέτρον μετρήσει τὸ Ζ. >έστι δὲ τὸ Ε; τὸ Ζ >άρα τὸ Ε μετρήσει, τὸ μείζον τὸ >έλασσον; <ὅπερ ἐστὶν ἀδύνατον. οὐκ >άρα μείζον τι τοῦ Ε μεγέθους [μέγεθος] τὰ Α, Β, Γ μετρεῖ; τὸ Ε >άρα τῶν Α, Β, Γ τὸ μέγιστον κοινὸν μέτρον ἐστίν, ἔαν μὴ μετρήῃ τὸ Δ τὸ Γ, ἔαν δὲ μετρήῃ, αὐτὸ τὸ Δ. Τριῶν >άρα μεγεθῶν συμμέτρων δοθέντων τὸ μέγιστον κοινὸν μέτρον η<ύρῃται [<ὅπερ >έδει δεῖχαι].

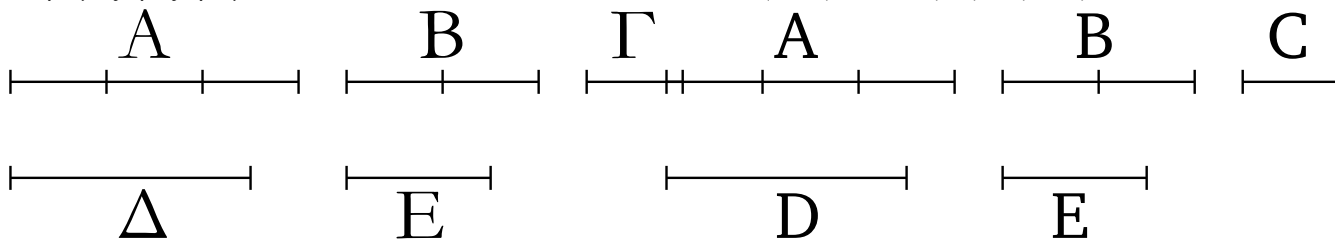
Πόρισμα.

Ἐκ δὴ τούτου φανερόν, <ὅτι, ἔαν μέγεθος τρία μεγέθη μετρήῃ, καὶ τὸ μέγιστον αὐτῶν κοινὸν μέτρον μετρήσει.

Ὅμοίως δὴ καὶ ἐπὶ πλειόνων τὸ μέγιστον κοινὸν μέτρον ληφθήσεται, καὶ τὸ πόρισμα προθωρήσει. <ὅπερ >έδει δεῖχαι.

5.

Τὰ σύμμετρα μεγέθη πρὸς >άλληλα λόγον >έθει, <ὄν ἀριθμὸς πρὸς ἀριθμόν.



>Ἐστω σύμμετρα μεγέθη τὰ Α, Β; λέγω, <ὅτι τὸ Α πρὸς τὸ Β λόγον >έθει, <ὄν ἀριθμὸς πρὸς ἀριθμόν.

Ἐπεὶ γὰρ σύμμετρά ἐστὶ τὰ Α, Β, μετρήσει τι αὐτὰ μέγεθος. μετρεῖτω, καὶ >έστω τὸ Γ. καὶ ὅσάκις τὸ Γ τὸ Α μετρεῖ, τοσαῦται μονάδες >έστωσαν ἐν τῷ Δ, ὅσάκις δὲ τὸ Γ τὸ Β μετρεῖ, τοσαῦται μονάδες >έστωσαν ἐν τῷ Ε.

Ἐπεὶ οὖν τὸ Γ τὸ Α μετρεῖ κατὰ τὰς ἐν τῷ Δ μονάδας, μετρεῖ δὲ καὶ ἡ μονὰς τὸν Δ κατὰ τὰς ἐν αὐτῷ μονάδας, ἰσάκις >άρα ἡ μονὰς τὸν Δ μετρεῖ ἀριθμόν καὶ τὸ Γ μέγεθος τὸ Α; >έστιν >άρα ὡς τὸ Γ πρὸς τὸ Α, ο<ύτως ἡ μονὰς πρὸς

measurable. For if A, B, C are commensurable then some magnitude will measure them which will clearly also measure A and B . Hence, it will also measure D , the greatest common measure of A and B [Prop. 10.3 corr.]. And it also measures C . Hence, the aforementioned magnitude will measure (both) C and D . Thus, C and D are commensurable [Def. 10.1]. Therefore, let their greatest common measure be taken [Prop. 10.3], and let it be E . Therefore, since E measures D , but D measures (both) A and B , E will thus also measure A and B . And it also measures C . Thus, E measures A, B, C . Thus, E is a common measure of A, B, C . So I say that (it is) also (the) greatest (common measure). For, if possible, let F be some magnitude greater than E , and let it measure A, B, C . And since F measures A, B, C , it will thus also measure A and B , and will (thus) measure the greatest common measure of A and B [Prop. 10.3 corr.]. And D is the greatest common measure of A and B . Thus, F measures D . And it also measures C . Thus, F measures (both) C and D . Thus, F will also measure the greatest common measure of C and D [Prop. 10.3 corr.]. And it is E . Thus, F will measure E , the greater (measuring) the lesser. The very thing is impossible. Thus, some [magnitude] greater than the magnitude E cannot measure A, B, C . Thus, if D does not measure C then E is the greatest common measure of A, B, C . And if it does measure (C) then D itself (is the greatest common measure).

Thus, the greatest common measure of three given commensurable magnitudes has been found. [(Which is) the very thing it was required to show.]

Corollary

So (it is) clear, from this, that if a magnitude measures three magnitudes then it will also measure their greatest common measure.

So, similarly, the greatest common measure of more (magnitudes) can also be taken, and the (above) corollary will go forward. (Which is) the very thing it was required to show.

Proposition 5

Commensurable magnitudes have to one another the ratio which (some) number (has) to (some) number.

Let A and B be commensurable magnitudes. I say that A has to B the ratio which (some) number (has) to (some) number.

For if A and B are commensurable (magnitudes) then some magnitude will measure them. Let it (so) measure (them), and let it be C . And as many times as C measures A , so many units let there be in D . And as many times as C measures B , so many units let there be in E .

Therefore, since C measures A according to the units in D , and a unit also measures D according to the units in it, a unit

τὸν Δ; ἀνάπαλιν >άρα, ὥς τὸ Α πρὸς τὸ Γ, ο<ύτωξ ὁ Δ πρὸς τὴν μονάδα. πάλιν ἐπεὶ τὸ Γ τὸ Β μετρεῖ κατὰ τὰς ἐν τῷ Ε μονάδας, μετρεῖ δὲ καὶ ἡ μονὰξ τὸν Ε κατὰ τὰς ἐν αὐτῷ μονάδας, ἰσάκιξ >άρα ἡ μονὰξ τὸν Ε μετρεῖ καὶ τὸ Γ τὸ Β; >έστιν >άρα ὥς τὸ Γ πρὸς τὸ Β, ο<ύτωξ ἡ μονὰξ πρὸς τὸν Ε. ἐδείθθη δὲ καὶ ὥς τὸ Α πρὸς τὸ Γ, ὁ Δ πρὸς τὴν μονάδα; δι> >ίσου >άρα ἐστὶν ὥς τὸ Α πρὸς τὸ Β, ο<ύτωξ ὁ Δ ἀριθμὸξ πρὸς τὸν Ε.

Τὰ >άρα σύμμετρα μεγέθη τὰ Α, Β πρὸς >άλληλα λόγον >έθει, <ὸν ἀριθμὸξ ὁ Δ πρὸς ἀριθμὸν τὸν Ε; <όπερ >έδει δεῖχαι.

thus measures the number D as many times as the magnitude C (measures) A . Thus, as C is to A , so a unit (is) to D [Def. 7.20].[†] Thus, inversely, as A (is) to C , so D (is) to a unit [Prop. 5.7 corr.]. Again, since C measures B according to the units in E , and a unit also measures E according to the units in it, a unit thus measures E the same number of times that C (measures) B . Thus, as C is to B , so a unit (is) to E [Def. 7.20]. And it was also shown that as A (is) to C , so D (is) to a unit. Thus, via equality, as A is to B , so the number D (is) to the (number) E [Prop. 5.22].

Thus, the commensurable magnitudes A and B have to one another the ratio which the number D (has) to the number E . (Which is) the very thing it was required to show.

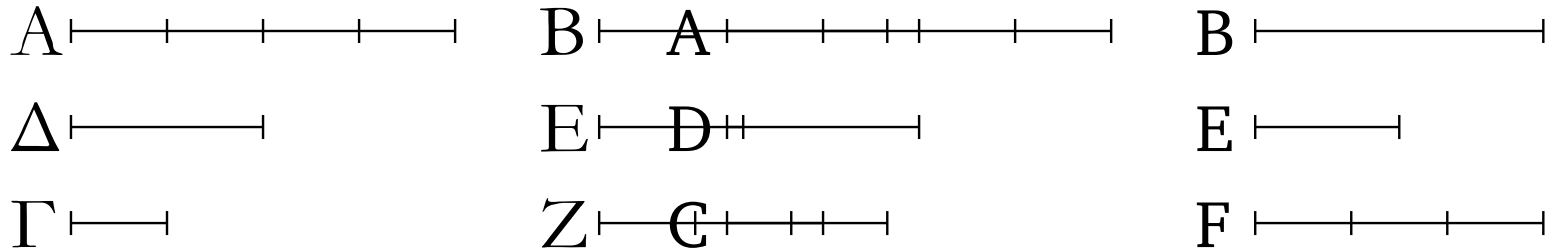
[†] There is a slight logical gap here, since Def. 7.20 applies to four numbers, rather than two number and two magnitudes.

6.

Proposition 6

Ἐὰν δύο μεγέθη πρὸς >άλληλα λόγον >έθῃ, <ὸν ἀριθμὸξ πρὸς ἀριθμὸν, σύμμετρα >έσται τὰ μεγέθη.

If two magnitudes have to one another the ratio which (some) number (has) to (some) number then the magnitudes will be commensurable.



Δύο γὰρ μεγέθη τὰ Α, Β πρὸς >άλληλα λόγον ἐθέτω, <ὸν ἀριθμὸξ ὁ Δ πρὸς ἀριθμὸν τὸν Ε; λέγω, <ὅτι σύμμετρά ἐστι τὰ Α, Β μεγέθη.

<Όσαι γάρ εἰσιν ἐν τῷ Δ μονάδες, εἰς τοσαῦτα >ῖσα διηρήσθω τὸ Α, καὶ ἐνὶ αὐτῶν >ῖσον >έστω τὸ Γ; <όσαι δὲ εἰσιν ἐν τῷ Ε μονάδες, ἐκ τοσούτων μεγεθῶν >ῖσων τῷ Γ συγκείσθω τὸ Ζ.

Ἐπεὶ ο>ύν, <όσαι εἰσιν ἐν τῷ Δ μονάδες, τοσαῦτά εἰσι καὶ ἐν τῷ Α μεγέθη >ῖσα τῷ Γ, <ὁ >άρα μέρος ἐστὶν ἡ μονὰξ τοῦ Δ, τὸ αὐτὸ μέρος ἐστὶ καὶ τὸ Γ τοῦ Α; >έστιν >άρα ὥς τὸ Γ πρὸς τὸ Α, ο<ύτωξ ἡ μονὰξ πρὸς τὸν Δ. μετρεῖ δὲ ἡ μονὰξ τὸν Δ ἀριθμὸν; μετρεῖ >άρα καὶ τὸ Γ τὸ Α. καὶ ἐπεὶ ἐστὶν ὥς τὸ Γ πρὸς τὸ Α, ο<ύτωξ ἡ μονὰξ πρὸς τὸν Δ [ἀριθμὸν], ἀνάπαλιν >άρα ὥς τὸ Α πρὸς τὸ Γ, ο<ύτωξ ὁ Δ ἀριθμὸξ πρὸς τὴν μονάδα. πάλιν ἐπεὶ, <όσαι εἰσιν ἐν τῷ Ε μονάδες, τοσαῦτά εἰσι καὶ ἐν τῷ Ζ >ῖσα τῷ Γ, >έστιν >άρα ὥς τὸ Γ πρὸς τὸ Ζ, ο<ύτωξ ἡ μονὰξ πρὸς τὸν Ε [ἀριθμὸν]. ἐδείθθη δὲ καὶ ὥς τὸ Α πρὸς τὸ Γ, ο<ύτωξ ὁ Δ πρὸς τὴν μονάδα; δι> >ίσου >άρα ἐστὶν ὥς τὸ Α πρὸς τὸ Ζ, ο<ύτωξ ὁ Δ πρὸς τὸν Ε. ἀλλ> ὥς ὁ Δ πρὸς τὸν Ε, ο<ύτωξ ἐστὶ τὸ Α πρὸς τὸ Β; καὶ ὥς >άρα τὸ Α πρὸς τὸ Β, ο<ύτωξ καὶ πρὸς τὸ Ζ. τὸ Α >άρα πρὸς ἑκάτερον τῶν Β, Ζ τὸν αὐτὸν >έθει λόγον; >ῖσον >άρα ἐστὶ τὸ Β τῷ Ζ. μετρεῖ δὲ τὸ Γ τὸ Ζ; μετρεῖ >άρα καὶ τὸ Β. ἀλλὰ μὴν καὶ τὸ Α; τὸ Γ >άρα τὰ Α, Β μετρεῖ. σύμμετρον >άρα ἐστὶ τὸ Α τῷ Β.

Ἐὰν >άρα δύο μεγέθη πρὸς >άλληλα, καὶ τὰ ἐχῇξ.

For let the two magnitudes A and B have to one another the ratio which the number D (has) to the number E . I say that the magnitudes A and B are commensurable.

For, as many units as there are in D , let A be divided into so many equal (divisions). And let C be equal to one of them. And as many units as there are in E , let F be the sum of so many magnitudes equal to C .

Therefore, since as many units as there are in D , so many magnitudes equal to C are also in A , therefore whichever part a unit is of D , C is also the same part of A . Thus, as C is to A , so a unit (is) to D [Def. 7.20]. And a unit measures the number D . Thus, C also measures A . And since as C is to A , so a unit (is) to the [number] D , thus, inversely, as A (is) to C , so the number D (is) to a unit [Prop. 5.7 corr.]. Again, since as many units as there are in E , so many (magnitudes) equal to C are also in F , thus as C is to F , so a unit (is) to the [number] E [Def. 7.20]. And it was also shown that as A (is) to C , so D (is) to a unit. Thus, via equality, as A is to F , so D (is) to E [Prop. 5.22]. But, as D (is) to E , so A is to B . And thus as A (is) to B , so (it) also is to F [Prop. 5.11]. Thus, A has the same ratio to each of B and F . Thus, B is equal to F [Prop. 5.9]. And C measures F . Thus, it also measures B . But, in fact, (it) also (measures) A . Thus, C measures (both) A and B . Thus, A is commensurable with B [Def. 10.1].

Thus, if two magnitudes ...to one another, and so on

Πόρισμα.

Corollary

Ἐκ δὴ τούτου φανερόν, <ὅτι, ἔαν >ῶσι δύο ἀριθμοί, ὥς οἱ Δ, Ε, καὶ εὐθεῖα, ὥς ἡ Α, δύνατόν ἐστι ποιῆσαι ὥς ὁ Δ ἀριθμὸξ πρὸς τὸν Ε ἀριθμὸν, ο<ύτωξ τὴν εὐθεῖαν πρὸς εὐθεῖαν. ἔαν δὲ καὶ τῶν Α, Ζ μέση ἀνάλογον ληφθῇ, ὥς ἡ Β, >έσται ὥς

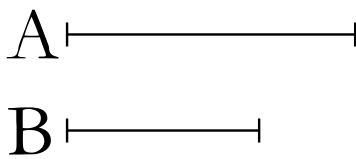
So it is clear, from this, that if there are two numbers, like D and E , and a straight-line, like A , then it is possible to contrive that as the number D (is) to the number E , so the straight-line (is) to (another) straight-line (i.e., F). And if

ἢ A πρὸς τὴν Z , οὐτως τὸ ἀπὸ τῆς A πρὸς τὸ ἀπὸ τῆς B , τουτέστιν ὡς ἡ πρώτη πρὸς τὴν τρίτην, οὐτως τὸ ἀπὸ τῆς πρώτης πρὸς τὸ ἀπὸ τῆς δευτέρας τὸ <ὅμοιον καὶ ὁμοίως ἀναγραφόμενον. ἀλλ> ὡς ἡ A πρὸς τὴν Z , οὐτως ἐστὶν ὁ Δ ἀριθμὸς πρὸς τὸν E ἀριθμόν; γέγονεν >άρα καὶ ὡς ὁ Δ ἀριθμὸς πρὸς τὸν E ἀριθμόν, οὐτως τὸ ἀπὸ τῆς A εὐθείας πρὸς τὸ ἀπὸ τῆς B εὐθείας; <ὅπερ >έδει δεῖχαι.

7.

Τὰ ἀσύμμετρα μεγέθη πρὸς >άλληλα λόγον οὐκ >έθει, <ὃν ἀριθμὸς πρὸς ἀριθμόν.

>Ἐστω ἀσύμμετρα μεγέθη τὰ A , B ; λέγω, <ὅτι τὸ A πρὸς τὸ B λόγον οὐκ >έθει, <ὃν ἀριθμὸς πρὸς ἀριθμόν.

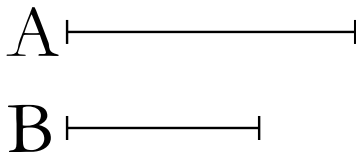


Εἰ γὰρ >έθει τὸ A πρὸς τὸ B λόγον, <ὃν ἀριθμὸς πρὸς ἀριθμόν, σύμμετρον >έσται τὸ A τῷ B . οὐκ >έστι δέ; οὐκ >άρα τὸ A πρὸς τὸ B λόγον >έθει, <ὃν ἀριθμὸς πρὸς ἀριθμόν.

Τὰ >άρα ἀσύμμετρα μεγέθη πρὸς >άλληλα λόγον οὐκ >έθει, καὶ τὰ ἐχῆξ.

8.

Ἐὰν δύο μεγέθη πρὸς >άλληλα λόγον μὴ >έθῃ, <ὃν ἀριθμὸς πρὸς ἀριθμόν, ἀσύμμετρα >έσται τὰ μεγέθη.



Δύο γὰρ μεγέθη τὰ A , B πρὸς >άλληλα λόγον μὴ ἐθέτω, <ὃν ἀριθμὸς πρὸς ἀριθμόν; λέγω, <ὅτι ἀσύμμετρά ἐστι τὰ A , B μεγέθη.

Εἰ γὰρ >έσται σύμμετρα, τὸ A πρὸς τὸ B λόγον <έχει, <ὃν ἀριθμὸς πρὸς ἀριθμόν. οὐκ >έθει δέ. ἀσύμμετρα >άρα ἐστὶ τὰ A , B μεγέθη.

Ἐὰν >άρα δύο μεγέθη πρὸς >άλληλα, καὶ τὰ ἐχῆξ.

9.

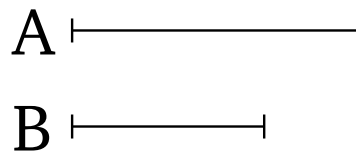
Τὰ ἀπὸ τῶν μήκει συμμέτρων εὐθειῶν τετράγωνα πρὸς >άλληλα λόγον >έθει, <ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν; καὶ τὰ τετράγωνα τὰ πρὸς >άλληλα λόγον >έθοντα, <ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, καὶ τὰς πλευρὰς <έχει μήκει συμμέτρους. τὰ δὲ ἀπὸ τῶν μήκει ἀσύμμετρων εὐθειῶν τετράγωνα πρὸς >άλληλα λόγον οὐκ >έθει, <ὃν περ τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν; καὶ τὰ τετράγωνα τὰ πρὸς >άλληλα λόγον μὴ >έθοντα, <ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν,

the mean proportion, (say) B , is taken of A and F , then as A is to F , so the (square) on A (will be) to the (square) on B . That is to say, as the first (is) to the third, so the (figure) on the first (is) to the similar, and similarly described, (figure) on the second [Prop. 6.19 corr.]. But, as A (is) to F , so the number D is to the number E . Thus, it has also been contrived that as the number D (is) to the number E , so the (figure) on the straight-line A (is) to the (similar figure) on the straight-line B . (Which is) the very thing it was required to show.

Proposition 7

Incommensurable magnitudes do not have to one another the ratio which (some) number (has) to (some) number.

Let A and B be incommensurable magnitudes. I say that A does not have to B the ratio which (some) number (has) to (some) number.

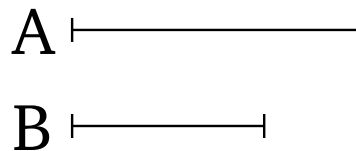


For if A has to B the ratio which (some) number (has) to (some) number then A will be commensurable with B [Prop. 10.6]. But it is not. Thus, A does not have to B the ratio which (some) number (has) to (some) number.

Thus, incommensurable numbers do not have to one another, and so on

Proposition 8

If two magnitudes do not have to one another the ratio which (some) number (has) to (some) number then the magnitudes will be incommensurable.



For let the two magnitudes A and B not have to one another the ratio which (some) number (has) to (some) number. I say that the magnitudes A and B are incommensurable.

For if they are commensurable, A will have to B the ratio which (some) number (has) to (some) number [Prop. 10.5]. But it does not have (such a ratio). Thus, the magnitudes A and B are incommensurable.

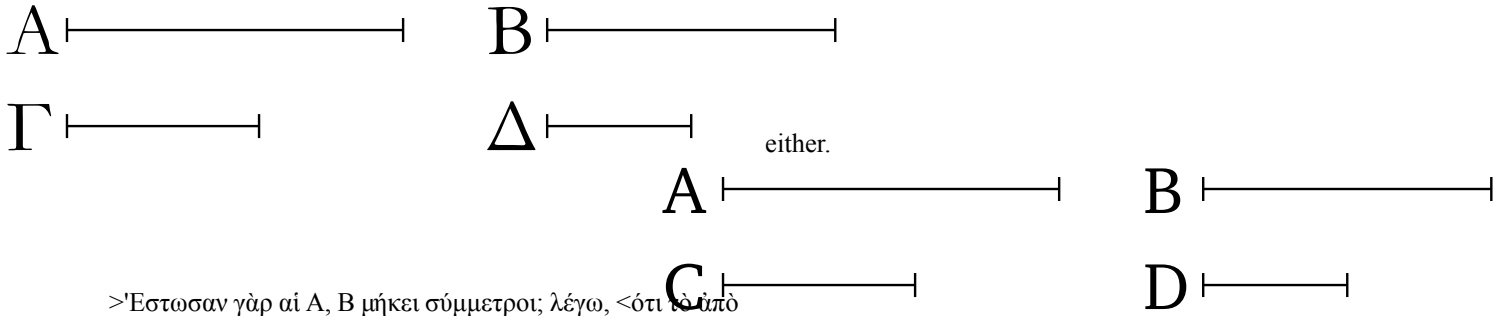
Thus, if two magnitudes ... to one another, and so on

Proposition 9

Squares on straight-lines (which are) commensurable in length have to one another the ratio which (some) square number (has) to (some) square number. And squares having to one another the ratio which (some) square number (has) to (some) square number will also have sides (which are) commensurable in length. But squares on straight-lines (which are) incommensurable in length do not have to one another the ratio which (some) square number (has) to (some) square number. And squares not having to one another the ratio

οὐδὲ τὰς πλευρὰς <έχει μήκει συμμέτρουξ.

which (some) square number (has) to (some) square number will not have sides (which are) commensurable in length



>Ἐστωσαν γὰρ αἱ A, B μήκει σύμμετροι; λέγω, <ὅτι τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς B τετράγωνον λόγον >έθει, <ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. Ἐπεὶ γὰρ σύμμετρος ἐστὶν ἡ A τῇ B μήκει, ἡ A >άρα πρὸς τὴν B λόγον >έθει, <ὃν ἀριθμὸς πρὸς ἀριθμόν. ἐθέτω, <ὃν ὁ Γ πρὸς τὸν Δ . ἐπεὶ ο>ὖν ἐστὶν ὡς ἡ A πρὸς τὴν B , ο<ύτωξ ὁ Γ πρὸς τὸν Δ , ἀλλὰ τοῦ μὲν τῆς A πρὸς τὴν B λόγου διπλασίων ἐστὶν ὁ τοῦ ἀπὸ τῆς A τετραγώνου πρὸς τὸ ἀπὸ τῆς B τετράγωνον; τὰ γὰρ <όμοια σήματα ἐν διπλασίονι λόγῳ ἐστὶ τῶν ὁμολόγων πλευρῶν; τοῦ δὲ τοῦ Γ [ἀριθμοῦ] πρὸς τὸν Δ [ἀριθμόν] λόγου διπλασίων ἐστὶν ὁ τοῦ ἀπὸ τοῦ Γ τετραγώνου πρὸς τὸν ἀπὸ τοῦ Δ τετράγωνον; δύο γὰρ τετραγώνων ἀριθμῶν ε<ῖξ μέσοξ ἀνάλογόν ἐστιν ἀριθμὸξ, καὶ ὁ τετράγωνος πρὸς τὸν τετράγωνον [ἀριθμόν] διπλασίονα λόγον >έθει, >ἤπερ ἡ πλευρὰ πρὸς τὴν πλευράν; >ἐστὶν >άρα καὶ ὡς τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς B τετράγωνον, ο<ύτωξ ὁ ἀπὸ τοῦ Γ τετράγωνος [ἀριθμοῦ] πρὸς τὸν ἀπὸ τοῦ Δ [ἀριθμοῦ] τετράγωνον [ἀριθμόν].

Ἀλλὰ δὴ >έστω ὡς τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς B , ο<ύτωξ ὁ ἀπὸ τοῦ Γ τετράγωνος πρὸς τὸν ἀπὸ τοῦ Δ [τετράγωνον]; λέγω, <ὅτι σύμμετρος ἐστὶν ἡ A τῇ B μήκει. Ἐπεὶ γὰρ ἐστὶν ὡς τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς B [τετράγωνον], ο<ύτωξ ὁ ἀπὸ τοῦ Γ τετράγωνος πρὸς τὸν ἀπὸ τοῦ Δ [τετράγωνον], ἀλλ> ὁ μὲν τοῦ ἀπὸ τῆς A τετραγώνου πρὸς τὸ ἀπὸ τῆς B [τετράγωνον] λόγος διπλασίων ἐστὶ τοῦ τῆς A πρὸς τὴν B λόγου, ὁ δὲ τοῦ ἀπὸ τοῦ Γ [ἀριθμοῦ] τετραγώνου [ἀριθμοῦ] πρὸς τὸν ἀπὸ τοῦ Δ [ἀριθμοῦ] τετράγωνον [ἀριθμόν] λόγος διπλασίων ἐστὶ τοῦ τοῦ Γ [ἀριθμοῦ] πρὸς τὸν Δ [ἀριθμόν] λόγου, >ἐστὶν >άρα καὶ ὡς ἡ A πρὸς τὴν B , ο<ύτωξ ὁ Γ [ἀριθμοῦ] πρὸς τὸν Δ [ἀριθμόν]. ἡ A >άρα πρὸς τὴν B λόγον >έθει, <ὃν ἀριθμὸξ ὁ Γ πρὸς ἀριθμόν τὸν Δ ; σύμμετροξ >άρα ἐστὶν ἡ A τῇ B μήκει.

Ἀλλὰ δὴ ἀσύμμετροξ >έστω ἡ A τῇ B μήκει; λέγω, <ὅτι τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς B [τετράγωνον] λόγον οὐκ >έθει, <ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν.

Εἰ γὰρ >έθει τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς B [τετράγωνον] λόγον, <ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, σύμμετροξ >έσται ἡ A τῇ B . οὐκ >έστι δέ; οὐκ >άρα τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς B [τετράγωνον] λόγον >έθει, <ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν.

Πάλιν δὴ τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς B [τετράγωνον] λόγον μὴ ἐθέτω, <ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν; λέγω, <ὅτι ἀσύμμετρος ἐστὶν ἡ A τῇ B μήκει.

Εἰ γὰρ ἐστὶ σύμμετροξ ἡ A τῇ B , <έχει τὸ ἀπὸ τῆς A

For let A and B be (straight-lines which are) commensurable in length. I say that the square on A has to the square on B the ratio which (some) square number (has) to (some) square number.

For since A is commensurable in length with B , A thus has to B the ratio which (some) number (has) to (some) number [Prop. 10.5]. Let it have (that) which C (has) to D . Therefore, since as A is to B , so C (is) to D . But the (ratio) of the square on A to the square on B is the square of the ratio of A to B . For similar figures are in the squared ratio of (their) corresponding sides [Prop. 6.20 corr.]. And the (ratio) of the square on C to the square on D is the square of the ratio of the [number] C to the [number] D . For there exists one number in mean proportion to two square numbers, and (one) square (number) has to the (other) square [number] a squared ratio with respect to (that) the side (of the former) has to the side (of the latter) [Prop. 8.11]. And, thus, as the square on A is to the square on B , so the square [number] on the (number) C (is) to the square [number] on the [number] D .[†]

And so let the square on A be to the (square) on B as the square (number) on C (is) to the [square] (number) on D . I say that A is commensurable in length with B .

For since as the square on A is to the [square] on B , so the square (number) on C (is) to the [square] (number) on D . But, the ratio of the square on A to the (square) on B is the square of the (ratio) of A to B [Prop. 6.20 corr.]. And the (ratio) of the square [number] on the [number] C to the square [number] on the [number] D is the square of the ratio of the [number] C to the [number] D [Prop. 8.11]. Thus, as A is to B , so the [number] C also (is) to the [number] D . A , thus, has to B the ratio which the number C has to the number D . Thus, A is commensurable in length with B [Prop. 10.6].[‡]

And so let A be incommensurable in length with B . I say that the square on A does not have to the [square] on B the ratio which (some) square number (has) to (some) square number.

For if the square on A has to the [square] on B the ratio which (some) square number (has) to (some) square number then A will be commensurable (in length) with B . But it is not. Thus, the square on A does not have to the [square] on the B the ratio which (some) square number (has) to (some) square number.

So, again, let the square on A not have to the [square] on B the ratio which (some) square number (has) to (some) square number. I say that A is incommensurable in length with B . For if A is commensurable (in length) with B then the

πρὸς τὸ ἀπὸ τῆς B λόγον, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. οὐκ >έθει δέ; οὐκ >άρα σύμμετρος ἐστὶν ἡ A τῇ B μήκει.
Τὰ >άρα ἀπὸ τῶν μήκει συμμέτρων, καὶ τὰ ἐχῆς.

(square) on A will have to the (square) on B the ratio which (some) square number (has) to (some) square number. But it does not have (such a ratio). Thus, A is not commensurable in length with B .

Thus, (squares) on (straight-lines which are) commensurable in length, and so on

Πόρισμα.

Καὶ φανερόν ἐκ τῶν δεδειγμένων >έσται, <ὅτι αἱ μήκει σύμμετροι πάντως καὶ δυνάμει, αἱ δὲ δυνάμει οὐ πάντως καὶ μήκει.

Corollary

And it will be clear, from (what) has been demonstrated, that (straight-lines) commensurable in length (are) always also (commensurable) in square, but (straight-lines commensurable) in square (are) not always also (commensurable) in length.

† There is an unstated assumption here that if $\alpha : \beta :: \gamma : \delta$ then $\alpha^2 : \beta^2 :: \gamma^2 : \delta^2$.

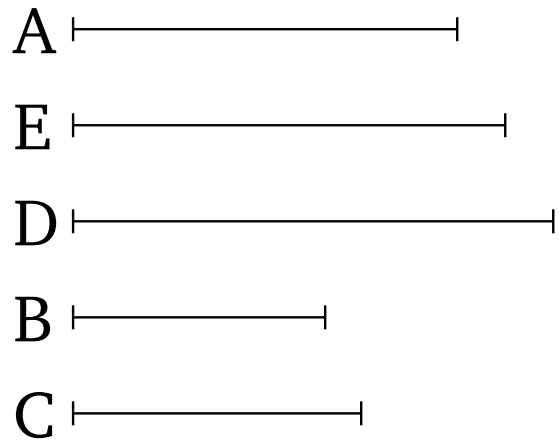
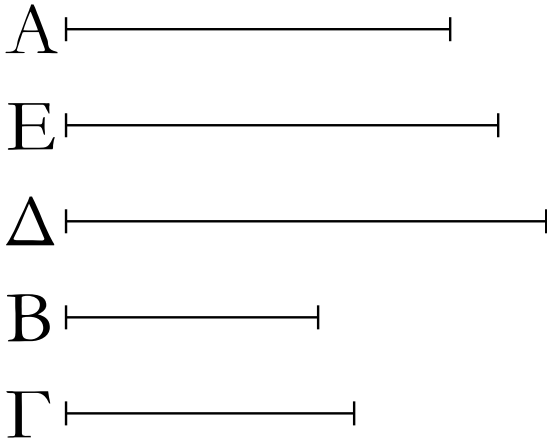
‡ There is an unstated assumption here that if $\alpha^2 : \beta^2 :: \gamma^2 : \delta^2$ then $\alpha : \beta :: \gamma : \delta$.

10.

Τῇ προτεθείσῃ εὐθείᾳ προσευρεῖν δύο εὐθείας ἀσυμμέτρους, τὴν μὲν μήκει μόνον, τὴν δὲ καὶ δυνάμει.

Proposition 10†

To find two straight-lines incommensurable with a given straight-line, the one (incommensurable) in length only, the other also (incommensurable) in square.



>Ἐστω ἡ προτεθείσα εὐθεῖα ἡ A ; δεῖ δὴ τῇ A προσευρεῖν δύο εὐθείας ἀσυμμέτρους, τὴν μὲν μήκει μόνον, τὴν δὲ καὶ δυνάμει.

Let A be the given straight-line. So it is required to find two straight-lines incommensurable with A , the one (incommensurable) in length only, the other also (incommensurable) in square.

Ἐκκεῖσθωσαν γὰρ δύο ἀριθμοὶ οἱ B, Γ πρὸς ἀλλήλους λόγον μὴ >έθοντε, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, τουτέστι μὴ <όμοιοι ἐπίπεδοι, καὶ γερονέτω ὡς ὁ B πρὸς τὸν Γ , ο<ύτωξ τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς Δ τετράγωνον; ἐμάθομεν γάρ; σύμμετρον >άρα τὸ ἀπὸ τῆς A τῷ ἀπὸ τῆς Δ . καὶ ἐπεὶ ὁ B πρὸς τὸν Γ λόγον οὐκ >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, οὐδ' >άρα τὸ ἀπὸ τῆς A πρὸς τὸ ἀπὸ τῆς Δ λόγον >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν; ἀσύμμετρος >άρα ἐστὶν ἡ A τῇ Δ μήκει. εἰλήφθω τῶν A, Δ μέση ἀνάλογον ἡ E ; >έστιν >άρα ὡς ἡ A πρὸς τὴν Δ , ο<ύτωξ τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς E . ἀσύμμετρος δὲ ἐστὶν ἡ A τῇ Δ μήκει; ἀσύμμετρον >άρα ἐστὶ καὶ τὸ ἀπὸ τῆς A τετράγωνον τῷ ἀπὸ τῆς E τετραγώνῳ; ἀσύμμετρος >άρα ἐστὶν ἡ A τῇ E δυνάμει.

For let two numbers, B and C , not having to one another the ratio which (some) square number (has) to (some) square number—that is to say, not (being) similar plane (numbers)—be taken. And let it be contrived that as B (is) to C , so the square on A (is) to the square on D . For we learned (how to do this) [Prop. 10.6 corr.]. Thus, the (square) on A (is) commensurable with the (square) on D [Prop. 10.6]. And since B does not have to C the ratio which (some) square number (has) to (some) square number, the (square) on A thus does not have to the (square) on D the ratio which (some) square number (has) to (some) square number either. Thus, A is incommensurable in length with D [Prop. 10.9]. Let the (straight-line) E (which is) in mean proportion to A and D be taken [Prop. 6.13]. Thus, as A is to D , so the square on A (is) to the (square) on E [Def. 5.9]. And A is incommensurable in length with D . Thus, the square on A is also incommensurable with the square on E [Prop. 10.11]. Thus, A is incommensurable in square with E .

Τῇ >άρα προτεθείσῃ εὐθείᾳ τῇ A προσεύρηται δύο εὐθεῖαι ἀσύμμετροι αἱ Δ, E , μήκει μὲν μόνον ἡ Δ , δυνάμει δὲ καὶ μήκει δηλαδὴ ἡ E [<ὅπερ >έδει δεῖχαι].

Thus, two straight-lines, D and E , (which are) incommensurable with the given straight-line A , be found, the one,

D , (incommensurable) in length only, the other, E , (incommensurable) in square, and, clearly, also in length. [(Which is) the very thing it was required to show.]

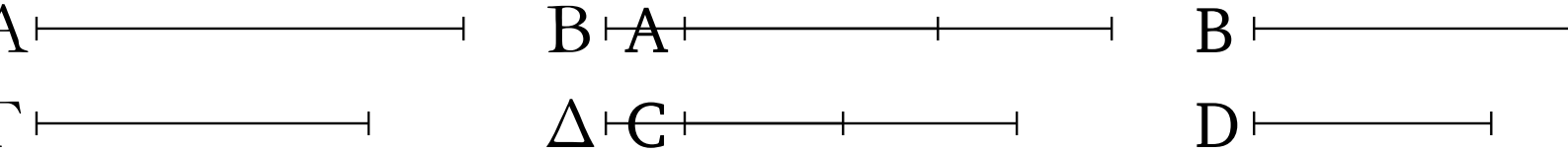
† This whole proposition is regarded by Heiberg as an interpolation into the original text.

11.

Proposition 11

Ἐὰν τέσσαρα μεγέθη ἀνάλογον $>\eta|$, τὸ δὲ πρῶτον τῷ δευτέρῳ σύμμετρον $>\eta|$, καὶ τὸ τρίτον τῷ τετάρτῳ σύμμετρον $>\epsilon\sigma\tau\alpha\iota$; κ>ὰν τὸ πρῶτον τῷ δευτέρῳ ἀσύμμετρον $>\eta|$, καὶ τὸ τρίτον τῷ τετάρτῳ ἀσύμμετρον $>\epsilon\sigma\tau\alpha\iota$.

If four magnitudes are proportional, and the first is commensurable with the second, then the third will also be commensurable with the fourth. And if the first is incommensurable with the second, then the third will also be incommensurable with the fourth.



>Ἐστωσαν τέσσαρα μεγέθη ἀνάλογον τὰ A, B, Γ, Δ , ὥς τὸ A πρὸς τὸ B , ο<ύτωξ τὸ Γ πρὸς τὸ Δ , τὸ A δὲ τῷ B σύμμετρον $>\epsilon\sigma\tau\omega$; λέγω, <ὅτι καὶ τὸ Γ τῷ Δ σύμμετρον $>\epsilon\sigma\tau\alpha\iota$.

Ἐπεὶ γὰρ σύμμετρον ἐστὶ τὸ A τῷ B , τὸ A >άρα πρὸς τὸ B λόγον >έθει, <ὃν ἀριθμὸς πρὸς ἀριθμόν. καὶ ἐστὶν ὥς τὸ A πρὸς τὸ B , ο<ύτωξ τὸ Γ πρὸς τὸ Δ ; καὶ τὸ Γ >άρα πρὸς τὸ Δ λόγον >έθει, <ὃν ἀριθμὸς πρὸς ἀριθμόν; σύμμετρον >άρα ἐστὶ τὸ Γ τῷ Δ .

Ἀλλὰ δὴ τὸ A τῷ B ἀσύμμετρον $>\epsilon\sigma\tau\omega$; λέγω, <ὅτι καὶ τὸ Γ τῷ Δ ἀσύμμετρον $>\epsilon\sigma\tau\alpha\iota$. ἐπεὶ γὰρ ἀσύμμετρον ἐστὶ τὸ A τῷ B , τὸ A >άρα πρὸς τὸ B λόγον οὐκ >έθει, <ὃν ἀριθμὸς πρὸς ἀριθμόν. καὶ ἐστὶν ὥς τὸ A πρὸς τὸ B , ο<ύτωξ τὸ Γ πρὸς τὸ Δ ; οὐδὲ τὸ Γ >άρα πρὸς τὸ Δ λόγον >έθει, <ὃν ἀριθμὸς πρὸς ἀριθμόν; ἀσύμμετρον >άρα ἐστὶ τὸ Γ τῷ Δ .

Ἐὰν >άρα τέσσαρα μεγέθη, καὶ τὰ ἐχῇξ.

Let A, B, C, D be four proportional magnitudes, (such that) as A (is) to B , so C (is) to D . And let A be commensurable with B . I say that C will also be commensurable with D .

For since A is commensurable with B , A thus has to B the ratio which (some) number (has) to (some) number [Prop. 10.5]. And as A is to B , so C (is) to D . Thus, C also has to D the ratio which (some) number (has) to (some) number. Thus, C is commensurable with D [Prop. 10.6].

And so let A be incommensurable with B . I say that C will also be incommensurable with D . For since A is incommensurable with B , A thus does not have to B the ratio which (some) number (has) to (some) number [Prop. 10.7]. And as A is to B , so C (is) to D . Thus, C does not have to D the ratio which (some) number (has) to (some) number either. Thus, C is incommensurable with D [Prop. 10.8].

Thus, if four magnitudes, and so on

12.

Proposition 12

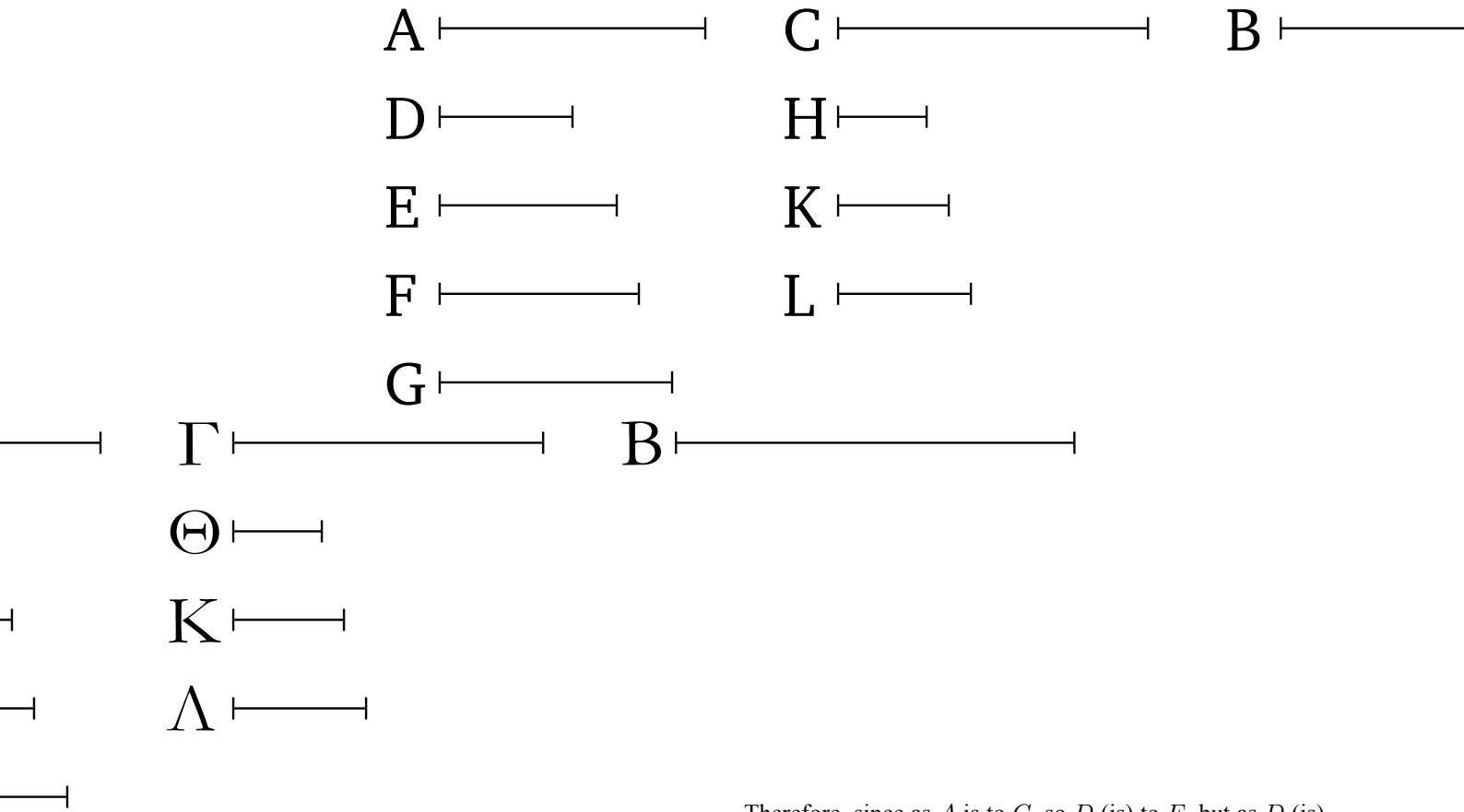
Τὰ τῷ αὐτῷ μεγέθει σύμμετρα καὶ ἀλλήλοισι ἐστὶ σύμμετρα. Ἐκάτερον γὰρ τῶν A, B τῷ Γ >έστω σύμμετρον. λέγω, <ὅτι καὶ τὸ A τῷ B ἐστὶ σύμμετρον.

Ἐπεὶ γὰρ σύμμετρον ἐστὶ τὸ A τῷ Γ , τὸ A >άρα πρὸς τὸ Γ λόγον >έθει, <ὃν ἀριθμὸς πρὸς ἀριθμόν. ἐθέτω, <ὃν ὁ Δ πρὸς τὸν E . πάλιν, ἐπεὶ σύμμετρον ἐστὶ τὸ Γ τῷ B , τὸ Γ >άρα πρὸς τὸ B λόγον >έθει, <ὃν ἀριθμὸς πρὸς ἀριθμόν. ἐθέτω, <ὃν ὁ Z πρὸς τὸν H . καὶ λόγων δοθέντων ὁποσωνοῦν τοῦ τε, <ὃν >έθει ὁ Δ πρὸς τὸν E , καὶ ὁ Z πρὸς τὸν H εἰλήφθωσαν ἀριθμοὶ ἐχῇξ ἐν τοῖς δοθεῖσι λόγοις οἱ Θ, K, Λ ; <ώστε ε>ῖναι ὥς μὲν τὸν Δ πρὸς τὸν E , ο<ύτωξ τὸν Θ πρὸς τὸν K , ὥς δὲ τὸν Z πρὸς τὸν H , ο<ύτωξ τὸν K πρὸς τὸν Λ .

(Magnitudes) commensurable with the same magnitude are also commensurable with one another.

For let A and B each be commensurable with C . I say that A is also commensurable with B .

For since A is commensurable with C , A thus has to C the ratio which (some) number (has) to (some) number [Prop. 10.5]. Let it have (the ratio) which D (has) to E . Again, since C is commensurable with B , C thus has to B the ratio which (some) number (has) to (some) number [Prop. 10.5]. Let it have (the ratio) which F (has) to G . And for any multitude whatsoever of given ratios—(namely,) those which D has to E , and F to G —let the numbers H, K, L (which are) continuously (proportional) in the(se) given ratios be taken [Prop. 8.4]. Hence, as D is to E , so H (is) to K , and as F (is) to G , so K (is) to L .



Ἐπεὶ οὖν ἐστὶν ὥς τὸ A πρὸς τὸ Γ , οὕτως ὁ Δ πρὸς τὸν E , ἀλλ' ὥς ὁ Δ πρὸς τὸν E , οὕτως ὁ Θ πρὸς τὸν K , >ἐστὶν > ἄρα καὶ ὥς τὸ A πρὸς τὸ Γ , οὕτως ὁ Θ πρὸς τὸν K . πάλιν, ἐπεὶ ἐστὶν ὥς τὸ Γ πρὸς τὸ B , οὕτως ὁ Z πρὸς τὸν H , ἀλλ' ὥς ὁ Z πρὸς τὸν H , [οὕτως] ὁ K πρὸς τὸν Λ , καὶ ὥς > ἄρα τὸ Γ πρὸς τὸ B , οὕτως ὁ K πρὸς τὸν Λ . >ἐστὶ δὲ καὶ ὥς τὸ A πρὸς τὸ Γ , οὕτως ὁ Θ πρὸς τὸν K ; δι' >ἴσου > ἄρα ἐστὶν ὥς τὸ A πρὸς τὸ B , οὕτως ὁ Θ πρὸς τὸν Λ . τὸ A > ἄρα πρὸς τὸ B λόγον >έθει, <ὃν ἀριθμὸς ὁ Θ πρὸς ἀριθμὸν τὸν Λ ; σύμμετρον > ἄρα ἐστὶ τὸ A τῷ B .
Τὰ > ἄρα τῷ αὐτῷ μεγέθει σύμμετρα καὶ ἀλλήλοις ἐστὶ σύμμετρα; <ὅπερ >έδει δεῖξαι.

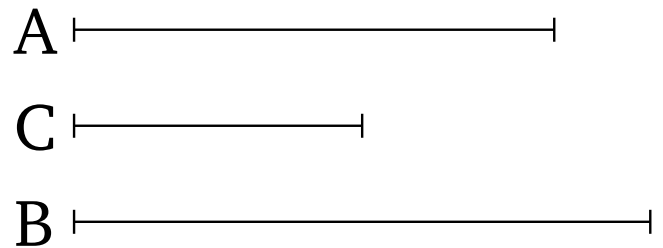
13.

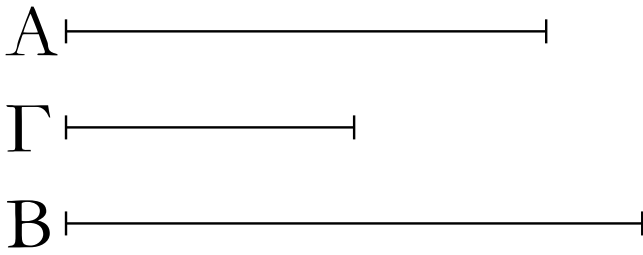
Ἐὰν >ῃ| δύο μεγέθη σύμμετρα, τὸ δὲ <έτερον αὐτῶν μεγέθει τινὶ ἀσύμμετρον >ῃ|, καὶ τὸ λοιπὸν τῷ αὐτῷ| ἀσύμμετρον >έσται.

Therefore, since as A is to C , so D (is) to E , but as D (is) to E , so H (is) to K , thus also as A is to C , so H (is) to K [Prop. 5.11]. Again, since as C is to B , so F (is) to G , but as F (is) to G , [so] K (is) to L , thus also as C (is) to B , so K (is) to L [Prop. 5.11]. And also as A is to C , so H (is) to K . Thus, via equality, as A is to B , so H (is) to L [Prop. 5.22]. Thus, A has to B the ratio which the number H (has) to the number L . Thus, A is commensurable with B [Prop. 10.6]. Thus, (magnitudes) commensurable with the same magnitude are also commensurable with one another. (Which is) the very thing it was required to show.

Proposition 13

If two magnitudes are commensurable, and one of them is incommensurable with some magnitude, then the remaining (magnitude) will also be incommensurable with it.





>Ἐστω δύο μεγέθη σύμμετρα τὰ A, B , τὸ δὲ <έτερον αὐτῶν τὸ A >ἄλλῳ τινὶ τῷ Γ ἀσύμμετρον >έστω; λέγω, <ὅτι καὶ τὸ λοιπὸν τὸ B τῷ Γ ἀσύμμετρον ἔστιν.

Εἰ γάρ ἐστι σύμμετρον τὸ B τῷ Γ , ἀλλὰ καὶ τὸ A τῷ B σύμμετρον ἔστιν, καὶ τὸ A >ἄρα τῷ Γ σύμμετρον ἔστιν. ἀλλὰ καὶ ἀσύμμετρον; <ὅπερ ἀδύνατον. οὐκ >ἄρα σύμμετρον ἔστι τὸ B τῷ Γ ; ἀσύμμετρον >ἄρα.

Ἐὰν >ἄρα >ῃ δύο μεγέθη σύμμετρα, καὶ τὰ ἐχῇξ.

Let A and B be two commensurable magnitudes, and let one of them, A , be incommensurable with some other (magnitude), C . I say that the remaining (magnitude), B , is also incommensurable with C .

For if B is commensurable with C , but A is also commensurable with B , A is thus also commensurable with C [Prop. 10.12]. But, (it is) also incommensurable (with C). The very thing (is) impossible. Thus, B is not commensurable with C . Thus, (it is) incommensurable.

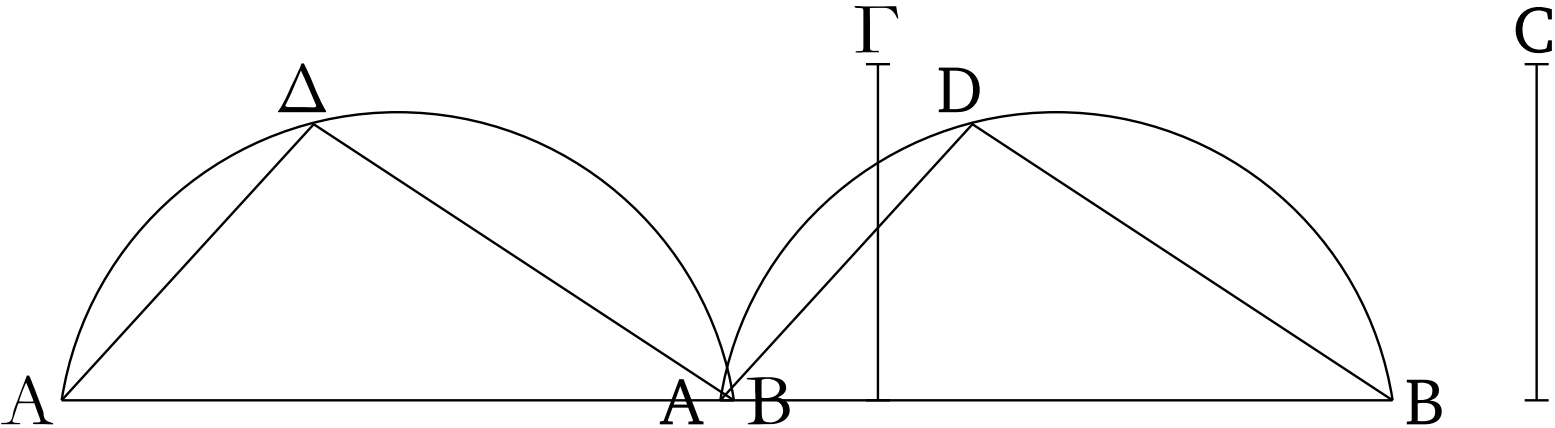
Thus, if two magnitudes are commensurable, and so on

Λήμμα.

Δύο δοθεισῶν εὐθειῶν ἀνίσων εὐρεῖν, τίνι μείζον δύναται ἡ μείζων τῇξ ἐλάσσονος.

Lemma

For two given unequal straight-lines, to find by (the square on) which (straight-line) the square on the greater (straight-line is) larger than (the square on) the lesser.[†]



>Ἐστωσαν αἱ δοθεῖσαι δύο >άνισοι εὐθεῖαι αἱ AB, Γ , <ῶν μείζων >έστω ἡ AB ; δεῖ δὲ εὐρεῖν, τίνι μείζον δύναται ἡ AB τῇξ Γ .

Γεγράφθω ἐπὶ τῇξ AB ἡμικύκλιον τὸ ADB , καὶ εἰς αὐτὸ ἐνηρμόσθω τῇ Γ >ῖση ἡ AD , καὶ ἐπεζεύθθω ἡ DB . φανερόν δὲ, <ὅτι ὀρθή ἐστιν ἡ ὑπὸ ADB γωνία, καὶ <ὅτι ἡ AB τῇξ AD , τουτέστι τῇξ Γ , μείζον δύναται τῇ DB .

Ὁμοίωξ δὲ καὶ δύο δοθεισῶν εὐθειῶν ἡ δυναμένη αὐτὰς εὐρίσκεται οὕτως.

>Ἐστωσαν αἱ δοθεῖσαι δύο εὐθεῖαι αἱ AD, DB , καὶ δέον >έστω εὐρεῖν τὴν δυναμένην αὐτὰς. κείσθωσαν γάρ, <ώστε ὀρθὴν γωνίαν περιέθην τὴν ὑπὸ AD, DB , καὶ ἐπεζεύθθω ἡ AB ; φανερόν πάλιν, <ὅτι ἡ AD, DB δυναμένη ἐστὶν ἡ AB ; <ὅπερ >έδει δεῖξαι.

Let AB and C be the two given unequal straight-lines, and let AB be the greater of them. So it is required to find by (the square on) which (straight-line) the square on AB (is) greater than (the square on) C .

Let the semi-circle ADB be described on AB . And let AD , equal to C , be inserted into it [Prop. 4.1]. And let DB be joined. So (it is) clear that the angle ADB is a right-angle [Prop. 3.31], and that the square on AB (is) greater than (the square on) AD —that is to say, (the square on) C —by (the square on) DB [Prop. 1.47].

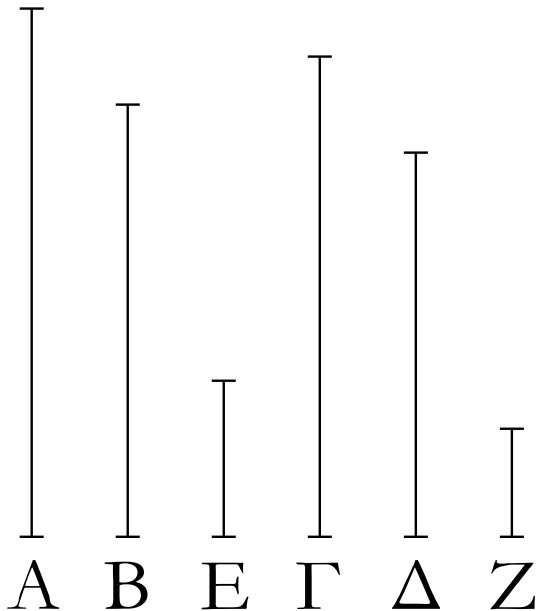
And, similarly, the square-root of (the sum of the squares on) two given straight-lines is also found likewise.

Let AD and DB be the two given straight-lines. And let it be necessary to find the square-root of (the sum of the squares on) them. For let them be laid down such as to encompass a right-angle—(namely), that (angle encompassed) by AD and DB . And let AB be joined. (It is) again clear that AB is the square-root of (the sum of the squares on) AD and DB [Prop. 1.47]. (Which is) the very thing it was required to show.

[†] That is, if α and β are the lengths of two given straight-lines, with α being greater than β , to find a straight-line of length γ such that $\alpha^2 = \beta^2 + \gamma^2$. Similarly, we can also find γ such that $\gamma^2 = \alpha^2 + \beta^2$.

Ἐάν τέσσαρες εὐθεῖαι ἀνάλογον >ῶσιν, δύνηται δὲ ἡ πρώτη τῆς δευτέρας μείζον τῷ ἀπὸ συμμετρου ἑαυτῇ [μήκει], καὶ ἡ τρίτη τῆς τετάρτης μείζον δυνήσεται τῷ ἀπὸ συμμετρου ἑαυτῇ [μήκει]. καὶ ἔαν ἡ πρώτη τῆς δευτέρας μείζον δύνηται τῷ ἀπὸ ἀσυμμετρου ἑαυτῇ [μήκει], καὶ ἡ τρίτη τῆς τετάρτης μείζον δυνήσεται τῷ ἀπὸ ἀσυμμετρου ἑαυτῇ [μήκει].

>Ἐστωσαν τέσσαρες εὐθεῖαι ἀνάλογον αἱ A, B, Γ, Δ , ὥς ἡ A πρὸς τὴν B , ο<ύτωξ ἡ Γ πρὸς τὴν Δ , καὶ ἡ A μὲν τῆς B μείζον δυνάσθω τῷ ἀπὸ τῆς E , ἡ δὲ Γ τῆς Δ μείζον δυνάσθω τῷ ἀπὸ τῆς Z ; λέγω, <ὅτι, ε>ίτε σύμμετρόξ ἐστὶν ἡ A τῇ E , σύμμετρόξ ἐστὶ καὶ ἡ Γ τῇ Z , ε>ίτε ἀσύμμετρόξ ἐστὶν ἡ A τῇ E , ἀσύμμετρόξ ἐστὶ καὶ ὁ Γ τῇ Z .

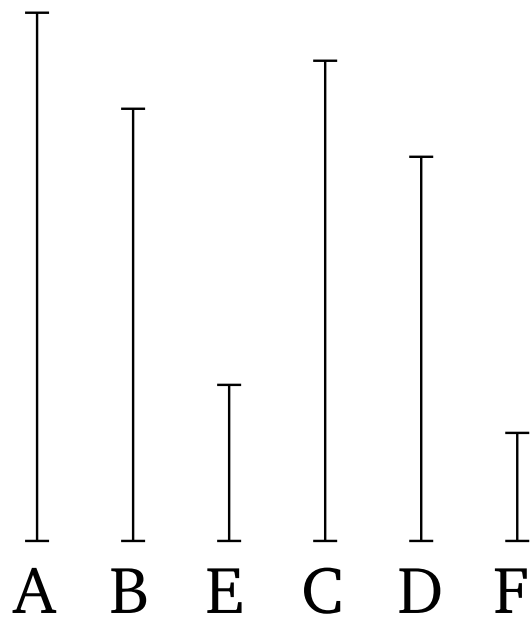


Ἐπεὶ γάρ ἐστιν ὥς ἡ A πρὸς τὴν B , ο<ύτωξ ἡ Γ πρὸς τὴν Δ , >ἐστὶν >άρα καὶ ὥς τὸ ἀπὸ τῆς A πρὸς τὸ ἀπὸ τῆς B , ο<ύτωξ τὸ ἀπὸ τῆς Γ πρὸς τὸ ἀπὸ τῆς Δ . ἀλλὰ τῷ μὲν ἀπὸ

If four straight-lines are proportional, and the square on the first is greater than (the square on) the second by the (square) on (some straight-line) commensurable [in length] with the first, then the square on the third will also be greater than (the square on) the fourth by the (square) on (some straight-line) commensurable [in length] with the third. And if the square on the first is greater than (the square on) the second by the (square) on (some straight-line) incommensurable [in length] with the first, then the square on the third will also be greater than (the square on) the fourth by the (square) on (some straight-line) incommensurable [in length] with the third.

Let A, B, C, D be four proportional straight-lines, (such that) as A (is) to B , so C (is) to D . And let the square on A be greater than (the square on) B by the (square) on E , and let the square on C be greater than (the square on) D by the (square) on F . I say that A is either commensurable (in length) with E , and C is also commensurable with F , or A is incommensurable (in length) with E , and C is also

incommensurable with F .



For since as A is to B , so C (is) to D , thus as the (square) on A is to the (square) on B , so the (square) on C (is) to the

τῆς A ἴσα ἐστὶ τὰ ἀπὸ τῶν E, B , τῶ δὲ ἀπὸ τῆς Γ ἴσα ἐστὶ τὰ ἀπὸ τῶν Δ, Z . >έστιν >άρα ὥς τὰ ἀπὸ τῶν E, B πρὸς τὸ ἀπὸ τῆς B , ο<ύτως τὰ ἀπὸ τῶν Δ, Z πρὸς τὸ ἀπὸ τῆς Δ ; διελόντι >άρα ἐστὶν ὥς τὸ ἀπὸ τῆς E πρὸς τὸ ἀπὸ τῆς B , ο<ύτως τὸ ἀπὸ τῆς Z πρὸς τὸ ἀπὸ τῆς Δ ; >έστιν >άρα καὶ ὥς ἡ E πρὸς τὴν B , ο<ύτως ἡ Z πρὸς τὴν Δ ; ἀνάπαλιν >άρα ἐστὶν ὥς ἡ B πρὸς τὴν E , ο<ύτως ἡ Δ πρὸς τὴν Z . >έστι δὲ καὶ ὥς ἡ A πρὸς τὴν B , ο<ύτως ἡ Γ πρὸς τὴν Δ ; δι>ίςου >άρα ἐστὶν ὥς ἡ A πρὸς τὴν E , ο<ύτως ἡ Γ πρὸς τὴν Z . ε>ίτε ο>υν σύμμετρος ἐστὶν ἡ A τῇ E , συμμετρος ἐστὶ καὶ ἡ Γ τῇ Z , ε>ίτε ἀσύμμετρος ἐστὶν ἡ A τῇ E , ἀσύμμετρος ἐστὶ καὶ ἡ Γ τῇ Z .
Ἐὰν >άρα, καὶ τὰ ἐχῇς.

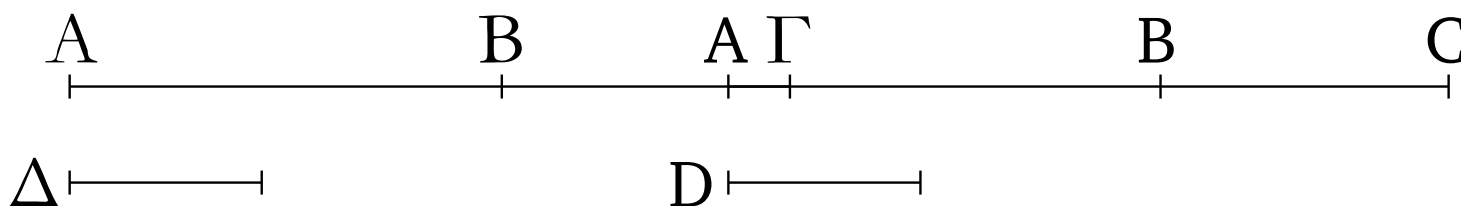
(square) on D [Prop. 6.22]. But the (sum of the squares) on E and B is equal to the (square) on A , and the (sum of the squares) on D and F is equal to the (square) on C . Thus, as the (sum of the squares) on E and B is to the (square) on B , so the (sum of the squares) on D and F (is) to the (square) on D . Thus, via separation, as the (square) on E is to the (square) on B , so the (square) on F (is) to the (square) on D [Prop. 5.17]. Thus, also, as E is to B , so F (is) to D [Prop. 6.22]. Thus, inversely, as B is to E , so D (is) to F [Prop. 5.7 corr.]. But, as A is to B , so C also (is) to D . Thus, via equality, as A is to E , so C (is) to F [Prop. 5.22]. Therefore, A is either commensurable (in length) with E , and C is also commensurable with F , or A is incommensurable (in length) with E , and C is also incommensurable with F [Prop. 10.11].
Thus, if, and so on

15.

Proposition 15

Ἐὰν δύο μεγέθη σύμμετρα συντεθῇ, καὶ τὸ <όλον ἐκατέρῳ αὐτῶν σύμμετρον >έσται; κ>ὰν τὸ <όλον ἐνὶ αὐτῶν σύμμετρον >ῇ, καὶ τὰ ἐκ ἀρθῆς μεγέθη σύμμετρα >έσται.
Συγκείσθω γὰρ δύο μεγέθη σύμμετρα τὰ $AB, B\Gamma$; λέγω, <ὅτι καὶ <όλον τὸ AG ἐκατέρῳ τῶν $AB, B\Gamma$ ἐστὶ σύμμετρον.

If two commensurable magnitudes are added together then the whole will also be commensurable with each of them. And if the whole is commensurable with one of them then the original magnitudes will also be commensurable (with one another).
For let the two commensurable magnitudes AB and BC be laid down together. I say that the whole AC is also commensurable with each of AB and BC .



Ἐπεὶ γὰρ σύμμετρα ἐστὶ τὰ $AB, B\Gamma$, μετρήσει τι αὐτὰ μέγεθος. μετρεῖτω, καὶ >έστω τὸ Δ . ἐπεὶ ο>υν τὸ Δ τὰ $AB, B\Gamma$ μετρεῖ, καὶ <όλον τὸ AG μετρήσει. μετρεῖ δὲ καὶ τὰ $AB, B\Gamma$. τὸ Δ >άρα τὰ $AB, B\Gamma, AG$ μετρεῖ; σύμμετρον >άρα ἐστὶ τὸ AG ἐκατέρῳ τῶν $AB, B\Gamma$.
Ἀλλὰ δὴ τὸ AG >έστω σύμμετρον τῷ AB ; λέγω δὴ, <ὅτι καὶ τὰ $AB, B\Gamma$ σύμμετρα ἐστὶν.
Ἐπεὶ γὰρ σύμμετρα ἐστὶ τὰ AG, AB , μετρήσει τι αὐτὰ μέγεθος. μετρεῖτω, καὶ >έστω τὸ Δ . ἐπεὶ ο>υν τὸ Δ τὰ GA, AB μετρεῖ, καὶ λοιπὸν >άρα τὸ $B\Gamma$ μετρήσει. μετρεῖ δὲ καὶ τὸ AB ; τὸ Δ >άρα τὰ $AB, B\Gamma$ μετρήσει; σύμμετρα >άρα ἐστὶ τὰ $AB, B\Gamma$.
Ἐὰν >άρα δύο μεγέθη, καὶ τὰ ἐχῇς.

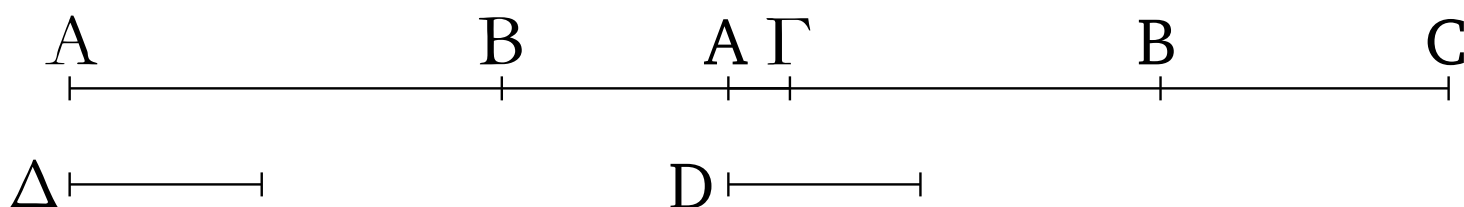
For since AB and BC are commensurable, some magnitude will measure them. Let it (so) measure (them), and let it be D . Therefore, since D measures (both) AB and BC , it will also measure the whole AC . And it also measures AB and BC . Thus, D measures AB, BC , and AC . Thus, AC is commensurable with each of AB and BC [Def. 10.1].
And so let AC be commensurable with AB . I say that AB and BC are also commensurable.
For since AC and AB are commensurable, some magnitude will measure them. Let it (so) measure (them), and let it be D . Therefore, since D measures (both) CA and AB , it will thus also measure the remainder BC . And it also measures AB . Thus, D will measure (both) AB and BC . Thus, AB and BC are commensurable [Def. 10.1].
Thus, if two magnitudes, and so on

16.

Proposition 16

Ἐὰν δύο μεγέθη ἀσύμμετρα συντεθῇ, καὶ τὸ <όλον ἐκατέρῳ αὐτῶν ἀσύμμετρον >έσται; κ>ὰν τὸ <όλον ἐνὶ αὐτῶν ἀσύμμετρον >ῇ, καὶ τὰ ἐκ ἀρθῆς μεγέθη ἀσύμμετρα >έσται.

If two incommensurable magnitudes are added together then the whole will also be incommensurable with each of them. And if the whole is incommensurable with one of them then the original magnitudes will also be incommensurable (with one another).



Συγκείσθω γὰρ δύο μεγέθη ἀσύμμετρα τὰ AB , $BΓ$; λέγω, <ὅτι καὶ <όλον τὸ $AΓ$ ἑκατέρῳ τῶν AB , $BΓ$ ἀσύμμετρόν ἐστιν.

Εἰ γὰρ μὴ ἐστὶν ἀσύμμετρα τὰ $ΓA$, AB , μετρήσει τι [αὐτὰ] μέγεθος. μετρεῖτω, εἰ δυνατόν, καὶ >έστω τὸ Δ . ἐπεὶ ο>ὖν τὸ Δ τὰ $ΓA$, AB μετρεῖ, καὶ λοιπὸν >άρα τὸ $BΓ$ μετρήσει. μετρεῖ δὲ καὶ τὸ AB ; τὸ Δ >άρα τὰ AB , $BΓ$ μετρεῖ. σύμμετρα >άρα ἐστὶ τὰ AB , $BΓ$; ὑπέκειντο δὲ καὶ ἀσύμμετρα; <όπερ ἐστὶν ἀδύνατον. οὐκ >άρα τὰ $ΓA$, AB μετρήσει τι μέγεθος; ἀσύμμετρα >άρα ἐστὶ τὰ $ΓA$, AB . ὁμοίωξ δὴ δείχομεν, <ὅτι καὶ τὰ $AΓ$, $ΓB$ ἀσύμμετρά ἐστιν. τὸ $AΓ$ >άρα ἑκατέρῳ τῶν AB , $BΓ$ ἀσύμμετρόν ἐστιν.

Ἀλλὰ δὴ τὸ $AΓ$ ἐνὶ τῶν AB , $BΓ$ ἀσύμμετρον >έστω. >έστω δὴ πρότερον τῷ AB ; λέγω, <ὅτι καὶ τὰ AB , $BΓ$ ἀσύμμετρά ἐστιν. εἰ γὰρ >έσται σύμμετρα, μετρήσει τι αὐτὰ μέγεθος. μετρεῖτω, καὶ >έστω τὸ Δ . ἐπεὶ ο>ὖν τὸ Δ τὰ AB , $BΓ$ μετρεῖ, καὶ <όλον >άρα τὸ $AΓ$ μετρήσει. μετρεῖ δὲ καὶ τὸ AB ; τὸ Δ >άρα τὰ $ΓA$, AB μετρεῖ. σύμμετρα >άρα ἐστὶ τὰ $ΓA$, AB ; ὑπέκειντο δὲ καὶ ἀσύμμετρα; <όπερ ἐστὶν ἀδύνατον. οὐκ >άρα τὰ AB , $BΓ$ μετρήσει τι μέγεθος; ἀσύμμετρα >άρα ἐστὶ τὰ AB , $BΓ$.

Ἐὰν >άρα δύο μεγέθη, καὶ τὰ ἐχῇξ.

For let the two incommensurable magnitudes AB and BC be laid down together. I say that that the whole AC is also incommensurable with each of AB and BC .

For if CA and AB are not incommensurable then some magnitude will measure [them]. If possible, let it (so) measure (them), and let it be D . Therefore, since D measures (both) CA and AB , it will thus also measure the remainder BC . And it also measures AB . Thus, D measures (both) AB and BC . Thus, AB and BC are commensurable [Def. 10.1]. But they were also assumed (to be) incommensurable. The very thing is impossible. Thus, some magnitude cannot measure (both) CA and AB . Thus, CA and AB are incommensurable [Def. 10.1]. So, similarly, we can show that AC and CB are also incommensurable. Thus, AC is incommensurable with each of AB and BC .

And so let AC be incommensurable with one of AB and BC . So let it, first of all, be incommensurable with AB . I say that AB and BC are also incommensurable. For if they are commensurable then some magnitude will measure them. Let it (so) measure (them), and let it be D . Therefore, since D measures (both) AB and BC , it will thus also measure the whole AC . And it also measures AB . Thus, D measures (both) CA and AB . Thus, CA and AB are commensurable [Def. 10.1]. But they were also assumed (to be) incommensurable. The very thing is impossible. Thus, some magnitude cannot measure (both) AB and BC . Thus, AB and BC are incommensurable [Def. 10.1].

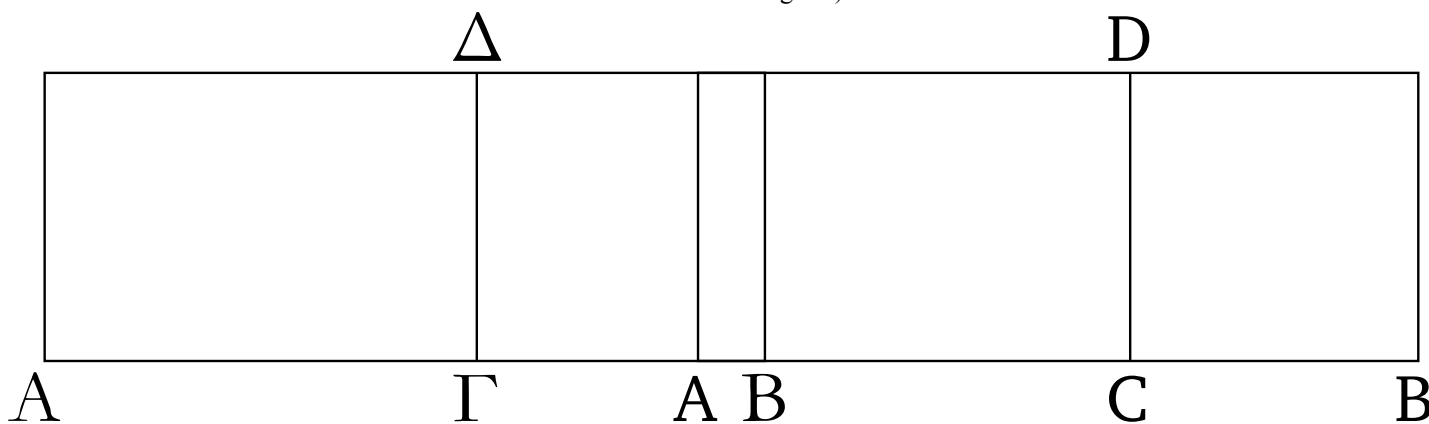
Thus, if two...magnitudes, and so on

Λήμμα.

Ἐὰν παρὰ τινα εὐθεῖαν παραβλήθῃ παραλληλόγραμμον ἑλλειπόν ε>ίδει τετραγώνῳ, τὸ παραβληθὲν >ῖσον ἐστὶ τῷ ὑπὸ τῶν ἐκ τῆς παραβολῆς γενομένων τμημάτων τῆς εὐθείας.

Lemma

If a parallelogram,[†] falling short by a square figure, is applied to some straight-line then the applied (parallelogram) is equal (in area) to the (rectangle contained) by the pieces of the straight-line created via the application (of the parallelogram).



Παρὰ γὰρ εὐθεῖαν τὴν AB παραβελήσθω παραλληλόγραμμον τὸ AD ἑλλειπόν ε>ίδει τετραγώνῳ τῷ ΔB ; λέγω, <ὅτι >ῖσον ἐστὶ τὸ AD τῷ ὑπὸ τῶν AG , GB .

Καὶ ἐστὶν αὐτόθεν φανερόν; ἐπεὶ γὰρ τετράγωνόν ἐστὶ τὸ ΔB , >ῖση ἐστὶν ἡ $\Delta Γ$ τῇ $ΓB$, καὶ ἐστὶ τὸ AD τὸ ὑπὸ τῶν AG , $ΓA$, τουτέστι τὸ ὑπὸ τῶν AG , $ΓB$.

For let the parallelogram AD , falling short by the square figure DB , be applied to the straight-line AB . I say that AD is equal to the (rectangle contained) by AC and CB . And it is immediately obvious. For since DB is a square, DC is equal to CB . And AD is the (rectangle contained) by AC and CD —that is to say, by AC and CB .

Ἐὰν >άρα παρά τινα εὐθεΐαν, καὶ τὰ ἐχῆζ.

Thus, if ...to some straight-line, and so on

† Note that this lemma only applies to rectangular parallelograms.

17.

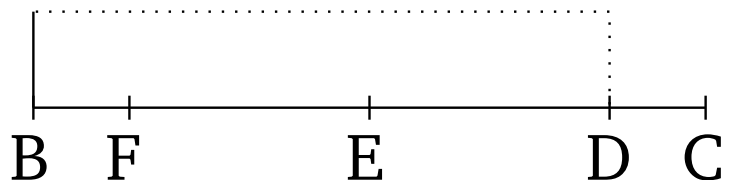
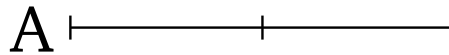
Ἐὰν >ῶσι δύο εὐθεΐαι >άνισοι, τῷ δὲ τετράτῳ μέρει τοῦ ἀπὸ τῆς ἐλάσσονος >ίσον παρὰ τὴν μείζονα παραβληθῇ| ἐλλείπον ε>ίδει τετραγώνῳ καὶ εἰς σύμμετρα αὐτὴν διαιρῇ| μήκει, ἢ μείζον τῆς ἐλάσσονος μείζον δυνήσεται τῷ ἀπὸ συμμέτου ἐαυτῇ| [μήκει]. καὶ ἔαν ἡ μείζων τῆς ἐλάσσονος μείζον δύνηται τῷ ἀπὸ συμμέτρου ἐαυτῇ| [μήκει], τῷ δὲ τετράτῳ τοῦ ἀπὸ τῆς ἐλάσσονος >ίσον παρὰ τὴν μείζονα παραβληθῇ| ἐλλείπον ε>ίδει τετραγώνῳ, εἰς σύμμετρα αὐτὴν διαιρεῖ μήκει.

>Ἐστωσαν δύο εὐθεΐαι >άνισοι αἱ A , $B\Gamma$, <ῶν μείζων ἡ $B\Gamma$, τῷ δὲ τετράτῳ μέρει τοῦ ἀπὸ ἐλάσσονος τῆς A , τουτέστι τῷ ἀπὸ τῆς ἡμισείας τῆς A , >ίσον παρὰ τὴν $B\Gamma$ παραβεβλήσθω ἐλλείπον ε>ίδει τετραγώνῳ, καὶ >έστω τὸ ὑπὸ τῶν $B\Delta$, $\Delta\Gamma$, σύμμετρος δὲ >έστω ἡ $B\Delta$ τῇ $\Delta\Gamma$ μήκει, λέγω, <ὅτι ἡ $B\Gamma$ τῆς A μείζον δύναται τῷ ἀπὸ συμμέτρου ἐαυτῇ|.

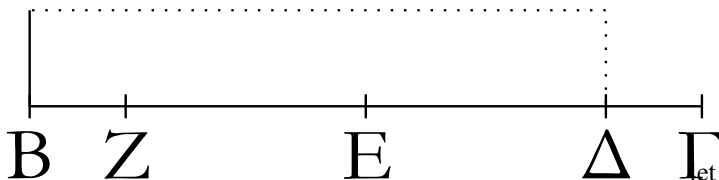
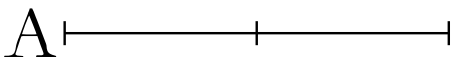
Proposition 17†

If there are two unequal straight-lines, and a (rectangle) equal to the fourth part of the (square) on the lesser, falling short by a square figure, is applied to the greater, and divides it into (parts which are) commensurable in length, then the square on the greater will be larger than (the square on) the lesser by the (square) on (some straight-line) commensurable [in length] with the greater. And if the square on the greater is larger than (the square on) the lesser by the (square) on (some straight-line) commensurable [in length] with the greater, and a (rectangle) equal to the fourth (part) of the (square) on the lesser, falling short by a square figure, is applied to the greater, then it divides it into (parts which are) commensurable in length.

Let A and BC be two unequal straight-lines, of which (let) BC (be) the greater. And let a (rectangle) equal to the fourth part of the (square) on the lesser, A —that is, (equal) to the (square) on half of A —falling short by a square figure, be applied to BC . And let it be the (rectangle contained) by BD and DC [see previous lemma]. And let BD be commensurable in length with DC . I say that that the square on BC is greater than the (square on) A by (the square on some straight-line) commensurable (in length) with (BC).



For let BC be cut in half at the point E [Prop. 1.10]. And



Τεμήσθω γὰρ ἡ $B\Gamma$ δίθᾳ κατὰ τὸ E σημεῖον, καὶ κείσθω τῇ ΔE >ίση ἡ EZ . λοιπὴ >άρα ἡ $\Delta\Gamma$ >ίση ἐστὶ τῇ BZ . καὶ ἐπεὶ εὐθεΐα ἡ $B\Gamma$ τέμνεται εἰς μὲν >ίσα κατὰ τὸ E , εἰς δὲ >άνισα κατὰ τὸ Δ , τὸ >άρα ὑπὸ $B\Delta$, $\Delta\Gamma$ περιεθόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τῆς $E\Delta$ τετραγώνου >ίσον ἐστὶ τῷ ἀπὸ τῆς $E\Gamma$ τετραγώνῳ; καὶ τὰ τετραπλάσια; τὸ >άρα τετράκις ὑπὸ τῶν $B\Delta$, $\Delta\Gamma$ μετὰ τοῦ τετραπλασίου τοῦ ἀπὸ τῆς ΔE >ίσον ἐστὶ τῷ τετράκις ἀπὸ τῆς $E\Gamma$ τετραγώνῳ. ἀλλὰ τῷ μὲν τετραπλασίῳ τοῦ ὑπὸ τῶν $B\Delta$, $\Delta\Gamma$ >ίσον ἐστὶ τὸ ἀπὸ τῆς A τετράγωνον, τῷ δὲ τετραπλασίῳ τοῦ ἀπὸ τῆς ΔE >ίσον ἐστὶ τὸ ἀπὸ τῆς ΔZ τετράγωνον; διπλασίων γάρ ἐστιν ἡ ΔZ τῆς

Let EF be made equal to DE [Prop. 1.3]. Thus, the remainder DC is equal to BF . And since the straight-line BC has been cut into equal (pieces) at E , and into unequal (pieces) at D , the rectangle contained by BD and DC , plus the square on ED , is thus equal to the square on EC [Prop. 2.5]. (The same) also (for) the quadruples. Thus, four times the (rectangle contained) by BD and DC , plus the quadruple of the (square) on DE , is equal to four times the square on EC . But, the square on A is equal to the quadruple of the (rectangle contained) by BD and DC , and the square on DF is equal to the quadruple of the (square) on DE . For DF is double DE . And the square on BC is equal to the quadru-

ΔΕ. τῷ δὲ τετραπλασίῳ τοῦ ἀπὸ τῆς ΕΓ >ίσον ἐστὶ τὸ ἀπὸ τῆς ΒΓ τετράγωνον· διπλασίῳ γάρ ἐστι πάλιν ἡ ΒΓ τῆς ΓΕ. τὰ >άρα ἀπὸ τῶν Α, ΔΖ τετράγωνα >ίσα ἐστὶ τῷ ἀπὸ τῆς ΒΓ τετράγωνῳ· <ώστε τὸ ἀπὸ τῆς ΒΓ τοῦ ἀπὸ τῆς Α μείζον ἐστὶ τῷ ἀπὸ τῆς ΔΖ· ἡ ΒΓ >άρα τῆς Α μείζον δύναται τῇ ΔΖ. δεικτέον, <ὅτι καὶ σύμμετρος ἐστὶν ἡ ΒΓ τῇ ΔΖ. ἐπεὶ γὰρ σύμμετρος ἐστὶν ἡ ΒΔ τῇ ΔΓ μήκει, σύμμετρος >άρα ἐστὶ καὶ ἡ ΒΓ τῇ ΓΔ μήκει. ἀλλὰ ἡ ΓΔ ταῖς ΓΔ, ΒΖ ἐστὶ σύμμετρος μήκει· >ίση γάρ ἐστὶν ἡ ΓΔ τῇ ΒΖ. καὶ ἡ ΒΓ >άρα σύμμετρος ἐστὶ ταῖς ΒΖ, ΓΔ μήκει· <ώστε καὶ λοιπῇ τῇ ΖΔ σύμμετρος ἐστὶν ἡ ΒΓ μήκει· ἡ ΒΓ >άρα τῆς Α μείζον δύναται τῷ ἀπὸ συμμέτρου ἐαυτῇ.

Ἀλλὰ δὴ ἡ ΒΓ τῆς Α μείζον δύνασθω τῷ ἀπὸ συμμέτρου ἐαυτῇ, τῷ δὲ τετάρτῳ τοῦ ἀπὸ τῆς Α >ίσον παρὰ τὴν ΒΓ παραβεβλήσθω ἐλλείπον ε>ίδει τετραγώνῳ, καὶ >έστω τὸ ὑπὸ τῶν ΒΔ, ΔΓ. δεικτέον, <ὅτι σύμμετρος ἐστὶν ἡ ΒΔ τῇ ΔΓ μήκει.

Τῶν γὰρ αὐτῶν κατασκευασθέντων ὁμοίως δείχομεν, <ὅτι ἡ ΒΓ τῆς Α μείζον δύναται τῷ ἀπὸ τῆς ΖΔ. δύναται δὲ ἡ ΒΓ τῆς Α μείζον τῷ ἀπὸ συμμέτρου ἐαυτῇ. σύμμετρος >άρα ἐστὶν ἡ ΒΓ τῇ ΖΔ μήκει· <ώστε καὶ λοιπῇ συναμφοτέρῳ τῇ ΒΖ, ΔΓ σύμμετρος ἐστὶν ἡ ΒΓ μήκει. ἀλλὰ συναμφοτέροξ ἡ ΒΖ, ΔΓ σύμμετρος ἐστὶ τῇ ΔΓ [μήκει]. <ώστε καὶ ἡ ΒΓ τῇ ΓΔ σύμμετρος ἐστὶ μήκει· καὶ διελόντι >άρα ἡ ΒΔ τῇ ΔΓ ἐστὶ σύμμετρος μήκει.

Ἐὰν >άρα >ῶσι δύο εὐθεῖαι >άνισοι, καὶ τὰ ἐχθῆ.

ple of the (square) on EC . For, again, BC is double CE . Thus, the (sum of the) squares on A and DF is equal to the square on BC . Hence, the (square) on BC is greater than the (square) on A by the (square) on DF . Thus, BC is greater in square than A by DF . It must also be shown that BC is commensurable (in length) with DF . For since BD is commensurable in length with DC , BC is thus also commensurable in length with CD [Prop. 10.15]. But, CD is commensurable in length with CD plus BF . For CD is equal to BF [Prop. 10.6]. Thus, BC is also commensurable in length with BF plus CD [Prop. 10.12]. Hence, BC is also commensurable in length with the remainder FD [Prop. 10.15]. Thus, the square on BC is greater than (the square on) A by the (square) on (some straight-line) commensurable (in length) with (BC) .

And so let the square on BC be greater than the (square on) A by the (square) on (some straight-line) commensurable (in length) with (BC) . And let a (rectangle) equal to the fourth (part) of the (square) on A , falling short by a square figure, be applied to BC . And let it be the (rectangle contained) by BD and DC . It must be shown that BD is commensurable in length with DC .

For, similarly, by the same construction, we can show that the square on BC is greater than the (square on) A by the (square) on FD . And the square on BC is greater than the (square on) A by the (square) on (some straight-line) commensurable (in length) with (BC) . Thus, BC is commensurable in length with FD . Hence, BC is also commensurable in length with the remaining sum of BF and DC [Prop. 10.15]. But, the sum of BF and DC is commensurable [in length] with DC [Prop. 10.6]. Hence, BC is also commensurable in length with CD [Prop. 10.12]. Thus, via separation, BD is also commensurable in length with DC [Prop. 10.15].

Thus, if there are two unequal straight-lines, and so on

† This proposition states that if $\alpha x - x^2 = \beta^2/4$ (where $\alpha = BC$, $x = DC$, and $\beta = A$) then α and $\sqrt{\alpha^2 - \beta^2}$ are commensurable when $\alpha - x$ are x are commensurable, and vice versa.

18.

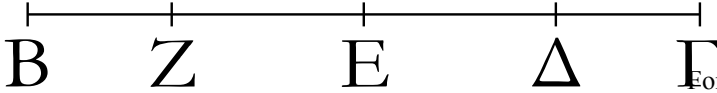
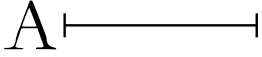
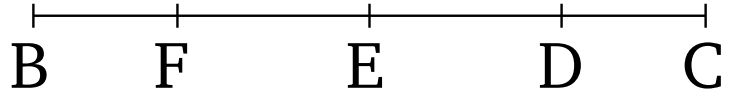
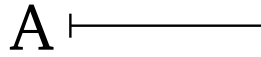
Ἐὰν >ῶσι δύο εὐθεῖαι >άνισοι, τῷ δὲ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ἐλάσσονος >ίσον παρὰ τὴν μείζονα παραβληθῇ ἐλλείπον ε>ίδει τετραγώνῳ, καὶ εἰς ἀσύμμετρα αὐτὴν διαιρῇ [μήκει], ἡ μείζον τῆς ἐλάσσονος μείζον δυνήσεται τῷ ἀπὸ ἀσυμμέτρου ἐαυτῇ. καὶ ἔαν ἡ μείζον τῆς ἐλάσσονος μείζον δύνηται τῷ ἀπὸ ἀσυμμέτρου ἐαυτῇ, τῷ δὲ τετάρτῳ τοῦ ἀπὸ τῆς ἐλάσσονος >ίσον παρὰ τὴν μείζονα παραβληθῇ ἐλλείπον ε>ίδει τετραγώνῳ, εἰς ἀσύμμετρα αὐτὴν διαιρεῖ [μήκει].

>Ἐστώσαν δύο εὐθεῖαι >άνισοι αἱ Α, ΒΓ, <ῶν μείζον ἡ ΒΓ, τῷ δὲ τετάρτῳ [μέρει] τοῦ ἀπὸ τῆς ἐλάσσονος τῆς Α >ίσον παρὰ τὴν ΒΓ παραβεβλήσθω ἐλλείπον ε>ίδει τετραγώνῳ, καὶ >έστω τὸ ὑπὸ τῶν ΒΔΓ, ἀσύμμετρος δὲ >έστω ἡ ΒΔ τῇ ΔΓ μήκει· λέγω, <ὅτι ἡ ΒΓ τῆς Α μείζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἐαυτῇ.

Proposition 18†

If there are two unequal straight-lines, and a (rectangle) equal to the fourth part of the (square) on the lesser, falling short by a square figure, is applied to the greater, and divides it into (parts which are) incommensurable [in length], then the square on the greater will be larger than the (square on the) lesser by the (square) on (some straight-line) incommensurable (in length) with the greater. And if the square on the greater is larger than the (square on the) lesser by the (square) on (some straight-line) incommensurable (in length) with the greater, and a (rectangle) equal to the fourth (part) of the (square) on the lesser, falling short by a square figure, is applied to the greater, then it divides it into (parts which are) incommensurable [in length].

Let A and BC be two unequal straight-lines, of which (let) BC (be) the greater. And let a (rectangle) equal to the fourth [part] of the (square) on the lesser, A , falling short by a square figure, be applied to BC . And let it be the (rectangle contained) by BDC . And let BD be incommensurable in length with DC . I say that the square on BC is greater than the (square on) A by the (square) on (some straight-line) incommensurable (in length) with (BC) .



Τῶν γὰρ αὐτῶν κατασκευασθέντων τῶ| πρότερον ὁμοίωξ δείχομεν, <ὅτι ἡ ΒΓ τῆς Α μείζον δύναται τῶ| ἀπὸ τῆς ΖΔ. δεικτέον [ο>ύν], <ὅτι ἀσύμμετρόξ ἐστὶν ἡ ΒΓ τῇ| ΔΖ μήκει. ἐπεὶ γὰρ ἀσύμμετρόξ ἐστὶν ἡ ΒΔ τῇ| ΔΓ μήκει, ἀσύμμετροξ >άρα ἐστὶ καὶ ἡ ΒΓ τῇ| ΓΔ μήκει. ἀλλὰ ἡ ΔΓ σύμμετρόξ ἐστὶ συναμφοτέραιξ ταῖξ ΒΖ, ΔΓ; καὶ ἡ ΒΓ >άρα ἀσύμμετρόξ ἐστὶ συναμφοτέραιξ ταῖξ ΒΖ, ΔΓ. <ώστε καὶ λοιπῇ| τῇ| ΖΔ ἀσύμμετρόξ >ἐστὶν ἡ ΒΓ μήκει. καὶ ἡ ΒΓ τῆς Α μείζον δύναται τῶ| ἀπὸ τῆς ΖΔ; ἡ ΒΓ >άρα τῆς Α μείζον δύναται τῶ| ἀπὸ ἀσυμμέτρου ἑαυτῇ|.

Δυνάσθω δὴ πάλιν ἡ ΒΓ τῆς Α μείζον τῶ| ἀπὸ ἀσυμμέτρου ἑαυτῇ|, τῶ| δὲ τετάρτῳ τοῦ ἀπὸ τῆς Α >ίσον παρὰ τὴν ΒΓ παραβεβλήσθω ἐλλείπον >ἰδει τετραγώνῳ, καὶ >έστω τὸ ὑπὸ τῶν ΒΔ, ΔΓ. δεικτέον, <ὅτι ἀσύμμετρόξ ἐστὶν ἡ ΒΔ τῇ| ΔΓ μήκει.

Τῶν γὰρ αὐτῶν κατασκευασθέντων ὁμοίωξ δείχομεν, <ὅτι ἡ ΒΓ τῆς Α μείζον δύναται τῶ| ἀπὸ τῆς ΖΔ. ἀλλὰ ἡ ΒΓ τῆς Α μείζον δύναται τῶ| ἀπὸ ἀσυμμέτρου ἑαυτῇ|. ἀσύμμετροξ >άρα ἐστὶν ἡ ΒΓ τῇ| ΖΔ μήκει; <ώστε καὶ λοιπῇ| συναμφοτέρῳ τῇ| ΒΖ, ΔΓ ἀσύμμετρόξ ἐστὶν ἡ ΒΓ. ἀλλὰ συναμφοτέροξ ἡ ΒΖ, ΔΓ τῇ| ΔΓ σύμμετρόξ ἐστὶ μήκει; καὶ ἡ ΒΓ >άρα τῇ| ΔΓ ἀσύμμετρόξ ἐστὶ μήκει; <ώστε καὶ διελόντι ἡ ΒΔ τῇ| ΔΓ ἀσύμμετρόξ ἐστὶ μήκει.

Ἐὰν >άρα >ῶσι δύο εὐθεῖαι, καὶ τὰ ἐχῆξ.

For, similarly, by the same construction as before, we can show that the square on BC is greater than the (square on) A by the (square) on FD . [Therefore] it must be shown that BC is incommensurable in length with DF . For since BD is incommensurable in length with DC , BC is thus also incommensurable in length with CD [Prop. 10.16]. But, DC is commensurable (in length) with the sum of BF and DC [Prop. 10.6]. And, thus, BC is incommensurable (in length) with the sum of BF and DC [Prop. 10.13]. Hence, BC is also incommensurable in length with the remainder FD [Prop. 10.16]. And the square on BC is greater than the (square on) A by the (square) on FD . Thus, the square on BC is greater than the (square on) A by the (square) on (some straight-line) incommensurable (in length) with (BC).

So, again, let the square on BC be greater than the (square on) A by the (square) on (some straight-line) incommensurable (in length) with (BC). And let a (rectangle) equal to the fourth [part] of the (square) on A , falling short by a square figure, be applied to BC . And let it be the (rectangle contained) by BD and DC . It must be shown that BD is incommensurable in length with DC .

For, similarly, by the same construction, we can show that the square on BC is greater than the (square) on A by the (square) on FD . But, the square on BC is greater than the (square) on A by the (square) on (some straight-line) incommensurable (in length) with (BC). Thus, BC is incommensurable in length with FD . Hence, BC is also incommensurable (in length) with the remaining sum of BF and DC [Prop. 10.16]. But, the sum of BF and DC is commensurable in length with DC [Prop. 10.6]. Thus, BC is also incommensurable in length with DC [Prop. 10.13]. Hence, via separation, BD is also incommensurable in length with DC [Prop. 10.16].

Thus, if there are two ...straight-lines, and so on

† This proposition states that if $\alpha x - x^2 = \beta^2/4$ (where $\alpha = BC$, $x = DC$, and $\beta = A$) then α and $\sqrt{\alpha^2 - \beta^2}$ are incommensurable when $\alpha - x$ are x are incommensurable, and vice versa.

19.

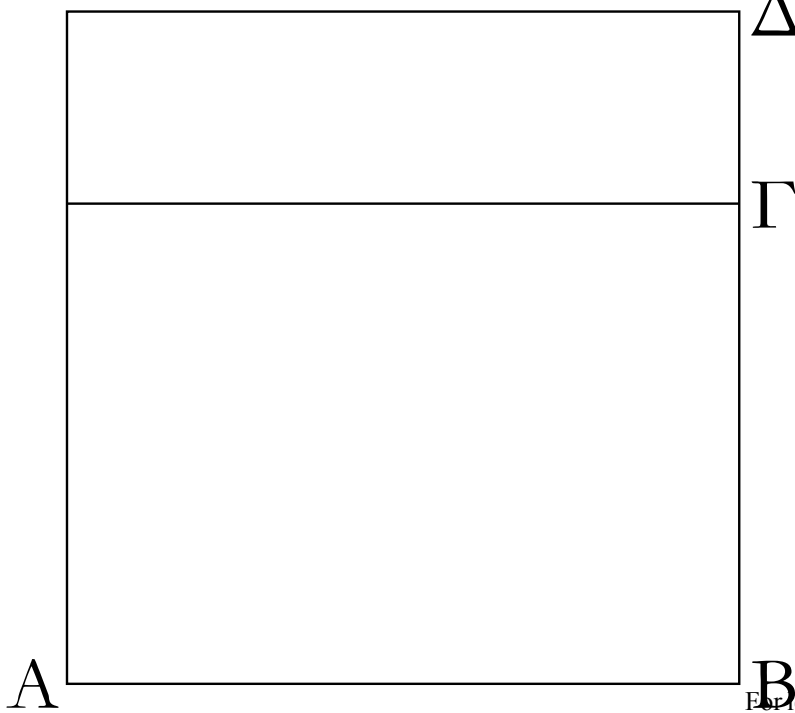
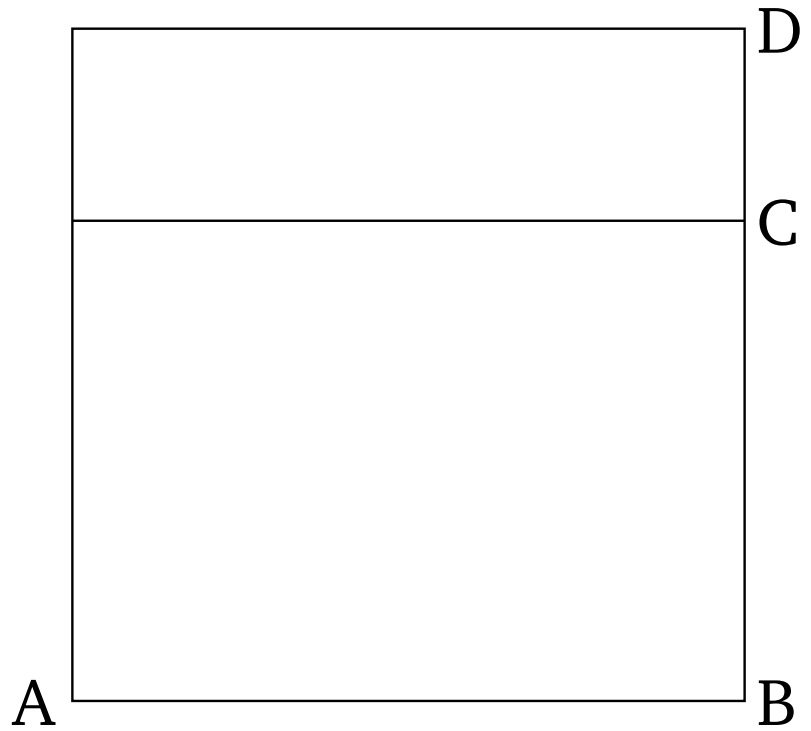
Τὸ ὑπὸ <ρητῶν μήκει συμμέτρων εὐθειῶν περιεχόμενον ὀρθογώνιον <ρητόν ἐστιν.

Ὑπὸ γὰρ <ρητῶν μήκει συμμέτρων εὐθειῶν τῶν AB , $BΓ$ ὀρθογώνιον περιεθέσθω τὸ $ΑΓ$; λέγω, <ὅτι <ρητόν ἐστι τὸ $ΑΓ$.

Proposition 19

The rectangle contained by rational straight-lines (which are) commensurable in length is rational.

For let the rectangle AC be enclosed by the rational straight-lines AB and BC (which are) commensurable in length. I say that AC is rational.



Ἀναγεγράφω γὰρ ἀπὸ τῆς AB τετράγωνον τὸ AD ; <ρητὸν >άρα ἐστὶ τὸ AD . καὶ ἐπεὶ σύμμετρόξ ἐστὶν ἡ AB τῇ $BΓ$ μήκει, >ίση δέ ἐστιν ἡ AB τῇ $BΔ$, σύμμετροξ >άρα ἐστὶν ἡ $BΔ$ τῇ $BΓ$ μήκει. καὶ ἐστὶν ὥξ ἡ $BΔ$ πρὸξ τὴν $BΓ$, ο<ύτωξ τὸ $ΔA$ πρὸξ τὸ $ΑΓ$. σύμμετρον >άρα ἐστὶ τὸ $ΔA$ τῷ $ΑΓ$. <ρητὸν δέ τὸ $ΔA$; <ρητὸν >άρα ἐστὶ καὶ τὸ $ΑΓ$.
Τὸ >άρα ὑπὸ <ρητῶν μήκει συμμέτρων, καὶ τὰ ἐχῆξ.

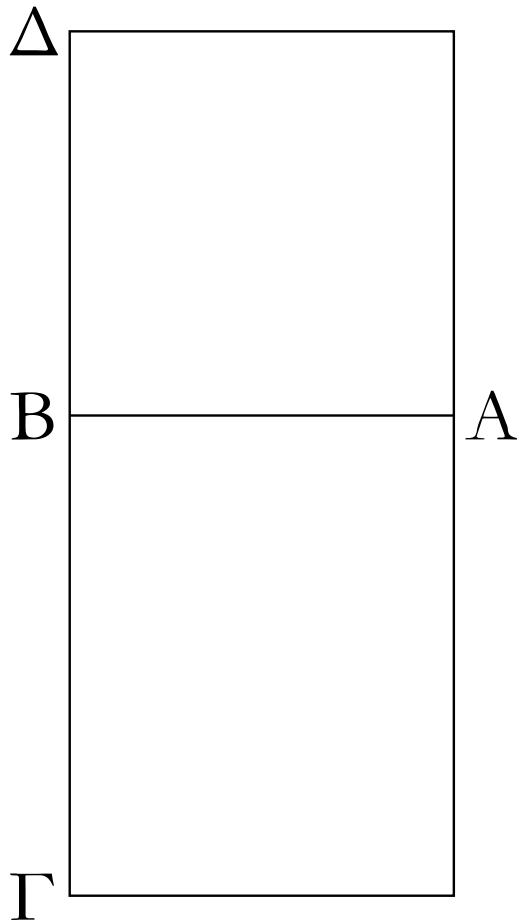
20.

Ἐὰν <ρητὸν παρὰ <ρητὴν παραβληθῇ, πλάτοξ ποιεῖ <ρητὴν καὶ σύμμετρον τῇ, παρ> <ὴν παράκειται, μήκει.

For let the square AD be described on AB . AD is thus rational [Def. 10.4]. And since AB is commensurable in length with BC , and AB is equal to BD , BD is thus commensurable in length with BC . And as BD is to BC , so DA (is) to AC [Prop. 6.1]. Thus, DA is commensurable with AC [Prop. 10.11]. And DA (is) rational. Thus, AC is also rational [Def. 10.4]. Thus, the ...by rational straight-lines ...commensurable, and so on

Proposition 20

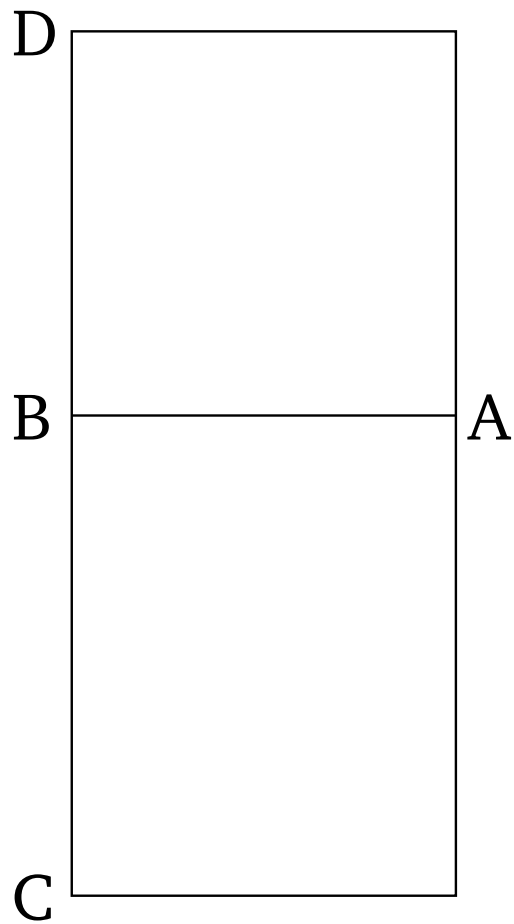
If a rational (area) is applied to a rational (straight-line) then it produces as breadth a (straight-line which is) rational, and commensurable in length with the (straight-line) to which it is applied.



<Ρητὸν γὰρ τὸ ΑΓ παρὰ <ρητὴν τὴν ΑΒ παραβεβλήσθω πλάτος ποιοῦν τὴν ΒΓ; λέγω, <ὅτι <ρητὴ ἐστὶν ἡ ΒΓ καὶ σύμμετρος τῇ ΒΑ μήκει.

Ἀναγεγράφθω γὰρ ἀπὸ τῆς ΑΒ τετράγωνον τὸ ΑΔ; <ρητὸν >άρα ἐστὶ τὸ ΑΔ. <ρητὸν δὲ καὶ τὸ ΑΓ; σύμμετρον >άρα ἐστὶ τὸ ΔΑ τῷ ΑΓ. καὶ ἐστὶν ὥς τὸ ΔΑ πρὸς τὸ ΑΓ, οὕτως ἡ ΔΒ πρὸς τὴν ΒΓ. σύμμετρος >άρα ἐστὶ καὶ ἡ ΔΒ τῇ ΒΓ; >ίση δὲ ἡ ΔΒ τῇ ΒΑ; σύμμετρος >άρα καὶ ἡ ΑΒ τῇ ΒΓ. <ρητὴ δὲ ἐστὶν ἡ ΑΒ; <ρητὴ >άρα ἐστὶ καὶ ἡ ΒΓ καὶ σύμμετρος τῇ ΑΒ μήκει.

Ἐὰν >άρα <ρητὸν παρὰ <ρητὴν παραβληθῇ, καὶ τὰ ἐχθῇξ.



For let the rational (area) AC be applied to the rational (straight-line) AB , producing the (straight-line) BC as breadth. I say that BC is rational, and commensurable in length with BA .

For let the square AD be described on AB . AD is thus rational [Def. 10.4]. And AC (is) also rational. DA is thus commensurable with AC . And as DA is to AC , so DB (is) to BC [Prop. 6.1]. Thus, DB is also commensurable (in length) with BC [Prop. 10.11]. And DB (is) equal to BA . Thus, AB (is) also commensurable (in length) with BC . And AB is rational. Thus, BC is also rational, and commensurable in length with AB [Def. 10.3].

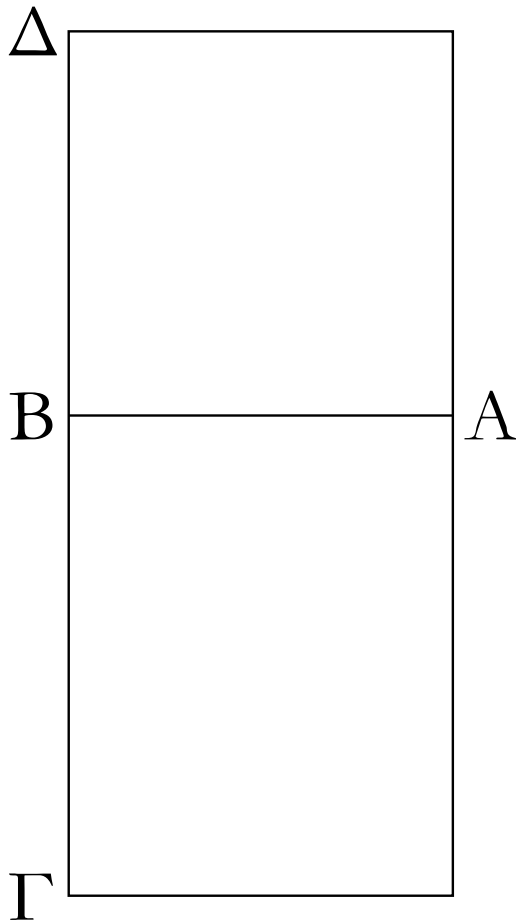
Thus, if a rational (area) is applied to a rational (straight-line), and so on

21.

Proposition 21

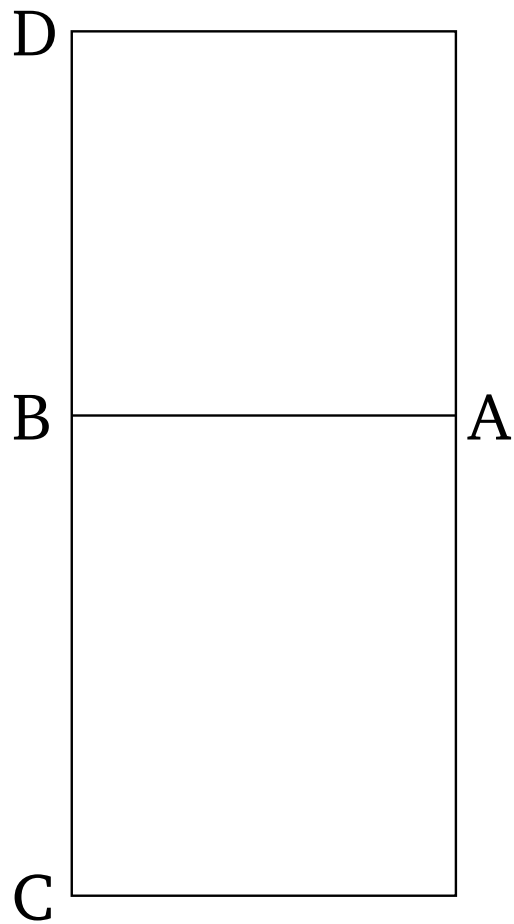
Τὸ ὑπὸ <ρητῶν δυνάμει μόνον συμμέτρων εὐθειῶν περιεχόμενον ὀρθογώνιον >άλογόν ἐστιν, καὶ ἡ δυναμένη αὐτὸ >άλογός ἐστιν, καλεῖσθω δὲ μέση.

The rectangle contained by rational straight-lines (which are) commensurable in square only is irrational, and its square-root is irrational—let it be called medial.[†]



Ὑπὸ γὰρ <ρητῶν δυνάμει μόνον συμμετρῶν εὐθειῶν τῶν AB , $BΓ$ ὀρθογώνιον περιεθέσθω τὸ $AΓ$; λέγω, <ὅτι> ἄλογόν ἐστι τὸ $AΓ$, καὶ ἡ δυνάμενι αὐτὸ >αλογόξ ἐστιν, καλείσθω δὲ μέση.

Ἀναγεγράφθω γὰρ ἀπὸ τῆς AB τετράγωνον τὸ AD ; <ρητὸν> ἄρα ἐστὶ τὸ AD . καὶ ἐπεὶ ἀσύμμετρόξ ἐστιν ἡ AB τῇ $BΓ$ μήκει; δυνάμει γὰρ μόνον ὑπόκεινται σύμμετροι; >ίση δὲ ἡ AB τῇ BD , ἀσύμμετροξ ἄρα ἐστὶ καὶ ἡ DB τῇ $BΓ$ μήκει. καὶ ἐστὶν ὥς ἡ DB πρὸς τὴν $BΓ$, ο<ύτωξ τὸ AD πρὸς τὸ $AΓ$; ἀσύμμετρον ἄρα [ἐστὶ] τὸ DA τῷ $AΓ$. <ρητὸν δὲ τὸ DA ; >άλογον ἄρα ἐστὶ τὸ $AΓ$; <ὥστε καὶ ἡ δυνάμενι τὸ $AΓ$ [τουτέστιν ἡ >ίσον αὐτῷ] τετράγωνον δυνάμενι] >αλογόξ ἐστιν, καλείσθω δε μέση; <ὅπερ> >έδει δεῖξαι.



For let the rectangle AC be contained by the rational straight-lines AB and BC (which are) commensurable in square only. I say that AC is irrational, and its square-root is irrational—let it be called medial.

For let the square AD be described on AB . AD is thus rational [Def. 10.4]. And since AB is incommensurable in length with BC . For they were assumed to be commensurable in square only. And AB (is) equal to BD . DB is thus also incommensurable in length with BC . And as DB is to BC , so AD (is) to AC [Prop. 6.1]. Thus, DA [is] incommensurable with AC [Prop. 10.11]. And DA (is) rational. Thus, AC is irrational [Def. 10.4]. Hence, its square-root [that is to say, the square-root of the square equal to it] is also irrational [Def. 10.4]—let it be called medial. (Which is) the very thing it was required to show.

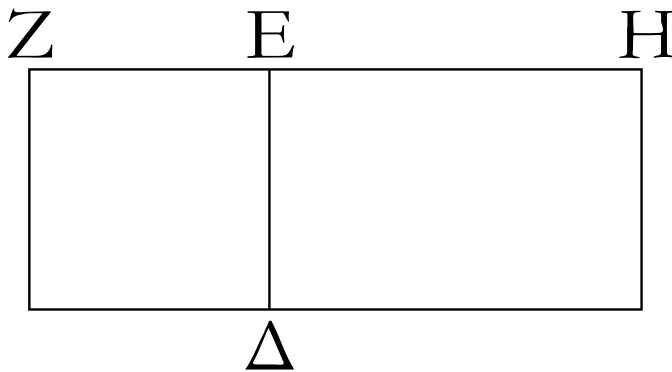
† Thus, a medial straight-line has a length expressible as $k^{1/4}$.

Λήμμα.

Ἐὰν >ῶσι δύο εὐθεῖαι, >έστιν ὥς ἡ πρώτη πρὸς τὴν δευτέραν, ο<ύτωξ τὸ ἀπὸ τῆς πρώτης πρὸς τὸ ὑπὸ τῶν δύο εὐθειῶν.

Lemma

If there are two straight-lines then as the first is to the second, so the (square) on the first (is) to the (rectangle contained) by the two straight-lines.



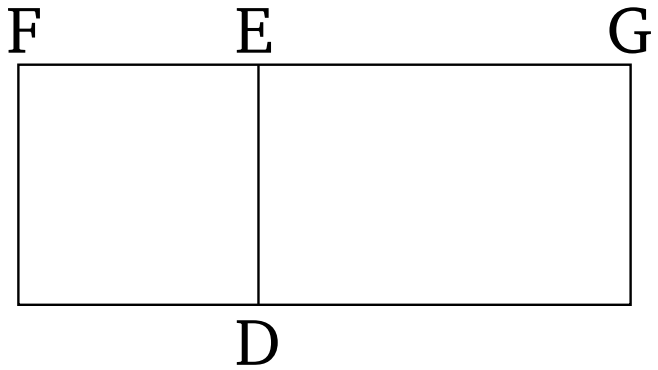
>Ἐστωσαν δύο εὐθεῖαι αἱ ZE, EH. λέγω, <ὅτι ἐστὶν ὥς ἡ ZE πρὸς τὴν EH, ο<ύτωξ τὸ ἀπὸ τῆς ZE πρὸς τὸ ὑπὸ τῶν ZE, EH.

Ἀναγεγράφθω γὰρ ἀπὸ τῆς ZE τετράγωνον τὸ ΔΖ, καὶ συμπληρώσθω τὸ ΗΔ. ἐπεὶ ο>ὖν ἐστὶν ὥς ἡ ZE πρὸς τὴν EH, ο<ύτωξ τὸ ΖΔ πρὸς τὸ ΔΗ, καὶ ἐστὶ τὸ μὲν ΖΔ τὸ ἀπὸ τῆς ZE, τὸ δὲ ΔΗ τὸ ὑπὸ τῶν ΔΕ, ΕΗ, τουτέστι τὸ ὑπὸ τῶν ZE, ΕΗ, >ἐστὶν >ἀρα ὥς ἡ ZE πρὸς τὴν EH, ο<ύτωξ τὸ ἀπὸ τῆς ZE πρὸς τὸ ὑπὸ τῶν ZE, ΕΗ. ὁμοίως δὲ καὶ ὥς τὸ ὑπὸ τῶν ΗΕ, ΕΖ πρὸς τὸ ἀπὸ τῆς ΕΖ, τουτέστιν ὥς τὸ ΗΔ πρὸς τὸ ΖΔ, ο<ύτωξ ἡ ΗΕ πρὸς τὴν ΕΖ; <όπερ >έδει δεῖχαι.

22.

Τὸ ἀπὸ μέσης παρὰ <ρητὴν παραβαλλόμενον πλάτος ποιεῖ <ρητὴν καὶ ἀσύμμετρον τῇ, παρ> <ὴν παράκειται, μήκει.

>Ἐστω μέση μὲν ἡ Α, <ρητὴ δὲ ἡ ΓΒ, καὶ τῷ ἀπὸ τῆς Α >ῖσον παρὰ τὴν ΒΓ παραβεβλήσθω θωρίον ὀρθογώνιον τὸ ΒΔ πλάτος ποιοῦν τὴν ΓΔ; λέγω, <ὅτι <ρητὴ ἐστὶν ἡ ΓΔ καὶ ἀσύμμετρος τῇ ΓΒ μήκει.



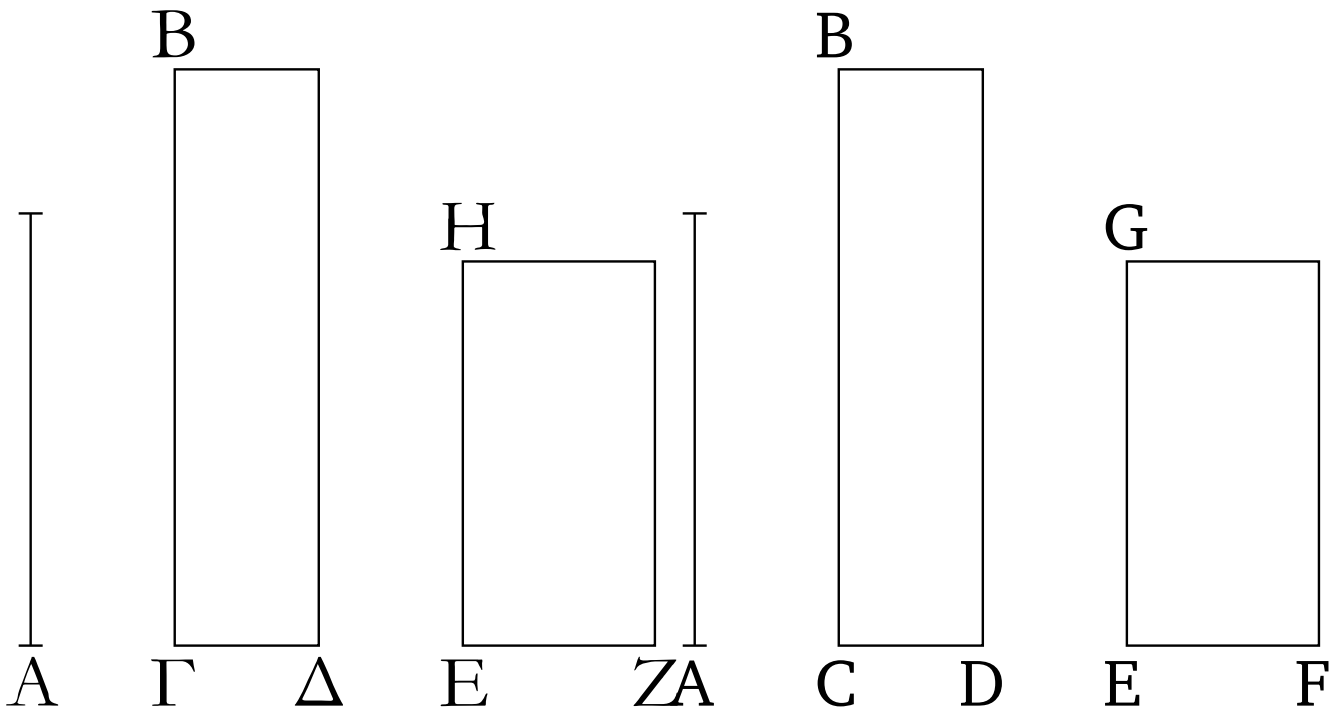
Let FE and EG be two straight-lines. I say that as FE is to EG , so the (square) on FE (is) to the (rectangle contained) by FE and EG .

For let the square DF be described on FE . And let GD be completed. Therefore, since as FE is to EG , so FD (is) to DG [Prop. 6.1], and FD is the (square) on FE , and DG the (rectangle contained) by DE and EG —that is to say, the (rectangle contained) by FE and EG —thus as FE is to EG , so the (square) on FE (is) to the (rectangle contained) by FE and EG . And also, similarly, as the (rectangle contained) by GE and EF is to the (square on) EF —that is to say, as GD (is) to FD —so GE (is) to EF . (Which is) the very thing it was required to show.

Proposition 22

The square on a medial (straight-line), being applied to a rational (straight-line), produces as breadth a (straight-line which is) rational, and incommensurable in length with the (straight-line) to which it is applied.

Let A be a medial (straight-line), and CB a rational (straight-line), and let the rectangular area BD , equal to the (square) on A , be applied to BC , producing CD as breadth. I say that CD is rational, and incommensurable in length with CB .



Ἐπεὶ γὰρ μέση ἐστὶν ἡ A , δύναται θωρίον περιεχόμενον ὑπὸ <ρητῶν δυνάμει μόνον συμμέτρων. δυνάσθω τὸ HZ . δύναται δὲ καὶ τὸ $B\Delta$; >ῖσον >άρα ἐστὶ τὸ $B\Delta$ τῷ HZ . >ἐστι δὲ αὐτῶν καὶ ἰσογώνιον; τῶν δὲ >ῖσων τε καὶ ἰσογωνίων παραλληλογράμμων ἀντιπεπόνθασιν αἱ πλευραὶ αἱ περὶ τὰς >ῖσας γωνίας; ἀνάλογον >άρα ἐστὶν ὡς ἡ $B\Gamma$ πρὸς τὴν EH , ο<ύτωξ ἡ EZ πρὸς τὴν $\Gamma\Delta$. >ἐστὶν >άρα καὶ ὡς τὸ ἀπὸ τῆς $B\Gamma$ πρὸς τὸ ἀπὸ τῆς EH , ο<ύτωξ τὸ ἀπὸ τῆς EZ πρὸς τὸ ἀπὸ τῆς $\Gamma\Delta$. σύμμετρον δὲ ἐστὶ τὸ ἀπὸ τῆς ΓB τῷ ἀπὸ τῆς EH ; <ρητὴ γάρ ἐστὶν ἑκατέρα αὐτῶν; σύμμετρον >άρα ἐστὶ καὶ τὸ ἀπὸ τῆς EZ τῷ ἀπὸ τῆς $\Gamma\Delta$. <ρητὸν δὲ ἐστὶ τὸ ἀπὸ τῆς EZ ; <ρητὸν >άρα ἐστὶ καὶ τὸ ἀπὸ τῆς $\Gamma\Delta$; <ρητὴ >άρα ἐστὶν ἡ $\Gamma\Delta$. καὶ ἐπεὶ ἀσύμμετροξ ἐστὶν ἡ EZ τῇ EH μήκει; δυνάμει γὰρ μόνον εἰσὶ σύμμετροι; ὡς δὲ ἡ EZ πρὸς τὴν EH , ο<ύτωξ τὸ ἀπὸ τῆς EZ πρὸς τὸ ὑπὸ τῶν ZE , EH , ἀσύμμετρον >άρα [ἐστὶ] τὸ ἀπὸ τῆς EZ τῷ ὑπὸ τῶν ZE , EH . ἀλλὰ τῷ μὲν ἀπὸ τῆς EZ σύμμετρόν ἐστὶ τὸ ἀπὸ τῆς $\Gamma\Delta$; <ρηταὶ γὰρ εἰσι δυνάμει; τῷ δὲ ὑπὸ τῶν ZE , EH σύμμετρόν ἐστὶ τὸ ὑπὸ τῶν $\Delta\Gamma$, ΓB ; >ῖσα γὰρ ἐστὶ τῷ ἀπὸ τῆς A ; ἀσύμμετρον >άρα ἐστὶ καὶ τὸ ἀπὸ τῆς $\Gamma\Delta$ τῷ ὑπὸ τῶν $\Delta\Gamma$, ΓB . ὡς δὲ τὸ ἀπὸ τῆς $\Gamma\Delta$ πρὸς τὸ ὑπὸ τῶν $\Delta\Gamma$, ΓB , ο<ύτωξ ἐστὶν ἡ $\Delta\Gamma$ πρὸς τὴν ΓB ; ἀσύμμετροξ >άρα ἐστὶν ἡ $\Delta\Gamma$ τῇ ΓB μήκει. <ρητὴ >άρα ἐστὶν ἡ $\Gamma\Delta$ καὶ ἀσύμμετροξ τῇ ΓB μήκει; <όπερ >έδει δεῖξαι.

For since A is medial, the square on it is equal to a (rectangular) area contained by rational (straight-lines which are) commensurable in square only [Prop. 10.21]. Let the square on (A) be equal to GF . And the square on (A) is also equal to BD . Thus, BD is equal to GF . And (BD) is also equiangular with (GF) . And for equal and equiangular parallelograms, the sides about the equal angles are reciprocally proportional [Prop. 6.14]. Thus, proportionally, as BC is to EG , so EF (is) to CD . And, also, as the (square) on BC is to the (square) on EG , so the (square) on EF (is) to the (square) on CD [Prop. 6.22]. And the (square) on CB is commensurable with the (square) on EG . For they are each rational. Thus, the (square) on EF is also commensurable with the (square) on CD [Prop. 10.11]. And the (square) on EF is rational. Thus, the (square) on CD is also rational [Def. 10.4]. Thus, CD is rational. And since EF is incommensurable in length with EG . For they are commensurable in square only. And as EF (is) to EG , so the (square) on EF (is) to the (rectangle contained) by FE and EG [see previous lemma]. The (square) on EF [is] thus incommensurable with the (rectangle contained) by FE and EG [Prop. 10.11]. But, the (square) on CD is commensurable with the (square) on EF . For they are rational in square. And the (rectangle contained) by DC and CB is commensurable with the (rectangle contained) by FE and EG . For they are (both) equal to the (square) on A . Thus, the (square) on CD is also incommensurable with the (rectangle contained) by DC and CB [Prop. 10.13]. And as the (square) on CD (is) to the (rectangle contained) by DC and CB , so DC is to CB [see previous lemma]. Thus, DC is incommensurable in length with CB [Prop. 10.11]. Thus, CD is rational, and incommensurable in length with CB . (Which is) the very thing it was required to show.

† Literally, “rational”.

23.

Ἡ τῇ μέση σύμμετροξ μέση ἐστίν.

Proposition 23

A (straight-line) commensurable with a medial (straight-

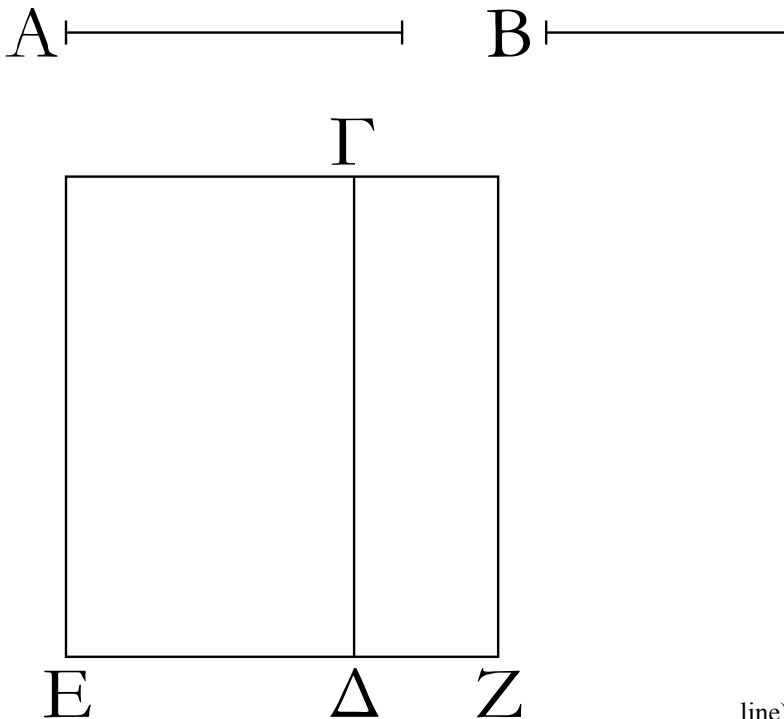
>Ἐστω μέση ἡ A , καὶ τῇ A σύμμετρος >έστω ἡ B ; λέγω,
<ὅτι καὶ ἡ B μέση ἐστίν.

Ἐκκείσθω γάρ <ρητὴ ἡ $\Gamma\Delta$, καὶ τῷ μὲν ἀπὸ τῆς A >ίσον
παρὰ τὴν $\Gamma\Delta$ παραβεβλήσθω θωρίον ὀρθογώνιον τὸ $\Gamma\Theta$
πλάτος ποιῶν τὴν $\Theta\Delta$; <ρητὴ >άρα ἐστὶν ἡ $\Theta\Delta$ καὶ ἀσύμμετρος
τῇ $\Gamma\Delta$ μήκει. τῷ δὲ ἀπὸ τῆς B >ίσον παρὰ τὴν $\Gamma\Delta$
παραβεβλήσθω θωρίον ὀρθογώνιον τὸ $\Gamma\Lambda$ πλάτος ποιῶν
τὴν $\Lambda\Delta$. ἐπεὶ ο>ὖν σύμμετρος ἐστὶν ἡ A τῇ B , σύμμετρον
ἐστὶ καὶ τὸ ἀπὸ τῆς A τῷ ἀπὸ τῆς B . ἀλλὰ τῷ μὲν ἀπὸ
τῆς A >ίσον ἐστὶ τὸ $\Theta\Gamma$, τῷ δὲ ἀπὸ τῆς B >ίσον ἐστὶ τὸ $\Lambda\Gamma$;
σύμμετρον >άρα ἐστὶ τὸ $\Theta\Gamma$ τῷ $\Lambda\Gamma$. καὶ ἐστὶν ὥς τὸ $\Theta\Gamma$ πρὸς
τὸ $\Lambda\Gamma$, ο<ύτως ἡ $\Theta\Delta$ πρὸς τὴν $\Lambda\Delta$; σύμμετρος >άρα ἐστὶν
ἡ $\Theta\Delta$ τῇ $\Lambda\Delta$ μήκει. <ρητὴ δὲ ἐστὶν ἡ $\Theta\Delta$ καὶ ἀσύμμετρος
τῇ $\Lambda\Delta$ μήκει; <ρητὴ >άρα ἐστὶ καὶ ἡ $\Lambda\Delta$ καὶ ἀσύμμετρος
τῇ $\Lambda\Gamma$ μήκει; αἱ $\Gamma\Delta$, $\Lambda\Delta$ >άρα <ρηταὶ εἰσι δυνάμει μόνον
σύμμετροι. ἡ δὲ τὸ ὑπὸ <ρητῶν δυνάμει μόνον συμμέτρων
δυναμένη μέση ἐστίν. ἡ >άρα τὸ ὑπὸ τῶν $\Gamma\Delta$, $\Lambda\Delta$ δυναμένη
μέση ἐστίν; καὶ δύναται τὸ ὑπὸ τῶν $\Gamma\Delta$, $\Lambda\Delta$ ἡ B ; μέση >άρα
ἐστὶν ἡ B .

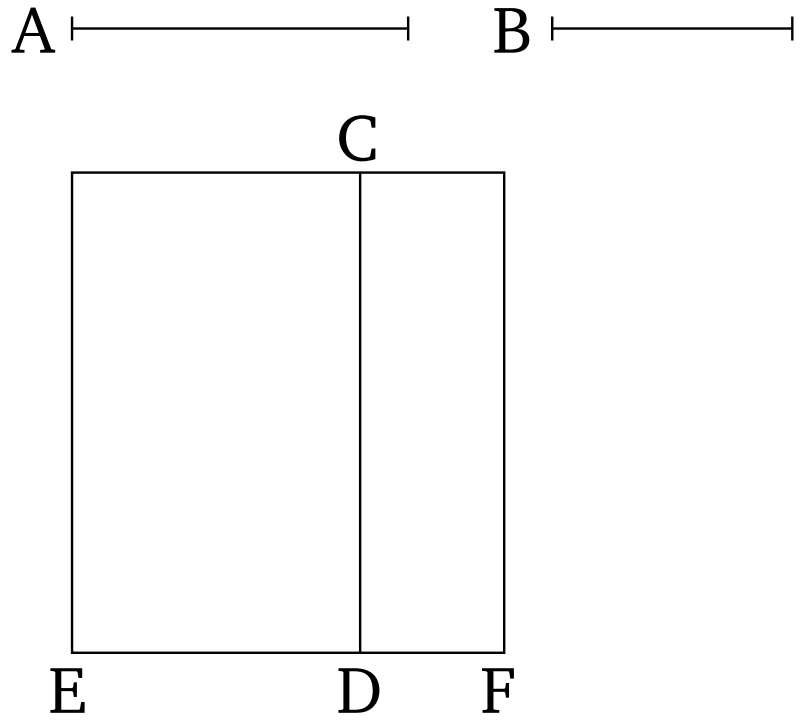
line) is medial.

Let A be a medial (straight-line), and let B be commensurable with A . I say that B is also a medial (straight-line).

Let the rational (straight-line) CD be set out, and let the rectangular area CE , equal to the (square) on A , be applied to CD , producing ED as width. ED is thus rational, and incommensurable in length with CD [Prop. 10.22]. And let the rectangular area CF , equal to the (square) on B , be applied to CD , producing DF as width. Therefore, since A is commensurable with B , the (square) on A is also commensurable with the (square) on B . But, EC is equal to the (square) on A , and CF is equal to the (square) on B . Thus, EC is commensurable with CF . And as EC is to CF , so ED (is) to DF [Prop. 6.1]. Thus, ED is commensurable in length with DF [Prop. 10.11]. And ED is rational, and incommensurable in length with CD . DF is thus also rational [Def. 10.3], and incommensurable in length with DC [Prop. 10.13]. Thus, CD and DF are rational, and commensurable in square only. And the square-root of a (rectangle contained) by rational (straight-lines which are) commensurable in square only is medial [Prop. 10.21]. Thus, the square-root of the (rectangle contained) by CD and DF is medial. And the square on B is equal to the (rectangle contained) by CD and DF . Thus, B is a medial (straight-



line).



Πόρισμα.

Ἐκ δὴ τούτου φανερόν, <ὅτι τὸ τῶ| μέσῳ θωρήν σύμμετρον μέσον ἐστίν.

Corollary

And (it is) clear, from this, that an (area) commensurable with a medial area[†] is medial.

[†] A medial area is equal to the square on some medial straight-line. Hence, a medial area is expressible as $k^{1/2}$.

24.

Τὸ ὑπὸ μέσων μήκει συμμέτρων εὐθειῶν περιεχόμενον ὀρθογώνιον μέσον ἐστίν.

Ὑπὸ γὰρ μέσων μήκει συμμέτρων εὐθειῶν τῶν AB, BΓ περιεθέσθω ὀρθογώνιον τὸ AΓ; λέγω, <ὅτι τὸ AΓ μέσον ἐστίν.

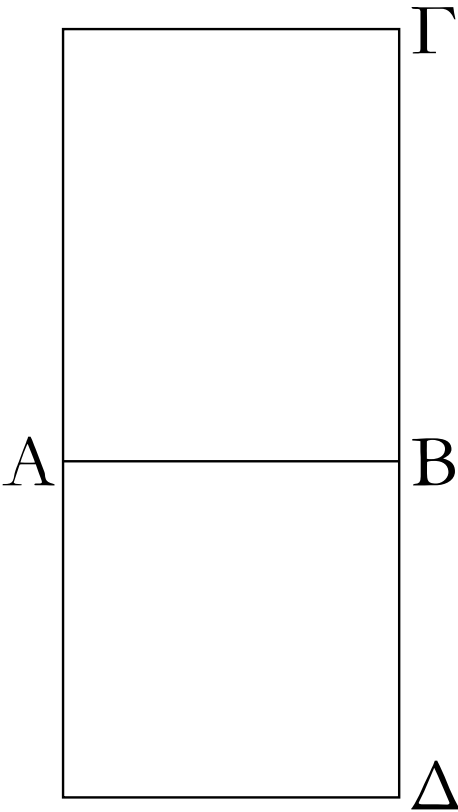
Ἀναγεγράφθω γὰρ ἀπὸ τῆς AB τετράγωνον τὸ AD; μέσον >άρα ἐστὶ τὸ AD. καὶ ἐπεὶ σύμμετρος ἐστὶν ἡ AB τῇ BΓ μήκει, >ίση δὲ ἡ AB τῇ BD, σύμμετρος >άρα ἐστὶ καὶ ἡ ΔB τῇ BΓ μήκει; <ὥστε καὶ τὸ ΔA τῶ| AΓ σύμμετρον ἐστίν. μέσον δὲ τὸ ΔA; μέσον >άρα καὶ τὸ AΓ; <όπερ >έδει δεῖχαι.

Proposition 24

A rectangle contained by medial straight-lines (which are) commensurable in length is medial.

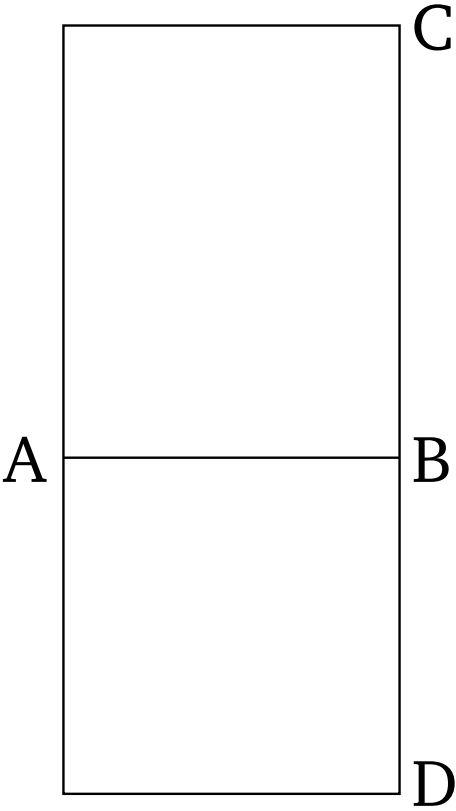
For let the rectangle AC be contained by the medial straight-lines AB and BC (which are) commensurable in length. I say that AC is medial.

For let the square AD be described on AB. AD is thus medial [see previous footnote]. And since AB is commensurable in length with BC, and AB (is) equal to BD, DB is thus also commensurable in length with BC. Hence, DA is also commensurable with AC [Props. 6.1, 10.11]. And DA (is) medial. Thus, AC (is) also medial [Prop. 10.23 corr.]. (Which is) the very thing it was required to show.



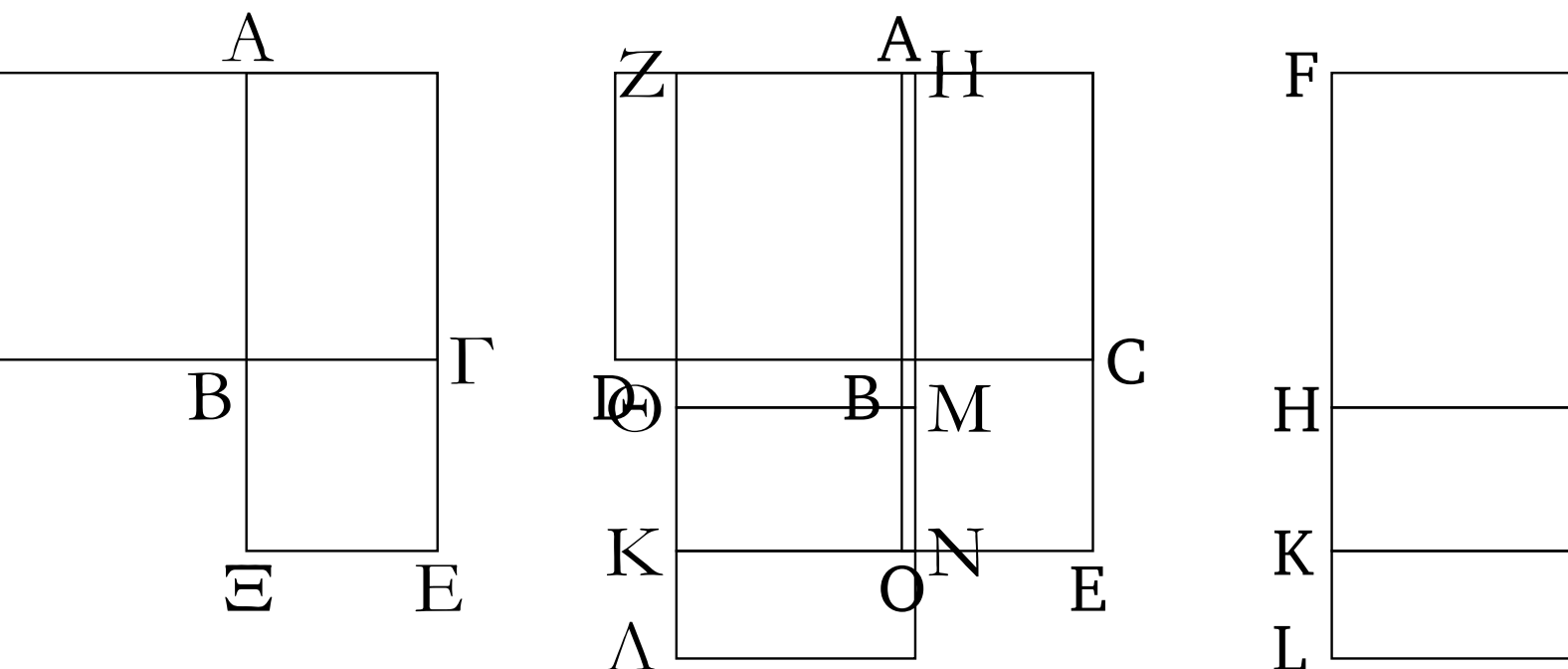
25.

Τὸ ὑπὸ μέσων δυνάμει μόνον συμμετρων εὐθειῶν περιεχόμενον ὀρθογώνιον >ήτοι <ρητὸν >ἢ μέσον ἐστίν.



Proposition 25

The rectangle contained by medial straight-lines (which are) commensurable in square only is either rational or medial.



Ὑπὸ γὰρ μέσων δυνάμει μόνον συμμετρῶν εὐθειῶν τῶν AB , $BΓ$ ὀρθογώνιον περιθέσθω τὸ $AΓ$; λέγω, <ὅτι τὸ $AΓ$ >ήτοι <ρητὸν >ή μέσον ἐστίν.

Ἀναγεγράψθω γὰρ ἀπὸ τῶν AB , $BΓ$ τετράγωνα τὰ AD , BE ; μέσον >άρα ἐστὶν ἑκάτερον τῶν AD , BE . καὶ ἐκκείσθω <ρητὴ ἢ ZH , καὶ τῷ μὲν AD >ῖσον παρὰ τὴν ZH παραβεβλήσθω ὀρθογώνιον παραλληλόγραμμον τὸ $HΘ$ πλάτος ποιῶν τὴν $ZΘ$, τῷ δὲ $AΓ$ >ῖσον παρὰ τὴν $ΘM$ παραβεβλήσθω ὀρθογώνιον παραλληλόγραμμον τὸ MK πλάτος ποιῶν τὴν $ΘK$, καὶ >έτι τῷ BE >ῖσον ὁμοίως παρὰ τὴν KN παραβεβλήσθω τὸ $NΛ$ πλάτος ποιῶν τὴν KL ; ἐπ' εὐθείᾳ >άρα εἰσὶν αἱ $ZΘ$, $ΘK$, KL . ἐπεὶ οὖν μέσον ἐστὶν ἑκάτερον τῶν AD , BE , καὶ ἐστὶν >ῖσον τὸ μὲν AD τῷ $HΘ$, τὸ δὲ BE τῷ $NΛ$, μέσον >άρα καὶ ἑκάτερον τῶν $HΘ$, $NΛ$. καὶ παρὰ <ρητὴν τὴν ZH παράκειται, <ρητὴ >άρα ἐστὶν ἑκάτερα τῶν $ZΘ$, KL καὶ ἀσύμμετρος τῇ ZH μήκει. καὶ ἐπεὶ σύμμετρον ἐστὶ τὸ AD τῷ BE , σύμμετρον >άρα ἐστὶ καὶ τὸ $HΘ$ τῷ $NΛ$. καὶ ἐστὶν ὥς τὸ $HΘ$ πρὸς τὸ $NΛ$, οὕτως ἡ $ZΘ$ πρὸς τὴν KL ; σύμμετρος >άρα ἐστὶν ἡ $ZΘ$ τῇ KL μήκει. αἱ $ZΘ$, KL >άρα <ρηταὶ εἰσι μήκει σύμμετροι; <ρητὸν >άρα ἐστὶ τὸ ὑπὸ τῶν $ZΘ$, KL . καὶ ἐπεὶ >ῖση ἐστὶν ἡ μὲν $ΔB$ τῇ BA , ἡ δὲ XB τῇ $BΓ$, >έστιν >άρα ὥς ἡ $ΔB$ πρὸς τὴν $BΓ$, οὕτως ἡ AB πρὸς τὴν BX . ἀλλ' ὥς μὲν ἡ $ΔB$ πρὸς τὴν $BΓ$, οὕτως τὸ $ΔA$ πρὸς τὸ $AΓ$; ὥς δὲ ἡ AB πρὸς τὴν BX , οὕτως τὸ $AΓ$ πρὸς τὸ $ΓX$; >έστιν >άρα ὥς τὸ $ΔA$ πρὸς τὸ $AΓ$, οὕτως τὸ $AΓ$ πρὸς τὸ $ΓX$. >ῖσον δὲ ἐστὶ τὸ μὲν AD τῷ $HΘ$, τὸ δὲ $AΓ$ τῷ MK , τὸ δὲ $ΓX$ τῷ $NΛ$; >έστιν >άρα ὥς τὸ $HΘ$ πρὸς τὸ MK , οὕτως τὸ MK πρὸς τὸ $NΛ$; >έστιν >άρα καὶ ὥς ἡ $ZΘ$ πρὸς τὴν $ΘK$, οὕτως ἡ $ΘK$ πρὸς τὴν KL ; τὸ >άρα ὑπὸ τῶν $ZΘ$, KL >ῖσον ἐστὶ τῷ ἀπὸ τῆς $ΘK$. <ρητὸν δὲ τὸ ὑπὸ τῶν $ZΘ$, KL ; <ρητὸν >άρα ἐστὶ καὶ τὸ ἀπὸ τῆς $ΘK$; <ρητὴ >άρα ἐστὶν ἡ $ΘK$. καὶ εἰ μὲν σύμμετρόξ ἐστὶ τῇ ZH μήκει, <ρητὸν ἐστὶ τὸ $ΘN$; εἰ δὲ ἀσύμμετρόξ ἐστὶ τῇ ZH μήκει, αἱ $KΘ$, $ΘM$ <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι; μέσον >άρα τὸ $ΘN$. τὸ $ΘN$ >άρα >ήτοι <ρητὸν >ή μέσον ἐστίν. >ῖσον δὲ τὸ $ΘN$ τῷ $AΓ$; τὸ $AΓ$ >άρα >ήτοι <ρητὸν >ή μέσον ἐστίν.

Τὸ >άρα ὑπὸ μέσων δυνάμει μόνον συμμετρῶν, καὶ τὰ ἐχῆξ.

For let the rectangle AC be contained by the medial straight-lines AB and BC (which are) commensurable in square only. I say that AC is either rational or medial.

For let the squares AD and BE be described on (the straight-lines) AB and BC (respectively). AD and BE are thus each medial. And let the rational (straight-line) FG be laid out. And let the rectangular parallelogram GH , equal to AD , be applied to FG , producing FH as breadth. And let the rectangular parallelogram MK , equal to AC , be applied to HM , producing HK as breadth. And, finally, let NL , equal to BE , have similarly been applied to KN , producing KL as breadth. Thus, FH , HK , and KL are in a straight-line. Therefore, since AD and BE are each medial, and AD is equal to GH , and BE to NL , GH and NL (are) thus each also medial. And they are applied to the rational (straight-line) FG . FH and KL are thus each rational, and incommensurable in length with FG [Prop. 10.22]. And since AD is commensurable with BE , GH is thus also commensurable with NL . And as GH is to NL , so FH (is) to KL [Prop. 6.1]. Thus, FH is commensurable in length with KL [Prop. 10.11]. Thus, FH and KL are rational (straight-lines which are) commensurable in length. Thus, the (rectangle contained) by FH and KL is rational [Prop. 10.19]. And since DB is equal to BA , and OB to BC , thus as DB is to BC , so AB (is) to BO . But, as DB (is) to BC , so DA (is) to AC [Props. 6.1]. And as AB (is) to BO , so AC (is) to CO [Prop. 6.1]. Thus, as DA is to AC , so AC (is) to CO . And AD is equal to GH , and AC to MK , and CO to NL . Thus, as GH is to MK , so MK (is) to NL . Thus, also, as FH is to HK , so HK (is) to KL [Props. 6.1, 5.11]. Thus, the (rectangle contained) by FH and KL is equal to the (square) on HK [Prop. 6.17]. And the (rectangle contained) by FH and KL (is) rational. Thus, the (square) on HK is also rational. Thus, HK is rational. And if it is commensurable in length with FG then HN is rational [Prop. 10.19]. And if it is incommensurable in length with FG then KH and HM are rational (straight-lines which are) commensurable in square only: thus, HN is medial [Prop. 10.21]. Thus, HN is either rational or medial. And HN (is) equal to AC . Thus, AC is either rational

or medial.

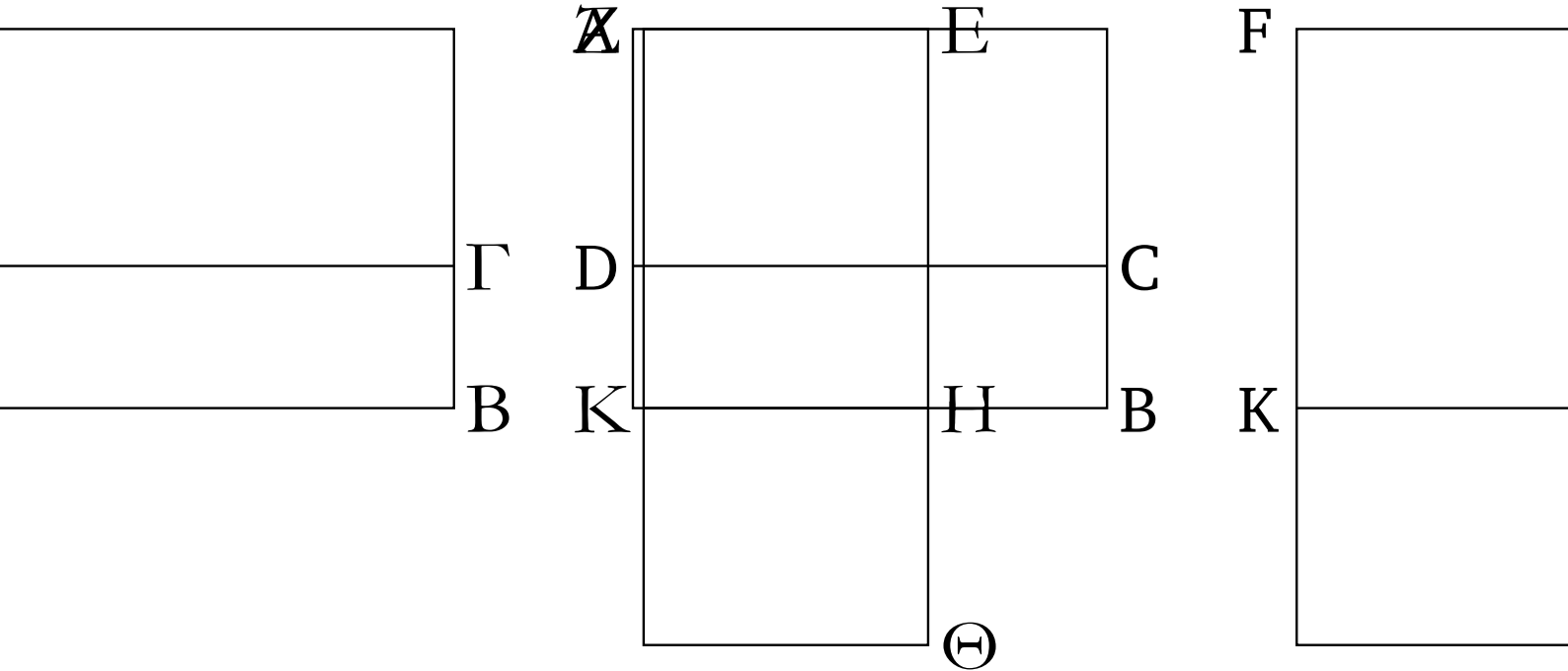
Thus, the ...by medial straight-lines (which are) commensurable in square only, and so on

26.

Μέσον μέσου οὐθ ὑπερέθει <ρητῶ|.

Proposition 26

A medial (area) does not exceed a medial (area) by a rational (area).[†]



Εἰ γὰρ δυνατόν, μέσον τὸ AB μέσου τοῦ AG ὑπερέθετω <ρητῶ| τῶ| ΔB , καὶ ἐκκείσθω <ρητὴ ἢ EZ , καὶ τῶ| AB >ίσον παρὰ τὴν EZ παραβελήσθω παραλληλόγραμμον ὀρθογώνιον τὸ $Z\Theta$ πλάτος ποιούν τὴν $E\Theta$, τῶ| δὲ AG >ίσον ἀφηρήσθω τὸ ZH ; λοιπὸν >άρα τὸ $B\Delta$ λοιπῶ| τῶ| $K\Theta$ ἐστὶν >ίσον. <ρητὸν δὲ ἐστὶ τὸ ΔB ; <ρητὸν >άρα ἐστὶ καὶ τὸ $K\Theta$. ἐπεὶ ο>ὖν μέσον ἐστὶν ἑκάτερον τῶν AB , AG , καὶ ἐστὶ τὸ μὲν AB τῶ| $Z\Theta$ >ίσον, τὸ δὲ AG τῶ| ZH , μέσον >άρα καὶ ἑκάτερον τῶν $Z\Theta$, ZH . καὶ παρὰ <ρητὴν τὴν EZ παράκειται; <ρητὴ >άρα ἐστὶν ἑκάτερα τῶν ΘE , $E\Delta$ καὶ ἀσύμμετρος τῇ EZ μήκει. καὶ ἐπεὶ <ρητὸν ἐστὶ τὸ ΔB καὶ ἐστὶν >ίσον τῶ| $K\Theta$, <ρητὸν >άρα ἐστὶ καὶ τὸ $K\Theta$. καὶ παρὰ <ρητὴν τὴν EZ παράκειται; <ρητὴ >άρα ἐστὶν ἢ $H\Theta$ καὶ σύμμετρος τῇ EZ μήκει. ἀλλὰ καὶ ἡ $E\Delta$ <ρητὴ ἐστὶ καὶ ἀσύμμετρος τῇ EZ μήκει; ἀσύμμετρος >άρα ἐστὶν ἢ $E\Delta$ τῇ $H\Theta$ μήκει. καὶ ἐστὶν ὥς ἡ $E\Delta$ πρὸς τὴν $H\Theta$, ο<ύτωξ τὸ ἀπὸ τῆς $E\Delta$ πρὸς τὸ ὑπὸ τῶν $E\Delta$, $H\Theta$; ἀσύμμετρον >άρα ἐστὶ τὸ ἀπὸ τῆς $E\Delta$ τῶ| ὑπὸ τῶν $E\Delta$, $H\Theta$. ἀλλὰ τῶ| μὲν ἀπὸ τῆς $E\Delta$ σύμμετρά ἐστὶ τὰ ἀπὸ τῶν $E\Delta$, $H\Theta$ τετράγωνα; <ρητὰ γὰρ ἀμφοτέρω; τῶ| δὲ ὑπὸ τῶν $E\Delta$, $H\Theta$ σύμμετρόν ἐστὶ τὸ διξ ὑπὸ τῶν $E\Delta$, $H\Theta$; διπλάσιον γὰρ ἐστὶν αὐτοῦ; ἀσύμμετρα >άρα ἐστὶ τὰ ἀπὸ τῶν $E\Delta$, $H\Theta$ τῶ| διξ ὑπὸ τῶν $E\Delta$, $H\Theta$; καὶ συναμφοτέρω >άρα τὰ τε ἀπὸ τῶν $E\Delta$, $H\Theta$ καὶ τὸ διξ ὑπὸ τῶν $E\Delta$, $H\Theta$, <όπερ ἐστὶ τὸ ἀπὸ τῆς $E\Theta$, ἀσύμμετρόν ἐστὶ τοῖξ ἀπὸ τῶν $E\Delta$, $H\Theta$. <ρητὰ δὲ τὰ ἀπὸ τῶν $E\Delta$, $H\Theta$; >άλογον >άρα τὸ ἀπὸ τῆς $E\Theta$. >άλογοξ >άρα ἐστὶν ἢ $E\Theta$. ἀλλὰ καὶ <ρηρὴ; <όπερ ἐστὶν ἀδύνατον.

Μέσον >άρα μέσου οὐθ ὑπερέθει <ρητῶ|; <όπερ >έδει δεῖχαι.

For, if possible, let the medial (area) AB exceed the medial (area) AC by the rational (area) DB . And let the rational (straight-line) EF be laid down. And let the rectangular parallelogram FH , equal to AB , be applied to to EF , producing EH as breadth. And let FG , equal to AC , be cut off (from FH). Thus, the remainder BD is equal to the remainder KH . And DB is rational. Thus, KH is also rational. Therefore, since AB and AC are each medial, and AB is equal to FH , and AC to FG , FH and FG are thus each also medial. And they are applied to the rational (straight-line) EF . Thus, HE and EG are each rational, and incommensurable in length with EF [Prop. 10.22]. And since DB is rational, and is equal to KH , KH is thus also rational. And (KH) is applied to the rational (straight-line) EF . GH is thus rational, and commensurable in length with EF [Prop. 10.20]. But, EG is also rational, and incommensurable in length with EF . Thus, EG is incommensurable in length with GH [Prop. 10.13]. And as EG is to GH , so the (square) on EG (is) to the (rectangle contained) by EG and GH [Prop. 10.13 lem.]. Thus, the (square) on EG is incommensurable with the (rectangle contained) by EG and GH [Prop. 10.11]. But, the (sum of the) squares on EG and GH is commensurable with the (square) on EG . For (EG and GH are) both rational. And twice the (rectangle contained) by EG and GH is commensurable with the (rectangle contained) by EG and GH [Prop. 10.6]. For (the former) is double the latter. Thus, the (sum of the squares) on EG and GH is incommensurable with twice the (rectangle contained) by EG and GH [Prop. 10.13]. And thus the sum of the (squares) on EG and GH plus twice the (rectangle contained) by EG and GH , that is the (square) on EH [Prop. 2.4], is incommensurable with the (sum of the

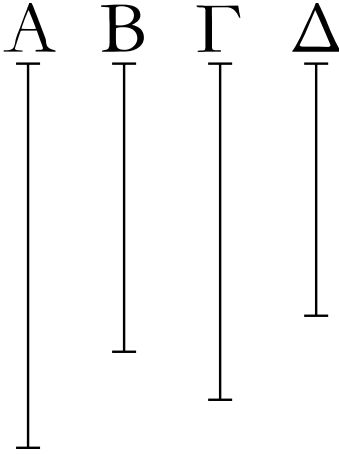
squares) on EG and GH [Prop. 10.16]. And the (sum of the squares) on EG and GH (is) rational. Thus, the (square) on EH is irrational [Def. 10.4]. Thus, EH is irrational [Def. 10.4]. But, (it is) also rational. The very thing is impossible.

Thus, a medial (area) does not exceed a medial (area) by a rational (area). (Which is) the very thing it was required to show.

† In other words, $\sqrt{k} - \sqrt{k'} \neq k''$.

27.

Μέσαζ εὑρεῖν δυνάμει μόνον συμμετρουνξ <ρητὸν περιεθούσαζ> To find (two) medial (straight-lines), containing a rational (area), (which are) commensurable in square only.

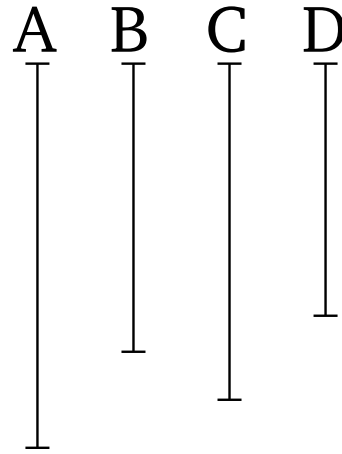


Ἐκκεῖσθωσαν δύο <ρηται δυνάμει μόνον σύμμετροι αἱ A , B , καὶ εἰλήφθω τῶν A , B μέση ἀνάλογον ἡ Γ , καὶ γεγονέτω ὥς ἡ A πρὸς τὴν B , οὕτως ἡ Γ πρὸς τὴν Δ .

Καὶ ἐπεὶ αἱ A , B <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι, τὸ >άρα ὑπὸ τῶν A , B , τουτέστι τὸ ἀπὸ τῆς Γ , μέσον ἐστίν. μέση >άρα ἡ Γ . καὶ ἐπεὶ ἐστὶν ὥς ἡ A πρὸς τὴν B , [οὕτως] ἡ Γ πρὸς τὴν Δ , αἱ δὲ A , B δυνάμει μόνον [εἰσὶ] σύμμετροι, καὶ αἱ Γ , Δ >άρα δυνάμει μόνον εἰσὶ σύμμετροι. καὶ ἐστὶ μέση ἡ Γ ; μέση >άρα καὶ ἡ Δ . αἱ Γ , Δ >άρα μέσαι εἰσι δυνάμει μόνον σύμμετροι. λέγω, <ὅτι καὶ <ρητὸν περιέθουσιν. ἐπεὶ γάρ ἐστὶν ὥς ἡ A πρὸς τὴν B , οὕτως ἡ Γ πρὸς τὴν Δ , ἐναλλάξ >άρα ἐστὶν ὥς ἡ A πρὸς τὴν Γ , ἡ B πρὸς τὴν Δ . ἀλλ' ὥς ἡ A πρὸς τὴν Γ , ἡ Γ πρὸς τὴν B ; καὶ ὥς >άρα ἡ Γ πρὸς τὴν B , οὕτως ἡ B πρὸς τὴν Δ ; τὸ >άρα ὑπὸ τῶν Γ , Δ >ῖσον ἐστὶ τῷ ἀπὸ τῆς B . <ρητὸν δὲ τὸ ἀπὸ τῆς B ; <ρητὸν >άρα [ἐστὶ] καὶ τὸ ὑπὸ τῶν Γ , Δ .

Εὐρηνται >άρα μέσαι δυνάμει μόνον σύμμετροι <ρητὸν περιέθουσαι; <όπερ >έδει δεῖχαι.

Proposition 27



Let the two rational (straight-lines) A and B , (which are) commensurable in square only, be laid down. And let C —the mean proportional (straight-line) to A and B —be taken [Prop. 6.13]. And let it be contrived that as A (is) to B , so C (is) to D [Prop. 6.12].

And since the rational (straight-lines) A and B are commensurable in square only, the (rectangle contained) by A and B —that is to say, the (square) on C [Prop. 6.17]—is thus medial [Prop. 10.21]. Thus, C is medial [Prop. 10.21]. And since as A is to B , [so] C (is) to D , and A and B [are] commensurable in square only, C and D are thus also commensurable in square only [Prop. 10.11]. And C is medial. Thus, D is also medial [Prop. 10.23]. Thus, C and D are medial (straight-lines which are) commensurable in square only. I say that they also contain a rational (area). For since as A is to B , so C (is) to D , thus, alternately, as A is to C , so B (is) to D [Prop. 5.16]. But, as A (is) to C , (so) C (is) to B . And thus as C (is) to B , so B (is) to D [Prop. 5.11]. Thus, the (rectangle contained) by C and D is equal to the (square) on B [Prop. 6.17]. And the (square) on B (is) rational. Thus, the (rectangle contained) by C and D [is] also rational.

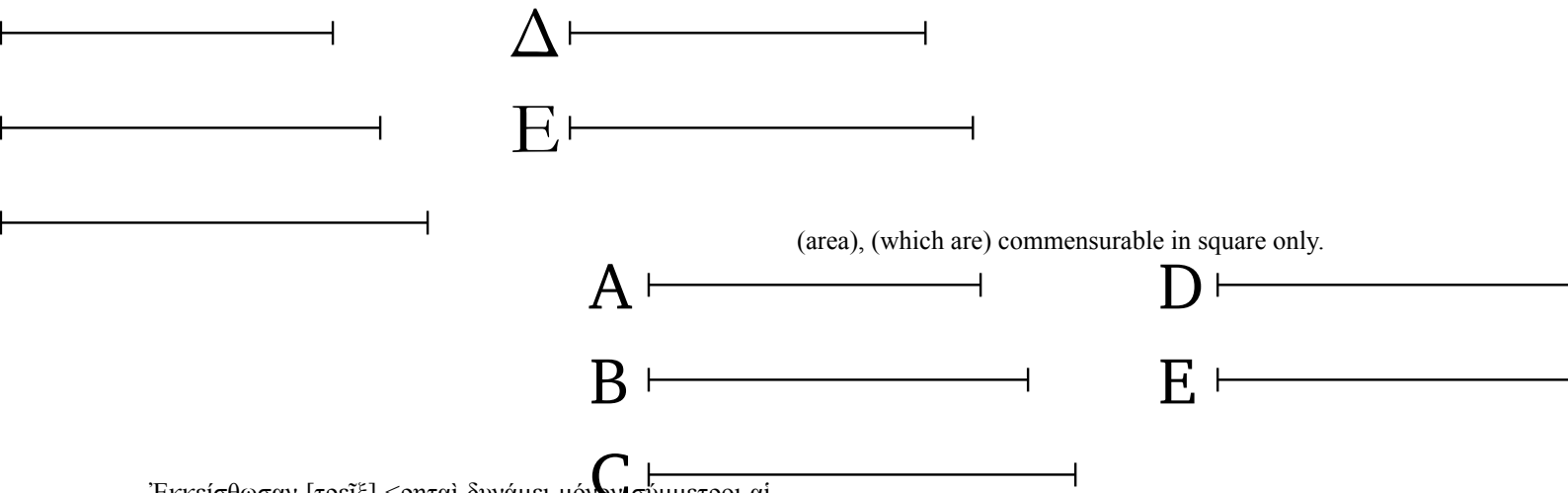
Thus, (two) medial (straight-lines, C and D), containing a rational (area), (which are) commensurable in square only, be found.† (Which is) the very thing it was required to show.

† C and D have lengths $k^{1/4}$ and $k^{3/4}$ times that of A , respectively, where the length of B is $k^{1/2}$ times that of A .

28.

Μέσαζ εὑρεῖν δυνάμει μόνον συμμετρουνξ μέσον περιεθούσαζ> To find (two) medial (straight-lines), containing a medial

Proposition 28



Ἐκκείσθωσαν [τρεῖς] <ρηταὶ δυνάμει μόνον σύμμετροι αἱ A, B, Γ , καὶ εἰλήφθω τῶν A, B μέση ἀνάλογον ἡ Δ , καὶ γεγονέτω ὥς ἡ B πρὸς τὴν Γ , ἡ Δ πρὸς τὴν E . Ἐπεὶ αἱ A, B <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι, τὸ >άρα ὑπὸ τῶν A, B , τουτέστι τὸ ἀπὸ τῆς Δ , μέσον ἐστίν. μέση >άρα ἡ Δ . καὶ ἐπεὶ αἱ B, Γ δυνάμει μόνον εἰσι σύμμετροι, καὶ ἐστὶν ὥς ἡ B πρὸς τὴν Γ , ἡ Δ πρὸς τὴν E , καὶ αἱ Δ, E >άρα δυνάμει μόνον εἰσι σύμμετροι. μέση δὲ ἡ Δ ; μέση >άρα καὶ ἡ E ; αἱ Δ, E >άρα μέσαι εἰσι δυνάμει μόνον σύμμετροι. λέγω δὴ, <ὅτι καὶ μέσον περιέθουσιν. ἐπεὶ γάρ ἐστὶν ὥς ἡ B πρὸς τὴν Γ , ἡ Δ πρὸς τὴν E , ἐναλλάξ >άρα ὥς ἡ B πρὸς τὴν Δ , ἡ Γ πρὸς τὴν E . ὥς δὲ ἡ B πρὸς τὴν Δ , ἡ Δ πρὸς τὴν A ; καὶ ὥς <άρα ἡ Δ πρὸς τὴν A , ἡ Γ πρὸς τὴν E ; τὸ >άρα ὑπὸ τῶν A, Γ >ῖσον ἐστὶ τῷ ὑπὸ τῶν Δ, E . μέσον δὲ τὸ ὑπὸ τῶν A, Γ ; μέσον >άρα καὶ τὸ ὑπὸ τῶν Δ, E . Ἐ<ύρηνται >άρα μέσαι δυνάμει μόνον σύμμετροι μέσον περιέθουσας; <ὅπερ >έδει δεῖξαι.

Let the [three] rational (straight-lines) A, B , and C , (which are) commensurable in square only, be laid down. And let, D , the mean proportional (straight-line) to A and B , be taken [Prop. 6.13]. And let it be contrived that as B (is) to C , (so) D (is) to E [Prop. 6.12].

Since the rational (straight-lines) A and B are commensurable in square only, the (rectangle contained) by A and B —that is to say, the (square) on D [Prop. 6.17]—is medial [Prop. 10.21]. Thus, D (is) medial [Prop. 10.21]. And since B and C are commensurable in square only, and as B is to C , (so) D (is) to E , D and E are thus commensurable in square only [Prop. 10.11]. And D (is) medial. E (is) thus also medial [Prop. 10.23]. Thus, D and E are medial (straight-lines which are) commensurable in square only. So, I say that they also enclose a medial (area). For since as B is to C , (so) D (is) to E , thus, alternately, as B (is) to D , (so) C (is) to E [Prop. 5.16]. And as B (is) to D , (so) D (is) to A . And thus as D (is) to A , (so) C (is) to E . Thus, the (rectangle contained) by A and C is equal to the (rectangle contained) by D and E [Prop. 6.16]. And the (rectangle contained) by A and C is medial [Prop. 10.21]. Thus, the (rectangle contained) by D and E (is) also medial. Thus, (two) medial (straight-lines, D and E), containing a medial (area), (which are) commensurable in square only, be found. (Which is) the very thing it was required to show.

† D and E have lengths $k^{1/4}$ and $k^{1/2}/k^{1/4}$ times that of A , respectively, where the lengths of B and C are $k^{1/2}$ and $k^{1/2}$ times that of A , respectively.

Λήμμα 1.

Lemma I

Εὑρεῖν δύο τετραγώνουζ ἀριθμούς, <ώστε καὶ τὸν συγκείμενον> To find two square numbers such that the sum of them is also square.



Ἐκκείσθωσαν δύο ἀριθμοὶ οἱ $AB, B\Gamma$, >έστωσαν δὲ >ήτοι >ἀρτιοὶ >ἢ περιττοί. καὶ ἐπεὶ, ἔαν τε ἀπὸ ἀρτίου >ἀρτιόξ ἀφαιρεθῇ, ἔαν τε ἀπὸ περισσοῦ περισσόξ, ὁ λοιπὸξ >ἀρτιόξ ἐστίν, ὁ λοιπὸξ >άρα ὁ $A\Gamma$ >ἀρτιόξ ἐστίν. τεμῆσθω ὁ $A\Gamma$ δίθα κατὰ τὸ Δ . >έστωσαν δὲ καὶ οἱ $AB, B\Gamma$ >ήτοι <όμοιοι ἐπίπεδοι >ἢ τετραγώνοι, ο<ὶ καὶ αὐτοὶ <όμοιοι εἰσιν ἐπίπεδοι; ὁ >άρα ἐκ τῶν $AB, B\Gamma$ μετὰ τοῦ ἀπὸ [τοῦ] $\Gamma\Delta$ τετραγώνου >ῖσοξ ἐστὶ τῷ ἀπὸ τοῦ $B\Delta$ τετραγώνου. καὶ ἐστὶ τετράγωνοξ ὁ ἐκ τῶν $AB, B\Gamma$, ἐπειδήπερ ἐδείχθη, <ὅτι, ἔαν δύο <όμοιοι ἐπίπεδοι πολλαπλασιάσαντεξ ἀλλήλουξ ποιῶσι

Let the two numbers AB and BC be laid down. And let them be either (both) even or (both) odd. And since, if an even (number) is subtracted from an even (number), or if an odd (number) is subtracted from an odd (number), then the remainder is even [Props. 9.24, 9.26], the remainder AC is thus even. Let AC be cut in half at D . And let AB and BC also be either similar plane (numbers), or square (numbers)—which are themselves also similar plane (numbers). Thus, the (number created) from (multiplying) AB and BC , plus the square on CD , is equal to the square on

τινα, ὁ γενόμενος τετράγωνός ἐστιν. ἐκύρηνται ἄρα δύο τετράγωνοι ἀριθμοὶ <ὅ τε ἐκ τῶν AB , $BΓ$ καὶ ὁ ἀπὸ τοῦ $ΓΔ$, οὐκ ἔστιν ἀντιστοιχῶν ποιοῦσι τὸν ἀπὸ τοῦ $ΒΔ$ τετράγωνον.

Καὶ φανερόν, <ὅτι ἐκύρηνται πάλιν δύο τετράγωνοι <ὅ τε ἀπὸ τοῦ $ΒΔ$ καὶ ὁ ἀπὸ τοῦ $ΓΔ$, <ὥστε τὴν ὑπεροθὴν αὐτῶν τὸν ὑπὸ AB , $BΓ$ εἶναι τετράγωνον, <ὅταν οἱ AB , $BΓ$ ὁμοιοὶ >ῶσιν ἐπίπεδοι. <ὅταν δὲ μὴ >ῶσιν ὁμοιοὶ ἐπίπεδοι, ἐκύρηνται δύο τετράγωνοι <ὅ τε ἀπὸ τοῦ $ΒΔ$ καὶ ὁ ἀπὸ τοῦ $ΔΓ$, <ὣν ἡ ὑπεροθὴ ὁ ὑπὸ τῶν AB , $BΓ$ οὐκ >ἐστι τετράγωνος; <ὅπερ >εἶδει δεῖχαι.

BD [Prop. 2.6]. And the (number created) from (multiplying) AB and BC is square—inasmuch as it was shown that if two similar plane (numbers) make some (number) by multiplying one another then the (number so) created is square [Prop. 9.1]. Thus, two square numbers be found—(namely,) the (number created) from (multiplying) AB and BC , and the (square) on CD —which, (when) added (together), make the square on BD .

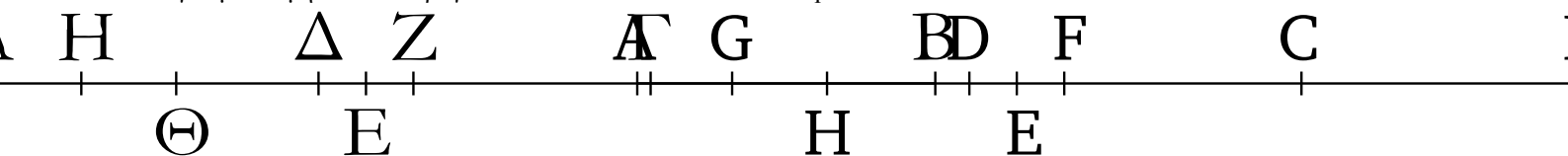
And (it is) clear that two square (numbers) have again been found—(namely,) the (square) on BD , and the (square) on CD —such that their difference—(namely,) the (rectangle) contained by AB and BC —is square whenever AB and BC are similar plane (numbers). But, when they are not similar plane numbers, two square (numbers) be found—(namely,) the (square) on BD , and the (square) on DC —between which the difference—(namely,) the (rectangle) contained by AB and BC —is not square. (Which is) the very thing it was required to show.

Λήμμα 2.

Εὐρεῖν δύο τετραγώνους ἀριθμοὺς, <ὥστε τὸν ἐκ αὐτῶν συγκείμενον μὴ εἶναι τετράγωνον.

Lemma II

To find two square numbers such that the sum of them is not square.



>Ἐστω γὰρ ὁ ἐκ τῶν AB , $BΓ$, ὡς >έφαμεν, τετράγωνος, καὶ >ἀρτιος ὁ $ΓΑ$, καὶ τεμήσθω ὁ $ΓΑ$ δίῃ τῷ $Δ$. φανερόν δὲ, <ὅτι ὁ ἐκ τῶν AB , $BΓ$ τετράγωνος μετὰ τοῦ ἀπὸ [τοῦ] $ΓΔ$ τετραγώνου >ῖσοξ ἐστὶ τῷ ἀπὸ [τοῦ] $ΒΔ$ τετραγώνῳ. ἀφηρήσθω μονάξ ἡ $ΔΕ$; ὁ >ἄρα ἐκ τῶν AB , $BΓ$ μετὰ τοῦ ἀπὸ [τοῦ] $ΓΕ$ ἐλάσσων ἐστὶ τοῦ ἀπὸ [τοῦ] $ΒΔ$ τετραγώνου. λέγω οὖν, <ὅτι ὁ ἐκ τῶν AB , $BΓ$ τετράγωνος μετὰ τοῦ ἀπὸ [τοῦ] $ΓΕ$ οὐκ >ἐστὶ τετράγωνος.

Εἰ γὰρ >ἐστὶ τετράγωνος, >ἦτοι >ῖσοξ ἐστὶ τῷ ἀπὸ [τοῦ] BE >ἢ ἐλάσσων τοῦ ἀπὸ [τοῦ] BE , οὐκέτι δὲ καὶ μείζων, <ἵνα μὴ τμηθῇ ἡ μονάξ. >ἔστω, εἰ δυνατόν, πρότερον ὁ ἐκ τῶν AB , $BΓ$ μετὰ τοῦ ἀπὸ $ΓΕ$ >ῖσοξ τῷ ἀπὸ BE , καὶ >ἔστω τῇ $ΔΕ$ μονάδος διπλασίῳ ὁ HA . ἐπεὶ οὖν <ὅλος ὁ $ΑΓ$ <όλου τοῦ $ΓΔ$ ἐστὶ διπλασίῳ, <ὣν ὁ $ΑΗ$ τοῦ $ΔΕ$ ἐστὶ διπλασίῳ, καὶ λοιπὸς >ἄρα ὁ $ΗΓ$ λοιποῦ τοῦ $ΕΓ$ ἐστὶ διπλασίῳ; δίῃ >ἄρα τέτμηται ὁ $ΗΓ$ τῷ $Ε$. ὁ >ἄρα ἐκ τῶν HB , $BΓ$ μετὰ τοῦ ἀπὸ $ΓΕ$ >ῖσοξ ἐστὶ τῷ ἀπὸ BE τετραγώνῳ. ἀλλὰ καὶ ὁ ἐκ τῶν AB , $BΓ$ μετὰ τοῦ ἀπὸ $ΓΕ$ >ῖσοξ ὑπόκειται τῷ ἀπὸ [τοῦ] BE τετραγώνῳ; ὁ >ἄρα ἐκ τῶν HB , $BΓ$ μετὰ τοῦ ἀπὸ $ΓΕ$ >ῖσοξ ἐστὶ τῷ ἐκ τῶν AB , $BΓ$ μετὰ τοῦ ἀπὸ $ΓΕ$. καὶ κοινοῦ ἀφαιρεθέντος τοῦ ἀπὸ $ΓΕ$ συνάγεται ὁ AB >ῖσοξ τῷ HB ; <ὅπερ >άτοπον. οὐκ >ἄρα ὁ ἐκ τῶν AB , $BΓ$ μετὰ τοῦ ἀπὸ [τοῦ] $ΓΕ$ >ῖσοξ ἐστὶ τῷ ἀπὸ BE . λέγω δὲ, <ὅτι οὐδὲ ἐλάσσων τοῦ ἀπὸ BE . εἰ γὰρ δυνατόν, >ἔστω τῷ ἀπὸ BZ >ῖσοξ, καὶ τοῦ $ΔΖ$ διπλασίῳ ὁ $ΘΑ$. καὶ συναθροίσεται πάλιν διπλασίῳ ὁ $ΘΓ$ τοῦ $ΓΖ$; <ὥστε καὶ τὸν $ΓΘ$ δίῃ τεμήσθαι κατὰ τὸ Z , καὶ διὰ τοῦτο τὸν ἐκ τῶν $ΘB$, $BΓ$ μετὰ τοῦ ἀπὸ $ZΓ$ >ῖσον γίνεσθαι τῷ ἀπὸ BZ . ὑπόκειται δὲ καὶ ὁ ἐκ τῶν AB , $BΓ$ μετὰ τοῦ ἀπὸ $ΓΕ$ >ῖσοξ τῷ ἀπὸ BZ . <ὥστε καὶ ὁ ἐκ τῶν $ΘB$, $BΓ$ μετὰ τοῦ ἀπὸ $ΓΖ$ >ῖσοξ >ἐστὶ τῷ ἐκ τῶν AB , $BΓ$ μετὰ τοῦ ἀπὸ $ΓΕ$; <ὅπερ >άτοπον. οὐκ >ἄρα ὁ ἐκ τῶν AB , $BΓ$ μετὰ τοῦ ἀπὸ $ΓΕ$ >ῖσοξ ἐστὶ [τῷ] ἐλάσσωνι τοῦ ἀπὸ BE . ἐδείκθη δέ, <ὅτι οὐδὲ [αὐτῷ] τῷ ἀπὸ BE . οὐκ >ἄρα ὁ

For let the (number created) from (multiplying) AB and BC , as we said, be square. And (let) CA (be) even. And let CA be cut in half at D . So it is clear that the square (number created) from (multiplying) AB and BC , plus the square on CD , is equal to the square on BD [see previous lemma]. Let the unit DE be subtracted (from BD). Thus, the (number created) from (multiplying) AB and BC , plus the (square) on CE , is less than the square on BD . I say, therefore, that the square (number created) from (multiplying) AB and BC , plus the (square) on CE , is not square. For if it is square, it is either equal to the (square) on BE , or less than the (square) on BE , but cannot any more be greater (than the square on BE), lest the unit be divided. First of all, if possible, let the (number created) from (multiplying) AB and BC , plus the (square) on CE , be equal to the (square) on BE . And let GA be double the unit DE . Therefore, since the whole of AC is double the whole of CD , of which AG is double DE , the remainder GC is thus double the remainder EC . Thus, GC has been cut in half at E . Thus, the (number created) from (multiplying) GB and BC , plus the (square) on CE , is equal to the square on BE [Prop. 2.6]. But, the (number created) from (multiplying) AB and BC , plus the (square) on CE , was also assumed (to be) equal to the square on BE . Thus, the (number created) from (multiplying) GB and BC , plus the (square) on CE , is equal to the (number created) from (multiplying) AB and BC , plus the (square) on CE . And subtracting the (square) on CE from both, AB is inferred (to be) equal to GB . The very thing is absurd. Thus, the (number created) from (multiplying) AB and BC , plus the (square) on CE , is not equal to the (square) on BE . So I say that (it is) not less than the (square) on BE either. For, if possible, let it be equal to the (square) on BF . And (let) HA (be) double DF . And

ἐκ τῶν AB , $BΓ$ μετὰ τοῦ ἀπὸ $ΓΕ$ τετράγωνός ἐστιν. <ὅπερ >έδει δεῖξαι.

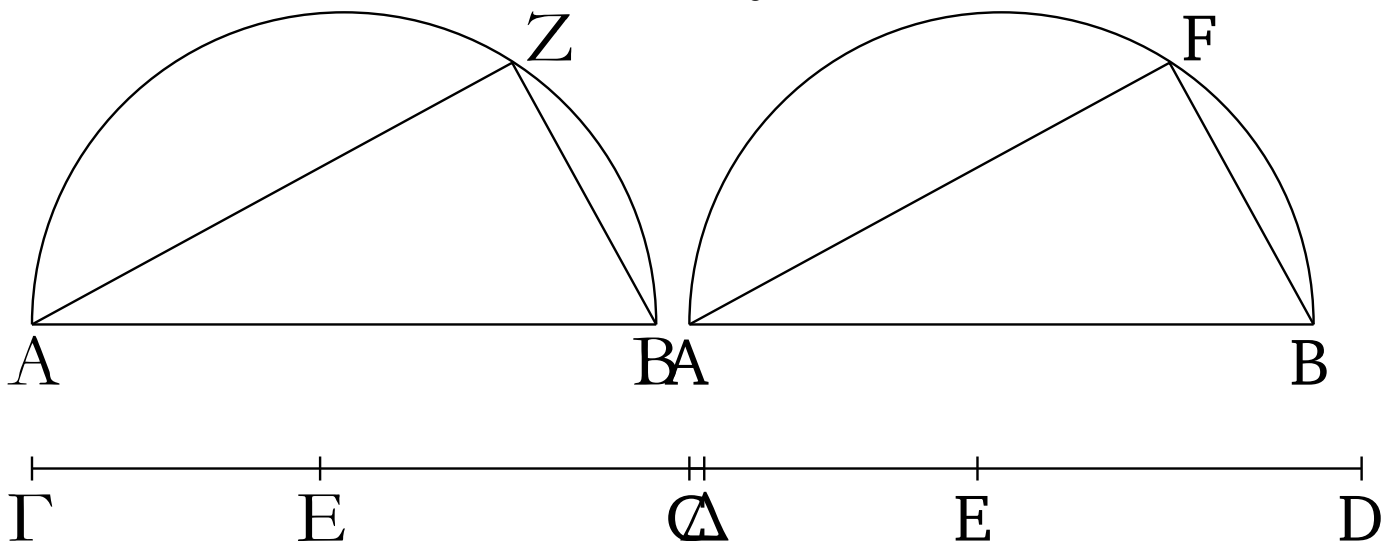
it can again be inferred that HC (is) double CF . Hence, CH has also been cut in half at F . And, on account of this, the (number created) from (multiplying) HB and BC , plus the (square) on FC , becomes equal to the (square) on BF [Prop. 2.6]. And the (number created) from (multiplying) AB and BC , plus the (square) on CE , was also assumed (to be) equal to the (square) on BF . Hence, the (number created) from (multiplying) HB and BC , plus the (square) on CF , will also be equal to the (number created) from (multiplying) AB and BC , plus the (square) on CE . The very thing is absurd. Thus, the (number created) from (multiplying) AB and BC , plus the (square) on CE , is not equal to less than the (square) on BE . And it was shown that (is it) not equal to the (square) on BE either. Thus, the (number created) from (multiplying) AB and BC , plus the square on CE , is not square. (Which is) the very thing it was required to show.

29.

Εὐρεῖν δύο <ρητὰς δυνάμει μόνον συμμετρουν>, <ώστε τὴν μείζονα τῆς ἐλάσσονος μείζον δύνασθαι τῷ ἀπὸ συμμετρου ἐαυτῇ> μήκει.

Proposition 29

To find two rational (straight-lines which are) commensurable in square only, such that the square on the greater is larger than the (square on the) lesser by the (square) on (some straight-line which is) commensurable in length with the greater.



Ἐκκεῖσθω γάρ τις <ρητὴ ἢ AB καὶ δύο τετράγωνοι ἀριθμοὶ οἱ $ΓΔ$, $ΔΕ$, <ώστε τὴν ὑπεροθὴν αὐτῶν τὸν $ΓΕ$ μή ε>ῖναι τετράγωνον, καὶ γεγράφθω ἐπὶ τῆς AB ἡμικύκλιον τὸ AZB , καὶ πεποιήσθω ὥς ὁ $ΔΓ$ πρὸς τὸν $ΓΕ$, ο<ύτωξ τὸ ἀπὸ τῆς BA τετράγωνον πρὸς τὸ ἀπὸ τῆς AZ τετράγωνον, καὶ ἐπεζεύθθω ἡ ZB .

Ἐπεὶ [ο>ύν] ἐστὶν ὥς τὸ ἀπὸ τῆς BA πρὸς τὸ ἀπὸ τῆς AZ , ο<ύτωξ ὁ $ΔΓ$ πρὸς τὸν $ΓΕ$, τὸ ἀπὸ τῆς BA >άρα πρὸς τὸ ἀπὸ τῆς AZ λόγον >έθει, <ὄν ἀριθμὸς ὁ $ΔΓ$ πρὸς ἀριθμὸν τὸν $ΓΕ$; σύμμετρον >άρα ἐστὶ τὸ ἀπὸ τῆς BA τῷ ἀπὸ τῆς AZ . <ρητὸν δὲ τὸ ἀπὸ τῆς AB ; <ρητὸν >άρα καὶ τὸ ἀπὸ τῆς AZ ; <ρητὴ >άρα καὶ ἡ AZ . καὶ ἐπεὶ ὁ $ΔΓ$ πρὸς τὸν $ΓΕ$ λόγον οὐκ >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν, οὐδὲ τὸ ἀπὸ τῆς BA >άρα πρὸς τὸ ἀπὸ τῆς AZ λόγον >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν; ἀσύμμετροξ >άρα ἐστὶν ἡ AB τῇ AZ μήκει; αἱ BA , AZ >άρα <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι. καὶ ἐπεὶ [ἐστὶν] ὥς ὁ $ΔΓ$ πρὸς τὸν $ΓΕ$, ο<ύτωξ τὸ ἀπὸ τῆς BA πρὸς τὸ ἀπὸ τῆς AZ , ἀναστρέψαντι >άρα ὥς ὁ $ΓΔ$ πρὸς τὸν $ΔΕ$,

For let some rational (straight-line) AB be laid down, and two square numbers, CD and DE , such that the difference between them, CE , is not square [Prop. 10.28 lem. I]. And let the semi-circle AFB be drawn on AB . And let it be contrived that as DC (is) to CE , so the square on BA (is) to the square on AF [Prop. 10.6 corr.]. And let FB be joined. [Therefore,] since as the (square) on BA is to the (square) on AF , so DC (is) to CE , the (square) on BA thus has to the (square) on AF the ratio which the number DC (has) to the number CE . Thus, the (square) on BA is commensurable with the (square) on AF [Prop. 10.6]. And the (square) on AB (is) rational [Def. 10.4]. Thus, the (square) on AF (is) also rational. Thus, AF (is) also rational. And since DC does not have to CE the ratio which (some) square number (has) to (some) square number, the (square) on BA thus does not have to the (square) on AF the ratio which (some) square number has to (some) square number either. Thus, AB is incommensurable in length with AF [Prop. 10.9]. Thus, the rational (straight-lines) BA and AF are commen-

οὐτως τὸ ἀπὸ τῆς AB πρὸς τὸ ἀπὸ τῆς BZ . ὁ δὲ $\Gamma\Delta$ πρὸς τὸν ΔE λόγον >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν; καὶ τὸ ἀπὸ τῆς AB >άρα πρὸς τὸ ἀπὸ τῆς BZ λόγον >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν; σύμμετρος >άρα ἐστὶν ἡ AB τῇ BZ μήκει. καὶ ἐστὶ τὸ ἀπὸ τῆς AB >ίσον τοῖς ἀπὸ τῶν AZ , ZB ; ἡ AB >άρα τῆς AZ μείζον δύναται τῇ BZ συμμέτρῳ ἑαυτῇ].

Εὐρίηται >άρα δύο <ρηταὶ δυνάμει μόνον σύμμετροι αἱ BA , AZ , <ώστε τὴν μείζονα τὴν AB τῆς ἐλάσσονος τῆς AZ μείζον δύνασθαι τῷ ἀπὸ τῆς BZ συμμέτρου ἑαυτῇ] μήκει, <όπερ >έδει δεῖχαι.

surable in square only. And since as DC [is] to CE , so the (square) on BA (is) to the (square) on AF , thus, via conversion, as CD (is) to DE , so the (square) on AB (is) to the (square) on BF [Props. 5.19 corr., 3.31, 1.47]. And CD has to DE the ratio which (some) square number (has) to (some) square number. Thus, the (square) on AB also has to the (square) on BF the ratio which (some) square number has to (some) square number. AB is thus commensurable in length with BF [Prop. 10.9]. And the (square) on AB is equal to the (sum of the squares) on AF and FB [Prop. 1.47]. Thus, the square on AB is greater than (the square on) AF by (the square on) BF , (which is) commensurable (in length) with (AB).

Thus, two rational (straight-lines), BA and AF , commensurable in square only, be found such that the square on the greater, AB , is larger than (the square on) the lesser, AF , by the (square) on BF , (which is) commensurable in length with (AB).[†] (Which is) the very thing it was required to show.

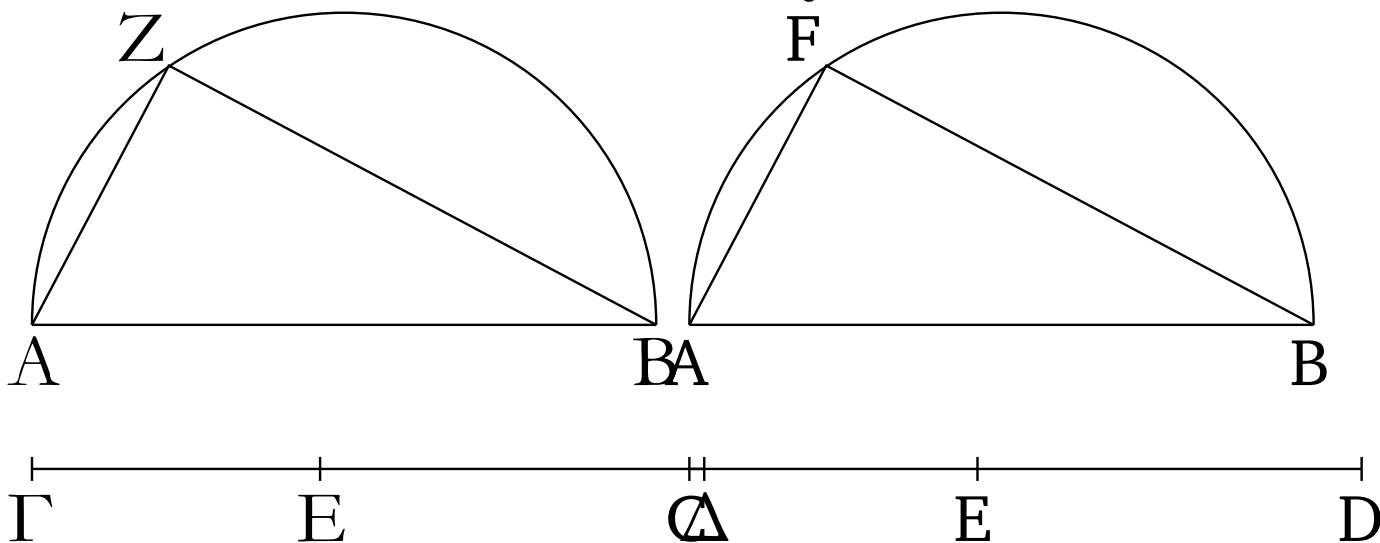
[†] BA and AF have lengths 1 and $\sqrt{1-k^2}$ times that of AB , respectively, where $k = \sqrt{DE/CD}$.

30.

Εὐρεῖν δύο <ρητὰς δυνάμει μόνον συμμέτρους, <ώστε τὴν μείζονα τῆς ἐλάσσονος μείζον δύνασθαι τῷ ἀπὸ ἀσύμμετρου ἑαυτῇ] μήκει.

Proposition 30

To find two rational (straight-lines which are) commensurable in square only, such that the square on the greater is larger than the (the square on) lesser by the (square) on (some straight-line which is) incommensurable in length with the greater.



Ἐκκεῖσθω <ρητὴ ἡ AB καὶ δύο τετράγωνοι ἀριθμοὶ οἱ ΓE , $E\Delta$, <ώστε τὸν συγκείμενον ἔχ ὠτῶν τὸν $\Gamma\Delta$ μὴ εἶναι τετράγωνον, καὶ γεγράφθω ἐπὶ τῆς AB ἡμικύκλιον τὸ AZB , καὶ πεποιήσθω ὥς ὁ $\Delta\Gamma$ πρὸς τὸν ΓE , οὐτως τὸ ἀπὸ τῆς BA πρὸς τὸ ἀπὸ τῆς AZ , καὶ ἐπεζεύθθω ἡ ZB .

Ὅμοιως δὲ δείχομεν τῷ πρὸς τοῦτον, <ὅτι αἱ BA , AZ <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι. καὶ ἐπεὶ ἐστὶν ὥς ὁ $\Delta\Gamma$ πρὸς τὸν ΓE , οὐτως τὸ ἀπὸ τῆς BA πρὸς τὸ ἀπὸ τῆς AZ , ἀναστρέψαντι >άρα ὥς ὁ $\Gamma\Delta$ πρὸς τὸν ΔE , οὐτως τὸ ἀπὸ τῆς AB πρὸς τὸ ἀπὸ τῆς BZ . ὁ δὲ $\Gamma\Delta$ πρὸς τὸν ΔE λόγον οὐκ >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν; οὐδ' >άρα τὸ ἀπὸ τῆς AB πρὸς τὸ ἀπὸ τῆς BZ λόγον >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν; ἀσύμμετρος >άρα ἐστὶν ἡ AB τῇ BZ μήκει. καὶ δύναται ἡ AB τῆς AZ μείζον τῷ ἀπὸ τῆς ZB ἀσύμμετρου ἑαυτῇ].

Let the rational (straight-line) AB be laid out, and the two square numbers, CE and ED , such that the sum of them, CD , is not square [Prop. 10.28 lem. II]. And let the semicircle AFB be drawn on AB . And let it be contrived that as DC (is) to CE , so the (square) on BA (is) to the (square) on AF [Prop. 10.6 corr]. And let FB be joined.

So, similarly to the (proposition) before this, we can show that BA and AF are rational (straight-lines which are) commensurable in square only. And since as DC is to CE , so the (square) on BA (is) to the (square) on AF , thus, via conversion, as CD (is) to DE , so the (square) on AB (is) to the (square) on BF [Props. 5.19 corr., 3.31, 1.47]. And CD does not have to DE the ratio which (some) square number (has) to (some) square number. Thus, the (square) on AB does not have to the (square) on BF the ratio which (some)

Αἱ AB , AZ >άρα <ρηταί εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ AB τῆς AZ μείζον δύναται τῷ ἀπὸ τῆς ZB ἀσύμμετρου ἑαυτῇ μήκει; <όπερ >έδει δεῖξαι.

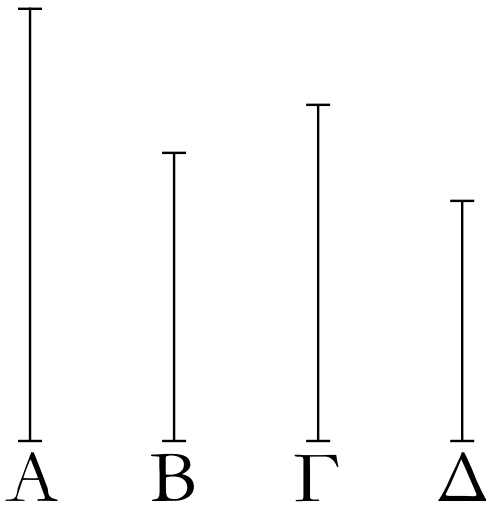
square number has to (some) square number either. Thus, AB is incommensurable in length with BF [Prop. 10.9]. And the square on AB is greater than the (square on) AF by the (square) on FB [Prop. 1.47], (which is) incommensurable (in length) with (AB) .

Thus, AB and AF are rational (straight-lines which are) commensurable in square only, and the square on AB is greater than (the square on) AF by the (square) on FB , (which is) incommensurable (in length) with (AB) .[†] (Which is) the very thing it was required to show.

[†] AB and AF have lengths 1 and $1/\sqrt{1+k^2}$ times that of AB , respectively, where $k = \sqrt{DE/CE}$.

31.

Εὐρεῖν δύο μέσαζ δυνάμει μόνον συμμέτρουζ <ρητὸν περιεθούσαζ, <ώστε τὴν μείζονα τῆς ἐλάσσονοζ μείζον δύνασθαι τῷ ἀπὸ συμμέτρου ἑαυτῇ μήκει.

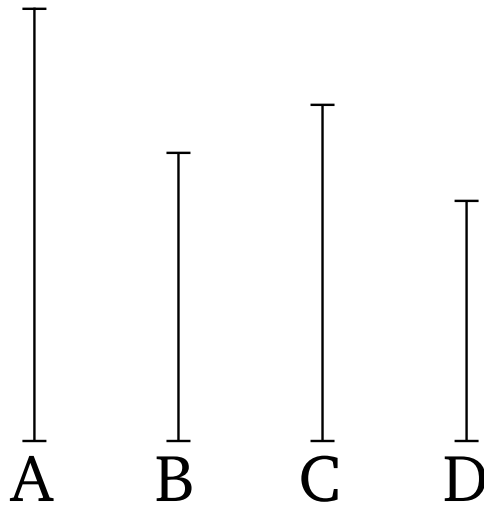


Ἐκκείσθωσαν δύο <ρηται δυνάμει μόνον σύμμετροι αἱ A , B , <ώστε τὴν A μείζονα ο>ῦσαν τῆς ἐλάσσονοζ τῆς B μείζον δύνασθαι τῷ ἀπὸ συμμέτρου ἑαυτῇ μήκει. καὶ τῷ ὑπὸ τῶν A , B >ίσον >έστω τὸ ἀπὸ τῆς Γ . μέσον δὲ τὸ ὑπὸ τῶν A , B ; μέσον >άρα καὶ τὸ ἀπὸ τῆς Γ ; μέση >άρα καὶ ἡ Γ . τῷ δὲ ἀπὸ τῆς B >ίσον >έστω τὸ ὑπὸ τῶν Γ , Δ ; <ρητὸν δὲ τὸ ἀπὸ τῆς B ; <ρητὸν >άρα καὶ τὸ ὑπὸ τῶν Γ , Δ . καὶ ἐπεὶ ἔστιν ὥς ἡ A πρὸς τὴν B , ο<ύτωξ τὸ ὑπὸ τῶν A , B πρὸς τὸ ἀπὸ τῆς B , ἀλλὰ τῷ μὲν ὑπὸ τῶν A , B >ίσον ἔστι τὸ ἀπὸ τῆς Γ , τῷ δὲ ἀπὸ τῆς B >ίσον τὸ ὑπὸ τῶν Γ , Δ , ὥς >άρα ἡ A πρὸς τὴν B , ο<ύτωξ τὸ ἀπὸ τῆς Γ πρὸς τὸ ὑπὸ τῶν Γ , Δ . ὥς δὲ τὸ ἀπὸ τῆς Γ πρὸς τὸ ὑπὸ τῶν Γ , Δ , ο<ύτωξ ἡ Γ πρὸς τὴν Δ ; καὶ ὥς >άρα ἡ A πρὸς τὴν B , ο<ύτωξ ἡ Γ πρὸς τὴν Δ . σύμμετροξ δὲ ἡ A τῇ B δυνάμει μόνον; σύμμετροξ >άρα καὶ ἡ Γ τῇ Δ δυνάμει μόνον. καὶ ἔστι μέση ἡ Γ ; μέση >άρα καὶ ἡ Δ . καὶ ἐπεὶ ἔστιν ὥς ἡ A πρὸς τὴν B , ἡ Γ πρὸς τὴν Δ , ἡ δὲ A τῆς B μείζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ, καὶ ἡ Γ >άρα τῆς Δ μείζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ. Ε<ύρηνται >άρα δύο μέσαι δυνάμει μόνον σύμμετροι αἱ Γ , Δ <ρητὸν περιέθουσαι, καὶ ἡ Γ τῆς Δ μείζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ μήκει.

Ὅμοίωξ δὴ δεῖθῇσεται καὶ τῷ ἀπὸ ἀσύμμετρου, <όταν ἡ A τῆς B μείζον δύνηται τῷ ἀπὸ ἀσύμμετρου ἑαυτῇ.

Proposition 31

To find two medial (straight-lines), commensurable in square only, (and) containing a rational (area), such that the square on the greater is larger than the (square on the) lesser by the (square) on (some straight-line) commensurable in length with the greater.



Let two rational (straight-lines), A and B , commensurable in square only, be laid out, such that the square on the greater A is larger than the (square on the) lesser B by the (square) on (some straight-line) commensurable in length with (A) [Prop. 10.29]. And let the (square) on C be equal to the (rectangle contained) by A and B . And the (rectangle contained) by A and B (is) medial [Prop. 10.21]. Thus, the (square) on C (is) also medial. Thus, C (is) also medial [Prop. 10.21]. And let the (rectangle contained) by C and D be equal to the (square) on B . And the (square) on B (is) rational. Thus, the (rectangle contained) by C and D (is) also rational. And since as A is to B , so the (rectangle contained) by A and B (is) to the (square) on B [Prop. 10.21 lem.], but the (square) on C is equal to the (rectangle contained) by A and B , and the (rectangle contained) by C and D to the (square) on B , thus as A (is) to B , so the (square) on C (is) to the (rectangle contained) by C and D . And as the (square) on C (is) to the (rectangle contained) by C and D , so C (is) to D [Prop. 10.21 lem.]. And thus as A (is) to B , so C (is) to D . And A is commensurable in square only with B . Thus, C (is) also commensurable in square only with D [Prop. 10.11]. And C is medial. Thus, D (is) also medial [Prop. 10.23]. And since as A is to B , (so) C (is) to D , and the square on A is greater than (the square on) B by the (square) on (some straight-line) commensurable (in

length) with (A) , the square on C is thus also greater than (the square on) D by the (square) on (some straight-line) commensurable (in length) with (C) [Prop. 10.14].

Thus, two medial (straight-lines), C and D , commensurable in square only, (and) containing a rational (area), be found. And the square on C is greater than (the square on) D by the (square) on (some straight-line) commensurable in length with (C) .[†]

So, similarly, (the proposition) can also be demonstrated for (some straight-line) incommensurable (in length with C), provided that the square on A is greater than (the square on B) by the (square) on (some straight-line) incommensurable (in length) with (A) [Prop. 10.30].[‡]

[†] C and D have lengths $(1 - k^2)^{1/4}$ and $(1 - k^2)^{3/4}$ times that of A , respectively, where k is defined in the footnote to Prop. 10.29.

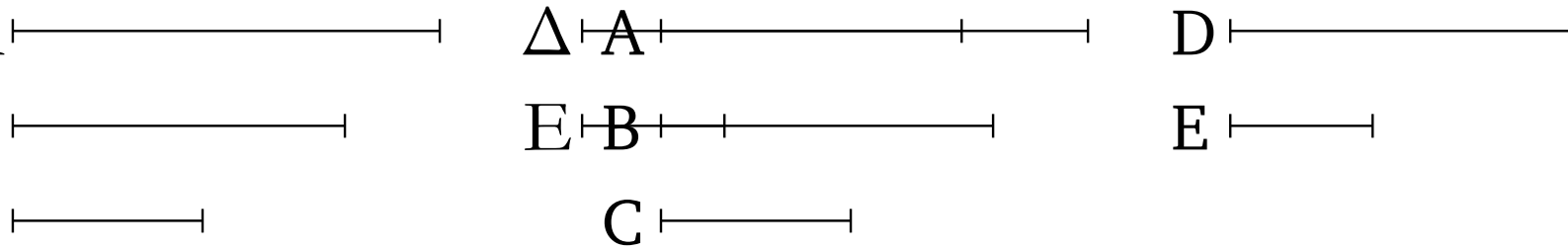
[‡] C and D would have lengths $1/(1 + k^2)^{1/4}$ and $1/(1 + k^2)^{3/4}$ times that of A , respectively, where k is defined in the footnote to Prop. 10.30.

32.

Proposition 32

Εὐρεῖν δύο μέσας δυνάμει μόνον συμμετρούς μέσον περιεθούσας, ὥστε τὴν μείζονα τῆς ἐλάσσονος μείζον δύνασθαι τῷ ἀπὸ συμμετροῦ ἑαυτῇ.

Find two medial (straight-lines), commensurable in square only, (and) containing a medial (area), such that the square on the greater is larger than the (square on the) lesser by the (square) on (some straight-line) commensurable (in length) with the greater.



Ἐκκεῖσθωσαν τρεῖς <ρηταὶ δυνάμει μόνον σύμμετροι αἱ A, B, Γ , <ώστε τὴν A τῆς Γ μείζον δύνασθαι τῷ ἀπὸ συμμετροῦ ἑαυτῇ], καὶ τῷ μὲν ὑπὸ τῶν A, B >ῖσον >έστω τὸ ἀπὸ τῆς Δ . μέσον >άρα τὸ ἀπὸ τῆς Δ ; καὶ ἡ Δ >άρα μέση ἐστίν. τῷ δὲ ὑπὸ τῶν B, Γ >ῖσον >έστω τὸ ὑπὸ τῶν Δ, E . καὶ ἐπεὶ ἐστὶν ὡς τὸ ὑπὸ τῶν A, B πρὸς τὸ ὑπὸ τῶν B, Γ , ο<ύτωξ ἡ A πρὸς τὴν Γ , ἀλλὰ τῷ μὲν ὑπὸ τῶν A, B >ῖσον ἐστὶ τὸ ἀπὸ τῆς Δ , τῷ δὲ ὑπὸ τῶν B, Γ >ῖσον τὸ ὑπὸ τῶν Δ, E , >έστιν >άρα ὡς ἡ A πρὸς τὴν Γ , ο<ύτωξ τὸ ἀπὸ τῆς Δ πρὸς τὸ ὑπὸ τῶν Δ, E . ὡς δὲ τὸ ἀπὸ τῆς Δ πρὸς τὸ ὑπὸ τῶν Δ, E , ο<ύτωξ ἡ Δ πρὸς τὴν E ; καὶ ὡς >άρα ἡ A πρὸς τὴν Γ , ο<ύτωξ ἡ Δ πρὸς τὴν E . σύμμετροξ δὲ ἡ A τῇ Γ δυνάμει [μόνον]. σύμμετροξ >άρα καὶ ἡ Δ τῇ E δυνάμει μόνον. μέση δὲ ἡ Δ ; μέση >άρα καὶ ἡ E . καὶ ἐπεὶ ἐστὶν ὡς ἡ A πρὸς τὴν Γ , ἡ Δ πρὸς τὴν E , ἡ δὲ A τῆς Γ μείζον δύνανται τῷ ἀπὸ συμμετροῦ ἑαυτῇ], καὶ ἡ Δ >άρα τῆς E μείζον δυνήσεται τῷ ἀπὸ συμμετροῦ ἑαυτῇ]. λέγω δὴ, <ότι καὶ μέσον ἐστὶ τὸ ὑπὸ τῶν Δ, E . ἐπεὶ γὰρ >ῖσον ἐστὶ τὸ ὑπὸ τῶν B, Γ τῷ ὑπὸ τῶν Δ, E , μέσον δὲ τὸ ὑπὸ τῶν B, Γ [αἱ γὰρ B, Γ <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι], μέσον >άρα καὶ τὸ ὑπὸ τῶν Δ, E . Ε<ύρηνται >άρα δύο μέσαι δυνάμει μόνον σύμμετροι αἱ Δ, E μέσον περιεθούσαι, <ώστε τὴν μείζονα τῆς ἐλάσσονος μείζον δύνασθαι τῷ ἀπὸ συμμετροῦ ἑαυτῇ]. Ὁμοίωξ δὴ πάλιν διεθθήσεται καὶ τῷ ἀπὸ ἀσύμμετρον, <όταν ἡ A τῆς Γ μείζον δύνῃται τῷ ἀπὸ ἀσύμμετρον ἑαυτῇ.

Let three rational (straight-lines), A, B and C , commensurable in square only, be laid out such that the square on A is greater than (the square on) C by the (square) on (some straight-line) commensurable (in length) with (A) [Prop. 10.29]. And let the (square) on D be equal to the (rectangle contained) by A and B . Thus, the (square) on D (is) medial. Thus, D is also medial [Prop. 10.21]. And let the (rectangle contained) by D and E be equal to the (rectangle contained) by B and C . And since as the (rectangle contained) by A and B is to the (rectangle contained) by B and C , so A (is) to C [Prop. 10.21 lem.], but the (square) on D is equal to the (rectangle contained) by A and B , and the (rectangle contained) by D and E to the (rectangle contained) by B and C , thus as A is to C , so the (square) on D (is) to the (rectangle contained) by D and E . And as the (square) on D (is) to the (rectangle contained) by D and E , so D (is) to E [Prop. 10.21 lem.]. And thus as A (is) to C , so D (is) to E . And A (is) commensurable in square [only] with C . Thus, D (is) also commensurable in square only with E [Prop. 10.11]. And D (is) medial. Thus, E (is) also medial [Prop. 10.23]. And since as A is to C , (so) D (is) to E , and the square on A is greater than (the square on) C by the (square) on (some straight-line) commensurable (in length) with (A) , the square on D will thus also be greater than (the square on) E by the (square) on (some straight-line) commensurable (in length) with (D) [Prop. 10.14]. So, I also say that the (rectangle contained) by D and E is medial. For since the (rectangle contained) by B and C is equal to the (rectangle contained) by D and E , and the (rectan-

gle contained) by B and C (is) medial [for B and C are rational (straight-lines which are) commensurable in square only] [Prop. 10.21], the (rectangle contained) by D and E (is) thus also medial.

Thus, two medial (straight-lines), D and E , commensurable in square only, (and) containing a medial (area), be found such that the square on the greater is larger than the (square on the) lesser by the (square) on (some straight-line) commensurable (in length) with the greater.[†]

So, similarly, (the proposition) can again also be demonstrated for (some straight-line) incommensurable (in length with the greater), provided that the square on A is greater than (the square on) C by the (square) on (some straight-line) incommensurable (in length) with (A) [Prop. 10.30].[‡]

[†] D and E have lengths $k^{1/4}$ and $k^{1/4}\sqrt{1-k^2}$ times that of A , respectively, where the length of B is $k^{1/2}$ times that of A , and k is defined in the footnote to Prop. 10.29.

[‡] D and E would have lengths $k^{1/4}$ and $k^{1/4}/\sqrt{1+k^2}$ times that of A , respectively, where the length of B is $k^{1/2}$ times that of A , and k is defined in the footnote to Prop. 10.30.

Λήμμα.

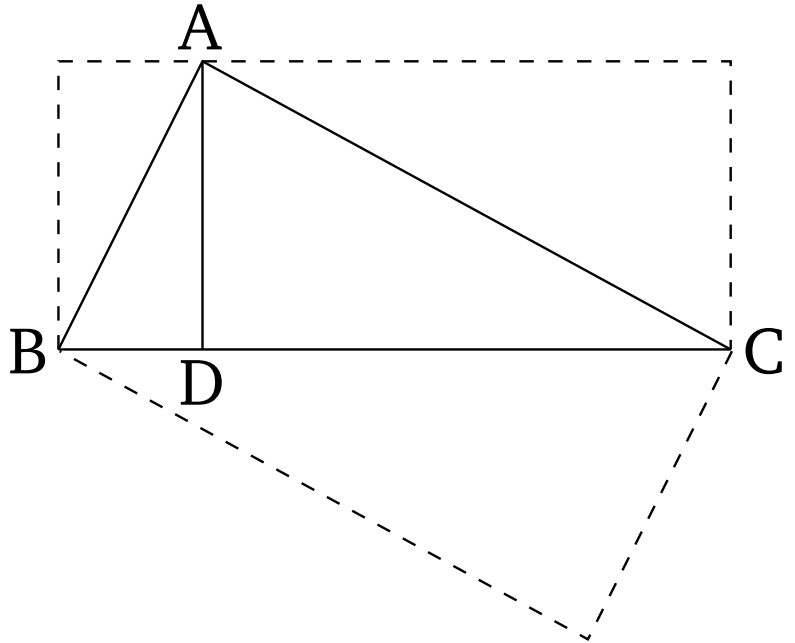
>Ἐστω τρίγωνον ὀρθογώνιον τὸ $AB\Gamma$ ὀρθὴν >έθον τὴν A , καὶ >ήθθω κάθετος ἡ AD ; λέγω, <ὅτι τὸ μὲν ὑπὸ τῶν $\Gamma B\Delta$ >ίσον ἐστὶ τῷ ὑπὸ τῆς BA , τὸ δὲ ὑπὸ τῶν $B\Gamma A$ >ίσον τῷ ὑπὸ τῆς ΓA , καὶ τὸ ὑπὸ τῶν $B\Delta$, $\Delta\Gamma$ >ίσον τῷ ὑπὸ τῆς AD , καὶ >έτι τὸ ὑπὸ τῶν $B\Gamma$, AD >ίσον [ἐστὶ] τῷ ὑπὸ τῶν BA , $A\Gamma$.

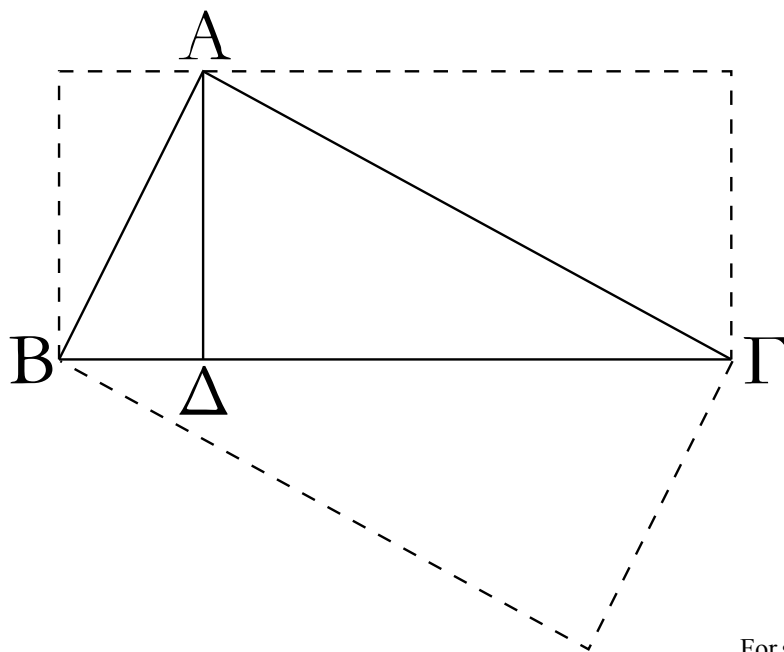
Καὶ πρῶτον, <ὅτι τὸ ὑπὸ τῶν $\Gamma B\Delta$ >ίσον [ἐστὶ] τῷ ὑπὸ τῆς BA .

Lemma

Let ABC be a right-angled triangle having the (angle) A a right-angle. And let the perpendicular AD be drawn. I say that the (rectangle contained) by CBD is equal to the (square) on BA , and the (rectangle contained) by BCD (is) equal to the (square) on CA , and the (rectangle contained) by BD and DC (is) equal to the (square) on AD , and, further, the (rectangle contained) by BC and AD [is] equal to the (rectangle contained) by BA and AC .

And, first of all, (let us prove) that the (rectangle contained) by CBD [is] equal to the (square) on BA .





Ἐπεὶ γὰρ ἐν ὀρθογωνίῳ τριγώνῳ ἀπὸ τῆς ὀρθῆς γωνίας ἐπὶ τὴν βάσιν κάθετος >ῆται ἡ AD , τὰ ABD , ADC >άρα τρίγωνα <όμοιά ἐστι τῷ > τε <ὅλῳ τῷ ABG καὶ ἀλλήλοις. καὶ ἐπεὶ <όμοιον ἐστι τὸ ABG τρίγωνον τῷ ABD τριγώνῳ, >έστιν >άρα ὥς ἡ GB πρὸς τὴν BA , ο<ύτως ἡ BA πρὸς τὴν BD ; τὸ >άρα ὑπὸ τῶν GBD >ίσον ἐστὶ τῷ ἀπὸ τῆς AB .

Διὰ τὰ αὐτὰ δὴ καὶ τὸ ὑπὸ τῶν BGD >ίσον ἐστὶ τῷ ἀπὸ τῆς AG .

Καὶ ἐπεὶ, ἔαν ἐν ὀρθογωνίῳ τριγώνῳ ἀπὸ τῆς ὀρθῆς γωνίας ἐπὶ τὴν βάσιν κάθετος ᾗθῃ, ἡ ᾗθεῖσα τῶν τῆς βάσεως τμημάτων μέση ἀνάλογόν ἐστιν, >έστιν >άρα ὥς ἡ BA πρὸς τὴν DA , ο<ύτως ἡ AD πρὸς τὴν DC ; τὸ >άρα ὑπὸ τῶν BAD , DC >ίσον ἐστὶ τῷ ἀπὸ τῆς DA .

Λέγω, <ὅτι καὶ τὸ ὑπὸ τῶν BG , AD >ίσον ἐστὶ τῷ ὑπὸ τῶν BA , AG . ἐπεὶ γὰρ, ὥς >έφαμεν, <όμοιον ἐστὶ τὸ ABG τῷ ABD , >έστιν >άρα ὥς ἡ BG πρὸς τὴν GA , ο<ύτως ἡ BA πρὸς τὴν AD . τὸ >άρα ὑπὸ τῶν BG , AD >ίσον ἐστὶ τῷ ὑπὸ τῶν BA , AG ; <όπερ >έδει δεῖχαι.

For since AD has been drawn from the right-angle in a right-angled triangle, perpendicular to the base, ABD and ADC are thus triangles (which are) similar to the whole, ABC , and to one another [Prop. 6.8]. And since triangle ABC is similar to triangle ABD , thus as CB is to BA , so BA (is) to BD [Prop. 6.4]. Thus, the (rectangle contained) by CBD is equal to the (square) on AB [Prop. 6.17].

So, for the same (reasons), the (rectangle contained) by BCD is also equal to the (square) on AC .

And since if a (straight-line) is drawn from the right-angle in a right-angled triangle, perpendicular to the base, the (straight-line so) drawn is the mean proportional to the pieces of the base [Prop. 6.8 corr.], thus as BD is to DA , so AD (is) to DC . Thus, the (rectangle contained) by BD and DC is equal to the (square) on DA [Prop. 6.17].

I also say that the (rectangle contained) by BC and AD is equal to the (rectangle contained) by BA and AC . For since, as we said, ABC is similar to ABD , thus as BC is to CA , so BA (is) to AD [Prop. 6.4]. Thus, the (rectangle contained) by BC and AD is equal to the (rectangle contained) by BA and AC [Prop. 6.16]. (Which is) the very thing it was required to show.

33.

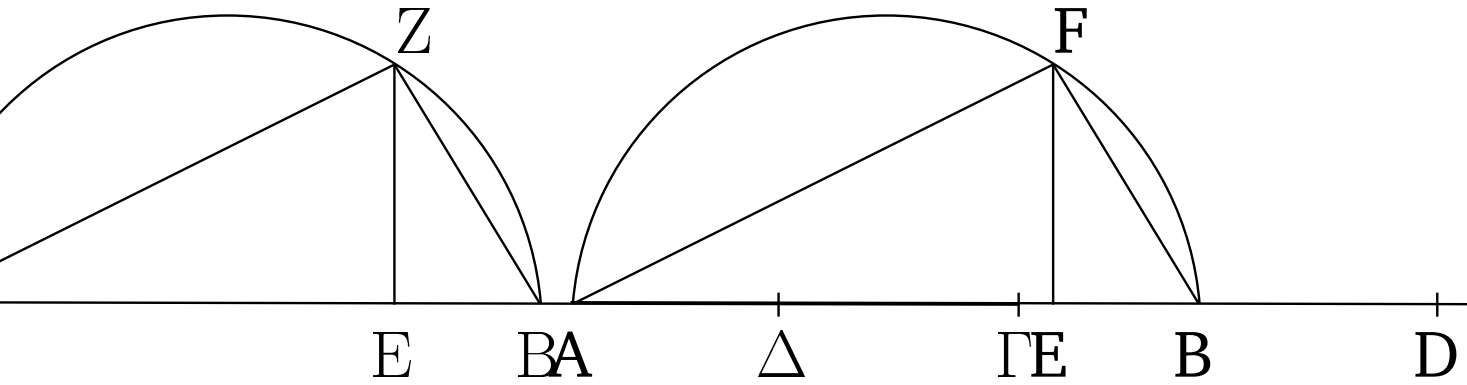
Εὐρεῖν δύο εὐθείας δυνάμει ἀσυμμέτρους ποιούσας τὸ μὲν συγκείμενον ἐκ τῶν ἀπ> αὐτῶν τετραγώνων <ρητόν, τὸ δ> ὑπ> αὐτῶν μέσον.

Ἐκκεῖσθωσαν δύο <ρηταὶ δυνάμει μόνον σύμμετροι αἱ AB , BG , <ώστε τὴν μείζονα τὴν AB τῆς ἐλάσσονος τῆς BG μείζον δύνασθαι τῷ ἀπὸ ἀσυμμέτρου ἐαντῇ, καὶ τετμήσθω ἡ BG διῖα κατὰ τὸ Δ , καὶ τῷ ἀφ> ὀποτέρας τῶν BD , DG >ίσον παρὰ τὴν AB παραβεβλήσθω παραλληλόγραμμον ἑλλείπον ε>ίδει τετραγώνῳ, καὶ >έστω τὸ ὑπὸ τῶν AEB , καὶ γεγράφθω ἐπὶ τῆς AB ημικύκλιον τὸ AZB , καὶ >ῆθῃ τῇ AB πρὸς ὀρθᾷς ἡ EZ , καὶ ἐπεξεύθθωσαν αἱ AZ , ZB .

Proposition 33

To find two straight-lines (which are) incommensurable in square, making the sum of the squares on them rational, and the (rectangle contained) by them medial.

Let the two rational (straight-lines) AB and BC , (which are) commensurable in square only, be laid out such that the square on the greater, AB , is larger than (the square on) the lesser, BC , by the (square) on (some straight-line which is) incommensurable (in length) with (AB) [Prop. 10.30]. And let BC be cut in half at D . And let a parallelogram equal to the (square) on either of BD or DC , (and) falling short by a square figure, be applied to AB [Prop. 6.28], and let it be the (rectangle contained) by AEB . And let the semicircle AFB be drawn on AB . And let EF be drawn at right-angles to AB . And let AF and FB be joined.



Καὶ ἐπεὶ [δύο] εὐθεῖαι >άνισοί εἰσιν αἱ AB, BΓ, καὶ ἡ AB τῆς BΓ μείζον δύναται τῷ ἀπὸ ἀσύμμετρου ἑαυτῇ, τῷ δὲ τετάρτῳ τοῦ ἀπὸ τῆς BΓ, τουτέστι τῷ ἀπὸ τῆς ἡμισείας αὐτῆς, >ίσον παρὰ τὴν AB παραβέβληται παραλληλόγραμμον ἑλλείπον >ίδει τετραγώνῳ καὶ ποιεῖ τὸ ὑπὸ τῶν AEB, ἀσύμμετρος >άρα ἐστὶν ἡ AE τῇ EB. καὶ ἐστὶν ὥς ἡ AE πρὸς EB, ο<ύτωξ τὸ ὑπὸ τῶν BA, AE πρὸς τὸ ὑπὸ τῶν AB, BE, >ίσον δὲ τὸ μὲν ὑπὸ τῶν BA, AE τῷ ἀπὸ τῆς AZ, τὸ δὲ ὑπὸ τῶν AB, BE τῷ ἀπὸ τῆς BZ; ἀσύμμετρον >άρα ἐστὶ τὸ ἀπὸ τῆς AZ τῷ ἀπὸ τῆς BZ; αἱ AZ, ZB >άρα δυνάμει εἰσὶν ἀσύμμετροι. καὶ ἐπεὶ ἡ AB <ρητὴ ἐστὶν, <ρητὸν >άρα ἐστὶ καὶ τὸ ἀπὸ τῆς AB; <ώστε καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AZ, ZB <ρητὸν ἐστὶν. καὶ ἐπεὶ πάλιν τὸ ὑπὸ τῶν AE, EB >ίσον ἐστὶ τῷ ἀπὸ τῆς EZ, ὑπόκειται δὲ τὸ ὑπὸ τῶν AE, EB καὶ τῷ ἀπὸ τῆς BA >ίσον, >ίση >άρα ἐστὶν ἡ ZE τῇ BA; διπλὴ >άρα ἡ BΓ τῆς ZE; <ώστε καὶ τὸ ὑπὸ τῶν AB, BΓ σύμμετρόν ἐστι τῷ ὑπὸ τῶν AB, EZ. μέσον δὲ τὸ ὑπὸ τῶν AB, BΓ; μέσον >άρα καὶ τὸ ὑπὸ τῶν AB, EZ. >ίσον δὲ τὸ ὑπὸ τῶν AB, EZ τῷ ὑπὸ τῶν AZ, ZB; μέσον >άρα καὶ τὸ ὑπὸ τῶν AZ, ZB. ἐδείθθη δὲ καὶ <ρητὸν τὸ συγκείμενον ἐκ τῶν ἀπ> αὐτῶν τετραγώνων.

Ε<ύρηνται >άρα δύο εὐθεῖαι δυνάμει ἀσύμμετροι αἱ AZ, ZB ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ> αὐτῶν τετραγώνων <ρητὸν, τὸ δὲ ὑπ> αὐτῶν μέσον; <όπερ >έδει δεῖχαι.

And since AB and BC are [two] unequal straight-lines, and the square on AB is greater than (the square on) BC by the (square) on (some straight-line which is) incommensurable (in length) with (AB) . And a parallelogram, equal to one quarter of the (square) on BC —that is to say, (equal) to the (square) on half of it—(and) falling short by a square figure, has been applied to AB , and makes the (rectangle contained) by AEB . AE is thus incommensurable (in length) with EB [Prop. 10.18]. And as AE is to EB , so the (rectangle contained) by BA and AE (is) to the (rectangle contained) by AB and BE . And the (rectangle contained) by BA and AE (is) equal to the (square) on AF , and the (rectangle contained) by AB and BE to the (square) on BF [Prop. 10.32 lem.]. The (square) on AF is thus incommensurable with the (square) on FB [Prop. 10.11]. Thus, AF and FB are incommensurable in square. And since AB is rational, the (square) on AB is also rational. Hence, the sum of the (squares) on AF and FB is also rational [Prop. 1.47]. And, again, since the (rectangle contained) by AE and EB is equal to the (square) on EF , and the (rectangle contained) by AE and EB was assumed (to be) equal to the (square) on BD , FE is thus equal to BD . Thus, BC is double FE . And hence the (rectangle contained) by AB and BC is commensurable with the (rectangle contained) by AB and EF [Prop. 10.6]. And the (rectangle contained) by AB and BC (is) medial [Prop. 10.21]. Thus, the (rectangle contained) by AB and EF (is) also medial [Prop. 10.23 corr.]. And the (rectangle contained) by AB and EF (is) equal to the (rectangle contained) by AF and FB [Prop. 10.32 lem.]. Thus, the (rectangle contained) by AF and FB (is) also medial. And the sum of the squares on them was also shown (to be) rational.

Thus, the two straight-lines, AF and FB , (which are) incommensurable in square, be found, making the sum of the squares on them rational, and the (rectangle contained) by them medial. (Which is) the very thing it was required to show.

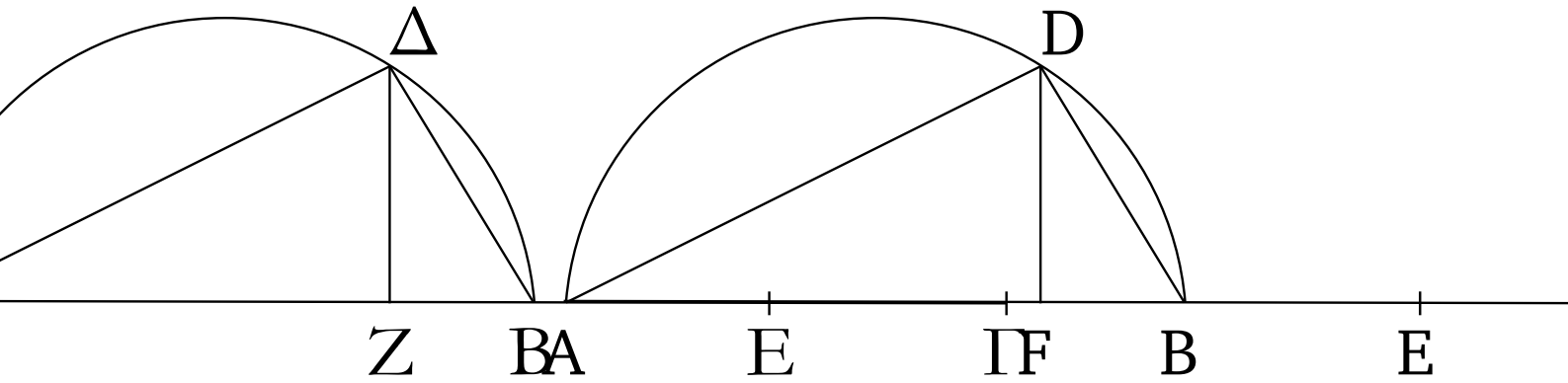
† AF and FB have lengths $\sqrt{[1 + k/(1 + k^2)^{1/2}]/2}$ and $\sqrt{[1 - k/(1 + k^2)^{1/2}]/2}$ times that of AB , respectively, where k is defined in the footnote to Prop. 10.30.

34.

Εὑρεῖν δύο εὐθεῖας δυνάμει ἀσύμμετρος ποιοῦσας τὸ μὲν συγκείμενον ἐκ τῶν ἀπ> αὐτῶν τετραγώνων μέσον, τὸ δ> ὑπ> αὐτῶν <ρητὸν.

Proposition 34

To find two straight-lines (which are) incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them rational.



Ἐκκεῖσθωσαν δύο μέσαι δυνάμει μόνον σύμμετροι αἱ AB , $BΓ$ <ρητὸν περιέθουσαι τὸ ὑπ> αὐτῶν, <ώστε τὴν AB τῆς $BΓ$ μείζον δύνασθαι τῷ ἀπὸ ἀσύμμετρου ἐαυτῇ, καὶ γεγράφθω ἐπὶ τῆς AB τὸ $ΑΔΒ$ ἡμικύκλιον, καὶ τετμήσθω ἡ $BΓ$ διῖθα κατὰ τὸ E , καὶ παραβεβλήσθω παρὰ τὴν AB τῷ ἀπὸ τῆς BE >ῖσον παραλληλόγραμμον ἐλλείπον ε>ῖδει τετραγώνῳ τὸ ὑπὸ τῶν AZB ; ἀσύμμετρος >άρα [ἐστίν] ἡ AZ τῇ ZB μήκει. καὶ >ήθθω ἀπὸ τοῦ Z τῇ AB πρὸς ὀρθὰς ἡ $ZΔ$, καὶ ἐπεζεύθωσαν αἱ $ΑΔ$, $ΔΒ$.

Ἐπεὶ ἀσύμμετρόξ ἐστίν ἡ AZ τῇ ZB , ἀσύμμετρον >άρα ἐστὶ καὶ τὸ ὑπὸ τῶν BA , AZ τῷ ὑπὸ τῶν AB , BZ . >ῖσον δὲ τὸ μὲν ὑπὸ τῶν BA , AZ τῷ ἀπὸ τῆς $ΑΔ$, τὸ δὲ ὑπὸ τῶν AB , BZ τῷ ἀπὸ τῆς $ΔΒ$; ἀσύμμετρον >άρα ἐστὶ καὶ τὸ ἀπὸ τῆς $ΑΔ$ τῷ ἀπὸ τῆς $ΔΒ$. καὶ ἐπεὶ μέσον ἐστὶ τὸ ἀπὸ τῆς AB , μέσον >άρα καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν $ΑΔ$, $ΔΒ$. καὶ ἐπεὶ διπλῇ ἐστίν ἡ $BΓ$ τῆς $ΔΖ$, διπλάσιον >άρα καὶ τὸ ὑπὸ τῶν AB , $BΓ$ τοῦ ὑπὸ τῶν AB , $ZΔ$. <ρητὸν δὲ τὸ ὑπὸ τῶν AB , $BΓ$; <ρητὸν >άρα καὶ τὸ ὑπὸ τῶν AB , $ZΔ$. τὸ δὲ ὑπὸ τῶν AB , $ZΔ$ >ῖσον τῷ ὑπὸ τῶν $ΑΔ$, $ΔΒ$; <ώστε καὶ τὸ ὑπὸ τῶν $ΑΔ$, $ΔΒ$ <ρητὸν ἐστίν.

E <ύρηνται >άρα δύο εὐθεῖαι δυνάμει ἀσύμμετροι αἱ $ΑΔ$, $ΔΒ$ ποιοῦσαι τὸ [μὲν] συγκείμενον ἐκ τῶν ἀπ> αὐτῶν τετραγώνων μέσον, τὸ δ> ὑπ> αὐτῶν <ρητὸν; <όπερ >έδει δεῖχαι.

Let the two medial (straight-lines) AB and BC , (which are) commensurable in square only, be laid out having the (rectangle contained) by them rational, (and) such that the square on AB is greater than (the square on) BC by the (square) on (some straight-line) incommensurable (in length) with (AB) [Prop. 10.31]. And let the semi-circle ADB be drawn on AB . And let BC be cut in half at E . And let a (rectangular) parallelogram equal to the (square) on BE , (and) falling short by a square figure, be applied to AB , (and let it be) the (rectangle contained by) AFB [Prop. 6.28]. Thus, AF [is] incommensurable in length with FB [Prop. 10.18]. And let FD be drawn from F at right-angles to AB . And let AD and DB be joined.

Since AF is incommensurable (in length) with FB , the (rectangle contained) by BA and AF is thus also incommensurable with the (rectangle contained) by AB and BF [Prop. 10.11]. And the (rectangle contained) by BA and AF (is) equal to the (square) on AD , and the (rectangle contained) by AB and BF to the (square) on DB [Prop. 10.32 lem.]. Thus, the (square) on AD is also incommensurable with the (square) on DB . And since the (square) on AB is medial, the sum of the (squares) on AD and DB (is) thus also medial [Props. 3.31, 1.47]. And since BC is double DF [see previous proposition], the (rectangle contained) by AB and BC (is) thus also double the (rectangle contained) by AB and FD . And the (rectangle contained) by AB and BC (is) rational. Thus, the (rectangle contained) by AB and FD (is) also rational [Prop. 10.6, Def. 10.4]. And the (rectangle contained) by AB and FD (is) equal to the (rectangle contained) by AD and DB [Prop. 10.32 lem.]. And hence the (rectangle contained) by AD and DB is rational.

Thus, two straight-lines, AD and DB , (which are) incommensurable in square, be found, making the sum of the squares on them medial, and the (rectangle contained) by them rational.[†] (Which is) the very thing it was required to show.

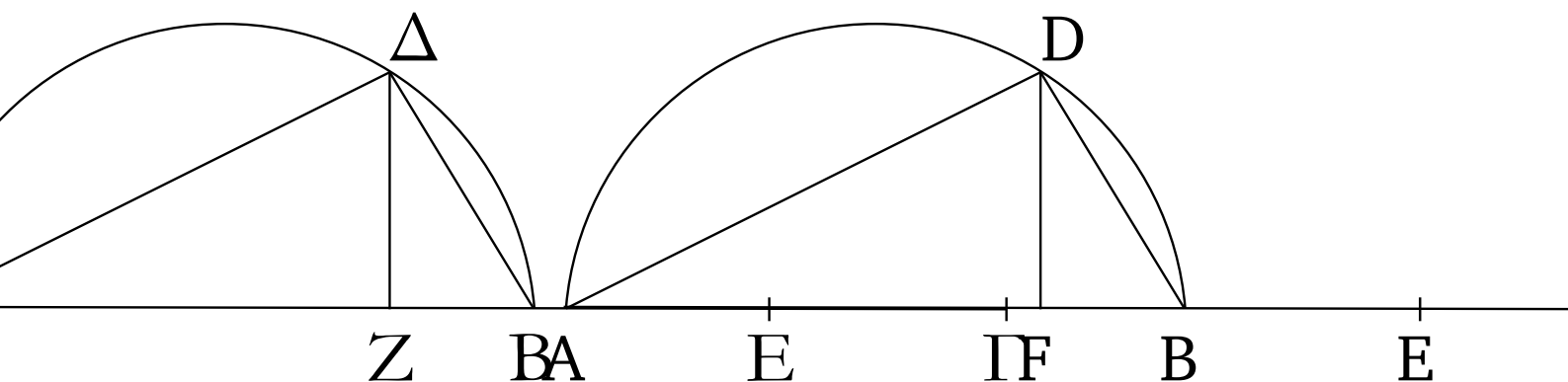
[†] AD and DB have lengths $\sqrt{[(1+k^2)^{1/2} + k]/[2(1+k^2)]}$ and $\sqrt{[(1+k^2)^{1/2} - k]/[2(1+k^2)]}$ times that of AB , respectively, where k is defined in the footnote to Prop. 10.29.

35.

Εὐρεῖν δύο εὐθεῖας δυνάμει ἀσύμμετρος ποιοῦσας τὸ τε συγκείμενον ἐκ τῶν ἀπ> αὐτῶν τετραγώνων μέσον καὶ τὸ ὑπ> αὐτῶν μέσον καὶ >έτι ἀσύμμετρον τῷ συγκειμένῳ ἐκ τῶν ἀπ> αὐτῶν τετραγώνῳ.

Proposition 35

To find two straight-lines (which are) incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them medial, and, moreover, incommensurable with the sum of the squares on them.



Ἐκκεῖσθωσαν δύο μέσαι δυνάμει μόνον σύμμετροι αἱ ΑΒ, ΒΓ μέσον περιέθουσαι, <ώστε τὴν ΑΒ τῆς ΒΓ μείζον δύνασθαι τῷ| ἀπὸ ἀσυμμέτρου ἑαυτῇ|, καὶ γεγράφθω ἐπὶ τῆς ΑΒ ἡμικύκλιον τὸ ΑΔΒ, καὶ τὰ λοιπὰ γεγονέντω τοις ἐπάνω ὁμοίωξ.

Και ἐπει ἀσύμμετροξ ἐστιν ἡ ΑΖ τῇ ΖΒ μήκει, ἀσύμμετροξ ἐστι καὶ ἡ ΑΔ τῇ ΔΒ δυνάμει. καὶ ἐπει μέσον ἐστὶ τὸ ἀπὸ τῆς ΑΒ, μέσον >άρα καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΑΔ, ΔΒ. καὶ ἐπει τὸ ὑπὸ τῶν ΑΖ, ΖΒ >ῖσον ἐστὶ τῷ ἄφ> ἐκατέρᾳξ τῶν ΒΕ, ΔΖ, >ῖση >άρα ἐστὶν ἡ ΒΕ τῇ ΔΖ; διπλῇ >άρα ἡ ΒΓ τῆς ΖΔ; <ώστε καὶ τὸ ὑπὸ τῶν ΑΒ, ΒΓ διπλασίον ἐστι τοῦ ὑπὸ τῶν ΑΒ, ΖΔ. μέσον δὲ τὸ ὑπὸ τῶν ΑΒ, ΒΓ; μέσον >άρα καὶ τὸ ὑπὸ τῶν ΑΒ, ΖΔ. καὶ ἐστὶν >ῖσον τῷ ὑπὸ τῶν ΑΔ, ΔΒ; μέσον >άρα καὶ τὸ ὑπὸ τῶν ΑΔ, ΔΒ. καὶ ἐπει ἀσύμμετροξ ἐστὶν ἡ ΑΒ τῇ ΒΓ μήκει, σύμμετροξ δὲ ἡ ΓΒ τῇ ΒΕ, ἀσύμμετροξ >άρα καὶ ἡ ΑΒ τῇ ΒΕ μήκει; <ώστε καὶ τὸ ἀπὸ τῆς ΑΒ τῷ ὑπὸ τῶν ΑΒ, ΒΕ ἀσύμμετρον ἐστίν. ἀλλὰ τῷ μὲν ἀπὸ τῆς ΑΒ >ῖσα ἐστὶ τὰ ἀπὸ τῶν ΑΔ, ΔΒ, τῷ δὲ ὑπὸ τῶν ΑΒ, ΒΕ >ῖσον ἐστὶ τὸ ὑπὸ τῶν ΑΒ, ΖΔ, τουτέστι τὸ ὑπὸ τῶν ΑΔ, ΔΒ; ἀσύμμετρον >άρα ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΑΔ, ΔΒ τῷ ὑπὸ τῶν ΑΔ, ΔΒ.

Εὐφρηνται >άρα δύο εὐθείαι αἱ ΑΔ, ΔΒ δυνάμει ἀσύμμετροι
 ποιοῦσαι τὸ τε συγκεκριμένον ἐκ τῶν ἀπ> αὐτῶν μέσον καὶ τὸ
 ὑπ> αὐτῶν μέσον καὶ >έτι ἀσύμμετρον τῷ| συγκεκριμένῳ ἐκ
 τῶν ἀπ> αὐτῶν τετραγώνων; <όπερ >έδει δεῖξαι.

Let the two medial (straight-lines) AB and BC , (which are) commensurable in square only, be laid out containing a medial (area), such that the square on AB is greater than (the square on) BC by the (square) on (some straight-line) incommensurable (in length) with (AB) [Prop. 10.32]. And let the semi-circle ADB be drawn on AB . And let the remainder (of the figure) be generated similarly to the above (proposition).

And since AF is incommensurable in length with FB [Prop. 10.18], AD is also incommensurable in square with DB [Prop. 10.11]. And since the (square) on AB is medial, the sum of the (squares) on AD and DB (is) thus also medial [Props. 3.31, 1.47]. And since the (rectangle contained) by AF and FB is equal to the (square) on each of BE and DF , BE is thus equal to DF . Thus, BC (is) double FD . And hence the (rectangle contained) by AB and BC is double the (rectangle contained) by AB and FD . And the (rectangle contained) by AB and BC (is) medial. Thus, the (rectangle contained) by AB and FD (is) also medial. And it is equal to the (rectangle contained) by AD and DB [Prop. 10.32 lem.]. Thus, the (rectangle contained) by AD and DB (is) also medial. And since AB is incommensurable in length with BC , and CB (is) commensurable (in length) with BE , AB (is) thus also incommensurable in length with BE [Prop. 10.13]. And hence the (square) on AB is also incommensurable with the (rectangle contained) by AB and BE [Prop. 10.11]. But the (sum of the squares) on AD and DB is equal to the (square) on AB [Prop. 1.47]. And the (rectangle contained) by AB and FD —that is to say, the (rectangle contained) by AD and DB —is equal to the (rectangle contained) by AB and BE . Thus, the sum of the (squares) on AD and DB is incommensurable with the (rectangle contained) by AD and DB .

Thus, two straight-lines, AD and DB , (which are) incommensurable in square, be found, making the sum of the (squares) on them medial, and the (rectangle contained) by them medial, and, moreover, incommensurable with the sum of the squares on them.[†] (Which is) the very thing it was required to show.

[†] AD and DB have lengths $k'^{1/4}\sqrt{[1+k/(1+k^2)^{1/2}]/2}$ and $k'^{1/4}\sqrt{[1-k/(1+k^2)^{1/2}]/2}$ times that of AB , respectively, where k and k' are defined in the footnote to Prop. 10.32.

36.

Ἐὰν δύο <ρηταὶ δυνάμει μόνον σύμμετροι συντεθῶσιν, ἡ <όλη> ἀλογόξ ἐστιν, καλεῖσθω δὲ ἐκ δύο ὀνομάτων.

Proposition 36

If two rational (straight-lines which are) commensurable in square only are added together then the whole (straight-line) is irrational—let it be called a binomial (straight-line).[†]



Συγκείσθωσαν γὰρ δύο <ρηταὶ δυνάμει μόνον σύμμετροι αἱ AB, BΓ; λέγω, <ὅτι <ὅλη ἢ AΓ >άλογός ἐστιν.

Ἐπεὶ γὰρ ἀσύμμετρόξ ἐστιν ἡ AB τῇ BΓ μήκει; δυνάμει γὰρ μόνον εἰσὶ σύμμετροι; ὥς δὲ ἡ AB πρὸς τὴν BΓ, ο<ύτωξ τὸ ὑπὸ τῶν ABΓ πρὸς τὸ ἀπὸ τῆς BΓ, ἀσύμμετρον >άρα ἐστὶ τὸ ὑπὸ τῶν AB, BΓ τῷ ἀπὸ τῆς BΓ. ἀλλὰ τῷ μὲν ὑπὸ τῶν AB, BΓ σύμμετρον ἐστὶ τὸ διξ ὑπὸ τῶν AB, BΓ, τῷ δὲ ἀπὸ τῆς BΓ σύμμετρά ἐστὶ τὰ ἀπὸ τῶν AB, BΓ; αἱ γὰρ AB, BΓ <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι; ἀσύμμετρον >άρα ἐστὶ τὸ διξ ὑπὸ τῶν AB, BΓ τοῖξ ἀπὸ τῶν AB, BΓ. καὶ συνθέντι τὸ διξ ὑπὸ τῶν AB, BΓ μετὰ τῶν ἀπὸ τῶν AB, BΓ, τουτέστι τὸ ἀπὸ τῆς AΓ, ἀσύμμετρον ἐστὶ τῷ συγκειμένῳ ἐκ τῶν ἀπὸ τῶν AB, BΓ; <ρητὸν δὲ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AB, BΓ; >άλογον >άρα [ἐστὶ] τὸ ἀπὸ τῆς AΓ; <ώστε καὶ ἡ AΓ >άλογός ἐστιν, καλείσθω δὲ ἐκ δύο ὀνομάτων; <όπερ >έδει δεῖχαι.

For let the two rational (straight-lines), AB and BC , (which are) commensurable in square only, be laid down together. I say that the whole (straight-line), AC , is irrational. For since AB is incommensurable in length with BC —for they are commensurable in square only—and as AB (is) to BC , so the (rectangle contained) by ABC (is) to the (square) on BC , the (rectangle contained) by AB and BC is thus incommensurable with the (square) on BC [Prop. 10.11]. But, twice the (rectangle contained) by AB and BC is commensurable with the (rectangle contained) by AB and BC [Prop. 10.6]. And (the sum of) the (squares) on AB and BC is commensurable with the (square) on BC —for the rational (straight-lines) AB and BC are commensurable in square only [Prop. 10.15]. Thus, twice the (rectangle contained) by AB and BC is incommensurable with (the sum of) the (squares) on AB and BC [Prop. 10.13]. And, via composition, twice the (rectangle contained) by AB and BC , plus (the sum of) the (squares) on AB and BC —that is to say, the (square) on AC [Prop. 2.4]—is incommensurable with the sum of the (squares) on AB and BC [Prop. 10.16]. And the sum of the (squares) on AB and BC (is) rational. Thus, the (square) on AC [is] irrational [Def. 10.4]. Hence, AC is also irrational [Def. 10.4]—let it be called a binomial (straight-line).[†] (Which is) the very thing it was required to show.

[†] Literally, “from two names”.

[‡] Thus, a binomial straight-line has a length expressible as $1 + k^{1/2}$ [or, more generally, $\rho(1 + k^{1/2})$, where ρ is rational—the same proviso applies to the definitions in the following propositions]. The binomial and the corresponding apotome, whose length is expressible as $1 - k^{1/2}$ (see Prop. 10.73), are the positive roots of the quartic $x^4 - 2(1 + k)x^2 + (1 - k)^2 = 0$.

37.

Proposition 37

Ἐὰν δύο μέσαι δυνάμει μόνον σύμμετροι συντεθῶσι <ρητὸν περιέθουσαι, ἡ <ὅλη >άλογός ἐστιν, καλείσθω δὲ ἐκ δύο μέσων πρώτη.

If two medial (straight-lines), commensurable in square only, which contain a rational (area), are added together then the whole (straight-line) is irrational—let it be called a first bimedian (straight-line).[†]



Συγκείσθωσαν γὰρ δύο μέσαι δυνάμει μόνον σύμμετροι αἱ AB, BΓ <ρητὸν περιέθουσαι; λέγω, <ὅτι <ὅλη ἢ AΓ >άλογός ἐστιν.

Ἐπεὶ γὰρ ἀσύμμετρόξ ἐστιν ἡ AB τῇ BΓ μήκει; καὶ τὰ ἀπὸ τῶν AB, BΓ >άρα ἀσύμμετρά ἐστι τῷ διξ ὑπὸ τῶν AB, BΓ; καὶ συνθέντι τὰ ἀπὸ τῶν AB, BΓ μετὰ τοῦ διξ ὑπὸ τῶν AB, BΓ, <όπερ ἐστὶ τὸ ἀπὸ τῆς AΓ, ἀσύμμετρον ἐστὶ τῷ ὑπὸ τῶν AB, BΓ. <ρητὸν δὲ τὸ ὑπὸ τῶν AB, BΓ; ὑπόκειται γὰρ αἱ AB, BΓ <ρητὸν περιέθουσαι; >άλογον >άρα τὸ ἀπὸ τῆς AΓ; >άλογός >άρα ἡ AΓ, καλείσθω δὲ ἐκ δύο μέσων πρώτη; <όπερ >έδει δεῖχαι.

For let the two medial (straight-lines), AB and BC , commensurable in square only, (and) containing a rational (area), be laid down together. I say that the whole (straight-line), AC , is irrational.

For since AB is incommensurable in length with BC , (the sum of) the (squares) on AB and BC is thus also incommensurable with twice the (rectangle contained) by AB and BC [see previous proposition]. And, via composition, (the sum of) the (squares) on AB and BC , plus twice the (rectangle contained) by AB and BC —that is, the (square) on AC [Prop. 2.4]—is incommensurable with the (rectangle contained) by AB and BC [Prop. 10.16]. And the (rectangle contained) by AB and BC (is) rational—for AB and BC were assumed to enclose a rational (area). Thus, the (square) on AC (is) irrational. Thus, AC (is) irrational [Def. 10.4]—let it be called a first bimedian (straight-line).[‡] (Which is) the very thing it was required to show.

[†] Literally, “first from two medials”.

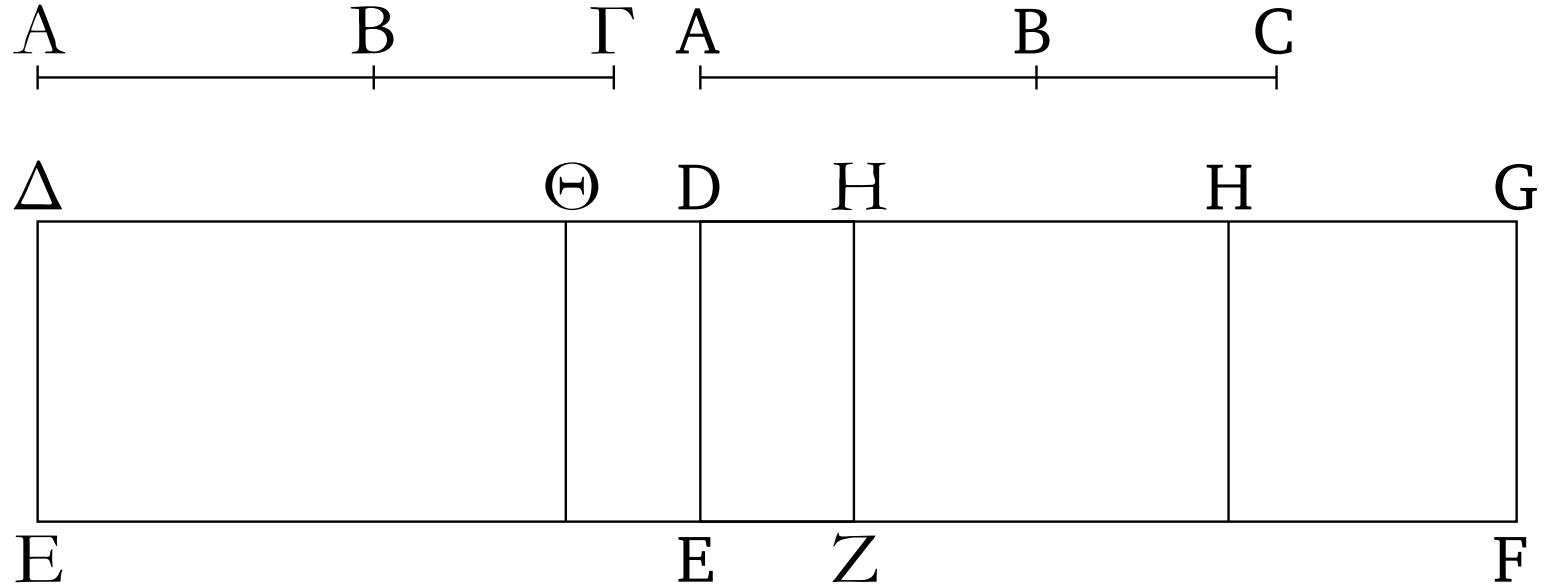
‡ Thus, a first bimedial straight-line has a length expressible as $k^{1/4} + k^{3/4}$. The first bimedial and the corresponding first apotome of a medial, whose length is expressible as $k^{1/4} - k^{3/4}$ (see Prop. 10.74), are the positive roots of the quartic $x^4 - 2\sqrt{k}(1+k)x^2 + k(1-k)^2 = 0$.

38.

Proposition 38

Ἐὰν δύο μέσαι δυνάμει μόνον σύμμετροι συντεθῶσι μέσον περιέθουσαι, ἡ <ὅλη> ἀλογόξ ἐστίν, καλείσθω δὲ ἐκ δύο μέσων δευτέρα.

If two medial (straight-lines), commensurable in square only, which contain a medial (area), are added together then the whole (straight-line) is irrational—let it be called a second bimedial (straight-line).



Συγκείσθωσαν γὰρ δύο μέσαι δυνάμει μόνον σύμμετροι αἱ AB , BC μέσον περιέθουσαι; λέγω, <ὅτι> ἀλογόξ ἐστίν ἡ AC . Ἐκκείσθω γὰρ <ρητὴ ἡ DE , καὶ τῷ ἀπὸ τῆς AC > ἴσον παρὰ τὴν DE παραβελήσθω τὸ DZ πλάτος ποιῶν τὴν DH . καὶ ἐπεὶ τὸ ἀπὸ τῆς AC > ἴσον ἐστὶ τοῖς τε ἀπὸ τῶν AB , BC καὶ τῷ διζῷ ὑπὸ τῶν AB , BC , παραβελήσθω δὴ τοῖς ἀπὸ τῶν AB , BC παρὰ τὴν DE > ἴσον τὸ $EΘ$; λοιπὸν >άρα τὸ $ΘZ$ > ἴσον ἐστὶ τῷ διζῷ ὑπὸ τῶν AB , BC . καὶ ἐπεὶ μέση ἐστὶν ἑκάτερα τῶν AB , BC , μέσα >άρα ἐστὶ καὶ τὰ ἀπὸ τῶν AB , BC . μέσον δὲ ὑπόκειται καὶ τὸ διζῷ ὑπὸ τῶν AB , BC . καὶ ἐστὶ τοῖς μὲν ἀπὸ τῶν AB , BC > ἴσον τὸ $EΘ$, τῷ δὲ διζῷ ὑπὸ τῶν AB , BC > ἴσον τὸ $ΘZ$; μέσον >άρα ἑκάτερον τῶν $EΘ$, $ΘZ$. καὶ παρὰ <ρητὴν τὴν DE παράκειται; <ρητὴ> >άρα ἐστὶν ἑκάτερα τῶν $ΔΘ$, $ΘH$ καὶ ἀσύμμετροξ τῇ DE μήκει. ἐπεὶ ο>ύν ἀσύμμετρόξ ἐστὶν ἡ AB τῇ BC μήκει, καὶ ἐστὶν ὥξ ἡ AB πρὸξ τὴν BC , ο<ύτωξ τὸ ἀπὸ τῆς AB πρὸξ τὸ ὑπὸ τῶν AB , BC , ἀσύμμετρον >άρα ἐστὶ τὸ ἀπὸ τῆς AB τῷ ὑπὸ τῶν AB , BC . ἀλλὰ τῷ μὲν ἀπὸ τῆς AB σύμμετρον ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AB , BC τετραγώνων, τῷ δὲ ὑπὸ τῶν AB , BC σύμμετρον ἐστὶ τὸ διζῷ ὑπὸ τῶν AB , BC . ἀσύμμετρον >άρα ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AB , BC τῷ διζῷ ὑπὸ τῶν AB , BC . ἀλλὰ τοῖς μὲν ἀπὸ τῶν AB , BC > ἴσον ἐστὶ τὸ $EΘ$, τῷ δὲ διζῷ ὑπὸ τῶν AB , BC > ἴσον ἐστὶ τὸ $ΘZ$. ἀσύμμετρον >άρα ἐστὶ τὸ $EΘ$ τῷ $ΘZ$; <ώστε καὶ ἡ $ΔΘ$ τῇ $ΘH$ ἐστὶν ἀσύμμετροξ μήκει. αἱ $ΔΘ$, $ΘH$ >άρα <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι. <ώστε ἡ DH > ἀλογόξ ἐστίν. <ρητὴ δὲ ἡ DE ; τὸ δὲ ὑπὸ ἀλόγου καὶ <ρητῆς περιεχόμενον ὀρθογώνιον >αλογόν ἐστίν; >αλογον >άρα ἐστὶ τὸ DZ θωρίον, καὶ ἡ δυνάμενη [αὐτὸ] >αλογόξ ἐστίν. δύναται δὲ τὸ DZ ἡ AC ; >αλογόξ >άρα ἐστὶν ἡ AC , καλείσθω δὲ ἐκ δύο μέσων δευτέρα. <όπερ> εἶδει δεῖχαι.

For let the two medial (straight-lines), AB and BC , commensurable in square only, (and) containing a medial (area), be laid down together [Prop. 10.28]. I say that AC is irrational.

For let the rational (straight-line) DE be laid down, and let (the rectangle) DF , equal to the (square) on AC , be applied to DE , making DG as breadth [Prop. 1.44]. And since the (square) on AC is equal to (the sum of) the (squares) on AB and BC , plus twice the (rectangle contained) by AB and BC [Prop. 2.4], so let (the rectangle) EH , equal to (the sum of) the squares on AB and BC , be applied to DE . The remainder HF is thus equal to twice the (rectangle contained) by AB and BC . And since AB and BC are each medial, (the sum of) the squares on AB and BC is thus also medial.‡ And twice the (rectangle contained) by AB and BC was also assumed (to be) medial. And EH is equal to (the sum of) the squares on AB and BC , and HF (is) equal to twice the (rectangle contained) by AB and BC . Thus, EH and HF (are) each medial. And they were applied to the rational (straight-line) DE . Thus, DH and HG are each rational, and incommensurable in length with DE [Prop. 10.22]. Therefore, since AB is incommensurable in length with BC , and as AB is to BC , so the (square) on AB (is) to the (rectangle contained) by AB and BC [Prop. 10.21 lem.], the (square) on AB is thus incommensurable with the (rectangle contained) by AB and BC [Prop. 10.11]. But, the sum of the squares on AB and BC is commensurable with the (square) on AB [Prop. 10.15], and twice the (rectangle contained) by AB and BC is commensurable with the (rectangle contained) by AB and BC [Prop. 10.6]. Thus, the sum of the (squares) on AB and BC is incommensurable with twice the (rectangle contained) by AB and BC [Prop. 10.13]. But, EH is equal to (the sum of) the squares on AB and BC , and HF is equal to twice

the (rectangle) contained by AB and BC . Thus, EH is incommensurable with HF . Hence, DH is also incommensurable in length with HG [Props. 6.1, 10.11]. Thus, DH and HG are rational (straight-lines which are) commensurable in square only. Hence, DG is irrational [Prop. 10.36]. And DE (is) rational. And the rectangle contained by irrational and rational (straight-lines) is irrational [Prop. 10.20]. The area DF is thus irrational, and (so) the square-root [of it] is irrational [Def. 10.4]. And AC is the square-root of DF . AC is thus irrational—let it be called a second bimedral (straight-line).[§] (Which is) the very thing it was required to show.

[†] Literally, “second from two medials”.

[‡] Since, by hypothesis, the squares on AB and BC are commensurable—see Props. 10.15, 10.23.

[§] Thus, a second bimedral straight-line has a length expressible as $k^{1/4} + k'^{1/2}/k^{1/4}$. The second bimedral and the corresponding second apotome of a medial, whose length is expressible as $k^{1/4} - k'^{1/2}/k^{1/4}$ (see Prop. 10.75), are the positive roots of the quartic $x^4 - 2[(k + k')/\sqrt{k}]x^2 + [(k - k')^2/k] = 0$.

39.

Proposition 39

Ἐάν δύο εὐθεῖαι δυνάμει ἀσύμμετροι συντεθῶσι ποιῶσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων <ρητόν, τὸ δ' ὑπὸ αὐτῶν μέσον, ἢ <όλη εὐθεῖα >ἀλογόξ ἐστιν, καλείσθω δὲ μείζων.

If two straight-lines (which are) incommensurable in square, making the sum of the squares on them rational, and the (rectangle contained) by them medial, are added together then the whole straight-line is irrational—let it be called a major (straight-line).



Συγκείσθωσαν γὰρ δύο εὐθεῖαι δυνάμει ἀσύμμετροι αἱ AB , $BΓ$ ποιῶσαι τὰ προκείμενα; λέγω, <ὅτι >ἀλογόξ ἐστιν ἡ $ΑΓ$.

Ἐπεὶ γὰρ τὸ ὑπὸ τῶν AB , $BΓ$ μέσον ἐστίν, καὶ τὸ διζ [>άρα] ὑπὸ τῶν AB , $BΓ$ μέσον ἐστίν. τὸ δὲ συγκείμενον ἐκ τῶν ἀπὸ τῶν AB , $BΓ$ <ρητόν; ἀσύμμετρον >άρα ἐστὶ τὸ διζ ὑπὸ τῶν AB , $BΓ$ τῷ συγκειμένῳ ἐκ τῶν ἀπὸ τῶν AB , $BΓ$; <ώστε καὶ τὰ ἀπὸ τῶν AB , $BΓ$ μετὰ τοῦ διζ ὑπὸ τῶν AB , $BΓ$, <όπερ ἐστὶ τὸ ἀπὸ τῆς $ΑΓ$, ἀσύμμετρόν ἐστι τῷ συγκειμένῳ ἐκ τῶν ἀπὸ τῶν AB , $BΓ$ [$<$ ρητόν δὲ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AB , $BΓ$]; >ἀλογον >άρα ἐστὶ τὸ ἀπὸ τῆς $ΑΓ$. <ώστε καὶ ἡ $ΑΓ$ >ἀλογόξ ἐστιν, καλείσθω δὲ μείζων. <όπερ >έδει δεῖξαι.

For let the two straight-lines, AB and BC , incommensurable in square, and fulfilling the prescribed (conditions), be laid down together [Prop. 10.33]. I say that AC is irrational. For since the (rectangle contained) by AB and BC is medial, twice the (rectangle contained) by AB and BC is [thus] also medial [Props. 10.6, 10.23 corr.]. And the sum of the (squares) on AB and BC (is) rational. Thus, twice the (rectangle contained) by AB and BC is incommensurable with the sum of the (squares) on AB and BC [Def. 10.4]. Hence, (the sum of) the squares on AB and BC , plus twice the (rectangle contained) by AB and BC —that is, the (square) on AC [Prop. 2.4]—is also incommensurable with the sum of the (squares) on AB and BC [Prop. 10.16] [and the sum of the (squares) on AB and BC (is) rational]. Thus, the (square) on AC is irrational. Hence, AC is also irrational [Def. 10.4]—let it be called a major (straight-line).[†] (Which is) the very thing it was required to show.

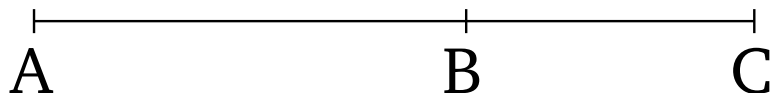
[†] Thus, a major straight-line has a length expressible as $\sqrt{[1 + k/(1 + k^2)^{1/2}]/2} + \sqrt{[1 - k/(1 + k^2)^{1/2}]/2}$. The major and the corresponding minor, whose length is expressible as $\sqrt{[1 + k/(1 + k^2)^{1/2}]/2} - \sqrt{[1 - k/(1 + k^2)^{1/2}]/2}$ (see Prop. 10.76), are the positive roots of the quartic $x^4 - 2x^2 + k^2/(1 + k^2) = 0$.

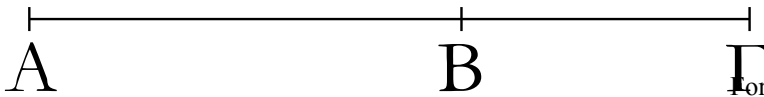
40.

Proposition 40

Ἐάν δύο εὐθεῖαι δυνάμει ἀσύμμετροι συντεθῶσι ποιῶσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον, τὸ δ' ὑπὸ αὐτῶν <ρητόν, ἢ <όλη εὐθεῖα >ἀλογόξ ἐστιν, καλείσθω δὲ <ρητόν καὶ μέσον δυναμένη.

If two straight-lines (which are) incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them rational, are added together then the whole straight-line is irrational—let it be called the square-root of a rational plus a medial (area).





Συγκείσθωσαν γὰρ δύο εὐθεῖαι δυνάμει ἀσύμμετροι αἱ AB, BG ποιοῦσαι τὰ προκείμενα; λέγω, <ὅτι >ἀλογόξ ἐστὶν ἡ AG.

Ἐπεὶ γὰρ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AB, BG μέσον ἐστίν, τὸ δὲ διζ ὑπὸ τῶν AB, BG <ρητόν, ἀσύμμετρον >ἀρα ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AB, BG τῷ διζ ὑπὸ τῶν AB, BG; <ὥστε καὶ τὸ ἀπὸ τῆς AG ἀσύμμετρόν ἐστι τῷ διζ ὑπὸ τῶν AB, BG. <ρητόν δὲ τὸ διζ ὑπὸ τῶν AB, BG; >ἀλογον >ἀρα τὸ ἀπὸ τῆς AG. >ἀλογοξ >ἀρα ἡ AG, καλεῖσθω δὲ <ρητόν καὶ μέσον δυναμένη. <ὅπερ >έδει δεῖξαι.

For let the two straight-lines, AB and BC , incommensurable in square, (and) fulfilling the prescribed (conditions), be laid down together [Prop. 10.34]. I say that AC is irrational.

For since the sum of the (squares) on AB and BC is medial, and twice the (rectangle contained) by AB and BC (is) rational, the sum of the (squares) on AB and BC is thus incommensurable with twice the (rectangle contained) by AB and BC . Hence, the (square) on AC is also incommensurable with twice the (rectangle contained) by AB and BC [Prop. 10.16]. And twice the (rectangle contained) by AB and BC (is) rational. The (square) on AC (is) thus irrational. Thus, AC (is) irrational [Def. 10.4]—let it be called the square-root of a rational plus a medial (area).[†] (Which is) the very thing it was required to show.

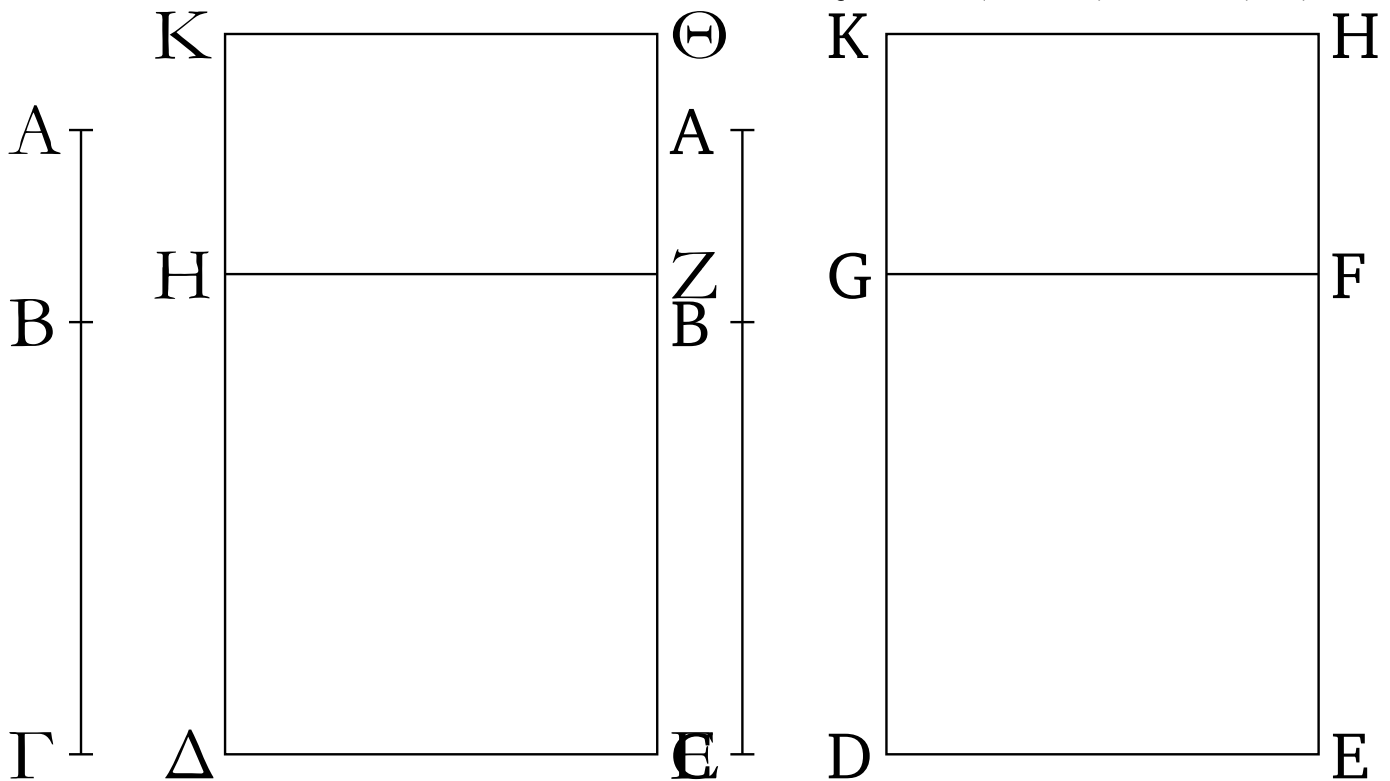
[†] Thus, the square-root of a rational plus a medial (area) has a length expressible as $\sqrt{[(1+k^2)^{1/2}+k]/[2(1+k^2)]}+\sqrt{[(1+k^2)^{1/2}-k]/[2(1+k^2)]}$. This and the corresponding irrational with a minus sign, whose length is expressible as $\sqrt{[(1+k^2)^{1/2}+k]/[2(1+k^2)]}-\sqrt{[(1+k^2)^{1/2}-k]/[2(1+k^2)]}$ (see Prop. 10.77), are the positive roots of the quartic $x^4 - (2/\sqrt{1+k^2})x^2 + k^2/(1+k^2)^2 = 0$.

41.

Proposition 41

Ἐὰν δύο εὐθεῖαι δυνάμει ἀσύμμετροι συντεθῶσι ποιοῦσαι τὸ τε συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον καὶ τὸ ὑπ' αὐτῶν μέσον καὶ >έτι ἀσύμμετρον τῷ συγκειμένῳ ἐκ τῶν ἀπ' αὐτῶν τετραγώνων, ἡ <ὅλη εὐθεῖα >ἀλογόξ ἐστὶν, καλεῖσθω δὲ δύο μέσα δυναμένη.

If two straight-lines (which are) incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them medial, and, moreover, incommensurable with the sum of the squares on them, are added together then the whole straight-line is irrational—let it be called the square-root of (the sum of) two medial (areas).



Συγκείσθωσαν γὰρ δύο εὐθεῖαι δυνάμει ἀσύμμετροι αἱ AB, BG ποιοῦσαι τὰ προκείμενα; λέγω, <ὅτι ἡ AG >ἀλογόξ ἐστὶν.

Ἐκκείσθω <ρητὴ ἡ ΔΕ, καὶ παραβελήσθω παρὰ τὴν ΔΕ τοῖς μὲν ἀπὸ τῶν AB, BG >ῖσον τὸ ΔΖ, τῷ δὲ διζ ὑπὸ τῶν AB, BG >ῖσον τὸ ΗΘ; <όλον >ἀρα τὸ ΘΘ >ῖσον ἐστὶ τῷ ἀπὸ

For let the two straight-lines, AB and BC , incommensurable in square, (and) fulfilling the prescribed (conditions), be laid down together [Prop. 10.35]. I say that AC is irrational.

Let the rational (straight-line) DE be laid out, and let (the rectangle) DF , equal to (the sum of) the (squares) on AB

τῆς ΑΓ τετραγώνῳ. καὶ ἐπεὶ μέσον ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΑΒ, ΒΓ, καὶ ἐστὶν >ῖσον τῷ ΔΖ, μέσον >άρα ἐστὶ καὶ τὸ ΔΖ. καὶ παρὰ <ρητὴν τὴν ΔΕ παράκειται; <ρητὴ >άρα ἐστὶν ἡ ΔΗ καὶ ἀσύμμετρος τῇ ΔΕ μήκει. διὰ τὰ αὐτὰ δὴ καὶ ἡ ΗΚ <ρητὴ ἐστὶ καὶ ἀσύμμετρος τῇ ΗΖ, τουτέστι τῇ ΔΕ, μήκει. καὶ ἐπεὶ ἀσύμμετρά ἐστι τὰ ἀπὸ τῶν ΑΒ, ΒΓ τῷ διζ ὑπὸ τῶν ΑΒ, ΒΓ, ἀσύμμετρόν ἐστι τὸ ΔΖ τῷ ΗΘ; <ώστε καὶ ἡ ΔΗ τῇ ΗΚ ἀσύμμετρόξ ἐστὶν. καὶ εἰσι <ρηταί; αἱ ΔΗ, ΗΚ >άρα <ρηταί εἰσι δυνάμει μόνον σύμμετροι; >άλογος >άρα ἐστὶν ἡ ΔΚ ἡ καλουμένη ἐκ δύο ὀνομάτων. <ρητὴ δὲ ἡ ΔΕ; >άλογον >άρα ἐστὶ τὸ ΘΑ καὶ ἡ δυναμένη αὐτὸ >άλογός ἐστιν. δύνανται δὲ τὸ ΘΑ ἡ ΑΓ; >άλογος >άρα ἐστὶν ἡ ΑΓ, καλεῖσθω δὲ δύο μέσα δυναμένη. <όπερ >έδει δεῖξαι.

and BC , and (the rectangle) GH , equal to twice the (rectangle contained) by AB and BC , be applied to DE . Thus, the whole of DH is equal to the square on AC [Prop. 2.4]. And since the sum of the (squares) on AB and BC is medial, and is equal to DF , DF is thus also medial. And it is applied to the rational (straight-line) DE . Thus, DG is rational, and incommensurable in length with DE [Prop. 10.22]. So, for the same (reasons), GK is also rational, and incommensurable in length with GF —that is to say, DE . And since (the sum of) the (squares) on AB and BC is incommensurable with twice the (rectangle contained) by AB and BC , DF is incommensurable with GH . Hence, DG is also incommensurable (in length) with GK [Props. 6.1, 10.11]. And they are rational. Thus, DG and GK are rational (straight-lines which are) commensurable in square only. Thus, DK is irrational, and that (straight-line which is) called binomial [Prop. 10.36]. And DE (is) rational. Thus, DH is irrational, and its square-root is irrational [Def. 10.4]. And AC (is) the square-root of HD . Thus, AC is irrational—let it be called the square-root of (the sum of) two medial (areas).[†] (Which is) the very thing it was required to show.

[†] Thus, the square-root of (the sum of) two medial (areas) has a length expressible as $k^{1/4} \left(\sqrt{[1 + k/(1 + k^2)^{1/2}]/2} + \sqrt{[1 - k/(1 + k^2)^{1/2}]/2} \right)$. This and the corresponding irrational with a minus sign, whose length is expressible as $k^{1/4} \left(\sqrt{[1 + k/(1 + k^2)^{1/2}]/2} - \sqrt{[1 - k/(1 + k^2)^{1/2}]/2} \right)$ (see Prop. 10.78), are the positive roots of the quartic $x^4 - 2k^{1/2}x^2 + k'k^2/(1 + k^2) = 0$.

Λήμμα.

<Ὅτι δὲ αἱ εἰρημέναι >άλογοι μοναθῶς διαιροῦνται εἰς τὰς εὐθείας, ἐχ <ὦν σύγκεινται ποιουσῶν τὰ προκείμενα ε>ἶδη, δείχομεν >ἤδη προεκθέμενοι λημμάτων τοιοῦτον;

Lemma

We will now demonstrate that the aforementioned irrational (straight-lines) are uniquely divided into the straight-lines of which they are the sum, and which produce the prescribed types, (after) setting forth the following lemma.



Ἐκκεῖσθω εὐθεῖα ἡ ΑΒ καὶ τετμήσθω ἡ <όλη εἰς >άνισα καθ' ἐκάτερον τῶν Γ, Δ, ὑποκεῖσθω δὲ μείζων ἡ ΑΓ τῇς ΔΒ; λέγω, <ὅτι τὰ ἀπὸ τῶν ΑΓ, ΓΒ μείζονά ἐστι τῶν ἀπὸ τῶν ΑΔ, ΔΒ.

Τετμήσθω γὰρ ἡ ΑΒ δίχα κατὰ τὸ Ε. καὶ ἐπεὶ μείζων ἐστὶν ἡ ΑΓ τῇς ΔΒ, κοινὴ ἀφηρήσθω ἡ ΔΓ; λοιπὴ >άρα ἡ ΑΔ λοιπῇς τῇς ΓΒ μείζων ἐστίν. >ῖση δὲ ἡ ΑΕ τῇ ΕΒ; ἐλάττων >άρα ἡ ΔΕ τῇς ΕΓ; τὰ Γ, Δ >άρα σημεῖα οὐκ >ῖσον ἀπέθουσι τῇς διχοτομίας. καὶ ἐπεὶ τὸ ὑπὸ τῶν ΑΓ, ΓΒ μετὰ τοῦ ἀπὸ τῇς ΕΓ >ῖσον ἐστὶ τῷ ἀπὸ τῇς ΕΒ, ἀλλὰ μὴν καὶ τὸ ὑπὸ τῶν ΑΔ, ΔΒ μετὰ τοῦ ἀπὸ ΔΕ >ῖσον ἐστὶ τῷ ἀπὸ τῇς ΕΒ, τὸ >άρα ὑπὸ τῶν ΑΓ, ΓΒ μετὰ τοῦ ἀπὸ τῇς ΕΓ >ῖσον ἐστὶ τῷ ὑπὸ τῶν ΑΔ, ΔΒ μετὰ τοῦ ἀπὸ τῇς ΔΕ; <ὦν τὸ ἀπὸ τῇς ΔΕ >έλασσόν ἐστὶ τοῦ ἀπὸ τῇς ΕΓ; καὶ λοιπὸν >άρα τὸ ὑπὸ τῶν ΑΓ, ΓΒ >έλασσόν ἐστὶ τοῦ ὑπὸ τῶν ΑΔ, ΔΒ. <ώστε καὶ τὸ διζ ὑπὸ τῶν ΑΓ, ΓΒ >έλασσόν ἐστὶ τοῦ διζ ὑπὸ τῶν ΑΔ, ΔΒ. καὶ λοιπὸν >άρα τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΑΓ, ΓΒ μείζον ἐστὶ τοῦ συγκειμένου ἐκ τῶν ἀπὸ τῶν ΑΔ, ΔΒ. <όπερ >έδει δεῖξαι.

Let the straight-line AB be laid out, and let the whole (straight-line) be cut into unequal parts at each of the (points) C and D . And let AC be assumed (to be) greater than DB . I say that (the sum of) the (squares) on AC and CB is greater than (the sum of) the (squares) on AD and DB .

For let AB be cut in half at E . And since AC is greater than DB , let DC be subtracted from both. Thus, the remainder AD is greater than the remainder CB . And AE (is) equal to EB . Thus, DE (is) less than EC . Thus, points C and D are not equally far from the point of bisection. And since the (rectangle contained) by AC and CB , plus the (square) on EC , is equal to the (square) on EB [Prop. 2.5], but, moreover, the (rectangle contained) by AD and DB , plus the (square) on DE , is also equal to the (square) on EB [Prop. 2.5], the (rectangle contained) by AC and CB , plus the (square) on EC , is thus equal to the (rectangle contained) by AD and DB , plus the (square) on DE . And, of these, the (square) on DE is less than the (square) on EC . And, thus, the remaining (rectangle contained) by AC and CB is less than the (rectangle contained) by AD and DB . And, hence, twice the (rectangle contained) by AC and CB is less than twice the (rectangle contained) by AD and DB . And thus the remaining sum of the (squares) on AC and CB is greater than the sum of the (squares) on AD and DB .[†]

(Which is) the very thing it was required to show.

† Since, $AC^2 + CB^2 + 2 AC CB = AD^2 + DB^2 + 2 AD DB = AB^2$.

42.

Proposition 42

Ἡ ἐκ δύο ὀνομάτων κατὰ <ἐν μόνον σημείον διαιρεῖται εἰς τὰ ὀνόματα.

A binomial (straight-line) can be divided into its (component) terms at one point only.†



>Ἐστω ἐκ δύο ὀνομάτων ἡ AB διηρημένη εἰς τὰ ὀνόματα κατὰ τὸ Γ ; αἱ AG , GB >ἀρα <ρηταί εἰσι δυνάμει μόνον σύμμετροι. λέγω, <ὅτι ἡ AB κατ> >ἄλλο σημείον οὐ διαιρεῖται εἰς δύο <ρητὰς δυνάμει μόνον συμμέτρουξ. Εἰ γὰρ δυνατόν, διηρήσθω καὶ κατὰ τὸ Δ , <ὥστε καὶ τὰς AD , DB <ρητὰς ε>ἶναι δυνάμει μόνον συμμέτρουξ. φανερόν δὲ, <ὅτι ἡ AG τῇ DB οὐκ >έστιν ἡ αὐτή. εἰ γὰρ δυνατόν, >έστω. >έσται δὲ καὶ ἡ AD τῇ GB ἡ αὐτή; καὶ >έσται ὡς ἡ AG πρὸς τὴν GB , οὕτως ἡ BD πρὸς τὴν DA , καὶ >έσται ἡ AB κατὰ τὸ αὐτὸ τῇ κατὰ τὸ Γ διαιρέσει διαιρεθεῖσα καὶ κατὰ τὸ Δ ; <ὅπερ οὐθ' ὑπόκειται. οὐκ >ἀρα ἡ AG τῇ DB ἐστὶν ἡ αὐτή. διὰ δὲ τοῦτο καὶ τὰ Γ , Δ σημεῖα οὐκ >ίσον ἀπέθουσι τῇ διχοτομίᾳ. <ὦ> >ἀρα διαφέρει τὰ ἀπὸ τῶν AG , GB τῶν ἀπὸ τῶν AD , DB , τοῦτω διαφέρει καὶ τὸ διζ' ὑπὸ τῶν AD , DB τοῦ διζ' ὑπὸ τῶν AG , GB διὰ τὸ καὶ τὰ ἀπὸ τῶν AG , GB μετὰ τοῦ διζ' ὑπὸ τῶν AD , DB καὶ τὰ ἀπὸ τῶν AD , DB μετὰ τοῦ διζ' ὑπὸ τῶν AD , DB >ίσα ε>ἶναι τῷ ἀπὸ τῇ AB . ἄλλὰ τὰ ἀπὸ τῶν AG , GB τῶν ἀπὸ τῶν AD , DB διαφέρει <ρητῷ>; <ρητὰ γὰρ ἀμφοτέρω; καὶ τὸ διζ' >ἀρα ὑπὸ τῶν AD , DB τοῦ διζ' ὑπὸ τῶν AG , GB διαφέρει <ρητῷ> μέσα >όντα; <ὅπερ >άτοπον; μέσον γὰρ μέσου οὐθ' ὑπερέθει <ρητῷ>. Οὐθ' >ἀρα ἡ ἐκ δύο ὀνομάτων κατ> >ἄλλο καὶ >ἄλλο σημείον διαιρεῖται; καθ> <ἐν >ἀρα μόνον; <ὅπερ >έδει δεῖχαι.

Let AB be a binomial (straight-line) which has been divided into its (component) terms at C . AC and CB are thus rational (straight-lines which are) commensurable in square only [Prop. 10.36]. I say that AB cannot be divided at another point into two rational (straight-lines which are) commensurable in square only.

For, if possible, let it also be divided at D , such that AD and DB are also rational (straight-lines which are) commensurable in square only. So, (it is) clear that AC is not the same as DB . For, if possible, let it be (the same). So, AD will also be the same as CB . And as AC will be to CB , so BD (will be) to DA . And AB will (thus) also be divided at D in the same (manner) as the division at C . The very opposite was assumed. Thus, AC is not the same as DB . So, on account of this, points C and D are not equally far from the point of bisection. Thus, by whatever (amount the sum of) the (squares) on AC and CB differs from (the sum of) the (squares) on AD and DB , twice the (rectangle contained) by AD and DB also differs from twice the (rectangle contained) by AC and CB by this (same amount)—on account of both (the sum of) the (squares) on AC and CB , plus twice the (rectangle contained) by AC and CB , and (the sum of) the (squares) on AD and DB , plus twice the (rectangle contained) by AD and DB , being equal to the (square) on AB [Prop. 2.4]. But, (the sum of) the (squares) on AC and CB differs from (the sum of) the (squares) on AD and DB by a rational (area). For (they are) both rational (areas). Thus, twice the (rectangle contained) by AD and DB also differs from twice the (rectangle contained) by AC and CB by a rational (area, despite both) being medial (areas) [Prop. 10.21]. The very thing is absurd. For a medial (area) cannot exceed a medial (area) by a rational (area) [Prop. 10.26].

Thus, a binomial (straight-line) cannot be divided (into its component terms) at different points. Thus, (it can be so divided) at one point only. (Which is) the very thing it was required to show.

† In other words, $k + k'^{1/2} = k'' + k'''^{1/2}$ has only one solution: i.e., $k'' = k$ and $k''' = k'$. Likewise, $k^{1/2} + k'^{1/2} = k''^{1/2} + k'''^{1/2}$ has only one solution: i.e., $k'' = k$ and $k''' = k'$ (or, equivalently, $k'' = k'$ and $k''' = k$).

43.

Proposition 43

Ἡ ἐκ δύο μέσων πρώτη καθ> <ἐν μόνον σημείον διαιρεῖται.

A first bimedral (straight-line) can be divided (into its component terms) at one point only.†



>Ἐστω ἐκ δύο μέσων πρώτη ἡ AB διηρημένη κατὰ τὸ Γ , <ὥστε τὰς AG , GB μέσα ε>ἶναι δυνάμει μόνον συμμέτρουξ <ρητὸν περιεθούσαξ; λέγω, <ὅτι ἡ AB κατ> >ἄλλο σημείον

Let AB be a first bimedral (straight-line) which has been divided at C , such that AC and CB are medial (straight-lines), commensurable in square only, (and) containing a rational

οὐ διαιρεῖται.

Εἰ γὰρ δυνατόν διηρήσθω καὶ κατὰ τὸ Δ, <ὥστε καὶ τὰς ΑΔ, ΔΒ μέσαξ ε>ῖναι δυνάμει μόνον συμμέτρουξ <ρητὸν περιεθούσαξ. ἐπεὶ ο>ῦν, <ῶ| διαφέρει τὸ διξ ὑπὸ τῶν ΑΔ, ΔΒ τοῦ διξ ὑπὸ τῶν ΑΓ, ΓΒ, τοῦτω διαφέρει τὰ ἀπὸ τῶν ΑΓ, ΓΒ τῶν ἀπὸ τῶν ΑΔ, ΔΒ, <ρητῶ| δὲ διαφέρει τὸ διξ ὑπὸ τῶν ΑΔ, ΔΒ τοῦ διξ ὑπὸ τῶν ΑΓ, ΓΒ; <ρητὰ γὰρ ἀμφοτέρα; <ρητῶ| >άρα διαφέρει καὶ τὰ ἀπὸ τῶν ΑΓ, ΓΒ τῶν ἀπὸ τῶν ΑΔ, ΔΒ μέσα >όντα; <όπερ >άτοπον.

Οὐκ >άρα ἢ ἐκ δύο μέσων πρώτη κατ> >άλλο καὶ >άλλο σημεῖον διαιρεῖται εἰς τὰ ὀνόματα; καθ> <ἐν >άρα μόνον; <όπερ >έδει δεῖχαι.

(area) [Prop. 10.37]. I say that AB cannot be (so) divided at another point.

For, if possible, let it also be divided at D , such that AD and DB are also medial (straight-lines), commensurable in square only, (and) containing a rational (area). Since, therefore, by whatever (amount) twice the (rectangle contained) by AD and DB differs from twice the (rectangle contained) by AC and CB , (the sum of) the (squares) on AC and CB differs from (the sum of) the (squares) on AD and DB by this (same amount) [Prop. 10.41 lem.]. And twice the (rectangle contained) by AD and DB differs from twice the (rectangle contained) by AC and CB by a rational (area). For (they are) both rational (areas). (The sum of) the (squares) on AC and CB thus differs from (the sum of) the (squares) on AD and DB by a rational (area, despite both) being medial (areas). The very thing is absurd [Prop. 10.26]. Thus, a first bimedral (straight-line) cannot be divided into its (component) terms at different points. Thus, (it can be so divided) at one point only. (Which is) the very thing it was required to show.

† In other words, $k^{1/4} + k^{3/4} = k'^{1/4} + k'^{3/4}$ has only one solution: i.e., $k' = k$.

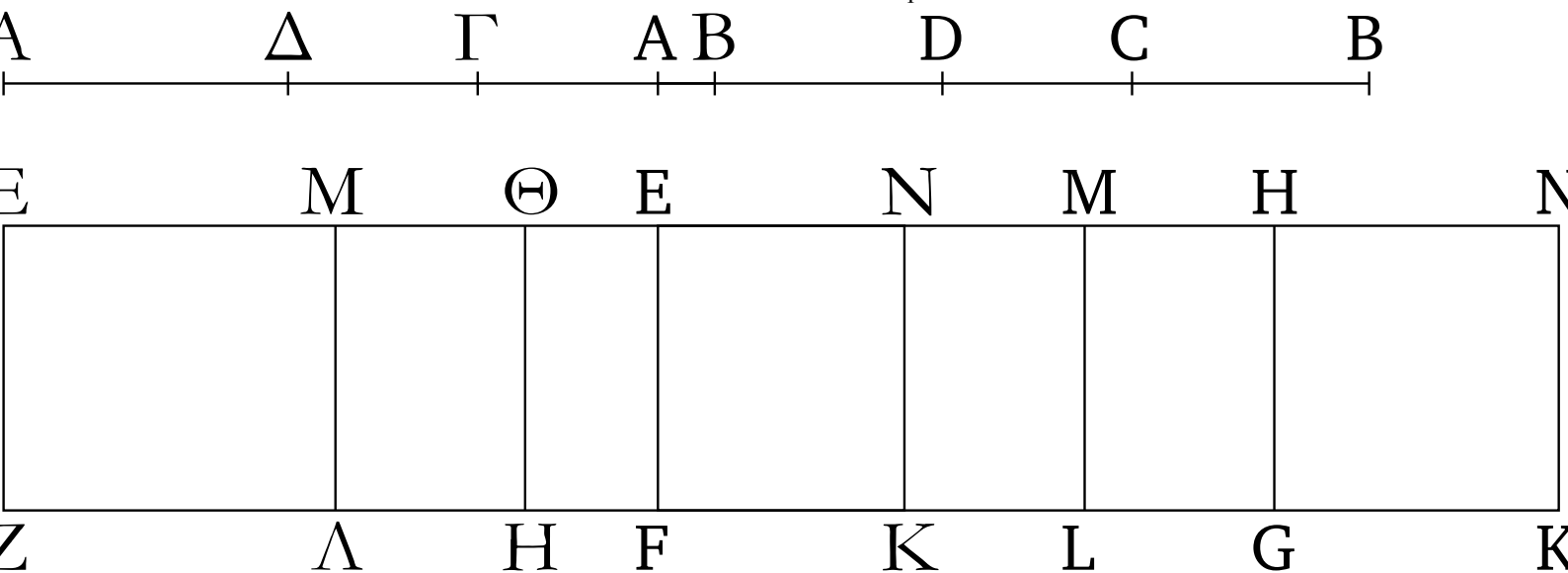
44.

Ἡ ἐκ δύο μέσων δευτέρα καθ> <ἐν μόνον σημεῖον διαιρεῖται. >Ἐστω ἐκ δύο μέσων δευτέρα ἢ ΑΒ διηρημένη κατὰ τὸ Γ, <ὥστε τὰς ΑΓ, ΓΒ μέσαξ ε>ῖναι δυνάμει μόνον συμμέτρουξ μέσον περιεθούσαξ; φανερόν δὴ, <ὅτι τὸ Γ οὐκ >έστι κατὰ τῆς διθотоμίας, <ὅτι οὐκ εἰσὶ μήκει σύμμετροι. λέγω, <ὅτι ἢ ΑΒ κατ> >άλλο σημεῖον οὐ διαιρεῖται.

Proposition 44

A second bimedral (straight-line) can be divided (into its component terms) at one point only.†

Let AB be a second bimedral (straight-line) which has been divided at C , so that AC and BC are medial (straight-lines), commensurable in square only, (and) containing a medial (area) [Prop. 10.38]. So, (it is) clear that C is not (located) at the point of bisection, since (AC and BC) are not commensurable in length. I say that AB cannot be (so) divided at another point.



Εἰ γὰρ δυνατόν, διηρήσθω καὶ κατὰ τὸ Δ, <ὥστε τὴν ΑΓ τῇ| ΔΒ μὴ ε>ῖναι τὴν αὐτήν, ἀλλὰ μείζονα καθ> ὑπόθεσιν τὴν ΑΓ; δῆλον δὴ, <ὅτι καὶ τὰ ἀπὸ τῶν ΑΔ, ΔΒ, ὡς ἐπάνω ἐδείξαμεν, ἐλάσσονα τῶν ἀπὸ τῶν ΑΓ, ΓΒ; καὶ τὰς ΑΔ, ΔΒ μέσαξ ε>ῖναι δυνάμει μόνον συμμέτρουξ μέσον περιεθούσαξ. καὶ ἐκκείσθω <ρητὴ ἢ ΕΖ, καὶ τῶ| μὲν ἀπὸ τῆς ΑΒ >ῖσον παρὰ τὴν ΕΖ παραλληλόγραμμον ὀρθογώνιον παραβεβλήσθω τὸ ΕΚ, τοῖξ δὲ ἀπὸ τῶν ΑΓ, ΓΒ >ῖσον ἀφηρήσθω τὸ ΕΗ; λοιπὸν >άρα τὸ ΘΚ >ῖσον ἐστὶ τῶ| διξ

For, if possible, let it also have been (so) divided at D , so that AC is not the same as DB , but AC (is), by hypothesis, greater. So, (it is) clear that (the sum of) the (squares) on AD and DB is also less than (the sum of) the (squares) on AC and CB , as we showed above [Prop. 10.41 lem.]. And AD and DB are medial (straight-lines), commensurable in square only, (and) containing a medial (area). And let the rational (straight-line) EF be laid down. And let the rectangular parallelogram EK , equal to the (square) on AB ,

ὑπὸ τῶν ΑΓ, ΓΒ. πάλιν δὲ τοῖς ἀπὸ τῶν ΑΔ, ΔΒ, <ἀπερ ἐλάσσονα ἐδείκθη τῶν ἀπὸ τῶν ΑΓ, ΓΒ, >ίσον ἀφηρήσθω τὸ ΕΛ; καὶ λοιπὸν >άρα τὸ ΜΚ >ίσον τῷ διζ ὑπὸ τῶν ΑΔ, ΔΒ. καὶ ἐπεὶ μέσα ἐστὶ τὰ ἀπὸ τῶν ΑΓ, ΓΒ, μέσον >άρα [καὶ] τὸ ΕΗ. καὶ παρὰ <ρητὴν τὴν ΕΖ παράκειται; <ρητὴ >άρα ἐστὶν ἡ ΕΘ καὶ ἀσύμμετρος τῇ ΕΖ μήκει. διὰ τὰ αὐτὰ δὲ καὶ ἡ ΘΝ <ρητὴ ἐστὶ καὶ ἀσύμμετρος τῇ ΕΖ μήκει. καὶ ἐπεὶ αἱ ΑΓ, ΓΒ μέσα εἰσὶ δυνάμει μόνον σύμμετροι, ἀσύμμετρος >άρα ἐστὶν ἡ ΑΓ τῇ ΓΒ μήκει. ὥς δὲ ἡ ΑΓ πρὸς τὴν ΓΒ, ο<ύτωξ τὸ ἀπὸ τῆς ΑΓ πρὸς τὸ ὑπὸ τῶν ΑΓ, ΓΒ; ἀσύμμετρον >άρα ἐστὶ τὸ ἀπὸ τῆς ΑΓ τῷ ὑπὸ τῶν ΑΓ, ΓΒ. ἀλλὰ τῷ μὲν ἀπὸ τῆς ΑΓ σύμμετρά ἐστὶ τὰ ἀπὸ τῶν ΑΓ, ΓΒ; δυνάμει γάρ εἰσι σύμμετροι αἱ ΑΓ, ΓΒ. τῷ δὲ ὑπὸ τῶν ΑΓ, ΓΒ σύμμετρόν ἐστὶ τὸ διζ ὑπὸ τῶν ΑΓ, ΓΒ. καὶ τὰ ἀπὸ τῶν ΑΓ, ΓΒ >άρα ἀσύμμετρά ἐστὶ τῷ διζ ὑπὸ τῶν ΑΓ, ΓΒ. ἀλλὰ τοῖς μὲν ἀπὸ τῶν ΑΓ, ΓΒ >ίσον ἐστὶ τὸ ΕΗ, τῷ δὲ διζ ὑπὸ τῶν ΑΓ, ΓΒ >ίσον τὸ ΘΚ; ἀσύμμετρον >άρα ἐστὶ τὸ ΕΗ τῷ ΘΚ; <ώστε καὶ ἡ ΕΘ τῇ ΘΝ ἀσύμμετρόξ ἐστὶ μήκει. καὶ εἰσι <ρηταί; αἱ ΕΘ, ΘΝ >άρα <ρηταί εἰσι δυνάμει μόνον σύμμετροι. ἔαν δὲ δύο <ρηταὶ δυνάμει μόνον σύμμετροι συντεθῶσιν, ἡ <όλη >αλογόξ ἐστὶν ἡ καλουμένη ἐκ δύο ὀνομάτων; ἡ ΕΝ >άρα ἐκ δύο ὀνομάτων ἐστὶ διηρημένη κατὰ τὸ Θ. κατὰ τὰ αὐτὰ δὲ δευθῆσονται καὶ αἱ ΕΜ, ΜΝ <ρηται δυνάμει μόνον σύμμετροι; καὶ >έσται ἡ ΕΝ ἐκ δύο ὀνομάτων κατ>άλλο καὶ >άλλο διηρημένη τὸ τε Θ καὶ τὸ Μ, καὶ οὐκ >έστιν ἡ ΕΘ τῇ ΜΝ ἢ αὐτῇ, <ότι τὰ ἀπὸ τῶν ΑΓ, ΓΒ μείζονά ἐστι τῶν ἀπὸ τῶν ΑΔ, ΔΒ. ἀλλὰ τὰ ἀπὸ τῶν ΑΔ, ΔΒ μείζονά ἐστι τοῦ διζ ὑπὸ ΑΔ, ΔΒ; πολλῶν >άρα καὶ τὰ ἀπὸ τῶν ΑΓ, ΓΒ, τουτέστι τὸ ΕΗ, μείζον ἐστὶ τοῦ διζ ὑπὸ τῶν ΑΔ, ΔΒ, τουτέστι τοῦ ΜΚ; <ώστε καὶ ἡ ΕΘ τῇ ΜΝ μείζων ἐστίν. ἡ >άρα ΕΘ τῇ ΜΝ οὐκ >έστιν ἡ αὐτῇ; <όπερ >έδει δεῖχαι.

be applied to EF . And let EG , equal to (the sum of) the (squares) on AC and CB , be cut off (from EK). Thus, the remainder, HK , is equal to twice the (rectangle contained) by AC and CB [Prop. 2.4]. So, again, let EL , equal to (the sum of) the (squares) on AD and DB —which was shown (to be) less than (the sum of) the (squares) on AC and CB —be cut off (from EK). And, thus, the remainder, MK , (is) equal to twice the (rectangle contained) by AD and DB . And since (the sum of) the (squares) on AC and CB is medial, EG (is) thus [also] medial. And it is applied to the rational (straight-line) EF . Thus, EH is rational, and incommensurable in length with EF [Prop. 10.22]. So, for the same (reasons), HN is also rational, and incommensurable in length with EF . And since AC and CB are medial (straight-lines which are) commensurable in square only, AC is thus incommensurable in length with CB . And as AC (is) to CB , so the (square) on AC (is) to the (rectangle contained) by AC and CB [Prop. 10.21 lem.]. Thus, the (square) on AC is incommensurable with the (rectangle contained) by AC and CB [Prop. 10.11]. But, (the sum of) the (squares) on AC and CB is commensurable with the (square) on AC . For, AC and CB are commensurable in square [Prop. 10.15]. And twice the (rectangle contained) by AC and CB is commensurable with the (rectangle contained) by AC and CB [Prop. 10.6]. And thus (the sum of) the squares on AC and CB is incommensurable with twice the (rectangle contained) by AC and CB [Prop. 10.13]. But, EG is equal to (the sum of) the (squares) on AC and CB , and HK equal to twice the (rectangle contained) by AC and CB . Thus, EG is incommensurable with HK . Hence, EH is also incommensurable in length with HN [Props. 6.1, 10.11]. And (they are) rational (straight-lines). Thus, EH and HN are rational (straight-lines which are) commensurable in square only. And if two rational (straight-lines which are) commensurable in square only are added together then the whole (straight-line) is that irrational called binomial [Prop. 10.36]. Thus, EN is a binomial (straight-line) which has been divided (into its component terms) at H . So, according to the same (reasoning), EM and MN can be shown (to be) rational (straight-lines which are) commensurable in square only. And EN will (thus) be a binomial (straight-line) which has been divided (into its component terms) at the different (points) H and M (which is absurd [Prop. 10.42]). And EH is not the same as MN , since (the sum of) the (squares) on AC and CB is greater than (the sum of) the (squares) on AD and DB . But, (the sum of) the (squares) on AD and DB is greater than twice the (rectangle contained) by AD and DB [Prop. 10.59 lem.]. Thus, (the sum of) the (squares) on AC and CB —that is to say, EG —is also much greater than twice the (rectangle contained) by AD and DB —that is to say, MK . Hence, EH is also greater than MN [Prop. 6.1]. Thus, EH is not the same as MN . (Which is) the very thing it was required to show.

† In other words, $k^{1/4} + k'^{1/2}/k^{1/4} = k''^{1/4} + k'''^{1/2}/k''^{1/4}$ has only one solution: i.e., $k'' = k$ and $k''' = k'$.

45.

Ἡ μείζων κατὰ τὸ αὐτὸ μόνον σημεῖον διαιρεῖται.

Proposition 45

A major (straight-line) can only be divided (into its component terms) at the same point.†



>Ἐστω μείζων ἢ AB διηρημένη κατὰ τὸ Γ , <ώστε τὰς AG , GB δυνάμει ἀσυμμέτρουξ ϵ >ἵναι ποιούσας τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν AG , GB τετραγώνων <ρητόν, τὸ δ > ὑπὸ τῶν AG , GB μέσον; λέγω, <ὅτι ἡ AB κατ> >ἄλλο σημεῖον οὐ διαιρεῖται.

Εἰ γὰρ δυνατόν, διηρήσθω καὶ κατὰ τὸ Δ , <ώστε καὶ τὰς $A\Delta$, ΔB δυνάμει ἀσυμμέτρουξ ϵ >ἵναι ποιούσας τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν $A\Delta$, ΔB <ρητόν, τὸ δ > ὑπ> αὐτῶν μέσον. καὶ ἐπεὶ, <ὦ| διαφέρει τὰ ἀπὸ τῶν AG , GB τῶν ἀπὸ τῶν $A\Delta$, ΔB , τοῦτω διαφέρει καὶ τὸ διζ ὑπὸ τῶν $A\Delta$, ΔB τοῦ διζ ὑπὸ τῶν AG , GB , ἀλλὰ τὰ ἀπὸ τῶν AG , GB τῶν ἀπὸ τῶν $A\Delta$, ΔB ὑπερέθει <ρητῶ|; <ρητὰ γὰρ ἀμφοτέρω; καὶ τὸ διζ ὑπὸ τῶν $A\Delta$, ΔB >ἄρα τοῦ διζ ὑπὸ τῶν AG , GB ὑπερέθει <ρητῶ| μέσα >όντα; <όπερ ἐστὶν ἀδύνατον. οὐκ >ἄρα ἡ μείζων κατ> >ἄλλο καὶ >ἄλλο σημεῖον διαιρεῖται; κατὰ τὸ αὐτὸ >ἄρα μόνον διαιρεῖται; <όπερ >έδει δεῖχαι.

Let AB be a major (straight-line) which has been divided at C , so that AC and CB are incommensurable in square, making the sum of the squares on AC and CB rational, and the (rectangle contained) by AC and CD medial [Prop. 10.39]. I say that AB cannot be (so) divided at another point.

For, if possible, let it also be divided at D , such that AD and DB are also incommensurable in square, making the sum of the (squares) on AD and DB rational, and the (rectangle contained) by them medial. And since, by whatever (amount the sum of) the (squares) on AC and CB differs from (the sum of) the (squares) on AD and DB , twice the (rectangle contained) by AD and DB also differs from twice the (rectangle contained) by AC and CB by this (same amount). But, (the sum of) the (squares) on AC and CB exceeds (the sum of) the (squares) on AD and DB by a rational (area). For (they are) both rational (areas). Thus, twice the (rectangle contained) by AD and DB also exceeds twice the (rectangle contained) by AC and CB by a rational (area), (despite both) being medial (areas). The very thing is impossible [Prop. 10.26]. Thus, a major (straight-line) cannot be divided (into its component terms) at different points. Thus, it can only be (so) divided at the same (point). (Which is) the very thing it was required to show.

† In other words, $\sqrt{[1 + k/(1 + k^2)^{1/2}]/2} + \sqrt{[1 - k/(1 + k^2)^{1/2}]/2} = \sqrt{[1 + k'/(1 + k'^2)^{1/2}]/2} + \sqrt{[1 - k'/(1 + k'^2)^{1/2}]/2}$ has only one solution: i.e., $k' = k$.

46.

Proposition 46

Ἡ <ρητόν καὶ μέσον δυναμένη καθ> <έν μόνον σημεῖον διαιρεῖται.

The square-root of a rational plus a medial (area) can be divided (into its component terms) at one point only.†



>Ἐστω <ρητόν καὶ μέσον δυναμένη ἢ AB διηρημένη κατὰ τὸ Γ , <ώστε τὰς AG , GB δυνάμει ἀσυμμέτρουξ ϵ >ἵναι ποιούσας τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν AG , GB μέσον, τὸ δὲ διζ ὑπὸ τῶν AG , GB <ρητόν; λέγω, <ὅτι ἡ AB κατ> >ἄλλο σημεῖον οὐ διαιρεῖται.

Εἰ γὰρ δυνατόν, διηρήσθω καὶ κατὰ τὸ Δ , <ώστε καὶ τὰς $A\Delta$, ΔB δυνάμει ἀσυμμέτρουξ ϵ >ἵναι ποιούσας τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν $A\Delta$, ΔB μέσον, τὸ δὲ διζ ὑπὸ τῶν $A\Delta$, ΔB <ρητόν. ἐπεὶ ο>ὖν, <ὦ| διαφέρει τὸ διζ ὑπὸ τῶν AG , GB τοῦ διζ ὑπὸ τῶν $A\Delta$, ΔB , τοῦτω διαφέρει καὶ τὰ ἀπὸ τῶν $A\Delta$, ΔB τῶν ἀπὸ τῶν AG , GB , τὸ δὲ διζ ὑπὸ τῶν AG , GB τοῦ διζ ὑπὸ τῶν $A\Delta$, ΔB ὑπερέθει <ρητῶ|, καὶ τὰ ἀπὸ τῶν $A\Delta$, ΔB >ἄρα τῶν ἀπὸ τῶν AG , GB ὑπερέθει <ρητῶ| μέσα >όντα; <όπερ ἐστὶν ἀδύνατον. οὐκ >ἄρα ἡ <ρητόν καὶ μέσον δυναμένη κατ> >ἄλλο καὶ >ἄλλο σημεῖον διαιρεῖται. κατὰ <έν >ἄρα σημεῖον διαιρεῖται; <όπερ >έδει δεῖχαι.

Let AB be the square-root of a rational plus a medial (area) which has been divided at C , so that AC and CB are incommensurable in square, making the sum of the (squares) on AC and CB medial, and twice the (rectangle contained) by AC and CB rational [Prop. 10.40]. I say that AB cannot be (so) divided at another point.

For, if possible, let it also be divided at D , so that AD and DB are also incommensurable in square, making the sum of the (squares) on AD and DB medial, and twice the (rectangle contained) by AD and DB rational. Therefore, since by whatever (amount) twice the (rectangle contained) by AC and CB differs from twice the (rectangle contained) by AD and DB , (the sum of) the (squares) on AD and DB also differs from (the sum of) the (squares) on AC and CB by this (same amount). And twice the (rectangle contained) by AC and CB exceeds twice the (rectangle contained) by AD and DB by a rational (area). (The sum of) the (squares) on AD and DB thus also exceeds (the sum of) the (squares) on AC and CB by a rational (area), (despite both) being medial (areas). The very thing is impossible [Prop. 10.26]. Thus, the square-root of a rational plus a medial (area) cannot be divided (into its component terms) at different points. Thus, it can be (so) divided at one point (only). (Which is)

the very thing it was required to show.

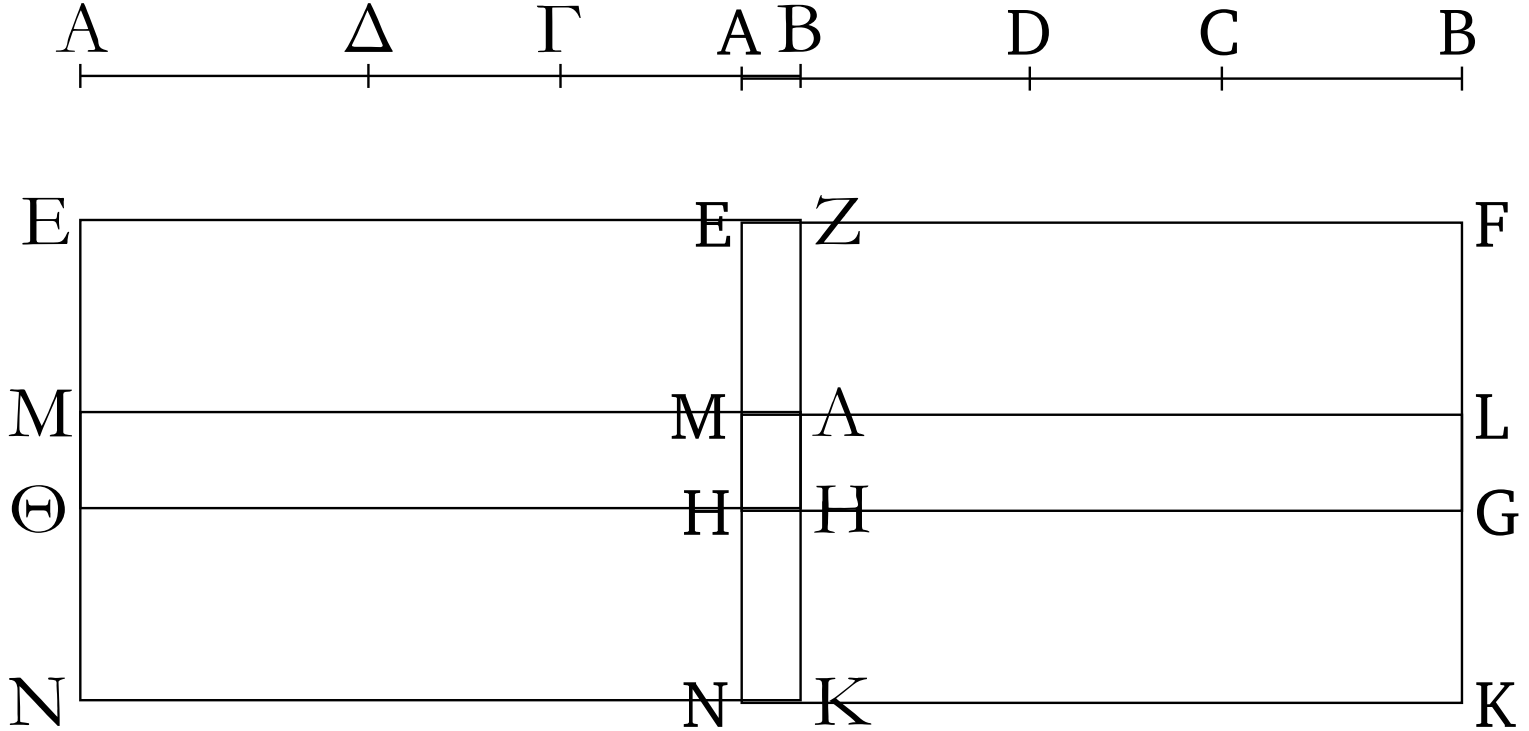
† In other words, $\sqrt{[(1+k^2)^{1/2} + k]/[2(1+k^2)]} + \sqrt{[(1+k^2)^{1/2} - k]/[2(1+k^2)]} = \sqrt{[(1+k'^2)^{1/2} + k']/[2(1+k'^2)]} + \sqrt{[(1+k'^2)^{1/2} - k']/[2(1+k'^2)]}$ has only one solution: i.e., $k' = k$.

47.

Proposition 47

Ἡ δύο μέσα δυναμένη καθ' <ἐν μόνον σημείον διαιρεῖται.

The square-root of (the sum of) two medial (areas) can be divided (into its component terms) at one point only.†



>Ἐστω [δύο μέσα δυναμένη] ἡ AB διηρημένη κατὰ τὸ Γ, <ώστε τὰς ΑΓ, ΓΒ δυνάμει ἀσύμμετρον >εἶναι ποιοῦσαξ τὸ τε συγκείμενον ἐκ τῶν ἀπὸ τῶν ΑΓ, ΓΒ μέσον καὶ τὸ ὑπὸ τῶν ΑΓ, ΓΒ μέσον καὶ >ἐτι ἀσύμμετρον τῶν συγκειμένων ἐκ τῶν ἀπ' αὐτῶν. λέγω, <ὅτι ἡ ΑΒ κατ' >ἄλλο σημείον οὐ διαιρεῖται ποιοῦσα τὰ προκείμενα.

Εἰ γὰρ δυνατόν, διηρήσθω κατὰ τὸ Δ, <ώστε πάλιν δηλονότι τὴν ΑΓ τῇ ΔΒ μὴ εἶναι τὴν αὐτήν, ἀλλὰ μείζονα καθ' >ὑπόθεσιν τὴν ΑΓ, καὶ ἐκκείσθω <ρητὴ ἡ ΕΖ, καὶ παραβεβλήσθω παρὰ τὴν ΕΖ τοῖς μὲν ἀπὸ τῶν ΑΓ, ΓΒ >ἴσον τὸ ΕΗ, τῶν δὲ διξ ὑπὸ τῶν ΑΓ, ΓΒ >ἴσον τὸ ΘΚ; <όλον >ἄρα τὸ ΕΚ >ἴσον ἐστὶ τῶν ἀπὸ τῆς ΑΒ τετραγώνω. πάλιν δὲ παραβεβλήσθω παρὰ τὴν ΕΖ τοῖς ἀπὸ τῶν ΑΔ, ΔΒ >ἴσον τὸ ΕΛ; λοιπὸν >ἄρα τὸ διξ ὑπὸ τῶν ΑΔ, ΔΒ λοιπῶν τῶν ΜΚ >ἴσον ἐστίν. καὶ ἐπεὶ μέσον ὑπόκειται τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΑΓ, ΓΒ, μέσον >ἄρα ἐστὶ καὶ τὸ ΕΗ. καὶ παρὰ <ρητὴν τὴν ΕΖ παράκειται; <ρητὴ >ἄρα ἐστὶν ἡ ΘΕ καὶ ἀσύμμετροξ τῇ ΕΖ μήκει. διὰ τὰ αὐτὰ δὲ καὶ ἡ ΘΝ <ρητὴ ἐστὶ καὶ ἀσύμμετροξ τῇ ΕΖ μήκει. καὶ ἐπεὶ ἀσύμμετρον ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΑΓ, ΓΒ τῶν διξ ὑπὸ τῶν ΑΓ, ΓΒ, καὶ τὸ ΕΗ >ἄρα τῶν ΗΝ ἀσύμμετρον ἐστίν; <ώστε καὶ ἡ ΕΘ τῇ ΘΝ ἀσύμμετρόξ ἐστίν. καὶ εἰσι <ρηταί; αἱ ΕΘ, ΘΝ >ἄρα <ρηταί εἰσι δυνάμει μόνον σύμμετροι; ἡ ΕΝ >ἄρα ἐκ δύο ὀνομάτων ἐστὶ διηρημένη κατὰ τὸ Θ. ὁμοίως δὲ δείχομεν, <ὅτι καὶ κατὰ τὸ Μ διήλρηται. καὶ οὐκ >ἐστὶν ἡ ΕΘ τῇ ΜΝ ἡ αὐτή; ἡ >ἄρα ἐκ δύο ὀνομάτων κατ' >ἄλλο καὶ >ἄλλο σημείον διήλρηται; <ὅπερ ἐστὶν >ἀτοπον. οὐκ >ἄρα ἡ δύο μέσα δυναμένη κατ' ἄλλο καὶ >ἄλλο σημείον διαιρεῖται; καθ' <ἐν >ἄρα μόνον [σημεῖον] διαιρεῖται.

Let AB be [the square-root of (the sum of) two medial (areas)] which has been divided at C , such that AC and CB are incommensurable in square, making the sum of the (squares) on AC and CB medial, and the (rectangle contained) by AC and CB medial, and, moreover, incommensurable with the sum of the (squares) on (AC and CB) [Prop. 10.41]. I say that AB cannot be divided at another point fulfilling the prescribed (conditions).

For, if possible, let it be divided at D , such that AC is again manifestly not the same as DB , but AC (is), by hypothesis, greater. And let the rational (straight-line) EF be laid down. And let EG , equal to (the sum of) the (squares) on AC and CB , and HK , equal to twice the (rectangle contained) by AC and CB , be applied to EF . Thus, the whole of EK is equal to the square on AB [Prop. 2.4]. So, again, let EL , equal to (the sum of) the (squares) on AD and DB , be applied to EF . Thus, the remainder—twice the (rectangle contained) by AD and DB —is equal to the remainder, MK . And since the sum of the (squares) on AC and CB was assumed (to be) medial, EG is also medial. And it is applied to the rational (straight-line) EF . HE is thus rational, and incommensurable in length with EF [Prop. 10.22]. So, for the same (reasons), HN is also rational, and incommensurable in length with EF . And since the sum of the (squares) on AC and CB is incommensurable with twice the (rectangle contained) by AC and CB , EG is thus also incommensurable with GN . Hence, EH is also incommensurable with HN [Props. 6.1, 10.11]. And they are (both) rational (straight-lines). Thus, EH and HN are rational

(straight-lines which are) commensurable in square only. Thus, EN is a binomial (straight-line) which has been divided (into its component terms) at H [Prop. 10.36]. So, similarly, we can show that it has also been (so) divided at M . And EH is not the same as MN . Thus, a binomial (straight-line) has been divided (into its component terms) at different points. The very thing is absurd [Prop. 10.42]. Thus, the square-root of (the sum of) two medial (areas) cannot be divided (into its component terms) at different points. Thus, it can be (so) divided at one [point] only.

† In other words, $k'^{1/4} \sqrt{[1 + k/(1 + k^2)^{1/2}]/2} + k'^{1/4} \sqrt{[1 - k/(1 + k^2)^{1/2}]/2} = k''^{1/4} \sqrt{[1 + k''/(1 + k''^2)^{1/2}]/2} + k''^{1/4} \sqrt{[1 - k''/(1 + k''^2)^{1/2}]/2}$ has only one solution: i.e., $k'' = k$ and $k''' = k'$.

<Όροι δεύτεροι.

5. Ὑποκειμένης <ρητῆς καὶ τῆς ἐκ δύο ὀνομάτων διηρημένης εἰς τὰ ὀνόματα, <ῆξ τὸ μείζον >όνομα τοῦ ἐλάσσονος μείζον δύναται τῷ ἀπὸ συμμετροῦ ἑαυτῇ| μήκει, ἔαν μὲν τὸ μείζον >όνομα σύμμετρον >ῆ| μήκει τῇ ἐκκειμένη <ρητῇ|, καλείσθω [ῆ <όλη] ἐκ δύο ὀνομάτων πρώτη.
6. Ἐὰν δὲ τὸ ἐλάσσον >όνομα σύμμετρον >ῆ| μήκει τῇ ἐκκειμένη <ρητῇ|, καλείσθω ἐκ δύο ὀνομάτων δευτέρα.
7. Ἐὰν δὲ μηδέτερον τῶν ὀνομάτων σύμμετρον >ῆ| μήκει τῇ ἐκκειμένη <ρητῇ|, καλείσθω ἐκ δύο ὀνομάτων τρίτη.
8. Πάλιν δὲ ἔαν τὸ μείζον >όνομα [τοῦ ἐλάσσονος] μείζον δύνηται τῷ ἀπὸ ἀσύμμετροῦ ἑαυτῇ| μήκει, ἔαν μὲν τὸ μείζον >όνομα σύμμετρον >ῆ| μήκει τῇ ἐκκειμένη <ρητῇ|, καλείσθω ἐκ δύο ὀνομάτων τετάρτη.
9. Ἐὰν δὲ τὸ >έλασσον, πέμπτη.
10. Ἐὰν δὲ μηδέτερον, <έκτη.

Definitions II

5. Given a rational (straight-line), and a binomial (straight-line) which has been divided into its (component) terms, of which the square on the greater term is larger than (the square on) the lesser by the (square) on (some straight-line) commensurable in length with (the greater) then, if the greater term is commensurable in length with the rational (straight-line previously) laid out, let [the whole] (straight-line) be called a first binomial (straight-line).
6. And if the lesser term is commensurable in length with the rational (straight-line previously) laid out then let (the whole straight-line) be called a second binomial (straight-line).
7. And if neither of the terms is commensurable in length with the rational (straight-line previously) laid out then let (the whole straight-line) be called a third binomial (straight-line).
8. So, again, if the square on the greater term is larger than (the square on) [the lesser] by the (square) on (some straight-line) incommensurable in length with (the greater) then, if the greater term is commensurable in length with the rational (straight-line previously) laid out, let (the whole straight-line) be called a fourth binomial (straight-line).
9. And if the lesser (term is commensurable), a fifth (binomial straight-line).
10. And if neither (term is commensurable), a sixth (binomial straight-line).

48.

Εὐρεῖν τὴν ἐκ δύο ὀνομάτων πρώτην.

Ἐκκείσθωσαν δύο ἀριθμοὶ οἱ ΑΓ, ΓΒ, <ώστε τὸν συγκείμενον ἔχ αὐτῶν τὸν ΑΒ πρὸς μὲν τὸν ΒΓ λόγον >έθειν, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, πρὸς δὲ τὸν ΓΑ λόγον μὴ >έθειν, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, καὶ ἐκκείσθω τις <ρητὴ ἢ Δ, καὶ τῇ Δ σύμμετρος >έστω μήκει ἢ ΕΖ. <ρητὴ >άρα ἐστὶ καὶ ἢ ΕΖ. καὶ γεγενέτω ὥς ὁ ΒΑ ἀριθμὸς πρὸς τὸν ΑΓ, ο<ύτως τὸ ἀπὸ τῆς ΕΖ πρὸς τὸ ἀπὸ τῆς ΖΗ. ὁ δὲ ΑΒ πρὸς τὸν ΑΓ λόγον >έθει, <ὄν ἀριθμὸς πρὸς ἀριθμόν; καὶ τὸ ἀπὸ τῆς ΕΖ >άρα πρὸς τὸ ἀπὸ τῆς ΖΗ λόγον >έθει, <ὄν ἀριθμὸς πρὸς ἀριθμόν; <ώστε σύμμετρον ἐστὶ τὸ ἀπὸ τῆς ΕΖ τῷ ἀπὸ τῆς ΖΗ. καὶ ἐστὶ <ρητὴ ἢ ΕΖ; <ρητὴ >άρα καὶ ἢ ΖΗ. καὶ ἐπεὶ ὁ ΒΑ πρὸς τὸν ΑΓ λόγον οὐκ >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, οὐδὲ τὸ ἀπὸ τῆς ΕΖ >άρα πρὸς τὸ ἀπὸ τῆς ΖΗ λόγον >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν; ἀσύμμετρος >άρα ἐστὶν ἢ ΕΖ τῇ ΖΗ μήκει. αἱ ΕΖ, ΖΗ >άρα <ρηταὶ εἰσι δυνάμει

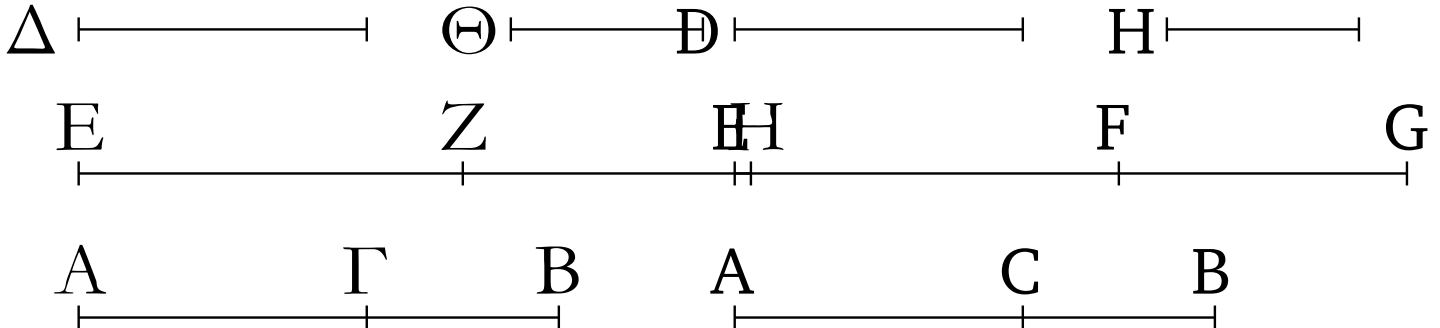
Proposition 48

To find a first binomial (straight-line).

Let two numbers AC and CB be laid down such that their sum AB has to BC the ratio which (some) square number (has) to (some) square number, and does not have to CA the ratio which (some) square number (has) to (some) square number [Prop. 10.28 lem. I]. And let some rational (straight-line) D be laid down. And let EF be commensurable in length with D . EF is thus also rational [Def. 10.3]. And let it be contrived that as the number BA (is) to AC , so the (square) on EF (is) to the (square) on FG [Prop. 10.6 corr.]. And AB has to AC the ratio which (some) number (has) to (some) number. Thus, the (square) on EF also has to the (square) on FG the ratio which (some) number (has) to (some) number. Hence, the (square) on EF is commensurable with the (square) on FG [Prop. 10.6]. And EF is rational. Thus, FG (is) also rational. And since BA does not have to AC the ratio which (some) square number (has) to (some) square number, thus the (square) on EF does not

μόνον σύμμετροι; ἐκ δύο >άρα ὀνομάτων ἐστὶν ἡ ΕΗ. λέγω,
<ὅτι καὶ πρώτη.

have to the (square) on FG the ratio which (some) square number (has) to (some) square number either. Thus, EF is incommensurable in length with FG [Prop. 10.9]. EF and FG are thus rational (straight-lines which are) commensurable in square only. Thus, EG is a binomial (straight-line) [Prop. 10.36]. I say that (it is) also a first (binomial straight-line).



Ἐπεὶ γάρ ἐστιν ὡς ὁ ΒΑ ἀριθμὸς πρὸς τὸν ΑΓ, ο<ύτως τὸ ἀπὸ τῆς ΕΖ πρὸς τὸ ἀπὸ τῆς ΖΗ, μείζων δὲ ὁ ΒΑ τοῦ ΑΓ, μείζων >άρα καὶ τὸ ἀπὸ τῆς ΕΖ τοῦ ἀπὸ τῆς ΖΗ. >έστω ο>ὖν τῷ ἀπὸ τῆς ΕΖ >ῖσα τὰ ἀπὸ τῶν ΖΗ, Θ. καὶ ἐπεὶ ἐστὶν ὡς ὁ ΒΑ πρὸς τὸν ΑΓ, ο<ύτως τὸ ἀπὸ τῆς ΕΖ πρὸς τὸ ἀπὸ τῆς ΖΗ, ἀναστρέψαντι >άρα ἐστὶν ὡς ὁ ΑΒ πρὸς τὸν ΒΓ, ο<ύτως τὸ ἀπὸ τῆς ΕΖ πρὸς τὸ ἀπὸ τῆς Θ. ὁ δὲ ΑΒ πρὸς τὸν ΒΓ λόγον >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. καὶ τὸ ἀπὸ τῆς ΕΖ >άρα πρὸς τὸ ἀπὸ τῆς Θ λόγον >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. σύμμετρος >άρα ἐστὶν ἡ ΕΖ τῇ Θ μήκει; ἡ ΕΖ >άρα τῆς ΖΗ μείζων δύναται τῷ ἀπὸ συμέτρου ἐαυτῇ. καὶ εἰσι <ρηταὶ αἱ ΕΖ, ΖΗ, καὶ σύμμετρος ἡ ΕΖ τῇ Δ μήκει. Ἡ ΕΗ >άρα ἐκ δύο ὀνομάτων ἐστὶ πρώτη; <όπερ >έδει δεῖχαι.

For since as the number BA is to AC , so the (square) on EF (is) to the (square) on FG , and BA (is) greater than AC , the (square) on EF (is) thus also greater than the (square) on FG [Prop. 5.14]. Therefore, let (the sum of) the (squares) on FG and H be equal to the (square) on EF . And since as BA is to AC , so the (square) on EF (is) to the (square) on FG , thus, via conversion, as AB is to BC , so the (square) on EF (is) to the (square) on H [Prop. 5.19 corr.]. And AB has to BC the ratio which (some) square number (has) to (some) square number. Thus, the (square) on EF also has to the (square) on H the ratio which (some) square number (has) to (some) square number. Thus, EF is commensurable in length with H [Prop. 10.9]. Thus, the square on EF is greater than (the square on) FG by the (square) on (some straight-line) commensurable (in length) with (EF). And EF and FG are rational (straight-lines). And EF (is) commensurable in length with D .

Thus, EG is a first binomial (straight-line) [Def. 10.5].[†] (Which is) the very thing it was required to show.

[†] If the rational straight-line has unit length then the length of a first binomial straight-line is $k + k\sqrt{1 - k'^2}$. This, and the first apotome, whose length is $k - k\sqrt{1 - k'^2}$ [Prop. 10.85], are the roots of $x^2 - 2kx + k^2k'^2 = 0$.

49.

Εὐρεῖν τὴν ἐκ δύο ὀνομάτων δευτέραν.

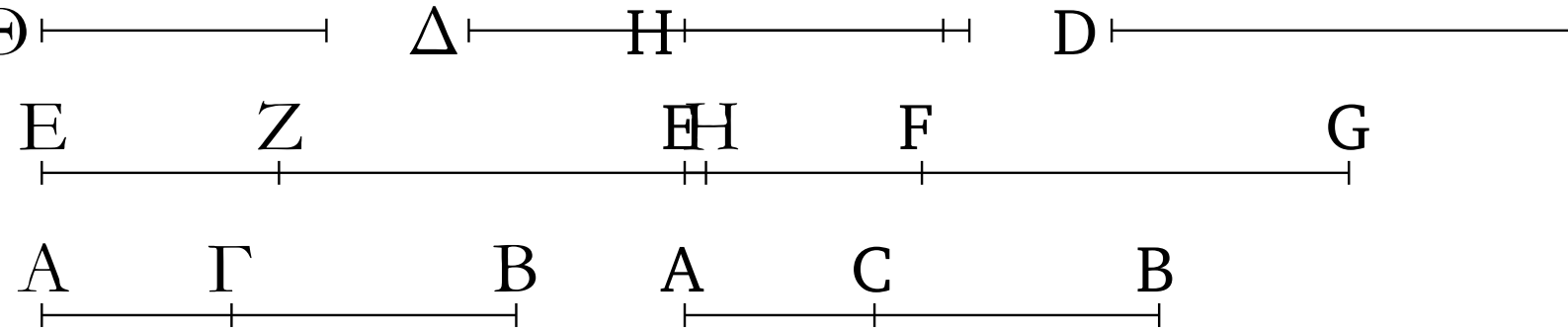
Ἐκκεῖσθωσαν δύο ἀριθμοὶ οἱ ΑΓ, ΓΒ, <ώστε τὸν συγκείμενον ἐχ αὐτῶν τὸν ΑΒ πρὸς μὲν τὸν ΒΓ λόγον >έθειν, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, πρὸς δὲ τὸν ΑΓ λόγον μὴ >έθειν, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, καὶ ἐκκεῖσθω <ρητὴ ἡ Δ, καὶ τῇ Δ σύμμετρος >έστω ἡ ΕΖ μήκει; <ρητὴ >άρα ἐστὶν ἡ ΕΖ. γεγονέτω δὴ καὶ ὡς ὁ ΓΑ ἀριθμὸς πρὸς τὸν ΑΒ, ο<ύτως τὸ ἀπὸ τῆς ΕΖ πρὸς τὸ ἀπὸ τῆς ΖΗ; σύμμετρον >άρα ἐστὶ τὸ ἀπὸ τῆς ΕΖ τῷ ἀπὸ τῆς ΖΗ. <ρητὴ >άρα ἐστὶ καὶ ἡ ΖΗ. καὶ ἐπεὶ ὁ ΓΑ ἀριθμὸς πρὸς τὸν ΑΒ λόγον οὐκ >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, οὐδὲ τὸ ἀπὸ τῆς ΕΖ πρὸς τὸ ἀπὸ τῆς ΖΗ λόγον >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. ἀσύμμετρος >άρα ἐστὶν ἡ ΕΖ τῇ ΖΗ μήκει; αἱ ΕΖ, ΖΗ >άρα <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι; ἐκ δύο >άρα ὀνομάτων ἐστὶν ἡ ΕΗ. δεικτέον δὴ, <ὅτι καὶ δευτέρα.

Proposition 49

To find a second binomial (straight-line).

Let the two numbers AC and CB be laid down such that their sum AB has to BC the ratio which (some) square number (has) to (some) square number, and does not have to AC the ratio which (some) square number (has) to (some) square number [Prop. 10.28 lem. I]. And let the rational (straight-line) D be laid down. And let EF be commensurable in length with D . EF is thus a rational (straight-line). So, let it also be contrived that as the number CA (is) to AB , so the (square) on EF (is) to the (square) on FG [Prop. 10.6 corr.]. Thus, the (square) on EF is commensurable with the (square) on FG [Prop. 10.6]. Thus, FG is also a rational (straight-line). And since the number CA does not have to AB the ratio which (some) square number (has) to (some) square number, the (square) on EF does not have to the (square) on FG the ratio which (some) square number (has) to (some) square number either. Thus, EF is incommensurable in length with FG [Prop. 10.9]. EF and FG are thus rational (straight-lines which are) commensu-

table in square only. Thus, EG is a binomial (straight-line) [Prop. 10.36]. So, we must show that (it is) also a second (binomial straight-line).



Ἐπεὶ γὰρ ἀνάπαλιν ἐστὶν ὥς ὁ BA ἀριθμὸς πρὸς τὸν $ΑΓ$, οὐτως τὸ ἀπὸ τῆς HZ πρὸς τὸ ἀπὸ τῆς ZE , μείζων δὲ ὁ BA τοῦ $ΑΓ$, μείζων ἄρα [καὶ] τὸ ἀπὸ τῆς HZ τοῦ ἀπὸ τῆς ZE . >έστω τῷ ἀπὸ τῆς HZ >ῖσα τὰ ἀπὸ τῶν EZ , $Θ$; ἀναστρέψαντι ἄρα ἐστὶν ὥς ὁ AB πρὸς τὸν $ΒΓ$, οὐτως τὸ ἀπὸ τῆς ZH πρὸς τὸ ἀπὸ τῆς $Θ$. ἀλλ' ὁ AB πρὸς τὸν $ΒΓ$ λόγον >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν; καὶ τὸ ἀπὸ τῆς ZH ἄρα πρὸς τὸ ἀπὸ τῆς $Θ$ λόγον >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν. σύμμετρος ἄρα ἐστὶν ἡ ZH τῇ $Θ$ μήκει; <ὥστε ἡ ZH τῆς ZE μείζων δύναται τῷ ἀπὸ συμέτρου ἐαυτῇ. καὶ εἰσι <ρηταὶ αἱ ZH , ZE δυνάμει μόνον σύμμετροι, καὶ τὸ EZ >έλασσον >όνομα τῇ ἐκκειμένῃ <ρητῇ σύμμετρόν ἐστι τῇ $Δ$ μήκει.

Ἡ EH ἄρα ἐκ δύο ὀνομάτων ἐστὶ δευτέρα; <όπερ >έδει δεῖχαι.

For since, inversely, as the number BA is to AC , so the (square) on GF (is) to the (square) on FE [Prop. 5.7 corr.], and BA (is) greater than AC , the (square) on GF (is) thus [also] greater than the (square) on FE [Prop. 5.14]. Let (the sum of) the (squares) on EF and H be equal to the (square) on GF . Thus, via conversion, as AB is to BC , so the (square) on FG (is) to the (square) on H [Prop. 5.19 corr.]. But, AB has to BC the ratio which (some) square number (has) to (some) square number. Thus, the (square) on FG also has to the (square) on H the ratio which (some) square number (has) to (some) square number. Thus, FG is commensurable in length with H [Prop. 10.9]. Hence, the square on FG is greater than (the square on) FE by the (square) on (some straight-line) commensurable in length with (FG). And FG and FE are rational (straight-lines which are) commensurable in square only. And the lesser term EF is commensurable in length with the rational (straight-line) D (previously) laid down.

Thus, EG is a second binomial (straight-line) [Def. 10.6].[†] (Which is) the very thing it was required to show.

[†] If the rational straight-line has unit length then the length of a second binomial straight-line is $k/\sqrt{1-k'^2} + k$. This, and the second apotome, whose length is $k/\sqrt{1-k'^2} - k$ [Prop. 10.86], are the roots of $x^2 - (2k/\sqrt{1-k'^2})x + k^2 [k'^2/(1-k'^2)] = 0$.

50.

Εὐρεῖν τὴν ἐκ δύο ὀνομάτων τρίτην.

Ἐκκείσθωσαν δύο ἀριθμοὶ οἱ $ΑΓ$, $ΓΒ$, <ὥστε τὸν συγκείμενον ἐξ αὐτῶν τὸν AB πρὸς μὲν τὸν $ΒΓ$ λόγον >έθειν, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν, πρὸς δὲ τὸν $ΑΓ$ λόγον μὴ >έθειν, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν. ἐκκείσθω δὲ τις καὶ >ἄλλος μὴ τετράγωνος ἀριθμὸς ὁ $Δ$, καὶ πρὸς ἑκάτερον τῶν BA , $ΑΓ$ λόγον μὴ ἐθέτω, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν; καὶ ἐκκείσθω τις <ρητὴ εὐθεῖα ἡ E , καὶ γερονέτω ὥς ὁ $Δ$ πρὸς τὸν AB , οὐτως τὸ ἀπὸ τῆς E πρὸς τὸ ἀπὸ τῆς ZH ; σύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς E τῷ ἀπὸ τῆς ZH . καὶ ἐστὶ <ρητὴ ἡ E ; <ρητὴ ἄρα ἐστὶ καὶ ἡ ZH . καὶ ἐπεὶ ὁ $Δ$ πρὸς τὸν AB λόγον οὐκ >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν, οὐδὲ τὸ ἀπὸ τῆς E πρὸς τὸ ἀπὸ τῆς ZH λόγον >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν; ἀσύμμετρος ἄρα ἐστὶν ἡ E τῇ ZH μήκει. γερονέτω δὲ πάλιν ὥς ἡ BA ἀριθμὸς πρὸς τὸν $ΑΓ$, οὐτως τὸ ἀπὸ τῆς ZH πρὸς τὸ ἀπὸ τῆς $HΘ$; σύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς ZH τῷ ἀπὸ τῆς $HΘ$. <ρητὴ δὲ ἡ ZH ; <ρητὴ ἄρα καὶ ἡ $HΘ$. καὶ ἐπεὶ ὁ BA πρὸς τὸν $ΑΓ$ λόγον οὐκ >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν, οὐδὲ τὸ ἀπὸ τῆς ZH πρὸς τὸ ἀπὸ τῆς $ΘH$ λόγον

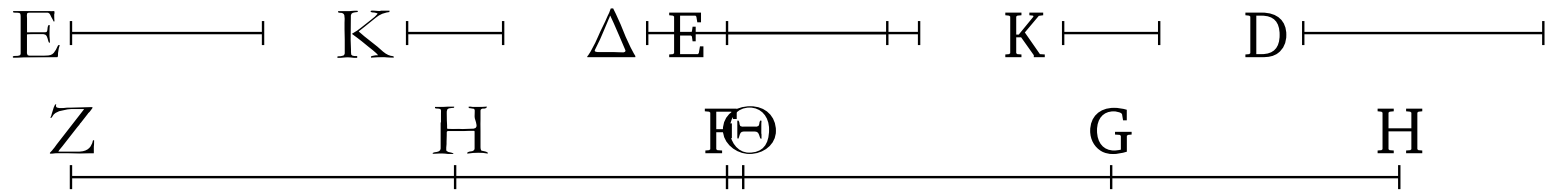
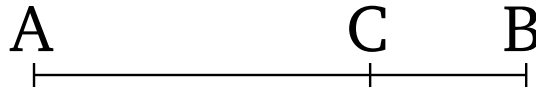
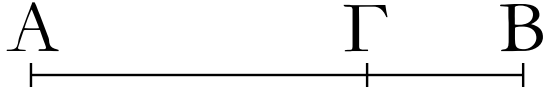
Proposition 50

To find a third binomial (straight-line).

Let the two numbers AC and CB be laid down such that their sum AB has to BC the ratio which (some) square number (has) to (some) square number, and does not have to AC the ratio which (some) square number (has) to (some) square number. And let some other non-square number D also be laid down, and let it not have to each of BA and AC the ratio which (some) square number (has) to (some) square number. And let some rational straight-line E be laid down, and let it be contrived that as D (is) to AB , so the (square) on E (is) to the (square) on FG [Prop. 10.6 corr.]. Thus, the (square) on E is commensurable with the (square) on FG [Prop. 10.6]. And E is a rational (straight-line). Thus, FG is also a rational (straight-line). And since D does not have to AB the ratio which (some) square number has to (some) square number, the (square) on E does not have to the (square) on FG the ratio which (some) square number (has) to (some) square number either. E is thus incommensurable in length with FG [Prop. 10.9]. So, again, let it be contrived that as the number BA (is) to AC , so the (square) on FG (is) to the (square) on GH [Prop. 10.6 corr.]. Thus, the (square) on FG is commensurable with the (square) on

>έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν; ἀσύμμετρος >άρα ἐστὶν ἡ ΖΗ τῇ ΗΘ μήκει. αἱ ΖΗ, ΗΘ >άρα <ρηταί εἰσι δυνάμει μόνον σύμμετροι; ἡ ΖΘ >άρα ἐκ δύο ὀνομάτων ἐστίν. λέγω δὴ, <ὅτι καὶ τρίτη.

GH [Prop. 10.6]. And FG (is) a rational (straight-line). Thus, GH (is) also a rational (straight-line). And since BA does not have to AC the ratio which (some) square number (has) to (some) square number, the (square) on FG does not have to the (square) on HG the ratio which (some) square number (has) to (some) square number either. Thus, FG is incommensurable in length with GH [Prop. 10.9]. FG and GH are thus rational (straight-lines which are) commensurable in square only. Thus, FH is a binomial (straight-line) [Prop. 10.36]. So, I say that (it is) also a third (binomial straight-line).



Ἐπεὶ γάρ ἐστιν ὡς ὁ Δ πρὸς τὸν ΑΒ, ο<ύτως τὸ ἀπὸ τῆς Ε πρὸς τὸ ἀπὸ τῆς ΖΗ, ὡς δὲ ὁ ΒΑ πρὸς τὸν ΑΓ, ο<ύτως τὸ ἀπὸ τῆς ΖΗ πρὸς τὸ ἀπὸ τῆς ΗΘ, δι> >ἴσου >άρα ἐστὶν ὡς ὁ Δ πρὸς τὸν ΑΓ, ο<ύτως τὸ ἀπὸ τῆς Ε πρὸς τὸ ἀπὸ τῆς ΗΘ. ὁ δὲ Δ πρὸς τὸν ΑΓ λόγον οὐκ >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν; οὐδὲ τὸ ἀπὸ τῆς Ε >άρα πρὸς τὸ ἀπὸ τῆς ΗΘ λόγον >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν; ἀσύμμετρος >άρα ἐστὶν ἡ Ε τῇ ΗΘ μήκει. καὶ ἐπεὶ ἐστὶν ὡς ὁ ΒΑ πρὸς τὸν ΑΓ, ο<ύτως τὸ ἀπὸ τῆς ΖΗ πρὸς τὸ ἀπὸ τῆς ΗΘ, μείζον >άρα τὸ ἀπὸ τῆς ΖΗ τοῦ ἀπὸ τῆς ΗΘ. >έστω ο>ὖν τῷ ἀπὸ τῆς ΖΗ >ἴσα τὰ ἀπὸ τῶν ΗΘ, Κ; ἀναστρέψαντι >άρα [ἐστίν] ὡς ὁ ΑΒ πρὸς τὸν ΒΓ, ο<ύτως τὸ ἀπὸ τῆς ΖΗ πρὸς τὸ ἀπὸ τῆς Κ. ὁ δὲ ΑΒ πρὸς τὸν ΒΓ λόγον >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν; καὶ τὸ ἀπὸ τῆς ΖΗ >άρα πρὸς τὸ ἀπὸ τῆς Κ λόγον >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν; σύμμετρος >άρα [ἐστίν] ἡ ΖΗ τῇ Κ μήκει. ἡ ΖΗ >άρα τῆς ΗΘ μείζον δύναται τῷ ἀπὸ συμέτρου ἑαυτῇ. καὶ εἰσιν αἱ ΖΗ, ΗΘ <ρηταί δυνάμει μόνον σύμμετροι, καὶ οὐδετέρα αὐτῶν σύμμετρος ἐστὶ τῇ Ε μήκει.

Ἡ ΖΘ >άρα ἐκ δύο ὀνομάτων ἐστὶ τρίτη; <όπερ >έδει δεῖξαι.

For since as D is to AB , so the (square) on E (is) to the (square) on FG , and as BA (is) to AC , so the (square) on FG (is) to the (square) on GH , thus, via equality, as D (is) to AC , so the (square) on E (is) to the (square) on GH [Prop. 5.22]. And D does not have to AC the ratio which (some) square number (has) to (some) square number. Thus, the (square) on E does not have to the (square) on GH the ratio which (some) square number (has) to (some) square number either. Thus, E is incommensurable in length with GH [Prop. 10.9]. And since as BA is to AC , so the (square) on FG (is) to the (square) on GH , the (square) on FG (is) thus greater than the (square) on GH [Prop. 5.14]. Therefore, let (the sum of) the (squares) on GH and K be equal to the (square) on FG . Thus, via conversion, as AB [is] to BC , so the (square) on FG (is) to the (square) on K [Prop. 5.19 corr.]. And AB has to BC the ratio which (some) square number (has) to (some) square number. Thus, the (square) on FG also has to the (square) on K the ratio which (some) square number (has) to (some) square number. Thus, FG [is] commensurable in length with K [Prop. 10.9]. Thus, the square on FG is greater than (the square on) GH by the (square) on (some straight-line) commensurable (in length) with (FG) . And FG and GH are rational (straight-lines which are) commensurable in square only, and neither of them is commensurable in length with E .

Thus, FH is a third binomial (straight-line) [Def. 10.7].[†] (Which is) the very thing it was required to show.

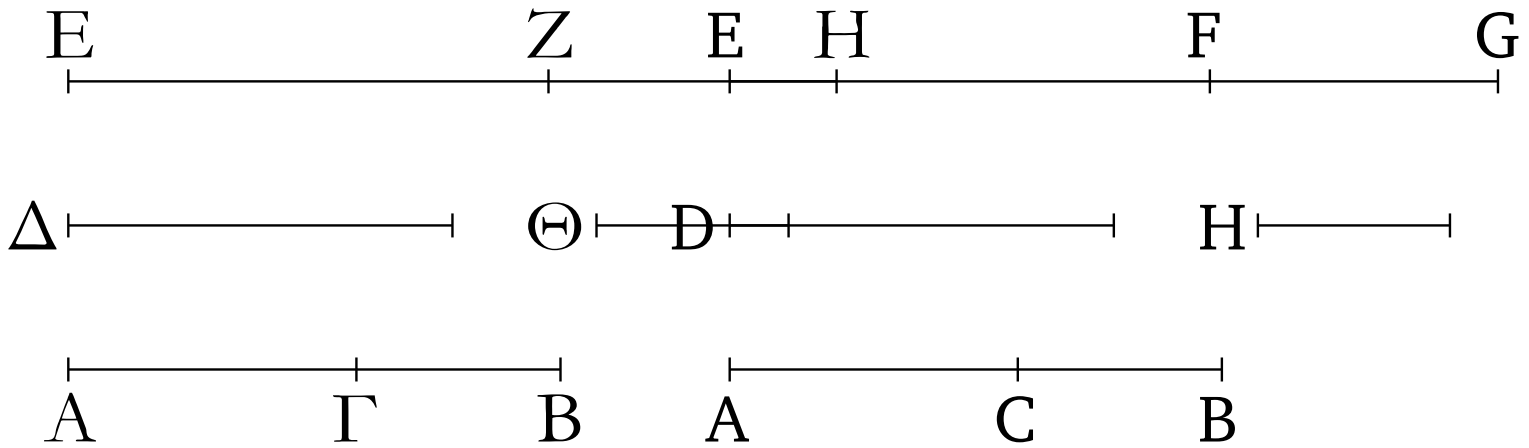
[†] If the rational straight-line has unit length then the length of a third binomial straight-line is $k^{1/2} (1 + \sqrt{1 - k'^2})$. This, and the third apotome, whose length is $k^{1/2} (1 - \sqrt{1 - k'^2})$ [Prop. 10.87], are the roots of $x^2 - 2k^{1/2}x + k k'^2 = 0$.

51.

Proposition 51

Εὐρεῖν τὴν ἐκ δύο ὀνομάτων τετάρτην.

To find a fourth binomial (straight-line).



Ἐκκείσθωσαν δύο ἀριθμοὶ οἱ ΑΓ, ΓΒ, <ώστε τὸν ΑΒ
πρὸς τὸν ΒΓ λόγον μὴ >έθειν μήτε μὴν πρὸς τὸν ΑΓ,
<ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. καὶ
ἐκκείσθω <ρητὴ ἢ Δ, καὶ τῇ Δ σύμμετρος >έστω μήκει
ἢ ΕΖ; <ρητὴ >άρα ἐστὶ καὶ ἡ ΕΖ. καὶ γεγονέτω ὥς ὁ ΒΑ
ἀριθμὸς πρὸς τὸν ΑΓ, ο<ύτωξ τὸ ἀπὸ τῆς ΕΖ πρὸς τὸ ἀπὸ
τῆς ΖΗ; σύμμετρον >άρα ἐστὶ τὸ ἀπὸ τῆς ΕΖ τῷ ἀπὸ τῆς
ΖΗ; <ρητὴ >άρα ἐστὶ καὶ ἡ ΖΗ. καὶ ἐπεὶ ὁ ΒΑ πρὸς τὸν ΑΓ
λόγον οὐκ >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον
ἀριθμόν, οὐδὲ τὸ ἀπὸ τῆς ΕΖ πρὸς τὸ ἀπὸ τῆς ΖΗ λόγον
>έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν;
ἀσύμμετρος >άρα ἐστὶν ἡ ΕΖ τῇ ΖΗ μήκει. αἱ ΕΖ, ΖΗ >άρα
<ρηταὶ εἰσι δυνάμει μόνον σύμμετροι; <ώστε ἡ ΕΗ ἐκ δύο
ὀνομάτων ἐστίν. λέγω δὴ, <ὅτι καὶ τετάρτη.

Ἐπεὶ γάρ ἐστιν ὥς ὁ ΒΑ πρὸς τὸν ΑΓ, ο<ύτωξ τὸ ἀπὸ
τῆς ΕΖ πρὸς τὸ ἀπὸ τῆς ΖΗ [μείζων δὲ ὁ ΒΑ τοῦ ΑΓ],
μείζων >άρα τὸ ἀπὸ τῆς ΕΖ τοῦ ἀπὸ τῆς ΖΗ. >έστω ο>ύν
τῷ ἀπὸ τῆς ΕΖ >ῖσα τὰ ἀπὸ τῶν ΖΗ, Θ; ἀναστρέψαντι
>άρα ὥς ὁ ΑΒ ἀριθμὸς πρὸς τὸν ΒΓ, ο<ύτωξ τὸ ἀπὸ τῆς
ΕΖ πρὸς τὸ ἀπὸ τῆς Θ. ὁ δὲ ΑΒ πρὸς τὸν ΒΓ λόγον οὐκ
>έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν;
οὐδ< >άρα τὸ ἀπὸ τῆς ΕΖ πρὸς τὸ ἀπὸ τῆς Θ λόγον
>έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν.
ἀσύμμετρος >άρα ἐστὶν ἡ ΕΖ τῇ Θ μήκει; ἡ ΕΖ >άρα τῆς
ΗΖ μείζων δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ]. καὶ εἰσιν αἱ
ΕΖ, ΖΗ <ρηταὶ δυνάμει μόνον σύμμετροι, καὶ ἡ ΕΖ τῇ Δ
σύμμετρος ἐστὶ μήκει.

Ἡ ΕΗ >άρα ἐκ δύο ὀνομάτων ἐστὶ τετάρτη; <όπερ >έδει
δεῖξαι.

Let the two numbers AC and CB be laid down such
that AB does not have to BC , or to AC either, the ratio
which (some) square number (has) to (some) square number
[Prop. 10.28 lem. I]. And let the rational (straight-line) D
be laid down. And let EF be commensurable in length with
 D . Thus, EF is also a rational (straight-line). And let it be
contrived that as the number BA (is) to AC , so the (square)
on EF (is) to the (square) on FG [Prop. 10.6 corr.]. Thus,
the (square) on EF is commensurable with the (square) on
 FG [Prop. 10.6]. Thus, FG is also a rational (straight-line).
And since BA does not have to AC the ratio which (some)
square number (has) to (some) square number, the (square)
on EF does not have to the (square) on FG the ratio which
(some) square number (has) to (some) square number ei-
ther. Thus, EF is incommensurable in length with FG
[Prop. 10.9]. Thus, EF and FG are rational (straight-lines
which are) commensurable in square only. Hence, EG is a
binomial (straight-line) [Prop. 10.36]. So, I say that (it is)
also a fourth (binomial straight-line).

For since as BA is to AC , so the (square) on EF (is) to
the (square) on FG [and BA (is) greater than AC], the
(square) on EF (is) thus greater than the (square) on FG
[Prop. 5.14]. Therefore, let (the sum of) the squares on FG
and H be equal to the (square) on EF . Thus, via conversion,
as the number AB (is) to BC , so the (square) on EF (is) to
the (square) on H [Prop. 5.19 corr.]. And AB does not have
to BC the ratio which (some) square number (has) to (some)
square number. Thus, the (square) on EF does not have to
the (square) on H the ratio which (some) square number
(has) to (some) square number either. Thus, EF is incom-
mensurable in length with H [Prop. 10.9]. Thus, the square
on EF is greater than (the square on) GF by the (square)
on (some straight-line) incommensurable (in length) with
(EF). And EF and FG are rational (straight-lines which
are) commensurable in square only. And EF is commen-
surable in length with D .

Thus, EG is a fourth binomial (straight-line) [Def. 10.8].[†]
(Which is) the very thing it was required to show.

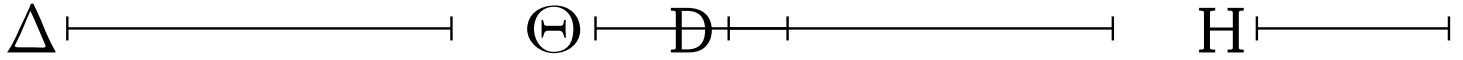
[†] If the rational straight-line has unit length then the length of a fourth binomial straight-line is $k(1 + 1/\sqrt{1+k'})$. This, and the fourth apotome, whose
length is $k(1 - 1/\sqrt{1+k'})$ [Prop. 10.88], are the roots of $x^2 - 2kx + k^2k'/(1+k') = 0$.

52.

Εὐρεῖν τὴν ἐκ δύο ὀνομάτων πέμπτην.

Proposition 52

To find a fifth binomial straight-line.



Ἐκκεῖσθωσαν δύο ἀριθμοὶ οἱ ΑΓ, ΓΒ, <ώστε τὸν ΑΒ πρὸς ἑκάτερον αὐτῶν λόγον μὴ >έθειν, <ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν, καὶ ἐκκεῖσθω <ρητὴ τις εὐθεῖα ἢ Δ, καὶ τῇ Δ σύμμετρος >έστω [μήκει] ἢ ΕΖ; <ρητὴ >άρα ἢ ΕΖ. καὶ γεγενέτω ὡς ὁ ΓΑ πρὸς τὸν ΑΒ, ο<ύτως τὸ ἀπὸ τῆς ΕΖ πρὸς τὸ ἀπὸ τῆς ΖΗ. ὁ δὲ ΓΑ πρὸς τὸν ΑΒ λόγον οὐκ >έθει, <ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν; οὐδὲ τὸ ἀπὸ τῆς ΕΖ >άρα πρὸς τὸ ἀπὸ τῆς ΖΗ λόγον >έθει, <ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν. αἱ ΕΖ, ΖΗ >άρα <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι; ἐκ δύο >άρα ὀνομάτων ἐστὶν ἢ ΕΗ. λέγω δὴ, <ὅτι καὶ πέμπτη. Ἐπεὶ γάρ ἐστιν ὡς ὁ ΓΑ πρὸς τὸν ΑΒ, ο<ύτως τὸ ἀπὸ τῆς ΕΖ πρὸς τὸ ἀπὸ τῆς ΖΗ, ἀνάπαλιν ὡς ὁ ΒΑ πρὸς τὸν ΑΓ, ο<ύτως τὸ ἀπὸ τῆς ΖΗ πρὸς τὸ ἀπὸ τῆς ΖΕ; μείζον >άρα τὸ ἀπὸ τῆς ΗΖ τοῦ ἀπὸ τῆς ΖΕ. >έστω ο<ὺν τῷ ἀπὸ τῆς ΗΖ >ῖσα τὰ ἀπὸ τῶν ΕΖ, Θ; ἀναστρέψαντι >άρα ἐστὶν ὡς ὁ ΑΒ ἀριθμὸς πρὸς τὸν ΒΓ, ο<ύτως τὸ ἀπὸ τῆς ΗΖ πρὸς τὸ ἀπὸ τῆς Θ. ὁ δὲ ΑΒ πρὸς τὸν ΒΓ λόγον οὐκ >έθει, <ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν; οὐδ< >άρα τὸ ἀπὸ τῆς ΖΗ πρὸς τὸ ἀπὸ τῆς Θ λόγον >έθει, <ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν. ἀσύμμετρος >άρα ἐστὶν ἢ ΖΗ τῇ Θ μήκει; <ώστε ἢ ΖΗ τῆς ΖΕ μείζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ. καὶ εἰσιν αἱ ΗΖ, ΖΕ <ρηταὶ δυνάμει μόνον σύμμετροι, καὶ τὸ ΕΖ >έλαττον >όνομα σύμμετρόν ἐστι τῇ ἐκκειμένῃ <ρητῇ τῇ Δ μήκει. Ἡ ΕΗ >άρα ἐκ δύο ὀνομάτων ἐστὶ πέμπτη; <όπερ >έδει δεῖχαι.

Let the two numbers AC and CB be laid down such that AB does not have to either of them the ratio which (some) square number (has) to (some) square number [Prop. 10.38 lem.]. And let some rational straight-line D be laid down. And let EF be commensurable [in length] with D . Thus, EF (is) a rational (straight-line). And let it be contrived that as CA (is) to AB , so the (square) on EF (is) to the (square) on FG [Prop. 10.6 corr.]. And CA does not have to AB the ratio which (some) square number (has) to (some) square number. Thus, the (square) on EF does not have to the (square) on FG the ratio which (some) square number (has) to (some) square number either. Thus, EF and FG are rational (straight-lines which are) commensurable in square only [Prop. 10.9]. Thus, EG is a binomial (straight-line) [Prop. 10.36]. So, I say that (it is) also a fifth (binomial straight-line).

For since as CA is to AB , so the (square) on EF (is) to the (square) on FG , inversely, as BA (is) to AC , so the (square) on FG (is) to the (square) on FE [Prop. 5.7 corr.]. Thus, the (square) on GF (is) greater than the (square) on FE [Prop. 5.14]. Therefore, let (the sum of) the (squares) on EF and H be equal to the (square) on GF . Thus, via conversion, as the number AB is to BC , so the (square) on GF (is) to the (square) on H [Prop. 5.19 corr.]. And AB does not have to BC the ratio which (some) square number (has) to (some) square number. Thus, the (square) on FG does not have to the (square) on H the ratio which (some) square number (has) to (some) square number either. Thus, FG is incommensurable in length with H [Prop. 10.9]. Hence, the square on FG is greater than (the square on) FE by the (square) on (some straight-line) incommensurable (in length) with (FG). And GF and FE are rational (straight-lines which are) commensurable in square only. And the lesser term EF is commensurable in length with the rational (straight-line previously) laid down, D .

Thus, EG is a fifth binomial (straight-line).[†] (Which is) the very thing it was required to show.

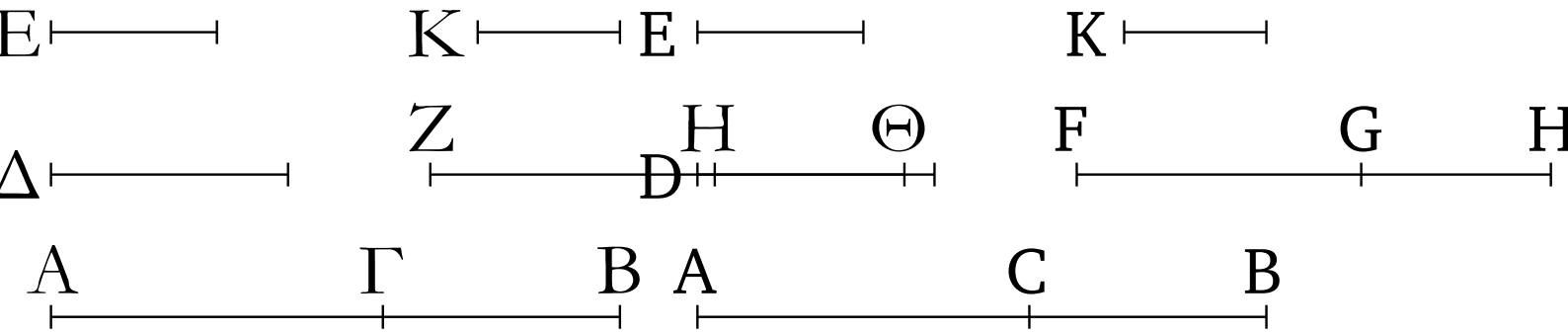
[†] If the rational straight-line has unit length then the length of a fifth binomial straight-line is $k(\sqrt{1+k'} + 1)$. This, and the fifth apotome, whose length is $k(\sqrt{1+k'} - 1)$ [Prop. 10.89], are the roots of $x^2 - 2k\sqrt{1+k'}x + k^2k' = 0$.

53.

Εὐρεῖν τὴν ἐκ δύο ὀνομάτων <έκτην.

Proposition 53

To find a sixth binomial (straight-line).



Ἐκκείσθωσαν δύο ἀριθμοὶ οἱ ΑΓ, ΓΒ, <ώστε τὸν ΑΒ πρὸς ἑκάτερον αὐτῶν λόγον μὴ >έθειν, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν; >έστω δὲ καὶ <έτεροξ ἀριθμὸς ὁ Δ μὴ τετράγωνος >ὄν μηδὲ πρὸς ἑκάτερον τῶν ΒΑ, ΑΓ λόγον >έθων, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν; καὶ ἐκκείσθω τις <ρητὴ εὐθεῖα ἡ Ε, καὶ γεγενέτω ὡς ὁ Δ πρὸς τὸν ΑΒ, ο<ύτωξ τὸ ἀπὸ τῆς Ε πρὸς τὸ ἀπὸ τῆς ΖΗ; σύμμετρον >άρα τὸ ἀπὸ τῆς Ε τῷ ἀπὸ τῆς ΖΗ. καὶ ἐστι <ρητὴ ἡ Ε; <ρητὴ >άρα καὶ ἡ ΖΗ. καὶ ἐπεὶ οὐκ >έθει ὁ Δ πρὸς τὸν ΑΒ λόγον, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, οὐδὲ τὸ ἀπὸ τῆς Ε >άρα πρὸς τὸ ἀπὸ τῆς ΖΗ λόγον >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν; ἀσύμμετροξ >άρα ἡ Ε τῇ ΖΗ μήκει. γεγενέτω δὴ πάλιν ὡς ὁ ΒΑ πρὸς τὸν ΑΓ, ο<ύτωξ τὸ ἀπὸ τῆς ΖΗ πρὸς τὸ ἀπὸ τῆς ΗΘ. σύμμετρον >άρα τὸ ἀπὸ τῆς ΖΗ τῷ ἀπὸ τῆς ΗΘ. <ρητὸν >άρα τὸ ἀπὸ τῆς ΗΘ; <ρητὴ >άρα ἡ ΗΘ. καὶ ἐπεὶ ὁ ΒΑ πρὸς τὸν ΑΓ λόγον οὐκ >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, οὐδὲ τὸ ἀπὸ τῆς ΖΗ πρὸς τὸ ἀπὸ τῆς ΗΘ λόγον >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν; ἀσύμμετροξ >άρα ἐστὶν ἡ ΖΗ τῇ ΗΘ μήκει. αἱ ΖΗ, ΗΘ >άρα <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι; ἐκ δύο >άρα ὀνομάτων ἐστὶν ἡ ΖΘ. δεικτέον δὴ, <ὅτι καὶ <έκτη.

Ἐπεὶ γάρ ἐστιν ὡς ὁ Δ πρὸς τὸν ΑΒ, ο<ύτωξ τὸ ἀπὸ τῆς Ε πρὸς τὸ ἀπὸ τῆς ΖΗ, >έστι δὲ καὶ ὡς ὁ ΒΑ πρὸς τὸν ΑΓ, ο<ύτωξ τὸ ἀπὸ τῆς ΖΗ πρὸς τὸ ἀπὸ τῆς ΗΘ, δι<>ίς οὖν >άρα ἐστὶν ὡς ὁ Δ πρὸς τὸν ΑΓ, ο<ύτωξ τὸ ἀπὸ τῆς Ε πρὸς τὸ ἀπὸ τῆς ΗΘ. ὁ δὲ Δ πρὸς τὸν ΑΓ λόγον οὐκ >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν; οὐδὲ τὸ ἀπὸ τῆς Ε >άρα πρὸς τὸ ἀπὸ τῆς ΗΘ λόγον >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν; ἀσύμμετροξ >άρα ἐστὶν ἡ Ε τῇ ΗΘ μήκει. ἐδείκθη δὲ καὶ τῇ ΖΗ ἀσύμμετροξ; ἑκατέρα >άρα τῶν ΖΗ, ΗΘ ἀσύμμετρόξ ἐστι τῇ Ε μήκει. καὶ ἐπεὶ ἐστὶν ὡς ὁ ΒΑ πρὸς τὸν ΑΓ, ο<ύτωξ τὸ ἀπὸ τῆς ΖΗ πρὸς τὸ ἀπὸ τῆς ΗΘ, μείζον >άρα τὸ ἀπὸ τῆς ΖΗ τοῦ ἀπὸ τῆς ΗΘ. >έστω ο>ὖν τῷ ἀπὸ [τῆς] ΖΗ >ίσα τὰ ἀπὸ τῶν ΗΘ, Κ; ἀναστρέψαντι >άρα ὡς ὁ ΑΒ πρὸς ΒΓ, ο<ύτωξ τὸ ἀπὸ ΖΗ πρὸς τὸ ἀπὸ τῆς Κ. ὁ δὲ ΑΒ πρὸς τὸν ΒΓ λόγον οὐκ >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν; <ώστε οὐδὲ τὸ ἀπὸ ΖΗ πρὸς τὸ ἀπὸ τῆς Κ λόγον >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. ἀσύμμετροξ >άρα ἐστὶν ἡ ΖΗ τῇ Κ μήκει; ἡ ΖΗ >άρα τῆς ΗΘ μείζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ. καὶ εἰσιν αἱ ΖΗ, ΗΘ <ρηταὶ δυνάμει μόνον σύμμετροι, καὶ οὐδετέρα αὐτῶν σύμμετρόξ ἐστι μήκει τῇ ἐκκειμένῃ <ρητῇ τῇ Ε.

Ἡ ΖΘ >άρα ἐκ δύο ὀνομάτων ἐστὶν <έκτη; <όπερ >έδει δεῖχαι.

Let the two numbers AC and CB be laid down such that AB does not have to each of them the ratio which (some) square number (has) to (some) square number. And let D also be another number, which is not square, and does not have to each of BA and AC the ratio which (some) square number (has) to (some) square number either [Prop. 10.28 lem. I]. And let some rational straight-line E be laid down. And let it be contrived that as D (is) to AB , so the (square) on E (is) to the (square) on FG [Prop. 10.6 corr.]. Thus, the (square) on E (is) commensurable with the (square) on FG [Prop. 10.6]. And E is rational. Thus, FG (is) also rational. And since D does not have to AB the ratio which (some) square number (has) to (some) square number, the (square) on E thus does not have to the (square) on FG the ratio which (some) square number (has) to (some) square number either. Thus, E (is) incommensurable in length with FG [Prop. 10.9]. So, again, let it have be contrived that as BA (is) to AC , so the (square) on FG (is) to the (square) on GH [Prop. 10.6 corr.]. The (square) on FG (is) thus commensurable with the (square) on HG [Prop. 10.6]. The (square) on HG (is) thus rational. Thus, HG (is) rational. And since BA does not have to AC the ratio which (some) square number (has) to (some) square number, the (square) on FG does not have to the (square) on GH the ratio which (some) square number (has) to (some) square number either. Thus, FG is incommensurable in length with GH [Prop. 10.9]. Thus, FG and GH are rational (straight-lines which are) commensurable in square only. Thus, FH is a binomial (straight-line) [Prop. 10.36]. So, we must show that (it is) also a sixth (binomial straight-line).

For since as D is to AB , so the (square) on E (is) to the (square) on FG , and also as BA is to AC , so the (square) on FG (is) to the (square) on GH , thus, via equality, as D is to AC , so the (square) on E (is) to the (square) on GH [Prop. 5.22]. And D does not have to AC the ratio which (some) square number (has) to (some) square number. Thus, the (square) on E does not have to the (square) on GH the ratio which (some) square number (has) to (some) square number either. E is thus incommensurable in length with GH [Prop. 10.9]. And (E) was also shown (to be) incommensurable (in length) with FG . Thus, FG and GH are each incommensurable in length with E . And since as BA is to AC , so the (square) on FG (is) to the (square) on GH , the (square) on FG (is) thus greater than the (square) on GH [Prop. 5.14]. Therefore, let (the sum of) the (squares) on GH and K be equal to the (square) on FG . Thus, via conversion, as AB (is) to BC , so the (square) on FG (is) to the (square) on K [Prop. 5.19 corr.]. And AB does not have to BC the ratio which (some) square number (has) to (some) square number. Hence, the (square) on FG does

not have to the (square) on K the ratio which (some) square number (has) to (some) square number either. Thus, FG is incommensurable in length with K [Prop. 10.9]. The square on FG is thus greater than (the square on) GH by the (square) on (some straight-line which is) incommensurable (in length) with (FG). And FG and GH are rational (straight-lines which are) commensurable in square only, and neither of them is commensurable in length with the rational (straight-line) E (previously) laid down.

Thus, FH is a sixth binomial (straight-line) [Def. 10.10].[†]
(Which is) the very thing it was required to show.

† If the rational straight-line has unit length then the length of a sixth binomial straight-line is $\sqrt{k} + \sqrt{k'}$. This, and the sixth apotome, whose length is $\sqrt{k} - \sqrt{k'}$ [Prop. 10.90], are the roots of $x^2 - 2\sqrt{k}x + (k - k') = 0$.

Λήμμα.

>Ἐστω δύο τετράγωνα τὰ AB, ΒΓ καὶ κείσθωσαν <ώστε ἐπ> εὐθείᾳξ ε>ῖναι τὴν ΔΒ τῇ| BE; ἐπ> εὐθείᾳξ >άρα ἐστὶ καὶ ἡ ΖΒ τῇ| BH. καὶ συμπεπληρώσθω τὸ ΑΓ παραλληλόγραμνον, λέγω, <ὅτι τετράγωνόν ἐστι τὸ ΑΓ, καὶ <ὅτι τῶν AB, ΒΓ μέσον ἀνάλογόν ἐστι τὸ ΔΗ, καὶ >έτι τῶν ΑΓ, ΓΒ μέσον ἀνάλογόν ἐστι τὸ ΔΓ.

Ἐπεὶ γὰρ $\angle$ ισὴ ἐστὶν ἡ μὲν ΔΒ τῇ| ΒΖ, ἡ δὲ ΒΕ τῇ| ΒΗ, $\angle$ όλη $\angle$ άρα ἡ ΔΕ $\angle$ όλη τῇ| ΖΗ ἐστὶν $\angle$ ισὴ. ἄλλ $\angle$ ἡ μὲν ΔΕ ἑκατέρω τῶν ΑΘ, ΚΓ ἐστὶν $\angle$ ισὴ, ἡ δὲ ΖΗ ἑκατέρω τῶν ΑΚ, ΘΓ ἐστὶν $\angle$ ισὴ; καὶ ἑκατέρω $\angle$ άρα τῶν ΑΘ, ΚΓ ἑκατέρω τῶν ΑΚ, ΘΓ ἐστὶν $\angle$ ισὴ. ἰσόπλευρον $\angle$ άρα ἐστὶ τὸ ΑΓ παραλληλόγραμμον; $\angle$ έστι δὲ καὶ ὀρθογώνιον; τετράγωνον $\angle$ άρα ἐστὶ τὸ ΑΓ.

Καὶ ἐπεὶ ἐστὶν ὥς ἡ ΖΒ πρὸς τὴν ΒΗ, ο<ύτως ἡ ΔΒ πρὸς τὴν ΒΕ, ἄλλ> ὥς μὲν ἡ ΖΒ πρὸς τὴν ΒΗ, ο<ύτως τὸ ΑΒ πρὸς τὸ ΔΗ, ὥς δὲ ἡ ΔΒ πρὸς τὴν ΒΕ, ο<ύτως τὸ ΔΗ πρὸς τὸ ΒΓ, καὶ ὥς >άρα τὸ ΑΒ πρὸς τὸ ΔΗ, ο<ύτως τὸ ΔΗ πρὸς τὸ ΒΓ. τῶν ΑΒ, ΒΓ >άρα μέσον ἀνάλογόν ἐστὶ τὸ ΔΗ.

Λέγω δὴ, <ὅτι καὶ τῶν ΑΓ, ΒΒ μέσον ἀνάλογόν [ἐστὶ] τὸ ΔΓ.

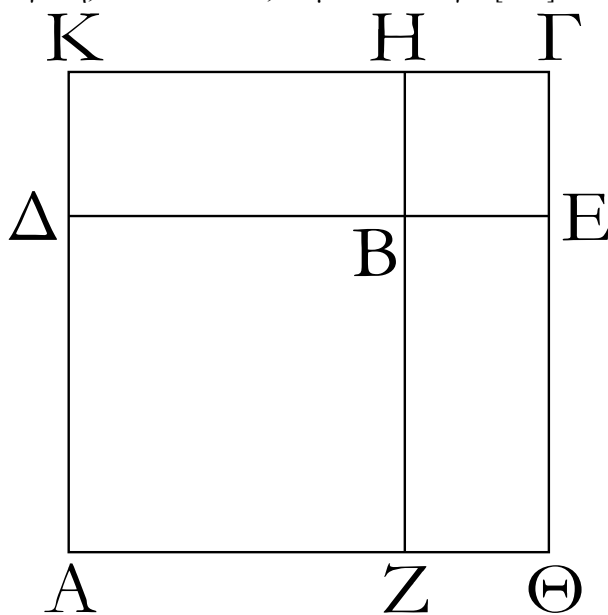
Lemma

Let AB and BC be two squares, and let them be laid down such that DB is straight-on to BE . FB is, thus, also straight-on to BG . And let the parallelogram AC be completed. I say that AC is a square, and that DG is the mean proportional to AB and BC , and, moreover, DC is the mean proportional to AC and CB .

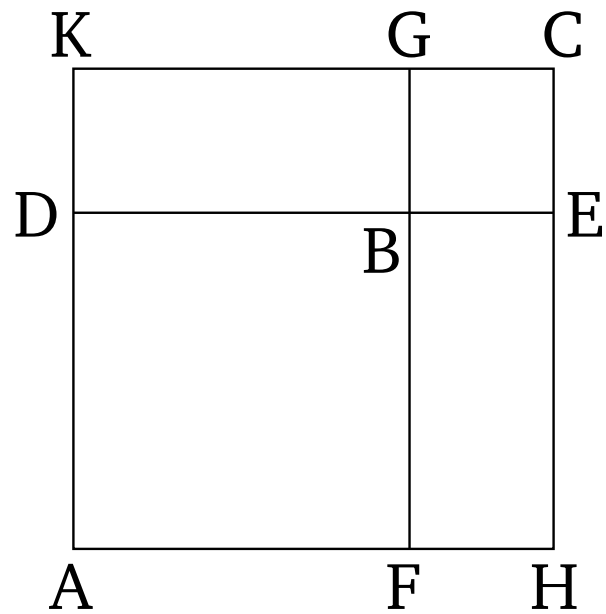
For since DB is equal to BF , and BE to BG , the whole of DE is thus equal to the whole of FG . But DE is equal to each of AH and KC , and FG is equal to each of AK and HC [Prop. 1.34]. Thus, AH and KC are also equal to AK and HC , respectively. Thus, the parallelogram AC is equilateral. And (it is) also right-angled. Thus, AC is a square.

And since as FB is to BG , so DB (is) to BE , but as FB (is) to BG , so AB (is) to DG , and as DB (is) to BE , so DG (is) to BC [Prop. 6.1], thus also as AB (is) to DG , so DG (is) to BC [Prop. 5.11]. Thus, DG is the mean proportional to AB and BC .

So I also say that DC [is] the mean proportional to AC and



CB.



Ἐπεὶ γὰρ ἐστὶν ὥς ἡ AD πρὸς τὴν DK , οὕτως ἡ KH πρὸς τὴν HF ; >ίση γὰρ [ἐστὶν] ἑκατέρα ἑκατέρᾳ; καὶ συνθέντι ὥς ἡ AK πρὸς KD , οὕτως ἡ KG πρὸς GH , ἀλλ' ὥς μὲν ἡ AK πρὸς KD , οὕτως τὸ AG πρὸς τὸ GD , ὥς δὲ ἡ KG πρὸς GH , οὕτως τὸ AG πρὸς GB , καὶ ὥς >άρα τὸ AG πρὸς $ΔΓ$, οὕτως τὸ $ΔΓ$ πρὸς τὸ BF . τῶν AG , GB >άρα μέσον ἀνάλογόν ἐστι τὸ $ΔΓ$; <ὰ προέκειτο δεῖχαι.

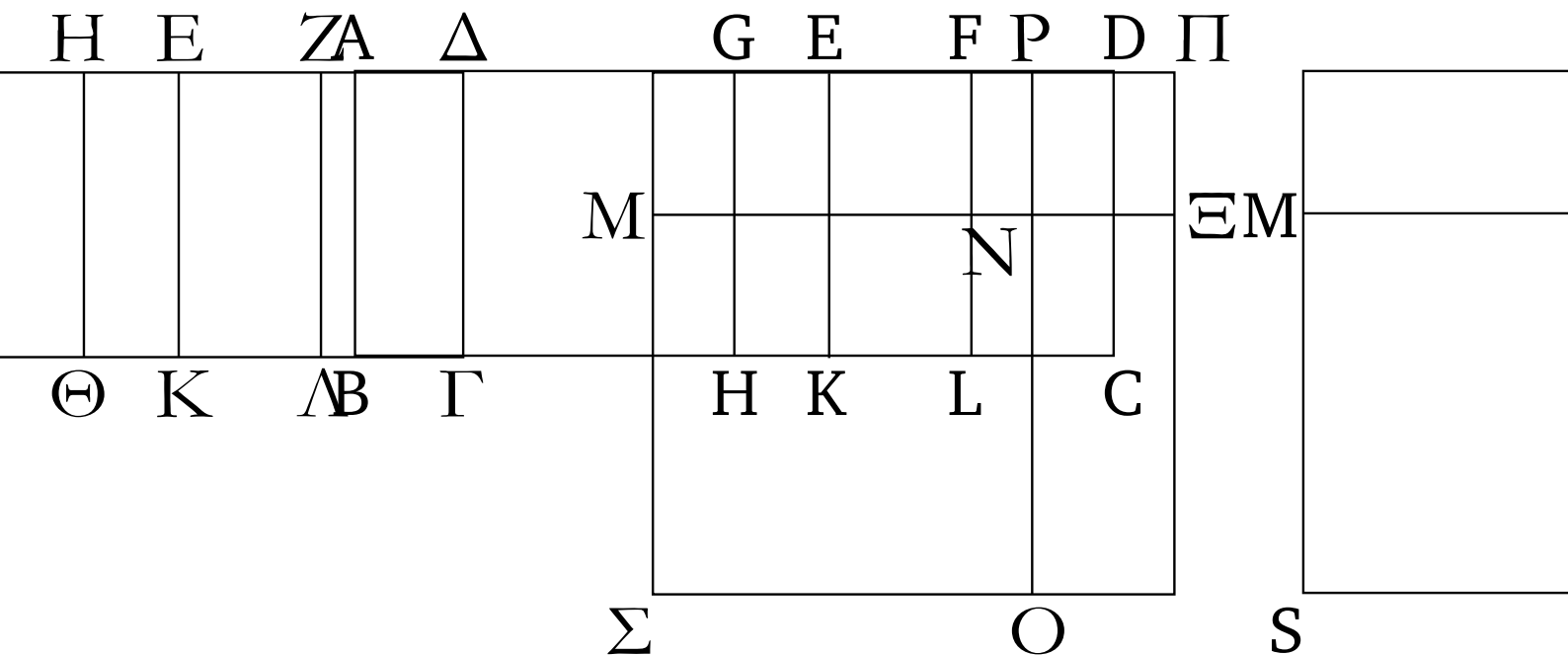
For since as AD is to DK , so KG (is) to GC . For [they are] respectively equal. And, via composition, as AK (is) to KD , so KC (is) to CG [Prop. 5.18]. But as AK (is) to KD , so AC (is) to CD , and as KC (is) to CG , so DC (is) to CB [Prop. 6.1]. Thus, also, as AC (is) to DC , so DC (is) to BC [Prop. 5.11]. Thus, DC is the mean proportional to AC and CB . Which (is the very thing) it was prescribed to show.

54.

Proposition 54

Ἐὰν θωρίον περιέθῃται ὑπὸ <ρητῆς καὶ τῆς ἐκ δύο ὀνομάτων πρώτης, ἡ τὸ θωρίον δυναμένη >αλογόξ ἐστὶν ἡ καλουμένη ἐκ δύο ὀνομάτων.

If an area is contained by a rational (straight-line) and a first binomial (straight-line) then the square-root of the area is the irrational (straight-line which is) called binomial.[†]



Θωρίον γὰρ τὸ AG περιέθεσθαι ὑπὸ <ρητῆς τῆς AB καὶ τῆς ἐκ δύο ὀνομάτων πρώτης τῆς AD ; λέγω, <ὅτι ἡ τὸ AG θωρίον δυναμένη >αλογόξ ἐστὶν ἡ καλουμένη ἐκ δύο ὀνομάτων.

Ἐπεὶ γὰρ ἐκ δύο ὀνομάτων ἐστὶ πρώτη ἡ AD , διηρήσθω εἰς τὰ ὀνόματα κατὰ τὸ E , καὶ >έστω τὸ μείζον >όνομα τὸ AE . φανερόν δὲ, <ὅτι αἱ AE , ED <ρηταὶ εἰσι δυνάμει

For let the area AC be contained by the rational (straight-line) AB and by the first binomial (straight-line) AD . I say that square-root of area AC is the irrational (straight-line which is) called binomial.

For since AD is a first binomial (straight-line), let it be divided into its (component) terms at E , and let AE be the greater term. So, (it is) clear that AE and ED are rational

μόνον σύμμετροι, καὶ ἡ AE τῆς ED μείζον δύναται τῷ ἀπὸ συμέτρου ἑαυτῇ, καὶ ἡ AE σύμμετρόξ ἐστὶ τῇ ἐκκειμένῃ <ρητῇ τῇ AB μήκει. τεμήσθω δὲ ἡ ED δίθα κατὰ τὸ Z σημεῖον. καὶ ἐπεὶ ἡ AE τῆς ED μείζον δύναται τῷ ἀπὸ συμέτρου ἑαυτῇ, ἔαν >άρα τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ἐλάσσονος, τουτέστι τῷ ἀπὸ τῆς EZ , >ίσον παρὰ τὴν μείζονα τὴν AE παραβληθῇ ἐλλείπον >ίδει τετραγώνω, εἷς σύμμετρα αὐτὴν διαιρεῖ. παραβλήσθω οὖν παρὰ τὴν AE τῷ ἀπὸ τῆς EZ >ίσον τὸ ὑπὸ AH , HE ; σύμμετροξ >άρα ἐστὶν ἡ AH τῇ EH μήκει. καὶ >ήθωσαν ἀπὸ τῶν H , E , Z ὁποτέρᾳ τῶν AB , $ΓΔ$ παράλληλοι αἱ $HΘ$, EK , $ZΛ$; καὶ τῷ μὲν $AΘ$ παραλληλογράμῳ >ίσον τετράγωνον συνεστάτω τὸ $ΣΝ$, τῷ δὲ HK >ίσον τὸ $ΝΠ$, καὶ κείσθω <ώστε ἐπ> εὐθείᾳς >ῖναι τὴν MN τῇ NX ; ἐπ> εὐθείᾳξ >άρα ἐστὶ καὶ ἡ PN τῇ NO . καὶ συμπεπληρώσθω τὸ $ΣΠ$ παραλληλόγραμμον; τετράγωνον >άρα ἐστὶ τὸ $ΣΠ$. καὶ ἐπεὶ τὸ ὑπὸ τῶν AH , HE >ίσον ἐστὶ τῷ ἀπὸ τῆς EZ , >έστιν >άρα ὥς ἡ AH πρὸς EZ , ο<ύτωξ ἡ ZE πρὸς EH ; καὶ ὥς >άρα τὸ $AΘ$ πρὸς $ΕΛ$, τὸ $ΕΛ$ πρὸς KH ; τῶν $AΘ$, HK >άρα μέσον ἀνάλογόν ἐστὶ τὸ $ΕΛ$. ἀλλὰ τὸ μὲν $AΘ$ >ίσον ἐστὶ τῷ $ΣΝ$, τὸ δὲ HK >ίσον τῷ $ΝΠ$; τῶν $ΣΝ$, $ΝΠ$ >άρα μέσον ἀνάλογόν ἐστὶ τὸ $ΕΛ$. >έστι δὲ τῶν αὐτῶν τῶν $ΣΝ$, $ΝΠ$ μέσον ἀνάλογον καὶ τὸ MP ; >ίσον >άρα ἐστὶ τὸ $ΕΛ$ τῷ MP ; <ώστε καὶ τῷ OX >ίσον ἐστίν. >έστι δὲ καὶ τὰ $AΘ$, HK τοῖς $ΣΝ$, $ΝΠ$ >ίσα; <όλον>άρα τὸ $ΑΓ$ >ίσον ἐστὶν <όλω τῷ $ΣΠ$, τουτέστι τῷ ἀπὸ τῆς MX τετραγώνω; τὸ $ΑΓ$ >άρα δύναται ἡ MX . λέγω, <ότι ἡ MX ἐκ δύο ὀνομάτων ἐστίν. Ἐπεὶ γὰρ σύμμετρόξ ἐστὶν ἡ AH τῇ HE , σύμμετρόξ ἐστὶ καὶ ἡ AE ἑκατέρᾳ τῶν AH , HE . ὑπόκειται δὲ καὶ ἡ AE τῇ AB σύμμετροξ; καὶ αἱ AH , HE >άρα τῇ AB σύμμετροί εἰσιν. καὶ ἐστὶ <ρητὴ ἡ AB ; <ρητὴ >άρα ἐστὶ καὶ ἑκάτερα τῶν AH , HE ; <ρητὸν >άρα ἐστὶν ἑκάτερον τῶν $AΘ$, HK , καὶ ἐστὶ σύμμετρον τὸ $AΘ$ τῷ HK . ἀλλὰ τὸ μὲν $AΘ$ τῷ $ΣΝ$ >ίσον ἐστίν, τὸ δὲ HK τῷ $ΝΠ$; καὶ τὰ $ΣΝ$, $ΝΠ$ >άρα, τουτέστι τὰ ἀπὸ τῶν MN , NX , <ρητά ἐστὶ καὶ σύμμετρα. καὶ ἐπεὶ ἀσύμμετρόξ ἐστὶν ἡ AE τῇ $ΕΔ$ μήκει, ἀλλ> ἡ μὲν AE τῇ AH ἐστὶ σύμμετροξ, ἡ δὲ $ΔΕ$ τῇ EZ σύμμετροξ, ἀσύμμετροξ >άρα καὶ ἡ AH τῇ EZ ; <ώστε καὶ τὸ $AΘ$ τῷ $ΕΛ$ ἀσύμμετρόν ἐστίν. ἀλλὰ τὸ μὲν $AΘ$ τῷ $ΣΝ$ ἐστὶν >ίσον, τὸ δὲ $ΕΛ$ τῷ MP ; καὶ τὸ $ΣΝ$ >άρα τῷ MP ἀσύμμετρόν ἐστίν. ἀλλ> ὥς τὸ $ΣΝ$ πρὸς MP , ἡ ON πρὸς τὴν NP ; ἀσύμμετροξ >άρα ἐστὶν ἡ ON τῇ NP . >ίση δὲ ἡ μὲν ON τῇ MN , ἡ δὲ NP τῇ NX ; ἀσύμμετροξ >άρα ἐστὶν ἡ MN τῇ NX . καὶ ἐστὶ τὸ ἀπὸ τῆς MN σύμμετρον τῷ ἀπὸ τῆς NX , καὶ <ρητὸν ἑκάτερον; αἱ MN , NX >άρα <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι. Ἡ MX >άρα ἐκ δύο ὀνομάτων ἐστὶ καὶ δύναται τὸ $ΑΓ$; <όπερ >έδει δεῖχαι.

(straight-lines which are) commensurable in square only, and that the square on AE is greater than (the square on) ED by the (square) on (some straight-line) commensurable (in length) with (AE) , and that AE is commensurable (in length) with the rational (straight-line) AB (first) laid out [Def. 10.5]. So, let ED be cut in half at point F . And since the square on AE is greater than (the square on) ED by the (square) on (some straight-line) commensurable (in length) with (AE) , thus if a (rectangle) equal to the fourth part of the (square) on the lesser (term)—that is to say, the (square) on EF —falling short by a square figure, is applied to the greater (term) AE , then it divides it into (terms which are) commensurable (in length) [Prop. 10.17]. Therefore, let the (rectangle contained) by AG and GE , equal to the (square) on EF , be applied to AE . AG is thus commensurable in length with EG . And let GH , EK , and FL be drawn from (points) G , E , and F (respectively), parallel to either of AB or CD . And let the square SN , equal to the parallelogram AH , be constructed, and (the square) NQ , equal to (the parallelogram) GK [Prop. 2.14]. And let MN be laid down so as to be straight-on to NO . RN is thus also straight-on to NP . And let the parallelogram SQ be completed. SQ is thus a square [Prop. 10.53 lem.]. And since the (rectangle contained) by AG and GE is equal to the (square) on EF , thus as AG is to EF , so FE (is) to EG [Prop. 6.17]. And thus as AH (is) to EL , (so) EL (is) to KG [Prop. 6.1]. Thus, EL is the mean proportional to AH and GK . But, AH is equal to SN , and GK (is) equal to NQ . EL is thus the mean proportional to SN and NQ . And MR is also the mean proportional to the same—(namely), SN and NQ [Prop. 10.53 lem.]. EL is thus equal to MR . Hence, it is also equal to PO [Prop. 1.43]. And AH plus GK is equal to SN plus NQ . Thus, the whole of AC is equal to the whole of SQ —that is to say, to the square on MO . Thus, MO (is) the square-root of (area) AC . I say that MO is a binomial (straight-line).

For since AG is commensurable (in length) with GE , AE is also commensurable (in length) with each of AG and GE [Prop. 10.15]. And AE was also assumed (to be) commensurable (in length) with AB . Thus, AG and GE are also commensurable (in length) with AB [Prop. 10.12]. And AB is rational. AG and GE are thus each also rational. Thus, AH and GK are each rational (areas), and AH is commensurable with GK [Prop. 10.19]. But, AH is equal to SN , and GK to NQ . SN and NQ —that is to say, the (squares) on MN and NO (respectively)—are thus also rational and commensurable. And since AE is incommensurable in length with ED , but AE is commensurable (in length) with AG , and DE (is) commensurable (in length) with EF , AG (is) thus also incommensurable (in length) with EF [Prop. 10.13]. Hence, AH is also incommensurable with EL [Props. 6.1, 10.11]. But, AH is equal to SN , and EL to MR . Thus, SN is also incommensurable with MR . But, as SN (is) to MR , (so) PN (is) to NR [Prop. 6.1]. PN is thus incommensurable (in length) with NR [Prop. 10.11]. And PN (is) equal to MN , and NR to NO . Thus, MN is incommensurable (in length) with NO . And the (square) on MN is commensurable with the (square) on NO , and each (is) rational. MN and NO are thus rational (straight-lines which are) commensurable in square only.

Thus, MO is (both) a binomial (straight-line) [Prop. 10.36], and the square-root of AC . (Which is) the very thing it was required to show.

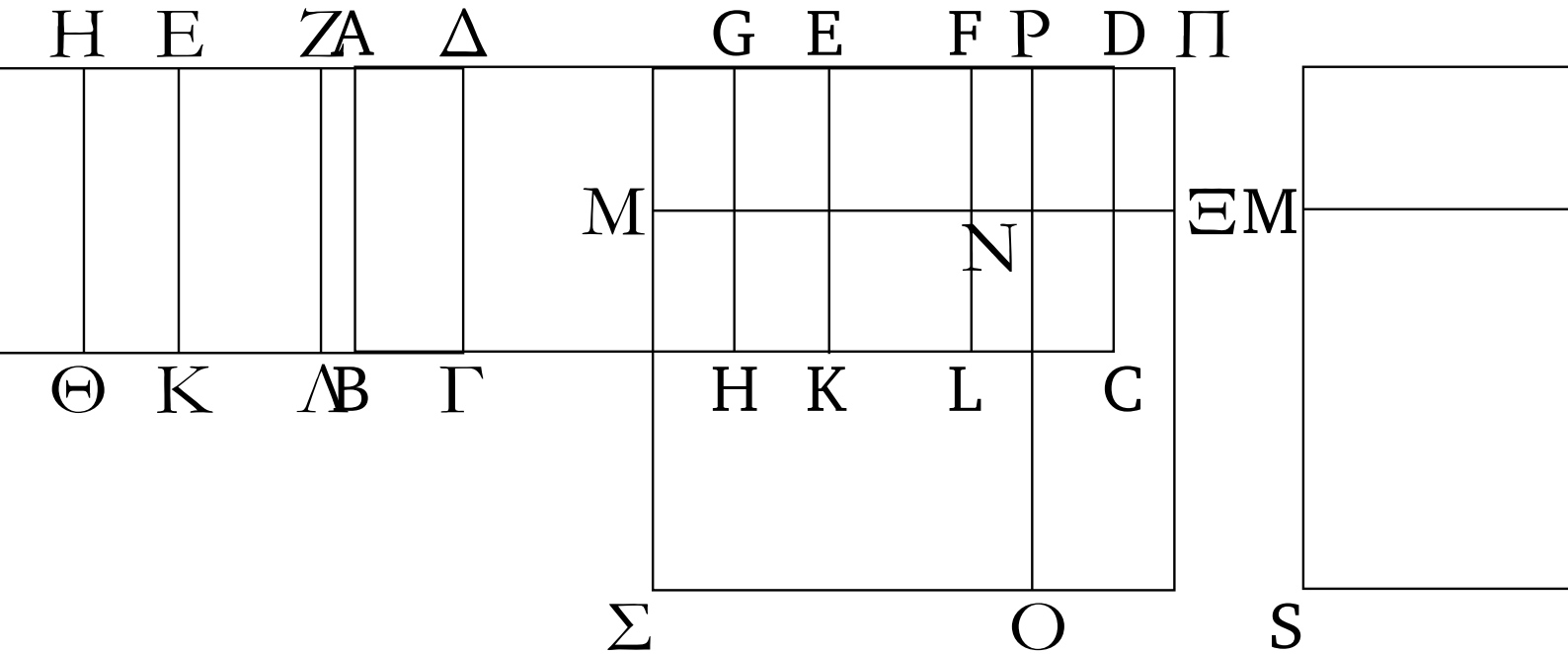
† If the rational straight-line has unit length then this proposition states that the square-root of a first binomial straight-line is a binomial straight-line: i.e., a first binomial straight-line has a length $k + k\sqrt{1 - k'^2}$ whose square-root can be written $\rho(1 + \sqrt{k'})$, where $\rho = \sqrt{k(1 + k')/2}$ and $k'' = (1 - k')/(1 + k')$. This is the length of a binomial straight-line (see Prop. 10.36), since ρ is rational.

55.

Proposition 55

Ἐάν θωρίον περιέθῃται ὑπὸ <ρητῆς καὶ τῆς ἐκ δύο ὀνομάτων δευτέρας, ἡ τὸ θωρίον δυναμένη >ἀλογόξ ἐστὶν ἡ καλουμένη ἐκ δύο μέσων πρώτη.

If an area is contained by a rational (straight-line) and a second binomial (straight-line) then the square-root of the area is the irrational (straight-line which is) called first bimedral.†



Περιεθέσθω γὰρ θωρίον τὸ $AB\Gamma\Delta$ ὑπὸ <ρητῆς τῆς AB καὶ τῆς ἐκ δύο ὀνομάτων δευτέρας τῆς AD ; λέγω, <ὅτι ἡ τὸ AG θωρίον δυναμένη ἐκ δύο μέσων πρώτη ἐστίν.

Ἐπεὶ γὰρ ἐκ δύο ὀνομάτων δευτέρα ἐστὶν ἡ AD , διηρήσθω εἰς τὰ ὀνόματα κατὰ τὸ E , <ὥστε τὸ μείζον >ὄνομα εἶναι τὸ AE ; αἱ AE , ED >ἀρα <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ AE τῆς ED μείζον δύναται τῷ ἀπὸ συμέτρου ἑαυτῇ, καὶ τὸ >έλαττον >ὄνομα ἡ ED σύμμετρόν ἐστι τῇ AB μήκει. τεμήσθω ἡ ED δίθα κατὰ τὸ Z , καὶ τῷ ἀπὸ τῆς EZ >ἴσον παρὰ τὴν AE παραβελήσθω ἐλλείπον εἶδει τετραγώνῳ τὸ ὑπὸ τῶν AHE ; σύμμετροξ >ἀρα ἡ AH τῇ HE μήκει. καὶ διὰ τῶν H , E , Z παράλληλοι >ῆθωσαν ταῖς AB , $\Gamma\Delta$ αἱ $H\Theta$, EK , $Z\Lambda$, καὶ τῷ μὲν $A\Theta$ παραλληλογράμῳ >ἴσον τετράγωνον συνεστάτω τὸ SN , τῷ δὲ HK >ἴσον τετράγωνον τὸ NI , καὶ κείσθω <ὥστε ἐπ> εὐθείας εἶναι τὴν MN τῇ NX ; ἐπ> εὐθείας >ἀρα [ἐστὶ] καὶ ἡ PN τῇ NO . καὶ συμπληρώσθω τὸ $\Sigma\Pi$ τετράγωνον; φανερόν δὲ ἐκ τοῦ προδεδειγμένου, <ὅτι τὸ MP μέσον ἀνάλογόν ἐστι τῶν SN , NI , καὶ >ἴσον τῷ EL , καὶ <ὅτι τὸ AG θωρίον δύναται ἡ MX . δεικτέον δὲ, <ὅτι ἡ MX ἐκ δύο μέσων ἐστὶ πρώτη.

Ἐπεὶ ἀσύμμετρόξ ἐστὶν ἡ AE τῇ ED μήκει, σύμμετροξ δὲ ἡ ED τῇ AB , ἀσύμμετροξ >ἀρα ἡ AE τῇ AB . καὶ ἐπεὶ σύμμετρόξ ἐστὶν ἡ AH τῇ EH , σύμμετρόξ ἐστὶ καὶ ἡ AE ἑκάτερα τῶν AH , HE . ἀλλὰ ἡ AE ἀσύμμετροξ τῇ AB μήκει; καὶ αἱ AH , HE >ἀρα ἀσύμμετροί εἰσι τῇ AB . αἱ BA , AH , HE >ἀρα <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι; <ὥστε μέσον ἐστὶν ἑκάτερον τῶν $A\Theta$, HK . <ὥστε καὶ ἑκάτερον τῶν SN , NI μέσον ἐστίν. καὶ αἱ MN , NX >ἀρα μέσαι εἰσίν. καὶ

For let the area $ABCD$ be contained by the rational (straight-line) AB and by the second binomial (straight-line) AD . I say that the square-root of area AC is a first bimedral (straight-line).

For since AD is a second binomial (straight-line), let it be divided into its (component) terms at E , such that AE is the greater term. Thus, AE and ED are rational (straight-lines which are) commensurable in square only, and the square on AE is greater than (the square on) ED by the (square) on (some straight-line) commensurable (in length) with (AE) , and the lesser term ED is commensurable in length with AB [Def. 10.6]. Let ED be cut in half at F . And let the (rectangle contained) by AGE , equal to the (square) on EF , be applied to AE , falling short by a square figure. AG (is) thus commensurable in length with GE [Prop. 10.17]. And let GH , EK , and FL be drawn through (points) G , E , and F (respectively), parallel to AB and CD . And let the square SN , equal to the parallelogram AH , be constructed, and the square NQ , equal to GK . And let MN be laid down so as to be straight-on to NO . Thus, RN [is] also straight-on to NP . And let the square SQ be completed. So, (it is) clear from what has been previously demonstrated [Prop. 10.53 lem.] that MR is the mean proportional to SN and NQ , and (is) equal to EL , and that MO is the square-root of the area AC . So, we must show that MO is a first bimedral (straight-line).

Since AE is incommensurable in length with ED , and ED (is) commensurable (in length) with AB , AE (is) thus in-

ἐπεὶ σύμμετρος ἡ AH τῇ HE μήκει, σύμμετρόν ἐστι καὶ τὸ $A\Theta$ τῷ HK , τουτέστι τὸ ΣN τῷ NP , τουτέστι τὸ ἀπὸ τῆς MN τῷ ἀπὸ τῆς NX [$\epsilon\omega\sigma\tau\epsilon$ δυνάμει εἰσὶ σύμμετροι αἱ MN, NX]. καὶ ἐπεὶ ἀσύμμετρος ἐστὶν ἡ AE τῇ ED μήκει, ἀλλ' ἡ μὲν AE σύμμετρος ἐστὶ τῇ AH , ἡ δὲ ED τῇ EZ σύμμετρος, ἀσύμμετρος $\epsilon\alpha\rho\alpha$ ἡ AH τῇ EZ ; $\epsilon\omega\sigma\tau\epsilon$ καὶ τὸ $A\Theta$ τῷ EL ἀσύμμετρόν ἐστιν, τουτέστι τὸ ΣN τῷ MP , τουτέστιν ὁ ON τῇ NP , τουτέστιν ἡ MN τῇ NX ἀσύμμετρος ἐστὶ μήκει. ἐδείκθησαν δὲ αἱ MN, NX καὶ μέσαι $\epsilon\upsilon\sigma\sigma\alpha\iota$ καὶ δυνάμει σύμμετροι; αἱ MN, NX $\epsilon\alpha\rho\alpha$ μέσαι εἰσὶ δυνάμει μόνον σύμμετροι. λέγω δὴ, $\epsilon\omega\tau\iota$ καὶ $\epsilon\sigma\tau\iota$ περιέθουσιν. ἐπεὶ γὰρ ἡ DE ὑπόκειται ἑκατέρᾳ τῶν AB, EZ σύμμετρος, σύμμετρος $\epsilon\alpha\rho\alpha$ καὶ ἡ EZ τῇ EK . καὶ $\epsilon\sigma\tau\iota$ ἑκατέρᾳ αὐτῶν; $\epsilon\sigma\tau\iota$ $\epsilon\alpha\rho\alpha$ τὸ EL , τουτέστι τὸ MP ; τὸ δὲ MP ἐστὶ τὸ ὑπὸ τῶν MNX . ἔαν δὲ δύο μέσαι δυνάμει μόνον σύμμετροι συντεθῶσι $\epsilon\sigma\tau\iota$ περιέθουσαι, ἡ $\epsilon\omega\lambda\eta$ $\epsilon\alpha\lambda\omicron\gamma\omicron\zeta$ ἐστὶν, καλεῖται δὲ ἐκ δύο μέσων πρώτη; $\epsilon\omega\pi\epsilon\rho$ $\epsilon\delta\epsilon\iota$ δεῖχαι.

commensurable (in length) with AB [Prop. 10.13]. And since AG is commensurable (in length) with EG , AE is also commensurable (in length) with each of AG and GE [Prop. 10.15]. But, AE is incommensurable in length with AB . Thus, AG and GE are also (both) incommensurable (in length) with AB [Prop. 10.13]. Thus, BA , AG , and (BA , and) GE are (pairs of) rational (straight-lines which are) commensurable in square only. And, hence, each of AH and GK is a medial (area) [Prop. 10.21]. Hence, each of SN and NQ is also a medial (area). Thus, MN and NO are medial (straight-lines). And since AG (is) commensurable in length with GE , AH is also commensurable with GK —that is to say, SN with NQ —that is to say, the (square) on MN with the (square) on NO [hence, MN and NO are commensurable in square] [Props. 6.1, 10.11]. And since AE is incommensurable in length with ED , but AE is commensurable (in length) with AG , and ED commensurable (in length) with EF , AG (is) thus incommensurable (in length) with EF [Prop. 10.13]. Hence, AH is also incommensurable with EL —that is to say, SN with MR —that is to say, PN with NR —that is to say, MN is incommensurable in length with NO [Props. 6.1, 10.11]. But MN and NO have also been shown to be medial (straight-lines) which are commensurable in square. Thus, MN and NO are medial (straight-lines which are) commensurable in square only. So, I say that they also contain a rational (area). For since DE was assumed (to be) commensurable (in length) with each of AB and EF , EF (is) thus also commensurable with EK [Prop. 10.12]. And they (are) each rational. Thus, EL —that is to say, MR —(is) rational [Prop. 10.19]. And MR is the (rectangle contained) by MNO . And if two medial (straight-lines), commensurable in square only, which contain a rational (area), are added together, then the whole is (that) irrational (straight-line which is) called first bimedral [Prop. 10.37].

Thus, MO is a first bimedral (straight-line). (Which is) the very thing it was required to show.

[†] If the rational straight-line has unit length then this proposition states that the square-root of a second binomial straight-line is a first bimedral straight-line: i.e., a second binomial straight-line has a length $k/\sqrt{1-k'^2} + k$ whose square-root can be written $\rho(k''^{1/4} + k''^{3/4})$, where $\rho = \sqrt{(k/2)(1+k')/(1-k')}$ and $k'' = (1-k')/(1+k')$. This is the length of a first bimedral straight-line (see Prop. 10.37), since ρ is rational.

56.

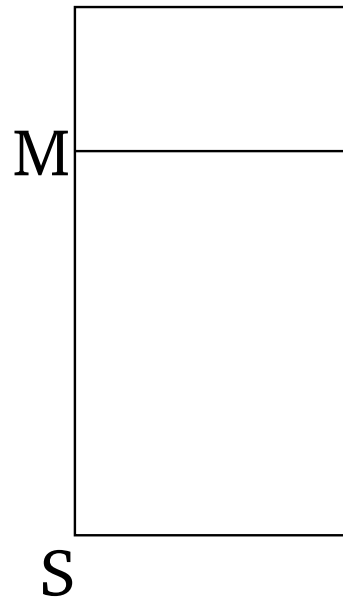
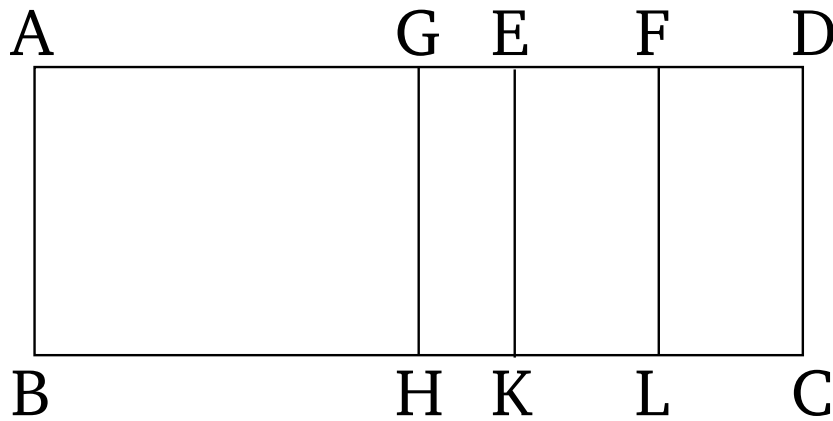
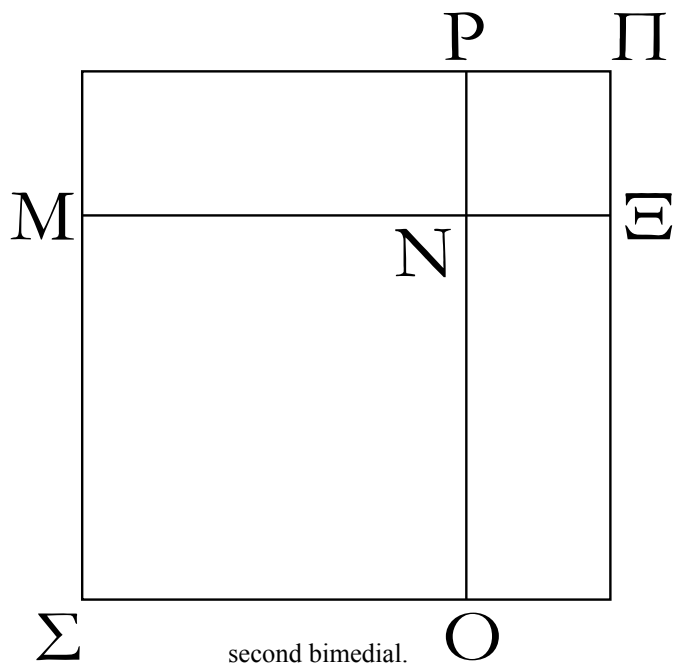
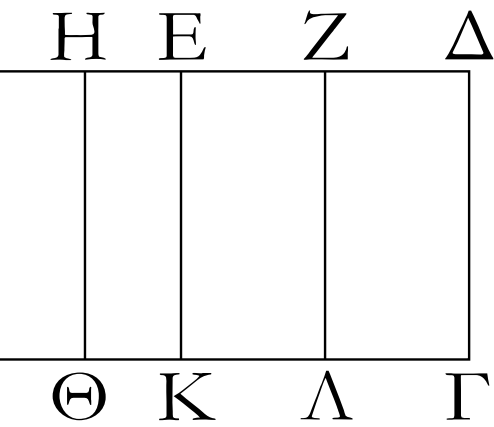
Ἐὰν θωρίον περιέθῃται ὑπὸ $\epsilon\sigma\tau\iota$ τῆς καὶ τῆς ἐκ δύο ὀνομάτων τρίτης, ἡ τὸ θωρίον δυναμένη $\epsilon\alpha\lambda\omicron\gamma\omicron\zeta$ ἐστὶν ἡ καλουμένη ἐκ δύο μέσων δευτέρα.

Θωρίον γὰρ τὸ $AB\Gamma\Delta$ περιέθῃτω ὑπὸ $\epsilon\sigma\tau\iota$ τῆς AB καὶ τῆς ἐκ δύο ὀνομάτων τρίτης τῆς $A\Delta$ διηρημένης εἰς τὰ ὀνόματα κατὰ τὸ E , $\epsilon\omega\sigma\tau\iota$ μείζον ἐστὶ τὸ AE ; λέγω, $\epsilon\omega\tau\iota$ ἡ τὸ $A\Gamma$ θωρίον δυναμένη $\epsilon\alpha\lambda\omicron\gamma\omicron\zeta$ ἐστὶν ἡ καλουμένη ἐκ δύο μέσων δευτέρα.

Proposition 56

If an area is contained by a rational (straight-line) and a third binomial (straight-line) then the square-root of the area is the irrational (straight-line which is) called second bimedral.[†]

For let the area $ABCD$ be contained by the rational (straight-line) AB and by the third binomial (straight-line) AD , which has been divided into its (component) terms at E , of which AE is the greater. I say that the square-root of area AC is the irrational (straight-line which is) called



Κατεσκευάσθω γὰρ τὰ αὐτὰ τοῖς πρότερον. καὶ ἐπεὶ ἐκ δύο ὀνομάτων ἐστὶ τρίτη ἡ ΑΔ, αἱ ΑΕ, ΕΔ >ἀρα <ρηται εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ ΑΕ τῆς ΕΔ μείζον δύναται τῷ ἀπὸ συμμέτρου ἐαυτῇ, καὶ οὐδετέρα τῶν ΑΕ, ΕΔ σύμμετρόξ [ἐστὶ] τῇ ΑΒ μήκει. ὁμοίωξ δὴ τοῖς προδεδειγμένοιξ δείχομεν, <ὅτι ἡ ΜΧ ἐστὶν ἡ τὸ ΑΓ θωρίον δυνάμενη, καὶ αἱ ΜΝ, ΝΧ μέσαι εἰσι δυνάμει μόνον σύμμετροι; <ὥστε ἡ ΜΧ ἐκ δύο μέσων ἐστίν. δεικτέον δὴ, <ὅτι καὶ δευτέρα.

[Ka'i] >epe'i >as'ummetr'oc >estin <h DE t h AB m'hkei, tout'esti t h EK, s'ummetroc d'e <h DE t h EZ, >as'ummetroc >'ara >est'in <h EZ t h EK m'hkei. ka'i e>isi <rhta'i; a<i ZE, EK >'ara <rhta'i e>isi dun'amei m'onon s'ummetroi. m'eson >'ara [>est'i] t'o EL, tout'esti t'o MR; ka'i peri'egetai <up'o t wn MNX; m'eson >'ara >est'i t'o <up'o t wn MNX.

Ἡ ΜΧ >ἀρα ἐκ δύο μέσων ἐστὶ δευτέρα; <ὅπερ >έδει δείχαι.

For let the same construction be made as previously. And since AD is a third binomial (straight-line), AE and ED are thus rational (straight-lines which are) commensurable in square only, and the square on AE is greater than (the square on) ED by the (square) on (some straight-line) commensurable (in length) with (AE) , and neither of AE and ED [is] commensurable in length with AB [Def. 10.7]. So, similarly to that which has been previously demonstrated, we can show that MO is the square-root of area AC , and MN and NO are medial (straight-lines which are) commensurable in square only. Hence, MO is bimedral. So, we must show that (it is) also second (bimedral).

[And] since DE is incommensurable in length with AB —that is to say, with EK —and DE (is) commensurable (in length) with EF , EF is thus incommensurable in length with EK [Prop. 10.13]. And they are (both) rational (straight-lines). Thus, FE and EK are rational (straight-lines which are) commensurable in square only. EL —that is to say, MR —[is] thus medial [Prop. 10.21]. And it is

contained by MNO . Thus, the (rectangle contained) by MNO is medial.

Thus, MO is a second bimedral (straight-line) [Prop. 10.38]. (Which is) the very thing it was required to show.

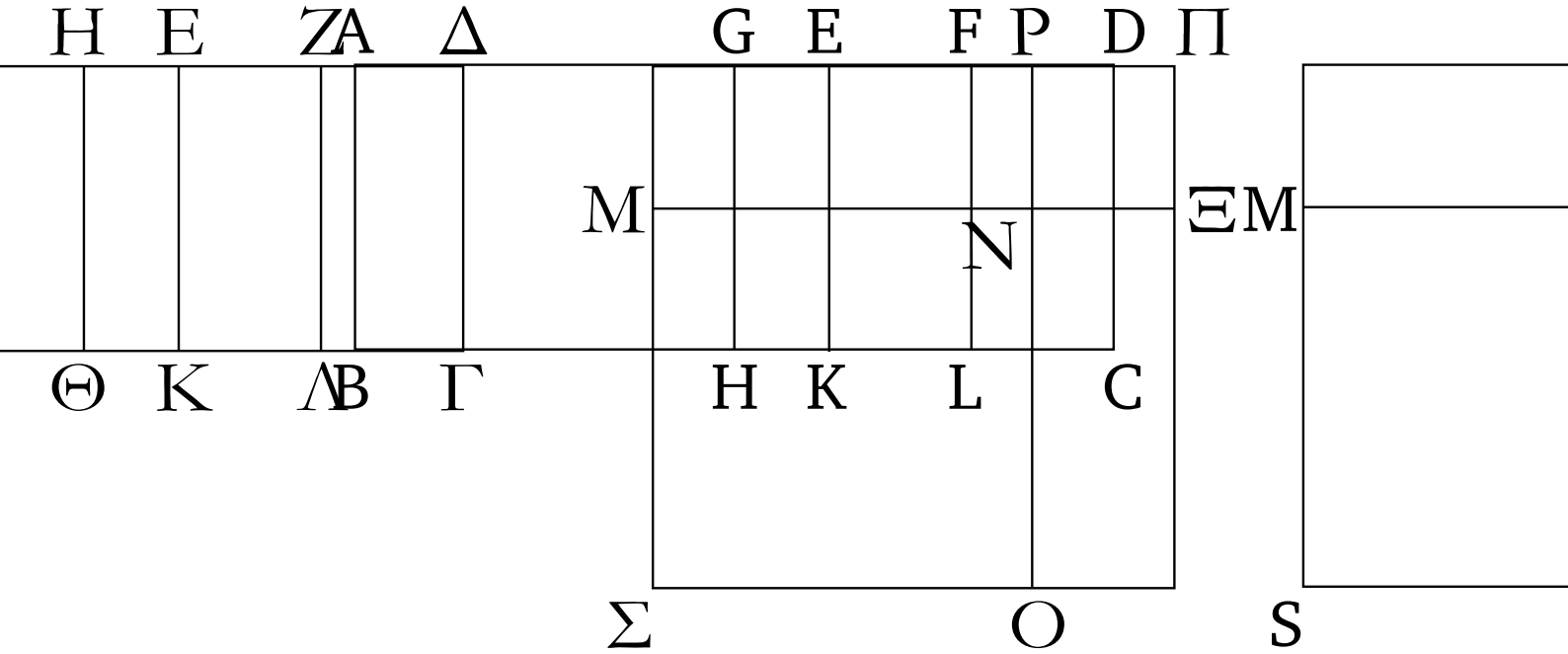
† If the rational straight-line has unit length then this proposition states that the square-root of a third binomial straight-line is a second bimedral straight-line: i.e., a third binomial straight-line has a length $k^{1/2}(1 + \sqrt{1 - k'^2})$ whose square-root can be written $\rho(k^{1/4} + k'^{1/2}/k^{1/4})$, where $\rho = \sqrt{(1 + k')/2}$ and $k'' = k(1 - k')/(1 + k')$. This is the length of a second bimedral straight-line (see Prop. 10.38), since ρ is rational.

57.

Proposition 57

Ἐάν θωρίον περιέθῃται ὑπὸ <ρητῆς καὶ τῆς ἐκ δύο ὀνομάτων τετάρτης, ἡ τὸ θωρίον δυναμένη >άλογός ἐστιν ἡ καλουμένη μείζων.

If an area is contained by a rational (straight-line) and a fourth binomial (straight-line) then the square-root of the area is the irrational (straight-line which is) called major.†



Θωρίον γὰρ τὸ ΑΓ περιέθεσθω ὑπὸ <ρητῆς τῆς ΑΒ καὶ τῆς ἐκ δύο ὀνομάτων τετάρτης τῆς ΑΔ διηρημένης εἰς τὰ ὀνόματα κατὰ τὸ Ε, <ὦν μείζων >έστω τὸ ΑΕ; λέγω, <ὅτι ἡ τὸ ΑΓ θωρίον δυναμένη >άλογός ἐστιν ἡ καλουμένη μείζων.

Ἐπεὶ γὰρ ἡ ΑΔ ἐκ δύο ὀνομάτων ἐστὶ τετάρτη, αἱ ΑΕ, ΕΔ >άρα <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ ΑΕ τῆς ΕΔ μείζων δύναται τῷ ἀπὸ ἀσύμμετρου ἐναντῇ, καὶ ἡ ΑΕ τῇ ΑΒ σύμμετρος [ἐστὶ] μήκει. τεμήσθω ἡ ΔΕ δίχα κατὰ τὸ Ζ, καὶ τῷ ἀπὸ τῆς ΕΖ >ἴσον παρὰ τὴν ΑΕ παραβλήσθω παραλληλόγραμμον τὸ ὑπὸ ΑΗ, ΗΕ; ἀσύμμετρος >άρα ἐστὶν ἡ ΑΗ τῇ ΗΕ μήκει. >ῆθωσαν παράλληλοι τῇ ΑΒ αἱ ΗΘ, ΕΚ, ΖΛ, καὶ τὰ λοιπὰ τὰ αὐτὰ τοῖς πρὸ τούτου γεγονέτω; φανερόν δὴ, <ὅτι ἡ τὸ ΑΓ θωρίον δυναμένη ἐστὶν ἡ ΜΧ. δεικτέον δὴ, <ὅτι ἡ ΜΧ >άλογός ἐστιν ἡ καλουμένη μείζων.

Ἐπεὶ ἀσύμμετρος ἐστὶν ἡ ΑΗ τῇ ΕΗ μήκει, ἀσύμμετρόν ἐστὶ καὶ τὸ ΑΘ τῷ ΗΚ, τουτέστι τὸ ΣΝ τῷ ΝΠ; αἱ ΜΝ, ΝΧ >άρα δυνάμει εἰσὶν ἀσύμμετροι. καὶ ἐπεὶ σύμμετρος ἐστὶν ἡ ΑΕ τῇ ΑΒ μήκει, <ρητόν ἐστι τὸ ΑΚ; καὶ ἐστὶν >ἴσον τοῖς ἀπὸ τῶν ΜΝ, ΝΧ; <ρητόν >άρα [ἐστὶ] καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΜΝ, ΝΧ. καὶ ἐπεὶ ἀσύμμετρος [ἐστὶν] ἡ ΔΕ τῇ ΑΒ μήκει, τουτέστι τῇ ΕΚ, ἀλλὰ ἡ ΔΕ σύμμετρος ἐστὶ τῇ ΕΖ, ἀσύμμετρος >άρα ἡ ΕΖ τῇ ΕΚ μήκει. αἱ ΕΚ, ΕΖ >άρα <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι; μέσον >άρα τὸ ΑΕ, τουτέστι τὸ ΜΡ. καὶ περιέθεται ὑπὸ τῶν ΜΝ, ΝΧ; μέσον >άρα ἐστὶ τὸ ὑπὸ τῶν ΜΝ, ΝΧ. καὶ <ρητόν

For let the area AC be contained by the rational (straight-line) AB and the fourth binomial (straight-line) AD , which has been divided into its (component) terms at E , of which let AE be the greater. I say that the square-root of AC is the irrational (straight-line which is) called major.

For since AD is a fourth binomial (straight-line), AE and ED are thus rational (straight-lines which are) commensurable in square only, and the square on AE is greater than (the square on) ED by the (square) on (some straight-line) incommensurable (in length) with (AE) , and AE [is] commensurable in length with AB [Def. 10.8]. Let DE be cut in half at F , and let the parallelogram (contained by) AG and GE , equal to the (square) on EF , (and falling short by a square figure) be applied to AE . AG is thus incommensurable in length with GE [Prop. 10.18]. Let GH , EK , and FL be drawn parallel to AB , and let the rest (of the construction) be made the same as the (proposition) before this. So, it is clear that MO is the square-root of area AC . So, we must show that MO is the irrational (straight-line which is) called major.

Since AG is incommensurable in length with EG , AH is also incommensurable with GK —that is to say, SN with NQ [Props. 6.1, 10.11]. Thus, MN and NO are incommensurable in square. And since AE is commensurable in length with AB , AK is rational [Prop. 10.19]. And it is equal to the (sum of the squares) on MN and NO . Thus, the sum of the (squares) on MN and NO [is] also ratio-

τὸ [συγκείμενον] ἐκ τῶν ἀπὸ τῶν MN , NX , καὶ εἰσιν ἀσύμμετροι αἱ MN , NX δυνάμει. ἔαν δὲ δύο εὐθεῖαι δυνάμει ἀσύμμετροι συντεθῶσι ποιῶσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων <ρητόν, τὸ δ' ὑπ' αὐτῶν μέσον, ἢ <όλη> ἀλογόξ ἐστίν, καλεῖται δὲ μείζων. Ἡ MX >άρα >ἀλογόξ ἐστίν ἢ καλουμένη μείζων, καὶ δύναται τὸ AG θωρίον; <όπερ >έδει δεῖχαι.

nal. And since DE [is] incommensurable in length with AB [Prop. 10.13]—that is to say, with EK —but DE is commensurable (in length) with EF , EF (is) thus incommensurable in length with EK [Prop. 10.13]. Thus, EK and EF are rational (straight-lines which are) commensurable in square only. LE —that is to say, MR —(is) thus medial [Prop. 10.21]. And it is contained by MN and NO . The (rectangle contained) by MN and NO is thus medial. And the [sum] of the (squares) on MN and NO (is) rational, and MN and NO are incommensurable in square. And if two straight-lines (which are) incommensurable in square, making the sum of the squares on them rational, and the (rectangle contained) by them medial, are added together, then the whole is the irrational (straight-line which is) called major [Prop. 10.39].

Thus, MO is the irrational (straight-line which is) called major. And (it is) the square-root of area AC . (Which is) the very thing it was required to show.

† If the rational straight-line has unit length then this proposition states that the square-root of a fourth binomial straight-line is a major straight-line: i.e., a fourth binomial straight-line has a length $k(1+1/\sqrt{1+k'})$ whose square-root can be written $\rho\sqrt{[1+k''/(1+k''^2)^{1/2}]/2} + \rho\sqrt{[1-k''/(1+k''^2)^{1/2}]/2}$, where $\rho = \sqrt{k}$ and $k''^2 = k'$. This is the length of a major straight-line (see Prop. 10.39), since ρ is rational.

58.

Ἐάν θωρίον περιέθῃται ὑπὸ <ρητῆς καὶ τῆς ἐκ δύο ὀνομάτων πέμπτῃς, ἢ τὸ θωρίον δυναμένη >ἀλογόξ ἐστίν ἢ καλουμένη <ρητόν καὶ μέσον δυναμένη.

Θωρίον γὰρ τὸ AG περιθέσθω ὑπὸ <ρητῆς τῆς AB καὶ τῆς ἐκ δύο ὀνομάτων πέμπτῃς τῆς AD διηρημένης εἰς τὰ ὀνόματα κατὰ τὸ E , <ώστε τὸ μείζον> ὄνομα εἶναι τὸ AE ; λέγω [δή], <ὅτι ἢ τὸ AG θωρίον δυναμένη >ἀλογόξ ἐστίν ἢ καλουμένη <ρητόν καὶ μέσον δυναμένη.

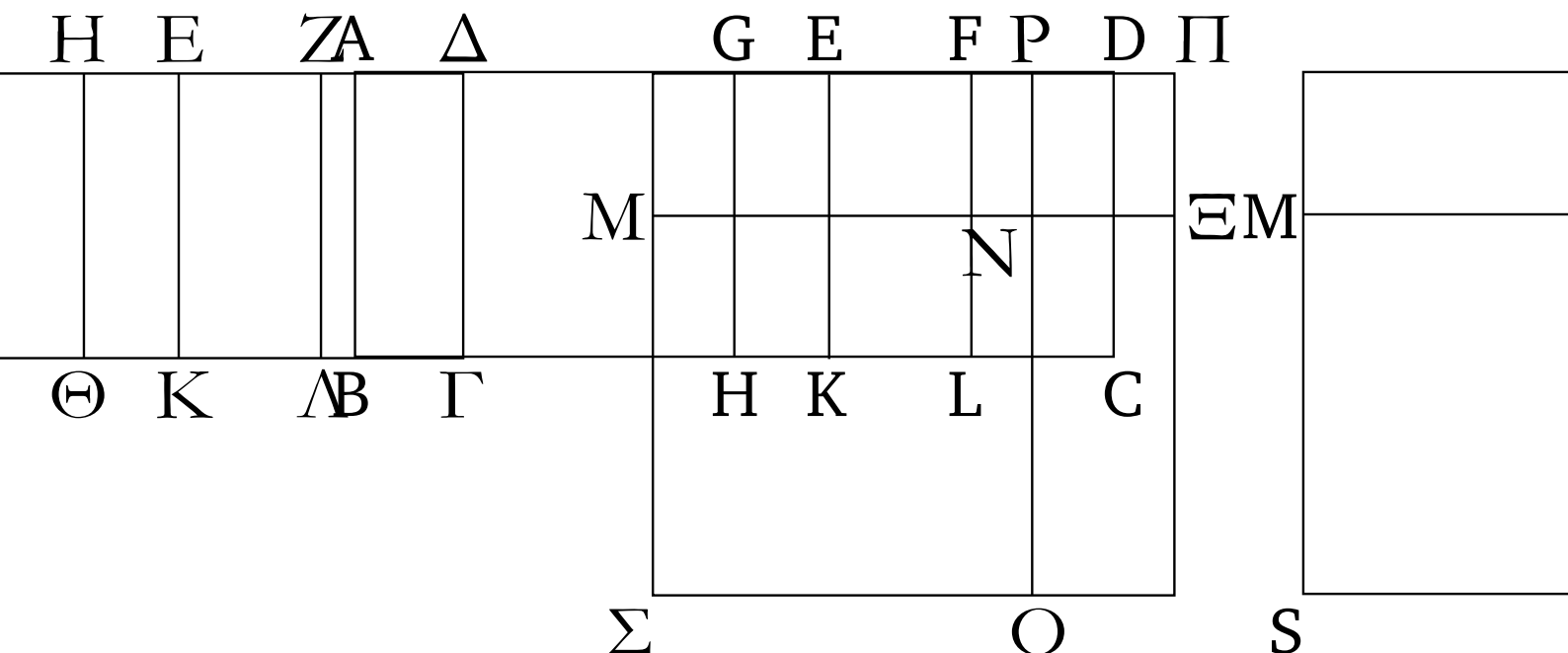
Κατεσκευάσθω γὰρ τὰ αὐτὰ τοῖς πρότερον δεδειγμένοις; φανερόν δὲ, <ὅτι ἢ τὸ AG θωρίον δυναμένη ἐστίν ἢ MX . δεικτέον δὲ, <ὅτι ἢ MX ἐστίν ἢ <ρητόν καὶ μέσον δυναμένη.

Proposition 58

If an area is contained by a rational (straight-line) and a fifth binomial (straight-line) then the square-root of the area is the irrational (straight-line which is) called the square-root of a rational plus a medial (area).†

For let the area AC be contained by the rational (straight-line) AB and the fifth binomial (straight-line) AD , which has been divided into its (component) terms at E , such that AE is the greater term. [So] I say that the square-root of area AC is the irrational (straight-line which is) called the square-root of a rational plus a medial (area).

For let the same construction be made as that shown previously. So, (it is) clear that MO is the square-root of area AC . So, we must show that MO is the square-root of a rational plus a medial (area).



Ἐπεὶ γὰρ ἀσύμμετρόξ ἐστίν ἢ AH τῇ HE , ἀσύμμετρον >άρα

For since AG is incommensurable (in length) with GE

ἐστὶ καὶ τὸ $ΑΘ$ τῷ $ΘΕ$, τουτέστι τὸ ἀπὸ τῆς MN τῷ ἀπὸ τῆς NX ; αἱ MN , NX >άρα δυνάμει εἰσὶν ἀσύμμετροι. καὶ ἐπεὶ ἡ $ΑΔ$ ἐκ δύο ὀνομάτων ἐστὶ πέμπτη, καὶ [ἐστὶν] >έλασσον αὐτῆς τμήμα τὸ $ΕΔ$, σύμμετρος >άρα ἡ $ΕΔ$ τῇ $ΑΒ$ μήκει. ἀλλὰ ἡ $ΑΕ$ τῇ $ΕΔ$ ἐστὶν ἀσύμμετρος; καὶ ἡ $ΑΒ$ >άρα τῇ $ΑΕ$ ἐστὶν ἀσύμμετρος μήκει [αἱ $ΒΑ$, $ΑΕ$ <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι]; μέσον >άρα ἐστὶ τὸ $ΑΚ$, τουτέστι τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν MN , NX . καὶ ἐπεὶ σύμμετρος ἐστὶν ἡ $ΔΕ$ τῇ $ΑΒ$ μήκει, τουτέστι τῇ $ΕΚ$, ἀλλὰ ἡ $ΔΕ$ τῇ $ΕΖ$ σύμμετρος ἐστὶν, καὶ ἡ $ΕΖ$ >άρα τῇ $ΕΚ$ σύμμετρος ἐστὶν. καὶ <ρητὴ ἡ $ΕΚ$; <ρητὸν >άρα καὶ τὸ $ΕΛ$, τουτέστι τὸ $ΜΡ$, τουτέστι τὸ ὑπὸ $ΜΝΧ$; αἱ MN , NX >άρα δυνάμει ἀσύμμετροί εἰσι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον, τὸ δ' ὑπ' αὐτῶν <ρητόν. Ἡ $ΜΧ$ >άρα <ρητόν καὶ μέσον δυναμένη ἐστὶ καὶ δύναται τὸ $ΑΓ$ θωρίον; <όπερ >έδει δεῖχαι.

[Prop. 10.18], AH is thus also incommensurable with HE —that is to say, the (square) on MN with the (square) on NO [Props. 6.1, 10.11]. Thus, MN and NO are incommensurable in square. And since AD is a fifth binomial (straight-line), and ED [is] its lesser segment, ED (is) thus commensurable in length with AB [Def. 10.9]. But, AE is incommensurable (in length) with ED . Thus, AB is also incommensurable in length with AE [BA and AE are rational (straight-lines which are) commensurable in square only] [Prop. 10.13]. Thus, AK —that is to say, the sum of the (squares) on MN and NO —is medial [Prop. 10.21]. And since DE is commensurable in length with AB —that is to say, with EK —but, DE is commensurable (in length) with EF , EF is thus also commensurable (in length) with EK [Prop. 10.12]. And EK (is) rational. Thus, EL —that is to say, MR —that is to say, the (rectangle contained) by MNO —(is) also rational [Prop. 10.19]. MN and NO are thus (straight-lines which are) incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them rational. Thus, MO is the square-root of a rational plus a medial (area) [Prop. 10.40]. And (it is) the square-root of area AC . (Which is) the very thing it was required to show.

† If the rational straight-line has unit length then this proposition states that the square-root of a fifth binomial straight-line is the square root of a rational plus a medial area: i.e., a fifth binomial straight-line has a length $k(\sqrt{1+k'}+1)$ whose square-root can be written

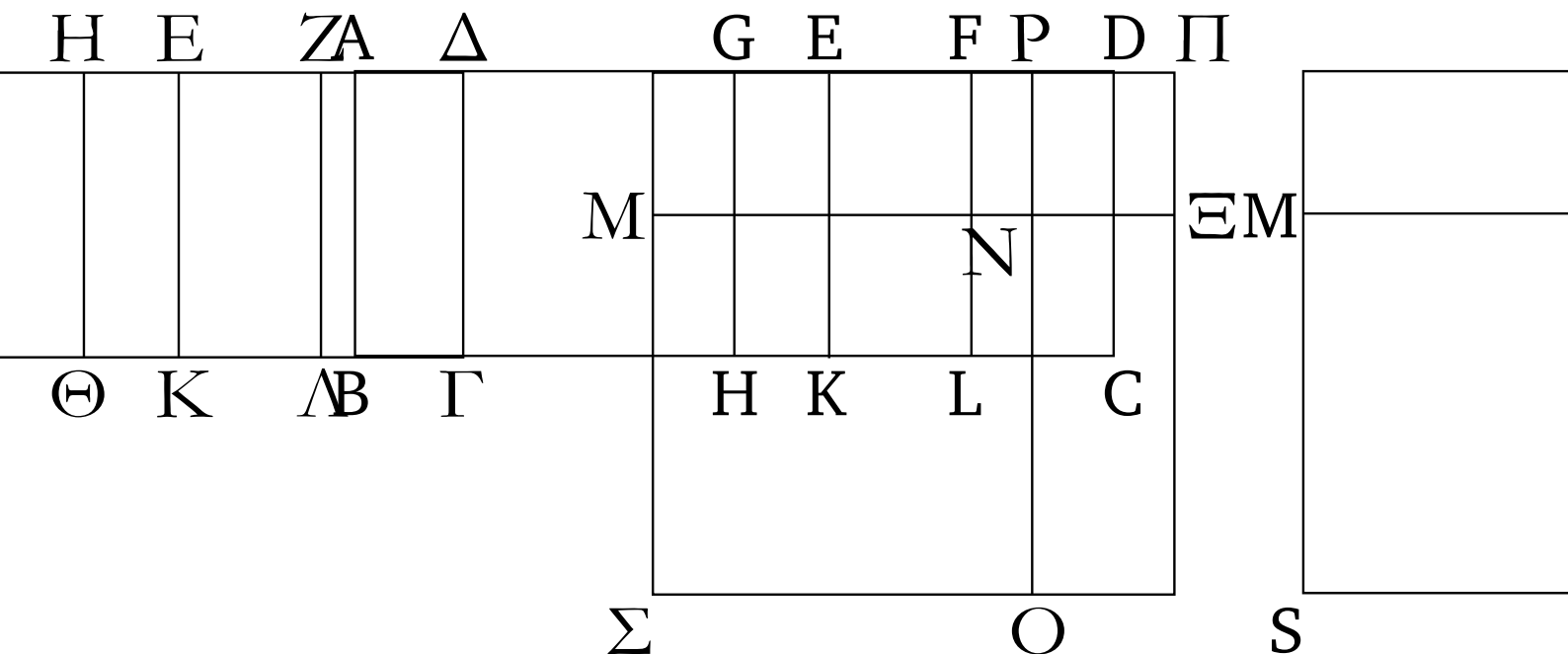
$\rho\sqrt{[(1+k''^2)^{1/2}+k'']/[2(1+k''^2)]} + \rho\sqrt{[(1+k''^2)^{1/2}-k'']/[2(1+k''^2)]}$, where $\rho = \sqrt{k(1+k''^2)}$ and $k''^2 = k'$. This is the length of the square root of a rational plus a medial area (see Prop. 10.40), since ρ is rational.

59.

Proposition 59

Ἐὰν θωρίον περιέθῃται ὑπὸ <ρητῆς καὶ τῆς ἐκ δύο ὀνομάτων <έκτης, ἡ τὸ θωρίον δυναμένη >άλογός ἐστὶν ἡ καλουμένη δύο μέσα δυναμένη.

If an area is contained by a rational (straight-line) and a sixth binomial (straight-line) then the square-root of the area is the irrational (straight-line which is) called the square-root of (the sum of) two medial (areas).†



Θωρίον γὰρ τὸ $ΑΒΓΔ$ περιέθεσθω ὑπὸ <ρητῆς τῆς $ΑΒ$ καὶ τῆς ἐκ δύο ὀνομάτων <έκτης τῆς $ΑΔ$ διηρημένης εἰς τὰ ὀνόματα κατὰ τὸ $Ε$, <ώστε τὸ μείζον >όνομα εἶναι τὸ $ΑΕ$; λέγω, <ὅτι ἡ τὸ $ΑΓ$ δυναμένη ἡ δύο μέσα δυναμένη ἐστὶν. Κατεσκευάσθω [γὰρ] τὰ αὐτὰ τοῖς προδεδειγμένοις. φανερόν

For let the area $ABCD$ be contained by the rational (straight-line) AB and the sixth binomial (straight-line) AD , which has been divided into its (component) terms at E , such that AE is the greater term. So, I say that the square-root of AC is the square-root of (the sum of) two

δή, <ὅτι [ή] τὸ ΑΓ δυνάμεν ἔστιν ἢ ΜΧ, καὶ <ὅτι ἀσύμμετρος medial (areas).

ἔστιν ἢ ΜΝ τῇ ΝΧ δυνάμει. καὶ ἐπεὶ ἀσύμμετρος ἔστιν ἢ ΕΑ τῇ ΑΒ μήκει, αἱ ΕΑ, ΑΒ >άρα <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι; μέσον >άρα ἔστι τὸ ΑΚ, τουτέστι τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΜΝ, ΝΧ. πάλιν, ἐπεὶ ἀσύμμετρος ἔστιν ἢ ΕΔ τῇ ΑΒ μήκει, ἀσύμμετρος >άρα ἔστι καὶ ἢ ΖΕ τῇ ΕΚ; αἱ ΖΕ, ΕΚ >άρα <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι; μέσον >άρα ἔστι τὸ ΕΛ, τουτέστι τὸ ΜΡ, τουτέστι τὸ ὑπὸ τῶν ΜΝΧ. καὶ ἐπεὶ ἀσύμμετρος ἢ ΑΕ τῇ ΕΖ, καὶ τὸ ΑΚ τῷ ΕΛ ἀσύμμετρον ἔστιν. ἀλλὰ τὸ μὲν ΑΚ ἔστι τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΜΝ, ΝΧ, τὸ δὲ ΕΛ ἔστι τὸ ὑπὸ τῶν ΜΝΧ; ἀσύμμετρον >άρα ἔστι τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΜΝΧ τῷ ὑπὸ τῶν ΜΝΧ. καὶ ἔστι μέσον ἐκάτερον αὐτῶν, καὶ αἱ ΜΝ, ΝΧ δυνάμει εἰσὶν ἀσύμμετροι. Ἡ ΜΧ >άρα δύο μέσα δυνάμεν ἔστι καὶ δύναται τὸ ΑΓ; <όπερ >έδει δεῖχαι.

[For] let the same construction be made as that shown previously. So, (it is) clear that MO is the square-root of AC , and that MN is incommensurable in square with NO . And since EA is incommensurable in length with AB [Def. 10.10], EA and AB are thus rational (straight-lines which are) commensurable in square only. Thus, AK —that is to say, the sum of the (squares) on MN and NO —is medial [Prop. 10.21]. Again, since ED is incommensurable in length with AB [Def. 10.10], FE is thus also incommensurable (in length) with EK [Prop. 10.13]. Thus, FE and EK are rational (straight-lines which are) commensurable in square only. Thus, EL —that is to say, MR —that is to say, the (rectangle contained) by MNO —is medial [Prop. 10.21]. And since AE is incommensurable (in length) with EF , AK is also incommensurable with EL [Props. 6.1, 10.11]. But, AK is the sum of the (squares) on MN and NO , and EL is the (rectangle contained) by MNO . Thus, the sum of the (squares) on MNO is incommensurable with the (rectangle contained) by MNO . And each of them is medial. And MN and NO are incommensurable in square.

Thus, MO is the square-root of (the sum of) two medial (areas) [Prop. 10.41]. And (it is) the square-root of AC . (Which is) the very thing it was required to show.

† If the rational straight-line has unit length then this proposition states that the square-root of a sixth binomial straight-line is the square root of the sum of two medial areas: i.e., a sixth binomial straight-line has a length $\sqrt{k} + \sqrt{k'}$ whose square-root can be written $k^{1/4} \left(\sqrt{[1 + k''/(1 + k''^2)^{1/2}]/2} + \sqrt{[1 - k''/(1 + k''^2)^{1/2}]/2} \right)$, where $k''^2 = (k - k')/k'$. This is the length of the square-root of the sum of two medial areas (see Prop. 10.41).

Λήμμα.

Ἐὰν εὐθεῖα γραμμὴ τμηθῇ εἰς >άνισα, τὰ ἀπὸ τῶν ἀνίσων τετράγωνα μείζονά ἐστι τοῦ διζῶν ὑπὸ τῶν ἀνίσων περιεθρομένου ὀρθογωνίου.

Lemma

If a straight-line is cut unequally then (the sum of) the squares on the unequal (parts) is greater than twice the rectangle contained by the unequal (parts).



>Ἐστω εὐθεῖα ἢ ΑΒ καὶ τετμήσθω εἰς >άνισα κατὰ τὸ Γ, καὶ >έστω μείζων ἢ ΑΓ; λέγω, <ὅτι τὰ ἀπὸ τῶν ΑΓ, ΓΒ μείζονά ἐστι τοῦ διζῶν ὑπὸ τῶν ΑΓ, ΓΒ.

Τετμήσθω γὰρ ἢ ΑΒ διὰ κατὰ τὸ Δ. ἐπεὶ ο>ὺν εὐθεῖα γραμμὴ τέτμηται εἰς μὲν >ῖσα κατὰ τὸ Δ, εἰς δὲ >άνισα κατὰ τὸ Γ, τὸ >άρα ὑπὸ τῶν ΑΓ, ΓΒ μετὰ τοῦ ἀπὸ ΓΔ >ῖσον ἔστι τῷ ἀπὸ ΑΔ; <ώστε τὸ ὑπὸ τῶν ΑΓ, ΓΒ >έλαττον ἔστι τοῦ ἀπὸ ΑΔ; τὸ >άρα διζῶν ὑπὸ τῶν ΑΓ, ΓΒ >έλαττον >ῖ διπλάσιόν ἐστι τοῦ ἀπὸ ΑΔ. ἀλλὰ τὰ ἀπὸ τῶν ΑΓ, ΓΒ διπλάσιά [ἐστι] τῶν ἀπὸ τῶν ΑΔ, ΔΓ; τὰ >άρα ἀπὸ τῶν ΑΓ, ΓΒ μείζονά ἐστι τοῦ διζῶν ὑπὸ τῶν ΑΓ, ΓΒ; <όπερ >έδει δεῖχαι.

Let AB be a straight-line, and let it be cut unequally at C , and let AC be greater (than CB). I say that (the sum of) the (squares) on AC and CB is greater than twice the (rectangle contained) by AC and CB .

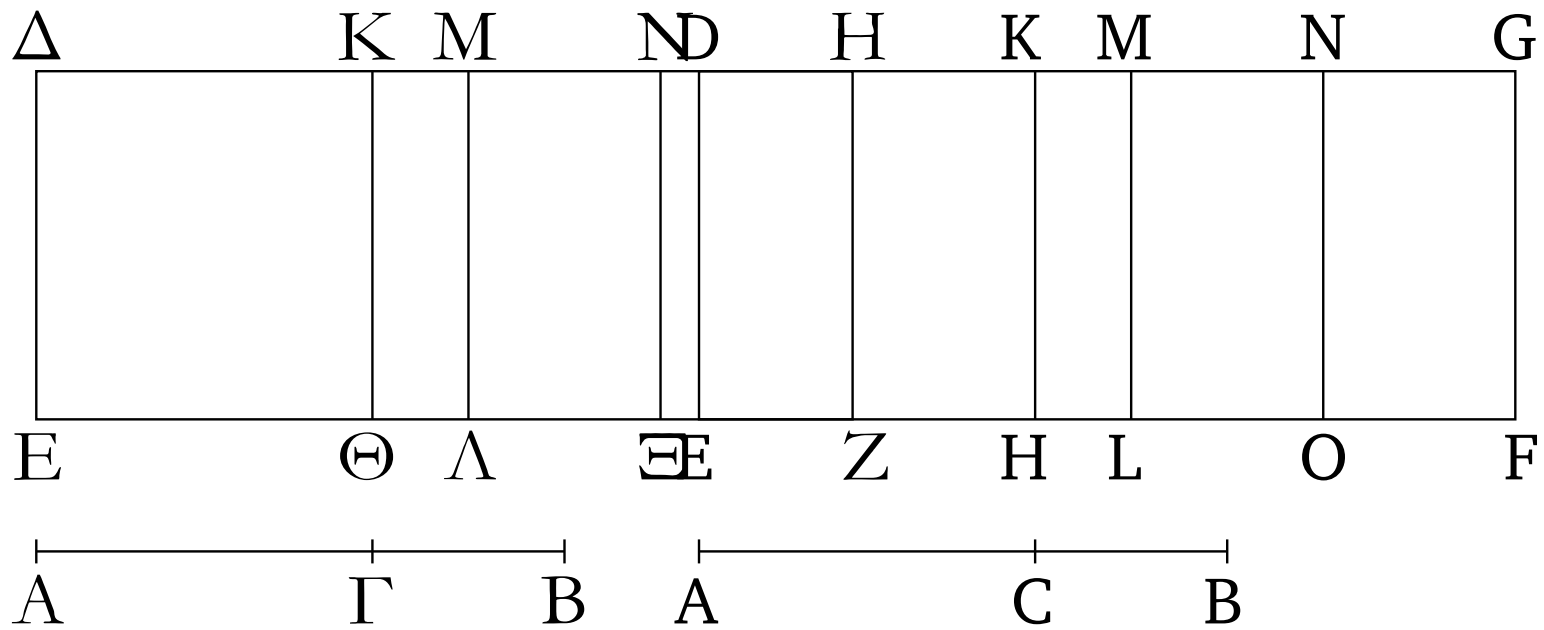
For let AB be cut in half at D . Therefore, since a straight-line has been cut into equal (parts) at D , and into unequal (parts) at C , the (rectangle contained) by AC and CB , plus the (square) on CD , is thus equal to the (square) on AD [Prop. 2.5]. Hence, the (rectangle contained) by AC and CB is less than the (square) on AD . Thus, twice the (rectangle contained) by AC and CB is less than double the (square) on AD . But, (the sum of) the (squares) on AC and CB [is] double (the sum of) the (squares) on AD and DC [Prop. 2.9]. Thus, (the sum of) the (squares) on AC and CB is greater than twice the (rectangle contained) by AC and CB . (Which is) the very thing it was required to show.

60.

Τὸ ἀπὸ τῆς ἐκ δύο ὀνομάτων παρὰ <ρητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων πρώτην.

Proposition 60

The square on a binomial (straight-line) applied to a rational (straight-line) produces as breadth a first binomial (straight-line).†



>Ἐστω ἐκ δύο ὀνομάτων ἡ AB διηρημένη εἰς τὰ ὀνόματα κατὰ τὸ Γ , <ώστε τὸ μείζον >ὄνομα εἶναι τὸ $ΑΓ$, καὶ ἐκκείσθω <ρητὴ ἡ $ΔΕ$, καὶ τῷ ἀπὸ τῆς AB >ἴσον παρὰ τὴν $ΔΕ$ παραβεβλήσθω τὸ $ΔΕΖΗ$ πλάτος ποιῶν τὴν $ΔΗ$; λέγω, <ὅτι ἡ $ΔΗ$ ἐκ δύο ὀνομάτων ἐστὶ πρώτη.

Παραβεβλήσθω γὰρ παρὰ τὴν $ΔΕ$ τῷ μὲν ἀπὸ τῆς $ΑΓ$ >ἴσον τὸ $ΔΘ$, τῷ δὲ ἀπὸ τῆς $ΒΓ$ >ἴσον τὸ $ΚΛ$; λοιπὸν >άρα τὸ διζ ὑπὸ τῶν $ΑΓ$, $ΒΓ$ >ἴσον ἐστὶ τῷ $ΜΖ$. τετμήσθω ἡ $ΜΗ$ διῖα κατὰ τὸ $Ν$, καὶ παράλληλος >ήθθω ἡ $ΝΧ$ [ἐκατέρω τῶν $ΜΛ$, $ΗΖ$]. ἐκάτερον >άρα τῶν $ΜΧ$, $ΝΖ$ >ἴσον ἐστὶ τῷ <άπαχ ὑπὸ τῶν $ΑΓΒ$. καὶ ἐπεὶ ἐκ δύο ὀνομάτων ἐστὶν ἡ AB διηρημένη εἰς τὰ ὀνόματα κατὰ τὸ Γ , αἱ $ΑΓ$, $ΒΓ$ >άρα <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι; τὰ >άρα ἀπὸ τῶν $ΑΓ$, $ΒΓ$ <ρητά ἐστὶ καὶ σύμμετρα ἀλλήλοις; <ώστε καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν $ΑΓ$, $ΒΓ$. καὶ ἐστὶν >ἴσον τῷ $ΔΛ$; <ρητὸν >άρα ἐστὶ τὸ $ΔΛ$. καὶ παρὰ <ρητὴν τὴν $ΔΕ$ παράκειται; <ρητὴ >άρα ἐστὶν ἡ $ΔΜ$ καὶ σύμμετρος τῇ $ΔΕ$ μήκει. πάλιν, ἐπεὶ αἱ $ΑΓ$, $ΒΓ$ <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι, μέσον >άρα ἐστὶ τὸ διζ ὑπὸ τῶν $ΑΓ$, $ΒΓ$, τουτέστι τὸ $ΜΖ$. καὶ παρὰ <ρητὴν τὴν $ΜΛ$ παράκειται; <ρητὴ >άρα καὶ ἡ $ΜΗ$ καὶ ἀσύμμετρος τῇ $ΜΛ$, τουτέστι τῇ $ΔΕ$, μήκει. >έστι δὲ καὶ ἡ $ΜΔ$ <ρητὴ καὶ τῇ $ΔΕ$ μήκει σύμμετρος; ἀσύμμετρος >άρα ἐστὶν ἡ $ΔΜ$ τῇ $ΜΗ$ μήκει. καὶ εἰσι <ρηταί; αἱ $ΔΜ$, $ΜΗ$ >άρα <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι; ἐκ δύο >άρα ὀνομάτων ἐστὶν ἡ $ΔΗ$. δεικτέον δὴ, <ὅτι καὶ πρώτη.

Ἐπεὶ τῶν ἀπὸ τῶν $ΑΓ$, $ΒΓ$ μέσον ἀνάλογόν ἐστὶ τὸ ὑπὸ τῶν $ΑΓΒ$, καὶ τῶν $ΔΘ$, $ΚΛ$ >άρα μέσον ἀνάλογόν ἐστὶ τὸ $ΜΧ$. >έστιν >άρα ὡς τὸ $ΔΘ$ πρὸς τὸ $ΜΧ$, οὕτως τὸ $ΜΧ$ πρὸς τὸ $ΚΛ$, τουτέστιν ὡς ἡ $ΔΚ$ πρὸς τὴν $ΜΝ$, ἡ $ΜΝ$ πρὸς τὴν $ΜΚ$; τὸ >άρα ὑπὸ τῶν $ΔΚ$, $ΚΜ$ >ἴσον ἐστὶ τῷ ἀπὸ τῆς $ΜΝ$. καὶ ἐπεὶ σύμμετρόν ἐστὶ τὸ ἀπὸ τῆς $ΑΓ$ τῷ ἀπὸ τῆς $ΒΓ$, σύμμετρόν ἐστὶ καὶ τὸ $ΔΘ$ τῷ $ΚΛ$; <ώστε καὶ ἡ $ΔΚ$ τῇ $ΚΜ$ σύμμετρόξ ἐστίν. καὶ ἐπεὶ μείζονά ἐστὶ τὰ ἀπὸ τῶν $ΑΓ$, $ΒΓ$ τοῦ διζ ὑπὸ τῶν $ΑΓ$, $ΒΓ$, μείζον >άρα καὶ τὸ $ΔΛ$ τοῦ $ΜΖ$; <ώστε καὶ ἡ $ΔΜ$ τῆς $ΜΗ$ μείζων ἐστίν. καὶ ἐστὶν >ἴσον τὸ ὑπὸ τῶν $ΔΚ$, $ΚΜ$ τῷ ἀπὸ τῆς $ΜΝ$, τουτέστι τῷ τετάρτῳ τοῦ ἀπὸ τῆς $ΜΗ$ - $\pi\tau\eta$, καὶ σύμμετρος ἡ $ΔΚ$ τῇ $ΚΜ$. ἔαν δὲ >ῶσι δύο εὐθεῖαι >άνισοι, τῷ δὲ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ἐλάσσονος >ἴσον παρὰ τὴν μείζονα παραβληθῇ; ἐλλείπον ε>ἶδει τετραγώνῳ καὶ εἰς σύμμετρα αὐτὴν διαιρῇ; ἡ μείζων τῆς ἐλάσσονος μείζον δύναται τῷ ἀπὸ συμέτρου ἑαυτῇ;

Let AB be a binomial (straight-line), having been divided into its (component) terms at C , such that AC is the greater term. And let the rational (straight-line) DE be laid down. And let the (rectangle) $DEFG$, equal to the (square) on AB , be applied to DE , producing DG as breadth. I say that DG is a first binomial (straight-line).

For let DH , equal to the (square) on AC , and KL , equal to the (square) on BC , be applied to DE . Thus, the remaining twice the (rectangle contained) by AC and CB is equal to MF [Prop. 2.4]. Let MG be cut in half at N , and let NO be drawn parallel [to each of ML and GF]. MO and NF are thus each equal to once the (rectangle contained) by ACB . And since AB is a binomial (straight-line), having been divided into its (component) terms at C , AC and CB are thus rational (straight-lines which are) commensurable in square only [Prop. 10.36]. Thus, the (squares) on AC and CB are rational, and commensurable with one another. And hence the sum of the (squares) on AC and CB (is rational) [Prop. 10.15], and is equal to DL . Thus, DL is rational. And it is applied to the rational (straight-line) DE . DM is thus rational, and commensurable in length with DE [Prop. 10.20]. Again, since AC and CB are rational (straight-lines which are) commensurable in square only, twice the (rectangle contained) by AC and CB —that is to say, MF —is thus medial [Prop. 10.21]. And it is applied to the rational (straight-line) ML . MG is thus also rational, and incommensurable in length with ML —that is to say, with DE [Prop. 10.22]. And MD is also rational, and commensurable in length with DE . Thus, DM is incommensurable in length with MG [Prop. 10.13]. And they are rational. DM and MG are thus rational (straight-lines which are) commensurable in square only. Thus, DG is a binomial (straight-line) [Prop. 10.36]. So, we must show that (it is) also a first (binomial straight-line).

Since the (rectangle contained) by ACB is the mean proportional to the squares on AC and CB [Prop. 10.53 lem.], MO is thus also the mean proportional to DH and KL . Thus, as DH is to MO , so MO (is) to KL —that is to say, as DK (is) to MN , (so) MN (is) to MK [Prop. 6.1]. Thus, the (rectangle contained) by DK and KM is equal to the (square) on MN [Prop. 6.17]. And since the (square) on

ἡ $\Delta M > \alpha\rho\alpha$ τῆς MH μείζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ|. καὶ εἰσι <ρηταὶ αἱ ΔM , MH , καὶ ἡ ΔM μείζον >ὄνομα ο>ῦσα σύμμετρος ἐστὶ τῇ ἐκκειμένη <ρητῇ| τῇ ΔE μήκει.
Ἡ $\Delta H > \alpha\rho\alpha$ ἐκ δύο ὀνομάτων ἐστὶ πρώτη; <ὅπερ >έδει δεῖχαι.

AC is commensurable with the (square) on CB , DH is also commensurable with KL . Hence, DK is also commensurable with KM [Props. 6.1, 10.11]. And since (the sum of) the squares on AC and CB is greater than twice the (rectangle contained) by AC and CB [Prop. 10.59 lem.], DL (is) thus also greater than MF . Hence, DM is also greater than MG [Props. 6.1, 5.14]. And the (rectangle contained) by DK and KM is equal to the (square) on MN —that is to say, to one quarter the (square) on MG . And DK (is) commensurable (in length) with KM . And if there are two unequal straight-lines, and a (rectangle) equal to the fourth part of the (square) on the lesser, falling short by a square figure, is applied to the greater, and divides it into (parts which are) commensurable (in length), then the square on the greater is larger than (the square on) the lesser by the (square) on (some straight-line) commensurable (in length) with the greater [Prop. 10.17]. Thus, the square on DM is greater than (the square on) MG by the (square) on (some straight-line) commensurable (in length) with (DM). And DM and MG are rational. And DM , which is the greater term, is commensurable in length with the (previously) laid down rational (straight-line) DE .

Thus, DG is a first binomial (straight-line) [Def. 10.5]. (Which is) the very thing it was required to show.

† In other words, the square of a binomial is a first binomial. See Prop. 10.54.

61.

Τὸ ἀπὸ τῆς ἐκ δύο μέσων πρώτης παρὰ <ρητὴν παραβαλλόμενης πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων δευτέραν.

>Ἐστω ἐκ δύο μέσων πρώτη ἡ AB διηρημένη εἰς τὰς μέσας κατὰ τὸ Γ , <ῶν μείζων ἡ AG , καὶ ἐκκείσθω <ρητὴ ἡ ΔE , καὶ παραβεβλήσθω παρὰ τὴν ΔE τῷ ἀπὸ τῆς AB >ῖσον παραλληλόγραμμον τὸ ΔZ πλάτος ποιοῦν τὴν ΔH ; λέγω, <ὅτι ἡ ΔH ἐκ δύο ὀνομάτων ἐστὶ δευτέρα.

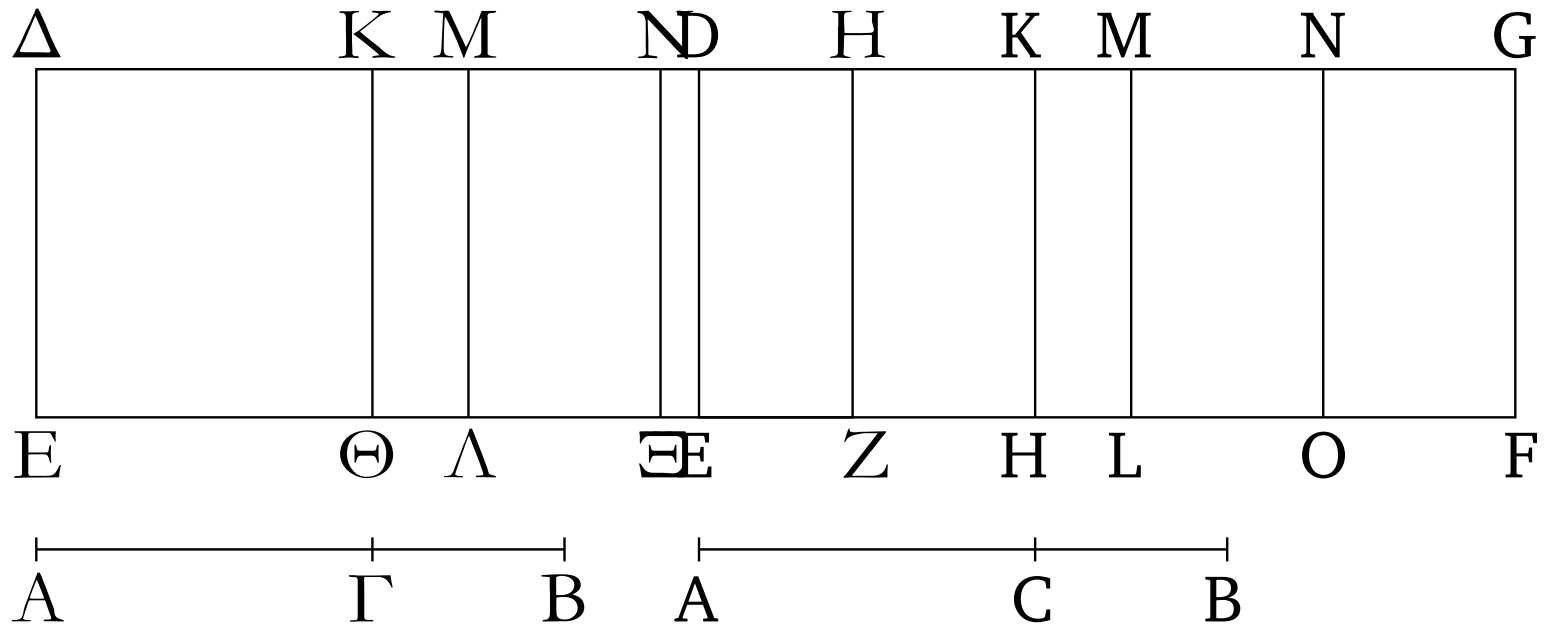
Κατεσκευάσθω γὰρ τὰ αὐτὰ τοῖς πρὸ τούτου. καὶ ἐπεὶ ἡ AB ἐκ δύο μέσων ἐστὶ πρώτη διηρημένη κατὰ τὸ Γ , αἱ AG , $GB > \alpha\rho\alpha$ μέσαι εἰσὶ δυνάμει μόνον σύμμετροι <ρητὸν περιέθουσai; <ῶστε καὶ τὰ ἀπὸ τῶν AG , GB μέσα ἐστίν. μέσον >ῖσον ἐστὶ τὸ $\Delta\Lambda$. καὶ παρὰ <ρητὴν τὴν ΔE παραβεβλήται; <ρητὴ >ῖσον ἐστὶν ἡ $M\Delta$ καὶ ἀσύμμετρος τῇ ΔE μήκει. πάλιν, ἐπεὶ <ρητὸν ἐστὶ τὸ διζ ὑπὸ τῶν AG , GB , <ρητὸν ἐστὶ καὶ τὸ MZ . καὶ παρὰ <ρητὴν τὴν $M\Lambda$ παράκειται; <ρητὴ >ῖσον [ἐστὶ] καὶ ἡ MH καὶ μήκει σύμμετρος τῇ $M\Lambda$, τουτέστι τῇ ΔE ; ἀσύμμετρος >ῖσον ἐστὶν ἡ ΔM τῇ MH μήκει. καὶ εἰσι <ρηταί; αἱ ΔM , $MH > \alpha\rho\alpha$ <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι; ἐκ δύο >ῖσον ὀνομάτων ἐστὶν ἡ ΔH . δεικτέον δὴ, <ὅτι καὶ δευτέρα.

Proposition 61

The square on a first binomial (straight-line) applied to a rational (straight-line) produces as breadth a second binomial (straight-line).[†]

Let AB be a first binomial (straight-line) having been divided into its (component) medial (straight-lines) at C , of which AC (is) the greater. And let the rational (straight-line) DE be laid down. And let the parallelogram DF , equal to the (square) on AB , be applied to DE , producing DG as breadth. I say that DG is a second binomial (straight-line).

For let the same construction be made as in the (proposition) before this. And since AB is a first binomial (straight-line), having been divided at C , AC and CB are thus medial (straight-lines) commensurable in square only, and containing a rational (area) [Prop. 10.37]. Hence, the (squares) on AC and CB are also medial [Prop. 10.21]. Thus, DL is medial [Props. 10.15, 10.23 corr.]. And it has been applied to the rational (straight-line) DE . MD is thus rational, and incommensurable in length with DE [Prop. 10.22]. Again, since twice the (rectangle contained) by AC and CB is rational, MF is also rational. And it is applied to the rational (straight-line) ML . Thus, MG [is] also rational, and commensurable in length with ML —that is to say, with DE [Prop. 10.20]. DM is thus incommensurable in length with MG [Prop. 10.13]. And they are rational. DM and MG are thus rational, and commensurable in square only. DG is thus a binomial (straight-line) [Prop. 10.36]. So, we must show that (it is) also a second (binomial straight-line).



Ἐπεὶ γὰρ τὰ ἀπὸ τῶν ΑΓ, ΓΒ μείζονά ἐστι τοῦ δι᾽ ὑπὸ τῶν ΑΓ, ΓΒ, μείζον > ἄρα καὶ τὸ ΔΛ τοῦ ΜΖ; < ὥστε καὶ ἡ ΔΜ τῆς ΜΗ. καὶ ἐπεὶ σύμμετρόν ἐστι τὸ ἀπὸ τῆς ΑΓ τῷ ἀπὸ τῆς ΓΒ, σύμμετρόν ἐστι καὶ τὸ ΔΘ τῷ ΚΛ; < ὥστε καὶ ἡ ΔΚ τῇ ΚΜ σύμμετρόξ ἐστιν. καὶ ἐστι τὸ ὑπὸ τῶν ΔΚΜ > ἴσον τῷ ἀπὸ τῆς ΜΝ; ἡ ΔΜ > ἄρα τῆς ΜΗ μείζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ. καὶ ἐστιν ἡ ΜΗ σύμμετροξ τῇ ΔΕ μήκει.

Ἡ ΔΗ > ἄρα ἐκ δύο ὀνομάτων ἐστὶ δευτέρα.

For since (the sum of) the squares on AC and CB is greater than twice the (rectangle contained) by AC and CB [Prop. 10.59], DL (is) thus also greater than MF . Hence, DM (is) also (greater) than MG [Prop. 6.1]. And since the (square) on AC is commensurable with the (square) on CB , DH is also commensurable with KL . Hence, DK is also commensurable (in length) with KM [Props. 6.1, 10.11]. And the (rectangle contained) by DKM is equal to the (square) on MN . Thus, the square on DM is greater than (the square on) MG by the (square) on (some straight-line) commensurable (in length) with (DM) [Prop. 10.17]. And MG is commensurable in length with DE .

Thus, DG is a second binomial (straight-line) [Def. 10.6].

† In other words, the square of a first binomial is a second binomial. See Prop. 10.55.

62.

Proposition 62

Τὸ ἀπὸ τῆς ἐκ δύο μέσων δευτέρας παρὰ < ρητὴν παραβαλλόμενης πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων τρίτην.

> Ἐστω ἐκ δύο μέσων δευτέρα ἡ ΑΒ διηρημένη εἰς τὰς μέσας κατὰ τὸ Γ, < ὥστε τὸ μείζον τμήμα εἶναι τὸ ΑΓ, < ρητὴ δέ τις > ἔστω ἡ ΔΕ, καὶ παρὰ τὴν ΔΕ τῷ ἀπὸ τῆς ΑΒ > ἴσον παραλληλόγραμμον παραβλήσθω τὸ ΔΖ πλάτος ποιῶν τὴν ΔΗ; λέγω, < ὅτι ἡ ΔΗ ἐκ δύο ὀνομάτων ἐστὶ τρίτη.

Κατεσκευάσθω τὰ αὐτὰ τοῖς προδεδειγμένοις. καὶ ἐπεὶ ἐκ δύο μέσων δευτέρα ἐστὶν ἡ ΑΒ διηρημένη κατὰ τὸ Γ, αἱ ΑΓ, ΓΒ > ἄρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι μέσον περιέθουσαι; < ὥστε καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΑΓ, ΓΒ μέσον ἐστίν. καὶ ἐστιν > ἴσον τῷ ΔΛ; μέσον > ἄρα καὶ τὸ ΔΛ. καὶ παράκειται παρὰ < ρητὴν τὴν ΔΕ; < ρητὴ > ἄρα ἐστὶ καὶ ἡ ΜΔ καὶ ἀσύμμετροξ τῇ ΔΕ μήκει. διὰ τὰ αὐτὰ δὴ καὶ ἡ ΜΗ < ρητὴ ἐστὶ καὶ ἀσύμμετροξ τῇ ΜΛ, τουτέστι τῇ ΔΕ, μήκει; < ρητὴ > ἄρα ἐστὶν ἑκάτερα τῶν ΔΜ, ΜΗ καὶ ἀσύμμετροξ τῇ ΔΕ μήκει. καὶ ἐπεὶ ἀσύμμετρόξ ἐστιν ἡ ΑΓ τῇ ΓΒ μήκει, ὥς δὲ ἡ ΑΓ πρὸς τὴν ΓΒ, οὕτως τὸ ἀπὸ τῆς ΑΓ πρὸς τὸ ὑπὸ τῶν ΑΓΒ, ἀσύμμετρον > ἄρα καὶ τὸ ἀπὸ τῆς ΑΓ τῷ ὑπὸ τῶν ΑΓΒ. < ὥστε καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΑΓ, ΓΒ τῷ δι᾽ ὑπὸ τῶν ΑΓΒ ἀσύμμετρόν ἐστιν, τουτέστι τὸ ΔΛ τῷ ΜΖ; < ὥστε καὶ ἡ ΔΜ τῇ ΜΗ ἀσύμμετρόξ ἐστιν. καὶ εἰσὶ < ρηταί; ἐκ δύο > ἄρα ὀνομάτων

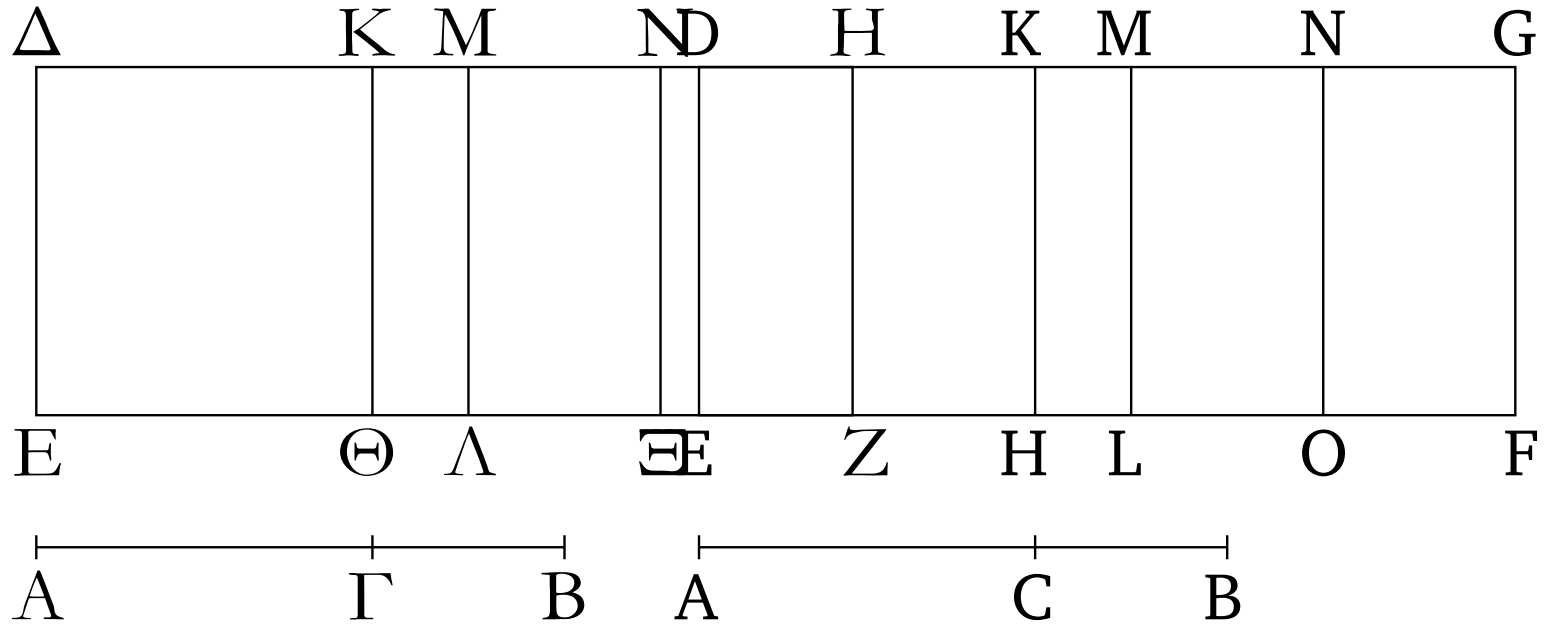
The square on a second binomial (straight-line) applied to a rational (straight-line) produces as breadth a third binomial (straight-line).†

Let AB be a second binomial (straight-line) having been divided into its (component) medial (straight-lines) at C , such that AC is the greater segment. And let DE be some rational (straight-line). And let the parallelogram DF , equal to the (square) on AB , be applied to DE , producing DG as breadth. I say that DG is a third binomial (straight-line).

Let the same construction be made as that shown previously. And since AB is a second binomial (straight-line), having been divided at C , AC and CB are thus medial (straight-lines) commensurable in square only, and containing a medial (area) [Prop. 10.38]. Hence, the sum of the (squares) on AC and CB is also medial [Props. 10.15, 10.23 corr.]. And it is equal to DL . Thus, DL (is) also medial. And it is applied to the rational (straight-line) DE . MD is thus also rational, and incommensurable in length with DE [Prop. 10.22]. So, for the same (reasons), MG is also rational, and incommensurable in length with ML —that is to say, with DE . Thus, DM and MG are each rational, and incommensurable in length with DE . And since AC is incommensurable in length with CB , and as AC (is) to CB , so the (square) on AC (is) to the (rectangle con-

ἐστὶν ἡ ΔΗ. δεικτέον [δὴ], <ὅτι καὶ τρίτη.

tained) by ACB [Prop. 10.21 lem.], the (square) on AC (is) also incommensurable with the (rectangle contained) by ACB [Prop. 10.11]. And hence the sum of the (squares) on AC and CB is incommensurable with twice the (rectangle contained) by ACB —that is to say, DL with MF [Props. 10.12, 10.13]. Hence, DM is also incommensurable (in length) with MG [Props. 6.1, 10.11]. And they are rational. DG is thus a binomial (straight-line) [Prop. 10.36]. [So] we must show that (it is) also a third (binomial straight-line).



Ὅμοιῶς δὴ τοῖς προτέροις ἐπιλογιούμεθα, <ὅτι μείζων ἐστὶν ἡ ΔΜ τῇς ΜΗ, καὶ σύμμετρος ἡ ΔΚ τῇ ΚΜ. καὶ ἐστὶ τὸ ὑπὸ τῶν ΔΚΜ >ῖσον τῷ ὑπὸ τῇς ΜΝ; ἡ ΔΜ >ῖρα τῇς ΜΗ μείζων δύναται τῷ ὑπὸ συμμέτρου ἑαυτῇ. καὶ οὐδετέρα τῶν ΔΜ, ΜΗ σύμμετρος ἐστὶ τῇ ΔΕ μήκει. Ἡ ΔΗ >ῖρα ἐκ δύο ὀνομάτων ἐστὶ τρίτη; <ὅπερ >έδει δεῖχαι.

So, similarly to the previous (propositions), we can conclude that DM is greater than MG , and DK (is) commensurable (in length) with KM . And the (rectangle contained) by DKM is equal to the (square) on MN . Thus, the square on DM is greater than (the square on) MG by the (square) on (some straight-line) commensurable (in length) with (DM) [Prop. 10.17]. And neither of DM and MG is commensurable in length with DE .

Thus, DG is a third binomial (straight-line) [Def. 10.7]. (Which is) the very thing it was required to show.

† In other words, the square of a second binomial is a third binomial. See Prop. 10.56.

63.

Τὸ ἀπὸ τῇς μείζονος παρὰ <ρητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων τετάρτην.

>Ἐστω μείζων ἡ ΑΒ διηρημένη κατὰ τὸ Γ, <ώστε μείζονα ε>ῖναι τὴν ΑΓ τῇς ΓΒ, <ρητὴ δὲ ἡ ΔΕ, καὶ τῷ ἀπὸ τῇς ΑΒ >ῖσον παρὰ τὴν ΔΕ παραβλήσθω τὸ ΔΖ παραλληλόγραμμον πλάτος ποιοῦν τὴν ΔΗ; λέγω, <ὅτι ἡ ΔΗ ἐκ δύο ὀνομάτων ἐστὶ τετάρτη.

Proposition 63

The square on a major (straight-line) applied to a rational (straight-line) produces as breadth a fourth binomial (straight-line).†

Let AB be a major (straight-line) having been divided at C , such that AC is greater than CB , and (let) DE (be) a rational (straight-line). And let the parallelogram DF , equal to the (square) on AB , be applied to DE , producing DG as breadth. I say that DG is a fourth binomial (straight-line).

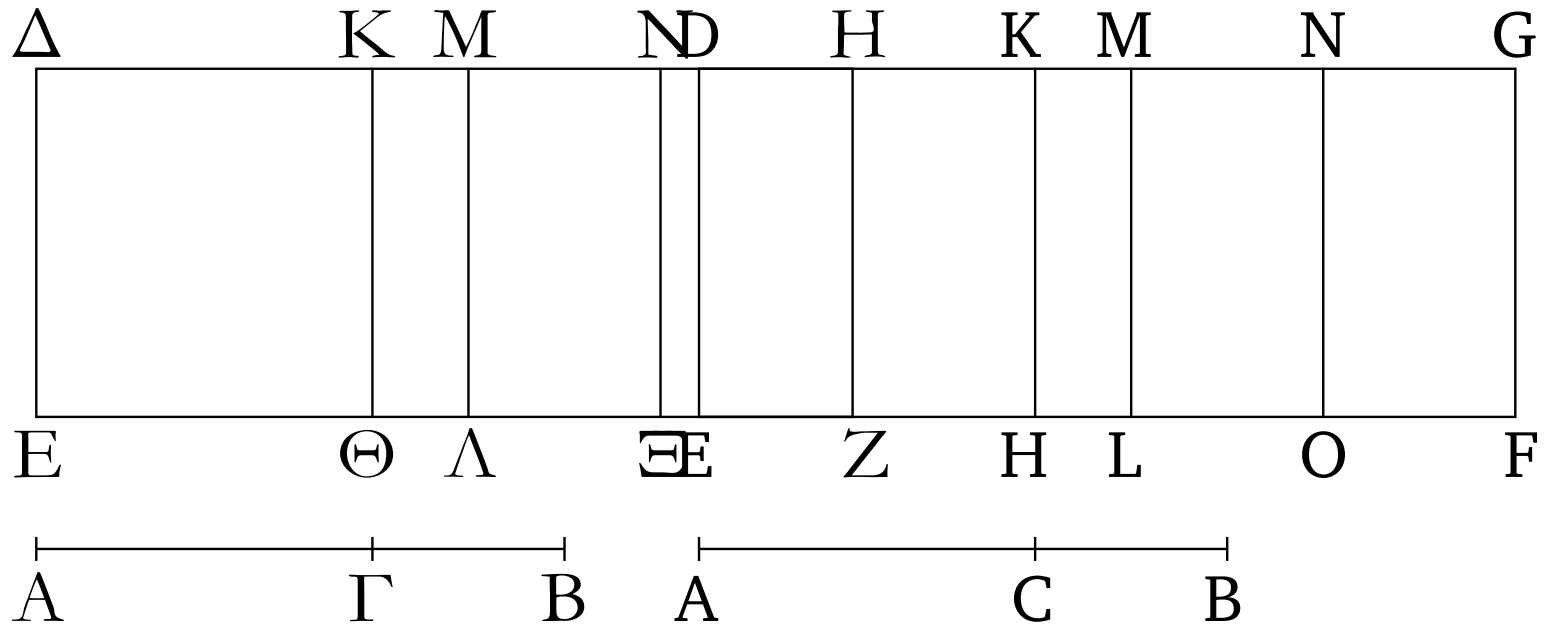
64.

Proposition 64

Τὸ ἀπὸ τῆς <ρητὸν καὶ μέσον δυναμένης παρὰ <ρητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων πέμπτῃ. >Ἐστω <ρητὸν καὶ μέσον δυναμένη ἡ AB διηρημένη εἰς τὰς εὐθείας κατὰ τὸ Γ , <ώστε μείζονα εἶναι τὴν AG , καὶ ἐκκείσθω <ρητὴ ἡ DE , καὶ τῷ ἀπὸ τῆς AB >ἴσον παρὰ τὴν DE παραβλήσθω τὸ DZ πλάτος ποιοῦν τὴν DH ; λέγω, <ὅτι ἡ DH ἐκ δύο ὀνομάτων ἐστὶ πέμπτῃ.

The square on the square-root of a rational plus a medial (area) applied to a rational (straight-line) produces as breadth a fifth binomial (straight-line).[†]

Let AB be the square-root of a rational plus a medial (area) having been divided into its (component) straight-lines at C , such that AC is greater. And let the rational (straight-line) DE be laid down. And let the (parallelogram) DF , equal to the (square) on AB , be applied to DE , producing DG as breadth. I say that DG is a fifth binomial straight-line.



Κατεσκευάσθω τὰ αὐτὰ τοῖς πρὸ τούτου. ἐπεὶ ο>ὖν <ρητὸν καὶ μέσον δυναμένη ἐστὶν ἡ AB διηρημένη κατὰ τὸ Γ , αἱ AG , GB >άρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ> αὐτῶν τετραγώνων μέσον, τὸ δ> ὑπ> αὐτῶν <ρητόν. ἐπεὶ ο>ὖν μέσον ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AG , GB , μέσον >άρα ἐστὶ τὸ $ΔA$; <ώστε <ρητὴ ἐστὶν ἡ $ΔM$ καὶ μήκει ἀσύμμετρος τῇ $ΔE$. πάλιν, ἐπεὶ <ρητόν ἐστὶ τὸ διζ ὑπὸ τῶν AGB , τουτέστι τὸ MZ , <ρητὴ >άρα ἡ MH καὶ σύμμετρος τῇ $ΔE$. ἀσύμμετρος >άρα ἡ $ΔM$ τῇ MH ; αἱ $ΔM$, MH >άρα <ρηταί εἰσι δυνάμει μόνον σύμμετροι; ἐκ δύο >άρα ὀνομάτων ἐστὶν ἡ DH . λέγω δὴ, <ὅτι καὶ πέμπτῃ.

Ὅμοιῶς γὰρ διεθθήσεται, <ὅτι τὸ ὑπὸ τῶν $ΔKM$ >ἴσον ἐστὶ τῷ ἀπὸ τῆς MN , καὶ ἀσύμμετρος ἡ $ΔK$ τῇ KM μήκει; ἡ $ΔM$ >άρα τῇ MH μείζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ. καὶ εἰσὶν αἱ $ΔM$, MH [<ρηταί] δυνάμει μόνον σύμμετροι, καὶ ἡ ἐλάσσων ἡ MH σύμμετρος τῇ $ΔE$ μήκει.

Ἡ DH >άρα ἐκ δύο ὀνομάτων ἐστὶ πέμπτῃ; <όπερ >έδει δεῖξαι.

Let the same construction be made as in the (propositions) before this. Therefore, since AB is the square-root of a rational plus a medial (area), having been divided at C , AC and CB are thus incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them rational [Prop. 10.40]. Therefore, since the sum of the (squares) on AC and CB is medial, DL is thus medial. Hence, DM is rational and incommensurable in length with DE [Prop. 10.22]. Again, since twice the (rectangle contained) by ACB —that is to say, MF —is rational, MG (is) thus rational and commensurable (in length) with DE [Prop. 10.20]. DM (is) thus incommensurable (in length) with MG [Prop. 10.13]. Thus, DM and MG are rational (straight-lines which are) commensurable in square only. Thus, DG is a binomial (straight-line) [Prop. 10.36]. So, I say that (it is) also a fifth (binomial straight-line).

For, similarly (to the previous propositions), it can be shown that the (rectangle contained) by DKM is equal to the (square) on MN , and DK (is) incommensurable in length with KM . Thus, the square on DM is greater than (the square on) MG by the (square) on (some straight-line) incommensurable (in length) with (DM) [Prop. 10.18]. And DM and MG are [rational] (straight-lines which are) commensurable in square only, and the lesser MG is commensurable in length with DE .

Thus, DG is a fifth binomial (straight-line) [Def. 10.9]. (Which is) the very thing it was required to show.

† In other words, the square of the square-root of a rational plus medial is a fifth binomial. See Prop. 10.58.

65.

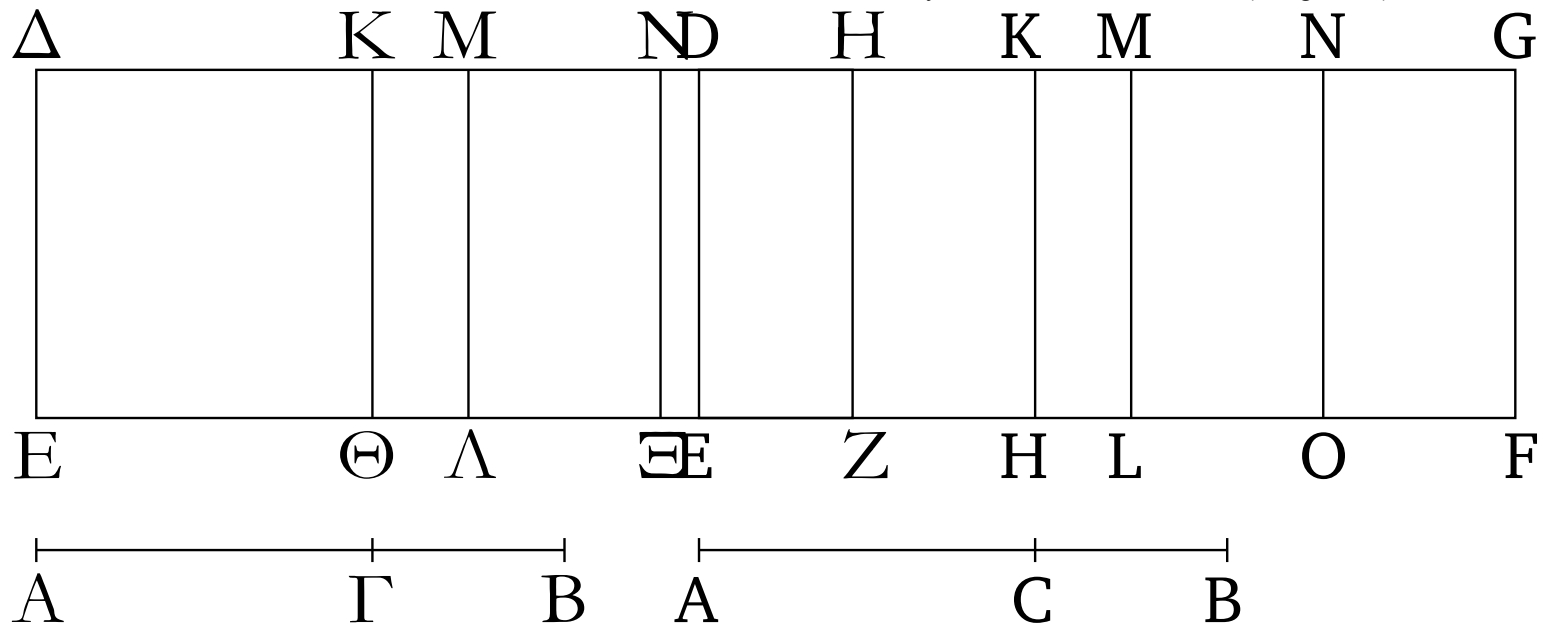
Proposition 65

Τὸ ἀπὸ τῆς δύο μέσα δυναμένης παρὰ <ρητὴν παραβαλλόμενον> πλάτοξ ποιεῖ τὴν ἐκ δύο ὀνομάτων <έκτην>.

>Ἐστω δύο μέσα δυναμένη ἡ AB διηρημένη κατὰ τὸ Γ , <ρητὴ δὲ >έστω ἡ DE , καὶ παρὰ τὴν DE τῶ| ἀπὸ τῆς AB >ῖσον παραβεβλήσθω τὸ ΔZ πλάτοξ ποιοῦν τὴν ΔH ; λέγω, <ὅτι ἡ ΔH ἐκ δύο ὀνομάτων ἐστὶν <έκτη>.

The square on the square-root of (the sum of) two medial (areas) applied to a rational (straight-line) produces as breadth a sixth binomial (straight-line).†

Let AB be the square-root of (the sum of) two medial (areas), having been divided at C . And let DE be a rational (straight-line). And let the (parallelogram) DF , equal to the (square) on AB , be applied to DE , producing DG as breadth. I say that DG is a sixth binomial (straight-line).



Κατεσκευάσθω γὰρ τὰ αὐτὰ τοῖς πρότερον. καὶ ἐπεὶ ἡ AB δύο μέσα δυναμένη ἐστὶ διηρημένη κατὰ τὸ Γ , αἱ AG , GB >ἀρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ τε συγκείμενον ἐκ τῶν ἀπ> αὐτῶν τετραγώνων μέσον καὶ τὸ ὑπ> αὐτῶν μέσον καὶ >έτι ἀσύμμετρον τὸ ἐκ τῶν ἀπ> αὐτῶν τετραγώνων συγκείμενον τῶ| ὑπ> αὐτῶν; <ώστε κατὰ τὰ προδεδειγμένα μέσον ἐστὶν ἑκάτερον τῶν $\Delta\Lambda$, MZ . καὶ παρὰ <ρητὴν τὴν DE παράκειται; <ρητὴ >ἀρα ἐστὶν ἑκάτερα τῶν ΔM , MH καὶ ἀσύμμετρος τῇ| DE μήκει. καὶ ἐπεὶ ἀσύμμετρόν ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AG , GB τῶ| διξ ὑπὸ τῶν AG , GB , ἀσύμμετρον >ἀρα ἐστὶ τὸ $\Delta\Lambda$ τῶ| MZ . ἀσύμμετρος >ἀρα καὶ ἡ ΔM τῇ| MH ; αἱ ΔM , MH >ἀρα <ρηταί εἰσι δυνάμει μόνον σύμμετροι; ἐκ δύο >ἀρα ὀνομάτων ἐστὶν ἡ ΔH . λέγω δὴ, <ὅτι καὶ <έκτη>.

Ὅμοίωξ δὴ πάλιν δεῖχομεν, <ὅτι τὸ ὑπὸ τῶν ΔKM >ῖσον ἐστὶ τῶ| ἀπὸ τῆς MN , καὶ <ὅτι ἡ ΔK τῇ| KM μήκει ἐστὶν ἀσύμμετρος; καὶ διὰ τὰ αὐτὰ δὴ ἡ ΔM τῇ| MH μεῖζον δύναται τῶ| ἀπὸ ἀσυμμέτρου ἐαυτῇ| μήκει. καὶ οὐδετέρα τῶν ΔM , MH σύμμετρόξ ἐστὶ τῇ| ἐκκειμένη <ρητῇ| τῇ| DE μήκει.

Ἡ ΔH >ἀρα ἐκ δύο ὀνομάτων ἐστὶν <έκτη>; <ὅπερ >έδει δεῖξαι.

For let the same construction be made as in the previous (propositions). And since AB is the square-root of (the sum of) two medial (areas), having been divided at C , AC and CB are thus incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them medial, and, moreover, the sum of the squares on them incommensurable with the (rectangle contained) by them [Prop. 10.41]. Hence, according to what has been previously demonstrated, DL and MF are each medial. And they are applied to the rational (straight-line) DE . Thus, DM and MG are each rational, and incommensurable in length with DE [Prop. 10.22]. And since the sum of the (squares) on AC and CB is incommensurable with twice the (rectangle contained) by AC and CB , DL is thus incommensurable with MF . Thus, DM (is) also incommensurable (in length) with MG [Props. 6.1, 10.11]. DM and MG are thus rational (straight-lines which are) commensurable in square only. Thus, DG is a binomial (straight-line) [Prop. 10.36]. So, I say that (it is) also a sixth (binomial straight-line).

So, similarly (to the previous propositions), we can again show that the (rectangle contained) by DKM is equal to the (square) on MN , and that DK is incommensurable in length with KM . And so, for the same (reasons), the square on DM is greater than (the square on) MG by the (square) on (some straight-line) incommensurable in length with (DM) [Prop. 10.18]. And neither of DM and MG is commensurable in length with the (previously) laid down rational (straight-line) DE .

Thus, DG is a sixth binomial (straight-line) [Def. 10.10].
(Which is) the very thing it was required to show.

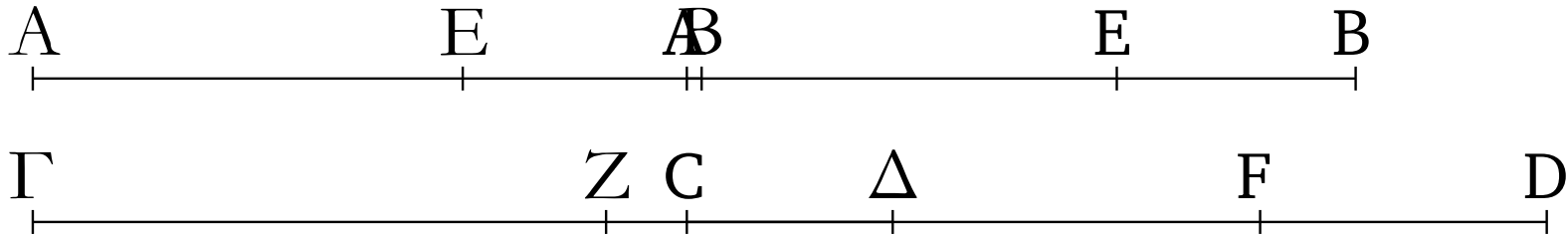
† In other words, the square of the square-root of two medials is a sixth binomial. See Prop. 10.59.

66.

Ἡ τῇ| ἐκ δύο ὀνομάτων μήκει σύμμετρος καὶ αὐτὴ ἐκ δύο ὀνομάτων ἐστὶ καὶ τῇ| τάχει ἢ αὐτῇ.
>Ἐστω ἐκ δύο ὀνομάτων ἡ AB , καὶ τῇ| AB μήκει σύμμετρος
>έστω ἡ $\Gamma\Delta$; λέγω, <ὅτι ἡ $\Gamma\Delta$ ἐκ δύο ὀνομάτων ἐστὶ καὶ τῇ|
τάχει ἢ αὐτῇ τῇ| AB .

Proposition 66

A (straight-line) commensurable in length with a binomial (straight-line) is itself also binomial, and the same in order. Let AB be a binomial (straight-line), and let CD be commensurable in length with AB . I say that CD is a binomial (straight-line), and (is) the same in order as AB .



Ἐπεὶ γὰρ ἐκ δύο ὀνομάτων ἐστὶν ἡ AB , διηρήσθω εἰς τὰ ὀνόματα κατὰ τὸ E , καὶ >έστω μείζον >όνομα τὸ AE ; αἱ AE , EB >άρα <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι. γεγονέντω ὥς ἡ AB πρὸς τὴν $\Gamma\Delta$, ο<ύτωξ ἡ AE πρὸς τὴν ΓZ ; καὶ λοιπὴ >άρα ἡ EB πρὸς λοιπὴν τὴν $Z\Delta$ ἐστίν, ὥς ἡ AB πρὸς τὴν $\Gamma\Delta$. σύμμετρος δὲ ἡ AB τῇ| $\Gamma\Delta$ μήκει; σύμμετρος >άρα ἐστὶ καὶ ἡ μὲν AE τῇ| ΓZ , ἡ δὲ EB τῇ| $Z\Delta$. καὶ εἰσι <ρηταὶ αἱ AE , EB ; <ρηταὶ >άρα εἰσι καὶ αἱ ΓZ , $Z\Delta$. καὶ ἐστὶν ὥς ἡ AE πρὸς ΓZ , ἡ EB πρὸς $Z\Delta$. ἐναλλάξ >άρα ἐστὶν ὥς ἡ AE πρὸς EB , ἡ ΓZ πρὸς $Z\Delta$. αἱ δὲ AE , EB δυνάμει μόνον [εἰσὶ] σύμμετροι; καὶ αἱ ΓZ , $Z\Delta$ >άρα δυνάμει μόνον εἰσὶ σύμμετροι. καὶ εἰσι <ρηταί; ἐκ δύο >άρα ὀνομάτων ἐστὶν ἡ $\Gamma\Delta$. λέγω δὴ, <ὅτι τῇ| τάχει ἐστὶν ἢ αὐτῇ τῇ| AB .

Ἡ γὰρ AE τῇξ EB μείζον δύναται >ήτοι τῷ| ἀπὸ συμέτρου ἐαυτῇ| >ἢ τῷ| ἀπὸ ἀσυμέτρου. εἰ μὲν ο>υν ἡ AE τῇξ EB μείζον δύναται τῷ| ἀπὸ συμέτρου ἐαυτῇ|, καὶ ἡ ΓZ τῇξ $Z\Delta$ μείζον δυνήσεται τῷ| ἀπὸ συμέτρου ἐαυτῇ|. καὶ εἰ μὲν σύμμετρος ἐστὶν ἡ AE τῇ| ἐκκειμένη <ρητῇ|, καὶ ἡ ΓZ σύμμετρος αὐτῇ| >έσται, καὶ διὰ τοῦτο ἑκάτερα τῶν AB , $\Gamma\Delta$ ἐκ δύο ὀνομάτων ἐστὶ πρώτη, τουτέστι τῇ| τάχει ἢ αὐτῇ. εἰ δὲ ἡ EB σύμμετρος ἐστὶ τῇ| ἐκκειμένη <ρητῇ|, καὶ ἡ $Z\Delta$ σύμμετρος ἐστὶν αὐτῇ|, καὶ διὰ τοῦτο πάλιν τῇ| τάχει ἢ αὐτῇ >έσται τῇ| AB ; ἑκάτερα γὰρ αὐτῶν >έσται ἐκ δύο ὀνομάτων δευτέρα. εἰ δὲ οὐδετέρα τῶν AE , EB σύμμετρος ἐστὶ τῇ| ἐκκειμένη <ρητῇ|, οὐδετέρα τῶν ΓZ , $Z\Delta$ σύμμετρος αὐτῇ| >έσται, καὶ ἐστὶν ἑκάτερα τρίτη. εἰ δὲ ἡ AE τῇξ EB μείζον δύναται τῷ| ἀπὸ ἀσυμέτρου ἐαυτῇ|, καὶ ἡ ΓZ τῇξ $Z\Delta$ μείζον δύναται τῷ| ἀπὸ ἀσυμέτρου ἐαυτῇ|. καὶ εἰ μὲν ἡ AE σύμμετρος ἐστὶ τῇ| ἐκκειμένη <ρητῇ|, καὶ ἡ ΓZ σύμμετρος ἐστὶν αὐτῇ|, καὶ ἐστὶν ἑκάτερα τετάρτη. εἰ δὲ ἡ EB , καὶ ἡ $Z\Delta$, καὶ >έσται ἑκάτερα πέμπτη. εἰ δὲ οὐδετέρα τῶν AE , EB , καὶ τῶν ΓZ , $Z\Delta$ οὐδετέρα σύμμετρος ἐστὶ τῇ| ἐκκειμένη <ρητῇ|, καὶ >έσται ἑκάτερα <έκτη.

<᾿Οστε ἡ τῇ| ἐκ δύο ὀνομάτων μήκει σύμμετρος ἐκ δύο ὀνομάτων ἐστὶ καὶ τῇ| τάχει ἢ αὐτῇ; <όπερ >έδει δεῖξαι.

For since AB is a binomial (straight-line), let it be divided into its (component) terms at E , and let AE be the greater term. AE and EB are thus rational (straight-lines which are) commensurable in square only [Prop. 10.36]. Let it be contrived that as AB (is) to CD , so AE (is) to CF [Prop. 6.12]. Thus, the remainder EB is also to the remainder FD , as AB (is) to CD [Props. 6.16, 5.19 corr.]. And AB (is) commensurable in length with CD . Thus, AE is also commensurable (in length) with CF , and EB with FD [Prop. 10.11]. And AE and EB are rational. Thus, CF and FD are also rational. And as AE is to CF , (so) EB (is) to FD [Prop. 5.11]. Thus, alternately, as AE is to EB , (so) CF (is) to FD [Prop. 5.16]. And AE and EB [are] commensurable in square only. Thus, CF and FD are also commensurable in square only [Prop. 10.11]. And they are rational. CD is thus a binomial (straight-line) [Prop. 10.36]. So, I say that it is the same in order as AB .

For the square on AE is greater than (the square on) EB by the (square) on (some straight-line) either commensurable or incommensurable (in length) with (AE). Therefore, if the square on AE is greater than (the square on) EB by the (square) on (some straight-line) commensurable (in length) with (AE) then the square on CF will also be greater than (the square on) FD by the (square) on (some straight-line) commensurable (in length) with (CF) [Prop. 10.14]. And if AE is commensurable (in length) with (some previously) laid down rational (straight-line) then CF will also be commensurable (in length) with it [Prop. 10.12]. And, on account of this, AB and CD are each first binomial (straight-lines) [Def. 10.5]—that is to say, the same in order. And if EB is commensurable (in length) with the (previously) laid down rational (straight-line) then FD is also commensurable (in length) with it [Prop. 10.12], and, again, on account of this, (CD) will be the same in order as AB . For each of them will be second binomial (straight-lines) [Def. 10.6]. And if neither of AE and EB is commensurable (in length) with the (previously) laid down rational (straight-line) then neither of CF and FD will be commensurable (in length) with it [Prop. 10.13], and each (of AB and CD) is a third (binomial straight-line) [Def. 10.7]. And if the square on AE is greater than (the square on) EB by the (square) on

(some straight-line) incommensurable (in length) with (AE) then the square on CF is also greater than (the square on) FD by the (square) on (some straight-line) incommensurable (in length) with (CF) [Prop. 10.14]. And if AE is commensurable (in length) with the (previously) laid down rational (straight-line) then CF is also commensurable (in length) with it [Prop. 10.12], and each (of AB and CD) is a fourth (binomial straight-line) [Def. 10.8]. And if EB (is commensurable in length with the previously laid down rational straight-line) then FD (is) also (commensurable in length with it), and each (of AB and CD) will be a fifth (binomial straight-line) [Def. 10.9]. And if neither of AE and EB (is commensurable in length with the previously laid down rational straight-line) then also neither of CF and FD is commensurable (in length) with the laid down rational (straight-line), and each (of AB and CD) will be a sixth (binomial straight-line) [Def. 10.10].

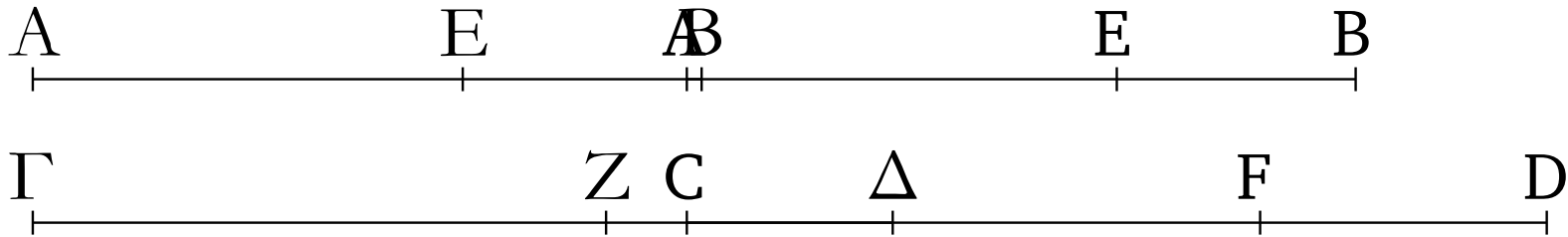
Hence, a (straight-line) commensurable in length with a binomial (straight-line) is a binomial (straight-line), and the same in order. (Which is) the very thing it was required to show.

67.

Proposition 67

Ἡ τῇ ἐκ δύο μέσων μήκει σύμμετρος καὶ αὐτὴ ἐκ δύο μέσων ἐστὶ καὶ τῇ τάχει ἢ αὐτῇ.

A (straight-line) commensurable in length with a bimedral (straight-line) is itself also bimedral, and the same in order.



>Ἐστω ἐκ δύο μέσων ἡ AB , καὶ τῇ AB σύμμετρος >έστω μήκει ἡ $\Gamma\Delta$; λέγω, <ὅτι ἡ $\Gamma\Delta$ ἐκ δύο μέσων ἐστὶ καὶ τῇ τάχει ἢ αὐτῇ τῇ AB .

Ἐπεὶ γὰρ ἐκ δύο μέσων ἐστὶν ἡ AB , διηρήσθω εἰς τὰς μέσας κατὰ τὸ E ; αἱ AE , EB >ἀρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι. καὶ γεγονένω ὥς ἡ AB πρὸς $\Gamma\Delta$, ἡ AE πρὸς ΓZ ; καὶ λοιπὴ >ἀρα ἡ EB πρὸς λοιπὴν τὴν $Z\Delta$ ἐστίν, ὥς ἡ AB πρὸς $\Gamma\Delta$. σύμμετρος δὲ ἡ AB τῇ $\Gamma\Delta$ μήκει; σύμμετρος >ἀρα καὶ ἑκάτερα τῶν AE , EB ἑκάτερα τῶν ΓZ , $Z\Delta$. μέσαι δὲ αἱ AE , EB ; μέσαι >ἀρα καὶ αἱ ΓZ , $Z\Delta$. καὶ ἐπεὶ ἐστίν ὥς ἡ AE πρὸς EB , ἡ ΓZ πρὸς $Z\Delta$, αἱ δὲ AE , EB δυνάμει μόνον σύμμετροί εἰσιν, καὶ αἱ ΓZ , $Z\Delta$ [>ἀρα] δυνάμει μόνον σύμμετροί εἰσιν, ἐδείκθην δὲ καὶ μέσαι; ἡ $\Gamma\Delta$ >ἀρα ἐκ δύο μέσων ἐστίν. λέγω δὲ, <ὅτι καὶ τῇ τάχει ἢ αὐτῇ ἐστὶ τῇ AB . Ἐπεὶ γὰρ ἐστίν ὥς ἡ AE πρὸς EB , ἡ ΓZ πρὸς $Z\Delta$, καὶ ὥς >ἀρα τὸ ἀπὸ τῆς AE πρὸς τὸ ὑπὸ τῶν AEB , ο<ύτωξ τὸ ἀπὸ τῆς ΓZ πρὸς τὸ ὑπὸ τῶν $\Gamma Z\Delta$; ἐναλλάξ ὥς τὸ ἀπὸ τῆς AE πρὸς τὸ ἀπὸ τῆς ΓZ , ο<ύτωξ τὸ ὑπὸ τῶν AEB πρὸς τὸ ὑπὸ τῶν $\Gamma Z\Delta$. σύμμετρον δὲ τὸ ἀπὸ τῆς AE τῷ ἀπὸ τῆς ΓZ ; σύμμετρον >ἀρα καὶ τὸ ὑπὸ τῶν AEB τῷ ὑπὸ τῶν $\Gamma Z\Delta$. ε>ἴτε ο>υν <ρητόν ἐστι τὸ ὑπὸ τῶν AEB , καὶ τὸ ὑπὸ τῶν $\Gamma Z\Delta$ <ρητόν ἐστίν [καὶ διὰ τοῦτό ἐστιν ἐκ δύο μέσων πρώτῃ]. ε>ἴτε μέσον, μέσον, καὶ ἐστίν ἑκάτερα δευτέρα. Καὶ διὰ τοῦτο >έσται ἡ $\Gamma\Delta$ τῇ AB τῇ τάχει ἢ αὐτῇ; <όπερ >έδει δεῖχαι.

Let AB be a bimedral (straight-line), and let CD be commensurable in length with AB . I say that CD is bimedral, and the same in order as AB .

For since AB is a bimedral (straight-line), let it be divided into its (component) medial (straight-lines) at E . Thus, AE and EB are medial (straight-lines which are) commensurable in square only [Props. 10.37, 10.38]. And let it be contrived that as AB (is) to CD , (so) AE (is) to CF [Prop. 6.12]. And thus as the remainder EB is to the remainder FD , so AB (is) to CD [Props. 5.19 corr., 6.16]. And AB (is) commensurable in length with CD . Thus, AE and EB are also commensurable (in length) with CF and FD , respectively [Prop. 10.11]. And AE and EB (are) medial. Thus, CF and FD (are) also medial [Prop. 10.23]. And since as AE is to EB , (so) CF (is) to FD , and AE and EB are commensurable in square only, CF and FD are [thus] also commensurable in square only [Prop. 10.11]. And they were also shown (to be) medial. Thus, CD is a bimedral (straight-line). So, I say that it is also the same in order as AB .

For since as AE is to EB , (so) CF (is) to FD , thus also as the (square) on AE (is) to the (rectangle contained) by AEB , so the (square) on CF (is) to the (rectangle contained) by CFD [Prop. 10.21 lem.]. Alternately, as the (square) on AE (is) to the (square) on CF , so the (rectangle contained) by AEB (is) to the (rectangle contained) by CFD [Prop. 5.16]. And the (square) on AE (is) com-

measurable with the (square) on CF . Thus, the (rectangle contained) by AEB (is) also commensurable with the (rectangle contained) by CFD [Prop. 10.11]. Therefore, either the (rectangle contained) by AEB is rational, and the (rectangle contained) by CFD is rational [and, on account of this, (AE and CD) are first bimedral (straight-lines)], or (the rectangle contained by AEB is) medial, and (the rectangle contained by CFD is) medial, and (AB and CD) are each second (bimedral straight-lines) [Props. 10.23, 10.37, 10.38].

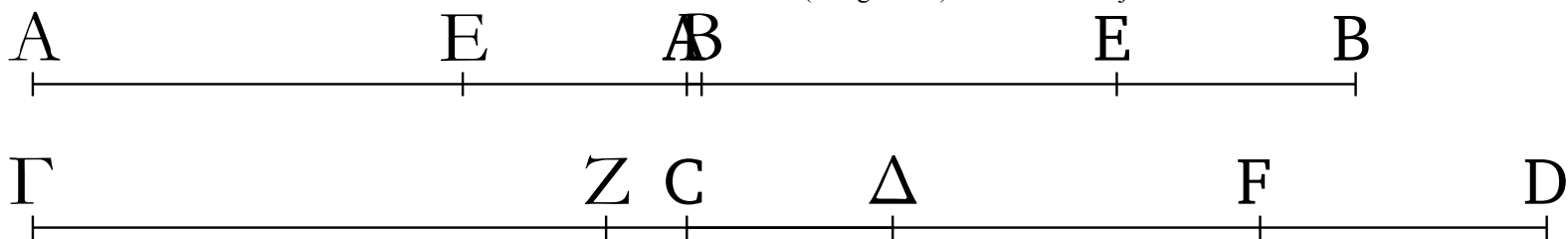
And, on account of this, CD will be the same in order as AB . (Which is) the very thing it was required to show.

68.

Ἡ τῇ μείζονι σύμμετρος καὶ αὐτὴ μείζων ἐστίν.

Proposition 68

A (straight-line) commensurable (in length) with a major (straight-line) is itself also major.



>Ἐστω μείζων ἡ AB , καὶ τῇ AB σύμμετρος >ἔστω ἡ $ΓΔ$; λέγω, <ὅτι ἡ $ΓΔ$ μείζων ἐστίν.

Διηρήσθω ἡ AB κατὰ τὸ E ; αἱ AE , EB >άρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων <ρητόν, τὸ δ' ὑπ' αὐτῶν μέσον; καὶ γεγονέτω τὰ αὐτὰ τοῖς πρότερον. καὶ ἐπεὶ ἐστὶν ὡς ἡ AB πρὸς τὴν $ΓΔ$, ο<ύτως <ή τε AE πρὸς τὴν $ΓΖ$ καὶ ἡ EB πρὸς τὴν $ΖΔ$, καὶ ὡς >άρα ἡ AE πρὸς τὴν $ΓΖ$, ο<ύτως ἡ EB πρὸς τὴν $ΖΔ$. σύμμετρος δὲ ἡ AB τῇ $ΓΔ$; σύμμετρος >άρα καὶ ἑκατέρω τῶν AE , EB ἑκατέρω τῶν $ΓΖ$, $ΖΔ$. καὶ ἐπεὶ ἐστὶν ὡς ἡ AE πρὸς τὴν $ΓΖ$, ο<ύτως ἡ EB πρὸς τὴν $ΖΔ$, καὶ ἐναλλάχ ὡς ἡ AE πρὸς EB , ο<ύτως ἡ $ΓΖ$ πρὸς $ΖΔ$, καὶ συνθέντι >άρα ἐστὶν ὡς ἡ AB πρὸς τὴν BE , ο<ύτως ἡ $ΓΔ$ πρὸς τὴν $ΔΖ$; καὶ ὡς >άρα τὸ ἀπὸ τῆς AB πρὸς τὸ ἀπὸ τῆς BE , ο<ύτως τὸ ἀπὸ τῆς $ΓΔ$ πρὸς τὸ ἀπὸ τῆς $ΔΖ$. ὁμοίως δὲ δείχομεν, <ὅτι καὶ ὡς τὸ ἀπὸ τῆς AB πρὸς τὸ ἀπὸ τῆς AE , ο<ύτως τὸ ἀπὸ τῆς $ΓΔ$ πρὸς τὸ ἀπὸ τῆς $ΓΖ$. καὶ ὡς >άρα τὸ ἀπὸ τῆς AB πρὸς τὰ ἀπὸ τῶν AE , EB , ο<ύτως τὸ ἀπὸ τῆς $ΓΔ$ πρὸς τὰ ἀπὸ τῶν $ΓΖ$, $ΖΔ$; καὶ ἐναλλάχ >άρα ἐστὶν ὡς τὸ ἀπὸ τῆς AB πρὸς τὸ ἀπὸ τῆς $ΓΔ$, ο<ύτως τὰ ἀπὸ τῶν AE , EB πρὸς τὰ ἀπὸ τῶν $ΓΖ$, $ΖΔ$. σύμμετρον δὲ τὸ ἀπὸ τῆς AB τῷ ἀπὸ τῆς $ΓΔ$; σύμμετρα >άρα καὶ τὰ ἀπὸ τῶν AE , EB τοῖς ἀπὸ τῶν $ΓΖ$, $ΖΔ$. καὶ ἐστὶ τὰ ἀπὸ τῶν AE , EB <άμα <ρητόν, καὶ τὰ ἀπὸ τῶν $ΓΖ$, $ΖΔ$ <άμα <ρητόν ἐστίν. ὁμοίως δὲ καὶ τὸ διζ ὑπὸ τῶν AE , EB σύμμετρόν ἐστι τῷ διζ ὑπὸ τῶν $ΓΖ$, $ΖΔ$. καὶ ἐστὶ μέσον τὸ διζ ὑπὸ τῶν AE , EB ; μέσον >άρα καὶ τὸ διζ ὑπὸ τῶν $ΓΖ$, $ΖΔ$. αἱ $ΓΖ$, $ΖΔ$ >άρα δυνάμει ἀσύμμετροί εἰσι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων <άμα <ρητόν, τὸ δὲ διζ ὑπ' αὐτῶν μέσον; <όλη >άρα ἡ $ΓΔ$ >αλογόξ ἐστὶν ἡ καλουμένη μείζων.

Ἡ >άρα τῇ μείζονι σύμμετρος μείζων ἐστίν; <όπερ >έδει δείχαι.

Let AB be a major (straight-line), and let CD be commensurable (in length) with AB . I say that CD is a major (straight-line).

Let AB be divided (into its component terms) at E . AE and EB are thus incommensurable in square, making the sum of the squares on them rational, and the (rectangle contained) by them medial [Prop. 10.39]. And let (the) same (things) be contrived as in the previous (propositions). And since as AB is to CD , so AE (is) to CF and EB to FD , thus also as AE (is) to CF , so EB (is) to FD [Prop. 5.11]. And AB (is) commensurable (in length) with CD . Thus, AE and EB (are) also commensurable (in length) with CF and FD , respectively [Prop. 10.11]. And since as AE is to CF , so EB (is) to FD , also, alternately, as AE (is) to EB , so CF (is) to FD [Prop. 5.16], and thus, via composition, as AB is to BE , so CD (is) to DF [Prop. 5.18]. And thus as the (square) on AB (is) to the (square) on BE , so the (square) on CD (is) to the (square) on DF [Prop. 6.20]. So, similarly, we can also show that as the (square) on AB (is) to the (square) on AE , so the (square) on CD (is) to the (square) on CF . And thus as the (square) on AB (is) to the (sum of) the (squares) on AE and EB , so the (square) on CD (is) to the (sum of) the (squares) on CF and FD . And thus, alternately, as the (square) on AB is to the (square) on CD , so the (sum of) the (squares) on AE and EB (is) to the (sum of) the (squares) on CF and FD [Prop. 5.16]. And the (square) on AB (is) commensurable with the (square) on CD . Thus, the (sum of) the (squares) on AE and EB (is) also commensurable with the (sum of) the (squares) on CF and FD [Prop. 10.11]. And the (squares) on AE and EB (added) together are rational. The (squares) on CF and FD (added) together (are) thus also rational. So, similarly, twice the (rectangle contained) by AE and EB is also commensurable with twice the (rectangle contained) by CF and FD . And twice the (rectangle contained) by AE and EB is medial. Therefore, twice the (rectangle con-

tained) by CF and FD (is) also medial [Prop. 10.23 corr.]. CF and FD are thus (straight-lines which are) incommensurable in square [Prop 10.13], simultaneously making the sum of the squares on them rational, and twice the (rectangle contained) by them medial. The whole, CD , is thus that irrational (straight-line) called major [Prop. 10.39].

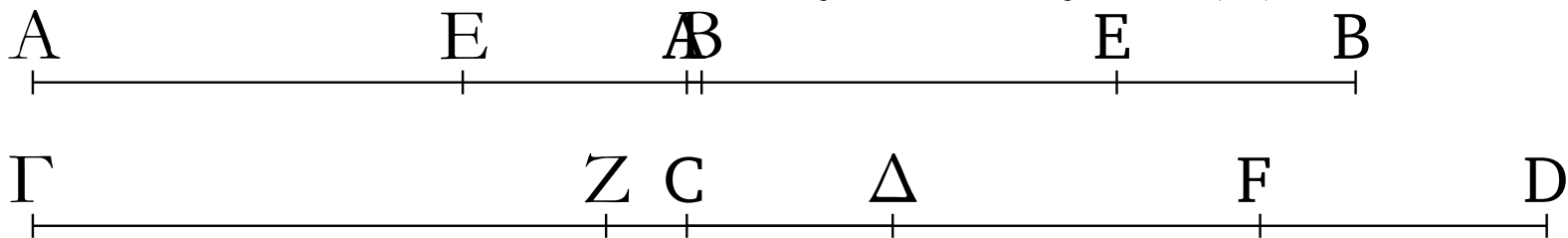
Thus, a (straight-line) commensurable (in length) with a major (straight-line) is major. (Which is) the very thing it was required to show.

69.

Ἡ τῇ| <ρητὸν καὶ μέσον δυναμένη σύμμετροξ [καὶ αὐτὴ| <ρητὸν καὶ μέσον δυναμένη ἐστίν.

Proposition 69

A (straight-line) commensurable (in length) with the square-root of a rational plus a medial (area) is [itself also] the square-root of a rational plus a medial (area).



>Ἐστω <ρητὸν καὶ μέσον δυναμένη ἡ AB , καὶ τῇ| AB σύμμετροξ >ἔστω ἡ $\Gamma\Delta$; δεικτέον, <ὅτι καὶ ἡ $\Gamma\Delta$ <ρητὸν καὶ μέσον δυναμένη ἐστίν.

Διηρήσθω ἡ AB εἰς τὰς εὐθείας κατὰ τὸ E ; αἱ AE , EB >ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον, τὸ δ' > ὑπ' αὐτῶν <ρητόν; καὶ τὰ αὐτὰ κατεσκευάσθω τοῖς πρότερον. ὁμοίως δὲ δείχομεν, <ὅτι καὶ αἱ ΓZ , $Z\Delta$ δυνάμει εἰσὶν ἀσύμμετροι, καὶ σύμμετρον τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν AE , EB τῶ| συγκειμένων ἐκ τῶν ἀπὸ τῶν ΓZ , $Z\Delta$, τὸ δὲ ὑπὸ AE , EB τῶ| ὑπὸ ΓZ , $Z\Delta$; <ώστε καὶ τὸ [μὲν] συγκείμενον ἐκ τῶν ἀπὸ τῶν ΓZ , $Z\Delta$ τετραγώνων ἐστὶ μέσον, τὸ δ' > ὑπὸ τῶν ΓZ , $Z\Delta$ <ρητόν.

<Ρητὸν >ἄρα καὶ μέσον δυναμένη ἐστὶν ἡ $\Gamma\Delta$; <όπερ >ἔδει δεῖξαι.

Let AB be the square-root of a rational plus a medial (area), and let CD be commensurable (in length) with AB . We must show that CD is also the square-root of a rational plus a medial (area).

Let AB be divided into its (component) straight-lines at E . AE and EB are thus incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them rational [Prop. 10.40]. And let the same construction be made as in the previous (propositions). So, similarly, we can show that CF and FD are also incommensurable in square, and that the sum of the (squares) on AE and EB (is) commensurable with the sum of the (squares) on CF and FD , and the (rectangle contained) by AE and EB with the (rectangle contained) by CF and FD . And hence the sum of the squares on CF and FD is medial, and the (rectangle contained) by CF and FD (is) rational. Thus, CD is the square-root of a rational plus a medial (area) [Prop. 10.40]. (Which is) the very thing it was required to show.

70.

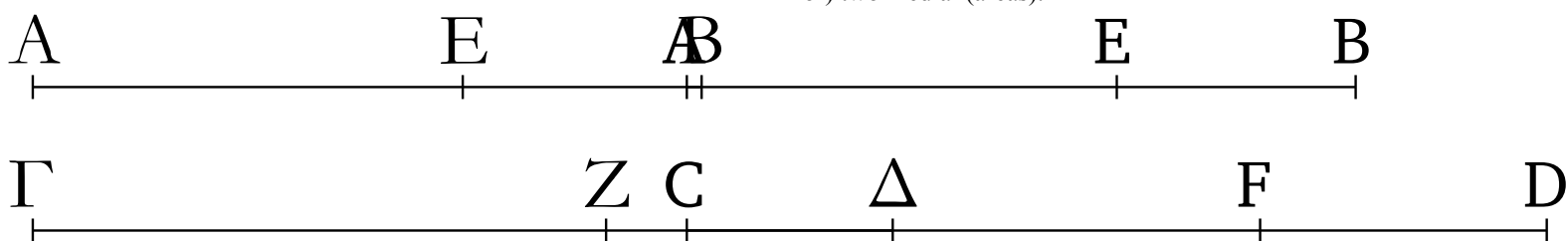
Ἡ τῇ| δύο μέσα δυναμένη σύμμετροξ δύο μέσα δυναμένη ἐστίν.

>Ἐστω δύο μέσα δυναμένη ἡ AB , καὶ τῇ| AB σύμμετροξ ἡ $\Gamma\Delta$; δεικτέον, <ὅτι καὶ ἡ $\Gamma\Delta$ δύο μέσα δυναμένη ἐστίν.

Proposition 70

A (straight-line) commensurable (in length) with the square-root of (the sum of) two medial (areas) is (itself also) the square-root of (the sum of) two medial (areas).

Let AB be the square-root of (the sum of) two medial (areas), and (let) CD (be) commensurable (in length) with AB . We must show that CD is also the square-root of (the sum of) two medial (areas).



Ἐπεὶ γὰρ δύο μέσα δυναμένη ἐστὶν ἡ AB , διηρήσθω εἰς τὰς εὐθείας κατὰ τὸ E ; αἱ AE , EB >ἄρα δυνάμει εἰσὶν

For since AB is the square-root of (the sum of) two medial (areas), let it be divided into its (component) straight-lines at

ἀσύμμετροι ποιοῦσαι τό τε συγκείμενον ἐκ τῶν ἀπ' αὐτῶν [τετραγώνων] μέσον καὶ τὸ ὑπ' αὐτῶν μέσον καὶ >έτι ἀσύμμετρον τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AE , EB τετραγώνων τῷ ὑπὸ τῶν AE , EB ; καὶ κατεσκευάσθω τὰ αὐτὰ τοῖς πρότερον. ὁμοίως δὲ δείχομεν, <ὅτι καὶ αἱ ΓZ , $Z\Delta$ δυνάμει εἰσὶν ἀσύμμετροι καὶ σύμμετρον τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν AE , EB τῷ συγκειμένῳ ἐκ τῶν ἀπὸ τῶν ΓZ , $Z\Delta$, τὸ δὲ ὑπὸ τῶν AE , EB τῷ ὑπὸ τῶν ΓZ , $Z\Delta$; <ὥστε καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΓZ , $Z\Delta$ τετραγώνων μέσον ἐστὶ καὶ τὸ ὑπὸ τῶν ΓZ , $Z\Delta$ μέσον καὶ >έτι ἀσύμμετρον τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΓZ , $Z\Delta$ τετραγώνων τῷ ὑπὸ τῶν ΓZ , $Z\Delta$.
 Ἡ >άρα $\Gamma\Delta$ δύο μέσα δυναμένη ἐστίν; <όπερ >έδει δεῖξαι.

71.

<Ρητοῦ καὶ μέσου συντιθεμένου τέσσαρες >άλογοι γίνονται >ήτοι ἐκ δύο ὀνομάτων >ή ἐκ δύο μέσων πρώτη >ή μείζων >ή <ρητὸν καὶ μέσον δυναμένη.

>Ἐστω <ρητὸν μὲν τὸ AB , μέσον δὲ τὸ $\Gamma\Delta$; λέγω, <ὅτι ἢ τὸ AD θωρίον δυναμένη >ήτοι ἐκ δύο ὀνομάτων ἐστίν >ή ἐκ δύο μέσων πρώτη >ή μείζων >ή <ρητὸν καὶ μέσον δυναμένη.

Τὸ γὰρ AB τοῦ $\Gamma\Delta$ >ήτοι μείζον ἐστίν >ή >έλασσον. >έστω πρότερον μείζον; καὶ ἐκκείσθω <ρητὴ ἢ EZ , καὶ παραβεβλήσθω παρὰ τὴν EZ τῷ AB >ἴσον τὸ EH πλάτοξ ποιοῦν τὴν $E\Theta$; τῷ δὲ $\Delta\Gamma$ >ἴσον παρὰ τὴν EZ παραβεβλήσθω τὸ ΘI πλάτοξ ποιοῦν τὴν ΘK . καὶ ἐπεὶ <ρητὸν ἐστὶ τὸ AB καὶ ἐστίν >ἴσον τῷ EH , <ρητὸν >άρα καὶ τὸ EH . καὶ παρὰ [<ρητὴν] τὴν EZ παραβεβλήται πλάτοξ ποιοῦν τὴν $E\Theta$; ἢ $E\Theta$ >άρα <ρητὴ ἐστὶ καὶ σύμμετρος τῇ EZ μήκει. πάλιν, ἐπεὶ μέσον ἐστὶ τὸ $\Gamma\Delta$ καὶ ἐστίν >ἴσον τῷ ΘI , μέσον >άρα ἐστὶ καὶ τὸ ΘI . καὶ παρὰ <ρητὴν τὴν EZ παράκειται πλάτοξ ποιοῦν τὴν ΘK ; <ρητὴ >άρα ἐστὶν ἢ ΘK καὶ ἀσύμμετρος τῇ EZ μήκει. καὶ ἐπεὶ μέσον ἐστὶ τὸ $\Gamma\Delta$, <ρητὸν δὲ τὸ AB , ἀσύμμετρον >άρα ἐστὶ τὸ AB τῷ $\Gamma\Delta$; <ὥστε καὶ τὸ EH ἀσύμμετρον ἐστὶ τῷ ΘI . ὥς δὲ τὸ EH πρὸς τὸ ΘI , ο<ύτως ἐστὶν ἢ $E\Theta$ πρὸς τὴν ΘK ; ἀσύμμετρος >άρα ἐστὶ καὶ ἢ $E\Theta$ τῇ ΘK μήκει. καὶ εἰσὶν ἀμφοτέραι <ρηταί; αἱ $E\Theta$, ΘK >άρα <ρηταί εἰσι δυνάμει μόνον σύμμετροι; ἐκ δύο >άρα ὀνομάτων ἐστὶν ἢ EK διηρημένη κατὰ τὸ Θ . καὶ ἐπεὶ μείζον ἐστὶ τὸ AB τοῦ $\Gamma\Delta$, >ἴσον δὲ τὸ μὲν AB τῷ EH , τὸ δὲ $\Gamma\Delta$ τῷ ΘI , μείζων >άρα καὶ τὸ EH τοῦ ΘI ; καὶ ἢ $E\Theta$ >άρα μείζων ἐστὶ τῇ ΘK . >ήτοι ο>ὐν ἢ $E\Theta$ τῇ ΘK μείζων δύναται τῷ ἀπὸ συμέτρου ἑαυτῇ μήκει >ή τῷ ἀπὸ ἀσυμέτρου. δυνάσθω πρότερον τῷ ἀπὸ συμέτρου ἑαυτῇ; καὶ ἐστὶν ἢ μείζων ἢ ΘE σύμμετρος τῇ ἐκκειμένη <ρητῇ τῇ EZ ; ἢ >άρα EK ἐκ δύο ὀνομάτων ἐστὶ πρώτη. <ρητὴ δὲ ἢ EZ ; ἔαν δὲ θωρίον περιέθῃται ὑπὸ <ρητῆς καὶ τῆς ἐκ δύο ὀνομάτων πρώτης, ἢ τὸ θωρίον δυναμένη ἐκ δύο ὀνομάτων ἐστίν. ἢ >άρα τὸ EI δυναμένη ἐκ δύο ὀνομάτων ἐστίν; <ὥστε καὶ ἢ τὸ AD δυναμένη ἐκ δύο ὀνομάτων ἐστίν. ἀλλὰ δὴ δυνάσθω ἢ $E\Theta$ τῇ ΘK μείζων τῷ ἀπὸ ἀσυμέτρου ἑαυτῇ; καὶ ἐστὶν ἢ μείζων ἢ $E\Theta$ σύμμετρος τῇ ἐκκειμένη <ρητῇ τῇ EZ μήκει; ἢ >άρα EK ἐκ δύο ὀνομάτων ἐστὶ τετάρτη. <ρητὴ δὲ ἢ

E . Thus, AE and EB are incommensurable in square, making the sum of the [squares] on them medial, and the (rectangle contained) by them medial, and, moreover, the sum of the (squares) on AE and EB incommensurable with the (rectangle) contained by AE and EB [Prop. 10.41]. And let the same construction be made as in the previous (propositions). So, similarly, we can show that CF and FD are also incommensurable in square, and (that) the sum of the (squares) on AE and EB (is) commensurable with the sum of the (squares) on CF and FD , and the (rectangle contained) by AE and EB with the (rectangle contained) by CF and FD . Hence, the sum of the squares on CF and FD is also medial, and the (rectangle contained) by CF and FD (is) medial, and, moreover, the sum of the squares on CF and FD (is) incommensurable with the (rectangle contained) by CF and FD .

Thus, CD is the square-root of (the sum of) two medial (areas) [Prop. 10.41]. (Which is) the very thing it was required to show.

Proposition 71

When a rational and a medial (area) are added together, four irrational (straight-lines) arise (as the square-roots of the total area)—either a binomial, or a first bimedial, or a major, or the square-root of a rational plus a medial (area).

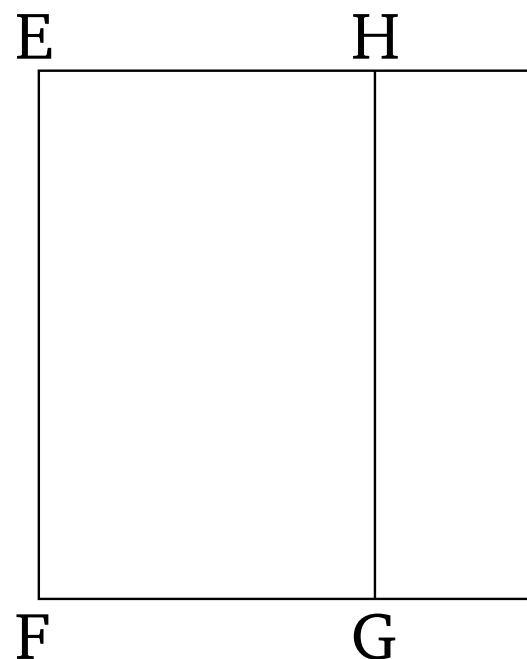
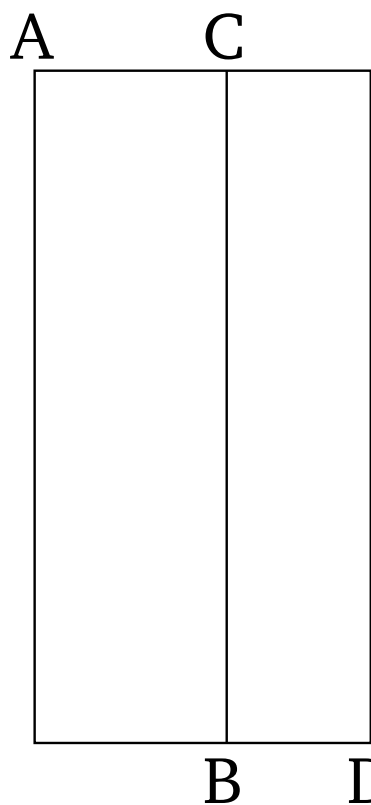
Let AB be a rational (area), and CD a medial (area). I say that the square-root of area AD is either binomial, or first bimedial, or major, or the square-root of a rational plus a medial (area).

For AB is either greater or less than CD . Let it, first of all, be greater. And let the rational (straight-line) EF be laid down. And let (the rectangle) EG , equal to AB , be applied to EF , producing EH as breadth. And let (the rectangle) HI , equal to DC , be applied to EF , producing HK as breadth. And since AB is rational, and is equal to EG , EG is thus also rational. And it has been applied to the [rational] (straight-line) EF , producing EH as breadth. EH is thus rational, and commensurable in length with EF [Prop. 10.20]. Again, since CD is medial, and is equal to HI , HI is thus also medial. And it is applied to the rational (straight-line) EF , producing HK as breadth. HK is thus rational, and incommensurable in length with EF [Prop. 10.22]. And since CD is medial, and AB rational, AB is thus incommensurable with CD . Hence, EG is also incommensurable with HI . And as EG (is) to HI , so EH is to HK [Prop. 6.1]. Thus, EH is also incommensurable in length with HK [Prop. 10.11]. And they are both rational. Thus, EH and HK are rational (straight-lines which are) commensurable in square only. EK is thus a binomial (straight-line), having been divided (into its component terms) at H [Prop. 10.36]. And since AB is greater than CD , and AB (is) equal to EG , and CD to HI , EG (is) thus also greater than HI . Thus, EH is also greater than HK [Prop. 5.14]. Therefore, the square on EH is greater than (the square on) HK either by the (square) on (some straight-line) commensurable in length with (EH), or by the (square) on (some straight-line) incommensurable (in length with EH). Let it, first of all, be greater by the (square) on (some straight-line) commensurable (in length with EH). And the greater (of the two components of EK) HE is

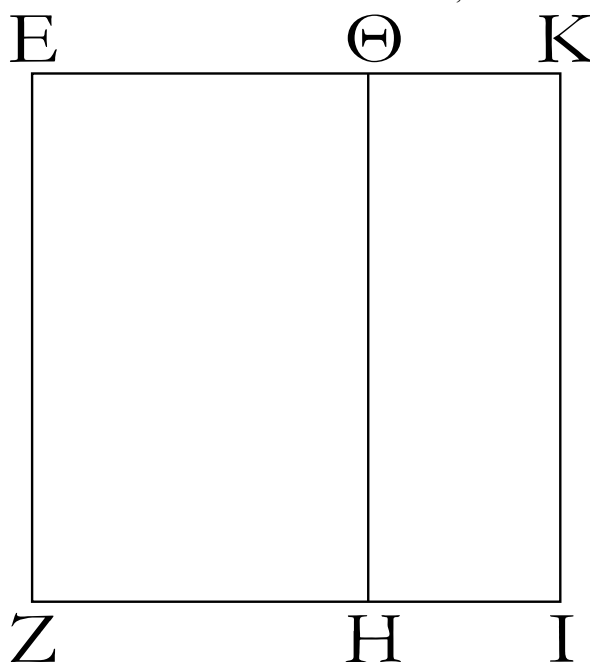
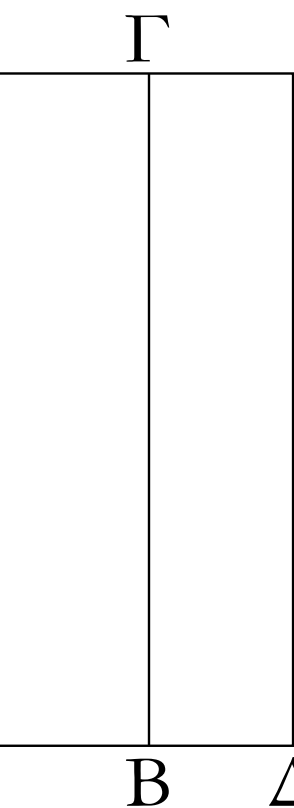
EZ ; ἔαν δὲ θωρίον περιέθῃται ὑπὸ <ρητῆς καὶ τῆς ἐκ δύο ὀνομάτων τετάρτης, ἡ τὸ θωρίον δυναμένη >άλογός ἐστιν ἡ καλουμένη μείζων. ἡ >άρα τὸ EI θωρίον δυναμένη μείζων ἐστίν; <ώστε καὶ ἡ τὸ AD δυναμένη μείζων ἐστίν. Ἀλλὰ δὴ >έστω >έλασσον τὸ AB τοῦ $\Gamma\Delta$; καὶ τὸ EH >άρα >έλασσόν ἐστι τοῦ ΘI ; <ώστε καὶ ἡ $E\Theta$ ἐλάσσων ἐστὶ τῆς ΘK . >ήτοι δὲ ἡ ΘK τῆς $E\Theta$ μείζων δύναται τῷ ἀπὸ συμμετρου ἐαυτῇ >ἢ τῷ ἀπὸ ἀσυμμετρου. δυνάσθω πρότερον τῷ ἀπὸ συμμετρου ἐαυτῇ μήκει; καὶ ἐστὶν ἡ ἐλάσσων ἡ $E\Theta$ σύμμετρος τῇ ἐκκειμένῃ <ρητῇ τῇ EZ μήκει; ἡ >άρα EK ἐκ δύο ὀνομάτων ἐστὶ δευτέρα. <ρητὴ δὲ ἡ EZ ; ἔαν δὲ θωρίον περιέθῃται ὑπὸ <ρητῆς καὶ τῆς ἐκ δύο ὀνομάτων δευτέρα, ἡ τὸ θωρίον δυναμένη ἐκ δύο μέσων ἐστὶ πρώτη. ἡ >άρα τὸ EI θωρίον δυναμένη ἐκ δύο μέσων ἐστὶ πρώτη; <ώστε καὶ ἡ τὸ AD δυναμένη ἐκ δύο μέσων ἐστὶ πρώτη. ἀλλὰ δὴ ἡ ΘK τῆς ΘE μείζων δυνάσθω τῷ ἀπὸ ἀσυμμετρου ἐαυτῇ. καὶ ἐστὶν ἡ ἐλάσσων ἡ $E\Theta$ σύμμετρος τῇ ἐκκειμένῃ <ρητῇ τῇ EZ ; ἡ >άρα EK ἐκ δύο ὀνομάτων ἐστὶ πέμπτη. <ρητὴ δὲ ἡ EZ ; ἔαν δὲ θωρίον περιέθῃται ὑπὸ <ρητῆς καὶ τῆς ἐκ δύο ὀνομάτων πέμπτης, ἡ τὸ θωρίον δυναμένη <ρητὸν καὶ μέσον δυναμένη ἐστίν. ἡ >άρα τὸ EI θωρίον δυναμένη <ρητὸν καὶ μέσον δυναμένη ἐστίν; <ώστε καὶ ἡ τὸ AD θωρίον δυναμένη <ρητὸν καὶ μέσον δυναμένη ἐστίν.

commensurable (in length) with the (previously) laid down (straight-line) EF . EK is thus a first binomial (straight-line) [Def. 10.5]. And EF (is) rational. And if an area is contained by a rational (straight-line) and a first binomial (straight-line) then the square-root of the area is a binomial (straight-line) [Prop. 10.54]. Thus, the square-root of EI is a binomial (straight-line). Hence the square-root of AD is also a binomial (straight-line). And, so, let the square on EH be greater than (the square on) HK by the (square) on (some straight-line) incommensurable (in length) with (EH). And the greater (of the two components of EK) EH is commensurable in length with the (previously) laid down rational (straight-line) EF . Thus, EK is a fourth binomial (straight-line) [Def. 10.8]. And EF (is) rational. And if an area is contained by a rational (straight-line) and a fourth binomial (straight-line) then the square-root of the area is the irrational (straight-line) called major [Prop. 10.57]. Thus, the square-root of area EI is a major (straight-line). Hence, the square-root of AD is also major.

And so, let AB be less than CD . Thus, EG is also less than HI . Hence, EH is also less than HK [Props. 6.1, 5.14]. And the square on HK is greater than (the square on) EH either by the (square) on (some straight-line) commensurable (in length) with (HK), or by the (square) on (some straight-line) incommensurable (in length) with (HK). Let it, first of all, be greater by the square on (some straight-line) commensurable in length with (HK). And the lesser (of the two components of EK) EH is commensurable in length with the (previously) laid down rational (straight-line) EF . Thus, EK is a second binomial (straight-line) [Def. 10.6]. And EF (is) rational. And if an area is contained by a rational (straight-line) and a second binomial (straight-line) then the square-root of the area is a first bimedial (straight-line) [Prop. 10.55]. Thus, the square-root of area EI is a first bimedial (straight-line). Hence, the square-root of AD is also a first bimedial (straight-line). And so, let the square on HK be greater than (the square on) HE by the (square) on (some straight-line) incommensurable (in length) with (HK). And the lesser (of the two components of EK) EH is commensurable (in length) with the (previously) laid down rational (straight-line) EF . Thus, EK is a fifth binomial (straight-line) [Def. 10.9]. And EF (is) rational. And if an area is contained by a rational (straight-line) and a fifth binomial (straight-line) then the square-root of the area is the square-root of a rational plus a medial (area) [Prop. 10.58]. Thus, the square-root of area EI is the square-root of a rational plus a medial (area). Hence, the square-root of area AD is also the square-root of a rational plus a medial (area).



Thus, when a rational and a medial area are added together,



<Ρητοῦ >άρα καὶ μέσου συντιθεμένου τέσσαρες >άλογοι γίνονται >ήτοι ἐκ δύο ὀνομάτων >ἢ ἐκ δύο μέσων πρώτη >ἢ μείζων >ἢ <ρητὸν καὶ μέσον δυναμένη; <όπερ >έδει δεῖλαι.

four irrational (straight-lines) arise (as the square-roots of the total area)—either a binomial, or a first bimedial, or a major, or the square-root of a rational plus a medial (area). (Which is) the very thing it was required to show.

62.

Δύο μέσων ἀσυμμέτρων ἀλλήλοις συντιθεμένων αἱ λοιπαὶ δύο >άλογοι γίνονται >ήτοι ἐκ δύο μέσων δευτέρα >ἢ [ἢ] δύο μέσα δυναμένη.

Proposition 72

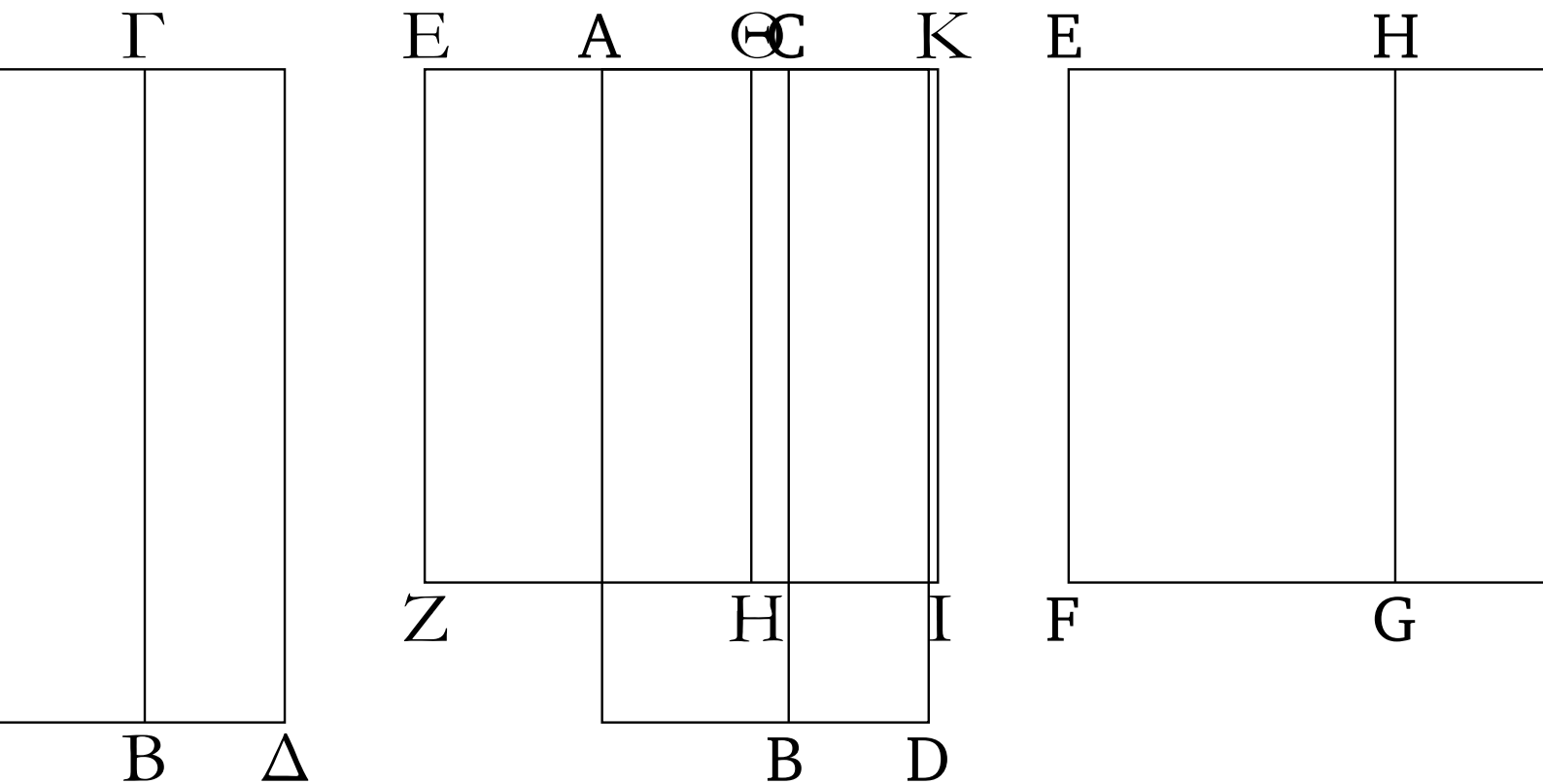
When two medial (areas which are) incommensurable with one another are added together, the remaining two irrational (straight-lines) arise (as the square-roots of the total area)—

Συγκείσθω γὰρ δύο μέσα ἀσύμμετρα ἀλλήλοις τὰ AB , $\Gamma\Delta$; λέγω, <ὅτι ἡ τὸ AD θωρίον δυναμένη >ήτοι ἐκ δύο μέσων ἐστὶ δευτέρα >ἢ δύο μέσα δυναμένη.
Τὸ γὰρ AB τοῦ $\Gamma\Delta$ >ήτοι μείζον ἐστὶν >ἢ >έλασσον. >έστω, εἰ τύθον, πρότερον μείζον τὸ AB τοῦ $\Gamma\Delta$; καὶ ἐκκείσθω <ρητὴ ἢ EZ , καὶ τῷ μὲν AB >ίσον παρὰ τὴν EZ παραβεβλήσθω τὸ EH πλάτοξ ποιοῦν τὴν EO , τῷ δὲ $\Gamma\Delta$ >ίσον τὸ $ΘΙ$ πλάτοξ ποιοῦν τὴν $ΘΚ$. καὶ ἐπεὶ μέσον ἐστὶν ἐκάτερον τῶν AB , $\Gamma\Delta$, μέσον >άρα καὶ ἐκάτερον τῶν EH , $ΘΙ$. καὶ παρὰ <ρητὴν τὴν ZE παράκειται πλάτοξ ποιοῦν τὰς EO , $ΘΚ$; ἐκατέρα >άρα τῶν EO , $ΘΚ$ <ρητὴ ἐστὶ καὶ ἀσύμμετρος τῇ EZ μήκει. καὶ ἐπεὶ ἀσύμμετρόν ἐστι τὸ AB τῷ $\Gamma\Delta$, καὶ ἐστὶν >ίσον τὸ μὲν AB τῷ EH , τὸ δὲ $\Gamma\Delta$ τῷ $ΘΙ$, ἀσύμμετρον >άρα ἐστὶ καὶ τὸ EH τῷ $ΘΙ$. ὥς δὲ τὸ EH πρὸς τὸ $ΘΙ$, ο<ύτωξ ἐστὶν ἡ EO πρὸς $ΘΚ$; ἀσύμμετρος >άρα ἐστὶν ἡ EO τῇ $ΘΚ$ μήκει. αἱ EO , $ΘΚ$ >άρα <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι; ἐκ δύο >άρα ὀνομάτων ἐστὶν ἡ EK . >ήτοι δὲ ἡ EO τῇ $ΘΚ$ μείζον δύναται τῷ ἀπὸ συμέτρου ἑαυτῇ >ἢ τῷ ἀπὸ ἀσυμέτρου. δυνάσθω πρότερον τῷ ἀπὸ συμέτρου ἑαυτῇ μήκει; καὶ οὐδετέρα τῶν EO , $ΘΚ$ σύμμετρος ἐστὶ τῇ ἐκκειμένη <ρητῇ τῇ EZ μήκει; ἡ EK >άρα ἐκ δύο ὀνομάτων ἐστὶ τρίτη. <ρητὴ δὲ ἡ EZ ; ἔαν δὲ θωρίον περιέθῃται ὑπὸ <ρητῆς καὶ τῇ ἐκ δύο ὀνομάτων τρίτης, ἡ τὸ θωρίον δυναμένη ἐκ δύο μέσων ἐστὶ δευτέρα; ἢ >άρα τὸ EI , τουτέστι τὸ AD , δυναμένη ἐκ δύο μέσων ἐστὶ δευτέρα. ἀλλὰ δὴ ἡ EO τῇ $ΘΚ$ μείζον δυνάσθω τῷ ἀπὸ ἀσυμέτρου ἑαυτῇ μήκει; καὶ ἀσύμμετρος ἐστὶν ἐκατέρα τῶν EO , $ΘΚ$ τῇ EZ μήκει; ἢ >άρα EK ἐκ δύο ὀνομάτων ἐστὶν <έκτη. ἔαν δὲ θωρίον περιέθῃται ὑπὸ <ρητῆς καὶ τῇ ἐκ δύο ὀνομάτων <έκτης, ἡ τὸ θωρίον δυναμένη ἢ δύο μέσα δυναμένη ἐστίν; <ώστε καὶ ἡ τὸ AD θωρίον δυναμένη ἢ δύο μέσα δυναμένη ἐστίν.

either a second bimedral, or the square-root of (the sum of) two medial (areas).

For let the two medial (areas) AB and CD , (which are) incommensurable with one another, be added together. I say that the square-root of area AD is either a second bimedral, or the square-root of (the sum of) two medial (areas).

For AB is either greater than or less than CD . By chance, let AB , first of all, be greater than CD . And let the rational (straight-line) EF be laid down. And let EG , equal to AB , be applied to EF , producing EH as breadth, and HI , equal to CD , producing HK as breadth. And since AB and CD are each medial, EG and HI (are) thus also each medial. And they are applied to the rational straight-line FE , producing EH and HK (respectively) as breadth. Thus, EH and HK are each rational (straight-lines which are) incommensurable in length with EF [Prop. 10.22]. And since AB is incommensurable with CD , and AB is equal to EG , and CD to HI , EG is thus also incommensurable with HI . And as EG (is) to HI , so EH is to HK [Prop. 6.1]. EH is thus incommensurable in length with HK [Prop. 10.11]. Thus, EH and HK are rational (straight-lines which are) commensurable in square only. EK is thus a binomial (straight-line) [Prop. 10.36]. And the square on EH is greater than (the square on) HK either by the (square) on (some straight-line) commensurable (in length) with (EH), or by the (square) on (some straight-line) incommensurable (in length with EH). Let it, first of all, be greater by the square on (some straight-line) commensurable in length with (EH). And neither of EH or HK is commensurable in length with the (previously) laid down rational (straight-line) EF . Thus, EK is a third binomial (straight-line) [Def. 10.7]. And EF (is) rational. And if an area is contained by a rational (straight-line) and a third binomial (straight-line) then the square-root of the area is a second bimedral (straight-line) [Prop. 10.56]. Thus, the square-root of EI —that is to say, of AD —is a second bimedral. And so, let the square on EH be greater than (the square) on HK by the (square) on (some straight-line) incommensurable in length with (EH). And EH and HK are each incommensurable in length with EF . Thus, EK is a sixth binomial (straight-line) [Def. 10.10]. And if an area is contained by a rational (straight-line) and a sixth binomial (straight-line) then the square-root of the area is the square-root of (the sum of) two medial (areas) [Prop. 10.59]. Hence, the square-root of area AD is also the square-root of (the sum of) two medial (areas).



[<Omo'iwc d'h de'ixomen, <'oti k>'an >'elaton > h| t'o AB to u GD, <h t'o AD qwr'ion dunam'enh >'h >ek d'uo m'eswn deut'era >est'in >'htoi d'uo m'esa dunam'enh].

Δύο >άρα μέσων ἀσυμμέτρων ἀλλήλοις συντιθεμένων αἰ λοιπαὶ δύο >άλογοι γίνονται >ήτοι ἐκ δύο μέσων δευτέρα >ἡ δύο μέσα δυναμένη.

Ἡ ἐκ δύο ὀνομάτων καὶ αἱ μετ' αὐτὴν >άλογοι ο>ύτε τῇ μέσῃ ο>ύτε ἀλλήλαις εἰσὶν αἱ αὐταί. τὸ μὲν γὰρ ἀπὸ μέσῃς παρὰ <ρητὴν παραβαλλόμενον πλάτος ποιεῖ <ρητὴν καὶ ἀσύμμετρον τῇ παρ> <ὴν παράκειται μήκει. τὸ δὲ ἀπὸ τῆς ἐκ δύο ὀνομάτων παρὰ <ρητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων πρώτην. τὸ δὲ ἀπὸ τῆς ἐκ δύο μέσων πρώτης παρὰ <ρητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων δευτέραν. τὸ δὲ ἀπὸ τῆς ἐκ δύο μέσων δευτέρας παρὰ <ρητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων τρίτην. τὸ δὲ ἀπὸ τῆς μείζονος παρὰ <ρητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων τετάρτην. τὸ δὲ ἀπὸ τῆς <ρητὸν καὶ μέσον δυναμένης παρὰ <ρητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων πέμπτην. τὸ δὲ ἀπὸ τῆς δύο μέσα δυναμένης παρὰ <ρητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων <έκτην. τὰ δ> εἰρημένα πλάτη διαφέρει τοῦ τε πρώτου καὶ ἀλλήλων, τοῦ μὲν πρώτου, <ὅτι <ρητὴ ἐστίν, ἀλλήλων δέ, <ὅτι τῇ τάχει οὐκ εἰσὶν αἱ αὐταί; <ὥστε καὶ αὐταὶ αἱ >άλογοι διαφέρουσιν ἀλλήλων.

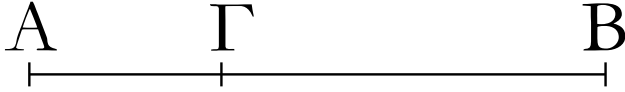
[So, similarly, we can show that, even if AB is less than CD , the square-root of area AD is either a second binomial or the square-root of (the sum of) two medial (areas).]

Thus, when two medial (areas which are) incommensurable with one another are added together, the remaining two irrational (straight-lines) arise (as the square-roots of the total area)—either a second binomial, or the square-root of (the sum of) two medial (areas).

A binomial (straight-line), and the (other) irrational (straight-lines) after it, are neither the same as a medial (straight-line) nor (the same) as one another. For the (square) on a medial (straight-line), applied to a rational (straight-line), produces as breadth a rational (straight-line which is) also incommensurable in length with (the straight-line) to which it is applied [Prop. 10.22]. And the (square) on a binomial (straight-line), applied to a rational (straight-line), produces as breadth a first binomial [Prop. 10.60]. And the (square) on a first binomial (straight-line), applied to a rational (straight-line), produces as breadth a second binomial [Prop. 10.61]. And the (square) on a second binomial (straight-line), applied to a rational (straight-line), produces as breadth a third binomial [Prop. 10.62]. And the (square) on a major (straight-line), applied to a rational (straight-line), produces as breadth a fourth binomial [Prop. 10.63]. And the (square) on the square-root of a rational plus a medial (area), applied to a rational (straight-line), produces as breadth a fifth binomial [Prop. 10.64]. And the (square) on the square-root of (the sum of) two medial (areas), applied to a rational (straight-line), produces as breadth a sixth binomial [Prop. 10.65]. And the aforementioned breadths differ from the first (breadth), and from one another—from the first, because it is rational—and from one another, because they are not the same in order. Hence, the (previously mentioned) irrational (straight-lines) themselves also differ

73.

Ἐὰν ἀπὸ <ρητῆς <ρητῆ ἀφαιρεθῇ| δυνάμει μόνον σύμμετρος ο>ῦσα τῇ| <όλῃ, ἢ λοιπὴ >άλογός ἐστιν; καλείσθω δὲ ἀποτομή.



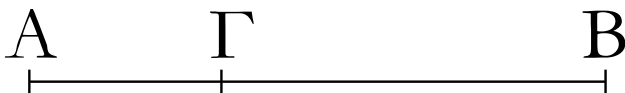
Ἀπὸ γὰρ <ρητῆς τῆς AB <ρητῆ ἀφαιρεθῶ ἡ BΓ δυνάμει μόνον σύμμετρος ο>ῦσα τῇ| <όλῃ; λέγω, <ὅτι ἡ λοιπὴ ἡ AΓ >άλογός ἐστιν ἡ καλουμένη ἀποτομή.

Ἐπεὶ γὰρ ἀσύμμετρος ἐστὶν ἡ AB τῇ| BΓ μήκει, καὶ ἐστὶν ὥς ἡ AB πρὸς τὴν BΓ, ο<ύτως τὸ ἀπὸ τῆς AB πρὸς τὸ ὑπὸ τῶν AB, BΓ, ἀσύμμετρον >άρα ἐστὶ τὸ ἀπὸ τῆς AB τῷ| ὑπὸ τῶν AB, BΓ. ἀλλὰ τῷ| μὲν ἀπὸ τῆς AB σύμμετρά ἐστι τὰ ἀπὸ τῶν AB, BΓ τετράγωνα, τῷ| δὲ ὑπὸ τῶν AB, BΓ σύμμετρον ἐστὶ τὸ διζ ὑπὸ τῶν AB, BΓ. καὶ ἐπειδὴ περ τὰ ἀπὸ τῶν AB, BΓ >ίσα ἐστὶ τῷ| διζ ὑπὸ τῶν AB, BΓ μετὰ τοῦ ἀπὸ ΓA, καὶ λοιπῷ| >άρα τῷ| ἀπὸ τῆς AΓ ἀσύμμετρά ἐστι τὰ ἀπὸ τῶν AB, BΓ. <ρητὰ δὲ τὰ ἀπὸ τῶν AB, BΓ; >άλογος >άρα ἐστὶν ἡ AΓ; καλείσθω δὲ ἀποτομή. <όπερ >έδει δεῖχαι.

† See footnote to Prop. 10.36.

74.

Ἐὰν ἀπὸ μέσης μέση ἀφαιρεθῇ| δυνάμει μόνον σύμμετρος ο>ῦσα τῇ| <όλῃ, μετὰ δὲ τῆς <όλῃς <ρητὸν περιέθουσα, ἢ λοιπὴ >άλογός ἐστιν; καλείσθω δὲ μέσης ἀποτομή πρώτη.



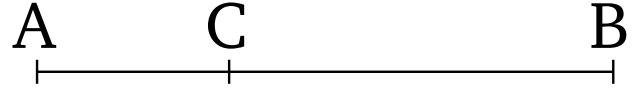
Ἀπὸ γὰρ μέσης τῆς AB μέση ἀφαιρεθῶ ἡ BΓ δυνάμει μόνον σύμμετρος ο>ῦσα τῇ| AB, μετὰ δὲ τῆς AB <ρητὸν ποιοῦσα τὸ ὑπὸ τῶν AB, BΓ; λέγω, <ὅτι ἡ λοιπὴ ἡ AΓ >άλογός ἐστιν; καλείσθω δὲ μέσης ἀποτομή πρώτη.

Ἐπεὶ γὰρ αἱ AB, BΓ μέσαι εἰσὶν, μέσα ἐστὶ καὶ τὰ ἀπὸ τῶν AB, BΓ. <ρητὸν δὲ τὸ διζ ὑπὸ τῶν AB, BΓ; ἀσύμμετρα >άρα τὰ ἀπὸ τῶν AB, BΓ τῷ| διζ ὑπὸ τῶν AB, BΓ; καὶ λοιπῷ| >άρα τῷ| ἀπὸ τῆς AΓ ἀσύμμετρον ἐστὶ τὸ διζ ὑπὸ τῶν AB, BΓ, ἐπεὶ κ>ὰν τὸ <όλον ἐνὶ αὐτῶν ἀσύμμετρον >ῃ|, καὶ τὰ ἐξ ἀρθῆς μεγέθη ἀσύμμετρα >έσται. <ρητὸν δὲ τὸ διζ ὑπὸ τῶν AB, BΓ; >άλογον >άρα τὸ ἀπὸ τῆς AΓ; >άλογος >άρα ἐστὶν ἡ AΓ; καλείσθω δὲ μέσης ἀποτομή πρώτη.

from one another.

Proposition 73

If a rational (straight-line), which is commensurable in square only with the whole, is subtracted from a(nother) rational (straight-line) then the remainder is an irrational (straight-line). Let it be called an apotome.

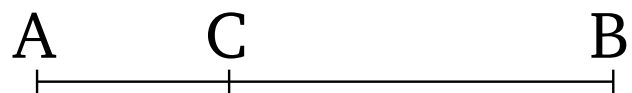


For let the rational (straight-line) BC , which commensurable in square only with the whole, be subtracted from the rational (straight-line) AB . I say that the remainder AC is that irrational (straight-line) called an apotome.

For since AB is incommensurable in length with BC , and as AB is to BC , so the (square) on AB (is) to the (rectangle contained) by AB and BC [Prop. 10.21 lem.], the (square) on AB is thus incommensurable with the (rectangle contained) by AB and BC [Prop. 10.11]. But, the (sum of the) squares on AB and BC is commensurable with the (square) on AB [Prop. 10.15], and twice the (rectangle contained) by AB and BC is commensurable with the (rectangle contained) by AB and BC [Prop. 10.6]. And, inasmuch as the (sum of the squares) on AB and BC is equal to twice the (rectangle contained) by AB and BC plus the (square) on CA [Prop. 2.7], the (sum of the squares) on AB and BC is thus also incommensurable with the remaining (square) on AC [Props. 10.13, 10.16]. And the (sum of the squares) on AB and BC is rational. AC is thus an irrational (straight-line) [Def. 10.4]. And let it be called an apotome.[†] (Which is) the very thing it was required to show.

Proposition 74

If a medial (straight-line), which is commensurable in square only with the whole, and which contains a rational (area) with the whole, is subtracted from a(nother) medial (straight-line) then the remainder is an irrational (straight-line). Let it be called a first apotome of a medial (straight-line).



For let the medial (straight-line) BC , which is commensurable in square only with AB , and which makes with AB the rational (rectangle contained) by AB and BC , be subtracted from the medial (straight-line) AB [Prop. 10.27]. I say that the remainder AC is an irrational (straight-line). Let it be called the first apotome of a medial (straight-line).

For since AB and BC are medial (straight-lines), the (sum of the squares) on AB and BC is also medial. And twice the (rectangle contained) by AB and BC (is) rational. The (sum of the squares) on AB and BC (is) thus incommensurable with twice the (rectangle contained) by AB and BC . Thus, twice the (rectangle contained) by AB and BC is also incommensurable with the remaining (square) on AC

[Prop. 2.7], since if the whole is incommensurable with one of the (constituent magnitudes) then the original magnitudes will also be incommensurable (with one another) [Prop. 10.16]. And twice the (rectangle contained) by AB and BC (is) rational. Thus, the (square) on AC is irrational. Thus, AC is an irrational (straight-line) [Def. 10.4]. Let it be called a first apotome of a medial (straight-line).[†]

[†] See footnote to Prop. 10.37.

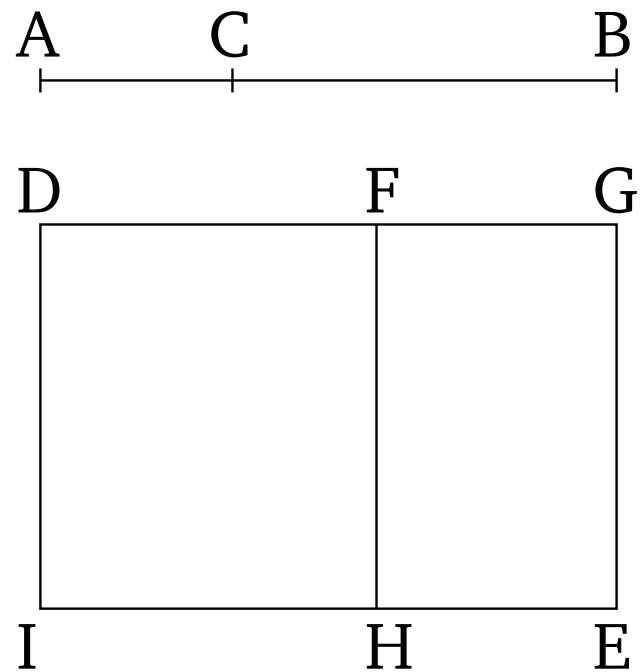
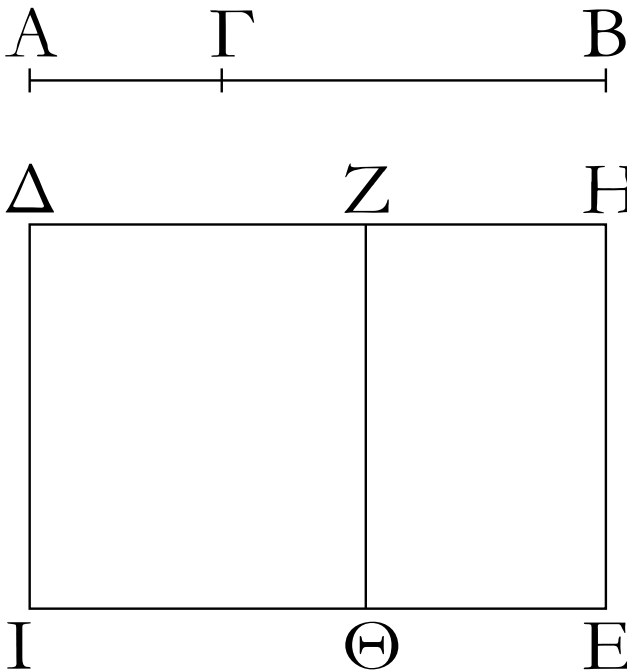
75.

Ἐὰν ἀπὸ μέσης μέση ἀφαιρεθῇ| δυνάμει μόνον σύμμετρος ο>ῦσα τῇ| <όλη, μετὰ δὲ τῆς <όλης μέσον περιέθουσα, ἡ λοιπὴ >άλογός ἐστιν; καλεῖσθω δὲ μέσης ἀποτομὴ δευτέρα. Ἀπὸ γὰρ μέσης τῆς AB μέση ἀφαιρεθῇ ἡ GB δυνάμει μόνον σύμμετρος ο>ῦσα τῇ| <όλη τῇ| AB , μετὰ δὲ τῆς <όλης τῆς AB μέσον περιέθουσα τὸ ὑπὸ τῶν AB , $BΓ$; λέγω, <ὅτι ἡ λοιπὴ ἡ $ΑΓ$ >άλογός ἐστιν; καλεῖσθω δὲ μέσης ἀποτομὴ δευτέρα.

Proposition 75

If a medial (straight-line), which is commensurable in square only with the whole, and which contains a medial (area) with the whole, is subtracted from a(nother) medial (straight-line) then the remainder is an irrational (straight-line). Let it be called a second apotome of a medial (straight-line).

For let the medial (straight-line) CB , which is commensurable in square only with the whole, AB , and which contains with the whole, AB , the medial (rectangle contained) by AB and BC , be subtracted from the medial (straight-line) AB [Prop. 10.28]. I say that the remainder AC is an irrational (straight-line). Let it be called a second apotome of a medial (straight-line).



Ἐκκεῖσθω γὰρ <ρητὴ ἡ $ΔΙ$, καὶ τοῖς μὲν ἀπὸ τῶν AB , $BΓ$ >ίσον παρὰ τὴν $ΔΙ$ παραβεβλήσθω τὸ $ΔΕ$ πλάτοξ ποιοῦν τὴν $ΔΗ$, τῷ δὲ διζ ὑπὸ τῶν AB , $BΓ$ >ίσον παρὰ τὴν $ΔΙ$ παραβεβλήσθω τὸ $ΔΘ$ πλάτοξ ποιοῦν τὴν $ΔΖ$; λοιπὸν >άρα τὸ $ΖΕ$ >ίσον ἐστὶ τῷ ἀπὸ τῆς $ΑΓ$. καὶ ἐπεὶ μέσα καὶ σύμμετρά ἐστι τὰ ἀπὸ τῶν AB , $BΓ$, μέσον >άρα καὶ τὸ $ΔΕ$. καὶ παρὰ <ρητὴν τὴν $ΔΙ$ παράκειται πλάτοξ ποιοῦν τὴν $ΔΗ$; <ρητὴ >άρα ἐστὶν ἡ $ΔΗ$ καὶ ἀσύμμετρος τῇ| $ΔΙ$ μήκει. πάλιν, ἐπεὶ μέσον ἐστὶ τὸ ὑπὸ τῶν AB , $BΓ$, καὶ τὸ διζ >άρα ὑπὸ τῶν AB , $BΓ$ μέσον ἐστίν. καὶ ἐστὶν >ίσον τῷ| $ΔΘ$; καὶ τὸ $ΔΘ$ >άρα μέσον ἐστίν. καὶ παρὰ <ρητὴν τὴν $ΔΙ$ παραβέβληται πλάτοξ ποιοῦν τὴν $ΔΖ$; <ρητὴ >άρα ἐστὶν ἡ $ΔΖ$ καὶ ἀσύμμετρος τῇ| $ΔΙ$ μήκει. καὶ ἐπεὶ αἱ AB , $BΓ$ δυνάμει μόνον σύμμετροί εἰσιν, ἀσύμμετρος >άρα ἐστὶν ἡ AB τῇ| $BΓ$ μήκει; ἀσύμμετρον >άρα καὶ τὸ ἀπὸ τῆς AB

For let the rational (straight-line) DI be laid down. And let DE , equal to the (sum of the squares) on AB and BC , be applied to DI , producing DG as breadth. And let DH , equal to twice the (rectangle contained) by AB and BC , be applied to DI , producing DF as breadth. The remainder FE is thus equal to the (square) on AC [Prop. 2.7]. And since the (squares) on AB and BC are medial and commensurable (with one another), DE (is) thus also medial [Props. 10.15, 10.23 corr.]. And it is applied to the rational (straight-line) DI , producing DG as breadth. Thus, DG is rational, and incommensurable in length with DI [Prop. 10.22]. Again, since the (rectangle contained) by AB and BC is medial, twice the (rectangle contained) by AB and BC is thus also medial [Prop. 10.23 corr.]. And it is equal to DH . Thus, DH is also medial. And it has

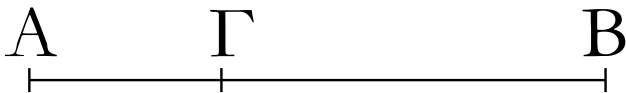
τετράγωνον τῷ ὑπὸ τῶν AB , $BΓ$. ἀλλὰ τῷ μὲν ἀπὸ τῆς AB σύμμετρά ἐστι τὰ ἀπὸ τῶν AB , $BΓ$, τῷ δὲ ὑπὸ τῶν AB , $BΓ$ σύμμετρον ἐστὶ τὸ διζ ὑπὸ τῶν AB , $BΓ$; ἀσύμμετρον >άρα ἐστὶ τὸ διζ ὑπὸ τῶν AB , $BΓ$ τοῖς ἀπὸ τῶν AB , $BΓ$. >ίσον δὲ τοῖς μὲν ἀπὸ τῶν AB , $BΓ$ τὸ ΔE , τῷ δὲ διζ ὑπὸ τῶν AB , $BΓ$ τὸ $\Delta\Theta$; ἀσύμμετρον >άρα [ἐστὶ] τὸ ΔE τῷ $\Delta\Theta$. ὥς δὲ τὸ ΔE πρὸς τὸ $\Delta\Theta$, ο<ύτωξ ἡ $H\Delta$ πρὸς τὴν ΔZ ; ἀσύμμετροξ >άρα ἐστὶν ἡ $H\Delta$ τῇ ΔZ . καὶ εἰσιν ἀμφοτέραι <ρηταί; αἱ >άρα $H\Delta$, ΔZ <ρηταί εἰσι δυνάμει μόνον σύμμετροι; ἡ ZH >άρα ἀποτομή ἐστίν. <ρητὴ δὲ ἡ ΔI ; τὸ δὲ ὑπὸ <ρητῆς καὶ ἀλόγου περιεχόμενον >άλογόν ἐστίν, καὶ ἡ δυναμένη αὐτὸ >άλογός ἐστιν. καὶ δύναται τὸ ZE ἡ AG ; ἡ AG >άρα >άλογός ἐστιν; καλείσθω δὲ μέσηξ ἀποτομὴ δευτέρα. <όπερ >έδει δεῖχαι.

been applied to the rational (straight-line) DI , producing DF as breadth. DF is thus rational, and incommensurable in length with DI [Prop. 10.22]. And since AB and BC are commensurable in square only, AB is thus incommensurable in length with BC . Thus, the square on AB (is) also incommensurable with the (rectangle contained) by AB and BC [Props. 10.21 lem., 10.11]. But, the (sum of the squares) on AB and BC is commensurable with the (square) on AB [Prop. 10.15], and twice the (rectangle contained) by AB and BC is commensurable with the (rectangle contained) by AB and BC [Prop. 10.6]. Thus, twice the (rectangle contained) by AB and BC is incommensurable with the (sum of the squares) on AB and BC [Prop. 10.13]. And DE is equal to the (sum of the squares) on AB and BC , and DH to twice the (rectangle contained) by AB and BC . Thus, DE [is] incommensurable with DH . And as DE (is) to DH , so GD (is) to DF [Prop. 6.1]. Thus, GD is incommensurable with DF [Prop. 10.11]. And they are both rational (straight-lines). Thus, GD and DF are rational (straight-lines which are) commensurable in square only. Thus, FG is an apotome [Prop. 10.73]. And DI (is) rational. And the (area) contained by a rational and an irrational (straight-line) is irrational [Prop. 10.20], and its square-root is irrational. And AC is the square-root of FE . Thus, AC is an irrational (straight-line) [Def. 10.4]. And let it be called the second apotome of a medial (straight-line).[†] (Which is) the very thing it was required to show.

[†] See footnote to Prop. 10.38.

76.

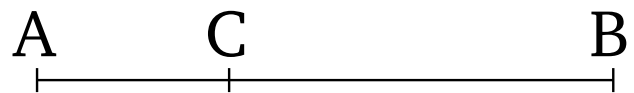
Ἐὰν ἀπὸ εὐθείας εὐθεῖα ἀφαιρεθῇ δυνάμει ἀσύμμετροξ ο>ύσα τῇ <όλη, μετὰ δὲ τῆς <όληξ ποιοῦσα τὰ μὲν ἀπ> αὐτῶν <άμα <ρητόν, τὸ δ> ὑπ> αὐτῶν μέσον, ἡ λοιπὴ >άλογός ἐστιν; καλείσθω δὲ ἐλάσσων.



Ἀπὸ γὰρ εὐθείας τῆς AB εὐθεῖα ἀφαιρήσθω ἡ $BΓ$ δυνάμει ἀσύμμετροξ ο>ύσα τῇ <όλη ποιοῦσα τὰ προκείμενα. λέγω, <ότι ἡ λοιπὴ ἡ AG >άλογός ἐστιν ἡ καλουμένη ἐλάσσων. Ἐπεὶ γὰρ τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν AB , $BΓ$ τετραγώνων <ρητόν ἐστίν, τὸ δὲ διζ ὑπὸ τῶν AB , $BΓ$ μέσον, ἀσύμμετρα >άρα ἐστὶ τὰ ἀπὸ τῶν AB , $BΓ$ τῷ διζ ὑπὸ τῶν AB , $BΓ$; καὶ ἀναστρέψαντι λοιπῷ τῷ ἀπὸ τῆς AG ἀσύμμετρά ἐστὶ τὰ ἀπὸ τῶν AB , $BΓ$. <ρητὰ δὲ τὰ ἀπὸ τῶν AB , $BΓ$; >άλογον >άρα τὸ ἀπὸ τῆς AG ; >άλογος >άρα ἡ AG ; καλείσθω δὲ ἐλάσσων. <όπερ >έδει δεῖχαι.

Proposition 76

If a straight-line, which is incommensurable in square with the whole, and with the whole makes the (squares) on them (added) together rational, and the (rectangle contained) by them medial, is subtracted from a(nother) straight-line then the remainder is an irrational (straight-line). Let it be called a minor (straight-line).



For let the straight-line BC , which is incommensurable in square with the whole, and fulfils the (other) prescribed (conditions), be subtracted from the straight-line AB [Prop. 10.33]. I say that the remainder AC is that irrational (straight-line) called minor.

For since the sum of the squares on AB and BC is rational, and twice the (rectangle contained) by AB and BC (is) medial, the (sum of the squares) on AB and BC is thus incommensurable with twice the (rectangle contained) by AB and BC . And, via conversion, the (sum of the squares) on AB and BC is incommensurable with the remaining (square) on AC [Props. 2.7, 10.16]. And the (sum of the squares) on AB and BC (is) rational. The (square) on AC (is) thus irrational. Thus, AC (is) an irrational (straight-line) [Def. 10.4]. Let it be called a minor (straight-line).[†] (Which is) the very thing it was required to show.

[†] See footnote to Prop. 10.39.

77.

Proposition 77

Ἐὰν ἀπὸ εὐθείας εὐθεῖα ἀφαιρεθῇ δυνάμει ἀσύμμετρος οὐκ ὅσα τῇ ὅλῃ, μετὰ δὲ τῇ ὅλῃ ποιοῦσα τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων μέσον, τὸ δὲ διζὺπὸ αὐτῶν <ρητόν, ἢ λοιπὴν >ἀλογόξ ἐστίν; καλείσθω δὲ ἡ μετὰ <ρητοῦ μέσον τὸ <όλον ποιοῦσα.



Ἀπὸ γὰρ εὐθείας τῆς AB εὐθεῖα ἀφηρήσθω ἡ BΓ δυνάμει ἀσύμμετρος οὐκ ὅσα τῇ AB ποιοῦσα τὰ προκείμενα; λέγω, <ὅτι ἡ λοιπὴ ἡ AΓ >ἀλογόξ ἐστίν ἢ προειρημένη.

Ἐπεὶ γὰρ τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν AB, BΓ τετραγώνων μέσον ἐστίν, τὸ δὲ διζὺπὸ τῶν AB, BΓ <ρητόν, ἀσύμμετρα >άρα ἐστὶ τὰ ἀπὸ τῶν AB, BΓ τῷ διζὺπὸ τῶν AB, BΓ; καὶ λοιπὸν >άρα τὸ ἀπὸ τῆς AΓ ἀσύμμετρον ἐστὶ τῷ διζὺπὸ τῶν AB, BΓ. καὶ ἐστὶ τὸ διζὺπὸ τῶν AB, BΓ <ρητόν; τὸ >άρα ἀπὸ τῆς AΓ >ἀλογόν ἐστίν; >ἀλογόξ >άρα ἐστὶν ἡ AΓ; καλείσθω δὲ ἡ μετὰ <ρητοῦ μέσον τὸ <όλον ποιοῦσα. <ὅπερ >έδει δεῖξαι.

If a straight-line, which is incommensurable in square with the whole, and with the whole makes the sum of the squares on them medial, and twice the (rectangle contained) by them rational, is subtracted from a(nother) straight-line then the remainder is an irrational (straight-line). Let it be called that which makes with a rational (area) a medial whole.



For let the straight-line BC , which is incommensurable in square with AB , and fulfils the (other) prescribed (conditions), be subtracted from the straight-line AB [Prop. 10.34]. I say that the remainder AC is the aforementioned irrational (straight-line).

For since the sum of the squares on AB and BC is medial, and twice the (rectangle contained) by AB and BC rational, the (sum of the squares) on AB and BC is thus incommensurable with twice the (rectangle contained) by AB and BC . Thus, the remaining (square) on AC is also incommensurable with twice the (rectangle contained) by AB and BC [Props. 2.7, 10.16]. And twice the (rectangle contained) by AB and BC is rational. Thus, the (square) on AC is irrational. Thus, AC is an irrational (straight-line) [Def. 10.4]. And let it be called that which makes with a rational (area) a medial whole.[†] (Which is) the very thing it was required to show.

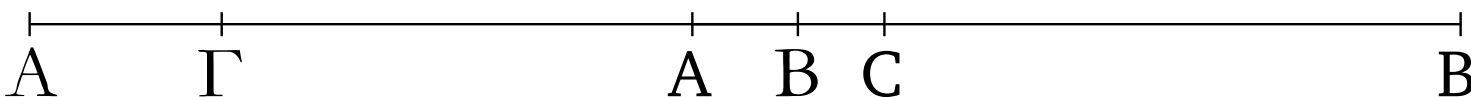
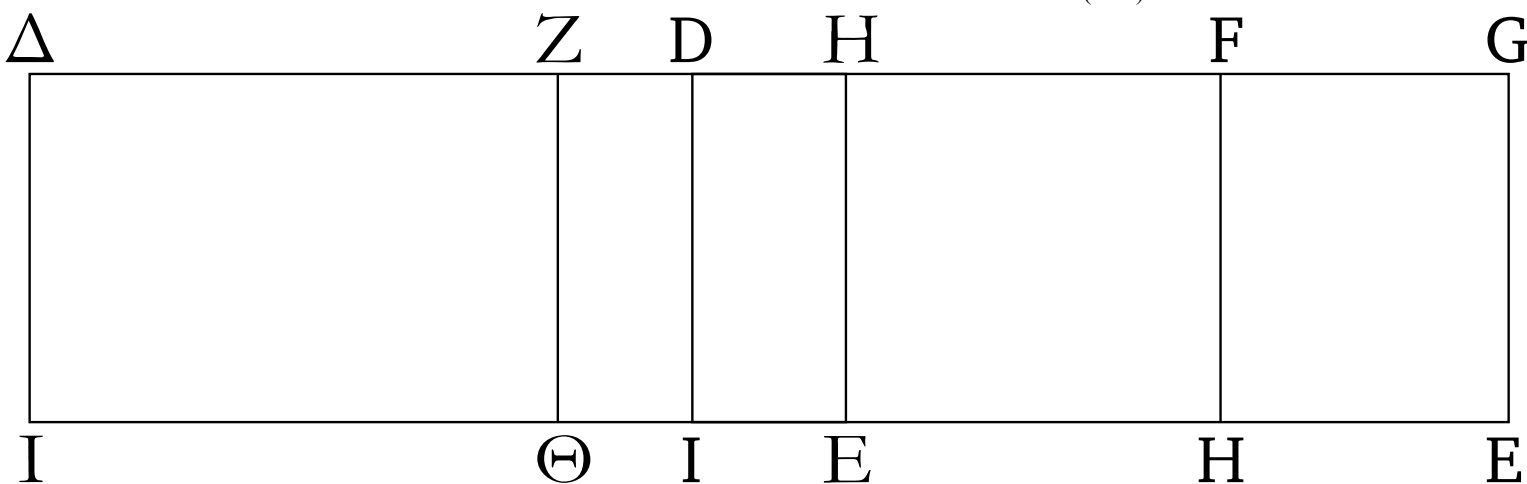
[†] See footnote to Prop. 10.40.

78.

Proposition 78

Ἐὰν ἀπὸ εὐθείας εὐθεῖα ἀφαιρεθῇ δυνάμει ἀσύμμετρος οὐκ ὅσα τῇ ὅλῃ, μετὰ δὲ τῇ ὅλῃ ποιοῦσα τὸ τε συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων μέσον τὸ τε διζὺπὸ αὐτῶν μέσον καὶ >έτι τὰ ἀπὸ αὐτῶν τετράγωνα ἀσύμμετρα τῷ διζὺπὸ αὐτῶν, ἢ λοιπὴν >ἀλογόξ ἐστίν; καλείσθω δὲ ἡ μετὰ μέσου μέσον τὸ <όλον ποιοῦσα.

If a straight-line, which is incommensurable in square with the whole, and with the whole makes the sum of the squares on them medial, and twice the (rectangle contained) by them medial, and, moreover, the (sum of the) squares on them incommensurable with twice the (rectangle contained) by them, is subtracted from a(nother) straight-line then the remainder is an irrational (straight-line). Let it be called that which makes with a medial (area) a medial whole.



Ἀπὸ γὰρ εὐθείας τῆς AB εὐθεῖα ἀφηρήσθω ἡ BΓ δυνάμει

For let the straight-line BC , which is incommensurable in

ἀσύμμετρος οὕσα τῇ AB ποιοῦσα τὰ προκείμενα; λέγω, <ὅτι ἡ λοιπὴ ἡ AG >ἀλογόξ ἐστιν ἡ καλουμένη ἡ μετὰ μέσου μέσον τὸ <όλον ποιοῦσα.

Ἐκκείσθω γὰρ <ρητὴ ἡ DI , καὶ τοῖς μὲν ἀπὸ τῶν AB , $BΓ$ >ίσον παρὰ τὴν DI παραβεβλήσθω τὸ $ΔΕ$ πλάτοξ ποιοῦν τὴν $ΔΗ$, τῷ δὲ διζ ὑπὸ τῶν AB , $BΓ$ >ίσον ἀφηρήσθω τὸ $ΔΘ$ [πλάτοξ ποιοῦν τὴν $ΔΖ$]. λοιπὸν >άρα τὸ $ΖΕ$ >ίσον ἐστὶ τῷ ἀπὸ τῆς AG ; <ώστε ἡ AG δύναται τὸ $ΖΕ$. καὶ ἐπεὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AB , $BΓ$ τετραγώνων μέσον ἐστὶ καὶ ἐστὶν >ίσον τῷ $ΔΕ$, μέσον >άρα [ἐστὶ] τὸ $ΔΕ$. καὶ παρὰ <ρητὴν τὴν DI παράκειται πλάτοξ ποιοῦν τὴν $ΔΗ$; <ρητὴ >άρα ἐστὶν ἡ $ΔΗ$ καὶ ἀσύμμετρος τῇ DI μήκει. πάλιν, ἐπεὶ τὸ διζ ὑπὸ τῶν AB , $BΓ$ μέσον ἐστὶ καὶ ἐστὶν >ίσον τῷ $ΔΘ$, τὸ >άρα $ΔΘ$ μέσον ἐστίν. καὶ παρὰ <ρητὴν τὴν DI παράκειται πλάτοξ ποιοῦν τὴν $ΔΖ$; <ρητὴ >άρα ἐστὶ καὶ ἡ $ΔΖ$ καὶ ἀσύμμετρος τῇ DI μήκει. καὶ ἐπεὶ ἀσύμμετρα ἐστὶ τὰ ἀπὸ τῶν AB , $BΓ$ τῷ διζ ὑπὸ τῶν AB , $BΓ$, ἀσύμμετρον >άρα καὶ τὸ $ΔΕ$ τῷ $ΔΘ$. ὥς δὲ τὸ $ΔΕ$ πρὸς τὸ $ΔΘ$, οὕτως ἐστὶ καὶ ἡ $ΔΗ$ πρὸς τὴν $ΔΖ$; ἀσύμμετρος >άρα ἡ $ΔΗ$ τῇ $ΔΖ$. καὶ εἰσὶν ἀμφοτέραι <ρηταί; αἱ $ΗΔ$, $ΔΖ$ >άρα <ρηταί εἰσι δυνάμει μόνον σύμμετροι. ἀποτομὴ >άρα ἐστὶν ἡ ZH ; <ρητὴ δὲ ἡ $ZΘ$. τὸ δὲ ὑπὸ <ρητῆς καὶ ἀποτομῆς περιεχόμενον [ὀρθογώνιον] >ἀλογόν ἐστιν, καὶ ἡ δυναμένη αὐτὸ >ἀλογόξ ἐστίν; καὶ δύναται τὸ $ΖΕ$ ἡ AG ; ἡ AG >άρα >ἀλογόξ ἐστίν; καλεῖσθω δὲ ἡ μετὰ μέσου μέσον τὸ <όλον ποιοῦσα. <όπερ >έδει δεῖξαι.

square AB , and fulfils the (other) prescribed (conditions), be subtracted from the (straight-line) AB [Prop. 10.35]. I say that the remainder AC is the irrational (straight-line) called that which makes with a medial (area) a medial whole. For let the rational (straight-line) DI be laid down. And let DE , equal to the (sum of the squares) on AB and BC , be applied to DI , producing DG as breadth. And let DH , equal to twice the (rectangle contained) by AB and BC , be subtracted (from DE) [producing DF as breadth]. Thus, the remainder FE is equal to the (square) on AC [Prop. 2.7]. Hence, AC is the square-root of FE . And since the sum of the squares on AB and BC is medial, and is equal to DE , DE [is] thus medial. And it is applied to the rational (straight-line) DI , producing DG as breadth. Thus, DG is rational, and incommensurable in length with DI [Prop. 10.22]. Again, since twice the (rectangle contained) by AB and BC is medial, and is equal to DH , DH is thus medial. And it is applied to the rational (straight-line) DI , producing DF as breadth. Thus, DF is also rational, and incommensurable in length with DI [Prop. 10.22]. And since the (sum of the squares) on AB and BC is incommensurable with twice the (rectangle contained) by AB and BC , DE (is) also incommensurable with DH . And as DE (is) to DH , so DG also is to DF [Prop. 6.1]. Thus, DG (is) incommensurable (in length) with DF [Prop. 10.11]. And they are both rational. Thus, GD and DF are rational (straight-lines which are) commensurable in square only. Thus, FG is an apotome [Prop. 10.73]. And FH (is) rational. And the [rectangle] contained by a rational (straight-line) and an apotome is irrational [Prop. 10.20], and its square-root is irrational. And AC is the square-root of FE . Thus, AC is irrational. Let it be called that which makes with a medial (area) a medial whole.[†] (Which is) the very thing it was required to show.

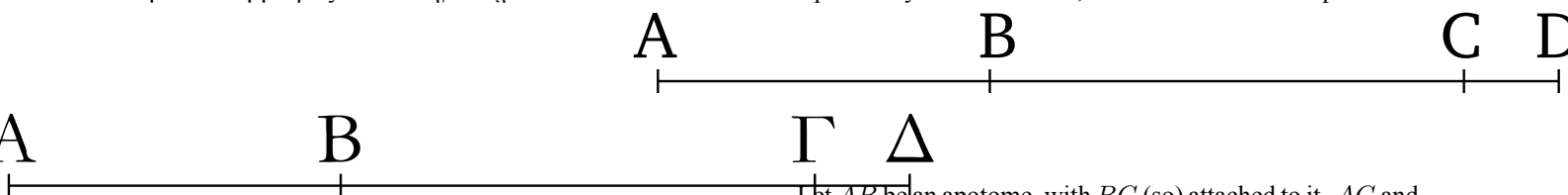
[†] See footnote to Prop. 10.41.

79.

Proposition 79

Τῇ ἀποτομῇ μία [μόνον] προσαρμόζει εὐθεῖα <ρητὴ δύναμει μόνον σύμμετρος οὕσα τῇ <όλη.

[Only] one rational straight-line, which is commensurable in square only with the whole, can be attached to an apotome.[†]



>Ἐστω ἀποτομὴ ἡ AB , προσαρμόζουσα δὲ αὐτῇ ἡ $BΓ$; αἱ AG , $ΓB$ >άρα <ρηταί εἰσι δυνάμει μόνον σύμμετροι; λέγω, <ὅτι τῇ AB ἑτέρα οὐ προσαρμόζει <ρητὴ δύναμει μόνον σύμμετρος οὕσα τῇ <όλη.

Εἰ γὰρ δυνατόν, προσαρμόζετω ἡ BD ; καὶ αἱ AD , $ΔB$ >άρα <ρηταί εἰσι δυνάμει μόνον σύμμετροι. καὶ ἐπεὶ, <ὡς ὑπερέθει τὰ ἀπὸ τῶν AD , $ΔB$ τοῦ διζ ὑπὸ τῶν AD , $ΔB$, τούτῳ ὑπερέθει καὶ τὰ ἀπὸ τῶν AG , $ΓB$ τοῦ διζ ὑπὸ τῶν AG , $ΓB$; τῷ γὰρ αὐτῷ τῷ ἀπὸ τῆς AB ἀμφοτέρα ὑπερέθει; ἐναλλάξ >άρα, <ὡς ὑπερέθει τὰ ἀπὸ τῶν AD , $ΔB$ τῶν ἀπὸ τῶν AG , $ΓB$, τούτῳ ὑπερέθει [καὶ] τὸ διζ ὑπὸ τῶν AD , $ΔB$ τοῦ διζ ὑπὸ τῶν AG , $ΓB$. τὰ δὲ ἀπὸ τῶν AD , $ΔB$ τῶν ἀπὸ τῶν AG , $ΓB$ ὑπερέθει <ρητῷ; <ρητὰ γὰρ ἀμφοτέρα. καὶ τὸ διζ >άρα ὑπὸ τῶν AD , $ΔB$ τοῦ διζ ὑπὸ τῶν AG , $ΓB$ ὑπερέθει <ρητῷ;

Let AB be an apotome, with BC (so) attached to it. AC and CB are thus rational (straight-lines which are) commensurable in square only [Prop. 10.73]. I say that another rational (straight-line), which is commensurable in square only with the whole, cannot be attached to AB .

For, if possible, let BD be (so) attached (to AB). Thus, AD and DB are also rational (straight-lines which are) commensurable in square only [Prop. 10.73]. And since by whatever (area) the (sum of the squares) on AD and DB exceeds twice the (rectangle contained) by AD and DB , the (sum of the squares) on AC and CB also exceeds twice the (rectangle contained) by AC and CB by this (same area). For both exceed by the same (area)—(namely), the (square) on AB [Prop. 2.7]. Thus, alternately, by whatever (area) the (sum of the squares) on AD and DB exceeds the (sum of the

<ὅπερ ἐστὶν ἀδύνατον; μέσα γὰρ ἀμφοτέρα, μέσον δὲ μέσου οὐθ' ὑπερέθει <ρητῶ|. τῇ| >ἄρα AB ἑτέρα οὐ προσαρμόζει <ρητῇ δυνάμει μόνον σύμμετρος ο>ῦσα τῇ| <όλη|. Μία >ἄρα μόνῃ τῇ| ἀποτομῇ| προσαρμόζει <ρητῇ δυνάμει μόνον σύμμετρος ο>ῦσα τῇ| <όλη; <ὅπερ >έδει δεῖχαι.

squares) on AC and CB , twice the (rectangle contained) by AD and DB [also] exceeds twice the (rectangle contained) by AC and CB by this (same area). And the (sum of the squares) on AD and DB exceeds the (sum of the squares) on AC and CB by a rational (area). For both (are) rational (areas). Thus, twice the (rectangle contained) by AD and DB also exceeds twice the (rectangle contained) by AC and CB by a rational (area). The very thing is impossible. For both are medial (areas) [Prop. 10.21], and a medial (area) cannot exceed a(nother) medial (area) by a rational (area) [Prop. 10.26]. Thus, another rational (straight-line), which is commensurable in square only with the whole, cannot be attached to AB .

Thus, only one rational (straight-line), which is commensurable in square only with the whole, can be attached to an apotome. (Which is) the very thing it was required to show.

† This proposition is equivalent to Prop. 10.42, with minus signs instead of plus signs.

80.

Τῇ| μέσῃ ἀποτομῇ| πρώτη μία μόνον προσαρμόζει εὐθεῖα μέση δυνάμει μόνον σύμμετρος ο>ῦσα τῇ| <όλη, μετὰ δὲ τῇξ <όληξ <ρητὸν περιέθουσα.

Proposition 80

Only one medial straight-line, which is commensurable in square only with the whole, and contains a rational (area) with the whole, can be attached to a first apotome of a medial (straight-line).[†]



>Ἐστω γὰρ μέσῃ ἀποτομῇ πρώτη ἡ AB, καὶ τῇ| AB προσαρμोजέτω ἡ BG; αἱ AG, ΓB >ἄρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι <ρητὸν περιέθουσαι τὸ ὑπὸ τῶν AG, ΓB; λέγω, <ὅτι τῇ| AB ἑτέρα οὐ προσαρμόζει μέση δυνάμει μόνον σύμμετρος ο>ῦσα τῇ| <όλη, μετὰ δὲ τῇξ <όληξ <ρητὸν περιέθουσα.

Εἰ γὰρ δυνατόν, προσαρμोजέτω καὶ ἡ ΔB; αἱ >ἄρα AD, ΔB μέσαι εἰσὶ δυνάμει μόνον σύμμετροι <ρητὸν περιέθουσαι τὸ ὑπὸ τῶν AD, ΔB. καὶ ἐπεὶ, <ὡ| ὑπερέθει τὰ ἀπὸ τῶν AD, ΔB τοῦ διζ ὑπὸ τῶν AD, ΔB, τοῦτ' ὑπερέθει καὶ τὰ ἀπὸ τῶν AG, ΓB τοῦ διζ ὑπὸ τῶν AG, ΓB; τῷ| γὰρ αὐτῷ [πάλιν] ὑπερέθουσι τῷ| ἀπὸ τῇξ AB; ἐναλλάξ >ἄρα, <ὡ| ὑπερέθει τὰ ἀπὸ τῶν AD, ΔB τῶν ἀπὸ τῶν AG, ΓB, τοῦτ' ὑπερέθει καὶ τὸ διζ ὑπὸ τῶν AD, ΔB τοῦ διζ ὑπὸ τῶν AG, ΓB. τὸ δὲ διζ ὑπὸ τῶν AD, ΔB τοῦ διζ ὑπὸ τῶν AG, ΓB ὑπερέθει <ρητῶ|; <ρητὰ γὰρ ἀμφοτέρα. καὶ τὰ ἀπὸ τῶν AD, ΔB >ἄρα τῶν ἀπὸ τῶν AG, ΓB [τετραγώνων] ὑπερέθει <ρητῶ|; <ὅπερ ἐστὶν ἀδύνατον; μέσα γὰρ ἐστὶν ἀμφοτέρα, μέσον δὲ μέσου οὐθ' ὑπερέθει <ρητῶ|.

Τῇ| >ἄρα μέσῃ ἀποτομῇ| πρώτη μία μόνον προσαρμόζει εὐθεῖα μέση δυνάμει μόνον σύμμετρος ο>ῦσα τῇ| <όλη, μετὰ δὲ τῇξ <όληξ <ρητὸν περιέθουσα; <ὅπερ >έδει δεῖχαι.

For let AB be a first apotome of a medial (straight-line), and let BC be (so) attached to AB . Thus, AC and CB are medial (straight-lines which are) commensurable in square only, containing a rational (area)—(namely, that contained) by AC and CB [Prop. 10.74]. I say that a(nother) medial (straight-line), which is commensurable in square only with the whole, and contains a rational (area) with the whole, cannot be attached to AB .

For, if possible, let DB also be (so) attached to AB . Thus, AD and DB are medial (straight-lines which are) commensurable in square only, containing a rational (area)—(namely, that) contained by AD and DB [Prop. 10.74]. And since by whatever (area) the (sum of the squares) on AD and DB exceeds twice the (rectangle contained) by AD and DB , the (sum of the squares) on AC and CB also exceeds twice the (rectangle contained) by AC and CB by this (same area). For [again] both exceed by the same (area)—(namely), the (square) on AB [Prop. 2.7]. Thus, alternately, by whatever (area) the (sum of the squares) on AD and DB exceeds the (sum of the squares) on AC and CB , twice the (rectangle contained) by AD and DB also exceeds twice the (rectangle contained) by AC and CB by this (same area). And twice the (rectangle contained) by AD and DB exceeds twice the (rectangle contained) by AC and CB by a rational (area). For both (are) rational (areas). Thus, the (sum of the squares) on AD and DB also exceeds the (sum of the) [squares] on AC and CB by a rational (area). The very thing is impossible. For both are medial (areas) [Props. 10.15, 10.23 corr.], and a medial (area) cannot exceed a(nother) medial (area) by a rational (area) [Prop. 10.26].

Thus, only one medial (straight-line), which is commensurable in square only with the whole, and contains a rational

(area) with the whole, can be attached to a first apotome of a medial (straight-line). (Which is) the very thing it was required to show.

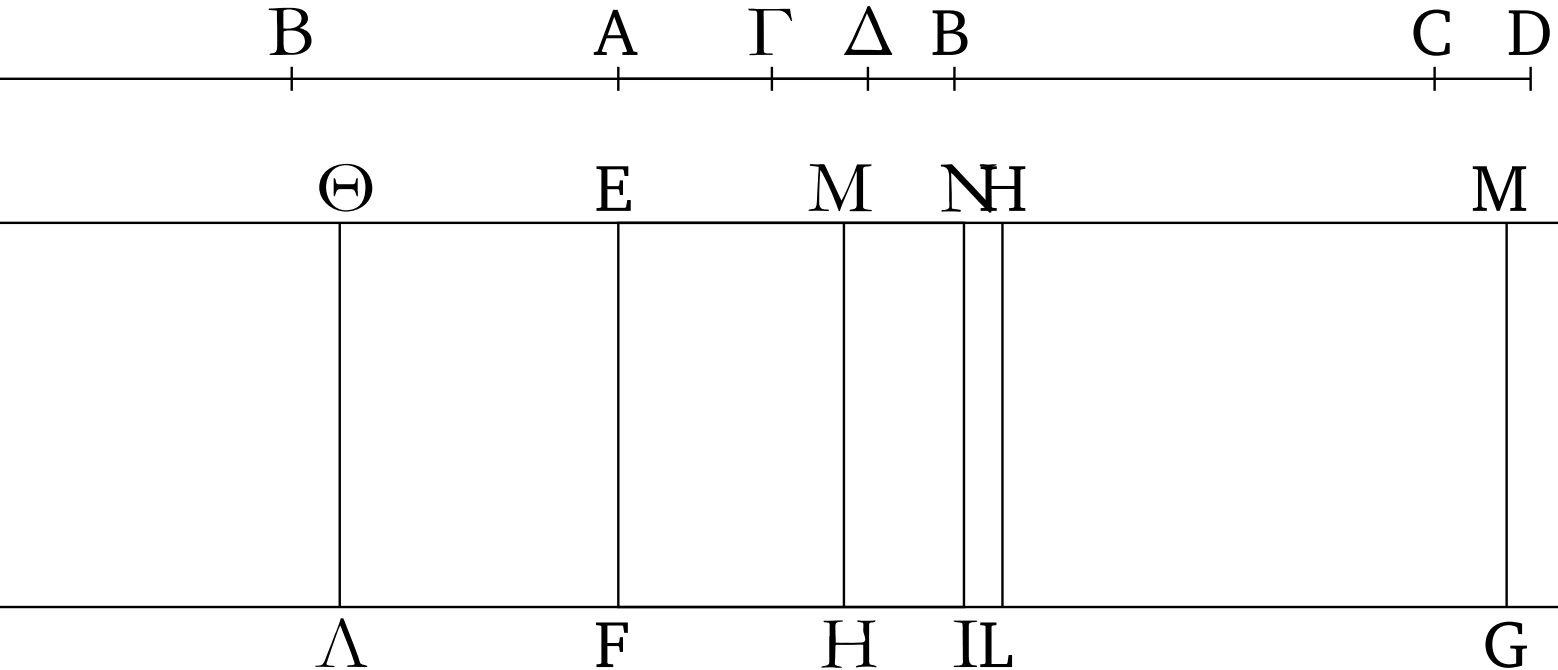
† This proposition is equivalent to Prop. 10.43, with minus signs instead of plus signs.

81.

Proposition 81

Τῇ μέσῃ ἀποτομῇ δευτέρα μία μόνον προσαρμόζει εὐθεῖα μέση δυνάμει μόνον σύμμετρος τῇ <όλη, μετὰ δὲ τῇ <όλης μέσον περιέθουσα.

Only one medial straight-line, which is commensurable in square only with the whole, and contains a medial (area) with the whole, can be attached to a second apotome of a medial (straight-line).[†]



>Ἐστω μέσῃ ἀποτομῇ δευτέρα ἡ AB καὶ τῇ AB προσαρμόζουσα ἡ BC ; αἱ >ἀρα AC , CB μέσαι εἰσὶ δυνάμει μόνον σύμμετροι μέσον περιέθουσαι τὸ ὑπὸ τῶν AC , CB ; λέγω, <ὅτι τῇ AB ἑτέρα οὐ προσαρμόσει εὐθεῖα μέση δυνάμει μόνον σύμμετρος ο>ύσα τῇ <όλη, μετὰ δὲ τῇ <όλης μέσον περιέθουσα.

Εἰ γὰρ δυνατόν, προσαρμοζέτω ἡ BD ; καὶ αἱ AD , DB >ἀρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι μέσον περιέθουσαι τὸ ὑπὸ τῶν AD , DB . καὶ ἐκκείσθω <ρητὴ ἡ EZ , καὶ τοῖς μὲν ἀπὸ τῶν AC , CB >ἴσον παρὰ τὴν EZ παραβεβλήσθω τὸ EH πλάτος ποιοῦν τὴν EM ; τῷ δὲ διζὺ ὑπὸ τῶν AC , CB >ἴσον ἀφηρήσθω τὸ $ΘH$ πλάτος ποιοῦν τὴν $ΘM$; λοιπὸν >ἀρα τὸ EL >ἴσον ἐστὶ τῷ ἀπὸ τῆς AB ; <ώστε ἡ AB δύναται τὸ EL . πάλιν δὲ τοῖς ἀπὸ τῶν AD , DB >ἴσον παρὰ τὴν EZ παραβεβλήσθω τὸ EI πλάτος ποιοῦν τὴν EN ; >ἐστὶ δὲ καὶ τὸ EL >ἴσον τῷ διζὺ ὑπὸ τῶν AD , DB . καὶ ἐπεὶ μέσαι εἰσὶν αἱ AC , CB , μέσαι >ἀρα ἐστὶ καὶ τὰ ἀπὸ τῶν AC , CB . καὶ ἐστὶν >ἴσα τῷ EH ; μέσον >ἀρα καὶ τὸ EH . καὶ παρὰ <ρητὴν τὴν EZ παράκειται πλάτος ποιοῦν τὴν EM ; <ρητὴ >ἀρα ἐστὶν ἡ EM καὶ ἀσύμμετρος τῇ EZ μήκει. πάλιν, ἐπεὶ μέσον ἐστὶ τὸ ὑπὸ τῶν AC , CB , καὶ τὸ διζὺ ὑπὸ τῶν AC , CB μέσον ἐστίν. καὶ ἐστὶν >ἴσον τῷ $ΘH$; καὶ τὸ $ΘH$ >ἀρα μέσον ἐστίν. καὶ παρὰ <ρητὴν τὴν EZ παράκειται πλάτος ποιοῦν τὴν $ΘM$; <ρητὴ >ἀρα ἐστὶ καὶ ἡ $ΘM$ καὶ ἀσύμμετρος τῇ EZ μήκει. καὶ ἐπεὶ αἱ AC , CB δυνάμει μόνον σύμμετροί εἰσιν, ἀσύμμετρος >ἀρα ἐστὶν ἡ AG τῇ GB μήκει. ὥς δὲ ἡ AG πρὸς τὴν GB , ο<ύτως ἐστὶ τὸ ἀπὸ τῆς AG πρὸς τὸ ὑπὸ τῶν AC ,

with BC (so) attached to AB . Thus, AC and CB are medial (straight-lines which are) commensurable in square only, containing a medial (area)—(namely, that contained) by AC and CB [Prop. 10.75]. I say that a(nother) medial straight-line, which is commensurable in square only with the whole, and contains a medial (area) with the whole, cannot be attached to AB .

For, if possible, let BD be (so) attached. Thus, AD and DB are also medial (straight-lines which are) commensurable in square only, containing a medial (area)—(namely, that contained) by AD and DB [Prop. 10.75]. And let the rational (straight-line) EF be laid down. And let EG , equal to the (sum of the squares) on AC and CB , be applied to EF , producing EM as breadth. And let HG , equal to twice the (rectangle contained) by AC and CB , be subtracted (from EG), producing HM as breadth. The remainder EL is thus equal to the (square) on AB [Prop. 2.7]. Hence, AB is the square-root of EL . So, again, let EI , equal to the (sum of the squares) on AD and DB be applied to EF , producing EN as breadth. And EL is also equal to the square on AB . Thus, the remainder HI is equal to twice the (rectangle contained) by AD and DB [Prop. 2.7]. And since AC and CB are (both) medial (straight-lines), the (sum of the squares) on AC and CB is also medial. And it is equal to EG . Thus, EG is also medial [Props. 10.15, 10.23 corr.]. And it is applied to the rational (straight-line) EF , producing EM as breadth. Thus, EM is rational, and incommensurable in

ΓΒ; ἀσύμμετρον >άρα ἐστὶ τὸ ἀπὸ τῆς ΑΓ τῷ ὑπὸ τῶν ΑΓ, ΓΒ. ἀλλὰ τῷ μὲν ἀπὸ τῆς ΑΓ σύμμετρά ἐστι τὰ ἀπὸ τῶν ΑΓ, ΓΒ, τῷ δὲ ὑπὸ τῶν ΑΓ, ΓΒ σύμμετρόν ἐστι τὸ διζ ὑπὸ τῶν ΑΓ, ΓΒ; ἀσύμμετρα >άρα ἐστὶ τὰ ἀπὸ τῶν ΑΓ, ΓΒ τῷ διζ ὑπὸ τῶν ΑΓ, ΓΒ. καὶ ἐστὶ τοῖς μὲν ἀπὸ τῶν ΑΓ, ΓΒ >ίσον τὸ ΕΗ, τῷ δὲ διζ ὑπὸ τῶν ΑΓ, ΓΒ >ίσον τὸ ΗΘ; ἀσύμμετρον >άρα ἐστὶ τὸ ΕΗ τῷ ΘΗ. ὥς δὲ τὸ ΕΗ πρὸς τὸ ΘΗ, οὕτως ἐστὶν ἡ ΕΜ πρὸς τὴν ΘΜ; ἀσύμμετρος >άρα ἐστὶν ἡ ΕΜ τῇ ΜΘ μήκει. καὶ εἰσιν ἀμφοτέραι <ρηταί; αἱ ΕΜ, ΜΘ >άρα <ρηταί εἰσι δυνάμει μόνον σύμμετροι; ἀποτομή >άρα ἐστὶν ἡ ΕΘ, προσαρμόζουσα δὲ αὐτῇ ἡ ΘΜ. ὁμοίως δὲ δείχομεν, <ὅτι καὶ ἡ ΘΝ αὐτῇ προσαρμόζει; τῇ >άρα ἀποτομῇ >άλλη καὶ >άλλη προσαρμόζει εὐθεῖα δυνάμει μόνον σύμμετρος ο>ῦσα τῇ <όλη; <όπερ ἐστὶν ἀδύνατον.

Τῇ >άρα μέσῃ ἀποτομῇ δευτέρα μία μόνον προσαρμόζει εὐθεῖα μέση δυνάμει μόνον σύμμετρος ο>ῦσα τῇ <όλη, μετὰ δὲ τῆς <όλης μέσον περιέθουσα; <όπερ >έδει δεῖχαι.

length with EF [Prop. 10.22]. Again, since the (rectangle contained) by AC and CB is medial, twice the (rectangle contained) by AC and CB is also medial [Prop. 10.23 corr.]. And it is equal to HG . Thus, HG is also medial. And it is applied to the rational (straight-line) EF , producing HM as breadth. Thus, HM is also rational, and incommensurable in length with EF [Prop. 10.22]. And since AC and CB are commensurable in square only, AC is thus incommensurable in length with CB . And as AC (is) to CB , so the (square) on AC is to the (rectangle contained) by AC and CB [Prop. 10.21 corr.]. Thus, the (square) on AC is incommensurable with the (rectangle contained) by AC and CB [Prop. 10.11]. But, the (sum of the squares) on AC and CB is commensurable with the (square) on AC , and twice the (rectangle contained) by AC and CB is commensurable with the (rectangle contained) by AC and CB [Prop. 10.6]. Thus, the (sum of the squares) on AC and CB is incommensurable with twice the (rectangle contained) by AC and CB [Prop. 10.13]. And EG is equal to the (sum of the squares) on AC and CB . And GH is equal to twice the (rectangle contained) by AC and CB . Thus, EG is incommensurable with HG . And as EG (is) to HG , so EM is to HM [Prop. 6.1]. Thus, EM is incommensurable in length with MH [Prop. 10.11]. And they are both rational (straight-lines). Thus, EM and MH are rational (straight-lines which are) commensurable in square only. Thus, EH is an apotome [Prop. 10.73], and HM (is) attached to it. So, similarly, we can show that HN (is) also (commensurable in square only with EN and is) attached to (EH). Thus, different straight-lines, which are commensurable in square only with the whole, are attached to an apotome. The very thing is impossible [Prop. 10.79].

Thus, only one medial straight-line, which is commensurable in square only with the whole, and contains a medial (area) with the whole, can be attached to a second apotome of a medial (straight-line). (Which is) the very thing it was required to show.

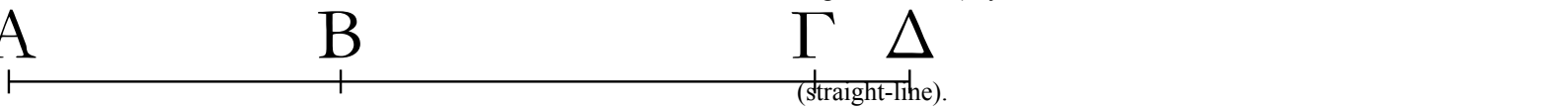
† This proposition is equivalent to Prop. 10.44, with minus signs instead of plus signs.

82.

Proposition 82

Τῇ ἐλάσσονι μία μόνον προσαρμόζει εὐθεῖα δυνάμει ἀσύμμετρος ο>ῦσα τῇ <όλη ποιοῦσα μετὰ τῆς <όλης τὸ μὲν ἐκ τῶν ἀπ> αὐτῶν τετραγώνων <ρητόν, τὸ δὲ διζ ὑπ> αὐτῶν μέσον.

Only one straight-line, which is incommensurable in square with the whole, and (together) with the whole makes the (sum of the) squares on them rational, and twice the (rectangle contained) by them medial, can be attached to a minor



>Ἐστω ἡ ἐλάσσων ἡ ΑΒ, καὶ τῇ ΑΒ προσαρμόζουσα >έστω ἡ ΒΓ; αἱ >άρα ΑΓ, ΓΒ δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ> αὐτῶν τετραγώνων <ρητόν, τὸ δὲ διζ ὑπ> αὐτῶν μέσον; λέγω, <ὅτι τῇ ΑΒ ἑτέρα εὐθεῖα οὐ προσαρμόσει τὰ αὐτὰ ποιοῦσα.

Εἰ γὰρ δυνατόν, προσαρμόζέτω ἡ ΒΔ; καὶ αἱ ΑΔ, ΔΒ >άρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὰ προειρημένα. καὶ ἐπεὶ, <ὦ ὑπερέθει τὰ ἀπὸ τῶν ΑΔ, ΔΒ τῶν ἀπὸ τῶν ΑΓ, ΓΒ, τοῦτο ὑπερέθει καὶ τὸ διζ ὑπὸ τῶν ΑΔ, ΔΒ τοῦ διζ ὑπὸ τῶν ΑΓ, ΓΒ, τὰ δὲ ἀπὸ τῶν ΑΔ, ΔΒ τετράγωνα τῶν ἀπὸ

Let AB be a minor (straight-line), and let BC be (so) attached to AB . Thus, AC and CB are (straight-lines which are) incommensurable in square, making the sum of the squares on them rational, and twice the (rectangle contained) by them medial [Prop. 10.76]. I say that another straight-line fulfilling the same (conditions) cannot be attached to AB .

For, if possible, let BD be (so) attached (to AB). Thus, AD and DB are also (straight-lines which are) incommen-

τῶν ΑΓ, ΓΒ τετραγώνων ὑπερέθει <ρητῶ>; <ρητὰ γάρ ἐστιν ἀμφοτέρω>; καὶ τὸ διζ ὑπὸ τῶν ΑΔ, ΔΒ >άρα τοῦ διζ ὑπὸ τῶν ΑΓ, ΓΒ ὑπερέθει <ρητῶ>; <όπερ ἐστὶν ἀδύνατον>; μέσα γάρ ἐστιν ἀμφοτέρω>.

Τῇ >άρα ἐλάσσονι μία μόνον προσαρμόζει εὐθεῖα δυνάμει ἀσύμμετρος ο>ῦσα τῇ <όλη καὶ ποιούσα τὰ μὲν ἀπ> αὐτῶν τετράγωνα <άμα <ρητόν, τὸ δὲ διζ ὑπ> αὐτῶν μέσον>; <όπερ >έδει δεῖξαι.

surable in square, fulfilling the (other) aforementioned (conditions) [Prop. 10.76]. And since by whatever (area) the (sum of the squares) on AD and DB exceeds the (sum of the squares) on AC and CB , twice the (rectangle contained) by AD and DB also exceeds twice the (rectangle contained) by AC and CB by this (same area) [Prop. 2.7]. And the (sum of the) squares on AD and DB exceeds the (sum of the) squares on AC and CB by a rational (area). For both are rational (areas). Thus, twice the (rectangle contained) by AD and DB also exceeds twice the (rectangle contained) by AC and CB by a rational (area). The very thing is impossible. For both are medial (areas) [Prop. 10.26].

Thus, only one straight-line, which is incommensurable in square with the whole, and (with the whole) makes the squares on them (added) together rational, and twice the (rectangle contained) by them medial, can be attached to a minor (straight-line). (Which is) the very thing it was required to show.

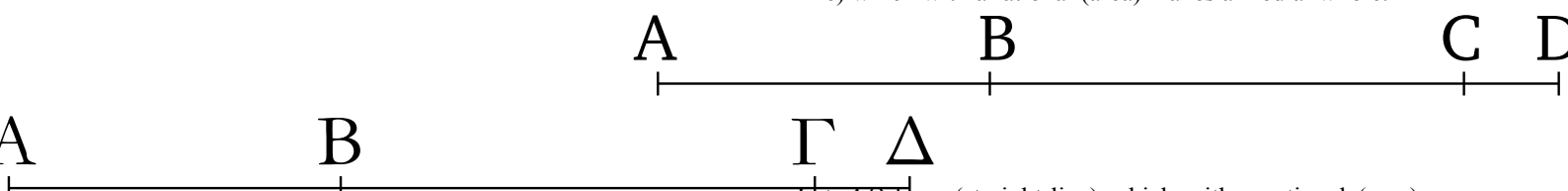
† This proposition is equivalent to Prop. 10.45, with minus signs instead of plus signs.

83.

Τῇ μετὰ <ρητοῦ μέσον τὸ <όλον ποιούση μία μόνον προσαρμόζει εὐθεῖα δυνάμει ἀσύμμετρος ο>ῦσα τῇ <όλη, μετὰ δὲ τῇ <όλη ποιούσα τὸ μὲν συγκείμενον ἐκ τῶν ἀπ> αὐτῶν τετραγώνων μέσον, τὸ δὲ διζ ὑπ> αὐτῶν <ρητόν.

Proposition 83

Only one straight-line, which is incommensurable in square with the whole, and (together) with the whole makes the sum of the squares on them medial, and twice the (rectangle contained) by them rational, can be attached to that (straight-line) which with a rational (area) makes a medial whole.†



>Ἐστω ἡ μετὰ <ρητοῦ μέσον τὸ <όλον ποιούσα ἡ ΑΒ, καὶ τῇ ΑΒ προσαρμοζέτω ἡ ΒΓ; αἱ >άρα ΑΓ, ΓΒ δυνάμει εἰσὶν ἀσύμμετροι ποιούσαι τὰ προκείμενα; λέγω, <ὅτι τῇ ΑΒ ἑτέρα οὐ προσαρμόσει τὰ αὐτὰ ποιούσα.

Εἰ γὰρ δυνατόν, προσαρμοζέτω ἡ ΒΔ; καὶ αἱ ΑΔ, ΔΒ >άρα εὐθεῖαι δυνάμει εἰσὶν ἀσύμμετροι ποιούσαι τὰ προκείμενα. ἐπεὶ οὖν, <ῶ ὑπερέθει τὰ ἀπὸ τῶν ΑΔ, ΔΒ τῶν ἀπὸ τῶν ΑΓ, ΓΒ, τοῦτω ὑπερέθει καὶ τὸ διζ ὑπὸ τῶν ΑΔ, ΔΒ τοῦ διζ ὑπὸ τῶν ΑΓ, ΓΒ ἀκολουθῶς τοῖς πρὸ αὐτοῦ, τὸ δὲ διζ ὑπὸ τῶν ΑΔ, ΔΒ τοῦ διζ ὑπὸ τῶν ΑΓ, ΓΒ ὑπερέθει <ρητῶ>; <ρητὰ γάρ ἐστιν ἀμφοτέρω>; καὶ τὰ ἀπὸ τῶν ΑΔ, ΔΒ >άρα τῶν ἀπὸ τῶν ΑΓ, ΓΒ ὑπερέθει <ρητῶ>; <όπερ ἐστὶν ἀδύνατον>; μέσα γάρ ἐστιν ἀμφοτέρω>.

Οὐκ >άρα τῇ ΑΒ ἑτέρα προσαρμόσει εὐθεῖα δυνάμει ἀσύμμετρος ο>ῦσα τῇ <όλη, μετὰ δὲ τῇ <όλη ποιούσα τὰ προειρημένα; μία >άρα μόνον προσαρμόσει; <όπερ >έδει δεῖξαι.

Let AB be a (straight-line) which with a rational (area) makes a medial whole, and let BC be (so) attached to AB . Thus, AC and CB are (straight-lines which are) incommensurable in square, fulfilling the (other) proscribed (conditions) [Prop. 10.77]. I say that another (straight-line) fulfilling the same (conditions) cannot be attached to AB .

For, if possible, let BD be (so) attached (to AB). Thus, AD and DB are also straight-lines (which are) incommensurable in square, fulfilling the (other) prescribed (conditions) [Prop. 10.77]. Therefore, analogously to the (propositions) before this, since by whatever (area) the (sum of the squares) on AD and DB exceeds the (sum of the squares) on AC and CB , twice the (rectangle contained) by AD and DB also exceeds twice the (rectangle contained) by AC and CB by this (same area). And twice the (rectangle contained) by AD and DB exceeds twice the (rectangle contained) by AC and CB by a rational (area). For they are (both) rational (areas). Thus, the (sum of the squares) on AD and DB also exceeds the (sum of the squares) on AC and CB by a rational (area). The very thing is impossible. For both are medial (areas) [Prop. 10.26].

Thus, another straight-line cannot be attached to AB , which is incommensurable in square with the whole, and fulfills the (other) aforementioned (conditions) with the whole. Thus, only one (such straight-line) can be (so) attached. (Which is) the very thing it was required to show.

† This proposition is equivalent to Prop. 10.46, with minus signs instead of plus signs.

84.

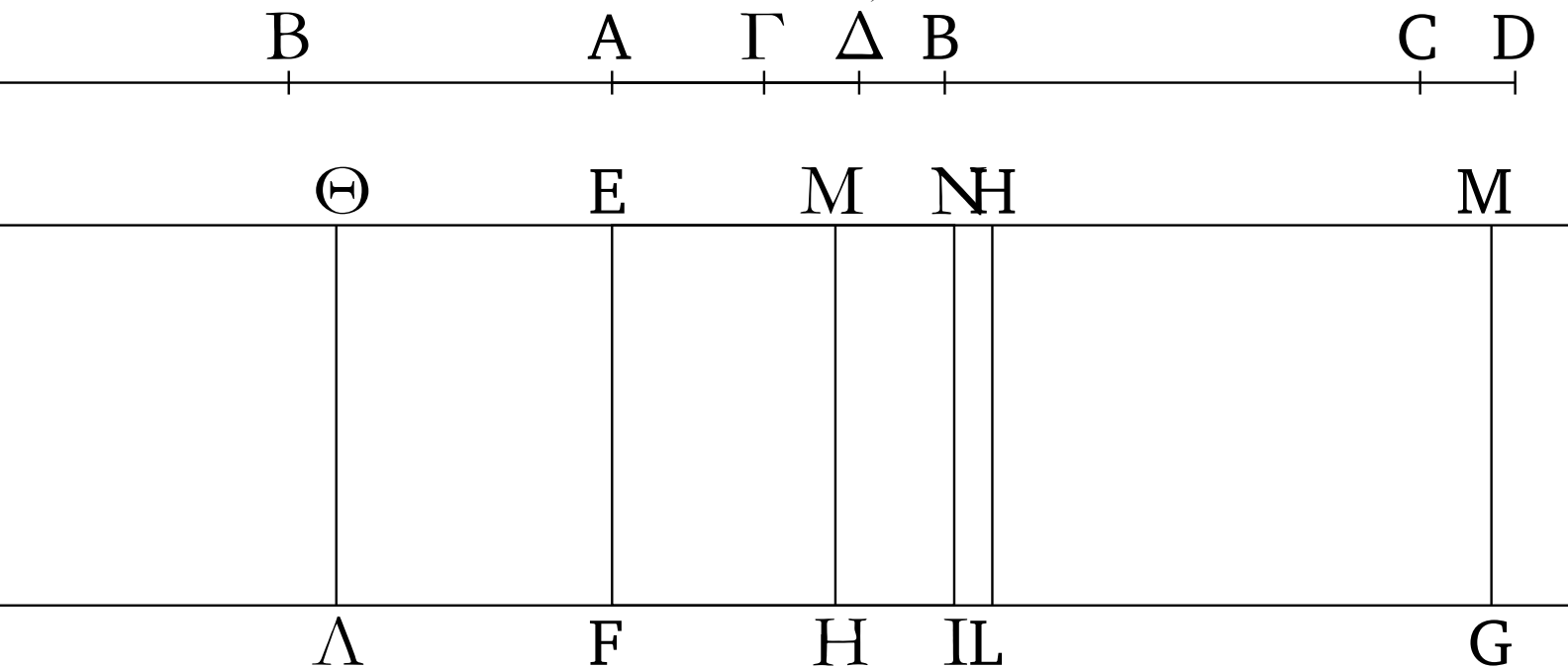
Proposition 84

Τῇ μετὰ μέσον μέσον τὸ <όλον ποιούσῃ μία μόνη προσαρμόζε-
εὐθεΐα δυνάμει ἀσύμμετρος ο>ῦσα τῇ <όλῃ, μετὰ δὲ
τῇ <όλῃ ποιούσα τὸ τε συγκείμενον ἐκ τῶν ἀπ> αὐτῶν
τετραγώνων μέσον τὸ τε δις ὑπ> αὐτῶν μέσον καὶ >ἐτι
ἀσύμμετρον τῷ συγκειμένῳ ἐκ τῶν ἀπ> αὐτῶν.

>Ἐστω ἡ μετὰ μέσον μέσον τὸ <όλον ποιούσα ἡ AB ,
προσαρμόζουσα δὲ αὐτῇ ἡ $BΓ$; αἱ >άρα $ΑΓ$, $ΓΒ$ δυνάμει
εἰσὶν ἀσύμμετροι ποιούσαι τὰ προειρημένα. λέγω, <ὅτι τῇ
 AB ἑτέρα οὐ προσαρμόσει ποιούσα προειρημένα.

Only one straight-line, which is incommensurable in square
with the whole, and (together) with the whole makes the
sum of the squares on them medial, and twice the (rectangle
contained) by them medial, and, moreover, incommensu-
rable with the sum of the (squares) on them, can be attached
to that (straight-line) which with a medial (area) makes a
medial whole.[†]

Let AB be a (straight-line) which with a medial (area)
makes a medial whole, BC being (so) attached to it. Thus,
 AC and CB are incommensurable in square, fulfilling the
(other) aforementioned (conditions) [Prop. 10.78]. I say that
a(nother) (straight-line) fulfilling the aforementioned (con-
ditions) cannot be attached to AB .



Εἰ γὰρ δυνατόν, προσαρμοζέτω ἡ BD , <ώστε καὶ τὰς AD ,
 $ΔB$ δυνάμει ἀσυμμέτρου ε>ἵναι ποιούσας τὰ τε ἀπὸ τῶν
 AD , $ΔB$ τετράγωνα <άμα μέσον καὶ τὸ δις ὑπὸ τῶν AD ,
 $ΔB$ μέσον καὶ >ἐτι τὰ ἀπὸ τῶν AD , $ΔB$ ἀσύμμετρα τῷ
δις ὑπὸ τῶν AD , $ΔB$; καὶ ἐκκείσθω <ρητὴ ἡ EZ , καὶ τοῖς
μὲν ἀπὸ τῶν $ΑΓ$, $ΓΒ$ >ἴσον παρὰ τὴν EZ παραβεβλήσθω
τὸ EH πλάτος ποιῶν τὴν EM , τῷ δὲ δις ὑπὸ τῶν $ΑΓ$, $ΓΒ$
>ἴσον παρὰ τὴν EZ παραβεβλήσθω τὸ $ΘH$ πλάτος ποιῶν
τὴν $ΘM$; λοιπὸν >άρα τὸ ἀπὸ τῆς AB >ἴσον ἐστὶ τῷ $ΕΛ$;
ἡ >άρα AB δύναται τὸ $ΕΛ$. πάλιν τοῖς ἀπὸ τῶν AD , $ΔB$
>ἴσον παρὰ τὴν EZ παραβεβλήσθω τὸ EI πλάτος ποιῶν
τὴν EN . >ἐστὶ δὲ καὶ τὸ ἀπὸ τῆς AB >ἴσον τῷ $ΕΛ$; λοιπὸν
>άρα τὸ δις ὑπὸ τῶν AD , $ΔB$ >ἴσον [ἐστὶ] τῷ $ΘI$. καὶ ἐπεὶ
μέσον ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν $ΑΓ$, $ΓΒ$ καὶ
ἐστὶν >ἴσον τῷ EH , μέσον >άρα ἐστὶ καὶ τὸ EH . καὶ παρὰ
<ρητὴν τὴν EZ παράκειται πλάτος ποιῶν τὴν EM ; <ρητὴ
>άρα ἐστὶν ἡ EM καὶ ἀσύμμετρος τῇ EZ μήκει. πάλιν,
ἐπεὶ μέσον ἐστὶ τὸ δις ὑπὸ τῶν $ΑΓ$, $ΓΒ$ καὶ ἐστὶν >ἴσον
τῷ $ΘH$, μέσον >άρα καὶ τὸ $ΘH$. καὶ παρὰ <ρητὴν τὴν EZ
παράκειται πλάτος ποιῶν τὴν $ΘM$; <ρητὴ >άρα ἐστὶν ἡ $ΘM$
καὶ ἀσύμμετρος τῇ EZ μήκει. καὶ ἐπεὶ ἀσύμμετρά ἐστι τὰ
ἀπὸ τῶν $ΑΓ$, $ΓΒ$ τῷ δις ὑπὸ τῶν $ΑΓ$, $ΓΒ$, ἀσύμμετρόν ἐστι
καὶ τὸ EH τῷ $ΘH$; ἀσύμμετρος >άρα ἐστὶ καὶ ἡ EM τῇ
 $MΘ$ μήκει. καὶ εἰσὶν ἀμφοτέραι <ρηταί; αἱ >άρα EM , $MΘ$
<ρηταί εἰσι δυνάμει μόνον σύμμετροι; ἀποτομὴ >άρα ἐστὶν

For, if possible, let BD be (so) attached. Hence, AD and
 DB are also (straight-lines which are) incommensurable in
square, making the squares on AD and DB (added) together
medial, and twice the (rectangle contained) by AD and DB
medial, and, moreover, the (sum of the squares) on AD and
 DB incommensurable with twice the (rectangle contained)
by AD and DB [Prop. 10.78]. And let the rational (straight-
line) EF be laid down. And let EG , equal to the (sum of
the squares) on AC and CB , be applied to EF , producing
 EM as breadth. And let HG , equal to twice the (rectangle
contained) by AC and CB , be applied to EF , producing
 HM as breadth. Thus, the remaining (square) on AB is
equal to EL [Prop. 2.7]. Thus, AB is the square-root of
 EL . Again, let EI , equal to the (sum of the squares) on
 AD and DB , be applied to EF , producing EN as breadth.
And the (square) on AB is also equal to EL . Thus, the re-
maining twice the (rectangle contained) by AD and DB [is]
equal to HI [Prop. 2.7]. And since the sum of the (squares)
on AC and CB is medial, and is equal to EG , EG is thus
also medial. And it is applied to the rational (straight-line)
 EF , producing EM as breadth. EM is thus rational, and
incommensurable in length with EF [Prop. 10.22]. Again,
since twice the (rectangle contained) by AC and CB is me-
dial, and is equal to HG , HG is thus also medial. And it is
applied to the rational (straight-line) EF , producing HM

ἡ ΕΘ, προσαρμόζουσα δὲ αὐτῇ ἡ ΘΜ. ὁμοίωξ δὲ δείχομεν, <ὅτι ἡ ΕΘ πάλιν ἀποτομή ἐστιν, προσαρμόζουσα δὲ αὐτῇ ἡ ΘΝ. τῇ >άρα ἀποτομῇ> ἄλλη καὶ >άλλη προσαρμόζει <ρητῇ δυνάμει μόνον σύμμετροξ ο>ύσα τῇ <όλη>; <όπερ ἐδείθθη ἀδύνατον. οὐκ >άρα τῇ ΑΒ ἐτέρα προσαρμόσει εὐθεῖα.

Τῇ >άρα ΑΒ μία μόνον προσαρμόζει εὐθεῖα δυνάμει ἀσύμμετροξ ο>ύσα τῇ <όλη>, μετὰ δὲ τῆξ <όληξ ποιούσα τὰ τε ἀπ> αὐτῶν τετράγωνα <άμα μέσον καὶ τὸ διξ ὑπ> αὐτῶν μέσον καὶ >έτι τὰ ἀπ> αὐτῶν τετράγωνα ἀσύμμετρα τῷ διξ ὑπ> αὐτῶν; <όπερ >έδει δεῖχαι.

as breadth. HM is thus rational, and incommensurable in length with EF [Prop. 10.22]. And since the (sum of the squares) on AC and CB is incommensurable with twice the (rectangle contained) by AC and CB , EG is also incommensurable with HG . Thus, EM is also incommensurable in length with MH [Props. 6.1, 10.11]. And they are both rational (straight-lines). Thus, EM and MH are rational (straight-lines which are) commensurable in square only. Thus, EH is an apotome [Prop. 10.73], with HM attached to it. So, similarly, we can show that EH is again an apotome, with HN attached to it. Thus, different rational (straight-lines), which are commensurable in square only with the whole, are attached to an apotome. The very thing was shown (to be) impossible [Prop. 10.79]. Thus, another straight-line cannot be (so) attached to AB .

Thus, only one straight-line, which is incommensurable in square with the whole, and (together) with the whole makes the squares on them (added) together medial, and twice the (rectangle contained) by them medial, and, moreover, the (sum of the) squares on them incommensurable with the (rectangle contained) by them, can be attached to AB . (Which is) the very thing it was required to show.

† This proposition is equivalent to Prop. 10.47, with minus signs instead of plus signs.

<Οροι τρίτοι.

11. Ὑποκειμένηξ <ρητῆξ καὶ ἀποτομῆξ, ἔαν μὲν ἡ <όλη τῆξ προσαρμοζούσηξ μείζον δύνηται τῷ ἀπὸ συμέτρου ἐαυτῇ μήκει, καὶ ἡ <όλη σύμμετροξ >ῇ τῇ ἐκκειμένη <ρητῇ μήκει, καλείσθω ἀποτομή πρώτη.

12. Ἐάν δὲ ἡ προσαρμόζουσα σύμμετροξ >ῇ τῇ ἐκκειμένη <ρητῇ μήκει, καὶ ἡ <όλη τῆξ προσαρμοζούσηξ μείζον δύνηται τῷ ἀπὸ συμέτρου ἐαυτῇ, καλείσθω ἀποτομή δευτέρα.

13. Ἐάν δὲ μηδετέρα σύμμετροξ >ῇ τῇ ἐκκειμένη <ρητῇ μήκει, ἡ δὲ <όλη τῆξ προσαρμοζούσηξ μείζον δύνηται τῷ ἀπὸ συμέτρου ἐαυτῇ, καλείσθω ἀποτομή τρίτη.

14. Πάλιν, ἔαν ἡ <όλη τῆξ προσαρμοζούσηξ μείζον δύνηται τῷ ἀπὸ ἀσυμέτρου ἐαυτῇ [μήκει], ἔαν μὲν ἡ <όλη σύμμετροξ >ῇ τῇ ἐκκειμένη <ρητῇ μήκει, καλείσθω ἀποτομή τετάρτη.

15. Ἐάν δὲ ἡ προσαρμόζουσα, πέμπτη.

16. Ἐάν δὲ μηδετέρα, <έκτη.

Definitions III

11. Given a rational (straight-line) and an apotome, if the square on the whole is greater than the (square on a straight-line) attached (to the apotome) by the (square) on (some straight-line) commensurable in length with (the whole), and the whole is commensurable in length with the (previously) laid down rational (straight-line), then let the (apotome) be called a first apotome.

12. And if the attached (straight-line) is commensurable in length with the (previously) laid down rational (straight-line), and the square on the whole is greater than (the square on) the attached (straight-line) by the (square) on (some straight-line) commensurable (in length) with (the whole), then let the (apotome) be called a second apotome.

13. And if neither of (the whole or the attached straight-line) is commensurable in length with the (previously) laid down rational (straight-line), and the square on the whole is greater than (the square on) the attached (straight-line) by the (square) on (some straight-line) commensurable (in length) with (the whole), then let the (apotome) be called a third apotome.

14. Again, if the square on the whole is greater than (the square on) the attached (straight-line) by the (square) on (some straight-line) incommensurable [in length] with (the whole), and the whole is commensurable in length with the (previously) laid down rational (straight-line), then let the (apotome) be called a fourth apotome.

15. And if the attached (straight-line is commensurable), a fifth (apotome).

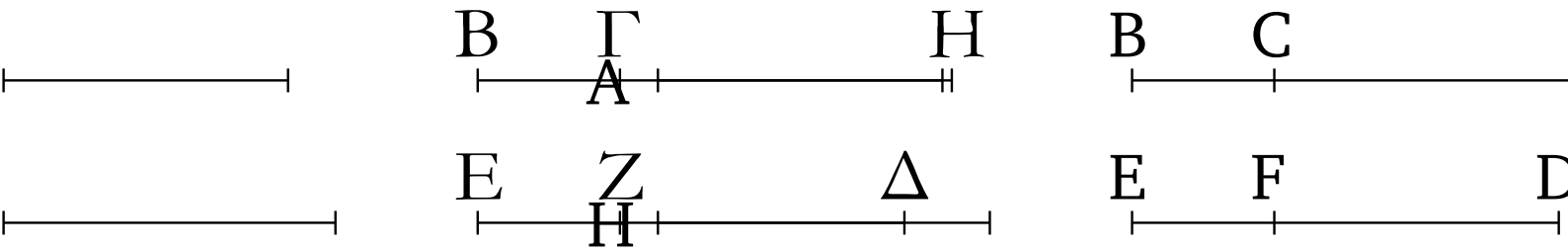
16. And if neither (the whole nor the attached straight-line is commensurable), a sixth (apotome).

85.

Εὐρεῖν τὴν πρώτην ἀποτομήν.

Proposition 85

To find a first apotome.



Ἐκκείσθω <ρητὴ ἢ A , καὶ τῇ A μήκει σύμμετρος >έστω ἢ BH ; <ρητὴ >άρα ἐστὶ καὶ ἡ BH . καὶ ἐκκείσθωσαν δύο τετράγωνοι ἀριθμοὶ οἱ ΔE , EZ , <ὦν ἢ ὑπεροθὴ ὁ $Z\Delta$ μὴ >έστω τετράγωνος; οὐδ' >άρα ὁ $E\Delta$ πρὸς τὸν ΔZ λόγον >έθει, <ὦν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. καὶ πεποιήσθω ὥς ὁ $E\Delta$ πρὸς τὸν ΔZ , ο<ύτωξ τὸ ἀπὸ τῆς BH τετράγωνον πρὸς τὸ ἀπὸ τῆς $H\Gamma$ τετράγωνον; σύμμετρον >άρα ἐστὶ τὸ ἀπὸ τῆς BH τῶ ἀπὸ τῆς $H\Gamma$. <ρητὸν δὲ τὸ ἀπὸ τῆς BH ; <ρητὸν >άρα καὶ τὸ ἀπὸ τῆς $H\Gamma$; <ρητὴ >άρα ἐστὶ καὶ ἡ $H\Gamma$. καὶ ἐπεὶ ὁ $E\Delta$ πρὸς τὸν ΔZ λόγον οὐκ >έθει, <ὦν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, οὐδ' >άρα τὸ ἀπὸ τῆς BH πρὸς τὸ ἀπὸ τῆς $H\Gamma$ λόγον >έθει, <ὦν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν; ἀσύμμετρος >άρα ἐστὶν ἡ BH τῇ $H\Gamma$ μήκει. καὶ εἰσιν ἀμφοτέραι <ρηταί; αἱ BH , $H\Gamma$ >άρα <ρηταί εἰσι δυνάμει μόνον σύμμετροι; ἢ >άρα $B\Gamma$ ἀποτομή ἐστίν. λέγω δὴ, <ότι καὶ πρώτη.

<Ὡς γὰρ μείζον ἐστὶ τὸ ἀπὸ τῆς BH τοῦ ἀπὸ τῆς $H\Gamma$, >έστω τὸ ἀπὸ τῆς Θ . καὶ ἐπεὶ ἐστὶν ὥς ὁ $E\Delta$ πρὸς τὸν $Z\Delta$, ο<ύτωξ τὸ ἀπὸ τῆς BH πρὸς τὸ ἀπὸ τῆς $H\Gamma$, καὶ ἀναστρέψαντι >άρα ἐστὶν ὥς ὁ ΔE πρὸς τὸν EZ , ο<ύτωξ τὸ ἀπὸ τῆς $H\Gamma$ πρὸς τὸ ἀπὸ τῆς Θ . ὁ δὲ ΔE πρὸς τὸν EZ λόγον >έθει, <ὦν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν; ἑκάτερος γὰρ τετράγωνός ἐστιν; καὶ τὸ ἀπὸ τῆς $H\Gamma$ >άρα πρὸς τὸ ἀπὸ τῆς Θ λόγον >έθει, <ὦν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν; σύμμετρος >άρα ἐστὶν ἡ BH τῇ Θ μήκει. καὶ δύναται ἡ BH τῆς $H\Gamma$ μείζον τῶ ἀπὸ τῆς Θ ; ἢ BH >άρα τῆς $H\Gamma$ μείζον δύναται τῶ ἀπὸ συμμέτρον ἑαυτῇ μήκει. καὶ ἐστὶν ἡ <όλη ἢ BH σύμμετρος τῇ ἐκκειμένῃ <ρητῇ μήκει τῇ A . ἢ $B\Gamma$ >άρα ἀποτομή ἐστὶ πρώτη. E -ύρηται >άρα ἡ πρώτη ἀποτομή ἢ $B\Gamma$; <όπερ >έδει εὐρεῖν.

Let the rational (straight-line) A be laid down. And let BG be commensurable in length with A . BG is thus also a rational (straight-line). And let two square numbers DE and EF be laid down, and let their difference FD be not square [Prop. 10.28 lem. I]. Thus, ED does not have to DF the ratio which (some) square number (has) to (some) square number. And let it be contrived that as ED (is) to DF , so the square on BG (is) to the square on GC [Prop. 10.6. corr.]. Thus, the (square) on BG is commensurable with the (square) on GC [Prop. 10.6]. And the (square) on BG (is) rational. Thus, the (square) on GC (is) also rational. Thus, GC is also rational. And since ED does not have to DF the ratio which (some) square number (has) to (some) square number, the (square) on BG thus does not have to the (square) on GC the ratio which (some) square number (has) to (some) square number either. Thus, BG is incommensurable in length with GC [Prop. 10.9]. And they are both rational (straight-lines). Thus, BG and GC are rational (straight-lines which are) commensurable in square only. Thus, BC is an apotome [Prop. 10.73]. So, I say that (it is) also a first (apotome).

Let the (square) on H be that (area) by which the (square) on BG is greater than the (square) on GC [Prop. 10.13 lem.]. And since as ED is to FD , so the (square) on BG (is) to the (square) on GC , thus, via conversion, as DE is to EF , so the (square) on GB (is) to the (square) on H [Prop. 5.19 corr.]. And DE has to EF the ratio which (some) square-number (has) to (some) square-number. For each is a square (number). Thus, the (square) on GB also has to the (square) on H the ratio which (some) square number (has) to (some) square number. Thus, BG is commensurable in length with H [Prop. 10.9]. And the square on BG is greater than (the square on) GC by the (square) on H . Thus, the square on BG is greater than (the square on) GC by the (square) on (some straight-line) commensurable in length with (BG). And the whole, BG , is commensurable in length with the (previously) laid down rational (straight-line) A . Thus, BC is a first apotome [Def. 10.11].

Thus, the first apotome BC has been found. (Which is) the very thing it was required to find.

† See footnote to Prop. 10.48.

86.

Εὐρεῖν τὴν δευτέραν ἀποτομήν.

Ἐκκείσθω <ρητὴ ἢ A καὶ τῇ A σύμμετρος μήκει ἢ $H\Gamma$. <ρητὴ >άρα ἐστὶν ἡ $H\Gamma$. καὶ ἐκκείσθωσαν δύο τετράγωνοι ἀριθμοὶ οἱ ΔE , EZ , <ὦν ἢ ὑπεροθὴ ὁ ΔZ μὴ >έστω τετράγωνος. καὶ πεποιήσθω ὥς ὁ $Z\Delta$ πρὸς τὸν ΔE , ο<ύτωξ τὸ ἀπὸ τῆς ΓH τετράγωνον πρὸς τὸ ἀπὸ τῆς $H\Gamma$ τετράγωνον. σύμμετρον >άρα ἐστὶ τὸ ἀπὸ τῆς ΓH τετράγωνον τῶ ἀπὸ τῆς $H\Gamma$ τετράγωνον. <ρητὸν δὲ τὸ ἀπὸ τῆς ΓH . <ρητὸν >άρα [ἐστὶ] καὶ τὸ ἀπὸ τῆς $H\Gamma$; <ρητὴ >άρα ἐστὶν ἡ BH .

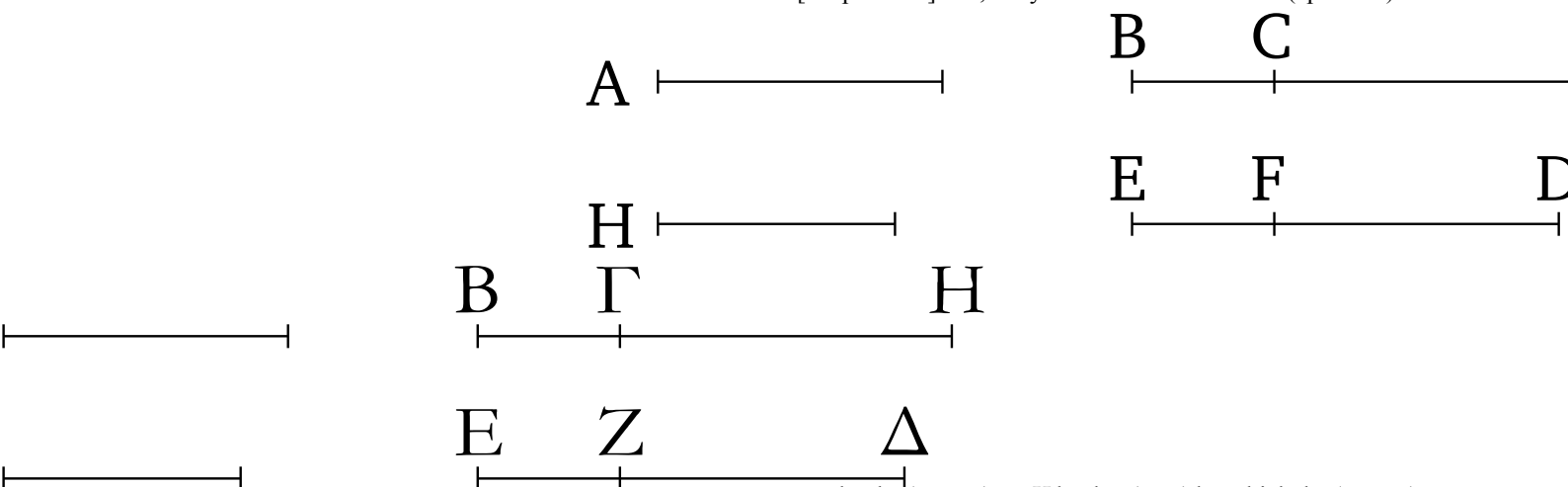
Proposition 86

To find a second apotome.

Let the rational (straight-line) A , and GC (which is) commensurable in length with A , be laid down. Thus, GC is a rational (straight-line). And let the two square numbers DE and EF be laid down, and let their difference DF be not square [Prop. 10.28 lem. I]. And let it be contrived that as FD (is) to DE , so the square on CG (is) to the square on GB [Prop. 10.6 corr.]. Thus, the square on CG is commensurable with the square on GB [Prop. 10.6]. And the

καὶ ἐπεὶ τὸ ἀπὸ τῆς ΗΓ τετραγώνων πρὸς τὸ ἀπὸ τῆς ΗΒ λόγον οὐκ >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, ἀσύμμετρος ἐστὶν ἡ ΓΗ τῇ ΗΒ μήκει. καὶ εἰσιν ἀμφοτέραι <ρηταί· αἱ ΓΗ, ΗΒ >άρα ρηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἡ ΒΓ >άρα ἀποτομή ἐστίν. λέγω δὴ, <ὅτι καὶ δευτέρα.

(square) on CG (is) rational. Thus, the (square) on GB [is] also rational. Thus, BG is a rational (straight-line). And since the square on GC does not have to the (square) on GB the ratio which (some) square number (has) to (some) square number, CG is incommensurable in length with GB [Prop. 10.9]. And they are both rational (straight-lines). Thus, CG and GB are rational (straight-lines which are) commensurable in square only. Thus, BC is an apotome [Prop. 10.73]. So, I say that it is also a second (apotome).



< Ωι γάρ μειζόν ἐστι τὸ ἀπὸ τῆς ΒΗ τοῦ ἀπὸ τῆς ΗΓ, > ἐστὼ
τὸ ἀπὸ τῆς Θ. ἐπεὶ ο>ὺν ἐστὶν ὥς τὸ ἀπὸ τῆς ΒΗ πρὸς τὸ
ἀπὸ τῆς ΗΓ, ο<ύτως ὁ ΕΔ ἀριθμὸς πρὸς τὸν ΔΖ ἀριθμὸν,
ἀναστρέψαντι >άρα ἐστὶν ὥς τὸ ἀπὸ τῆς ΒΗ πρὸς τὸ ἀπὸ
τῆς Θ, ο<ύτως ὁ ΔΕ πρὸς τὸν ΕΖ. καὶ ἐστὶν ἐκάτερος τῶν
ΔΕ, ΕΖ τετράγωνος; τὸ >άρα ἀπὸ τῆς ΒΗ πρὸς τὸ ἀπὸ τῆς
Θ λόγον >έθει, <ὸν τετράγωνος ἀριθμὸς πρὸς τετράγωνον
ἀριθμόν; σύμμετρος >άρα ἐστὶν ἡ ΒΗ τῇ Θ μήκει. καὶ
δύναται ἡ ΒΗ τῆς ΗΓ μειζον τῷ ἀπὸ τῆς Θ; ἡ ΒΗ >άρα
τῆς ΗΓ μειζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ μήκει.
καὶ ἐστὶν ἡ προσαρμόζουσα ἡ ΓΗ τῇ ἐκκειμένῃ <ρητῇ
σύμμετρος τῇ Α. ἡ ΒΓ >άρα ἀποτομή ἐστὶ δευτέρα.

Ε<ύρηται>άρα δευτέρα αποτομή ή ΒΓ; <όπερ>έδει δεῖχαι.

For let the (square) on H be that (area) by which the (square) on BG is greater than the (square) on GC [Prop. 10.1 lem.]. Therefore, since as the (square) on BG is to the (square) on GC , so the number ED (is) to the number DF , thus, also, via conversion, as the (square) on BG is to the (square) on H , so DE (is) to EF [Prop. 5.19 corr.]. And DE and EF are each square (numbers). Thus, the (square) on BG has to the (square) on H the ratio which (some) square number (has) to (some) square number. Thus, BG is commensurable in length with H [Prop. 10.9]. And the square on BG is greater than (the square on) GC by the (square) on H . Thus, the square on BG is greater than (the square on) GC by the (square) on (some straight-line) commensurable in length with (BG). And the attachment CG is commensurable (in length) with the (previously) laid down rational (straight-line) A . Thus, BC is a second apotome [Def. 10.12].[†]

Thus, the second apotome BC has been found. (Which is) the very thing it was required to show.

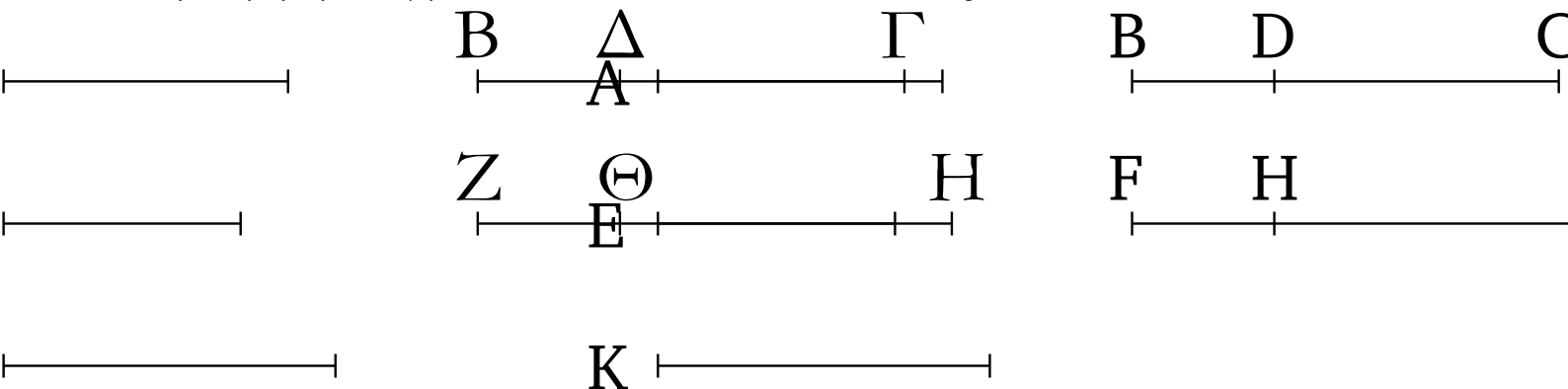
[†] See footnote to Prop. 10.49.

87.

Proposition 87

Εὐρεῖν τὴν τρίτην ἀποτομήν.

To find a third apotome.



Ἐκκεῖσθω <ρητὴ ἢ A , καὶ ἐκκεῖσθωσαν τρεῖς ἀριθμοὶ οἱ E , $B\Gamma$, $\Gamma\Delta$ λόγον μὴ >έθοντεξ πρὸς ἀλλήλουξ, <ὄν τετράγωνοξ ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν, ὁ δὲ ΓB πρὸς τὸν $B\Delta$ λόγον ἐθέτω, <ὄν τετράγωνοξ ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν, καὶ πεποιήσθω ὡς μὲν ὁ E πρὸς τὸν $B\Gamma$, ο<ύτωξ τὸ ἀπὸ τῆξ A τετράγωνον πρὸς τὸ ἀπὸ τῆξ ZH τετράγωνον, ὡς δὲ ὁ $B\Gamma$ πρὸς τὸν $\Gamma\Delta$, ο<ύτωξ τὸ ἀπὸ τῆξ ZH τετράγωνον πρὸς τὸ ἀπὸ τῆξ $H\Theta$. ἐπεὶ ο>ὖν ἐστὶν ὡς ὁ E πρὸς τὸν $B\Gamma$, ο<ύτωξ τὸ ἀπὸ τῆξ A τετράγωνον πρὸς τὸ ἀπὸ τῆξ ZH τετράγωνον, σύμμετρον >άρα ἐστὶ τὸ ἀπὸ τῆξ A τετράγωνον τῷ ἀπὸ τῆξ ZH τετραγώνῳ. <ρητὸν δὲ τὸ ἀπὸ τῆξ A τετράγωνον. <ρητὸν >άρα καὶ τὸ ἀπὸ τῆξ ZH ; <ρητὴ >άρα ἐστὶν ἡ ZH . καὶ ἐπεὶ ὁ E πρὸς τὸν $B\Gamma$ λόγον οὐκ >έθει, <ὄν τετράγωνοξ ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν, οὐδ> >άρα τὸ ἀπὸ τῆξ A τετράγωνον πρὸς τὸ ἀπὸ τῆξ ZH [τετράγωνον] λόγον >έθει, <ὄν τετράγωνοξ ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν; ἀσύμμετροξ >άρα ἐστὶν ἡ A τῇ ZH μήκει. πάλιν, ἐπεὶ ἐστὶν ὡς ὁ $B\Gamma$ πρὸς τὸν $\Gamma\Delta$, ο<ύτωξ τὸ ἀπὸ τῆξ ZH τετράγωνον πρὸς τὸ ἀπὸ τῆξ $H\Theta$, σύμμετρον >άρα ἐστὶ τὸ ἀπὸ τῆξ ZH τῷ ἀπὸ τῆξ $H\Theta$. <ρητὸν δὲ τὸ ἀπὸ τῆξ ZH ; <ρητὸν >άρα καὶ τὸ ἀπὸ τῆξ $H\Theta$; <ρητὴ >άρα ἐστὶν ἡ $H\Theta$. καὶ ἐπεὶ ὁ $B\Gamma$ πρὸς τὸν $\Gamma\Delta$ λόγον οὐκ >έθει, <ὄν τετράγωνοξ ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν, οὐδ> >άρα τὸ ἀπὸ τῆξ ZH πρὸς τὸ ἀπὸ τῆξ $H\Theta$ λόγον >έθει, <ὄν τετράγωνοξ ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν; ἀσύμμετροξ >άρα ἐστὶν ἡ ZH τῇ $H\Theta$ μήκει. καὶ εἰσιν ἀμφοτέραι <ρηταί; αἱ ZH , $H\Theta$ >άρα <ρηταί εἰσι δυνάμει μόνον σύμμετροι; ἀποτομὴ >άρα ἐστὶν ἡ $Z\Theta$. λέγω δὴ, <ότι καὶ τρίτη.

Ἐπεὶ γάρ ἐστὶν ὡς μὲν ὁ E πρὸς τὸν $B\Gamma$, ο<ύτωξ τὸ ἀπὸ τῆξ A τετράγωνον πρὸς τὸ ἀπὸ τῆξ ZH , ὡς δὲ ὁ $B\Gamma$ πρὸς τὸν $\Gamma\Delta$, ο<ύτωξ τὸ ἀπὸ τῆξ ZH πρὸς τὸ ἀπὸ τῆξ ΘH , δι> >ίσου >άρα ἐστὶν ὡς ὁ E πρὸς τὸν $\Gamma\Delta$, ο<ύτωξ τὸ ἀπὸ τῆξ A πρὸς τὸ ἀπὸ τῆξ ΘH . ὁ δὲ E πρὸς τὸν $\Gamma\Delta$ λόγον οὐκ >έθει, <ὄν τετράγωνοξ ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν; οὐδ> >άρα τὸ ἀπὸ τῆξ A πρὸς τὸ ἀπὸ τῆξ $H\Theta$ λόγον >έθει, <ὄν τετράγωνοξ ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν; ἀσύμμετροξ >άρα ἢ A τῇ $H\Theta$ μήκει. οὐδετέρα >άρα τῶν ZH , $H\Theta$ σύμμετρόξ ἐστὶ τῇ ἐκκειμένῃ <ρητῇ τῇ A μήκει. <ῶ ο>ὖν μεῖζόν ἐστὶ τὸ ἀπὸ τῆξ ZH τοῦ ἀπὸ τῆξ $H\Theta$, >έστω τὸ ἀπὸ τῆξ K . ἐπεὶ ο>ὖν ἐστὶν ὡς ὁ $B\Gamma$ πρὸς τὸν $\Gamma\Delta$, ο<ύτωξ τὸ ἀπὸ τῆξ ZH πρὸς τὸ ἀπὸ τῆξ $H\Theta$, ἀναστρέψαντι >άρα ἐστὶν ὡς ὁ $B\Gamma$ πρὸς τὸν $B\Delta$, ο<ύτωξ τὸ ἀπὸ τῆξ ZH τετράγωνον πρὸς τὸ ἀπὸ τῆξ K . ὁ δὲ $B\Gamma$ πρὸς τὸν $B\Delta$ λόγον >έθει, <ὄν τετράγωνοξ ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν. καὶ τὸ ἀπὸ τῆξ ZH >άρα πρὸς τὸ ἀπὸ τῆξ K λόγον >έθει, <ὄν τετράγωνοξ ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν. σύμμετρόξ >άρα ἐστὶν ἡ ZH τῇ K μήκει, καὶ δύναται ἡ ZH τῆξ $H\Theta$ μεῖζον τῷ ἀπὸ συμέτρου ἐαυτῇ. καὶ οὐδετέρα τῶν ZH , $H\Theta$ σύμμετρόξ ἐστὶ τῇ ἐκκειμένῃ <ρητῇ τῇ A μήκει; ἡ $Z\Theta$ >άρα ἀποτομὴ ἐστὶ τρίτη.

E <ύρηται >άρα ἢ τρίτῃ ἀποτομῇ ἡ $Z\Theta$; <όπερ >έδει δεῖχαι.

Let the rational (straight-line) A be laid down. And let the three numbers, E , BC , and CD , not having to one another the ratio which (some) square number (has) to (some) square number, be laid down. And let CB have to BD the ratio which (some) square number (has) to (some) square number. And let it be contrived that as E (is) to BC , so the square on A (is) to the square on FG , and as BC (is) to CD , so the square on FG (is) to the (square) on GH [Prop. 10.6 corr.]. Therefore, since as E is to BC , so the square on A (is) to the square on FG , the square on A is thus commensurable with the square on FG [Prop. 10.6]. And the square on A (is) rational. Thus, the (square) on FG (is) also rational. Thus, FG is a rational (straight-line). And since E does not have to BC the ratio which (some) square number (has) to (some) square number, the square on A thus does not have to the [square] on FG the ratio which (some) square number (has) to (some) square number either. Thus, A is incommensurable in length with FG [Prop. 10.9]. Again, since as BC is to CD , so the square on FG is to the (square) on GH , the square on FG is thus commensurable with the (square) on GH [Prop. 10.6]. And the (square) on FG (is) rational. Thus, the (square) on GH (is) also rational. Thus, GH is a rational (straight-line). And since BC does not have to CD the ratio which (some) square number (has) to (some) square number, the (square) on FG thus does not have to the (square) on GH the ratio which (some) square number (has) to (some) square number either. Thus, FG is incommensurable in length with GH [Prop. 10.9]. And both are rational (straight-lines). FG and GH are thus rational (straight-lines which are) commensurable in square only. Thus, FH is an apotome [Prop. 10.73]. So, I say that (it is) also a third (apotome).

For since as E is to BC , so the square on A (is) to the (square) on FG , and as BC (is) to CD , so the (square) on FG (is) to the (square) on HG , thus, via equality, as E is to CD , so the (square) on A (is) to the (square) on HG [Prop. 5.22]. And E does not have to CD the ratio which (some) square number (has) to (some) square number. Thus, the (square) on A does not have to the (square) on GH the ratio which (some) square number (has) to (some) square number either. A (is) thus incommensurable in length with GH [Prop. 10.9]. Thus, neither of FG and GH is commensurable in length with the (previously) laid down rational (straight-line) A . Therefore, let the (square) on K be that (area) by which the (square) on FG is greater than the (square) on GH [Prop. 10.13 lem.]. Therefore, since as BC is to CD , so the (square) on FG (is) to the (square) on GH , thus, via conversion, as BC is to BD , so the square on FG (is) to the square on K [Prop. 5.19 corr.]. And BC has to BD the ratio which (some) square number (has) to (some) square number. Thus, the (square) on FG also has to the (square) on K the ratio which (some) square number (has) to (some) square number. FG is thus commensurable in length with K [Prop. 10.9]. And the square on FG is (thus) greater than (the square on) GH by the (square) on (some straight-line) commensurable (in length) with (FG). And neither of FG and GH is commensurable in length with the (previously) laid down rational (straight-line) A . Thus, FH is a third apotome [Def. 10.13].

Thus, the third apotome FH has been found. (Which is) very thing it was required to show.

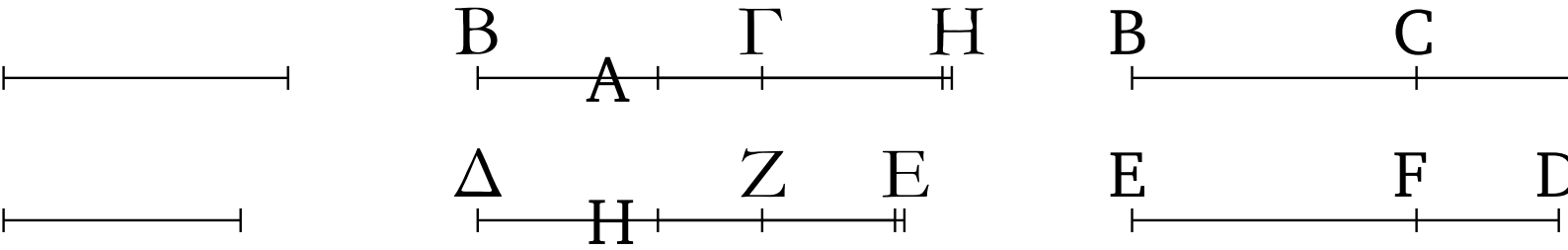
† See footnote to Prop. 10.50.

88.

Proposition 88

Εὐρεῖν τὴν τετάρτην ἀποτομήν.

To find a fourth apotome.



Ἐκκείσθω <ρητὴ ἡ A καὶ τῇ A μήκει σύμμετρος ἡ BH ; <ρητὴ >άρα ἐστὶ καὶ ἡ BH . καὶ ἐκκείσθωσαν δύο ἀριθμοὶ οἱ ΔZ , ZE , <ώστε τὸν ΔE <όλον πρὸς ἑκάτερον τῶν ΔZ , EZ λόγον μὴ >έθειν, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. καὶ πεποιήσθω ὥς ὁ ΔE πρὸς τὸν EZ , ο<ύτως τὸ ἀπὸ τῆς BH τετράγωνον πρὸς τὸ ἀπὸ τῆς $HΓ$; σύμμετρον >άρα ἐστὶ τὸ ἀπὸ τῆς BH τῷ ἀπὸ τῆς $HΓ$. <ρητὸν δὲ τὸ ἀπὸ τῆς BH ; <ρητὸν >άρα καὶ τὸ ἀπὸ τῆς $HΓ$; <ρητὴ >άρα ἐστὶν ἡ $HΓ$. καὶ ἐπεὶ ὁ ΔE πρὸς τὸν EZ λόγον οὐκ >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, οὐδ> >άρα τὸ ἀπὸ τῆς BH πρὸς τὸ ἀπὸ τῆς $HΓ$ λόγον >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν; ἀσύμμετρος >άρα ἐστὶν ἡ BH τῇ $HΓ$ μήκει. καὶ εἰσιν ἀμφοτέραι <ρηταί; αἱ BH , $HΓ$ >άρα <ρηταί εἰσι δυνάμει μόνον σύμμετροι; ἀποτομή >άρα ἐστὶν ἡ $BΓ$. [λέγω δὴ, <ὅτι καὶ τετάρτη.]

< Ωι ο>ὖν μείζον ἐστὶ τὸ ἀπὸ τῆς BH τοῦ ἀπὸ τῆς $HΓ$, >έστω τὸ ἀπὸ τῆς Θ . ἐπεὶ ο>ὖν ἐστὶν ὥς ὁ ΔE πρὸς τὸν EZ , ο<ύτως τὸ ἀπὸ τῆς BH πρὸς τὸ ἀπὸ τῆς $HΓ$, καὶ ἀναστρέψαντι >άρα ἐστὶν ὥς ὁ $E\Delta$ πρὸς τὸν ΔZ , ο<ύτως τὸ ἀπὸ τῆς $H\Gamma$ πρὸς τὸ ἀπὸ τῆς Θ . ὁ δὲ $E\Delta$ πρὸς τὸν ΔZ λόγον οὐκ >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν; οὐδ> >άρα τὸ ἀπὸ τῆς $H\Gamma$ πρὸς τὸ ἀπὸ τῆς Θ λόγον >έθει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν; ἀσύμμετρος >άρα ἐστὶν ἡ BH τῇ Θ μήκει. καὶ δύναται ἡ BH τῆς $HΓ$ μείζον τῷ ἀπὸ τῆς Θ ; ἡ >άρα BH τῆς $HΓ$ μείζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ. καὶ ἐστὶν <ὅλη ἡ BH σύμμετρος τῇ ἐκκειμένῃ <ρητῇ μήκει τῇ A . ἡ >άρα $BΓ$ ἀποτομή ἐστὶ τετάρτη.

Ε<ὐρηται >άρα ἡ τετάρτη ἀποτομή; <όπερ >έδει δεῖξαι.

Let the rational (straight-line) A , and BG (which is) commensurable in length with A , be laid down. Thus, BG is also a rational (straight-line). And let the two numbers DF and FE be laid down such that the whole, DE , does not have to each of DF and EF the ratio which (some) square number (has) to (some) square number. And let it be contrived that as DE (is) to EF , so the square on BG (is) to the (square) on GC [Prop. 10.6 corr.]. The (square) on BG is thus commensurable with the (square) on GC [Prop. 10.6]. And the (square) on BG (is) rational. Thus, the (square) on GC (is) also rational. Thus, GC (is) a rational (straight-line). And since DE does not have to EF the ratio which (some) square number (has) to (some) square number, the (square) on BG thus does not have to the (square) on GC the ratio which (some) square number (has) to (some) square number either. Thus, BG is incommensurable in length with GC [Prop. 10.9]. And they are both rational (straight-lines). Thus, BG and GC are rational (straight-lines which are) commensurable in square only. Thus, BC is an apotome [Prop. 10.73]. [So, I say that (it is) also a fourth (apotome).]

Now, let the (square) on H be that (area) by which the (square) on BG is greater than the (square) on GC [Prop. 10.13 lem.].

Therefore, since as DE is to EF , so the (square) on BG (is) to the (square) on GC , thus, also, via conversion, as ED is to DF , so the (square) on GB (is) to the (square) on H [Prop. 5.19 corr.]. And ED does not have to DF the ratio which (some) square number (has) to (some) square number. Thus, the (square) on GB does not have to the (square) on H the ratio which (some) square number (has) to (some) square number either. Thus, BG is incommensurable in length with H [Prop. 10.9]. And the square on BG is greater than (the square on) GC by the (square) on H . Thus, the square on BG is greater than (the square) on GC by the (square) on (some straight-line) incommensurable (in length) with (BG). And the whole, BG , is commensurable in length with the (previously) laid down rational (straight-line) A . Thus, BC is a fourth apotome [Def. 10.14].†

Thus, a fourth apotome has been found. (Which is) the very thing it was required to show.

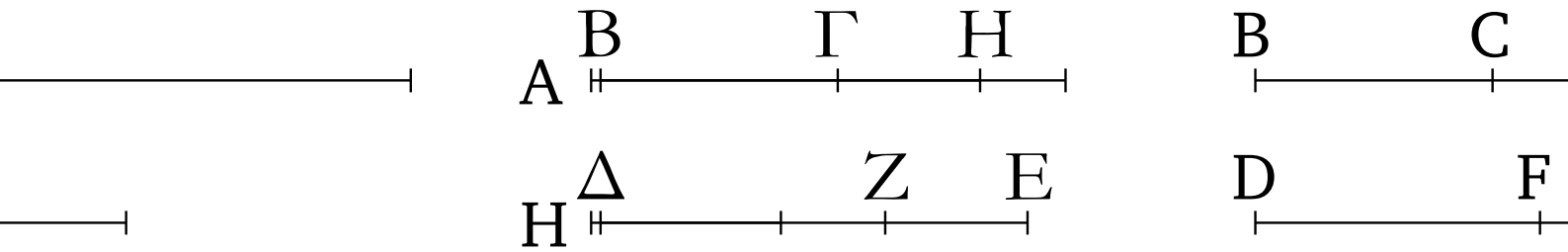
† See footnote to Prop. 10.51.

89.

Proposition 89

Εὐρεῖν τὴν πέμπτην ἀποτομήν.

To find a fifth apotome.



Ἐκκείσθω <ρητὴ ἡ A , καὶ τῇ A μήκει σύμμετρος >έστω ἡ ΓH ; <ρητὴ >άρα [έστιν] ἡ ΓH . καὶ ἐκκείσθωσαν δύο ἀριθμοὶ οἱ ΔZ , ZE , <ώστε τὸν ΔE πρὸς ἐκάτερον τῶν ΔZ , ZE λόγον πάλιν μὴ >έθειν, <ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν; καὶ πεποιήσθω ὥς ὁ ZE πρὸς τὸν $\text{E}\Delta$, ο<ύτωξ τὸ ἀπὸ τῆς ΓH πρὸς τὸ ἀπὸ τῆς HB . <ρητὸν >άρα καὶ τὸ ἀπὸ τῆς HB ; <ρητὴ >άρα ἐστὶ καὶ ἡ BH . καὶ ἐπεὶ ἐστὶν ὥς ὁ ΔE πρὸς τὸν EZ , ο<ύτωξ τὸ ἀπὸ τῆς BH πρὸς τὸ ἀπὸ τῆς $\text{H}\Gamma$, ὁ δὲ ΔE πρὸς τὸν EZ λόγον οὐκ >έθει, <ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν, οὐδ> >άρα τὸ ἀπὸ τῆς BH πρὸς τὸ ἀπὸ τῆς $\text{H}\Gamma$ λόγον >έθει, <ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν; ἀσύμμετρος >άρα ἐστὶν ἡ BH τῇ $\text{H}\Gamma$ μήκει. καὶ εἰσὶν ἀμφοτέραι <ρηταί; αἱ BH , $\text{H}\Gamma$ >άρα <ρηταί εἰσι δυνάμει μόνον σύμμετροι; ἡ $\text{B}\Gamma$ >άρα ἀποτομή ἐστίν. λέγω δὴ, <ὅτι καὶ πέμπτη.

<Ὡς γὰρ μείζον ἐστὶ τὸ ἀπὸ τῆς BH τοῦ ἀπὸ τῆς $\text{H}\Gamma$, >έστω τὸ ἀπὸ τῆς Θ . ἐπεὶ ο>ύν ἐστὶν ὥς τὸ ἀπὸ τῆς BH πρὸς τὸ ἀπὸ τῆς $\text{H}\Gamma$, ο<ύτωξ ὁ ΔE πρὸς τὸν EZ , ἀναστρέψαντι >άρα ἐστὶν ὥς ὁ $\text{E}\Delta$ πρὸς τὸν ΔZ , ο<ύτωξ τὸ ἀπὸ τῆς BH πρὸς τὸ ἀπὸ τῆς Θ , ὁ δὲ $\text{E}\Delta$ πρὸς τὸν ΔZ λόγον οὐκ >έθει, <ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν; οὐδ> >άρα τὸ ἀπὸ τῆς BH πρὸς τὸ ἀπὸ τῆς Θ λόγον >έθει, <ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν; ἀσύμμετρος >άρα ἐστὶν ἡ BH τῇ Θ μήκει. καὶ δύναται ἡ BH τῆς $\text{H}\Gamma$ μείζον τῷ ἀπὸ τῆς Θ ; ἡ HB >άρα τῆς $\text{H}\Gamma$ μείζον δύναται τῷ ἀπὸ ἀσύμμετρου ἐαντὶ| μήκει. καὶ ἐστὶν ἡ προσαρμόζουσα ἡ ΓH σύμμετρος τῇ ἐκκειμένῃ <ρητῇ τῇ A μήκει; ἡ >άρα $\text{B}\Gamma$ ἀποτομή ἐστὶ πέμπτη.

E <ύρηται >άρα ἡ πέμπτη ἀποτομή ἡ $\text{B}\Gamma$; <όπερ >έδει δείχαι.

Let the rational (straight-line) A be laid down, and let CG be commensurable in length with A . Thus, CG [is] a rational (straight-line). And let the two numbers DF and FE be laid down such that DE again does not have to each of DF and FE the ratio which (some) square number (has) to (some) square number. And let it be contrived that as FE (is) to ED , so the (square) on CG (is) to the (square) on GB . Thus, the (square) on GB (is) also rational [Prop. 10.6]. Thus, BG is also rational. And since as DE is to EF , so the (square) on BG (is) to the (square) on GC . And DE does not have to EF the ratio which (some) square number (has) to (some) square number. The (square) on BG thus does not have to the (square) on GC the ratio which (some) square number (has) to (some) square number either. Thus, BG is incommensurable in length with GC [Prop. 10.9]. And they are both rational (straight-lines). BG and GC are thus rational (straight-lines which are) commensurable in square only. Thus, BC is an apotome [Prop. 10.73]. So, I say that (it is) also a fifth (apotome).

For, let the (square) on H be that (area) by which the (square) on BG is greater than the (square) on GC [Prop. 10.13 lem.].

Therefore, since as the (square) on BG (is) to the (square) on GC , so DE (is) to EF , thus, via conversion, as ED is to DF , so the (square) on BG (is) to the (square) on H [Prop. 5.19 corr.]. And ED does not have to DF the ratio which (some) square number (has) to (some) square number. Thus, the (square) on BG does not have to the (square) on H the ratio which (some) square number (has) to (some) square number either. Thus, BG is incommensurable in length with H [Prop. 10.9]. And the square on BG is greater than (the square on) GC by the (square) on H . Thus, the square on GB is greater than (the square on) GC by the (square) on (some straight-line) incommensurable in length with (GB) . And the attachment CG is commensurable in length with the (previously) laid down rational (straight-line) A . Thus, BC is a fifth apotome [Def. 10.15].[†]

Thus, the fifth apotome BC has been found. (Which is) the very thing it was required to show.

[†] See footnote to Prop. 10.52.

90.

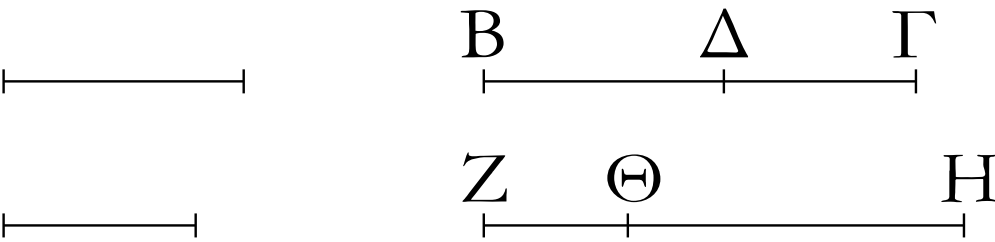
Εὐρεῖν τὴν <έκτην ἀποτομήν.

Ἐκκείσθω <ρητὴ ἡ A καὶ τρεῖς ἀριθμοὶ οἱ E , $\text{B}\Gamma$, $\Gamma\Delta$ λόγον μὴ >έθοντεξ πρὸς ἀλλήλους, <ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν; >έτι δὲ καὶ ὁ ΓB πρὸς τὸν $\text{B}\Delta$ λόγον μὴ έθετώ, <ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν; καὶ πεποιήσθω ὥς μὲν ὁ E πρὸς τὸν $\text{B}\Gamma$, ο<ύτωξ τὸ ἀπὸ τῆς A πρὸς τὸ ἀπὸ τῆς ZH , ὥς δὲ ὁ $\text{B}\Gamma$ πρὸς τὸν $\Gamma\Delta$, ο<ύτωξ τὸ ἀπὸ τῆς ZH πρὸς τὸ ἀπὸ τῆς $\text{H}\Theta$.

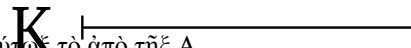
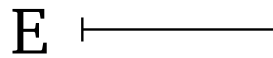
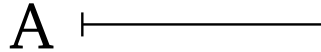
Proposition 90

To find a sixth apotome.

Let the rational (straight-line) A , and the three numbers E , BC , and CD , not having to one another the ratio which (some) square number (has) to (some) square number, be laid down. Furthermore, let CB also not have to BD the ratio which (some) square number (has) to (some) square number. And let it be contrived that as E (is) to BC , so the (square) on A (is) to the (square) on FG , and as BC (is) to CD , so the (square) on FG (is) to the (square) on GH .



[Prop. 10.6 corr.].



Ἐπεὶ οὖν ἐστὶν ὥς ὁ E πρὸς τὸν $B\Gamma$, οὕτως τὸ ἀπὸ τῆς A πρὸς τὸ ἀπὸ τῆς $Z\eta$, σύμμετρον ἄρα τὸ ἀπὸ τῆς A τῷ ἀπὸ τῆς $Z\eta$. <ρητὸν δὲ τὸ ἀπὸ τῆς A ; <ρητὸν ἄρα καὶ τὸ ἀπὸ τῆς $Z\eta$; <ρητὴ ἄρα ἐστὶ καὶ ἡ $Z\eta$. καὶ ἐπεὶ ὁ E πρὸς τὸν $B\Gamma$ λόγον οὐκ ἔχει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, οὐδ> ἄρα τὸ ἀπὸ τῆς A πρὸς τὸ ἀπὸ τῆς $Z\eta$ λόγον ἔχει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν; ἀσύμμετρος ἄρα ἐστὶν ἡ A τῇ $Z\eta$ μήκει. πάλιν, ἐπεὶ ἐστὶν ὥς ὁ $B\Gamma$ πρὸς τὸν $\Gamma\Delta$, οὕτως τὸ ἀπὸ τῆς $Z\eta$ πρὸς τὸ ἀπὸ τῆς $\Theta\theta$, σύμμετρον ἄρα τὸ ἀπὸ τῆς $Z\eta$ τῷ ἀπὸ τῆς $\Theta\theta$. <ρητὸν δὲ τὸ ἀπὸ τῆς $Z\eta$; <ρητὸν ἄρα καὶ τὸ ἀπὸ τῆς $\Theta\theta$; <ρητὴ ἄρα καὶ ἡ $\Theta\theta$. καὶ ἐπεὶ ὁ $B\Gamma$ πρὸς τὸν $\Gamma\Delta$ λόγον οὐκ ἔχει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, οὐδ> ἄρα τὸ ἀπὸ τῆς $Z\eta$ πρὸς τὸ ἀπὸ τῆς $\Theta\theta$ λόγον ἔχει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν; ἀσύμμετρος ἄρα ἐστὶν ἡ $Z\eta$ τῇ $\Theta\theta$ μήκει. καὶ εἰσιν ἀμφοτέραι <ρηταί; αἱ $Z\eta$, $\Theta\theta$ ἄρα <ρηταί εἰσι δυνάμει μόνον σύμμετροι; ἡ ἄρα $Z\theta$ ἀποτομή ἐστίν. λέγω δὴ, <ὅτι καὶ <έκτη.

Ἐπεὶ γάρ ἐστὶν ὥς μὲν ὁ E πρὸς τὸν $B\Gamma$, οὕτως τὸ ἀπὸ τῆς A πρὸς τὸ ἀπὸ τῆς $Z\eta$, ὥς δὲ ὁ $B\Gamma$ πρὸς τὸν $\Gamma\Delta$, οὕτως τὸ ἀπὸ τῆς $Z\eta$ πρὸς τὸ ἀπὸ τῆς $\Theta\theta$, δι> ἴσου ἄρα ἐστὶν ὥς ὁ E πρὸς τὸν $\Gamma\Delta$, οὕτως τὸ ἀπὸ τῆς A πρὸς τὸ ἀπὸ τῆς $\Theta\theta$. ὁ δὲ E πρὸς τὸν $\Gamma\Delta$ λόγον οὐκ ἔχει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν; οὐδ> ἄρα τὸ ἀπὸ τῆς A πρὸς τὸ ἀπὸ τῆς $\Theta\theta$ λόγον ἔχει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν; ἀσύμμετρος ἄρα ἐστὶν ἡ A τῇ $\Theta\theta$ μήκει; οὐδετέρα ἄρα τῶν $Z\eta$, $\Theta\theta$ σύμμετρος ἐστὶ τῇ A <ρητῇ μήκει. <ὦ! οὖν μείζον ἐστὶ τὸ ἀπὸ τῆς $Z\eta$ τοῦ ἀπὸ τῆς $\Theta\theta$, ἔστω τὸ ἀπὸ τῆς K . ἐπεὶ οὖν ἐστὶν ὥς ὁ $B\Gamma$ πρὸς τὸν $\Gamma\Delta$, οὕτως τὸ ἀπὸ τῆς $Z\eta$ πρὸς τὸ ἀπὸ τῆς $\Theta\theta$, ἀναστρέψαντι ἄρα ἐστὶν ὥς ὁ ΓB πρὸς τὸν $B\Delta$, οὕτως τὸ ἀπὸ τῆς $Z\eta$ πρὸς τὸ ἀπὸ τῆς K . ὁ δὲ ΓB πρὸς τὸν $B\Delta$ λόγον οὐκ ἔχει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν; οὐδ> ἄρα τὸ ἀπὸ τῆς $Z\eta$ πρὸς τὸ ἀπὸ τῆς K λόγον ἔχει, <ὄν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν; ἀσύμμετρος ἄρα ἐστὶν ἡ $Z\eta$ τῇ K μήκει. καὶ

Therefore, since as E is to BC , so the (square) on A (is) to the (square) on FG , the (square) on A (is) thus commensurable with the (square) on FG [Prop. 10.6]. And the (square) on A (is) rational. Thus, the (square) on FG (is) also rational. Thus, FG is also a rational (straight-line). And since E does not have to BC the ratio which (some) square number (has) to (some) square number, the (square) on A thus does not have to the (square) on FG the ratio which (some) square number (has) to (some) square number either. Thus, A is incommensurable in length with FG [Prop. 10.9]. Again, since as BC is to CD , so the (square) on FG (is) to the (square) on GH , the (square) on FG (is) thus commensurable with the (square) on GH [Prop. 10.6]. And the (square) on FG (is) rational. Thus, the (square) on GH (is) also rational. Thus, GH (is) also rational. And since BC does not have to CD the ratio which (some) square number (has) to (some) square number, the (square) on FG thus does not have to the (square) on GH the ratio which (some) square (number) has to (some) square (number) either. Thus, FG is incommensurable in length with GH [Prop. 10.9]. And both are rational (straight-lines). Thus, FG and GH are rational (straight-lines which are) commensurable in square only. Thus, FH is an apotome [Prop. 10.73]. So, I say that (it is) also a sixth (apotome). For since as E is to BC , so the (square) on A (is) to the (square) on FG , and as BC (is) to CD , so the (square) on FG (is) to the (square) on GH , thus, via equality, as E is to CD , so the (square) on A (is) to the (square) on GH [Prop. 5.22]. And E does not have to CD the ratio which (some) square number (has) to (some) square number. Thus, the (square) on A does not have to the (square) GH the ratio which (some) square number (has) to (some) square number either. A is thus incommensurable in length with GH [Prop. 10.9]. Thus, neither of FG and GH is commensurable in length with the rational (straight-line) A . Therefore, let the (square) on K be that (area) by which the (square) on

δύναται ἡ ZH τῆς $H\Theta$ μείζον τῶν ἀπὸ τῆς K ; ἡ ZH > ἄρα τῆς $H\Theta$ μείζον δύναται τῶν ἀπὸ ἀσυμμέτρου ἐαντὶ μήκει. καὶ οὐδετέρω τῶν ZH , $H\Theta$ σύμμετρόξ ἐστι τῇ ἐκκειμένῃ <ρητῇ μήκει τῇ A . ἡ > ἄρα $Z\Theta$ ἀποτομή ἐστὶν <έκτη. E < ὀρηται > ἄρα ἡ <έκτη ἀποτομή ἡ $Z\Theta$; <όπερ > ἐδει δεῖχαι.

FG is greater than the (square) on GH [Prop. 10.13 lem.]. Therefore, since as BC is to CD , so the (square) on FG (is) to the (square) on GH , thus, via conversion, as CB is to BD , so the (square) on FG (is) to the (square) on K [Prop. 5.19 corr.]. And CB does not have to BD the ratio which (some) square number (has) to (some) square number. Thus, the (square) on FG does not have to the (square) on K the ratio which (some) square number (has) to (some) square number either. FG is thus incommensurable in length with K [Prop. 10.9]. And the square on FG is greater than (the square on) GH by the (square) on K . Thus, the square on FG is greater than (the square on) GH by the (square) on (some straight-line) incommensurable in length with (FG). And neither of FG and GH is commensurable in length with the (previously) laid down rational (straight-line) A . Thus, FH is a sixth apotome [Def. 10.16]. Thus, the sixth apotome FH has been found. (Which is) the very thing it was required to show.

† See footnote to Prop. 10.53.

91.

Ἐὰν θωρίον περιέθῃται ὑπὸ <ρητῆς καὶ ἀποτομῆς πρώτης, ἡ τὸ θωρίον δυναμένη ἀπορομή ἐστὶν.

Περιεθέσθω γὰρ θωρίον τὸ AB ὑπὸ <ρητῆς τῆς AG καὶ ἀποτομῆς πρώτης τῆς AD ; λέγω, <ὅτι ἡ τὸ AB θωρίον δυναμένη ἀποτομή ἐστὶν.

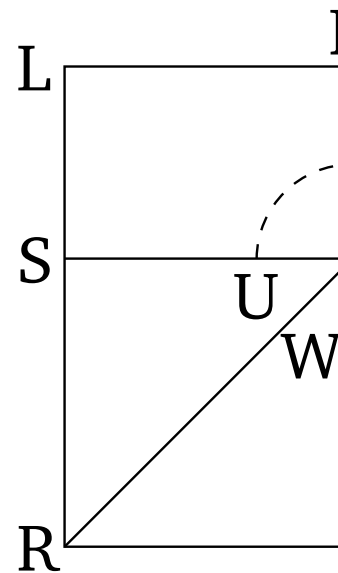
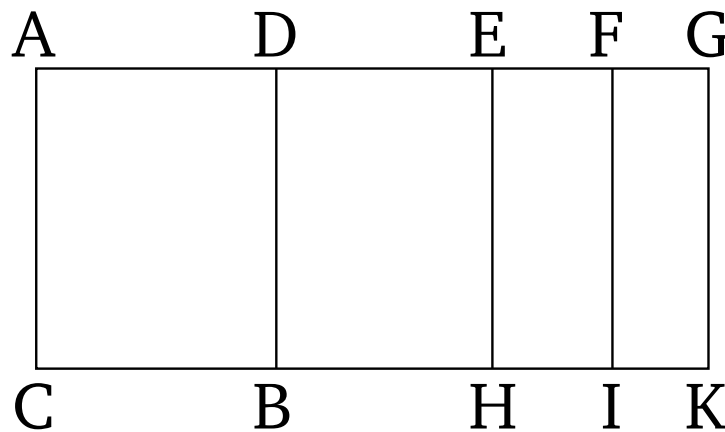
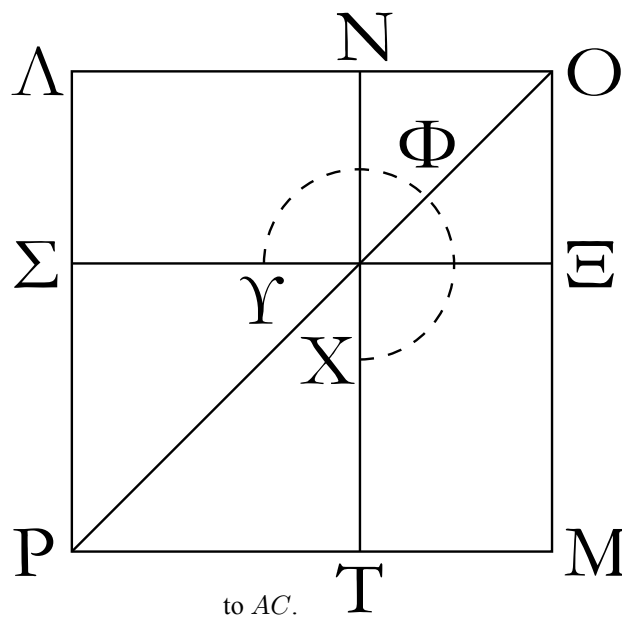
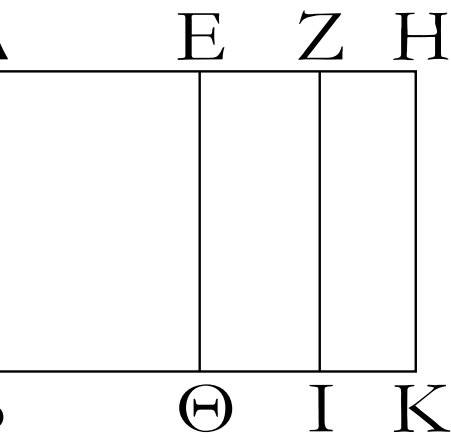
Ἐπεὶ γὰρ ἀποτομή ἐστὶ πρώτη ἡ AD , > ἐστὼ αὐτῇ προσαρμόζουσα ἡ ΔH ; αἱ AH , $H\Delta$ > ἄρα <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι. καὶ <ὅλη ἡ AH σύμμετρόξ ἐστὶ τῇ ἐκκειμένῃ <ρητῇ τῇ AG , καὶ ἡ AH τῆς $H\Delta$ μείζον δύναται τῶν ἀπὸ συμέτρου ἐαντὶ μήκει; ἔαν > ἄρα τῶν τετάρτῳ μέρει τοῦ ἀπὸ τῆς ΔH > ἴσον παρὰ τὴν AH παραβληθῇ ἑλλείπον > ἴδει τετραγώνῳ, εἷς σύμμετρα αὐτὴν διαιρεῖ. τετμήσθω ἡ ΔH δίθω κατὰ τὸ E , καὶ τῶν ἀπὸ τῆς EH > ἴσον παρὰ τὴν AH παραβεβλήσθω ἑλλείπον > ἴδει τετραγώνῳ, καὶ > ἐστὼ τὸ ὑπὸ τῶν AZ , ZH ; σύμμετροξ > ἄρα ἐστὶν ἡ AZ τῇ ZH . καὶ διὰ τῶν E , Z , H σημείων τῇ AG παράλληλοι > ἡθθωσαν αἱ $E\Theta$, ZI , HK .

Proposition 91

If an area is contained by a rational (straight-line) and a first apotome then the square-root of the area is an apotome.

For let the area AB be contained by the rational (straight-line) AC and the first apotome AD . I say that the square-root of area AB is an apotome.

For since AD is a first apotome, let DG be its attachment. Thus, AG and DG are rational (straight-lines which are) commensurable in square only [Prop. 10.73]. And the whole, AG , is commensurable (in length) with the (previously) laid down rational (straight-line) AC , and the square on AG is greater than (the square on) GD by the (square) on (some straight-line) commensurable in length with (AG) [Def. 10.11]. Thus, if (an area) equal to the fourth part of the (square) on DG is applied to AG , falling short by a square figure, then it divides (AG) into (parts which are) commensurable (in length) [Prop. 10.17]. Let DG be cut in half at E . And let (an area) equal to the (square) on EG be applied to AG , falling short by a square figure. And let it be the (rectangle contained) by AF and FG . AF is thus commensurable (in length) with FG . And let EH , FI , and GK be drawn through points E , F , and G (respectively), parallel



Καὶ ἐπεὶ σύμμετρος ἐστὶν ἡ ΑΖ τῇ ΖΗ μήκει, καὶ ἡ ΑΗ > ἄρα ἑκατέρᾳ τῶν ΑΖ, ΖΗ σύμμετρος ἐστὶ μήκει. ἀλλὰ ἡ ΑΗ σύμμετρος ἐστὶ τῇ ΑΓ; καὶ ἑκατέρα > ἄρα τῶν ΑΖ, ΖΗ σύμμετρος ἐστὶ τῇ ΑΓ μήκει. καὶ ἐστὶ <ρητὴ ἡ ΑΓ; <ρητὴ > ἄρα καὶ ἑκατέρᾳ τῶν ΑΖ, ΖΗ; <ὥστε καὶ ἑκάτερον τῶν ΑΙ, ΖΚ <ρητόν ἐστιν. καὶ ἐπεὶ σύμμετρος ἐστὶν ἡ ΔΕ τῇ ΕΗ μήκει, καὶ ἡ ΔΗ > ἄρα ἑκατέρᾳ τῶν ΔΕ, ΕΗ σύμμετρος ἐστὶ μήκει. <ρητὴ δὲ ἡ ΔΗ καὶ ἀσύμμετρος τῇ ΑΓ μήκει; <ρητὴ > ἄρα καὶ ἑκατέρᾳ τῶν ΔΕ, ΕΗ καὶ ἀσύμμετρος τῇ ΑΓ μήκει; ἑκάτερον > ἄρα τῶν ΔΘ, ΕΚ μέσον ἐστίν.

Κεῖσθω δὴ τῷ μὲν ΑΙ > ἴσον τετράγωνον τὸ ΛΜ, τῷ δὲ ΖΚ > ἴσον τετράγωνον ἀφελήσθω κοινὴν γωνίαν > ἔθον αὐτῶ| τὴν ὑπὸ ΛΟΜ τὸ ΝΧ; περὶ τὴν αὐτὴν > ἄρα διάμετρον ἐστὶ τὰ ΛΜ, ΝΧ τετράγωνα. > ἔστω αὐτῶν διάμετρος ἡ ΟΡ, καὶ καταγεγράφθω τὸ σθῆμα. ἐπεὶ ο> ὦν > ἴσον ἐστὶ τὸ ὑπὸ τῶν ΑΖ, ΖΗ περιεχόμενον ὀρθογώνιον τῷ ἀπὸ τῆς ΕΗ τετραγώνῳ, > ἐστὶν > ἄρα ὡς ἡ ΑΖ πρὸς τὴν ΕΗ, ο< ὅτως ἡ ΕΗ πρὸς τὴν ΖΗ. ἀλλ> ὡς μὲν ἡ ΑΖ πρὸς τὴν ΕΗ, ο< ὅτως τὸ ΑΙ πρὸς τὸ ΕΚ, ὡς δὲ ἡ ΕΗ πρὸς τὴν ΖΗ, ο< ὅτως ἐστὶ τὸ ΕΚ πρὸς τὸ ΚΖ; τῶν > ἄρα ΑΙ, ΚΖ μέσον ἀνάλογόν ἐστι τὸ ΕΚ. > ἐστὶ δὲ καὶ τῶν ΛΜ, ΝΧ μέσον ἀνάλογον τὸ ΜΝ, ὡς ἐν τοῖς > ἐμπροσθεν ἐδείχθη, καὶ ἐστὶ τὸ [μὲν] ΑΙ τῷ ΛΜ τετραγώνῳ > ἴσον, τὸ δὲ ΚΖ τῷ ΝΧ; καὶ τὸ ΜΝ > ἄρα τῷ ΕΚ

And since AF is commensurable in length with FG , AG is thus also commensurable in length with each of AF and FG [Prop. 10.15]. But AG is commensurable (in length) with AC . Thus, each of AF and FG is also commensurable in length with AC [Prop. 10.12]. And AC is a rational (straight-line). Thus, AF and FG (are) each also rational (straight-lines). Hence, AI and FK are also each rational (areas) [Prop. 10.19]. And since DE is commensurable in length with EG , DG is thus also commensurable in length with each of DE and EG [Prop. 10.15]. And DG (is) rational, and incommensurable in length with AC . DE and EG (are) thus each rational, and incommensurable in length with AC [Prop. 10.13]. Thus, DH and EK are each medial (areas) [Prop. 10.21].

So let the square LM , equal to AI , be laid down. And let the square NO , equal to FK , be subtracted (from LM), having with it the common angle LPM . Thus, the squares LM and NO are about the same diagonal [Prop. 6.26]. Let PR be their (common) diagonal, and let the (rest of the) figure be drawn. Therefore, since the rectangle contained by AF and FG is equal to the square EG , thus as AF is to EG , so EG (is) to FG [Prop. 6.17]. But, as AF (is) to EG ,

>ίσον ἐστίν. ἀλλὰ τὸ μὲν EK τῷ $\Delta\Theta$ ἐστίν >ίσον, τὸ δὲ MN τῷ ΛX ; τὸ >άρα ΔK >ίσον ἐστὶ τῷ $Y\Phi\Theta$ γνώμονι καὶ τῷ NX . >έστι δὲ καὶ τὸ AK >ίσον τοῖς ΛM , NX τετραγώνοις; λοιπὸν >άρα τὸ AB >ίσον ἐστὶ τῷ ST . τὸ δὲ ST τὸ ἀπὸ τῆς ΛN ἐστὶ τετράγωνον; τὸ >άρα ἀπὸ τῆς ΛN τετράγωνον >ίσον ἐστὶ τῷ AB ; ἡ ΛN >άρα δύναται τὸ AB . λέγω δὴ, <ὅτι ἡ ΛN ἀποτομή ἐστίν.

Ἐπεὶ γὰρ <ρητὸν ἐστὶν ἑκάτερον τῶν AI , ZK , καὶ ἐστὶν >ίσον τοῖς ΛM , NX , καὶ ἑκάτερον >άρα τῶν ΛM , NX <ρητὸν ἐστίν, τουτέστι τὸ ἀπὸ ἑκατέρᾳ τῶν AO , ON ; καὶ ἑκατέρα >άρα τῶν AO , ON <ρητὴ ἐστίν. πάλιν, ἐπεὶ μέσον ἐστὶ τὸ $\Delta\Theta$ καὶ ἐστὶν >ίσον τῷ ΛX , μέσον >άρα ἐστὶ καὶ τὸ ΛX . ἐπεὶ ο>ὖν τὸ μὲν ΛX μέσον ἐστίν, τὸ δὲ NX <ρητὸν, ἀσύμμετρον >άρα ἐστὶ τὸ ΛX τῷ NX ; ὥς δὲ τὸ ΛX πρὸς τὸ NX , ο<ύτως ἐστὶν ἡ AO πρὸς τὴν ON ; ἀσύμμετρος >άρα ἐστὶν ἡ AO τῇ ON μήκει. καὶ εἰσιν ἀμφοτέραι <ρηταί; αἱ AO , ON >άρα <ρηταί εἰσι δυνάμει μόνον σύμμετροι; ἀποτομή >άρα ἐστὶν ἡ ΛN . καὶ δύναται τὸ AB θωρίον; ἡ >άρα τὸ AB θωρίον δυνάμενη ἀποτομή ἐστίν.

Ἐὰν >άρα θωρίον περιέθῃται ὑπὸ <ρητῆς καὶ τὰ ἐχθῆς.

so AI (is) to EK , and as EG (is) to FG , so EK is to KF [Prop. 6.1]. Thus, EK is the mean proportional to AI and KF [Prop. 5.11]. And MN is also the mean proportional to LM and NO , as shown before [Prop. 10.53 lem.]. And AI is equal to the square LM , and KF to NO . Thus, MN is also equal to EK . But, EK is equal to DH , and MN to LO [Prop. 1.43]. Thus, DK is equal to the gnomon UVW and NO . And AK is also equal to (the sum of) the squares LM and NO . Thus, the remainder AB is equal to ST . And ST is the square on LN . Thus, the square on LN is equal to AB . Thus, LN is the square-root of AB . So, I say that LN is an apotome.

For since AI and FK are each rational (areas), and are equal to LM and NO (respectively), thus LM and NO —that is to say, the (squares) on each of LP and PN (respectively)—are also each rational (areas). Thus, LP and PN are also each rational (straight-lines). Again, since DH is a medial (area), and is equal to LO , LO is thus also a medial (area). Therefore, since LO is medial, and NO rational, LO is thus incommensurable with NO . And as LO (is) to NO , so LP is to PN [Prop. 6.1]. LP is thus incommensurable in length with PN [Prop. 10.11]. And they are both rational (straight-lines). Thus, LP and PN are rational (straight-lines which are) commensurable in square only. Thus, LN is an apotome [Prop. 10.73]. And it is the square-root of area AB . Thus, the square-root of area AB is an apotome.

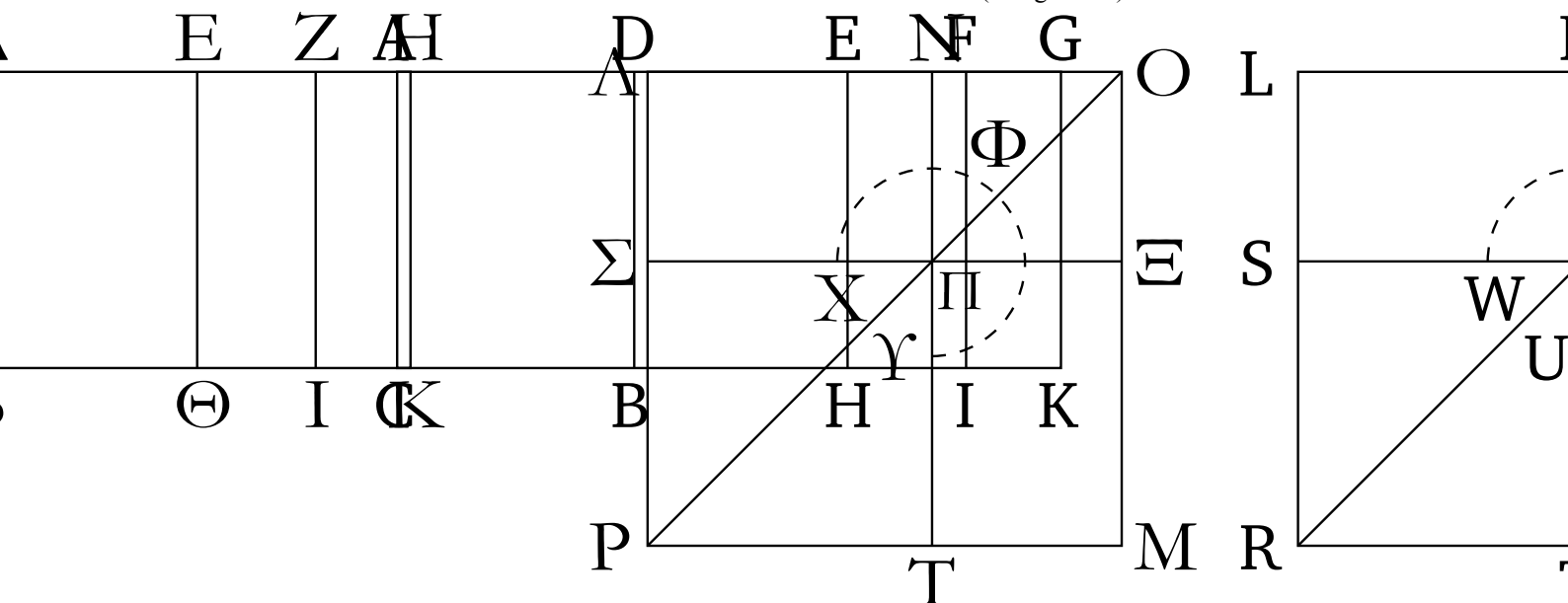
Thus, if an area is contained by a rational (straight-line), and so on

92.

Proposition 92

Ἐὰν θωρίον περιέθῃται ὑπὸ <ρητῆς καὶ ἀποτομῆς δευτέρας, ἡ τὸ θωρίον δυνάμενη μέση ἀποτομή ἐστὶ πρώτη.

If an area is contained by a rational (straight-line) and a second apotome then the square-root of the area is a first apotome of a medial (straight-line).



Θωρίον γὰρ τὸ AB περιέθεσθω ὑπὸ <ρητῆς τῆς AG καὶ ἀποτομῆς δευτέρας τῆς AD ; λέγω, <ὅτι ἡ τὸ AB θωρίον δυνάμενη μέση ἀποτομή ἐστὶ πρώτη.

>Ἐστω γὰρ τῇ AD προσαρμόζουσα ἡ DH ; αἱ >άρα AH , HA <ρηταί εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ προσαρμόζουσα ἡ DH σύμμετρος ἐστὶ τῇ ἐκκειμένῃ <ρητῇ τῇ AG , ἡ δὲ <όλη ἡ AH τῆς προσαρμόζουσας τῆς HA μείζον δύναται τῷ ἀπὸ συμμέτρου ἐαυτῇ μήκει. ἐπεὶ ο>ὖν ἡ AH τῆς HA μείζον

For let the area AB be contained by the rational (straight-line) AC and the second apotome AD . I say that the square-root of area AB is the first apotome of a medial (straight-line).

For let DG be an attachment to AD . Thus, AG and GD are rational (straight-lines which are) commensurable in square only [Prop. 10.73], and the attachment DG is commensurable (in length) with the (previously) laid down rational

δύναται τῷ ἀπὸ συμμετρου ἑαυτῇ, ἔαν >άρα τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ΗΔ >ίσον παρὰ τὴν ΑΗ παραβληθῇ| ἔλλειπον ε>ίδει τετραγώνῳ, εἷς σύμμετρα αὐτὴν διαιρεῖ. τετμήσθω ο>ὺν ἡ ΔΗ δίθῃ κατὰ τὸ Ε; καὶ τῷ ἀπὸ τῆς ΕΗ >ίσον παρὰ τὴν ΑΗ παραβεβλήσθω ἔλλειπον ε>ίδει τετραγώνῳ, καὶ >έστω τὸ ὑπὸ τῶν ΑΖ, ΖΗ; σύμμετρος >άρα ἐστὶν ἡ ΑΖ τῇ ΖΗ μήκει. καὶ ἡ ΑΗ >άρα ἑκατέρα τῶν ΑΖ, ΖΗ σύμμετρος ἐστὶ μήκει. <ρητὴ δὲ ἡ ΑΗ καὶ ἀσύμμετρος τῇ ΑΓ μήκει; καὶ ἑκατέρα >άρα τῶν ΑΖ, ΖΗ <ρητὴ ἐστὶ καὶ ἀσύμμετρος τῇ ΑΓ μήκει; ἑκάτερον >άρα τῶν ΑΙ, ΖΚ μέσον ἐστίν. πάλιν, ἐπεὶ σύμμετρος ἐστὶν ἡ ΔΕ τῇ ΕΗ, καὶ ἡ ΔΗ >άρα ἑκατέρα τῶν ΔΕ, ΕΗ σύμμετρος ἐστίν. ἄλλ> ἡ ΔΗ σύμμετρος ἐστὶ τῇ ΑΓ μήκει [<ρητὴ >άρα καὶ ἑκατέρα τῶν ΔΕ, ΕΗ καὶ σύμμετρος τῇ ΑΓ μήκει]. ἑκάτερον >άρα τῶν ΔΘ, ΕΚ <ρητόν ἐστιν.

Συνεστάτω ο>ὺν τῷ μὲν ΑΙ >ίσον τετράγωνον τὸ ΑΜ, τῷ δὲ ΖΚ >ίσον ἀφηρήσθω τὸ ΝΧ περὶ τὴν αὐτὴν γωνίαν >ὸν τῷ ΑΜ τὴν ὑπὸ τῶν ΛΟΜ; περὶ τὴν αὐτὴν >άρα ἐστὶ διάμετρον τὰ ΑΜ, ΝΧ τετράγωνα. >έστω αὐτῶν διάμετρος ἡ ΟΡ, καὶ καταγεγράφθω τὸ σθῆμα. ἐπεὶ ο>ὺν τὰ ΑΙ, ΖΚ μέσα ἐστὶ καὶ ἐστὶν >ίσα τοῖς ἀπὸ τῶν ΛΟ, ΟΝ, καὶ τὰ ἀπὸ τῶν ΛΟ, ΟΝ [>άρα] μέσα ἐστίν; καὶ αἱ ΛΟ, ΟΝ >άρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι. καὶ ἐπεὶ τὸ ὑπὸ τῶν ΑΖ, ΖΗ >ίσον ἐστὶ τῷ ἀπὸ τῆς ΕΗ, >έστιν >άρα ὥς ἡ ΑΖ πρὸς τὴν ΕΗ, ο<ύτω ἡ ΕΗ πρὸς τὴν ΖΗ; ἄλλ> ὥς μὲν ἡ ΑΖ πρὸς τὴν ΕΗ, ο<ύτω ἡ ΕΗ πρὸς τὸ ΕΚ; ὥς δὲ ἡ ΕΗ πρὸς τὴν ΖΗ, ο<ύτω [ἐστὶ] τὸ ΕΚ πρὸς τὸ ΖΚ; τῶν >άρα ΑΙ, ΖΚ μέσον ἀνάλογον ἐστὶ τὸ ΕΚ. >έστι δὲ καὶ τῶν ΑΜ, ΝΧ τετραγώνων μέσον ἀνάλογον τὸ ΜΝ; καὶ ἐστὶν >ίσον τὸ μὲν ΑΙ τῷ ΑΜ, τὸ δὲ ΖΚ τῷ ΝΧ; καὶ τὸ ΜΝ >άρα >ίσον ἐστὶ τῷ ΕΚ. ἄλλὰ τῷ μὲν ΕΚ >ίσον [ἐστὶ] τὸ ΔΘ, τῷ δὲ ΜΝ >ίσον τὸ ΛΧ; <όλον >άρα τὸ ΔΚ >ίσον ἐστὶ τῷ ΥΦΘ γνώμονι καὶ τῷ ΝΧ. ἐπεὶ ο>ὺν <όλον τὸ ΑΚ >ίσον ἐστὶ τοῖς ΑΜ, ΝΧ, <ὦν τὸ ΔΚ >ίσον ἐστὶ τῷ ΥΦΘ γνώμονι καὶ τῷ ΝΧ, λοιπὸν >άρα τὸ ΑΒ >ίσον ἐστὶ τῷ ΤΣ. τὸ δὲ ΤΣ ἐστὶ τὸ ἀπὸ τῆς ΑΝ; τὸ ἀπὸ τῆς ΑΝ >άρα >ίσον ἐστὶ τῷ ΑΒ θωρίῳ; ἡ ΑΝ >άρα δύναται τὸ ΑΒ θωρίον. λέγω [δὴ], <ότι ἡ ΑΝ μέσηξ ἀποτομή ἐστὶ πρώτη.

Ἐπεὶ γὰρ <ρητόν ἐστὶ τὸ ΕΚ καὶ ἐστὶν >ίσον τῷ ΛΧ, <ρητόν >άρα ἐστὶ τὸ ΛΧ, τουτέστι τὸ ὑπὸ τῶν ΛΟ, ΟΝ. μέσον δὲ ἐδείκθη τὸ ΝΧ; ἀσύμμετρον >άρα ἐστὶ τὸ ΛΧ τῷ ΝΧ; ὥς δὲ τὸ ΛΧ πρὸς τὸ ΝΧ, ο<ύτω ἐστὶν ἡ ΛΟ πρὸς ΟΝ; αἱ ΛΟ, ΟΝ >άρα ἀσύμμετροί εἰσι μήκει. αἱ >άρα ΛΟ, ΟΝ μέσαι εἰσὶ δυνάμει μόνον σύμμετροι <ρητόν περιέθουσαι; ἡ ΑΝ >άρα μέσηξ ἀποτομή ἐστὶ πρώτη; καὶ δύναται τὸ ΑΒ θωρίον.

Ἡ >άρα τὸ ΑΒ θωρίον δυναμένη μέσηξ ἀποτομή ἐστὶ πρώτη; <όπερ >έδει δεῖχαι.

(straight-line) AC , and the square on the whole, AG , is greater than (the square on) the attachment, GD , by the (square) on (some straight-line) commensurable in length with (AG) [Def. 10.12]. Therefore, since the square on AG is greater than (the square on) GD by the (square) on (some straight-line) commensurable (in length) with (AG) , thus if (an area) equal to the fourth part of the (square) on GD is applied to AG , falling short by a square figure, then it divides (AG) into (parts which are) commensurable (in length) [Prop. 10.17]. Therefore, let DG be cut in half at E . And let (an area) equal to the (square) on EG be applied to AG , falling short by a square figure. And let it be the (rectangle contained) by AF and FG . Thus, AF is commensurable in length with FG . AG is thus also commensurable in length with each of AF and FG [Prop. 10.15]. And AG (is) a rational (straight-line), and incommensurable in length with AC . AF and FG are thus also each rational (straight-lines), and incommensurable in length with AC [Prop. 10.13]. Thus, AI and FK are each medial (areas) [Prop. 10.21]. Again, since DE is commensurable (in length) with EG , thus DG is also commensurable (in length) with each of DE and EG [Prop. 10.15]. But, DG is commensurable in length with AC [thus, DE and EG are also each rational, and commensurable in length with AC]. Thus, DH and EK are each rational (areas) [Prop. 10.19]. Therefore, let the square LM , equal to AI , be constructed. And let NO , equal to FK , which is about the same angle LPM as LM , be subtracted (from LM). Thus, the squares LM and NO are about the same diagonal [Prop. 6.26]. Let PR be their (common) diagonal, and let the (rest of the) figure be drawn. Therefore, since AI and FK are medial (areas), and are equal to the (squares) on LP and PN (respectively), [thus] the (squares) on LP and PN are also medial. Thus, LP and PN are also medial (straight-lines which are) commensurable in square only.[†] And since the (rectangle contained) by AF and FG is equal to the (square) on EG , thus as AF is to EG , so EG (is) to FG [Prop. 10.17]. But, as AF (is) to EG , so AI (is) to EK . And as EG (is) to FG , so EK [is] to FK [Prop. 6.1]. Thus, EK is the mean proportional to AI and FK [Prop. 5.11]. And MN is also the mean proportional to the squares LM and NO [Prop. 10.53 lem.]. And AI is equal to LM , and FK to NO . Thus, MN is also equal to EK . But, DH [is] equal to EK , and LO equal to MN [Prop. 1.43]. Thus, the whole (of) DK is equal to the gnomon UVW and NO . Therefore, since the whole (of) AK is equal to LM and NO , of which DK is equal to the gnomon UVW and NO , the remainder AB is thus equal to TS . And TS is the (square) on LN . Thus, the (square) on LN is equal to the area AB . LN is thus the square-root of area AB . [So], I say that LN is the first apotome of a medial (straight-line).

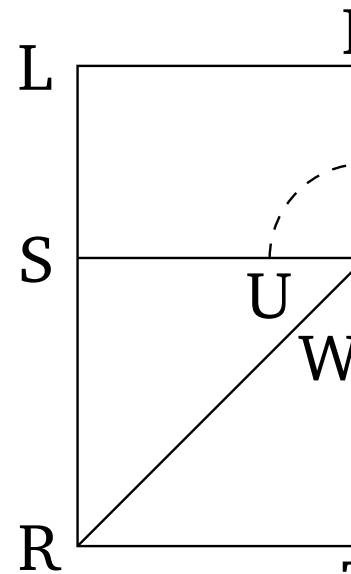
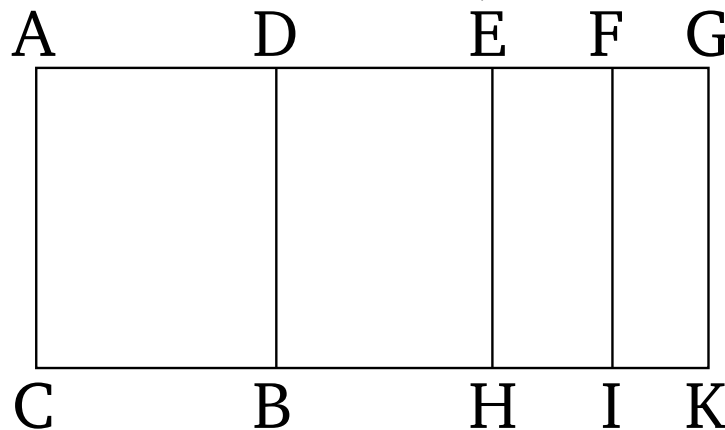
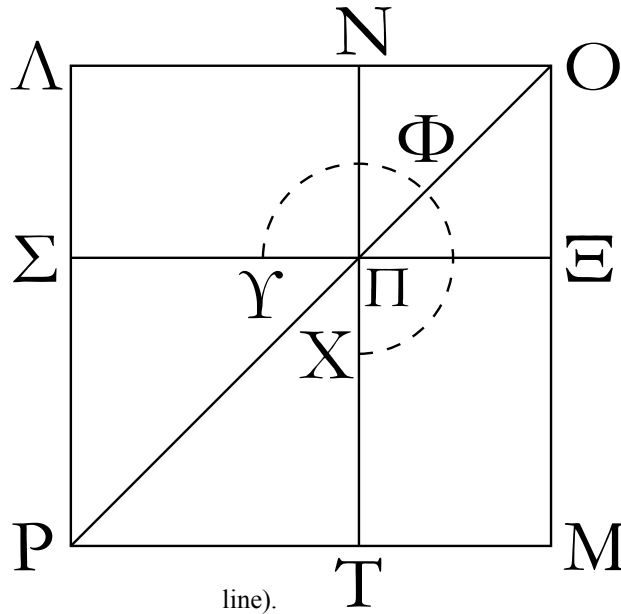
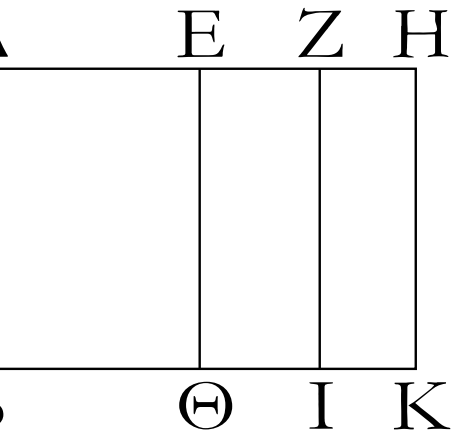
For since EK is a rational (area), and is equal to LO , LO —that is to say, the (rectangle contained) by LP and PN —is thus a rational (area). And NO was shown (to be) a medial (area). Thus, LO is incommensurable with NO . And as LO (is) to NO , so LP is to PN [Prop. 6.1]. Thus, LP and PN are incommensurable in length [Prop. 10.11]. LP and PN are thus medial (straight-lines which are) commensurable in square only, and which contain a rational (area). Thus, LN is the first apotome of a medial (straight-line) [Prop. 10.74]. And it is the square-root of area AB .

Thus, the square root of area AB is the first apotome of a medial (straight-line). (Which is) the very thing it was required to show.

† There is an error in the argument here. It should just say that LP and PN are commensurable in square, rather than in square only, since LP and PN are only shown to be incommensurable in length later on.

93.

Ἐάν θωρίον περιέθῃται ὑπὸ <ρητῆς καὶ ἀποτομῆς τρίτης, ἢ τὸ θωρίον δυναμένη μέσῃ ἀποτομῇ ἐστὶ δευτέρα. Θωρίον γὰρ τὸ AB περιέθῃται ὑπὸ <ρητῆς τῆς AG καὶ ἀποτομῆς τρίτης τῆς AD ; λέγω, <ὅτι ἡ τὸ AB θωρίον δυναμένη μέσῃ ἀποτομῇ ἐστὶ δευτέρα.



>Ἐστω γὰρ τῇ AD προσαρμόζουσα ἡ $ΔΗ$; αἱ AH , $HΔ$ >ἀρα <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι, καὶ οὐδετέρα τῶν AH , $HΔ$ σύμμετρόξ ἐστὶ μήκει τῇ ἐκκειμένῃ <ρητῇ τῇ AG , ἢ δὲ <όλη ἡ AH τῆς προσαρμοζούσης τῆς $ΔΗ$ μείζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ. ἐπεὶ ο>ὖν ἡ AH τῆς $HΔ$ μείζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ, ἔαν >ἀρα τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς $ΔΗ$ >ἴσον παρὰ τὴν AH παραβληθῇ ἔλλείπον ε>ἰδει τετραγώνῳ, εἰς σύμμετρα αὐτὴν διελεῖ. τεμήσθω ο>ὖν ἡ $ΔΗ$ δίθα κατὰ τὸ E , καὶ τῷ

Proposition 93

If an area is contained by a rational (straight-line) and a third apotome then the square-root of the area is a second apotome of a medial (straight-line).

For let the area AB be contained by the rational (straight-line) AC and the third apotome AD . I say that the square-root of area AB is the second apotome of a medial (straight-

For let DG be an attachment to AD . Thus, AG and GD are rational (straight-lines which are) commensurable in square only [Prop. 10.73], and neither of AG and GD is commensurable in length with the (previously) laid down rational (straight-line) AC , and the square on the whole, AG , is greater than (the square on) the attachment, DG , by the (square) on (some straight-line) commensurable (in length) with (AG) [Def. 10.13]. Therefore, since the square on

ἀπὸ τῆς EH >ἴσον παρὰ τὴν AH παραβεβλήσθω ἐλλείπων ϵ >ἴδει τετραγώνω, καὶ >έστω τὸ ὑπὸ τῶν AZ , ZH . καὶ >ήθθωσαν διὰ τῶν E , Z , H σημείων τῇ AG παράλληλοι αἱ $E\Theta$, ZI , HK ; σύμμετροι >άρα εἰσὶν αἱ AZ , ZH ; σύμμετρον >άρα καὶ τὸ AI τῷ ZK . καὶ ἐπεὶ αἱ AZ , ZH σύμμετροι εἰσι μήκει, καὶ ἡ AH >άρα ἑκάτερα τῶν AZ , ZH σύμμετρόξ ἐστὶ μήκει. <ρητὴ δὲ ἡ AH καὶ ἀσύμμετροξ τῇ AG μήκει; <ώστε καὶ αἱ AZ , ZH . ἑκάτερον >άρα τῶν AI , ZK μέσον ἐστίν. πάλιν, ἐπεὶ σύμμετρόξ ἐστὶν ἡ DE τῇ EH μήκει, καὶ ἡ DH >άρα ἑκάτερα τῶν DE , EH σύμμετρόξ ἐστὶ μήκει. <ρητὴ δὲ ἡ HD καὶ ἀσύμμετροξ τῇ AG μήκει; <ρητὴ >άρα καὶ ἑκάτερα τῶν DE , EH καὶ ἀσύμμετροξ τῇ AG μήκει; ἑκάτερον >άρα τῶν $\Delta\Theta$, EK μέσον ἐστίν. καὶ ἐπεὶ αἱ AH , HD δυνάμει μόνον σύμμετροί εἰσιν, ἀσύμμετροξ >άρα ἐστὶ μήκει ἡ AH τῇ HD . ἀλλ> ἡ μὲν AH τῇ AZ σύμμετρόξ ἐστὶ μήκει ἡ δὲ DH τῇ EH ; ἀσύμμετροξ >άρα ἐστὶν ἡ AZ τῇ EH μήκει. ὥς δὲ ἡ AZ πρὸς τὴν EH , ο<ύτωξ ἐστὶ τὸ AI πρὸς τὸ EK ; ἀσύμμετρον >άρα ἐστὶ τὸ AI τῷ EK .

Συνεστάτω ο>ὖν τῷ μὲν AI >ἴσον τετραγώνον τὸ ΛM , τῷ δὲ ZK >ἴσον ἀφῆ|ρήσθω τὸ NX περὶ τὴν αὐτὴν γωνίαν >ὄν τῷ ΛM ; περὶ τὴν αὐτὴν >άρα διάμετρόν ἐστὶ τὰ ΛM , NX . >έστω αὐτῶν διάμετροξ ἡ OP , καὶ καταγεγράφθω τὸ σθῆμα. ἐπεὶ ο>ὖν τὸ ὑπὸ τῶν AZ , ZH >ἴσον ἐστὶ τῷ ἀπὸ τῆς EH , >έστιν >άρα ὡς ἡ AZ πρὸς τὴν EH , ο<ύτωξ ἡ EH πρὸς τὴν ZH . ἀλλ> ὡς μὲν ἡ AZ πρὸς τὴν EH , ο<ύτωξ ἐστὶ τὸ AI πρὸς τὸ EK ; ὡς δὲ ἡ EH πρὸς τὴν ZH , ο<ύτωξ ἐστὶ τὸ EK πρὸς τὸ ZK ; καὶ ὡς >άρα τὸ AI πρὸς τὸ EK , ο<ύτωξ τὸ EK πρὸς τὸ ZK ; τῶν >άρα AI , ZK μέσον ἀνάλογόν ἐστὶ τὸ EK . >έστι δὲ καὶ τῶν ΛM , NX τετραγώνων μέσον ἀνάλογον τὸ MN ; καὶ ἐστὶν >ἴσον τὸ μὲν AI τῷ ΛM , τὸ δὲ ZK τῷ NX ; καὶ τὸ EK >άρα >ἴσον ἐστὶ τῷ MN . ἀλλὰ τὸ μὲν MN >ἴσον ἐστὶ τῷ ΛX , τὸ δὲ EK >ἴσον [ἐστὶ] τῷ $\Delta\Theta$; καὶ <όλον>άρα τὸ ΔK >ἴσον ἐστὶ τῷ $\Upsilon\Phi\Theta$ γνώμονι καὶ τῷ NX . >έστι δὲ καὶ τὸ AK >ἴσον τοῖς ΛM , NX ; λοιπὸν >άρα τὸ AB >ἴσον ἐστὶ τῷ ST , τουτέστι τῷ ἀπὸ τῆς ΛN τετραγώνω; ἡ ΛN >άρα δύναται τὸ AB θωρίον. λέγω, <ὅτι ἡ ΛN μέσηξ ἀποτομὴ ἐστὶ δευτέρα.

Ἐπεὶ γὰρ μέσα ἐδείθθη τὰ AI , ZK καὶ ἐστὶν >ἴσα τοῖς ἀπὸ τῶν ΛO , ON , μέσον >άρα καὶ ἑκάτερον τῶν ἀπὸ τῶν ΛO , ON ; μέση >άρα ἑκάτερα τῶν ΛO , ON . καὶ ἐπεὶ σύμμετρόν ἐστὶ τὸ AI τῷ ZK , σύμμετρον >άρα καὶ τὸ ἀπὸ τῆς ΛO τῷ ἀπὸ τῆς ON . πάλιν, ἐπεὶ ἀσύμμετρον ἐδείθθη τὸ AI τῷ EK , ἀσύμμετρον >άρα ἐστὶ καὶ τὸ ΛM τῷ MN , τουτέστι τὸ ἀπὸ τῆς ΛO τῷ ὑπὸ τῶν ΛO , ON ; <ώστε καὶ ἡ ΛO ἀσύμμετρόξ ἐστὶ μήκει τῇ ON ; αἱ ΛO , ON >άρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι. λέγω δὲ, <ὅτι καὶ μέσον περιέθουσιν.

Ἐπεὶ γὰρ μέσον ἐδείθθη τὸ EK καὶ ἐστὶν >ἴσον τῷ ὑπὸ τῶν ΛO , ON , μέσον >άρα ἐστὶ καὶ τὸ ὑπὸ τῶν ΛO , ON ; <ώστε αἱ ΛO , ON μέσαι εἰσὶ δυνάμει μόνον σύμμετροι μέσον περιέθουσαι. ἡ ΛN >άρα μέσηξ ἀποτομὴ ἐστὶ δευτέρα; καὶ δύναται τὸ AB θωρίον.

Ἡ >άρα τὸ AB θωρίον δυναμένη μέσηξ ἀποτομὴ ἐστὶ δευτέρα; <όπερ >έδει δεῖχαι.

AG is greater than (the square on) GD by the (square) on (some straight-line) commensurable (in length) with (AG), thus if (an area) equal to the fourth part of the square on DG is applied to AG , falling short by a square figure, then it divides (AG) into (parts which are) commensurable (in length) [Prop. 10.17]. Therefore, let DG be cut in half at E . And let (an area) equal to the (square) on EG be applied to AG , falling short by a square figure. And let it be the (rectangle contained) by AF and FG . And let EH , FI , and GK be drawn through points E , F , and G (respectively), parallel to AC . Thus, AF and FG are commensurable (in length). AI (is) thus also commensurable with FK [Props. 6.1, 10.11]. And since AF and FG are commensurable in length, AG is thus also commensurable in length with each of AF and FG [Prop. 10.15]. And AG (is) rational, and incommensurable in length with AC . Hence, AF and FG (are) also (rational, and incommensurable in length with AC) [Prop. 10.13]. Thus, AI and FK are each medial (areas) [Prop. 10.21]. Again, since DE is commensurable in length with EG , DG is also commensurable in length with each of DE and EG [Prop. 10.15]. And GD (is) rational, and incommensurable in length with AC . Thus, DE and EG (are) each also rational, and incommensurable in length with AC [Prop. 10.13]. DH and EK are thus each medial (areas) [Prop. 10.21]. And since AG and GD are commensurable in square only, AG is thus incommensurable in length with GD . But, AG is commensurable in length with AF , and DG with EG . Thus, AF is incommensurable in length with EG [Prop. 10.13]. And as AF (is) to EG , so AI is to EK [Prop. 6.1]. Thus, AI is incommensurable with EK [Prop. 10.11].

Therefore, let the square LM , equal to AI , be constructed. And let NO , equal to FK , which is about the same angle as LM , be subtracted (from LM). Thus, LM and NO are about the same diagonal [Prop. 6.26]. Let PR be their (common) diagonal, and let the (rest of the) figure be drawn. Therefore, since the (rectangle contained) by AF and FG is equal to the (square) on EG , thus as AF is to EG , so EG (is) to FG [Prop. 6.17]. But, as AF (is) to EG , so AI is to EK [Prop. 6.1]. And as EG (is) to FG , so EK is to FK [Prop. 6.1]. And thus as AI (is) to EK , so EK (is) to FK [Prop. 5.11]. Thus, EK is the mean proportional to AI and FK . And MN is also the mean proportional to the squares LM and NO [Prop. 10.53 lem.]. And AI is equal to LM , and FK to NO . Thus, EK is also equal to MN . But, MN is equal to LO , and EK [is] equal to DH [Prop. 1.43]. And thus the whole of DK is equal to the gnomon UVW and NO . And AK (is) also equal to LM and NO . Thus, the remainder AB is equal to ST —that is to say, to the square on LN . Thus, LN is the square-root of area AB . I say that LN is the second apotome of a medial (straight-line).

For since AI and FK were shown (to be) medial (areas), and are equal to the (squares) on LP and PN (respectively), the (squares) on each of LP and PN (are) thus also medial. Thus, LP and PN (are) each medial (straight-lines). And since AI is commensurable with FK [Props. 6.1, 10.11], the (square) on LP (is) thus also commensurable with the (square) on PN . Again, since AI was shown (to be) incommensurable with EK , LM is thus also incommensurable with MN —that is to say, the (square) on LP with the (rectangle contained) by LP and PN . Hence, LP is also incom-

mensurable in length with PN [Props. 6.1, 10.11]. Thus, LP and PN are medial (straight-lines which are) commensurable in square only. So, I say that they also contain a medial (area).

For since EK was shown (to be) a medial (area), and is equal to the (rectangle contained) by LP and PN , the (rectangle contained) by LP and PN is thus also medial. Hence, LP and PN are medial (straight-lines which are) commensurable in square only, and which contain a medial (area). Thus, LN is the second apotome of a medial (straight-line) [Prop. 10.75]. And it is the square-root of area AB .

Thus, the square-root of area AB is the second apotome of a medial (straight-line). (Which is) the very thing it was required to show.

94.

Ἐάν θωρίον περιέθῃται ὑπὸ <ρητῆς καὶ ἀποτομῆς τετάρτης, ἢ τὸ θωρίον δυναμένη ἐλάσσων ἐστίν.

Θωρίον γάρ τὸ AB περιθέσθω ὑπὸ <ρητῆς τῆς AG καὶ ἀποτομῆς τετάρτης τῆς AD ; λέγω, <ὅτι ἢ τὸ AB θωρίον δυναμένη ἐλάσσων ἐστίν.

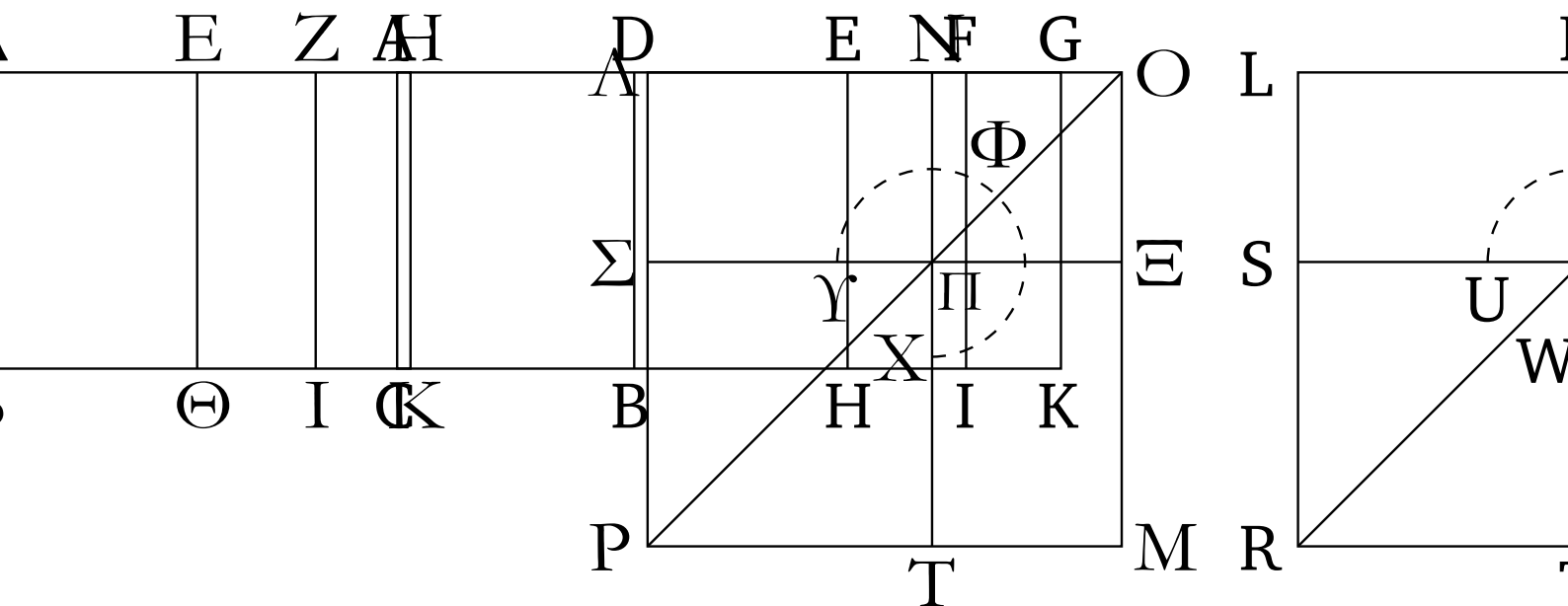
>Ἐστω γὰρ τῇ AD προσαρμόζουσα ἡ $ΔΗ$; αἱ >άρα AH , HA <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ AH σύμμετρος ἐστὶ τῇ ἐκκειμένῃ <ρητῇ τῇ AG μήκει, ἡ δὲ <όλη ἢ AH τῆς προσαρμοζούσης τῆς $ΔΗ$ μείζον δύναται τῷ ἀπὸ ἀσύμμετρου ἑαυτῇ μήκει. ἐπεὶ ο>ὗν ἡ AH τῆς HA μείζον δύναται τῷ ἀπὸ ἀσύμμετρου ἑαυτῇ μήκει, ἔαν >άρα τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς $ΔΗ$ >ίσον παρὰ τὴν AH παραβληθῇ ἐλλείπον >ἰδει τετραγώνῳ, εἰς ἀσύμμετρα αὐτὴν διελεί. τεμήσθω ο>ὗν ἡ $ΔΗ$ δίθῃ κατὰ τὸ E , καὶ τῷ ἀπὸ τῆς EH >ίσον παρὰ τὴν AH παραβεβλήσθω ἐλλείπον >ἰδει τετραγώνῳ, καὶ >έστω τὸ ὑπὸ τῶν AZ , ZH ; ἀσύμμετρος >άρα ἐστὶ μήκει ἡ AZ τῇ ZH . >ήθθωσαν ο>ὗν διὰ τῶν E , Z , H παράλληλοι ταῖς AG , BD αἱ $EΘ$, ZI , HK . ἐπεὶ ο>ὗν <ρητὴ ἐστὶν ἡ AH καὶ σύμμετρος τῇ AG μήκει, <ρητὸν >άρα ἐστὶν <όλον τὸ AK . πάλιν, ἐπεὶ ἀσύμμετρος ἐστὶν ἡ $ΔΗ$ τῇ AG μήκει, καὶ εἰσιν ἀμφοτέραι <ρηταί, μέσον >άρα ἐστὶ τὸ $ΔK$. πάλιν, ἐπεὶ ἀσύμμετρος ἐστὶν ἡ AZ τῇ ZH μήκει, ἀσύμμετρον >άρα καὶ τὸ AI τῷ ZK .

Proposition 94

If an area is contained by a rational (straight-line) and a fourth apotome then the square-root of the area is a minor (straight-line).

For let the area AB be contained by the rational (straight-line) AC and the fourth apotome AD . I say that the square-root of area AB is a minor (straight-line).

For let DG be an attachment to AD . Thus, AG and DG are rational (straight-lines which are) commensurable in square only [Prop. 10.73], and AG is commensurable in length with the (previously) laid down rational (straight-line) AC , and the square on the whole, AG , is greater than (the square on) the attachment, DG , by the square on (some straight-line) incommensurable in length with (AG) [Def. 10.14]. Therefore, since the square on AG is greater than (the square on) GD by the (square) on (some straight-line) incommensurable in length with (AG), thus if (some area), equal to the fourth part of the (square) on DG , is applied to AG , falling short by a square figure, then it divides (AG) into (parts which are) incommensurable (in length) [Prop. 10.18]. Therefore, let DG be cut in half at E , and let (some area), equal to the (square) on EG , be applied to AG , falling short by a square figure, and let it be the (rectangle contained) by AF and FG . Thus, AF is incommensurable in length with FG . Therefore, let EH , FI , and GK be drawn through E , F , and G (respectively), parallel to AC and BD . Therefore, since AG is rational, and commensurable in length with AC , the whole (area) AK is thus rational [Prop. 10.19]. Again, since DG is incommensurable in length with AC , and both are rational (straight-lines), DK is thus a medial (area) [Prop. 10.21]. Again, since AF is incommensurable in length with FG , AI (is) thus also incommensurable with FK [Props. 6.1, 10.11].



Συνεστάτω οὖν τῷ μὲν AI ἴσον τετράγωνον τὸ AM , τῷ δὲ ZK ἴσον ἀφηρήσθω περὶ τὴν αὐτὴν γωνίαν τὴν ὑπὸ τῶν LOM τὸ NX . περὶ τὴν αὐτὴν ᾗρα διάμετόν ἐστι τὰ AM , NX τετράγωνα. ἔστω αὐτῶν διάμετοξ ἡ OP , καὶ καταγεγράφθω τὸ σθῆμα. ἐπεὶ οὖν τὸ ὑπὸ τῶν AZ , ZH ἴσον ἐστὶ τῷ ἀπὸ τῆς EH , ἀνάλογον ᾗρα ἐστὶν ὡς ἡ AZ πρὸς τὴν EH , οὕτως ἡ EH πρὸς τὴν ZH . ἀλλ' ὡς μὲν ἡ AZ πρὸς τὴν EH , οὕτως ἐστὶ τὸ AI πρὸς τὸ EK , ὡς δὲ ἡ EH πρὸς τὴν ZH , οὕτως ἐστὶ τὸ EK πρὸς τὸ ZK ; τῶν ᾗρα AI , ZK μέσον ἀνάλογόν ἐστι τὸ EK . ἔστι δὲ καὶ τῶν AM , NX τετραγώνων μέσον ἀνάλογόν τὸ MN , καὶ ἐστὶν ἴσον τὸ μὲν AI τῷ AM , τὸ δὲ ZK τῷ NX ; καὶ τὸ EK ᾗρα ἴσον ἐστὶ τῷ MN . ἀλλὰ τῷ μὲν EK ἴσον ἐστὶ τὸ $\Delta\Theta$, τῷ δὲ MN ἴσον ἐστὶ τὸ ΛX ; ὅλον ᾗρα τὸ ΔK ἴσον ἐστὶ τῷ $Y\Phi\Theta$ γνώμονι καὶ τῷ NX . ἐπεὶ οὖν ὅλον τὸ ΔK ἴσον ἐστὶ τοῖς AM , NX τετραγώνοις, ὧν τὸ ΔK ἴσον ἐστὶ τῷ $Y\Phi\Theta$ γνώμονι καὶ τῷ NX τετραγώνῳ, λοιπὸν ᾗρα τὸ AB ἴσον ἐστὶ τῷ ST , τουτέστι τῷ ἀπὸ τῆς ΛN τετραγώνῳ; ἡ ΛN ᾗρα δύναται τὸ AB θωρίον. λέγω, ὅτι ἡ ΛN ἀλογόξ ἐστὶν ἡ καλουμένη ἐλάσσων.

Ἐπεὶ γὰρ ρητόν ἐστι τὸ AK καὶ ἐστὶν ἴσον τοῖς ἀπὸ τῶν AO , ON τετράγωνοις, τὸ ᾗρα συγκείμενον ἐκ τῶν ἀπὸ τῶν AO , ON ρητόν ἐστίν. πάλιν, ἐπεὶ τὸ ΔK μέσον ἐστίν, καὶ ἐστὶν ἴσον τὸ ΔK τῷ διζ ὑπὸ τῶν AO , ON , τὸ ᾗρα διζ ὑπὸ τῶν AO , ON μέσον ἐστίν. καὶ ἐπεὶ ἀσύμμετρον ἐδείκθη τὸ AI τῷ ZK , ἀσύμμετρον ᾗρα καὶ τὸ ἀπὸ τῆς AO τετράγωνον τῷ ἀπὸ τῆς ON τετραγώνῳ. αἱ AO , ON ᾗρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων ρητόν, τὸ δὲ διζ ὑπὸ αὐτῶν μέσον. ἡ ΛN ᾗρα ἀλογόξ ἐστὶν ἡ καλουμένη ἐλάσσων; καὶ δύναται τὸ AB θωρίον.

Ἡ ᾗρα τὸ AB θωρίον δυναμένη ἐλάσσων ἐστίν; ὅπερ >έδει δεῖχαι.

Therefore, let the square LM , equal to AI , be constructed. And let NO , equal to FK , (and) about the same angle, LPM , be subtracted (from LM). Thus, the squares LM and NO are about the same diagonal [Prop. 6.26]. Let PR be their (common) diagonal, and let the (rest of the) figure be drawn. Therefore, since the (rectangle contained) by AF and FG is equal to the (square) on EG , thus, proportionally, as AF is to EG , so EG (is) to FG [Prop. 6.17]. But, as AF (is) to EG , so AI is to EK , and as EG (is) to FG , so EK is to FK [Prop. 6.1]. Thus, EK is the mean proportional to AI and FK [Prop. 5.11]. And MN is also the mean proportional to the squares LM and NO [Prop. 10.13 lem.], and AI is equal to LM , and FK to NO . EK is thus also equal to MN . But, DH is equal to EK , and LO is equal to MN [Prop. 1.43]. Thus, the whole of DK is equal to the gnomon UVW and NO . Therefore, since the whole of AK is equal to the (sum of the) squares LM and NO , of which DK is equal to the gnomon UVW and the square NO , the remainder AB is thus equal to ST —that is to say, to the square on LN . Thus, LN is the square-root of area AB . I say that LN is the irrational (straight-line which is) called minor.

For since AK is rational, and is equal to the (sum of the) squares LP and PN , the sum of the (squares) on LP and PN is thus rational. Again, since DK is medial, and DK is equal to twice the (rectangle contained) by LP and PN , thus twice the (rectangle contained) by LP and PN is medial. And since AI was shown (to be) incommensurable with FK , the square on LP (is) thus also incommensurable with the square on PN . Thus, LP and PN are (straight-lines which are) incommensurable in square, making the sum of the squares on them rational, and twice the (rectangle contained) by them medial. LN is thus the irrational (straight-line) called minor [Prop. 10.76]. And it is the square-root of area AB .

Thus, the square-root of area AB is a minor (straight-line). (Which is) the very thing it was required to show.

95.

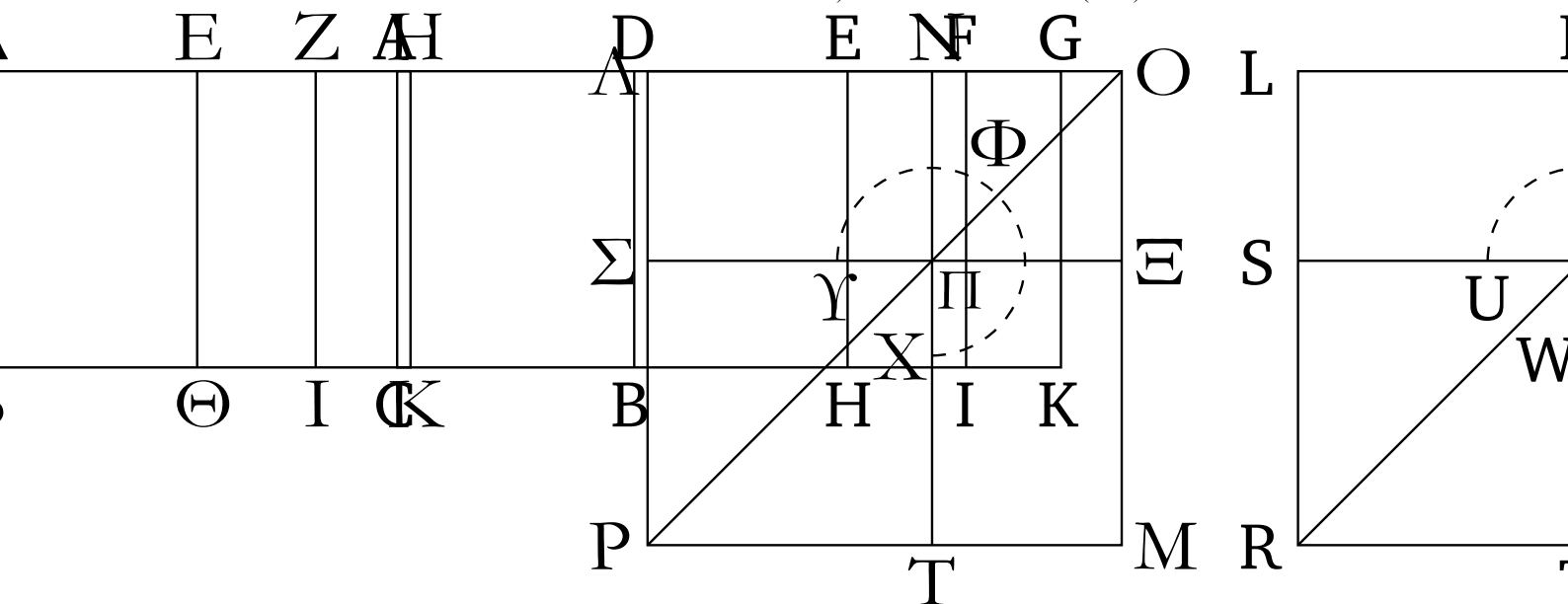
Proposition 95

Ἐὰν θωρίον περιέθῃται ὑπὸ ρητῆς καὶ ἀποτομῆς πέμπτης, ἡ τὸ θωρίον δυναμένη [ἡ] μετὰ ρητοῦ μέσον τὸ ὅλον

If an area is contained by a rational (straight-line) and a fifth apotome then the square-root of the area is that (straight-

ποιοῦσά ἐστιν.

line) which with a rational (area) makes a medial whole.



Θωρίον γὰρ τὸ AB περιθέσθω ὑπὸ <ρητῆς τῆς AG καὶ ἀποτομῆς πέμπτῃς τῆς AD ; λέγω, <ὅτι ἡ τὸ AB θωρίον δυναμένη [ή] μετὰ <ρητοῦ μέσον τὸ <όλον ποιοῦσά ἐστιν. >Ἐστω γὰρ τῇ AD προσαρμόζουσα ἡ ΔH ; αἱ >άρα AH , HA <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ προσαρμόζουσα ἡ HA σύμμετρόξ ἐστι μήκει τῇ ἐκκειμένη <ρητῇ τῇ AG , ἡ δὲ <όλη ἡ AH τῆς προσαρμόζουσῆς τῆς ΔH μεῖζον δύναται τῷ ἀπὸ ἀσύμμετρου ἐαυτῇ. ἔαν >άρα τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ΔH >ίσον παρὰ τὴν AH παραβληθῇ ἔλλειπον ε>ίδει τετραγώνῳ, εἰς ἀσύμμετρα αὐτὴν διελεῖ. τεμήσθω ο>ὖν ἡ ΔH δίθῃ κατὰ τὸ E σημεῖον, καὶ τῷ ἀπὸ τῆς EH >ίσον παρὰ τὴν AH παραβελήσθω ἔλλειπον ε>ίδει τετραγώνῳ καὶ >έστω τὸ ὑπὸ τῶν AZ , ZH ; ἀσύμμετροξ >άρα ἐστὶν ἡ AZ τῇ ZH μήκει. καὶ ἐπεὶ ἀσύμμετρόξ ἐστὶν ἡ AH τῇ GA μήκει, καὶ εἰσὶν ἀμφοτέραι <ρηταί, μέσον >άρα ἐστὶ τὸ AK . πάλιν, ἐπεὶ <ρητὴ ἐστὶν ἡ ΔH καὶ σύμμετροξ τῇ AG μήκει, <ρητόν ἐστι τὸ ΔK .

Συνεστάτω ο>ὖν τῷ μὲν AI >ίσον τετράγωνον τὸ AM , τῷ δὲ ZK >ίσον τετράγωνον ἀφηρήσθω τὸ NX περὶ τὴν αὐτὴν γωνίαν τὴν ὑπὸ ΛOM ; περὶ τὴν αὐτὴν >άρα διάμετρόν ἐστι τὰ AM , NX τετράγωνα. >έστω αὐτῶν διάμετροξ ἡ OP , καὶ καταγεγράφθω τὸ σθῆμα. ὁμοίωξ δὲ δείχομεν, <ὅτι ἡ ΛN δύναται τὸ AB θωρίον. λέγω, <ὅτι ἡ ΛN ἡ μετὰ <ρητοῦ μέσον τὸ <όλον ποιοῦσά ἐστιν.

Ἐπεὶ γὰρ μέσον ἐδείχθη τὸ AK καὶ ἐστὶν >ίσον τοῖς ἀπὸ τῶν ΛO , ON , τὸ >άρα συγκείμενον ἐκ τῶν ἀπὸ τῶν ΛO , ON μέσον ἐστίν. πάλιν, ἐπεὶ <ρητόν ἐστι τὸ ΔK καὶ ἐστὶν >ίσον τῷ διξ ὑπὸ τῶν ΛO , ON , καὶ αὐτὸ <ρητόν ἐστιν. καὶ ἐπεὶ ἀσύμμετρόν ἐστι τὸ AI τῷ ZK , ἀσύμμετρον >άρα ἐστὶ καὶ τὸ ἀπὸ τῆς ΛO τῷ ἀπὸ τῆς ON ; αἱ ΛO , ON >άρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ> αὐτῶν τετραγώνων μέσον, τὸ δὲ διξ ὑπ> αὐτῶν <ρητόν. ἡ λοιπὴ >άρα ἡ ΛN >άλογόξ ἐστὶν ἡ καλουμένη μετὰ <ρητοῦ μέσον τὸ <όλον ποιοῦσα; καὶ δύναται τὸ AB θωρίον.

Ἡ τὸ AB >άρα θωρίον δυναμένη μετὰ <ρητοῦ μέσον τὸ <όλον ποιοῦσά ἐστιν; <όπερ >έδει δεῖχαι.

For let the area AB be contained by the rational (straight-line) AC and the fifth apotome AD . I say that the square-root of area AB is that (straight-line) which with a rational (area) makes a medial whole.

For let DG be an attachment to AD . Thus, AG and DG are rational (straight-lines which are) commensurable in square only [Prop. 10.73], and the attachment GD is commensurable in length the the (previously) laid down rational (straight-line) AC , and the square on the whole, AG , is greater than (the square on) the attachment, DG , by the (square) on (some straight-line) incommensurable (in length) with (AG) [Def. 10.15]. Thus, if (some area), equal to the fourth part of the (square) on DG , is applied to AG , falling short by a square figure, then it divides (AG) into (parts which are) incommensurable (in length) [Prop. 10.18]. Therefore, let DG be divided in half at point E , and let (some area), equal to the (square) on EG , be applied to AG , falling short by a square figure, and let it be the (rectangle contained) by AF and FG . Thus, AF is incommensurable in length with FG . And since AG is incommensurable in length with CA , and both are rational (straight-lines), AK is thus a medial (area) [Prop. 10.21]. Again, since DG is rational, and commensurable in length with AC , DK is a rational (area) [Prop. 10.19].

Therefore, let the square LM , equal to AI , be constructed. And let the square NO , equal to FK , (and) about the same angle, LPM , be subtracted (from NO). Thus, the squares LM and NO are about the same diagonal [Prop. 6.26]. Let PR be their (common) diagonal, and let (the rest of) the figure be drawn. So, similarly (to the previous propositions), we can show that LN is the square-root of area AB . I say that LN is that (straight-line) which with a rational (area) makes a medial whole.

For since AK was shown (to be) a medial (area), and is equal to (the sum of) the squares on LP and PN , the sum of the (squares) on LP and PN is thus medial. Again, since DK is rational, and is equal to twice the (rectangle contained) by LP and PN , (the latter) is also rational. And since AI is incommensurable with FK , the (square) on LP is thus also incommensurable with the (square) on PN .

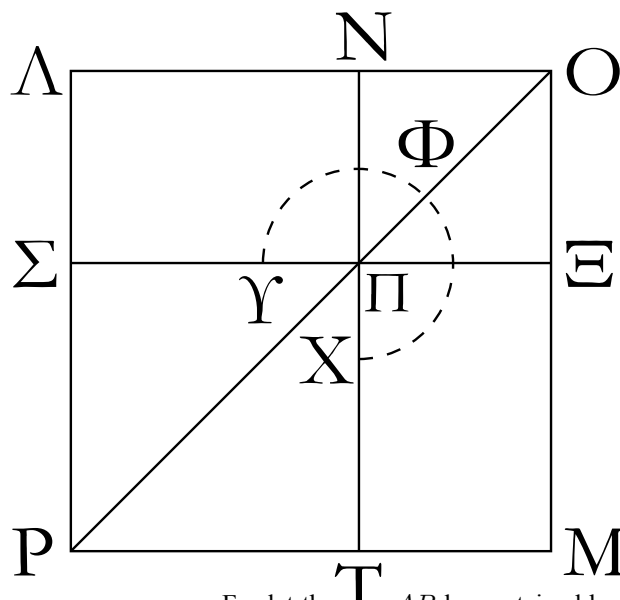
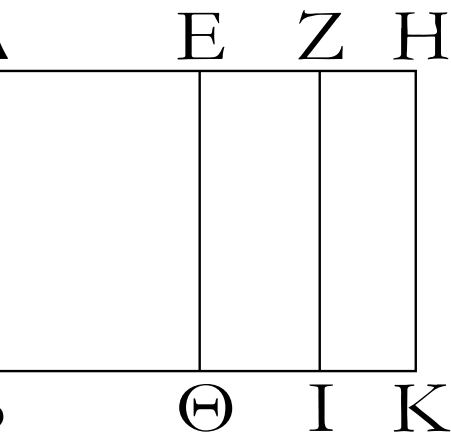
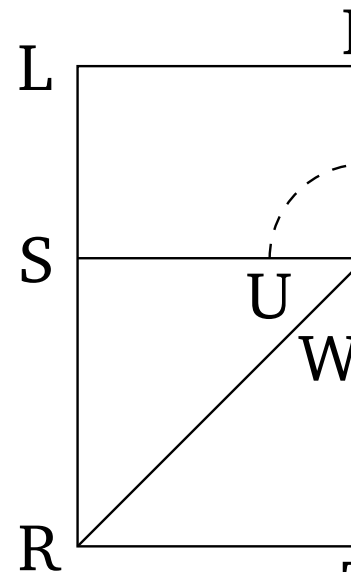
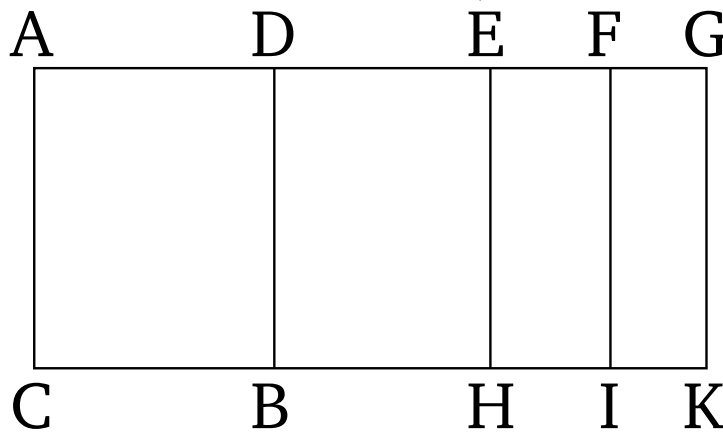
Thus, LP and PN are (straight-lines which are) incommensurable in square, making the sum of the squares on them medial, and twice the (rectangle contained) by them rational. Thus, the remainder LN is the irrational (straight-line) called that which with a rational (area) makes a medial whole [Prop. 10.77]. And it is the square-root of area AB . Thus, the square-root of area AB is that (straight-line) which with a rational (area) makes a medial whole. (Which is) the very thing it was required to show.

96.

Ἐάν θωρίον περιέθῃται ὑπὸ <ρητῆς καὶ ἀποτομῆς <έκτης, ἢ τὸ θωρίον δυναμένη μετὰ μέσου μέσον τὸ <όλον ποιοῦσά ἐστιν.

Proposition 96

If an area is contained by a rational (straight-line) and a sixth apotome then the square-root of the area is that (straight-line) which with a medial (area) makes a medial whole.



Θωρίον γάρ τὸ AB περιεθέσθω ὑπὸ <ρητῆς τῆς AG καὶ ἀποτομῆς <έκτης τῆς AD ; λέγω, <ότι ἡ τὸ AB θωρίον δυναμένη [ἢ] μετὰ μέσου μέσον τὸ <όλον ποιοῦσά ἐστιν.

>Ἐστω γάρ τῃ AD προσαρμόζουσα ἡ $ΔH$; αἱ >άρα AH , $HΔ$ <ρηταί εἰσι δυνάμει μόνον σύμμετροι, καὶ οὐδετέρα αὐτῶν σύμμετρόξ ἐστι τῇ ἐκκειμένη <ρητῇ τῇ AG μήκει, ἢ δὲ <όλη ἡ AH τῆξ προσαρμοζούσης τῆξ $ΔH$ μεῖζον δύναται τῷ ἀπὸ ἀσύμμετρου ἑαυτῇ μήκει. ἐπεὶ ο>ὖν ἡ

For let the area AB be contained by the rational (straight-line) AC and the sixth apotome AD . I say that the square-root of area AB is that (straight-line) which with a medial (area) makes a medial whole.

For let DG be an attachment to AD . Thus, AG and GD are rational (straight-lines which are) commensurable in square only [Prop. 10.73], and neither of them is commensurable in length with the (previously) laid down rational (straight-line) AC , and the square on the whole, AG , is greater than

ΑΗ τῆς ΗΔ μείζον δύναται τῷ ἀπὸ ἀσύμμετρου ἑαυτῇ μήκει, ἔαν >άρα τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ΔΗ >ίσον παρὰ τὴν ΑΗ παραβληθῇ ἑλλείπον ε>ίδει τετραγώνῳ, εἰς ἀσύμμετρα αὐτὴν διελεί. τεμήσθω ο>ὺν ἡ ΔΗ δίθῃ κατὰ τὸ Ε [σημεῖον], καὶ τῷ ἀπὸ τῆς ΕΗ >ίσον παρὰ τὴν ΑΗ παραβεβλήσθω ἑλλείπον ε>ίδει τετραγώνῳ, καὶ >έστω τὸ ὑπὸ τῶν ΑΖ, ΖΗ; ἀσύμμετρος >άρα ἐστὶν ἡ ΑΖ τῇ ΖΗ μήκει. ὥς δὲ ἡ ΑΖ πρὸς τὴν ΖΗ, ο<ύτωξ ἐστὶ τὸ ΑΙ πρὸς τὸ ΖΚ; ἀσύμμετρον >άρα ἐστὶ τὸ ΑΙ τῷ ΖΚ. καὶ ἐπεὶ αἱ ΑΗ, ΑΓ <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι, μέσον ἐστὶ τὸ ΑΚ. πάλιν, ἐπεὶ αἱ ΑΓ, ΔΗ <ρηταὶ εἰσι καὶ ἀσύμμετροι μήκει, μέσον ἐστὶ καὶ τὸ ΔΚ. ἐπεὶ ο>ὺν αἱ ΑΗ, ΗΔ δυνάμει μόνον σύμμετροί εἰσιν, ἀσύμμετρος >άρα ἐστὶν ἡ ΑΗ τῇ ΗΔ μήκει. ὥς δὲ ἡ ΑΗ πρὸς τὴν ΗΔ, ο<ύτωξ ἐστὶ τὸ ΑΚ πρὸς τὸ ΚΔ; ἀσύμμετρον >άρα ἐστὶ τὸ ΑΚ τῷ ΚΔ.

Συνεστάτω ο>ὺν τῷ μὲν ΑΙ >ίσον τετράγωνον τὸ ΛΜ, τῷ δὲ ΖΚ >ίσον ἀφηρήσθω περὶ τὴν αὐτὴν γωνίαν τὸ ΝΧ; περὶ τὴν αὐτὴν >άρα διάμετρον ἐστὶ τὰ ΛΜ, ΝΧ τετράγωνα. >έστω αὐτῶν διάμετρος ἡ ΟΡ, καὶ καταγεγράφθω τὸ σθῆμα. ὁμοίως δὲ τοῖς ἐπάνω δείχμεν, <ὅτι ἡ ΑΝ δύναται τὸ ΑΒ θωρίον. λέγω, <ὅτι ἡ ΑΝ [ἡ] μετὰ μέσου μέσον τὸ <όλον ποιοῦσά ἐστιν.

Ἐπεὶ γὰρ μέσον ἐδείθθη τὸ ΑΚ καὶ ἐστὶν >ίσον τοῖς ἀπὸ τῶν ΛΟ, ΟΝ, τὸ >άρα συγκείμενον ἐκ τῶν ἀπὸ τῶν ΛΟ, ΟΝ μέσον ἐστίν. πάλιν, ἐπεὶ μέσον ἐδείθθη τὸ ΔΚ καὶ ἐστὶν >ίσον τῷ διζ ὑπὸ τῶν ΛΟ, ΟΝ, καὶ τὸ διζ ὑπὸ τῶν ΛΟ, ΟΝ μέσον ἐστίν. καὶ ἐπεὶ ἀσύμμετρον ἐδείθθη τὸ ΑΚ τῷ ΔΚ, ἀσύμμετρα [>άρα] ἐστὶ καὶ τὰ ἀπὸ τῶν ΛΟ, ΟΝ τετράγωνα τῷ διζ ὑπὸ τῶν ΛΟ, ΟΝ. καὶ ἐπεὶ ἀσύμμετρον ἐστὶ τὸ ΑΙ τῷ ΖΚ, ἀσύμμετρον >άρα καὶ τὸ ἀπὸ τῆς ΛΟ τῷ ἀπὸ τῆς ΟΝ; αἱ ΛΟ, ΟΝ >άρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τό τε συγκείμενον ἐκ τῶν ἀπ> αὐτῶν τετραγώνων μέσον καὶ τὸ διζ ὑπ> αὐτῶν μέσον >έτι τε τὰ ἀπ> αὐτῶν τετράγωνα ἀσύμμετρα τῷ διζ ὑπ> αὐτῶν. ἡ >άρα ΑΝ >αλογόξ ἐστὶν ἡ καλουμένη μετὰ μέσου μέσον τὸ <όλον ποιοῦσα; καὶ δύναται τὸ ΑΒ θωρίον.

Ἡ >άρα τὸ θωρίον δυναμένη μετὰ μέσου μέσον τὸ <όλον ποιοῦσά ἐστίν; <όπερ >έδει δεῖχαι.

(the square on) the attachment, DG , by the (square) on (some straight-line) incommensurable in length with (AG) [Def. 10.16]. Therefore, since the square on AG is greater than (the square on) GD by the (square) on (some straight-line) incommensurable in length with (AG) , thus if (some area), equal to the fourth part of square on DG , is applied to AG , falling short by a square figure, then it divides (AG) into (parts which are) incommensurable (in length) [Prop. 10.18]. Therefore, let DG be cut in half at [point] E . And let (some area), equal to the (square) on EG , be applied to AG , falling short by a square figure. And let it be the (rectangle contained) by AF and FG . AF is thus incommensurable in length with FG . And as AF (is) to FG , so AI is to FK [Prop. 6.1]. Thus, AI is incommensurable with FK [Prop. 10.11]. And since AG and AC are rational (straight-lines which are) commensurable in square only, AK is a medial (area) [Prop. 10.21]. Again, since AC and DG are rational (straight-lines which are) incommensurable in length, DK is also a medial (area) [Prop. 10.21]. Therefore, since AG and GD are commensurable in square only, AG is thus incommensurable in length with GD . And as AG (is) to GD , so AK is to KD [Prop. 6.1]. Thus, AK is incommensurable with KD [Prop. 10.11].

Therefore, let the square LM , equal to AI , be constructed. And let NO , equal to FK , (and) about the same angle, be subtracted (from LM). Thus, the squares LM and NO are about the same diagonal [Prop. 6.26]. Let PR be their (common) diagonal, and let (the rest of) the figure be drawn. So, similarly to the above, we can show that LN is the square-root of area AB . I say that LN is that (straight-line) which with a medial (area) makes a medial whole.

For since AK was shown (to be) a medial (area), and is equal to the (sum of the) squares on LP and PN , the sum of the (squares) on LP and PN is medial. Again, since DK was shown (to be) a medial (area), and is equal to twice the (rectangle contained) by LP and PN , twice the (rectangle contained) by LP and PN is also medial. And since AK was shown (to be) incommensurable with DK , [thus] the (sum of the) squares on LP and PN is also incommensurable with twice the (rectangle contained) by LP and PN . And since AI is incommensurable with FK , the (square) on LP (is) thus also incommensurable with the (square) on PN . Thus, LP and PN are (straight-lines which are) incommensurable in square, making the sum of the squares on them medial, and twice the (rectangle contained) by medial, and, furthermore, the (sum of the) squares on them incommensurable with twice the (rectangle contained) by them. Thus, LN is the irrational (straight-line) called that which with a medial (area) makes a medial whole [Prop. 10.78]. And it is the square-root of area AB .

Thus, the square-root of area (AB) is that (straight-line) which with a medial (area) makes a medial whole. (Which is) the very thing it was required to show.

97.

Τὸ ἀπὸ ἀποτομῆς παρὰ <ρητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν πρώτην.

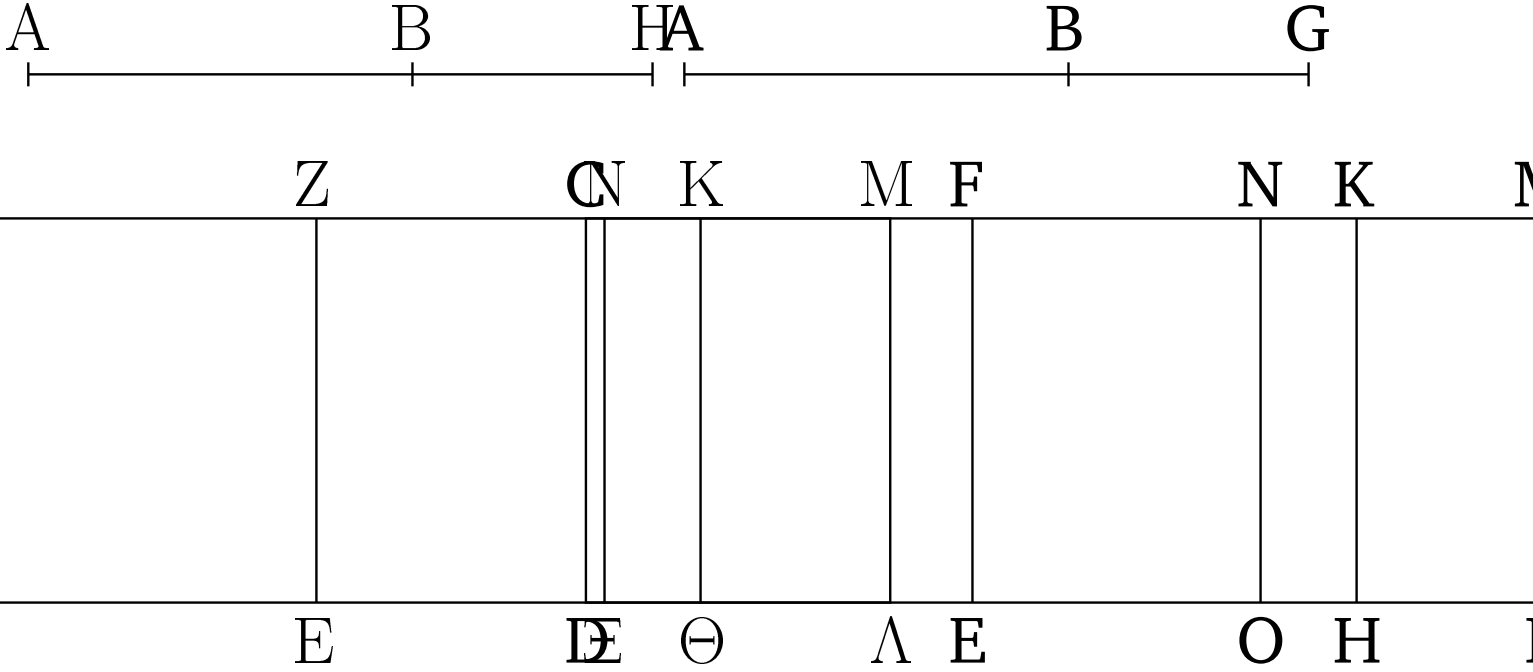
>Ἐστω ἀποτομή ἡ ΑΒ, <ρητὴ δὲ ἡ ΓΔ, καὶ τῷ ἀπὸ τῆς ΑΒ >ίσον παρὰ τὴν ΓΔ παραβεβλήσθω τὸ ΓΕ πλάτος ποιοῦν τὴν

Proposition 97

The (square) on an apotome, applied to a rational (straight-line), produces a first apotome as breadth.

Let AB be an apotome, and CD a rational (straight-line). And let CE , equal to the (square) on AB , be applied to CD ,

ΓΖ; λέγω, <ὅτι ἡ ΓΖ ἀποτομή ἐστι πρώτη.

producing CF as breadth. I say that CF is a first apotome.

>Ἐστω γὰρ τῇ| AB προσαρμόζουσα ἡ BH ; αἱ >άρα AH , HB <ρηταί εἰσι δυνάμει μόνον σύμμετροι. καὶ τῷ| μὲν ἀπὸ τῆς AH >ίσον παρὰ τὴν $\Gamma\Delta$ παραβεβλήσθω τὸ $\Gamma\Theta$, τῷ| δὲ ἀπὸ τῆς BH τὸ $ΚΛ$. <όλον >άρα τὸ $\Gamma\Lambda$ >ίσον ἐστὶ τοῖς ἀπὸ τῶν AH , HB ; <ὦν τὸ $\Gamma\Theta$ >ίσον ἐστὶ τῷ| ἀπὸ τῆς AB ; λοιπὸν >άρα τὸ $ΖΛ$ >ίσον ἐστὶ τῷ| δις ὑπὸ τῶν AH , HB . τετμήσθω ἡ ZM δίθω κατὰ τὸ N σημεῖον, καὶ >ήθθω διὰ τοῦ N τῇ| $\Gamma\Delta$ παράλληλος ἡ NX ; ἐκάτερον >άρα τῶν ZX , ΛN >ίσον ἐστὶ τῷ| ὑπὸ τῶν AH , HB . καὶ ἐπεὶ τὰ ἀπὸ τῶν AH , HB <ρητά ἐστιν, καὶ ἐστὶ τοῖς ἀπὸ τῶν AH , HB >ίσον τὸ ΔM , <ρητὸν >άρα ἐστὶ τὸ ΔM . καὶ παρὰ <ρητὴν τὴν $\Gamma\Delta$ παραβεβλήται πλάτος ποιοῦν τὴν ΓM ; <ρητὴ >άρα ἐστὶν ἡ ΓM καὶ σύμμετρος τῇ| $\Gamma\Delta$ μήκει. πάλιν, ἐπεὶ μέσον ἐστὶ τὸ δις ὑπὸ τῶν AH , HB , καὶ τῷ| δις ὑπὸ τῶν AH , HB >ίσον τὸ $ΖΛ$, μέσον >άρα τὸ $ΖΛ$. καὶ παρὰ <ρητὴν τὴν $\Gamma\Delta$ παράκειται πλάτος ποιοῦν τὴν ZM ; <ρητὴ >άρα ἐστὶν ἡ ZM καὶ ἀσύμμετρος τῇ| $\Gamma\Delta$ μήκει. καὶ ἐπεὶ τὰ μὲν ἀπὸ τῶν AH , HB <ρητά ἐστιν, τὸ δὲ δις ὑπὸ τῶν AH , HB μέσον, ἀσύμμετρα >άρα ἐστὶ τὰ ἀπὸ τῶν AH , HB τῷ| δις ὑπὸ τῶν AH , HB . καὶ τοῖς μὲν ἀπὸ τῶν AH , HB >ίσον ἐστὶ τὸ $\Gamma\Lambda$, τῷ| δὲ δις ὑπὸ τῶν AH , HB τὸ $ΖΛ$; ἀσύμμετρον >άρα ἐστὶ τὸ ΔM τῷ| $ΖΛ$. ὥς δὲ τὸ ΔM πρὸς τὸ $ΖΛ$, ο<ύτως ἐστὶν ἡ ΓM πρὸς τὴν ZM . ἀσύμμετρος >άρα ἐστὶν ἡ ΓM τῇ| ZM μήκει. καὶ εἰσιν ἀμφοτέραι <ρηταί; αἱ >άρα ΓM , MZ <ρηταί εἰσι δυνάμει μόνον σύμμετροι; ἡ ΓZ >άρα ἀποτομή ἐστὶν. λέγω δὴ, <ὅτι καὶ πρώτη.

Ἐπεὶ γὰρ τῶν ἀπὸ τῶν AH , HB μέσον ἀνάλογόν ἐστι τὸ ὑπὸ τῶν AH , HB , καὶ ἐστὶ τῷ| μὲν ἀπὸ τῆς AH >ίσον τὸ $\Gamma\Theta$, τῷ| δὲ ἀπὸ τῆς BH >ίσον τὸ $ΚΛ$, τῷ| δὲ ὑπὸ τῶν AH , HB τὸ $N\Lambda$, καὶ τῶν $\Gamma\Theta$, $ΚΛ$ >άρα μέσον ἀνάλογόν ἐστι τὸ $N\Lambda$; >έστιν >άρα ὡς τὸ $\Gamma\Theta$ πρὸς τὸ $N\Lambda$, ο<ύτως τὸ $N\Lambda$ πρὸς τὸ $ΚΛ$. ἄλλ> ὡς μὲν τὸ $\Gamma\Theta$ πρὸς τὸ $N\Lambda$, ο<ύτως ἐστὶν ἡ ΓK πρὸς τὴν NM ; ὡς δὲ τὸ $N\Lambda$ πρὸς τὸ $ΚΛ$, ο<ύτως ἐστὶν ἡ NM πρὸς τὴν KM ; τὸ >άρα ὑπὸ τῶν ΓK , KM >ίσον ἐστὶ τῷ| ἀπὸ τῆς NM , τουτέστι τῷ| τετάρτῳ μέρει τοῦ ἀπὸ τῆς ZM . καὶ ἐπεὶ σύμμετρόν ἐστι τὸ ἀπὸ τῆς AH τῷ| ἀπὸ τῆς HB , σύμμετρόν [ἐστὶ] καὶ τὸ $\Gamma\Theta$ τῷ| $ΚΛ$. ὥς δὲ τὸ $\Gamma\Theta$ πρὸς τὸ $ΚΛ$, ο<ύτως ἡ ΓK πρὸς τὴν KM ; σύμμετρος >άρα ἐστὶν

For let BG be an attachment to AB . Thus, AG and GB are rational (straight-lines which are) commensurable in square only [Prop. 10.73]. And let CH , equal to the (square) on AG , and KL , (equal) to the (square) on BG , be applied to CD . Thus, the whole of CL is equal to the (sum of the squares) on AG and GB , of which CE is equal to the (square) on AB . The remainder FL is thus equal to twice the (rectangle contained) by AG and GB [Prop. 2.7]. Let FM be cut in half at point N . And let NO be drawn through N , parallel to CD . Thus, FO and LN are each equal to the (rectangle contained) by AG and GB . And since the (sum of the squares) on AG and GB is rational, and DM is equal to the (sum of the squares) on AG and GB , DM is thus rational. And it has been applied to the rational (straight-line) CD , producing CM as breadth. Thus, CM is rational, and commensurable in length with CD [Prop. 10.20]. Again, since twice the (rectangle contained) by AG and GB is medial, and FL (is) equal to twice the (rectangle contained) by AG and GB , FL (is) thus a medial (area). And it is applied to the rational (straight-line) CD , producing FM as breadth. FM is thus rational, and incommensurable in length with CD [Prop. 10.22]. And since the (sum of the squares) on AG and GB is rational, and twice the (rectangle contained) by AG and GB medial, the (sum of the squares) on AG and GB is thus incommensurable with twice the (rectangle contained) by AG and GB . And CL is equal to the (sum of the squares) on AG and GB , and FL to twice the (rectangle contained) by AG and GB . DM is thus incommensurable with FL . And as DM (is) to FL , so CM is to FM [Prop. 6.1]. CM is thus incommensurable in length with FM [Prop. 10.11]. And both are rational (straight-lines). Thus, CM and MF are rational (straight-lines which are) commensurable in square only. CF is thus an apotome [Prop. 10.73]. So, I say that (it is) also a first (apotome).

For since the (rectangle contained) by AG and GB is the mean proportional to the (squares) on AG and GB [Prop. 10.21 lem.], and CH is equal to the (square) on AG ,

ἡ ΓΚ τῇ ΚΜ. ἐπεὶ οὖν δύο εὐθεῖαι >άνισοι εἰσιν αἱ ΓΜ, ΜΖ, καὶ τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ΖΜ >ίσον παρὰ τὴν ΓΜ παραβέβληται ἐλλείπον ε>ίδει τετραγώνῳ τὸ ὑπὸ τῶν ΓΚ, ΚΜ, καὶ ἐστὶ σύμμετρος ἡ ΓΚ τῇ ΚΜ, ἡ >άρα ΓΜ τῆς ΜΖ μείζον δύναται τῷ ἀπὸ συμέτρου ἐαυτῇ μήκει. καὶ ἐστὶν ἡ ΓΜ σύμμετρος τῇ ἐκκειμένη <ρητῇ τῇ ΓΔ μήκει, ἡ >άρα ΓΖ ἀποτομή ἐστὶ πρώτη.

Τὸ >άρα ἀπὸ ἀποτομῆς παρὰ <ρητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν πρώτην; <όπερ >έδει δεῖξαι.

and KL equal to the (square) on BG , and NL to the (rectangle contained) by AG and GB , NL is thus also the mean proportional to CH and KL . Thus, as CH is to NL , so NL (is) to KL . But, as CH (is) to NL , so CK is to NM , and as NL (is) to KL , so NM is to KM [Prop. 6.1]. Thus, the (rectangle contained) by CK and KM is equal to the (square) on NM —that is to say, to the fourth part of the (square) on FM [Prop. 6.17]. And since the (square) on AG is commensurable with the (square) on GB , CH [is] also commensurable with KL . And as CH (is) to KL , so CK (is) to KM [Prop. 6.1]. CK is thus commensurable (in length) with KM [Prop. 10.11]. Therefore, since CM and MF are two unequal straight-lines, and the (rectangle contained) by CK and KM , equal to the fourth part of the (square) on FM , has been applied to CM , falling short by a square figure, and CK is commensurable (in length) with KM , the square on CM is thus greater than (the square on) MF by the (square) on (some straight-line) commensurable in length with (CM) [Prop. 10.17]. And CM is commensurable in length with the (previously) laid down rational (straight-line) CD . Thus, CF is a first apotome [Def. 10.15].

Thus, the (square) on an apotome, applied to a rational (straight-line), produces a first apotome as breadth. (Which is) the very thing it was required to show.

98.

Proposition 98

Τὸ ἀπὸ μέσης ἀποτομῆς πρώτης παρὰ <ρητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν δευτέραν.

>Ἐστω μέσης ἀποτομῆς πρώτη ἡ ΑΒ, <ρητὴ δὲ ἡ ΓΔ, καὶ τῷ ἀπὸ τῆς ΑΒ >ίσον παρὰ τὴν ΓΔ παραβέβλησθω τὸ ΓΕ πλάτος ποιοῦν τὴν ΓΖ; λέγω, <ότι ἡ ΓΖ ἀποτομή ἐστὶ δευτέρα.

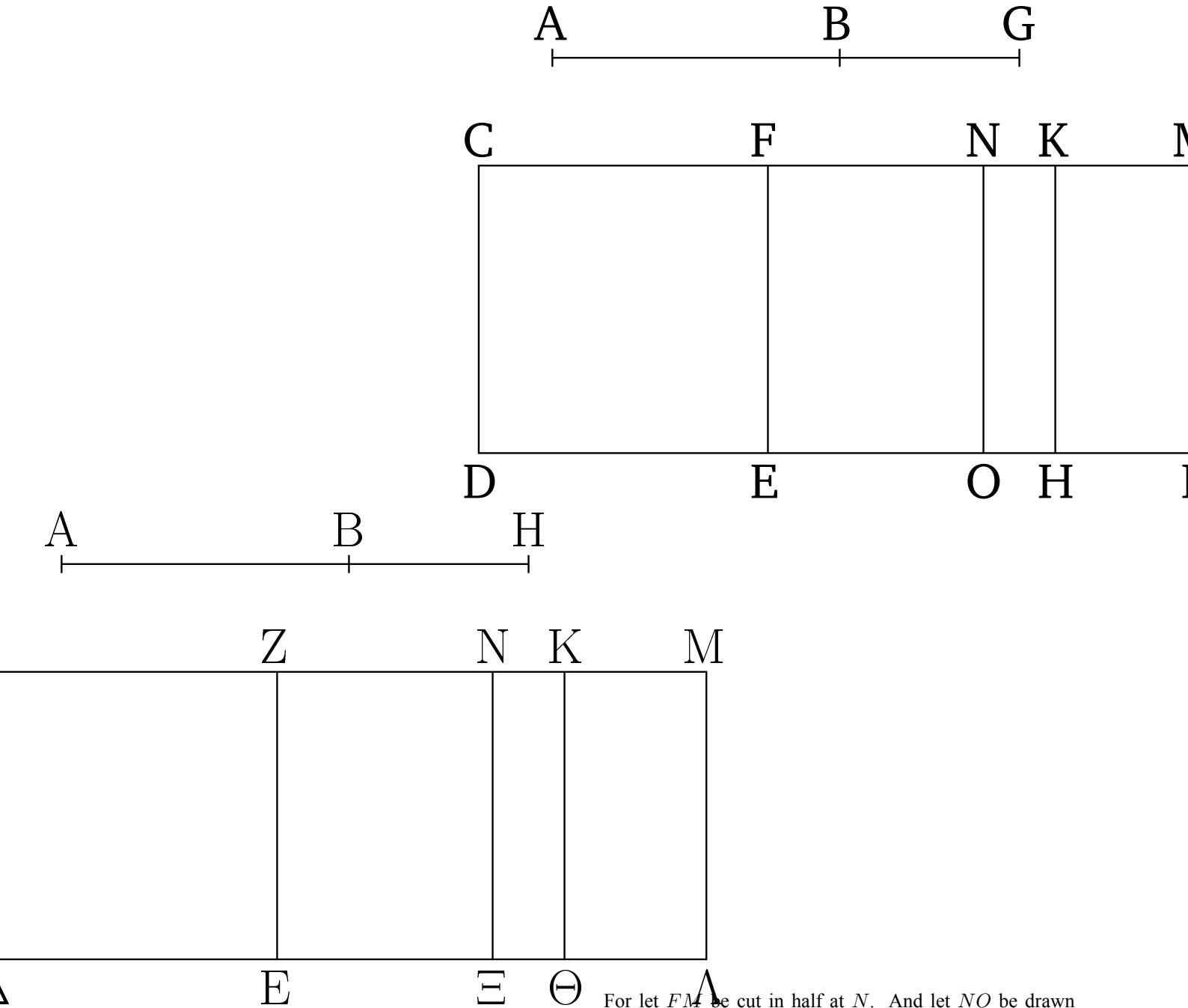
>Ἐστω γάρ τῇ ΑΒ προσαρμόζουσα ἡ ΒΗ; αἱ >άρα ΑΗ, ΗΒ μέσαι εἰσὶ δυνάμει μόνον σύμμετροι <ρητὸν περιέθουσαι. καὶ τῷ μὲν ἀπὸ τῆς ΑΗ >ίσον παρὰ τὴν ΓΔ παραβέβλησθω τὸ ΓΘ πλάτος ποιοῦν τὴν ΓΚ, τῷ δὲ ἀπὸ τῆς ΗΒ >ίσον τὸ ΚΛ πλάτος ποιοῦν τὴν ΚΜ; <όλον >άρα τὸ ΓΛ >ίσον ἐστὶ τοῖς ἀπὸ τῶν ΑΗ, ΗΒ; μέσον >άρα καὶ τὸ ΓΛ. καὶ παρὰ <ρητὴν τὴν ΓΔ παράκειται πλάτος ποιοῦν τὴν ΓΜ; <ρητὴ >άρα ἐστὶν ἡ ΓΜ καὶ ἀσύμμετρος τῇ ΓΔ μήκει. καὶ ἐπεὶ τὸ ΓΛ >ίσον ἐστὶ τοῖς ἀπὸ τῶν ΑΗ, ΗΒ, <ὢν τὸ ἀπὸ τῆς ΑΒ >ίσον ἐστὶ τῷ ΓΕ, λοιπὸν >άρα τὸ διζ ὑπὸ τῶν ΑΗ, ΗΒ >ίσον ἐστὶ τῷ ΖΑ. <ρητὸν δὲ [ἐστὶ] τὸ διζ ὑπὸ τῶν ΑΗ, ΗΒ; <ρητὸν >άρα τὸ ΖΑ. καὶ παρὰ <ρητὴν τὴν ΖΕ παράκειται πλάτος ποιοῦν τὴν ΖΜ; <ρητὴ >άρα ἐστὶ καὶ ἡ ΖΜ καὶ σύμμετρος τῇ ΓΔ μήκει. ἐπεὶ οὖν τὰ μὲν ἀπὸ τῶν ΑΗ, ΗΒ, τουτέστι τὸ ΓΛ, μέσον ἐστίν, τὸ δὲ διζ ὑπὸ τῶν ΑΗ, ΗΒ, τουτέστι τὸ ΖΑ, <ρητὸν ἀσύμμετρον >άρα ἐστὶ τὸ ΓΛ τῷ ΖΑ. ὥς δὲ τὸ ΓΛ πρὸς τὸ ΖΑ, ο<ύτως ἐστὶν ἡ ΓΜ πρὸς τὴν ΖΜ; ἀσύμμετρος >άρα ἡ ΓΜ τῇ ΖΜ μήκει. καὶ εἰσιν ἀμφοτέραι <ρηταί; αἱ >άρα ΓΜ, ΜΖ <ρηταί εἰσι δυνάμει μόνον σύμμετροι; ἡ ΓΖ >άρα ἀποτομή ἐστὶν. λέγω δὴ, <ότι καὶ δευτέρα.

The (square) on a first apotome of a medial (straight-line), applied to a rational (straight-line), produces a second apotome as breadth.

Let AB be a first apotome of a medial (straight-line), and CD a rational (straight-line). And let CE , equal to the (square) on AB , be applied to CD , producing CF as breadth. I say that CF is a second apotome.

For let BG be an attachment to AB . Thus, AG and GB are medial (straight-lines which are) commensurable in square only, containing a rational (area) [Prop. 10.74]. And let CH , equal to the (square) on AG , be applied to CD , producing CK as breadth, and KL , equal to the (square) on GB , producing KM as breadth. Thus, the whole of CL is equal to the (sum of the squares) on AG and GB . Thus, CL (is) also a medial (area) [Props. 10.15, 10.23 corr.]. And it is applied to the rational (straight-line) CD , producing CM as breadth. CM is thus rational, and incommensurable in length with CD [Prop. 10.22]. And since CL is equal to the (sum of the squares) on AG and GB , of which the (square) on AB is equal to CE , the remainder, twice the (rectangle contained) by AG and GB , is thus equal to FL [Prop. 2.7]. And twice the (rectangle contained) by AG and GB [is] rational. Thus, FL (is) rational. And it is applied to the rational (straight-line) FE , producing FM as breadth. FM is thus also rational, and commensurable in length with CD [Prop. 10.20]. Therefore, since the (sum of the squares) on AG and GB —that is to say, CL —is medial, and twice the (rectangle contained) by AG and GB —that is to say, FL —(is) rational, CL is thus incommensurable with FL . And as CL (is) to FL , so CM is to FM [Prop. 6.1]. Thus, CM (is) incommensurable in length with FM [Prop. 10.11]. And they are both rational (straight-lines). Thus, CM and MF are rational (straight-lines which

are) commensurable in square only. CF is thus an apotome [Prop. 10.73]. So, I say that (it is) also a second (apotome).



Τετμήσθω γὰρ ἡ ZM δίθρα κατὰ τὸ N , καὶ >ήθθω διὰ τοῦ N τῇ $\Gamma\Delta$ παράλληλοξ ἡ NX ; ἐκάτερον >άρα τῶν ZX , $ΝΛ$ >ίσον ἐστὶ τῷ ὑπὸ τῶν AH , HB . καὶ ἐπεὶ τῶν ἀπὸ τῶν AH , HB τετραγώνων μέσον ἀνάλογόν ἐστι τὸ ὑπὸ τῶν AH , HB , καὶ ἐστὶν >ίσον τὸ μὲν ἀπὸ τῆς AH τῷ $\Gamma\Theta$, τὸ δὲ ὑπὸ τῶν AH , HB τῷ $ΝΛ$, τὸ δὲ ἀπὸ τῆς BH τῷ $ΚΛ$, καὶ τῶν $\Gamma\Theta$, $ΚΛ$ >άρα μέσον ἀνάλογόν ἐστι τὸ $ΝΛ$; >έστιν >άρα ὡς τὸ $\Gamma\Theta$ πρὸς τὸ $ΝΛ$, ο<ύτωξ τὸ $ΝΛ$ πρὸς τὸ $ΚΛ$. ἀλλ> ὡς μὲν τὸ $\Gamma\Theta$ πρὸς τὸ $ΝΛ$, ο<ύτωξ ἐστὶν ἡ $\GammaΚ$ πρὸς τὴν NM , ὡς δὲ τὸ $ΝΛ$ πρὸς τὸ $ΚΛ$, ο<ύτωξ ἐστὶν ἡ NM πρὸς τὴν MK ; ὡς >άρα ἡ $\GammaΚ$ πρὸς τὴν NM , ο<ύτωξ ἐστὶν ἡ NM πρὸς τὴν KM ; τὸ >άρα ὑπὸ τῶν $\GammaΚ$, KM >ίσον ἐστὶ τῷ ἀπὸ τῆς NM , τουτέστι τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ZM [καὶ ἐπεὶ σύμμετρόν ἐστι τὸ ἀπὸ τῆς AH τῷ ἀπὸ τῆς BH , σύμμετρόν ἐστι καὶ τὸ $\Gamma\Theta$ τῷ $ΚΛ$, τουτέστιν ἡ $\GammaΚ$ τῇ KM]. ἐπεὶ ο>ὗν δύο εὐθεῖαι >άνισοί εἰσιν αἱ ΓM , MZ , καὶ τῷ

For let FM be cut in half at N . And let NO be drawn through (point) N , parallel to CD . Thus, FO and NL are each equal to the (rectangle contained) by AG and GB . And since the (rectangle contained) by AG and GB is the mean proportional to the squares on AG and GB [Prop. 10.21 lem.], and the (square) on AG is equal to CH , and the (rectangle contained) by AG and GB to NL , and the (square) on BG to KL , NL is thus also the mean proportional to CH and KL . Thus, as CH is to NL , so NL (is) to KL [Prop. 5.11]. But, as CH (is) to NL , so CK is to NM , and as NL (is) to KL , so NM is to MK [Prop. 6.1]. Thus, as CK (is) to NM , so NM is to KM [Prop. 5.11]. The (rectangle contained) by CK and KM is thus equal to the (square) on NM [Prop. 6.17]—that is to say, to the fourth part of the (square) on FM [and since the (square) on AG is commensurable with the (square) on BG , CH is also commensurable with KL —that is to say, CK with KM].

τετάρτῳ μέρει τοῦ ἀπὸ τῆς MZ ὕψους παρὰ τὴν μείζονα τὴν $ΓΜ$ παραβέβληται ἔλλειπον εἰς τετραγώνῳ τὸ ὑπὸ τῶν $ΓΚ, ΚΜ$ καὶ εἰς σύμμετρα αὐτὴν διαιρεῖ, ἢ ὅρα $ΓΜ$ τῆς MZ μείζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ μήκει. καὶ ἔστιν ἢ προσαρμόζουσα ἢ ZM σύμμετρος μήκει τῇ ἐκκειμένῃ <ρητῇ τῇ $ΓΔ$; ἢ ὅρα $ΓΖ$ ἀποτομή ἐστὶ δευτέρα.

Τὸ ὅρα ἀπὸ μέσης ἀποτομῆς πρώτης παρὰ <ρητὴν παραβαλλόμενῃ πλάτοξ ποιεῖ ἀποτομὴν δευτέραν; <όπερ >έδει δεῖχαι.

Therefore, since CM and MF are two unequal straight-lines, and the (rectangle contained) by CK and KM , equal to the fourth part of the (square) on MF , has been applied to the greater CM , falling short by a square figure, and divides it into commensurable (parts), the square on CM is thus greater than (the square on) MF by the (square) on CF (straight-line) commensurable in length with (CM) [Prop. 10.17]. The attachment FM is also commensurable in length with the (previously) laid down rational (straight-line) CD . CF is thus a second apotome [Def. 10.16].

Thus, the (square) on a first apotome of a medial (straight-line), applied to a rational (straight-line), produces a second apotome as breadth. (Which is) the very thing it was required to show.

99.

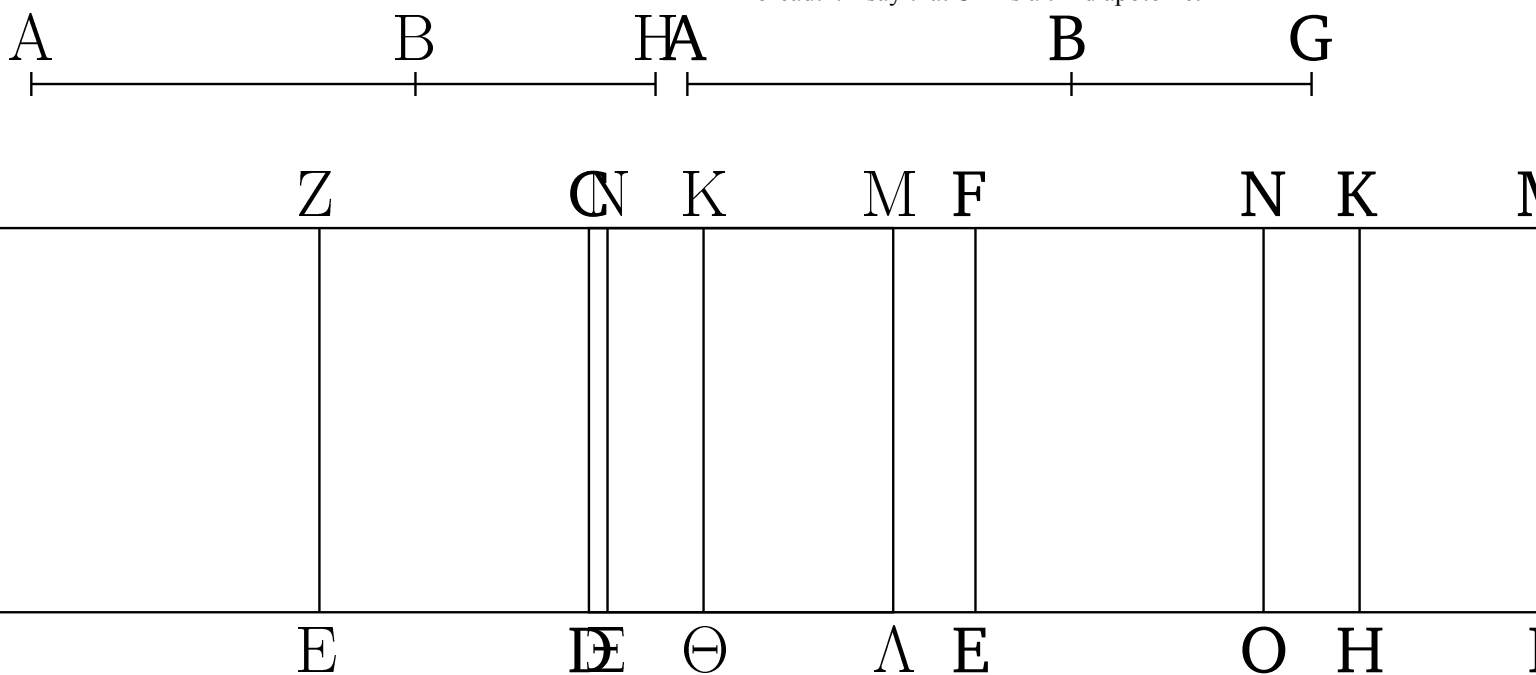
Proposition 99

Τὸ ἀπὸ μέσης ἀποτομῆς δευτέρας παρὰ <ρητὴν παραβαλλόμενῃ πλάτοξ ποιεῖ ἀποτομὴν τρίτην.

>Ἐστω μέσης ἀποτομῆς δευτέρα ἡ AB , <ρητὴ δὲ ἡ $ΓΔ$, καὶ τῷ ἀπὸ τῆς AB ὕψους παρὰ τὴν $ΓΔ$ παραβελήσθω τὸ $ΓΕ$ πλάτοξ ποιοῦν τὴν $ΓΖ$; λέγω, <ὅτι ἡ $ΓΖ$ ἀποτομή ἐστὶ τρίτη.

The (square) on a second apotome of a medial (straight-line), applied to a rational (straight-line), produces a third apotome as breadth.

Let AB be the second apotome of a medial (straight-line), and CD a rational (straight-line). And let CE , equal to the (square) on AB , be applied to CD , producing CF as breadth. I say that CF is a third apotome.



>Ἐστω γὰρ τῇ AB προσαρμόζουσα ἡ BH ; αἱ ὅρα AH, HB μέσαι εἰσὶ δυνάμει μόνον σύμμετροι μέσον περιέθουσαι. καὶ τῷ μὲν ἀπὸ τῆς AH ὕψους παρὰ τὴν $ΓΔ$ παραβελήσθω τὸ $ΓΘ$ πλάτοξ ποιοῦν τὴν $ΓΚ$, τῷ δὲ ἀπὸ τῆς BH ὕψους παρὰ τὴν $ΚΘ$ παραβελήσθω τὸ $ΚΛ$ πλάτοξ ποιοῦν τὴν $ΚΜ$; <όλον >όρα τὸ $ΓΛ$ ὕψους ἐστὶ τοῖς ἀπὸ τῶν AH, HB [καὶ ἐστὶ μέσα τὰ ἀπὸ τῶν AH, HB]; μέσον ὅρα καὶ τὸ $ΓΛ$. καὶ παρὰ <ρητὴν τὴν $ΓΔ$ παραβέβληται πλάτοξ ποιοῦν τὴν $ΓΜ$; <ρητὴ ὅρα ἐστὶν ἡ $ΓΜ$ καὶ ἀσύμμετρος τῇ $ΓΔ$ μήκει. καὶ ἐπεὶ <όλον τὸ $ΓΛ$ ὕψους ἐστὶ τοῖς ἀπὸ τῶν AH, HB , <ὢν τὸ $ΓΕ$ ὕψους ἐστὶ τῷ ἀπὸ τῆς AB , λοιπὸν ὅρα τὸ $ΛΖ$ ὕψους ἐστὶ τῷ δι᾽ ὑπὸ τῶν AH, HB . τεμήσθω οὖν ἡ ZM δι᾽ ἡ κατὰ τὸ N σημεῖον, καὶ τῇ $ΓΔ$ παράλληλος ἢ $ΝΧ$; ἑκάτερον ὅρα τῶν $ZX, ΝΛ$ ὕψους ἐστὶ τῷ ὑπὸ τῶν AH, HB . μέσον δὲ τὸ ὑπὸ τῶν AH, HB ; μέσον ὅρα ἐστὶ καὶ τὸ $ΖΛ$. καὶ παρὰ <ρητὴν τὴν EZ παράκειται πλάτοξ ποιοῦν τὴν ZM ;

For let BG be an attachment to AB . Thus, AG and GB are medial (straight-lines which are) commensurable in square only, containing a medial (area) [Prop. 10.75]. And let CH , equal to the (square) on AG , be applied to CD , producing CK as breadth. And let KL , equal to the (square) on BG , be applied to KH , producing KM as breadth. Thus, the whole of CL is equal to the (sum of the squares) on AG and GB [and the (sum of the squares) on AG and GB is medial]. CL (is) thus also medial [Props. 10.15, 10.23 corr.]. And it has been applied to the rational (straight-line) CD , producing CM as breadth. Thus, CM is rational, and incommensurable in length with CD [Prop. 10.22]. And since the whole of CL is equal to the (sum of the squares) on AG and GB , of which CE is equal to the (square) on AB , the remainder LF is thus equal to twice the (rectangle contained) by AG and GB [Prop. 2.7]. Therefore, let FM be cut in

<ρητὴ >άρα καὶ ἡ ZM καὶ ἀσύμμετρος τῇ $\Gamma\Delta$ μήκει. καὶ ἐπεὶ αἱ AH , HB δυνάμει μόνον εἰσὶ σύμμετροι, ἀσύμμετρος >άρα [ἐστὶ] μήκει ἡ AH τῇ HB ; ἀσύμμετρον >άρα ἐστὶ καὶ τὸ ἀπὸ τῆς AH τῷ ὑπὸ τῶν AH , HB . ἀλλὰ τῷ μὲν ἀπὸ τῆς AH σύμμετρά ἐστι τὰ ἀπὸ τῶν AH , HB , τῷ δὲ ὑπὸ τῶν AH , HB τὸ δις ὑπὸ τῶν AH , HB ; ἀσύμμετρα >άρα ἐστὶ τὰ ἀπὸ τῶν AH , HB τῷ δις ὑπὸ τῶν AH , HB . ἀλλὰ τοῖς μὲν ἀπὸ τῶν AH , HB >ίσον ἐστὶ τὸ $\Gamma\Lambda$, τῷ δὲ δις ὑπὸ τῶν AH , HB >ίσον ἐστὶ τὸ $Z\Lambda$; ἀσύμμετρον >άρα ἐστὶ τὸ $\Gamma\Lambda$ τῷ $Z\Lambda$. ὥς δὲ τὸ $\Gamma\Lambda$ πρὸς τὸ $Z\Lambda$, ο<ύτωξ ἐστὶν ἡ ΓM πρὸς τὴν ZM ; ἀσύμμετρος >άρα ἐστὶν ἡ ΓM τῇ ZM μήκει. καὶ εἰσιν ἀμφοτέραι <ρηταί; αἱ >άρα ΓM , MZ <ρηταί εἰσι δυνάμει μόνον σύμμετροι; ἀποτομὴ >άρα ἐστὶν ἡ ΓZ . λέγω δὴ, <ότι καὶ τρίτη.

Ἐπεὶ γὰρ σύμμετρόν ἐστι τὸ ἀπὸ τῆς AH τῷ ἀπὸ τῆς HB , σύμμετρον >άρα καὶ τὸ $\Gamma\Theta$ τῷ $K\Lambda$; <ώστε καὶ ἡ ΓK τῇ KM . καὶ ἐπεὶ τῶν ἀπὸ τῶν AH , HB μέσον ἀνάλογόν ἐστι τὸ ὑπὸ τῶν AH , HB , καὶ ἐστὶ τῷ μὲν ἀπὸ τῆς AH >ίσον τὸ $\Gamma\Theta$, τῷ δὲ ἀπὸ τῆς HB >ίσον τὸ $K\Lambda$, τῷ δὲ ὑπὸ τῶν AH , HB >ίσον τὸ $N\Lambda$, καὶ τῶν $\Gamma\Theta$, $K\Lambda$ >άρα μέσον ἀνάλογόν ἐστι τὸ $N\Lambda$; >έστιν >άρα ὡς τὸ $\Gamma\Theta$ πρὸς τὸ $N\Lambda$, ο<ύτωξ τὸ $N\Lambda$ πρὸς τὸ $K\Lambda$. ἀλλ> ὡς μὲν τὸ $\Gamma\Theta$ πρὸς τὸ $N\Lambda$, ο<ύτωξ ἐστὶν ἡ ΓK πρὸς τὴν NM , ὡς δὲ τὸ $N\Lambda$ πρὸς τὸ $K\Lambda$, ο<ύτωξ ἐστὶν ἡ NM πρὸς τὴν KM ; ὡς >άρα ἡ ΓK πρὸς τὴν MN , ο<ύτωξ ἐστὶν ἡ MN πρὸς τὴν KM ; τὸ >άρα ὑπὸ τῶν ΓK , KM >ίσον ἐστὶ τῷ [ἀπὸ τῆς MN , τουτέστι τῷ] τετάρτῳ μέρει τοῦ ἀπὸ τῆς ZM . ἐπεὶ ο>ύν δύο εὐθεῖαι >άνισοι εἰσιν αἱ ΓM , MZ , καὶ τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ZM >ίσον παρὰ τὴν ΓM παραβέβληται ἐλλείπον ε>ίδει τετραγώνῳ καὶ εἰς σύμμετρα αὐτὴν διαιρεῖ, ἡ ΓM >άρα τῆς MZ μείζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ. καὶ οὐδετέρα τῶν ΓM , MZ σύμμετρόξ ἐστὶ μήκει τῇ ἐκκειμένη <ρητῇ τῇ $\Gamma\Delta$; ἡ >άρα ΓZ ἀποτομὴ ἐστὶ τρίτη.

Τὸ >άρα ἀπὸ μέσηξ ἀποτομῆξ δευτέραξ παρὰ <ρητὴν παραβαλλόμενον πλάτοξ ποιεῖ ἀποτομὴν τρίτην; <όπερ >έδει δεῖχαι.

half at point N . And let NO be drawn parallel to CD . Thus, FO and NL are each equal to the (rectangle contained) by AG and GB . And the (rectangle contained) by AG and GB (is) medial. Thus, FL is also medial. And it is applied to the rational (straight-line) EF , producing FM as breadth. FM is thus rational, and incommensurable in length with CD [Prop. 10.22]. And since AG and GB are commensurable in square only, AG [is] thus incommensurable in length with GB . Thus, the (square) on AG is also incommensurable with the (rectangle contained) by AG and GB [Props. 6.1, 10.11]. But, the (sum of the squares) on AG and GB is commensurable with the (square) on AG , and twice the (rectangle contained) by AG and GB with the (rectangle contained) by AG and GB . The (sum of the squares) on AG and GB is thus incommensurable with twice the (rectangle contained) by AG and GB [Prop. 10.13]. But, CL is equal to the (sum of the squares) on AG and GB , and FL is equal to twice the (rectangle contained) by AG and GB . Thus, CL is incommensurable with FL . And as CL (is) to FL , so CM is to FM [Prop. 6.1]. CM is thus incommensurable in length with FM [Prop. 10.11]. And they are both rational (straight-lines). Thus, CM and MF are rational (straight-lines which are) commensurable in square only. CF is thus an apotome [Prop. 10.73]. So, I say that (it is) also a third (apotome).

For since the (square) on AG is commensurable with the (square) on GB , CH (is) thus also commensurable with KL . Hence, CK (is) also (commensurable in length) with KM [Props. 6.1, 10.11]. And since the (rectangle contained) by AG and GB is the mean proportional to the (squares) on AG and GB [Prop. 10.21 lem.], and CH is equal to the (square) on AG , and KL equal to the (square) on GB , and NL equal to the (rectangle contained) by AG and GB , NL is thus also the mean proportional to CH and KL . Thus, as CH is to NL , so NL (is) to KL . But, as CH (is) to NL , so CK is to NM , and as NL (is) to KL , so NM (is) to KM [Prop. 6.1]. Thus, as CK (is) to MN , so MN is to KM [Prop. 5.11]. Thus, the (rectangle contained) by CK and KM is equal to the [(square) on MN —that is to say, to the] fourth part of the (square) on FM [Prop. 6.17]. Therefore, since CM and MF are two unequal straight-lines, and (some area), equal to the fourth part of the (square) on FM , has been applied to CM , falling short by a square figure, and divides it into commensurable (parts), the square on CM is thus greater than (the square on) MF by the (square) on (some straight-line) commensurable (in length) with (CM) [Prop. 10.17]. And neither of CM and MF is commensurable in length with the (previously) laid down rational (straight-line) CD . CF is thus a third apotome [Def. 10.13].

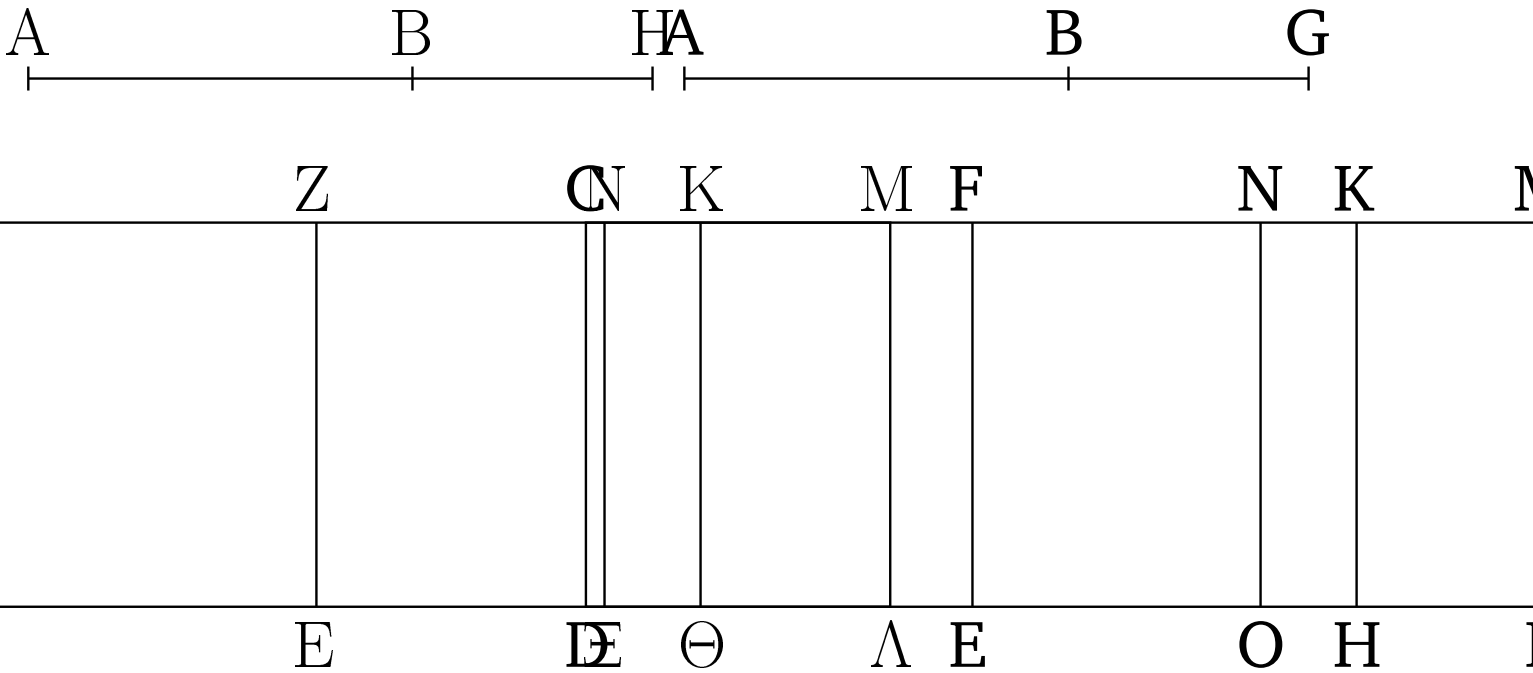
Thus, the (square) on a second apotome of a medial (straight-line), applied to a rational (straight-line), produces a third apotome as breadth. (Which is) the very thing it was required to show.

100.

Τὸ ἀπὸ ἐλάσσονοξ παρὰ <ρητὴν παραβαλλόμενον πλάτοξ ποιεῖ ἀποτομὴν τετάρτην.

Proposition 100

The (square) on a minor (straight-line), applied to a rational (straight-line), produces a fourth apotome as breadth.



>Ἐστω ἐλάσσων ἡ AB , <ρητὴ δὲ ἡ $ΓΔ$, καὶ τῷ ἀπὸ τῆς AB >ίσον παρὰ <ρητὴν τὴν $ΓΔ$ παραβεβλήσθω τὸ $ΓΕ$ πλάτοξ ποιοῦν τὴν $ΓΖ$; λέγω, <ὅτι ἡ $ΓΖ$ ἀποτομή ἐστὶ τετάρτη.

>Ἐστω γάρ τῃ AB προσαρμόζουσα ἡ BH ; αἱ >άρα AH , HB δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν AH , HB τετραγώνων <ρητόν, τὸ δὲ διζὺ ὑπὸ τῶν AH , HB μέσον. καὶ τῷ μὲν ἀπὸ τῆς AH >ίσον παρὰ τὴν $ΓΔ$ παραβεβλήσθω τὸ $ΓΘ$ πλάτοξ ποιοῦν τὴν $ΓΚ$, τῷ δὲ ἀπὸ τῆς BH >ίσον τὸ $ΚΛ$ πλάτοξ ποιοῦν τὴν $ΚΜ$; <όλον >άρα τὸ $ΓΛ$ >ίσον ἐστὶ τοῖς ἀπὸ τῶν AH , HB . καὶ ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AH , HB <ρητόν; <ρητόν >άρα ἐστὶ καὶ τὸ $ΓΛ$. καὶ παρὰ <ρητὴν τὴν $ΓΔ$ παράκειται πλάτοξ ποιοῦν τὴν $ΓΜ$; <ρητὴ >άρα καὶ ἡ $ΓΜ$ καὶ σύμμετρος τῇ $ΓΔ$ μήκει. καὶ ἐπεὶ <όλον τὸ $ΓΛ$ >ίσον ἐστὶ τοῖς ἀπὸ τῶν AH , HB , <ὦν τὸ $ΓΕ$ >ίσον ἐστὶ τῷ ἀπὸ τῆς AB , λοιπὸν >άρα τὸ $ΖΛ$ >ίσον ἐστὶ τῷ διζὺ ὑπὸ τῶν AH , HB . τετμήσθω ο>ὺν ἡ ZM δίῃα κατὰ τὸ N σημεῖον, καὶ >ήθθω διὰ τοῦ N ὀποτέρῃ τῶν $ΓΔ$, $ΜΛ$ παράλληλοξ ἡ NX ; ἐκάτερον >άρα τῶν ZX , $NΛ$ >ίσον ἐστὶ τῷ ὑπὸ τῶν AH , HB . καὶ ἐπεὶ τὸ διζὺ ὑπὸ τῶν AH , HB μέσον ἐστὶ καὶ ἐστὶν >ίσον τῷ $ΖΛ$, καὶ τὸ $ΖΛ$ >άρα μέσον ἐστίν. καὶ παρὰ <ρητὴν τὴν ZE παράκειται πλάτοξ ποιοῦν τὴν ZM ; <ρητὴ >άρα ἐστὶν ἡ ZM καὶ ἀσύμμετρος τῇ $ΓΔ$ μήκει. καὶ ἐπεὶ τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν AH , HB <ρητόν ἐστίν, τὸ δὲ διζὺ ὑπὸ τῶν AH , HB μέσον, ἀσύμμετρα [>άρα] ἐστὶ τὰ ἀπὸ τῶν AH , HB τῷ διζὺ ὑπὸ τῶν AH , HB . >ίσον δὲ [ἐστὶ] τὸ $ΓΛ$ τοῖς ἀπὸ τῶν AH , HB , τῷ δὲ διζὺ ὑπὸ τῶν AH , HB >ίσον τὸ $ΖΛ$; ἀσύμμετρον >άρα [ἐστὶ] τὸ $ΓΛ$ τῷ $ΖΛ$. ὥς δὲ τὸ $ΓΛ$ πρὸς τὸ $ΖΛ$, ο<ύτωξ ἐστὶν ἡ $ΓΜ$ πρὸς τὴν MZ ; ἀσύμμετρος >άρα ἐστὶν ἡ $ΓΜ$ τῇ MZ μήκει. καὶ εἰσὶν ἀμφοτέραι <ρηταί; αἱ >άρα $ΓΜ$, MZ <ρηταί εἰσι δυνάμει μόνον σύμμετροι; ἀποτομή >άρα ἐστὶν ἡ $ΓΖ$. λέγω [δή], <ὅτι καὶ τετάρτη.

Ἐπεὶ γὰρ αἱ AH , HB δυνάμει εἰσὶν ἀσύμμετροι, ἀσύμμετρον >άρα καὶ τὸ ἀπὸ τῆς AH τῷ ἀπὸ τῆς HB . καὶ ἐστὶ τῷ μὲν ἀπὸ τῆς AH >ίσον τὸ $ΓΘ$, τῷ δὲ ἀπὸ τῆς HB >ίσον τὸ $ΚΛ$; ἀσύμμετρον >άρα ἐστὶ τὸ $ΓΘ$ τῷ $ΚΛ$. ὥς δὲ τὸ $ΓΘ$ πρὸς τὸ $ΚΛ$, ο<ύτωξ ἐστὶν ἡ $ΓΚ$ πρὸς τὴν $ΚΜ$; ἀσύμμετρος >άρα ἐστὶν ἡ $ΓΚ$ τῇ $ΚΜ$ μήκει. καὶ ἐπεὶ τῶν ἀπὸ τῶν AH , HB μέσον ἀνάλογόν ἐστὶ τὸ ὑπὸ τῶν AH , HB , καὶ ἐστὶν >ίσον τὸ μὲν ἀπὸ τῆς AH τῷ $ΓΘ$, τὸ δὲ ἀπὸ τῆς HB τῷ $ΚΛ$,

Let AB be a minor (straight-line), and CD a rational (straight-line). And let CE , equal to the (square) on AB , be applied to the rational (straight-line) CD , producing CF as breadth. I say that CF is a fourth apotome.

For let BG be an attachment to AB . Thus, AG and GB are incommensurable in square, making the sum of the squares on AG and GB rational, and twice the (rectangle contained) by AG and GB medial [Prop. 10.76]. And let CH , equal to the (square) on AG , be applied to CD , producing CK as breadth, and KL , equal to the (square) on BG , producing KM as breadth. Thus, the whole of CL is equal to the (sum of the squares) on AG and GB . And the sum of the (squares) on AG and GB is rational. CL is thus also rational. And it is applied to the rational (straight-line) CD , producing CM as breadth. Thus, CM (is) also rational, and commensurable in length with CD [Prop. 10.20]. And since the whole of CL is equal to the (sum of the squares) on AG and GB , of which CE is equal to the (square) on AB , the remainder FL is thus equal to twice the (rectangle contained) by AG and GB [Prop. 2.7]. Therefore, let FM be cut in half at point N . And let NO be drawn through N , parallel to either of CD or ML . Thus, FO and NL are each equal to the (rectangle contained) by AG and GB . And since twice the (rectangle contained) by AG and GB is medial, and is equal to FL , FL is thus also medial. And it is applied to the rational (straight-line) FE , producing FM as breadth. Thus, FM is rational, and incommensurable in length with CD [Prop. 10.22]. And since the sum of the (squares) on AG and GB is rational, and twice the (rectangle contained) by AG and GB medial, the (sum of the squares) on AG and GB is [thus] incommensurable with twice the (rectangle contained) by AG and GB . And CL (is) equal to the (sum of the squares) on AG and GB , and FL equal to twice the (rectangle contained) by AG and GB . CL [is] thus incommensurable with FL . And as CL (is) to FL , so CM is to MF [Prop. 6.1]. CM is thus incommensurable in length with MF [Prop. 10.11]. And both are rational (straight-lines). Thus, CM and MF are rational (straight-lines which are) commensurable in square only. CF is thus an apotome [Prop. 10.73]. [So], I say that (it is) also a fourth (apotome).

τὸ δὲ ὑπὸ τῶν AH , HB τῶν NA , τῶν $\gamma\alpha\rho\alpha$ $\Gamma\Theta$, $ΚΛ$ μέσον ἀνάλογόν ἐστι τὸ NA ; $\gamma\epsilon\sigma\tau\iota\nu$ $\gamma\alpha\rho\alpha$ ὡς τὸ $\Gamma\Theta$ πρὸς τὸ NA , οὕτως τὸ NA πρὸς τὸ $ΚΛ$. ἀλλ' ὡς μὲν τὸ $\Gamma\Theta$ πρὸς τὸ NA , οὕτως ἐστὶν ἡ $\GammaΚ$ πρὸς τὴν NM , ὡς δὲ τὸ NA πρὸς τὸ $ΚΛ$, οὕτως ἐστὶν ἡ NM πρὸς τὴν KM ; ὡς $\gamma\alpha\rho\alpha$ ἡ $\GammaΚ$ πρὸς τὴν MN , οὕτως ἐστὶν ἡ MN πρὸς τὴν KM ; τὸ $\gamma\alpha\rho\alpha$ ὑπὸ τῶν $\GammaΚ$, KM $\gamma\iota\sigma\omega\nu$ ἐστὶ τῶν ἀπὸ τῆς MN , τουτέστι τῶν τετάρτων μέρει τοῦ ἀπὸ τῆς ZM . ἐπεὶ οὖν δύο εὐθεῖαι $\gamma\alpha\nu\iota\sigma\omega\iota$ εἰσιν αἱ ΓM , MZ , καὶ τῶν τετάρτων μέρει τοῦ ἀπὸ τῆς MZ $\gamma\iota\sigma\omega\nu$ παρὰ τὴν ΓM παραβέβληται ἐλλείπον $\epsilon\gamma\iota\delta\epsilon\iota$ τετραγώνῳ τὸ ὑπὸ τῶν $\GammaΚ$, KM καὶ εἰς ἀσύμμετρα αὐτὴν διαιρεῖ, ἡ $\gamma\alpha\rho\alpha$ ΓM τῆς MZ μείζον δύνатаι τῶν ἀπὸ ἀσυμμέτρου ἐαυτῇ. καὶ ἐστὶν ὅλη ἡ ΓM σύμμετρος μήκει τῇ ἐκκειμένη $\kappa\rho\eta\tau\eta$ τῇ $\GammaΔ$; ἡ $\gamma\alpha\rho\alpha$ ΓZ ἀποτομή ἐστὶ τετάρτη. Τὸ $\gamma\alpha\rho\alpha$ ἀπὸ ἐλάσσονος καὶ τὰ ἐχῆξ.

For since AG and GB are incommensurable in square, the (square) on AG (is) thus also incommensurable with the (square) on GB . And CH is equal to the (square) on AG , and KL equal to the (square) on GB . Thus, CH is incommensurable with KL . And as CH (is) to KL , so CK is to KM [Prop. 6.1]. CK is thus incommensurable in length with KM [Prop. 10.11]. And since the (rectangle contained) by AG and GB is the mean proportional to the (squares) on AG and GB [Prop. 10.21 lem.], and the (square) on AG is equal to CH , and the (square) on GB to KL , and the (rectangle contained) by AG and GB to NL , NL is thus the mean proportional to CH and KL . Thus, as CH is to NL , so NL (is) to KL . But, as CH (is) to NL , so CK is to NM , and as NL (is) to KL , so NM is to KM [Prop. 6.1]. Thus, as CK (is) to MN , so MN is to KM [Prop. 5.11]. The (rectangle contained) by CK and KM is thus equal to the (square) on MN —that is to say, to the fourth part of the (square) on FM [Prop. 6.17]. Therefore, since CM and MF are two unequal straight-lines, and the (rectangle contained) by CK and KM , equal to the fourth part of the (square) on MF , has been applied to CM , falling short by a square figure, and divides it into incommensurable (parts), the square on CM is thus greater than (the square on) MF by the (square) on (some straight-line) incommensurable (in length) with (CM) [Prop. 10.18]. And the whole of CM is commensurable in length with the (previously) laid down rational (straight-line) CD . Thus, CF is a fourth apotome [Def. 10.14].

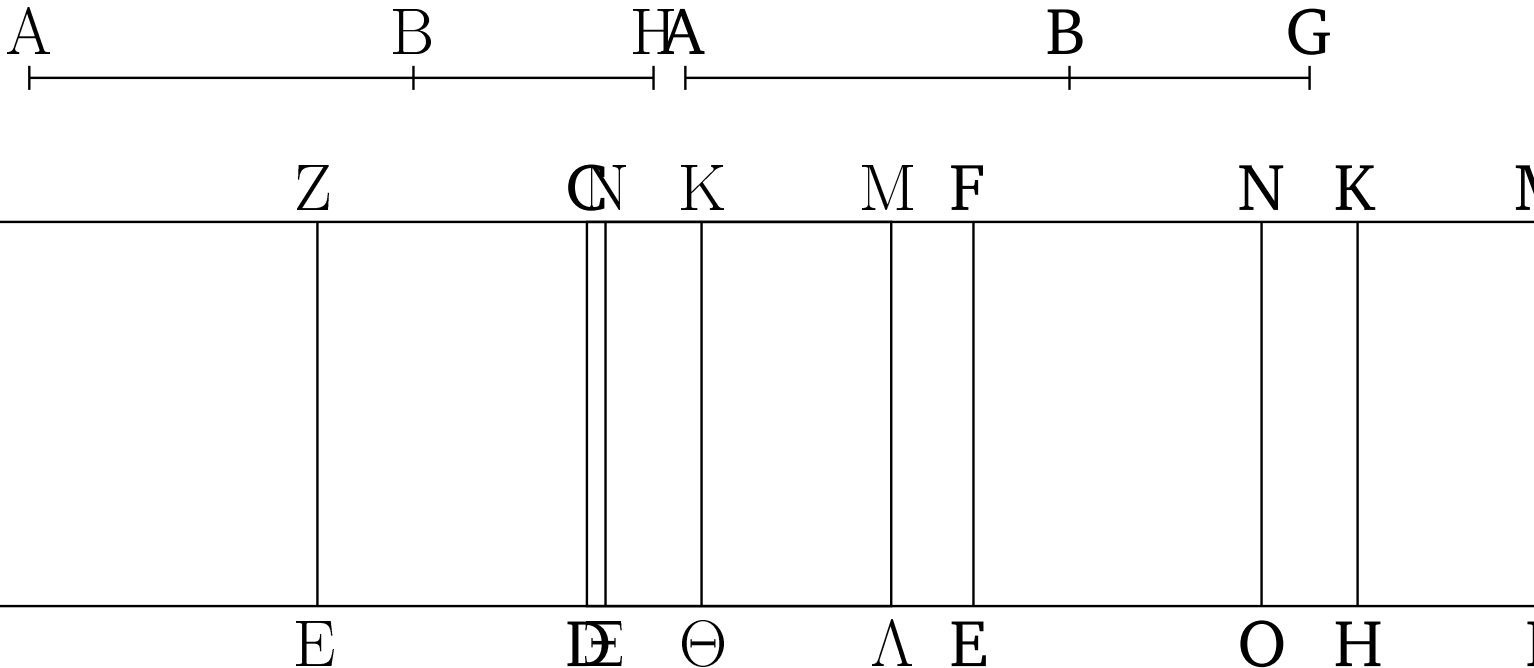
Thus, the (square) on a minor, and so on ...

101.

Τὸ ἀπὸ τῆς μετὰ $\kappa\rho\eta\tau\omicron\upsilon$ μέσον τὸ ὅλον ποιούσης παρὰ $\kappa\rho\eta\tau\eta$ ν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν πέμπτην.

Proposition 101

The (square) on that (straight-line) which with a rational (area) makes a medial whole, applied to a rational (straight-line), produces a fifth apotome as breadth.



$\gamma\epsilon\sigma\tau\omega$ ἡ μετὰ $\kappa\rho\eta\tau\omicron\upsilon$ μέσον τὸ ὅλον ποιούσα ἡ AB , $\kappa\rho\eta\tau\eta$ δὲ ἡ $\GammaΔ$, καὶ τῶν ἀπὸ τῆς AB $\gamma\iota\sigma\omega\nu$ παρὰ τὴν $\GammaΔ$ παραβέβλησθω τὸ ΓE πλάτος ποιούν τὴν ΓZ ; λέγω, ὅτι ἡ ΓZ ἀποτομή ἐστὶ πέμπτη.
 $\gamma\epsilon\sigma\tau\omega$ γὰρ τῇ AB προσαρμόζουσα ἡ BH ; αἱ $\gamma\alpha\rho\alpha$ AH ,

Let AB be that (straight-line) which with a rational (area) makes a medial whole, and CD a rational (straight-line). And let CE , equal to the (square) on AB , be applied to CD , producing CF as breadth. I say that CF is a fifth apotome. Let BG be an attachment to AB . Thus, the straight-lines

HB εὐθεῖαι δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον, τὸ δὲ διζ' ὑπ' αὐτῶν <ρητόν, καὶ τῷ μὲν ἀπὸ τῆς AH >ίσον παρὰ τὴν ΓΔ παραβεβλήσθω τὸ ΓΘ, τῷ δὲ ἀπὸ τῆς HB <ίσον τὸ ΚΛ; <όλον >άρα τὸ ΓΛ >ίσον ἐστὶ τοῖς ἀπὸ τῶν AH, HB. τὸ δὲ συγκείμενον ἐκ τῶν ἀπὸ τῶν AH, HB <άμα μέσον ἐστίν; μέσον >άρα ἐστὶ τὸ ΓΛ. καὶ παρὰ <ρητὴν τὴν ΓΔ παράκειται πλάτος ποιοῦν τὴν ΓΜ; <ρητὴ >άρα ἐστὶν ἡ ΓΜ καὶ ἀσύμμετρος τῇ ΓΔ. καὶ ἐπεὶ <όλον τὸ ΓΛ >ίσον ἐστὶ τοῖς ἀπὸ τῶν AH, HB, <ὦν τὸ ΓΕ >ίσον ἐστὶ τῷ ἀπὸ τῆς AB, λοιπὸν >άρα τὸ ΖΛ >ίσον ἐστὶ τῷ διζ' ὑπὸ τῶν AH, HB. τετμήσθω οὖν ἡ ΖΜ δίθα κατὰ τὸ Ν, καὶ >ήθθω διὰ τοῦ Ν ὁποτέρᾳ τῶν ΓΔ, ΜΛ παράλληλος ἡ ΝΧ; ἐκάτερον >άρα τῶν ΖΧ, ΝΛ >ίσον ἐστὶ τῷ ὑπὸ τῶν AH, HB, καὶ ἐπεὶ τὸ διζ' ὑπὸ τῶν AH, HB <ρητόν ἐστι καὶ [ἐστίν] >ίσον τῷ ΖΛ, <ρητόν >άρα ἐστὶ τὸ ΖΛ. καὶ παρὰ <ρητὴν τὴν ΕΖ παράκειται πλάτος ποιοῦν τὴν ΖΜ; <ρητὴ >άρα ἐστὶν ἡ ΖΜ καὶ σύμμετρος τῇ ΓΔ μήκει. καὶ ἐπεὶ τὸ μὲν ΓΛ μέσον ἐστίν, τὸ δὲ ΖΛ <ρητόν, ἀσύμμετρον >άρα ἐστὶ τὸ ΓΛ τῷ ΖΛ. ὥς δὲ τὸ ΓΛ πρὸς τὸ ΖΛ, οὕτως ἡ ΓΜ πρὸς τὴν ΜΖ; ἀσύμμετρος >άρα ἐστὶν ἡ ΓΜ τῇ ΜΖ μήκει. καὶ εἰσιν ἀμφοτέραι <ρηταί; αἱ >άρα ΓΜ, ΜΖ <ρηταί εἰσι δυνάμει μόνον σύμμετροι; ἀποτομή >άρα ἐστὶν ἡ ΓΖ. λέγω δὴ, <ότι καὶ πέμπτη.

Ὅμοιώς γὰρ δείχομεν, <ότι τὸ ὑπὸ τῶν ΓΚΜ >ίσον ἐστὶ τῷ ἀπὸ τῆς ΝΜ, τουτέστι τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ΖΜ. καὶ ἐπεὶ ἀσύμμετρόν ἐστι τὸ ἀπὸ τῆς AH τῷ ἀπὸ τῆς HB, >ίσον δὲ τὸ μὲν ἀπὸ τῆς AH τῷ ΓΘ, τὸ δὲ ἀπὸ τῆς HB τῷ ΚΛ, ἀσύμμετρον >άρα τὸ ΓΘ τῷ ΚΛ. ὥς δὲ τὸ ΓΘ πρὸς τὸ ΚΛ, οὕτως ἡ ΓΚ πρὸς τὴν ΚΜ; ἀσύμμετρος >άρα ἡ ΓΚ τῇ ΚΜ μήκει. ἐπεὶ οὖν δύο εὐθεῖαι >άνισοι εἰσιν αἱ ΓΜ, ΜΖ, καὶ τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ΖΜ >ίσον παρὰ τὴν ΓΜ παραβεβλήται ἐλλείπον ε>ίδει τετραγώνῳ καὶ εἰς ἀσύμμετρα αὐτὴν διαιρεῖ, ἡ >άρα ΓΜ τῆς ΜΖ μείζον δύναται τῷ ἀπὸ ἀσύμμετρου ἑαυτῇ. καὶ ἐστὶν ἡ προσαρμόζουσα ἡ ΖΜ σύμμετρος τῇ ἐκκειμένη <ρητῇ τῇ ΓΔ; ἡ >άρα ΓΖ ἀποτομή ἐστὶ πέμπτη; <όπερ >έδει δεῖχαι.

AG and *GB* are incommensurable in square, making the sum of the squares on them medial, and twice the (rectangle contained) by them rational [Prop. 10.77]. And let *CH*, equal to the (square) on *AG*, be applied to *CD*, and *KL*, equal to the (square) on *GB*. The whole of *CL* is thus equal to the (sum of the squares) on *AG* and *GB*. And the sum of the (squares) on *AG* and *GB* together is medial. Thus, *CL* is medial. And it has been applied to the rational (straight-line) *CD*, producing *CM* as breadth. *CM* is thus rational, and incommensurable (in length) with *CD* [Prop. 10.22]. And since the whole of *CL* is equal to the (sum of the squares) on *AG* and *GB*, of which *CE* is equal to the (square) on *AB*, the remainder *FL* is thus equal to twice the (rectangle contained) by *AG* and *GB* [Prop. 2.7]. Therefore, let *FM* be cut in half at *N*. And let *NO* be drawn through *N*, parallel to either of *CD* or *ML*. Thus, *FO* and *NL* are each equal to the (rectangle contained) by *AG* and *GB*. And since twice the (rectangle contained) by *AG* and *GB* is rational, and [is] equal to *FL*, *FL* is thus rational. And it is applied to the rational (straight-line) *EF*, producing *FM* as breadth. Thus, *FM* is rational, and commensurable in length with *CD* [Prop. 10.20]. And since *CL* is medial, and *FL* rational, *CL* is thus incommensurable with *FL*. And as *CL* (is) to *FL*, so *CM* (is) to *MF* [Prop. 6.1]. *CM* is thus incommensurable in length with *MF* [Prop. 10.11]. And both are rational. Thus, *CM* and *MF* are rational (straight-lines which are) commensurable in square only. *CF* is thus an apotome [Prop. 10.73]. So, I say that (it is) also a fifth (apotome).

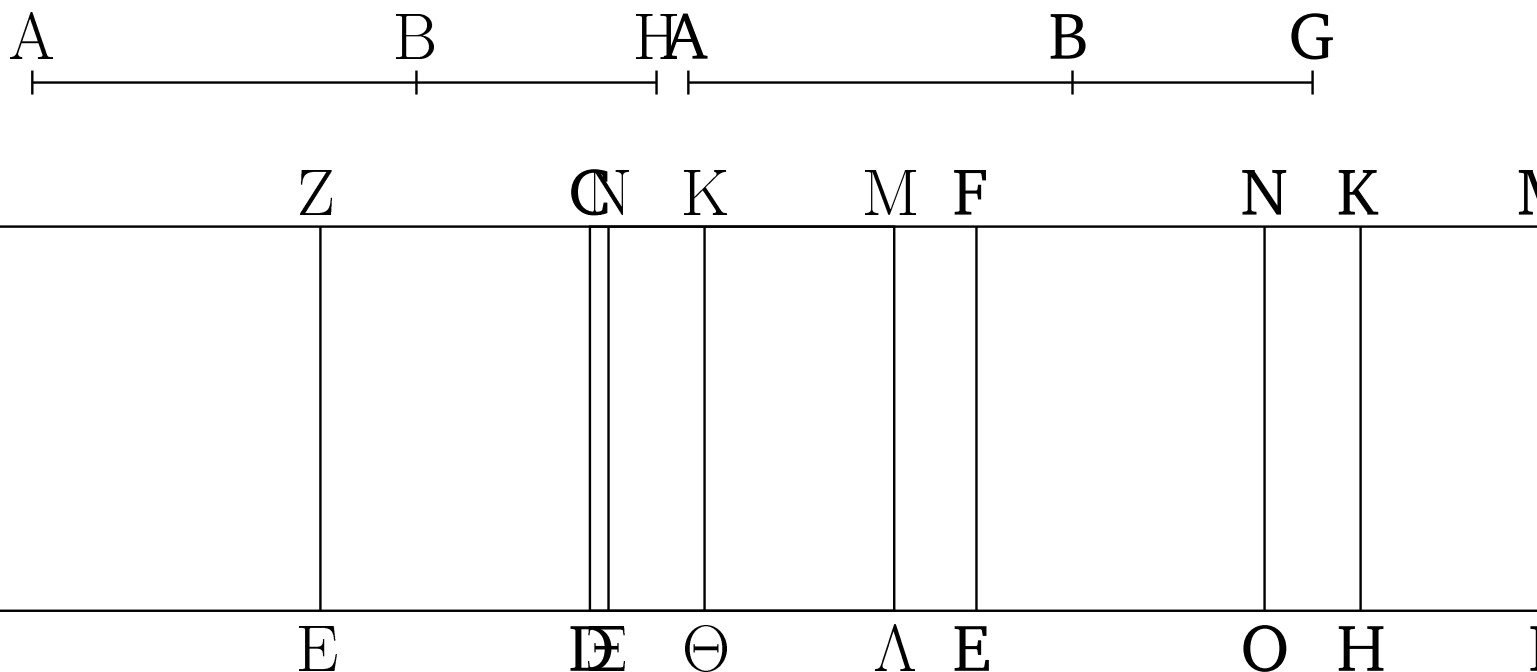
For, similarly (to the previous propositions), we can show that the (rectangle contained) by *CKM* is equal to the (square) on *NM*—that is to say, to the fourth part of the (square) on *FM*. And since the (square) on *AG* is incommensurable with the (square) on *GB*, and the (square) on *AG* (is) equal to *CH*, and the (square) on *GB* to *KL*, *CH* (is) thus incommensurable with *KL*. And as *CH* (is) to *KL*, so *CK* (is) to *KM* [Prop. 6.1]. Thus, *CK* (is) incommensurable in length with *KM* [Prop. 10.11]. Therefore, since *CM* and *MF* are two unequal straight-lines, and (some area), equal to the fourth part of the (square) on *FM*, has been applied to *CM*, falling short by a square figure, and divides it into incommensurable (parts), the square on *CM* is thus greater than (the square on) *MF* by the (square) on (some straight-line) incommensurable (in length) with (*CM*) [Prop. 10.18]. And the attachment *FM* is commensurable with the (previously) laid down rational (straight-line) *CD*. Thus, *CF* is a fifth apotome [Def. 10.15]. (Which is) the very thing it was required to show.

102.

Τὸ ἀπὸ τῆς μετὰ μέσου μέσον τὸ <όλον ποιούσης παρὰ <ρητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομήν <έκτην.

Proposition 102

The (square) on that (straight-line) which with a medial (area) makes a medial whole, applied to a rational (straight-line), produces a sixth apotome as breadth.



>Ἐστω ἡ μετὰ μέσου μέσον τὸ <όλον ποιῶσα ἡ AB , <ρητὴ δὲ ἡ $\Gamma\Delta$, καὶ τῶ| ἀπὸ τῆς AB >ίσον παρὰ τὴν $\Gamma\Delta$ παραβεβλήσθω τὸ ΓE πλάτοξ ποιοῦν τὴν ΓZ ; λέγω, <ὅτι ἡ ΓZ ἀποτομή ἐστὶν <έκτη.

>Ἐστω γὰρ τῇ| AB προσαρμόζουσα ἡ BH ; αἱ >άρα AH , HB δυνάμει εἰσὶν ἀσύμμετροι ποιῶσαι τὸ τε συγκείμενον ἐκ τῶν ἀπ> αὐτῶν τετραγώνων μέσον καὶ τὸ διζ ὑπὸ τῶν AH , HB μέσον καὶ ἀσύμμετρον τὰ ἀπὸ τῶν AH , HB τῶ| διζ ὑπὸ τῶν AH , HB . παραβεβλήσθω ο>ὺν παρὰ τὴν $\Gamma\Delta$ τῶ| μὲν ἀπὸ τῆς AH >ίσον τὸ $\Gamma\Theta$ πλάτοξ ποιοῦν τὴν ΓK , τῶ| δὲ ἀπὸ τῆς BH τὸ $K\Lambda$; <όλον >άρα τὸ $\Gamma\Lambda$ >ίσον ἐστὶ τοῖς ἀπὸ τῶν AH , HB ; μέσον >άρα [ἐστὶ] καὶ τὸ $\Gamma\Lambda$. καὶ παρὰ <ρητὴν τὴν $\Gamma\Delta$ παράκειται πλάτοξ ποιοῦν τὴν ΓM ; <ρητὴ >άρα ἐστὶν ἡ ΓM καὶ ἀσύμμετρος τῇ| $\Gamma\Delta$ μήκει. ἐπεὶ ο>ὺν τὸ $\Gamma\Lambda$ >ίσον ἐστὶ τοῖς ἀπὸ τῶν AH , HB , <ὦν τὸ ΓE >ίσον τῶ| ἀπὸ τῆς AB , λοιπὸν >άρα τὸ $Z\Lambda$ >ίσον ἐστὶ τῶ| διζ ὑπὸ τῶν AH , HB . καὶ ἐστὶ τὸ διζ ὑπὸ τῶν AH , HB μέσον; καὶ τὸ $Z\Lambda$ >άρα μέσον ἐστίν. καὶ παρὰ <ρητὴν τὴν ZE παράκειται πλάτοξ ποιοῦν τὴν ZM ; <ρητὴ >άρα ἐστὶν ἡ ZM καὶ ἀσύμμετρος τῇ| $\Gamma\Delta$ μήκει. καὶ ἐπεὶ τὰ ἀπὸ τῶν AH , HB ἀσύμμετρα ἐστὶ τῶ| διζ ὑπὸ τῶν AH , HB , καὶ ἐστὶ τοῖς μὲν ἀπὸ τῶν AH , HB >ίσον τὸ $\Gamma\Lambda$, τῶ| δὲ διζ ὑπὸ τῶν AH , HB >ίσον τὸ $Z\Lambda$, ἀσύμμετρος >άρα [ἐστὶ] τὸ $\Gamma\Lambda$ τῶ| $Z\Lambda$. ὥς δὲ τὸ $\Gamma\Lambda$ πρὸς τὸ $Z\Lambda$, ο<ύτως ἐστὶν ἡ ΓM πρὸς τὴν MZ ; ἀσύμμετρος >άρα ἐστὶν ἡ ΓM τῇ| MZ μήκει. καὶ εἰσὶν ἀμφοτέραι <ρηταί. αἱ ΓM , MZ >άρα <ρηταί εἰσι δυνάμει μόνον σύμμετροι; ἀποτομή >άρα ἐστὶν ἡ ΓZ . λέγω δὴ, <ὅτι καὶ <έκτη.

Ἐπεὶ γὰρ τὸ $Z\Lambda$ >ίσον ἐστὶ τῶ| διζ ὑπὸ τῶν AH , HB , τετμήσθω δίθω ἡ ZM κατὰ τὸ N , καὶ >ήθθω διὰ τοῦ N τῇ| $\Gamma\Delta$ παράλληλοξ ἡ NX ; ἐκάτερον >άρα τῶν ZX , $N\Lambda$ >ίσον ἐστὶ τῶ| ὑπὸ τῶν AH , HB . καὶ ἐπεὶ αἱ AH , HB δυνάμει εἰσὶν ἀσύμμετροι, ἀσύμμετρον >άρα ἐστὶ τὸ ἀπὸ τῆς AH τῶ| ἀπὸ τῆς HB . ἀλλὰ τῶ| μὲν ἀπὸ τῆς AH >ίσον ἐστὶ τὸ $\Gamma\Theta$, τῶ| δὲ ἀπὸ τῆς HB >ίσον ἐστὶ τὸ $K\Lambda$; ἀσύμμετρον >άρα ἐστὶ τὸ $\Gamma\Theta$ τῶ| $K\Lambda$. ὥς δὲ τὸ $\Gamma\Theta$ πρὸς τὸ $K\Lambda$, ο<ύτως ἐστὶν ἡ ΓK πρὸς τὴν KM ; ἀσύμμετρος >άρα ἐστὶν ἡ ΓK τῇ| KM . καὶ ἐπεὶ τῶν ἀπὸ τῶν AH , HB μέσον ἀνάλογόν ἐστὶ τὸ ὑπὸ τῶν AH , HB , καὶ ἐστὶ τῶ| μὲν ἀπὸ τῆς AH >ίσον τὸ $\Gamma\Theta$, τῶ| δὲ ἀπὸ τῆς HB >ίσον τὸ $K\Lambda$, τῶ| δὲ ὑπὸ τῶν AH , HB >ίσον τὸ $N\Lambda$, καὶ τῶν >άρα $\Gamma\Theta$, $K\Lambda$ μέσον ἀνάλογόν ἐστὶ τὸ $N\Lambda$;

Let AB be that (straight-line) which with a medial (area) makes a medial whole, and CD a rational (straight-line). And let CE , equal to the (square) on AB , be applied to CD , producing CF as breadth. I say that CF is a sixth apotome. For let BG be an attachment to AB . Thus, AG and GB are incommensurable in square, making the sum of the squares on them medial, and twice the (rectangle contained) by AG and GB medial, and the (sum of the squares) on AG and GB incommensurable with twice the (rectangle contained) by AG and GB [Prop. 10.78]. Therefore, let CH , equal to the (square) on AG , be applied to CD , producing CK as breadth, and KL , equal to the (square) on BG . Thus, the whole of CL is equal to the (sum of the squares) on AG and GB . CL [is] thus also medial. And it is applied to the rational (straight-line) CD , producing CM as breadth. Thus, CM is rational, and incommensurable in length with CD [Prop. 10.22]. Therefore, since CL is equal to the (sum of the squares) on AG and GB , of which CE (is) equal to the (square) on AB , the remainder FL is thus equal to twice the (rectangle contained) by AG and GB [Prop. 2.7]. And twice the (rectangle contained) by AG and GB (is) medial. Thus, FL is also medial. And it is applied to the rational (straight-line) FE , producing FM as breadth. FM is thus rational, and incommensurable in length with CD [Prop. 10.22]. And since the (sum of the squares) on AG and GB is incommensurable with twice the (rectangle contained) by AG and GB , and CL equal to the (sum of the squares) on AG and GB , and FL equal to twice the (rectangle contained) by AG and GB , CL [is] thus incommensurable with FL . And as CL (is) to FL , so CM is to MF [Prop. 6.1]. Thus, CM is incommensurable in length with MF [Prop. 10.11]. And they are both rational. Thus, CM and MF are rational (straight-lines which are) commensurable in square only. CF is thus an apotome [Prop. 10.73]. So, I say that (it is) also a sixth (apotome).

For since FL is equal to twice the (rectangle contained) by AG and GB , let FM be cut in half at N , and let NO be drawn through N , parallel to CD . Thus, FO and NL are each equal to the (rectangle contained) by AG and GB . And since AG and GB are incommensurable in square, the

>έστιν >άρα ὥς τὸ ΓΘ πρὸς τὸ ΝΛ, ο<ύτωξ τὸ ΝΛ πρὸς τὸ ΚΛ. καὶ διὰ τὰ αὐτὰ ἡ ΓΜ τῆξ ΜΖ μείζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ|. καὶ οὐδετέρα αὐτῶν σύμμετροξ ἐστι τῇ| ἐκκειμένη <ρητῇ| τῇ| ΓΔ; ἡ ΓΖ >άρα ἀποτομή ἐστιν <έκτη; <όπερ >έδει δεῖχαι.

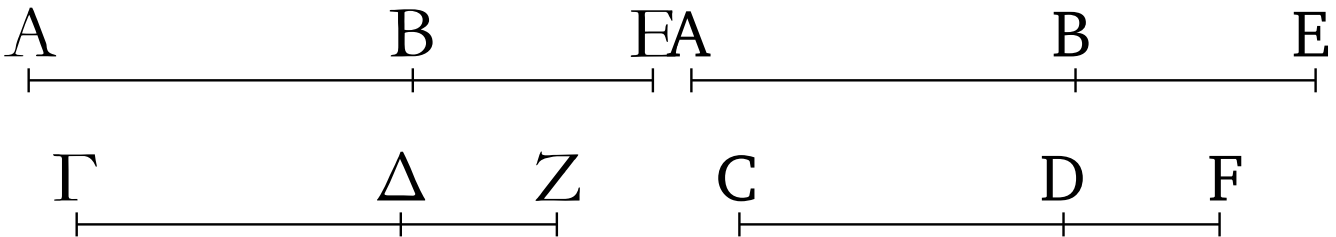
(square) on AG is thus incommensurable with the (square) on GB . But, CH is equal to the (square) on AG , and KL is equal to the (square) on GB . Thus, CH is incommensurable with KL . And as CH (is) to KL , so CK is to KM [Prop. 6.1]. Thus, CK is incommensurable (in length) with KM [Prop. 10.11]. And since the (rectangle contained) by AG and GB is the mean proportional to the (squares) on AG and GB [Prop. 10.21 lem.], and CH is equal to the (square) on AG , and KL equal to the (square) on GB , and NL equal to the (rectangle contained) by AG and GB , NL is thus also the mean proportional to CH and KL . Thus, as CH is to NL , so NL (is) to KL . And for the same (reasons as the preceding propositions), the square on CM is greater than (the square on) MF by the (square) on (some straight-line) incommensurable (in length) with (CM) [Prop. 10.18]. And neither of them is commensurable with the (previously) laid down rational (straight-line) CD . Thus, CF is a sixth apotome [Def. 10.16]. (Which is) the very thing it was required to show.

103.

Proposition 103

Ἡ τῇ| ἀποτομῇ| μήκει σύμμετροξ ἀποτομή ἐστι καὶ τῇ| τάχει ἡ αὐτή.
>Ἐστω ἀποτομή ἡ AB , καὶ τῇ| AB μήκει σύμμετροξ >έστω ἡ $ΓΔ$; λέγω, <ότι καὶ ἡ $ΓΔ$ ἀποτομή ἐστι καὶ τῇ| τάχει ἡ αὐτῇ| AB .

A (straight-line) commensurable in length with an apotome is an apotome, and (is) the same in order.
Let AB be an apotome, and let CD be commensurable in length with AB . I say that CD is also an apotome, and (is) the same in order as AB .



Ἐπεὶ γὰρ ἀποτομή ἐστιν ἡ AB , >έστω αὐτῇ| προσαρμοζουσα ἡ BE ; αἱ AE , EB >άρα <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι. καὶ τῷ| τῆξ AB πρὸς τὴν $ΓΔ$ λόγῳ ὁ αὐτὸς γεγονέντω ὁ τῆξ BE πρὸς τὴν $ΔΖ$; καὶ ὥς <έν >άρα πρὸς <έν, πάντα [ἐστὶ] πρὸς πάντα; >έστιν >άρα καὶ ὥς <όλη ἡ AE πρὸς <όλην τὴν $ΓΖ$, ο<ύτωξ ἡ AB πρὸς τὴν $ΓΔ$. σύμμετροξ δὲ ἡ AB τῇ| $ΓΔ$ μήκει; σύμμετροξ >άρα καὶ ἡ AE μὲν τῇ| $ΓΖ$, ἡ δὲ BE τῇ| $ΔΖ$. καὶ αἱ AE , EB <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι; καὶ αἱ $ΓΖ$, $ΖΔ$ >άρα <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι [ἀποτομή >άρα ἐστὶν ἡ $ΓΔ$. λέγω δὴ, <ότι καὶ τῇ| τάχει ἡ αὐτῇ| AB].

Ἐπεὶ ο>ύν ἐστιν ὥς ἡ AE πρὸς τὴν $ΓΖ$, ο<ύτωξ ἡ BE πρὸς τὴν $ΔΖ$, ἐναλλάχ >άρα ἐστὶν ὥς ἡ AE πρὸς τὴν EB , ο<ύτωξ ἡ $ΓΖ$ πρὸς τὴν $ΖΔ$. >ήτοι δὴ ἡ AE τῆξ EB μείζον δύναται τῷ| ἀπὸ συμέτρου ἑαυτῇ| >ἡ τῷ| ἀπὸ ἀσυμμέτρου. εἰ μὲν ο>ύν ἡ AE τῆξ EB μείζον δύναται τῷ| ἀπὸ συμέτρου ἑαυτῇ|, καὶ ἡ $ΓΖ$ τῆξ $ΖΔ$ μείζον δυνήσεται τῷ| ἀπὸ συμέτρου ἑαυτῇ|. καὶ εἰ μὲν σύμμετροξ ἐστὶν ἡ AE τῇ| ἐκκειμένη <ρητῇ| μήκει, καὶ ἡ $ΓΖ$, εἰ δὲ ἡ BE , καὶ ἡ $ΔΖ$, εἰ δὲ οὐδετέρα τῶν AE , EB , καὶ οὐδετέρα τῶν $ΓΖ$, $ΖΔ$. εἰ δὲ ἡ AE [τῆξ EB] μείζον δύναται τῷ| ἀπὸ ἀσυμμέτρου ἑαυτῇ|, καὶ ἡ $ΓΖ$ τῆξ $ΖΔ$ μείζον δυνήσεται τῷ| ἀπὸ ἀσυμμέτρου ἑαυτῇ|. καὶ εἰ μὲν σύμμετροξ ἐστὶν ἡ AE τῇ| ἐκκειμένη <ρητῇ| μήκει, καὶ ἡ $ΓΖ$, εἰ δὲ ἡ BE , καὶ ἡ $ΔΖ$, εἰ δὲ οὐδετέρα τῶν AE , EB , οὐδετέρα τῶν $ΓΖ$, $ΖΔ$.

For since AB is an apotome, let BE be an attachment to it. Thus, AE and EB are rational (straight-lines which are) commensurable in square only [Prop. 10.73]. And let it be contrived that the (ratio) of BE to DF is the same as the ratio of AB to CD [Prop. 6.12]. Thus, also, as one is to one, (so) all [are] to all [Prop. 5.12]. And thus as the whole AE is to the whole CF , so AB (is) to CD . And AB (is) commensurable in length with CD . AE (is) thus also commensurable (in length) with CF , and BE with DF [Prop. 10.11]. And AE and BE are rational (straight-lines which are) commensurable in square only. Thus, CF and FD are also rational (straight-lines which are) commensurable in square only [Prop. 10.13]. [CD is thus an apotome. So, I say that (it is) also the same in order as AB .] Therefore, since as AE is to CF , so BE (is) to DF , thus, alternately, as AE is to EB , so CF (is) to FD [Prop. 5.16]. So, the square on AE is greater than (the square on) EB either by the (square) on (some straight-line) commensurable, or by the (square) on (some straight-line) incommensurable, (in length) with (AE). Therefore, if the (square) on AE is greater than (the square on) EB by the (square) on (some straight-line) commensurable (in length) with (AE) then the square on CF will also be greater than (the square on) FD by the (square) on (some straight-line) commensurable (in length) with (CF) [Prop. 10.14]. And if AE

Ἀποτομή >άρα ἐστὶν ἡ ΓΔ καὶ τῇ| τάχει ἡ αὐτὴ τῇ| AB; <όπερ >έδει δεῖχαι.

is commensurable in length with a (previously) laid down rational (straight-line) then so (is) CF [Prop. 10.12], and if BE (is commensurable), so (is) DF , and if neither of AE or EB (are commensurable), neither (are) either of CF or FD [Prop. 10.13]. And if the (square) on AE is greater [than (the square on) EB] by the (square) on (some straight-line) incommensurable (in length) with (AE) then the (square) on CF will also be greater than (the square on) FD by the (square) on (some straight-line) incommensurable (in length) with (CF) [Prop. 10.14]. And if AE is commensurable in length with a (previously) laid down rational (straight-line), so (is) CF [Prop. 10.12], and if BE (is commensurable), so (is) DF , and if neither of AE or EB (are commensurable), neither (are) either of CF or FD [Prop. 10.13].

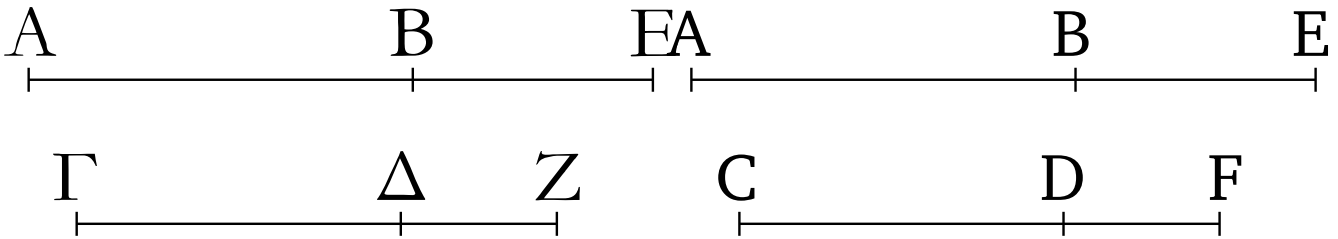
Thus, CD is an apotome, and (is) the same in order as AB [Defs. 10.11—10.16]. (Which is) the very thing it was required to show.

104.

Proposition 104

Ἡ τῇ| μέσηξ ἀποτομῇ| σύμμετροξ μέσηξ ἀποτομή ἐστι καὶ τῇ| τάχει ἡ αὐτή.

A (straight-line) commensurable (in length) with an apotome of a medial (straight-line) is an apotome of a medial (straight-line), and (is) the same in order.



>Ἐστω μέσηξ ἀποτομή ἡ AB, καὶ τῇ| AB μήκει σύμμετροξ >έστω ἡ ΓΔ; λέγω, <ὅτι καὶ ἡ ΓΔ μέσηξ ἀποτομή ἐστι καὶ τῇ| τάχει ἡ αὐτὴ τῇ| AB.

Ἐπεὶ γὰρ μέσηξ ἀποτομή ἐστὶν ἡ AB, >έστω αὐτῇ| προσαρμοζόμενη ἡ EB. αἱ AE, EB >άρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι. καὶ γεγονέτω ὥς ἡ AB πρὸς τὴν ΓΔ, ο<ύτωξ ἡ BE πρὸς τὴν ΔΖ; σύμμετροξ >άρα [ἐστὶ] καὶ ἡ AE τῇ| ΓΖ, ἡ δὲ BE τῇ| ΔΖ. αἱ δὲ AE, EB μέσαι εἰσὶ δυνάμει μόνον σύμμετροι; καὶ αἱ ΓΖ, ΖΔ >άρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι; μέσηξ >άρα ἀποτομή ἐστὶν ἡ ΓΔ. λέγω δὴ, <ὅτι καὶ τῇ| τάχει ἐστὶν ἡ αὐτὴ τῇ| AB.

Ἐπεὶ [γάρ] ἐστὶν ὥς ἡ AE πρὸς τὴν EB, ο<ύτωξ ἡ ΓΖ πρὸς τὴν ΖΔ [ἀλλ> ὥς μὲν ἡ AE πρὸς τὴν EB, ο<ύτωξ τὸ ἀπὸ τῆς AE πρὸς τὸ ὑπὸ τῶν AE, EB, ὥς δὲ ἡ ΓΖ πρὸς τὴν ΖΔ, ο<ύτωξ τὸ ἀπὸ τῆς ΓΖ πρὸς τὸ ὑπὸ τῶν ΓΖ, ΖΔ], >έστιν >άρα καὶ ὥς τὸ ἀπὸ τῆς AE πρὸς τὸ ὑπὸ τῶν AE, EB, ο<ύτωξ τὸ ἀπὸ τῆς ΓΖ πρὸς τὸ ὑπὸ τῶν ΓΖ, ΖΔ [καὶ ἐναλλάξ ὥς τὸ ἀπὸ τῆς AE πρὸς τὸ ἀπὸ τῆς ΓΖ, ο<ύτωξ τὸ ὑπὸ τῶν AE, EB πρὸς τὸ ὑπὸ τῶν ΓΖ, ΖΔ]. σύμμετρον δὲ τὸ ἀπὸ τῆς AE τῷ| ἀπὸ τῆς ΓΖ; σύμμετρον >άρα ἐστὶ καὶ τὸ ὑπὸ τῶν AE, EB τῷ| ὑπὸ τῶν ΓΖ, ΖΔ. ε>ῖτε ο>ύν <ρητόν ἐστι τὸ ὑπὸ τῶν AE, EB, <ρητόν >έσται καὶ τὸ ὑπὸ τῶν ΓΖ, ΖΔ, ε>ῖτε μέσον [ἐστὶ] τὸ ὑπὸ τῶν AE, EB, μέσον [ἐστὶ] καὶ τὸ ὑπὸ τῶν ΓΖ, ΖΔ.

Μέσηξ >άρα ἀποτομή ἐστὶν ἡ ΓΔ καὶ τῇ| τάχει ἡ αὐτὴ τῇ| AB; <όπερ >έδει δεῖχαι.

Let AB be an apotome of a medial (straight-line), and let CD be commensurable in length with AB . I say that CD is also an apotome of a medial (straight-line), and (is) the same in order as AB .

For since AB is an apotome of a medial (straight-line), let EB be an attachment to it. Thus, AE and EB are medial (straight-lines which are) commensurable in square only [Props. 10.74, 10.75]. And let it be contrived that as AB is to CD , so BE (is) to DF [Prop. 6.12]. Thus, AE [is] also commensurable (in length) with CF , and BE with DF [Props. 5.12, 10.11]. And AE and EB are medial (straight-lines which are) commensurable in square only. CF and FD are thus also medial (straight-lines which are) commensurable in square only [Props. 10.23, 10.13]. Thus, CD is an apotome of a medial (straight-line) [Props. 10.74, 10.75]. So, I say that it is also the same in order as AB .

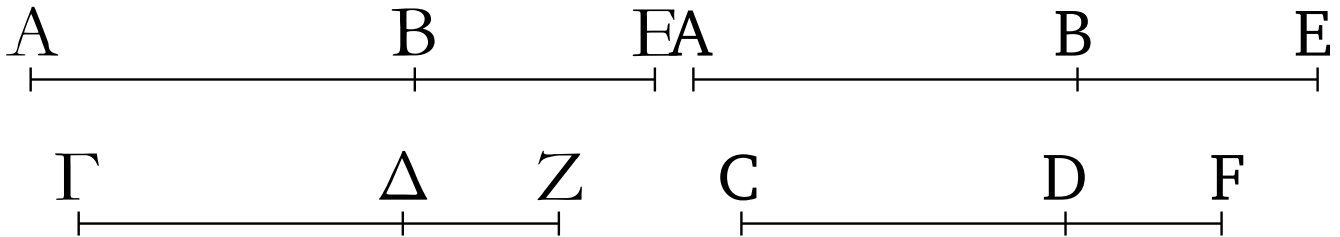
[For] since as AE is to EB , so CF (is) to FD [Props. 5.12, 5.16] [but as AE (is) to EB , so the (square) on AE (is) to the (rectangle contained) by AE and EB , and as CF (is) to FD , so the (square) on CF (is) to the (rectangle contained) by CF and FD], thus as the (square) on AE is to the (rectangle contained) by AE and EB , so the (square) on CF also (is) to the (rectangle contained) by CF and FD [Prop. 10.21 lem.] [and, alternately, as the (square) on AE (is) to the (square) on CF , so the (rectangle contained) by AE and EB (is) to the (rectangle contained) by CF and FD]. And the (square) on AE (is) commensurable with the (square) on CF . Thus, the (rectangle contained) by AE and EB is also commensurable with the (rectangle contained)

by CF and FD [Props. 5.16, 10.11]. Therefore, either the (rectangle contained) by AE and EB is rational, and the (rectangle contained) by CF and FD will also be rational [Def. 10.4], or the (rectangle contained) by AE and EB [is] medial, and the (rectangle contained) by CF and FD [is] also medial [Prop. 10.23 corr.].

Therefore, CD is the apotome of a medial (straight-line), and is the same in order as AB [Props. 10.74, 10.75]. (Which is) the very thing it was required to show.

105.

Ἡ τῇ ἐλάσσονι σύμμετρος ἐλάσσων ἐστίν.
>Ἐστω γὰρ ἐλάσσων ἡ AB καὶ τῇ AB σύμμετρος ἡ $\Gamma\Delta$;
λέγω, <ὅτι καὶ ἡ $\Gamma\Delta$ ἐλάσσων ἐστίν.



Γεγονέτω γὰρ τὰ αὐτά; καὶ ἐπεὶ αἱ AE , EB δυνάμει εἰσὶν ἀσύμμετροι, καὶ αἱ ΓZ , $Z\Delta$ >άρα δυνάμει εἰσὶν ἀσύμμετροι. ἐπεὶ ο>ὗν ἐστὶν ὥς ἡ AE πρὸς τὴν EB , ο<ύτωξ ἡ ΓZ πρὸς τὴν $Z\Delta$, >ἐστὶν >άρα καὶ ὥς τὸ ἀπὸ τῆς AE πρὸς τὸ ἀπὸ τῆς EB , ο<ύτωξ τὸ ἀπὸ τῆς ΓZ πρὸς τὸ ἀπὸ τῆς $Z\Delta$. συνθέντι >άρα ἐστὶν ὥς τὰ ἀπὸ τῶν AE , EB πρὸς τὸ ἀπὸ τῆς EB , ο<ύτωξ τὰ ἀπὸ τῶν ΓZ , $Z\Delta$ πρὸς τὸ ἀπὸ τῆς $Z\Delta$ [καὶ ἐναλλάχ]; σύμμετρον δὲ ἐστὶ τὸ ἀπὸ τῆς BE τῷ ἀπὸ τῆς ΔZ ; σύμμετρον >άρα καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AE , EB τετραγώνων τῷ συγκείμενῳ ἐκ τῶν ἀπὸ τῶν ΓZ , $Z\Delta$ τετραγώνων. <ρητὸν δὲ ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AE , EB τετραγώνων; <ρητὸν >άρα ἐστὶ καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΓZ , $Z\Delta$ τετραγώνων. πάλιν, ἐπεὶ ἐστὶν ὥς τὸ ἀπὸ τῆς AE πρὸς τὸ ὑπὸ τῶν AE , EB , ο<ύτωξ τὸ ἀπὸ τῆς ΓZ πρὸς τὸ ὑπὸ τῶν ΓZ , $Z\Delta$, σύμμετρον δὲ τὸ ἀπὸ τῆς AE τετράγωνον τῷ ἀπὸ τῆς ΓZ τετραγώνῳ, σύμμετρον >άρα ἐστὶ καὶ τὸ ὑπὸ τῶν AE , EB τῷ ὑπὸ τῶν ΓZ , $Z\Delta$. μέσον δὲ τὸ ὑπὸ τῶν AE , EB ; μέσον >άρα καὶ τὸ ὑπὸ τῶν ΓZ , $Z\Delta$; αἱ ΓZ , $Z\Delta$ >άρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ> αὐτῶν τετραγώνων <ρητὸν, τὸ δ> ὑπ> αὐτῶν μέσον.

Ἐλάσσων >άρα ἐστὶν ἡ $\Gamma\Delta$; <όπερ >έδει δεῖχαι.

Proposition 105

A (straight-line) commensurable (in length) with a minor (straight-line) is a minor (straight-line).

For let AB be a minor (straight-line), and (let) CD (be) commensurable (in length) with AB . I say that CD is also a minor (straight-line).

For let the same things be contrived (as in the former proposition). And since AE and EB are (straight-lines which are) incommensurable in square [Prop. 10.76], CF and FD are thus also (straight-lines which are) incommensurable in square [Prop. 10.13]. Therefore, since as AE is to EB , so CF (is) to FD [Props. 5.12, 5.16], thus also as the (square) on AE is to the (square) on EB , so the (square) on CF (is) to the (square) on FD [Prop. 6.22]. Thus, via composition, as the (sum of the squares) on AE and EB is to the (square) on EB , so the (sum of the squares) on CF and FD (is) to the (square) on FD [Prop. 5.18], [also alternately]. And the (square) on BE is commensurable with the (square) on DF [Prop. 10.104]. The sum of the squares on AE and EB (is) thus also commensurable with the sum of the squares on CF and FD [Prop. 5.16, 10.11]. And the sum of the (squares) on AE and EB is rational [Prop. 10.76]. Thus, the sum of the (squares) on CF and FD is also rational [Def. 10.4]. Again, since as the (square) on AE is to the (rectangle contained) by AE and EB , so the (square) on CF (is) to the (rectangle contained) by CF and FD [Prop. 10.21 lem.], and the square on AE (is) commensurable with the square on CF , the (rectangle contained) by AE and EB is thus also commensurable with the (rectangle contained) by CF and FD . And the (rectangle contained) by AE and EB (is) medial [Prop. 10.76]. Thus, the (rectangle contained) by CF and FD (is) also medial [Prop. 10.23 corr.]. CF and FD are thus (straight-lines which are) incommensurable in square, making the sum of the squares on them rational, and the (rectangle contained) by them medial.

Thus, CD is a minor (straight-line) [Prop. 10.76]. (Which is) the very thing it was required to show.

106.

Ἡ τῇ μετὰ <ρητοῦ μέσον τὸ <όλον ποιοῦση σύμμετρος μετὰ <ρητοῦ μέσον τὸ <όλον ποιοῦσά ἐστιν.
>Ἐστω μετὰ <ρητοῦ μέσον τὸ <όλον ποιοῦσα ἡ AB καὶ

Proposition 106

A (straight-line) commensurable (in length) with a (straight-line) which with a rational (area) makes a medial whole is a (straight-line) which with a rational (area) makes a medial

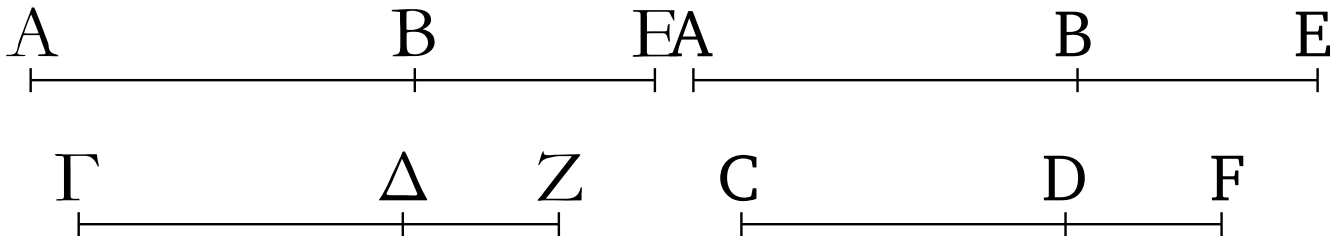
τῇ| AB σύμμετροξ ἡ $\Gamma\Delta$; λέγω, <ὅτι καὶ ἡ $\Gamma\Delta$ μετὰ <ρητοῦ μέσον τὸ <όλον ποιούσά ἐστιν.

>Ἐστω γὰρ τῇ| AB προσαρμόζουσα ἡ BE ; αἱ AE , EB >άρα δυνάμει εἰσὶν ἀσύμμετροι ποιῶσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν AE , EB τετραγώνων μέσον, τὸ δ> ὑπ> αὐτῶν <ρητόν. καὶ τὰ αὐτὰ κατασκευάσθω. ὁμοίωξ δὴ δείχομεν τοῖξ πρότερον, <ὅτι αἱ ΓZ , $Z\Delta$ ἐν τῷ| αὐτῷ| λόγῳ εἰσὶ ταῖξ AE , EB , καὶ σύμμετρόν ἐστι τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AE , EB τετραγώνων τῷ| συγκειμένῳ ἐκ τῶν ἀπὸ τῶν ΓZ , $Z\Delta$ τετραγώνων, τὸ δὲ ὑπὸ τῶν AE , EB τῷ| ὑπὸ τῶν ΓZ , $Z\Delta$; <ώστε καὶ αἱ ΓZ , $Z\Delta$ δυνάμει εἰσὶν ἀσύμμετροι ποιῶσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν ΓZ , $Z\Delta$ τετραγώνων μέσον, τὸ δ> ὑπ> αὐτῶν <ρητόν.

whole.

Let AB be a (straight-line) which with a rational (area) makes a medial whole, and (let) CD (be) commensurable (in length) with AB . I say that CD is also a (straight-line) which with a rational (area) makes a medial (whole).

For let BE be an attachment to AB . Thus, AE and EB are (straight-lines which are) incommensurable in square, making the sum of the squares on AE and EB medial, and the (rectangle contained) by them rational [Prop. 10.77]. And let the same construction be made (as in the previous propositions). So, similarly to the previous (propositions), we can show that CF and FD are in the same ratio as AE and EB , and the sum of the squares on AE and EB is commensurable with the sum of the squares on CF and FD , and the (rectangle contained) by AE and EB with the (rectangle contained) by CF and FD . Hence, CF and FD are also (straight-lines which are) incommensurable in square, making the sum of the squares on CF and FD medial, and the (rectangle contained) by them rational.



Ἡ $\Gamma\Delta$ >άρα μετὰ <ρητοῦ μέσον τὸ <όλον ποιούσά ἐστιν; <όπερ >έδει δείχαι.

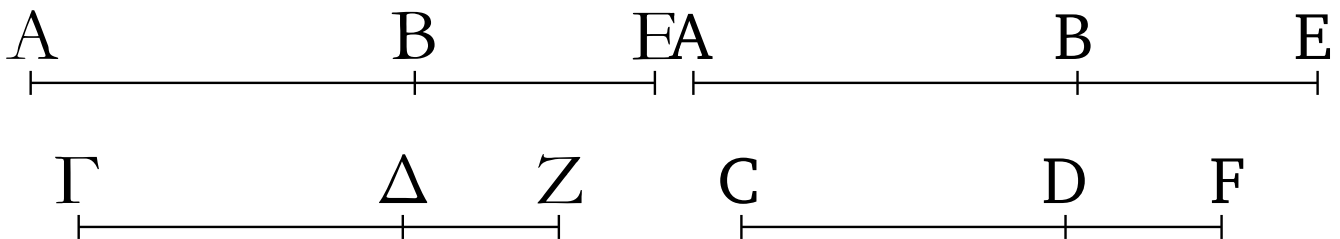
CD is thus a (straight-line) which with a rational (area) makes a medial whole [Prop. 10.77]. (Which is) the very thing it was required to show.

107.

Proposition 107

Ἡ τῇ| μετὰ μέσου μέσον τὸ <όλον ποιούση σύμμετροξ καὶ αὐτὴ μετὰ μέσου μέσον τὸ <όλον ποιούσά ἐστιν.

A (straight-line) commensurable (in length) with a (straight-line) which with a medial (area) makes a medial whole is itself also a (straight-line) which with a medial (area) makes a medial whole.



>Ἐστω μετὰ μέσου μέσον τὸ <όλον ποιούσα ἡ AB , καὶ τῇ| AB >έστω σύμμετροξ ἡ $\Gamma\Delta$; λέγω, <ὅτι καὶ ἡ $\Gamma\Delta$ μετὰ μέσου μέσον τὸ <όλον ποιούσά ἐστιν.

>Ἐστω γὰρ τῇ| AB προσαρμόζουσα ἡ BE , καὶ τὰ αὐτὰ κατασκευάσθω; αἱ AE , EB >άρα δυνάμει εἰσὶν ἀσύμμετροι ποιῶσαι τὸ τε συγκείμενον ἐκ τῶν ἀπ> αὐτῶν τετραγώνων μέσον καὶ τὸ ὑπ> αὐτῶν μέσον καὶ >έτι ἀσύμμετρον τὸ συγκείμενον ἐκ τῶν ἀπ> αὐτῶν τετραγώνων τῷ| ὑπ> αὐτῶν. καὶ εἰσιν, ὡξ ἐδείθθη, αἱ AE , EB σύμμετροι ταῖξ ΓZ , $Z\Delta$, καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AE , EB τετραγώνων τῷ| συγκειμένῳ ἐκ τῶν ἀπὸ τῶν ΓZ , $Z\Delta$, τὸ δὲ ὑπὸ τῶν AE , EB τῷ| ὑπὸ τῶν ΓZ , $Z\Delta$; καὶ αἱ ΓZ , $Z\Delta$ >άρα δυνάμει εἰσὶν ἀσύμμετροι ποιῶσαι τὸ τε συγκείμενον ἐκ τῶν ἀπ> αὐτῶν τετραγώνων μέσον καὶ τὸ ὑπ> αὐτῶν μέσον καὶ >έτι ἀσύμμετρον τὸ συγκείμενον ἐκ τῶν ἀπ> αὐτῶν [τετραγώνων] τῷ| ὑπ> αὐτῶν.

Ἡ $\Gamma\Delta$ >άρα μετὰ μέσου μέσον τὸ <όλον ποιούσά ἐστιν;

Let AB be a (straight-line) which with a medial (area) makes a medial whole, and let CD be commensurable (in length) with AB . I say that CD is also a (straight-line) which with a medial (area) makes a medial whole.

For let BE be an attachment to AB . And let the same construction be made (as in the previous propositions). Thus, AE and EB are (straight-lines which are) incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them medial, and, further, the sum of the squares on them incommensurable with the (rectangle contained) by them [Prop. 10.78]. And, as was shown (previously), AE and EB are commensurable (in length) with CF and FD (respectively), and the sum of the squares on AE and EB with the sum of the squares on CF and FD , and the (rectangle contained) by AE and EB with the (rectangle contained) by CF and FD . Thus, CF and FD are also (straight-lines which are) incommen-

δυναμένη ἐλάσσων ἐστίν; <όπερ >έδει δεῖχαι.

109.

Ἀπὸ μέσου <ρητοῦ ἀφαιρουμένου >ἄλλαι δύο >άλογοι γίνονται >ήτοι μέσηξ ἀποτομή πρώτη >ἢ μετὰ <ρητοῦ μέσον τὸ <όλον ποιοῦσα.

Ἀπὸ γὰρ μέσου τοῦ ΒΓ <ρητὸν ἀφηρήσθω τὸ ΒΔ. λέγω, <ὅτι ἡ τὸ λοιπὸν τὸ ΕΓ δυναμένη μία δύο ἀλόγων γίνεται >ήτοι μέσηξ ἀποτομή πρώτη >ἢ μετὰ <ρητοῦ μέσον τὸ <όλον ποιοῦσα.

Ἐκκείσθω γὰρ <ρητὴ ἡ ΖΗ, καὶ παραβεβλήσθω ὁμοίωξ τὰ θωρία. >έστι δὲ ἀκολουθῶξ <ρητὴ μὲν ἡ ΖΘ καὶ ἀσύμμετροξ τῇ ΖΗ μήκει, <ρητὴ δὲ ἡ ΚΖ καὶ σύμμετροξ τῇ ΖΗ μήκει; αἱ ΖΘ, ΖΚ >άρα <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι; ἀποτομή >άρα ἐστὶν ἡ ΚΘ, προσαρμόζουσα δὲ ταύτῃ ἡ ΖΚ. >ήτοι δὲ ἡ ΘΖ τῇξ ΖΚ μείζον δύνатаι τῷ ἀπὸ συμμέτρου ἑαυτῇ >ἢ τῷ ἀπὸ ἀσυμμέτρου.

Thus, KH is a first apotome [Def. 10.1]. And the square-root of an (area) contained by a rational (straight-line) and a first apotome is an apotome [Prop. 10.91]. Thus, the square-root of LH —that is to say, (of) EC —is an apotome.

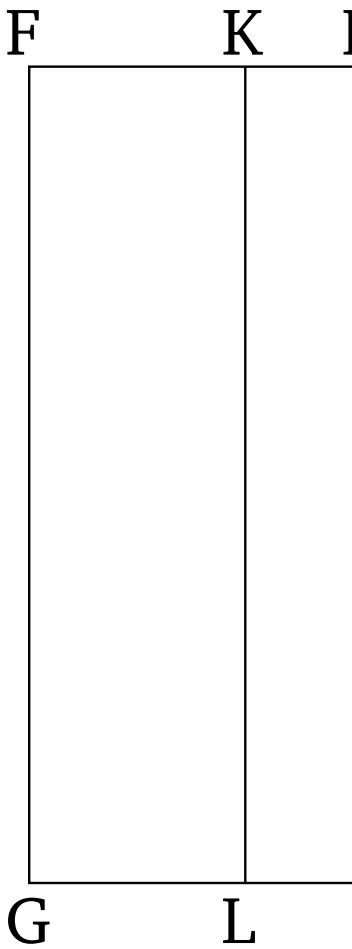
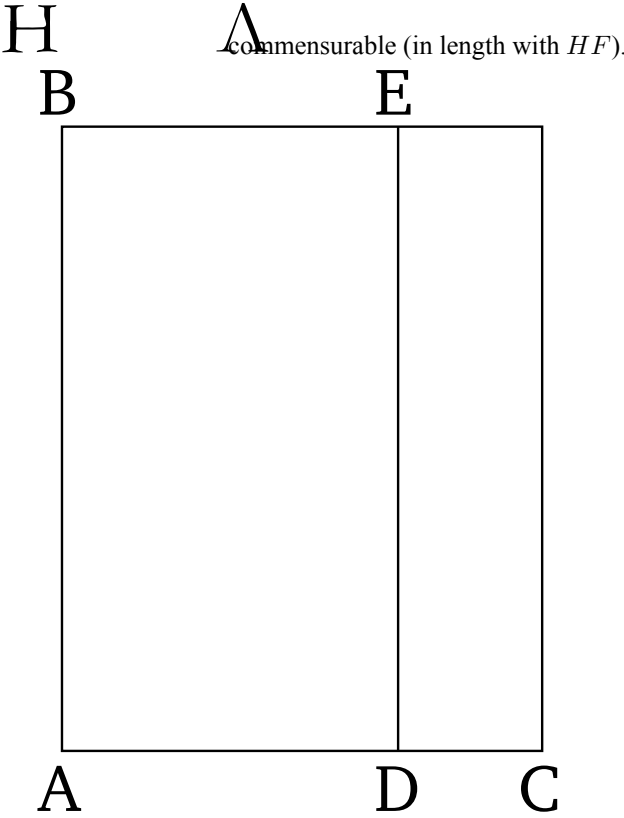
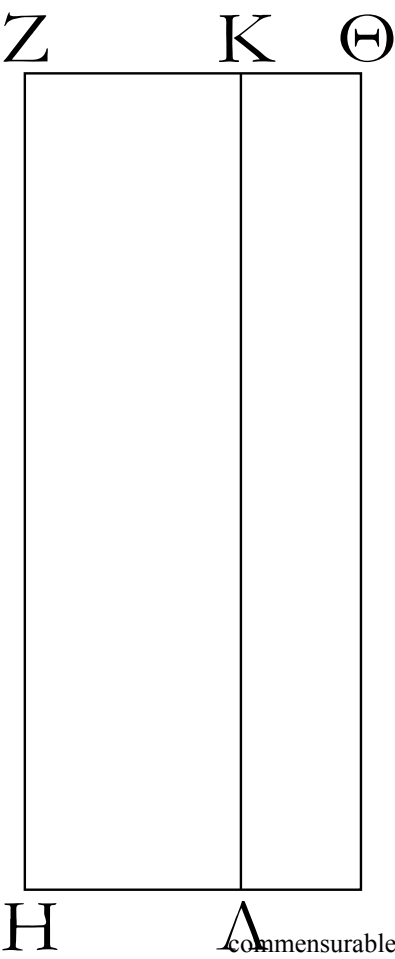
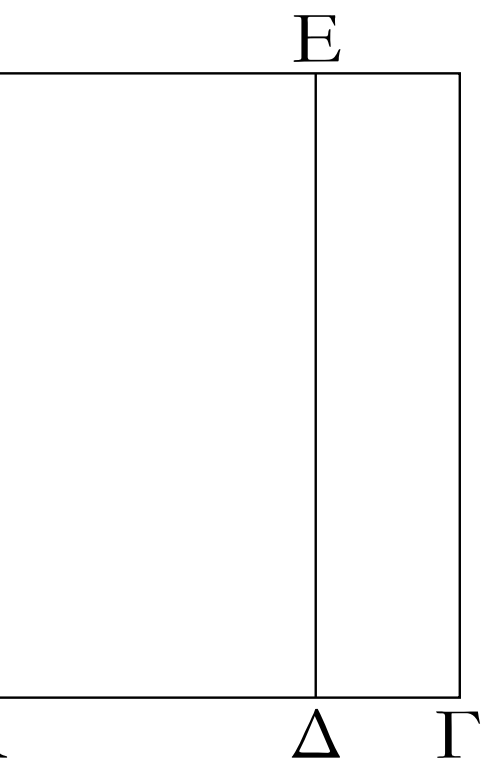
And if the square on HF is greater than (the square on) FK by the (square) on (some straight-line which is) incommensurable (in length) with (HF) , and (since) the whole of FH is commensurable in length with the (previously) laid down rational (straight-line) FG , KH is a fourth apotome [Prop. 10.14]. And the square-root of an (area) contained by a rational (straight-line) and a fourth apotome is a minor (straight-line) [Prop. 10.94]. (Which is) the very thing it was required to show.

Proposition 109

A rational (area) being subtracted from a medial (area), two other irrational (straight-lines) arise (as the square-root of the remaining area)—either a first apotome of a medial (straight-line), or that (straight-line) which with a rational (area) makes a medial whole.

For let the rational (area) BD be subtracted from the medial (area) BC . I say that one of two irrational (straight-lines) arise (as) the square-root of the remaining (area), EC —either a first apotome of a medial (straight-line), or that (straight-line) which with a rational (area) makes a medial whole.

For let the rational (straight-line) FG be laid down, and let similar areas (to the preceding proposition) have been applied (to it). So, accordingly, FH is rational, and incommensurable in length with FG , and KF (is) also rational, and commensurable in length with FG . Thus, FH and FK are rational (straight-lines which are) commensurable in square only [Prop. 10.13]. KH is thus an apotome [Prop. 10.73], and FK an attachment to it. So, the square on HF is greater than (the square on) FK either by the (square) on (some straight-line) commensurable (in length) with (HF) , or by the (square) on (some straight-line) in-



Εἰ μὲν οὖν ἡ ΘΖ τῇ ΖΚ μείζον δύναται τῷ ἀπὸ συμμετροῦ
ἑαυτῇ, καὶ ἔστιν ἡ προσαρμόζουσα ἡ ΖΚ σύμμετρος τῇ | Therefore, if the square on HF is greater than (the square

ἐκκειμένη <ρητῇ| μήκει τῇ| ZH , ἀποτομή δευτέρα ἐστὶν ἡ $KΘ$. <ρητὴ δὲ ἡ ZH ; <ώστε ἡ τὸ $ΛΘ$, τουτέστι τὸ $ΕΓ$, δυναμένη μέσηξ ἀποτομὴ πρώτη ἐστίν.

Εἰ δὲ ἡ $ΘΖ$ τῆξ ZK μείζον δύναται τῷ| ἀπὸ ἀσυμμέτρου, καὶ ἐστὶν ἡ προσαρμόζουσα ἡ ZK σύμμετροξ τῇ| ἐκκειμένη <ρητῇ| μήκει τῇ| ZH , ἀποτομὴ πέμπτη ἐστὶν ἡ $KΘ$; <ώστε ἡ τὸ $ΕΓ$ δυναμένη μετὰ <ρητοῦ μέσον τὸ <όλον ποιοῦσα ἐστίν; <όπερ >έδει δεῖξαι.

on) FK by the (square) on (some straight-line) commensurable (in length) with (HF), and (since) the attachment FK is commensurable in length with the (previously) laid down rational (straight-line) FG , KH is a second apotome [Def. 10.12]. And FG (is) rational. Hence, the square-root of LH —that is to say, (of) EC —is a first apotome of a medial (straight-line) [Prop. 10.92].

And if the square on HF is greater than (the square on) FK by the (square) on (some straight-line) incommensurable (in length with HF), and (since) the attachment FK is commensurable in length with the (previously) laid down rational (straight-line) FG , KH is a fifth apotome [Def. 10.15]. Hence, the square-root of EC is that (straight-line) which with a rational (area) makes a medial whole [Prop. 10.95]. (Which is) the very thing it was required to show.

110.

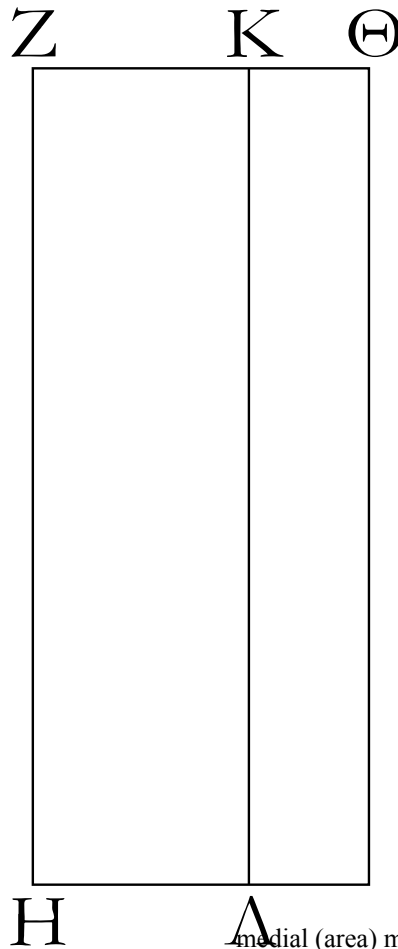
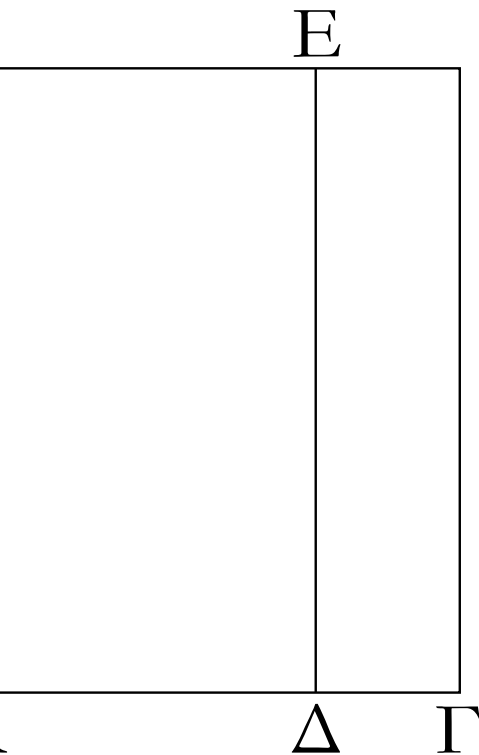
Ἀπὸ μέσου μέσου ἀφαιρουμένου ἀσυμμέτρου τῷ| <όλω αἱ λοιπαὶ δύο >άλογοι γίνονται >ήτοι μέσηξ ἀποτομὴ δευτέρα >ἡ μετὰ μέσου μέσον τὸ <όλον ποιοῦσα.

Ἀφηρήσθω γὰρ ὥξ ἐπὶ τῶν προκειμένων καταγραφῶν ἀπὸ μέσου τοῦ $BΓ$ μέσον τὸ $ΒΔ$ ἀσύμμετρον τῷ| <όλω; λέγω, <ότι ἡ τὸ $ΕΓ$ δυναμένη μία ἐστὶ δύο ἀλόγων >ήτοι μέσηξ ἀποτομὴ δευτέρα >ἡ μετὰ μέσου μέσον τὸ <όλον ποιοῦσα.

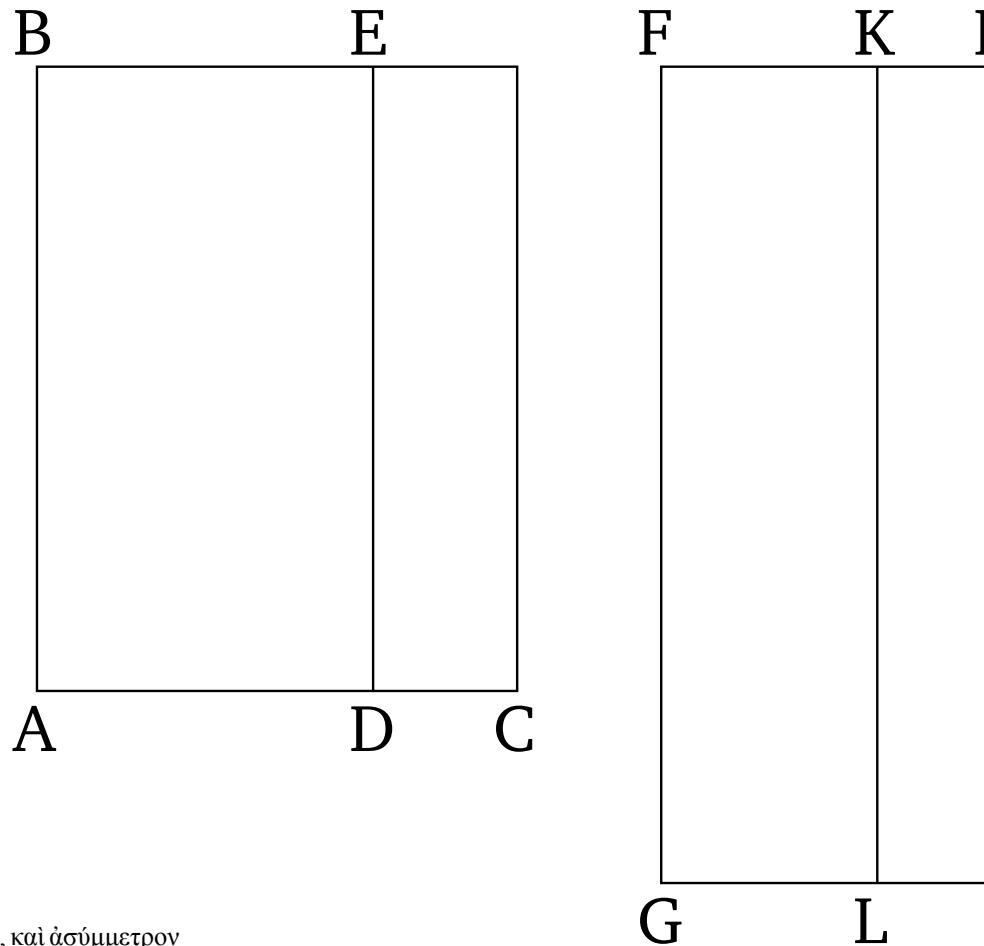
Proposition 110

A medial (area), incommensurable with the whole, being subtracted from a medial (area), the two remaining irrational (straight-lines) arise (as) the (square-root of the area)—either a second apotome of a medial (straight-line), or that (straight-line) which with a medial (area) makes a medial whole.

For, as in the previous figures, let the medial (area) BD , incommensurable with the whole, be subtracted from the medial (area) BC . I say that the square-root of EC is one of two irrational (straight-lines)—either a second apotome of a medial (straight-line), or that (straight-line) which with a



medial (area) makes a medial whole.



Ἐπεὶ γὰρ μέσον ἐστὶν ἑκάτερον τῶν ΒΓ, ΒΔ, καὶ ἀσύμμετρον τὸ ΒΓ τῷ ΒΔ, >έσται ἀκολούθως <ρητὴ ἑκάτερα τῶν ΖΘ, ΖΚ καὶ ἀσύμμετρος τῇ ΖΗ μήκει. καὶ ἐπεὶ ἀσύμμετρόν ἐστι τὸ ΒΓ τῷ ΒΔ, τοῦτέστι τὸ ΗΘ τῷ ΗΚ, ἀσύμμετρος καὶ ἡ ΘΖ τῇ ΖΚ; αἱ ΖΘ, ΖΚ >άρα <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι; ἀποτομὴ >άρα ἐστὶν ἡ ΚΘ [προσαρμόζουσα δὲ ἡ ΖΚ. >ήτοι δὴ ἡ ΖΘ τῆς ΖΚ μείζον δύνανται τῷ ἀπὸ συμμέτρου >ἡ τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ].

Εἰ μὲν δὴ ἡ ΖΘ τῆς ΖΚ μείζον δύνανται τῷ ἀπὸ συμμέτρου ἑαυτῇ, καὶ οὐθετέρα τῶν ΖΘ, ΖΚ σύμμετρος ἐστὶ τῇ ἐκκειμένη <ρητῇ μήκει τῇ ΖΗ, ἀποτομὴ τρίτη ἐστὶν ἡ ΚΘ. <ρητὴ δὲ ἡ ΚΛ, τὸ δ> ὑπὸ <ρητῆς καὶ ἀποτομῆς τρίτης περιεχόμενον ὀρθογώνιον >άλογόν ἐστιν, καὶ ἡ δυναμένη αὐτὸ >άλογός ἐστιν, καλεῖται δὲ μέση ἀποτομὴ δευτέρα; <ώστε ἡ τὸ ΛΘ, τοῦτέστι τὸ ΕΓ, δυναμένη μέση ἀποτομὴ ἐστὶ δευτέρα.

Εἰ δὲ ἡ ΖΘ τῆς ΖΚ μείζον δύνανται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ [μήκει], καὶ οὐθετέρα τῶν ΘΖ, ΖΚ σύμμετρος ἐστὶ τῇ ΖΗ μήκει, ἀποτομὴ <έκτη ἐστὶν ἡ ΚΘ. τὸ δ> ὑπὸ <ρητῆς καὶ ἀποτομῆς <έκτης ἡ δυναμένη ἐστὶ μετὰ μέσου μέσον τὸ <όλον ποιοῦσα. ἡ τὸ ΛΘ >άρα, τοῦτέστι τὸ ΕΓ, δυναμένη μετὰ μέσου μέσον τὸ <όλον ποιοῦσά ἐστιν; <όπερ >έδει δεῖχαι.

For since BC and BD are each medial (areas), and BC (is) incommensurable with BD , accordingly, FH and FK will each be rational (straight-lines), and incommensurable in length with FG [Prop. 10.22]. And since BC is incommensurable with BD —that is to say, GH with GK — HF (is) also incommensurable (in length) with FK [Props. 6.1, 10.11]. Thus, FH and FK are rational (straight-lines which are) commensurable in square only. KH is thus as apotome [Prop. 10.73], [and FK an attachment (to it). So, the square on FH is greater than (the square on) FK either by the (square) on (some straight-line) commensurable, or by the (square) on (some straight-line) incommensurable, (in length) with (FH) .]

So, if the square on FH is greater than (the square on) FK by the (square) on (some straight-line) commensurable (in length) with (FH) , and (since) neither of FH and FK is commensurable in length with the (previously) laid down rational (straight-line) FG , KH is a third apotome [Def. 10.3]. And KL (is) rational. And the rectangle contained by a rational (straight-line) and a third apotome is irrational, and the square-root of it is that irrational (straight-line) called a second apotome of a medial (straight-line) [Prop. 10.93]. Hence, the square-root of LH —that is to say, (of) EC —is a second apotome of a medial (straight-line).

And if the square on FH is greater than (the square on) FK by the (square) on (some straight-line) incommensurable [in length] with (FH) , and (since) neither of HF and FK is commensurable in length with FG , KH is a sixth apotome [Def. 10.16]. And the square-root of the (rectangle contained) by a rational (straight-line) and a sixth apotome is that (straight-line) which with a medial (area) makes a

Ἡ >άρα ἀποτομή οὐκ >έστιν ἡ αὐτὴ τῇ| ἐκ δύο ὀνομάτων;
<όπερ >έδει δεῖχαι.

[Πόρισμα.]

Ἡ ἀποτομή καὶ αἱ μετ> αὐτὴν >άλογοι ο>ύτε τῇ| μέση ο>ύτε ἀλλήλαις εἰσὶν αἱ αὐταί.

Τὸ μὲν γὰρ ἀπὸ μέσης παρὰ <ρητὴν παραβαλλόμενον πλάτος ποιεῖ <ρητὴν καὶ ἀσύμμετρον τῇ|, παρ> <ὴν παράκειται μήκει, τὸ δὲ ἀπὸ ἀποτομῆς παρὰ <ρητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν πρώτην, τὸ δὲ ἀπὸ μέσης ἀποτομῆς πρώτης παρὰ <ρητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν δευτέραν, τὸ δὲ ἀπὸ μέσης ἀποτομῆς δευτέρας παρὰ <ρητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν τρίτην, τὸ δὲ ἀπὸ ἐλάσσονος παρὰ <ρητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν τετάρτην, τὸ δὲ ἀπὸ τῆς μετὰ <ρητοῦ μέσον τὸ <όλον ποιούσης παρὰ <ρητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν πέμπτην, τὸ δὲ ἀπὸ τῆς μετὰ μέσου μέσον τὸ <όλον ποιούσης παρὰ <ρητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν <έκτην. ἐπεὶ ο>ὖν τὰ εἰρημένα πλάτη διαφέρει τοῦ τε πρώτου καὶ ἀλλήλων, τοῦ μὲν πρώτου, <ότι <ρητὴ ἐστίν, ἀλλήλων δὲ, ἐπεὶ τῇ| τάχει οὐκ εἰσὶν αἱ αὐταί, δῆλον, ὥς καὶ αὐταὶ αἱ >άλογοι διαφέρουσιν ἀλλήλων. καὶ ἐπεὶ δέδεικται ἡ ἀποτομή οὐκ ο>ὔσα ἡ αὐτὴ τῇ| ἐκ δύο ὀνομάτων, ποιοῦσι δὲ πλάτη παρὰ <ρητὴν παραβαλλόμενα αἱ μετὰ τὴν ἀποτομὴν ἀποτομὰς ἀκολουθῶς ἐκάστη τῇ| τάχει τῇ| καθ> αὐτὴν, αἱ δὲ μετὰ τὴν ἐκ δύο ὀνομάτων τὰς ἐκ δύο ὀνομάτων καὶ αὐταὶ τῇ| τάχει ἀκολουθῶς, <έτεροι >άρα εἰσὶν αἱ μετὰ τὴν ἀποτομὴν καὶ <έτεροι αἱ μετὰ τὴν ἐκ δύο ὀνομάτων, ὥς ε>ἶναι τῇ| τάχει πάσαξ ἀλόγουξ ιγ,

mainder GF is thus commensurable in length with DF [Prop. 10.15]. [Therefore, since DF is commensurable with GF , and DF is rational, GF is thus also rational. Therefore, since DF is commensurable in length with GF ,] DF (is) incommensurable in length with EF . Thus, FG is also incommensurable in length with EF [Prop. 10.13]. GF and FE [are] thus rational (straight-lines which are) commensurable in square only. Thus, EG is an apotome [Prop. 10.73]. But, (it is) also rational. The very thing is impossible. Thus, an apotome is not the same as a binomial. (Which is) the very thing it was required to show.

[Corollary]

The apotome and the irrational (straight-lines) after it are neither the same as a medial (straight-line) nor (the same) as one another.

For the (square) on a medial (straight-line), applied to a rational (straight-line), produces as breadth a rational (straight-line which is) incommensurable in length with the (straight-line) to which (the area) is applied [Prop. 10.22]. And the (square) on an apotome, applied to a rational (straight-line), produces as breadth a first apotome [Prop. 10.97]. And the (square) on a first apotome of a medial (straight-line), applied to a rational (straight-line), produces as breadth a second apotome [Prop. 10.98]. And the (square) on a second apotome of a medial (straight-line), applied to a rational (straight-line), produces as breadth a third apotome [Prop. 10.99]. And (square) on a minor (straight-line), applied to a rational (straight-line), produces as breadth a fourth apotome [Prop. 10.100]. And (square) on that (straight-line) which with a rational (area) produces a medial whole, applied to a rational (straight-line), produces as breadth a fifth apotome [Prop. 10.101]. And (square) on that (straight-line) which with a medial (area) produces a medial whole, applied to a rational (straight-line), produces as breadth a sixth apotome [Prop. 10.102]. Therefore, since the aforementioned breadths differ from the first (breadth), and from one another—from the first, because it is rational, and from one another since they are not the same in order—clearly, the irrational (straight-lines) themselves also differ from one another. And since it has been shown that an apotome is not the same as a binomial [Prop. 10.111], and (that) the (irrational straight-lines) after the apotome, being applied to a rational (straight-line), produce as breadth, each according to its own (order), apotomes, and (that) the (irrational straight-lines) after the binomial themselves also (produce as breadth), according (to their) order, binomials, the (irrational straight-lines) after the apotome are thus different, and the (irrational straight-lines) after the binomial (are also) different, so that there are, in order, 13 irrational (straight-lines) in all:

Μέσην,
 Ἐκ δύο ὀνομάτων,
 Ἐκ δύο μέσων πρώτην,
 Ἐκ δύο μέσων δευτέραν,
 Μείζονα,
 μῆ<ρητὸν καὶ μέσον δυναμένην,
 Δύο μέσα δυναμένην,
 Ἀποτομήν,
 Μέσηξ ἀποτομήν πρώτην,
 Μέσηξ ἀποτομήν δευτέραν,
 Ἐλάσσονα,
 Μετὰ <ρητοῦ μέσον τὸ <όλον ποιοῦσαν,
 Μετὰ μέσου μέσον τὸ <όλον ποιοῦσαν.

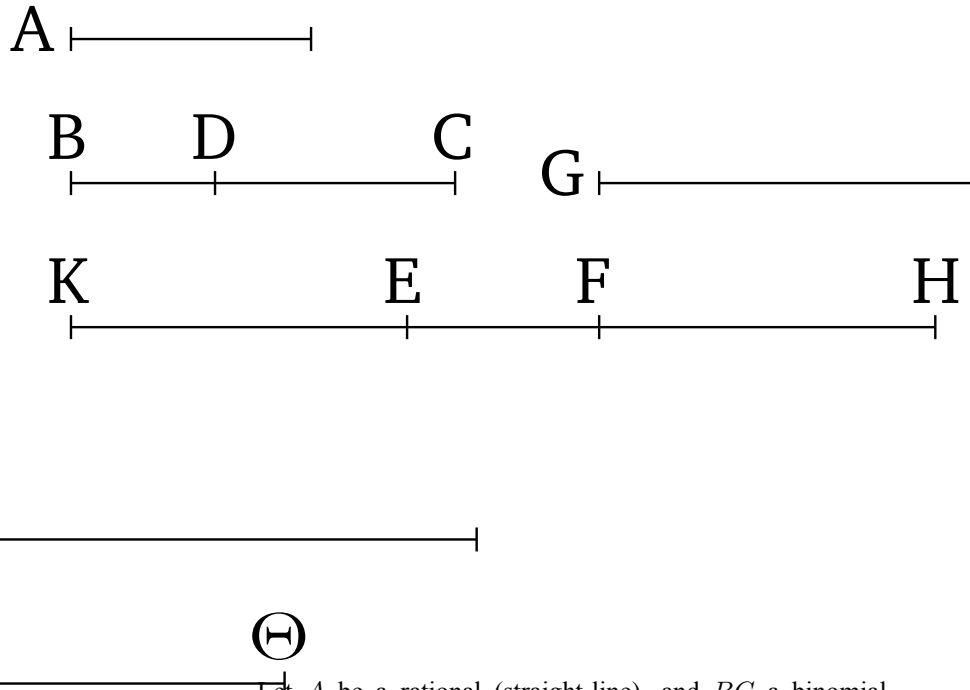
Medial,
 Binomial,
 First bimedral,
 Second bimedral,
 Major,
 Square-root of a rational plus a medial (area),
 Square-root of (the sum of) two medial (areas),
 Apotome,
 First apotome of a medial,
 Second apotome of a medial,
 Minor,
 That which with a rational (area) produces a medial whole,
 That which with a medial (area) produces a medial whole.

112.

Τὸ ἀπὸ <ρητῆς παρὰ τὴν ἐκ δύο ὀνομάτων παραβαλλόμενον πλάτος ποιεῖ ἀποτομήν, <ῆξ τὰ ὀνόματα σύμμετρά ἐστι τοῖς τῆς ἐκ δύο ὀνομάτων ὀνόμασι καὶ >έτι ἐν τῷ αὐτῷ λόγῳ, καὶ >έτι ἡ γινομένη ἀποτομή τὴν αὐτὴν <έχει τάχιν τῇ ἐκ δύο ὀνομάτων.

Proposition 112[†]

The (square) on a rational (straight-line), applied to a binomial (straight-line), produces as breadth an apotome whose terms are commensurable (in length) with the terms of the binomial, and, furthermore, in the same ratio. Moreover, the created apotome will have the same order as the binomial.



>Ἐστω <ρητὴ μὲν ἡ Α, ἐκ δύο ὀνομάτων δὲ ἡ ΒΓ, <ῆξ μείζων >όνομα >έστω ἡ ΔΓ, καὶ τῷ ἀπὸ τῆς Α >ίσον >έστω τὸ ὑπὸ τῶν ΒΓ, ΕΖ; λέγω, <ὅτι ἡ ΕΖ ἀποτομή ἐστίν, <ῆξ τὰ ὀνόματα σύμμετρά ἐστι τοῖς ΓΔ, ΔΒ, καὶ ἐν τῷ αὐτῷ λόγῳ, καὶ >έτι ἡ ΕΖ τὴν αὐτὴν <έχει τάχιν τῇ ΒΓ.

>Ἐστω γὰρ πάλιν τῷ ἀπὸ τῆς Α >ίσον τὸ ὑπὸ τῶν ΒΔ, Η. ἐπεὶ ο>ὖν τὸ ὑπὸ τῶν ΒΓ, ΕΖ >ίσον ἐστὶ τῷ ὑπὸ τῶν ΒΔ, Η, >έστιν >άρα ὡς ἡ ΓΒ πρὸς τὴν ΒΔ, ο<ύτως ἡ Η πρὸς τὴν ΕΖ. μείζων δὲ ἡ ΓΒ τῆς ΒΔ; μείζων >άρα ἐστὶ καὶ ἡ Η τῆς ΕΖ. >έστω τῇ Η >ίση ἡ ΕΘ; >έστιν >άρα ὡς ἡ ΓΒ πρὸς τὴν ΒΔ, ο<ύτως ἡ ΘΕ πρὸς τὴν ΕΖ; διελόντι >άρα ἐστὶν ὡς ἡ ΓΔ πρὸς τὴν ΒΔ, ο<ύτως ἡ ΘΖ πρὸς τὴν ΖΕ. γεγενέτω ὡς ἡ ΘΖ πρὸς τὴν ΖΕ, ο<ύτως ἡ ΖΕ πρὸς τὴν ΚΕ; καὶ <όλη >άρα ἡ ΘΚ πρὸς <όλην τὴν ΚΖ ἐστίν, ὡς ἡ ΖΚ πρὸς ΚΕ; ὡς γὰρ <ἐν τῶν ἡγουμένων πρὸς <ἐν τῶν ἐπομένων, ο<ύτως <άπαντα τὰ ἡγούμενα πρὸς <άπαντα τὰ ἐπόμενα. ὡς δὲ ἡ ΖΚ πρὸς ΚΕ, ο<ύτως ἐστὶν ἡ ΓΔ πρὸς τὴν ΔΒ; καὶ ὡς >άρα ἡ ΘΚ πρὸς ΚΖ, ο<ύτως ἡ ΓΔ πρὸς τὴν ΔΒ. σύμμετρον δὲ

Let A be a rational (straight-line), and BC a binomial (straight-line), of which let DC be the greater term. And let the (rectangle contained) by BC and EF be equal to the (square) on A . I say that EF is an apotome whose terms are commensurable (in length) with CD and DB , and in the same ratio, and, moreover, that EF will have the same order as BC .

For, again, let the (rectangle contained) by BD and G be equal to the (square) on A . Therefore, since the (rectangle contained) by BC and EF is equal to the (rectangle contained) by BD and G , thus as CB is to BD , so G (is) to EF [Prop. 6.16]. And CB (is) greater than BD . Thus, G is also greater than EF [Props. 5.16, 5.14]. Let EH be equal to G . Thus, as CB is to BD , so HE (is) to EF . Thus, via separation, as CD is to BD , so HF (is) to FE [Prop. 5.17]. Let it be contrived that as HF (is) to FE , so FK (is) to KE . And, thus, the whole HK is to the whole KF , as FK (is) to KE . For as one of the leading (proportional magnitudes is) to one of the following, so all of the leading (magnitudes)

τὸ ἀπὸ τῆς ΓΔ τῶ| ἀπὸ τῆς ΔΒ; σύμμετρον >άρα ἐστὶ καὶ τὸ ἀπὸ τῆς ΘΚ τῶ| ἀπὸ τῆς ΚΖ. καὶ ἐστὶν ὥς τὸ ἀπὸ τῆς ΘΚ πρὸς τὸ ἀπὸ τῆς ΚΖ, ο<ύτωξ ἡ ΘΚ πρὸς τὴν ΚΕ, ἐπεὶ αἱ τρεῖς αἱ ΘΚ, ΚΖ, ΚΕ ἀνάλογόν εἰσιν. σύμμετροξ >άρα ἡ ΘΚ τῇ| ΚΕ μήκει. <ώστε καὶ ἡ ΘΕ τῇ| ΕΚ σύμμετρόξ ἐστὶ μήκει. καὶ ἐπεὶ τὸ ἀπὸ τῆς Α >ῖσον ἐστὶ τῶ| ὑπὸ τῶν ΕΘ, ΒΔ, <ρητὸν δέ ἐστὶ τὸ ἀπὸ τῆς Α, <ρητὸν >άρα ἐστὶ καὶ τὸ ὑπὸ τῶν ΕΘ, ΒΔ. καὶ παρὰ <ρητὴν τὴν ΒΔ παράκειται, <ρητὴ >άρα ἐστὶν ἡ ΕΘ καὶ σύμμετροξ τῇ| ΒΔ μήκει; <ώστε καὶ ἡ σύμμετροξ αὐτῇ| ἡ ΕΚ <ρητὴ ἐστὶ καὶ σύμμετροξ τῇ| ΒΔ μήκει. ἐπεὶ ο>ὺν ἐστὶν ὥς ἡ ΓΔ πρὸς ΔΒ, ο<ύτωξ ἡ ΖΚ πρὸς ΚΕ, αἱ δὲ ΓΔ, ΔΒ δυνάμει μόνον εἰσὶ σύμμετροι, καὶ αἱ ΖΚ, ΚΕ δυνάμει μόνον εἰσὶ σύμμετροι. <ρητὴ δέ ἐστὶν ἡ ΚΕ; <ρητὴ >άρα ἐστὶ καὶ ἡ ΖΚ. αἱ ΖΚ, ΚΕ >άρα <ρηταὶ δυνάμει μόνον εἰσὶ σύμμετροι; ἀποτομή >άρα ἐστὶν ἡ ΕΖ. >Ἦτοι δὲ ἡ ΓΔ τῆς ΔΒ μείζον δύναται τῶ| ἀπὸ συμέτρου ἐαυτῇ| >ἡ τῶ| ἀπὸ ἀσυμέτρου.

Εἰ μὲν ο>ὺν ἡ ΓΔ τῆς ΔΒ μείζον δύναται τῶ| ἀπὸ συμέτρου [ἐαυτῇ|], καὶ ἡ ΖΚ τῆς ΚΕ μείζον δυνήσεται τῶ| ἀπὸ συμέτρου ἐαυτῇ|. καὶ εἰ μὲν σύμμετρόξ ἐστὶν ἡ ΓΔ τῇ| ἐκκειμένη <ρητῇ| μήκει, καὶ ἡ ΖΚ; εἰ δὲ ἡ ΒΔ, καὶ ἡ ΚΕ; εἰ δὲ οὐδετέρα τῶν ΓΔ, ΔΒ, καὶ οὐδετέρα τῶν ΖΚ, ΚΕ.

Εἰ δὲ ἡ ΓΔ τῆς ΔΒ μείζον δύναται τῶ| ἀπὸ ἀσυμέτρου ἐαυτῇ|, καὶ ἡ ΖΚ τῆς ΚΕ μείζον δυνήσεται τῶ| ἀπὸ ἀσυμέτρου ἐαυτῇ|. καὶ εἰ μὲν ἡ ΓΔ σύμμετρόξ ἐστὶ τῇ| ἐκκειμένη <ρητῇ| μήκει, καὶ ἡ ΖΚ; εἰ δὲ ἡ ΒΔ, καὶ ἡ ΚΕ; εἰ δὲ οὐδετέρα τῶν ΓΔ, ΔΒ, καὶ οὐδετέρα τῶν ΖΚ, ΚΕ; <ώστε ἀποτομή ἐστὶν ἡ ΖΕ, <ῆξ τὰ ὀνόματα τὰ ΖΚ, ΚΕ σύμμετρά ἐστὶ τοῖς τῆς ἐκ δύο ὀνομάτων ὀνόμασι τοῖς ΓΔ, ΔΒ καὶ ἐν τῶ| αὐτῶ| λόγῳ, καὶ τὴν αὐτῇν τάχιν >έθει τῇ| ΒΓ; <όπερ >έδει δεῖχαι.

are to all of the following [Prop. 5.12]. And as FK (is) to KE , so CD is to DB [Prop. 5.11]. And, thus, as HK (is) to KF , so CD is to DB [Prop. 5.11]. And the (square) on CD (is) commensurable with the (square) on DB [Prop. 10.36]. The (square) on HK is thus also commensurable with the (square) on KF [Props. 6.22, 10.11]. And as the (square) on HK is to the (square) on KF , so HK (is) to KE , since the three (straight-lines) HK , KF , and KE are proportional [Def. 5.9]. HK is thus commensurable in length with KE [Prop. 10.11]. Hence, HE is also commensurable in length with EK [Prop. 10.15]. And since the (square) on A is equal to the (rectangle contained) by EH and BD , and the (square) on A is rational, the (rectangle contained) by EH and BD is thus also rational. And it is applied to the rational (straight-line) BD . Thus, EH is rational, and commensurable in length with BD [Prop. 10.20]. And, hence, the (straight-line) commensurable (in length) with it, EK , is also rational [Def. 10.3], and commensurable in length with BD [Prop. 10.12]. Therefore, since as CD is to DB , so FK (is) to KE , and CD and DB are (straight-lines which are) commensurable in square only, FK and KE are also commensurable in square only [Prop. 10.11]. And KE is rational. Thus, FK is also rational. FK and KE are thus rational (straight-lines which are) commensurable in square only. Thus, EF is an apotome [Prop. 10.73].

And the square on CD is greater than (the square on) DB either by the (square) on (some straight-line) commensurable, or by the (square) on (some straight-line) incommensurable, (in length) with (CD) .

Therefore, if the square on CD is greater than (the square on) DB by the (square) on (some straight-line) commensurable (in length) with $[CD]$ then the square on FK will also be greater than (the square on) KE by the (square) on (some straight-line) commensurable (in length) with (FK) [Prop. 10.14]. And if CD is commensurable in length with a (previously) laid down rational (straight-line), (so) also (is) FK [Props. 10.11, 10.12]. And if BD (is commensurable), (so) also (is) KE [Prop. 10.12]. And if neither of CD or DB (is commensurable), neither also (are) either of FK or KE .

And if the square on CD is greater than (the square on) DB by the (square) on (some straight-line) incommensurable (in length) with (CD) then the square on FK will also be greater than (the square on) KE by the (square) on (some straight-line) incommensurable (in length) with (FK) [Prop. 10.14]. And if CD is commensurable in length with a (previously) laid down rational (straight-line), (so) also (is) FK [Props. 10.11, 10.12]. And if BD (is commensurable), (so) also (is) KE [Prop. 10.12]. And if neither of CD or DB (is commensurable), neither also (are) either of FK or KE . Hence, FE is an apotome whose terms, FK and KE , are commensurable (in length) with the terms, CD and DB , of the binomial, and in the same ratio. And (FE) has the same order as BC [Defs. 10.5—10.10]. (Which is) the very thing it was required to show.

† Heiberg considers this proposition, and the succeeding ones, to be relatively early interpolations into the original text.

113.

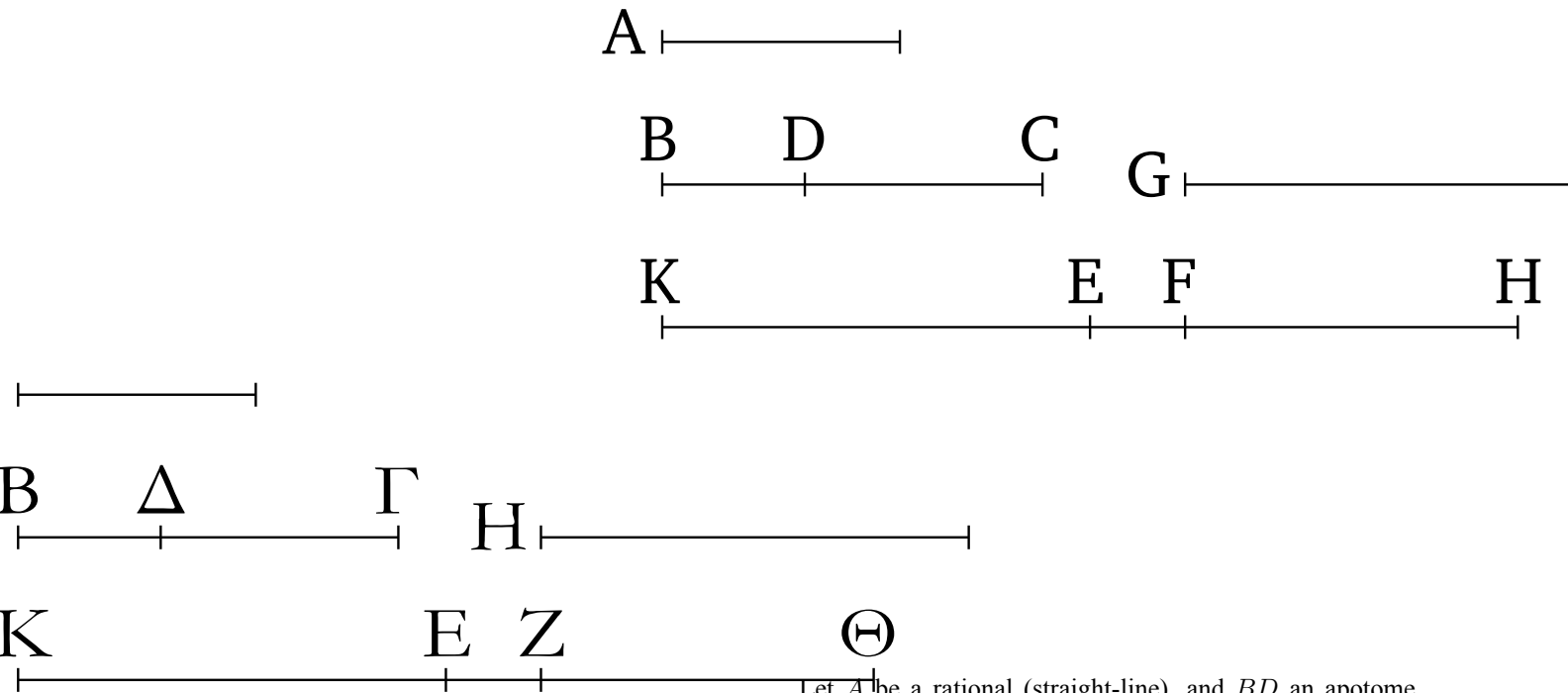
Τὸ ἀπὸ <ρητῆς παρὰ ἀποτομῆν παραβαλλόμενον πλάτοξ ποιεῖ τὴν ἐκ δύο ὀνομάτων, <ῆξ τὰ ὀνόματα σύμμετρά ἐστι

Proposition 113

The (square) on a rational (straight-line), applied to an apotome, produces as breadth a binomial whose terms are com-

τοῖς τῆς ἀποτομῆς ὀνόμασι καὶ ἐν τῷ αὐτῷ λόγῳ, >έτι δὲ ἡ γινομένη ἐκ δύο ὀνομάτων τὴν αὐτὴν τάχιν >έθει τῇ ἀποτομῇ|.

measurable with the terms of the apotome, and in the same ratio. Moreover, the created binomial has the same order as the apotome.



>Ἐστω <ρητὴ μὲν ἡ A , ἀποτομὴ δὲ ἡ BD , καὶ τῷ ἀπὸ τῆς A >ῖσον >έστω τὸ ὑπὸ τῶν BD , $KΘ$, <ώστε τὸ ἀπὸ τῆς A <ρητῆς παρὰ τὴν BD ἀποτομὴν παραβαλλόμενον πλάτος ποιεῖ τὴν $KΘ$; λέγω, <ὅτι ἐκ δύο ὀνομάτων ἐστὶν ἡ $KΘ$, <ἢ τὰ ὀνόματα σύμμετρά ἐστι τοῖς τῆς BD ὀνόμασι καὶ ἐν τῷ αὐτῷ λόγῳ, καὶ >έτι ἡ $KΘ$ τὴν αὐτὴν >έθει τάχιν τῇ BD .

>Ἐστω γὰρ τῇ BD προσαρμόζουσα ἡ $ΔΓ$; αἱ $BΓ$, $ΓΔ$ >άρα <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι. καὶ τῷ ἀπὸ τῆς A >ῖσον >έστω καὶ τὸ ὑπὸ τῶν $BΓ$, H . <ρητὸν δὲ τὸ ἀπὸ τῆς A ; <ρητὸν >άρα καὶ τὸ ὑπὸ τῶν $BΓ$, H . καὶ παρὰ <ρητὴν τὴν $BΓ$ παραβέβληται; <ρητὴ >άρα ἐστὶν ἡ H καὶ σύμμετρος τῇ $BΓ$ μήκει. ἐπεὶ οὖν τὸ ὑπὸ τῶν $BΓ$, H >ῖσον ἐστὶ τῷ ὑπὸ τῶν BD , $KΘ$, ἀνάλογον >άρα ἐστὶν ὡς ἡ $ΓB$ πρὸς BD , οὕτως ἡ $KΘ$ πρὸς H . μείζων δὲ ἡ $BΓ$ τῆς BD ; μείζων >άρα καὶ ἡ $KΘ$ τῆς H . κείσθω τῇ H >ῖση ἡ KE ; σύμμετρος >άρα ἐστὶν ἡ KE τῇ $BΓ$ μήκει. καὶ ἐπεὶ ἐστὶν ὡς ἡ $ΓB$ πρὸς BD , οὕτως ἡ $ΘK$ πρὸς KE , ἀναστρέψαντι >άρα ἐστὶν ὡς ἡ $BΓ$ πρὸς τὴν $ΓΔ$, οὕτως ἡ $KΘ$ πρὸς $ΘE$. γεγονένω ὡς ἡ $KΘ$ πρὸς $ΘE$, οὕτως ἡ $ΘZ$ πρὸς ZE ; καὶ λοιπὴ >άρα ἡ KZ πρὸς $ZΘ$ ἐστὶν, ὡς ἡ $KΘ$ πρὸς $ΘE$, τουτέστιν [ὡς] ἡ $BΓ$ πρὸς $ΓΔ$. αἱ δὲ $BΓ$, $ΓΔ$ δυνάμει μόνον [εἰσὶ] σύμμετροι; καὶ αἱ KZ , $ZΘ$ >άρα δυνάμει μόνον εἰσὶ σύμμετροι; καὶ ἐπεὶ ἐστὶν ὡς ἡ $KΘ$ πρὸς $ΘE$, ἡ KZ πρὸς $ZΘ$, ἄλλ' ὡς ἡ $KΘ$ πρὸς $ΘE$, ἡ $ΘZ$ πρὸς ZE , καὶ ὡς >άρα ἡ KZ πρὸς $ZΘ$, ἡ $ΘZ$ πρὸς ZE ; <ώστε καὶ ὡς ἡ πρώτη πρὸς τὴν τρίτην, τὸ ἀπὸ τῆς πρώτης πρὸς τὸ ἀπὸ τῆς δευτέρας; καὶ ὡς >άρα ἡ KZ πρὸς ZE , οὕτως τὸ ἀπὸ τῆς KZ πρὸς τὸ ἀπὸ τῆς $ZΘ$. σύμμετρον δὲ ἐστὶ τὸ ἀπὸ τῆς KZ τῷ ἀπὸ τῆς $ZΘ$; αἱ γὰρ KZ , $ZΘ$ δυνάμει εἰσὶ σύμμετροι; σύμμετρος >άρα ἐστὶ καὶ ἡ KZ τῇ ZE μήκει; <ώστε ἡ KZ καὶ τῇ KE σύμμετρος [ἐστὶ] μήκει. <ρητὴ δὲ ἐστὶν ἡ KE καὶ σύμμετρος τῇ $BΓ$ μήκει. <ρητὴ >άρα καὶ ἡ KZ καὶ σύμμετρος τῇ $BΓ$ μήκει. καὶ ἐπεὶ ἐστὶν ὡς ἡ $BΓ$ πρὸς $ΓΔ$, οὕτως ἡ KZ πρὸς $ZΘ$, ἐναλλάξ ὡς ἡ $BΓ$ πρὸς KZ , οὕτως ἡ $ΔΓ$ πρὸς $ZΘ$. σύμμετρος δὲ ἡ $BΓ$ τῇ KZ ; σύμμετρος >άρα καὶ ἡ $ZΘ$ τῇ $ΓΔ$ μήκει. αἱ $BΓ$, $ΓΔ$ δὲ <ρηταὶ

Let A be a rational (straight-line), and BD an apotome. And let the (rectangle contained) by BD and KH be equal to the (square) on A , such that the square on the rational (straight-line) A , applied to the apotome BD , produces KH as breadth. I say that KH is a binomial whose terms are commensurable with the terms of BD , and in the same ratio, and, moreover, that KH has the same order as BD .

For let DC be an attachment to BD . Thus, BC and CD are rational (straight-lines which are) commensurable in square only [Prop. 10.73]. And let the (rectangle contained) by BC and G also be equal to the (square) on A . And the (square) on A (is) rational. The (rectangle contained) by BC and G (is) thus also rational. And it has been applied to the rational (straight-line) BC . Thus, G is rational, and commensurable in length with BC [Prop. 10.20]. Therefore, since the (rectangle contained) by BC and G is equal to the (rectangle contained) by BD and KH , thus, proportionally, as CB is to BD , so KH (is) to G [Prop. 6.16]. And BC (is) greater than BD . Thus, KH (is) also greater than G [Prop. 5.16, 5.14]. Let KE be made equal to G . KE is thus commensurable in length with BC . And since as CB is to BD , so HK (is) to KE , thus, via conversion, as BC (is) to CD , so KH (is) to HE [Prop. 5.19 corr.]. Let it be contrived that as KH (is) to HE , so HF (is) to FE . And thus the remainder KF is to FH , as KH (is) to HE —that is to say, [as] BC (is) to CD [Prop. 5.19]. And BC and CD [are] commensurable in square only. KF and FH are thus also commensurable in square only [Prop. 10.11]. And since as KH is to HE , (so) KF (is) to FH , but as KH (is) to HE , (so) HF (is) to FE , thus, also as KF (is) to FH , (so) HF (is) to FE [Prop. 5.11]. And hence as the first (is) to the third, so the (square) on the first (is) to the (square) on the second [Def. 5.9]. And thus as KF (is) to FE , so the (square) on KF (is) to the (square) on FH . And the (square) on KF is commensurable with the (square) on FH . For KF and FH are commensurable in square. Thus, KF is also commensu-

εἰσι δυνάμει μόνον σύμμετροι; καὶ αἱ KZ , $Z\Theta$ >άρα <ρηταί εἰσι δυνάμει μόνον σύμμετροι; ἐκ δύο ὀνομάτων ἐστὶν >άρα ἢ $K\Theta$.

Εἰ μὲν ο>ὐν ἡ $B\Gamma$ τῆς $\Gamma\Delta$ μείζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ, καὶ ἡ KZ τῆς $Z\Theta$ μείζον δυνήσεται τῷ ἀπὸ συμμέτρου ἑαυτῇ. καὶ εἰ μὲν σύμμετρόξ ἐστὶν ἡ $B\Gamma$ τῇ ἐκκειμένῃ <ρητῇ μήκει, καὶ ἡ KZ , εἰ δὲ ἡ $\Gamma\Delta$ σύμμετρόξ ἐστὶ τῇ ἐκκειμένῃ <ρητῇ μήκει, καὶ ἡ $Z\Theta$, εἰ δὲ οὐδετέρα τῶν $B\Gamma$, $\Gamma\Delta$, οὐδετέρα τῶν KZ , $Z\Theta$.

Εἰ δὲ ἡ $B\Gamma$ τῆς $\Gamma\Delta$ μείζον δύναται τῷ ἀπὸ ἀσυνμέτρου ἑαυτῇ, καὶ ἡ KZ τῆς $Z\Theta$ μείζον δυνήσεται τῷ ἀπὸ ἀσυνμέτρου ἑαυτῇ. καὶ εἰ μὲν σύμμετρόξ ἐστὶν ἡ $B\Gamma$ τῇ ἐκκειμένῃ <ρητῇ μήκει, καὶ ἡ KZ , εἰ δὲ ἡ $\Gamma\Delta$, καὶ ἡ $Z\Theta$, εἰ δὲ οὐδετέρα τῶν $B\Gamma$, $\Gamma\Delta$, οὐδετέρα τῶν KZ , $Z\Theta$.

Ἐκ δύο >άρα ὀνομάτων ἐστὶν ἡ $K\Theta$, <ῆξ τὰ ὀνόματα τὰ KZ , $Z\Theta$ σύμμετρά [ἐστὶ] τοῖς τῆς ἀποτομῆς ὀνόμασι τοῖς $B\Gamma$, $\Gamma\Delta$ καὶ ἐν τῷ αὐτῷ λόγῳ, καὶ >έτι ἡ $K\Theta$ τῇ $B\Gamma$ τὴν αὐτὴν <έχει τάχιν; <όπερ >έδει δεῖχαι.

table in length with FE [Prop. 10.11]. Hence, KF [is] also commensurable in length with KE [Prop. 10.15]. And KE is rational, and commensurable in length with BC . Thus, KF (is) also rational, and commensurable in length with BC [Prop. 10.12]. And since as BC is to CD , (so) KF (is) to FH , alternately, as BC (is) to KF , so DC (is) to FH [Prop. 5.16]. And BC (is) commensurable (in length) with KF . Thus, FH (is) also commensurable in length with CD [Prop. 10.11]. And BC and CD are rational (straight-lines which are) commensurable in square only. KF and FH are thus also rational (straight-lines which are) commensurable in square only [Def. 10.3, Prop. 10.13]. Thus, KH is a binomial [Prop. 10.36].

Therefore, if the square on BC is greater than (the square on) CD by the (square) on (some straight-line) commensurable (in length) with (BC), then the square on KF will also be greater than (the square on) FH by the (square) on (some straight-line) commensurable (in length) with (KF) [Prop. 10.14]. And if BC is commensurable in length with a (previously) laid down rational (straight-line), (so) also (is) KF [Prop. 10.12]. And if CD is commensurable in length with a (previously) laid down rational (straight-line), (so) also (is) FH [Prop. 10.12]. And if neither of BC or CD (are commensurable), neither also (are) either of KF or FH [Prop. 10.13].

And if the square on BC is greater than (the square on) CD by the (square) on (some straight-line) incommensurable (in length) with (BC) then the square on KF will also be greater than (the square on) FH by the (square) on (some straight-line) incommensurable (in length) with (KF) [Prop. 10.14]. And if BC is commensurable in length with a (previously) laid down rational (straight-line), (so) also (is) KF [Prop. 10.12]. And if CD is commensurable, (so) also (is) FH [Prop. 10.12]. And if neither of BC or CD (are commensurable), neither also (are) either of KF or FH [Prop. 10.13].

KH is thus a binomial whose terms, KF and FH , [are] commensurable (in length) with the terms, BC and CD , of the apotome, and in the same ratio. Moreover, KH will have the same order as BC [Defs. 10.5—10.10]. (Which is) the very thing it was required to show.

114.

Ἐὰν θωρίον περιέθῃται ὑπὸ ἀποτομῆς καὶ τῆς ἐκ δύο ὀνομάτων, <ῆξ τὰ ὀνόματα σύμμετρά τέ ἐστὶ τοῖς τῆς ἀποτομῆς ὀνόμασι καὶ ἐν τῷ αὐτῷ λόγῳ, ἡ τὸ θωρίον δυναμένη <ρητὴ ἐστὶν.

Περιθέσθω γὰρ θωρίον τὸ ὑπὸ τῶν AB , $\Gamma\Delta$ ὑπὸ ἀποτομῆς τῆς AB καὶ τῆς ἐκ δύο ὀνομάτων τῆς $\Gamma\Delta$, <ῆξ μείζον >όνομα >έστω τὸ ΓE , καὶ >έστω τὰ ὀνόματα τῆς ἐκ δύο ὀνομάτων τὰ ΓE , $E\Delta$ σύμμετρά τε τοῖς τῆς ἀποτομῆς ὀνόμασι τοῖς AZ , ZB καὶ ἐν τῷ αὐτῷ λόγῳ, καὶ >έστω ἡ τὸ ὑπὸ τῶν AB , $\Gamma\Delta$ δυναμένη ἡ H ; λέγω, <ὅτι <ρητὴ ἐστὶν ἡ H .

Ἐκκείσθω γὰρ <ρητὴ ἡ Θ , καὶ τῷ ἀπὸ τῆς Θ >ῖσον παρὰ τὴν $\Gamma\Delta$ παραβεβλήσθω πλάτος ποιοῦν τὴν KL ; ἀποτομῇ >άρα ἐστὶν ἡ KL , <ῆξ τὰ ὀνόματα >έστω τὰ KM , ML σύμμετρα τοῖς τῆς ἐκ δύο ὀνομάτων ὀνόμασι τοῖς ΓE , $E\Delta$ καὶ ἐν τῷ αὐτῷ λόγῳ. ἀλλὰ καὶ αἱ ΓE , $E\Delta$ σύμμετροί τε εἰσι ταῖς AZ , ZB καὶ ἐν τῷ αὐτῷ λόγῳ; >έστιν >άρα ὡς ἡ AZ πρὸς τὴν ZB , ο<ύτωξ ἡ KM πρὸς ML . ἐναλλάχ >άρα ἐστὶν ὡς

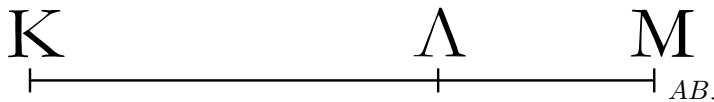
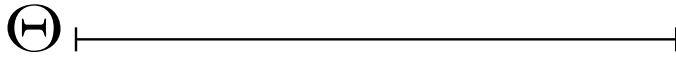
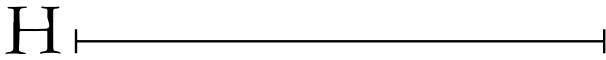
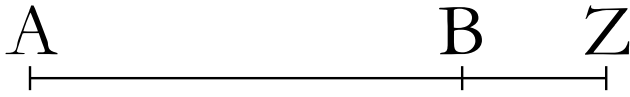
Proposition 114

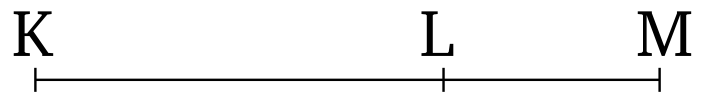
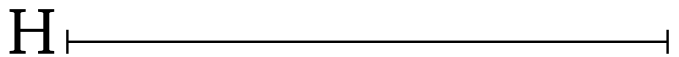
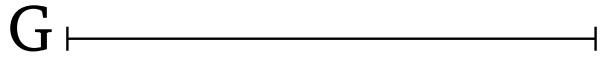
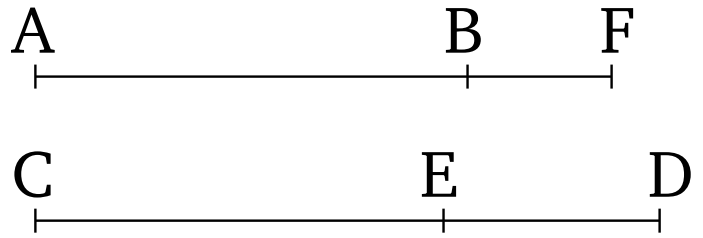
If an area is contained by an apotome, and a binomial whose terms are commensurable with, and in the same ratio as, the terms of the apotome then the square-root of the area is a rational (straight-line).

For let an area, the (rectangle contained) by AB and CD , be contained by the apotome AB , and the binomial CD , of which let the greater term be CE . And let the terms of the binomial, CE and ED , be commensurable with the terms of the apotome, AF and FB (respectively), and in the same ratio. And let the square-root of the (rectangle contained) by AB and CD be G . I say that G is a rational (straight-line). For let the rational (straight-line) H be laid down. And let (some rectangle), equal to the (square) on H , be applied to CD , producing KL as breadth. Thus, KL is an apotome, of which let the terms, KM and ML , be commensurable with the terms of the binomial, CE and ED (respectively), and in the same ratio [Prop. 10.112]. But, CE and ED are

ἡ AZ πρὸς τὴν KM , ὁ $\acute{\alpha}$ ντων ἡ BZ πρὸς τὴν ΛM ; καὶ λοιπὴ $\acute{\alpha}$ ρα ἡ AB πρὸς λοιπὴν τὴν KL ἐστὶν ὡς ἡ AZ πρὸς KM . σύμμετρος δὲ ἡ AZ τῇ KM ; σύμμετρος $\acute{\alpha}$ ρα ἐστὶ καὶ ἡ AB τῇ KL . καὶ ἐστὶν ὡς ἡ AB πρὸς KL , ὁ $\acute{\alpha}$ ντων τὸ ὑπὸ τῶν $\Gamma\Delta$, AB πρὸς τὸ ὑπὸ τῶν $\Gamma\Delta$, KL ; σύμμετρον $\acute{\alpha}$ ρα ἐστὶ καὶ τὸ ὑπὸ τῶν $\Gamma\Delta$, AB τῷ ὑπὸ τῶν $\Gamma\Delta$, KL . $\acute{\alpha}$ ισον δὲ τὸ ὑπὸ τῶν $\Gamma\Delta$, KL τῷ ἀπὸ τῆς Θ ; σύμμετρον $\acute{\alpha}$ ρα ἐστὶ τὸ ὑπὸ τῶν $\Gamma\Delta$, AB τῷ ἀπὸ τῆς Θ . τῷ δὲ ὑπὸ τῶν $\Gamma\Delta$, AB $\acute{\alpha}$ ισον ἐστὶ τὸ ἀπὸ τῆς H ; σύμμετρον $\acute{\alpha}$ ρα ἐστὶ τὸ ἀπὸ τῆς H τῷ ἀπὸ τῆς Θ . ϵ ρητὸν δὲ τὸ ἀπὸ τῆς Θ ; ϵ ρητὸν $\acute{\alpha}$ ρα ἐστὶ καὶ τὸ ἀπὸ τῆς H ; ϵ ρητὴ $\acute{\alpha}$ ρα ἐστὶν ἡ H . καὶ δύνатаι τὸ ὑπὸ τῶν $\Gamma\Delta$, AB .

also commensurable with AF and FB (respectively), and in the same ratio. Thus, as AF is to FB , so KM (is) to ML . Thus, alternately, as AF is to KM , so BF (is) to LM [Prop. 5.16]. Thus, the remainder AB is also to the remainder KL as AF (is) to KM [Prop. 5.19]. And AF (is) commensurable with KM [Prop. 10.12]. AB is thus also commensurable with KL [Prop. 10.11]. And as AB is to KL , so the (rectangle contained) by CD and AB (is) to the (rectangle contained) by CD and KL [Prop. 6.1]. Thus, the (rectangle contained) by CD and AB is also commensurable with the (rectangle contained) by CD and KL [Prop. 10.11]. And the (rectangle contained) by CD and KL (is) equal to the (square) on H . Thus, the (rectangle contained) by CD and AB is commensurable with the (square) on H . And the (square) on G is equal to the (rectangle contained) by CD and AB . The (square) on G is thus commensurable with the (square) on H . And the (square) on H (is) rational. Thus, the (square) on G is also rational. G is thus rational. And it is the square-root of the (rectangle contained) by CD and





Ἐὰν >άρα θωρίον περιέθῃται ὑπὸ ἀποτομῆς καὶ τῆς ἐκ δύο ὀνομάτων, <ῆς τὰ ὀνόματα σύμμετρά ἐστι τοῖς τῆς ἀποτομῆς ὀνόμασι καὶ ἐν τῷ αὐτῷ λόγῳ, ἢ τὸ θωρίον δυναμένη <ρητὴ ἐστίν.

Thus, if an area is contained by an apotome, and a binomial whose terms are commensurable with, and in the same ratio as, the terms of the apotome, then the square-root of the area is a rational (straight-line).

Πόρισμα.

Καὶ γέγονεν ἡμῖν καὶ διὰ τοῦτου φανερόν, <ὅτι δυνατόν ἐστι <ρητὸν θωρίον ὑπὸ ἀλόγων εὐθειῶν περιέθεσθαι. <ὅπερ >έδει δεῖξαι.

Corollary

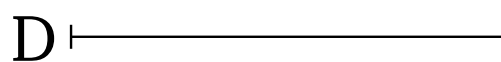
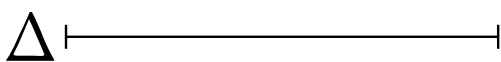
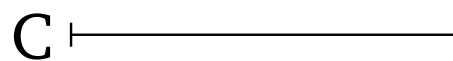
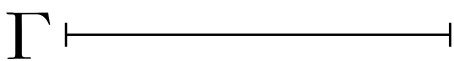
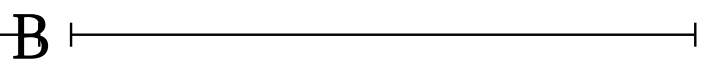
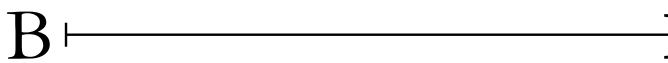
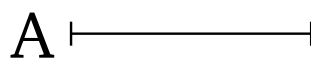
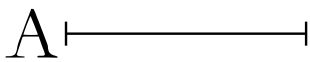
And it has also been made clear to us, through this, that it is possible for a rational area to be contained by irrational straight-lines. (Which is) the very thing it was required to show.

115.

Ἀπὸ μέσης >άπειροι >άλογοι γίνονται, καὶ οὐδεμία οὐδεμιᾷ τῶν πρότερον ἢ αὐτῇ.
>Ἐστω μέση ἢ Α; λέγω, <ὅτι ἀπὸ τῆς Α >άπειροι >άλογοι γίνονται, καὶ οὐδεμία οὐδεμιᾷ τῶν πρότερον ἢ αὐτῇ.

Proposition 115

An infinite (series) of irrational (straight-lines) can be created from a medial (straight-line), and none of them is the same as any of the preceding (straight-lines).
Let A be a medial (straight-line). I say that an infinite (series) of irrational (straight-lines) can be created from A , and that none of them is the same as any of the preceding (straight-lines).



Ἐκκεῖσθω <ρητὴ ἢ Β, καὶ τῷ ὑπὸ τῶν Β, Α >ῖσον >έστω τὸ ἀπὸ τῆς Γ; >άλογος >άρα ἐστίν ἢ Γ; τὸ γὰρ ὑπὸ ἀλόγου καὶ <ρητῆς >άλογόν ἐστιν. καὶ οὐδεμιᾷ τῶν πρότερον ἢ αὐτῇ; τὸ γὰρ ἀπ> οὐδεμιᾶς τῶν πρότερον παρὰ <ρητὴν παραβαλλόμενον πλάτος ποιεῖ μέσην. πάλιν δὲ τῷ ὑπὸ τῶν Β, Γ >ῖσον >έστω τὸ ἀπὸ τῆς Δ; >άλογον >άρα ἐστὶ τὸ ἀπὸ τῆς Δ. >άλογος >άρα ἐστίν ἢ Δ; καὶ οὐδεμιᾷ τῶν πρότερον ἢ αὐτῇ; τὸ γὰρ ἀπ> οὐδεμιᾶς τῶν πρότερον παρὰ <ρητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν Γ. ὁμοίως δὲ

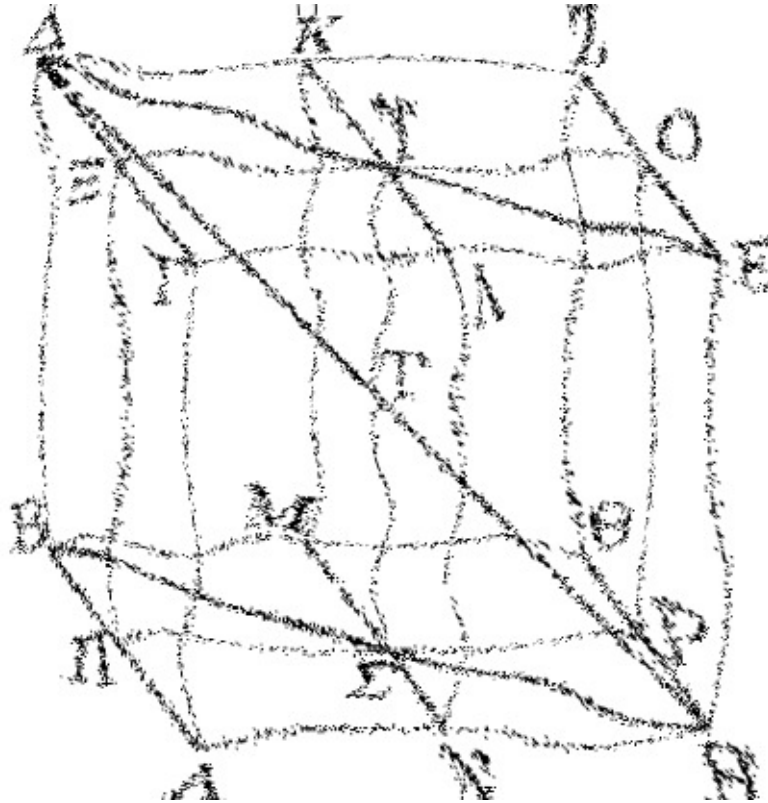
Let the rational (straight-line) B be laid down. And let the (square) on C be equal to the (rectangle contained) by B and A . Thus, C is irrational [Def. 10.4]. For an (area contained) by an irrational and a rational (straight-line) is irrational [Prop. 10.20]. And (C is) not the same as any of the preceding (straight-lines). For the (square) on none of the preceding (straight-lines), applied to a rational (straight-line), produces a medial (straight-line) as breadth. So, again, let the (square) on D be equal to the (rectangle contained) by

τῆς τοιαύτης τάξεως ἐπ' ἀπειρον προβαινούσης φανερόν,
 <ὅτι ἀπὸ τῆς μέσης ἀπειροὶ ἀλόγοι γίνονται, καὶ οὐδεμία
 οὐδεμίᾳ τῶν πρότερον ἢ αὐτῇ; >ὅπερ >έδει δεῖχαι.

B and *C*. Thus, the (square) on *D* is irrational [Prop. 10.20].
D is thus irrational [Def. 10.4]. And (*D* is) not the same as
 any of the preceding (straight-lines). For the (square) on
 none of the preceding (straight-lines), applied to a rational
 (straight-line), produces *C* as breadth. So, similarly, this
 arrangement being advanced to infinity, it is clear that an
 infinite (series) of irrational (straight-lines) can be created
 from a medial (straight-line), and that none of them is the
 same as any of the preceding (straight-lines). (Which is) the
 very thing it was required to show.

ELEMENTS BOOK 11

Elementary Stereometry



<Οροι.

1. Στερεόν ἐστι τὸ μῆκος καὶ πλάτος καὶ βάθος >έθον.
2. Στερεοῦ δὲ πέραξ ἐπιφάνεια.
3. Εὐθεῖα πρὸς ἐπίπεδον ὀρθή ἐστιν, <ὅταν πρὸς πάσας τὰς ἀπτομένας αὐτῆς εὐθείας καὶ ο>ύσας ἐν τῷ [ὑποκειμένῳ] ἐπιπέδῳ ὀρθὰς ποιῇ| γωνίαξ.
4. Ἐπίπεδον πρὸς ἐπίπεδον ὀρθόν ἐστιν, <ὅταν αἱ τῇ| κοινῇ| τομῇ| τῶν ἐπιπέδων πρὸς ὀρθὰς ἀγόμεναι εὐθεῖαι ἐν ἐνὶ τῶν ἐπιπέδων τῷ| λοιπῷ| ἐπιπέδῳ πρὸς ὀρθὰς >ῶσιν.
5. Εὐθείαξ πρὸς ἐπίπεδον κλίσιξ ἐστίν, <ὅταν ἀπὸ τοῦ μετεώρου πέρατος τῆς εὐθείας ἐπὶ τὸ ἐπίπεδον κάθετοξ ἄθθῇ|, καὶ ἀπὸ τοῦ γενομένου σημείου ἐπὶ τὸ ἐν τῷ| ἐπιπέδῳ πέραξ τῆς εὐθείας εὐθεῖα ἐπιζευθθῇ|, ἡ περιεθομένη γωνία ὑπὸ τῆς ἄθθεισης καὶ τῆς ἐφεστώσης.
6. Ἐπιπέδου πρὸς ἐπίπεδον κλίσιξ ἐστίν ἡ περιεθομένη ὀχρεῖα γωνία ὑπὸ τῶν πρὸς ὀρθὰς τῇ| κοινῇ| τομῇ| ἀγομένων πρὸς τῷ| αὐτῷ| σημείῳ ἐν ἑκατέρῳ τῶν ἐπιπέδων.
7. Ἐπίπεδον πρὸς ἐπίπεδον ὁμοίως κεκλίσθαι λέγεται καὶ <έτερον πρὸς <έτερον, <ὅταν αἱ εἰρημέναι τῶν κλίσεων γωνίαι >ίσαι ἀλλήλαιξ >ῶσιν.
8. Παράλληλα ἐπίπεδά ἐστι τὰ ἀσύμπτωτα.
9. <Ὅμοια στερεὰ σθῆματά ἐστι τὰ ὑπὸ ὁμοίων ἐπιπέδων περιεθόμενα >ίσων τὸ πλῆθος.
10. >ἴσα δὲ καὶ <όμοια στερεὰ σθῆματά ἐστι τὰ ὑπὸ ὁμοίων ἐπιπέδων περιεθόμενα >ίσων τῷ| πλήθει καὶ τῷ| μεγέθει.
11. Στερεὰ γωνία ἐστίν ἡ ὑπὸ πλειόνων >ῇ δύο γραμμῶν ἀπτομένων ἀλλήλων καὶ μὴ ἐν τῇ| αὐτῇ| ἐπιφανείᾳ οὐσῶν πρὸς πάσαιξ ταῖξ γραμμαῖξ κλίσιξ. >άλλωξ; στερεὰ γωνία ἐστίν ἡ ὑπὸ πλειόνων >ῇ δύο γωνιῶν ἐπιπέδων περιεθομένη μὴ οὐσῶν ἐν τῷ| αὐτῷ| ἐπιπέδῳ πρὸς ἐνὶ σημείῳ συνισταμένων.
12. Πυραμίξ ἐστι σθῆμα στερεὸν ἐπιπέδοιξ περιθόμενον ἀπὸ ἐνὸς ἐπιπέδου πρὸς ἐνὶ σημείῳ συνεστῶξ.
13. Πρίσμα ἐστὶ σθῆμα στερεὸν ἐπιπέδοιξ περιεχόμενον, <ὦν δύο τὰ ἀπεναντίον >ίσα τε καὶ <όμοιά ἐστι καὶ παράλληλα, τὰ δὲ λοιπὰ παραλληλόγραμμα.
14. Σφαῖρά ἐστιν, <ὅταν ἡμικυκλίου μενούσης τῆς διαμέτρου περιενεθθὲν τὸ ἡμικύκλιον εἰς τὸ αὐτὸ πάλιν ἀποκατασταθῇ|, <όθεν >ήρχατο φέρεσθαι, τὸ περιληφθὲν σθῆμα.
15. >Ἀχων δὲ τῆς σφαίρας ἐστίν ἡ μένουσα εὐθεῖα, περὶ <ῇν τὸ ἡμικύκλιον στρέφεται.
16. Κέντρον δὲ τῆς σφαίρας ἐστὶ τὸ αὐτό, <ὃ καὶ τοῦ ἡμικυκλίου.
17. Διάμετροξ δὲ τῆς σφαίρας ἐστίν εὐθεῖα τιξ διὰ τοῦ κέντρου ἡγμένη καὶ περατουμένη ἐφ> ἑκάτερα τὰ μέρη ὑπὸ τῆς ἐπιφανείας τῆς σφαίρας.
18. Κῶνόξ ἐστίν, <ὅταν ὀρθογωνίου τριγώνου μενούσης μιᾶξ πλευρᾶξ τῶν περὶ τὴν ὀρθὴν γωνίαν περιενεθθὲν τὸ τρίγωνον εἰς τὸ αὐτὸ πάλιν ἀποκατασταθῇ|, <όθεν >ήρχατο φέρεσθαι, τὸ περιληφθὲν σθῆμα. κ>ὰν μὲν ἡ μένουσα εὐθεῖα >ίση >ῇ| τῇ| λοιπῇ| [τῇ|] περὶ τὴν ὀρθὴν περιφερομένη, ὀρθογώνιοξ >έσται ὁ κῶνοξ, ἔαν δὲ ἐλάττων, ἀμβλυγώνιοξ, ἔαν δὲ μείζων, ὀξυγώνιοξ.
19. >Ἀχων δὲ τοῦ κώνου ἐστίν ἡ μένουσα εὐθεῖα, περὶ <ῇν τὸ τρίγωνον στρέφεται.
20. Βάσιξ δὲ ὁ κύκλος ὁ ὑπὸ τῆς περιφερομένης εὐθείας γραφόμενος.
21. Κύλινδρόξ ἐστίν, <ὅταν ὀρθογωνίου παραλληλογράμμου μενούσης μιᾶξ πλευρᾶξ τῶν περὶ τὴν ὀρθὴν γωνίαν περιενεθθὲν τὸ παραλληλόγραμμον εἰς τὸ αὐτὸ πάλιν ἀποκατασταθῇ|,

<όθεν >ήρχατο φέρεσθαι, τὸ περιληφθὲν σθῆμα.

22. >'Αχων δὲ τοῦ κυλίνδρου ἐστὶν ἡ μένουσα εὐθεΐα, περὶ <ὣν τὸ παραλληλόγραμμον στρέφεται.

23. Βάσειξ δὲ οἱ κύκλοι οἱ ὑπὸ τῶν ἀπεναντίον περιεχομένων δύο πλευρῶν γραφόμενοι.

24. <Ὅμοιοι κῶνοι καὶ κύλινδροί εισιν, <ῶν ο<ί τε >άχονεξ καὶ αἱ διάμετροι τῶν βάσεων ἀνάλογόν εισιν.

25. Κύβοξ ἐστὶ σθῆμα στερεὸν ὑπὸ <ἐχ τετραγώνων >ίσων περιεχόμενον.

26. Ὀκτάεδρόν ἐστὶ σθῆμα στερεὸν ὑπὸ ὀκτὼ τριγώνων >ίσων καὶ ἰσοπλεύρων περιεχόμενον.

27. Εἰκοσάεδρόν ἐστὶ σθῆμα στερεὸν ὑπὸ ε>ίκοσι τριγώνων >ίσων καὶ ἰσοπλεύρων περιεχόμενον.

28. Δωδεκάεδρόν ἐστὶ σθῆμα στερεὸν ὑπὸ δώδεκα πενταγώνων >ίσων καὶ ἰσοπλεύρων καὶ ἰσογωνίων περιεχόμενον.

Definitions

1. A solid is a (figure) having length and breadth and depth.

2. The extremity of a solid (is) a surface.

3. A straight-line is at right-angles to a plane when it makes right-angles with all of the straight-lines joined to it which are also in the plane.

4. A plane is at right-angles to a(nother) plane when (all of) the straight-lines drawn in one of the planes, at right-angles to the common section of the planes, are at right-angles to the remaining plane.

5. The inclination of a straight-line to a plane is the angle contained by the drawn and standing (straight-lines), when a perpendicular is lead to the plane from the end of the (standing) straight-line raised (out of the plane), and a straight-line is (then) joined from the point (so) generated to the end of the (standing) straight-line (lying) in the plane.

6. The inclination of a plane to a(nother) plane is the acute angle contained by the (straight-lines), (one) in each of the planes, drawn at right-angles to the common segment (of the planes), at the same point.

7. A plane is said to be similarly inclined to a plane, as another to another, when the aforementioned angles of inclination are equal to one another.

8. Parallel planes are those which do not meet (one another).

9. Similar solid figures are those contained by equal numbers of similar planes (which are similarly arranged).

10. But equal and similar solid figures are those contained by similar planes equal in number and in magnitude (which are similarly arranged).

11. A solid angle is the inclination (constituted) by more than two lines joining one another (at the same point), and not being in the same surface, to all of the lines. Otherwise, a solid angle is that contained by more than two plane angles, not being in the same plane, and constructed at one point.

12. A pyramid is a solid figure, contained by planes, (which is) constructed from one plane to one point.

13. A prism is a solid figure, contained by planes, of which the two opposite (planes) are equal, similar, and parallel, and the remaining (planes) are parallelograms.

14. A sphere is the figure enclosed when, the diameter of a semicircle remaining (fixed), the semicircle is carried around, and again established at the same (position) from which it began to be moved.

15. And the axis of the sphere is the fixed straight-line about which the semicircle is turned.

16. And the center of the sphere is the same as that of the

semicircle.

17. And the diameter of the sphere is any straight-line which is drawn through the center and terminated in both directions by the surface of the sphere.

18. A cone is the figure enclosed when, one of the sides of a right-angled triangle about the right-angle remaining (fixed), the triangle is carried around, and again established at the same (position) from which it began to be moved. And if the fixed straight-line is equal to the remaining (straight-line) about the right-angle, (which is) carried around, then the cone will be right-angled, and if less, obtuse-angled, and if greater, acute-angled.

19. And the axis of the cone is the fixed straight-line about which the triangle is turned.

20. And the base (of the cone is) the circle described by the (remaining) straight-line (about the right-angle which is) carried around (the axis).

21. A cylinder is the figure enclosed when, one of the sides of a right-angled parallelogram about the right-angle remaining (fixed), the parallelogram is carried around, and again established at the same (position) from which it began to be moved.

22. And the axis of the cylinder is the stationary straight-line about which the parallelogram is turned.

23. And the bases (of the cylinder are) the circles described by the two opposite sides (which are) carried around.

24. Similar cones and cylinders are those for which the axes and the diameters of the bases are proportional.

25. A cube is a solid figure contained by six equal squares.

26. An octahedron is a solid figure contained by eight equal and equilateral triangles.

27. An icosahedron is a solid figure contained by twenty equal and equilateral triangles.

28. A dodecahedron is a solid figure contained by twelve equal, equilateral, and equiangular pentagons.

1.

Εὐθεία γραμμῆς μέρος μὲν τι οὐκ >έστιν ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ, μέρος δέ τι ἐν μετεωροτέρῳ.

Εἰ γὰρ δυνατόν, εὐθεία γραμμῆς τῆς ΑΒΓ μέρος μὲν τι τὸ ΑΒ >έστω ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ, μέρος δέ τι τὸ ΒΓ ἐν μετεωροτέρῳ.

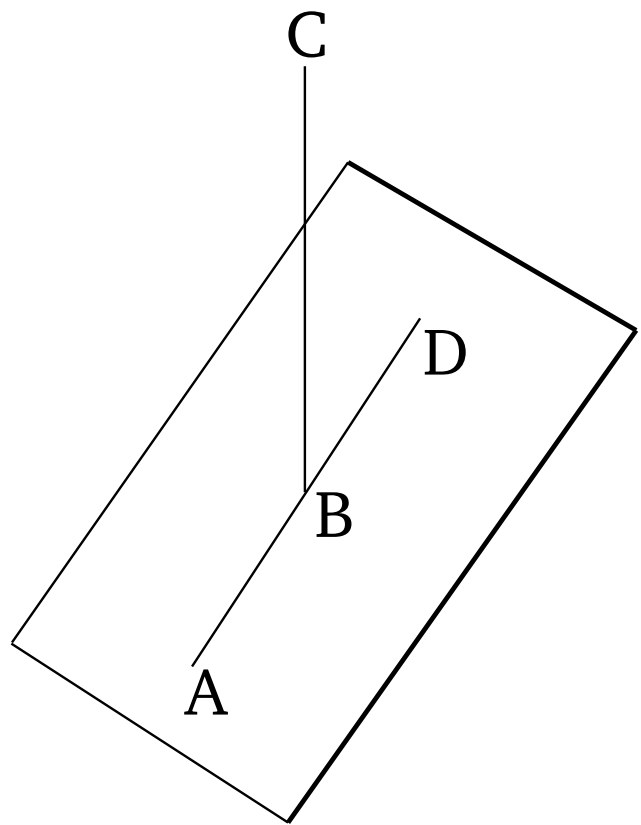
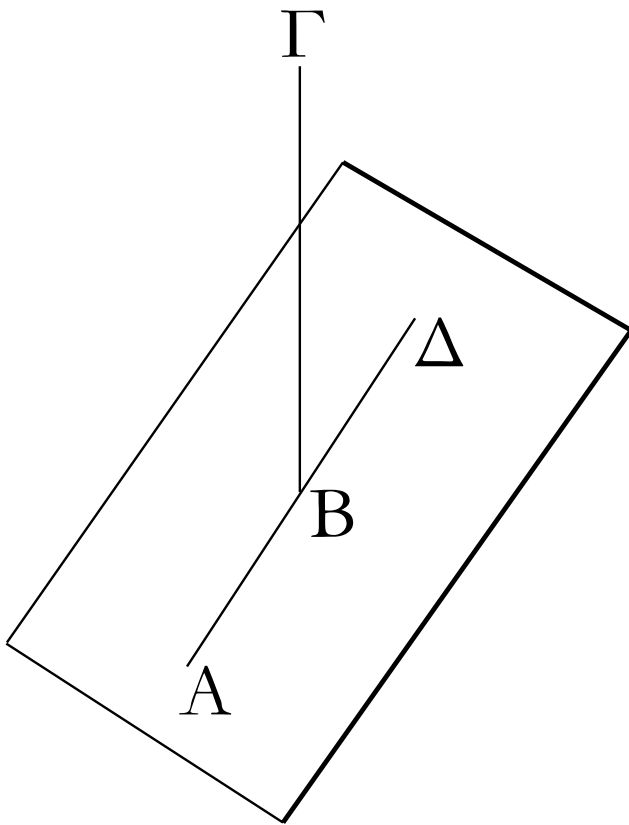
>Ἔσται δὴ τις τῇ ΑΒ συννεθῇ εὐθεῖα ἐπ' εὐθείᾳ ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ. >έστω ἡ ΒΔ; δύο >άρα εὐθειῶν τῶν ΑΒΓ, ΑΒΔ κοινὸν τμήμα ἐστὶν ἡ ΑΒ; <όπερ ἐστὶν ἀδύνατον, ἐπειδὴ περ ἕαν κέντρῳ τῷ Β καὶ διαστήματι τῷ ΑΒ κύκλον γράψωμεν, αἱ διάμετροι ἀνίσουσ' ἀπολήγονται τοῦ κύκλου περιφερείας.

Proposition 1[†]

Some part of a straight-line cannot be in a reference plane, and some part in a more elevated (plane).

For, if possible, let some part, AB , of the straight-line ABC be in a reference plane, and some part, BC , in a more elevated (plane).

In the reference plane, there will be some straight-line continuous with, and straight-on to, AB .[‡] Let it be BD . Thus, AB is a common segment of the two (different) straight-lines ABC and ABD . The very thing is impossible, inasmuch as if we draw a circle with center B and radius AB then the diameters (ABD and ABC) will cut off unequal circumferences of the circle.



Εὐθείαξ ὥρα γραμμῆξ μέρος μὲν τι οὐκ ἔστιν ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ, τὸ δὲ ἐν μετεωροτέρῳ; <ὅπερ ἔδει δεῖχαι.

Thus, some part of a straight-line cannot be in a reference plane, and (some part) in a more elevated (plane). (Which is) the very thing it was required to show.

† The proofs of the first three propositions in this book are not at all rigorous. Hence, these three propositions should properly be regarded as additional axioms.

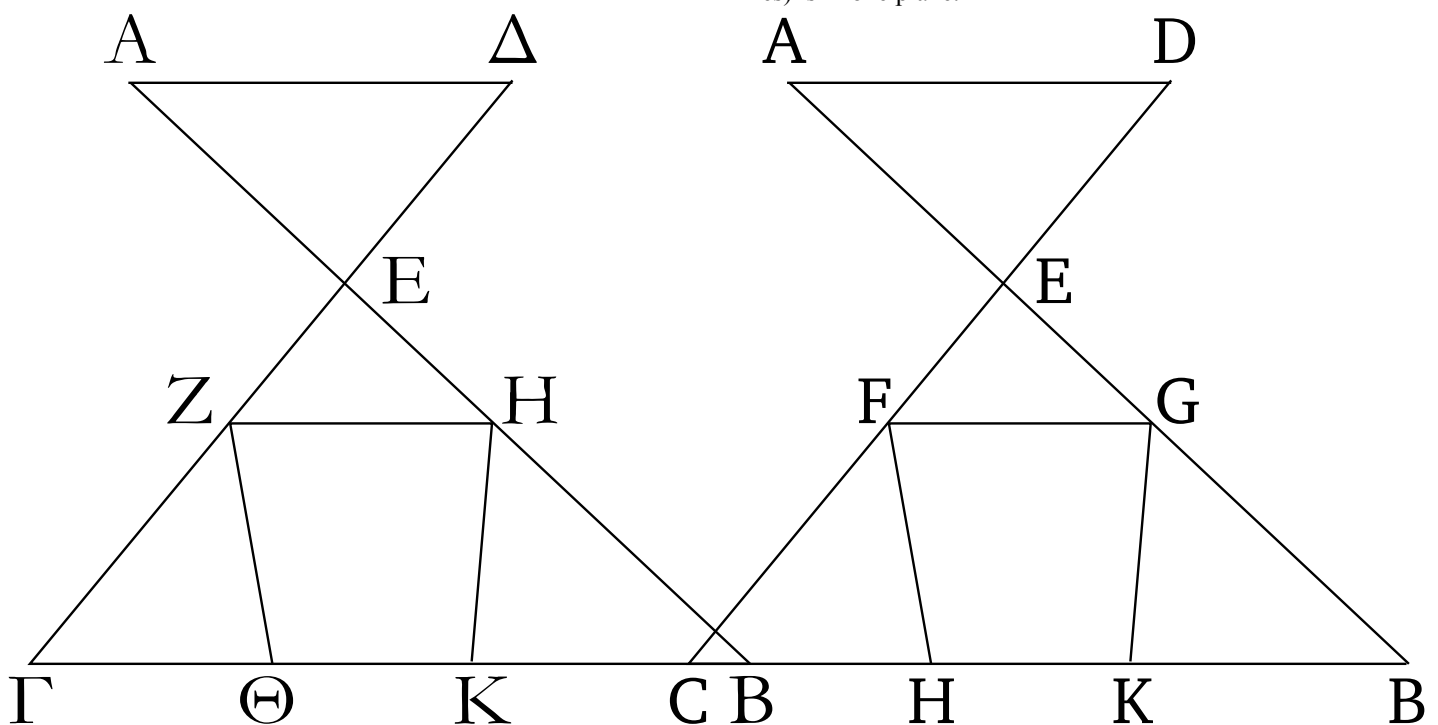
‡ This assumption essentially presupposes the validity of the proposition under discussion.

2.

Proposition 2

Ἐὰν δύο εὐθεῖαι τέμνωσιν ἀλλήλας, ἐν ἐνί εἰσιν ἐπιπέδῳ, καὶ πᾶν τρίγωνον ἐν ἐνί ἔστιν ἐπιπέδῳ.

If two straight-lines cut one another then they are in one plane, and every triangle (formed using segments of both lines) is in one plane.



Δύο γὰρ εὐθεῖαι αἱ AB , $\Gamma\Delta$ τεμνέτωσαν ἀλλήλας κατὰ τὸ E σημεῖον. λέγω, <ὅτι αἱ AB , $\Gamma\Delta$ ἐν ἐνὶ εἰσιν ἐπιπέδῳ, καὶ πᾶν τρίγωνον ἐν ἐνὶ ἐστὶν ἐπιπέδῳ.

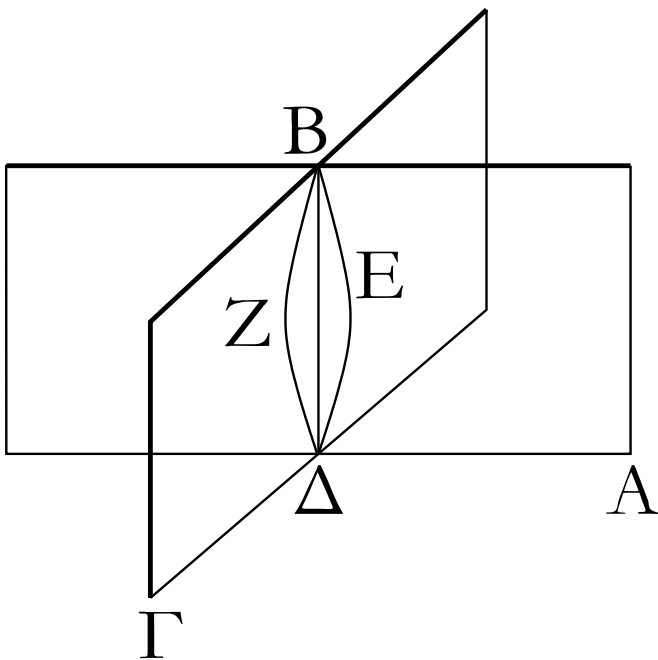
Εἰλήφθω γὰρ ἐπὶ τῶν $ΕΓ$, $ΕΒ$ τυθόντα σημεῖα τὰ Z , H , καὶ ἐπεζεύθωσαν αἱ ΓB , ZH , καὶ διήθωσαν αἱ $Z\Theta$, HK ; λέγω πρῶτον, <ὅτι τὸ $ΕΓΒ$ τρίγωνον ἐν ἐνὶ ἐστὶν ἐπιπέδῳ. εἰ γὰρ ἐστὶ τοῦ $ΕΓΒ$ τριγώνου μέρος >ῆτοι τὸ $Z\Theta\Gamma$ >ῆ τὸ HBK ἐν τῷ ὑποκειμένῳ [ἐπιπέδῳ], τὸ δὲ λοιπὸν ἐν >άλλῳ, >έσται καὶ μιᾶς τῶν $ΕΓ$, $ΕΒ$ εὐθειῶν μέρος μὲν τι ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ, τὸ δὲ ἐν ἄλλῳ. εἰ δὲ τοῦ $ΕΓΒ$ τριγώνου τὸ $Z\Gamma B H$ μέρος >ῆ ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ, τὸ δὲ λοιπὸν ἐν >άλλῳ, >έσται καὶ ἀμφοτέρων τῶν $ΕΓ$, $ΕΒ$ εὐθειῶν μέρος μὲν τι ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ, τὸ δὲ ἐν >άλλῳ; <όπερ >άτοπον ἐδείχθη. τὸ >άρα $ΕΓΒ$ τρίγωνον ἐν ἐνὶ ἐστὶν ἐπιπέδῳ. ἐν <ῶ δὲ ἐστὶ τὸ $ΕΓΒ$ τρίγωνον, ἐν τούτῳ καὶ ἑκατέρα τῶν $ΕΓ$, $ΕΒ$, ἐν <ῶ δὲ ἑκατέρα τῶν $ΕΓ$, $ΕΒ$, ἐν τούτῳ καὶ αἱ AB , $\Gamma\Delta$. αἱ AB , $\Gamma\Delta$ >άρα εὐθεῖαι ἐν ἐνὶ εἰσιν ἐπιπέδῳ, καὶ πᾶν τρίγωνον ἐν ἐνὶ ἐστὶν ἐπιπέδῳ; <όπερ >έδει δεῖχαι.

For let the two straight-lines AB and CD have cut one another at point E . I say that AB and CD are in one plane, and that every triangle (formed using segments of both lines) is in one plane.

For let the random points F and G be taken on EC and EB (respectively). And let CB and FG be joined, and let FH and GK be drawn across. I say, first of all, that triangle ECB is in one (reference) plane. For if part of triangle ECB , either FHC or GBK , is in the reference [plane], and the remainder in a different (plane) then a part of one the straight-lines EC and EB will also be in the reference plane, and (a part) in a different (plane). And if the part $FCBG$ of triangle ECB is in the reference plane, and the remainder in a different (plane) then parts of both of the straight-lines EC and EB will also be in the reference plane, and (parts) in a different (plane). The very thing was shown to be absurd [Prop. 11.1]. Thus, triangle ECB is in one plane. And in whichever (plane) triangle ECB is (found), in that (plane) EC and EB (will) each also (be found). And in whichever (plane) EC and EB (are) each (found), in that (plane) AB and CD (will) also (be found) [Prop. 11.1]. Thus, the straight-lines AB and CD are in one plane, and every triangle (formed using segments of both lines) is in one plane. (Which is) the very thing it was required to show.

3.

Ἐὰν δύο ἐπίπεδα τεμνῇ >άλληλα, ἡ κοινὴ αὐτῶν τομὴ εὐθεῖα ἐστίν.

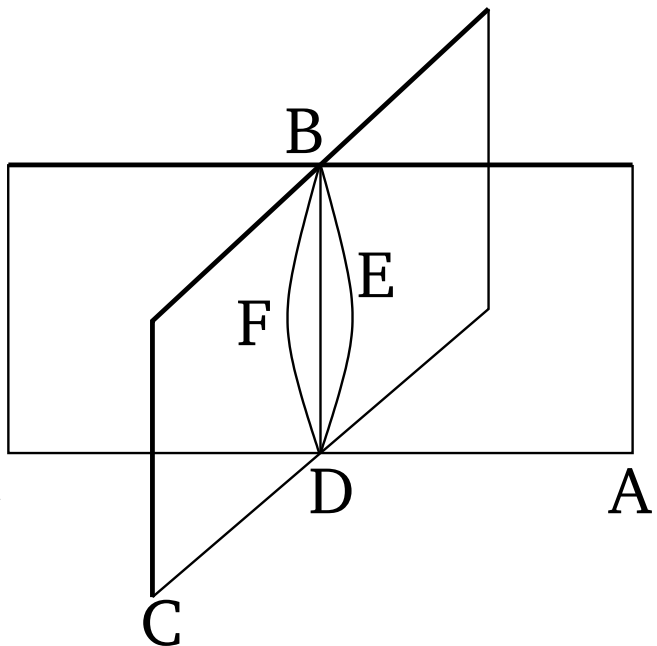


Δύο γὰρ ἐπίπεδα τὰ AB , $B\Gamma$ τεμνέτω >άλληλα, κοινὴ δὲ αὐτῶν τομὴ >έστω ἡ ΔB γραμμὴ; λέγω, <ὅτι ἡ ΔB γραμμὴ εὐθεῖα ἐστίν.

Εἰ γὰρ μή, ἐπεζεύθω ἀπὸ τοῦ Δ ἐπὶ τὸ B ἐν μὲν τῷ AB ἐπιπέδῳ εὐθεῖα ἡ ΔEB , ἐν δὲ τῷ $B\Gamma$ ἐπιπέδῳ εὐθεῖα ἡ ΔZB . >έσται δὴ δύο εὐθειῶν τῶν ΔEB , ΔZB τὰ αὐτὰ πέρατα, καὶ περιέχουσι δηλαδὴ θωρίον; <όπερ >άτοπον. ο>ύκ >άρα αἱ ΔEB , ΔZB εὐθεῖαι εἰσιν. ὁμοίως δὲ δείχομεν, <ὅτι οὐδὲ >άλλη τις ἀπὸ τοῦ Δ ἐπὶ τὸ B ἐπιζευγνυμένη εὐθεῖα >έσται πλὴν τῆς ΔB κοινῆς τομῆς τῶν AB , $B\Gamma$ ἐπιπέδων.

Proposition 3

If two planes cut one another then their common section is a straight-line.



For let the two planes AB and BC cut one another, and let their common section be the line DB . I say that the line DB is straight.

For, if not, let the straight-line DEB be joined from D to B in the plane AB , and the straight-line DFB in the plane BC . So two straight-lines, DEB and DFB , will have the same ends, and they will clearly enclose an area. The very thing (is) absurd. Thus, DEB and DFB are not straight-lines. So, similarly, we can show than no other straight-line can be joined from D to B except DB , the common section

Ἐάν >άρα δύο ἐπίπεδα τέμνῃ >άλληλα, ἡ κοινὴ αὐτῶν τομὴ εὐθεΐα ἐστίν; <ὅπερ >έδει δεῖξαι.

4.

Ἐάν εὐθεΐα δύο εὐθειῶν τεμνοῦσαι ἀλλήλας πρὸς ὀρθὰς ἐπὶ τῇ κοινῇ τομῇ ἐπισταθῇ, καὶ τῷ δι' αὐτῶν ἐπιπέδῳ πρὸς ὀρθὰς >έσται.

Εὐθεΐα γάρ τις ἡ EZ δύο εὐθειῶν ταῖς AB , $\Gamma\Delta$ τεμνοῦσαι ἀλλήλας κατὰ τὸ E σημεῖον ἀπὸ τοῦ E πρὸς ὀρθὰς ἐφεστάτω; λέγω, <ὅτι ἡ EZ καὶ τῷ διὰ τῶν AB , $\Gamma\Delta$ ἐπιπέδῳ πρὸς ὀρθὰς ἐστίν.

Ἀπειλήφθωσαν γὰρ αἱ AE , EB , ΓE , ED >ίσαι ἀλλήλαις, καὶ διήθω τις διὰ τοῦ E , ὡς >έθυεν, ἡ $HE\Theta$, καὶ ἐπεξεύθωσαν αἱ AD , ΓB , καὶ >έτι ἀπὸ τυθόντος τοῦ Z ἐπεξεύθωσαν αἱ ZA , ZH , $Z\Delta$, $Z\Gamma$, $Z\Theta$, ZB .

of the planes AB and BC .

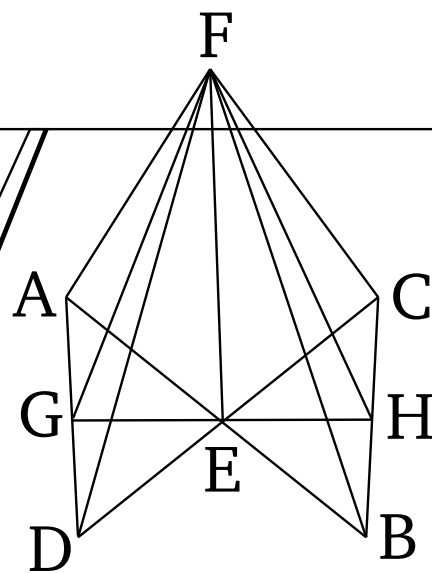
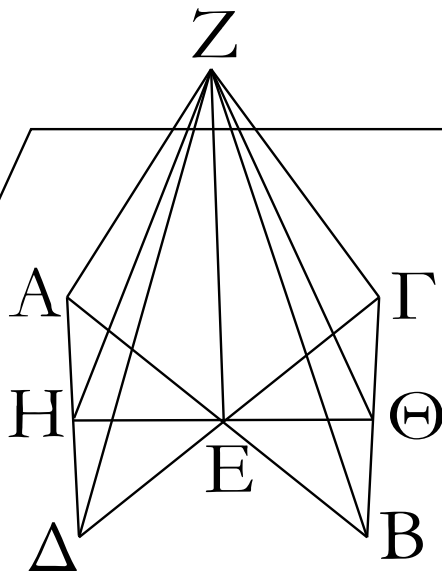
Thus, if two planes cut one another then their common section is a straight-line. (Which is) the very thing it was required to show.

Proposition 4

If a straight-line is set up at right-angles to two straight-lines cutting one another, at the common point of section, then it will also be at right-angles to the plane (passing) through them (both).

For let some straight-line EF have (been) set up at right-angles to two straight-lines, AB and CD , cutting one another at point E , at E . I say that EF is also at right-angles to the plane (passing) through AB and CD .

For let AE , EB , CE and ED be cut off from (the two straight-lines so as to be) equal to one another. And let GEH be drawn, at random, through E (in the plane passing through AB and CD). And let AD and CB be joined. And, furthermore, let FA , FG , FD , FC , FH , and FB be joined from the random (point) F (on EF).



Καὶ ἐπεὶ δύο αἱ AE , ED δυοὶ ταῖς ΓE , EB >ίσαι εἰσὶ καὶ γωνίαξ >ίσαξ περιέθουσιν, βάσιξ >άρα ἡ AD βάσει τῇ ΓB >ίση ἐστίν, καὶ τὸ AED τρίγωνον τῷ ΓEB τριγώνῳ >ίσον >έσται; <ὥστε καὶ γωνία ἡ ὑπὸ ΔAE γωνία τῇ ὑπὸ $EB\Gamma$ >ίση [ἐστίν]. >έστι δὲ καὶ ἡ ὑπὸ AEH γωνία τῇ ὑπὸ $BE\Theta$ >ίση. δύο δὲ τριγώνα ἐστὶ τὰ AHE , $BE\Theta$ τὰξ δύο γωνίαξ δυοὶ γωνίαξ >ίσαξ >έθοντα ἑκατέραν ἑκατέρῳ καὶ μίαν πλευρὰν μιᾷ | πλευρᾷ | >ίσην τὴν πρὸξ ταῖξ >ίσαιξ γωνίαξ τὴν AE τῇ EB ; καὶ τὰξ λοιπαῖξ >άρα πλευρὰξ ταῖξ λοιπαῖξ πλευραῖξ >ίσαξ <έχουσιν. >ίση >άρα ἡ μὲν HE τῇ $E\Theta$, ἡ δὲ AH τῇ $B\Theta$. καὶ ἐπεὶ >ίση ἐστίν ἡ AE τῇ EB , κοινὴ δὲ καὶ πρὸξ ὀρθὰς ἡ ZE , βάσιξ >άρα ἡ ZA βάσει τῇ ZB ἐστίν >ίση. διὰ τὰ αὐτὰ δὲ καὶ ἡ $Z\Gamma$ τῇ $Z\Delta$ ἐστίν >ίση. καὶ ἐπεὶ >ίση ἐστίν ἡ AD τῇ ΓB , >έστι δὲ καὶ ἡ ZA τῇ ZB >ίση, δύο δὲ αἱ ZA , AD δυοὶ ταῖξ ZB , $B\Gamma$ >ίσαι εἰσὶν ἑκατέρα ἑκατέρῳ; καὶ βάσιξ ἡ $Z\Delta$ βάσει τῇ $Z\Gamma$ ἐδείθθη >ίση; καὶ γωνία >άρα ἡ ὑπὸ ZAD γωνία τῇ ὑπὸ $ZB\Gamma$ >ίση ἐστίν. καὶ ἐπεὶ πάλιν ἐδείθθη ἡ AH τῇ $B\Theta$ >ίση, ἀλλὰ μὴν καὶ ἡ ZA τῇ ZB >ίση, δύο δὲ αἱ ZA , AH δυοὶ ταῖξ ZB , $B\Theta$ >ίσαι εἰσίν. καὶ γωνία ἡ ὑπὸ ZAH

For since the two (straight-lines) AE and ED are equal to the two (straight-lines) CE and EB , and they enclose equal angles [Prop. 1.15], the base AD is thus equal to the base CB , and triangle AED will be equal to triangle CEB [Prop. 1.4]. Hence, the angle DAE [is] equal to the angle EBC . And the angle AEG (is) also equal to the angle BEH [Prop. 1.15]. So AGE and BEH are two triangles having two angles equal to two angles, respectively, and one side equal to one side—(namely), those by the equal angles, AE and EB . Thus, they will also have the remaining sides equal to the remaining sides [Prop. 1.26]. Thus, GE (is) equal to EH , and AG to BH . And since AE is equal to EB , and FE is common and at right-angles, the base FA is thus equal to the base FB [Prop. 1.4]. So, for the same (reasons), FC is also equal to FD . And since AD is equal to CB , and FA is also equal to FB , the two (straight-lines) FA and AD are equal to the two (straight-lines) FB and BC , respectively. And the base FD was shown (to be) equal to the base FC . Thus, the angle FAD is also equal

ἐδείθθη >ίση τῇ| ὑπὸ ΖΒΘ; βάσις >άρα ἢ ΖΗ βάσει τῇ| ΖΘ ἐστίν >ίση. καὶ ἐπεὶ πάλιν >ίση ἐδείθθη ἢ ΗΕ τῇ| ΕΘ, κοινὴ δὲ ἢ ΕΖ, δύο δὴ αἱ ΗΕ, ΕΖ δυοὶ ταῖς ΘΕ, ΕΖ >ίσαι εἰσίν; καὶ βάσις ἢ ΖΗ βάσει τῇ| ΖΘ >ίση; γωνία >άρα ἢ ὑπὸ ΗΕΖ γωνία τῇ| ὑπὸ ΘΕΖ >ίση ἐστίν. ὀρθὴ >άρα ἑκάτερα τῶν ὑπὸ ΗΕΖ, ΘΕΖ γωνιῶν. ἢ ΖΕ >άρα πρὸς τὴν ΗΘ τυθόντως διὰ τοῦ Ε ἀθθεῖσαν ὀρθὴ ἐστίν. ὁμοίως δὲ δείχομεν, <ὅτι ἢ ΖΕ καὶ πρὸς πάσας τὰς ἀπτομένας αὐτῆς εὐθείας καὶ ο>ύσας ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ὀρθὰς ποιήσει γωνίας. εὐθεῖα δὲ πρὸς ἐπίπεδον ὀρθὴ ἐστίν, <όταν πρὸς πάσας τὰς ἀπτομένας αὐτῆς εὐθείας καὶ ο>ύσας ἐν τῷ αὐτῷ ἐπιπέδῳ ὀρθὰς ποιῇ| γωνίας; ἢ ΖΕ >άρα τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς ἐστίν. τὸ δὲ ὑποκείμενον ἐπίπεδόν ἐστι τὸ διὰ τῶν ΑΒ, ΓΔ εὐθειῶν. ἢ ΖΕ >άρα πρὸς ὀρθὰς ἐστὶ τῷ διὰ τῶν ΑΒ, ΓΔ ἐπιπέδῳ.

Ἐὰν >άρα εὐθεῖα δύο εὐθείαις τεμνούσαις ἀλλήλας πρὸς ὀρθὰς ἐπὶ τῆς κοινῆς τομῆς ἐπισταθῇ, καὶ τῷ δι> αὐτῶν ἐπιπέδῳ πρὸς ὀρθὰς >έσται; <όπερ >έδει δεῖχαι.

to the angle FBC [Prop. 1.8]. And, again, since AG was shown (to be) equal to BH , but FA (is) also equal to FB , the two (straight-lines) FA and AG are equal to the two (straight-lines) FB and BH (respectively). And the angle FAG was shown (to be) equal to the angle FBH . Thus, the base FG is equal to the base FH [Prop. 1.4]. And, again, since GE was shown (to be) equal to EH , and EF (is) common, the two (straight-lines) GE and EF are equal to the two (straight-lines) HE and EF (respectively). And the base FG (is) equal to the base FH . Thus, the angle GEF is equal to the angle HEF [Prop. 1.8]. Each of the angles GEF and HEF (are) thus right-angles [Def. 1.10]. Thus, FE is at right-angles to GH , which was drawn at random through E (in the reference plane passing through AB and AC). So, similarly, we can show that FE will make right-angles with all straight-lines joined to it which are in the reference plane. And a straight-line is at right-angles to a plane when it makes right-angles with all straight-lines joined to it which are in the plane [Def. 11.3]. Thus, FE is at right-angles to the reference plane. And the reference plane is that (passing) through the straight-lines AB and CD . Thus, FE is at right-angles to the plane (passing) through AB and CD .

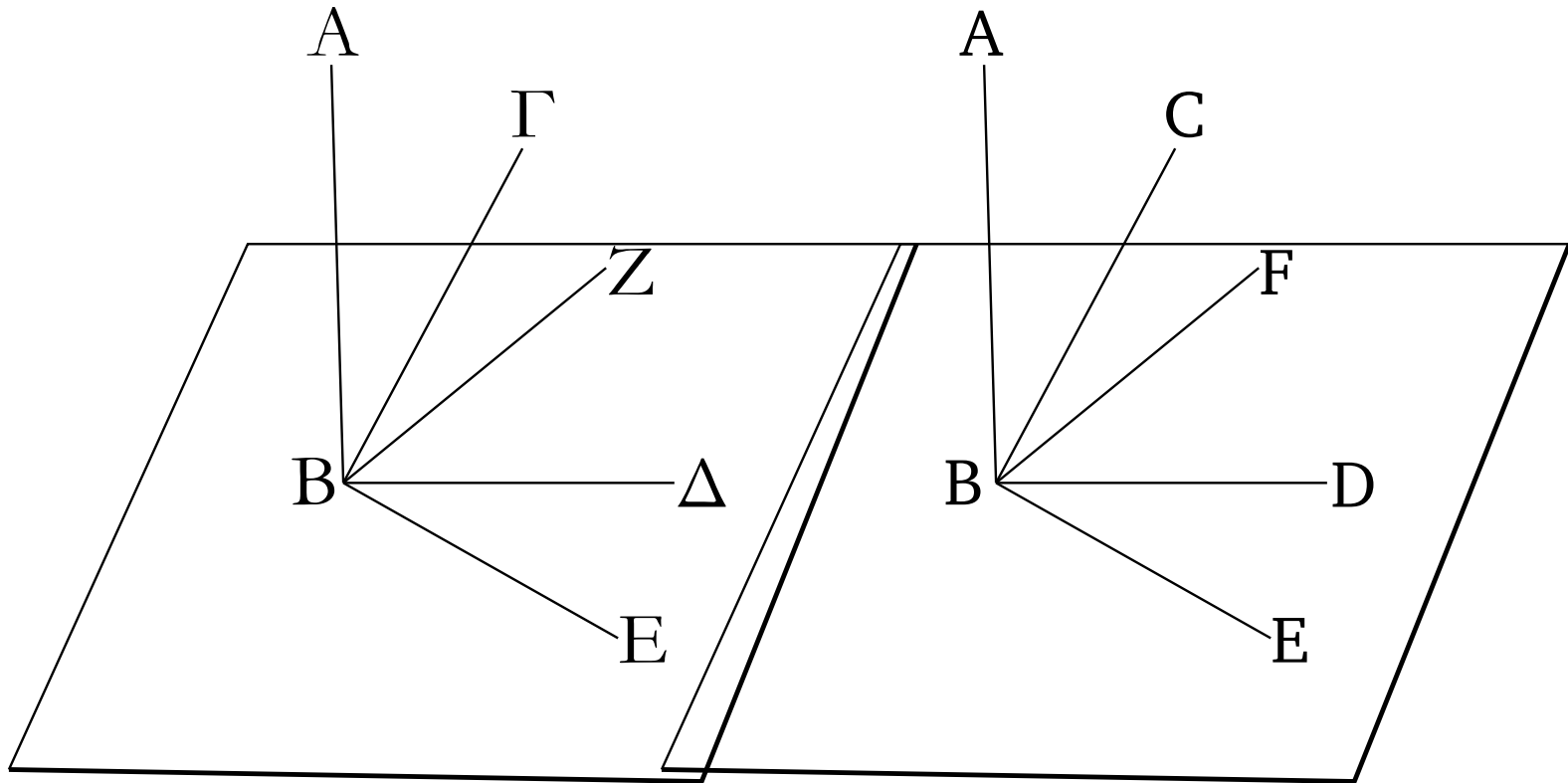
Thus, if a straight-line is set up at right-angles to two straight-lines cutting one another, at the common point of section, then it will also be at right-angles to the plane (passing) through them (both). (Which is) the very thing it was required to show.

5.

Ἐὰν εὐθεῖα τρισὶν εὐθείαις ἀπτομέναις ἀλλήλων πρὸς ὀρθὰς ἐπὶ τῆς κοινῆς τομῆς ἐπισταθῇ, αἱ τρεῖς εὐθεῖαι ἐν ἐνὶ εἰσιν ἐπιπέδῳ.

Proposition 5

If a straight-line is set up at right-angles to three straight-lines cutting one another, at the common point of section, then the three straight-lines are in one plane.



Εὐθεῖα γάρ τις ἢ ΑΒ τρισὶν εὐθείαις ταῖς ΒΓ, ΒΔ, ΒΕ πρὸς ὀρθὰς ἐπὶ τῆς κατὰ τὸ Β ἀφῆς ἐφεστάτω; λέγω, <ὅτι αἱ ΒΓ,

For let some straight-line AB be set up at right-angles to three straight-lines BC , BD , and BE , at the (common)

BA, BE ἐν ἐνί εἰσιν ἐπιπέδῳ.

μη γάρ, ἀλλ' εἰ δυνατόν, >έστωσαν αἱ μὲν BA, BE ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ, ἡ δὲ BF ἐν μετεωροτέρῳ, καὶ ἐκβεβλήσθω τὸ διὰ τῶν AB, BF ἐπίπεδον; κοινὴν δὲ τομὴν ποιήσῃ ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ εὐθεῖαν. ποιείτω τὴν BZ . ἐν ἐνί >άρα εἰσιν ἐπιπέδῳ τῷ διηγμένῳ διὰ τῶν AB, BF αἱ τρεῖς εὐθεῖαι αἱ AB, BF, BZ . καὶ ἐπεὶ ἡ AB ὀρθή ἐστι πρὸς ἑκατέραν τῶν BA, BE , καὶ τῷ διὰ τῶν BA, BE >άρα ἐπιπέδῳ ὀρθή ἐστὶν ἡ AB . τὸ δὲ διὰ τῶν BA, BE ἐπίπεδον τὸ ὑποκείμενον ἐστίν; ἡ AB >άρα ὀρθή ἐστι πρὸς τὸ ὑποκείμενον ἐπίπεδον. <ὥστε καὶ πρὸς πάσας τὰς ἀπτομένας αὐτῆς εὐθείας καὶ >ύσας ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ὀρθὰς ποιήσει γωνίαξ ἡ AB . <ἀπτεται δὲ αὐτῆς ἡ BZ ὅσα ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ; ἡ >άρα ὑπὸ ABZ γωνία ὀρθή ἐστίν. ὑπόκειται δὲ καὶ ἡ ὑπὸ ABF ὀρθή; >ίση >άρα ἡ ὑπὸ ABZ γωνία τῇ ὑπὸ ABF . καὶ εἰσιν ἐν ἐνί ἐπιπέδῳ; <ὅπερ ἐστὶν ἀδύνατον. οὐκ >άρα ἡ BF εὐθεῖα ἐν μετεωροτέρῳ ἐστὶν ἐπιπέδῳ; αἱ τρεῖς >άρα εὐθεῖαι αἱ BF, BA, BE ἐν ἐνί εἰσιν ἐπιπέδῳ.

Ἐὰν >άρα εὐθεῖα τρισὶν εὐθείαις ἀπτομέναις ἀλλήλων ἐπὶ τῆς ἀφῆς πρὸς ὀρθὰς ἐπισταθῇ, αἱ τρεῖς εὐθεῖαι ἐν ἐνί εἰσιν ἐπιπέδῳ; <ὅπερ >έδει δεῖχαι.

point of section B . I say that BC, BD , and BE are in one plane.

For (if) not, and if possible, let BD and BE be in the reference plane, and BC in a more elevated (plane). And let the plane through AB and BC be produced. So it will make a straight-line as a common section with the reference plane [Def. 11.3]. Let it make BF . Thus, the three straight-lines AB, BC , and BF are in one plane—(namely), that drawn through AB and BC . And since AB is at right-angles to each of BD and BE , AB is thus also at right-angles to the plane (passing) through BD and BE [Prop. 11.4]. And the plane (passing) through BD and BE is the reference plane. Thus, AB is at right-angles to the reference plane. Hence, AB will also make right-angles with all straight-lines joined to it which are also in the reference plane [Def. 11.3]. And BF , which is in the reference plane, is joined to it. Thus, the angle ABF is a right-angle. And ABC was also assumed to be a right-angle. Thus, angle ABF (is) equal to ABC . And they are in one plane. The very thing is impossible. Thus, BC is not in a more elevated plane. Thus, the three straight-lines BC, BD , and BE are in one plane.

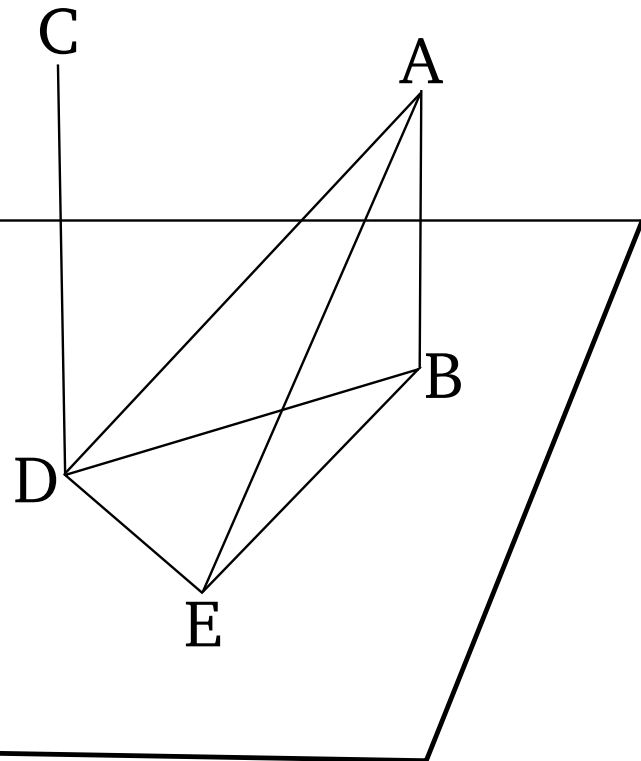
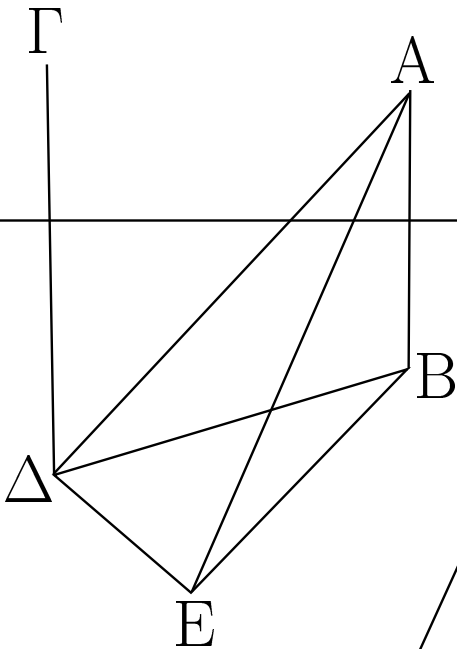
Thus, if a straight-line is set up at right-angles to three straight-lines cutting one another, at the (common) point of section, then the three straight-lines are in one plane. (Which is) the very thing it was required to show.

6.

Ἐὰν δύο εὐθεῖαι τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰς >ῶσιν, παράλληλοι >έσονται αἱ εὐθεῖαι.

Proposition 6

If two straight-lines are at right-angles to the same plane then the straight-lines will be parallel.[†]



Δύο γάρ εὐθεῖαι αἱ $AB, \Gamma\Delta$ τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς >έστωσαν; λέγω, <ὅτι παράλληλός ἐστιν ἡ AB τῇ $\Gamma\Delta$.

Συμβαλλέτωσαν γάρ τῷ ὑποκειμένῳ ἐπιπέδῳ κατὰ τὰ B, Δ σημεῖα, καὶ ἐπεζεύθθω ἡ BD εὐθεῖα, καὶ >ήθθω τῇ BD πρὸς ὀρθὰς ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ἡ DE , καὶ κείσθω τῇ AB >ίση ἡ DE , καὶ ἐπεζεύθθωσαν αἱ BE, AE, AD .

For let the two straight-lines AB and CD be at right-angles to a reference plane. I say that AB is parallel to CD .

For let them meet the reference plane at points B and D (respectively). And let the straight-line BD be joined. And let DE be drawn at right-angles to BD in the reference plane. And let DE be made equal to AB . And let BE, AE , and AD be joined.

Καὶ ἐπεὶ ἡ AB ὀρθή ἐστι πρὸς τὸ ὑποκείμενον ἐπίπεδον, καὶ πρὸς πάσας [$>$ άρα] τὰς ἀπτομένας αὐτῆς εὐθείας καὶ ο $>$ ύσας ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ὀρθὰς ποιήσει γωνίας. $<$ ἀπτεται δὲ τῆς AB ἑκάτερα τῶν BD , BE ο $>$ ύσα ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ; ὀρθή $>$ άρα ἐστὶν ἑκάτερα τῶν ὑπὸ ABD , ABE γωνιῶν. διὰ τὰ αὐτὰ δὴ καὶ ἑκάτερα τῶν ὑπὸ $ΓΔΒ$, $ΓΔΕ$ ὀρθή ἐστίν. καὶ ἐπεὶ $>$ ίση ἐστὶν ἡ AB τῇ $ΔΕ$, κοινὴ δὲ ἡ $ΒΔ$, δύο δὴ αἱ AB , $ΒΔ$ δυοὶ ταῖς $ΕΔ$, $ΔΒ$ $>$ ίσαι εἰσίν; καὶ γωνίας ὀρθὰς περιέθουσιν; βάσις $>$ άρα ἡ $ΑΔ$ βάσει τῇ $ΒΕ$ ἐστὶν $>$ ίση. καὶ ἐπεὶ $>$ ίση ἐστὶν ἡ AB τῇ $ΔΕ$, ἀλλὰ καὶ ἡ $ΑΔ$ τῇ $ΒΕ$, δύο δὴ αἱ AB , $ΒΕ$ δυοὶ ταῖς $ΕΔ$, $ΔΑ$ $>$ ίσαι εἰσίν; καὶ βάσις αὐτῶν κοινὴ ἡ $ΑΕ$; γωνία $>$ άρα ἡ ὑπὸ ABE γωνία τῇ ὑπὸ $ΕΔΑ$ ἐστὶν $>$ ίση. ὀρθὴ δὲ ἡ ὑπὸ ABE ; ὀρθὴ $>$ άρα καὶ ἡ ὑπὸ $ΕΔΑ$; ἡ $ΕΔ$ $>$ άρα πρὸς τὴν $ΔΑ$ ὀρθή ἐστίν. $>$ έστι δὲ καὶ πρὸς ἑκατέραν τῶν $ΒΔ$, $ΔΓ$ ὀρθή. ἡ $ΕΔ$ $>$ άρα τρισὶν εὐθείαις ταῖς $ΒΔ$, $ΔΑ$, $ΔΓ$ πρὸς ὀρθὰς ἐπὶ τῆς ἀφῆς ἐφέστηκεν; αἱ τρεῖς $>$ άρα εὐθεῖαι αἱ $ΒΔ$, $ΔΑ$, $ΔΓ$ ἐν ἐνὶ εἰσιν ἐπιπέδῳ. ἐν $<$ ῶ δὲ αἱ $ΔΒ$, $ΔΑ$, ἐν τούτῳ καὶ ἡ AB ; πᾶν γὰρ τρίγωνον ἐν ἐνὶ ἐστὶν ἐπιπέδῳ; αἱ $>$ άρα AB , $ΒΔ$, $ΔΓ$ εὐθεῖαι ἐν ἐνὶ εἰσιν ἐπιπέδῳ. καὶ ἐστὶν ὀρθὴ ἑκάτερα τῶν ὑπὸ $ABΔ$, $ΒΔΓ$ γωνιῶν; παράλληλος $>$ άρα ἐστὶν ἡ AB τῇ $ΓΔ$. Ἐὰν $>$ άρα δύο εὐθεῖαι τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰς $>$ ῶσιν, παράλληλοι $>$ έσσονται αἱ εὐθεῖαι; $<$ όπερ $>$ έδει δεῖχαι.

And since AB is at right-angles to the reference plane, it will [thus] also make right-angles with all straight-lines joined to it which are in the reference plane [Def. 11.3]. And BD and BE , which are in the reference plane, are each joined to AB . Thus, each of the angles ABD and ABE are right-angles. So, for the same (reasons), each of the angles CDB and CDE are also right-angles. And since AB is equal to DE , and BD (is) common, the two (straight-lines) AB and BD are equal to the two (straight-lines) ED and DB (respectively). And they contain right-angles. Thus, the base AD is equal to the base BE [Prop. 1.4]. And since AB is equal to DE , and AD (is) also (equal) to BE , the two (straight-lines) AB and BE are thus equal to the two (straight-lines) ED and DA (respectively). And their base AE (is) common. Thus, angle ABE is equal to angle EDA [Prop. 1.8]. And ABE (is) a right-angle. Thus, EDA (is) also a right-angle. ED is thus at right-angles to DA . And it is also at right-angles to each of BD and DC . Thus, ED is standing at right-angles to the three straight-lines BD , DA , and DC at the (common) point of section. Thus, the three straight-lines BD , DA , and DC are in one plane [Prop. 11.5]. And in which(ever) plane DB and DA (are found), in that (plane) AB (will) also (be found). For every triangle is in one plane [Prop. 11.2]. And each of the angles ABD and BDC is a right-angle. Thus, AB is parallel to CD [Prop. 1.28]. Thus, if two straight-lines are at right-angles to the same plane then the straight-lines will be parallel. (Which is) the very thing it was required to show.

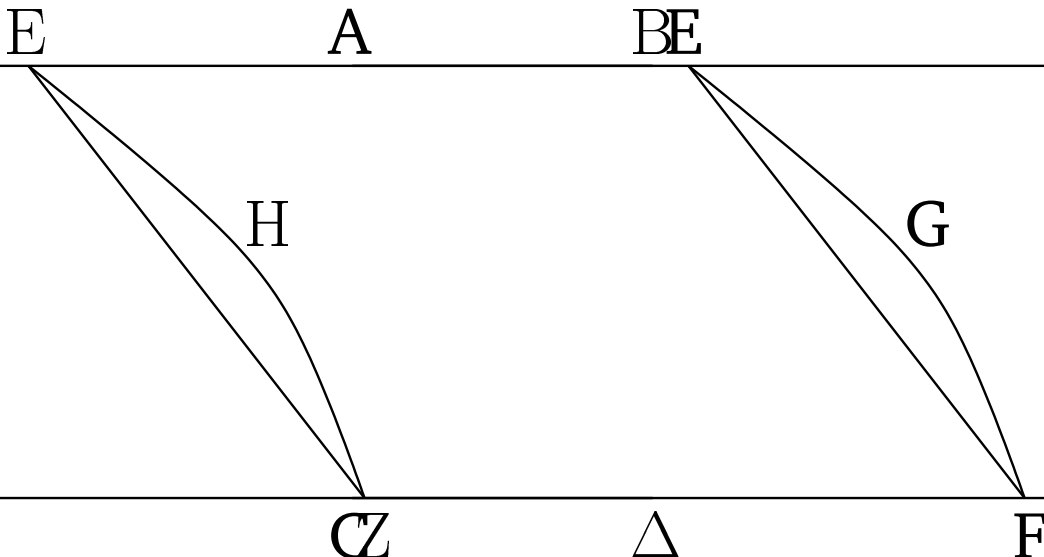
† In other words, the two straight-lines lie in the same plane, and never meet when produced in either direction.

7.

Proposition 7

Ἐὰν $>$ ῶσι δύο εὐθεῖαι παράλληλοι, ληφθῇ δὲ ἐφ' ἑκατέρας αὐτῶν τυθόντα σημεῖα, ἡ ἐπὶ τὰ σημεῖα ἐπιζευγνυμένη εὐθεῖα ἐν τῷ αὐτῷ ἐπιπέδῳ ἐστὶ ταῖς παραλλήλοις.

If there are two parallel straight-lines, and random points are taken on each of them, then the straight-line joining the two points is in the same plane as the parallel (straight-lines).



$>$ Ἐστωσαν δύο εὐθεῖαι παράλληλοι αἱ AB , $ΓΔ$, καὶ εἰλήφθω ἐφ' ἑκατέρας αὐτῶν τυθόντα σημεῖα τὰ E , Z ; λέγω, $<$ ότι ἡ ἐπὶ τὰ E , Z σημεῖα ἐπιζευγνυμένη εὐθεῖα ἐν τῷ αὐτῷ ἐπιπέδῳ ἐστὶ ταῖς παραλλήλοις. μὴ γάρ, ἀλλ' εἰ δυνατόν, $>$ έστω ἐν μετεωροτέρῳ ὥς ἡ $ΕΗΖ$, καὶ διήθθω διὰ τῆς $ΕΗΖ$ ἐπίπεδον; τομὴν δὴ ποιήσει ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ εὐθεῖαν. ποιεῖτω ὥς τὴν $ΕΖ$; δύο $>$ άρα εὐθεῖαι αἱ $ΕΗΖ$, $ΕΖ$ θωρίον περιέχουσιν; $<$ όπερ ἐστὶν

Let AB and CD be two parallel straight-lines, and let the random points E and F be taken on each of them (respectively). I say that the straight-line joining points E and F is in the same (reference) plane as the parallel (straight-lines). For (if) not, and if possible, let it be in a more elevated (plane), such as EGF . And let a plane be drawn through EGF . So it will make a straight cutting in the reference plane [Prop. 11.3]. Let it make EF . Thus, two straight-lines

ἀδύνατον. οὐκ >άρα ἢ ἀπὸ τοῦ E ἐπὶ τὸ Z ἐπιζευγνυμένη εὐθεῖα ἐν μετεωροτέρῳ ἐστὶν ἐπιπέδῳ; ἐν τῷ διὰ τῶν AB , $ΓB$ >άρα παραλλήλων ἐστὶν ἐπιπέδῳ ἢ ἀπὸ τοῦ E ἐπὶ τὸ Z ἐπιζευγνυμένη εὐθεῖα.

Ἐὰν >άρα >ῶσι δύο εὐθεῖαι παράλληλοι, ληφθῇ δὲ ἐφ' ἑκατέρᾳ αὐτῶν τυθόντα σημεῖα, ἢ ἐπὶ τὰ σημεῖα ἐπιζευγνυμένη εὐθεῖα ἐν τῷ αὐτῷ ἐπιπέδῳ ἐστὶ ταῖς παραλλήλοις; <ὅπερ >έδει δεῖχαι.

8.

Ἐὰν >ῶσι δύο εὐθεῖαι παράλληλοι, ἢ δὲ ἑτέρα αὐτῶν ἐπιπέδῳ τινὶ πρὸς ὀρθὰς >ῇ, καὶ ἡ λοιπὴ τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰς >έσται.

>Ἐστωσαν δύο εὐθεῖαι παράλληλοι αἱ AB , $ΓΔ$, ἢ δὲ ἑτέρα αὐτῶν ἡ AB τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς >έστω; λέγω, <ὅτι καὶ ἡ λοιπὴ ἡ $ΓΔ$ τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰς >έσται.

Συμβαλλέτωσαν γὰρ αἱ AB , $ΓΔ$ τῷ ὑποκειμένῳ ἐπιπέδῳ κατὰ τὰ B , $Δ$ σημεῖα, καὶ ἐπεζεύθω ἡ $ΒΔ$; αἱ AB , $ΓΔ$, $ΒΔ$ >άρα ἐν ἐνὶ εἰσὶν ἐπιπέδῳ. >ῆθω τῇ BA πρὸς ὀρθὰς ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ἡ $ΔΕ$, καὶ κείσθω τῇ AB >ῖσις ἡ $ΔΕ$, καὶ ἐπεζεύθωσαν αἱ BE , AE , $ΑΔ$.

(with the same end-points), EGF and EF , will enclose an area. The very thing is impossible. Thus, the straight-line joining E to F is not in a more elevated plane. The straight-line joining E to F is thus in the plane through the parallel (straight-lines) AB and CD .

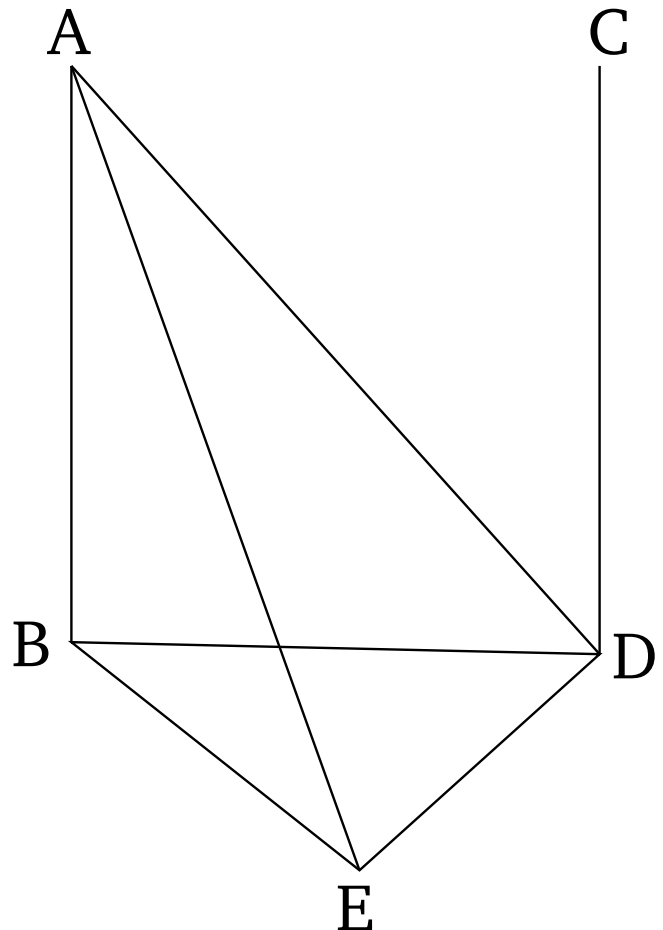
Thus, if there are two parallel straight-lines, and random points are taken on each of them, then the straight-line joining the two points is in the same plane as the parallel (straight-lines). (Which is) the very thing it was required to show.

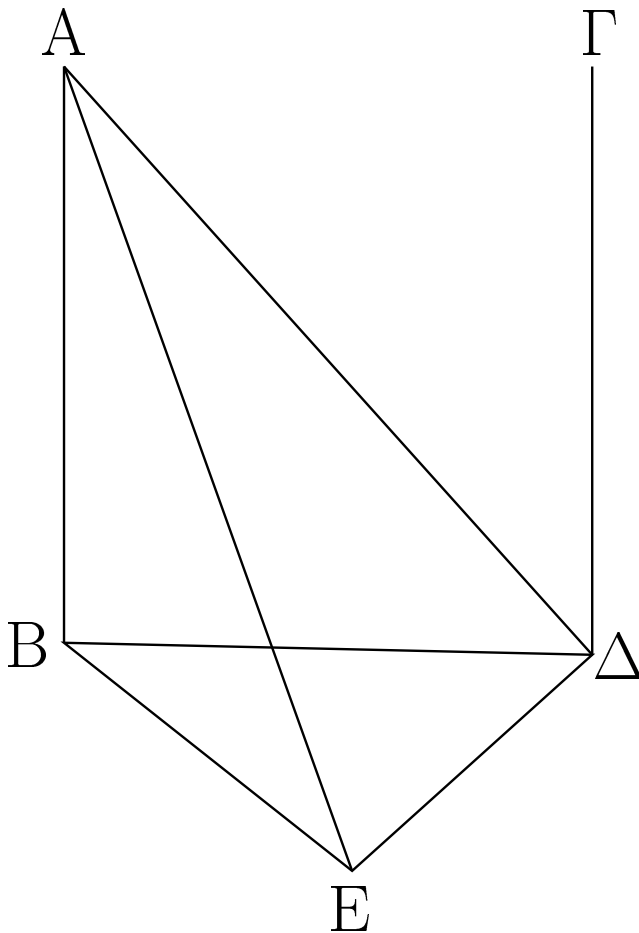
Proposition 8

If two straight-lines are parallel, and one of them is at right-angles to some plane, then the remaining (one) will also be at right-angles to the same plane.

Let AB and CD be two parallel straight-lines, and let one of them, AB , be at right-angles to a reference plane. I say that the remaining (one), CD , will also be at right-angles to the same plane.

For let AB and CD meet the reference plane at points B and D (respectively). And let BD be joined. AB , CD , and BD are thus in one plane [Prop. 11.7]. Let DE be drawn at right-angles to BD in the reference plane, and let DE be made equal to AB , and let BE , AE , and AD be joined.





Καὶ ἐπεὶ ἡ AB ὀρθή ἐστι πρὸς τὸ ὑποκείμενον ἐπίπεδον, καὶ πρὸς πάσας >άρα τὰς ἀπτομένας αὐτῆς εὐθείας καὶ ο>ύσαζ ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθάξ ἐστὶν ἡ AB ; ὀρθή >άρα [ἐστὶν] ἑκάτερα τῶν ὑπὸ $AB\Delta$, ABE γωνιῶν. καὶ ἐπεὶ εἰς παραλλήλουζ τὰς AB , $\Gamma\Delta$ εὐθεῖα ἐμπέπτωκεν ἡ $B\Delta$, αἱ >άρα ὑπὸ $AB\Delta$, $\Gamma\Delta B$ γωνίαι δυσὶν ὀρθαῖς >ίσαι εἰσὶν. ὀρθή δὲ ἡ ὑπὸ $AB\Delta$; ὀρθή >άρα καὶ ἡ ὑπὸ $\Gamma\Delta B$; ἡ $\Gamma\Delta$ >άρα πρὸς τὴν $B\Delta$ ὀρθή ἐστὶν. καὶ ἐπεὶ >ίση ἐστὶν ἡ AB τῇ ΔE , κοινὴ δὲ ἡ $B\Delta$, δύο δὴ αἱ AB , $B\Delta$ δυσὶ ταῖς $E\Delta$, ΔB >ίσαι εἰσὶν; καὶ γωνία ἡ ὑπὸ $AB\Delta$ γωνία τῇ ὑπὸ $E\Delta B$ >ίση; ὀρθή γὰρ ἑκάτερα; βάσις >άρα ἡ $\Delta\Delta$ βάσει τῇ BE >ίση. καὶ ἐπεὶ >ίση ἐστὶν ἡ μὲν AB τῇ ΔE , ἡ δὲ BE τῇ $\Delta\Delta$, δύο δὴ αἱ AB , BE δυσὶ ταῖς $E\Delta$, $\Delta\Delta$ >ίσαι εἰσὶν ἑκάτερα ἑκατέρῃ. καὶ βάσις αὐτῶν κοινὴ ἡ AE ; γωνία >άρα ἡ ὑπὸ ABE γωνία τῇ ὑπὸ $E\Delta\Delta$ ἐστὶν >ίση. ὀρθή δὲ ἡ ὑπὸ ABE ; ὀρθή >άρα καὶ ἡ ὑπὸ $E\Delta\Delta$; ἡ $E\Delta$ >άρα πρὸς τὴν $\Delta\Delta$ ὀρθή ἐστὶν. >έστι δὲ καὶ πρὸς τὴν ΔB ὀρθή; ἡ $E\Delta$ >άρα καὶ τῷ διὰ τῶν $B\Delta$, $\Delta\Delta$ ἐπιπέδῳ ὀρθή ἐστὶν. καὶ πρὸς πάσας >άρα τὰς ἀπτομένας αὐτῆς εὐθείας καὶ ο>ύσαζ ἐν τῷ διὰ τῶν $B\Delta\Delta$ ἐπιπέδῳ ὀρθάξ ποιήσει γωνίαξ ἡ $E\Delta$. ἐν δὲ τῷ διὰ τῶν $B\Delta\Delta$ ἐπιπέδῳ ἐστὶν ἡ $\Delta\Gamma$, ἐπειδὴ περ ἐν τῷ διὰ τῶν $B\Delta\Delta$ ἐπιπέδῳ ἐστὶν αἱ AB , $B\Delta$, ἐν <ῶ| δὲ αἱ AB , $B\Delta$, ἐν τούτῳ ἐστὶ καὶ ἡ $\Delta\Gamma$. ἡ $E\Delta$ >άρα τῇ $\Delta\Gamma$ πρὸς ὀρθάξ ἐστὶν; <ώστε καὶ ἡ $\Gamma\Delta$ τῇ ΔE πρὸς ὀρθάξ ἐστὶν. >έστι δὲ καὶ ἡ $\Gamma\Delta$ τῇ $B\Delta$ πρὸς ὀρθάξ. ἡ $\Gamma\Delta$ >άρα δύο εὐθείαις τεμνοῦσαι ἀλλήλας ταῖς ΔE , ΔB ἀπὸ τῆς κατὰ τὸ Δ τομῆς πρὸς ὀρθάξ ἐφέστηκεν; <ώστε ἡ $\Gamma\Delta$ καὶ τῷ διὰ τῶν ΔE , ΔB ἐπιπέδῳ πρὸς ὀρθάξ ἐστὶν. τὸ δὲ διὰ τῶν ΔE , ΔB ἐπίπεδον τὸ ὑποκείμενόν ἐστιν; ἡ $\Gamma\Delta$ >άρα τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθάξ ἐστὶν.

Ἐὰν >άρα >ῶσι δύο εὐθεῖαι παράλληλοι, ἡ δὲ μία αὐτῶν ἐπιπέδῳ τινὶ πρὸς ὀρθάξ >ῇ|, καὶ ἡ λοιπὴ τῷ αὐτῷ ἐπιπέδῳ

And since AB is at right-angles to the reference plane, AB is thus also at right-angles to all of the straight-lines joined to it which are in the reference plane [Def. 11.3]. Thus, the angles ABD and ABE [are] each right-angles. And since the straight-line BD has met the parallel (straight-lines) AB and CD , the (sum of the) angles ABD and CDB is thus equal to two right-angles [Prop. 1.29]. And ABD (is) a right-angle. Thus, CDB (is) also a right-angle. CD is thus at right-angles to BD . And since AB is equal to DE , and BD (is) common, the two (straight-lines) AB and BD are equal to the two (straight-lines) ED and DB (respectively). And angle ABD (is) equal to angle EDB . For each (is) a right-angle. Thus, the base AD (is) equal to the base BE [Prop. 1.4]. And since AB is equal to DE , and BE to AD , the two (sides) AB , BE are equal to the two (sides) ED , DA , respectively. And their base AE is common. Thus, angle ABE is equal to angle EDA [Prop. 1.8]. And ABE (is) a right-angle. EDA (is) thus also a right-angle. Thus, ED is at right-angles to AD . And it is also at right-angles to DB . Thus, ED is also at right-angles to the plane through BD and DA [Prop. 11.4]. And ED will thus make right-angles with all of the straight-lines joined to it which are also in the plane through BDA . And DC is in the plane through BDA , inasmuch as AB and BD are in the plane through BDA [Prop. 11.2], and in which (ever plane) AB and BD (are found), DC is also (found). Thus, ED is at right-angles to DC . Hence, CD is also at right-angles to DE . And CD is also at right-angles to BD . Thus, CD is standing at right-angles to two straight-lines, DE and DB , which meet one another, at the (point) of section, D . Hence, CD is also at right-angles to the plane through DE and DB [Prop. 11.4]. And the plane through DE and DB is the reference (plane).

πρὸς ὀρθὰς >έσται; <όπερ >έδει δεῖχαι.

CD is thus at right-angles to the reference plane.

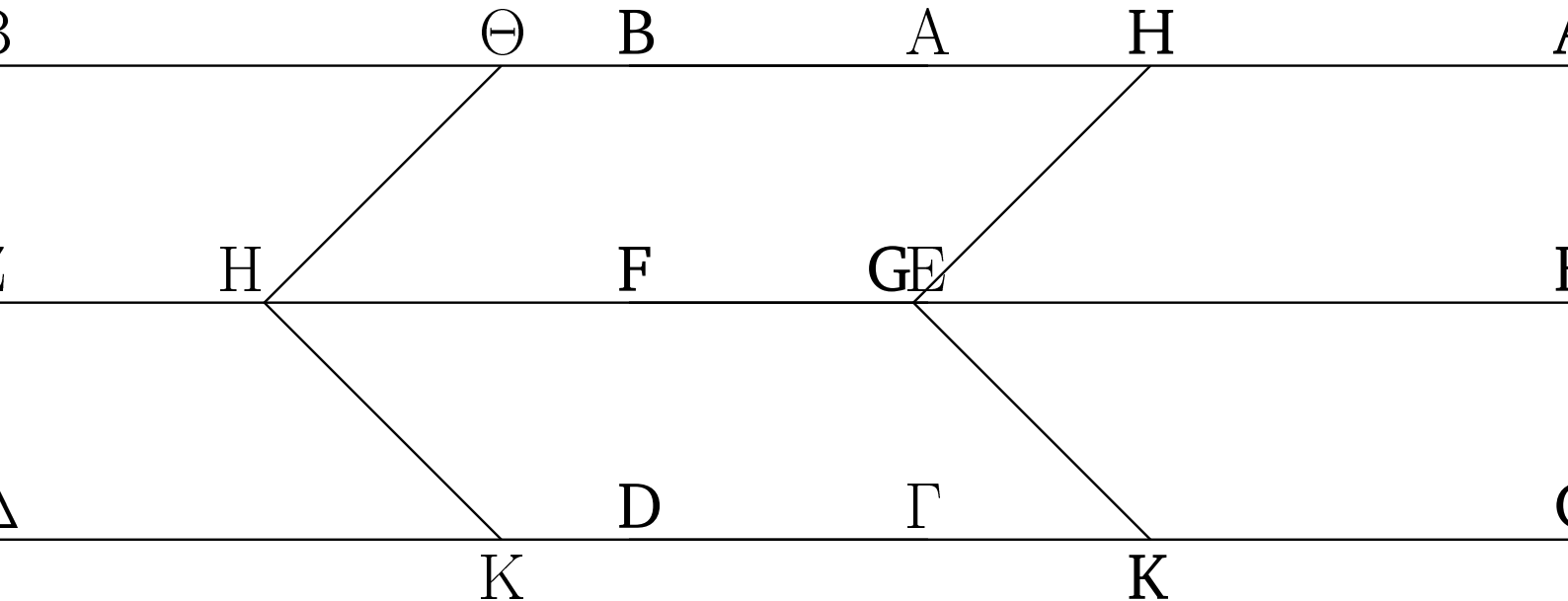
Thus, if two straight-lines are parallel, and one of them is at right-angles to some plane, then the remaining (one) will also be at right-angles to the same plane. (Which is) the very thing it was required to show.

9.

Proposition 9

Αἱ τῇ αὐτῇ εὐθείᾳ παράλληλοι καὶ μὴ ο>ῦσαι αὐτῇ ἐν τῷ αὐτῷ ἐπιπέδῳ καὶ ἀλλήλαις εἰσι παράλληλοι.

(Straight-lines) parallel to the same straight-line, and which are not in the same plane as it, are also parallel to one another.



>Ἐστω γὰρ ἑκατέρα τῶν AB , $ΓΔ$ τῇ EZ παράλληλος μὴ ο>ῦσαι αὐτῇ ἐν τῷ αὐτῷ ἐπιπέδῳ; λέγω, <ὅτι παράλληλός ἐστιν ἡ AB τῇ $ΓΔ$.

Εἰλήφθω γὰρ ἐπὶ τῇ EZ τυθὸν σημεῖον τὸ H , καὶ ἀπ' αὐτοῦ τῇ EZ ἐν μὲν τῷ διὰ τῶν EZ , AB ἐπιπέδῳ πρὸς ὀρθὰς >ήθθω ἡ $HΘ$, ἐν δὲ τῷ διὰ τῶν ZE , $ΓΔ$ τῇ EZ πάλιν πρὸς ὀρθὰς >ήθθω ἡ HK .

Καὶ ἐπεὶ ἡ EZ πρὸς ἑκατέραν τῶν $HΘ$, HK ὀρθή ἐστιν, ἡ EZ >άρα καὶ τῷ διὰ τῶν $HΘ$, HK ἐπιπέδῳ πρὸς ὀρθὰς ἐστιν. καὶ ἐστιν ἡ EZ τῇ AB παράλληλος; καὶ ἡ AB >άρα τῷ διὰ τῶν $ΘHK$ ἐπιπέδῳ πρὸς ὀρθὰς ἐστιν. διὰ τὰ αὐτὰ δὴ καὶ ἡ $ΓΔ$ τῷ διὰ τῶν $ΘHK$ ἐπιπέδῳ πρὸς ὀρθὰς ἐστιν; ἑκατέρα >άρα τῶν AB , $ΓΔ$ τῷ διὰ τῶν $ΘHK$ ἐπιπέδῳ πρὸς ὀρθὰς ἐστιν. ἔαν δὲ δύο εὐθεῖαι τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰς >ῶσιν, παράλληλοί εἰσιν αἱ εὐθεῖαι; παράλληλος >άρα ἐστιν ἡ AB τῇ $ΓΔ$; <όπερ >έδει δεῖχαι.

For let AB and CD each be parallel to EF , not being in the same plane as it. I say that AB is parallel to CD .

For let some point G be taken at random on EF . And from it let GH be drawn at right-angles to EF in the plane through EF and AB . And let GK be drawn, again at right-angles to EF , in the plane through FE and CD .

And since EF is at right-angles to each of GH and GK , EF is thus also at right-angles to the plane through GH and GK [Prop. 11.4]. And EF is parallel to AB . Thus, AB is also at right-angles to the plane through HGK [Prop. 11.8]. So, for the same (reasons), CD is also at right-angles to the plane through HGK . Thus, AB and CD are each at right-angles to the plane through HGK . And if two straight-lines are at right-angles to the same plane then the straight-lines are parallel [Prop. 11.6]. Thus, AB is parallel to CD . (Which is) the very thing it was required to show.

10.

Proposition 10

Ἐὰν δύο εὐθεῖαι ἀπτόμεναι ἀλλήλων παρὰ δύο εὐθείας ἀπτομένας ἀλλήλων >ῶσι μὴ ἐν τῷ αὐτῷ ἐπιπέδῳ, <ῖσαι γωνίας περιέχουσιν.

Δύο γὰρ εὐθεῖαι αἱ AB , $ΒΓ$ ἀπτόμεναι ἀλλήλων παρὰ δύο εὐθείας τὰς $ΔΕ$, $ΕΖ$ ἀπτομένας ἀλλήλων >έστωσαν μὴ ἐν τῷ αὐτῷ ἐπιπέδῳ; λέγω, <ὅτι >ῖση ἐστὶν ἡ ὑπὸ $ABΓ$ γωνία τῇ ὑπὸ $ΔΕΖ$.

Ἀπειλήφθωσαν γὰρ αἱ BA , $ΒΓ$, $ΕΔ$, $ΕΖ$ >ῖσαι ἀλλήλαις, καὶ ἐπέξευθθωσαν αἱ $ΑΔ$, $ΓΖ$, BE , $ΑΓ$, $ΔΖ$.

Καὶ ἐπεὶ ἡ BA τῇ $ΕΔ$ >ῖση ἐστὶ καὶ παράλληλος, καὶ ἡ $ΑΔ$ >άρα τῇ BE >ῖση ἐστὶ καὶ παράλληλος. διὰ τὰ αὐτὰ δὴ καὶ ἡ $ΓΖ$ τῇ BE >ῖση ἐστὶ καὶ παράλληλος; ἑκατέρα >άρα τῶν $ΑΔ$, $ΓΖ$ τῇ BE >ῖση ἐστὶ καὶ παράλληλος. αἱ δὲ τῇ

If two straight-lines joined to one another are (respectively) parallel to two straight-lines joined to one another, (but are) not in the same plane, then they will contain equal angles.

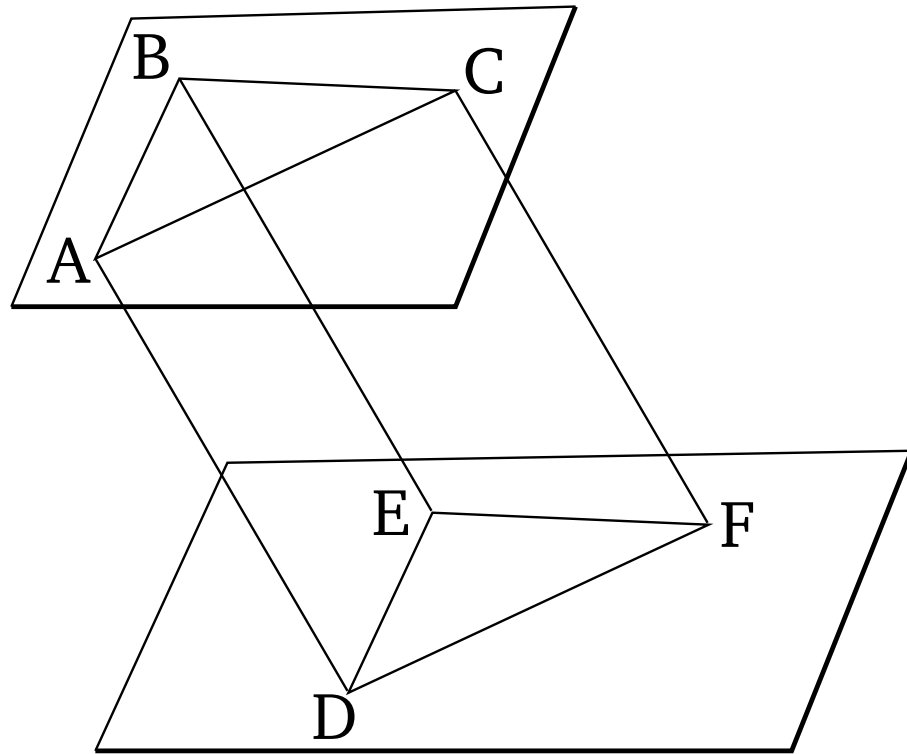
For let the two straight-lines joined to one another, AB and BC , be (respectively) parallel to the two straight-lines joined to one another, DE and EF , (but) not in the same plane. I say that angle ABC is equal to (angle) DEF .

For let BA , BC , ED , and EF be cut off (so as to be, respectively) equal to one another. And let AD , CF , BE , AC , and DF be joined.

And since BA is equal and parallel to ED , AD is thus also equal and parallel to BE [Prop. 1.33]. So, for the same reasons, CF is also equal and parallel to BE . Thus, AD

αὐτῇ| εὐθείᾳ παράλληλοι καὶ μὴ οὕσαι αὐτῇ| ἐν τῷ| αὐτῷ| ἐπιπέδῳ καὶ ἀλλήλαις εἰσὶ παράλληλοι; παράλληλος >άρα ἐστὶν ἡ AD τῇ| ΓZ καὶ >ίση. καὶ ἐπιζευγνύουσιν αὐτάς αἱ AG , ΔZ ; καὶ ἡ AG >άρα τῇ| ΔZ >ίση ἐστὶ καὶ παράλληλος. καὶ ἐπεὶ δύο αἱ AB , BF δυοὶ ταῖς ΔE , EZ >ίσαι εἰσὶν, καὶ βάσις ἡ AG βάσει τῇ| ΔZ >ίση, γωνία >άρα ἡ ὑπὸ ABF γωνία τῇ| ὑπὸ ΔEZ ἐστὶν >ίση.

and CF are each equal and parallel to BE . And straight-lines parallel to the same straight-line, and which are not in the same plane as it, are also parallel to one another [Prop. 11.9]. Thus, AD is parallel and equal to CF . And AC and DF join them. Thus, AC is also equal and parallel to DF [Prop. 1.33]. And since the two (straight-lines) AB and BC are equal to the two (straight-lines) DE and EF (respectively), and the base AC (is) equal to the base DF , the angle ABC is thus equal to the (angle) DEF [Prop. 1.8].



Ἐὰν >άρα δύο εὐθεῖαι ἀπτόμεναι ἀλλήλων παρὰ δύο εὐθείας ἀπτομένας ἀλλήλων >ῶσι μὴ ἐν τῷ| αὐτῷ| ἐπιπέδῳ, <ίσαξ γωνίαξ περιέχουσιν; <όπερ >έδει δεῖχαι.

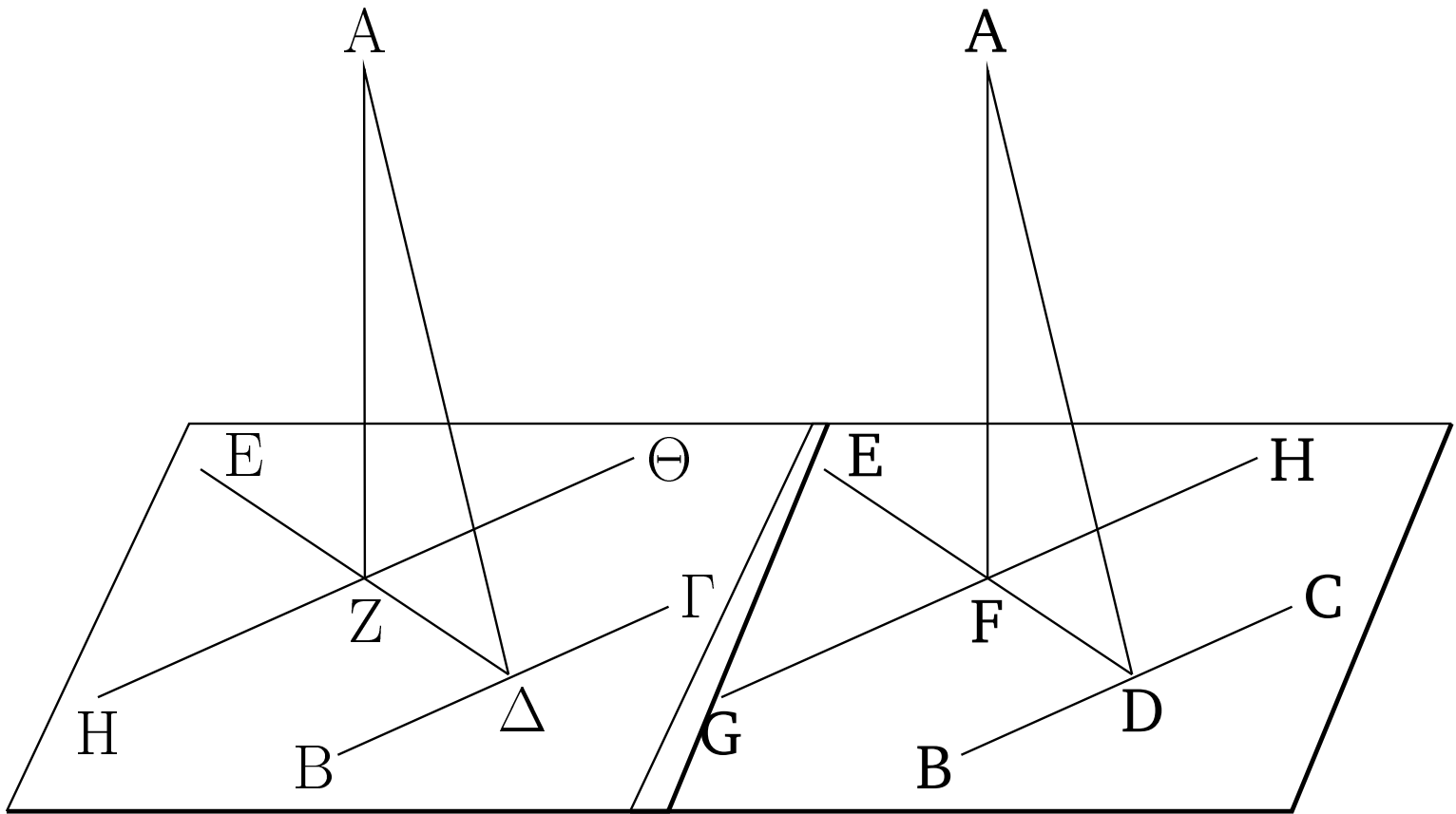
Thus, if two straight-lines joined to one another are (respectively) parallel to two straight-lines joined to one another, (but are) not in the same plane, then they will contain equal angles. (Which is) the very thing it was required to show.

11.

Ἀπὸ τοῦ δοθέντος σημείου μετῴρου ἐπὶ τὸ δοθὲν ἐπίπεδον κάθετον εὐθεῖαν γραμμὴν ἀγαγεῖν.

Proposition 11

To draw a perpendicular straight-line from a given raised point to a given plane.



>Ἐστω τὸ μὲν δοθὲν σημεῖον μετέωρον τὸ Α, τὸ δὲ δοθὲν ἐπίπεδον τὸ ὑποκείμενον; δεῖ δὴ ἀπὸ τοῦ Α σημείου ἐπὶ τὸ ὑποκείμενον ἐπίπεδον κάθετον εὐθεῖαν γραμμὴν ἀγαγεῖν. Διήθθω γάρ τις ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ εὐθεῖα, ὡς >έτυθεν, ἡ ΒΓ, καὶ >ήθθω ἀπὸ τοῦ Α σημείου ἐπὶ τὴν ΒΓ κάθετος ἡ ΑΔ. εἰ μὲν οὖν ἡ ΑΔ κάθετός ἐστι καὶ ἐπὶ τὸ ὑποκείμενον ἐπίπεδον, γεγονός >άν ε>ίη τὸ ἐπιταθέν. εἰ δὲ οὐ, >ήθθω ἀπὸ τοῦ Δ σημείου τῇ ΒΓ ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς ἡ ΔΕ, καὶ >ήθθω ἀπὸ τοῦ Α ἐπὶ τὴν ΔΕ κάθετος ἡ ΑΖ, καὶ διὰ τοῦ Ζ σημείου τῇ ΒΓ παράλληλος >ήθθω ἡ ΗΘ.

Καὶ ἐπεὶ ἡ ΒΓ ἐκατέρᾳ τῶν ΔΑ, ΔΕ πρὸς ὀρθὰς ἐστίν, ἡ ΒΓ >άρα καὶ τῷ διὰ τῶν ΕΔΑ ἐπιπέδῳ πρὸς ὀρθὰς ἐστίν. καὶ ἐστὶν αὐτῇ παράλληλος ἡ ΗΘ; ἔαν δὲ >ῶσι δύο εὐθεῖαι παράλληλοι, ἡ δὲ μία αὐτῶν ἐπιπέδῳ τινὶ πρὸς ὀρθὰς >ῇ, καὶ ἡ λοιπὴ τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰς >έσται; καὶ ἡ ΗΘ >άρα τῷ διὰ τῶν ΕΔ, ΔΑ ἐπιπέδῳ πρὸς ὀρθὰς ἐστίν. καὶ πρὸς πάσας >άρα τὰς ἀπτομένας αὐτῇ εὐθείας καὶ ο>ύσαξ ἐν τῷ διὰ τῶν ΕΔ, ΔΑ ἐπιπέδῳ ὀρθή ἐστὶν ἡ ΗΘ. <ἀπτεται δὲ αὐτῇ ἡ ΑΖ ο>ύσα ἐν τῷ διὰ τῶν ΕΔ, ΔΑ ἐπιπέδῳ; ἡ ΗΘ >άρα ὀρθή ἐστὶ πρὸς τὴν ΖΑ; <ώστε καὶ ἡ ΖΑ ὀρθή ἐστὶ πρὸς τὴν ΘΗ. >έστι δὲ ἡ ΑΖ καὶ πρὸς τὴν ΔΕ ὀρθή; ἡ ΑΖ >άρα πρὸς ἐκατέραν τῶν ΗΘ, ΔΕ ὀρθή ἐστίν. ἔαν δὲ εὐθεῖα δυσὶν εὐθείαις τεμνόμεναι ἀλλήλας ἐπὶ τῇς τομῇς πρὸς ὀρθὰς ἐπισταθῇ, καὶ τῷ δι> αὐτῶν ἐπιπέδῳ πρὸς ὀρθὰς >έσται; ἡ ΖΑ >άρα τῷ διὰ τῶν ΕΔ, ΗΘ ἐπιπέδῳ πρὸς ὀρθὰς ἐστίν. τὸ δὲ διὰ τῶν ΕΔ, ΗΘ ἐπίπεδόν ἐστι τὸ ὑποκείμενον; ἡ ΑΖ >άρα τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς ἐστίν.

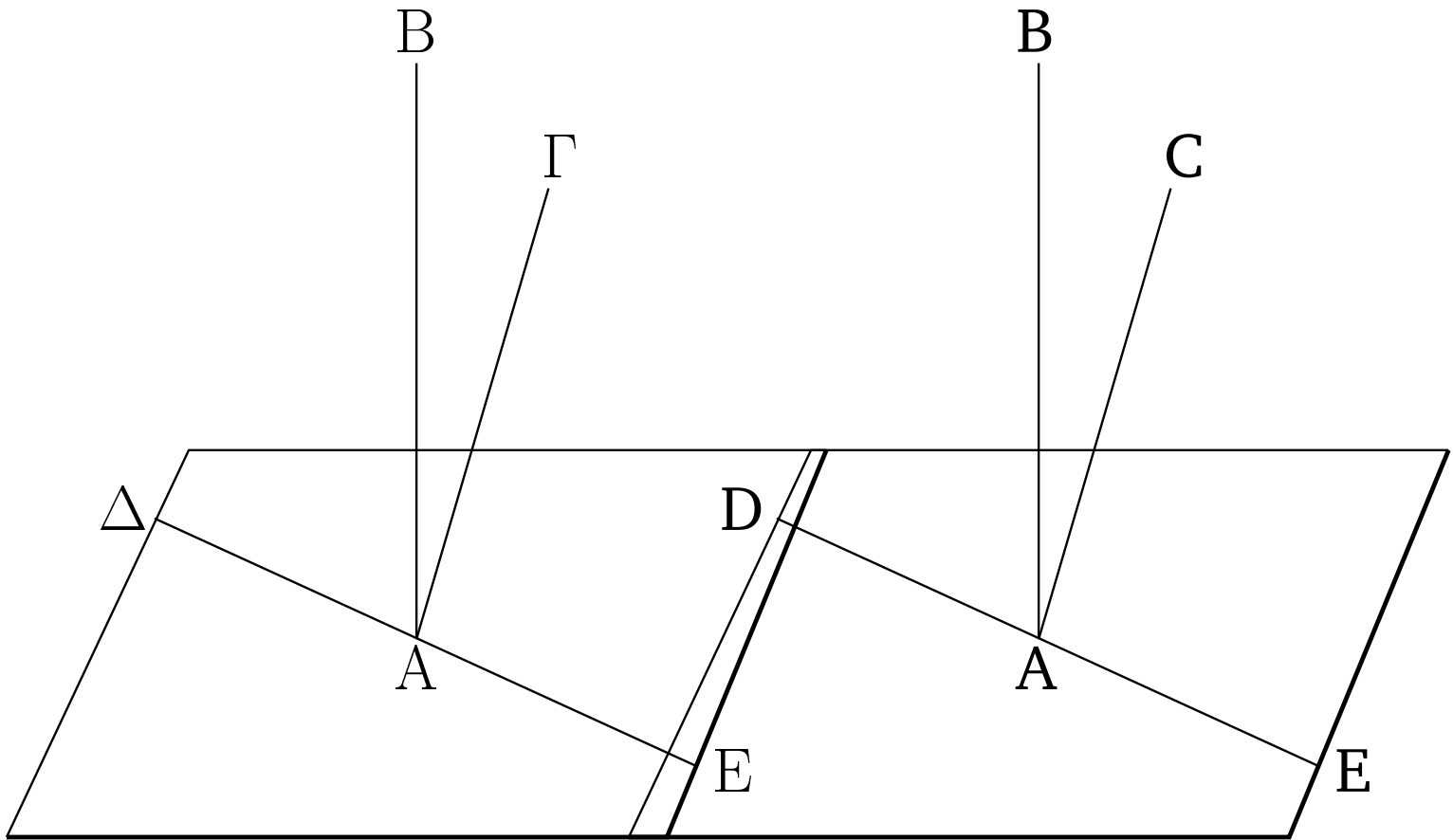
Ἀπὸ τοῦ >άρα δοθέντος σημείου μετέωρου τοῦ Α ἐπὶ τὸ ὑποκείμενον ἐπίπεδον κάθετος εὐθεῖα γραμμὴ >ῇκται ἡ ΑΖ; <όπερ >έδει ποιῆσαι.

Let A be the given raised point, and the given plane the reference (plane). So, it is required to draw a perpendicular straight-line from point A to the reference plane.

Let some random straight-line BC be drawn across in the reference plane, and let the (straight-line) AD be drawn from point A perpendicular to BC [Prop. 1.12]. If, therefore, AD is also perpendicular to the reference plane then that which was prescribed will have occurred. And, if not, let DE be drawn in the reference plane from point D at right-angles to BC [Prop. 1.11], and let the (straight-line) AF be drawn from A perpendicular to DE [Prop. 1.12], and let GH be drawn through point F , parallel to BC [Prop. 1.31].

And since BC is at right-angles to each of DA and DE , BC is thus also at right-angles to the plane through EDA [Prop. 11.4]. And GH is parallel to it. And if two straight-lines are parallel, and one of them is at right-angles to some plane, then the remaining (straight-line) will also be at right-angles to the same plane [Prop. 11.8]. Thus, GH is also at right-angles to the plane through ED and DA . And GH is thus at right-angles to all of the straight-lines joined to it which are also in the plane through ED and AD [Def. 11.3]. And AF , which is in the plane through ED and DA , is joined to it. Thus, GH is at right-angles to FA . Hence, FA is also at right-angles to HG . And AF is also at right-angles to DE . Thus, AF is at right-angles to each of GH and DE . And if a straight-line is set up at right-angles to two straight-lines cutting one another, at the point of section, then it will also be at right-angles to the plane through them [Prop. 11.4]. Thus, FA is at right-angles to the plane through ED and GH . And the plane through ED and GH is the reference (plane). Thus, AF is at right-angles to the reference plane.

Thus, the straight-line AF has been drawn from the given



Εἰ γὰρ δυνατόν, ἀπὸ τοῦ αὐτοῦ σημείου τοῦ Α τῷ ὑποκειμένῳ ἐπιπέδῳ δύο εὐθεῖαι αἱ ΑΒ, ΒΓ πρὸς ὀρθὰς ἀνεστάτωσαν ἐπὶ τὰ αὐτὰ μέρη, καὶ διήθθω τὸ διὰ τῶν ΒΑ, ΑΓ ἐπιπέδον; τομὴν δὴ ποιήσει διὰ τοῦ Α ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ εὐθεῖαν. ποιεῖτω τὴν ΔΑΕ; αἱ ὅρα ΑΒ, ΑΓ, ΔΑΕ εὐθεῖαι ἐν ἐνὶ εἰσιν ἐπιπέδῳ. καὶ ἐπεὶ ἡ ΓΑ τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς ἐστίν, καὶ πρὸς πάσαξ ὅρα τὰς ἀπτομένας αὐτῇ εὐθείας καὶ οὕσαξ ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ὀρθὰς ποιήσει γωνίαξ. ἄπτεται δὲ αὐτῇ ἡ ΔΑΕ οὕσα ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ; ἡ ὅρα ὑπὸ ΓΑΕ γωνία ὀρθή ἐστίν. διὰ τὰ αὐτὰ δὴ καὶ ἡ ὑπὸ ΒΑΕ ὀρθή ἐστίν; ὅρα ἡ ὑπὸ ΓΑΕ τῇ ὑπὸ ΒΑΕ καὶ εἰσιν ἐν ἐνὶ ἐπιπέδῳ; ὅπερ ἐστὶν ἀδύνατον.

Οὐκ ὅρα ἀπὸ τοῦ αὐτοῦ σημείου τῷ αὐτῷ ἐπιπέδῳ δύο εὐθεῖαι πρὸς ὀρθὰς ἀνασταθῇσονται ἐπὶ τὰ αὐτὰ μέρη; ὅπερ ἔδει δεῖχαι.

For, if possible, let the two straight-lines AB and AC be set up at the same point A at right-angles to the reference plane, on the same side. And let the plane through BA and AC be drawn. So it will make a straight cutting (passing) through (point) A in the reference plane [Prop. 11.3]. Let it have made DAE . Thus, AB , AC , and DAE are straight-lines in one plane. And since CA is at right-angles to the reference plane, it will thus also make right-angles with all of the straight-lines joined to it which are also in the reference plane [Def. 11.3]. And DAE , which is in the reference plane, is joined to it. Thus, angle CAE is a right-angle. So, for the same (reasons), BAE is also a right-angle. Thus, CAE (is) equal to BAE . And they are in one plane. The very thing is impossible.

Thus, two (different) straight-lines cannot be set up at the same point at right-angles to the same plane, on the same side. (Which is) the very thing it was required to show.

14.

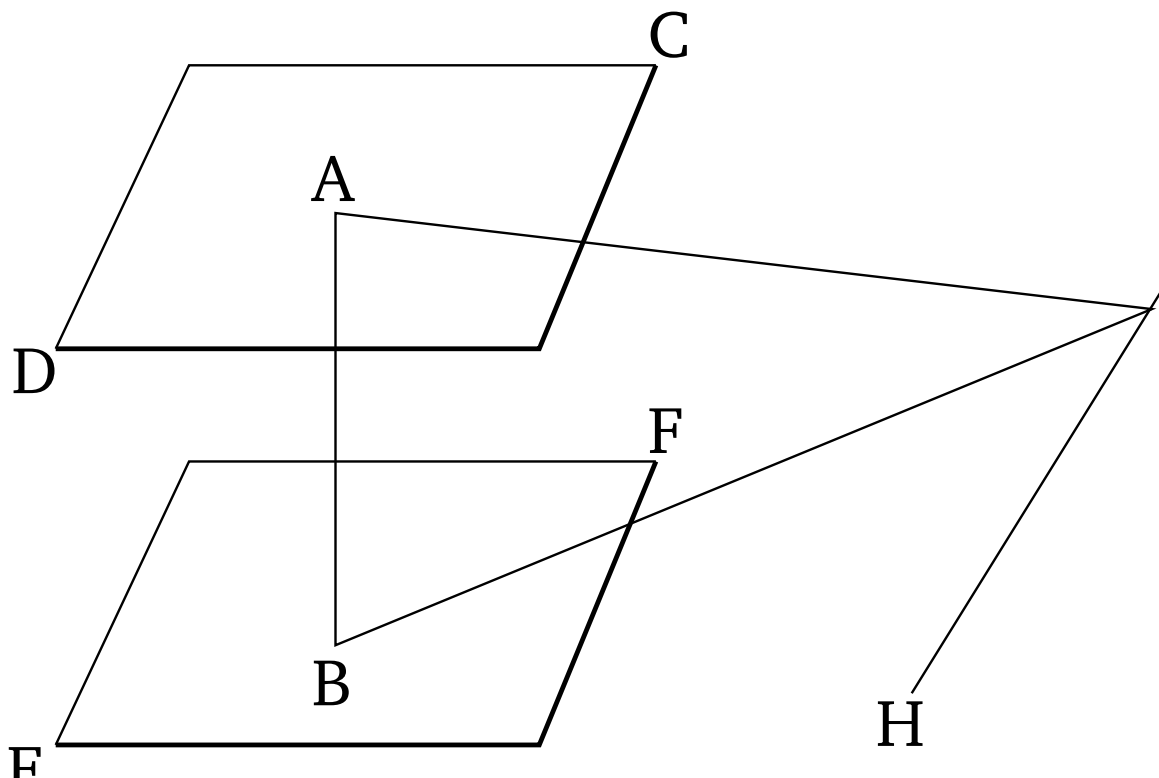
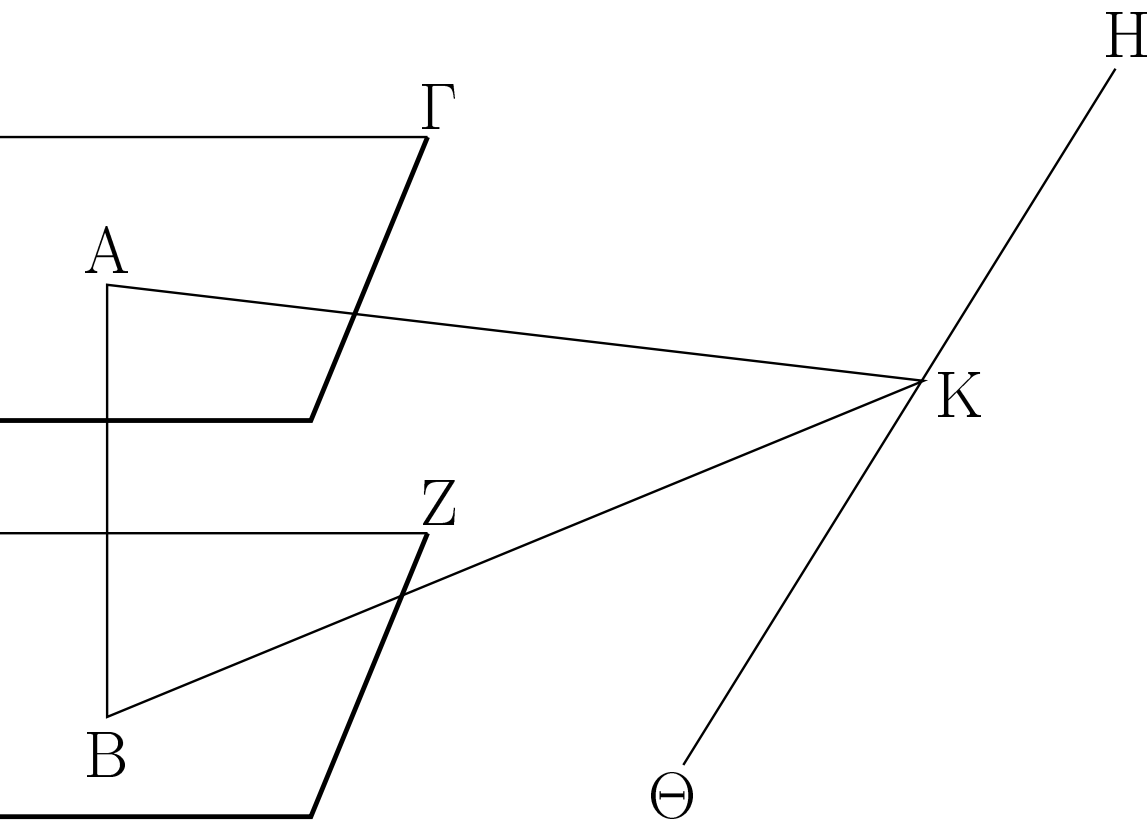
Πρὸς ἅ ἐπίπεδα ἡ αὐτὴ εὐθεῖα ὀρθή ἐστίν, παράλληλα ἔσται τὰ ἐπίπεδα.

Εὐθεῖα γάρ τις ἡ ΑΒ πρὸς ἑκάτερον τῶν ΓΔ, ΕΖ ἐπιπέδων πρὸς ὀρθὰς ἔστω; λέγω, ὅτι παράλληλά ἐστι τὰ ἐπίπεδα.

Proposition 14

Planes to which the same straight-line is at right-angles will be parallel planes.

For let some straight-line AB be at right-angles to each of the planes CD and EF . I say that the planes are parallel.



Εἰ γὰρ μὴ, ἐκβαλλόμενα συμπεσοῦνται. συμπιπτεύωσαν;
 ποιήσουσι δὴ κοινὴν τομὴν εὐθεΐαν. ποιείτωσαν τὴν
 $H\Theta$, καὶ εἰλήφθω ἐπὶ τῇ $H\Theta$ τυθὸν σημεῖον τὸ K , καὶ
 ἐπέξευθθωσαν αἱ AK , BK .
 Καὶ ἐπεὶ ἡ AB ὀρθὴ ἐστὶ πρὸς τὸ EZ ἐπίπεδον, καὶ πρὸς τὴν
 BK > ἄρα εὐθεΐαν οὕτως ἐν τῷ EZ ἐκβληθέντι ἐπιπέδῳ
 ὀρθὴ ἐστὶν ἡ AB ; ἡ > ἄρα ὑπὸ ABK γωνία ὀρθὴ ἐστίν. διὰ

For, if not, being produced, they will meet. Let them have
 met. So they will make a straight-line as a common section
 [Prop. 11.3]. Let them have made GH . And let some ran-
 dom point K be taken on GH . And let AK and BK be
 joined.
 And since AB is at right-angles to the plane EF , AB is

τὰ αὐτὰ δὴ καὶ ἡ ὑπὸ BAK ὀρθή ἐστιν. τριγώνου δὲ τοῦ ABK αἱ δύο γωνίαι αἱ ὑπὸ ABK , BAK δυσὶν ὀρθαῖς εἰσιν >ίσαι; <ὅπερ ἐστὶν ἀδύνατον. οὐκ >άρα τὰ $\Gamma\Delta$, EZ ἐπίπεδα ἐκβαλλόμενα συμπεσοῦνται; παράλληλα >άρα ἐστὶ τὰ $\Gamma\Delta$, EZ ἐπίπεδα.

Πρὸς <ὰ ἐπίπεδα >άρα ἡ αὐτὴ εὐθεῖα ὀρθή ἐστιν, παράλληλά ἐστὶ τὰ ἐπίπεδα; <ὅπερ >έδει δεῖχαι.

thus also at right-angles to BK , which is a straight-line in the produced plane EF [Def. 11.3]. Thus, angle ABK is a right-angle. So, for the same (reasons), BAK is also a right-angle. So the (sum of the) two angles ABK and BAK in the triangle ABK is equal to two right-angles. The very thing is impossible [Prop. 1.17]. Thus, planes CD and EF , being produced, will not meet. Planes CD and EF are thus parallel [Def. 11.8].

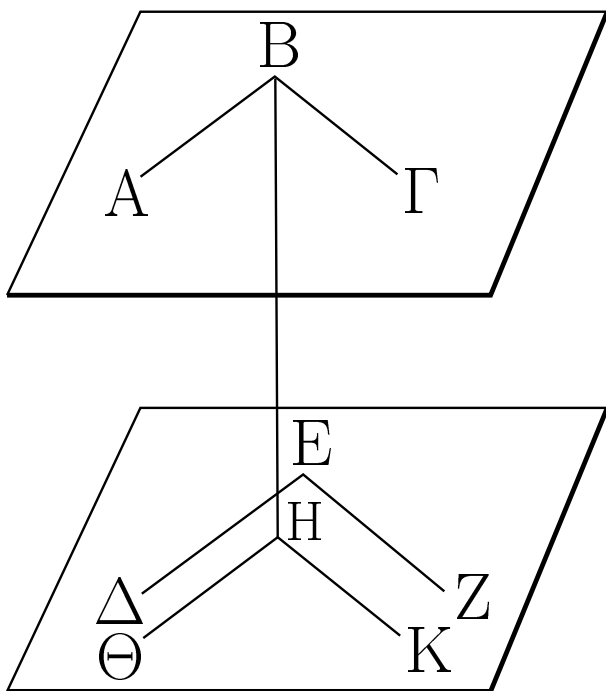
Thus, planes to which the same straight-line is at right-angles are parallel planes. (Which is) the very thing it was required to show.

15.

Ἐὰν δύο εὐθεῖαι ἀπτόμεναι ἀλλήλων παρὰ δύο εὐθείας ἀπτομένας ἀλλήλων >ῶσι μὴ ἐν τῷ αὐτῷ ἐπιπέδῳ ο>ῦσαι, παράλληλά ἐστὶ τὰ δι> αὐτῶν ἐπίπεδα.

Δύο γὰρ εὐθεῖαι ἀπτόμεναι ἀλλήλων αἱ AB , $B\Gamma$ παρὰ δύο εὐθείας ἀπτομένας ἀλλήλων τὰς ΔE , EZ >έστωσαν μὴ ἐν τῷ αὐτῷ ἐπιπέδῳ ο>ῦσαι; λέγω, <ὅτι ἐκβαλλόμενα τὰ διὰ τῶν AB , $B\Gamma$, ΔE , EZ ἐπίπεδα οὐ συμπεσεῖται ἀλλήλοις.

>Ἡθθω γὰρ ἀπὸ τοῦ B σημείου ἐπὶ τὸ διὰ τῶν ΔE , EZ ἐπίπεδον κάθετος ἡ BH καὶ συμβαλλέτω τῷ ἐπιπέδῳ κατὰ τὸ H σημεῖον, καὶ διὰ τοῦ H τῇ μὲν ED παράλληλος >ῆθθω ἡ $H\Theta$, τῇ δὲ EZ ἡ HK .



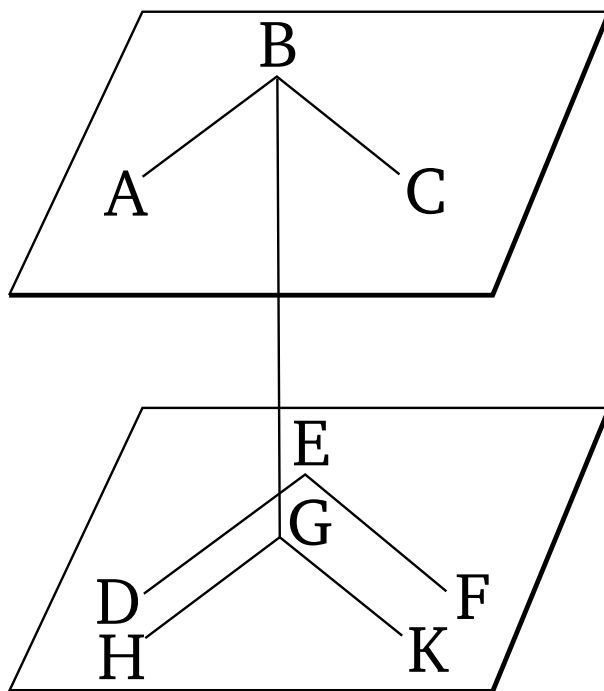
Καὶ ἐπεὶ ἡ BH ὀρθή ἐστι πρὸς τὸ διὰ τῶν ΔE , EZ ἐπίπεδον, καὶ πρὸς πάσας >άρα τὰς ἀπτομένας αὐτῆς εὐθείας καὶ ο>ύσας ἐν τῷ διὰ τῶν ΔE , EZ ἐπιπέδῳ ὀρθὰς ποιήσει γωνίας. <ἀπτεται δὲ αὐτῆς ἑκάτερα τῶν $H\Theta$, HK ο>ῦσα ἐν τῷ διὰ τῶν ΔE , EZ ἐπιπέδῳ; ὀρθὴ >άρα ἐστὶν ἑκάτερα τῶν ὑπὸ $BH\Theta$, BHK γωνιῶν. καὶ ἐπεὶ παράλληλός ἐστιν ἡ BA τῇ $H\Theta$, αἱ >άρα ὑπὸ HBA , $BH\Theta$ γωνίαι δυσὶν ὀρθαῖς >ίσαι εἰσιν. ὀρθὴ δὲ ἡ ὑπὸ $BH\Theta$; ὀρθὴ >άρα καὶ ἡ ὑπὸ HBA ; ἡ HB >άρα τῇ BA πρὸς ὀρθὰς ἐστὶν. διὰ τὰ αὐτὰ δὴ ἡ HB καὶ τῇ $B\Gamma$ ἐστὶ πρὸς ὀρθὰς. ἐπεὶ ο>ὖν εὐθεῖα ἡ HB δυσὶν εὐθείαις ταῖς BA , $B\Gamma$ τεμνόμεναι ἀλλήλας πρὸς ὀρθὰς ἐφέστηκεν, ἡ HB >άρα καὶ τῷ διὰ τῶν BA , $B\Gamma$ ἐπιπέδῳ πρὸς ὀρθὰς ἐστὶν.

Proposition 15

If two straight-lines joined to one another are parallel (respectively) to two straight-lines joined to one another, which are not in the same plane, then the planes through them are parallel (to one another).

For let the two straight-lines joined to one another, AB and BC , be parallel to the two straight-lines joined to one another, DE and EF (respectively), not being in the same plane. I say that the planes through AB , BC and DE , EF will not meet one another (when) produced.

For let BG be drawn from point B perpendicular to the plane through DE and EF [Prop. 11.11], and let it meet the plane at point G . And let GH be drawn through G parallel to ED , and GK (parallel) to EF [Prop. 1.31].



And since BG is at right-angles to the plane through DE and EF , it will thus also make right-angles with all of the straight-lines joined to it, which are also in the plane through DE and EF [Def. 11.3]. And each of GH and GK , which are in the plane through DE and EF , are joined to it. Thus, each of the angles BGH and BGK are right-angles. And since BA is parallel to GH [Prop. 11.9], the (sum of the) angles GBA and BGH is equal to two right-angles [Prop. 1.29]. And BGH (is) a right-angle. GBA (is) thus also a right-angle. Thus, GB is at right-angles to BA . So, for the same (reasons), GB is also at right-angles to BC . Therefore, since the straight-line GB has been set

[διὰ τὰ αὐτὰ δὴ ἡ BH καὶ $\tau\omega$] διὰ τῶν $H\Theta$, HK ἐπιπέδῳ πρὸς ὀρθὰς ἐστίν. τὸ δὲ διὰ τῶν $H\Theta$, HK ἐπιπέδον ἐστὶ τὸ διὰ τῶν ΔE , EZ ; ἡ BH > ἄρα $\tau\omega$] διὰ τῶν ΔE , EZ ἐπιπέδῳ ἐστὶ πρὸς ὀρθὰς. ἐδείκθη δὲ ἡ HB καὶ $\tau\omega$] διὰ τῶν AB , $B\Gamma$ ἐπιπέδῳ πρὸς ὀρθὰς]. πρὸς < ἃ δὲ ἐπίπεδα ἡ αὐτὴ εὐθεῖα ὀρθὴ ἐστίν, παράλληλά ἐστὶ τὰ ἐπίπεδα; παράλληλον > ἄρα ἐστὶ τὸ διὰ τῶν AB , $B\Gamma$ ἐπίπεδον $\tau\omega$] διὰ τῶν ΔE , EZ .

Ἐὰν > ἄρα δύο εὐθεῖαι ἀπτόμεναι ἀλλήλων παρὰ δύο εὐθείας ἀπτομένας ἀλλήλων > ὥς μὴ ἐν $\tau\omega$] αὐτῷ ἐπιπέδῳ, παράλληλά ἐστὶ τὰ δι> αὐτῶν ἐπίπεδα; < ὅπερ > ἐδει δεῖχαι.

up at right-angles to two straight-lines, BA and BC , cutting one another, GB is thus at right-angles to the plane through BA and BC [Prop. 11.4]. [So, for the same (reasons), BG is also at right-angles to the plane through GH and GK . And the plane through GH and GK is the (plane) through DE and EF . And it was also shown that GB is at right-angles to the plane through AB and BC .] And planes to which the same straight-line is at right-angles are parallel planes [Prop. 11.14]. Thus, the plane through AB and BC is parallel to the (plane) through DE and EF .

Thus, if two straight-lines joined to one another are parallel (respectively) to two straight-lines joined to one another, which are not in the same plane, then the planes through them are parallel (to one another). (Which is) the very thing it was required to show.

16.

Ἐὰν δύο ἐπίπεδα παράλληλα ὑπὸ ἐπιπέδου τινὸς τέμνηται, αἱ κοιναὶ αὐτῶν τομαὶ παράλληλοί εἰσιν.

Δύο γὰρ ἐπίπεδα παράλληλα τὰ AB , $\Gamma\Delta$ ὑπὸ ἐπιπέδου τοῦ $EZH\Theta$ τεμνέσθω, κοιναὶ δὲ αὐτῶν τομαὶ > ἐστῶσαν αἱ EZ , $H\Theta$; λέγω, < ὅτι παράλληλός ἐστίν ἡ EZ τῇ $H\Theta$.

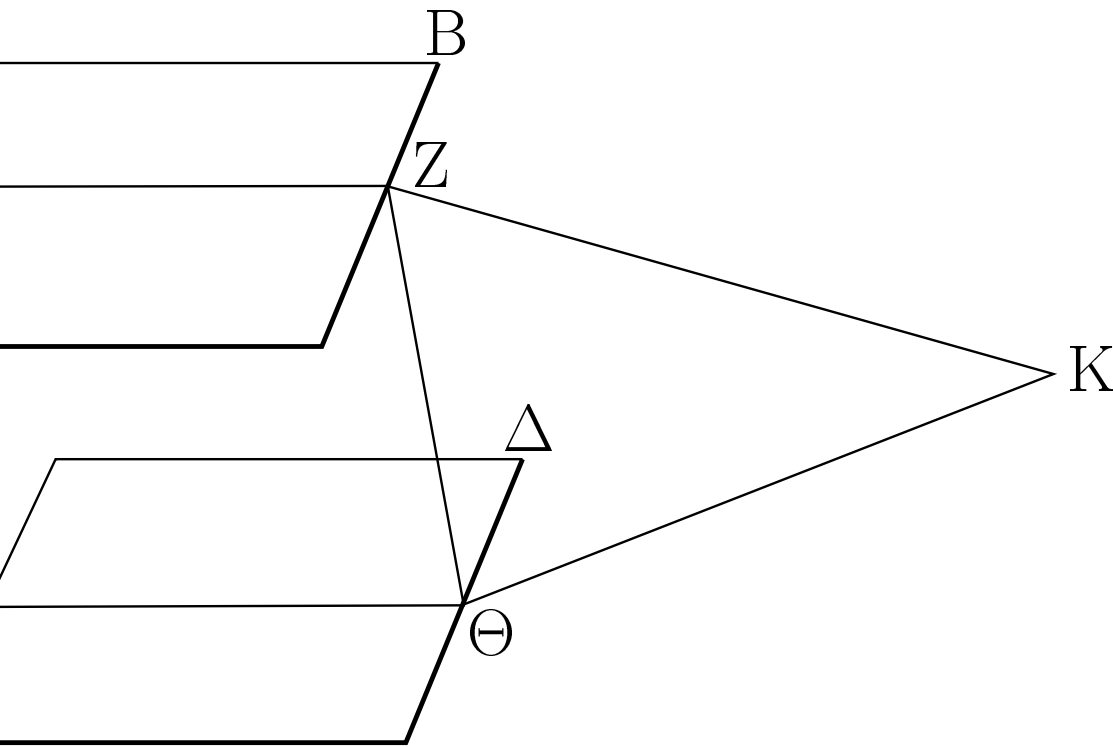
Εἰ γὰρ μὴ, ἐκβαλλόμεναι αἱ EZ , $H\Theta$ > ἤτοι ἐπὶ τὰ Z , Θ μέρη > ἢ ἐπὶ τὰ E , H συμπεσοῦνται. ἐκβεβλήσθωσαν ὥς ἐπὶ τὰ Z , Θ μέρη καὶ συμπίπτῃωσαν πρότερον κατὰ τὸ K . καὶ ἐπεὶ ἡ EZK ἐν $\tau\omega$] AB ἐστίν ἐπιπέδῳ, καὶ πάντα > ἄρα τὰ ἐπὶ τῇ EZK σημεία ἐν $\tau\omega$] AB ἐστίν ἐπιπέδῳ. < ἐν δὲ τῶν ἐπὶ τῇ EZK εὐθείᾳ σημείων ἐστὶ τὸ K ; τὸ K > ἄρα ἐν $\tau\omega$] AB ἐστίν ἐπιπέδῳ. διὰ τὰ αὐτὰ δὴ τὸ K καὶ ἐν $\tau\omega$] $\Gamma\Delta$ ἐστίν ἐπιπέδῳ; τὰ AB , $\Gamma\Delta$ > ἄρα ἐπίπεδα ἐκβαλλόμενα συμπεσοῦνται. οὐ συμπίπτουσι δὲ διὰ τὸ παράλληλα ὑποκεῖσθαι; οὐκ > ἄρα αἱ EZ , $H\Theta$ εὐθεῖαι ἐκβαλλόμεναι ἐπὶ τὰ Z , Θ μέρη συμπεσοῦνται. ὁμοίως δὲ δείχομεν, < ὅτι αἱ EZ , $H\Theta$ εὐθεῖαι οὐδέ ἐπὶ τὰ E , H μέρη ἐκβαλλόμεναι συμπεσοῦνται. αἱ δὲ ἐπὶ μηδέτερα τὰ μέρη συμπίπτουσαι παράλληλοί εἰσιν. παράλληλός > ἄρα ἐστίν ἡ EZ τῇ $H\Theta$.

Proposition 16

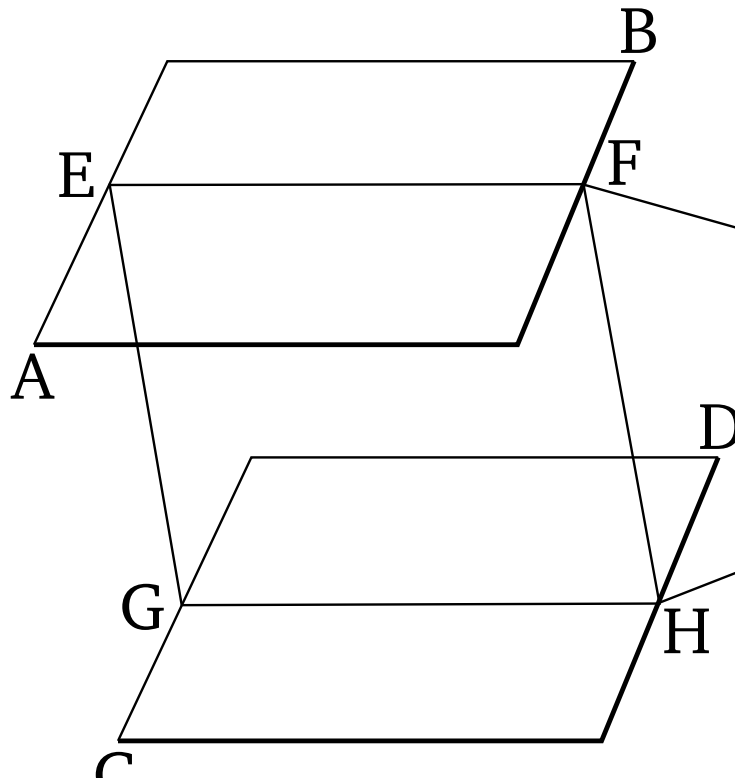
If two parallel planes are cut by some plane then their common sections are parallel.

For let the two parallel planes AB and CD be cut by the plane $EFGH$. And let EF and GH be their common sections. I say that EF is parallel to GH .

For, if not, being produced, EF and GH will meet either in the direction of F , H , or of E , G . Let them be produced, as in the direction of F , H , and let them, first of all, have met at K . And since EFK is in the plane AB , all of the points on EFK are thus also in the plane AB [Prop. 11.1]. And K is one of the points on EFK . Thus, K is in the plane AB . So, for the same (reasons), K is also in the plane CD . Thus, the planes AB and CD , being produced, will meet. But they do not meet, on account of being (initially) assumed (to be mutually) parallel. Thus, the straight-lines EF and GH , being produced in the direction of F , H , will not meet. So, similarly, we can show that the straight-lines EF and GH , being produced in the direction of E , G , will not meet either. And (straight-lines in one plane which), being produced, do not meet in either direction are parallel



[Def. 1.23]. EF is thus parallel to GH .



Ἐάν >άρα δύο ἐπίπεδα παράλληλα >πρὸς ἐπιπέδου τινὸς τέμνηται, αἱ κοιναὶ αὐτῶν τομαὶ παράλληλοι εἰσιν; <όπερ >έδει δεῖχαι.

Thus, if two parallel planes are cut by some plane then their common sections are parallel. (Which is) the very thing it was required to show.

17.

Ἐάν δύο εὐθεῖαι ὑπὸ παραλλήλων ἐπιπέδων τέμνωνται, εἰς τοὺς αὐτοὺς λόγους τμηθήσονται.
Δύο γὰρ εὐθεῖαι αἱ AB , $\Gamma\Delta$ ὑπὸ παραλλήλων ἐπιπέδων τῶν $H\Theta$, $K\Lambda$, MN τεμνέσθωσαν κατὰ τὰ A , E , B , Γ , Z , Δ σημεῖα; λέγω, <ὅτι ἐστὶν ὡς ἡ AE εὐθεῖα πρὸς τὴν EB , ο<ύτως ἡ ΓZ

Proposition 17

If two straight-lines are cut by parallel planes then they will be cut in the same ratios.
For let the two straight-lines AB and CD be cut by the parallel planes GH , KL , and MN at the points A , E , B , and C , F , D (respectively). I say that as the straight-line AE is

πρὸς τὴν ΖΔ.

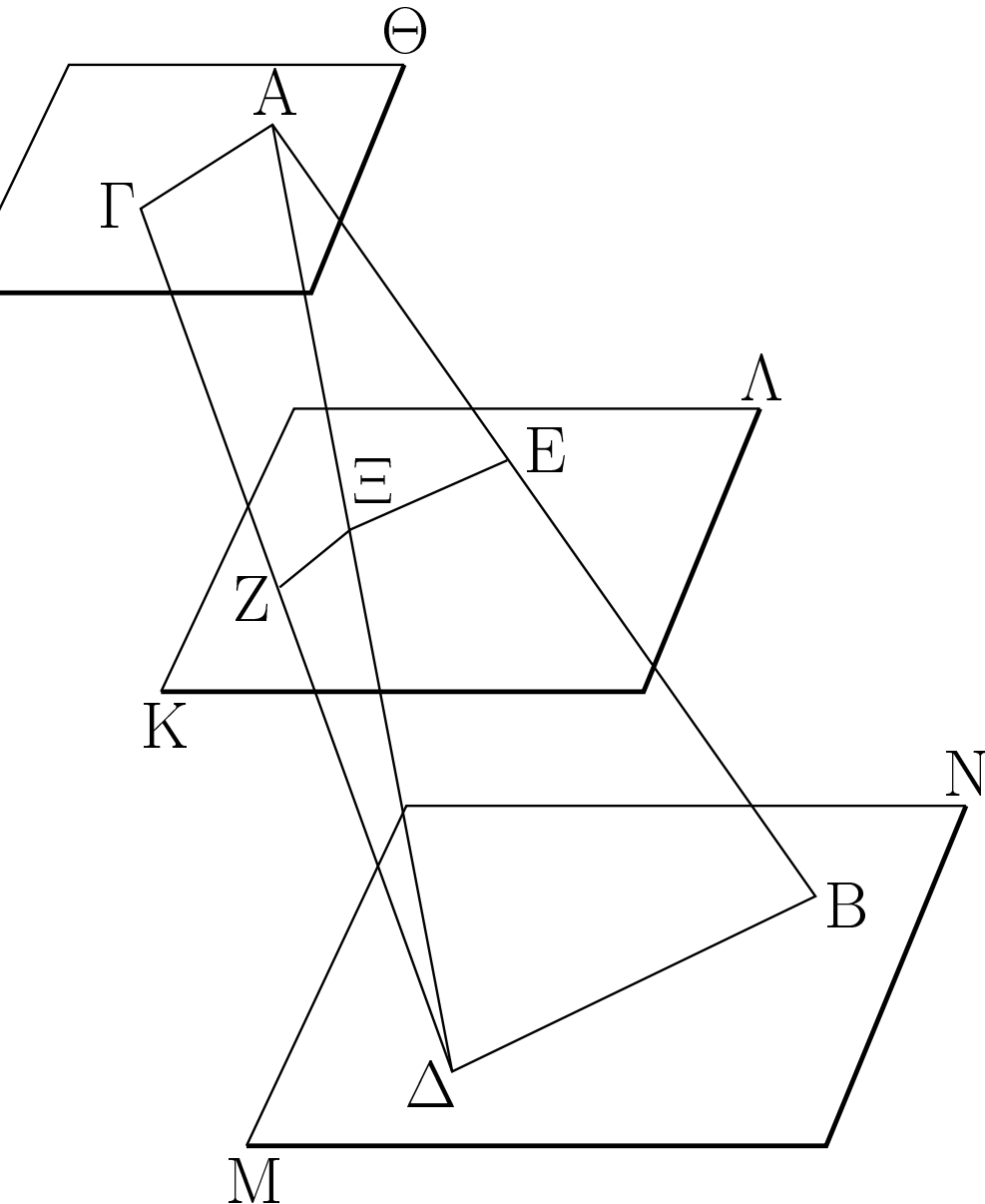
Ἐπεξεύθθωσαν γὰρ αἱ ΑΓ, ΒΔ, ΑΔ, καὶ συμβαλλέτω ἡ ΑΔ τῷ ΚΛ ἐπιπέδῳ κατὰ τὸ Χ σημεῖον, καὶ ἐπεξεύθθωσαν αἱ ΕΧ, ΧΖ.

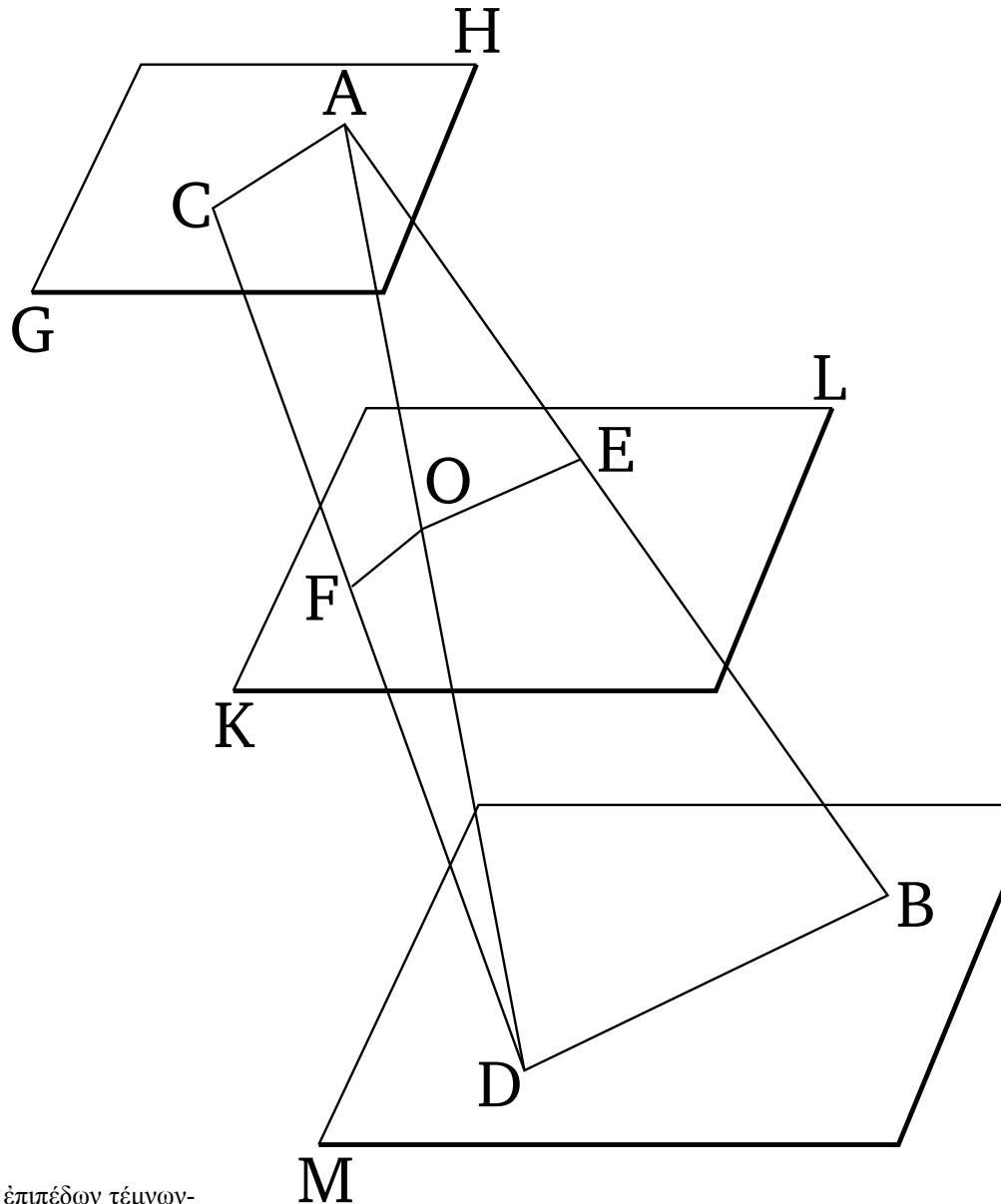
Καὶ ἐπεὶ δύο ἐπίπεδα παράλληλα τὰ ΚΛ, ΜΝ ὑπὸ ἐπιπέδου τοῦ ΕΒΔΧ τέμνεται, αἱ κοιναὶ αὐτῶν τομαὶ αἱ ΕΧ, ΒΔ παράλληλοί εἰσιν. διὰ τὰ αὐτὰ δὲ ἐπεὶ δύο ἐπίπεδα παράλληλα τὰ ΗΘ, ΚΛ ὑπὸ ἐπιπέδου τοῦ ΑΧΖΓ τέμνεται, αἱ κοιναὶ αὐτῶν τομαὶ αἱ ΑΓ, ΧΖ παράλληλοί εἰσιν. καὶ ἐπεὶ τριγώνου τοῦ ΑΒΔ παρὰ μίαν τῶν πλευρῶν τὴν ΒΔ εὐθεῖα >ῆκται ἡ ΕΧ, ἀνάλογον >άρα ἐστὶν ὡς ἡ ΑΕ πρὸς ΕΒ, ο<ύτως ἡ ΑΧ πρὸς ΧΔ. πάλιν ἐπεὶ τριγώνου τοῦ ΑΔΓ παρὰ μίαν τῶν πλευρῶν τὴν ΑΓ εὐθεῖα >ῆκται ἡ ΧΖ, ἀνάλογόν ἐστιν ὡς ἡ ΑΧ πρὸς ΧΔ, ο<ύτως ἡ ΓΖ πρὸς ΖΔ. ἐδείκθη δὲ καὶ ὡς ἡ ΑΧ πρὸς ΧΔ, ο<ύτως ἡ ΑΕ πρὸς ΕΒ; καὶ ὡς >άρα ἡ ΑΕ πρὸς ΕΒ, ο<ύτως ἡ ΓΖ πρὸς ΖΔ.

to EB , so CF (is) to FD .

For let AC , BD , and AD be joined, and let AD meet the plane KL at point O , and let EO and OF be joined.

And since two parallel planes KL and MN are cut by the plane $EBDO$, their common sections EO and BD are parallel [Prop. 11.16]. So, for the same (reasons), since two parallel planes GH and KL are cut by the plane $AOFC$, their common sections AC and OF are parallel [Prop. 11.16]. And since the straight-line EO has been drawn parallel to one of the sides BD of triangle ABD , thus, proportionally, as AE is to EB , so AO (is) to OD [Prop. 6.2]. Again, since the straight-line OF has been drawn parallel to one of the sides AC of triangle ADC , proportionally, as AO is to OD , so CF (is) to FD [Prop. 6.2]. And it was also shown that as AO (is) to OD , so AE (is) to EB . And thus as AE (is) to EB , so CF (is) to FD [Prop. 5.11].





Ἐὰν >άρα δύο εὐθεῖαι ὑπὸ παραλλήλων ἐπιπέδων τέμνων-
ται, εἰς τοὺς αὐτοὺς λόγους τμηθήσονται; <όπερ >έδει
δειχαι.

Thus, if two straight-lines are cut by parallel planes then
they will be cut in the same ratios. (Which is) the very thing
it was required to show.

18.

Proposition 18

Ἐὰν εὐθεῖα ἐπιπέδῳ τινὶ πρὸς ὀρθὰς >ῆ|, καὶ πάντα τὰ δι>
αὐτῆς ἐπίπεδα τῶ| αὐτῶ| ἐπιπέδῳ πρὸς ὀρθὰς >έσται.

Εὐθεῖα γάρ τις ἢ AB τῶ| ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς
>έστω; λέγω, <ὅτι καὶ πάντα τὰ διὰ τῆς AB ἐπίπεδα τῶ|
ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς ἔστιν.

Ἐκβεβλήσθω γὰρ διὰ τῆς AB ἐπίπεδον τὸ DE , καὶ >έστω
κοινὴ τομὴ τοῦ DE ἐπιπέδου καὶ τοῦ ὑποκειμένου ἢ GE , καὶ
εἰλήφθω ἐπὶ τῆς GE τυθὸν σημεῖον τὸ Z , καὶ ἀπὸ τοῦ Z τῇ|
 GE πρὸς ὀρθὰς >ήθθω ἐν τῶ| DE ἐπιπέδῳ ἢ ZH .

Καὶ ἐπεὶ ἡ AB πρὸς τὸ ὑποκείμενον ἐπίπεδον ὀρθή ἔστιν,
καὶ πρὸς πάσας >άρα τὰς ἀπτομένας αὐτῆς εὐθείας καὶ
ο>ύσας ἐν τῶ| ὑποκειμένῳ ἐπιπέδῳ ὀρθή ἔστιν ἢ AB ; <ώστε
καὶ πρὸς τὴν GE ὀρθή ἔστιν; ἢ >άρα ὑπὸ ABZ γωνία ὀρθή
ἔστιν. >έστι δὲ καὶ ἡ ὑπὸ HZB ὀρθή; παράλληλοξ >άρα
ἔστιν ἢ AB τῇ| ZH . ἢ δὲ AB τῶ| ὑποκειμένῳ ἐπιπέδῳ πρὸς
ὀρθὰς ἔστιν; καὶ ἡ ZH >άρα τῶ| ὑποκειμένῳ ἐπιπέδῳ πρὸς
ὀρθὰς ἔστιν. καὶ ἐπίπεδον πρὸς ἐπίπεδον ὀρθόν ἔστιν,

If a straight-line is at right-angles to some plane then all of
the planes (passing) through it will also be at right-angles to
the same plane.

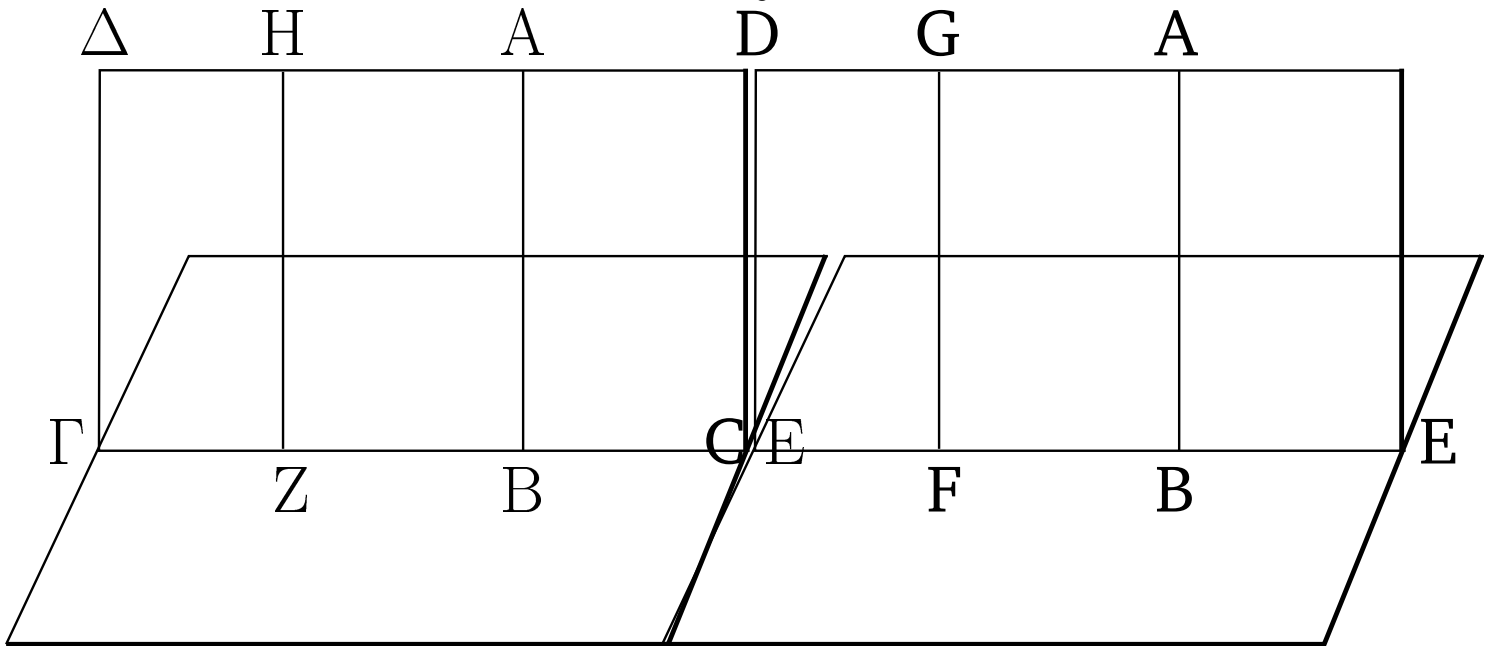
For let some straight-line AB be at right-angles to a refer-
ence plane. I say that all of the planes (passing) through AB
are also at right-angles to the reference plane.

For let the plane DE be produced through AB . And let CE
be the common section of the plane DE and the reference
(plane). And let some random point F be taken on CE . And
let FG be drawn from F , at right-angles to CE , in the plane
 DE [Prop. 1.11].

And since AB is at right-angles to the reference plane, AB
is thus also at right-angles to all of the straight-lines joined to
it which are also in the reference plane [Def. 11.3]. Hence,
it is also at right-angles to CE . Thus, angle ABF is a right-
angle. And GFB is also a right-angle. Thus, AB is paral-
lel to FG [Prop. 1.28]. And AB is at right-angles to the

<όταν αἱ τῇ κοινῇ τομῇ τῶν ἐπιπέδων πρὸς ὀρθὰς ἀγόμεναι εὐθεῖαι ἐν ἐνὶ τῶν ἐπιπέδων τῶ λοιπῷ ἐπιπέδῳ πρὸς ὀρθὰς >ῶσιν. καὶ τῇ κοινῇ τομῇ τῶν ἐπιπέδων τῇ ΓΕ ἐν ἐνὶ τῶν ἐπιπέδων τῶ ΔΕ πρὸς ὀρθὰς ἀθροῖσα ἡ ΖΗ ἐδείχθη τῶ ὑποκείμενῳ ἐπιπέδῳ πρὸς ὀρθὰς; τὸ >άρα ΔΕ ἐπίπεδον ὀρθὸν ἐστὶ πρὸς τὸ ὑποκείμενον. ὁμοίως δὲ δειθθήσεται καὶ πάντα τὰ διὰ τῆς ΑΒ ἐπίπεδα ὀρθὰ τυγθανόντα πρὸς τὸ ὑποκείμενον ἐπίπεδον.

reference plane. Thus, FG is also at right-angles to the reference plane [Prop. 11.8]. And a plane is at right-angles to a(nother) plane when the straight-lines drawn at right-angles to the common section of the planes, (and lying) in one of the planes, are at right-angles to the remaining plane [Def. 11.4]. And FG , (which was) drawn at right-angles to the common section of the planes, CE , in one of the planes, DE , was shown to be at right-angles to the reference plane. Thus, plane DE is at right-angles to the reference (plane). So, similarly, it can be shown that all of the planes (passing) at random through AB (are) at right-angles to the reference plane.



Ἐὰν >άρα εὐθεῖα ἐπίπεδῳ τινὶ πρὸς ὀρθὰς >ῇ, καὶ πάντα τὰ δι' αὐτῆς ἐπίπεδα τῶ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰς >έσται; <όπερ >έδει δεῖχαι.

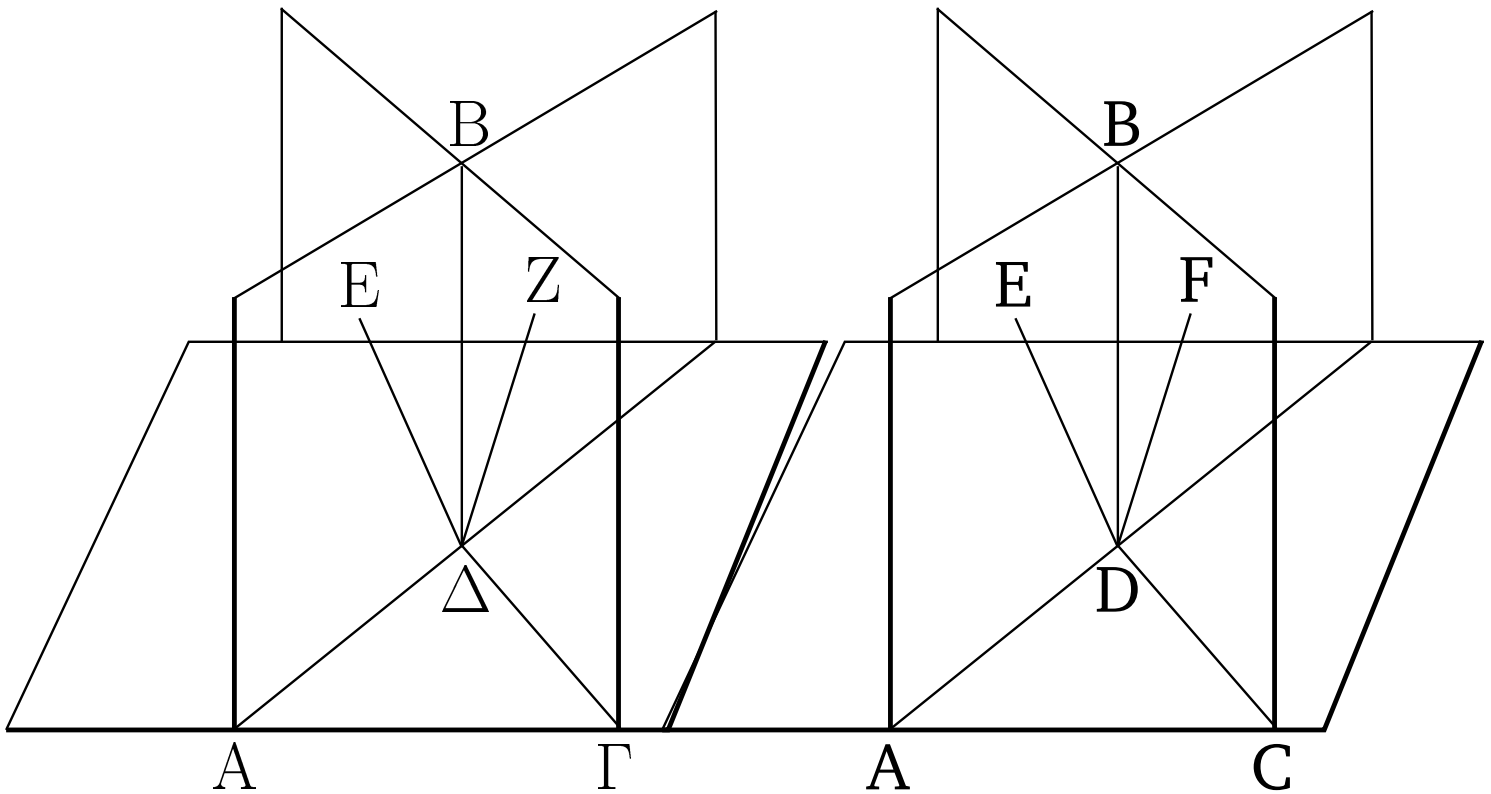
Thus, if a straight-line is at right-angles to some plane then all of the planes (passing) through it will also be at right-angles to the same plane. (Which is) the very thing it was required to show.

19.

Ἐὰν δύο ἐπίπεδα τέμνοντα >ἀλλήλα ἐπιπέδῳ τινὶ πρὸς ὀρθὰς >ῇ, καὶ ἡ κοινὴ αὐτῶν τομὴ τῶ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰς >έσται.

Proposition 19

If two planes cutting one another are at right-angles to some plane then their common section will also be at right-angles to the same plane.



Δύο γὰρ ἐπίπεδα τὰ AB , $B\Gamma$ τῶ| ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς >έστω, κοινὴ δὲ αὐτῶν τομὴ >έστω ἡ BD ; λέγω, <ὅτι ἡ BD τῶ| ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς ἐστίν.

μὴ γάρ, καὶ >ήθωσαν ἀπὸ τοῦ Δ σημείου ἐν μὲν τῶ| AB ἐπιπέδῳ τῇ| AD εὐθείᾳ πρὸς ὀρθὰς ἡ DE , ἐν δὲ τῶ| $B\Gamma$ ἐπιπέδῳ τῇ| ΓD πρὸς ὀρθὰς ἡ DZ .

Καὶ ἐπεὶ τὸ AB ἐπίπεδον ὀρθὸν ἐστὶ πρὸς τὸ ὑποκείμενον, καὶ τῇ| κοινῇ| αὐτῶν τομῇ| τῇ| AD πρὸς ὀρθὰς ἐν τῶ| AB ἐπιπέδῳ >ῆκται ἡ DE , ἡ DE >άρα ὀρθὴ ἐστὶ πρὸς τὸ ὑποκείμενον ἐπίπεδον. ὁμοίως δὲ δείχομεν, <ὅτι καὶ ἡ DZ ὀρθὴ ἐστὶ πρὸς τὸ ὑποκείμενον ἐπίπεδον. ἀπὸ τοῦ αὐτοῦ >άρα σημείου τοῦ Δ τῶ| ὑποκειμένῳ ἐπιπέδῳ δύο εὐθεῖα πρὸς ὀρθὰς ἀνεσταμέναι εἰσὶν ἐπὶ τὰ αὐτὰ μέρη; <ὅπερ ἐστὶν ἀδύνατον. οὐκ >άρα τῶ| ὑποκειμένῳ ἐπιπέδῳ ἀπὸ τοῦ Δ σημείου ἀνασταθήσεται πρὸς ὀρθὰς πλὴν τῆς DB κοινῆς τομῆς τῶν AB , $B\Gamma$ ἐπιπέδων.

Ἐὰν >άρα δύο ἐπίπεδα τέμνοντα >άλληλα ἐπιπέδῳ τινὶ πρὸς ὀρθὰς >ῆ|, καὶ ἡ κοινὴ αὐτῶν τομὴ τῶ| αὐτῶ| ἐπιπέδῳ πρὸς ὀρθὰς >έσται; <ὅπερ >έδει δεῖχαι.

For let the two planes AB and BC be at right-angles to a reference plane, and let their common section be BD . I say that BD is at right-angles to the reference plane.

For (if) not, let DE also be drawn from point D , in the plane AB , at right-angles to the straight-line AD , and DF , in the plane BC , at right-angles to CD .

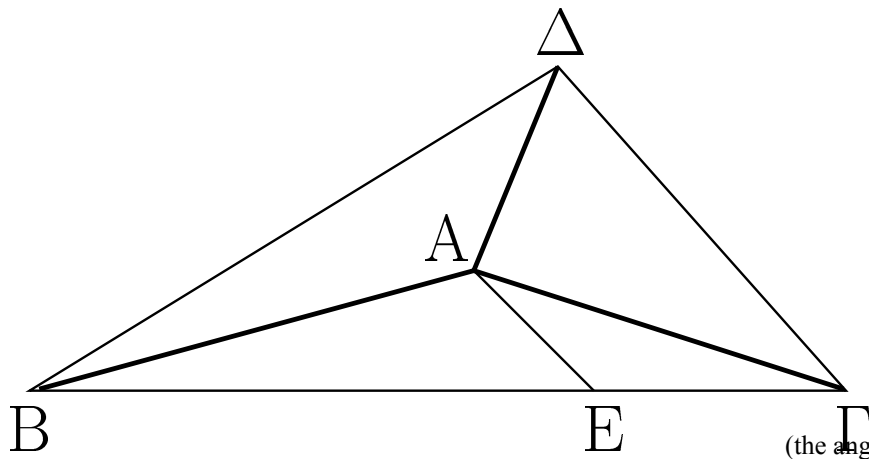
And since the plane AB is at right-angles to the reference (plane), and DE has been drawn at right-angles to their common section AD , in the plane AB , DE is thus at right-angles to the reference plane [Def. 11.4]. So, similarly, we can show that DF is also at right-angles to the reference plane. Thus, two (different) straight-lines are set up, at the same point D , at right-angles to the reference plane, on the same side. The very thing is impossible [Prop. 11.13]. Thus, no (other straight-line) except the common section DB of the planes AB and BC can be set up at point D , at right-angles to the reference plane.

Thus, if two planes cutting one another are at right-angles to some plane then their common section will also be at right-angles to the same plane. (Which is) the very thing it was required to show.

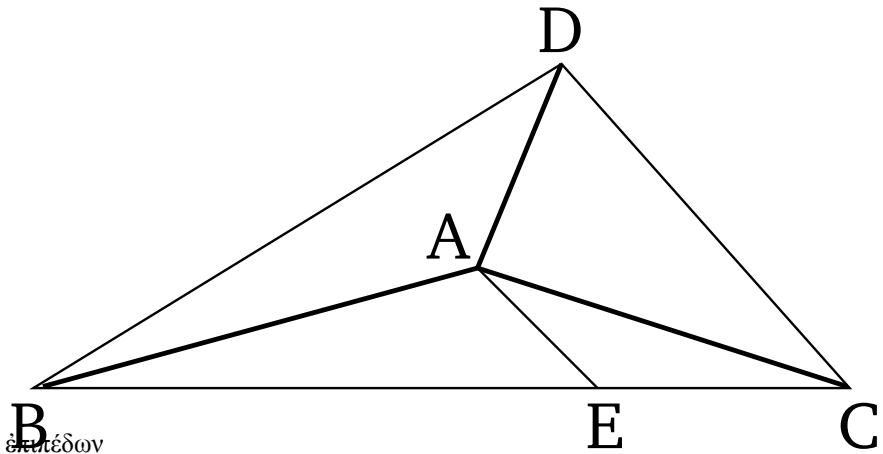
20.

Proposition 20

Ἐὰν στερεὰ γωνία ὑπὸ τριῶν γωνιῶν ἐπιπέδων περιέθῃται, If a solid angle is contained by three plane angles then (the δύο ὅποιοι οὖν τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμενα sum of) any two (angles) is greater than the remaining (one),



(the angles) being taken up in any (possible way).



Στερεὰ γὰρ γωνία ἢ πρὸς τῷ A ὑπὸ τριῶν γωνιῶν ἐπιπέδων τῶν ὑπὸ BAC, ΓΑΔ, ΔΑΒ περιεθέσθω; λέγω, <ὅτι τῶν ὑπὸ BAC, ΓΑΔ, ΔΑΒ γωνιῶν δύο ὅποιοι οὖν τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι.

Εἰ μὲν οὖν αἱ ὑπὸ BAC, ΓΑΔ, ΔΑΒ γωνίαι >ίσαι ἀλλήλαις εἰσίν, φανερόν, <ὅτι δύο ὅποιοι οὖν τῆς λοιπῆς μείζονές εἰσιν. εἰ δὲ οὐ>, >έστω μείζων ἡ ὑπὸ BAC, καὶ συνεστάτω πρὸς τῇ AB εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ A τῇ ὑπὸ ΔΑΒ γωνίᾳ ἐν τῷ διὰ τῶν BAC ἐπιπέδῳ >ίση ἡ ὑπὸ BAE, καὶ κείσθω τῇ AD >ίση ἡ AE, καὶ διὰ τοῦ E σημείου διαθεῖσθαι ἡ BEΓ τεμνέτω τὰς AB, AG εὐθείας κατὰ τὰ B, Γ σημεία, καὶ ἐπεζεύθωσαν αἱ ΔB, ΔΓ.

Καὶ ἐπεὶ >ίση ἐστὶν ἡ ΔA τῇ AE, κοινὴ δὲ ἡ AB, δύο δυσὶν >ίσαι; καὶ γωνία ἡ ὑπὸ ΔΑΒ γωνία τῇ ὑπὸ BAE >ίση; βάσις >άρα ἡ ΔB βάσει τῇ BE ἐστὶν >ίση. καὶ ἐπεὶ δύο αἱ ΔA, ΔΓ τῆς BΓ μείζονές εἰσιν, <ὢν ἡ ΔB τῇ BE ἐδείχθη >ίση, λοιπὴ >άρα ἡ ΔΓ λοιπῆς τῆς ΕΓ μείζων ἐστίν. καὶ ἐπεὶ >ίση ἐστὶν ἡ ΔA τῇ AE, κοινὴ δὲ ἡ AG, καὶ βάσις ἡ ΔΓ βάσει τῆς ΕΓ μείζων ἐστίν, γωνία >άρα ἡ ὑπὸ ΔΑΓ γωνία τῆς ὑπὸ ΕΑΓ μείζων ἐστίν. ἐδείχθη δὲ καὶ ἡ ὑπὸ ΔΑΒ τῇ ὑπὸ BAE >ίση; αἱ >άρα ὑπὸ ΔΑΒ, ΔΑΓ τῆς ὑπὸ BAC μείζονές εἰσιν. ὁμοίως δὲ δείχομεν, <ὅτι καὶ αἱ λοιπαὶ σύνδου λαμβανόμεναι τῆς λοιπῆς μείζονές εἰσιν.

Ἐὰν >άρα στερεὰ γωνία ὑπὸ τριῶν γωνιῶν ἐπιπέδων περιέθῃται, <ὅτι δύο ὅποιοι οὖν τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι <ὅπερ >έδει δεῖξαι.

For let the solid angle A be contained by the three plane angles BAC , CAD , and DAB . I say that (the sum of) any two of the angles BAC , CAD , and DAB is greater than the remaining (one), (the angles) being taken up in any (possible way).

For if the angles BAC , CAD , and DAB are equal to one another then (it is) clear that (the sum of) any two is greater than the remaining (one). But, if not, let BAC be greater (than CAD or DAB). And let (angle) BAE , equal to the angle DAB , be constructed in the plane through BAC , on the straight-line AB , at the point A on it. And let AE be made equal to AD . And BEC being drawn across through point E , let it cut the straight-lines AB and AC at points B and C (respectively). And let DB and DC be joined.

And since DA is equal to AE , and AB (is) common, the two (straight-lines AD and AB are) equal to the two (straight-lines EA and AB , respectively). And angle DAB (is) equal to angle BAE . Thus, the base DB is equal to the base BE [Prop. 1.4]. And since the (sum of the) two (straight-lines) BD and DC is greater than BC [Prop. 1.20], of which DB was shown (to be) equal to BE , the remainder DC is thus greater than the remainder EC . And since DA is equal to AE , but AC (is) common, and the base DC is greater than the base EC , the angle DAC is thus greater than the angle EAC [Prop. 1.25]. And DAB was also shown (to be) equal to BAE . Thus, (the sum of) DAB and DAC is greater than BAC . So, similarly, we can also show that the remaining (angles), being taken in pairs, are greater than the remaining (one).

Thus, if a solid angle is contained by three plane angles then (the sum of) any two (angles) is greater than the remaining (one), (the angles) being taken up in any (possible way).

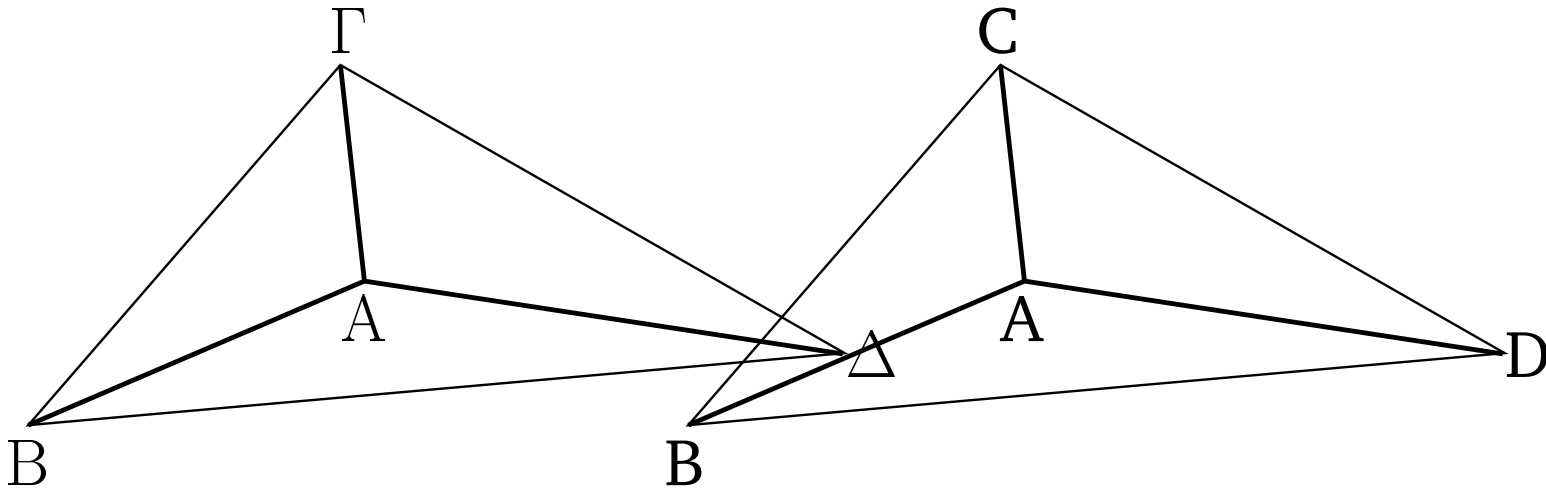
(Which is) the very thing it was required to show.

21.

<Ἄπανα στερεὰ γωνία ὑπὸ ἐλασσόνων [$>\eta$] τεσσάρων ὀρθῶν γωνιῶν ἐπιπέδων περιέθεται.

Proposition 21

Any solid angle is contained by plane angles (whose sum is) less [than] four right-angles.[†]



>Ἐστω στερεὰ γωνία ἡ πρὸς τῷ A περιεθομένη ὑπὸ ἐπιπέδων γωνιῶν τῶν ὑπὸ BAC, ΓAC, ΔAB; λέγω, <ὅτι αἱ ὑπὸ BAC, ΓAC, ΔAB τεσσάρων ὀρθῶν ἐλάσσονές εἰσιν.

Εἰλήφθω γὰρ ἐφ' ἐκάστης τῶν AB, AC, AD τυθόντα σημεῖα τὰ B, Γ, Δ, καὶ ἐπεζεύθωσαν αἱ BG, ΓΔ, ΔB. καὶ ἐπεὶ στερεὰ γωνία ἡ πρὸς τῷ B ὑπὸ τριῶν γωνιῶν ἐπιπέδων περιέθεται τῶν ὑπὸ ΓBA, ABA, ΓBA, δύο ὅποιοι οὖν τῇ λοιπῇ μείζονές εἰσιν; αἱ > ἄρα ὑπὸ ΓBA, ABA τῇ ὑπὸ ΓBA μείζονές εἰσιν. διὰ τὰ αὐτὰ δὲ καὶ αἱ μὲν ὑπὸ BGA, AΓA τῇ ὑπὸ BGA μείζονές εἰσιν, αἱ δὲ ὑπὸ ΓΔA, AΔB τῇ ὑπὸ ΓΔB μείζονές εἰσιν; αἱ < ἐχ' ἄρα γωνίαι αἱ ὑπὸ ΓBA, ABA, BGA, AΓA, ΓΔA, AΔB τριῶν τῶν ὑπὸ ΓBA, BGA, ΓΔB μείζονές εἰσιν. ἀλλὰ αἱ τρεῖς αἱ ὑπὸ ΓBA, BΔΓ, BΓA δυσὶν ὀρθαῖς > ἴσαι εἰσιν; αἱ < ἐχ' ἄρα αἱ ὑπὸ ΓBA, ABA, BGA, AΓA, ΓΔA, AΔB δύο ὀρθῶν μείζονές εἰσιν. καὶ ἐπεὶ ἐκάστου τῶν ABΓ, AΓA, AΔB τριγώνων αἱ τρεῖς γωνίαι δυσὶν ὀρθαῖς > ἴσαι εἰσιν, αἱ > ἄρα τῶν τριῶν τριγώνων ἑννέα γωνίαι αἱ ὑπὸ ΓBA, AΓB, BAC, AΓA, ΓΔA, ΓAΔ, AΔB, ΔBA, BAA < ἐχ' ὀρθαῖς > ἴσαι εἰσιν, <ὧν αἱ ὑπὸ ABΓ, BΓA, AΓA, ΓΔA, AΔB, ΔBA < ἐχ' γωνίαι δύο ὀρθῶν εἰσι μείζονες; λοιπαὶ > ἄρα αἱ ὑπὸ BAC, ΓAC, ΔAB τρεῖς [γωνίαι] περιέθουσιν τὴν στερεὰν γωνίαν τεσσάρων ὀρθῶν ἐλάσσονές εἰσιν.

<Ἄπανα > ἄρα στερεὰ γωνία ὑπὸ ἐλασσόνων [$>\eta$] τεσσάρων ὀρθῶν γωνιῶν ἐπιπέδων περιέθεται; ὅπερ > ἐδει δεῖχαι.

Let the solid angle A be contained by the plane angles BAC , CAD , and DAB . I say that (the sum of) BAC , CAD , and DAB is less than four right-angles.

For let the random points B , C , and D be taken on each of (the straight-lines) AB , AC , and AD (respectively). And let BC , CD , and DB be joined. And since the solid angle at B is contained by the three plane angles CBA , ABD , and CBD , (the sum of) any two is greater than the remaining (one) [Prop. 11.20]. Thus, (the sum of) CBA and ABD is greater than CBD . So, for the same (reasons), (the sum of) BCA and ACD is also greater than BCD , and (the sum of) CDA and ADB is greater than CDB . Thus, the (sum of the) six angles CBA , ABD , BCA , ACD , CDA , and ADB is greater than the (sum of the) three (angles) CBD , BCD , and CDB . But, the (sum of the) three (angles) CBD , BDC , and BCD is equal to two right-angles [Prop. 1.32]. Thus, the (sum of the) six angles CBA , ABD , BCA , ACD , CDA , and ADB is greater than two right-angles. And since the (sum of the) three angles of each of the triangles ABC , ACD , and ADB is equal to two right-angles, the (sum of the) nine angles CBA , ACB , BAC , ACD , CDA , CAD , ADB , DBA , and BAD of the three triangles is equal to six right-angles, of which the (sum of the) six angles ABC , BCA , ACD , CDA , ADB , and DBA is greater than two right-angles. Thus, the (sum of the) remaining three [angles] BAC , CAD , and DAB , containing the solid angle, is less than four right-angles.

Thus, any solid angle is contained by plane angles (whose sum is) less [than] four right-angles. (Which is) the very thing it was required to show.

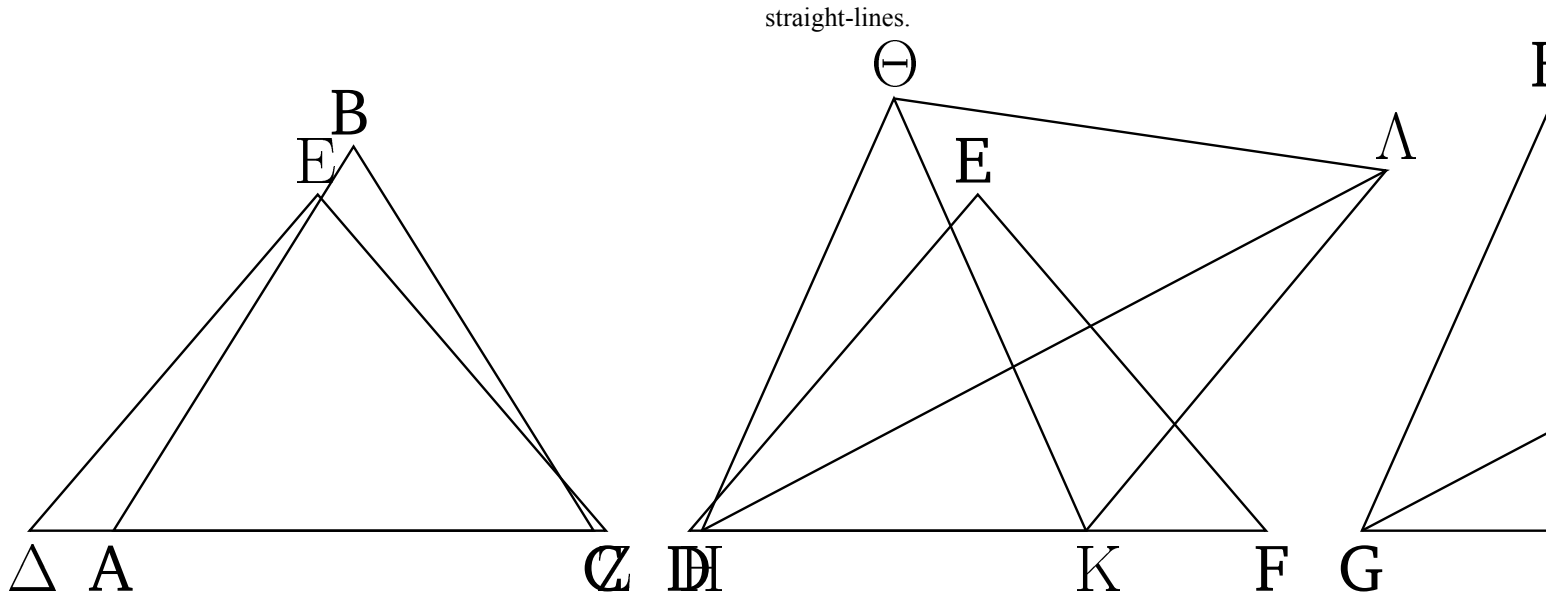
[†] This proposition is only proved for the case of a solid angle contained by three plane angles. However, the generalization to a solid angle contained by more than three plane angles is straightforward.

22.

Ἐὰν <ὧς τρεῖς γωνίαι ἐπίπεδοι, <ὧν αἱ δύο τῇ λοιπῇ μείζονές εἰσι πάντῃ μεταλαμβάνομεναι, περιέθωσι δὲ αὐτὰς > ἴσαι εὐθεῖαι, δυνατόν ἐστιν ἐκ τῶν ἐπιζευγνυουσῶν τὰς > ἴσας εὐθείας τρίγωνον συστήσασθαι.

Proposition 22

If there are three plane angles, of which (the sum of any) two is greater than the remaining (one), (the angles) being taken up in any (possible way), and if equal straight-lines contain them, then it is possible to construct a triangle from (the straight-lines created by) joining the (ends of the) equal



>Ἐστωσαν τρεῖς γωνίαι ἐπίπεδοι αἱ ὑπὸ ΑΒΓ, ΔΕΖ, ΗΘΚ, <ῶν αἱ δύο τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι, αἱ μὲν ὑπὸ ΑΒΓ, ΔΕΖ τῆς ὑπὸ ΗΘΚ, αἱ δὲ ὑπὸ ΔΕΖ, ΗΘΚ τῆς ὑπὸ ΑΒΓ, καὶ >έτι αἱ ὑπὸ ΗΘΚ, ΑΒΓ τῆς ὑπὸ ΔΕΖ, καὶ >έστωσαν >ίσαι αἱ ΑΒ, ΒΓ, ΔΕ, ΕΖ, ΗΘ, ΘΚ εὐθεῖαι, καὶ ἐπεξεύθωσαν αἱ ΑΓ, ΔΖ, ΗΚ; λέγω, <ὅτι δυνατόν ἐστιν ἐκ τῶν >ίσων ταῖς ΑΓ, ΔΖ, ΗΚ τρίγωνον συστήσασθαι, τούτέστιν <ὅτι τῶν ΑΓ, ΔΖ, ΗΚ δύο ὁποιοιοῦν τῆς λοιπῆς μείζονές εἰσιν.

Εἰ μὲν ο>ὺν αἱ ὑπὸ ΑΒΓ, ΔΕΖ, ΗΘΚ γωνίαι >ίσαι ἀλλήλαις εἰσίν, φανερόν, <ὅτι καὶ τῶν ΑΓ, ΔΖ, ΗΚ >ίσων γινομένων δυνατόν ἐστιν ἐκ τῶν >ίσων ταῖς ΑΓ, ΔΖ, ΗΚ τρίγωνον συστήσασθαι. εἰ δὲ ο>ύ, >έστωσαν >άνισοι, καὶ συνεστάτω πρὸς τῇ ΘΚ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ Θ τῇ ὑπὸ ΑΒΓ γωνίᾳ >ίση ἢ ὑπὸ ΚΘΛ; καὶ κείσθω μὲν τῶν ΑΒ, ΒΓ, ΔΕ, ΕΖ, ΗΘ, ΘΚ >ίση ἢ ΘΛ, καὶ ἐπεξεύθωσαν αἱ ΚΛ, ΗΛ. καὶ ἐπεὶ δύο αἱ ΑΒ, ΒΓ δυσὶ ταῖς ΚΘ, ΘΛ >ίσαι εἰσίν, καὶ γωνία ἢ πρὸς τῷ Β γωνία τῇ ὑπὸ ΚΘΛ >ίση, βάσεις >άρα ἢ ΑΓ βάσει τῇ ΚΛ >ίση. καὶ ἐπεὶ αἱ ὑπὸ ΑΒΓ, ΗΘΚ τῆς ὑπὸ ΔΕΖ μείζονές εἰσιν, >ίση δὲ ἢ ὑπὸ ΑΒΓ τῇ ὑπὸ ΚΘΛ, ἢ >άρα ὑπὸ ΗΘΛ τῆς ὑπὸ ΔΕΖ μείζων ἐστίν. καὶ ἐπεὶ δύο αἱ ΗΘ, ΘΛ δύο ταῖς ΔΕ, ΕΖ >ίσαι εἰσίν, καὶ γωνία ἢ ὑπὸ ΗΘΛ γωνία τῆς ὑπὸ ΔΕΖ μείζων, βάσεις >άρα ἢ ΗΛ βάσεως τῆς ΔΖ μείζων ἐστίν. ἀλλὰ αἱ ΗΚ, ΚΛ τῆς ΗΛ μείζονές εἰσιν. πολλῶν >άρα αἱ ΗΚ, ΚΛ τῆς ΔΖ μείζονές εἰσιν. >ίση δὲ ἢ ΚΛ τῇ ΑΓ; αἱ ΑΓ, ΗΚ >άρα τῆς λοιπῆς τῆς ΔΖ μείζονές εἰσιν. ὁμοίως δὲ δείχομεν, <ὅτι καὶ αἱ μὲν ΑΓ, ΔΖ τῆς ΗΚ μείζονές εἰσιν, καὶ >έτι αἱ ΔΖ, ΗΚ τῆς ΑΓ μείζονές εἰσιν. δυνατόν >άρα ἐστίν ἐκ τῶν >ίσων ταῖς ΑΓ, ΔΖ, ΗΚ τρίγωνον συστήσασθαι; <ὅπερ >έδει δεῖξαι.

straight-lines.

Let ABC , DEF , and GHK be three plane angles, of which the sum of any two is greater than the remaining (one), (the angles) being taken up in any (possible way)—(that is), ABC and DEF (greater) than GHK , DEF and GHK (greater) than ABC , and, further, GHK and ABC (greater) than DEF . And let AB , BC , DE , EF , GH , and HK be equal straight-lines. And let AC , DF , and GK be joined. I say that that it is possible to construct a triangle out of (straight-lines) equal to AC , DF , and GK —that is to say, that (the sum of) any two of AC , DF , and GK is greater than the remaining (one).

Now, if the angles ABC , DEF , and GHK are equal to one another then (it is) clear that, (with) AC , DF , and GK also becoming equal, it is possible to construct a triangle from (straight-lines) equal to AC , DF , and GK . And if not, let them be unequal, and let KHL , equal to angle ABC , be constructed on the straight-line HK , at the point H on it. And let HL be made equal to one of AB , BC , DE , EF , GH , and HK . And let KL and GL be joined. And since the two (straight-lines) AB and BC are equal to the two (straight-lines) KH and HL (respectively), and the angle at B (is) equal to KHL , the base AC is thus equal to the base KL [Prop. 1.4]. And since (the sum of) ABC and GHK is greater than DEF , and ABC equal to KHL , GHL is thus greater than DEF . And since the two (straight-lines) GH and HL are equal to the two (straight-lines) DE and EF (respectively), and angle GHL (is) greater than DEF , the base GL is thus greater than the base DF [Prop. 1.24]. But, (the sum of) GK and KL is greater than GL [Prop. 1.20]. Thus, (the sum of) GK and KL is much greater than DF . And KL (is) equal to AC . Thus, (the sum of) AC and GK is greater than the remaining (straight-line) DF . So, similarly, we can show that (the sum of) AC and DF is greater than GK , and, further, that (the sum of) DF and GK is greater than AC . Thus, it is possible to construct a triangle from (straight-lines) equal to AC , DF , and GK . (Which is) the very thing it was required to show.

23.

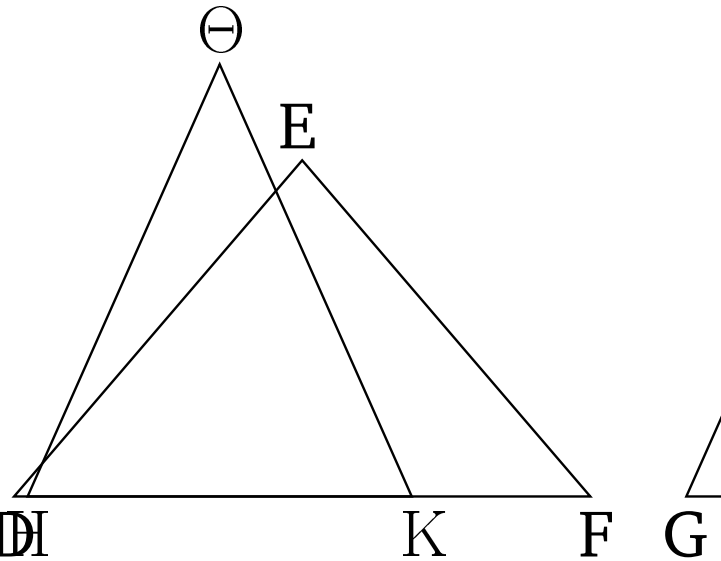
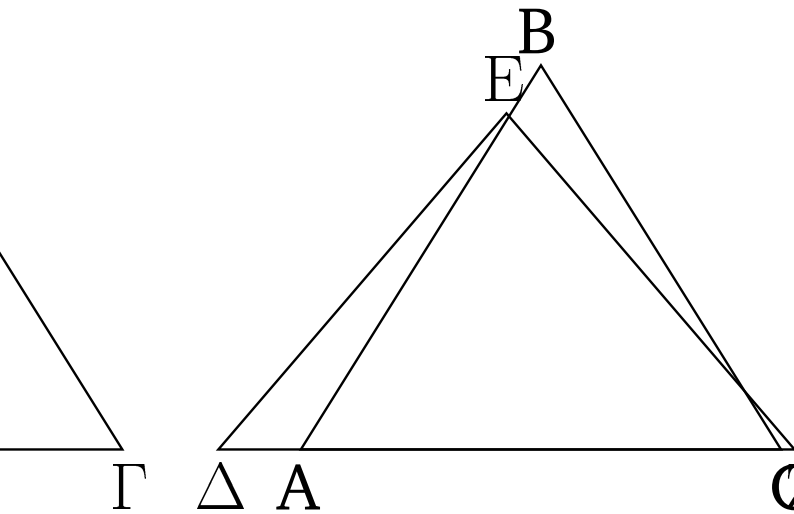
Proposition 23

Ἐκ τριῶν γωνιῶν ἐπιπέδων, <ῶν αἱ δύο τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι, στερεὰν γωνίαν συστήσασθαι;

To construct a solid angle from three (given) plane angles, (the sum of) two of which is greater than the remaining (one,

δεῖ δὴ τὰς τρεῖς τεσσάρων ὀρθῶν ἐλάσσονα εἶναι.

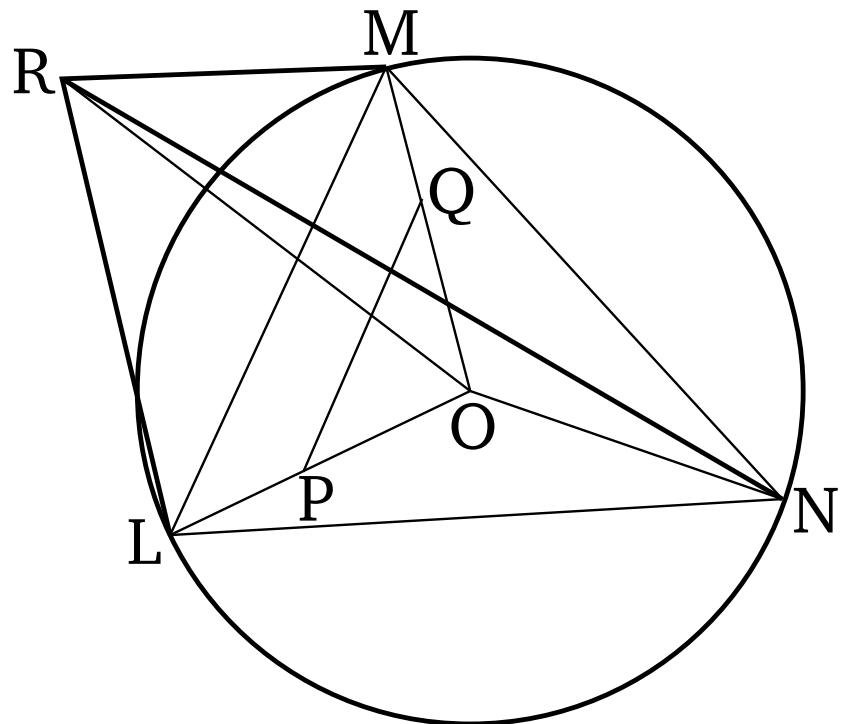
the angles) being taken up in any (possible way). So, it is necessary for the (sum of the) three (angles) to be less than four right-angles [Prop. 11.21].



>Ἐστωσαν αἱ δοθεῖσαι τρεῖς γωνίαι ἐπίπεδοι αἱ ὑπὸ ΑΒΓ, ΔΕΖ, ΗΘΚ, <ῶν αἱ δύο τῆς λοιπῆς μείζονες >έστωσαν πάντῃ μεταλαμβανόμεναι, >έτι δὲ αἱ τρεῖς τεσσάρων ὀρθῶν ἐλάσσονες; δεῖ δὴ ἐκ τῶν >ίσων ταῖς ὑπὸ ΑΒΓ, ΔΕΖ, ΗΘΚ στερεὰν γωνίαν συστήσασθαι.

Ἀπειλήφθωσαν >ίσαι αἱ ΑΒ, ΒΓ, ΔΕ, ΕΖ, ΗΘ, ΘΚ, καὶ ἐπέζεύθωσαν αἱ ΑΓ, ΔΖ, ΗΚ; δυνατόν >άρα ἐστὶν ἐκ τῶν >ίσων ταῖς ΑΓ, ΔΖ, ΗΚ τρίγωνον συστήσασθαι. συνεστάτω τὸ ΛΜΝ, <ώστε >ίσην εἶναι τὴν μὲν ΑΓ τῇ ΛΜ, τὴν δὲ ΔΖ τῇ ΜΝ, καὶ >έτι τὴν ΗΚ τῇ ΝΛ, καὶ περιγεγράφθω περὶ τὸ ΛΜΝ τρίγωνον κύκλος ὁ ΛΜΝ, καὶ εἰλήφθω αὐτοῦ τὸ κέντρον καὶ >έστω τὸ Χ, καὶ ἐπέζεύθωσαν αἱ ΛΧ, ΜΧ, ΝΧ;

Let ABC , DEF , and GHK be the three given plane angles, of which let (the sum of) two be greater than the remaining (one, the angles) being taken up in any (possible way), and, further, (let) the (sum of the) three (be) less than four right-angles. So, it is necessary to construct a solid angle from (plane angles) equal to ABC , DEF , and GHK . Let AB , BC , DE , EF , GH , and HK be cut off (so as to be) equal (to one another). And let AC , DF , and GK be joined. It is, thus, possible to construct a triangle from (straight-lines) equal to AC , DF , and GK [Prop. 11.22]. Let (such a triangle), LMN , have been constructed, such that AC is equal to LM , DF to MN , and, further, GK to NL . And let the circle LMN be circumscribed about triangle LMN [Prop. 4.5]. And let its center be found, and let it be (at) O . And let LO , MO , and NO be joined.



ἀπὸ τῆς XP , καὶ ἐπεξεύθωσαν αἱ PA , PM , PN .

Καὶ ἐπεὶ ἡ PX ὀρθὴ ἐστὶ πρὸς τὸ τοῦ AMN κύκλου ἐπίπεδον, καὶ πρὸς ἐκάστην $>$ άρα τῶν AX , MX , NX ὀρθὴ ἐστὶν ἡ PX . καὶ ἐπεὶ $>$ ίση ἐστὶν ἡ AX τῇ XM , κοινὴ δὲ καὶ πρὸς ὀρθὰς ἡ XP , βάσις $>$ άρα ἡ PA βάσει τῇ PM ἐστὶν $>$ ίση. διὰ τὰ αὐτὰ δὲ καὶ ἡ PN ἐκατέρα τῶν PA , PM ἐστὶν $>$ ίση; αἱ τρεῖς $>$ άρα αἱ PA , PM , PN $>$ ίσαι ἀλλήλαις εἰσίν. καὶ ἐπεὶ $<$ ῶ μείζον ἐστὶ τὸ ἀπὸ τῆς AB τοῦ ἀπὸ τῆς AX , ἐκείνῳ $>$ ίσον ὑπόκειται τὸ ἀπὸ τῆς XP , τὸ $>$ άρα ἀπὸ τῆς AB $>$ ίσον ἐστὶ τοῖς ἀπὸ τῶν AX , XP . τοῖς δὲ ἀπὸ τῶν AX , XP $>$ ίσον ἐστὶ τὸ ἀπὸ τῆς AP ; ὀρθὴ γὰρ ἡ ὑπὸ AXP ; τὸ $>$ άρα ἀπὸ τῆς AB $>$ ίσον ἐστὶ τῷ ἀπὸ τῆς PA ; $>$ ίση $>$ άρα ἡ AB τῇ PA . ἀλλὰ τῇ μὲν AB $>$ ίση ἐστὶν ἐκάστη τῶν $BΓ$, $ΔΕ$, $ΕΖ$, $ΗΘ$, $ΘΚ$, τῇ δὲ PA $>$ ίση ἐκατέρα τῶν PM , PN ; ἐκάστη $>$ άρα τῶν AB , $BΓ$, $ΔΕ$, $ΕΖ$, $ΗΘ$, $ΘΚ$ ἐκάστη τῶν PA , PM , PN $>$ ίση ἐστίν. καὶ ἐπεὶ δύο αἱ AP , PM δυοὶ ταῖς AB , $BΓ$ $>$ ίσαι εἰσίν, καὶ βάσις ἡ AM βάσει τῇ $ΑΓ$ ὑπόκειται $>$ ίση, γωνία $>$ άρα ἡ ὑπὸ APM γωνία τῇ ὑπὸ $ABΓ$ ἐστὶν $>$ ίση. διὰ τὰ αὐτὰ δὲ καὶ ἡ μὲν ὑπὸ MPN τῇ ὑπὸ $ΔΕΖ$ ἐστὶν $>$ ίση, ἡ δὲ ὑπὸ APN τῇ ὑπὸ $ΗΘΚ$.

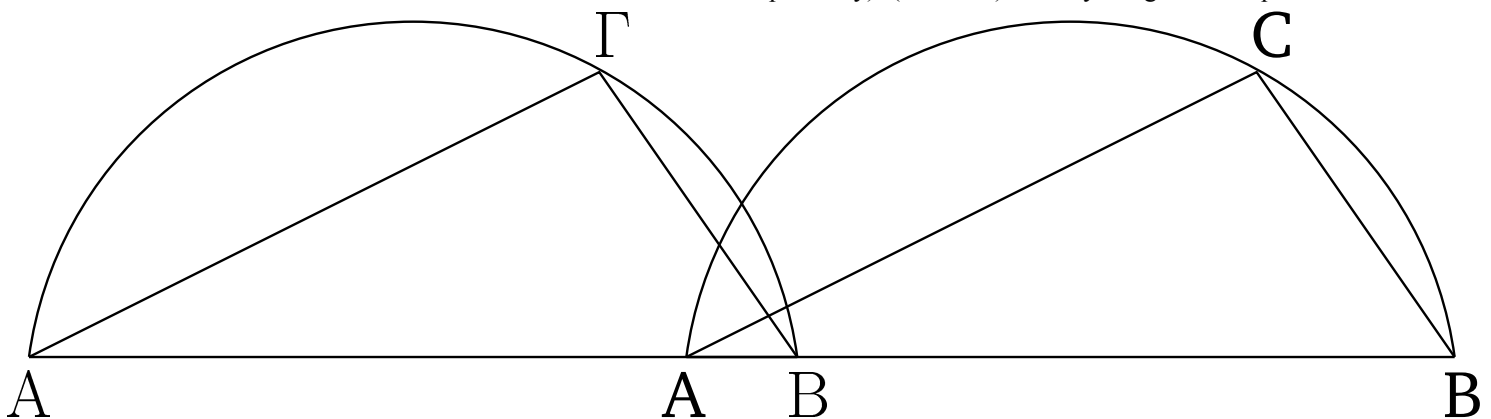
Ἐκ τριῶν $>$ άρα γωνιῶν ἐπιπέδων τῶν ὑπὸ APM , MPN , APN , $<$ αἱ εἰσίν $>$ ίσαι τρισὶ ταῖς δοθείσαις ταῖς ὑπὸ $ABΓ$, $ΔΕΖ$, $ΗΘΚ$, στερεὰ γωνία συνέσταται ἡ πρὸς τῷ P περιεχομένη ὑπὸ τῶν APM , MPN , APN γωνιῶν; $<$ όπερ $>$ έδει ποιῆσαι.

not less than LO . And it was shown (to be) not equal either. Thus, AB (is) greater than LO .

So let OR be set up at point O at right-angles to the plane of circle LMN [Prop. 11.12]. And let the (square) on OR be equal to that (area) by which the square on AB is greater than the (square) on LO [Prop. 11.23 lem.]. And let RL , RM , and RN be joined.

And since RO is at right-angles to the plane of circle LMN , RO is thus also at right-angles to each of LO , MO , and NO . And since LO is equal to OM , and OR is common and at right-angles, the base RL is thus equal to the base RM [Prop. 1.4]. So, for the same (reasons), RN is also equal to each of RL and RM . Thus, the three (straight-lines) RL , RM , and RN are equal to one another. And since the (square) on OR was assumed to be equal to that (area) by which the (square) on AB is greater than the (square) on LO , the (square) on AB is thus equal to the (sum of the squares) on LO and OR . And the (square) on LR is equal to the (sum of the squares) on LO and OR . For LOR (is) a right-angle [Prop. 1.47]. Thus, the (square) on AB is equal to the (square) on RL . Thus, AB (is) equal to RL . But, each of BC , DE , EF , GH , and HK is equal to AB , and each of RM and RN equal to RL . Thus, each of AB , BC , DE , EF , GH , and HK is equal to each of RL , RM , and RN . And since the two (straight-lines) LR and RM are equal to the two (straight-lines) AB and BC (respectively), and the base LM was assumed (to be) equal to the base AC , the angle LRM is thus equal to the angle ABC [Prop. 1.8]. So, for the same (reasons), MRN is also equal to DEF , and LRN to GHK .

Thus, the solid angle R , contained by the angles LRM , MRN , and LRN , has been constructed out of the three plane angles LRM , MRN , and LRN , which are equal to the three given (plane angles) ABC , DEF , and GHK (respectively). (Which is) the very thing it was required to do.



Λήμμα.

Lemma

$<$ Ὅν δὲ τρόπον, $<$ ῶ μείζον ἐστὶ τὸ ἀπὸ τῆς AB τοῦ ἀπὸ τῆς AX , ἐκείνῳ $>$ ίσον λαβεῖν $>$ έστι τὸ ἀπὸ τῆς XP , δείχομεν οὕτως. ἐκκείσθωσαν αἱ AB , AX εὐθεῖαι, καὶ $>$ έστω μείζων ἡ AB , καὶ γεγράφθω ἐπ' αὐτῆς ἡμικύκλιον τὸ $ABΓ$, καὶ εἰς τὸ $ABΓ$ ἡμικύκλιον ἐνηρμόσθω τῇ AX εὐθείᾳ μὴ μείζονι οὕση τῆς AB διαμέτρου $>$ ίση ἡ $ΑΓ$, καὶ ἐπεξεύθω ἡ $ΓΒ$. ἐπεὶ οὖν ἐν ἡμικυκλίῳ τῷ $ΑΓΒ$ γωνία ἐστὶν ἡ ὑπὸ $ΑΓΒ$, ὀρθὴ $>$ άρα ἐστὶν ἡ ὑπὸ $ΑΓΒ$. τὸ $>$ άρα ἀπὸ τῆς AB $>$ ίσον ἐστὶ τοῖς ἀπὸ τῶν $ΑΓ$, $ΓΒ$. $<$ ώστε τὸ ἀπὸ τῆς AB τοῦ ἀπὸ τῆς $ΑΓ$ μείζον ἐστὶ τῷ ἀπὸ τῆς $ΓΒ$. $>$ ίση δὲ ἡ $ΑΓ$ τῇ AX . τὸ $>$ άρα ἀπὸ τῆς AB τοῦ ἀπὸ τῆς AX μείζον ἐστὶ τῷ

And we can demonstrate, thusly, in which manner to take the (square) on OR equal to that (area) by which the (square) on AB is greater than the (square) on LO . Let the straight-lines AB and LO be set out, and let AB be greater, and let the semicircle ABC be drawn around it. And let AC , equal to the straight-line LO , which is not greater than the diameter AB , be inserted into the semicircle ABC [Prop. 4.1]. And let CB be joined. Therefore, since the angle ACB is in the semicircle ACB , ACB is thus a right-angle [Prop. 3.31]. Thus, the (square) on AB is equal to the (sum of the) squares on AC and CB [Prop. 1.47]. Hence, the (square) on AB is

ἀπὸ τῆς ΓΒ. ἔαν οὖν τῇ ΒΓ >ίσην τὴν ΧΡ ἀπολάβωμεν, >έσται τὸ ἀπὸ τῆς ΑΒ τοῦ ἀπὸ τῆς ΛΧ μείζον τῷ ἀπὸ τῆς ΧΡ; <όπερ προέκειτο ποιῆσαι.

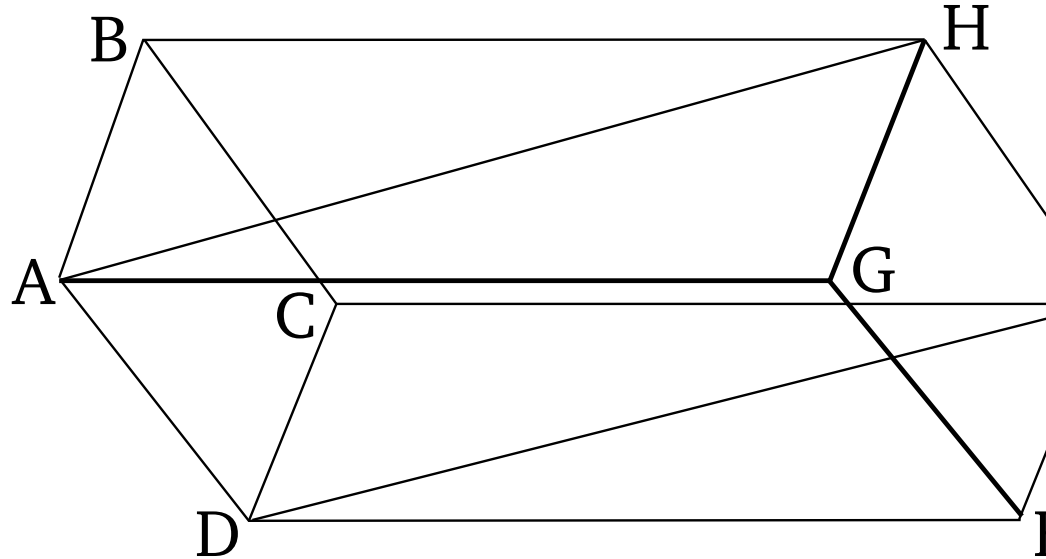
greater than the (square) on AC by the (square) on CB . And AC (is) equal to LO . Thus, the (square) on AB is greater than the (square) on LO by the (square) on CB . Therefore, if we take OR equal to BC then the (square) on AB will be greater than the (square) on LO by the (square) on OR . (Which is) the very thing it was prescribed to do.

24.

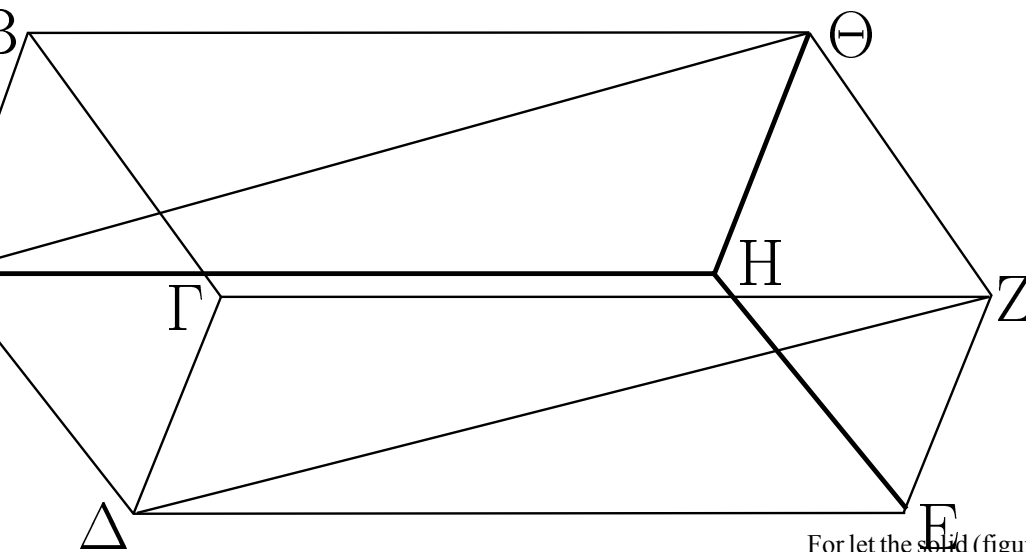
Proposition 24

Ἐάν στερεὸν ὑπὸ παραλλήλων ἐπιπέδων περιέθῃται, τὰ ἀπεναντίον αὐτοῦ ἐπίπεδα >ίσα τε καὶ παραλληλόγραμμά

If a solid (figure) is contained by (six) parallel planes then its opposite planes are both equal and parallelogrammic.



ἔστιν.



Στερεὸν γὰρ τὸ ΓΔΘΗ ὑπὸ παραλλήλων ἐπιπέδων περιέθῃται τῶν ΑΓ, ΗΖ, ΑΘ, ΔΖ, ΒΖ, ΑΕ; λέγω, <ὅτι τὰ ἀπεναντίον αὐτοῦ ἐπίπεδα >ίσα τε καὶ παραλληλόγραμμά ἐστιν.

Ἐπεὶ γὰρ δύο ἐπίπεδα παράλληλα τὰ ΒΗ, ΓΕ ὑπὸ ἐπιπέδου τοῦ ΑΓ τέμνεται, αἱ κοιναὶ αὐτῶν τομαὶ παράλληλοί εἰσιν. παράλληλος >άρα ἐστὶν ἡ ΑΒ τῇ ΔΓ. πάλιν, ἐπεὶ δύο ἐπίπεδα παράλληλα τὰ ΒΖ, ΑΕ ὑπὸ ἐπιπέδου τοῦ ΑΓ τέμνεται, αἱ κοιναὶ αὐτῶν τομαὶ παράλληλοί εἰσιν. παράλληλος >άρα ἐστὶν ἡ ΒΓ τῇ ΑΔ. ἐδείκθη δὲ καὶ ἡ ΑΒ τῇ ΔΓ παράλληλος; παραλληλόγραμμον >άρα ἐστὶ τὸ ΑΓ. ὁμοίως δὲ δείχομεν, <ὅτι καὶ <έκαστον τῶν ΔΖ, ΖΗ, ΗΒ, ΒΖ, ΑΕ παραλληλόγραμμον ἐστὶν.

Ἐπεξεύθωσαν αἱ ΑΘ, ΔΖ. καὶ ἐπεὶ παράλληλός ἐστιν ἡ μὲν ΑΒ τῇ ΔΓ, ἡ δὲ ΒΘ τῇ ΓΖ, δύο δὲ αἱ ΑΒ, ΒΘ ἀπτόμεναι ἀλλήλων παρὰ δύο εὐθείας τὰς ΔΓ, ΓΖ ἀπτόμεναις ἀλλήλων

For let the solid (figure) $CDHG$ be contained by the parallel planes AC , GF , and AH , DF , and BF , AE . I say that its opposite planes are both equal and parallelogrammic.

For since the two parallel planes BG and CE are cut by the plane AC , their common sections are parallel [Prop. 11.16]. Thus, AB is parallel to DC . Again, since the two parallel planes BF and AE are cut by the plane AC , their common sections are parallel [Prop. 11.16]. Thus, BC is parallel to AD . And AB was also shown (to be) parallel to DC . Thus, AC is a parallelogram. So, similarly, we can also show that DF , FG , GB , BF , and AE are each parallelograms.

Let AH and DF be joined. And since AB is parallel to DC , and BH to CF , so the two (straight-lines) joining one another, AB and BH , are parallel to the two straight-lines joining one another, DC and CF (respectively), not (being) in the same plane. Thus, they will contain equal an-

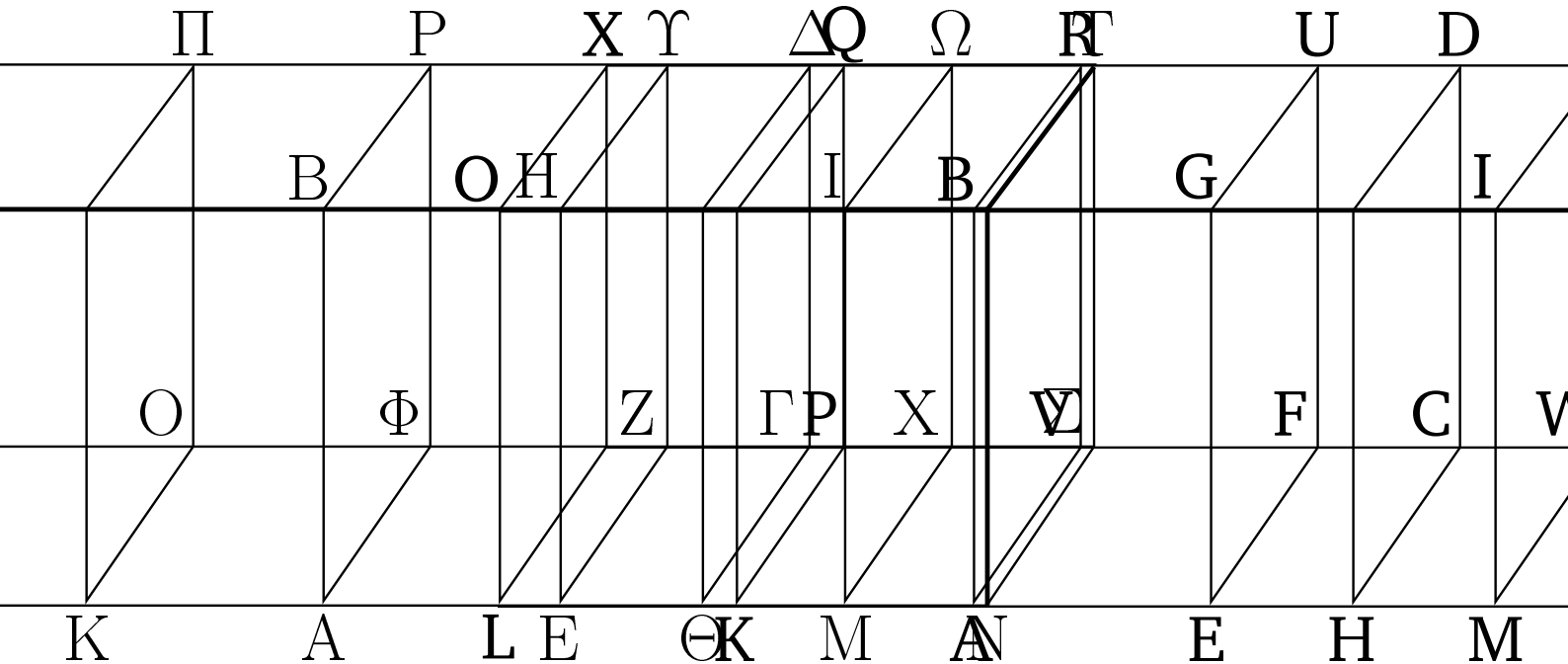
εἰσὶν οὐκ ἐν τῷ αὐτῷ ἐπιπέδῳ; >ίσαξ> ἄρα γωνίαξ περιέχουσιν
>ίση >ἄρα ἢ ὑπὸ $AB\Theta$ γωνία τῇ ὑπὸ $\Delta\Gamma Z$. καὶ ἐπεὶ δύο αἱ
 AB , $B\Theta$ δυσὶ ταῖς $\Delta\Gamma$, ΓZ >ίσαι εἰσὶν, καὶ γωνία ἢ ὑπὸ
 $AB\Theta$ γωνία τῇ ὑπὸ $\Delta\Gamma Z$ ἐστὶν >ίση, βάσιξ >ἄρα ἢ $A\Theta$
βάσει τῇ ΔZ ἐστὶν >ίση, καὶ τὸ $AB\Theta$ τρίγωνον τῷ $\Delta\Gamma Z$
τριγώνῳ >ίσον ἐστίν. καὶ ἐστὶ τοῦ μὲν $AB\Theta$ διπλάσιον
τὸ BH παραλληλόγραμμον, τοῦ δὲ $\Delta\Gamma Z$ διπλάσιον τὸ ΓE
παραλληλόγραμμον; >ίσον >ἄρα τὸ BH παραλληλόγραμμον
τῷ ΓE παραλληλογράμμῳ; ὁμοίωξ δὴ δείχομεν, <ὅτι καὶ τὸ
μὲν $A\Gamma$ τῷ HZ ἐστὶν >ίσον, τὸ δὲ AE τῷ BZ .
Ἐὰν >ἄρα στερεὸν ὑπὸ παραλλήλων ἐπιπέδων περιέθῃται,
τὰ ἀπεναντίον αὐτοῦ ἐπίπεδα >ίσα τε καὶ παραλληλόγραμμά
ἐστὶν; <ὅπερ >έδει δεῖχαι.

25.

Ἐὰν στερεὸν παραλληλεπίπεδον ἐπιπέδῳ τμηθῇ παραλλήλῳ
>όντι τοῖς ἀπεναντίον ἐπιπέδοις, >έσται ὡς ἡ βάσιξ πρὸς
τὴν βάσιν, οὕτωξ τὸ στερεὸν πρὸς τὸ στερεόν.

Proposition 25

If a parallelepiped solid is cut by a plane which is parallel to
the opposite planes (of the parallelepiped) then as the base
(is) to the base, so the solid will be to the solid.



Στερεὸν γὰρ παραλληλεπίπεδον τὸ $AB\Gamma\Delta$ ἐπιπέδῳ τῷ ZH
τετμήσθω παραλλήλῳ >όντι τοῖς ἀπεναντίον ἐπιπέδοις τοῖς
 PA , $\Delta\Theta$; λέγω, <ὅτι ἐστὶν ὡς ἡ $AEZF$ βάσιξ πρὸς τὴν $E\Theta\Gamma Z$
βάσιν, οὕτωξ τὸ $ABZY$ στερεὸν πρὸς τὸ $EH\Gamma\Delta$ στερεόν.
Ἐκβεβλήσθω γὰρ ἡ $A\Theta$ ἐκ> ἑκάτερα τὰ μέρη, καὶ κείσθωσαν
τῇ μὲν AE >ίσαι ὁσαυδηποτοῦν αἱ AK , KL , τῇ δὲ $E\Theta$ >ίσαι
ὁσαυδηποτοῦν αἱ ΘM , MN , καὶ συμπληρώσθω τὰ AO ,
 $K\Phi$, $\Theta\Theta$, $M\Xi$ παραλληλόγραμμα καὶ τὰ AP , KP , ΔM , MT
στερεά.
Καὶ ἐπεὶ >ίσαι εἰσὶν αἱ AK , KA , AE εὐθεῖαι ἀλλήλαιξ, >ίσα
ἐστὶ καὶ τὰ μὲν AO , $K\Phi$, AZ παραλληλόγραμμα ἀλλήλοισξ,
τὰ δὲ KX , KB , AH ἀλλήλοισξ καὶ >έτι τὰ $\Lambda\Psi$, $K\Pi$, AP
ἀλλήλοισξ; ἀπεναντίον γάρ. διὰ τὰ αὐτὰ δὴ καὶ τὰ μὲν
 $E\Gamma$, $\Theta\Theta$, $M\Xi$ παραλληλόγραμμα >ίσα εἰσὶν ἀλλήλοισξ, τὰ
δὲ ΘH , ΘI , IN >ίσα εἰσὶν ἀλλήλοισξ, καὶ >έτι τὰ $\Delta\Theta$, $M\Omega$,
 NT ; τρία >ἄρα ἐπίπεδα τῶν $\Lambda\Pi$, KP , AY στερεῶν τρισὶν
ἐπιπέδοις ἐστὶν >ίσα. ἀλλὰ τὰ τρία τρισὶ τοῖς ἀπεναντίον
ἐστὶν >ίσα; τὰ >ἄρα τρία στερεὰ τὰ $\Lambda\Pi$, KP , AY >ίσα
ἀλλήλοισξ ἐστὶν. διὰ τὰ αὐτὰ δὴ καὶ τὰ τρία στερεὰ τὰ
 $E\Delta$, ΔM , MT >ίσα ἀλλήλοισξ ἐστὶν; ὁσαπλασίον >ἄρα ἐστὶν

For let the parallelepiped solid $ABCD$ be cut by the plane
 FG which is parallel to the opposite planes RA and DH . I
say that as the base $AEFV$ (is) to the base $EHCF$, so the
solid $ABFU$ (is) to the solid $EGCD$.

For let AH be produced in each direction. And let any num-
ber whatsoever (of lengths), AK and KL , be made equal
to AE , and any number whatsoever (of lengths), HM and
 MN , equal to EH . And let the parallelograms LP , KV ,
 HW , and MS be completed, and the solids LQ , KR , DM ,
and MT .

And since the straight-lines LK , KA , and AE are equal to
one another, the parallelograms LP , KV , and AF are also
equal to one another, and KO , KB , and AG (are equal)
to one another, and, further, LX , KQ , and AR (are equal)
to one another. For (they are) opposite [Prop. 11.24]. So,
for the same (reasons), the parallelograms EC , HW , and
 MS are also equal to one another, and HG , HI , and IN
(are equal to one another). Thus, three planes of (one of)
the solids LQ , KR , and AU are equal to the (correspond-

ἡ AZ βάσις τῆς AZ βάσεως, τοσαυταπλάσιόν ἐστι καὶ τὸ AY στερεὸν τοῦ AY στερεοῦ. διὰ τὰ αὐτὰ δὴ ὅσαπλασίον ἐστὶν ἡ NZ βάσις τῆς $Z\Theta$ βάσεως, τοσαυταπλάσιόν ἐστι καὶ τὸ NY στερεὸν τοῦ ΘY στερεοῦ. καὶ εἰ $>$ ἴση ἐστὶν ἡ AZ βάσις τῇ NZ βάσει, $>$ ἴσον ἐστὶ καὶ τὸ AY στερεὸν τῷ NY στερεῷ, καὶ εἰ ὑπερέθει ἡ AZ βάσις τῆς NZ βάσεως, ὑπερέθει καὶ τὸ AY στερεὸν τοῦ NY στερεοῦ, καὶ εἰ ἐλλείπει, ἐλλείπει. τεσσάρων δὲ $>$ ὄντων μεγεθῶν, δύο μὲν βάσεων τῶν AZ , $Z\Theta$, δύο δὲ στερεῶν τῶν AY , $Y\Theta$, $\varepsilon >$ ἴληπται ἰσάκις πολλαπλάσια τῆς μὲν AZ βάσεως καὶ τοῦ AY στερεοῦ $<$ ἢ τε AZ βάσις καὶ τὸ AY στερεόν, τῆς δὲ ΘZ βάσεως καὶ τοῦ ΘY στερεοῦ $<$ ἢ τε NZ βάσις καὶ τὸ NY στερεόν, καὶ δέδεικται, $<$ ὅτι εἰ ὑπερέθει ἡ AZ βάσις τῆς ZN βάσεως, ὑπερέθει καὶ τὸ AY στερεὸν τοῦ NY [στερεοῦ], καὶ εἰ $>$ ἴση, $>$ ἴσον, καὶ εἰ ἐλλείπει, ἐλλείπει. $>$ ἐστὶν $>$ ἄρα ὥς ἡ AZ βάσις πρὸς τὴν $Z\Theta$ βάσιν, $<$ ὅτωξ τὸ AY στερεὸν πρὸς τὸ $Y\Theta$ στερεόν; $<$ ὅπερ $>$ ἐδει δεῖχαι.

ing) three planes (of the others). But, the three planes (in one of the solids) are equal to the three opposite planes [Prop. 11.24]. Thus, the three solids LQ , KR , and AU are equal to one another [Def. 11.10]. So, for the same (reasons), the three solids ED , DM , and MT are also equal to one another. Thus, as many multiples as the base LF is of the base AF , so many multiples is the solid LU also of the solid AU . So, for the same (reasons), as many multiples as the base NF is of the base FH , so many multiples is the solid NU also of the solid HU . And if the base LF is equal to the base NF then the solid LU is also equal to the solid NU .[†] And if the base LF exceeds the base NF then the solid LU also exceeds the solid NU . And if (LF) is less than (NF) then (LU) is (also) less than (NU) . So, there are four magnitudes, the two bases AF and FH , and the two solids AU and UH , and equal multiples be taken of the base AF and the solid AU —(namely), the base LF and the solid LU —and of the base FH and the solid HU —(namely), the base NF and the solid NU . And it has been shown that if the base LF exceeds the base NF then the solid LU also exceeds the [solid] NU , and if (LF) is equal (to NF) then (LU) is equal (to NU), and if (LF) is less than (NF) then (LU) is less than (NU) . Thus, as the base AF is to the base FH , so the solid AU (is) to the solid UH [Def. 5.5]. (Which is) the very thing it was required to show.

[†] Here, Euclid assumes that $LF \cong NF$ implies $LU \cong NU$. This is easily demonstrated.

26.

Πρὸς τῇ δοθείσῃ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῇ δοθείσῃ στερεᾷ γωνία $>$ ἴσην στερεὰν γωνίαν συστήσασθαι. $>$ Ἐστω ἡ μὲν δοθεῖσα εὐθεῖα ἡ AB , τὸ δὲ πρὸς αὐτῇ δοθὲν σημεῖον τὸ A , ἡ δὲ δοθεῖσα στερεὰ γωνία ἡ πρὸς τῷ Δ περιεθρομένη ὑπὸ τῶν ὑπὸ $E\Delta\Gamma$, $E\Delta Z$, $Z\Delta\Gamma$ γωνιῶν ἐπιπέδων; δεῖ δὴ πρὸς τῇ AB εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ A τῇ πρὸς τῷ Δ στερεᾷ γωνία $>$ ἴσην στερεὰν γωνίαν συστήσασθαι.

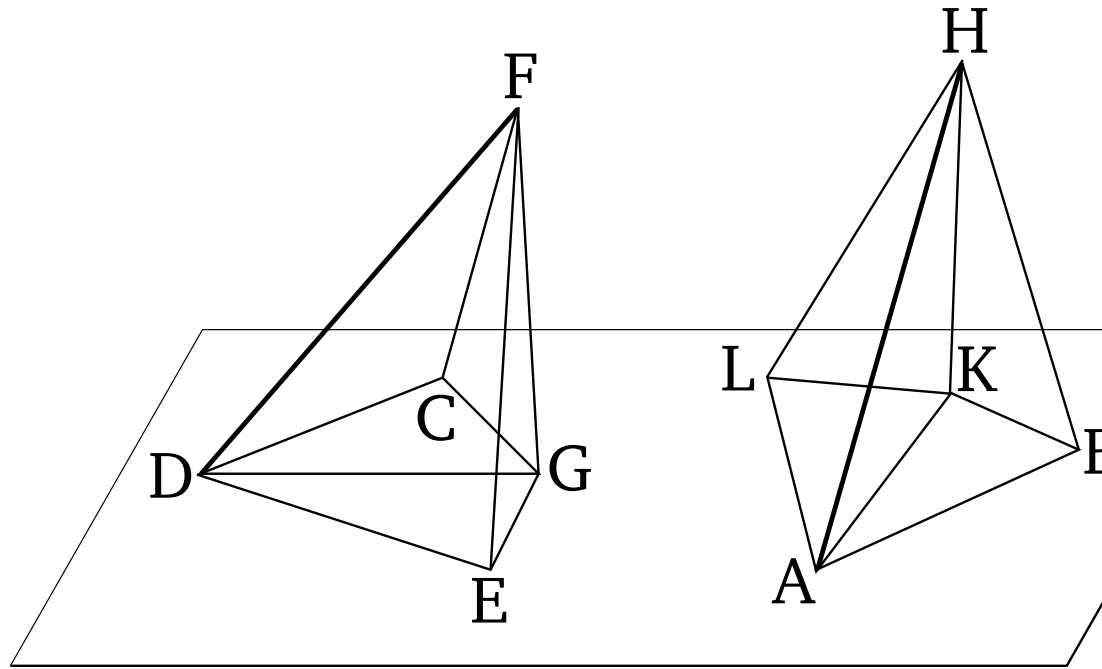
Εἰλήφθω γὰρ ἐπὶ τῆς ΔZ τυθὸν σημεῖον τὸ Z , καὶ $>$ ἴθθω ἀπὸ τοῦ Z ἐπὶ τὸ διὰ τῶν $E\Delta$, $\Delta\Gamma$ ἐπίπεδον κάθετοξ ἡ ZH , καὶ συμβαλλέτω τῷ ἐπιπέδῳ κατὰ τὸ H , καὶ ἐπεξεύθθω ἡ ΔH , καὶ συνεστάτω πρὸς τῇ AB εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ A τῇ μὲν ὑπὸ $E\Delta\Gamma$ γωνία $>$ ἴση ἡ ὑπὸ BAL , τῇ δὲ ὑπὸ $E\Delta H$ $>$ ἴση ἡ ὑπὸ BAK , καὶ κείσθω τῇ ΔH $>$ ἴση ἡ AK , καὶ ἀνεστάτω ἀπὸ τοῦ K σημείου τῷ διὰ τῶν BAL ἐπιπέδῳ πρὸς ὀρθὰξ ἡ $K\Theta$, καὶ κείσθω $>$ ἴση τῇ HZ ἡ $K\Theta$, καὶ ἐπεξεύθθω ἡ ΘA ; λέγω, $<$ ὅτι ἡ πρὸς τῷ A στερεὰ γωνία περιεθρομένη ὑπὸ τῶν BAL , $BA\Theta$, ΘAL γωνιῶν $>$ ἴση ἐστὶ τῇ πρὸς τῷ Δ στερεᾷ γωνία τῇ περιεθρομένη ὑπὸ τῶν $E\Delta\Gamma$,

Proposition 26

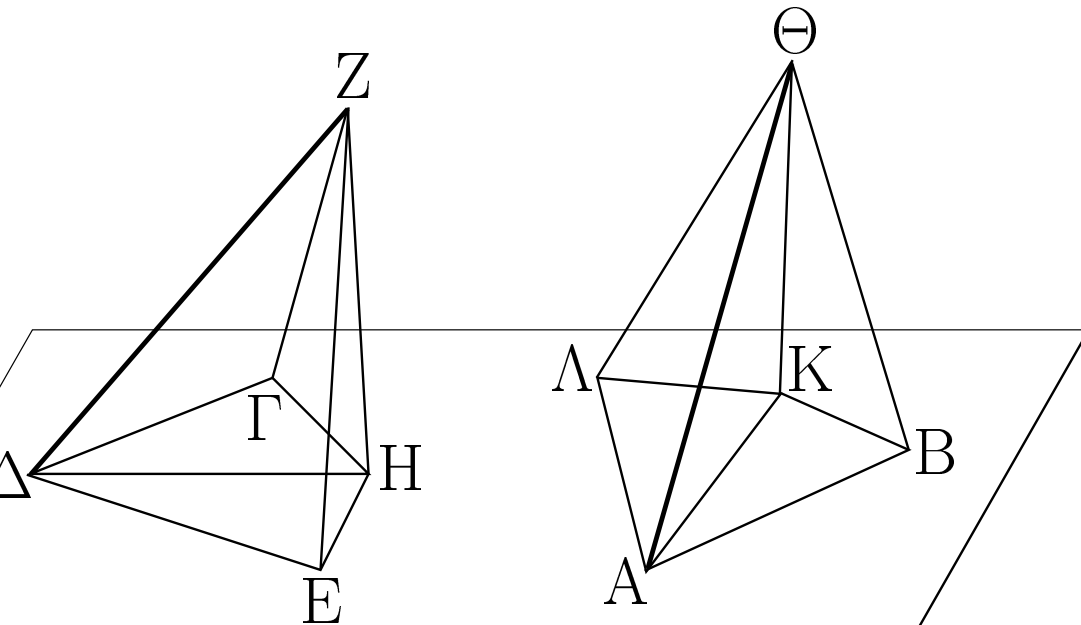
To construct a solid angle equal to a given solid angle on a given straight-line, and at a given point on it.

Let AB be the given straight-line, and A the given point on it, and D the given solid angle, contained by the plane angles EDC , EDF , and FDC . So, it is necessary to construct a solid angle equal to the solid angle D on the straight-line AB , and at the point A on it.

For let some random point F be taken on DF , and let FG be drawn from F perpendicular to the plane through ED and DC [Prop. 11.11], and let it meet the plane at G , and let DG be joined. And let BAL , equal to the angle EDC , and BAK , equal to EDG , be constructed on the straight-line AB at the point A on it [Prop. 1.23]. And let AK be made equal to DG . And let KH be set up at the point K at right-angles to the plane through BAL [Prop. 11.12]. And let KH be made equal to GF . And let HA be joined. I say that the solid angle at A , contained by the (plane) angles BAL , BAH , and HAL , is equal to the solid angle at D , contained by the (plane) angles EDC , EDF , and FDC .



ΕΔΖ, ΖΔΓ γωνιών.



For let AB and DE be cut off (so as to be) equal, and let HB , KB , FE , and GE be joined. And since FG is at right-angles to the reference plane (EDC), it will also make right-angles with all of the straight-lines joined to it which are also in the reference plane [Def. 11.3]. Thus, the angles FGD and FGE are right-angles. So, for the same (reasons), the angles HKA and HKB are also right-angles. And since the two (straight-lines) KA and AB are equal to the two (straight-lines) GD and DE , respectively, and they contain equal angles, the base KB is thus equal to the base GE [Prop. 1.4]. And KH is also equal to GF . And they contain right-angles (with the respective bases). Thus, HB (is) also equal to FE [Prop. 1.4]. Again, since the two (straight-lines) AK and KH are equal to the two (straight-lines) DG and GF (respectively), and they contain right-angles, the base AH is thus equal to the base FD [Prop. 1.4]. And AB (is) also equal to DE . So, the two (straight-lines) HA and AB are equal to the two (straight-lines) DF and DE (respectively). And the base HB (is)

Ἀπειλήφθωσαν γὰρ $>$ ίσαι αἱ AB , DE , καὶ ἐπεζεύθωσαν αἱ $ΘB$, KB , ZE , HE . καὶ ἐπεὶ ἡ ZH ὀρθή ἐστι πρὸς τὸ ὑποκείμενον ἐπίπεδον, καὶ πρὸς πάσαζ $>$ άρα τὰς ἀπτομένας αὐτῆς εὐθείαζ καὶ ο $>$ ύσαζ ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ὀρθὰς ποιήσει γωνίαζ; ὀρθή $>$ άρα ἐστὶν ἑκάτερα τῶν ὑπὸ ZHA , ZHE γωνιῶν. διὰ τὰ αὐτὰ δὴ καὶ ἑκάτερα τῶν ὑπὸ $ΘKA$, $ΘKB$ γωνιῶν ὀρθή ἐστιν. καὶ ἐπεὶ δύο αἱ KA , AB δύο ταῖς HA , DE $>$ ίσαι εἰσὶν ἑκάτερα ἑκατέρᾳ, καὶ γωνίαζ $>$ ίσαζ περιέθουσιν, βάσιζ $>$ άρα ἡ KB βάσει τῇ HE $>$ ιση ἐστίν. $>$ έστι δὲ καὶ ἡ $KΘ$ τῇ HZ $>$ ιση; καὶ γωνίαζ ὀρθὰς περιέθουσιν; $>$ ιση $>$ άρα καὶ ἡ $ΘB$ τῇ ZE . πάλιν ἐπεὶ δύο αἱ AK , $KΘ$ δυοὶ ταῖς $ΔH$, HZ $>$ ίσαι εἰσὶν, καὶ γωνίαζ ὀρθὰς περιέθουσιν, βάσιζ $>$ άρα ἡ $AΘ$ βάσει τῇ $ΖΔ$ $>$ ιση ἐστίν. $>$ έστι δὲ καὶ ἡ AB τῇ DE $>$ ιση; δύο δὴ αἱ $ΘA$, AB δύο ταῖς $ΔZ$, DE $>$ ίσαι εἰσὶν. καὶ βάσιζ ἡ $ΘB$ βάσει τῇ ZE $>$ ιση; γωνία $>$ άρα ἡ ὑπὸ $BAΘ$ γωνία τῇ ὑπὸ $EΔZ$ ἐστὶν $>$ ιση. διὰ τὰ αὐτὰ δὴ καὶ ἡ ὑπὸ $ΘAΔ$ τῇ ὑπὸ $ΖΔΓ$ ἐστὶν $>$ ιση. $>$ έστι δὲ καὶ ἡ ὑπὸ BAA τῇ ὑπὸ $EΔΓ$ $>$ ιση.

Πρὸς ὅρα τῇ δοθείσῃ εὐθείᾳ τῇ AB καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ A τῇ δοθείσῃ στερεᾷ γωνίᾳ τῇ πρὸς τῷ Δ ὅση συνέσταται; ὅπερ ἔδει ποιῆσαι.

equal to the base FE . Thus, the angle BAH is equal to the angle EDF [Prop. 1.8]. So, for the same (reasons), HAL is also equal to FDC . And BAL is also equal to EDC .

Thus, (a solid angle) has been constructed, equal to the given solid angle at D , on the given straight-line AB , at the given point A on it. (Which is) the very thing it was required to do.

27.

Proposition 27

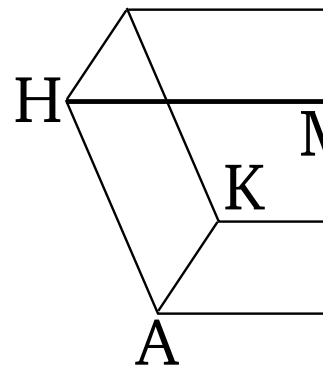
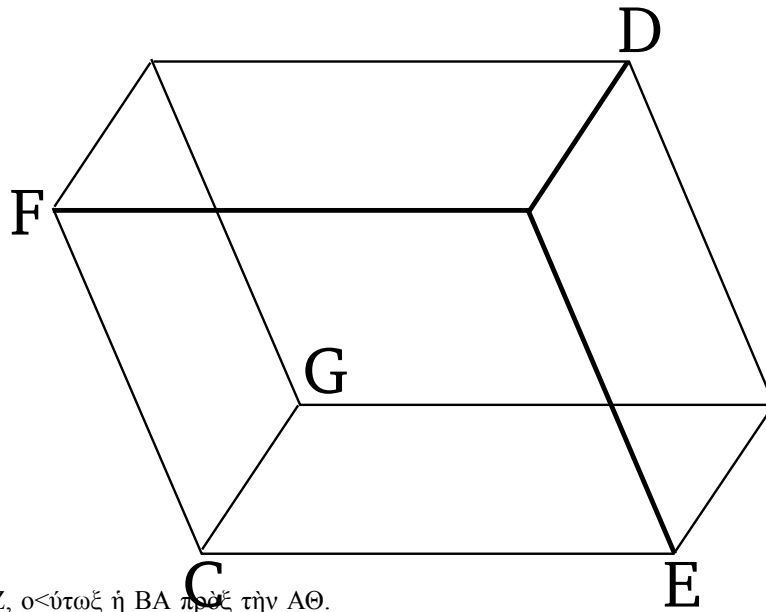
Ἀπὸ τῆς δοθείσης εὐθείας τῷ δοθέντι στερεῷ παραλληλεπίπεδον ὁμοίων τε καὶ ὁμοίως κείμενον στερεὸν παραλληλεπίπεδον ἀναγράψαι.

>Ἐστω ἡ μὲν δοθεῖσα εὐθεῖα ἡ AB , τὸ δὲ δοθὲν στερεὸν παραλληλεπίπεδον τὸ $\Gamma\Delta$; δεῖ δὲ ἀπὸ τῆς δοθείσης εὐθείας τῆς AB τῷ δοθέντι στερεῷ παραλληλεπίπεδῳ τῷ $\Gamma\Delta$ ὁμοίων τε καὶ ὁμοίως κείμενον στερεὸν παραλληλεπίπεδον ἀναγράψαι.

Συνεστάτω γὰρ πρὸς τῇ AB εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ A τῇ πρὸς τῷ Γ στερεᾷ γωνίᾳ ὅση ἡ περιεθρομένη ὑπὸ τῶν $BA\Theta$, ΘAK , KAB , ὥστε ὅσην εἶναι τὴν μὲν ὑπὸ $BA\Theta$ γωνίαν τῇ ὑπὸ $ΕΓΖ$, τὴν δὲ ὑπὸ BAK τῇ ὑπὸ $ΕΓΗ$, τὴν δὲ ὑπὸ $KA\Theta$ τῇ ὑπὸ $ΗΓΖ$; καὶ γεγόνετω ὥς μὲν ἡ $ΕΓ$ πρὸς τὴν $ΓΗ$, οὕτως ἡ BA πρὸς τὴν AK , ὥς δὲ ἡ $ΗΓ$ πρὸς τὴν $ΓΖ$, οὕτως ἡ KA πρὸς τὴν $A\Theta$. καὶ δι' ὅσου ὅρα

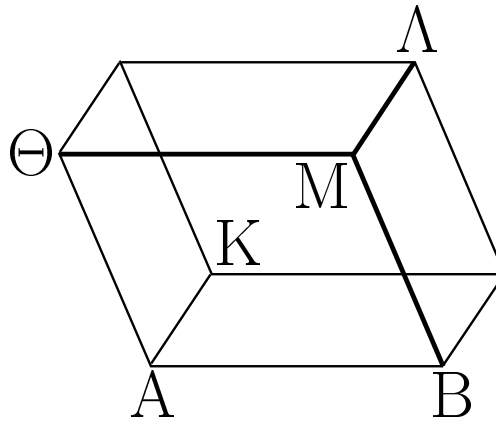
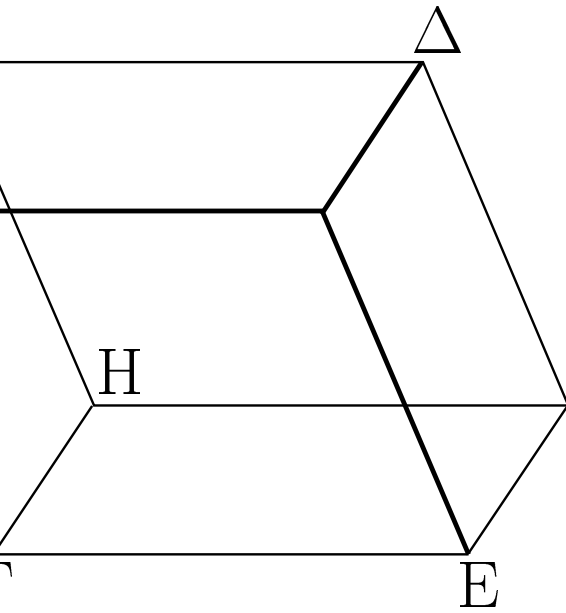
To describe a parallelepiped solid similar, and similarly laid out, to a given parallelepiped solid on a given straight-line. Let the given straight-line be AB , and the given parallelepiped solid CD . So, it is necessary to describe a parallelepiped solid similar, and similarly laid out, to the given parallelepiped solid CD on the given straight-line AB .

For, let a (solid angle) contained by the (plane angles) BAH , HAK , and KAB be constructed, equal to solid angle at C , on the straight-line AB at the point A on it [Prop. 11.26], such that angle BAH is equal to ECF , and BAK to ECG , and KAH to GCF . And let it be contrived that as EC (is) to CG , so BA (is) to AK , and as GC (is) to CF , so KA (is) to AH [Prop. 6.12]. And thus, via equality, as EC is to CF , so BA (is) to AH [Prop. 5.22]. And let the parallelogram HB be completed, and the solid AL .



ἔστιν ὥς ἡ $ΕΓ$ πρὸς τὴν $ΓΖ$, οὕτως ἡ BA πρὸς τὴν $A\Theta$. καὶ συμπληρώσω τὸ ΘB παραλληλόγραμμον καὶ τὸ AL στερεόν.

And since as EC is to CG , so BA (is) to AK , and the sides about the equal angles ECG and BAK are (thus) propor-



Καὶ ἐπεὶ ἐστὶν ὡς ἡ ΕΓ πρὸς τὴν ΓΗ, οὕτως ἡ ΒΑ πρὸς τὴν ΑΚ, καὶ περὶ ἴσαξ γωνίαξ τὰξ ὑπὸ ΕΓΗ, ΒΑΚ αἱ πλευραὶ ἀνάλογόν εἰσιν, ὁμοίων ἄρα ἐστὶ τὸ ΗΕ παραλληλόγραμμον τῷ ΚΒ παραλληλογράμμῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ μὲν ΚΘ παραλληλόγραμμον τῷ ΗΖ παραλληλόγραμμῳ, καὶ τὸ ΖΕ τῷ ΘΒ; τρία ἄρα παραλληλόγραμμά ἐστιν ὅμοιά ἐστιν. ἀλλὰ τὰ μὲν τρία τρισὶ τοῖξ ἀπεναντίον ἴσα τέ ἐστὶ καὶ ὁμοία, τὰ δὲ τρία τρισὶ τοῖξ ἀπεναντίον ἴσα τέ ἐστὶ καὶ ὁμοία; ὅλον ἄρα τὸ ΓΔ στερεὸν ὅλῳ τῷ ΑΛ στερεῷ ὁμοίων ἐστὶν. Ἀπὸ τῆξ δοθείσης ἄρα εὐθείας τῆξ ΑΒ τῷ δοθέντι στερεῷ παραλληλεπίεδῳ τῷ ΓΔ ὁμοίων τε καὶ ὁμοίωξ κείμενον ἀναγράφεται τὸ ΑΛ; ὅπερ >έδει ποιῆσαι.

tional, the parallelogram GE is thus similar to the parallelogram KB . So, for the same (reasons), the parallelogram KH is also similar to the parallelogram GF , and, further, FE (is similar) to HB . Thus, three of the parallelograms of solid CD are similar to three of the parallelograms of solid AL . But, the (former) three are equal and similar to the three opposite, and the (latter) three are equal and similar to the three opposite. Thus, the whole solid CD is similar to the whole solid AL [Def. 11.9].

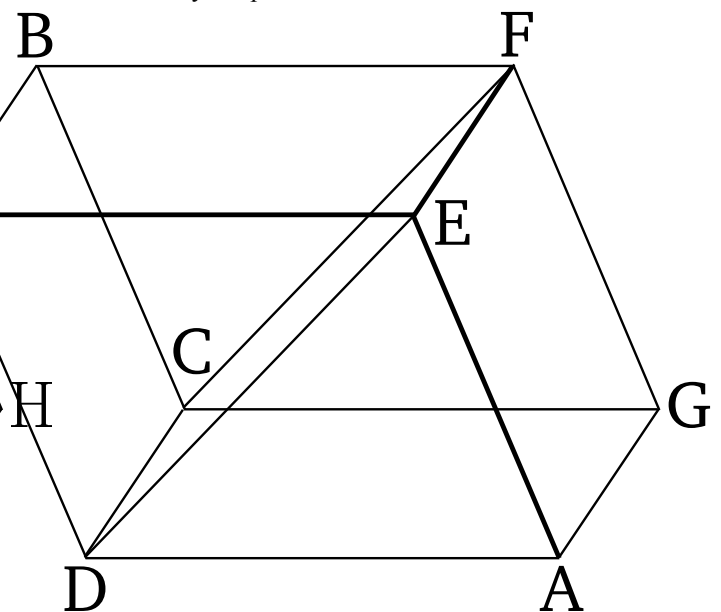
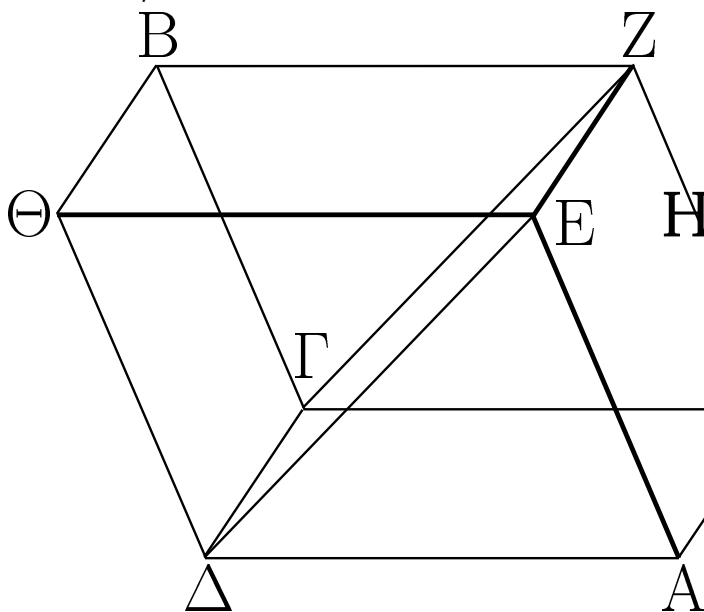
Thus, AL , similar, and similarly laid out, to the given parallelepiped solid CD , has been described on the given straight-lines AB . (Which is) the very thing it was required to do.

28.

Ἐὰν στερεὸν παραλληλεπίπεδον ἐπιπέδῳ τμηθῇ κατὰ τὰξ διαγωνίουξ τῶν ἀπεναντίον ἐπιπέδων, δίθα τμηθήσεται τὸ στερεὸν ὑπὸ τοῦ ἐπιπέδου.

Proposition 28

If a parallelepiped solid is cut by a plane (passing) through the diagonals of (a pair of) opposite planes then the solid will be cut in half by the plane.



Στερεὸν γὰρ παραλληλεπίπεδον τὸ ΑΒ ἐπιπέδῳ τῷ ΓΔΕΖ τετμήσθω κατὰ τὰξ διαγωνίουξ τῶν ἀπεναντίον ἐπιπέδων τὰξ ΓΖ, ΔΕ; λέγω, ὅτι δίθα τμηθήσεται τὸ ΑΒ στερεὸν ὑπὸ

For let the parallelepiped solid AB be cut by the plane $CDEF$ (passing) through the diagonals of the opposite planes CF and DE .[†] I say that the solid AB will be cut

τοῦ ΓΔΕΖ ἐπιπέδου.

Ἐπεὶ γὰρ ὁ ἴσος ἐστὶ τὸ μὲν ΓΗΖ τρίγωνον τῷ ΓΖΒ τριγώνῳ, τὸ δὲ ΑΔΕ τῷ ΔΕΘ, ὁ ἴσος δὲ καὶ τὸ μὲν ΓΑ παραλληλόγραμμον τῷ ΕΒ ὁ ἴσος; ἀπεναντίον γάρ; τὸ δὲ ΗΕ τῷ ΓΘ, καὶ τὸ πρίσμα ὅρα τὸ περιεχόμενον ὑπὸ δύο μὲν τριγώνων τῶν ΓΗΖ, ΑΔΕ, τριῶν δὲ παραλληλογράμμων τῶν ΗΕ, ΑΓ, ΓΕ ὁ ἴσος ἐστὶ τῷ πρίσματι τῷ περιεχομένῳ ὑπὸ δύο μὲν τριγώνων τῶν ΓΖΒ, ΔΕΘ, τριῶν δὲ παραλληλογράμμων τῶν ΓΘ, ΒΕ, ΓΕ; ὑπὸ γὰρ ὁ ἴσος ἐπιπέδων περιέθονται τῷ τε πλήθει καὶ τῷ μεγέθει. ὥστε ὅλον τὸ ΑΒ στερεὸν διῆται τέμνεται ὑπὸ τοῦ ΓΔΕΖ ἐπιπέδου; ὅπερ ἔδει δεῖξαι.

in half by the plane $CDEF$.

For since triangle CGF is equal to triangle CFB , and ADE (is equal) to DEH [Prop. 1.34], and parallelogram CA is also equal to EB —for (they are) opposite [Prop. 11.24]—and GE (equal) to CH , thus the prism contained by the two triangles CGF and ADE , and the three parallelograms GE , AC , and CE , is also equal to the prism contained by the two triangles CFB and DEH , and the three parallelograms CH , BE , and CE . For they are contained by planes (which are) equal in number and in magnitude [Def. 11.10].[†] Thus, the whole of solid AB is cut in half by the plane $CDEF$. (Which is) the very thing it was required to show.

[†] Here, it is assumed that the two diagonals lie in the same plane. The proof is easily supplied.

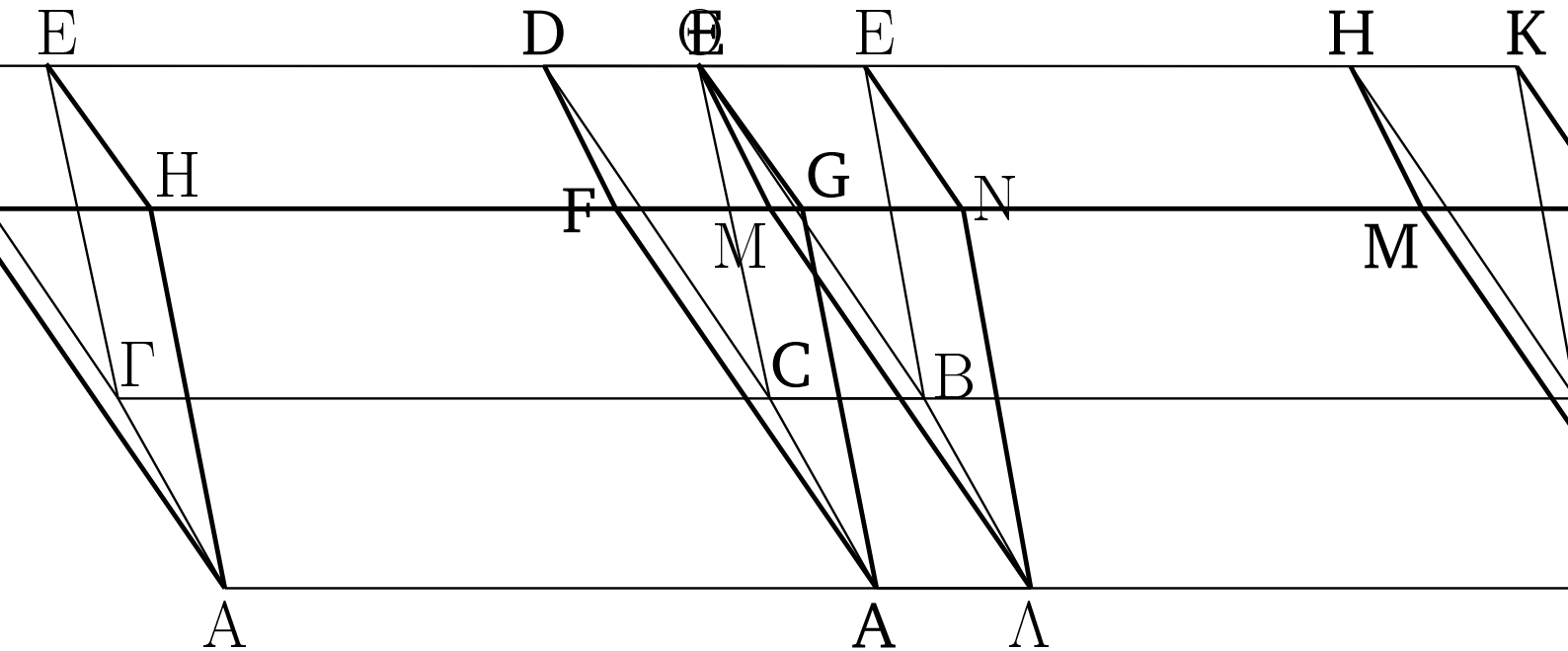
[‡] However, strictly speaking, the prisms are not similarly arranged, being mirror images of one another.

29.

Proposition 29

Τὰ ἐπὶ τῇ αὐτῇ βάσει ὄντα στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι ἐπὶ τῶν αὐτῶν εἰσιν εὐθειῶν, ἴσα ἀλλήλοις ἐστίν.

Parallelepiped solids which are on the same base, and (have) the same height, and in which the (ends of the straight-lines) standing up are on the same straight-lines, are equal to one another.



Ἐστω ἐπὶ τῇ αὐτῇ βάσει τῇ ΑΒ στερεὰ παραλληλεπίπεδα τὰ ΓΜ, ΓΝ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι αἱ ΑΗ, ΑΖ, ΛΜ, ΛΝ, ΓΔ, ΓΕ, ΒΘ, ΒΚ ἐπὶ τῶν αὐτῶν εὐθειῶν ὥστε τῶν ΖΝ, ΔΚ; λέγω, ὅτι ὁ ἴσος ἐστὶ τὸ ΓΜ στερεὸν τῷ ΓΝ στερεῷ.

Ἐπεὶ γὰρ παραλληλόγραμμον ἐστὶν ἑκάτερον τῶν ΓΘ, ΓΚ, ἴση ἐστὶν ἡ ΓΒ ἑκατέρᾳ τῶν ΔΘ, ΕΚ; ὥστε καὶ ἡ ΔΘ τῇ ΕΚ ἐστὶν ἴση. κοινὴ ἀφηρήσθω ἡ ΕΘ; λοιπὴ ὅρα ἡ ΔΕ λοιπῇ τῇ ΘΚ ἐστὶν ἴση. ὥστε καὶ τὸ μὲν ΔΓΕ τρίγωνον τῷ ΘΒΚ τριγώνῳ ὁ ἴσος ἐστίν, τὸ δὲ ΔΗ παραλληλόγραμμον τῷ ΘΝ παραλληλογράμμῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ ΑΖΗ τρίγωνον τῷ ΜΑΝ τριγώνῳ ὁ ἴσος ἐστίν. ὥστε δὲ καὶ τὸ μὲν ΓΖ παραλληλόγραμμον τῷ ΒΜ παραλληλογράμμῳ ὁ ἴσος, τὸ δὲ ΓΗ τῷ ΒΝ; ἀπεναντίον γάρ; καὶ τὸ πρίσμα ὅρα τὸ περιεχόμενον ὑπὸ δύο μὲν τριγώνων τῶν ΑΖΗ, ΔΓΕ, τριῶν δὲ παραλληλογράμμων τῶν ΑΔ, ΔΗ, ΓΗ ὁ ἴσος ἐστὶ τῷ πρίσματι τῷ περιεχομένῳ ὑπὸ δύο μὲν τριγώνων τῶν ΜΑΝ, ΘΒΚ, τριῶν δὲ παραλληλογράμμων τῶν ΒΜ, ΘΝ, ΒΝ. κοινὸν προσκείσθω τὸ στερεόν, ὅ

For let the parallelepiped solids CM and CN be on the same base AB , and (have) the same height, and let the (ends of the straight-lines) standing up in them, AG , AF , LM , LN , CD , CE , BH , and BK , be on the same straight-lines, FN and DK . I say that solid CM is equal to solid CN .

For since CH and CK are each parallelograms, CB is equal to each of DH and EK [Prop. 1.34]. Hence, DH is also equal to EK . Let EH be subtracted from both. Thus, the remainder DE is equal to the remainder HK . Hence, triangle DCE is also equal to triangle HBK [Props. 1.4, 1.8], and parallelogram DG to parallelogram HN [Prop. 1.36]. So, for the same (reasons), triangle AFG is also equal to triangle MLN . And parallelogram CF is also equal to parallelogram BM , and CG to BN [Prop. 11.24]. For they are opposite. Thus, the prism contained by the two triangles AFG and DCE , and the three parallelograms AD , DG , and CG , is equal to the prism contained by the two triangles MLN and HBK , and the three parallelograms BM , HN , and BN . Let the solid whose base (is) parallelogram

βάσις μὲν τὸ AB παραλληλόγραμμον, ἀπεναντίον δὲ τὸ $HEOM$; <ὅλον> ἄρα τὸ GM στερεὸν παραλληλεπίπεδον <ὅλῳ τῷ GN στερεῷ παραλληλεπίπεδῳ> ἴσον ἐστίν.

Τὰ > ἄρα ἐπὶ τῇ αὐτῇ βάσει > ὄντα στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ <ὑψοξ, <ὧν αἱ ἐφεστῶσαι ἐπὶ τῶν αὐτῶν εἰσιν εὐθειῶν, > ἴσα ἀλλήλοις ἐστίν; <ὅπερ> δεῖ δεῖξαι.

AB , and (whose) opposite (face is) $GEHM$, be added to both (prisms). Thus, the whole parallelepiped solid CM is equal to the whole parallelepiped solid CN .

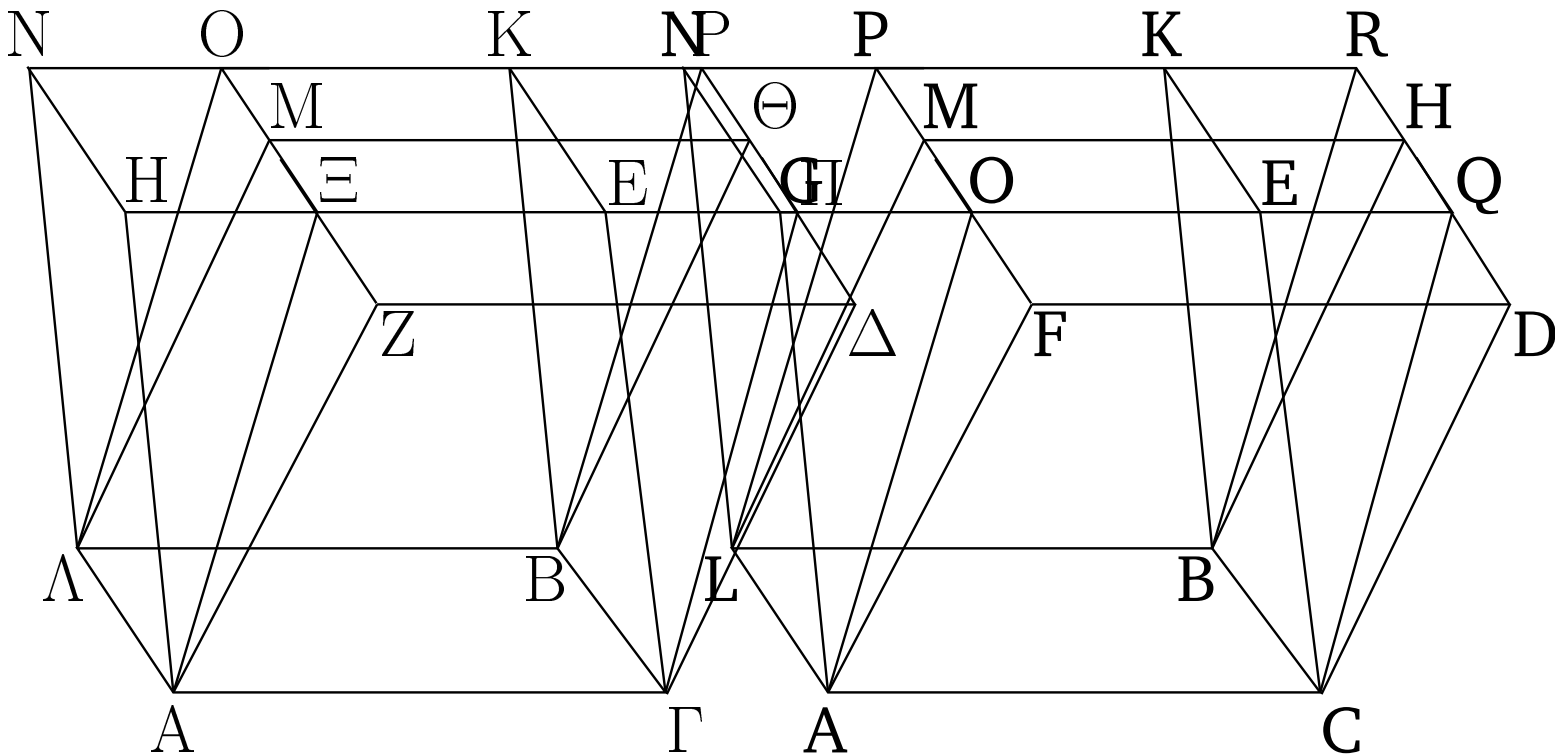
Thus, parallelepiped solids which are on the same base, and (have) the same height, and in which the (ends of the straight-lines) standing up (are) on the same straight-lines, are equal to one another. (Which is) the very thing it was required to show.

30.

Τὰ ἐπὶ τῇ αὐτῇ βάσει > ὄντα στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ <ὑψοξ, <ὧν αἱ ἐφεστῶσαι οὐκ εἰσιν ἐπὶ τῶν αὐτῶν εὐθειῶν, > ἴσα ἀλλήλοις ἐστίν.

Proposition 30

Parallelepiped solids which are on the same base, and (have) the same height, and in which the (ends of the straight-lines) standing up are not on the same straight-lines, are equal to one another.



> Ἐστω ἐπὶ τῇ αὐτῇ βάσει τῇ AB στερεὰ παραλληλεπίπεδα τὰ GM , GN ὑπὸ τὸ αὐτὸ <ὑψοξ, <ὧν αἱ ἐφεστῶσαι αἱ AZ , AH , AM , AN , $ΓΔ$, $ΓΕ$, $ΒΘ$, $ΒΚ$ μὴ > ἐστῶσαν ἐπὶ τῶν αὐτῶν εὐθειῶν; λέγω, <ὅτι> ἴσον ἐστὶ τὸ GM στερεὸν τῷ GN στερεῷ.

Ἐκβεβλήσθωσαν γὰρ αἱ NK , $ΔΘ$ καὶ συμπιπτέτωσαν ἀλλήλαις κατὰ τὸ P , καὶ > ἐτι ἐκβεβλήσθωσαν αἱ ZM , HE ἐπὶ τὰ O , $Π$, καὶ ἐπεξεύθθωσαν αἱ AX , $ΛΟ$, $ΓΠ$, BP . > ἴσον δὴ ἐστὶ τὸ GM στερεόν, ο<ὅ βάσις μὲν τὸ $ΑΓΒΛ$ παραλληλόγραμμον, ἀπεναντίον δὲ τὸ $ΖΔΘΜ$, τῷ $ΓΟ$ στερεῷ, ο<ὅ βάσις μὲν τὸ $ΑΓΒΛ$ παραλληλόγραμμον, ἀπεναντίον δὲ τὸ $ΧΠΡΟ$; ἐπὶ τε γὰρ τῇ αὐτῇ βάσει εἰσι τῇ $ΑΓΒΛ$ καὶ ὑπὸ τὸ αὐτὸ <ὑψοξ, <ὧν αἱ ἐφεστῶσαι αἱ AZ , AX , AM , $ΛΟ$, $ΓΔ$, $ΓΠ$, $ΒΘ$, BP ἐπὶ τῶν αὐτῶν εἰσιν εὐθειῶν τῶν ZO , $ΔΡ$. ἀλλὰ τὸ $ΓΟ$ στερεόν, ο<ὅ βάσις μὲν ἐστὶ τὸ $ΑΓΒΛ$ παραλληλόγραμμον, ἀπεναντίον δὲ τὸ $ΧΠΡΟ$, > ἴσον ἐστὶ τῷ GN στερεῷ, ο<ὅ βάσις μὲν τὸ $ΑΓΒΛ$ παραλληλόγραμμον, ἀπεναντίον δὲ τὸ $HEKN$; ἐπὶ τε γὰρ πάλιν τῇ αὐτῇ βάσει εἰσι τῇ $ΑΓΒΛ$ καὶ ὑπὸ τὸ αὐτὸ <ὑψοξ, <ὧν αἱ ἐφεστῶσαι αἱ AH , AX , $ΓΕ$, $ΓΠ$, AN , $ΛΟ$, $ΒΚ$, BP ἐπὶ τῶν αὐτῶν εἰσιν εὐθειῶν τῶν $ΗΠ$, NP . <ὥστε καὶ τὸ GM στερεὸν > ἴσον ἐστὶ τῷ GN στερεῷ. Τὰ > ἄρα ἐπὶ τῇ αὐτῇ βάσει στερεὰ παραλληλεπίπεδα

Let the parallelepiped solids CM and CN be on the same base, AB , and (have) the same height, and let the (ends of the straight-lines) standing up in them, AF , AG , LM , LN , CD , CE , BH , and BK , not be on the same straight-lines. I say that the solid CM is equal to the solid CN .

For let NK and DH be produced, and let them have joined one another at R . And, further, let FM and GE be produced to P and Q (respectively). And let AO , LP , CQ , and BR be joined. So, solid CM , whose base (is) parallelogram $ACBL$, and opposite (face) $FDHM$, is equal to solid CP , whose base (is) parallelogram $ACBL$, and opposite (face) $OQRP$. For they are on the same base, $ACBL$, and (have) the same height, and the (ends of the straight-lines) standing up in them, AF , AO , LM , LP , CD , CQ , BH , and BR , are on the same straight-lines, FP and DR [Prop. 11.29]. But, solid CP , whose base is parallelogram $ACBL$, and opposite (face) $OQRP$, is equal to solid CN , whose base (is) parallelogram $ACBL$, and opposite (face) $GEKN$. For, again, they are on the same base, $ACBL$, and (have) the same height, and the (ends of the straight-lines) standing up in them, AG , AO , CE , CQ , LN , LP , BK , and BR , are on the same straight-lines, GQ and NR

καὶ ὑπὸ τὸ αὐτὸ <ύψος, <ὧν αἱ ἐφεστῶσαι οὐκ εἰσὶν ἐπὶ τῶν αὐτῶν εὐθειῶν, >ῖσα ἀλλήλοις ἐστίν; <ὅπερ >έδει δεῖξαι.

[Prop. 11.29]. Hence, solid CM is also equal to solid CN . Thus, parallelepiped solids (which are) on the same base, and (have) the same height, and in which the (ends of the straight-lines) standing up are not on the same straight-lines, are equal to one another. (Which is) the very thing it was required to show.

31.

Τὰ ἐπὶ >ῖσων βάσεων >όντα στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ <ύψος >ῖσα ἀλλήλοις ἐστίν.

>Ἐστω ἐπὶ >ῖσων βάσεων τῶν AB , $\Gamma\Delta$ στερεὰ παραλληλεπίπεδα τὰ AE , ΓZ ὑπὸ τὸ αὐτὸ <ύψος. λέγω, <ὅτι >ῖσον ἐστὶ τὸ AE στερεὸν τῷ ΓZ στερεῷ|.

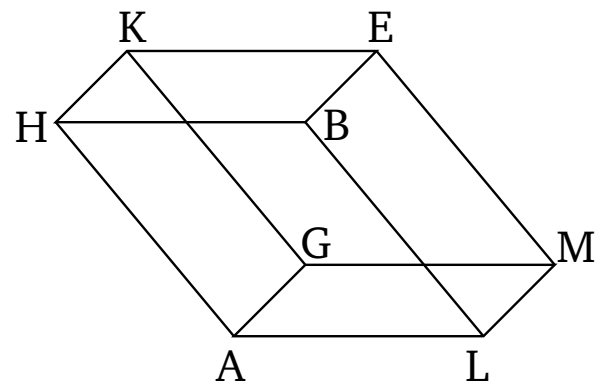
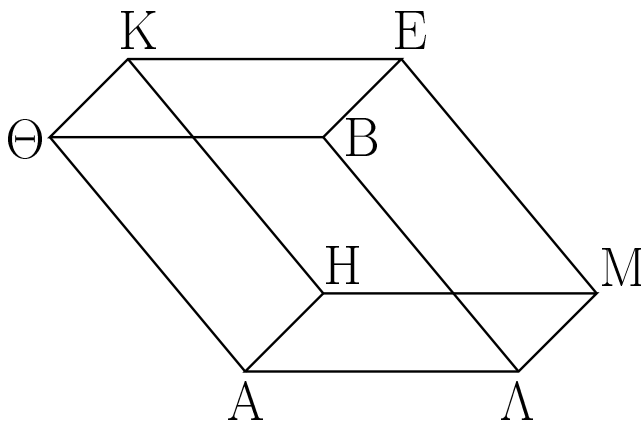
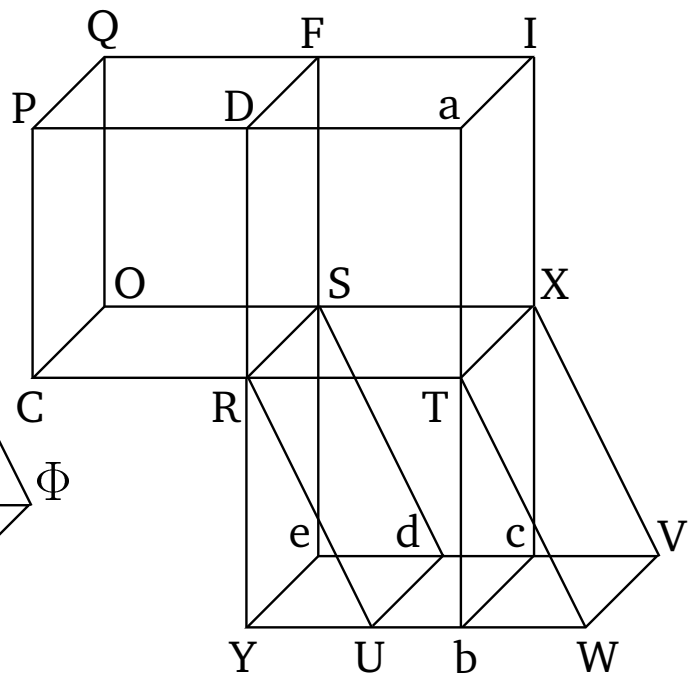
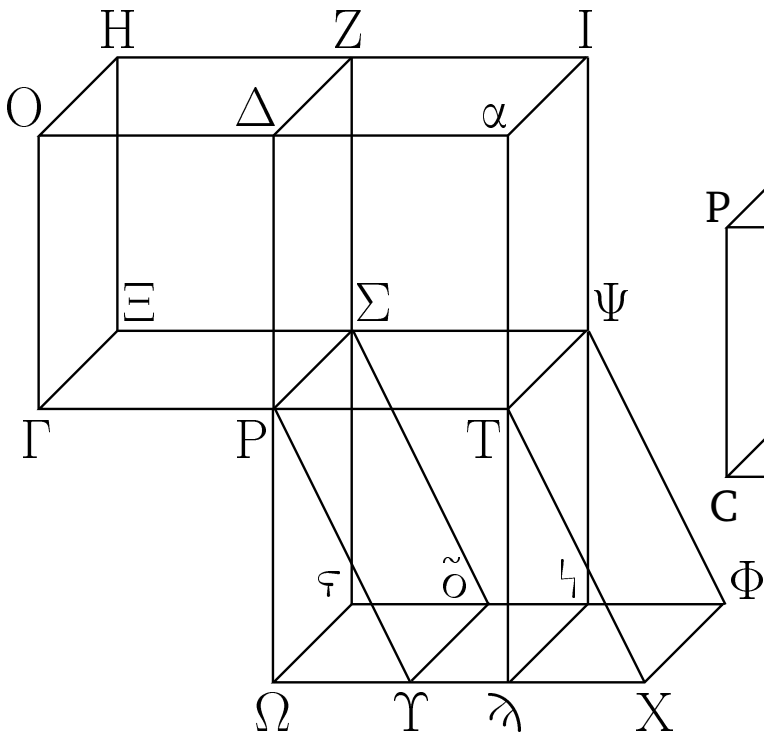
>Ἐστωσαν δὴ πρότερον αἱ ἐφεστηκυῖαι αἱ ΘK , BE , AH , ΛM , OP , ΔZ , ΓX , $P\Sigma$ πρὸς ὀρθὰς ταῖς AB , $\Gamma\Delta$ βάσεσιν, καὶ ἐκβεβλήσθω ἐπ' εὐθείᾳ τῇ ΓP εὐθεῖα ἡ PT , καὶ συνεστάτω πρὸς τῇ PT εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ P τῇ ὑπὸ $\Lambda\Lambda B$ γωνίᾳ >ῖση ἡ ὑπὸ TPY , καὶ κείσθω τῇ μὲν AL >ῖση ἡ PT , τῇ δὲ AB >ῖση ἡ PY , καὶ συμπληρώσθω <ῆ τε $P\Theta$ βάσις καὶ τὸ ΨY στερεόν.

Proposition 31

Parallelepiped solids which are on equal bases, and (have) the same height, are equal to one another.

Let the parallelepiped solids AE and CF be on the equal bases AB and CD (respectively), and (have) the same height. I say that solid AE is equal to solid CF .

So, let the (straight-lines) standing up, HK , BE , AG , LM , PQ , DF , CO , and RS , first of all, be at right-angles to the bases AB and CD . And let RT be produced in a straight-line with CR . And let (angle) TRU , equal to angle ALB , be constructed on the straight-line RT , at the point R on it [Prop. 1.23]. And let RT be made equal to AL , and RU to LB . And let the base RW , and the solid XU , be completed.

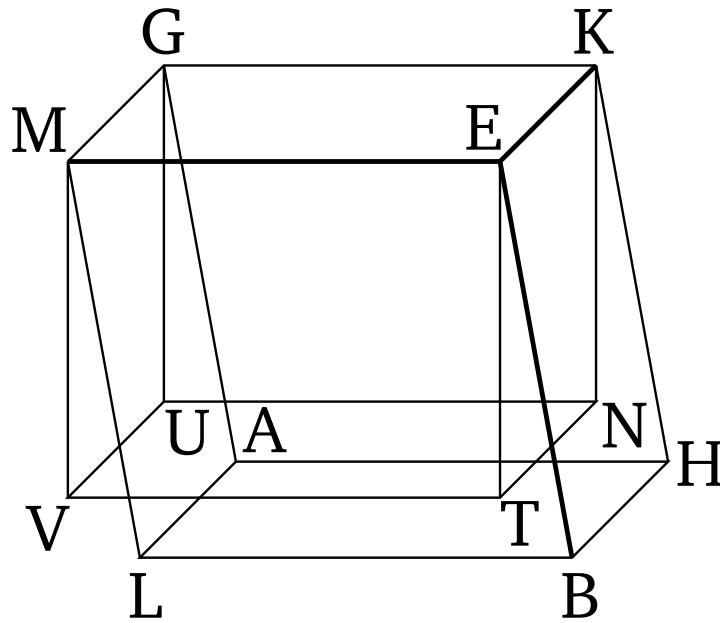


Καὶ ἐπεὶ δύο αἱ TP , PY δυοὶ ταῖς AL , AB >ῖσαι εἰσὶν, καὶ γωνίαξ >ῖσαξ περιέθουσιν, >ῖσον >άρα καὶ <όμοιον

And since the two (straight-lines) TR and RU are equal to the two (straight-lines) AL and LB (respectively), and they

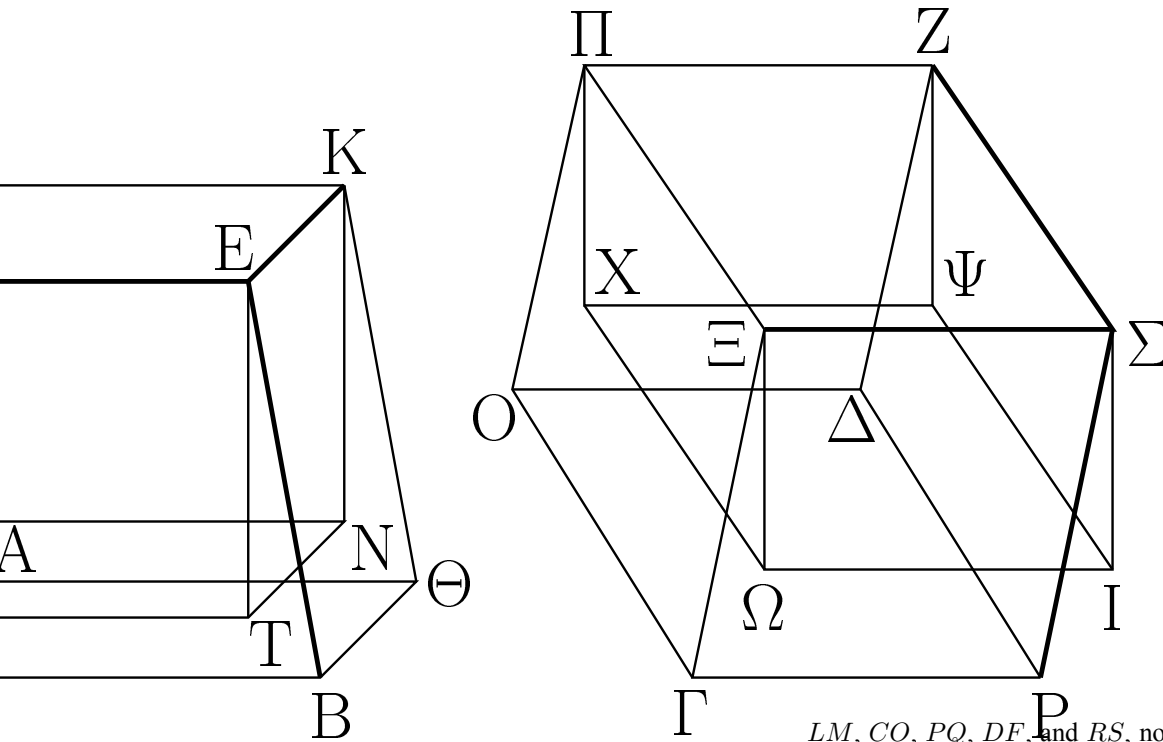
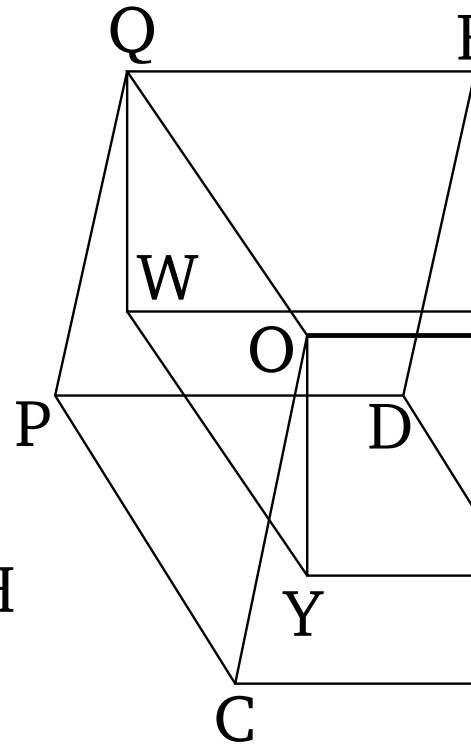
τὸ ΡΘ παραλληλόγραμμον τῷ ΘΛ παραλληλογράμῳ. καὶ ἐπεὶ πάλιν >ίση μὲν ἢ ΑΛ τῇ ΡΤ, ἢ δὲ ΛΜ τῇ ΡΣ, καὶ γωνίαξ ὀρθὰξ περιέθουσιν, >ίσον >άρα καὶ <όμοιόν ἐστι τὸ ΡΨ παραλληλόγραμμον τῷ ΑΜ παραλληλογράμῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ ΛΕ τῷ ΣΥ >ίσον τέ ἐστι καὶ <όμοιον; τρία >άρα παραλληλόγραμμα τοῦ ΑΕ στερεοῦ τρισὶ παραλληλογράμοιξ τοῦ ΨΥ στερεοῦ >ίσα τέ ἐστι καὶ <όμοια. ἀλλὰ τὰ μὲν τρία τρισὶ τοῖξ ἀπεναντίον >ίσα τέ ἐστι καὶ <όμοια, τὰ δὲ τρία τρισὶ τοῖξ ἀπεναντίον; <όλον >άρα τὸ ΑΕ στερεὸν παραλληλεπίπεδον <όλω τῷ ΨΥ στερεῷ παραλληλεπίπεδῳ >ίσον ἐστίν. διήθθωσαν αἱ ΔΡ, ΘΥ καὶ συμπιπτεύωσαν ἀλλήλαιξ κατὰ τὸ Ω, καὶ διὰ τοῦ Τ τῇ ΔΩ παράλληλοξ >ήθθω ἢ αΤ, καὶ ἐκβεβλήσθω ἢ ΟΔ κατὰ τὸ α, καὶ συμπεπληρώσθω τὰ ΩΨ, ΡΙ στερεά. >ίσον δὴ ἐστι τὸ ΨΩ στερεόν, ο<ῷ βάσιξ μὲν ἐστι τὸ ΡΨ παραλληλόγραμμον, ἀπεναντίον δὲ τὸ Ω, τῷ ΨΥ στερεῷ, ο<ῷ βάσιξ μὲν τὸ ΡΨ παραλληλόγραμμον, ἀπεναντίον δὲ τὸ ΥΦ; ἐπὶ τε γὰρ τῇξ αὐτῇξ βάσεώξ εἰσι τῇξ ΡΨ καὶ ὑπὸ τὸ αὐτὸ <ύψοξ, <ὼν αἱ ἐφεστῶσαι αἱ ΡΩ, ΡΥ, Τ, ΤΘ, Σ, Σ, Υ, ΥΦ >εῖ t wn a>ut wn e>isin e>ujei wn t wn WQ, F. >all'a t'o YU stere'on t w| AE >estin >ison; ka'i t'o YW >ara stere'on t w| AE stere w| >estin >ison. ka'i >epe'i >ison >est'i t'o RUQT parallhlogrammon t w| WT parallhlogrammw|; >ep'i te g'ar t hc a>ut hc b'ase'wc e>isi t hc RT ka'i >en ta ic a>uta ic parall'hloic ta ic RT, WQ; >all'a t'o RUQT t w| GD >estin >ison, >epe'i ka'i t w| AB, ka'i t'o WT >ara parallhlogrammon t w| GD >estin >ison. >allo d'e t'o DT; >estin >ara <wc <h GD b'asic pr'oc t'hn DT, o<utwc <h WT pr'oc t'hn DT. ka'i >epe'i stere'on parallhlep'ipedon t'o GI >epip'edw| t w| RZ t'etmhtai parall'hlw| >onti to ic >apenant'ion >epip'edoic, >estin <wc <h GD b'asic pr'oc t'hn DT b'asin, o<utwc t'o GZ stere'on pr'oc t'o RI stere'on. di'a t'a a>ut'a d'h, >epe'i stere'on parallhlep'ipedon t'o WI >epip'edw| t w| RY t'etmhtai parall'hlw| >onti to ic >apenant'ion >epip'edoic, >estin <wc <h WT b'asic pr'oc t'hn TL b'asin, o<utwc t'o WY stere'on pr'oc t'o RI. >all> <wc <h GD b'asic pr'oc t'hn DT, o<utwc <h WT pr'oc t'hn DT; ka'i <wc >ara t'o GZ stere'on pr'oc t'o RI stere'on, o<utwc t'o WY stere'on pr'oc t'o RI. <ek'ateron >ara t wn GZ, WY stere wn pr'oc t'o RI t'on a>ut'on >eqei l'ogon; >ison >ara >est'i t'o GZ stere'on t w| WY stere w|. >all'a t'o WY t w| AE >ede'iqjh >ison; ka'i t'o AE >ara

contain equal angles, parallelogram *RW* is thus equal and similar to parallelogram *HL* [Prop. 6.14]. And, again, since *AL* is equal to *RT*, and *LM* to *RS*, and they contain right-angles, parallelogram *RX* is thus equal and similar to parallelogram *AM* [Prop. 6.14]. So, for the same (reasons), *LE* is also equal and similar to *SU*. Thus, three parallelograms of solid *AE* are equal and similar to three parallelograms of solid *XU*. But, the three (faces of the former solid) are equal and similar to the three opposite (faces), and the three (faces of the latter solid) to the three opposite (faces) [Prop. 11.24]. Thus, the whole parallelepiped solid *AE* is equal to the whole parallelepiped solid *XU* [Def. 11.10]. Let *DR* and *WU* be drawn across, and let them have met one another at *Y*. And let *aTb* be drawn through *T* parallel to *DY*. And let *PD* be produced to *a*. And let the solids *YX* and *RI* be completed. So, solid *XY*, whose base is parallelogram *RX*, and opposite (face) *Yc*, is equal to solid *XU*, whose base (is) parallelogram *RX*, and opposite (face) *UV*. For they are on the same base *RX*, and (have) the same height, and the (ends of the straight-lines) standing up in them, *RY*, *RU*, *Tb*, *TW*, *Se*, *Sd*, *Xc* and *XV*, are on the same straight-lines, *YW* and *eV* [Prop. 11.29]. But, solid *XU* is equal to *AE*. Thus, solid *XY* is also equal to solid *AE*. And since parallelogram *RUWT* is equal to parallelogram *YT*. For they are on the same base *RT*, and between the same parallels *RT* and *YW* [Prop. 1.35]. But, *RUWT* is equal to *CD*, since (it is) also (equal) to *AB*. Parallelogram *YT* is thus also equal to *CD*. And *DT* is another (parallelogram). Thus, as base *CD* is to *DT*, so *YT* (is) to *DT* [Prop. 5.7]. And since the parallelepiped solid *CI* has been cut by the plane *RF*, which is parallel to the opposite planes (of *CI*), as base *CD* is to base *DT*, so solid *CF* (is) to solid *RI* [Prop. 11.25]. So, for the same (reasons), since the parallelepiped solid *YI* has been cut by the plane *RX*, which is parallel to the opposite planes (of *YI*), as base *YT* is to base *TD*, so solid *YX* (is) to solid *RI* [Prop. 11.25]. But, as base *CD* (is) to *DT*, so *YT* (is) to *DT*. And, thus, as solid *CF* (is) to solid *RI*, so solid *YX* (is) to solid *RI*. Thus, solids *CF* and *YX* each have the same ratio to *RI* [Prop. 5.11]. Thus, solid *CF* is equal to solid *YX* [Prop. 5.9]. But, *YX* was show (to be) equal to *AE*. Thus, *AE* is also equal to *CF*.



t w| GZ >estin >'ison.

And so let the (straight-lines) standing up, AG , HK , BE ,



μη >έστωσαν δὴ αἱ ἐφεστηκυῖαι αἱ AH , ΘK , BE , ΛM , ΓX , $O\Pi$, ΔZ , $P\Sigma$ πρὸς ὀρθὰς ταῖς AB , $\Gamma\Delta$ βάσεσιν; λέγω πάλιν, <ὅτι <ίσον τὸ AE στερεὸν τῷ ΓZ στερεῷ>. >ήθωσαν γὰρ ἀπὸ τῶν K , E , H , M , Π , Z , X , Σ σημείων ἐπὶ τὸ ὑποκείμενον ἐπίπεδον κάθετοι αἱ KN , ET , HY , $M\Phi$, $\Pi\Theta$, $Z\Psi$, $X\Omega$, ΣI , καὶ συμβαλλέτωσαν τῷ ἐπιπέδῳ κατὰ τὰ N , T , Y , Φ , Θ , Ψ , Ω , I σημεία, καὶ ἐπεζεύθωσαν αἱ NT , NU , $Y\Phi$, $T\Phi$, $\Theta\Psi$, $\Theta\Omega$, ΩI , $I\Psi$. >ίσον δὴ ἔστι τὸ $K\Phi$ στερεὸν τῷ ΠI στερεῷ; ἐπὶ τε γὰρ >ίσων βάσεων εἰσι τῶν KM , $\Pi\Sigma$ καὶ ὑπὸ τὸ αὐτὸ <ύψος, <ὧν αἱ ἐφεστῶσαι πρὸς ὀρθὰς εἰσι ταῖς βάσεσιν. ἀλλὰ τὸ μὲν $K\Phi$ στερεὸν τῷ AE στερεῷ ἔστιν >ίσον, τὸ δὲ ΠI τῷ ΓZ ; ἐπὶ τε γὰρ τῆς αὐτῆς βάσεώς εἰσι καὶ ὑπὸ τὸ

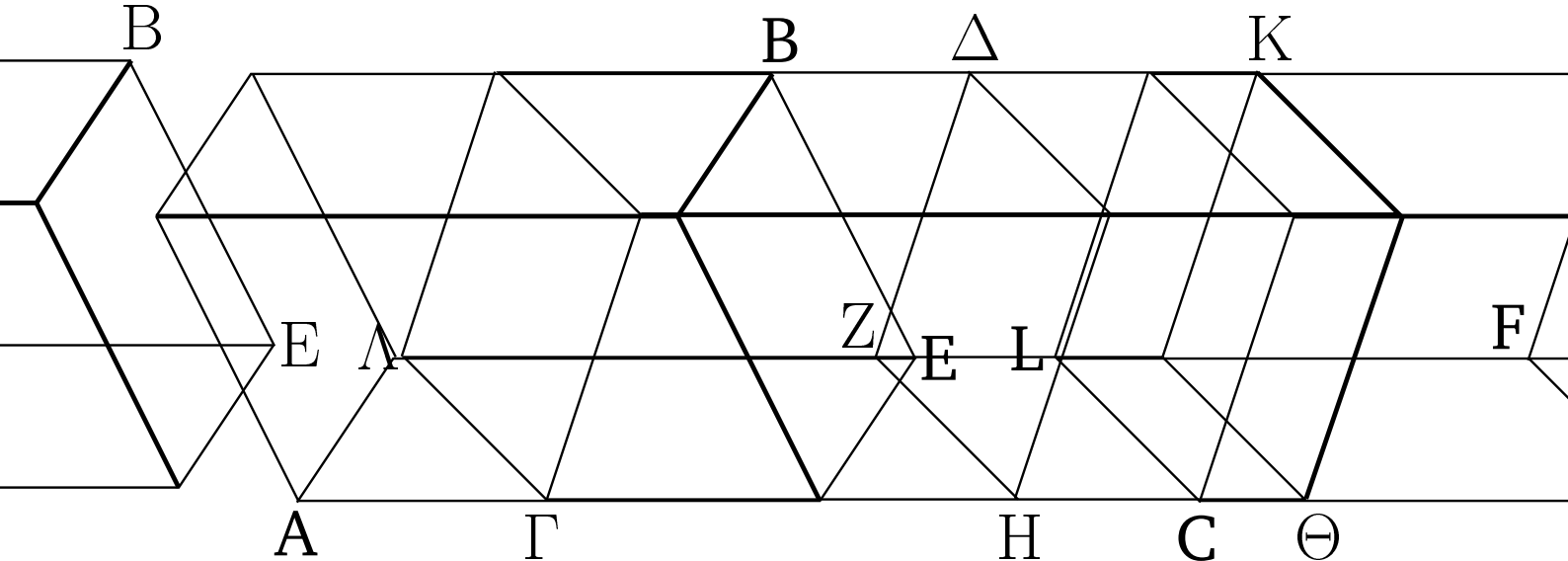
LM , CO , PQ , DF , and RS , not be at right-angles to the bases AB and CD . Again, I say that solid AE (is) equal to solid CF . For let KN , ET , GU , MV , QW , FX , OY , and SI be drawn from points K , E , G , M , Q , F , O , and S (respectively) perpendicular to the reference plane (i.e., the plane of the bases AB and CD), and let them have met the plane at points N , T , U , V , W , X , Y , and I (respectively). And let NT , NU , UV , TV , WX , WY , YI , and IX be joined. So solid KV is equal to solid QI . For they are on the equal bases KM and QS , and (have) the same height, and the (straight-lines) standing up in them are at right-angles to their bases (see first part of proposition). But, solid KV is equal to solid AE , and QI to CF . For they are on the

αὐτὸ <ὑψοξ, <ῶν αἱ ἐφεστῶσαι ο>ὐκ εἰσὶν ἐπὶ τῶν αὐτῶν εὐθειῶν. καὶ τὸ $AE > \alpha\rho\alpha$ στερεὸν τῷ GZ στερεῷ ἐστὶν >ῖσον.

Τὰ >ῥα ἐπὶ >ῖσων βάσεων >όντα στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ <ὑψοξ>ῖσα ἀλλήλοις ἐστίν; <ὅπερ> ἐδει δεῖχαι.

32.

Τὰ ὑπὸ τὸ αὐτὸ <ὑψοξ>όντα στερεὰ παραλληλεπίπεδα πρὸς >ἀλλήλα ἐστὶν ὡς αἱ βάσεις.



>Ἐστω ὑπὸ τὸ αὐτὸ <ὑψοξ> στερεὰ παραλληλεπίπεδα τὰ AB , $\Gamma\Delta$; λέγω, <ὅτι τὰ AB , $\Gamma\Delta$ στερεὰ παραλληλεπίπεδα πρὸς >ἀλλήλα ἐστὶν ὡς αἱ βάσεις, τουτέστιν <ὅτι ἐστὶν ὡς ἡ AE βάσις πρὸς τὴν GZ βάσιν, ο<ὕτως τὸ AB στερεὸν πρὸς τὸ $\Gamma\Delta$ στερεόν.

Παραβεβλήσθω γὰρ παρὰ τὴν ZH τῷ AE >ῖσον τὸ $Z\Theta$, καὶ ἀπὸ βάσεως μὲν τῆς $Z\Theta$, <ὑψουξ> δὲ τοῦ αὐτοῦ τῷ $\Gamma\Delta$ στερεὸν παραλληλεπίπεδον συμπληρώσθω τὸ HK . >ῖσον δὴ ἐστὶ τὸ AB στερεὸν τῷ HK στερεῷ; ἐπὶ τε γὰρ >ῖσων βάσεων εἰσι τῶν AE , $Z\Theta$ καὶ ὑπὸ τὸ αὐτὸ <ὑψοξ>. καὶ ἐπεὶ στερεὸν παραλληλεπίπεδον τὸ ΓK ἐπιπέδῳ τῷ ΔH τέμνεται παραλλήλῳ >όντι τοῖς ἀπεναντίον ἐπιπέδοις, >έστιν >ῥα ὡς ἡ GZ βάσις πρὸς τὴν $Z\Theta$ βάσιν, ο<ὕτως τὸ $\Gamma\Delta$ στερεὸν πρὸς τὸ $\Delta\Theta$ στερεόν. >ῖση δὲ ἡ μὲν $Z\Theta$ βάσις τῇ AE βάσει, τὸ δὲ HK στερεὸν τῷ AB στερεῷ; >έστιν >ῥα καὶ ὡς ἡ AE βάσις πρὸς τὴν GZ βάσιν, ο<ὕτως τὸ AB στερεὸν πρὸς τὸ $\Gamma\Delta$ στερεόν.

Τὰ >ῥα ὑπὸ τὸ αὐτὸ <ὑψοξ>όντα στερεὰ παραλληλεπίπεδα πρὸς >ἀλλήλα ἐστὶν ὡς αἱ βάσεις; <ὅπερ> ἐδει δεῖχαι.

33.

Τὰ <ὅμοια> στερεὰ παραλληλεπίπεδα πρὸς >ἀλλήλα ἐν τριπλασίονι λόγῳ εἰσι τῶν ὁμολόγων πλευρῶν.

>Ἐστω <ὅμοια> στερεὰ παραλληλεπίπεδα τὰ AB , $\Gamma\Delta$, ὁμολογῶς δὲ >έστω ἡ AE τῇ GZ ; λέγω, <ὅτι τὸ AB στερεὸν πρὸς τὸ $\Gamma\Delta$ στερεὸν τριπλασίονα λόγον>έθει, >ἢπερ ἡ AE πρὸς τὴν GZ .

Ἐκβεβλήσθωσαν γὰρ ἐπ' εὐθείας ταῖς AE , HE , ΘE αἱ EK , EL , EM , καὶ κείσθω τῇ μὲν GZ >ῖση ἡ EK , τῇ δὲ ZN >ῖση ἡ EL , καὶ >έτι τῇ $ΖΡ$ >ῖση ἡ EM , καὶ συμπληρώσθω τὸ $ΚΛ$ παραλληλόγραμμον καὶ τὸ KO στερεόν.

same base, and (have) the same height, and the (straight-lines) standing up in them are not on the same straight-lines [Prop. 11.30]. Thus, solid AE is also equal to solid CF .

Thus, parallelepiped solids which are on equal bases, and (have) the same height, are equal to one another. (Which is) the very thing it was required to show.

Proposition 32

Parallelepiped solids which (have) the same height are to one another as their bases.

Let AB and CD be parallelepiped solids (having) the same height. I say that the parallelepiped solids AB and CD are to one another as their bases. That is to say, as base AE is to base CF , so solid AB (is) to solid CD .

For let FH , equal to AE , be applied to FG (in the angle FGH equal to angle LCG) [Prop. 1.45]. And let the parallelepiped solid GK , (having) the same height as CD , be completed on the base FH . So solid AB is equal to solid GK . For they are on the equal bases AE and FH , and (have) the same height [Prop. 11.31]. And since the parallelepiped solid CK has been cut by the plane DG , which is parallel to the opposite planes (of CK), thus as the base CF is to the base FH , so the solid CD (is) to the solid DH [Prop. 11.25]. And base FH (is) equal to base AE , and solid GK to solid AB . And thus as base AE is to base CF , so solid AB (is) to solid CD .

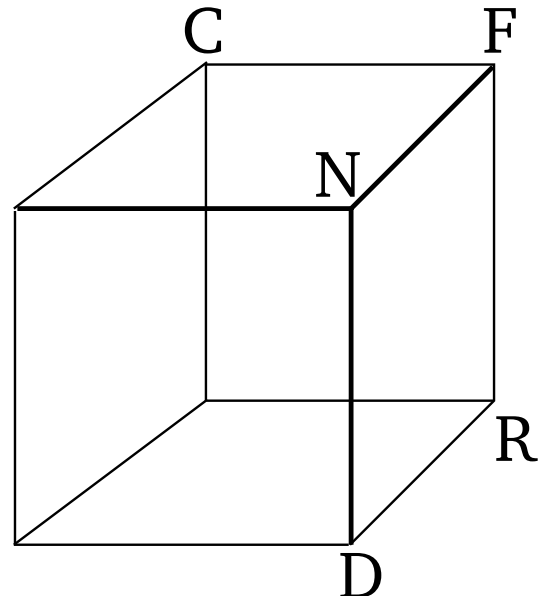
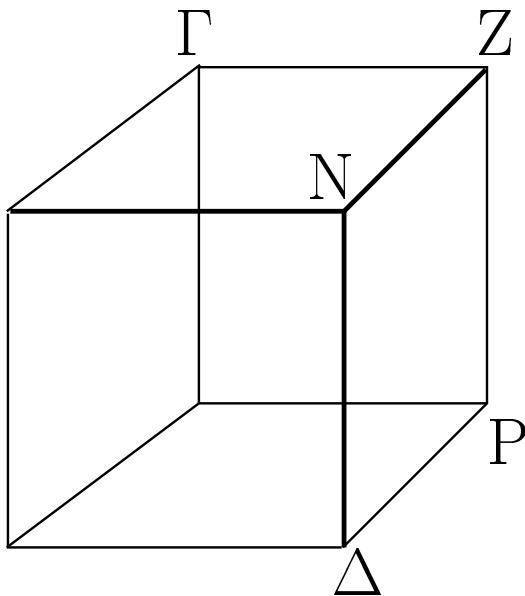
Thus, parallelepiped solids which (have) the same height are to one another as their bases. (Which is) the very thing it was required to show.

Proposition 33

Similar parallelepiped solids are to one another as the cubed ratio of their corresponding sides.

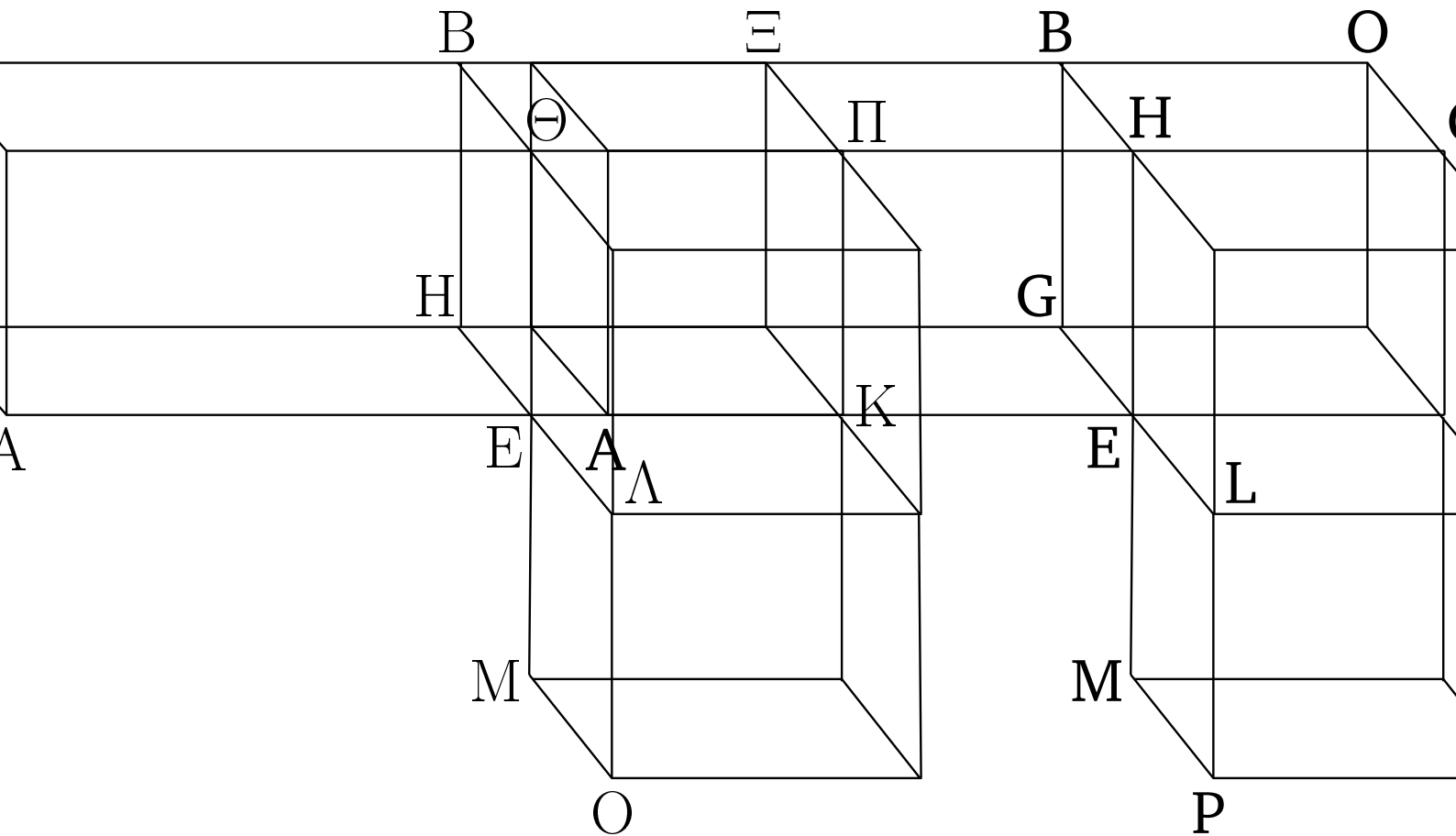
Let AB and CD be similar parallelepiped solids, and let AE correspond to CF . I say that solid AB has to solid CD the cubed ratio that AE (has) to CF .

For let EK , EL , and EM be produced in a straight-line with AE , GE , and HE (respectively). And let EK be made equal to CF , and EL equal to FN , and, further, EM equal to FR . And let the parallelogram KL be completed, and the solid KP .



Καὶ ἐπεὶ δύο αἱ KE , EL δυοὶ ταῖς $ΓΖ$, ZN ὀρθοὶ εἰσὶν, ἀλλὰ καὶ γωνία ἡ ὑπὸ KEA γωνία τῇ ὑπὸ $ΓΖΝ$ ἔστιν ὀρθή, ἐπειδὴ περ καὶ ἡ ὑπὸ AEH τῇ ὑπὸ $ΓΖΝ$ ἔστιν ὀρθή διὰ τὴν ὁμοιότητα τῶν AB , $ΓΔ$ στερεῶν, ὅσον ὅρα ἔστι [καὶ ὁμοίων] τὸ $ΚΛ$ παραλληλόγραμμον τῷ $ΓΝ$ παραλληλόγραμμῳ. διὰ τὰ αὐτὰ δὲ καὶ τὸ μὲν $ΚΜ$ παραλληλόγραμμον ὅσον ἔστι καὶ ὁμοίον τῷ $ΓΡ$ [παραλληλόγραμμῳ] καὶ ἔτι τὸ $ΕΟ$ τῷ $ΔΖ$; τρία ὅρα παραλληλόγραμμα τοῦ $ΚΟ$ στερεοῦ τρισὶ παραλληλόγραμμοις τοῦ $ΓΔ$ στερεοῦ ὅσα ἔστι καὶ ὁμοία. ἀλλὰ τὰ μὲν τρία τρισὶ τοῖς ἀπεναντίον ὅσα ἔστι καὶ ὁμοία, τὰ δὲ τρία τρισὶ τοῖς ἀπεναντίον ὅσα ἔστι καὶ ὁμοία; ὅλον ὅρα τὸ $ΚΟ$ στερεὸν ὅλῳ τῷ $ΓΔ$ στερεῷ ὅσον ἔστι καὶ ὁμοίον. συμπληρώσθω τὸ $ΗΚ$ παραλληλόγραμμον, καὶ ἀπὸ βάσεων μὲν τῶν $ΗΚ$, $ΚΛ$ παραλληλόγραμμων, ὡς τοῦ αὐτοῦ τῷ $ΑΒ$ στερεῷ συμπληρώσθω τὰ $ΕΧ$, $ΑΠ$. καὶ ἐπεὶ διὰ τὴν ὁμοιότητα τῶν $ΑΒ$, $ΓΔ$ στερεῶν ἔστιν ὡς ἡ $ΑΕ$ πρὸς τὴν $ΓΖ$, οὕτως ἡ $ΕΗ$ πρὸς τὴν $ΖΝ$, καὶ ἡ $ΕΘ$ πρὸς τὴν $ΖΡ$, ὅση δὲ ἡ μὲν $ΓΖ$ τῇ $ΕΚ$, ἡ δὲ $ΖΝ$ τῇ $ΕΛ$, ἡ δὲ $ΖΡ$ τῇ $ΕΜ$, ὅστις ὅρα ὡς ἡ $ΑΕ$ πρὸς τὴν $ΕΚ$, οὕτως ἡ $ΗΕ$ πρὸς τὴν $ΕΛ$ καὶ ἡ $ΘΕ$ πρὸς τὴν $ΕΜ$. ἀλλ' ὡς μὲν ἡ $ΑΕ$ πρὸς τὴν $ΕΚ$, οὕτως τὸ $ΑΗ$ [παραλληλόγραμμον] πρὸς τὸ $ΗΚ$ παραλληλόγραμμον, ὡς δὲ ἡ $ΗΕ$ πρὸς τὴν $ΕΛ$, οὕτως τὸ $ΗΚ$ πρὸς τὸ $ΚΛ$, ὡς δὲ ἡ $ΘΕ$ πρὸς τὴν $ΕΜ$, οὕτως τὸ $ΠΕ$ πρὸς τὸ $ΚΜ$; καὶ ὡς ὅρα τὸ $ΑΗ$ παραλληλόγραμμον πρὸς τὸ $ΗΚ$, οὕτως τὸ $ΗΚ$ πρὸς τὸ $ΚΛ$ καὶ τὸ $ΠΕ$ πρὸς τὸ $ΚΜ$. ἀλλ' ὡς μὲν τὸ $ΑΗ$ πρὸς τὸ $ΗΚ$, οὕτως τὸ $ΑΒ$ στερεὸν πρὸς τὸ $ΕΧ$ στερεόν, ὡς δὲ τὸ $ΗΚ$ πρὸς τὸ $ΚΛ$, οὕτως τὸ $ΧΕ$ στερεὸν πρὸς τὸ $ΠΑ$ στερεόν, ὡς δὲ τὸ $ΠΕ$ πρὸς τὸ $ΚΜ$, οὕτως τὸ $ΠΑ$ στερεὸν πρὸς τὸ $ΚΟ$ στερεόν; καὶ ὡς ὅρα τὸ $ΑΒ$ στερεὸν πρὸς τὸ $ΕΧ$, οὕτως τὸ $ΕΧ$ πρὸς τὸ $ΠΑ$ καὶ τὸ $ΠΑ$ πρὸς τὸ $ΚΟ$. ἔαν δὲ τέσσαρα μεγέθη κατὰ τὸ συνεθὲς ἀνάλογον ᾗ, τὸ πρῶτον πρὸς τὸ τέταρτον τριπλασίονα λόγον ᾗ, ᾗ ἢ ᾗ, τὸ πρῶτον πρὸς τὸ δεῦτερον; τὸ $ΑΒ$ ὅρα στερεὸν πρὸς τὸ $ΚΟ$ τριπλασίονα λόγον ᾗ ἢ ᾗ, ᾗ ἢ ᾗ, τὸ $ΑΒ$ πρὸς τὸ $ΕΧ$. ἀλλ' ὡς τὸ $ΑΒ$ πρὸς τὸ $ΕΧ$, οὕτως τὸ $ΑΗ$ παραλληλόγραμμον πρὸς τὸ $ΗΚ$ καὶ ἡ $ΑΕ$ εὐθεῖα πρὸς τὴν $ΕΚ$; ὥστε καὶ τὸ $ΑΒ$ στερεὸν πρὸς τὸ $ΚΟ$ τριπλασίονα λόγον ᾗ ἢ ᾗ, ᾗ ἢ ᾗ, ἡ δὲ $ΕΚ$ εὐθεῖα τῇ $ΓΖ$; καὶ τὸ $ΑΒ$ ὅρα στερεὸν πρὸς τὸ $ΓΔ$ στερεὸν τριπλασίονα λόγον ᾗ ἢ ᾗ, ᾗ ἢ ᾗ, ἡ ὁμόλογος αὐτοῦ πλευρὰ ἡ $ΑΕ$ πρὸς τὴν ὁμόλογον πλευρὰν τὴν $ΓΖ$.

And since the two (straight-lines) KE and EL are equal to the two (straight-lines) CF and FN , but angle KEL is also equal to angle CFN , inasmuch as AEG is also equal to CFN , on account of the similarity of the solids AB and CD , parallelogram KL is thus equal [and similar] to parallelogram CN . So, for the same (reasons), parallelogram KM is also equal and similar to [parallelogram] CR , and, further, EP to DF . Thus, three parallelograms of solid KP are equal and similar to three parallelograms of solid CD . But the three (former parallelograms) are equal and similar to the three opposite (parallelograms), and the three (latter parallelograms) are equal and similar to the three opposite (parallelograms) [Prop. 11.24]. Thus, the whole of solid KP is equal and similar to the whole of solid CD [Def. 11.10]. Let parallelogram GK be completed. And let the solids EO and LQ , with bases the parallelograms GK and KL (respectively), and with the same height as AB , be completed. And since, on account of the similarity of solids AB and CD , as AE is to CF , so EG (is) to FN , and EH to FR [Defs. 6.1, 11.9], and CF (is) equal to EK , and FN to EL , and FR to EM , thus as AE is to EK , so GE (is) to EL , and HE to EM . But, as AE (is) to EK , so [parallelogram] AG (is) to parallelogram GK , and as GE (is) to EL , so GK (is) to KL , and as HE (is) to EM , so QE (is) to KM [Prop. 6.1]. And thus as parallelogram AG (is) to GK , so GK (is) to KL , and QE (is) to KM . But, as AG (is) to GK , so solid AB (is) to solid EO , and as GK (is) to KL , so solid OE (is) to solid QL , and as QE (is) to KM , so solid QL (is) to solid KP [Prop. 11.32]. And, thus, as solid AB is to EO , so EO (is) to QL , and QL to KP . And if four magnitudes are continuously proportional then the first has to the fourth the cubed ratio that (it has) to the second [Def. 5.10]. Thus, solid AB has to KP the cubed ratio which AB (has) to EO . But, as AB (is) to EO , so parallelogram AG (is) to GK , and the straight-line AE to EK [Prop. 6.1]. Hence, solid AB also has to KP the cubed ratio that AE (has) to EK . And solid KP (is) equal to solid CD , and straight-line EK to CF . Thus, solid AB also has to solid CD the cubed ratio which its corresponding side AE (has) to the corresponding side CF .



Τὰ >άρα <όμοια στερεὰ παραλληλεπίπεδα ἐν τριπλασίονι λόγῳ ἐστὶ τῶν ὁμολόγων πλευρῶν; <όπερ >έδει δεῖξαι.

Thus, similar parallelepiped solids are to one another as the cubed ratio of their corresponding sides. (Which is) the very thing it was required to show.

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, <ὅτι ἔαν τέσσαρες εὐθεῖαι ἀνάλογον >ῶσιν, >έσται ὡς ἡ πρώτη πρὸς τὴν τετάρτην, ο<ύτω τὸ ἀπὸ τῆς πρώτης στερεὸν παραλληλεπίπεδον πρὸς τὸ ἀπὸ τῆς δευτέρας τὸ <όμοιον καὶ ὁμοίως ἀναγραφόμενον, ἐπείπερ καὶ ἡ πρώτη πρὸς τὴν τετάρτην τριπλασίονα λόγον >έθει >ήπερ πρὸς τὴν δευτέραν.

Corollary

So, (it is) clear, from this, that if four straight-lines are (continuously) proportional then as the first is to the fourth, so the parallelepiped solid on the first will be to the similar, and similarly described, parallelepiped solid on the second, since the first also has to the fourth the cubed ratio that (it has) to the second.

34.

Τῶν >ίσων στερεῶν παραλληλεπιπέδων ἀντιτεπόνθασιν αἱ βάσεις τοῖς <ύψεσιν; καὶ <ῶν στερεῶν παραλληλεπιπέδων ἀντιτεπόνθασιν αἱ βάσεις τοῖς <ύψεσιν, <ίσα ἐστὶν ἐκεῖνα. >Ἐστω >ίσα στερεὰ παραλληλεπίπεδα τὰ AB, ΓΔ; λέγω, <ὅτι τῶν AB, ΓΔ στερεῶν παραλληλεπιπέδων ἀντιτεπόνθασιν αἱ βάσεις τοῖς <ύψεσιν, καὶ ἐστὶν ὡς ἡ ΕΘ βάσις πρὸς τὴν ΝΠ βάσιν, ο<ύτωξ τὸ τοῦ ΓΔ στερεοῦ <ύψοξ πρὸς τὸ τοῦ AB στερεοῦ <ύψοξ. >Ἐστώσαν γὰρ πρότερον αἱ ἐφεστηκυῖαι αἱ AH, EZ, ΛB, ΘK, ΓM, NX, OΔ, ΠP πρὸς ὀρθὰς ταῖς βάσεσιν αὐτῶν; λέγω, <ὅτι ἐστὶν ὡς ἡ ΕΘ βάσις πρὸς τὴν ΝΠ βάσιν, ο<ύτωξ ἡ ΓM πρὸς τὴν AH. Εἰ μὲν ο>ὖν >ίση ἐστὶν ἡ ΕΘ βάσις τῇ ΝΠ βάσει, >έστι δὲ καὶ τὸ AB στερεὸν τῷ ΓΔ στερεῷ >ίσον, >έσται καὶ ἡ ΓM τῇ AH >ίση. τὰ γὰρ ὑπὸ τὸ αὐτὸ <ύψοξ στερεὰ παραλληλεπίπεδα πρὸς >άλληλά ἐστιν ὡς αἱ βάσεις. καὶ >έσται ὡς ἡ ΕΘ βάσις πρὸς τὴν ΝΠ, ο<ύτωξ ἡ ΓM πρὸς τὴν AH, καὶ φανερόν, <ὅτι τῶν AB, ΓΔ στερεῶν παραλληλεπιπέδων ἀντιτεπόνθασιν αἱ βάσεις τοῖς <ύψεσιν.

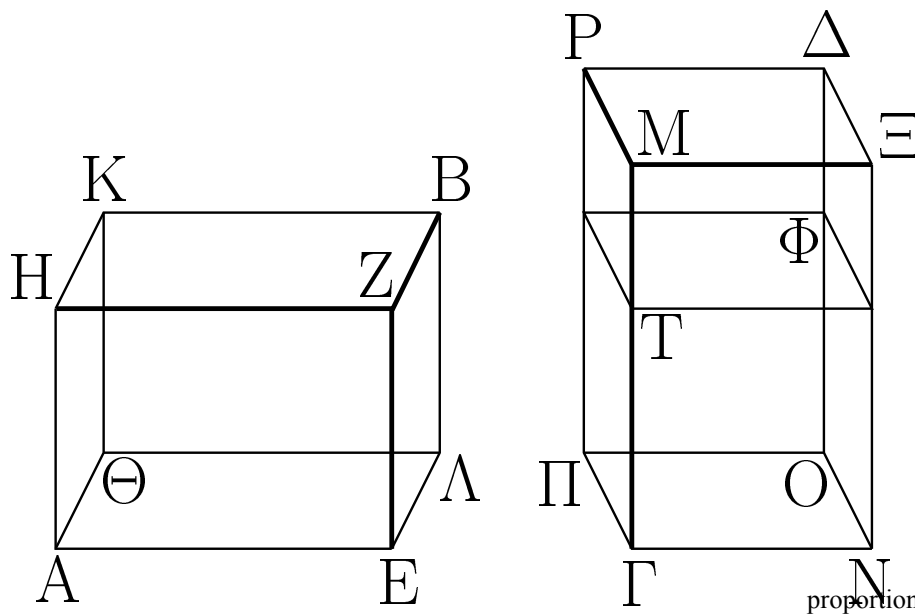
Proposition 34[†]

The bases of equal parallelepiped solids are reciprocally proportional to their heights. And those parallelepiped solids whose bases are reciprocally proportional to their heights are equal.

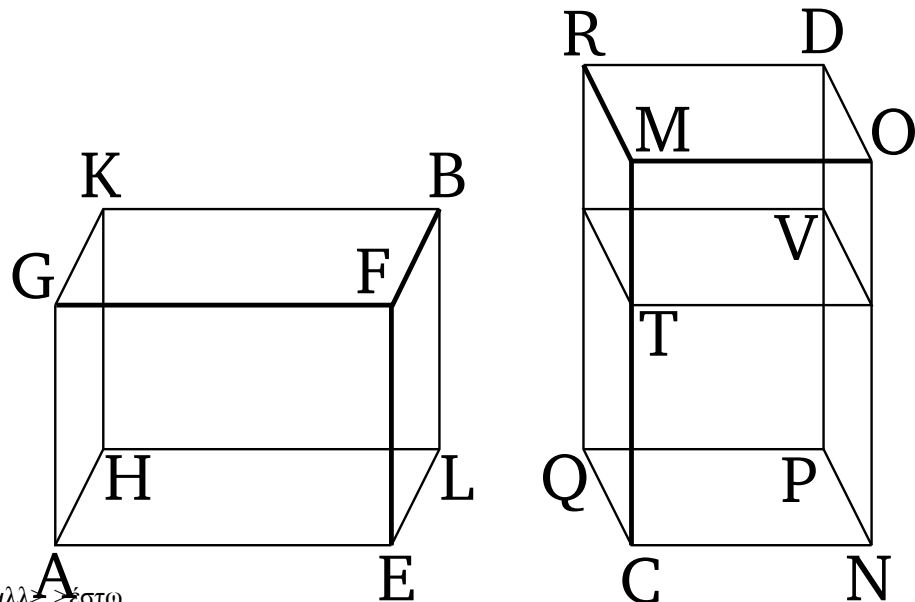
Let AB and CD be equal parallelepiped solids. I say that the bases of the parallelepiped solids AB and CD are reciprocally proportional to their heights, and (so) as base EH is to base NQ , so the height of solid CD (is) to the height of solid AB .

For, first of all, let the (straight-lines) standing up, AG , EF , LB , HK , CM , NO , PD , and QR , be at right-angles to their bases. I say that as base EH is to base NQ , so CM (is) to AG .

Therefore, if base EH is equal to base NQ , and solid AB is also equal to solid CD , CM will also be equal to AG . For parallelepiped solids of the same height are to one another as their bases [Prop. 11.32]. And as base EH (is) to NQ , so CM will be to AG . And (so it is) clear that the bases of the parallelepiped solids AB and CD are reciprocally



proportional to their heights.



μη >έστω δὴ >ίση ἢ ΕΘ βάσει τῇ ΝΠ βάσει, ἀλλ' >έστω
μείζων ἢ ΕΘ. >έστι δὲ καὶ τὸ ΑΒ στερεὸν τῷ ΓΔ στερεῷ
>ίσον; μείζων >άρα ἐστὶ καὶ ἡ ΓΜ τῇς ΑΗ. κείσθω οὖν
τῇ ΑΗ >ίση ἡ ΓΤ, καὶ συμπληρώσθω ἀπὸ βάσεως μὲν
τῇς ΝΠ, <ύψους δὲ τοῦ ΓΤ, στερεὸν παραλληλεπίπεδον τὸ
ΦΓ. καὶ ἐπεὶ >ίσον ἐστὶ τὸ ΑΒ στερεὸν τῷ ΓΔ στερεῷ,
>έχθωεν δὲ τὸ ΓΦ, τὰ δὲ >ίσα πρὸς τὸ αὐτὸ τὸν αὐτὸν >έθει
λόγον, >έστιν >άρα ὡς τὸ ΑΒ στερεὸν πρὸς τὸ ΓΦ στερεόν,
οὕτως τὸ ΓΔ στερεὸν πρὸς τὸ ΓΦ στερεόν. ἀλλ' ὡς μὲν
τὸ ΑΒ στερεὸν πρὸς τὸ ΓΦ στερεόν, οὕτως ἡ ΕΘ βάσις
πρὸς τὴν ΝΠ βάσιν; ἴσο" ὑψὲ γὰρ τὰ ΑΒ, ΓΦ στερεά; ὡς δὲ
τὸ ΓΔ στερεὸν πρὸς τὸ ΓΦ στερεόν, οὕτως ἡ ΜΠ βάσις
πρὸς τὴν ΤΠ βάσιν καὶ ἡ ΓΜ πρὸς τὴν ΓΤ; καὶ ὡς >άρα ἡ
ΕΘ βάσις πρὸς τὴν ΝΠ βάσιν, οὕτως ἡ ΜΓ πρὸς τὴν ΓΤ.
>ίση δὲ ἡ ΓΤ τῇ ΑΗ; καὶ ὡς >άρα ἡ ΕΘ βάσις πρὸς τὴν
ΝΠ βάσιν, οὕτως ἡ ΜΓ πρὸς τὴν ΑΗ. τῶν ΑΒ, ΓΔ >άρα
στερεῶν παραλληλεπίπεδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς
<ύψεσιν.

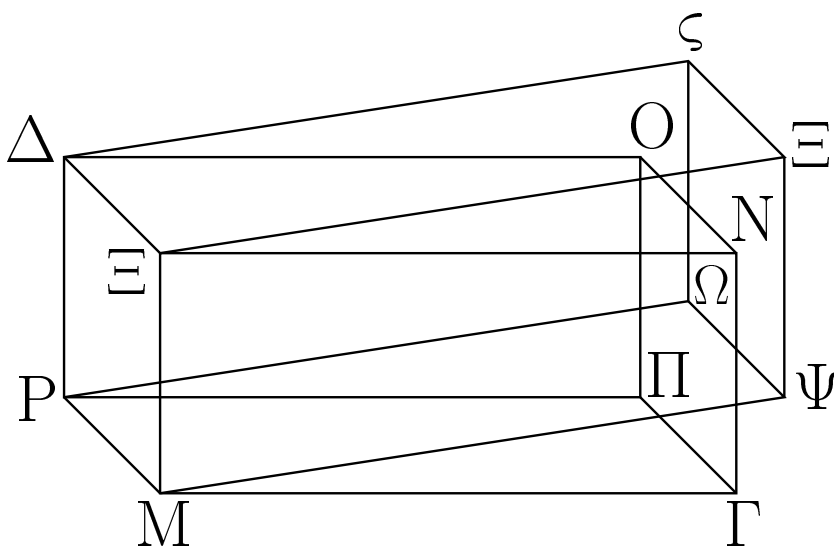
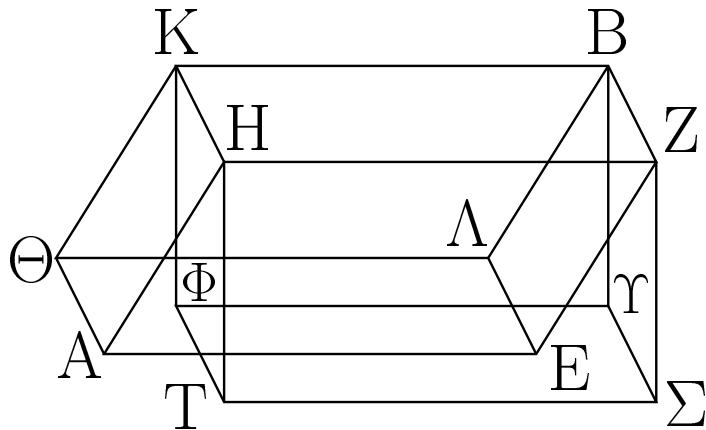
Πάλιν δὴ τῶν ΑΒ, ΓΔ στερεῶν παραλληλεπίπεδων ἀντιπεπόνθασιν αἱ
βάσεις τοῖς <ύψεσιν, καὶ >έστω ὡς ἡ ΕΘ βάσις πρὸς τὴν
ΝΠ βάσιν, οὕτως τὸ τοῦ ΓΔ στερεοῦ <ύψος πρὸς τὸ τοῦ
ΑΒ στερεοῦ <ύψος; λέγω, <ότι >ίσον ἐστὶ τὸ ΑΒ στερεὸν
τῷ ΓΔ στερεῷ].

So let base EH not be equal to base NQ , but let EH be
greater. And solid AB is also equal to solid CD . Thus, CM
is also greater than AG . Therefore, let CT be made equal
to AG . And let the parallelepiped solid VC be completed
on the base NQ , with height CT . And since solid AB is
equal to solid CD , and CV (is) extrinsic (to them), and
equal (magnitudes) have the same ratio to the same (magnit-
ude) [Prop. 5.7], thus as solid AB is to solid CV , so solid
 CD (is) to solid CV . But, as solid AB (is) to solid CV , so
base EH (is) to base NQ . For the solids AB and CV (are)
of equal height [Prop. 11.32]. And as solid CD (is) to solid
 CV , so base MQ (is) to base TQ [Prop. 11.25], and CM
to CT [Prop. 6.1]. And, thus, as base EH is to base NQ ,
so MC (is) to AG . And CT (is) equal to AG . And thus
as base EH (is) to base NQ , so MC (is) to AG . Thus, the
bases of the parallelepiped solids AB and CD are reciprocally
proportional to their heights.

So again, let the bases of the parallelepiped solids AB and
 CD be reciprocally proportional to their heights, and let
base EH be to base NQ , as the height of solid CD (is)
to the height of solid AB . I say that solid AB is equal to
solid CD . [For] let the (straight-lines) standing up again

>Ἐστωσαν [γάρ] πάλιν αἱ ἐφεστηκυῖαι πρὸς ὀρθὰς ταῖς βάσεσιν. καὶ εἰ μὲν >ῖση ἐστὶν ἡ ΕΘ βάσις τῇ ΝΠ βάσει, καὶ ἐστὶν ὡς ἡ ΕΘ βάσις πρὸς τὴν ΝΠ βάσιν, οὕτως τὸ τοῦ ΓΔ στερεοῦ <ύψος πρὸς τὸ τοῦ ΑΒ στερεοῦ <ύψος, >ῖσον >άρα ἐστὶ καὶ τὸ τοῦ ΓΔ στερεοῦ <ύψος τῷ τοῦ ΑΒ στερεοῦ <ύψει. τὰ δὲ ἐπὶ >ῖσων βάσεων στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ <ύψος >ῖσα ἀλλήλοις ἐστίν; >ῖσον >άρα ἐστὶ τὸ ΑΒ στερεὸν τῷ ΓΔ στερεῷ].

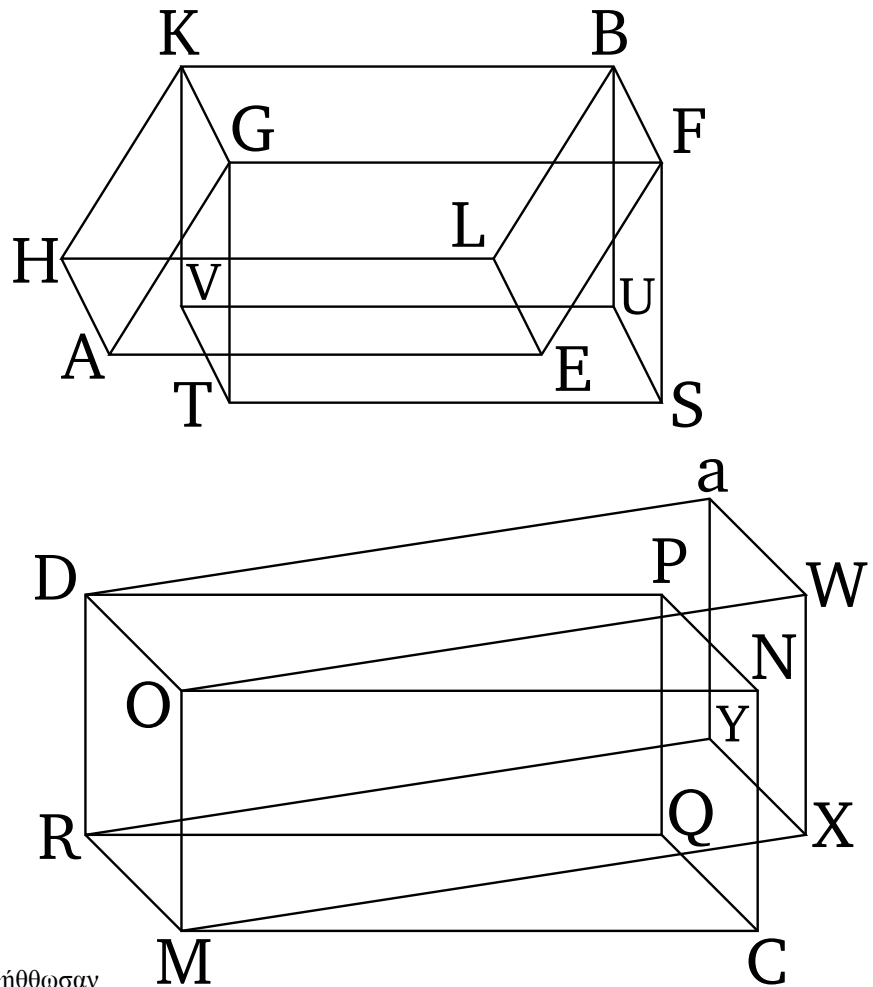
μὴ >έστω δὲ ἡ ΕΘ βάσις τῇ ΝΠ [βάσει] >ῖση, ἀλλ' >έστω μείζων ἡ ΕΘ; μείζων >άρα ἐστὶ καὶ τὸ τοῦ ΓΔ στερεοῦ <ύψος τοῦ τοῦ ΑΒ στερεοῦ <ύψους, τουτέστιν ἡ ΓΜ τῇς ΑΗ. κείσθω τῇ ΑΗ >ῖση πάλιν ἡ ΓΤ, καὶ συμπληρώσθω ὁμοίως τὸ ΓΦ στερεόν. ἐπεὶ ἐστὶν ὡς ἡ ΕΘ βάσις πρὸς τὴν ΝΠ βάσιν, οὕτως ἡ ΜΓ πρὸς τὴν ΑΗ, >ῖση δὲ ἡ ΑΗ τῇ ΓΤ, >έστιν >άρα ὡς ἡ ΕΘ βάσις πρὸς τὴν ΝΠ βάσιν, οὕτως ἡ ΓΜ πρὸς τὴν ΓΤ. ἀλλ' ὡς μὲν ἡ ΕΘ [βάσις] πρὸς τὴν ΝΠ βάσιν, οὕτως τὸ ΑΒ στερεὸν πρὸς τὸ ΓΦ στερεόν; ἴσο" ὑψὴ γάρ ἐστὶ τὰ ΑΒ, ΓΦ στερεά; ὡς δὲ ἡ ΓΜ πρὸς τὴν ΓΤ, οὕτως <ή τε ΜΠ βάσις πρὸς τὴν ΠΤ βάσιν καὶ τὸ ΓΔ στερεὸν πρὸς τὸ ΓΦ στερεόν. καὶ ὡς >άρα τὸ ΑΒ στερεὸν πρὸς τὸ ΓΦ στερεόν, οὕτως τὸ ΓΔ στερεὸν πρὸς τὸ ΓΦ στερεόν; ἐκότερον >άρα τῶν ΑΒ, ΓΔ πρὸς τὸ ΓΦ τὸν αὐτὸν >έθει λόγον. >ῖσον >άρα ἐστὶ τὸ ΑΒ στερεὸν τῷ ΓΔ στερεῷ].



μὴ >έστωσαν δὲ αἱ ἐφεστηκυῖαι αἱ ΖΕ, ΒΛ, ΗΑ, ΚΘ, ΧΝ,

be at right-angles to the bases. And if base EH is equal to base NQ , and as base EH is to base NQ , so the height of solid CD (is) to the height of solid AB , the height of solid CD is thus also equal to the height of solid AB . And parallelepiped solids on equal bases, and also with the same height, are equal to one another [Prop. 11.31]. Thus, solid AB is equal to solid CD .

So, let base EH not be equal to [base] NQ , but let EH be greater. Thus, the height of solid CD is also greater than the height of solid AB , that is to say CM (greater) than AG . Let CT again be made equal to AG , and let the solid CV be similarly completed. Since as base EH is to base NQ , so MC (is) to AG , and AG (is) equal to CT , thus as base EH (is) to base NQ , so CM (is) to CT . But, as [base] EH (is) to base NQ , so solid AB (is) to solid CV . For solids AB and CV are of equal heights [Prop. 11.32]. And as CM (is) to CT , so (is) base MQ to base QT [Prop. 6.1], and solid CD to solid CV [Prop. 11.25]. And thus as solid AB (is) to solid CV , so solid CD (is) to solid CV . Thus, AB and CD each have the same ratio to CV . Thus, solid AB is equal to solid CD [Prop. 5.9].



ΔΟ, ΜΓ, ΡΠ πρὸς ὀρθὰς ταῖς βάσεσιν αὐτῶν, καὶ >ήθθωσαν ἀπὸ τῶν Ζ, Η, Β, Κ, Χ, Μ, Ρ, Δ σημείων ἐπὶ τὰ διὰ τῶν ΕΘ, ΝΠ ἐπίπεδα κάθετοι καὶ συμβαλλέτωσαν τοῖς ἐπιπέδοις κατὰ τὰ Σ, Τ, Υ, Φ, Θ, Ψ, Ω, Ξ, καὶ συμπληρώσθω τὰ ΖΦ, ΧΩ στερεά; λέγω, <ὅτι καὶ ο<ύτωξ >ίσων >όντων τῶν ΑΒ, ΓΔ στερεῶν ἀντιπεπόνθασιν αἱ βάσεις τοῖς <ύψεσιν, καὶ ἔστιν ὡς ἡ ΕΘ βάσιν πρὸς τὴν ΝΠ βάσιν, ο<ύτωξ τὸ τοῦ ΓΔ στερεοῦ <ύψος πρὸς τὸ τοῦ ΑΒ στερεοῦ <ύψος.

Ἐπεὶ >ίσον ἔστι τὸ ΑΒ στερεὸν τῷ ΓΔ στερεῷ, ἀλλὰ τὸ μὲν ΑΒ τῷ ΒΤ ἔστιν >ίσον; ἐπὶ τε γὰρ τῆς αὐτῆς βάσεως εἰσι τῆς ΖΚ καὶ ὑπὸ τὸ αὐτὸ <ύψος; τὸ δὲ ΓΔ στερεὸν τῷ ΔΨ ἔστιν >ίσον; ἐπὶ τε γὰρ πάλιν τῆς αὐτῆς βάσεως εἰσι τῆς ΡΧ καὶ ὑπὸ τὸ αὐτὸ <ύψος; καὶ τὸ ΒΤ >άρα στερεὸν τῷ ΔΨ στερεῷ >ίσον ἔστιν. >έστιν >άρα ὡς ἡ ΖΚ βάσις πρὸς τὴν ΡΧ βάσιν, ο<ύτωξ τὸ τοῦ ΔΨ στερεοῦ <ύψος πρὸς τὸ τοῦ ΒΤ στερεοῦ <ύψος. >ίση δὲ ἡ μὲν ΖΚ βάσις τῇ ΕΘ βάσει, ἡ δὲ ΡΧ βάσις τῇ ΝΠ βάσει; >έστιν >άρα ὡς ἡ ΕΘ βάσις πρὸς τὴν ΝΠ βάσιν, ο<ύτωξ τὸ τοῦ ΔΨ στερεοῦ <ύψος πρὸς τὸ τοῦ ΒΤ στερεοῦ <ύψος. τὰ δ' αὐτὰ <ύψη ἔστι τῶν ΔΨ, ΒΤ στερεῶν καὶ τῶν ΔΓ, ΒΑ; >έστιν >άρα ὡς ἡ ΕΘ βάσις πρὸς τὴν ΝΠ βάσιν, ο<ύτωξ τὸ τοῦ ΔΓ στερεοῦ <ύψος πρὸς τὸ τοῦ ΑΒ στερεοῦ <ύψος. τῶν ΑΒ, ΓΔ >άρα στερεῶν παραλληλεπιδῶν ἀντιπεπόνθασιν αἱ βάσεις τοῖς <ύψεσιν.

Πάλιν δὲ τῶν ΑΒ, ΓΔ στερεῶν παραλληλεπιδῶν ἀντιπεπόνθασιν αἱ βάσεις τοῖς <ύψεσιν, καὶ >έστω ὡς ἡ ΕΘ βάσις πρὸς τὴν ΝΠ βάσιν, ο<ύτωξ τὸ τοῦ ΓΔ στερεοῦ <ύψος πρὸς τὸ τοῦ ΑΒ στερεοῦ <ύψος; λέγω, <ὅτι >ίσον ἔστι τὸ ΑΒ στερεὸν τῷ ΓΔ στερεῷ.

Τῶν γὰρ αὐτῶν κατασκευασθέντων, ἐπεὶ ἔστιν ὡς ἡ ΕΘ βάσις πρὸς τὴν ΝΠ βάσιν, ο<ύτωξ τὸ τοῦ ΓΔ στερεοῦ <ύψος

So, let the (straight-lines) standing up, FE , BL , GA , KH , ON , DP , MC , and RQ , not be at right-angles to their bases. And let perpendiculars be drawn to the planes through EH and NQ from points F , G , B , K , O , M , R , and D , and let them have joined the planes at (points) S , T , U , V , W , X , Y , and a (respectively). And let the solids FV and OY be completed. In this case, also, I say that the solids AB and CD being equal, their bases are reciprocally proportional to their heights, and (so) as base EH is to base NQ , so the height of solid CD (is) to the height of solid AB .

Since solid AB is equal to solid CD , but AB is equal to BT . For they are on the same base FK , and (have) the same height [Props. 11.29, 11.30]. And solid CD is equal to DX . For, again, they are on the same base RO , and (have) the same height [Props. 11.29, 11.30]. Solid BT is thus also equal to solid DX . Thus, as base FK (is) to base OR , so the height of solid DX (is) to the height of solid BT (see first part of proposition). And base FK (is) equal to base EH , and base OR to NQ . Thus, as base EH is to base NQ , so the height of solid DX (is) to the height of solid BT . And solids DX , BT are the same height as (solids) DC , BA (respectively). Thus, as base EH is to base NQ , so the height of solid DC (is) to the height of solid AB . Thus, the bases of the parallelepiped solids AB and CD are reciprocally proportional to their heights.

So, again, let the bases of the parallelepiped solids AB and CD be reciprocally proportional to their heights, and (so) let base EH be to base NQ , as the height of solid CD (is) to the height of solid AB . I say that solid AB is equal to

πρὸς τὸ τοῦ AB στερεοῦ <ύψος, >ίση δὲ ἡ μὲν EO βάσις τῇ ZK βάσει, ἡ δὲ NP τῇ XP , >έστιν >άρα ὥς ἡ ZK βάσις πρὸς τὴν XP βάσιν, ο<ύτωξ τὸ τοῦ $\Gamma\Delta$ στερεοῦ <ύψος πρὸς τὸ τοῦ AB στερεοῦ <ύψος. τὰ δ' αὐτὰ <ύψη ἐστί τῶν AB , $\Gamma\Delta$ στερεῶν καὶ τῶν BT , $\Delta\Psi$; >έστιν >άρα ὥς ἡ ZK βάσις πρὸς τὴν XP βάσιν, ο<ύτωξ τὸ τοῦ $\Delta\Psi$ στερεοῦ <ύψος πρὸς τὸ τοῦ BT στερεοῦ <ύψος. τῶν BT , $\Delta\Psi$ >άρα στερεῶν παραλληλεπίπεδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς <ύψεσιν; >ίσον >άρα ἐστί τὸ BT στερεὸν τῷ $\Delta\Psi$ στερεῷ. ἀλλὰ τὸ μὲν BT τῷ BA >ίσον ἐστίν; ἐπὶ τε γὰρ τῆς αὐτῆς βάσεως [εἰσι] τῆς ZK καὶ ὑπὸ τὸ αὐτὸ <ύψος. τὸ δὲ $\Delta\Psi$ στερεὸν τῷ $\Delta\Gamma$ στερεῷ >ίσον ἐστίν. καὶ τὸ AB >άρα στερεὸν τῷ $\Gamma\Delta$ στερεῷ ἐστίν >ίσον; <όπερ >έδει δεῖξαι.

solid CD .

For, with the same construction (as before), since as base EH is to base NQ , so the height of solid CD (is) to the height of solid AB , and base EH (is) equal to base FK , and NQ to OR , thus as base FK is to base OR , so the height of solid CD (is) to the height of solid AB . And solids AB , CD are the same height as (solids) BT , DX (respectively). Thus, as base FK is to base OR , so the height of solid DX (is) to the height of solid BT . Thus, the bases of the parallelepiped solids BT and DX are reciprocally proportional to their heights. Thus, solid BT is equal to solid DX (see first part of proposition). But, BT is equal to BA . For [they are] on the same base FK , and (have) the same height [Props. 11.29, 11.30]. And solid DX is equal to solid DC [Props. 11.29, 11.30]. Thus, solid AB is also equal to solid CD . (Which is) the very thing it was required to show.

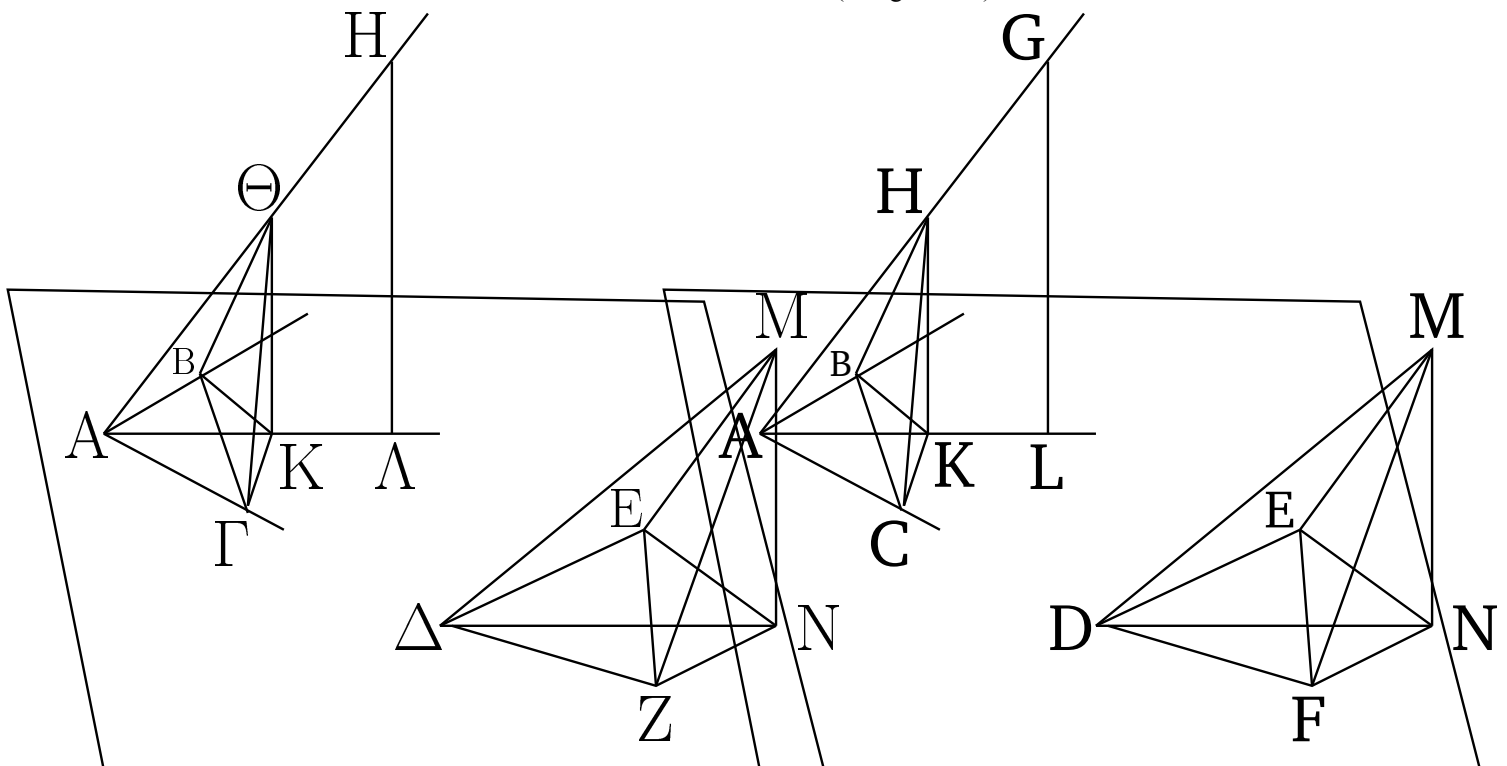
† This proposition assumes that (a) if two parallelepipeds are equal, and have equal bases, then their heights are equal, and (b) if the bases of two equal parallelepipeds are unequal, then that solid which has the lesser base has the greater height.

35.

Ἐὰν >ῶσι δύο γωνίαί ἐπίπεδοι >ίσαι, ἐπὶ δὲ τῶν κορυφῶν αὐτῶν μετέωροι εὐθεῖαι ἐπισταθῶσιν >ίσαξ γωνίαξ περιέθουσιν αὐτὰ μετὰ τῶν ἐξ ἀρθῆξ εὐθειῶν ἑκατέραν ἑκατέρω, ἐπὶ δὲ τῶν μετεώρων ληφθῇ τυθόντα σημεῖα, καὶ ἀπ' αὐτῶν ἐπὶ τὰ ἐπίπεδα, ἐν ο<ίξ εἰσιν αἱ ἐξ ἀρθῆξ γωνίαί, κάθετοι ἀθῶσιν, ἀπὸ δὲ τῶν γενομένων σημείων ἐν τοῖς ἐπίπεδοις ἐπὶ τὰς ἐξ ἀρθῆξ γωνίαξ ἐπιζευθῶσιν εὐθεῖαι, >ίσαξ γωνίαξ περιέχουσι μετὰ τῶν μετεώρων.

Proposition 35

If there are two equal plane angles, and raised straight-lines are stood on the apexes of them, containing equal angles respectively with the original straight-lines (forming the angles), and random points are taken on the raised (straight-lines), and perpendiculars are drawn from them to the planes in which the original angles are, and straight-lines are joined from the points created in the planes to the (vertices of the) original angles, then they will enclose equal angles with the raised (straight-lines).



>Ἐστωσαν δύο γωνίαί εὐθύγραμμοι >ίσαι αἱ ὑπὸ BAC , EDZ , ἀπὸ δὲ τῶν A , Δ σημείων μετέωροι εὐθεῖαι ἐφεστάτωσαν αἱ AH , ΔM >ίσαξ γωνίαξ περιέθουσιν μετὰ τῶν ἐξ ἀρθῆξ εὐθειῶν ἑκατέραν ἑκατέρω, τὴν μὲν ὑπὸ $M\Delta E$ τῇ ὑπὸ HAB , τὴν δὲ ὑπὸ $M\Delta Z$ τῇ ὑπὸ $H\Delta\Gamma$, καὶ εἰλήφθω ἐπὶ τῶν AH , ΔM τυθόντα σημεῖα τὰ H , M , καὶ >ήθθωσαν ἀπὸ τῶν

Let BAC and EDF be two equal rectilinear angles. And let the raised straight-lines AG and DM be stood on points A and D , containing equal angles respectively with the original straight-lines. (That is) MDE (equal) to GAB , and MDF (to) GAC . And let the random points G and M be taken on AG and DM (respectively). And let the GL

H, M σημείων ἐπὶ τὰ διὰ τῶν $BA\Gamma$, EAZ ἐπίπεδα κάθετοι αἱ HA, MN , καὶ συμβαλλέτωσαν τοῖς ἐπιπέδοις κατὰ τὰ Λ, N , καὶ ἐπεζεύθωσαν αἱ AA, ND ; λέγω, <ὅτι >ἴση ἐστὶν ἡ ὑπὸ $HA\Lambda$ γωνία τῇ ὑπὸ MDN γωνία.

Κείσθω τῇ ΔM >ἴση ἡ $A\Theta$, καὶ >ῆθω διὰ τοῦ Θ σημείου τῇ HA παράλληλος ἡ ΘK . ἡ δὲ HA κάθετός ἐστιν ἐπὶ τὸ διὰ τῶν $BA\Gamma$ ἐπίπεδον; καὶ ἡ ΘK >άρα κάθετός ἐστιν ἐπὶ τὸ διὰ τῶν $BA\Gamma$ ἐπίπεδον. >ῆθωσαν ἀπὸ τῶν K, N σημείων ἐπὶ τὰς $AG, \Delta Z, AB, \Delta E$ εὐθείας κάθετοι αἱ $K\Gamma, NZ, KB, NE$, καὶ ἐπεζεύθωσαν αἱ $\Theta\Gamma, \Gamma B, MZ, ZE$. ἐπεὶ τὸ ἀπὸ τῆς ΘA >ἴσον ἐστὶ τοῖς ἀπὸ τῶν $\Theta K, KA$, τῷ δὲ ἀπὸ τῆς KA >ἴσα ἐστὶ τὰ ἀπὸ τῶν $K\Gamma, \Gamma A$, καὶ τὸ ἀπὸ τῆς ΘA >άρα >ἴσον ἐστὶ τοῖς ἀπὸ τῶν $\Theta K, K\Gamma, \Gamma A$. τοῖς δὲ ἀπὸ τῶν $\Theta K, K\Gamma$ >ἴσον ἐστὶ τὸ ἀπὸ τῆς $\Theta\Gamma$; τὸ >άρα ἀπὸ τῆς ΘA >ἴσον ἐστὶ τοῖς ἀπὸ τῶν $\Theta\Gamma, \Gamma A$. ὀρθὴ >άρα ἐστὶν ἡ ὑπὸ $\Theta\Gamma A$ γωνία. διὰ τὰ αὐτὰ δὴ καὶ ἡ ὑπὸ ΔZM γωνία ὀρθὴ ἐστὶν. >ἴση >άρα ἐστὶν ἡ ὑπὸ $A\Gamma\Theta$ γωνία τῇ ὑπὸ ΔZM . >έστι δὲ καὶ ἡ ὑπὸ $\Theta A\Gamma$ τῇ ὑπὸ $M\Delta Z$ >ἴση. δύο δὴ τρίγωνά ἐστι τὰ $M\Delta Z, \Theta A\Gamma$ δύο γωνίας δυσὶ γωνίαις >ἴσας >έθοντα ἑκατέραν ἑκατέρᾳ καὶ μίαν πλευρὰν μιᾷ πλευρᾷ >ἴσην τὴν ὑποτείνουσιν ὑπὸ μίαν τῶν >ἴσων γωνιῶν τὴν ΘA τῇ $M\Delta$; καὶ τὰς λοιπὰς >άρα πλευρὰς ταῖς λοιπαῖς πλευραῖς >ἴσας <έχει ἑκατέραν ἑκατέρᾳ. >ἴση >άρα ἐστὶν ἡ $A\Gamma$ τῇ ΔZ . ὁμοίως δὲ δείχομεν, <ὅτι καὶ ἡ AB τῇ ΔE ἐστὶν >ἴση. ἐπεὶ ο>ὖν >ἴση ἐστὶν ἡ μὲν $A\Gamma$ τῇ ΔZ , ἡ δὲ AB τῇ ΔE , δύο δὴ αἱ $\Gamma A, AB$ δυσὶ ταῖς $Z\Delta, \Delta E$ >ἴσαι εἰσίν. ἀλλὰ καὶ γωνία ἡ ὑπὸ ΓAB γωνία τῇ ὑπὸ $Z\Delta E$ ἐστὶν >ἴση; βάσις >άρα ἡ $B\Gamma$ βάσει τῇ EZ >ἴση ἐστὶ καὶ τὸ τρίγωνον τῷ τριγώνῳ καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις; >ἴση >άρα ἡ ὑπὸ $A\Gamma B$ γωνία τῇ ὑπὸ $\Delta Z E$. >έστι δὲ καὶ ὀρθὴ ἡ ὑπὸ $A\Gamma K$ ὀρθὴ τῇ ὑπὸ $\Delta Z N$ >ἴση; καὶ λοιπὴ >άρα ἡ ὑπὸ $B\Gamma K$ λοιπὴ τῇ ὑπὸ EZN ἐστὶν >ἴση. διὰ τὰ αὐτὰ δὴ καὶ ἡ ὑπὸ $\Gamma B K$ τῇ ὑπὸ $Z E N$ ἐστὶν >ἴση. δύο δὴ τρίγωνά ἐστι τὰ $B\Gamma K, EZN$ [τὰς] δύο γωνίας δυσὶ γωνίαις >ἴσας >έθοντα ἑκατέραν ἑκατέρᾳ καὶ μίαν πλευρὰν μιᾷ πλευρᾷ >ἴσην τὴν πρὸς ταῖς >ἴσαις γωνίαις τὴν $B\Gamma$ τῇ EZ ; καὶ τὰς λοιπὰς >άρα πλευρὰς ταῖς λοιπαῖς πλευραῖς >ἴσας <έχουσιν. >ἴση >άρα ἐστὶν ἡ ΓK τῇ ZN . >έστι δὲ καὶ ἡ $A\Gamma$ τῇ ΔZ >ἴση; δύο δὴ αἱ $A\Gamma, \Gamma K$ δυσὶ ταῖς $\Delta Z, ZN$ >ἴσαι εἰσίν; καὶ ὀρθὰς γωνίας περιέθουσιν. βάσις >άρα ἡ AK βάσει τῇ ΔN >ἴση ἐστὶν. καὶ ἐπεὶ >ἴση ἐστὶν ἡ $A\Theta$ τῇ ΔM , >ἴσον ἐστὶ καὶ τὸ ἀπὸ τῆς $A\Theta$ τῷ ἀπὸ τῆς ΔM . ἀλλὰ τῷ μὲν ἀπὸ τῆς $A\Theta$ >ἴσα ἐστὶ τὰ ἀπὸ τῶν $AK, K\Theta$; ὀρθὴ γὰρ ἡ ὑπὸ $AK\Theta$; τῷ δὲ ἀπὸ τῆς ΔM >ἴσα τὰ ἀπὸ τῶν $\Delta N, NM$; ὀρθὴ γὰρ ἡ ὑπὸ ΔNM ; τὰ >άρα ἀπὸ τῶν $AK, K\Theta$ >ἴσα ἐστὶ τοῖς ἀπὸ τῶν $\Delta N, NM$, <ὦν τὸ ἀπὸ τῆς AK >ἴσον ἐστὶ τῷ ἀπὸ τῆς ΔN ; λοιπὸν >άρα τὸ ἀπὸ τῆς $K\Theta$ >ἴσον ἐστὶ τῷ ἀπὸ τῆς NM ; >ἴση >άρα ἡ ΘK τῇ MN . καὶ ἐπεὶ δύο αἱ $\Theta A, AK$ δυσὶ ταῖς $M\Delta, \Delta N$ >ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ, καὶ βάσις ἡ ΘK βάσει τῇ MN ἔδειθθαι >ἴση, γωνία >άρα ἡ ὑπὸ ΘAK γωνία τῇ ὑπὸ $M\Delta N$ ἐστὶν >ἴση.

Ἐὰν >άρα >ῶσι δύο γωνίαι ἐπίπεδοι >ἴσαι καὶ τὰ ἐχῆς τῆς προτάσεως [<ὅπερ >έδει δεῖχαι].

and MN be drawn from points G and M perpendicular to the planes through BAC and EDF (respectively). And let them have joined the planes at points L and N (respectively). And let LA and ND be joined. I say that angle GAL is equal to angle MDN .

Let AH be made equal to DM . And let HK be drawn through point H parallel to GL . And GL is perpendicular to the plane through BAC . Thus, HK is also perpendicular to the plane through BAC [Prop. 11.8]. And let KC, NF, KB , and NE be drawn from points K and N perpendicular to the straight-lines AC, DF, AB , and DE . And let HC, CB, MF , and FE be joined. Since the (square) on HA is equal to the (sum of the squares) on HK and KA [Prop. 1.47], and the (sum of the squares) on KC and CA is equal to the (square) on KA [Prop. 1.47], thus the (square) on HA is equal to the (sum of the squares) on HK, KC , and CA . And the (square) on HC is equal to the (sum of the squares) on HK and KC [Prop. 1.47]. Thus, the (square) on HA is equal to the (sum of the squares) on HC and CA . Thus, angle HCA is a right-angle [Prop. 1.48]. So, for the same (reasons), angle DFM is also a right-angle. Thus, angle ACH is equal to (angle) DFM . And HAC is also equal to MDF . So, MDF and HAC are two triangles having two angles equal to two angles, respectively, and one side equal to one side—(namely), that subtending one of the equal angles—(that is), HA (equal) to MD . Thus, they will also have the remaining sides equal to the remaining sides, respectively [Prop. 1.26]. Thus, AC is equal to DF . So, similarly, we can show that AB is also equal to DE . Therefore, since AC is equal to DF , and AB to DE , so the two (straight-lines) CA and AB are equal to the two (straight-lines) FD and DE (respectively). But, angle CAB is also equal to angle FDE . Thus, base BC is equal to base EF , and triangle (ACB) to triangle (DFE) , and the remaining angles to the remaining angles (respectively) [Prop. 1.4]. Thus, angle ACB (is) equal to DFE . And the right-angle ACK is also equal to the right-angle DFN . Thus, the remainder BCK is equal to the remainder EFN . So, for the same (reasons), CBK is also equal to FEN . So, BCK and EFN are two triangles having two angles equal to two angles, respectively, and one side equal to one side—(namely), that by the equal angles—(that is), BC (equal) to EF . Thus, they will also have the remaining sides equal to the remaining sides (respectively) [Prop. 1.26]. Thus, CK is equal to FN . And AC (is) also equal to DF . So, the two (straight-lines) AC and CK are equal to the two (straight-lines) DF and FN (respectively). And they enclose right-angles. Thus, base AK is equal to base DN [Prop. 1.4]. And since AH is equal to DM , the (square) on AH is also equal to the (square) on DM . But, the (sum of the squares) on AK and KH is equal to the (square) on AH . For angle AKH (is) a right-angle [Prop. 1.47]. And the (sum of the squares) on DN and NM (is) equal to the square on DM . For angle DNM (is) a right-angle [Prop. 1.47]. Thus, the (sum of the squares) on AK and KH is equal to the (sum of the squares) on DN and NM , of which the (square) on AK is equal to the (square) on DN . Thus, the remaining (square) on KH is equal to the (square) on NM . Thus, HK (is) equal to MN . And since the two (straight-lines) HA and AK are equal to the two (straight-lines) MD and DN , respectively, and base HK

was shown (to be) equal to base MN , angle HAK is thus equal to angle MDN [Prop. 1.8].

Thus, if there are two equal plane angles, and so on of the proposition. [(Which is) the very thing it was required to show].

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, <ὅτι, ἐάν >ῶσι δύο γωνίαι ἐπίπεδοι >ίσαι, ἐπισταθῶσι δὲ ἐπ' αὐτῶν μετέωροι εὐθεῖαι >ίσαι >ίσαξ γωνίαξ περιέθουσαι μετὰ τῶν ἐχ ἄρθῃξ εὐθειῶν ἑκατέραν ἑκατέρα, αἱ ἀπ' αὐτῶν κάθετοι ἀγόμεναι ἐπὶ τὰ ἐπίπεδα, ἐν οἷς εἰσὶν αἱ ἐχ ἄρθῃξ γωνίαι, >ίσαι ἀλλήλαιξ εἰσὶν. <ὅπερ >έδει δεῖχαι.

Corollary

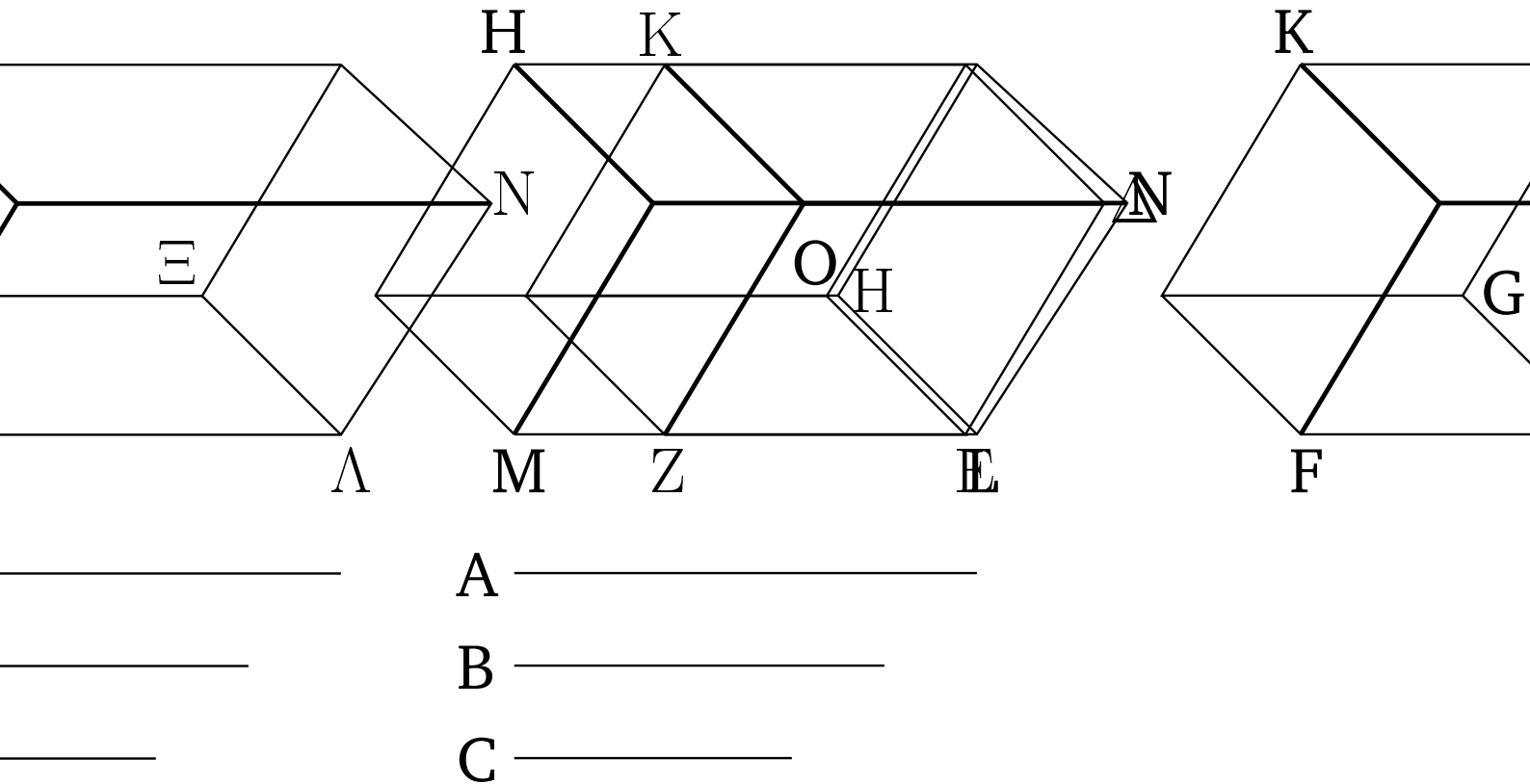
So, it is clear, from this, that if there are two equal plane angles, and equal raised straight-lines are stood on them (at their apexes), containing equal angles respectively with the original straight-lines (forming the angles), then the perpendiculars drawn from (the raised ends of) them to the planes in which the original angles lie are equal to one another. (Which is) the very thing it was required to show.

36.

Ἐάν τρεῖς εὐθεῖαι ἀνάλογον >ῶσιν, τὸ ἐκ τῶν τριῶν στερεὸν παραλληλεπίπεδον >ῖσον ἐστὶ τῷ ἀπὸ τῆς μέσης στερεῷ παραλληλεπίπεδῳ ἰσοπλευρῷ μὲν, ἰσογωνίῳ δὲ τῷ προειρημένῳ.

Proposition 36

If three straight-lines are (continuously) proportional then the parallelepiped solid (formed) from the three (straight-lines) is equal to the equilateral parallelepiped solid on the middle (straight-line which is) equiangular to the aforementioned (parallelepiped solid).



>Ἐστωσαν τρεῖς εὐθεῖαι ἀνάλογον αἱ A, B, Γ , ὥς ἡ A πρὸς τὴν B , οὕτως ἡ B πρὸς τὴν Γ ; λέγω, <ὅτι τὸ ἐκ τῶν A, B, Γ στερεὸν >ῖσον ἐστὶ τῷ ἀπὸ τῆς B στερεῷ ἰσοπλευρῷ μὲν, ἰσογωνίῳ δὲ τῷ προειρημένῳ.

Ἐκκείσθω στερεὰ γωνία ἡ πρὸς τῷ E περιεθομένη ὑπὸ τῶν ὑπὸ $\Delta E H, H E Z, Z E \Delta$, καὶ κείσθω τῇ μὲν B >ῖση ἑκάστη τῶν $\Delta E, H E, E Z$, καὶ συμπληρώσθω τὸ $E K$ στερεὸν παραλληλεπίπεδον, τῇ δὲ A >ῖση ἡ ΛM , καὶ συνεστάτω πρὸς τῇ ΛM εὐθεῖα καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ Λ τῇ πρὸς τῷ E στερεᾷ γωνία >ῖση στερεὰ γωνία ἡ περιεθομένη ὑπὸ τῶν $\Lambda X, X \Lambda M, M \Lambda N$, καὶ κείσθω τῇ μὲν B >ῖση ἡ ΛX , τῇ δὲ Γ >ῖση ἡ ΛN . καὶ ἐπεὶ ἐστὶν ὥς ἡ A πρὸς τὴν B , οὕτως ἡ B πρὸς τὴν Γ , >ῖση δὲ

Let A, B , and C be three (continuously) proportional straight-lines, (such that) as A (is) to B , so B (is) to C . I say that the (parallelepiped) solid (formed) from A, B , and C is equal to the equilateral solid on B (which is) equiangular with the aforementioned (solid).

Let the solid angle at E , contained by DEG, GEF , and FED , be set out. And let DE, GE , and EF each be made equal to B . And let the parallelepiped solid $E K$ be completed. And (let) LM (be made) equal to A . And let the solid angle contained by NLO, OLM , and MLN be constructed on the straight-line LM , and at the point L on it, (so as to be) equal to the solid angle E [Prop. 11.23]. And let LO be made equal to B , and LN equal to C . And since

ή μὲν A τῇ $ΛΜ$, ή δὲ B ἑκατέρα τῶν $ΛΧ$, $ΕΔ$, ή δὲ $Γ$ τῇ $ΛΝ$, >έστιν >άρα ὡς ή $ΛΜ$ πρὸς τὴν $ΕΖ$, ο<ύτωξ ή $ΔΕ$ πρὸς τὴν $ΛΝ$. καὶ περὶ >ίσαξ γωνίαξ τὰξ ὑπὸ $ΝΑΜ$, $ΔΕΖ$ αὶ πλευραὶ ἀντιπεπόνθασιν; >ίσον >άρα ἐστὶ τὸ MN παραλληλόγραμμον τῷ $ΔΖ$ παραλληλογραμμάμμω. καὶ ἐπεὶ δύο γωνίαι ἐπίπεδοι εὐθύγραμμοι >ίσαι εἰσὶν αὶ ὑπὸ $ΔΕΖ$, $ΝΑΜ$, καὶ ἐπ> αὐτῶν μετέωροι εὐθεῖαι ἐφεστᾶσιν αὶ $ΛΧ$, $ΕΗ$ >ίσαι τε ἀλλήλαιξ καὶ >ίσαξ γωνίαξ περιέθουσai μετὰ τῶν ἐχ ἀρθῆξ εὐθειῶν ἑκατέραν ἑκατέρα, αὶ >άρα ἀπὸ τῶν H , X σημείων κάθετοι ἀγόμεναι ἐπὶ τὰ διὰ τῶν $ΝΑΜ$, $ΔΕΖ$ ἐπίπεδα >ίσαι ἀλλήλαιξ εἰσὶν; <ώστε τὰ $ΛΘ$, $ΕΚ$ στερεὰ ὑπὸ τὸ αὐτὸ <ύψοξ ἐστίν. τὰ δὲ ἐπὶ >ίσων βάσεων στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ <ύψοξ >ίσα ἀλλήλοιξ ἐστίν; >ίσον >άρα ἐστὶ τὸ $ΘΛ$ στερεὸν τῷ $ΕΚ$ στερεῷ|. καὶ ἐστὶ τὸ μὲν $ΛΘ$ τὸ ἐκ τῶν A , B , $Γ$ στερεόν, τὸ δὲ $ΕΚ$ τὸ ἀπὸ τῆξ B στερεόν; τὸ >άρα ἐκ τῶν A , B , $Γ$ στερεὸν παραλληλεπίπεδον >ίσον ἐστὶ τῷ ἀπὸ τῆξ B στερεῷ| ἰσοπλεύρῳ μὲν, ἰσογωνίῳ δὲ τῷ προειρημένῳ; <όπερ >έδει δεῖχαι.

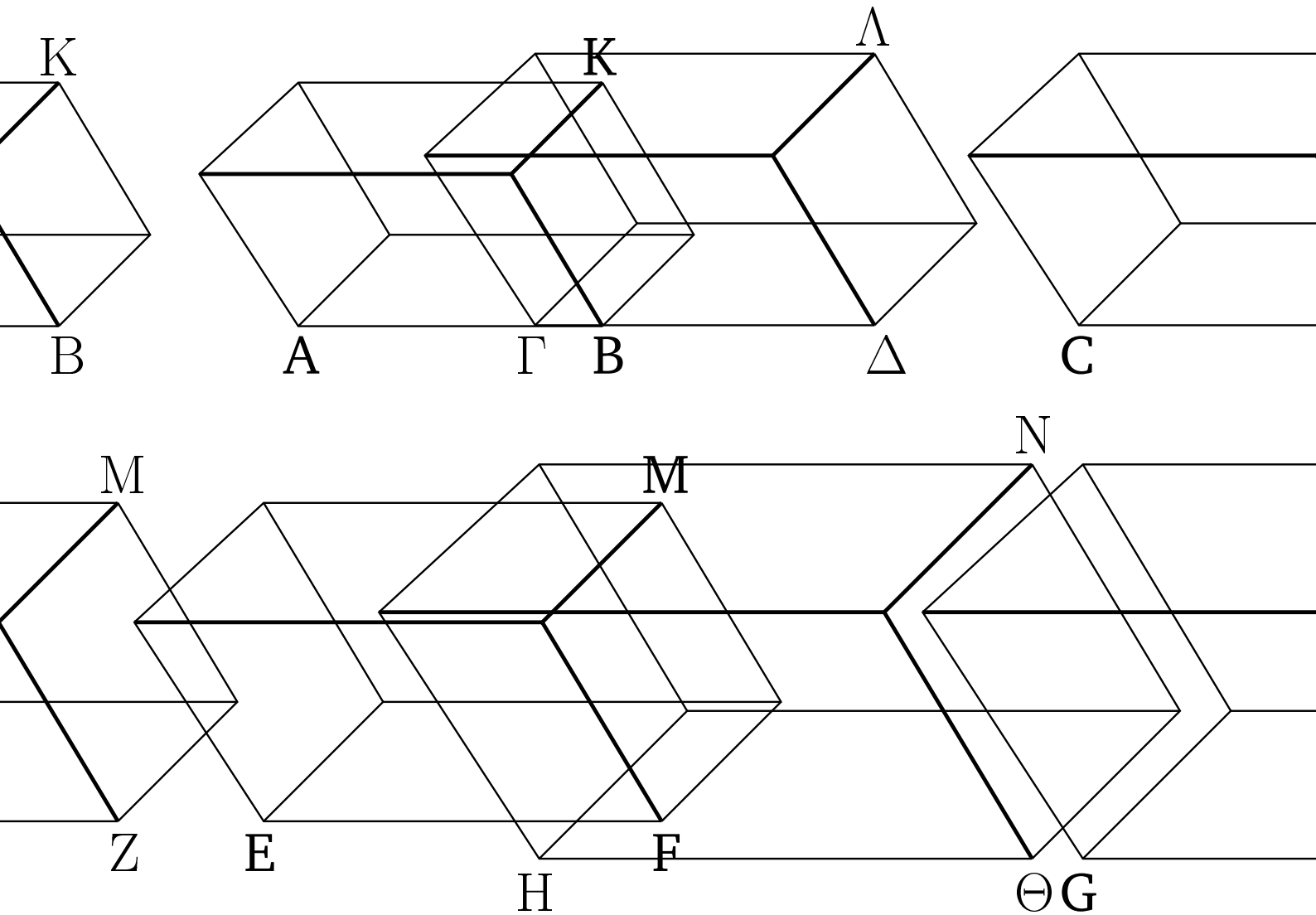
as A (is) to B , so B (is) to C , and A (is) equal to LM , and B to each of LO and ED , and C to LN , thus as LM (is) to EF , so DE (is) to LN . And (so) the sides around the equal angles NLM and DEF are reciprocally proportional. Thus, parallelogram MN is equal to parallelogram DF [Prop. 6.14]. And since the two plane rectilinear angles DEF and NLM are equal, and the raised straight-lines stood on them (at their apexes), LO and EG , are equal to one another, and contain equal angles respectively with the original straight-lines (forming the angles), the perpendiculars drawn from points G and O to the planes through NLM and DEF (respectively) are thus equal to one another [Prop. 11.35 corr.]. Thus, the solids LH and EK (have) the same height. And parallelepiped solids on equal bases, and with the same height, are equal to one another [Prop. 11.31]. Thus, solid HL is equal to solid EK . And LH is the solid (formed) from A , B , and C , and EK the solid on B . Thus, the parallelepiped solid (formed) from A , B , and C is equal to the equilateral solid on B (which is) equiangular with the aforementioned (solid). (Which is) the very thing it was required to show.

37.

Proposition 37[†]

Ἐὰν τέσσαρες εὐθεῖαι ἀνάλογον >ῶσιν, καὶ τὰ ἀπ> αὐτῶν στερεὰ παραλληλεπίπεδα <όμοιά τε καὶ ὁμοίωξ ἀναγραφόμενα ἀνάλογον >έσται; καὶ ἔαν τὰ ἀπ> αὐτῶν στερεὰ παραλληλεπίπεδα <όμοιά τε καὶ ὁμοίωξ ἀναγραφόμενα ἀνάλογον >ῇ|, καὶ αὐταὶ αὶ εὐθεῖαι ἀνάλογον >έσονται.

If four straight-lines are proportional then the similar, and similarly described, parallelepiped solids on them will also be proportional. And if the similar, and similarly described, parallelepiped solids on them are proportional then the straight-lines themselves will be proportional.



>Ἐστῶσαν τέσσαρες εὐθεῖαι ἀνάλογον αἱ AB , $\Gamma\Delta$, EZ , $H\Theta$, ὥς ἡ AB πρὸς τὴν $\Gamma\Delta$, οὐτως ἡ EZ πρὸς τὴν $H\Theta$, καὶ ἀναγεγράφωσαν ἀπὸ τῶν AB , $\Gamma\Delta$, EZ , $H\Theta$ ὁμοίως τε καὶ ὁμοίως κείμενα στερεὰ παραλληλεπίπεδα τὰ KA , $\Lambda\Gamma$, ME , NH ; λέγω, ὅτι ἐστὶν ὥς τὸ KA πρὸς τὸ $\Lambda\Gamma$, οὐτως τὸ ME πρὸς τὸ NH .

Ἐπεὶ γὰρ ὁμοίον ἐστὶ τὸ KA στερεὸν παραλληλεπίπεδον τῷ $\Lambda\Gamma$, τὸ KA ἄρα πρὸς τὸ $\Lambda\Gamma$ τριπλασίονα λόγον ἔχει ἢ πρὸς τὴν $\Gamma\Delta$. διὰ τὰ αὐτὰ δὲ καὶ τὸ ME πρὸς τὸ NH τριπλασίονα λόγον ἔχει ἢ πρὸς τὴν $H\Theta$. καὶ ἐστὶν ὥς ἡ AB πρὸς τὴν $\Gamma\Delta$, οὐτως ἡ EZ πρὸς τὴν $H\Theta$. καὶ ὥς ἄρα τὸ AK πρὸς τὸ $\Lambda\Gamma$, οὐτως τὸ ME πρὸς τὸ NH .

Ἀλλὰ δὴ ἔστω ὥς τὸ AK στερεὸν πρὸς τὸ $\Lambda\Gamma$ στερεόν, οὐτως τὸ ME στερεὸν πρὸς τὸ NH ; λέγω, ὅτι ἐστὶν ὥς ἡ AB εὐθεῖα πρὸς τὴν $\Gamma\Delta$, οὐτως ἡ EZ πρὸς τὴν $H\Theta$.

Ἐπεὶ γὰρ πάλιν τὸ KA πρὸς τὸ $\Lambda\Gamma$ τριπλασίονα λόγον ἔχει ἢ πρὸς τὴν $\Gamma\Delta$, ἔχει δὲ καὶ τὸ ME πρὸς τὸ NH τριπλασίονα λόγον ἢ πρὸς τὴν $H\Theta$, καὶ ἐστὶν ὥς τὸ KA πρὸς τὸ $\Lambda\Gamma$, οὐτως τὸ ME πρὸς τὸ NH , καὶ ὥς ἄρα ἡ AB πρὸς τὴν $\Gamma\Delta$, οὐτως ἡ EZ πρὸς τὴν $H\Theta$.

Ἐὰν ἄρα τέσσαρες εὐθεῖαι ἀνάλογον ᾗσι καὶ τὰ ἐχρῆς τῆς προτάσεως; ὅπερ ἔδει δεῖξαι.

Let AB , CD , EF , and GH , be four proportional straight-lines, (such that) as AB (is) to CD , so EF (is) to GH . And let the similar, and similarly laid out, parallelepiped solids KA , LC , ME and NG be described on AB , CD , EF , and GH (respectively). I say that as KA is to LC , so ME (is) to NG .

For since the parallelepiped solid KA is similar to LC , KA thus has to LC the cubed ratio that AB (has) to CD [Prop. 11.33]. So, for the same (reasons), ME also has to NG the cubed ratio that EF (has) to GH [Prop. 11.33]. And since as AB is to CD , so EF (is) to GH , thus, also, as AK (is) to LC , so ME (is) to NG .

And so let solid AK be to solid LC , as solid ME (is) to NG . I say that as straight-line AB is to CD , so EF (is) to GH .

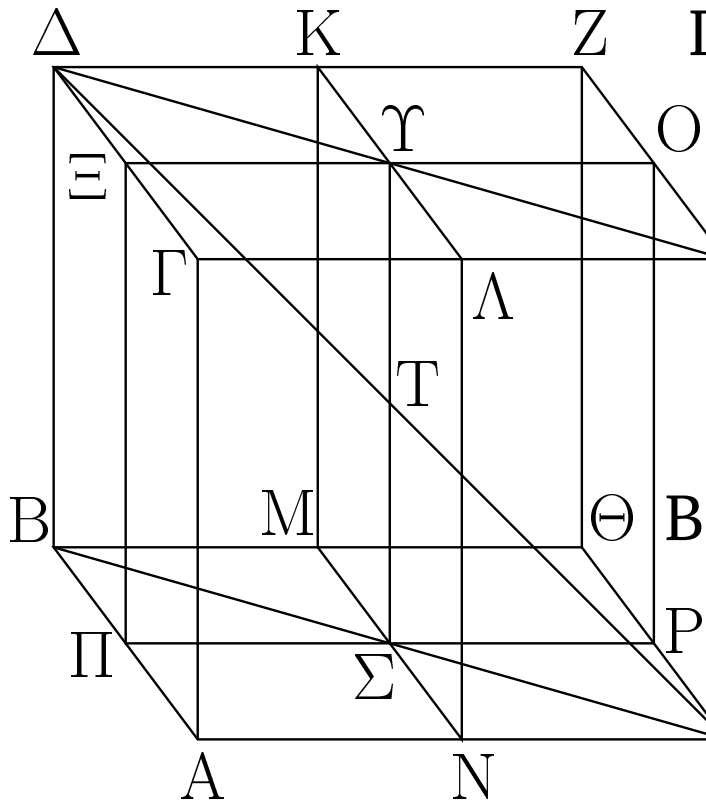
For, again, since KA has to LC the cubed ratio that AB (has) to CD [Prop. 11.33], and ME also has to NG the cubed ratio that EF (has) to GH [Prop. 11.33], and as KA is to LC , so ME (is) to NG , thus, also, as AB (is) to CD , so EF (is) to GH .

Thus, if four straight-lines are proportional, and so on of the proposition. (Which is) the very thing it was required to show.

† This proposition assumes that if two ratios are equal then the cube of the former is also equal to the cube of the latter, and vice versa.

38.

Ἐάν κύβου τῶν ἀπεναντίων ἐπιπέδων αἱ πλευραὶ δίθῃ τμηθῶσιν, διὰ δὲ τῶν τομῶν ἐπίπεδα ἐκβληθῇ, ἡ κοινὴ τομὴ τῶν ἐπιπέδων καὶ ἡ τοῦ κύβου διάμετρος δίθῃ τέμνουσιν ἀλλήλας.



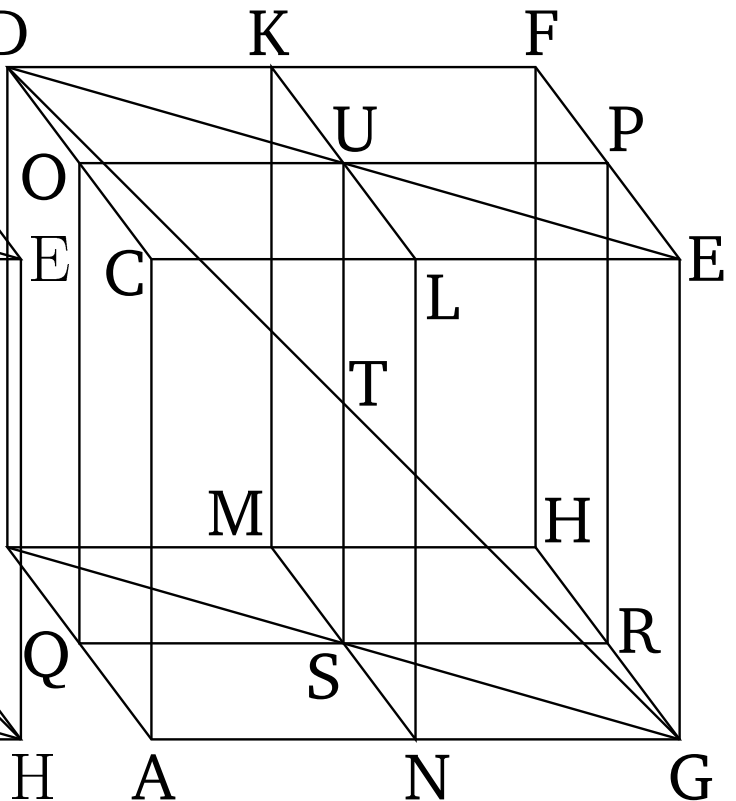
Κύβου γὰρ τοῦ ΑΖ τῶν ἀπεναντίων ἐπιπέδων τῶν ΓΖ, ΑΘ αἱ πλευραὶ δίθῃ τετμήσθωσαν κατὰ τὰ Κ, Λ, Μ, Ν, Χ, Π, Ο, Ρ σημεῖα, διὰ δὲ τῶν τομῶν ἐπίπεδα ἐκβεβλήσθω τὰ ΚΝ, ΧΡ, κοινὴ δὲ τομὴ τῶν ἐπιπέδων >έστω ἡ ΥΣ, τοῦ δὲ ΑΖ κύβου διαγώνιος ἡ ΔΗ. λέγω, <ὅτι> ἵση ἐστὶν ἡ μὲν ΥΤ τῇ ΤΣ, ἡ δὲ ΔΤ τῇ ΤΗ.

Ἐπεξεύθθωσαν γὰρ αἱ ΔΥ, ΥΕ, ΒΣ, ΣΗ. καὶ ἐπεὶ παράλληλός ἐστιν ἡ ΔΧ τῇ ΟΕ, αἱ ἐναλλάχ γωνίαι αἱ ὑπὸ ΔΧΥ, ΥΟΕ >ίσαι ἀλλήλαις εἰσίν. καὶ ἐπεὶ ἵση ἐστὶν ἡ μὲν ΔΧ τῇ ΟΕ, ἡ δὲ ΧΥ τῇ ΥΟ, καὶ γωνία >ίσα ἐπερίθουσιν, βάσις >άρα ἡ ΔΥ τῇ ΥΕ ἐστὶν ἵση, καὶ τὸ ΔΧΥ τρίγωνον τῷ ΟΥΕ τρίγωνῳ ἐστὶν ἴσον καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις >ίσαι; >ίση >άρα ἡ ὑπὸ ΧΥΔ γωνία τῇ ὑπὸ ΟΥΕ γωνία. διὰ δὲ τοῦτο εὐθεῖα ἐστὶν ἡ ΔΥΕ. διὰ τὰ αὐτὰ δὴ καὶ ΒΣΗ εὐθεῖα ἐστὶν, καὶ ἵση ἡ ΒΣ τῇ ΣΗ. καὶ ἐπεὶ ἡ ΓΑ τῇ ΔΒ >ίση ἐστὶ καὶ παράλληλος, καὶ ἡ ΔΒ >άρα τῇ ΕΗ >ίση τέ ἐστὶ καὶ παράλληλος. καὶ ἐπιζευγνύουσιν αὐτὰς εὐθεῖαι αἱ ΔΕ, ΒΗ; παράλληλος >άρα ἐστὶν ἡ ΔΕ τῇ ΒΗ. >ίση >άρα ἡ μὲν ὑπὸ ΕΔΤ γωνία τῇ ὑπὸ ΒΗΤ; ἐναλλάχ γάρ; ἡ δὲ ὑπὸ ΔΤΥ τῇ ὑπὸ ΗΤΣ. δύο δὲ τρίγωνα ἐστὶ τὰ ΔΤΥ, ΗΤΣ τὰς δύο γωνίας ταῖς δυσὶ γωνίαις >ίσας >έθοντα καὶ μίαν πλευρὰν μιᾷ πλευρᾷ >ίσην τὴν ὑποτείνουσιν ὑπὸ μίαν τῶν >ίσων γωνιῶν τὴν ΔΥ τῇ ΗΣ; ἡμίσειαι γάρ εἰσι τῶν ΔΕ, ΒΗ; καὶ τὰς λοιπὰς πλευρὰς ταῖς λοιπαῖς πλευραῖς >ίσας <έχει. >ίση >άρα ἡ μὲν ΔΤ τῇ ΤΗ, ἡ δὲ ΥΤ τῇ ΤΣ.

Ἐάν >άρα κύβου τῶν ἀπεναντίων ἐπιπέδων αἱ πλευραὶ δίθῃ τμηθῶσιν, διὰ δὲ τῶν τομῶν ἐπίπεδα ἐκβληθῇ, ἡ κοινὴ τομὴ τῶν ἐπιπέδων καὶ ἡ τοῦ κύβου διάμετρος δίθῃ τέμνουσιν ἀλλήλας; <όπερ> ἐδει δεῖξαι.

Proposition 38

If the sides of the opposite planes of a cube are cut in half, and planes are produced through the pieces, then the common section of the (latter) planes and the diameter of the cube cut one another in half.



For let the opposite planes CF and AH of the cube AF be cut in half at the points K, L, M, N, O, Q, P , and R . And let the planes KN and OR be produced through the pieces. And let US be the common section of the planes, and DG the diameter of cube AF . I say that UT is equal to TS , and DT to TG .

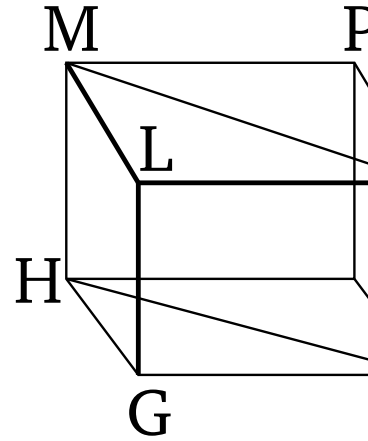
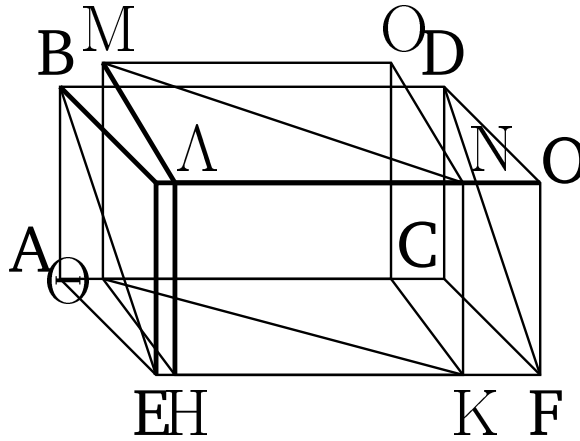
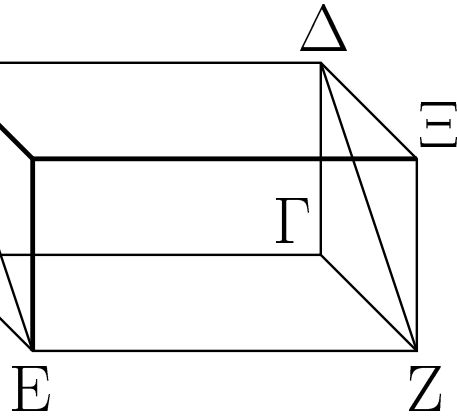
For let DU, UE, BS , and SG be joined. And since DO is parallel to PE , the alternate angles DOU and UPE are equal to one another [Prop. 1.29]. And since DO is equal to PE , and OU to UP , and they contain equal angles, base DU is thus equal to base UE , and triangle DOU is equal to triangle PUE , and the remaining angles (are) equal to the remaining angles [Prop. 1.4]. Thus, angle ODU (is) equal to angle PUE . So, for this (reason), DUE is a straight-line [Prop. 1.14]. So, for the same (reason), BSG is also a straight-line, and BS equal to SG . And since CA is equal and parallel to DB , but CA is also equal and parallel to EG , DB is thus also equal and parallel to EG [Prop. 11.9]. And the straight-lines DE and BG join them. DE is thus parallel to BG [Prop. 1.33]. Thus, angle EDT (is) equal to BGT . For (they are) alternate [Prop. 1.29]. And (angle) DTU (is equal) to GTS [Prop. 1.15]. So, DTU and GTS are two triangles having two angles equal to two angles, and one side equal to one side—(namely), that subtended by one of the equal angles—that is, DU (equal) to GS . For they are halves of DE and BG (respectively). (Thus), they will also have the remaining sides equal to the remaining sides [Prop. 1.26]. Thus, DT (is) equal to TG , and UT to TS .

Thus, if the sides of the opposite planes of a cube are cut in half, and planes are produced through the pieces, then the

common section of the (latter) planes and the diameter of the cube cut one another in half. (Which is) the very thing it was required to show.

39.

Ἐάν >ῆ| δύο πρίσματα ἰσο"υψῆ, καὶ τὸ μὲν >έθῆ| βάσιν παραλληλόγραμμον, τὸ δὲ τρίγωνον, διπλάσιον δὲ >ῆ| τὸ παραλληλόγραμμον τοῦ τριγώνου, >ῖσα >έσται τὰ πρίσματα.



>Ἐστω δύο πρίσματα ἰσο"υψῆ τὰ $ABΓΔEZ$, $HΘΚΛΜΝ$, καὶ τὸ μὲν ἐθέτω βάσιν τὸ AZ παραλληλόγραμμον, τὸ δὲ τὸ $HΘΚ$ τρίγωνον, διπλάσιον δὲ >έστω τὸ AZ παραλληλόγραμμον τοῦ $HΘΚ$ τριγώνου; λέγω, <ὅτι >ῖσον ἐστὶ τὸ $ABΓΔEZ$ πρίσμα τῷ $HΘΚΛΜΝ$ πρίσματι.

Συμπεπληρώσθω γὰρ τὰ AX , HO στερεά. ἐπεὶ διπλάσιόν ἐστὶ τὸ AZ παραλληλόγραμμον τοῦ $HΘΚ$ τριγώνου, >έστι δὲ καὶ τὸ $ΘΚ$ παραλληλόγραμμον διπλάσιον τοῦ $HΘΚ$ τριγώνου, >ῖσον >άρα ἐστὶ τὸ AZ παραλληλόγραμμον τῷ $ΘΚ$ παραλληλόγραμμῳ. τὰ δὲ ἐπὶ >ῖσων βάσεων >όντα στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ <ύψοξ >ῖσα ἀλλήλοις ἐστίν; >ῖσον >άρα ἐστὶ τὸ AX στερεὸν τῷ HO στερεῷ. καὶ ἐστὶ τοῦ μὲν AX στερεοῦ <ήμισυ τὸ $ABΓΔEZ$ πρίσμα, τοῦ δὲ HO στερεοῦ <ήμισυ τὸ $HΘΚΛΜΝ$ πρίσμα; >ῖσον >άρα ἐστὶ τὸ $ABΓΔEZ$ πρίσμα τῷ $HΘΚΛΜΝ$ πρίσματι. Ἐάν >άρα >ῆ| δύο πρίσματα ἰσο"υψῆ, καὶ τὸ μὲν >έθῆ| βάσιν παραλληλόγραμμον, τὸ δὲ τρίγωνον, διπλάσιον δὲ >ῆ| τὸ παραλληλόγραμμον τοῦ τριγώνου, >ῖσα >έσται τὰ πρίσματα; <όπερ >έδει δεῖχαι.

Proposition 39

If there are two equal height prisms, and one has a parallelogram, and the other a triangle, (as a) base, and the parallelogram is double the triangle, then the prisms will be equal.

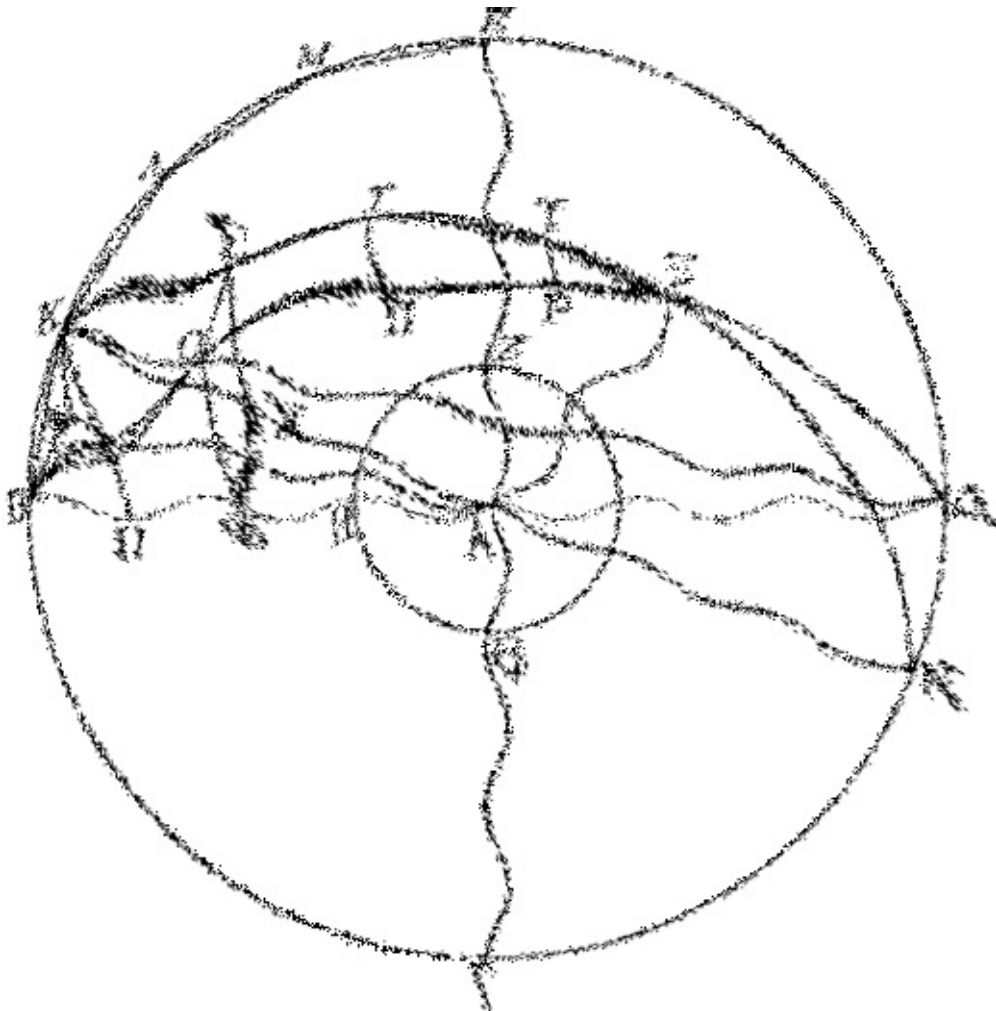
Let $ABCDEF$ and $GHKLMN$ be two equal height prisms, and let the former have the parallelogram AF , and the latter the triangle GHK , as a base. And let parallelogram AF be twice triangle GHK . I say that prism $ABCDEF$ is equal to prism $GHKLMN$.

For let the solids AO and GP be completed. Since parallelogram AF is double triangle GHK , and parallelogram HK is also double triangle GHK [Prop. 1.34], parallelogram AF is thus equal to parallelogram HK . And parallelepiped solids which are on equal bases, and (have) the same height, are equal to one another [Prop. 11.31]. Thus, solid AO is equal to solid GP . And prism $ABCDEF$ is half of solid AO , and prism $GHKLMN$ half of solid GP [Prop. 11.28]. Prism $ABCDEF$ is thus equal to prism $GHKLMN$.

Thus, if there are two equal height prisms, and one has a parallelogram, and the other a triangle, (as a) base, and the parallelogram is double the triangle, then the prisms are equal. (Which is) the very thing it was required to show.

ELEMENTS BOOK 12[†]

Proportional Stereometry[†]



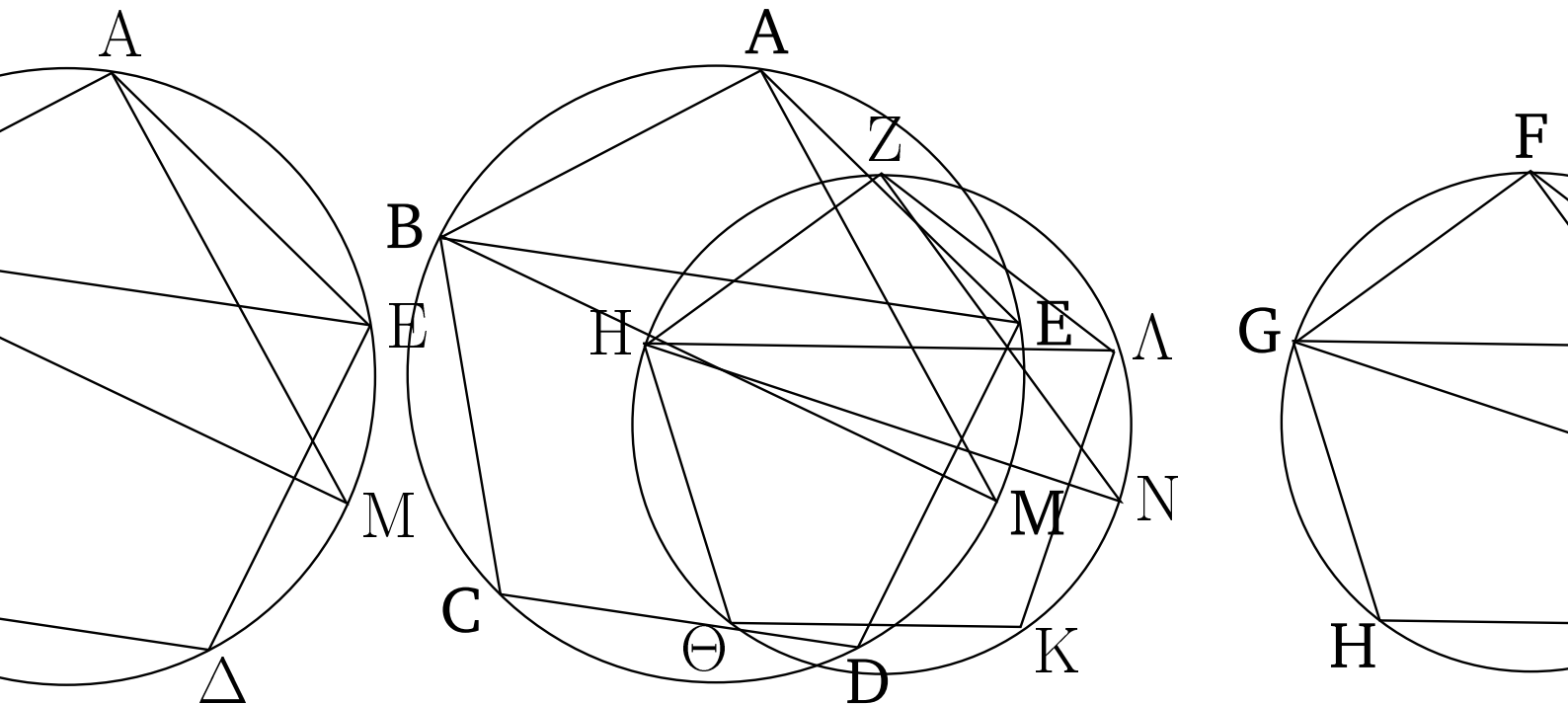
[†]The novel feature of this book is the use of the so-called method of exhaustion (see Prop. 10.1), a precursor to integration which is generally attributed to Eudoxus of Cnidus.

1.

Τὰ ἐν τοῖς κύκλοις ὁμοία πολύγωνα πρὸς ἀλλήλα ἐστὶν ὡς τὰ ἀπὸ τῶν διαμέτρων τετράγωνα.

Proposition 1

Similar polygons (inscribed) in circles are to one another as the squares on the diameters (of the circles).



>Ἐστωσαν κύκλοι οἱ ΑΒΓ, ΖΗΘ, καὶ ἐν αὐτοῖς ὁμοία πολύγωνα >έστω τὰ ΑΒΓΔΕ, ΖΗΘΚΛ, διαμέτροι δὲ τῶν κύκλων >έστωσαν ΒΜ, ΗΝ; λέγω, <ὅτι ἐστὶν ὡς τὸ ἀπὸ τῆς ΒΜ τετράγωνον πρὸς τὸ ἀπὸ τῆς ΗΝ τετράγωνον, ο<ύτως τὸ ΑΒΓΔΕ πολύγωνον πρὸς τὸ ΖΗΘΚΛ πολύγωνον.

Ἐπεζεύθωσαν γὰρ αἱ ΒΕ, ΑΜ, ΗΛ, ΖΝ. καὶ ἐπεὶ ὁμοῖον τὸ ΑΒΓΔΕ πολύγωνον τῷ ΖΗΘΚΛ πολυγώνῳ, >ῖση ἐστὶ καὶ ἡ ὑπὸ ΒΑΕ γωνία τῇ ὑπὸ ΗΖΛ, καὶ ἐστὶν ὡς ἡ ΒΑ πρὸς τὴν ΑΕ, ο<ύτως ἡ ΗΖ πρὸς τὴν ΖΛ. δύο δὲ τρίγωνα ἐστὶ τὰ ΒΑΕ, ΗΖΛ μίαν γωνίαν μῖα γωνία >ῖσην >έθοντα τὴν ὑπὸ ΒΑΕ τῇ ὑπὸ ΗΖΛ, περὶ δὲ τὰς >ῖσα γωνίας τὰς πλευράς ἀνάλογον; ἰσογώνιον >άρα ἐστὶ τὸ ΑΒΕ τρίγωνον τῷ ΖΗΛ τριγώνῳ. >ῖση >άρα ἐστὶν ἡ ὑπὸ ΑΕΒ γωνία τῇ ὑπὸ ΖΛΗ. ἀλλ' ἡ μὲν ὑπὸ ΑΕΒ τῇ ὑπὸ ΑΜΒ ἐστὶν >ῖση; ἐπὶ γὰρ τῆς αὐτῆς περιφερείας βεβήκασιν; ἡ δὲ ὑπὸ ΖΛΗ τῇ ὑπὸ ΖΝΗ; καὶ ἡ ὑπὸ ΑΜΒ >άρα τῇ ὑπὸ ΖΝΗ ἐστὶν >ῖση. >έστι δὲ καὶ ὀρθὴ ἡ ὑπὸ ΒΑΜ ὀρθὴ τῇ ὑπὸ ΗΖΝ >ῖση; καὶ ἡ λοιπὴ >άρα τῇ λοιπῇ ἐστὶν >ῖση. ἰσογώνιον >άρα ἐστὶ τὸ ΑΒΜ τρίγωνον τῷ ΖΗΝ τριγώνῳ. ἀνάλογον >άρα ἐστὶν ὡς ἡ ΒΜ πρὸς τὴν ΗΝ, ο<ύτως ἡ ΒΑ πρὸς τὴν ΗΖ. ἀλλὰ τοῦ μὲν τῆς ΒΜ πρὸς τὴν ΗΝ λόγον διπλασίον ἐστὶν ὁ τοῦ ἀπὸ τῆς ΒΜ τετραγώνου πρὸς τὸ ἀπὸ τῆς ΗΝ τετράγωνον, τοῦ δὲ τῆς ΒΑ πρὸς τὴν ΗΖ διπλασίον ἐστὶν ὁ τοῦ ΑΒΓΔΕ πολυγώνου πρὸς τὸ ΖΗΘΚΛ πολύγωνον; καὶ ὡς >άρα τὸ ἀπὸ τῆς ΒΜ τετράγωνον πρὸς τὸ ἀπὸ τῆς ΗΝ τετράγωνον, ο<ύτως τὸ ΑΒΓΔΕ πολύγωνον πρὸς τὸ ΖΗΘΚΛ πολύγωνον.

Τὰ >άρα ἐν τοῖς κύκλοις ὁμοία πολύγωνα πρὸς ἀλλήλα ἐστὶν ὡς τὰ ἀπὸ τῶν διαμέτρων τετράγωνα; ὁπερ >έδει δεῖχαι.

Let ABC and FGH be circles, and let $ABCDE$ and $FGHKL$ be similar polygons (inscribed) in them (respectively), and let BM and GN be the diameters of the circles (respectively). I say that as the square on BM is to the square on GN , so polygon $ABCDE$ (is) to polygon $FGHKL$.

For let BE , AM , GL , and FN be joined. And since polygon $ABCDE$ (is) similar to polygon $FGHKL$, angle BAE is also equal to (angle) GFL , and as BA is to AE , so GF (is) to FL [Def. 6.1]. So, BAE and GFL are two triangles having one angle equal to one angle, (namely), BAE (equal) to GFL , and the sides around the equal angles proportional. Triangle ABE is thus equiangular with triangle FGL [Prop. 6.6]. Thus, angle AEB is equal to (angle) FLG . But, AEB is equal to AMB , and FLG to FNG , for they stand on the same circumference [Prop. 3.27]. Thus, AMB is also equal to FNG . And the right-angle BAM is also equal to the right-angle GFN [Prop. 3.31]. Thus, the remaining (angle) is also equal to the remaining (angle) [Prop. 1.32]. Thus, triangle ABM is equiangular with triangle FGN . Thus, proportionally, as BM is to GN , so BA (is) to GF [Prop. 6.4]. But, the (ratio) of the square on BM to the square on GN is the square of the ratio of BM to GN , and the (ratio) of polygon $ABCDE$ to polygon $FGHKL$ is the square of the (ratio) of BA to GF [Prop. 6.20]. And, thus, as the square on BM (is) to the square on GN , so polygon $ABCDE$ (is) to polygon $FGHKL$.

Thus, similar polygons (inscribed) in circles are to one another as the squares on the diameters (of the circles). (Which is) the very thing it was required to show.

2.

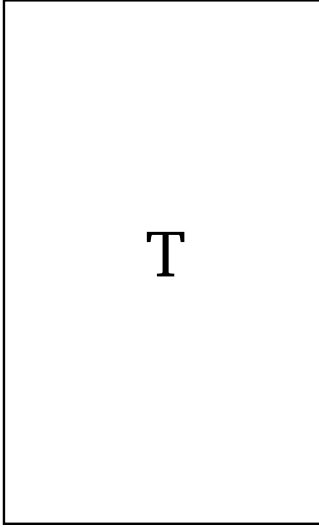
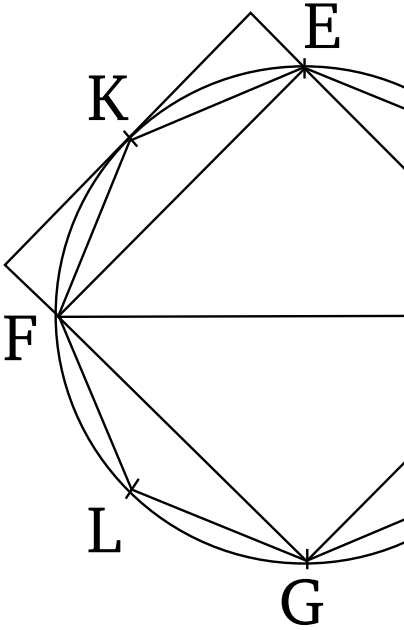
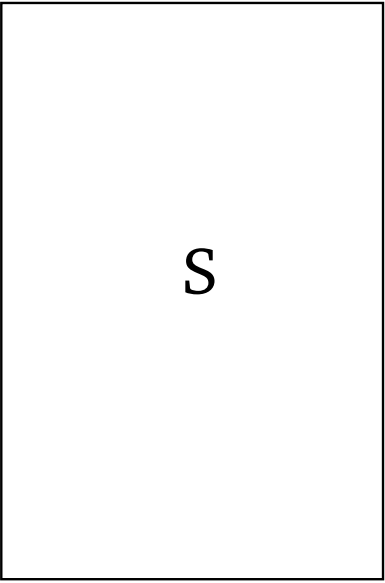
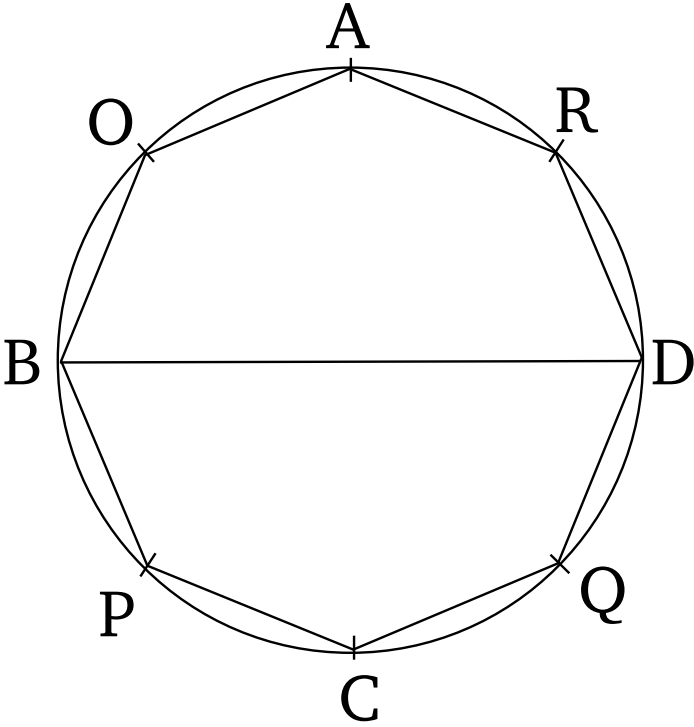
Proposition 2

Οἱ κύκλοι πρὸς ἀλλήλους εἰσὶν ὡς τὰ ἀπὸ τῶν διαμέτρων

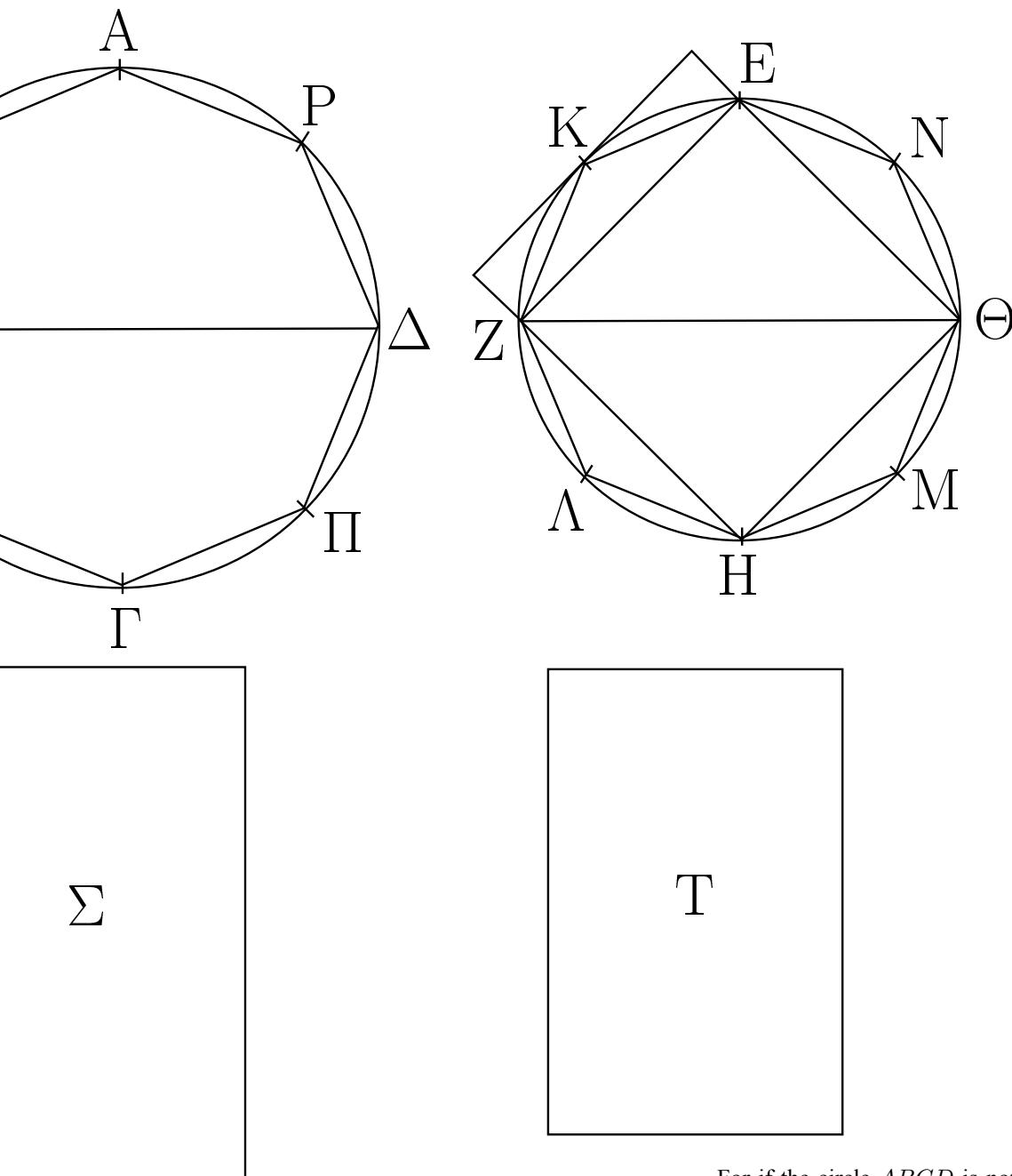
Circles are to one another as the squares on (their) diameters.

τετράγωνα.
>Ἐστωσαν κύκλοι οἱ $AB\Gamma\Delta$, $EZH\Theta$, διαμέτροι δὲ αὐτῶν
[>ἔστωσαν] αἱ $B\Delta$, $Z\Theta$; λέγω, <ὅτι ἔστιν ὡς ὁ $AB\Gamma\Delta$ κύκλος
πρὸς τὸν $EZH\Theta$ κύκλον, ο<ύτως τὸ ἀπὸ τῆς $B\Delta$ τετράγωνον

Let $ABCD$ and $EFGH$ be circles, and [let] BD and FH
[be] their diameters. I say that as circle $ABCD$ is to circle
 $EFGH$, so the square on BD (is) to the square on FH .



πρὸς τὸ ἀπὸ τῆς $Z\Theta$ τετράγωνον.



Εἰ γὰρ μὴ ἔστιν ὥς ὁ ΑΒΓΔ κύκλος πρὸς τὸν ΕΖΗΘ, οὐτως
τὸ ἀπὸ τῆς ΒΔ τετράγωνον πρὸς τὸ ἀπὸ τῆς ΖΘ, ἔσται
ὥς τὸ ἀπὸ τῆς ΒΔ πρὸς τὸ ἀπὸ τῆς ΖΘ, οὐτως ὁ ΑΒΓΔ
κύκλος ᾗτοι πρὸς ἑλάσσον τι τοῦ ΕΖΗΘ κύκλου θωρίον
ᾗ πρὸς μείζον. ἔστω πρότερον πρὸς ἑλάσσον τὸ Σ. καὶ
ἐγγεγράφθω εἰς τὸν ΕΖΗΘ κύκλον τετράγωνον τὸ ΕΖΗΘ.
τὸ δὴ ἐγγεγραμμένον τετράγωνον μείζον ἔστιν ᾗ τὸ ἥμισυ
τοῦ ΕΖΗΘ κύκλου, ἐπειδήπερ ἔαν διὰ τῶν Ε, Ζ, Η, Θ
σημείων ἐφαπτομένας [εὐθείας] τοῦ κύκλου ἀγάγωμεν, τοῦ
περιγραφομένου περὶ τὸν κύκλον τετραγώνου ἥμισυ ἔστι
τὸ ΕΖΗΘ τετράγωνον, τοῦ δὲ περιγραφέντος τετραγώνου
ἐλάττων ἔστιν ὁ κύκλος; ὥστε τὸ ΕΖΗΘ ἐγγεγραμμένομην
τετράγωνον μείζον ἔστι τοῦ ἡμίσεως τοῦ ΕΖΗΘ κύκλου.
τετμήσθωσαν δὶθα αἱ ΕΖ, ΖΗ, ΗΘ, ΘΕ περιφέρειαι κατὰ τὰ
Κ, Λ, Μ, Ν σημεία, καὶ ἐπεζεύθωσαν αἱ ΕΚ, ΚΖ, ΖΛ,
ΛΗ, ΗΜ, ΜΘ, ΘΝ, ΝΕ; καὶ ἕκαστον ἄρα τῶν ΕΚΖ,
ΖΛΗ, ΗΜΘ, ΘΝΕ τριγώνων μείζον ἔστιν ᾗ τὸ ἥμισυ
τοῦ καθ' ἑαυτὸ τμήματος τοῦ κύκλου, ἐπειδήπερ ἔαν
διὰ τῶν Κ, Λ, Μ, Ν σημείων ἐφαπτομένας τοῦ κύκλου

For if the circle $ABCD$ is not to the (circle) $EFGH$, as
the square on BD (is) to the (square) on FH , then as the
(square) on BD (is) to the (square) on FH , so circle $ABCD$
will be to some area either less than, or greater than, circle
 $EFGH$. Let it, first of all, be (in that ratio) to (some) lesser
(area), S . And let the square $EFGH$ be inscribed in cir-
cle $EFGH$ [Prop. 4.6]. So the inscribed square is greater
than half of circle $EFGH$, inasmuch as if we draw tan-
gents to the circle through the points E, F, G , and H , then
square $EFGH$ is half of the square circumscribed about
the circle [Prop. 1.47], and the circle is less than the cir-
cumscribed square. Hence, the inscribed square $EFGH$
is greater than half of circle $EFGH$. Let the circumfer-
ences EF, FG, GH , and HE be cut in half at points $K,$
 L, M , and N (respectively), and let $EK, KF, FL, LG,$
 GM, MH, HN , and NE be joined. And, thus, each of the
triangles EKF, FLG, GMH , and HNE is greater than
half of the segment of the circle about it, inasmuch as if we
draw tangents to the circle through points K, L, M , and
 N , and complete the parallelograms on the straight-lines

ἀγάγωμεν καὶ ἀναπληρώσωμεν τὰ ἐπὶ τῶν EZ , ZH , $H\Theta$, ΘE εὐθειῶν παραλληλόγραμμα, <έκαστον τῶν EKZ , $Z\Lambda H$, $H\Lambda\Theta$, ΘNE τριγώνων >ήμισυ >έσται τοῦ καθ> ἐαυτὸ παραλληλογράμμου, ἀλλὰ τὸ καθ> ἐαυτὸ τμήμα >έλαττον ἐστὶ τοῦ παραλληλογράμμου; <ὥστε <έκαστον τῶν EKZ , $Z\Lambda H$, $H\Lambda\Theta$, ΘNE τριγώνων μείζον ἐστὶ τοῦ ἡμίσεως τοῦ καθ> ἐαυτὸ τμήματος τοῦ κύκλου. τέμνοντες δὴ τὰς ὑπολειπομένας περιφερείας δίθαι καὶ ἐπιζευγνύντες εὐθείαις καὶ τοῦτο αἰ ποιοῦντες καταλείβομεν τινα ἀποτμήματα τοῦ κύκλου, <ὰ >έσται ἐλάσσονα τῆς ὑπεροθῆς, <ῆ| ὑπερέθει ὁ $EZH\Theta$ κύκλος τοῦ Σ θωρίου. ἐδείθη γὰρ ἐν τῷ πρώτῳ θεωρήματι τοῦ δεκάτου βιβλίου, <ὅτι δύο μεγεθῶν ἀνίσων ἐκκειμένων, ἔαν ἀπὸ τοῦ μείζονος ἀφαιρεθῇ| μείζον >ῆ τὸ <ήμισυ καὶ τοῦ καταλειπομένου μείζον >ῆ τὸ <ήμισυ, καὶ τοῦτο αἰ γίγνεται, λειφθήσεται τι μέγεθος, <ὸ >έσται >έλασσον τοῦ ἐκκειμένου ἐλάσσονος μεγέθους. λειφθῶ ο>ν, καὶ >έστω τὰ ἐπὶ τῶν EK , KZ , $Z\Lambda$, ΛH , $H\Lambda$, $\Lambda\Theta$, ΘN , NE τμήματα τοῦ $EZH\Theta$ κύκλου ἐλάττονα τῆς ὑπεροθῆς, <ῆ| ὑπερέθει ὁ $EZH\Theta$ κύκλος τοῦ Σ θωρίου. λοιπὸν >άρα τὸ $EKZ\Lambda H\Lambda\Theta N$ πολύγωνον μείζον ἐστὶ τοῦ Σ θωρίου. ἐγγεγράφθω καὶ εἰς τὸν $AB\Gamma\Delta$ κύκλον τῷ $EKZ\Lambda H\Lambda\Theta N$ πολυγώνῳ <όμοιον πολύγωνον τὸ $AXBO\Gamma\Pi\Delta P$; >έστιν >άρα ὡς τὸ ἀπὸ τῆς BA τετράγωνον πρὸς τὸ ἀπὸ τῆς $Z\Theta$ τετράγωνον, ο<ύτως τὸ $AXBO\Gamma\Pi\Delta P$ πολύγωνον πρὸς τὸ $EKZ\Lambda H\Lambda\Theta N$ πολύγωνον. ἀλλὰ καὶ ὡς τὸ ἀπὸ τῆς BA τετράγωνον πρὸς τὸ ἀπὸ τῆς $Z\Theta$, ο<ύτως ὁ $AB\Gamma\Delta$ κύκλος πρὸς τὸ Σ θωρίον; καὶ ὡς >άρα ὁ $AB\Gamma\Delta$ κύκλος πρὸς τὸ Σ θωρίον, ο<ύτως τὸ $AXBO\Gamma\Pi\Delta P$ πολύγωνον πρὸς τὸ $EKZ\Lambda H\Lambda\Theta N$ πολύγωνον; ἐναλλάξ >άρα ὡς ὁ $AB\Gamma\Delta$ κύκλος πρὸς τὸ ἐν αὐτῷ πολύγωνον, ο<ύτως τὸ Σ θωρίον πρὸς τὸ $EKZ\Lambda H\Lambda\Theta N$ πολύγωνον. μείζον δὲ ὁ $AB\Gamma\Delta$ κύκλος τοῦ ἐν αὐτῷ πολυγώνου; μείζον >άρα καὶ τὸ Σ θωρίον τοῦ $EKZ\Lambda H\Lambda\Theta N$ πολυγώνου. ἀλλὰ καὶ >έλαττον; <όπερ ἐστὶν ἀδύνατον. οὐκ >άρα ἐστὶν ὡς τὸ ἀπὸ τῆς BA τετράγωνον πρὸς τὸ ἀπὸ τῆς $Z\Theta$, ο<ύτως ὁ $AB\Gamma\Delta$ κύκλος πρὸς >έλασσόν τι τοῦ $EZH\Theta$ κύκλου θωρίου. ὁμοίως δὴ δείχομεν, <ὅτι οὐδὲ ὡς τὸ ἀπὸ $Z\Theta$ πρὸς τὸ ἀπὸ BA , ο<ύτως ὁ $EZH\Theta$ κύκλος πρὸς >έλασσόν τι τοῦ $AB\Gamma\Delta$ κύκλου θωρίου.

Λέγω δὴ, <ὅτι οὐδὲ ὡς τὸ ἀπὸ τῆς BA πρὸς τὸ ἀπὸ τῆς $Z\Theta$, ο<ύτως ὁ $AB\Gamma\Delta$ κύκλος πρὸς μείζον τι τοῦ $EZH\Theta$ κύκλου θωρίου.

Εἰ γὰρ δυνατόν, >έστω πρὸς μείζον τὸ Σ . ἀνάπαλιν >άρα [ἐστὶν] ὡς τὸ ἀπὸ τῆς $Z\Theta$ τετράγωνον πρὸς τὸ ἀπὸ τῆς BA , ο<ύτως τὸ Σ θωρίον πρὸς τὸν $AB\Gamma\Delta$ κύκλον. ἀλλ> ὡς τὸ Σ θωρίον πρὸς τὸν $AB\Gamma\Delta$ κύκλον, ο<ύτως ὁ $EZH\Theta$ κύκλος πρὸς >έλαττον τι τοῦ $AB\Gamma\Delta$ κύκλου θωρίου; καὶ ὡς >άρα τὸ ἀπὸ τῆς $Z\Theta$ πρὸς τὸ ἀπὸ τῆς BA , ο<ύτως ὁ $EZH\Theta$ κύκλος πρὸς >έλασσόν τι τοῦ $AB\Gamma\Delta$ κύκλου θωρίου; <όπερ ἀδύνατον ἐδείθη. οὐκ >άρα ἐστὶν ὡς τὸ ἀπὸ τῆς BA τετράγωνον πρὸς τὸ ἀπὸ τῆς $Z\Theta$, ο<ύτως ὁ $AB\Gamma\Delta$ κύκλος πρὸς μείζον τι τοῦ $EZH\Theta$ κύκλου θωρίου. ἐδείθη δέ, <ὅτι οὐδὲ πρὸς >έλασσον; >έστιν >άρα ὡς τὸ ἀπὸ τῆς BA τετράγωνον πρὸς τὸ ἀπὸ τῆς $Z\Theta$, ο<ύτως ὁ $AB\Gamma\Delta$ κύκλος πρὸς τὸν $EZH\Theta$ κύκλον.

Οἱ >άρα κύκλοι πρὸς ἀλλήλους εἰσὶν ὡς τὰ ἀπὸ τῶν διαμέτρων τετράγωνα; <όπερ >έδει δεῖχαι.

EF , FG , GH , and HE , then each of the triangles EKF , FLG , GMH , and HNE will be half of the parallelogram about it, but the segment about it is less than the parallelogram. Hence, each of the triangles EKF , FLG , GMH , and HNE is greater than half of the segment of the circle about it. So, by cutting the circumferences remaining behind in half, and joining straight-lines, and doing this continually, we will (eventually) leave behind some segments of the circle whose (sum) will be less than the excess by which circle $EFGH$ exceeds the area S . For we showed in the first theorem of the tenth book that if two unequal magnitudes are laid out, and if (a part) greater than a half is subtracted from the greater, and (if from) the remainder (a part) greater than a half (is subtracted), and this happens continually, then some magnitude will (eventually) be left which will be less than the lesser laid out magnitude [Prop. 10.1]. Therefore, let the (segments) be left, and let the (sum of the) segments of the circle $EFGH$ on EK , KF , FL , LG , GM , MH , HN , and NE be less than the excess by which circle $EFGH$ exceeds area S . Thus, the remaining polygon $EKFLGMHN$ is greater than area S . And let the polygon $AOBPCQDR$, similar to the polygon $EKFLGMHN$, be inscribed in circle $ABCD$. Thus, as the square on BD is to the square on FH , so polygon $AOBPCQDR$ (is) to polygon $EKFLGMHN$ [Prop. 12.1]. But, also, as the square on BD (is) to the square on FH , so circle $ABCD$ (is) to area S . And, thus, as circle $ABCD$ (is) to area S , so polygon $AOBPCQDR$ (is) to polygon $EKFLGMHN$ [Prop. 5.11]. Thus, alternately, as circle $ABCD$ (is) to the polygon (inscribed) within it, so area S (is) to polygon $EKFLGMHN$ [Prop. 5.16]. And circle $ABCD$ (is) greater than the polygon (inscribed) within it. Thus, area S is also greater than polygon $EKFLGMHN$. But, (it is) also less. The very thing is impossible. Thus, the square on BD is not to the (square) on FH , as circle $ABCD$ (is) to some area less than circle $EFGH$. So, similarly, we can show that the (square) on FH (is) not to the (square) on BD as circle $EFGH$ (is) to some area less than circle $ABCD$ either.

So, I say that neither (is) the (square) on BD to the (square) on FH , as circle $ABCD$ (is) to some area greater than circle $EFGH$.

For, if possible, let it be (in that ratio) to (some) greater (area), S . Thus, inversely, as the square on FH [is] to the (square) on DB , so area S (is) to circle $ABCD$ [Prop. 5.7 corr.]. But, as area S (is) to circle $ABCD$, so circle $EFGH$ (is) to some area less than circle $ABCD$ (see lemma). And, thus, as the (square) on FH (is) to the (square) on BD , so circle $EFGH$ (is) to some area less than circle $ABCD$ [Prop. 5.11]. The very thing was shown (to be) impossible. Thus, as the square on BD is to the (square) on FH , so circle $ABCD$ (is) not to some area greater than circle $EFGH$. And it was shown that neither (is it in that ratio) to (some) lesser (area). Thus, as the square on BD is to the (square) on FH , so circle $ABCD$ (is) to circle $EFGH$.

Thus, circles are to one another as the squares on (their) diameters. (Which is) the very thing it was required to show.

Λήμμα.

Lemma

Λέγω δὴ, <ὅτι τοῦ Σ θωρίου μείζονος >όντοξ τοῦ ΕΖΗΘ κύκλου ἐστὶν ὡς τὸ Σ θωρίον πρὸξ τὸν ΑΒΓΔ κύκλον, ο<ύτωξ ὁ ΕΖΗΘ κύκλος πρὸξ >έλαττόν τι τοῦ ΑΒΓΔ κύκλου θωρίον.

Γεγονέτω γάρ ὡς τὸ Σ θωρίον πρὸξ τὸν ΑΒΓΔ κύκλον, ο<ύτωξ ὁ ΕΖΗΘ κύκλος πρὸξ τὸ Τ θωρίον. λέγω, <ὅτι >έλαττόν ἐστι τὸ Τ θωρίον τοῦ ΑΒΓΔ κύκλου. ἐπεὶ γάρ ἐστὶν ὡς τὸ Σ θωρίον πρὸξ τὸν ΑΒΓΔ κύκλον, ο<ύτωξ ὁ ΕΖΗΘ κύκλος πρὸξ τὸ Τ θωρίον, ἐναλλάξ ἐστὶν ὡς τὸ Σ θωρίον πρὸξ τὸν ΕΖΗΘ κύκλον, ο<ύτωξ ὁ ΑΒΓΔ κύκλος πρὸξ τὸ Τ θωρίον. μείζον δὲ τὸ Σ θωρίον τοῦ ΕΖΗΘ κύκλου; μείζων >άρα καὶ ὁ ΑΒΓΔ κύκλος τοῦ Τ θωρίου. <ώστε ἐστὶν ὡς τὸ Σ θωρίον πρὸξ τὸν ΑΒΓΔ κύκλον, ο<ύτωξ ὁ ΕΖΗΘ κύκλος πρὸξ >έλαττόν τι τοῦ ΑΒΓΔ κύκλου θωρίον; <όπερ >έδει δεῖχαι.

3.

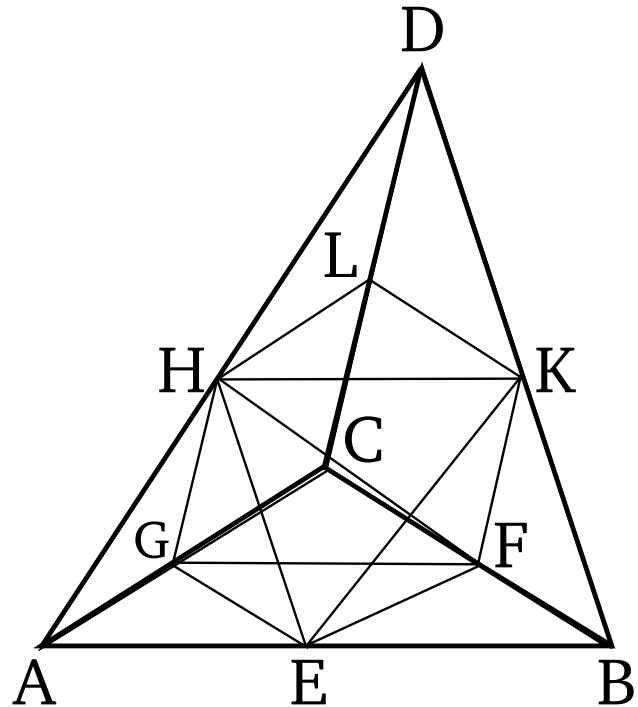
Πᾶσα πυραμὶξ τρίγωνον >έθουσα βάσιν διαιρεῖται εἰς δύο πυραμίδαξ >ίσαξ τε καὶ ὁμοίαξ ἀλλήλαιξ καὶ [ὁμοίαξ] τῇ <όλῃ τριγώνουξ ἐθουσαξ βάσειξ καὶ εἰς δύο πρίσματα >ίσα; καὶ τὰ δύο πρίσματα μείζονά ἐστιν >ῆ τὸ <ήμισυ τῆξ <όληξ πυραμίδοξ.

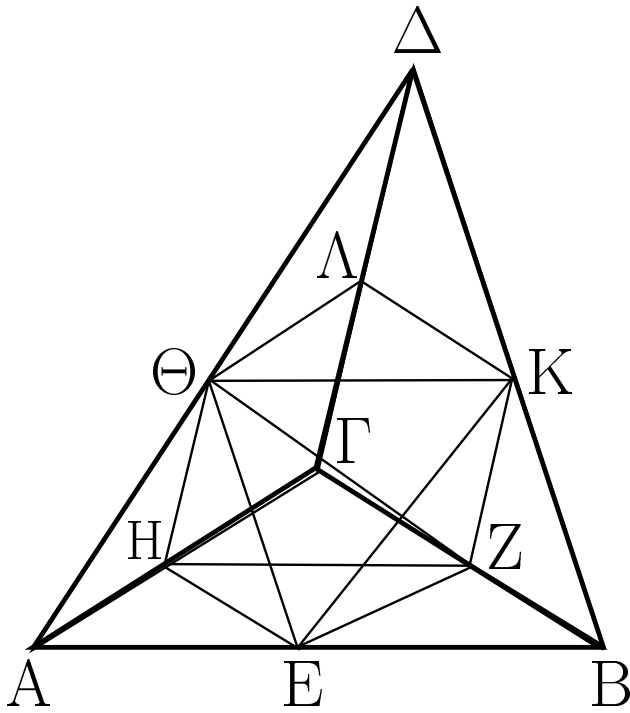
So, I say that, area S being greater than circle $EFGH$, as area S is to circle $ABCD$, so circle $EFGH$ (is) to some area less than circle $ABCD$.

For let it be contrived that as area S (is) to circle $ABCD$, so circle $EFGH$ (is) to area T . I say that area T is less than circle $ABCD$. For since as area S is to circle $ABCD$, so circle $EFGH$ (is) to area T , alternately, as area S is to circle $EFGH$, so circle $ABCD$ (is) to area T [Prop. 5.16]. And area S (is) greater than circle $EFGH$. Thus, circle $ABCD$ (is) also greater than area T [Prop. 5.14]. Hence, as area S is to circle $ABCD$, so circle $EFGH$ (is) to some area less than circle $ABCD$. (Which is) the very thing it was required to show.

Proposition 3

Any pyramid having a triangular base is divided into two pyramids having triangular bases (which are) equal, similar to one another, and [similar] to the whole, and into two equal prisms. And the (sum of the) two prisms is greater than half of the whole pyramid.





>Ἐστω πυραμὶς, <ἣς βάσις μὲν ἐστὶ τὸ $AB\Gamma$ τρίγωνον, κορυφή δὲ τὸ Δ σημεῖον; λέγω, <ὅτι ἡ $AB\Gamma\Delta$ πυραμὶς διαιρεῖται εἰς δύο πυραμίδας >ῖσας ἀλλήλαις τριγώνουξ βάσειξ ἐθούσας καὶ ὁμοίαξ τῇ <όλῃ καὶ εἰς δύο πρίσματα >ῖσα; καὶ τὰ δύο πρίσματα μείζονά ἐστιν >ῆ τὸ <ἡμισυ τῆς <όλης πυραμίδος.

Τετμήσθωσαν γὰρ αἱ AB , $B\Gamma$, ΓA , AD , ΔB , $\Delta \Gamma$ δίθῃ κατὰ τὰ E , Z , H , Θ , K , Λ σημεία, καὶ ἐπεζεύθωσαν αἱ ΘE , $E H$, $H \Theta$, ΘK , $K \Lambda$, $\Lambda \Theta$, $K Z$, $Z H$. ἐπεὶ >ῖση ἐστὶν ἡ μὲν $A E$ τῇ $E B$, ἡ δὲ $A \Theta$ τῇ $\Delta \Theta$, παράλληλος >ἄρα ἐστὶν ἡ $E \Theta$ τῇ ΔB . διὰ τὰ αὐτὰ δὴ καὶ ἡ ΘK τῇ $A B$ παράλληλός ἐστιν. παραλληλόγραμμον >ἄρα ἐστὶ τὸ $\Theta E B K$; >ῖση >ἄρα ἐστὶν ἡ ΘK τῇ $E B$. ἀλλὰ ἡ $E B$ τῇ $E A$ ἐστὶν >ῖση; καὶ ἡ $A E$ >ἄρα τῇ ΘK ἐστὶν >ῖση. >έστι δὲ καὶ ἡ $A \Theta$ τῇ $\Delta \Theta$ >ῖση; δύο δὴ αἱ $E A$, $A \Theta$ δυσὶ ταῖς $K \Theta$, $\Delta \Theta$ >ῖσαι εἰσὶν ἑκατέρα ἑκατέρα; καὶ γωνία ἡ ὑπὸ $E A \Theta$ γωνία τῇ ὑπὸ $K \Theta \Delta$ >ῖση; βάσις >ἄρα ἡ $E \Theta$ βάσει τῇ $K \Delta$ ἐστὶν >ῖση. >ῖσον >ἄρα καὶ <ὁμοίον ἐστὶ τὸ $A E \Theta$ τρίγωνον τῷ $\Theta K \Delta$ τριγώνῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ $A \Theta H$ τρίγωνον τῷ $\Theta \Lambda \Delta$ τριγώνῳ >ῖσον τέ ἐστι καὶ <ὁμοίον. καὶ ἐπεὶ δύο εὐθεῖαι ἀπτόμεναι ἀλλήλων αἱ $E \Theta$, ΘH παρὰ δύο εὐθείας ἀπτομένας ἀλλήλων τὰς $K \Delta$, $\Delta \Lambda$ εἰσὶν οὐκ ἐν τῷ αὐτῷ ἐπιπέδῳ ο>ύσαι, >ῖσας γωνίας περιέχουσιν. >ῖση >ἄρα ἐστὶν ἡ ὑπὸ $E \Theta H$ γωνία τῇ ὑπὸ $K \Delta \Lambda$ γωνία. καὶ ἐπεὶ δύο εὐθεῖαι αἱ $E \Theta$, ΘH δυσὶ ταῖς $K \Delta$, $\Delta \Lambda$ >ῖσαι εἰσὶν ἑκατέρα ἑκατέρα, καὶ γωνία ἡ ὑπὸ $E \Theta H$ γωνία τῇ ὑπὸ $K \Delta \Lambda$ ἐστὶν >ῖση, βάσις >ἄρα ἡ $E H$ βάσει τῇ $K \Lambda$ [ἐστὶν] >ῖση; >ῖσον >ἄρα καὶ <ὁμοίον ἐστὶ τὸ $E \Theta H$ τρίγωνον τῷ $K \Delta \Lambda$ τριγώνῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ $A E H$ τρίγωνον τῷ $\Theta K \Lambda$ τριγώνῳ >ῖσον τε καὶ <ὁμοίον ἐστὶν. ἡ >ἄρα πυραμὶς, <ἣς βάσις μὲν ἐστὶ τὸ $A E H$ τρίγωνον, κορυφή δὲ τὸ Θ σημεῖον, >ῖση καὶ ὁμοία ἐστὶ πυραμίδι, <ἣς βάσις μὲν ἐστὶ τὸ $\Theta K \Lambda$ τρίγωνον, κορυφή δὲ τὸ Δ σημεῖον. καὶ ἐπεὶ τριγώνου τοῦ $A \Delta B$ παρὰ μίαν τῶν πλευρῶν τὴν $A B$ >ῆται ἡ ΘK , ἰσογώνιον ἐστὶ τὸ $A \Delta B$ τρίγωνον τῷ $\Delta \Theta K$ τριγώνῳ, καὶ τὰς πλευρὰς ἀνάλογον >έθουσιν; <ὁμοίον >ἄρα ἐστὶ τὸ $A \Delta B$ τρίγωνον τῷ $\Delta \Theta K$ τριγώνῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ μὲν $\Delta B \Gamma$ τρίγωνον τῷ $\Delta K \Lambda$ τριγώνῳ <ὁμοίον ἐστὶν, τὸ δὲ $\Delta \Lambda \Gamma$ τῷ $\Delta \Lambda \Theta$. καὶ ἐπεὶ δύο εὐθεῖαι ἀπτόμεναι ἀλλήλων αἱ $B A$, $A \Gamma$

Let there be a pyramid whose base is triangle ABC , and (whose) apex (is) point D . I say that pyramid $ABCD$ is divided into two pyramids having triangular bases (which are) equal to one another, and similar to the whole, and into two equal prisms. And the (sum of the) two prisms is greater than half of the whole pyramid.

For let AB , BC , CA , AD , DB , and DC be cut in half at points E , F , G , H , K , and L (respectively). And let HE , EG , GH , HK , KL , LH , KF , and FG be joined. Since AE is equal to EB , and AH to DH , EH is thus parallel to DB [Prop. 6.2]. So, for the same (reasons), HK is also parallel to AB . Thus, $HEBK$ is a parallelogram. Thus, HK is equal to EB [Prop. 1.34]. But, EB is equal to EA . Thus, AE is also equal to HK . And AH is also equal to HD . So the two (straight-lines) EA and AH are equal to the two (straight-lines) KH and HD , respectively. And angle EAH (is) equal to angle KHD [Prop. 1.29]. Thus, base EH is equal to base KD [Prop. 1.4]. Thus, triangle AEH is equal and similar to triangle HKD [Prop. 1.4]. So, for the same (reasons), triangle AHG is also equal and similar to triangle HLD . And since EH and HG are two straight-lines joining one another (which are respectively) parallel to two straight-lines joining one another, KD and DL , not being in the same plane, they will contain equal angles [Prop. 11.10]. Thus, angle EHG is equal to angle KDL . And since the two straight-lines EH and HG are equal to the two straight-lines KD and DL , respectively, and angle EHG is equal to angle KDL , base EG [is] thus equal to base KL [Prop. 1.4]. Thus, triangle EHG is equal and similar to triangle KDL . So, for the same (reasons), triangle AEG is also equal and similar to triangle HKL . Thus, the pyramid whose base is triangle AEG , and apex the point H , is equal and similar to the pyramid whose base is triangle HKL , and apex the point D [Def. 11.10]. And since HK has been drawn parallel to one of the sides, AB , of triangle ADB , triangle ADB is equiangular to triangle DHK [Prop. 1.29], and they have proportional sides. Thus, triangle ADB is similar to triangle DHK [Def. 6.1]. So, for the same (reasons), triangle DBC is also similar to tri-

παρὰ δύο εὐθείας ἀπτομένας ἀλλήλων τὰς ΚΘ, ΘΛ εἰσιν οὐκ ἐν τῷ αὐτῷ ἐπιπέδῳ, >ίσαζ γωνίας περιέχουσιν. >ίση >άρα ἐστὶν ἡ ὑπὸ ΒΑΓ γωνία τῇ ὑπὸ ΚΘΛ. καὶ ἐστὶν ὡς ἡ ΒΑ πρὸς τὴν ΑΓ, οὕτως ἡ ΚΘ πρὸς τὴν ΘΛ; <όμοιον >άρα ἐστὶ τὸ ΑΒΓ τρίγωνον τῷ ΘΚΛ τριγώνῳ. καὶ πυραμὶς >άρα, <ἥξ βάσις μὲν ἐστὶ τὸ ΑΒΓ τρίγωνον, κορυφή δὲ τὸ Δ σημεῖον, ὁμοία ἐστὶ πυραμίδι, <ἥξ βάσις μὲν ἐστὶ τὸ ΘΚΛ τρίγωνον, κορυφή δὲ τὸ Δ σημεῖον. ἀλλὰ πυραμὶς, <ἥξ βάσις μὲν [ἐστὶ] τὸ ΘΚΛ τρίγωνον, κορυφή δὲ τὸ Δ σημεῖον, ὁμοία ἐδείθθη πυραμίδι, <ἥξ βάσις μὲν ἐστὶ τὸ ΑΕΗ τρίγωνον, κορυφή δὲ τὸ Θ σημεῖον. ἐκατέρα >άρα τῶν ΑΕΗΘ, ΘΚΛΔ πυραμίδων ὁμοία ἐστὶ τῇ <όλη τῇ ΑΒΓΔ πυραμίδι.

Καὶ ἐπεὶ >ίση ἐστὶν ἡ ΒΖ τῇ ΖΓ, διπλάσιόν ἐστὶ τὸ ΕΒΖΗ παραλληλόγραμμον τοῦ ΗΖΓ τριγώνου. καὶ ἐπεὶ, ἔαν >ῇ δύο πρίσματα ἰσο"υψῇ, καὶ τὸ μὲν >έθῃ βάσιν παραλληλόγραμμοιο, τὸ δὲ τρίγωνον, διπλάσιον δὲ >ῇ τὸ παραλληλόγραμμον τοῦ τριγώνου, >ίσα ἐστὶ τὰ πρίσματα, >ίσον >άρα ἐστὶ τὸ πρίσμα τὸ περιεχόμενον ὑπὸ δύο μὲν τριγώνων τῶν ΒΚΖ, ΕΘΗ, τριῶν δὲ παραλληλογράμμων τῶν ΕΒΖΗ, ΕΒΚΘ, ΘΚΖΗ τῷ πρισματι τῷ περιεχομένῳ ὑπὸ δύο μὲν τριγώνων τῶν ΗΖΓ, ΘΚΛ, τριῶν δὲ παραλληλογράμμων τῶν ΚΖΓΛ, ΛΓΗΘ, ΘΚΖΗ. καὶ φανερόν, <ὅτι ἐκάτρων τῶν πρισμάτων, ο<ὕ τε βάσις τὸ ΕΒΖΗ παραλληλόγραμμον, ἀπεναντίον δὲ ἡ ΘΚ εὐθεῖα, καὶ ο<ὕ βάσις τὸ ΗΖΓ τρίγωνον, ἀπεναντίον δὲ τὸ ΘΚΛ τρίγωνον, μεῖζόν ἐστὶν ἐκατέρα τῶν πυραμίδων, <ὦν βάσεις μὲν τὰ ΑΕΗ, ΘΚΛ τρίγωνα, κορυφαὶ, δὲ τὰ Θ, Δ σημεῖα, ἐπειδὴ περ [καί] ἔαν ἐπιζεύχωμεν τὰς ΕΖ, ΕΚ εὐθείας, τὸ μὲν πρίσμα, ο<ὕ βάσις τὸ ΕΒΖΗ παραλληλόγραμμον, ἀπεναντίον δὲ ἡ ΘΚ εὐθεῖα, μεῖζόν ἐστὶ τῇς πυραμίδος, <ἥξ βάσις τὸ ΕΒΖ τρίγωνον, κορυφή δὲ τὸ Κ σημεῖον. ἀλλ' >ῇ πυραμὶς, <ἥξ βάσις τὸ ΕΒΖ τρίγωνον, κορυφή δὲ τὸ Κ σημεῖον, >ίση ἐστὶ πυραμίδι, <ἥξ βάσις τὸ ΑΕΗ τρίγωνον, κορυφή δὲ τὸ Θ σημεῖον; ὑπὸ γὰρ >ίσων καὶ ὁμοίων ἐπιπέδων περιέθονται. <ώστε καὶ τὸ πρίσμα, ο<ὕ βάσις μὲν τὸ ΕΒΖΗ παραλληλόγραμμον, ἀπεναντίον δὲ ἡ ΘΚ εὐθεῖα, μεῖζόν ἐστὶ πυραμίδος, <ἥξ βάσις μὲν τὸ ΑΕΗ τρίγωνον, κορυφή δὲ τὸ Θ σημεῖον. >ίσον δὲ τὸ μὲν πρίσμα, ο<ὕ βάσις τὸ ΕΒΖΗ παραλληλόγραμμον, ἀπεναντίον δὲ ἡ ΘΚ εὐθεῖα, τῷ πρισματι, ο<ὕ βάσις μὲν τὸ ΗΖΓ τρίγωνον, ἀπεναντίον δὲ τὸ ΘΚΛ τρίγωνον; ἡ δὲ πυραμὶς, <ἥξ βάσις τὸ ΑΕΗ τρίγωνον, κορυφή δὲ τὸ Θ σημεῖον, >ίση ἐστὶ πυραμίδι, <ἥξ βάσις τὸ ΘΚΛ τρίγωνον, κορυφή δὲ τὸ Δ σημεῖον. τὰ >άρα εἰρημένα δύο πρίσματα μεῖζονά ἐστὶ τῶν εἰρημένων δύο πυραμίδων, <ὦν βάσεις μὲν τὰ ΑΕΗ, ΘΚΛ τρίγωνα, κορυφαὶ δὲ τὰ Θ, Δ σημεῖα.

Ἡ >άρα <όλη πυραμὶς, <ἥξ βάσις τὸ ΑΒΓ τρίγωνον, κορυφή δὲ τὸ Δ σημεῖον, διή|ρηται ε>ῖξ τε δύο πυραμίδαξ >ίσαξ ἀλλήλαιξ [καὶ ὁμοίαξ τῇ] <όλη] καὶ εἰς δύο πρίσματα >ίσα, καὶ τὰ δύο πρίσματα μεῖζονά ἐστὶν >ῇ τὸ <ῆμισυ τῇς <όλης πυραμίδος; <όπερ >έδει δεῖχαι.

4.

Ἐὰν >ῶσι δύο πυραμίδεξ ὑπὸ τὸ αὐτὸ <ύψοξ τριγώνουξ >έθουσαι βάσεις, διαιρεθῇ δὲ ἐκατέρα αὐτῶν ε>ῖξ τε δύο πυραμίδαξ >ίσαξ ἀλλήλαιξ καὶ ὁμοίαξ τῇ <όλη καὶ εἰς δύο

angle DKL , and ADC to DLH . And since two straight-lines joining one another, BA and AC , are parallel to two straight-lines joining one another, KH and HL , not in the same plane, they will contain equal angles [Prop. 11.10]. Thus, angle BAC is equal to (angle) KHL . And as BA is to AC , so KH (is) to HL . Thus, triangle ABC is similar to triangle HKL [Prop. 6.6]. And, thus, the pyramid whose base is triangle ABC , and apex the point D , is similar to the pyramid whose base is triangle HKL , and apex the point D [Def. 11.9]. But, the pyramid whose base [is] triangle HKL , and apex the point D , was shown (to be) similar to the pyramid whose base is triangle AEG , and apex the point H . Thus, each of the pyramids $AEHG$ and $HKLD$ is similar to the whole pyramid $ABCD$.

And since BF is equal to FC , parallelogram $EBFG$ is double triangle GFC [Prop. 1.41]. And since, if two prisms (have) equal heights, and the former has a parallelogram as a base, and the latter a triangle, and the parallelogram (is) double the triangle, then the prisms are equal [Prop. 11.39], the prism contained by the two triangles BKF and EHG , and the three parallelograms $EBFG$, $EBKH$, and $HKFG$, is thus equal to the prism contained by the two triangles GFC and HKL , and the three parallelograms $KFCL$, $LCGH$, and $HKFG$. And (it is) clear that each of the prisms whose base (is) parallelogram $EBFG$, and opposite (side) straight-line HK , and whose base (is) triangle GFC , and opposite (plane) triangle HKL , is greater than each of the pyramids whose bases are triangles AEG and HKL , and apexes the points H and D (respectively), inasmuch as, if we [also] join the straight-lines EF and EK then the prism whose base (is) parallelogram $EBFG$, and opposite (side) straight-line HK , is greater than the pyramid whose base (is) triangle EBF , and apex the point K . But the pyramid whose base (is) triangle EBF , and apex the point K , is equal to the pyramid whose base is triangle AEG , and apex point H . For they are contained by equal and similar planes. And, hence, the prism whose base (is) parallelogram $EBFG$, and opposite (side) straight-line HK , is greater than the pyramid whose base (is) triangle AEG , and apex the point H . And the prism whose base is parallelogram $EBFG$, and opposite (side) straight-line HK , (is) equal to the prism whose base (is) triangle GFC , and opposite (plane) triangle HKL . And the pyramid whose base (is) triangle AEG , and apex the point H , is equal to the pyramid whose base (is) triangle HKL , and apex the point D . Thus, the (sum of the) aforementioned two prisms is greater than the (sum of the) aforementioned two pyramids, whose bases (are) triangles AEG and HKL , and apexes the points H and D (respectively).

Thus, the whole pyramid, whose base (is) triangle ABC , and apex the point D , has been divided into two pyramids (which are) equal to one another [and similar to the whole], and into two equal prisms. And the (sum of the) two prisms is greater than half of the whole pyramid. (Which is) the very thing it was required to show.

Proposition 4

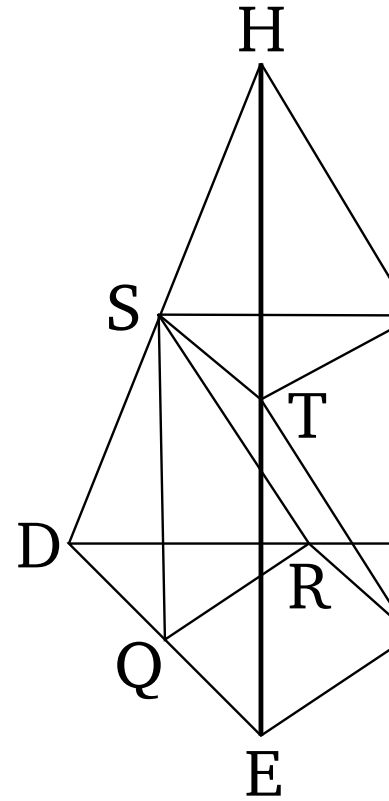
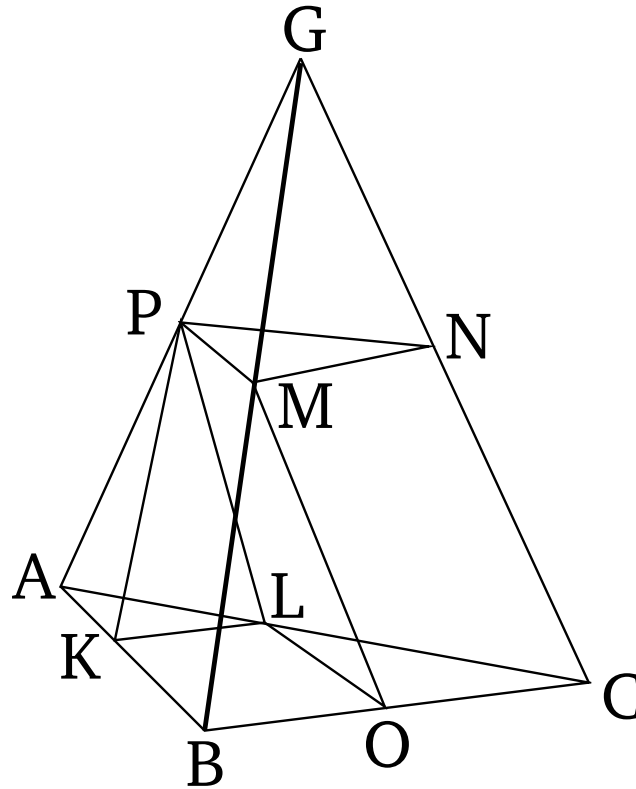
If there are two pyramids with the same height, having triangular bases, and each of them is divided into two pyramids equal to one another, and similar to the whole, and into

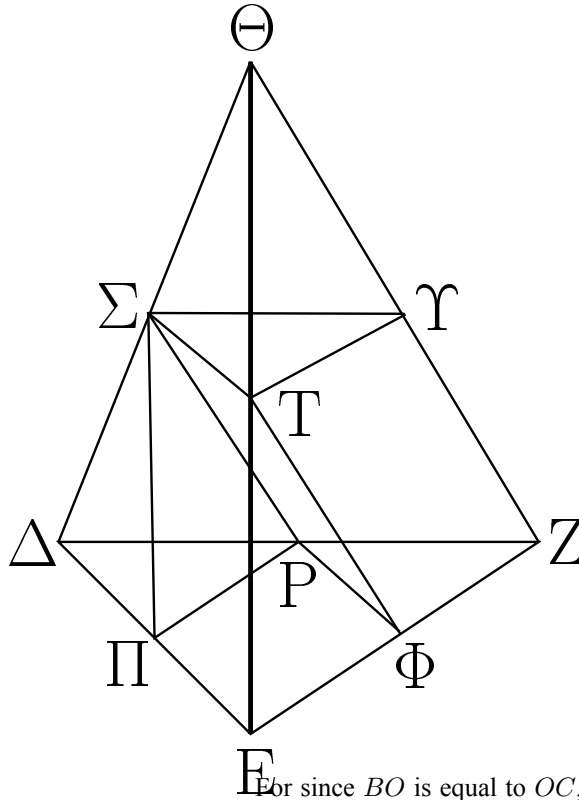
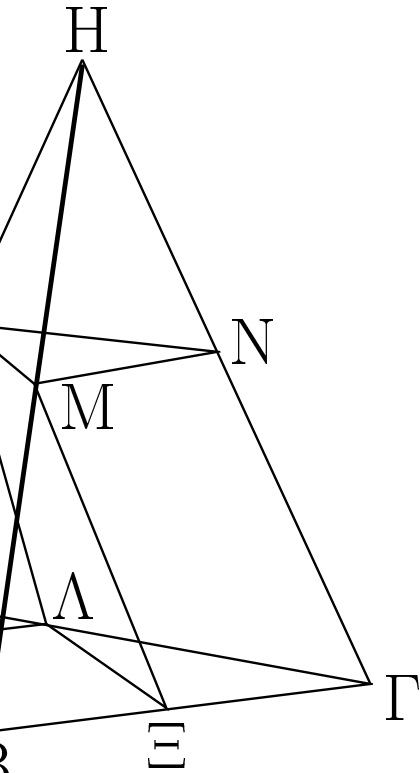
πρίσματα >ίσα, >έσται ὥς ἡ τῆς μιᾶς πυραμίδος βάσις πρὸς τὴν τῆς ἐτέρας πυραμίδος βάσιν, οὕτως τὰ ἐν τῇ μιᾷ πυραμίδι πρίσματα πάντα πρὸς τὰ ἐν τῇ ἐτέρᾳ πυραμίδι πρίσματα πάντα ἰσοπληθῆ.

>Ἐστωσαν δύο πυραμίδες ὑπὸ τὸ αὐτὸ <ύψος τριγώνου >έθουσαι βάσεις τὰς $AB\Gamma$, ΔEZ , κορυφὰς δὲ τὰ H , Θ σημεία, καὶ διηρήσθω ἑκάτερα αὐτῶν >ίς τε δύο πυραμίδα >ίσα >ἀλλήλαις καὶ ὁμοία τῇ <όλῃ καὶ εἰς δύο πρίσματα >ίσα; λέγω, <ότι ἐστὶν ὥς ἡ $AB\Gamma$ βάσις πρὸς τὴν ΔEZ βάσιν, οὕτως τὰ ἐν τῇ $AB\Gamma H$ πυραμίδι πρίσματα πάντα πρὸς τὰ ἐν τῇ $\Delta EZ\Theta$ πυραμίδι πρίσματα ἰσοπληθῆ.

two equal prisms then as the base of one pyramid (is) to the base of the other pyramid, so (the sum of) all the prisms in one pyramid will be to (the sum of all) the equal number of prisms in the other pyramid.

Let there be two pyramids with the same height, having the triangular bases ABC and DEF , (with) apexes the points G and H (respectively). And let each of them be divided into two pyramids equal to one another, and similar to the whole, and into two equal prisms [Prop. 12.3]. I say that as base ABC is to base DEF , so (the sum of) all the prisms in pyramid $ABCG$ (is) to (the sum of) all the equal number of prisms in pyramid $DEFH$.





Ἐπεὶ γὰρ ὅτι ἐστὶν ἡ μὲν BX τῇ $XΓ$, ἡ δὲ $ΑΛ$ τῇ $ΛΓ$, παράλληλος ὅρα ἐστὶν ἡ $ΛX$ τῇ AB καὶ ὁμοίον τὸ $ABΓ$ τρίγωνον τῷ $ΛXΓ$ τριγώνῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ $ΔΕΖ$ τρίγωνον τῷ $ΡΦΖ$ τριγώνῳ ὁμοίον ἐστίν. καὶ ἐπεὶ διπλασίον ἐστὶν ἡ μὲν $BΓ$ τῇ $ΓX$, ἡ δὲ EZ τῇ $ZΦ$, ὅτι ἐστὶν ὅρα ὡς ἡ $BΓ$ πρὸς τὴν $ΓX$, οὕτως ἡ EZ πρὸς τὴν $ZΦ$. καὶ ἀναγέγραπται ἀπὸ μὲν τῶν $BΓ$, $ΓX$ ὁμοία τε καὶ ὁμοίως κείμενα εὐθύγραμμα τὰ $ABΓ$, $ΛXΓ$, ἀπὸ δὲ τῶν EZ , $ZΦ$ ὁμοία τε καὶ ὁμοίως κείμενα [εὐθύγραμμα] τὰ $ΔΕΖ$, $ΡΦΖ$; ὅτι ἐστὶν ὅρα ὡς τὸ $ABΓ$ τρίγωνον πρὸς τὸ $ΛXΓ$ τρίγωνον, οὕτως τὸ $ΔΕΖ$ τρίγωνον πρὸς τὸ $ΡΦΖ$ τρίγωνον; ἐναλλάξ ὅρα ἐστὶν ὡς τὸ $ABΓ$ τρίγωνον πρὸς τὸ $ΔΕΖ$ [τρίγωνον], οὕτως τὸ $ΛXΓ$ [τρίγωνον] πρὸς τὸ $ΡΦΖ$ τρίγωνον. ἀλλ' ὡς τὸ $ΛXΓ$ τρίγωνον πρὸς τὸ $ΡΦΖ$ τρίγωνον, οὕτως τὸ πρίσμα, ὃ βάσις μὲν [ἐστὶ] τὸ $ΛXΓ$ τρίγωνον, ἀπεναντίον δὲ τὸ OMN , πρὸς τὸ πρίσμα, ὃ βάσις μὲν τὸ $ΡΦΖ$ τρίγωνον, ἀπεναντίον δὲ τὸ $ΣΤΥ$; καὶ ὡς ὅρα τὸ $ABΓ$ τρίγωνον πρὸς τὸ $ΔΕΖ$ τρίγωνον, οὕτως τὸ πρίσμα, ὃ βάσις μὲν τὸ $ΛXΓ$ τρίγωνον, ἀπεναντίον δὲ τὸ OMN , πρὸς τὸ πρίσμα, ὃ βάσις μὲν τὸ $ΡΦΖ$ τρίγωνον, ἀπεναντίον δὲ τὸ $ΣΤΥ$. ὡς δὲ τὰ εἰρημένα πρίσματα πρὸς ἄλληλα, οὕτως τὸ πρίσμα, ὃ βάσις μὲν τὸ $KBXΛ$ παραλληλόγραμμον, ἀπεναντίον δὲ ἡ OM εὐθεῖα, πρὸς τὸ πρίσμα, ὃ βάσις μὲν τὸ $ΠΕΦΡ$ παραλληλόγραμμον, ἀπεναντίον δὲ ἡ $ΣΤ$ εὐθεῖα. καὶ τὰ δύο ὅρα πρίσματα, ὃ βάσις μὲν τὸ $KBXΛ$ παραλληλόγραμμον, ἀπεναντίον δὲ ἡ OM , καὶ ὃ βάσις μὲν τὸ $ΛXΓ$, ἀπεναντίον δὲ τὸ OMN , πρὸς τὰ πρίσματα, ὃ βάσις μὲν τὸ $ΠΕΦΡ$, ἀπεναντίον δὲ ἡ $ΣΤ$ εὐθεῖα, καὶ ὃ βάσις μὲν τὸ $ΡΦΖ$ τρίγωνον, ἀπεναντίον δὲ τὸ $ΣΤΥ$. καὶ ὡς ὅρα ἡ $ABΓ$ βάσις πρὸς τὴν $ΔΕΖ$ βάσιν, οὕτως τὰ εἰρημένα δύο πρίσματα πρὸς τὰ εἰρημένα δύο πρίσματα.

Καὶ ὁμοίως, ἔαν διαιρεθῶσιν αἱ $OMNH$, $ΣΤΥΘ$ πυραμίδες ἐξ τε δύο πρίσματα καὶ δύο πυραμίδαξ, ὅτι ἐστὶ ὡς ἡ OMN βάσις πρὸς τὴν $ΣΤΥ$ βάσιν, οὕτως τὰ ἐν τῇ $OMNH$ πυραμίδι δύο πρίσματα πρὸς τὰ ἐν τῇ $ΣΤΥΘ$ πυραμίδι δύο

For since BO is equal to OC , and AL to LC , LO is thus parallel to AB , and triangle ABC similar to triangle LOC [Prop. 12.3]. So, for the same (reasons), triangle DEF is also similar to triangle RVF . And since BC is double CO , and EF (double) FV , thus as BC (is) to CO , so EF (is) to FV . And the similar, and similarly laid out, rectilinear (figures) ABC and LOC be described on BC and CO (respectively), and the similar, and similarly laid out, [rectilinear] (figures) DEF and RVF on EF and FV (respectively). Thus, as triangle ABC is to triangle LOC , so triangle DEF (is) to triangle RVF [Prop. 6.22]. Thus, alternately, as triangle ABC is to [triangle] DEF , so [triangle] LOC (is) to triangle RVF [Prop. 5.16]. But, as triangle LOC (is) to triangle RVF , so the prism whose base [is] triangle LOC , and opposite (plane) PMN , (is) to the prism whose base (is) triangle RVF , and opposite (plane) STU (see lemma). And, thus, as triangle ABC (is) to triangle DEF , so the prism whose base (is) triangle LOC , and opposite (plane) PMN , (is) to the prism whose base (is) triangle RVF , and opposite (plane) STU . And as the aforementioned prisms (are) to one another, so the prism whose base (is) parallelogram $KBOL$, and opposite (side) straight-line PM , (is) to the prism whose base (is) parallelogram $QEV R$, and opposite (side) straight-line ST [Props. 11.39, 12.3]. Thus, also, (is) the (sum of the) two prisms—that whose base (is) parallelogram $KBOL$, and opposite (side) PM , and that whose base (is) LOC , and opposite (plane) PMN —to (the sum of) the (two) prisms—that whose base (is) $QEV R$, and opposite (side) straight-line ST , and that whose base (is) triangle RVF , and opposite (plane) STU [Prop. 5.12]. And, thus, as base ABC (is) to base DEF , so the (sum of the first) aforementioned two prisms (is) to the (sum of the second) aforementioned two prisms.

And, similarly, if pyramids $PMNG$ and $STUH$ are divided into two prisms, and two pyramids, as base PMN (is) to base STU , so (the sum of) the two prisms in pyramid $PMNG$ will be to (the sum of) the two prisms in pyra-

πρίσματα. ἀλλ' ὥς ἡ OMN βάσις πρὸς τὴν $ΣΤΥ$ βάσιν, οὕτως ἡ $ABΓ$ βάσις πρὸς τὴν $ΔΕΖ$ βάσιν; >ἴσον γὰρ ἑκάτερον τῶν OMN , $ΣΤΥ$ τριγώνων ἐκατέρω τῶν $ΛΧΓ$, $ΡΦΖ$. καὶ ὥς >άρα ἡ $ABΓ$ βάσις πρὸς τὴν $ΔΕΖ$ βάσιν, οὕτως τὰ τέσσαρα πρίσματα πρὸς τὰ τέσσαρα πρίσματα. ὁμοίως δὲ κ' ἂν τὰς ὑπολειπομένας πυραμίδας διέλωμεν εἰς τε δύο πυραμίδας καὶ εἰς δύο πρίσματα, >έσται ὥς ἡ $ABΓ$ βάσις πρὸς τὴν $ΔΕΖ$ βάσιν, οὕτως τὰ ἐν τῇ $ABΓΗ$ πυραμίδι πρίσματα πάντα πρὸς τὰ ἐν τῇ $ΔΕΖΘ$ πυραμίδι πρίσματα πάντα ἰσοπληθῆ; <όπερ >έδει δεῖχαι.

Λήμμα.

<Ὅτι δὲ ἔστιν ὥς τὸ $ΛΧΓ$ τρίγωνον πρὸς τὸ $ΡΦΖ$ τρίγωνον, οὕτως τὸ πρίσμα, οὕβ βάσις τὸ $ΛΧΓ$ τρίγωνον, ἀπεναντίον δὲ τὸ OMN , πρὸς τὸ πρίσμα, οὕβ βάσις μὲν τὸ $ΡΦΖ$ [τρίγωνον], ἀπεναντίον δὲ τὸ $ΣΤΥ$, οὕτω δεικτέον. Ἐπὶ γὰρ τῇ αὐτῇ καταγραφῇ νενοήσθωσαν ἀπὸ τῶν H , $Θ$ κάθετοι ἐπὶ τὰ $ABΓ$, $ΔΕΖ$ ἐπίπεδα, >ἴσαι δηλαδὴ τυγθάνουσαι διὰ τὸ ἰσο"υψεῖς ὑποκεῖσθαι τὰς πυραμίδας. καὶ ἐπεὶ δύο εὐθεῖαι <ἡ τε $HΓ$ καὶ ἡ ἀπὸ τοῦ H κάθετος ὑπὸ παραλλήλων ἐπιπέδων τῶν $ABΓ$, OMN τέμνονται, εἰς τοὺς αὐτοὺς λόγους τμηθήσονται. καὶ τέμνεται ἡ $HΓ$ δίθια ὑπὸ τοῦ OMN ἐπιπέδου κατὰ τὸ N ; καὶ ἡ ἀπὸ τοῦ H >άρα κάθετος ἐπὶ τὸ $ABΓ$ ἐπίπεδον δίθια τμηθήσεται ὑπὸ τοῦ OMN ἐπιπέδου. διὰ τὰ αὐτὰ δὲ καὶ ἡ ἀπὸ τοῦ $Θ$ κάθετος ἐπὶ τὸ $ΔΕΖ$ ἐπίπεδον δίθια τμηθήσεται ὑπὸ τοῦ $ΣΤΥ$ ἐπιπέδου. καὶ εἰσιν >ἴσαι αἱ ἀπὸ τῶν H , $Θ$ κάθετοι ἐπὶ τὰ $ABΓ$, $ΔΕΖ$ ἐπίπεδα; >ἴσαι >άρα καὶ αἱ ἀπὸ τῶν OMN , $ΣΤΥ$ τριγώνων ἐπὶ τὰ $ABΓ$, $ΔΕΖ$ κάθετοι. ἰσο"υψῆ >άρα [έστι] τὰ πρίσματα, <ὥν βάσεις μὲν εἰσι τὰ $ΛΧΓ$, $ΡΦΖ$ τρίγωνα, ἀπεναντίον δὲ τὰ OMN , $ΣΤΥ$. <ώστε καὶ τὰ στερεὰ παραλληλεπίπεδα τὰ ἀπὸ τῶν εἰρημένων πρισμάτων ἀναγραφόμενα ἰσο"υψῆ καὶ πρὸς >ἀλλήλα [εἰσιν] ὥς αἱ βάσεις; καὶ τὰ ἡμίση >άρα ἔστιν ὥς ἡ $ΛΧΓ$ βάσις πρὸς τὴν $ΡΦΖ$ βάσιν, οὕτως τὰ εἰρημένα πρίσματα πρὸς >ἀλλήλα; <όπερ >έδει δεῖχαι.

5.

Αἱ ὑπὸ τὸ αὐτὸ <ύψος ο>ῦσαι πυραμίδες καὶ τριγώνουζ >έθουσαι βάσεις πρὸς ἀλλήλας εἰσιν ὥς αἱ βάσεις.

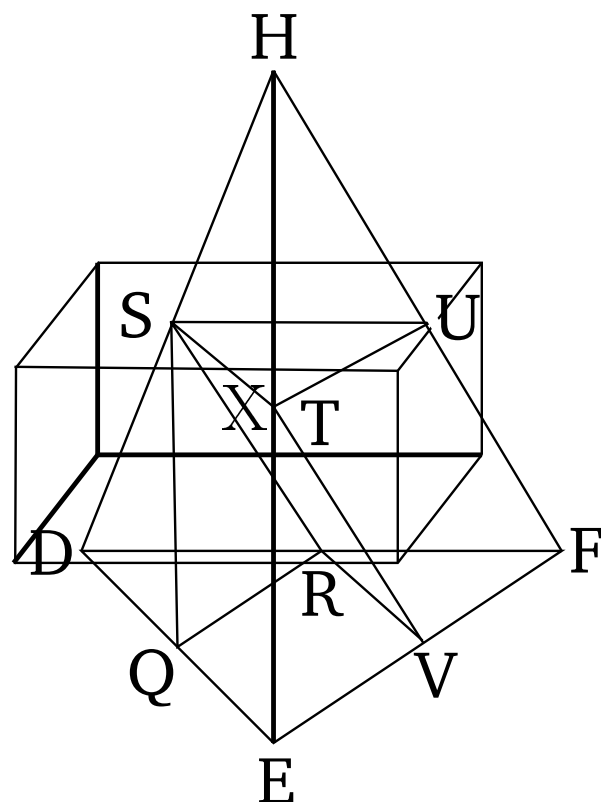
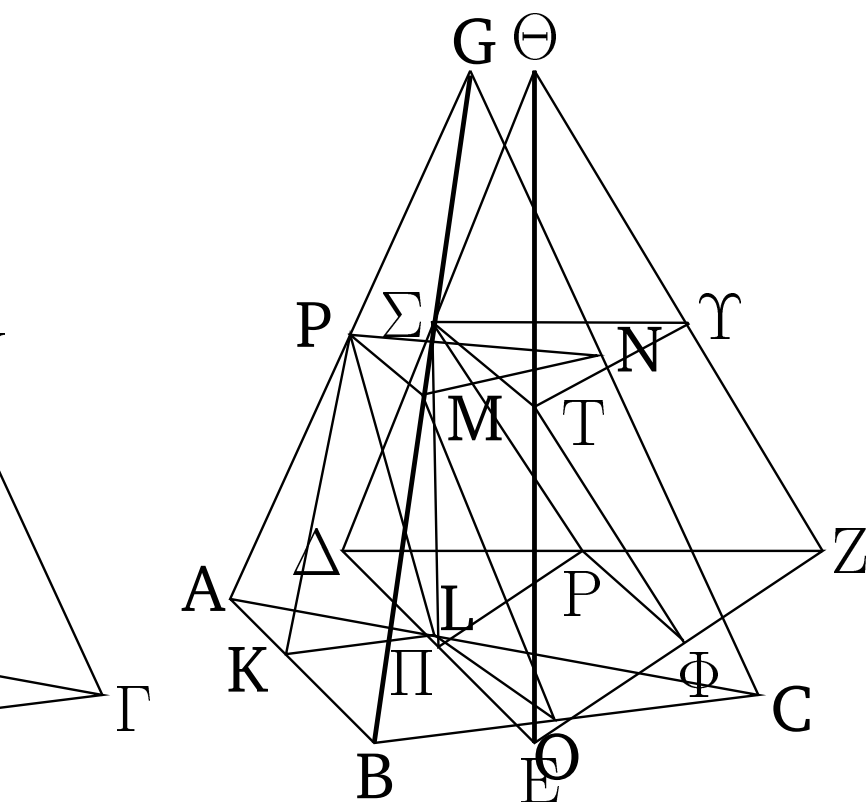
mid $STUH$. But, as base PMN (is) to base STU , so base ABC (is) to base DEF . For the triangles PMN and STU (are) equal to LOC and RVF , respectively. And, thus, as base ABC (is) to base DEF , so (the sum of) the four prisms (is) to (the sum of) the four prisms [Prop. 5.12]. So, similarly, even if we divide the pyramids left behind into two pyramids and into two prisms, as base ABC (is) to base DEF , so (the sum of) all the prisms in pyramid $ABCG$ will be to (the sum of) all the equal number of prisms in pyramid $DEFH$. (Which is) the very thing it was required to show.

Lemma

And one may show, as follows, that as triangle LOC is to triangle RVF , so the prism whose base (is) triangle LOC , and opposite (plane) PMN , (is) to the prism whose base (is) [triangle] RVF , and opposite (plane) STU . For, in the same figure, let perpendiculars be conceived (drawn) from (points) G and H to the planes ABC and DEF (respectively). These clearly turn out to be equal, on account of the pyramids being assumed (to be) of equal height. And since two straight-lines, GC and the perpendicular from G , are cut by the parallel planes ABC and PMN they will be cut in the same ratios [Prop. 11.17]. And GC was cut in half by the plane PMN at N . Thus, the perpendicular from G to the plane ABC will also be cut in half by the plane PMN . So, for the same (reasons), the perpendicular from H to the plane DEF will also be cut in half by the plane STU . And the perpendiculars from G and H to the planes ABC and DEF (respectively) are equal. Thus, the perpendiculars from the triangles PMN and STU to ABC and DEF (respectively, are) also equal. Thus, the prisms whose bases are triangles LOC and RVF , and opposite (sides) PMN and STU (respectively), [are] of equal height. And, hence, the parallelepiped solids described on the aforementioned prisms [are] of equal height and (are) to one another as their bases [Prop. 11.32]. Likewise, the halves (of the solids) [Prop. 11.28]. Thus, as base LOC is to base RVF , so the aforementioned prisms (are) to one another. (Which is) the very thing it was required to show.

Proposition 5

Pyramids which are of the same height, and have triangular bases, are to one another as their bases.



>Ἐστωσαν ὑπὸ τὸ αὐτὸ <ὕψος πυραμίδες, <ὧν βάσεις μὲν τὰ ABΓ, ΔΕΖ τρίγωνα, κορυφαὶ δὲ τὰ Η, Θ σημεῖα; λέγω, <ὅτι ἐστὶν ὡς ἡ ABΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, ο<ὕτως ἡ ABΓΗ πυραμὶς πρὸς τὴν ΔΕΖΘ πυραμίδα.

Εἰ γὰρ μὴ ἐστὶν ὡς ἡ ABΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, ο<ὕτως ἡ ABΓΗ πυραμὶς πρὸς τὴν ΔΕΖΘ πυραμίδα, >έσται ὡς ἡ ABΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, ο<ὕτως ἡ ABΓΗ πυραμὶς >ήτοι πρὸς >έλασσόν τι τῆς ΔΕΖΘ πυραμίδος στερεὸν >ἢ πρὸς μείζον. >έστω πρότερον πρὸς >έλασσον τὸ Θ, καὶ διηρήσθω ἡ ΔΕΖΘ πυραμὶς ε>ἰς τε δύο πυραμίδας >ἴσας ἀλλήλαις καὶ ὁμοίαις τῇ <όλῃ καὶ εἰς δύο πρίσματα >ἴσα; τὰ δὲ δύο πρίσματα μείζονά ἐστιν >ἢ τὸ <ήμισυ τῆς <όλης πυραμίδος. καὶ πάλιν αἱ ἐκ τῆς διαιρέσεως γινόμεναι πυραμίδες ὁμοίως διηρήσθωσαν, καὶ τοῦτο αἰεὶ γινέσθω, <έως ο<ὅ λειφθῶσι τινες πυραμίδες ἀπὸ τῆς ΔΕΖΘ πυραμίδος, α<ὶ εἰσιν ἐλάττωες τῆς ὑπεροθῆς, <ἢ ὑπερέθει ἡ ΔΕΖΘ πυραμὶς τοῦ Θ στερεοῦ. λελείφθωσαν καὶ >έστωσαν λόγου <ένεκεν αἱ ΔΠΡΣ, ΣΤΥΘ; λοιπὰ >άρα τὰ ἐν τῇ ΔΕΖΘ πυραμίδι πρίσματα μείζονά ἐστι τοῦ Θ στερεοῦ. διηρήσθω καὶ ἡ ABΓΗ πυραμὶς ὁμοίως καὶ ἰσοπληθῶς τῇ ΔΕΖΘ πυραμίδι; >έστιν >άρα ὡς ἡ ABΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, ο<ὕτως τὰ ἐν τῇ ABΓΗ πυραμίδι πρίσματα πρὸς τὰ ἐν τῇ ΔΕΖΘ πυραμίδι πρίσματα, ἀλλὰ καὶ ὡς ἡ ABΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, ο<ὕτως ἡ ABΓΗ πυραμὶς πρὸς τὸ Θ στερεόν; καὶ ὡς >άρα ἡ ABΓΗ πυραμὶς πρὸς τὸ Θ στερεόν, ο<ὕτως τὰ ἐν τῇ ABΓΗ πυραμίδι πρίσματα πρὸς τὰ ἐν τῇ ΔΕΖΘ πυραμίδι πρίσματα; ἐναλλάχ >άρα ὡς ἡ ABΓΗ πυραμὶς πρὸς τὰ ἐν αὐτῇ πρίσματα, ο<ὕτως τὸ Θ στερεὸν πρὸς τὰ ἐν τῇ ΔΕΖΘ πυραμίδι πρίσματα. μείζων δὲ ἡ ABΓΗ πυραμὶς τῶν ἐν αὐτῇ πρισματῶν; μείζων >άρα καὶ τὸ Θ στερεὸν τῶν ἐν τῇ ΔΕΖΘ πυραμίδι πρισματῶν. ἀλλὰ καὶ >έλαττον; <όπερ ἐστὶν ἀδύνατον. οὐκ >άρα ἐστὶν ὡς ἡ ABΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, ο<ὕτως ἡ ABΓΗ πυραμὶς πρὸς >έλασσόν τι τῆς ΔΕΖΘ πυραμίδος στερεόν. ὁμοίως δὲ δεῖσθαι, <ὅτι οὐδὲ ὡς ἡ ΔΕΖ βάσις πρὸς τὴν ABΓ βάσιν, ο<ὕτως

Let there be pyramids of the same height whose bases (are) the triangles ABC and DEF , and apexes the points G and H (respectively). I say that as base ABC is to base DEF , so pyramid $ABCG$ (is) to pyramid $DEFH$.

For if base ABC is not to base DEF , as pyramid $ABCG$ (is) to pyramid $DEFH$, then base ABC will be to base DEF , as pyramid $ABCG$ (is) to some solid either less than, or greater than, pyramid $DEFH$. Let it, first of all, be (in this ratio) to (some) lesser (solid), W . And let pyramid $DEFH$ be divided into two pyramids equal to one another, and similar to the whole, and into two equal prisms. So, the (sum of the) two prisms is greater than half of the whole pyramid [Prop. 12.3]. And, again, let the pyramids generated by the division be similarly divided, and let this be done continually until some pyramids are left from pyramid $DEFH$ which (when added together) are less than the excess by which pyramid $DEFH$ exceeds the solid W [Prop. 10.1]. Let them be left, and, for the sake of argument, let them be $DQRS$ and $STUH$. Thus, the (sum of the) remaining prisms within pyramid $DEFH$ is greater than solid W . Let pyramid $ABCG$ also be divided similarly, and a similar number of times, as pyramid $DEFH$. Thus, as base ABC is to base DEF , so the (sum of the) prisms within pyramid $ABCG$ (is) to the (sum of the) prisms within pyramid $DEFH$ [Prop. 12.4]. But, also, as base ABC (is) to base DEF , so pyramid $ABCG$ (is) to solid W . And, thus, as pyramid $ABCG$ (is) to solid W , so the (sum of the) prisms within pyramid $ABCG$ (is) to the (sum of the) prisms within pyramid $DEFH$ [Prop. 5.11]. Thus, alternately, as pyramid $ABCG$ (is) to the (sum of the) prisms within it, so solid W (is) to the (sum of the) prisms within pyramid $DEFH$ [Prop. 5.16]. And pyramid $ABCG$ (is) greater than the (sum of the) prisms within it. Thus, solid W (is) also greater than the (sum of the) prisms within pyramid $DEFH$ [Prop. 5.14]. But, (it is) also less. This very thing is impossible. Thus, as base ABC is to base DEF ,

ἡ ΔΕΖΘ πυραμὶς πρὸς >ἐλαττόν τι τῆς ΑΒΓΗ πυραμίδος στερεόν.

Λέγω δὴ, <ὅτι οὐκ >έστιν οὐδὲ ὡς ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, ο<ύτωξ ἡ ΑΒΓΗ πυραμὶς πρὸς μείζον τι τῆς ΔΕΖΘ πυραμίδος στερεόν.

Εἰ γὰρ δυνατόν, >έστω πρὸς μείζον τὸ Θ; ἀνάπαλιν >άρα ἔστιν ὡς ἡ ΔΕΖ βάσις πρὸς τὴν ΑΒΓ βάσιν, ο<ύτωξ τὸ Θ στερεὸν πρὸς τὴν ΑΒΓΗ πυραμίδα. ὡς δὲ τὸ Θ στερεὸν πρὸς τὴν ΑΒΓΗ πυραμίδα, ο<ύτωξ ἡ ΔΕΖΘ πυραμὶς πρὸς >έλασσόν τι τῆς ΑΒΓΗ πυραμίδος, ὡς >έμπροσθεν ἐδείκθη; καὶ ὡς >άρα ἡ ΔΕΖ βάσις πρὸς τὴν ΑΒΓ βάσιν, ο<ύτωξ ἡ ΔΕΖΘ πυραμὶς πρὸς >έλασσόν τι τῆς ΑΒΓΗ πυραμίδος; <όπερ >άτοπον ἐδείκθη. οὐκ >άρα ἔστιν ὡς ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, ο<ύτωξ ἡ ΑΒΓΗ πυραμὶς πρὸς μείζον τι τῆς ΔΕΖΘ πυραμίδος στερεόν. ἐδείκθη δέ, <ὅτι οὐδὲ πρὸς >έλασσον. >έστιν >άρα ὡς ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, ο<ύτωξ ἡ ΑΒΓΗ πυραμὶς πρὸς τὴν ΔΕΖΘ πυραμίδα; <όπερ >έδει δεῖχαι.

so pyramid $ABCG$ (is) not to some solid less than pyramid $DEFH$. So, similarly, we can show that base DEF is not to base ABC , as pyramid $DEFH$ (is) to some solid less than pyramid $ABCG$ either.

So, I say that neither is base ABC to base DEF , as pyramid $ABCG$ (is) to some solid greater than pyramid $DEFH$.

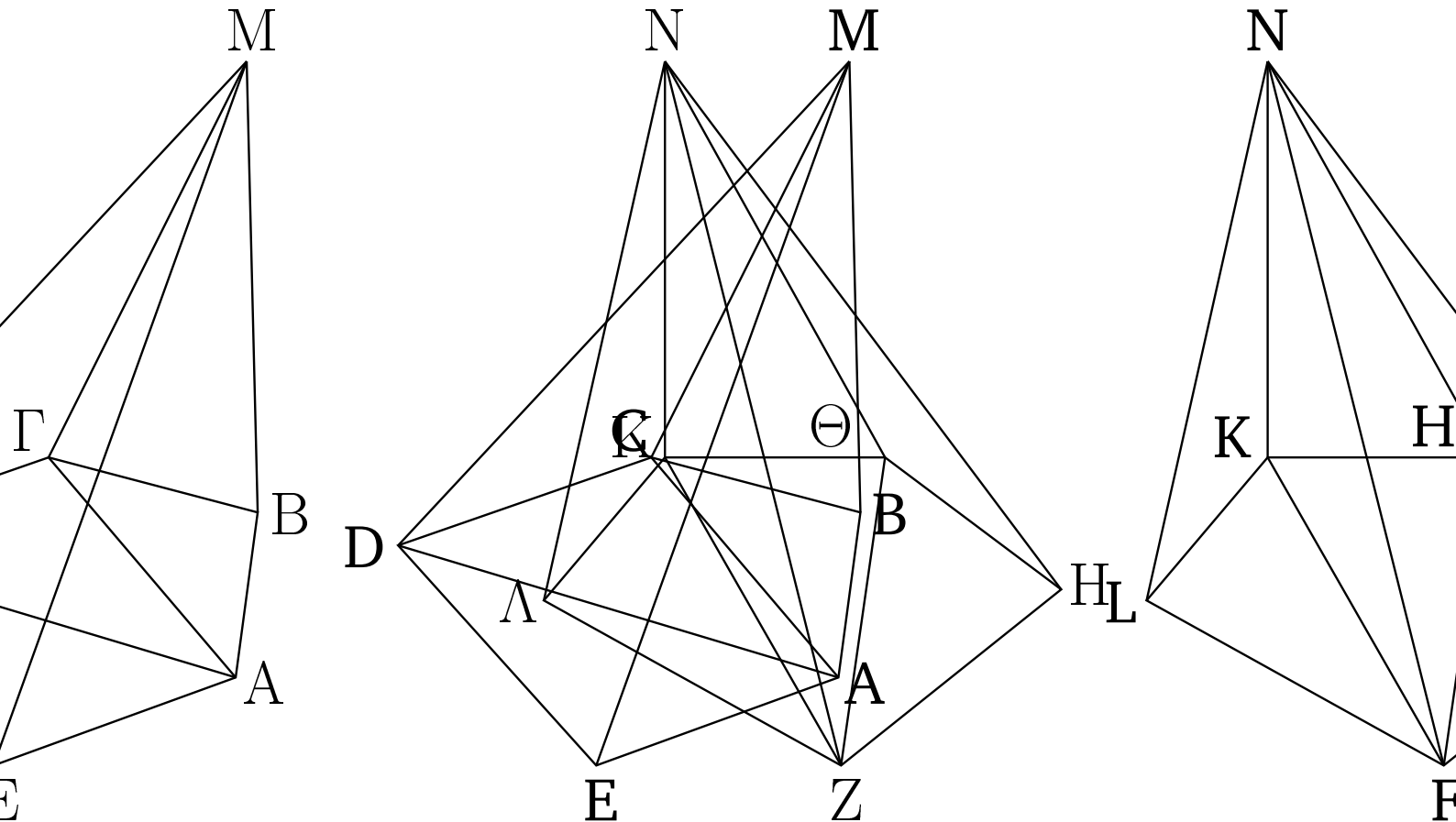
For, if possible, let it be (in this ratio) to some greater (solid), W . Thus, inversely, as base DEF (is) to base ABC , so solid W (is) to pyramid $ABCG$ [Prop. 5.7. corr.]. And as solid W (is) to pyramid $ABCG$, so pyramid $DEFH$ (is) to some (solid) less than pyramid $ABCG$, as shown before [Prop. 12.2 lem.]. And, thus, as base DEF (is) to base ABC , so pyramid $DEFH$ (is) to some (solid) less than pyramid $ABCG$ [Prop. 5.11]. The very thing was shown (to be) absurd. Thus, base ABC is not to base DEF , as pyramid $ABCG$ (is) to some solid greater than pyramid $DEFH$. And, it was shown that neither (is it in this ratio) to a lesser (solid). Thus, as base ABC is to base DEF , so pyramid $ABCG$ (is) to pyramid $DEFH$. (Which is) the very thing it was required to show.

6.

Αἱ ὑπὸ τὸ αὐτὸ <ύψοξ ο>ῦσαι πυραμίδες καὶ πολυγώνουξ >έθουσαι βάσεις πρὸς ἀλλήλας εἰσὶν ὡς αἱ βάσεις.

Proposition 6

Pyramids which are of the same height, and have polygonal bases, are to one another as their bases.



>Εστωσαν ὑπὸ τὸ αὐτὸ <ύψοξ πυραμίδες, <ὧν [αἱ] βάσεις μὲν τὰ ΑΒΓΔΕ, ΖΗΘΚΛ πολύγωνα, κορυφαὶ δὲ τὰ Μ, Ν σημεῖα; λέγω, <ὅτι ἔστιν ὡς ἡ ΑΒΓΔΕ βάσις πρὸς τὴν ΖΗΘΚΛ βάσιν, ο<ύτωξ ἡ ΑΒΓΔΕΜ πυραμὶς πρὸς τὴν ΖΗΘΚΛΝ πυραμίδα.

Ἐπεξεύθωσαν γὰρ αἱ ΑΓ, ΑΔ, ΖΘ, ΖΚ. ἐπεὶ ο>ὦν δύο πυραμίδες εἰσὶν αἱ ΑΒΓΜ, ΑΓΔΜ τριγώνουξ >έθουσαι

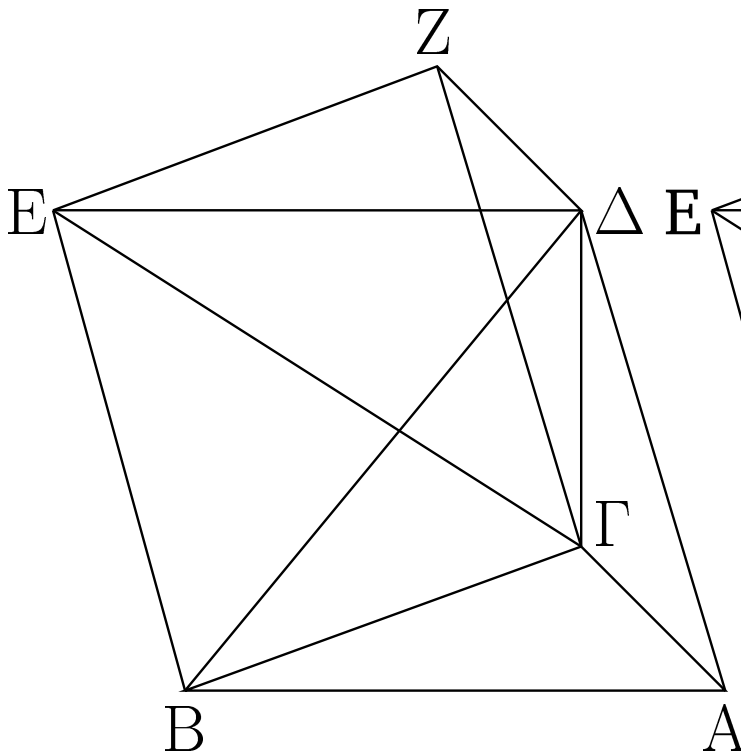
Let there be pyramids of the same height whose bases (are) the polygons $ABCDE$ and $FGHKL$, and apexes the points M and N (respectively). I say that as base $ABCDE$ is to base $FGHKL$, so pyramid $ABCDEM$ (is) to pyramid $FGHKLN$.

For let AC , AD , FH , and FK be joined. Therefore, since $ABCM$ and $ACDM$ are two pyramids having triangular

βάσεις καὶ <ὕψος> ἴσον, πρὸς ἀλλήλας εἰσὶν ὡς αἱ βάσεις; >έστιν >άρα ὡς ἡ $AB\Gamma$ βάσις πρὸς τὴν $A\Gamma\Delta$ βάσιν, ο<ύτωξ ἡ $AB\Gamma M$ πυραμὶς πρὸς τὴν $A\Gamma\Delta M$ πυραμίδα. καὶ συνθέντι ὡς ἡ $AB\Gamma\Delta$ βάσις πρὸς τὴν $A\Gamma\Delta$ βάσιν, ο<ύτωξ ἡ $AB\Gamma\Delta M$ πυραμὶς πρὸς τὴν $A\Gamma\Delta M$ πυραμίδα. ἀλλὰ καὶ ὡς ἡ $A\Gamma\Delta$ βάσις πρὸς τὴν $A\Delta E$ βάσιν, ο<ύτωξ ἡ $A\Gamma\Delta M$ πυραμὶς πρὸς τὴν $A\Delta E M$ πυραμίδα. δι> >ἴσου >άρα ὡς ἡ $AB\Gamma\Delta$ βάσις πρὸς τὴν $A\Delta E$ βάσιν, ο<ύτωξ ἡ $AB\Gamma\Delta M$ πυραμὶς πρὸς τὴν $A\Delta E M$ πυραμίδα. καὶ συνθέντι πάλιν, ὡς ἡ $AB\Gamma\Delta E$ βάσις πρὸς τὴν $A\Delta E$ βάσιν, ο<ύτωξ ἡ $AB\Gamma\Delta E M$ πυραμὶς πρὸς τὴν $A\Delta E M$ πυραμίδα. ὁμοίως δὲ δειθθήσεται, <ὅτι καὶ ὡς ἡ $ZH\Theta K\Lambda$ βάσις πρὸς τὴν $ZH\Theta$ βάσιν, ο<ύτωξ καὶ ἡ $ZH\Theta K\Lambda N$ πυραμὶς πρὸς τὴν $ZH\Theta N$ πυραμίδα. καὶ ἐπεὶ δύο πυραμίδες εἰσὶν αἱ $A\Delta E M$, $ZH\Theta N$ τριγώνουξ >έθουσαι βάσεις καὶ <ὕψος> ἴσον, >έστιν >άρα ὡς ἡ $A\Delta E$ βάσις πρὸς τὴν $ZH\Theta$ βάσιν, ο<ύτωξ ἡ $A\Delta E M$ πυραμὶς πρὸς τὴν $ZH\Theta N$ πυραμίδα. ἀλλ> ὡς ἡ $A\Delta E$ βάσις πρὸς τὴν $AB\Gamma\Delta E$ βάσιν, ο<ύτωξ >ἦν ἡ $A\Delta E M$ πυραμὶς πρὸς τὴν $AB\Gamma\Delta E M$ πυραμίδα. καὶ δι> >ἴσου >άρα ὡς ἡ $AB\Gamma\Delta E$ βάσις πρὸς τὴν $ZH\Theta$ βάσιν, ο<ύτωξ ἡ $AB\Gamma\Delta E M$ πυραμὶς πρὸς τὴν $ZH\Theta N$ πυραμίδα. ἀλλὰ μὴν καὶ ὡς ἡ $ZH\Theta$ βάσις πρὸς τὴν $ZH\Theta K\Lambda$ βάσιν, ο<ύτωξ >ἦν καὶ ἡ $ZH\Theta N$ πυραμὶς πρὸς τὴν $ZH\Theta K\Lambda N$ πυραμίδα, καὶ δι> >ἴσου >άρα ὡς ἡ $AB\Gamma\Delta E$ βάσις πρὸς τὴν $ZH\Theta K\Lambda$ βάσιν, ο<ύτωξ ἡ $AB\Gamma\Delta E M$ πυραμὶς πρὸς τὴν $ZH\Theta K\Lambda N$ πυραμίδα; <ὅπερ >έδει δεῖξαι.

7.

Πᾶν πρίσμα τρίγωνον >έθον βάσιν διαιρεῖται εἰς τρεῖς πυραμίδαξ >ἴσαξ ἀλλήλαιξ τριγώνουξ βάσειξ ἐθούσαξ.



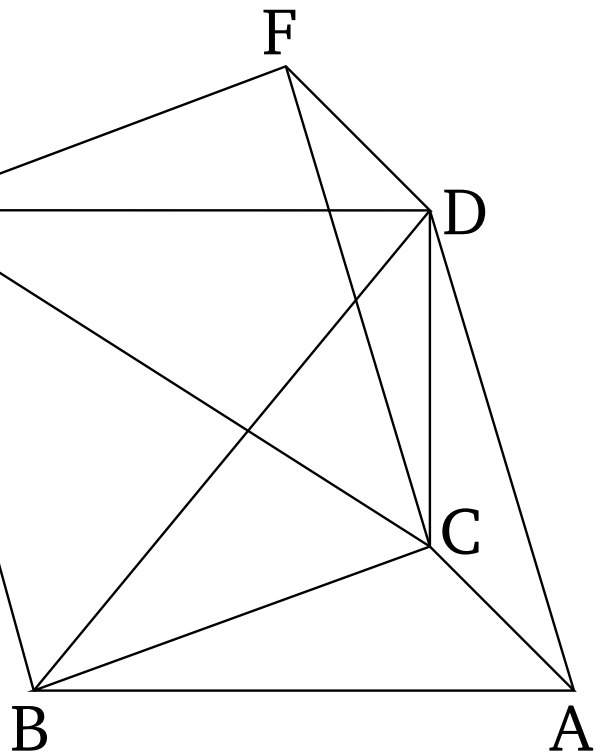
>Ἐστω πρίσμα, ο<ὅ βάσις μὲν τὸ $AB\Gamma$ τρίγωνον, ἀπεναντίον δὲ τὸ $\Delta E Z$; λέγω, <ὅτι τὸ $AB\Gamma\Delta E Z$ πρίσμα διαιρεῖται εἰς τρεῖς πυραμίδαξ >ἴσαξ ἀλλήλαιξ τριγώνουξ ἐθούσαξ βάσειξ.

Ἐπεξεύθωσαν γὰρ αἱ BA , EG , GA . ἐπεὶ παραλληλόγραμμον

bases and equal height, they are to one another as their bases [Prop. 12.5]. Thus, as base ABC is to base ACD , so pyramid $ABCM$ (is) to pyramid $ACDM$. And, via composition, as base $ABCD$ (is) to base ACD , so pyramid $ABCDM$ (is) to pyramid $ACDM$ [Prop. 5.18]. But, as base ACD (is) to base ADE , so pyramid $ACDM$ (is) also to pyramid $ADEM$ [Prop. 12.5]. Thus, via equality, as base $ABCD$ (is) to base ADE , so pyramid $ABCDM$ (is) to pyramid $ADEM$ [Prop. 5.22]. And, again, via composition, as base $ABCDE$ (is) to base ADE , so pyramid $ABCDEM$ (is) to pyramid $ADEM$ [Prop. 5.18]. So, similarly, it can also be shown that as base $FGHKL$ (is) to base FGH , so pyramid $FGHKLN$ (is) also to pyramid $FGHN$. And since $ADEM$ and $FGHN$ are two pyramids having triangular bases and equal height, thus as base ADE (is) to base FGH , so pyramid $ADEM$ (is) to pyramid $FGHN$ [Prop. 12.5]. But, as base ADE (is) to base $ABCDE$, so pyramid $ADEM$ (was) to pyramid $ABCDEM$. Thus, via equality, as base $ABCDE$ (is) to base FGH , so pyramid $ABCDEM$ (is) also to pyramid $FGHN$ [Prop. 5.22]. But, furthermore, as base FGH (is) to base $FGHKL$, so pyramid $FGHN$ was also to pyramid $FGHKLN$. Thus, via equality, as base $ABCDE$ (is) to base $FGHKL$, so pyramid $ABCDEM$ (is) also to pyramid $FGHKLN$ [Prop. 5.22]. (Which is) the very thing it was required to show.

Proposition 7

Any prism having a triangular base is divided into three pyramids having triangular bases (which are) equal to one another.



Let there be a prism whose base (is) triangle ABC , and opposite (plane) DEF . I say that prism $ABCDEF$ is divided into three pyramids having triangular bases (which are) equal to one another.

For let BD , EC , and CD be joined. Since $ABED$ is a par-

ἔστι τὸ $ABED$, διάμετρος δὲ αὐτοῦ ἔστιν ἡ BD , >ίσον >άρα ἔστι τὸ ABD τρίγωνον τῷ EBD τρίγωνῳ; καὶ ἡ πυραμὶς >άρα, <ἧς βάσις μὲν τὸ ABD τρίγωνον, κορυφὴ δὲ τὸ Γ σημεῖον, >ίση ἔστι πυραμίδι, <ἧς βάσις μὲν ἔστι τὸ DEB τρίγωνον, κορυφὴ δὲ τὸ Γ σημεῖον. ἀλλὰ ἡ πυραμὶς, <ἧς βάσις μὲν ἔστι τὸ DEB τρίγωνον, κορυφὴ δὲ τὸ Γ σημεῖον, ἡ αὐτὴ ἔστι πυραμίδι, <ἧς βάσις μὲν ἔστι τὸ $EB\Gamma$ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον; ὑπὸ γὰρ τῶν αὐτῶν ἐπιπέδων περιέθεται. καὶ πυραμὶς >άρα, <ἧς βάσις μὲν ἔστι τὸ ABD τρίγωνον, κορυφὴ δὲ τὸ Γ σημεῖον, >ίση ἔστι πυραμίδι, <ἧς βάσις μὲν ἔστι τὸ $EB\Gamma$ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον. πάλιν, ἐπεὶ παραλληλόγραμμον ἔστι τὸ $Z\Gamma BE$, διάμετρος δὲ ἔστιν αὐτοῦ ἡ GE , >ίσον ἔστι τὸ GEZ τρίγωνον τῷ ΓBE τρίγωνῳ. καὶ πυραμὶς >άρα, <ἧς βάσις μὲν ἔστι τὸ $B\Gamma E$ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον, >ίση ἔστι πυραμίδι, <ἧς βάσις μὲν ἔστι τὸ EGZ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον. ἡ δὲ πυραμὶς, <ἧς βάσις μὲν ἔστι τὸ $B\Gamma E$ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον, >ίση ἐδείκθη πυραμίδι, <ἧς βάσις μὲν ἔστι τὸ ABD τρίγωνον, κορυφὴ δὲ τὸ Γ σημεῖον; καὶ πυραμὶς >άρα, <ἧς βάσις μὲν ἔστι τὸ GEZ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον, >ίση ἔστι πυραμίδι, <ἧς βάσις μὲν [ἔστι] τὸ ABD τρίγωνον, κορυφὴ δὲ τὸ Γ σημεῖον; διήρηται >άρα τὸ $AB\Gamma\Delta EZ$ πρίσμα εἰς τρεῖς πυραμίδας >ίσας ἀλλήλαις τριγώνουζ ἐθούσας βάσεις.

Καὶ ἐπεὶ πυραμὶς, <ἧς βάσις μὲν ἔστι τὸ ABD τρίγωνον, κορυφὴ δὲ τὸ Γ σημεῖον, ἡ αὐτὴ ἔστι πυραμίδι, <ἧς βάσις τὸ ΓAB τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον; ὑπὸ γὰρ τῶν αὐτῶν ἐπιπέδων περιέθονται; ἡ δὲ πυραμὶς, <ἧς βάσις τὸ ABD τρίγωνον, κορυφὴ δὲ τὸ Γ σημεῖον, τρίτον ἐδείκθη τοῦ πρίσματος, οὐδὲν βάσις τὸ $AB\Gamma$ τρίγωνον, ἀπεναντίον δὲ τὸ ΔEZ , καὶ ἡ πυραμὶς >άρα, <ἧς βάσις τὸ $AB\Gamma$ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον, τρίτον ἔστι τοῦ πρίσματος τοῦ >έθοντος βάσις τὴν αὐτὴν τὸ $AB\Gamma$ τρίγωνον, ἀπεναντίον δὲ τὸ ΔEZ .

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, <ὅτι πᾶσα πυραμὶς τρίτον μέρος ἔστι τοῦ πρίσματος τοῦ τὴν αὐτὴν βάσιν >έθοντος αὐτῇ καὶ <ύψος >ίσον; <ὅπερ >έδει δεῖξαι.

8.

Αἱ <όμοιαι πυραμίδες καὶ τριγώνουζ >έθουσαι βάσεις ἐν τριπλασίονι λόγῳ εἰσι τῶν ὁμολόγων πλευρῶν.

>Ἐστῶσαν <όμοιαι καὶ ὁμοίωζ κείμεναι πυραμίδες, <ὦν βάσεις μὲν εἰσι τὰ $AB\Gamma$, ΔEZ τρίγωνα, κορυφαὶ δὲ τὰ H , Θ σημεία; λέγω, <ὅτι ἡ $AB\Gamma H$ πυραμὶς πρὸς τὴν $\Delta EZ\Theta$ πυραμίδα τριπλασίονα λόγον >έθει >ήπερ ἡ $B\Gamma$ πρὸς τὴν

allelogram, and BD is its diagonal, triangle ABD is thus equal to triangle EBD [Prop. 1.34]. And, thus, the pyramid whose base (is) triangle ABD , and apex the point C , is equal to the pyramid whose base is triangle DEB , and apex the point C [Prop. 12.5]. But, the pyramid whose base is triangle DEB , and apex the point C , is the same as the pyramid whose base is triangle EBC , and apex the point D . For they are contained by the same planes. And, thus, the pyramid whose base is ABD , and apex the point C , is equal to the pyramid whose base is EBC and apex the point D . Again, since $FCBE$ is a parallelogram, and CE is its diagonal, triangle CEF is equal to triangle CBE [Prop. 1.34]. And, thus, the pyramid whose base is triangle BCE , and apex the point D , is equal to the pyramid whose base is triangle ECF , and apex the point D [Prop. 12.5]. And the pyramid whose base is triangle BCE , and apex the point D , was shown (to be) equal to the pyramid whose base is triangle ABD , and apex the point C . Thus, the pyramid whose base is triangle CEF , and apex the point D , is also equal to the pyramid whose base [is] triangle ABD , and apex the point C . Thus, the prism $ABCDEF$ has been divided into three pyramids having triangular bases (which are) equal to one another.

And since the pyramid whose base is triangle ABD , and apex the point C , is the same as the pyramid whose base is triangle CAB , and apex the point D . For they are contained by the same planes. And the pyramid whose base (is) triangle ABD , and apex the point C , was shown (to be) a third of the prism whose base is triangle ABC , and opposite (plane) DEF , thus the pyramid whose base is triangle ABC , and apex the point D , is also a third of the pyramid having the same base, triangle ABC , and opposite (plane) DEF .

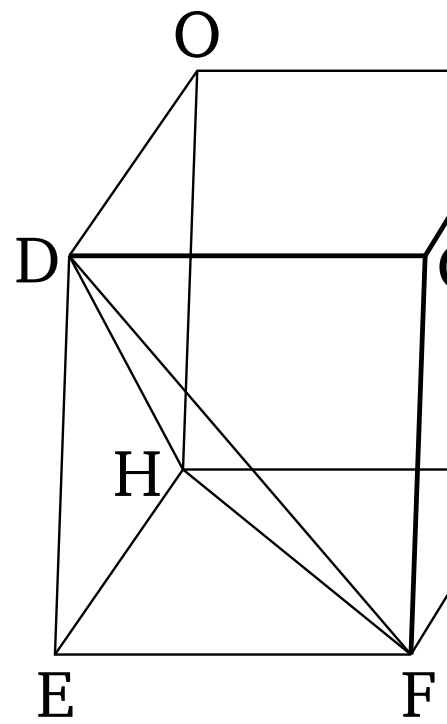
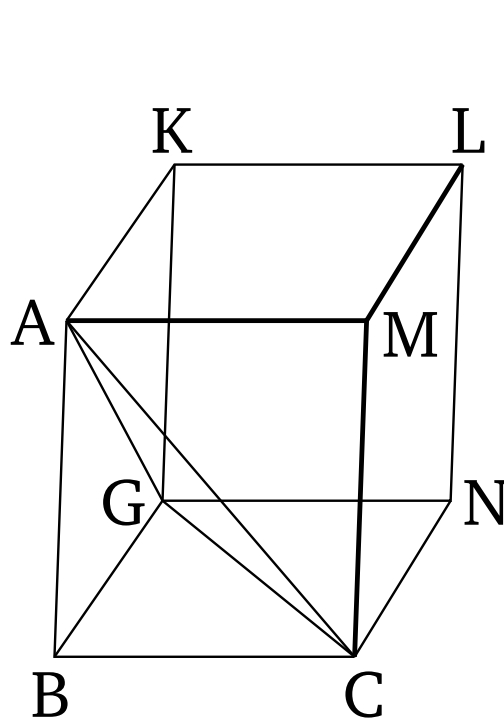
Corollary

And, from this, (it is) clear that any pyramid is the third part of the prism having the same base as it, and an equal height. (Which is) the very thing it was required to show.

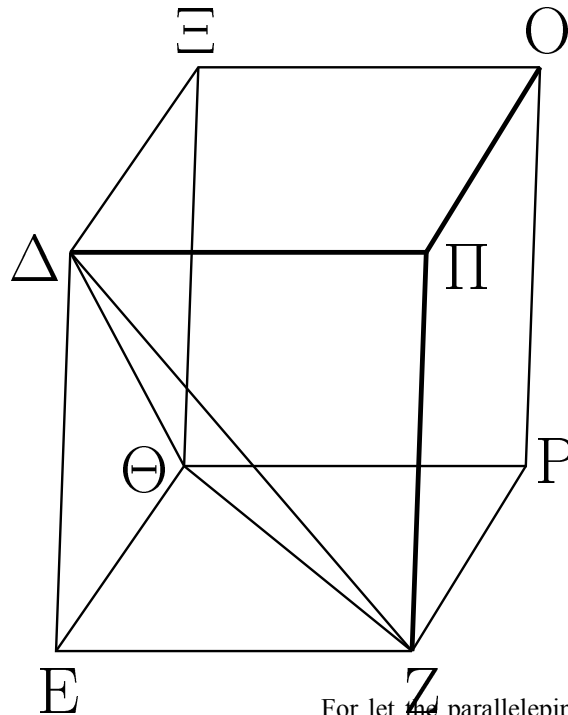
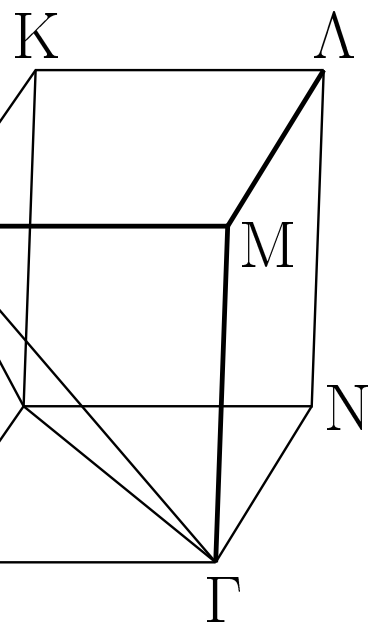
Proposition 8

Similar pyramids which also have triangular bases are in the cubed ratio of their corresponding sides.

Let there be similar, and similarly laid out, pyramids whose bases are triangles ABC and DEF , and apexes the points G and H (respectively). I say that pyramid $ABCG$ has to pyramid $DEFH$ the cubed ratio of that BC (has) to EF .



EZ.



Συμπεληρώσθω γὰρ τὰ $BHML$, $EΘΠΟ$ στερεὰ παραλληλεπίπεδα, καὶ ἐπεὶ ὁμοία ἐστὶν ἡ $ABΓH$ πυραμὶς τῇ $ΔΕΖΘ$ πυραμίδι, ὁμοία ἔστιν ἡ μὲν ὑπὸ $ABΓ$ γωνία τῇ ὑπὸ $ΔΕΖ$ γωνίᾳ, ἡ δὲ ὑπὸ HBG τῇ ὑπὸ $ΘΕΖ$, ἡ δὲ ὑπὸ ABH τῇ ὑπὸ $ΔΕΘ$, καὶ ἐστὶν ὥς ἡ AB πρὸς τὴν $ΔΕ$, οὕτως ἡ $BΓ$ πρὸς τὴν $ΕΖ$, καὶ ἡ BH πρὸς τὴν $ΕΘ$. καὶ ἐπεὶ ἐστὶν ὥς ἡ AB πρὸς τὴν $ΔΕ$, οὕτως ἡ $BΓ$ πρὸς τὴν $ΕΖ$, καὶ περὶ ὁμοίων γωνιῶν αἱ πλευραὶ ἀνάλογόν εἰσιν, ὁμοίων ἔστι τὸ BM παραλληλόγραμμον τῷ $ΕΠ$ παραλληλογράμμῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ μὲν BN τῷ $ΕΡ$ ὁμοίον ἐστὶ, τὸ δὲ BK τῷ $ΕΧ$; τὰ τρία ἄρα τὰ MB , BK , BN τρισὶ τοῖς $ΕΠ$, $ΕΧ$, $ΕΡ$ ὁμοία ἐστὶν. ἀλλὰ τὰ μὲν τρία τὰ MB , BK , BN τρισὶ τοῖς ἀπεναντίον ὅμοια τε καὶ ὁμοία ἐστὶν, τὰ δὲ τρία τὰ $ΕΠ$, $ΕΧ$, $ΕΡ$ τρισὶ τοῖς ἀπεναντίον ὅμοια τε καὶ ὁμοία ἐστὶν. τὰ $BHML$, $EΘΠΟ$ ἄρα στερεὰ ὑπὸ

For let the parallelepiped solids $BGML$ and $EHQP$ be completed. And since pyramid $ABCH$ is similar to pyramid $DEFH$, angle ABC is thus equal to angle DEF , and GBC to HEF , and ABG to DEH . And as AB is to DE , so BC (is) to EF , and BG to EH [Def. 11.9]. And since as AB is to DE , so BC (is) to EF , and (so) the sides around equal angles are proportional, parallelogram BM is thus similar to parallelogram EQ . So, for the same (reasons), BN is also similar to ER , and BK to EO . Thus, the three (parallelograms) MB , BK , and BN are similar to the three (parallelograms) EQ , EO , ER (respectively). But, the three (parallelograms) MB , BK , and BN are (both) equal and similar to the three opposite (parallelograms), and the three (parallelograms) EQ , EO , and ER are (both) equal and similar to the three opposite (parallelograms) [Prop. 11.24]. Thus, the solids $BGML$

ὁμοίων ἐπιπέδων >ίσων τὸ πληθὸς περιέθεται. <ὁμοιον >άρα ἐστὶ τὸ $BHML$ στερεὸν τῷ $EΘΠΟ$ στερεῷ. τὰ δὲ <ὁμοια στερεὰ παραλληλεπίπεδα ἐν τριπλασίονι λόγῳ ἐστὶ τῶν ὁμολόγων πλευρῶν. τὸ $BHML$ >άρα στερεὸν πρὸς τὸ $EΘΠΟ$ στερεὸν τριπλασίονα λόγον >έθει >ήπερ ἡ ὁμόλογος πλευρὰ ἢ $BΓ$ πρὸς τὴν ὁμόλογον πλευρὰν τὴν $EΖ$. ὥς δὲ τὸ $BHML$ στερεὸν πρὸς τὸ $EΘΠΟ$ στερεόν, ο<ύτως ἢ $ABΓΗ$ πυραμὶς πρὸς τὴν $ΔΕΖΘ$ πυραμίδα, ἐπειδήπερ ἡ πυραμὶς <έκτον μέρος ἐστὶ τοῦ στερεοῦ διὰ τὸ καὶ τὸ πρίσμα <ήμισυ >ὸν τοῦ στερεοῦ παραλληλεπίπεδου τριπλασίον ε>ῖναι τῆς πυραμίδος. καὶ ἡ $ABΓΗ$ >άρα πυραμὶς πρὸς τὴν $ΔΕΖΘ$ πυραμίδα τριπλασίονα λόγον >έθει >ήπερ ἢ $BΓ$ πρὸς τὴν $EΖ$; <όπερ >έδει δεῖχαι.

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, <ότι καὶ αἱ πολυγώνους >έθουσαι βάσεις <ὁμοια πυραμίδες πρὸς ἀλλήλας ἐν τριπλασίονι λόγῳ εἰσὶ τῶν ὁμολόγων πλευρῶν. διαιρεθεισῶν γὰρ αὐτῶν εἰς τὰς ἐν αὐταῖς πυραμίδας τριγώνους βάσεις ἐθούσας τῷ καὶ τὰ <ὁμοια πολύγωνα τῶν βάσεων εἰς <ὁμοια τρίγωνα διαιρεῖσθαι καὶ >ίσα τῷ πλήθει καὶ ὁμόλογα τοῖς <όλοις >έσται ὡς [ἡ] ἐν τῇ ἐτέρᾳ μία πυραμὶς τριγώνων >έθουσα βάσιν πρὸς τὴν ἐν τῇ ἐτέρᾳ μίαν πυραμίδα τριγώνων >έθουσαν βάσιν, ο<ύτως καὶ <άπασαι αἱ ἐν τῇ ἐτέρᾳ πυραμίδι πυραμίδες τριγώνους >έθουσαι βάσεις πρὸς τὰς ἐν τῇ ἐτέρᾳ πυραμίδι πυραμίδας τριγώνους βάσεις ἐθούσας, τουτέστιν αὐτὴ ἡ πολύγωνος βάσιν >έθουσα πυραμὶς πρὸς τὴν πολύγωνον βάσιν >έθουσαν πυραμίδα. ἢ δὲ τρίγωνον βάσιν >έθουσα πυραμὶς πρὸς τὴν τρίγωνον βάσιν >έθουσαν ἐν τριπλασίονι λόγῳ ἐστὶ τῶν ὁμολόγων πλευρῶν; καὶ ἡ πολύγωνος >άρα βάσιν >έθουσα πρὸς τὴν ὁμοίαν βάσιν >έθουσαν τριπλασίονα λόγον >έθει >ήπερ ἡ πλευρὰ πρὸς τὴν πλευράν.

9.

Τῶν >ίσων πυραμίδων καὶ τριγώνους βάσεις ἐθουσῶν ἀντιπεπόνθασιν αἱ βάσεις τοῖς <ύψεσιν; καὶ <ὼν πυραμίδων τριγώνους βάσεις ἐθουσῶν ἀντιπεπόνθασιν αἱ βάσεις τοῖς <ύψεσιν, >ίσαι εἰσὶν ἐκεῖναι.

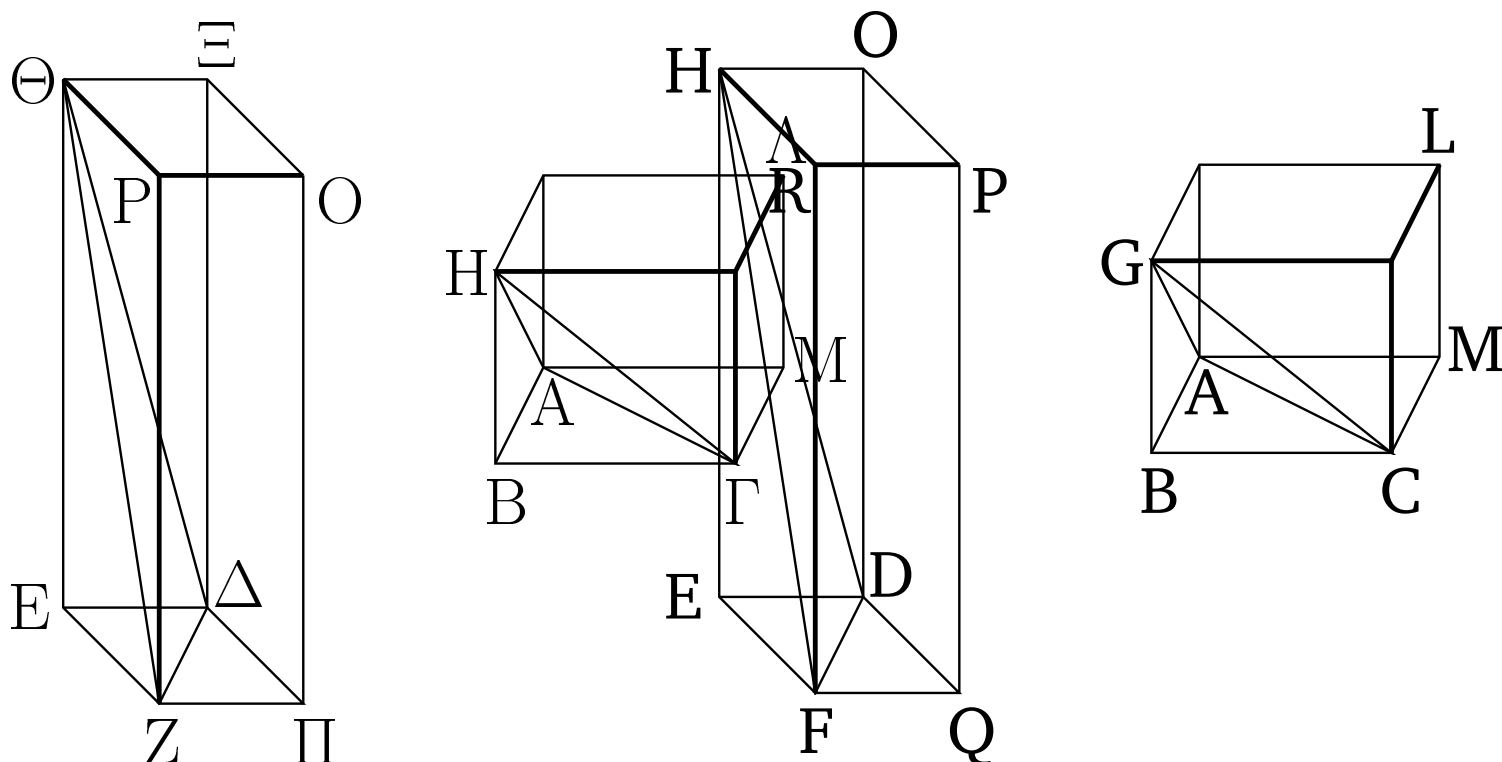
and $EHQP$ are contained by equal numbers of similar (and similarly laid out) planes. Thus, solid $BGML$ is similar to solid $EHQP$ [Def. 11.9]. And similar parallelepiped solids are in the cubed ratio of corresponding sides [Prop. 11.33]. Thus, solid $BGML$ has to solid $EHQP$ the cubed ratio that the corresponding side BC (has) to the corresponding side EF . And as solid $BGML$ (is) to solid $EHQP$, so pyramid $ABCG$ (is) to pyramid $DEFH$, inasmuch as the pyramid is the sixth part of the solid, on account of the prism, being half of the parallelepiped solid [Prop. 11.28], also being three times the pyramid [Prop. 12.7]. Thus, pyramid $ABCG$ also has to pyramid $DEFH$ the cubed ratio that BC (has) to EF . (Which is) the very thing it was required to show.

Corollary

So, from this, (it is) also clear that similar pyramids having polygonal bases (are) to one another as the cubed ratio of their corresponding sides. For, dividing them into the pyramids (contained) within them which have triangular bases, with the similar polygons of the bases also being divided into similar triangles (which are) both equal in number, and corresponding, to the wholes [Prop. 6.20]. As one pyramid having a triangular base in the former (pyramid having a polygonal base is) to one pyramid having a triangular base in the latter (pyramid having a polygonal base), so (the sum of) all the pyramids having triangular bases in the former pyramid will also be to (the sum of) all the pyramids having triangular bases in the latter pyramid [Prop. 5.12]—that is to say, the (former) pyramid itself having a polygonal base to the (latter) pyramid having a polygonal base. And a pyramid having a triangular base is to a (pyramid) having a triangular base in the cubed ratio of corresponding sides [Prop. 12.8]. Thus, a (pyramid) having a polygonal base also has to to a (pyramid) having a similar base the cubed ratio of a (corresponding) side to a (corresponding) side.

Proposition 9

The bases of equal pyramids which also have triangular bases are reciprocally proportional to their heights. And those pyramids which have triangular bases whose bases are reciprocally proportional to their heights are equal.



>Ἐστωσαν γὰρ >ῖσαι πυραμίδες τριγώνου βάσεις >έθουσαι τὰς ABΓ, ΔΕΖ, κορυφὰς δὲ τὰ Η, Θ σημεία; λέγω, <ὅτι τῶν ABΓΗ, ΔΕΖΘ πυραμίδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς <ύψεσιν, καὶ ἔστιν ὡς ἡ ABΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, ο<ύτως τὸ τῆς ΔΕΖΘ πυραμίδος <ύψος πρὸς τὸ τῆς ABΓΗ πυραμίδος <ύψος.

Συμπεληρώσω γὰρ τὰ BHΜΛ, ΕΘΠΟ στερεὰ παραλληλεπίπεδα, καὶ ἐπεὶ >ῖση ἔστιν ἡ ABΓΗ πυραμὶς τῇ ΔΕΖΘ πυραμίδι, καὶ ἔστι τῆς μὲν ABΓΗ πυραμίδος ἔχαπλάσιον τὸ BHΜΛ στερεόν, τῆς δὲ ΔΕΖΘ πυραμίδος ἔχαπλάσιον τὸ ΕΘΠΟ στερεόν, >ῖσον >άρα ἔστι τὸ BHΜΛ στερεόν τῷ ΕΘΠΟ στερεῷ. τῶν δὲ >ῖσων στερεῶν παραλληλεπιπύδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς <ύψεσιν; >έστιν >άρα ὡς ἡ BM βάσις πρὸς τὴν ΕΠ βάσιν, ο<ύτως τὸ τοῦ ΕΘΠΟ στερεοῦ <ύψος πρὸς τὸ τοῦ BHΜΛ στερεοῦ <ύψος. ἀλλ> ὡς ἡ BM βάσις πρὸς τὴν ΕΠ, ο<ύτως τὸ ABΓ τρίγωνον πρὸς τὸ ΔΕΖ τρίγωνον. καὶ ὡς >άρα τὸ ABΓ τρίγωνον πρὸς τὸ ΔΕΖ τρίγωνον, ο<ύτως τὸ τοῦ ΕΘΠΟ στερεοῦ <ύψος πρὸς τὸ τοῦ BHΜΛ στερεοῦ <ύψος. ἀλλὰ τὸ μὲν τοῦ ΕΘΠΟ στερεοῦ <ύψος τὸ αὐτὸ ἔστι τῷ τῆς ΔΕΖΘ πυραμίδος <ύψει, τὸ δὲ τοῦ BHΜΛ στερεοῦ <ύψος τὸ αὐτὸ ἔστι τῷ τῆς ABΓΗ πυραμίδος <ύψει; >έστιν >άρα ὡς ἡ ABΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, ο<ύτως τὸ τῆς ΔΕΖΘ πυραμίδος <ύψος πρὸς τὸ τῆς ABΓΗ πυραμίδος <ύψος. τῶν ABΓΗ, ΔΕΖΘ >άρα πυραμίδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς <ύψεσιν.

Ἀλλὰ δὴ τῶν ABΓΗ, ΔΕΖΘ πυραμίδων ἀντιπεπονηθέντων αἱ βάσεις τοῖς <ύψεσιν, καὶ >έστω ὡς ἡ ABΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, ο<ύτως τὸ τῆς ΔΕΖΘ πυραμίδος <ύψος πρὸς τὸ τῆς ABΓΗ πυραμίδος <ύψος; λέγω, <ὅτι >ῖση ἔστιν ἡ ABΓΗ πυραμὶς τῇ ΔΕΖΘ πυραμίδι.

Τῶν γὰρ αὐτῶν κατασκευασθέντων, ἐπεὶ ἔστιν ὡς ἡ ABΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, ο<ύτως τὸ τῆς ΔΕΖΘ πυραμίδος <ύψος πρὸς τὸ τῆς ABΓΗ πυραμίδος <ύψος, ἀλλ> ὡς ἡ ABΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, ο<ύτως τὸ BM παραλληλόγραμνον πρὸς τὸ ΕΠ παραλληλόγραμμον, καὶ ὡς >άρα τὸ BM παραλληλόγραμνον πρὸς τὸ ΕΠ παραλληλόγραμμον, ο<ύτως τὸ τῆς ΔΕΖΘ πυραμίδος <ύψος πρὸς τὸ τῆς ABΓΗ πυραμίδος

For let there be (two) equal pyramids having the triangular bases ABC and DEF , and apexes the points G and H (respectively). I say that the bases of the pyramids $ABCG$ and $DEFH$ are reciprocally proportional to their heights, and (so) that as base ABC is to base DEF , so the height of pyramid $DEFH$ (is) to the height of pyramid $ABCG$.

Let the parallelepiped solids $BGML$ and $EHQP$ be completed. And since pyramid $ABCG$ is equal to pyramid $DEFH$, and solid $BGML$ is six times pyramid $ABCG$ (see previous proposition), and solid $EHQP$ (is) six times pyramid $DEFH$, solid $BGML$ is thus equal to solid $EHQP$. And the bases of equal parallelepiped solids are reciprocally proportional to their heights [Prop. 11.34]. Thus, as base BM is to base EQ , so the height of solid $EHQP$ (is) to the height of solid $BGML$. But, as base BM (is) to base EQ , so triangle ABC (is) to triangle DEF [Prop. 1.34]. And, thus, as triangle ABC (is) to triangle DEF , so the height of solid $EHQP$ (is) to the height of solid $BGML$ [Prop. 5.11]. But, the height of solid $EHQP$ is the same as the height of pyramid $DEFH$, and the height of solid $BGML$ is the same as the height of pyramid $ABCG$. Thus, as base ABC is to base DEF , so the height of pyramid $DEFH$ (is) to the height of pyramid $ABCG$. Thus, the bases of pyramids $ABCG$ and $DEFH$ are reciprocally proportional to their heights.

And so, let the bases of pyramids $ABCG$ and $DEFH$ be reciprocally proportional to their heights, and (thus) let base ABC be to base DEF , as the height of pyramid $DEFH$ (is) to the height of pyramid $ABCG$. I say that pyramid $ABCG$ is equal to pyramid $DEFH$.

For, with the same construction, since as base ABC is to base DEF , so the height of pyramid $DEFH$ (is) to the height of pyramid $ABCG$, but as base ABC (is) to base DEF , so parallelogram BM (is) to parallelogram EQ [Prop. 1.34], thus as parallelogram BM (is) to parallelogram EQ , so the height of pyramid $DEFH$ (is) also to the height of pyramid $ABCG$ [Prop. 5.11]. But, the height of

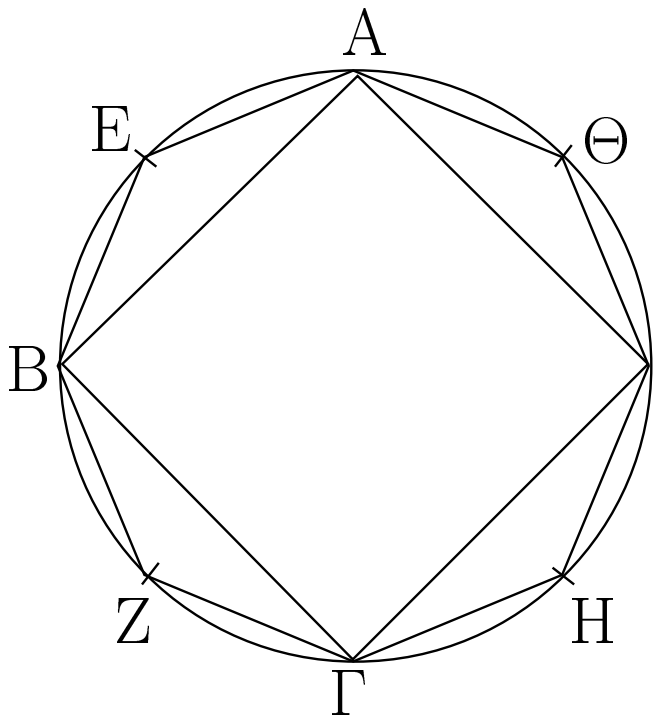
<ύψος. ἀλλὰ τὸ [μὲν] τῆς ΔΕΖΘ πυραμίδος <ύψος τὸ αὐτὸ ἐστὶ τῷ τοῦ ΕΘΠΟ παραλληλεπιπέδου <ύψει, τὸ δὲ τῆς ΑΒΓΗ πυραμίδος <ύψος τὸ αὐτὸ ἐστὶ τῷ τοῦ ΒΗΜΛ παραλληλεπιπέδου <ύψει; >ἐστὶν >άρα ὥς ἡ ΒΜ βάσις πρὸς τὴν ΕΠ βάσιν, ο-ύτως τὸ τοῦ ΕΘΠΟ παραλληλεπιπέδου <ύψος πρὸς τὸ τοῦ ΒΗΜΛ παραλληλεπιπέδου <ύψος. <ὦν δὲ στερεῶν παραλληλεπιπέδων ἀντιτεπόνθασιν αἱ βάσεις τοῖς <ύψεσιν, >ίσα ἐστὶν ἐκεῖνα; >ίσον >άρα ἐστὶ τὸ ΒΗΜΛ στερεὸν παραλληλεπίπεδον τῷ ΕΘΠΟ στερεῷ παραλληλεπιπέδῳ. καὶ ἐστὶ τοῦ μὲν ΒΗΜΛ <έκτον μέρος ἡ ΑΒΓΗ πυραμὶς, τοῦ δὲ ΕΘΠΟ παραλληλεπιπέδου <έκτον μέρος ἡ ΔΕΖΘ πυραμὶς; >ίση >άρα ἡ ΑΒΓΗ πυραμὶς τῇ ΔΕΖΘ πυραμίδι.

Τῶν >άρα >ίσων πυραμίδων καὶ τριγώνου βάσεις ἐθουσῶν ἀντιτεπόνθασιν αἱ βάσεις τοῖς <ύψεσιν; καὶ <ὦν πυραμίδων τριγώνου βάσεις ἐθουσῶν ἀντιτεπόνθασιν αἱ βάσεις τοῖς <ύψεσιν, >ίσαι εἰσὶν ἐκεῖναι; <όπερ >έδει δεῖχαι.

10.

Πᾶς κῶνος κυλίνδρου τρίτον μέρος ἐστὶ τοῦ τὴν αὐτὴν βάσιν >έθοντοξ αὐτῷ καὶ <ύψος >ίσον.

Ἐθέτω γὰρ κῶνος κυλίνδρῳ βάσιν τε τὴν αὐτὴν τὸν ΑΒΓΔ κύκλον καὶ <ύψος >ίσον; λέγω, <ὅτι ὁ κῶνος τοῦ κυλίνδρου τρίτον ἐστὶ μέρος, τουτέστιν <ὅτι ὁ κύλινδρος τοῦ κώνου τριπλασίων ἐστίν.



Εἰ γὰρ μὴ ἐστὶν ὁ κύλινδρος τοῦ κώνου τριπλασίων, >έσται ὁ κύλινδρος τοῦ κώνου >ήτοι μείζων >ἢ τριπλασίων >ἢ ἐλάσσων >ἢ τριπλασίων. >έστω πρότερον μείζων >ἢ τριπλασίων, καὶ ἐγγεγράφθω εἰς τὸν ΑΒΓΔ κύκλον τετράγωνον τὸ ΑΒΓΔ; τὸ δὴ ΑΒΓΔ τετράγωνον μείζον ἐστὶν >ἢ τὸ <ἡμισυ τοῦ ΑΒΓΔ κύκλου. καὶ ἀνεστάτω ἀπὸ τοῦ ΑΒΓΔ τετραγώνου πρίσμα ἰσο"υψὲς τῷ κυλίνδρῳ. τὸ δὴ ἀνιστάμενον πρίσμα μείζον ἐστὶν >ἢ τὸ <ἡμισυ τοῦ κυλίνδρου, ἐπεὶ δὴ περὶ τὸν ΑΒΓΔ κύκλον τετράγωνον περιγράψωμεν, τὸ ἐγγεγραμμένον εἰς τὸν ΑΒΓΔ κύκλον τετράγωνον <ἡμισυ ἐστὶ τοῦ περιγεγραμμένου; καὶ ἐστὶ τὰ ἀπ' αὐτῶν ἀνιστάμενα στερεὰ παραλληλεπίπεδα πρίσματα ἰσο"υψη; τὰ δὲ ὑπὸ τὸ αὐτὸ <ύψος >όντα στερεὰ παραλληλεπίπεδα

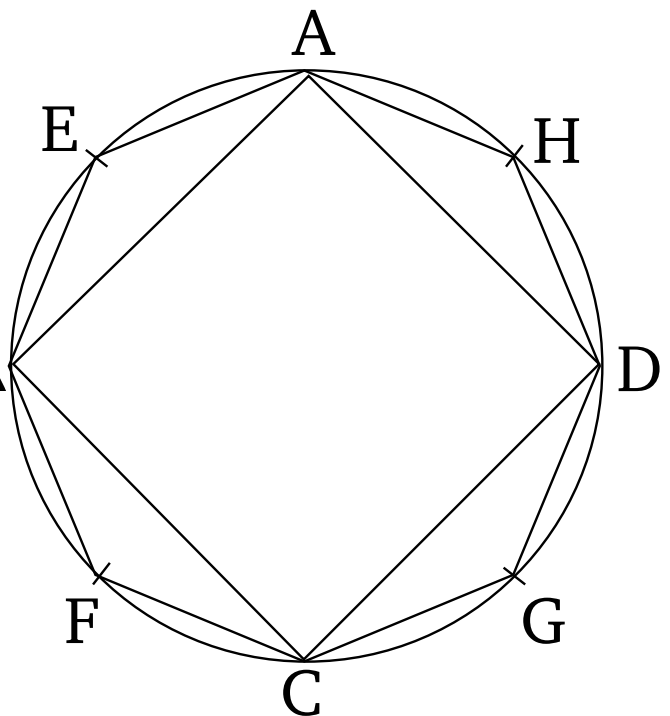
pyramid $DEFH$ is the same as the height of parallelepiped $EHQP$, and the height of pyramid $ABCG$ is the same as the height of parallelepiped $BGML$. Thus, as base BM is to base EQ , so the height of parallelepiped $EHQP$ (is) to the height of parallelepiped $BGML$. And those parallelepiped solids whose bases are reciprocally proportional to their heights are equal [Prop. 11.34]. Thus, the parallelepiped solid $BGML$ is equal to the parallelepiped solid $EHQP$. And pyramid $ABCG$ is a sixth part of $BGML$, and pyramid $DEFH$ a sixth part of parallelepiped $EHQP$. Thus, pyramid $ABCG$ is equal to pyramid $DEFH$.

Thus, the bases of equal pyramids which also have triangular bases are reciprocally proportional to their heights. And those pyramids having triangular bases whose bases are reciprocally proportional to their heights are equal. (Which is) the very thing it was required to show.

Proposition 10

Every cone is the third part of the cylinder which has the same base as it, and an equal height.

For let there be a cone (with) the same base as a cylinder, (namely) the circle $ABCD$, and an equal height. I say that the cone is the third part of the cylinder—that is to say, that the cylinder is three times the cone.



For if the cylinder is not three times the cone then the cylinder will be either more than three times, or less than three times, (the cone). Let it, first of all, be more than three times (the cone). And let the square $ABCD$ be inscribed in circle $ABCD$ [Prop. 4.6]. So, square $ABCD$ is more than half of circle $ABCD$ [Prop. 12.2]. And let a prism of equal height to the cylinder be set up on square $ABCD$. So, the prism set up is more than half of the cylinder, inasmuch as if we also circumscribe a square around circle $ABCD$ [Prop. 4.7] then the square inscribed in circle $ABCD$ is half of the circumscribed (square). And the solids set up on them are parallelepiped prisms of equal height. And parallelepiped solids having the same height are to one another as their

πρὸς >ἀλλήλα ἔστιν ὡς αἱ βάσεις; καὶ τὸ ἐπὶ τοῦ ΑΒΓΔ >ἄρα τετραγώνου ἀνασταθὲν πρίσμα <ἥμισυ ἔστι τοῦ ἀνασταθέντος πρίσματος ἀπὸ τοῦ περὶ τὸν ΑΒΓΔ κύκλον περιγραφέντος τετραγώνου; καὶ ἔστιν ὁ κύλινδρος ἐλάττων τοῦ πρίσματος τοῦ ἀνασταθέντος ἀπὸ τοῦ περὶ τὸν ΑΒΓΔ κύκλον περιγραφέντος τετραγώνου; τὸ >ἄρα πρίσμα τὸ ἀνασταθὲν ἀπὸ τοῦ ΑΒΓΔ τετραγώνου ἰσο"υψὲς τῷ κύλινδρῳ μείζον ἔστι τοῦ ἡμίσεως τοῦ κύλινδρου. τεμήσθωσαν αἱ ΑΒ, ΒΓ, ΓΔ, ΔΑ περιφέρειαι δίθῃ κατὰ τὰ Ε, Ζ, Η, Θ σημεία, καὶ ἐπεζεύθωσαν αἱ ΑΕ, ΕΒ, ΒΖ, ΖΓ, ΓΗ, ΗΔ, ΔΘ, ΘΑ; καὶ <ἐκαστον >ἄρα τῶν ΑΕΒ, ΒΖΓ, ΓΗΔ, ΔΘΑ τριγώνων μείζον ἔστιν >ἢ τὸ <ἥμισυ τοῦ καθ> ἑαυτοῦ τμήματος τοῦ ΑΒΓΔ κύκλου, ὡς >ἐμπροσθεν ἐδείκνυμεν. ἀνεστάτω ἐφ' ἐκάστου τῶν ΑΕΒ, ΒΖΓ, ΓΗΔ, ΔΘΑ τριγώνων πρίσματα ἰσο"υψῆ τῷ κύλινδρῳ; καὶ <ἐκαστον >ἄρα τῶν ἀνασταθέντων πρισματῶν μείζον ἔστιν >ἢ τὸ <ἥμισυ μέρος τοῦ καθ> ἑαυτοῦ τμήματος τοῦ κύλινδρου, ἐπειδήπερ ἔαν διὰ τῶν Ε, Ζ, Η, Θ σημείων παραλλήλους ταῖς ΑΒ, ΒΓ, ΓΔ, ΔΑ ἀγάγωμεν, καὶ συμπληρώσωμεν τὰ ἐπὶ τῶν ΑΒ, ΒΓ, ΓΔ, ΔΑ παραλληλόγραμμα, καὶ ἀπ' αὐτῶν ἀναστήσωμεν στερεὰ παραλληλεπίπεδα ἰσο"υψῆ τῷ κύλινδρῳ, ἐκάστου τῶν ἀνασταθέντων ἡμίση ἔστι τὰ πρίσματα τὰ ἐπὶ τῶν ΑΕΒ, ΒΖΓ, ΓΗΔ, ΔΘΑ τριγώνων; καὶ ἔστι τὰ τοῦ κύλινδρου τμήματα ἐλάττονα τῶν ἀνασταθέντων στερεῶν παραλληλεπιπέδων; <ὥστε καὶ τὰ ἐπὶ τῶν ΑΕΒ, ΒΖΓ, ΓΗΔ, ΔΘΑ τριγώνων πρίσματα μείζονα ἔστιν >ἢ τὸ <ἥμισυ τῶν καθ> ἑαυτὰ τοῦ κύλινδρου τμημάτων. τέμνοντες δὴ τὰς ὑπολειπομένας περιφερείας δίθῃ καὶ ἐπιζευγύντες εὐθείας καὶ ἀνιστάντες ἐφ' ἐκάστου τῶν τριγώνων πρίσματα ἰσο"υψῆ τῷ κύλινδρῳ καὶ τοῦτο αἰ ποιοῦντες καταλείβομεν τινα ἀποτμήματα τοῦ κύλινδρου, <ἂ> ἔσται ἐλάττονα τῆς ὑπεροχῆς, <ἢ> ὑπερέθει ὁ κύλινδρος τοῦ τριπλασίου τοῦ κώνου. λελείφθω, καὶ >έστω τὰ ΑΕ, ΕΒ, ΒΖ, ΖΓ, ΓΗ, ΗΔ, ΔΘ, ΘΑ; λοιπὸν >ἄρα τὸ πρίσμα, ο<ὗ βάσις μὲν τὸ ΑΕΒΖΓΗΔΘ πολύγωνον, <ὕψος δὲ τὸ αὐτὸ τῷ κύλινδρῳ>, μείζον ἔστιν >ἢ τριπλάσιον τοῦ κώνου. ἀλλὰ τὸ πρίσμα, ο<ὗ βάσις μὲν ἔστι τὸ ΑΕΒΖΓΗΔΘ πολύγωνον, <ὕψος δὲ τὸ αὐτὸ τῷ κύλινδρῳ>, τριπλάσιον ἔστι τῆς πυραμίδος, <ἢ> βάσις μὲν ἔστι τὸ ΑΕΒΖΓΗΔΘ πολύγωνον, κορυφὴ δὲ ἡ αὐτὴ τῷ κώνῳ; καὶ ἡ πυραμὶς >ἄρα, <ἢ> βάσις μὲν ἔστι τὸ ΑΕΒΖΓΗΔΘ πολύγωνον, κορυφὴ δὲ ἡ αὐτὴ τῷ κώνῳ>, μείζον ἔστι τοῦ κώνου τοῦ βάσιν >έθοντες τὸν ΑΒΓΔ κύκλον. ἀλλὰ καὶ ἐλάττων; ἐμπεριέθεται γὰρ ὑπ' αὐτοῦ; <ὅπερ ἔστιν ἀδύνατον. οὐκ >ἄρα ἔστιν ὁ κύλινδρος τοῦ κώνου μείζων >ἢ τριπλάσιος. Λέγω δὴ, <ὅτι οὐδὲ ἐλάττων ἔστιν >ἢ τριπλάσιος ὁ κύλινδρος τοῦ κώνου. Εἰ γὰρ δυνατόν, >έστω ἐλάττων >ἢ τριπλάσιος ὁ κύλινδρος τοῦ κώνου; ἀνάπαλιν >ἄρα ὁ κώνος τοῦ κύλινδρου μείζων ἔστιν >ἢ τρίτον μέρος. ἐγγεγράφθω δὴ εἰς τὸν ΑΒΓΔ κύκλον τετράγωνον τὸ ΑΒΓΔ; τὸ ΑΒΓΔ >ἄρα τετράγωνον μείζον ἔστιν >ἢ τὸ <ἥμισυ τοῦ ΑΒΓΔ κύκλου. καὶ ἀνεστάτω ἀπὸ τοῦ ΑΒΓΔ τετραγώνου πυραμὶς τὴν αὐτὴν κορυφὴν >έθουσα τῷ κώνῳ; ἡ >ἄρα ἀνασταθεῖσα πυραμὶς μείζων ἔστιν >ἢ τὸ <ἥμισυ μέρος τοῦ κώνου>, ἐπειδήπερ, ὡς <ἐμπροσθεν ἐδείκνυμεν, <ὅτι ἔαν περὶ τὸν κύκλον τετράγωνον περιγράψωμεν, >έσται τὸ ΑΒΓΔ τετράγωνον <ἥμισυ τοῦ περὶ τὸν κύκλον περιγεγραμμένου τετραγώνου; καὶ ἔαν ἀπὸ τῶν τετραγώνων στερεὰ παραλληλεπίπεδα ἀναστήσωμεν ἰσο"υψῆ τῷ κώνῳ, >ἂ καὶ καλεῖται πρίσματα, >έσται τὸ ἀνασταθὲν ἀπὸ τοῦ ΑΒΓΔ τετραγώνου <ἥμισυ τοῦ ἀνασταθέντος ἀπὸ τοῦ περὶ τὸν κύκλον περιγραφέντος

bases [Prop. 11.32]. And, thus, the prism set up on square $ABCD$ is half of the prism set up on the square circumscribed about circle $ABCD$. And the cylinder is less than the prism set up on the square circumscribed about circle $ABCD$. Thus, the prism set up on square $ABCD$ of the same height as the cylinder is more than half of the cylinder. Let the circumferences AB , BC , CD , and DA be cut in half at points E , F , G , and H . And let AE , EB , BF , FC , CG , GD , DH , and HA be joined. And thus each of the triangles AEB , BFC , CGD , and DHA is more than half of the segment of circle $ABCD$ about it, as was shown previously [Prop. 12.2]. Let prisms of equal height to the cylinder be set up on each of the triangles AEB , BFC , CGD , and DHA . And each of the prisms set up is greater than the half part of the segment of the cylinder about it—inasmuch as if we draw (straight-lines) parallel to AB , BC , CD , and DA through points E , F , G , and H (respectively), and complete the parallelograms on AB , BC , CD , and DA , and set up parallelepiped solids of equal height to the cylinder on them, then the prisms on triangles AEB , BFC , CGD , and DHA are each half of the set up (parallelepipeds). And the segments of the cylinder are less than the set up parallelepiped solids. Hence, the prisms on triangles AEB , BFC , CGD , and DHA are also greater than half of the segments of the cylinder about them. So (if) the remaining circumferences are cut in half, and straight-lines are joined, and prisms of equal height to the cylinder are set up on each of the triangles, and this is done continually, then we will (eventually) leave some segments of the cylinder whose (sum) is less than the excess by which the cylinder exceeds three times the cone [Prop. 10.1]. Let them be left, and let them be AE , EB , BF , FC , CG , GD , DH , and HA . Thus, the remaining prism whose base (is) polygon $AEBFCGDH$, and height the same as the cylinder, is greater than three times the cone. But, the prism whose base is polygon $AEBFCGDH$, and height the same as the cylinder, is three times the pyramid whose base is polygon $AEBFCGDH$, and apex the same as the cone [Prop. 12.7 corr.]. And thus the pyramid whose base [is] polygon $AEBFCGDH$, and apex the same as the cone, is greater than the cone having (as) base circle $ABCD$. But (it is) also less. For it is encompassed by it. The very thing (is) impossible. Thus, the cylinder is not more than three times the cone.

So, I say that neither (is) the cylinder less than three times the cone.

For, if possible, let the cylinder be less than three times the cone. Thus, inversely, the cone is greater than the third part of the cylinder. So, let the square $ABCD$ be inscribed in circle $ABCD$ [Prop. 4.6]. Thus, square $ABCD$ is greater than half of circle $ABCD$. And let a pyramid having the same apex as the cone be set up on square $ABCD$. Thus, the pyramid set up is greater than the half part of the cone, inasmuch as we showed previously that if we circumscribe a square about the circle [Prop. 4.7] then the square $ABCD$ will be half of the square circumscribed about the circle [Prop. 12.2]. And if we set up on the squares parallelepiped solids—which are also called prisms—of the same height as the cone, then the (prism) set up on square $ABCD$ will be half of the (prism) set up on the square circumscribed about the circle. For they are to one another as their bases

τετραγώνου; πρὸς >ἀλλήλα γάρ εἰσιν ὡς αἱ βάσεις. <ὥστε καὶ τὰ τρίτα; καὶ πυραμῖς >άρα, <ἥξ βάσις τὸ ΑΒΓΔ τετράγωνον, <ἡμισύ ἐστι τῆς πυραμίδος τῆς ἀνασταθείσης ἀπὸ τοῦ περὶ τὸν κύκλον περιγραφέντος τετραγώνου. καὶ ἐστὶ μείζων ἢ πυραμῖς ἢ ἀνασταθείσα ἀπὸ τοῦ περὶ τὸν κύκλον τετραγώνου τοῦ κώνου; ἐμπεριέθει γὰρ αὐτόν. ἢ >άρα πυραμῖς, <ἥξ βάσις τὸ ΑΒΓΔ τετράγωνον, κορυφή δὲ ἢ αὐτὴ τῶ| κώνῳ, μείζων ἐστὶν >ἢ τὸ <ἡμισυ τοῦ κώνου. τετμήσθωσαν αἱ ΑΒ, ΒΓ, ΓΔ, ΔΑ περιφέρειαι δίθῃ κατὰ τὰ Ε, Ζ, Η, Θ σημεῖα, καὶ ἐπεζεύθωσαν αἱ ΑΕ, ΕΒ, ΒΖ, ΖΓ, ΓΗ, ΗΔ, ΔΘ, ΘΑ; καὶ <έκαστον >άρα τῶν ΑΕΒ, ΒΖΓ, ΓΗΔ, ΔΘΑ τριγώνων μείζον ἐστὶν >ἢ τὸ <ἡμισυ μέρος τοῦ καθ> ἐαυτὸν τμήματος τοῦ ΑΒΓΔ κύκλου. καὶ ἀνεστάτωσαν ἐφ> ἐκάστου τῶν ΑΕΒ, ΒΖΓ, ΓΗΔ, ΔΘΑ τριγώνων πυραμίδες τὴν αὐτὴν κορυφὴν >έθουσαι τῶ| κώνῳ; καὶ ἐκάστη >άρα τῶν ἀνασταθεισῶν πυραμίδων κατὰ τὸν αὐτὸν τρόπον μείζων ἐστὶν >ἢ τὸ <ἡμισυ μέρος τοῦ καθ> ἐαυτὴν τμήματος τοῦ κώνου. τέμνοντες δὴ τὰς ὑπολειπομένας περιφερείας δίθῃ καὶ ἐπιζευγνύντες εὐθείαις καὶ ἀνιστάντες ἐφ> ἐκάστου τῶν τριγώνων πυραμίδα τὴν αὐτὴν κορυφὴν >έθουσιν τῶ| κώνῳ καὶ τοῦτο αἰεὶ ποιοῦντες καταλείπομένῃ τινι ἀποτμήματι τοῦ κώνου, <ὰ >έσται ἐλάττωνα τῆς ὑπεροθῆς, <ἢ| ὑπερέθει ὁ κώνος τοῦ τρίτου μέρους τοῦ κυλίνδρου. λελειφθῶ, καὶ >έστω τὰ ἐπὶ τῶν ΑΕ, ΕΒ, ΒΖ, ΖΓ, ΓΗ, ΗΔ, ΔΘ, ΘΑ; λοιπὴ >άρα ἢ πυραμῖς, <ἥξ βάσις μὲν ἐστὶ τὸ ΑΕΒΖΓΗΔΘ πολύγωνον, κορυφή δὲ ἢ αὐτὴ τῶ| κώνῳ, μείζων ἐστὶν >ἢ τρίτον μέρος τοῦ κυλίνδρου. ἀλλ> ἢ πυραμῖς, <ἥξ βάσις μὲν ἐστὶ τὸ ΑΕΒΖΓΗΔΘ πολύγωνον, κορυφή δὲ ἢ αὐτὴ τῶ| κώνῳ, τρίτον ἐστὶ μέρος τοῦ πρίσματος, ο<ὺ βάσις μὲν ἐστὶ τὸ ΑΕΒΖΓΗΔΘ πολύγωνον, <ύψος δὲ τὸ αὐτὸ τῶ| κυλίνδρῳ; τὸ >άρα πρίσμα, ο<ὺ βάσις μὲν ἐστὶ τὸ ΑΕΒΖΓΗΔΘ πολύγωνον, <ύψος δὲ τὸ αὐτὸ τῶ| κυλίνδρῳ, μείζον ἐστὶ τοῦ κυλίνδρου, ο<ὺ βάσις ἐστὶν ὁ ΑΒΓΔ κύκλος. ἀλλὰ καὶ >έλαττον; ἐμπεριέθεται γὰρ ὑπ> αὐτοῦ; <όπερ ἐστὶν ἀδύνατον. οὐκ >άρα ὁ κύλινδρος τοῦ κώνου ἐλάττων ἐστὶν >ἢ τριπλάσιος. ἐδείθῃ δέ, <ὅτι οὐδὲ μείζων >ἢ τριπλάσιος; τριπλάσιος >άρα ὁ κύλινδρος τοῦ κώνου; <ὥστε ὁ κώνος τρίτον ἐστὶ μέρος τοῦ κυλίνδρου. Πᾶς >άρα κώνος κυλίνδρου τρίτον μέρος ἐστὶ τοῦ τὴν αὐτὴν βάσιν >έθοντος αὐτῶ| καὶ <ύψος >ίσον; <όπερ >έδει δείχαι.

11.

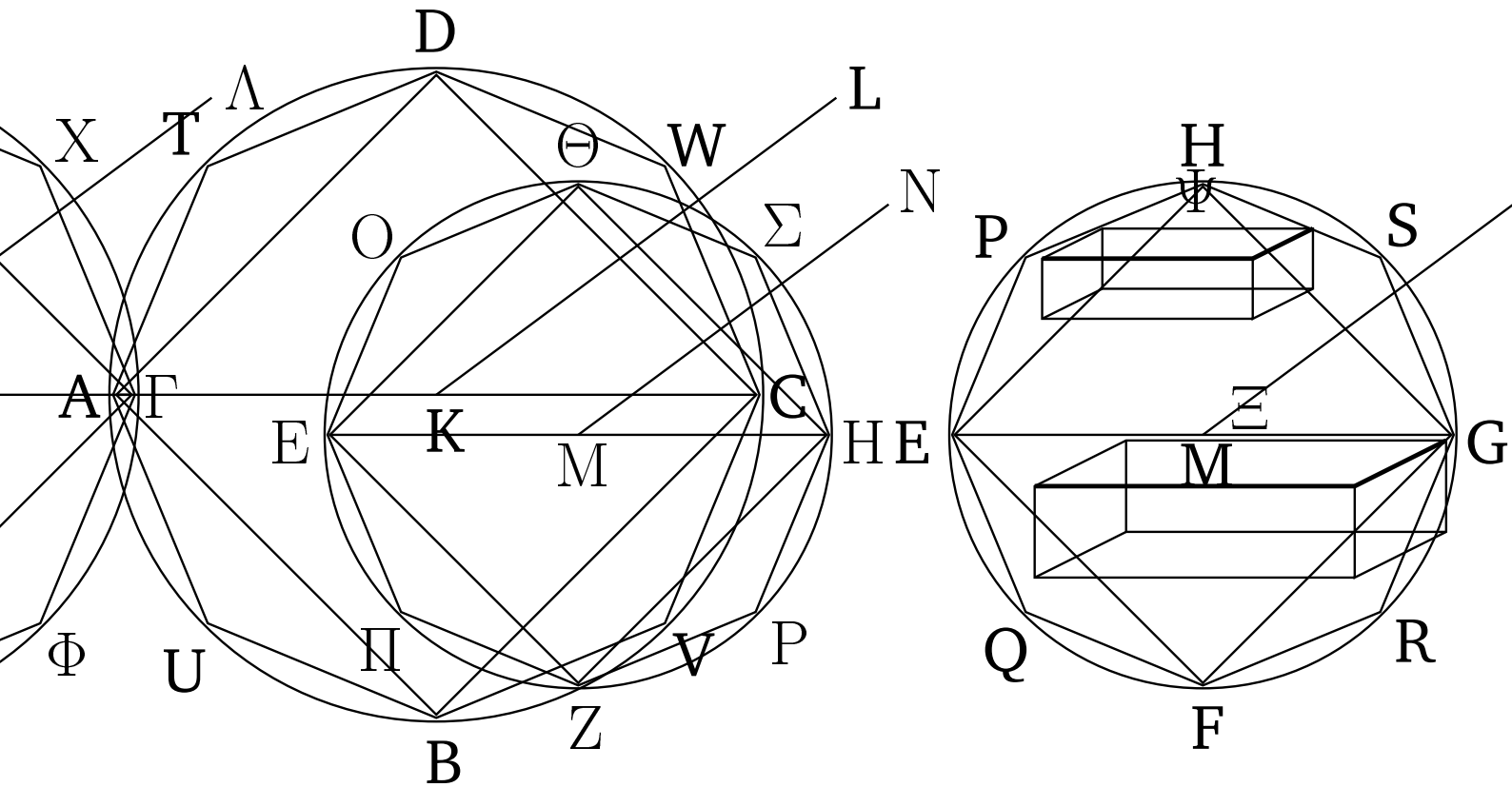
Οἱ ὑπο τὸ αὐτὸ <ύψος >όντες κῶνοι καὶ κύλινδροι πρὸς ἀλλήλους εἰσιν ὡς αἱ βάσεις.
>Εστῶσαν ὑπὸ τὸ αὐτὸ <ύψος κῶνοι καὶ κύλινδροι, <ὼν βάσεις μὲν [εἰσιν] οἱ ΑΒΓΔ, ΕΖΗΘ κύκλοι, >άχονες δὲ οἱ ΚΛ, ΜΝ, διαμέτροι δὲ τῶν βάσεων αἱ ΑΓ, ΕΗ; λέγω, <ὅτι ἐστὶν ὡς ὁ ΑΒΓΔ κύκλος πρὸς τὸν ΕΖΗΘ κύκλον, ο<ὕτως ὁ ΑΛ κώνος πρὸς τὸν ΕΝ κώνον.

[Prop. 11.32]. Hence, (the same) also (goes for) the thirds. Thus, the pyramid whose base is square $ABCD$ is half of the pyramid set up on the square circumscribed about the circle [Prop. 12.7 corr.]. And the pyramid set up on the square circumscribed about the circle is greater than the cone. For it encompasses it. Thus, the pyramid whose base is square $ABCD$, and apex the same as the cone, is greater than half of the cone. Let the circumferences AB , BC , CD , and DA be cut in half at points E , F , G , and H (respectively). And let AE , EB , BF , FC , CG , GD , DH , and HA be joined. And, thus, each of the triangles AEB , BFC , CGD , and DHA is greater than the half part of the segment of circle $ABCD$ about it [Prop. 12.2]. And let pyramids having the same apex as the cone be set up on each of the triangles AEB , BFC , CGD , and DHA . And, thus, in the same way, each of the pyramids set up is more than the half part of the segment of the cone about it. So, (if) the remaining circumferences are cut in half, and straight-lines are joined, and pyramids having the same apex as the cone are set up on each of the triangles, and this is done continually, then we will (eventually) leave some segments of the cone whose (sum) is less than the excess by which the cone exceeds the third part of the cylinder [Prop. 10.1]. Let them be left, and let them be the (segments) on AE , EB , BF , FC , CG , GD , DH , and HA . Thus, the remaining pyramid whose base is polygon $AEBFCGDH$, and apex the same as the cone, is greater than the third part of the cylinder. But, the pyramid whose base is polygon $AEBFCGDH$, and apex the same as the cone, is the third part of the prism whose base is polygon $AEBFCGDH$, and height the same as the cylinder [Prop. 12.7 corr.]. Thus, the prism whose base is polygon $AEBFCGDH$, and height the same as the cylinder, is greater than the cylinder whose base is circle $ABCD$. But, (it is) also less. For it is encompassed by it. The very thing is impossible. Thus, the cylinder is not less than three times the cone. And it was shown that neither (is it) greater than three times (the cone). Thus, the cylinder (is) three times the cone. Hence, the cone is the third part of the cylinder. Thus, every cone is the third part of the cylinder which has the same base as it, and an equal height. (Which is) the very thing it was required to show.

Proposition 11

Cones and cylinders having the same height are to one another as their bases.

Let there be cones and cylinders of the same height whose bases [are] the circles $ABCD$ and $EFGH$, axes KL and MN , and diameters of the bases AC and EG (respectively). I say that as circle $ABCD$ is to circle $EFGH$, so cone AL (is) to cone EN .



Εἰ γὰρ μή, >έσται ὡς ὁ $AB\Gamma\Delta$ κύκλος πρὸς τὸν $EZH\Theta$ κύκλον, οὕτως ὁ AL κώνος >ήτοι πρὸς >έλασσόν τι τοῦ EN κώνου στερεὸν >ή πρὸς μείζον. >έστω πρότερον πρὸς >έλασσον τὸ X , καὶ <ῶ> >έλασσόν ἐστι τὸ X στερεὸν τοῦ EN κώνου, ἐκείνῳ >ῖσον >έστω τὸ Ψ στερεόν; ὁ EN κώνος >άρα >ῖσος ἐστὶ τοῖς X , Ψ στερεοῖς. ἐγγεγράφθω εἰς τὸν $EZH\Theta$ κύκλον τετράγωνον τὸ $EZH\Theta$; τὸ >άρα τετράγωνον μείζον ἐστὶν >ή τὸ <ήμισυ τοῦ κύκλου. ἀνεστάτω ἀπὸ τοῦ $EZH\Theta$ τετραγώνου πυραμὶς ἰσο"υψὴς τῷ κώνῳ; ἢ >άρα ἀνασταθεῖσα πυραμὶς μείζων ἐστὶν >ή τὸ <ήμισυ τοῦ κώνου, ἐπειδὴ περ ἕαν περιγράψωμεν περὶ τὸν κύκλον τετράγωνον, καὶ ἀπ' αὐτοῦ ἀναστήσωμεν πυραμίδα ἰσο"υψὴς τῷ κώνῳ, ἢ ἐγγραφεῖσα πυραμὶς <ήμισυ ἐστὶ τῆς περιγραφείσης, πρὸς ἀλλήλας γὰρ εἰσιν ὡς αἱ βάσεις; ἐλάττων δὲ ὁ κώνος τῆς περιγραφείσης πυραμίδος. τεμήσθωσαν αἱ EZ , ZH , $H\Theta$, ΘE περιφέρειαι διῖθα κατὰ τὰ O , Π , P , Σ σημεία, καὶ ἐπεζεύθθωσαν αἱ ΘO , OE , EP , PZ , ZP , PH , $H\Sigma$, $\Sigma\Theta$. <έκαστον >άρα τῶν ΘOE , EPZ , ZPH , $HZ\Theta$ τριγώνων μείζον ἐστὶν >ή τὸ <ήμισυ τοῦ καθ' ἑαυτὸ τμήματος τοῦ κύκλου. ἀνεστάτω ἐφ' ἑκάστου τῶν ΘOE , EPZ , ZPH , $HZ\Theta$ τριγώνων πυραμὶς ἰσο"υψὴς τῷ κώνῳ; καὶ ἑκάστη >άρα τῶν ἀνασταθεισῶν πυραμίδων μείζων ἐστὶν >ή τὸ <ήμισυ τοῦ καθ' ἑαυτὴν τμήματος τοῦ κώνου. τέμνοντες δὴ τὰς ὑπολειπομένας περιφερείας διῖθα καὶ ἐπιζευγνύντες εὐθείας καὶ ἀνιστάντες ἐπὶ ἑκάστου τῶν τριγώνων πυραμίδας ἰσο"υψεῖς τῷ κώνῳ καὶ αἱ τοῦτο ποιούντες καταλείψομεν τινα ἀποτμήματα τοῦ κώνου, <ά >έσται ἐλάσσονα τοῦ Ψ στερεοῦ. λείψομεν, καὶ >έστω τὰ ἐπὶ τῶν ΘOE , EPZ , ZPH , $HZ\Theta$; λοιπὴ >άρα ἢ πυραμὶς, <ῆς βάσις τὸ $\Theta OE\Pi ZP H\Sigma$ πολύγωνον, <ύψος δὲ τὸ αὐτὸ τῷ κώνῳ, μείζων ἐστὶ τοῦ X στερεοῦ. ἐγγεγράφθω καὶ εἰς τὸν $AB\Gamma\Delta$ κύκλον τῷ $\Theta OE\Pi ZP H\Sigma$ πολυγώνῳ ὁμοίον τε καὶ ὁμοίως κείμενον πολύγωνον τὸ $\Delta TAYB\Phi\Gamma\Theta$, καὶ ἀνεστάτω ἐπ' αὐτοῦ πυραμὶς ἰσο"υψὴς τῷ AL κώνῳ. ἐπεὶ οὖν ἐστὶν ὡς τὸ ἀπὸ τῆς AG πρὸς τὸ ἀπὸ τῆς EH , οὕτως τὸ

For if not, then as circle $ABCD$ (is) to circle $EFGH$, so cone AL will be to some solid either less than, or greater than, cone EN . Let it, first of all, be (in this ratio) to (some) lesser (solid), O . And let solid X be equal to that (magnitude) by which solid O is less than cone EN . Thus, cone EN is equal to (the sum of) solids O and X . Let the square $EFGH$ be inscribed in circle $EFGH$ [Prop. 4.6]. Thus, the square is greater than half of the circle [Prop. 12.2]. Let a pyramid of the same height as the cone be set up on square $EFGH$. Thus, the pyramid set up is greater than half of the cone, inasmuch as, if we circumscribe a square about the circle [Prop. 4.7], and set up on it a pyramid of the same height as the cone, then the inscribed pyramid is half of the circumscribed pyramid. For they are to one another as their bases [Prop. 12.6]. And the cone (is) less than the circumscribed pyramid. Let the circumferences EF , FG , GH , and HE be cut in half at points P , Q , R , and S . And let HP , PE , EQ , QF , FR , RG , GS , and SH be joined. Thus, each of the triangles HPE , EQF , FRG , and GSH is greater than half of the segment of the circle about it [Prop. 12.2]. Let pyramids of the same height as the cone be set up on each of the triangles HPE , EQF , FRG , and GSH . And, thus, each of the pyramids set up is greater than half of the segment of the cone about it [Prop. 12.10]. So, (if) the remaining circumferences are cut in half, and straight-lines are joined, and pyramids of equal height to the cone are set up on each of the triangles, and this is done continually, then we will (eventually) leave some segments of the cone (the sum of) which is less than solid X [Prop. 10.1]. Let them be left, and let them be the (segments) on HPE , EQF , FRG , and GSH . Thus, the remaining pyramid whose base is polygon $HPEQFRGS$, and height the same as the cone, is greater than solid O [Prop. 6.18]. And let the polygon $DTAUBVCW$, similar, and similarly laid out, to polygon $HPEQFRGS$, be inscribed in circle $ABCD$. And on it let

ΔΤΑΥΒΦΓΘ πολύγωνον πρὸς τὸ ΘΟΕΠΖΡΗΣ πολύγωνον, ὥς δὲ τὸ ἀπὸ τῆς ΑΓ πρὸς τὸ ἀπὸ τῆς ΕΗ, οὕτως ὁ ΑΒΓΔ κύκλος πρὸς τὸν ΕΖΗΘ κύκλον, καὶ ὥς >άρα ὁ ΑΒΓΔ κύκλος πρὸς τὸν ΕΖΗΘ κύκλον, οὕτως τὸ ΔΤΑΥΒΦΓΘ πολύγωνον πρὸς τὸ ΘΟΕΠΖΡΗΣ πολύγωνον. ὥς δὲ ὁ ΑΒΓΔ κύκλος πρὸς τὸν ΕΖΗΘ κύκλον, οὕτως ὁ ΑΛ κῶνος πρὸς τὸ Χ στερεόν, ὥς δὲ τὸ ΔΤΑΥΒΦΓΘ πολύγωνον πρὸς τὸ ΘΟΕΠΖΡΗΣ πολύγωνον, οὕτως ἡ πυραμὶς, <ἥς βάσις μὲν τὸ ΔΤΑΥΒΦΓΘ πολύγωνον, κορυφή δὲ τὸ Λ σημεῖον, πρὸς τὴν πυραμίδα, <ἥς βάσις μὲν τὸ ΘΟΕΠΖΡΗΣ πολύγωνον, κορυφή δὲ τὸ Ν σημεῖον. καὶ ὥς >άρα ὁ ΑΛ κῶνος πρὸς τὸ Χ στερεόν, οὕτως ἡ πυραμὶς, <ἥς βάσις μὲν τὸ ΔΤΑΥΒΦΓΘ πολύγωνον, κορυφή δὲ τὸ Λ σημεῖον, πρὸς τὴν πυραμίδα, <ἥς βάσις μὲν τὸ ΘΟΕΠΖΡΗΣ πολύγωνον, κορυφή δὲ τὸ Ν σημεῖον; ἐναλλάχ >άρα ἐστὶν ὥς ὁ ΑΛ κῶνος πρὸς τὴν ἐν αὐτῷ πυραμίδα, οὕτως τὸ Χ στερεόν πρὸς τὴν ἐν τῷ ΕΝ κώνῳ πυραμίδα. μείζων δὲ ὁ ΑΛ κῶνος τῆς ἐν αὐτῷ πυραμίδος; μείζων >άρα καὶ τὸ Χ στερεόν τῆς ἐν τῷ ΕΝ κώνῳ πυραμίδος. ἀλλὰ καὶ >έλασσον; <όπερ >άτοπον. οὐκ >άρα ἐστὶν ὥς ὁ ΑΒΓΔ κύκλος πρὸς τὸν ΕΖΗΘ κύκλον, οὕτως ὁ ΑΛ κῶνος πρὸς >έλασσόν τι τοῦ ΕΝ κώνου στερεόν. ὁμοίως δὲ δείχομεν, <ὅτι οὐδὲ ἐστὶν ὥς ὁ ΕΖΗΘ κύκλος πρὸς τὸν ΑΒΓΔ κύκλον, οὕτως ὁ ΕΝ κῶνος πρὸς >έλασσόν τι τοῦ ΑΛ κώνου στερεόν.

Λέγω δὴ, <ὅτι οὐδὲ ἐστὶν ὥς ὁ ΑΒΓΔ κύκλος πρὸς τὸν ΕΖΗΘ κύκλον, οὕτως ὁ ΑΛ κῶνος πρὸς μείζον τι τοῦ ΕΝ κώνου στερεόν.

Εἰ γὰρ δυνατόν, <έστω πρὸς μείζον τὸ Χ; ἀνάπαλιν >άρα ἐστὶν ὥς ὁ ΕΖΗΘ κύκλος πρὸς τὸν ΑΒΓΔ κύκλον, οὕτως τὸ Χ στερεόν πρὸς τὸν ΑΛ κῶνον. ἀλλ' ὥς τὸ Χ στερεόν πρὸς τὸν ΑΛ κῶνον, οὕτως ὁ ΕΝ κῶνος πρὸς >έλασσόν τι τοῦ ΑΛ κώνου στερεόν; καὶ ὥς >άρα ὁ ΕΖΗΘ κύκλος πρὸς τὸν ΑΒΓΔ κύκλον, οὕτως ὁ ΕΝ κῶνος πρὸς >έλασσόν τι τοῦ ΑΛ κώνου στερεόν; <όπερ ἀδύνατον ἐδείκθη. οὐκ >άρα ἐστὶν ὥς ὁ ΑΒΓΔ κύκλος πρὸς τὸν ΕΖΗΘ κύκλον, οὕτως ὁ ΑΛ κῶνος πρὸς μείζον τι τοῦ ΕΝ κώνου στερεόν. ἐδείκθη δέ, <ὅτι οὐδὲ πρὸς >έλασσον; >έστιν >άρα ὥς ὁ ΑΒΓΔ κύκλος πρὸς τὸν ΕΖΗΘ κύκλον, οὕτως ὁ ΑΛ κῶνος πρὸς τὸν ΕΝ κῶνον.

Ἀλλ' ὥς ὁ κῶνος πρὸς τὸν κῶνον, ὁ κύλινδρος πρὸς τὸν κύλινδρον; τριπλασίῳ γὰρ ἑκάτερος ἑκατέρου. καὶ ὥς >άρα ὁ ΑΒΓΔ κύκλος πρὸς τὸν ΕΖΗΘ κύκλον, οὕτως οἱ ἐπ' αὐτῶν ἴσο" ὑπεῖξ.

Οἱ >άρα ὑπὸ τὸ αὐτὸ <ύψος >όντεξ κῶνοι καὶ κύλινδροι πρὸς ἀλλήλους εἰσὶν ὥς αἱ βάσεις; <όπερ >έδει δεῖχαι.

12.

Οἱ <όμοιοι κῶνοι καὶ κύλινδροι πρὸς ἀλλήλους ἐν τριπλασίῳ λόγῳ εἰσὶ τῶν ἐν ταῖς βάσεσι διαμέτρων.

>Εστωσαν <όμοιοι κῶνοι καὶ κύλινδροι, <ὼν βάσεις μὲν οἱ ΑΒΓΔ, ΕΖΗΘ κύκλοι, διάμετροι δὲ τῶν βάσεων αἱ ΒΔ, ΖΘ, >άχονες δὲ τῶν κώνων καὶ κυλίνδρων οἱ ΚΛ, ΜΝ; λέγω, <ὅτι ὁ κῶνος, ο<ὕ βάσις μὲν [ἐστὶν] ὁ ΑΒΓΔ κύκλος, κορυφή δὲ τὸ Λ σημεῖον, πρὸς τὸν κῶνον, ο<ὕ βάσις μὲν [ἐστὶν] ὁ ΕΖΗΘ κύκλος, κορυφή δὲ τὸ Ν σημεῖον, τριπλασίονα λόγον >έθει >ήπερ ἡ ΒΔ πρὸς τὴν ΖΘ.

a pyramid of the same height as cone AL be set up. Therefore, since as the (square) on AC is to the (square) on EG , so polygon $DTAUBVCW$ (is) to polygon $HPEQFRGS$ [Prop. 12.1], and as the (square) on AC (is) to the (square) on EG , so circle $ABCD$ (is) to circle $EFGH$ [Prop. 12.2], thus as circle $ABCD$ (is) to circle $EFGH$, so polygon $DTAUBVCW$ also (is) to polygon $HPEQFRGS$. And as circle $ABCD$ (is) to circle $EFGH$, so cone AL (is) to solid O . And as polygon $DTAUBVCW$ (is) to polygon $HPEQFRGS$, so the pyramid whose base is polygon $DTAUBVCW$, and apex the point L , (is) to the pyramid whose base is polygon $HPEQFRGS$, and apex the point N [Prop. 12.6]. And, thus, as cone AL (is) to solid O , so the pyramid whose base is $DTAUBVCW$, and apex the point L , (is) to the pyramid whose base is polygon $HPEQFRGS$, and apex the point N [Prop. 5.11]. Thus, alternately, as cone AL is to the pyramid within it, so solid O (is) to the pyramid within cone EN [Prop. 5.16]. But, cone AL (is) greater than the pyramid within it. Thus, solid O (is) also greater than the pyramid within cone EN [Prop. 5.14]. But, (it is) also less. The very thing (is) absurd. Thus, circle $ABCD$ is not to circle $EFGH$, as cone AL (is) to some solid less than cone EN . So, similarly, we can show that neither is circle $EFGH$ to circle $ABCD$, as cone EN (is) to some solid less than cone AL .

So, I say that neither is circle $ABCD$ to circle $EFGH$, as cone AL (is) to some solid greater than cone EN .

For, if possible, let it be (in this ratio) to (some) greater (solid), O . Thus, inversely, as circle $EFGH$ is to circle $ABCD$, so solid O (is) to cone AL [Prop. 5.7 corr.]. But, as solid O (is) to cone AL , so cone EN (is) to some solid less than cone AL [Prop. 12.2 lem.]. And, thus, as circle $EFGH$ (is) to circle $ABCD$, so cone EN (is) to some solid less than cone AL . The very thing was shown (to be) impossible. Thus, circle $ABCD$ is not to circle $EFGH$, as cone AL (is) to some solid greater than cone EN . And, it was shown that neither (is it in this ratio) to (some) lesser (solid). Thus, as circle $ABCD$ is to circle $EFGH$, so cone AL (is) to cone EN .

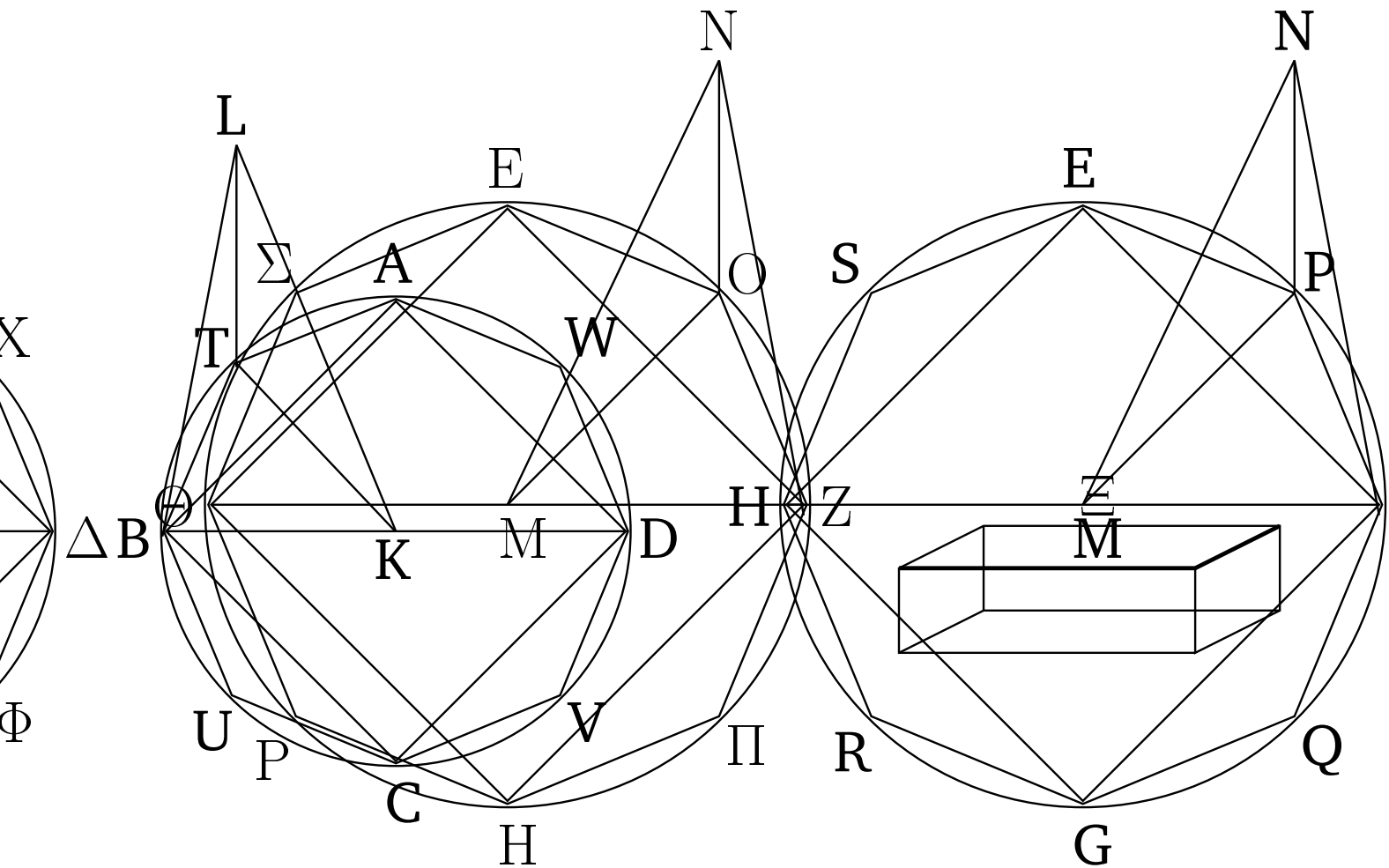
But, as the cone (is) to the cone, (so) the cylinder (is) to the cylinder. For each (is) three times each [Prop. 12.10]. Thus, circle $ABCD$ (is) also to circle $EFGH$, as (the ratio of the cylinders) on them (having) the same height.

Thus, cones and cylinders having the same height are to one another as their bases. (Which is) the very thing it was required to show.

Proposition 12

Similar cones and cylinders are to one another in the cubed ratio of the diameters of their bases.

Let there be similar cones and cylinders of which the bases (are) the circles $ABCD$ and $EFGH$, the diameters of the bases (are) BD and FH , and the axes of the cones and cylinders (are) KL and MN (respectively). I say that the cone whose base [is] circle $ABCD$, and apex the point L , has to the cone whose base [is] circle $EFGH$, and apex the point N , the cubed ratio that BD (has) to FH .



Εἰ γὰρ μὴ >έθει ὁ ΑΒΓΔΛ κῶνος πρὸς τὸν ΕΖΗΘΝ κῶνον
 πριπλασία λόγον >ήπερ ἢ ΒΔ πρὸς τὴν ΖΘ, <έχει ὁ
 ΑΒΓΔΛ κῶνος >ή πρὸς >έλασσόν τι τοῦ ΕΖΗΘΝ κῶνου
 στερεὸν τριπλασία λόγον >ή πρὸς μείζον. ἐθέτω πρότερον
 πρὸς >έλασσον τὸ Χ, καὶ ἐγγεγράφω εἰς τὸν ΕΖΗΘ κύκλον
 τετράγωνον τὸ ΕΖΗΘ; τὸ >άρα ΕΖΗΘ τετράγωνον μείζον
 ἐστὶν >ή τὸ <ήμισυ τοῦ ΕΖΗΘ κύκλου. καὶ ἀνεστάτω
 ἐπὶ τοῦ ΕΖΗΘ τετραγώνου πυραμὶς τὴν αὐτὴν κορυφὴν
 >έθουσα τῷ κῶνῳ; ἢ >άρα ἀνασταθείσα πυραμὶς μείζων
 ἐστὶν >ή τὸ <ήμισυ μέρος τοῦ κῶνου. τεμήσθωσαν δὲ
 αἱ ΕΖ, ΖΗ, ΗΘ, ΘΕ περιφέρειαι δίθα κατὰ τὰ Ο, Π, Ρ,
 Σ σημεῖα, καὶ ἐπεζεύθωσαν αἱ ΕΟ, ΟΖ, ΖΠ, ΠΗ, ΗΡ,
 ΡΘ, ΘΣ, ΣΕ. καὶ <έκαστον >άρα τῶν ΕΟΖ, ΖΠΗ, ΗΡΘ,
 ΘΣΕ τριγώνων μείζον ἐστὶν >ή τὸ <ήμισυ μέρος τοῦ καθ>
 ἑαυτοῦ τμήματος τοῦ ΕΖΗΘ κύκλου. καὶ ἀνεστάτω ἐφ>
 ἑκάστου τῶν ΕΟΖ, ΖΠΗ, ΗΡΘ, ΘΣΕ τριγώνων πυραμὶς τὴν
 αὐτὴν κορυφὴν >έθουσα τῷ κῶνῳ; καὶ ἑκάστη >άρα τῶν
 ἀνασταθεισῶν πυραμίδων μείζων ἐστὶν >ή τὸ <ήμισυ μέρος
 τοῦ καθ> ἑαυτὴν τμήματος τοῦ κῶνου. τέμνοντες δὲ τὰς
 ὑπολειπομένας περιφερείας δίθα καὶ ἐπιζευγνύντες εὐθείας
 καὶ ἀνιστάντες ἐφ> ἑκάστου τῶν τριγώνων πυραμίδα τὴν
 αὐτὴν κορυφὴν ἐθούσας τῷ κῶνῳ καὶ τοῦτο αἰε ποιοῦντες
 καταλείψομεν τινα ἀποτμήματα τοῦ κῶνου, <ὰ >έσται
 ἐλάσσονα τῆς ὑπεροθῆς, <ή| ὑπερέθει ὁ ΕΖΗΘΝ κῶνος τοῦ
 Χ στερεοῦ. λελείφθω, καὶ >έστω τὰ ἐπὶ τῶν ΕΟ, ΟΖ, ΖΠ,
 ΠΗ, ΗΡ, ΡΘ, ΘΣ, ΣΕ; λοιπὴ >άρα ἡ πυραμὶς, <ῆς βάσις μὲν
 ἐστὶ τὸ ΕΟΖΠΗΡΘΣ πολύγωνον, κορυφὴ δὲ τὸ Ν σημεῖον,
 μείζων ἐστὶ τοῦ Χ στερεοῦ. ἐγγεγράφω καὶ εἰς τὸν ΑΒΓΔ
 κύκλον τῷ ΕΟΖΠΗΡΘΣ πολυγώνῳ <όμοιόν τε καὶ ὁμοίως
 κείμενον πολύγωνον τὸ ΑΤΒΥΓΦΔΘ, καὶ ἀνεστάτω ἐπὶ

For if cone $ABCDL$ does not have to cone $EFGHN$ the
 cubed ratio that BD (has) to FH then cone $ABCDL$ will
 have the cubed ratio to some solid either less than, or greater
 than, cone $EFGHN$. Let it, first of all, have (such a ratio)
 to (some) lesser (solid), O . And let the square $EFGH$ be in-
 scribed in circle $EFGH$ [Prop. 4.6]. Thus, square $EFGH$
 is greater than half of circle $EFGH$ [Prop. 12.2]. And let
 a pyramid having the same apex as the cone be set up on
 square $EFGH$. Thus, the pyramid set up is greater than the
 half part of the cone [Prop. 12.10]. So, let the circumfer-
 ences EF , FG , GH , and HE be cut in half at points P , Q ,
 R , and S (respectively). And let EP , PF , FQ , QG , GR ,
 RH , HS , and SE be joined. And, thus, each of the trian-
 gles EPF , FQG , GRH , and HSE is greater than the half
 part of the segment of circle $EFGH$ about it [Prop. 12.2].
 And let a pyramid having the same apex as the cone be set
 up on each of the triangles EPF , FQG , GRH , and HSE .
 And thus each of the pyramids set up is greater than the half
 part of the segment of the cone about it [Prop. 12.10]. So,
 (if) the the remaining circumferences are cut in half, and
 straight-lines are joined, and pyramids having the same apex
 as the cone are set up on each of the triangles, and this is
 done continually, then we will (eventually) leave some seg-
 ments of the cone whose (sum) is less than the excess by
 which cone $EFGHN$ exceeds solid O [Prop. 10.1]. Let
 them be left, and let them be the (segments) on EP , PF ,
 FQ , QG , GR , RH , HS , and SE . Thus, the remaining
 pyramid whose base is polygon $EPFQGRHS$, and apex
 the point N , is greater than solid O . And let the polygon
 $ATBUCVDW$, similar, and similarly laid out, to polygon

τοῦ ΑΤΒΥΓΦΔΘ πολυγώνου πυραμίδι τὴν αὐτὴν κορυφὴν >έθουσα τῷ κώνῳ, καὶ τῶν μὲν περιεθόντων τὴν πυραμίδα, <ἥξ βάσις μὲν ἐστὶ τὸ ΑΤΒΥΓΦΔΘ πολύγωνον, κορυφὴ δὲ τὸ Α σημεῖον, <ἐν τρίγωνον >έστω τὸ ΑΒΤ, τῶν δὲ περιεθόντων τὴν πυραμίδα, <ἥξ βάσις μὲν ἐστὶ τὸ ΕΟΖΠΗΡΘΣΑΤΒΥΓΦΔΘ, καὶ ἀπὸ τοῦ Α σημεῖον. <ἐν τρίγωνον >έστω τὸ ΝΖΟ, καὶ ἐπεξεύθωσαν αἱ ΚΤ, ΜΟ. καὶ ἐπεὶ <ὁμοιόζ ἐστὶν ὁ ΑΒΓΔΛ κώνος τῷ ΕΖΗΘΝ κώνῳ, >έστιν >άρα ὡς ἡ ΒΔ πρὸς τὴν ΖΘ, ο<ύτωξ ὁ ΚΛ >άχων πρὸς τὸν ΜΝ >άχονα. ὡς δὲ ἡ ΒΔ πρὸς τὴν ΖΘ, ο<ύτωξ ἡ ΒΚ πρὸς τὴν ΖΜ; καὶ ὡς >άρα ἡ ΒΚ πρὸς τὴν ΖΜ, ο<ύτωξ ἡ ΚΛ πρὸς τὴν ΜΝ. καὶ ἐναλλάχ ὡς ἡ ΒΚ πρὸς τὴν ΚΛ, ο<ύτωξ ἡ ΖΜ πρὸς τὴν ΜΝ. καὶ περὶ >ίσαξ γωνίαξ τὰξ ὑπὸ ΒΚΛ, ΖΜΝ αἱ πλευραὶ ἀνάλογόν εἰσιν; <ὁμοιον >άρα ἐστὶ τὸ ΒΚΛ τρίγωνον τῷ ΖΜΝ τριγώνῳ. πάλιν, ἐπεὶ ἐστὶν ὡς ἡ ΒΚ πρὸς τὴν ΚΤ, ο<ύτωξ ἡ ΖΜ πρὸς τὴν ΜΟ, καὶ περὶ >ίσαξ γωνίαξ τὰξ ὑπὸ ΒΚΤ, ΖΜΟ, ἐπειδήπερ, <ὁ μέρος ἐστὶν ἡ ὑπὸ ΒΚΤ γωνία τῶν πρὸς τῷ Κ κέντρῳ τεσσάρων ὀρθῶν, τὸ αὐτὸ μέρος ἐστὶ καὶ ἡ ὑπὸ ΖΜΟ γωνία τῶν πρὸς τῷ Μ κέντρῳ τεσσάρων ὀρθῶν; ἐπεὶ οὖν περὶ >ίσαξ γωνίαξ αἱ πλευραὶ ἀνάλογόν εἰσιν, <ὁμοιον >άρα ἐστὶ τὸ ΒΚΤ τρίγωνον τῷ ΖΜΟ τριγώνῳ. πάλιν, ἐπεὶ ἐδείθθη ὡς ἡ ΒΚ πρὸς τὴν ΚΛ, ο<ύτωξ ἡ ΖΜ πρὸς τὴν ΜΝ, >ίση δὲ ἡ μὲν ΒΚ τῇ ΚΤ, ἡ δὲ ΖΜ τῇ ΜΟ, >έστιν >άρα ὡς ἡ ΤΚ πρὸς τὴν ΚΛ, ο<ύτωξ ἡ ΟΜ πρὸς τὴν ΜΝ. καὶ περὶ >ίσαξ γωνίαξ τὰξ ὑπὸ ΤΚΛ, ΟΜΝ; ὀρθαὶ γάρ; αἱ πλευραὶ ἀνάλογόν εἰσιν; <ὁμοιον >άρα ἐστὶ τὸ ΑΚΤ τρίγωνον τῷ ΝΜΟ τριγώνῳ. καὶ ἐπεὶ διὰ τὴν ὁμοιότητα τῶν ΑΚΒ, ΝΜΖ τριγώνων ἐστὶν ὡς ἡ ΑΒ πρὸς τὴν ΒΚ, ο<ύτωξ ἡ ΝΖ πρὸς τὴν ΖΜ, διὰ δὲ τὴν ὁμοιότητα τῶν ΒΚΤ, ΖΜΟ τριγώνων ἐστὶν ὡς ἡ ΚΒ πρὸς τὴν ΒΤ, ο<ύτωξ ἡ ΜΖ πρὸς τὴν ΖΟ, δι> >ίσου >άρα ὡς ἡ ΑΒ πρὸς τὴν ΒΤ, ο<ύτωξ ἡ ΝΖ πρὸς τὴν ΖΟ. πάλιν, ἐπεὶ διὰ τὴν ὁμοιότητα τῶν ΑΤΚ, ΝΟΜ τριγώνων ἐστὶν ὡς ἡ ΑΤ πρὸς τὴν ΤΚ, ο<ύτωξ ἡ ΝΟ πρὸς τὴν ΟΜ, διὰ δὲ τὴν ὁμοιότητα τῶν ΤΚΒ, ΟΜΖ τριγώνων ἐστὶν ὡς ἡ ΚΤ πρὸς τὴν ΤΒ, ο<ύτωξ ἡ ΜΟ πρὸς τὴν ΟΖ, δι> >ίσου >άρα ὡς ἡ ΑΤ πρὸς τὴν ΤΒ, ο<ύτωξ ἡ ΝΟ πρὸς τὴν ΟΖ. ἐδείθθη δὲ καὶ ὡς ἡ ΤΒ πρὸς τὴν ΒΛ, ο<ύτωξ ἡ ΟΖ πρὸς τὴν ΖΝ. δι> >ίσου >άρα ὡς ἡ ΤΛ πρὸς τὴν ΑΒ, ο<ύτωξ ἡ ΟΝ πρὸς τὴν ΝΖ. τῶν ΑΤΒ, ΝΟΖ >άρα τριγώνων ἀνάλογόν εἰσιν αἱ πλευραὶ; ἰσογώνια >άρα ἐστὶ τὰ ΑΤΒ, ΝΟΖ τρίγωνα; <ώστε καὶ <ὁμοια. καὶ πυραμίδεξ >άρα, <ἥξ βάσις μὲν τὸ ΒΚΤ τρίγωνον, κορυφὴ δὲ τὸ Α σημεῖον, ὁμοία ἐστὶ πυραμίδι, <ἥξ βάσις μὲν τὸ ΖΜΟ τρίγωνον, κορυφὴ δὲ τὸ Ν σημεῖον; ὑπὸ γὰρ <ὁμοίων ἐπιπέδων περιέθονται >ίσων τὸ πλῆθος. αἱ δὲ <ὁμοια πυραμίδεξ καὶ τριγώνουξ >έθουσαι βάσειξ ἐν τριπλασίονι λόγῳ εἰσὶ τῶν ὁμολόγων πλευρῶν. ἡ >άρα ΒΚΤΛ πυραμίδεξ πρὸς τὴν ΖΜΟΝ πυραμίδα τριπλασίονα λόγον >έθει >ήπερ ἡ ΒΚ πρὸς τὴν ΖΜ. ὁμοίωξ δὲ ἐπιεγγύντεξ ἀπὸ τῶν Α, Θ, Δ, Φ, Γ, Υ ἐπὶ τὸ Κ εὐθείαξ καὶ ἀπὸ τῶν Ε, Σ, Θ, Ρ, Η, Π ἐπὶ τὸ Μ καὶ ἀνιστάντεξ ἐφ> ἐκάστου τῶν τριγώνων πυραμίδαξ τὴν αὐτὴν κορυφὴν ἐθούσαξ τοῖξ κώνοιξ δείχομεν, <ὅτι καὶ ἐκάστη τῶν ὁμοταγῶν πυραμίδων πρὸξ ἐκάστην ὁμοταγῇ πυραμίδα τριπλασίονα λόγον <έχει >ήπερ ἡ ΒΚ ὁμόλογοξ πλευρά πρὸξ τὴν ΖΜ ὁμόλογον πλευράν, τουτέστιν >ήπερ ἡ ΒΔ πρὸξ τὴν ΖΘ. καὶ ὡς <ἐν τῶν ἡγουμένων πρὸξ <ἐν τῶν ἐπομένων, ο<ύτωξ <άπαντα τὰ ἡγούμενα πρὸξ <άπαντα τὰ ἐπόμενα; >έστιν >άρα καὶ ὡς ἡ ΒΚΤΛ πυραμίδεξ πρὸξ τὴν ΖΜΟΝ πυραμίδα, ο<ύτωξ ἡ <όλη πυραμίδεξ, <ἥξ βάσις τὸ ΑΤΒΥΓΦΔΘ πολύγωνον, κορυφὴ δὲ τὸ Α σημεῖον,

EPFQGRHS, be inscribed in circle *ABCD* [Prop. 6.18]. And let a pyramid having the same apex as the cone be set up on polygon *ATBUCVDW*. And let *LBT* be one of the triangles containing the pyramid whose base is polygon *ATBUCVDW*, and apex the point *L*. And let *NFP* be one of the triangles containing the pyramid whose base is triangle *EPFQGRHS*, and apex the point *N*. And let *KT* and *MP* be joined. And since cone *ABCDL* is similar to cone *EFGHN*, thus as *BD* is to *FH*, so axis *KL* (is) to axis *MN* [Def. 11.24]. And as *BD* (is) to *FH*, so *BK* (is) to *FM*. And, thus, as *BK* (is) to *FM*, so *KL* (is) to *MN*. And, alternately, as *BK* (is) to *KL*, so *FM* (is) to *MN* [Prop. 5.16]. And the sides around the equal angles *BKL* and *FMN* are proportional. Thus, triangle *BKL* is similar to triangle *FMN* [Prop. 6.6]. Again, since as *BK* (is) to *KT*, so *FM* (is) to *MP*, and (they are) about the equal angles *BKT* and *FMP*, inasmuch as whatever part angle *BKT* is of the four right-angles at the center *K*, angle *FMP* is also the same part of the four right-angles at the center *M*. Therefore, since the sides about equal angles are proportional, triangle *BKT* is thus similar to triangle *FMP* [Prop. 6.6]. Again, since it was shown that as *BK* (is) to *KL*, so *FM* (is) to *MN*, and *BK* (is) equal to *KT*, and *FM* to *PM*, thus as *TK* (is) to *KL*, so *PM* (is) to *MN*. And the sides about the equal angles *TKL* and *PMN*—for (they are both) right-angles—are proportional. Thus, triangle *LKT* (is) similar to triangle *NMP* [Prop. 6.6]. And since, on account of the similarity of triangles *LKB* and *NMF*, as *LB* (is) to *BK*, so *NF* (is) to *FM*, and, on account of the similarity of triangles *BKT* and *FMP*, as *KB* (is) to *BT*, so *MF* (is) to *FP* [Def. 6.1], thus, via equality, as *LB* (is) to *BT*, so *NF* (is) to *FP* [Prop. 5.22]. Again, since, on account of the similarity of triangles *LTK* and *NPM*, as *LT* (is) to *TK*, so *NP* (is) to *PM*, and, on account of the similarity of triangles *TKB* and *PMF*, as *KT* (is) to *TB*, so *MP* (is) to *PF*, thus, via equality, as *LT* (is) to *TB*, so *NP* (is) to *PF* [Prop. 5.22]. And it was shown that as *TB* (is) to *BL*, so *PF* (is) to *FN*. Thus, via equality, as *TL* (is) to *LB*, so *PN* (is) to *NF* [Prop. 5.22]. Thus, the sides of triangles *LTB* and *NPF* are proportional. Thus, triangles *LTB* and *NPF* are equiangular [Prop. 6.5]. And, hence, (they are) similar [Def. 6.1]. And, thus, the pyramid whose base is triangle *BKT*, and apex the point *L*, is similar to the pyramid whose base is triangle *FMP*, and apex the point *N*. For they are contained by equal numbers of similar planes [Def. 11.9]. And similar pyramids which also have triangular bases are in the cubed ratio of corresponding sides [Prop. 12.8]. Thus, pyramid *BKTL* has to pyramid *FMPN* the cubed ratio that *BK* (has) to *FM*. So, similarly, joining straight-lines from (points) *A*, *W*, *D*, *V*, *C*, and *U* to (center) *K*, and from (points) *E*, *S*, *H*, *R*, *G*, and *Q* to (center) *M*, and setting up pyramids having the same apexes as the cones on each of the triangles (so formed), we can also show that each of the pyramids (on base *ABCD* taken) in order will have to each of the pyramids (on base *EFGH* taken) in order the cubed ratio that the corresponding side *BK* (has) to the corresponding side *FM*—that is to say, that *BD* (has) to *FH*. And (for two sets of proportional magnitudes) as one of the leading (magnitudes is) to one of the following, so (the sum of) all of the leading (magnitudes is) to (the sum of) all of the following (mag-

πρὸς τὴν <όλην πυραμίδα, <ἥξ βάσις μὲν τὸ ΕΟΖΠΗΡΘΣ πολύγωνον, κορυφή δὲ τὸ Ν σημείον; <ώστε καὶ πυραμίς, <ἥξ βάσις μὲν τὸ ΑΤΒΥΓΦΔΘ, κορυφή δὲ τὸ Λ, πρὸς τὴν πυραμίδα, <ἥξ βάσις [μὲν] τὸ ΕΟΖΠΗΡΘΣ πολύγωνον, κορυφή δὲ τὸ Ν σημείον, τριπλασίονα λόγον >έθει >ἥπερ ἢ ΒΔ πρὸς τὴν ΖΘ. ὑπόκειται δὲ καὶ ὁ κῶνος, ο<ῦ βάσις [μὲν] ὁ ΑΒΓΔ κύκλοξ, κορυφή δὲ τὸ Λ σημείον, πρὸς τὸ Χ στερεὸν τριπλασίονα λόγον >έθων >ἥπερ ἢ ΒΔ πρὸς τὴν ΖΘ; >έστιν >άρα ὡς ὁ κῶνος, ο<ῦ βάσις μὲν ἔστιν ὁ ΑΒΓΔ κύκλοξ, κορυφή δὲ τὸ Λ, πρὸς τὸ Χ στερεόν, ο<ύτωξ ἢ πυραμίς, <ἥξ βάσις μὲν τὸ ΑΤΒΥΓΦΔΘ [πολύγωνον], κορυφή δὲ τὸ Λ, πρὸς τὴν πυραμίδα, <ἥξ βάσις μὲν ἔστι τὸ ΕΟΖΠΗΡΘΣ πολύγωνον, κορυφή δὲ τὸ Ν; ἐναλλάξ >άρα, ὡς ὁ κῶνος, ο<ῦ βάσις μὲν ὁ ΑΒΓΔ κύκλοξ, κορυφή δὲ τὸ Λ, πρὸς τὴν ἐν αὐτῷ πυραμίδα, <ἥξ βάσις μὲν τὸ ΑΤΒΥΓΦΔΘ πολύγωνον, κορυφή δὲ τὸ Λ, ο<ύτωξ τὸ Χ [στερεόν] πρὸς τὴν πυραμίδα, <ἥξ βάσις μὲν ἔστι τὸ ΕΟΖΠΗΡΘΣ πολύγωνον, κορυφή δὲ τὸ Ν. μείζων δὲ ὁ εἰρημένος κῶνος τῆς ἐν αὐτῷ πυραμίδος; ἐμπεριέθει γὰρ αὐτήν. μείζων >άρα καὶ τὸ Χ στερεὸν τῆς πυραμίδος, <ἥξ βάσις μὲν ἔστι τὸ ΕΟΖΠΗΡΘΣ πολύγωνον, κορυφή δὲ τὸ Ν. ἀλλὰ καὶ >έλαττον; <όπερ ἔστιν ἀδύνατον. οὐκ >άρα ὁ κῶνος, ο<ῦ βάσις ὁ ΑΒΓΔ κύκλοξ, κορυφή δὲ τὸ Λ [σημείον], πρὸς >έλαττόν τι τοῦ κώνου στερεόν, ο<ῦ βάσις μὲν ὁ ΕΖΗΘ κύκλοξ, κορυφή δὲ τὸ Ν σημείον, τριπλασίονα λόγον >έθει >ἥπερ ἢ ΒΔ πρὸς τὴν ΖΘ. ὁμοίως δὲ δείχομεν, <ὅτι οὐδὲ ὁ ΕΖΗΘ κῶνος πρὸς >έλαττόν τι τοῦ ΑΒΓΔ κώνου στερεὸν τριπλασίονα λόγον >έθει >ἥπερ ἢ ΖΘ πρὸς τὴν ΒΔ.

Λέγω δὴ, <ὅτι οὐδὲ ὁ ΑΒΓΔ κῶνος πρὸς μείζον τι τοῦ ΕΖΗΘ κώνου στερεὸν τριπλασίονα λόγον >έθει >ἥπερ ἢ ΒΔ πρὸς τὴν ΖΘ.

Εἰ γὰρ δυνατόν, ἐθέτω πρὸς μείζον τὸ Χ. ἀνάπαλιν >άρα τὸ Χ στερεὸν πρὸς τὸν ΑΒΓΔ κῶνον τριπλασίονα λόγον >έθει >ἥπερ ἢ ΖΘ πρὸς τὴν ΒΔ. ὡς δὲ τὸ Χ στερεὸν πρὸς τὸν ΑΒΓΔ κῶνον, ο<ύτωξ ὁ ΕΖΗΘ κῶνος πρὸς >έλαττόν τι τοῦ ΑΒΓΔ κώνου στερεόν. καὶ ὁ ΕΖΗΘ >άρα κῶνος πρὸς >έλαττόν τι τοῦ ΑΒΓΔ κώνου στερεόν τριπλασίονα λόγον >έθει >ἥπερ ἢ ΖΘ πρὸς τὴν ΒΔ; <όπερ ἀδύνατον ἐδείκθη. οὐκ >άρα ὁ ΑΒΓΔ κῶνος πρὸς μείζον τι τοῦ ΕΖΗΘ κώνου στερεόν τριπλασίονα λόγον >έθει >ἥπερ ἢ ΒΔ πρὸς τὴν ΖΘ. ἐδείκθη δέ, <ὅτι οὐδὲ πρὸς >έλαττον. ὁ ΑΒΓΔ >άρα κῶνος πρὸς τὸν ΕΖΗΘ κῶνον τριπλασίονα λόγον >έθει >ἥπερ ἢ ΒΔ πρὸς τὴν ΖΘ.

Ὡς δὲ ὁ κῶνος πρὸς τὸν κῶνον, ὁ κύλινδρος πρὸς τὸν κύλινδρον; τριπλασίος γὰρ ὁ κύλινδρος τοῦ κώνου ὁ ἐπὶ τῆς αὐτῆς βάσεως τῷ κώνῳ καὶ ἰσο"υψῆς αὐτῷ. καὶ ὁ κύλινδρος >άρα πρὸς τὸν κύλινδρον τριπλασίονα λόγον >έθει >ἥπερ ἢ ΒΔ πρὸς τὴν ΖΘ.

Οἱ >άρα ὁμοιοὶ κῶνοι καὶ κύλινδροι πρὸς ἀλλήλους ἐν τριπλασίονι λόγῳ εἰσι τῶν ἐν ταῖς βάσεσι διαμέτρων; <όπερ >έδει δεῖχαι.

13.

Ἐὰν κύλινδρος ἐπιπέδῳ τμηθῇ παραλλήλῳ >όντι τοῖς ἀπεναντίον ἐπιπέδοις, >έσται ὡς ὁ κύλινδρος πρὸς τὸν κύλινδρον, ο<ύτωξ ὁ >άχων πρὸς τὸν >άχωνα.

nitudes) [Prop. 5.12]. And, thus, as pyramid $BKTL$ (is) to pyramid $FMPN$, so the whole pyramid whose base is polygon $ATBUCVDW$, and apex the point L , (is) to the whole pyramid whose base is polygon $EPFQGRHS$, and apex the point N . And, hence, the pyramid whose base is polygon $ATBUCVDW$, and apex the point L , has to the pyramid whose base is polygon $EPFQGRHS$, and apex the point N , the cubed ratio that BD (has) to FH . And it was also assumed that the cone whose base is circle $ABCD$, and apex the point L , has to solid O the cubed ratio that BD (has) to FH . Thus, as the cone whose base is circle $ABCD$, and apex the point L , is to solid O , so the pyramid whose base (is) [polygon] $ATBUCVDW$, and apex the point L , (is) to the pyramid whose base is polygon $EPFQGRHS$, and apex the point N . Thus, alternately, as the cone whose base (is) circle $ABCD$, and apex the point L , (is) to the pyramid within it whose base (is) the polygon $ATBUCVDW$, and apex the point L , so the [solid] O (is) to the pyramid whose base is polygon $EPFQGRHS$, and apex the point N [Prop. 5.16]. And the aforementioned cone (is) greater than the pyramid within it. For it encompasses it. Thus, solid O (is) also greater than the pyramid whose base is polygon $EPFQGRHS$, and apex the point N . But, (it is) also less. The very thing is impossible. Thus, the cone whose base (is) circle $ABCD$, and apex the [point] L , does not have to some solid less than the cone whose base (is) circle $EFGH$, and apex the point N , the cubed ratio that BD (has) to EH . So, similarly, we can show that neither does cone $EFGHN$ have to some solid less than cone $ABCDL$ the cubed ratio that FH (has) to BD .

So, I say that neither does cone $ABCDL$ have to some solid greater than cone $EFGHN$ the cubed ratio that BD (has) to FH .

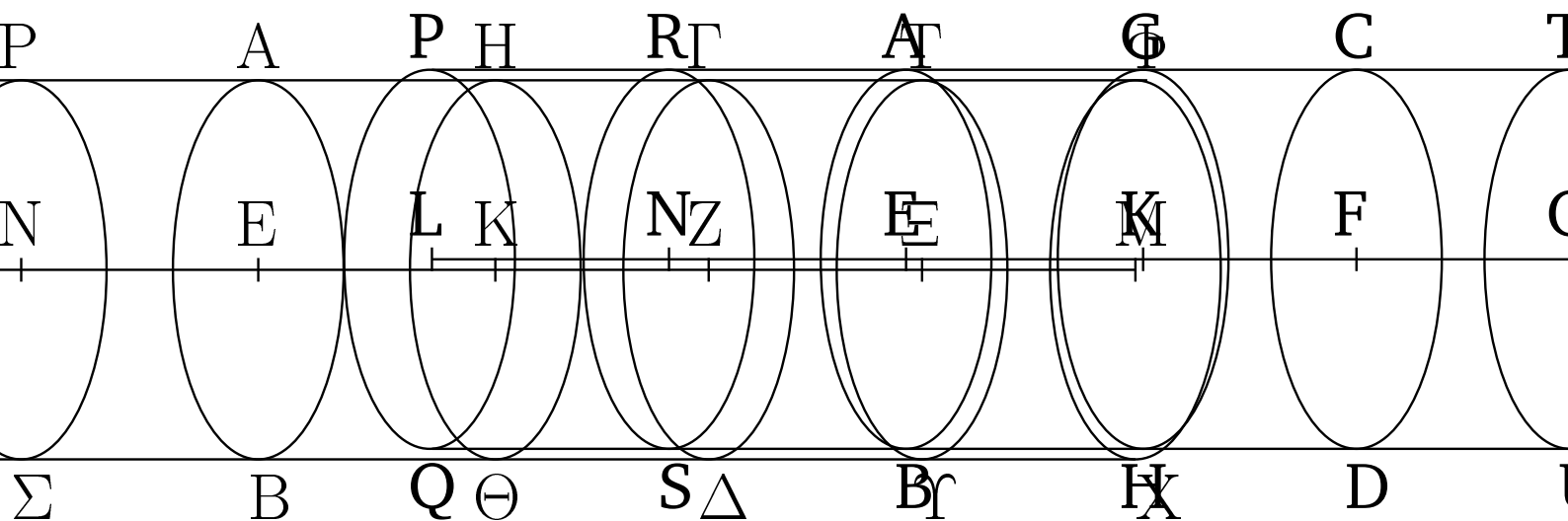
For, if possible, let it have (such a ratio) to a greater (solid), O . Thus, inversely, solid O has to cone $ABCDL$ the cubed ratio that FH (has) to BD [Prop. 5.7 corr.]. And as solid O (is) to cone $ABCDL$, so cone $EFGHN$ (is) to some solid less than cone $ABCDL$ [12.2 lem.]. Thus, cone $EFGHN$ also has to some solid less than cone $ABCDL$ the cubed ratio that FH (has) to BD . The very thing was shown (to be) impossible. Thus, cone $ABCDL$ does not have to some solid greater than cone $EFGHN$ the cubed ratio than BD (has) to FH . And it was shown that neither (does it have such a ratio) to a lesser (solid). Thus, cone $ABCDL$ has to cone $EFGHN$ the cubed ratio that BD (has) to FG .

And as the cone (is) to the cone, so the cylinder (is) to the cylinder. For a cylinder is three times a cone on the same base as the cone, and of the same height as it [Prop. 12.10]. Thus, the cylinder also has to the cylinder the cubed ratio that BD (has) to FH .

Thus, similar cones and cylinders are in the cubed ratio of the diameters of their bases. (Which is) the very thing it was required to show.

Proposition 13

If a cylinder is cut by a plane which is parallel to the opposite planes (of the cylinder) then as the cylinder (is) to the cylinder, so the axis will be to the axis.



Κύλινδροι γὰρ ὁ ΑΔ ἐπιπέδῳ τῷ ΗΘ τετμήσθω παραλλήλῳ
>όντι τοῖς ἀπεναντίον ἐπιπέδοις τοῖς ΑΒ, ΓΔ, καὶ συμβαλλέτω
τῷ >άχονι τὸ ΗΘ ἐπίπεδον κατὰ τὸ Κ σημεῖον· λέγω, <ὅτι
ἐστὶν ὡς ὁ ΒΗ κύλινδρος πρὸς τὸν ΗΔ κύλινδρον, ο<ύτωξ
ὁ ΕΚ >άχων πρὸς τὸν ΚΖ >άχωνα.

Ἐκβεβλήσθω γὰρ ὁ ΕΖ >άχων ἐφ' ἐκάτερα τὰ μέρη ἐπὶ
τὰ Λ, Μ σημεῖα, καὶ ἐκκείσθωσαν τῷ ΕΚ >άχονι >ῖσοι
ὁσοιδηποτοῦν οἱ ΕΝ, ΝΛ, τῷ δὲ ΖΚ >ῖσοι ὁσοιδηποτοῦν
οἱ ΖΧ, ΧΜ, καὶ νοείσθω ὁ ἐπὶ τοῦ ΛΜ >άχονος κύλινδρος
ὁ ΟΘ, ο<ύ βάσειξ οἱ ΟΠ, ΦΘ κύκλοι. καὶ ἐκβεβλήσθω
διὰ τῶν Ν, Χ σημείων ἐπίπεδα παράλληλα τοῖς ΑΒ, ΓΔ
καὶ ταῖξ βάσειξ τοῦ ΟΘ κύλινδρου καὶ ποιείτωσαν τοῖξ
ΡΣ, ΤΥ κύκλους περὶ τὰ Ν, Χ κέντρα. καὶ ἐπεὶ οἱ ΑΝ,
ΝΕ, ΕΚ >άχονεξ >ῖσοι εἰσὶν ἀλλήλοιξ, οἱ >άρα ΠΡ, ΡΒ,
ΒΗ κύλινδροι πρὸς ἀλλήλους εἰσὶν ὡς αἱ βάσειξ. >ῖσαι δὲ
εἰσὶν αἱ βάσειξ· >ῖσοι >άρα καὶ οἱ ΠΡ, ΡΒ, ΒΗ κύλινδροι
ἀλλήλοιξ. ἐπεὶ ο<ύν οἱ ΑΝ, ΝΕ, ΕΚ >άχονεξ >ῖσοι
εἰσὶν ἀλλήλοιξ, εἰσὶ δὲ καὶ οἱ ΠΡ, ΡΒ, ΒΗ κύλινδροι
>ῖσοι ἀλλήλοιξ, καὶ ἐστὶν >ῖσον τὸ πλῆθοξ τῷ πλῆθει,
ὁσαπλασίῳν >άρα ὁ ΚΛ >άχων τοῦ ΕΚ >άχονοξ, τοσαυταπλασίῳν
>έσται καὶ ὁ ΠΗ κύλινδρος τοῦ ΗΒ κύλινδρου. διὰ τὰ
αὐτὰ δὴ καὶ ὁσαπλασίῳν ἐστὶν ὁ ΜΚ >άχων τοῦ ΚΖ
>άχονοξ, τοσαυταπλασίῳν ἐστὶ καὶ ὁ ΘΗ κύλινδρος τοῦ
ΗΔ κύλινδρου. καὶ εἰ μὲν >ῖσοξ ἐστὶν ὁ ΚΛ >άχων τῷ ΚΜ
>άχονι, >ῖσοξ >έσται καὶ ὁ ΠΗ κύλινδρος τῷ ΗΘ κύλινδρῳ,
εἰ δὲ μείζων ὁ >άχων τοῦ >άχονοξ, μείζων καὶ ὁ κύλινδρος
τοῦ κύλινδρου, καὶ εἰ ἐλάσσων, ἐλάσσων. τεσσάρων δὴ
μεγεθῶν >όντων, ἀχόνων μὲν τῶν ΕΚ, ΚΖ, κύλινδρων δὲ
τῶν ΒΗ, ΗΔ, ε<ἰληπται ἰσάκιξ πολλαπλάσια, τοῦ μὲν ΕΚ
>άχονοξ καὶ τοῦ ΒΗ κύλινδρου <ὅ τε ΑΚ >άχων καὶ ὁ ΠΗ
κύλινδρος, τοῦ δὲ ΚΖ >άχονοξ καὶ τοῦ ΗΔ κύλινδρου <ὅ
τε ΚΜ >άχων καὶ ὁ ΗΘ κύλινδρος, καὶ δέδεικται, <ὅτι
εἰ ὑπερέθει ὁ ΚΛ >άχων τοῦ ΚΜ >άχονοξ, ὑπερέθει καὶ
ὁ ΠΗ κύλινδρος τοῦ ΗΘ κύλινδρου, καὶ εἰ >ῖσοξ, >ῖσοξ,
καὶ εἰ ἐλάσσων, ἐλάσσων. >έστιν >άρα ὡς ὁ ΕΚ >άχων
πρὸς τὸν ΚΖ >άχωνα, ο<ύτωξ ὁ ΒΗ κύλινδρος πρὸς τὸν
ΗΔ κύλινδρον· <ὅπερ >έδει δεῖχαι.

For let the cylinder AD be cut by the plane GH which is par-
allel to the opposite planes (of the cylinder), AB and CD .
And let the plane GH have met the axis at point K . I say
that as cylinder BG is to cylinder GD , so axis EK (is) to
axis KF .

For let axis EF be produced in each direction to points L
and M . And let any number whatsoever (of lengths), EN
and NL , equal to axis EK , be set out (on the axis EL), and
any number whatsoever (of lengths), FO and OM , equal to
(axis) FK , (on the axis KM). And let the cylinder PW ,
whose bases (are) the circles PQ and VW , be conceived
on axis LM . And let planes parallel to AB , CD , and the
bases of cylinder PW , be produced through points N and
 O , and let them have made the circles RS and TU around
the centers N and O (respectively). And since axes LN ,
 NE , and EK are equal to one another, the cylinders QR ,
 RB , and BG are to one another as their bases [Prop. 12.11].
But the bases are equal. Thus, the cylinders QR , RB , and
 BG (are) also equal to one another. Therefore, since the
axes LN , NE , and EK are equal to one another, and the
cylinders QR , RB , and BG are also equal to one another,
and the number (of the former) is equal to the number (of the
latter), thus as many multiples as axis KL is of axis EK , so
many multiples is cylinder QG also of cylinder GB . And
so, for the same (reasons), as many multiples as axis MK
is of axis KF , so many multiples is cylinder WG also of
cylinder GD . And if axis KL is equal to axis KM then
cylinder QG will also be equal to cylinder GW , and if the
axis (is) greater than the axis then the cylinder (will also be)
greater than the cylinder, and if (the axis is) less then (the
cylinder will also be) less. So, there are four magnitudes—
the axes EK and KF , and the cylinders BG and GD —and
equal multiples be taken of axis EK and cylinder BG —
(namely), axis LK and cylinder QG —and of axis KF and
cylinder GD —(namely), axis KM and cylinder GW . And
it has been shown that if axis KL exceeds axis KM then
cylinder QG also exceeds cylinder GW , and if (the axes are)
equal then (the cylinders are) equal, and if (KL is) less then
(QG is) less. Thus, as axis EK is to axis KF , so cylinder
 BG (is) to cylinder GD [Def. 5.5]. (Which is) the very thing
it was required to show.

14.

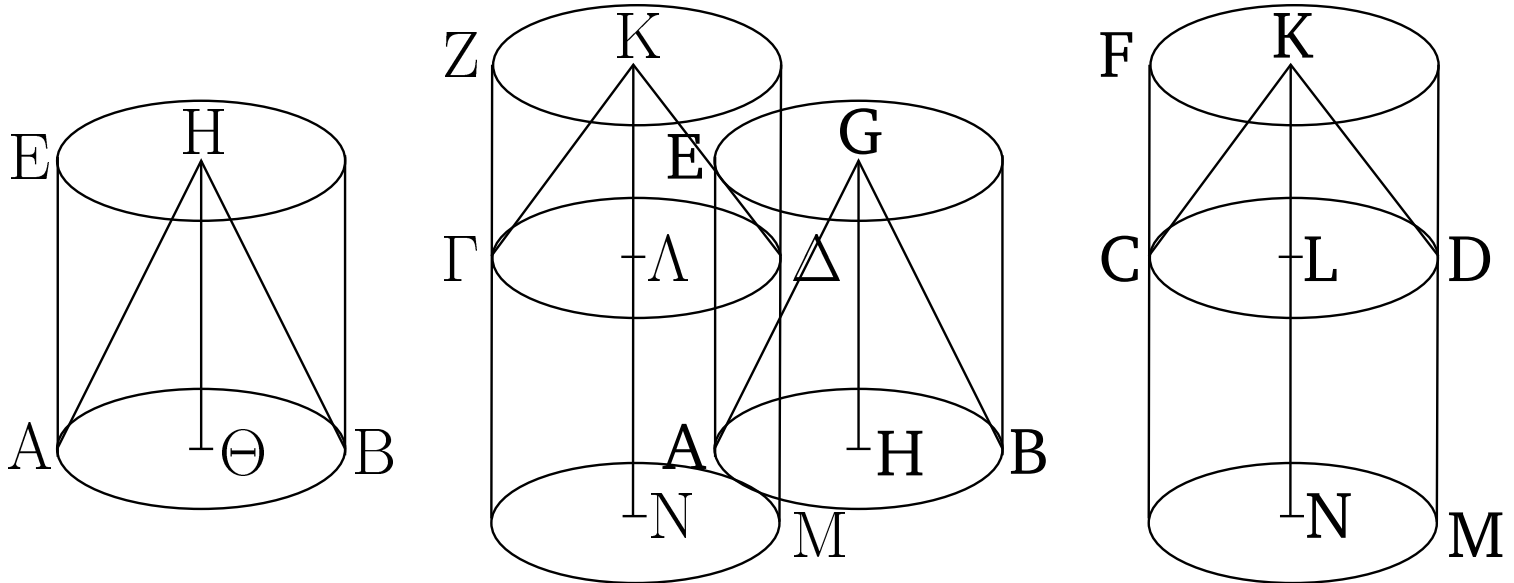
Proposition 14

Οἱ ἐπὶ >ῖσων βάσεων >όντεξ κῶνοι καὶ κύλινδροι πρὸξ

Cones and cylinders which are on equal bases are to one

ἀλλήλους εἰσὶν ὡς τὰ ὑψη.

another as their heights.



>Ἐστωσαν γὰρ ἐπὶ >ίσων βάσεων τῶν AB , $\Gamma\Delta$ κύκλων κύλινδροι οἱ EB , $Z\Delta$; λέγω, <ὅτι ἐστὶν ὡς ὁ EB κύλινδρος πρὸς τὸν $Z\Delta$ κύλινδρον, ο<ύτως ὁ $H\Theta$ >άχων πρὸς τὸν $K\Lambda$ >άχονα.

Ἐκβεβλήσθω γὰρ ὁ $K\Lambda$ >άχων ἐπὶ τὸ N σημεῖον, καὶ κείσθω τῷ $H\Theta$ >άχονι >ίσοξ ὁ ΛN , καὶ περὶ >άχονα τὸν ΛN κύλινδρος νενοήσθω ὁ ΓM . ἐπεὶ ο>ὖν οἱ EB , ΓM κύλινδροι ὑπὸ τὸ αὐτὸ <ύψος εἰσὶν, πρὸς ἀλλήλους εἰσὶν ὡς αἱ βάσεις. >ἴσαι δέ εἰσὶν αἱ βάσεις ἀλλήλαι; >ἴσοι >άρα εἰσὶ καὶ οἱ EB , ΓM κύλινδροι. καὶ ἐπεὶ κύλινδρος ὁ ZM ἐπιπέδῳ τέμνεται τῷ $\Gamma\Delta$ παραλλήλῳ >όντι τοῖς ἀπεναντίον ἐπιπέδοις, >έστιν >άρα ὡς ὁ ΓM κύλινδρος πρὸς τὸν $Z\Delta$ κύλινδρον, ο<ύτως ὁ ΛN >άχων πρὸς τὸν $K\Lambda$ >άχονα. >ίσοξ δέ ἐστὶν ὁ μὲν ΓM κύλινδρος τῷ EB κύλινδρῳ, ὁ δὲ ΛN >άχων τῷ $H\Theta$ >άχονι; >έστιν >άρα ὡς ὁ EB κύλινδρος πρὸς τὸν $Z\Delta$ κύλινδρον, ο<ύτως ὁ $H\Theta$ >άχων πρὸς τὸν $K\Lambda$ >άχονα. ὡς δὲ ὁ EB κύλινδρος πρὸς τὸν $Z\Delta$ κύλινδρον, οὕτως ὁ ABH κῶνος πρὸς τὸν $\Gamma\Delta K$ κῶνον. καὶ ὡς >άρα ὁ $H\Theta$ >άχων πρὸς τὸν $K\Lambda$ >άχονα, ο<ύτως ὁ ABH κῶνος πρὸς τὸν $\Gamma\Delta K$ κῶνον καὶ ὁ EB κύλινδρος πρὸς τὸν $Z\Delta$ κύλινδρον; <όπερ >έδει δεῖχαι.

For let EB and FD be cylinders on equal bases, (namely) the circles AB and CD (respectively). I say that as cylinder EB is to cylinder FD , so axis GH (is) to axis KL .

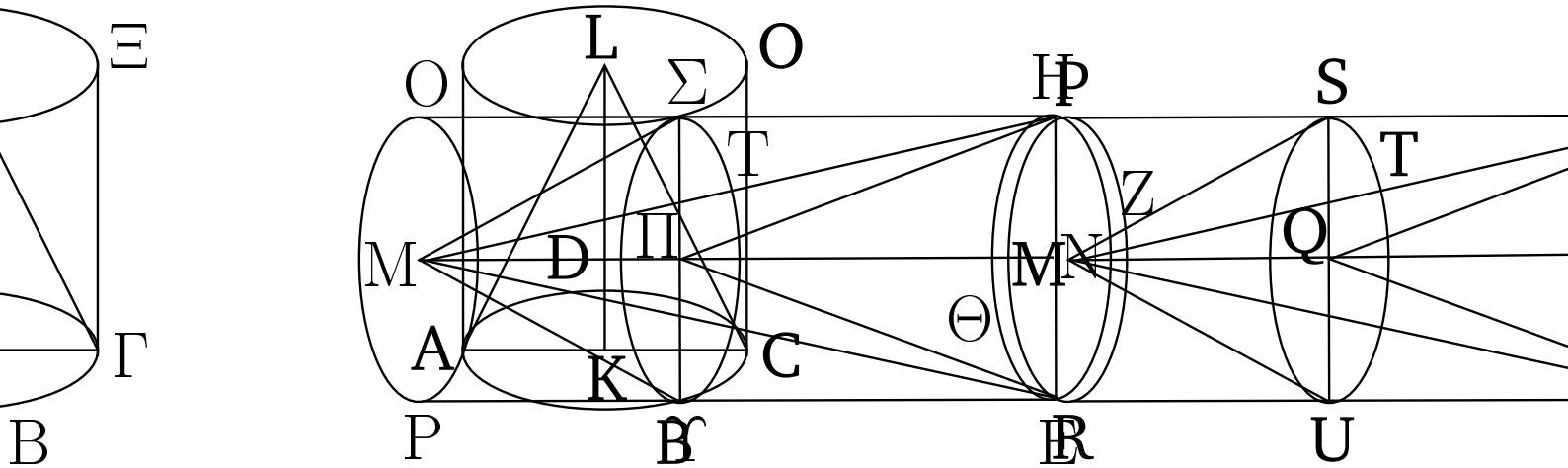
For let the axis KL be produced to point N . And let LN be made equal to axis GH . And let the cylinder CM be conceived about axis LN . Therefore, since cylinders EB and CM have the same height they are to one another as their bases [Prop. 12.11]. And the bases are equal to one another. Thus, cylinders EB and CM are also equal to one another. And since cylinder FM has been cut by the plane CD , which is parallel to its opposite planes, thus as cylinder CM is to cylinder FD , so axis LN (is) to axis KL [Prop. 12.13]. And cylinder CM is equal to cylinder EB , and axis LN to axis GH . Thus, as cylinder EB is to cylinder FD , so axis GH (is) to axis KL . And as cylinder EB (is) to cylinder FD , so cone ABG (is) to cone CDK [Prop. 12.10]. Thus, also, as axis GH (is) to axis KL , so cone ABG (is) to cone CDK , and cylinder EB to cylinder FD . (Which is) the very thing it was required to show.

15.

Τῶν >ίσων κῶνων καὶ κυλίνδρων ἀντιπεπόνθασιν αἱ βάσεις τοῖς <ύψεσιν; καὶ <ῶν κῶνων καὶ κυλίνδρων ἀντιπεπόνθασιν αἱ βάσεις τοῖς <ύψεσιν, >ἴσοι εἰσὶν ἐκεῖνοι.

Proposition 15

The bases of equal cones and cylinders are reciprocally proportional to their heights. And, those cones and cylinders whose bases (are) reciprocally proportional to their heights are equal.



>Ἐστωσαν >ῖσοι κῶνοι καὶ κύλινδροι, <ῶν βάσεις μὲν οἱ ΑΒΓΔ, ΕΖΗΘ κύκλοι, διάμετροι δὲ αὐτῶν αἱ ΑΓ, ΕΗ, <άχονες δὲ οἱ ΚΛ, ΜΝ, ο<ίτινες καὶ <ύψη εἰσὶ τῶν κῶνων >ῆ κυλίνδρων, καὶ συμπεπληρώσθωσαν οἱ ΑΧ, ΕΟ κύλινδροι. λέγω, <ὅτι τῶν ΑΧ, ΕΟ κυλίνδρων ἀντιπεπόνθασιν αἱ βάσεις τοῖς <ύψεσιν, καὶ ἔστιν ὡς ἡ ΑΒΓΔ βάσις πρὸς τὴν ΕΖΗΘ βάσιν, ο<ύτως τὸ ΜΝ <ύψος πρὸς τὸ ΚΛ <ύψος. Τὸ γὰρ ΑΚ <ύψος τῷ ΜΝ <ύψει >ῆτοι >ῖσον ἔστιν >ῆ ο>ύ. >ἔστω πρότερον >ῖσον. >ἔστι δὲ καὶ ὁ ΑΧ κύλινδρος τῷ ΕΟ κύλινδρῳ >ῖσος. οἱ δὲ ὑπὸ τὸ αὐτὸ <ύψος >όντεξ κῶνοι καὶ κύλινδροι πρὸς ἀλλήλους εἰσὶν ὡς αἱ βάσεις; >ῖση >άρα καὶ ἡ ΑΒΓΔ βάσις τῇ ΕΖΗΘ βάσει. <ώστε καὶ ἀντιπέπονθεν, ὡς ἡ ΑΒΓΔ βάσις πρὸς τὴν ΕΖΗΘ βάσιν, ο<ύτως τὸ ΜΝ <ύψος πρὸς τὸ ΚΛ <ύψος. ἀλλὰ δὴ μὴ >ἔστω τὸ ΑΚ <ύψος τῷ ΜΝ >ῖσον, ἀλλ> >ἔστω μείζον τὸ ΜΝ, καὶ ἀφρηθήσθω ἀπὸ τοῦ ΜΝ <ύψους τῷ ΚΛ >ῖσον τὸ ΠΝ, καὶ διὰ τοῦ Π σημείου τεμήσθω ὁ ΕΟ κύλινδρος ἐπιπέδῳ τῷ ΤΥΣ παραλλήλῳ τοῖς τῶν ΕΖΗΘ, ΡΟ κύκλων ἐπιπέδοις, καὶ ἀπὸ βάσεως μὲν τοῦ ΕΖΗΘ κύκλου, <ύψους δὲ τοῦ ΝΠ κύλινδρος νενοήσθω ὁ ΕΣ. καὶ ἐπεὶ >ῖσος ἔστιν ὁ ΑΧ κύλινδρος τῷ ΕΟ κύλινδρῳ, >ἔστιν >άρα ὡς ὁ ΑΧ κύλινδρος πρὸς τὸν ΕΣ κύλινδρον, ο<ύτως ὁ ΕΟ κύλινδρος πρὸς τὸν ΕΣ κύλινδρον. ἀλλ> ὡς μὲν ὁ ΑΧ κύλινδρος πρὸς τὸν ΕΣ κύλινδρον, ο<ύτως ἡ ΑΒΓΔ βάσις πρὸς τὴν ΕΖΗΘ; ὑπὸ γὰρ τὸ αὐτὸ <ύψος εἰσὶν οἱ ΑΧ, ΕΣ κύλινδροι; ὡς δὲ ὁ ΕΟ κύλινδρος πρὸς τὸν ΕΣ, ο<ύτως τὸ ΜΝ <ύψος πρὸς τὸ ΠΝ <ύψος; ὁ γὰρ ΕΟ κύλινδρος ἐπιπέδῳ τέμνεται παραλλήλῳ >όντι τοῖς ἀπεναντίον ἐπιπέδοις. >ἔστιν >άρα καὶ ὡς ἡ ΑΒΓΔ βάσις πρὸς τὴν ΕΖΗΘ βάσιν, ο<ύτως τὸ ΜΝ <ύψος πρὸς τὸ ΠΝ <ύψος. >ῖσον δὲ τὸ ΠΝ <ύψος τῷ ΚΛ <ύψει; >ἔστιν >άρα ὡς ἡ ΑΒΓΔ βάσις πρὸς τὴν ΕΖΗΘ βάσιν, ο<ύτως τὸ ΜΝ <ύψος πρὸς τὸ ΚΛ <ύψος. τῶν >άρα ΑΧ, ΕΟ κυλίνδρων ἀντιπεπόνθασιν αἱ βάσεις τοῖς <ύψεσιν. Ἀλλὰ δὴ τῶν ΑΧ, ΕΟ κυλίνδρων ἀντιπεπονθέτωσαν αἱ βάσεις τοῖς <ύψεσιν, καὶ >ἔστω ὡς ἡ ΑΒΓΔ βάσις πρὸς τὴν ΕΖΗΘ βάσιν, ο<ύτως τὸ ΜΝ <ύψος πρὸς τὸ ΚΛ <ύψος; λέγω, <ὅτι >ῖσος ἔστιν ὁ ΑΧ κύλινδρος τῷ ΕΟ κύλινδρῳ. Τῶν γὰρ αὐτῶν κατασκευασθέντων ἐπεὶ ἔστιν ὡς ἡ ΑΒΓΔ βάσις πρὸς τὴν ΕΖΗΘ βάσιν, ο<ύτως τὸ ΜΝ <ύψος πρὸς τὸ ΚΛ <ύψος, >ῖσον δὲ τὸ ΚΛ <ύψος τῷ ΠΝ <ύψει, >ἔσται >άρα ὡς ἡ ΑΒΓΔ βάσις πρὸς τὴν ΕΖΗΘ βάσιν, ο<ύτως τὸ ΜΝ <ύψος πρὸς τὸ ΠΝ <ύψος. ἀλλ> ὡς μὲν ἡ ΑΒΓΔ βάσις πρὸς τὴν ΕΖΗΘ βάσιν, ο<ύτως ὁ ΑΧ κύλινδρος πρὸς τὸν ΕΣ κύλινδρον; ὑπὸ γὰρ τὸ αὐτὸ <ύψος εἰσὶν; ὡς δὲ τὸ ΜΝ <ύψος πρὸς τὸ ΠΝ [<ύψος], ο<ύτως ὁ ΕΟ κύλινδρος πρὸς τὸν ΕΣ κύλινδρον; >ἔστιν >άρα ὡς ὁ ΑΧ κύλινδρος πρὸς

Let there be equal cones and cylinders whose bases are the circles $ABCD$ and $EFGH$, and the diameters of (the bases) AC and EG , and (whose) axes (are) KL and MN , which are also the heights of the cones and cylinders (respectively). And let the cylinders AO and EP be completed. I say that the bases of cylinders AO and EP are reciprocally proportional to their heights, and (so) as base $ABCD$ is to base $EFGH$, so height MN (is) to height KL .

For height LK is either equal to height MN , or not. Let it, first of all, be equal. And cylinder AO is also equal to cylinder EP . And cones and cylinders having the same height are to one another as their bases [Prop. 12.11]. Thus, base $ABCD$ (is) also equal to base $EFGH$. And, hence, reciprocally, as base $ABCD$ (is) to base $EFGH$, so height MN (is) to height KL . And so, let height LK not be equal to MN , but let MN be greater. And let QN , equal to KL , be cut off from height MN . And let the cylinder EP be cut, through point Q , by the plane TUS (which is) parallel to the planes of the circles $EFGH$ and RP . And let cylinder ES be conceived, with base the circle $EFGH$, and height NQ . And since cylinder AO is equal to cylinder EP , thus, as cylinder AO (is) to cylinder ES , so cylinder EP (is) to cylinder ES [Prop. 5.7]. But, as cylinder AO (is) to cylinder ES , so base $ABCD$ (is) to base $EFGH$. For cylinders AO and ES (have) the same height [Prop. 12.11]. And as cylinder EP (is) to (cylinder) ES , so height MN (is) to height QN . For cylinder EP has been cut by a plane which is parallel to its opposite planes [Prop. 12.13]. And, thus, as base $ABCD$ is to base $EFGH$, so height MN (is) to height QN [Prop. 5.11]. And height QN (is) equal to height KL . Thus, as base $ABCD$ is to base $EFGH$, so height MN (is) to height KL . Thus, the bases of cylinders AO and EP are reciprocally proportional to their heights.

And, so, let the bases of cylinders AO and EP be reciprocally proportional to their heights, and (thus) let base $ABCD$ be to base $EFGH$, as height MN (is) to height KL . I say that cylinder AO is equal to cylinder EP .

For, with the same construction, since as base $ABCD$ is to base $EFGH$, so height MN (is) to height KL , and height KL (is) equal to height QN , thus, as base $ABCD$ (is) to base $EFGH$, so height MN will be to height QN . But, as base $ABCD$ (is) to base $EFGH$, so cylinder AO (is) to cylinder ES . For they are the same height [Prop. 12.11]. And as height MN (is) to [height] QN , so cylinder EP (is) to cylinder ES [Prop. 12.13]. Thus, as cylinder AO

τὸν ΕΣ κύλινδρον, οὐτως ὁ ΕΟ κύλινδρος πρὸς τὸν ΕΣ.
>ἴσος >ἀρα ὁ ΑΧ κύλινδρος τῷ ΕΟ κυλίνδρῳ. ὡσαύτως
δὲ καὶ ἐπὶ τῶν κώνων; <ὅπερ >έδει δεῖξαι.

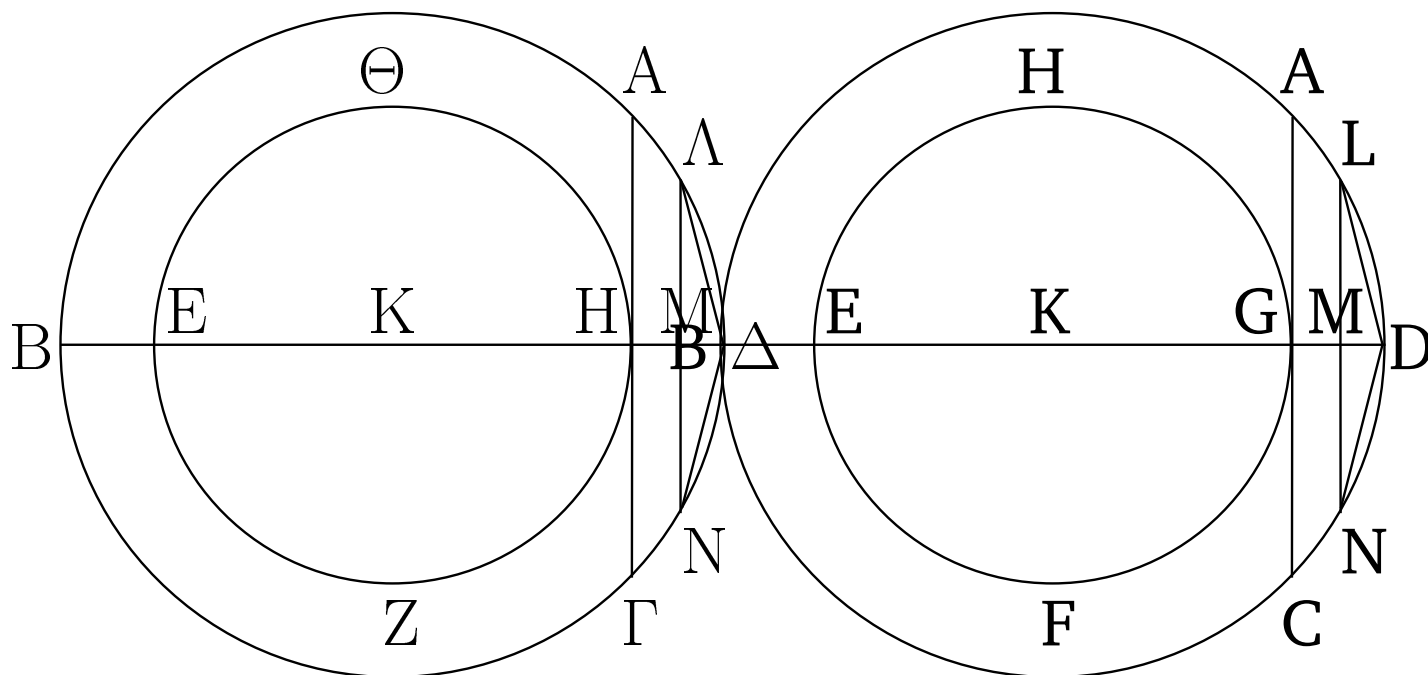
is to cylinder ES , so cylinder EP (is) to (cylinder) ES [Prop. 5.11]. Thus, cylinder AO (is) equal to cylinder EP [Prop. 5.9]. In the same manner, (the proposition can) also (be demonstrated) for the cones. (Which is) the very thing it was required to show.

16.

Δύο κύκλων περὶ τὸ αὐτὸ κέντρον >όντων εἰς τὸν μείζονα
κύκλον πολύγωνον ἰσόπλευρόν τε καὶ ἄρτιόπλευρον ἐγγράψαι
μὴ ψαῦον τοῦ ἐλάσσονος κύκλου.

Proposition 16

There being two circles about the same center, to inscribe an equilateral and even-sided polygon in the greater circle, not touching the lesser circle.



>Ἐστωσαν οἱ δοθέντες δύο κύκλοι οἱ ΑΒΓΔ, ΕΖΗΘ περὶ τὸ
αὐτὸ κέντρον τὸ Κ; δεῖ δὴ εἰς τὸν μείζονα κύκλον τὸν ΑΒΓΔ
πολύγωνον ἰσόπλευρόν τε καὶ ἄρτιόπλευρον ἐγγράψαι μὴ
ψαῦον τοῦ ΕΖΗΘ κύκλου.

>Ἦθω γὰρ διὰ τοῦ Κ κέντρου εὐθεῖα ἡ ΒΚΔ, καὶ ἀπὸ
τοῦ Η σημείου τῇ ΒΔ εὐθείᾳ πρὸς ὀρθὰς >ήθω ἡ ΗΑ
καὶ διήθω ἐπὶ τὸ Γ; ἡ ΑΓ >ἀρα ἐφάπτεται τοῦ ΕΖΗΘ
κύκλου. τέμνοντες δὴ τὴν ΒΑΔ περιφέρειαν δίθω καὶ τὴν
ἡμίσειαν αὐτῆς δίθω καὶ τοῦτο αἰ ποιοῦντες καταλείψομεν
περιφέρειαν ἐλάσσονα τῆς ΑΔ. λείψω, καὶ >έστω ἡ ΛΔ,
καὶ ἀπὸ τοῦ Λ ἐπὶ τὴν ΒΔ κάθετος >ήθω ἡ ΛΜ καὶ διήθω
ἐπὶ τὸ Ν, καὶ ἐπεζεύθωσαν αἱ ΛΔ, ΔΝ; >ἴση >ἀρα ἐστὶν
ἡ ΛΔ τῇ ΔΝ. καὶ ἐπεὶ παράλληλός ἐστιν ἡ ΑΝ τῇ ΑΓ,
ἡ δὲ ΑΓ ἐφάπτεται τοῦ ΕΖΗΘ κύκλου, ἡ ΑΝ >ἀρα οὐκ
ἐφάπτεται τοῦ ΕΖΗΘ κύκλου; πολλῶν >ἀρα αἱ ΛΔ, ΔΝ
οὐκ ἐφάπτονται τοῦ ΕΖΗΘ κύκλου. ἔαν δὴ τῇ ΛΔ εὐθείᾳ
>ἴσας κατὰ τὸ συνεθὲς ἐναρμόσωμεν εἰς τὸν ΑΒΓΔ κύκλον,
ἐγγραφήσεται εἰς τὸν ΑΒΓΔ κύκλον πολύγωνον ἰσόπλευρόν
τε καὶ ἄρτιόπλευρον μὴ ψαῦον τοῦ ἐλάσσονος κύκλου τοῦ
ΕΖΗΘ; <ὅπερ >έδει ποιῆσαι.

Let $ABCD$ and $EFGH$ be the given two circles, about the
same center, K . So, it is necessary to inscribe an equilateral
and even-sided polygon in the greater circle $ABCD$, not
touching circle $EFGH$.

Let the straight-line BKD be drawn through the center K .
And let GA be drawn, at right-angles to the straight-line
 BD , through point G , and let it be drawn through to C .
Thus, AC touches circle $EFGH$ [Prop. 3.16 corr.]. So,
(by) cutting circumference BAD in half, and the half of
it in half, and doing this continually, we will (eventually)
leave a circumference less than AD [Prop. 10.1]. Let it
be left, and let it be LD . And let LM be drawn, from
 L , perpendicular to BD , and let it be drawn through to
 N . And let LD and DN be joined. Thus, LD is equal
to DN [Props. 3.3, 1.4]. And since LN is parallel to AC
[Prop. 1.28], and AC touches circle $EFGH$, LN thus does
not touch circle $EFGH$. Thus, even more so, LD and DN
do not touch circle $EFGH$. And if we continuously insert
(straight-lines) equal to straight-line LD into circle $ABCD$
[Prop. 4.1] then an equilateral and even-sided polygon, not
touching the lesser circle $EFGH$, will be inscribed in circle
 $ABCD$.[†] (Which is) the very thing it was required to do.

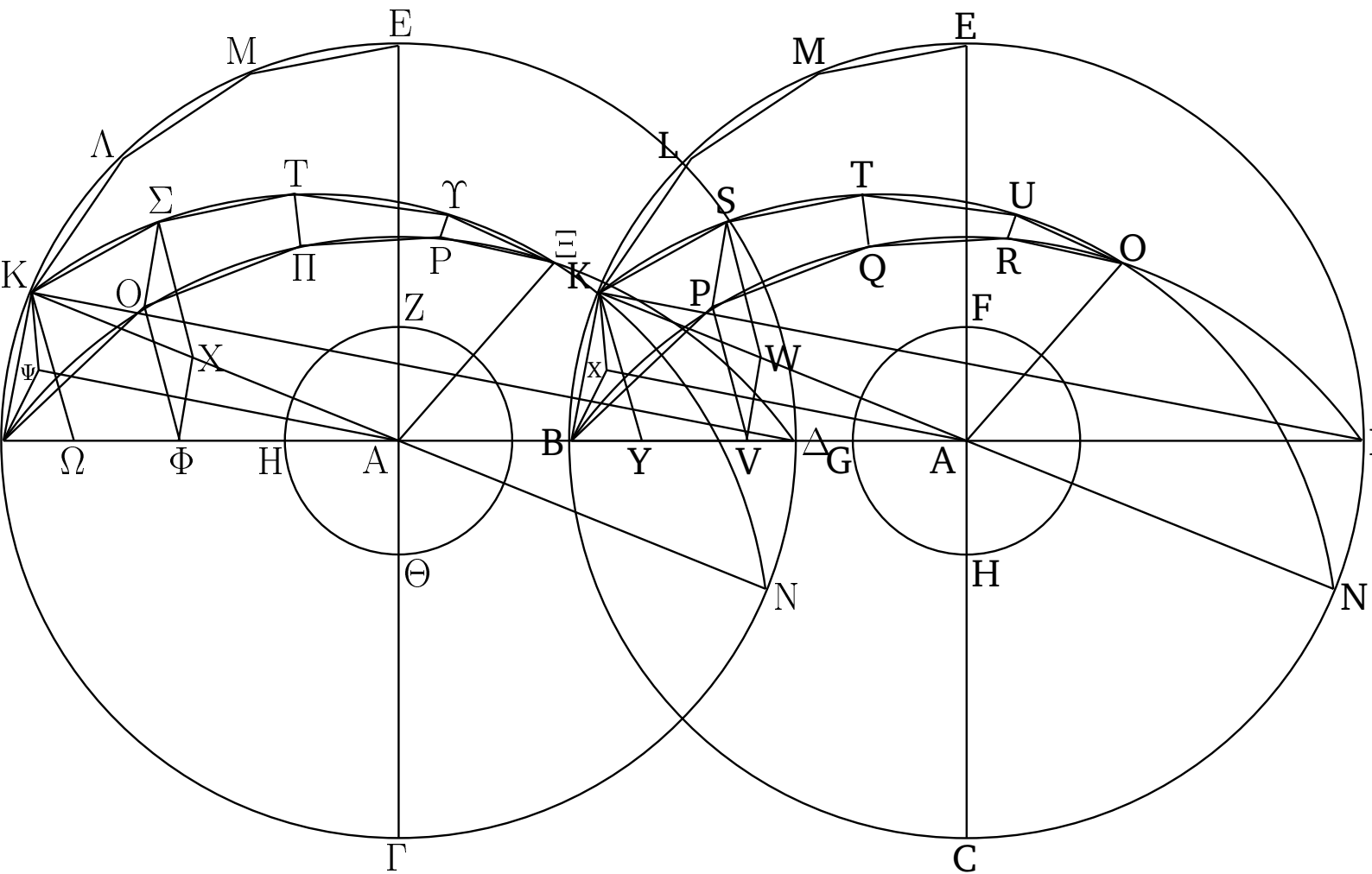
[†] Note that the chord of the polygon, LN , does not touch the inner circle either.

17.

Δύο σφαιρῶν περὶ τὸ αὐτὸ κέντρον οὐσῶν εἰς τὴν μείζονα
σφαῖραν στερεὸν πολυέδρον ἐγγράψαι μὴ ψαῦον τῆς ἐλάσσονος
σφαίρας κατὰ τὴν ἐπιφάνειαν.

Proposition 17

There being two spheres about the same center, to inscribe a
polyhedral solid in the greater sphere, not touching the lesser
sphere on its surface.



Νεοήσθωσαν δύο σφαῖραι περὶ τὸ αὐτὸ κέντρον τὸ Α; δεῖ δὴ εἰς τὴν μείζονα σφαῖραν στερεὸν πολύεδρον ἐγγράψαι μὴ ψαῖον τῆς ἐλάσσονος σφαίρας κατὰ τὴν ἐπιφάνειαν. Τετμήσθωσαν αἱ σφαῖραι ἐπιπέδῳ τινὶ διὰ τοῦ κέντρου; ἔσονται δὴ αἱ τομαὶ κύκλοι, ἐπειδὴ περ μενούσης τῆς διαμέτρου καὶ περιφερομένου τοῦ ἡμικυκλίου ἐγγίγνεται ἡ σφαῖρα; <ὥστε καὶ καθ' οὐκ ἴσως> ἂν θέσωμεν τὸ ἡμικύκλιον, τὸ δι' αὐτοῦ ἐκβαλλόμενον ἐπίπεδον ποιήσει ἐπὶ τῆς ἐπιφανείας τῆς σφαίρας κύκλον. καὶ φανερόν, <ὅτι καὶ μέγιστον, ἐπειδὴ περ ἡ διάμετρος τῆς σφαίρας, <ἢ τις ἐστὶ καὶ τοῦ ἡμικυκλίου διάμετρος δηλαδὴ καὶ τοῦ κύκλου, μείζων ἐστὶ πασῶν τῶν εἰς τὸν κύκλον> ἢ τὴν σφαῖραν διαγομένων [εὐθειῶν]. ἔστω οὖν ἐν μὲν τῇ μείζονι σφαίρᾳ κύκλος ὁ ΒΓΔΕ, ἐν δὲ τῇ ἐλάσσονι σφαίρᾳ κύκλος ὁ ΖΗΘ, καὶ ἡθώσαν αὐτῶν δύο διαμέτροι πρὸς ὀρθὰς ἀλλήλαις αἱ ΒΔ, ΓΕ, καὶ δύο κύκλων περὶ τὸ αὐτὸ κέντρον ὄντων τῶν ΒΓΔΕ, ΖΗΘ εἰς τὸν μείζονα κύκλον τὸν ΒΓΔΕ πολύγωνον ἰσόπλευρον καὶ ἀρτιόπλευρον ἐγγεγράφθω μὴ ψαῖον τοῦ ἐλάσσονος κύκλου τοῦ ΖΗΘ, οὐ πλευραὶ ἔστωσαν ἐν τῷ ΒΕ τεταρτημορίῳ αἱ ΒΚ, ΚΛ, ΛΜ, ΜΕ, καὶ ἐπιευθεῖσα ἡ ΚΑ διήθθω ἐπὶ τὸ Ν, καὶ ἀνεστάτω ἀπὸ τοῦ Α σημείου τῷ τοῦ ΒΓΔΕ κύκλου ἐπιπέδῳ πρὸς ὀρθὰς ἡ ΑΧ καὶ συμβαλλέτω τῇ ἐπιφανείᾳ τῆς σφαίρας κατὰ τὸ Χ, καὶ διὰ τῆς ΑΧ καὶ ἑκατέρα τῶν ΒΔ, ΚΝ ἐπίπεδα ἐκβεβλήσθω; ποιήσουσι δὴ διὰ τὰ εἰρημένα ἐπὶ τῆς ἐπιφανείας τῆς σφαίρας μεγίστους κύκλους. ποιείτωσαν, ὧν ἡμικύκλια ἔστω ἐπὶ τῶν ΒΔ, ΚΝ διαμέτρων τὰ ΒΧΔ, ΚΧΝ. καὶ ἐπεὶ ἡ ΧΑ ὀρθὴ ἐστὶ πρὸς τὸ τοῦ ΒΓΔΕ κύκλου ἐπίπεδον, καὶ πάντα ἄρα

Let two spheres be conceived about the same center, A. So, it is necessary to inscribe a polyhedral solid in the greater sphere, not touching the lesser sphere on its surface.

Let the spheres be cut by some plane through the center. So, the sections will be circles, inasmuch as a sphere is generated by the diameter remaining behind, and a semi-circle being carried around [Def. 11.14]. And, hence, whatever position we conceive (of for) the semi-circle, the plane produced through it will make a circle on the surface of the sphere. And (it is) clear that (it is) also a great (circle), inasmuch as the diameter of the sphere, which is also manifestly the diameter of the semi-circle and the circle, is greater than all of the (other) [straight-lines] drawn across in the circle or the sphere [Prop. 3.15]. Therefore, let BCDE be the circle in the greater sphere, and FGH the circle in the lesser sphere. And let two diameters of them be drawn at right-angles to one another, (namely), BD and CE. And there being two circles about the same center—(namely), BCDE and FGH—let an equilateral and even-sided polygon be inscribed in the greater circle, BCDE, not touching the lesser circle, FGH [Prop. 12.16], of which let the sides in the quadrant BE be BK, KL, LM, and ME. And, KA being joined, let it be drawn across to N. And let AO be set up at point A, at right-angles to the plane of circle BCDE. And let it meet the surface of the (greater) sphere at O. And let planes be produced through AO and each of BD and KN. So, according to the aforementioned (discussion), they will make great circles on the surface of the (greater) sphere. Let them make (great circles), of which let BOD

τὰ διὰ τῆς ΧΑ ἐπίπεδά ἐστιν ὀρθὰ πρὸς τὸ τοῦ ΒΓΔΕ κύκλου ἐπίπεδον; <ὥστε καὶ τὰ ΒΧΔ, ΚΧΝ ἡμικύκλια ὀρθὰ ἐστὶ πρὸς τὸ τοῦ ΒΓΔΕ κύκλου ἐπίπεδον. καὶ ἐπεὶ >ῖσα ἐστὶ τὰ ΒΕΔ, ΒΧΔ, ΚΧΝ ἡμικύκλια; ἐπὶ γὰρ >ῖσαν εἰσὶ διαμέτρων τῶν ΒΔ, ΚΝ; >ῖσα ἐστὶ καὶ τὰ ΒΕ, ΒΧ, ΚΧ τεταρτημόρια ἀλλήλοισι. >όσαι >άρα εἰσὶν ἐν τῷ ΒΕ τεταρτημορίῳ πλευραὶ τοῦ πολυγώνου, τοσαῦται εἰσὶ καὶ ἐν τοῖς ΒΧ, ΚΧ τεταρτημορίοις >ῖσαι ταῖς ΒΚ, ΚΛ, ΛΜ, ΜΕ εὐθείαις. ἐγγεγράφωσαν καὶ >έστωσαν αἱ ΒΟ, ΟΠ, ΠΡ, ΡΧ, ΚΣ, ΣΤ, ΤΥ, ΥΧ, καὶ ἐπεζεύθωσαν αἱ ΣΟ, ΤΠ, ΥΡ, καὶ ἀπὸ τῶν Ο, Σ ἐπὶ τὸ τοῦ ΒΓΔΕ κύκλου ἐπίπεδον κάθετοι >ήθωσαν; πεσοῦνται δὲ ἐπὶ τὰς κοινὰς τομὰς τῶν ἐπιπέδων τὰς ΒΔ, ΚΝ, ἐπειδὴ περ καὶ τὰ τῶν ΒΧΔ, ΚΧΝ ἐπίπεδα ὀρθὰ ἐστὶ πρὸς τὸ τοῦ ΒΓΔΕ κύκλου ἐπίπεδον. πιπτέωσαν, καὶ >έστωσαν αἱ ΟΦ, ΣΘ, καὶ ἐπεζεύθω ἡ ΘΦ. καὶ ἐπεὶ ἐν >ῖσοις ἡμικυκλίοις τοῖς ΒΧΔ, ΚΧΝ >ῖσαι ἀπειλημμέναι εἰσὶν αἱ ΒΟ, ΚΣ, καὶ κάθετοι ἡγμέναι εἰσὶν αἱ ΟΦ, ΣΘ, >ῖση [>άρα] ἐστὶν ἡ μὲν ΟΦ τῇ ΣΘ, ἡ δὲ ΒΦ τῇ ΚΘ. >έστι δὲ καὶ <όλη ἡ ΒΑ <όλη τῇ ΚΑ >ῖση; καὶ λοιπὴ >άρα ἡ ΦΑ λοιπῇ τῇ ΘΑ ἐστὶν >ῖση; >έστιν >άρα ὥς ἡ ΒΦ πρὸς τὴν ΦΑ, ο<ύτως ἡ ΚΘ πρὸς τὴν ΘΑ; παράλληλος >άρα ἐστὶν ἡ ΘΦ τῇ ΚΒ. καὶ ἐπεὶ ἑκατέρω τῶν ΟΦ, ΣΘ ὀρθὴ ἐστὶ πρὸς τὸ τοῦ ΒΓΔΕ κύκλου ἐπίπεδον, παράλληλος >άρα ἐστὶν ἡ ΟΦ τῇ ΣΘ. ἐδείκθη δὲ αὐτῇ καὶ >ῖση; καὶ αἱ ΟΦ, ΣΟ >άρα >ῖσαι εἰσὶ καὶ παράλληλοι. καὶ ἐπεὶ παράλληλός ἐστιν ἡ ΘΦ τῇ ΣΟ, ἀλλὰ ἡ ΘΦ τῇ ΚΒ ἐστὶ παράλληλος, καὶ ἡ ΣΟ >άρα τῇ ΚΒ ἐστὶ παράλληλος. καὶ ἐπιζευγνύουσιν αὐτὰς αἱ ΒΟ, ΚΣ; τὸ ΚΒΟΣ >άρα τετράπλευρον ἐν ἐνὶ ἐστὶν ἐπίπεδῳ, ἐπειδὴ περ, ἔαν >ῶσι δύο εὐθεῖαι παράλληλοι, καὶ ἐφ> ἑκατέρω αὐτῶν ληφθῇ τυθόντα σημεῖα, ἡ ἐπὶ τὰ σημεῖα ἐπιζευγνυμένη εὐθεῖα ἐν τῷ αὐτῷ ἐπίπεδῳ ἐστὶ ταῖς παραλλήλοισι. διὰ τὰ αὐτὰ δὲ καὶ ἑκάτερον τῶν ΣΟΠΤ, ΤΠΡΥ τετραπλεύρων ἐν ἐνὶ ἐστὶν ἐπίπεδῳ. >έστι δὲ καὶ τὸ ΥΡΧ τρίγωνον ἐν ἐνὶ ἐπίπεδῳ. ἔαν δὲ νοήσωμεν ἀπὸ τῶν Ο, Σ, Π, Τ, Ρ, Υ σημείων ἐπὶ τὸ Α ἐπιζευγνυμένας εὐθείας, συσταθήσεται τι σῆμα στερεὸν πολυέδρον ματαχὺ τῶν ΒΧ, ΚΧ περιφερειῶν ἐκ πυραμίδων συγκείμενον, <ὢν βάσεις μὲν τὰ ΚΒΟΣ, ΣΟΠΤ, ΤΠΡΥ τετράπλευρα καὶ τὸ ΥΡΧ τρίγωνον, κορυφὴ δὲ τὸ Α σημεῖον. ἔαν δὲ καὶ ἐπὶ ἐκάστης τῶν ΚΛ, ΛΜ, ΜΕ πλευρῶν καθάπερ ἐπὶ τῆς ΒΚ τὰ αὐτὰ κατασκευάσωμεν καὶ >έτι τῶν λοιπῶν τριῶν τεταρτημορίων, συσταθήσεται τι σῆμα πολυέδρον ἐγγεγραμμένον εἰς τὴν σφαῖραν πυραμίσι περιεχόμενον, <ὢν βάσις [μὲν] τὰ εἰρημένα τετράπλευρα καὶ τὸ ΥΡΧ τρίγωνον καὶ τὰ ὁμοταγῇ αὐτοῖς, κορυφὴ δὲ τὸ Α σημεῖον.

Λέγω <ὅτι τὸ εἰρημένον πολυέδρον οὐκ ἐφάπεται τῆς ἐλάσσονος σφαίρας κατὰ τὴν ἐπιφάνειαν, ἐφ> <ῆς ἐστὶν ὁ ΖΗΘ κύκλος.

>Ἡθω ἀπὸ τοῦ Α σημείου ἐπὶ τὸ τοῦ ΚΒΟΣ τετραπλεύρου ἐπίπεδον κάθετος ἡ ΑΨ καὶ συμβαλλέτω τῷ ἐπίπεδῳ κατὰ τὸ Ψ σημεῖον, καὶ ἐπεζεύθωσαν αἱ ΨΒ, ΨΚ. καὶ ἐπεὶ ἡ ΑΨ ὀρθὴ ἐστὶ πρὸς τὸ τοῦ ΚΒΟΣ τετραπλεύρου ἐπίπεδον, καὶ πρὸς πάσας >άρα τὰς ἀπτομένας αὐτῇ εὐθείας καὶ ο>ύσας ἐν τῷ τοῦ τετραπλεύρου ἐπίπεδῳ ὀρθὴ ἐστὶν. ἡ ΑΨ >άρα ὀρθὴ ἐστὶ πρὸς ἑκατέραν τῶν ΒΨ, ΨΚ. καὶ ἐπεὶ >ῖση ἐστὶν ἡ ΑΒ τῇ ΑΚ, <ῖσον ἐστὶ καὶ τὸ ἀπὸ τῆς ΑΒ τῷ ἀπὸ τῆς ΑΚ. καὶ ἐστὶ τῷ μὲν ἀπὸ τῆς ΑΒ >ῖσα τὰ ἀπὸ τῶν ΑΨ, ΨΒ; ὀρθὴ γὰρ ἡ πρὸς τῷ Ψ; τῷ δὲ ἀπὸ τῆς ΑΚ >ῖσα τὰ ἀπὸ τῶν ΑΨ, ΨΚ. τὰ >άρα ἀπὸ τῶν ΑΨ, ΨΒ >ῖσα ἐστὶ τοῖς ἀπὸ τῶν ΑΨ, ΨΚ. κοινὸν ἀφηρήσθω τὸ ἀπὸ τῆς ΑΨ; λοιπὸν >άρα τὸ

and KON be semi-circles on the diameters BD and KN (respectively). And since OA is at right-angles to the plane of circle $BCDE$, all of the planes through OA are thus also at right-angles to the plane of circle $BCDE$ [Prop. 11.18]. And, hence, the semi-circles BOD and KON are also at right-angles to the plane of circle $BCDE$. And since semi-circles BED , BOD , and KON are equal—for (they are) on the equal diameters BD and KN [Def. 3.1]—the quadrants BE , BO , and KO are also equal to one another. Thus, as many sides of the polygon as are in quadrant BE , so many are also in quadrants BO and KO equal to the straight-lines BK , KL , LM , and ME . Let them be inscribed, and let them be BP , PQ , QR , RO , KS , ST , TU , and UO . And let SP , TQ , and UR be joined. And let perpendiculars be drawn from P and S to the plane of circle $BCDE$ [Prop. 11.11]. So, they will fall on the common sections of the planes BD and KN (with $BCDE$), inasmuch as the planes of BOD and KON are also at right-angles to the plane of circle $BCDE$ [Def. 11.4]. Let them have fallen, and let them be PV and SW . And let WV be joined. And since BP and KS are equal (circumferences) having been cut off in the equal semi-circles BOD and KON [Def. 3.28], and PV and SW are perpendiculars having been drawn (from them), PV is [thus] equal to SW , and BV to KW [Props. 3.27, 1.26]. And the whole of BA is also equal to the whole of KA . And, thus, as BV is to VA , so KW (is) to WA . WV is thus parallel to KB [Prop. 6.2]. And since PV and SW are each at right-angles to the plane of circle $BCDE$, PV is thus parallel to SW [Prop. 11.6]. And it was also shown (to be) equal to it. And, thus, WV and SP are equal and parallel [Prop. 1.33]. And since WV is parallel to SP , but WV is parallel to KB , SP is thus also parallel to KB [Prop. 11.1]. And BP and KS join them. Thus, the quadrilateral $KBPS$ is in one plane, inasmuch as if there are two parallel straight-lines, and a random point is taken on each of them, then the straight-line joining the points is in the same plane as the parallel (straight-lines) [Prop. 11.7]. So, for the same (reasons), each of the quadrilaterals $SPQT$ and $TQRU$ is also in one plane. And triangle URO is also in one plane [Prop. 11.2]. So, if we conceive straight-lines joining points P , S , Q , T , R , and U to A then some solid polyhedral figure will be constructed between the circumferences BO and KO , being composed of pyramids whose bases (are) the quadrilaterals $KBPS$, $SPQT$, $TQRU$, and the triangle URO , and apex the point A . And if we also make the same construction on each of the sides KL , LM , and ME , just as on BK , and, further, (repeat the construction) in the remaining three quadrants, then some polyhedral figure which has been inscribed in the sphere will be constructed, being contained by pyramids whose bases (are) the aforementioned quadrilaterals, and triangle URO , and the (quadrilaterals and triangles) similarly arranged to them, and apex the point A .

So, I say that the aforementioned polyhedron will not touch the lesser sphere on the surface on which the circle FGH is (situated).

Let the perpendicular (straight-line) AX be drawn from point A to the plane $KBPS$, and let it meet the plane at point X [Prop. 11.11]. And let XB and XK be joined. And since AX is at right-angles to the plane of quadrilateral $KBPS$, it is thus also at right-angles to all of the straight-

ἀπὸ τῆς ΒΨ λοιπῶ| τῶ| ἀπὸ τῆς ΨΚ >ίσον ἐστίν; >ίση >άρα ἡ ΒΨ τῇ ΨΚ. ὁμοίωξ δὴ δείχομεν, <ὅτι καὶ αἱ ἀπὸ τοῦ Ψ ἐπὶ τὰ Ο, Σ ἐπιζευγνύμεναι εὐθεῖαι >ίσαι εἰσὶν ἑκατέρᾳ τῶν ΒΨ, ΨΚ. ὁ >άρα κέντρον τῶ| Ψ καὶ διαστήματι ἐνὶ τῶν ΨΒ, ΨΚ γραφόμενος κύκλος <ήχει καὶ διὰ τῶν Ο, Σ, καὶ >έσται ἐν κύκλῳ τὸ ΚΒΟΣ τετράπλευρον.

Καὶ ἐπεὶ μείζων ἐστὶν ἡ ΚΒ τῇς ΘΦ, >ίση δὲ ἡ ΘΦ τῇ ΣΟ, μείζων >άρα ἡ ΚΒ τῇς ΣΟ. >ίση δὲ ἡ ΚΒ ἑκατέρᾳ τῶν ΚΣ, ΒΟ; καὶ ἑκατέρᾳ >άρα τῶν ΚΣ, ΒΟ τῇς ΣΟ μείζων ἐστίν. καὶ ἐπεὶ ἐν κύκλῳ τετράπλευρόν ἐστι τὸ ΚΒΟΣ, καὶ >ίσαι αἱ ΚΒ, ΒΟ, ΚΣ, καὶ ἐλάττων ἡ ΟΣ, καὶ ἐκ τοῦ κέντρου τοῦ κύκλου ἐστὶν ἡ ΒΨ, τὸ >άρα ἀπὸ τῆς ΚΒ τοῦ ἀπὸ τῆς ΒΨ μείζον ἐστὶν >ἢ διπλάσιον. >ήθθω ἀπὸ τοῦ Κ ἐπὶ τὴν ΒΦ κάθετοξ ἡ ΚΩ. καὶ ἐπεὶ ἡ ΒΔ τῇς ΔΩ ἐλάττων ἐστὶν >ἢ διπλῇ, καὶ ἐστὶν ὡς ἡ ΒΔ πρὸς τὴν ΔΩ, ο<ύτωξ τὸ ὑπὸ τῶν ΔΒ, ΒΩ πρὸς τὸ ὑπὸ [τῶν] ΔΩ, ΩΒ, ἀναγραφομένου ἀπὸ τῆς ΒΩ τετραγώνου καὶ συμπληρουμένου τοῦ ἐπὶ τῇς ΩΔ παραλληλογράμμου καὶ τὸ ὑπὸ ΔΒ, ΒΩ >άρα τοῦ ὑπὸ ΔΩ, ΩΒ >ελαττόν ἐστὶν >ἢ διπλάσιον. καὶ ἐστὶ τῇς ΚΔ ἐπιζευγνυμένηξ τὸ μὲν ὑπὸ ΔΒ, ΒΩ >ίσον τῶ| ἀπὸ τῆς ΒΚ, τὸ δὲ ὑπὸ τῶν ΔΩ, ΩΒ >ίσον τῶ| ἀπὸ τῆς ΚΩ; τὸ >άρα ἀπὸ τῆς ΚΒ τοῦ ἀπὸ τῆς ΚΩ >ελασσόν ἐστὶν >ἢ διπλάσιον. ἀλλὰ τὸ ἀπὸ τῆς ΚΒ τοῦ ἀπὸ τῆς ΒΨ μείζον ἐστὶν >ἢ διπλάσιον; μείζων >άρα τὸ ἀπὸ τῆς ΚΩ τοῦ ἀπὸ τῆς ΒΨ. καὶ ἐπεὶ >ίση ἐστὶν ἡ ΒΑ τῇ ΚΑ, >ίσον ἐστὶ τὸ ἀπὸ τῆς ΒΑ τῶ| ἀπὸ τῆς ΑΚ. καὶ ἐστὶ τῶ| μὲν ἀπὸ τῆς ΒΑ >ίσα τὰ ἀπὸ τῶν ΒΨ, ΨΑ, τῶ| δὲ ἀπὸ τῆς ΚΑ >ίσα τὰ ἀπὸ τῶν ΚΩ, ΩΑ; τὰ >άρα ἀπὸ τῶν ΒΨ, ΨΑ >ίσα ἐστὶ τοῖξ ἀπὸ τῶν ΚΩ, ΩΑ, <ὢν τὸ ἀπὸ τῆς ΚΩ μείζον τοῦ ἀπὸ τῆς ΒΨ; λοιπὸν >άρα τὸ ἀπὸ τῆς ΩΑ >ελασσόν ἐστὶ τοῦ ἀπὸ τῆς ΨΑ. μείζων >άρα ἡ ΑΨ τῇς ΑΩ; πολλῶ| >άρα ἡ ΑΨ μείζων ἐστὶ τῇς ΑΗ. καὶ ἐστὶν ἡ μὲν ΑΨ ἐπὶ μίαν τοῦ πολυέδρου βάσιν, ἡ δὲ ΑΗ ἐπὶ τὴν τῇς ἐλάσσονοξ σφαίραξ ἐπιφάνειαν; <ώστε τὸ πολυέδρον οὐ ψαύσει τῇς ἐλάσσονοξ σφαίραξ κατὰ τὴν ἐπιφάνειαν. Δύο >άρα σφαιρῶν περὶ τὸ αὐτὸ κέντρον οὐσῶν εἰξ τὴν μείζονα σφαῖραν στερεὸν πολυέδρον ἐγγέγραπται μὴ ψαῦον τῇς ἐλάσσονοξ σφαίραξ κατὰ τὴν ἐπιφάνειαν; <όπερ >έδει ποιῆσαι.

lines joined to it which are also in the plane of the quadrilateral [Def. 11.3]. Thus, AX is at right-angles to each of BX and XK . And since AB is equal to AK , the (square) on AB is also equal to the (square) on AK . And the (sum of the squares) on AX and XB is equal to the (square) on AB . For the angle at X (is) a right-angle [Prop. 1.47]. And the (sum of the squares) on AX and XK is equal to the (square) on AK [Prop. 1.47]. Thus, the (sum of the squares) on AX and XB is equal to the (sum of the squares) on AX and XK . Let the (square) on AX be subtracted from both. Thus, the remaining (square) on BX is equal to the remaining (square) on XK . Thus, BX (is) equal to XK . So, similarly, we can show that the straight-lines joined from X to P and S are equal to each of BX and XK . Thus, a circle drawn (in the plane of the quadrilateral) with center X , and radius one of XB or XK , will also pass through P and S , and the quadrilateral $KBPS$ will be inside the circle.

And since KB is greater than WV , and WV (is) equal to SP , KB (is) thus greater than SP . And KB (is) equal to each of KS and BP . Thus, KS and BP are each greater than SP . And since quadrilateral $KBPS$ is in a circle, and KB , BP , and KS are equal (to one another), and PS (is) less (than them), and BX is the radius of the circle, the (square) on KB is thus greater than double the (square) on BX .[†] Let the perpendicular KY be drawn from K to BV .[‡] And since BD is less than double DY , and as BD is to DY , so the (rectangle contained) by DB and BY (is) to the (rectangle contained) by DY and YB —a square being described on BY , and a (rectangular) parallelogram (with short side equal to BY) completed on YD —the (rectangle contained) by DB and BY is thus also less than double the (rectangle contained) by DY and YB . And, KD being joined, the (rectangle contained) by DB and BY is equal to the (square) on BK , and the (rectangle contained) by DY and YB equal to the (square) on KY [Props. 3.31, 6.8 corr.]. Thus, the (square) on KB is less than double the (square) on KY . But, the (square) on KB is greater than double the (square) on BX . Thus, the (square) on KY (is) greater than the (square) on BX . And since BA is equal to KA , the (square) on BA is equal to the (square) on KA . And the (sum of the squares) on BX and XA is equal to the (square) on BA , and the (sum of the squares) on KY and YA (is) equal to the (square) on KA [Prop. 1.47]. Thus, the (sum of the squares) on BX and XA is equal to the (sum of the squares) on KY and YA , of which the (square) on KY (is) greater than the (square) on BX . Thus, the remaining (square) on YA is less than the (square) on XA . Thus, AX (is) greater than AY . Thus, AX is much greater than AG .[§] And AX is (a perpendicular) on one of the bases of the polyhedron, and AG (is a perpendicular) on the surface of the lesser sphere. Hence, the polyhedron will not touch the lesser sphere on its surface.

Thus, there being two spheres about the same center, a polyhedral solid has been inscribed in the greater sphere which does not touch the lesser sphere on its surface. (Which is) the very thing it was required to do.

[†] Since KB , BP , and KS are greater than the sides of an inscribed square, which are each of length $\sqrt{2} BX$.

[‡] Note that points Y and V are actually identical.

[§] This conclusion depends on the fact that the chord of the polygon in proposition 12.16 does not touch the inner circle.

Πόρισμα.

Ἐὰν δὲ καὶ εἰς ἑτέραν σφαῖραν τῷ ἐν τῇ ΒΓΔΕ σφαίρᾳ στερεῷ πολυέδρῳ ὁμοιον στερεὸν πολυέδρον ἐγγραφῇ, τὸ ἐν τῇ ΒΓΔΕ σφαίρᾳ στερεὸν πολυέδρον πρὸς τὸ ἐν τῇ ἑτέρᾳ σφαίρᾳ στερεὸν πολυέδρον τριπλασίονα λόγον >έθει, >ἢπερ ἡ τῆς ΒΓΔΕ σφαίραξ διάμετρος πρὸς τὴν τῆς ἑτέρας σφαίραξ διάμετρον. διαιρεθέντων γὰρ τῶν στερεῶν εἰς τὰς ὁμοιοπληθεῖς καὶ ὁμοιοταγεῖς πυραμίδαξ >έσονται αἱ πυραμίδες ὁμοιοι. αἱ δὲ ὁμοιοι πυραμίδες πρὸς ἀλλήλας ἐν τριπλασίονι λόγῳ εἰσὶ τῶν ὁμολόγων πλευρῶν; ἢ >άρα πυραμῖξ, <ἥξ βάσις μὲν ἐστὶ τὸ ΚΒΟΣ τετράπλευρον, κορυφή δὲ τὸ Α σημείον, πρὸς τὴν ἐν τῇ ἑτέρᾳ σφαίρᾳ ὁμοιοταγῇ πυραμίδα τριπλασίονα λόγον >έθει, >ἢπερ ἡ ὁμολογὸς πλευρὰ πρὸς τὴν ὁμολογὸν πλευράν, τουτέστιν >ἢπερ ἡ ΑΒ ἐκ τοῦ κέντρου τῆς σφαίραξ τῆς περὶ κέντρον τὸ Α πρὸς τὴν ἐκ τοῦ κέντρου τῆς ἑτέρας σφαίραξ. ὁμοίως καὶ ἐκάστη πυραμῖξ τῶν ἐν τῇ περὶ κέντρον τὸ Α σφαίρᾳ πρὸς ἐκάστην ὁμοιοταγῇ πυραμίδα τῶν ἐν τῇ ἑτέρᾳ σφαίρᾳ τριπλασίονα λόγον <έχει, >ἢπερ ἡ ΑΒ πρὸς τὴν ἐκ τοῦ κέντρου τῆς ἑτέρας σφαίραξ. καὶ ὥς <ἐν τῶν ἡγουμένων πρὸς <ἐν τῶν ἐπομένων, οὕτως <άπαντα τὰ ἡγούμενα πρὸς <άπαντα τὰ ἐπόμενα; <ώστε <όλον τὸ ἐν τῇ περὶ κέντρον τὸ Α σφαίρᾳ στερεὸν πολυέδρον πρὸς <όλον τὸ ἐν τῇ ἑτέρᾳ [σφαίρᾳ] στερεὸν πολυέδρον τριπλασίονα λόγον <έχει, >ἢπερ ἡ ΑΒ πρὸς τὴν ἐκ τοῦ κέντρου τῆς ἑτέρας σφαίραξ, τουτέστιν >ἢπερ ἡ ΒΔ διάμετρος πρὸς τὴν τῆς ἑτέρας σφαίραξ διάμετρον; <όπερ >έδει δεῖχαι.

18.

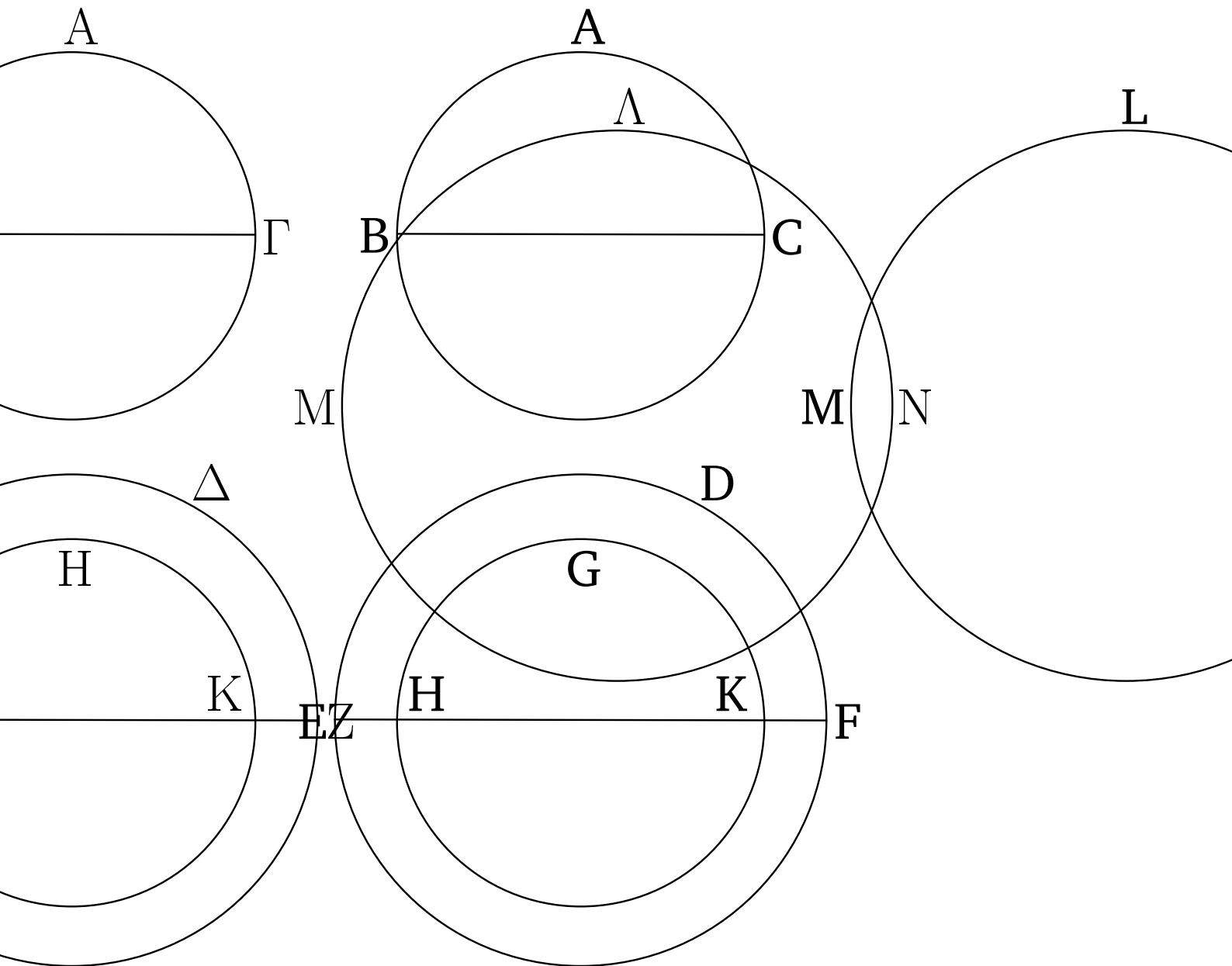
Αἱ σφαῖραι πρὸς ἀλλήλας ἐν τριπλασίονι λόγῳ εἰσὶ τῶν ἰδίων διαμέτρων.

Corollary

And, also, if a similar polyhedral solid to that in sphere $BCDE$ is inscribed in another sphere then the polyhedral solid in sphere $BCDE$ has to the polyhedral solid in the other sphere the cubed ratio that the diameter of sphere $BCDE$ has to the diameter of the other sphere. For if the solids are divided into similarly numbered, and similarly situated, pyramids, then the pyramids will be similar. And similar pyramids are in the cubed ratio of corresponding sides [Prop. 12.8 corr.]. Thus, the pyramid whose base is quadrilateral $KBPS$, and apex the point A , will have to the similarly situated pyramid in the other sphere the cubed ratio that a corresponding side (has) to a corresponding side. That is to say, that of radius AB of the sphere about center A to the radius of the other sphere. And, similarly, each pyramid in the sphere about center A will have to each similarly situated pyramid in the other sphere the cubed ratio that AB (has) to the radius of the other sphere. And as one of the leading (magnitudes is) to one of the following (in two sets of proportional magnitudes), so (the sum of) all the leading (magnitudes is) to (the sum of) all of the following (magnitudes) [Prop. 5.12]. Hence, the whole polyhedral solid in the sphere about center A will have to the whole polyhedral solid in the other [sphere] the cubed ratio that (radius) AB (has) to the radius of the other sphere. That is to say, that diameter BD (has) to the diameter of the other sphere. (Which is) the very thing it was required to show.

Proposition 18

Spheres are to one another in the cubed ratio of their respective diameters.



Νενοήσθωσαν σφαῖραι αἱ $AB\Gamma$, ΔEZ , διάμετροι δὲ αὐτῶν αἱ $B\Gamma$, EZ ; λέγω, <ὅτι ἡ $AB\Gamma$ σφαῖρα πρὸς τὴν ΔEZ σφαῖραν τριπλασίονα λόγον >έθει >ήπερ ἡ $B\Gamma$ πρὸς τὴν EZ .

Εἰ γὰρ μὴ ἡ $AB\Gamma$ σφαῖρα πρὸς τὴν ΔEZ σφαῖραν τριπλασίονα λόγον >έθει >ήπερ ἡ $B\Gamma$ πρὸς τὴν EZ , <έχει >άρα ἡ $AB\Gamma$ σφαῖρα πρὸς ἐλάσσονά τινα τῆς ΔEZ σφαίρας τριπλασίονα λόγον >ἡ πρὸς μείζονα >ήπερ ἡ $B\Gamma$ πρὸς τὴν EZ . ἐθέτω πρότερον πρὸς ἐλάσσονα τὴν $H\Theta K$, καὶ νενοήσθω ἡ ΔEZ τῇ $H\Theta K$ περὶ τὸ αὐτὸ κέντρον, καὶ ἐγγεγράφθω εἰς τὴν μείζονα σφαῖραν τὴν ΔEZ στερεὸν πολυέδρον μὴ ψαῦον τῆς ἐλάσσονος σφαίρας τῆς $H\Theta K$ κατὰ τὴν ἐπιφάνειαν, ἐγγεγράφθω δὲ καὶ εἰς τὴν $AB\Gamma$ σφαῖραν τῷ ἐν τῇ ΔEZ σφαίρᾳ στερεῷ πολυέδρῳ <όμοιον στερεὸν πολυέδρον; τὸ >άρα ἐν τῇ $AB\Gamma$ στερεὸν πολυέδρον πρὸς τὸ ἐν τῇ ΔEZ στερεὸν πολυέδρον τριπλασίονα λόγον >έθει >ήπερ ἡ $B\Gamma$ πρὸς τὴν EZ . >έθει δὲ καὶ ἡ $AB\Gamma$ σφαῖρα πρὸς τὴν $H\Theta K$ σφαῖραν τριπλασίονα λόγον >ήπερ ἡ $B\Gamma$ πρὸς τὴν EZ ; >έστιν >άρα ὥς ἡ $AB\Gamma$ σφαῖρα πρὸς τὴν $H\Theta K$ σφαῖραν, ο<ύτωξ τὸ ἐν τῇ $AB\Gamma$ σφαίρᾳ στερεὸν πολυέδρον πρὸς τὸ ἐν τῇ ΔEZ σφαίρᾳ στερεὸν πολυέδρον; ἐναλλάχ[>άρα] ὥς ἡ $AB\Gamma$ σφαῖρα πρὸς τὸ ἐν αὐτῇ πολυέδρον, ο<ύτωξ ἡ $H\Theta K$ σφαῖρα πρὸς τὸ ἐν τῇ ΔEZ σφαίρᾳ στερεὸν πολυέδρον.

Let the spheres ABC and DEF be conceived, and (let) their diameters (be) BC and EF (respectively). I say that sphere ABC has to sphere DEF the cubed ratio that BC (has) to EF .

For if sphere ABC does not have to sphere DEF the cubed ratio that BC (has) to EF then sphere ABC will have to some (sphere) either less than, or greater than, sphere DEF the cubed ratio that BC (has) to EF . Let it, first of all, have (such a ratio) to a lesser (sphere), GHK . And let DEF be conceived about the same center as GHK . And let a polyhedral solid be inscribed in the greater sphere DEF , not touching the lesser sphere GHK on its surface [Prop. 12.17]. And let a polyhedral solid, similar to the polyhedral solid in sphere DEF , have also been inscribed in sphere ABC . Thus, the polyhedral solid in sphere ABC has to the polyhedral solid in sphere DEF the cubed ratio that BC (has) to EF [Prop. 12.17 corr.]. And sphere ABC also has to sphere GHK the cubed ratio that BC (has) to EF . Thus, as sphere ABC is to sphere GHK , so the polyhedral solid in sphere ABC (is) to the polyhedral solid in sphere DEF . [Thus], alternately, as sphere ABC (is) to the polygon within it, so sphere GHK (is) to the polyhedral solid

μείζων δὲ ἡ $AB\Gamma$ σφαῖρα τοῦ ἐν αὐτῇ πολυέδρου; μείζων >άρα καὶ ἡ $H\Theta K$ σφαῖρα τοῦ ἐν τῇ ΔEZ σφαίρα πολυέδρου. ἀλλὰ καὶ ἐλάττω; ἐμπεριέθεται γὰρ ὑπ' αὐτοῦ. οὐκ >άρα ἡ $AB\Gamma$ σφαῖρα πρὸς ἐλάσσονα τῇ ΔEZ σφαίρα τριπλασίονα λόγον >έθει >ήπερ ἡ $B\Gamma$ διάμετρος πρὸς τὴν EZ . ὁμοίως δὲ δείχομεν, <ὅτι οὐδὲ ἡ ΔEZ σφαῖρα πρὸς ἐλάσσονα τῇ $AB\Gamma$ σφαίρα τριπλασίονα λόγον >έθει >ήπερ ἡ EZ πρὸς τὴν $B\Gamma$. Λέγω δὴ, <ὅτι οὐδὲ ἡ $AB\Gamma$ σφαῖρα πρὸς μείζονά τινα τῇ ΔEZ σφαίρα τριπλασίονα λόγον >έθει >ήπερ ἡ $B\Gamma$ πρὸς τὴν EZ .

Εἰ γὰρ δυνατόν, ἐθέτω πρὸς μείζονα τὴν ΛMN ; ἀνάπαλιν >άρα ἡ ΛMN σφαῖρα πρὸς τὴν $AB\Gamma$ σφαῖραν τριπλασίονα λόγον >έθει >ήπερ ἡ EZ διάμετρος πρὸς τὴν $B\Gamma$ διάμετρον. ὥς δὲ ἡ ΛMN σφαῖρα πρὸς τὴν $AB\Gamma$ σφαῖραν, ο<ύτωξ ἡ ΔEZ σφαῖρα πρὸς ἐλάσσονά τινα τῇ $AB\Gamma$ σφαίρα, ἐπειδήπερ μείζων ἐστὶν ἡ ΛMN τῇ ΔEZ , ὥς >έμπροσθεν ἐδείκθη. καὶ ἡ ΔEZ >άρα σφαῖρα πρὸς ἐλάσσονά τινα τῇ $AB\Gamma$ σφαίρα τριπλασίονα λόγον >έθει >ήπερ ἡ EZ πρὸς τὴν $B\Gamma$; <ὅπερ ἀδύνατον ἐδείκθη. οὐκ >άρα ἡ $AB\Gamma$ σφαῖρα πρὸς μείζονά τινα τῇ ΔEZ σφαίρα τριπλασίονα λόγον >έθει >ήπερ ἡ $B\Gamma$ πρὸς τὴν EZ . ἐδείκθη δέ, <ὅτι οὐδὲ πρὸς ἐλάσσονα. ἡ >άρα $AB\Gamma$ σφαῖρα πρὸς τὴν ΔEZ σφαῖραν τριπλασίονα λόγον >έθει >ήπερ ἡ $B\Gamma$ πρὸς τὴν EZ ; <ὅπερ >έδει δεῖχαι.

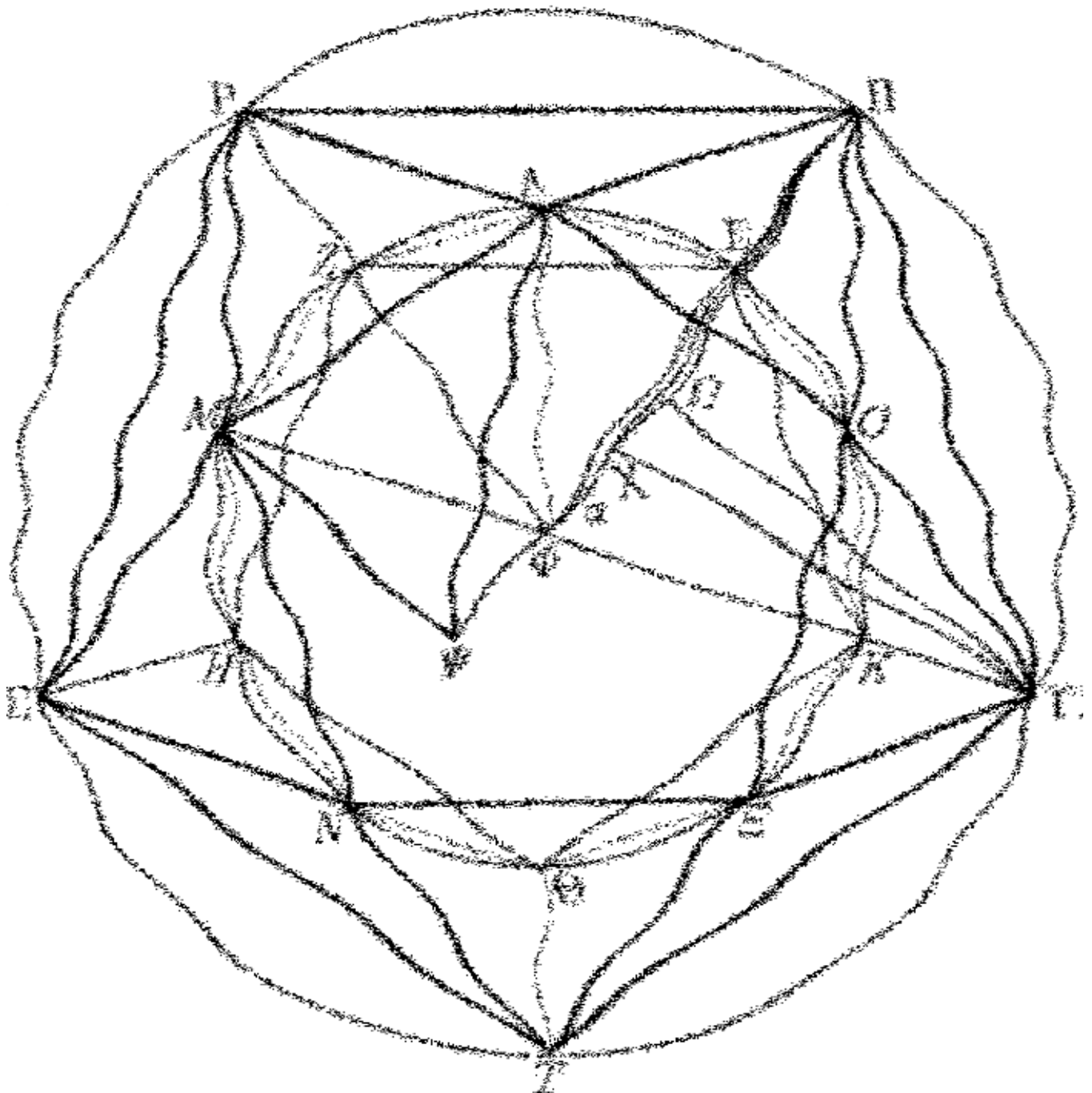
within sphere DEF [Prop. 5.16]. And sphere ABC (is) greater than the polyhedron within it. Thus, sphere GHK (is) also greater than the polyhedron within sphere DEF [Prop. 5.14]. But, (it is) also less. For it is encompassed by it. Thus, sphere ABC does not have to (a sphere) less than sphere DEF the cubed ratio that diameter BC (has) to EF . So, similarly, we can show that sphere DEF does not have to (a sphere) less than sphere ABC the cubed ratio that EF (has) to BC either.

So, I say that sphere ABC does not have to some (sphere) greater than sphere DEF the cubed ratio that BC (has) to EF either.

For, if possible, let it have (the cubed ratio) to a greater (sphere), LMN . Thus, inversely, sphere LMN (has) to sphere ABC the cubed ratio that diameter EF (has) to diameter BC [Prop. 5.7 corr.]. And as sphere LMN (is) to sphere ABC , so sphere DEF (is) to some (sphere) less than sphere ABC , inasmuch as LMN is greater than DEF , as was shown before [Prop. 12.2 lem.]. And, thus, sphere DEF has to some (sphere) less than sphere ABC the cubed ratio that EF (has) to BC . The very thing was shown (to be) impossible. Thus, sphere ABC does not have to some (sphere) greater than sphere DEF the cubed ratio that BC (has) to EF . And it was shown that neither (does it have such a ratio) to a lesser (sphere). Thus, sphere ABC has to sphere DEF the cubed ratio that BC (has) to EF . (Which is) the very thing it was required to show.

ELEMENTS BOOK 13

The Platonic Solids[†]

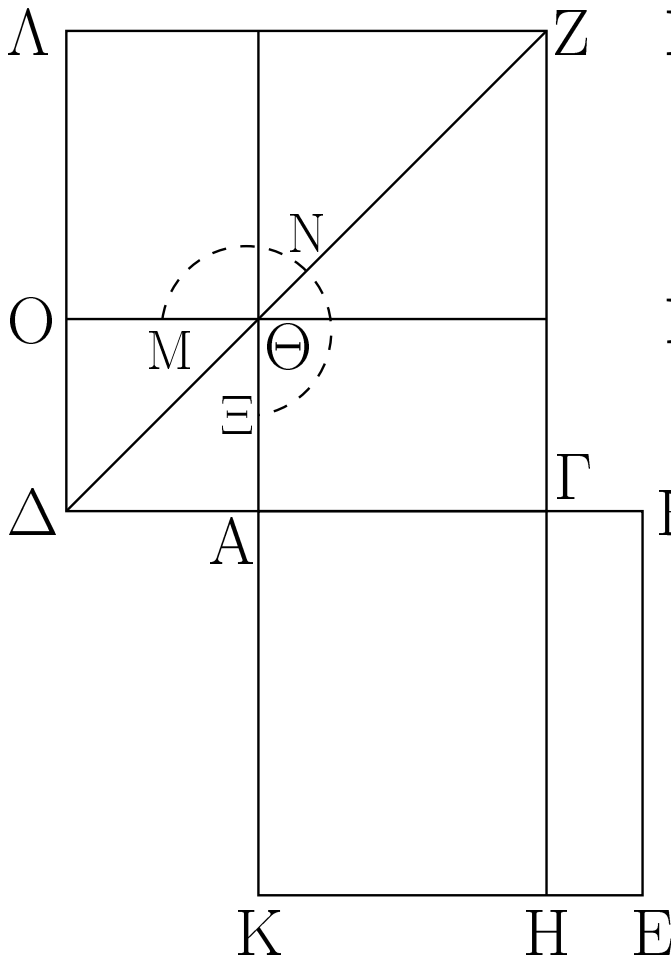


[†]The five regular solids—the cube, tetrahedron (i.e., pyramid), octahedron, icosahedron, and dodecahedron—were probably discovered by the school of Pythagoras. They are generally termed “Platonic” solids because they feature prominently in Plato’s famous dialogue *Timaeus*. Many of the theorems contained in this book—particularly those which pertain to the last two solids—are ascribed to Theaetetus of Athens.

1.

Proposition 1

Ἐάν εὐθεῖα γραμμὴ >ἄκρον καὶ μέσον λόγον τμηθῇ, τὸ μείζον τμήμα προσλαβὼν τὴν ἡμίσειαν τῆς <όλης πενταπλάσιον square on the greater piece, added to half of the whole, is δύναται τοῦ ἀπὸ τῆς ἡμισείας τετραγώνου. five times the square on the half.



Εὐθεῖα γὰρ γραμμὴ ἡ AB >ἄκρον καὶ μέσον λόγον τετμήσθω κατὰ τὸ Γ σημεῖον, καὶ >έστω μείζον τμήμα τὸ AG , καὶ ἐκβεβλήσθω ἐπ' εὐθείᾳ τῇ GA εὐθεῖα ἡ AD , καὶ κείσθω τῆς AB ἡμίσεια ἡ AD ; λέγω, <ὅτι πενταπλάσιόν ἐστι τὸ ἀπὸ τῆς GA τοῦ ἀπὸ τῆς DA .

Ἀναγεγράφθωσαν γὰρ ἀπὸ τῶν AB , AD τετράγωνα τὰ AE , AD , καὶ καταγεγράφθω ἐν τῷ AD τὸ σθῆμα, καὶ διήθθω ἡ ZG ἐπὶ τὸ H . καὶ ἐπεὶ ἡ AB >ἄκρον καὶ μέσον λόγον τέτμηται κατὰ τὸ Γ , τὸ >ἄρα ὑπὸ τῶν ABG >ἴσον ἐστὶ τῷ ἀπὸ τῆς AG . καὶ ἐστὶ τὸ μὲν ὑπὸ τῶν ABG τὸ GE , τὸ δὲ ἀπὸ τῆς AG τὸ $Z\Theta$; >ἴσον >ἄρα τὸ GE τῷ $Z\Theta$. καὶ ἐπεὶ διπλῆ ἐστὶν ἡ BA τῆς AD , >ἴση δὲ ἡ μὲν BA τῇ KA , ἡ δὲ AD τῇ $A\Theta$, διπλῇ >ἄρα καὶ ἡ KA τῆς $A\Theta$. ὥς δὲ ἡ KA πρὸς τὴν $A\Theta$, ο<ὕτως τὸ $ΓΚ$ πρὸς τὸ $\Gamma\Theta$; διπλάσιον >ἄρα τὸ $ΓΚ$ τοῦ $\Gamma\Theta$. εἰσι δὲ καὶ τὰ $\Lambda\Theta$, $\Theta\Gamma$ διπλάσια τοῦ $\Gamma\Theta$. >ἴσον >ἄρα τὸ $ΚΓ$ τοῖς $\Lambda\Theta$, $\Theta\Gamma$. ἐδείκθη δὲ καὶ τὸ GE τῷ ΘZ >ἴσον; <όλον >ἄρα τὸ AE τετράγωνον >ἴσον ἐστὶ τῷ MNX γνόμωνι. καὶ ἐπεὶ διπλῆ ἐστὶν ἡ BA τῆς AD , τετραπλάσιόν ἐστι τὸ ἀπὸ τῆς BA τοῦ ἀπὸ τῆς AD , τουτέστι τὸ AE τοῦ $\Delta\Theta$. >ἴσον δὲ τὸ AE τῷ MNX γνόμωνι; καὶ ὁ MNX >ἄρα γνόμων τετραπλάσιός ἐστι τοῦ AO ; <όλον >ἄρα τὸ ΔZ πενταπλάσιόν ἐστι τοῦ AO . καὶ ἐστὶ τὸ μὲν ΔZ τὸ ἀπὸ τῆς $\Delta\Gamma$, τὸ δὲ AO τὸ ἀπὸ τῆς DA ; τὸ >ἄρα ἀπὸ τῆς GA πενταπλάσιόν ἐστι τοῦ ἀπὸ τῆς DA .

Ἐάν >ἄρα εὐθεῖα >ἄκρον καὶ μέσον λόγον τμηθῇ, τὸ μείζον τμήμα προσλαβὼν τὴν ἡμίσειαν τῆς <όλης πενταπλάσιον

For let the straight-line AB be cut in extreme and mean ratio at point C , and let AC be the greater piece. And let the straight-line AD be produced in a straight-line with CA . And let AD be made (equal to) half of AB . I say that the (square) on CD is five times the (square) on DA .

For let the squares AE and DF be described on AB and DC (respectively). And let the figure in DF be drawn. And let FC be drawn across to G . And since AB has been cut in extreme and mean ratio at C , the (rectangle contained) by ABC is thus equal to the (square) on

δύναται τοῦ ἀπὸ τῆς ἡμισείας τετραγώνου; <ὅπερ >ἔδει δεῖξαι.

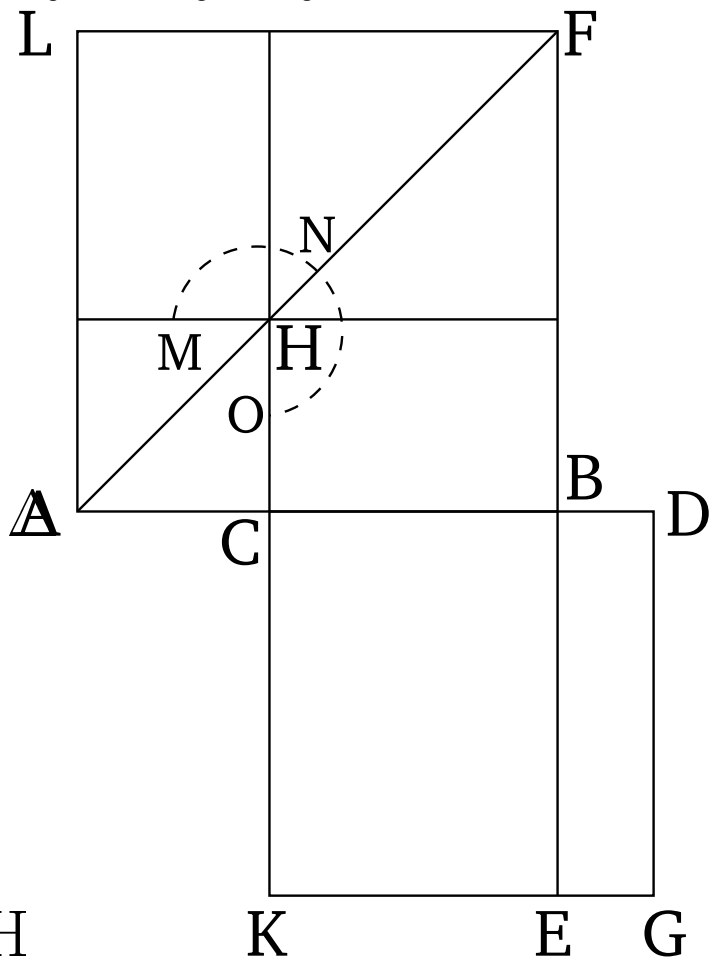
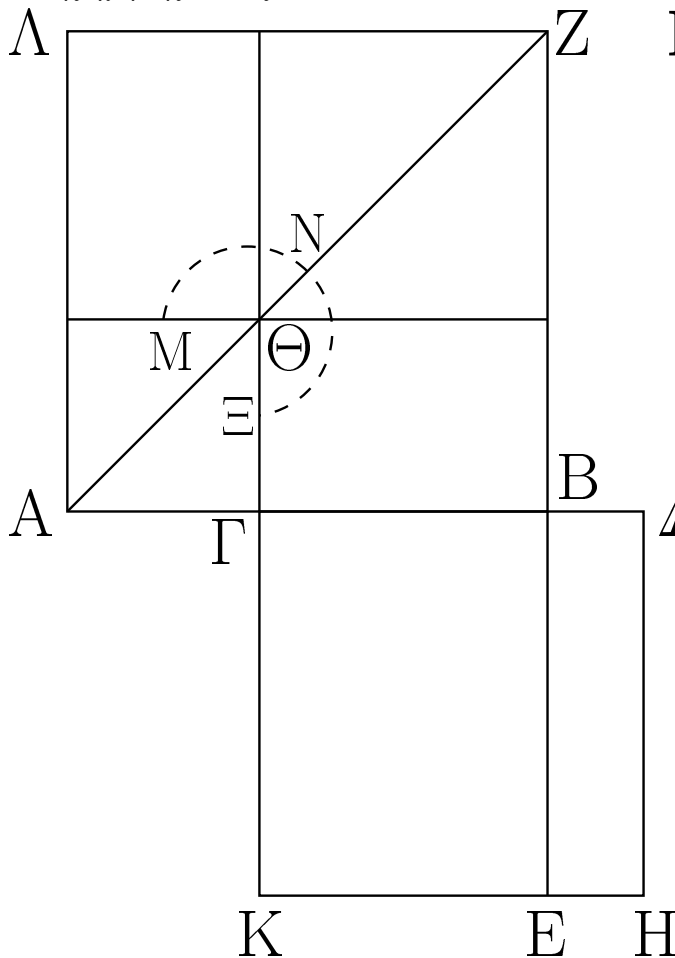
Thus, if a straight-line is cut in extreme and mean ratio then the square on the greater piece, added to half of the whole, is five times the square on the half. (Which is) the very thing it was required to show.

2.

Ἐὰν εὐθεῖα γραμμὴ τμήματοξ ἑαυτῆς πενταπλάσιον δύνῃται, τῆς διπλασίας τοῦ εἰρημένου τμήματοξ >ἄκρον καὶ μέσον λόγον τεμνομένης τὸ μείζον τμήμα τὸ λοιπὸν μέρος ἐστὶ τῆς ἐκ ἄρθῆς εὐθείας.

Proposition 2

If the square on a straight-line is five times the (square) on a piece of it, and double the aforementioned piece is cut in extreme and mean ratio, then the greater piece is the remaining part of the original straight-line.



Εὐθεῖα γὰρ γραμμὴ ἡ ΑΒ τμήματοξ ἑαυτῆς τοῦ ΑΓ πενταπλάσιον δυνάσθω, τῆς δὲ ΑΓ διπλῆ >έστω ἡ ΓΔ. λέγω, <ὅτι τῆς ΓΔ >ἄκρον καὶ μέσον λόγον τεμνομένης τὸ μείζον τμήμα ἐστὶν ἡ ΓΒ.

Ἀναγεγράφθω γὰρ ἄφ' ἑκατέρᾳ τῶν ΑΒ, ΓΔ τετράγωνα τὰ ΑΖ, ΓΗ, καὶ καταγεγράφθω ἐν τῷ ΑΖ τὸ σθῆμα, καὶ διήθθω ἡ ΒΕ. καὶ ἐπεὶ πενταπλάσιόν ἐστι τὸ ἀπὸ τῆς ΒΑ τοῦ ἀπὸ τῆς ΑΓ, πενταπλάσιόν ἐστι τὸ ΑΖ τοῦ ΑΘ. τετραπλάσιοξ >ἄρα ὁ ΜΝΧ γνῶμων τοῦ ΑΘ. καὶ ἐπεὶ διπλῆ ἐστὶν ἡ ΔΓ τῆς ΓΑ, τετραπλάσιον >ἄρα ἐστὶ τὸ ἀπὸ ΔΓ τοῦ ἀπὸ ΓΑ, τουτέστι τὸ ΓΗ τοῦ ΑΘ. ἐδείθθη δὲ καὶ ὁ ΜΝΧ γνῶμων τετραπλάσιοξ τοῦ ΑΘ; >ἴσοξ >ἄρα ὁ ΜΝΧ γνῶμων τῷ ΓΗ. καὶ ἐπεὶ διπλῆ ἐστὶν ἡ ΔΓ τῆς ΓΑ, >ἴση δὲ ἡ μὲν ΔΓ τῇ ΓΚ, ἡ δὲ ΑΓ τῇ ΓΘ, [διπλῆ >ἄρα καὶ ἡ ΚΓ τῆς ΓΘ], διπλάσιον >ἄρα καὶ τὸ ΚΒ τοῦ ΒΘ. εἰσὶ δὲ καὶ τὰ ΛΘ, ΘΒ τοῦ ΘΒ διπλάσια; >ἴσον >ἄρα τὸ ΚΒ τοῖς ΛΘ, ΘΒ. ἐδείθθη δὲ καὶ <όλοξ ὁ ΜΝΧ γνῶμων <όλω τῷ ΓΗ >ἴσοξ; καὶ λοιπὸν >ἄρα τὸ ΘΖ τῷ ΒΗ ἐστὶν >ἴσον. καὶ ἐστὶ τὸ μὲν ΒΗ τὸ ὑπὸ τῶν ΓΔΒ; >ἴση γὰρ ἡ ΓΔ τῇ ΔΗ; τὸ δὲ ΘΖ τὸ ἀπὸ τῆς ΓΒ; τὸ >ἄρα ὑπὸ τῶν

For let the square on the straight-line AB be five times the (square) on the piece of it, AC . And let CD be double AC . I say that if CD is cut in extreme and mean ratio then the greater piece is CB .

For let the squares AF and CG be described on each of AB and CD (respectively). And let the figure in AF be drawn. And let BE be drawn across. And since the (square) on BA is five times the (square) on AC , AF is five times AH . Thus, gnomon MNO (is) four times AH . And since DC is double CA , the (square) on DC is thus four times the (square) on CA —that is to say, CG (is four times) AH . And the gnomon MNO was also shown (to be) four times AH . Thus, gnomon MNO (is) equal to CG . And since DC is double CA , and DC (is) equal to CK , and AC to CH , [KC (is) thus also double CH], (and) KB (is) also double BH [Prop. 6.1]. And LH plus HB is also double HB [Prop. 1.43]. Thus, KB (is) equal to LH plus HB . And the whole gnomon MNO was also shown (to be) equal to the whole of CG . Thus, the remainder HF is

ΓΔΒ >ἴσον ἐστὶ τῷ ἀπὸ τῆς ΓΒ. >ἐστὶν >άρα ὡς ἡ ΔΓ πρὸς τὴν ΓΒ, ο<ύτωξ ἡ ΓΒ πρὸς τὴν ΒΔ. μείζων δὲ ἡ ΔΓ τῆς ΓΒ; μείζων >άρα καὶ ἡ ΓΒ τῆς ΒΔ. τῆς ΓΔ >άρα εὐθείαξ >άκρον καὶ μέσον λόγον τεμνομένης τὸ μείζον τμήμα ἐστὶν ἡ ΓΒ. Ἐὰν >άρα εὐθεῖα γραμμὴ τμήματοξ ἐαυτῆς πενταπλάσιον δύνῃται, τῆς διπλασίας τοῦ εἰρημένου τμήματοξ >άκρον καὶ μέσον λόγον τεμνομένης τὸ μείζον τμήμα τὸ λοιπὸν μέρος ἐστὶ τῆς ἐξ ἄρθῆς εὐθείας; <όπερ >έδει δεῖχαι.

Λήμμα.

<Ὅτι δὲ ἡ διπλὴ τῆς ΑΓ μείζων ἐστὶ τῆς ΒΓ, ο<ύτωξ δεικτέον.

Εἰ γὰρ μή, >έστω, εἰ δυνατόν, ἡ ΒΓ διπλὴ τῆς ΓΑ. τετραπλάσιον >άρα τὸ ἀπὸ τῆς ΒΓ τοῦ ἀπὸ τῆς ΓΑ; πενταπλάσια >άρα τὰ ἀπὸ τῶν ΒΓ, ΓΑ τοῦ ἀπὸ τῆς ΓΑ. ὑπόκειται δὲ καὶ τὸ ἀπὸ τῆς ΒΑ πενταπλάσιον τοῦ ἀπὸ τῆς ΓΑ; τὸ >άρα ἀπὸ τῆς ΒΑ >ἴσον ἐστὶ τοῖξ ἀπὸ τῶν ΒΓ, ΓΑ; <όπερ ἀδύνατον. οὐκ >άρα ἡ ΓΒ διπλασία ἐστὶ τῆς ΑΓ. ὁμοίωξ δὲ δείχομεν, <ότι οὐδὲ ἡ ἐλάττων τῆς ΓΒ διπλασίον ἐστὶ τῆς ΓΑ; πολλῶ γὰρ [μείζον] τὸ >άτοπον.

Ἡ >άρα τῆς ΑΓ διπλὴ μείζων ἐστὶ τῆς ΒΓ; <όπερ >έδει δεῖχαι.

3.

Ἐὰν εὐθεῖα γραμμὴ >άκρον καὶ μέσον λόγον τμηθῇ, τὸ >έλασσον τμήμα προσλαβὼν τὴν ἡμίσειαν τοῦ μείζονος τμήματοξ πενταπλάσιον δύνῃται τοῦ ἀπὸ τῆς ἡμισείας τοῦ μείζονος τμήματοξ τετραγώνου.

also equal to (the remainder) BG . And BG is the (rectangle contained) by CDB . For CD (is) equal to DG . And HF (is) the square on CB . Thus, the (rectangle contained) by CDB is equal to the (square) on CB . Thus, as DC is to CB , so CB (is) to BD [Prop. 6.17]. And DC (is) greater than CB (see lemma). Thus, CB (is) also greater than BD [Prop. 5.14]. Thus, if the straight-line CD is cut in extreme and mean ratio then the greater piece is CB .

Thus, if the square on a straight-line is five times the (square) on a piece of itself, and double the aforementioned piece is cut in extreme and mean ratio, then the greater piece is the remaining part of the original straight-line. (Which is) the very thing it was required to show.

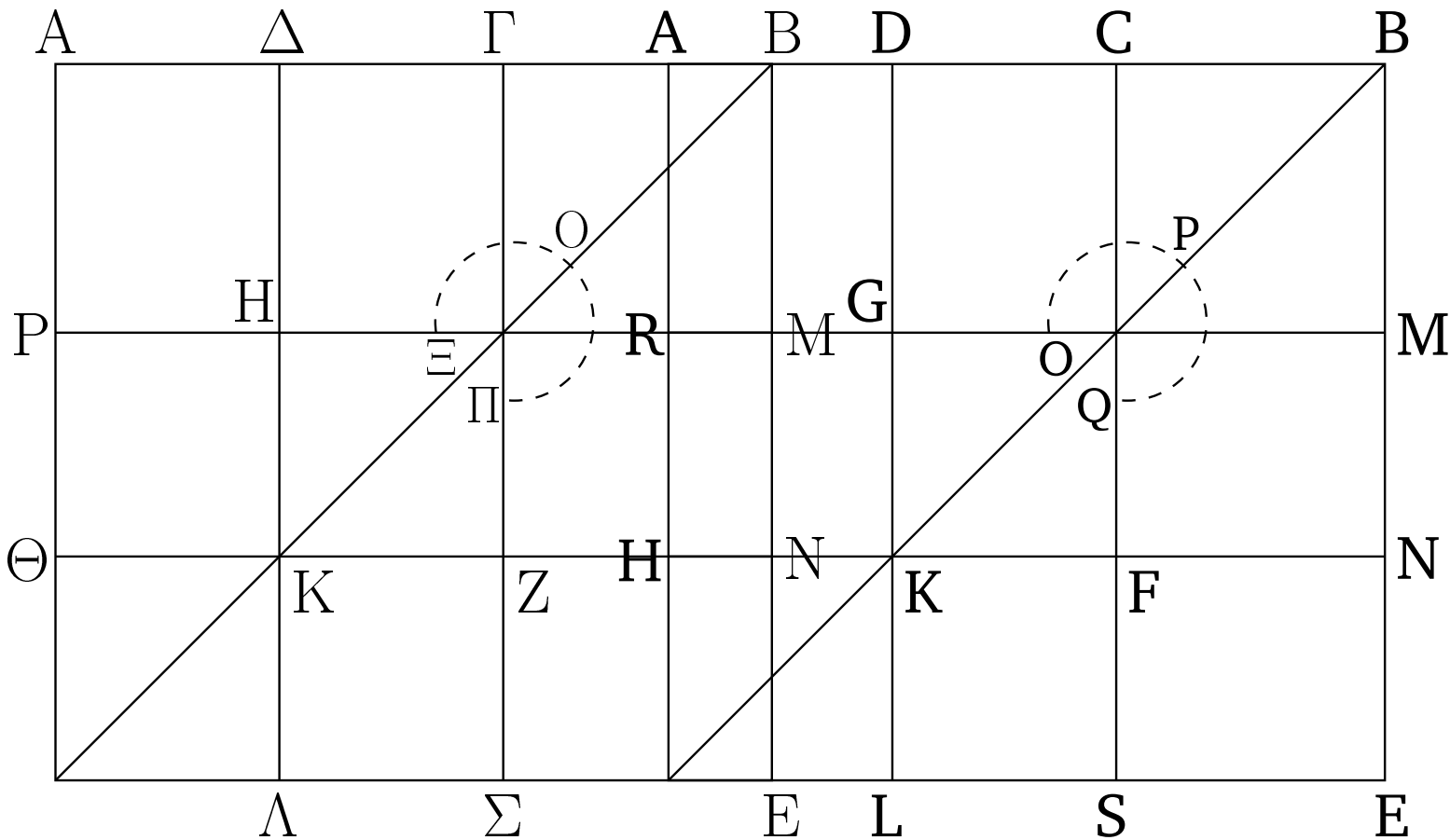
Lemma

And it can be shown that double AC (i.e., DC) is greater than BC , as follows.

For if (double AC is) not (greater than BC), if possible, let BC be double CA . Thus, the (square) on BC (is) four times the (square) on CA . Thus, the (sum of) the (squares) on BC and CA (is) five times the (square) on CA . And the (square) on BA was assumed (to be) five times the (square) on CA . Thus, the (square) on BA is equal to the (sum of) the (squares) on BC and CA . The very thing (is) impossible [Prop. 2.4]. Thus, CB is not double AC . So, similarly, we can show that a (straight-line) less than CB is not double AC either. For (in this case) the absurdity is much [greater]. Thus, double AC is greater than CB . (Which is) the very thing it was required to show.

Proposition 3

If a straight-line is cut in extreme and mean ratio then the square on the lesser piece added to half of the greater piece is five times the square on half of the greater piece.



Εὐθεῖα γάρ τις ἡ AB >ἄκρον καὶ μέσον λόγον τετμήσθω κατὰ τὸ Γ σημεῖον, καὶ >έστω μείζον τμήμα τὸ AG , καὶ τετμήσθω ἡ AG δίθα κατὰ τὸ Δ ; λέγω, <ὅτι πενταπλάσιόν ἐστι τὸ ἀπὸ τῆς BD τοῦ ἀπὸ τῆς $\Delta\Gamma$.

Ἀναγεγράφθω γὰρ ἀπὸ τῆς AB τετράγωνον τὸ AE , καὶ καταγεγράφθω διπλοῦν τὸ σθῆμα. ἐπεὶ διπλῇ ἐστὶν ἡ AG τῆς $\Delta\Gamma$, τετραπλάσιον >ἄρα τὸ ἀπὸ τῆς AG τοῦ ἀπὸ τῆς $\Delta\Gamma$, τουτέστι τὸ $P\Sigma$ τοῦ ZH . καὶ ἐπεὶ τὸ ὑπὸ τῶν $AB\Gamma$ >ἴσον ἐστὶ τῷ ἀπὸ τῆς AG , καὶ ἐστὶ τὸ ὑπὸ τῶν $AB\Gamma$ τὸ ΓE , τὸ >ἄρα ΓE >ἴσον ἐστὶ τῷ $P\Sigma$. τετραπλάσιον δὲ τὸ $P\Sigma$ τοῦ ZH ; τετραπλάσιον >ἄρα καὶ τὸ ΓE τοῦ ZH . πάλιν ἐπεὶ >ἴση ἐστὶν ἡ AD τῇ $\Delta\Gamma$, >ἴση ἐστὶ καὶ ἡ ΘK τῇ KZ . <ὥστε καὶ τὸ HZ τετράγωνον >ἴσον ἐστὶ τῷ $\Theta\Lambda$ τετραγώνῳ. >ἴση >ἄρα ἡ HK τῇ KL , τουτέστιν ἡ MN τῇ NE ; <ὥστε καὶ τὸ MZ τῷ ZE ἐστὶν >ἴσον. ἀλλὰ τὸ MZ τῷ ΓH ἐστὶν >ἴσον; καὶ τὸ ΓH >ἄρα τῷ ZE ἐστὶν >ἴσον. κοινὸν προσκείσθω τὸ ΓN ; ὁ >ἄρα $XO\Pi$ γνῶμων >ἴσοξ ἐστὶ τῷ ΓE . ἀλλὰ τὸ ΓE τετραπλάσιον ἐδείκθη τοῦ HZ ; καὶ ὁ $XO\Pi$ >ἄρα γνῶμων τετραπλάσιός ἐστι τοῦ ZH τετραγώνου. ὁ $XO\Pi$ >ἄρα γνῶμων καὶ τὸ ZH τετράγωνον πενταπλάσιός ἐστι τοῦ ZH . ἀλλὰ ὁ $XO\Pi$ γνῶμων καὶ τὸ ZH τετράγωνόν ἐστι τὸ ΔN . καὶ ἐστὶ τὸ μὲν ΔN τὸ ἀπὸ τῆς ΔB , τὸ δὲ HZ τὸ ἀπὸ τῆς $\Delta\Gamma$. τὸ >ἄρα ἀπὸ τῆς ΔB πενταπλάσιόν ἐστι τοῦ ἀπὸ τῆς $\Delta\Gamma$; <ὅπερ >έδει δεῖχαι.

For let some straight-line AB be cut in extreme and mean ratio at point C . And let AC be the greater piece. And let AC be cut in half at D . I say that the (square) on BD is five times the (square) on DC .

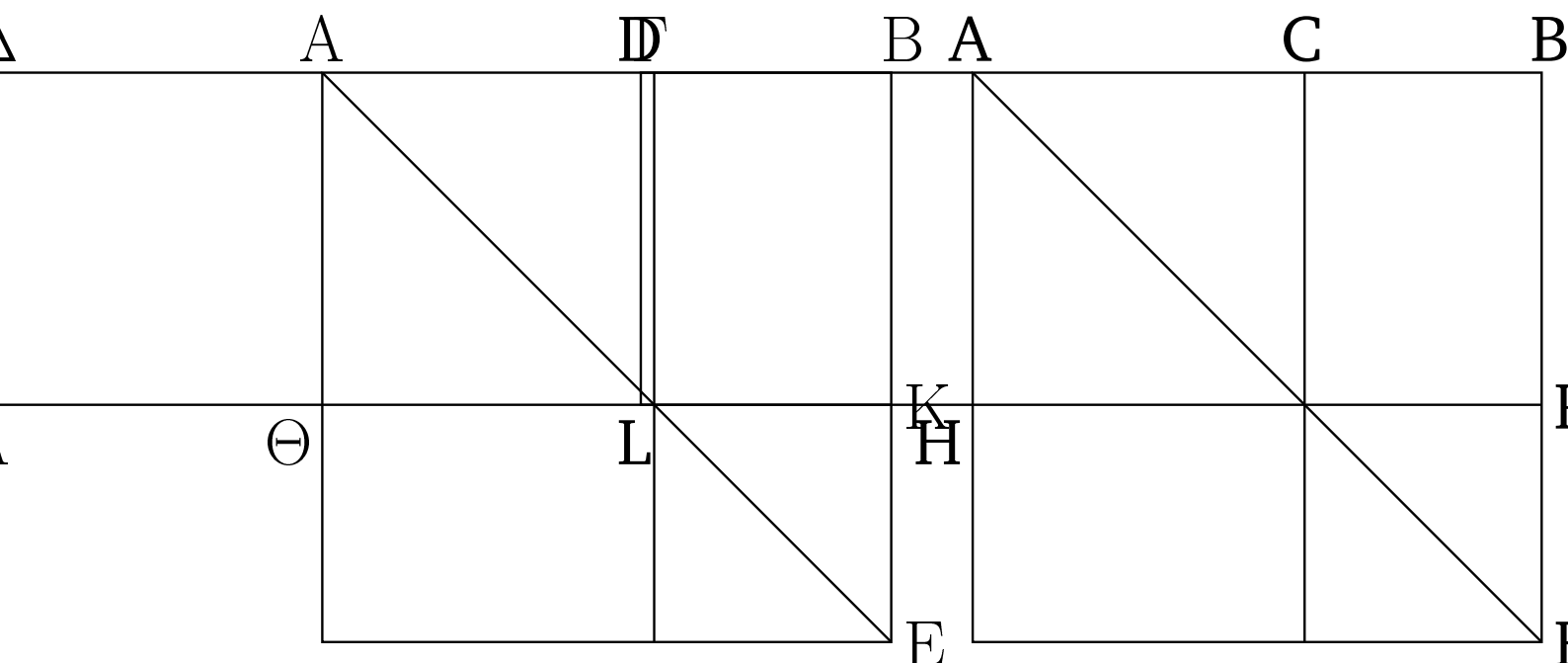
For let the square AE be described on AB . And let the figure be drawn double. Since AC is double DC , the (square) on AC (is) thus four times the (square) on DC —that is to say, RS (is four times) FG . And since the (rectangle contained) by ABC is equal to the (square) on AC [Def. 6.3, Prop. 6.17], and CE is the (rectangle contained) by ABC , CE is thus equal to RS . And RS (is) four times FG . Thus, CE (is) also four times FG . Again, since AD is equal to DC , HK is also equal to KF . Hence, square GF is also equal to square HL . Thus, GK (is) equal to KL —that is to say, MN to NE . Hence, MF is also equal to FE . But, MF is equal to CG . Thus, CG is also equal to FE . Let CN be added to both. Thus, gnomon OPQ is equal to CE . But, CE was shown (to be) equal to four times GF . Thus, gnomon OPQ is also four times square FG . Thus, gnomon OPQ plus square FG is five times FG . But, gnomon OPQ plus square FG is (square) DN . And DN is the (square) on DB , and GF the (square) on DC . Thus, the (square) on DB is five times the (square) on DC . (Which is) the very thing it was required to show.

4.

Ἐὰν εὐθεῖα γραμμὴ >ἄκρον καὶ μέσον λόγον τμηθῇ, τὸ ἀπὸ τῆς <όλης καὶ τοῦ ἐλάσσονος τμήματοξ, τὰ συναμφότερα τετράγωνα, τριπλάσιά ἐστι τοῦ ἀπὸ τοῦ μείζονος τμήματοξ τετραγώνου.

Proposition 4

If a straight-line is cut in extreme and mean ratio then the sum of the squares on the whole and the lesser piece is three times the square on the greater piece.



Εὐθεΐα γὰρ γραμμὴ ἡ $AB >$ ἄκρον καὶ μέσον λόγον τετμήσθω κατὰ τὸ Γ σημεῖον, καὶ $>$ ἔστω μείζον τμήμα ἡ AG , καὶ τῇ $AG >$ ἴση [κείσθω] ἡ AD . λέγω, $<$ ὅτι ἡ DB εὐθεΐα $>$ ἄκρον καὶ μέσον λόγον τέτμηται κατὰ τὸ A , καὶ τὸ μείζον τμήμα ἔστιν ἡ $\epsilon\gamma$ ἄρθῃς εὐθεΐα ἡ AB .

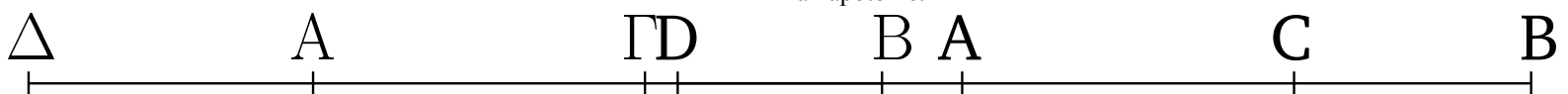
Αναγεγράφθω γὰρ ἀπὸ τῆς AB τετραγώνων τὸ AE, καὶ
 καταγεγράφθω τὸ σθῆμα. ἐπεὶ ἡ AB >άκρον καὶ μέσον
 λόγον τέτμηται κατὰ τὸ Γ, τὸ >άρα ὑπὸ ABΓ >ῖσον ἐστὶ τῷ|
 ἀπὸ AΓ. καὶ ἐστὶ τὸ μὲν ὑπὸ ABΓ τὸ ΓΕ, τὸ δὲ ἀπὸ τῆς AΓ
 τὸ ΓΘ; >ῖσον >άρα τὸ ΓΕ τῷ| ΘΓ. ἀλλὰ τῷ| μὲν ΓΕ >ῖσον
 ἐστὶ τὸ ΘΕ, τῷ| δὲ ΘΓ >ῖσον τὸ ΔΘ; καὶ τὸ ΔΘ >άρα >ῖσον
 ἐστὶ τῷ| ΘΕ [κοινὸν προσκείσθω τὸ ΘΒ]. <όλον >άρα τὸ ΔΚ
 <όλω τῷ| AE ἐστὶν >ῖσον. καὶ ἐστὶ τὸ μὲν ΔΚ τὸ ὑπὸ τῶν
 ΒΔ, ΔΑ; >ῖση γὰρ ἡ ΑΔ τῇ| ΔΛ; τὸ δὲ AE τὸ ἀπὸ τῆς AB;
 τὸ >άρα ὑπὸ τῶν ΒΔΑ >ῖσον ἐστὶ τῷ| ἀπὸ τῆς AB. >έστιν
 >άρα ὡς ἡ ΔΒ πρὸς τὴν ΒΑ, ο<ύτως ἡ ΒΑ πρὸς τὴν ΑΔ.
 μείζων δὲ ἡ ΔΒ τῆς ΒΑ; μείζων >άρα καὶ ἡ ΒΑ τῆς ΑΔ.
 Ἡ >άρα ΔΒ >άκρον καὶ μέσον λόγον τέτμηται κατὰ τὸ Α,
 καὶ τὸ μείζων τμημᾶ ἐστὶν ἡ AB; <όπερ >έδει δεῖξαι.

For let the straight-line AB be cut in extreme and mean ratio at point C . And let AC be the greater piece. And let AD be [made] equal to AC . I say that the straight-line DB has been cut in extreme and mean ratio at A , and that the original straight-line AB is the greater piece.

For let the square AE be described on AB , and let the (remainder of the) figure be drawn. And since AB has been cut in extreme and mean ratio at C , the (rectangle contained) by ABC is thus equal to the (square) on AC [Def. 6.3, Prop. 6.17]. And CE is the (rectangle contained) by ABC , and CH the (square) on AC . But, HE is equal to CE [Prop. 1.43], and DH equal to HC . Thus, DH is also equal to HE . [Let HB be added to both.] Thus, the whole of DK is equal to the whole of AE . And DK is the (rectangle contained) by BD and DA . For AD (is) equal to DL . And AE (is) the (square) on AB . Thus, the (rectangle contained) by BDA is equal to the (square) on AB . Thus, as DB (is) to BA , so BA (is) to AD [Prop. 6.17]. And DB (is) greater than BA . Thus, BA (is) also greater than AD [Prop. 5.14]. Thus, DB has been cut in extreme and mean ratio at A , and the greater piece is AB . (Which is) the very thing it was required to show.

6.

Ἐὰν εὐθεῖα <ρητη> ἄκρον καὶ μέσον λόγον τμηθῇ, ἑκάτερον τῶν τμημάτων >ἀλογόξ ἐστὶν ἢ καλουμένη ἀποτομή.



>Ἐστω εὐθεῖα <ρητή ἡ AB καὶ τετμήσθω >άκρον καὶ μέσον λόγον κατὰ τὸ Γ, καὶ >έστω μείζον τμήμα ἡ ΑΓ; λέγω, <ὅτι ἑκατέρᾳ τῶν ΑΓ, ΓΒ >άλογός ἐστιν ἡ καλουμένη ἀποτομή. Ἐκβεβλήσθω γάρ ἡ ΒΑ, καὶ κείσθω τῆς ΒΑ ἡμίσεια ἡ ΑΔ. ἐπεὶ ο>ὗν εὐθεῖα ἡ ΑΒ τέτμηται >άκρον καὶ μέσον λόγον κατὰ τὸ Γ, καὶ τῷ| μείζονι τμήματι τῷ| ΑΓ πρόσκειται ἡ ΑΔ ἡμίσεια ο>ὔσα τῆς ΑΒ, τὸ >άρα ἀπὸ ΓΔ τοῦ ἀπὸ ΔΑ πενταπλάσιόν ἐστιν. τὸ >άρα ἀπὸ ΓΔ πρὸς τὸ ἀπὸ ΔΑ λόγον >έθει, <ὄν ἀριθμὸς πρὸς ἀριθμὸν; σύμμετρον >άρα τὸ ἀπὸ

Proposition 6

If a rational straight-line is cut in extreme and mean ratio then each of the pieces is that irrational (straight-line) called an apotome.

Let AB be a rational straight-line cut in extreme and mean ratio at C , and let AC be the greater piece. I say that AC and CB is each that irrational (straight-line) called an apotome. For let BA be produced, and let AD be made (equal) to half of BA . Therefore, since the straight-line AB has been cut in extreme and mean ratio at C , and AD , which is half of AB , has been added to the greater piece AC , the (square) on CD is thus five times the (square) on DA [Prop. 13.1]. Thus, the (square) on CD has to the (square) on DA the ratio

ΓΔ τῷ ἀπὸ ΔΑ. <ρητὸν δὲ τὸ ἀπὸ ΔΑ; <ρητὴ γάρ [ἐστίν] ἡ ΔΑ ἡμίσεια οὐσα τῆς ΑΒ <ρητῆς οὐσηξ; <ρητὸν >άρα καὶ τὸ ἀπὸ ΓΔ; <ρητὴ >άρα ἐστὶ καὶ ἡ ΓΔ. καὶ ἐπεὶ τὸ ἀπὸ ΓΔ πρὸς τὸ ἀπὸ ΔΑ λόγον οὐκ >έθει, <ὸν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν, ἀσύμμετρος >άρα μήκει ἡ ΓΔ τῇ ΔΑ; αἱ ΓΔ, ΔΑ >άρα <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι; ἀποτομὴ >άρα ἐστὶν ἡ ΑΓ. πάλιν, ἐπεὶ ἡ ΑΒ >άκρον καὶ μέσον λόγον τέμνεται, καὶ τὸ μείζον τμημά ἐστίν ἡ ΑΓ, τὸ >άρα ὑπὸ ΑΒ, ΒΓ τῷ ἀπὸ ΑΓ >ίσον ἐστίν. τὸ >άρα ἀπὸ τῆς ΑΓ ἀποτομῆς παρὰ τὴν ΑΒ <ρητὴν παραβληθὲν πλάτος ποιεῖ τὴν ΒΓ. τὸ δὲ ἀπὸ ἀποτομῆς παρὰ <ρητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν πρώτην; ἀποτομὴ >άρα πρώτη ἐστὶν ἡ ΓΒ. ἐδείκθη δὲ καὶ ἡ ΓΑ ἀποτομή. Ἐὰν >άρα εὐθεῖα <ρητὴ >άκρον καὶ μέσον λόγον τμηθῇ, ἐκάτερον τῶν τμημάτων >αλογός ἐστίν ἡ καλουμένη ἀποτομή; <όπερ >έδει δεῖξαι.

7.

Ἐὰν πενταγώνου ἰσοπλεύρου αἱ τρεῖς γωνίαι >ήτοι αἱ κατὰ τὸ ἐχῆξ >ἡ αἱ μὴ κατὰ τὸ ἐχῆξ >ίσαι >ῶσιν, ἰσογώνιον >έσται τὸ πεντάγωνον.

Πενταγώνου γὰρ ἰσοπλεύρου τοῦ ΑΒΓΔΕ αἱ τρεῖς γωνίαι πρότερον αἱ κατὰ τὸ ἐχῆξ αἱ πρὸς τοῖς Α, Β, Γ >ίσαι ἀλλήλαις >έστωσαν; λέγω, <ὅτι ἰσογώνιον ἐστὶ τὸ ΑΒΓΔΕ πεντάγωνον.

Ἐπεξεύθωσαν γὰρ αἱ ΑΓ, ΒΕ, ΖΔ. καὶ ἐπεὶ δύο αἱ ΓΒ, ΒΑ δυοὶ ταῖς ΒΑ, ΑΕ >ίσαι εἰσὶν ἑκατέρα ἑκατέρα, καὶ γωνία ἡ ὑπὸ ΓΒΑ γωνία τῇ ὑπὸ ΒΑΕ ἐστὶν >ίση, βάσις >άρα ἡ ΑΓ βάσει τῇ ΒΕ ἐστὶν >ίση, καὶ τὸ ΑΒΓ τρίγωνον τῷ ΑΒΕ τριγώνῳ >ίσον, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις >ίσαι >έσονται, ὅφρ' <ὰς αἱ >ίσαι πλευραὶ ὑποτείνουσιν, ἡ μὲν ὑπὸ ΒΓΑ τῇ ὑπὸ ΒΕΑ, ἡ δὲ ὑπὸ ΑΒΕ τῇ ὑπὸ ΓΑΒ; <ώστε καὶ πλευρὰ ἡ ΑΖ πλευρᾷ τῇ ΒΖ ἐστὶν >ίση. ἐδείκθη δὲ καὶ <ὅλη ἡ ΑΓ <όλη τῇ ΒΕ >ίση; καὶ λοιπὴ >άρα ἡ ΖΓ λοιπῇ τῇ ΖΕ ἐστὶν >ίση. >έστι δὲ καὶ ἡ ΓΔ τῇ ΔΕ >ίση. δύο δὲ αἱ ΖΓ, ΓΔ δυοὶ ταῖς ΖΕ, ΕΔ >ίσαι εἰσὶν; καὶ βάσις αὐτῶν κοινὴ ἡ ΖΔ; γωνία >άρα ἡ ὑπὸ ΖΓΔ γωνία τῇ ὑπὸ ΖΕΔ ἐστὶν >ίση. ἐδείκθη δὲ καὶ ἡ ὑπὸ ΒΓΑ τῇ ὑπὸ ΑΕΒ >ίση; καὶ <ὅλη >άρα ἡ ὑπὸ ΒΓΔ <όλη τῇ ὑπὸ ΑΕΔ >ίση. ἀλλ' ἡ ὑπὸ ΒΓΔ >ίση ὑπόκειται ταῖς πρὸς τοῖς Α, Β γωνίαις; καὶ ἡ ὑπὸ ΑΕΔ >άρα ταῖς πρὸς τοῖς Α, Β γωνίαις >ίση ἐστίν. ὁμοίως δὲ δείχομεν, <ὅτι καὶ ἡ ὑπὸ ΓΔΕ γωνία >ίση ἐστὶ ταῖς πρὸς τοῖς Α, Β, Γ γωνίαις; ἰσογώνιον >άρα ἐστὶ τὸ ΑΒΓΔΕ πεντάγωνον.

which a number (has) to a number. The (square) on CD (is) thus commensurable with the (square) on DA [Prop. 10.6]. And the (square) on DA (is) rational. For DA [is] rational, being half of AB , which is rational. Thus, the (square) on CD (is) also rational [Def. 10.4]. Thus, CD is also rational. And since the (square) on CD does not have to the (square) on DA the ratio which a square number (has) to a square number, CD (is) thus incommensurable in length with DA [Prop. 10.9]. Thus, CD and DA are rational (straight-lines which are) commensurable in square only. Thus, AC is an apotome [Prop. 10.73]. Again, since AB has been cut in extreme and mean ratio, and AC is the greater piece, the (rectangle contained) by AB and BC is thus equal to the (square) on AC [Def. 6.3, Prop. 6.17]. Thus, the (square) on the apotome AC , applied to the rational (straight-line) AB , makes BC as width. And the (square) on an apotome, applied to a rational (straight-line), makes a first apotome as width [Prop. 10.97]. Thus, CB is a first apotome. And CA was also shown (to be) an apotome.

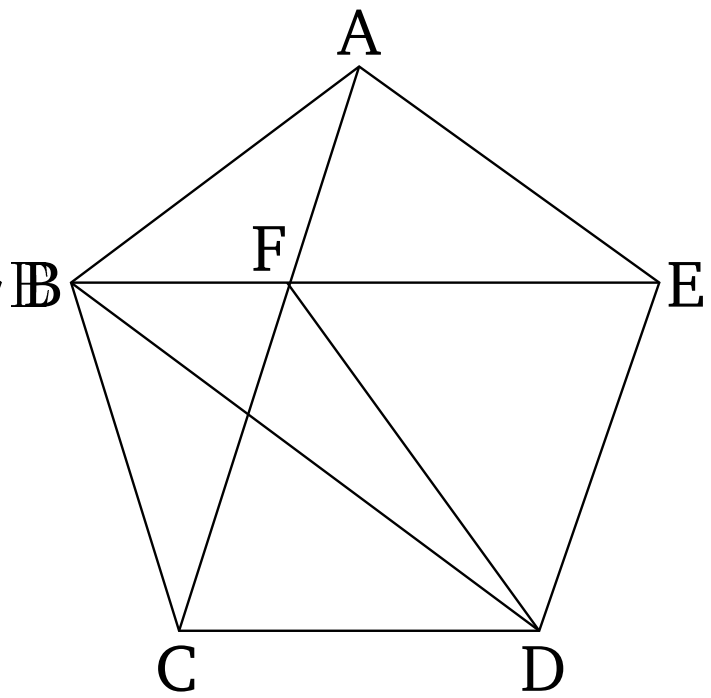
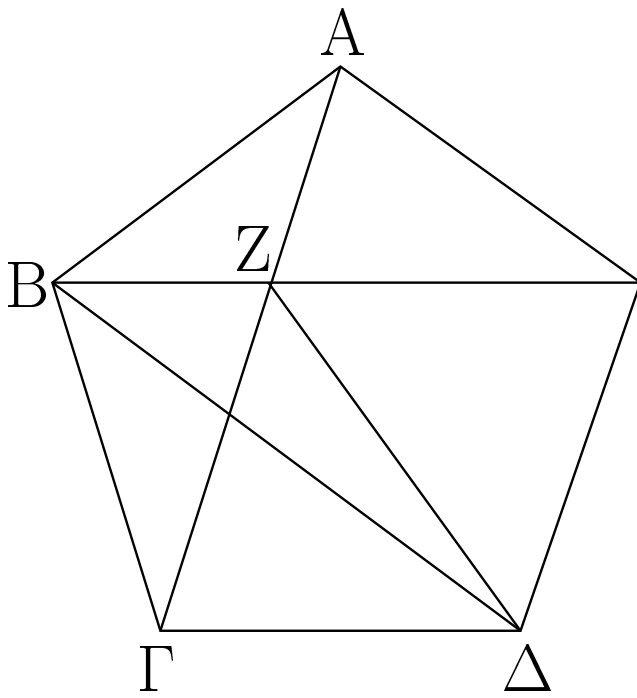
Thus, if a rational straight-line is cut in extreme and mean ratio then each of the pieces is that irrational (straight-line) called an apotome.

Proposition 7

If three angles, either consecutive or not consecutive, of an equilateral pentagon are equal then the pentagon will be equiangular.

For let three angles of the equilateral pentagon $ABCDE$ —first of all, the consecutive (angles) at A , B , and C —be equal to one another. I say that pentagon $ABCDE$ is equiangular.

For let AC , BE , and FD be joined. And since the two (straight-lines) CB and BA are equal to the two (straight-lines) BA and AE , respectively, and angle CBA is equal to angle BAE , base AC is thus equal to base BE , and triangle ABC equal to triangle ABE , and the remaining angles will be equal to the remaining angles which the equal sides subtend [Prop. 1.4], (that is), BCA (equal) to BEA , and ABE to CAB . And hence side AF is also equal to side BF [Prop. 1.6]. And the whole of AC was also shown (to be) equal to the whole of BE . Thus, the remainder FC is also equal to the remainder FE . And CD is also equal to DE . So, the two (straight-lines) FC and CD are equal to the two (straight-lines) FE and ED (respectively). And FD is their common base. Thus, angle FCD is equal to angle FED [Prop. 1.8]. And BCA was also shown (to be) equal to AEB . And thus the whole of BCD (is) equal to the whole of AED . But, (angle) BCD was assumed (to be) equal to the angles at A and B . Thus, (angle) AED is also equal to the angles at A and B . So, similarly, we can show that angle CDE is also equal to the angles at A , B , C . Thus, pentagon $ABCDE$ is equiangular.



Ἀλλὰ δὴ μὴ $\angle$ ίσαι αἱ κατὰ τὸ ἐχθίζ γωνίαι, ἀλλ' $\angle$ ίσαι αἱ πρὸς τοῖς Α, Γ, Δ σημείοις; λέγω, $\angle$ ότι καὶ οὕτως ἰσογώνιον ἐστὶ τὸ ΑΒΓΔΕ πεντάγωνον. Ἐπεξεύθθω γὰρ ἡ ΒΔ. καὶ ἐπεὶ δύο αἱ ΒΑ, ΑΕ δυσὶ ταῖς ΒΓ, ΓΔ $\angle$ ίαι εἰσὶ καὶ γωνίαξ $\angle$ ίαι περιέθουσιν, βάσιξ $\angle$ άρα ἡ ΒΕ βάσει τῇ ΒΔ $\angle$ ίση ἐστίν, καὶ τὸ ΑΒΕ τρίγωνον τῷ ΒΓΔ τριγώνῳ $\angle$ ίση ἐστίν, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαξ $\angle$ ίαι $\angle$ έσσονται, ὅφ' $\angle$ ὰξ αἱ $\angle$ ίαι πλευραὶ ὑποτείνουσιν; $\angle$ ίση $\angle$ άρα ἐστὶν ἡ ὑπὸ ΑΕΒ γωνία τῇ ὑπὸ ΓΔΒ. $\angle$ έστι δὲ καὶ ἡ ὑπὸ ΒΕΔ γωνία τῇ ὑπὸ ΒΔΕ $\angle$ ίση, ἐπεὶ καὶ πλευρὰ ἡ ΒΕ πλευρᾷ τῇ ΒΔ ἐστὶν $\angle$ ίση. καὶ $\angle$ όλη $\angle$ άρα ἡ ὑπὸ ΑΕΔ γωνία $\angle$ όλη τῇ ὑπὸ ΓΔΕ ἐστὶν $\angle$ ίση. ἀλλὰ ἡ ὑπὸ ΓΔΕ ταῖς πρὸς τοῖς Α, Γ γωνίαξ ὑπόκειται $\angle$ ίση; καὶ ἡ ὑπὸ ΑΕΔ $\angle$ άρα γωνία ταῖς πρὸς τοῖς Α, Γ $\angle$ ίση ἐστίν. διὰ τὰ αὐτὰ δὴ καὶ ἡ ὑπὸ ΑΒΓ $\angle$ ίση ἐστὶ ταῖς πρὸς τοῖς Α, Γ, Δ γωνίαξ. ἰσογώνιον $\angle$ άρα ἐστὶ τὸ ΑΒΓΔΕ πεντάγωνον; $\angle$ όπερ $\angle$ έδει δεῖξαι.

And so let consecutive angles not be equal, but let the (angles) at points Α, C, and D be equal. I say that pentagon $ABCDE$ is also equiangular in this case.

For let BD be joined. And since the two (straight-lines) BA and AE are equal to the (straight-lines) BC and CD , and they contain equal angles, base BE is thus equal to base BD , and triangle ABE is equal to triangle BCD , and the remaining angles will be equal to the remaining angles which the equal sides subtend [Prop. 1.4]. Thus, angle AEB is equal to (angle) CDB . And angle BED is also equal to (angle) BDE , since side BE is also equal to side BD [Prop. 1.5]. Thus, the whole angle AED is also equal to the whole (angle) CDE . But, (angle) CDE was assumed (to be) equal to the angles at Α and C. Thus, angle AED is also equal to the (angles) at Α and C. So, for the same (reasons), (angle) ABC is also equal to the angles at Α, C, and D. Thus, pentagon $ABCDE$ is equiangular. (Which is) the very thing it was required to show.

8.

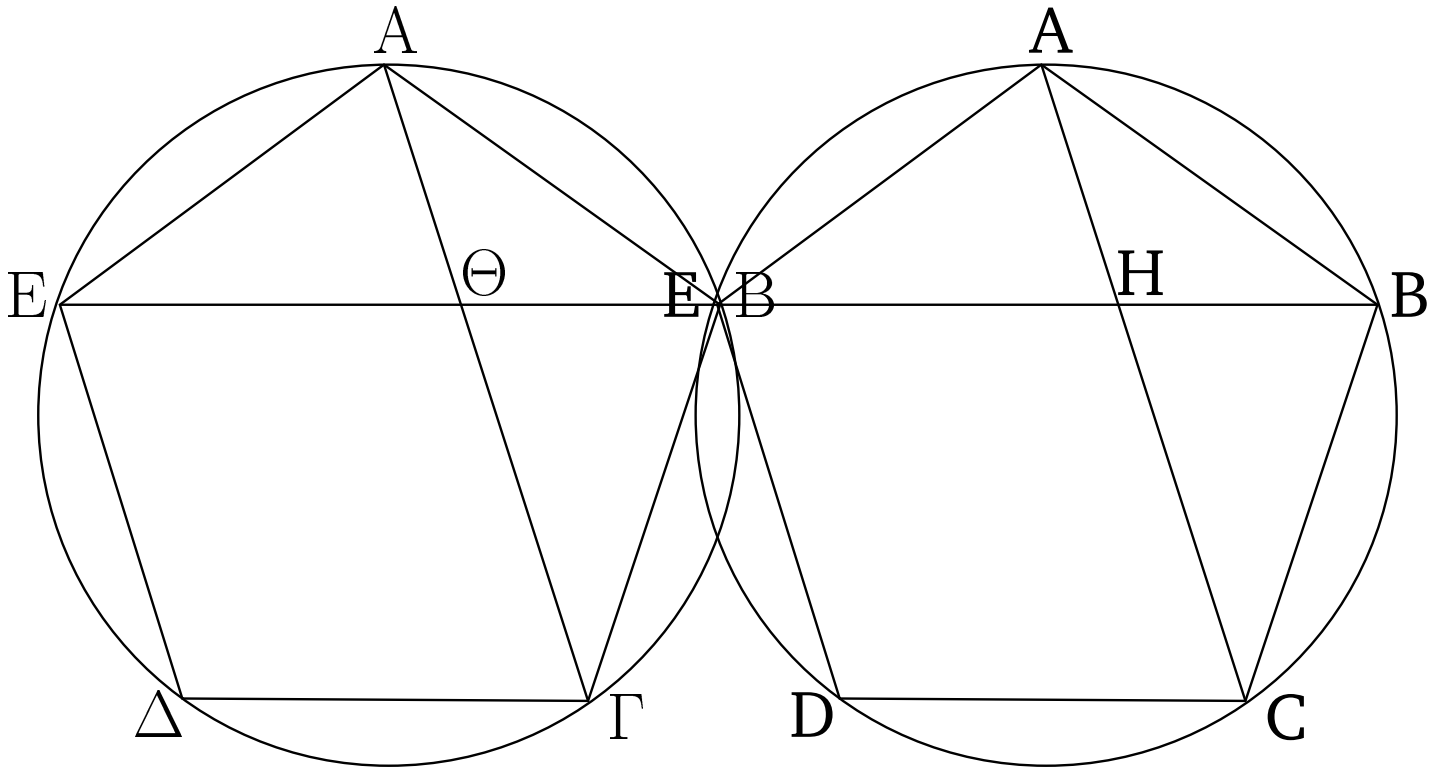
Ἐὰν πενταγώνου ἰσοπλεύρου καὶ ἰσογώνου τὰξ κατὰ τὸ ἐχθίζ δύο γωνίαξ ὑποτείνωσιν εὐθεῖαι, $\angle$ άκρον καὶ μέσον λόγον τέμνουσιν ἀλλήλαξ, καὶ τὰ μείζονα αὐτῶν τμήματα $\angle$ ίση ἐστὶ τῇ τοῦ πενταγώνου πλευρᾷ.

Πενταγώνου γὰρ ἰσοπλεύρου καὶ ἰσογώνου τοῦ ΑΒΓΔΕ δύο γωνίαξ τὰξ κατὰ τὸ ἐχθίζ τὰξ πρὸς τοῖς Α, Β ὑποτείνετωσιν εὐθεῖαι αἱ ΑΓ, ΒΕ τέμνουσαι ἀλλήλαξ κατὰ τὸ Θ σημεῖον; λέγω, $\angle$ ότι ἑκάτερα αὐτῶν $\angle$ άκρον καὶ μέσον λόγον τέμνεται κατὰ τὸ Θ σημεῖον, καὶ τὰ μείζονα αὐτῶν τμήματα $\angle$ ίση ἐστὶ τῇ τοῦ πενταγώνου πλευρᾷ.

Proposition 8

If straight-lines subtend two consecutive angles of an equilateral and equiangular pentagon then they cut one another in extreme and mean ratio, and their greater pieces are equal to the sides of the pentagon.

For let the two straight-lines, AC and BE , cutting one another at point H , have subtended two consecutive angles, at Α and Β (respectively), of the equilateral and equiangular pentagon $ABCDE$. I say that each of them has been cut in extreme and mean ratio at point H , and that their greater pieces are equal to the sides of the pentagon.



Περιγεγράφω γὰρ περὶ τὸ $ABΓΔΕ$ πεντάγωνον κύκλος ὁ $ABΓΔΕ$. καὶ ἐπεὶ δύο εὐθεῖαι αἱ EA , AB δυοὶ ταῖς AB , $BΓ$ >ίσαι εἰσὶ καὶ γωνίαξ >ίσαξ περιέθουσιν, βάσιξ >άρα ἢ BE βάσει τῇ $ΑΓ$ >ίση ἐστίν, καὶ τὸ ABE τρίγωνον τῷ $ABΓ$ τριγώνῳ >ίσον ἐστίν, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαξ >ίσαι >έσονται ἑκατέρα ἑκατέρα, ὅφ <ὰς αἱ >ίσαι πλευραὶ ὑποτείνουσιν. >ίση >άρα ἐστίν ἡ ὑπὸ $BAΓ$ γωνία τῇ ὑπὸ ABE ; διπλῇ >άρα ἡ ὑπὸ $AΘE$ τῇ ὑπὸ $BAΘ$. >έστι δὲ καὶ ἡ ὑπὸ $EAΓ$ τῇ ὑπὸ $BAΓ$ διπλῇ, ἐπειδὴ περ καὶ περιφέρεια ἡ $ΕΔΓ$ περιφερείας τῇ $ΓΒ$ ἐστὶ διπλῇ; >ίση >άρα ἡ ὑπὸ $ΘAE$ γωνία τῇ ὑπὸ $AΘE$; <ώστε καὶ ἡ $ΘE$ εὐθεῖα τῇ EA , τουτέστι τῇ AB ἐστὶν >ίση. καὶ ἐπεὶ >ίση ἐστὶν ἡ BA εὐθεῖα τῇ AE , >ίση ἐστὶ καὶ γωνία ἡ ὑπὸ ABE τῇ ὑπὸ AEB . ἀλλὰ ἡ ὑπὸ ABE τῇ ὑπὸ $BAΘ$ ἐδείκθη >ίση; καὶ ἡ ὑπὸ BEA >άρα τῇ ὑπὸ $BAΘ$ ἐστὶν >ίση. καὶ κοινὴ τῶν δύο τριγώνων τοῦ τε ABE καὶ τοῦ $ABΘ$ ἐστὶν ἡ ὑπὸ ABE ; λοιπὴ >άρα ἡ ὑπὸ BAE γωνία λοιπῇ τῇ ὑπὸ $AΘB$ ἐστὶν >ίση; ἰσογώνιον >άρα ἐστὶ τὸ ABE τρίγωνον τῷ $ABΘ$ τριγώνῳ; ἀνάλογον >άρα ἐστὶν ὥς ἡ EB πρὸς τὴν BA , ο<ύτωξ ἡ AB πρὸς τὴν $BΘ$. >ίση δὲ ἡ BA τῇ $ΕΘ$; ὥς >άρα ἡ BE πρὸς τὴν $ΕΘ$, ο<ύτωξ ἡ $ΕΘ$ πρὸς τὴν $ΘB$. μείζων δὲ ἡ BE τῇ $ΕΘ$; μείζων >άρα καὶ ἡ $ΕΘ$ τῇ $ΘB$. ἡ BE <άρα >άκρον καὶ μέσον λόγον τέτμηται κατὰ τὸ $Θ$, καὶ τὸ μείζον τμήμα τὸ $ΘE$ >ίσον ἐστὶ τῇ τοῦ πενταγώνου πλευρᾷ. ὁμοίωξ δὲ δείχομεν, <ὅτι καὶ ἡ $ΑΓ$ >άκρον καὶ μέσον λόγον τέτμηται κατὰ τὸ $Θ$, καὶ τὸ μείζον αὐτῆς τμήμα ἡ $ΓΘ$ >ίσον ἐστὶ τῇ τοῦ πενταγώνου πλευρᾷ; <ὅπερ >έδει δεῖχαι.

9.

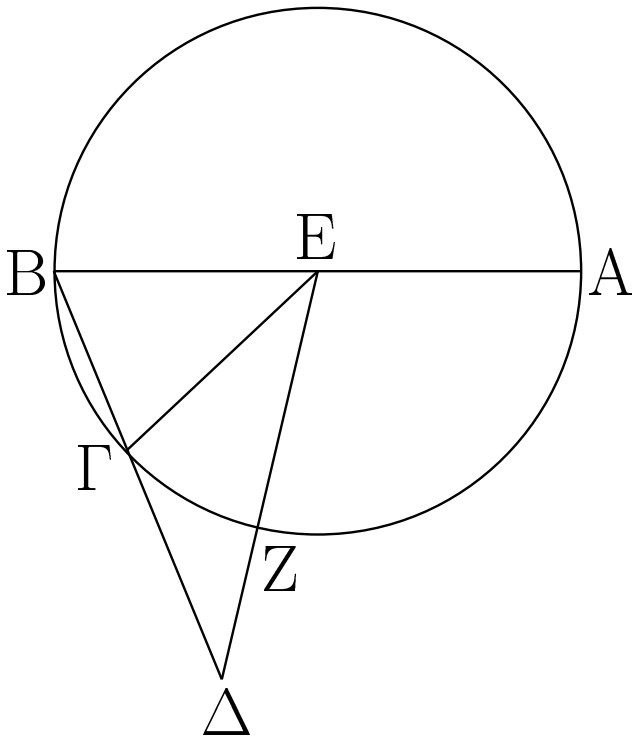
Ἐὰν ἡ τοῦ ἑξαγώνου πλευρὰ καὶ ἡ τοῦ δεκαγώνου τῶν εἰς τὸν αὐτὸν κύκλον ἐγγραφομένων συντεθῶσιν, ἡ <όλη εὐθεῖα >άκρον καὶ μέσον λόγον τέτμηται, καὶ τὸ μείζον αὐτῆς τμήμα ἐστὶν ἡ τοῦ ἑξαγώνου πλευρὰ.

For let the circle $ABCDE$ be circumscribed about pentagon $ABCDE$ [Prop. 4.14]. And since the two straight-lines EA and AB are equal to the two (straight-lines) AB and BC (respectively), and they contain equal angles, the base BE is thus equal to the base AC , and triangle ABE is equal to triangle ABC , and the remaining angles will be equal to the remaining angles, respectively, which the equal sides subtend [Prop. 1.4]. Thus, angle BAC is equal to (angle) ABE . Thus, (angle) AHE (is) double (angle) BAH [Prop. 1.32]. And EAC is also double BAC , inasmuch as circumference EDC is also double circumference CB [Props. 3.28, 6.33]. Thus, angle HAE (is) equal to (angle) AHE . Hence, straight-line HE is also equal to (straight-line) EA —that is to say, to (straight-line) AB [Prop. 1.6]. And since straight-line BA is equal to AE , angle ABE is also equal to AEB [Prop. 1.5]. But, ABE was shown (to be) equal to BAH . Thus, BEA is also equal to BAH . And (angle) ABE is common to the two triangles ABE and ABH . Thus, the remaining angle BAE is equal to the remaining (angle) AHB [Prop. 1.32]. Thus, triangle ABE is equiangular to triangle ABH . Thus, proportionally, as EB is to BA , so AB (is) to BH [Prop. 6.4]. And BA (is) equal to EH . Thus, as BE (is) to EH , so EH (is) to HB . And BE (is) greater than EH . EH (is) thus also greater than HB [Prop. 5.14]. Thus, BE has been cut in extreme and mean ratio at H , and the greater piece HE is equal to the side of the pentagon. So, similarly, we can show that AC has also been cut in extreme and mean ratio at H , and that its greater piece CH is equal to the side of the pentagon. (Which is) the very thing it was required to show.

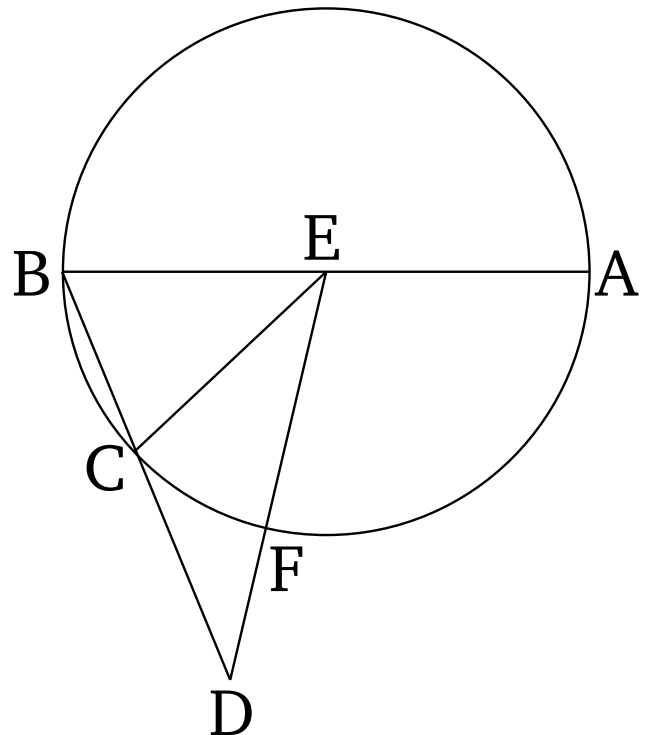
Proposition 9

If the side of a hexagon and of a decagon inscribed in the same circle are added together then the whole straight-line has been cut in extreme and mean ratio (at the junction point), and its greater piece is the side of the hexagon.[†]

>Ἐστω κύκλος ὁ $AB\Gamma$, καὶ τῶν εἰς τὸν $AB\Gamma$ κύκλον ἐγγραφομένων σθημάτων, δεκαγώνου μὲν >έστω πλευρὰ ἡ $B\Gamma$, ἑξαγώνου δὲ ἡ $\Gamma\Delta$, καὶ >έστωσαν ἐπ' εὐθείᾳ; λέγω, <ὅτι ἡ <όλη εὐθεῖα ἡ $B\Delta$ >ἄκρον καὶ μέσον λόγον τέτμηται, καὶ τὸ μείζον αὐτῆς τμημὰ ἐστὶν ἡ $\Gamma\Delta$.



Let ABC be a circle. And of the figures inscribed in circle ABC , let BC be the side of a decagon, and CD (the side) of a hexagon. And let them be (laid down) straight-on (to one another). I say that the whole straight-line BD has been cut in extreme and mean ratio (at C), and that CD is its greater piece.



Εἰλήφθω γὰρ τὸ κέντρον τοῦ κύκλου τὸ E σημεῖον, καὶ ἐπέξευθθωσαν αἱ EB , EG , ED , καὶ διήθθω ἡ BE ἐπὶ τὸ A . ἐπεὶ δεκαγώνου ἰσοπλευρὸν πλευρὰ ἐστὶν ἡ $B\Gamma$, πενταπλασίον >ἄρα ἡ $A\Gamma B$ περιφέρεια τῆς $B\Gamma$ περιφερείας; τετραπλασίον >ἄρα ἡ $A\Gamma$ περιφέρεια τῆς ΓB . ὥς δὲ ἡ $A\Gamma$ περιφέρεια πρὸς τὴν ΓB , ο<ύτωξ ἡ ὑπὸ AEG γωνία πρὸς τὴν ὑπὸ $ΓEB$; τετραπλασίον >ἄρα ἡ ὑπὸ AEG τῆς ὑπὸ $ΓEB$. καὶ ἐπεὶ >ἴση ἡ ὑπὸ $EB\Gamma$ γωνία τῇ ὑπὸ $ΕΓB$, ἡ >ἄρα ὑπὸ AEG γωνία διπλασία ἐστὶ τῆς ὑπὸ $ΕΓB$. καὶ ἐπεὶ >ἴση ἐστὶν ἡ EG εὐθεῖα

For let the center of the circle, point E , be found [Prop. 3.1], and let EB , EC , and ED be joined, and let BE be drawn across to A . Since BC is a side on an equilateral decagon, circumference ACB (is) thus five times circumference BC . Thus, circumference AC (is) four times CB . And as circumference AC (is) to CB , so angle AEC (is) to CEB [Prop. 6.33]. Thus, (angle) AEC (is) four times CEB . And since angle EBC (is) equal to ECB [Prop. 1.5], an-

τῇ ΓΔ; ἑκατέρα γὰρ αὐτῶν >ίση ἐστὶ τῇ τοῦ ἑξαγώνου πλευρᾷ τοῦ εἰς τὸν ΑΒΓ κύκλον [ἐγγραφομένου]; >ίση ἐστὶ καὶ ἡ ὑπὸ ΓΕΔ γωνία τῇ ὑπὸ ΓΔΕ γωνίᾳ; διπλασία >άρα ἡ ὑπὸ ΕΓΒ γωνία τῇ ὑπὸ ΕΔΓ. ἀλλὰ τῇ ὑπὸ ΕΓΒ διπλασία ἐδείκθη ἡ ὑπὸ ΑΕΓ; τετραπλασία >άρα ἡ ὑπὸ ΑΕΓ τῇ ὑπὸ ΕΔΓ. ἐδείκθη δὲ καὶ τῇ ὑπὸ ΒΕΓ τετραπλασία ἡ ὑπὸ ΑΕΓ; >ίση >άρα ἡ ὑπὸ ΕΔΓ τῇ ὑπὸ ΒΕΓ. κοινὴ δὲ τῶν δύο τριγώνων, τοῦ τε ΒΕΓ καὶ τοῦ ΒΕΔ, ἡ ὑπὸ ΕΒΔ γωνία; καὶ λοιπὴ >άρα ἡ ὑπὸ ΒΕΔ τῇ ὑπὸ ΕΓΒ ἐστὶν >ίση; ἰσογώνιον >άρα ἐστὶ τὸ ΕΒΔ τρίγωνον τῷ ΕΒΓ τριγώνῳ. ἀνάλογον >άρα ἐστὶν ὡς ἡ ΑΒ πρὸς τὴν ΒΕ, οὕτως ἡ ΕΒ πρὸς τὴν ΒΓ. >ίση δὲ ἡ ΕΒ τῇ ΓΔ. >έστιν >άρα ὡς ἡ ΒΔ πρὸς τὴν ΔΓ, οὕτως ἡ ΔΓ πρὸς τὴν ΓΒ. μείζων δὲ ἡ ΒΔ τῇ ΔΓ; μείζων >άρα καὶ ἡ ΔΓ τῇ ΓΒ. ἡ ΒΔ >άρα εὐθεῖα >άκρον καὶ μέσον λόγον τέτμηται [κατὰ τὸ Γ], καὶ τὸ μείζων τμήμα αὐτῇ ἐστὶν ἡ ΔΓ; <όπερ >έδει δεῖχαι.

gle AEC is thus double ECB [Prop. 1.32]. And since straight-line EC is equal to CD —for each of them is equal to the side of the hexagon [inscribed] in circle ABC [Prop. 4.15 corr.]—angle CED is also equal to angle CDE [Prop. 1.5]. Thus, angle ECB (is) double EDC [Prop. 1.32]. But, AEC was shown (to be) double ECB . Thus, AEC (is) four times EDC . And AEC was also shown (to be) four times BEC . Thus, EDC (is) equal to BEC . And angle EBD (is) common to the two triangles BEC and BED . Thus, the remaining (angle) BED is equal to the (remaining angle) ECB [Prop. 1.32]. Thus, triangle EBD is equiangular to triangle EBC . Thus, proportionally, as DB is to BE , so EB (is) to BC [Prop. 6.4]. And EB (is) equal to CD . Thus, as BD is to DC , so DC (is) to CB . And BD (is) greater than DC . Thus, DC (is) also greater than CB [Prop. 5.14]. Thus, the straight-line BD has been cut in extreme and mean ratio [at C], and DC is its greater piece. (Which is), the very thing it was required to show.

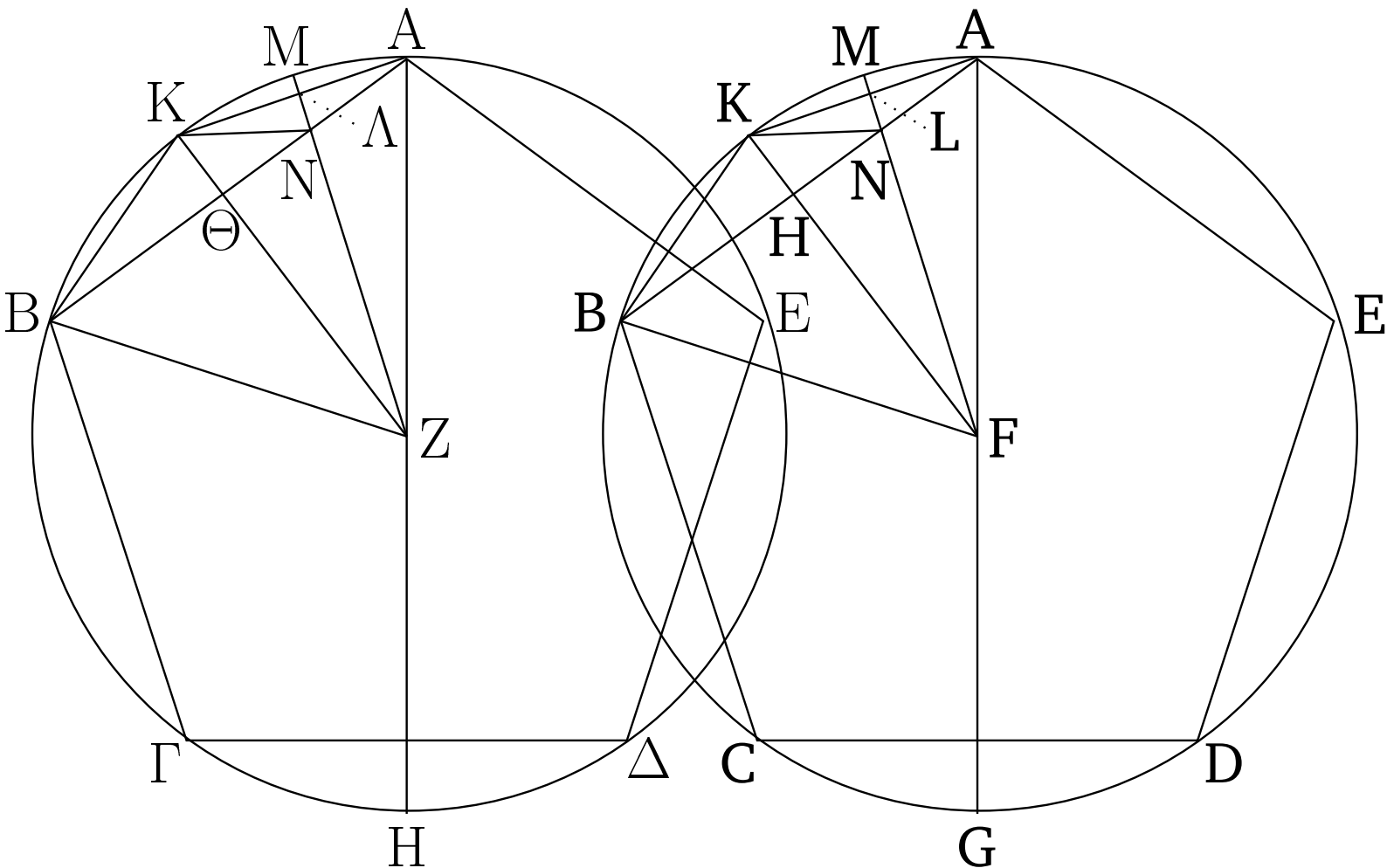
† If the circle is of unit radius then the side of the hexagon is 1, whereas the side of the decagon is $(1/2)(\sqrt{5} - 1)$.

10.

Ἐὰν εἰς κύκλον πεντάγωνον ἰσόπλευρον ἐγγραφῇ, ἡ τοῦ πεντάγωνου πλευρὰ δύναται τὴν τε τοῦ ἑξαγώνου καὶ τὴν τοῦ δεκαγώνου τῶν εἰς τὸν αὐτὸν κύκλον ἐγγραφομένων.

Proposition 10

If an equilateral pentagon is inscribed in a circle then the square on the side of the pentagon is (equal to) the (sum of the squares) on the (sides) of the hexagon and of the decagon inscribed in the same circle.†



>Ἐστω κύκλος ὁ ΑΒΓΔΕ, καὶ εἰς τὸ ΑΒΓΔΕ κύκλον πεντάγωνον ἰσόπλευρον ἐγγεγράφθω τὸ ΑΒΓΔΕ. λέγω,

Let $ABCDE$ be a circle. And let the equilateral pentagon $ABCDE$ be inscribed in circle $ABCDE$. I say that the

<ὅτι ἡ τοῦ ΑΒΓΔΕ πενταγώνου πλευρὰ δύναται τὴν τε τοῦ ἑξαγώνου καὶ τὴν τοῦ δεκαγώνου πλευρὰν τῶν εἰς τὸν ΑΒΓΔΕ κύκλον ἐγγραφομένων.

Εἰλήφθω γάρ τὸ κέντρον τοῦ κύκλου τὸ Ζ σημείον, καὶ ἐπεζευθῆσα ἡ ΑΖ διήθῃ ἐπὶ τὸ Η σημείον, καὶ ἐπεζεύθῃ ἡ ΖΒ, καὶ ἀπὸ τοῦ Ζ ἐπὶ τὴν ΑΒ κάθετοξ >ήθῃ ἡ ΖΘ, καὶ διήθῃ ἐπὶ τὸ Κ, καὶ ἐπεζεύθῃσαν αἱ ΑΚ, ΚΒ, καὶ πάλιν ἀπὸ τοῦ Ζ ἐπὶ τὴν ΑΚ κάθετοξ >ήθῃ ἡ ΖΛ, καὶ διήθῃ ἐπὶ τὸ Μ, καὶ ἐπεζεύθῃ ἡ ΚΝ.

Ἐπεὶ >ῖση ἐστὶν ἡ ΑΒΓΗ περιφέρεια τῇ ΑΕΔΗ περιφερείᾳ, <ὦν ἡ ΑΒΓ τῇ ΑΕΔ ἐστὶν >ῖση, λοιπὴ >άρα ἡ ΓΗ περιφέρεια λοιπῇ τῇ ΗΔ ἐστὶν >ῖση. πενταγώνου δὲ ἡ ΓΔ; δεκαγώνου >άρα ἡ ΓΗ. καὶ ἐπεὶ >ῖση ἐστὶν ἡ ΖΑ τῇ ΖΒ, καὶ κάθετοξ ἡ ΖΘ, >ῖση >άρα καὶ ἡ ὑπὸ ΑΖΚ γωνία τῇ ὑπὸ ΚΖΒ. <ώστε καὶ περιφέρεια ἡ ΑΚ τῇ ΚΒ ἐστὶν >ῖση; διπλῇ >άρα ἡ ΑΒ περιφέρεια τῇ ΒΚ περιφερείᾳ; δεκαγώνου >άρα πλευρὰ ἐστὶν ἡ ΑΚ εὐθεῖα. διὰ τὰ αὐτὰ δὴ καὶ ἡ ΑΚ τῇ ΚΜ ἐστὶ διπλῇ. καὶ ἐπεὶ διπλῇ ἐστὶν ἡ ΑΒ περιφέρεια τῇ ΒΚ περιφερείᾳ, >ῖση δὲ ἡ ΓΔ περιφέρεια τῇ ΑΒ περιφερείᾳ, διπλῇ >άρα καὶ ἡ ΓΔ περιφέρεια τῇ ΒΚ περιφερείᾳ. >έστι δὲ ἡ ΓΔ περιφέρεια καὶ τῇ ΓΗ διπλῇ; >ῖση >άρα ἡ ΓΗ περιφέρεια τῇ ΒΚ περιφερείᾳ. ἀλλὰ ἡ ΒΚ τῇ ΚΜ ἐστὶ διπλῇ, ἐπεὶ καὶ ἡ ΚΑ; καὶ ἡ ΓΗ >άρα τῇ ΚΜ ἐστὶ διπλῇ. ἀλλὰ μὴν καὶ ἡ ΓΒ περιφέρεια τῇ ΒΚ περιφερείᾳ ἐστὶ διπλῇ; >ῖση γάρ ἡ ΓΒ περιφέρεια τῇ ΒΑ. καὶ <όλη >άρα ἡ ΗΒ περιφέρεια τῇ ΒΜ ἐστὶ διπλῇ; <ώστε καὶ γωνία ἡ ὑπὸ ΗΖΒ γωνίαξ τῇ ὑπὸ ΒΖΜ [ἐστὶ] διπλῇ. >έστι δὲ ἡ ὑπὸ ΗΖΒ καὶ τῇ ὑπὸ ΖΑΒ διπλῇ; >ῖση γάρ ἡ ὑπὸ ΖΑΒ τῇ ὑπὸ ΑΒΖ. καὶ ἡ ὑπὸ ΒΖΝ >άρα τῇ ὑπὸ ΖΑΒ ἐστὶν >ῖση. κοινὴ δὲ τῶν δύο τριγώνων, τοῦ τε ΑΒΖ καὶ τοῦ ΒΖΝ, ἡ ὑπὸ ΑΒΖ γωνία; λοιπὴ >άρα ἡ ὑπὸ ΑΖΒ λοιπῇ τῇ ὑπὸ ΒΝΖ ἐστὶν >ῖση; >ῖσογώνιον >άρα ἐστὶ τὸ ΑΒΖ τρίγωνον τῷ ΒΖΝ τριγώνῳ. ἀνάλογον >άρα ἐστὶν ὡς ἡ ΑΒ εὐθεῖα πρὸς τὴν ΒΖ, ο<ύτωξ ἡ ΖΒ πρὸς τὴν ΒΝ; τὸ >άρα ὑπὸ τῶν ΑΒΝ >ῖσον ἐστὶ τῷ ἀπὸ ΒΖ. πάλιν ἐπεὶ >ῖση ἐστὶν ἡ ΑΛ τῇ ΑΚ, κοινὴ δὲ καὶ πρὸς ὀρθὰξ ἡ ΑΝ, βάσιξ >άρα ἡ ΚΝ βάσει τῇ ΑΝ ἐστὶν >ῖση; καὶ γωνία >άρα ἡ ὑπὸ ΑΚΝ γωνία τῇ ὑπὸ ΛΑΝ ἐστὶν >ῖση. ἀλλὰ ἡ ὑπὸ ΛΑΝ τῇ ὑπὸ ΚΒΝ ἐστὶν >ῖση; καὶ ἡ ὑπὸ ΑΚΝ >άρα τῇ ὑπὸ ΚΒΝ ἐστὶν >ῖση. καὶ κοινὴ τῶν δύο τριγώνων τοῦ τε ΑΚΒ καὶ τοῦ ΑΚΝ ἡ πρὸξ τῷ Α. λοιπὴ >άρα ἡ ὑπὸ ΑΚΒ λοιπῇ τῇ ὑπὸ ΚΝΑ ἐστὶν >ῖση; ῖσογώνιον >άρα ἐστὶ τὸ ΚΒΑ τρίγωνον τῷ ΚΝΑ τριγώνῳ. ἀνάλογον >άρα ἐστὶν ὡς ἡ ΒΑ εὐθεῖα πρὸς τὴν ΑΚ, ο<ύτωξ ἡ ΚΑ πρὸς τὴν ΑΝ; τὸ >άρα ὑπὸ τῶν ΒΑΝ >ῖσον ἐστὶ τῷ ἀπὸ τῇξ ΑΚ. ἐδείθῃ δὲ καὶ τὸ ὑπὸ τῶν ΑΒΝ >ῖσον τῷ ἀπὸ τῇξ ΒΖ; τὸ >άρα ὑπὸ τῶν ΑΒΝ μετὰ τοῦ ὑπὸ ΒΑΝ, <όπερ ἐστὶ τὸ ἀπὸ τῇξ ΒΑ, >ῖσον ἐστὶ τῷ ἀπὸ τῇξ ΒΖ μετὰ τοῦ ἀπὸ τῇξ ΑΚ. καὶ ἐστὶν ἡ μὲν ΒΑ πενταγώνου πλευρὰ, ἡ δὲ ΒΖ ἑξαγώνου, ἡ δὲ ΑΚ δεκαγώνου.

Ἡ >άρα τοῦ πενταγώνου πλευρὰ δύναται τὴν τε τοῦ ἑξαγώνου καὶ τὴν τοῦ δεκαγώνου τῶν εἰς τὸν αὐτὸν κύκλον ἐγγραφομένων <όπερ >έδει δεῖχαι.

square on the side of pentagon $ABCDE$ is the (sum of the squares) on the sides of the hexagon and of the decagon inscribed in circle $ABCDE$.

For let the center of the circle, point F , be found [Prop. 3.1]. And, AF being joined, let it be drawn across to point G . And let FB be joined. And let FH be drawn from F perpendicular to AB . And let it be drawn across to K . And let AK and KB be joined. And, again, let FL have been drawn from F perpendicular to AK . And let it be drawn across to M . And let KN be joined.

Since circumference $ABCG$ is equal to circumference $AEDG$, of which ABC is equal to AED , the remaining circumference CG is thus equal to the remaining (circumference) GD . And CD (is the side) of the pentagon. CG (is) thus (the side) of the decagon. And since FA is equal to FB , and FH is perpendicular (to AB), angle AFK (is) thus also equal to KFB [Props. 1.5, 1.26]. Hence, circumference AK is also equal to KB [Prop. 3.26]. Thus, circumference AB (is) double circumference BK . Thus, straight-line AK is the side of the decagon. So, for the same (reasons, circumference) AK is also double KM . And since circumference AB is double circumference BK , and circumference CD (is) equal to circumference AB , circumference CD (is) thus also double circumference BK . And circumference CD is also double CG . Thus, circumference CG (is) equal to circumference BK . But, BK is double KM , since KA (is) also (double KM). Thus, (circumference) CG is also double KM . But, indeed, circumference CB is also double circumference BK . For circumference CB (is) equal to BA . Thus, the whole circumference GB is also double BM . Hence, angle GFB [is] also double angle BFM [Prop. 6.33]. And GFB (is) also double FAB . For FAB (is) equal to ABF . Thus, BFN is also equal to FAB . And angle ABF (is) common to the two triangles ABF and BFN . Thus, the remaining (angle) AFB is equal to the remaining (angle) BNF [Prop. 1.32]. Thus, triangle ABF is equiangular to triangle BFN . Thus, proportionally, as straight-line AB (is) to BF , so FB (is) to BN [Prop. 6.4]. Thus, the (rectangle contained) by ABN is equal to the (square) on BF [Prop. 6.17]. Again, since AL is equal to LK , and LN is common and at right-angles (to KA), base KN is thus equal to base AN [Prop. 1.4]. And, thus, angle LKN is equal to angle LAN . But, LAN is equal to KBN [Props. 3.29, 1.5]. Thus, LKN is also equal to KBN . And the (angle) at A (is) common to the two triangles AKB and AKN . Thus, the remaining (angle) AKB is equal to the remaining (angle) KNA [Prop. 1.32]. Thus, triangle KBA is equiangular to triangle KNA . Thus, proportionally, as straight-line BA is to AK , so KA (is) to AN [Prop. 6.4]. Thus, the (rectangle contained) by BAN is equal to the (square) on AK [Prop. 6.17]. And the (rectangle contained) by ABN was also shown (to be) equal to the (square) on BF . Thus, the (rectangle contained) by ABN plus the (rectangle contained) by BAN , which is the (square) on BA [Prop. 2.2], is equal to the (square) on BF plus the (square) on AK . And BA is the side of the pentagon, and BF (the side) of the hexagon [Prop. 4.15 corr.], and AK (the side) of the decagon.

Thus, the square on the side of the pentagon (inscribed in a circle) is (equal to) the (sum of the squares) on the (sides) of the hexagon and of the decagon inscribed in the same circle.

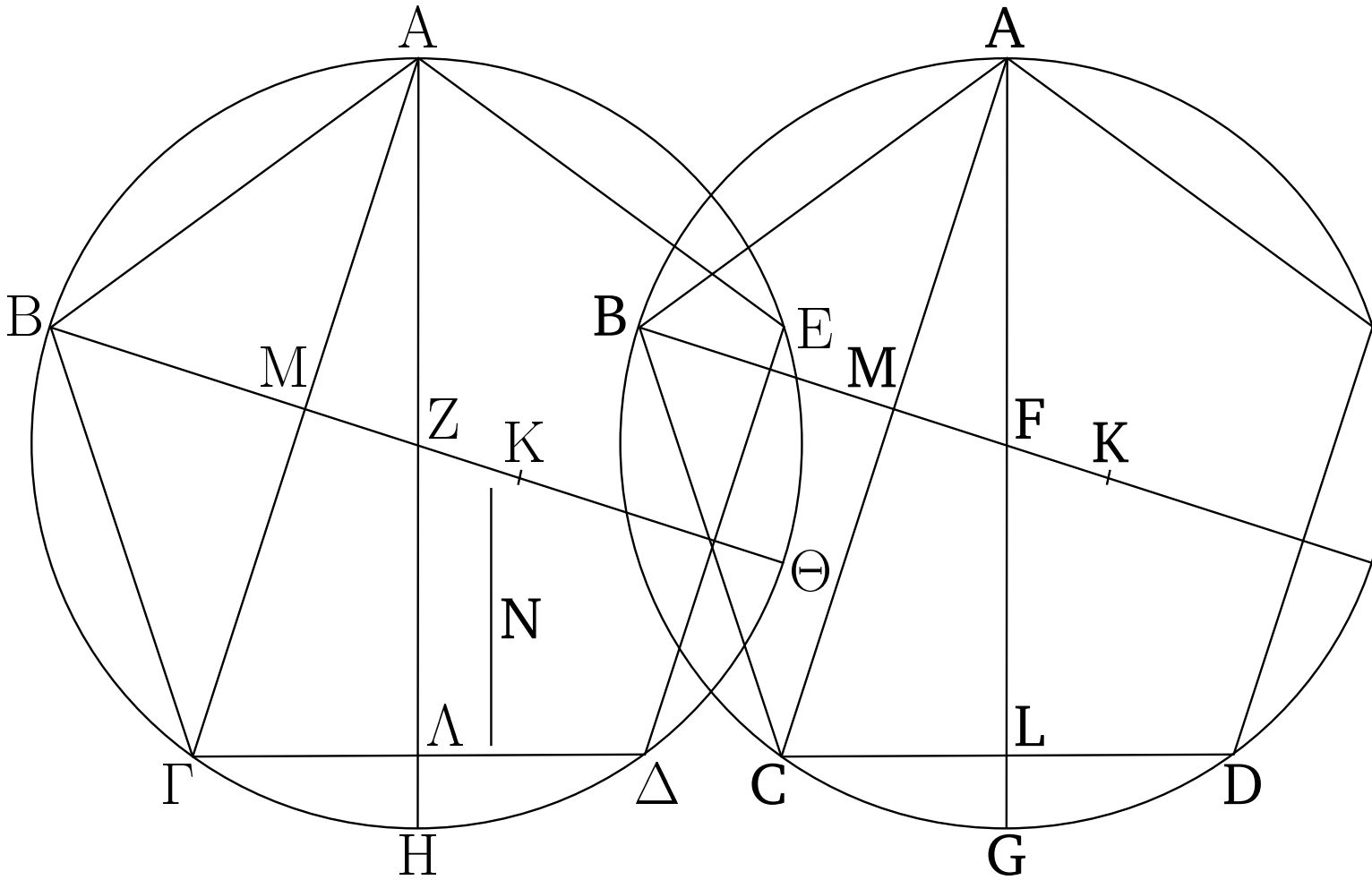
† If the circle is of unit radius then the side of the pentagon is $(1/2) \sqrt{10 - 2\sqrt{5}}$.

11.

Ἐάν εἰς κύκλον <ρητὴν>έθοντα τὴν διάμετρον πεντάγωνον ἰσόπλευρον ἐγγεγράφῃ, ἡ τοῦ πενταγώνου πλευρὰ >άλογός ἐστιν ἢ καλουμένη ἐλάσσων.

Proposition 11

If an equilateral pentagon is inscribed in a circle which has a rational diameter then the side of the pentagon is that irrational (straight-line) called minor.



Εἰς γὰρ κύκλον τὸν ΑΒΓΔΕ <ρητὴν>έθοντα τὴν διάμετρον πεντάγωνον ἰσόπλευρον ἐγγεγράφθω τὸ ΑΒΓΔΕ; λέγω, <ὅτι ἡ τοῦ [ΑΒΓΔΕ] πενταγώνου πλευρὰ >άλογός ἐστιν ἢ καλουμένη ἐλάσσων.

Εἰλήφθω γὰρ τὸ κέντρον τοῦ κύκλου τὸ Ζ σημεῖον, καὶ ἐπεζεύθωσαν αἱ ΑΖ, ΖΒ καὶ διήθωσαν ἐπὶ τὰ Η, Θ σημεῖα, καὶ ἐπεζεύθω ἡ ΑΓ, καὶ κείσθω τῆς ΑΖ τέταρτον μέρος ἡ ΖΚ. <ρητὴ δὲ ἡ ΑΖ; <ρητὴ >άρα καὶ ἡ ΖΚ. >έστι δὲ καὶ ἡ ΒΖ <ρητὴ; <όλη >άρα ἡ ΒΚ <ρητὴ ἐστίν. καὶ ἐπεὶ >ίση ἐστὶν ἡ ΑΓΗ περιφέρεια τῇ ΑΔΗ περιφέρειᾳ, <ὦν ἡ ΑΒΓ τῇ ΑΕΔ ἐστίν >ίση, λοιπὴ >άρα ἡ ΓΗ λοιπῇ τῇ ΗΔ ἐστίν >ίση. καὶ ἔαν ἐπιζεύχωμεν τὴν ΑΔ, συνάγονται ὀρθαὶ αἱ πρὸς τῷ Λ γωνίαι, καὶ διπλὴ ἡ ΓΔ τῆς ΓΑ. διὰ τὰ αὐτὰ δὲ καὶ αἱ πρὸς τῷ Μ ὀρθαὶ εἰσιν, καὶ διπλὴ ἡ ΑΓ τῆς ΓΜ. ἐπεὶ ο>ύν >ίση ἐστὶν ἡ ὑπὸ ΑΛΓ γωνία τῇ ὑπὸ ΑΜΖ, κοινὴ δὲ τῶν δύο τριγώνων τοῦ τε ΑΓΛ καὶ τοῦ ΑΜΖ ἡ ὑπὸ ΑΛΓ, λοιπὴ >άρα ἡ ὑπὸ ΑΓΛ λοιπῇ τῇ ὑπὸ ΜΖΑ ἐστίν >ίση; ἰσογώνιον >άρα ἐστὶ τὸ ΑΓΛ τρίγωνον τῷ ΑΜΖ τριγώνῳ; ἀνάλογον >άρα ἐστὶν ὡς ἡ ΛΓ πρὸς ΓΑ, ο<ύτως ἡ ΜΖ πρὸς ΖΑ; καὶ τῶν ἡγουμένων τὰ διπλάσια; ὡς >άρα ἡ τῆς ΛΓ διπλὴ πρὸς τὴν ΓΑ, ο<ύτως ἡ τῆς ΜΖ διπλὴ πρὸς τὴν ΖΑ. ὡς δὲ ἡ τῆς ΜΖ διπλὴ πρὸς τὴν ΖΑ, ο<ύτως ἡ ΜΖ πρὸς τὴν ἡμίσειαν τῆς ΖΑ; καὶ ὡς >άρα ἡ τῆς ΛΓ διπλὴ πρὸς

For let the equilateral pentagon $ABCDE$ be inscribed in the circle $ABCDE$ which has a rational diameter. I say that the side of pentagon $[ABCDE]$ is that irrational (straight-line) called minor.

For let the center of the circle, point F , be found [Prop. 3.1]. And let AF and FB be joined. And let them be drawn across to points G and H (respectively). And let AC be joined. And let FK made (equal) to the fourth part of AF . And AF (is) rational. FK (is) thus also rational. And BF is also rational. Thus, the whole of BK is rational. And since circumference ACG is equal to circumference ADG , of which ABC is equal to AED , the remainder CG is thus equal to the remainder GD . And if we join AD then the angles at L are inferred (to be) right-angles, and CD (is inferred to be) double CL [Prop. 1.4]. So, for the same (reasons), the (angles) at M are also right-angles, and AC (is) double CM . Therefore, since angle ALC (is) equal to AMF , and (angle) LAC (is) common to the two triangles ACL and AMF , the remaining (angle) ACL is thus equal to the remaining (angle) MFA [Prop. 1.32]. Thus, triangle ACL is equiangular to triangle AMF . Thus, proportionally, as LC (is) to CA , so MF (is) to FA [Prop. 6.4]. And (we can take) the doubles of the leading (magnitudes).

τὴν ΓΑ, οὐτως ἢ ΜΖ πρὸς τὴν ἡμίσειαν τῆς ΖΑ; καὶ τῶν ἐπομένων τὰ ἡμίσεια; ὥς >άρα ἢ τῆς ΛΓ διπλὴ πρὸς τὴν ἡμίσειαν τῆς ΓΑ, οὐτως ἢ ΜΖ πρὸς τὸ τέτατρον τῆς ΖΑ. καὶ ἐστὶ τῆς μὲν ΛΓ διπλὴ ἢ ΔΓ, τῆς δὲ ΓΑ ἡμίσεια ἢ ΓΜ, τῆς δὲ ΖΑ τέτατρον μέροξ ἢ ΖΚ; >έστιν >άρα ὥς ἢ ΔΓ πρὸς τὴν ΓΜ, οὐτως ἢ ΜΖ πρὸς τὴν ΖΚ. συνθέντι καὶ ὥς συναμφοτέροξ ἢ ΔΓΜ πρὸς τὴν ΓΜ, οὐτως ἢ ΜΚ πρὸς ΚΖ; καὶ ὥς >άρα τὸ ἀπὸ συναμφοτέρου τῆς ΔΓΜ πρὸς τὸ ἀπὸ ΓΜ, οὐτως τὸ ἀπὸ ΜΚ πρὸς τὸ ἀπὸ ΚΖ. καὶ ἐπεὶ τῆς ὑπὸ δύο πλευρὰς τοῦ πενταγώνου ὑποτείνουσῃς, οἷον τῆς ΑΓ, >άκρον καὶ μέσον λόγον τεμνομένης τὸ μείζον τμήμα >ίσον ἐστὶ τῇ τοῦ πενταγώνου πλευρᾷ, τουτέστι τῇ ΔΓ, τὸ δὲ μείζον τμήμα προσλαβὼν τὴν ἡμίσειαν τῆς <όλῃς πενταπλάσιον δύναται τοῦ ἀπὸ τῆς ἡμισείας τῆς <όλῃς, καὶ ἐστὶν <όλῃς τῆς ΑΓ ἡμίσεια ἢ ΓΜ, τὸ >άρα ἀπὸ τῆς ΔΓΜ ὥς μιᾷ πενταπλάσιον ἐστὶ τοῦ ἀπὸ τῆς ΓΜ. ὥς δὲ τὸ ἀπὸ τῆς ΔΓΜ ὥς μιᾷ πρὸς τὸ ἀπὸ τῆς ΓΜ, οὐτως ἐδειθῇ τὸ ἀπὸ τῆς ΜΚ πρὸς τὸ ἀπὸ τῆς ΚΖ; πενταπλάσιον >άρα τὸ ἀπὸ τῆς ΜΚ τοῦ ἀπὸ τῆς ΚΖ. <ρητὸν δὲ τὸ ἀπὸ τῆς ΚΖ; <ρητὴ γὰρ ἢ διάμετροξ; <ρητὸν >άρα καὶ τὸ ἀπὸ τῆς ΜΚ; <ρητὴ >άρα ἐστὶν ἢ ΜΚ [δυνάμει μόνον]. καὶ ἐπεὶ τετραπλασία ἐστὶν ἢ ΒΖ τῆς ΖΚ, πενταπλασία >άρα ἐστὶν ἢ ΒΚ τῆς ΚΖ; εἰκοσιπενταπλάσιον >άρα τὸ ἀπὸ τῆς ΒΚ τοῦ ἀπὸ τῆς ΚΖ. πενταπλάσιον δὲ τὸ ἀπὸ τῆς ΜΚ τοῦ ἀπὸ τῆς ΚΖ; πενταπλάσιον >άρα τὸ ἀπὸ τῆς ΒΚ τοῦ ἀπὸ τῆς ΚΜ; τὸ >άρα ἀπὸ τῆς ΒΚ πρὸς τὸ ἀπὸ ΚΜ λόγον οὐκ >έθει, <ὸν τετράγωνοξ ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν; ἀσύμμετροξ >άρα ἐστὶν ἢ ΒΚ τῇ ΚΜ μήκει. καὶ ἐστὶ <ρητὴ ἑκατέρα αὐτῶν. αἱ ΒΚ, ΚΜ >άρα <ρηταὶ εἰσι δυνάμει μόνον σύμμετροι. ἔαν δὲ ἀπὸ <ρητῆς <ρητὴ ἀφαιρεθῇ δυνάμει μόνον σύμμετροξ ο>ῦσα τῇ <όλῃ, ἢ λοιπὴ >αλογόξ ἐστὶν ἀποτομή; ἀποτομή >άρα ἐστὶν ἢ ΜΒ, προσαρμόζουσα δὲ αὐτῇ ἢ ΜΚ. λέγω δὴ, <ὅτι καὶ τετάρτη. <ῶ δὴ μείζον ἐστὶ τὸ ἀπὸ τῆς ΒΚ τοῦ ἀπὸ τῆς ΚΜ, ἐκείνω >ίσον >έστω τὸ ἀπὸ τῆς Ν; ἢ ΒΚ >άρα τῆς ΚΜ μείζον δύναται τῇ Ν. καὶ ἐπεὶ σύμμετροξ ἐστὶν ἢ ΚΖ τῇ ΖΒ, καὶ συνθέντι σύμμετροξ ἐστὶ ἢ ΚΒ τῇ ΖΒ. ἀλλὰ ἢ ΒΖ τῇ ΒΘ σύμμετροξ ἐστὶν; καὶ ἢ ΒΚ >άρα τῇ ΒΘ σύμμετροξ ἐστὶν. καὶ ἐπεὶ πενταπλάσιον ἐστὶ τὸ ἀπὸ τῆς ΒΚ τοῦ ἀπὸ τῆς ΚΜ, τὸ >άρα ἀπὸ τῆς ΒΚ πρὸς τὸ ἀπὸ τῆς ΚΜ λόγον >έθει, <ὸν ε pr'oc <'en. >anast'eyanti >'ara t'o >ap'o t hc BK pr'oc t'o >ap'o t hc N l'ogon >'eqei, <'on ē pr'oc d̄, ο>uq <'on tetr'agwnoc pr'oc tetr'agwnon; >as'ummetroc >'ara >est'in <h BK t h| N; <h BK >'ara t hc KM me izon d'unatai t w| >ap'o >asumm'etrou <eat h|. >epe'i o> un <'olh <h BK t hc prosarmozo'ushe t hc KM me izon d'unatai t w| >ap'o >asumm'etrou <eat h|, ka'i <'olh <h BK s'ummet'roc >esti t h| >ekkeim'enh| <rht h| t h| BJ, >apotom'h >'ara tet'arth >est'in <h MB. t'o d'e <up'o <rht hc ka'i >apotom hc tet'arthc perieq'omenon >orjog'wnion >'alog'on >est'in, ka'i <h dunam'enh a>ut'o >'alog'oc >estin, kale itai d'e >el'attwn. d'unatai d'e t'o <up'o t wn JBM <h AB di'a t'o >epizeugnum'enhc t hc AJ >isog'wnion g'inesjai t'o ABJ tr'igwnon t w| ABM trig'wnw| ka'i e> inai <wc t'hn JB pr'oc t'hn BA, o<'utwc t'hn AB pr'oc t'hn BM.

Ἡ >άρα ΑΒ τοῦ πενταγώνου πλευρὰ >αλογόξ ἐστὶν ἢ καλουμένη ἐλάττων; <όπερ >έδει δεῖχαι.

Thus, as double LC (is) to CA , so double MF (is) to FA . And as double MF (is) to FA , so MF (is) to half of FA . And, thus, as double LC (is) to CA , so MF (is) to half of FA . And (we can take) the halves of the following (magnitudes). Thus, as double LC (is) to half of CA , so MF (is) to the fourth of FA . And DC is double LC , and CM half of CA , and FK the fourth part of FA . Thus, as DC is to CM , so MF (is) to FK . Via composition, as the sum of DCM (i.e., DC and CM) (is) to CM , so MK (is) to KF [Prop. 5.18]. And, thus, as the (square) on the sum of DCM (is) to the (square) on CM , so the (square) on MK (is) to the (square) on KF . And since the greater piece of a (straight-line) subtending two sides of a pentagon, such as AC , (which is) cut in extreme and mean ratio is equal to the side of the pentagon [Prop. 13.8]—that is to say, to DC —and the square on the greater piece added to half of the whole is five times the (square) on half of the whole [Prop. 13.1], and CM (is) half of the whole, AC , thus the (square) on DCM , (taken) as one, is five times the (square) on CM . And the (square) on DCM , (taken) as one, (is) to the (square) on CM , so the (square) on MK was shown (to be) to the (square) on KF . Thus, the (square) on MK (is) five times the (square) on KF . And the square on KF (is) rational. For the diameter (is) rational. Thus, the (square) on MK (is) also rational. Thus, MK is rational [in square only]. And since BF is four times FK , BK is thus five times KF . Thus, the (square) on BK (is) twenty-five times the (square) on KF . And the (square) on MK (is) five times the square on KF . Thus, the (square) on BK (is) five times the (square) on KM . Thus, the (square) on BK does not have to the (square) on KM the ratio which a square number (has) to a square number. Thus, BK is incommensurable in length with KM [Prop. 10.9]. And each of them is a rational (straight-line). Thus, BK and KM are rational (straight-lines which are) commensurable in square only. And if from a rational (straight-line) a rational (straight-line) is subtracted, which is commensurable in square only with the whole, then the remainder is that irrational (straight-line called) an apotome [Prop. 10.73]. Thus, MB is an apotome, and MK its attachment. So, I say that (it is) also a fourth (apotome). So, let the (square) on N be (made) equal to that (magnitude) by which the (square) on BK is greater than the (square) on KM . Thus, the square on BK is greater than the (square) on KM by the (square) on N . And since KF is commensurable (in length) with FB then, via composition, KB is also commensurable (in length) with FB [Prop. 10.15]. But, BF is commensurable (in length) with BH . Thus, BK is also commensurable (in length) with BH [Prop. 10.12]. And since the (square) on BK is five times the (square) on KM , the (square) on BK thus has to the (square) on KM the ratio which 5 (has) to one. Thus, via conversion, the (square) on BK has to the (square) on N the ratio which 5 (has) to 4 [Prop. 5.19 corr.], which is not (that) of a square (number) to a square (number). BK is thus incommensurable (in length) with N [Prop. 10.9]. Thus, the square on BK is greater than the (square) on KM by the (square) on (some straight-line which is) incommensurable (in length) with (BK). Therefore, since the square on the whole, BK , is greater than the (square) on the attachment, KM , by the (square) on (some straight-line

which is) incommensurable (in length) with (BK) , and the whole, BK , is commensurable (in length) with the (previously) laid down rational (straight-line) BH , MB is thus a fourth apotome [Def. 10.14]. And the rectangle contained by a rational (straight-line) and a fourth apotome is irrational, and its square-root is that irrational (straight-line) called minor [Prop. 10.94]. And the square on AB is the rectangle contained by HBM , on account of joining AH , (so that) triangle ABH becomes equiangular with triangle ABM [Prop. 6.8], and (proportionally) as HB is to BA , so AB (is) to BM .

Thus, the side AB of the pentagon is that irrational (straight-line) called minor.[†] (Which is) the very thing it was required to show.

[†] If the circle has unit radius then the side of the pentagon is $(1/2)\sqrt{10 - 2\sqrt{5}}$. However, this length can be written in the “minor” form (see Prop. 10.94) $(\rho/\sqrt{2})\sqrt{1 + k/\sqrt{1 + k^2}} - (\rho/\sqrt{2})\sqrt{1 - k/\sqrt{1 + k^2}}$, with $\rho = \sqrt{5}/2$ and $k = 2$.

12.

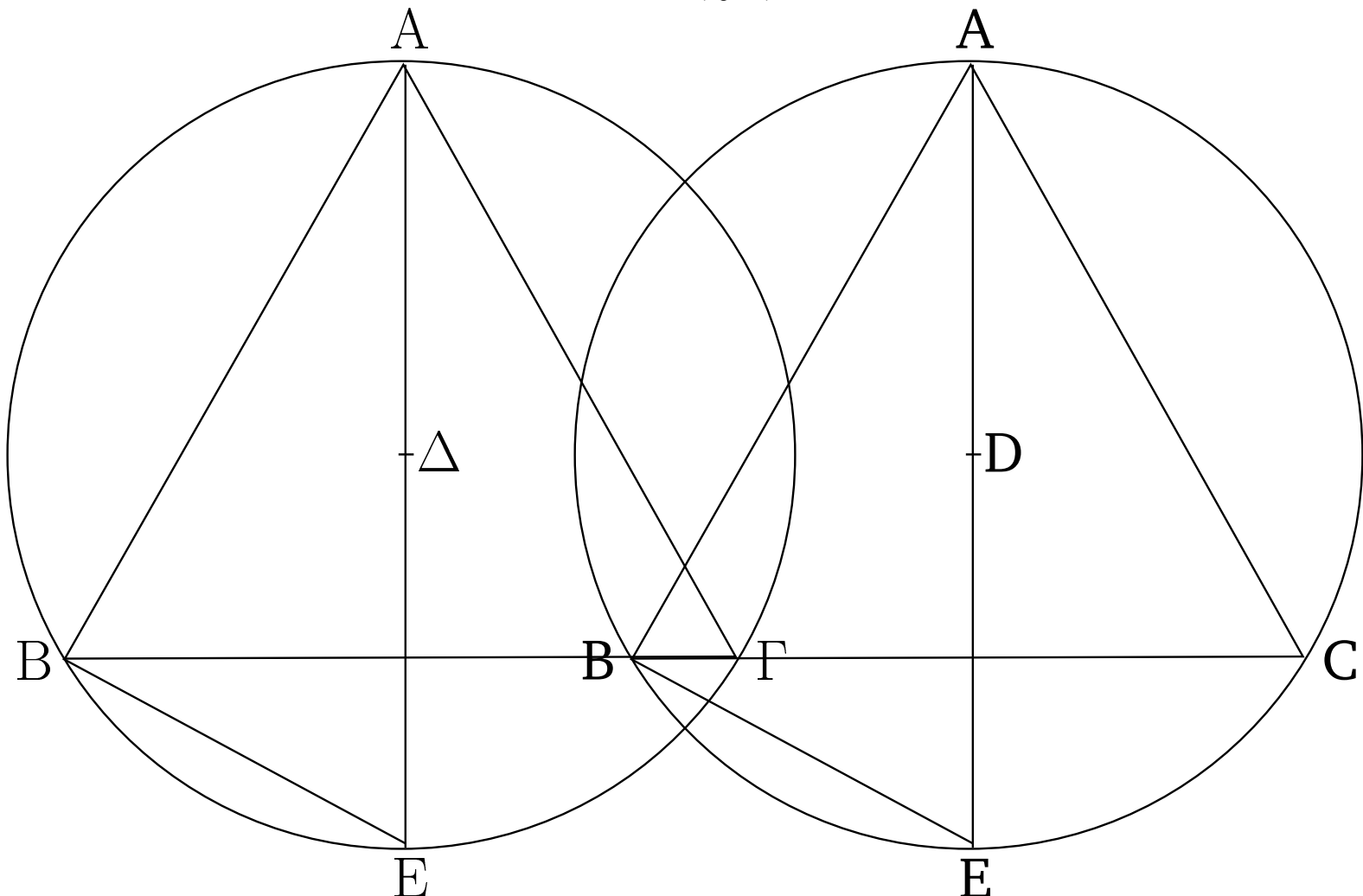
Ἐὰν εἰς κύκλον τρίγωνον ἰσόπλευρον ἐγγραφή, ἡ τοῦ τριγώνου πλευρὰ δυνάμει τριπλασίων ἐστὶ τῆς ἐκ τοῦ κέντρου τοῦ κύκλου.

Ἐστω κύκλος ὁ $AB\Gamma$, καὶ εἰς αὐτὸν τρίγωνον ἰσόπλευρον ἐγγεγράφθω τὸ $AB\Gamma$; λέγω, <ὅτι τοῦ $AB\Gamma$ τριγώνου μία πλευρὰ δυνάμει τριπλασίων ἐστὶ τῆς ἐκ τοῦ κέντρου τοῦ $AB\Gamma$ κύκλου.

Proposition 12

If an equilateral triangle is inscribed in a circle then the square on the side of the triangle is three times the (square) on the radius of the circle.

Let there be a circle ABC , and let the equilateral triangle ABC have been inscribed in it [Prop. 4.2]. I say that the square on one side of triangle ABC is three times the (square) on the radius of circle ABC .



Εἰλήφθω γὰρ τὸ κέντρον τοῦ $AB\Gamma$ κύκλου τὸ Δ , καὶ

For let the center, D , of circle ABC be found [Prop. 3.1].

ἐπιζευθθεῖσα ἡ AD διήθθω ἐπὶ τὸ E , καὶ ἐπεζεύθθω ἡ BE .
 Καὶ ἐπεὶ ἰσόπλευρόν ἐστι τὸ $ABΓ$ τρίγωνον, ἡ BEG >ἄρα
 περιφέρεια τρίτον μέρος ἐστὶ τῆς τοῦ $ABΓ$ κύκλου περιφερείας
 ἢ >ἄρα BE περιφέρεια <έκτον ἐστὶ μέρος τῆς τοῦ κύκλου
 περιφερείας; ἐχαγόνου >ἄρα ἐστὶν ἡ BE εὐθεῖα; >ίση >ἄρα
 ἐστὶ τῇ ἐκ τοῦ κέντρου τῇ DE . καὶ ἐπεὶ διπλῇ ἐστὶν ἡ AE
 τῆς DE , τετραπλάσιον ἐστὶ τὸ ἀπὸ τῆς AE τοῦ ἀπὸ τῆς ED ,
 τουτέστι τοῦ ἀπὸ τῆς BE . >ίσον δὲ τὸ ἀπὸ τῆς AE τοῖς ἀπὸ
 τῶν AB , BE ; τὰ >ἄρα ἀπὸ τῶν AB , BE τετραπλάσιά ἐστι
 τοῦ ἀπὸ τῆς BE . διελόντι >ἄρα τὸ ἀπὸ τῆς AB τριπλάσιόν
 ἐστὶ τοῦ ἀπὸ BE . >ίση δὲ ἡ BE τῇ DE ; τὸ >ἄρα ἀπὸ τῆς AB
 τριπλάσιόν ἐστὶ τοῦ ἀπὸ τῆς DE .
 Ἡ >ἄρα τοῦ τριγώνου πλευρὰ δυνάμει τριπλασία ἐστὶ τῆς
 ἐκ τοῦ κέντρου [τοῦ κύκλου]; <όπερ >έδει δεῖχαι.

And AD (being) joined, let it be drawn across to E . And let
 BE be joined.

And since triangle ABC is equilateral, circumference
 BEC is thus the third part of the circumference of circle ABC .
 Thus, circumference BE is the sixth part of
 the circumference of the circle. Thus, straight-line BE is
 (the side) of a hexagon. Thus, it is equal to the radius DE
 [Prop. 4.15 corr.]. And since AE is double DE , the (square)
 on AE is four times the (square) on ED —that is to say, of
 the (square) on BE . And the (square) on AE (is) equal to
 the (sum of the squares) on AB and BE [Props. 3.31, 1.47].
 Thus, the (sum of the squares) on AB and BE is four times
 the (square) on BE . Thus, via separation, the (square) on
 AB is three times the (square) on BE . And BE (is) equal
 to DE . Thus, the (square) on AB is three times the (square)
 on DE .

Thus, the square on the side of the triangle is three times the
 (square) on the radius [of the circle]. (Which is) the very
 thing it was required to show.

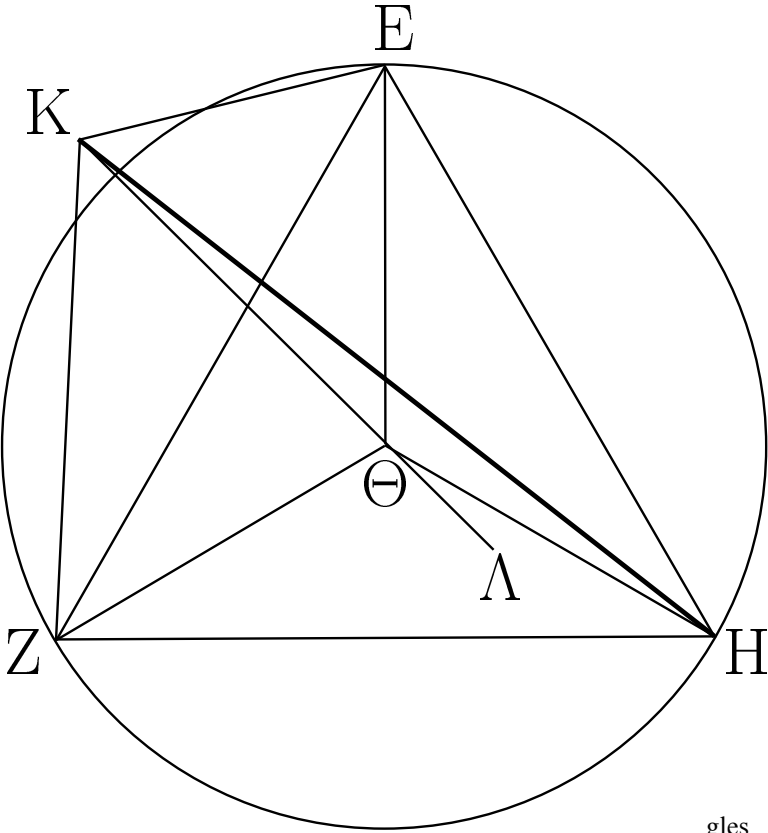
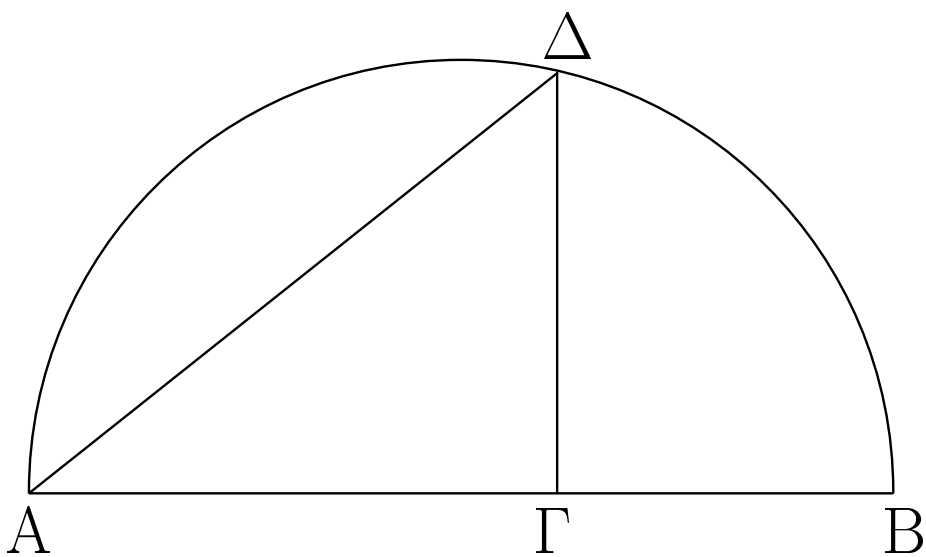
13.

Proposition 13

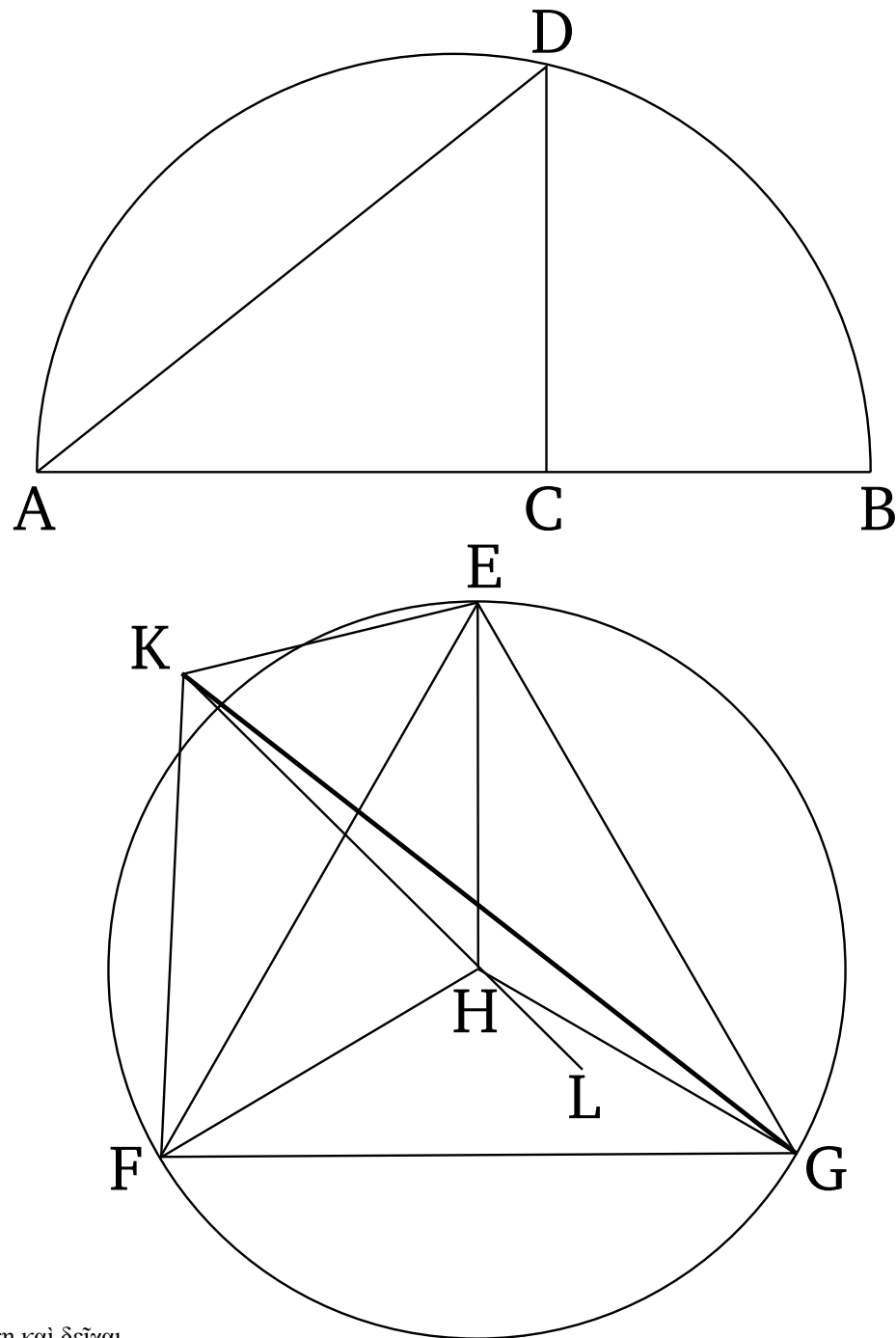
Πυραμίδα συστήσασθαι καὶ σφαῖρα περιλαβεῖν τῇ δοθείσῃ
 καὶ δεῖχαι, <ὅτι ἡ τῆς σφαίρας διάμετρος δυνάμει ἡμιολία
 ἐστὶ τῆς πλευρᾶς τῆς πυραμίδος.
 Ἐκκείσθω ἡ τῆς δοθείσης σφαίρας διάμετρος ἡ AB , καὶ
 τετμήσθω κατὰ τὸ $Γ$ σημεῖον, <ώστε διπλασίαν ε>ῖναι τὴν
 AG τῆς GB ; καὶ γεγράθω ἐπὶ τῆς AB ἡμικύκλιον τὸ $AΔB$,
 καὶ >ήθθω ἀπὸ τοῦ $Γ$ σημείου τῇ AB πρὸς ὀρθὰς ἡ $ΓΔ$,
 καὶ ἐπεζεύθθω ἡ $ΔA$; καὶ ἐκκείσθω κύκλος ὁ EZH >ίσην
 >έθων τὴν ἐκ τοῦ κέντρου τῇ $ΔΓ$, καὶ ἐγγεγράθω εἰς τὸν
 EZH κύκλον τρίγωνον ἰσόπλευρον τὸ EZH ; καὶ εἰλήθω
 τὸ κέντρον τοῦ κύκλου τὸ $Θ$ σημεῖον, καὶ ἐπεζεύθθωσαν
 αἱ $EΘ$, $ΘZ$, $ΘH$; καὶ ἀνεστάτω ἀπὸ τοῦ $Θ$ σημείου τῶν
 EZH κύκλου ἐπιπέδῳ πρὸς ὀρθὰς ἡ $ΘK$, καὶ ἀφῆρήσθω ἀπὸ
 τῆς $ΘK$ τῇ AG εὐθείᾳ >ίση ἡ $ΘK$, καὶ ἐπεζεύθθωσαν αἱ KE ,
 KZ , KH . καὶ ἐπεὶ ἡ $KΘ$ ὀρθή ἐστι πρὸς τὸ τοῦ EZH κύκλου
 ἐπίπεδον, καὶ πρὸς πάσας >ἄρα τὰς ἀπτομένας αὐτῆς
 εὐθείας καὶ >ύσας ἐν τῷ τοῦ EZH κύκλου ἐπιπέδῳ ὀρθὰς
 ποιήσει γωνίας. <ἀπτεται δὲ αὐτῆς ἐκάστη τῶν $ΘE$, $ΘZ$,
 $ΘH$; ἡ $ΘK$ >ἄρα πρὸς ἐκάστη τῶν $ΘE$, $ΘZ$, $ΘH$ ὀρθή ἐστίν.
 καὶ ἐπεὶ >ίση ἐστὶν ἡ μὲν AG τῇ $ΘK$, ἡ δὲ $ΓΔ$ τῇ $ΘE$, καὶ
 ὀρθὰς γωνίας περιέθουσιν, βάσις >ἄρα ἡ $ΔA$ βάσει τῇ KE
 ἐστὶν >ίση. διὰ τὰ αὐτὰ δὴ καὶ ἑκατέρα τῶν KZ , KH τῇ $ΔA$
 ἐστὶν >ίση; αἱ τρεῖς >ἄρα αἱ KE , KZ , KH >ίσαι ἀλλήλαις
 εἰσίν. καὶ ἐπεὶ διπλῇ ἐστὶν ἡ AG τῆς GB , τριπλῇ >ἄρα ἡ AB
 τῆς BG . ὥς δὲ ἡ AB πρὸς τὴν BG , ο<ύτως τὸ ἀπὸ τῆς AD
 πρὸς τὸ ἀπὸ τῆς $ΔΓ$, ὥς ἐχῆς δειθῆσεται. τριπλάσιον >ἄρα
 τὸ ἀπὸ τῆς AD τοῦ ἀπὸ τῆς $ΔΓ$. >έστι δὲ καὶ τὸ ἀπὸ τῆς ZE
 τοῦ ἀπὸ τῆς $EΘ$ τριπλάσιον, καὶ ἐστὶν >ίση ἡ $ΔΓ$ τῇ $EΘ$;
 >ίση >ἄρα καὶ ἡ $ΔA$ τῇ EZ . ἀλλὰ ἡ $ΔA$ ἐκάστη τῶν KE ,
 KZ , KH ἔδειθθη >ίση; καὶ ἐκάστη >ἄρα τῶν EZ , ZH , HE
 ἐκάστη τῶν KE , KZ , KH ἐστὶν >ίση; ἰσόπλευρα >ἄρα ἐστὶ
 τὰ τέσσαρα τρίγωνα τὰ EZH , KEZ , KZH , KEH . πυραμὶς
 >ἄρα συνέσταται ἐκ τεσσάρων τριγώνων ἰσοπλῶν, <ἥς
 βάσις μὲν ἐστὶ τὸ EZH τρίγωνον, κορυφή δὲ τὸ K σημεῖον.

To construct a (regular) pyramid (*i.e.*, a tetrahedron), and to
 enclose (it) in a given sphere, and to show that the square
 on the diameter of the sphere is one and a half times the
 (square) on the side of the pyramid.

Let the diameter AB of the given sphere be laid out, and let
 it be cut at point C such that AC is double CB [Prop. 6.10].
 And let the semi-circle ADB be drawn on AB . And let CD
 be drawn from point C at right-angles to AB . And let DA
 be joined. And let the circle EFG be laid down having a
 radius equal to DC , and let the equilateral triangle EFG
 be inscribed in circle EFG [Prop. 4.2]. And let the center
 of the circle, point H , be found [Prop. 3.1]. And let EH ,
 HF , and HG be joined. And let HK be set up, at point
 H , at right-angles to the plane of circle EFG [Prop. 11.12].
 And let HK , equal to the straight-line AC , be cut off from
 HK . And let KE , KF , and KG be joined. And since KH
 is at right-angles to the plane of circle EFG , it will thus
 also make right-angles with all of the straight-lines joining
 it (which are) also in the plane of circle EFG [Def. 11.3].
 And HE , HF , and HG each join it. Thus, HK is at right-
 angles to each of HE , HF , and HG . And since AC is equal
 to HK , and CD to HE , and they contain right-angles, the
 base DA is thus equal to the base KE [Prop. 1.4]. So, for
 the same (reasons), KF and KG is each equal to DA . Thus,
 the three (straight-lines) KE , KF , and KG are equal to one
 another. And since AC is double CB , AB (is) thus triple
 BC . And as AB (is) to BC , so the (square) on AD (is)
 to the (square) on DC , as will be shown later [see lemma].
 Thus, the (square) on AD (is) three times the (square) on
 DC . And the (square) on FE is also three times the (square)
 on EH [Prop. 13.12], and DC is equal to EH . Thus, DA
 (is) also equal to EF . But, DA was shown (to be) equal to
 each of KE , KF , and KG . Thus, EF , FG , and GE are
 equal to KE , KF , and KG , respectively. Thus, the four
 triangles EFG , KEF , KFG , and KEG are equilateral.
 Thus, a pyramid, whose base is triangle EFG , and apex the
 point K , has been constructed from four equilateral trian-



gles.



Δεῖ δὴ αὐτὴν καὶ σφαῖρα περιλαβεῖν τῇ δοθείσῃ καὶ δεῖξαι,
 <ὅτι ἡ τῆς σφαίρας διάμετρος ἡμολία ἐστὶ δυνάμει τῆς
 πλευρᾶς τῆς πυραμίδος.

Ἐκβεβλήσθω γάρ ἐπ' εὐθείᾳ τῇ ΚΘ εὐθεῖα ἡ ΘΛ, καὶ
 κείσθω τῇ ΓΒ >ῖση ἡ ΘΛ. καὶ ἐπεὶ ἐστὶν ὥς ἡ ΑΓ πρὸς
 τὴν ΓΔ, ο<ύτωξ ἡ ΓΔ πρὸς τὴν ΓΒ, >ῖση δὲ ἡ μὲν ΑΓ
 τῇ ΚΘ, ἡ δὲ ΓΔ τῇ ΘΕ, ἡ δὲ ΓΒ τῇ ΘΛ, >έστιν >άρα
 ὥς ἡ ΚΘ πρὸς τὴν ΘΕ, ο<ύτωξ ἡ ΕΘ πρὸς τὴν ΘΛ; τὸ
 >άρα ὑπὸ τῶν ΚΘ, ΘΛ >ῖσον ἐστὶ τῷ ὑπὸ τῆς ΕΘ. καὶ
 ἐστὶν ὀρθὴ ἑκάτερα τῶν ὑπὸ ΚΘΕ, ΕΘΛ γωνιῶν; τὸ >άρα
 ἐπὶ τῆς ΚΛ γραφόμενον ἡμικύκλιον <ήχει καὶ διὰ τοῦ Ε
 [ἐπειδὴ περ ἔαν ἐπιζεύχωμεν τὴν ΕΛ, ὀρθὴ γίνεται ἡ ὑπὸ
 ΛΕΚ γωνία διὰ τὸ ἰσογώνιον γίνεσθαι τὸ ΕΛΚ τρίγωνον
 ἑκατέρῳ τῶν ΕΛΘ, ΕΘΚ τριγώνων]. ἔαν δὴ μενούσῃς
 τῆς ΚΛ περιενεθῇ τὸ ἡμικύκλιον εἰς τὸ αὐτὸ πάλιν
 ἀποκατασταθῇ, <όθεν >ήρχατο φέρεσθαι, <ήχει καὶ διὰ
 τῶν Ζ, Η σημείων ἐπιζευγνυμένων τῶν ΖΛ, ΛΗ καὶ ὀρθῶν
 ὁμοίως γινομένων τῶν πρὸς τοῖς Ζ, Η γωνιῶν; καὶ >έσται

So, it is also necessary to enclose it in the given sphere, and
 to show that the square on the diameter of the sphere is one
 and a half times the (square) on the side of the pyramid.

For let the straight-line HL be produced in a straight-line
 with KH , and let HL be made equal to CB . And since as
 AC (is) to CD , so CD (is) to CB [Prop. 6.8 corr.], and AC
 (is) equal to KH , and CD to HE , and CB to HL , thus
 as KH is to HE , so EH (is) to HL . Thus, the (rectan-
 gle contained) by KH and HL is equal to the (square) on
 EH [Prop. 6.17]. And each of the angles KHE and EHL
 is a right-angle. Thus, the semi-circle drawn on KL will
 also pass through E [inasmuch as if we join EL then the
 angle LEK becomes a right-angle, on account of triangle
 ELK becoming equiangular to each of the triangles ELH
 and EHK [Props. 6.8, 3.31]]. So, if KL remains (fixed),
 and the semi-circle is carried around, and again established
 at the same (position) from which it began to be moved, it

ἡ πυραμὶς σφαῖρα περιελημμένη τῇ δοθείσῃ. ἡ γὰρ ΚΛ τῆς σφαίρας διάμετρος ὅση ἐστὶ τῇ τῆς δοθείσης σφαίρας διαμέτρῳ τῇ ΑΒ, ἐπειδὴ περ τῇ μὲν ΑΓ ὅση κεῖται ἡ ΚΘ, τῇ δὲ ΓΒ ἡ ΘΛ.

Λέγω δὴ, <ὅτι ἡ τῆς σφαίρας διάμετρος ἡμολία ἐστὶ δυνάμει τῆς πλευρᾶς τῆς πυραμίδος.

Ἐπεὶ γὰρ διπλὴ ἐστὶν ἡ ΑΓ τῆς ΓΒ, τριπλὴ ὅρα ἐστὶν ἡ ΑΒ τῆς ΒΓ; ἀναστρέψαντι ἡμολία ὅρα ἐστὶν ἡ ΒΑ τῆς ΑΓ. ὥς δὲ ἡ ΒΑ πρὸς τὴν ΑΓ, οὕτως τὸ ἀπὸ τῆς ΒΑ πρὸς τὸ ἀπὸ τῆς ΑΔ [ἐπειδὴ περ ἐπιζευγμένης τῆς ΔΒ ἐστὶν ὥς ἡ ΒΑ πρὸς τὴν ΑΔ, οὕτως ἡ ΔΑ πρὸς τὴν ΑΓ διὰ τὴν ὁμοιότητα τῶν ΔΑΒ, ΔΑΓ τριγώνων, καὶ εἶναι ὥς τὴν πρώτην πρὸς τὴν τρίτην, οὕτως τὸ ἀπὸ τῆς τρώτης πρὸς τὸ ἀπὸ τῆς δευτέρας]. ἡμολιον ὅρα καὶ τὸ ἀπὸ τῆς ΒΑ τοῦ ἀπὸ τῆς ΑΔ. καὶ ἐστὶν ἡ μὲν ΒΑ ἡ τῆς δοθείσης σφαίρας διάμετρος, ἡ δὲ ΑΔ ὅση τῇ πλευρᾷ τῆς πυραμίδος.

Ἦ ὅρα τῆς σφαίρας διάμετρος ἡμολία ἐστὶ τῆς πλευρᾶς τῆς πυραμίδος; <ὅπερ >εἰδει δεῖχαι.

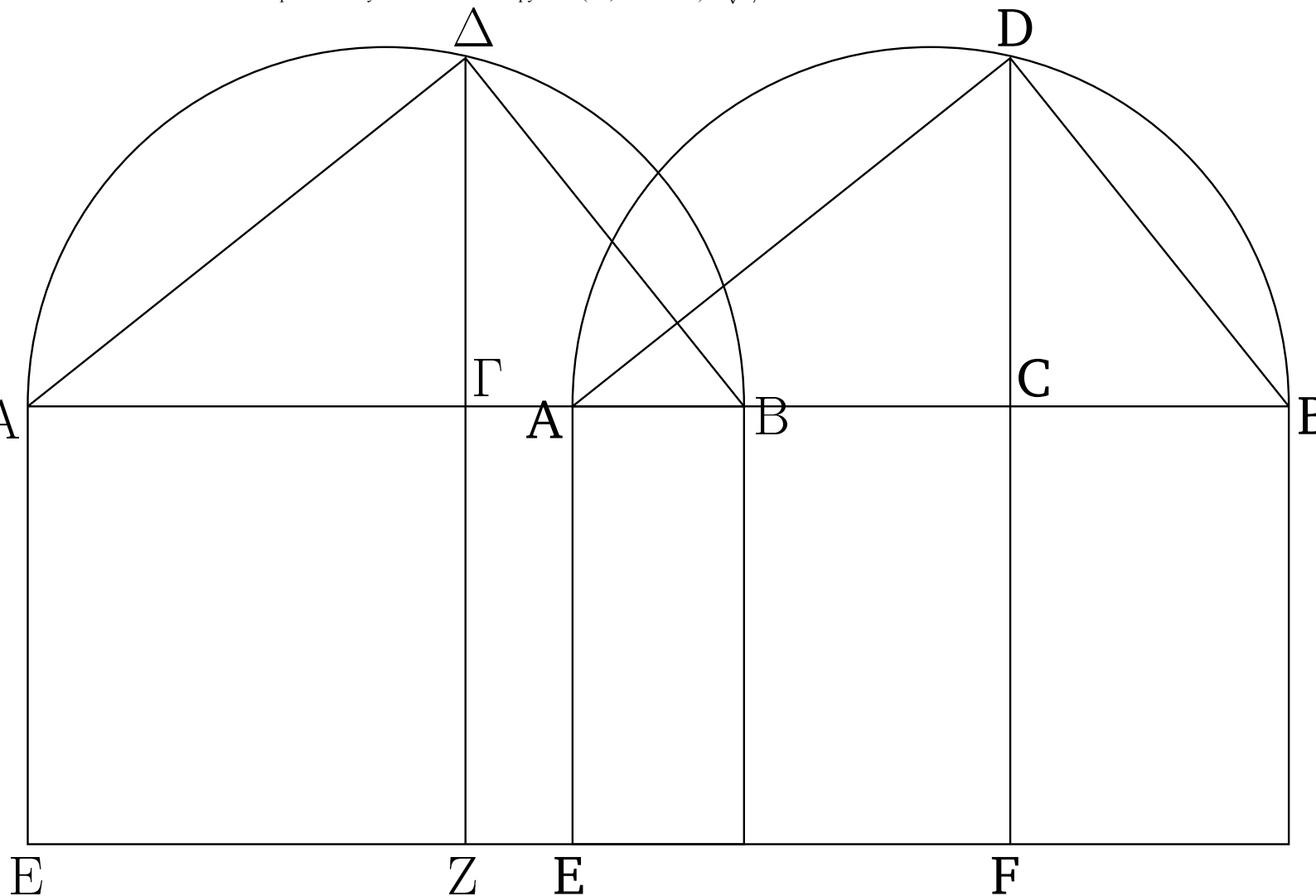
will also pass through points *F* and *G*, (because) if *FL* and *LG* are joined, the angles at *F* and *G* will similarly become right-angles. And the pyramid will be enclosed by the given sphere. For the diameter, *KL*, of the sphere is equal to the diameter, *AB*, of the given sphere—inasmuch as *KH* was made equal to *AC*, and *HL* to *CB*.

So, I say that the square on the diameter of the sphere is one and a half times the (square) on the side of the pyramid.

For since *AC* is double *CB*, *AB* is thus triple *BC*. Thus, via conversion, *BA* is one and a half times *AC*. And as *BA* (is) to *AC*, so the (square) on *BA* (is) to the (square) on *AD* [inasmuch as if *DB* is joined then as *BA* is to *AD*, so *DA* (is) to *AC*, on account of the similarity of triangles *DAB* and *DAC*. And as the first is to the third (of four proportional magnitudes), so the (square) on the first (is) to the (square) on the second.] Thus, the (square) on *BA* (is) also one and a half times the (square) on *AD*. And *BA* is the diameter of the given sphere, and *AD* (is) equal to the side of the pyramid.

Thus, the square on the diameter of the sphere is one and a half times the (square) on the side of the pyramid.[†] (Which is) the very thing it was required to show.

[†] If the radius of the sphere is unity then the side of the pyramid (*i.e.*, tetrahedron) is $\sqrt{8/3}$.



Λήμμα.

Lemma

Δεικτέον, <ὅτι ἐστὶν ὥς ἡ ΑΒ πρὸς τὴν ΒΓ, οὕτως τὸ ἀπὸ

It must be shown that as *AB* is to *BC*, so the (square) on

τῆς AD πρὸς τὸ ἀπὸ τῆς $\Delta\Gamma$.

Ἐκκείσθω γὰρ ἡ τοῦ ἡμικυκλίου καταγραφὴ, καὶ ἐπεζεύθω ἡ ΔB , καὶ ἀναγεγράφθω ἀπὸ τῆς AG τετράγωνον τὸ EG , καὶ συμπληρώσθω τὸ ZB παραλληλόγραμμον. ἐπεὶ οὖν διὰ τὸ ἰσογώνιον εἶναι τὸ ΔAB τρίγωνον τῷ ΔAG τριγώνῳ ἐστὶν ὡς ἡ BA πρὸς τὴν AD , οὐτωὶς ἡ ΔA πρὸς τὴν AG , τὸ >άρα ὑπὸ τῶν BA , AG >ἴσον ἐστὶ τῷ ἀπὸ τῆς AD . καὶ ἐπεὶ ἐστὶν ὡς ἡ AB πρὸς τὴν BG , οὐτωὶς τὸ EB πρὸς τὸ BZ , καὶ ἐστὶ τὸ μὲν EB τὸ ὑπὸ τῶν BA , AG ; >ἴση γὰρ ἡ EA τῇ AG ; τὸ δὲ BZ τὸ ὑπὸ τῶν AG , GB , ὡς >άρα ἡ AB πρὸς τὴν BG , οὐτωὶς τὸ ὑπὸ τῶν BA , AG πρὸν τὸ ὑπὸ τῶν AG , GB . καὶ ἐστὶ τὸ μὲν ὑπὸ τῶν BA , AG >ἴσον τῷ ἀπὸ τῆς AD , τὸ δὲ ὑπὸ τῶν AGB >ἴσον τῷ ἀπὸ τῆς $\Delta\Gamma$; ἡ γὰρ $\Delta\Gamma$ κάθετος τῶν τῆς βάσεως τμημάτων τῶν AG , GB μέση ἀνάλογόν ἐστι διὰ τὸ ὀρθὴν εἶναι τὴν ὑπὸ ADB . ὡς >άρα ἡ AB πρὸς τὴν BG , οὐτωὶς τὸ ἀπὸ τῆς AD πρὸς τὸ ἀπὸ τῆς $\Delta\Gamma$; <όπερ >έδει δεῖξαι.

14.

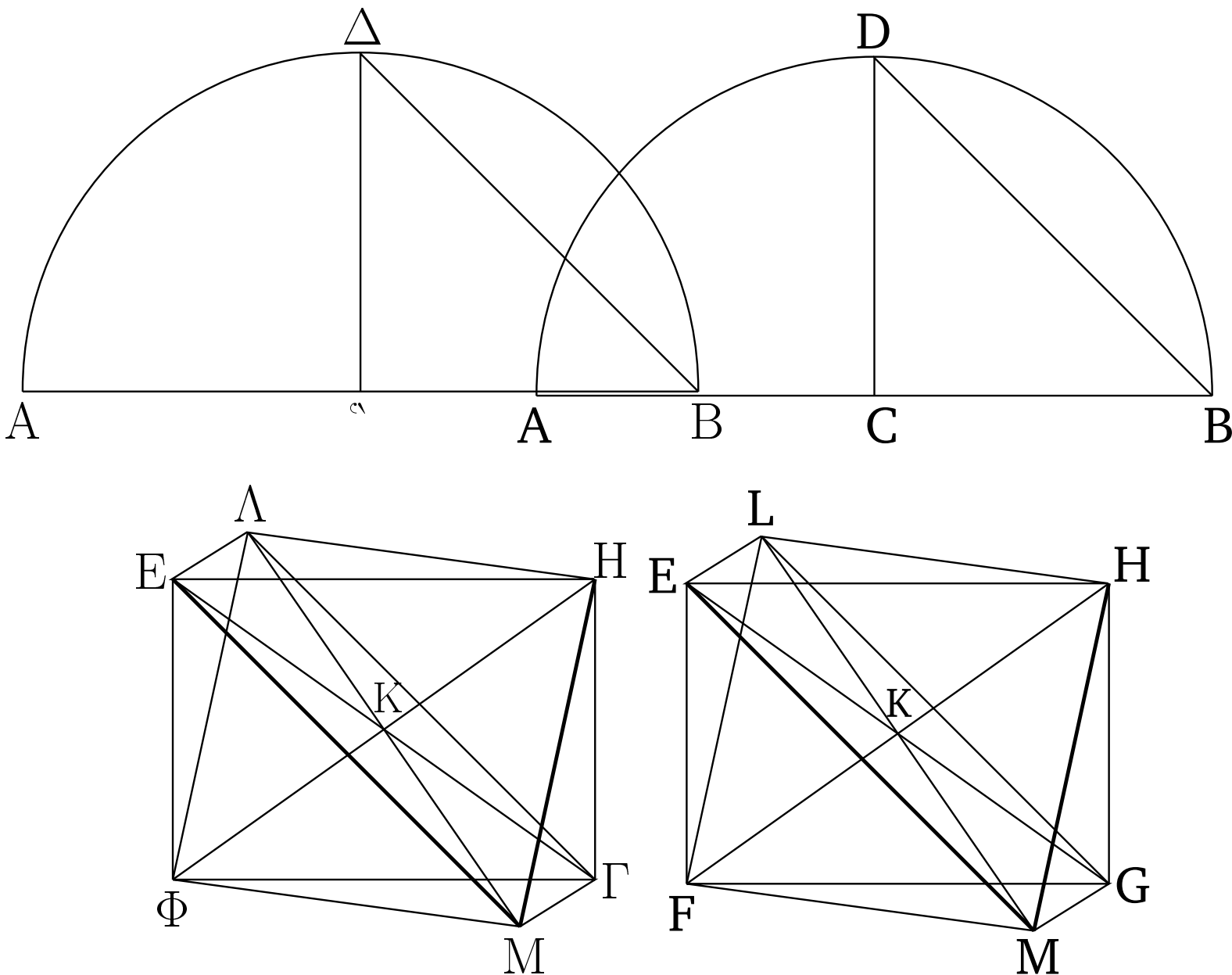
Ὀκτάεδρον συστήσασθαι καὶ σφαῖρα περιλαβεῖν, <ἥ| καὶ τὰ πρότερα, καὶ δεῖξαι, <ὅτι ἡ τῆς σφαίρας διάμετρος δυνάμει διπλασία ἐστὶ τῆς πλευρᾶς τοῦ ὀκταέδρου.

AD (is) to the (square) on DC .

For, let the figure of the semi-circle be set out, and let DB have been joined. And let the square EC be described on AC . And let the parallelogram FB be completed. Therefore, since, on account of triangle DAB being equiangular to triangle DAC [Props. 6.8, 6.4], (proportionally) as BA is to AD , so DA (is) to AC , the (rectangle contained) by BA and AC is thus equal to the (square) on AD [Prop. 6.17]. And since as AB is to BC , so EB (is) to BF [Prop. 6.1]. And EB is the (rectangle contained) by BA and AC —for EA (is) equal to AC . And BF the (rectangle contained) by AC and CB . Thus, as AB (is) to BC , so the (rectangle contained) by BA and AC (is) to the (rectangle contained) by AC and CB . And the (rectangle contained) by BA and AC is equal to the (square) on AD , and the (rectangle contained) by ACB (is) equal to the (square) on DC . For the perpendicular DC is the mean proportional to the pieces of the base, AC and CB , on account of ADB being a right-angle [Prop. 6.8 corr.]. Thus, as AB (is) to BC , so the (square) on AD (is) to the (square) on DC . (Which is) the very thing it was required to show.

Proposition 14

To construct an octahedron, and to enclose (it) in a (given) sphere, like in the preceding (proposition), and to show that the square on the diameter of the sphere is double the (square) on the side of the octahedron.



Ἐκκείσθω ἡ τῆς δοθείσης σφαίρας διάμετρος ἡ AB , καὶ τετμήσθω διῖθα κατὰ τὸ Γ , καὶ γεγράφθω ἐπὶ τῆς AB ἡμικύκλιον τὸ ADB , καὶ >ήθθω ἀπὸ τοῦ Γ τῇ AB πρὸς ὀρθὰς ἡ $\Gamma\Delta$, καὶ ἐπεζεύθω ἡ DB , καὶ ἐκκείσθω τετράγωνον τὸ $EZH\Theta$ >ίσην >έθον ἐκάστην τῶν πλευρῶν τῇ ΔB , καὶ ἐπεζεύθωσαν αἱ ΘZ , EH , καὶ ἀνεστάτω ἀπὸ τοῦ K σημείου τῷ τοῦ $EZH\Theta$ τετραγώνου ἐπιπέδῳ πρὸς ὀρθὰς εὐθεΐα ἡ KL καὶ διήθθω ἐπὶ τὰ <έτερα μέρη τοῦ ἐπιπέδου ὡς ἡ KM , καὶ ἀφηρήσθω ἀφ' ἐκατέρας τῶν KL , KM μιᾷ τῶν EK , ZK , HK , ΘK >ίση ἐκάτερα τῶν KL , KM , καὶ ἐπεζεύθωσαν αἱ LE , LZ , LH , $L\Theta$, ME , MZ , MH , $M\Theta$.

Καὶ ἐπεὶ >ίση ἐστὶν ἡ KE τῇ $K\Theta$, καὶ ἐστὶν ὀρθὴ ἡ ὑπὸ $EK\Theta$ γωνία, τὸ >άρα ἀπὸ τῆς ΘE διπλάσιόν ἐστι τοῦ ἀπὸ τῆς EK . πάλιν, ἐπεὶ >ίση ἐστὶν ἡ AK τῇ KE , καὶ ἐστὶν ὀρθὴ ἡ ὑπὸ AKE γωνία, τὸ >άρα ἀπὸ τῆς EL διπλάσιόν ἐστι τοῦ ἀπὸ EK . ἐδείκθη δὲ καὶ τὸ ἀπὸ τῆς ΘE διπλάσιον τοῦ ἀπὸ τῆς EK ; τὸ >άρα ἀπὸ τῆς LE >ίσον ἐστὶ τῷ ἀπὸ τῆς $E\Theta$; >ίση >άρα ἐστὶν ἡ LE τῇ $E\Theta$. διὰ τὰ αὐτὰ δὴ καὶ ἡ $L\Theta$ τῇ ΘE ἐστὶν >ίση; ἰσόπλευρον >άρα ἐστὶ τὸ $LE\Theta$ τρίγωνον. ὁμοίως δὲ δείχομεν, <ὅτι καὶ <έκαστον τῶν λοιπῶν τριγώνων, <ὦν βάσεις μὲν εἰσιν αἱ τοῦ $EZH\Theta$ τετραγώνου πλευραὶ, κορυφαὶ δὲ τὰ Λ , M

Let the diameter AB of the given sphere be laid out, and let it have been cut in half at C . And let the semi-circle ADB be drawn on AB . And let CD be drawn from C at right-angles to AB . And let DB be joined. And let the square $EFGH$, having each of its sides equal to DB , be laid out. And let HF and EG be joined. And let the straight-line KL be set up, at point K , at right-angles to the plane of square $EFGH$ [Prop. 11.12]. And let it be drawn across on the other side of the plane, like KM . And let KL and KM , equal to one of EK , FK , GK , and HK , be cut off from KL and KM , respectively. And let LE , LF , LG , LH , ME , MF , MG , and MH be joined.

And since KE is equal to KH , and angle EKH is a right-angle, the (square) on the HE is thus double the (square) on EK [Prop. 1.47]. Again, since LK is equal to KE , and angle LKE is a right-angle, the (square) on EL is thus double the (square) on EK [Prop. 1.47]. And the (square) on HE was also shown (to be) double the (square) on EK . Thus, the (square) on LE is equal to the (square) on EH . Thus, LE is equal to EH . So, for the same (reasons), LH is also equal to HE . Triangle LEH is thus equilateral. So, similarly, we can show that each of the remaining trian-

σημεῖα, ἰσόπλευρόν ἐστιν; ὀκτάεδρον >άρα συνέσταται ὑπὸ ὀκτὼ τριγώνων ἰσοπλεύρων περιεχόμενον.

Δεῖ δὴ αὐτὸ καὶ σφαῖρα περιλαβεῖν τῇ| δοθείσῃ καὶ δεῖξαι, <ὅτι ἡ τῆς σφαίρας διάμετρος δυνάμει διπλασίον ἐστὶ τῆς τοῦ ὀκταέδρου πλευρᾶς.

Ἐπεὶ γὰρ αἱ τρεῖς αἱ AK , KM , KE >ῖσαι ἀλλήλαις εἰσίν, τὸ >άρα ἐπὶ τῆς AM γραφόμενον ἡμικύκλιον <ήχει καὶ διὰ τοῦ E . καὶ διὰ τὰ αὐτά, ἔαν μενούσῃ τῆς AM περιενεθθῇ τὸ ἡμικύκλιον εἰς τὸ αὐτὸ ἀποκατασταθῇ|, <όθεν >ήρχατο φέρεσθαι, <ήχει καὶ διὰ τῶν Z , H , Θ σημείων, καὶ >έσται σφαῖρα περιελημμένη τὸ ὀκτάεδρον. λέγω δὴ, <ὅτι καὶ τῇ| δοθείσῃ. ἐπεὶ γὰρ >ῖση ἐστὶν ἡ AK τῇ| KM , κοινὴ δὲ ἡ KE , καὶ γωνίαξ ὀρθὰξ περιέθουσιν, βάσιξ >άρα ἡ AE βάσει τῇ| EM ἐστὶν >ῖση. καὶ ἐπεὶ ὀρθὴ ἐστὶν ἡ ὑπὸ LEM γωνία, ἐν ἡμικυκλίῳ γάρ; τὸ >άρα ἀπὸ τῆς AM διπλασίον ἐστὶ τοῦ ἀπὸ τῆς AE . πάλιν, ἐπεὶ >ῖση ἐστὶν ἡ AG τῇ| GB , διπλασία ἐστὶν ἡ AB τῆς BG . ὥς δὲ ἡ AB πρὸς τὴν BG , ο<ύτωξ τὸ ἀπὸ τῆς AB πρὸς τὸ ἀπὸ τῆς BD ; διπλασίον >άρα ἐστὶ τὸ ἀπὸ τῆς AB τοῦ ἀπὸ τῆς BD . ἐδείθθη δὲ καὶ τὸ ἀπὸ τῆς AM διπλασίον τοῦ ἀπὸ τῆς AE . καὶ ἐστὶν >ῖσον τὸ ἀπὸ τῆς AB τῷ| ἀπὸ τῆς AE ; >ῖση γὰρ κεῖται ἡ $E\Theta$ τῇ| AB . >ῖσον >άρα καὶ τὸ ἀπὸ τῆς AB τῷ| ἀπὸ τῆς AM ; >ῖση >άρα ἡ AB τῇ| AM . καὶ ἐστὶν ἡ AB ἡ τῆς δοθείσης σφαίρας διάμετρος; ἡ AM >άρα >ῖση ἐστὶ τῇ| τῆς δοθείσης σφαίρας διαμέτρω.

Περιεῖληπται >άρα τὸ ὀκτάεδρον τῇ| δοθείσῃ σφαίρᾳ. καὶ συναποδέδεικται, <ὅτι ἡ τῆς σφαίρας διάμετρος δυνάμει διπλασίον ἐστὶ τῆς τοῦ ὀκταέδρου πλευρᾶς; <όπερ >έδει δεῖξαι.

gles, whose bases are the sides of the square $EFGH$, and apexes the points L and M , are equilateral. Thus, an octahedron contained by eight equilateral triangles has been constructed.

So, it is also necessary to enclose it by the given sphere, and to show that the square on the diameter of the sphere is double the (square) on the side of the octahedron.

For since the three (straight-lines) LK , KM , and KE are equal to one another, the semi-circle drawn on LM will thus also pass through E . And, for the same (reasons), if LM remains (fixed), and the semi-circle is carried around, and again established at the same (position) from which it began to be moved, then it will also pass through points F , G , and H , and the octahedron will be enclosed by a sphere. So, I say that (it is) also (enclosed) by the given (sphere). For since LK is equal to KM , and KE (is) common, and they contain right-angles, the base LE is thus equal to the base EM [Prop. 1.4]. And since angle LEM is a right-angle—for (it is) in a semi-circle [Prop. 3.31]—the (square) on LM is thus double the (square) on LE [Prop. 1.47]. Again, since AC is equal to CB , AB is double BC . And as AB (is) to BC , so the (square) on AB (is) to the (square) on BD [Prop. 6.8, Def. 5.9]. Thus, the (square) on AB is double the (square) on BD . And the (square) on LM was also shown (to be) double the (square) on LE . And the (square) on DB is equal to the (square) on LE . For EH was made equal to DB . Thus, the (square) on AB (is) also equal to the (square) on LM . Thus, AB (is) equal to LM . And AB is the diameter of the given sphere. Thus, LM is equal to the diameter of the given sphere.

Thus, the octahedron has been enclosed by the given sphere, and it has been simultaneously proved that the square on the diameter of the sphere is double the (square) on the side of the octahedron.[†] (Which is) the very thing it was required to show.

[†] If the radius of the sphere is unity then the side of octahedron is $\sqrt{2}$.

15.

Κύβον συστήσασθαι καὶ σφαῖρα περιλαβεῖν, <ῆ| καὶ τὴν πυραμίδα, καὶ δεῖξαι, <ὅτι ἡ τῆς σφαίρας διάμετρος δυνάμει τριπλασίον ἐστὶ τῆς τοῦ κύβου πλευρᾶς.

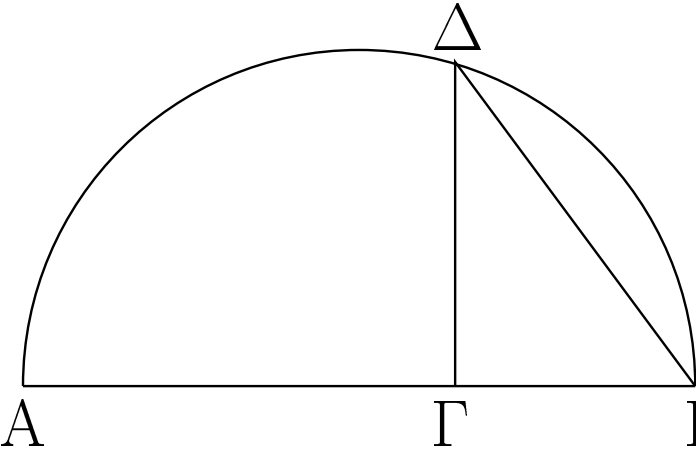
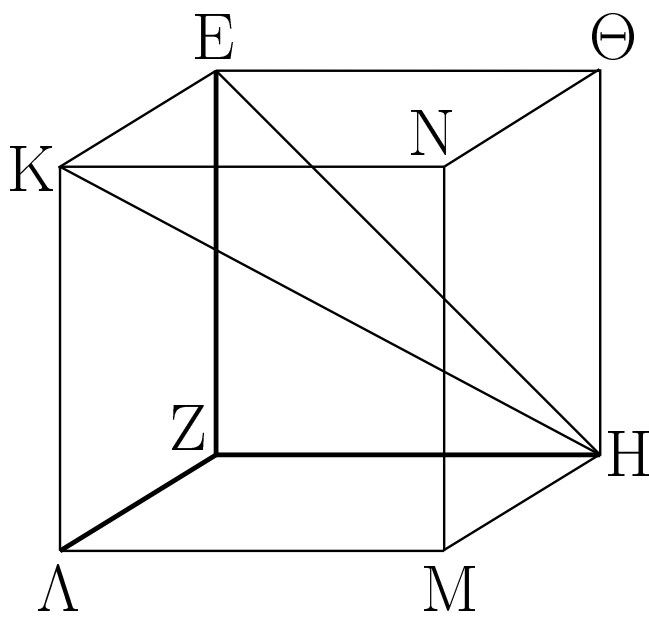
Ἐκκείσθω ἡ τῆς δοθείσης σφαίρας διάμετρος ἡ AB καὶ τεμήσθω κατὰ τὸ Γ <ώστε διπλὴν ε>ῖναι τὴν AG τῆς GB , καὶ γεγράφθω ἐπὶ τῆς AB ἡμικύκλιον τὸ $A\Delta B$, καὶ ἀπὸ τοῦ Γ τῇ| AB πρὸς ὀρθὰξ >ήθθω ἡ $\Gamma\Delta$, καὶ ἐπεζεύθθω ἡ ΔB , καὶ ἐκκείσθω τετράγωνον τὸ $EZH\Theta$ >ῖσην >έθον τὴν πλευρὰν τῇ| ΔB , καὶ ἀπὸ τῶν E , Z , H , Θ τῷ| τοῦ $EZH\Theta$ τετραγώνου ἐπιπέδω πρὸς ὀρθὰξ >ήθθωσαν αἱ EK , $Z\Lambda$, HM , ΘN , καὶ ἀφηρήσθω ἀπὸ ἐκάστης τῶν EK , $Z\Lambda$, HM , ΘN μιᾷ| τῶν EZ , ZH , $H\Theta$, ΘE >ῖση ἐκάστη τῶν EK , $Z\Lambda$, HM , ΘN , καὶ ἐπεζεύθθωσαν αἱ $K\Lambda$, ΛM , MN , NK ; κύβοξ >άρα συνέσταται ὁ ZN ὑπὸ <ἐξ τετραγώνων >ῖσων περιεθόμενος. Δεῖ δὴ αὐτὸν καὶ σφαῖρα περιλαβεῖν τῇ| δοθείσῃ καὶ δεῖξαι, <ὅτι ἡ τῆς σφαίρας διάμετρος δυνάμει τριπλασία ἐστὶ τῆς πλευρᾶς τοῦ κύβου.

Proposition 15

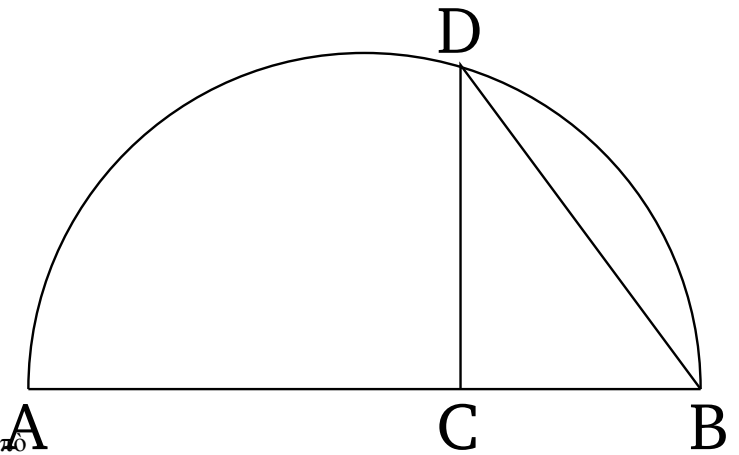
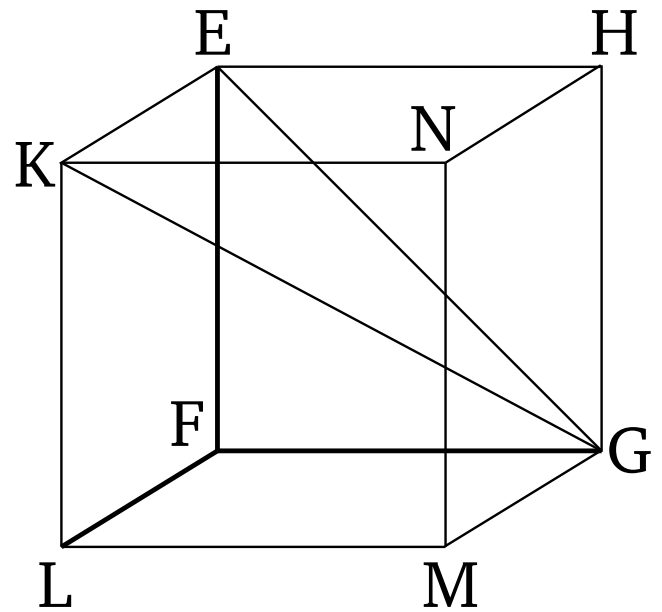
To construct a cube, and to enclose (it) in a sphere, like in the (case of the) pyramid, and to show that the square on the diameter of the sphere is three times the (square) on the side of the cube.

Let the diameter AB of the given sphere be laid out, and let it be cut at C such that AC is double CB . And let the semi-circle ADB be drawn on AB . And let CD be drawn from C at right-angles to AB . And let DB be joined. And let the square $EFGH$, having (its) side equal to DB , be laid out. And let EK , FL , GM , and HN be drawn from (points) E , F , G , and H , (respectively), at right-angles to the plane of square $EFGH$. And let EK , FL , GM , and HN , equal to one of EF , FG , GH , and HE , be cut off from EK , FL , GM , and HN , respectively. And let KL , LM , MN , and NK be joined. Thus, a cube contained by six equal squares has been constructed.

So, it is also necessary to enclose it by the given sphere, and to show that the square on the diameter of the sphere is three



times the (square) on the side of the cube.



Ἐπεξεύθθωσαν γὰρ αἱ KH , EH . καὶ ἐπεὶ ὀρθὴ ἐστὶν ἡ ὑπὸ KEH γωνία διὰ τὸ καὶ τὴν KE ὀρθὴν εἶναι πρὸς τὸ EH ἐπίπεδον δηλαδὴ καὶ πρὸς τὴν EH εὐθεΐαν, τὸ >άρα ἐπὶ τῇ KH γραφόμενον ἡμικύκλιον <ήχει καὶ διὰ τοῦ E σημείου. πάλιν, ἐπεὶ ἡ HZ ὀρθὴ ἐστὶ πρὸς ἑκατέραν τῶν $ZΛ$, ZE , καὶ πρὸς τὸ ZK >άρα ἐπίπεδον ὀρθὴ ἐστὶν ἡ HZ ; <ώστε καὶ ἔαν ἐπιζεύχωμεν τὴν ZK , ἡ HZ ὀρθὴ >έσται καὶ πρὸς τὴν ZK ; καὶ διὰ τοῦτο πάλιν τὸ ἐπὶ τῇ HK γραφόμενον ἡμικύκλιον <ήχει καὶ διὰ τοῦ Z . ὁμοίως καὶ διὰ τῶν λοιπῶν τοῦ κύβου σημείων <ήχει. ἔαν δὴ μενούσῃ τῇ KH περιενεθῇ τὸ ἡμικύκλιον εἰς τὸ αὐτὸ ἀποκατασταθῇ, <όθεν >ήρχατο φέρεσθαι, >έσται σφαῖρα περιελημμένη ὁ κύβος. λέγω δὴ, <ότι καὶ τῇ δοθείσῃ. ἐπεὶ γὰρ >ῖση ἐστὶν ἡ HZ τῇ ZE , καὶ ἐστὶν ὀρθὴ ἡ πρὸς τῷ Z γωνία, τὸ >άρα ἀπὸ τῇ EH διπλάσιόν ἐστι τοῦ ἀπὸ τῇ EZ . >ῖση δὲ ἡ EZ τῇ EK ; τὸ >άρα ἀπὸ τῇ EH διπλάσιόν ἐστι τοῦ ἀπὸ τῇ EK ; <ώστε τὰ ἀπὸ τῶν HE , EK , τουτέστι τὸ ἀπὸ τῇ HK , τριπλάσιόν ἐστι τοῦ ἀπὸ τῇ EK . καὶ ἐπεὶ τριπλασίον ἐστὶν ἡ AB τῇ BF , ὥς δὲ ἡ AB πρὸς τὴν BF , ο<ύτως τὸ ἀπὸ τῇ AB πρὸς τὸ ἀπὸ τῇ BD , τριπλάσιον >άρα τὸ ἀπὸ τῇ AB τοῦ ἀπὸ τῇ BD . ἐδειθῇ δὲ καὶ τὸ ἀπὸ τῇ HK τοῦ ἀπὸ τῇ KE τριπλάσιον. καὶ κεῖται >ῖση ἡ KE τῇ $ΔB$; >ῖση >άρα καὶ ἡ KH τῇ AB . καὶ ἐστὶν ἡ AB τῇ δοθείσῃ σφαίραξ διάμετρος; καὶ ἡ KH >άρα >ῖση ἐστὶ τῇ τῇ δοθείσῃ σφαίραξ διαμέτρῳ. Τῇ δοθείσῃ >άρα σφαῖρα περιεληπτὰ ὁ κύβος; καὶ συναποδείκνυται <ότι ἡ τῇ σφαίραξ διάμετρος δυνάμει τριπλασίον ἐστὶ τῇ τοῦ κύβου πλευρᾷ; <όπερ >έδει δεῖχαι.

For let KG and EG be joined. And since angle KEG is a right-angle—on account of KE also being at right-angles to the plane EG , and manifestly also to the straight-line EG [Def. 11.3]—the semi-circle drawn on KG will thus also pass through point E . Again, since GF is at right-angles to each of FL and FE , GF is thus also at right-angles to the plane FK [Prop. 11.4]. Hence, if we also join FK then GF will also be at right-angles to FK . And, again, on account of this, the semi-circle drawn on GK will also pass through point F . Similarly, it will also pass through the remaining (angular) points of the cube. So, if KG remains (fixed), and the semi-circle is carried around, and again established at the same (position) from which it began to be moved, then the cube will be enclosed by a sphere. So, I say that (it is) also (enclosed) by the given (sphere). For since GF is equal to FE , and the angle at F is a right-angle, the (square) on EG is thus double the (square) on EF [Prop. 1.47]. And EF (is) equal to EK . Thus, the (square) on EG is double the (square) on EK . Hence, the (sum of the squares) on GE and EK —that is to say, the (square) on GK [Prop. 1.47]—is three times the (square) on EK . And since AB is three times BC , and as AB (is) to BC , so the (square) on AB (is) to the (square) on BD [Prop. 6.8, Def. 5.9], the (square) on AB (is) thus three times the (square) on BD . And the (square) on GK was also shown (to be) three times the (square) on KE . And

KE was made equal to DB . Thus, KG (is) also equal to AB . And AB is the radius of the given sphere. Thus, KG is also equal to the diameter of the given sphere.

Thus, the cube has been enclosed by the given sphere. And it has simultaneously been shown that the square on the diameter of the sphere is three times the (square) on the side of the cube.[†] (Which is) the very thing it was required to show.

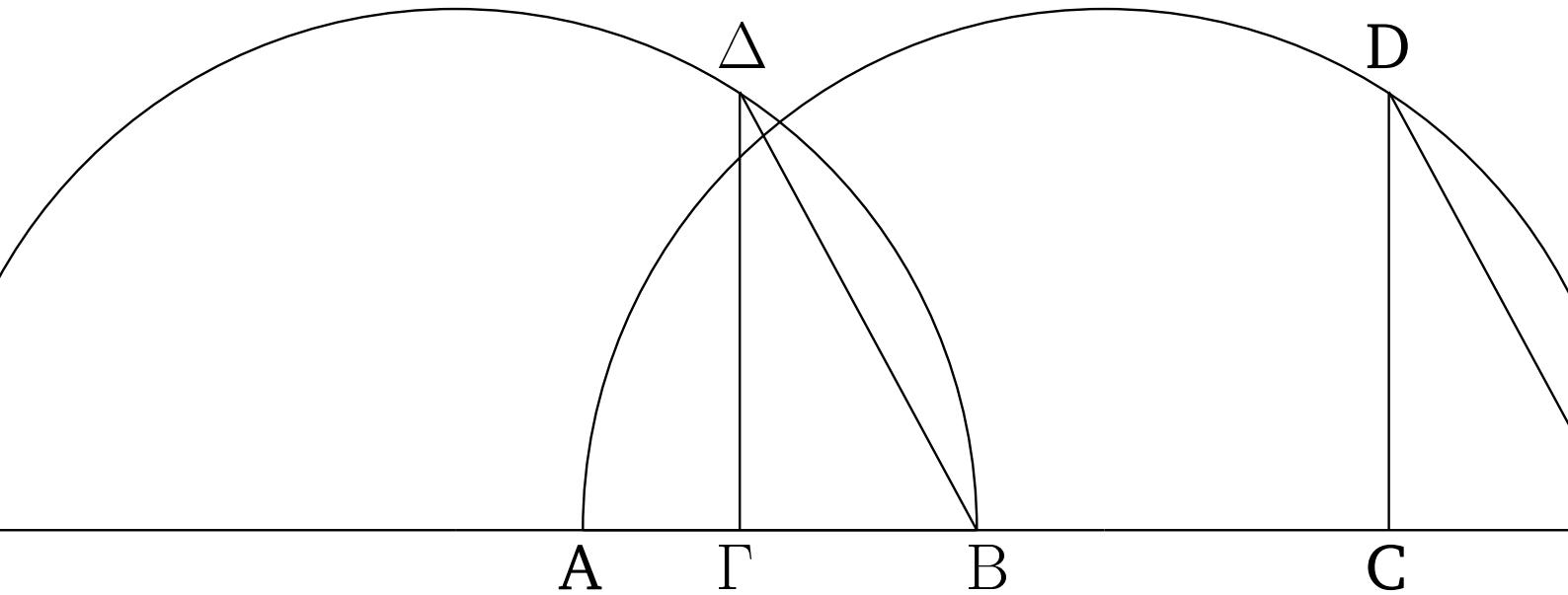
[†] If the radius of the sphere is unity then the side of the cube is $\sqrt{4/3}$.

16.

Εἰκοσάεδρον συστήσασθαι καὶ σφαῖρα περιλαβεῖν, <ἥ| καὶ τὰ προειρημένα σθήματα, καὶ δεῖξαι, <ὅτι ἡ τοῦ εἰκοσαέδρου πλευρὰ >ἀλογόξ ἐστὶν ἡ καλουμένη ἐλάττων.

Proposition 16

To construct an icosahedron, and to enclose (it) in a sphere, like the aforementioned figures, and to show that the side of the icosahedron is that irrational (straight-line) called minor.



Ἐκκείσθω ἡ τῆς δοθείσης σφαίρας διάμετρος ἡ AB καὶ τετμήσθω κατὰ τὸ Γ <ὥστε τετραπλῆν ε>ῖναι τὴν $A\Gamma$ τῆς ΓB , καὶ γεγράφθω ἐπὶ τῆς AB ἡμικύκλιον τὸ $A\Delta B$, καὶ >ῆθθω ἀπὸ τοῦ Γ τῇ AB πρὸς ὀρθὰς γωνίαξ εὐθεῖα γραμμὴ ἡ $\Gamma\Delta$, καὶ ἐπεζεύθθω ἡ ΔB , καὶ ἐκκείσθω κύκλος ὁ $EZH\Theta K$, ο<ῦ ἡ ἐν τοῦ κέντρου >ῖση >έστω τῇ ΔB , καὶ ἐγγεγράφθω εἰς τὸν $EZH\Theta K$ κύκλον πεντάγωνον ἰσοπλευρόν τε καὶ ἰσογώνιον τὸ $EZH\Theta K$, καὶ τετμήσθωσαν αἱ EZ , ZH , $H\Theta$, ΘK , KE περιφέρειαι δίθω κατὰ τὸ Λ , M , N , X , O σημεία, καὶ ἐπεζεύθθωσαν αἱ ΛM , MN , NX , XO , $O\Lambda$, EO . >ῖσοπλευρον >άρα ἐστὶ καὶ τὸ $\Lambda MNXO$ πεντάγωνον, καὶ δεκαγώνου ἡ EO εὐθεῖα. καὶ ἀνεστάτωσαν >ἀπὸ τῶν E , Z , H , Θ , K σημείων τῶ| τοῦ κύκλου ἐπιπέδω πρὸς ὀρθὰς γωνίαξ εὐθεῖαι αἱ $E\Pi$, ZP , $H\Sigma$, ΘT , KY >ῖσαι ο>ῦσαι τῇ ἐκ τοῦ κέντρου τοῦ $EZH\Theta K$ κύκλου, καὶ ἐπεζεύθθωσαν αἱ ΠP , $P\Sigma$, ΣT , TY , $Y\Pi$, $\Pi\Lambda$, ΛP , $P M$, $M\Sigma$, ΣN , $N T$, $T X$, $X Y$, $Y O$, $O\Pi$.

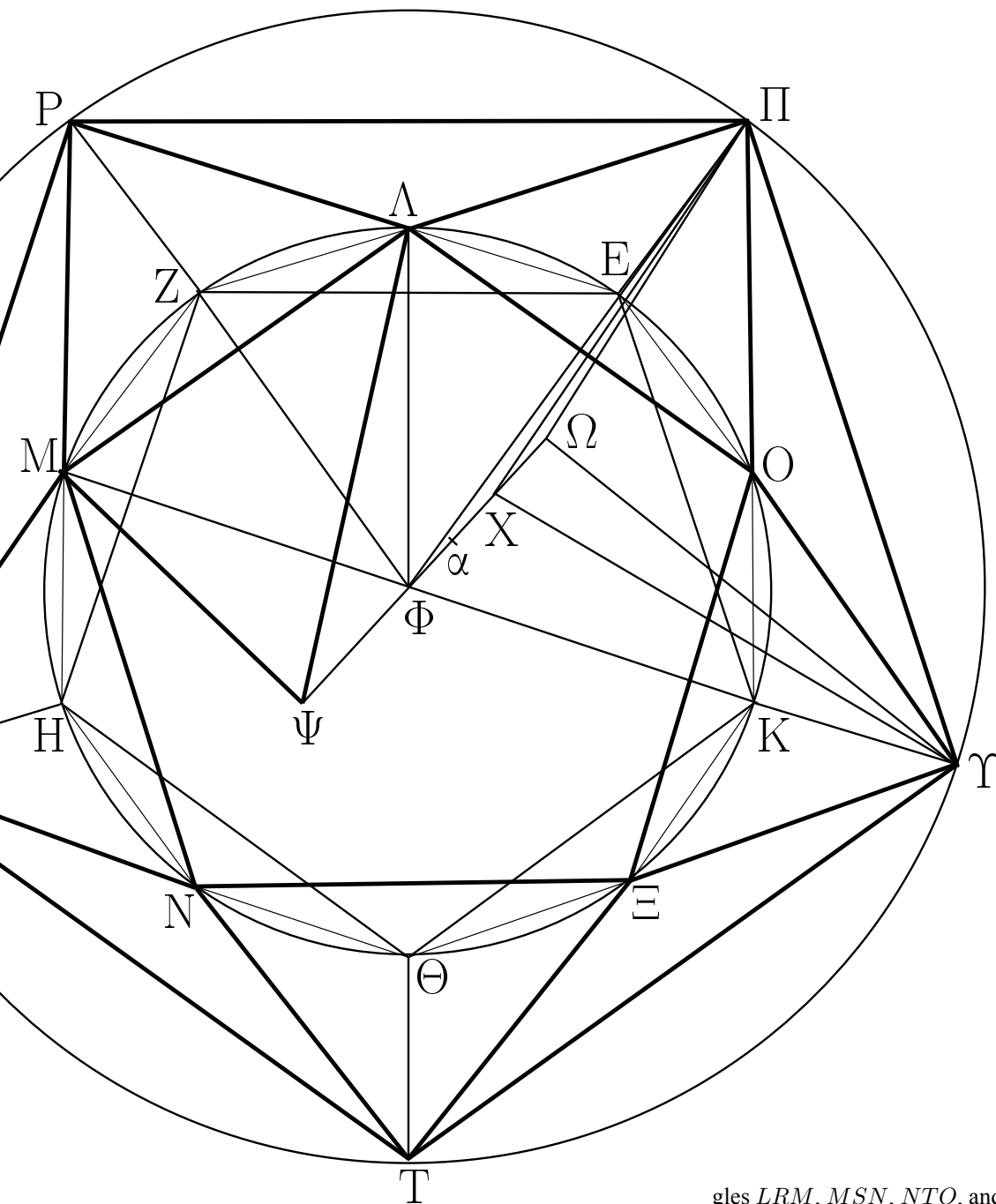
Καὶ ἐπεὶ ἐκάτερα τῶν $E\Pi$, KY τῶ| αὐτῶ| ἐπιπέδω πρὸς ὀρθὰς ἐστὶν, παράλληλος >άρα ἐστὶν ἡ $E\Pi$ τῇ KY . >έστι δὲ αὐτῇ| καὶ >ῖση; αἱ δὲ τὰς >ῖσαξ τε καὶ παραλλήλουξ ἐπιζευγνύουσαι ἐπὶ τὰ αὐτὰ μέρη εὐθεῖαι >ῖσαι τε καὶ παράλληλοί εἰσιν. ἡ ΠY >άρα τῇ $E K$ >ῖση τε καὶ παράλληλός ἐστιν. πενταγώνου δὲ ἰσοπλευροῦ ἡ $E K$; πενταγώνου >άρα ἰσοπλευροῦ καὶ ἡ ΠY τοῦ εἰς τὸν $EZH\Theta K$ κύκλον ἐγγραφομένου. διὰ τὰ αὐτὰ δὴ καὶ ἐκάστη τῶν ΠP , $P\Sigma$, ΣT , $T Y$ πενταγώνου ἐστὶν ἰσοπλευροῦ τοῦ εἰς τὸν $EZH\Theta K$ κύκλον ἐγγραφομένου; ἰσοπλευρον >άρα τὸ $\Pi P\Sigma T Y$ πεντάγωνον, καὶ ἐπεὶ ἑξαγώνου μὲν ἐστὶν ἡ ΠE , δεκαγώνου δὲ ἡ EO ,

Let the diameter AB of the given sphere be laid out, and let it be cut at C such that AC is four times CB [Prop. 6.10]. And let the semi-circle ADB be drawn on AB . And let the straight-line CD be drawn from C at right-angles to AB . And let DB be joined. And let the circle $EFGHK$ be set down, and let its radius be equal to DB . And let the equilateral and equiangular pentagon $EFGHK$ be inscribed in circle $EFGHK$ [Prop. 4.11]. And let the circumferences EF , FG , GH , HK , and KE be cut in half at points L , M , N , O , and P (respectively). And let LM , MN , NO , OP , PL , and EP be joined. Thus, pentagon $LMNOP$ is also equilateral, and EP (is) the side of the decagon (inscribed in the circle). And let the straight-lines EQ , FR , GS , HT , and KU , which are equal to the radius of circle $EFGHK$, have been set up at right-angles to the plane of the circle, at points E , F , G , H , and K (respectively). And let QR , RS , ST , TU , UQ , QL , LR , RM , MS , SN , NT , TO , OU , UP , and PQ be joined.

And since EQ and KU are each at right-angles to the same plane, EQ is thus parallel to KU [Prop. 11.6]. And it is also equal to it. And straight-lines joining equal and parallel (straight-lines) on the same side are (themselves) equal and parallel [Prop. 1.33]. Thus, QU is equal and parallel to $E K$. And $E K$ (is the side) of an equilateral pentagon (inscribed in circle $EFGHK$). Thus, QU (is) also the side of an equilateral pentagon inscribed in circle $EFGHK$. So, for the same (reasons), QR , RS , ST , and TU are also the sides of

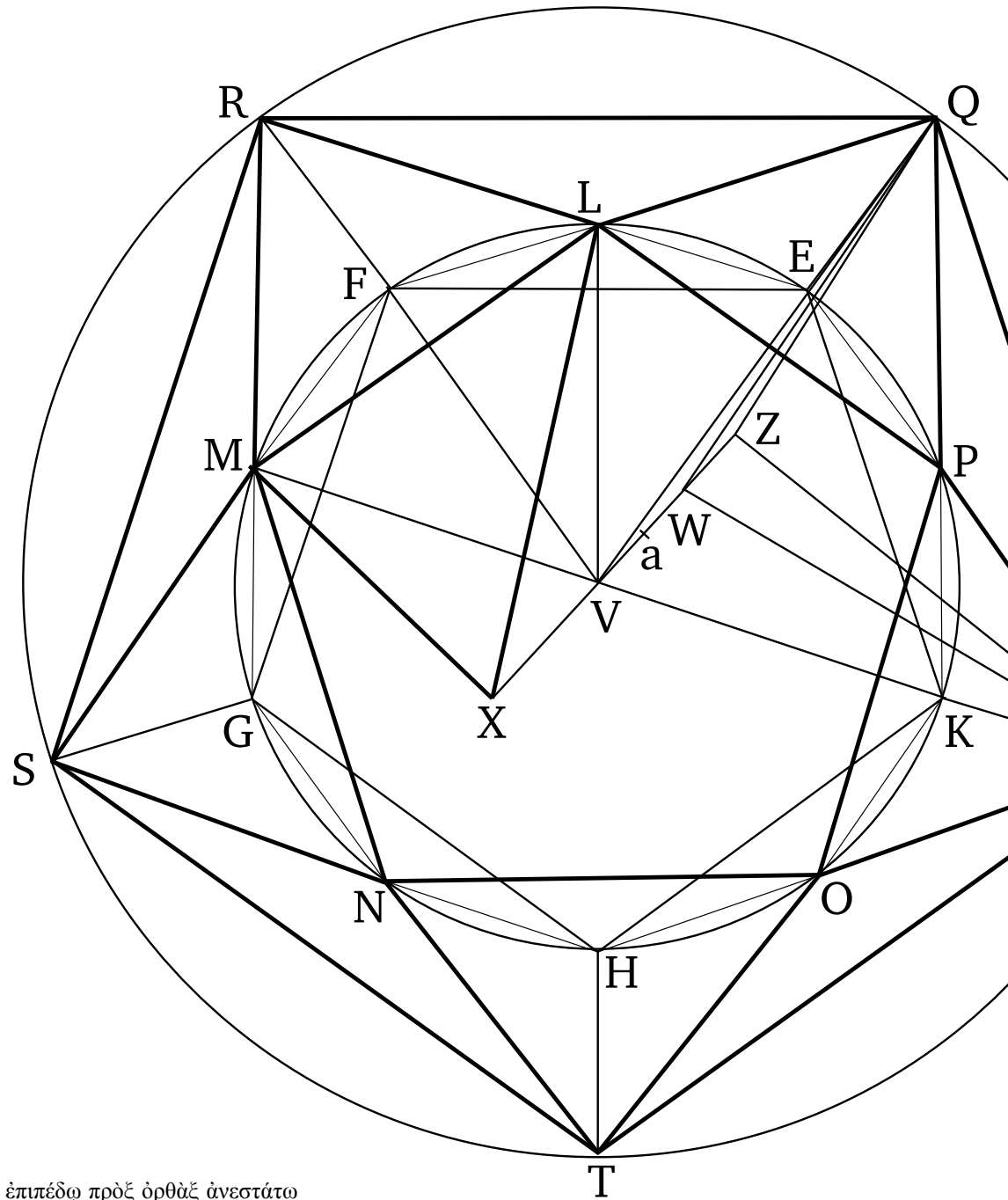
καὶ ἐστὶν ὀρθή ἡ ὑπὸ ΠΕΟ, πενταγώνου > ἄρα ἐστὶν ἡ ΠΟ; ἡ γὰρ τοῦ πενταγώνου πλευρὰ δύναται τὴν τε τοῦ ἑξαγώνου καὶ τὴν τοῦ δεκαγώνου τῶν εἰς τὸν αὐτὸν κύκλον ἐγγραφομένων. διὰ τὰ αὐτὰ δὴ καὶ ἡ ΟΥ πενταγώνου ἐστὶ πλευρὰ. > ἐστὶ δὲ καὶ ἡ ΠΥ πενταγώνου; ἰσόπλευρον > ἄρα ἐστὶ τὸ ΠΟΥ τρίγωνον. διὰ τὰ αὐτὰ δὴ καὶ < ἕκαστον τῶν ΠΛΡ, ΡΜΣ, ΣΝΤ, ΤΧΥ ἰσόπλευρόν ἐστιν. καὶ ἐπεὶ πενταγώνου ἐδείκθη ἑκάτερα τῶν ΠΛ, ΠΟ, > ἐστὶ δὲ καὶ ἡ ΛΟ πενταγώνου, ἰσόπλευρον > ἄρα ἐστὶ τὸ ΠΛΟ τρίγωνον. διὰ τὰ αὐτὰ δὴ καὶ < ἕκαστον τῶν ΛΡΜ, ΜΣΝ, ΝΤΧ, ΧΥΟ τριγώνων ἰσόπλευρόν ἐστιν.

an equilateral pentagon inscribed in circle $EFGHK$. Pentagon $QRSTU$ (is) thus equilateral. And side QE is (the side) of a hexagon (inscribed in circle $EFGHK$), and EP (the side) of a decagon, and (angle) QEP is a right-angle, thus QP is (the side) of a pentagon (inscribed in the same circle). For the square on the side of a pentagon is (equal to the sum of) the (squares) on (the sides of) a hexagon and a decagon inscribed in the same circle [Prop. 13.10]. So, for the same (reasons), PU is also the side of a pentagon. And QU is also (the side) of a pentagon. Thus, triangle QPU is equilateral. So, for the same (reasons), (triangles) QLR , RMS , SNT , and TOU are each also equilateral. And since QL and QP were each shown (to be the sides) of a pentagon, and LP is also (the side) of a pentagon, triangle QLP is thus equilateral. So, for the same (reasons), trian-



Εἰλήφθω τὸ κέντρον τοῦ ΕΖΗΘΚ κύκλου τὸ Φ σημεῖον; καὶ

gles LRM , MSN , NTO , and OUP are each also equilateral.



ἀπὸ τοῦ Φ τῷ τοῦ κύκλου ἐπιπέδῳ πρὸς ὀρθὰς ἀνεστάτω ἡ ΦΩ, καὶ ἐκβεβλήσθω ἐπὶ τὰ <έτερα μέρη ὡς ἡ ΦΨ, καὶ ἀφηρήσθω ἑκαγώνου μὲν ἡ ΦΘ, δεκαγώνου δὲ ἑκατέρω τῶν ΦΨ, ΘΩ, καὶ ἐπεζεύθωσαν αἱ ΠΩ, ΠΘ, ΥΩ, ΕΦ, ΛΦ, ΛΨ, ΨΜ.

Καὶ ἐπεὶ ἑκατέρω τῶν ΦΘ, ΠΕ τῷ τοῦ κύκλου ἐπιπέδῳ πρὸς ὀρθὰς ἐστὶν, παράλληλοξ >άρα ἐστὶν ἡ ΦΘ τῇ ΠΕ. εἰσὶ δὲ καὶ >ίσαι; καὶ αἱ ΕΦ, ΠΘ >άρα >ίσαι τε καὶ παράλληλοί εἰσιν. ἑκαγώνου δὲ ἡ ΕΦ; ἑκαγώνου >άρα καὶ ἡ ΠΘ. καὶ ἐπεὶ ἑκαγώνου μὲν ἐστὶν ἡ ΠΘ, δεκαγώνου δὲ ἡ ΘΩ, καὶ ὀρθή ἐστὶν ἡ ὑπὸ ΠΘΩ γωνία, πενταγώνου >άρα ἐστὶν ἡ ΠΩ. διὰ τὰ αὐτὰ δὴ καὶ ἡ ΥΩ πενταγώνου ἐστίν, ἐπειδήπερ, ἔαν ἐπιζεύχωμεν τὰς ΦΚ, ΘΥ, >ίσαι καὶ ἀπεναντίον >έσσονται, καὶ ἐστὶν ἡ ΦΚ ἐκ τοῦ κέντρου ο>ῦσα ἑκαγώνου. ἑκαγώνου >άρα καὶ ἡ ΘΥ. δεκαγώνου δὲ ἡ ΘΩ, καὶ ὀρθή ἡ ὑπὸ ΥΘΩ; πενταγώνου >άρα ἡ ΥΩ. >έστι δὲ καὶ ἡ ΠΥ πενταγώνου; ἰσόπλευρον >άρα ἐστὶ τὸ ΠΥΩ τρίγωνον. διὰ τὰ αὐτὰ δὴ καὶ <έκαστον τῶν λοιπῶν τριγώνων, <ὼν βάσεις μὲν εἰσιν αἱ ΠΡ, ΡΣ, ΣΤ, ΤΥ εὐθεῖαι,

Let the center, point V , of circle $EFGHK$ be found [Prop. 3.1]. And let VZ be set up, at (point) V , at right-angles to the plane of the circle. And let it be produced on the other side (of the circle), like VX . And let VW be cut off (from XZ so as to be equal to the side) of a hexagon, and each of VX and WZ (so as to be equal to the side) of a decagon. And let QZ , QW , UZ , EV , LV , LX , and XM be joined.

And since VW and QE are each at right-angles to the plane of the circle, VW is thus parallel to QE [Prop. 11.6]. And they are also equal. EV and QW are thus equal and parallel (to one another) [Prop. 1.33]. And EV (is the side) of a hexagon. Thus, QW (is) also (the side) of a hexagon. And since QW is (the side) of a hexagon, and WZ (the side) of a decagon, and angle QWZ is a right-angle [Def. 11.3, Prop. 1.29], QZ is thus (the side) of a pentagon [Prop. 13.10]. So, for the same (reasons), UZ is also (the side) of a pentagon—inasmuch as, if we join VK and

κορυφή δὲ τὸ Ω σημεῖον, ἰσόπλευρόν ἐστιν. πάλιν, ἐπεὶ ἐξαγώνου μὲν ἡ $\Phi\Lambda$, δεκαγώνου δὲ ἡ $\Phi\Psi$, καὶ ὀρθή ἐστιν ἡ ὑπὸ $\Lambda\Phi\Psi$ γωνία, πενταγώνου >άρα ἐστὶν ἡ $\Lambda\Psi$. διὰ τὰ αὐτὰ δὲ ἔαν ἐπιζεύχωμεν τὴν $M\Phi$ οὕσαν ἐξαγώνου, συνάγεται καὶ ἡ $M\Psi$ πενταγώνου. >έστι δὲ καὶ ἡ ΛM πενταγώνου; ἰσόπλευρον >άρα ἐστὶ τὸ $\Lambda M\Psi$ τρίγωνον. ὁμοίως δὲ δειθθήσεται, <ὅτι καὶ <έκαστον τῶν λοιπῶν τριγώνων, <ὧν βάσεις μὲν εἰσιν αἱ MN , NX , XO , OL , κορυφή δὲ τὸ Ψ σημεῖον, ἰσόπλευρόν ἐστιν. συνέσταται >άρα εἰκοσάεδρον ὑπὸ >έικοσι τριγώνων ἰσοπλεύρων περιεχόμενον.

Δεῖ δὲ αὐτὸ καὶ σφαῖρα περιλαβεῖν τῇ δοθείσῃ καὶ δεῖξαι, <ὅτι ἡ τοῦ εἰκοσαέδρου πλευρὰ >άλογός ἐστιν ἡ καλουμένη ἐλάσσων.

Ἐπεὶ γὰρ ἐξαγώνου ἐστὶν ἡ $\Phi\Theta$, δεκαγώνου δὲ ἡ $\Theta\Omega$, ἡ $\Phi\Omega$ >άρα >άκρον καὶ μέσον λόγον τέτμηται κατὰ τὸ Θ , καὶ τὸ μείζον αὐτῆς τμημὰ ἐστὶν ἡ $\Phi\Theta$; >έστιν >άρα ὡς ἡ $\Omega\Phi$ πρὸς τὴν $\Phi\Theta$, οὕτως ἡ $\Phi\Theta$ πρὸς τὴν $\Theta\Omega$. >ίση δὲ ἡ μὲν $\Phi\Theta$ τῇ ΦE , ἡ δὲ $\Theta\Omega$ τῇ $\Phi\Psi$; >έστιν >άρα ὡς ἡ $\Omega\Phi$ πρὸς τὴν ΦE , οὕτως ἡ $E\Phi$ πρὸς τὴν $\Phi\Psi$. καὶ εἰσιν ὀρθαὶ αἱ ὑπὸ $\Omega\Phi E$, $E\Phi\Psi$ γωνίαι; ἔαν >άρα ἐπιζεύχωμεν τὴν $E\Omega$ εὐθείαν, ὀρθή >έσται ἡ ὑπὸ $\Psi E\Omega$ γωνία διὰ τὴν ὁμοιότητα τῶν $\Psi E\Omega$, $\Phi E\Omega$ τριγώνων. διὰ τὰ αὐτὰ δὲ ἐπεὶ ἐστὶν ὡς ἡ $\Omega\Phi$ πρὸς τὴν $\Phi\Theta$, οὕτως ἡ $\Phi\Theta$ πρὸς τὴν $\Theta\Omega$, >ίση δὲ ἡ μὲν $\Omega\Phi$ τῇ $\Psi\Theta$, ἡ δὲ $\Phi\Theta$ τῇ $\Theta\Pi$, >έστιν >άρα ὡς ἡ $\Psi\Theta$ πρὸς τὴν $\Theta\Pi$, οὕτως ἡ $\Pi\Theta$ πρὸς τὴν $\Theta\Omega$. καὶ διὰ τοῦτο πάλιν ἔαν ἐπιζεύχωμεν τὴν $\Pi\Psi$, ὀρθή >έσται ἡ πρὸς τῷ Π γωνία; τὸ >άρα ἐπὶ τῇ $\Psi\Omega$ γραφόμενον ἡμικύκλιον <ήχει καὶ διὰ τοῦ Π . καὶ ἔαν μενούσης τῇ $\Psi\Omega$ περιενεθῇ τὸ ἡμικύκλιον εἰς τὸ αὐτὸ πάλιν ἀποκατασταθῇ, <όθεν >ήρχατο φέρεσθαι, <ήχει καὶ διὰ τοῦ Π καὶ τῶν λοιπῶν σημείων τοῦ εἰκοσαέδρου, καὶ >έσται σφαῖρα περιειλημμένη τὸ εἰκοσάεδρον. λέγω δὲ, <ὅτι καὶ τῇ δοθείσῃ. τετμήσθω γὰρ ἡ $\Phi\Theta$ δίθῃ κατὰ τὸ α . καὶ ἐπεὶ εὐθεῖα γραμμὴ ἡ $\Phi\Omega$ >άκρον καὶ μέσον λόγον τέτμηται κατὰ τὸ Θ , καὶ τὸ >έλασσον αὐτῆς τμημὰ ἐστὶν ἡ $\Omega\Theta$, ἡ >άρα $\Omega\Theta$ προσλαβοῦσα τὴν ἡμίσειαν τοῦ μείζονος τμηματοῦ τὴν $\Theta\alpha$ πενταπλάσιον δύναται τοῦ ἀπὸ τῆς ἡμίσειας τοῦ μείζονος τμηματοῦ; πενταπλάσιον >άρα ἐστὶ τὸ ἀπὸ τῆς $\Omega\alpha$ τοῦ ἀπὸ τῆς $\alpha\Theta$. καὶ ἐστὶ τῆς μὲν $\Omega\alpha$ διπλῇ ἡ $\Omega\Psi$, τῆς δὲ $\alpha\Theta$ διπλῇ ἡ $\Phi\Theta$; πενταπλάσιον >άρα ἐστὶ τὸ ἀπὸ τῆς $\Omega\Psi$ τοῦ ἀπὸ τῆς $\Phi\Theta$. καὶ ἐπεὶ τετραπλῇ ἐστὶν ἡ $\Lambda\Gamma$ τῇ ΓB , πενταπλῇ >άρα ἐστὶν ἡ AB τῇ $B\Gamma$. ὡς δὲ ἡ AB πρὸς τὴν $B\Gamma$, οὕτως τὸ ἀπὸ τῆς AB πρὸς τὸ ἀπὸ τῆς $B\Delta$; πενταπλάσιον >άρα ἐστὶ τὸ ἀπὸ τῆς AB τοῦ ἀπὸ τῆς $B\Delta$. ἐδειθθὲ δὲ καὶ τὸ ἀπὸ τῆς $\Omega\Psi$ πενταπλάσιον τοῦ ἀπὸ τῆς $\Phi\Theta$. καὶ ἐστὶν >ίση ἡ ΔB τῇ $\Phi\Theta$; ἑκάτερα γὰρ αὐτῶν >ίση ἐστὶ τῇ ἐκ τοῦ κέντρου τοῦ $EZH\Theta K$ κύκλου; >ίση >άρα καὶ ἡ AB τῇ $\Psi\Omega$. καὶ ἐστὶν ἡ AB ἡ τῇ δοθείσῃ σφαίραξ διάμετρος; καὶ ἡ $\Psi\Omega$ >άρα >ίση ἐστὶ τῇ τῇ δοθείσῃ σφαίραξ διαμέτρῳ; τῇ >άρα δοθείσῃ σφαίρα περιειληπτὰ τὸ εἰκοσάεδρον.

Λέγω δὲ, <ὅτι ἡ τοῦ εἰκοσαέδρου πλευρὰ >άλογός ἐστιν ἡ καλουμένη ἐλάττων. ἐπεὶ γὰρ <ρητή ἐστὶν ἡ τῇ σφαίραξ διάμετρος, καὶ ἐστὶ δυνάμει πενταπλάσιον τῇ ἐκ τοῦ κέντρου τοῦ $EZH\Theta K$ κύκλου, <ρητή >άρα ἐστὶ καὶ ἡ ἐκ τοῦ κέντρου τοῦ $EZH\Theta K$ κύκλου; <ώστε καὶ ἡ διάμετρος αὐτοῦ <ρητή ἐστὶν. ἔαν δὲ εἰς κύκλον <ρητὴν >έθοντα τὴν διάμετρον πεντάγωνον ἰσόπλευρον ἐγγραφῇ, ἡ τοῦ πενταγώνου πλευρὰ >άλογός ἐστιν ἡ καλουμένη ἐλάττων. ἡ δὲ τοῦ $EZH\Theta K$ πενταγώνου πλευρὰ ἡ τοῦ εἰκοσαέδρου ἐστίν. ἡ >άρα τοῦ εἰκοσαέδρου πλευρὰ >άλογός ἐστιν ἡ καλουμένη ἐλάττων.

WU then they will be equal and opposite. And VK , being (equal) to the radius (of the circle), is (the side) of a hexagon [Prop. 4.15 corr.]. Thus, WU (is) also the side of a hexagon. And WZ (is the side) of a decagon, and (angle) UWZ (is) a right-angle. Thus, UZ (is the side) of a pentagon [Prop. 13.10]. And QU is also (the side) of a pentagon. Triangle QUZ is thus equilateral. So, for the same (reasons), each of the remaining triangles, whose bases are the straight-lines QR , RS , ST , and TU , and apexes the point Z , are also equilateral. Again, since VL (is the side) of a hexagon, and VX (the side) of a decagon, and angle LVX is a right-angle, LX is thus (the side) of a pentagon [Prop. 13.10]. So, for the same (reasons), if we join MV , which is (the side) of a hexagon, MX is also inferred (to be the side) of a pentagon. And LM is also (the side) of a pentagon. Thus, triangle LMX is equilateral. So, similarly, it can be shown that each of the remaining triangles, whose bases are the (straight-lines) MN , NO , OP , and PL , and apexes the point X , are also equilateral. Thus, an icosahedron contained by twenty equilateral triangles has been constructed.

So, it is also necessary to enclose it in the given sphere, and to show that the side of the icosahedron is that irrational (straight-line) called minor.

For, since VW is (the side) of a hexagon, and WZ (the side) of a decagon, VZ has thus been cut in extreme and mean ratio at W , and VW is its greater piece [Prop. 13.9]. Thus, as ZV is to VW , so VW (is) to WZ . And VW (is) equal to VE , and WZ to VX . Thus, as ZV is to VE , so EV (is) to VX . And angles ZVE and EVX are right-angles. Thus, if we join straight-line EZ then angle XEZ will be a right-angle, on account of the similarity of triangles XEZ and VEZ . [Prop. 6.8]. So, for the same (reasons), since as ZV is to VW , so VW (is) to WZ , and ZV (is) equal to XW , and VW to WQ , thus as XW is to WQ , so QW (is) to WZ . And, again, on account of this, if we join QX then the angle at Q will be a right-angle [Prop. 6.8]. Thus, the semi-circle drawn on XZ will also pass through Q [Prop. 3.31]. And if XZ remains fixed, and the semi-circle is carried around, and again established at the same (position) from which it began to be moved, then it will also pass through (point) Q , and (through) the remaining (angular) points of the icosahedron. And the icosahedron will be enclosed by a sphere. So, I say that (it is) also (enclosed) by the given (sphere). For let VW be cut in half at a . And since the straight-line VZ has been cut in extreme and mean ratio at W , and ZW is its lesser piece, then the square on ZW added to half of the greater piece, Wa , is five times the (square) on half of the greater piece [Prop. 13.3]. Thus, the (square) on Za is five times the (square) on aW . And ZX is double Za , and VW double aW . Thus, the (square) on ZX is five times the (square) on WV . And since AC is four times CB , AB is thus five times BC . And as AB (is) to BC , so the (square) on AB (is) to the (square) on BD [Prop. 6.8, Def. 5.9]. Thus, the (square) on AB is five times the (square) on BD . And the (square) on ZX was also shown (to be) five times the (square) on VW . And DB is equal to VW . For each of them is equal to the radius of circle $EFGHK$. Thus, AB (is) also equal to XZ . And AB is the diameter of the given sphere. Thus, XZ is equal to the diameter of the given sphere. Thus, the icosahedron has

been enclosed by the given sphere.

So, I say that the side of the icosahedron is that irrational (straight-line) called minor. For since the diameter of the sphere is rational, and the square on it is five times the (square) on the radius of circle $EFGHK$, the radius of circle $EFGHK$ is thus also rational. Hence, its diameter is also rational. And if an equilateral pentagon is inscribed in a circle having a rational diameter then the side of the pentagon is that irrational (straight-line) called minor [Prop. 13.11]. And the side of pentagon $EFGHK$ is (the side) of the icosahedron. Thus, the side of the icosahedron is that irrational (straight-line) called minor.

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, <ὅτι ἡ τῆς σφαίρας διάμετρος δυνάμει πενταπλασίον ἐστὶ τῆς ἐκ τοῦ κέντρου τοῦ κύκλου, ἀφ' οὗ τὸ εἰκοσάεδρον ἀναγράφεται, καὶ <ὅτι ἡ τῆς σφαίρας διάμετρος σύγκειται >ἐκ τε τῆς τοῦ ἑξαγώνου καὶ δύο τῶν τοῦ δεκαγώνου τῶν εἰς τὸν αὐτὸν κύκλον ἐγγραφομένων. <ὅπερ >έδει δεῖχαι.

† If the radius of the sphere is unity then the radius of the circle is $2/\sqrt{5}$, and the sides of the hexagon, decagon, and pentagon/icosahedron are $2/\sqrt{5}$, $1 - 1/\sqrt{5}$, and $(1/\sqrt{5})\sqrt{10 - 2\sqrt{5}}$, respectively.

Corollary

So, (it is) clear, from this, that the square on the diameter of the sphere is five times the square on the radius of the circle from which the icosahedron has been described, and that the diameter of the sphere is the sum of (the side) of the hexagon, and two of (the sides) of the decagon, inscribed in the same circle.†

17.

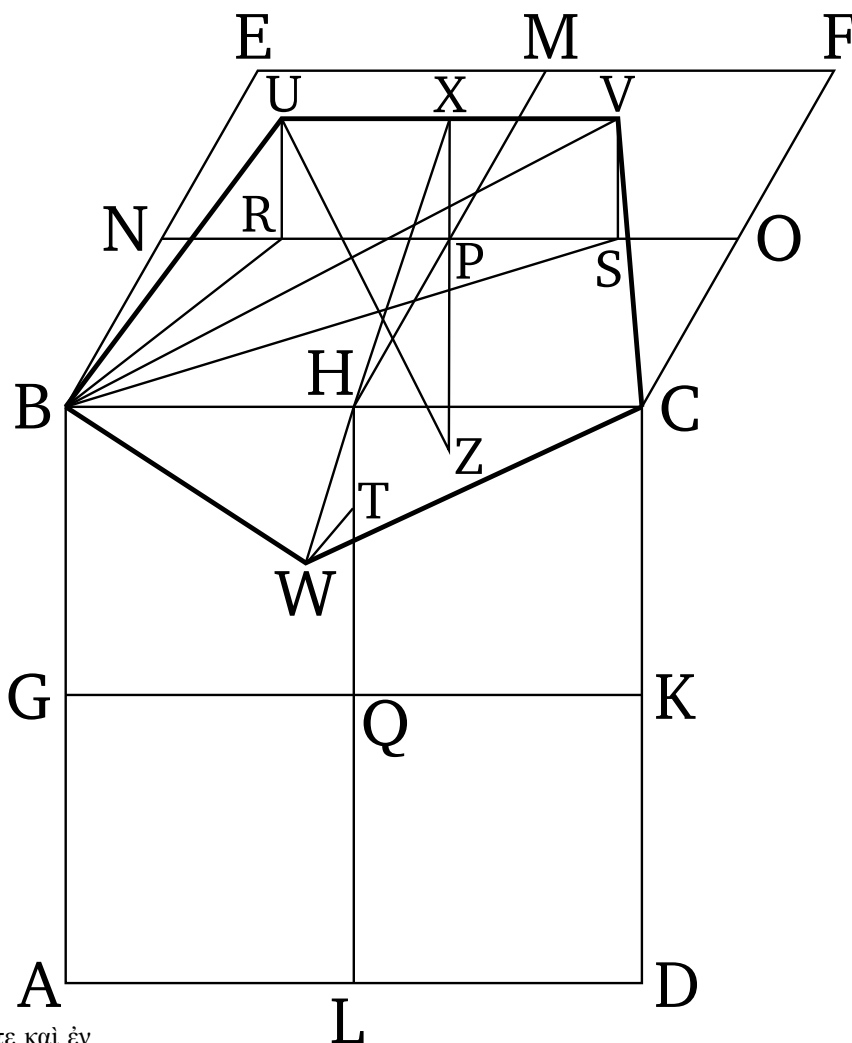
Δωδεκάεδρον συστήσασθαι καὶ σφαῖρα περιλαβεῖν, <ἥ| καὶ τὰ προειρημένα σθήματα, καὶ δεῖχαι, <ὅτι ἡ τοῦ δωδεκαέδρου πλευρὰ >ἀλογόξ ἐστὶν ἡ καλουμένη ἀποτομή.

Ἐκκείσθωσαν τοῦ προειρημένου κύβου δύο ἐπίπεδα πρὸς ὀρθὰς ἀλλήλοις τὰ $AB\Gamma\Delta$, ΓBEZ , καὶ τετμήσθω ἐκάστη τῶν AB , $B\Gamma$, $\Gamma\Delta$, ΔA , EZ , EB , $Z\Gamma$ πλευρῶν δίθῃ κατὰ τὰ H , Θ , K , Λ , M , N , X , καὶ ἐπεζεύθωσαν αἱ HK , $\Theta\Lambda$, $M\Theta$, NX , καὶ τετμήσθω ἐκάστη τῶν NO , OX , $\Theta\Pi$ >ἄκρον καὶ μέσον λόγον κατὰ τὰ P , Σ , T σημεία, καὶ >έστω αὐτῶν μείζονα τμήματα τὰ PO , OS , ΠP , καὶ ἀνεστάτωσαν ἀπὸ τῶν P , Σ , T σημείων τοῖς τοῦ κύβου ἐπιπέδοις πρὸς ὀρθὰς ἐπὶ τὰ ἐκτὸς μέρη τοῦ κύβου αἱ PY , $\Sigma\Phi$, $T\Theta$, καὶ κείσθωσαν >ίσαι ταῖς PO , OS , ΠP , καὶ ἐπεζεύθωσαν αἱ YB , $B\Theta$, $\Theta\Gamma$, $\Gamma\Phi$, ΦY .

Proposition 17

To construct a dodecahedron, and to enclose (it) in a sphere, like the aforementioned figures, and to show that the side of the dodecahedron is that irrational (straight-line) called an apotome.

Let two planes of the aforementioned cube [Prop. 13.15], $ABCD$ and $CBEF$, (which are) at right-angles to one another, be laid out. And let the sides AB , BC , CD , DA , EF , EB , and FC have each been cut in half at points G , H , K , L , M , N , and O (respectively). And let GK , HL , MH , and NO be joined. And let NP , PO , and HQ have each been cut in extreme and mean ratio at points R , S , and T (respectively). And let their greater pieces be RP , PS , and TQ (respectively). And let RU , SV , and TW be set up on the exterior side of the cube, at points R , S , and T (respectively), at right-angles to the planes of the cube. And let them be made equal to RP , PS , and TQ . And let UB ,



Λέγω, <ὅτι τὸ ΥΒΘΓΦ πεντάγωνον ἰσόπλευρόν τε καὶ ἐν ἐνὶ ἐπιπέδῳ καὶ >έτι ἰσογώνιον ἔστιν. ἐπεξεύθθωσαν γὰρ αἱ ΠΒ, ΣΒ, ΦΒ. καὶ ἐπεὶ εὐθεῖα ἡ ΝΟ >ἄκρον καὶ μέσον λόγον τέτμηται κατὰ τὸ Ρ, καὶ τὸ μείζον τμήμα ἔστιν ἡ ΡΟ, τὰ >άρα ἀπὸ τῶν ΟΝ, ΝΡ τριπλάσιά ἐστι τοῦ ἀπὸ τῆς ΡΟ. >ἴση δὲ ἡ μὲν ΟΝ τῇ ΝΒ, ἡ δὲ ΟΡ τῇ ΡΥ; τὰ >άρα ἀπὸ τῶν ΒΝ, ΝΡ τριπλάσιά ἐστι τοῦ ἀπὸ τῆς ΡΥ. τοῖς δὲ ἀπὸ τῶν ΒΝ, ΝΡ τὸ ἀπὸ τῆς ΒΡ ἔστιν >ἴσον; τὸ >άρα ἀπὸ τῆς ΒΡ τριπλάσιόν ἐστι τοῦ ἀπὸ τῆς ΡΥ; <ὥστε τὰ ἀπὸ τῶν ΒΡ, ΡΥ τετραπλάσιά ἐστι τοῦ ἀπὸ τῆς ΡΥ. τοῖς δὲ ἀπὸ τῶν ΒΡ, ΡΥ >ἴσον ἔστι τὸ ἀπὸ τῆς ΒΥ; τὸ >άρα >ἀπὸ τῆς ΒΥ τετραπλάσιόν ἐστι τοῦ ἀπὸ τῆς ΥΡ; διπλὴ >άρα ἔστιν ἡ ΒΥ τῆς ΡΥ. >έστι δὲ καὶ ἡ ΦΥ τῆς ΥΡ διπλῇ, ἐπειδὴ περ καὶ ἡ ΣΡ τῆς ΟΡ, τουτέστι τῆς ΡΥ, ἔστι διπλῇ; >ἴση >άρα ἡ ΒΥ τῇ ΥΦ. ὁμοίως δὲ δεῖθῃσεται, <ὅτι καὶ ἐκάστη τῶν ΒΘ, ΘΓ, ΓΦ ἐκατέρω τῶν ΒΥ, ΥΦ ἔστιν >ἴση. ἰσόπλευρον >άρα ἔστι τὸ ΒΥΦΓΘ πεντάγωνον. λέγω δὲ, <ὅτι καὶ ἐν ἐνὶ ἔστιν ἐπιπέδῳ. >ήθθω γὰρ ἀπὸ τοῦ Ο ἐκατέρω τῶν ΡΥ, ΣΦ παράλληλος ἐπὶ τὰ ἐκτὸς τοῦ κύβου μέρη ἡ ΟΨ, καὶ ἐπεξεύθθωσαν αἱ ΨΘ, ΘΘ; λέγω, <ὅτι ἡ ΨΘΘ εὐθεῖα ἔστιν. ἐπεὶ γὰρ ἡ ΘΠ >ἄκρον καὶ μέσον λόγον τέτμηται κατὰ τὸ Τ, καὶ τὸ μείζον αὐτῆς τμήμα ἔστιν ἡ ΠΤ, >έστιν >άρα ὡς ἡ ΘΠ πρὸς τὴν ΠΤ, ο<ύτως ἡ ΠΤ πρὸς τὴν ΤΘ. >ἴση δὲ ἡ μὲν ΘΠ τῇ ΘΟ, ἡ δὲ ΠΤ ἐκατέρω τῶν ΤΘ, ΟΨ; >έστιν >άρα ὡς ἡ ΘΟ πρὸς τὴν ΟΨ, ο<ύτως ἡ ΘΤ πρὸς τὴν ΤΘ. καὶ ἔστι παράλληλος ἡ μὲν ΘΟ τῇ ΤΘ; ἐκατέρα γὰρ αὐτῶν τῷ ΒΔ ἐπιπέδῳ πρὸς ὀρθὰς ἔστιν; ἡ δὲ ΤΘ τῇ ΟΨ; ἐκατέρα γὰρ αὐτῶν τῷ ΒΖ ἐπιπέδῳ πρὸς ὀρθὰς ἔστιν. ἔαν δὲ δύο τρίγωνα συντεθῇ κατὰ μίαν γωνίαν, ὡς τὰ ΨΘΘ,

I say that the pentagon $UBWCV$ is equilateral, and in one plane, and, further, equiangular. For let RB , SB , and VB be joined. And since the straight-line NP has been cut in extreme and mean ratio at R , and RP is the greater piece, the (sum of the squares) on PN and NR is thus three times the (square) on RP [Prop. 13.4]. And PN (is) equal to NB , and PR to RU . Thus, the (sum of the squares) on BN and NR is three times the (square) on RU . And the (square) on BR is equal to the (sum of the squares) on BN and NR [Prop. 1.47]. Thus, the (square) on BR is three times the (square) on RU . Hence, the (sum of the squares) on BR and RU is four times the (square) on RU . And the (square) on BU is equal to the (sum of the squares) on BR and RU [Prop. 1.47]. Thus, the (square) on BU is four times the (square) on RU . Thus, BU is double RU . And VU is also double UR , inasmuch as SR is also double PR —that is to say, RU . Thus, BU (is) equal to UV . So, similarly, it can be shown that each of BW , WC , CV is equal to each of BU and UV . Thus, pentagon $BUVCW$ is equilateral. So, I say that it is also in one plane. For let PX be drawn from P , parallel to each of RU and SV , on the exterior side of the cube. And let XH and HW be joined. I say that XHW is a straight-line. For since HQ has been cut in extreme and mean ratio at T , and QT is its greater piece, thus as HQ is to QT , so QT (is) to TH . And HQ (is) equal to HP , and QT to each of TW and PX . Thus, as HP is to PX , so WT (is) to TH . And HP is parallel to TW . For of each of them is at right-angles to the plane

ΘΤΘ, τὰς δύο πλευρὰς ταῖς δυνὶν ἀνάλογον >έθοντα, <ώστε τὰς ὁμολόγουζ αὐτῶν πλευρὰς καὶ παραλλήλους >ῖναι, αἱ λοιπαὶ εὐθεῖαι ἐπ> εὐθείας >έσσονται; ἐπ> εὐθείας >άρα ἐστὶν ἡ ΨΘ τῇ ΘΘ. πᾶσα δὲ εὐθεῖα ἐν ἐνὶ ἐστὶν ἐπιπέδῳ; ἐν ἐνὶ >άρα ἐπιπέδῳ ἐστὶ τὸ ΥΒΘΓΦ πεντάγωνον.

Λέγω δὴ, <ὅτι καὶ ἰσογώνιον ἐστὶν.

Ἐπεὶ γὰρ εὐθεῖα γραμμὴ ἡ ΝΟ >άκρον καὶ μέσον λόγον τέτμηται κατὰ τὸ Ρ, καὶ τὸ μείζον τμήμα ἐστὶν ἡ ΟΡ [>ἐστὶν >άρα ὡς συναμφοτέροζ ἡ ΝΟ, ΟΡ πρὸς τὴν ΟΝ, ο<ύτωζ ἡ ΝΟ πρὸς τὴν ΟΡ], >ίση δὲ ἡ ΟΡ τῇ ΟΣ [>ἐστὶν >άρα ὡς ἡ ΣΝ πρὸς τὴν ΝΟ, ο<ύτωζ ἡ ΝΟ πρὸς τὴν ΟΣ], ἡ ΝΣ >άρα >άκρον καὶ μέσον λόγον τέτμηται κατὰ τὸ Ο, καὶ τὸ μείζον τμήμα ἐστὶν ἡ ΝΟ; τὰ >άρα ἀπὸ τῶν ΝΣ, ΣΟ τριπλάσιά ἐστὶ τοῦ ἀπὸ τῆς ΝΟ. >ίση δὲ ἡ μὲν ΝΟ τῇ ΝΒ, ἡ δὲ ΟΣ τῇ ΣΦ; τὰ >άρα ἀπὸ τῶν ΝΣ, ΣΦ τετράγωνα τριπλάσιά ἐστὶ τοῦ ἀπὸ τῆς ΝΒ; <ώστε τὰ ἀπὸ τῶν ΦΣ, ΣΝ, ΝΒ τετραπλάσιά ἐστὶ τοῦ ἀπὸ τῆς ΝΒ. τοῖς δὲ ἀπὸ τῶν ΣΝ, ΝΒ >ίσον ἐστὶ τὸ ἀπὸ τῆς ΣΒ; τὰ >άρα ἀπὸ τῶν ΒΣ, ΣΦ, τουτέστι τὸ ἀπὸ τῆς ΒΦ [ὁρθὴ γὰρ ἡ ὑπὸ ΦΣΒ γωνία], τετραπλάσιόν ἐστὶ τοῦ ἀπὸ τῆς ΝΒ; διπλῇ >άρα ἐστὶν ἡ ΦΒ τῇς ΒΝ. >έστι δὲ καὶ ἡ ΒΓ τῇς ΒΝ διπλῇ; >ίση >άρα ἐστὶν ἡ ΒΦ τῇ ΒΓ. καὶ ἐπεὶ δύο αἱ ΒΥ, ΥΦ δυοὶ ταῖς ΒΘ, ΘΓ >ίσαι εἰσίν, καὶ βάσις ἡ ΒΦ βάσει τῇ ΒΓ >ίση, γωνία >άρα ἡ ὑπὸ ΒΥΦ γωνία τῇ ὑπὸ ΒΘΓ ἐστὶν >ίση. ὁμοίωζ δὴ δείχομεν, <ὅτι καὶ ἡ ὑπὸ ΥΦΓ γωνία >ίση ἐστὶ τῇ ὑπὸ ΒΘΓ; αἱ >άρα ὑπὸ ΒΘΓ, ΒΥΦ, ΥΦΓ τρεῖς γωνίαι >ίσαι ἀλλήλαις εἰσίν. ἔαν δὲ πενταγώνου ἰσοπλευροῦ αἱ τρεῖς γωνίαι >ίσαι ἀλλήλαις >ῶσιν, ἰσογώνιον >έσται τὸ πεντάγωνον; ἰσογώνιον >άρα ἐστὶ τὸ ΒΥΦΓΘ πεντάγωνον. ἐδειθθῇ δὲ καὶ ἰσόπλευρον; τὸ >άρα ΒΥΦΓΘ πεντάγωνον ἰσόπλευρόν ἐστι καὶ ἰσογώνιον, καὶ ἐστὶν ἐπὶ μιᾷ τοῦ κύβου πλευρᾷ τῆς ΒΓ. ἔαν >άρα ἐφ> ἐκάστης τῶν τοῦ κύβου δώδεκα πλευρῶν τὰ αὐτὰ κατασκευάσωμεν, συσταθήσεται τι σῆμα στερεὸν ὑπὸ δώδεκα πενταγόνων ἰσοπλευρῶν τε καὶ ἰσογώνιων περιεχόμενον, <ὃ καλεῖται δωδεκάεδρον.

Δεῖ δὴ αὐτὸ καὶ σφαῖρα περιλαβεῖν τῇ δοθείσῃ καὶ δεῖχαι, <ὅτι ἡ τοῦ δωδεκαέδρου πλευρὰ >αλογόζ ἐστὶν ἡ καλουμένη ἀποτομή.

Ἐκβεβλήσθω γὰρ ἡ ΨΟ, καὶ >έστω ἡ ΨΩ; συμβάλλει >άρα ἡ ΟΩ τῇ τοῦ κύβου διαμέτρῳ, καὶ δίθα τέμνουσιν ἀλλήλας; τοῦτο γὰρ δέδεικται ἐν τῷ παρατελεύτῳ θεωρήματι τοῦ ἐνδεκάτου βιβλίου. τεμνέτωσαν κατὰ τὸ Ω; τὸ Ω >άρα κέντρον ἐστὶ τῆς σφαίρας τῆς περιλαμβανούσης τὸν κύβον, καὶ ἡ ΩΟ ἡμίσεια τῆς πλευρᾷ τοῦ κύβου. ἐπεξεύθθω δὴ ἡ ΥΩ. καὶ ἐπεὶ εὐθεῖα γραμμὴ ἡ ΝΣ >άκρον καὶ μέσον λόγον τέτμηται κατὰ τὸ Ο, καὶ τὸ μείζον αὐτῆς τμήμα ἐστὶν ἡ ΝΟ, τὰ >άρα ἀπὸ τῶν ΝΣ, ΣΟ τριπλάσιά ἐστὶ τοῦ ἀπὸ τῆς ΝΟ. >ίση δὲ ἡ μὲν ΝΣ τῇ ΨΩ, ἐπειδήπερ καὶ ἡ μὲν ΝΟ τῇ ΟΩ ἐστὶν >ίση, ἡ δὲ ΨΟ τῇ ΟΣ. ἀλλὰ μὴν καὶ ἡ ΟΣ τῇ ΨΥ, ἐπεὶ καὶ τῇ ΡΟ; τὰ >άρα ἀπὸ τῶν ΩΨ, ΨΥ τριπλάσιά ἐστὶ τοῦ ἀπὸ τῆς ΝΟ. τοῖς δὲ ἀπὸ τῶν ΩΨ, ΨΥ >ίσον ἐστὶ τὸ ἀπὸ τῆς ΥΩ; τὸ >άρα ἀπὸ τῆς ΥΩ τριπλάσιόν ἐστὶ τοῦ ἀπὸ τῆς ΝΟ. >έστι δὲ καὶ ἡ ἐκ τοῦ κέντρου τῆς σφαίρας τῆς περιλαμβανούσης τὸν κύβον δυνάμει τριπλασιῶν τῆς ἡμισείας τῆς τοῦ κύβου πλευρᾷ; προδέδεικται γὰρ κύβον συστήσασθαι καὶ σφαῖρα περιλαβεῖν καὶ δεῖχαι, <ὅτι ἡ τῆς σφαίρας διάμετρος δυνάμει τριπλασιῶν ἐστὶ τῆς πλευρᾷ τοῦ κύβου. εἰ δὲ <όλη τῆς <όλης, καὶ [ἡ] ἡμίσεια τῆς ἡμισείας; καὶ ἐστὶν ἡ ΝΟ ἡμίσεια τῆς τοῦ κύβου πλευρᾷ; ἡ >άρα ΥΩ >ίση ἐστὶ τῇ ἐκ τοῦ κέντρου τῆς σφαίρας τῆς περιλαμβανούσης τὸν κύβον. καὶ ἐστὶ τὸ Ω κέντρον

BD [Prop. 11.6]. And *TH* (is parallel) to *PX*. For each of them is at right-angles to the plane *BF* [Prop. 11.6]. And if two triangles, like *XPH* and *HTW*, having two sides proportional to two sides, are placed together at a single angle such that their corresponding sides are also parallel then the remaining sides will be straight-on (to one another) [Prop. 6.32]. Thus, *XH* is straight-on to *HW*. And every straight-line is in one plane [Prop. 11.1]. Thus, pentagon *UBWCV* is in one plane.

So, I say that it is also equiangular.

For since the straight-line *NP* has been cut in extreme and mean ratio at *R*, and *PR* is the greater piece [thus as the sum of *NP* and *PR* is to *PN*, so *NP* (is) to *PR*], and *PR* (is) equal to *PS* [thus as *SN* is to *NP*, so *NP* (is) to *PS*], *NS* has thus also been cut in extreme and mean ratio at *P*, and *NP* is the greater piece [Prop. 13.5]. Thus, the (sum of the squares) on *NS* and *SP* is three times the (square) on *NP* [Prop. 13.4]. And *NP* (is) equal to *NB*, and *PS* to *SV*. Thus, the (sum of the) squares on *NS* and *SV* is three times the (square) on *NB*. Hence, the (sum of the squares) on *VS*, *SN*, and *NB* is four times the (square) on *NB*. And the (square) on *SB* is equal to the (sum of the squares) on *SN* and *NB* [Prop. 1.47]. Thus, the (sum of the squares) on *BS* and *SV*—that is to say, the (square) on *BV* [for angle *VSΒ* (is) a right-angle]—is four times the (square) on *NB* [Def. 11.3, Prop. 1.47]. Thus, *VB* is double *BN*. And *BC* (is) also double *BN*. Thus, *BV* is equal to *BC*. And since the two (straight-lines) *BU* and *UV* are equal to the two (straight-lines) *BW* and *WC* (respectively), and the base *BV* (is) equal to the base *BC*, angle *BUV* is thus equal to angle *BWC* [Prop. 1.8]. So, similarly, we can show that angle *UVC* is equal to angle *BWC*. Thus, the three angles *BWC*, *BUV*, and *UVC* are equal to one another. And if three angles of an equilateral pentagon are equal to one another then the pentagon is equiangular [Prop. 13.7]. Thus, pentagon *BUVCW* is equiangular. And it was also shown (to be) equilateral. Thus, pentagon *BUVCW* is equilateral and equiangular, and it is on one of the sides, *BC*, of the cube. Thus, if we make the same construction on each of the twelve sides of the cube then some solid figure contained by twelve equilateral and equiangular pentagons will be constructed, which is called a dodecahedron.

So, it is necessary to enclose it in the given sphere, and to show that the side of the dodecahedron is that irrational (straight-line) called an apotome.

For let *XP* be produced, and let (the produced straight-line) be *XZ*. Thus, *PZ* meets the diameter of the cube, and they cut one another in half. For, this has been proved in the penultimate theorem of the eleventh book [Prop. 11.38]. Let them cut (one another) at *Z*. Thus, *Z* is the center of the sphere enclosing the cube, and *ZP* (is) half the side of the cube. So, let *UZ* be joined. And since the straight-line *NS* has been cut in extreme and mean ratio at *P*, and its greater piece is *NP*, the (sum of the squares) on *NS* and *SP* is thus three times the (square) on *NP* [Prop. 13.4]. And *NS* (is) equal to *XZ*, inasmuch as *NP* is also equal to *PZ*, and *XP* to *PS*. But, indeed, *PS* (is) also (equal) to *XU*, since (it is) also (equal) to *RP*. Thus, the (sum of the squares) on *ZX* and *XU* is three times the (square) on *NP*. And the (square) on *UZ* is equal to the (sum of the squares) on *ZX* and *XU* [Prop. 1.47]. Thus, the (square)

τῆς σφαίρας τῆς περιλαμβανούσης τὸν κύβον; τὸ $Y >$ ἄρα σημείον πρὸς τῇ ἐπιφανείᾳ ἐστὶ τῆς σφαίρας. ὁμοίωξ δὴ δεῖχομεν, <ὅτι καὶ ἐκάστη τῶν λοιπῶν γωνιῶν τοῦ δωδεκαέδρου πρὸς τῇ ἐπιφανείᾳ ἐστὶ τῆς σφαίρας; περιεῖληπται > ἄρα τὸ δωδεκαέδρον τῇ δοθείσῃ σφαίρᾳ.

Λέγω δὴ, <ὅτι ἡ τοῦ δωδεκαέδρου πλευρὰ > ἀλογόξ ἐστὶν ἡ καλουμένη ἀποτομή.

Ἐπεὶ γὰρ τῆς $NO >$ ἄκρον καὶ μέσον λόγον τετμημένης τὸ μείζον τμήμα ἐστὶν ὁ PO , τῆς δὲ $OX >$ ἄκρον καὶ μέσον λόγον τετμημένης τὸ μείζον τμήμα ἐστὶν ἡ OS , <ὅληξ > ἄρα τῆς $NX >$ ἄκρον καὶ μέσον λόγον τετμημένης τὸ μείζον τμήμα ἐστὶν ἡ PS . [$\text{ο} < \text{ιον}$ ἐπεὶ ἐστὶν ὡς ἡ NO πρὸς τὴν OP , ἡ OP πρὸς τὴν PN , καὶ τὰ διπλάσια; τὰ γὰρ μέρη τοῖς ἰσάκις πολλαπλασίοις τὸν αὐτὸν > ἔθει λόγον; ὡς > ἄρα ἡ NX πρὸς τὴν PS , ὁ < ὅτωξ ἡ PS πρὸς συναμφοτέρων τὴν NP , ΣX . μείζων δὲ ἡ NX τῆς PS ; μείζων > ἄρα καὶ ἡ PS συναμφοτέρου τῆς NP , ΣX ; ἡ $NX >$ ἄρα > ἄκρον καὶ μέσον λόγον τέτμηται, καὶ τὸ μείζον αὐτῆς τμήμα ἐστὶν ἡ PS .] > ἴση δὲ ἡ PS τῇ $Y\Phi$; τῆς > ἄρα $NX >$ ἄκρον καὶ μέσον λόγον τετμημένης τὸ μείζον τμήμα ἐστὶν ἡ $Y\Phi$. καὶ ἐπεὶ < ρητὴ ἐστὶν τῆς σφαίρας διάμετρος καὶ ἐστὶ δυνάμει τριπλασίον τῆς τοῦ κύβου πλευρᾶς, < ρητὴ > ἄρα ἐστὶν ἡ NX πλευρὰ < ὅσσα τοῦ κύβου. ἔαν δὲ < ρητὴ γραμμὴ > ἄκρον καὶ μέσον λόγον τμηθῇ, ἐκάτερον τῶν τμημάτων > ἀλογόξ ἐστὶν ἀποτομή.

Ἡ $Y\Phi >$ ἄρα πλευρὰ < ὅσσα τοῦ δωδεκαέδρου > ἀλογόξ ἐστὶν ἀποτομή.

on UZ is three times the (square) on NP . And the square on the radius of the sphere enclosing the cube is also three times the (square) on half the side of the cube. For it has previously been demonstrated (how to) construct the cube, and to enclose (it) in a sphere, and to show that the square on the diameter of the sphere is three times the (square) on the side of the cube [Prop. 13.15]. And if the (square on the) whole (is three times) the (square on the) whole, then the (square on the) half (is) also (three times) the (square on the) half. And NP is half of the side of the cube. Thus, UZ is equal to the radius of the sphere enclosing the cube. And Z is the center of the sphere enclosing the cube. Thus, point U is on the surface of the sphere. So, similarly, we can show that each of the remaining angles of the dodecahedron is also on the surface of the sphere. Thus, the dodecahedron has been enclosed by the given sphere.

So, I say that the side of the dodecahedron is that irrational straight-line called an apotome.

For since RP is the greater piece of NP , which has been cut in extreme and mean ratio, and PS is the greater piece of PO , which has been cut in extreme and mean ratio, RS is thus the greater piece of the whole of NO , which has been cut in extreme and mean ratio. [Thus, since as NP is to PR , (so) PR (is) to RN , and (the same is also true) of the doubles. For parts have the same ratio as similar multiples (taken in corresponding order) [Prop. 5.15]. Thus, as NO (is) to RS , so RS (is) to the sum of NR and SO . And NO (is) greater than RS . Thus, RS (is) also greater than the sum of NR and SO [Prop. 5.14]. Thus, NO has been cut in extreme and mean ratio, and RS is its greater piece.] And RS (is) equal to UV . Thus, UV is the greater piece of NO , which has been cut in extreme and mean ratio. And since the diameter of the sphere is rational, and the square on it is three times the (square) on the side of the cube, NO , which is the side of the cube, is thus rational. And if a rational (straight)-line is cut in extreme and mean ratio then each of the pieces is the irrational (straight-line called) an apotome. Thus, UV , which is the side of the dodecahedron, is the irrational (straight-line called) an apotome [Prop. 13.6].

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, <ὅτι τῆς τοῦ κύβου πλευρᾶς > ἄκρον καὶ μέσον λόγον τετμημένης τὸ μείζον τμήμα ἐστὶν ἡ τοῦ δωδεκαέδρου πλευρὰ. <ὅπερ > ἔδει δεῖχαι.

Corollary

So, (it is) clear, from this, that the side of the dodecahedron is the greater piece of the side of the cube, when it is cut in extreme and mean ratio.[†] (Which is) the very thing it was required to show.

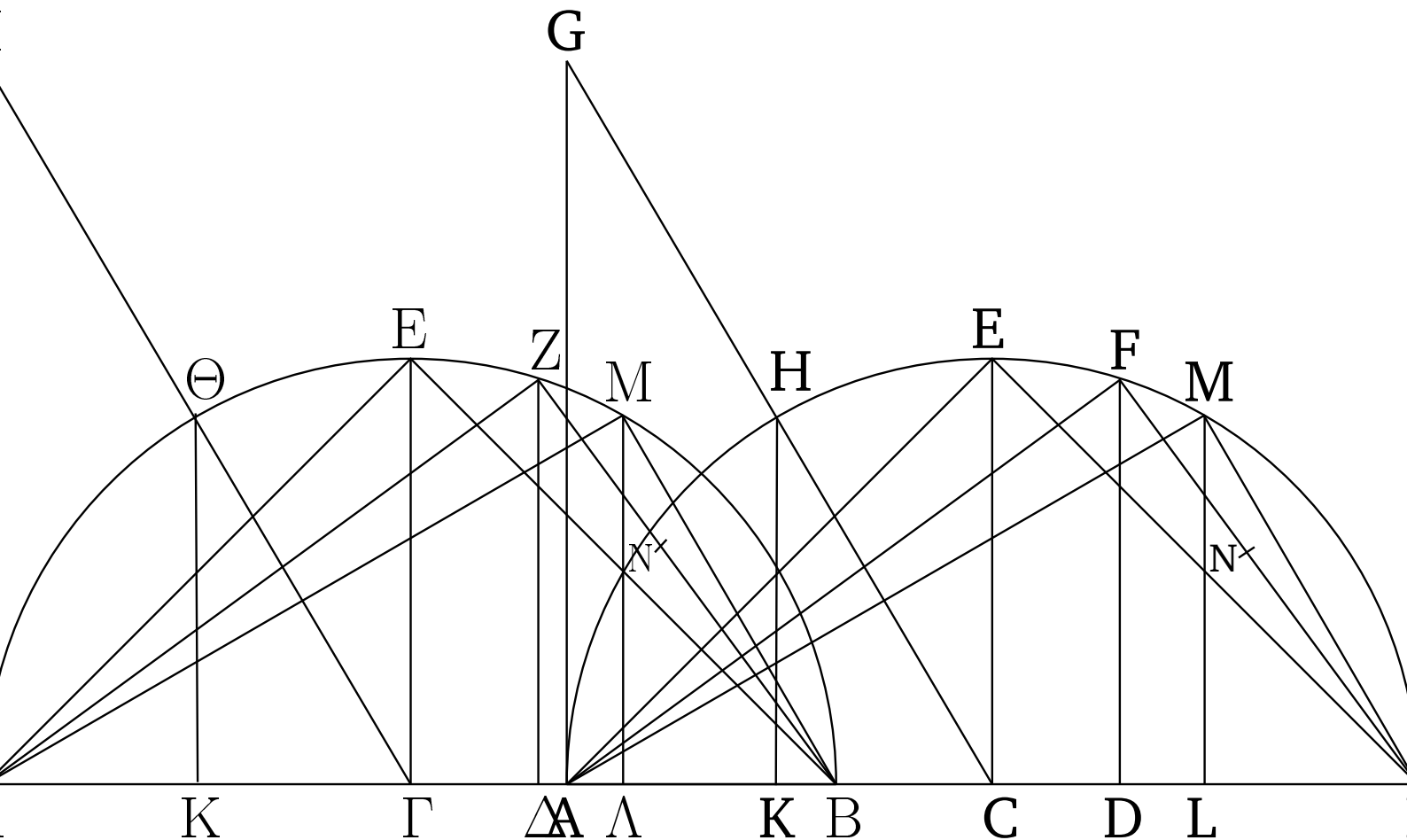
[†] If the radius of the circumscribed sphere is unity then the side of the cube is $\sqrt{4/3}$, and the side of the dodecahedron is $(1/3)(\sqrt{15} - \sqrt{3})$.

18.

Τὰς πλευρὰς τῶν πέντε σθημάτων ἐκθέσθαι καὶ συγκρῖναι πρὸς ἀλλήλας.

Proposition 18

To set out the sides of the five (aforementioned) figures, and to compare (them) with one another.[†]



Ἐκκείσθω ἡ τῆς δοθείσης σφαίρας διάμετρος ἡ AB , καὶ τετμήσθω κατὰ τὸ Γ <ώστε >ίσην εἶναι τὴν $A\Gamma$ τῇ ΓB , κατὰ δὲ τὸ Δ <ώστε διπλασίονα εἶναι τὴν $A\Delta$ τῆς ΔB , καὶ γεγράφθω ἐπὶ τῆς AB ἡμικύκλιον τὸ AEB , καὶ ἀπὸ τῶν Γ, Δ τῇ AB πρὸς ὀρθὰς >ήθθωσαν αἱ $\Gamma E, \Delta Z$, καὶ ἐπεζεύθωσαν αἱ AZ, ZB, EB . καὶ ἐπεὶ διπλῇ ἐστὶν ἡ $A\Delta$ τῆς ΔB , τριπλῇ >άρα ἐστὶν ἡ AB τῆς ΔB . ἀναστρέψαντι ἡμολία >άρα ἐστὶν ἡ BA τῆς $A\Delta$. ὥς δὲ ἡ BA πρὸς τὴν $A\Delta$, ο<ύτως τὸ ἀπὸ τῆς BA πρὸς τὸ ἀπὸ τῆς AZ ; ἰσογώνιον γάρ ἐστι τὸ AZB τρίγωνον τῷ $AZ\Delta$ τριγώνῳ; ἡμολιον >άρα ἐστὶ τὸ ἀπὸ τῆς BA τοῦ ἀπὸ τῆς AZ . >έστι δὲ καὶ ἡ τῆς σφαίρας διάμετρος δυνάμει ἡμολία τῆς πλευρᾶς τῆς πυραμίδος. καὶ ἐστὶν ἡ AB ἡ τῆς σφαίρας διάμετρος; ἡ AZ >άρα >ίση ἐστὶ τῇ πλευρᾷ τῆς πυραμίδος.

Πάλιν, ἐπεὶ διπλασίον ἐστὶν ἡ $A\Delta$ τῆς ΔB , τριπλῇ >άρα ἐστὶν ἡ AB τῆς ΔB . ὥς δὲ ἡ AB πρὸς τὴν ΔB , ο<ύτως τὸ ἀπὸ τῆς AB πρὸς τὸ ἀπὸ τῆς BZ ; τριπλάσιον >άρα ἐστὶ τὸ ἀπὸ τῆς AB τοῦ ἀπὸ τῆς BZ . >έστι δὲ καὶ ἡ τῆς σφαίρας διάμετρος δυνάμει τριπλασίον τῆς τοῦ κύβου πλευρᾶς. καὶ ἐστὶν ἡ AB ἡ τῆς σφαίρας διάμετρος; ἡ BZ >άρα τοῦ κύβου ἐστὶ πλευρά.

Καὶ ἐπεὶ >ίση ἐστὶν ἡ $A\Gamma$ τῇ ΓB , διπλῇ >άρα ἐστὶν ἡ AB τῆς ΓB . ὥς δὲ ἡ AB πρὸς τὴν ΓB , ο<ύτως τὸ ἀπὸ τῆς AB πρὸς τὸ ἀπὸ τῆς BE ; διπλάσιον >άρα ἐστὶ τὸ ἀπὸ τῆς AB τοῦ ἀπὸ τῆς BE . >έστι δὲ καὶ ἡ τῆς σφαίρας διάμετρος δυνάμει διπλασίον τῆς τοῦ ὀκταέδρου πλευρᾶς. καὶ ἐστὶν ἡ AB ἡ τῆς δοθείσης σφαίρας διάμετρος; ἡ BE >άρα τοῦ ὀκταέδρου ἐστὶ πλευρά.

>Ἡθθω δὴ ἀπὸ τοῦ A σημείου τῇ AB εὐθείᾳ πρὸς ὀρθὰς ἡ AH , καὶ κείσθω ἡ AH >ίση τῇ AB , καὶ ἐπεζεύθω ἡ $H\Gamma$, καὶ ἀπὸ τοῦ Θ ἐπὶ τὴν AB κάθετος >ήθθω ἡ ΘK . καὶ ἐπεὶ

Let the diameter, AB , of the given sphere be laid out. And let it be cut at C , such that AC is equal to CB , and at D , such that AD is double DB . And let the semi-circle AEB be drawn on AB . And let CE and DF be drawn from C and D (respectively), at right-angles to AB . And let AF, FB , and EB be joined. And since AD is double DB , AB is thus triple BD . Thus, via conversion, BA is one and a half times AD . And as BA (is) to AD , so the (square) on BA (is) to the (square) on AF [Def. 5.9]. For triangle AFB is equiangular to triangle AFD [Prop. 6.8]. Thus, the (square) on BA is one and a half times the (square) on AF . And the square on the diameter of the sphere is also one and a half times the (square) on the side of the pyramid [Prop. 13.13]. And AB is the diameter of the sphere. Thus, AF is equal to the side of the pyramid.

Again, since AD is double DB , AB is thus triple BD . And as AB (is) to BD , so the (square) on AB (is) to the (square) on BF [Prop. 6.8, Def. 5.9]. Thus, the (square) on AB is three times the (square) on BF . And the square on the diameter of the sphere is also three times the (square) on the side of the cube [Prop. 13.15]. And AB is the diameter of the sphere. Thus, BF is the side of the cube.

And since AC is equal to CB , AB is thus double BC . And as AB (is) to BC , so the (square) on AB (is) to the (square) on BE [Prop. 6.8, Def. 5.9]. Thus, the (square) on AB is double the (square) on BE . And the square on the diameter of the sphere is also double the (square) on the side of the octagon [Prop. 13.14]. And AB is the diameter of the given sphere. Thus, BE is the side of the octagon.

So let AG be drawn from point A at right-angles to the straight-line AB . And let AG be made equal to AB . And

διπλῇ ἐστὶν ἡ HA τῆς AG ; $>$ ἴση γάρ ἡ HA τῇ AB ; ὥς δὲ ἡ HA πρὸς τὴν AG , $ο<$ ύτως ἡ $ΘK$ πρὸς τὴν $KΓ$, διπλῇ $>$ άρα καὶ ἡ $ΘK$ τῆς $KΓ$. τετραπλάσιον $>$ άρα ἐστὶ τὸ ἀπὸ τῆς $ΘK$ τοῦ ἀπὸ τῆς $KΓ$; τὰ $>$ άρα ἀπὸ τῶν $ΘK$, $KΓ$, $<$ όπερ ἐστὶ τὸ ἀπὸ τῆς $ΘΓ$, πενταπλάσιόν ἐστι τοῦ ἀπὸ τῆς $KΓ$. $>$ ἴση δὲ ἡ $ΘΓ$ τῇ $ΓB$; πενταπλάσιον $>$ άρα ἐστὶ τὸ ἀπὸ τῆς $BΓ$ τοῦ ἀπὸ τῆς $ΓK$. καὶ ἐπεὶ διπλῇ ἐστὶν ἡ AB τῆς $ΓB$, $<$ ὢν ἡ AD τῆς AB ἐστὶ διπλῇ, λοιπὴ $>$ άρα ἡ BD λοιπῆς τῆς $ΔΓ$ ἐστὶ διπλῇ. τριπλῇ $>$ άρα ἡ $BΓ$ τῆς $ΓΔ$; ἐνναπλάσιον $>$ άρα τὸ ἀπὸ τῆς $BΓ$ τοῦ ἀπὸ τῆς $ΓΔ$. πενταπλάσιον δὲ τὸ ἀπὸ τῆς $BΓ$ τοῦ ἀπὸ τῆς $ΓK$; μείζον $>$ άρα τὸ ἀπὸ τῆς $ΓK$ τοῦ ἀπὸ τῆς $ΓΔ$. μείζον $>$ άρα ἐστὶν ἡ $ΓK$ τῆς $ΓΔ$. κείσθω τῇ $ΓK$ $>$ ἴση ἡ $ΓΛ$, καὶ ἀπὸ τοῦ $Λ$ τῇ AB πρὸς ὀρθὰς $>$ ήθθω ἡ $ΛM$, καὶ ἐπεζεύθθω ἡ MB . καὶ ἐπεὶ πενταπλάσιόν ἐστι τὸ ἀπὸ τῆς $BΓ$ τοῦ ἀπὸ τῆς $ΓK$, καὶ ἐστὶ τῆς μὲν $BΓ$ διπλῇ ἡ AB , τῆς δὲ $ΓK$ διπλῇ ἡ $ΚΛ$, πενταπλάσιον $>$ άρα ἐστὶ τὸ ἀπὸ τῆς AB τοῦ ἀπὸ τῆς $ΚΛ$. $>$ έστι δὲ καὶ ἡ τῆς σφαίρας διάμετρος δυνάμει πενταπλάσιον τῆς ἐκ τοῦ κέντρου τοῦ κύκλου, ἄφ' $ο<$ ύ τὸ εἰκοσάεδρον ἀναγράφεται. καὶ ἐστὶν ἡ AB ἡ τῆς σφαίρας διάμετρος; ἡ $ΚΛ$ $>$ άρα ἐκ τοῦ κέντρου ἐστὶ τοῦ κύκλου, ἄφ' $ο<$ ύ τὸ εἰκοσάεδρον ἀναγράφεται; ἡ $ΚΛ$ $>$ άρα ἐξαγώνου ἐστὶ πλευρὰ τοῦ εἰρημένου κύκλου. καὶ ἐπεὶ ἡ τῆς σφαίρας διάμετρος σύγκειται $>$ έκ τε τῆς τοῦ ἐξαγώνου καὶ δύο τῶν τοῦ δεκαγώνου τῶν εἰς τὸν εἰρημένον κύκλον ἐγγραφομένων, καὶ ἐστὶν ἡ μὲν AB ἡ τῆς σφαίρας διάμετρος, ἡ δὲ $ΚΛ$ ἐξαγώνου πλευρά, καὶ $>$ ἴση ἡ AK τῇ $ΛB$, ἑκάτερα $>$ άρα τῶν AK , $ΛB$ δεκαγώνου ἐστὶ πλευρὰ τοῦ ἐγγραφομένου εἰς τὸν κύκλον, ἄφ' $ο<$ ύ τὸ εἰκοσάεδρον ἀναγράφεται. καὶ ἐπεὶ δεκαγώνου μὲν ἡ $ΛB$, ἐξαγώνου δὲ ἡ $ΜΛ$; $>$ ἴση γάρ ἐστὶ τῇ $ΚΛ$, ἐπεὶ καὶ τῇ $ΘK$; $>$ ἴσον γὰρ ἀπέθουσιν ἀπὸ τοῦ κέντρου; καὶ ἐστὶν ἑκάτερα τῶν $ΘK$, $ΚΛ$ διπλασίων τῆς $KΓ$; πενταγώνου $>$ άρα ἐστὶν ἡ MB . ἡ δὲ τοῦ πενταγώνου ἐστὶν ἡ τοῦ εἰκοσάεδρου; εἰκοσάεδρου $>$ άρα ἐστὶν ἡ MB .

Καὶ ἐπεὶ ἡ ZB κύβου ἐστὶ πλευρά, τετμήσθω $>$ άκρον καὶ μέσον λόγον κατὰ τὸ N , καὶ $>$ έστω μείζον τμήμα τὸ NB ; ἡ NB $>$ άρα δωδεκαέδρου ἐστὶ πλευρά.

Καὶ ἐπεὶ ἡ τῆς σφαίρας διάμετρος ἐδείθθη τῆς μὲν AZ πλευρᾶς τῆς πυραμίδος δυνάμει ἡμιολία, τῆς δὲ τοῦ ὀκταέδρου τῆς BE δυνάμει διπλασίον, τῆς δὲ τοῦ κύβου τῆς ZB δυνάμει τριπλασίον, $ο<$ ίων $>$ άρα ἡ τῆς σφαίρας διάμετρος δυνάμει $<$ έχ. τοιούτων ἡ μὲν τῆς πυραμίδος τεσσάρων, ἡ δὲ τοῦ ὀκταέδρου τριῶν, ἡ δὲ τοῦ κύβου δύο. ἡ μὲν $>$ άρα τῆς πυραμίδος πλευρὰ τῆς μὲν τοῦ ὀκταέδρου πλευρᾶς δυνάμει ἐστὶν ἐπίτριτοξ, τῆς δὲ τοῦ κύβου δυνάμει διπλῇ, ἡ δὲ τοῦ ὀκταέδρου τῆς τοῦ κύβου δυνάμει ἡμιολία. αἱ μὲν $ο>$ ύν εἰρημέναι τῶν τριῶν σθημάτων πλευραί, λέγω δὴ πυραμίδος καὶ ὀκταέδρου καὶ κύβου, πρὸς ἀλλήλας εἰσὶν ἐν λόγοις $<$ ρητοῖς. αἱ δὲ λοιπαὶ δύο, λέγω δὴ $<$ ή τε τοῦ εἰκοσάεδρου καὶ ἡ τοῦ δωδεκαέδρου, $ο>$ ύτε πρὸς ἀλλήλας $ο>$ ύτε πρὸς τὰς προειρημένας εἰσὶν ἐν λόγοις $<$ ρητοῖς; $>$ άλογοι γάρ εἰσιν, ἡ μὲν ἐλάττων, ἡ δὲ ἀποτομή.

$<$ Ότι μείζον ἐστὶν ἡ τοῦ εἰκοσάεδρου πλευρὰ ἡ MB τῆς τοῦ δωδεκαέδρου τῆς NB , δείχομεν $ο<$ ύτως.

Ἐπεὶ γὰρ ἰσογώνιον ἐστὶ τὸ $ΔAB$ τρίγωνον τῷ ZAB τριγώνῳ, ἀνάλογόν ἐστιν ὥς ἡ $ΔB$ πρὸς τὴν BZ , $ο<$ ύτως ἡ BZ πρὸς τὴν BA . καὶ ἐπεὶ τρεῖς εὐθεῖαι ἀνάλογόν εἰσιν, $>$ έστιν ὥς ἡ πρώτη πρὸς τὴν τρίτην, $ο<$ ύτως τὸ ἀπὸ τῆς πρώτης πρὸς τὸ ἀπὸ τῆς δευτέρας; $>$ έστιν $>$ άρα ὥς ἡ $ΔB$ πρὸς τὴν BA , $ο<$ ύτως τὸ ἀπὸ τῆς $ΔB$ πρὸς τὸ ἀπὸ τῆς BZ ; ἀνάπαλιν $>$ άρα ὥς ἡ AB πρὸς τὴν BD , $ο<$ ύτως τὸ ἀπὸ τῆς ZB πρὸς τὸ ἀπὸ

let GC be joined. And let HK be drawn from H , perpendicular to AB . And since GA is double AC . For GA (is) equal to AB . And as GA (is) to AC , so HK (is) to KC [Prop. 6.4]. HK (is) thus also double KC . Thus, the (square) on HK is four times the (square) on KC . Thus, the (sum of the squares) on HK and KC , which is the (square) on HC [Prop. 1.47], is five times the (square) on KC . And HC (is) equal to CB . Thus, the (square) on BC (is) five times the (square) on CK . And since AB is double CB , of which AD is double DB , the remainder BD is thus double the remainder DC . BC (is) thus triple CD . The (square) on BC (is) thus nine times the (square) on CD . And the (square) on BC (is) five times the (square) on CK . Thus, the (square) on CK (is) greater than the (square) on CD . CK is thus greater than CD . Let CL be made equal to CK . And let LM be drawn from L at right-angles to AB . And let MB be joined. And since the (square) on BC is five times the (square) on CK , and AB is double BC , and KL double CK , the (square) on AB is thus five times the (square) on KL . And the square on the diameter of the sphere is also five times the (square) on the radius of the circle from which the icosahedron has been described [Prop. 13.16 corr.]. And AB is the diameter of the sphere. Thus, KL is the radius of the circle from which the icosahedron has been described. Thus, KL is (the side) of the hexagon (inscribed) in the aforementioned circle [Prop. 4.15 corr.]. And since the diameter of the sphere is composed of (the side) of the hexagon, and two of (the sides) of the decagon, inscribed in the aforementioned circle, and AB is the diameter of the sphere, and KL the side of the hexagon, and AK (is) equal to LB , thus AK and LB are each sides of the decagon inscribed in the circle from which the icosahedron has been described. And since LB is (the side) of the decagon. And ML (is the side) of the hexagon—for (it is) equal to KL , since (it is) also (equal) to HK , for they are equally far from the center. And HK and KL are each double KC . MB is thus (the side) of the pentagon (inscribed in the circle) [Props. 13.10, 1.47]. And (the side) of the pentagon is (the side) of the icosahedron [Prop. 13.16]. Thus, MB is (the side) of the icosahedron.

And since FB is the side of the cube, let it be cut in extreme and mean ratio at N , and let NB be the greater piece. Thus, NB is the side of the dodecahedron [Prop. 13.17 corr.].

And since the (square) on the diameter of the sphere was shown (to be) one and a half times the square on the side, AF , of the pyramid, and twice the square on (the side), BE , of the octagon, and three times the square on (the side), FB , of the cube, thus, of whatever (parts) the (square) on the diameter of the sphere (makes) six, of such (parts) the (square) on (the side) of the pyramid (makes) four, and (the square) on (the side) of the octagon three, and (the square) on (the side) of the cube two. Thus, the (square) on the side of the pyramid is one and a third times the square on the side of the octagon, and double the square on (the side) of the cube. And the (square) on (the side) of the octahedron is one and a half times the square on (the side) of the cube. Therefore, the aforementioned sides of the three figures—I mean, of the pyramid, and of the octahedron, and of the cube—are in rational ratios to one another. And (the sides of) the remaining two (figures)—I mean, of the icosahedron, and of the dodecahedron—are neither in rational ratios to one

τῆς ΒΔ. τριπλῇ δὲ ἡ ΑΒ τῆς ΒΔ; τριπλάσιον > ἄρα τὸ ἀπὸ τῆς ΖΒ τοῦ ἀπὸ τῆς ΒΔ. > ἐστὶ δὲ καὶ τὸ ἀπὸ τῆς ΑΔ τοῦ ἀπὸ τῆς ΔΒ τετραπλάσιον; διπλῇ γὰρ ἡ ΑΔ τῆς ΔΒ; μείζων > ἄρα τὸ ἀπὸ τῆς ΑΔ τοῦ ἀπὸ τῆς ΖΒ; μείζων > ἄρα ἡ ΑΔ τῆς ΖΒ; πολλῶν > ἄρα ἡ ΑΛ τῆς ΖΒ μείζων ἐστίν. καὶ τῆς μὲν ΑΛ > ἄκρον καὶ μέσον λόγον τεμνομένης τὸ μείζων τμήμα ἐστὶν ἡ ΚΛ, ἐπειδὴ περ ἡ μὲν ΛΚ ἔχων ἄκρον ἐστίν, ἡ δὲ ΚΑ δεκαγώνου; τῆς δὲ ΖΒ > ἄκρον καὶ μέσον λόγον τεμνομένης τὸ μείζων τμήμα ἐστὶν ἡ ΝΒ; μείζων > ἄρα ἡ ΚΛ τῆς ΝΒ. > ἴση δὲ ἡ ΚΛ τῇ ΛΜ; μείζων > ἄρα ἡ ΛΜ τῆς ΝΒ [τῆς δὲ ΛΜ μείζων ἐστὶν ἡ ΜΒ]. πολλῶν > ἄρα ἡ ΜΒ πλευρὰ οὐ > ὅσα τοῦ εἰκοσαέδρου μείζων ἐστὶ τῆς ΝΒ πλευρᾶς οὐ > ὅσης τοῦ δωδεκαέδρου; < ὅπερ > ἐδει δεῖχαι.

another, nor to the (sides) of the aforementioned (three figures). For they are irrational (straight-lines): (namely), a minor [Prop. 13.16], and an apotome [Prop. 13.17]. (And), we can show that the side, MB , of the icosahedron is greater than the (side), NB , of the dodecahedron, as follows. For, since triangle FDB is equiangular to triangle FAB [Prop. 6.8], proportionally, as DB is to BF , so BF (is) to BA [Prop. 6.4]. And since three straight-lines are (continually) proportional, as the first (is) to the third, so the (square) on the first (is) to the (square) on the second [Def. 5.9, Prop. 6.20 corr.]. Thus, as DB is to BA , so the (square) on DB (is) to the (square) on BA . Thus, inversely, as AB (is) to BD , so the (square) on FB (is) to the (square) on BD . And AB (is) triple BD . Thus, the (square) on FB (is) three times the (square) on BD . And the (square) on AD is also four times the (square) on DB . For AD (is) double DB . Thus, the (square) on AD (is) greater than the (square) on FB . Thus, AD (is) greater than FB . Thus, AL is much greater than FB . And KL is the greater piece of AL , which is cut in extreme and mean ratio—inasmuch as LK is (the side) of the hexagon, and KA (the side) of the decagon [Prop. 13.9]. And NB is the greater piece of FB , which is cut in extreme and mean ratio. Thus, KL (is) greater than NB . And KL (is) equal to LM . Thus, LM (is) greater than NB [and MB is greater than LM]. Thus, MB , which is (the side) of the icosahedron, is much greater than NB , which is (the side) of the dodecahedron. (Which is) the very thing it was required to show.

† If the radius of the given sphere is unity then the sides of the pyramid (*i.e.*, tetrahedron), octahedron, cube, icosahedron, and dodecahedron, respectively, satisfy the following inequality: $\sqrt{8/3} > \sqrt{2} > \sqrt{4/3} > (1/\sqrt{5}) \sqrt{10 - 2\sqrt{5}} > (1/3)(\sqrt{15} - \sqrt{3})$.

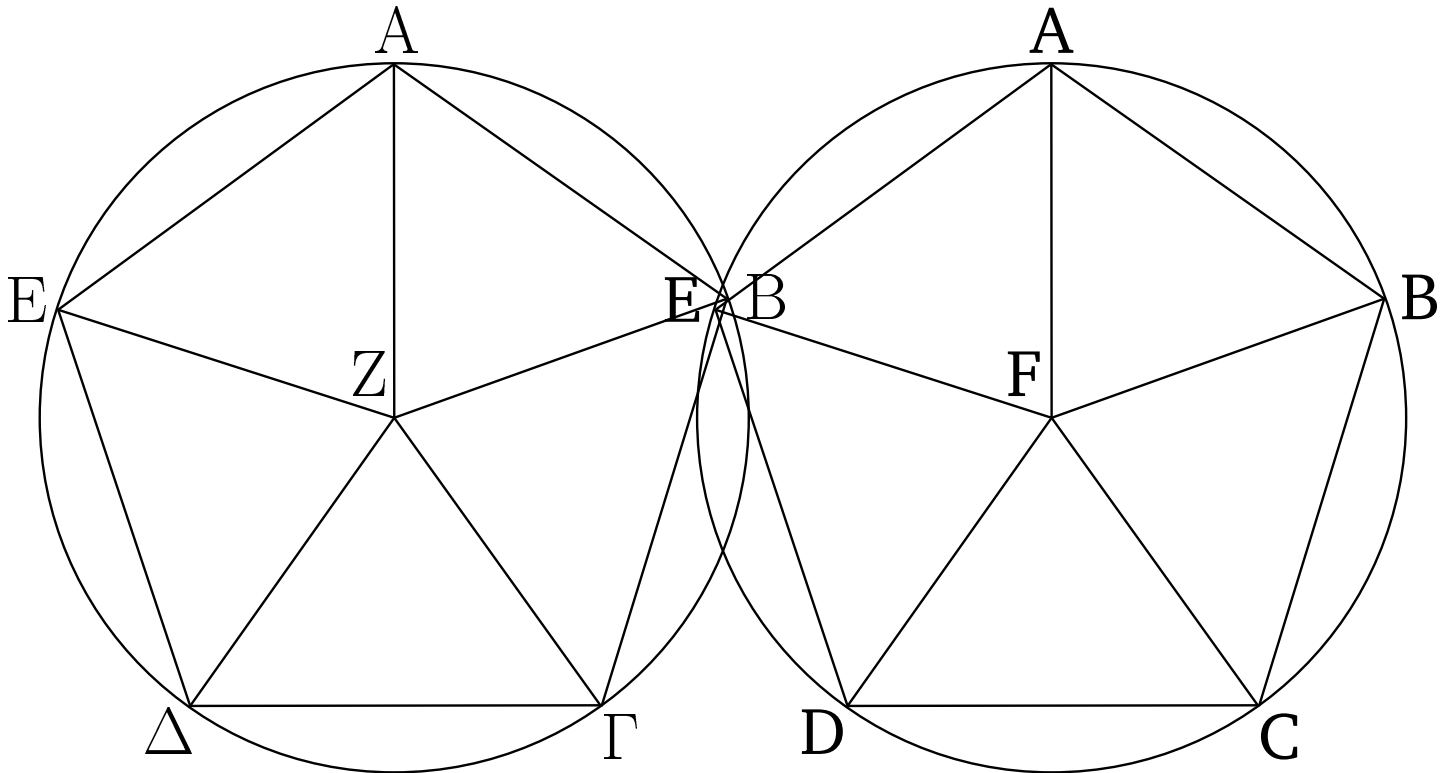
Λέγω δὴ, <ὅτι παρὰ τὰ εἰρημένα πέντε σήματα οὐ συσταθήσεται, I say that, beside the five aforementioned figures, no <έτερον σῆμα περιεχόμενον ὑπὸ ἰσοπλευρῶν τε καὶ ἰσογωνίων> other (solid) figure can be constructed (which is) contained > ἴσων ἀλλήλοις. by equilateral and equiangular (planes), equal to one another.

Ὑπὸ μὲν γὰρ δύο τριγώνων > ἡ <ὅλως ἐπιπέδων στερεὰ γωνία οὐ συνίσταται. ὑπὸ δὲ τριῶν τριγώνων ἡ τῆς πυραμίδος, For a solid angle cannot be constructed from two triangles, or indeed (two) planes (of any sort) [Def. 11.11]. And (the solid angle) of the pyramid (is constructed) from three (equiangular) triangles, and (that) of the octahedron from four (triangles), and (that) of the icosahedron from (five) triangles. And a solid angle cannot be (made) from six equilateral and equiangular triangles set up together at one point. For, since the angles of a equilateral triangle are (each) two-thirds of a right-angle, the (sum of the) six (plane) angles (containing the solid angle) will be four right-angles. The very thing (is) impossible. For every solid angle is contained by (plane angles whose sum is) less than four right-angles [Prop. 11.21]. So, for the same (reasons), a solid angle cannot be constructed from more than six plane angles (equal to two-thirds of a right-angle) either. And the (solid) angle of a cube is contained by three squares. And (a solid angle contained) by four (squares is) impossible. For, again, the (sum of the plane angles containing the solid angle) will be four right-angles. And (the solid angle) of a dodecahedron (is contained) by three equilateral and equiangular pentagons. And (a solid angle contained) by four (equiangular pentagons is) impossible. For, the angle of an equilateral pentagon being one and one-fifth of right-angle, four (such) angles will be greater (in sum) than four right-angles. The

Οὐκ > ἄρα παρὰ τὰ εἰρημένα πέντε σήματα <έτερον σῆμα στερεὸν συσταθήσεται ὑπὸ ἰσοπλευρῶν τε καὶ ἰσογωνίων περιεχόμενον; < ὅπερ > ἐδει δεῖχαι.

very thing (is) impossible. And, on account of the same absurdity, a solid angle cannot be constructed from any other (equiangular) polygonal figures either.

Thus, beside the five aforementioned figures, no other solid figure can be constructed (which is) contained by equilateral and equiangular (planes). (Which is) the very thing it was required to show.



Λήμμα.

Lemma

<Ὅτι δὲ ἡ τοῦ ἰσοπλεύρου καὶ ἰσογώνιου πενταγώνου γωνία ὀρθή ἐστι καὶ πέμπτου, οὕτω δεικτέον.

>Ἐστω γὰρ πεντάγωνον ἰσόπλευρον καὶ ἰσογώνιον τὸ ΑΒΓΔΕ, καὶ περιγεγράφθω περὶ αὐτὸ κύκλος ὁ ΑΒΓΔΕ, καὶ εἰλήφθω αὐτοῦ τὸ κέντρον τὸ Ζ, καὶ ἐπεζεύθωσαν αἱ ΖΑ, ΖΒ, ΖΓ, ΖΔ, ΖΕ. διὸ αἱ >άρα τέμνουσι τὰς πρὸς τοῖς Α, Β, Γ, Δ, Ε τοῦ πενταγώνου γωνίας. καὶ ἐπεὶ αἱ πρὸς τῷ Ζ πέντε γωνίαι τέσσαρσιν ὀρθαῖς >ίσαι εἰσὶ καὶ εἰσιν >ίσαι, μία >άρα αὐτῶν, ὥς ἡ ὑπὸ ΑΖΒ, μιᾶς ὀρθῆς ἐστὶ παρὰ πέμπτου; λοιπαὶ >άρα αἱ ὑπὸ ΖΑΒ, ΑΒΖ μιᾶς εἰσιν ὀρθῆς καὶ πέμπτου. >ίση δὲ ἡ ὑπὸ ΖΑΒ τῇ ὑπὸ ΖΒΓ; καὶ <ὅλη >άρα ἡ ὑπὸ ΑΒΓ τοῦ πενταγώνου γωνία μιᾶς ἐστὶν ὀρθῆς καὶ πέμπτου; <ὅπερ >έδει δεῖχαι.

It can be shown that the angle of an equilateral and equiangular pentagon is one and one-fifth of a right-angle, as follows.

For let $ABCDE$ be an equilateral and equiangular pentagon, and let the circle $ABCDE$ be circumscribed about it [Prop. 4.14]. And let its center, F , be found [Prop. 3.1]. And let FA , FB , FC , FD , and FE be joined. Thus, they cut the angles of the pentagon in half at (points) A , B , C , D , and E [Prop. 1.4]. And since the five angles at F are equal (in sum) to four right-angles, and are also equal (to one another), (any) one of them, like AFB , is thus one less a fifth of a right-angle. Thus, the (sum of the) remaining (angles in triangle ABF), FAB and ABF , is one plus a fifth of a right-angle [Prop. 1.32]. And FAB (is) equal to FBC . Thus, the whole angle, ABC , of the pentagon is also one and one-fifth of a right-angle. (Which is) the very thing it was required to show.

GREEK-ENGLISH LEXICON

ABBREVIATIONS: *act* - active; *adj* - adjective; *adv* - adverb; *conj* - conjunction; *fut* - future; *gen* - genitive; *imperat* - imperative; *impf* - imperfect; *ind* - indeclinable; *indic* - indicative; *intr* - intransitive; *mid* - middle; *neut* - neuter; *no* - noun; *par* - particle; *part* - participle; *pass* - passive; *perf* - perfect; *pre* - preposition; *pres* - present; *pro* - pronoun; *sg* - singular; *tr* - transitive; *vb* - verb.

>άγω, >άχω, >ήγαγον, >ήθα, >ήγμαι, >ήθην : *vb*, lead, draw (a line).

ἀδύνατος -ον : *adj*, impossible.

ἀεί : *adv*, always, for ever.

αἰρέω, αἰρήσω, εἰ | ἵλον, <ή|ρηκα, <ή|ρημαι, ή|ρέθην : *vb*, grasp.

αἰτέω, αἰτήσω, >ήτησα, >ή|τηκα, >ή|τημαι, ή|τήθη : *vb*, postulate.

α>ίτημα -ατοξ, τό : *no*, postulate.

ἀκόλουθος -ον : *adj*, analogous, consequent on, in conformity with.

>άκροξ -α -ον : *adj*, outermost, end, extreme.

ἄλλά : *conj*, but, otherwise.

>άλογος -ον : *adj*, irrational.

<άμα : *adv*, at once, at the same time, together.

ἀμβλυγώνιος -ον : *adj*, obtuse-angled; τὸ ἀμβλυγώνιον, *no*, obtuse angle.

ἀμβλύξ -εῖα -ύ : *adj*, obtuse.

ἀμφοτέροξ -α -ον : *pro*, both.

ἀναγράφω : *vb*, describe (a figure); see γράφω.

ἀναλογία, ή : *no*, proportion, (geometric) progression.

ἀνάλογος -ον : *adj*, proportional.

ἀνάπαλιν : *adv*, inverse(ly).

αναπληρόω : *vb*, fill up.

ἀναστρέφω : *vb*, turn upside down, convert (ratio); see στρέφω.

ἀναστροφή, ή : *no*, turning upside down, conversion (of ratio).

ἀνθυφαιρέω : *vb*, take away in turn; see αἰρέω.

ἀνίστημι : *vb*, set up; see >ίστημι.

>άνισος -ον : *adj*, unequal, uneven.

ἀντιπάσθω : *vb*, be reciprocally proportional; see πάσθω.

>άχων -ονοξ, ό : *vb*, axis.

<άπαχ : *adv*, once.

<άπαξ, <άπασα, <άπαν : *adj*, quite all, the whole.

>άπειρος -ον : *adj*, infinite.

ἀπεναντίον : *ind*, opposite.

ἀπέθω : *vb*, be far from, be away from; see >έθω.

ἀπλατήξ -έξ : *adj*, without breadth.

ἀπόδειξις -εωξ, ή : *no*, proof.

ἀποκαθίστημι : *vb*, re-establish, restore; see >ίστημι.

ἀπολαμβάνω : *vb*, take from, subtract from, cut off from; see λαμβάνω.

ἀποτέμνω : *vb*, cut off, subtend.

>ap'otmhma -atoc, t'ο : *no*, piece cut off, segment.

ἀποτομή, ή : *vb*, piece cut off, apotome.

<άπτω, <άψω, <ήψα, —, <ήμμαι, — : *vb*, touch, join, meet.

ἀπώτεροξ -α -ον : *adj*, further off.

>άρα : *par*, thus, as it seems (inferential).

ἀριθμός, ό : *no*, number.

ἄρτιάκιξ : *adv*, an even number of times.

αρτιόπλευρος -ον : *adj*, having a even number of sides.

>άρθω, >άρχω, >ήρχα, >ήρθα, >ήργμαι, >ήρθην : *vb*, rule; *mid.*, begin.

ἀσύμμετροξ -ον : *adj*, incommensurable.

ἀσύμπτωτος -ον : *adj*, not touching, not meeting.

>άρτιος -α -ον : *adj*, even, perfect.

>άρμητος -ον : *adj*, uncut.

ἀτόπος -ον : *adj*, absurd, paradoxical.

αὐτόθεν : *adv*, immediately, obviously.

ἀφαίρεω : *vb*, take from, subtract from, cut off from; see αἰρέω.

ἀφή, ή : *no*, point of contact.

βάθος -εοξ, τό : *no*, depth, height.

βαίνω, -βήσομαι, -έβην, βέβηκα, —, — : *vb*, walk; *perf*, stand (of angle).

βάλλω, βαλῶ, >έβαλον, βέβληκα, βέβλημαι, έβλήθην : *vb*, throw.

βάσις -εωξ, ή : *no*, base (of a triangle).

γάρ : *conj*, for (explanatory).

γίγνομαι, γενήσομαι, έγενόμην, γέγονα, γεγένημαι, — : *vb*, happen, become.

γνώμων -ονοξ, ή : *no*, gnomon.

γραμμή, ή : *no*, line.

γράφω, γράψω, >έγρα[ψ/φ]α, γέγραφα, γέγραμμαι, έραψάμην : *vb*, draw (a figure).

γωνία, ή : *no*, angle.

δεῖ : *vb*, be necessary; δεῖ, it is necessary; >έδει, it was necessary; δέον, being necessary.

δείκνυμι, δείχω, >έδειχα, δέδειθα, δέδειγμαι, έδείθην : *vb*, show, demonstrate.

δεικτέον : *ind*, one must show.

δείξις -εωξ, ή : *no*, proof.

δεκαγώνος -ον : *adj*, ten-sided; τὸ δεκαγώνον, *no*, decagon.

δέθομαι, δέχομαι, έδεχάμην, —, δέδεγμαι, έδέθην : *vb*, receive, accept.

δή : *conj*, so (explanatory).

δηλαδή : *ind*, quite clear, manifest.

δήλος -η -ον : *adj*, clear.

δηλονότι : *adv*, manifestly.

διάγω : *vb*, carry over, draw through, draw across; see >άγω.

διαγώνιος -ον : *adj*, diagonal.

διαλείπω : *vb*, leave an interval between.

διάμετροξ -ον : *adj*, diametrical; ή διάμετροξ, *no*, diameter, diagonal.

διαίρεσις -εωξ, ή : *no*, division, separation.

διαίρεω : *vb*, divide (in two); διαρεθέντος -η -ον, *adj*, separated (ratio); see αἰρέω.

διάστημα -ατοξ, τό : *no*, radius.

διαφέρω : *vb*, differ; see φέρω.

δίδωμι, δώσω, >έδωκα, δέδωκα, δέδομαι, έδόθην : *vb*, give.

διμοίρος -ον : *adj*, two-thirds.

διπλασιάζω : *vb*, double.

διπλάσιος -α -ον : *adj*, double, twofold.

διπλάσιων -ον : *adj*, double, twofold.

διπλοῦξ -ή -οῦν : *adj*, double.

δίξ : *adv*, twice.

διθα : *adv*, in two, in half.

διθορομία, ή : *no*, point of bisection.

δυάς -άδοξ, ή : *no*, the number two, dyad.

δύναμαι : *vb*, be able, be capable, generate, square, be when squared; δυναμένη, ή, *no*, square-root (of area)—i.e., straight-line whose square is equal to a given area.

δύναμις -εωξ, ή : *no*, power (usually 2nd power when used in mathematical sence, hence), square.

δυνατός -ή -όν : *adj*, possible.

δωδεκάεδρος -ον : *adj*, twelve-sided.

εαυτοῦ -ήξ -οῦ : *adj*, of him/her/it/self, his/her/its/own.

- ἐγγίων -ον : *adj*, nearer, nearest.
 ἐγγράφω : *vb*, inscribe; see γράφω.
 ε>ἶδος -εοξ, τό : *no*, figure, form, shape.
 εἰκοσάεδρος -ον : *adj*, twenty-sided.
 ε>ίρω/λέγω, ἐρῶ/ερέω, ε>ἵπον, ε>ίρηκα, ε>ίρημαι, ἐρρήθην : *vb*, say, speak; *per pass part*, εἰρημένος -η -ον, *adj*, said, aforementioned.
 ε>ἴτε ...ε>ἵτε : *ind*, either ...or.
 <έκαστος -η -ον : *pro*, each, every one.
 ἐκατέροξ -α -ον : *pro*, each (of two).
 ἐκβάλλω : *vb*, produce (a line); see βάλλω.
 ἐκθέω : *vb*, set out.
 >έκκειμαι : *vb*, be set out, be taken; see κείμαι.
 ἐκτίθημι : *vb*, set out; see τίθημι.
 ἐκτός : *pre* + *gen*, outside, external.
 ἐλάσσων/ἐλάττων -ον : *adj*, less, lesser.
 ἐλάχιστος -η -ον : *adj*, least.
 ἐλλείπω : *vb*, be less than, fall short of.
 ἐμπίπτω : *vb*, meet (of lines), fall on; see πίπτω.
 >έμπροσθεν : *adv*, in front.
 ἐναλλάχ : *adv*, alternate(ly).
 ἐναρμόζω : *vb*, insert; *perf indic pass 3rd sg*, ἐνήρμοσται.
 ἐνδέθομαι : *vb*, admit, allow.
 <ένεκεν : *ind*, on account of, for the sake of.
 ἐνναπλάσιος -α -ον : *adj*, nine-fold, nine-times.
 >έννοια, ἡ : *no*, notion.
 ἐνπεριέθω : *vb*, encompass; see >έθω.
 ἐνπίπτω : see ἐμπίπτω.
 ἐντός : *pre* + *gen*, inside, interior, within, internal.
 ἐξάγωνος -ον : *adj*, hexagonal; τὸ ἐξάγωνον, *no*, hexagon.
 ἐξαπλάσιος -α -ον : *adj*, sixfold.
 ἐξῆξ : *adv*, in order, successively, consecutively.
 >έχωθεν : *adv*, outside, extrinsic.
 ἐπάνω : *adv*, above.
 ἐπαφή, ἡ : *no*, point of contact.
 ἐπεί : *conj*, since (causal).
 ἐπειδήπερ : *ind*, inasmuch as, seeing that.
 ἐπιζεύγνυμι, ἐπιζεύγω, ἐπέζευχα, —, ἐπέζευγμαι, ἐπέζεύθην : *vb*, join (by a line).
 ἐπιλογίζομαι : *vb*, conclude.
 ἐπινοέω : *vb*, think of, contrive.
 ἐπιπέδος -ον : *adj*, level, flat, plane; τὸ ἐπιπέδον, *no*, plane.
 ἐπισκέπτομαι : *vb*, investigate.
 ἐπίσκεψιξ -εωξ, ἡ : *no*, inspection, investigation.
 ἐπιτάσσω : *vb*, put upon, enjoin; τὸ ἐπιταθέν, *no*, the (thing) prescribed; see τάσσω.
 ἐπίτριτος -ον : *adj*, one and a third times.
 ἐπιφάνεια, ἡ : *no*, surface.
 >έπομαι : *vb*, follow.
 >έρθομαι, ἐλεύσομαι, >ήλθον, ἐλήλυθα, —, — : *vb*, come, go.
 >έσθατος -η -ον : *adj*, outermost, uttermost, last.
 ἑτερόμηκηξ -εξ : *adj*, oblong; τὸ ἑτερόμηκεξ, *no*, rectangle.
 <έτερος -α -ον : *adj*, other (of two).
 >έτι : *par*, yet, still, besides.
 εὐθύγραμμος -ον : *adj*, rectilinear; τὸ εὐθύγραμμον, *no*, rectilinear figure.
 εὐθύξ -εἶα -ύ : *adj*, straight; ἡ εὐθεῖα, *no*, straight-line; ἐπ>εὐθεῖαξ, in a straight-line, straight-on.
 εὐρίσκω, εὐρήσκω, η<ύρον, ε<ύρεκα, ε<ύρημαι, εὐρέθην : *vb*, find.
 ἐφάπτω : *vb*, bind to; *mid*, touch; ἡ ἐφαπτομένη, *no*, tangent; see <άπτω.
 ἐφαρμόζω, ἐφαρμόσω, ἐφήρμοσα, ἐφήμοκα, ἐφήμοσμαι, ἐφήμόσθην : *vb*, coincide; *pass*, be applied.
 ἐφεχῆξ : *adv*, in order, adjacent.
 ἐφίστημι : *vb*, set, stand, place upon; see >ίστημι.
 >έθω, <έχω, >έσθον, >έσθηκα, -έσθημαι, — : *vb*, have.
 ἡγέομαι, ἡγήσομαι, ἡγησάμην, ἡγῆμαι, —, ἡγήθην : *vb*, lead.
 >ήδη : *ind*, already, now.
 <ήκω, <ήχω, —, —, —, — : *vb*, have come, be present.
 ἡμικύκλιον, τό : *no*, semi-circle.
 ἡμιόλιος -α -ον : *adj*, containing one and a half, one and a half times.
 <ήμισυξ -εἰα -υ : *adj*, half.
 >ήπερ = >ή + περ : *conj*, than, than indeed.
 >ήτοι ...>ή : *par*, surely, either ...or; in fact, either ...or.
 θέσις -εωξ, ἡ : *no*, placing, setting, position.
 θεωρημα -ατος, τό : *no*, theorem.
 >ίδιος -α -ον : *adj*, one's own.
 ισάκιξ : *adv*, the same number of times; ισάκιξ πολλαπλάσια, the same multiples, equal multiples.
 ισογώνιος -ον : *adj*, equiangular.
 ισόπλευρος -ον : *adj*, equilateral.
 ισοπληθής -έξ : *adj*, equal in number.
 >ίσοξ -η -ον : *adj*, equal; ἐχ>ίσου, equally, evenly.
 ισοσκελής -έξ : *adj*, isosceles.
 >ίστημι, στήσω, >έστησα, —, —, ἐστάθην : *vb tr*, stand (something).
 >ίστημι, στήσω, >έστην, <έστηκα, <έσταμαι, ἐσταθην : *vb intr*, stand up (oneself); Note: perfect *I have stood up* can be taken to mean present *I am standing*.
 ἰσο"υψής -έξ : *adj*, of equal height.
 καθάπερ : *ind*, according as, just as.
 κάθετος -ον : *adj*, perpendicular.
 καθόλου : *adv*, on the whole, in general.
 καλέω : *vb*, call.
 κἀκεινοξ = καὶ ἐκεῖνοξ .
 κ>άν = καὶ >άν : *ind*, even if, and if.
 καταγραφῆ, ἡ : *no*, diagram, figure.
 καταγράφω : *vb*, describe/draw, inscribe (a figure); see γράφω.
 κατακολουθέω : *vb*, follow after.
 καταλείπω : *vb*, leave behind; see λείπω; τὰ καταλειπόμενα, *no*, remainder.
 κατάλληλος -ον : *adj*, in succession, in corresponding order.
 καταμετρέω : *vb*, measure (exactly).
 καταντάω : *vb*, come to, arrive at.
 κατασκευάζω : *vb*, furnish, construct.
 κείμαι, κείσομαι, —, —, —, — : *vb*, have been placed, lie, be made; see τίθημι.
 κέντρον, τό : *no*, center.
 κλάω : *vb*, break off, inflect.
 κλίνω, κλίνω, >έκλινα, κέκλικα, κέκλιμαι, ἐκλίθην : *vb*, lean, incline.
 κλίσιξ -εωξ, ἡ : *no*, inclination, bending.
 κοῦλος -η -ον : *adj*, hollow, concave.
 κορυφή, ἡ : *no*, top, summit, apex; κατὰ κορυφήν, vertically opposite (of angles).
 κρίνω, κρίνω, >έκρινα, κέκρικα, κέκριμαι, ἐκρίθην : *vb*, judge.
 κύβοξ, ό : *no*, cube.

κύκλος, ὁ : *no*, circle.

κύλινδρος, ὁ : *no*, cylinder.

κυρτός -ή -όν : *adj*, convex.

κῶνος, ὁ : *no*, cone.

λαμβάνω, λήψομαι, >έλαβον, >έληφα >έλημμαι, ἐλήφθην : *vb*, take.

λέγω : *vb*, say; *pres pass part*, λεγόμενος -η -ον, *adj*, so-called; see >είρω.

λείπω, λείψω, >έλιπον, λέλοιπα, λέλειμμαι, ἐλείφθην : *vb*, leave, leave behind.

λημμάτιον, τό : *no*, diminutive of λήμμα.

λήμμα -ατος, τό : *no*, lemma.

λήψις -εως, ἡ : *no*, taking, catching.

λόγος, ὁ : *no*, ratio, proportion, argument.

λοιπός -ή -όν : *adj*, remaining.

μανθάνω, μαθήσομαι, >έμαθον, μεμάθηκα, —, — : *vb*, learn.

μέγεθος -εως, τό : *no*, magnitude, size.

μείζων -ον : *adj*, greater.

μένω, μενῶ, >έμεινα, μεμένηκα, —, — : *vb*, stay, remain.

μέρος -ους, τό : *no*, part, direction, side.

μέσος -η -ον : *adj*, middle, mean, medial; ἐκ δύο μέσων, bimedial.

μεταλαμβάνω : *vb*, take up.

μεταξύ : *adv*, between.

μετέωρος -ον : *adj*, raised off the ground.

μετρέω : *vb*, measure.

μέτρον, τό : *no*, measure.

μηδεῖς, μηδεμία, μηδέν : *adj*, not even one, (neut.) nothing.

μηδέποτε : *adv*, never.

μηδέτερος -α -ον : *pro*, neither (of two).

μήκος -εως, τό : *no*, length.

μήν : *par*, truly, indeed.

μονάς -άδος, ἡ : *no*, unit, unity.

μοναθός -ή -όν : *adj*, unique.

μοναθῶς : *adv*, uniquely.

μόνος -η -ον : *adj*, alone.

νοέω, —, νόησα, νενόηκα, νενόημαι, ἐνόηθην : *vb*, apprehend, conceive.

ο<τις -α -ον : *pre*, such as, of what sort.

ὀκτάεδρος -ον : *adj*, eight-sided.

<όλος -η -ον : *adj*, whole.

ὁμογενής -έξ : *adj*, of the same kind.

<όμοιος -α -ον : *adj*, similar.

ὁμοιοπληθής -έξ : *adj*, similar in number.

ὁμοιοταγής -έξ : *adj*, similarly arranged.

ὁμοιότης -ητος, ἡ : *no* similarity.

ὁμοίως : *adv*, similarly.

ὁμόλογος -ον : *adj*, corresponding, homologous.

ὁμοταγής -έξ : *adj*, ranged in the same row or line.

ὁμώνυμος -ον : *adj*, having the same name.

>όνομα -ατος, τό : *no*, name; ἐκ δύο ὀνομάτων, binomial.

ὀχυγώνιος -ον : *adj*, acute-angled; τὸ ὀχυγώνιον, *no*, acute angle.

ὀχύς -εῖα -ύ : *adj*, acute.

ὅποιοσὺν = ὅποῖός -α -ον + ο>ύν : *adj*, of whatever kind, any kind whatsoever.

ὀπόσος -η -ον : *pro*, as many, as many as.

ὀποσοσδηποτοῦν = ὀπόσος -η -ον + δὴ + ποτέ + ο>ύν : *adj*, of whatever number, any number whatsoever.

ὀποσοσὺν = ὀπόσος -η -ον + ο>ύν : *adj*, of whatever number, any number whatsoever.

ὀπότερος -α -ον : *pro*, either (of two), which (of two).

ὀρθογώνιον, τό : *no*, rectangle, right-angle.

ὀρθός -ή -όν : *adj*, straight, right-angled, perpendicular; πρὸς ὀρθὰς γωνίας, at right-angles.

>όρος, ὁ : *no*, boundary, definition, term (of a ratio).

ὅσαδηποτοῦν = <όσα + δὴ + ποτέ + ο>ύν : *ind*, any number whatsoever.

ὀσάκις : *ind*, as many times as, as often as.

ὅσαπλάσιος -ον : *pro*, as many times as.

<όσος -η -ον : *pro*, as many as.

<όσπερ, <ήπερ, <όπερ : *pro*, the very man who, the very thing which.

<όστις, <ήτις, <ό τι : *pro*, anyone who, anything which.

<όταν : *adv*, when, whenever.

ὅτιοῦν : *ind*, whatsoever.

οὐδεῖς, οὐδεμία, οὐδέν : *pro*, not one, nothing.

οὐδέτερος -α -ον : *pro*, not either.

οὐθέτερος : see οὐδέτερος.

οὐθέν : *ind*, nothing.

ο>ύν : *adv*, therefore, in fact.

ο<ύτωξ : *adv*, thusly, in this case.

πάλιν : *adv*, back, again.

πάντως : *adv*, in all ways.

παρὰ : *prep* + *acc*, parallel to.

παραβάλλω : *vb*, apply (a figure); see βάλλω.

παραβολή, ἡ : *no*, application.

παρακείμει : *vb*, lie beside, apply (a figure); see κείμει.

παραλλάσσω, παραλλάχω, —, παρήλλαθα, —, — : *vb*, miss, fall awry.

παραλληλεπίεδος, -ον : *adj*, with parallel surfaces; τὸ παραλληλεπίεδον, *no*, parallelepiped.

παραλληλόγραμμος -ον : *adj*, bounded by parallel lines; τὸ παραλληλόγραμμον, *no*, parallelogram.

παράλληλος -ον : *adj*, parallel; τὸ παράλληλον, *no*, parallel, parallel-line.

παραπλήρωμα -ατος, τό : *no*, complement (of a parallelogram).

παρατέλυτος -ον : *adj*, penultimate.

παρέκ : *prep* + *gen*, except.

παρεμπίπτω : *vb*, insert; see πίπτω.

πάσθω, πείσομαι, >έπαθον, πέπονθα, —, — : *vb*, suffer.

πεντάγωνος -ον : *adj*, pentagonal; τὸ πεντάγωνον, *no*, pentagon.

πενταπλάσιος -α -ον : *adj*, five-fold, five-times.

πεντεκαίδεκάγωνον, τό : *no*, fifteen-sided figure.

πεπερασμένος -η -ον : *adj*, finite, limited; see περαίνω.

περαίνω, περανῶ, ἐπέρανα, —, πεπεράσμαι, ἐπερανάνθην : *vb*, bring to end, finish, complete; *pass*, be finite.

πέρας -ατος, τό : *no*, end, extremity.

περατώ, —, —, —, — : *vb*, bring to an end.

περιγράφω : *vb*, circumscribe; see γράφω.

περιέθω : *vb*, encompass, surround, contain, comprise; see >έθω.

περιλαμβάνω : *vb*, enclose; see λαμβάνω.

περισσάκις : *adv*, an odd number of times.

περισσός -ή -όν : *adj*, odd.

περιφέρεια, ἡ : *no*, circumference.

περιφέρω : *vb*, carry round; see φέρω.

πηλικότης -ητος, ἡ : *no*, magnitude, size.

πίπτω, πεσοῦμαι, >έπεσον, ἐπέτωκα, —, — : *vb*, fall.

πλάτος -εως, τό : *no*, breadth, width.

πλείων -ον : *adj*, more, several.

πλευρά, ἡ : *no*, side.

πληθὸς -εὼς, τό : *no*, great number, multitude, number.

πλήν : *adv & prep + gen*, more than.

ποιόξ -ά -όν : *adj*, of a certain nature, kind, quality, type.

πολλαπλασιάζω : *vb*, multiply.

πολλαπλασιασμόξ, ό : *no*, multiplication.

πολλαπλάσιον, τό : *no*, multiple.

πολύεδροξ -ον : *adj*, polyhedral; τό πολυέδρον, *no*, polyhedron.

πολύγωνοξ -ον : *adj*, polygonal; τό πολυγώνον, *no*, polygon.

πολύπλευροξ -ον : *adj*, multilateral.

πόρισμα -ατοξ, τό : *no*, corollary.

ποτέ : *ind*, at some time.

πρίσμα -ατοξ, τό : *no*, prism.

προβαίνω : *vb*, step forward, advance.

προδείκνυμι : *vb*, show previously; see δείκνυμι.

προεκτίθημι : *vb*, set forth beforehand; see τίθημι.

προερέω : *vb*, say beforehand; *perf pass part*, προειρημένοξ -η -ον, *adj*, aforementioned; see ε>ίρω.

προσαναπληρώω : *vb*, fill up, complete.

προσαναγράφω : *vb*, complete (tracing of); see γράφω.

προσαρμόζω : *vb*, fit to, attach to.

προσεκβάλλω : *vb*, produce (a line); see ἐκβάλλω.

προσευρίσκω : *vb*, find besides, find; see εὐρίσκω.

προσλαμβάνω : *vb*, add.

πρόκειμαι : *vb*, set before, prescribe; see κείμει.

πρόσκειμαι : *vb*, be laid on, have been added to; see κείμει.

προσπίπτω : *vb*, fall on, fall toward, meet; see πίπτω.

προτασιξ -εὼς, ή : *no*, proposition.

προστάσσω : *vb*, prescribe, enjoin; τὸ τροσταθθέν, *no*, the thing prescribed; see τάσσω.

προστίθημι : *vb*, add; see τίθημι.

πρότεροξ -α -ον : *adj*, first (comparative), before, former.

προτίθημι : *vb*, assign; see τίθημι.

προθωρέω : *vb*, go/come forward, advance.

πρώτοξ -α -ον : *adj*, first, prime.

πυραμίξ -ίδοξ, ή : *no*, pyramid.

<ρητόξ -ή -όν : *adj*, expressible, rational.

<ρομβοειδήξ -έξ : *adj*, rhomboidal; τὸ <ρομβοειδέξ, *no*, rhomboid.

<ρόμβοξ, ό : *no*, rhombus.

σημείον, τό : *no*, point.

σκαληνόξ -ή -όν : *adj*, scalene.

στερεόξ -ά -όν : *adj*, solid; τὸ στερεόν, *no*, solid, solid body.

στοιχείον, τό : *no*, element.

στρέφω, -στρέψω, >έστρεψα, —, έσταμμαι, έστάφηγ : *vb*, turn.

σύνκειμαι : *vb*, lie together, be the sum of, be composed; συγκείμενοξ -η -ον, *adj*, composed (ratio), compounded; see κείμει.

σύγκρίνω : *vb*, compare; see κρίνω.

συμβαίνω : *vb*, come to pass, happen, follow; see βαίνω.

συμβάλλω : *vb*, throw together, meet; see βάλλω.

σύμμετροξ -ον : *adj*, commensurable.

σύμπαξ -αντοξ, ό : *no*, sum, whole.

συνπίπτω : *vb*, meet together (of lines); see πίπτω.

συνπληρόω : *vb*, complete (a figure), fill in.

συνάγω : *vb*, conclude, infer; see >άγω.

συναμφότεροι -αι -α : *adj*, both together; ό συναμφότεροξ, *no*, sum (of two things).

συναποδείκνυμι : *no*, demonstrate together; see δείκνυμι.

συναφή, ή : *no*, point of junction.

σύνδυο, οί, αί, τά : *no*, two together, in pairs.

συνεθήςξ -έξ : *adj*, continuous; κατὰ τὸ συνεθεξ, continuously.

σύνθεσιξ -εὼς, ή : *no*, putting together, composition.

σύνθετοξ -ον : *adj*, composite.

συ[ν]ίστημι : *vb*, construct (a figure), set up together; *perf imperat pass 3rd sg*, συνεστάτω; see >ίστημι.

συντίθημι : *vb*, put together, add together, compound (ratio); see τίθημι.

σθέσιξ -εὼς, ή : *no*, state, condition.

σθήμα -ατοξ, τό : *no*, figure.

σφαίρα -αξ, ή : *no*, sphere.

τάχιξ -εὼς, ή : *no*, arrangement, order.

ταράσσω, ταράχω, —, —, τετάραγμαι, έταράθην : *vb*, stir, trouble, disturb; τεταραγμένοξ -η -ον, *adj*, disturbed, perturbed.

τάσσω, τάχω, >έταχα, τέταθα, τέταγμαι, έτάθην : *vb*, arrange, draw up.

τέλειοξ -α -ον : *adj*, perfect.

τέμνω, τεμνώ, >έτεμον, -τέμνηκα, τέμνημαι, έμήθην : *vb*, cut; *pres/fut indic act 3rd sg*, τέμει.

τεταρτημοριον, τό : *no*, quadrant.

τετράγωνοξ -ον : *adj*, square; τὸ τετράγώνον, *no*, square.

τετράκιξ : *adv*, four times.

τετραπλάσιοξ -α -ον : *adj*, quadruple.

τετράπλευροξ -ον : *adj*, quadrilateral.

τετραπλόοξ -η -ον : *adj*, fourfold.

τίθημι, θήσω, >έθηκα, τέθηκα, κείμει, έτέθην : *vb*, place, put.

τμήμα -ατοξ, τό : *no*, part cut off, piece, segment.

τοίνυν : *par*, accordingly.

τοιούτοξ -αύτη -ούτο : *pro*, such as this.

τομεύξ -έωξ, ό : *no*, sector (of circle).

τομή, ή : *no*, cutting, stump, piece.

τόποξ, ό : *no*, place, space.

τοσαντάκιξ : *adv*, so many times.

τοσανταπλάσιοξ -α -ον : *pro*, so many times.

τοσουτοξ -αύτη -ούτο : *pro*, so many.

τουτέστι = τουτ> >έστι : *par*, that is to say.

τραπέζιον, τό : *no*, trapezium.

τρίγωνοξ -ον : *adj*, triangular; τὸ τρίγώνον, *no*, triangle.

τριπλάσιοξ -α -ον : *adj*, triple, threefold.

τρίπλευροξ -ον : *adj*, trilateral.

τριπλ -όοξ -η -ον : *adj*, triple.

τρόποξ, ό : *no*, way.

τυγθάνω, τεύχομαι, >έτυθον, τετύθηκα, τέτευγμαι, έτεύθην : *vb*, hit, happen to be at (a place).

υ>πάρθω : *vb*, begin, be, exist; see >άρθω.

υ>πεχαίρεσιξ -εὼς, ή : *no*, removal.

υ>περβάλλω : *vb*, overshoot, exceed; see βάλλω.

υ>περοθή, ή : *no*, excess, difference.

υ>περέθω : *vb*, exceed; see >έθω.

υ>πόθεσιξ -εὼς, ή : *no*, hypothesis.

υ>πόκειμαι : *vb*, underlie, be assumed (as hypothesis); see κείμει.

υ>πολείπω : *vb*, leave remaining.

υ>ποτείνω, υ>ποτενῶ, υ>πέτεινα, υ>ποτέτακα, υ>ποτέταμαι, υ>πετάθην : *vb*, subtend.

<ύψοξ -εὼς, τό : *no*, height.

φανερόξ -ά -όν : *adj*, visible, manifest.

φημι, φήσω, >έφηγ, —, —, — : *vb*, say; >έφαμεν, we said.

φέρω, ο>ίσω, >ήνεγκον, ένήνοθα, ένήνεγμαι, ήνέθην : *vb*, carry.

θώριον, τό : *no*, place, spot, area, figure.

θωρίζ : *pre + gen*, apart from.

ψάω : *vb*, touch.

ώξ : *par*, as, like, for instance.

ώξ <έτυθεν : *par*, at random.

ώσαύτωξ : *adv*, in the same manner, just so.

<ώστε : *conj*, so that (causal), hence.